OSTERWALDER–SCHRADER AXIOMS FOR THE THREE-DIMENSIONAL ABELIAN HIGGS MODEL AT FIXED PHOTON MASS

Abstract. King constructed the continuum, infinite-volume limit of the three-dimensional $U(1)$ Higgs model along Neumann–Dirichlet cubes at block factor 5 [13, 14]. For his printed state at fixed photon mass we prove the Osterwalder–Schrader axioms: regularity E0″ with absolute constants, invariance under the full Euclidean group, reflection positivity about every plane, symmetry, and the measure with its ergodic decomposition, on whose components the cluster axiom holds in all its forms. The existence of King's cube limits, the existence clause of his Theorem 2.1, is proved from his renormalisation-group analysis in a ratio form in which the vacuum energies cancel; rotation invariance is proved by comparing King's cubes with periodic tori carrying a smooth mass wall on two lattices whose orientations differ by arctan(3/4). With real scalar sources the same holds jointly for the curvature and the renormalised Wick square on a tube of small Wick amplitudes, without a Euclidean clause. The curvature results hold on King's whole coupling range; the Wick results hold under a one-sided sign clause of the Wick leg, which we prove on explicit coupling regions. The limit of vanishing photon mass is left to later work.

1. Introduction and main results

1.1. **King's construction.** King [13, 14] constructed the three-dimensional $U(1)$ Higgs model, scalar electrodynamics with a massive photon, as a continuum and infinite-volume limit of lattice models. The lattice theory lives on $\Lambda_k = \varepsilon_k \mathbb{Z}^3$ with spacings $\varepsilon_k = 5^{-k}$; its action has a photon A in Feynman form with mass $\mu_0 > 0$, an N-component scalar ϕ with quartic coupling $\lambda > 0$ and bare mass $m > 0$, minimally coupled with charge $e > 0$, and one mass counterterm $\delta m^2_{\varepsilon_k}$. The counterterm is King's free-boundary one: it is computed with infinite-lattice propagators, so it is one number per spacing, the same at every site of every box and every torus. The ultraviolet problem is settled in [13] with Bałaban's renormalisation group [1, 2, 3, 4] at block factor 5: on a finite volume the effective actions converge as $k \to \infty$. The infinite-volume problem is treated in [14] with the correlation inequalities of Brydges, Fröhlich and Seiler [6]. King's finite-volume states are Neumann–Dirichlet states on the cubes $C_J = [-MJ, MJ)^3$, $M \geq 1$ his block integer: the photon gradient terms are kept only inside the cube (Neumann), and the scalar field vanishes on a boundary layer (Dirichlet). The limit is iterated, first $k \to \infty$ at fixed J, then $J \to \infty$; the second limit exists because the cube states are monotone in J on a cone of observables, by [6].

This paper proves the Osterwalder–Schrader axioms for King's printed state at fixed photon mass, with full Euclidean invariance, and the corresponding statements for the joint field of the curvature and the renormalised Wick square on a tube of small Wick amplitudes. It is self-contained in the following sense. The existence of King's cube limits, the existence clause of his Theorem 2.1 [14], is proved in the existence part of the paper (section 5, and section 6 for the version with scalar sources): King's route, the comparison of two spacings at a common block scale, is run in a ratio form in which the vacuum energies cancel, so that the convergence of the vacuum energies, which his printed argument uses and which is not available on the cube, is not needed. The passage from the cube limits to the axioms in [14] is asserted at three places, and at each we proceed differently. First, King's Neumann–Dirichlet cube form is not reflection symmetric about any plane, and its natural symmetrisation is not reflection positive (an exact counterexample at $e = 0$, Remark 9.7), so his reflection positivity goes through a modified box state; that this box state is reflection positive is his Proposition 3.3, which he derives from [6, Cor. 5.4], whose hypotheses do not cover the finite-box Gaussian of the modified action. Proposition 3.3 is true, and it is proved here (Lemma 9.3) rather than derived from [6, Cor. 5.4]. Second, the inequality by which he passes from the box states back to the cube limit holds in the reverse direction for the class of observables his argument needs: weakly, at every coupling, for the box state with the site-independent counterterm, and strictly at $e = 0$

(Lemma 9.6); it is replaced by a two-sided sandwich between cube states with a margin of order μ_0^{-1} (Lemmas 9.9 and 9.10). Third, the continuum subsequences through which he reaches the limit theory are built here (Remark 14.15). With scalar sources, finally, his Theorem 2.1 needs a vacuum counterterm that his Wick ordering does not contain; we work with a scheme a and prove the existence of the cube limits with sources in that scheme.

The authors' earlier paper [24] constructs torus limits of the same model at block factor 2 and proves the Osterwalder–Schrader axioms with cubic symmetry for them; its abstract measure theory (the reconstruction of the measure, its ergodic decomposition and the weak clustering of the components) is used here as it stands, cited by label. Rotation invariance is proved in the rotation part of this paper, whose main theorem (Theorem 15.1) precedes the main theorem of the paper: King's cubes are compared with periodic tori carrying a smooth mass wall, the walled tori on two lattices whose orientations differ by $\arctan(3/4)$ are compared, and King's state is identified as the lower envelope of the walled tori through a heavy-shell decoupling estimate on King's block-spin flow with the shell mass inside the matter covariance.

1.2. **The two cube conventions.** King's cube state $\mathrm{ND}^K_{k,J}$ (Definition 2.2) has its photon variables on $X_J = C_J \cap \Lambda_k$ and its scalar alive on $\{x \in X_J : \ x_\nu < MJ - \varepsilon_k \text{ for all } \nu\}$, that is, killed on the three top face layers T_J of the cube. The comparison cube state $\mathrm{ND}^{(a)}_{k,J}$ (Definition 2.1) has the scalar alive on all of X_J. The Wick-tube lemmas and the lattice comparisons of the rotation part are written for the comparison convention, while King's printed state is on his own. The two are interlaced: for observables of the cone class, with supports at ℓ^∞-distance > 1 from the complement of the cube,

$$\langle\Theta\rangle^K_{k,J} \ \le \ \langle\Theta\rangle^{(a)}_{k,J} \ \le \ \langle\Theta\rangle^K_{k,J+1},$$

and reversely for the opposite class (Lemma 4.2). Hence both families have the same J-limit at fixed k; the comparison family's fixed-cube limits inferior and superior are squeezed to King's printed state Z^K_∞ as $J \to \infty$, and every continuum-subsequence limit of the comparison family equals Z^K_∞ (Corollary 4.4). What does not transfer is the existence of the fixed-J limit $k \to \infty$ itself: on one convention it does not imply it on the other. This is why the existence theorems are proved on King's convention, where the printed state lives and where King's block-spin flow runs; the Wick tube, proved on the comparison convention, is carried to King's convention by a transfer theorem (Theorem 21.3).

1.3. **The main theorem.** Sources are real 2-forms $f_j \in C^\infty_c(\mathbb{R}^3;\Lambda^2)$ with coefficients z_j and real scalar test functions $h_j \in C^\infty_c(\mathbb{R}^3)$ with coefficients w_j; $F(f)$ is the curvature paired with f, $W^a(h)$ the renormalised Wick square in the scheme a, and $g(z) = \sum_j (\mathrm{Re}\, z_j) f_j$, $k(\zeta) = \sum_j (\mathrm{Re}\, w_j) h_j = k_+ - k_-$, with the norms $\|\cdot\|_2$, $|\cdot|_4$, $|\cdot|_w$ and the tube $\mathcal{D}_{\sigma_+,H}$ of Definitions 2.3 and 2.5. The Schwinger functions S_n, $S_{n,p}$ are the Taylor coefficients of the limit functionals at 0. A scheme a is admissible if $a = z$, the reference scheme, or $z_k - a_k$ converges (Definition 3.6). The following summary is restated with every clause in full and proved as Theorem 21.1.

Theorem 1.1 (Main Theorem, summary)**.** *Let $e > 0$, $\lambda, m, \mu_0 > 0$ and $M \ge 1$ be arbitrary.*

(A) The curvature field. *King's printed state $Z^K_\infty(f,0) = \lim_J \lim_k \langle e^{F(f)}\rangle^K_{k,J}$ exists for every real 2-form f, extends to an entire function of the coefficients of every finite family, and satisfies:*

(i) (OS0) $|Z^K_\infty(\sum_j z_j f_j, 0)| \le e^{\|g(z)\|_2^2}$ *for all $z \in \mathbb{C}^N$, with no dependence on μ_0, e, λ, m or M, and $|Z^K_\infty(\sum_j z_j f_j, 0)| \le \exp\big(\frac{1}{2}\mu_0^{-2}\|\sum_j (\mathrm{Re}\, z_j) d^* f_j\|_2^2\big)$; for real test functions $|S_n(f_1,\dots,f_n)| \le (\sqrt{2}e)^n (n!)^{1/2} \prod_j \|f_j\|_2 \le (e\pi/2)^n (n!)^{1/2} \prod_j |f_j|_4$; the S_n are tempered distributions and satisfy the regularity axiom* E0″ *with $s = 4$, the constant $\sigma_n = (e\pi)^n (n!)^{1/2}$ for complex $f_j \in \mathcal{S}$, and every $\beta > \frac{1}{2}$.*

(ii) (OS1) *Invariance under the full Euclidean group $\mathbb{R}^3 \rtimes O(3)$: under all translations of $\mathbb{R}^3$, under the cubic group O_h about every point, and under all rotations, F transforming as a 2-form; the S_n are Euclidean covariant.*

(iii) (OS2) *Reflection positivity about every plane* $\{x \cdot n = c\}$, n *a unit vector,* $c \in \mathbb{R}$, *in matrix form for exponentials with arbitrary complex curvature coefficients and in polynomial form.*
(iv) (OS3) S_n *is real and symmetric, and* $S_n = 0$ *for odd* n.
(v) (The measure) *For every even* $r \geq 14$ *there is exactly one Borel probability measure* μ_F *on* $\mathcal{H}^{-r}(\mathbb{R}^3; \Lambda^2)$ *with moments* S_n *and characteristic functional* $Z^K_\infty(\mathrm{i}\,\cdot, 0)$ *on* $\mathcal{S}$*; its image on* H^{-r}_{loc} *is determined by the characteristic functional and by the moments. The laws of the localised lattice fields* χF_{Λ_k} *under* $\mathrm{ND}^K_{k,J}$ *converge weakly on* $H^{-r}_{\mathrm{per}}(Q')$, Q' *a box containing* $\operatorname{supp}\chi$, *along the full sequence* $k \to \infty$ *at every fixed* $J \geq J_0(\chi)$, *and then as* $J \to \infty$, *to the law of* χF *under* μ_F.
(vi) (OS4 on the ergodic components) μ_F *is* $\mathbb{R}^3 \rtimes O(3)$*-invariant and* $\mu_F = \int \mu_\xi \, d\nu(\xi)$ *with translation-ergodic components;* ν*-almost every component is* $\mathbb{R}^3 \rtimes O(3)$*-invariant, reflection positive about every plane, clusters in the mean and under diagonal separation, has a unique joint vacuum in every direction, satisfies* OS4 *in all its forms for every choice of the time direction, and has real symmetric moments with a factorial growth bound with component constants; the decomposition is* O_h*-invariant, each component being a mixture of* $k \in \{1, 2, 3, 6\}$ *ergodic measures. The vanishing of the odd moments is not claimed for a component.*

(B) The joint field on the Wick tube. *Let the couplings satisfy the sign clause* (T3) *of Hypothesis 3.11,* $\sigma_1(M_1) > 0$ *for some* $M_1 \in (0, m)$*; in particular, let them lie in one of the explicit regions of Proposition 22.15. Let* a *be an admissible scheme. Then for every family of sources supported in* $Q_R = [-R, R]^3$ *the joint functional of* (F, W^a) *has a limit* G_∞ *on the tube* $\mathcal{D}_{\sigma_+, H}$, *with*

$$\sigma_+ = s_1, \qquad H = \tfrac{s_1}{16}, \qquad \kappa = \tfrac{C_U}{s_1},$$

and K_- *determined by the cell constant* B_{cell} *(Equation (30)), where* $s_1 \in (0, \sigma_1(M_1))$ *is the amplitude of the extensive bound and* C_U *its constant (Equation (28) and proposition 12.6); the constants depend on* $(e, \lambda, m, \mu_0, M, a)$ *and on the auxiliary choices* M_1, χ_0, s_1, *and on nothing else, and they are not uniform in* μ_0. *The joint Schwinger functions* $S_{n,p}$ *satisfy* (i)–(vi) *in the following form:*

(i) $|G_\infty(\zeta)| \leq \exp\big(\|g\|_2^2 + \kappa\|k_+\|_{L^1} + K_-|k_-|_w\big)$ *on* $\mathcal{D}_{\sigma_+, H}$, *and for real test functions* $|S_{n,p}| \leq e^{K'\sigma_0}(\sqrt{2}e)^n (n!)^{1/2} (e/\sigma_0)^p \, p! \prod_i \|f_i\|_2 \prod_j |h_j|_w$ *with* $\sigma_0 = H/4$ *and* $K' = \kappa\pi^2 + K_-$ *(a factor* 2^{n+p} *for complex test functions);* E0″ *holds with* $s = 4$ *and every* $\beta > 1$.
(ii) *Translations and* O_h *about every point hold as identities on the tube,* W^a *transforming as a scalar; there is no Euclidean clause.*
(iii) *Reflection positivity about every coordinate plane for families whose Wick amplitudes satisfy* $\mathsf{a}(\zeta_i) < \sigma_W = \sigma_+/8$, *in particular for all purely imaginary Wick coefficients, and the polynomial form.*
(iv) $S_{n,p} = 0$ *for odd* n.
(v), (vi) *As in* (A) *for the measure on* $\mathcal{H}^{-r}(\mathbb{R}^3; \Lambda^2 \oplus \mathbb{R})$, $r \geq 14$ *even, with the scheme-shifted lattice field in the convergence clause, and with the weak form of* OS4 *(ergodicity under* $\mathbb{R}^3$ *and clustering in coordinate directions) on* ν*-almost every component; there is no Euclidean clause.*

The same statements hold for the tube object G^K_∞ *of King's printed state on his own convention (Theorem 21.3 and proposition 21.4).*

The μ_0-free bound in (A)(i) is absolute: it depends on none of the parameters of the model. The μ_0^{-2} bound is the finite-volume Gaussian bound carried to the limit. Part (B) is the headline form of the Wick leg; the extension of (B) to arbitrary Wick amplitudes, under the hypothesis (W2) at every amplitude, with the only Euclidean clause of the Wick leg, is stated separately as Theorem 23.7 among the open problems.

1.4. **The massless limit.** The photon mass μ_0 is fixed throughout: King's construction, and every statement above, is at $\mu_0 > 0$, and the constants of the Wick leg are not uniform in μ_0. The limit of vanishing photon mass, along a family whose counterterm and scheme are frozen

at a reference mass, is not treated in this paper; it is listed among the open problems as later work.

1.5. **The coupling range.** The curvature leg holds on all of King's range: $e > 0$ and $\lambda, m, \mu_0 > 0$ arbitrary. Every smallness condition in [13, 14, 1, 2, 3, 4] is a cap on the running couplings at a stopping scale that is chosen after the point (e, λ, m, μ_0) and the datum, so the quantifier over the couplings is the one over the sources, and no sentence "e, λ sufficiently small" occurs (Lemma 22.1 and remark 22.16). No relation $\mu_0 \le m$ is used or needed. The only coupling condition in the paper is the one-sided sign clause (T3) of the Wick leg, $\sigma_1(M_1) > 0$ for some $M_1 \in (0, m)$; it fails exactly when m minimises King's free-boundary bare coefficient α_∞ on $(0, m]$, and to one-loop order it holds if and only if

$$e^2 < e_c^2(\lambda, m, \mu_0) = \max\Big\{ \frac{8\pi(m+\mu_0)^2}{m+2\mu_0}\Big(1 + \frac{\lambda(N+2)}{2\pi m}\Big),\ 4\pi(m+\mu_0)\Big(1 + \frac{\lambda(N+2)}{\pi m}\Big)\Big\}$$

(Equation (49)), so that a half-line of couplings fails it at one loop. Beyond one loop, (T3) is proved on explicit coupling regions, at $\mu_0 = m$ on a region containing $0 \le \lambda, e^2 \le 0.1m$ and at every $\mu_0 \in (0, m]$ on explicit boxes (Proposition 22.15); the Wick leg is stated on those regions, and more generally wherever (T3) holds. The mass clause in Bałaban's upper bound, a cap on the dimensionless scalar mass at the stopping scales where his Proposition 3.1 is used, is part of that proposition's statement and is one more one-sided cap on the stopping scale, not a condition on the couplings (Remark 22.2).

1.6. **What is proved where.** The inputs of the main theorem are collected in Section 3: the existence theorems for King's cube limits without and with scalar sources (Theorems 3.2 and 3.5), proved in the existence part; the rotation theorem (Theorem 3.13), proved in the rotation part; Bałaban's torus stability bounds and the c-number lemma in the form the Wick leg uses (Proposition 3.8 and lemma 3.9); and the coupling condition (T3). Everything else is proved in the sections between. Open are only the optional extensions: the Wick sector at arbitrary amplitudes ((W2), Form (A), Theorem 23.7), the relation of King's state to the torus states of [24] and its uniqueness and ergodicity, the limit of vanishing photon mass, and non-triviality; they are collected in the final section.

1.7. **Structure of the paper.** The model, King's cube, the comparison cube, the sources and the standing lemmas (the correlation inequalities of [6], with the two inequalities that are only sketched there proved in full, Gaussian domination, cutting and the elementary facts) come first, then the inputs with their provenance. The interlacing of the two cube conventions (Lemma 4.2 and corollary 4.4) and its three geometry margins follow, and then the existence part: King's existence theorem on the cube (Section 5) and its version with scalar sources. The curvature sector is then built up in the order of the axioms: regularity with the absolute constants (Theorems 7.4 and 7.6), translations, the cubic group and the rotation clause (Theorems 8.5 and 8.7), reflection positivity through the modified box and its passage to the limit with OS3 and the polynomial form (Lemma 9.3, theorem 10.6, and propositions 10.9 and 10.10), and the measure with its ergodic decomposition (Theorem 11.1). The Wick tube is treated in two sections, the torus inputs and the extensive bound (Propositions 12.2 and 12.6) and the uniform tube bound with its limits (Theorem 13.10 and lemma 13.11); continuum subsequences and the states they define, with no existence input, form a section of their own (Theorem 14.9). The rotation part follows: the geometry of the two lattices and the walled tori, the admissible schemes and the full object, the torus comparison and its reduction to one-sided bounds at one spacing, the closing estimate and the invariance theorems, and the heavy shell in two sections; a short section collects the corrections to King's papers that it uses. The main theorem is assembled in Theorem 21.1, and the coupling range is established in Lemma 22.1 and proposition 22.15. The paper ends with the open problems.

2. The model, King's cube and the standing lemmas

2.1. **Couplings, lattices and the action.** Throughout, $e > 0$ and $\lambda, m, \mu_0 > 0$ are the charge, the quartic coupling, the bare scalar mass and the photon mass; no further condition on the couplings is imposed, since every smallness condition of [13, 14, 1, 2, 3, 4] is a cap on the running couplings at a stopping scale chosen after the point (e, λ, m, μ_0) and after the datum (the section on the coupling range). The block factor is 5: the spacings are $\varepsilon_k := 5^{-k}$, the lattices $\Lambda_k := \varepsilon_k \mathbb{Z}^3$ with the standard frame (e_α), and $M \in \mathbb{N}$ is King's large-block integer, fixed once and for all. King's cubes are $C_J := [-MJ, MJ)^3$, $J \in \mathbb{N}$, and $X_J := C_J \cap \Lambda_k$ (the spacing understood from the context). A *King torus* is $\mathbb{R}^3/(\ell\mathbb{Z})^3$ with $\ell \in M\mathbb{N}$ even; Λ_k descends to it, with $\ell\, 5^k$ sites per side.

The scalar field is $\phi(x) = r_x e^{\mathrm{i}\vartheta_x} \in \mathbb{C}$ and the photon field a real $A_\alpha(x)$ for every site x and direction α; the bond $\langle x, x + \varepsilon e_\alpha\rangle$ carries $A_\alpha(x)$ and the reversed bond $-A_\alpha(x)$. King's action [14, (2.5)] at spacing ε is

$$S = \tfrac{1}{2}\sum_{\alpha,\beta}\sum_{x} \varepsilon \big(A_\alpha(x+\varepsilon e_\beta) - A_\alpha(x)\big)^2 + \frac{\mu_0^2}{2}\sum_x \varepsilon^3 |A(x)|^2 + \tfrac{1}{2}\sum_\beta\sum_x \varepsilon \left|e^{\mathrm{i}e\varepsilon A_\beta(x)}\phi(x+\varepsilon e_\beta) - \phi(x)\right|^2$$
$$+ \sum_x \varepsilon^3 \Big[\tfrac{1}{2}\big(m^2 + \delta m_\varepsilon^2\big) r_x^2 + \lambda r_x^4\Big],$$

the photon in Feynman form with mass μ_0, and the hopping term of a bond $\langle x, y\rangle$, $y = x + \varepsilon e_\beta$, equal to $\frac{\varepsilon}{2}(r_x^2 + r_y^2) - \varepsilon r_x r_y \cos(\vartheta_x - e\varepsilon A_\beta(x) - \vartheta_y)$: a *diagonal part* and a *cosine part*. The counterterm δm_ε^2 is King's free-boundary counterterm [14, p. 330], fixed by his two-point graphs of order at most four in (e, λ) at zero momentum with the propagators of the infinite lattice $\varepsilon\mathbb{Z}^3$: it is *one number per spacing*, the same at every site of every box and torus. The Wick constant is the free one, $\mathsf{W}_\varepsilon := (-\Delta_{\varepsilon\mathbb{Z}^3} + m^2)^{-1}(0,0)$. This *free convention* is used on cubes and tori alike unless said otherwise; King's own torus convention appears only inside the torus inputs of the Wick leg.

2.2. **The states.**

Definition 2.1 (Neumann–Dirichlet boxes, killed sites, the comparison cube)**.** For a finite set $X \subset \Lambda_k$ the *Neumann–Dirichlet state* $\langle\cdot\rangle_{k,X}^{\mathrm{ND}}$ is the normalised measure $Z^{-1}e^{-S}\prod_{x\in X} d^2\phi(x) \prod_{x\in X,\alpha} dA_\alpha(x)$ with $\phi = 0$ at every site outside X (Dirichlet condition: a bond leaving X contributes only its diagonal part $\frac{\varepsilon}{2}r_x^2$ at its end $x \in X$), photon gradient terms only for pairs $x, x + \varepsilon e_\beta$ both in X, and the photon mass term on X (Neumann condition). For $U \subset X$ the state *with the matter killed on* U replaces the single-site measure $d^2\phi(x)$ times its weight by the point mass at $\phi(x) = 0$ for $x \in U$; every hopping term with an endpoint in U then reduces to its diagonal part at the other endpoint, and the photon variables are unchanged. The *comparison cube* is $\mathrm{ND}_{k,J}^{(a)} := \langle\cdot\rangle_{k,X_J}^{\mathrm{ND}}$, written $\langle\cdot\rangle_{k,J}^{(a)}$: the scalar is alive on all of X_J, with Dirichlet planes $x_\mu = -MJ - \varepsilon_k$ and $x_\mu = MJ$.

Definition 2.2 (King's RG cube)**.** Let $\mathrm{alive}_J^K := \{x \in X_J : \ x_\mu < MJ - \varepsilon_k \ \forall\mu\}$ and $T_J := X_J \setminus \mathrm{alive}_J^K$, the three top face layers. *King's RG cube* $\mathrm{ND}_{k,J}^K$, written $\langle\cdot\rangle_{k,J}^K$, is the Neumann–Dirichlet state on X_J with the matter killed on T_J: the photon variables $A_\alpha(x)$, $x \in X_J$, their gradient terms for pairs in X_J and their mass term on X_J are those of the comparison cube; the scalar is alive on alive_J^K and $\phi = 0$ on every other site of Λ_k, in particular on T_J. Its Dirichlet planes are $x_\mu = -MJ - \varepsilon_k$ and $x_\mu = MJ - \varepsilon_k$, at separation $2MJ$; the Neumann image planes of the photon, $x_\mu = -MJ - \varepsilon_k/2$ and $x_\mu = MJ - \varepsilon_k/2$, are the same in both conventions.

The two conventions differ exactly by the matter on T_J; the photon Gaussian is the same. King's RG cube is the box [14, (3.3)–(3.4)] with the Dirichlet planes of [14, (3.9)] and the photon reflection planes of [14, (3.7)], the geometry of his renormalisation group; his Theorem 3.2 and his modified box [14, (3.38)] are written in the comparison convention. The two families are interlaced in Section 4; the existence theorems of this paper are posed on ND^K, while the rotation part and the cell lemmas of the Wick leg are written for the comparison family. The

periodic state $\langle\cdot\rangle_{k,T}$ on a King torus T has the action summed over the finite graph $\Lambda_k/(\ell\mathbb{Z})^3$, in the free convention.

BFS measures. A *BFS measure* on a finite set V of sites is a probability measure

$$Z^{-1}\exp\Big(\sum_{\{x,y\}\in B} c_{xy}\, r_x r_y \cos\big(\vartheta_x-\vartheta_y-\ell_{xy}(A)\big)\Big)\prod_{x\in V} d\rho_x(r_x)\, d\vartheta_x\; d\gamma(A),$$

with B a set of pairs of sites, $c_{xy}\ge 0$, each ρ_x a finite measure on $[0,\infty)$ with $\int e^{\alpha r^2}d\rho_x<\infty$ for all $\alpha\ge 0$ (point masses at 0, that is killed sites, included), γ a non-degenerate centred Gaussian measure on $\mathbb{R}^m$ (the photon variables), and linear functionals ℓ_{xy}. Every state above is a BFS measure: $c_{xy}=\varepsilon$ for the bonds present, the single-site measures contain $e^{-\varepsilon^3[\frac12(m^2+\delta m_\varepsilon^2)r^2+\lambda r^4]}$ and the diagonal parts $\frac{\varepsilon}{2}r_x^2$ of all bonds at x (those leading out of the box or into a killed site included); the photon form is positive definite because $\mu_0>0$, and the exponential moments are finite because $\lambda>0$. So is every measure obtained from such a state by multiplying some c_{xy} or some photon gradient terms by $t\in[0,1]$, adding $s|\ell(A)|^2$ ($s\ge 0$) to the action, multiplying single-site measures by $e^{\varepsilon^3 q(x)r^2}$ with real q, or killing sites. A BFS measure is *of King type* if $c_{xy}\in[0,\varepsilon]$ and each ρ_x is a point mass at 0 or has the density $r\,e^{-\varepsilon^3[\frac12(m^2+\delta m_\varepsilon^2)r^2+\lambda r^4]-\varepsilon^3 w(x)r^2-\frac{\varepsilon}{2}n_x r^2}$ with $w(x)\ge 0$, $n_x\in\{0,\dots,6\}$; every state above, and every measure obtained from one by the operations just listed with $q\le 0$, is of King type. Every state above is invariant under the *charge conjugation* $(A,\vartheta)\mapsto(-A,-\vartheta)$, which changes the sign of every field strength and fixes r.

2.3. **Sources, schemes and the objects.**

Definition 2.3 (sources, schemes, classes, transport)**.** A *curvature source* is a real 2-form $f\in C_c^\infty(\mathbb{R}^3;\Lambda^2)$, a *scalar source* a real $h\in C_c^\infty(\mathbb{R}^3)$. For the plaquette $p=(x;\alpha<\beta)$ with centre $c(p)$ let $F(p):=\varepsilon^{-1}(A_\alpha(x)+A_\beta(x+\varepsilon e_\alpha)-A_\alpha(x+\varepsilon e_\beta)-A_\beta(x))$ and

$$F_{\Lambda_k}(f):=\sum_p \varepsilon^3 f_{c(p)}(e_\alpha,e_\beta)F(p),\qquad \mathsf{S}_{\Lambda_k}(h):=\sum_x \varepsilon^3 h(x),\qquad :\phi^2:(h):=\sum_x \varepsilon^3 h(x)\big(|\phi(x)|^2-2\mathsf{W}_\varepsilon\big),$$

the sums over the plaquettes (corners in the box), respectively the sites, of the region. A *scheme* is a real sequence $a=(a_k)$, a_k depending only on ε_k and the couplings; $W^a(h):=\sum_x\varepsilon^3 h(x)(|\phi(x)|^2-2\mathsf{W}_\varepsilon-a_k)$. For a finite family $\zeta=(z_j,f_j;w_j,h_j)_j$ with complex coefficients put $\Theta(\zeta):=\exp(\sum_j z_j F(f_j)+\sum_j w_j{:}\phi^2{:}(h_j))$ and, for a Neumann–Dirichlet state on a box X (killed sites allowed),

$$Z^a_{k,X}(\zeta):=\Big\langle\exp\Big(\sum_j z_j F(f_j)+\sum_j w_j W^a(h_j)\Big)\Big\rangle^{\mathrm{ND}}_{k,X}=e^{-\sum_j w_j a_k \mathsf{S}_{\Lambda_k}(h_j)}\big\langle\Theta(\zeta)\big\rangle^{\mathrm{ND}}_{k,X};$$

$Z^a_{k,J}:=Z^a_{k,X_J}$ for the comparison cube, $Z^{a,K}_{k,J}$ for King's RG cube, and $N^a_{k,X}(0,q):=e^{-(a_k+2\mathsf{W}_\varepsilon)\mathsf{S}(q)}\langle e^{\sum_x\varepsilon^3 q(x)r_x^2}\rangle^{\mathrm{N}}_{k,}$ for one real scalar source. At $h=0$ the multiplier is 1 and a is dropped: $Z_{k,J}(z):=\langle e^{\sum_j z_j F(f_j)}\rangle^{(a)}_{k,J}$, $Z^K_{k,J}(z):=\langle e^{\sum_j z_j F(f_j)}\rangle^K_{k,J}$ for $z\in\mathbb{C}^n$. The *cone* is $\mathcal{K}:=\{(f,g): f \text{ a real 2-form},\ g\ge 0\}$ with the observable $\Theta(\mathrm{i}f,g)$ (*class* $\mathcal{I}$); *class* $\mathcal{D}$ consists of the observables $\Theta(f,-g)$, $(f,g)\in\mathcal{K}$. Transport: $(O\cdot h)(x):=h(O^{-1}x)$, $(O\cdot f)_x(u,v):=f_{O^{-1}x}(O^{-1}u,O^{-1}v)$ for $O\in O(3)$; $\tau_b h:=h(\cdot-b)$, $(\tau_b f)_x:=f_{x-b}$. For $\nu\in\{1,2,3\}$, $c\in\mathbb{R}$ let $\rho=\rho_{\nu,c}$, $x\mapsto x-2(x_\nu-c)e_\nu$; $\theta h:=h\circ\rho$, $\theta f:=\rho^* f$, $(\rho^* f)_x(u,v):=f_{\rho x}(\rho u,\rho v)$ (the components $f_{\mu\nu}$, $\mu\ne\nu$, change sign); $\theta\zeta^*:=(\bar z_j,\rho^* f_j;\bar w_j,h_j\circ\rho)_j$, and $\zeta\sqcup\zeta'$ is the union of two families. The *amplitude* is $\mathsf{a}(\zeta):=\sum_j|\operatorname{Re}w_j|\,\|h_j\|_\infty$. The *reference scheme* is $z=a^\sharp$, $a^\sharp_k:=e^2X_{\varepsilon_k}$, X_ε the coefficient of e^2 in the perturbative expansion of $\langle|\phi(0)|^2\rangle$ on $\varepsilon\mathbb{Z}^3$ at $\lambda=0$ with the order-e^2 part of δm_ε^2 inserted.

By charge conjugation $\langle\Theta(\mathrm{i}f,g)\rangle=\langle\cos F(f)\,e^{:\phi^2:(g)}\rangle$ is real for every state above, and $Z^a_{k,X}(-z,w)=Z^a_{k,X}(z,w)$.

Remark 2.4 (schemes). The Wick leg is stated for the reference scheme z, or for an admissible scheme a (definition 3.6: z_k-a_k converges, hence is bounded). If the fixed-cube limits exist in

$(0,\infty)$ for both a and z at a real scalar source, the scheme is admissible: for $0 \le h \in C_c^\infty$ with $\int h > 0$ supported well inside some C_J, $Z^z_{k,J}(0,h)/Z^a_{k,J}(0,h) = e^{-(z_k-a_k)\mathsf{S}_{\Lambda_k}(h)}$ converges to a number in $(0,\infty)$, and $\mathsf{S}_{\Lambda_k}(h) \to \int h > 0$. What the Wick leg uses is only $\sup_k |z_k - a_k| < \infty$.

Definition 2.5 (norms and the tube)**.** For a family with sources $f_1,\dots,f_n$, $h_1,\dots,h_p$ supported in $Q_R := [-R,R]^3$ and $\zeta = (z,w) \in \mathbb{C}^{n+p}$ put $g(\zeta) := \sum_i (\operatorname{Re} z_i) f_i$ and $k(\zeta) := \sum_j (\operatorname{Re} w_j) h_j = k_+ - k_-$ (pointwise positive and negative parts). Lattice norms: $\|g\|^2_{2,k} := \sum_p \varepsilon_k^3 g_{c(p)}(e_\alpha, e_\beta)^2$ over the plaquettes of Λ_k, $\|q\|_{1,k} := \sum_{x\in\Lambda_k} \varepsilon_k^3 |q(x)|$. Continuum norms: $\|g\|_2^2 := \sum_{\alpha<\beta} \|g_{\alpha\beta}\|^2_{L^2}$; $|q|_w := \sup_x (1+|x|^2)^2 |q(x)|$; $|q|_4 := \sup_{x,|\gamma|\le 4} (1+|x|^2)^2 |D^\gamma q(x)|$ and $|g|_4 := (\sum_{\alpha<\beta} |g_{\alpha\beta}|_4^2)^{1/2}$, as in [24]. For $\sigma, H > 0$ the *tube* is

$$\mathcal{D}_{\sigma,H} := \big\{ \zeta \in \mathbb{C}^{n+p} : \ \|k_+(\zeta)\|_\infty < \sigma/2, \ \ \|k_-(\zeta)\|_\infty < H/2 \big\}.$$

Since $s \mapsto \|(\sum_j s_j h_j)_\pm\|_\infty = \sup_x (\pm \sum_j s_j h_j(x))_+$ is a supremum of convex functions of $s = \operatorname{Re} w$, hence convex and continuous, $\mathcal{D}_{\sigma,H}$ is the preimage under $\zeta \mapsto \operatorname{Re} w$ of the intersection of two open convex sublevel sets: open, convex, hence connected; it contains 0 and every ζ with purely imaginary w; $\|k(\zeta)\|_\infty < \min(\sigma, H)/2$ implies $\zeta \in \mathcal{D}_{\sigma,H}$; its real points form an open subset of $\mathbb{R}^{n+p}$. The tube of the Wick leg is $\mathcal{D}_{\sigma_+,H}$, with σ_+ and H fixed there.

The objects. For real 2-forms $f_1,\dots,f_n$ let $J_0(f)$ be the least J such that every support lies at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_J$. King's *cube limits* and *printed state* are

$$Z^K_J(z) := \lim_{k\to\infty} Z^K_{k,J}(z) \quad (J \ge J_0(f)), \qquad Z^K_\infty\Big(\sum_j z_j f_j, 0\Big) := \lim_{J\to\infty} Z^K_J(z),$$

whenever the limits exist: their existence at real z is theorem 3.2, and their existence at every $z \in \mathbb{C}^n$, entire in z, is derived from it in the curvature-sector section. The object is scheme-free. For general sources in a scheme a the *tube object* is $G_J(\zeta) := \lim_k Z^a_{k,J}(\zeta)$, $G_\infty(\zeta) := \lim_J G_J(\zeta)$, $\zeta \in \mathcal{D}_{\sigma_+,H}$, whenever these limits exist (the Wick-leg sections); where the full object $Z^a_\infty := \lim_J \lim_k Z^a_{k,J}$ of the optional extension (theorem 23.7) exists it agrees with G_∞ on the tube, being the same iterated limit.

2.4. **Elementary facts.**

Lemma 2.6 (elementary facts)**.** (E1) *For $\varepsilon \le 1$ and any translate Γ of $\varepsilon\mathbb{Z}^3$: $\sum_{x\in\Gamma} \varepsilon^3 (1+|x|^2)^{-4} \le C_0 := (5/2)^4\pi^2/8 < 49$ and $\sum_{x\in\Gamma} \varepsilon^3 (1+|x|^2)^{-2} \le C_0' := (5/2)^2\pi^2 < 62$.* (E2) *$\|q\|_{L^1} \le \pi^2 |q|_w$, $\|q\|_{L^2} \le \frac{\pi}{2\sqrt{2}} |q|_4$, $\|q\|_\infty \le |q|_w \le |q|_4$, $\|g\|_2 \le \frac{\pi}{2\sqrt{2}} |g|_4$; $|\cdot|_w$ is a norm, so $|k(\zeta)|_w \le \sum_j |\operatorname{Re} w_j| |h_j|_w$ and $|k_\pm|_w \le |k|_w$.* (E3) *Sampling is evaluation at points: it commutes with pointwise positive and negative parts, and a sampled sup is at most the continuum sup, so $\zeta \in \mathcal{D}_{\sigma,H}$ for the continuum sources implies the same bounds for the sampled ones. For continuous compactly supported g, q: $\|g\|_{2,k} \to \|g\|_2$, $\|q\|_{1,k} \to \|q\|_{L^1}$, and for every k: $\|g\|^2_{2,k} \le C_0 |g|_4^2$, $\|q\|_{1,k} \le C_0' |q|_w$.* (E4) *$n^{n/2} \le e^{n/2} (n!)^{1/2}$.*

Proof. (E1) For y in the cube of side ε centred at $x \in \Gamma$, $|x-y|^2 \le \frac{3}{4}$ and $|y|^2 \le 2|x|^2 + 2|x-y|^2 \le 2|x|^2 + \frac{3}{2}$, so $1+|y|^2 \le \frac{5}{2}(1+|x|^2)$ and $(1+|x|^2)^{-r} \le (5/2)^r (1+|y|^2)^{-r}$. Integrate over the cube and sum over the disjoint cubes: $\sum_{x\in\Gamma} \varepsilon^3 (1+|x|^2)^{-r} \le (5/2)^r \int_{\mathbb{R}^3} (1+|y|^2)^{-r} dy$, with $\int (1+|y|^2)^{-4} = 4\pi \cdot \frac{\pi}{32} = \frac{\pi^2}{8}$ and $\int (1+|y|^2)^{-2} = 4\pi \cdot \frac{\pi}{4} = \pi^2$. (E2) $\|q\|_{L^1} \le |q|_w \int (1+|y|^2)^{-2}$, $\|q\|^2_{L^2} \le |q|_4^2 \int (1+|y|^2)^{-4}$; the rest is immediate. (E3) Riemann sums for the convergences; the plaquette centres of one orientation form a translate of $\varepsilon_k\mathbb{Z}^3$, so (E1) bounds $\sum_{p:(\alpha,\beta)} \varepsilon_k^3 g_{c(p)}(e_\alpha, e_\beta)^2$ by $C_0 |g_{\alpha\beta}|_4^2$, and the three orientations sum to $C_0 |g|_4^2$; the last bound is (E1) with $r = 2$. (E4) is $n^n \le e^n n!$. □

Lattice-to-continuum. On a box X let $\langle A, B\rangle_k := \sum_{x\in X} \sum_\alpha \varepsilon^3 A_\alpha(x) B_\alpha(x)$, and for a real 2-form g whose non-vanishing plaquettes have their corners in X define $d_k^* g$ by $F_{\Lambda_k}(g) = \langle d_k^* g, A\rangle_k$. Each component $(d_k^* g)_\gamma$ is a backward difference quotient of the sampled components $g_{\gamma\beta}$, $\beta \ne \gamma$;

for smooth compactly supported g it converges uniformly to $(d^*g)_\gamma$, up to the orientation sign convention of the components, at the rate $O(\varepsilon_k)$ of Taylor's theorem. Hence

$$\|d_k^*g\|_{2,k} \to \|d^*g\|_{L^2}, \qquad \sup_k \|d_k^*g\|_{2,k} < \infty, \qquad \|g\|_{2,k} \to \|g\|_2 \qquad (k \to \infty). \tag{1}$$

2.5. **Correlation inequalities.** For a BFS measure, a real linear functional F of A and $g \geq 0$ on V put $\Theta_{F,g} := e^{\mathrm{i}F} e^{\sum_x g_x r_x^2}$, $\Theta^-_{F,g} := e^F e^{-\sum_x g_x r_x^2}$, and $\langle X; Y\rangle := \langle XY\rangle - \langle X\rangle\langle Y\rangle$. Since $F(f)$ is a real linear functional of A and $e^{:\phi^2:(g)}$ a positive constant times $e^{\sum_x \varepsilon^3 g(x) r_x^2}$, class $\mathcal{I}$ consists of multiples of observables $\Theta_{F,g}$ and class $\mathcal{D}$ of multiples of observables $\Theta^-_{F,g}$.

Lemma 2.7 (correlation inequalities)**.** *Let $\langle\cdot\rangle$ be a BFS measure, F a real linear functional of A, $g \geq 0$, $\Theta = \Theta_{F,g}$, $\Theta^- = \Theta^-_{F,g}$. Then $\langle\Theta\rangle$ is real, and*

(a) *$\langle\Theta; |\ell(A)|^2\rangle \leq 0$ for every complex linear functional ℓ of A; hence adding $s|\ell(A)|^2$ to the action, $s \geq 0$, does not decrease $\langle\Theta\rangle$, and deleting a photon gradient term does not increase it;*

(b) *$\langle\Theta; r_x r_y \cos(\vartheta_x - \vartheta_y - \ell_{xy}(A))\rangle \geq 0$; hence decreasing c_{xy} does not increase $\langle\Theta\rangle$;*

(c) *$\langle\Theta; r_x^2\rangle \geq 0$; hence multiplying ρ_x by e^{-sr^2}, $s \geq 0$, does not increase $\langle\Theta\rangle$, and killing a set U of sites does not increase it, provided $g = 0$ on U or the measure is of King type;*

(d) *for Θ^- the three inequalities and the three monotonicity statements hold with the opposite signs;*

(e) *if the pairs in B and the Gaussian γ do not couple the fields at the sites of $X \subset V$ (and the photon variables attached to them) to the others, the measure is a product, and for an observable depending only on the fields in X the expectation equals the expectation in the X-factor.*

Proof. (a)–(c) are [6, Thms. 6.1, 6.2(1),(2)], whose hypotheses are those of a BFS measure: an arbitrary finite lattice, $c_{xy} \geq 0$, finite single-site measures on $[0,\infty)$ with $\int e^{\alpha r^2} d\rho_x < \infty$ for all $\alpha \geq 0$ (a point mass at 0 is one), an arbitrary centred Gaussian with covariance $C \geq 0$, and $g \geq 0$ (so $g = 0$ is included). Theorem 6.1 states $\langle P(r)\cos(n\cdot\vartheta + f\cdot A); Q(r)\cos(m\cdot\vartheta + h\cdot A)\rangle \geq 0$ for polynomials P, Q in the r_x with positive coefficients; (a), (b) follow as in the printed proof of 6.2(1),(2): $e^{\sum g_x r_x^2}$ is a limit of such polynomials, $e^{\mathrm{i}F}$ may be replaced by $\cos F$ because the expectation and both observables are invariant under $(A,\vartheta) \mapsto (-A,-\vartheta)$ (which also gives the realness), and $|\ell(A)|^2 = \ell_1(A)^2 + \ell_2(A)^2$ with $\ell_r(A)^2 = \lim_{t\downarrow 0} t^{-2}[1-\cos(t\ell_r(A))]$; (c) is Theorem 6.1 with $Q = r_x^2$ and $m = 0$, $h = 0$. Each monotonicity statement follows by differentiating in the parameter, every interpolating measure being a BFS measure. Killing: if $g = 0$ on U, set $c_{xy} = 0$ on every bond at U (not increasing, by (b)); the measure is then a product (e) of the matter at U, which carries no observable and cancels, and of the state with the matter killed on U (the diagonal parts at the other ends of the bonds into U are already in their single-site measures). For King type in general, multiply ρ_x, $x \in U$, by $e^{-s\varepsilon^3 r^2}$ (not increasing) and let $s \to \infty$ by Lemma 2.9.

(d) is stated in [6, Thm. 6.2(3),(4)] for the photon and the hopping inequality with a sketched proof, and its r^2-inequality is [14, (3.28)], stated there without proof; we prove all three. Let μ be the unnormalised measure and $\omega' = (r', \vartheta', A')$ an independent copy of $\omega = (r, \vartheta, A)$; then $(\int d\mu)^2\langle\Theta^-; Y\rangle = \iint \Theta^-(\omega)(Y(\omega) - Y(\omega'))\, d\mu(\omega) d\mu(\omega')$. Write $r = u - v$, $r' = u + v$ (so $u_x \geq |v_x|$), $\vartheta = \vartheta_+ + \vartheta_-$, $\vartheta' = \vartheta_+ - \vartheta_-$, $A = a_+ + a_-$, $A' = a_+ - a_-$. Then $a_\pm = (A \pm A')/2$ are independent with the common centred Gaussian law of covariance $C/2$, and the radial law $\prod_x d\rho_x(u_x - v_x) d\rho_x(u_x + v_x)$ on $\{u \geq |v|\}$ is invariant under $v_x \mapsto -v_x$ at each site. With $\chi_{xy} := \vartheta_{+,x} - \vartheta_{+,y} - \ell_{xy}(a_+)$ and $\psi_{xy} := \vartheta_{-,x} - \vartheta_{-,y} - \ell_{xy}(a_-)$ the unprimed phase is $\chi + \psi$, the primed one $\chi - \psi$, and

$$\begin{aligned} r_x r_y \cos(\chi+\psi) + r'_x r'_y \cos(\chi-\psi) &= 2(u_x u_y + v_x v_y)\cos\chi\cos\psi + 2(u_x v_y + v_x u_y)\sin\chi\sin\psi, \\ r_x r_y \cos(\chi+\psi) - r'_x r'_y \cos(\chi-\psi) &= -2(u_x v_y + v_x u_y)\cos\chi\cos\psi - 2(u_x u_y + v_x v_y)\sin\chi\sin\psi, \end{aligned}$$

$r_x^2 - r_x'^2 = -4u_x v_x$, $\ell(A)^2 - \ell(A')^2 = 4\ell(a_+)\ell(a_-)$ for real ℓ, and $\Theta^-(\omega) = e^{F(a_+)} e^{F(a_-)} e^{-\sum_x g_x(u_x^2+v_x^2)} e^{2\sum_x g_x u_x v_x}$. Expand $\exp(\sum_b c_b[\text{first line}])$ and $e^{2\sum g_x u_x v_x}$ in power series (absolutely convergent at fixed spacing). Every term of the integrand is a sign σ times a monomial in (u, v) with non-negative coefficient, times the even radial factor $e^{-\sum g_x(u_x^2+v_x^2)}$, times $V(\vartheta_+, a_+)V(\vartheta_-, a_-)$ with one real function V (a product of $e^{F(\cdot)}$, of factors $\cos\chi_b$ or $\sin\chi_b$, and of $\ell(\cdot)$ for the photon slot). The photon integral of $V(\vartheta_+, a_+)V(\vartheta_-, a_-)$ factorises because a_+, a_- are independent and identically distributed. The angular integral factorises by the following argument: per site $d\vartheta\, d\vartheta' = 2\, d\vartheta_+ d\vartheta_-$, and $[0, 2\pi)^2$ in the (ϑ, ϑ')-variables is a fundamental domain of the lattice L generated by (π, π) and $(\pi, -\pi)$ in the $(\vartheta_+, \vartheta_-)$-plane, which contains $(2\pi\mathbb{Z})^2$ with index 2; $V(\vartheta_+, \cdot)V(\vartheta_-, \cdot)$ is L-invariant, because the simultaneous shift of $\vartheta_{+,x}$ and $\vartheta_{-,x}$ by π at one site changes every χ_b and ψ_b with $b \ni x$ by π and leaves $\cos\chi_b \cos\psi_b$ and $\sin\chi_b \sin\psi_b$ unchanged; hence $\int_{[0,2\pi)^2} VV\, d\vartheta\, d\vartheta' = 2\int_{L\text{-domain}} VV\, d\vartheta_+ d\vartheta_- = \int_{[0,2\pi)^2} VV\, d\vartheta_+ d\vartheta_- = (\int_0^{2\pi} V)^2$. The radial integral of a monomial vanishes if its degree in some v_x is odd, by the $v \mapsto -v$ symmetry, and is non-negative otherwise, since $u_x \geq |v_x|$. So every term has the sign σ: $\sigma = -1$ for $Y = r_x^2$ (from $-4u_x v_x$) and for the hopping term (second line), $\sigma = +1$ for $Y = \ell(A)^2$ (for complex ℓ, real and imaginary parts). Hence $\langle \Theta^-; r_x^2 \rangle \leq 0$, $\langle \Theta^-; r_x r_y \cos(\cdot) \rangle \leq 0$, $\langle \Theta^-; |\ell(A)|^2 \rangle \geq 0$. Monotonicity and killing follow as in (a)–(c), with (d)'s hopping inequality in place of (b) and Lemma 2.9, which covers Θ^-, for King type. (e) is clear. □

Lemma 2.8 (current representation and Gaussian domination)**.** *Let $\langle\cdot\rangle$ be a BFS measure whose Gaussian has covariance C. Integrating out the matter at fixed A, the matter integral is $\Psi(A) = \sum_q w(q) \cos\langle \kappa(q), A\rangle$, q running over the integer-valued divergence-free currents on B, $w(q) \geq 0$, $\sum_q w(q) < \infty$, $\langle \kappa(q), A\rangle := \sum_{xy} q_{xy}\ell_{xy}(A)$. Let $\mathbb{E}^\sharp$ be the probability measure on currents with weights proportional to $w(q)e^{-\langle \kappa(q), C\kappa(q)\rangle/2}$. Then for $v \in \mathbb{R}^m$, $s \in \mathbb{R}$,*

$$\langle e^{s\langle v, A\rangle} \rangle = e^{s^2\langle v, Cv\rangle/2}\, \mathbb{E}^\sharp \cos\big(s\langle Cv, \kappa\rangle\big) \leq e^{s^2\langle v, Cv\rangle/2}.$$

Proof. Each bond factor expands as $e^{c_{xy} r_x r_y \cos t} = \sum_{q\in\mathbb{Z}} I_q(c_{xy} r_x r_y) e^{\mathrm{i}qt}$ with $I_q \geq 0$, $I_{-q} = I_q$, $\sum_q I_q(c) = e^c$; after the radial integrals the multiple series converges absolutely, dominated by $\prod_x e^{\frac{1}{2}\sum_y c_{xy} r_x^2}$, integrable against the single-site measures. Integrating the angles forces $\sum_y q_{xy} = 0$ at every site (at a killed site $r_x = 0$ and $I_q(0) = \delta_{q0}$, so the currents vanish on its bonds); pairing q with $-q$ gives Ψ with $w(q) \geq 0$. For a centred Gaussian γ with covariance C and $a \in \mathbb{C}^m$, $\int e^{\langle a, A\rangle} d\gamma = e^{\langle a, Ca\rangle/2}$; with $a = sv \pm \mathrm{i}\kappa$ and the average of the two signs, $\int e^{s\langle v, A\rangle} \cos\langle \kappa, A\rangle\, d\gamma = e^{s^2\langle v, Cv\rangle/2 - \langle \kappa, C\kappa\rangle/2} \cos(s\langle v, C\kappa\rangle)$. Sum over q with the weights $w(q)$, divide by the same sum at $s = 0$ (positive), and use $|\cos| \leq 1$. □

The lemma applies to every state of this paper at every k, J, $\mu_0 > 0$, with or without a real scalar source absorbed in the single-site measures (the photon covariance is unchanged).

Lemma 2.9 (killing)**.** *Let $\langle\cdot\rangle$ be a BFS measure of King type, U a set of alive sites, $\chi : U \to (0, \infty)$, and $\langle\cdot\rangle_{U,s}$ the measure with ρ_x multiplied by $e^{-s\varepsilon^3\chi(x) r_x^2}$ for $x \in U$. Let Θ be continuous with $|\Theta(A, \phi)| \leq e^{\ell(A)} e^{C\sum_x r_x^2}$ for a real linear functional ℓ and $C \geq 0$. Then $\langle\Theta\rangle_{U,s} \to \langle\Theta\rangle_U$ as $s \to \infty$, $\langle\cdot\rangle_U$ the measure with ρ_x replaced by the point mass at 0 for $x \in U$ (the state with the matter killed on U).*

Proof. The unnormalised density is $e^{-Q(A)/2} dA \prod_{x \text{ alive}} w_x(r_x) r_x dr_x d\vartheta_x \prod_b e^{c_b r_x r_y \cos(\cdot)} \prod_{x\in U} e^{-s\varepsilon^3\chi(x) r_x^2}$, Q positive definite, killed sites carrying no variables. Substitute $r_x = (s\varepsilon^3\chi(x))^{-1/2} u_x$ for $x \in U$; the Jacobians cancel between numerator and denominator. *Domination for $s \geq s_0$:* at $x \in U$, $w_x(r_x) \leq e^{\varepsilon^3|\delta m_\varepsilon^2| r_x^2/2}$ (the other exponents in w_x are ≤ 0), $|\Theta|$ contributes $e^{Cr_x^2}$, and each of the at most six bond weights at x is at most $e^{\frac{\varepsilon}{2} r_x^2 + \frac{\varepsilon}{2} r_y^2}$ ($0 \leq c_b \leq \varepsilon$); in the variable u_x these factors together are $\leq e^{u_x^2/2}$ once $s \geq s_0(\varepsilon, m, \delta m_\varepsilon^2, C, \min\chi)$, so the U-factor is at most $e^{-u_x^2/2} u_x$, integrable on $[0, \infty) \times [0, 2\pi)$; at an alive $y \notin U$ the factor is at most $w_y(r_y) e^{(C+3\varepsilon) r_y^2} r_y$, integrable since $\lambda > 0$; the photon factor $e^{\ell(A)} e^{-Q(A)/2}$ is integrable since Q is positive definite. *Pointwise limit:* at fixed $(u, \vartheta, A, r_{V\setminus U})$, $r_x \to 0$ for $x \in U$, so $\Theta \to \Theta|_{\phi_U = 0}$ by continuity, every bond

weight with an endpoint in U and every $w_x(r_x)$, $x \in U$, tends to 1, and the U-factor tends to $e^{-u_x^2} u_x\, du_x d\vartheta_x$, whose integral cancels between numerator and denominator. By dominated convergence the ratio tends to $\int \Theta|_{\phi_U=0} d\mu_U / \int d\mu_U$, where μ_U has the cosines at U removed, the weights w_y (which already contain all diagonals at y) and the photon unchanged, and the point mass at U: that is $\langle\cdot\rangle_U$, with a positive denominator. □

Lemma 2.10 (cutting)**.** *Let $\langle\cdot\rangle_Y$ be a Neumann–Dirichlet box state on Λ_k, possibly with killed sites, or a periodic state on a King torus, and let X be a set of sites of Y such that X and its nearest neighbours lie in one fundamental cell (a clause in torus language: for a box state it has no content beyond $X \subset Y$). Let $\langle\cdot\rangle_X^{\mathrm{ND}}$ be the Neumann–Dirichlet state on X whose single-site measures (killed sites and the diagonal parts of the bonds leaving X included) are those of $\langle\cdot\rangle_Y$. Let $(f,g) \in \mathcal{K}$ be such that every plaquette and every site at which f, respectively g, does not vanish lies in X. Then, at every lattice spacing,*

$$\langle\Theta(\mathrm{i}f,g)\rangle_X^{\mathrm{ND}} \le \langle\Theta(\mathrm{i}f,g)\rangle_Y, \qquad \langle\Theta(f,-g)\rangle_X^{\mathrm{ND}} \ge \langle\Theta(f,-g)\rangle_Y,$$

and the same for Z^a. The first inequality uses Lemma 2.7(a),(b),(e) only; the second uses (d),(e). If no site of X is killed, $\langle\cdot\rangle_X^{\mathrm{ND}}$ is the state of Definition 2.1.

Proof. Delete from Y every photon gradient term coupling a site of X to a site outside X, and the cosine parts of all hopping terms between X and its complement, keeping their diagonal parts $\frac{\varepsilon}{2}(r_x^2 + r_y^2)$. By Lemma 2.7(a),(b) this does not increase the first expectation, by (d) it does not decrease the second, and every intermediate measure is a BFS measure. The result is a product (Lemma 2.7(e)) whose X-factor is $\langle\cdot\rangle_X^{\mathrm{ND}}$: the diagonal parts of the cut bonds are exactly the Dirichlet terms and the remaining photon terms the Neumann ones of X; the observable depends on the X-variables only. The free Wick constant, δm_ε^2 and the scheme multiplier are the same on both sides. □

With $Y = X_{J+1}$, $X = X_J$ this is King's monotonicity in the volume [14, Thm. 3.2, (3.31)–(3.33)] between comparison cubes, in the form his proof proves: with one Wick constant for all J, the free subtraction King himself switches to at the start of that proof [14, p. 333]. With the Dirichlet Wick constant of [14, (2.13)], which increases with J, the c-number factor moves against the monotone factor and the monotonicity does not follow; that form is not used. The box version with killed sites and the monotonicity of King's RG cubes are in Section 4.

2.6. **Algebra of the scheme, covariance, reflection, continuity.**

Lemma 2.11 (algebra of the scheme)**.** *Let a be a scheme and Z^a the functional of a state above on a box at spacing ε_k.*

(a) *For real coefficients w the multiplier $e^{-\sum_j w_j a_k \mathsf{S}(h_j)}$ is a positive constant depending only on k, w and $\mathsf{S}_{\Lambda_k}(h_j)$. Hence the inequalities of Lemmas 2.7 and 2.10 between two states on the same lattice with the same sources hold for Z^a in the same direction.*

(d) $|Z^a(\zeta)| \le Z^a(\mathrm{Re}\,\zeta)$, and $\ln Z^a$ is convex in the real coefficients.

(e) $Z^a(f,h)\, Z^a(-f,-h) \ge 1$ for real (f,h).

(f) For real f and real $k_1, k_2 \ge 0$: $Z^a(f, k_1 - k_2) \le Z^a(2f, -2k_2)^{1/2}\, Z^a(0, 2k_1)^{1/2}$.

Proof. (a) is immediate. (d) $|\Theta(\zeta)| = \Theta(\mathrm{Re}\,\zeta)$ and the multiplier has modulus $e^{-\sum_j (\mathrm{Re}\, w_j) a_k \mathsf{S}(h_j)}$; convexity is Hölder's inequality. (e) $\langle e^X\rangle\langle e^{-X}\rangle \ge 1$ by Cauchy–Schwarz, and the multipliers cancel. (f) Cauchy–Schwarz gives $\langle e^{F + :\phi^2:(k_1) - :\phi^2:(k_2)}\rangle \le \langle e^{2F - 2:\phi^2:(k_2)}\rangle^{1/2} \langle e^{2:\phi^2:(k_1)}\rangle^{1/2}$, and $e^{-a_k \mathsf{S}(k_1 - k_2)} = (e^{2a_k \mathsf{S}(k_2)})^{1/2} (e^{-2a_k \mathsf{S}(k_1)})^{1/2}$. □

Lemma 2.12 (passive covariance)**.** *Let $O \in O(3)$, Λ a lattice of spacing ε with frame (f_α), and X a box or a torus for Λ. Equip $O\Lambda$ with the frame (Of_α) and OX with the transported boundary conditions. Then $\langle\Theta(z\,O\cdot f, w\,O\cdot h)\rangle_{O\Lambda, OX} = \langle\Theta(zf, wh)\rangle_{\Lambda, X}$ and $\mathsf{S}_{O\Lambda}(O\cdot h) = \mathsf{S}_\Lambda(h)$, likewise for families and for Z^a. On a torus the periodic model does not change if the frame (f_α) is replaced by $(\pm f_{\pi(\alpha)})$, π a permutation.*

Proof. The map $\phi'(Ox) = \phi(x)$, $A'_\alpha(Ox) = A_\alpha(x)$, components taken in the frames (f_α) and (Of_α), is a bijection of the configuration spaces preserving Lebesgue measure; the action is

written in the frame, and δm_ε^2, W_ε, a_k depend on ε only; plaquettes, centres and field strengths are transported, and $(O\cdot f)_{Oc}(Of_\alpha, Of_\beta) = f_c(f_\alpha, f_\beta)$. A permutation of the frame relabels the components; reversing f_α replaces $A_\alpha(x)$ by $-A_\alpha(x-\varepsilon f_\alpha)$, a bijection of the torus variables preserving the action, which changes the sign of both $F(p)$ and $f_{c(p)}(f_\alpha, f_\beta)$ and preserves the set of plaquette centres. □

Lemma 2.13 (the reflection sandwich at fixed spacing, comparison convention)**.** *Let* $\sigma_3 = \mathrm{diag}(1,1,-1)$ *and let* $(f,g) \in \mathcal{K}$ *be supported at* ℓ^∞*-distance* > 1 *from* $\mathbb{R}^3 \setminus C_{J-1}$*. Then for every* $k \geq 0$

$$Z^a_{k,J-1}(\mathrm{i}f, g) \leq Z^a_{k,J}(\mathrm{i}\,\sigma_3\cdot f, \sigma_3\cdot g) \leq Z^a_{k,J+1}(\mathrm{i}f, g),$$

and the reversed inequalities hold for $Z^a(f,-g)$*.*

Proof. σ_3 preserves Λ_k and S_{Λ_k} and maps X_J onto $X_+ := (C_J + \varepsilon_k e_3) \cap \Lambda_k$. By Lemma 2.12, $Z^a_{k,J}(\mathrm{i}\sigma_3\cdot f, \sigma_3\cdot g)$ is the functional with sources (f,g) of the Neumann–Dirichlet state on X_+ in the frame $(e_1, e_2, -e_3)$. Written in the frame (e_α), with $A'_3(x) = -A_3(x-\varepsilon e_3)$, this is the BFS measure R in which ϕ, A_1, A_2 live on X_+ and A_3 on $X_+ - \varepsilon e_3 = X_J$, with Dirichlet conditions for ϕ on X_+, photon gradient and mass terms within these sets, and the hopping term in direction 3 the standard one of the bond $\langle y-\varepsilon e_3, y\rangle$; the plaquette terms $(x;\alpha,3)$ pick up two sign changes and keep their centres, so the observable is $\Theta(\mathrm{i}f,g)$ in the standard frame. R is obtained from the Neumann–Dirichlet state on X_{J+1} by deleting photon gradient terms and hopping cosines and factorising, since $X_+ \cup X_J \subset X_{J+1}$, and the state on X_{J-1} is obtained from R in the same way, since $X_{J-1} \subset X_J \cap X_+$; the diagonal parts of the cut bonds are the Dirichlet terms of the smaller state, and the sources stay inside X_{J-1}. Lemma 2.7(a),(b),(e), respectively (d),(e), give the directions, and Lemma 2.11(a) carries them to Z^a. □

Lemma 2.14 (continuity of convex functions)**.** *Let* U *be a non-empty open convex subset of a Fréchet space and* $\Phi : U \to \mathbb{R}$ *a convex Borel function. Then* Φ *is continuous on* U*.*

Proof. U is a Baire space and $U = \bigcup_{j\in\mathbb{N}}\{\Phi \leq j\}$, so some $S := \{\Phi \leq j\}$ is non-meagre in U; being Borel, S has the Baire property [12]: $O \setminus \mathcal{M} \subset S$ for a non-empty open $O \subset U$ and a meagre $\mathcal{M}$. Pick $x_0 \in O$ and a balanced open neighbourhood W of 0 with $x_0 + W + W \subset O$. For $w \in W$ and $y := x_0 + w$, $y + W \subset O$, and the point reflection $\iota(x) := 2y - x$ maps $y+W$ onto itself and preserves meagre sets; so $S \cap (y+W)$ and $\iota(S \cap (y+W))$ are comeagre in $y+W$ and intersect: there is $u \in S$ with $2y - u \in S$, and $\Phi(y) \leq \frac{1}{2}(\Phi(u) + \Phi(2y-u)) \leq j$. Hence $\Phi \leq j$ on $x_0 + W$. A convex function on an open convex set bounded above near one point is bounded above near every point $y \in U$ (choose $z \in U$ with y strictly between x_0 and z, and take the convex combinations of $x_0 + W$ with z), and a convex function locally bounded above is continuous [22]. □

3. The inputs

This section states, in the form in which the later sections consume them, the results on which the theorems of this paper rest, and says where each is proved. Three are theorems of this paper proved in later parts: the existence of King's cube limits, without and with scalar sources (theorems 3.2 and 3.5; the existence part, sections 5 and 6), and the rotation theorem (theorem 3.13; the rotation part, section 15, whose main theorem is theorem 15.1). Two are the torus inputs of the Wick leg, Bałaban's stability bounds and the c-number lemma (proposition 3.8 and lemma 3.9), established in appendix F. One is a condition on the couplings, the sign clause (T3) (Hypothesis 3.11), proved on explicit coupling regions in proposition 22.15; it is the only condition on the couplings anywhere in the paper. The optional extension of the Wick leg to arbitrary amplitudes, together with the hypotheses it needs ((W2), (R_W), (FA^K)), is stated in the final section (section 23); no clause of the main theorem uses them. From the literature, the paper uses the correlation inequalities of [6], the chessboard estimate of [8], the reconstruction theorems of [17, 18], Bałaban's torus analysis [1, 2, 3, 4] and King's papers [13, 14], each at the place where it is applied; the general lemmas on the measure, its ergodic decomposition and weak clustering are taken from the authors' earlier paper on the torus limits [24] (remark 3.16).

3.1. **Objects and conventions.** The two cube conventions of Definitions 2.2 and 2.1 are King's convention ND^K (the scalar field killed on the three top face layers T_J of the cube C_J, the convention of the state King prints) and the comparison convention $\mathrm{ND}^{(a)}$ (the scalar alive on all of X_J), in which the tube results (Theorem 13.10, Lemma 13.11) and the lattice comparisons of the rotation part are written. The two are interlaced by Lemma 4.2. Which convention a statement is posed on is part of it:

- The existence theorems theorems 3.2 and 3.5 are posed on King's convention ND^K, on which King's renormalisation-group flow runs (remark 5.3). By Corollary 4.4 the existence of the fixed-cube limits on one convention does not imply it on the other; the comparison family's fixed-cube limits are reached through that corollary, and the statements proved for the comparison family under the assumption that they exist (definition 3.1) serve as templates for King's convention.
- The rotation theorem theorem 3.13 is proved in the rotation part for King's printed state Z^K_∞ on King's convention (theorem 15.1) and is carried to the limit states of the comparison family through Corollary 4.4; its Wick-sector form is posed on the comparison family (theorems 18.9 and 18.11).
- Proposition 3.8 and lemma 3.9 and (T3) are statements about King's tori (Definition 12.1) and about c-numbers.

Throughout, $Z^a_{k,J}(z,w)$ denotes the generating functional of Definition 2.3 with curvature coefficients z and Wick coefficients w in the scheme a, on the convention in question (written $Z^{a,K}_{k,J}$ when the convention ND^K is to be stressed); $z = a^\sharp$ is the reference scheme of Remark 2.4; $\langle\,\cdot\,\rangle^K_{k,J}$ is King's cube expectation at spacing ε_k.

Definition 3.1 (convergence of the comparison family at fixed cubes)**.** The comparison family is *convergent at fixed cubes* if for every finite family $f_1,\dots,f_n$ of real 2-forms in $C^\infty_c(\mathbb{R}^3;\Lambda^2)$, every $J \geq J_0(f)$ and every $x \in \mathbb{R}^n$ the limit $\lim_k \langle e^{\sum_j x_j F(f_j)}\rangle^{(a)}_{k,J}$ exists and lies in $(0,\infty)$.

This is the clause of theorem 3.2 read for the family $\mathrm{ND}^{(a)}$. It is not a result of this paper and is not needed for the main theorem: the statements below that assume it are proved for the comparison family as templates, and their forms for King's printed state are proved under theorem 3.2; the continuum-subsequence section (theorem 14.9) removes the dependence on the fixed-cube limits altogether.

3.2. **Existence of King's cube limits.**

Theorem 3.2 (King's cube limits, curvature)**.** *For every finite family $f_1,\dots,f_n$ of real 2-forms in $C^\infty_c(\mathbb{R}^3;\Lambda^2)$, every $J \geq J_0(f)$, where $J_0(f)$ is the least J such that every* $\operatorname{supp} f_j$ *lies at ℓ^∞-distance greater than 1 from $\mathbb{R}^3 \setminus C_J$, and every $z \in \mathbb{R}^n$, the limit*

$$\lim_{k\to\infty} \langle e^{\sum_j z_j F(f_j)}\rangle^K_{k,J}$$

exists and lies in $(0,\infty)$.

Proved in section 5 (theorem 5.9), from King's renormalisation-group analysis of the cube.

Remark 3.3 (on theorem 3.2). (i) *Existence clause only.* Theorem 3.2 is used only through the existence of the fixed-cube limit at real coefficients: no constants clause, no uniformity in J and no locally uniform convergence in z enter from it. Every bound used on the limit is proved here, with absolute constants, in Theorem 7.4; the monotonicity in the cube, the J-limit, the entire extension and the translations are proved in Lemma 4.2, Theorem 7.4, Lemma 7.7 and Theorem 8.7. (ii) *Origin.* This is the existence clause of King's Theorem 2.1 [14] at zero scalar source, for the state he prints. The proof in section 5 follows King's route, the comparison of two spacings at a common block scale, in a ratio form in which the vacuum energies cancel; what it uses of Bałaban's block-factor-5 analysis [1, 2, 3, 4] is stated there. The mass clause of [2, Prop. 3.1], upstream of it, is part of that proposition's statement and is cited under it (Remark 22.2); it is not a condition of this paper. (iii) *Consumers.* The whole curvature

sector (Theorems 7.4, 8.7, 10.6, 11.1), the identification of the LIM states with King's state (Corollary 14.13) and the main theorem (Theorem 21.1).

Remark 3.4 (the cube). The cube on which the existence theorem is proved is King's RG cube C_J^K of definition 2.2, with King's killing of the scalar on the top layers T_J; it is the only cube on which King's block-spin flow exists (remark 5.3). Every placement of a comparison-cube result on King's cube goes through Lemma 4.2 and its corollaries.

Theorem 3.5 (King's cube limits with scalar sources)**.** *Let $z = a^\sharp$ be the reference scheme and let $s_W := s_1$ be the amplitude of Proposition 12.6. For every finite family of real 2-forms $f_j \in C_c^\infty(\mathbb{R}^3;\Lambda^2)$ and real functions $h_i \in C_c^\infty(\mathbb{R}^3)$, every $J \geq J_0(f,h)$ and every real coefficient vector (z,w) whose positive scalar part $k_+ := \big(\sum_i w_i h_i\big)_+$ satisfies $2\|k_+\|_\infty < s_W$, the condition of the tube $\mathcal{D}_{\sigma_+,H}$ with $\sigma_+ = s_W$ (the negative scalar part is unrestricted), the limit $\lim_{k\to\infty} Z^z_{k,J}(z,w)$ exists and lies in $(0,\infty)$.*

Proved in section 6 (theorem 6.17 and corollary 6.18), by the route of section 5 with the scheme c-number carried along.

Definition 3.6 (admissible schemes)**.** A scheme a (Remark 2.4) is *admissible* if $a = z$ or $\lim_k(z_k - a_k)$ exists. For an admissible scheme theorem 3.5 holds for $Z^a_{k,J}$ in place of $Z^z_{k,J}$ (Lemma 2.11(a)), and $\sup_k |z_k - a_k| < \infty$.

Remark 3.7 (on theorem 3.5). (i) At zero Wick coefficient theorem 3.5 is theorem 3.2. (ii) *Origin.* Theorem 3.5 is *not* King's Theorem 2.1 with scalar sources: as printed, in King's Wick ordering, that statement fails, and the state with scalar sources exists only after a vacuum counterterm, which is what the scheme supplies (Remark 2.4). The threshold on the positive scalar part is the one at which the positive-source bound of the Wick leg is available (Proposition 13.2); what happens above it is the subject of the optional extension (Hypothesis 23.1). (iii) One instance of theorem 3.5 is named separately because the statements on continuum subsequences need it alone: ($\mathrm{B}_{\mathrm{cell}}$) (lemma 14.4; one cube, one negative Wick source). (iv) *Consumers.* The Wick tube (Proposition 12.6, Theorem 13.10), its transfer to King's convention (Theorem 21.3), the Wick sector of Theorem 14.9, and Form (B) of the main theorem (Theorem 21.1, Proposition 21.4).

3.3. The torus inputs of the Wick leg.

Proposition 3.8 (torus stability at two scalar masses)**.** *For the scalar masses $M = m$ and $M = M_1$ (the auxiliary mass fixed in Proposition 12.6) there are constants $E_\pm(M) < \infty$, a tail K_1 and, for each $k \geq K_1$, a $J_*(k)$ such that the source-free partition functions of King's tori T (Definition 12.1) at spacing ε_k and at scalar mass M satisfy, in King's torus convention,*

$$e^{-E_-(M)|T|} \leq Z^{(M)}_{k,T} \leq e^{E_+(M)|T|} \qquad (k \geq K_1,\ J \geq J_*(k)).$$

These are Bałaban's lower and upper stability bounds, [1, (1.14)] and [2, (1.3)], at a fixed scalar mass, read at the two masses m and M_1 and in King's torus normalisation of the counterterm; the form used here is established in Appendix F (theorem F.1). Proposition 3.1 of [2], behind the upper bound, is stated under a cap on the dimensionless scalar mass at the scales at which it is applied; it is cited under that cap, one more one-sided cap on the stopping scale and no condition on the bare mass (Remark 22.2). Only a tail in k and, for each k, large J are used (Proposition 12.6, Theorem 13.10).

Lemma 3.9 (the c-number input)**.** *Let $\alpha_k^\infty(M) = M^2 + \delta m_k^2(M)$ be King's free-boundary bare coefficient at scalar mass M and spacing ε_k, in the reference scheme z. For the auxiliary mass M_1 of Proposition 12.6, the quantities*

$$\sigma_1(M_1) := \liminf_{k\to\infty} \tfrac{1}{2}\big(\alpha_k^\infty(m) - \alpha_k^\infty(M_1)\big) \qquad \text{and} \qquad \beta_1(M_1)$$

are finite, where $\beta_1(M_1)$ is the second constant entering Proposition 12.6.

The proof is given in Appendix F (theorem F.159).

Remark 3.10 (on lemma 3.9). (i) This is a statement about lattice Feynman integrals: the difference of two mass counterterms at the same photon mass. The limit $\alpha_\infty(M) = \lim_k \alpha_k^\infty(M)$ and its description as a function of the couplings are discussed in Lemma 22.1. (ii) *Consumers.* Proposition 12.6 (the constants $\sigma_+ = s_1$, H, κ of the tube) and the coupling range, Lemma 22.1.

Hypothesis 3.11 ((T3): the coupling condition)**.** $\sigma_1(M_1) > 0$ for some $M_1 \in (0, m)$. The couplings (e, λ, m, μ_0) are said to lie in the *coupling region* of this paper when (T3) holds. By Proposition 22.15 the coupling region contains the explicit regions listed there: at $\mu_0 = m$ a region containing $0 \le \lambda, e^2 \le 0.1m$, and at every $\mu_0 \in (0, m]$ explicit boxes.

This is the condition on the couplings under which the Wick leg of the main theorem is stated; the curvature leg carries no condition on the couplings.

Remark 3.12 (on (T3)). (i) (T3) is the only condition on the couplings anywhere in this paper, and it is one-sided: it fails exactly on the set $E_W^<$ of couplings (e, λ, m, μ_0) at which m minimises $M \mapsto \alpha_\infty(M)$ on $(0, m]$ (Lemma 22.1). At one loop it holds if and only if

$$e^2 < e_c^2(\lambda, m, \mu_0) = \max\Big\{ \frac{8\pi(m+\mu_0)^2}{m+2\mu_0}\Big(1 + \frac{\lambda(N+2)}{2\pi m}\Big),\ 4\pi(m+\mu_0)\Big(1 + \frac{\lambda(N+2)}{\pi m}\Big)\Big\},$$

N the number of real components of the scalar field (equation (49)); so at one loop a half-line of values of e^2 fails. Beyond one loop, (T3) is *proved* on explicit coupling regions (Proposition 22.15): at $\mu_0 = m$ on a region containing $0 \le \lambda, e^2 \le 0.1m$, and at every $\mu_0 \in (0, m]$ on explicit boxes; the main theorem is stated on the region where it holds. The two-sided variant and the set $E_W \subseteq E_W^<$ are described in Lemma 22.1. (ii) *Consumers.* Proposition 12.6 and Lemma 22.1.

3.4. **The rotation theorem.** For a family f of real 2-forms and $J \ge J_0(f)$ let L_J^f be the set of locally uniform limits on $\mathbb{C}^N$ of subsequences of $(a_{k,J}^{(a)})_k$, $a_{k,J}^{(a)}(z) = \langle e^{\sum_j z_j F(f_j)} \rangle_{k,J}^{(a)}$ (the comparison family; by Lemma 7.3 and Montel's theorem the set is non-empty).

Theorem 3.13 (rotation invariance of the comparison family's limit states)**.** *For every finite family f of real 2-forms in $C_c^\infty(\mathbb{R}^3; \Lambda^2)$ and every choice $\zeta_J \in L_J^f$, $J \ge J_0(f)$, the limit $Z_\infty^{\mathrm{LIM}}(\sum_j z_j f_j, 0) := \lim_{J\to\infty} \zeta_J(z)$ exists locally uniformly on $\mathbb{C}^N$, is entire and does not depend on the choices. The functional $Z_\infty^{\mathrm{LIM}}(\cdot, 0)$ is invariant under the Euclidean group $\mathbb{R}^3 \rtimes O(3)$: $Z_\infty^{\mathrm{LIM}}(\gamma \cdot f, 0) = Z_\infty^{\mathrm{LIM}}(f, 0)$ for every Euclidean motion γ and every complex coefficient vector, F transforming as a 2-form.*

Proved in the rotation part: theorem 15.1 gives the $\mathbb{R}^3 \rtimes O(3)$-invariance of King's printed state $Z_\infty^K(\cdot, 0)$, and by Corollary 4.4(iii), with the fixed-cube limits of theorem 3.2, every J-limit of subsequential fixed-cube limits of the comparison family exists and equals $Z_\infty^K(\cdot, 0)$, which is the statement as posed. King's cubes are compared with periodic tori carrying a smooth mass wall, the walled tori on two lattices whose orientations differ by $\arctan(3/4)$ are compared (theorem 17.1), and King's state is identified as the lower envelope of the walled tori (theorem 18.4) through a heavy-shell decoupling estimate on King's block-spin flow (theorem 20.2).

Remark 3.14 (on theorem 3.13). (i) The theorem is posed for the comparison family and for the J-limit of its subsequential fixed-cube limits; its proof goes through King's printed state, whose invariance theorem 15.1 proves directly, and Corollary 4.4 identifies $Z_\infty^{\mathrm{LIM}}(\cdot, 0)$ with King's printed state $Z_\infty^K(\cdot, 0)$, which is thereby $\mathbb{R}^3 \rtimes O(3)$-invariant (Theorem 8.7). (ii) *Consumers.* The $O(3)$ clause of the curvature sector (Theorem 8.7), the Euclidean clause of the measure's ergodic components (Theorem 11.1(c)), and the $O(3)$ form of the LIM theorem (Theorem 14.9).

Remark 3.15 (the Wick tube on King's convention). Theorem 13.10 is proved for the comparison convention $\mathrm{ND}^{(a)}$. Its transfer to King's convention — the (k, J)-uniform tube bound (31) for $Z_{k,J}^{a,K}$ and the fixed-cube limits on the tube — is Theorem 21.3, with the kill step of Remark 4.8, the cell statement of Lemma 4.7 and the direction check for the negative leg as its convention-sensitive steps; Form (B) of the main theorem then holds on King's convention (Proposition 21.4).

3.5. **The optional extension.** The extension of the Wick leg to arbitrary amplitudes (Form (A), theorem 23.7) is stated in the final section, with the three statements it needs: (W2) (Hypothesis 23.1; the fixed-cube limits and the positive-source bound at every amplitude), (R_W) (Hypothesis 23.3; its Euclidean clause) and (FA^K) (Hypothesis 23.5; its King-convention form). No clause of theorem 21.1 uses them. What this paper proves of (W2) is its cone instance below the threshold $\sigma_1(M_1)$ (corollary 13.3), which is all that Form (B) needs.

3.6. **Auxiliary statements of the section on continuum subsequences.** The statements on continuum subsequences (Theorem 14.9) are formulated in terms of three named properties proved in that section: the cell bound ($\mathrm{B}_{\mathrm{cell}}$) (lemma 14.4; the boundedness form of theorem 3.5 at one cube and one negative Wick source), the scheme bound (SCH) (lemma 14.5; $a = z$ or $\sup_k |z_k - a_k| < \infty$) and the independence of the subsequence (UNIQ) (proposition 14.6; any two continuum subsequences give the same LIM state). The first two hold for every admissible scheme under (T3), and the third follows from the rotation theorem; the three are collected in Proposition 14.12. The limit of vanishing photon mass is not treated in this paper (section 23).

3.7. **Results of the earlier paper used as inputs.**

Remark 3.16 (results of the torus paper used as inputs). The following results of the authors' earlier paper on the torus limits [24] are used under the hypotheses they carry there, which are matched to King's state at the places where they are applied (Theorem 11.1, Proposition 22.15): its abstract chain from a limit functional with the properties of Theorem 7.4 and the reflection positivity of Theorem 10.6 to the measure, its ergodic decomposition and weak clustering — Theorems 10.10, 10.19, 10.20, Propositions 10.4, 10.16, 10.17(iii), Lemmas 10.3, 10.5, Corollaries 10.43(iv), 10.44, Theorem 10.12, Definition 2.12, Proposition 8.3, Proposition 7.9(a), Lemmas 7.3, 7.7 and the lemmas of its Appendix G (the two-loop majorants behind Proposition 22.15). Of [6], Theorems 6.1 and 6.2(1),(2) are cited as printed; items (3),(4) of its Theorem 6.2, a sketch in print, are a lemma of this paper (Lemma 2.7). The chessboard estimate is Theorem 4.1/4.2 of [8], applied after its hypotheses are checked (Proposition 12.2). Nothing of [13, 14] is used as a result: King's existence theorem is proved in sections 5 and 6, and his reflection-positivity argument for the modified box and his passage to subsequences are replaced by proofs in this paper.

4. The two cube conventions: interlacing and margins

King's printed state lives on King's convention $\mathrm{ND}^K_{k,J}$ (definition 2.2); the rotation part and the Wick-tube estimates of this paper use the comparison convention $\mathrm{ND}^{(a)}_{k,J}$ (definition 2.1). We show that the two families are interlaced in J (lemma 4.2), draw the consequences for the limits (corollaries 4.3 and 4.4), and show that the comparisons of a cube with a reflected cube, a larger box and a tiling by cells carry over to King's convention with the printed margins (lemmas 4.5 to 4.7).

4.1. **The two conventions and the support condition.** Throughout, $\varepsilon = \varepsilon_k = 5^{-k}$, $X_J = C_J \cap \Lambda_k = \{-MJ, \dots, MJ - \varepsilon\}^3$, $T_J = \{x \in X_J : \ x_\mu = MJ - \varepsilon \text{ for some } \mu\}$ and $\mathrm{alive}^K_J = X_J \setminus T_J = \{-MJ, \dots, MJ - 2\varepsilon\}^3$. Both states carry the same photon variables $A_\alpha(x)$, $x \in X_J$, with the Neumann form of X_J, so their photon Gaussians are identical and every plaquette observable is the same function of the shared variables. They differ exactly by the matter on T_J: alive in the comparison convention (scalar Dirichlet planes $x_\mu = -MJ - \varepsilon$, $x_\mu = MJ$), killed in King's (planes $x_\mu = -MJ - \varepsilon$, $x_\mu = MJ - \varepsilon$, $2MJ$ apart, a multiple of every block scale; cf. remark 3.4). A bond with one end alive and the other killed or outside the site set contributes its diagonal part $\frac{\varepsilon}{2} r_x^2$ at the alive end, so an alive site carries six diagonals in either convention. In the proof of his Theorem 3.2 and in his box form (3.38), King [14] works in the comparison convention: every site of the region is alive there.

For (f, g) with f a real 2-form and $g \geq 0$ a scalar, both in C_c^∞, we write as in definition 2.3 $\Theta(\mathrm{i}f, g) = e^{\mathrm{i}F(f) + W(g)}$ (class $\mathcal{I}$), $\Theta(f, -g) = e^{F(f) - W(g)}$ (class $\mathcal{D}$), $W(g) = \sum_x \varepsilon^3 g(x)(|\phi(x)|^2 - 2\mathsf{W}_\varepsilon)$ (free Wick constant), and $Z^a_{k,J}(\mathrm{i}f, g) = e^{-a_k \mathsf{S}(g)} \langle \Theta(\mathrm{i}f, g) \rangle$, $Z^a_{k,J}(f, -g) = e^{a_k \mathsf{S}(g)} \langle \Theta(f, -g) \rangle$,

$\mathsf{S}(g) = \sum_x \varepsilon^3 g(x)$; a superscript K ($Z^{a,K}_{k,J}$) marks King's convention. All these expectations are real. The sources lie inside X_J if

(S) every plaquette $p = (x;\alpha,\beta)$ of Λ_k with $f_{c(p)}(e_\alpha, e_\beta) \neq 0$ has its four corners in X_J, and every site $x \in \Lambda_k$ with $g(x) \neq 0$ lies in $\mathrm{alive}^K_J = X_J \setminus T_J$.

If $\operatorname{supp} f \cup \operatorname{supp} g \subset \{y : \operatorname{dist}_\infty(y, \mathbb{R}^3 \setminus C_J) > 1\}$, as every hypothesis of this paper demands of its sources, then (S) holds: a site with $g(x) \neq 0$ has $-MJ + 1 < x_\mu < MJ - 1 \le MJ - \varepsilon$, so $x \in \mathrm{alive}^K_J$; a plaquette with $f_{c(p)} \neq 0$ has its centre at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_J$ and its corners within $\varepsilon/2 \le \frac{1}{2}$ of it, hence in X_J. Also (S) for X_{J_0} implies (S) for every $J \ge J_0$.

4.2. **Cut-outs and the interlacing lemma.**

Lemma 4.1 (cut-outs from a box with killed sites)**.** *Let $\langle\cdot\rangle_Y$ be the state of definition 2.1 on a finite set $V \subset \Lambda_k$ of sites with the matter killed on $T \subset V$, and $X \subset V$ any subset. Let (f,g), $g \ge 0$, satisfy* (S_X)*: every plaquette with $f \neq 0$ has its four corners in X and every site with $g \neq 0$ lies in X (g on $X \cap T$ is immaterial). Then deleting from $\langle\cdot\rangle_Y$ the photon gradient terms coupling X to $V \setminus X$ and the cosine parts of the hopping terms between X and $V \setminus X$ yields a product whose X-factor is $\langle\cdot\rangle^{(a),X\cap T}_X$, the state of definition 2.1 on X with the matter killed on $X \cap T$; consequently*

$$\langle\Theta(\mathrm{i}f,g)\rangle^{(a),X\cap T}_X \;\le\; \langle\Theta(\mathrm{i}f,g)\rangle_Y, \qquad \langle\Theta(f,-g)\rangle^{(a),X\cap T}_X \;\ge\; \langle\Theta(f,-g)\rangle_Y,$$

and the same for Z^a. In particular, (a) *if $X \cap T = \emptyset$ the factor is $\langle\cdot\rangle^{(a)}_X$;* (b) *if $X = \prod_\mu \{a_\mu, \dots, b_\mu\}$ is a box and $X \cap T = \{x \in X : \ x_\mu = b_\mu \text{ for some } \mu\}$, the factor is the state of King's convention on X.*

Proof. Each deletion is in the cone direction: deleting a photon gradient term does not increase $\langle\Theta_{F,g}\rangle$ and does not decrease $\langle\Theta^-_{F,g}\rangle$ (lemma 2.7(a),(d)), and setting a hopping coefficient c_{xy} to 0 likewise (lemma 2.7(b),(d)); every intermediate measure is a BFS measure. After all deletions nothing couples the X-variables to the rest, so the measure is a product (lemma 2.7(e)). In the X-factor the photon is the Neumann photon of X; a site of $X \cap T$ carries its point mass; an alive $x \in X \setminus T$ carries the single-site weight times $\frac{\varepsilon}{2} r_x^2$ for each of its six bonds, since in $\langle\cdot\rangle_Y$ every bond at an alive x contributes $\frac{\varepsilon}{2} r_x^2$ at x whether its other end is alive in V (the diagonal part of $\frac{\varepsilon}{2}|e^{\mathrm{i}e\varepsilon A}\phi_y - \phi_x|^2$), killed, or outside V ($\phi = 0$ there), and the cut keeps all diagonals; the cosines are those of the bonds within X between alive sites. That is the state of definition 2.1 on X killed on $X \cap T$. By (S_X) the observable is a function of the X-factor's variables (at $x \in X \cap T$ the factor $e^{\varepsilon^3 g(x) r_x^2}$ is 1 in both states), and the c-numbers $e^{\mp 2\mathsf{W}_\varepsilon \mathsf{S}(g)}$, $e^{\mp a_k \mathsf{S}(g)}$ are the same positive constants on both sides (lemma 2.11(a); the counterterm is site-independent). Hence $\langle\Theta\rangle^{(a),X\cap T}_X$ is the expectation of Θ in the fully deleted measure, which is $\le \langle\Theta\rangle_Y$ on $\mathcal{I}$ and $\ge$ on $\mathcal{D}$. □

The clause of lemma 2.10 that X and its neighbours lie in one fundamental cell is a torus clause; for a box it has no content, since a bond leaving V already carries its Dirichlet diagonal in $\langle\cdot\rangle_Y$. Every sandwich $C_{J-1} \subset \cdot \subset C_{J+1}$ of this paper, read in King's convention, is an instance of lemma 4.1.

Lemma 4.2 (interlacing of the two cube families)**.** *Fix $k \ge 0$, $J \ge 1$ and (f,g), $g \ge 0$, satisfying* (S) *for X_J. Then*

$$\langle\Theta(\mathrm{i}f,g)\rangle^K_{k,J} \;\le\; \langle\Theta(\mathrm{i}f,g)\rangle^{(a)}_{k,J} \;\le\; \langle\Theta(\mathrm{i}f,g)\rangle^K_{k,J+1}, \qquad \langle\Theta(f,-g)\rangle^K_{k,J} \;\ge\; \langle\Theta(f,-g)\rangle^{(a)}_{k,J} \;\ge\; \langle\Theta(f,-g)\rangle^K_{k,J+1},$$

and the same for the scheme functionals $Z^a_{k,J}(\mathrm{i}f,g)$, $Z^a_{k,J}(f,-g)$ of the three states.

Proof. Step 0 (one observable, one multiplier). By (S) the plaquettes with non-zero coefficient have all corners in X_J, whose photon variables are present in all three states, so $F(f)$ is one real linear functional F of the shared variables; the sites with $g \neq 0$ lie in alive^K_J, alive in all three states; W_ε, δm^2_ε, $\mathsf{S}(g)$ are the same for the three regions. Hence, with $g'_x := \varepsilon^3 g(x) \ge 0$,

$\Theta(\mathrm{i}f,g) = e^{-2\mathsf{W}_\varepsilon \mathsf{S}(g)}\Theta_{F,g'}$ and $\Theta(f,-g) = e^{2\mathsf{W}_\varepsilon \mathsf{S}(g)}\Theta^-_{F,g'}$, where $\Theta_{F,g'} = e^{\mathrm{i}F}e^{\sum_x g'_x r_x^2}$, $\Theta^-_{F,g'} = e^{F}e^{-\sum_x g'_x r_x^2}$ are the observables of lemma 2.7, and the scheme multiplier $e^{\mp a_k \mathsf{S}(g)}$ is one positive constant for all three states (lemma 2.11(a)). It suffices to treat $\langle\Theta_{F,g'}\rangle$, $\langle\Theta^-_{F,g'}\rangle$.

Step 1 (kill step on $\mathcal{I}$). For $s \geq 0$ let $\langle\cdot\rangle_s$ be $\langle\cdot\rangle^{(a)}_{k,J}$ with the single-site measure at each $x \in T_J$ multiplied by $e^{-s\varepsilon^3 r_x^2}$, a BFS measure, and $\varphi(s) := \langle\Theta_{F,g'}\rangle_s = N(s)/D(s)$, $N(s) = \int \Theta\, e^{-s\varepsilon^3\sum_{T_J} r_x^2}d\mu_0$, $D(s) = \int e^{-s\varepsilon^3\sum_{T_J} r_x^2}d\mu_0$, μ_0 the unnormalised measure at $s=0$. The s-derivative of either integrand is bounded in modulus by $\varepsilon^3\sum_{T_J} r_x^2$ times the integrand at $s=0$, integrable against the quartic weight $e^{-\lambda\varepsilon^3 r^4}$ and the Gaussian; so differentiation under the integral sign is justified, $D(s) > 0$, and

$$\varphi'(s) = -\varepsilon^3 \sum_{x\in T_J} \big(\langle\Theta\, r_x^2\rangle_s - \langle\Theta\rangle_s\langle r_x^2\rangle_s\big) = -\varepsilon^3 \sum_{x\in T_J} \langle\Theta; r_x^2\rangle_s \;\leq\; 0$$

by lemma 2.7(c). So $\varphi(s) \leq \varphi(0) = \langle\Theta\rangle^{(a)}_{k,J}$. Since $\Theta_{F,g'}$ is continuous with $|\Theta_{F,g'}| = e^{\sum_x g'_x r_x^2}$ and $\langle\cdot\rangle^{(a)}_{k,J}$ is of King type, lemma 2.9 gives $\varphi(s) \to \langle\Theta\rangle_{T_J}$, the state with the matter killed on T_J, which is $\langle\cdot\rangle^K_{k,J}$. Hence $\langle\Theta\rangle^K_{k,J} \leq \langle\Theta\rangle^{(a)}_{k,J}$.

Step 2 (cut step). Apply lemma 4.1 with $V := X_{J+1}$, $T := T_{J+1}$ (so $\langle\cdot\rangle_Y = \langle\cdot\rangle^K_{k,J+1}$), $X := X_J$: $X_J \subset X_{J+1}$, and $X_J \cap T_{J+1} = \emptyset$ since every site of T_{J+1} has a coordinate $M(J+1)-\varepsilon \geq MJ > MJ-\varepsilon$. (At $k=0$, $M=1$ the outward neighbours of X_J lie on T_{J+1} and are killed in $\langle\cdot\rangle_Y$; harmless, since the bond from an alive $x \in X_J$ to such a neighbour already carries $\frac{\varepsilon}{2}r_x^2$ at x, the Dirichlet diagonal of the comparison cube.) By lemma 4.1(a) the X_J-factor is $\langle\cdot\rangle^{(a)}_{k,J}$, and (S) is ($\mathrm{S}_{X_J}$): $\langle\Theta(\mathrm{i}f,g)\rangle^{(a)}_{k,J} \leq \langle\Theta(\mathrm{i}f,g)\rangle^K_{k,J+1}$, $\langle\Theta(f,-g)\rangle^{(a)}_{k,J} \geq \langle\Theta(f,-g)\rangle^K_{k,J+1}$.

Step 3 (kill step on $\mathcal{D}$). With $\langle\cdot\rangle_s$ as in Step 1 and $\varphi^-(s) := \langle\Theta^-_{F,g'}\rangle_s$ (differentiation justified as before, now with $|\Theta^-| \leq e^{F(A)}$ integrable against the Gaussian), $(\varphi^-)'(s) = -\varepsilon^3\sum_{x\in T_J}\langle\Theta^-; r_x^2\rangle_s \geq 0$ by lemma 2.7(d), so $\varphi^-(s) \geq \varphi^-(0) = \langle\Theta^-\rangle^{(a)}_{k,J}$; as $|\Theta^-_{F,g'}| \leq e^{F(A)}$ with F real linear, lemma 2.9 (with $\ell = F$, $C=0$) gives $\varphi^-(s) \to \langle\Theta^-\rangle^K_{k,J}$. Hence $\langle\Theta(f,-g)\rangle^K_{k,J} \geq \langle\Theta(f,-g)\rangle^{(a)}_{k,J}$.

Step 4 (Z^a). The multiplier of Step 0 is one positive constant. □

Nothing in the proof uses a reflection symmetry of either cube (neither has one), the counterterm beyond its site-independence, or the renormalisation group. Since the observable vanishes at the killed sites in every kill step of this section, each such step also follows with no limit: set $c_{xy} = 0$ on every bond at the set U to be killed, one bond at a time (lemma 2.7(b),(d)); the measure is then a product (lemma 2.7(e)) whose U-factor cancels, and the other factor is the state with U killed.

4.3. **Monotonicity of the K family and the limits.**

Corollary 4.3 (the K family is monotone in J)**.** *Under* (S) *for X_J: $\langle\Theta(\mathrm{i}f,g)\rangle^K_{k,J} \leq \langle\Theta(\mathrm{i}f,g)\rangle^K_{k,J+1}$ and $\langle\Theta(f,-g)\rangle^K_{k,J} \geq \langle\Theta(f,-g)\rangle^K_{k,J+1}$, and the same for $Z^{a,K}$; hence for all $J \geq J_0$ whenever the sources satisfy* (S) *for X_{J_0}.*

Proof. Chain the two inequalities of lemma 4.2. (For the comparison family this is lemma 4.1(a) with $V = X_{J+1}$, $X = X_J$, i.e. the first inequality of lemma 2.10.) □

The proof of Theorem 3.2 of [14] first replaces the J-dependent Dirichlet Wick subtraction (2.13) by the free, J-independent one and then proves the monotonicity in J in the comparison convention (lemma 4.1(a)); by corollary 4.3 that statement holds between cubes of King's convention too, with the kill step (Step 1) not printed there. The form with the Dirichlet constant follows in neither convention and is not used here; at $g=0$ no Wick constant enters.

Corollary 4.4 (the limits)**.** *Let $f = (f_1,\dots,f_N)$ be real 2-forms in C_c^∞ with* $\operatorname{supp} f_j \subset \{\operatorname{dist}_\infty(\cdot, \mathbb{R}^3 \setminus C_{J_0}) > 1\}$*; for $J \geq J_0$, $z \in \mathbb{C}^N$ put $a^K_{k,J}(z) := \langle e^{\sum_j z_j F(f_j)}\rangle^K_{k,J}$, $a^{(a)}_{k,J}(z) :=$*

$\langle e^{\sum_j z_j F(f_j)}\rangle^{(a)}_{k,J}$, $g(z) := \sum_j (\operatorname{Re} z_j) f_j$. *At the cone points* $z = \mathrm{i}t$, $t \in \mathbb{R}^N$, *both are real with modulus* ≤ 1, *and* (S) *holds for* X_J.

(i) (One J-limit at fixed k; no hypothesis.) *For every* k *and* t, $J \mapsto a^K_{k,J}(\mathrm{i}t)$ *and* $J \mapsto a^{(a)}_{k,J}(\mathrm{i}t)$ *are nondecreasing on* $J \ge J_0$, *bounded by* 1 *and interlaced; hence* $\lim_J a^K_{k,J}(\mathrm{i}t) = \lim_J a^{(a)}_{k,J}(\mathrm{i}t) \in [-1,1]$. *The same holds for* $\langle \Theta(\mathrm{i}f, g)\rangle$, $g \ge 0$, *with a common limit in* $(-\infty, \infty]$.

(ii) (King's cube limits.) *By theorem 3.2, posed on King's convention, for every such family, every* $J \ge J_0$ *and every* $t \in \mathbb{R}^N$, $\lim_k a^K_{k,J}(t)$ *exists in* $(0,\infty)$. *Then:* (α) $a^K_{k,J} \to Z^K_J$ *locally uniformly on* $\mathbb{C}^N$, Z^K_J *is entire and* $|Z^K_J(z)| \le e^{\|g(z)\|^2_2}$; ($\beta$) $Z^K_J \to Z^K_\infty$ *locally uniformly on* $\mathbb{C}^N$, Z^K_∞ *is entire and* $|Z^K_\infty(z)| \le e^{\|g(z)\|^2_2}$; ($\gamma$) *for all* $J \ge J_0$ *and* $t \in \mathbb{R}^N$,

$$Z^K_J(\mathrm{i}t) \;\le\; \liminf_k a^{(a)}_{k,J}(\mathrm{i}t) \;\le\; \limsup_k a^{(a)}_{k,J}(\mathrm{i}t) \;\le\; Z^K_{J+1}(\mathrm{i}t),$$

hence $\lim_J \liminf_k a^{(a)}_{k,J}(\mathrm{i}t) = \lim_J \limsup_k a^{(a)}_{k,J}(\mathrm{i}t) = Z^K_\infty(\mathrm{i}t)$.

(iii) (The subsequential limits of the comparison family.) *Let* $\mathcal{S} = (k_m)$ *be an increasing sequence of spacings along which* $a^{(a)}_{k_m,J} \to Z^{\mathcal{S}}_J$ *locally uniformly on* $\mathbb{C}^N$ *for every* $J \ge J_0$ *and every such family (a continuum subsequence of the comparison family, definition 14.1; they exist with no hypothesis, lemma 14.7). Then* $Z^{\mathcal{S}}_J \to Z^K_\infty$ *locally uniformly on* $\mathbb{C}^N$ *as* $J \to \infty$*: every subsequential fixed-cube limit of the comparison family, followed by* $J \to \infty$, *is King's printed state. In particular, if the fixed-cube limits of the comparison family exist along the full sequence,* $\lim_J \lim_k a^{(a)}_{k,J} = Z^K_\infty$.

Proof. (i) Monotonicity is corollary 4.3 for the K family and lemma 4.1(a) for the comparison family; lemma 4.2 interlaces them; two nondecreasing sequences with $a_J \le b_J \le a_{J+1}$ have the same limit, finite or $+\infty$.

(ii)(α) This is the first step of the proof of theorem 7.4(b) run on the K family. (1) $a^K_{k,J}$ is entire (Gaussian decay in A, $e^{-\lambda\varepsilon^3 r^4}$ decay in r at the alive sites). (2) The local bound of lemma 7.3 holds for the K family with the same constant: $\operatorname{Re}\sum_j z_j F(f_j) = F(g(z))$, so $|a^K_{k,J}(z)| \le \langle e^{F(g(z))}\rangle^K_{k,J}$; lemma 2.8(a) with $s = 1$ and $v := d^*_k g(z)$ (defined by $F_{\Lambda_k}(g(z)) = \langle v, A\rangle_k$) applies to any BFS measure whose Gaussian has covariance C and gives $\langle e^{\langle v,A\rangle}\rangle \le e^{\langle v, Cv\rangle/2}$; the K cube is a BFS measure whose Gaussian is the Neumann photon Gaussian of X_J, the same as the comparison cube's, so its covariance is the C of lemma 7.1, a statement about this C and the plaquettes with corners in X_J: $\langle v, Cv\rangle_k \le 2\|g(z)\|^2_{2,k}$. Hence $|a^K_{k,J}(z)| \le e^{\|g(z)\|^2_{2,k}} \le e^{3C_0|g(z)|^2_4}$ uniformly in $k \ge 0$, $J \ge J_0$, by (E3) of lemma 2.6: the plaquette centres of each of the three orientations form a translate of $\varepsilon_k\mathbb{Z}^3$, each contributing at most $C_0|g|^2_4$ to $\|g\|^2_{2,k}$ by (E1), and the three orientations give $3C_0$. Nothing in this chain sees the matter beyond the state being a BFS measure. (3) Theorem 3.2 gives convergence at every real point. (4) $\mathbb{R}^N$ is a set of uniqueness for entire functions on $\mathbb{C}^N$; by Montel's theorem every subsequence has a locally uniformly convergent sub-subsequence, all limits are entire and agree on $\mathbb{R}^N$, hence everywhere [10]. So $a^K_{k,J} \to Z^K_J$ locally uniformly, Z^K_J is entire, and the bound passes to the limit, $|Z^K_J(z)| \le \lim_k e^{\|g(z)\|^2_{2,k}} = e^{\|g(z)\|^2_2}$ (lemma 2.6), a J-free bound.

(β) By corollary 4.3 at every k and (α), $J \mapsto Z^K_J(\mathrm{i}t)$ is nondecreasing and ≤ 1, so $Z^K_\infty(\mathrm{i}t) := \lim_J Z^K_J(\mathrm{i}t)$ exists; $\{Z^K_J\}_{J\ge J_0}$ is locally uniformly bounded by the J-free bound of (α); $(\mathrm{i}\mathbb{R})^N$ is a maximal totally real subspace, hence a set of uniqueness for entire functions; Montel's theorem and the identity theorem give $Z^K_J \to Z^K_\infty$ locally uniformly, Z^K_∞ entire, $|Z^K_\infty(z)| \le e^{\|g(z)\|^2_2}$. (This is the proof of theorem 7.4(b) with lemma 2.10 replaced by corollary 4.3.)

(γ) lemma 4.2 at every k, with $k \to \infty$ at the two ends (limits by (α)) and lim inf, lim sup in the middle, gives the sandwich; (β) gives the limits.

(iii) lemma 4.2 along $\mathcal{S}$, with the full-sequence limits of (α) at the ends, gives $Z^K_J(\mathrm{i}t) \le Z^{\mathcal{S}}_J(\mathrm{i}t) \le Z^K_{J+1}(\mathrm{i}t)$, so $\lim_J Z^{\mathcal{S}}_J(\mathrm{i}t) = Z^K_\infty(\mathrm{i}t)$ by (β). $\{Z^{\mathcal{S}}_J\}_J$ is locally bounded by the J-free

bound $e^{\|g(z)\|_2^2}$ (lemma 7.3 passed to the limit), so by Montel's theorem and the uniqueness set $(\mathrm{i}\mathbb{R})^N$ it converges locally uniformly to an entire function (lemma 14.8), which agrees with Z_∞^K on $(\mathrm{i}\mathbb{R})^N$, hence on $\mathbb{C}^N$. If the fixed-cube limits exist along the full sequence, that sequence is a continuum subsequence. □

Statements (ii)–(iii) also hold with the two conventions exchanged, for $J \geq J_0 + 1$, when the comparison family is convergent at fixed cubes (definition 3.1; read lemma 4.2 as $a^{(a)}_{k,J-1} \leq a^K_{k,J} \leq a^{(a)}_{k,J}$); the exchanged (iii) needs lemmas 14.7 and 14.8 for the K family, whose proofs use only the local bound and the photon-only bound $\langle F(u-u')^2\rangle \leq 2\|u-u'\|^2_{2,k}$, both transferred by the covariance identity in (α).

What does not follow. At fixed J, lemma 4.2 yields exactly (γ) and nothing on whether $\lim_k a^{(a)}_{k,J}(\mathrm{i}t)$ exists: the bounds $Z^K_J(\mathrm{i}t)$, $Z^K_{J+1}(\mathrm{i}t)$ differ in general (the cubes differ by M, so the gap does not shrink with k), and nothing constrains the oscillation of $a^{(a)}_{k,J}(\mathrm{i}t)$ in k between them. So the existence of the fixed-cube limits on one convention does not imply it on the other by these inputs, and likewise with scalar sources: the existence theorem must be proved on the family on which the renormalisation-group construction is carried out; the other is a comparison family with the same infinite-volume state by (ii)–(iii). No counterexample is claimed.

4.4. **Three margins in King's convention.** Let $\sigma_3 = \mathrm{diag}(1,1,-1)$, $X := X_J$, $X_+ := \sigma_3 X = (C_J + \varepsilon e_3) \cap \Lambda_k$ (third coordinates $\{-MJ+\varepsilon, \dots, MJ\}$), $\sigma_3 T_J = \{x \in X_+ :\ x_1 = MJ-\varepsilon$ or $x_2 = MJ-\varepsilon$ or $x_3 = -MJ+\varepsilon\}$ and $A_\sigma := \sigma_3(\mathrm{alive}^K_J) = \{x :\ x_1, x_2 \in \{-MJ, \dots, MJ-2\varepsilon\},\ x_3 \in \{-MJ+2\varepsilon, \dots, MJ\}\}$.

Lemma 4.5 (the reflection sandwich)**.** *Let $(f,g) \in \mathcal{K}$ be supported at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_{J_0}$.* (a) *If $M \geq 2\varepsilon_k$ (i.e. $k \geq 1$, or $M \geq 2$), then for every $J \geq J_0 + 1$*

$$Z^{a,K}_{k,J-1}(\mathrm{i}f, g) \ \leq\ Z^{a,K}_{k,J}(\mathrm{i}\,\sigma_3\cdot f, \sigma_3\cdot g) \ \leq\ Z^{a,K}_{k,J+1}(\mathrm{i}f, g).$$

(b) *For every $k \geq 0$, $M \geq 1$ and $J \geq J_0 + 2$ the same holds with $J \mp 2$ in place of $J \mp 1$.* (c) *For $k = 0$, $M = 1$ the one-step sandwich* (a) *is not obtainable by deletions and kills: A_σ contains alive sites ($x_3 = MJ$) killed in King's cube of index $J+1$, and* alive^K_{J-1} *contains sites ($x_3 = -MJ+\varepsilon$) killed in the reflected state. Whether the inequality itself holds there is left open.*

Proof. Step 1 (transport). By lemma 8.4, $Z^{a,K}_{k,J}(\mathrm{i}\sigma_3\cdot f, \sigma_3\cdot g) = e^{-a_k \mathsf{S}(g)}\langle\Theta(\mathrm{i}f,g)\rangle_R$, with R the state on X_+ for the frame $(e_1, e_2, -e_3)$ killed on $\sigma_3 T_J$ ($\mathsf{S}(\sigma_3\cdot g) = \mathsf{S}(g)$). In the frame (e_α), as in lemma 2.13, R is the BFS measure with A_1, A_2 on X_+ and A_3 on $X_+ - \varepsilon e_3 = X$ (gradient pairs within these sets, masses on them), matter alive exactly on A_σ with point masses on $\sigma_3 T_J$ and $\phi = 0$ outside X_+, all six diagonals at each alive site, and the standard cosine on every bond with both ends in A_σ (the flipped bond with $A_3'(x) = -A_3(x - \varepsilon e_3)$ gives the standard cosine of $\langle x - \varepsilon e_3, x\rangle$); $F(f)$, $W(g)$ are the standard ones (the plaquettes $(x;\alpha,3)$ pick up two sign changes and keep their centres). R is of King type. The sources satisfy (S) for X_{J-1} (resp. X_{J-2}), so by the inclusions below Step 0 of lemma 4.2 applies to all states compared.

Step 2 (upper end). If $A_\sigma \subset \mathrm{alive}^K_{J'}$ and $X_+ \cup X \subset X_{J'}$: from $\langle\cdot\rangle^K_{k,J'}$ kill every site of $\mathrm{alive}^K_{J'} \setminus A_\sigma$ (lemma 2.7(c), lemma 2.9, which allows the already killed $T_{J'}$), leaving exactly the cosines of bonds within A_σ, i.e. R's; then delete every photon gradient term except the pairs within X_+ for A_1, A_2 and within X for A_3 (lemma 2.7(a); these are gradient pairs of $X_{J'}$). The result is a product (lemma 2.7(e)) whose factor on R's variables is R (photon, alive set with all six diagonals by the bookkeeping of lemma 4.1, cosines); the other factor cancels. Hence $\langle\Theta(\mathrm{i}f,g)\rangle_R \leq \langle\Theta(\mathrm{i}f,g)\rangle^K_{k,J'}$. The inclusions hold iff $MJ \leq MJ' - 2\varepsilon$ (direction 3; the rest is trivial): for $J' = J+1$ iff $M \geq 2\varepsilon$, for $J' = J+2$ always.

Step 3 (lower end). If $\mathrm{alive}^K_{J''} \subset A_\sigma$ and $X_{J''} \subset X \cap X_+$: from R kill every site of $A_\sigma \setminus \mathrm{alive}^K_{J''}$ (lemma 2.7(c), lemma 2.9) and delete every photon gradient term coupling an $X_{J''}$-variable to another variable of R (lemma 2.7(a)). The $X_{J''}$-factor has the Neumann photon of $X_{J''}$ (gradient pairs within $X_+ \cap X_{J''} = X \cap X_{J''} = X_{J''}$) and the matter alive on $\mathrm{alive}^K_{J''}$ with all diagonals,

killed on $T_{J''}$: it is $\langle\cdot\rangle^K_{k,J''}$; the rest cancels. Hence $\langle\Theta(\mathrm{i}f,g)\rangle^K_{k,J''} \le \langle\Theta(\mathrm{i}f,g)\rangle_R$. The inclusions hold iff $-MJ'' \ge -MJ + 2\varepsilon$: for $J'' = J-1$ iff $M \ge 2\varepsilon$, for $J'' = J-2$ always.

Step 4. Steps 2–3 with $J\pm1$ give (a), with $J\pm2$ give (b); the multipliers agree (lemma 2.11(a)). For (c): at $k=0$, $M=1$ the top of A_σ in direction 3 is J while that of alive^K_{J+1} is $J-1$, and the bottom of alive^K_{J-1} is $-J+1 = -MJ+\varepsilon$, the killed layer of R; deletions and kills never revive a killed site. □

In the curvature sector ($g=0$) the limit sentences of theorem 8.5 run on (a) for $k \ge 1$ or on (b) for all k, with $k\to\infty$ by corollary 4.4(ii) and $J\to\infty$ by corollary 4.3; with scalar sources they need the K-family tube limits, supplied by theorem 21.3 (remark 4.9).

Lemma 4.6 (the upper box needs no change)**.** *Let $\Omega \subset B$ be lattice boxes at spacing ε such that B contains every site within ℓ^∞-distance N lattice steps of Ω, $N \ge 1$ (the hypothesis of lemma 9.10, whose mass condition forces $N \ge 1$), and let $\langle\cdot\rangle^K_B$ be the state of King's convention on B (killed on its top face layers T_B). Then* (a) *$\Omega \cap T_B = \emptyset$;* (b) *the second part of lemma 9.10 holds with $\langle\cdot\rangle^K_B$ in place of the comparison state on B, with the same proof, and its first part is a photon statement, unchanged;* (c) *consequently the sandwich in the proof of theorem 10.6 holds for King's cubes with the printed cubes and margins (20): for every k and every ζ admitted there, $Z^{a,K}_{k,J}(\zeta) \le \Phi_k(\zeta) \le Z^{a,K}_{k,J+2n}(\zeta)$, Φ_k the functional of the box state of definition 9.2 on $\Omega_{J,k}$; the lower end is lemma 9.9 with Step 1 of lemma 4.2 prepended. (By corollary 4.3 the cube of index $J+2n+1$ would also do; it is not needed.)*

Proof. (a) For $x \in \Omega$ and each μ, $x + N\varepsilon e_\mu \in B$, so $x_\mu < \max_B x_\mu$ and $x \notin T_B$. (b) The proof of lemma 9.10 starts from the state on B and removes all hopping cosines touching $B \setminus \Omega$; the matter on Ω is then the Dirichlet matter of Ω (diagonals of the bonds leaving Ω kept, each such bond ending in B as $N \ge 1$), coupled to A only through the bonds within Ω, and the sites of $B \setminus \Omega$ decouple. For $\langle\cdot\rangle^K_B$ the cosines touching T_B are already absent; remove the remaining cosines touching $B \setminus \Omega$ (lemma 2.7(b) on $\mathcal{I}$, (d) on $\mathcal{D}$). By (a) every site of Ω is alive, each bond at $x \in \Omega$ carries $\frac{\varepsilon}{2} r_x^2$ at x (its other end alive, killed or outside), and the cosines within Ω are present; so the matter on Ω is the same Dirichlet matter, and the sites of $B \setminus \Omega$, alive or killed, decouple and cancel. The rest of the proof concerns only the photon law of B, the same in both conventions; the first part uses only the photon forms and a cutoff supported in B. (c) Lower end: lemma 9.9 gives $\langle Y\rangle^{(a)}_{C_J} \le \langle Y\rangle_{\Omega_{J,k}}$ on $\mathcal{I}$ (its proof lands on the Dirichlet matter of C_J, the comparison convention), and Step 1 of lemma 4.2 gives $\langle Y\rangle^K_{C_J} \le \langle Y\rangle^{(a)}_{C_J}$, (S) holding since the sources of the enlarged family and their reflections are supported at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_{J_0}$, $C_{J_0} \subset C_J$. Upper end: (b) with $B = X_{J+2n}$, whose hypothesis is the second margin of (20), a statement about photon sites. The multipliers agree (lemma 2.11(a)). □

For the cell bound we use the cells of definition 13.4: $\ell_c = 2ML_c \ge 10$; the cells $c = \prod_\mu [w_\mu, w_\mu + \ell_c)$ of the tiling $\mathcal{T}_s$, $s \in \{0,1\}^3$, $w_\mu \in W_\mu(s_\mu)$, $W_\mu(0) = \ell_c\mathbb{Z}$, $W_\mu(1) = ML_c + \ell_c\mathbb{Z}$ (so $w_\mu, w_\mu+\ell_c, MJ \in \mathbb{Z}$); the good sets $G_s = \{x : \mathrm{dist}(x_\mu, W_\mu(s_\mu)) > 2\ \forall\mu\}$ and cutoffs χ_s with $\mathrm{supp}\,\chi_s \subset G_s$; $X_c := c \cap X_J$; $J_1(R) = \min\{J : MJ \ge R + \ell_c\}$ for sources supported in $Q_R = [-R,R]^3$; B_{cell}, defined on the comparison cell C_{L_c}. N^K_{k,C_J} is the normalised scalar-source functional of $\langle\cdot\rangle^K_{k,J}$ and $N^{(a),T}_{k,X}$ that of the comparison state on X killed on $T \subset X$.

Lemma 4.7 (the cell lemmas hold as printed)**.** (a) *For a tiling $\mathcal{T}_s$ and any lattice function $q \ge 0$ on X_J: $N^K_{k,C_J}(0,-q) \le \prod_i N^{(a),X_i\cap T_J}_{k,X_i}(0,-q|_{X_i})$, over the site sets $X_i = c \cap X_J$ of the cells meeting C_J.* (b) *For a full cell $c \subset C_J$: $X_c \cap T_J \ne \emptyset$ iff $w_\mu + \ell_c = MJ$ for some μ.* (c) *Let $q \ge 0$ be any lattice function supported in Q_R, $J \ge J_1(R)$, $s \in \{0,1\}^3$ and $c \in \mathcal{T}_s$ with $(\chi_s q)|_{X_c} \not\equiv 0$. Then $-MJ < w_\mu$ and $w_\mu + \ell_c < MJ - 2$ for every μ: c is a full cell inside C_J with $X_c \cap T_J = \emptyset$, and $(\chi_s q)|_{X_c}$ is supported at ℓ^∞-distance > 2 from $\mathbb{R}^3 \setminus c$.* (d) *Hence lemmas 13.6 to 13.8 are unchanged and proposition 13.9 holds for $\langle\cdot\rangle^K_{k,J}$ as printed, under its hypotheses and with its constant K_-: $\ln N^K_{k,C_J}(0,-q) \le K_-|q|_w$ for every k, every $J \ge J_1(R)$ and every lattice $0 \le q \le H$ supported in Q_R.* (e) *(Direction.) For a cell c and $\emptyset \ne T \subset X_c$:*

$N_{k,c}^{(a),T}(0,-\tilde{q}) \geq N_{k,c}^{(a)}(0,-\tilde{q})$ *for* $\tilde{q} \geq 0$ *vanishing on* T. *So* B_{cell}, *a bound on a comparison cell, does not bound a source-carrying cell of King's type, and* (c) *is load-bearing.*

Proof. (a) The proof of lemma 13.5 deletes the cross-cell photon gradient terms and hopping cosines, each in the $\mathcal{D}$ direction (lemma 2.7(d)); by lemma 2.7(e) and the factor identification of lemma 4.1 with $V = X_J$, $T = T_J$, $X = X_i$, the product has the factors $N_{k,X_i}^{(a),X_i\cap T_J}$; the observable $e^{-\sum \varepsilon^3 q r^2}$ and the multiplier $e^{a_k \mathsf{S}(q)} = \prod_i e^{a_k \mathsf{S}(q|_{X_i})}$ factorise, and at the sites of T_J the observable factor is 1 in every state. (b) The sites of a full cell have $x_\mu \in \{w_\mu, \dots, w_\mu + \ell_c - \varepsilon\}$, and $x \in T_J$ iff $w_\mu + \ell_c - \varepsilon = MJ - \varepsilon$ for some μ. (c) Let $x \in X_c$ with $(\chi_s q)(x) \neq 0$. Then $x \in G_s$; since $w_\mu, w_\mu + \ell_c \in W_\mu(s_\mu)$ and $w_\mu \leq x_\mu < w_\mu + \ell_c$, this gives $w_\mu + 2 < x_\mu < w_\mu + \ell_c - 2$; and $q(x) \neq 0$ gives $|x_\mu| \leq R$. Hence $w_\mu + \ell_c < R + \ell_c - 2 \leq MJ - 2$ (as $MJ \geq R + \ell_c$) and $w_\mu > -R - \ell_c + 2 \geq -MJ + 2$. So $c \subset (-MJ, MJ)^3$, $w_\mu + \ell_c \neq MJ$, hence $X_c \cap T_J = \emptyset$ by (b), and $\mathrm{dist}_\infty(x, \mathbb{R}^3 \setminus c) = \min_\mu \min(x_\mu - w_\mu, w_\mu + \ell_c - x_\mu) > 2$. (d) In the chain of proposition 13.9 the Hölder step of lemma 13.8 is an inequality within one state, convention-free; then (a) with the tiling $\mathcal{T}_s$ and $q_s := 8\chi_s q$. A factor with $q_s|_{X_i} \equiv 0$ equals 1 whatever is killed on X_i (this covers the clipped cells at the faces of C_J and any cell flush with them); a factor with $q_s|_{X_i} \not\equiv 0$ is, by (c), the comparison-cell functional $N_{k,c}^{(a)}(0, -q_s|_c)$ of a full cell $c \subset C_J$ with $q_s|_c$ supported at distance > 2 from $\mathbb{R}^3 \setminus c$ and $0 \leq q_s \leq 8H = H'$: exactly the hypotheses of lemma 13.7 (hence of lemma 13.6: translation to C_{L_c}, B_{cell}, proposition 13.2 on the comparison box), as printed; lemma 13.8 sums the cells. (A cell flush with a face, $w_\mu + \ell_c = MJ$, can meet $\mathrm{supp}\, q$ but never $\mathrm{supp}(\chi_s q)$; the chain feeds the cutoff sources, not q, to (a).) (e) Step 3 of lemma 4.2 applied to the comparison cell with $U = T$: $s \mapsto \langle \Theta^- \rangle_s$ is nondecreasing (lemma 2.7(d)) and tends to the killed state (lemma 2.9). □

Remark 4.8 (one-line kill steps). Proposition 13.1 (cube ≤ torus on $\mathcal{I}$), proposition 13.2 (box ≤ cube ≤ torus), the lower end of lemma 9.9, and the comparison of the cube with the box state of definition 9.2 on the cone (remark 9.8) are each proved for the comparison cube; for King's cube prepend Step 1 of lemma 4.2 ($\langle\cdot\rangle_{k,J}^K \leq \langle\cdot\rangle_{k,J}^{(a)}$ on $\mathcal{I}$), and for the $\mathcal{D}$ versions (cube ≥ torus for negative sources) Step 3. The same applies to a box of King's type (a killed face layer) in proposition 13.2. Nothing beyond lemma 4.2 is needed.

Remark 4.9 (the Wick sector). With scalar sources $g \geq 0$, corollary 4.4(i) holds as stated. The analogues of corollary 4.4(ii)–(iii) on the Wick tube need the fixed-cube limits and the (k, J)-uniform bound (31) for both families. For King's family the bound is the chain of theorem 13.10(i) run on ND^K, whose convention-sensitive steps are the cube-versus-torus comparison (corollary 13.3, via remark 4.8) and the cell bound (lemma 4.7); the assembly of that chain and the existence of the K-family tube limits are carried out in theorem 21.3, after which the tube statements hold on King's convention (remark 13.12, proposition 21.4).

The existence part

5. King's existence theorem on the cube

The existence theorem for King's cube limits (theorem 3.2), and its form with scalar sources that the Wick leg consumes (theorem 3.5), are proved in this section and the next from estimates on King's renormalisation-group integrals on the cube C_J^K. The argument is the one King gives for his Theorem 2.1 [14, Sect. 3.1] and [13, Sect. 3.2]: the generating functional is approximated, up to a small-field remainder, by an integral of the effective action after j renormalisation steps, and two spacings whose effective actions live on the same unit lattice are compared. We write it as a Cauchy property of the *normalised* functional, in a ratio form in which every field-independent constant of the effective action cancels; this removes from King's route the convergence of the vacuum energies, which his printed argument uses and which is not available on the cube.

The inputs are of two kinds. Estimates taken from Bałaban's block-factor-5 analysis [1, 2, 3, 4] on tori, and their transfer to King's cube, are stated as propositions, each followed by a description of its content; their proofs are collected in an appendix. Three of these statements

are used at several places and are therefore stated once, with a precise formulation, in section 5.2 ((BAL$_5$)[int], (REN$_{\mathrm{orb}}$)) and section 6.1 ((K$_h$)). Everything else in the two sections is proved in place.

5.1. **Renormalisation-group integrals on King's cube.** Fix the couplings, a cube C_J and King's RG cube C_J^K of definition 2.2. In this section and the next the index k is the spacing index ($\varepsilon_k = 5^{-k}$, as throughout), and j counts renormalisation steps, $0 \le j \le k - J_*$ for a fixed J_* (King's "J" of [13, Thm. 3.1]); the block scale after j steps is $\eta := 5^j\varepsilon_k = 5^{j-k} \le 1$, and the unit lattice after j steps is $\eta^{-1}(C_J \cap \eta\mathbb{Z}^3)$. Since the side $2MJ$ of C_J is an integer, the faces of C_J lie on block planes at every scale.

Sources. A real source is $s = (u, h)$ with $u = u_x := \sum_i x_i f_i$ a real combination of 2-forms $f_i \in C_c^\infty(\mathbb{R}^3; \Lambda^2)$ sampled at the plaquette centres in the lattice's own frame, and $h \in C_c^\infty(\mathbb{R}^3)$ real; all supports lie at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_J$ (so $J \ge J_0$). Put $X(s) := F(u) + {:}\phi^2{:}(h)$, with the free Wick constant of definition 2.3; X is real and linear in s. The unnormalised functional is

$$Z_k(s) \;:=\; \int e^{-S}\, e^{X(s)} \prod dA \prod d^2\phi$$

over the variables of C_J^K at spacing ε_k (any fixed, s-independent normalisation of the measure; it cancels everywhere below), and $N_k(s) := Z_k(s)/Z_k(0) = \langle e^{X(s)}\rangle_{k,J}^K$. At $h = 0$ we write $a_k(x) := N_k(u_x, 0)$, which is the $a_{k,J}^K(x)$ of corollary 4.4.

King's flow on the cube. The block-spin transformations of [13, Sect. 2] are run on C_J^K with block factor 5: the photon field is averaged over the $5^j\varepsilon_k$-blocks that tile X_J, with the Neumann condition of definition 2.2; the scalar field is averaged with King's Dirichlet weight [14, (3.8)–(3.9)] (in the corrected form in which the averaging blocks have their top layers on the Dirichlet planes) at the finest step and with the uniform weight above it. The Dirichlet planes of C_J^K are $2MJ$ apart, a multiple of every block side, which is what makes this flow possible on C_J^K and not on the comparison cube (remark 5.3). Let $S_k^{(j)}(A_j, \phi_j; s)$ be King's effective action after j steps [13, (3.14)], [14, (3.12)], with the source vertices [13, (3.20)–(3.21)]; let χ_j be King's small-field indicator, [13, (3.2)] times the face factor [14, (5.41)], a function of (A_j, ϕ_j) alone; and

$$I_k^{(j)}(s) \;:=\; \int dA_j\, d\phi_j\; \chi_j\; e^{-S_k^{(j)}(A_j,\phi_j;s)}.$$

The *vacuum constant* $v_{k,j} := S_k^{(j)}(0, 0; 0)$ is finite; it contains King's E_0, E_1, the normalisation constants $\ln Z_j$, $\ln Z_j(0)$ of the block-spin kernels and every field- and source-free graph. The *normalised action* is $S_{N,k}^{(j)}(\cdot; s) := S_k^{(j)}(\cdot; s) - v_{k,j}$ for every s (the field-free, source-dependent part is kept), and $\hat I_k^{(j)}(s) := \int \chi_j e^{-S_{N,k}^{(j)}(\cdot;s)} = e^{v_{k,j}} I_k^{(j)}(s)$. The flow is a device of the proof: the state $\langle\cdot\rangle_{k,J}^K$ does not depend on it.

Diagonal pairs. Fix $\alpha \in (0, \frac12)$ and put $j(k) := \lfloor(1-\alpha)k\rfloor$, $\eta_k := 5^{j(k)-k}$; then $\eta_k \le 5^{-\alpha k}$ and $k - j(k) \le \alpha k + 1$. The *diagonal pair* at k consists of the k-model after $j(k)$ steps and the $(k+1)$-model after $j(k)+1$ steps. Both have the block scale η_k ($5^{j+1}\varepsilon_{k+1} = 5^j\varepsilon_k$), hence the same unit lattice and the same indicator $\chi_{j(k)}$; this is the situation of [13, p. 656] ("the fields A_k, ϕ_k have the same bounds in each integral"). No relation between $j(k+1)$ and $j(k)+1$ is used anywhere.

Bałaban's decomposition. In Bałaban's block-spin representation [1, 2], $Z_k(s) = \int d\mu_{k,s}$ with $d\mu_{k,s} = e^{X(s)} d\mu_{k,0}$ a positive measure on the fields and on the block variables of the scales $\le j$, and $P = P_j \in [0, 1]$ is the product of all small-field factors of those scales, a function of the block and fluctuation variables alone. The identity $Z_k(s) = \int P\, d\mu_{k,s} + \int (1-P)\, d\mu_{k,s}$ is exact; the first term is the *small-field part*, the second the *large-field remainder*.

5.2. **Two statements from the renormalisation-group analysis.**

Lemma 5.1 ((BAL$_5$)[int]: Bałaban's interaction-term bound at a nonzero background)**.** *In the setting of* [2, Sect. 2] *at block factor $L = 5$, in unit-lattice form at the step k, $t = 5^k\varepsilon$: let*

$\Omega_A = B^k(\Lambda_2^{(k-1)\prime}) \supset \Omega_C = B^k(\Lambda_2^{(k)})$ be the two propagator domains of [2, Prop. 2.6]*, let $\tilde{B}$ be the background field of* [2, p. 579]*, regular on Ω_A, let $G_k(\Omega, \tilde{B})$ be the scalar propagator of the step on Ω in the background $\tilde{B}$, and write the interaction terms of the stage of* [2, p. 579]*, localised in $B^k(\Lambda_7^{(k)})$, as a sum of per-label full renormalised classes with scalar lines $G_k(\Omega_A, \tilde{B})$, external scalar fields $E_A = a_k G_k(\Omega_A, \tilde{B}) f$ with $\|f\|_\infty \le C p \lambda(t)^{-1/4}$, and other external fields of norms $\le C p^a \lambda(t)^{-b}$ (appendix G.2). Replace $G_k(\Omega_A, \tilde{B})$ by $G_k(\Omega_C, B^{(k+1),\eta})$ in every scalar line and in every E_A. For every $\kappa > 0$ there are $t_0(\kappa) > 0$ and $C_\kappa < \infty$, depending on κ, on Bałaban's parameters $(R, \varrho, b_0, \tilde{p}, \bar{n}, \alpha_0)$ and on the couplings only, such that for $t \le t_0(\kappa)$ the sum over all localisations of the absolute values of all terms containing at least one difference of propagators or of external fields is at most $C_\kappa\, t^\kappa\, |\Lambda_7^{(k)}|$. The constants do not depend on k, on the domains or on $\tilde{B}$, and $\tilde{B} = 0$ is included.*

The proof is given in appendix G.2 (theorem G.15). This is the interaction-term half of Bałaban's domain-change estimate [2, Prop. 2.6], at every background. The terms with exactly one difference line, grouped line by line, carry their counterterm partners but not the complete renormalised classes of the cut graph, and a lone divergent two-photon-leg subgraph is not small per graph; the proof expands every line and every external field in the difference and regroups the terms exactly into contractions of *full* classes of the cut components, to which the renormalisation bound [3, Prop. 1, (1.33)] applies with constants free of the domains and of the background. Every use of the upper stability bound [2, (1.3)] and of [13, Thm. 3.1, (3.4)] at block factor 5 passes through this statement.

Lemma 5.2 ($(\mathrm{REN}_{\mathrm{orb}})$: orbit regrouping of cut renormalised families)**.** *Let $\mathcal{R} = \sum_{\Gamma \in \mathcal{G}} c_\Gamma E(\Gamma)$ be a renormalised family of* [3, Sects. 1–3] *with $\mathcal{G}$ a finite union of renormalised classes (definition G.1), in which the objects (lines or external legs) of certain kinds are decomposed as $o = o' - \Delta o$ with symmetric kernels Δo, and expand multilinearly, $\mathcal{R} = \sum_\tau 2^{-|S_L|} c_\Gamma \prod_{o \in S}(-\Delta o) \prod_{o \notin S} o'$ over the oriented cut terms $\tau = (\Gamma, S, \theta)$ of definition G.2. Call two cut terms equivalent when one arises from the other by finitely many of Bałaban's three completions of* [3, pp. 420, 429–432] *(the counterterm partner; the two-vector-leg class; the permutation of the external scalar legs of a divergent $\phi\phi A$ subgraph) applied at divergent subgraphs containing no line of S, the objects of S carried along to the image slots. Then:* (i) *the equivalence classes (orbits) partition the cut terms, and every move keeps the label multiset and the number and kinds of the objects of S;* (ii) *a connected line set of Γ avoiding S is a subgraph of the cut graph $\Gamma \setminus S$ with the same vertices, slots, slot types, order and degree* [3, (2.1)–(2.3)]*, and cutting commutes with the moves, so that the cut graphs of one orbit form exactly one renormalised class of the cut graph, the product of the classes of its connected components;* (iii) *at fixed labels, after the sum over the separable indices of the cut kernels, the sum of the cut terms of an orbit O is $(-1)^{|S|} 2^{-|S_L|} r_O^{-1} c_* \prod_K E(\mathrm{Cl}(K);$ its fields$)$, the product over the components K of the class sums with the cut kernels as external fields, c_* a constant and $r_O \ge 1$ the number of relabellings of identical S-objects preserving the class;* (iv) *the sum over all orbits is the full expansion;* (v) *if $\mathcal{G}$ is a per-label full class, the cut terms with a given cut datum (components with labels and orders, stubs and fields at each component, incidence of the S-objects) correspond s_δ-to-one to the tuples of members of the per-label full classes of the components, members without internal line and of order 0 included; each such group is a union of orbits and is closed under the exchange of the scalar legs of every two-scalar-leg or $\phi\phi A$ subgraph.*

The proof is given in appendix G.1 (lemmas G.3 and G.5). This is Bałaban's regrouping step in the proof of [3, Prop. 1, p. 432] ("if we remove a line in any graph of the class, then for the obtained graph we can find in this class all graphs necessary to form a new renormalized class"), made explicit for decomposed objects, with the sharper conclusion that one orbit gives exactly one class of the cut graph. It is needed wherever the renormalisation bound is applied to a family some of whose lines have been cut, in particular in the comparison of two propagator domains and in the telescoping of the cube's large-field remainders; at a nonzero background, where the bound of [3, Prop. 1] is available for per-label full classes, it is used in the form (v).

Remark 5.3 (the cube). The propositions below are stated for King's RG cube C_J^K, on which King's Dirichlet weight is compatible with the image group of the two Dirichlet planes (they are $2MJ$ apart). On the comparison cube $\mathrm{ND}^{(a)}$ the planes are $2MJ+\varepsilon_k$ apart, no partition into consecutive blocks is invariant under the image group, and the flow does not exist; this is why theorem 3.2 is posed on King's convention, and why the fixed-cube limits of the comparison family are reached only through corollary 4.4. The cube of the existence theorem is C_J^K itself (remark 3.4).

5.3. **The inputs at zero scalar source.**

Proposition 5.4 ($(\mathrm{H0rel})_C$: the small-field representation on King's cube, zero scalar source)**.** *Fix $J \geq J_0$, a family $f_1, \dots, f_N$ and $x \in \mathbb{R}^N$, and let $s \in \{u_x, 0\}$. There are $\eta_1 > 0$, k_0 and $C, c, \sigma > 0$, depending on the couplings, on J, on f and x, and on nothing else (not on k, j), such that for all $k \geq k_0$ and all j with $J_* \leq j \leq k - J_*$ and $\eta = 5^{j-k} \leq \eta_1$:*

$$e^{-C\eta^\sigma} I_k^{(j)}(s) \;\leq\; Z_k(s) \;\leq\; e^{C\eta^\sigma} I_k^{(j)}(s) + \rho(\eta)\, Z_k(0), \qquad \rho(\eta) := e^{-c\,p(\eta)^2 + C}, \tag{2}$$

where $p(\eta) = b_0(1 + \ln \eta^{-1})^p$, $p \geq 1$, is Bałaban's small-field threshold. Moreover, at $s = 0$ the two terms of the upper line are the two parts of Bałaban's decomposition: $\int P\, d\mu_{k,0} \leq e^{C\eta^\sigma} I_k^{(j)}(0)$ and $\int (1-P)\, d\mu_{k,0} \leq \rho(\eta) Z_k(0)$ ((R_0)).

The proof is given in Appendix A. This is [13, Thm. 3.1] on the cube, which [14, Sect. 5.2] asserts ("the proof follows by modifying suitably the material in [Ba 1] and [Ba 2]"), in the form in which the remainder is measured against $Z_k(0)$ rather than against an absolute constant, and with constants uniform in the step j (the form the diagonal pairs need, since $\eta_k \to 0$). The lower line is the weaker of the two readings of the sign in [13, (3.3)]. No lower bound on $Z_k(0)$ appears. Of the statements of section 5.2, (BAL_5)[int] (through Bałaban's torus bounds) and $(\mathrm{REN}_{\mathrm{orb}})$ (through the comparison of propagator domains and the telescoping of the large-field remainders) enter.

Proposition 5.5 ($(\mathrm{K\text{-}pos})_N$: one-step bound on the normalised action, zero scalar source)**.** *With J, f, x and $s \in \{u_x, 0\}$ as above there are $\gamma \in (0,1)$, $\beta, \sigma > 0$ and $C_3 < \infty$, depending on the couplings, J, f, x only, such that for all $k \geq k_0$, all $J_* \leq j \leq k - J_*$ with $\eta = 5^{j-k} \leq \eta_1$, and all (A_j, ϕ_j) in the support of χ_j,*

$$\left| S_{N,k}^{(j)}(A_j, \phi_j; s) - S_{N,k+1}^{(j+1)}(A_j, \phi_j; s) \right| \;\leq\; C_3\big(5^{-\gamma j}\eta^{-\beta} + \eta^\sigma\big). \tag{3}$$

The proof is given in Appendix B. This is [13, Thm. 3.4] at $n = 1$ on the cube, for the field- and source-dependent part of the effective action: the two members of a diagonal pair live on one unit lattice with one indicator χ_j, and the statement compares them there. It is assembled from six pieces: the photon quadratic form (an even-restriction identity reduces the cube to the doubled torus, where [13, Prop. 3.10] applies); the scalar quadratic form at zero background (the Dirichlet symbol) and with background (a two-leg tree graph); the background determinant (one-loop graphs with a remainder, [13, (3.94)–(3.98)], of which the middle line of (3.98) is false as printed and is replaced); the graphs with curvature vertices; the source-free field-dependent graphs away from the faces and near them (the photon face terms and the scalar boundary remainders of [14, pp. 331–332]); and the block-vertex classes. For the curvature-vertex and the source-free field-dependent classes, the localisations whose tree length R exceeds the small-field threshold p are bounded by cutting the localised graph into clusters of mutual distance $> 3\sqrt{d}$ and reducing each cluster in its own box of side $O(p)$, with the cut lines as external lines of both clusters (corollary B.36(b) and lemma B.42; the zone-B tails by theorem B.51(a) and theorem B.58(d)). The vacuum constant $v_{k,j}$ is not part of the statement.

Lemma 5.6 (bounds for free)**.** *Put $\beta_f := C_0 \sum_i |f_i|_4^2$ with C_0 as in lemma 2.6. For all $k \geq 0$, $J \geq J_0$ and $x \in \mathbb{R}^N$,*

$$e^{-\beta_f |x|^2} \;\leq\; a_k(x) \;\leq\; e^{\beta_f |x|^2}.$$

Proof. By theorem 7.6(a), $a_k(x) \leq \exp(\|u_x\|_{2,k}^2)$, and $\|u_x\|_{2,k} \leq \sum_i |x_i| \|f_i\|_{2,k} \leq |x| (\sum_i \|f_i\|_{2,k}^2)^{1/2} \leq |x|\, \beta_f^{1/2}$ by lemma 2.6(E3). The lower bound is $a_k(x) a_k(-x) \geq 1$, lemma 2.11(e). □

5.4. **The Cauchy property in the spacing.**

Theorem 5.7 (the Cauchy property, ratio form)**.** *Fix $J \geq J_0$, f, $x \in \mathbb{R}^N$, and assume (2) and (3) for $s \in \{u_x, 0\}$ with the constants of propositions 5.4 and 5.5. Choose $\alpha \in (0, \min\{\frac{1}{2}, \gamma/(\gamma+\beta)\})$ and the diagonal pairs $j(k) = \lfloor (1-\alpha)k \rfloor$, $\eta_k = 5^{j(k)-k}$. Put*

$$b_k := C\eta_k^{\sigma}, \qquad \theta_k := C_3\big(5^{-\gamma j(k)}\eta_k^{-\beta} + \eta_k^{\sigma}\big), \qquad \rho_k := \rho(\eta_k), \qquad r_k(0) := 2\rho_k, \qquad r_k(u_x) := 2\rho_k e^{\beta_f |x|^2}.$$

Then there is k_1 such that for all $k \geq k_1$

$$\big|\ln a_{k+1}(x) - \ln a_k(x)\big| \ \leq\ 4b_k + 2\theta_k + r_k(u_x) + r_k(0). \tag{4}$$

The right side is summable in k, and its tail from k on is at most $C'5^{-\kappa k}$ for some $\kappa > 0$. Hence $a_k(x)$ converges, and its limit lies in $[e^{-\beta_f|x|^2}, e^{\beta_f|x|^2}]$.

Proof. Fix $k \geq k_0$ so large that $j(k) \geq J_*$, $\eta_k \leq \eta_1$ and $r_k(u_x) \leq 1$ (possible by Step 6), and put $j := j(k)$, $b := b_k$, $\theta := \theta_k$. For $s \in \{u_x, 0\}$ write $Z := Z_k(s)$, $I := I_k^{(j)}(s)$, $\hat{I} := \hat{I}_k^{(j)}(s)$ for the k-model after j steps, and $Z' := Z_{k+1}(s)$, $I' := I_{k+1}^{(j+1)}(s)$, $\hat{I}' := \hat{I}_{k+1}^{(j+1)}(s)$ for the $(k+1)$-model after $j+1$ steps. Both members have the block scale η_k and the indicator χ_j, and (2) applies to both, since $j+1 \leq (k+1) - J_*$; their remainders are $R := \rho_k Z_k(0)$ and $R' := \rho_k Z_{k+1}(0)$.

Step 1 (the remainders relative to Z). For $s = 0$, $R/Z = \rho_k = r_k(0)/2$ and $R'/Z' = \rho_k$. For $s = u_x$, $Z = a_k(x)Z_k(0) \geq e^{-\beta_f|x|^2}Z_k(0)$ by lemma 5.6, so $R/Z \leq \rho_k e^{\beta_f|x|^2} = r_k(u_x)/2$, and likewise $R'/Z' \leq r_k(u_x)/2$. In both cases $R/Z,\ R'/Z' \leq r_k(s)/2 \leq \frac{1}{2}$. No lower bound on $Z_k(0)$ and no bound on $Z_k(u_x)$ beyond lemma 5.6 is used.

Step 2 (one model). From $e^{-b}I \leq Z \leq e^{b}I + R$: $I \leq e^{b}Z$ and $I \geq e^{-b}(Z-R) = e^{-b}Z(1-R/Z)$. So $I \in [e^{-b}Z(1-R/Z),\ e^{b}Z]$, and in the same way $I' \in [e^{-b}Z'(1-R'/Z'),\ e^{b}Z']$. In particular $I, I' > 0$.

Step 3 (across the pair). On the support of χ_j, (3) gives $|S_{N,k}^{(j)}(\cdot;s) - S_{N,k+1}^{(j+1)}(\cdot;s)| \leq \theta$. The two integrals $\hat{I}$, $\hat{I}'$ run over the same variables with the same χ_j and have non-negative integrands, so $\hat{I}'/\hat{I} \in [e^{-\theta}, e^{\theta}]$. Since $I = e^{-v_{k,j}}\hat{I}$ and $I' = e^{-v_{k+1,j+1}}\hat{I}'$,

$$I'/I \ \in\ e^{\Delta}\,[e^{-\theta}, e^{\theta}], \qquad \Delta := v_{k,j} - v_{k+1,j+1},$$

and Δ does not depend on s: the vacuum constant is the same number in the numerator and in the denominator of a_k.

Step 4 (ratio of partition functions). By Steps 2 and 3,

$$\frac{Z'}{Z} = \frac{Z'}{I'}\cdot\frac{I'}{\hat{I}}\cdot\frac{\hat{I}}{I}\cdot\frac{I}{Z} \ \in\ e^{\Delta}\Big[e^{-2b-\theta}\big(1-\tfrac{R}{Z}\big),\ e^{2b+\theta}\big(1-\tfrac{R'}{Z'}\big)^{-1}\Big].$$

With $|\ln(1-t)| \leq 2t$ for $0 \leq t \leq \frac{1}{2}$ and Step 1, $|\ln(1-R/Z)|, |\ln(1-R'/Z')| \leq r_k(s)$, hence

$$\big|\ln(Z'/Z) - \Delta\big| \ \leq\ 2b + \theta + r_k(s).$$

Step 5 (ratio of ratios). $\ln\dfrac{a_{k+1}(x)}{a_k(x)} = \Big[\ln\dfrac{Z_{k+1}(u_x)}{Z_k(u_x)} - \Delta\Big] - \Big[\ln\dfrac{Z_{k+1}(0)}{Z_k(0)} - \Delta\Big]$, and Step 4 at $s = u_x$ and at $s = 0$ gives (4). (If the constants of (2), (3) differ between the two sources, take the larger ones.)

Step 6 (summability). Since $k - j(k) \geq \alpha k$, $\eta_k \leq 5^{-\alpha k}$, so b_k and $C_3\eta_k^{\sigma}$ decay geometrically. Since $j(k) \geq (1-\alpha)k - 1$ and $k - j(k) \leq \alpha k + 1$,

$$5^{-\gamma j(k)}\eta_k^{-\beta} \ \leq\ 5^{\gamma+\beta}\,5^{-(\gamma(1-\alpha)-\beta\alpha)k},$$

which decays geometrically because $\alpha < \gamma/(\gamma+\beta)$. Finally $p(\eta_k)^2 \geq b_0^2(1+\alpha k\ln 5)^2 \geq c_1 k^2$ for $p \geq 1$, so $\rho_k \leq e^{-cc_1k^2+C}$; thus $r_k(u_x) \leq 1$ for large k, $\sum_k r_k(s) < \infty$, and the tail of $\sum_k \rho_k$ is eventually below any geometric tail. The series in (4) converges with a geometric tail, so $(\ln a_k(x))_k$ is Cauchy, $a_k(x)$ converges, and the limit lies in the closed interval of lemma 5.6. □

Remark 5.8 (what the argument does not use). (i) No convergence, and no bound, of the vacuum constants $v_{k,j}$ (King's E_0, E_1, the kernel normalisations, the face vacuum energies) enters: they cancel in Step 3. King's printed route [14, p. 329] goes through the convergence of Z_k itself, which

requires exactly those constants to converge. (ii) No upper stability bound, no lower bound on $Z_k(0)$ and no bound on $Z_k(u_x)$ beyond lemma 5.6 is used; the remainder of (2) is compared with $Z_k(0)$ only. (iii) No reflection positivity, no Ward identity, and no subsequence. (iv) The all-n form of [13, (3.13)] (comparing the k-model with the $(k+n)$-model at a fixed stopping scale η for every n) also gives the Cauchy property, provided the constants of (2) and (3) are taken at fixed η with no growth in k: then $|\ln a_{k+n} - \ln a_k| \le 4b(\eta) + 2\theta_k(\eta) + 4\rho(\eta)(e^{\beta_f|x|^2}+1)$ for every n, and one lets $k\to\infty$ and then $\eta\to 0$. The diagonal form above uses $n=1$ only, at the price of constants uniform in j; the two forms need different readings of the inputs, and neither implies the other.

Theorem 5.9 (King's existence theorem on the cube: the curvature field)**.** *For every finite family $f_1,\dots,f_N$ of real 2-forms in $C_c^\infty(\mathbb{R}^3;\Lambda^2)$, every $J \ge J_0(f)$ and every $x\in\mathbb{R}^N$ the limit*

$$Z_J^K(x) = \lim_{k\to\infty} \langle e^{\sum_i x_i F(f_i)}\rangle_{k,J}^K$$

exists and lies in $[e^{-\beta_f|x|^2}, e^{\beta_f|x|^2}] \subset (0,\infty)$, with β_f as in lemma 5.6; the convergence is geometric. That is, theorem 3.2 holds. Consequently Z_J^K and Z_∞^K exist locally uniformly on $\mathbb{C}^N$ and are entire, with the bounds and the monotonicity of theorem 7.6(b).

Proof. Propositions 5.4 and 5.5 supply the hypotheses of theorem 5.7 at every x; the theorem gives the limit and its bounds; theorem 7.6(b) (that is, corollary 4.4(ii)) gives the rest. The constants k_0, η_1 and the rates may depend on x and on f; existence needs no uniformity in x. □

Remark 5.10 (the comparison family). Theorem 5.9 says nothing about the fixed-cube limits of the comparison family $\mathrm{ND}^{(a)}$ (remark 5.3); for that family corollary 4.4(ii)(γ) squeezes the limits inferior and superior at fixed J between Z_J^K and Z_{J+1}^K, and corollary 4.4(iii) identifies every subsequential limit, followed by $J\to\infty$, with Z_∞^K. At fixed J every inequality available for the comparison family bounds its oscillation in k only by the difference of two distinct limits, which tends to zero only as $J\to\infty$; a Cauchy statement at fixed J for that family would need the small-field analysis at its faces, where no flow of the kind used above exists.

6. King's existence theorem on the cube with scalar sources

This section proves theorem 3.5, the existence theorem with scalar sources in the form the Wick leg consumes: in the reference scheme z, the fixed-cube limits of the generating functional with curvature and scalar sources exist, locally uniformly and holomorphically, on the domain of coefficient vectors whose positive scalar part is below a threshold, and they are positive at real points; the negative scalar part is unrestricted. The route is the Cauchy argument of section 5 with the scheme c-number carried along (theorem 6.14), applied at real points with a small non-negative scalar source, followed by a normal-family argument on the domain. The domain is bounded by Hölder's inequality from three one-sided estimates: the curvature bound of lemma 5.6, a positive-source bound (proposition 6.10), and a negative-source bound (theorem 6.16), the last obtained from positive-source statements at a heavier scalar mass by an exact mass shift and a correlation inequality. The notation is that of section 5.1.

6.1. **Setting; the source-dependent graphs.** Throughout, the scheme is the reference scheme $z = a^\sharp$ of definition 2.3; write z_k for its k-th term. Put $\mathsf{S}_k(h) := \sum_{x\in\Lambda_k\cap C_J} \varepsilon_k^3 h(x)$ and $c_k(h) := z_k \mathsf{S}_k(h)$, which is linear in h. For a real source $s=(u,h)$ the normalised functional in the scheme is

$$n_k(s) \;:=\; e^{-c_k(h)}\, N_k(s) \;=\; e^{-z_k\mathsf{S}_k(h)}\,\langle e^{F(u)+:\phi^2:(h)}\rangle_{k,J}^K \;=\; Z_{k,J}^{z,K}(s),$$

the object of theorem 3.5. For a finite family $f_1,\dots,f_n$ of real 2-forms and $h_1,\dots,h_p$ of real functions, all supported at ℓ^∞-distance >1 from $\mathbb{R}^3\setminus C_J$, and $\zeta=(z,w)\in\mathbb{C}^{n+p}$, $n_k(\zeta) := e^{-\sum_j w_j c_k(h_j)}\langle e^{\sum_i z_i F(f_i)+\sum_j w_j:\phi^2:(h_j)}\rangle_{k,J}^K$ (here z_i are coefficients and z_k the scheme term; no confusion will arise). Every n_k is entire in ζ, $|n_k(\zeta)|\le n_k(\mathrm{Re}\,\zeta)$, and $\ln n_k$ is convex on real points (lemma 2.11(d)): a log-Laplace transform minus a linear function.

A second scalar mass $M > 0$ will be used with the *same* e, λ, μ_0, lattice, cube and boundary conditions; $\langle\cdot\rangle^{(M)}$ denotes King's cube state at mass M, $\alpha_k(M) := M^2 + \delta m^2_{\varepsilon_k}(M)$ King's free-boundary bare coefficient at mass M (as in lemma 3.9), $u_k(M) := \frac{1}{2}(\alpha_k(M) - \alpha_k(m))$, and $s_-(M) := \liminf_k u_k(M)$. Every statement of section 5 and of this section holds at every point of King's range with point-dependent constants, in particular at the point (e, λ, M, μ_0); we use this at $M = m$ and at one heavier M_2.

Lemma 6.1 ((K_h): one-step convergence of the source-dependent graphs on King's cube)**.** *Let $P_k^{(j),1}$ be the interaction part of King's effective action after j steps on C_J^K [13, (3.17)], written as a sum over connected renormalised graphs H, and let $\mathcal{N}_h$ be the family of its members that depend on the scalar source h: the field-dependent part of every graph with an h-vertex, and every field-independent h-dependent graph except the h-linear ones of the three lowest orders $(\alpha, \beta) \in \{(0,0), (0,1), (2,0)\}$ in (e^2, λ). Then for every fixed real $h \in C_c^\infty$ supported at distance > 1 from $\mathbb{R}^3 \setminus C_J$ and every member of $\mathcal{N}_h$, for $J_* \le j \le k - J_*$: the localisation bound* [13, (3.39)] *holds for its localisations at distance from the faces larger than the small-field scale; the bound* [13, (3.38)] *with the restricted degree holds with the factor $(1 + k - j)^{p_0}$, summed over the localisations sharing the cube of an h-vertex, and for each single localisation at distance > 1 from the faces; and the sum over positions of the difference between the member at (k, j) and the member at $(k + 1, j + 1)$, on the common unit lattice, is bounded by $C\big[5^{-\gamma' j}\eta^{-\beta'} + \eta^{\sigma'}\big]\,|C_J|$ with C independent of j and k.*

This is the cube form, on C_J^K with King's Dirichlet weight, of the corresponding statement on King's tori; the h-free graphs, those with a curvature vertex and no h-vertex and the source-free ones, are covered on the cube by proposition 5.5. The localisations of a member of $\mathcal{N}_h$ with tree length $R \le p$ lie at distance > 1 from the faces, where the bounds reduce to those of the free flow on the torus (proposition 6.4); a localisation with $R > p$ carries the factor $e^{-\delta R}$ once the localised graph is cut into clusters of mutual distance $> 3\sqrt{d}$, each reduced in its own box of side $O(p)$, the clusters containing the source vertex lying away from the faces and the clusters meeting the Dirichlet faces being source-free (proposition 6.5). The proof is given in section 6.2.

6.2. **Proof of lemma 6.1.** We work in the unit units of depth j of appendix B.5: the live sites are T_η, F denotes the scalar Dirichlet planes, $t(x) := \mathrm{dist}(x, F)$, and a localised graph H carries King's boxes $\square \subset \square'' \subset \square'$ and $\Omega = \square' \cap T_\eta$, in zone I if $\square' \cap F = \emptyset$ and in zone B otherwise. The localisation is by centres, as in the proof of proposition 5.5: a tuple τ has the point set $P(\tau)$, $N := |P(\tau)| \le N_*$ with N_* depending on the order $\bar{n}$ of the expansion only, and the tree length $R = R(\tau) := \mathrm{dist}(P(\tau))$. The cut by clusters is that of lemma B.33: $c_0 := \frac{13}{3}\sqrt{3}$, $R_0 := 3\sqrt{3}$, the clusters K are the components of the c_0-proximity graph on $P(\tau)$, every line whose ends lie in different clusters is cut and kept whole as an external line of both cluster graphs H_K, and H_K is reduced with its own boxes $\square_K \subset \square''_K \subset \square'_K$ and gauge $B_K = A_j(x_0^K)$; put $\Omega_K := \square'_K \cap T_\eta$. A cluster is in zone I if $\Omega_K \cap F = \emptyset$ (then $\square'_K$ is not truncated and $\Omega_K = \square'_K$) and in zone B otherwise. The two members of the diagonal pair, the k-model at depth j and the $(k+1)$-model at depth $j+1$, live on the same unit lattice; every bound below holds for each of them separately with the same constants, and the two are compared only in zone I.

The h-vertex is King's vertex [13, (3.21)]: the smooth function h sampled at the points of the lattice's own frame, the coupling $\theta\eta^2$ in the units of depth j, and at most two fields $\phi^{(j)}$. In King's count of the coupling power $\rho(H) = \sum_v \rho_v$ of a graph [13, p. 665], every vertex has $\rho_v - s_v/4 \ge 0$, s_v its number of external scalar fields, and the h-vertex has $\rho_h - s_h/4 \ge 2 - \frac{1}{2} = \frac{3}{2}$; hence

$$\rho(H) - \tfrac{1}{4}s(H) \;\ge\; \tfrac{3}{2} \qquad \text{for every member of } \mathcal{N}_h \text{ with external fields,} \tag{5}$$

$s(H)$ its number of external scalar legs, and $\rho(H) \ge 0$ for the field-independent members. The Wick constant of [13, (3.21)] is field-free and h-linear of the lowest order, so it enters only the excluded family $\mathcal{T}_h$ of definition D.1 and no member of $\mathcal{N}_h$ depends on it.

The range. Let $\eta_2 \le \eta_1$ be so small that for $\eta \le \eta_2$, with $p = p(\eta)$ and M the block integer of C_J,

$$\eta^{-1} \;\ge\; 4p + 8M + 2c_0 N_*^2 + 8, \qquad p \;\ge\; c_0 N_*^2, \qquad C_\flat\, e\, \eta^{1/2}\, p^2 \;\le\; \tfrac{1}{2}, \tag{6}$$

where $C_\flat$ is the constant of lemma 6.2(e) below times the constant of the difference bound [13, (3.45)]. All three hold for η small because $p(\eta)$ grows slower than every power of η^{-1} and faster than every logarithm. The first condition contains the one of appendix B.5, so J_* is taken with $5^{-J_*} \le \eta_2$.

Lemma 6.2 (the source vertices and the clusters)**.** *Let $\eta \le \eta_2$, let H be a member of $\mathcal{N}_h$, s an h-vertex of H with centre z_s, and τ a tuple of H. Then:* (a) $t(z_s) \ge \eta^{-1} - 3$. (b) *If τ is in zone B, then $R(\tau) \ge \eta^{-1}/2$.* (c) *If τ is in zone B, then τ has at least two clusters.* (d) *Every cluster K containing an h-vertex has $\operatorname{dist}(\square_K, F) \ge 2p + M + 1$, $\Omega_K = \square'_K$ and $\operatorname{dist}(\square_K, \partial\Omega_K) \ge 2p$: it is a zone-I cluster. Hence every zone-B cluster is source-free.* (e) *Every $\square'_K$ has side $\le c_0 N_* + 4p + 6M \le C_{\mathrm{box}}\, p$. The same holds with a curvature vertex in place of the h-vertex.*

Proof. (a) $h(x) \ne 0$ forces ℓ^∞-$\operatorname{dist}(x, \mathbb{R}^3 \setminus C_J) > 1$ in physical units, that is $> \eta^{-1}$ unit lengths; the planes F lie within one fine spacing $5^{-j} \le 1$ of ∂C_J, and the vertex lies in $\operatorname{supp}\chi_{z_s}$, within $\frac{2}{3}\sqrt{3} < 2$ of its centre. (b) A tree of total length R joins the points of $P(\tau)$, so all of them lie within R of z_s; $\square$ lies in the sup-norm cube of half-side $R+M$ about z_s, the large blocks within $2p$ of it in that of half-side $R+2p+2M$, and $\square'$ in that of half-side $R+2p+3M$. If $\square'$ meets F, then $t(z_s) \le R+2p+3M$, and with (a) and (6) $R \ge \eta^{-1} - 2p - 3M - 3 \ge \eta^{-1}/2$. (c) With one cluster $\operatorname{diam} P(\tau) \le c_0(N-1)$ (lemma B.33(c)), so $R(\tau) \le (N-1)\operatorname{diam} P(\tau) \le c_0(N_*-1)^2 < \eta^{-1}/2$ by (6), against (b). (d) If $s \in K$, every point of P_K lies within $c_0(N-1)$ of z_s, so $\square_K$ lies in the cube of half-side $c_0 N + M$ about z_s and $\square'_K$ in that of half-side $c_0 N + 2p + 3M$. By (a) and (6), $t(z_s) - (c_0 N + 2p + 3M) \ge \eta^{-1} - 3 - c_0 N - 2p - 3M > 2p + M + 1$. So $\square'_K$ misses F, it is not truncated, it contains every large block within $2p$ of $\square_K$, and $\operatorname{dist}(\square_K, F) \ge t(z_s) - c_0 N - M \ge 2p + M + 1$. (e) $\square_K$ has side $\le \operatorname{diam} P_K + 2M \le c_0 N_* + 2M$ and $\square'_K$ adds $2p + 2M$ per side; truncation at ∂T_η only shrinks it. The curvature source f is supported at the same distance from $\mathbb{R}^3 \setminus C_J$, so the same arithmetic applies to its vertex. $\square$

Lemma 6.3 (cut lines vanish on the Dirichlet planes)**.** *Let ℓ be a scalar, dressed or external cut line of lemma B.33 with its end u in a cluster K, in either model, at the background $A^{(j)}$ or at zero background.* (a) *For every evaluation point u' of the u-end with $t(u') \le \frac{1}{4}$ and every operation O_v at the far end (a difference or Hölder quotient of order $\le 1+\alpha$, or the identity) whose evaluation set E_v lies at distance $\ge \frac{1}{2}$ from u',*

$$|O_v \ell(u', \cdot)| \;\le\; C \min(1, t(u'))\, e^{-\delta \operatorname{dist}(u', E_v)}.$$

(b) *For a scalar or dressed cut line and evaluation points u', v' of its two ends with $|u' - v'| \ge 1$ and $t(u'), t(v') \le \frac{1}{4}$,*

$$|\ell(u', v')| \;\le\; C \min(1, t(u'))\min(1, t(v'))\, e^{-\delta |u' - v'|}.$$

The constants are those of lemma B.34(b).

Proof. Each of the three kinds of line vanishes when its u-end lies on F: a scalar line by the Dirichlet condition of the cube propagator, a dressed line and an external line through their first factor, which is such a propagator. Let $u'_F \in F$ be the foot of u' and telescope along the $t(u')/5^{-j}$ bonds of the normal segment from u'_F to u'; the parallel transporters along a bond are unimodular, so $|O_v \ell(u', \cdot)| \le t(u') \max_b |O_v D_b \ell(\cdot, \cdot)|$, the maximum over the bond differences D_b on that segment. Every such bond lies within $\frac{1}{4}$ of u', hence at distance $\ge \frac{1}{4}$ from E_v, and the mixed bound of lemma B.34(b) ($C^{1,\alpha} \otimes C^{1,\alpha}$ at unit scale in both arguments, decay $e^{-\delta_1 |u-v|}$, no distance condition) bounds each by $C e^{-\delta \operatorname{dist}(u', E_v)}$. This is (a). For (b), $\ell(u', v'_F) = 0$ by the Dirichlet condition in the second variable (for a dressed line through its last factor); telescoping at v' along its normal segment gives $|\ell(u', v')| \le t(v') \max_b |D_b \ell(u', \cdot)|$ over bonds b within $\frac{1}{4}$ of v', each at distance $\ge 1 - \frac{1}{4} - 5^{-j} \ge \frac{1}{2}$ from u', and (a) with $O_v := D_b$ bounds each by

$C\min(1,t(u'))e^{-\delta|u'-v'|}$. Where $t>\frac14$ at an end, $\min(1,t)\ge\frac14$ and lemma B.34(b) alone gives the bound. □

Proposition 6.4 (the branch $R\le p$: zone I)**.** *Let H be a member of $\mathcal{N}_h$ and τ a localisation of H in zone I; by lemma B.20(i) this covers every localisation with $R\le p$. Then, for each model, on* $\operatorname{supp}\chi_j$*,*

$$|E^{(j)}(H;\tau)| \;\le\; C(1+k-j)^{p_0}\,\eta^{\rho(H)}\,p^{n_1}\,e^{-\delta R}, \tag{7}$$

which is [13, (3.39)]*, and across the diagonal pair, on the common unit lattice,*

$$|E'^{(j+1)}(H;\tau)-E^{(j)}(H;\tau)| \;\le\; C\,\eta^{\rho(H)-1/2}\,p^{n_1}\left[5^{-\gamma j}+\eta^{3+\sigma+1/2}\right]e^{-\delta R}. \tag{8}$$

The constants depend on the point of King's range, J, $\bar n$, $\|h\|_{C^1}$ and, for members with curvature vertices, $\max\|f_{\mu\nu}\|_{C^{2,\alpha}}$*; not on j, k, τ.*

Proof. By lemma B.21, which uses only that the localisation is in zone I, $E(H)=E(H^{\mathrm{free}})+\mathcal{R}$ with H^{free} the same graph on the free f_D flow, every term of $\mathcal{R}$ carrying an image line of any prescribed degree with the factor $e^{-\delta_0 p/2}\le C\eta^{3+\sigma+1/2}$, and the free-boundary counterterm renormalising H^{free} exactly. For H^{free} we run King's proof of [13, Thm. 3.5, Prop. 3.6] as in the proof of theorem B.37, with the following points for the h-vertex. *Census:* every connected component of an ordered prefix that contains the h-vertex has positive degree, the vertex [13, (3.21)] carrying the coupling power 2 against at most two fields, except the field-independent h-linear graphs of the three lowest orders P_0 [13, Thm. 3.5], which form the family $\mathcal{T}_h$ of definition D.1 and are not in $\mathcal{N}_h$; the h-free components obey the degree count of theorem B.24, with the Ward rows by theorem B.30, and the components with block points are convergent by lemma B.41. *Bound:* the ordered multiscale bound theorem B.40(a) with the root held at the h-vertex gives (7); the power of η is $\rho(H)$ by [13, p. 665], the fields on $\operatorname{supp}\chi_j$ being bounded by [13, (3.2)], which is the content of (5), and the logarithm comes from the infrared sums as in theorem B.37. *Comparison:* theorem B.40(b), one factor at a time [13, pp. 663–665]: the lines by proposition B.9, the h-vertex by the sampling of the same smooth h at points of the two frames at distance $\le\sqrt3\,\varepsilon_k$, whose difference is $\le C\varepsilon_k\|h\|_{C^1}\le C5^{-\gamma j}$, and the finer slices through the finest index sum; each error term is the graph with one element weakened, which costs the factor $\eta^{-1/2}$ in (8). The identities [13, (3.43)–(3.55)] act on the h-vertex only through the phases at its scalar ends, which the gauge transformation [13, (3.46)] carries. Members with curvature vertices: lemma B.22(b)–(d) as in theorem B.37. The h-vertex is bounded by $\|h\|_\infty$, and only through it does h enter the constants of (7). □

Proposition 6.5 (the branch $R>p$: the cut by clusters)**.** *Let $\eta\le\eta_2$, let H be a member of $\mathcal{N}_h$ and τ a tuple of H with at least two clusters. By lemma 6.2(c) this covers every zone-B tuple, and by lemma B.33(c) and (6) every tuple with $R>p$. Then, for each model separately, on* $\operatorname{supp}\chi_j$*,*

$$|E_N^{(j)}(H;\tau)| \;\le\; C(1+k-j)^{p_0}\,\eta^{\rho(H)}\,p^{n_1}\,e^{-\delta\,\operatorname{dist}(P(\tau))}, \tag{9}$$

E_N the field-dependent part (the whole value for a field-independent member); this is [13, (3.38)] *at the tuple. Moreover, for every unit cube y_* and $0<\delta'\le\delta/2$,*

$$\sum_{\substack{\tau\ \text{in zone B}\\ \text{with an } h\text{-vertex in } y_*}} |E_N^{(j)}(H;\tau)|\,e^{\delta' R(\tau)} \;\le\; C(1+k-j)^{p_0}\,\eta^{\rho(H)}\,p^{n_1}\,e^{-\delta/(8\eta)}. \tag{10}$$

The constants depend on the point of King's range, J, $\bar n$, $\|h\|_\infty$ and, for members with curvature vertices, $\max\|f_{\mu\nu}\|_{C^{2,\alpha}}$*.*

Proof. At a tuple with at least two clusters every connected component of every cluster graph carries a cut-line leg (corollary B.36(c)), so the vacuum members and the component-wise cancellation of [13, p. 667] do not arise, and $E(H;A^{(j)};\tau)$ and $E(H;0;\tau)$ are bounded separately; at zero background the cut lines are the cube propagators at $A=0$ and $B_K=0$. Fix one of the two terms.

Step 0: regularised cut lines. Replace every cut line ℓ by $\ell^\flat$; by lemma B.34(a) this changes nothing.

Step 1: reduction per cluster. Each H_K is reduced by [13, (3.43)–(3.55)] with its own box Ω_K, centre x_0^K, gauge B_K and field $\tilde{A}_K = A^{(j)} - B_K$ on Ω_K (lemma B.33), its cut lines keeping the background and receiving the unitary factors [13, (3.46)] at their ends. The expansion of the reduced propagator in $\tilde{A}_K$ converges on Ω_K: $e_j \sup_{\Omega_K} |\tilde{A}_K| \le C e_j p \operatorname{diam} \Omega_K \le C_\flat e \eta^{1/2} p^2 \le \frac{1}{2}$ by the difference bound [13, (3.45)], lemma 6.2(e) and (6): this is the reduction of the branch $R \le p$ on a box of side $O(p)$, run inside each cluster. By lemma B.33(a) the partition, the boxes, the gauges and the cut status of every line are the same for every member of every family met in the nesting and for every position, so renormalisation families are formed inside each H_K. A zone-I cluster, in particular every cluster containing a source vertex (lemma 6.2(d)), satisfies $\operatorname{dist}(\square_K, \partial\Omega_K) \ge 2p$ and $\Omega_K \cap F = \emptyset$, so lemma B.21 applies to H_K: it is its free f_D-flow version plus terms carrying an image line of any prescribed degree, with the cut lines as external legs; the vertices of the expansion in $\tilde{A}_K$ that lie in $\square'_K \setminus \square_K$ are joined to a localised vertex of K by a path of lines one of which is long, and that line supplies a credit of any order, as in step (2) of the proof of theorem B.30 run inside H_K (theorem B.35). A zone-B cluster is source-free and is rewritten by definition B.44 within Ω_K.

Step 2: the cut-line factors. Every evaluation point of a cut-line end in an output term is joined to a localised anchor of its own cluster by at most $C(\bar{n})$ lines of that cluster's rewritten units (theorem B.35), and by lemma B.34(b) the phase-dressed cut line satisfies $|O_u O_v \ell^{\flat,\mathrm{ph}}| \le C e^{-\delta_2 d(K,K')} \times$(transfer factors on the lines of those paths), for all operations O_u, O_v of order $\le 1+\alpha$ at its ends, and $\le C$ crudely. The quotient graph (clusters as nodes, cut lines as edges) is connected because H is (lemma B.33(d)); bound the cut lines of one spanning tree T of it by the first bound and the others by C.

Step 3: factorisation. The output term is bounded by $\prod_K \mathfrak{E}_K \times \prod_{\ell\in T} C e^{-\delta_2 d(K,K')} \times C^{\#\text{cut lines}}$, where $\mathfrak{E}_K$ depends on the positions of K only; the transfer factors cost one extra decay draw per line of a cluster, within the budget of theorem B.35.

Step 4: the cluster factors. Zone-I clusters. The cluster graph is a free f_D-flow graph with its cut lines as undifferentiated external legs of unit scale (lemma B.34(b)). For the cluster containing the h-vertex, the census is that of proposition 6.4; since the cluster carries a cut leg, every component has an external leg and the excepted field-independent family does not occur. For the source-free zone-I clusters, the degree count holds at every tuple by theorem B.24 and theorem B.35, and block points by lemma B.41. The multiscale bound theorem B.40(a) with the root held, summed over the root in Ω_K (volume $\le Cp^3$ by lemma B.33(c)), gives $\sum_{\text{positions of } K} \mathfrak{E}_K \le C\eta^{\rho_K} p^{n_K} (1+k-j)^{p_{0,K}}$, with ρ_K the coupling power of the vertices of K. *Zone-B clusters.* The cluster is source-free (lemma 6.2(d)); apply theorem B.51(a) and theorem B.58 with no vertex held to the renormalised family of H_K, localised at P_K with the box Ω_K of side $O(p)$; the hypothesis "at least one external field" is the cut leg. The proof of theorem B.51 uses an external scalar leg only in the following ways, each of which a cut line satisfies: through its sup and $(1,\alpha)$-norms on Ω_K up to the faces, in the Taylor and transport steps, the Hölder splits and the ordered induction (lemma B.34(b)); and, when the charging rule of lemma B.50 charges the leg at a $D' = -1$ remainder vertex v, through the vanishing factor $m_\alpha(v;1)$ of lemma B.49(ii), which a cut line supplies by lemma 6.3(a). A cut line joining two zone-B clusters may be charged at both ends, by the charging rules of the two clusters; then lemma 6.3(b) supplies both factors. After the borrowing of lemma B.50 the cut-line factor left in the output term is $\ell/[m_\beta(u';1) m_\beta(v';1)]$ (or with one factor), an undifferentiated evaluation bounded by $C \min(1, t(u'))^{1-\beta} \min(1, t(v'))^{1-\beta} e^{-\delta \operatorname{dist}} \le C e^{-\delta \operatorname{dist}}$ by lemma 6.3, which Steps 2–3 treat like every other cut-line factor. The resulting bound is again $\le C\eta^{\rho_K} p^{n_K} (1+k-j)^{p_{0,K}}$, the logarithm coming from the infrared sums inside Ω_K.

Step 5: the tree. By lemma B.33(d), $\prod_{\ell\in T} e^{-\delta_2 d(K,K')} \le e^{\delta_2 c_0 N} e^{-\delta_2 \operatorname{dist}(P(\tau))}$.

Step 6: exponents. $\rho(H) = \sum_K \rho_K$, the count being vertex-local (an unlocalised mass-counterterm vertex on a cut dressed line keeps its own coupling power); $n_1 := \sum_K n_K$ and

$p_0 := \sum_K p_{0,K}$ plus the dressed-line factor of lemma B.34(b); the number of clusters is $\le N_*$ and $\delta := \delta_2$. This is (9).

The root sum. On the tuples of (10), $R(\tau) \ge \eta^{-1}/2$ by lemma 6.2(b). The number of tuples of H with a point in y_* and $R \in [n, n+1)$ is $\le C(n+1)^{3N_*}$, and $e^{\delta' R} e^{-\delta R} \le e^{-\delta R/2}$. So the sum is $\le C(1+k-j)^{p_0} \eta^{\rho(H)} p^{n_1} \sum_{n \ge \eta^{-1}/2-1} (n+1)^{3N_*} e^{-\delta n/2} \le C'(1+k-j)^{p_0} \eta^{\rho(H)} p^{n_1} e^{-\delta/(8\eta)}$, since $(n+1)^{3N_*} e^{-\delta n/4} \le C$ and $\eta^{-1} \ge 8$. Members with curvature vertices: the curvature vertex lies in a zone-I cluster (lemma 6.2(d)), is invariant under the reduction (lemma B.22(b)) and is bounded through lemma B.22(c). □

Proof of lemma 6.1. Let H be a member of $\mathcal{N}_h$ and $J_* \le j \le k - J_*$, so that $\eta \le \eta_2$. *First clause.* (7) is [13, (3.39)] for the localisations in zone I, which are those at distance from the faces larger than the small-field scale. *Second clause.* A tuple with one cluster has $R \le c_0(N_*-1)^2 \le p$, lies in zone I by lemma B.20(i), and obeys (7); a tuple with at least two clusters obeys (9). Summed over the localisations sharing the cube y_* of an h-vertex: the zone-I tuples by $\sum_n (n+1)^{3N_*} e^{-(\delta-\delta')n} \le C$, the zone-B tuples by (10). This is [13, (3.38)] with the factor $(1+k-j)^{p_0}$. *Third clause.* Split the localisations at $R = p$. Those with $R \le p$ are in zone I; sum (8) by lemma B.42 (branch $R \le p$): the result is $\le C\eta^{\rho(H)-1/2} p^{n_1} (1+k)^{p_0} \eta^{-3} |C_J| [5^{-\gamma' j} + \eta^{3+\sigma+1/2}]$, whose first term is $\le C 5^{-\gamma' j} \eta^{-\beta'} |C_J|$ for every $\gamma' < \gamma$ and $\beta' > \frac{7}{2} - \rho(H)$, the powers $p^{n_1}(1+k)^{p_0}$ being absorbed on the diagonal sequence as in lemma B.42, and whose second term is $\le C\eta^{\rho(H)+\sigma} p^{n_1} (1+k)^{p_0} |C_J| \le C\eta^{\sigma'} |C_J|$ for every $\sigma' < \sigma$. Those with $R > p$ have at least two clusters; bound the two members of the pair separately by (9), with the roots at the h-vertex, $\le C\eta^{-3}|C_J|$ of them. For the zone-I tuples among them the factor $e^{-(\delta-\delta')p}$ of lemma B.42 (branch $R > p$) is $\le C\eta^{3+\sigma}$, since $p(\eta)$ grows faster than every logarithm, and the sum is $\le C\eta^{-3}|C_J| \eta^{\rho(H)} p^{n_1} (1+k-j)^{p_0} \eta^{3+\sigma} \le C\eta^{\sigma'} |C_J|$; for the zone-B tuples (10) gives $\le C\eta^{-3}|C_J| \eta^{\rho(H)} p^{n_1} (1+k-j)^{p_0} e^{-\delta/(8\eta)} \le C_N \eta^N |C_J|$ for every N. In both cases $\rho(H) \ge 0$, and the external fields on $\operatorname{supp} \chi_j$ are accounted for in $\rho(H)$ by (5). Adding the three parts gives the third clause with C independent of j and k. The constants depend on the point of King's range, so the lemma holds at every point of the range, in particular at the second mass M_2 of section 6.1. □

6.3. **The inputs with scalar sources.** For a source $s = (u, h)$ put $\mathsf{S}(s) := \sum_i |x_i| \max_{\mu\nu} \|f_{i,\mu\nu}\|_{C^{2,\alpha}} + \|h\|_\infty$, a size; a compact family of sources is contained in $\{\mathsf{S}(s) \le S_0\}$.

Proposition 6.6 ((U$^+$): the small-field lines at a non-negative scalar source)**.** *Fix $J \ge J_0$, $\alpha \in (0, \frac{1}{2})$ and $S_0 > 0$. There are k_0 and $C, \sigma > 0$, depending on the couplings, J, α, S_0 and on nothing else, such that for every real source $s = (u, h)$ with $h \ge 0$ and $\mathsf{S}(s) \le S_0$, every $k \ge k_0$ and both members $(k', j') \in \{(k, j(k)), (k+1, j(k)+1)\}$ of the diagonal pair,*

$$Z_{k'}(s) \;\ge\; e^{-C\eta_k^\sigma} I_{k'}^{(j')}(s) \qquad \text{and} \qquad \int P \, d\mu_{k',s} \;\le\; e^{C\eta_k^\sigma} I_{k'}^{(j')}(s). \tag{11}$$

The proof is given in Appendix C. These are the two small-field lines of (2) with the scalar source inserted into the small-field analysis through the vertex [13, (3.21)]; the large-field remainder at $s \ne 0$ is not part of the statement, and is supplied instead by lemma 6.12. The source enters the small-field steps only through its vertex, and the sign $h \ge 0$ is used only through the sign of the Wick-ordered source term in the large-field region, which is not examined here.

Proposition 6.7 ((B^h): one-step bound with the scheme c-number)**.** *Fix $J \ge J_0$ and a compact family Σ of real sources. For $\sigma \in \Sigma \cup \{0\}$ and h_σ its scalar part, there is a sequence $\theta_k(\sigma) \le \theta_k$, with θ_k independent of $\sigma \in \Sigma$ and decaying geometrically, such that for $k \ge k_0$, on the support of $\chi_{j(k)}$,*

$$\left| S_{N,k}^{(j(k))}(\cdot;\sigma) + c_k(h_\sigma) - S_{N,k+1}^{(j(k)+1)}(\cdot;\sigma) - c_{k+1}(h_\sigma) \right| \;\le\; \theta_k(\sigma). \tag{12}$$

The proof given in Appendix D. The field-dependent and the h-nonlinear parts are proposition 5.5, the graphs with curvature vertices at $s = (u, h)$, and (K$_h$). The field-independent h-linear part of the three lowest orders is a c-number $\bar\beta_k^{(j)} \int h$ up to a localisation error, and

$c_k(h)$ is exactly the c-number that makes $\bar\beta_k^{(j)} + z_k$ converge along the diagonal pairs at the rate $5^{-\gamma' j}\eta^{-\beta'} + \eta^{\sigma'}$; this is the sense in which z is the scheme adapted to King's flow. At $\sigma = 0$, (12) is (3) with $j = j(k)$.

Lemma 6.8 (c-numbers)**.** (a) $|z_k| \le C(1+k)$. (b) *For every* $M > 0$, $\Delta_k(M) := 2(\mathsf{W}_{\varepsilon_k}(m) - \mathsf{W}_{\varepsilon_k}(M)) + z_k(m) - z_k(M)$ *is bounded in* k, *where* $\mathsf{W}_\varepsilon(M) = (-\Delta_{\varepsilon\mathbb{Z}^3} + M^2)^{-1}(0,0)$ *and* $z_k(M)$ *is the reference scheme at mass* M.

The proof is given in Appendix E. Both are statements about lattice Feynman integrals: (a) is the logarithmic growth of the order-e^2 coefficient of $\langle|\phi|^2\rangle$ with the counterterm inserted; in (b) the Wick difference is a convergent lattice integral in three dimensions, and the scheme difference converges.

Proposition 6.9 ((NEG$_\infty$): the bare coefficient grows with the mass)**.** *For every point of King's range* ($e, \lambda \ge 0$, $m, \mu_0 > 0$) *and every* $M > m$, $u_\infty(M) := \lim_k u_k(M)$ *exists and*

$$2u_\infty(M) \;\ge\; (M-m)\Big(M + m + \frac{4\lambda}{\pi} - \frac{e^2}{4\pi}\Big) - \Big(\frac{e^4}{4\pi^2} + \frac{8\lambda^2}{\pi^2}\Big)\ln\frac{M}{m};$$

in particular $s_-(M) = u_\infty(M) \to \infty$ *as* $M \to \infty$, *and* $s_-(M_2) \ge \frac12$ *for* $M_2 := \max\{m,\ \frac{e^2}{4\pi} + (\frac{e^4}{4\pi^2} + \frac{8\lambda^2}{\pi^2})/m\} + 1$.

The proof is given in Appendix E. The content is the exact continuum limit of King's free-boundary counterterm at all five of its orders and the k-uniform bounds on its derivatives in the mass; only $\sup_M s_-(M) = \infty$ is used below.

Proposition 6.10 ((W$^\sharp$)[s_W]: the positive-source bound)**.** *Assume* (I2) *and* (I3) *(propositions 12.2 and 12.6;* (I2) *follows from* (T)*). Then with* $s_W := s_1$ *(the amplitude of proposition 13.2) and* $\kappa = C_U/s_1$: *for every* k, J *and every lattice function* $0 \le q \le s_W$ *on* X_J,

$$\ln n_k(0,q) \;\le\; \kappa\,\|q\|_{1,k}.$$

In particular $\sup_k n_k(0,q) < \infty$ *for every* $0 \le q \in C_c^\infty$ *with* $\sup q \le s_W$ *and* $\operatorname{supp} q$ *at distance* > 1 *from* $\mathbb{R}^3 \setminus C_J$.

Proof. $e^{\sum_x \varepsilon^3 q(x) r_x^2}$ is a class-$\mathcal{I}$ observable, so by Step 1 of lemma 4.2 (killing the matter on T_J does not increase it; remark 4.8) $n_k(0,q) \le N^z_{k,C_J}(0,q)$, the comparison-cube functional, which proposition 13.2 at $X = X_J$ bounds by $e^{\kappa\|q\|_{1,k}}$; the scheme multiplier is common (lemma 2.11(a)). For $q \in C_c^\infty$, $\|q\|_{1,k} \le C_0'|q|_w$ by lemma 2.6(E3). □

Any other threshold with the same property serves below in place of s_1; the appendix contains a second instance, obtained from Bałaban's torus bounds through the comparison of the cube with King's tori.

6.4. **Three lemmas on positive and negative scalar sources.** For a mass M let $Y_k^{(M)}(u,h) := e^{-c_k(h)}\langle e^{F(u)+:\phi^2:(h)}\rangle^{(M)}$ denote the functional of the mass-M state *in the conventions of the mass* m (free Wick constant $\mathsf{W}_{\varepsilon_k}(m)$ and scheme $z_k = z_k(m)$), so that $Y_k^{(m)} = n_k$ and $Y_k^{(M)}(u,h) = e^{-\Delta_k(M)\mathsf{S}_k(h)} n_k^{(M)}(u,h)$, with $n_k^{(M)}$ the functional in the mass-M conventions and $\Delta_k(M)$ as in lemma 6.8(b). Write $r^2(q) := \sum_x \varepsilon_k^3 q(x) r_x^2$ over the live sites.

Lemma 6.11 (mass shift and the correlation inequality)**.** *Let* $M > 0$, $k \ge 0$, *and let* $0 \le q$ *be a lattice function on the live sites with* $q(x) \le u_k(M)$ *for all* x. *Then*

$$n_k(0,-q) \;\le\; \frac{1}{Y_k^{(M)}(0,q)}, \qquad \textit{and} \qquad \frac{1}{n_k(0,q)} \;\le\; n_k(0,-q) \;\le\; \frac{1}{Y_k^{(M)}(0,q)}, \quad \textit{hence} \quad Y_k^{(M)}(0,q) \le n_k(0,q).$$

Proof. Mass shift. In King's action the scalar mass enters only through $\sum_x \varepsilon^3 \frac12 (M^2 + \delta m_\varepsilon^2(M)) r_x^2$: δm_ε^2 is one number per spacing and mass (definition 2.3), the Dirichlet terms $\frac{\varepsilon}{2} r_x^2$ do not depend on M, and the photon does not see M. Hence, with $u := u_k(M)$ and $r^2(1) := \sum_x \varepsilon_k^3 r_x^2$, $e^{-S^{(m)}} = e^{-S^{(M)}} e^{u\, r^2(1)}$, and for every Φ integrable against both measures $\langle\Phi\rangle^{(m)} =$

$\langle \Phi e^{ur^2(1)}\rangle^{(M)} / \langle e^{ur^2(1)}\rangle^{(M)}$; the quartic term makes $e^{\pm r^2(\cdot)}$ integrable. *Correlation inequality.* Put $g_1 := u\mathbf{1} - q \geq 0$ and $g_2 := q \geq 0$. The mass-M state is a BFS measure, and so is the state with its single-site measures multiplied by $e^{t\, r^2(g_1)}$, $t \in [0,1]$ (definition 2.1 and the paragraph on BFS measures following it: the exponential moments stay finite because $\lambda > 0$). By lemma 2.7(c), multiplying the single-site measures by $e^{-t r^2(g_1)}$ does not increase $\langle e^{r^2(g_2)}\rangle$; read from $t = 1$ down to $t = 0$, $\langle e^{r^2(g_2)}\rangle_{t=1}^{(M)} \geq \langle e^{r^2(g_2)}\rangle_{t=0}^{(M)}$, that is

$$\langle e^{r^2(g_1)} e^{r^2(g_2)}\rangle^{(M)} \;\geq\; \langle e^{r^2(g_1)}\rangle^{(M)} \langle e^{r^2(g_2)}\rangle^{(M)}.$$

Conclusion. With $\Phi = e^{-r^2(q)}$: $\langle e^{-r^2(q)}\rangle^{(m)} = \langle e^{r^2(g_1)}\rangle^{(M)} / \langle e^{r^2(g_1+g_2)}\rangle^{(M)} \leq 1/\langle e^{r^2(q)}\rangle^{(M)}$. Multiply both sides by $e^{(2\mathsf{W}_{\varepsilon_k}(m)+z_k)\mathsf{S}_k(q)}$: the left side becomes $n_k(0,-q)$ and the right side $1/Y_k^{(M)}(0,q)$, both in the conventions of the mass m. The left inequality of the second display is lemma 2.11(e), and the last inequality follows. □

Lemma 6.12 (the large-field remainder at a source, from the one at zero source)**.** *Let σ be a real source and $p > 1$, and assume (11) at σ and* (R_0) *of proposition 5.4 for the member (k,j), with $a := C\eta^\sigma$ and $\rho := \rho(\eta)$. Then*

$$e^{-a}\, I_k^{(j)}(\sigma) \;\leq\; Z_k(\sigma) \;\leq\; e^{a}\, I_k^{(j)}(\sigma) + \rho^{1-1/p}\, \langle e^{X(p\sigma)}\rangle_k^{1/p}\, Z_k(0).$$

Proof. The lower line is the first inequality of (11). For the upper line, $Z_k(\sigma) = \int P\, d\mu_{k,\sigma} + \int (1-P) e^{X(\sigma)} d\mu_{k,0}$. The first term is at most $e^a I_k^{(j)}(\sigma)$ by (11). For the second, Hölder's inequality for the positive measure $(1-P)d\mu_{k,0}$ with the functions 1 and $e^{X(\sigma)}$, then $1 - P \leq 1$ in the second factor and $pX(\sigma) = X(p\sigma)$ (linearity), give

$$\int (1-P) e^{X(\sigma)} d\mu_{k,0} \;\leq\; \Big[\int (1-P) d\mu_{k,0}\Big]^{1-1/p} \Big[\int e^{X(p\sigma)} d\mu_{k,0}\Big]^{1/p} \;\leq\; \big[\rho Z_k(0)\big]^{1-1/p} \big[Z_k(0) \langle e^{X(p\sigma)}\rangle_k\big]^{1/p},$$

using (R_0) for the first bracket. □

At a negative scalar source the same step would need a bound on $\langle e^{X(p\sigma)}\rangle$ at a negative source of larger amplitude, which is what is to be proved; theorem 6.16 avoids this through the mass shift.

Lemma 6.13 (positivity of the limit at a positive source)**.** *Fix a mass M, a function $0 \leq q \in C_c^\infty$ as in proposition 6.10, and $\bar{w} \in (0,1]$. Assume that at mass M, in its own conventions, the hypotheses of theorem 6.14 below hold at $\sigma = (0, wq)$ for every $w \in [0, \bar{w}]$ and at $\sigma = 0$, with sequences b_k, $\theta_k(wq)$, $r_k(0)$ and $e^{-c_k(wq)} r_k(wq)$ dominated by w-independent summable sequences, for all $k \geq k_0$ with k_0 independent of w. Then $\inf_{k \geq k_0} n_k^{(M)}(0,q) > 0$.*

Proof. Steps 1–4 of the proof of theorem 6.14 use the hypotheses at $\{wq, 0\}$ only and give the lower half of (15) with w-independent summable $\bar{\delta}_k$, $\bar{\epsilon}_k$: $n_{k+1}(wq) \geq (1 - \bar{\delta}_k) n_k(wq) - \bar{\epsilon}_k$. By induction, $n_k(wq) \geq \prod_{i=k_1}^{k-1} (1 - \bar{\delta}_i)\, n_{k_1}(wq) - \sum_{i \geq k_1} \bar{\epsilon}_i$ for $k \geq k_1$. Choose $k_1 \geq k_0$ with $\prod_{i \geq k_1} (1 - \bar{\delta}_i) \geq \frac{1}{2}$ and $\sum_{i \geq k_1} \bar{\epsilon}_i \leq \frac{1}{8}$. At this fixed k_1, $w \mapsto n_{k_1}(wq)$ is continuous (a finite-dimensional integral) and equals 1 at $w = 0$; choose $w_0 \in (0, \bar{w}]$ with $n_{k_1}(w_0 q) \geq \frac{1}{2}$. Then $n_k(w_0 q) \geq \frac{1}{8}$ for all $k \geq k_1$. Since $\ln n_k(wq)$ is convex in w and vanishes at $w = 0$, $n_k(w_0 q) \leq n_k(q)^{w_0}$, so $n_k(q) \geq 8^{-1/w_0}$ for $k \geq k_1$; the finitely many $k_0 \leq k < k_1$ have $n_k(q) > 0$. □

6.5. The Cauchy argument with a *c*-number.

Theorem 6.14 (the Cauchy property with the scheme *c*-number)**.** *Fix a real source $s = (u, h)$ and let σ range over $\{s, 0\}$. Suppose there are k_0 and non-negative sequences b_k, $r_k(\sigma)$, $\theta_k(\sigma)$ such that for all $k \geq k_0$:*

(U) *for every σ and both members (k', j') of the diagonal pair at k, $e^{-b_k} I_{k'}^{(j')}(\sigma) \leq Z_{k'}(\sigma) \leq e^{b_k} I_{k'}^{(j')}(\sigma) + r_k(\sigma) Z_{k'}(0)$, with $\sum_k b_k < \infty$, $r_k(0) \leq \frac{1}{2}$, $\sum_k r_k(0) < \infty$ and $\sum_k e^{C_c(h)(k+2)} r_k(s) < \infty$, where $|c_k(h)| \leq C_c(h)(1+k)$ (lemma 6.8(a));*

(B) *on the support of $\chi_{j(k)}$, (12) holds at σ with $\sum_k \theta_k(\sigma) < \infty$.*

Put $\Theta_k := \theta_k(s)+\theta_k(0)$, $\delta_k := (4b_k+\Theta_k+2r_k(0))\, e^{4b_k+\Theta_k+2r_k(0)}$ *and* $\epsilon_k := (e^{-c_k(h)}+e^{-c_{k+1}(h)})\, r_k(s)$. *Then for all* $k \geq k_0$*:*

(i) $n_k(s) \leq \mathsf{M}(s) := \exp(\sum_{i\geq k_0} \delta_i)\big(n_{k_0}(s) + \sum_{i\geq k_0} \epsilon_i\big) < \infty$;
(ii) $|n_{k+1}(s) - n_k(s)| \leq \delta_k \mathsf{M}(s) + \epsilon_k$, *so* $n_k(s)$ *converges to a limit* $n(s) \in [0, \mathsf{M}(s)]$;
(iii) *if* (U) *and* (B) *hold at* $-s$ *as well,* $n_k(s) \geq 1/\mathsf{M}(-s)$ *and* $n(s) \in [1/\mathsf{M}(-s), \mathsf{M}(s)] \subset (0, \infty)$.

If b_k, Θ_k *decay geometrically and* $r_k(\sigma) \leq e^{-ck^2+Ck}$, *then* $|n_k(s) - n(s)| \leq C' 5^{-\kappa k}$ *for some* $\kappa > 0$.

Proof. Fix $k \geq k_0$, $j := j(k)$, and let primes denote the $(k+1)$-model after $j+1$ steps.

Step 1 (one model, (U) *only).* For one member write $Z(\sigma)$, $I(\sigma)$, $b = b_k$, $r(\sigma) = r_k(\sigma)$. From (U) at $\sigma = 0$, $(1-r(0))Z(0) \leq e^b I(0)$ and $Z(0) \geq e^{-b} I(0)$, so $I(0) > 0$. With $\varrho(\sigma) := I(\sigma)/I(0)$ and $N(\sigma) = Z(\sigma)/Z(0)$,

$$e^{-2b}(1 - r(0))\, \varrho(\sigma) \;\leq\; N(\sigma) \;\leq\; e^{2b}\varrho(\sigma) + r(\sigma)\,: \tag{13}$$

the upper bound from $Z(\sigma) \leq e^b I(\sigma) + r(\sigma) Z(0)$ and $1/Z(0) \leq e^b/I(0)$, the lower one from $Z(\sigma) \geq e^{-b} I(\sigma)$ and $1/Z(0) \geq e^{-b}(1 - r(0))/I(0)$. No lower bound on $Z(\sigma)$ for $\sigma \neq 0$ is used: the remainder enters relative to $Z(0)$ only.

Step 2 (the vacuum constant). $\varrho(\sigma) = \hat{I}(\sigma)/\hat{I}(0)$, since $I = e^{-v}\hat{I}$ with one v for all σ; likewise for the primed member.

Step 3 (across the pair, (B) *only).* On the support of χ_j, (B) at $\sigma = s$ reads $S'_N(\cdot; s) = S_N(\cdot; s) + c_k(h) - c_{k+1}(h) + E$ with $|E| \leq \theta_k(s)$; the integrals $\hat{I}(s)$, $\hat{I}'(s)$ run over the same variables with the same χ_j, so $\hat{I}'(s) \in e^{c_{k+1}(h) - c_k(h)}[e^{-\theta_k(s)}, e^{\theta_k(s)}]\, \hat{I}(s)$. At $\sigma = 0$ ($c_k(0) = 0$), $\hat{I}'(0) \in [e^{-\theta_k(0)}, e^{\theta_k(0)}]\, \hat{I}(0)$. With $\varrho := \varrho(s)$ for the k-model and ϱ' for the $(k+1)$-model,

$$e^{-c_{k+1}(h)}\varrho' \;\in\; [e^{-\Theta_k}, e^{\Theta_k}]\; e^{-c_k(h)}\varrho. \tag{14}$$

Step 4 (two-sided recursion). Write $n := n_k(s)$, $n' := n_{k+1}(s)$, $r_0 := r_k(0)$, $r_s := r_k(s)$, $\Theta := \Theta_k$. Apply (13) to the primed member, multiply by $e^{-c_{k+1}(h)}$, use (14), and then (13) for the k-model in the forms $\varrho \leq e^{2b}(1 - r_0)^{-1} N(s)$ and $\varrho \geq e^{-2b}(N(s) - r_s)$:

$$\begin{aligned} n' &\leq e^{2b}\, e^{-c_{k+1}}\varrho' + e^{-c_{k+1}} r_s \;\leq\; e^{2b+\Theta} e^{-c_k}\varrho + e^{-c_{k+1}} r_s \;\leq\; e^{4b+\Theta}(1 - r_0)^{-1}\, n + e^{-c_{k+1}} r_s, \\ n' &\geq e^{-2b}(1 - r_0)\, e^{-c_{k+1}}\varrho' \;\geq\; e^{-2b-\Theta}(1 - r_0)\, e^{-c_k}\varrho \;\geq\; e^{-4b-\Theta}(1 - r_0)\big(n - e^{-c_k} r_s\big). \end{aligned}$$

For $t \leq \frac{1}{2}$, $(1-t)^{-1} \leq e^{2t}$ and $1 - t \geq e^{-2t}$; with $e^y - 1 \leq y e^y$ and $1 - e^{-y} \leq y$ this gives $e^{4b+\Theta}(1 - r_0)^{-1} - 1 \leq \delta_k$ and $1 - e^{-4b-\Theta}(1 - r_0) \leq \delta_k$, hence

$$(1 - \delta_k)\, n_k(s) - \epsilon_k \;\leq\; n_{k+1}(s) \;\leq\; (1 + \delta_k)\, n_k(s) + \epsilon_k. \tag{15}$$

Step 5 (boundedness and convergence). By induction from the upper half of (15), $n_k \leq \prod_{i=k_0}^{k-1}(1 + \delta_i)\big(n_{k_0} + \sum_{i<k} \epsilon_i\big) \leq \mathsf{M}(s)$. Here $\sum_i \delta_i < \infty$ because b_i, Θ_i, $r_i(0)$ are summable (so bounded, and $\delta_i \leq C(4b_i + \Theta_i + 2r_i(0))$), and $\sum_i \epsilon_i \leq 2\sum_i e^{C_c(h)(i+2)} r_i(s) < \infty$ by (U) and $e^{-c_i(h)} \leq e^{C_c(h)(1+i)}$. Then (15) gives $|n_{k+1} - n_k| \leq \delta_k n_k + \epsilon_k \leq \delta_k \mathsf{M}(s) + \epsilon_k$, summable; so $(n_k(s))$ is Cauchy. This is (i), (ii).

Step 6 (positivity). $X(-s) = -X(s)$ and $c_k(-h) = -c_k(h)$, so $n_k(s) n_k(-s) = \langle e^X \rangle \langle e^{-X} \rangle \geq 1$ (lemma 2.11(e)). Steps 1–5 at $-s$ give $n_k(-s) \leq \mathsf{M}(-s)$, hence $n_k(s) \geq 1/\mathsf{M}(-s)$, and the limit inherits both bounds. The rate follows by summing the bound of (ii) from k on. □

Remark 6.15 (why the ratio is not taken). At $h = 0$ and $c_k \equiv 0$ this is theorem 5.7 with lemma 5.6 replaced by Step 6. The logarithmic form of theorem 5.7 needs $r_k(\sigma) Z(0)/Z(\sigma)$ small, that is a lower bound on $Z_k(s)$ uniform enough in k; at $h \neq 0$ the only candidates are lemma 2.11(e), which needs the upper bound at $-s$ the theorem is to produce, and Jensen's inequality, which needs a renormalised one-point bound that is not available. The additive comparison (13) avoids both: the remainder is compared with $Z(0)$, and the lower bound on $n_k(s)$ is produced at the end. Nothing in the proof uses the cube or any relation between $j(k+1)$ and $j(k) + 1$.

6.6. **The negative-source bound.**

Theorem 6.16 ((UB$^-$): uniform bound at a negative scalar source)**.** *Assume* (I2) *and* (I3). *Let* $0 \le q \in C_c^\infty(\mathbb{R}^3)$ *be supported at* ℓ^∞*-distance* > 1 *from* $\mathbb{R}^3 \backslash C_J$. *Then* $\sup_k n_k(0,-q) < \infty$. *Moreover, for every real* 2*-form* u *with the same support condition,* $n_k(u,-q) \le n_k(2u,0)^{1/2} n_k(0,-2q)^{1/2}$, *so* $\sup_k n_k(u,-q) < \infty$.

Proof. By proposition 6.9 choose $M_2 > m$ with $s_-(M_2) > \sup q$. Propositions 5.4, 6.6 and 6.7 and lemma 6.8 hold at the point (e, λ, M_2, μ_0) as well; $\Delta := \sup_k |\Delta_k(M_2)| < \infty$.

Step 1. Choose $\bar w \in (0, \frac12]$ with $2\bar w \sup q < s_W$, and $k_1 \ge k_0$ with $u_k(M_2) \ge \sup q$ for $k \ge k_1$ (possible since $\sup q < \liminf_k u_k(M_2)$). Then for $k \ge k_1$ and $w \in [0, \bar w]$, both $q \le u_k(M_2)$ and $2wq \le u_k(M_2)$, so lemma 6.11 applies at q and at $2wq$.

Step 2 (an a priori bound at M_2). For $k \ge k_1$, $w \in [0,\bar w]$: by lemma 6.11, $Y_k^{(M_2)}(0,2wq) \le n_k(0,2wq)$; by convexity of $t \mapsto \ln n_k(0,tq)$ with value 0 at $t=0$, $n_k(0,2wq) \le \max\{1, n_k(0,2\bar w q)\}$; and by proposition 6.10, $n_k(0,2\bar w q) \le e^{\kappa \|2\bar w q\|_{1,k}} \le e^{\kappa C_0' |2\bar w q|_w}$. Hence, in the conventions of the mass M_2,

$$n_k^{(M_2)}(0,2wq) = e^{\Delta_k(M_2)\mathsf{S}_k(2wq)}\, Y_k^{(M_2)}(0,2wq) \;\le\; e^{\Delta\, \mathsf{S}_k(2q)} \max\{1, e^{\kappa C_0' |2\bar w q|_w}\} \;=:\; \bar{\mathsf{M}},$$

uniformly in w and k ($\mathsf{S}_k(2q) \le C_0' |2q|_w$).

Step 3 (the hypotheses of lemma 6.13 at M_2). The family $\{wq : 0 \le w \le \bar w\}$ is compact. At $\sigma = (0, wq)$: (U) holds by lemma 6.12 at mass M_2 with $p=2$, from (11) (proposition 6.6 at M_2; $wq \ge 0$) and (R_0) (proposition 5.4 at M_2), with $b_k = C\eta_k^\sigma$ and, in the conventions of M_2, $e^{-c_k(wq)} r_k(wq) = \rho_k^{1/2}\, n_k^{(M_2)}(0,2wq)^{1/2} \le \rho_k^{1/2} \bar{\mathsf{M}}^{1/2}$ by Step 2, a w-independent sequence; since $\rho_k \le e^{-ck^2+C}$ and $e^{c_k(wq)} \le e^{C_c(1+k)}$ with C_c uniform on the family, $\sum_k e^{C_c(k+2)} r_k(wq) < \infty$ uniformly in w. At $\sigma = 0$, (U) is (2) with $r_k(0) = \rho_k \le \frac12$ for $k \ge k_0$. (B) at $\sigma \in \{wq, 0\}$ is proposition 6.7 at M_2 on the compact family, with w-independent θ_k.

Step 4 (positivity at M_2). Lemma 6.13 gives $c := \inf_{k \ge k_1} n_k^{(M_2)}(0,q) > 0$, hence $Y_k^{(M_2)}(0,q) \ge e^{-\Delta \mathsf{S}_k(q)} c$ for $k \ge k_1$.

Step 5 (the mass shift). By lemma 6.11 at q, $n_k(0,-q) \le 1/Y_k^{(M_2)}(0,q) \le e^{\Delta \mathsf{S}_k(q)}/c \le e^{\Delta C_0' |q|_w}/c$ for $k \ge k_1$; the finitely many $k < k_1$ are finite numbers. The mixed bound is Cauchy–Schwarz, $\langle e^{F(u) - :\phi^2:(q)} \rangle \le \langle e^{2F(u)} \rangle^{1/2} \langle e^{-2:\phi^2:(q)} \rangle^{1/2}$, with the c-numbers distributed ($e^{c_k(q)} = (e^{c_k(2q)})^{1/2}$); the first factor is bounded by lemma 5.6. □

6.7. **Existence on the cube with scalar sources.**

Theorem 6.17 (King's existence theorem on the cube with scalar sources, reference scheme)**.** *Assume* (I2) *and* (I3) *(so that proposition 6.10 holds with $s_W = s_1$). Let $f_1, \dots, f_n$ be real* 2*-forms and $h_1, \dots, h_p \ge 0$ non-negative real functions, all in C_c^∞ and supported at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_J$, $J \ge J_0$. For $\zeta = (z,w) \in \mathbb{C}^{n+p}$ put $k_1(\zeta) := \sum_j (\operatorname{Re} w_j)_+ h_j$, $k_2(\zeta) := \sum_j (\operatorname{Re} w_j)_- h_j$ (both ≥ 0, smooth, with $\sum_j (\operatorname{Re} w_j) h_j = k_1 - k_2$), and*

$$D_J(s_W) \;:=\; \big\{\zeta \in \mathbb{C}^{n+p} : \sup_x 2k_1(\zeta)(x) < s_W\big\}.$$

Then $D_J(s_W)$ is an open convex tube, containing every ζ with $\operatorname{Re} w_j \le 0$ for all j and every ζ with w purely imaginary, and

(i) *$n_k \to F_J$ locally uniformly on $D_J(s_W)$, and F_J is holomorphic there;*
(ii) *$F_J(\zeta) \in (0,\infty)$ for every real $\zeta \in D_J(s_W)$.*

Proof. The domain. For each x, $\operatorname{Re} w \mapsto 2k_1(\zeta)(x) = \sum_j 2(\operatorname{Re} w_j)_+ h_j(x)$ is convex, since $h_j(x) \ge 0$ and $t \mapsto t_+$ is convex; so $\operatorname{Re} w \mapsto \sup_x 2k_1(\zeta)(x)$ is a finite convex function on $\mathbb{R}^p$, hence continuous, and $D_J(s_W)$ is the preimage under $\zeta \mapsto \operatorname{Re} w$ of an open convex set: open, convex, connected. Write $D := D_J(s_W)$ and $D_{\mathbb{R}} := D \cap \mathbb{R}^{n+p}$, an open convex subset of $\mathbb{R}^{n+p}$.

Step 1 (local boundedness). Let $Q \subset D$ be compact. $\operatorname{Re} Q$ is a compact subset of the open set $D_{\mathbb{R}}$, so it is covered by finitely many closed coordinate boxes $B \subset D_{\mathbb{R}}$. For $\zeta \in Q$ with $\operatorname{Re}\zeta \in B$, $|n_k(\zeta)| \le n_k(\operatorname{Re}\zeta) \le \max_{v \in \mathrm{vert}(B)} n_k(v)$, by lemma 2.11(d) and the convexity of $\ln n_k$ on B. At a

vertex $v = (x, y) \in D_{\mathbb{R}}$ put $k_1 := k_1(v)$, $k_2 := k_2(v)$; then $X(v) = F(u_x) + {:}\phi^2{:}(k_1) - {:}\phi^2{:}(k_2)$ and the c-number is $e^{-c_k(k_1)+c_k(k_2)}$. Hölder's inequality with exponents $(4, 2, 4)$, with the c-number distributed as $e^{-c_k(k_1)} = (e^{-c_k(2k_1)})^{1/2}$, $e^{c_k(k_2)} = (e^{-c_k(-4k_2)})^{1/4}$, gives

$$n_k(v) \;\le\; a_k(4x)^{1/4}\, n_k(0, 2k_1)^{1/2}\, n_k(0, -4k_2)^{1/4}.$$

The three factors are bounded uniformly in k: the first by lemma 5.6; the second by proposition 6.10, since $2k_1$ is a non-negative smooth function with $\sup 2k_1 < s_W$ ($v \in D_{\mathbb{R}}$) and the required support; the third by theorem 6.16 at $q = 4k_2$. Hence $\sup_k \sup_Q |n_k| < \infty$.

Step 2 (convergence on a real open set). Choose $\delta > 0$ with $4\delta \sup_x \sum_j h_j(x) < s_W$ and put $U := \{\zeta = (x, y) \in \mathbb{R}^{n+p} : \; 0 < y_j < \delta \;\forall j\}$, a non-empty open subset of $\mathbb{R}^{n+p}$ contained in $D_{\mathbb{R}}$. At $\zeta \in U$, $k_2(\zeta) = 0$ and $s := (u_x, k_1(\zeta))$ has a non-negative scalar part; the sources $\{k_1(\zeta) : \zeta \in \bar{U}\}$ form a compact family. Apply theorem 6.14(i),(ii) at s. Hypothesis (U) at $\sigma = 0$ is (2) on both members, with $r_k(0) = \rho_k$. Hypothesis (U) at $\sigma = s$ is lemma 6.12 with $p = 2$, from (11) (proposition 6.6) and (R_0), with $r_k(s) = \rho_k^{1/2} \langle e^{X(2s)} \rangle_k^{1/2}$; here $\langle e^{X(2s)} \rangle_k = e^{c_k(2k_1)} n_k(2u_x, 2k_1) \le e^{c_k(2k_1)} a_k(4x)^{1/2} n_k(0, 4k_1)^{1/2}$ by Cauchy–Schwarz, bounded by $e^{C_c(1+k)}$ times a constant (lemma 5.6; proposition 6.10, as $\sup 4k_1 < s_W$), so $\sum_k e^{C_c(k+2)} r_k(s) \le C \sum_k e^{2C_c(k+2)} \rho_k^{1/2} < \infty$ since $\rho_k \le e^{-ck^2+C}$. Hypothesis (B) at $\sigma \in \{s, 0\}$ is proposition 6.7. Hence $n_k(\zeta)$ converges for every $\zeta \in U$.

Step 3 (Vitali). By Step 1 the family $\{n_k\}$ is locally bounded on the connected domain D; by Montel's theorem every subsequence has a further subsequence converging locally uniformly to a holomorphic function; any two such limits agree on U (Step 2), hence on D, since a holomorphic function vanishing on a non-empty open subset of $\mathbb{R}^{n+p}$ has all its derivatives equal to zero at a point of it and vanishes on the connected D. So the whole sequence converges locally uniformly on D to a holomorphic F_J: (i).

Step 4 (positivity at real points). On $D_{\mathbb{R}}$ every n_k is positive and $\ln n_k$ is convex, so $F_J \ge 0$ there and $g := \ln F_J : D_{\mathbb{R}} \to [-\infty, \infty)$ satisfies $g(\lambda a + (1-\lambda)b) \le \lambda g(a) + (1-\lambda) g(b)$ (the limit of the inequalities for $\ln n_k$; all values are $< \infty$). For $\zeta \in U$ the point $-\zeta$ lies in $D_{\mathbb{R}}$ ($k_1(-\zeta) = 0$), both $n_k(\zeta)$ and $n_k(-\zeta)$ converge by (i), and $n_k(\zeta) n_k(-\zeta) \ge 1$ (lemma 2.11(e)), so $F_J(\zeta) F_J(-\zeta) \ge 1$ and $g > -\infty$ on U. If $g(t) = -\infty$ for some $t \in D_{\mathbb{R}}$, pick $t' \in U$ and, $D_{\mathbb{R}}$ being open, a point $t'' \in D_{\mathbb{R}}$ beyond t' on the ray from t through t', so that $t' = \lambda t + (1-\lambda) t''$ with $\lambda \in (0, 1)$; then $g(t') \le \lambda g(t) + (1-\lambda) g(t'') = -\infty$, a contradiction. Hence $F_J > 0$ on $D_{\mathbb{R}}$: (ii). □

Corollary 6.18 (King's cube limits with scalar sources in the reference scheme, where they are consumed)**.** *Under the hypotheses of theorem 6.17:*

(a) *For every finite family of real 2-forms f_i and real functions h_j (of either sign) in C_c^∞ with the support condition, and every real coefficient vector (x, y) with $2\|(\sum_j y_j h_j)_+\|_\infty < s_W$, the limit $\lim_k Z_{k,J}^{z,K}(x, y)$ exists and lies in $(0, \infty)$. In particular theorem 3.5 for the reference scheme holds at every real point of the tube $\mathcal{D}_{\sigma_+, H}$ with $\sigma_+ = s_1$ (there $2\|k_+\|_\infty < \sigma_+ = s_W$), and at every real coefficient vector whose scalar source $\sum_j y_j h_j$ is non-positive, with no restriction on its amplitude.*

(b) *For $0 \le q$ as in theorem 6.16, the comparison-convention functional satisfies $\sup_k N_{k,C_J}^z(0, -q) \le \sup_k n_k(0, -q) < \infty$.*

Proof. (a) Put $\mathsf{k} := \sum_j y_j h_j$ and let k_+ be its pointwise positive part. Choose $\chi \in C_c^\infty$ with $0 \le \chi \le 1$, $\chi = 1$ on a neighbourhood of $\operatorname{supp} \mathsf{k}$ and $\operatorname{supp} \chi$ at distance > 1 from $\mathbb{R}^3 \setminus C_J$, and $\delta > 0$ with $2(\|\mathsf{k}_+\|_\infty + \delta) < s_W$; put $h^+ := (\|\mathsf{k}_+\|_\infty + \delta)\chi$ and $h^- := h^+ - \mathsf{k}$. Both are smooth, non-negative ($h^+ \ge \mathsf{k}_+ \ge \mathsf{k}$ on $\operatorname{supp} \mathsf{k}$, and $h^+ \ge 0 = \mathsf{k}$ off it), compactly supported with the support condition, and $\mathsf{k} = h^+ - h^-$. Then $Z_{k,J}^{z,K}(x, y)$ for the given family equals $n_k(\zeta')$ for the family $(f_1, \dots, f_n; h^+, h^-)$ at $\zeta' = (x; 1, -1)$, for which $k_1(\zeta') = h^+$ and $\sup 2k_1(\zeta') = 2(\|\mathsf{k}_+\|_\infty + \delta) < s_W$; so $\zeta' \in D_J(s_W)$ and theorem 6.17 applies. (b) $e^{-:\phi^2:(q)}$ is a class-$\mathcal{D}$ observable; killing the matter on T_J does not decrease its expectation (Step 3 of lemma 4.2, the $\mathcal{D}$ direction), and the scheme multiplier is common. □

Remark 6.19 (other schemes). For a scheme a, $Z^{a,K}_{k,J}(x,y) = e^{-(a_k-z_k)\sum_j y_j \mathsf{S}_k(h_j)} Z^{z,K}_{k,J}(x,y)$. At a coefficient vector with $\sum_j y_j \int h_j \neq 0$ the limit for a therefore exists if and only if $a_k - z_k$ converges: on the domain of corollary 6.18(a), theorem 3.5 for a is the convergence of the scheme difference, in agreement with remark 2.4, and corollary 6.18(b) holds for a whenever $\sup_k |a_k - z_k| < \infty$.

Remark 6.20 (the cell bound). ($\mathrm{B}_{\mathrm{cell}}$) (lemma 14.4) is the boundedness of $N^a_{k,C_{L_c}}(0, -2H'\chi_0)$ on one comparison-convention cell. When the cell is a King cube, that is $\ell_c = 2MJ'$ for some J' up to a translation by a lattice vector (lemma 2.12), and χ_0 is supported at ℓ^∞-distance > 1 from the cell's complement, corollary 6.18(b) and remark 6.19 give ($\mathrm{B}_{\mathrm{cell}}$) for every scheme with $\sup_k |a_k - z_k| < \infty$, and corollary 6.18(a) gives the convergence as well.

Remark 6.21 (what is and is not claimed). Beyond the statements of section 5.2 and lemma 6.1, the only input of this section is the threshold s_W of proposition 6.10, supplied by (I2) and (I3). Nothing is claimed at a non-negative scalar source of amplitude $\geq s_W/2$: there the positive-source bound is the content of (W2) above the threshold (remark 23.2). The negative scalar sources are unrestricted, because the mass shift of lemma 6.11 turns a negative source of the mass-m model into a positive source of a heavier model, where the small-field analysis applies with no threshold on the amplitude, and proposition 6.9 supplies a heavy enough mass for every amplitude.

The axioms

7. Regularity in the curvature sector

This section proves the regularity axiom E0″ for the curvature field, for the comparison convention $\mathrm{ND}^{(a)}$ (definition 2.1) and for King's convention ND^K (definition 2.2), on which King's printed state Z^K_∞ lives. Gaussian domination (lemma 2.8) and a bound on the Neumann photon form of the cube (lemma 7.1) give a bound on every lattice generating functional, uniform in the spacing, the cube and the scalar sector, with absolute constants (remark 7.5); under the existence hypothesis it passes to the limits by Montel's theorem, and Cauchy's estimates turn it into E0″ with $s = 4$. The section closes with the curvature-sector properties of the fixed-cube limits of the comparison family (locally uniform convergence, monotonicity in the cube, positivity, continuity and translations), proved with cone-direction correlation inequalities only (lemma 7.7).

Throughout, $f_1, \dots, f_n$ are real 2-forms in $C^\infty_c(\mathbb{R}^3; \Lambda^2)$ with supports at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_{J_0}$, $J \geq J_0$, $z \in \mathbb{C}^n$, and $g(z) := \sum_j (\mathrm{Re}\, z_j) f_j$, so that $\mathrm{Re} \sum_j z_j F(f_j) = F(g(z))$ (lemma 2.11(d)). For a box X, $\langle A, B\rangle_k$ and $d^*_k g$ are as in (1): $d^*_k g$ is the lattice 1-form with $F_{\Lambda_k}(g) = \langle d^*_k g, A\rangle_k$, defined whenever every plaquette at which $g \neq 0$ has its four corners in X. The support condition implies this for $X = X_J$: the plaquette centre is at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_J$ and the corners are within $\varepsilon/2 \leq \frac{1}{2}$ of it.

7.1. The Neumann cube photon form.

Lemma 7.1 (the Neumann cube photon form). *Let $X = C_J \cap \Lambda_k$, let C be the photon covariance of the cube states on X (the same for $\mathrm{ND}^{(a)}_{k,J}$ and $\mathrm{ND}^K_{k,J}$) with respect to $\langle\cdot,\cdot\rangle_k$, let g be a real 2-form on the plaquettes $p = (x; \alpha < \beta)$ all four of whose corners lie in X, and $v := d^*_k g$. Then*

$$\langle v, Cv\rangle_k \;\leq\; \min\big\{\, 2\|g\|^2_{2,k},\ \mu_0^{-2}\|v\|^2_{2,k} \,\big\}.$$

Proof. The Gaussian density of the photon field is $e^{-Q(A)/2}$ with

$$Q(A) = \langle A, C^{-1}A\rangle_k = \sum_{\alpha,\beta} \sum_{x,\, x+\varepsilon e_\beta \in X} \varepsilon\, (\partial_\beta A_\alpha(x))^2 + \mu_0^2 \|A\|^2_{2,k}, \qquad \partial_\beta A_\alpha(x) := A_\alpha(x+\varepsilon e_\beta) - A_\alpha(x)\,:$$

the Neumann rule keeps exactly the gradient terms of pairs inside X. By Legendre duality, $\langle v, Cv\rangle = \sup_A \big[2\langle v, A\rangle - Q(A)\big]$, attained at $A = Cv$. *Second bound.* $Q(A) \geq \mu_0^2 \|A\|^2$, so $C \leq \mu_0^{-2}$ as forms. *First bound.* Let P_g be the set of plaquettes on which $g \neq 0$. For $p = (x; \alpha <$

$\beta) \in P_g$ both gradient terms $(\partial_\alpha A_\beta(x))^2$ (pair $(x, x+\varepsilon e_\alpha)$, component β) and $(\partial_\beta A_\alpha(x))^2$ (pair $(x, x+\varepsilon e_\beta)$, component α) are present in Q, because the pairs lie in X. With $F(p) = \varepsilon^{-1}(\partial_\alpha A_\beta(x) - \partial_\beta A_\alpha(x))$,

$$\|dA\|^2_{P_g} := \sum_{p\in P_g} \varepsilon^3 F(p)^2 = \sum_{p\in P_g} \varepsilon\,(\partial_\alpha A_\beta(x) - \partial_\beta A_\alpha(x))^2 \le 2\sum_{p\in P_g} \varepsilon\big[(\partial_\alpha A_\beta(x))^2 + (\partial_\beta A_\alpha(x))^2\big] \le 2Q(A),$$

where the last step is a counting statement: the ordered triple (site x, component β, direction $\alpha \neq \beta$) determines the plaquette $(x;\{\alpha,\beta\})$, so distinct plaquettes use distinct gradient terms of Q and no term is counted twice. Also $\langle v, A\rangle_k = F_{\Lambda_k}(g) = \sum_{p\in P_g} \varepsilon^3 g_p F(p) =: \langle g, dA\rangle_{P_g}$. Hence

$$\langle v, Cv\rangle \le \sup_A \big[2\langle g, dA\rangle_{P_g} - \tfrac12 \|dA\|^2_{P_g}\big] \le \sup_{B\in\mathbb{R}^{P_g}} \big[2\langle g, B\rangle_{P_g} - \tfrac12 \|B\|^2_{P_g}\big] = 2\|g\|^2_{2,k},$$

the last supremum being attained at $B = 2g$. The inequality $Q(A) - \frac12\|dA\|^2_{P_g} \ge 0$ used in the middle is in fact the identity

$$Q(A) - \tfrac12\|dA\|^2_{P_g} = \tfrac12 \sum_{p\in P_g} \varepsilon\,(\partial_\alpha A_\beta(x) + \partial_\beta A_\alpha(x))^2 + \big(\text{gradient terms of } Q \text{ used by no } p \in P_g\big) + \mu_0^2\|A\|^2_{2,k}.$$

□

Remark 7.2 (on the constant 2). On a torus the Hodge decomposition gives the constant 1 in place of 2 ([24, Lemma 7.3]); on the Neumann cube no Hodge argument is available, and the constant 2 appears to be nearly sharp: the exact quantity $\rho := \sup_g \operatorname{Var} F(g)/\|g\|^2_{2,k}$, which depends on ε and μ_0 only through $\mu_0\varepsilon$, exceeds 1.94 on a cube of 28^3 sites at $\mu_0\varepsilon = 10^{-4}$ and increases with the cube at small $\mu_0\varepsilon$. Whether it tends to 2 is not proved; nothing below depends on it. The second bound is the cube form of the two-point bound $0 \le S_2(f) \le \mu_0^{-2}\|d^*f\|^2_{L^2}$; it is the worse of the two at small μ_0.

7.2. **The comparison convention.**

Lemma 7.3 (standalone local bound)**.** *For either convention and every compact $K \subset \mathbb{C}^n$,*

$$\sup_{k\ge 0}\ \sup_{J\ge J_0}\ \sup_{z\in K} \big|\langle e^{\sum_j z_j F_{\Lambda_k}(f_j)}\rangle_{k,J}\big| \ \le\ \exp\Big(3C_0 \sup_{z\in K}|g(z)|_4^2\Big) < \infty,$$

with C_0 the constant of lemma 2.6(E1)*. No hypothesis and no correlation inequality is used.*

Proof. $|\langle e^{\sum z_j F(f_j)}\rangle| \le \langle e^{F(g(z))}\rangle$. Both cube states are BFS measures whose Gaussian is the Neumann photon Gaussian of X_J with covariance C (the matter on T_J does not touch the photon variables), so lemma 2.8 with $s = 1$ and $v = d_k^* g(z)$ gives $\langle e^{F(g(z))}\rangle \le e^{\langle v, Cv\rangle_k/2} \le e^{\|g(z)\|^2_{2,k}}$ by lemma 7.1. The plaquette centres of each of the three orientations form a translate of $\varepsilon_k\mathbb{Z}^3$, and $|g_{c(p)}(e_\alpha, e_\beta)| \le (1+|c(p)|^2)^{-2}|g|_4$, so by (E1) each orientation contributes at most $C_0|g|_4^2$ to $\|g\|^2_{2,k}$ and $\|g(z)\|^2_{2,k} \le 3C_0|g(z)|_4^2$ for every k (this is the count behind (E3) of lemma 2.6). Finally $|g(z)|_4 \le \sum_j |\operatorname{Re} z_j||f_j|_4$ is bounded on K. □

The existence theorem theorem 3.2 is posed on King's convention. For the comparison family the analogous clause is its convergence at fixed cubes (definition 3.1): for every such family, every $J \ge J_0$ and every real z, $\lim_k \langle e^{\sum_j z_j F(f_j)}\rangle^{(a)}_{k,J}$ exists in $(0,\infty)$. It is not needed for the main theorem (the two clauses do not imply one another by the inputs of this paper, corollary 4.4); theorem 7.4 is stated under it because the rotation part and the tube estimates are written for the comparison family, and theorem 7.6 is its form for King's printed state, where the limits exist by theorem 3.2.

Theorem 7.4 (regularity, comparison convention)**.** *Let $f_1, \dots, f_n$, J_0, $g(z)$ be as above.*

(a) (Finite volume; no hypothesis.) *For all $k \ge 0$, $J \ge J_0$, $\mu_0 > 0$, $z \in \mathbb{C}^n$, and in the presence of any real scalar source $h \in C_c^\infty$,*

$$\left|\frac{\langle e^{\sum_j z_j F_{\Lambda_k}(f_j)} e^{:\phi^2:(h)}\rangle^{(a)}_{k,J}}{\langle e^{:\phi^2:(h)}\rangle^{(a)}_{k,J}}\right| \ \le\ \frac{\langle e^{F_{\Lambda_k}(g(z))} e^{:\phi^2:(h)}\rangle^{(a)}_{k,J}}{\langle e^{:\phi^2:(h)}\rangle^{(a)}_{k,J}} \ \le\ \exp\Big(\min\big\{\|g(z)\|^2_{2,k},\ \tfrac12\mu_0^{-2}\|d_k^* g(z)\|^2_{2,k}\big\}\Big).$$

(b) (Limits; if the comparison family is convergent at fixed cubes.) $Z_J(\sum_j z_j f_j, 0) := \lim_k \langle e^{\sum_j z_j F(f_j)}\rangle^{(a)}_{k,J}$ *exists locally uniformly on* $\mathbb{C}^n$ *and is entire;* $Z_\infty(\sum_j z_j f_j, 0) := \lim_J Z_J(\sum_j z_j f_j, 0)$ *exists locally uniformly on* $\mathbb{C}^n$ *and is entire; and for* $J \geq J_0$

$$|Z_J(\textstyle\sum_j z_j f_j, 0)|,\ |Z_\infty(\sum_j z_j f_j, 0)| \ \leq\ \exp\Big(\min\big\{\|g(z)\|_2^2,\ \tfrac{1}{2}\mu_0^{-2}\|d^*g(z)\|_{L^2}^2\big\}\Big).$$

(c) (Schwinger functions; if the comparison family is convergent at fixed cubes.) $S_n(f_1,\dots,f_n) := \partial_{z_1}\cdots\partial_{z_n} Z_\infty(\sum_j z_j f_j, 0)\big|_{z=0}$ *satisfies, for real* f_i,

$$|S_n(f_1,\dots,f_n)| \ \leq\ (\sqrt{2}\,e)^n (n!)^{1/2}\prod_{i=1}^n \|f_i\|_2 \ \leq\ (e\pi/2)^n (n!)^{1/2}\prod_{i=1}^n |f_i|_4.$$

S_n extends to a continuous symmetric multilinear form on $\mathcal{S}(\mathbb{R}^3;\Lambda^2_{\mathbb{C}})^n$ and defines a tempered distribution; for complex test functions the bound carries the factor 2^n. This is the axiom E0″ *of* [18] *with $s=4$ and $\sigma_n := (e\pi)^n(n!)^{1/2} \leq \alpha_\beta (n!)^\beta$ for every $\beta > \frac{1}{2}$ (and for no α at $\beta = \frac{1}{2}$).*

Proof. (a) The first inequality is $\operatorname{Re}\sum_j z_j F(f_j) = F(g(z))$. The measure $e^{:\phi^2:(h)} d\langle\cdot\rangle^{(a)}_{k,J}$, normalised, is a BFS measure whose single-site weights carry the extra factor $e^{\varepsilon^3 h(x) r_x^2}$ and whose photon covariance is the same C; lemma 2.8 with $s=1$ and $v = d_k^* g(z)$ bounds the middle expression by $e^{\langle v, Cv\rangle_k/2}$, and lemma 7.1 gives both terms of the minimum.

(b) *Step 1: $k\to\infty$ at fixed $J \geq J_0$.* The functions $\Phi_k(z) := \langle e^{\sum_j z_j F(f_j)}\rangle^{(a)}_{k,J}$ are entire on $\mathbb{C}^n$: finite-dimensional integrals with Gaussian decay in A ($\mu_0 > 0$) and $e^{-\lambda\varepsilon^3 r^4}$ decay in r, converging absolutely and locally uniformly in z. By lemma 7.3 they are locally uniformly bounded; by the convergence at fixed cubes they converge at every real point. $\mathbb{R}^n$ is a set of uniqueness for entire functions on $\mathbb{C}^n$ (a holomorphic function vanishing on $\mathbb{R}^n$ has all its Taylor coefficients at 0 equal to 0). By Montel's theorem every subsequence has a locally uniformly convergent sub-subsequence; all subsequential limits are entire and agree on $\mathbb{R}^n$, hence everywhere [10]; so $\Phi_k \to Z_J(\cdot,0)$ locally uniformly on $\mathbb{C}^n$ and $Z_J(\cdot,0)$ is entire. The bound of (a) at $h=0$ passes to the limit: $|Z_J(\sum z_j f_j, 0)| \leq \lim_k e^{\|g(z)\|^2_{2,k}} = e^{\|g(z)\|_2^2}$ by (E3) of lemma 2.6, and likewise $|Z_J(\sum z_j f_j,0)| \leq \lim_k \exp(\frac{1}{2}\mu_0^{-2}\|d_k^* g(z)\|^2_{2,k}) = \exp(\frac{1}{2}\mu_0^{-2}\|d^*g(z)\|^2_{L^2})$ by (1). Both bounds are free of J.

Step 2: monotonicity in J on the imaginary axes. For $t\in\mathbb{R}^n$ and $f := \sum_j t_j f_j$, $\langle e^{\mathrm{i}F(f)}\rangle^{(a)}_{k,J}$ is real (the state is invariant under $(A,\vartheta)\mapsto(-A,-\vartheta)$, which maps $F(f)$ to $-F(f)$) and of modulus ≤ 1, and $J \mapsto \langle e^{\mathrm{i}F(f)}\rangle^{(a)}_{k,J}$ is nondecreasing on $J \geq J_0$ at fixed k: this is the first inequality of lemma 2.10 with $Y = C_{J+1}\cap\Lambda_k$, $X = C_J\cap\Lambda_k$ and the class-$\mathcal{I}$ observable $\Theta(\mathrm{i}f,0)$, whose plaquettes lie in X by the support condition (its proof uses lemma 2.7(a),(b) only, and the site-independence of the counterterm). Letting $k\to\infty$ (Step 1), $J\mapsto Z_J(\mathrm{i}f,0)$ is nondecreasing and bounded by 1, hence convergent, for every $t\in\mathbb{R}^n$.

Step 3: $J\to\infty$. $\{Z_J(\cdot,0)\}_{J\geq J_0}$ is locally uniformly bounded on $\mathbb{C}^n$ by the J-free bound of Step 1, and converges on $(\mathrm{i}\mathbb{R})^n$ (Step 2), a totally real subspace of maximal dimension, hence a set of uniqueness for entire functions (at a point $\mathrm{i}t_0$ the restriction to $\mathrm{i}t_0 + \mathrm{i}\mathbb{R}^n$ determines the Taylor series). Montel's theorem and the identity theorem as in Step 1 give $Z_J(\cdot,0)\to Z_\infty(\cdot,0)$ locally uniformly on $\mathbb{C}^n$, $Z_\infty(\cdot,0)$ entire, and both bounds of Step 1 pass to the limit.

(c) By multilinearity normalise $\|f_i\|_2 = 1$. On the polydisc $|z_i|\leq r$, $\|g(z)\|_2 \leq \sum_i |\operatorname{Re} z_i| \leq nr$, so $|Z_\infty(\sum z_j f_j,0)| \leq e^{n^2 r^2}$ by (b). Cauchy's estimate for the mixed first derivative in the n distinct variables gives $|S_n| \leq r^{-n}\sup_{|z_i|=r}|Z_\infty| \leq r^{-n} e^{n^2 r^2}$; at $r^2 = 1/(2n)$ this is $(2en)^{n/2} = (2e)^{n/2} n^{n/2} \leq (\sqrt{2}\,e)^n (n!)^{1/2}$ by (E4) of lemma 2.6. Then (E2), $\|f_i\|_2 \leq \frac{\pi}{2\sqrt{2}}|f_i|_4$, gives the second bound. S_n is symmetric and multilinear (derivatives of an entire function of a linear combination) and continuous for the L^2 norms, so it extends by density from C_c^∞ to $\mathcal{S}$; for complex $f_i = f_i' + \mathrm{i}f_i''$ with real parts, multilinearity writes $S_n(f_1,\dots,f_n)$ as a sum of 2^n terms $\mathrm{i}^m S_n(f_1^\sharp,\dots,f_n^\sharp)$, each bounded by $(e\pi/2)^n(n!)^{1/2}\prod|f_i^\sharp|_4 \leq (e\pi/2)^n(n!)^{1/2}\prod|f_i|_4$, since $|\operatorname{Re} f|_4, |\operatorname{Im} f|_4 \leq |f|_4$; the kernel theorem gives the tempered distribution. The axiom E0″ of

[18, p. 287] asks for $|S_n(f_1,\dots,f_n)| \le \sigma_n \prod_i |f_i|_s$ on $\mathcal{S}$ with $\sigma_n \le \alpha(n!)^\beta$; here $s=4$ ($|\cdot|_4$ is the norm that [24] identifies with that of [18] at $s=4$), and $\sigma_n = (e\pi)^n(n!)^{1/2} \le \alpha_\beta(n!)^\beta$ exactly when $(e\pi)^n/(n!)^{\beta-1/2}$ is bounded, that is for every $\beta > \frac12$. □

Remark 7.5 (the constants are absolute). The bound $e^{\|g\|_2^2}$ of theorem 7.4 and the constants $(\sqrt{2}\,e)^n(n!)^{1/2}$, $(e\pi/2)^n(n!)^{1/2}$, σ_n of E0″ depend on none of e, λ, m, μ_0 and M: the only inputs are Gaussian domination, the Neumann form's constant 2 and the elementary facts (E1)–(E4). The second bound of (a) and (b) is the one that carries μ_0^{-2}. The proof of (b) uses, besides the existence clause, only [6, Thm. 6.2(1),(2)] through lemma 2.10, Gaussian domination and lemma 7.1; it uses neither [6, Thm. 6.2(3),(4)] (lemma 2.7(d)) nor any local bound resting on that reversed inequality.

7.3. **King's convention.** For the K family write $a^K_{k,J}(z) := \langle e^{\sum_j z_j F(f_j)}\rangle^K_{k,J}$ as in corollary 4.4, and Z^K_J, Z^K_∞ for its limits.

Theorem 7.6 (regularity of King's printed state)**.** *Let $f=(f_1,\dots,f_n)$, J_0, $g(z)$ be as above.*

(a) (No hypothesis.) *For all $k \ge 0$, $J \ge J_0$, $z \in \mathbb{C}^n$,*

$$|a^K_{k,J}(z)| \;\le\; \langle e^{F(g(z))}\rangle^K_{k,J} \;\le\; \exp\Big(\min\big\{\|g(z)\|^2_{2,k},\ \tfrac12\mu_0^{-2}\|d_k^* g(z)\|^2_{2,k}\big\}\Big),$$

and for every real 2-form u whose plaquettes have their corners in X_J: $\langle F(u)\rangle^K_{k,J}=0$ and $\langle F(u)^2\rangle^K_{k,J} \le \langle v, Cv\rangle_k \le 2\|u\|^2_{2,k}$, $v=d_k^*u$.

(b) (Limits.) *$Z^K_J(z) := \lim_k a^K_{k,J}(z)$ exists locally uniformly on $\mathbb{C}^n$ and is entire; $Z^K_\infty(z) := \lim_J Z^K_J(z)$ exists locally uniformly on $\mathbb{C}^n$ and is entire; $|Z^K_J(z)|, |Z^K_\infty(z)| \le e^{\|g(z)\|_2^2}$ and also $\le \exp(\frac12\mu_0^{-2}\|d^*g(z)\|^2_{L^2})$; $J \mapsto Z^K_J(\mathrm{i}t)$ is nondecreasing on $J \ge J_0$ with values in $[0,1]$ for $t \in \mathbb{R}^n$; and $Z^K_\infty(x) \ge 1$ at every real $x \in \mathbb{R}^n$.*

(c) (Schwinger functions.) *The Schwinger functions $S^K_n(f_1,\dots,f_n) := \partial_{z_1}\cdots\partial_{z_n} Z^K_\infty(0)$ satisfy $|S^K_n| \le (\sqrt{2}\,e)^n(n!)^{1/2}\prod_i\|f_i\|_2 \le (e\pi/2)^n(n!)^{1/2}\prod_i|f_i|_4$ for real f_i; the lattice Schwinger functions $S_n^{K,(k,J)} := \partial_{z_1}\cdots\partial_{z_n} a^K_{k,J}(0)$ satisfy the same bound at every (k,J) with $\|f_i\|_{2,k}$ in place of $\|f_i\|_2$, and $S_n^{K,(k,J)} \to S^K_n$ as $k\to\infty$ and then $J\to\infty$. Each S^K_n extends to a continuous symmetric multilinear form on $\mathcal{S}(\mathbb{R}^3;\Lambda^2_{\mathbb{C}})^n$ with the same bound, carrying the factor 2^n for complex test functions: E0″ holds with $s=4$ and $\sigma_n=(e\pi)^n(n!)^{1/2} \le \alpha_\beta(n!)^\beta$ for every $\beta > \frac12$.*

Proof. (a) The two inequalities are the bound of lemma 7.3 for the K family, with lemma 7.1 kept in both forms: $\mathrm{Re}\sum z_jF(f_j) = F(g(z))$; $\langle\cdot\rangle^K_{k,J}$ is a BFS measure (the killed sites carry point masses, which lemma 2.8 allows: at a killed site $I_q(0)=\delta_{q0}$ and the currents vanish on its bonds), and its Gaussian is the Neumann photon Gaussian of X_J, the same as the comparison cube's, so lemma 2.8 with $s=1$, $v=d_k^*g(z)$ and lemma 7.1 give $\langle e^{F(g(z))}\rangle^K_{k,J} \le e^{\langle v,Cv\rangle_k/2} \le \exp(\frac12\min\{2\|g\|^2_{2,k}, \mu_0^{-2}\|v\|^2_{2,k}\})$. *Second-moment clause.* $\langle F(u)\rangle^K_{k,J}=0$ because the state is invariant under $(A,\vartheta)\mapsto(-A,-\vartheta)$ and $F(u)$ is odd. For real s, $e^x+e^{-x} \ge 2+x^2$ pointwise and $e^{\pm sF(u)}$ are integrable, so $2+s^2\langle F(u)^2\rangle^K_{k,J} \le \langle e^{sF(u)}\rangle^K_{k,J} + \langle e^{-sF(u)}\rangle^K_{k,J} \le 2e^{s^2\langle v,Cv\rangle_k/2}$ (lemma 2.8 at $\pm s$; $\pm v$ give the same quadratic form). Hence $\langle F(u)^2\rangle^K_{k,J} \le 2(e^{s^2\langle v,Cv\rangle_k/2}-1)/s^2 \to \langle v,Cv\rangle_k$ as $s\downarrow 0$, and lemma 7.1 applies since the corners lie in X_J.

(b) The two limits, their entireness and the J-free bound $e^{\|g(z)\|_2^2}$ are corollary 4.4(ii)(α),(β): the proof of theorem 7.4(b) run on the K family, with the local bound of (a) (uniform in k, J by $\|g\|^2_{2,k} \le 3C_0|g|_4^2$), theorem 3.2 for the convergence at real points, Montel's theorem with the uniqueness set $\mathbb{R}^n$, and then, for $J\to\infty$, the monotonicity of corollary 4.3 on the cone in place of lemma 2.10 and the uniqueness set $(\mathrm{i}\mathbb{R})^n$. Three additions. (1) *Values in* $[0,1]$. $a^K_{k,J}(\mathrm{i}t)$ is real by the symmetry used in (a) and of modulus ≤ 1; it is ≥ 0 by the current representation of lemma 2.8: with $\Psi(A) = \sum_q w(q)\cos\langle\kappa(q),A\rangle$, $w(q)\ge 0$, and the Gaussian identity at $a = \mathrm{i}v \pm \mathrm{i}\kappa$,

$$\langle e^{\mathrm{i}\langle v,A\rangle}\rangle^K_{k,J} = e^{-\langle v,Cv\rangle_k/2}\,\mathbb{E}^\sharp\cosh\langle Cv,\kappa\rangle \;\ge\; 0, \qquad v = d_k^*\textstyle\sum_j t_j f_j,$$

the killed sites being covered as in (a). Monotonicity in J is corollary 4.3 at every k (condition (S) holds for X_{J_0} by the support condition), passed to the limit $k \to \infty$. (2) *The μ_0^{-2} bound.* $\|d_k^* g\|_{2,k} \to \|d^* g\|_{L^2}$ by (1), so the second bound of (a) passes to Z_J^K and then to Z_∞^K exactly as the first. (3) $Z_\infty^K(x) \geq 1$. For real x, $a_{k,J}^K(x) = \langle e^{F(g(x))}\rangle_{k,J}^K \geq e^{\langle F(g(x))\rangle_{k,J}^K} = 1$ by Jensen's inequality and (a); the inequality is closed under both limits.

(c) Let $\|f_j\|_{2,k} > 0$ for all j (otherwise $F_{\Lambda_k}(f_j) = 0$ and $S_n^{K,(k,J)} = 0$). Cauchy's estimates on the polydisc $|z_j| \leq \rho_j := \rho/(n\|f_j\|_{2,k})$ with the bound (a): there $\|g(z)\|_{2,k} \leq \sum_j |z_j| \|f_j\|_{2,k} \leq \rho$, so $|a_{k,J}^K| \leq e^{\rho^2}$ and $|S_n^{K,(k,J)}| \leq e^{\rho^2} \prod_j \rho_j^{-1} = e^{\rho^2} n^n \rho^{-n} \prod_j \|f_j\|_{2,k}$; at $\rho^2 = n/2$ the prefactor is $e^{n/2} 2^{n/2} n^{n/2} \leq e^{n/2} 2^{n/2} e^{n/2} (n!)^{1/2} = (\sqrt{2}\, e)^n (n!)^{1/2}$ by (E4). The Taylor coefficients at 0 converge under the locally uniform convergences of (b) (Weierstrass), and $\|f_j\|_{2,k} \to \|f_j\|_2$ by (E3), which gives the bound for S_n^K; (E2) gives the $|\cdot|_4$ form. The 2^n factor for complex test functions, the extension to $\mathcal{S}$ and the growth class σ_n are as in theorem 7.4(c). □

7.4. **The curvature sector of the comparison family's limits.** For the comparison family and an admissible scheme a, the general-source functionals $Z_J^a(\zeta) = \lim_k Z_{k,J}^a(\zeta)$, $\zeta \in \mathbb{C}^{2N}$, have, under (W2) (the optional extension), the following properties: (i) locally uniform convergence to an entire Z_J^a; (ii) on the cone, $J \mapsto Z_J^a(\mathrm{i}f, g)$ nondecreasing and $J \mapsto Z_J^a(f, -g)$ nonincreasing; (iii) the limit $Z_\infty^a = \lim_J Z_J^a$, locally uniform and entire, positive at real sources, with $Z_J^a \leq Z_\infty^a$ on the cone; (iv) $\Phi^a = \ln Z_\infty^a$ finite, convex and continuous on the Fréchet space $E_\rho = C_c^\infty(\bar{B}_\rho; \Lambda^2) \times C_c^\infty(\bar{B}_\rho)$; (v) translation invariance; (vi) the change of scheme. At $h = 0$ these conclusions hold under the existence clause alone; the natural local bound, $|Z_J(\zeta)| \leq Z_J(\mathrm{Re}\,\zeta) \leq Z_{J_0}(\mathrm{Re}\,\zeta)$, would use the nonincreasing clause of (ii), that is lemma 2.7(d). The following lemma states the curvature sector with that use removed; the Wick sector on the tube is lemma 13.11.

Lemma 7.7 (the comparison family's limits, curvature sector, with cone-direction inequalities only)**.** *Assume that the comparison family is convergent at fixed cubes (definition 3.1). For every finite family $f_1, \dots, f_N$ of real 2-forms in C_c^∞ supported at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_{J_0}$, with $Z_{k,J}(\sum_j z_j f_j, 0) := \langle e^{\sum_j z_j F(f_j)}\rangle_{k,J}^{(a)}$ and $g(z) = \sum_j (\mathrm{Re}\, z_j) f_j$:*

(i) *$Z_{k,J}(\cdot, 0) \to Z_J(\cdot, 0)$ locally uniformly on $\mathbb{C}^N$ as $k \to \infty$, and $Z_J(\cdot, 0)$ is entire, for every $J \geq J_0$;*

(ii) *$J \mapsto Z_J(\mathrm{i}f, 0)$ is nondecreasing on $J \geq J_0$ for every real $f = \sum_j t_j f_j$;*

(iii) *for all $k \geq 0$, $J \geq J_0$ and $z \in \mathbb{C}^N$,*

$$|Z_{k,J}(\textstyle\sum_j z_j f_j, 0)| \leq \langle e^{F(g(z))}\rangle_{k,J}^{(a)} \leq e^{\frac{1}{2}\langle v, Cv\rangle_k} \leq \exp\left(\tfrac{1}{2}\mu_0^{-2}\|d_k^* g(z)\|_{2,k}^2\right) \leq \exp\left(\tfrac{1}{2}\mu_0^{-2} \sup_{k'} \|d_{k'}^* g(z)\|_{2,k'}^2\right), \quad v = c$$

a bound uniform in k and J and locally uniform in z; consequently $Z_\infty(\cdot, 0) := \lim_J Z_J(\cdot, 0)$ exists locally uniformly on $\mathbb{C}^N$, is entire, satisfies $Z_\infty(f, 0) > 0$ for real f, and $Z_J(\mathrm{i}f, 0) \leq Z_\infty(\mathrm{i}f, 0)$ for real f;

(iv) *for every $\rho > 0$, $\Phi(f) := \ln Z_\infty(f, 0)$ is finite, convex and continuous on the Fréchet space $C_c^\infty(\bar{B}_\rho; \Lambda^2)$;*

(v) *$Z_\infty(\tau_b f, 0) = Z_\infty(f, 0)$ for all $b \in \mathbb{R}^3$ and all complex coefficient vectors, τ_b acting on every f_j.*

Clauses (i), (iii)–(v) *use, besides the convergence at fixed cubes, only* [6, Thm. 6.2(1),(2)] *through lemma 2.10, lemma 2.8, lemma 2.11*(d), (1), *lemma 2.14, Montel's theorem and the identity theorem. The nonincreasing clause* (ii) *at $h = 0$, $J \mapsto Z_J(f, 0)$ nonincreasing for real f, holds as well and uses lemma 2.7*(d)*; it is not consumed by the curvature sector of this paper. Clause* (vi) *is void at $h = 0$, no scheme entering.*

Proof. (i) is Step 1 of the proof of theorem 7.4(b), which uses lemma 7.3, the convergence at fixed cubes and Montel's theorem. (ii) is its Step 2. (iii) The first inequality is lemma 2.11(d), the second is lemma 2.8 with $s = 1$ (the cube state is a BFS measure), the third is $C \leq \mu_0^{-2}$ as forms (the photon form is bounded below by $\mu_0^2 \|A\|_{2,k}^2$, the second bound of lemma 7.1), and the supremum over k' is finite by (1). The bound passes to Z_J by (i) and makes $\{Z_J(\cdot, 0)\}_{J \geq J_0}$ locally bounded on $\mathbb{C}^N$; with the convergence on $(\mathrm{i}\mathbb{R})^N$ from (ii) and the bound $|Z_J(\mathrm{i}f, 0)| \leq 1$,

Montel's theorem and the identity theorem give the locally uniform convergence to an entire $Z_\infty(\cdot,0)$ as in Step 3 of theorem 7.4(b). Positivity: $Z_{k,J}(f,0) \geq e^{\langle F(f)\rangle^{(a)}_{k,J}} = 1$ for real f by Jensen's inequality and $\langle F(f)\rangle^{(a)}_{k,J} = 0$ (the symmetry $(A,\vartheta) \mapsto (-A,-\vartheta)$), closed under both limits. The last clause is (ii) in the limit.

(iv) Φ is finite by (iii). Convexity: for $f, f' \in C^\infty_c(\bar B_\rho;\Lambda^2)$ and $t \in [0,1]$ apply (i) and (iii) to the family (f,f') at the real coefficients $(t,1-t)$; at fixed k,J, $\ln Z_{k,J}(\cdot,0)$ is convex in the coefficients by Hölder's inequality, and convexity survives the two pointwise limits. Borel measurability: fix J with $MJ > \rho+1$, so that every $f \in C^\infty_c(\bar B_\rho;\Lambda^2)$ is supported at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_J$; at each k, $\ln Z_{k,J}(f,0)$ depends on f only through the finitely many sampled values $f_{c(p)}(e_\alpha, e_\beta)$, continuous functionals of f in the C^∞ topology, and the finite-dimensional integral is continuous in them (dominated convergence); so $\Phi = \lim_J \lim_k$ of continuous functions is Borel (of Baire class ≤ 2). $C^\infty_c(\bar B_\rho;\Lambda^2)$ is a closed subspace of $C^\infty(\bar B_{\rho+1};\Lambda^2)$, hence Fréchet. lemma 2.14 gives the continuity.

(v) is proposition 8.1, whose proof uses the lattice sandwich between C_{J-1} and C_{J+1} (first inequality of lemma 2.10), $k \to \infty$ by (i) (the convergence at fixed cubes covers the translated family), $J \to \infty$ by (iii), the identity theorem with the uniqueness set $(\mathrm{i}\mathbb{R})^N$, the continuity of (iv) and the density of $\bigcup_k 5^{-k}\mathbb{Z}^3$. □

For King's convention the same clauses hold by theorem 3.2: (i) and (iii) are theorem 7.6(b), with the local bound of theorem 7.6(a); (ii) is corollary 4.3; (iv) follows from theorem 7.6(b) by the argument above, word for word; (v) is corollary 8.3. In every clause the inequalities used are the cone directions of lemma 2.7(a),(b) and of lemma 4.2; the reversed inequalities of lemma 2.7(d) enter the curvature sector of this paper nowhere.

8. Translations, the cubic group, and rotations

This section proves the clause OS1 of King's printed state as far as the lattice allows: invariance under all translations and under the cubic group O_h about every point; full $O(3)$ invariance comes from the rotation theorem (theorem 3.13, proved in the rotation part, theorem 15.1) and is stated at the end. No cube is invariant under a non-trivial translation or a reflection; what replaces exactness is a sandwich between a smaller and a larger cube of the same family, in the direction of the cone inequalities of lemmas 2.10 and 4.1, followed by the limits $k \to \infty$, $J \to \infty$. Coordinate permutations alone are exact (lemma 8.4).

Conventions. We use the notation of definition 2.3 and corollary 4.4: for a finite family $f = (f_1,\dots,f_N)$ of real 2-forms in $C^\infty_c(\mathbb{R}^3;\Lambda^2)$ and $z \in \mathbb{C}^N$, $a^K_{k,J}(z) = \langle e^{\sum_j z_j F(f_j)}\rangle^K_{k,J}$ and $a^{(a)}_{k,J}(z) = \langle e^{\sum_j z_j F(f_j)}\rangle^{(a)}_{k,J}$ are the lattice functionals, Z^K_J, Z^K_∞ and Z_J, Z_∞ their limits where these exist, and $g(z) = \sum_j (\operatorname{Re} z_j) f_j$. Since F_{Λ_k} is linear in the source (extended $\mathbb{C}$-linearly), a lattice functional depends on a family only through its total source, and we use the same symbol for a coefficient vector and for its total source: $a^K_{k,J}(z) = Z^K_{k,J}(\sum_j z_j f_j)$ with $Z^K_{k,J}(u) := \langle e^{F(u)}\rangle^K_{k,J}$ for a complex-valued 2-form u; in particular, for a real 2-form u, $Z^K_{k,J}(\mathrm{i}u)$ is the one-member functional at the coefficient i, and the same convention is carried to Z^K_J, Z^K_∞ (so $Z^K_\infty(\mathrm{i}u_t) = Z^{K,f}_\infty(\mathrm{i}t)$ for $u_t = \sum_j t_j f_j$). When several families are in play we write $a^{K,f}_{k,J}$, $Z^{K,f}_J$, $Z^{K,f}_\infty$. $J_0(f)$ is the least J such that every $\operatorname{supp} f_j$ lies at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_J$; for $J \geq J_0(f)$ the support condition (S) holds for X_J. The transport $(O\cdot f)_x(u,v) = f_{O^{-1}x}(O^{-1}u, O^{-1}v)$ and the translation $(\tau_b f)_x = f_{x-b}$ are those of definition 2.3; $O\cdot(O'\cdot f) = (OO')\cdot f$, and both are linear in f. Translations and the cubic group act on the Schwinger functions by this signed push-forward of 2-forms, as in the torus paper [24]: for an improper O no extra sign appears.

8.1. Translations on the comparison convention.

Proposition 8.1 (translations, comparison convention)**.** *Assume that the comparison family is convergent at fixed cubes (definition 3.1), so that by lemma 7.7(i),(iii) the limits $Z_J(z) = \lim_k a^{(a)}_{k,J}(z)$ and $Z_\infty(z) = \lim_J Z_J(z)$ exist locally uniformly on $\mathbb{C}^N$ for every family and are*

entire. Then for every finite family f of real 2-forms, every $b \in \mathbb{R}^3$ and every $z \in \mathbb{C}^N$,

$$Z_\infty\Big(\sum_j z_j \tau_b f_j\Big) = Z_\infty\Big(\sum_j z_j f_j\Big).$$

Proof. Step 1 (lattice covariance). Let $k_0 \geq 0$, $b \in \Lambda_{k_0}$ and $k \geq k_0$. Then τ_b maps Λ_k onto itself, and the substitution $A'_\alpha(x) := A_\alpha(x+b)$, $\phi'(x) := \phi(x+b)$ is a bijection of the configurations of the comparison state on X_J onto those of the comparison state on $X_J - b$ (definition 2.1 on the translated box) which carries the Lebesgue measure and the action to themselves: the action is a sum of local terms with site-independent coefficients (δm_ε^2, W_ε depend on ε only), and the boundary conditions are transported. The plaquette p corresponds to $p - b$ with the same orientation and $F(p)(A) = F(p-b)(A')$, so $F_{\Lambda_k}(\tau_b f)(A) = F_{\Lambda_k}(f)(A')$. Hence, for all coefficients,

$$Z_{k,X_J}\Big(\sum_j z_j \tau_b f_j\Big) = Z_{k,X_J - b}\Big(\sum_j z_j f_j\Big) \qquad (b \in \Lambda_{k_0},\ k \geq k_0), \tag{16}$$

the right side being the functional of the comparison state on $X_J - b$.

Step 2 (interlacing of the translated box). For $|b|_\infty \leq M$: $X_{J-1} \subset X_J - b \subset X_{J+1}$, since $-MJ - b_\mu \leq -M(J-1)$, $MJ - b_\mu \geq M(J-1)$, $-MJ - b_\mu \geq -M(J+1)$ and $MJ - b_\mu \leq M(J+1)$ for every μ.

Step 3 (the sandwich on the imaginary axes). Let $J \geq J_0(f) + 1$, $t \in \mathbb{R}^N$ and $u_t = \sum_j t_j f_j$, so that $(u_t, 0) \in \mathcal{K}$ is supported at distance > 1 from $\mathbb{R}^3 \setminus C_{J-1}$. Apply the first inequality of lemma 2.10 twice: with $Y = X_J - b$ and $X = X_{J-1}$, and with $Y = X_{J+1}$ and $X = X_J - b$. The support condition of the lemma holds because the plaquettes carrying u_t have their corners in X_{J-1}; no site is killed, so each cut-out factor is the comparison state on the smaller box (the Dirichlet terms of the smaller state are the diagonal parts of the cut bonds, and where the boundaries of two of the boxes touch they coincide). With (16),

$$a^{(a)}_{k,J-1}(\mathrm{i}t) \ \leq\ a^{(a),\tau_b f}_{k,J}(\mathrm{i}t) \ \leq\ a^{(a)}_{k,J+1}(\mathrm{i}t) \qquad (k \geq k_0,\ J \geq J_0(f) + 1).$$

Let $k \to \infty$: the two ends converge by lemma 7.7(i) for the family f at $J \mp 1 \geq J_0(f)$, and the middle term by lemma 7.7(i) for the family $\tau_b f$, which is admissible at J because $\operatorname{supp} f_j + b$ lies at distance > 1 from $\mathbb{R}^3 \setminus C_J$ (as $C_{J-1} + b \subset C_J$). Then let $J \to \infty$ by lemma 7.7(iii): $Z^{\tau_b f}_\infty(\mathrm{i}t) = Z^f_\infty(\mathrm{i}t)$ for all $t \in \mathbb{R}^N$.

Step 4 (all coefficients). $D(z) := Z^{\tau_b f}_\infty(z) - Z^f_\infty(z)$ is entire on $\mathbb{C}^N$ and vanishes on $(\mathrm{i}\mathbb{R})^N$, a totally real subspace of maximal real dimension N; every partial derivative of D at 0 vanishes ($\partial_{y_j} D(\mathrm{i}y) = \mathrm{i}\,\partial_{z_j} D$), so $D \equiv 0$. Thus the identity holds for $b \in \Lambda_{k_0}$, $|b|_\infty \leq M$, and all $z \in \mathbb{C}^N$.

Step 5 (all lattice translations). A vector $b \in \bigcup_k \Lambda_k$ lies in some Λ_{k_0} and is a finite sum $b_1 + \cdots + b_n$ of vectors $b_i \in \Lambda_{k_0}$ with $|b_i|_\infty \leq M$ (each coordinate splits into steps $\pm M$ and a remainder in $\varepsilon_{k_0}\mathbb{Z}$ of modulus $< M$). Steps 3–4 hold for every family, in particular for the translated families $\tau_{b_1 + \cdots + b_{i-1}} f$; composing gives the identity for b.

Step 6 (all $b \in \mathbb{R}^3$ at real coefficients). Fix $\rho > 0$ and let $E_\rho := C^\infty_c(\bar{B}_\rho; \Lambda^2)$ with its C^∞ topology, a Fréchet space. On E_ρ the function $\Phi(u) := \ln Z_\infty(u)$ at real u is finite ($Z_\infty(u) \geq 1$ by Jensen's inequality, as $\langle F(u)\rangle = 0$ by charge conjugation), convex (a pointwise limit of the convex functions $u \mapsto \ln\langle e^{F(u)}\rangle^{(a)}_{k,J}$) and Borel (each lattice functional depends continuously on finitely many values of u), hence continuous by lemma 2.14; this is lemma 7.7(iv). For $|b| \leq \rho' - \rho$ the map $b \mapsto \tau_b u \in E_{\rho'}$ is continuous, $\bigcup_k \Lambda_k$ is dense, and Φ on $E_{\rho'}$ restricted to E_ρ is the same pointwise limit; so Step 5 gives $Z_\infty(\tau_b u) = Z_\infty(u)$ for all $b \in \mathbb{R}^3$ and real $u \in E_\rho$, and exhausting over ρ for all real $u \in C^\infty_c$.

Step 7 (complex coefficients). For a family f and $b \in \mathbb{R}^3$, both sides of the asserted identity are entire in z and agree on $\mathbb{R}^N$ by Step 6 (applied to $u = \sum_j x_j f_j$), a set of uniqueness; hence they agree on $\mathbb{C}^N$. □

Only the first inequality of lemma 2.10 enters: the sandwich is taken on the cone, and the passage off it is the identity theorem. The translation clause for the joint field on the tube, by the same sandwich at the cone points of the tube, is lemma 13.11(v).

8.2. **The translation sandwich on King's convention.** On King's convention the translated cube has its killed sites on $T_J - b$, which neither contains nor is contained in $T_{J\pm1}$, so lemma 2.10 does not compare it with $C^K_{J\pm1}$ directly; one kill step (lemma 2.7(c) through lemma 2.9) and one cut-out with killed sites (lemma 4.1) at each end do.

Lattice covariance on King's convention. For $b \in \Lambda_{k_0}$ and $k \geq k_0$ let $\langle\cdot\rangle^{K,-b}_{k,J}$ denote the comparison state on $X_J - b$ with the matter killed on $T_J - b$ (King's cube translated by $-b$). The substitution of Step 1 above is a bijection of the configurations on X_J killed on T_J onto those on $X_J - b$ killed on $T_J - b$, carrying the point masses to the point masses and the action to the action (the kill set translates with the box). Hence, for every finite family and all coefficients,

$$Z^{a,K}_{k,J}\Big(\sum_j z_j \tau_b f_j, \sum_j w_j \tau_b h_j\Big) = Z^{a,K,-b}_{k,J}\Big(\sum_j z_j f_j, \sum_j w_j h_j\Big) \qquad (b \in \Lambda_{k_0},\ k \geq k_0), \tag{17}$$

the right side being the scheme functional of $\langle\cdot\rangle^{K,-b}_{k,J}$ with the same multiplier ($\mathsf{S}_{\Lambda_k}(\tau_b h) = \mathsf{S}_{\Lambda_k}(h)$).

Lemma 8.2 (translation sandwich, King's convention)**.** *Let $(f,g) \in \mathcal{K}$ be supported at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_{J_0}$, and let $b \in \Lambda_{k_0}$ with $|b|_\infty \leq M$. Then for every $k \geq k_0$, every $M \geq 1$ and every $J \geq J_0 + 1$,*

$$Z^{a,K}_{k,J-1}(\mathrm{i}f, g) \ \leq\ Z^{a,K}_{k,J}(\mathrm{i}\,\tau_b f, \tau_b g) \ \leq\ Z^{a,K}_{k,J+1}(\mathrm{i}f, g),$$

and the reversed inequalities hold for $(f, -g)$.

Proof. By (17) the middle term is $Z^{a,K,-b}_{k,J}(\mathrm{i}f, g)$; the multipliers of the three terms are the same positive constant (lemma 2.11(a); S is translation invariant), so it suffices to prove the inequalities for $\langle\Theta(\mathrm{i}f, g)\rangle$, and the reversed ones for $\Theta(f, -g)$; all these expectations are real.

Geometry. Here $b_\mu \in \varepsilon_{k_0}\mathbb{Z}$, $|b_\mu| \leq M$, and all sets are subsets of Λ_k. (G1) $\mathrm{alive}^K_J - b = \prod_\mu\{-MJ - b_\mu, \dots, MJ - 2\varepsilon - b_\mu\} \subset \mathrm{alive}^K_{J+1} = \{-MJ - M, \dots, MJ + M - 2\varepsilon\}^3$, since $-MJ - b_\mu \geq -MJ - M$ and $MJ - 2\varepsilon - b_\mu \leq MJ + M - 2\varepsilon$. (G2) $X_J - b \subset X_{J+1}$: the same arithmetic with ε in place of 2ε. (G3) $\mathrm{alive}^K_{J-1} = \{-MJ + M, \dots, MJ - M - 2\varepsilon\}^3 \subset \mathrm{alive}^K_J - b$, since $-MJ + M \geq -MJ - b_\mu$ and $MJ - M - 2\varepsilon \leq MJ - 2\varepsilon - b_\mu$. (G4) $X_{J-1} \subset X_J - b$, likewise. All four inclusions hold with margin $M - |b_\mu| \geq 0$ on every face, at every spacing, including $k = 0$, $M = 1$, $|b_\mu| = M$; non-strict inclusions are all that lemma 4.1 needs.

Sources. By (S) for X_{J-1}, every plaquette carrying f has its corners in X_{J-1} and every site carrying g lies in alive^K_{J-1}; by (G1)–(G4) these lie in the site sets and alive sets of all three states. So the observable is one function of variables present and alive in all three states, and its scalar factor equals 1 at every site killed below.

Upper end. Start from $\langle\cdot\rangle^K_{k,J+1}$, the comparison state on X_{J+1} killed on T_{J+1}, a BFS measure of King type (definition 2.2). (i) Kill $U_+ := \mathrm{alive}^K_{J+1} \setminus (\mathrm{alive}^K_J - b)$. These are alive sites of the state, $g = 0$ on U_+ (by (G3), $U_+ \cap \mathrm{alive}^K_{J-1} = \emptyset$), and $\Theta(\mathrm{i}f, g)$ is continuous with $|\Theta| = e^{\sum_x g_x r_x^2}$; so by lemma 2.7(c), through lemma 2.9 (which allows the already killed sites T_{J+1}), killing U_+ does not increase $\langle\Theta(\mathrm{i}f, g)\rangle$, and by lemma 2.7(d) with lemma 2.9 it does not decrease $\langle\Theta(f, -g)\rangle$. The result is the comparison state on X_{J+1} killed on $T' := T_{J+1} \cup U_+ = X_{J+1} \setminus (\mathrm{alive}^K_J - b)$ (by (G1); nothing is revived). (ii) Apply lemma 4.1 with $V = X_{J+1}$, $T = T'$ and $X = X_J - b \subset V$ (by (G2)); the lemma allows any $T \subset V$ and any $X \subset V$. The cut-out factor is the comparison state on $X_J - b$ killed on $(X_J - b) \cap T' = (X_J - b) \setminus (\mathrm{alive}^K_J - b) = T_J - b$, that is $\langle\cdot\rangle^{K,-b}_{k,J}$; the support condition (S_X) holds for $X = X_J - b$ by *Sources*. Hence $\langle\Theta(\mathrm{i}f, g)\rangle^{K,-b}_{k,J} \leq \langle\Theta(\mathrm{i}f, g)\rangle^{(a),T'}_{X_{J+1}} \leq \langle\Theta(\mathrm{i}f, g)\rangle^K_{k,J+1}$, and $\geq$ for class $\mathcal{D}$.

Lower end. Start from $\langle\cdot\rangle^{K,-b}_{k,J}$, the comparison state on $X_J - b$ killed on $T_J - b$ (of King type, a translate of a state of definition 2.2). (i) Kill $U_- := (\mathrm{alive}^K_J - b) \setminus \mathrm{alive}^K_{J-1}$, alive sites at which $g = 0$ by *Sources*; as above, $\langle\Theta(\mathrm{i}f, g)\rangle$ does not increase and $\langle\Theta(f, -g)\rangle$ does not decrease, and the result is the comparison state on $X_J - b$ killed on $T'' := (X_J - b) \setminus \mathrm{alive}^K_{J-1}$ (by (G3)). (ii) lemma 4.1 with $V = X_J - b$, $T = T''$ and $X = X_{J-1} \subset V$ (by (G4)): the factor is the

comparison state on X_{J-1} killed on $X_{J-1}\cap T'' = X_{J-1}\setminus \mathrm{alive}^K_{J-1} = T_{J-1}$, that is $\langle\cdot\rangle^K_{k,J-1}$; (S_X) for X_{J-1} is the hypothesis. Hence $\langle\Theta(\mathrm{i}f,g)\rangle^K_{k,J-1} \le \langle\Theta(\mathrm{i}f,g)\rangle^{K,-b}_{k,J}$, and $\ge$ for class $\mathcal{D}$.

There is no edge case because a translation by $b\in\Lambda_{k_0}$, $|b|_\infty \le M$, moves King's cube rigidly, with the alive sets nested at margins $M-|b_\mu|\ge 0$ at every spacing; the reflection of lemma 4.5 shifts the cube by one lattice layer instead, which is why its one-step form needs $M\ge 2\varepsilon_k$. □

The class-$\mathcal{I}$ clause uses lemma 2.7(a),(b),(c),(e) and lemma 2.9 only; the class-$\mathcal{D}$ clause uses lemma 2.7(d) in addition. Only the class-$\mathcal{I}$ clause is used in this section.

Corollary 8.3 (translations of King's printed state)**.** *For every finite family f of real 2-forms, every $b\in\mathbb{R}^3$ and every $z\in\mathbb{C}^N$, $Z^K_\infty(\sum_j z_j\tau_b f_j) = Z^K_\infty(\sum_j z_j f_j)$. Moreover, for all real 2-forms $u,u'\in C^\infty_c$,*

$$|Z^K_\infty(\mathrm{i}u) - Z^K_\infty(\mathrm{i}u')| \le \sqrt{2}\,\|u-u'\|_2,$$

the L^2 norm of a 2-form being computed in a basis (e_c) of Λ^2 orthonormal for the norm formula.

Proof. Write $u_t=\sum_j t_j f_j$ for $t\in\mathbb{R}^N$. (1) *$b\in\Lambda_{k_0}$, $|b|_\infty\le M$, on the imaginary axes.* For $J\ge J_0(f)+1$ and $k\ge k_0$, lemma 8.2 at $(u_t,0)\in\mathcal{K}$ gives $a^K_{k,J-1}(\mathrm{i}t)\le a^{K,\tau_b f}_{k,J}(\mathrm{i}t)\le a^K_{k,J+1}(\mathrm{i}t)$. Let $k\to\infty$: the ends converge by theorem 7.6(b) for f at $J\mp 1\ge J_0(f)$; the middle term by theorem 7.6(b) for the family $\tau_b f$, which is admissible at J since $\operatorname{supp} f_j + b$ lies at distance >1 from $\mathbb{R}^3\setminus C_J$ ($C_{J-1}+b\subset C_J$ for $|b|_\infty\le M$), so that theorem 3.2, posed for every finite family, applies to it. Hence $Z^K_{J-1}(\mathrm{i}t)\le Z^{K,\tau_b f}_J(\mathrm{i}t)\le Z^K_{J+1}(\mathrm{i}t)$, and $J\to\infty$ (corollary 4.4(ii)(β) at both ends) gives $Z^{K,\tau_b f}_\infty(\mathrm{i}t) = Z^{K,f}_\infty(\mathrm{i}t)$ for all $t\in\mathbb{R}^N$. (2) *All coefficients, same b.* $D(z) := Z^{K,\tau_b f}_\infty(z) - Z^{K,f}_\infty(z)$ is entire on $\mathbb{C}^N$ (theorem 7.6(b) for both families) and vanishes on $(\mathrm{i}\mathbb{R})^N$, a totally real subspace of maximal real dimension, so every partial derivative of D at 0 vanishes and $D\equiv 0$. (3) *All $b\in\bigcup_k\Lambda_k$*: as in Step 5 of proposition 8.1, by composition over the translated families. (4) *The Lipschitz bound.* For real 2-forms u,u' whose plaquettes have corners in X_J, at every (k,J), $|\langle e^{\mathrm{i}F(u)}\rangle^K_{k,J} - \langle e^{\mathrm{i}F(u')}\rangle^K_{k,J}| \le \langle|F(u-u')|\rangle^K_{k,J} \le \langle F(u-u')^2\rangle^{K\,1/2}_{k,J} \le \sqrt{2}\|u-u'\|_{2,k}$, by $|e^{\mathrm{i}x}-e^{\mathrm{i}y}|\le|x-y|$, the linearity of F_{Λ_k} and the second-moment clause of theorem 7.6(a). Both sides have limits as $k\to\infty$ and $J\to\infty$ (theorem 7.6(b) for the one-member families at the coefficient i; $\|u-u'\|_{2,k}\to\|u-u'\|_2$ by (1)). (5) *All $b_0\in\mathbb{R}^3$.* Pick $b_n\to b_0$ with $b_n\in\bigcup_k\Lambda_k$. For real $u\in C^\infty_c$, $\tau_{b_n}u\to\tau_{b_0}u$ in L^2 (uniform continuity and compact support), so by (4) and (3), $Z^K_\infty(\mathrm{i}\tau_{b_0}u) = \lim_n Z^K_\infty(\mathrm{i}\tau_{b_n}u) = Z^K_\infty(\mathrm{i}u)$. With $u=u_t$ this is $Z^{K,\tau_{b_0}f}_\infty(\mathrm{i}t) = Z^{K,f}_\infty(\mathrm{i}t)$ for all t, and (2) extends it to all $z\in\mathbb{C}^N$. □

Nothing here uses a reflection or any property of the limit beyond theorem 7.6; King's derivation of translation invariance from the uniqueness of his limit [14, Thm. 2.3(i)] is not used.

8.3. The cubic group.

Lemma 8.4 (passive covariance with killed sites)**.** *Let O, Λ, (f_α) and X be as in lemma 2.12, and let $U\subset X$ be a set of killed sites. Then the identities of lemma 2.12 hold between the state on X with the matter killed on U and the state on OX (frame (Of_α), transported boundary conditions and wall) with the matter killed on OU:*

$$\big\langle\Theta(z\,O\cdot f, w\,O\cdot h)\big\rangle_{O\Lambda,OX;\,OU} = \big\langle\Theta(zf,wh)\big\rangle_{\Lambda,X;\,U},\qquad \mathsf{S}_{O\Lambda}(O\cdot h) = \mathsf{S}_\Lambda(h).$$

Proof. The configuration-space bijection $\phi'(Ox)=\phi(x)$, $A'_\alpha(Ox)=A_\alpha(x)$ of lemma 2.12 carries $\prod_{x\in U}\delta_0(d\phi(x))\,d\nu$ to $\prod_{x\in OU}\delta_0(d\phi'(Ox))\,d\nu$ (the point mass at $\phi(x)=0$ goes to the point mass at $\phi'(Ox)=0$), and the diagonal parts $\frac{\varepsilon}{2}r^2$ at the alive ends of the bonds into U are frame-free; the rest of that proof is unchanged. □

Theorem 8.5 (cubic symmetry of King's printed state)**.** *For all real 2-forms $f_1,\dots,f_N\in C^\infty_c(\mathbb{R}^3;\Lambda^2)$, all $z\in\mathbb{C}^N$, all $O\in O_h$ (the signed permutation matrices) and all $b\in\mathbb{R}^3$,*

$$Z^K_\infty\Big(\sum_j z_j\,\tau_b O\cdot f_j\Big) = Z^K_\infty\Big(\sum_j z_j f_j\Big).$$

Hence the Schwinger functions S_n^K of theorem 7.6(c) are $\mathbb{R}^3 \rtimes O_h$-covariant, F transforming as a 2-form.

Proof. "(k, J) exact" means an identity of the lattice functionals for all complex coefficients and all k, J for which both families have their plaquettes in X_J.

Step 1 (permutations are exact). Let P be a permutation matrix, $Pe_\alpha = e_{\pi(\alpha)}$. Then $P\Lambda_k = \Lambda_k$, $PX_J = X_J$ and $PT_J = T_J$ (the condition "$x_\mu = MJ - \varepsilon$ for some μ" is permutation invariant), hence P alive$_J^K$ = alive$_J^K$. By lemma 8.4 with $O = P$, $X = X_J$, $U = T_J$, $\langle \Theta(zP \cdot f)\rangle_{P\Lambda_k, PX_J;\, PT_J}$ in the frame $(e_{\pi(\alpha)})$ equals $\langle \Theta(zf)\rangle_{k,J}^K$, and the state on the left is $\langle\cdot\rangle_{k,J}^K$ itself: the Dirichlet rule is a condition on the site set X_J, the kill set is T_J, the Neumann rule keeps the same set of (bond, direction) pairs relabelled, the mass term runs over the same bonds, and the hopping and single-site terms are frame-free. The plaquette set and the centres are the same; for a transposed pair $(x; \beta, \alpha)$ both $F(p)$ and $f_{c(p)}(e_\beta, e_\alpha)$ change sign by antisymmetry, so their product is unchanged. Thus $a_{k,J}^{K,P\cdot f}(z) = a_{k,J}^{K,f}(z)$ is (k, J) exact. The distance-> 1 interior $(-MJ + 1, MJ - 1)^3$ of C_J is P-invariant, so $J_0(P \cdot f) = J_0(f)$, and theorem 7.6(b) for both families gives $Z_\infty^{K,P\cdot f} = Z_\infty^{K,f}$ on $\mathbb{C}^N$.

Step 2 (the reflection σ_3 on the imaginary axes). Let $\sigma_3 = \mathrm{diag}(1, 1, -1)$, $t \in \mathbb{R}^N$, $u_t = \sum_j t_j f_j$, $J_0 = J_0(f)$. lemma 4.5(b), for every $k \geq 0$, every $M \geq 1$ and every $J \geq J_0 + 2$, gives

$$a_{k,J-2}^{K,f}(\mathrm{i}t) \;\leq\; a_{k,J}^{K,\sigma_3\cdot f}(\mathrm{i}t) \;\leq\; a_{k,J+2}^{K,f}(\mathrm{i}t),$$

since $\sigma_3 \cdot u_t = \sum_j t_j \sigma_3 \cdot f_j$. The family $\sigma_3 \cdot f$ is admissible at J, the interior being σ_3-symmetric, so $J_0(\sigma_3 \cdot f) = J_0(f)$. Let $k \to \infty$ (theorem 7.6(b) for f at $J \mp 2$ and for $\sigma_3 \cdot f$ at J), then $J \to \infty$ (corollary 4.4(ii)(β) at both ends): $Z_\infty^{K,\sigma_3\cdot f}(\mathrm{i}t) = Z_\infty^{K,f}(\mathrm{i}t)$ for all $t \in \mathbb{R}^N$. The one-step sandwich of lemma 4.5(a) would serve whenever $M \geq 2\varepsilon_k$, that is for all but finitely many k, which the k-limit ignores; the two-step form is used because it needs no case distinction, and the J-limit absorbs the extra step. This is the use King makes of the correlation inequalities in the proof of his Theorem 3.2 [14], with the kill steps added; no reversed inequality enters.

Step 3 (σ_1, σ_2). $\sigma_\nu = P\sigma_3 P^{-1}$ with P the transposition of the axes ν and 3. Since the transport is a left action, for every real 2-form u,

$$Z_\infty^K(\mathrm{i}\sigma_\nu\cdot u) = Z_\infty^K(\mathrm{i}P\cdot(\sigma_3 P^{-1}\cdot u)) = Z_\infty^K(\mathrm{i}\sigma_3\cdot(P^{-1}\cdot u)) = Z_\infty^K(\mathrm{i}P^{-1}\cdot u) = Z_\infty^K(\mathrm{i}u),$$

by Step 1 for the family $\sigma_3 P^{-1} \cdot u$, Step 2 for the family $P^{-1} \cdot u$, and Step 1 again; each step is a statement about every family.

Step 4 (the group, on the imaginary axes). Every $O \in O_h$ is a product of a permutation matrix and some of $\sigma_1, \sigma_2, \sigma_3$, and if $Z_\infty^K(\mathrm{i}O \cdot u) = Z_\infty^K(\mathrm{i}u)$ for all real u holds for O_1 and O_2, it holds for $O_1 O_2$. By Steps 1–3, $Z_\infty^{K,O\cdot f}(\mathrm{i}t) = Z_\infty^{K,f}(\mathrm{i}t)$ for every $O \in O_h$, every family f and every $t \in \mathbb{R}^N$.

Step 5 (all coefficients). $D(z) := Z_\infty^{K,O\cdot f}(z) - Z_\infty^{K,f}(z)$ is entire on $\mathbb{C}^N$ (theorem 7.6(b); $J_0(O \cdot f) = J_0(f)$) and vanishes on $(\mathrm{i}\mathbb{R})^N$, a set of uniqueness; so $D \equiv 0$. No splitting of sources is needed, there being no scalar source.

Step 6 (about every point). O_h about b is $\tau_b O \tau_{-b}$, and translations are corollary 8.3 (every family, every $b \in \mathbb{R}^3$, all coefficients): $Z_\infty^K(\sum_j z_j \tau_b O \cdot f_j) = Z_\infty^K(\sum_j z_j O \cdot f_j) = Z_\infty^K(\sum_j z_j f_j)$.

Step 7 (Schwinger functions). Differentiating the identity of entire functions at $z = 0$ gives $S_n^K(\tau_b O \cdot f_1, \ldots, \tau_b O \cdot f_n) = S_n^K(f_1, \ldots, f_n)$ for all n and $(b, O) \in \mathbb{R}^3 \rtimes O_h$. □

Remark 8.6 (what cubic symmetry needs). King's cube is exactly symmetric under the coordinate permutations (Step 1) but not under the reflections; there the sandwich of lemma 4.5 replaces exactness, as that of lemma 2.13 does on the comparison convention. So O_h-invariance of King's printed state follows from the uniqueness of the monotone cube limit (corollary 4.3 with theorem 7.6(b)) together with the cut sandwich, and does not need the uniqueness of the infinite-volume state. The remark of the torus paper [24] that O_h-invariance "would follow from the uniqueness of the infinite-volume limit, which is not known" is superseded by theorem 8.5. Only class-$\mathcal{I}$ clauses enter, and no scheme or Wick constant.

8.4. **Rotations.**

Theorem 8.7 (rotations)**.** *Z^K_∞ is $\mathbb{R}^3 \rtimes O(3)$-invariant: for every finite family f, every $z \in \mathbb{C}^N$ and every $(b,R) \in \mathbb{R}^3 \rtimes O(3)$, $Z^K_\infty(\sum_j z_j \tau_b R \cdot f_j) = Z^K_\infty(\sum_j z_j f_j)$, and the Schwinger functions S^K_n are Euclidean covariant, F transforming as a 2-form.*

Proof. By theorem 18.12 (the rotation theorem theorem 3.13 on King's convention: the torus comparison theorem 17.1 with the closing estimate theorem 18.4), $Z^K_\infty(\cdot,0)$ is $O(3)$-invariant for all complex coefficients; translations are corollary 8.3; the Schwinger functions follow by differentiation at $z=0$. By corollary 4.4(iii), with the fixed-cube limits of theorem 3.2, the limit state $Z^{\mathrm{LIM}}_\infty(\cdot,0)$ of the comparison family equals $Z^K_\infty(\cdot,0)$ and is therefore $\mathbb{R}^3 \rtimes O(3)$-invariant as well. □

The rotation theorem is proved in the rotation part (section 15, theorem 15.1). Nothing in corollary 8.3 and theorem 8.5 depends on it. Its consequences for reflection positivity about every plane and for the components of the measure are drawn in theorem 11.1(c).

9. Reflection positivity I: the modified box

King's Neumann–Dirichlet cube is not reflection symmetric about any plane, and the natural symmetrisation of its photon form is not reflection positive on gauge-invariant functionals (remark 9.7). Reflection positivity is therefore proved first for the modified box state of [14, (3.38)] and then carried to the cube states by a two-sided comparison on the cone: the cube C_J sits inside the box (lemma 9.9), and the box sits inside a larger cube once a margin proportional to μ_0^{-1} is allowed (lemma 9.10). The passage to the limits is in theorem 10.6. Throughout, $\varepsilon = \varepsilon_k$, the lattice is $\Lambda_k \subset \mathbb{R}^d$ with $d=3$ (the certificates of lemma 9.6 are also given for $d=2$), a *lattice box* is a product $\Omega = \prod_\mu \{a_\mu, a_\mu+\varepsilon, \dots, b_\mu\} \subset \Lambda_k$, Ω^* is the set of its internal bonds (both ends in Ω) and Ω^{**} the set of its plaquettes (all four corners in Ω).

9.1. **The lattice reflection and the modified box.**

Definition 9.1 (mid-planes and the lattice reflection)**.** The *mid-planes* are the planes $\{x_\nu = c\}$ with $c \in D := \bigcup_{k\ge 0} \varepsilon_k(\mathbb{Z}+\frac12)$. If $c \in \varepsilon_{k_0}(\mathbb{Z}+\frac12)$ then $c \in \varepsilon_k(\mathbb{Z}+\frac12)$ for every $k \ge k_0$, because $5^j(m+\frac12) = 5^j m + \frac{5^j-1}{2} + \frac12$ and 5^j is odd; the same set D is obtained from any tail $k \ge k_1$ of the spacings. D is dense in $\mathbb{R}$ but not all of $\mathbb{R}$.

Fix ν, a mid-plane $c \in \varepsilon_k(\mathbb{Z}+\frac12)$ and a lattice box $\Omega \subset \Lambda_k$ with $\rho\Omega = \Omega$, $\rho = \rho_{\nu,c}$ the reflection of definition 2.3. On the fields put $(\hat\rho\phi)(x) := \phi(\rho x)$ and $(\hat\rho A)(b) := A(\rho b)$, the image bond taken with its orientation: $(\hat\rho A)_\alpha(x) = A_\alpha(\rho x)$ for $\alpha \ne \nu$ and $(\hat\rho A)_\nu(x) = -A_\nu(\rho x - \varepsilon e_\nu)$. Then $\hat\rho$ is a signed permutation of the variables attached to Ω and Ω^* (it preserves Lebesgue measure and maps Ω^*, Ω^{**} to themselves); the *crossing bonds* $B_0 := \{\langle x^-, x^- + \varepsilon e_\nu\rangle : \ x^-_\nu = c - \varepsilon/2\}$ are mapped to their reversals, $(\hat\rho A)(b) = -A(b)$ for $b \in B_0$. Let $\Omega_\pm := \{x \in \Omega : \ \pm(x_\nu - c) > 0\}$ (no site lies on the plane), B_+ the bonds with both ends in Ω_+, $\mathfrak{B}_+$ the Borel functions of $(\phi|_{\Omega_+}, A|_{B_+})$, and

$$\theta G := \overline{G \circ \hat\rho} \qquad (G \in \mathfrak{B}_+).$$

On the smeared fields, $F_{\Lambda_k}(f)\circ\hat\rho = F_{\Lambda_k}(\rho^* f)$ and $W^a_{\Lambda_k}(h)\circ\hat\rho = W^a_{\Lambda_k}(h\circ\rho)$. Indeed, $F(p)\circ\hat\rho = \sigma_p F(\rho p)$ with $\sigma_p = -1$ exactly when $\nu \in \{\alpha,\beta\}$ (the orientation of the image plaquette, whose base point is shifted back one layer), the centres satisfy $c(\rho p) = \rho c(p)$, and $(\rho^* f)_{c(\rho p)}(e_\alpha, e_\beta) = f_{c(p)}(\rho e_\alpha, \rho e_\beta) = \sigma_p f_{c(p)}(e_\alpha, e_\beta)$; reindexing $p \mapsto \rho p$ gives the first identity. The scalar case is a relabelling of sites, and the c-numbers $2\mathsf{W}_\varepsilon \mathsf{S}(h)$ and $a_k \mathsf{S}(h)$ are invariant because ρ permutes Λ_k ($2(x_\nu - c) \in \varepsilon\mathbb{Z}$). Consequently, for families ζ_i whose lattice observables $\Theta(\zeta_i)$ lie in $\mathfrak{B}_+$,

$$\theta(\Theta(\zeta_i))\,\Theta(\zeta_j) = \Theta(\theta\zeta_i^* \sqcup \zeta_j), \qquad m(\theta\zeta_i^* \sqcup \zeta_j) = \overline{m(\zeta_i)}\, m(\zeta_j) \tag{18}$$

for the scheme multiplier $m(\zeta) := e^{-\sum_j w_j a_k \mathsf{S}(h_j)}$ (a_k real), so that for any state $\langle\cdot\rangle$ on the box

$$\sum_{i,j} \bar\gamma_i \gamma_j\, m(\theta\zeta_i^* \sqcup \zeta_j)\langle\Theta(\theta\zeta_i^* \sqcup \zeta_j)\rangle = \sum_{i,j} \overline{(\gamma_i m(\zeta_i))}\,(\gamma_j m(\zeta_j))\,\big\langle\theta(\Theta(\zeta_i))\,\Theta(\zeta_j)\big\rangle. \tag{19}$$

Support. If the sources of ζ_i are supported in $\{x_\nu \geq c+\delta\}$ with $\delta > 0$, then for $\varepsilon_k < 2\delta$ every site carrying a sampled h and every corner of every plaquette carrying a sampled f (centre in the support, corners at $x_\nu \geq c+\delta-\varepsilon_k/2 > c$) lies in Ω_+, so $\Theta(\zeta_i) \in \mathfrak{B}_+$. The splitting $h = h^+ - h^-$ of definition 10.1 is taken with cutoffs supported in $\{x_\nu \geq c+\delta/2\}$, so that $h^\pm$ are supported in $\{x_\nu > c\}$ as well.

Definition 9.2 (the modified box state)**.** Let $\Omega \subset \Lambda_k$ be a lattice box, $F(p) = \varepsilon^{-1}\sum_{b\in\partial p} A(b)$, $\Omega(y)$ the indicator of Ω, and x^* the nearest neighbours of x in Λ_k. King's modified action [14, (3.38)] on Ω is

$$\begin{aligned} S^\varepsilon_\Omega(A,\phi) = {}& \tfrac12 \sum_{p\in\Omega^{**}} \varepsilon^d |F(p)|^2 + \tfrac12\mu_0^2 \sum_{b\in\Omega^*} \varepsilon^d |A(b)|^2 + \tfrac12 \sum_{x\in\Omega} \varepsilon^d \Big| \sum_{y\in x^*} \varepsilon^{-1}\Omega(y) A(\langle x,y\rangle)\Big|^2 \\ & + \tfrac12 \sum_{\langle x,y\rangle} \varepsilon^{d-2}\big[(\Omega r)(x)^2 + (\Omega r)(y)^2 - 2(\Omega r)(x)(\Omega r)(y)\cos(\vartheta_x - e\varepsilon A(\langle x,y\rangle) - \vartheta_y)\big] \\ & + \sum_{x\in\Omega} \varepsilon^d\big[\tfrac12 m^2 r_x^2 + \lambda r_x^4 + \tfrac12 \delta m_\varepsilon^2\, r_x^2\big], \end{aligned}$$

with the *constant* free-boundary counterterm δm_ε^2 of definition 2.2 at every site, and with the free Wick constant W_ε in the Wick observables. (King prints a site-dependent counterterm in the last term; it is not used here, and the statements below are made for the constant one.) The photon variables are the internal bonds Ω^* only. The matter part is exactly the Dirichlet matter of the Neumann–Dirichlet state on $X = \Omega$: cosines on the internal bonds, and the diagonal $\frac12\varepsilon^{d-2} r_x^2$ for every bond leaving Ω. The *modified box state* $\langle\cdot\rangle^{(3.38)}_{k,\Omega}$ is the normalised measure $e^{-S^\varepsilon_\Omega}\prod_{x\in\Omega} d^2\phi_x \prod_{b\in\Omega^*} dA_b$; it is a BFS measure (the Gaussian is non-degenerate since $\mu_0 > 0$, the single-site measures have all exponential moments since $\lambda > 0$). We write $Z^{a,(3.38)}_{k,\Omega}(\zeta) := m(\zeta)\langle\Theta(\zeta)\rangle^{(3.38)}_{k,\Omega}$ with the same observables and multiplier as in definition 2.3.

The third term is the graph divergence of (Ω,Ω^*): with $d_0 : \mathbb{R}^\Omega \to \mathbb{R}^{\Omega^*}$, $(d_0\chi)(\langle x, x+\varepsilon e_\alpha\rangle) = \varepsilon^{-1}(\chi(x+\varepsilon e_\alpha) - \chi(x))$, one has $\sum_{y\in x^*}\Omega(y)A(\langle x,y\rangle) = -\varepsilon(d_0^T A)(x)$ (only internal bonds contribute). With $d_1 A := (\varepsilon F(p))_{p\in\Omega^{**}}$, the *unitary form* on the box and the modified form are

$$Q^u_\Omega(A) := \tfrac12 \sum_{p\in\Omega^{**}} \varepsilon^d |F(p)|^2 + \tfrac12\mu_0^2 \sum_{b\in\Omega^*} \varepsilon^d |A(b)|^2, \qquad Q^{(3.38)}_\Omega = Q^u_\Omega + \tfrac12\varepsilon^d \|d_0^T A\|^2,$$

i.e. the modified form is the unitary form plus a gauge-fixing term depending on A only through $d_0^T A$. The modified action is invariant under $\hat\rho$ for every mid-plane c with $\rho\Omega = \Omega$: every term is written through $A(\langle x,y\rangle)$ and nearest-neighbour pairs, which ρ permutes, $(d_0^T\hat\rho A)(x) = (d_0^T A)(\rho x)$, and δm_ε^2 is site independent.

The boxes. Fix $c \in D$, $k \geq k_0(c)$, $J \geq J_0$ such that every source of the enlarged family of definition 10.1 and its reflection is supported at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_{J_0}$, an integer $n \geq 1$, and put $a_k := c + \varepsilon_k/2 \in \varepsilon_k\mathbb{Z}$ and

$$\Omega_{J,k} := (C_{J+n} + a_k e_\nu) \cap \Lambda_k, \qquad Mn \;\geq\; |c| + 2 + R(\mu_0), \qquad R(\mu_0) := \frac{\pi\sqrt{3}}{2\mu_0}. \tag{20}$$

$C_{J+n}\cap\Lambda_k$ is symmetric in direction ν about $-\varepsilon_k/2$ (sites $x_\nu \in \{-M(J+n),\dots,M(J+n)-\varepsilon_k\}$, and $M(J+n) \in \varepsilon_k\mathbb{Z}$), so $\Omega_{J,k}$ is a lattice box symmetric about $\{x_\nu = c\}$ with an even number $2M(J+n)5^k$ of layers. Under the margin condition of (20), for every k: (i) C_J plus one layer is contained in $\Omega_{J,k}$ (this needs $Mn \geq |c| + \frac32\varepsilon_k$); (ii) C_{J+2n} contains every site at ℓ^∞-distance $\leq R(\mu_0) + \varepsilon_k$ from $\Omega_{J,k}$ (this needs $Mn \geq |c| + R(\mu_0) + \frac32\varepsilon_k$). We fix such an $n = n(\mu_0, c)$, independent of k and J. The two margins are exactly the hypotheses of lemmas 9.9 and 9.10 below; by lemma 4.6 they serve King's convention as they stand.

9.2. **Reflection positivity of the modified box.**

Lemma 9.3 (lattice reflection positivity of the modified box)**.** *Let $\mu_0 > 0$, c a mid-plane and Ω a lattice box with $\rho\Omega = \Omega$, $\rho = \rho_{\nu,c}$.*

(a) *For every gauge-invariant $G \in \mathfrak{B}_+$ that is bounded, or of the form $\Theta(\zeta)$ with sources supported in Ω_+, or more generally with $\langle |G|^2\rangle^{(3.38)}_{k,\Omega} < \infty$: $\langle \theta(G)G\rangle^{(3.38)}_{k,\Omega} \geq 0$.*

(b) *$(G, G') \mapsto \langle \theta(G')G\rangle^{(3.38)}_{k,\Omega}$ is a Hermitian positive semidefinite form on such G; hence $\sum_{i,j} \bar\gamma_i\gamma_j \langle \theta(G_i)G_j\rangle^{(3.38)}_{k,\Omega} \geq 0$, and by (19) and the support clause of definition 9.1, $\sum_{i,j} \bar\gamma_i\gamma_j Z^{a,(3.38)}_{k,\Omega}(\theta\zeta_i^* \sqcup \zeta_j) \geq 0$ for families ζ_i with sources supported in $\{x_\nu \geq c + \delta\}$, $\varepsilon_k < 2\delta$.*

Proof. Step 1: gauge-invariant expectations of the modified form equal those of the unitary form. The graph (Ω, Ω^*) is connected, $\ker d_0$ consists of the constants, $\mathbb{R}^{\Omega^*} = \operatorname{ran} d_0 \oplus \ker d_0^T$ orthogonally, and $d_1 d_0 = 0$. Write $A = A^T + d_0\chi$ with $d_0^T A^T = 0$ and $\chi \perp 1$; $(A^T, \chi) \mapsto A$ is a linear bijection with constant Jacobian. Then $Q^u_\Omega(A) = Q^u_\Omega(A^T) + \frac{\mu_0^2\varepsilon^d}{2}\|d_0\chi\|^2$ (orthogonality and $d_1 d_0 = 0$) and $Q^{(3.38)}_\Omega(A) = Q^u_\Omega(A^T) + q(\chi)$ with $q(\chi) = \frac{\mu_0^2\varepsilon^d}{2}\|d_0\chi\|^2 + \frac{\varepsilon^d}{2}\|d_0^T d_0\chi\|^2$; both χ-forms are positive definite on $1^\perp$ because $\mu_0 > 0$. A gauge transformation is $A \mapsto A + d_0\chi$ together with $\vartheta \mapsto \vartheta - e\chi$: the hopping phase $\vartheta_x - e\varepsilon A_b - e(\chi(y) - \chi(x)) - \vartheta_y$ equals $(\vartheta_x + e\chi_x) - e\varepsilon A_b - (\vartheta_y + e\chi_y)$ and the Dirichlet diagonals carry no A, so for gauge-invariant G the matter integral $\Psi(A) := \int \prod_x d^2\phi_x\, (Ge^{-S_m})(A, \phi)$ satisfies $\Psi(A^T + d_0\chi) = \Psi(A^T)$ (rotate ϕ site by site; $\prod d^2\phi$ is invariant). Hence $\int dA\, e^{-Q^{(3.38)}_\Omega}\Psi = \int dA^T e^{-Q^u_\Omega(A^T)}\Psi(A^T) \int d\chi\, e^{-q(\chi)}$, and the same with Q^u_Ω in place of $Q^{(3.38)}_\Omega$ and a different finite χ-factor; the factors cancel in the normalised expectation. Fubini applies, since $|G| \leq Ce^{a\sum_b |A(b)|}e^{C\sum_x r_x^2}$ for $G = \Theta(\zeta)$ (trivially for bounded G) is dominated by the Gaussian and by $e^{-\lambda\varepsilon^d r^4}$. This is the mechanism of [6, Thm. 5.1] and of [24, Lemma 7.7], carried to the box graph.

Step 2: the unitary box measure is reflection positive on all of $\mathfrak{B}_+$. Let $S^u_\Omega := Q^u_\Omega + S_m$, write $\omega = (\omega_+, \omega_-, a)$ with $a = (a_b)_{b\in B_0}$ the crossing-bond variables, and put $\mathsf{M} := \mu_0^2\varepsilon^d$. The terms of S^u_Ω are of four kinds. Single-site terms (local matter and Dirichlet diagonals), hopping cosines on bonds of $B_\pm$, masses of bonds of $B_\pm$, and plaquettes with all corners in Ω_+ or all in Ω_- involve one side only; by $\hat\rho$-invariance the minus-side terms are the $\hat\rho$-transport of the plus-side terms. The crossing masses are $\frac{\mathsf{M}}{2}a_b^2$, $b \in B_0$. A crossing plaquette $p = (x^-; \alpha, \nu)$, $\alpha \neq \nu$, is present iff $x^-, x^- + \varepsilon e_\alpha \in \Omega$ (then all four corners are in Ω by symmetry); with $x^+ := x^- + \varepsilon e_\nu$, $b = \langle x^-, x^+\rangle$ and $b' = \langle x^- + \varepsilon e_\alpha, x^+ + \varepsilon e_\alpha\rangle$ it reads $\varepsilon F(p) = \pm[A_\alpha(x^-) + a_{b'} - A_\alpha(x^+) - a_b]$, the overall sign immaterial after squaring. A crossing hopping term is $-\varepsilon^{d-2} r_{x^-} r_{x^+} \cos(\vartheta_{x^-} - e\varepsilon a_b - \vartheta_{x^+})$, $b \in B_0$.

Split each crossing variable: $\int_{\mathbb{R}} g(a_b) e^{-\frac{\mathsf{M}}{2}a_b^2} da_b = \sqrt{2\pi/\mathsf{M}}\, \mathbb{E}\, g(\alpha_b - \alpha'_b)$ with α_b, α'_b independent $N(0, \frac{1}{2\mathsf{M}})$, whose difference is $N(0, \frac{1}{\mathsf{M}})$. Put $\xi := (\omega_+, \alpha)$ and $\xi' := (\omega_-^\rho, \alpha')$, where ω_-^ρ is the $\hat\rho$-image of the minus data transported to the plus side ($(\omega_-^\rho)_\alpha(x^+) = A_\alpha(x^-)$ for the tangential bonds, $\phi^\rho(x^+) = \phi(x^-)$, the ν-bonds of B_- carried with the sign of definition 9.1); by $\hat\rho$-invariance the minus-side one-sided terms are the plus-side terms evaluated on ω_-^ρ, and the a priori measures of ξ and ξ' coincide. For a crossing plaquette put $X_p(\xi) := A_\alpha(x^+) - \alpha_{b'} + \alpha_b$, a function of plus variables and α (note $A_\alpha(x^+) \in B_+$ since $x^+, x^+ + \varepsilon e_\alpha \in \Omega_+$), and the same function on ξ', $X_p(\xi') = A_\alpha(x^-) - \alpha'_{b'} + \alpha'_b$; then $\varepsilon F(p) = \pm[X_p(\xi') - X_p(\xi)]$ and

$$\tfrac{\varepsilon^d}{2}F(p)^2 = \tfrac{\varepsilon^{d-2}}{2}X_p(\xi)^2 + \tfrac{\varepsilon^{d-2}}{2}X_p(\xi')^2 - \varepsilon^{d-2}X_p(\xi)X_p(\xi').$$

For a crossing hopping term put $\psi_b(\xi) := \phi(x^+)e^{\mathrm{i}e\varepsilon\alpha_b}$, $\psi_b(\xi') = \phi(x^-)e^{\mathrm{i}e\varepsilon\alpha'_b}$; then $r_{x^-}r_{x^+}\cos(\vartheta_{x^-} - e\varepsilon(\alpha_b - \alpha'_b) - \vartheta_{x^+}) = \operatorname{Re}[\psi_b(\xi')\overline{\psi_b(\xi)}] = \sum_{i=1,2}\psi_b^i(\xi)\psi_b^i(\xi')$. Hence, with S' the action minus the crossing masses,

$$e^{-S'} = \mathcal{G}(\xi)\,\mathcal{G}(\xi')\,\exp\Big(\sum_\iota \gamma_\iota Y_\iota(\xi)Y_\iota(\xi')\Big), \qquad \mathcal{G}(\xi) := e^{-S_+(\xi) - \frac{\varepsilon^{d-2}}{2}\sum_p X_p(\xi)^2},$$

where S_+ collects the one-sided plus terms and $\{(\gamma_\iota, Y_\iota)\} = \{(\varepsilon^{d-2}, X_p)\}_p \cup \{(\varepsilon^{d-2}, \psi_b^i)\}_{b,i}$: all $\gamma_\iota > 0$, all Y_ι real. For $G \in \mathfrak{B}_+$ we have $G(\omega) = G(\xi)$ and $\theta G(\omega) = \overline{G(\xi')}$, the same function on the transported data (G depends neither on a nor on α, α'). Expanding the exponential,

$$Z\,\langle \theta(G)G\rangle^u_{k,\Omega} = K \sum_{n\in\mathbb{N}^I} \prod_\iota \frac{\gamma_\iota^{n_\iota}}{n_\iota!} \Big| \int d\xi\, G(\xi)\mathcal{G}(\xi) \prod_\iota Y_\iota(\xi)^{n_\iota} \Big|^2 \;\geq 0, \qquad K > 0,$$

the two integrals of each term being complex conjugates because measures and functions coincide on the two copies. The interchange of sum and integral is justified by $\sum_n \prod_\iota \gamma_\iota^{n_\iota} |Y_\iota Y_\iota'|^{n_\iota}/n_\iota! \leq \exp(\frac{1}{2}\sum_\iota \gamma_\iota(Y_\iota^2 + Y_\iota'^2))$: the plaquette parts cancel against $\mathcal{G}$ exactly, $|\psi_b| = r_{x^+}$, and $\int |G| e^{-S_+(\xi) + \frac{\varepsilon^{d-2}}{2}\sum_b r_{x^+}^2} d\xi <$ ∞ by $\lambda > 0$ (a quartic term at every plus site), $\mu_0 > 0$ (a mass on every B_+ bond and the α-Gaussians) and the growth bound on G. This is the mechanism of [24, Lemma 7.8(ii)] with one plane in place of a pair and the Dirichlet diagonals as single-site terms; the classical form is the positive cross-term criterion of [9, 8]. The Hermitian property $\overline{\langle\theta(G')G\rangle} = \langle (G'\circ\hat\rho)\bar G\rangle = \langle G'\,\overline{G\circ\hat\rho}\rangle = \langle \theta(G)G'\rangle$ follows from the $\hat\rho$-invariance of the measure.

Step 3: combination and L^2 extension. For gauge-invariant $G \in \mathfrak{B}_+$, bounded or equal to $\Theta(\zeta)$, the product $\theta(G)G$ is gauge invariant ($\hat\rho$ maps gauge transformations to gauge transformations: $\hat\rho\, d_0\chi = d_0(\chi\circ\rho)$), so $\langle\theta(G)G\rangle^{(3.38)}_{k,\Omega} = \langle\theta(G)G\rangle^u_{k,\Omega} \geq 0$ by Steps 1–2. For gauge-invariant $G \in \mathfrak{B}_+$ with $\langle |G|^2\rangle < \infty$ (for instance polynomials in the plus-side smeared fields, proposition 10.10), $G_n := G\,\mathbf{1}\{|G| \leq n\}$ is bounded, gauge invariant and in $\mathfrak{B}_+$, so $\langle\theta(G_n)G_n\rangle \geq 0$; moreover $|\theta(G_n)G_n| \leq |\theta G||G|$ and $\langle|\theta G||G|\rangle \leq \langle|G|^2\rangle^{1/2}\langle|\theta G|^2\rangle^{1/2} = \langle|G|^2\rangle$ by Cauchy–Schwarz and $\hat\rho$-invariance, so dominated convergence gives (a). The modified measure is itself $\hat\rho$-invariant (definition 9.2), so the computation of the Hermitian property in Step 2 holds verbatim for it on L^2; the form is sesquilinear, Hermitian and nonnegative on the diagonal, hence positive semidefinite (apply (a) to $G = \sum_i \gamma_i G_i$). With (19) and the support clause this is (b). □

Remark 9.4 (on King's Proposition 3.3). Lemma 9.3 is the statement of [14, Prop. 3.3], which King derives from [6, Cor. 5.4]. That corollary (with [6, Lemma 5.2, Thm. 5.3]) concerns the lattice actions of [6, §5], whose photon Gaussian is translation invariant with a reflection-positive boundary term; King's finite-box Gaussian with its masked divergence is not one of them, the obstruction being the box geometry of the gauge-fixing term. So his Proposition 3.3 is true, and it is proved here rather than derived from [6, Cor. 5.4]. The gauge-fixing term $\|d_0^T A\|^2$ is not itself compatible with the mechanism of Step 2 (at a site next to the plane it produces cross terms $Y(\xi)Z(\xi') + Z(\xi)Y(\xi')$ of indefinite sign); on the torus it drops out for gauge-invariant observables, on King's cube it does not reduce to a divergence at the boundary (lemma 9.5), and this is why the photon form has to be changed before reflecting (remark 9.7).

9.3. **The same box: the boundary identity and the reversed inequality.** In this subsection we use lattice units, $\varepsilon = 1$ (every term of the three photon forms carries the same weight $\frac{1}{2}\varepsilon^{d-2}$), and write $\mu := \mu_0\varepsilon$. For a lattice box Ω and $A \in \mathbb{R}^{\Omega^*}$ let $K^{\mathrm{int}}_\Omega(A) := \sum_\mu \sum_\nu \sum |A_\mu(x + e_\nu) - A_\mu(x)|^2$, the sum over pairs with both bonds $(x,\mu), (x+e_\nu,\mu)$ in Ω^* (the componentwise Neumann gradient form on the internal bonds), and $N_\partial(A) := \sum |A_b|^2$, the sum over the normal crossing pairs: for $b = (x,\mu) \in \Omega^*$ one term for each of $(x \pm e_\mu, \mu)$ that is not in Ω^*. Split $N_\partial = N_\partial^{\mathrm{up}} + N_\partial^{\mathrm{low}}$ according to whether the missing partner is $(x + e_\mu, \mu)$ (i.e. $x + 2e_\mu \notin \Omega$) or $(x - e_\mu, \mu)$ (i.e. $x - e_\mu \notin \Omega$).

Lemma 9.5 (the boundary identity)**.** *For every lattice box Ω and every $A \in \mathbb{R}^{\Omega^*}$,*

$$\|d_1 A\|^2_{\Omega^{**}} + \|d_0^T A\|^2 = K^{\mathrm{int}}_\Omega(A) + N_\partial(A).$$

Consequently $Q^{(3.38)}_\Omega = \frac{1}{2}\big[K^{\mathrm{int}}_\Omega + N_\partial + \mu^2\|\cdot\|^2_{\Omega^*}\big]$: the modified photon form is the internal Neumann–Feynman form plus the positive boundary term N_∂. (By the proof of lemma 9.6(a), the marginal of the Neumann–Dirichlet Gaussian of Ω on the internal bonds has the precision $M^{(3.38)} - N_\partial^{\mathrm{low}} - P_{12}P_{22}^{-1}P_{21}$: the modified form minus the squares of the near-normal bonds and minus the Schur term of the far-face variables.)

Proof. Extend A by 0 to a 1-form a on $\mathbb{Z}^d$. For finitely supported 1-forms, $\sum_p |da(p)|^2 + \sum_x |d^*a(x)|^2 = \sum_\mu \sum_\nu \sum_x |a_\mu(x+e_\nu) - a_\mu(x)|^2$ (the Hodge Laplacian on $\mathbb{Z}^d$ is the componentwise Laplacian; summation by parts). On the left, $d^*a(x)$ is the masked divergence for $x \in \Omega$ and 0 outside, since no internal bond has an end outside Ω; a plaquette $p \notin \Omega^{**}$ meeting Ω^* has exactly one internal bond b (two parallel internal edges put all four corners in Ω; two adjacent ones put three corners in Ω, and a product of intervals then contains the fourth), so $|da(p)|^2 = |A_b|^2$. On the right, pairs with both bonds internal give K_Ω^{int}; a pair $(x,\mu),(x+e_\nu,\mu)$ with exactly one internal bond contributes $|A_b|^2$; for $\nu \neq \mu$ the partner is non-internal iff $x + e_\nu \notin \Omega$ (as $x, x+e_\mu \in \Omega$, and in a box $x + e_\nu \in \Omega \iff x + e_\mu + e_\nu \in \Omega$), i.e. iff the plaquette spanned by b and e_ν lies outside Ω^{**}; these pairs are in bijection with the plaquettes counted on the left and cancel; for $\nu = \mu$ they are the terms of N_∂. □

Lemma 9.6 (the same box: the second inequality of King's (3.42) is reversed)**.** *Let Ω be a lattice box, $\langle\cdot\rangle_\Omega^{\mathrm{ND}}$ the Neumann–Dirichlet state on $X = \Omega$ (variables $A_\mu(x)$, $x \in \Omega$, all outgoing bonds; photon form $Q_\Omega^{\mathrm{ND}} = \frac{1}{2}[K_\Omega + \mu^2\|\cdot\|^2]$ of definition 2.1) and $\langle\cdot\rangle_\Omega^{(3.38)}$ the modified box state of definition 9.2 (variables Ω^*); both carry the same Dirichlet matter on Ω (constant δm_ε^2, free Wick constant), coupled to A through the internal bonds only. Let $\Sigma_\Omega^{\mathrm{ND}}$ and Σ_Ω^u be the covariances of $F = (d_1 A(p))_{p\in\Omega^{**}}$ under the two photon Gaussians. Then:*

(a) $\Sigma_\Omega^u = \Sigma_\Omega^{(3.38)} \leq \Sigma_\Omega^{\mathrm{ND}}$ *on* $\mathbb{R}^{\Omega^{**}}$*;*

(b) *for every gauge-invariant $Y \in \mathcal{I}$ with sources in Ω and every $e \geq 0$: $\langle Y\rangle_\Omega^{\mathrm{ND}} \leq \langle Y\rangle_\Omega^{(3.38)}$, and $\langle X\rangle_\Omega^{\mathrm{ND}} \geq \langle X\rangle_\Omega^{(3.38)}$ for $X \in \mathcal{D}$. This is the reverse of the second inequality of* [14, (3.42)]*, whose middle term is the modified box state on C_{J+1}; it holds weakly, for the modified state with the constant δm_ε^2 (nothing is asserted for King's printed site-dependent counterterm);*

(c) *King's inequality fails strictly in the Gaussian sector: at $e = 0$ and $Y = e^{\mathrm{i}t\langle v,F\rangle}$ it reads $v^T\Sigma_\Omega^{\mathrm{ND}}v \leq v^T\Sigma_\Omega^u v$, while $\Delta_\Omega := \Sigma_\Omega^{\mathrm{ND}} - \Sigma_\Omega^u \geq 0$ is nonzero. In exact rational arithmetic the inertia (negative, zero, positive) of Δ_Ω is $(0,1,8)$ on the 4×4 box ($d=2$) for $\mu^2 \in \{\frac{1}{20},\frac{1}{4},1\}$ and $(0,3,13)$ on the $4\times2\times2$ box ($d=3$) for $\mu^2 \in \{\frac{1}{20},\frac{1}{4}\}$; single-plaquette certificates are $\Delta_{pp} = \mathrm{Var}^{\mathrm{ND}} F(p) - \mathrm{Var}^{(3.38)} F(p) = \frac{14285248}{44314767}$ at $p = ((0,0);0,1)$ on 4×4, $\mu^2 = \frac{1}{4}$, and $\frac{121897601600}{395580222961}$ at $p = ((0,0,0);1,2)$ on $4\times2\times2$, $\mu^2 = \frac{1}{20}$.*

Proof. (a) Write the Neumann–Dirichlet precision $P := K_\Omega + \mu^2$ in blocks, the internal variables Ω^* first and the outgoing boundary variables second. Its internal block is $P_{11} = K_\Omega^{\mathrm{int}} + N_\partial^{\mathrm{up}} + \mu^2$: pairs of K_Ω with both bonds internal give K_Ω^{int}; a pair $(x,\mu),(x+e_\nu,\mu)$ with exactly one internal bond is impossible for $\nu \neq \mu$ (box convexity, as in lemma 9.5) and for $\nu = \mu$ is (x,μ) internal with $(x+e_\mu,\mu)$ outgoing, i.e. $x + 2e_\mu \notin \Omega$ (the other assignment would need $x + e_\mu \notin \Omega$ and $x + e_\mu \in \Omega$ at once); its diagonal contribution to the internal block is $|A_\mu(x)|^2$, which is N_∂^{up}. Hence, by lemma 9.5, $M^{(3.38)} := K_\Omega^{\mathrm{int}} + N_\partial + \mu^2 = P_{11} + N_\partial^{\mathrm{low}} \geq P_{11}$. The marginal covariance of the internal variables under the Neumann–Dirichlet Gaussian is the inverse of the Schur complement, $(P^{-1})_{11} = (P_{11} - P_{12}P_{22}^{-1}P_{21})^{-1} \geq P_{11}^{-1} \geq (M^{(3.38)})^{-1}$, by operator monotonicity of the inverse on positive definite matrices and $P_{12}P_{22}^{-1}P_{21} \geq 0$. Since $F = d_1 A$ on Ω^{**} uses internal bonds only, $\Sigma_\Omega^{\mathrm{ND}} = D_1(P^{-1})_{11}D_1^T \geq D_1(M^{(3.38)})^{-1}D_1^T = \Sigma_\Omega^{(3.38)}$, and $\Sigma_\Omega^{(3.38)} = \Sigma_\Omega^u$ by Step 0 in the proof of lemma 9.10 (the field strength depends only on the $\ker d_0^T$-component, whose law is the same under both Gaussians, Step 1 of lemma 9.3). (b) The mechanism of the second part of lemma 9.10 with $B = \Omega$ and no hopping removal: both states have the Dirichlet matter of Ω coupled to Ω^*, and for gauge-invariant Y with sources in Ω the expectation depends on the photon law only through the centred Gaussian law of F on $\operatorname{ran} d_1$; the two F-laws are ordered by (a); realise them, regularise, apply lemma 2.7(a) (resp. (d)) by interpolation and let the regularisation go to zero exactly as there. The larger covariance has the smaller $\mathcal{I}$-expectation, so $\langle Y\rangle_\Omega^{\mathrm{ND}} \leq \langle Y\rangle_\Omega^{(3.38)}$. (c) At $e = 0$ the matter decouples from A and is identical in the two states, so for $Y = e^{\mathrm{i}t\langle v,F\rangle}$ both sides equal $\exp(-\frac{t^2}{2}v^T\Sigma v)$ with the respective Σ; King's $\leq$ becomes $v^T\Sigma_\Omega^{\mathrm{ND}}v \leq v^T\Sigma_\Omega^u v$, which is false for every v with $v^T\Delta_\Omega v > 0$, for instance $v = e_p$

at the plaquettes listed. The inertias and the certificates are obtained from the two covariance matrices in exact rational arithmetic (symmetric elimination with diagonal pivoting over $\mathbb{Q}$). □

Scope. (i) King's passage from the box states to the cube limit on [14, p. 336] needs the sandwich for the box functionals with purely imaginary curvature coefficients, i.e. for the whole class $\mathcal{I}$; (c) refutes the second inequality on that class exactly. (ii) For the scalar observable $\Theta(0, 2h_+)$ of the printed display, (b) shows that the inequality can hold only with equality; at $e = 0$ it does (identical decoupled matter), and at $e > 0$ a strict failure is to be expected (the Wick square couples to F at order e^2) but is not certified here. Both clauses are for the modified state with the constant δm_ε^2: if King's site-dependent counterterm differs from the free-boundary number, the two matter measures differ, the mechanism of (b) does not apply, and neither clause is proved for his printed objects; (c) alone is independent of the counterterm, the matter decoupling from A at $e = 0$.

Remark 9.7 (what is false in the printed route). (i) King's Neumann–Dirichlet cube state is not reflection symmetric about any plane: the site set $C_J \cap \Lambda_k$ is symmetric in direction ν about $x_\nu = -\varepsilon_k/2$, the set of ν-bonds about $x_\nu = 0$, and no reflection maps the variable set to itself (lemma 2.13 accordingly obtains the reflection symmetry of the limit from a cut sandwich, not from a lattice symmetry). (ii) The natural symmetrisation, the Feynman gradient form on the internal bonds $K_\Omega^{\mathrm{int}} + \mu^2$ (the modified form without N_∂; reflection symmetric), is *not* reflection positive on gauge-invariant functionals. This is exact at $e = 0$: on the 4×4 box at $\mu^2 = \frac{1}{20}$ the reflection kernel $R_{pq} = \sigma_p \Sigma[\rho p, q]$ of the plus-side plaquettes is negative definite (inertia $(3, 0, 0)$), and on $4 \times 2 \times 2$ at $\mu^2 = \frac{1}{20}$ it has inertia $(2, 2, 2)$ with $R_{pp} = \mathrm{Cov}(\theta F(p), F(p)) < 0$ for $p = ((2, 0, 0); 0, 1)$. Indeed, at $e = 0$ the matter decouples, and the gauge-invariant plus-side functionals $G_1 = e^{\mathrm{i}t\langle v, F_+\rangle}$, $G_2 = 1$ give the matrix $(\langle \theta(G_i) G_j\rangle)$ with determinant $e^{-t^2 v^T \Sigma_{++} v}(e^{t^2 v^T R v} - 1) < 0$ for every $t > 0$ when $v^T R v < 0$, since $\langle \theta(G_1) G_1\rangle = \exp(-t^2 v^T \Sigma_{++} v + t^2 v^T R v)$ and $\langle \theta(G_1)\rangle = \langle G_1\rangle = \exp(-\frac{1}{2} t^2 v^T \Sigma_{++} v)$ by the reflection symmetry $\Sigma_{--} = \Sigma_{++}$. By continuity at fixed spacing the symmetrised cube state is not reflection positive for small $e > 0$ either. A proof must work at every coupling, so reflection positivity of King's own cube form, or of this symmetrisation, at fixed spacing is not available as a route: any route through cube states at fixed spacing has to change the photon form, as the modified box does (a reflection-symmetric, reflection-positive modification whose values converge to the same limit, theorem 10.6). Nothing is claimed about the symmetrised form at a given larger e or about other proofs; the route taken here needs the comparison of lemmas 9.9 and 9.10, and its reflecting plane must be a lattice mid-plane for the half-bond mechanism of lemma 9.3 (King's cubes have an even number of layers). (iii) Accordingly, the second inequality of [14, (3.42)] is false as printed in the following sense: false for the class of observables that the argument on [14, p. 336] needs (exactly, lemma 9.6(c)), and never strictly true for the modified state with the constant counterterm (lemma 9.6(b)). Its valid first half is lemma 9.9; the invalid half is replaced by lemma 9.10 with the margin $R(\mu_0)$ of (20), $n(\mu_0, c)$ cube steps instead of one.

Remark 9.8 (the box form is the larger one). King describes the modified box action as obtained from the cube action by omitting some terms. By lemma 9.5 the relation is the reverse: on one box the modified photon form is the internal Neumann–Dirichlet form P_{11} plus the positive term $N_\partial^{\mathrm{low}}$ and without the outgoing boundary variables, so the modified state is the one with the *larger* form, the smaller field-strength covariance, and the larger $\mathcal{I}$-expectations: $\langle Y\rangle_C^{\mathrm{ND}} \le \langle Y\rangle_C^{(3.38)}$ on the cone for every box C (lemma 9.6(b)), the direction that lemma 2.10 requires for consistency with lemmas 9.9 and 9.10. For King's convention the same comparison holds with Step 1 of lemma 4.2 prepended (remark 4.8): $\langle Y\rangle_{k,J}^{K} \le \langle Y\rangle_{k,J}^{(a)} \le \langle Y\rangle_{k,X_J}^{(3.38)}$ on $\mathcal{I}$.

9.4. **The sandwich.**

Lemma 9.9 (the lower half)**.** *Let $C \subset \Omega$ be lattice boxes with C plus one layer contained in Ω, so that the variables $A_\mu(x)$, $x \in C$, of the Neumann–Dirichlet state on C (the outgoing bonds) are internal bonds of Ω. Then*

(a) $Q_\Omega^{(3.38)} \ge Q_C^{\mathrm{ND}} \oplus \frac{\mu^2}{2} I_{\mathrm{rest}}$ *as quadratic forms on* $\mathbb{R}^{\Omega^*} = \mathbb{R}^{C\text{-variables}} \oplus \mathbb{R}^{\mathrm{rest}}$;

(b) *for sources such that every plaquette and every site at which they do not vanish lies in* C*,*

$$\langle Y\rangle_C^{\mathrm{ND}} \le \langle Y\rangle_\Omega^{(3.38)} \quad (Y\in\mathcal{I}), \qquad \langle X\rangle_C^{\mathrm{ND}} \ge \langle X\rangle_\Omega^{(3.38)} \quad (X\in\mathcal{D}),$$

and the same for Z^a *(equal positive multipliers, lemma 2.11(a)). The* $\mathcal{D}$ *half uses lemma 2.7(d); the* $\mathcal{I}$ *half only (a), (b), (e).*

Proof. (a) K_Ω^{int} contains every gradient pair of K_C (both base points in C, both bonds internal to Ω); the remaining pairs and N_∂ are nonnegative; the mass term splits. (b) Start from $\langle\cdot\rangle_\Omega^{(3.38)}$ and remove the hopping cosines across ∂C and all cosines not internal to C; by lemma 2.7(b),(d) $\langle Y\rangle$ does not increase and $\langle X\rangle$ does not decrease, every intermediate measure being a BFS measure. The matter is now the Dirichlet matter of C: the diagonal parts $\frac12 r^2$ of the cut bonds stay and are exactly the Dirichlet terms, as in the proof of lemma 2.10; it is coupled to A only through bonds internal to C, which are C-variables, and the matter of $\Omega\setminus C$ is decoupled and cancels (lemma 2.7(e)). Then pass from the form $Q_\Omega^{(3.38)}$ to the smaller form $Q_C^{\mathrm{ND}}\oplus\frac{\mu^2}{2}I_{\mathrm{rest}}$: by (a) the difference is positive semidefinite, a finite sum $\sum_i s_i|\ell_i(A)|^2$ with $s_i\ge0$, and adding each term does not decrease $\langle Y\rangle$ and does not increase $\langle X\rangle$ (lemma 2.7(a),(d)). So the state with the larger form, the modified one, has the larger $\langle Y\rangle$ and the smaller $\langle X\rangle$. The state with the smaller form factorises as $\langle\cdot\rangle_C^{\mathrm{ND}}$ times an independent Gaussian on the rest, which cancels. □

For King's convention the lower end is used in the form $\langle Y\rangle_{k,J}^K \le \langle Y\rangle_{k,J}^{(a)} \le \langle Y\rangle_\Omega^{(3.38)}$ on $\mathcal{I}$ (Step 1 of lemma 4.2, then lemma 9.9; remark 4.8), and the upper end needs no change (lemma 4.6).

Lemma 9.10 (the upper half, with a margin)**.** *Let* $\Omega\subset B$ *be lattice boxes at spacing* ε *such that* B *contains every site within* ℓ^∞*-distance* N *lattice steps of* Ω*, and let* $\mu=\mu_0\varepsilon$ *satisfy* $d(\pi/2N)^2\le\mu^2$*, i.e.*

$$N\varepsilon \;\ge\; \frac{\pi\sqrt{d}}{2\mu_0} \qquad \big(=R(\mu_0)=\tfrac{\pi\sqrt3}{2\mu_0}\ \text{in } d=3\big).$$

Let Σ_X *denote the covariance of* $(\varepsilon F(p))_{p\in\Omega^{**}}$ *under the photon Gaussian of the state on* X*. Then*

(a) $\Sigma_B^{\mathrm{ND}}\le\Sigma_\Omega^{(3.38)}=\Sigma_\Omega^u$ *on* $\mathbb{R}^{\Omega^{**}}$*;*

(b) *for sources supported in* Ω *as in lemma 9.9,*

$$\langle Y\rangle_\Omega^{(3.38)} \le \langle Y\rangle_B^{\mathrm{ND}} \quad (Y\in\mathcal{I}), \qquad \langle X\rangle_\Omega^{(3.38)} \ge \langle X\rangle_B^{\mathrm{ND}} \quad (X\in\mathcal{D}),$$

and the same for Z^a*. Only the* $\mathcal{I}$ *half is used in the sequel.*

Proof. (a), in lattice units. Step 0: $\Sigma_\Omega^{(3.38)}=\Sigma_\Omega^u$, because $d_1A=d_1A^T$ depends only on the $\ker d_0^T$-component, whose law $\propto e^{-Q_\Omega^u(A^T)}$ is the same under both Gaussians (Step 1 of lemma 9.3). Step 1 (variational identity): for a Gaussian with density $\propto e^{-\frac12 A^TM_XA}$, $\langle g,\Sigma_Xg\rangle=\mathrm{Var}\langle g,d_1A\rangle=\sup_A[2\langle g,d_1A\rangle-A^TM_XA]$, the supremum attained at $A=M_X^{-1}d_1^Tg$. So it suffices to show that for every A on the B-variables there is $A'\in\mathbb{R}^{\Omega^*}$ with $d_1A'=d_1A$ on Ω^{**} and $\|d_1A'\|_{\Omega^{**}}^2+\mu^2\|A'\|_{\Omega^*}^2\le K_B(A)+\mu^2\|A\|_B^2$; take $A':=A|_{\Omega^*}$ (the plaquettes of Ω^{**} use only Ω^*-bonds). Step 2 (cutoff): let $\psi(x)=\prod_\nu\phi_\nu(x_\nu)$ with $\phi_\nu=1$ on the ν-range of Ω and on the first layer outside, $\phi_\nu(t)=\cos(\pi(t-1)/2N)$ at the t-th layer outside for $1\le t\le N+1$ (so $\phi_\nu(N+1)=0$), and 0 beyond; then $0\le\psi\le1$, $\psi=1$ on Ω and its neighbours, $\mathrm{supp}\,\psi\subset B$. Put $a:=\psi A$ on $\mathbb{Z}^d$ (the bond $\langle x,x+e_\mu\rangle$ weighted by $\psi(x)$; $A:=0$ off the B-variables). Step 3 (Hodge on $\mathbb{Z}^d$): $\|d_1A\|_{\Omega^{**}}^2=\|d_1A'\|_{\Omega^{**}}^2\le\sum_p|da(p)|^2\le\sum_p|da|^2+\sum_x|d^*a|^2=\sum_\mu\sum_\nu\sum_x|a_\mu(x+e_\nu)-a_\mu(x)|^2$, since the bonds of plaquettes in Ω^{**} have base points in Ω, where $\psi=1$. Step 4 (discrete IMS identity): for finitely supported ψf and a direction e,

$$\sum_x|\psi(x+e)f(x+e)-\psi(x)f(x)|^2=\sum_x\psi(x)\psi(x+e)|f(x+e)-f(x)|^2+\sum_x\psi(x)\big(2\psi(x)-\psi(x+e)-\psi(x-e)\big)f(x)^2,$$

exactly (per bond the difference of the two sides is $\psi'(\psi'-\psi)f'^2+\psi(\psi-\psi')f^2$; collect by site). With $f=A_\mu$ and $e=e_\nu$: $\psi\psi'\le1$, and $\psi\psi'>0$ only for pairs with both base points in B, which

are gradient pairs of K_B; so the first sums, over μ,ν, total at most $K_B(A)$. The potential is $V(x)=\sum_\nu\big(\prod_{\kappa\neq\nu}\phi_\kappa^2\big)\phi_\nu(-\Delta\phi_\nu)(x_\nu)$. Step 5 (the profile): with $\vartheta:=\pi/2N$, $\phi(-\Delta\phi)=0$ on the range of Ω (all neighbours equal 1), $=1-\cos\vartheta\le\vartheta^2/2$ at $t=1$, $=2\cos^2(\vartheta(t-1))(1-\cos\vartheta)\le\vartheta^2$ for $2\le t\le N$ (using $\phi(N+1)=\cos(\vartheta N)=0$ at $t=N$), and $=0$ for $t\ge N+1$; hence $V=0$ on Ω and $V\le d\vartheta^2$ everywhere. Step 6: $\|d_1A'\|^2_{\Omega^{**}}\le K_B(A)+d\vartheta^2\sum_{x\in B\setminus\Omega}|A(x)|^2$; the variables with base point in $B\setminus\Omega$ and those of Ω^* (base point in Ω) are disjoint subsets of the B-variables, so $\|d_1A'\|^2_{\Omega^{**}}+\mu^2\|A'\|^2_{\Omega^*}\le K_B(A)+\mu^2\|A\|^2_B$ as soon as $d\vartheta^2\le\mu^2$.

(b). Start from $\langle\cdot\rangle_B^{\mathrm{ND}}$ and remove all hopping cosines touching $B\setminus\Omega$ (lemma 2.7(b): $\mathcal{I}$ does not increase; (d): $\mathcal{D}$ does not decrease). The matter is now the Dirichlet matter of Ω (the diagonals of the bonds leaving Ω kept; every such bond ends in B since $N\ge1$), coupled to A on Ω^*, and the sites of $B\setminus\Omega$ decouple. This matter is identical to that of the modified box state: the same cosines on Ω^*, the same diagonals $\frac12 r_x^2$ for every bond leaving Ω, the same single-site weights with the constant δm_ε^2 and the same Wick constant. For a gauge-invariant observable with sources in Ω the matter integral at fixed A, $\Psi_Y(A)=\int\prod_{x\in\Omega}d^2\phi_x\,Y\,e^{-S_m}$, is invariant under $A\mapsto A+d_0\chi$, $\chi\in\mathbb{R}^\Omega$ (as in Step 1 of lemma 9.3), hence a function of the class of $A|_{\Omega^*}$ modulo $\operatorname{ran}d_0=\ker d_1$ (a box has $H^1=0$: the cycle space of a grid graph is generated by its 4-cycles), hence a function $\tilde\Psi_Y(d_1A|_{\Omega^{**}})$ of the field strengths on Ω^{**}. So $\langle Y\rangle=\mathbb{E}[\tilde\Psi_Y(F)]/\mathbb{E}[\tilde\Psi_1(F)]$ depends on the photon law only through the centred Gaussian law of $F=(d_1A(p))_{p\in\Omega^{**}}$, with covariance Σ_B^{ND} on the Neumann–Dirichlet side after the removal and Σ_Ω^u on the modified side (Step 0). Both laws live on $\operatorname{ran}d_1$. Let $T:\mathbb{R}^{\Omega^{**}}\to\mathbb{R}^{\Omega^*}$ be a linear map with $d_1T=\mathrm{id}$ on $\operatorname{ran}d_1$, extended linearly to all of $\mathbb{R}^{\Omega^{**}}$ (any extension serves, both laws being supported on $\operatorname{ran}d_1$), and realise each law as the photon Gaussian of $A'':=TF\in\mathbb{R}^{\Omega^*}$, covariance $T\Sigma T^T$: then $d_1A''=F$ almost surely, the gauge-invariant expectations are reproduced, and $T\Sigma_B^{\mathrm{ND}}T^T\le T\Sigma_\Omega^uT^T$ by (a). These covariances are degenerate; regularise to $T\Sigma T^T+\delta I$, $\delta>0$, which are non-degenerate, so the regularised states are BFS measures, the ordering of the covariances is preserved, and the law of $F''=d_1A''$ is $N(0,\Sigma+\delta d_1d_1^T)$, still ordered. Lemma 2.7(a) by interpolation (the difference of the precision matrices is positive semidefinite by operator monotonicity of the inverse, hence a finite sum of terms $s|\ell(A)|^2$, $s\ge0$) gives $\langle Y\rangle_{\Sigma^u+\delta}\le\langle Y\rangle_{\Sigma^{\mathrm{ND}}+\delta}$ for $Y\in\mathcal{I}$, and the reverse for $\mathcal{D}$ by lemma 2.7(d). *The limit* $\delta\downarrow0$. For the $\mathcal{I}$ half, $\tilde\Psi_Y$ is bounded and continuous: $|\Psi_Y(A)|\le\int e^{r^2(g)}e^{\sum\varepsilon^{d-2}r_xr_y}\prod_x d\rho_x<\infty$ independently of A ($|e^{\mathrm{i}F}|=1$; the hopping bound $e^{\varepsilon^{d-2}r_xr_y}\le e^{\frac12\varepsilon^{d-2}(r_x^2+r_y^2)}$ cancels the diagonals and $\lambda>0$ does the rest), and so is $\tilde\Psi_1$, with $\tilde\Psi_1>0$; the Gaussians $N(0,\Sigma+\delta d_1d_1^T)$ converge weakly to $N(0,\Sigma)$, so numerator and denominator converge and the inequality passes to the limit. For the $\mathcal{D}$ half (not used below), $X=\Theta(f,-g)$ gives $\tilde\Psi_X(F)=e^{\langle v,F\rangle}\tilde\Psi_X^0(F)$ with v the plaquette vector of f and $0<\tilde\Psi_X^0\le e^{2\mathsf{W}_\varepsilon\mathsf{S}(g)}\tilde\Psi_1\le C$ bounded and continuous, so $\tilde\Psi_X$ is continuous but unbounded; under $N(0,\Sigma+\delta d_1d_1^T)$ one has $\mathbb{E}[e^{2\langle v,F''\rangle}]=e^{2v^T(\Sigma+\delta d_1d_1^T)v}\le e^{2v^T\Sigma v+2\|d_1^Tv\|^2}$ for $\delta\le1$, so $\{\tilde\Psi_X(F'')\}_{0<\delta\le1}$ is bounded in L^2, hence uniformly integrable, and weak convergence with uniform integrability and continuity gives $\mathbb{E}[\tilde\Psi_X(F'')]\to\mathbb{E}[\tilde\Psi_X(F)]$. Altogether $\langle Y\rangle_\Omega^{(3.38)}\le\langle Y\rangle^{\text{after removal}}\le\langle Y\rangle_B^{\mathrm{ND}}$, and the reverse for $\mathcal{D}$. □

Remark 9.11 (why a margin, and why a fixed photon mass). Part (a) of lemma 9.10 fails without a margin: for $B=\Omega$ the comparison is reversed (lemma 9.6), and with a single layer ($N=1$) the difference $\Sigma_\Omega^u-\Sigma_B^{\mathrm{ND}}$ has a negative eigenvalue already at moderate μ. No fixed margin works at $\mu_0=0$: there Σ_Ω^u is the projector onto exact 2-forms, while $\mathrm{Var}_B\langle g,dA\rangle>\|g\|^2$ for a Coulomb potential whose gradient leaks out of B. So lemma 9.10 is a statement at fixed $\mu_0>0$ with $R(\mu_0)\to\infty$ as $\mu_0\to0$; a limit of vanishing photon mass inherits reflection positivity, if it exists, as a pointwise limit of positive semidefinite matrices. On one box the three states are ordered on $\mathcal{I}$ as $\langle Y\rangle_\Omega^{\mathrm{ND}}\le\langle Y\rangle_\Omega^{(3.38)}\le\langle Y\rangle_B^{\mathrm{ND}}$ (lemmas 9.6 and 9.10), as the monotonicity of lemma 2.10 requires. The sandwich with the boxes (20) is assembled in the proof of theorem 10.6; for King's convention its lower end carries the kill step of remark 4.8 and its upper end is unchanged (lemma 4.6).

10. Reflection positivity II: the limit, OS3 and the polynomial form

The lattice statements of the previous section are now carried to the limits $k \to \infty$ and $J \to \infty$. The scheme of proof is the same for the curvature sector and for the Wick tube: uniform local bounds for the modified-box functionals at complex coefficients (lemma 10.3), Montel's theorem in k at fixed J, the two-sided sandwich of lemmas 9.9 and 9.10 on the cone as $J \to \infty$ with the identity theorem off it, and a translation for the planes that are not mid-planes. The sector-specific input is the convergence of the cube functionals themselves: the convergence at fixed cubes (definition 3.1; on King's convention theorem 3.2) in the curvature sector, and on the Wick tube the positive-leg bound of corollary 13.3 together with theorem 3.5 (proposition 10.5). The arbitrary-amplitude form of the Wick sector, which needs (W2), is the optional extension of theorem 23.8; the one lemma written for it here, lemma 10.2, is marked as such and is not used by the main theorem.

10.1. **The enlarged family.**

Definition 10.1 (the enlarged family, its tube and its bounded domain)**.** Fix $\nu \in \{1,2,3\}$, a mid-plane $c \in D$ (definition 9.1), and finite families $\zeta_1, \dots, \zeta_N$ of real sources $f_{il} \in C_c^\infty(\mathbb{R}^3; \Lambda^2)$, $h_{il} \in C_c^\infty(\mathbb{R}^3)$ supported in $\{x_\nu \geq c + \delta\}$ for some $\delta > 0$, with complex coefficients z_{il}, w_{il}. Split each scalar source as $h_{il} = h_{il}^+ - h_{il}^-$ with $h_{il}^- := \|h_{il}\|_\infty \chi_{il}$, where $\chi_{il} \in C_c^\infty$, $0 \leq \chi_{il} \leq 1$, $\chi_{il} = 1$ on $\operatorname{supp} h_{il}$ and $\operatorname{supp} \chi_{il} \subset \{x_\nu \geq c + \delta/2\}$, and $h_{il}^+ := h_{il} + h_{il}^- \geq 0$. The *enlarged family* is the family of real sources

$$\mathcal{F} := \{\rho^* f_{il}\} \cup \{f_{il}\} \cup \{h_{il}^\pm \circ \rho\} \cup \{h_{il}^\pm\}, \qquad \text{coefficient space } \mathbb{C}^m,\ m = m_F + m_h,$$

whose scalar members $\tilde{h}_j \geq 0$ are all nonnegative. For a real coefficient vector $\xi \in \mathbb{R}^m$ the total source is $(f_\xi, k_1(\xi) - k_2(\xi))$ with $k_1(\xi) := \sum_j (\xi_j)_+ \tilde{h}_j$, $k_2(\xi) := \sum_j (\xi_j)_- \tilde{h}_j$, both smooth and nonnegative ($x_\pm := \max(\pm x, 0)$). Let J_0 be the least J such that every member of $\mathcal{F}$ is supported at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_J$; for $J \geq J_0$, $n = n(\mu_0, c)$ and the boxes $\Omega_{J,k}$ of (20) put

$$\Phi_k(\zeta) := Z^{a,(3.38)}_{k,\Omega_{J,k}}(\zeta), \qquad \zeta \in \mathbb{C}^m,$$

an entire function (a finite-dimensional integral with Gaussian decay in A and $e^{-\lambda\varepsilon^3 r^4}$ decay in r). The *matrix points* $\zeta_{ij}^* \in \mathbb{C}^m$ carry $\bar{z}_{il}$ on $\rho^* f_{il}$, z_{jl} on f_{jl}, $\pm\bar{w}_{il}$ on $h_{il}^\pm \circ \rho$, $\pm w_{jl}$ on $h_{jl}^\pm$ and 0 elsewhere; their total source is that of $\theta\zeta_i^* \sqcup \zeta_j$, so $\Phi_k(\zeta_{ij}^*)$ is the (i,j) entry of lemma 9.3(b). Since reflected and unreflected members have disjoint supports, $\|h^+\|_\infty \leq 2\|h\|_\infty$ and $\|h^-\|_\infty = \|h\|_\infty$,

$$\|k_1(\operatorname{Re}\zeta_{ij}^*)\|_\infty \ \leq\ 2\max\big(\mathsf{a}(\zeta_i), \mathsf{a}(\zeta_j)\big). \tag{21}$$

(i) The *Wick tube* of $\mathcal{F}$ is $T_\mathcal{F} := \{\zeta \in \mathbb{C}^m :\ 2\|k_1(\operatorname{Re}\zeta)\|_\infty < \sigma_+/2\}$, with $\sigma_+ = s_1$ the amplitude of (I2) (definition 12.5, (30)). It is open, convex ($\xi \mapsto \|k_1(\xi)\|_\infty$ is a supremum of convex functions), connected, invariant under $z \mapsto -z$ on the curvature coordinates, contains 0, and by (21) contains every ζ_{ij}^* as soon as $\mathsf{a}(\zeta_i) < \sigma_+/8 =: \sigma_W$ for every i.

(ii) The *bounded domain*: let $B_1 \subset \mathbb{R}^m$ be the open convex hull of the unit ℓ^∞-balls about 0 and about the real parts $\operatorname{Re}\zeta_{ij}^*$; let $\beta_j := 2\sup_{\xi \in B_1} |\xi_j|$ for the scalar coordinates (so $\beta_j = 2(|\operatorname{Re} w_{il}| + 1)$ for the coordinate of $h_{il}^\pm$ or of $h_{il}^\pm \circ \rho$), $B'' := \prod_j [0, \beta_j]$ over the scalar coordinates, and $q_v := \sum_j v_j \tilde{h}_j$ for $v \in \operatorname{vert}(B'')$, the *vertex sources*. All q_v are pointwise $\leq q_\zeta := \sum_{i,l} 2(|\operatorname{Re} w_{il}| + 1)(h_{il}^+ + h_{il}^- + h_{il}^+ \circ \rho + h_{il}^- \circ \rho)$. Put $\mathcal{T}_1 := \{\zeta \in \mathbb{C}^m : \operatorname{Re}\zeta \in B_1\}$: open, convex, connected, containing every ζ_{ij}^* and the unit polydisc about 0, so that $\mathcal{T}_1 \cap T_\mathcal{F}$ is a non-empty open set. For $\xi \in B_1$ both $2\xi_+$ and ξ_+ lie in B''.

(iii) The *uniqueness set* $U \subset \mathbb{C}^m$ is the set of coefficient vectors whose curvature coefficients are purely imaginary and whose scalar coefficients are positive reals. It is a non-empty, relatively open subset of the totally real subspace $(\mathrm{i}\mathbb{R})^{m_F} \times \mathbb{R}^{m_h}$ of maximal real dimension m (it is not open in $\mathbb{C}^m$), hence a set of uniqueness for holomorphic functions on any connected open set meeting it: a holomorphic function vanishing on a relatively open piece of a maximal totally real subspace has all derivatives zero there [10]. $U \cap \mathcal{T}_1 \neq \emptyset$ and $U \cap T_\mathcal{F} \neq \emptyset$ (small positive scalar coefficients). On U the total source is $(\mathrm{i}f_\zeta, g_\zeta) \in \mathcal{K}$,

so $\Theta(\zeta) \in \mathcal{I}$; the scheme multipliers are positive reals, equal for the Neumann–Dirichlet and the modified states; and every lattice value is real (charge conjugation, definition 2.3: $\overline{Z^a(\mathrm{i}f,g)} = Z^a(-\mathrm{i}f,g) = Z^a(\mathrm{i}f,g)$).

Every lattice functional, and hence every limit below, depends on a family only through its total sources; the splitting is therefore immaterial at the matrix points and at the Taylor coefficients at 0.

10.2. **Uniform local bounds.** The following local bound is used only in the optional extension (theorem 23.8); it is stated here because its domain $\mathcal{T}_1$ is what makes "finitely many instances of (W2)" literally true there. The fixed-cube limits $Z^a_J := \lim_k Z^a_{k,J}$ exist for all complex coefficients and are entire under (W2)(a) alone: it gives convergence at the real points, the vertex argument of lemma 10.3 with lemma 2.11(d) gives local boundedness in k, and Montel's theorem with the uniqueness set $\mathbb{R}^m$ does the rest.

Lemma 10.2 (uniform local bound; optional extension)**.** *Assume* (W2)*(a) and* (W2)*(b) at the finitely many sources q_v, $v \in \mathrm{vert}(B'')$. Then $\{Z^a_J\}_{J\ge J_0}$ is uniformly bounded on $\mathcal{T}_1$, $Z^a_J \to Z^a_\infty$ locally uniformly on $\mathcal{T}_1$, Z^a_∞ is holomorphic on $\mathcal{T}_1$, and $Z^a_J \le Z^a_\infty$ on $U \cap \mathcal{T}_1$.*

Proof. For real $\xi \in B_1$, lemma 2.11(f) gives $Z^a_J(\xi) \le Z^a_J(2f_\xi, -2k_2(\xi))^{1/2} Z^a_J(0, 2k_1(\xi))^{1/2}$. The first factor is nonincreasing in J: at fixed k this is the second inequality of lemma 2.10 with lemma 2.11(a), and it passes to $k \to \infty$ by (W2)(a); (W2)(b) is not used there. Hence it is $\le Z^a_{J_0}(2f_\xi, -2k_2(\xi))$, and $\xi \mapsto \ln Z^a_{J_0}(2f_\xi, -2k_2(\xi))$ is bounded on the compact $\bar{B}_1$: the coefficient map is piecewise linear, so the image of $\bar{B}_1$ lies in a real box B', on which $\ln Z^a_{J_0}$ is convex (lemma 2.11(d)) and so bounded by its maximum over the vertices of B', each finite by (W2)(a). For the second factor, $2\xi_+ \in B''$ and $\ln Z^a_J(0,\cdot)$ is convex in the coefficients, so $Z^a_J(0, 2k_1(\xi)) \le \max_{v\in\mathrm{vert}(B'')} Z^a_J(0,q_v) \le \max_v \sup_{J'\ge J_{q_v}} Z^a_{J'}(0,q_v) =: C_{B''} < \infty$ by (W2) at the q_v (note $J \ge J_0 \ge J_{q_v}$). By lemma 2.11(d), $|Z^a_J(\zeta)| \le Z^a_J(\mathrm{Re}\,\zeta)$, so $\sup_{J\ge J_0}\sup_{\mathcal{T}_1}|Z^a_J| < \infty$. On $U \cap \mathcal{T}_1$ the sources lie in $\mathcal{K}$, so $J \mapsto Z^a_J(\zeta)$ is nondecreasing (first inequality of lemma 2.10 at fixed k, then $k \to \infty$) and bounded, hence convergent. $U \cap \mathcal{T}_1$ is a set of uniqueness for holomorphic functions on the connected $\mathcal{T}_1$ (definition 10.1(iii)); Montel's theorem gives locally uniform convergence on $\mathcal{T}_1$ to a holomorphic limit, which is Z^a_∞ wherever the latter is defined (the same J-limit), and the monotone limit on $U \cap \mathcal{T}_1$ gives $Z^a_J \le Z^a_\infty$ there. Under (W2)(b) for every $0 \le q \in C^\infty_c$ the same holds on all of $\mathbb{C}^m$. □

Lemma 10.3 (uniform local bounds for the modified-box functionals)**.** *Let $\mathcal{F}$, Φ_k, $\mathcal{T}_1$, $T_{\mathcal{F}}$ be as in definition 10.1.*

(A) (Optional extension.) *Under* (W2)*(a) and* (W2)*(b) at the q_v, $v \in \mathrm{vert}(B'')$: there is $C < \infty$, independent of k and of $J \ge J_0$, with $\limsup_k \sup_{\mathcal{T}_1} |\Phi_k| \le C$; and at fixed J, $\sup_k \sup_Q |\Phi_k| < \infty$ for every compact $Q \subset \mathbb{C}^m$ (this part uses* (W2)*(a) only).*

(A_∞) (Optional extension.) *Under* (W2) *for every $0 \le q \in C^\infty_c$: the same with $\mathcal{T}_1$ replaced by any compact $Q \subset \mathbb{C}^m$, with $C = C_Q$.*

(T) (The tube.) *Under theorem 3.5,* (I2), (I3) *and the correlation inequalities of lemma 2.7(a)–(e) only (no* (W2)*): the same for every compact $Q \subset T_{\mathcal{F}}$ and all $J \ge J_0$, with no further threshold on J.*

(C) (Curvature sector, $h \equiv 0$; no hypothesis.) $|\Phi_k(z)| \le \langle e^{F(f_{\mathrm{Re}\,z})}\rangle^{(3.38)}_{k,\Omega_{J,k}} \le \exp(\tfrac12 \mathrm{Var}^{(3.38)} F(f_{\mathrm{Re}\,z}))$ *and* $\mathrm{Var}^{(3.38)} F(f) \le \sum_p \varepsilon^3 f^2_{c(p)} \to \|f\|^2_{L^2}$, *uniformly in k and J.*

Proof. $|\Phi_k(\zeta)| \le \Phi_k(\mathrm{Re}\,\zeta)$ by the proof of lemma 2.11(d) ($|\Theta(\zeta)| = \Theta(\mathrm{Re}\,\zeta)$ and the modulus of the multiplier), which is valid for any probability measure with the same real multiplier, in particular for the modified box state, which is not one of the states of definitions 2.1 and 2.2. At a real ξ the proof of lemma 2.11(f) (Cauchy–Schwarz with the multiplier identity, again valid for any such measure) gives

$$\Phi_k(\xi) \;\le\; Z^{a,(3.38)}_{k,\Omega_{J,k}}(2f_\xi, -2k_2(\xi))^{1/2}\; Z^{a,(3.38)}_{k,\Omega_{J,k}}(0, 2k_1(\xi))^{1/2}. \tag{22}$$

First factor (class $\mathcal{D}$). By the $\mathcal{D}$ half of lemma 9.9 with $C = C_J \cap \Lambda_k$ (its hypothesis is (20)(i)) it is $\le Z^a_{k,J}(2f_\xi, -2k_2) \le Z^a_{k,J_0}(2f_\xi, -2k_2)$, the second step being the second inequality of lemma 2.10

at fixed k. The map $\xi \mapsto \eta(\xi) :=$ (coefficient vector of $(2f_\xi, -2k_2(\xi))$) is continuous, so $\eta(\operatorname{Re} Q)$, resp. $\eta(\bar{B}_1)$, lies in a real box B'; $\eta \mapsto \ln Z^a_{k,J_0}(\eta)$ is convex (lemma 2.11(d)), so $\sup_\xi Z^a_{k,J_0}(\eta(\xi)) \le \max_{v\in\operatorname{vert}(B')} Z^a_{k,J_0}(v)$, finitely many values, each convergent in k by theorem 3.5 (the sources here have negative Wick coefficients, on which the theorem has no amplitude restriction; in versions (A), (A$_\infty$) by (W2)(a)), and convergence in k implies boundedness in k. So the first factor is bounded uniformly in k, ξ and $J \ge J_0$. *Second factor* (class $\mathcal{I}$). By the $\mathcal{I}$ half of lemma 9.10 with $B = C_{J+2n} \cap \Lambda_k$ and $N := \lceil R(\mu_0)/\varepsilon_k \rceil$ (hypothesis (20)(ii)) it is $\le Z^a_{k,J+2n}(0, 2k_1(\xi))$. (A): for $\xi \in B_1$, $2\xi_+ \in B''$, so by convexity it is $\le \max_{v\in\operatorname{vert}(B'')} Z^a_{k,J+2n}(0, q_v)$; each of these finitely many values converges in k by (W2)(a) (hence is bounded in k, which is the fixed-J bound Montel's theorem needs, and the $\limsup_k$ bound passes to any subsequential limit) with limit $Z^a_{J+2n}(0, q_v) \le \sup_{J'\ge J_{q_v}} Z^a_{J'}(0, q_v) < \infty$: this is (W2) at q_v, used directly ($J + 2n \ge J_0 \ge J_{q_v}$), uniformly in J; the finitely many instances q_v are all of (W2) that version (A) uses. (A$_\infty$): the same with a box $B'' \supset 2(\operatorname{Re} Q)_+$ depending on Q. (T): for $\xi \in \operatorname{Re} Q$, $Q \subset T_{\mathcal{F}}$, one has $0 \le 2k_1(\xi) \le \sigma_+/2 \le s_1$ on the lattice (the sampled supremum is at most the continuum one), so corollary 13.3, $\ln N^a_{k,C_J}(0, q) \le \kappa\|q\|_{1,k}$ for $0 \le q \le s_1$ on $C_J \cap \Lambda_k$, uniformly in k and J, applied with $N^a_{k,X}(0, q) = Z^a_{k,X}(0, q)$ (the same functional with the same scheme multiplier, so no J-dependent factor intervenes), gives $Z^a_{k,J+2n}(0, 2k_1(\xi)) \le \exp(\kappa\|2k_1(\xi)\|_{1,k}) \le \exp\big(2\kappa \sum_j |\xi_j| \sup_{k'} \|\tilde{h}_j\|_{1,k'}\big)$, bounded uniformly in k, J and $\xi \in \operatorname{Re} Q$ because $\|\tilde{h}_j\|_{1,k'} \to \|\tilde{h}_j\|_{L^1}$ (Riemann sums, (1)). No lower bound on J beyond J_0 enters: corollary 13.3 holds for every k and J. *(C).* At $h \equiv 0$ the first inequality is lemma 2.11(d) as above and the second is lemma 2.8 with $s = 1$, the modified box state being a BFS measure (definition 9.2). Its photon Gaussian has the unitary form for gauge-invariant observables (Step 1 of lemma 9.3), and $\operatorname{Var}^u\langle v, d_1 A\rangle = \varepsilon^{-1}\langle d_1^T v, (d_1^T d_1 + \mu_0^2\varepsilon^2)^{-1} d_1^T v\rangle \le \varepsilon^{-1}\|v\|^2$; with $v_p = \varepsilon^2 f_{c(p)}$ this is $\operatorname{Var} F(f) \le \sum_p \varepsilon^3 f^2_{c(p)}$. □

Remark 10.4 (one instance of (W2) in place of the vertex set). For $v \in \operatorname{vert}(B'')$ one has $0 \le q_v \le q_\zeta$ pointwise, and for $p = 1+\eta > 1$, $p' = p/(p-1)$, $q_v = \frac{1}{p}(pq_\zeta) - \frac{1}{p'}\big(p'(q_\zeta - q_v)\big)$. Hölder's inequality with the multiplier bookkeeping $e^{-a_k \mathsf{S}(q_v)} = (e^{-a_k \mathsf{S}(pq_\zeta)})^{1/p} (e^{a_k \mathsf{S}(p'(q_\zeta - q_v))})^{1/p'}$ gives, for every state, $Z^a(0, q_v) \le Z^a(0, pq_\zeta)^{1/p} Z^a(0, -p'(q_\zeta - q_v))^{1/p'}$. The second factor is of class $\mathcal{D}$, nonincreasing in J (lemma 2.10, then $k \to \infty$ by (W2)(a)), hence $\le Z^a_{J_0}(0, -p'(q_\zeta - q_v)) < \infty$, one real point of theorem 3.5; the first is bounded uniformly in J by (W2)(b) at the single source $(1+\eta)q_\zeta$. So (W2) at $(1+\eta)q_\zeta$, for any one $\eta > 0$, implies the finitely many instances used in lemmas 10.2 and 10.3(A). Only this direction is proved; no converse is claimed. Pointwise domination $q_v \le q_\zeta$ alone gives nothing, since $Z^a(0, \cdot)$ is not monotone in q: the Wick square has no sign, and the scheme multiplier $e^{-a_k \mathsf{S}(q)}$ has no sign control.

10.3. **King's state on the Wick tube.**

Proposition 10.5 (the Wick-tube object: existence and translations)**.** *Let a be an admissible scheme (definition 3.6), and assume* (I2) *and* (I3)*. For a finite family $\mathcal{F}'$ of real sources with nonnegative scalar members $\tilde{h}_j \ge 0$ (any family is brought to this form by the splitting of definition 10.1) let $T_{\mathcal{F}'}$ be its Wick tube, as in definition 10.1(i).*

(a) *$Z^a_J \to Z^a_{\infty,T}$ locally uniformly on $T_{\mathcal{F}'}$ as $J \to \infty$; $Z^a_{\infty,T}$ is holomorphic on $T_{\mathcal{F}'}$; $Z^a_{\infty,T} = G_\infty$ on $T_{\mathcal{F}'} \cap \mathcal{D}_{\sigma_+,H}$ and $Z^a_{\infty,T} = Z^a_\infty$ wherever the latter exists (the same J-limit of the same Z^a_J); $|Z^a_{\infty,T}(\zeta)| \le Z^a_{\infty,T}(\operatorname{Re}\zeta)$; $Z^a_{\infty,T}(\xi) > 0$ for real $\xi \in T_{\mathcal{F}'}$ with $-\xi \in T_{\mathcal{F}'}$; and $Z^a_{\infty,T}(-z, w) = Z^a_{\infty,T}(z, w)$.*

(b) (Translations.) *$Z^a_{\infty,T}(\tau_b \cdot \zeta) = Z^a_{\infty,T}(\zeta)$ for all $b \in \mathbb{R}^3$ and $\zeta \in T_{\mathcal{F}'}$, where τ_b acts on all sources; the translated family has the same tube.*

The proof uses, besides theorem 3.5, only corollary 13.3 (hence (I2)*,* (I3) *and* [6, Thm. 6.2(1),(2)]*), the second inequality of lemma 2.10 (hence lemma 2.7(d)), lemmas 2.11 and 2.14, Montel's theorem and the identity theorem;* (W2) *is not used.*

Proof. (a) *Local boundedness* of $\{Z^a_J\}_{J\ge J_0}$ on $T_{\mathcal{F}'}$: $|Z^a_J(\zeta)| \le Z^a_J(\operatorname{Re}\zeta) \le Z^a_J(2f,-2k_2)^{1/2} Z^a_J(0,2k_1)^{1/2} \le Z^a_{J_0}(2f,-2k_2)^{1/2}\exp(\kappa\|k_1\|_{L^1})$, by lemma 2.11(d),(f), the nonincreasing $\mathcal{D}$-factor (second inequality of lemma 2.10 at fixed k, then $k\to\infty$), and corollary 13.3 in the limit $k\to\infty$ for the $\mathcal{I}$-factor, valid since $2\|k_1\|_\infty < \sigma_+/2 \le s_1$; the first factor is bounded on compacts by the vertex argument of lemma 10.3 with theorem 3.5 at C_{J_0}. *Convergence on* $U\cap T_{\mathcal{F}'}$: there the values are real, nondecreasing in J (first inequality of lemma 2.10, then $k\to\infty$) and bounded, since $Z^a_J(\mathrm{i}f_\zeta, g_\zeta) \le Z^a_J(0,g_\zeta)$ by lemma 2.11(d) and $Z^a_J(0,g_\zeta)\le \exp(\kappa\|g_\zeta\|_{L^1})$ by corollary 13.3 in the limit $k\to\infty$ ($0\le g_\zeta \le \sigma_+/4 \le s_1$ there). Montel's theorem on the connected open $T_{\mathcal{F}'}$ with the uniqueness set $U\cap T_{\mathcal{F}'}$ gives locally uniform convergence to a holomorphic $Z^a_{\infty,T}$; on $T_{\mathcal{F}'}\cap\mathcal{D}_{\sigma_+,H}$ this is the limit G_∞ of theorem 13.10(i), the same argument on the other tube. The bound $|Z^a_{\infty,T}(\zeta)| \le Z^a_{\infty,T}(\operatorname{Re}\zeta)$ and the evenness in z pass from the lattice (lemma 2.11(d); the invariance $(A,\vartheta)\mapsto(-A,-\vartheta)$ of every state (charge conjugation, definition 2.3), which maps $F(f)\mapsto -F(f)$ and fixes r; $T_{\mathcal{F}'}$ is invariant under $z\mapsto -z$). Positivity at symmetric real points is lemma 2.11(e), $Z^a(\xi)Z^a(-\xi)\ge 1$, in the limit.

(b) *Lattice translations.* Let $b\in 5^{-k_1}\mathbb{Z}^3$, $|b|_\infty \le M$, $k\ge k_1$. τ_b preserves Λ_k and S_{Λ_k}, so $Z^a_{k,J}(\tau_b\cdot\zeta) = Z^a_{k,(C_J-b)\cap\Lambda_k}(\zeta)$ (the action is translation covariant on Λ_k). As sets of sites, $C_{J-1}\cap\Lambda_k \subset (C_J-b)\cap\Lambda_k \subset C_{J+1}\cap\Lambda_k$, so the first inequality of lemma 2.10 (stated for any finite site set) gives, for $\zeta\in U$ and $J\ge J_0+1$ (so that every plaquette and site at which the sources do not vanish lies in C_{J-1}), $Z^a_{k,J-1}(\zeta) \le Z^a_{k,J}(\tau_b\cdot\zeta) \le Z^a_{k,J+1}(\zeta)$. Let $k\to\infty$ and then $J\to\infty$ on $U\cap T_{\mathcal{F}'}$, where the outer limits converge by (a): $Z^a_{\infty,T}(\tau_b\cdot\zeta) = Z^a_{\infty,T}(\zeta)$ there, hence on all of $T_{\mathcal{F}'}$ by the identity theorem (both sides are holomorphic on the connected $T_{\mathcal{F}'}$, the translated family having the same supremum norms). Iterating, the identity holds for all $b\in\bigcup_k 5^{-k}\mathbb{Z}^3$. *All* $b\in\mathbb{R}^3$. For $\rho>0$ let $E_\rho := C^\infty_c(\bar B_\rho;\Lambda^2)\times C^\infty_c(\bar B_\rho)$ (a Fréchet space) and $O_\rho := \{(f,h)\in E_\rho : 4\|h\|_\infty < \sigma_+/2\}$, open, convex and symmetric under $s\mapsto -s$. For $s=(f,h)\in O_\rho$ the one-member family with its splitting has the coefficient vector $(1,1,-1)$ in its tube ($2\|k_1\|_\infty = 2\|h^+\|_\infty \le 4\|h\|_\infty < \sigma_+/2$), so $\Phi_T(s) := \ln Z^a_{\infty,T}(f,h)$ is defined and finite on O_ρ by (a). Φ_T is convex (a pointwise limit of the functions $\ln Z^a_{k,J}$, convex by Hölder's inequality) and Borel (a pointwise limit of continuous functions of finitely many sampled values). By lemma 2.14, Φ_T is continuous on O_ρ. For $s\in O_\rho$ and $b\in\mathbb{R}^3$ pick $b_n\to b$ in $\bigcup_k 5^{-k}\mathbb{Z}^3$; then $\tau_{b_n}s, \tau_b s\in O_{\rho+|b|+1}$, $\tau_{b_n}s\to\tau_b s$ in $E_{\rho+|b|+1}$ (translations act continuously on C^∞_c), and $\Phi_T(\tau_{b_n}s)=\Phi_T(s)$; so $\Phi_T(\tau_b s)=\Phi_T(s)$. *Complex coefficients.* For a family $\mathcal{F}'$ the two holomorphic functions $\zeta\mapsto Z^a_{\infty,T}(\tau_b\cdot\zeta)$ and $\zeta\mapsto Z^a_{\infty,T}(\zeta)$ on $T_{\mathcal{F}'}$ agree at the real points of $\{\xi\in\mathbb{R}^m : 4\|h_\xi\|_\infty<\sigma_+/2\}\cap T_{\mathcal{F}'}$, where both limits exist and both equal the one-member value $\exp\Phi_T(f_\xi,h_\xi)$ (the same J-limit of the same Z^a_J). This intersection is open in $\mathbb{R}^m$ and contains 0 (the first set alone need not lie in the tube: with $\tilde h_1=\tilde h_2$ and $\xi=(1,-1)$, $h_\xi=0$ while $k_1=\tilde h_1$), hence is a set of uniqueness in the connected $T_{\mathcal{F}'}$; so the two functions agree on $T_{\mathcal{F}'}$. □

10.4. **The two limits and the passage to every plane.** Throughout this subsection $\mathcal{F}$, Φ_k, ζ^*_{ij}, $\mathcal{T}_1$, $T_{\mathcal{F}}$, U are those of definition 10.1 for a mid-plane $c\in D$, and "version" refers to lemma 10.3: (C) is the curvature sector when the comparison family is convergent at fixed cubes (the uniform local bound being theorem 7.4(b)), (T) the tube under theorem 3.5, (I2), (I3), and (A), (A_∞) the optional extension under (W2).

Step 5 ($k\to\infty$ at fixed J). By the fixed-J bound of lemma 10.3 and Montel's theorem there is a subsequence $k_m\to\infty$ along which $\Phi_{k_m}\to\Phi^J$ locally uniformly on $\mathbb{C}^m$, with Φ^J entire. No convergence of the full sequence is claimed or needed. By lemma 9.3(b), $\sum_{ij}\bar\gamma_i\gamma_j\Phi_k(\zeta^*_{ij})\ge 0$ for every $k\ge k_0(c,\delta)$ and $\gamma\in\mathbb{C}^N$; positive semidefiniteness of a Hermitian matrix is closed under entrywise limits, so $\sum_{ij}\bar\gamma_i\gamma_j\Phi^J(\zeta^*_{ij})\ge 0$. The uniform bounds of lemma 10.3 pass to Φ^J: $\sup_{\mathcal{T}_1}|\Phi^J|\le C$ in (A), $\sup_Q|\Phi^J|\le C_Q$ in (A_∞), (T) and (C), uniformly in $J\ge J_0$, whatever subsequence was taken. (The box $\Omega_{J,k}$ depends on J, so Φ^J does; the subsequence may depend on J.)

Step 6 ($J \to \infty$: sandwich on the cone, identity theorem off it). By the $\mathcal{I}$ halves of lemmas 9.9 and 9.10, whose hypotheses are (20)(i),(ii), for every k and every $\zeta \in U$,

$$Z^a_{k,J}(\zeta) \;\le\; \Phi_k(\zeta) \;\le\; Z^a_{k,J+2n}(\zeta). \tag{23}$$

Let $k \to \infty$ along the subsequence of Step 5. The outer terms converge along the full sequence (version (C): theorem 7.4(b); (T), (A), (A_∞): theorem 3.5, resp. (W2)(a), with Montel's theorem), so $Z^a_J \le \Phi^J \le Z^a_{J+2n}$ on U for every subsequential Φ^J. As $J \to \infty$ the outer terms converge on the respective uniqueness set: in (C) on $U = (\mathrm{i}\mathbb{R})^{m_F}$ to $Z_\infty(\cdot, 0)$ (theorem 7.4(b)); in (T) on $U \cap T_{\mathcal{F}}$ to $Z^a_{\infty,T}$ (proposition 10.5(a); there $J \mapsto Z^a_J(\zeta)$ is real, nondecreasing and bounded by $Z^a_J(0, g_\zeta) \le \exp(\kappa \|g_\zeta\|_{L^1})$, since $0 \le g_\zeta \le \sigma_+/4 \le s_1$); in (A) on $U \cap \mathcal{T}_1$ to Z^a_∞ (lemma 10.2); in (A_∞) on U to Z^a_∞ (lemma 10.2 with (W2)(b) for every q). Hence $\Phi^J \to Z^a_{\infty(,T)}$ pointwise on that set. By Step 5 the family $\{\Phi^J\}_J$ is locally bounded on the corresponding connected domain ($\mathbb{C}^m$, $T_{\mathcal{F}}$, $\mathcal{T}_1$, $\mathbb{C}^m$), on which the limit functional is holomorphic (theorem 7.4(b), proposition 10.5(a), lemma 10.2, lemma 10.2 on $\mathbb{C}^m$ respectively); by Montel's theorem every subsequence of (Φ^J) has a further subsequence converging locally uniformly to a holomorphic function, which agrees with the limit functional on the uniqueness set and hence on the whole domain [10]. So $\Phi^J \to Z^a_{\infty(,T)}$ locally uniformly on the domain, in particular at the finitely many matrix points ζ^*_{ij} (which lie in $\mathcal{T}_1$ by construction and in $T_{\mathcal{F}}$ by (21) when every $\mathsf{a}(\zeta_i) < \sigma_W$), where $\Phi^J(\zeta^*_{ij}) \to Z^a_{\infty(,T)}(\theta\zeta^*_i \sqcup \zeta_j)$ because the functionals depend on the family only through the total sources. Positivity passes to the limit:

$$\sum_{i,j} \bar\gamma_i \gamma_j \, Z^a_{\infty(,T)}(\theta\zeta^*_i \sqcup \zeta_j) \;\ge\; 0 \qquad \text{for the plane } \{x_\nu = c\},\ c \in D. \tag{24}$$

A side result: the modified-box values converge to the cube state, $\lim_J \lim_{k_m} Z^{a,(3.38)}_{k_m,\Omega_{J,k_m}}(\zeta) = Z^a_{\infty(,T)}(\zeta)$, on the respective domain and along any k-subsequence on which the fixed-J limits exist; this is the corrected form of King's passage from the modified boxes to the cube limit [14, (3.42)], with the margin $2n$ in place of one cube step (lemma 9.6 and remark 9.8). The reversal of King's second inequality is proved, weakly at every coupling and strictly at $e = 0$ (lemma 9.6); no step of the passage to the limit uses that inequality in King's direction.

Step 7 (every plane). Let $c \in \mathbb{R}$ be arbitrary and the sources supported in $\{x_\nu > c\}$. If $c \in D$, (24) is the result. Otherwise pick $c' \in D$ and $b := (c' - c)e_\nu$. The translated sources $\tau_b\zeta_i$ are supported in $\{x_\nu > c'\}$, and $\theta_{c'}(\tau_b\zeta)^* = \tau_b(\theta_c\zeta^*)$, because $\rho_{c'}(x + b) = \rho_c(x) + b$ for this b and the linear parts of $\rho_c, \rho_{c'}$ agree. By (24) for the plane c' and the translation invariance of the limit functional,

$$\sum \bar\gamma_i\gamma_j Z^a_{\infty(,T)}(\theta_c\zeta^*_i \sqcup \zeta_j) = \sum \bar\gamma_i\gamma_j Z^a_{\infty(,T)}\big(\tau_b(\theta_c\zeta^*_i \sqcup \zeta_j)\big) = \sum \bar\gamma_i\gamma_j Z^a_{\infty(,T)}\big(\theta_{c'}(\tau_b\zeta_i)^* \sqcup \tau_b\zeta_j\big) \;\ge\; 0.$$

The translation invariance is supplied in (C) by proposition 8.1, and in (T) by proposition 10.5(b) on $T_{\mathcal{F}}$ (the amplitude a is translation invariant, so the matrix points stay in the tube). In the optional extension a plane $c \notin D$ needs a non-lattice translation, which finitely many instances of (W2)(b) do not supply; there the step is carried by proposition 10.5(b) together with lemma 10.2 for $\mathcal{F}$ and for $\tau_b\mathcal{F}$ and the identity theorem on $\mathcal{T}_1$ (theorem 23.8).

10.5. **The theorems.**

Theorem 10.6 (reflection positivity, curvature sector, comparison convention)**.** *Assume that the comparison family is convergent at fixed cubes (definition 3.1). Let $\mu_0 > 0$, $\nu \in \{1, 2, 3\}$, $c \in \mathbb{R}$, and let $\zeta_1, \dots, \zeta_N$ be finite families of real 2-form sources $f_{il} \in C^\infty_c(\mathbb{R}^3; \Lambda^2)$ supported in the open half-space $\{x_\nu > c\}$, with arbitrary complex coefficients z_{il} and no scalar sources. Then for all $\gamma \in \mathbb{C}^N$*

$$\sum_{i,j=1}^{N} \bar\gamma_i\gamma_j \, Z_\infty\big(\theta\zeta^*_i \sqcup \zeta_j,\, 0\big) \;\ge\; 0.$$

Besides the convergence at fixed cubes the proof uses only [6, Thm. 6.2(1),(2)] *(through the first inequality of lemma 2.10 and the $\mathcal{I}$ halves of lemmas 9.9 and 9.10), theorem 7.4(b) with*

lemma 7.3, and lemma 2.8, lemma 2.11(d),(e), lemma 2.7(e), lemma 2.14; no inequality of class $\mathcal{D}$ enters.

Proof. At $h \equiv 0$ there is no Wick constant and no scheme, and $\mathcal{F}$ consists of the f_{il} and $\rho^* f_{il}$ with $U = (\mathrm{i}\mathbb{R})^m$. What the argument of Steps 5–7 needs of the cube functionals is: the existence of $Z_J(\cdot,0) = \lim_k Z_{k,J}(\cdot,0)$ locally uniformly on $\mathbb{C}^m$, entire; the monotonicity of $J \mapsto Z_J(\mathrm{i}f,0)$ on the cone (first inequality of lemma 2.10); the existence of $Z_\infty(\cdot,0) = \lim_J Z_J(\cdot,0)$, entire, locally uniformly on $\mathbb{C}^m$; and translation invariance. The first three are theorem 7.4(b), whose local bound is the μ_0-free bound of theorem 7.4(a) and lemma 7.3 rather than a local bound resting on the nonincreasing clause of lemma 7.7 and so on [6, Thm. 6.2(3),(4)]; the fourth is proposition 8.1, which uses lemma 2.14 with $\ln Z_\infty$ finite on all of E_ρ (upper bound from theorem 7.4(b), lower bound lemma 2.11(e)) and the identity theorem. The lattice steps lemmas 9.3, 9.9 and 9.10 include $g = 0$, and only their $\mathcal{I}$ halves are used; lemma 10.3(C) is the fixed-J and J-uniform bound; Steps 5–7 in version (C) give the claim. □

Theorem 10.7 (reflection positivity on the Wick tube, comparison convention)**.** *Let a be an admissible scheme, and assume* (I2) *and* (I3) *(the hypotheses of corollary 13.3); the $\mathcal{D}$-class steps use in addition lemma 2.7(d). Let $\sigma_+ = s_1$ be the amplitude of* (I2) *and $\sigma_W := \sigma_+/8$. Let $\mu_0 > 0$, ν, $c \in \mathbb{R}$, and let $\zeta_1, \dots, \zeta_N$ be finite families of real sources $f_{il} \in C_c^\infty(\mathbb{R}^3; \Lambda^2)$, $h_{il} \in C_c^\infty(\mathbb{R}^3)$ supported in $\{x_\nu > c\}$, with complex coefficients z_{il}, w_{il} such that $\mathsf{a}(\zeta_i) < \sigma_W$ for every i; in particular, all purely imaginary w_{il} with arbitrary z_{il} are admitted. Then the values $Z^a_{\infty,T}(\theta\zeta_i^* \sqcup \zeta_j) = \lim_J Z^a_J(\theta\zeta_i^* \sqcup \zeta_j)$ exist (proposition 10.5) and*

$$\sum_{i,j=1}^{N} \bar\gamma_i \gamma_j \, Z^a_{\infty,T}\big(\theta\zeta_i^* \sqcup \zeta_j\big) \;\geq\; 0 \qquad \text{for all } \gamma \in \mathbb{C}^N.$$

Where Z^a_∞ is defined, $Z^a_{\infty,T} = Z^a_\infty$ there. No lower bound on J beyond J_0 enters, and (W2) *is not used.*

Proof. With the splitting of definition 10.1 the matrix points lie in $T_\mathcal{F}$ by (21). Lemma 10.3(T) gives the bounds; Steps 5 and 6 in version (T), with proposition 10.5(a) for the limit on $U \cap T_\mathcal{F}$ and its holomorphy on $T_\mathcal{F}$, give (24) for $c \in D$; Step 7 with proposition 10.5(b) gives every plane. The $\mathcal{D}$-class steps are the first factor of (22) (the $\mathcal{D}$ half of lemma 9.9 and the second inequality of lemma 2.10) and the nonincreasing factor in proposition 10.5(a). □

The arbitrary-amplitude form, reflection positivity of Z^a_∞ at all complex coefficients under (W2), is theorem 23.8 in the optional extension; its proof is Steps 5–7 in version (A) for $c \in D$, where (W2) enters only at the finitely many q_v, and in version (A) or (A′) for $c \notin D$.

Lemma 10.8 (reflection positivity of King's printed state)**.** *Theorem 10.6 holds for Z^K_∞: for ν, c, $\zeta_1, \dots, \zeta_N$ as there, $\sum_{i,j} \bar\gamma_i \gamma_j Z^K_\infty(\theta\zeta_i^* \sqcup \zeta_j, 0) \geq 0$ for all $\gamma \in \mathbb{C}^N$; and propositions 10.9 and 10.10 below hold for $Z^K_\infty(\cdot,0)$. The inputs are theorem 3.2,* [6, Thm. 6.2(1),(2)]*, theorem 7.6(a),(b) in place of theorem 7.4, lemma 4.6(c) and remark 4.8 for the two places where the cube convention is seen, and corollary 8.3 for the translations; no inequality of class $\mathcal{D}$ enters.*

Proof. Lemmas 9.3 and 9.5 concern the modified box state alone and the lattice reflection, and lemma 10.3(C) bounds Φ_k on the modified photon form: these are convention-free, as is Step 5 (Montel in k; positivity passes). The two places where the Neumann–Dirichlet cube states enter are the ends of the sandwich (23), i.e. the $\mathcal{I}$ halves of lemmas 9.9 and 9.10; on King's convention they read $Z^K_{k,J}(\zeta) \leq \Phi_k(\zeta) \leq Z^K_{k,J+2n}(\zeta)$ for every k and $\zeta \in U = (\mathrm{i}\mathbb{R})^m$: the lower end is lemma 9.9 with Step 1 of lemma 4.2 prepended (remark 4.8: $\langle\cdot\rangle^K_{k,J} \leq \langle\cdot\rangle^{(a)}_{k,J} \leq \langle\cdot\rangle^{(3.38)}_{k,\Omega_{J,k}}$ on $\mathcal{I}$), the upper end is lemma 9.10 with $\langle\cdot\rangle^K_{k,J}$ on X_{J+2n} in place of the comparison state (lemma 4.6(b),(c): the box $\Omega_{J,k}$ does not meet the killed layers T_{J+2n}, and part (a) of lemma 9.10 is photon-only), under the hypotheses of (20). Here J_0 is the J_0 of the enlarged family (the sources and their reflections), which exceeds that of the sources alone when $c \neq 0$; theorem 7.6(b) is applied to the enlarged family at J and $J + 2n$, $J \geq J_0$. Along Step 5's subsequence the ends converge along

the full sequence by theorem 7.6(b), so $Z_J^K \le \Phi^J \le Z_{J+2n}^K$ on U for every subsequential Φ^J; as $J \to \infty$ both ends tend to Z_∞^K on U, $\{\Phi^J\}_J$ is locally bounded on $\mathbb{C}^m$ by lemma 10.3(C), and U is a set of uniqueness, so $\Phi^J \to Z_\infty^K$ locally uniformly on $\mathbb{C}^m$ as in Step 6, in particular at the matrix points; positivity passes to the limit. Step 7 uses corollary 8.3 in place of proposition 8.1. Propositions 10.9 and 10.10 use only the existence, holomorphy near 0 and lattice symmetries of the limit, and the locally uniform convergences just established; their proofs below apply verbatim with Z_∞^K, the charge-conjugation invariance being exact for $\langle\cdot\rangle_{k,J}^K$ as well. □

10.6. OS3 **and the polynomial form.**

Proposition 10.9 (symmetry, realness and charge conjugation)**.** *Let Z be one of: $Z_\infty(\cdot,0)$ if the comparison family is convergent at fixed cubes (resp. $Z_\infty^K(\cdot,0)$); $Z_{\infty,T}^a$ under the hypotheses of proposition 10.5; Z_∞^a under* (W2) *(optional extension). The Schwinger functions*

$$S_{n,p}(f_1,\dots,f_n;h_1,\dots,h_p) := \partial_{z_1}\cdots\partial_{z_n}\partial_{w_1}\cdots\partial_{w_p} Z\big(\textstyle\sum_i z_i f_i, \sum_l w_l h_l\big)\big|_0$$

(with $p=0$ and S_n in the curvature sector) are (a) symmetric multilinear in the f's and in the h's; (b) real for real test functions; (c) zero for odd n. The proof uses only that Z exists and is holomorphic near 0*, with the stated symmetries of the lattice functionals; the existence is the input of the respective sector.*

Proof. (a) Z depends on a family only through its total sources (so does every lattice functional, hence both limits), so its Taylor coefficients at 0 are symmetric multilinear forms in the f's and in the h's; $Z_{\infty,T}^a$ is holomorphic on a tube containing 0. The definition of $S_{n,p}$ does not depend on the family used to compute it (restrict a larger family to the given sources). (b) For real coefficients every lattice functional is real (real integrand and multiplier, lemma 2.11(a)), so Z is real at the real points near 0 (for $Z_{\infty,T}^a$ on $\mathbb{R}^m \cap T$, an open piece of $\mathbb{R}^m$ containing 0); a holomorphic function real on the real points near 0 has real Taylor coefficients at 0. (c) Every cube state and every modified box state is invariant under $(A,\vartheta) \mapsto (-A,-\vartheta)$ (the photon terms are even in A, the masked divergence included; $\cos(\vartheta_x - e\varepsilon A - \vartheta_y)$ is invariant; W^a depends on r only; Lebesgue measure is preserved), which maps $F_{\Lambda_k}(f) \mapsto -F_{\Lambda_k}(f)$ and fixes $W_{\Lambda_k}^a(h)$. Hence $Z_{k,X}(-z,w) = Z_{k,X}(z,w)$ for all k, X, and in both limits $Z(-z,w) = Z(z,w)$ (on the $z \mapsto -z$ invariant tube for $Z_{\infty,T}^a$). So $\partial_z^{(n)}\partial_w^{(p)} Z(0) = 0$ for odd n. This is the argument of [24, Lemma 9.7, Thm. 9.8] for the cube states. □

Proposition 10.10 (the polynomial form of reflection positivity)**.** *Let $P = \sum_\alpha c_\alpha \prod_i F(f_{\alpha i}) \prod_l W^a(h_{\alpha l})$ be a polynomial in the smeared fields with real sources supported in $\{x_\nu > c\}$, and $\theta P := \sum_\alpha \bar c_\alpha \prod_i F(\rho^* f_{\alpha i}) \prod_l W^a(h_{\alpha l}\circ\rho)$. Then $S(\theta(P)P) \ge 0$, where S assigns to a monomial the corresponding Schwinger function of proposition 10.9: the Taylor coefficient at* 0 *of $Z_{\infty,T}^a$ under the hypotheses of theorem 10.7, resp. of $Z_\infty(\cdot,0)$ (of $Z_\infty^K(\cdot,0)$), where at $h \equiv 0$ the W^a factors are absent.*

Proof. Let all sources of P be supported in $\{x_\nu \ge c+\delta\}$ and let $\mathcal{F}'$ be the family of all sources of P and their reflections. Its scalar members have no sign, so Steps 4–6 are applied to the split family $\mathcal{F}'_{\text{split}}$ (members $h_{\alpha l}^{\pm}$, $h_{\alpha l}^{\pm}\circ\rho$, all ≥ 0); its tube contains 0 ($k_1(0)=0$), and the unsplit functional is the composition of the split one with the linear map $w_{\alpha l} \mapsto (w_{\alpha l}^+, w_{\alpha l}^-) = (w_{\alpha l}, -w_{\alpha l})$, so by the chain rule the derivatives at 0 of the unsplit Φ_k, Φ^J, $Z_{\infty(,T)}^a$ are finite linear combinations of those of the split ones, and locally uniform convergence near 0 passes through the linear map. (i) *Lattice.* For $k \ge k_0$ the sampled P is a gauge-invariant polynomial in the plus-side smeared fields, lies in L^2 of the modified box state (Gaussian moments; $\lambda > 0$) and in $\mathfrak{B}_+$ (definition 9.1), and $\theta(P) = \overline{P\circ\hat\rho}$ by (18). By lemma 9.3(a) (its L^2 extension), $\langle\theta(P)P\rangle_{k,\Omega_{J,k}}^{(3.38)} \ge 0$. (ii) *Moments as derivatives.* $\langle\theta(P)P\rangle^{(3.38)} = \sum_{\alpha,\beta} \bar c_\alpha c_\beta \langle \prod_i F(\rho^* f_{\alpha i}) \prod_l W^a(h_{\alpha l}\circ\rho) \prod_i F(f_{\beta i}) \prod_l W^a(h_{\beta l})\rangle^{(3.38)}$, and each mixed moment is a mixed partial derivative at 0 of Φ_k for the family $\mathcal{F}'$ (differentiate under the integral; the scheme multiplier $e^{-wa_k \mathsf{S}(h)}$ is the $-a_k$ inside W^a). (iii) *Limits.* By Steps 5–6 for $\mathcal{F}'_{\text{split}}$, $\Phi_{k_m} \to \Phi^J$ and $\Phi^J \to Z_{\infty(,T)}^a$ locally uniformly on a neighbourhood of 0, so by

Weierstrass' theorem all derivatives at 0 converge, and $\sum \bar{c}_\alpha c_\beta\, \partial^{(\alpha,\beta)} Z^a_{\infty(,T)}(0) \geq 0$; the left side is $S(\theta(P)P)$ by the family-independence of the Schwinger functions (proposition 10.9(a)). Every plane: Step 7 (translation of the sources; S is translation invariant by proposition 10.5(b), resp. proposition 8.1 or corollary 8.3). The curvature sector is the case $h \equiv 0$ with no splitting. This is the form OS2(b) of [24, Def. 2.10, Thm. 9.4] for the cube states. □

Remark 10.11 (which form is used, and the amplitude constants). The main theorem (theorem 21.1) states reflection positivity of the Wick sector in the tube form of theorem 10.7, with propositions 10.9 and 10.10 for the Schwinger functions; its inputs for this clause are theorem 3.5, (I2), (I3) and [6, Thm. 6.2(1)–(4)], without (W2). This suffices for the consumer: the weak form of OS4 and the reconstruction of theorem 11.1 use reflection positivity for exponentials $e^{\mathrm{i}\Phi(f)}$ of the smeared fields, whose Wick coefficients are purely imaginary, so amplitude 0 only. Three positive, ε-free amplitude constants occur in the Wick sector: $\sigma_W = \sigma_+/8$ here; $H/2 = \sigma_+/32$, the negative-part radius of the tube $\mathcal{D}_{\sigma_+,H}$ of theorem 13.10 ($H = s_1/16$); and $\sigma_0 = H/4$ in the bound E0″ of theorem 13.10(ii). On $T_{\mathcal{F}} \cap \mathcal{D}_{\sigma_+,H}$ the objects $Z^a_{\infty,T}$ and G_∞ coincide (proposition 10.5(a)). On King's convention the tube statements of this section are read under (TW^K) (remark 13.12, proposition 21.4); the curvature-sector statement needs no such hypothesis (lemma 10.8). The margin $R(\mu_0)$ of lemma 9.10 enters the proof, not the conclusion: reflection positivity is proved at every fixed $\mu_0 > 0$, and in a limit of vanishing photon mass it passes to the limit functionals as a pointwise limit of positive semidefinite matrices, the polynomial form by Weierstrass' theorem.

11. The measure and weak clustering

This section builds the measure of King's printed state in the curvature sector and draws from the authors' earlier paper on the torus limits [24] the consequences of reflection positivity: the ergodic decomposition, clustering, the unique joint vacuum on a component, the growth of the components' moments, and weak OS4 on almost every component. The joint field with the Wick square is the subject of the tube sections. The measure is built from the characteristic functional, not from the lattice laws, which converge only locally (remark 11.2).

Conventions. $\mathcal{V} := \Lambda^2\mathbb{R}^3$ with its Euclidean fibre norm. For a real $u \in C^\infty_c(\mathbb{R}^3;\Lambda^2)$ put $\mathcal{Z}^K(u) := Z^K_\infty(\mathrm{i}u)$, the one-member functional at the coefficient i (theorem 7.6(b), with the total-source convention of theorem 8.5). Fix an even $r \geq 14$. $\mathcal{H}^{-r}(\mathbb{R}^3;\Lambda^2)$ is the Hermite–Sobolev space, the dual of the completion of $\mathcal{S}(\mathbb{R}^3;\Lambda^2)$ under $\|u\|_r^2 = \sum_{\alpha,c} \lambda_\alpha^r \langle h_{\alpha,c}, u\rangle^2$, where $(h_{\alpha,c})_{\alpha\in\mathbb{N}^3,\, c\leq 3}$ is the Hermite basis of $L^2(\mathbb{R}^3;\Lambda^2)$ and $\lambda_\alpha = 2|\alpha| + 4$ are the eigenvalues of $-\Delta + |x|^2 + 1$; $|\cdot|_4$ is the weighted C^4 norm of lemma 2.6, and $|u|_4 \leq C\|u\|_{10}$ on $\mathcal{S}$ [24, Lemma 10.5(a)]. $X := H^{-r}_{\mathrm{loc}}(\mathbb{R}^3;\Lambda^2)$ carries the actions $(\tau_a\Phi)(u) = \Phi(\tau_{-a}u)$, $(g\Phi)(u) = \Phi(g^{-1}\cdot u)$ for $g \in O_h$ (or $O(3)$), $(\theta_j\Phi)(u) = \Phi(\sigma_j \cdot u)$ for the coordinate reflections σ_j, with the signed transport of 2-forms of definition 2.3, and the local σ-algebras $\Sigma(U)$ generated by $\Phi \mapsto \Phi(u)$, $u \in C^\infty_c(U;\Lambda^2)$ real; $\theta_{j,c} := \tau_{ce_j}\theta_j\tau_{-ce_j}$ is the reflection in the plane $x_j = c$ and $\rho_{j,c*}u$ the reflected source (definition 9.1). These are the space, actions and σ-algebras of the abstract part of [24], restricted from $\Lambda^2 \oplus \mathbb{R}$ to the curvature summand. The *lattice curvature field* of King's cube is the random distribution $\Phi^K_{k,J} : u \mapsto F_{\Lambda_k}(u)$ under $\langle\cdot\rangle^K_{k,J}$. $J_0(u)$ is the least J with $\operatorname{supp} u$ at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_J$.

Theorem 11.1 (the measure, the local lattice laws, weak clustering)**.** *The limits below exist by theorem 3.2.*

(a) (The measure; uses theorem 7.6 and corollary 8.3 only.) *For every even $r \geq 14$ there is exactly one Borel probability measure $\mu^{K,H}$ on $\mathcal{H}^{-r}(\mathbb{R}^3;\Lambda^2)$ whose mixed moments on C^∞_c are the Schwinger functions S^K_n of King's printed state. Its characteristic functional on C^∞_c, indeed on $\mathcal{S}$, is $\mathcal{Z}^K$ (extended by continuity); its moments on $\mathcal{S}$ are the continuous extensions of the S^K_n and satisfy the bounds of theorem 7.6(c). Its image μ^K_S under the restriction map $\iota : \mathcal{H}^{-r} \to X$ is the unique Borel probability measure on X with characteristic functional $\mathcal{Z}^K$ on C^∞_c, and also the unique one whose mixed moments on C^∞_c are the S^K_n.* Local convergence of the lattice laws: *for every real $\chi \in C^\infty_c(\mathbb{R}^3)$ and every box Q containing*

$\operatorname{supp}\chi$ *in its interior, the laws* $\nu^K_{k,J,\chi}$ *of the localised lattice fields* $\chi\Phi^K_{k,J}$ *on* $H^{-r}_{\mathrm{per}}(Q;\Lambda^2)$ *converge weakly, along the full sequence* $k\to\infty$ *at every fixed* $J\geq J_0(\chi)$*, to a law* $\nu^K_{J,\chi}$*, and* $\nu^K_{J,\chi}\Rightarrow\nu^K_\chi$ *as* $J\to\infty$*, where* ν^K_χ *is the law of* $\chi\Phi$ *under* μ^K_S*. Nothing is asserted about the laws of the unlocalised lattice fields at fixed* J *(remark 11.2).*

(b) (Invariance and weak clustering; uses also theorem 8.5 and lemma 10.8.) μ^K_S *satisfies the hypotheses (H1)–(H3) of the abstract part of* [24]*, is invariant under* $\mathbb{R}^3\rtimes O_h$ *about every point, and the abstract theorems of* [24] *apply to it with their proofs:* $\mu^K_S=\int\mu_\xi\,d\nu(\xi)$ *with components that are translation-invariant and ergodic,* θ_j*-invariant, reflection positive about every coordinate plane and satisfy (H2)* [24, Thm. 10.10]*; clustering in the mean* [24, Thm. 10.19] *and under diagonal separation* [24, Thm. 10.20]*; the Osterwalder–Schrader reconstruction in each coordinate direction with a unique joint vacuum on every component* [24, Prop. 10.16, Prop. 10.17(iii)]*; the growth bound* $|S^\xi_N(u_1,\dots,u_N)|\leq\sigma^\xi_N\prod_i\|u_i\|_{r+2}$*,* $\sigma^\xi_N\leq\alpha_\beta(\xi)(N!)^\beta$ *for every* $\beta>1$*, for the components' Schwartz moments* [24, Cor. 10.43(iv)]*; and the cubic-invariant decomposition into mixtures of* $k(\zeta)\in\{1,2,3,6\}$ *ergodic measures* [24, Thm. 10.12]*. The same holds on* $\mathcal{H}^{-r}$ *for* $\mu^{K,H}$ [24, Lemma 10.5(d)]*.* ν*-almost every component satisfies* OS4 *in the weak form, ergodicity under the translation group* $\mathbb{R}^3$ [24, Def. 2.12(iii)].

(c) (The rotation corollary.) $\mathcal{Z}^K$ *is* $O(3)$*-invariant,* $\mathcal{Z}^K(R\cdot u)=\mathcal{Z}^K(u)$ *for all* $R\in O(3)$ *and real* u*, by theorem 8.7 (the rotation theorem theorem 3.13); the clauses below use only this invariance. Then for* ν*-a.e.* ξ*:* μ_ξ *is invariant under* $\mathbb{R}^3\rtimes O(3)$*;* $\mathbb{E}_{\mu_\xi}[\overline{F\circ\theta_{n,c}}\,F]\geq0$ *for every unit vector* n*, every* $c\in\mathbb{R}$ *and every* $F\in L^2(\Sigma(H^+_{n,c}),\mu_\xi)$*,* $H^+_{n,c}=\{x\cdot n>c\}$*; and* $\mathbb{E}_{\mu_\xi}[G'\,U(a)G]\to\mathbb{E}_{\mu_\xi}[G']\,\mathbb{E}_{\mu_\xi}[G]$ *as* $|a|\to\infty$ *in any manner, for all* $G,G'\in L^2(\mu_\xi)$*. Consequently* μ_ξ *satisfies* OS4 *in each of the forms of* [24, Def. 2.11, Def. 2.12] *(the cluster property on the positive-time algebra, ergodicity of the time translations, ergodicity under* $\mathbb{R}^3$*) for every choice of the time direction, coordinate or not, and in the reconstruction from* μ_ξ *in any direction* n *the joint fixed space is* $\ker H_n=\mathbb{C}\Omega_n$ [24, Cor. 10.44]*. The moments of* μ_ξ *are real and symmetric; the vanishing of its odd curvature moments is* not *claimed for a component.*

11.1. **Proof of (a).** *Step 1 (moment bounds, and where they hold).* For real 2-forms $f_1,\dots,f_n$ at distance >1 from $\mathbb{R}^3\setminus C_{J_0}$, theorem 7.6(c) gives, at every k and $J\geq J_0$, $|S^{K,(k,J)}_n|\leq(\sqrt{2}\,e)^n(n!)^{1/2}\prod_j\|f_j\|_{2,k}$ with $\|f\|^2_{2,k}\leq3C_0|f|^2_4$ uniformly in k (lemma 2.6), and in the limit the bound of theorem 7.6(c); by theorem 7.6(a), $\langle F(u)^2\rangle^K_{k,J}\leq2\|u\|^2_{2,k}$ and hence $S^K_2(u,u)\leq2\|u\|^2_2$. The lattice Schwinger functions are the Taylor coefficients at 0 of $a^K_{k,J}$, that is the moments of $\Phi^K_{k,J}$, and converge to the S^K_n. These bounds are uniform in k and $J\geq J_0$ for sources at distance >1 inside C_{J_0}; at a fixed J they are not available for sources reaching ∂C_J, and neither is the convergence in k: theorem 3.2 and theorem 7.6(a) are stated for sources whose plaquettes have their corners in X_J.

Step 2 (the measure). (i) *The functional on* $\mathcal{S}$*.* For real $u_1,\dots,u_N\in C^\infty_c$ and $c\in\mathbb{C}^N$, at every (k,J) with $J\geq J_0(u)$, $\sum_{l,m}\bar c_l c_m\,Z^K_{k,J}(\mathrm{i}(u_m-u_l))=\langle|\sum_m c_m e^{\mathrm{i}F(u_m)}|^2\rangle^K_{k,J}\geq0$ by the linearity of F_{Λ_k} in the source. The limits $k\to\infty$, then $J\to\infty$, exist at these points by theorem 7.6(b) for the family u, so $\mathcal{Z}^K$ is positive definite on C^∞_c with $\mathcal{Z}^K(0)=1$. By corollary 8.3's Lipschitz bound and (E2), $|\mathcal{Z}^K(u)-\mathcal{Z}^K(u')|\leq\sqrt{2}\|u-u'\|_2\leq\frac{\pi}{2}|u-u'|_4$. Since $|\cdot|_4$ is a continuous seminorm on $\mathcal{S}(\mathbb{R}^3;\Lambda^2)$ and C^∞_c is dense in $\mathcal{S}$, $\mathcal{Z}^K$ extends uniquely to a continuous function on $\mathcal{S}$, still positive definite with $\mathcal{Z}^K(0)=1$. By the Bochner–Minlos theorem [9, 11] ($\mathcal{S}$ is nuclear) there is exactly one Borel probability measure μ' on $\mathcal{S}'(\mathbb{R}^3;\Lambda^2)$, with its cylinder σ-algebra, such that $\int e^{\mathrm{i}\Phi(u)}\,d\mu'=\mathcal{Z}^K(u)$ for all real $u\in\mathcal{S}$.

(ii) *Moments on* C^∞_c*, and on* $\mathcal{S}$ *by uniform integrability.* For real $u_1,\dots,u_N\in C^\infty_c$ the characteristic function of $(\Phi(u_l))_l$ under μ' is $\varphi(t)=\mathcal{Z}^K(\sum_l t_l u_l)=Z^{K,u}_\infty(\mathrm{i}t)$, the restriction to $\mathrm{i}\mathbb{R}^N$ of the entire function $Z^{K,u}_\infty$ (theorem 7.6(b)). A characteristic function real-analytic near 0 has moments of all orders, with $\int\prod_l\Phi(u_l)\,d\mu'=(-\mathrm{i})^N\partial_{t_1}\cdots\partial_{t_N}\varphi(0)$ [16]; the chain rule gives

$\partial_{t_1}\cdots\partial_{t_N}\varphi(0) = \mathrm{i}^N\partial_{z_1}\cdots\partial_{z_N}Z^{K,u}_\infty(0)$, so the factors cancel and

(25)
$$\int\prod_{l=1}^{N}\Phi(u_l)\,d\mu' = S^K_N(u_1,\dots,u_N); \qquad \int\Phi(u)^2\,d\mu' = S^K_2(u,u) \le 2\|u\|_2^2 \le \tfrac{\pi^2}{4}|u|_4^2 \le C_2''\|u\|_{10}^2$$

by Step 1, (E2) and $|u|_4 \le C\|u\|_{10}$. *Extension to* $\mathcal{S}$. By theorem 7.6(c) each S^K_N extends to a continuous symmetric multilinear form on $\mathcal{S}^N$ with $|S^K_N| \le \sigma_N\prod_l|u_l|_4$. For $u_l\in\mathcal{S}$ take cut-offs $\chi_R = \chi(\cdot/R)$, $\chi\in C^\infty_c$, $\chi = 1$ on the unit ball, $0\le\chi\le 1$, $R\ge 1$: $\chi_R u_l \to u_l$ in $\mathcal{S}$, so $\Phi(\chi_R u_l)\to\Phi(u_l)$ for every $\Phi\in\mathcal{S}'$; and $|\chi_R u|_4 \le C_\chi|u|_4$ uniformly in $R\ge 1$ with $C_\chi = 2^4\|\chi\|_{C^4}$, by Leibniz with $|R^{-|\beta|}(D^\beta\chi)(\cdot/R)| \le \|\chi\|_{C^4}$ (the norm $|\cdot|_4$ carries derivatives up to order 4, so the cut-off does not pass through it with constant 1). Hence $\int\prod_l\Phi(\chi_R u_l)^2\,d\mu' = S^K_{2N}(\chi_R u_1,\chi_R u_1,\dots)\le\sigma_{2N}C_\chi^{2N}\prod_l|u_l|_4^2$ uniformly in R: the products $\prod_l\Phi(\chi_R u_l)$ are bounded in $L^2(\mu')$, hence uniformly integrable, and converge pointwise, so $\int\prod_l\Phi(u_l)\,d\mu' = \lim_R S^K_N(\chi_R u_1,\dots) = S^K_N(u_1,\dots,u_N)$ by the continuity of S^K_N on $\mathcal{S}^N$.

(iii) *The Hermite carrier.* Put $M(\Phi) := \sum_{\alpha,c}\lambda_\alpha^{-r}\Phi(h_{\alpha,c})^2$. By (25) with $\|h_{\alpha,c}\|_{10}^2 = \lambda_\alpha^{10}$, $\int M\,d\mu' \le 3C_2''\sum_\alpha\lambda_\alpha^{10-r} < \infty$ as soon as $r-10>3$ (there are $(n+1)(n+2)/2$ multi-indices with $|\alpha|=n$), that is for every even $r\ge 14$; so $\mu'(M<\infty)=1$. For $\Phi\in\mathcal{S}'$ and $u\in\mathcal{S}$ the Hermite expansion $u = \sum_{\alpha,c}\langle h_{\alpha,c},u\rangle\,h_{\alpha,c}$ converges in $\mathcal{S}$ [22, Thm. V.13], so $\Phi(u) = \sum_{\alpha,c}\langle h_{\alpha,c},u\rangle\,\Phi(h_{\alpha,c})$ and, on $\{M<\infty\}$, $|\Phi(u)|\le M(\Phi)^{1/2}\|u\|_r$ by Cauchy–Schwarz: Φ extends to $\Psi(\Phi)\in\mathcal{H}^{-r}$ with $\|\Psi(\Phi)\|^2_{-r} = M(\Phi)$ and $\Psi(\Phi)=\Phi$ on $\mathcal{S}$. The map $\Phi\mapsto\Psi(\Phi)$ (:= 0 off $\{M<\infty\}$) is measurable from the cylinder σ-algebra to the Borel σ-algebra of the separable Hilbert space $\mathcal{H}^{-r}$, its coordinates being evaluations. Put $\mu^{K,H} := \Psi_*\mu'$. Then $\int e^{\mathrm{i}\Psi(u)}\,d\mu^{K,H} = \mathcal{Z}^K(u)$ and $\int\prod_l\Psi(u_l)\,d\mu^{K,H} = S^K_N(u_1,\dots,u_N)$ for all $u,u_l\in\mathcal{S}$, by (ii). This is the carrier construction of [24, Lemma 10.5(b)] run on $\mathcal{S}'$.

(iv) *Uniqueness on* $\mathcal{H}^{-r}$. By [24, Lemma 10.5(c)], $\mu^{K,H}$ is the only Borel probability measure on $\mathcal{H}^{-r}$ whose mixed moments on C^∞_c exist and equal the S^K_n. The proof there uses only that the moments m_n of $Z=\sum_l t_l\Psi(u_l)$ satisfy $|m_n|\le c_S(n+1)K^n n!$ (then, by Cauchy–Schwarz for the odd absolute moments and the Taylor remainder bound $|y|^n/n!$, the characteristic function of Z is the sum of its Taylor series on every interval of length $<2/K$, hence determined by the moments on all of $\mathbb{R}$) and the Lusin–Souslin theorem on $\mathcal{H}^{-r}$ [12]. Our moments give the hypothesis with $c_S=1$: $m_n = S^K_n(g,\dots,g)$ for $g=\sum_l t_l u_l$, and $|m_n|\le(\sqrt{2}\,e)^n(n!)^{1/2}\|g\|_2^n \le (n+1)K^n n!$ with $K=\sqrt{2}\,e\|g\|_2$.

(v) *The law on* X *and its two uniqueness statements.* The map $\iota:\mathcal{H}^{-r}\to X$, $\iota\Psi := \Psi|_{C^\infty_c}$, is continuous, linear and injective [24, Lemma 10.5(b)]: $|\Psi(\chi f)|\le\|\Psi\|_{-r}\|\chi f\|_r \le C_\chi\|\Psi\|_{-r}\|f\|_{H^r}$ for a cut-off χ, and C^∞_c is dense in $\mathcal{S}$, which is dense in $\mathcal{H}^r$. Put $\mu^K_S := \iota_*\mu^{K,H}$; its characteristic functional on real C^∞_c is $\mathcal{Z}^K$. By the uniqueness part of [24, Lemma 10.3], which applies to any Borel probability measure on X (for a countable dense $(f_n)\subset C^\infty_c$ the sets $\{\Phi:(\Phi(f_1),\dots,\Phi(f_N))\in B\}$ form a π-system generating the Borel σ-algebra, by Lusin–Souslin, and the law of each such vector is determined by $t\mapsto\mathcal{Z}^K(\sum_k t_k f_k)$), μ^K_S is the only Borel probability measure on X with this characteristic functional. *Moment uniqueness on* X*:* if μ'' is any Borel probability measure on X whose mixed moments on C^∞_c are the S^K_n, the argument of (iv) applies verbatim on X with this π-system [24, Prop. 10.4], so $\mu''=\mu^K_S$. And $\mu^{K,H}$ is the only measure on $\mathcal{H}^{-r}$ with $\iota_*\mu^{K,H}=\mu^K_S$, since ι is a Borel isomorphism onto the Borel set $\iota\mathcal{H}^{-r}$. Nothing in Step 2 sees the lattice: the measure clauses of (a) depend on King's printed state only through $\mathcal{Z}^K$ and the S^K_n.

Step 2′ (local convergence of the lattice laws). Fix a real $\chi\in C^\infty_c(\mathbb{R}^3)$ and a box Q with $\operatorname{supp}\chi$ compact in the interior of Q (as in [24, Prop. 8.3]), and let $(e_{\alpha,c})_{\alpha\in\mathbb{Z}^3,\,c\le 3}$ be the Fourier basis of $L^2(Q;\Lambda^2)$, $|e_{\alpha,c}(x)| = |Q|^{-1/2}$. For a real Q-periodic smooth 2-form ψ put $\langle\chi\Phi^K_{k,J},\psi\rangle := \Phi^K_{k,J}(\chi\psi) = F_{\Lambda_k}(\chi\psi)$, where $\chi\psi\in C^\infty_c$ is supported in $\operatorname{supp}\chi$ (for the complex basis vectors, used only in (1), take real and imaginary parts); this identifies $\chi\Phi^K_{k,J}$ with an element of the real Hilbert space $H_\chi := H^{-r}_{\mathrm{per}}(Q;\Lambda^2)$, and $\nu^K_{k,J,\chi}$ denotes its law.

(1) *Second moments, uniformly.* For $J \geq J_0(\chi)$ and every k, every plaquette carrying $\chi e_{\alpha,c}$ has its centre in $\operatorname{supp}\chi$, hence its corners in X_J, so the second-moment clause of theorem 7.6(a) applies to the real and imaginary parts: $\mathbb{E}^K_{k,J}|F_{\Lambda_k}(\chi e_{\alpha,c})|^2 \leq 2\|\chi \operatorname{Re} e_{\alpha,c}\|^2_{2,k} + 2\|\chi \operatorname{Im} e_{\alpha,c}\|^2_{2,k} \leq 4\|\chi e_{\alpha,c}\|^2_{2,k}$. *Riemann-sum bound:* the plaquettes of one orientation have their centres on a translate of $\varepsilon_k\mathbb{Z}^3$; those carrying $\chi e_{\alpha,c}$ have centres in $\operatorname{supp}\chi$; the cubes of side $\varepsilon_k \leq 1$ about these centres are disjoint and lie in the $\frac{1}{2}$-neighbourhood $N_{1/2}(\operatorname{supp}\chi)$; so $\|\chi e_{\alpha,c}\|^2_{2,k} \leq 3\operatorname{vol}(N_{1/2}(\operatorname{supp}\chi))\,\|\chi\|^2_\infty/|Q| =: K_\chi$, uniformly in α, c, k and $J \geq J_0(\chi)$ (no derivative of χ enters). Hence, for $s_0 > \frac{3}{2}$,

(26)
$$\mathbb{E}^K_{k,J}\|\chi\Phi^K_{k,J}\|^2_{H^{-s_0}_{\mathrm{per}}(Q)} = \sum_{\alpha,c}(1+|\alpha|^2)^{-s_0}\,\mathbb{E}^K_{k,J}|F_{\Lambda_k}(\chi e_{\alpha,c})|^2 \;\leq\; 12K_\chi \sum_{\alpha\in\mathbb{Z}^3}(1+|\alpha|^2)^{-s_0} =: C'_\chi < \infty,$$

uniformly in k and $J \geq J_0(\chi)$ (the constants 4 and $12K_\chi$ are valid but not sharp).

(2) *Tightness.* The embedding $H^{-s_0}_{\mathrm{per}}(Q) \hookrightarrow H^{-r}_{\mathrm{per}}(Q)$ is compact for $r > s_0$ (Rellich on the torus); by Chebyshev, $\nu^K_{k,J,\chi}(\|\cdot\|_{H^{-s_0}} > \rho) \leq C'_\chi/\rho^2$, and the H^{-s_0}-ball of radius ρ is compact in H_χ; so $(\nu^K_{k,J,\chi})_{k\geq 0,\,J\geq J_0(\chi)}$ is tight on the separable Hilbert space H_χ, and Prokhorov's theorem [7] gives weak limit points along k at fixed J.

(3) *Identification at fixed J: the full sequence, and where theorem 3.2 enters.* For a real trigonometric polynomial ψ, $\mathbb{E}^K_{k,J}e^{\mathrm{i}\langle\chi\Phi^K_{k,J},\psi\rangle} = Z^K_{k,J}(\mathrm{i}\chi\psi) \to Z^K_J(\mathrm{i}\chi\psi)$ as $k\to\infty$, along the full sequence, by theorem 7.6(b) (that is corollary 4.4(ii)) for the one-member family $(\chi\psi)$ at $J \geq J_0(\chi) \geq J_0(\chi\psi)$: this is theorem 3.2 for that family on King's cube, together with the local bound of theorem 7.6(a) and Montel's theorem. A Borel probability measure on H_χ is determined by its characteristic functional on the dual $H^r_{\mathrm{per}}(Q;\Lambda^2)$, which is continuous there (dominated convergence), hence by its values on the dense set of real trigonometric polynomials. So every weak limit point of $(\nu^K_{k,J,\chi})_k$ is the one probability measure $\nu^K_{J,\chi}$ on H_χ with characteristic functional $\psi \mapsto Z^K_J(\mathrm{i}\chi\psi)$, and $\nu^K_{k,J,\chi} \Rightarrow \nu^K_{J,\chi}$ along the whole sequence $k\to\infty$ by tightness and the uniqueness of the limit point. Without theorem 3.2 the same argument gives the convergence along every continuum subsequence (theorem 14.9 and corollary 14.14), not along the full sequence; the full-sequence clause is what theorem 3.2 adds for King's printed state.

(4) $J\to\infty$. $\|\cdot\|^2_{H^{-s_0}_{\mathrm{per}}} = \sum_{\alpha,c}(1+|\alpha|^2)^{-s_0}|\langle\cdot,e_{\alpha,c}\rangle|^2$ is lower semicontinuous on H_χ (a supremum of finite sums of continuous functions), so by the portmanteau theorem [7] the bound (26) passes to $\nu^K_{J,\chi}$: $\int\|\cdot\|^2_{H^{-s_0}}\,d\nu^K_{J,\chi} \leq C'_\chi$ for every $J \geq J_0(\chi)$; hence $(\nu^K_{J,\chi})_J$ is tight on H_χ by the argument of (2), and $Z^K_J(\mathrm{i}\chi\psi) \to Z^K_\infty(\mathrm{i}\chi\psi) = \mathcal{Z}^K(\chi\psi)$ (theorem 7.6(b)) identifies every limit point: $\nu^K_{J,\chi} \Rightarrow \nu^K_\chi$, the measure with characteristic functional $\psi \mapsto \mathcal{Z}^K(\chi\psi)$.

(5) *ν^K_χ is the law of $\chi\Phi$ under μ^K_S.* For $\Phi \in X$, $\chi\Phi \in H^{-r}(\mathbb{R}^3;\Lambda^2)$ is supported in $\operatorname{supp}\chi$, compact in the interior of Q, so its periodisation is an element of H_χ with $\langle\chi\Phi,\psi\rangle = \Phi(\chi\psi)$ for Q-periodic smooth ψ; the map $\Phi \mapsto \chi\Phi$ is continuous from X to H_χ, since $|\Phi(\chi\psi)| \leq \|\tilde\chi\Phi\|_{H^{-r}}\|\chi\psi\|_{H^r} \leq c_\chi\|\tilde\chi\Phi\|_{H^{-r}}\|\psi\|_{H^r_{\mathrm{per}}(Q)}$ for a cut-off $\tilde\chi = 1$ on Q. And $\int e^{\mathrm{i}\langle\chi\Phi,\psi\rangle}\,d\mu^K_S = \int e^{\mathrm{i}\Phi(\chi\psi)}\,d\mu^K_S = \mathcal{Z}^K(\chi\psi)$ by Step 2(v).

(6) *Moments.* For every trigonometric polynomial ψ (indeed every $\psi \in H^r_{\mathrm{per}}(Q;\Lambda^2)$, by the uniform-integrability argument of Step 2(ii) with $\chi\psi_n \to \chi\psi$), all moments of $\langle\chi\Phi^K_{k,J},\psi\rangle = F_{\Lambda_k}(\chi\psi)$ are bounded uniformly in k and $J \geq J_0(\chi)$ (Step 1; $|\chi\psi|_4 \leq d_\chi\|\psi\|_{C^4}$ by Leibniz), so by uniform integrability the moments of $\nu^K_{J,\chi}$ are the Taylor coefficients of Z^K_J at $\chi\psi$, and those of ν^K_χ are the $S^K_n(\chi\psi,\dots,\chi\psi)$, consistently with (25). This proves (a). □

Remark 11.2 (why the convergence clause is local). The convergence $Z^K_{k,J}(\mathrm{i}u) \to Z^K_J(\mathrm{i}u)$ is available only for sources u whose plaquettes have their corners in X_J, that is for u at distance > 1 inside C_J, where theorem 3.2 speaks. For a u reaching ∂C_J the lattice functional is defined only after a convention for the plaquettes with a corner outside X_J (for instance, drop them), and for it no bound uniform in k and no convergence is asserted: the second-moment bound of theorem 7.6(a) carries the corner hypothesis, and the crude bound $\langle e^{F(\tilde u)}\rangle^K_{k,J} \leq e^{\frac{1}{2}\mu_0^{-2}\|d^*_k\tilde u\|^2_{2,k}}$

(lemma 2.8 with $C \le \mu_0^{-2}$ as forms) is not uniform in k for a source $\tilde u$ truncated at ∂C_J: $d_k^*\tilde u$ is of order $\|u\|_\infty/\varepsilon_k$ on the $O(\varepsilon_k^{-2}|\partial C_J|)$ bonds of the truncation layer, so $\|d_k^*\tilde u\|_{2,k}^2 = O(\varepsilon_k^{-1})$. In particular the fixed-J laws of the unlocalised lattice field on X are not asserted to be tight; nothing below depends on them.

11.2. **Proof of (b) and (c).** *Step 3 ((H1)–(H3)).* The hypotheses of the abstract part of [24] are: (H1) a Polish space with a jointly continuous $\mathbb{R}^3$-action and three involutive homeomorphisms θ_j with $\theta_j\tau_a = \tau_{\rho_j a}\theta_j$, and μ invariant under all of them; (H2) local σ-algebras $\Sigma(U)$ with monotonicity, $\Sigma(U) = \bigvee_{K\Subset U}\Sigma(K)$, $\bigvee_K \Sigma(K) = \mathcal{B}$, countable generation by bounded continuous functions, and covariance under $U(a)$ and Θ_j; (H3) $\mathbb{E}_\mu[\overline{F\circ\theta_{j,c}}\,F] \ge 0$ for $F \in L^2(\Sigma(H^+_{j,c}),\mu)$. On X with the actions of the Conventions, the structural parts of (H1) and all of (H2) are proved in [24, Prop. 10.4] by arguments that use only the space and the push-forward actions (joint continuity of the translations on H^{-s}_{loc}; the θ_j are linear homeomorphisms; Lusin–Souslin for $\bigvee_K\Sigma(K) = \mathcal{B}$; countable dense families $\arctan\Phi(f_n)$; $\operatorname{supp}\tau_a f = \operatorname{supp} f + a$, $\operatorname{supp}\sigma_j\cdot f = \sigma_j \operatorname{supp} f$); they hold verbatim for our X, that paper's space restricted to the curvature summand. *Invariance of μ_S^K:* by the uniqueness in Step 2(v) it suffices that $\mathcal{Z}^K(\tau_a u) = \mathcal{Z}^K(u)$ and $\mathcal{Z}^K(\sigma_j\cdot u) = \mathcal{Z}^K(u)$ for real u; the first is corollary 8.3 at the coefficient i, the second theorem 8.5 at $z = \mathrm{i}$ with $O = \sigma_j$, and the same theorem with $O = \tau_b g \tau_{-b}$ gives the O_h-invariance about every point. *(H3):* for $E_a = e^{\mathrm{i}\Phi(u_a)}$ with u_a real in $C_c^\infty(\{x_j > c\};\Lambda^2)$, $\overline{E_a\circ\theta_{j,c}}\,E_b = e^{\mathrm{i}\Phi(u_b - \rho_{j,c*}u_a)}$ and

$$\mathbb{E}_{\mu_S^K}\big[\overline{E_a\circ\theta_{j,c}}\,E_b\big] = \mathcal{Z}^K(u_b - \rho_{j,c*}u_a) = Z^K_\infty(\theta\zeta_a^* \sqcup \zeta_b), \qquad \zeta_a = (\mathrm{i}, u_a),$$

the conjugated coefficient $-\mathrm{i}$ sitting on the reflected source as in definition 9.1. By lemma 10.8, for the plane $x_j = c$ and every $c \in \mathbb{R}$, this Gram matrix is positive semidefinite. The exponential algebra is dense in $L^2(\Sigma(H^+_{j,c}),\mu_S^K)$ [24, Lemma J.2] (an algebra of bounded complex functions, closed under conjugation and containing the constants, is dense in L^2 of the σ-algebra it generates), and $\theta_{j,c}$ preserves μ_S^K, so (H3) follows as in [24, Prop. 10.4]. Finally [24, Lemma 10.5(d)] carries (H1)–(H3) and the O_h-invariance about every point from (X,μ_S^K) to $(\mathcal{H}^{-r},\mu^{K,H})$.

Step 4 (weak clustering). The standing hypothesis of the abstract part of [24] is that μ is any measure satisfying (H1)–(H3); so [24, Thm. 10.10] gives the ergodic decomposition $\mu_S^K = \int \mu_\xi\, d\nu$ with components translation-invariant and ergodic, θ_j-invariant, satisfying (H3) and (H2); [24, Thm. 10.19, Thm. 10.20] give clustering in the mean and under diagonal separation; [24, Prop. 10.16] is the reconstruction in direction j, and on an ergodic component the joint fixed space is $\mathbb{C}\Omega_j$ [24, Prop. 10.17(iii)]. The proof of [24, Cor. 10.43(iv)] uses only the E0″ bound in the form $\mathbb{E}_{\mu_S} Z_{\alpha,c}^{2n} \le c_S C_0^{2n}(2n)!\,|h_{\alpha,c}|_4^{2n}$ on $Z_{\alpha,c} = \Phi(h_{\alpha,c})$, together with [24, Lemma 10.5(a)], Jensen, Markov and Borel–Cantelli; theorem 7.6(c) supplies that bound on $\mathcal{S}$ through (25), with $(e\pi/2)^{2n}((2n)!)^{1/2} \le (e\pi/2)^{2n}(2n)!$, so $c_S = 1$, $C_0 = e\pi/2$, and the growth bound of (b) holds for our components with the same proof. [24, Thm. 10.12] assumes in addition that O_h acts on X by homeomorphisms with $\rho_j = \theta_j$, $g\tau_a g^{-1} = \tau_{ga}$, μ invariant under every g, and $G \mapsto G\circ g^{-1}$ mapping $\Sigma(U)$-measurable functions to $\Sigma(gU)$-measurable ones: structural for the push-forward action ($(g\Phi)(u) = \Phi(g^{-1}\cdot u)$ is a linear homeomorphism of X, the relations are identities on test functions, $\operatorname{supp} g\cdot u = g\operatorname{supp} u$) together with the invariance of Step 3; it gives the cubic-invariant decomposition with $k(\zeta) \in \{1,2,3,6\}$. Weak OS4 is [24, Def. 2.12(iii)], ergodicity for the action of $\mathbb{R}^3$, the ergodicity clause of the decomposition for ν-a.e. component.

Step 5 (the rotation corollary). [24, Cor. 10.44] assumes (H1)–(H3) and that $O(3)$ acts on X by homeomorphisms $x \mapsto Rx$, continuously in R for each x, with $\rho_j x = \theta_j x$ for the coordinate reflections, $R\tau_a R^{-1} = \tau_{Ra}$, $G \mapsto G\circ R^{-1}$ mapping $\Sigma(U)$-measurable functions to $\Sigma(RU)$-measurable ones, and μ invariant under every $R \in O(3)$. The structural parts hold for $(R\Phi)(u) = \Phi(R^{-1}\cdot u)$ on X: the action is by linear homeomorphisms, continuous in R, and covariant for the local σ-algebras, as noted in [24, Rem. 10.45(1)] for $H^{-s}_{\mathrm{loc}}(\mathbb{R}^3;\mathcal{V})$, and the push-forward respects the restriction to the Λ^2 summand. The invariance of μ_S^K is, by Step 2(v), $\mathcal{Z}^K(R\cdot u) = \mathcal{Z}^K(u)$ for real u, the hypothesis of (c), that is the $O(3)$-invariance of Z^K_∞ at imaginary coefficients, supplied by theorem 8.7; this section does not derive it. The conclusions are those quoted in (c), proved there from the R-invariance and (H3) (reflection positivity about every plane), the argument

of [24, Thm. 10.10] with a countable dense subgroup (the Euclidean invariance and reflection positivity of a.e. component), [24, Thm. 10.20] along $(1,1,1)/\sqrt{3}$ after a rotation (clustering in every manner), and [24, Prop. 10.41(a)] in a frame containing n ($\ker H_n = \mathbb{C}\Omega_n$). They contain nothing on OS3: the vanishing of the odd curvature moments of a component is not among them, and is not known for a single component.

Step 6 (curvature alone). Everything above runs on $X = H^{-r}_{\mathrm{loc}}(\mathbb{R}^3;\Lambda^2)$ for the curvature field alone: the regularity input is theorem 7.6, the reflection positivity input lemma 10.8, the invariance input theorem 8.5, the $O(3)$ input that of (c). The result is an ergodic decomposition of the law of the curvature field; nothing is claimed about the components of a joint law with the Wick square, and the single-axis statements of [24] that rest on a torus-specific domination are not claimed for King's state. □

Remark 11.3 (the hypotheses of the cited results, matched). For each result of [24] used above, the step that supplies its printed hypotheses. (H1)–(H3): Step 3 (structure from prop:discharge; invariance from corollary 8.3 and theorem 8.5; positivity from lemma 10.8 and lem:A:mc). lem:D:realize, uniqueness part (a measure on X with a given characteristic functional; Lusin–Souslin, lem:A:LS): Step 2(v). prop:discharge (the space and the push-forward actions; "Determination by S"): Steps 3 and 2(v). lem:D:hermite: (a) no hypothesis; (b) the two-point bound $\mathbb{E}\Phi(f)^2 \le C\|f\|^2_{10}$, Step 2(ii); (c) $|m_n| \le c_S(n+1)K^n n!$, $r \ge 14$ even, Step 2(iv) with $K = \sqrt{2}\,e\|g\|_2$, $c_S = 1$; (d) (H1)–(H3) on X, Step 3. The standing hypothesis "μ is any measure satisfying (H1)–(H3)" of thm:D:decomp, thm:D:mean, thm:D:diag, prop:D:recon, prop:D:fixed: Step 3; the ergodicity needed in prop:D:fixed(iii) holds on a component by thm:D:decomp. add:Ohdecomp (the O_h action and invariance): Step 4 and theorem 8.5. cor:D:schwinger(iv) (the E0″ bound on Hermite functions; lem:D:hermite(a)): theorem 7.6(c) through Step 2(ii), $c_S = 1$, $C_0 = e\pi/2$. p:cor:rotcor (the $O(3)$ action and invariance): Step 5, the structure from add:rotcond-rem(1), the invariance from the hypothesis of (c). def:OS4, add:OS4(i)–(iii): definitions, Step 5; add:OS4(iii) is the ergodicity clause of thm:D:decomp. prop:C-tight, mechanism only (uniformly bounded second moments of $\Phi(\chi e_\alpha)$): Step 2′(1). lem:A:LS, lem:A:mc: Steps 2(iv),(v), 3. The results of [24] stated for its own limit states (its clustering theorem, its Schwinger-function corollary as stated, its remark on components) are not cited; their abstract content is in the items above, and the odd-moment clause is not transferred to the components (clause (c)).

Remark 11.4 (the comparison family). The same conclusions hold for the limit states of the comparison family $\mathrm{ND}^{(a)}$: for every continuum subsequence $\mathcal{K}$ (definition 14.2) the limit state $Z^{\mathcal{K}}_\infty$ carries a measure with the properties of theorem 11.1(a) and (b), the local lattice laws converging along $\mathcal{K}$ instead of the full sequence, and (c) (theorem 14.9, clauses (v)–(vii)). By theorem 3.2 every such limit state equals Z^K_∞ (corollary 4.4(iii)), so for King's printed state the two routes to the measure agree. For the iterated limit of the comparison family itself, when it is convergent at fixed cubes (definition 3.1), see proposition 14.12(b); nothing beyond corollary 4.4(iii) is asserted about it here, the two existence clauses not being interchangeable by the inputs of this paper (corollary 4.4(iv)).

12. The Wick tube I: torus inputs and the extensive bound

The Wick leg controls the joint functional of the curvature and the Wick square on the tube $\mathcal{D}_{\sigma_+,H}$ of definition 2.5. Its positive leg (sources $q \ge 0$) and negative leg (sources $-q$) are proved in the next section from three inputs on *King's tori*: lattice reflection positivity (proposition 12.2), the chessboard estimate it yields (lemma 12.4), and an *extensive bound* (I2) for constant positive scalar sources in the fixed-spacing volume limit (definition 12.5). This section proves the first two and derives the third from the torus inputs proposition 3.8 and lemma 3.9 and the sign clause (T3) (proposition 12.6); a variant with the auxiliary mass on either side of m (proposition 12.11) serves the discussion of the coupling range; no continuum limit is taken here.

12.1. **King's tori, the two conventions and mid-plane pairs.**

Definition 12.1 (King's tori)**.** A *King torus* is $T = \mathbb{R}^3/(\ell\mathbb{Z})^3$ with $\ell \in M\mathbb{N}$ even; $|T| := \ell^3$ is its volume, and $T_J := \mathbb{R}^3/(2MJ\mathbb{Z})^3$, $J \in \mathbb{N}$, is the torus with fundamental cell C_J. Since $\ell\mathbb{Z}^3 \subset \mathbb{Z}^3 \subset \Lambda_k$, the lattice Λ_k descends to T, with $\ell\, 5^k$ sites per side, an even number, and $|T|/\varepsilon_k^3$ sites in all. The *periodic state* $\langle\cdot\rangle_{k,T}$ is King's action of Section 2 summed over the finite graph $\Lambda_k/(\ell\mathbb{Z})^3$, with periodic conditions for both fields [14, (4.1)], [13, (2.21)]: Feynman-form gradient terms for every pair of parallel neighbouring bonds, the photon mass μ_0^2, hopping terms on every bond, the single-site weights $e^{-\varepsilon^3[\frac{1}{2}(m^2+\delta m^2)r^2+\lambda r^4]}$. Two normalisations of the counterterm are used:

(i) the *free convention* of Section 2 ($\delta m^2 = \delta m_\varepsilon^2$, Wick constant W_ε; the convention of every cube state of this paper), with $N_{k,T}^a(0,q) := e^{-(a_k+2\mathsf{W}_\varepsilon)\mathsf{S}(q)}\langle e^{\sum_x \varepsilon^3 q(x) r_x^2}\rangle_{k,T}$ for a real function q on the sites of T, exactly as on boxes (definition 2.3), and

$$\alpha_{k,T}^a(u) := |T|^{-1} \ln N_{k,T}^a(0, u\mathbf{1}_T) \qquad (u \in \mathbb{R}),$$

the *free energy density* of a constant source;

(ii) *King's torus convention* [13, (2.4), (3.33), (3.35)]: the counterterm $\delta m_{k,T}^2(M)$ and the Wick constant $c_{k,T}(M)$ computed with the periodic propagators of T at scalar mass M, together with the field-independent constants $E(M) = E_0(M) + E_1(M)$ of King's torus action. $Z_{k,T}^{(M)} := \int e^{-S}$ is the source-free partition function of this model at scalar mass M (bare coefficient $M^2 + \delta m_{k,T}^2(M)$; proposition 3.8 is a statement about these numbers), and $N_{k,T}^{z,\mathrm{tor}}(0,q)$ the normalised source functional of the torus model at mass m in the reference scheme z, with $c_{k,T} = c_{k,T}(m)$ in place of W_ε.

In either convention the coefficients are the same at every site of T. A *mid-plane pair* is $\pi = \{x_\nu = c\} \cup \{x_\nu = c + \ell/2\}$ with $\nu \in \{1,2,3\}$ and $c \in \varepsilon_k(\mathbb{Z} + \frac{1}{2})$ (both planes lie between lattice planes, since $\ell 5^k$ is even); θ denotes the reflection of T in π, which exchanges the two slabs of $T \setminus \pi$ ("the two sides of π"), acting on the fields as in definition 9.1.

12.2. **Reflection positivity of King's tori.**

Proposition 12.2 (lattice reflection positivity of King's tori)**.** *Let T be a King torus, $k \geq 0$, π a mid-plane pair and θ the reflection in π. For every finite family $G_1, \dots, G_N$ of gauge-invariant Borel functions of the fields on one side of π with $\langle |G_i|^2\rangle_{k,T} < \infty$ and every $\gamma \in \mathbb{C}^N$,*

$$\sum_{i,l=1}^{N} \bar\gamma_i \gamma_l \, \langle \theta(G_i)\, G_l\rangle_{k,T} \;\geq\; 0.$$

The periodic state is invariant under θ and under all lattice translations. This holds in both conventions of definition 12.1: reflection positivity does not see the value of the (site-independent) counterterm.

Proof. The proof is the mechanism of lemma 9.3 with a pair of planes in place of one and without Dirichlet diagonals, in three steps.

Step 1: from the Feynman form to the Proca form. On the torus write $A = Z + d_0\chi$ with $d_0^T Z = 0$, $\chi \perp 1$ (Hodge decomposition on the finite periodic graph). The Feynman form equals $\frac{1}{2}\langle Z, (d_1^T d_1 + \mu_0^2)Z\rangle$ plus a quadratic form in χ, and the Proca form $\frac{1}{2}\langle A, (d_1^T d_1 + \mu_0^2)A\rangle$ equals the same Z-form plus another χ-form; both χ-forms are positive definite on $1^\perp$ because $\mu_0 > 0$. For gauge-invariant G the matter integral at fixed A is invariant under $A \mapsto A + d_0\chi$ (rotate ϕ site by site), so the χ-integrals factor and cancel in the normalised expectation: gauge-invariant expectations in the Feynman form equal those in the Proca form. Fubini's theorem applies by $\mu_0 > 0$, $\lambda > 0$ and a growth bound $|G| \leq Ce^{a\sum_b |A(b)|} e^{C\sum_x r_x^2}$, satisfied by bounded G and by the observables $\Theta(\zeta)$; so this step uses $\mu_0 > 0$, gauge invariance, $\lambda > 0$ and the growth bound, and nothing about the value of the counterterm.

Step 2: the Proca torus measure is reflection positive about π. In the Proca form the action couples the two sides of π only through the plaquettes and hopping terms crossing one of the two

planes of π (no gradient terms between parallel bonds across a plane; the mass term is local). Write each crossing bond variable as $a_b = \alpha_b - \alpha'_b$ with α_b, α'_b independent centred Gaussians of variance $(2\mu_0^2\varepsilon^3)^{-1}$, attached to the two sides. Then θ becomes the swap of the two data sets $\xi = (\omega_+, \alpha)$, $\xi' = (\omega_-^\theta, \alpha')$, whose a priori measures coincide by the θ-invariance of the one-sided terms, and exactly as in Step 2 of lemma 9.3 every crossing plaquette term is $\frac{\varepsilon}{2}X_p(\xi)^2 + \frac{\varepsilon}{2}X_p(\xi')^2 - \varepsilon X_p(\xi)X_p(\xi')$ and every crossing hopping term is $-\varepsilon\sum_{i=1,2}\psi_b^i(\xi)\psi_b^i(\xi')$, with the same real functions on both copies. So the Boltzmann factor is $\mathcal{G}(\xi)\mathcal{G}(\xi')\exp(\sum_\iota \gamma_\iota Y_\iota(\xi)Y_\iota(\xi'))$ with all $\gamma_\iota = \varepsilon > 0$ and all Y_ι real, and for G a function of one side $\theta(G)G = \overline{G(\xi')}G(\xi)$. Expanding the exponential, $Z\langle\theta(G)G\rangle$ is a sum with nonnegative coefficients of squared moduli $|\int d\xi\, G(\xi)\mathcal{G}(\xi)\prod_\iota Y_\iota(\xi)^{n_\iota}|^2$, the interchange of sum and integral being justified as in lemma 9.3 ($\lambda > 0$, $\mu_0 > 0$ and the growth bound). The Hermitian property $\overline{\langle\theta(G')G\rangle} = \langle\theta(G)G'\rangle$ follows from the θ-invariance of the measure, so $(G, G') \mapsto \langle\theta(G')G\rangle$ is a positive semidefinite Hermitian form on bounded gauge-invariant one-sided G.

Step 3: L^2 observables. For gauge-invariant one-sided G with $\langle|G|^2\rangle < \infty$ apply Step 2 to $G_n := G\mathbf{1}\{|G| \le n\}$; since $|\theta(G_n)G_n| \le |\theta G||G|$ and $\langle|\theta G||G|\rangle \le \langle|G|^2\rangle$ by Cauchy–Schwarz and θ-invariance, dominated convergence gives the inequality for $G = \sum_i \gamma_i G_i$, hence the form statement.

Invariance. Under θ and under every lattice translation each term of the periodic action, in either convention, goes to a term of the same kind with the same coefficient, and the a priori measure is invariant; the value of the site-independent bare coefficient enters nowhere above. □

Remark 12.3 (the torus paper and the literature). (i) The torus paper [24] proves proposition 12.2 for its lattice model, King's action on the torus with a constant bare coefficient b_T, for every even side, every direction, both kinds of plane pair and every b_T [24, Prop. 7.9(a)]; its printed scope is $\mu_0 \in (0, \mu_*]$, where $\mu_* = m$ is the photon mass at which that paper freezes its counterterm. The restriction plays no role in the proof, which uses no bound on μ_0: the two lemmas behind it there, the passage to the Proca form [24, Lemma 7.7] and reflection positivity of the Proca torus measure in a pair of planes [24, Lemma 7.8(ii)], are stated for every photon mass $\mu > 0$, with the inputs site-independent coefficients, $\lambda > 0$ and even sides. So the statement lifts to every $\mu_0 > 0$, as the proof above also shows directly. (ii) [6, Cor. 5.4] (reflection positivity "in every gauge", for covariances $C_F = C_u + dFd^* \ge 0$ with F of either sign) is the same mechanism: on the torus the Feynman covariance has this form with $F = -\mu_0^{-2}(\Delta_0 + \mu_0^2)^{-1} \le 0$ on functions orthogonal to the constants. It is not a literal instance (the region of [6, §5] is a bounded subset of the lattice with a boundary condition, and reflection positivity of periodic conditions is asserted there in a remark), and proposition 12.2 does not rest on it. (iii) The chessboard estimate uses proposition 12.2 for functions of the r_x only; nothing in this section uses reflection positivity of a cube or of a limit state.

12.3. **The chessboard estimate on King's tori.**

Lemma 12.4 (chessboard estimate)**.** *Let T be a King torus at spacing $\varepsilon = \varepsilon_k$, a a scheme, and q a real function on the sites of T. In the free convention,*

$$\ln N^a_{k,T}(0, q) \;\le\; \sum_x \varepsilon^3\, \alpha^a_{k,T}(q(x)), \tag{27}$$

and $u \mapsto \alpha^a_{k,T}(u)$ is convex on $\mathbb{R}$ with $\alpha^a_{k,T}(0) = 0$. In King's torus convention $u \mapsto |T|^{-1}\ln N^{z,\mathrm{tor}}_{k,T}(0, u\mathbf{1}_T)$ is likewise convex with value 0 at $u = 0$.

Proof. We apply the chessboard estimate [8, Thm. 4.2] (lattice-gauge form: [6, Thm. 2.10]), whose hypotheses we check: the site set of T is a rectangular lattice box with sides of even length $\ell 5^k$; the periodic state is invariant under translations modulo the sides and reflection positive with respect to the reflections in all planes perpendicular to the coordinate axes running midway between neighbouring sites (proposition 12.2; on the torus such a reflection is the reflection in a mid-plane pair, the antipodal plane $c + \ell/2$ being again a mid-plane). Take as the "spin at x"

the gauge-invariant variable r_x and

$$G_x := \exp\big(\varepsilon^3 q(x)(r_x^2 - 2\mathsf{W}_\varepsilon - a_k)\big), \qquad \text{so that} \qquad \Big\langle \prod_x G_x \Big\rangle_{k,T} = N^a_{k,T}(0,q).$$

The G_x are positive with all moments ($\lambda > 0$), and the Schwarz inequality for such unbounded gauge-invariant functions, all that [8, Thm. 4.1] uses beyond cyclic invariance, follows from the bounded case by truncation and monotone convergence. The conclusion is $|\langle\prod_x G_x\rangle| \le \prod_x \langle\prod_y G_x^{(y)}\rangle^{1/|\Lambda|}$, $G_x^{(y)}$ the copy of G_x carried to y by reflections, $|\Lambda| = |T|/\varepsilon^3$ the number of sites. Because W_ε, a_k and δm_ε^2 are the same at every site, $\prod_y G_x^{(y)} = \exp(q(x)\sum_y \varepsilon^3(r_y^2 - 2\mathsf{W}_\varepsilon - a_k))$ has expectation $N^a_{k,T}(0, q(x)\mathbf{1}_T) = e^{|T|\alpha^a_{k,T}(q(x))}$ in the same state; taking logarithms gives (27). Convexity: $|T|\alpha^a_{k,T}(u)$ is the logarithm of a Laplace transform in the linear parameter u plus a linear term, and vanishes at $u = 0$; the same computation gives the convexity statement in the torus convention. □

12.4. **Constant sources on large tori: the extensive bound.**

Definition 12.5 (the extensive bound (I2))**.** We say that (I2) (one-sided, in the volume limit at fixed spacing; also written (UVS$^+_\infty$)) holds for the scheme a in the free convention if there are $s_1 > 0$ and $C_U \in [0,\infty)$ such that

$$\sup_{k\ge 0} \limsup_{J\to\infty} \alpha^a_{k,T_J}(s_1) \;\le\; C_U. \tag{28}$$

The positive leg of the next section consumes exactly this, with $\kappa := C_U/s_1$ (by convexity and $\alpha(0) = 0$, $\limsup_J \alpha^a_{k,T_J}(u) \le \kappa u$ for $0 \le u \le s_1$). Throughout, $M_1 \in (0, m)$ is an auxiliary scalar mass, $\alpha_k^\infty(M) = M^2 + \delta m_k^2(M)$ is King's free-boundary bare coefficient on Λ_k at scalar mass M (lemma 3.9; it is the fixed-k volume limit of the torus coefficient $M^2 + \delta m^2_{k,T_J}(M)$, lemma 12.9), and, in the reference scheme z,

$$s_k^\infty(M_1) := \tfrac12\big(\alpha_k^\infty(m) - \alpha_k^\infty(M_1)\big), \qquad \sigma_1 = \sigma_1(M_1) := \liminf_{k\to\infty} s_k^\infty(M_1),$$

$$B^z_{k,T}(M) := E(M) - E(m) - \tfrac12\big(2c_{k,T} + z_k\big)\big(\alpha^{\mathrm{tor}}_{k,T}(m) - \alpha^{\mathrm{tor}}_{k,T}(M)\big)|T|,$$

$$b_k^\infty(M_1) := \lim_{J\to\infty} |T_J|^{-1} B^z_{k,T_J}(M_1), \qquad \beta_1 = \beta_1(M_1) := \limsup_{k\to\infty} b_k^\infty(M_1),$$

where $\alpha^{\mathrm{tor}}_{k,T}(M) := M^2 + \delta m^2_{k,T}(M)$ is the bare coefficient of King's torus convention and the limit defining b_k^∞ exists by lemma 12.9. The constant $\beta_1(M_1)$ is the second constant of lemma 3.9, which says that $\sigma_1(M_1)$ and $\beta_1(M_1)$ are finite; (T3) says that $\sigma_1(M_1) > 0$.

Proposition 12.6 ((I2) from the torus inputs)**.** *Assume* (T3) *(Hypothesis 3.11); the torus inputs are proposition 3.8 at the two masses m and M_1 and lemma 3.9. Then* (I2) *holds in the reference scheme z for every $s_1 \in (0, \sigma_1)$, with a constant $C_U = C_U(s_1) < \infty$ depending on $(e, \lambda, m, \mu_0, M)$, on M_1 and on s_1. For a scheme $a \ne z$ with $\sup_k |z_k - a_k| < \infty$ (an admissible scheme, definition 3.6) it holds with the constant $C_U + s_1 \sup_k |z_k - a_k|$.*

The proof assembles four lemmas on King's torus model at fixed spacing, stated in the form used and with their short proofs where elementary. Notation at fixed k (spacing ε) on $T = T_J$: $f^{\mathrm{tor}}(u) := \ln N^{z,\mathrm{tor}}_{k,T}(0, u\mathbf{1})$ (King's torus model at mass m, Wick constant $c_{k,T}$, scheme z) and $f^F(u) := \ln N^z_{k,T}(0, u\mathbf{1})$ (free convention); both are convex with $f(0) = 0$ (lemma 12.4). Write $\alpha^{\mathrm{tor}} := m^2 + \delta m^2_{k,T}(m)$, $\alpha^F := m^2 + \delta m_\varepsilon^2$, $\sigma_{k,T} := \frac12(\alpha^{\mathrm{tor}} - \alpha^F)$ and $\Delta c_T := \mathsf{W}_\varepsilon - c_{k,T}$ (the two *convention offsets*), and $s_{k,T}(M_1) := \frac12(\alpha^{\mathrm{tor}}_{k,T}(m) - \alpha^{\mathrm{tor}}_{k,T}(M_1))$.

Lemma 12.7 (mass shift)**.** *If $M'' > 0$ satisfies $\alpha^{\mathrm{tor}}_{k,T}(M'') = \alpha^{\mathrm{tor}}_{k,T}(m) - 2u$, then*

$$N^{z,\mathrm{tor}}_{k,T}(0, u\mathbf{1}) = e^B\, Z^{(M'')}_{k,T} / Z^{(m)}_{k,T}, \qquad B = E(M'') - E(m) - (2c_{k,T} + z_k)\, u\, |T|.$$

Proof. The source $u\sum_x \varepsilon^3 r_x^2$ shifts the bare coefficient of the torus model at mass m by $-2u$, that is to the bare coefficient of the model at mass M''; beyond the fixed terms, King's torus action [13, (2.4)] contains only the counterterm and the field-independent constants E_0, E_1, so the two Boltzmann factors differ by the constant $e^{E(M'')-E(m)}$, and the Wick and scheme multipliers $e^{-(2c_{k,T}+z_k)u|T|}$ of $N^{z,\mathrm{tor}}$ are bookkeeping. □

Lemma 12.8 (chord)**.** *If f is convex on $\mathbb{R}$ with $f(0)=0$, then $f(u)\le (u/s)f(s)$ for $0\le u\le s$, $s>0$.*

Proof. $u=(1-u/s)\cdot 0+(u/s)\cdot s$ and $f(0)=0$. □

Lemma 12.9 (the volume limit at fixed spacing)**.** *Fix k. As $J\to\infty$: $\alpha^{\mathrm{tor}}_{k,T_J}(M)\to\alpha^{\infty}_k(M)$ uniformly for $M\in[M_1,m]$; $s_{k,T_J}(M_1)\to s^{\infty}_k(M_1)$; $|T_J|^{-1}B^z_{k,T_J}(M_1)\to b^{\infty}_k(M_1)$, a finite limit; and the offsets vanish, $\sigma_{k,T_J}\to 0$ and $\Delta c_{T_J}\to 0$.*

The proof is given in Appendix F (appendix F.3); the reason the statement holds: $\delta m^2_{k,T}$ and E_1 ([13, (3.32)–(3.33), (3.35)]) are finite sums of lattice graphs with translation-invariant massive propagators $(m,\mu_0>0)$, which on the torus are exact image sums of the propagators of $\varepsilon\mathbb{Z}^3$, each decaying exponentially in $|x|/\varepsilon$ at fixed ε, so dominated convergence over the images gives the limits; $E_0/|T|$ is a Riemann sum of a smooth periodic log-determinant. No rate and no uniformity in the source are claimed or used. The boundedness in $J\ge J_*(k)$ of $\delta m^2_{k,T_J}$ and c_{k,T_J}, used in lemma 12.10, is a further clause of the same lemma, not one of those listed here. Of the clauses listed, Step 2 of the proof of proposition 12.6 uses the convergence of $s_{k,T_J}(M_1)$ and the $\limsup_J$ of $|T_J|^{-1}B^z_{k,T_J}(M_1)$, and Step 3 the two offset clauses.

Lemma 12.10 (the fixed-spacing extensive bound)**.** *Fix k and $v>0$. There is $C_k(v)<\infty$ such that for every King torus $T=T_J$ with $J\ge J_*(k)$, in either convention, $f(\pm v)\le C_k(v)|T|$; hence, by convexity, $|f(w)|\le C_k(v)\,|w|\,|T|/v$ for $|w|\le v$.*

Proof. Let $I(b):=\int_{\mathbb{C}} e^{-\varepsilon^3[\frac12 b|\phi|^2+\lambda|\phi|^4]}d^2\phi<\infty$ for real b ($\lambda>0$), α the bare coefficient and c the Wick constant of the convention in use, and Z_γ the photon Gaussian integral, finite because $\mu_0>0$ (at $\mu_0=0$ the constant mode is in the kernel of the Feynman form). The hopping term is ≥ 0, so dropping it bounds the numerator: $\int e^{-S+w\sum\varepsilon^3 r^2}\le Z_\gamma I(\alpha-2w)^{|T|/\varepsilon^3}$. The hopping term of a bond is $\le\varepsilon(r_x^2+r_y^2)$, and each site lies on six bonds, so the hopping part of the action is $\le 6\varepsilon^{-2}\sum_x\varepsilon^3 r_x^2=\sum_x\varepsilon^3\cdot\frac12(12\varepsilon^{-2})r_x^2$ and $\int e^{-S}\ge Z_\gamma I(\alpha+12\varepsilon^{-2})^{|T|/\varepsilon^3}$. Hence $\ln\langle e^{w\sum\varepsilon^3 r^2}\rangle\le\varepsilon^{-3}|T|[\ln I(\alpha-2w)-\ln I(\alpha+12\varepsilon^{-2})]$ and

$$f(w)\le\big\{\varepsilon^{-3}[\ln I(\alpha-2w)-\ln I(\alpha+12\varepsilon^{-2})]+(2|c|+|z_k|)|w|\big\}|T|.$$

In the free convention α and c do not depend on T; in the torus convention they converge as $J\to\infty$ and are bounded for $J\ge J_*(k)$ (the boundedness clause noted after lemma 12.9). So $C_k(v)$, the supremum over the sign and over $J\ge J_*(k)$ of the brace at $w=\pm v$, is finite. The last claim is convexity with $f(0)=0$: $f(w)\le(|w|/v)f(\pm v)$ and $f(w)\ge -f(-w)\ge-(|w|/v)f(\mp v)$. □

Proof of proposition 12.6. Step 1: the exact convention bridge. On the same torus the free-convention action differs from the torus-convention action by $\frac12(\alpha^F-\alpha^{\mathrm{tor}})\sum_x\varepsilon^3 r_x^2=-\sigma_{k,T}\sum_x\varepsilon^3 r_x^2$ plus field-independent constants. Hence $\langle e^{u\sum\varepsilon^3 r^2}\rangle^F=\langle e^{(u+\sigma)\sum\varepsilon^3 r^2}\rangle^{\mathrm{tor}}/\langle e^{\sigma\sum\varepsilon^3 r^2}\rangle^{\mathrm{tor}}$ with $\sigma=\sigma_{k,T}$, and inserting the Wick and scheme multipliers,

$$f^F(u)=f^{\mathrm{tor}}(u+\sigma)-f^{\mathrm{tor}}(\sigma)-2\,\Delta c_T\,u\,|T|. \tag{29}$$

Step 2: the bound in the torus convention. Let $0\le u<\sigma_1$ and $\eta>0$. By the definitions of σ_1 and β_1 there is $K_1=K_1(u,\eta)$, which we take at least as large as the tail of proposition 3.8, such that $s^{\infty}_k(M_1)>u$ and $b^{\infty}_k(M_1)<\beta_1+\eta$ for all $k\ge K_1$. Fix such a k. By lemma 12.9 there is $J_*\ge J_*(k)$ with $s:=s_{k,T_J}(M_1)>u$ for $J\ge J_*$. The chord (lemma 12.8) at this s gives $f^{\mathrm{tor}}(u)\le(u/s)f^{\mathrm{tor}}(s)\le[f^{\mathrm{tor}}(s)]_+$. Lemma 12.7 with $M''=M_1$ applies exactly, since $\alpha^{\mathrm{tor}}_{k,T_J}(M_1)=\alpha^{\mathrm{tor}}_{k,T_J}(m)-2s$ by the definition of s:

$$f^{\mathrm{tor}}(s)=B^z_{k,T_J}(M_1)+\ln Z^{(M_1)}_{k,T_J}-\ln Z^{(m)}_{k,T_J}\le B^z_{k,T_J}(M_1)+\big(E_+(M_1)+E_-(m)\big)|T_J|$$

by proposition 3.8, used as an upper bound at M_1 and as a lower bound at m. Divide by $|T_J|$ and let $J \to \infty$ (lemma 12.9 for the first term):

$$\limsup_{J\to\infty} |T_J|^{-1} f^{\mathrm{tor}}_{k,T_J}(u) \le \big[\beta_1 + \eta + E_+(M_1) + E_-(m)\big]_+ \qquad (k \ge K_1).$$

For each of the finitely many $k < K_1$, lemma 12.10 with any $v \ge u$ gives $\limsup_J |T_J|^{-1} f^{\mathrm{tor}}_{k,T_J}(u) \le C_k(v)u/v < \infty$. Fix $\eta := 1$ (any $\eta > 0$ serves); then

$$C^{\mathrm{tor}}(u) := \sup_{k\ge 0} \limsup_{J\to\infty} |T_J|^{-1} f^{\mathrm{tor}}_{k,T_J}(u) < \infty \qquad \text{for every } u \in [0,\sigma_1).$$

Step 3: transfer to the free convention. Fix $u \in [0,\sigma_1)$ and $p > 1$ with $pu < \sigma_1$, and let $p' = p/(p-1)$. Since $u+\sigma = \frac{1}{p}(pu) + \frac{1}{p'}(p'\sigma)$, convexity gives $f^{\mathrm{tor}}(u+\sigma) \le \frac{1}{p} f^{\mathrm{tor}}(pu) + \frac{1}{p'} f^{\mathrm{tor}}(p'\sigma)$, whatever the sign of $\sigma = \sigma_{k,T_J}$. Once $p'|\sigma_{k,T_J}| \le v$, which holds for all large J by lemma 12.9, lemma 12.10 gives $|f^{\mathrm{tor}}(p'\sigma)|, |f^{\mathrm{tor}}(\sigma)| \le C_k(v) p' |\sigma_{k,T_J}| |T_J|/v$. Inserting this into (29),

$$|T_J|^{-1} f^F_{k,T_J}(u) \;\le\; \tfrac{1}{p}|T_J|^{-1} f^{\mathrm{tor}}_{k,T_J}(pu) + \tfrac{2C_k(v)p'}{v}|\sigma_{k,T_J}| + 2|\Delta c_{T_J}|\, u \;\xrightarrow[J\to\infty]{\limsup}\; \tfrac{1}{p} C^{\mathrm{tor}}(pu),$$

because $\sigma_{k,T_J} \to 0$ and $\Delta c_{T_J} \to 0$ at fixed k (lemma 12.9; the factor $\frac{1}{p'} \le 1$ in the middle term has been dropped). Hence $\sup_k \limsup_J \alpha^z_{k,T_J}(u) \le \frac{1}{p} C^{\mathrm{tor}}(pu) < \infty$ for every $u \in [0,\sigma_1)$. Take $u = s_1 \in (0,\sigma_1)$ and $C_U := [\inf_{p>1:\, ps_1<\sigma_1} \frac{1}{p} C^{\mathrm{tor}}(ps_1)]_+$; this is (28) in the scheme z. For a scheme $a \ne z$, $N^a_{k,T}(0,q) = e^{-(a_k - z_k)\mathsf{S}(q)} N^z_{k,T}(0,q)$ gives $\alpha^a_{k,T}(u) = \alpha^z_{k,T}(u) + (z_k - a_k)u \le \alpha^z_{k,T}(u) + |z_k - a_k|\, u$, so (28) holds with $C_U + s_1 \sup_k |z_k - a_k|$. □

Proposition 12.11 (the auxiliary mass on either side of m)**.** *Let $M_1 > 0$, $M_1 \ne m$. Assume proposition 3.8 at m and at M_1, and lemma 3.9 and* (T3) *read at this M_1: $s^\infty_k(M_1) = \frac{1}{2}(\alpha^\infty_k(m) - \alpha^\infty_k(M_1))$, of either sign, has $\sigma_1(M_1) = \liminf_k s^\infty_k(M_1) > 0$, and $\beta_1(M_1) < \infty$. Then* (I2) *holds in the scheme z for every $s_1 \in (0, \sigma_1(M_1))$, and for $a \ne z$ with the shifted constant, exactly as in proposition 12.6.*

Proof. The proof of proposition 12.6 is repeated with lemma 12.9 stated on the interval $I := [\min(M_1, m), \max(M_1, m)]$, and we check that no step uses $M_1 < m$. Lemma 12.7 is an identity for any M'' with $\alpha^{\mathrm{tor}}(M'') = \alpha^{\mathrm{tor}}(m) - 2u$, whatever the order of M'' and m. Lemma 12.8 needs $0 \le u \le s$ only. Lemma 12.9: its proof uses only that the propagators are translation-invariant massive lattice covariances at fixed ε with masses in $I \cup \{\mu_0\}$, bounded below by $\min(M_1, m, \mu_0) > 0$, so it holds on I word for word; Step 2 uses it at the two masses m and M_1 only. Lemma 12.10 carries no mass order, and (29) is an identity. In Step 2 the sign clause is $s^\infty_k(M_1) > u \ge 0$, the chord is applied at $s = s_{k,T_J}(M_1) > u \ge 0$, lemma 12.7 at $M'' = M_1$ is exact by the definition of s, and proposition 3.8 is used as an upper bound at M_1 and a lower bound at m; none of this uses $M_1 < m$. Step 3 carries no mass. □

The two-sided variant matters exactly on the set $E^{\le}_W \setminus E_W$ of couplings at which m minimises α_∞ on $(0, m]$ but not on $(0, \infty)$ (lemma 22.1; for instance when $\alpha'_\infty(m) < 0$ and m minimises α_∞ on $(0, m]$): there (T3) fails for every $M_1 < m$ while some $M_1 > m$ has $\sigma_1(M_1) > 0$. The main results use proposition 12.6 with $M_1 \in (0, m)$, as (T3) is posed.

Remark 12.12 (the convention of proposition 3.8 and how much of it is used). (i) Proposition 3.8 is posed in King's torus convention (counterterm and Wick constant computed with the periodic propagators of the torus itself), not in the normalisation of Bałaban's own papers. This is what the proof needs: the offsets σ_{k,T_J}, Δc_{T_J} between King's torus convention and the free convention vanish in the volume limit at fixed spacing (lemma 12.9), and Step 3 of the proof of proposition 12.6 transfers the bound through them; between other normalisations they need not vanish. (ii) Only a tail of proposition 3.8 is used: Step 2 of that proof invokes it for $k \ge K_1$ and $J \ge J_*(k)$, the finitely many $k < K_1$ being handled by lemma 12.10 at fixed spacing, and the constants $E_\pm(M)$ are independent of k and of the torus. (iii) In the scheme z, C_U depends on the couplings, on M and on the auxiliary choices M_1, s_1, and on nothing else; neither it nor any other bound of this section is uniform in μ_0.

13. The Wick tube II: the uniform bound and its limits

This section proves the (k, J)-uniform bound (31) on the general-source functionals of the comparison cubes on the tube $\mathcal{D}_{\sigma_+,H}$ from the extensive bound (I2) (definition 12.5): the positive part of the Wick source is controlled through King's tori, the negative part through a tiling into cells of fixed size, each controlled by one instance of theorem 3.5 and the positive bound. Then come the limits on the tube, E0″, and the continuity and translation invariance of the tube object. States are those of the comparison convention $\mathrm{ND}^{(a)}$ (definition 2.1) unless King's convention is named; $N^a_{k,X}(0,q)$ is the functional of definition 2.3 on a finite $X \subset \Lambda_k$ or a King torus (definition 12.1), the torus with fundamental cell C_J being written $\mathbb{T}_J$ here (T_J is the top-layer set of King's RG cube), and $\alpha^a_{k,\mathbb{T}}(u) = |\mathbb{T}|^{-1} \ln N^a_{k,\mathbb{T}}(0, u\mathbf{1})$. $s_1 \in (0, \sigma_1)$ and C_U are the amplitude and constant of (I2), $\kappa := C_U/s_1$; the sources $f_1, \dots, f_n$, $h_1, \dots, h_p$ are in C^∞_c with supports in $Q_R = [-R, R]^3$; $g(\zeta)$, $k(\zeta) = k_+ - k_-$ and the norms are as in definition 2.5. The scheme is the reference scheme z, or an admissible scheme a (definition 3.6, so that $\sup_k |z_k - a_k| < \infty$); then C_U stands for $C_U + s_1 \sup_k |z_k - a_k|$ throughout (proposition 12.6).

13.1. The positive leg on cubes.

Proposition 13.1 (cube $\le$ torus at one bare mass)**.** *For every k, J and every lattice function $q \ge 0$ on X_J: $N^a_{k,C_J}(0,q) \le N^a_{k,\mathbb{T}_J}(0,q)$, both in the free convention.*

Proof. The periodic state on $\mathbb{T}_J$ has the sites of X_J with opposite faces identified; its action is the Neumann–Dirichlet action on C_J plus, for every wrap-around pair $\{x, y\}$, the photon gradient terms $\varepsilon(A_\alpha(y) - A_\alpha(x))^2$ and the hopping term $\frac{\varepsilon}{2}(r_x^2 + r_y^2) - \varepsilon r_x r_y \cos(\cdot)$, whose diagonal part is exactly the pair of Dirichlet terms the cube action carries at x, y for their outward bonds; so the cube state is the torus state with the wrap-around gradient terms and cosine parts deleted. For the class-$\mathcal{I}$ observable $e^{\sum \varepsilon^3 q r^2}$ ($\Theta_{F,g}$, $F = 0$, $g = \varepsilon^3 q \ge 0$) deleting a gradient term, a quadratic $|\ell(A)|^2$, does not increase the expectation (lemma 2.7(a)), nor does deleting a cosine part (lemma 2.7(b)); each intermediate measure is a BFS measure, and δm^2_ε, W_ε, a_k, $\mathsf{S}(q)$ agree on both sides. □

Proposition 13.2 (the positive leg in L^1 form, on every Neumann–Dirichlet box)**.** *Assume* (I2) *and* (I3) *(proposition 12.2, entering through the chessboard estimate lemma 12.4). Let $X \subset \Lambda_k$ be finite, $\langle\cdot\rangle^{\mathrm{ND}}_{k,X}$ the Neumann–Dirichlet state of definition 2.1 on X, and $0 \le q \le s_1$ a lattice function on X. Then*

$$\ln N^a_{k,X}(0,q) \;\le\; \kappa\, \|q\|_{1,k}, \qquad \kappa = C_U/s_1.$$

Proof. Choose J' with X and all its nearest neighbours in $C_{J'}$. (1) *Box $\le$ cube:* the first inequality of lemma 2.10 with $Y = X_{J'}$ and $(f, g) = (0, q)$ (lemma 2.7(a),(b)); its X-state is the state of definition 2.1 on X, since all six bonds at each site of X are present in Y and no site is killed; the scheme multiplier is common (lemma 2.11(a)). (2) *Cube $\le$ torus:* proposition 13.1 on $C_{J'}$, q extended by 0. (3) *Chessboard:* lemma 12.4 on $\mathbb{T}_{J'}$, $\ln N^a_{k,\mathbb{T}_{J'}}(0,q) \le \sum_x \varepsilon^3 \alpha^a_{k,\mathbb{T}_{J'}}(q(x))$. (4) *Chord:* $\alpha = \alpha^a_{k,\mathbb{T}_{J'}}$ is convex with $\alpha(0) = 0$, so $\alpha(u) \le (u/s_1)[\alpha(s_1)]_+$ for $0 \le u \le s_1$, and (3) is at most $s_1^{-1}[\alpha(s_1)]_+ \|q\|_{1,k}$ (the ℓ^1 norm of q is the same on X, $C_{J'}$, $\mathbb{T}_{J'}$). (5) The left side does not depend on J'; let $J' \to \infty$ at fixed k and use (I2): $\limsup_{J'} \alpha^a_{k,\mathbb{T}_{J'}}(s_1) \le C_U$ for every k. □

Corollary 13.3 (the positive leg on the cubes)**.** *Under the hypotheses of proposition 13.2, for $0 \le q \le s_1$ on X_J: $\ln N^a_{k,C_J}(0,q) \le \kappa \|q\|_{1,k}$, uniformly in k and J. In particular* (W2)*(b) holds for $\sup q \le s_1$: for every $0 \le q \in C^\infty_c$ with $\sup q \le s_1$ and every J with $\operatorname{supp} q$ at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_J$, $\lim_k Z^a_{k,J}(0,q)$ exists and $\sup_J \lim_k Z^a_{k,J}(0,q) \le e^{\kappa \|q\|_{L^1}}$.*

Proof. Proposition 13.2 at $X = X_J$. The limit exists by theorem 3.5 ($0 \le q \le s_1 = s_W$); the sampled source satisfies $0 \le q \le s_1$ (lemma 2.6(E3)) and $\|q\|_{1,k} \to \|q\|_{L^1}$. □

13.2. **Cells and the constants of the negative leg.**

Definition 13.4 (cells, tilings, cutoffs)**.** Let $L_c := \min\{L \in \mathbb{N} : ML \geq 5\}$, $\ell_c := 2ML_c \geq 10$; the *cell* $C_{L_c} = [-ML_c, ML_c)^3$ is a King cube. For $s \in \{0,1\}^3$ let $W_\mu(0) := \ell_c\mathbb{Z}$, $W_\mu(1) := ML_c + \ell_c\mathbb{Z}$, and let the tiling $\mathcal{T}_s$ consist of the cells $\prod_\mu [w_\mu, w_\mu + \ell_c)$, $w_\mu \in W_\mu(s_\mu)$; every cell is $C_{L_c} + b$ with $b \in ML_c\mathbb{Z}^3 \subset \Lambda_k$. The good sets $G_s := \{x : \operatorname{dist}(x_\mu, W_\mu(s_\mu)) > 2 \ \forall \mu\}$ are open and cover $\mathbb{R}^3$ (the 2-neighbourhoods of $W_\mu(0)$, $W_\mu(1)$ are disjoint as $ML_c \geq 5 > 4$). Fix a smooth partition of unity $0 \leq \chi_s \leq 1$, $\sum_s \chi_s = 1$, $\operatorname{supp}\chi_s \subset G_s$; so χ_s vanishes within distance 2 of every wall of $\mathcal{T}_s$. Fix $\chi_0 \in C_c^\infty$, $0 \leq \chi_0 \leq 1$, $\chi_0 = 1$ on $\{\operatorname{dist}_\infty(x, \mathbb{R}^3 \setminus C_{L_c}) \geq 2\}$, $\operatorname{supp}\chi_0 \subset \{\operatorname{dist}_\infty(x, \mathbb{R}^3 \setminus C_{L_c}) > 1\}$, and put $\chi_c := \tau_b\chi_0$ for the cell $c = C_{L_c} + b$. For sources supported in Q_R let $J_1(R) := \min\{J \in \mathbb{N} : MJ \geq R + \ell_c\}$; a cell meeting Q_R has $w_\mu \in [-R - \ell_c, R]$, hence lies in $[-R - \ell_c, R + \ell_c)^3 \subset C_J$ for $J \geq J_1(R)$, so every cell meeting Q_R is a full cell inside C_J and the cells clipped by the faces of C_J carry no source.

The constants of the Wick leg are

(30)

$$\sigma_+ := s_1, \qquad H' := \frac{s_1}{2}, \qquad H := \frac{H'}{8} = \frac{s_1}{16}, \qquad \kappa := \frac{C_U}{s_1}, \qquad B_{\text{cell}} := \sup_k \ln N^a_{k,C_{L_c}}(0, -2H'\chi_0),$$

$$C_{\text{cell}} := \tfrac{1}{2}B_{\text{cell}} + \kappa H'\ell_c^3, \qquad C_\Sigma := \frac{(2 + 6\ell_c^2)^2\pi^2}{\ell_c^3}, \qquad K_- := \frac{8\,[C_{\text{cell}}]_+\, C_\Sigma}{H'}.$$

B_{cell} is finite by theorem 3.5 at the single cube C_{L_c} and the single real source $-2H'\chi_0$, the supremum of a convergent sequence of positive numbers ($\operatorname{supp}\chi_0$ is at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_{L_c}$ as theorem 3.5 requires, and the theorem puts no amplitude restriction on a negative Wick source, a position-dependent increase of the scalar mass squared). This instance of theorem 3.5 is (B_{cell}) (lemma 14.4) and the only place where it enters the bound (31).

13.3. **The negative leg on cubes.**

Lemma 13.5 (wall insertion)**.** *Let $X_1, \dots, X_N$ be the site sets $c \cap X_J$ of the cells c of one tiling $\mathcal{T}_s$ that meet C_J, and $q \geq 0$ a lattice function on X_J. Then $N^a_{k,C_J}(0,-q) \leq \prod_{i=1}^N N^a_{k,X_i}(0, -q|_{X_i})$, with N^a_{k,X_i} the functional of the Neumann–Dirichlet state of definition 2.1 on X_i.*

Proof. $e^{-\sum \varepsilon^3 q r^2}$ is the class-$\mathcal{D}$ observable $\Theta^-_{F,g}$, $F = 0$, $g = \varepsilon^3 q \geq 0$. Delete, for every nearest-neighbour pair with ends in different cells, the three photon gradient terms and the cosine part of the hopping term, one at a time: by lemma 2.7(d) (the inequalities [6, Thm. 6.2(3),(4)], proved in Section 2) neither deletion decreases $\langle\Theta^-\rangle$, and each intermediate measure is a BFS measure. The result is a product over the cells (lemma 2.7(e)) whose factors are the Neumann–Dirichlet states on the X_i: the kept diagonal parts of the deleted bonds are their Dirichlet terms, the remaining gradient terms their Neumann terms, and δm_ε^2, W_ε, a_k are site-independent. The observable and the multiplier $e^{a_k \mathsf{S}(q)} = \prod_i e^{a_k \mathsf{S}(q|_{X_i})}$ factorise. □

Lemma 13.6 (one cell)**.** *Assume* (I2) *and* (I3). *Let $c = C_{L_c} + b$ be a cell ($b \in ML_c\mathbb{Z}^3$) and $0 \leq \tilde{q} \leq H'$ a lattice function supported in $\{x : \operatorname{dist}_\infty(x, \mathbb{R}^3 \setminus c) \geq 2\}$. Then $\ln N^a_{k,c}(0, -\tilde{q}) \leq C_{\text{cell}}$ for every k.*

Proof. $\chi_c = 1$ on $\operatorname{supp}\tilde{q}$, so $-\tilde{q} = k_1 - k_2$ with $k_1 := H'\chi_c - \tilde{q} \geq 0$, $k_2 := H'\chi_c$, and lemma 2.11(f) at $f = 0$ gives $N^a_{k,c}(0,-\tilde{q}) \leq N^a_{k,c}(0, -2H'\chi_c)^{1/2} N^a_{k,c}(0, 2k_1)^{1/2}$. *First factor:* exact lattice translation covariance ($b \in \Lambda_k$; the change of variables $x \mapsto x - b$ transports the boundary conditions, and the action, δm_ε^2, W_ε, a_k are site-independent) gives $N^a_{k,c}(0, -2H'\chi_c) = N^a_{k,C_{L_c}}(0, -2H'\chi_0) \leq e^{B_{\text{cell}}}$. *Second factor:* proposition 13.2 on the box $c \cap \Lambda_k$ with $q' := 2k_1$, $0 \leq q' \leq 2H' = s_1$, gives at most $\exp(\kappa\|q'\|_{1,k}) \leq \exp(2\kappa H'\ell_c^3)$, the cell having volume ℓ_c^3. Hence $\ln N^a_{k,c}(0,-\tilde{q}) \leq \frac{1}{2}B_{\text{cell}} + \kappa H'\ell_c^3$. □

Lemma 13.7 (chord on the cell)**.** *Under the hypotheses of lemma 13.6:* $\ln N^a_{k,c}(0,-\tilde{q}) \leq ([C_{\text{cell}}]_+/H')\sup\tilde{q}$.

Proof. Let $t := \sup \tilde{q}/H' \in (0,1]$, $\hat{q} := \tilde{q}/t$ ($0 \le \hat{q} \le H'$, same support). $\varphi(u) := \ln N^a_{k,c}(0, -u\hat{q})$ is convex (lemma 2.11(d)), $\varphi(0) = 0$, $\varphi(1) \le C_{\mathrm{cell}}$ (lemma 13.6); so $\varphi(t) \le t\varphi(1) \le t[C_{\mathrm{cell}}]_+$. □

Lemma 13.8 (eight families and the sum over cells)**.** (a) *For a lattice function $q \ge 0$ on X_J: $N^a_{k,C_J}(0,-q) \le \prod_{s\in\{0,1\}^3} N^a_{k,C_J}(0,-8\chi_s q)^{1/8}$. If q is supported in Q_R and $J \ge J_1(R)$, then for each s every cell of $\mathcal{T}_s$ meeting $\mathrm{supp}(\chi_s q)$ is a full cell inside C_J, on which $\chi_s q$ is supported at ℓ^∞-distance > 2 from the complement of the cell.* (b) *For any tiling of $\mathbb{R}^3$ by cubes of side ℓ_c: $\sum_c \sup_{x\in c}(1+|x|^2)^{-2} \le C_\Sigma$.*

Proof. (a) Hölder's inequality with eight exponents 8 in the one state $\langle\cdot\rangle^{\mathrm{ND}}_{k,C_J}$ for $e^{-\sum \varepsilon^3 qr^2} = \prod_s e^{-\sum \varepsilon^3 \chi_s q r^2}$; the multipliers match since $\mathsf{S}(q) = \sum_s \mathsf{S}(\chi_s q)$ at lattice points. Support: every cell meeting Q_R is a full cell inside C_J for $J \ge J_1(R)$ (definition 13.4), and a point of G_s in the cell $\prod_\mu [w_\mu, w_\mu + \ell_c)$ has $\mathrm{dist}(x_\mu, \{w_\mu, w_\mu+\ell_c\}) > 2$ for every μ. (b) For x, y in one cell $|y| \le |x| + \sqrt{3}\ell_c$, so $1 + |y|^2 \le (2 + 6\ell_c^2)(1+|x|^2)$ and $\sup_{x\in c}(1+|x|^2)^{-2} \le (2+6\ell_c^2)^2 \ell_c^{-3} \int_c (1+|y|^2)^{-2} dy$; sum over the disjoint cells, with $\int_{\mathbb{R}^3}(1+|y|^2)^{-2}dy = \pi^2$. □

Proposition 13.9 (the negative leg)**.** *Assume* (I2) *and* (I3)*. For every k, every $J \ge J_1(R)$ and every lattice function $q \ge 0$ supported in Q_R with $\sup q \le H$:*

$$\ln N^a_{k,C_J}(0,-q) \;\le\; K_- \, |q|_w .$$

Proof. By lemma 13.8(a), then lemma 13.5 for each tiling $\mathcal{T}_s$ (the clipped cells carry no source and contribute $\ln N^a_{k,X_i}(0,0) = 0$), then lemma 13.7, applicable since $8\chi_s q \le 8 \sup q \le H'$ and by the support clause of lemma 13.8(a):

$$\ln N^a_{k,C_J}(0,-q) \le \frac{1}{8}\sum_s \sum_{c\in\mathcal{T}_s} \ln N^a_{k,c}(0, -8\chi_s q|_c) \le \frac{1}{8}\sum_s\sum_c \frac{[C_{\mathrm{cell}}]_+}{H'} 8 \sup_c q \le \frac{[C_{\mathrm{cell}}]_+}{H'} |q|_w \sum_s \sum_c \sup_{x\in c}(1+|x|^2)^{-2}$$

using $\sup_c q \le |q|_w \sup_{x\in c}(1+|x|^2)^{-2}$ (a sampled supremum is at most the continuum one, lemma 2.6(E3)) and lemma 13.8(b) for each of the eight tilings. □

The output norm is $|q|_w$: the chord bound on a cell sees only $\sup_c q$, so no L^1 information inside a cell survives, and the tube bound carries $K_-|k_-|_w$ where the torus paper [24] has an L^1 norm.

13.4. **The uniform bound on the tube and its limits.** For a family with supports in Q_R and $\zeta = (z,w) \in \mathbb{C}^{n+p}$ put $G^{(k,C_J)}(\zeta) := Z^a_{k,J}(\sum_i z_i f_i, \sum_j w_j h_j)$, and let $\mathcal{D}_{\sigma_+,H}$ be the tube of definition 2.5 with $\sigma_+ = s_1$, $H = s_1/16$. Two facts from several complex variables [10] are used: (T2) a holomorphic function on a connected open $D \subset \mathbb{C}^m$ vanishing on a non-empty relatively open subset of $D \cap \Sigma$, Σ an affine totally real subspace of real dimension m (such as $\mathbb{R}^m$ or $(\mathrm{i}\mathbb{R})^{m_1} \times \mathbb{R}^{m_2}$), vanishes on D (all its derivatives vanish at such a point); (T3) a locally bounded family of holomorphic functions on D is normal, and if it converges on a set as in (T2) it converges locally uniformly on D to a holomorphic limit (Montel, Vitali).

Theorem 13.10 (the tube bound)**.** *Assume* (I2) *(which proposition 12.6 derives from* (T3) *and the torus inputs proposition 3.8 and lemma 3.9), for the reference scheme z or for an admissible scheme a (definition 3.6), with* (I3) = *proposition 12.2 and* (I4) = *lemma 2.7. The constants σ_+, H, κ, K_- of* (30) *depend on $(e, \lambda, m, \mu_0, M, a)$ and on the fixed auxiliary choices M_1, χ_0, $s_1 \in (0, \sigma_1)$, and on nothing else; no uniformity in μ_0 is claimed. For every family with supports in Q_R, every $k \ge 0$, $J \ge J_1(R)$ and $\zeta \in \mathcal{D}_{\sigma_+,H}$:*

$$|G^{(k,C_J)}(\zeta)| \;\le\; G^{(k,C_J)}(\mathrm{Re}\,\zeta) \;\le\; \exp\Big(\|g(\zeta)\|^2_{2,k} + \kappa \|k_+(\zeta)\|_{1,k} + K_- |k_-(\zeta)|_w \Big). \tag{31}$$

Consequently:

(i) (Limits on the whole tube.) *$G_J := \lim_k G^{(k,C_J)}$ exists locally uniformly on $\mathcal{D}_{\sigma_+,H}$ for every $J \ge J_1(R)$, $G_\infty := \lim_J G_J$ exists locally uniformly on $\mathcal{D}_{\sigma_+,H}$ (the limit along $J \ge J_1(R + \frac{1}{2})$, hence along any tail), both are holomorphic there, and*

$$|G_\infty(\zeta)| \le \exp\big(\|g(\zeta)\|_2^2 + \kappa \|k_+(\zeta)\|_{L^1} + K_- |k_-(\zeta)|_w \big) \qquad (\zeta \in \mathcal{D}_{\sigma_+,H}).$$

(W2) *is not used; where the full object* Z^a_∞ *of the optional extension exists,* $G_\infty = Z^a_\infty$ *on* $\mathcal{D}_{\sigma_+,H}$.

(ii) (E0″.) *With* $\sigma_0 := H/4$, $K' := \kappa\pi^2 + K_-$, *the mixed Schwinger functions* $S_{n,p}(f;h) := \partial_z^{(n)}\partial_w^{(p)}G_\infty(0)$ *satisfy, for real test functions,*

$$|S_{n,p}(f_1,\dots,f_n;h_1,\dots,h_p)| \le e^{K'\sigma_0}(\sqrt{2}e)^n(n!)^{1/2}(e/\sigma_0)^p\,p!\prod_i\|f_i\|_2\prod_j|h_j|_w \le \sigma_{n+p}\prod_i|f_i|_4\prod_j|h_j|_4,$$

$\sigma_N := e^{K'\sigma_0}C_1^N N!$, $C_1 := 2\max(e\pi/2, e/\sigma_0)$; *for complex test functions the first bound carries the factor* $2^{n+p} \le 2^N$, $N = n+p$, *absorbed in* C_1. *This is* E0″ *with* $s = 4$ *and every* $\beta > 1$; *the* $S_{n,p}$ *are tempered distributions.*

σ_1, C_U depend on (e,λ,m,μ_0,M_1) through the torus stability constants of proposition 3.8 at m, M_1, the c-number β_1 of lemma 3.9 and finitely many fixed-spacing constants; B_{cell} depends on everything, including μ_0, M, a. The curvature term of (31) rests on the constant 2 of lemma 7.1, numerically nearly sharp but not proved to be (remark 7.2).

Proof. The bound (31). The first inequality is lemma 2.11(d). Write $g = g(\zeta)$, $k = k(\zeta)$ (real). *Step 1* (curvature off): theorem 7.4(a) in the presence of the real scalar source k gives $N^a(g,k) \le e^{\|g\|^2_{2,k}}N^a(0,k)$; the supports of the f_i are at distance > 1 from $\mathbb{R}^3 \setminus C_J$ since $MJ \ge R + \ell_c > R+1$. *Step 2*: $N^a(0,k_+ - k_-) \le N^a(0,2k_+)^{1/2}N^a(0,-2k_-)^{1/2}$ (lemma 2.11(f) at $f = 0$). *Step 3* (positive leg): $\zeta \in \mathcal{D}_{\sigma_+,H}$ and (E3) give $0 \le 2k_+ \le s_1$ on the lattice, so $\frac12\ln N^a(0,2k_+) \le \kappa\|k_+\|_{1,k}$ by corollary 13.3. *Step 4* (negative leg): $0 \le 2k_- \le H$, supported in Q_R, $J \ge J_1(R)$, so $\frac12\ln N^a(0,-2k_-) \le \frac12 K_-|2k_-|_w = K_-|k_-|_w$ by proposition 13.9.

(i), $k \to \infty$ at fixed $J \ge J_1(R)$. Each $G^{(k,C_J)}$ is entire on $\mathbb{C}^{n+p}$. On a compact $K \subset \mathcal{D}_{\sigma_+,H}$ the family $\{G^{(k,C_J)}\}_k$ is uniformly bounded by (31) and lemma 2.6(E1)–(E3) ($\|g\|^2_{2,k} \le C_0|g|^2_4$, $\|k_+\|_{1,k} \le C_0'|k_+|_w$, $|k_\pm|_w \le \sum_j|\operatorname{Re} w_j||h_j|_w$). At the real points of $\mathcal{D}_{\sigma_+,H}$, a non-empty open subset of $\mathbb{R}^{n+p}$, it converges by theorem 3.5 (the positive scalar part is below $\sigma_+/2 < s_W$). By (T3) with $\Sigma = \mathbb{R}^{n+p}$, $G^{(k,C_J)} \to G_J$ locally uniformly on $\mathcal{D}_{\sigma_+,H}$, G_J is holomorphic, and (31) passes to the limit with $\|g\|_{2,k} \to \|g\|_2$, $\|k_+\|_{1,k} \to \|k_+\|_{L^1}$ (k_+ continuous with compact support) and the sampled $|k_-|_w$ at most the continuum one: $|G_J|$ is bounded on $\mathcal{D}_{\sigma_+,H}$ by the right side of the display in (i), uniformly in $J \ge J_1(R)$.

(i), $J \to \infty$ on a sub-tube. Split $h_j = h_j^+ - h_j^-$, $h_j^- := \|h_j\|_\infty\chi_j$, $h_j^+ := h_j + h_j^- \ge 0$, with $\chi_j \in C_c^\infty$, $0 \le \chi_j \le 1$, $\chi_j = 1$ on $\operatorname{supp} h_j$, $\operatorname{supp}\chi_j \subset Q_{R+1/2}$. Put $F_{k,J}(z,w^+,w^-) := Z^a_{k,J}(\sum_i z_i f_i, \sum_j(w_j^+h_j^+ + w_j^-h_j^-))$ and $\mathcal{D}'' := \{(z,w^+,w^-) \in \mathbb{C}^{n+2p} : |\operatorname{Re} w_j^\pm| < \delta_j\}$ with $\sum_j\delta_j(\|h_j^+\|_\infty + \|h_j^-\|_\infty) < H/2$. On $\mathcal{D}''$ the split family's scalar source has $\|k_\pm^{\mathrm{sp}}\|_\infty \le \|k^{\mathrm{sp}}\|_\infty < H/2 \le \sigma_+/2$, so (31) holds for it uniformly in k and $J \ge J_1(R+\frac12)$, and the previous paragraph gives $F_{k,J} \to F_J$ locally uniformly on $\mathcal{D}''$ with the limit bound. On $\Omega''_+ := \mathcal{D}'' \cap ((\mathrm{i}\mathbb{R})^n \times (0,\infty)^{2p})$ the sources lie in the cone $\mathcal{K}$ (imaginary curvature coefficients, nonnegative total scalar source), so $J \mapsto F_J$ is nondecreasing there (first inequality of lemma 2.10 with lemma 2.11(a) at fixed k, then $k \to \infty$) and bounded, hence convergent. Ω''_+ is a non-empty open subset of the maximal totally real subspace $(\mathrm{i}\mathbb{R})^n \times \mathbb{R}^{2p}$ and $\mathcal{D}''$ is convex, so (T3) gives $F_J \to F_\infty$ locally uniformly on $\mathcal{D}''$. Restricting to $w_j^+ = w_j$, $w_j^- = -w_j$ gives $G_J \to G_\infty$ locally uniformly on $\mathcal{D}' := \{(z,w) : |\operatorname{Re} w_j| < \delta_j\} \subset \mathcal{D}_{\sigma_+,H}$ (there $\|k_\pm\|_\infty \le \|k\|_\infty < \sum_j\delta_j\|h_j\|_\infty < H/2$).

(i), the whole tube. $\{G_J\}_{J\ge J_1(R+1/2)}$ is locally uniformly bounded on $\mathcal{D}_{\sigma_+,H}$ by the J-free limit bound; every Montel subsequential limit is holomorphic on $\mathcal{D}_{\sigma_+,H}$ and agrees with G_∞ on the open set $\mathcal{D}'$; $\mathcal{D}_{\sigma_+,H}$ is connected, so all subsequential limits coincide: $G_J \to G_\infty$ locally uniformly on $\mathcal{D}_{\sigma_+,H}$, and the bound passes to G_∞. Where Z^a_∞ exists it is the same iterated limit of the same Z^a_J.

(ii). Normalise $\|f_i\|_2 = 1$ and let $\rho_j := \sigma_0/(p|h_j|_w)$. On the polydisc $|w_j| \le \rho_j$, $z \in \mathbb{C}^n$: $\|k\|_\infty \le \sum_j|\operatorname{Re} w_j|\|h_j\|_\infty \le \sum_j\rho_j|h_j|_w = \sigma_0 < H/2 \le \sigma_+/2$, so the polydisc lies in $\mathcal{D}_{\sigma_+,H}$, where G_∞ exists with its bound by (i); the exponent there is $\le \kappa\pi^2|k_+|_w + K_-|k_-|_w \le K'|k|_w \le K'\sum_j|\operatorname{Re} w_j||h_j|_w \le K'\sigma_0$ by (E2). Cauchy's estimate for the mixed first w-derivatives at

$w=0$, z fixed: $|\partial_w^{(p)} G_\infty(z,0)| \le \prod_j \rho_j^{-1} \sup|G_\infty| \le (p/\sigma_0)^p \prod_j |h_j|_w\, e^{K'\sigma_0} e^{\|g(z)\|_2^2}$, and $p^p \le e^p p!$. The entire function $z \mapsto \partial_w^{(p)} G_\infty(z,0)$ is thus bounded by $Ae^{\|g(z)\|_2^2}$, and Cauchy's estimate in the z's with radius $(2n)^{-1/2}$, as in the proof of theorem 7.4(c), gives $(\sqrt{2}e)^n (n!)^{1/2}$. Then (E2) and $(n!)^{1/2} p! \le (n+p)!$ give the second bound; the extension to $\mathcal{S}$, to complex test functions (the factor $2^{n+p} \le 2^N$) and to tempered distributions is as in theorem 7.4(c), and $C_1^N/(N!)^{\beta-1} \to 0$ for every $\beta > 1$. □

Lemma 13.11 (the tube object: limit, continuity, translations)**.** *Assume the bound* (31)*, which theorem 13.10 derives from* (T3) *and the torus inputs (through* (I2)*). Write $\mathcal{D} = \mathcal{D}_{\sigma_+,H}$ for the tube of a family with supports in Q_R.*

(iii) (The J-limit on the tube.) *For $J \ge J_1(R+\frac{1}{2})$, which exceeds the least J_0 with the supports at ℓ^∞-distance >1 from $\mathbb{R}^3 \setminus C_J$ (as $MJ_1(R') - R' \ge \ell_c \ge 10 > 1$), $G_\infty = \lim_J G_J$ exists locally uniformly on $\mathcal{D}$, is holomorphic on $\mathcal{D}$, satisfies the bound of theorem 13.10*(i)*, is >0 at every real point of $\mathcal{D}$, and $G_J \le G_\infty$ at the cone points of $\mathcal{D}$ ($z \in (\mathrm{i}\mathbb{R})^n$, $w \in \mathbb{R}^p$, $k(\zeta) \ge 0$: the real scalar source $k(\zeta) \ge 0$ sits in the slot of the cone).*

(iv) (Continuity on the real tube.) *Let $E_\rho := C_c^\infty(\bar{B}_\rho;\Lambda^2) \times C_c^\infty(\bar{B}_\rho)$ with its C^∞ topology, a Fréchet space, and $U_\rho := \{(f,h) \in E_\rho : \|h_+\|_\infty < \sigma_+/2,\ \|h_-\|_\infty < H/2\}$, open and convex ($h \mapsto \|h_\pm\|_\infty$ is continuous and convex). Then $\Phi^a(f,h) := \ln G_\infty(f,h)$ is finite, convex and continuous on U_ρ.*

(v) (Translations on the tube.) *For every family and every $b \in \mathbb{R}^3$: $G_\infty(\tau_b \cdot \zeta) = G_\infty(\zeta)$ for all $\zeta \in \mathcal{D}$; the tube of the translated family is that of the family, since $k(\tau_b \cdot \zeta) = \tau_b k(\zeta)$ has the same supremum norms.*

Besides theorem 3.5 and (31) *the proofs use only the first inequality of lemma 2.10, lemmas 2.11 and 2.14, lattice covariance and (T2), (T3): not* (W2)*, and lemma 2.7(d) only inside* (31)*.*

Proof. (iii) Existence, holomorphy and the bound are the last three paragraphs of the proof of theorem 13.10(i), which use (31), theorem 3.5 for the k-limits and the first inequality of lemma 2.10. *Positivity at real points.* For real $\zeta \in \mathcal{D}$ pick $\epsilon > 0$ with $-\epsilon\zeta \in \mathcal{D}$ ($\mathcal{D}$ is an open neighbourhood of 0). $\varphi_J(t) := \ln G_J(t\zeta)$ is convex on $[-\epsilon, 1]$ (lemma 2.11(d) in the limit) with $\varphi_J(0) = 0$, so $0 \le \frac{\epsilon}{1+\epsilon}\varphi_J(1) + \frac{1}{1+\epsilon}\varphi_J(-\epsilon)$, i.e. $\varphi_J(1) \ge -\varphi_J(-\epsilon)/\epsilon \ge -C_\epsilon/\epsilon$, C_ϵ the limit bound of (31) at $-\epsilon\zeta$; hence $G_J(\zeta) \ge e^{-C_\epsilon/\epsilon}$ uniformly in J and $G_\infty(\zeta) > 0$. *Cone points.* There $G_J \le G_{J+1}$ (first inequality of lemma 2.10 with lemma 2.11(a) at fixed k, then $k \to \infty$; it needs $J \ge J_0$, which the range of (iii) ensures), so $G_J \le G_\infty$.

(iv) *Finite:* (iii) for the two-source family $(f;h)$ at $\zeta = (1,1)$, in $\mathcal{D}$ by the definition of U_ρ; the value is the pointwise iterated limit and does not depend on the family used. *Convex:* for $s, s' \in U_\rho$, $t \in [0,1]$, use the family $(f,f';h,h')$ at the real point $(t,1-t,t,1-t)$, in $\mathcal{D}$ because $(th + (1-t)h')_\pm \le th_\pm + (1-t)h'_\pm$ pointwise; the convexity of $\ln Z^a_{k,J}$ in the real coefficients (lemma 2.11(d)) passes to the limits. *Borel:* a pointwise limit of functions each depending continuously on finitely many values of the sources. *Continuous:* lemma 2.14 on the open convex U_ρ.

(v) *Lattice translations at cone points.* Let $b \in 5^{-k_0}\mathbb{Z}^3$, $|b|_\infty \le M$, and ζ a cone point of $\mathcal{D}$ ($k_- = 0$, $q := k(\zeta) \ge 0$, $\|q\|_\infty < \sigma_+/2$). Steps 1–3 of the proof of proposition 8.1 run unchanged with the real scalar source q in the slot of the cone: lattice covariance (τ_b preserves Λ_k and S_{Λ_k} for $k \ge k_0$) gives $Z^a_{k,C_J}(\tau_b\cdot\zeta) = Z^a_{k,C_J-b}(\zeta)$, the interlacing $X_{J-1} \subset (C_J - b) \cap \Lambda_k \subset X_{J+1}$ and the first inequality of lemma 2.10 twice (no killed site) give $Z^a_{k,C_{J-1}}(\zeta) \le Z^a_{k,C_J}(\tau_b\cdot\zeta) \le Z^a_{k,C_{J+1}}(\zeta)$ for $J \ge J_0 + 1$. Let $k \to \infty$ by theorem 13.10(i) for the three families (the translated family is supported at distance >1 from $\mathbb{R}^3 \setminus C_{J_0+1}$), then $J \to \infty$ by (iii), the three terms being cone points of their tubes: $G_\infty(\tau_b\cdot\zeta) = G_\infty(\zeta)$ at the cone points of $\mathcal{D}$. *All $\zeta \in \mathcal{D}$.* For the split family of the proof of theorem 13.10(i) and its translate, $F_\infty^{\tau_b} - F_\infty$ is holomorphic on the connected $\mathcal{D}''$ and vanishes on Ω''_+, whose points are cone points of the split family's tube and which is an open subset of the maximal totally real subspace $(\mathrm{i}\mathbb{R})^n \times \mathbb{R}^{2p}$; by (T2) it vanishes on $\mathcal{D}''$, and the diagonal gives $G_\infty^{\tau_b} = G_\infty$ on $\mathcal{D}'$. Then $G_\infty^{\tau_b} - G_\infty$ is holomorphic on the connected $\mathcal{D}$ ((iii) for both families) and vanishes on the open set $\mathcal{D}'$, hence on $\mathcal{D}$. *All $b \in \bigcup_k 5^{-k}\mathbb{Z}^3$:*

iterate; the tube is the same at every step. *All $b \in \mathbb{R}^3$ at real points.* For real $\zeta \in \mathcal{D}$ the source pair $s := (\sum_i z_i f_i, \sum_j w_j h_j)$ lies in some U_ρ, $\tau_b s \in U_{\rho'}$ for $|b| \le \rho' - \rho$, and $b \mapsto \tau_b s$ is continuous into $E_{\rho'}$; by (iv) and the density of $\bigcup_k 5^{-k}\mathbb{Z}^3$, $\Phi^a(\tau_b s) = \Phi^a(s)$. *Complex $\zeta \in \mathcal{D}$.* $G_\infty^{\tau_b} - G_\infty$ is holomorphic on $\mathcal{D}$ and vanishes on the real points of $\mathcal{D}$, a non-empty open subset of $\mathbb{R}^{n+p}$; (T2) gives the claim. □

Remark 13.12 (King's convention). On King's RG cube $\mathrm{ND}^K_{k,J}$ the negative leg holds as printed: by lemma 4.7, the wall insertion of lemma 13.5 runs on $\mathrm{ND}^K_{k,J}$ with factors that are comparison box states killed on $X_i \cap T_J$, and every cell on which a cutoff source $8\chi_s q$ does not vanish is a full cell inside C_J meeting no killed site, because χ_s vanishes within distance 2 of every cell wall; so lemmas 13.6 to 13.8 and proposition 13.9 apply with the printed K_-. The direction matters: killing sites of a cell increases a class-$\mathcal{D}$ functional (lemma 4.7(e)), so B_{cell} does not control a source-carrying cell with killed sites. For the positive leg, remark 4.8 prepends the interlacing inequality $\langle\cdot\rangle^K_{k,J} \le \langle\cdot\rangle^{(a)}_{k,J}$ on class $\mathcal{I}$ (Step 1 of lemma 4.2) to propositions 13.1 and 13.2, so corollary 13.3 holds for $N^K_{k,C_J}(0,q)$ at every (k,J) and, for the fixed-cube limits, in the form $\limsup_k Z^{a,K}_{k,J}(0,q) \le e^{\kappa\|q\|_{L^1}}$. The assembly of (31) on ND^K from these pieces, with the check that every step of the proof of theorem 13.10 runs in the right direction on King's convention, and the existence of the fixed-cube limits of $Z^{a,K}_{k,J}$ on the tube (theorem 13.10(i) on ND^K) are carried out in theorem 21.3; the limit statements of this section and of the Wick sector of Sections 8, 9 and 12 then hold on King's convention and the Wick sector of the main theorem is stated for King's printed state (proposition 21.4).

14. Continuum subsequences and LIM states

The axioms of the preceding sections rest on the existence of the fixed-cube limits $\lim_k$: on theorem 3.2 in the curvature sector, on theorem 3.5 on the Wick tube, both proved in the existence part. This section shows that the axioms hold without that input, for the limit state along any continuum subsequence of spacings, and identifies every such state with the iterated limit. The k-uniform bounds of lemma 7.3 and (31) make the fixed-cube functionals normal families, so one subsequence of spacings serves every cube and every family at once; along it the cube limit $J \to \infty$ exists because the cone functionals are monotone in the cube at fixed k (lemma 2.10). The limit state of such a subsequence (a *LIM state*) satisfies every axiom proved above, with no hypothesis in the curvature sector and, on the Wick tube, under (T3) for every admissible scheme (through two properties of the scheme, the cell bound and the scheme bound, proved below); theorem 3.2 (resp. theorem 3.5) adds exactly the identification with the iterated limit and the independence of the subsequence. The section is written on the comparison convention $\mathrm{ND}^{(a)}$ (definition 2.1), in which the tube lemmas and the rotation part are written; King's convention is corollary 14.14. Notation is that of definitions 2.3, 2.5 and 13.4 and (30): $a^{(a)}_{k,J}(z) = \langle e^{\sum_j z_j F(f_j)}\rangle^{(a)}_{k,J}$ for a family $f = (f_1, \dots, f_N)$ of real 2-forms; $Z^a_{k,J}(\zeta)$ for $\mathcal{F} = (f_1, \dots, f_n; h_1, \dots, h_p)$ and $\zeta = (z, w) \in \mathbb{C}^{n+p}$ in the scheme a; $g(\zeta)$, $k(\zeta) = k_+ - k_-$, $\mathsf{a}(\zeta)$; $U_J := \{x : \mathrm{dist}_\infty(x, \mathbb{R}^3 \setminus C_J) > 1\}$, $J_0(f)$ the least J with all supports in U_J, and $R(\mathcal{F})$ the least R with all supports in $Q_R = [-R, R]^3$. Two standard facts [10] are used by name: the *identity theorem* (a holomorphic function on a connected open $D \subset \mathbb{C}^m$ vanishing on a non-empty relatively open subset of $D \cap \Sigma$, Σ a maximal totally real affine subspace such as $\mathbb{R}^m$ or $(\mathrm{i}\mathbb{R})^{m_1} \times \mathbb{R}^{m_2}$, vanishes on D) and *Montel–Vitali* (a locally bounded family of holomorphic functions on D is normal, and converges locally uniformly on D if it converges on such a set).

14.1. Definitions, the cell bound and the scheme bound.

Definition 14.1 (continuum subsequences)**.** An increasing sequence $\mathcal{K} = (k_m)_{m \ge 1} \subset \mathbb{N}$ of spacings is a *continuum subsequence* (of the comparison family) if for every finite family f of real 2-forms and every $J \ge J_0(f)$ the sequence $(a^{(a)}_{k_m,J})_m$ converges locally uniformly on $\mathbb{C}^N$; the limit is written $Z^{\mathcal{K}}_J(\sum_j z_j f_j, 0)$ and depends on the family only through the total complex source, as each $a^{(a)}_{k,J}$ does. Fix a scheme a. $\mathcal{K}$ is a *continuum subsequence for the Wick sector*

(Form (B)) if it is a continuum subsequence and, for every finite family $\mathcal{F}$ of real 2-forms and scalars and every J with $MJ > R(\mathcal{F}) + \ell_c$, the sequence $(Z^{a}_{k_m,J})_m$ converges locally uniformly on the tube $\mathcal{D}_{\sigma_+,H}$ of $\mathcal{F}$; the limit is $Z^{a,\mathcal{K}}_J$. (The strict inequality replaces the threshold $J_1(R)$ of theorem 13.10 by at most one cube step; only tails in J are used.) Form (A), convergence on all of $\mathbb{C}^{n+p}$ for the same J, is the optional extension of theorem 23.7 and is not used here.

Definition 14.2 (LIM states)**.** The *LIM state* of a continuum subsequence $\mathcal{K}$ is $Z^{\mathcal{K}}_\infty := \lim_{J\to\infty} Z^{\mathcal{K}}_J$ on $\mathbb{C}^N$ (curvature sector), resp. $Z^{a,\mathcal{K}}_\infty := \lim_J Z^{a,\mathcal{K}}_J$ on the tube (Wick sector); lemma 14.8 gives existence. A subsequence of a continuum subsequence is a continuum subsequence with the same fixed-cube limits, hence the same LIM state.

Definition 14.3 (the limit state of the rotation theorem)**.** For a family f and $J \geq J_0(f)$ let L^f_J be the set of locally uniform limits on $\mathbb{C}^N$ of subsequences of $(a^{(a)}_{k,J})_k$. The rotation theorem (theorem 3.13) gives an entire $Z^{\mathrm{LIM}}_\infty(\cdot,0)$ to which *every* choice $\zeta_J \in L^f_J$ converges locally uniformly as $J\to\infty$, the subsequence being allowed to depend on J. With J-dependent subsequences one only has $\liminf_k a^{(a)}_{k,J}(iu) \leq \zeta_J(iu) \leq \limsup_k a^{(a)}_{k,J}(iu)$, and the J-limit is pinned only if the J-limits of the two bounds agree, which is the content of that theorem. A LIM state in the sense of definition 14.2 uses one $\mathcal{K}$ common to all J and all families, along which the fixed-cube limits are monotone in J (lemma 14.8), and needs no hypothesis; proposition 14.12(a) relates the two.

The curvature sector needs no hypothesis. On the Wick tube the bound (31) of theorem 13.10 is used; it rests on (T3) and the torus inputs (proposition 3.8, lemma 3.9, and Hypothesis 3.11, which give (I2) with amplitude $s_1 \in (0,\sigma_1)$ and constant C_U through proposition 12.6; (I3) is proposition 12.2 and (I4) is lemma 2.7) and on one instance of theorem 3.5, isolated as follows.

We say that the scheme a has the *cell bound* ($\mathrm{B_{cell}}$) if $B_{\mathrm{cell}} := \sup_k \ln N^a_{k,C_{L_c}}(0,-2H'\chi_0) < \infty$, for the one cell C_{L_c} (side $\ell_c \geq 10$) and the one negative Wick source $-2H'\chi_0$ of definition 13.4, and that it has the *scheme bound* (SCH) if either $a = z$, the reference scheme in which lemma 3.9 is computed, or $\sup_k |z_k - a_k| < \infty$. The cell bound is the boundedness form of theorem 3.5 at one cube and one source, and the only place where that theorem enters the proof of (31) (through lemmas 13.6 and 13.7 and proposition 13.9; K_- is built from B_{cell}). Both properties hold for the schemes of the main theorem:

Lemma 14.4 ($(\mathrm{B_{cell}})$, the cell bound)**.** *Let the couplings satisfy* (T3) *and let a be an admissible scheme (definition 3.6). Then a has the cell bound: $B_{\mathrm{cell}} = \sup_k \ln N^a_{k,C_{L_c}}(0,-2H'\chi_0) < \infty$.*

Proof. By definition 13.4 the cell $C_{L_c} = [-ML_c, ML_c)^3$ is a King cube and $\operatorname{supp}\chi_0$ lies at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_{L_c}$; the source $q := 2H'\chi_0$ is a non-negative function of $C^\infty_c(\mathbb{R}^3)$ with this support condition. Under (T3) the inputs (I2) and (I3) of the existence theorem with scalar sources hold (proposition 12.6, proposition 12.2), so corollary 6.18(b) applies and gives $\sup_k N^z_{k,C_{L_c}}(0,-q) < \infty$ for the comparison-convention functional in the reference scheme z. For a scheme a with $\sup_k |a_k - z_k| < \infty$ the two functionals differ by the factor $e^{-(a_k-z_k)\mathsf{S}_k(-q)}$ (remark 2.4), bounded uniformly in k since $\mathsf{S}_k(q) \to \int q$; this is remark 6.19, and an admissible scheme has $\sup_k |z_k - a_k| < \infty$ by definition 3.6. Hence $\sup_k N^a_{k,C_{L_c}}(0,-2H'\chi_0) < \infty$, and $B_{\mathrm{cell}} < \infty$ because each term is positive (remark 6.20). □

Lemma 14.5 ((SCH), the scheme bound)**.** *An admissible scheme has the scheme bound. If $a \neq z$ has the scheme bound, the constants of* (31) *hold with C_U replaced by $C_U + s_1 \sup_k |z_k - a_k|$.*

Proof. For an admissible scheme either $a = z$ or $\lim_k (z_k - a_k)$ exists, so $\sup_k |z_k - a_k| < \infty$ (definition 3.6). The replacement of the constant is the last clause of proposition 12.6, carried through theorem 13.10. □

Proposition 14.6 ((UNIQ), independence of the subsequence)**.** *Any two continuum subsequences $\mathcal{K}, \mathcal{K}'$ have the same LIM state: $Z^{\mathcal{K}}_\infty(\cdot,0) = Z^{\mathcal{K}'}_\infty(\cdot,0)$ for every family.*

Proof. This is proposition 14.12(a) below, from the rotation theorem theorem 3.13: every continuum subsequence's LIM state is the limit state $Z_\infty^{\mathrm{LIM}}(\cdot,0)$ of that theorem. It also follows from theorem 3.2 through corollary 4.4(iii) (corollary 14.14(b)): every LIM state of the comparison family is King's printed state Z_∞^K. □

The proofs of the axioms below hold for each LIM state separately and do not use (UNIQ); it enters only in the Euclidean clause (vii) of theorem 14.9, through proposition 14.12(a). Without the rotation theorem and the existence theorem one would only know that every LIM state lies on $(\mathrm{i}\mathbb{R})^N$ between $\lim_J \liminf_k a_{k,J}^{(a)}$ and $\lim_J \limsup_k a_{k,J}^{(a)}$ (both exist, by monotonicity in J), with equality of the LIM states if these coincide at every real argument, an entire function being determined on $(\mathrm{i}\mathbb{R})^N$.

14.2. **Existence of continuum subsequences and the cube limit.**

Lemma 14.7 (continuum subsequences exist)**.** (a) (Curvature sector; no hypothesis.) *Every increasing sequence of spacings has a continuum subsequence. For every continuum subsequence $\mathcal{K}$, family f and $J \geq J_0(f)$: $Z_J^{\mathcal{K}}(\cdot,0)$ is entire, $|Z_J^{\mathcal{K}}(\sum_j z_j f_j,0)| \leq e^{\|g(z)\|_2^2}$, $Z_J^{\mathcal{K}}(x,0)$ is real and ≥ 1 at real x, and $0 \leq Z_J^{\mathcal{K}}(\mathrm{i}t,0) \leq 1$ at imaginary* it. Several models at once: *for every countable set $\mathcal{M}$ of lattice models of King's type (Neumann–Dirichlet cube states with a site-independent real bare scalar coefficient, photon mass $\mu > 0$, $\lambda > 0$; for instance the models at the photon masses of a countable set, with the counterterm fixed) one $\mathcal{K}$ is a continuum subsequence for every member of $\mathcal{M}$.*

(b) (Wick sector, Form (B).) *Let the couplings satisfy* (T3) *and let the scheme a have the cell bound* ($\mathrm{B}_{\mathrm{cell}}$) *and the scheme bound* (SCH) *(both hold for every admissible scheme, lemmas 14.4 and 14.5). Then every increasing sequence of spacings has a continuum subsequence for the Wick sector, and for every such $\mathcal{K}$, family $\mathcal{F}$ and J with $MJ > R(\mathcal{F}) + \ell_c$, $Z_J^{a,\mathcal{K}}$ is holomorphic on $\mathcal{D}_{\sigma_+,H}$ and*

$$|Z_J^{a,\mathcal{K}}(\zeta)| \leq \exp\left(\|g(\zeta)\|_2^2 + \kappa\|k_+(\zeta)\|_{L^1} + K_-|k_-(\zeta)|_w\right) \qquad (\zeta \in \mathcal{D}_{\sigma_+,H}), \tag{32}$$

with the constants of (30). Several models at once: *for a countable set of models all satisfying* (31) *with one set of constants, one $\mathcal{K}$ serves all.*

Form (A), which needs in addition the existence of the limits at every real coefficient vector ((W2)(a)), is theorem 23.7. The subsequence is chosen in Step 2 below and nowhere else (remark 14.15).

Proof. (a). Inputs: lemma 7.3 ($\sup_{k\geq 0,\, J\geq J_0(f)} \sup_K |a_{k,J}^{(a)}| < \infty$ for every family and compact $K \subset \mathbb{C}^N$); lemma 7.1 at second order ($\langle F(u)^2\rangle_{k,J}^{(a)} \leq 2\|u\|_{2,k}^2$ for real u with plaquettes in X_J); charge conjugation $(A,\vartheta) \mapsto (-A,-\vartheta)$, a symmetry of every Neumann–Dirichlet state, so $\langle F(u)\rangle_{k,J}^{(a)} = 0$ and $\langle e^{\mathrm{i}F(u)}\rangle_{k,J}^{(a)}$ is real. *Step 1 (a countable core).* Let $\mathcal{B}$ be the countable set of finite unions of closed boxes with rational corners, for $B \in \mathcal{B}$ let $S_0(B)$ be a countable $\|\cdot\|_\infty$-dense subset of $C_c^\infty(\operatorname{int} B;\Lambda^2)$, and let $\mathcal{P}$ be the countable set of pairs (J,u'), u' a finite family from $\bigcup_B S_0(B)$, $J \geq J_0(u')$. *Step 2 (Montel and the diagonal).* For $(J,u') \in \mathcal{P}$, $(a_{k,J}^{u'})_k$ is locally bounded on $\mathbb{C}^N$ uniformly in k (lemma 7.3), hence normal; enumerate $\mathcal{P}$, extract successively, and let $\mathcal{K} = (k_m)$ be the diagonal sequence: $(a_{k_m,J}^{u'})_m$ converges locally uniformly for every $(J,u') \in \mathcal{P}$. *Step 3 (every family, imaginary coefficients).* Let f be a family, $J \geq J_0(f)$, and $B \in \mathcal{B}$ with $\bigcup_j \operatorname{supp} f_j \subset \operatorname{int} B \subset B \subset U_{J_0(f)}$ (so every $u' \in S_0(B)$ is admissible at every $J \geq J_0(f)$). For real 2-forms u, u', g supported in B and every k with $\varepsilon_k \leq 1$,

$$|\langle e^{\mathrm{i}F(u)}\rangle_{k,J}^{(a)} - \langle e^{\mathrm{i}F(u')}\rangle_{k,J}^{(a)}| \leq \langle F(u-u')^2\rangle_{k,J}^{(a)\,1/2} \leq \sqrt{2}\|u-u'\|_{2,k}, \qquad \|g\|_{2,k}^2 \leq V_B\|g\|_\infty^2, \quad V_B := 3\,\mathrm{vol}(N_{1/2}(B)),$$

the first by $|e^{\mathrm{i}x} - 1| \leq |x|$ and the second-order bound, the second because the cubes of side ε_k about the centres of the plaquettes of one orientation on which $g \neq 0$ are disjoint and lie in the closed ℓ^∞-neighbourhood $N_{1/2}(B)$. Given $t \in \mathbb{R}^N$ and $\eta > 0$ pick $u' \in S_0(B)$ with $\|\sum_j t_j f_j -$

$u'\|_\infty \le \eta$; then $|a^f_{k_m,J}(\mathrm{i}t) - a^f_{k_{m'},J}(\mathrm{i}t)| \le 2\sqrt{2V_B}\,\eta + |\langle e^{\mathrm{i}F(u')}\rangle^{(a)}_{k,J_{k_m}} - \langle e^{\mathrm{i}F(u')}\rangle^{(a)}_{k,J_{k_{m'}}}|$, and the last term tends to 0 by Step 2. So $(a^f_{k_m,J}(\mathrm{i}t))_m$ converges for every t. *Step 4 (all complex coefficients).* $(a^f_{k_m,J})_m$ is locally bounded on $\mathbb{C}^N$ and converges on the maximal totally real $(\mathrm{i}\mathbb{R})^N$; Montel–Vitali gives locally uniform convergence to an entire $Z^{\mathcal{K}}_J(\cdot,0)$. The bound $|a^{(a)}_{k,J}(z)| \le e^{\|g(z)\|^2_{2,k}}$ (theorem 7.4(a)) passes to the limit with $\|g\|_{2,k} \to \|g\|_2$ (1); realness at real points passes; $a^{(a)}_{k,J}(x) = \langle e^{F(u_x)}\rangle^{(a)}_{k,J} \ge e^{\langle F(u_x)\rangle^{(a)}_{k,J}} = 1$ by Jensen, and $[1,\infty)$ is closed; $0 \le a^{(a)}_{k,J}(\mathrm{i}t) \le 1$ by the current representation of lemma 2.8(a), $\langle e^{\mathrm{i}\langle v,A\rangle}\rangle^{(a)}_{k,J} = e^{-\langle v,Cv\rangle/2}\,\mathbb{E}^\sharp \cosh\langle Cv,\kappa\rangle \ge 0$, and $|e^{\mathrm{i}F}| = 1$. Dependence only on the total source holds at every (k,J), hence in the limit. *Step 5 (several models).* The constant of lemma 7.3, the bound $e^{\|g\|^2_{2,k}}$ and the Lipschitz constant $\sqrt{2}$ are model-free: theorem 7.4(a) is lemma 2.8(a) for a measure of the class of lemma 2.7 plus lemma 7.1 for the Neumann–Dirichlet photon covariance, valid for every real site-independent bare coefficient (the matter weight does not enter) and every $\mu > 0$. For countable $\mathcal{M}$ run Step 2 over $\mathcal{M}\times\mathcal{P}$ and Steps 3–4 member by member.

(b). Inputs: under (I2)–(I4) and $B_{\mathrm{cell}} < \infty$, for every family with supports in Q_R, every k, $J \ge J_1(R)$ and $\zeta \in \mathcal{D}_{\sigma_+,H}$, (31) gives $|Z^a_{k,J}(\zeta)| \le Z^a_{k,J}(\mathrm{Re}\,\zeta) \le \exp(\|g\|^2_{2,k} + \kappa\|k_+\|_{1,k} + K_-|k_-|_w)$, with (I2) from (T3) by proposition 12.6 and (SCH) for the scheme; on compacts of the tube the right side is bounded uniformly in k (lemma 2.6). *Steps 1–2.* As in (a), with $S^{\mathsf{W}}_0(B)$ a countable $\|\cdot\|_\infty$-dense subset of $C^\infty_c(\mathrm{int}\,B;\Lambda^2)\times C^\infty_c(\mathrm{int}\,B)$ and $\mathcal{P}^{\mathsf{W}}$ the countable set of pairs $(J,\mathcal{F}')$, $\mathcal{F}'$ a finite family from $\bigcup_B S^{\mathsf{W}}_0(B)$, $MJ > R(\mathcal{F}') + \ell_c$. For such a pair $J \ge J_1(R)$ for every $R \in (R(\mathcal{F}'), MJ - \ell_c]$, so (31) holds for $\mathcal{F}'$ at every k and $(Z^a_{k,J})_k$ is normal on the tube of $\mathcal{F}'$; the diagonal over $\mathcal{P}^{\mathsf{W}}$, together with that of (a), gives $\mathcal{K}$. *Step 3 (every family, imaginary coefficients).* Let $\mathcal{F} = (f_i; h_j)$, $MJ > R(\mathcal{F}) + \ell_c$, $R_J := MJ - \ell_c > R(\mathcal{F})$, so that all supports lie in the open cube $Q^\circ_{R_J}$ and $J \ge J_1(R_J)$; choose $B \in \mathcal{B}$ with the supports in $\mathrm{int}\,B \subset B \subset Q^\circ_{R_J}$, so that $(J,\mathcal{F}') \in \mathcal{P}^{\mathsf{W}}$ for every $\mathcal{F}'$ from $S^{\mathsf{W}}_0(B)$. For $(s,s') \in \mathbb{R}^{n+p}$ put $\sigma := (\sum_i s_i f_i, \sum_j s'_j h_j)$, so that $Z^a_{k,J}(\mathrm{i}s, \mathrm{i}s') = \langle e^{\mathrm{i}\Phi^a_k(\sigma)}\rangle^{(a)}_{k,J}$ with $\Phi^a_k(g,h) := F_{\Lambda_k}(g) + W^a_{\Lambda_k}(h)$. *Lipschitz bound.* For a real pair $q = (g_q, h_q)$ supported in Q_{R_J}, $(t_1,t_2) \mapsto Z^a_{k,J}(t_1 g_q, t_2 h_q)$ is holomorphic, and on the polydisc $|t_1| \le \|g_q\|^{-1}_{2,k}$, $|t_2| \le \sigma_0/|h_q|_w$ one has $\|k_\pm\|_\infty \le |\,\mathrm{Re}\,t_2|\,\|h_q\|_\infty \le \sigma_0 < H/2 < \sigma_+/2$, so the polydisc lies in the tube and (31) gives $|Z^a_{k,J}| \le e^{1+K'_{\mathrm{lat}}\sigma_0}$ there, $K'_{\mathrm{lat}} := \kappa C'_0 + K_-$ ($\|k_+\|_{1,k} \le C'_0|k_+|_w$ by lemma 2.6). The Taylor coefficients at 0 are the second moments of $F(g_q)$ and $W^a(h_q)$, and Cauchy's estimates give

$$\langle \Phi^a_k(q)^2\rangle^{(a)}_{k,J}{}^{1/2} \le C\,(\|g_q\|_{2,k} + |h_q|_w), \qquad C := \sqrt{2}\,e^{(1+K'_{\mathrm{lat}}\sigma_0)/2}\max(1,\sigma_0^{-1}),$$

uniformly in k, J and R_J; hence $|Z^a_{k,J}(\mathrm{i}\sigma) - Z^a_{k,J}(\mathrm{i}\sigma')| \le C(\|g_{\sigma-\sigma'}\|_{2,k} + |h_{\sigma-\sigma'}|_w)$ for real pairs supported in Q_{R_J}. *Density.* Given $\eta > 0$ pick $\sigma' \in S^{\mathsf{W}}_0(B)$ with $\|g_\sigma - g_{\sigma'}\|_\infty, \|h_\sigma - h_{\sigma'}\|_\infty \le \eta$; with V_B and $|h|_w \le (1+3R_J^2)^2\|h\|_\infty$ for h supported in Q_{R_J}, $|Z^a_{k,J}(\mathrm{i}\sigma) - Z^a_{k,J}(\mathrm{i}\sigma')| \le C(\sqrt{V_B} + (1+3R_J^2)^2)\eta$ for every k, while $Z^a_{k_m,J}(\mathrm{i}\sigma')$, the core functional at the point (i,i) of its tube, converges by Step 2. So $(Z^a_{k_m,J})_m$ converges on $(\mathrm{i}\mathbb{R})^{n+p}$. *Step 4 (the tube).* $(Z^a_{k_m,J})_m$ is locally bounded on the connected open $\mathcal{D}_{\sigma_+,H}$ and converges on the maximal totally real $(\mathrm{i}\mathbb{R})^{n+p} \subset \mathcal{D}_{\sigma_+,H}$; Montel–Vitali gives locally uniform convergence to a holomorphic $Z^{a,\mathcal{K}}_J$, and (31) passes to the limit with $\|g\|_{2,k} \to \|g\|_2$, $\|k_+\|_{1,k} \to \|k_+\|_{L^1}$ (Riemann sums; k_+ is continuous with compact support) and sampled $|k_-|_w \le$ continuum $|k_-|_w$: this is (32). *Step 5 (several models).* With one set of constants in (31) for all members, C is model-free and Step 2 runs over the product of the model set with $\mathcal{P}^{\mathsf{W}}$. □

Lemma 14.8 (the cube limit along $\mathcal{K}$ is a full limit)**.** *Let $\mathcal{K}$ be a continuum subsequence.* (Curvature sector.) *For every family, $Z^{\mathcal{K}}_J(\cdot,0) \to Z^{\mathcal{K}}_\infty(\cdot,0)$ locally uniformly on $\mathbb{C}^N$ as $J \to \infty$; the limit is entire, $|Z^{\mathcal{K}}_\infty(\sum_j z_j f_j, 0)| \le e^{\|g(z)\|^2_2}$, real and ≥ 1 at real points, in $[0,1]$ at imaginary points; $J \mapsto Z^{\mathcal{K}}_J(\mathrm{i}t,0)$ is nondecreasing, so $Z^{\mathcal{K}}_J \le Z^{\mathcal{K}}_\infty$ on $(\mathrm{i}\mathbb{R})^N$.* (Wick sector, Form (B).) *Under the hypotheses of lemma 14.7(b), $Z^{a,\mathcal{K}}_J \to Z^{a,\mathcal{K}}_\infty$ locally uniformly on $\mathcal{D}_{\sigma_+,H}$; the limit is*

holomorphic, obeys (32)*, is* >0 *at the real points of the tube, and* $Z_J^{a,\mathcal{K}} \le Z_\infty^{a,\mathcal{K}}$ *at the cone points* ($z \in (\mathrm{i}\mathbb{R})^n$*, w real,* $k(\zeta) \ge 0$*).*

Proof. Curvature sector. For $t \in \mathbb{R}^N$, $J \ge J_0(f)$ and every k: $a^{(a)}_{k,J}(\mathrm{i}t) \le a^{(a)}_{k,J+1}(\mathrm{i}t) \le 1$ (first inequality of lemma 2.10 with $Y = C_{J+1} \cap \Lambda_k$, $X = X_J$; the cut-out factor is the Neumann–Dirichlet state on C_J because δm_ε^2 is site-independent; realness by charge conjugation). Both sides converge along $\mathcal{K}$, so $J \mapsto Z_J^{\mathcal{K}}(\mathrm{i}t, 0)$ is nondecreasing, bounded by 1, convergent. $\{Z_J^{\mathcal{K}}\}_J$ is locally bounded on $\mathbb{C}^N$ by the J-free bound $e^{\|g(z)\|_2^2}$ and converges on $(\mathrm{i}\mathbb{R})^N$; Montel–Vitali gives the entire limit with the same bound; realness, ≥ 1 at real points and $[0,1]$ at imaginary points pass to the limit.

Wick sector. (1) *Split family.* Write $h_j = h_j^+ - h_j^-$, $h_j^- := \|h_j\|_\infty \chi_j$, $h_j^+ := h_j + h_j^- \ge 0$, with $\chi_j \in C_c^\infty$, $0 \le \chi_j \le 1$, $\chi_j = 1$ on $\operatorname{supp} h_j$, $\operatorname{supp}\chi_j \subset N_{1/2}(\operatorname{supp} h_j)$; put $F_{k,J}(z, w^+, w^-) := Z^a_{k,J}(\sum_i z_i f_i, \sum_j (w_j^+ h_j^+ + w_j^- h_j^-))$ and $\mathcal{D}'' := \{|\operatorname{Re} w_j^\pm| < \delta_j\}$ with $\sum_j \delta_j(\|h_j^+\|_\infty + \|h_j^-\|_\infty) < H/2$, a convex subset of the split family's tube. By lemma 14.7(b) for the split family, $F_{k_m,J} \to F_J^{\mathcal{K}}$ locally uniformly on $\mathcal{D}''$ for $MJ > R(\mathcal{F}) + \frac12 + \ell_c$, with (32). (2) On $\Omega_+ := \mathcal{D}'' \cap ((\mathrm{i}\mathbb{R})^n \times (0,\infty)^{2p})$ the observable has imaginary curvature coefficients and nonnegative total scalar source, so $J \mapsto F_{k,J}$ is nondecreasing at every k (first inequality of lemma 2.10 with lemma 2.11(a), the scheme multiplier being common); along $\mathcal{K}$, $J \mapsto F_J^{\mathcal{K}}$ is nondecreasing and bounded on Ω_+, hence convergent. (3) Ω_+ is a non-empty open subset of the maximal totally real $(\mathrm{i}\mathbb{R})^n \times \mathbb{R}^{2p}$ and $\mathcal{D}''$ is connected: $F_J^{\mathcal{K}} \to F_\infty^{\mathcal{K}}$ locally uniformly on $\mathcal{D}''$. (4) At $w^+ = w$, $w^- = -w$ the lattice functionals equal $Z^a_{k,J}(\zeta)$ exactly (equal total sources), so $Z_J^{a,\mathcal{K}}$ converges locally uniformly on $\mathcal{D}' := \{|\operatorname{Re} w_j| < \delta_j\} \times \mathbb{C}^n \subset \mathcal{D}_{\sigma_+,H}$; $\{Z_J^{a,\mathcal{K}}\}_J$ is locally bounded on $\mathcal{D}_{\sigma_+,H}$ by the J-free (32) and converges on the open $\mathcal{D}'$; Montel–Vitali on the connected tube gives $Z_J^{a,\mathcal{K}} \to Z_\infty^{a,\mathcal{K}}$, holomorphic, with (32). (5) *Positivity at real tube points.* For real $\zeta \in \mathcal{D}_{\sigma_+,H}$ pick $\epsilon > 0$ with $-\epsilon\zeta \in \mathcal{D}_{\sigma_+,H}$ and let $C_\epsilon := \sup_k(\|g\|^2_{2,k} + \kappa\|k_+\|_{1,k} + K_-|k_-|_w)|_{-\epsilon\zeta} < \infty$, the exponent of (31) at $-\epsilon\zeta$ (bounded in k by lemma 2.6, free of J). At every k, $t \mapsto \ln Z^a_{k,J}(t\zeta)$ is convex on $\mathbb{R}$ (Hölder, lemma 2.11(d)) and vanishes at $t = 0$, so $1 \le Z^a_{k,J}(\zeta)^{\epsilon/(1+\epsilon)} Z^a_{k,J}(-\epsilon\zeta)^{1/(1+\epsilon)} \le Z^a_{k,J}(\zeta)^{\epsilon/(1+\epsilon)} e^{C_\epsilon/(1+\epsilon)}$, i.e. $Z^a_{k,J}(\zeta) \ge e^{-C_\epsilon/\epsilon}$ for every k and J; the bound passes along $\mathcal{K}$ and then in J: $Z_\infty^{a,\mathcal{K}}(\zeta) \ge e^{-C_\epsilon/\epsilon} > 0$. (The product inequality of lemma 2.11(e) is not used: the tube is asymmetric, $H = \sigma_+/16$, and for real $\xi \in \mathcal{D}_{\sigma_+,H}$ with $\|k_+(\xi)\|_\infty \ge H/2$ the point $-\xi$ lies outside it, where definition 14.1 gives no limit.) (6) At a cone point of the original family $J \mapsto Z^a_{k,J}(\zeta)$ is nondecreasing at every k (first inequality of lemma 2.10 with lemma 2.11(a)); pass along $\mathcal{K}$. □

14.3. **The axioms for LIM states.**

Theorem 14.9 (LIM states of the comparison family at fixed photon mass)**.** *Let* $e > 0$, $\lambda, m, \mu_0 > 0$, $\varepsilon_k = 5^{-k}$, $C_J = [-MJ, MJ)^3$, *and the states* $\mathrm{ND}^{(a)}_{k,J}$ *of definition 2.1.*

Curvature sector (no hypothesis). *Every increasing sequence of spacings has a continuum subsequence. For every continuum subsequence* $\mathcal{K}$ *and every finite family of real* 2*-forms,*

$$Z_\infty^{\mathcal{K}}\Big(\sum_j z_j f_j, 0\Big) = \lim_{J\to\infty}\lim_{m\to\infty} \Big\langle e^{\sum_j z_j F_{\Lambda_{k_m}}(f_j)} \Big\rangle^{(a)}_{k_m,J}$$

exists locally uniformly in $z \in \mathbb{C}^N$*, is entire, and depends on the family only through the total source. Its Schwinger functions* $S_n^{\mathcal{K}}(f_1,\dots,f_n) := \partial_{z_1}\cdots\partial_{z_n} Z_\infty^{\mathcal{K}}(\sum_j z_j f_j, 0)|_{z=0}$ *satisfy:*

(i) (OS0.) $|Z_\infty^{\mathcal{K}}(\sum_j z_j f_j, 0)| \le e^{\|g(z)\|_2^2}$ *(no dependence on* μ_0, e, λ, m*) and* $\le \exp(\frac12 \mu_0^{-2} \|\sum_j (\operatorname{Re} z_j) d^* f_j\|_2^2)$*;* $|S_n^{\mathcal{K}}| \le (\sqrt{2}e)^n (n!)^{1/2} \prod_j \|f_j\|_2 \le (e\pi/2)^n (n!)^{1/2} \prod_j |f_j|_4$*; the* $S_n^{\mathcal{K}}$ *are tempered distributions, and* E0″ *holds with* $s = 4$ *and every* $\beta > \frac12$*.*

(ii) (OS1.) $S_n^{\mathcal{K}}$ *is invariant under all translations and under* O_h *about every point,* F *transforming as a* 2*-form.*

(iii) (OS2.) *Reflection positivity about every coordinate plane* $\{x_\nu = c\}$*: for exponentials* $E_a = e^{\sum_l z_{al} F(f_{al})}$ *with arbitrary complex* z_{al} *and* f_{al} *supported in* $\{x_\nu > c\}$*,* $(S(\overline{\theta_c E_a} E_b))_{ab}$ *is positive semidefinite, and* $S(\theta_c(P)P) \ge 0$ *for polynomials* P *in such fields.*

(iv) (OS3.) *$S_n^{\mathcal{K}}$ is real, symmetric, and zero for odd n.*

(v) (The measure.) *For every even $r \geq 14$ there is exactly one Borel probability measure $\mu_F^{\mathcal{K}}$ on $\mathcal{H}^{-r}(\mathbb{R}^3;\Lambda^2)$ whose moments on C_c^∞ are the $S_n^{\mathcal{K}}$; its characteristic functional is $Z_\infty^{\mathcal{K}}(\mathrm{i}\,\cdot,0)$ extended by continuity to $\mathcal{S}$, and its image on H^{-r}_{loc} is the unique measure there with this characteristic functional and the unique one with these moments. For every $\chi \in C_c^\infty$ and box Q containing* $\operatorname{supp}\chi$ *in its interior, the laws of $\chi F_{\Lambda_{k_m}}$ under $\mathrm{ND}^{(a)}_{k_m,J}$ converge weakly on $H^{-r}_{\mathrm{per}}(Q;\Lambda^2)$,* first as $m\to\infty$ along $\mathcal{K}$ *at every fixed $J \geq J_0(\chi)$ and then as $J\to\infty$, to the law of $\chi\Phi$ under $\mu_F^{\mathcal{K}}$.*

(vi) (Weak OS4.) *$\mu_F^{\mathcal{K}}$ is $\mathbb{R}^3 \rtimes O_h$-invariant and satisfies the clauses of theorem 11.1(b) with $\mu_F^{\mathcal{K}}$ in place of μ_S^K: it is an integral of translation-ergodic, reflection-positive, θ_j-invariant components clustering in the mean and under diagonal separation, with a unique joint vacuum per coordinate direction and the component growth bound; almost every component satisfies* OS4 *in the weak form; the O_h-invariant decomposition into mixtures of $k \in \{1,2,3,6\}$ ergodic measures exists.*

(vii) (Euclidean clause.) *$Z_\infty^{\mathcal{K}}$ does not depend on $\mathcal{K}$, equals the limit state Z_∞^{LIM} of definition 14.3, is $\mathbb{R}^3 \rtimes O(3)$-invariant for all complex coefficients (theorem 3.13), and the Euclidean forms of (ii) (full $\mathbb{R}^3 \rtimes O(3)$), (iii) (reflection positivity about every plane) and (vi) (almost every component $\mathbb{R}^3 \rtimes O(3)$-invariant, reflection positive about every plane, clustering in any manner, with a unique vacuum in every direction; the vanishing of a component's odd moments is not claimed) hold.*

Wick sector, Form (B). *Let the couplings satisfy* (T3) *and let the scheme $\mathfrak{a}$ have the cell bound* ($\mathrm{B}_{\mathrm{cell}}$) *and the scheme bound* (SCH) *(lemmas 14.4 and 14.5; every admissible scheme does). Every increasing sequence of spacings has a continuum subsequence for the Wick sector, and for every such $\mathcal{K}$ and family the LIM state $Z_\infty^{a,\mathcal{K}}$ exists locally uniformly on $\mathcal{D}_{\sigma_+,H}$, is holomorphic there, obeys (32), and (i)–(vi) hold for the joint Schwinger functions $S^{\mathcal{K}}_{n,p}(f;h) = \partial_z^{(n)}\partial_w^{(p)} Z_\infty^{a,\mathcal{K}}(0)$ with: in (i) the mixed bounds $|S^{\mathcal{K}}_{n,p}| \leq e^{K'\sigma_0}(\sqrt{2}e)^n (n!)^{1/2}(e/\sigma_0)^p\, p! \prod_i \|f_i\|_2 \prod_j |h_j|_w$, $\sigma_0 = H/4$, $K' = \kappa\pi^2 + K_-$, and* $\mathrm{E0}''$ *with $s=4$ and every $\beta>1$; in (ii) W^a a scalar, the identity holding on the tube; in (iii) exponentials with arbitrary complex curvature coefficients and Wick coefficient vectors $\zeta_1,\dots,\zeta_N$ of* amplitude $\mathfrak{a}(\zeta_i) < H/2$ *(in particular all purely imaginary Wick coefficients), and the polynomial form; in (iv) $S^{\mathcal{K}}_{n,p} = 0$ for odd n; in (v) the measures on $\mathcal{H}^{-r}(\mathbb{R}^3;\Lambda^2\oplus\mathbb{R})$ and H^{-r}_{loc}, the localised lattice fields $\chi(F_{\Lambda_k}, W^a_{\Lambda_k})$ converging along $\mathcal{K}$ and then in J; (vi) unchanged; and* no Euclidean clause. Schemes: *if a, a' both satisfy the hypotheses, every continuum subsequence has a further subsequence $\mathcal{K}'$ along which $a'_k - a_k \to \kappa^{\mathcal{K}'}$, and $Z_\infty^{a',\mathcal{K}'}(\zeta) = e^{-\kappa^{\mathcal{K}'}\sum_j w_j \int h_j} Z_\infty^{a,\mathcal{K}'}(\zeta)$: the two LIM states differ by the shift $W \mapsto W - \kappa^{\mathcal{K}'}\int(\cdot)$ of the Wick component, and joint cumulants of order ≥ 2 are scheme-free.*

Proof. Each clause is the corresponding proof for the iterated limit (theorems 7.4, 8.5, 10.6, 10.7 and 11.1 and propositions 8.1 and 10.9) with exactly one of three replacements: (α) a fixed-(k,J) identity or inequality between Neumann–Dirichlet functionals of explicit finite families is passed along $\mathcal{K}$ (both sides converge by lemma 14.7, which covers *every* family); (β) Montel on a k-uniform bound, the identification of the limit by convergence at real points being replaced by the choice of $\mathcal{K}$; (γ) a step about the limit object only, unchanged and not repeated here. The only uniqueness used is that of the monotone cube limit along $\mathcal{K}$ (lemma 14.8).

(i). lemmas 14.7 and 14.8 give existence, entireness and the first bound; the second, $\exp(\frac{1}{2}\mu_0^{-2}\|d_k^* g\|^2_{2,k})$ at every (k,J) (theorem 7.4(a)), passes along $\mathcal{K}$ and J with $\|d_k^* g\|_{2,k} \to \|d^* g\|_2$. The Schwinger bounds, the extension to $\mathcal{S}$, the kernel theorem and $\mathrm{E0}''$ are the arguments of theorem 7.4 (Wick sector: theorem 13.10(ii), on the polydisc inside the tube) run on the limit function with its bound (γ).

(ii), translations. For $b \in 5^{-k_0}\mathbb{Z}^3$, $|b|_\infty \leq M$, $k \geq k_0$: $Z^a_{k,C_J}(\tau_b\cdot\zeta) = Z^a_{k,C_J-b}(\zeta)$ (exact lattice covariance) and $C_{J-1}\cap\Lambda_k \subset (C_J - b)\cap\Lambda_k \subset C_{J+1}\cap\Lambda_k$, so for a cone point s supported in U_{J_0} and $J \geq J_0+1$, $Z^a_{k,C_{J-1}}(s) \leq Z^a_{k,C_J}(\tau_b\cdot s) \leq Z^a_{k,C_{J+1}}(s)$ (first inequality of lemma 2.10, twice). The three terms are functionals of the families s and $\tau_b\cdot s$ (supports in Q_{R+M}, the same tube),

so they converge along $\mathcal{K}$ (α), and lemma 14.8 sends both ends to $Z^{\mathcal{K}}_\infty(s)$: $Z^{\mathcal{K}}_\infty(\tau_b \cdot s) = Z^{\mathcal{K}}_\infty(s)$ at cone points, with no uniqueness of the k-limit. All coefficients (γ): the difference of the two limit functions is entire and vanishes on $(\mathrm{i}\mathbb{R})^N$ (curvature), resp. for the split family is holomorphic on the connected $\mathcal{D}''$ and vanishes on Ω_+, so the identity theorem, the diagonal (4) of lemma 14.8 and the identity theorem on the connected tube give it on $\mathcal{D}_{\sigma_+,H}$. All $b \in \bigcup_k 5^{-k}\mathbb{Z}^3$ by iteration; all $b \in \mathbb{R}^3$ at imaginary coefficients because the Lipschitz bounds of lemma 14.7, Step 3, pass to the limits ($|Z^{\mathcal{K}}_\infty(\mathrm{i}u,0) - Z^{\mathcal{K}}_\infty(\mathrm{i}u',0)| \le \sqrt{2}\|u-u'\|_2$; $|Z^{a,\mathcal{K}}_\infty(\mathrm{i}\sigma) - Z^{a,\mathcal{K}}_\infty(\mathrm{i}\sigma')| \le C(\|g_{\sigma-\sigma'}\|_2 + |h_{\sigma-\sigma'}|_w)$) and $\tau_{b_n} s \to \tau_{b_0} s$ in L^2 and in $|\cdot|_w$ for $s \in C^\infty_c$; complex coefficients by the identity theorem.

(ii), the cubic group. Coordinate permutations are exact at every (k,J) (lemma 2.12); pass along $\mathcal{K}$ and J (α). For the reflection σ_3 and a cone point (f,g) supported in U_{J-1}, at every k, $Z^a_{k,C_{J-1}}(\mathrm{i}f,g) \le Z^a_{k,C_J}(\mathrm{i}\sigma_3\cdot f, \sigma_3\cdot g) \le Z^a_{k,C_{J+1}}(\mathrm{i}f,g)$ (the frame flip of lemma 2.12 with the one-site shift $X_+ = (C_J + \varepsilon_k e_3)\cap\Lambda_k$ and two deletions in the cone direction, lemma 2.7(a),(b)); the middle term is the functional of the reflected family (same tube), so (α) and lemma 14.8 give $Z^{\mathcal{K}}_\infty(\mathrm{i}\sigma_3\cdot f, \sigma_3\cdot g) = Z^{\mathcal{K}}_\infty(\mathrm{i}f,g)$ on the cone (all of it at $g=0$; in Form (B) at cone points with $\|g\|_\infty < \sigma_+/2$): the reflections are not exact lattice symmetries; the sandwich replaces exactness and survives along $\mathcal{K}$. Conjugation $\sigma_\nu = P\sigma_3 P^{-1}$, group closure on the cone, the identity theorem off the cone (on $\mathbb{C}^N$ with $(\mathrm{i}\mathbb{R})^N$; on $\mathcal{D}''$ with Ω_+), invariance about every point by the translations, and differentiation are (γ), as in theorem 8.5.

(iii), reflection positivity. Fix ν, a mid-plane $c \in D$ (definition 9.1) and the enlarged family $\mathcal{F}$ of definition 10.1 with the modified-box functionals $\Phi_k(\zeta) = Z^{a,(3.38)}_{k,\Omega_{J,k}}(\zeta)$, $\zeta \in \mathbb{C}^m$, and support radius $R_{\mathcal{F}}$. The fixed-k steps (reflection positivity of the modified box, lemma 9.3; the sandwich lemmas lemmas 9.9 and 9.10) concern the photon form and the inequalities of lemma 2.7 only and are unchanged. *Curvature sector.* At $h=0$, $|\Phi_k(z)| \le e^{\|f_{\mathrm{Re}\, z}\|^2_{2,k}}$ by lemma 2.8(a) and the modified photon form, uniformly in k and J. (β) Extract from $\mathcal{K}$ a further $\mathcal{K}'$ along which $\Phi_k \to \Phi^J$ locally uniformly on $\mathbb{C}^m$; positive semidefiniteness of the Gram matrices at every $k \ge k_0(c,\delta)$ passes to Φ^J. (α) On U (imaginary curvature coefficients) $a^{(a)}_{k,C_J}(\zeta) \le \Phi_k(\zeta) \le a^{(a)}_{k,C_{J+2n}}(\zeta)$ at every k (lemmas 9.9 and 9.10, cone halves); along $\mathcal{K}'$ the ends converge to $Z^{\mathcal{K}}_J, Z^{\mathcal{K}}_{J+2n}$, so $Z^{\mathcal{K}}_J \le \Phi^J \le Z^{\mathcal{K}}_{J+2n}$ on U, and lemma 14.8 sends both ends to $Z^{\mathcal{K}}_\infty$. $\{\Phi^J\}_J$ is locally bounded on $\mathbb{C}^m$ (J-free bound) and U is a set of uniqueness, so $\Phi^J \to Z^{\mathcal{K}}_\infty$ locally uniformly, in particular at the matrix points, where $\Phi^J(\zeta^*_{ij}) \to Z^{\mathcal{K}}_\infty(\theta\zeta^*_i \sqcup \zeta_j)$ (dependence through the total source only); positivity passes. $\mathcal{K}'$ may depend on J; this is harmless, since the sandwich ends converge along all of $\mathcal{K}$ and the bound on Φ^J is J-free, so neither $\lim_J \Phi^J$ nor the positivity depends on the choice. Every plane: translate by $(c'-c)e_\nu$ and use (ii). The polynomial form: Weierstrass along $\mathcal{K}'$ and J (proposition 10.10). *Wick sector, amplitude $< H/2$, without theorem 3.5.* Let $\zeta_1,\dots,\zeta_N$ be families of real sources supported in $\{x_\nu > c\}$ with $\mathsf{a}(\zeta_i) < H/2$; the (i,j) Gram entry is $\Phi_k(\zeta^*_{ij})$, with total scalar source $\sum_l \bar{w}_{il}\, h_{il}\circ\rho + \sum_l w_{jl} h_{jl}$. The bound of lemma 10.3, which depends on theorem 3.5, is replaced by lemma 14.10 below, after which the argument is completed.

(iv). Symmetry: Taylor coefficients of a function of the total source. Realness: real at real points at every (k,J), closed under limits (Form (B): at the real points of the tube). Charge conjugation gives $Z^a_{k,J}(-z,w) = Z^a_{k,J}(z,w)$ at every (k,J), hence for $Z^{\mathcal{K}}_J$ and $Z^{\mathcal{K}}_\infty$ (the tube is invariant under $z \mapsto -z$); odd z-derivatives at 0 vanish.

(v), (vi). The moment bounds at (k,J) (theorem 7.4(a), resp. (31), with Cauchy's estimates) are uniform in k and J above the thresholds, and the lattice Schwinger functions converge along $\mathcal{K}$ then J (Weierstrass). The measure is built from the limit functional (γ): $\mathcal{Z}^{\mathcal{K}}(f) := Z^{\mathcal{K}}_\infty(\mathrm{i}f)$ on real $f \in C^\infty_c$ is normalised, positive definite (at every (k,J) the matrix $(Z^a_{k,J}(\mathrm{i}(f_m - f_l)))_{lm}$ is a Gram matrix; the entries converge along $\mathcal{K}$ then J) and Lipschitz for $\sqrt{2}\|\cdot\|_2$, resp. $C(\|\cdot\|_2 + |\cdot|_w)$, hence continuous on $\mathcal{S}$; Bochner–Minlos, moments by Lukacs [16] and uniform integrability, the Hermite carrier on $\mathcal{H}^{-r}$, $r \ge 14$ even, and the two uniqueness statements are the steps of theorem 11.1(a) with [24, Lemma 10.5(c), Lemma 10.3, Prop. 10.4]. *Local laws along $\mathcal{K}$* (((β); the one weakened clause): tightness of the laws of $\chi\Phi^a_{k,J}$ on $H^{-r}_{\mathrm{per}}(Q;\mathcal{V})$ is uniform in k

and $J \geq J(\chi)$ from the second-moment bounds (Prokhorov [7]); the characteristic functionals $Z^a_{k,J}(\mathrm{i}\chi\psi)$ converge *along* $\mathcal{K}$ to $Z^{a,\mathcal{K}}_J(\mathrm{i}\chi\psi)$; a probability measure on H^{-r}_{per} is determined by its characteristic functional on trigonometric polynomials; so the laws converge along $\mathcal{K}$, then in J (lemma 14.8), to the law of $\chi\Phi$ under the measure just built (the periodisation identity of theorem 11.1). (vi): $\mathbb{R}^3 \rtimes O_h$-invariance of the measure is (ii) at imaginary coefficients plus uniqueness of the characteristic functional; (H3) of [24] is (iii) at purely imaginary Wick coefficients ($\mathsf{a} = 0 < H/2$) with the density of the exponential algebra [24, Lemma J.2]; then [24, Thm. 10.10, Thm. 10.19, Thm. 10.20, Prop. 10.16, Cor. 10.43(iv), Thm. 10.12, Def. 2.12(iii)], stated for a measure satisfying (H1)–(H3) and matched as in remark 11.3.

Schemes. $\sup_k |a'_k - a_k| \leq \sup_k |a'_k - z_k| + \sup_k |z_k - a_k| < \infty$ by (SCH) for both. For a continuum subsequence $\mathcal{K}$ for a, Bolzano–Weierstrass gives $\mathcal{K}' \subset \mathcal{K}$ with $a'_k - a_k \to \kappa^{\mathcal{K}'} \in \mathbb{R}$. At every (k, J), $Z^{a'}_{k,J}(\zeta) = e^{-\sum_j w_j (a'_k - a_k)\mathsf{S}_{\Lambda_k}(h_j)} Z^a_{k,J}(\zeta)$ (remark 2.4) with $\mathsf{S}_{\Lambda_k}(h_j) \to \int h_j$; so along $\mathcal{K}'$ the a'-functionals converge wherever the a-functionals do, $\mathcal{K}'$ is a continuum subsequence for a' with $Z^{a',\mathcal{K}'}_J = e^{-\kappa^{\mathcal{K}'} \sum_j w_j \int h_j} Z^{a,\mathcal{K}'}_J$, and then $J \to \infty$. (For admissible a, a', $a'_k - a_k$ converges along the full sequence and $\kappa^{\mathcal{K}'} = \kappa$.) The cumulant statement is that of theorem 21.1. ($\mathrm{B}_{\mathrm{cell}}$) transfers between schemes with bounded difference: $B^{a'}_{\mathrm{cell}} \leq B^a_{\mathrm{cell}} + \sup_k |a'_k - a_k| \, \|2H'\chi_0\|_{1,k}$.

(vii). By proposition 14.12(a), $Z^{\mathcal{K}}_\infty = Z^{\mathrm{LIM}}_\infty$, which is $\mathbb{R}^3 \rtimes O(3)$-invariant for all $z \in \mathbb{C}^N$ by theorem 3.13. (ii) in Euclidean form follows by differentiation; for (iii), a plane P is $E\{x_\nu = c\}$ for a Euclidean motion E, $\theta_P = E\theta_{\nu,c}E^{-1}$, and $Z^{\mathcal{K}}_\infty(\theta_P \zeta_i^* \sqcup \zeta_j) = Z^{\mathcal{K}}_\infty(\theta_{\nu,c}(E^{-1}\zeta_i)^* \sqcup E^{-1}\zeta_j)$ by invariance at complex coefficients, the $E^{-1}\zeta_i$ being supported in $\{x_\nu > c\}$; (vi) is [24, Cor. 10.44] with $\mathcal{Z}^{\mathcal{K}}$ $O(3)$-invariant, as in theorem 11.1(c), with its proviso on the odd moments of a component. □

Lemma 14.10 (a k- and J-uniform bound for the modified-box functionals on the tube of the enlarged family)**.** *Under* (T3), ($\mathrm{B}_{\mathrm{cell}}$), (SCH), *for every k, every $J \geq \max(J_0, J_1(R_{\mathcal{F}}))$ and every $\zeta \in \mathcal{D}^{\mathcal{F}}_{\sigma_+,H}$ (the tube of $\mathcal{F}$ in the sense of definition 2.5),*

$$|\Phi_k(\zeta)| \leq \exp\big(2\|g(\zeta)\|^2_{2,k} + \kappa\|k_+(\zeta)\|_{1,k} + K_-|k_-(\zeta)|_w\big).$$

The curvature constant is 2, not the 1 of (31), because the Cauchy–Schwarz split below doubles the curvature source and theorem 7.4(a) is quadratic in it; nothing depends on its value.

Proof. $|\Phi_k(\zeta)| \leq \Phi_k(\xi)$, $\xi := \mathrm{Re}\,\zeta$ (lemma 2.11(d)). Let $k(\xi)$ be the total real scalar source at ξ and $k_1 := k(\xi)_+$, $k_2 := k(\xi)_-$ its *pointwise* parts, sampled on the lattice. Cauchy–Schwarz at the lattice level (lemma 2.11(f); the multiplier $e^{-a_k\mathsf{S}(k_1-k_2)}$ splits by linearity of S): $\Phi_k(\xi) \leq Z^{a,(3.38)}(2f_\xi, -2k_2)^{1/2} Z^{a,(3.38)}(0, 2k_1)^{1/2}$. *First factor* (class $\mathcal{D}$): the $\mathcal{D}$-half of lemma 9.9, which needs only that the sources vanish off X_J ($J \geq J_0$), gives $\leq Z^a_{k,C_J}(2f_\xi, -2k_2)$; theorem 7.4(a) with the real scalar source $-2k_2$ gives $\leq e^{\|2f_\xi\|^2_{2,k}} N^a_{k,C_J}(0, -2k_2)$ (the same scheme multiplier on both sides); proposition 13.9 with $q := 2k_2 \geq 0$, supported in $Q_{R_{\mathcal{F}}}$, $J \geq J_1(R_{\mathcal{F}})$ and $\sup q = 2\|k(\xi)_-\|_\infty < H$ (this is $\zeta \in \mathcal{D}^{\mathcal{F}}_{\sigma_+,H}$; the proposition allows $\sup q \leq H$) gives $\ln N^a_{k,C_J}(0, -2k_2) \leq K_-|2k_2|_w$. Hence $\frac{1}{2}\ln(\text{first factor}) \leq 2\|g(\zeta)\|^2_{2,k} + K_-|k_-(\zeta)|_w$, as $f_\xi = g(\zeta)$ and $k_2 = k_-(\zeta)$. *Second factor* (class $\mathcal{I}$): the $\mathcal{I}$-half of lemma 9.10 with $B = C_{J+2n} \cap \Lambda_k$ gives $\leq Z^a_{k,C_{J+2n}}(0, 2k_1)$, and corollary 13.3, uniform in k and in the cube, with $0 \leq 2k_1 \leq s_1$ on the lattice ($\|k_+(\zeta)\|_\infty < \sigma_+/2$, sampled $\leq$ continuum) gives $\ln Z^a_{k,C_{J+2n}}(0, 2k_1) \leq \kappa\|2k_1\|_{1,k}$, so $\frac{1}{2}\ln(\text{second factor}) \leq \kappa\|k_+(\zeta)\|_{1,k}$. Add. On compacts of the tube the right side is bounded uniformly in k (lemma 2.6). □

The lemma is (31) (with curvature constant 2) for the modified-box functional, transported through lemmas 9.9 and 9.10; the two factors live on C_J and C_{J+2n}, which is harmless as proposition 13.9 and corollary 13.3 are uniform in the cube, and the amplitude $H/2$ is exactly the tube condition $\|k(\xi)_-\|_\infty \leq H/2$ on the total source. *The matrix points lie in the tube:* at $\xi^*_{ij} = \mathrm{Re}\,\zeta^*_{ij}$ the two sums of the total scalar source have disjoint supports, so $\|k(\xi^*)_\pm\|_\infty \leq \max(\mathsf{a}(\zeta_i), \mathsf{a}(\zeta_j)) < H/2 \leq \sigma_+/2$, and by continuity a neighbourhood of the finitely many matrix points lies in $\mathcal{D}^{\mathcal{F}}_{\sigma_+,H}$. *End of the proof of (iii), Wick sector.* (β) By lemma 14.10, $(\Phi_k)_k$ is locally

bounded on the open connected $\mathcal{D}^{\mathcal{F}}_{\sigma_+,H}$; extract $\mathcal{K}' \subset \mathcal{K}$ with $\Phi_k \to \Phi^J$ locally uniformly there; Gram positivity passes ($\mathcal{K}'$ may depend on J; harmless as in the curvature sector). (α) Let U be the set of coefficient vectors with purely imaginary curvature coefficients and positive real scalar coefficients on the nonnegative $\tilde{h}_j$; $U \cap \mathcal{D}^{\mathcal{F}} \neq \emptyset$ ($k_- = 0$, k_+ small), and it is a relatively open subset of the maximal totally real $(\mathrm{i}\mathbb{R})^{m_F} \times \mathbb{R}^{m_h}$, a set of uniqueness in $\mathcal{D}^{\mathcal{F}}$. On U, $Z^a_{k,C_J}(\zeta) \le \Phi_k(\zeta) \le Z^a_{k,C_{J+2n}}(\zeta)$ at every k (lemmas 9.9 and 9.10, $\mathcal{I}$-halves); the ends are functionals of $\mathcal{F}$ at points of its tube, converging along $\mathcal{K}'$ (lemma 14.7(b), J above the threshold of $\mathcal{F}$) to $Z^{a,\mathcal{K}}_J, Z^{a,\mathcal{K}}_{J+2n}$, which tend to $Z^{a,\mathcal{K}}_\infty$ (lemma 14.8, locally uniformly on the tube of $\mathcal{F}$, in particular on $U \cap \mathcal{D}^{\mathcal{F}}$). $\{\Phi^J\}_J$ is locally bounded on $\mathcal{D}^{\mathcal{F}}$ (J-free), so $\Phi^J \to Z^{a,\mathcal{K}}_\infty$ locally uniformly there, in particular at the matrix points, where the limit is $Z^{a,\mathcal{K}}_\infty(\theta\zeta_i^* \sqcup \zeta_j)$; positivity passes. Every plane by (ii) (the amplitude is translation invariant); the polynomial form by Weierstrass.

Remark 14.11 (amplitudes in $[H/2, \sigma_+/8)$: the route through the fixed-cube limits, for a further subsequence). The proof of lemma 10.3 bounds the first factor on the Wick tube $T_{\mathcal{F}'}$ of definition 10.1 by the vertex argument at the cube C_{J_0}, which needs the fixed-cube limits in their boundedness form only. For amplitudes in $[H/2, \sigma_+/8)$ the matrix points lie in $T_{\mathcal{F}'}$ but in general outside $\mathcal{D}^{\mathcal{F}}_{\sigma_+,H}$, the only domain on which definition 14.1 gives convergence along $\mathcal{K}$; what follows is therefore stated here only for a further subsequence. Assume the boundedness in k of the functionals of the class-$\mathcal{D}$ families $(2f_\xi, -2k_2(\xi))$ at the one cube C_{J_0} (with the coefficient-sign split of definition 10.1). The vertex argument then gives local boundedness of $(Z^a_{k,J})_k$ and $(\Phi_k)_k$ on $T_{\mathcal{F}'}$ (first factors at C_{J_0} by that boundedness, second factors by corollary 13.3); Montel, with a diagonal over the countably many J, yields a further subsequence $\mathcal{K}' \subset \mathcal{K}$, depending on the family, along which $Z^a_{k,J} \to Z^{a,\mathcal{K}'}_J$ and $\Phi_k \to \Phi^J$ locally uniformly on $T_{\mathcal{F}'}$ (on $T_{\mathcal{F}'} \cap \mathcal{D}^{\mathcal{F}}$ the first limit is $Z^{a,\mathcal{K}}_J$). The sandwich step runs on $T_{\mathcal{F}'}$ with the uniqueness set $U \cap T_{\mathcal{F}'}$, the J-free local bound of proposition 10.5(a) on $T_{\mathcal{F}'}$ serving the Vitali step in J, and proposition 10.5(b) along $\mathcal{K}'$ for the remaining planes; so OS2 holds up to $\mathsf{a}(\zeta_i) < \sigma_W = \sigma_+/8$ for the LIM state of such a $\mathcal{K}'$ on $T_{\mathcal{F}'}$. Whether the convergence holds along $\mathcal{K}$ itself on $T_{\mathcal{F}'}$ is not claimed here; under the hypotheses of Form (A), theorem 23.7 gives OS2 along $\mathcal{K}$ with no amplitude restriction. The remark is used nowhere else.

14.4. **Identification, and King's printed state.**

Proposition 14.12 (identification)**.** (a) *For every continuum subsequence $\mathcal{K}$ and every family, $Z^{\mathcal{K}}_\infty(\cdot,0) = Z^{\mathrm{LIM}}_\infty(\cdot,0)$, the limit state of the rotation theorem (theorem 3.13); in particular all LIM states coincide,* (UNIQ) *holds, and they are $\mathbb{R}^3 \rtimes O(3)$-invariant. Theorem 3.13 does* not *give theorem 3.2.*

(b) *The following are equivalent: (i) the comparison family is convergent at fixed cubes (definition 3.1), that is the existence in $(0,\infty)$ of the fixed-cube limits of the comparison family at real coefficients; (ii) for every J and family, $a^{(a)}_{k,J} \to Z_J$ locally uniformly on $\mathbb{C}^N$ along the full sequence; (iii) every increasing sequence of spacings is a continuum subsequence. Under them $Z^{\mathcal{K}}_J = Z_J$ and $Z^{\mathcal{K}}_\infty = Z_\infty := \lim_J Z_J$ for every $\mathcal{K}$: every LIM state is the iterated limit of the comparison family.*

(c) *Assume* (T3) *and let a be admissible. Then* (SCH) *and* ($\mathrm{B}_{\mathrm{cell}}$) *hold (lemmas 14.4 and 14.5), every increasing sequence of spacings is a continuum subsequence for the Wick sector, and for every $\mathcal{K}$, $Z^{a,\mathcal{K}}_\infty = G_\infty$ of theorem 13.10 on $\mathcal{D}_{\sigma_+,H}$, which equals $Z^a_{\infty,T}$ of proposition 10.5 on $T_{\mathcal{F}'} \cap \mathcal{D}_{\sigma_+,H}$ (proposition 10.5 lives on the Wick tube $T_{\mathcal{F}'}$, a different domain). Once* (SCH) *and* ($\mathrm{B}_{\mathrm{cell}}$) *are given, the argument uses theorem 3.5 for a alone.*

Proof. (a) Let $\mathcal{K}$ be a continuum subsequence, f a family, $J \ge J_0(f)$. By definition 14.1, $Z^{\mathcal{K}}_J(\cdot,0)$ is the locally uniform limit of $(a^{(a)}_{k_m,J})_m$, so $Z^{\mathcal{K}}_J(\cdot,0) \in L^f_J$. By theorem 3.13 every choice $\zeta_J \in L^f_J$, $J \ge J_0(f)$, converges locally uniformly to $Z^{\mathrm{LIM}}_\infty(\cdot,0)$; take $\zeta_J := Z^{\mathcal{K}}_J(\cdot,0)$, which by lemma 14.8 converges to $Z^{\mathcal{K}}_\infty(\cdot,0)$, and limits are unique. Hence any two continuum subsequences have the same LIM state, and invariance is the second clause of theorem 3.13. That theorem pins the

J-limit of the subsequential limits, not the fixed-cube limits themselves, so it does not give theorem 3.2.

(b) (i)⇒(ii): $(a^{(a)}_{k,J})_k$ is locally bounded on $\mathbb{C}^N$ (lemma 7.3) and converges at every real point; $\mathbb{R}^N$ is a set of uniqueness for entire functions, and Montel–Vitali gives locally uniform convergence of the whole sequence to an entire Z_J. (ii)⇒(iii): every subsequence converges to the same limit. (iii)⇒(i): the full sequence is itself an increasing sequence, hence a continuum subsequence, so $a^{(a)}_{k,J}$ converges locally uniformly, and at real x the limit lies in $[1,\infty) \subset (0,\infty)$ (lemma 14.7, Step 4). The last clause is lemma 14.8.

(c) (SCH) and ($\mathrm{B}_{\mathrm{cell}}$) are lemmas 14.4 and 14.5; so (31) holds. $(Z^a_{k,J})_k$ is locally bounded on the tube by (31) and converges at the real points of the tube by theorem 3.5, a non-empty open subset of $\mathbb{R}^{n+p}$ and a set of uniqueness in the connected tube; Montel–Vitali: the whole sequence converges locally uniformly on the tube (theorem 13.10(i)), so every increasing sequence is a continuum subsequence for the Wick sector with $Z^{a,\mathcal{K}}_J = G_J$, and $Z^{a,\mathcal{K}}_\infty = G_\infty$ by lemma 14.8; on $T_{\mathcal{F}'} \cap \mathcal{D}_{\sigma_+,H}$ this is the same J-limit of the same G_J as $Z^a_{\infty,T}$ (proposition 10.5(a)). □

Corollary 14.13 (the iterated limits of the comparison family)**.** *Assume that the comparison family is convergent at fixed cubes (definition 3.1). Then every increasing sequence of spacings is a continuum subsequence, all LIM states coincide with the iterated limit* $Z_\infty(f,0) = \lim_J \lim_k \langle e^{F(f)}\rangle^{(a)}_{k,J}$ *of the comparison family, and the curvature sector (i)–(vi) of theorem 14.9 holds for it with the lattice laws of (v) converging along the* full *sequence of spacings; (vii) holds as well. For general sources assume* (T3) *and let a be admissible; then every increasing sequence is a continuum subsequence for the Wick sector, every LIM state is* G_∞ *(*$= Z^a_{\infty,T}$ *on* $T_{\mathcal{F}'} \cap \mathcal{D}_{\sigma_+,H}$*),* $\kappa^{\mathcal{K}} = \kappa$ *is independent of* $\mathcal{K}$*, and Form (B) of theorem 14.9 holds for* G_∞ *with the full-sequence lattice clause.*

Proof. Curvature sector: proposition 14.12(b) makes every increasing sequence a continuum subsequence with $Z^{\mathcal{K}}_\infty = Z_\infty$; (i)–(vi) are those of theorem 14.9, and in (v) the local laws converge along the full sequence because the characteristic functionals now do (step (β) with the full sequence); (vii) is proposition 14.12(a). Wick sector: proposition 14.12(c); the clauses are Form (B) of theorem 14.9 for G_∞, with OS2 at amplitude $< H/2$ by the route of lemma 14.10, which does not use theorem 3.5, and at amplitude $< \sigma_W = \sigma_+/8$ by the route of theorem 10.7; both hold ($H/2 = \sigma_+/32 < \sigma_+/8$: the route without the fixed-cube limits gives the smaller amplitude, and what it improves is the list of inputs); $\kappa^{\mathcal{K}} = \kappa$ by the scheme clause. □

Theorem 3.5 is the scheme-a analogue, with scalar sources, of the existence clause of King's theorem; it is not that theorem, which is false as printed with scalar sources in King's Wick ordering. With (W2) in addition one obtains Form (A) for the full object Z^a_∞ and, under (R_W), its Euclidean clause; this is the optional extension of theorem 23.7.

Corollary 14.14 (King's convention)**.** *Define continuum subsequences and LIM states of the K-family* $a^K_{k,J}(z) = \langle e^{\sum_j z_j F(f_j)}\rangle^K_{k,J}$ *(definition 2.2) by definitions 14.1 and 14.2 with* $\langle\cdot\rangle^K_{k,J}$ *in place of* $\langle\cdot\rangle^{(a)}_{k,J}$*.*

(a) (No hypothesis.) *Every increasing sequence of spacings has a continuum subsequence of the K-family; for every such* $\mathcal{K}$*, family and* $J \geq J_0(f)$ *the fixed-cube limits* $Z^{K,\mathcal{K}}_J$ *and the LIM state* $Z^{K,\mathcal{K}}_\infty = \lim_J Z^{K,\mathcal{K}}_J$ *exist, are entire and obey* $|Z^{K,\mathcal{K}}_J|, |Z^{K,\mathcal{K}}_\infty| \leq e^{\|g(z)\|^2_2}$*, with the other clauses of lemma 14.7(a) and lemma 14.8. Nothing beyond this is asserted for a K-family LIM state from the bounds alone.*

(b) (With theorem 3.2.) *Every increasing sequence of spacings is a continuum subsequence of the K-family, every K-family LIM state is King's printed state* Z^K_∞*, every LIM state of the comparison family is* Z^K_∞ *as well (corollary 4.4(iii)), and the curvature-sector axioms of theorem 21.1 hold for it, with the lattice laws of theorem 11.1(a) converging along the full sequence.*

With scalar sources, the corresponding statements on King's convention are supplied by theorem 21.3 and proposition 21.4.

Proof. (a) The proof of lemma 14.7(a) uses only the k- and J-uniform local bound of lemma 7.3 (Step 2), the second-order bound $\langle F(u)^2\rangle \le 2\|u\|_{2,k}^2$ of lemma 7.1 (Step 3), and the current representation and Jensen's inequality (Step 4); that of lemma 14.8 uses in addition the first inequality of lemma 2.10. All of these hold on King's convention: the bounds by theorem 7.6 (the second-order bound is photon-only and transfers by the covariance identity of lemma 4.2), the cut inequality by corollary 4.3. So the two lemmas hold for the K-family by the same proofs; the argument makes no use of theorem 3.2. (b) By theorem 3.2, corollary 4.4(ii)(α) gives, for every $J \ge J_0$ and family, locally uniform convergence of $a^K_{k,J}$ along the full sequence, so every increasing sequence is a continuum subsequence of the K-family, and by (β) every K-family LIM state is Z^K_∞; corollary 4.4(iii) is the statement for the comparison family; the axioms are the curvature sector of theorem 21.1, and the full-sequence lattice laws are theorem 11.1(a). □

Remark 14.15 (on the construction of the subsequences). Every subsequence of spacings used in this paper is constructed in Step 2 of the proof of lemma 14.7 from the k-uniform bounds of lemma 7.3 and (31) and Montel's theorem, or is a further extraction from one so constructed, and the identification of its limits uses only the monotone cube limit of lemma 14.8; no subsequence statement is cited from [14].

Rotation invariance

15. The rotation part: lattices, walled tori and the route

This and the following six sections form the rotation part of the paper. Its object is King's printed state on curvature sources, $z \mapsto Z^K_\infty(\sum_j z_j f_j, 0)$, the entire function of the coefficients of real 2-forms $f_j \in C^\infty_c$ whose derivatives at $z = 0$ are the curvature Schwinger functions (Theorem 3.2 and the curvature-sector section); its result, Theorem 15.1, is that this function is invariant under every Euclidean motion.

Rotation invariance is not automatic, since all lattices in King's construction have the same orientation. Theorem 2.4 of [14] states that on the tori D_J the lattices $\varepsilon_k\mathbb{Z}^3$ and $R(\varepsilon_k\mathbb{Z}^3)$, with R the rotation by $\theta_0 = \arctan(3/4)$ in a coordinate plane, have the same continuum limit; since θ_0/π is irrational these rotations generate a dense subgroup of $SO(3)$. As printed the statement is false in two ways (Proposition I.2, Remark 16.1): the source it pairs with the plaquettes is one scalar function on both lattices, which couples the field to two different 2-forms, and the Wick constant it uses misses an order-e^2 logarithm. We therefore call the printed statement *King's torus conjecture*, and we use instead the corrected statement, for 2-form sources sampled in each lattice's own frame and for the two lattices at the same spacing; it is the torus comparison (K_0) of Theorem 17.1, proved in section 17.3. Moreover, even granting the torus comparison, two steps remain between it and rotation invariance of King's state: the infinite-volume limit along tori, where correlation inequalities give no monotonicity, and its identification with the limit along cubes, which King writes can be expected from a cluster expansion [14, (2.24)]. The ultraviolet analysis of [13, 14] needs boundaries that are lattice hyperplanes [14, p. 326], and no bounded region has that property for both lattices (Proposition 20.11). Where to place the two-lattice comparison is therefore the whole question. The rotation part places it on periodic tori carrying a smooth mass wall, and proves $O(3)$ invariance of King's printed state in the curvature sector from three inputs, each stated in one line and proved where it is stated: the existence theorem (E_C) of Theorem 3.2, the torus comparison (K_0) (Theorem 17.1) and the heavy-shell estimates (HS), (DS) (Theorem 20.2 and proposition 20.1).

15.1. **The torus route.** On each lattice separately King's cubes are compared with tori carrying a *smooth mass wall*. Fix R in the generating set $\mathcal{R}_0 = \{R_{\mu\nu}(\pm\theta_0)\}$ of Definition 15.2, a pair $(f, g) \in \mathcal{K}$ (a real 2-form f and a function $g \ge 0$, both smooth with compact support) and a scheme a; at $g = 0$ nothing below depends on the scheme. For a walled torus $\mathfrak{c} = (J', w)$, the periodic model on $D_{J'}$ in which the scalar mass squared is raised by $2w$, with $w \ge 0$ smooth

and vanishing near the supports (Definition 15.6, Definition 18.1), the chain is

(33)
$$Z^a_{k,L}(\mathrm{i}\,R\cdot f, R\cdot g) \;=\; Z^a_{R^{-1}\Lambda_k,\,R^{-1}X_L}(\mathrm{i}f,g) \;\le\; T^a_{\mathfrak{c},R^{-1}\Lambda_k}(f,g) \;\xrightarrow[k\to\infty]{}\; \mathcal{W}^a_{\mathfrak{c}}(f,g) \;\xleftarrow[k\to\infty]{}\; T^a_{\mathfrak{c},\Lambda_k}(f,g).$$

Here $Z^a_{k,L}$ is the functional of the comparison cube $X_L = C_L \cap \Lambda_k$ on the axis lattice (Definition 2.3); the first equality is a passive rotation (Lemma 15.9(b)); the inequality is a correlation inequality on the rotated lattice $R^{-1}\Lambda_k$ (Lemma 15.8), for which $R^{-1}C_L$ is a cube with faces on lattice hyperplanes; $T^a_{\mathfrak{c},\Lambda}$ is the walled ratio of Definition 15.6; and the two continuum limits exist and agree by the torus comparison (K_0) (Theorem 17.1; at $g \neq 0$ its scheme form $(\mathrm{K})_a$, Definition 17.2). The comparison cubes on the left are squeezed, as $k \to \infty$ and then $L \to \infty$, to King's printed state by the interlacing of the two cube families (the section on the interlacing). What is missing is an upper bound for walled tori on the axis lattice by King's printed state:

(34) for every $\eta > 0$ there is an admissible $\mathfrak{c}$ with $\mathcal{W}_{\mathfrak{c}}(f,0) \le Z^K_\infty(\mathrm{i}f,0) + \eta$.

This is the closing estimate (TIGHT_0) of Theorem 18.4. It involves one lattice and one orientation, and the reverse inequality holds in the limit (Lemma 18.2), so (TIGHT_0) says that King's printed state is the lower envelope of the walled tori. With it, letting $L \to \infty$ in (33) gives $Z^K_\infty(\mathrm{i}\,R\cdot f,0) \le Z^K_\infty(\mathrm{i}f,0)$, and applying this to $R^{-1} \in \mathcal{R}_0$ gives equality. The passage from the dense subgroup to $O(3)$ uses the Lipschitz bound of Corollary 8.3 at imaginary coefficients and the cubic symmetry of King's printed state (Theorem 18.12), so no regularity of the Schwinger functions beyond that corollary is needed.

The closing estimate (TIGHT_0) does not follow from the correlation inequalities: Remark 18.8 gives a family of Gaussian functionals with all the other properties used in the proof whose analogue of King's state is not rotation invariant. We therefore isolate the local estimate that implies (TIGHT_0). In the walled torus let the sources sit in a box H and let U be a thick shell around H inside the heavy plateau, where the wall is at least 1. Raise the scalar mass squared on the shell further by $2K$, through a second smooth profile, and let $\Psi_k(n,K)$ be the resulting expectation at lattice spacing ε_k, with $\Psi_k(n,\infty)$ the expectation with the matter removed on the shell (Definition 19.3). The decoupling property (RES_0) of Definition 19.8 says that for one such configuration and one finite K

$$\liminf_{k\to\infty} \big[\Psi_k(n,K) - \Psi_k(n,\infty)\big] \le \eta,$$

with the width of the shell chosen so that a photon boundary term is below η as well. At every fixed spacing the bracket tends to 0 as $K \to \infty$ (Proposition 20.10); the content of (RES_0) is that the heavy-shell limit and the continuum limit can be interchanged, at one geometry and with no rate. A two-line sandwich (Lemma 19.7) shows that the heavy shell is bounded below by a higher wall, and once the matter on the shell is dead the photon on the far side can be pinned and then replaced by King's Neumann photon on a larger cube at a cost exponentially small in the width of the shell, uniformly in the spacing (Lemma 19.1); so (RES_0) implies (TIGHT_0) (Theorem 19.10); in the exterior-shell geometry the same chain is (DS) (Proposition 20.1). Under (K_0) the limit $\lim_k \Psi_k(n,K)$ exists for every finite K, and (RES_0) is equivalent to the statement that the monotone limit $\lim_K \lim_k \Psi_k(n,K)$ lies within η of $\limsup_k \Psi_k(n,\infty)$ (Theorem 20.8); when the dead-shell limit $\lim_k \Psi_k(n,\infty)$ exists, which is a further existence statement not needed for (RES_0), this is the continuum form (RES^0), in which no lattice spacing appears. In the curvature sector (RES_0) is proved at one configuration (Theorem 20.5): the shell mass $\varkappa$ is put inside the matter covariance of King's block-spin flow, and two estimates on that flow, a two-sided bound on the stopping-scale integral uniformly in $\varkappa$ and a $\varkappa$-Cauchy property of the reduced small-field action on one test set, and a disc bound for the heavy shell, which is established there, give the tail bound at one configuration. The two estimates form (HS) (Theorem 20.2); they come from a renormalisation-group analysis of King's walled torus model with the shell mass inside the matter covariance (Propositions 20.3 and 20.4).

The torus comparison is reduced to one-sided bounds at a single lattice spacing (Theorem 17.6). The passive map $S = R^{-1}\sigma_\nu$ is an involution that exchanges the two lattices and maps the torus D_J to the torus $R^{-1}D_J$. A forward bound of King's type on each of the two

tori therefore gives both directions of the comparison on D_J. Theorem 17.6 reduces (K_0) in the curvature sector to the one-sided bounds (F_0) and (V) of Definition 17.5 (bounds at one spacing, with a condition on their constants) together with the existence of the axis-lattice limits (Remark 17.9); these are what the proof of (K_0) along King's lines produces, with the large-field clause of King's upper bound in place of King's torus bound [14, (4.2)], which is vacuous at a non-zero wall (Remark 17.9, Remark 17.10). An alternative placement of the comparison, on a Neumann–Dirichlet box aligned with one lattice and a staircase for the other, is possible in principle and is not used here.

15.2. **The main theorem of the rotation part.** For $R \in O(3)$ and $b \in \mathbb{R}^3$, sources are transported as 2-forms, $R \cdot f$ and $\tau_b f$ (Definition 2.3).

Theorem 15.1 ($O(3)$ invariance of the curvature sector)**.** *King's printed state on curvature sources is $O(3)$-invariant: for every $R \in O(3)$, all real 2-forms $f_1, \dots, f_N \in C_c^\infty(\mathbb{R}^3; \Lambda^2)$ and all $z \in \mathbb{C}^N$,*

$$Z_\infty^K\Big(\sum_j z_j\, R\cdot f_j,\, 0\Big) = Z_\infty^K\Big(\sum_j z_j f_j,\, 0\Big),$$

so all curvature Schwinger functions are $O(3)$-covariant. With the translations of Corollary 8.3 the same holds for every Euclidean motion $x \mapsto Rx + b$, with $\tau_b R \cdot f_j$ in place of $R \cdot f_j$: the curvature sector is invariant under $\mathbb{R}^3 \rtimes O(3)$, and the curvature Schwinger functions are Euclidean covariant.

The proof is assembled at the end of the heavy-shell sections (after Theorem 20.5): the heavy-shell estimates (HS) (Theorem 20.2) give (RES_0) at one configuration (Theorem 20.5), (RES_0) gives the closing estimate (TIGHT_0) (Theorem 19.10, with the torus comparison (K_0) of Theorem 17.1 and the dead-shell chain (DS) of Proposition 20.1), and Theorem 18.12 gives the invariance from (E_C) (Theorem 3.2), (K_0) and (TIGHT_0). For the curvature two-point function the weaker form (TIGHT_2) of the closing estimate suffices (Corollary 18.13). The rotation part is organised as follows: this section sets up the lattices with a frame, the walled tori and the three lemmas (cutting, passive covariance, second moment) that every later argument uses; the next section treats Wick schemes and the full object on the comparison convention, which the scalar-source forms of the theorems need; then come the torus comparison, the closing hypothesis with the rotation theorems, the two heavy-shell sections, and the corrections to [13, 14] that the rotation part relies on.

15.3. **Geometry.**

Definition 15.2. Let $\theta_0 \in (0, \pi/2)$ with $\cos\theta_0 = \frac{4}{5}$, $\sin\theta_0 = \frac{3}{5}$. For $1 \le \mu < \nu \le 3$ let $R_{\mu\nu}(\theta)$ be the rotation with $Re_\mu = \cos\theta\, e_\mu + \sin\theta\, e_\nu$, $Re_\nu = -\sin\theta\, e_\mu + \cos\theta\, e_\nu$, $Re_\rho = e_\rho$ for $\rho \neq \mu, \nu$. Put $\mathcal{R}_0 := \{R_{\mu\nu}(\pm\theta_0) : 1 \le \mu < \nu \le 3\}$ and let G_0 be the group generated by $\mathcal{R}_0$. The spacings $\varepsilon_k = 5^{-k}$, the axis lattices $\Lambda_k = \varepsilon_k\mathbb{Z}^3$, King's large-block integer $M \ge 1$ [13, §2.3] and King's cubes $C_J = [-MJ, MJ)^3$, $J \in \mathbb{N}$ [14, (2.7)], are those of the first sections. The *King tori* of the rotation part are $D_J := \mathbb{R}^3/(50MJ\,\mathbb{Z})^3$ with fundamental cell $\Gamma_J := [-25MJ, 25MJ)^3$ [14, (4.1)]; D_J is a King torus in the sense of the section on the model ($\ell = 50MJ \in M\mathbb{N}$ is even), so Λ_k descends to it. We write $Q_b := [-b, b]^3$ (written Q_R in the first sections), $N_r(X) := \{y : \mathrm{dist}_\infty(y, X) \le r\}$, and $\bar{B}_\rho$ for the closed Euclidean ball of radius ρ about 0.

Lemma 15.3. (a) *For $R \in \mathcal{R}_0$ the matrix $5R$ has integer entries, $R^{-1} \in \mathcal{R}_0$, and $\sigma_\nu R_{\mu\nu}(\theta)\sigma_\nu = R_{\mu\nu}(-\theta)$, where σ_ν is the reflection $x_\nu \mapsto -x_\nu$.*

(b) *For $R \in \mathcal{R}_0$, $k \ge 0$ and $J \ge 1$ the lattices $R\Lambda_k$ and $R^{-1}\Lambda_k$ are invariant under translations by $50MJ\,\mathbb{Z}^3$, so they descend to D_J.*

(c) *For $R \in \mathcal{R}_0$, $|Rx|_\infty \le \frac{7}{5}|x|_\infty$, and $Q_{5b/7} \subset R^{-1}Q_b \subset Q_{7b/5}$ for $b > 0$.*

(d) *θ_0/π is irrational, and G_0 is dense in $SO(3)$.*

Proof. (a) The entries of R are $0, 1, \pm\frac{3}{5}, \pm\frac{4}{5}$, and $R_{\mu\nu}(\theta)^{-1} = R_{\mu\nu}(-\theta)$; conjugation by σ_ν reverses the orientation of the (x_μ, x_ν)-plane and fixes the third axis. (b) For $p \in 50MJ\,\mathbb{Z}^3$ we have $R^{\pm1}p = \frac{1}{5}(5R^{\pm1})p \in 10MJ\,\mathbb{Z}^3 \subset \Lambda_k$. Hence $R^{-1}\Lambda_k + p = R^{-1}(\Lambda_k + Rp) = R^{-1}\Lambda_k$,

and likewise for $R\Lambda_k$. (c) The sums of the absolute values of the entries in each row of R are at most $\frac{7}{5}$. If $Rx \in Q_b$ then $|x|_\infty = |R^{-1}(Rx)|_\infty \le \frac{7}{5}b$, since $R^{-1} \in \mathcal{R}_0$; if $|x|_\infty \le \frac{5}{7}b$ then $|Rx|_\infty \le b$. (d) If $\theta_0 \in \pi\mathbb{Q}$, then $e^{\mathrm{i}\theta_0} = (4+3\mathrm{i})/5$ is a root of unity, hence an algebraic integer. Its minimal polynomial over $\mathbb{Q}$ is $x^2 - \frac{8}{5}x + 1$, which does not have integer coefficients; this is a contradiction (compare [14, pp. 337–338]). Hence $\{n\theta_0\}_{n\in\mathbb{Z}}$ is dense modulo 2π, and the closure of $\{R_{\mu\nu}(\theta_0)^n\}_{n\in\mathbb{Z}}$ is the group of all rotations in the (x_μ, x_ν)-plane. Every element of $SO(3)$ is a product $R_{12}(\alpha)R_{23}(\beta)R_{12}(\gamma)$ (Euler angles), so $\overline{G_0} = SO(3)$. □

15.4. **Lattices with a frame.**

Definition 15.4 (lattices with a frame, plaquettes, sources)**.** A *lattice of spacing* ε_k is $\Lambda = O\Lambda_k$ with $O \in O(3)$, equipped with the frame $f_\alpha := Oe_\alpha$, $\alpha = 1,2,3$, and with the nearest-neighbour bonds $\langle x, x+\varepsilon_k f_\alpha\rangle$. (The frame vectors f_α and the 2-form sources f share a letter; the index tells them apart.) A field configuration on a set X of sites consists of $\phi(x) = r_x e^{\mathrm{i}\vartheta_x} \in \mathbb{C}$ and $A_\alpha(x) \in \mathbb{R}$, $x \in X$; the bond $\langle x, x+\varepsilon f_\alpha\rangle$ carries $A_{\langle x,x+\varepsilon f_\alpha\rangle} := A_\alpha(x)$ and the reversed bond carries $-A_\alpha(x)$ (Feynman form of the vector field, [14, (2.5)]). The action on Λ is King's action of the section on the model written in the frame, e_α replaced by f_α throughout, with the same free-boundary counterterm δm_ε^2 and the same free Wick constant W_ε, which depend on ε only. A plaquette is $p = (x;\alpha,\beta)$ with $\alpha < \beta$; its centre is $c(p) := x + \frac{\varepsilon}{2}(f_\alpha + f_\beta)$ and its field strength is

$$F(p) := \varepsilon^{-1}\big(A_\alpha(x) + A_\beta(x+\varepsilon f_\alpha) - A_\alpha(x+\varepsilon f_\beta) - A_\beta(x)\big) \qquad [14,\ (2.11)].$$

For a real 2-form f and a real function h, smooth and compactly supported (on a torus: smooth and periodic; a compactly supported source whose support lies in the interior of a fundamental cell is periodised from it), put

$$F_\Lambda(f) := \sum_p \varepsilon^3\, f_{c(p)}(f_\alpha, f_\beta)\, F(p), \qquad \mathsf{S}_\Lambda(h) := \sum_x \varepsilon^3 h(x), \qquad W_\Lambda(h) := \sum_x \varepsilon^3 h(x)\big(|\phi(x)|^2 - 2\mathsf{W}_\varepsilon\big),$$

the sums running over the plaquettes $p = (x;\alpha,\beta)$, respectively the sites, of the region (of one fundamental cell on a torus): the 2-form is *sampled in the lattice's own frame*. On Λ_k with the standard frame these are $F_{\Lambda_k}(f)$, $\mathsf{S}_{\Lambda_k}(h)$ and ${:}\phi^2{:}(h)$ of Definition 2.3. For a finite family $\zeta = (z_j, f_j; w_j, h_j)_j$ with complex coefficients, $\Theta(\zeta) := \exp(\sum_j z_j F_\Lambda(f_j) + \sum_j w_j W_\Lambda(h_j))$ as there, and $O\cdot\zeta := (z_j, O\cdot f_j; w_j, O\cdot h_j)_j$ with the transport of Definition 2.3. The cone $\mathcal{K}$ of Definition 2.3 is invariant under $O(3)$, under translations, under $(f,g) \mapsto (tf, tg)$, $t \ge 0$, and under sums.

Remark 15.5 (the frame convention). In [13, 14] the curvature source is written $F(g) = \sum_p \varepsilon^3 g(p) F(p)$ with one function g on all plaquettes [14, (2.12)]. On one lattice a 2-form sampled as above is a function on plaquettes, and King's proofs treat the source terms componentwise ("the sources $g_{\mu\nu}(x'), h(x') \ldots$ can be replaced by $g_{\mu\nu}(x), h(x)$", [13, p. 665]; the sentence is stated there without an argument). King's single-lattice statements at $h = 0$ are used in this sense throughout: the existence theorem (E_C) is stated, in Theorem 3.2, for real smooth 2-form sources sampled at plaquette centres in the lattice's own frame, and this existence clause of King's Theorem 2.1 of [14] is proved in section 5; the statements of [14, Thms. 2.2, 2.3, (3.40)–(3.44)] that the rotation part uses, all at $h = 0$, are established in the first sections from (E_C) and the correlation inequalities, [14, (3.42)–(3.43)] are not used, and the torus statement [13, Thm. 2.1] at a wall is part of the torus comparison (K_0) (Theorem 17.1). The two-lattice statement is different: Proposition I.2 shows that it fails for a single scalar function on all plaquettes, and (K_0) is formulated for 2-forms sampled in each lattice's own frame.

15.5. **Tori and walled tori.**

Definition 15.6 (periodic and walled torus states)**.** Let Λ be a lattice of spacing $\varepsilon = \varepsilon_k$ with a frame, and $d\nu := \prod_x \prod_\alpha dA_\alpha(x)\, d^2\phi(x)$ over the sites in question.

(a) *Boxes.* For a finite set X of sites of Λ, $\langle\cdot\rangle^{\mathrm{ND}}_{\Lambda,X}$ is the Neumann–Dirichlet state of Definition 2.1 written in the frame of Λ (Dirichlet condition for the scalar field outside X, Neumann condition for the vector field), possibly with the matter killed on a set $U \subset X$ in the sense

of that definition. On Λ_k these are the states $\langle\cdot\rangle^{\mathrm{ND}}_{k,X}$ of Definition 2.1, among them the comparison cube $\langle\cdot\rangle^{(a)}_{k,J}$ and King's RG cube $\langle\cdot\rangle^{K}_{k,J}$ (Definition 2.2).

(b) *Tori.* If Λ descends to D_J, $\langle\cdot\rangle_{\Lambda,D_J}$ is the normalised measure $Z^{-1}e^{-S}d\nu$ with the action S summed over the sites and bonds of the finite graph $\Lambda/(50MJ\,\mathbb{Z})^3$ (periodic conditions for both fields). On Λ_k this is the periodic state $\langle\cdot\rangle_{k,T}$ of the section on the model for the King torus $T = D_J$.

(c) *Walled tori.* A *walled configuration* is $\mathfrak{c} = (J', w)$ with $J' \in \mathbb{N}$ and $w \in C^\infty(D_{J'}; [0,\infty))$, the *wall*. If Λ descends to $D_{J'}$, $\langle\cdot\rangle_{\mathfrak{c},\Lambda}$ is the torus state of (b) with the density multiplied by $\prod_x \exp(-\varepsilon^3 w(x)|\phi(x)|^2)$. The wall raises the scalar mass squared by $2w$ and acts in the single-site measures. We write $(J', \psi, n) := (J', n\psi)$ for $n \geq 0$ and a fixed profile ψ.

(d) *Killed sites.* For any of the states (b), (c) and a set U of sites, the state *with the matter killed on* U is defined as in Definition 2.1: the single-site measure at $x \in U$ is replaced by the point mass at $\phi(x) = 0$, every hopping term with an endpoint in U reduces to its diagonal part at the other endpoint, and the photon variables are unchanged.

For every state $\langle\cdot\rangle_{\Lambda,X}$ above and a scheme a (Definition 2.3, Remark 2.4) the scheme functional is defined by the formula of Definition 2.3 with F_Λ, W_Λ, S_Λ in place of their axis-lattice versions,

$$Z^a_{\Lambda,X}(\zeta) := e^{-\sum_j w_j a_k \mathsf{S}_\Lambda(h_j)} \big\langle \Theta(\zeta) \big\rangle_{\Lambda,X};$$

on Λ_k we keep the notation $Z^a_{k,X}$ of Definition 2.3, and on the rotated lattice we write $Z^a_{R^{-1}\Lambda_k,X}$. All these states are invariant under the charge conjugation $(A,\vartheta) \mapsto (-A,-\vartheta)$, which maps $F_\Lambda(f)$ to $-F_\Lambda(f)$ and fixes r; hence $\langle\Theta(\mathrm{i}f,g)\rangle = \langle\cos F_\Lambda(f)\, e^{W_\Lambda(g)}\rangle$ is real for $(f,g) \in \mathcal{K}$.

Definition 15.7 (the unnormalised torus functional and the walled ratio)**.** For a lattice Λ of spacing ε_k descending to a torus T, and real smooth periodic sources G (a 2-form) and h (a function), the *unnormalised* generating functional is

$$\mathsf{Z}_{\Lambda,T}(G,h) := \int e^{-S(A,\phi)+F_\Lambda(G)+W_\Lambda(h)}\, d\nu,$$

with no field-independent constant in the action; for a pair $s = (G,h)$ we also write $\mathsf{Z}_{\Lambda,T}(s)$. Two lattices of the same spacing descending to T have the same number of sites in a fundamental cell, so any field-independent constant per site multiplies both by the same factor. For a walled configuration $\mathfrak{c} = (J', w)$, a lattice Λ descending to $D_{J'}$, a scheme a and $(f,g) \in \mathcal{K}$ with $w = 0$ on the support of g, the *walled ratio* is

$$T^a_{\mathfrak{c},\Lambda}(f,g) := e^{-a_k \mathsf{S}_\Lambda(g)} \big\langle \Theta(\mathrm{i}f,g) \big\rangle_{\mathfrak{c},\Lambda} = \frac{Z^a_{\Lambda,D_{J'}}(\mathrm{i}f, g-w)}{Z^a_{\Lambda,D_{J'}}(0,-w)}.$$

The second expression agrees with the first because

$$e^{-W_\Lambda(w)} = e^{2\mathsf{W}_\varepsilon \mathsf{S}_\Lambda(w)} \prod_x e^{-\varepsilon^3 w(x)|\phi(x)|^2},$$

so that the wall is the scalar source $-w$ up to the constant $e^{2\mathsf{W}_\varepsilon \mathsf{S}_\Lambda(w)}$, which cancels between numerator and denominator, as do the multipliers $e^{a_k \mathsf{S}_\Lambda(w)}$ of both. The same formulas define the walled ratio for the periodic model of Λ on any torus $T = \mathbb{R}^3/P$ to which Λ descends (defined as in Definition 15.6(b), with the finite graph Λ/P) and any smooth wall $w \geq 0$ on T vanishing on the support of g; this is used only for the transported tori of Lemma 15.9(b).

Every state of Definition 15.6, with or without killed sites, is a BFS measure of King type in the sense of the section on the model: $c_{xy} = \varepsilon$ for the bonds present, the single-site measures contain $e^{-\varepsilon^3[\frac12(m^2+\delta m_\varepsilon^2)r^2+\lambda r^4]}$, the diagonal parts $\frac{\varepsilon}{2} r_x^2$ of all bonds at x (including, for boxes, those leading out of X, and those leading into a killed site) and the wall factor $e^{-\varepsilon^3 w(x) r_x^2}$ with $w \geq 0$; the photon form is positive definite because $\mu_0 > 0$, and the exponential moments are finite because $\lambda > 0$. The same holds for every measure obtained from such a state by multiplying some c_{xy} by $t \in [0,1]$, multiplying photon gradient terms by $t \in [0,1]$, adding $s|\ell(A)|^2$ ($s \geq 0$) to the action, multiplying single-site measures by e^{-sr^2} ($s \geq 0$), or killing sites. Curvature sources

$F_\Lambda(f)$ are real linear functionals of A, and $e^{W_\Lambda(g)}$ is a positive constant times $e^{\sum_x \varepsilon^3 g(x) r_x^2}$; so Lemma 2.7, Lemma 2.8 and Lemma 2.9 apply to all of them, on every lattice with a frame.

15.6. **Cutting, passive covariance and the second moment.**

Lemma 15.8 (cutting, walled tori)**.** *Let $\langle\cdot\rangle_Y$ be a box, torus or walled-torus state of Definition 15.6 on a lattice Λ with a frame, possibly with killed sites, and let X be a set of sites of Λ such that X and its nearest neighbours lie in one fundamental cell. Let $\langle\cdot\rangle_X^{\mathrm{ND}}$ be the Neumann–Dirichlet state on X whose single-site measures (wall, killed sites and diagonal parts of the bonds leaving X included) are those of $\langle\cdot\rangle_Y$. Let $(f,g)\in\mathcal{K}$ be such that every plaquette and every site at which f, respectively g, does not vanish lies in X. Then, at every lattice spacing,*

$$\langle\Theta(\mathrm{i}f,g)\rangle_X^{\mathrm{ND}} \le \langle\Theta(\mathrm{i}f,g)\rangle_Y, \qquad \langle\Theta(f,-g)\rangle_X^{\mathrm{ND}} \ge \langle\Theta(f,-g)\rangle_Y,$$

and the same for Z^a. If the wall of Y vanishes on X and no site of X is killed, $\langle\cdot\rangle_X^{\mathrm{ND}} = \langle\cdot\rangle_{\Lambda,X}^{\mathrm{ND}}$ is the box state of Definition 15.6(a).

Proof. The wall and the killed sites are part of the single-site measures, and the frame plays no role, so the proof of Lemma 2.10 applies word for word. Delete from Y every photon gradient term coupling a site of X to a site outside X, and the cosine parts of all hopping terms between X and its complement, keeping their diagonal parts $\frac{\varepsilon}{2}(r_x^2+r_y^2)$; since X and its neighbours lie in one fundamental cell, these are bonds between distinct sites of the finite graph. By Lemma 2.7(a),(b) this does not increase the first expectation, by (d) it does not decrease the second, and every intermediate measure is a BFS measure. The resulting measure is a product (Lemma 2.7(e)); its X-factor is $\langle\cdot\rangle_X^{\mathrm{ND}}$, since the diagonal parts of the cut bonds are exactly the Dirichlet terms and the remaining photon terms are the Neumann ones, and the observable depends on the X-variables only. The free Wick constant, δm_ε^2 and the scheme multiplier $e^{\mp a_k \mathsf{S}_\Lambda(g)}$ are the same in both states. □

With $Y = X_{J+1}$ and $X = X_J$ on Λ_k this is King's monotonicity in the volume [14, Thm. 3.2, (3.31)–(3.33)] in the form his proof proves, with one Wick constant for all J, as discussed after Lemma 2.10; with Y a walled torus and X a cube of Λ inside the cell, it is the inequality of (33).

Lemma 15.9 (passive covariance and the algebra of the walled ratio)**.** *Let a be a scheme.*

(a) *For real coefficients the multiplier $e^{-\sum_j w_j a_k \mathsf{S}_\Lambda(h_j)}$ is a positive constant depending only on k, the coefficients and the numbers $\mathsf{S}_\Lambda(h_j)$. Hence the inequalities of Lemma 2.7 and Lemma 15.8 between two states on the same lattice with the same sources hold for Z^a, and for $T^a_{\mathfrak{c},\Lambda}$, in the same direction.*

(b) *Let $O\in O(3)$, let Λ be a lattice of spacing ε with frame (f_α), and let X be a box, a torus or a walled torus for Λ, possibly with the matter killed on U. Equip $O\Lambda$ with the frame (Of_α) and OX with the transported boundary conditions, the wall $O\cdot w$ and the killed set OU; for a torus $X=\mathbb{R}^3/P$ this means the torus $OX=\mathbb{R}^3/OP$, to which $O\Lambda$ descends. Then for every finite family ζ*

$$\big\langle\Theta(O\cdot\zeta)\big\rangle_{O\Lambda,OX} = \big\langle\Theta(\zeta)\big\rangle_{\Lambda,X}, \qquad \mathsf{S}_{O\Lambda}(O\cdot h)=\mathsf{S}_\Lambda(h), \qquad Z^a_{O\Lambda,OX}(O\cdot\zeta)=Z^a_{\Lambda,X}(\zeta).$$

In particular, for a walled configuration $\mathfrak{c}=(J',w)$ and (f,g) as in Definition 15.7, $T^a_{O\cdot\mathfrak{c},O\Lambda}(O\cdot f,O\cdot g)=T^a_{\mathfrak{c},\Lambda}(f,g)$, where $O\cdot\mathfrak{c}$ denotes the transported walled torus, the torus $OD_{J'}$ with the wall $O\cdot w$, and the left side is the walled ratio on that torus in the sense of Definition 15.7; $O\cdot\mathfrak{c}$ is a walled configuration $(J',O\cdot w)$ of Definition 15.6(c) when $OD_{J'}=D_{J'}$, as for a signed permutation matrix O. Moreover, on every torus the periodic model on Λ does not change if the frame (f_α) is replaced by $(\pm f_{\pi(\alpha)})$, for a permutation π and any choice of signs.

(c) *$T^a_{\mathfrak{c},\Lambda}(f,g)=e^{-a_k\mathsf{S}_\Lambda(g)}\langle\Theta(\mathrm{i}f,g)\rangle_{\mathfrak{c},\Lambda}$ is the normalised expectation of the walled torus state up to the multiplier of* (a)*, and at $g=0$ the walled ratio $T^a_{\mathfrak{c},\Lambda}(f,0)=\langle e^{\mathrm{i}F_\Lambda(f)}\rangle_{\mathfrak{c},\Lambda}$ does not depend on the scheme.*

Proof. (a) The first statement is immediate (compare the remark in [6, p. 281] that the inequalities are not affected by Wick ordering); for $T^{a}_{\mathfrak{c},\Lambda}$ the multiplier is $e^{-a_k \mathsf{S}_\Lambda(g)}$. (b) The map $\phi'(Ox) = \phi(x)$, $A'_\alpha(Ox) = A_\alpha(x)$, with components taken in the frames (f_α) and (Of_α), is a bijection of the configuration spaces preserving $d\nu$; this is the proof of Lemma 2.12, which covers boxes and tori, and for a torus $X = \mathbb{R}^3/P$ it carries the finite graph Λ/P onto $O\Lambda/OP$. The action is written in the frame, and δm^2_ε, W_ε and a_k depend on ε only; the wall term $\sum_x \varepsilon^3 w(x)|\phi(x)|^2$ becomes $\sum_x \varepsilon^3 (O\cdot w)(Ox)|\phi'(Ox)|^2$ since $(O\cdot w)(Ox) = w(x)$, and a point mass at $\phi(x) = 0$ becomes a point mass at $\phi'(Ox) = 0$. Plaquettes, their centres and field strengths are transported, and $(O\cdot f)_{Oc}(Of_\alpha, Of_\beta) = f_c(f_\alpha, f_\beta)$, $\sum_{Ox} \varepsilon^3 (O\cdot h)(Ox) = \sum_x \varepsilon^3 h(x)$. The clause on the walled ratio is the first identity for $\zeta = (\mathrm{i}, f; 1, g)$ and the walled torus $X = D_{J'}$, together with $\mathsf{S}_{O\Lambda}(O\cdot g) = \mathsf{S}_\Lambda(g)$ and the first expression of Definition 15.7 read on $OD_{J'}$. For the last statement, a permutation of the frame relabels the components; reversing f_α replaces $A_\alpha(x)$ by $-A_\alpha(x-\varepsilon f_\alpha)$, a bijection of the torus variables that preserves the action, and changes the sign of both $F(p)$ and $f_{c(p)}(f_\alpha, f_\beta)$ while preserving the set of plaquette centres. (c) The expression for $T^{a}_{\mathfrak{c},\Lambda}$ is its definition, and at $g = 0$ the multiplier is 1 and $W_\Lambda(0) = 0$. □

Lemma 15.10 (the characteristic function and the second moment of a photon functional)**.** *Let $\langle\cdot\rangle$ be a BFS measure whose Gaussian has covariance C, and let $\Psi(A) = \sum_q w(q)\cos\langle\kappa(q), A\rangle$ and $\mathbb{E}^\sharp$ be the current representation and the current measure of Lemma 2.8. Then for $v \in \mathbb{R}^m$:*

(a) $\langle e^{\mathrm{i}\langle v,A\rangle}\rangle = e^{-\langle v,Cv\rangle/2}\,\mathbb{E}^\sharp \cosh\langle Cv,\kappa\rangle$*; in particular* $\langle e^{\mathrm{i}\langle v,A\rangle}\rangle \in [0,1]$ *is real;*

(b) $\langle\langle v,A\rangle\rangle = 0$ *and* $\langle\langle v,A\rangle^2\rangle = \langle v,Cv\rangle - \mathbb{E}^\sharp\langle Cv,\kappa\rangle^2 \in [0, \langle v,Cv\rangle]$.

Proof. (a) For a centred Gaussian γ with covariance C and $a \in \mathbb{C}^m$, $\int e^{\langle a,A\rangle} d\gamma = e^{\langle a,Ca\rangle/2}$. With $a = \mathrm{i}(v \pm \kappa)$ and the average of the two signs,

$$\int e^{\mathrm{i}\langle v,A\rangle} \cos\langle\kappa, A\rangle\, d\gamma = e^{-\langle v,Cv\rangle/2 - \langle\kappa,C\kappa\rangle/2} \cosh\langle v, C\kappa\rangle,$$

since C is symmetric. Multiply by $w(q) \geq 0$, sum over the currents q (the series converges absolutely: $\cosh\langle v,C\kappa\rangle \leq e^{|\langle v,C\kappa\rangle|} \leq e^{\langle v,Cv\rangle/2+\langle\kappa,C\kappa\rangle/2}$ by the Cauchy–Schwarz inequality for the form C, so the terms are dominated by $w(q)$ and $\sum_q w(q) < \infty$), and divide by the same sum at $v = 0$, which is the positive normalisation $\sum_q w(q) e^{-\langle\kappa,C\kappa\rangle/2}$ of $\mathbb{E}^\sharp$; this is the formula. Its right side lies in $[0,\infty)$, and $|\langle e^{\mathrm{i}\langle v,A\rangle}\rangle| \leq 1$. (b) Replace v by tv, $t \in \mathbb{R}$, in (a). The left side is the characteristic function of the real random variable $\langle v, A\rangle$, which has all moments because $\langle e^{s\langle v,A\rangle}\rangle \leq e^{s^2\langle v,Cv\rangle/2}$ for every real s (Lemma 2.8); so it equals $1 + \mathrm{i}t\langle\langle v,A\rangle\rangle - \frac{t^2}{2}\langle\langle v,A\rangle^2\rangle + O(t^3)$. The right side is $e^{-t^2\langle v,Cv\rangle/2}\mathbb{E}^\sharp\cosh(t\langle Cv,\kappa\rangle)$, and $\mathbb{E}^\sharp\cosh(t\langle Cv,\kappa\rangle) = 1 + \frac{t^2}{2}\mathbb{E}^\sharp\langle Cv,\kappa\rangle^2 + O(t^4)$, the expansion under $\mathbb{E}^\sharp$ being justified by the domination $\cosh(t\langle Cv,\kappa\rangle) \leq e^{t^2\langle v,Cv\rangle/2+\langle\kappa,C\kappa\rangle/2}$ of (a), which shows that $\langle Cv,\kappa\rangle$ has all moments under $\mathbb{E}^\sharp$. The right side is real and even in t, so $\langle\langle v,A\rangle\rangle = 0$; comparing the terms of order t^2 gives $\langle\langle v,A\rangle^2\rangle = \langle v,Cv\rangle - \mathbb{E}^\sharp\langle Cv,\kappa\rangle^2$. The bounds follow since both $\langle\langle v,A\rangle^2\rangle$ and $\mathbb{E}^\sharp\langle Cv,\kappa\rangle^2$ are non-negative. □

The lemma applies to every state of Definition 15.6 and to every state of the first sections, with v the vector $d^*_\Lambda f$ defined by $F_\Lambda(f) = \langle d^*_\Lambda f, A\rangle$ for a real 2-form f whose plaquettes carry variables of the state; the photon Gaussian of a torus or walled torus is the periodic Feynman form with mass μ_0, and killed sites are covered as in Lemma 2.8.

16. Schemes, admissibility and the full object

This section fixes the scheme in which generating functionals with scalar sources are taken, defines W-admissibility of a scheme, and proves the full-object theorem (Theorem 16.6): the iterated limit of the comparison cubes in a W-admissible scheme exists for all curvature and scalar sources, is entire and translation invariant, and depends on the scheme only through one explicit factor. The curvature sector $h = 0$ of all this is scheme-free and is already in place (Theorem 7.6, Lemma 7.7); the rotation statements of the Wick sector (Theorem 18.9, Theorem 18.11) are stated for the object defined here.

16.1. **The Wick square.**

Remark 16.1 (a logarithm in the Wick square). With the free Wick constant W_ε alone, the expectation of the Wick square $:\phi^2:(h)$ of Definition 2.3 diverges at order e^2 in perturbation theory: for the model on $\varepsilon\mathbb{Z}^3$, at order $e^2\lambda^0$,

$$\big\langle :\phi^2:(h)\big\rangle = \frac{e^2}{4\pi^2}\,\ln(1/\varepsilon)\int h + O(1),$$

with a coefficient that depends neither on m nor on μ_0. We indicate the computation, in formal perturbation theory. In the normalisation of the action each real component of ϕ has propagator $(p^2+m^2)^{-1}$ at $e=0$. Write the one-loop correction to the inverse propagator as $\Sigma(p)$; the mass counterterm removes $\Sigma(0)$. The photon-exchange part is $-e^2\int \frac{d^3k}{(2\pi)^3}\frac{(2p-k)^2}{((p-k)^2+m^2)(k^2+\mu_0^2)}$; writing $(2p-k)^2 = 2p^2+2(p-k)^2-k^2$ and using $\int \frac{d^3k}{(2\pi)^3}\frac{1}{k^2(p-k)^2} = \frac{1}{8|p|}$, one finds $\Sigma(p)-\Sigma(0) = -\frac{e^2}{4}|p| + O(m+\mu_0)$ for $1 \ll |p| \ll \varepsilon^{-1}$, the lattice regularisation changing only momentum-independent terms and terms of order εp^2. Inserting,

$$\langle|\phi(0)|^2\rangle - 2\mathsf{W}_\varepsilon = -2\int \frac{d^3p}{(2\pi)^3}\frac{\Sigma(p)-\Sigma(0)}{(p^2+m^2)^2} + O(e^4) = \frac{e^2}{2}\cdot\frac{1}{2\pi^2}\ln(1/\varepsilon) + O(1)$$

at order e^2. Two consequences.

(a) For h with $\int h \neq 0$ the generating functionals with scalar sources in King's printed Wick ordering have no finite continuum limit beyond the curvature sector, at least in perturbation theory: for $h \geq 0$, $h \not\equiv 0$, Jensen's inequality gives $\langle e^{:\phi^2:(h)}\rangle \geq e^{\langle:\phi^2:(h)\rangle} \to \infty$. This affects [13, Thm. 2.1] and [14, Thms. 2.1–2.3, (3.43), (4.2)] for scalar sources with $\int h \neq 0$. In the cube King subtracts the Dirichlet constant [14, (2.13)] and, at distance > 1 from the boundary, the free one, which differ by a convergent amount only. The statement of [13, p. 677] that the sources are proportional to $(5^k\varepsilon)^\alpha$, $\alpha > 0$, in the small-field region does not hold for the marginal graph that produces the logarithm. We therefore allow an additional vacuum counterterm, linear in h and depending only on the lattice spacing: the schemes of Definition 16.2. Walled ratios (Definition 15.6) and the curvature sector $h = 0$ are unaffected, since the h-linear c-numbers cancel in them.

(b) King's proof of [14, Thm. 2.4] compares the effective action generated from spacing ε by a first block-spin step with rotated blocks with the effective action generated from the rotated lattice of spacing 5ε [14, (4.3)–(4.9)]. In a scheme a the two generating functionals carry the multipliers $e^{-wa_\varepsilon \mathsf{S}(h)}$ and $e^{-wa_{5\varepsilon}\mathsf{S}(h)}$. If a removes the logarithm, then $a_\varepsilon - a_{5\varepsilon} \to \frac{e^2}{4\pi^2}\ln 5$, so the comparison [14, (4.6)] must carry the c-number $\frac{e^2}{4\pi^2}\ln 5\int h$, and as printed it cannot hold for $\int h \neq 0$. Since a walled ratio contains the wall source $-w$ with $\int w > 0$, this enters the proof of walled comparisons even though their statement is scheme-free. In the equal-spacing formulation of the torus comparison (Theorem 17.1) the two lattices carry the same a_k, and the issue does not arise.

16.2. **Schemes.**

Definition 16.2 (schemes)**.** A *scheme* is a real sequence $a = (a_k)_{k\geq 0}$ such that a_k depends only on ε_k and on (e,λ,m,μ_0), and $Z^a_{k,X}(\zeta) = e^{-\sum_j w_j a_k \mathsf{S}_{\Lambda_k}(h_j)}\langle\Theta(\zeta)\rangle^{\mathrm{ND}}_{k,X}$ is the functional of Definition 2.3, the expectation of $\exp(\sum_j z_j F(f_j) + \sum_j w_j \sum_x \varepsilon_k^3 h_j(x)[|\phi(x)|^2 - 2\mathsf{W}_{\varepsilon_k} - a_k])$; the same multiplier defines $Z^a_{\Lambda,T}(\zeta)$ for the periodic and walled states of Definition 15.6 on a lattice Λ of spacing ε_k. Three schemes are used.

(i) The *printed scheme* $a \equiv 0$.

(ii) The *reference scheme* $z = a^\sharp$ of Definition 2.3: $a^\sharp_k = e^2 X_{\varepsilon_k}$, where X_ε is the coefficient of e^2 in the perturbative expansion of $\langle|\phi(0)|^2\rangle$ for the model on $\varepsilon\mathbb{Z}^3$ at $\lambda = 0$, with the order-e^2 part of δm^2_ε inserted.

(iii) The *self-normalised scheme*: fix J_0 and $0 \leq h_0 \in C^\infty_c(\mathbb{R}^3)$, $h_0 \not\equiv 0$, supported at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_{J_0}$, and put $a^{sn}_k := \mathsf{S}_{\Lambda_k}(h_0)^{-1}\ln Z^0_{k,J_0}(0,h_0)$ when $\mathsf{S}_{\Lambda_k}(h_0) > 0$,

and $a_k^{sn} := 0$ otherwise; here $Z^0_{k,J_0}(0,h_0) = \langle e^{:\phi^2:(h_0)}\rangle^{(a)}_{k,J}$ at $J = J_0$ is the functional of the comparison cube in the printed scheme.

At order e^2 the reference scheme removes the logarithm of Remark 16.1 identically on $\varepsilon\mathbb{Z}^3$: by translation invariance the order-$e^2\lambda^0$ part of $\langle:\phi^2:(h)\rangle$ there equals $e^2 X_\varepsilon \mathsf{S}(h)$.

16.3. W-admissible schemes.

Definition 16.3 ($(\mathrm{W})_a$: W-admissibility of the scheme a)**.** Let a be a scheme. $(\mathrm{W})_a$ is the statement (W2) of Hypothesis 23.1 for the scheme a, posed on the comparison convention $\mathrm{ND}^{(a)}$, in the following two clauses:

- $(\mathrm{W1})^{(a)}$ For every King cube C_J, every finite family of curvature sources f_j and scalar sources h_j supported at ℓ^∞-distance > 1 from $\mathbb{R}^3\backslash C_J$, and every $(z,w) \in \mathbb{R}^{2N}$, the limit $\lim_{k\to\infty} Z^a_{k,J}(\sum_j z_j f_j, \sum_j w_j h_j)$ exists in $(0,\infty)$.
- $(\mathrm{W2})^{(a)}$ For every $0 \le q \in C^\infty_c(\mathbb{R}^3)$, $\sup_{J\ge J_q} \lim_k Z^a_{k,J}(0,q) < \infty$, where J_q is the least J such that $\operatorname{supp} q$ has ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_J$.

A scheme with $(\mathrm{W})_a$ is called W-*admissible*. Below the tube threshold, $(\mathrm{W1})^{(a)}$ is the existence theorem with scalar sources, Theorem 3.5; at arbitrary amplitudes (W2) is a hypothesis of the optional extension of the Wick leg (Theorem 23.7), and so are the Wick-sector statements of this part that assume $(\mathrm{W})_a$. Two W-admissible schemes are compared in Lemma 16.4; by its part (b) they differ by a convergent sequence, so a W-admissible scheme is admissible in the sense of Definition 3.6 as soon as the reference scheme is.

$(\mathrm{W1})^{(a)}$ is the analogue in the scheme a of [14, Thm. 2.1], and $(\mathrm{W2})^{(a)}$ of the bound [14, (3.43)], which King asserts for the Neumann–Dirichlet cubes from (3.42), reflection positivity and ultra-violet stability by "a standard argument (see [GJ1])" (not printed) and which, in King's Wick ordering, fails for $\int h_+ > 0$ at order e^2 in perturbation theory (Remark 16.1). At $h = 0$ no multiplier enters: $(\mathrm{W1})^{(a)}$ at zero Wick coefficient is $(\mathrm{E}_C)^{(a)}$, the existence theorem (E_C) of Theorem 3.2 read for the comparison family, with 2-form sources read as in Remark 15.5, and $(\mathrm{W2})^{(a)}$ is not needed, the curvature sector being Lemma 7.7 on the comparison convention and Theorem 7.6 on King's.

Lemma 16.4 (comparison of schemes)**.** (a) *If a is* W*-admissible and $a'_k - a_k \to \kappa_{a'a} \in \mathbb{R}$, then a' is* W*-admissible, and for every family as in* $(\mathrm{W1})^{(a)}$, $\lim_k Z^{a'}_{k,J}(\zeta) = e^{-\kappa_{a'a}\sum_j w_j \int h_j} \lim_k Z^a_{k,J}(\zeta)$.
(b) *If a and a' are* W*-admissible, then $a'_k - a_k$ converges.*
(c) *If some scheme is* W*-admissible, then a^{sn} is* W*-admissible.*

Proof. (a) $Z^{a'}_{k,J}(\zeta) = e^{-\sum_j w_j (a'_k - a_k)\mathsf{S}_{\Lambda_k}(h_j)} Z^a_{k,J}(\zeta)$, and $\mathsf{S}_{\Lambda_k}(h_j) \to \int h_j$ (a Riemann sum of a smooth compactly supported function). (b) For $0 \le h \in C^\infty_c$ with $\int h > 0$, supported at distance > 1 from $\mathbb{R}^3 \setminus C_J$, the ratio $Z^{a'}_{k,J}(0,h)/Z^a_{k,J}(0,h) = e^{-(a'_k - a_k)\mathsf{S}_{\Lambda_k}(h)}$ converges to a number in $(0,\infty)$ by $(\mathrm{W1})^{(a)}$ and $(\mathrm{W1})^{(a')}$, and $\mathsf{S}_{\Lambda_k}(h) \to \int h > 0$; this is the argument of Remark 2.4. (c) If a is W-admissible, $a^{sn}_k - a_k = \mathsf{S}_{\Lambda_k}(h_0)^{-1} \ln Z^a_{k,J_0}(0,h_0)$ for all large k, which converges by $(\mathrm{W1})^{(a)}$ and $\mathsf{S}_{\Lambda_k}(h_0) \to \int h_0 > 0$; apply (a). □

Remark 16.5 (the class of W-admissible schemes). By Lemma 16.4, $(\mathrm{W})_a$ holds for some scheme if and only if it holds for a^{sn}, and W-admissible schemes form a single class modulo convergent sequences; the reference scheme $a^\sharp$ is W-admissible if and only if some scheme is and $a^\sharp - a^{sn}$ converges. If the logarithm of Remark 16.1 is present beyond perturbation theory, the printed scheme $a \equiv 0$ is not W-admissible. A proof of $(\mathrm{W})_a$ would carry the counterterm $a_k\mathsf{S}(h)$ as an additional h-linear vacuum vertex through King's renormalisation group. $(\mathrm{W})_a$ enters none of the curvature-sector results of this part: at $g = 0$ every walled quantity is scheme-free (Lemma 15.9), and King's printed state on curvature sources is Z^K_∞ of Theorem 7.6, under (E_C) alone.

16.4. **The full object in a W-admissible scheme.** For $\rho > 0$ let $\mathcal{E}_\rho := C_c^\infty(\bar B_\rho;\Lambda^2)\times C_c^\infty(\bar B_\rho)$ be the space of pairs $s = (f,h)$ of smooth real curvature and scalar sources supported in the closed ball $\bar B_\rho$, with the C^∞ topology. It is a closed subspace of $C^\infty(\bar B_{\rho+1};\Lambda^2)\times C^\infty(\bar B_{\rho+1})$, hence a Fréchet space, and it is invariant under $O(3)$ and under the translations τ_b in the sense that τ_b maps $\mathcal{E}_\rho$ into $\mathcal{E}_{\rho+|b|}$. (The 2-form space $E_\rho = C_c^\infty(\bar B_\rho;\Lambda^2)$ of the translation section is the first factor.) For a W-admissible scheme a and a finite family $\zeta = (z_j, f_j; w_j, h_j)_j$ with all supports at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_J$ write

$$Z_J^a(\zeta) := \lim_{k\to\infty} Z_{k,J}^a(\zeta),$$

the limit of the comparison cubes at fixed J, which exists at real coefficients by $(\mathrm{W1})^{(a)}$. We use the total-source convention of the translation section: a functional depends on a family only through its total sources $(\sum_j z_j f_j, \sum_j w_j h_j)$, and we write $Z_J^a(s)$, $Z_\infty^a(s)$ for $s \in \mathcal{E}_\rho$.

Theorem 16.6 (the full-object theorem)**.** *Let a be a* W*-admissible scheme,* $(\mathrm{W})_a$.

(i) *For C_J and families as in* $(\mathrm{W1})^{(a)}$, *$Z_{k,J}^a \to Z_J^a$ locally uniformly on $\mathbb{C}^{2N}$ as $k\to\infty$, and Z_J^a is entire.*

(ii) *For $(f,g)\in\mathcal{K}$ supported at ℓ^∞-distance > 1 from $\mathbb{R}^3\setminus C_{J_0}$, $J \mapsto Z_J^a(\mathrm{i}f,g)$ is nondecreasing and $J\mapsto Z_J^a(f,-g)$ is nonincreasing on $J \ge J_0$.*

(iii) *For every finite family of curvature and scalar sources the limit $Z_\infty^a(\zeta) := \lim_{J\to\infty} Z_J^a(\zeta)$ exists locally uniformly on $\mathbb{C}^{2N}$ and is entire; $Z_\infty^a(f,h) > 0$ for real sources; and $Z_J^a \le Z_\infty^a$ on $\mathcal{K}$.*

(iv) *For every $\rho > 0$, $\Phi^a(s) := \ln Z_\infty^a(s)$ is finite, convex and continuous on $\mathcal{E}_\rho$.*

(v) *$Z_\infty^a(\tau_b\cdot\zeta) = Z_\infty^a(\zeta)$ for all $b\in\mathbb{R}^3$, where τ_b acts on all sources of the family.*

(vi) *If a' is another* W*-admissible scheme, then $a'_k - a_k \to \kappa_{a'a}\in\mathbb{R}$ and $Z_\infty^{a'}(\zeta) = e^{-\kappa_{a'a}\sum_j w_j\int h_j} Z_\infty^a(\zeta)$.*

Proof. (i) At fixed k the functionals are entire in the coefficients: a finite-dimensional integral with Gaussian decay in A and decay $e^{-\lambda\varepsilon^3 r^4}$ in r. Fix a compact $K\subset\mathbb{C}^{2N}$ and a real box $Q\supset \operatorname{Re}K$. By Lemma 2.11(d), $\sup_K|Z_{k,J}^a| \le \sup_Q Z_{k,J}^a$, and since $\ln Z_{k,J}^a$ is convex in the real coefficients the latter is the maximum over the vertices of Q, at which the sequences converge by $(\mathrm{W1})^{(a)}$, hence are bounded. So the family $\{Z_{k,J}^a\}_k$ is locally bounded; by Montel's theorem and since $\mathbb{R}^{2N}$ is a set of uniqueness for entire functions on $\mathbb{C}^{2N}$, it converges locally uniformly, and the limit is entire [10]. (Compare [14, (3.40)–(3.44)]: Lemma 2.11(d) plays the role of (3.40); King's (3.42)–(3.43) are not used, $(\mathrm{W2})^{(a)}$ replacing (3.43).)

(ii) The sources of (f,g) lie inside X_J for $J\ge J_0$: the corners of a plaquette lie within ℓ^∞-distance $\varepsilon/2 \le \frac12$ of its centre, so every plaquette and site at which (f,g) does not vanish lies in C_{J_0}. Lemma 2.10 with $Y = X_{J+1}$, $X = X_J$ (no killed sites, so the cut-out factor is the comparison cube) and Lemma 2.11(a) give both inequalities at fixed k for $Z_{k,J}^a$; let $k\to\infty$ by (i).

(iii) Split each $h_j = h_j^+ - h_j^-$ with $h_j^- := \|h_j\|_\infty\chi_j$ and $h_j^+ := h_j + h_j^-$, where $\chi_j\in C_c^\infty$ takes values in $[0,1]$, equals 1 on $\operatorname{supp}h_j$ and is supported in $N_{1/2}(\operatorname{supp}h_j)$; both $h_j^\pm$ are smooth and non-negative. Let J_0 be such that all supports (those of the f_j, $h_j^\pm$) have ℓ^∞-distance > 1 from $\mathbb{R}^3\setminus C_{J_0}$, and for $J\ge J_0$ and $(z,w^+,w^-)\in\mathbb{C}^{3N}$ let

$$F_J(z,w^+,w^-) := Z_J^a\Big(\sum_j z_j f_j,\ \sum_j (w_j^+ h_j^+ + w_j^- h_j^-)\Big),$$

entire by (i). *Local bound.* For real $\zeta = (z,w^+,w^-)$ let $k_1 := \sum_j[(w_j^+)_+ h_j^+ + (w_j^-)_+ h_j^-]$ and $k_2 := \sum_j[(w_j^+)_- h_j^+ + (w_j^-)_- h_j^-]$, where $x_\pm = \max(\pm x, 0)$; both are smooth and non-negative, and the total scalar source is $k_1 - k_2$. By Lemma 2.11(f), passed to the limit $k\to\infty$ by (i),

$$F_J(\zeta) \le Z_J^a\Big(2\sum_j z_j f_j, -2k_2\Big)^{1/2} Z_J^a(0,2k_1)^{1/2}.$$

The first factor is nonincreasing in J by (ii), hence at most its value at J_0; the second is bounded uniformly in J by $(\mathrm{W2})^{(a)}$ applied to $q = 2k_1$ (whose support lies at distance > 1 from $\mathbb{R}^3\setminus C_{J_0}$

by the choice of J_0). By the convexity of $\ln F_J$ in the real coefficients and Lemma 2.11(d) (both inherited from $Z^a_{k,J}$ through (i)), the family $\{F_J\}_{J\ge J_0}$ is locally bounded on $\mathbb{C}^{3N}$. *Convergence on a set of uniqueness.* On $\Omega_+ := (\mathrm{i}\mathbb{R})^N \times (0,\infty)^{2N}$ the total sources are $(\mathrm{i}f, g)$ with f real and $g = \sum_j (w_j^+ h_j^+ + w_j^- h_j^-) \ge 0$, that is, they lie in $\mathcal{K}$; so $J \mapsto F_J$ is nondecreasing on Ω_+ by (ii) and bounded, hence convergent. Ω_+ is a non-empty open subset of the totally real subspace $(\mathrm{i}\mathbb{R})^N \times \mathbb{R}^{2N}$ of maximal real dimension, hence a set of uniqueness for entire functions on $\mathbb{C}^{3N}$. By Montel's theorem, $F_J \to F_\infty$ locally uniformly on $\mathbb{C}^{3N}$, with F_∞ entire [10]. Setting $w^+ = w$ and $w^- = -w$ gives the claim for the original family, since $\sum_j w_j h_j = \sum_j (w_j h_j^+ - w_j h_j^-)$; the limit is $\lim_J Z^a_J(\sum_j z_j f_j, \sum_j w_j h_j)$ and does not depend on the splitting. Positivity at real sources follows from Lemma 2.11(e), $Z^a_J(f,h) Z^a_J(-f,-h) \ge 1$, in the limit $J \to \infty$ (both factors are finite, so neither vanishes), and the last claim from (ii).

(iv) Φ^a is finite by the positivity in (iii), and convex as a pointwise limit of the convex functions $s \mapsto \ln Z^a_{k,J}(s)$. For J so large that $N_1(\bar{B}_\rho) \subset C_J$, and each k, the function $s \mapsto \ln Z^a_{k,J}(s)$ on $\mathcal{E}_\rho$ depends on s only through the finitely many sampled values $f_{c(p)}(e_\alpha, e_\beta)$, $h(x)$, continuous functionals of s in the C^∞ topology, and the finite-dimensional integral depends continuously on them (dominated convergence); so it is continuous, and $\Phi^a = \lim_J \lim_k$ of continuous functions is Borel. Lemma 2.14, applied on the open convex set $\mathcal{E}_\rho$ itself, gives the continuity.

(v) Let $b \in 5^{-k_0}\mathbb{Z}^3$ with $|b|_\infty \le M$. For $k \ge k_0$, τ_b preserves Λ_k and S_{Λ_k}, and the substitution $A'_\alpha(x) = A_\alpha(x+b)$, $\phi'(x) = \phi(x+b)$ of Step 1 of the proof of Proposition 8.1 carries the comparison state on X_J to the comparison state on $(C_J - b) \cap \Lambda_k$ together with both sources, the single-site weights, W_ε, δm^2_ε and a_k being site-independent; so $Z^a_{k,X_J}(\tau_b \cdot s) = Z^a_{k,(C_J-b)\cap\Lambda_k}(s)$, the right side being the functional of the comparison state on the translated box. As sets of sites, $C_{J-1} \cap \Lambda_k \subset (C_J - b) \cap \Lambda_k \subset C_{J+1} \cap \Lambda_k$. Let $s \in \mathcal{K}$ have all sources supported at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_{J_0}$ and let $J \ge J_0 + 1$; then, as in (ii), every plaquette and site at which s does not vanish lies in C_{J-1}. The first inequality of Lemma 2.10 (with $X = C_{J-1} \cap \Lambda_k$ in the state on $(C_J - b) \cap \Lambda_k$, then $X = (C_J - b) \cap \Lambda_k$ in the state on $C_{J+1} \cap \Lambda_k$; no site is killed, and the Dirichlet terms of each smaller state are the diagonal parts of the cut bonds) and Lemma 2.11(a) give

$$Z^a_{k,J-1}(s) \le Z^a_{k,J}(\tau_b \cdot s) \le Z^a_{k,J+1}(s) \qquad (k \ge k_0,\ J \ge J_0 + 1).$$

Let $k \to \infty$ by (i) (the translated family is as in $(\mathrm{W1})^{(a)}$ at J, since $C_{J-1} + b \subset C_J$ for $|b|_\infty \le M$) and then $J \to \infty$ by (iii): $Z^a_\infty(\tau_b \cdot s) = Z^a_\infty(s)$ on $\mathcal{K}$. The identity extends to all complex coefficients of a family by the argument of (iii) (both sides are entire in the split coefficients and agree on the set of uniqueness Ω_+), to all $b \in \bigcup_k 5^{-k}\mathbb{Z}^3$ by iteration over the translated families (each $b \in 5^{-k_0}\mathbb{Z}^3$ is a finite sum of vectors of $5^{-k_0}\mathbb{Z}^3$ with $|\cdot|_\infty \le M$), and to all $b \in \mathbb{R}^3$ at real coefficients by (iv), since $b \mapsto \tau_b \cdot s \in \mathcal{E}_{\rho'}$ is continuous for $|b| \le \rho' - \rho$ and $\bigcup_k 5^{-k}\mathbb{Z}^3$ is dense; then to complex coefficients by the identity theorem, $\mathbb{R}^{2N}$ being a set of uniqueness. (King states translation invariance as [14, Thm. 2.3(i)]; his argument is not used.) At $h = 0$, where $(\mathrm{W1})^{(a)}$ is $(\mathrm{E}_C)^{(a)}$, this is Proposition 8.1, whose proof is the one just given with the scalar sources absent.

(vi) Lemma 16.4(a),(b) and (iii). □

Definition 16.7 (the full object in the scheme a)**.** Let the couplings e, λ, m, μ_0 be fixed in King's range, together with the block factor 5 and the large-block integer M; let C_J be the King cubes, $\Lambda_k = \varepsilon_k \mathbb{Z}^3$, $\varepsilon_k = 5^{-k}$, and let $\langle\cdot\rangle^{(a)}_{k,J}$ be the comparison cube of Definition 2.1, with the free-boundary mass counterterm $\delta m^2_{\varepsilon_k}$ of the setting section, one number per spacing. Let a be a W-admissible scheme ($(\mathrm{W})_a$). For a finite family ζ of curvature and scalar sources (Definition 2.3) the *full object in the scheme a* is

$$Z^a_\infty(\zeta) := \lim_{J\to\infty} \lim_{k\to\infty} Z^a_{k,J}(\zeta) :$$

the inner limit is the cube limit along the spacings 5^{-k} at fixed μ_0 and fixed J, taken for the J at which the sources have ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_J$, and the outer limit runs along the cube family $(C_J)_J$. Both limits exist, locally uniformly in the coefficients, by Theorem 16.6(i),(iii),

and Z^a_∞ is the functional so named among the objects of the setting section, on the comparison convention. It is the object of the Wick-sector theorems of this part (Theorems 18.9 and 18.11) and of Form (A). At $h = 0$ the multiplier of Definition 16.2 is 1, so the object does not depend on the scheme: it is the iterated limit $\lim_J \lim_k Z_{k,J}(\sum_j z_j f_j, 0)$ of the comparison family, which exists under $(\mathrm{E}_C)^{(a)}$ (Lemma 7.7) and then equals King's printed state Z^K_∞ by Corollary 4.4(iii). The existence theorem (E_C) is posed on King's convention, and nothing is asserted about the fixed-cube limits of the comparison family; Corollary 4.4(ii)(γ) gives, at the cone points $z = \mathrm{i}t$,

$$Z^K_J(\mathrm{i}t) \le \liminf_k Z_{k,J}(\mathrm{i}t) \le \limsup_k Z_{k,J}(\mathrm{i}t) \le Z^K_{J+1}(\mathrm{i}t),$$

and both ends tend to $Z^K_\infty(\mathrm{i}t)$ as $J \to \infty$. Throughout this part, *King's printed state* means Z^K_∞ of Theorem 7.6, on King's convention ((E_C)); the curvature-sector statements of the part are posed for it.

Remark 16.8 (scope). The scheme changes Z^a_∞ only by the factor in Theorem 16.6(vi), which affects the one-point function of the Wick square; joint cumulants of order ≥ 2 do not depend on the scheme. The relation of the full object to infinite-volume limits along periodic boxes, King's expected identity [14, (2.24)], is never used.

16.5. **The general-source extension.** For scalar sources with $\int h \neq 0$ King's printed Wick ordering has no finite continuum limit beyond the curvature sector (Remark 16.1), and the object is the full object in a W-admissible scheme (Definition 16.7). For the reference scheme $z = a^\sharp$ the form of $(\mathrm{W})_a$ that carries the extension to such sources is the following.

Remark 16.9 (the reference scheme on the cone). $(\mathrm{W})_{a^\sharp}$ on the cone consists of the two clauses

($\mathrm{W}^\sharp$1) For every King cube C_J, every finite family of real 2-forms $f_j \in C^\infty_c(\mathbb{R}^3; \Lambda^2)$, sampled at the plaquette centres as in Definition 2.3, and of scalar sources $h_j \in C^\infty_c(\mathbb{R}^3)$, all supported at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_J$, and every real (z, w), the limit $\lim_k Z^{a^\sharp}_{k,J}(\sum_j z_j f_j, \sum_j w_j h_j)$ exists in $(0, \infty)$, locally uniformly in complex (z, w).

($\mathrm{W}^\sharp$2) For every real $0 \le q \in C^\infty_c(\mathbb{R}^3)$ with $\sup q < s_*(e)$, $\sup_{J \ge J_q} \limsup_k Z^{a^\sharp}_{k,J}(0, q) < \infty$; here $s_*(e) > 0$ is a threshold fixed by the bulk bounds of the scheme, positive for small λ outside an exceptional set of at most two values of e^2 for given (λ, μ_0, m).

The first clause is Theorem 3.5 with the threshold on the positive scalar part removed, that is clause (a) of (W2) for the reference scheme; the second is the bound of clause (b) of (W2) below a threshold, in the form proved on the tube in Proposition 6.10 and Corollary 13.3. With these two clauses in place of $(\mathrm{W})_a$, and with $(\mathrm{K})_a$ and $(\mathrm{TIGHT})_a$ for scalar sources (Definitions 17.2 and 18.5), the arguments of this and the following two sections apply on the corresponding cone of sources. None of this enters Theorem 15.1.

17. The torus comparison

The rotation enters the rotation part through a single statement, the torus comparison, posed on King's tori for the two lattices Λ_k and $R^{-1}\Lambda_k$. This section states it, reduces it by an exact symmetry to one-sided bounds at one lattice spacing, and proves it along the lines of [14].

17.1. **The torus comparison.** Throughout, for a lattice Λ of spacing ε_k descending to a King torus D_J (Definitions 15.4 and 15.6), $Z^a_{\Lambda,D_J}(\zeta) := e^{-\sum_j w_j a_k \mathsf{S}_\Lambda(h_j)} \langle \Theta(\zeta) \rangle_{\Lambda,D_J}$ is the functional of the periodic state in the scheme a (Definition 2.3), $\mathsf{Z}_{\Lambda,D_J}(s)$ the unnormalised functional of Definition 15.7, and $\langle\cdot\rangle_{\mathfrak{c},\Lambda}$ the walled state of the configuration $\mathfrak{c} = (J', w)$.

Theorem 17.1 ((K_0): the torus comparison, curvature sector)**.** *For every $R \in \mathcal{R}_0$, every walled configuration $\mathfrak{c} = (J', w)$ of the class of Remark 18.6 (the conditions of Theorem 17.13) and every real smooth $D_{J'}$-periodic 2-form G, the limits $\lim_k \langle e^{F(G)} \rangle_{\mathfrak{c},\Lambda_k}$ and $\lim_k \langle e^{F(G)} \rangle_{\mathfrak{c},R^{-1}\Lambda_k}$ exist and are equal.*

The proof is given at the end of the next subsection, which reduces the statement to one-sided bounds at a single lattice spacing; the bounds themselves are proved in section 17.3, and Remark 17.10 lists the inputs of that proof from the literature. With scalar sources the comparison is a property of a scheme a, which is not proved here (it is listed among the open problems, in the final section) and is used only by the optional extension of the Wick leg (Theorem 23.7):

Definition 17.2 ($(\mathrm{K})_a$: the torus comparison with scalar sources)**.** For every $R \in \mathcal{R}_0$, every King torus D_J, every finite family of real smooth D_J-periodic 2-forms G_j and functions h_j, and every $(z, w) \in \mathbb{R}^{2N}$, the limits

$$\lim_{k\to\infty} Z^a_{\Lambda_k, D_J}\Big(\sum_j z_j G_j, \sum_j w_j h_j\Big) \qquad \text{and} \qquad \lim_{k\to\infty} Z^a_{R^{-1}\Lambda_k, D_J}\Big(\sum_j z_j G_j, \sum_j w_j h_j\Big)$$

exist in $(0, \infty)$ and are equal.

(K_0) is stated on the sub-class of walled configurations described in Remark 18.6, which is the one on which the results that consume it use it. Here $R^{-1}\Lambda_k$ carries the frame $(R^{-1}e_\alpha)$ and descends to D_J by Lemma 15.3(b); both lattices have spacing ε_k and carry the same a_k, $\delta m^2_{\varepsilon_k}$ and $\mathsf{W}_{\varepsilon_k}$. Since $\mathcal{R}_0 = \mathcal{R}_0^{-1}$ (Lemma 15.3(a)), the statement for all $R \in \mathcal{R}_0$ is the same as the statement for the lattices $R\Lambda_k$ used in [14]. Read with a single scalar function on all plaquettes, as printed in [14, (2.12), Thm. 2.4], the comparison is false already at $e = 0$ (Proposition I.2); the 2-form reading, in which each lattice samples the source in its own frame, is consistent with King's proof, which treats the source components separately (Remark 15.5).

Lemma 17.3. (a) *Under* $(\mathrm{K})_a$ *the convergence holds locally uniformly on* $\mathbb{C}^{2N}$*, the two limits are the same entire function, and for every walled configuration* $\mathfrak{c}$ *and every* (f, g) *supported in one fundamental cell with* $w = 0$ *on* $\operatorname{supp} g$*,* $\lim_k T^a_{\mathfrak{c},\Lambda_k}(f, g)$ *and* $\lim_k T^a_{\mathfrak{c},R^{-1}\Lambda_k}(f, g)$ *exist and are equal. Under* (K_0) *the same holds, for the configurations of its class, for* $\langle e^{zF(G)}\rangle_{\mathfrak{c},\Lambda_k}$ *and* $\langle e^{zF(G)}\rangle_{\mathfrak{c},R^{-1}\Lambda_k}$ *at complex* z*.*

(b) $(\mathrm{K})_a$ *implies* (K_0)*, and* $(\mathrm{K})_a$ *implies* $(\mathrm{K})_{a'}$ *if* $a'_k - a_k$ *converges.*

(c) *If the statement of* $(\mathrm{K})_a$*, respectively* (K_0)*, holds for* $R = R_{12}(\theta_0)$*, it holds for every* $R \in \mathcal{R}_0$*.*

(d) *Under* $(\mathrm{K})_a$*, for every King torus* D_J *and every* $0 \le q \in C^\infty_c$ *supported in one fundamental cell,* $C_{D_J}(q) := \sup_k Z^a_{\Lambda_k, D_J}(0, q) < \infty$*.*

Proof. (a) As in the proof of Theorem 16.6(i): at fixed k the functionals are entire; by Lemma 2.11(d), whose proof applies verbatim to periodic states, $|Z^a(\zeta)| \le Z^a(\operatorname{Re}\zeta)$ and $\ln Z^a$ is convex in the real coefficients, so the convergence on $\mathbb{R}^{2N}$ gives local boundedness of the sequence on $\mathbb{C}^{2N}$; Montel's theorem gives convergent subsequences, whose limits agree on $\mathbb{R}^{2N}$, a set of uniqueness for entire functions on $\mathbb{C}^{2N}$ [10]; hence both sequences converge locally uniformly, to the same entire function. The walled ratios are quotients of such limits: with the family $(f; g, w)$ and the coefficients $(\mathrm{i}; 1, -1)$ in the numerator and $(0; 0, -1)$ in the denominator, $T^a_{\mathfrak{c},\Lambda}(f, g) = Z^a_{\Lambda,D_{J'}}(\mathrm{i}f, g - w)/Z^a_{\Lambda,D_{J'}}(0, -w)$ (Definition 15.6), and the denominators have positive limits. For (K_0) at complex z use $|\langle e^{zF(G)}\rangle| \le \langle e^{(\operatorname{Re} z)F(G)}\rangle$ and that (K_0) for all G includes all real multiples tG; the same argument applies. (b) $\langle e^{F(G)}\rangle_{\mathfrak{c},\Lambda} = Z^a_{\Lambda,D}(G, -w)/Z^a_{\Lambda,D}(0, -w)$, the wall being the scalar source $-w$ (Definition 15.6; the scheme multiplier cancels). For the second claim, the factors $e^{-\sum_j w_j(a'_k - a_k)\mathsf{S}_\Lambda(h_j)}$ have the same limit on both lattices, since $\mathsf{S}_{\Lambda_k}(h)$ and $\mathsf{S}_{R^{-1}\Lambda_k}(h)$ are Riemann sums of the smooth periodic function h over lattices descending to D_J and both tend to $\int_{\Gamma_J} h$. (c) Let $\sigma_2 = \operatorname{diag}(1, -1, 1)$. It preserves Λ_k and D_J, and changes the frame only by a sign, which does not change the periodic model (Lemma 2.12); moreover $\sigma_2 R_{12}(\theta_0)^{-1}\sigma_2 = R_{12}(\theta_0)$ (Lemma 15.3(a)). By Lemma 2.12 with $O = \sigma_2$ (for walled configurations Lemma 15.9(b), the wall becoming $\sigma_2 \cdot w$) the statement for $(R_{12}(\theta_0), \text{sources } s)$ is equivalent to the statement for $(R_{12}(-\theta_0), \sigma_2 \cdot s)$; as s ranges over all sources, so does $\sigma_2 \cdot s$. For the other planes, $R_{\mu\nu}(\pm\theta_0) = \Pi R_{12}(\pm\theta_0)\Pi^{-1}$ for a permutation matrix Π mapping e_1, e_2 to

e_μ, e_ν; apply Lemma 2.12 with $O = \Pi$, which preserves Λ_k and D_J and permutes frames. (d) A convergent sequence with a finite limit is bounded. □

17.2. **Reduction to one-sided bounds at one lattice spacing.** King's proof of his Theorem 2.4 compares the lattice of spacing ε with the rotated lattice of spacing 5ε, and obtains the reverse inequality by "switching around the argument" [14, p. 337]. We compare the two lattices at the *same* spacing, and obtain the reverse inequality from an exact symmetry.

Lemma 17.4 (the involution)**.** *Let $R = R_{\mu\nu}(\pm\theta_0) \in \mathcal{R}_0$, let $\sigma := \sigma_\nu$ be the reflection $x_\nu \mapsto -x_\nu$, and put $S := R^{-1}\sigma$. Then $S^2 = 1$, $S\Lambda_k = R^{-1}\Lambda_k$ and $S(R^{-1}\Lambda_k) = \Lambda_k$, and S maps $D_J = \mathbb{R}^3/P_J$, $P_J := (50MJ\mathbb{Z})^3$, onto the torus $R^{-1}D_J := \mathbb{R}^3/R^{-1}P_J$. Both lattices descend to both tori, and for all real smooth D_J-periodic sources $s = (G, h)$,*

$$\mathsf{Z}_{R^{-1}\Lambda_k, D_J}(s) = \mathsf{Z}_{\Lambda_k, R^{-1}D_J}(S \cdot s), \qquad \mathsf{Z}_{\Lambda_k, D_J}(s) = \mathsf{Z}_{R^{-1}\Lambda_k, R^{-1}D_J}(S \cdot s),$$
$$\mathsf{S}_{R^{-1}\Lambda_k, D_J}(h) = \mathsf{S}_{\Lambda_k, R^{-1}D_J}(S \cdot h).$$

Proof. By Lemma 15.3(a), $\sigma R \sigma = R^{-1}$, so $S^2 = R^{-1}(\sigma R^{-1}\sigma) = R^{-1}R = 1$; $\sigma\Lambda_k = \Lambda_k$ gives $S\Lambda_k = R^{-1}\Lambda_k$, and then $S(R^{-1}\Lambda_k) = S^2\Lambda_k = \Lambda_k$. Also $\sigma P_J = P_J$, so $SP_J = R^{-1}P_J$. The lattice $R^{-1}\Lambda_k$ is invariant under $R^{-1}P_J$ trivially, and Λ_k is invariant under $R^{-1}P_J$ because $R^{-1}p \in 10MJ\mathbb{Z}^3$ for $p \in P_J$ (Lemma 15.3(b)). The map S carries the frame $(R^{-1}e_\alpha)$ to $(SR^{-1}e_\alpha) = (\sigma e_\alpha)$ and the frame (e_α) to $(R^{-1}\sigma e_\alpha)$; these differ from the frames of Λ_k and $R^{-1}\Lambda_k$ only by signs, which do not change the periodic models (Lemma 2.12). Lemma 2.12 with $O = S$, applied to the unnormalised integrals (the bijection of configurations preserves $d\nu$), gives the identities. □

Definition 17.5 ((F), (V) and (C): one-sided bounds at one spacing)**.** Let $R \in \mathcal{R}_0$ and $J \in \mathbb{N}$. Write $T_1 := D_J$, $T_2 := R^{-1}D_J$. The following are properties of the pair (R, J). (F): for $i = 1, 2$ there are real numbers $v_k(T_i)$ and $c_k(T_i)$, $k \geq k_0(J)$, independent of the fields and the sources, with $|c_k(T_i)| \leq C\varepsilon_k^{-q}$ for some $q < \infty$ and $v_k(T_i)$ bounded, such that for every real smooth T_i-periodic source $s = (G, h)$,

$$\mathsf{Z}_{R^{-1}\Lambda_k, T_i}(s) \leq \big(1 + \delta_k(T_i, s)\big) \exp\big(- v_k(T_i) - c_k(T_i)\, \mathsf{S}_{\Lambda_k, T_i}(h)\big)\, \mathsf{Z}_{\Lambda_k, T_i}(s), \qquad \lim_{k\to\infty} \delta_k(T_i, s) = 0.$$

(V): $v_k(T_1) + v_k(T_2) \to 0$. (C): $c_k(T_1) - c_k(T_2) \to 0$. ($\mathrm{F}_0$) is (F) restricted to sources with $h = 0$ and to the sources $(G, \pm h_0)$, where $h_0 = -w$ is a wall; it involves no Wick scheme.

On each torus, (F) bounds the functional of the rotated lattice from above by that of the axis lattice. The rotated lattice is the one whose flow is King's standard block-spin flow in its own frame, and the axis lattice is the one run through King's hybrid flow; this is the direction of [14, (4.7)–(4.9)]. Since smooth periodic functions have rapidly convergent Riemann sums, $|\mathsf{S}_{\Lambda,T}(h) - \int_\Gamma h| \leq C_N(h)\varepsilon_k^N$ for every N and every lattice Λ of spacing ε_k descending to T, Γ a fundamental cell; with $N > q$ the error $c_k(\mathsf{S}_{\Lambda,T}(h) - \int h)$ tends to 0.

Theorem 17.6 (closing at one spacing)**.** *Let $R \in \mathcal{R}_0$, $J \in \mathbb{N}$, and assume* (F)*. Write $v = v_k(T_1)$, $v' = v_k(T_2)$, $c = c_k(T_1)$, $c' = c_k(T_2)$, and $\rho_k(s) := \mathsf{Z}_{R^{-1}\Lambda_k, D_J}(s)/\mathsf{Z}_{\Lambda_k, D_J}(s)$.*

(a) *For every real smooth D_J-periodic source $s = (G, h)$, $e^{v'+c'\int h}(1 + o(1)) \leq \rho_k(s) \leq e^{-v-c\int h}(1 + o(1))$ as $k \to \infty$.*

(b) *$v + v' \leq o(1)$. If* (V) *holds, then $c + c' \to 0$.*

(c) *If* (V) *holds, then $\rho_k(s) = e^{-v-c\int h}(1+o(1))$ for every s. Consequently, for sources s_0, s_1 with $\int h_0 = \int h_1$,*

$$\frac{\mathsf{Z}_{R^{-1}\Lambda_k, D_J}(s_1)}{\mathsf{Z}_{R^{-1}\Lambda_k, D_J}(s_0)} = \frac{\mathsf{Z}_{\Lambda_k, D_J}(s_1)}{\mathsf{Z}_{\Lambda_k, D_J}(s_0)}\,(1 + o(1)).$$

(d) *If* (V) *and* (C) *hold, then $c, c' \to 0$, and the identity in (c) holds for all s_0, s_1.*

No convergence of either family and no Wick scheme is assumed.

Proof. (a) The upper bound is (F) on T_1, with $c\,\mathsf{S}_{\Lambda_k, D_J}(h) = c\int h + o(1)$. Apply (F) on T_2 to the T_2-periodic source $S \cdot s$ and use Lemma 17.4: $\mathsf{Z}_{R^{-1}\Lambda_k, T_2}(S \cdot s) = \mathsf{Z}_{\Lambda_k, D_J}(s)$, $\mathsf{Z}_{\Lambda_k, T_2}(S \cdot s) =$

$\mathsf{Z}_{R^{-1}\Lambda_k,D_J}(s)$ and $\mathsf{S}_{\Lambda_k,T_2}(S\cdot h)=\mathsf{S}_{R^{-1}\Lambda_k,D_J}(h)=\int h+O(\varepsilon_k^N)$. This is the lower bound. (b) The interval in (a) is non-empty, so $(v+v')+(c+c')\int h\le o(1)$ for every s. Apply this to (G,h) and to $(G,-h)$ and add: $v+v'\le o(1)$. The same two inequalities give $|c+c'|\,|\int h|\le o(1)-(v+v')$; under (V) the right side tends to 0, and choosing h with $\int h\neq 0$ gives $c+c'\to 0$. (c) Under (V) and (b), the ratio of the two ends of the interval in (a) tends to 1. Dividing the statement for s_1 by the one for s_0, the prefactors cancel when $\int h_1=\int h_0$. (d) Now $c+c'\to 0$ and $c-c'\to 0$. □

The proof uses (F) only for the sources in question and their images under $h\mapsto -h$. In particular, under (F_0) and (V) the conclusions (a)–(c) hold for the sources $(G,\pm h_0)$ (where $\int h_0\neq 0$ unless $w=0$) and, with no condition on c,c', for the sources with $h=0$.

Corollary 17.7 (walled curvature ratios)**.** *Assume* (F_0) *and* (V) *for R and J'. Then for every walled configuration $\mathfrak{c}=(J',w)$ and every real smooth periodic 2-form G, $\langle e^{zF(G)}\rangle_{\mathfrak{c},R^{-1}\Lambda_k}-\langle e^{zF(G)}\rangle_{\mathfrak{c},\Lambda_k}\to 0$ locally uniformly in $z\in\mathbb{C}$. If moreover $\lim_k\langle e^{tF(G)}\rangle_{\mathfrak{c},\Lambda_k}$ exists for real t, then* (K_0) *holds for $\mathfrak{c}$ and G.*

Proof. With $h_0=-w$, $\langle e^{tF(G)}\rangle_{\mathfrak{c},\Lambda}=\mathsf{Z}_\Lambda(tG,h_0)/\mathsf{Z}_\Lambda(0,h_0)$ (the Wick constants cancel; Lemma 15.9(c)), a ratio of two sources with the same $\int h$. By Theorem 17.6(c), $u_k(t):=\langle e^{tF(G)}\rangle_{\mathfrak{c},R^{-1}\Lambda_k}/\langle e^{tF(G)}\rangle_{\mathfrak{c},\Lambda_k}\to 1$ for real t. The walled states are BFS measures (Definition 15.6), so by Lemma 2.8 and the symmetry $(A,\vartheta)\mapsto(-A,-\vartheta)$, $1\le\langle e^{tF(G)}\rangle\le e^{t^2\sigma_\Lambda/2}$ for real t, where $\sigma_\Lambda=\langle v,Cv\rangle$ is the variance of $F_\Lambda(G)=\langle v,A\rangle$ under the free periodic photon on the lattice; since the periodic photon form is at least μ_0^2 times the ℓ^2 norm, $\sigma_\Lambda\le\mu_0^{-2}\|\delta_\varepsilon G\|^2$, δ_ε the lattice codifferential, and this is bounded uniformly in k (a Riemann sum for $\|d^*G\|^2_{L^2(\Gamma_{J'})}$). Hence the difference of the two functions tends to 0 on $\mathbb{R}$, and it is bounded on every strip $|\operatorname{Re}z|\le r$ uniformly in k, because $|\langle e^{zF}\rangle|\le\langle e^{(\operatorname{Re}z)F}\rangle$. Vitali's theorem [10] gives locally uniform convergence to 0 on $\mathbb{C}$. The last statement follows by the same argument. □

Corollary 17.8 (the full comparison)**.** *Let a be a scheme with $|a_k|\le C(1+k)$. Assume* (F), (V) *and* (C) *for $R=R_{12}(\theta_0)$ and every J, and that for every King torus and every finite family of real smooth periodic sources the limit $\lim_k Z^a_{\Lambda_k,D_J}$ exists in $(0,\infty)$ for real coefficients. Then* (K)$_a$ *holds.*

Proof. $Z^a_{\Lambda,D_J}(s)=e^{-a_k\mathsf{S}_\Lambda(h)}\mathsf{Z}_\Lambda(s)/\mathsf{Z}_\Lambda(0)$, and the multipliers of the two lattices differ by $e^{-a_k(\mathsf{S}_{R^{-1}\Lambda_k}(h)-\mathsf{S}_{\Lambda_k}(h))}=1+O(k\varepsilon_k^N)$. Theorem 17.6(d) with $s_0=0$ gives $Z^a_{R^{-1}\Lambda_k,D_J}(s)/Z^a_{\Lambda_k,D_J}(s)\to 1$ for every real source, which with the assumed convergence gives (K)$_a$ for $R_{12}(\theta_0)$; Lemma 17.3(c) gives the other generators. □

Remark 17.9 (what this subsection supplies). Lemma 17.4, Theorem 17.6 and its corollaries are a reduction of the torus comparison to the one-sided bounds (F), (V), (C); the bounds themselves come from the renormalisation-group analysis. In the curvature sector, what Corollary 17.7 needs is (F_0) and (V) together with the existence of the axis-lattice limits, and these are what the proof of (K_0) along the lines of [14] produces (Remark 17.10). In that proof King's torus bound [14, (4.2)] is of no use, because at a non-zero wall it is vacuous; its place is taken by the large-field clause of King's upper bound [13, Thm. 3.1, (3.4)] at blocking factor 5, and at a mass-increasing wall by a polynomial lower bound for the walled partition function. Corollary 17.8 concerns scalar sources and is not used in the curvature sector (Theorem 15.1).

Proof of Theorem 17.1. By Lemma 17.3(c) it suffices to treat $R=R_{12}(\theta_0)$. Fix a walled configuration $\mathfrak{c}=(J',w)$ of the class of Remark 18.6 and a real smooth periodic 2-form G. By Corollary 17.7 the statement follows from (F_0) and (V) for R and J' together with the existence of $\lim_k\langle e^{tF(G)}\rangle_{\mathfrak{c},\Lambda_k}$ for real t. The existence of the axis-lattice limit is King's torus theorem [13, Thm. 2.1] with the wall as scalar source, read on the normalised ratio, in which the vacuum and the h-linear constants cancel (Theorem 17.122). The bounds (F_0) on the two tori T_1,T_2, in total form, with (V), come from the comparison, at equal spacing, of the effective actions of King's standard block-spin flow on the rotated lattice and of the hybrid flow on the axis lattice; they are proved in section 17.3, and the proof is completed there (Theorem 17.13 and the paragraph after Proposition 17.135). □

Remark 17.10 (the inputs of the proof of (K_0)). (K_0) is the corrected form of King's torus theorem [14, Thm. 2.4]: 2-form sources in each lattice's own frame, equal spacing, the free-boundary counterterm, walls allowed. The proof of section 17.3 follows King's lines. Passive rotations and Lemma 17.4 reduce the comparison to the two lattices on T_1, T_2. The effective actions of two block-spin flows, the standard flow of $R\Lambda_k$ in its own frame and a hybrid flow of Λ_k whose first step averages over rotated blocks, are compared at equal spacing on King's small-field domain, in total form (field-independent constants included). Inserting the comparison into King's upper bound [13, Thm. 3.1] for the rotated lattice and into the hybrid analogue of [14, Prop. 4.1] for the axis lattice gives a bound of type (F) on each torus, the large-field term absorbed by a polynomial lower bound on the partition function; Theorem 17.6 closes, with (V) automatic because the comparison is in total form. The inputs: Bałaban's renormalisation bound [3, Prop. 1], with one element weakened by a small degree, for the renormalised families of [13, §3.5] with lines from both flows; decay and regularity of the block-spin operators, [13, Thm. 3.3], [4], and of the hybrid kernels; King's upper bound, also on the torus skew in the rotated frame; the hybrid lower bound; the polynomial lower bound [1, (1.14)] at zero wall and at a mass-increasing wall; the existence of the walled values on the axis lattice, [13, Thm. 2.1] with the wall as scalar source, read on the normalised ratio; Reisz's theorems [20, 21] in a corrected three-dimensional form; the local limit law [15]; the bridge between King's torus counterterm [13, (3.33)] and the free-boundary one [14, p. 330]. The estimates of Bałaban's papers that the proof shares with the proof of (E_C) (section 5), stated in those papers without a printed argument, are among them [1, (2.33), (2.39)–(2.42), (3.20)–(3.21), (3.25), (3.34)–(3.36)] and the frame proof of its (1.14), [2, Prop. 2.6, Lemma 2.7, (3.5), (3.7)–(3.8)], [3, Prop. 1 with (1.33)], [4, Lemma 2.4, (1.8), (1.15), (1.16), (1.20)], [13, Thm. 3.1, Props. 3.6–3.9, Lemma 4.5] with the treatment of the sources there (pp. 665, 677), and the sentence of [14] that Bałaban's results apply to both models; the mass clause of [2, Prop. 3.1] is cited under it as in Remark 22.2. Statements false as printed are replaced: [13, Thm. 3.5(i), (4.21), (4.36)–(4.37)] ((4.33) itself holds, Proposition I.3; (3.78) fails on certain vacuum graphs whose contribution is a removed constant and is not used), the count (3.45) and the first inequality of (3.46) of [2], the gap value in [4, (2.27)], and the naturalness clause of Reisz's theorem.

Remark 17.11 (the content of (K_0)). (K_0) is [14, Thm. 2.4] with the corrections of Proposition I.2 and Remark 16.1, for arbitrary smooth periodic sources, walls included, and with free-boundary counterterms. By Corollaries 17.7 and 17.8 its two-lattice content is (F) and (V), plus (C) for scalar sources, and its one-lattice content is the existence of the torus limits on the axis lattice. The rotation enters the whole argument of the rotation part only through Theorem 17.1.

Remark 17.12 (what the torus comparison alone does not give). Given (K_0), the walled values of a configuration $\mathfrak{c}$ on D_J agree on the two lattices, and by passive covariance (Lemma 15.9(b)) the walled value of $R\cdot\mathfrak{c}$ on the rotated torus $R\,D_J$ for the axis lattice equals the walled value of $\mathfrak{c}$ on D_J for the rotated lattice. Rotation invariance of a limit along tori would therefore follow from (K_0) if the limit did not depend on the shape and orientation of the tori, which is the content of King's expected identity [14, (2.24)] between torus and cube limits. That identity is a uniqueness statement for the infinite-volume limit, and we do not use it. The walls of the next section replace it by a one-sided statement about one lattice, (TIGHT$_0$) (Theorem 18.4), which is proved from the heavy-shell estimates (Theorem 20.5, Theorem 19.10).

17.3. **Proof of the torus comparison.** This subsection proves the one-sided bounds (F_0) and (V) of Definition 17.5, in the total form in which the vacuum constants are absent and the source-linear constants tend to zero, for the walled configurations of the class of Remark 18.6; together with the existence of the axis-lattice limits (Theorem 17.122) and Corollary 17.7 this completes the proof of Theorem 17.1. Throughout, $R\in\mathcal{R}_0$ is fixed, $5R$ and $5R^{-1}$ are integer matrices (Lemma 15.3(a)), and we write

$$X_k:=\Lambda_k=\varepsilon_k\mathbb{Z}^3\quad\text{(the axis lattice)},\qquad Y_k:=R^{-1}\Lambda_k\quad\text{(the rotated lattice)},\qquad Y_{k-j}:=R^{-1}(5^j\varepsilon_k\mathbb{Z}^3),$$

so that $Y_{k-j}\subset X_k\cap Y_k$ for $j\ge 1$; the frame of X_k is (e_α) and that of Y_k is $(R^{-1}e_\alpha)$. The tori are $T_1=D_J=\mathbb{R}^3/P_J$, $P_J=50MJ\,\mathbb{Z}^3$, and $T_2=R^{-1}D_J=\mathbb{R}^3/R^{-1}P_J$ (Definition 17.5); the

involution is $S = R^{-1}\sigma_\nu$ of Lemma 17.4. In this subsection r denotes the number of block-spin steps, never the modulus $r_x = |\phi(x)|$, which does not occur.

Theorem 17.13 (the torus comparison at one configuration)**.** *Let the couplings be in King's range and $\mathfrak{c} = (J, \psi, n)$ a walled configuration with $J \in 5^{n_*}\mathbb{N}$, $J \geq J_0(\delta_r)$, $n > 0$, the wall condition $n\mathsf{S}_{X_k}(\psi) > \Delta_*|T_1|/2$ (Δ_* the constant of Proposition 17.121(b)) and block alignment (the source box and the torus unions of blocks of the final scale); here n_* is the number of large-field scales of the flow of Y_k, fixed by the couplings, and $J_0(\delta_r)$ the non-wrapping threshold of Lemma 17.120 at the coarsest spacing δ_r of the flow (a condition void for large k, since $\delta_r \to 0$). Then for every real smooth P_J-periodic 2-form G:*

(i) $\langle e^{zF(G)}\rangle_{\mathfrak{c},Y_k} - \langle e^{zF(G)}\rangle_{\mathfrak{c},X_k} \to 0$ *as $k \to \infty$, locally uniformly in $z \in \mathbb{C}$; in particular all moments $\langle F(G)^m\rangle$ differ by $o(1)$;*

(ii) *both families converge, locally uniformly on $\mathbb{C}$, to the same entire function $\mathcal{W}_{\mathfrak{c}}(zG)$, and all moments converge to the same limits;*

(iii) *the same holds with R replaced by any $R' \in \mathcal{R}_0$.*

At $e = 0$ the statement holds for every walled configuration with no condition: $\langle e^{zF(G)}\rangle_{\mathfrak{c},\Lambda} = e^{z^2 q_\Lambda(G)/2}$ exactly (Lemma 17.21), and $q_{X_k}(G), q_{Y_k}(G) \to q(G)$ (Lemma 17.22).

The three conditions on $\mathfrak{c}$ are those of Remark 18.6, and the configuration produced by Theorem 19.10 satisfies them (its wall height may be raised at will, Remark 19.11(iii), and J, the source box and the torus are chosen in Theorem 20.5). The proof is assembled in Proposition 17.134 and the paragraph after it, from the comparison of the effective actions of two block-spin flows at equal spacing (Theorem 17.62), King's upper bound for the rotated lattice (Proposition 17.96), the hybrid lower bound (Proposition 17.113) and a polynomial lower bound at the wall (Theorem 17.116), closed by Theorem 17.6.

Lemma 17.14 (admissibility)**.** (i) $P_J \subset M\mathbb{Z}^3 \cap R^{-1}(M\mathbb{Z}^3)$. (ii) $SP_J = R^{-1}P_J \subset M\mathbb{Z}^3 \cap R^{-1}(M\mathbb{Z}^3)$. (iii) *For $0 \leq j \leq k$, $P_J \subset 5^j\varepsilon_k M\mathbb{Z}^3 \cap R^{-1}(5^j\varepsilon_k M\mathbb{Z}^3)$, and the same for $R^{-1}P_J$: every block lattice of both flows below descends to both tori.*

Proof. (i) $50J\,R^{\mathrm{T}} = 10J \cdot 5R^{\mathrm{T}}$ is integral, so $RP_J = R^{-\mathrm{T}}P_J \subset 10MJ\,\mathbb{Z}^3$, and $R^{-1}(M\mathbb{Z}^3) \supset P_J$. (ii) $\sigma_\nu\mathbb{Z}^3 = \mathbb{Z}^3$ gives $SP_J = R^{-1}P_J$, and $R^{-1}P_J \subset 10MJ\,\mathbb{Z}^3$ as in (i). (iii) $5^{k-j} \in \mathbb{Z}$. □

The two block-spin flows.

Definition 17.15 (the standard flow of Y_k and the hybrid flow of X_k)**.** Fix k and $r \leq k$; $\delta_r := 5^r\varepsilon_k$ is the physical spacing of the coarse lattice Y_{k-r}. In the unit scale of level r write $\eta := 5^{1-r}$ and $\eta' := \eta/5 = 5^{-r}$. Blocks are centred at every level (Convention 17.17).

- *Standard flow on Y_k* (flow Y): Bałaban's and King's block-spin transformation [1, (2.1)–(2.11)], [13, (2.10)–(2.17)] in the frame of Y_k: centred cubes of 125 points, $Y_{k-j+1} \to Y_{k-j}$, with contours made of fine Y_k-bonds in the axial order f_1, f_2, f_3 of the frame inside each block.
- *Hybrid flow on X_k* (flow X): the first step averages over the centred tiles $y + \varepsilon B_0$, $y \in Y_{k-1}$, with
$$B_0 := \mathbb{Z}^3 \cap R^{-1}\big(-\tfrac{5}{2}, \tfrac{5}{2}\big)^3$$
(125 sites, a complete residue system of $\mathbb{Z}^3/5R^{-1}\mathbb{Z}^3$, Lemma 17.19), onto Y_{k-1}; this is [14, (4.3)–(4.4)], with the photon average read frame-covariantly, $(\hat{Q}A)_\alpha(y) = \frac{1}{125}\sum_{x\in B(y)} f_\alpha \cdot A(x)$, $f_\alpha = R^{-1}e_\alpha$ (Remark 17.18). Steps $2, \dots, r$ are the standard flow's steps $Y_{k-1} \to Y_{k-r}$ with the same blocks, except that every rotated bond of Y_{k-1} occurring in a contour is replaced by a fixed path of fine X_k-bonds ([14, p. 336]: "succeeding steps of the renormalization transformation in the usual way, that is without rotating the blocks").
- *Contours, rotation-paired.* Let $\Gamma^{\mathrm{st}}_{z,w}$ be King's coordinate staircase from z to w [13, (2.12)] and $\Gamma^{(r)}_{y,x}$ his composite contour [13, (2.11)], which the standard flow uses in Y's frame. The legs of a Y-frame staircase between two centres of Y_{k-1}-blocks are concatenations of the η-bonds $(u, u + 5\varepsilon f_\alpha)$, $u \in Y_{k-1}$. The hybrid composite contour from the level-r block centre y to $x' \in X_k$ is the same staircase with every η-bond $(u, u + 5\varepsilon f_\alpha)$ replaced by the *fixed* fine

axis staircase $u+\varepsilon\gamma_\alpha$ (and the reversed bond by its reverse), where, for $R=R_{12}(\theta_0)$ and $f_\alpha=R^{-1}e_\alpha$ with $5f_1=(4,3,0)$, $5f_2=(-3,4,0)$,

$$\gamma_1:\ e_1,e_2,e_1,e_2,e_1,e_2,e_1\ \text{(to }(4,3,0)),\qquad \gamma_2:\ e_2,-e_1,e_2,-e_1,e_2,-e_1,e_2\ \text{(to }(-3,4,0)),\qquad \gamma_3:\ e_3^5,$$

followed, inside the first-step tile, by a fine axis path of at most 5 bonds (Lemma 17.19(ii)), chosen once as a function of the displacement only and the same in every tile, so that the block and contour system of the flow is invariant under Y_{k-1}, in particular under the periods of both tori; this is the first-step contour of [14, (4.4)]. (For the other generators conjugate by the permutation and sign matrices of Lemma 17.3(c).) The gauge field along a composite contour is read by King's convention [13, (2.19)] (Convention 17.17(ii)).

Both flows use King's coefficients $a_j=a(1-5^{-2})/(1-5^{-2j})$ [13, p. 653], take r steps and land on the same coarse lattice Y_{k-r}/P with the same unit-scaled fields (A_r,ϕ_r) and the same small-field indicator χ_r [13, (3.2)]. The effective actions are written so that

$$\mathsf{Z}_{X_k,T}(s)=\int(dA_r)(d\phi_r)\,e^{-\hat{s}_T^{(r)}(A_r,\phi_r;s)},\qquad \mathsf{Z}_{Y_k,T}(s)=\int(dA_r)(d\phi_r)\,e^{-S_T^{(r)}(A_r,\phi_r;s)}$$

exactly, every c-number kept inside: this is [13, (3.14), (3.90)] with E_0 inserted from [13, (3.89)].

Lemma 17.16 (the paired contours)**.** *Every point of $u+\varepsilon\gamma_\alpha$ lies within 0.6ε of the segment $[u,u+5\varepsilon f_\alpha]$, and the two bound a planar region of area at most $5\varepsilon^2$. Each level-i piece of the hybrid composite contour lies inside its level-$(i+1)$ block and the in-tile piece inside its tile, so the contour lies inside the level-r block; in the unit scale of level r its length is at most 2.1 (the standard flow's at most 1.5); the Hausdorff distance between the level-i pieces of the two flows is at most η' for $i\geq 1$ and at most $2\sqrt{13}\,\eta'$ in the first-step cell; and the closed polygon formed by the two composite contours and the matching segment $[x',\tau x']$ (Definition 17.28) bounds a piecewise planar surface of area at most $1.5\,\eta'+33\,\eta'^2$.*

Proof. A finite check. The staircase γ_1 alternates e_1 and e_2 steps from 0 to $(4,3,0)$ along the segment of slope $3/4$; its vertices $(1,0),(1,1),(2,1),(2,2),(3,2),(3,3),(4,3)$ have distances ≤ 0.6 from the segment (the maximum, $3/5$, at $(1,0)$ and $(3,3)$), and the region between them is a union of triangles of total area ≤ 5; γ_2 is γ_1 rotated by a right angle, and γ_3 is the segment itself. The centres of the blocks are at distance $\geq\frac{5}{2}\varepsilon$ from the block faces, against the deviation 0.6ε, which gives the containments; the lengths are 7 fine bonds per η-bond for $\alpha=1,2$ and 5 for $\alpha=3$, against 5 for the rotated bond, plus the in-tile piece of at most 5 bonds, which gives 2.1 against 1.5 per unit length of staircase in the unit scale; the Hausdorff distances and the area are the sums over the pieces of the two levels, the first-step cell contributing $\max_{B_0}|x|\leq\sqrt{13}$ (Lemma 17.19(ii)). □

Convention 17.17 (centred blocks; the contour convention)**.** (i) King's blocks are corner-rooted; both flows here use blocks centred at the sites of the next coarse lattice. Since 5 is odd, centred cubes and centred tiles are inversion-symmetric. King's blocks are translates of these, and the printed estimates use only that blocks are translates of one cube and that contours stay inside blocks with bounded length (tiling, nesting, site in block with bounded diameter, all blocks translates of one cube); the four properties hold for centred blocks at block factor 5 by exact enumeration, the fluctuation-integral estimates of [1, 2, 3, 4] being insensitive to the root of the block once the central symmetry of the fixed staircases is given. With centred blocks the inversion about any site of the coarsest lattice is an exact symmetry of both flows at every scale, which removes every first-moment term from both effective actions. (ii) The gauge field along a composite contour is read in King's convention [13, (2.19)]: if $A=\sum_{j<r}A'^{(j),\varepsilon}+A^{(r),\varepsilon}$ is the decomposition of the fine field into the lifts of the integrated fluctuations [13, (2.18)], then $A(\Gamma^{(r)}_{y,x})=\sum_{j=0}^{r-1}A'^{(j),\varepsilon}(\Gamma^{(j+1)}_{x_{j+1},x})+A^{(r),\varepsilon}(\Gamma^{(r)}_{y,x})$: the stage-$j$ fluctuation is integrated only along the part of the contour inside the $(j+1)$-block. This is what iterating the one-step transformations produces, and it is a definition in all three prints, [13, (2.19)], [1, (3.34)], [3, (1.3)]. Under the other reading (one fine field along the whole composite contour) a staircase and the straight rotated bond with the same displacement carry different order-e^2 self-energies, by $0.0265\,e^2$ per

unit length, an extensive amount; under [13, (2.19)] the mismatch is $O(e^2\varepsilon)$ per contour and its total vanishes.

Remark 17.18 (the photon average is frame-covariant). [14, (4.3)] prints $(\hat{Q}A_\mu)(y) = 5^{-d}\sum_{x\in B(y)} A_\mu(x)$, which puts the axis component A_μ on the rotated bond direction. For a constant field B it produces the coarse field $R^{-1}B$ while the fine contours see B, so King's constant-background step [13, (3.43)–(3.46)] is inconsistent with it. With the frame-covariant average of Definition 17.15, $\hat{Q}^*\hat{Q} = \hat{P}\otimes\mathbf{1}$ because $\sum_\alpha f_\alpha f_\alpha^{\mathrm{T}} = \mathbf{1}$, the hybrid photon covariances are the scalar ones times the identity, and a constant background passes through every step unchanged. This repair is used throughout.

Lemma 17.19 (the tile average is a block-spin average). *Let $B(y) := y + \varepsilon B_0$ for $y \in Y_{k-1}$.*

(i) *B_0 is a complete residue system of $\mathbb{Z}^3/5R^{-1}\mathbb{Z}^3$ with $|B_0| = 125$; the tiles $B(y)$, $y \in Y_{k-1}$, partition X_k, and for every period lattice P of Lemma 17.14 the tiles $B(y)$, $y \in Y_{k-1}/P$, partition X_k/P.*

(ii) *$B_0 = -B_0$, $\max_{x\in B_0}|x|^2 = 13$, $|x|_\infty \le 3$ on B_0, and every $x \in B_0$ is joined to 0 by a nearest-neighbour path inside B_0, monotone in each coordinate, of length $|x|_1 \le 5$.*

(iii) *With $\hat{Q}$ and $Q(A)$ of Definition 17.15 and the pairings $\langle u, v\rangle_\Lambda = \sum_x s^3 u(x)\cdot v(x)$ of the two lattices, $\hat{Q}\hat{Q}^* = \mathbf{1}$ on coarse bond fields and $Q(A)Q(A)^* = \mathbf{1}$ on coarse scalar fields, for every bond field A. Hence $\hat{Q}^*\hat{Q}$ and $Q(A)^*Q(A)$ are orthogonal projections, the Gaussian block-spin kernels $\exp[-\frac{a}{2}\|A_1 - \hat{Q}A\|^2]$, $\exp[-\frac{a}{2}\|\phi_1 - Q(A)\phi\|^2]$ have the normalisations of* [13, (2.10)–(2.14)]*, and the first hybrid step is King's renormalisation transformation with $B(y)$ in place of the cube.*

(iv) *The composite of the first step with the $j-1$ standard steps that follow it is the equal-weight average over the composite blocks $B^{(j)}(y)$, $y \in Y_{k-j}$ (unions of 125^{j-1} tiles), which partition X_k and have diameter $\le C5^j\varepsilon$; the composite Gaussian kernel has King's coefficient a_j, as for the standard flow.*

Proof. (i) The open cube $R^{-1}(-\frac{5}{2},\frac{5}{2})^3$ is a fundamental domain of the lattice $5R^{-1}\mathbb{Z}^3$, and no point of $\mathbb{Z}^3$ lies on its boundary: for $R = R_{12}(\theta_0)$ the boundary planes are $4x_1 + 3x_2 = \pm\frac{25}{2}$, $-3x_1 + 4x_2 = \pm\frac{25}{2}$, $x_3 = \pm\frac{5}{2}$, none of which has an integer solution. A fundamental domain whose boundary carries no lattice point contains exactly one point of every coset, so B_0 is a complete residue system and $|B_0| = |\det 5R^{-1}| = 125$. Hence every $x \in \mathbb{Z}^3$ is uniquely $x = 5R^{-1}m + b$ with $m \in \mathbb{Z}^3$, $b \in B_0$, i.e. $\varepsilon x = y + \varepsilon b$ with $y = 5\varepsilon R^{-1}m \in Y_{k-1}$ unique: the tiles partition X_k. For admissible P, $P \subset Y_{k-1}$ (Lemma 17.14(iii) with $j = 1$), so P-translates of tiles are tiles and the partition descends. (ii) The cube is inversion-symmetric, so $B_0 = -B_0$. The rest is a finite check over the 125 points: $|x|^2 \le 13$ with equality at $(\pm3, 0, \pm2)$, $(0, \pm3, \pm2)$; $|x|_\infty \le 3$; for each x a coordinate-monotone path from 0 stays in B_0, the longest having $|x|_1 = 5$ steps. (iii) For a coarse bond field ψ on Y_{k-1} the adjoint of $\hat{Q}$ with respect to the pairings is $(\hat{Q}^*\psi)(x) = \sum_\alpha \psi_\alpha(y(x))f_\alpha$, $y(x)$ the centre of the tile containing x, because $(5\varepsilon)^3/125 = \varepsilon^3$; then $(\hat{Q}\hat{Q}^*\psi)_\alpha(y) = \frac{1}{125}\sum_{x\in B(y)} f_\alpha \cdot \sum_\beta \psi_\beta(y) f_\beta = \psi_\alpha(y)$. For the scalar field, $(Q(A)\phi)(y) = \frac{1}{125}\sum_{x\in B(y)} U(A(\Gamma_{yx}))\phi(x)$ with $|U| = 1$ has adjoint $(Q(A)^*\chi)(x) = \overline{U(A(\Gamma_{y(x)x}))}\chi(y(x))$, and $Q(A)Q(A)^* = \mathbf{1}$. An operator with $QQ^* = \mathbf{1}$ has Q^*Q an orthogonal projection, and the Gaussian integrals $\int dA_1 \exp[-\frac{a}{2}\|A_1 - \hat{Q}A\|^2]$, $\int d\phi_1 \exp[-\frac{a}{2}\|\phi_1 - Q(A)\phi\|^2]$ are constants independent of A and ϕ, which is all that [13, (2.10)–(2.14)] uses of the block. (iv) The composition of equal-weight averages over successive partitions is the equal-weight average over the composite blocks; the diameter bound is (ii) at step 1 and the cube's diameter at the later steps; the coefficient a_j is King's computation [13, p. 653], which uses only the form of the kernels. □

Lemma 17.20 (flux through the sliver). *For two lattice curves with the same endpoints, within Hausdorff distance w and of length $\le \ell$, $|\int_\gamma A - \int_{\gamma'} A| \le \|dA\|_\infty \operatorname{area}(\Sigma)$ for a surface Σ they bound together with the matching segment, and in particular $\le C\|dA\|_\infty w\ell$. In the unit scale of level r the paired systems of Definition 17.15 bound a surface of area $\le 1.5\eta' + 33\eta'^2$*

(Lemma 17.16), so the phases $U(A(\Gamma))$ of the two flows at matched endpoints differ by at most $C\eta'\|dA\|_\infty$, a relative η' against the Wilson-line phase.

Proof. Stokes' theorem on the lattice (the sum of the plaquette fluxes over Σ). The pairing is a condition, not a convenience: if the enclosed loops did not shrink with the scale, the flux $\sim e_r p A \ell^2$ would be of the same order as the Wilson-line phase $e_r A\ell$ at the coarse momenta $p\ell = O(1)$ at which the block-spin comparison is made, and Theorem 17.56 would fail; [14, (4.4)] asks only that the contour "lie entirely inside $B(y)$" and leaves its shape open. □

The Cameron–Martin transfer of the curvature source.

Lemma 17.21 (Cameron–Martin transfer)**.** *Let Λ be a lattice of spacing $s = \varepsilon_k$ with frame (f_α) descending to a torus T, γ_Λ the normalised Gaussian photon measure of the action, with covariance $C_\Lambda = (-\Delta_\Lambda + \mu_0^2)^{-1} \otimes \mathbf{1}_3$ for the pairing $\langle u, v\rangle_\Lambda = \sum_x s^3 \sum_\alpha u_\alpha(x) v_\alpha(x)$, and write $\mathsf{Z}_{\Lambda,T}(G,h) = \int d\gamma_\Lambda(A)\, e^{F_\Lambda(G)} W_\Lambda(A;h)$ with W_Λ the matter integral, which depends on A only through the bond phases. Then $F_\Lambda(G) = \langle A, \ell_\Lambda\rangle_\Lambda$ with $\ell_{\Lambda,\beta} := \sum_{\alpha\neq\beta} \partial^*_\alpha G_{\alpha\beta}$, $G_{\alpha\beta}(x) := G_{c(x;\alpha\beta)}(f_\alpha, f_\beta)$, $\partial^*_\alpha u(x) := s^{-1}(u(x - sf_\alpha) - u(x))$, and for real z*

$$\mathsf{Z}_{\Lambda,T}(zG,h) = e^{z^2 q_\Lambda(G)/2}\, \mathsf{Z}_\Lambda[z a_\Lambda](T;h), \qquad q_\Lambda(G) := \langle \ell_\Lambda, C_\Lambda \ell_\Lambda\rangle_\Lambda, \quad a_\Lambda := C_\Lambda \ell_\Lambda,$$

where $\mathsf{Z}_\Lambda[b](T;h) := \int d\gamma_\Lambda(A)\, W_\Lambda(A+b;h)$. Counterterms, Wick constant and wall are untouched. At $e = 0$, W_Λ does not depend on A, so $\langle e^{zF(G)}\rangle_{\mathfrak{c},\Lambda} = e^{z^2 q_\Lambda(G)/2}$ exactly.

Proof. $\sum_{\alpha<\beta} G_{\alpha\beta}(\partial_\alpha A_\beta - \partial_\beta A_\alpha) = \sum_{\alpha\neq\beta} G_{\alpha\beta}\partial_\alpha A_\beta$ by antisymmetry, and summation by parts on the torus gives $\langle A, \ell_\Lambda\rangle$. Completing the square, $e^{z\langle A,\ell\rangle} d\gamma(A) = e^{z^2\langle \ell, C\ell\rangle/2} d\gamma(A - zC\ell)$; substitute $A \mapsto A + zC\ell$. Both sides are finite ($\mu_0 > 0$, $\lambda > 0$, finite lattice). □

Lemma 17.22 (the Gaussian factor is frame-free)**.** *With $q(G) := \int_T \langle d^*G, (-\Delta + \mu_0^2)^{-1} d^*G\rangle\, dx$: $|q_\Lambda(G) - q(G)| \le C(\mu_0) s^2 \|G\|^2_{H^3(T)} + C_N(G) s^N$ for every N and every lattice $\Lambda = O(s\mathbb{Z}^3)$ descending to T.*

Proof. In Fourier variables on Λ/P, with $k_\alpha := k \cdot f_\alpha$, $\sigma_\alpha(k) := \frac{2}{s}\sin\frac{sk_\alpha}{2}$ and $u_{\alpha\beta}(y) := G_y(f_\alpha, f_\beta)$, Poisson summation on the coset of the plaquette centres gives $\hat\ell_{\Lambda,\beta}(k) = -\mathrm{i} e^{\mathrm{i}sk_\beta/2} \sum_\alpha \sigma_\alpha(k)[\hat u_{\alpha\beta}(k) + r_{\alpha\beta}(k)]$ with alias terms $|r| \le C_N s^N$ on the Brillouin zone, and $q_\Lambda = |T|^{-1} \sum_{k\in B_\Lambda \cap P^*} \sum_\beta |\hat\ell_{\Lambda,\beta}(k)|^2 / (\sum_\gamma \sigma_\gamma(k)^2 + \mu_0^2)$. In the continuum, $q = |T|^{-1} \sum_{k\in P^*} (|k|^2 + \mu_0^2)^{-1} \sum_\beta |\sum_\alpha k_\alpha \hat u_{\alpha\beta}(k)|^2$, and $\sum_\beta |\sum_\alpha k_\alpha G(f_\alpha, f_\beta)|^2 = \sum_\beta |G(k, f_\beta)|^2 = |\iota_k G|^2$ is frame-free because $\sum_\alpha k_\alpha f_\alpha = k$. Finally $|\sigma_\alpha - k_\alpha| \le s^2|k|^3/24$, $|k|^2 - s^2|k|^4/12 \le \sum \sigma_\gamma^2 \le |k|^2$, $\sum \sigma_\gamma^2 \ge \frac{4}{\pi^2}|k|^2$ on B_Λ, and the continuum terms with $k \notin B_\Lambda$ sum to $O(s^N)$. □

Lemma 17.23 (the shift field is regular)**.** *Let $a^{\mathrm{c}} := (-\Delta + \mu_0^2)^{-1} d^*G$, a smooth co-exact 1-form on T. For every bond $(x, x + sf_\beta)$ of Λ, $|a_{\Lambda,\beta}(x) - (a^{\mathrm{c}} \cdot f_\beta)(x + \frac{s}{2} f_\beta)| \le C(G, \mu_0) s^2$, and the same for lattice differences of any fixed order. In King's unit scaling at step r the background $z a_\Lambda$ is regular in the sense of* [13, Def. 3.2] *for $\delta_r \le \delta_0(z, G, e)$, and so is $A^{(r)} + z a_\Lambda$ whenever $A^{(r)}$ is.*

Proof. $\hat a_{\Lambda,\beta} = (\sum_\gamma \sigma_\gamma^2 + \mu_0^2)^{-1} \hat\ell_{\Lambda,\beta}$; compare coefficient by coefficient with the sampled continuum field as in Lemma 17.22. In unit scaling ($A_u(x_u) = \delta_r^{1/2} A(\delta_r x_u)$, $e_u = e\delta_r^{1/2}$, [13, (2.20)]) a derivative of $z a_\Lambda$ is $\le |z| \delta_r^{3/2} (\|\nabla a^{\mathrm{c}}\|_\infty + C s^2)$, while [13, Def. 3.2] allows $C e^{\beta-1} \delta_r^{(\beta-1)/2}$ with a small $\beta > 0$. □

Remark 17.24 (what the transfer does for the comparison). After Lemma 17.21 the curvature source has disappeared from the photon: the model is King's with a scalar source and a fixed smooth external background $z a_\Lambda$ in the matter coupling only. Running the matter averages in the total field, the background enters exactly where King's coarse background $A^{(r)}$ enters, through $A^{(r)} + z a_\Lambda$, regular and small at every scale below $\delta_0(z, G)$ (Lemma 17.23); this is the content of the identities [13, (3.82)–(3.87)], which express the effective action at a background as the action at zero background with the background inserted in the lines and vertices. The

two backgrounds a_{X_k} (axis bonds) and a_{Y_k} (rotated bonds) are samples of one continuum 1-form a^{c}, so replacing one by the other at matched points is the same step as King's replacement of $g_{\mu\nu}(x')$ by $g_{\mu\nu}(x)$ [13, p. 665]. The comparison of the effective actions below is therefore carried out at "$h = 0$ with background", and no frame-paired source vertex [13, (3.20)] appears.

Lemma 17.25 (lattice sums)**.** *For $h \in C^\infty(T)$, every lattice $\Lambda = O(s\mathbb{Z}^3) \supset P$ and every N: $|\mathsf{S}_\Lambda(h) - \int_T h| \le C_N(h) s^N$, with $C_N(h)$ independent of O.*

Proof. $\mathsf{S}_\Lambda(h) = \sum_{k\in\Lambda^*} \hat h(k)$ by Poisson summation on Λ/P ($P \subset \Lambda$ gives $\Lambda^* \subset P^*$); the term $k = 0$ is $\int h$; every $k \in \Lambda^* \setminus 0$ has $|k| \ge 2\pi/s$; $|\hat h(k)| \le \|\Delta^M h\|_{L^1} |k|^{-2M}$ and $\sum_{k\in\Lambda^*\setminus 0} |k|^{-2M} \le C_M s^{2M}$ for $2M > 3$. □

The number of steps, the effective actions, the matching.

Lemma 17.26 (the choice of r)**.** *Let $\gamma_*, \beta_*, \sigma > 0$ be the exponents of* [13, Thm. 3.4]*, β_1 the field-norm exponent of step (5) of the proof of Theorem 17.56, θ_1 the gain exponent of Theorem 17.39, $\gamma := \min(\gamma_*, \theta_1, \frac12)$ and $\beta := \max(\beta_*, \beta_1 + \frac12, 5)$. Fix $0 < \varrho < \min\{\gamma/(\gamma+\beta), \frac19\}$ and put $k - r(k) := \lfloor \varrho k \rfloor$. Then, as $k \to \infty$: $\delta_r \to 0$; $5^{-\gamma r}\delta_r^{-\beta}(1+k)^p \to 0$ for every p; $5^{3k-5r} \to 0$; $\delta_r^{-2}(\varepsilon_k/\delta_r)^{1/4}\log(1/\varepsilon_k)^2 \to 0$; and $\varepsilon_k^{-q} e^{-c\delta_r^{-1}} \to 0$ for all $q, c > 0$. Moreover $p(\delta_r)^2 \ge b_0^2(1 + \varrho k \ln 5 - \ln 5)^{2\mathrm{p}}$ grows at least quadratically in k, where $p(\delta) = b_0(1 + \log\delta^{-1})^{\mathrm{p}}$, $\mathrm{p} \ge 1$, is King's small-field function* [13, (3.1)]*.*

Proof. $\delta_r = 5^{-\lfloor \varrho k\rfloor}$. The second quantity is $5^{-\gamma k + (\gamma+\beta)\lfloor\varrho k\rfloor}(1+k)^p$; the third $\le 5^{-2k+5\varrho k+5}$; the fourth $\le 5^{2\varrho k+2-(1-\varrho)k/4}(k\ln 5)^2$, and $2\varrho < (1-\varrho)/4$ iff $\varrho < \frac19$; the fifth because $\delta_r^{-1} \ge 5^{\varrho k-1}$ grows exponentially. The last claim is [13, (3.1)] with $\mathrm{p} \ge 1$. King's own choice [13, (3.12)], $\varrho = \gamma/(2\beta+\gamma)$, satisfies every condition. □

Definition 17.27 (effective actions; vertices)**.** For the standard flow the r-step effective action is [13, (3.14)] rewritten as [13, (3.90)]:

$$S^{(r)} = \tfrac12\langle A_r, \Delta^{(r)} A_r\rangle + \ln N_r + \tfrac12\langle \phi_r, \Delta^{(r)}(A^{(r)})\phi_r\rangle + \ln N_r(0) - \ln\big[Z_r(A^{(r)}) Z_r(0)^{-1}\big] + P^{(r)}(A_r, \phi_r; s),$$

where $P^{(r)}$ is the Taylor polynomial of order $\bar n = 13$ in the running couplings of King's E'_r [13, (3.17)], a finite sum of connected graphs with the vertices [13, (3.18)–(3.26)]. The hybrid $\hat s^{(r)}$ is the same expression for the hybrid flow. A curvature source enters only through the Cameron–Martin background (Remark 17.24); a wall enters through the source vertex [13, (3.21)], the *h-vertex*, $\theta(5^r\varepsilon)^2 \sum_x \eta^3 h(x) \sum_i [(\phi'_i + \phi_i^{(r)})(x)^2 - (-\Delta^{\eta,\Omega} + m^2(5^r\varepsilon)^2)^{-1}(x,x)]$, whose subtraction is in [13] the coincident torus propagator at the running scale and here, in the free-boundary convention which both flows inherit from the fine action by exact integration, the corresponding infinite-lattice coincident propagator, the same number for both lattices because X_k and Y_k are isometric; the two readings differ by the image sum $\sum_{\gamma\in P\setminus 0}$ of the running propagator, a c-number times $\mathsf{S}(h)$, which is part of the bridge Proposition 17.121. Everything below is written in the free-boundary reading.

Definition 17.28 (matching; slices)**.** In the unit scale of level r let $\Gamma_r = R^{-1}\mathbb{Z}^3$ (the coarse lattice), $\Gamma_1 = R^{-1}(\eta\mathbb{Z}^3)$ ($= Y_{k-1}$ rescaled), $\Lambda_X = \eta'\mathbb{Z}^3$ and $\Lambda_Y = R^{-1}(\eta'\mathbb{Z}^3)$. For $x \in \Gamma_1$ the first-step cells are the tile $x + \eta' B_0 \subset \Lambda_X$ and the centred cube $Q_Y(x) \subset \Lambda_Y$, 125 sites each. Fix bijections $\tau_x : x + \eta' B_0 \to Q_Y(x)$; they glue to a bijection $\tau : \Lambda_X \to \Lambda_Y$, and on every admissible torus to $X_k/P \to Y_k/P$, with $|\tau x' - x'| \le c_\tau \eta'$, $c_\tau = \sqrt{13} + 2\sqrt3 < 7.1$. Points $x'' = \tau x'$ are *matched*. For a flow $F \in \{X, Y\}$ let $S_0^F := G_1^F$ be the first fluctuation covariance and $S_j^F := H_j^F C_j^F H_j^{F*}$ the $(j+1)$-st, lifted to the fine lattice; then $G_r^F = \sum_{j<r} S_j^F$ exactly, and on the infinite lattice $C^F = G_n^F + \Gamma_n^F$ with $\Gamma_n^F := H_n^F (A_n^F)^{-1} H_n^{F*}$ (Woodbury). Slice j has length scale $\ell_j := 5^j\eta'$. The *coarse slices* $j \ge r$ are the further steps of the same flow; for $F = X$ these are standard rotated steps on $Y_{k-r}, Y_{k-r-1}, \dots$

Lemma 17.29 (vertex matching; frame-free contractions)**.** (i) $\sum_{y\in\Lambda_Y^n} \eta'^{3n} g(y) = \sum_{x\in\Lambda_X^n} \eta'^{3n} g(\tau x)$: *every Y-graph is, exactly, a sum over X-vertices with τ-transported kernels.* (ii) *A smooth vertex*

function w (couplings, the Cameron–Martin background, partitions of unity) sampled at x' and at $\tau x'$ differs by $\le c_\tau \eta' \|\nabla w\|_\infty$. (iii) *Vector indices are contracted by the Euclidean inner product and derivative lines are compared as vectors; the current vertex $eA\cdot j$ ([13, (3.22)], $m=1$) and the seagull $e^2|A|^2|\phi|^2$ (the $m=2$ member of [13, (3.23)]) are frame-free;* [13, (3.22)] *with $m=2$ is the bond-diagonal derivative vertex $\frac{e^2\eta}{2}\sum_b A_b^2 D_b\phi\cdot U\phi(b_+)$ and is not frame-free; the vertices with $m\ge 3$ in (3.22) and $m\ge 4$ in (3.23) carry explicit factors η'^{m-2} or more and are bounded per flow.*

Proof. (i) τ is a bijection of Λ_X onto Λ_Y and both lattices have the cell volume η'^3 per site, so $y=\tau x$ is an exact change of the summation variable in each factor; on an admissible torus the same holds for the bijection $X_k/P\to Y_k/P$. (ii) The mean value theorem along the segment from x' to $\tau x'$. (iii) In the unit scale of level r the hopping term on the fine lattice of spacing η' is $\sum_b \eta'\big[|\phi(b_-)|^2+|\phi(b_+)|^2-2\,\mathrm{Re}\,\big(\bar\phi(b_+)e^{\mathrm{i}e_r\eta' A_b}\phi(b_-)\big)\big]$, the bond integral of the unit-scaled field being $\eta' A_b$ [13, (2.20)] and $e_r=e(5^r\varepsilon)^{1/2}$ the running charge; its expansion in powers of A_b carries $(e_r\eta')^m/m!$ at order m, one explicit η' per power of A, which is how [13, (3.22)–(3.23)] are organised. At $m=1$ the vertex is $\sum_x \eta'^3 e_r A(x)\cdot j(x)$ with $j=\sum_\alpha j_\alpha f_\alpha$ the lattice current, $j_\alpha(x)=2\,\mathrm{Im}(\bar\phi(x+\eta' f_\alpha)\phi(x))/\eta'$; at $m=2$ the member of (3.23) is $\sum_x \eta'^3 e_r^2|A(x)|^2|\phi(x)|^2$. The pairing $A\cdot j$ and the norm $|A|^2$ are the same functions of the vectors in every orthonormal frame; comparing the X-version with the τ-transported Y-version therefore compares the vectors A, j, ϕ at matched points only. The member of (3.22) with $m=2$ contracts $A\otimes A\otimes D\phi$ with the rank-three tensor $\sum_\alpha f_\alpha\otimes f_\alpha\otimes f_\alpha$, which is not $O(3)$-invariant (there is no non-zero isotropic symmetric tensor of rank three), so the two versions differ at order one on the same vectors; its tadpole is handled by the summation by parts of Lemma 17.67. The vertices with $m\ge 3$, $m\ge 4$ carry η'^{m-2} relative to the two order-e_r^2 vertices, and each flow's own line bounds [13, (3.63)–(3.65)] bound them without any comparison. □

Lemma 17.30 (counterterms take the same value along both flows)**.** *Let g be a counterterm graph of δm^2 or E_1 (free boundary conditions), an infinite-lattice graph with full lines C localised at one vertex. Expanding each of its lines in the slices of flow F* [13, (3.76)]*, the sum over all slice assignments equals g for both flows. Mixed assignments take other values; they occur only inside a telescoping sum whose endpoints are the pure assignments.*

Proof. $\sum_j S_j^F = C^F$ pointwise; C^Y is the image of C^X under the isometry R^{-1}, and the localised vertex sum of an infinite-lattice graph is invariant under that isometry. □

The equal-spacing propagator package. Unit scale of level $r\ge 2$, infinite lattice, $A=0$; $V_\eta = R^{-1}[-\frac{\pi}{\eta},\frac{\pi}{\eta}]^3$, $V_1=R^{-1}[-\pi,\pi]^3$; u_{r-1} is the common symbol of the $(r-1)$-step rotated centred average; $u_X(q)=\frac{1}{125}\sum_{x\in B_0}\cos(\eta' q\cdot x)$ is the tile symbol and $u_Y(q)=\prod_\alpha W(\eta' q\cdot f_\alpha)$, $W(t)=\frac15\sum_{j=-2}^2\cos jt$, the cube symbol; $\Delta'_X(q)=4\eta'^{-2}\sum_\mu \sin^2\frac{\eta' q_\mu}{2}+m_r^2$ and Δ'_Y its rotated analogue, $m_r=m\delta_r$ the running mass; L_2^F are the 125 alias classes; A^F is the coarse multiplier of flow F (the symbol of $\Delta^{F(r)}$), $A^F\le a$; Δ_R is the rotated Laplacian at spacing η; $J_s:=\int_{V_\eta}|u_{r-1}(q)||q|^{-s}dq$, bounded uniformly in r for $s=1$. Constants depend only on a and are uniform in r, k, the torus and the mass $m_r^2\ge 0$.

Proposition 17.31 (free core at equal spacing)**.** $A^F(p)^{-1}=a_{r-1}^{-1}+\sum_{l_1\in\Gamma_r^*/\Gamma_1^*}u_{r-1}(p+l_1)^2[\eta^2 a^{-1}+S_F(p+l_1)]$ *with $S_F(q):=\sum_{l\in L_2^F}u_F(q+l)^2/\Delta'_F(q+l)$, and $\sup_{V_\eta}|S_X-S_Y|\le 0.198\,\eta^2$. Hence for all $r\ge 2$, all p and all $m_r\ge 0$,*

$$|A^X(p)-A^Y(p)|\le 0.20\,\eta^2 A^X(p)A^Y(p)\le 0.20\,a\,\eta^2 A^Y(p),$$

and the same for the photon (three copies at mass $\mu_0\delta_r$).

Proof. The formula is the Woodbury identity for a composite average whose symbol is $u_{r-1}u_F$; the terms η^2/a cancel in the difference and $\sum_{l_1}u_{r-1}^2=1$ gives the sup bound. Scale to $\eta=1$, $\eta'=\frac15$. *Zones:* on V_1, $\Delta'_X\ge 0.93718|q|^2+m^2$ and $\Delta'_Y\ge 0.96753|q|^2+m^2$ (monotonicity of $\sin x/x$), while $\Delta'_F\le|q|^2+m^2$ and $(|q|^2+m^2)-\Delta'_F\le\eta'^2|q|^4/12$. *Second moments:* $1-u_X\le$

$\frac{\eta'^2}{2}q^{\mathrm{T}}M_0q$ with $M_0=\mathrm{diag}(\frac{52}{25},\frac{52}{25},2)$ (exact enumeration of the tile), so $1-u_X^2\le\frac{52}{25}\eta'^2|q|^2$; $1-W(t)\le t^2$ gives $1-u_Y^2\le 2\eta'^2|q|^2$. The main terms ($l=0$) then differ by at most 0.0924. *Aliases:* every non-zero alias $q+l$ has a frame component of modulus in $[\pi,5\pi]$, where $100\sum\sin^2(r_\mu/10)\ge 9.549$ (concavity of $t\mapsto\sin^2(\sqrt{t}/10)$ on $[0,25\pi^2]$); with Parseval $\sum_l u_F(q+l)^2=1$, $0\le \mathrm{Al}_F\le(1-u_F^2)/9.549\le 0.1047$. Total 0.198. Finally $A^F\le a_{r-1}\le a$. (The true constant is about 0.005: both first moments vanish by inversion symmetry and the second moments differ by 4%.) □

Lemma 17.32 (external lines). *For $x'\in x+\eta' B_0$, $x''=\tau x'\in Q_Y(x)$ ($x\in\Gamma_1$) and $z\in\Gamma_r$,*

$$H^F(x^F,z)=(2\pi)^{-3}\int_{V_\eta}dq\,e^{\mathrm{i}q\cdot(x-z)}u_{r-1}(q)M^F(q)A^F(q),\qquad M^F(q):=\sum_{l\in L_2^F}\frac{e^{\mathrm{i}(q+l)\cdot r_F}u_F(q+l)}{\Delta'_F(q+l)},$$

with $r_X=x'-x$, $r_Y=x''-x$, $x^X=x'$, $x^Y=x''$, and

$$|H^X(x',z)-H^Y(x'',z)|\le C_EJ_1\,\eta,\qquad |H^X(x',z)-H^Y(x'',z)|\le C(C_EJ_1\eta)^{1/2}e^{-\delta|x-z|/2},$$

$C_E=345a+0.2a(C_*/20+6.9a)$, $C_*=\max\{43.6(1+\sqrt{3}\pi)^2,\frac{\pi^2}{4}a(1+\frac{1}{\pi})^2\}$.

Proof. The representation: $H=D^{-1}Q^*\Delta^{(r)}$ (Woodbury), Q^* has symbol u_Fu_{r-1}; decompose the fine Brillouin zone into the translates $V_\eta+l$, $l\in L_2^F$, and use the Γ_1^*-periodicity of u_{r-1}, A^F and $e^{-\mathrm{i}q\cdot z}$, with $e^{\mathrm{i}l\cdot x}=1$. *Bounds on V_η:* for both flows $|M^F-\Delta_R^{-1}|\le 172.1\,\eta|q|^{-1}$. For Y: $|e^{\mathrm{i}q\cdot s}u_Y(q)-1|\le|q||s|+(1-u_Y)\le 0.910\,\eta|q|$ ($|s|\le 2\sqrt{3}\eta'$, $1-u_Y\le\eta'^2|q|^2$), divided by $\Delta'_Y\ge\frac{4}{\pi^2}|q|^2$ gives $2.25\,\eta|q|^{-1}$; $|\Delta_Y'^{-1}-\Delta_R^{-1}|\le\min\{0.51\eta^2,\frac{\pi^2}{4}|q|^{-2}\}\le 1.12\,\eta|q|^{-1}$; the 124 aliases have $\Delta'_Y(q+l)\ge 4/\eta^2$ and $|u_Y|\le 1$, so their sum is $\le 31\eta^2\le 168.7\,\eta|q|^{-1}$ on V_η. For X the same three bounds are $2.37+0.73+169$. Hence $|M^X-M^Y|\le 345\,\eta|q|^{-1}$. *Multipliers:* $E:=M^XA^X-M^YA^Y=(M^X-M^Y)A^X+M^Y(A^X-A^Y)$; the first term is $\le 345a\eta|q|^{-1}$; for the second use Proposition 17.31, $|M^Y|\le\Delta_R^{-1}+172.1\eta|q|^{-1}$ and $\Delta_R^{-1}A^Y\le C_*(1+|q|)^{-2}$ (on V_1 from $A^Y(q)^{-1}\ge u_{r-1}^2u_Y^2\Delta_Y'^{-1}$ with $u_{r-1}^2\ge(2/\pi)^6$, $u_Y\ge 0.95$, $\Delta'_Y/\Delta_R\le\pi^2/4$; off V_1 from $|q|\ge\pi$), so $|M^Y(A^X-A^Y)|\le 0.2a(C_*/20+6.9a)\eta|q|^{-1}$. Integrate $|E|\le C_E\eta|q|^{-1}$ against $|u_{r-1}|$. The decaying form is the geometric mean with the separate decay bound $|H^F(x^F,z)|\le Ce^{-\delta|x^F-z|}$ of each chain (Combes–Thomas for the coercive coarse operators [25]) and $|x^F-x|\le 2\sqrt{13}\eta'$. □

Lemma 17.33 (gradients and Hölder bounds at $A=0$). *$|\nabla H^F(x,z)|\le C_\nabla$ for the forward difference of each flow, hence $|H^F(x,z)-H^F(y,z)|\le C|x-y|^{1/2}(e^{-\delta|x-z|/2}+e^{-\delta|y-z|/2})$.*

Proof. The difference multiplies the alias-l summand by $d_F(q+l)$ with $|d_F(Q)|\le\min(|Q|,2\sqrt{3}/\eta')$; the main term is $\le C|q|(1+|q|)^{-2}$ and the aliases give $537a\,\eta$ integrated against $|u_{r-1}|$ with $J_0\le C\log^3(1/\eta)$ (Lemma 17.63), $\eta\log^3(1/\eta)\le C$. Geometric mean with the decay. □

Lemma 17.34 (coarse covariances). *Let $B^F:=(A^F)^{-1}$ and $\Gamma^F:=\overline{H^FB^FH^{F*}}$.* (i) *$C^F=G_r^F+\Gamma^F$ exactly (Woodbury).* (ii) *$\Gamma^F(x,y)=(2\pi)^{-3}\int_{V_1}B^F(p)\hat{h}_x^F(p)\overline{\hat{h}_y^F(p)}dp$ with $\hat{h}_x^F(p):=\sum_z H^F(x,z)e^{-\mathrm{i}p\cdot z}$; $B^F(p)\le C_B|p|^{-2}$ on V_1 and $\sup|\Gamma^F|\le C_1$.* (iii) *$|\Gamma^X(x',y')-\Gamma^Y(x'',y'')|\le C_\Gamma\eta^{1/2}$ for matched pairs.* (iv) *$|\Gamma^F(x,y)-\Gamma^F(x,w)|\le C_2|y-w|^{1/2}$.*

Proof. (ii) Parseval on Γ_r; $A^F(p)^{-1}=a_{r-1}^{-1}+\sum_{l_1}u_{r-1}(p+l_1)^2[\eta^2a^{-1}+S_F(p+l_1)]$ with $S_F(q)\le\frac{\pi^2}{4|q|^2}+31\eta^2$; the term $l_1=0$ gives $\frac{\pi^2}{4|p|^2}+C$ and the others are summable. (iii) $|B^X-B^Y|\le 0.198\,\eta^2$ pointwise (Proposition 17.31), $|\hat{h}^X_{x'}-\hat{h}^Y_{x''}|\le C'_h\eta^{1/2}$ (Lemma 17.32, decaying form); three differences. (iv) Lemma 17.33. □

Lemma 17.35 (scale pieces $j\ge 2$). *For $j\ge 2$, in the unit scale of level j ($\eta_j=5^{1-j}$) and matched points, $|S_j^X(x',y')-S_j^Y(x'',y'')|\le C\eta_j^{1/2}e^{-\delta|x'-y'|}$. In the unit scale of level r ($j<r$): $\le C(5\eta'/\ell_j)^{1/2}\ell_j^{-1}e^{-\delta|x'-y'|/\ell_j}$. The same holds for the photon.*

Proof. At level j: $|(A_j^X-A_j^Y)(z,w)|\le C\eta_je^{-\delta|z-w|/2}$ (geometric mean of the sup bound $0.20a^2\eta_j^2$ with the decay of each kernel); $C_j^F=(A_j^F+bQ^*Q)^{-1}$ decays for both chains; $C_j^X-C_j^Y=$

$C_j^X(A_j^Y - A_j^X)C_j^Y$. Then $S_j^X - S_j^Y \circ \tau = (H_j^X - H_j^Y \circ \tau)C_j^X H_j^{X*} + H_j^Y \circ \tau(C_j^X - C_j^Y)H_j^{X*} + H_j^Y \circ \tau\, C_j^Y(H_j^X - H_j^Y \circ \tau)^*$; bound the three terms by Lemma 17.32 at level j and sum the exponentials. Rescale with [13, (2.20)]: scalar kernels scale as ℓ^{-1} in three dimensions. □

Corollary 17.36 (uniform comparison form)**.** *Let $D_j := S_j^X - S_j^Y \circ \tau$ in the unit scale of level r and $\theta := \frac{1}{2}$.* (a) *Fine slices $0 \le j \le r-1$: $|D_j(x', y')| \le C_D \eta'^{\theta} \ell_j^{-1-\theta} e^{-\delta|x'-y'|/\ell_j}$.* (b) *Coarse slices and remainders $j \ge r$: $|D_j| \le C\eta^{1/2}\ell_j^{-1}e^{-\delta|x'-y'|/\ell_j}$ and $|\Gamma_n^X - \Gamma_n^Y \circ \tau| \le C\eta^{1/2}$ at level n.* (c) *Given Proposition 17.72(a) for the hybrid flow and* [13, Thm. 3.3] *for the standard flow, the derivative, Hölder and contour versions hold: the contour lines* [13, (3.27)] *along the two paired contour systems differ by the kernel difference along the contour plus the flux of the slice kernel through the sliver between the paired partial contours, a gain $(\eta'/\ell_j)^{1/2}$ on slice j; the double contour lines, both ends on contours, have the same relative gain against their size $C\ell_j$.*

Proof. (a) For $j \ge 2$ this is Lemma 17.35. For $j \in \{0, 1\}$ bound $|S_j^X|$ and $|S_j^Y \circ \tau|$ separately by $C\ell_j^{-1}e^{-\delta|\cdot|/\ell_j}$ and use $\ell_j \le 5\eta'$, so $1 \le 5^{\theta}(\eta'/\ell_j)^{\theta}$: there is no gain on these lines, and the factor η'^{θ} is paid for by a degree loss θ (King's own device at the finest slice, [13, p. 675]: "When $j = 0$, we bound each term separately"). (b) Beyond level r both flows apply the same rotated steps; only the input B^F differs, by the relative amount $0.2a\eta^2$ (Proposition 17.31), and every coarse multiplier is built from it by the same formulas with non-negative weights. (c) Interpolation of (a), (b) with the separate bounds of each flow; for the contours, Lemmas 17.16 and 17.20: the sliver has area $\le 1.5\eta'\ell_{j+1} + 33\eta'^2$, against the slice kernel's gradient $\le C\ell_j^{-2}$, plus the end segment $c_\tau\eta'$ and Riemann errors; for $j \in \{0, 1\}$ there is no gain and the factor is paid by a degree loss, as in (a). □

Corollary 17.37 (the full propagators of the two lattices)**.** *On the infinite lattices, for matched pairs and $u = x' - y'$ with $|u| \ge 25\varepsilon$ (physical units),*

$$|C^X(x' - y') - C^Y(x'' - y'')| \le C_6\, \varepsilon^{1/2}|u|^{-3/2},$$

and the same for the photon and for G_r^F, Γ^F separately. For all u the difference is $\le 2C_0(|u| + \varepsilon)^{-1}$.

Proof. Pick $n \ge 2$ with $\delta_n \le |u| < 5\delta_n$ and write $C^F = \sum_{j<n} S_j^F + \Gamma_n^F$. The slices $j \in \{0, 1\}$ are each $\le C\varepsilon^{-1}e^{-\delta_*|u|/(5\varepsilon)} \le C'\varepsilon^{1/2}|u|^{-3/2}$; the slices $2 \le j < n$ differ by $\le C\delta_j^{-1}(5\varepsilon/\delta_j)^{1/2}e^{-\delta|u|/\delta_j}$ and sum to $\le C\varepsilon^{1/2}|u|^{-3/2}$; the remainder differs by $\le C_\Gamma\delta_n^{-1}(5\varepsilon/\delta_n)^{1/2}$. Pairs stay matched at every level because τ is defined by the first-step cells. □

Since $C^Y(x'' - y'') = C^X(R(x'' - y''))$, Corollary 17.37 is a quantitative rotation estimate for the massive lattice Green's function, with relative error $(\varepsilon/|u|)^{1/2}$, obtained from the two block-spin decompositions alone.

The comparison principle and the two-flow bound. A *renormalised family* is a finite sum of King's renormalised graphs [13, §3.5], Bałaban's classes of [3, §§2–3] (Definition G.1), localised as in [3, (1.33)]. It is in *per-flow normal form* when, for each flow separately and before any element is mixed, every counterterm graph is expanded in the lines of the flow [13, (3.76)], every Taylor subtraction [13, (3.83)] and Ward-identity rewriting has been carried out with that flow's exact identities, and the exact cancellations between members with identical element sets ([3, (3.6)/(3.7), (2.21b)/(2.21g), (2.18)]) are co-switched, the identified elements of both members being switched together. Its *elements* are lines, vertex functions and external fields.

Proposition 17.38 (comparison principle)**.** *Let $\mathcal{R}$ be a renormalised family in per-flow normal form with $n_{\mathcal{R}}$ elements and $E^F(\mathcal{R})$ its value along flow F. Then, with $\theta = \min(\frac{1}{2}, \theta_1)$ and θ_1 of Theorem 17.39,*

$$|E^X(\mathcal{R}) - E^Y(\mathcal{R})| \le C n_{\mathcal{R}}\, p_r^2 \delta_r^{-1/2}\, \eta'^{\theta}(1 + k)^p \cdot [\textit{right side of } [3, (1.33)] \textit{ for } \mathcal{R}],$$

the prefactor $p_r^2\delta_r^{-1/2}$ being that of the contour elements at $e > 0$ (absent at $e = 0$).

Proof. By Lemma 17.29(i) both values are sums over X-vertices. Order the elements consistently across members and telescope: $E^X - E^Y = \sum_i E[e_1,\dots,e_{i-1}$ in X-version; $e_i \to e_i^X - e_i^Y$; $e_{i+1},\dots$ in Y-version]. In the i-th term the element $e_i^X - e_i^Y$, divided by $C\eta'^{\theta}$, obeys the bounds of its type weakened by the degree θ: fine lines by Corollary 17.36(a), coarse lines by (b) ($\eta^{1/2} \le \sqrt{5}\eta'^{1/2}$), vertex functions by Lemma 17.29(ii) with $\eta' = \eta'^{\theta}\eta'^{1-\theta}$, external fields $H^F\phi_r$ and their covariant derivatives by Lemma 17.32 and its derivative and Hölder versions; at $e > 0$, contour lines [13, (3.27)] by Corollary 17.36(c) (gain $(\eta'/\ell_j)^{1/2}$), double contour lines by Proposition 17.72(b) (the same relative gain, size $C\ell_j$), contour vertices [13, (3.24)–(3.26)] by Proposition 17.72(b) (relative gain $\eta^{1/2}$, size $Ce_r p_r^2$) and the constant-background phases [13, (3.46)] by Proposition 17.72(b) ($\le e\, p_r \mu_0^{-1} c_r \delta_r^{-1/2}\eta'$), whence the prefactor. The weakened-element bound for each term is Theorem 17.39, whose proof runs this telescoping with the subtraction rule described there. □

Theorem 17.39 (the two-flow comparison bound)**.** *Let $\alpha \in (0,1)$ be the Hölder exponent of* [13, (3.8)] *and put $\theta_1 := \alpha^2/(4(1+\alpha)^2)$. There are $p = p(\bar n)$ and C such that for every renormalised family $\mathcal{R}$ of order $\le \bar n$ in per-flow normal form (the h-families of Lemma 17.60 other than $\mathcal{T}_h$, and the families of step (5) of the proof of Theorem 17.56 at $e > 0$ after the constant-background reduction of Lemma 17.40), for $0 < \theta \le \theta_1$, uniformly in r, k, the torus and the masses, the two-flow difference obeys the bound of Proposition 17.38: in every term of its telescoping, the element replaced by the difference $e_i^X - e_i^Y$, which obeys the bounds of its type weakened by the degree θ,* [3, (1.33)] *holds with the extra factor $C(1+k)^p\eta'^{\theta}$ and the prefactor stated there. The exponent θ_1 does not depend on $\bar n$, and any $\theta > 0$ with any polynomial factor suffices for the consumers (Lemma 17.26).*

The proof is given at the end of this part, after the constant-background reduction (Lemma 17.40), which it uses at $e > 0$. Its inputs are Proposition 17.72, [13, Thm. 3.3] for the standard flow, [3, Prop. 2.1, (2.8)] with the ordering device (2.7) there, the zero-momentum counterterm conditions [13, (3.33)] through order four with the vanishing of the one-loop vacuum-polarisation moments [3, (2.23)–(2.28)], the local limit law for the simple random walk [15, Thm. 2.5.6], and the h-census of Lemma 17.60. In outline: every line is written as one flow-independent lattice function sliced by the heat kernel; degrees are read per ordering of the scale-summed elements; each unit (a self-energy, vacuum-polarisation or triangle insertion) is bundled with one of its legs into a single element, and one subtraction rule is applied, in each flow's own frame, to every connected component of degree zero that forms in an initial segment, its local terms being common numbers of both flows; after the rule every component containing the weakened element has degree $\ge \min(\alpha, 1)$, and the scale sums give the gain η'^{θ_1}.

Lemma 17.40 (the constant-background reduction)**.** *In the localised form* [13, (3.55)] *every family has its external vertices in a cube D of side $\le Cp_r$. Let D'' be the union of the level-r blocks within p of D, $x_0 \in D$, $B := A_r(x_0)$ and $\tilde A := A^{(r)} - B$ on all of T. For each flow F the reduction runs on the infinite fine lattice Λ_F of level r with the P_T-periodic lifts A_{lift}, $\tilde A_{\mathrm{lift}}$ (no periodic cell is formed); F_T is the lifted fundamental cube centred at D, $F_1 := \{\mathrm{dist}(\cdot, D'') \le p/3\}$, and the vertex region F_v is the union of the level-r blocks within $p/3 - 3$ of D''. The sup-systole of P_T in the unit scale of level r is $s_T \ge 40MJ\,5^{\lfloor \varrho k\rfloor} \ge Cp_r$ for $k \ge k_0$.*

(0) (P) *For $x, y \in F_T$, $G_r(T, A^{(r)})(x,y) = \sum_{\gamma\in P_T} G_r(\Lambda_F, A_{\mathrm{lift}})(x, y+\gamma)$, and likewise for $D_A G_r$, Q_r and H^F; the image part δ^{im} (the terms $\gamma \ne 0$) obeys $|(\delta^{\mathrm{im}} f)(x)|, |(D_A\delta^{\mathrm{im}} f)(x)| \le Ce^{-\delta_0 s_T/4}\|f\|_\infty$ for $\operatorname{supp} f \subset B(x, s_T/4)$.* (G) *On Λ_F the constant B is removed exactly by the gauge function $B\cdot x$, leaving King's phases* [13, (3.46)] *at the vertices.* (R) *$|\tilde A_r(w)| \le p_r(d_T(w, x_0) + 1)$ and $|\partial\tilde A^{(r)}| \le Cp_r$ on all of T, so $\tilde A_{\mathrm{lift}}$, $\vartheta\tilde A_{\mathrm{lift}}$ and $B_{\mathrm{bg}} := (1-\vartheta)\tilde A_{\mathrm{lift}}$ are regular.* (N) *If A_1, A_2 are regular on Λ_F and $A_1 = A_2$ on the bonds inside F_1, then $\delta := G_r(\Lambda_F, A_1) - G_r(\Lambda_F, A_2)$ obeys $|\delta(x,y)| \le Ce^{-\delta_1(|x-y| + \mathrm{dist}(x, F_1^c) + \mathrm{dist}(y, F_1^c))}$ with C independent of F_1, and the same for $D_A\delta$ when the bond at x lies inside F_1; in particular $G_r(\Lambda_F, B_{\mathrm{bg}}) = G_r(\Lambda_F, 0) + \delta^{\mathrm{bg}}$.*

(1) *For each flow* $F \in \{X, \tau Y\}$ *and each renormalised family* $\mathcal{R}$ *at background* $A^{(r)}$ *on* $\operatorname{supp}\chi_r$*:* $\mathcal{R}^F(A^{(r)}) = \sum_{q\le q_0}\sum_{w_1,\dots,w_q} A_r(w_1)\cdots A_r(w_q)\,\mathcal{R}^F_{q;w}(A=0)\prod_v U^F(B(\Gamma_v)) + \mathcal{R}^F_{\delta G} + \mathcal{R}^F_{\mathrm{rem}}$*, where* $\mathcal{R}^F_{q;w}$ *is a renormalised family in per-flow normal form whose lines are the* $A=0$ *lines of the flow* ($G^F_r(0) = C - \Gamma^F_r(0)$, H^F, $(A^F)^{-1}$) *and whose vertices are* [13, (3.18)–(3.26)] *at* $B=0$ *with the explicit factors* η *of the pin and contour vertices; the coarse-field factors* $A_r(w_i)$ *are the same functions of the common coarse field for both flows;* $U^F(B(\Gamma_v)) = \exp[\mathrm{i}e(5^r\varepsilon)^{1/2}qB(\Gamma^F_v)]$ *are the constant-background phases* [13, (3.46)]*;* $\mathcal{R}^F_{\delta G}$ *collects the terms with a line* δ^{im} *or* δ^{bg} *or with a vertex outside* F_v*, and* $\mathcal{R}^F_{\mathrm{rem}}$ *the Taylor remainders of* [13, (3.48), (3.52)].

(2) $|\mathcal{R}^F_{\delta G}| + |\mathcal{R}^F_{\mathrm{rem}}| \le C(\delta_r^\sigma + e^{-\delta' p(\delta_r)})|T|$ *per flow.*

(3) *The two-flow difference of the leading sum is controlled element by element: the* $A=0$ *lines by Corollary 17.36 and the difference elements* $\Delta\Gamma$, ΔH, $\Delta(A^{-1})$ *of size* $O(\eta'^{1/2})$*; for a constant field* B *the line integral depends only on the endpoints,* $B(\Gamma^F_v) = B\cdot(c(v)-v)$*, so* $U^X(B(\Gamma_v))$ *and* $U^{\tau Y}(B(\Gamma_{\tau v}))$ *differ by the phase of* $e(5^r\varepsilon)^{1/2}B\cdot(\tau v - v)$*,* $|\tau v - v| \le c_\tau\eta'$*, i.e. by* $\le e p_r \mu_0^{-1} c_\tau \delta_r^{-1/2}\eta'$ *([13, (3.2)]); the contour vertices* [13, (3.24)–(3.26)] *at* $B=0$ *differ entrywise by a relative* $\eta^{1/2}$ *and the contour lines, one-ended and double, obey the slice-*j *bounds in both flows with mismatch* $(\eta'/\ell_j)^{1/2}$ *(Proposition 17.72(b)); every connected subgraph containing a contour line has Bałaban's degree* ≥ 1*; the explicit factors* $A_r(w_i)$ *are common.*

Proof. (1) is King's expansion [13, (3.43)–(3.55)] applied to each flow separately; it uses only the gauge covariance of the flow's lines (the hybrid flow is gauge covariant: its block average is the frame-covariant photon average of Remark 17.18 and Lemma 17.19, so a constant B is mapped to the coarse constant $R^{-1}B$ for the Y flow and B for the X flow, and the gauge transformation $A^{(r)} \mapsto A^{(r)} - B$ acts on both by the same unitary at each vertex), the universal-cover form (0) of the localisation, and the Taylor expansion around $\vartheta\tilde{A}_{\mathrm{lift}} = 0$ with background B_{bg}, [13, (3.47)–(3.54)], exact algebra in which the lines become $A=0$ lines and the powers of $\tilde{A}$ become explicit coarse-field factors sampled at block points. The paired contours (Lemma 17.16) make the contour phases of the two flows differ by the flux through an area $O(\eta')$ (Lemma 17.20), so that the contour vertices and lines are difference elements with the gains of (3) rather than $O(1)$ mismatches. (0): the block systems of both flows, tiles and fixed staircases included, are P_T-invariant (Lemma 17.14(iii) and the in-tile convention of Definition 17.15), so the operator $M_{A_{\mathrm{lift}}} + a_r Q_F^* Q_F$ on Λ_F commutes with P_T; its inverse has a summable kernel (the bounds of (2)), its periodisation is a right inverse of the torus operator, which is positive definite, and so equals its inverse; for $|x-y| \le s_T/2$ and $\gamma \ne 0$, $|x-y-\gamma| \ge |\gamma|/2$, whence the image bound. (G): on Λ_F the gauge function is single-valued and both flows are gauge covariant for every gauge function. (R): [13, (3.2)] on all of T, $\sum_z \nabla H^{\mathrm{ph}}(x,z) = 0$ with the r-uniform decay of ∇H^{ph}, and the product rule with $|\nabla\vartheta| \le 3/p$. (N): the random-walk representation of Proposition 17.72(a) with one domain: chains whose cells lie with a fixed margin inside F_1 are the same for A_1 and A_2, every other chain of length n has $\sqrt{3}M_{\mathrm{w}}n \ge \operatorname{dist}(x, F_1^c) + \operatorname{dist}(\operatorname{supp} f, F_1^c) - c$ and $\sqrt{3}M_{\mathrm{w}}n \ge |x - \operatorname{supp} f| - c$, and the average of the two gives the rate $\delta_1 = (2\sqrt{3}M_{\mathrm{w}})^{-1}$. Vertices in F_v see $B_{\mathrm{bg}} = 0$ on every bond and in-block contour they touch, so King's "taking $A' = \theta\tilde{A}$ and $B = 0$" [13, p. 662] holds there literally. (2) is King's bound [13, p. 665] with the kernel bounds at the regular fields of (R): a δ^{bg} line with an end within p of D gains $e^{-\delta_1 p/3}$, otherwise the string from an external vertex gives $e^{-\delta_2 p}$; a vertex outside F_v gives $e^{-\delta_2(4p/3-3)}$ by tree decay; a δ^{im} line gives $e^{-\delta_0 s_T/4}$; half of every exponent pays the vertex sums, whence $\delta' = \delta_2/6$, and $e^{-\delta' p(\delta_r)} \le C\delta_r^\sigma$ since p > 1; the Taylor remainders are $\le (Ce_r p_r^2)^{\bar{n}+1} \le C\delta_r^\sigma$. The universal-cover form replaces the periodic-cell reading of [14, p. 337]: the periodised restriction of $\tilde{A}$ to a cell of side $\asymp p_r$ jumps by $\asymp p_r^2$ across the seam and is not regular, so the kernel bounds do not apply to it. (3) is as stated; the inversion symmetry about the coarse site used in the oddness rule requires the centred block convention for both flows (Convention 17.17(i)), and the frame-covariant average keeps it for the tiles. □

Corollary 17.41 (the one-flow instance)**.** *For one flow F (standard or hybrid), a renormalised family of order $\le \bar n$ with lines from the slices $j = 0, \dots, k$ of that flow (*[13, (3.76)]*, each slice obeying* [13, (3.63)–(3.65)]*; the coarse lines of depths r and $r+n$ of the same flow included) and one element weakened by degree $\theta < \alpha_0$ obeys* [3, (1.33)] *with an extra factor $C(1+k)^p$. Consequently King's estimate* [13, Prop. 3.6]*, asserted on p. 665 without a printed argument, holds for both flows.*

Proof. The multiscale bound on the point-split graph (the two summation points of a block product as vertices, the block kernel a line of index r): the slice lines satisfy the slice bounds with $d_l = -1$, a contour line of [13, (3.26)–(3.27)] with $d_l = -1 + \#(\text{contour ends})$ ([13, (3.63)–(3.65), Prop. 3.7, Thm. 3.3] for the standard flow, Proposition 17.72(a) for the hybrid flow); the per-flow renormalised structure gives degree $\ge \alpha_0 > 0$ for every connected all-scale-summed subgraph except the trees [3, (2.4)] and the counterterm-attachment subgraphs of [13, (3.78)] with degree 0, each of which costs a factor $(1+r)$ (King's "possible addition of some power of $\ln(L^k\varepsilon)^{-1}$", p. 666, is $(1+k-r)^p$; both are $\le (1+k)^p$, Lemma 17.26). The weakened line is the replacement $c_{l_0} \mapsto c_{l_0}\ell_j^{-\theta}$: the degree of every subgraph containing l_0 drops by $\theta < \alpha_0$ and stays positive, and no new zero-degree subgraph appears; a weakened vertex changes its exponent by $-\theta$; a weakened external field changes a norm only; the Hölder remainders of [13, (3.83)] raise d_l by α on the line they act on. This is King's own weakened-element induction [13, (3.67)–(3.70)], with no two-flow input; the strict positivity of the degrees after Bałaban's resummed rewriting is [3, (3.5)], and no cancellation for a fixed ordering of the lines is used. □

Proof of the two-flow comparison bound. The proof of Theorem 17.39 occupies the rest of this part. Its steps: the lines of every family are written as flow-independent lattice functions sliced by the heat kernel, plus coarse elements (this paragraph); degrees are read per ordering of the scale-summed elements, so that only the connected components of initial segments matter (Lemma 17.42); each flow's expression is put in a normal form and one subtraction rule is applied, in each flow's own frame, to every component of degree ≤ 0 that forms (Definition 17.47); after the rule every component containing the weakened element has degree $\ge \min(\alpha, 1)$ (Lemma 17.48); every element type obeys a two-flow difference law with a positive exponent (Lemmas 17.52 to 17.54); and the telescoping over the elements, together with the scale sums, gives the bound with the exponent θ_1 (Lemma 17.55 and the proof at the end).
The all-C form and the slices. After the constant-background reduction (Lemma 17.40; at $e = 0$ nothing is to be reduced) every line of a family is an $A = 0$ line of its flow on the infinite lattice: $G_r^F = C^F - \Gamma^F$ (Lemma 17.34(i)), where C^F is the free propagator of the flow's lattice, $C^X(x,y) = C(x-y)$ and $C^Y(x'',y'') = C(R^{-1}(x''-y''))$ with *one* lattice function $C = (-\Delta_{\eta'} + m_r^2)^{-1}$ on $\eta'\mathbb{Z}^3$ (for the photon, $D = (-\Delta_{\eta'} + \mu_0^2\delta_r^2)^{-1} \otimes \mathbf{1}_3$). Expanding every line of every graph in C and Γ gives C/Γ *monomials*, an exact identity; in each monomial the C-lines are the same lattice function evaluated in each flow's own frame, the vertices are matched by Lemma 17.29 and the counterterm numbers are the same (Lemma 17.30), while Γ^F, H^F, $B^F = (A^F)^{-1}$ and the external fields depend on the flow. Write C as the resolvent of the continuous-time simple random walk on $\eta'\mathbb{Z}^3$ with rate η'^{-2} per direction, $C(u) = \eta'^{-3}\int_0^\infty e^{-m_r^2 s}P_s(u)\,ds$, and slice it in the time variable:

$$C^{(0)} := \eta'^{-3}\int_0^{\ell_1^2}, \qquad C^{(j)} := \eta'^{-3}\int_{\ell_j^2}^{\ell_{j+1}^2} \ (1 \le j \le r-1), \qquad C^{\mathrm{top}} := \eta'^{-3}\int_1^\infty,$$

all of $e^{-m_r^2 s}P_s(u)\,ds$ ($\ell_r = 1$). Gaussian and Poisson tail bounds for the walk give, for every $n \ge 0$,

$$|\partial^n C^{(j)}(u)| \le c_n \ell_j^{-1-n} e^{-\delta|u|/\ell_j} \quad (0 \le j \le r-1), \qquad |\partial^n C^{\mathrm{top}}(u)| \le c_n e^{-c' m_r |u|}(1+|u|)^{-1}, \tag{35}$$

∂ a lattice difference; a derivative is charged to its line. The decomposition is the same for both flows as lattice functions; the Y flow evaluates it at $R^{-1}(\tau x - \tau y)$, and in X-coordinates the Y-version obeys (35) with $\delta/2$, since $|R^{-1}(\tau x - \tau y)| \ge |x-y| - 2c_\tau\eta'$. The heat-kernel slices replace the block-spin slices of Lemma 17.35 in the comparison of the all-C lines; both decompositions are exact, and the heat-kernel one is flow-independent as a lattice function, which is what the

subtraction rule below needs. (The gain $\eta'^{1/2}$ of Corollary 17.36(a) is recovered, and improved, by Lemma 17.52.)

Elements. The *elements* of a monomial, with their per-flow bounds in the unit scale of level r, are: (a) the fine slice lines $C^{(j)}$, $D^{(j)}$, $0 \le j \le r-1$, with their lattice derivatives, bounded by (35); (a′) the top pieces; (b) the coarse elements Γ^F, H^F, B^F (Lemma 17.34, Lemma 17.32), bounded and decaying at the coarse mass, with one derivative and an α-Hölder gradient ([13, Thm. 3.3] for the standard flow, Proposition 17.72(a) for the hybrid flow); (c) the external fields $E^F \in \{H^F\phi_r, H^F_A A_r$, their covariant derivatives, the Cameron–Martin background$\}$, bounded by the field norms [13, (3.2)] on $\operatorname{supp}\chi_r$ and $C^{1,\alpha}$ in the same sense; (d) the vertex functions w (couplings, partitions of unity, the background), smooth with $\|\nabla w\| \le C$ (Lemma 17.29(ii)); (e) at $e > 0$ the contour elements of Proposition 17.72(b): contour lines [13, (3.27)], double contour lines, contour vertices [13, (3.24)–(3.26)] and the constant-background phases [13, (3.46)]; (f) the counterterm numbers δm^2, E_1, the same for both flows. Their matched differences, X-version at x against Y-version at τx: (b) $|\Gamma^X - \Gamma^Y \circ \tau| \le C_\Gamma \eta^{1/2}$ (Lemma 17.34(iii)), $|B^X - B^Y| \le 0.2\eta^2$ (Proposition 17.31), coarse slices $\le C\eta^{1/2}\ell_j^{-1} e^{-\delta|x-y|/\ell_j}$ (Corollary 17.36(b)); (c) $|E^X(x) - E^Y(\tau x)| \le C\eta^{1/2}\|E\|$ (Lemma 17.32, decaying form, applied to the lift); (d) $|w(x) - w(\tau x)| \le c_\tau \eta' \|\nabla w\|$; (e) contour lines: relative gain $(\eta'/\ell_j)^{1/2}$ on slice j, also with one lattice derivative; contour vertices: relative gain $\eta^{1/2}$ against their size $Ce_r p_r^2$; phases: $\le e\, p_r \mu_0^{-1} c_\tau \delta_r^{-1/2} \eta'$ (Proposition 17.72(b)); the prefactor $p_r^2 \delta_r^{-1/2}$ of the theorem comes from here and from nowhere else. Gradients of (b) and (c) are compared in Lemma 17.53.

The structures. With the vertices [13, (3.18)–(3.23)] at $B = 0$ (Lemma 17.40(1)), the connected all-C line sets of non-positive degree that occur are the following (the list is proved in Lemma 17.45). SE: the order-e^2 scalar self-energy, lines $\{C(x,y), D(x,y)\}$, two current vertices, both derivatives internal. VP: the order-e^2 vacuum polarisation, lines $\{C(x,y), C(x,y)\}$, two derivatives on the two lines, together with its seagull tadpole and derivative tadpole ([13, (3.22)] with $m = 2$), which enter only through the Ward sum of Lemma 17.51. TAD: the λ-tadpole $C^{(j)}(0)$ and the seagull scalar tadpole $D^{(j)}(0)$, numbers, identical for both flows and cancelled exactly, slice by slice, by their counterterm shares ([13, (3.33)], Lemma 17.30). T: the order-e^3 triangle $\{C(a,b), C(b,c), D(a,c)\}$, three current vertices. SC: the order-e^3 seagull–current loop $\{C(s,b), D(s,b)\}$. W: the one-particle-irreducible two-point structures of order four: scalar (the λ^2 sunset; the e^4 rainbow, crossed, insertion and seagull graphs) and photon (the e^4 self-energy insertion, vertex correction, seagull graphs and the double seagull). L24: Bałaban's graph [3, (2.4)], a single scalar line between two bare current vertices carrying both their derivatives, $\partial\partial C$, with two external photon legs. CT: the counterterm share c_σ of a self-energy or tadpole at slice assignment σ (a one-site insertion, $|c_\sigma| \le C\ell_{\sigma_{\max}}^{-1}$, a loop line of dimension -1 at scale $\sigma_{\max}$), and the order-four share of dimension 0. The $e^2\lambda$ two-point members are numbers or vanish identically; structures with an odd number of photon legs vanish by charge conjugation; the $(2,1)$-structure at order $e\lambda$ vanishes; four-leg loops have positive degree; contour vertices carry the explicit weight $\ge \frac{1}{2}$ of [13, (3.30)] at their site.

Degrees. For a connected set H of elements (lines with their endpoints; a compound, defined in Definition 17.47, counts with the vertices and line dimensions it contains) put

$$D(H) := \sum_{l \in H} d_l + \sum_{v \in V(H)} a_v + 3(|V(H)| - 1), \qquad d_l = -1 - (\text{number of lattice derivatives on } l),$$

a_v the vertex weight: an explicit factor η'^a, $a \ge 0$, at v counts as $\ell^a_{j(v)}$ with $j(v) := \min\{j_l : l \ni v\}$; a derivative acting on an external field costs nothing, since the externals have bounded gradients. This is Bałaban's degree [3, (2.1)–(2.3)] with every derivative charged to its line. The *bare degree* $D_0(H)$ is the same count with the derivative placement of the normal form (NF2) below and before any derivative is transferred. All-C degrees are integers; the only non-integer degrees are vertex weights $\alpha \in (0,1)$ produced by second-order subtractions with an external leg (Lemma 17.54(i)), so every degree is $n + m\alpha$ with $m \ge 0$.

Lemma 17.42 (the multiscale bound per ordering)**.** *Let G be a connected set of elements of a monomial, each scale-summed element l carrying the bound $c_l \ell_{j_l}^{d_l} e^{-\delta|x_l - y_l|/\ell_{j_l}}$ at its scale j_l (fixed elements carry a bounded decaying kernel and impose no condition), and fix one ordering class of* [3, (2.7)]*: an ordering $l(1), \dots, l(m)$ of the scale-summed elements with the scale sums restricted to $j_{l(1)} \le \cdots \le j_{l(m)}$, ties assigned to one class. Suppose that every connected component of every initial segment $G_i := \{l(1), \dots, l(i)\}$ has $D \ge 0$, and $D \ge \theta$ if it contains a distinguished element e_0 whose constant carries the factor $(\eta'/\ell_{j(e_0)})^\theta$; let n_0 be the number of such components with $D = 0$. Then the sum over the scales in the class and over all vertices but one is $\le C(1+r)^{n_0} \prod_l c_l$, and with the distinguished element $\le C\eta'^\theta (1+r)^{n_0} \prod_l c_l$, $C = C(G)$. Only the connected components of initial segments are tested.*

Proof. This is the proof of [3, Prop. 2.1] read for one ordering class, which is how Bałaban runs it: the finest element $l(1)$ is summed first, shrinking its line to a point with the exponent $D(G_1)$; the sum over $j_1 \le j_2$ is geometric when $D(G_1) > 0$ and costs a factor $(1+r)$ when $D(G_1) = 0$; the merged vertex receives the weight $D(G_1)$, and the induction continues with $G_2, G_3, \dots$, at each stage testing only "the zero-degree subgraphs met along the way", the connected components of the initial segments. The same per-ordering positivity is King's [13, p. 665]. With the distinguished element, every component containing it has exponent $D - \theta > 0$ for the geometric sums, and the factor $\eta'^\theta \ell_{j(e_0)}^{-\theta}$ is absorbed by the merged vertex weight $\ell_{j'}^{D-\theta} \le \ell_{j(v)}^{D-\theta}$ at every later stage, so η'^θ survives to the end. A compound enters as a line of its stated dimension at the scale of its leg; its internal scale sums are done in Lemma 17.54(i) uniformly in the positions of its outer vertices, so the first step applies to it verbatim. □

Lemma 17.43 (common moments)**.** *Let $H(x,t)$ be a kernel built in a flow's own frame from C, D, their lattice derivatives, lattice Taylor remainders, lattice sums, the counterterm numbers and the bond shifts of Lemma 17.44 (no externals, no coarse elements). Then for $a = 0, 1$,*

$$\sum_{t \in \Lambda_X} H^Y(\tau x, \tau t)\, (\tau t - \tau x)^{\otimes a} = R^{\otimes a} \sum_{t \in \Lambda_X} H^X(x,t)\, (t-x)^{\otimes a},$$

the indices compared frame-free (Lemma 17.29(iii)): zero moments are the same numbers and first moments the same vectors in both flows. An O_h-invariant moment is isotropic: a rank-one or rank-three tensor vanishes, a rank-two one is $c\,\delta_{\mu\nu}$. On the tori the kernels are periodisations of the infinite-lattice ones and the moments agree up to $O(e^{-cm_r \cdot 50 M J})$ (Theorem 17.57(b)).

Proof. H^Y is the conjugate of H^X by the isometry R (one lattice function evaluated in two frames) and $\tau : \Lambda_X \to \Lambda_Y$ is a bijection (Definition 17.28), so the left side is $\sum_{v \in R\eta'\mathbb{Z}^3} H(R^{-1}v) v^{\otimes a} = R^{\otimes a} \sum_{u \in \eta'\mathbb{Z}^3} H(u) u^{\otimes a}$. The lattice model is O_h-invariant (King's action on $\varepsilon\mathbb{Z}^3$ with bond fields, in midpoint coordinates for bond kernels), so an O_h-invariant tensor of odd rank vanishes under $-\mathbf{1} \in O_h$ and one of rank two is a multiple of $\delta_{\mu\nu}$. □

Lemma 17.44 (the current vertex and the placement of its derivative)**.** *The current vertex* [13, (3.22)] *($m = 1$) on the bond $b = (x, x + \eta' e_\mu)$ is $e_r A_b\, [\phi(b_+) \cdot q\phi(b_-)]/\eta'$ with q antisymmetric. For two fields a, b (a leg and an internal line after Wick contraction),*

$$(D_\mu a)(b) \cdot q\, b(b_-) = -a(b_+) \cdot q\, (D_\mu b)(b) + \eta'^{-1} \big[(a \cdot qb)(b_+) - (a \cdot qb)(b_-)\big],$$

and summed against A_b over the bonds the last term is $-\sum_x (\partial_\mu^* A_\mu)(x)\, (a \cdot qb)(x)$, the first term of [3, (2.8)]*; if the vertex function g at the vertex is position dependent, the second term of* [3, (2.8)]*, $\sum_x (a \cdot qb)(x) \sum_\mu (\partial_\mu^{\eta'} g)(x) A_\mu(x - \eta' e_\mu)$, appears as well. Hence the one lattice derivative of a current vertex may be written on either of its two scalar fields, the other being read at the shifted end of the bond, at the cost of a term in which the derivative acts on the vertex's photon field (and one with an explicit factor η' from $\partial^{\eta'} g = \eta' \nabla g + O(\eta'^2)$). The identity is exact in each flow's own frame; a shifted evaluation point is matched by τ with c_τ replaced by $c_\tau + 1$ in every matched-point bound. A current vertex has exactly two scalar fields and one photon field; the seagull $e_r^2 |A|^2 |\phi|^2$ and the quartic $\lambda_r |\phi|^4$ carry no derivative.*

Proof. $a(b_+) = a(b_-) + \eta'(D_\mu a)(b)$ and $b(b_-) = b(b_+) - \eta'(D_\mu b)(b)$ give $a(b_+)\cdot qb(b_-)/\eta' = (D_\mu a)\cdot qb(b_-) + (a\cdot qb)(b_-)/\eta' = -a(b_+)\cdot q(D_\mu b) + (a\cdot qb)(b_+)/\eta'$; subtract, and sum by parts over the bonds $(x, x+\eta' e_\mu)$. For $a = b = \phi$ the remainder is $\phi\cdot q\phi/\eta' = 0$, which is why the one-field form of the vertex has no such term; for two distinct fields it does not vanish, since the photon covariance is $\delta_{\mu\nu}$ times a scalar and not transverse. □

Lemma 17.45 (the connected all-C sets of non-positive bare degree)**.** *Let H be a connected set of all-C lines with every current-vertex derivative placed on an internal line of its unit when the vertex belongs to a unit (the placement (NF2) below), and let n_e, n_s, n_λ be the numbers of its current, seagull and quartic vertices, n_{ct} the number of its counterterm insertions, E the number of its slots (legs of $V(H)$ not on lines of H, external fields included) and x the number of current-vertex derivatives of $V(H)$ placed outside H. Then*

$$D_0(H) = \tfrac{1}{2}n_e + n_s + n_\lambda + n_{\mathrm{ct}} + \tfrac{1}{2}E - 3 + x.$$

Decompose H into its two-edge-connected blocks P joined by $\rho-1$ bridges b. Then $D_0(H) = \sum_P[\omega(P)+\mathrm{out}(P)]+\sum_b(2-\mathrm{in}(b)) \geq \rho-2$, where $\omega(P)$ is the degree of P with every derivative of its vertices placed inside P, $\mathrm{out}(P)$ the number of derivatives of vertices of P placed outside P and $\mathrm{in}(b)\leq 2$ the number of derivatives on the bridge, so that $\sum_P \mathrm{out}(P) \geq \sum_b \mathrm{in}(b)$; and $\omega(P)\geq -1$ for every block with legs, with equality exactly for the units SE and VP (the tadpoles being dropped), for a counterterm insertion and, by convention, for a bare current vertex ($\omega=-1$, $\mathrm{out}=1$). Consequently $D_0(H)\leq 0$ forces one of: $\rho=1$ and H is a unit (-1) or one of T, SC, W, the double seagull, a λ^2 or e^4 or $e^2\lambda$ two-point structure (0); $\rho=2$ and H is two blocks with $\omega=-1$ and every non-internal derivative on the bridge: U–l–U' (l undifferentiated), U–∂l–v, v–$\partial\partial l$–v' (= L24), and with a counterterm insertion ct–l–U, ct–∂l–v, ct–l–ct; and, in the local-term monomials of the rule below, the same with a vertex of weight -1 (the vacuum-polarisation leftover of Lemma 17.51) joined by an undifferentiated photon line to a vacuum-polarisation compound whose leg is that line, or to another such vertex. Unions of blocks at a cut vertex (a quartic or seagull vertex) have both slots at one vertex and are common numbers. Every member is a two-point structure except T and SC (three slots) and the one-point numbers.

Proof. Count: $\sum_l d_l = -I - n_\partial$ with I the number of lines and n_∂ the number of internal derivatives, $n_\partial = n_e - x$; the scalar legs give $2I_\phi + E_\phi = 2n_e + 2n_s + 4n_\lambda + 2n_{\mathrm{ct}}$ and the photon legs $2I_A + E_A = n_e + 2n_s$; $|V| = n_e + n_s + n_\lambda + n_{\mathrm{ct}}$, and the counterterm insertions carry the vertex weights $\sum_v a_v = -n_{\mathrm{ct}}$. Substituting in $-I - n_\partial + \sum_v a_v + 3(|V|-1)$ gives the formula. Blocks and bridges: $3(|V(H)|-1) = \sum_P 3(|V(P)|-1)+3(\rho-1)$ because the block–bridge graph is a tree; a scalar bridge has $d_b = -1-\mathrm{in}(b)$ and a photon bridge carries no derivative in the placement (NF2); a derivative of a vertex of P placed on a bridge counts $+1$ in $\mathrm{out}(P)$ and -1 on the bridge, and every derivative on a bridge is the own derivative of one of its end vertices, each of which lies in a block, whence $\sum_P \mathrm{out}(P) \geq \sum_b \mathrm{in}(b)$ and $D_0(H) \geq -\rho+2(\rho-1) = \rho-2$. The formula applied to a block with all derivatives internal gives $\omega(P) = \frac{1}{2}n_e + n_s + n_\lambda + \frac{1}{2}E - 3$ for a block without insertions, and a counterterm insertion alone is a block with $E=2$ and $\omega=-1$; hence $\omega(P)\geq -1$ with the stated equality cases ($E=2$: $(n_e,n_s,n_\lambda)=(2,0,0)$; $E=3$: a single current vertex, whose derivative is necessarily outside), and the enumeration of the pairs (P,P') with $\omega=-1$ and $\mathrm{in}(b)\leq 2$ gives the $\rho=2$ list; $\rho\geq 3$ is impossible. The connected sets of bare degree 0 with all derivatives internal are those with $\frac{1}{2}n_e + n_s + n_\lambda + \frac{1}{2}E = 3$: $E=2$ with $(4,0,0)$, $(0,0,2)$, $(2,0,1)$, $(0,2,0)$, $(2,1,0)$, $(0,1,1)$; $E=3$ with $(3,0,0)$, $(1,1,0)$, $(1,0,1)$; $E=4$ with $(2,0,0)$, which is L24 (two current vertices joined by one scalar line carrying both derivatives; joined by a photon line both derivatives are outside and the degree is 2). Integrality: $E \equiv n_e \pmod 2$. □

Corollary 17.46 (placement at the terminals; the derivative count)**.** (i) *In a connected set with $D_0(H)\leq 0$ no vertex of H places its own derivative outside H.* (ii) *After the rule of Definition 17.47 every line carries at most one lattice derivative at each end (its end vertex's own derivative or one transferred derivative, never both), hence at most two in all. The leg of a compound carries at most one as formed; when the compound is the differenced factor of (R2b) at one of its outer vertices, its leg carries the difference (Lemma 17.54(iv)) and then at most*

two; the middle line of a double compound carries none; a photon line carries only transferred derivatives, at most one per end; a contour line receives at most one, at its bond end.

Proof. (i) With $x \geq 1$ the formula gives $D_0(H) \geq -1 + x \geq 0$, with equality only if the set with internal placement is a unit; but a unit's vertices have their derivatives inside by (NF2). (ii) Own derivatives: a current vertex has one, placed on one of its two scalar lines. Transferred derivatives land on a line l at a vertex w of the component H being processed, $l \notin H$ later than H, $D(H) = 0$ at formation, hence $D_0(H) \leq 0$ and, by (i), w's own derivative is inside H: l carries no own derivative of w. A second transfer onto l at w would come from a component $H' \ni w$ formed before l enters the segment; $H' \supseteq H$ (nested initial segments), so $D(H') \geq D_0(H') + 1 \geq 1$ unless H' is a bare unit, which is never a component (Lemma 17.48(b)); so H' is not processed. The leg of a compound is attached to the unit at a vertex whose own derivative is internal and at which no rule acts (the unit's lines are not elements before the compound enters, and afterwards they are inside it), so the leg carries at most one derivative, at its far end; the middle line of a double compound has two such ends. Photon lines carry no own derivative (Lemma 17.44). For contour lines see (iii) of the rule. □

Definition 17.47 (normal form and the subtraction rule)**.** In each flow, in its own frame, before any element is mixed:

(NF1) Tadpoles and their counterterm shares cancel exactly per slice (TAD): they are dropped.

(NF2) At every vertex of a unit (SE, VP, T, SC) whose derivative sits on a leg, the derivative is written on the internal scalar line by Lemma 17.44; the remainder term, in which the derivative acts on the vertex's photon field, is the unit's own photon line, so it stays inside the unit (for SE the kernel is the sum of the pieces $\partial\partial C \cdot D$, $\partial C \cdot \partial D$, $C \cdot \partial\partial D$, each with $\sum_l d_l = -4$). After this a unit's leg carries at most the far vertex's derivative.

(NF3) The seagull and derivative tadpoles of the order-e^2 vacuum polarisation are eliminated through the exact Ward sum of Lemma 17.51; what remains at a vacuum-polarisation unit is the bubble for every slice assignment plus one common vertex weight.

(NF4) The telescoping is done per ordering class of [3, (2.7)]. The two classes that differ only by swapping the two internal lines of a vacuum-polarisation unit are merged before telescoping (identical element sets; Lemma 17.42 is applied per class in absolute value, a factor 2 per such unit), so a vacuum-polarisation slice assignment is the unordered pair $\{j_1, j_2\}$ and its kernel the line-symmetrised one. The two Wick contractions of a self-energy between the same two legs lie in one class and are one element (their sum is the symmetrised kernel of Lemma 17.50).

Fix an ordering of the scale-summed elements and process the initial segments $G_1 \subset G_2 \subset \dots$. Let $\sigma_{\max}(U)$ be the coarsest scale of a unit U.

(R1) *Compounds.* A unit U (SE, VP; order two) or a degree-zero one-loop piece (T, SC; order one) one of whose vertices y has all its non-U elements later than $\sigma_{\max}(U)$ (an external field counts as later) is bundled into the *compound* (U, F_y): the own-frame Taylor subtraction, about U's other vertex x (in leg-site coordinates, the vertex sites shifted by the bonds of the currents), of the product F_y of the elements at y, taken on the earliest line f of F_y (or on the external if F_y has no line), to second order for SE, VP and to first order for T, SC; one leg per unit, preferring a derivative-free, unclaimed leg. In the reduced ordering the compound is one element at the position of f with the vertex set $V(U) \cup V(f)$ and the line dimensions of U and f: $D(\{(U, f)\}) = 2 + d_f \in \{1, 0\}$ (SE, VP), $= 3 + d_f \geq 1$ (T, SC); with an external leg $D = \alpha$ resp. 1 (Lemma 17.54). Two units forced onto the same leg l form the *double compound* (U_1, l, U_2), one element at l's position with $D = D_0(U_1 \cup l \cup U_2) = 0$. The local terms of the subtraction are exact and taken in the flow's own frame: for SE the zeroth-order term $c_\sigma f(x)g(x)$ cancels the co-switched counterterm share of the same σ ([13, (3.33)] through order four; Lemma 17.30) and the first-order term vanishes by Lemma 17.50; for VP the zeroth-order term enters the Ward sum of Lemma 17.51 and the first-order term vanishes by Lemma 17.50; for T the local term is $f(x) \sum_z T''(x, z) g(z)$ with the two-point kernel $T''(x, z) := \sum_y T(x; y, z)$ of dimension -3, a degree-zero line

processed by (R2); for SC the local term vanishes or carries an explicit factor η' (the bond shift). A unit none of whose vertices qualifies (both legs earlier than $\sigma_{\max}$, the *split* unit) is not an element: its lines are ordinary elements and its counterterm share stands alone as a loop line of dimension -1 at scale $\sigma_{\max}$ in a separate monomial.

(R2) *Components of degree zero.* Whenever a newly formed connected component H of the current initial segment has $D(H) = 0$ (degrees after the earlier transfers), apply the first available clause.

(a) If H contains a bare current vertex w, or a vertex w of a split unit, whose derivative sits on a line of H, whose other scalar field is not in H and whose photon field is an external field or a photon line not in H: rewrite the vertex by Lemma 17.44. This gives two monomials, (i) with the derivative on the other scalar field and (ii) with ∂^* on the photon field; in both $D(H)$ rises to 1, the recipient (later than H, since H is a component) changing $d \mapsto d-1$, or at no cost if it is an external field; the term with $\partial^{\eta'} g$ carries an explicit η', a vertex weight 1. Each vertex is rewritten at most once. If w's photon field is a line inside H, the clause is not available (the move would leave $D(H) = 0$): this happens exactly for the outer vertices of the two-point structures W and for the leg-carrying vertices of a triangle, which go to (b) and (R1).

(b) Else, if H is a two-point structure (exactly two vertices x, t of H carry elements outside H or externals): the first-order subtraction at the terminal t about x (either terminal may be chosen),

$$\textstyle\sum_t H(x,t)G(t) = c\,G(x) + \sum_t H(x,t)[G(t) - G(x)], \qquad c := \sum_t H(x,t),$$

G the product of the later elements at t, Leibniz-expanded so that exactly one factor is differenced. c is a common number of both flows (Lemma 17.43), bounded by Lemma 17.42 applied per flow to H with one vertex fixed ($|\sum_\sigma c_\sigma| \le C(1+r)^{n_0+1}$); for the sunset and the scalar e^4 structures it is 0 exactly with the co-switched order-four counterterm share, for the e^4 vacuum polarisation it is $c\,\delta_{\mu\nu}$, for a triangle kernel T'' it is 0 exactly (the summand is odd under the lattice inversion through a site: the current on a bond goes to minus the current on the inverted bond, the propagators are even and the three lines are distinguishable), and for a double compound standing alone it is 0 exactly (translation invariance of the outer unit's kernel). $D(H)$ rises to 1 (the factor $|t-x|$ absorbed on a line of H); the differenced factor changes $d \mapsto d-1$ if it is a line (for a compound, its leg, Lemma 17.54(iv)), at no cost if it is an external, a coarse element or a vertex function; the Y-displacement $|\tau t - \tau x - (t-x)| \le 2c_\tau \eta'$ against the Y-gradient is an explicit-η' vertex weight, a degree-zero component with gain $\eta' \ge \eta'^{\theta}$, counted among the logarithms.

(c) Else H has at most one vertex with outside elements or externals and is a common number (Lemma 17.43 with $a = 0$; both flows evaluate the same closed lattice sum), bounded per flow by Lemma 17.42: a vertex weight 0, never a switched element.

(R3) *Local-term monomials.* The monomials produced by the zeroth-order terms of (R1) and (R2b) (the unit or structure collapsed to a vertex of weight 0, or -1 for the vacuum-polarisation leftover of Lemma 17.51) are processed by the same rules; their degree-zero components are L24-type lines, the weight classes of Lemma 17.45, and, when a triangle inside a W is collapsed, the kernels $\{\partial C, D\}$ and $\{C, \partial C\}$ between the collapsed point and the other terminal, all handled by (R2b) with common numbers.

(iii) At $e > 0$ the recipient of (R2a)(ii) may be a contour line [13, (3.27)] whose bond end sits at w; it receives this one derivative at its bond end and no other, since its contour end is the site of a contour vertex, which carries the weight $\ge \frac{1}{2}$ and so never lies in a degree-zero component. Contour vertices are never recipients of any clause, for the same reason, and the constant-background phases [13, (3.46)] are not photon fields. The lifted external background at w is an external field with the gradient bound $\le Cp_r$ (Lemma 17.40(0)(R)).

The rule is Bałaban's own device for the graph (2.4), "the first few $G_1, G_2^{(2)}, \dots$ are graphs (2.4)... replaced by graphs with degree $+1$" by the integration by parts [3, (2.8)], applied to

every degree-zero component that forms in an initial segment, with composite lines (compounds) allowed and with the first-order subtraction (R2b) for the components without a usable bare vertex. Two features are forced by the two flows: the units are bundled with a leg as one element, so that no line of a unit is ever processed alone inside an isolated unit (processing the lone $\partial\partial C$ of a self-energy with both legs later would lose the gain, since the Taylor order at the slot would be fixed before the second unit line enters), and the local terms are common numbers of both flows, so that no first moment of a flow-dependent object is ever mixed between the frames.

Lemma 17.48 (degree bookkeeping)**.** *Fix an all-C component G of a monomial, a derivative placement, an ordering of its scale-summed elements, and apply (NF1)–(NF4) and (R1)–(R3). Then in the reduced ordering every connected component H of every initial segment satisfies*

$$D(H) \geq 1, \quad \text{or} \quad D(H) = n + m\alpha \ (n \geq 0,\ m \geq 1;\ H \text{ contains a compound on an external leg}), \quad \text{or} \quad H \text{ is an}$$

(a Y-displacement term of (R2b) or a $\partial^{\eta'} g$ term of (R2a), with $D(H) = 1 - \theta$ after the gain $\eta' \geq \eta'^{\theta}$ is taken out), with at most $n_0 \leq \bar{n}$ zero-degree components in the per-flow count (local vertices of weight 0, explicit-η' components). In particular $D(H) - \theta \geq \min(\alpha, 1) - \theta > 0$ for $\theta \leq \theta_1$ whenever H contains the weakened element.

Proof. (a) Degrees are a property of the multilinear form and flow-independent: both flows have the same lines, the same placements and the same transfers, since (NF2) and the rule are applied in each flow's own frame to the same combinatorial objects.

(b) *No component has $D < 0$.* By Lemma 17.45 a connected line set with $D_0 < 0$ is a unit U. In the reduced ordering U's lines are either inside a compound, and then U alone is not an element, or U is split, and then both legs f, g of U are earlier than $\sigma_{\max}(U)$: the first initial segment containing both lines of U contains f and g, attached at U's vertices, so the component is $\{f, g, U, \dots\}$ with $D \geq d_f + d_g - 4 + 9 \geq 1$ ($d_f, d_g \geq -2$ by Corollary 17.46(i)); the segment containing only U's first line, the lone $\partial\partial C$ of a split self-energy with $D = 0$ when it precedes both legs, is a degree-zero component covered by (c). Transfers never create $D < 0$: a transfer lands on an element later than its source component H_0 and attached to a vertex of H_0, so every later component containing the recipient contains H_0, where the $+1$ and the -1 cancel (this is [3, (2.8)–(2.9)]: "the degrees of these graphs are $\geq D(G_0)$"); the components containing H_0 but not the recipient rose by 1; no other component changed. Loop closure after a transfer is harmless for the same reason: if the recipient line later closes a loop, the component $H \cup l$ has $D(H \cup l) \geq D_0(H \cup l) \geq -1$, with -1 only for a bare unit; then H is the first line of a split unit and l its other line, the coarsest, so both legs are earlier and attached when l enters, and the component is $\{f, g, U, \dots\}$ with $D \geq 1$ as above.

(c) *Components with $D(H) = 0$ at formation.* Since every transfer inside H with recipient inside H is neutral and one with recipient outside is $+1$, $D(H) \geq D_0(H)$, so $D_0(H) \in \{-1, 0\}$; -1 is excluded by (b), so H's line set is in the list of Lemma 17.45 (compounds expanded; for a derived structure the degree is that of its line set). Case by case:

- $H = (U, \partial f)$ with the far vertex w of f bare, or $H =$ L24, or $H = \{\partial f, \mathrm{ct}_\sigma\}$ (a split form), or $H = (U, \partial f)$ with the derivative transferred onto f earlier: a bare vertex $w \in H$ has its derivative on a line of H, its other scalar field outside H and its photon field outside H (external, or a photon line later than H); (R2a) raises D to 1. The lone $\partial\partial C$ of a split self-energy: (R2a) at a vertex of the split unit (its photon is the unit's D line, outside the one-line component), or, equivalently, (R2b) with $c = \sum_y \partial\partial C(x, y) = 0$ exactly (a lattice total derivative), the recipient being the leg at a terminal, which precedes the second unit line in the split regime; then $\{\partial\partial C^{+1}, \nabla f\}$ has $D = 2$ and $\{\partial\partial C^{+1}, \nabla f, D\}$ has $D = 1$.
- $H = T''$ (the triangle's two-point local kernel): the terminals' photon field is the triangle's own D line, inside T'', so (R2a) is not available; (R2b) applies with $c = 0$ exactly (inversion oddness); $D \mapsto 1$.
- $H =$ W, scalar (outer current vertices bare, derivatives internal): an outer vertex's photon is a line inside W; (R2b) with $c = 0$ exactly against the co-switched order-four counterterm share; $D \mapsto 1$. $H =$ W, photon, or the double seagull, the sunset, the $e^2\lambda$ members: (R2b) with the common number c; $D \mapsto 1$. When two triangles inside a W are both bundled (the

crossed and vertex-correction graphs: each triangle's outer vertex carries an external leg or a later line, so each qualifies under (R1)), their line sets overlap and the two compounds merge into one element whose line set is all of W: a two-point degree-zero component when it stands alone, handled by (R2b). The collapsed local terms inside W, $\{\partial C, D\}$ and $\{C, \partial C\}$ between the collapsed point and the other terminal, are degree-zero two-point kernels of the SC and VP type: (R2b) with the common number; $D \mapsto 1$.

- Chains U_1–l–U_2, ct–l–U, ct–l–ct with $D = 0$: each U_i is bundled on the chain line l or on its outer leg (if neither vertex of U_i qualifies, its outer leg is earlier and attached, and then H contains it with $D \geq 1$); if U_1 bundles on l and U_2 on its outer leg g, $\{(U_1, l), (U_2, g)\}$ has $D = (2 + d_l) + (2 + d_g) \geq 1$; if both bundle on l, H is the double compound element, which with two units is never alone (its outer legs are earlier) and in chains of three or more units can stand alone (the middle unit forced onto a leg already claimed by a neighbour while both outer legs of the double compound are later): then it is a two-point degree-zero component with no bare vertex and (R2b) applies, with $c = 0$ exactly; $D \mapsto 1$. A counterterm share stands alone only when its unit is split, so $\{f, g, \mathrm{ct}\} \geq 5 + d_f + d_g \geq 1$, and $\{\partial f, \mathrm{ct}\}$ alone never forms since g is earlier too.
- H one-point (both slots at one vertex: the seagull with two external photons and a hanging self-energy; the all-C components of vacuum monomials cut at their Γ-lines): a common number, (R2c).
- The weight classes of the local-term monomials, a weight-(-1) vertex joined by an undifferentiated photon line to a vacuum-polarisation compound whose leg is that line, or to another such vertex ($-1 + (-4 - 1) + 3 \cdot 2 = 0$, $-1 - 1 - 1 + 3 = 0$): two-point structures whose outer vertices carry external or later photon fields; (R2b) with the common number; $D \mapsto 1$.

Nothing else occurs: the list of Lemma 17.45 is exhausted (T and SC are bundled when a vertex qualifies, and otherwise never alone: $\{f_{\mathrm{earlier}}, T\}$ has $D = d_f + 3 \geq 1$).

(d) *Components with $D = \alpha$.* A compound (U, E) on an external leg alone; any component containing it and a line has $D \geq \alpha + 1$.

(e) *Termination and the count.* Each application of (R2) raises one freshly formed component and rewrites one vertex or differences one factor; a vertex is rewritten at most once (after (R2a) at w every later component containing the recipient contains H_0 with $D \geq 1$); (R2b) is applied once per formed two-point component; the local-term monomials have fewer vertices. The per-flow zero-degree components are the weight-0 local vertices and the explicit-η' components, at most one per rule application, so $n_0 \leq \bar{n}$. □

Remark 17.49 (the enumeration). The case analysis of Lemma 17.48(c) is a finite statement for each graph, and an exhaustive enumeration, in exact integer arithmetic, confirms it for every graph with at most two units among the two- and three-point structures through order four (the isolated self-energy and vacuum polarisation, the tree (2.4), the triangle and the seagull–current loop with their leg currents, a unit between two currents, the rainbow, crossed, insertion, vertex-correction and seagull graphs, the chains U–l–U and U–∂l–w–l'–U with bare and unit vertices, the collapsed local-term graphs and the weight classes): for every derivative placement allowed by Lemma 17.44 and every ordering of the scale-summed lines, walking the initial segments and applying the rule as stated (both monomials of (R2a) followed separately), no component of negative degree forms, every degree-zero component falls under one of the clauses, the rule terminates, and no line carries more than two lattice derivatives; the same holds for the chains of three and four units, the rings and the counterterm monomials, the double compound standing alone in a fraction of the orderings of the three-unit chain and being handled by (R2b). The proof for every family of order $\leq \bar{n}$ is Lemma 17.45 together with (b)–(e), which use only the block arithmetic and the nesting of the initial segments. The statement for chains of any number of units is degree arithmetic: a third unit cannot join a set of bare degree ≤ 0 (Lemma 17.45, $\rho \leq 2$), and each further unit enters as a compound at a later leg or inside a component of degree ≥ 1.

Lemma 17.50 (the first moments of the units vanish)**.** *In each flow's own frame:* (i) *for every slice assignment σ the self-energy kernel k_σ, as it enters any family (the sum of the two Wick*

contractions between the same two legs, in leg-site coordinates), has $F_\sigma := \sum_u k_\sigma(u)\, u = 0$*;* (ii) *for every unordered slice assignment* $\sigma = \{j_1, j_2\}$ *of the vacuum polarisation, the line-symmetrised kernel* $\Pi_{\sigma,\mu\nu} := \Pi_{(j_1,j_2),\mu\nu} + \Pi_{(j_2,j_1),\mu\nu}$ *has* $F_{\sigma,\mu\nu\rho} := \sum_y \Pi_{\sigma,\mu\nu}(x,y)(y-x)_\rho = 0$*, in site and in bond-midpoint coordinates alike. (For a labelled assignment* (j_1, j_2) *with* $j_1 \neq j_2$ *the first moment is not zero; this is why (NF4) merges the two classes.)*

Proof. Why it is needed: the compound's expansion $\sum_y k_\sigma(x,y)f(y) = c_\sigma f(x) + F_\sigma \cdot \nabla f(x) + \sum_y k_\sigma(x,y) T_x f(y)$ removes the first-order term. For a slice-line leg a term $F_\sigma \cdot \nabla f(x)$ would be harmless even if $F_\sigma \neq 0$ ($F^Y = RF^X$ by Lemma 17.43). For an external or coarse leg it would not: $F^X \cdot \nabla E^X(x)$ and $F^X \cdot R^{-1} \nabla E^Y(\tau x)$ differ by $F^X \cdot (1 - R^{-1}) \nabla E(x)$ up to small terms, a frame term of order $|F|\|E\|$ with no gain in η'. (i) After (NF2) the self-energy at slice σ is a bilinear form $B_\sigma(\phi,\phi) = \sum_{x',y'} \phi(x') \cdot \tilde{k}_\sigma(x',y') \phi(y')$ in the two leg fields, x', y' the leg sites (vertex sites shifted by the bonds of the currents), with $\tilde{k}_\sigma$ proportional to the identity in the $O(2)$ indices ($q^2 = -1$; no q-part by charge conjugation). In every family the self-energy sits between two leg elements g at x' and f at y' through both Wick contractions, which have identical element sets; their sum is $\sum g(x')[\tilde{k}_\sigma(x',y') + \tilde{k}_\sigma(y',x')] f(y')$. The kernel $k_\sigma := \tilde{k}_\sigma + \tilde{k}_\sigma^T$ is translation invariant and symmetric, so $\kappa_\sigma(u) := k_\sigma(x', x'+u)$ is even and $F_\sigma = 0$, whatever the rewriting of Lemma 17.44 did inside the kernel; the expansion is about the leg site, so no shifted-site moment appears. (ii) The bubble at labelled slices is $\Pi_{(j_1,j_2),\mu\nu}(x,y) \propto C^{(j_1)}(x + e_\mu, y + e_\nu) C^{(j_2)}(x,y) - C^{(j_1)}(x + e_\mu, y) C^{(j_2)}(x, y + e_\nu)$; under $-\mathbf{1} \in O_h$ the two factor positions swap slices, so the labelled kernel is O_h-covariant only up to $j_1 \leftrightarrow j_2$ and the symmetrised one is the O_h-covariant object, a kernel on bonds covariant in midpoint coordinates, $\Pi_\sigma(Rm, Rm') = R \Pi_\sigma(m,m') R^T$ for $R \in O_h$. Its midpoint first moment is an O_h-invariant rank-three tensor, hence 0; its zero moment is $c_\sigma \delta_{\mu\nu}$. In site coordinates $m' - m = (y - x) + (e_\nu - e_\mu)\eta'/2$, so $F_{\sigma,\mu\nu\rho} = 0 - c_\sigma \delta_{\mu\nu} (e_\nu - e_\mu)_\rho \eta'/2 = 0$. □

Lemma 17.51 (the vacuum polarisation)**.** (a) *With every line sliced, the order-*e^2 *photon two-point function at zero momentum vanishes in each flow,* [3, (2.23)–(2.28)] *(valid with free boundary conditions) and* [13, §3.5]*:* $\sum_\sigma c_{\sigma,\mu\nu} + \sum_j c_j^{\mathrm{sg}} \delta_{\mu\nu} + c_{\mu\nu}^{\mathrm{dt}} + c_{\mu\nu}^{\mathrm{top}} = 0$*, where* $c_\sigma := \sum_y \Pi_\sigma(x,y)$ *is the bubble's zero moment,* c_j^{sg} *the seagull tadpole,* c^{dt} *the derivative tadpole and* c^{top} *the top pieces, all common numbers, all sums finite (*$\sum_\sigma |c_\sigma| \leq C\eta'^{-1}$ *at fixed* r*).* (b) *The seagull and derivative tadpoles are single-vertex structures of degree* -1 *and may not appear as elements; (a) replaces them by* $-\sum_\sigma c_\sigma - c^{\mathrm{top}}$*. With the local terms* $+c_\sigma f(x) g(x)$ *of the bundled bubbles ((R1);* f*,* g *the photon legs) the complete local content at a vacuum-polarisation vertex is* $[-\sum_{\sigma \text{ unbundled}} c_\sigma - c^{\mathrm{top}}] f(x) g(x)$*, a slice assignment being unbundled iff* $\sigma_{\max} > J := \max(j_f, j_g)$*, plus the remainders of the bundled bubbles and the unbundled bubbles themselves, which are never alone (*$\{f, \Pi_\sigma\}$ *has* $D = 1$*). Since* $|c_\sigma| \leq C \ell_{\sigma_{\min}} / \ell^2_{\sigma_{\max}}$ *(placement* $(1,1)$*) and* $\leq C \ell^2_{\sigma_{\min}} / \ell^3_{\sigma_{\max}}$ *(placement* $(2,0)$*, using* $\sum_u \partial\partial C^{(j_1)}(u) = 0$*),*

$$\Big| \sum_{\sigma_{\max} > J} c_\sigma \Big| \leq C \sum_{j_2 > J} \ell_{j_2}^{-2} \sum_{j_1 \leq j_2} \ell_{j_1} \leq C \ell_J^{-1}, \qquad |c^{\mathrm{top}}| \leq C :$$

a vertex weight -1 *at scale* $J \geq j(x)$ *plus a weight* 0*. Photon legs carry no own derivative, so* $\{f, \mathrm{weight}\}$ *has* $D = 1$ *and* $\{f, g, \mathrm{weight}\}$ *has* $D = 3$*. The Ward identity is used once, per flow, for the sum over all* σ*; its fine end,* $\sum_{\sigma_{\max} \leq J} c_\sigma$*, of order* η'^{-1}*, is exactly what the bundled remainders carry, and only the coarse end survives as a weight. With external photon legs the leftover is* 0 *exactly.* (c) *The order-*e^4 *vacuum polarisation, and every degree-zero two-point structure handled by (R2b), needs no identity:* $c_{\mu\nu} := \sum_t \Pi_W^{(4)}(x,t)$ *is a common number (Lemma 17.43), isotropic, and* $|\sum_\sigma c_\sigma| \leq C(1+r)^{n_0+1}$ *by Lemma 17.42 applied per flow to* W *with one vertex fixed (all proper components of* W *are already of degree* ≥ 0 *after the rule;* $D(W) = 0$ *gives one logarithm). The local term* $c\, A^F(x) \cdot A^F(x)$ *is a weight-*0 *vertex term common to both flows; its difference is telescoped over the externals.* (d) *In the scalar sector the bundled self-energy's zeroth-order term cancels against* ct_σ *exactly; the sunset and the scalar* e^4 *structures give* $c = 0$ *exactly with the co-switched order-four share.*

Lemma 17.52 (the slice law)**.** *Let $g_j := C^{(j)} - C_c^{(j)}$ with $C_c^{(j)}(v) := \int_{\ell_j^2}^{\ell_{j+1}^2} e^{-m_r^2 s}(4\pi s)^{-3/2} e^{-|v|^2/4s} ds$ the isotropic continuum slice, $3 \le j \le r-1$. Then for every $n \ge 0$ lattice derivatives*

$$|\partial^n g_j(u)| \le C_n (\eta'/\ell_j)^{1/2^n} \ell_j^{-1-n} e^{-\delta_n |u|/\ell_j},$$

and consequently, for matched points $|u' - R^{-1}u| \le 4c_\tau \eta'$ (bond shifts included),

$$|\partial^n C^{(j)}(u) - R^{\otimes n} \partial^n C^{(j)}(u')| \le C (\eta'/\ell_j)^{1/2^n} \ell_j^{-1-n} e^{-\delta_n |u|/(2\ell_j)},$$

the same for the photon slices and, with ℓ_j replaced by 1 and the mass decay, for the top piece. For $j \le 2$ there is no gain and the factor $(\eta'/\ell_j)^\theta$ is paid by a degree loss θ (Corollary 17.36(a), King's device at the finest slices [13, p. 675]*). Constants are uniform in $m_r \in (0,1]$ and r.*

Proof. Value clause $n = 0$: the coordinates of the continuous-time walk are independent one-dimensional walks [15, Prop. 1.2.2], and the local limit law [15, Thm. 2.5.6] gives, for $|u| \le s/(8C_0\eta')$ and $s \ge \ell_3^2$, $|\eta'^{-3} P_s(u) - (4\pi s)^{-3/2} e^{-|u|^2/4s}| \le c(\eta'/\sqrt{s})(4\pi s)^{-3/2} e^{-|u|^2/8s}$; integrated over $s \in [\ell_j^2, \ell_{j+1}^2]$ this is $\le c(\eta'/\ell_j)\ell_j^{-1} e^{-\delta|u|/\ell_j}$, and beyond that range both kernels are bounded by their Gaussian and Poisson tails, $\le C_N(\eta'/\ell_j)^N e^{-\delta|u|/\ell_j}$; the geometric mean with the decay of each kernel halves the rate and keeps the factor. Derivatives by interpolation: for a lattice function h and $\rho = N\eta'$, $\partial_\mu h(u) = [h(u+\rho e_\mu) - h(u)]/\rho - \rho^{-1}\sum_{i=1}^{N-1}(\rho - i\eta')\partial_\mu\partial_\mu h(u + i\eta' e_\mu)\eta'$, so $|\partial_\mu h(u)| \le 2\sup|h|/\rho + \rho \sup|\partial\partial h|$ over the segment. With $h = \partial^n g_j$, the crude bound $|\partial^{n+2} g_j| \le |\partial^{n+2} C^{(j)}| + |\partial^{n+2} C_c^{(j)}| \le C\ell_j^{-3-n} e^{-\delta|u|/\ell_j}$ from (35) and the smoothness of C_c, and $\rho := \ell_j \epsilon_n^{1/2}$ with $\epsilon_n := C_n(\eta'/\ell_j)^{1/2^n}$ (so $\rho \ge \eta'$; round to a multiple of η'), one gets $|\partial^{n+1} g_j| \le C\epsilon_n^{1/2}\ell_j^{-2-n}$, the decay rate kept since the segment has length $\le \ell_j$; induction on n. Matched points: $\partial^n C^{(j)}(u) - R^{\otimes n}\partial^n C^{(j)}(u') = \partial^n g_j(u) + [\partial^n C_c^{(j)}(u) - R^{\otimes n}\partial^n C_c^{(j)}(u')] - R^{\otimes n}\partial^n g_j(u')$; the middle term is the rotation covariance of the isotropic C_c, $\nabla^n C_c(Rv) = R^{\otimes n}\nabla^n C_c(v)$, plus the displacement $|R^{-1}u - u'| \le 4c_\tau\eta'$ against $|\nabla^{n+1} C_c| \le C\ell_j^{-2-n}$, and the lattice-difference-versus-derivative error of C_c, $\le C\eta'\ell_j^{-2-n}$; both are inside the claim. (The true law is $(\eta'/\ell_j)^2$ for the values and η'/ℓ_j for first differences; the exponents $1/2^n$ are far from sharp and suffice, since every gain is compared with $\theta_1 \le \frac{1}{16}$.) □

Lemma 17.53 (gradients of externals and coarse elements)**.** *For $E \in \{H^F\phi_r, H_A^F A_r,$ their covariant derivatives, the background$\}$ and for $\Gamma^F(\cdot, z)$, $H^F(\cdot, z)$: $|E^X(x) - E^Y(\tau x)| \le C\eta^{1/2}\|E\|$ (Lemma 17.32, Lemma 17.34(iii)), and both are restrictions of $C^{1,\alpha}$ functions g^X, g^Y on $\mathbb{R}^3$ with $\|\nabla g^F\|_\infty \le C_\nabla \|E\|$ and $[\nabla g^F]_\alpha \le C\|E\|$ ([13, Thm. 3.3, (3.8)] for the standard flow, Proposition 17.72(a) for the hybrid flow; for the lifts, the Fourier representation of Lemma 17.32 with smooth symbols). Then, with $\kappa := \alpha/(2(1+\alpha))$, $|\nabla g^X(x) - \nabla g^Y(x)| \le C\eta^\kappa \|E\|$ for all $x \in \mathbb{R}^3$, in particular at x and at τx. Consequently the second-order own-frame Taylor remainders $T_x^F E^F(y) := E^F(y) - E^F(x) - (y-x)\cdot\nabla g^F(x)$ (in the Y frame at τx, τy) differ by*

$$|T_x^X E^X(y) - T_{\tau x}^Y E^Y(\tau y)| \le C\|E\| \min\{|u|^{1+\alpha} + \eta'^{1+\alpha},\ \eta^{1/2} + |u|\eta^\kappa + \eta'\}, \qquad u = y - x.$$

For a fine slice line $f = \partial^a C^{(j_f)}(\cdot - z)$ in place of E the same holds with $\eta^{1/2}$, η^κ, η' replaced by $(\eta'/\ell_{j_f})^{1/2^a}\ell_{j_f}^{-1-a}$, $(\eta'/\ell_{j_f})^{1/2^{a+1}}\ell_{j_f}^{-2-a}$, $\eta'\ell_{j_f}^{-2-a}$, times $e^{-\delta\,\mathrm{dist}/(2\ell_{j_f})}$, and with the remainder bound $c|u|^{1+\alpha}\ell_{j_f}^{-2-a-\alpha}$ (Lemma 17.52; no interpolation needed).

Proof. $h := g^X - g^Y$ has $\|h\|_\infty \le C\eta^{1/2}\|E\|$ (at lattice points by the value difference and $|g^Y(\tau x) - g^Y(x)| \le C_\nabla c_\tau \eta'\|E\|$, between them by the Lipschitz bound) and $[\nabla h]_\alpha \le 2C\|E\|$; for a unit vector e and $\rho > 0$, $|e\cdot\nabla h(x)| \le |h(x+\rho e) - h(x)|/\rho + [\nabla h]_\alpha \rho^\alpha \le 2C\eta^{1/2}\|E\|/\rho + 2C\|E\|\rho^\alpha$; choose $\rho = \eta^{1/(2(1+\alpha))}$. The first remainder bound is the sum of the two remainders ($|\tau y - \tau x| \le |u| + 2c_\tau\eta'$); the second is the value law at y and x, the gradient law at x and the displacement $|(\tau y - \tau x) - (y - x)| \le 2c_\tau\eta'$ against $\|\nabla g^Y\|$. □

Lemma 17.54 (the compound as a weakened element)**.** *Let $B = (U, f)$ be a compound of (R1): U a self-energy or vacuum polarisation with slice assignment σ between the leg sites*

x and y, placement-summed and symmetrised kernel $k_\sigma(x,y) = \kappa_\sigma(x-y)$ with $|\kappa_\sigma(u)| \le C\ell_{j_1}^{d_1}\ell_{j_2}^{d_2}e^{-\delta|u|/\ell_{\sigma_{\min}}}$, $d_1+d_2=-4$ (the product of two slices decays at the finest line); $f = \partial^a C^{(j_f)}(y-w)$, $a\in\{0,1,2\}$ ($a\le 1$ for a compound as formed, Corollary 17.46(ii); $a=2$ when the compound is the differenced factor of (R2b), see (iv)), $\sigma_{\max}\le j_f$; K_σ^F the own-frame second-order remainder sums $K_\sigma^X(x,w) := \sum_y \kappa_\sigma(x-y)T_x^X f^X(y)$, $K_\sigma^Y(\tau x,\tau w) := \sum_{y'\in\Lambda_Y}\kappa_\sigma(R^{-1}(\tau x - y'))T_{\tau x}^Y f^Y(y')$, with lattice differences in the Taylor polynomials. Put $s := \alpha/(1+\alpha)$, $\kappa := \alpha/(2(1+\alpha))$.

(i) *$|\sum_{\sigma:\sigma_{\max}\le j_f} K_\sigma(x,w)| \le C\ell_{j_f}^{-2-a}e^{-\delta|x-w|/(4\ell_{j_f})}$ per flow: the compound is a line of dimension $-2-a$ at scale j_f. With an external or coarse leg E: $|\sum_{\sigma_{\max}\le j} K_\sigma(x)| \le C\ell_j^\alpha\|E\|$, a vertex weight α at scale $\sigma_{\max}$, summed geometrically.*

(ii) *$\sum_{\sigma_{\max}\le j_f}|K_\sigma^X(x,w) - K_\sigma^Y(\tau x,\tau w)| \le C(1+r)\eta'^{\theta_U}\ell_{j_f}^{-2-a-\theta_U}e^{-\delta|x-w|/(4\ell_{j_f})}$ with $\theta_U := \min\{\theta, s/2^{a+1}\}$; with an external or coarse leg $\le C(1+r)\eta'^{\theta_U}\|E\|$ with $\theta_U := \min\{\theta,\kappa s\}$. No frame vector appears.*

(iii) *For the double compound (U_1,l,U_2) with $l=\partial^a C^{(j_l)}$, $a\in\{0,1\}$: per flow a line of dimension $-3-a$ at scale j_l; the difference gains $(\eta'/\ell_{j_l})^{s/8}$, with one logarithm per unit. The same holds when l is a coarse element Γ^F: per flow $|\sum_{\sigma_1,\sigma_2}(U_1,\Gamma,U_2)(x,w)| \le Ce^{-m_r|x-w|/2}$ (dimension 0 at the top scale), and the difference is $\le C(1+r)^2\eta^{\kappa s}e^{-m_r|x-w|/4}$.*

(iv) *First-order forms (the compounds (T,f), (SC,f) and the brackets of (R2b) whose differenced factor is a line carrying $a\le 2$ derivatives): gain $(\eta'/\ell_{j_f})^{1/2^{a+2}} \ge (\eta'/\ell_{j_f})^{1/16}$ (slice leg) resp. $\eta^{1/4}$ (external leg), with one logarithm. When the differenced factor of (R2b) is a compound (U,f), the bracket is bounded by (i)–(ii) with $a+1\le 2$ in place of a: per flow $|t-x|$ times a line of dimension $-3-a$ at scale j_f, and the gain η'^{θ_U} with $\theta_U \ge \min\{\theta, s/8\}$.*

Proof. (i) $\sum_u \eta'^3|\kappa_\sigma(u)||u|^{1+\alpha} \le C\ell_{j_1}^{d_1}\ell_{j_2}^{d_2}\ell_{\sigma_{\min}}^{4+\alpha} \le C\ell_{\sigma_{\max}}^\alpha$ (the sum is dominated by $|u|\sim\ell_{\sigma_{\min}}$, where the coarser line is bounded by $\ell_{\sigma_{\max}}^{d_{\max}}$; more precisely it is $\le C\ell_{\sigma_{\min}}^\alpha(\ell_{\sigma_{\min}}/\ell_{\sigma_{\max}})^{|d_{\max}|}$, geometric in $\sigma_{\min}$ at fixed $\sigma_{\max}$); $|T_x f(y)| \le C\ell_{j_f}^{-2-a-\alpha}|u|^{1+\alpha}$ with the leg's decay $e^{-\delta|y-w|/(2\ell_{j_f})} \le e^{-\delta|x-w|/(4\ell_{j_f})}e^{\delta|u|/(4\ell_{j_f})}$, the last factor absorbed by the unit's decay at $\ell_{\sigma_{\min}}\le\ell_{j_f}$; $\sum_{\sigma_{\max}\le j_f}\sum_{\sigma_{\min}\le\sigma_{\max}}\ell_{\sigma_{\min}}^\alpha(\ell$ $C\ell_{j_f}^\alpha$. External leg: the same with $\|E\|$. (ii) Split $K_\sigma^X - K_\sigma^Y = \sum_y[\kappa_\sigma(x-y) - \kappa_\sigma(R^{-1}(\tau x - \tau y))]T_x^X f^X(y) + \sum_y \kappa_\sigma(R^{-1}(\tau x-\tau y))[T_x^X f^X(y) - T_{\tau x}^Y f^Y(\tau y)] =: A_\sigma + B_\sigma$. A_σ: telescoping the product of the unit's lines with Lemma 17.52 ($n\le 2$) for the differenced line and (35) for the other gives $|\Delta\kappa_\sigma| \le C(\eta'/\ell_{\sigma_{\min}})^{1/4}\ell_{j_1}^{d_1}\ell_{j_2}^{d_2}e^{-\delta|u|/(2\ell_{\sigma_{\min}})}$ (with the 125^θ-form for $\sigma_{\min}\le 2$), so as in (i) $\sum_{\sigma_{\max}\le j_f} A_\sigma \le C\eta'^\theta\ell_{j_f}^{\alpha-\theta}\ell_{j_f}^{-2-a-\alpha}$ for $\theta\le\frac14$, $\theta<\alpha$. B_σ: the bracket is $\le\min\{P,Q\}$ with $P := C\ell_{j_f}^{-2-a-\alpha}(|u|^{1+\alpha}+\eta'^{1+\alpha})$ and, by Lemma 17.53 for a slice leg, $Q := b_0+b_1|u|+b_2$ with $b_0 = C(\eta'/\ell_f)^{1/2^a}\ell_f^{-1-a}$, $b_1 = C(\eta'/\ell_f)^{1/2^{a+1}}\ell_f^{-2-a}$, $b_2 = C\eta'\ell_f^{-2-a}$ ($\ell_f := \ell_{j_f}$). Since $(1+\alpha)(1-s)=1$, $\min\{P,Q\} \le P^{1-s}Q^s \le C\gamma_f\ell_f^{-2-a}(|u|+\eta')(1+|u|/\ell_f)^s$ with $\gamma_f := (\eta'/\ell_f)^{s/2^{a+1}}$ (the ℓ_f-exponent is $-(2+a+\alpha)(1-s)-(1+a)s = -2-a$); then $B_\sigma \le C\gamma_f\ell_f^{-2-a}e^{\cdots}\sum_u\eta'^3|\kappa_\sigma(u)|(|u|+\eta')$, and the first-order sum of a degree-(-1) kernel is $\le C$ per σ, hence $\le C(1+r)$ after the σ-sum: one logarithm. External leg: $b_0 = C\eta^{1/2}\|E\|$, $b_1 = C\eta^\kappa\|E\|$, $b_2 = C\eta'\|E\|$, $\gamma_f = \eta^{\kappa s} \le 5\eta'^{\kappa s}$. (iii) Slice l: $|T_xT_w l(y,z)| \le \ell_l^{-3-a-2\alpha}|u_1|^{1+\alpha}|u_2|^{1+\alpha}$ (mixed second-order lattice remainder, from (35) at order $a+4$) gives, summed, ℓ_l^{-3-a} per flow. Difference: $P = \ell_l^{-3-a-2\alpha}|u_1|^{1+\alpha}|u_2|^{1+\alpha}$ and $Q \le C(\eta'/\ell_l)^{1/8}\ell_l^{-1-a}(1+|u_1|/\ell_l)(1+|u_2|/\ell_l)$, since the differenced factors $l, \nabla l, \nabla\nabla l$ carry $n \le a+2 \le 3$ derivatives (Lemma 17.52, exponent $\frac18$) plus the displacements; $P^{1-s}Q^s \le C(\eta'/\ell_l)^{s/8}\ell_l^{-3-a}|u_1||u_2|(\dots)^s$ as $(3+a+2\alpha)(1-s)+(1+a)s = 3+a$; two first-order sums give $(1+r)^2$. Coarse $l = \Gamma^F = H^F B^F H^{F*}$ (Lemma 17.34): T_xT_w acts on the two H-factors separately, each $C^{1,\alpha}$ in its first variable with $|T_xH^X(\cdot,p)(y)| \le C|y-x|^{1+\alpha}e^{-\delta|x-p|}$, and $|B^F(p,q)| \le Ce^{-\delta|p-q|}$, so $|T_xT_w\Gamma^F(y,z)| \le C|u_1|^{1+\alpha}|u_2|^{1+\alpha}e^{-\delta|x-w|/2}$; the difference uses $|\Gamma^X - \Gamma^Y\circ\tau| \le C\eta^{1/2}$ and Lemma 17.53 for each H-factor, $Q \le C\eta^\kappa(1+|u_1|)(1+|u_2|)$,

whence $P^{1-s}Q^s \le C\eta^{\kappa s}|u_1||u_2|$. (iv) First-order forms: $P = \ell_f^{-2-a}|u|$ (slice; $C\|E\||u|$ external), Q as above with $n \le a+1$ in the gradient; $P^{1/2}Q^{1/2} \le C(\eta'/\ell_f)^{1/2^{a+2}}\ell_f^{-1-a}|u|^{1/2}(\dots)$ resp. $C\eta^{1/4}\|E\||u|^{1/2}(\dots)$, and the first-order sum converges per σ with the $(1+r)$ after the σ-sum. For the brackets of (R2b) the switched factor $\Delta[g(t)-g(x)]$ is $\le |t-x|$ times the slice law for ∇g plus $2c_\tau\eta' \sup|\nabla g^Y|$: the degree-one term with its gain and the explicit-η' term (degree zero, one logarithm). When the differenced factor is a compound (U,f) at an outer vertex $t \in \{x,w\}$, its lattice difference is the compound with the differenced leg: at $t=w$ directly, and at the unit vertex $t=x$ by the translation invariance of κ_σ, since shifting the summation variable gives $K_\sigma(x+\eta' e_\mu,w) - K_\sigma(x,w) = \eta' \sum_y \kappa_\sigma(x-y)\, T_x(\partial_\mu f)(y)$ exactly, the lattice differences commuting (and likewise on Λ_Y with the step $\eta' Re_\mu$). So the difference of the bracket is a sum of $|t-x|/\eta'$ terms $\eta'\,[K_\sigma^X - K_\sigma^Y]$ with the leg $\partial_\mu f$, $a+1 \le 2$, each bounded by (ii) (Lemma 17.53 for the slice leg with $a+1$, Lemma 17.52 at $n \le 3$; the unit lines keep $n \le 2$): per flow $|t-x|\,\ell_{j_f}^{-3-a}$, gain η'^{θ_U} with $\theta_U = \min\{\theta, s/2^{a+2}\} \ge \min\{\theta, s/8\}$, one logarithm; the Y-displacement term is explicit-η' as before. □

The telescoping. After (NF1)–(NF4) and (R1)–(R3) in each flow's own frame, $E^F(\mathcal{R})$ for one ordering class is a finite sum of multilinear expressions in the elements: slice lines and top pieces (with at most two lattice derivatives, Corollary 17.46), compounds (U,f) and (U_1,l,U_2) (own-frame remainder sums, Lemma 17.54), first-order compounds (T,f), (SC,f), the brackets $[G(t)-G(x)]$ of (R2b) with one differenced factor, the coarse elements Γ^F, H^F, B^F, the externals and vertex functions at τ-matched points (shifts included), the contour elements at $e>0$, and the common numbers (the zero moments c, the order-four shares, the vacuum-polarisation leftover weight, the one-point structures), which are never switched. The two flows' expressions have the same combinatorics. Order the elements consistently (compounds first, then lines, coarse elements, externals, contour elements) and telescope:

$$E^X(\mathcal{R}) - E^Y(\mathcal{R}) = \sum_i T_i, \qquad T_i := \big[e_1,\dots,e_{i-1} \text{ in the } Y\circ\tau \text{ version};\ e_i \to e_i^X - e_i^Y\circ\tau;\ e_{i+1},\dots \text{ in the } X \text{ versio}$$

an identity, every term being the same multilinear form with elements substituted, the co-switched pairs (a self-energy and its counterterm share, the two Wick contractions of Lemma 17.50, a scalar W and its order-four share) switched together; at most $n_{\mathcal{R}}$ terms per class, and finitely many classes per family. Compounds are switched first so that a unit and its leg are compared as a whole and no mixed-frame first moment is ever written; externals are switched late so that the only frame-sensitive quantity left is the difference of the two flows' own-frame remainders, which Lemma 17.53 bounds. Identities (the counterterm cancellations, the Ward sum, the moment subtractions) are used only at the two pure endpoints, in each flow's own frame; the mixed terms are bounded in absolute value.

Lemma 17.55 (every telescoping term is small)**.** *For every ordering class and every term T_i, with $\theta \le \theta_1 = \alpha^2/(4(1+\alpha)^2)$,*

$$|T_i| \le C(1+k)^p \eta'^{\theta} \cdot [\textit{the per-flow bound } [3,\ (1.33)] \textit{ of the monomial}], \qquad p = p(\bar n),$$

with the prefactor $p_r^2\delta_r^{-1/2}$ when e_i is a contour element.

Proof. Apply Lemma 17.42 to $|T_i|$, the absolute value of a multilinear form whose degrees are those of Lemma 17.48(a). (i) *Per-flow bounds* of every element in X-coordinates, whichever flow it comes from: lines by (35), with $\delta/2$ for the $Y\circ\tau$ versions; compounds by Lemma 17.54(i),(iii), lines of dimension $-2-a$, resp. $-3-a$ (0 at the top scale for a Γ middle), at their leg's scale, or vertex weights α for external legs; first-order compounds by Lemma 17.54(iv) (degree-one objects); the brackets of (R2b) by $|G(t)-G(x)| \le |t-x| \sup|\nabla g|$ plus the explicit-η' displacement term; coarse elements, externals and vertex functions by their table; the common numbers by Lemma 17.43 and Lemma 17.42 per flow ($\le C(1+r)^{n_0+1}$ each; the vacuum-polarisation leftover $\le C\ell_J^{-1}$ as a weight -1); at $e>0$ the contour elements by Proposition 17.72(b), a contour line that received a derivative being a line of dimension -1 at its slice (the contour sum carries the weight η per bond against $\sim \ell_j/\eta$ bonds within the decay range, [13, (3.63)–(3.64)] with the

contour clause of [13, Prop. 3.7]). Decay is re-anchored at merged vertices as in the first step of Lemma 17.42. (ii) *The switched element*, divided by $C\eta'^{\theta}$, obeys the bound of its type weakened by $\ell_j^{-\theta}$ at its scale (or unweakened if it is fixed): a line with $n \le 2$ derivatives by Lemma 17.52, gain $(\eta'/\ell_j)^{1/4} \ge (\eta'/\ell_j)^{\theta}$ (top piece: $\eta'^{1/4}$); a compound by Lemma 17.54(ii),(iii), gain $\eta'^{\theta_U}\ell_{j_f}^{-\theta_U}$ with $\theta_U \ge \theta$; a first-order compound or a bracket by Lemma 17.54(iv), gain $\ge (\eta'/\ell_j)^{1/16}$ resp. $\eta^{1/4}$ (a bracket whose differenced factor is a compound: $\eta'^{\theta_U}\ell_{j_f}^{-\theta_U}$ with $\theta_U \ge \min\{\theta, s/8\}$); a coarse element by $C\eta^{1/2}$; an external by $C\eta^{1/2}\|E\|$, its gradient as a recipient by $C\eta^{\kappa}\|E\|$ (Lemma 17.53); a vertex function by $c_\tau\eta'\|\nabla w\|$, and the $\partial^{\eta'}g$ term of (R2a), of size $O(\eta')$ in both flows, by $O(\eta') \le \eta'^{\theta}\ell_{j(v)}^{1-\theta}$; at $e > 0$ a contour element by Proposition 17.72(b): $(\eta'/\ell_j)^{1/2}$ for contour lines, also with the one lattice derivative a recipient contour line carries, $\eta^{1/2}$ for contour vertices, $e\,p_r\mu_0^{-1}c_\tau\delta_r^{-1/2}\eta'$ for the phases, whence the prefactor; contour vertices are never recipients (Definition 17.47(iii)). For $j \le 2$ there is no gain and the factor is paid by the degree loss θ, as in Corollary 17.36(a). (iii) *Degrees.* By Lemma 17.48 every component of every initial segment of the reduced ordering has $D \ge 1$, or $D \in \alpha + \mathbb{N}_0$, or is an explicit-$\eta'$ component with $D = 1-\theta$ after η'^{θ} is taken out; fixed elements impose no condition. Hence $D(H)-\theta > 0$ for every component containing the switched element, and the scale sums of Lemma 17.42 converge geometrically there. (iv) *Logarithms.* $n_0 \le \bar n$ zero-degree components, $(1+r)$ per compound, $(1+r)^{n_0+1}$ per common number, $(1+r)^2$ per double compound, and the $(1+r)^{n_0}$ of Lemma 17.42 itself; with $\le \bar n$ elements, $p(\bar n) \le 3\bar n + 2$ suffices, and $(1+r) \le (1+k)$. Lemma 17.42 then gives $|T_i| \le C(1+k)^p \prod_l c_l$ with $\prod_l c_l = \eta'^{\theta} \cdot$ [per-flow constants]; the localisation [13, (3.55)] costs at most p_r^3 per coarse line, absorbed in the field norms of step (5) of the proof of Theorem 17.56. □

The exponent θ_1. In every term the switched element carries $\eta'^{\theta'}\ell_j^{-\theta'}$ with its own exponent θ', and Lemma 17.42 at the stage where the component $H \ni e_i$ is shrunk sums $\sum_{j \le j'} \ell_j^{D(H)-\theta}\eta'^{\theta} \le C\ell_{j'}^{D(H)-\theta}\eta'^{\theta}$, geometric because $D(H) - \theta > 0$; the remaining factor is the merged vertex's weight, and since all later components containing it again have $D-\theta > 0$ the factor η'^{θ} survives. (When $D(H) = 0$ the sum $\sum_j (\eta'/\ell_j)^{\theta} = \sum_j 5^{-j\theta}$ is $O(1)$ and the gain is lost; Lemma 17.48 excludes this.) With $s = \alpha/(1+\alpha)$, $\kappa = \alpha/(2(1+\alpha))$, the gains are: a switched line with $n \le 2$ derivatives, $1/2^n \ge \frac14$; the top piece, $\frac14$; a compound with a slice leg carrying $a \le 2$ derivatives ($a = 2$ only for the differenced compound of (R2b)), $s/2^{a+1} \ge s/8$; with an external or coarse leg, κs; the double compound, $s/8$ (slice middle) or κs (coarse middle); the first-order forms, $1/2^{a+2} \ge \frac{1}{16}$ (slice) or $\frac14$ (external); coarse elements, $\frac12$; vertex functions and explicit-η' terms, 1; contour lines, $\frac12$; contour vertices, $\frac12$; phases, 1 with the prefactor. The arithmetic: $\kappa s = \alpha^2/(2(1+\alpha)^2) = 2\theta_1$; $s/4 - \theta_1 = \alpha/(4(1+\alpha)^2) > 0$; $s/8 - \theta_1 = \alpha(1-\alpha)/(8(1+\alpha)^2) > 0$ for $\alpha < 1$; $\frac{1}{16} - \theta_1 = (1-\alpha)(1+3\alpha)/(16(1+\alpha)^2) > 0$; and $\min(\alpha, \frac12) - \theta_1 > 0$ because $\theta_1 < \alpha/4 \le \alpha$ and $\theta_1 < \frac{1}{16}$. So $\theta_1 := \alpha^2/(4(1+\alpha)^2)$ is at most every gain exponent (the weakest two being $s/8$ and $2\theta_1$), and every component containing the switched element keeps the margin $D - \theta_1 \ge \min(\alpha, 1) - \theta_1 > 0$. Nothing is nested: no compound is the leg of another (the double compound is one element, and the recipients of (R2) receive a single gradient), so θ_1 does not depend on $\bar n$. The consumers (Lemma 17.26, Proposition 17.38) use only $\theta_1 > 0$ and a polynomial $(1+k)^p$, so the exact value plays no role; what matters is that the set of element types and derivative counts is finite, that every type has a difference law with some positive exponent, and that the local terms are common numbers.

Proof of Theorem 17.39. Sum Lemma 17.55 over the $\le n_{\mathcal R}$ terms of each ordering class and over the finitely many classes ([3, (2.7)]; their number depends on the family only). The sectors: $e > 0$. The constant-background reduction (Lemma 17.40) is flow-independent algebra; after it the new elements are the contour elements of Proposition 17.72(b). Every connected subgraph containing a contour vertex has degree $\ge \frac12$ ([13, (3.30)]; Lemma 17.40(3)), so no new structure of degree ≤ 0 appears and Lemma 17.48 is unchanged; a contour line may be the recipient of (R2a)(ii) at its bond end, and this is the one-derivative clause of Proposition 17.72(b); the

phases are Lipschitz in the vertex position with the bound $e\,p_r\mu_0^{-1}\delta_r^{-1/2}$, inside the prefactor. e enters only through $e_r^2 = e^2\delta_r$, so the bound holds at every $e > 0$. *The sectors with legs at $A = 0$.* Scalar two-point: orders λ, λ^2, e^2, e^4 with the counterterm cancellations of Lemma 17.51(d), $e^2\lambda$ trivial; the $(2,1)$-sector at order e^3: T and SC as first-order compounds (no Ward identity of the vertex class is used: in the per-flow normal form the identities of this sector are identities on all-C expressions, applied in each flow alone, and a reduction of this sector to a "mixed zero-moment" bound for current–current subgraphs with lines from different flows is not used, the mixed zero moment being a lattice-scale quantity of order e_r^2/η'); the vacuum polarisation at e^2 through Lemma 17.51(a),(b) and at e^4 through (c); L24; everything else of positive degree ([3, Cor. 2.3, (2.17)]). *The vacuum sector.* All-C vacuum monomials cancel against E_1 exactly (Lemma 17.30, [13, (3.35)]); the Γ-monomials, cut at their Γ-lines, leave all-C components with at least two legs, renormalised by the rule with the Γ-lines as legs and recipients (one-point components are common numbers, (R2c)); Lemma 17.55 applies per coarse site, times $|T|\delta_r^{-3}$ sites (Theorem 17.57(e)). *The source vertex.* The h-vertex has degree 2 (Lemma 17.60); the h-families other than $\mathcal{T}_h$ contain no new structure of degree ≤ 0. Hence $|E^X(\mathcal{R}) - E^Y(\mathcal{R})| \le Cn_{\mathcal{R}}\,p_r^2\delta_r^{-1/2}\eta'^{\theta}(1+k)^p \cdot$ [right side of [3, (1.33)]] for $0 < \theta \le \theta_1$, uniformly in r, k, the torus (the tori enter through periodisation, Lemma 17.43) and the masses, and in every term of the telescoping the switched element obeys the bound of its type weakened by θ, which is the statement. □

The comparison of the effective actions at equal spacing.

Theorem 17.56 (equal-spacing comparison, total form)**.** *Let $T \in \{T_1, T_2\}$, G a real smooth P-periodic 2-form, $z \in \mathbb{R}$, and $r = r(k)$ as in Lemma 17.26. Then for $k \ge k_0(zG)$,*

$$\sup_{\operatorname{supp}\chi_r} \big|\hat{s}_T^{(r)}(A_r,\phi_r;zG,0) - S_T^{(r)}(A_r,\phi_r;zG,0)\big| \le \vartheta_k|T|, \qquad \vartheta_k := C\big(5^{-\gamma r}\delta_r^{-\beta}(1+k)^p + \delta_r^{\sigma}\big) \to 0,$$

every c-number included: there is no vacuum constant and no source-linear constant at $h = 0$.

Theorem 17.57 (the vacuum constant)**.** *Let $v_k(T) := S_T^{(r)}(0,0;0) - \hat{s}_T^{(r)}(0,0;0)$. Then:* (a) *$v_k = v_k^{\mathrm{G}} + v_k^{\mathrm{I}}$ with $v^{\mathrm{G}} = \ln N_r^Y - \ln \hat{N}_r + \ln N_r^Y(0) - \ln \hat{N}_r(0)$ and $|v_k^{\mathrm{G}}(T)| \le 12.5\,a\,5^{3k-5r}|T|$;* (b) *$E_1$ cancels exactly in v^{I}, and $v_k^{\mathrm{I}}(T) = |T|\Delta p_k + \varpi_k(T)$ with Δp_k a difference of infinite-lattice vacuum densities, the same for both tori, and $|\varpi_k(T)| \le C\varepsilon^{-q}e^{-c\delta_r^{-1}}$;* (c) *the order-$\lambda$ part obeys $|v_\lambda^{\mathrm{I}}| \le C\lambda\eta^{1/2}\delta_r^{-2}|T| + |\varpi_k|$;* (d) *the order-$\lambda^2e^0$ part obeys $|v_{\lambda^2}^{\mathrm{I}}| \le C\lambda^2\delta_r^{-2}(\varepsilon/\delta_r)^{1/4}\log(1/\varepsilon)^2|T| + |\varpi_k|$;* (e) *$|v_k^{\mathrm{I}}(T)| \le \vartheta_k|T|$. Hence $v_k(T) \to 0$ on each torus.*

Proof. (a) [13, (3.89)] gives $E_0 = \ln Z_r + \ln Z_r(0) + \ln N_r + \ln N_r(0)$; inserted in [13, (3.14)] at $A_r = \phi_r = 0$ this leaves $\ln N_r + \ln N_r(0) + P^{(r)}(0)$ for each flow, with $\ln N_r = -\frac{3}{2}\sum_{p\in T^{(r)*}} \ln(\Delta_{\mu_0}^{(r)}(p)/2\pi)$ and $\ln N_r(0) = -\frac{N}{2}\sum_p \ln(\Delta_m^{(r)}(p)/2\pi)$ over the dual of the common coarse torus. By Proposition 17.31, $|\ln(A^X(p)/A^Y(p))| \le 0.20\,a\eta^2$ at every p and every mass; sum over $|T|\delta_r^{-3}$ momenta and $3 + N = 5$ components with the factor $\frac{1}{2}$; $\eta^2\delta_r^{-3} = 25 \cdot 5^{-2r}5^{3(k-r)}$. (b) $E_1 = |T|e_1(\varepsilon)$ is the same c-number for both lattices and passes through every block-spin step unchanged. The remaining vertices at zero coarse fields are [13, (3.18), (3.19), (3.22), (3.23)] and the products (3.25)×(3.26), (3.26)×(3.26), with fluctuation kernels as lines; each torus kernel is the P-periodisation of its infinite-lattice kernel, decaying at rate δ_*/δ_r in unit scale, with prefactor $\le C\eta'^{-3}$; expanding every torus line as infinite-lattice line plus winding remainder, a graph with $n \le 2\bar{n}$ lines differs from its infinite-lattice value by $\le C^n\eta'^{-3n}|T|\delta_r^{-3}\sum_{\gamma\neq 0} e^{-\delta_*|\gamma|/2} \le C\varepsilon^{-q}e^{-c\delta_r^{-1}}$, the shortest period being $50MJ \ge 50\delta_r^{-1}$ in unit scale. The infinite-lattice values are $|T|p_k^F$ by Y_{k-r}-periodicity of both flows. (c) The order-λe^0 vertices are (3.18) and the order-λ part of (3.19), and $\delta m_1^2 = -4(N+2)\lambda C(0)$ (the vanishing of the one-particle-irreducible two-point function at zero momentum at order λ, [13, (3.33)]). With Gaussian moments, $P^F(0)|_\lambda = \lambda N(N+2)\sum_x[(C(0) - G_T^F(x,x))^2 - C(0)^2]$; $C(0) - G_r^F(x,x) = \Gamma^F(x,x)$ by Lemma 17.34(i); the terms $-C(0)^2$ give equal totals; re-index the Y-sum by τ and use Lemma 17.34(ii),(iii):

$|\Gamma^X(x',x')^2 - \Gamma^Y(\tau x',\tau x')^2| \le 2C_1C_\Gamma\eta^{1/2}\delta_r^{-2}$. (d) With $V_1 := \lambda\sum{:}|\phi|^4{:}_C$ there are no tadpoles at order λ^2; $\frac12\delta m_2^2 = 16(N+2)\lambda^2 c_3$, $c_3 = \sum_y C(y)^3$; writing ${:}F{:}_C = e^{-\frac12\Delta_\Gamma}{:}F{:}_G$ with $\Gamma = C - G$, the order-λ^2e^0 part of v^{I} is the Y-minus-X difference of $N(N+2)\lambda^2[16T_1 - 24T_2 + 16T_3 - 4T_4 - 4(N+2)T_5]$ with $T_1 = \sum_{x,y} C_{xy}^3(\Gamma_{xy} - \Gamma_{xx})$, $T_2 = \sum C_{xy}^2\Gamma_{xy}^2$, $T_3 = \sum C_{xy}\Gamma_{xy}^3$, $T_4 = \sum\Gamma_{xy}^4$, $T_5 = \sum\Gamma_{xx}\Gamma_{yy}(C_{xy} - \Gamma_{xy})^2$, up to $O(\varepsilon^{-q}e^{-c\delta_r^{-1}})$ (Proposition 17.58); the pure-C terms have equal totals. Each difference is bounded, in the three regions $|u| \le \rho := (\varepsilon\delta_r)^{1/2}$, $\rho < |u| \le R_\varepsilon := (40/m)\ln(1/\varepsilon)$ and $|u| > R_\varepsilon$, by the pointwise bounds $|C_{xy}| \le C_0(|u|+\varepsilon)^{-1}$, $|\Gamma_{xy}| \le C_1'\min(\delta_r^{-1}, (|u|+\varepsilon)^{-1})$, $|\Gamma_{xy} - \Gamma_{xx}| \le C_2\delta_r^{-1}(|u|/\delta_r)^{1/2}$, the matched-pair bounds of Corollary 17.37 and Lemma 17.34(iii), and Combes–Thomas decay beyond R_ε; the result is $\le C\delta_r^{-2}(\varepsilon/\delta_r)^{1/4}\ln(1/\varepsilon)^2$ per unit volume. (e) Group the vacuum graphs, with E_1 expanded in each flow's slices (Lemma 17.30), into renormalised vacuum families in per-flow normal form; all lines are $A = 0$ lines on the infinite lattice; the vacuum graphs reduce to families with at least two legs, to which Proposition 17.38 applies per coarse site; multiply by $|T|\delta_r^{-3}$ and use Lemma 17.26. □

Proposition 17.58 (the order-λ^2 identity)**.** *The decomposition of $v^{\mathrm{I}}_{\lambda^2}$ into $T_1,\dots,T_5$ in the proof of Theorem 17.57(d) is an exact identity; in particular the counterterm δm_2^2 enters only through T_1, in the form $\sum C^3[\Gamma(x,y) - \Gamma(x,x)]$.*

Proof. Wick's theorem for the two Gaussians with covariances C and $G = C - \Gamma$ at second order in V_1, with ${:}|\phi|^4{:}_C$ re-expanded in G-normal order: the connected contractions of two quartic vertices with n lines of type C and $4-n$ of type Γ, $n = 0,\dots,4$, give T_4, T_3, T_2, T_1 and the pure-C term, with the combinatorial weights 4, 16, 24, 16; the counterterm $\frac12\delta m_2^2|\phi|^2$ contracted with the quartic vertex gives the $-c_3\sum\Gamma(x,x)$ half of T_1; the two one-loop pieces of the re-expansion contracted together give T_5. The form is the one in which each telescoping term is small (Remark 17.59): with an unexpanded counterterm the intermediate terms would be of order $\lambda^2|T|\delta_r^{-1}$ and small only because $\sum_y \varepsilon^3(C^{X3} - (C^Y\circ\tau)^3) = 0$. □

Remark 17.59 (why manifest forms). The comparison must be applied to families in which every counterterm is expanded in the flow's own lines, and the same applies to Ward-identity cancellations, which hold for pure slice assignments only: King's "replace one by one every factor" [13, p. 665] is correct on such manifest forms only (Lemma 17.30).

Proof of Theorem 17.56. Fix T and work on $\operatorname{supp}\chi_r$, with the bounds [13, (3.2)]: $|A_r| \le p_r(\mu_0\delta_r)^{-1}$, $|\partial A_r| \le p_r$, $|\phi_r| \le p_r(\lambda\delta_r)^{-1/4}$, $|D_{A_r}\phi_r| \le p_r$, $p_r := p(\delta_r)$. Write both effective actions in the form of Definition 17.27, with the curvature source carried by the Cameron–Martin background (Remark 17.24), and compare term by term; every bound is per coarse site, times $|T|\delta_r^{-3}$ sites.

(1) *Photon quadratic form.* Three copies of the scalar coarse operator at mass μ_0, and $|\frac12\langle A_r, (\hat\Delta^{(r)} - \Delta^{(r)})A_r\rangle| \le 0.3a^2\eta^2p_r^2(\mu_0\delta_r)^{-2}|T|\delta_r^{-3}$ by Proposition 17.31 (King's (3.92) at equal spacing).
(2) *Normalisations.* Theorem 17.57(a).
(3) *Scalar quadratic form at $A^{(r)} = 0$.* As (1). Its $A^{(r)}$-dependent part has the form of a graph in $P^{(r)}$ with two external scalar fields [13, p. 669] and is treated with (5).
(4) *The background determinant $B3^F := -\ln[Z_r^F(A^F)Z_r^F(0)^{-1}]$.* By the scale factorisation of the determinant, valid for both flows by Lemma 17.19(iii),(iv), $B3^F = \frac12\sum_{j<r}\sum_x[g_{j,x}^F(A^F) - g_{j,x}^F(0)]$ with $g_{j,x}^F = \operatorname{Tr}_N[\ln K_j^F](x,x)$; the depths $j \le 1$ are bounded per flow (per-site gauge and charge-conjugation parity), at cost $\le CNe^2p_r^2\delta_r^{-2}5^{-r}|T|$; the depths $j \ge 2$ live on the common lattice Y_{k-j} and are compared per site by an exact matched re-indexing, the telescoping of the resolvents along a cascade of exponential weights, a kernel sweep with the kernel bounds of Proposition 17.72(a) and [13, Thm. 3.3], and gradients compared as vectors, giving, for every $s_0 > 0$, $|B3^X - B3^Y| \le C_{s_0}Ne^2p_r^c(1+k)^p\delta_r^{-2}(5^{-\theta_D r} + \delta_r^{s_0})|T| + C\varepsilon^{-q}e^{-c/\delta_r} \le \vartheta_k|T|$, $\theta_D = \alpha/4$, by Lemma 17.26. King's [13, (3.94)–(3.98)] is not used: the per-p middle line of (3.98) is false as printed, its seagull and bubble classes each diverging like h^{-1} and cancelling only between classes.

(5) *The interaction $P^{(r)}$.* For each flow apply King's reduction [13, (3.43)–(3.57)] in the form of Lemma 17.40: the constant-background gauge transformation (Remark 17.18); the localisation in universal-cover form; the expansions (3.47)–(3.54) around $\vartheta A^{(r)} = 0$; the contour phases (Convention 17.17(ii), Lemma 17.23). After the reduction the background sits in the explicit factors of (3.55) and in the special lines δ^{im}, δ^{bg}, and every compared line is an $A = 0$ line of Corollary 17.36; the contour vertices (3.24)–(3.26) and the phases (3.46) are compared by Proposition 17.72(b), with relative gains $\eta^{1/2}$ and η' and the prefactor $p_r^2\delta_r^{-1/2}$; the explicit factors $A_r(w_1)\cdots A_r(w_q)$ are the same coarse field for both flows; the special lines, the terms with a vertex outside the vertex region and the Taylor remainders are bounded per flow by Lemma 17.40(2). Group the graphs into renormalised families in per-flow normal form; this is possible for both flows with the same combinatorics because each flow satisfies the identities used: counterterms (Lemma 17.30), inversion about coarse sites (Convention 17.17(i)), gauge covariance, and [13, (3.87)], an identity for each lattice's own full propagator. Proposition 17.38 bounds each family difference by $Cp_r^2\delta_r^{-1/2}\eta'^{\theta}(1+k)^p$ times [3, (1.33)]; the localisation sums give C per coarse site, the field norms on $\operatorname{supp}\chi_r$ give at most $Cp_r^{\bar n}\delta_r^{-\beta_1}$ with β_1 fixed by [13, (3.2)] and $\bar n$, and the number of families depends on $\bar n$ only.

Summing, $|\hat s^{(r)} - S^{(r)}| \le C(\eta'^{\theta}\delta_r^{-\beta}(1+k)^p + \delta_r^{\sigma})|T|$ with $\eta^2 \le 25\eta'^{\theta}$ and the prefactor $p_r^2\delta_r^{-1/2}$ absorbed in $\delta_r^{-\beta}$ ($\beta \ge \beta_1 + \frac12$ in Lemma 17.26); with Lemma 17.26 this is $\vartheta_k|T| \to 0$. The vacuum c-number is Theorem 17.57(e). On T_1 the Y-flow lives on a torus skew in its own frame; this enters only through periodisation at $A = 0$ and deck descent at $A \neq 0$ (Lemma 17.118). □

Walls: the source vertex.

Lemma 17.60 (the h-census)**.** *In King's degree ([13, (3.30)–(3.31)]; $d = 3$: a scalar end counts $-\frac12$, a differentiated end $-\frac32$, each factor η counts $+1$, $D(G) = \sum_v D_G(v) - 3$) the h-vertex [13, (3.21)] has $D = 3 - \frac12\cdot 2 = 2$ with both legs internal. Inserting one h-vertex into a scalar line of a graph G_0 gives $D(G_0 + h) = D(G_0) + 2$. Field-dependent h-free families have $D \ge -1$ with one exception, the current tadpole (one vertex [13, (3.22)] with $m = 1$ on a scalar loop, the photon leg external), which has $D = -2$: its value at $A = 0$ vanishes ($\operatorname{Tr} q = 0$), and in a background every insertion adds an external photon leg and raises the degree by 1. So every non-vanishing field-dependent h-family has $D \ge 1$. The only h-families with $D \le 0$ are field-independent and linear in h: the free tadpole ($D = -1$; its residue at zero coarse fields, $-N\sum \eta^3 h\,\Gamma^F(x,x)$, is Term 0 of Proposition 17.71), the order-λ figure-eight with its δm_1^2 partner, and the order-e^2 exchange and seagull tadpoles with their δm_2^2 partner ($D = 0$). Call this set $\mathcal{T}_h$. The tree-level term $\delta_r^2\sum\eta^3 h|\phi^{(r)}|^2$ has $D = 0$ but no internal line; it is compared by Lemma 17.32 directly and gives $O(|T|\varepsilon\delta_r^{-2}p_r^2) \to 0$. The $X/\tau Y$ difference of the h-vertex is a smooth vertex function sampled at x' and $\tau x'$ (Lemma 17.29(ii)), an explicit factor η'; $\sum_i \phi_i^2$ is frame-free.*

Proof. Degree arithmetic from the vertex list [13, (3.18)–(3.26)] (the seagull of order two included) and the minimal degree -1 of the field-dependent h-free families read from Bałaban's list [3, (2.4)]; the exhaustive enumeration of the h-linear graphs of order ≤ 2 in the couplings, with the merged counterterm vertices, gives $\mathcal{T}_h$. For a one-particle-reducible family a bridge line adds $-1 + 3 = +2$ to the degree, so its degree is at least the minimum over its one-particle-irreducible parts. □

Lemma 17.61 (the source-linear constant is the same on both tori)**.** *Let $f^F(x)$ be the infinite-lattice one-point density of $\mathcal{T}_h$ along flow F (one h-vertex at x, all lines infinite-lattice kernels); it is Y_{k-r}-periodic. Let $\bar f^F$ be its cell average and $c_k := -(\bar f^X - \bar f^Y)$, a number independent of T. For $T \in \{T_1, T_2\}$ the $\mathcal{T}_h$-part of $\hat s_T - S_T$ at zero coarse fields equals $-c_k\mathsf{S}_{X_k}(h) + \rho_T(h)$ with $|\rho_T(h)| \le C(\varepsilon^{-q}e^{-c\delta_r^{-1}} + C_N\varepsilon^{-q}\delta_r^N)\|h\|_{C^N} \to 0$. Moreover $|c_k| \le C\varepsilon^{-q}$.*

Proof. Periodisation as in Theorem 17.57(b), with one h-vertex summed over T. Write $x = z + t$ with $z \in Y_{k-r}/P$ and t in a cell; the Riemann sum of the smooth periodic h over the coarse lattice is exact to $O(C_N\delta_r^N)$, so $\sum_x \varepsilon^3 h(x) f^F(x) = \bar f^F \int h + O(\|f^F\|_\infty \delta_r^N)$, with $\|f^F\|_\infty \le C\varepsilon^{-q}$

since each line is $\le C\eta'^{-3}$; and $\mathsf{S}_{X_k}(h) = \int h + O(C_N\varepsilon^N)$ (Lemma 17.25). Take N large: with $\delta_r = 5^{-\lfloor \varrho k\rfloor}$, $N > (3+q-3\varrho)/\varrho$ suffices. □

Theorem 17.62 (comparison with a wall)**.** *For $T \in \{T_1, T_2\}$ and $s = (zG, h)$ real, on* supp χ_r*: $\big|\hat{s}_T^{(r)}(\cdot; s) - S_T^{(r)}(\cdot; s) + c_k \mathsf{S}_{X_k}(h)\big| \le \vartheta_k(s)|T| \to 0$, with c_k of Lemma 17.61.*

Proof. Split $\hat{s}(G,h) - S(G,h) = [\hat{s}(G,0) - S(G,0)] + [\Delta_h \hat{s} - \Delta_h S]$. The first bracket is Theorem 17.56. Since [13, (3.17)] depends on h only through the h-vertex, the second bracket consists of the h-families of the two Taylor polynomials plus Taylor remainders bounded per flow by $C\delta_r^{\sigma}|T|$ [13, p. 665]. The $\mathcal{T}_h$ part is $-c_k \mathsf{S}_{X_k}(h) + o(1)$ by Lemma 17.61; its G-dependence through the Cameron–Martin background is an h-family with background insertions, which by gauge invariance of each flow acts through the curvature of the background and so has positive degree. Every other h-family has $D \ge 1$ (Lemma 17.60); group them in per-flow normal form with the same combinatorics in both flows, telescope element by element (lines: Corollary 17.36; external fields: Lemma 17.32; the h-vertex: Lemma 17.29(ii)), apply Proposition 17.38, and sum as in the proof of Theorem 17.56. The exponents of Lemma 17.26 are source-independent ([13, Thm. 3.4]: "C depends on g, h"). □

The source-linear constant vanishes. Throughout, unit scale of level r (Definitions 17.27 and 17.28): fine lattices $\Lambda_X = \eta'\mathbb{Z}^3$, $\Lambda_Y = R^{-1}(\eta'\mathbb{Z}^3)$, $\eta' = 5^{-r}$, $\eta = 5\eta'$; τ the cell-wise bijection with $|\tau x' - x'| \le c_\tau \eta'$; for $F \in \{X, Y\}$ the full propagator C^F (mass $m_r = m\delta_r$; $C^Y = C^X \circ R$), the photon propagator D_C^F (three scalar copies, frame-free), the r-step fluctuation kernels $G^F = \sum_{j<r} S_j^F$ and D^F, and the coarse covariances $\Gamma^F = C^F - G^F = H^F B^F H^{F*}$, $\Gamma^F_{\mathrm{ph}} = D_C^F - D^F$ (Lemma 17.34(i)). Running couplings $\lambda_r = \lambda\delta_r$, $e_r^2 = e^2\delta_r$; the h-vertex at zero coarse fields is $\delta_r^2 \sum_x \eta'^3 h(x)[|\phi'(x)|^2 - C^F(x,x)]$, $N = 2$ real components, with the free-boundary subtraction $C^F(x,x) = C(0)$ common to both flows. A unit-scale density d corresponds to the physical density $\delta_r^{-1} d$. Constants depend on a, m, μ_0, N, τ only. $L_\eta := C\log^3(1/\eta)$.

Kernel toolbox (u the displacement of the two arguments; "both flows" $= F \in \{X, Y\}$). (Q1) $|\nabla^j C(u)| \le C_j(|u| + \eta')^{-1-j} e^{-cm_r|u|}$, $j \le 3$, lattice differences along frame directions; the same for D_C; constants uniform in $\eta' \le 1$ and in the mass ($m_r \in [0, m]$ for C, $\mu_0\delta_r \in [0, \mu_0]$ for D_C; c halved for $j = 3$). This is the standard bound for the massive free lattice Laplacian: scale to $\eta' = 1$, $\mu = m_r\eta'$, symbol $E(q) + \mu^2$ with $E(q) = \sum_\nu 4\sin^2(q_\nu/2)$; by the lattice symmetries let $|u_1| = |u|_\infty$, and shift the q_1-contour by $\mathrm{i}\kappa$, $\kappa = \min(\mu/2, 1)$, where $\mathrm{Re}[E(q + \mathrm{i}\kappa e_1) + \mu^2] \ge \frac{1}{2}(E(q) + \mu^2)$ ($\cosh\kappa - 1 \le \kappa^2$; the vertical sides cancel by periodicity): factor $e^{-\kappa|u_1|}$; on the shifted contour the integrand with j differences obeys $|\partial_{q_1}^b(\cdot)| \le C_b(E + \mu^2)^{(j-2-b)/2} \le C_b'|q|^{j-2-b}$ uniformly in the mass ($j \le 2$), and a cutoff at $|q| \sim |u_1|^{-1}$ with $j+2$ integrations by parts in q_1 on the high part gives $|u_1|^{-1-j}$ ($j = 3$: an extra factor $1 + \mu|u_1|$, absorbed by halving c). Both factors come from one integral, so the product of the two bounds holds. (Q2) Matched-pair differences of C-type kernels: Corollary 17.37 gives $|C^X(y', z') - C^Y(\tau y', \tau z')| \le C_6 \eta'^{1/2}|u|^{-3/2}$ for $|u| \ge 25\eta'$ (the same for D_C); for a kernel carrying $j \le 2$ differences, interpolation of this bound against the per-flow bounds $|\nabla^{j+1} C| \le C(|u| + \eta')^{-2-j}$ (difference quotients at an optimised step) gives the relative gains $(\eta'/|u|)^{1/4}$ ($j = 1$) and $(\eta'/|u|)^{1/8}$ ($j = 2$) on the per-flow size $(|u| + \eta')^{-1-j}$; inside $|u| < 25\eta'$ the difference is bounded by the sum of the two per-flow sizes. (Q3) Fluctuation kernels per flow: $|S_j^F(x,y)| \le C\ell_j^{-1} e^{-\delta|x-y|/\ell_j}$, $|\nabla S_j^F| \le C\ell_j^{-2} e^{-\delta|x-y|/\ell_j}$, $|\nabla S_j^F(x,y) - \nabla S_j^F(x',y)| \le C_{\alpha,r}\ell_j^{-2-\alpha}|x - x'|^\alpha e^{-\delta \min(|x-y|, |x'-y|)/\ell_j}$, $\ell_j = 5^j\eta'$; hybrid flow: Proposition 17.72(a) at $A = 0$, with $C_{\alpha,r} \le C_\alpha \log^{3\alpha}(2 + \ell_j/\eta')$ on slice j; standard flow: [4, Lemmas 2.1, 2.2, 2.4] and [13, Thm. 3.3] at $A = 0$. Hence $|\nabla^j G^F(u)| \le C(|u| + \eta')^{-1-j} e^{-\delta'|u|}$, $j = 0, 1$. (Q4) Coarse covariances per flow: $\sup|\Gamma^F| \le C_1$ (Lemma 17.34(ii)); $|\Gamma^F(x,y)|, |\nabla_y \Gamma^F(x,y)| \le C(1 + |x-y|)^{-1} e^{-cm_r|x-y|}$, uniformly in r and $m_r \in [0, m]$: for $|x-y| \ge 1$ the triangle inequality between (Q1) ($j = 0, 1$) and the fixed-rate decay $|\nabla_y^i G^F(x,y)| \le C_G e^{-\delta'|x-y|}$, $i \le 1$, which (Q3) gives by summing the slices ($\ell_j \le \frac{1}{5}$), with $e^{-\delta'\rho} \le c_{\delta'}\rho^{-2} e^{-(\delta'/2m)m_r\rho}$; for $|x - y| < 1$ it is C_1 and the Lipschitz bound with $C_\nabla' := (2\pi)^{-3} C_B C_h C_{\nabla h} \int_{V_1} |p|^{-2} dp$ from Lemma 17.34(ii) and the bounds $|\hat{h}_x| \le C_h$, $|\nabla_y \hat{h}_y| \le C_{\nabla h}$ of the proof of Lemma 17.64. Hölder $|\Gamma^F(x,y) - \Gamma^F(x,w)| \le C_2|y-w|^{1/2}$ (Lemma 17.34(iv))

and Lipschitz with constant $\sqrt{3}C'_\nabla$. The same for Γ_{ph} (mass $\mu_0\delta_r$). (Q5) Second differences, both flows: $|\nabla\nabla H^F(x,z)| \le L_\eta$ and $|\nabla_y\nabla_y\Gamma^F(x,y)| \le CL_\eta$ (Lemma 17.64). (Q6) Matched-pair differences without derivatives: $|\Gamma^X(x',y') - \Gamma^Y(\tau x',\tau y')| \le C_\Gamma\eta^{1/2}$ (Lemma 17.34(iii)), and $\le C_6\eta'^{1/2}|u|^{-3/2}$ for $|u| \ge 25\eta'$ (Corollary 17.37, for G^F_r and Γ^F separately); $|G^X(x',y') - G^Y(\tau x',\tau y')| \le C_6\eta'^{1/2}|u|^{-3/2}$ ($|u| \ge 25\eta'$) and $\le 2C_0(|u|+\eta')^{-1}$ always; the same for the photon kernels. (Q7) Vertex matching: Lemma 17.29(i)–(iii). (Q8) Summation by parts on the infinite lattice and the lattice Leibniz rule $\nabla_\mu(FG)(y) = (\nabla_\mu F)(y)G(y+\eta' e_\mu) + F(y)(\nabla_\mu G)(y)$.

Lemma 17.63 (the block-average symbol)**.** *The $(r-1)$-step centred cube average is the composition of 5-point averages along each rotated axis, whose symbols telescope, $\prod_j \sin(5^{j+1}t/2)/(5\sin(5^j t/2)) = \sin(5^{r-1}t/2)/(5^{r-1}\sin(t/2))$; hence $u_{r-1}(q) = \prod_\alpha \sin(q_\alpha/2)/(\eta^{-1}\sin(\eta q_\alpha/2))$, $q_\alpha = q\cdot f_\alpha$, on V_η. Consequently $J_0 := \int_{V_\eta}|u_{r-1}(q)|\,dq \le (C\log(1/\eta))^3$ (the L^1 norm of the Dirichlet kernel per axis), $J_1 \le C$, and $\sum_{l_1}|u_{r-1}(p+l_1)| \le C\log^3(1/\eta)$, $\sum_{l_1\neq 0}|u_{r-1}(p+l_1)|/|p+l_1| \le C$, uniformly in $p \in V_1$.*

Proof. The telescoping of the Dirichlet kernels along each axis; the L^1 norm of the Dirichlet kernel of order n is $\asymp \log n$. □

Lemma 17.64 (second differences of the coarse elements, both flows)**.** *$|\nabla\nabla H^F(x,z)| \le CL_\eta$ for every pair of frame directions, $F \in \{X,Y\}$; and, uniformly in $m_r \ge 0$, $|\nabla_y\nabla_y\Gamma^F(x,y)| \le CL_\eta$ and $|\nabla_y\nabla_w\Gamma^F(y,w)| \le C$. The same for Γ_{ph}.*

Proof. By Lemma 17.32, $H^F(x,z) = (2\pi)^{-3}\int_{V_\eta} e^{\mathrm{i}q\cdot(x-z)}u_{r-1}(q)M^F(q)A^F(q)\,dq$ with $M^F(q) = \sum_l e^{\mathrm{i}(q+l)\cdot r_F}u_F(q+l)/\Delta'_F(q+l)$. A second lattice difference in the fine variable multiplies the alias-l summand by $d_F(q+l)^2$, $|d_F(Q)| \le |Q|$; since $\Delta'_F(Q) \ge c|Q|^2$ on every alias cell and $\sum_l u_F(q+l)^2 = 1$ (Proposition 17.31), $|\sum_l d_F^2 u_F/\Delta'_F| \le C$, and the integral against $|u_{r-1}|$ costs $J_0 \le CL_\eta$ (Lemma 17.63). For Γ write $\Gamma(x,y) = (2\pi)^{-3}\int_{V_1} B(p)\hat h_x(p)\overline{\hat h_y(p)}\,dp$, $\hat h_y(p) = \sum_z H(y,z)e^{-\mathrm{i}pz}$, which by Poisson summation equals $A^F(p)\sum_{Q=p+l_1+l} e^{\mathrm{i}Q\cdot y}u_{r-1}(p+l_1)u_F(Q)/\Delta'_F(Q)$; a fine difference in y multiplies the Q-summand by $d_F(Q)$. Hence $|\hat h_y| \le C$, $|\nabla_y\hat h_y| \le C[|p|A^F/\Delta'_F + a\sum_{l_1\neq 0}|u_{r-1}(p+l_1)|/|p+l_1|] \le C$ and $|\nabla_y\nabla_y\hat h_y| \le Ca\sum_{l_1}|u_{r-1}(p+l_1)|\sum_l|u_F| \le CL_\eta$, uniformly in $p \in V_1$ and $m_r \ge 0$ (Lemma 17.63). With $B^F(p) \le C_B|p|^{-2}$ (Lemma 17.34(ii)), integrable on V_1 in three dimensions, both bounds follow. □

Lemma 17.65 (matched-pair comparison of gradient vectors)**.** *Let g on Λ_X and h on Λ_Y satisfy, on a region, $|g(x') - h(\tau x')| \le A$ at matched pairs, $|\nabla g|, |\nabla h| \le C_\nabla$, and the Hölder modulus $|\nabla g(y) - \nabla g(y+v)| \le C_H|v|^\alpha$ (likewise h), $0 < \alpha \le 1$. With $\nabla^X g(y') := \sum_\mu(\nabla_\mu g)(y')e_\mu$ and $\nabla^Y h(\tau y') := \sum_\mu(\nabla^Y_\mu h)(\tau y')f_\mu$ the lattice gradient vectors, for every $\rho \in \eta'\mathbb{N}$*

$$|\nabla^X g(y') - \nabla^Y h(\tau y')| \le 3\big[2A/\rho + 2C_H\rho^\alpha + 4c_\tau C_\nabla\eta'/\rho\big].$$

(a) *For Γ-type kernels, $A = C_\Gamma\eta^{1/2}$ (Q6), $C_H = CL_\eta$ with $\alpha = 1$ (Q5): $\rho = \eta^{1/4}$ gives $|\nabla^X\Gamma^X(x',\cdot)(y') - \nabla^Y\Gamma^Y(\tau x',\cdot)(\tau y')| \le C\eta^{1/4}L_\eta =: A_1$.* (b) *Decaying form: $A = C_6\eta'^{1/2}|u|^{-3/2}$ for $|u| \ge 25\eta'$ gives $\le C\eta'^{1/4}|u|^{-3/4}L_\eta$.* (c) *Fluctuation kernels slice by slice: on slice j, $A_j = C\eta'^{1/2}\ell_j^{-3/2}e^{-\delta|u|/\ell_j}$, $C_{H,j} = C_\alpha\log^{3\alpha}(2+\ell_j/\eta')\,\ell_j^{-2-\alpha}$, $C_{\nabla,j} = C\ell_j^{-2}$; with $\alpha = \frac12$ and $\rho = \ell_j(\eta'/\ell_j)^{1/3}$, $|\nabla^X S^X_j(x',y') - \nabla^Y S^Y_j(\tau x',\tau y')| \le C_\vartheta\ell_j^{-2}(\eta'/\ell_j)^\vartheta e^{-\delta|u|/(2\ell_j)}$ for every $\vartheta < \frac16$ and every u, and, summing over $j < r$, $|\nabla^X G^X(x',y') - \nabla^Y G^Y(\tau x',\tau y')| \le C\eta'^{1/6}\log(1/\eta)\,(|u|+\eta')^{-13/6}e^{-\delta'|u|}$.*

Proof. Lattice Taylor expansion: for $v \in \Lambda_Y - \Lambda_Y$, $h(\tau y'+v) - h(\tau y') = \nabla^Y h(\tau y')\cdot v + \varrho$, $|\varrho| \le 3C_H|v|^{1+\alpha}$ (sum the increments along a lattice path of at most $3|v|/\eta'$ bonds). Take $v_\mu := \tau(y'+\rho e_\mu) - \tau y'$, $|v_\mu - \rho e_\mu| \le 2c_\tau\eta'$; then $e_\mu\cdot\nabla^Y h(\tau y') = \rho^{-1}[h(\tau(y'+\rho e_\mu)) - h(\tau y')] + O(C_H\rho^\alpha) + O(C_\nabla c_\tau\eta'/\rho)$, and likewise $e_\mu\cdot\nabla^X g(y') = \rho^{-1}[g(y'+\rho e_\mu) - g(y')] + O(C_H\rho^\alpha)$. Subtract: the bracket differences are matched-pair differences, each $\le A$; sum over the three components. In (c) the two main terms balance at the stated ρ, giving the exponent $\alpha/(2(1+\alpha)) = \frac16$, the η' term is smaller, and the sum over j is dominated by $\ell_j \sim |u|$. □

Lemma 17.66 (order λ: the h-inserted figure-eight)**.** $f_\lambda^F(x) = s_\lambda\, 4N(N{+}2)\lambda_r \sum_{y\in\Lambda_F} \eta'^3 G^F(x,y)^2\Gamma^F(y,y)$ *with a sign* s_λ *common to both flows, and* $|f_\lambda^X(x') - f_\lambda^Y(\tau x')| \le C\lambda_r\eta^{1/2}$ *(unit scale), i.e.* $\le C\lambda\eta^{1/2}$ *in physical units.*

Proof. The order-λ vertices are [13, (3.18)], $\lambda_r\sum_y \eta'^3|\phi(y)|^4$, and the order-$\lambda$ part of (3.19), $\frac12\delta m_1^2\sum_y\eta'^3|\phi(y)|^2$ with $\delta m_1^2 = -4(N+2)\lambda_r C(0)$. The connected pairing of $|\phi(x)|^2$ with $|\phi(y)|^4$ gives $4N(N+2)G(x,y)^2G(y,y)$, with $|\phi(y)|^2$ it gives $2NG(x,y)^2$; the sum is $4N(N+2)\lambda_r G(x,y)^2[G(y,y)-C(0)] = -4N(N+2)\lambda_r G(x,y)^2\Gamma(y,y)$ (Lemma 17.34(i)). Re-index the Y-sum by τ (Q7) and telescope: the term with $[\Gamma^X(y',y') - \Gamma^Y(\tau y',\tau y')]$ is bounded by $C_\Gamma\eta^{1/2}$ times the convergent sum $\sum_{y'}\eta'^3 G^Y(\tau x',\tau y')^2 \le C$; the term with $[G^X(x',y')^2 - G^Y(\tau x',\tau y')^2]\Gamma^X(y',y')$ is bounded through (Q6) and (Q3): for $|u|<25\eta'$ the at most $25^3\cdot 4\pi/3$ sites give $\le C\eta'$, and for $|u|\ge 25\eta'$, $\sum \eta'^3 C_6\eta'^{1/2}|u|^{-3/2}\cdot 2C|u|^{-1}e^{-\delta'|u|} \le C\eta'^{1/2}$. Convert with $\lambda_r\delta_r^{-1} = \lambda$. □

Lemma 17.67 (order e^2: kernel form)**.** *Let* $\Pi^F[L_c,L_d](y,z)$ *be the one-loop self-energy kernel with scalar loop line* L_c *and photon line* L_d *(exchange part from two current vertices,* [13, (3.22)] *with* $m=1$*; seagull part from the* A^2 *vertex),* $\Pi_C := \Pi[C,D_C]$, $\delta m_2^2 := -\hat\Pi_C(0)$ *(the zero-momentum reading of* [13, (3.33)] *at order* e^2 *in the free-boundary convention, the same number for both flows), and* $\Pi_{C,R}(y,z) := \Pi_C(y,z) + \delta m_2^2\eta'^{-3}\delta_{yz}$, *so that* $\sum_z \eta'^3\Pi_{C,R}(y,z) = 0$. *Then, with a sign* s_e *common to both flows,*

$$f_{e^2}^F(x) = s_e\Big\{\sum_{y,z}\eta'^6 G^F(x,y)\,\Pi^F[G^F,D^F](y,z)\,G^F(z,x) + \delta m_2^2\sum_y \eta'^3 G^F(x,y)^2\Big\},$$

$|\Pi_C(y,z)| \le Ce_r^2(|u|+\eta')^{-4}e^{-cm_r|u|}$ *(Q1), and* $|\Pi_C^X(y',z') - \Pi_C^Y(\tau y',\tau z')| \le Ce_r^2(\eta'/|u|)^{1/8}|u|^{-4}$ *for* $|u|\ge 25\eta'$, $u = y-z$ *(telescoping the two lines, each carrying at most two differences, with (Q2)). In each flow the exchange part equals a finite sum of terms* $s_\iota\sum_{w,w'}\eta'^6 L_a(x,w^+)L_b(x,w'^+)L_c(w^+,w'^+)L_d$ *with* $w^+\in\{w, w+\eta' e_\mu\}$, $L_a,L_b,L_c\in\{G,\nabla_\mu G\}$, $L_d\in\{D,\nabla_\mu D\}$, *exactly two differences per term and at most one per line, both along the same* μ *(a Euclidean inner product of two gradient vectors, Lemma 17.29(iii)); the* A^2 *part is frame-free.*

Proof. Wick algebra at zero coarse fields; the graph with both h-legs at one current vertex vanishes because the current one-point function $\operatorname{Im} G(b_+,b_-)$ is zero. At each current vertex $\sum_s(-1)^s G(x,b_{\rm ext})G_c(b_{\rm loop},\cdot) = \eta'[(\nabla_\mu G)(x,w)G_c(w+\eta' e_\mu,\cdot) - G(x,w+\eta' e_\mu)(\nabla_\mu G_c)(w,\cdot)]$; in the (loop, loop) type the loop line carries two differences, which (Q8) moves onto $G(x,\cdot)$ or $D(\cdot,w')$ with arguments shifted by at most η'. The shifts are frame-dependent ($w+\eta' e_\mu$ against $\tau w+\eta' f_\mu$); the comparison charges one per-flow Taylor step per shifted argument, a relative gain η'/ρ that integrates to $C\eta'(1+\log(1/\eta'))$ and is absorbed. □

Lemma 17.68 (the C/Γ split of the order-e^2 density)**.** *With* $G = C-\Gamma$, $D = D_C - \Gamma_{\rm ph}$ *and* $\Pi_r := \Pi[G,D]$, $f_{e^2}^F = s_e\{Z^F + A^F(x) + B^F(x)\}$ *exactly, where* $Z^F := \sum\eta'^6 C(x,y)\Pi_{C,R}(y,z)C(z,x)$, $A^F(x) := \sum\eta'^6[G(x,y)G(x,z) - C(x,y)C(x,z)]\Pi_{C,R}(y,z)$ *and* $B^F(x) := \sum\eta'^6 G(x,y)G(x,z)[\Pi_r - \Pi_C](y,z)$. (i) Z^F *is independent of* x *and* $Z^X = Z^Y$ *(the all-C part cancels exactly; it contains the logarithm of Remark 16.1).* (ii) $A^F(x) = \sum_{y\ne z}\eta'^6\Pi_C(y,z)[\Gamma(x,y)-\Gamma(x,z)]\{[C(x,y) - C(x,z)] - \frac12[\Gamma(x,y)-\Gamma(x,z)]\}$. (iii) $\Pi_r - \Pi_C = -\Pi[\Gamma,D_C] - \Pi[C,\Gamma_{\rm ph}] + \Pi[\Gamma,\Gamma_{\rm ph}] - \Pi^{\rm sg}[\Gamma_{\rm ph}]$: *every term has at least one internal* Γ*-line. The counterterm never appears alone: it is bound into* $\Pi_{C,R}$*, which occurs only in* Z *and* A. *If King's scheme differs from the zero-momentum reading by a constant* $\varrho = O(e_r^2)$, $f_{e^2}^F$ *acquires* $\varrho\sum_y\eta'^3 G(x,y)^2$, *whose two-flow difference is* $\le C|\varrho|\eta'^{1/2}$ *by the argument of Lemma 17.66.*

Proof. (i) is Lemma 17.30's argument. (ii) $GG - CC = -\Gamma C - C\Gamma + \Gamma\Gamma$; symmetrise in (y,z) using $\Pi(y,z) = \Pi(z,y)$ and the zero sum of $\Pi_{C,R}$ in z. (iii) is multilinearity. □

Lemma 17.69 (the external-Γ part)**.** $|A^F(x)| \le Ce_r^2(1+m_r^{-1})$ *and* $|A^X(x') - A^Y(\tau x')| \le Ce_r^2\big[\eta^{1/4}L_\eta\log(1/\eta') + \eta'^{1/8}(1+\log(1/\eta'))^2(1+m_r^{-1})\big]$.

Proof. With $\rho = |x-y| \le \rho' = |x-z|$ and $u = y - z$: per flow $\mathsf{a} := |\Gamma(x,y) - \Gamma(x,z)| \le C\min(|u|,1)(1+\rho)^{-1}e^{-c'm_r\rho}$ for $|u| \le \rho/2$, $\le 2C(1+\rho)^{-1}e^{-c'm_r\rho}$ and $\le \sqrt{3}C'_\nabla|u|$ always ((Q4), $c' = c/2$); $\mathsf{a}_C := |C(x,y) - C(x,z)| \le C|u|(\rho+\eta')^{-2}$ for $|u| \le \rho/2$, $\le 2C(\rho+\eta')^{-1}$ always (Q1). Difference elements: $\delta\Pi \le Ce_r^2(\eta'/|u|)^{1/8}|u|^{-4}$ (Lemma 17.67); $\delta\mathsf{a} \le C\min\{\eta^{1/2};\ \eta'^{1/2}(\rho^{-3/2} + \rho'^{-3/2})$ $(\rho,\rho' \ge 25\eta');\ A_1|u| + L_\eta|u|^2 + \eta'\}$ ((Q6), Corollary 17.37, Lemma 17.65(a) with the lattice Taylor expansion of $\Gamma(x,\cdot)$ in each flow); $\delta\mathsf{a}_C \le C(\eta'/\rho)^{1/2}(\rho+\eta')^{-1}$ (Q2). Telescope (ii) of Lemma 17.68 and sum over the three regions $|u| \le \rho/2$; $|u| > \rho/2$, $\rho \le 1$; $|u| > \rho/2$, $\rho > 1$, bounding lattice sums by integrals and using $\sum_y \eta'^3(\rho+\eta')^{-2}e^{-cm_r\rho} \le C/m_r$, $\sum_y \eta'^3\rho^{-3}e^{-cm_r\rho} \le C\log(1/m_r)$; the regions $\rho, |u| < 25\eta'$ are bounded by the per-flow sizes. The $\delta\Pi$ terms give $C\eta'^{1/8}(1+\log(1/\eta'))(1+m_r^{-1})$, the $\delta\mathsf{a}$ terms $C[\eta^{1/4}L_\eta\log(1/\eta') + \eta'^{1/2}\log(1/\eta')m_r^{-1/2}]$, the $\delta\mathsf{a}_C$ terms $C\eta'^{1/2}(1+\log(1/\eta'))^2$, and $\log(1/m_r) \le \log(1/\eta')$ absorbs the remaining logarithms. □

Lemma 17.70 (the internal-Γ part)**.** *$|B^F(x)| \le Ce_r^2$ and $|B^X(x') - B^Y(\tau x')| \le Ce_r^2\eta'^{1/6}(1+\log(1/\eta'))^2$.*

Proof. By Lemmas 17.67 and 17.68(iii), B^F is a finite sum of terms with externals $L_a, L_b \in \{G, \nabla G\}$, an internal pair with at least one Γ-type line and at most one difference per line, plus the A^2 terms with the photon tadpole $\Gamma_{\rm ph}(w,w)$. Per-flow sizes: $|L_a| \le C(\rho+\eta')^{-1-j_a}e^{-\delta'\rho}$ (Q3); a C-type internal line with j differences $\le C(|u|+\eta')^{-1-j}$, a Γ-type one $\le C(1+|u|)^{-1}$ (Q1, Q4). The total scaling dimension is $6 - (1+j_a) - (1+j_b) - (1+j_C) \ge 1 > 0$ and every sub-region converges, whence the size bound; the region where ρ, ρ' or $|u|$ is $\le M\eta'$ contributes $\le CM\eta'(1+\log(1/\eta'))$. Telescope over the lines. Outside that region the difference elements are $\delta G \le (\eta'/\rho)^{1/2}C\rho^{-1}$ (Q6), $\delta\nabla G \le (\eta'/\rho)^{1/6}\log(1/\eta)\,C\rho^{-2}$ (Lemma 17.65(c)), C-type internal $\le (\eta'/|u|)^{1/2^{j+1}}C|u|^{-1-j}$, $j \le 1$ (Q2), Γ-type internal $\delta\Gamma \le C\eta^{1/2}$, $\delta\nabla\Gamma \le A_1$ (Lemma 17.65(a)). The gains $(\eta'/\rho)^\theta$ are integrated against the size integrand region by region: $\int \rho^2\rho^{-1}(\eta'/\rho)^{1/2}d\rho \le C\eta'^{1/2}$, $\int \rho^2\rho^{-2}(\eta'/\rho)^{1/6}\log(1/\eta)\,d\rho \le C\eta'^{1/6}\log(1/\eta)$, $\int |u|^2|u|^{-1-j}(\eta'/|u|)^{1/2^{j+1}}d|u| \le C\eta'^{1/4}$, and the Γ-type gain $C\eta^{1/4}L_\eta(1+|u|)$ is confined by the externals' decay; the joint sub-regions have the margin 1 of the power counting and every gain exponent is ≤ 1, so each is $\le C\eta'^{1/6}(1+\log(1/\eta'))^2$. Inside the near region the difference is at most the sum of the two per-flow sizes, $\le C\eta'(1+\log(1/\eta'))$. The photon-tadpole difference is $\le C_\Gamma\eta^{1/2}$ (Q6), and the $\eta'^2(\nabla G)^2$ term has size $\le Ce_r^2\eta'$. □

Proposition 17.71 (the source-linear constant vanishes)**.** *With c_k as in Lemma 17.61 and $r = k - \lfloor\varrho k\rfloor$, $\varrho < \frac{1}{9}$, for $k \ge k_0(a,m,\mu_0,\varrho)$*

$$|c_k| \le NC_\Gamma\,\eta^{1/2}\delta_r^{-1} + C_1\lambda\,\eta^{1/2} + C_2e^2\,\eta'^{1/8}(1+\delta_r^{-1})(1+\log(1/\eta'))^4 \le C\,5^{-(1-9\varrho)k/8}(1+k)^4,$$

with constants depending on a, m, μ_0, N, τ only (not on J, ψ, n, the torus or h). Hence $c_k \to 0$.

Proof. Both densities are Y_{k-r}-periodic (Lemma 17.61) and the cell average of f^Y equals the average of $f^Y\circ\tau$ over the matching X-cell (Lemma 17.29(i)), so $|\bar f^X - \bar f^Y| \le \sup_{x'}|f^X(x') - f^Y(\tau x')|$. By Lemma 17.60, $f^F = f_0^F + f_\lambda^F + f_{e^2}^F$. *Term 0*: at zero coarse fields $\langle|\phi'(x)|^2\rangle = NG^F(x,x) = N(C(0) - \Gamma^F(x,x))$, so the free tadpole leaves the residue $f_0^F(x) = -N\Gamma^F(x,x)$ (the subtraction in [13, (3.21)] removes the full-propagator expectation, not this), with matched-pair difference $\le NC_\Gamma\eta^{1/2}$ (Q6); in physical units Γ carries δ_r^{-1}, and $\eta^{1/2}\delta_r^{-1} = 5^{1/2}5^{-(k-3\lfloor\varrho k\rfloor)/2} \le C5^{-k/3}$. The order-$\lambda$ term is Lemma 17.66, $\eta^{1/2} \le C5^{-4k/9}$; the order-$e^2$ term is Lemmas 17.68 to 17.70 with $e_r^2\delta_r^{-1} = e^2$ and $m_r^{-1} = m^{-1}\delta_r^{-1}$, and $\eta'^{1/8}\delta_r^{-1} \le C5^{-(1-9\varrho)k/8}$ because $\varrho < \frac{1}{9}$. The rate is not optimal; only $c_k \to 0$ is used. □

The inputs of the forward bound. Throughout $T \in \{T_1, T_2\}$ and $r = r(k)$ is the choice of Lemma 17.26; $p(\delta) = b_0(1+\log\delta^{-1})^{\rm p}$ is King's small-field function.

Proposition 17.72 (kernel bounds of the hybrid flow at a non-zero gauge field)**.** *Let $r \ge 2$ and let A be regular in the sense of* [13, Def. 3.2] *on Λ_X or on Λ_X/P for a torus P fitted by both lattices, with $e_r^\beta \le c_0(a)$.*

(a) (kernel bounds) *The operators G_r, $\Delta^{(r)}$, $C^{(r)}$ of the hybrid flow with the paired contours of Definition 17.15 obey the bounds of* [13, Thm. 3.3, (3.5)–(3.8)] *in the periodic form (on the infinite lattice and on tori fitted by both lattices, each in its own quotient distance), with constants depending on a, M, α only, except the Hölder constant of (3.8), which is $C_\alpha \log^{3\alpha}(1/\eta')$ (Remark 17.83); the differences between two tori, or between a torus and the infinite lattice, with a common fundamental region F that is a union of $(r+1)$-blocks on which the backgrounds agree up to a collar, obey the same bounds with the distance taken in the smaller $d_\wedge$ of the two quotient metrics and the additional factor $e^{-\delta(\mathrm{dist}(\{x,y\},\partial F)+\mathrm{dist}(\mathrm{supp}\, f,\partial F))}$ for G_r and $e^{-\delta(\mathrm{dist}(z,\partial F)+\mathrm{dist}(w,\partial F))}$ for $\Delta^{(r)}$ and $C^{(r)}$ (Lemma 17.79), in place of the Neumann corrections; the next-step precision $\Delta^{(r)}+a5^{-2}\hat{Q}^*\hat{Q}$ obeys $(25\gamma_0^{-1}+a_r^{-1})^{-1} \le \Delta^{(r)}+a5^{-2}\hat{Q}^*\hat{Q} \le a_r + a/25$, with $\gamma_0=\gamma_0(a)$ the coercivity constant of Lemma 17.75, and has an exponentially decaying kernel, and so has its restriction to every subset Λ of the coarse lattice, in particular to every union of big blocks, whose inverse $\hat{C}_\Lambda^{(r)}$ of* [14, (5.53)] *obeys* [14, (5.54)] *and the two-torus clause (Proposition 17.80); the fine-slice lines obey* [13, (3.63)–(3.65)] *in the covariant form at regular A (covariant lattice derivatives and transported Hölder quotients,* [3, (2.10)–(2.11)]*) and in the plain form at the reduced background of step (5) of the proof of Theorem 17.56; and the level-r elements obey $|H^X(x,z)|, |D_A H^X(x,z)| \le Ce^{-\delta_0|x-z|}$ with the Hölder clause, $|B^X(z,w)| = |(A^X)^{-1}(z,w)| \le C(1+|z-w|)^{-1}$ and $|\Gamma^X(x,y)| \le C(1+|x-y|)^{-1}$ with the fine-variable regularity of H^X, B^X and Γ^X decaying exponentially only at the coarse mass; all constants are uniform in r, k and the torus. The same holds at every level $j \le r$ with constants uniform in j, and, in the form in which the hybrid lower bound uses it, at the levels $j \in \{0,1\}$. The same statements hold for the standard flow with King's uniform constants (*[13, Thm. 3.3]*, proved in* [4]*).*

(b) (gains) *After the constant-background reduction of Lemma 17.40 the difference elements $\Delta\Gamma := \Gamma^X - \Gamma^Y \circ \tau$, ΔH, $\Delta(A^{-1})$ at regular A obey the bounds of Corollary 17.36 with the factor $\eta'^{1/2}$; the contour vertices* [13, (3.24)–(3.26)] *at the reduced background differ entrywise by a relative $\eta^{1/2}$ against their size $Ce_r p_r^2$; the constant-background phases* [13, (3.46)] *at matched vertices differ by $\le e\, p_r \mu_0^{-1} c_\tau \delta_r^{-1/2}\eta'$; and the per-slice pieces carried by the contour vertex* [13, (3.26)–(3.27)]*, with one end or both ends on contours, obey the slice-j bounds* [13, (3.63)–(3.65)] *in both flows with a relative mismatch $(\eta'/\ell_j)^{1/2}$, and, for a one-ended line with one lattice derivative at its lattice end, with the relative mismatch $(\eta'/\ell_j)^{\theta_\partial}$, $\theta_\partial := \alpha/(2(2+\alpha)) \ge \theta_1$ with α the Hölder exponent of* [13, (3.8)] *(Lemma 17.89).*

At $A=0$ the proposition is the propagator package (Proposition 17.31, lemmas 17.32 to 17.35, and corollaries 17.36 and 17.37). At $A \neq 0$ the proof, which occupies the rest of this subsection, is the programme of [14, §5.3] written out: coercivity of $-\Delta_A + m_r^2 + a_r Q(A)^*Q(A)$ at a regular background by a cover with axis cubes carrying a local gauge for the constant part of the field (the tile properties of Lemma 17.19 and the paired contours of Lemma 17.16 are what enters: blocks partition the lattice, have bounded diameter, carry contours of bounded length inside the block, and $QQ^* = \mathbf{1}$ for every A, so that Q^*Q is the projection onto the block-constant mode); Combes–Thomas decay in L^2 form [25]; the domain-difference lemma for the two-torus clause; Bałaban's random walk [4, §2] with periodic cells fitted by both lattices ([14, (5.55)–(5.56)]); the L^p and Hölder bounds of [4, Lemma 2.2] with the $A=0$ input, the slice clauses stated covariantly; and, for (b), King's constant-background reduction [13, (3.43)–(3.55)] applied to both flows on their infinite lattices in the universal-cover form of Lemma 17.40, after which every difference element at matched points is of one of seven types (an $A=0$ line, with the gain $\eta'^{1/2}$ of Corollary 17.36; a contour line, with the kernel difference along the contour plus the flux of the slice kernel through the sliver of Lemma 17.20; a double contour line; a contour vertex; an external lifted field, Lemma 17.32; a constant-background phase; a hopping vertex, Lemma 17.29, unchanged), the extension of the lattice kernels off the lattice being the cubic-Hermite cardinal interpolation in the first argument, which decays like the lattice kernel.

The hybrid operators at a gauge field. Throughout the proof of Proposition 17.72 we work in the unit scale of level $r \ge 2$ ([13, (2.20)]): the physical spacing $\delta_r = 5^r\varepsilon_k$ is the unit, the fine spacing

is $\eta' = 5^{-r}$, $\eta = 5\eta'$, and $\ell_j := 5^j\eta'$ ($\ell_r = 1$). We write $L := 5$ for the block factor where a formula is clearer with it, $d = 3$, and $f_\alpha = R^{-1}e_\alpha$ for the frame of Y_k. The coarse lattice $\Gamma_r = R^{-1}\mathbb{Z}^3$ is common to both flows; $\Gamma_j := R^{-1}(\ell_j\mathbb{Z}^3)$ for $0 \le j \le r$, so that $\Gamma_0 = \Lambda_Y = R^{-1}(\eta'\mathbb{Z}^3)$ is the fine lattice of the flow Y and $\Gamma_1 = R^{-1}(\eta\mathbb{Z}^3)$ is Y_{k-1} rescaled; $\Lambda_X = \eta'\mathbb{Z}^3$ is the fine lattice of the flow X (Definition 17.28). Since $5R^{-1}$ is an integer matrix (Lemma 15.3(a)), $\Gamma_1 \subset \Lambda_X \cap \Lambda_Y$.

For $1 \le j \le r$ and $z \in \Gamma_j$ the *level-j block* of the flow F is

$$B_j^F(z) := \Lambda_F \cap \big(z + R^{-1}(-\tfrac{\ell_j}{2}, \tfrac{\ell_j}{2})^3\big).$$

For $F = Y$ this is a cube of 5^{3j} points aligned with (f_α); for $F = X$ it is the digitised rotated cube, with no lattice point on its boundary (Lemma 17.19(i)), and $B_1^X(z) = z + \eta' B_0$ is the tile. Both are nested, $B_{j+1}^F(z) = \bigcup_{s\in\{-2,\dots,2\}^3} B_j^F(z + \ell_j R^{-1}s)$, and for $x \in B_r^F(z)$ we write $x_j = x_j(x)$ for the centre of the level-j block containing x ($x_r = z$). Blocks are centred at the sites of the next coarse lattice (Convention 17.17(i)): the root of every block is its centre, in both flows. The matching $\tau : \Lambda_X \to \Lambda_Y$ of Definition 17.28 maps the tile $B_1^X(x)$ onto the cube $B_1^Y(x)$ for every $x \in \Gamma_1$, with $|\tau x' - x'| \le c_\tau\eta'$, $c_\tau = \sqrt{13} + 2\sqrt{3} < 7.1$; points $x'' = \tau x'$ are matched. The composite contours are those of Definition 17.15: for the flow Y King's [13, (2.12)], $\Gamma_{z,x}^{Y,(r)} = \Gamma_{z,x_{r-1}}^Y \cup \cdots \cup \Gamma_{x_1,x}^Y$, where $\Gamma_{u,v}^Y$ is the L-path that changes the f_3-coordinate first, then f_2, then f_1; for the flow X, $\Gamma_{z,x'}^{X,(r)} = \Gamma_{z,x_{r-1}}^X \cup \cdots \cup \Gamma_{x_2,x_1}^X \cup \Gamma_{x_1,x'}^X$, where for $i \ge 1$ the level-i piece Γ_{x_{i+1},x_i}^X is the L-path Γ_{x_{i+1},x_i}^Y with every η-bond $(u, u + \eta f_\alpha)$, $u \in \Gamma_1$, replaced by the fixed staircase $u + \eta'\gamma_\alpha$ (the reversed bond by the reversed staircase), and $\Gamma_{x_1,x'}^X$ is the fixed in-tile path of at most 5 bonds. The gauge field along a composite contour is read in King's convention [13, (2.19)] (Convention 17.17(ii)).

Backgrounds. A background is a real bond field A on Λ_F, identified with a site vector field by $A_{(x,x+\eta' f_\alpha)} = A_\alpha(x)$ ([13, p. 652]); $U(A(\Gamma)) := \exp[\mathrm{i}e_r A(\Gamma)]$ with $A(\Gamma) = \sum_{b\in\Gamma}\eta' A_b$ (oriented) and the running charge $e_r = e\,\delta_r^{1/2}$ ([13, (3.22)]). *Regular* means [13, Def. 3.2]: $|\partial_\mu A_\nu(x)| \le Ce_r^{\beta-1}$ for all x, μ, ν and some $\beta > 0$. The small-field class at level r is [13, (3.2)]: a coarse field A_r on Γ_r with $|A_r| \le p_r(\mu_0\delta_r)^{-1}$, $|\partial A_r| \le p_r$, $p_r = p(\delta_r)$, lifted to the flow F by [13, (2.18)], $A^{(r),F} := a_r G_r^{F,\mathrm{ph}} Q_F^* A_r$ (the photon fluctuation propagator and the frame-covariant photon average of Remark 17.18), plus the Cameron–Martin shift za_F of Lemma 17.21. Both are regular for $\delta_r \le \delta_0$ ([13, (3.45)]; Lemma 17.21).

The pinned operators. For a flow F and a background A on Λ_F,

$$Q_F(A)\phi(z) := \sum_{x\in B_r^F(z)} \eta'^3\, U(A(\Gamma_{z,x}^{F,(r)}))^{-1}\phi(x) \quad (z \in \Gamma_r), \qquad M_A := -\Delta_A + m_r^2, \qquad G_r^F(A) := \big(M_A + a_r Q_F(A)^*$$

with $-\Delta_A = D_A^* D_A$ the covariant Laplacian of the fine lattice, $(D_A\phi)(b) = \eta'^{-1}(U(-A_b)\phi(b_+) - \phi(b_-))$, $m_r = m\delta_r$, $a_r = a(1 - 5^{-2})(1 - 5^{-2r})^{-1}$, and the pairings $\langle u, v\rangle = \sum \eta'^3 \bar{u}v$ on Λ_F and $\sum_z \bar{u}v$ on Γ_r (unit block volume, $|B_r^F|\eta'^3 = 1$). This is [13, (2.11), (2.13)] in unit scale; the convention U^{-1} makes $Q_F(A)\phi$ transform with the coarse gauge function (Lemma 17.74). Put

$$C_A := M_A^{-1}, \qquad u^F(A) := C_A Q_F(A)^*, \qquad A^F(A) := \big(a_r^{-1} + Q_F(A) C_A Q_F(A)^*\big)^{-1}, \qquad B^F(A) := A^F(A)^{-1},$$

$$H^F(A) := a_r G_r^F(A) Q_F(A)^* = u^F(A) A^F(A), \qquad \Gamma^F(A) := C_A - G_r^F(A) = u^F(A) A^F(A) u^F(A)^* = H^F B^F H^{F*}.$$

By the Woodbury identity these relations are exact for every A. $A^F(A)$ is King's effective Laplacian $\Delta^{(r)}(A) = a_r - a_r^2 Q_F G_r Q_F^*$ ([13, (2.14)]; indeed $a_r(1 - a_r Q_F G_r Q_F^*) = a_r(1 + a_r Q_F C_A Q_F^*)^{-1} = A^F(A)$), H^F is the lift operator of [13, (2.18)] and Lemma 17.32, and $C^{(r)}(A) = (A^F(A) + aL^{-2}Q^{(1)}(A)^* Q^{(1)}(A))^{-1}$ is the next fluctuation covariance ([13, (2.16)]), $Q^{(1)}$ the one-step average of the coarse lattice. The photon kernels are three copies of the scalar kernels at $A = 0$ and mass $\mu_0\delta_r$ ([13, p. 654]; Remark 17.18, Proposition 17.31): no photon kernel sees a background, so the photon sector of Proposition 17.72 is the $A = 0$ package of Proposition 17.31, lemmas 17.32 to 17.35, and corollary 17.36.

Lemma 17.73 (the pieces of the composite contours)**.** *Let $z \in \Gamma_r$, $x' \in B_r^X(z)$, $x'' = \tau x'$, and let x_i ($1 \le i \le r$) be the common roots.*

(i) *For $i \geq 1$ the displacement is $x_i - x_{i+1} = \ell_i R^{-1} s$ with $s \in \{-2,\dots,2\}^3$, and every leg of $\Gamma^Y_{x_{i+1},x_i}$ is a concatenation of $|s_\alpha| 5^{i-1}$ η-bonds between points of Γ_1. The level-i piece has length $\leq 6\ell_i$ in the flow Y and $\leq \frac{7}{5}\cdot 6\ell_i = 8.4\,\ell_i$ in the flow X; the level-0 pieces have length $\leq 6\eta'$ (Y) and $\leq 5\eta'$ (X). The partial contours $\Gamma^{F,(j+1)}_{x_{j+1},x}$ of* [13, (2.19)] *(the pieces of levels $j, j-1,\dots,0$) have length $\leq 7.5\,\ell_j$ (Y) and $\leq 10.5\,\ell_j$ (X); the composite contours have length ≤ 1.5 (Y) and ≤ 2.1 (X).*

(ii) *Every level-i piece lies inside the open rotated cube $x_{i+1} + R^{-1}(-\frac{\ell_{i+1}}{2}, \frac{\ell_{i+1}}{2})^3$, hence inside $B^F_{i+1}(x_{i+1})$, and the level-0 piece inside its first-step cell; so every piece lies in its block and $\Gamma^{F,(r)}_{z,x} \subset B^F_r(z)$.*

(iii) *The two composite contours $\Gamma^{X,(r)}_{z,x'}$ and $\Gamma^{Y,(r)}_{z,x''}$ pass through the same points $x_{r-1},\dots,x_1$ and through the same endpoints of η-bonds; their level-i pieces are within Hausdorff distance η' for $i \geq 1$, the level-0 pieces lie in the ball of radius $\sqrt{13}\eta'$ about x_1, and the closed polygon $\Gamma^{X,(r)}_{z,x'} \cup [x', x''] \cup (\Gamma^{Y,(r)}_{z,x''})^{-1}$ bounds a piecewise planar surface $\Sigma(z,x')$, a union of the η-bond slivers of Lemma 17.16 and one polygon inside that ball, of area $\leq 1.5\eta' + 33\eta'^2$. The sub-polygon formed by the partial contours up to level $j+1$ and the matching segment bounds a surface of area $\leq 1.5\eta'\ell_{j+1} + 33\eta'^2$.*

Proof. (i) $x_i = \ell_i R^{-1} m_i$ with $m_i \in \mathbb{Z}^3$ the frame coordinates of the block centre and, by nesting, $m_i - 5m_{i+1} =: s \in \{-2,\dots,2\}^3$ (the level-i block of x is one of the 125 sub-blocks of its level-$(i+1)$ block; there are no ties since no lattice point lies on a block face). The L-path from x_{i+1} to x_i has legs $\ell_i s_\alpha f_\alpha = 5^{i-1} s_\alpha \cdot \eta f_\alpha$, each a run of $|s_\alpha| 5^{i-1}$ η-bonds starting at a point of Γ_1 (because $x_{i+1} \in \Gamma_{i+1} \subset \Gamma_1$ and $\eta f_\alpha \in \Gamma_1$). Lengths: $\sum_\alpha |s_\alpha|\ell_i \leq 6\ell_i$; the staircases γ_1, γ_2 have 7 fine steps for an η-bond of length $5\eta'$ and γ_3 has 5; the in-tile piece has at most 5 bonds (Lemma 17.19(ii)) and the level-0 Y-piece at most 6. Summing, $\sum_{i=1}^{j} 8.4\ell_i + 5\eta' \leq 8.4 \cdot \frac{5}{4}\ell_j(1-5^{-j}) + 5\eta' < 10.5\,\ell_j$ and $6\sum_{i=0}^{j}\ell_i \leq 7.5\,\ell_j$; for $j = r-1$ this gives 2.1 and 1.5 (Lemma 17.16). (ii) The endpoints of every η-bond of $\Gamma^Y_{x_{i+1},x_i}$ are points of Γ_1 inside the open cube $x_{i+1} + R^{-1}(-\frac{\ell_{i+1}}{2}, \frac{\ell_{i+1}}{2})^3$; in the frame coordinates $(f_\alpha \cdot (\cdot - x_{i+1}))_\alpha$ they lie in $\eta\mathbb{Z}^3 \cap (-\frac{\ell_{i+1}}{2}, \frac{\ell_{i+1}}{2})^3$, and since $\frac{\ell_{i+1}}{2} = \frac{5^{i+1}}{2}\eta'$ is an odd multiple of $\frac{\eta'}{2}$ while the points are multiples of $5\eta'$, each coordinate is at distance $\geq \frac{5}{2}\eta'$ from the faces. The segment between two such points keeps that distance, and the staircase deviates from the segment by $\leq 0.6\eta' < \frac{5}{2}\eta'$ (Lemma 17.16), so it stays in the open cube; its points lie in Λ_X, hence in $B^X_{i+1}(x_{i+1})$. The in-tile piece lies in the tile (Lemma 17.19(ii)), which is a sub-block. (iii) The Hausdorff distances and the containment of the level-0 pieces are Lemma 17.16 ($|x' - x_1| \leq \sqrt{13}\eta'$, $|x'' - x_1| \leq 2\sqrt{3}\eta'$, both level-0 pieces monotone inside their cells); the two composite contours pass through the same points $x_{r-1},\dots,x_1$ and through the same η-bond endpoints, so the closed polygon decomposes into the η-bond slivers (area $\leq 5\eta'^2$ each, at most $1.5/\eta = 0.3/\eta'$ of them, total $\leq 1.5\eta'$) and the polygon $\Gamma^X_{x_1,x'} \cup [x', x''] \cup (\Gamma^Y_{x_1,x''})^{-1}$ inside the ball of radius $\sqrt{13}\eta'$, of perimeter $\leq (5 + 7.1 + 6)\eta'$, which the cone from x_1 spans with area $\leq \frac{1}{2}\cdot 18.1 \cdot \sqrt{13}\,\eta'^2 < 33\eta'^2$. For the partial contours up to level $j+1$ the Y-length is $\leq 7.5\ell_j = 1.5\ell_{j+1}$, so there are $\leq 1.5\ell_{j+1}/\eta$ η-bonds and the slivers have total area $\leq 5\eta'^2 \cdot 1.5\ell_{j+1}/\eta = 1.5\eta'\ell_{j+1}$. □

Lemma 17.74 (block-spin algebra for any contour system)**.** *For every background A on Λ_X and every system of contours $\Gamma_{z,x} \subset B^X_r(z)$ from the root z to the point x:*

(i) *$Q_X(A)Q_X(A)^* = \mathbf{1}$, and $P_X(A) := Q_X(A)^* Q_X(A) = \sum_z w_z w_z^\dagger$ with $w_z(x) := U(A(\Gamma_{z,x}))\mathbf{1}_{B^X_r(z)}(x)$ is an orthogonal projection ($\|w_z\| = 1$, orthogonal supports);*

(ii) *for a gauge function λ on Λ_X, $Q_X(A+\partial\lambda)(e^{\mathrm{i}e_r\lambda}\phi) = e^{\mathrm{i}e_r\lambda|_{\Gamma_r}} Q_X(A)\phi$, hence $G^X_r(A+\partial\lambda) = e^{\mathrm{i}e_r\lambda} G^X_r(A) e^{-\mathrm{i}e_r\lambda}$, and likewise for H^X, A^X, B^X, Γ^X, $C^{(r)}$ with the coarse phases;*

(iii) *for a constant field $A \equiv B_0$, $U(B_0(\Gamma_{z,x})) = e^{\mathrm{i}e_r B_0\cdot(x-z)}$ for every contour, and $G^X_r(B_0)(x,y) = e^{\mathrm{i}e_r B_0\cdot x} G^X_r(0)(x,y) e^{-\mathrm{i}e_r B_0\cdot y}$ on Λ_X (on a torus: the twisted periodisation of Lemma 17.81(c));*

(iv) *with the contours of Definition 17.15, $Q_X(A) = Q^{\mathrm{rot}}_{r-1}(A) \circ \hat{Q}(A)$ under the convention* [13, (2.19)]*, and the composite blocks are unions of tiles (Lemma 17.19(iv)).*

Proof. (i) is Lemma 17.19(iii),(iv): $|U| = 1$ and $|B_r^X(z)|\eta'^3 = 1$. (ii) $A(\Gamma_{z,x}) \mapsto A(\Gamma_{z,x}) + \lambda(x) - \lambda(z)$ for any path from z to x, so $U(A(\Gamma_{z,x}))^{-1}e^{\mathrm{i}e_r\lambda(x)} = e^{\mathrm{i}e_r\lambda(z)}U(A(\Gamma_{z,x}))^{-1}$, and $-\Delta_{A+\partial\lambda} = e^{\mathrm{i}e_r\lambda}(-\Delta_A)e^{-\mathrm{i}e_r\lambda}$. (iii) A constant field is $\partial(B_0\cdot x)$ on Λ_X. (iv) The level-j pieces are the contours of the steps of the flow Y with the bonds re-drawn; iterating the one-step transformations produces exactly the phases of [13, (2.19)] along the composite contour (Convention 17.17(ii); [1, (3.33)–(3.34)]). □

The pairing of the contours is a property of Definition 17.15, not a further hypothesis; a hybrid transformation whose first-step contour were a fine-axis shortest path from the level-r root across the whole block would be a different transformation, for which the contour phases of the two flows differ at order one (Lemma 17.20).

Coercivity and decay in L^2 form. The proof of Proposition 17.72(a) follows [4, §2] and [14, §5.3] with two changes: the lower bound on the quadratic form is proved directly at a regular background by a cover with axis cubes (so that no small-domain reduction is needed for the L^2 theory), and the local cells of the random walk are periodic cells fitted by both lattices ([14, (5.55)]), on which the $A = 0$ kernels are the twisted periodisations of the explicit kernels of Lemmas 17.32 and 17.35. On the tori of the chain the periods lie in $10\mathbb{Z}^3$ in the unit scale of every level $r \le k$ ($P_J = 50MJ\,5^{k-r}\mathbb{Z}^3$ and $R^{-1}P_J = 10MJ\,5^{k-r}(5R^{-1})\mathbb{Z}^3$, Lemma 17.14(iii)), with sup-systole $s_\infty(P) \ge 50$ for $r \le k-1$.

Lemma 17.75 (coercivity at a regular background)**.** *Let Ω be Λ_X or Λ_X/P with P fitted by both lattices and $s_\infty(P) \ge 10$ in unit scale. There are $c_0(a) > 0$ and $\gamma_0(a) > 0$ such that for every regular A on Ω with $e_r^\beta \le c_0$ and every ϕ,*

$$\langle\phi, (-\Delta_A + m_r^2 + a_r P_X(A))\phi\rangle \;\ge\; \gamma_0\|\phi\|^2.$$

Explicitly $\gamma_0 = \frac{1}{2m_c}\min\{\frac{1}{8}, \frac{a'}{64(8a'+1)}\}$ with $a' = a(1-5^{-2}) \le a_r \le a$ and m_c the multiplicity of the cover below; $m_c = 8$ when $P \subset 2\mathbb{Z}^3$, as on the tori of the chain.

Proof. Cover. Let $\{\Delta_i\}$ be the axis cubes $y_i + [0,4)^3$ with $y_i \in \Lambda_c := 2\mathbb{Z}^3 + P$ ($P = 0$ on Λ_X); Λ_c is P-invariant, so the family descends to a finite family on the torus, and a cube does not wrap since every non-zero period has $|\gamma|_\infty \ge 10$. Every point, every bond and every block lies in at most m_c cubes, $m_c = 8$ if $P \subset 2\mathbb{Z}^3$ and $m_c \le 64$ in general ($\Lambda_c \subset \mathbb{Z}^3$). Every Δ_i contains a full block: the covering radius of $\Gamma_r = R^{-1}\mathbb{Z}^3$ is $\frac{\sqrt{3}}{2}$, so some $z_i \in \Gamma_r$ has $|z_i - c_i| \le \frac{\sqrt{3}}{2}$ (c_i the centre of Δ_i), and $B_r^X(z_i) \subset \{|x - z_i| < \frac{\sqrt{3}}{2}\} \subset \{|x - c_i| < \sqrt{3}\} \subset \Delta_i$ because $\sqrt{3} < 2$. Dropping the non-negative terms of the quadratic form that belong to bonds joining different cubes, to the mass and to the pins of the other blocks, and dividing by the multiplicity,

$$\langle\phi, (-\Delta_A + m_r^2 + a_r P_X(A))\phi\rangle \;\ge\; \tfrac{1}{m_c}\sum_i \Big[\sum_{b\subset\Delta_i} \eta'^3|D_A\phi(b)|^2 + a_r|\langle w_{z_i}, \phi\rangle|^2\Big] =: \tfrac{1}{m_c}\sum_i q_i(\phi).$$

Local gauge. On the cube Δ_i (contractible) write $A = A_0^{(i)} + A'^{(i)}$ with $A_0^{(i)} := A(c_i)$ constant and, by regularity along lattice paths of ℓ^1-length ≤ 12, $|A'^{(i)}| \le 12\,Ce_r^{\beta-1}$ on Δ_i. The gauge function $\lambda_i(x) = A_0^{(i)}\cdot x$ removes $A_0^{(i)}$ from both terms of q_i (Lemma 17.74(ii),(iii), applied to the bonds and the block inside Δ_i), so $q_i(\phi) = q_i^{A'}(\tilde\phi)$ with $\tilde\phi = e^{-\mathrm{i}e_r\lambda_i}\phi$, $|\tilde\phi| = |\phi|$. *Perturbation.* $D_{A'}\phi(b) = U(-A'_b)D_0\phi(b) + \eta'^{-1}(U(-A'_b)-1)\phi(b_-)$, hence $\|D_{A'}\phi\|_{\Delta_i} \ge \|D_0\phi\|_{\Delta_i} - e_r|A'|_\infty\|\phi\|_{\Delta_i}$ and $\|D_{A'}\phi\|^2 \ge \frac{3}{4}\|D_0\phi\|^2 - 3e_r^2|A'|_\infty^2\|\phi\|^2$. For the pin, $w_z(A') - w_z(0)$ has sup norm $\le e_r|A'(\Gamma)| \le 2.1\,e_r|A'|_\infty$ (Lemma 17.73(i)), so $\big||\langle w_z(A'),\phi\rangle|^2 - |\langle w_z(0),\phi\rangle|^2\big| \le (4.2e_r|A'|_\infty + 4.41e_r^2|A'|_\infty^2)\|\phi\|_B^2$. *The free cube.* With $\psi := \tilde\phi - c$ on Δ_i, c the mean, $\sum_{b\subset\Delta_i}\eta'^3|D_0\tilde\phi(b)|^2 \ge \lambda_1\|\psi\|_{\Delta_i}^2$ with $\lambda_1 = 4\eta'^{-2}\sin^2(\pi\eta'/8) \ge \frac{1}{4}$ (the first Neumann eigenvalue of the cube of $4/\eta'$ points per side; $\sin t \ge \frac{2}{\pi}t$), and, since $|B_i|\eta'^3 = 1$, $|\langle w_{z_i}, \tilde\phi\rangle|^2 = |c + \langle \mathbf{1}_{B_i}, \psi\rangle|^2 \ge (1-t)|c|^2 - (\frac{1}{t} - 1)\|\psi\|_{B_i}^2$ for $0 < t < 1$. Choosing $\frac{1}{t} - 1 = \frac{\lambda_1}{2a_r}$,

$$\lambda_1\|\psi\|^2 + a_r|\langle w_{z_i}, \tilde\phi\rangle|^2 \;\ge\; \tfrac{\lambda_1}{2}\|\psi\|^2 + \tfrac{a_r\lambda_1}{2a_r+\lambda_1}|c|^2 \;\ge\; \gamma_1\|\tilde\phi\|_{\Delta_i}^2, \qquad \gamma_1 := \min\Big\{\tfrac{\lambda_1}{2}, \tfrac{a_r\lambda_1}{64(2a_r+\lambda_1)}\Big\},$$

using $\|\tilde\phi\|^2_{\Delta_i} = \|\psi\|^2 + 64|c|^2$. *Assembly.* $q_i(\phi) \ge \frac34\gamma_1\|\phi\|^2_{\Delta_i} - (3e_r^2|A'|^2 + a_r(4.2e_r|A'| + 4.41e_r^2|A'|^2))\|\phi\|^2_{\Delta_i} \ge \frac12\gamma_1\|\phi\|^2_{\Delta_i}$ once $e_r^\beta \le c_0(a)$, and $\sum_i \|\phi\|^2_{\Delta_i} \ge \|\phi\|^2$. With $\lambda_1 \ge \frac14$, $\gamma_1 \ge \min\{\frac18, \frac{a'}{64(8a'+1)}\}$. □

This is [14, (5.59)–(5.60)] ("cover □ by cubes $\{\Delta_i\}$ on the η-lattice with sides parallel to the coordinate planes … each cube contains fully at least one block") written out, with the local gauge step that [14, p. 345] leaves to "$A = B + A'$ with B constant and A' small" made explicit; it is also [4, (1.8)] for the hybrid flow. No Poincaré inequality on the digitised rotated cube is needed: only axis cubes of the fine lattice enter.

Corollary 17.76 (L^2 bounds)**.** *Under the hypotheses of Lemma 17.75, $T := -\Delta_A + m_r^2 + a_r P_X(A) \ge \gamma_0$, so $\|G_r^X(A)\| \le \gamma_0^{-1}$, $\|D_A G_r^X(A)^{1/2}\| \le 1$, $\|G_r^X(A)^{1/2} D_A^*\| \le 1$, $\|D_A G_r^X(A) D_A^*\| \le 1$ and $0 \le G_r^X(A) \le \gamma_0^{-1}$ ([4, Lemma 2.1], [14, Lemma 5.7]). The same holds, on Λ_X and on a torus with $s_\infty(P) \ge 10L$ in unit scale (true on the tori of the chain for $r \le k-1$), for the operator with the level-$(r+1)$ blocks, $-\Delta_A + m_r^2 + a_{r+1}L^{-2}P_{r+1}(A)$ (the pin at scale L), with γ_0 replaced by $\gamma_0 L^{-2}$.*

Proof. $D_A^* D_A \le T$. For the $(r+1)$-blocks repeat the proof with cubes of side $4L$, corners in $2L\mathbb{Z}^3 + P$ (multiplicity 8 if $P \subset 10\mathbb{Z}^3$, as for P_J and $R^{-1}P_J$; periods with $|\gamma|_\infty \ge 10L$) and blocks of L^3 unit cells ([4, p. 588]: "replace $\min\{\pi^2, a_k\}$ by $\min\{\pi^2 L^{-2}, a_{k+1}L^{-2}\}$"). □

Proposition 17.77 (Combes–Thomas decay)**.** *Under the hypotheses of Lemma 17.75 there are $\theta_0, c_1 > 0$ depending on a only such that for all f, f' supported in unit cubes,*

$$|\langle f, G_r^X(A) f'\rangle|,\ |\langle f, D_A G_r^X(A) f'\rangle|,\ |\langle f, G_r^X(A) D_A^* f'\rangle| \ \le\ c_1 e^{-\theta_0 \operatorname{dist}(\operatorname{supp} f, \operatorname{supp} f')} \|f\|_2 \|f'\|_2,$$

dist *the quotient distance of Ω. Hence ([13, (3.6)]) $|A^X(A)(z,w)| = |a_r\delta_{zw} - a_r^2\langle w_z, G_r^X(A) w_w\rangle| \le Ce^{-\theta_0|z-w|}$. The same holds for the operator with the $(r+1)$-pin of Corollary 17.76, with θ_0, c_1 depending on a and L.*

Proof. For a weight ψ with $|\psi(x) - \psi(x')| \le |x - x'|$ and $0 \le t \le \theta_0$ put $T_\psi := e^{t\psi} T e^{-t\psi}$. On Λ_X one may take the linear weight $\psi = \hat n \cdot x$; on a torus $\vartheta \cdot x$ is not a function, and $\psi = \min\{\operatorname{dist}(\cdot, \operatorname{supp} f'), R\}$ is used, with the quotient distance; the estimates below use only the Lipschitz bound on bonds and inside blocks. The hopping part changes by a form-bounded perturbation: $e^{t\psi} D_A e^{-t\psi} = D_A + B_t$, with B_t the bond multiplication by $\eta'^{-1}(e^{-t(\psi(b_+)-\psi(b_-))} - 1)$ composed with the covariant shift (bond functions weighted at b_-), of norm $\|B_t\| \le \sqrt3\, t e^{t\eta'}$ (every site is the end b_+ of three bonds); with $e^{t\psi} D_A^* e^{-t\psi} = (D_A + B_{-t})^*$ this gives $|\langle\phi, (T_\psi - T)\phi'\rangle| \le Ct(\|D_A\phi\|\|\phi'\| + \|\phi\|\|D_A\phi'\|) + Ct^2\|\phi\|\|\phi'\|$ for $t\eta' \le 1$. The pin changes by $\sum_z w_z(x)\overline{w_z(y)}(e^{t(\psi(x)-\psi(y))} - 1)$ with $|\psi(x) - \psi(y)| \le |x - y| \le \sqrt3$ inside a block, an operator of norm $\le 2\sqrt3\, t e^{\sqrt3 t}$ (one rank-one piece per block, orthogonal supports). Hence $\|T^{-1/2}(T_\psi - T)T^{-1/2}\| \le Ct\gamma_0^{-1/2} + Ct^2\gamma_0^{-1} \le \frac12$ for $t \le \theta_0(a)$, using $\|D_A T^{-1/2}\| \le 1$ (Corollary 17.76); so $T_\psi = T^{1/2}(1 + T^{-1/2}(T_\psi - T)T^{-1/2})T^{1/2}$ is invertible with $\|T_\psi^{-1}\| \le 2\gamma_0^{-1}$. Then $\langle f, G_r^X f'\rangle = \langle e^{-t\psi} f, T_\psi^{-1} e^{t\psi} f'\rangle$, since $G_r^X = e^{-t\psi} T_\psi^{-1} e^{t\psi}$; choose $t = \theta_0$ and $\psi = \min\{\operatorname{dist}(\cdot, \operatorname{supp} f'), R\}$ with $R \ge \operatorname{dist}(\operatorname{supp} f, \operatorname{supp} f')$ (or, on Λ_X, $\psi = \hat n \cdot x$ with $\hat n$ the unit vector from $\operatorname{supp} f'$ to $\operatorname{supp} f$), and use $\sup_{\operatorname{supp} f} e^{-t\psi} \sup_{\operatorname{supp} f'} e^{t\psi} = e^{-\theta_0 \operatorname{dist}(\operatorname{supp} f, \operatorname{supp} f')}$ for the distance weight ($\psi = 0$ on $\operatorname{supp} f'$), $\le e^{-\theta_0(\operatorname{dist} - 2\sqrt3)}$ for the linear one. The derivative versions use $\|D_A T_\psi^{-1}\|, \|T_\psi^{-1} D_A^*\| \le C$ (the same algebra). For A^X: w_z, w_w have unit norm and supports in blocks of diameter $\le \sqrt3$. The same proof with the $(r+1)$-pin (blocks of diameter $\le \sqrt3 L$, Corollary 17.76) gives the bounds for that operator. □

Corollary 17.78 ([13, (3.6)] for $C^{(r)}$)**.** $|C^{(r),X}(A)(z,w)| \le Ce^{-\delta_0 d(z,w)}$ *in the quotient distance of the domain. For two domains Ω_1, Ω_2 as in Lemma 17.79 (a common fundamental region F, a union of $(r+1)$-blocks, on which the backgrounds agree up to a collar of width W_0) and $y, y' \in \Gamma_r \cap F$,*

$$|C^{(r)}(\Omega_1, A_1; y, y') - C^{(r)}(\Omega_2, A_2; y, y')| \le Ce^{-\delta(d_\wedge(y,y') + \operatorname{dist}(y,\partial F) + \operatorname{dist}(y',\partial F))},$$

$d_\wedge$ the smaller of the two quotient distances; the same for $\Delta^{(r)} = A^X$.

Proof. [4, §3] writes $C^{(r)}(\Omega, A; y, y')$ through the Gaussian integral (3.1), factorises it over the next-level blocks after the gauge transformation "$\psi(y) \to U(A(\Gamma_{y,z}))\psi(y)$" (p. 587), and arrives at the identity (3.8): $C^{(r)}$ is a bounded combination of $\langle q_r(y), G_r(\Omega, \Lambda, A) q_r(y')\rangle$ and $\langle q_{r+1}(y), \cdot\, q_{r+1}(y')\rangle$, with $q_r(y)$, $q_{r+1}(y)$ the block modes (3.7) of unit and $L^{-d/2}$ norm, where $G_r(\Omega, \Lambda, A)$ is the fluctuation propagator pinned on the $(r+1)$-blocks (3.5). The two inputs of this algebra are that the blocks partition the lattice and that every contour runs from the block root to the point, both true for the hybrid flow (Lemma 17.74(i),(iv), Lemma 17.19); the decay of $G_r(\Omega, \Lambda, A)$ in the L^2-weighted form is Proposition 17.77 for the $(r+1)$-pin. This is what [14, pp. 345–346] says: "To prove Proposition 5.6, we use the method presented in Chap. 3 of [Ba4]. This method requires only L^2-bounds on the operators $G_k(\Omega, A)$." The two-domain bound is not [4, (3.9)–(3.10)], which change the pin region inside one domain: on each domain (3.6) gives $C^{(r)} = V^* G_r(\Omega, \Omega^{(r)}, A) V - (\beta/a_r) P + a_r^{-1}$ with $G_r(\Omega, \Omega^{(r)}, A)$ the $(r+1)$-pinned operator (3.5), $V = Q_r^*(1 - \beta P)$, $\beta = a_{r+1} L^{-2}/a_r$ ($Q_r = Q_X(A)$, $Q = Q^{(1)}(A)$, $P = Q^* Q$), whose local terms and block functions agree on the common region, so that $\delta C^{(r)}(y, y') = \langle v(y), \delta G\, v(y')\rangle$ there; δG for the $(r+1)$-pin obeys the weighted bound (36) of the geometric resolvent identity $\langle f', (G_2 - G_1) f\rangle = \langle f', G_1 [H_2, \chi] G_2 f\rangle$ with Proposition 17.77 on each side (Lemma 17.79). □

Lemma 17.79 (domain difference)**.** *Let $\Omega_1 = \Lambda_X/P_1$ be a torus fitted by both lattices with $s_\infty(P_1) \geq 10L$ in unit scale, and F a fundamental region of P_1 that is a union of $(r+1)$-blocks. Let Ω_2 be Λ_X, or a torus Λ_X/P_2 fitted by both lattices into which F embeds (the projection is injective on F); F is identified with its images. Let A_i be regular on Ω_i with $e_r^\beta \leq c_0(a)$, and let $A_1 = A_2$ on every bond with both ends in $F_0 := \{x \in F : \operatorname{dist}(x, \partial F) \geq W_0\}$, for a fixed $W_0 \geq 0$. Put $H_i := -\Delta_{A_i} + m_r^2 + a_{r+1} L^{-2} P_{r+1}(A_i)$ on Ω_i (the $(r+1)$-pin, $P_{r+1} = Q_{r+1}^* Q_{r+1}$ with the level-$(r+1)$ blocks) and $G_i := H_i^{-1}$. Let d_i be the quotient metric of Ω_i and $d_\wedge := \min\{d_1, d_2\}$ on F; distances to ∂F and to F_0^c are taken in Λ_X, for a lift of F. There are $C, \delta > 0$ depending on a, L, W_0 only such that for f, f' supported in unit cubes in F*

(36)
$$|\langle f', (G_1 - G_2) f\rangle| \leq C \exp\big[-\delta\big(d_\wedge(\operatorname{supp} f, \operatorname{supp} f') + \operatorname{dist}(\operatorname{supp} f, \partial F) + \operatorname{dist}(\operatorname{supp} f', \partial F)\big)\big] \|f\| \|f'\|,$$

and, for $y, y' \in \Gamma_r \cap F$,

$$|C^{(r)}(\Omega_1, A_1; y, y') - C^{(r)}(\Omega_2, A_2; y, y')| \leq C \exp\big[-\delta\big(d_\wedge(y, y') + \operatorname{dist}(y, \partial F) + \operatorname{dist}(y', \partial F)\big)\big]. \tag{37}$$

The same bound (37) holds for $\Delta^{(r)} = A^X$, from (36) at the r-pin. In particular $\delta C^{(r)}$ and $\delta \Delta^{(r)}$ carry the factor $e^{-\delta \operatorname{dist}(\{y, y'\}, \partial F)}$ of [13, Thm. 3.3]*, in the metric $d_\wedge$.*

The metric is forced: for y near one face of F and y' near the opposite face, $d_1(y, y') = O(1)$, so $C^{(r)}(\Omega_1, A_1; y, y')$ is of order one, while $C^{(r)}(\Omega_2, A_2; y, y')$ is small when $\Omega_2 = \Lambda_X$; no bound of the form (37) with the distance $|y - y'|$ of the lift in place of $d_\wedge$ can hold. The kernel bounds on each domain separately are in the quotient distance of that domain (Proposition 17.77).

Proof. Step 0 (decay on each domain). By Proposition 17.77 at the $(r+1)$-pin, with the distance weight on a torus and the linear weight on Λ_X, for f, f' supported in unit cubes and $i = 1, 2$, $|\langle f, G_i f'\rangle|$, $|\langle f, D_{A_i} G_i f'\rangle|$, $|\langle f, G_i D_{A_i}^* f'\rangle| \leq c_1 e^{-\theta_0 d_i(\operatorname{supp} f, \operatorname{supp} f')} \|f\| \|f'\|$, $\theta_0 = \theta_0(a, L)$.

Step 1 (locality). Put $r_1 := 1 + \sqrt{3} L$ and $S := \{x \in F_0 : \operatorname{dist}(x, F_0^c) \geq r_1\}$. For u supported in S, $H_1 u = H_2 u$ (both supported in F_0, identified). Indeed $(H_i u)(x)$ involves u at the bond neighbours of x and on the $(r+1)$-block of x, the bond variables of the bonds meeting $\operatorname{supp} u$ (both ends in F_0), and the block functions w_z of the blocks meeting S; such a block lies in F (F is a union of $(r+1)$-blocks) within $\sqrt{3} L$ of S, hence in F_0, where $A_1 = A_2$; its contours lie inside it (Lemma 17.74), so the contour phases of w_z agree.

Step 2 (geometric resolvent identity). Let $\chi(x) := \min\{1, \max\{0, \operatorname{dist}(x, F_0^c) - r_1\}\}$ (Lipschitz constant 1), so $\operatorname{supp} \chi \subset S$ and $\chi = 1$ on $S'' := \{x : \operatorname{dist}(x, F_0^c) \geq r_1 + 1\}$. Let f, f' be supported in S''. Then $\chi G_2 f$ is supported in S, so by Step 1 $H_1(\chi G_2 f) = H_2(\chi G_2 f) = \chi f + [H_2, \chi] G_2 f = f + [H_2, \chi] G_2 f$. The function $[H_2, \chi] G_2 f$ vanishes at points farther than r_1 from the set where χ is not locally constant, so it is supported in F_0 and is a function on Ω_1. Applying G_1 and

pairing with f' ($\chi f' = f'$, χ real),

$$\langle f', G_2 f\rangle - \langle f', G_1 f\rangle = \langle f', G_1[H_2,\chi]G_2 f\rangle. \tag{38}$$

Step 3 (the commutator). With $(D_A\phi)(b) = \eta'^{-1}(U(-A_b)\phi(b_+) - \phi(b_-))$, $D_A(\chi u)(b) = \chi(b_-)(D_A u)(b)+(Mu)(b)$, $(Mu)(b) := \eta'^{-1}(\chi(b_+)-\chi(b_-))U(-A_b)u(b_+)$; taking adjoints, $\chi D_A^* - D_A^*\chi_- = M^*$, hence $[-\Delta_A,\chi] = D_A^*M - M^*D_A$. Since $|\chi(b_+)-\chi(b_-)| \le \eta'$ and every site is the end b_+ of three bonds, $\|M\| \le \sqrt{3}$. For the pin, $P_{r+1} = \sum_z w_z w_z^*$ with $\|w_z\| = 1$ and $\operatorname{supp} w_z \subset B_{r+1}(z)$ (Lemma 17.74), so $[P_{r+1},\chi] = \sum_z \big(w_z w_z^*(\chi-\chi(z)) - (\chi-\chi(z))w_z w_z^*\big)$, each term acting in $\ell^2(B_{r+1}(z))$ with norm $\le 2\sup_{B_{r+1}(z)}|\chi-\chi(z)| \le \sqrt{3}L$, and zero unless χ is non-constant on the block; m_r^2 commutes with χ. All these terms live on bonds and blocks meeting the slab $\{r_1 \le \operatorname{dist}(\cdot,F_0^c) \le r_1+1\}$, hence on the set $Y := \{x : \frac12 \le \operatorname{dist}(x,F_0^c) \le r_1+2+\sqrt{3}L\} \subset F_0$ (the pieces lie within $\sqrt{3}L+\eta'$ of the slab and $r_1 - \sqrt{3}L - \eta' > \frac12$), on whose bonds $D_{A_1} = D_{A_2}$.

Step 4 (bound of (38)). Write $\langle f', G_1 D^* M G_2 f\rangle = \langle DG_1 f', MG_2 f\rangle$, $\langle f', G_1 M^* DG_2 f\rangle = \langle MG_1 f', DG_2 f\rangle$ and the pin terms as $\langle G_1 f', [w_z w_z^*,\chi]G_2 f\rangle$. Split Y into unit cubes Q (every bond and block meets at most $(\sqrt{3}L+2)^3$ of them) and apply the Cauchy–Schwarz inequality cube by cube, with $\|\mathbf{1}_Q DG_2 f\| = \sup\{|\langle h, DG_2 f\rangle| : h \in \ell^2(Q), \|h\| = 1\}$ and the same without D, and Step 0 on Ω_1 for the left factor and on Ω_2 for the right factor. Since $\operatorname{dist}(x,\partial F) \le \operatorname{dist}(x,F_0^c) + W_0$, $Y \subset K := \{x \in F : \operatorname{dist}(x,\partial F) \le W_0 + r_2\}$ with $r_2 := 2r_1 + 2$, and

$$|\langle f', (G_2 - G_1)f\rangle| \le C_L \sum_{Q\subset K} e^{-\theta_0(d_1(\operatorname{supp} f', Q)+d_2(Q,\operatorname{supp} f))}\|f\|\|f'\|. \tag{39}$$

Step 5 (geometry). For $x \in F$ and $q \in F$, a shortest lattice path in Ω_i from x to q, lifted from x, either stays in the lift of F (the projection is injective on F), and then has length $\ge |x-q| \ge \operatorname{dist}(x,\partial F) - \operatorname{dist}(q,\partial F)$, or it leaves it, and then has length $\ge \operatorname{dist}(x,\partial F)$. Hence $d_i(x,q) \ge \operatorname{dist}(x,\partial F) - \operatorname{dist}(q,\partial F)$, and for $q \in K$, $d_i(x,q) \ge \operatorname{dist}(x,\partial F) - W_0 - r_2 - 1$ (the 1 for the unit cubes). Splitting $\theta_0(d_1+d_2) = \frac{\theta_0}{2}d_1 + \frac{\theta_0}{2}(d_1+d_2)$ in (39),

$$\theta_0\big(d_1(\operatorname{supp} f', Q)+d_2(Q,\operatorname{supp} f)\big) \ge \tfrac{\theta_0}{2}d_1(\operatorname{supp} f', Q) + \tfrac{\theta_0}{2}\big(\operatorname{dist}(\operatorname{supp} f,\partial F)+\operatorname{dist}(\operatorname{supp} f',\partial F)\big) - \theta_0(W_0+r_2+3),$$

and $\sum_Q e^{-(\theta_0/2)d_1(\operatorname{supp} f', Q)} \le C(\theta_0)$, since the unit cubes of a fundamental region at d_1-distance in $[n, n+1)$ from a unit cube number at most $C(n+2)^3$ (choose for each the lift nearest to a fixed lift of $\operatorname{supp} f'$). So, for f, f' supported in S'',

$$\text{(ii)} \qquad |\langle f', (G_1 - G_2)f\rangle| \le C' e^{-(\theta_0/2)(\operatorname{dist}(\operatorname{supp} f,\partial F)+\operatorname{dist}(\operatorname{supp} f',\partial F))}\|f\|\|f'\|.$$

Since $\operatorname{dist}(x,F_0^c) \ge \operatorname{dist}(x,\partial F) - W_0$, a unit cube not contained in S'' lies within $r_3 := W_0 + r_1 + 3$ of ∂F. If $\operatorname{supp} f$ is such a cube, then by the first inequality of this step $d_\wedge(\operatorname{supp} f, \operatorname{supp} f') \ge \operatorname{dist}(\operatorname{supp} f', \partial F) - r_3 - 2$, so $\frac{\theta_0}{2}(\operatorname{dist}(\operatorname{supp} f,\partial F) + \operatorname{dist}(\operatorname{supp} f',\partial F)) \le \frac{\theta_0}{2}d_\wedge(\operatorname{supp} f, \operatorname{supp} f') + \theta_0(r_3+1)$, and (ii) follows from (i) below with C' replaced by $\max\{C', 2c_1 e^{\theta_0(r_3+1)}\}$; the same if $\operatorname{supp} f'$ is such a cube. So (ii) holds for all f, f' supported in unit cubes in F.

Step 6 (proof of (36)). By Step 0, (i) $|\langle f', (G_1 - G_2)f\rangle| \le |\langle f', G_1 f\rangle| + |\langle f', G_2 f\rangle| \le 2c_1 e^{-\theta_0 d_\wedge(\operatorname{supp} f, \operatorname{supp} f')}\|f\|\|f'\|$. Then $\min\{\text{(i)},\text{(ii)}\} \le (\text{i})^{1/2}(\text{ii})^{1/2}$, whose exponent is $\frac{\theta_0}{2}d_\wedge + \frac{\theta_0}{4}(\operatorname{dist}(\operatorname{supp} f,\partial F) + \operatorname{dist}(\operatorname{supp} f',\partial F))$; this is (36) with $\delta = \theta_0/4$ and $C = (2c_1C')^{1/2}$.

Step 7 (proof of (37)). Take $\Lambda = \Omega^{(r)}$ in [4, (3.5)]; then $B^r(\Lambda) = \Omega$ and $G_r(\Omega_i, \Omega_i^{(r)}, A_i) = G_i$. In the conventions above ($Q_r Q_r^* = \mathbf{1}$, $QQ^* = \mathbf{1}$, $P = Q^*Q$, $P_{r+1} = Q_r^* P Q_r$, with $Q_r = Q_X(A)$ and $Q = Q^{(1)}(A)$), [4, (3.6)] reads

$$C^{(r)}(\Omega_i, A_i) = V_i^* G_i V_i - \tfrac{\beta}{a_r}P(A_i) + a_r^{-1}, \qquad V_i := Q_r^*(A_i)\big(1 - \beta P(A_i)\big), \qquad \beta := \frac{a_{r+1}L^{-2}}{a_r},$$

with $a_{r+1} = aa_r/(aL^{-2} + a_r)$; it is obtained by integrating the coarse variable out of the Gaussian of [4, (3.1)], is an algebraic identity between bounded operators, and holds on Λ_X as on a torus. The vector $v_i(y) := V_i\delta_y$ is supported in the $(r+1)$-block of y, is built from contours inside that block (Lemma 17.74), and has $\|v_i(y)\| \le 1 + \beta \le 2$. *Case 1:* the $(r+1)$-blocks of y and y' lie in F_0. Then $v_1(y) = v_2(y) =: v(y)$, the same for y', and $P(A_1; y, y') = P(A_2; y, y')$, since $A_1 = A_2$ there; so $\delta C^{(r)}(y,y') = \langle v(y), (G_1 - G_2)v(y')\rangle$.

Splitting $v(y)$, $v(y')$ into at most $(\sqrt{3}L+2)^3$ unit-cube pieces and applying (36) to each pair, $|\delta C^{(r)}(y,y')| \le 4(\sqrt{3}L+2)^6 e^{6\delta\sqrt{3}L} C\, e^{-\delta(d_\wedge(y,y')+\mathrm{dist}(y,\partial F)+\mathrm{dist}(y',\partial F))}$. *Case 2:* a $(r+1)$-block of y or y' meets $F\setminus F_0$, say that of y, so $\mathrm{dist}(y,\partial F)\le r_4 := W_0+\sqrt{3}L$. Then $|\delta C^{(r)}| \le |C^{(r)}(\Omega_1)|+|C^{(r)}(\Omega_2)| \le 2C_0 e^{-\delta_0 d_\wedge(y,y')}$ by the single-domain decay of Corollary 17.78 on each domain (in its quotient distance), and $d_\wedge(y,y') \ge \mathrm{dist}(y',\partial F) - r_4$ by Step 5, so $e^{-\delta_0 d_\wedge} \le e^{\delta_0 r_4} e^{-(\delta_0/2)(d_\wedge(y,y')+\mathrm{dist}(y,\partial F)+\mathrm{dist}(y',\partial F))}$. Both cases give (37) with $\min\{\delta,\delta_0/2\}$ in place of δ. For $\Delta^{(r)} = a_r - a_r^2 Q_r G_r^X Q_r^*$ ([13, (2.14)], the r-pin) the same argument applies with w_z in place of $v(y)$, with (36) at the r-pin (Steps 0–6 with $r_1 = 1+\sqrt{3}$, Lemma 17.75). □

Proposition 17.80 ([14, Prop. 5.6] for the hybrid flow)**.** *Let $X := A^X(A) + aL^{-2}P(A)$ on the coarse lattice Γ_r of the domain, $P = Q^{(1)}(A)^* Q^{(1)}(A)$. Then $(L^2\gamma_0^{-1} + a_r^{-1})^{-1} \le X \le a_r + aL^{-2}$ and $|X(y,y')| \le Ce^{-\theta d(y,y')}$; for every subset Λ of the coarse lattice, in particular every union of big blocks, the same bounds hold for the compression $X|_\Lambda$, and $\hat C_\Lambda^{(r)} := (X|_\Lambda)^{-1}$ ([14, (5.53)]) obeys $|\hat C_\Lambda^{(r)}(y,y')| \le Ce^{-t_* d(y,y')}$ ([14, (5.54)]), $|\hat C_\Lambda^{(r)}(y,y') - C^{(r)}(y,y')| \le Ce^{-(t_*/6)(d(y,y')+d(y,\Lambda^c)+d(y',\Lambda^c))}$, and, under the hypotheses of Lemma 17.79 with $\Lambda\subset F$, the two-domain bound (37) in the metric $d_\wedge$. The constants depend on a, L (and W_0) only.*

Proof. Upper bound: $A^X = (a_r^{-1} + Q_r C_A Q_r^*)^{-1} \le a_r$ and $P\le 1$. Lower bound: [4, (3.6)] in the form of Step 7 above gives $X^{-1} = C^{(r)} = V^* G_{r+1} V - (\beta/a_r)P + a_r^{-1} \le \|V\|^2\|G_{r+1}\| + a_r^{-1}$ with $\|V\|\le 1$ and $\|G_{r+1}\| \le L^2\gamma_0^{-1}$ (Corollary 17.76, the $(r+1)$-pin). Decay of X: Proposition 17.77 for A^X and the finite range of P (its kernel vanishes unless y,y' lie in one L-block, of diameter $<\sqrt{3}L$). Compression to Λ keeps both bounds and the entries. Then, for a 1-Lipschitz ψ on Λ, $e^{t\psi}X|_\Lambda e^{-t\psi} - X|_\Lambda$ has entries bounded by $Ct\, d(y,y') e^{-(\theta/2)d(y,y')}$ for $t\le\theta/2$, hence norm $\le Ct$ (Schur test), so $e^{t\psi}X|_\Lambda e^{-t\psi}$ is invertible with inverse of norm $\le 2(L^2\gamma_0^{-1}+a_r^{-1})$ for $t\le t_*$, t_* depending on a,L only; with $\psi = \min\{d(\cdot,y'), d(y,y')\}$ this gives $|\hat C_\Lambda^{(r)}(y,y')| \le Ce^{-t_* d(y,y')}$ (the entry is bounded by the operator norm on the counting measure of Γ_r). The difference with $C^{(r)}$ is $\hat C_\Lambda^{(r)} X_{\Lambda\Lambda^c} C^{(r)}_{\Lambda^c\Lambda}$ (the block identity for the inverse of a compression), whose decay follows from the three factors' decay and the geometric mean as in Step 6; the two-domain bound is the resolvent identity on $\ell^2(\Lambda)$ with (37) for $\Delta^{(r)}$ ($\delta P = 0$ off the collar), Step 5 and Step 6's interpolation. □

The $A=0$ input and the bounds on a cell.

Lemma 17.81 ($A=0$ bounds for the hybrid flow)**.** *On Λ_X, with S_j^X the slice kernels of Definition 17.28 ($G_r^X(0) = \sum_{j<r} S_j^X$ exactly):*

(a) $|S_j^X(x,y)| \le C\ell_j^{-1} e^{-\delta|x-y|/\ell_j}$, $|\nabla_x S_j^X(x,y)| \le C\ell_j^{-2} e^{-\delta|x-y|/\ell_j}$, *and for $0<\alpha<1$*
$$|\nabla_x S_j^X(x,y) - \nabla_x S_j^X(x',y)| \le C_\alpha \log^{3\alpha}(2+\ell_j/\eta')\, \ell_j^{-2-\alpha}|x-x'|^\alpha e^{-\delta\min(|x-y|,|x'-y|)/\ell_j}$$
(the logarithm is the slice's own, $\le \log^{3\alpha}(1/\eta')$);

(b) *consequently $\|G_r^X(0)f\|_\infty + \|\nabla G_r^X(0)f\|_\infty \le C\|f\|_\infty$ with C uniform in r, $[\nabla G_r^X(0)f]_\alpha \le C_\alpha \log^{3\alpha}(1/\eta')\|f\|_\infty$, and $\|G_r^X(0)f\|_q \le C\|f\|_p$ for $0 \le \frac1p - \frac1q < \frac23$, with the exponential localisation $e^{-\delta_0\,\mathrm{dist}(x,\mathrm{supp}\,f)}$ ([4, (2.16)–(2.17)]);*

(c) *on a periodic cell $\square = \Lambda_X/P$ fitted by both lattices, of side $\ge 2M$, and for a constant field B_0,*
$$G_r^X(\square,B_0)(x,y) = e^{\mathrm{i}e_r B_0\cdot(x-y)} \sum_{\gamma\in P} e^{\mathrm{i}e_r B_0\cdot\gamma} G_r^X(0)(x,y+\gamma)$$
(twisted periodisation), and (b) holds on $\square$ for $G_r^X(\square,B_0)$ with constants independent of B_0 and M, the Hölder constant carrying the same $\log^{3\alpha}(1/\eta')$.

Proof. (a) In the unit scale of level j, $S_j^X = H_j^X C_j^X H_j^{X*}$ (Definition 17.28). The decay $|H_j^X(x,z)| \le Ce^{-\delta|x-z|}$ and the gradient bound $|\nabla H_j^X| \le C_\nabla$ are Lemmas 17.32 and 17.33 (the decay is the Combes–Thomas decay of the coercive coarse operators, [25]); the decay of the coarse kernel $C_j^X = (A_j^X + bQ^*Q)^{-1}$ is the one used in the proof of Lemma 17.35 (its

symbol is analytic and bounded on a strip because $A_j^X + bQ^*Q \geq c$). *The Hölder derivative.* The second difference of H^X in x multiplies the alias-l summand of M^X in the representation of Lemma 17.32 by $d(q+l)^2$ with $|d(Q)| \leq |Q|$; since $\Delta'_X(Q) \geq c|Q|^2$ on each alias cell and $\sum_l u_X(q+l)^2 = 1$, the alias sum is $\leq C\sum_l |u_X(q+l)| \leq C$, and integrating against $|u_{r-1}|$ over V_η costs $J_0 \leq C\log^3(1/\eta)$ (Lemma 17.63). So $|\nabla\nabla H^X| \leq \Lambda_\eta := C\log^3(1/\eta)$ and, by interpolation with $|\nabla H^X| \leq C_\nabla$,

$$|\nabla H^X(x,z) - \nabla H^X(x',z)| \leq \min\{2C_\nabla, \Lambda_\eta|x-x'|\} \leq (2C_\nabla)^{1-\alpha}\Lambda_\eta^\alpha |x-x'|^\alpha \leq C_\alpha \log^{3\alpha}(1/\eta)|x-x'|^\alpha$$

for every $\alpha < 1$; the geometric mean with the decay gives the localised form. Composing with C_j^X and H_j^{X*} (exponentially decaying kernels) and rescaling by [13, (2.20)] (scalar kernels scale as ℓ^{-1} in $d=3$, derivatives as ℓ^{-2}) gives (a); at level j the logarithm is $\log(1/\eta_j)$ with $\eta_j = 5^{1-j}$, i.e. $\leq \log(2+\ell_j/\eta')$. *Summation.* $\|S_j^X f\|_\infty \leq C\ell_j^{-1}\ell_j^3\|f\|_\infty$, $\|\nabla S_j^X f\|_\infty \leq C\ell_j\|f\|_\infty$, $[\nabla S_j^X f]_\alpha \leq C_\alpha \log^{3\alpha}(2+\ell_j/\eta')\,\ell_j^{1-\alpha}\|f\|_\infty$; summing over $j<r$ converges for $\alpha<1$, the Hölder term to $\leq C_\alpha \sum_{j<r} 5^{-(r-j)(1-\alpha)}(1+j\ln 5)^{3\alpha} \leq C_\alpha \log^{3\alpha}(1/\eta')$ (this is [4, (2.38)–(2.39)] with the slice kernels in place of $G_jQ_j^*C^{(j)}Q_jG_j$). The $L^p \to L^q$ bounds follow from the kernel bound $|G_r^X(0)(x,y)| \leq C\min(\eta'^{-1}, |x-y|^{-1})e^{-\delta|x-y|}$ by Young's inequality and the Riesz–Thorin theorem ([4, p. 583]). (c) Both sides are the inverse of $-\Delta_{B_0} + m_r^2 + a_rP_X(B_0)$ on P-periodic functions: $P \subset \Gamma_r$ makes the hybrid block system P-periodic, and $e^{\mathrm{i}e_rB_0\cdot x}$ conjugates the operator to the $A=0$ operator on functions with the twisted periodicity $\psi(x+\gamma) = e^{-\mathrm{i}e_rB_0\cdot\gamma}\psi(x)$, whose kernel is the twisted image sum ([14, (B.3)] with phases). The images with $\gamma \neq 0$ contribute $\leq Ce^{-\delta_0(2M-\sqrt{3})}$ to every bound in (b), uniformly in B_0. □

Lemma 17.82 ([4, Lemma 2.2], [14, Lemma 5.8], for the hybrid flow)**.** *Let $\square$ be a periodic cell fitted by both lattices of side $2M$, and $\tilde{A} = A_0 + A'$ regular on $\square$, A_0 constant, with $|A'|_\infty, |\partial A'|_\infty \leq c'e_r^{\beta-1}$ and A' supported away from the boundary of the fundamental region. Then for $e_r^\beta \leq c_0(a,M)$, with c_0, c_1, c_2 uniform in r,*

$$\|G_r^X(\square,\tilde{A})f\|_{1,0} \leq c_1\|f\|_\infty, \qquad \|G_r^X(\square,\tilde{A})f\|_{1,\alpha} \leq c_1C_\alpha\log^{3\alpha}(1/\eta')\|f\|_\infty,$$

$$\|G_r^X(\square,\tilde{A})f\|_q,\ \|D_{\tilde{A}}G_r^Xf\|_q,\ \|G_r^XD_{\tilde{A}}^*f\|_q \leq c_2\|f\|_p$$

for $1 \leq p,q \leq \infty$ with $\frac{1}{p} - \frac{1}{q} < \frac{2}{3}$, where $\|\cdot\|_{1,\alpha}$ is the covariant Hölder norm [4, (2.14)] *with the field $\tilde{A}$ and $\|\cdot\|_{1,0}$ the sup norm plus the sup of the covariant derivative.*

Proof. [4, (2.24)–(2.26), (2.31)–(2.33)]: $G_r(\square,\tilde{A}) = \sum_{n\geq 0} G_r(\square,A_0)[V_rG_r(\square,A_0)]^n$ with $V_r = V_r(A',A_0)$ the vertex operator of [1, (3.16)] (= [13, (3.53)]); V_r is a first-order differential operator with coefficients $O(e_r|A'|, e_r|\partial A'|)$ ([4, (2.32)], using that A' vanishes near the boundary of the fundamental region) plus the pin part $a_r[Q(A_0)^*F_{2,r}(A',A_0) + F_{2,r}^*Q(A_0) + F_{2,r}^*F_{2,r}]$ with $(F_{2,r}(A',A_0)\phi)(y) = \sum_{x\in B(y)} \eta'^3(U(A'(\Gamma_{y,x}))-1)U(A_0(\Gamma_{y,x}))^{-1}\phi(x)$, $\|F_{2,r}\|_{\infty\to\infty} \leq \sup|U(A'(\Gamma))-1| \leq 2.1\,e_r|A'|_\infty$ by Lemma 17.73(i). Hence $\|V_rg\|_\infty \leq C(M)e_r^\beta\|g\|_{1,0}$. By Lemma 17.81(b),(c), $\|G_r(\square,A_0)g\|_{1,0} \leq c\|g\|_\infty$ with c uniform in r and A_0, and $\|G_r(\square,A_0)g\|_{1,\alpha} \leq cC_\alpha\log^{3\alpha}(1/\eta')\|g\|_\infty$ (the constant background is a twisted periodisation of the $A=0$ kernel, and the Hölder norm (2.14) with field A_0 is the plain Hölder norm after the local gauge $e^{\mathrm{i}e_rA_0\cdot x}$ on each unit cube). *Order of summation.* Put $g_0 := f$, $g_n := V_rG_r(\square,A_0)g_{n-1}$. Then $\|g_n\|_\infty \leq C(M)e_r^\beta\|G_r(\square,A_0)g_{n-1}\|_{1,0} \leq (C(M)c\,e_r^\beta)^n\|f\|_\infty$, so for $C(M)c\,e_r^\beta \leq \frac{1}{2}$, a condition uniform in r, the sum $g := \sum_n g_n$ converges in the sup norm with $\|g\|_\infty \leq 2\|f\|_\infty$, and $G_r(\square,\tilde{A})f = \sum_n G_r(\square,A_0)g_n = G_r(\square,A_0)g$ (the cell is finite, every operator is bounded, and the sums may be exchanged). Hence $\|G_r(\square,\tilde{A})f\|_{1,0} \leq 2c\|f\|_\infty$ and $\|G_r(\square,\tilde{A})f\|_{1,\alpha} \leq 2cC_\alpha\log^{3\alpha}(1/\eta')\|f\|_\infty$: the Hölder constant enters once, at the end, and never the smallness condition. (Summing in $\|\cdot\|_{1,\alpha}$, as [4, (2.33)] may with its uniform constant, would turn the condition into $e_r^\beta\log^{3\alpha}(1/\eta') \leq c_0$, which depends on r.) The norms (2.14) with the fields A_0 and $\tilde{A}$ differ on the cell by at most

$Ce_r^\beta\|\cdot\|_{1,0}$ ($D_{\tilde A}=U(A')D_{A_0}+F_{1,r}(A')$ and $|U(A'(\Gamma_{y,x}))-1|\le Ce_r|A'|_\infty|x-y|$ for the bounded-length paths of (2.14)), which changes c_1 by a factor $1+O(e_r^\beta)$. The $L^p\to L^q$ bounds are summed in the same order, with $D_{\tilde A}=U(A')D_{A_0}+F_{1,r}(A')$. □

Remark 17.83 (the Hölder constant). In Proposition 17.72(a) the Hölder clause [13, (3.8)] of $G_r^X(A)$, and the Hölder clauses of $H^X(A)$ and $\Gamma^X(A)$, hold with the constant $C_\alpha\log^{3\alpha}(1/\eta')$, $0<\alpha<1$ (Lemma 17.81, Lemma 17.82, and the random walk below); on slice j the constant is $C_\alpha\log^{3\alpha}(2+\ell_j/\eta')$ (Lemma 17.81(a)). The sup, decay and one-derivative bounds (3.6)–(3.7) and the $\|\cdot\|_{1,0}$ bounds have constants uniform in r, and so does every smallness condition. [13, (3.8)] asks for some $0<\alpha<1$ with some constant ([13, p. 656]). Every use of the clause in this section is either as a gain, where $\eta'^\alpha\log^{3\alpha}(1/\eta')\le C_{\alpha,\alpha'}\eta'^{\alpha'}$ and, slice by slice, $(\eta'/\ell_j)^\alpha\log^{3\alpha}(2+\ell_j/\eta')\le C_{\alpha,\alpha'}(\eta'/\ell_j)^{\alpha'}$ for $\alpha'<\alpha$ (Theorem 17.39, whose exponent θ_1 is fixed by α alone), or inside a per-flow bound that already allows a factor $(1+k)^p$ (Corollary 17.41), where $\log^{3\alpha}(1/\eta')=(r\ln 5)^{3\alpha}\le C(1+k)^{3\alpha}$.

Proof of Proposition 17.72(a). Random walk. Let Ω be Λ_X or a torus fitted by both lattices with $P\subset M\mathbb{Z}^3$, A regular on Ω. For $j\in\mathbb{Z}^3$ let $\square_j$ be the periodic cell of side $2M$ "centred" at Mj and fitted by both lattices (which is possible iff $25\mid 2M$, as the tori of the chain allow; [14, p. 345]: "blocks $\{\square_j\}$ which are small scale versions of D_J, that is they can be fitted by both the usual lattice and the rotated lattice"), h_j the partition of unity of [4, §2] ($h_j=1$ on the M-cube around Mj, supported in the $\frac43M$-cube, $\sum_j h_j^2=1$), and $\tilde A_j=A_0^{(j)}+\theta_jA'^{(j)}$ with $A_0^{(j)}=A(Mj)$ and the cutoff θ_j of [4, p. 575], so that $\tilde A_j=A$ on $\operatorname{supp}h_j$, is regular with constant $c'=O(Mc)$, and is constant near the boundary of the fundamental region of $\square_j$. With $G_0:=\sum_j h_jG_r^X(\square_j,\tilde A_j)h_j$ one computes, exactly as [4, (2.3)–(2.9)] and [14, (5.8)–(5.12)],

$$(-\Delta_A+m_r^2+a_rP_X(A))G_0=\mathbf1-\mathcal R,\qquad \mathcal R=\sum_j K_jG_r^X(\square_j,\tilde A_j)h_j,$$

$$(K_j\phi)(x)=\mathbf1_{\square_j}(x)\Big[\sum_{b\in\mathrm{st}(x)}(\partial h_j)(b)(D_A\phi)(b)+(\Delta h_j)(x)\phi(x)-(R_r(A,\partial h_j)\phi)(x)\Big],$$

where $R_r(A,\partial h)\phi(x)=\sum_{x'\in B(z(x))}\eta'^3U(A(\Gamma_{z,x}))U(A(\Gamma_{z,x'}))^{-1}(\partial h)(\Gamma_{x,x'})\phi(x')$ collects the commutator of the pin with the multiplication by h_j ([4, (2.8)]); $(\partial h)(\Gamma_{x,x'})=h(x')-h(x)$ for any path, so $|R_r(A,\partial h_j)\phi(x)|\le\sup|\partial h_j|\cdot2.1\cdot\sup_B|\phi|$ by Lemma 17.73(i). The three identities used — the product rule for $-\Delta_A$ ([4, (2.5)]), the commutator of the pin with a smooth function ([4, (2.8)]), and the agreement of the cell operator with the global one on $\operatorname{supp}h_j$ ([4, (2.6)]; for a periodic cell this holds because $\operatorname{supp}h_j$ stays at distance $\ge\sqrt3$ from the boundary of the fundamental region, so no block and no bond of $\operatorname{supp}h_j$ wraps) — hold for any block system with contours from the root to the point. Hence $G_r^X(\Omega,A)=\sum_{n\ge0}G_0\mathcal R^n$ in L^2 (Corollary 17.76 gives $\|\mathcal R\|_{2\to2}\le O(M^{-1})$ as in [4], since $\sup|\partial h_j|=O(M^{-1})$ and $\sup|\Delta h_j|=O(M^{-2})$), i.e. the path representation [4, (2.13)], [14, (5.56)].

Bounds. [4, (2.18)–(2.22)] verbatim: for x,x' in $\operatorname{supp}h_{\omega_0}$ and f supported in $\square_{\omega_n}$, each factor $\|K_{\omega_i}G_r^X(\square_{\omega_i},\tilde A_{\omega_i})h_{\omega_i}\|_{\infty\to\infty}$ is $\le O(M^{-1})c_1$ by Lemma 17.82 and the bound on K_j (c_1 the uniform $\|\cdot\|_{1,0}$ constant of Lemma 17.82: K_j takes one covariant derivative of its argument, so the choice of M below is uniform in r), the first factor is bounded in $\|\cdot\|_{1,\alpha}$ by Lemma 17.82 (the only place where $C_\alpha\log^{3\alpha}(1/\eta')$ enters, and only in the clause (3.8)), the number of paths of length n is $\le2^d(3^d)^{n-1}2^d$, and $n\ge M^{-1}\operatorname{dist}(\{x,x'\},\operatorname{supp}f)-2$; for M fixed with $3^dc_1O(1)M^{-1}\le e^{-1}$ the sum converges and gives [13, (3.7)–(3.8)] with $\delta_0=(2M)^{-1}$. The tail with $n>n_0$ is handled with the L^p bounds (2.17) as in [4, (2.21)]. The bounds (3.6) for $\Delta^{(r)}$ and $C^{(r)}$ are Proposition 17.77 and Corollary 17.78; the bounds for the next-step precision $\Delta^{(r)}+aL^{-2}\hat Q^*\hat Q$ and its restricted inverses $\hat C_\Lambda^{(r)}$ are Proposition 17.80.

Two-torus clause. For two domains Ω_1,Ω_2 sharing a fundamental region F as in Lemma 17.79 (or a torus against Λ_X), the representations coincide term by term for the paths whose cells do not meet the boundary of F, and the difference is bounded, as on [4, p. 579], by the same sum

restricted to paths meeting the boundary; such a path of n cells also joins x to $\operatorname{supp} f$ in its own domain, so $\sqrt{3}Mn \geq \max\{d_i(\{x,x'\},\operatorname{supp} f), \operatorname{dist}(\{x,x'\},\partial F)+\operatorname{dist}(\operatorname{supp} f,\partial F)\}-c$ for $i=1,2$, and the average of the two exponents gives the factor $e^{-(2\sqrt{3}M)^{-1}(d_\wedge(\{x,x'\},\operatorname{supp} f)+\operatorname{dist}(\{x,x'\},\partial F)+\operatorname{dist}(\operatorname{supp} f,\partial F))}$ for the difference of the G_r^X clauses (3.7)–(3.8), with $d_\wedge \leq d_i$ the smaller of the two quotient distances. For $\Delta^{(r)}$, $C^{(r)}$ and $\hat{C}_\Lambda^{(r)}$ the two-torus clause is Lemma 17.79 and Proposition 17.80. The metric $d_\wedge$ cannot be replaced by the distance of the lift (the remark after Lemma 17.79); the uses of the clause in this section (Lemma 17.40(N), the kernel sweep of Theorem 17.62, and Lemma 17.120(IV)) need only the decay on each torus in its own quotient distance together with the boundary factor, which is what is proved.

The remaining kernels. $H^X(A) = a_r G_r^X(A) Q_X(A)^*$: apply (3.7)–(3.8) to $f = w_z$ ($\|w_z\|_\infty = 1$, support in a block of diameter $\leq \sqrt{3}$). $B^X(A) = a_r^{-1} + Q_X(A) C_A Q_X(A)^*$: by the diamagnetic inequality $|C_A(x,y)| \leq C_0(x,y) = (-\Delta + m_r^2)^{-1}(x,y)$ entrywise ($-\Delta_A + m_r^2 = c - T_A$ with $|T_A| = T_0$ entrywise and $(c-T_0)^{-1} = \sum c^{-n-1} T_0^n \geq 0$), so $|B^X(z,w)| \leq a_r^{-1}\delta_{zw} + \sum_{x\in B(z), y\in B(w)} \eta'^6 C_0(x-y) \leq C(1+|z-w|)^{-1}$ ($d=3$). $\Gamma^X(A) = H^X B^X H^{X*}$: the fine-variable regularity is that of H^X, and $|\Gamma^X(x,y)| \leq \sum_{z,w} C e^{-\delta_0|x-z|}(1+|z-w|)^{-1} e^{-\delta_0|y-w|} \leq C(1+|x-y|)^{-1}$. $B^X(A)$ and $\Gamma^X(A)$ are not unit-decaying: $\Gamma^X(x,y) \approx C_A(x,y)$ for $|x-y| \gg 1$, at the coarse mass $m_r = m\delta_r \to 0$; [13, Thm. 3.3] never claims otherwise (it concerns G_r, $\Delta^{(r)}$, $C^{(r)}$). Where a renormalised family carries a line Γ^F or $(A^F)^{-1}$ at the top scale, either the coarse-slice decomposition of Corollary 17.36(b) is used, or, in the localised form [13, (3.55)], all vertex positions lie in the region F_v of Lemma 17.40, of diameter $\leq Cp_r$, where a line bounded by $C(1+|x-y|)^{-1}$ costs at most a factor Cp_r^3, absorbed in the field-norm polynomial of Proposition 17.38.

The slice clauses. Nothing in the argument above uses the level: the same proof at level $j \leq r$, in the unit scale of level j, gives (3.6)–(3.8) for the step-j operators G_j, $\Delta^{(j)}$, $C^{(j)}$, H_j of the hybrid flow with constants uniform in j (the first step, with the tiles as blocks, is covered by Lemma 17.75 and Lemma 17.81 with B_1^X in place of B_r^X; the steps $j \geq 2$ are the standard rotated steps acting on composite blocks, Lemma 17.74(iv)). King's derivation of the slice bounds [13, (3.63)–(3.65)] from Theorem 3.3 ([13, Prop. 3.7]: the slice $S_j = H_j C_j H_j^*$ at a regular background, the one- and two-ended contour clauses summed along contours of bounded length) therefore applies verbatim to the hybrid flow, with the covariant lattice derivatives and transported Hölder quotients of [3, (2.10)–(2.11)] at regular A; after the constant-background reduction of Lemma 17.40 the compared lines are the $A=0$ slices of Lemma 17.81, so there the plain form holds. For the contour clauses the contour lengths are those of Lemma 17.73(i).

Levels 0 *and* 1. In the form in which the hybrid lower bound (Proposition 17.113) consumes (a) at its first steps (precision bounds, precision decay, sup-decay of $C^{(0)}$ and $C^{(1)}$), (a) holds also at $j \in \{0,1\}$ by the same proofs, which use $r \geq 2$ nowhere: at $j=1$ the level-1 block is the tile, inside the ball of radius $\frac{\sqrt{3}}{2}$ about its centre, with an in-tile contour of length ≤ 1 in the unit scale of level 1, so Lemma 17.75, Corollary 17.76, Proposition 17.77 and Lemma 17.79 read verbatim (the first Neumann eigenvalue of the cube is then $100\sin^2(\pi/40) > \frac{1}{4}$), and the fine spacing $\eta' = \frac{1}{5}$ is a constant; at $j=0$ the precision $-\Delta_A + m_0^2 + aL^{-2}\hat{Q}^*\hat{Q}$ has finite range and its lower bound is the $(r+1)$-pin clause of Corollary 17.76 at $r=0$.

The Neumann variant. [13, Thm. 3.3] is printed for Neumann boundary conditions on a rectangular parallelepiped of large blocks. For the hybrid flow such a domain is a digitised rotated parallelepiped, on which neither the reflection representation [4, (2.42)] nor the axis-cube cover near the boundary is available; the Neumann variant is not claimed and not used: [14, §§4–5] decouple with periodic cells in the random walk, the localisation of Lemma 17.40 runs on the infinite lattice in the universal-cover form, and the only Neumann object of the chain, the boundary factor of δG_r, is replaced by the two-torus clause above and the same-domain clause Lemma 17.40(N).

The standard flow. For the flow Y every statement of (a) is [13, Thm. 3.3], proved in [4] on $\eta'\mathbb{Z}^3$ in the frame of Y_k, with King's uniform constants; its two-torus clause and restricted inverses are [14, Prop. 5.6] and Lemma 17.79, whose proof uses nothing specific to the hybrid blocks. This completes the proof of (a). □

The continuum extension of the Y kernels. The comparisons of Lemmas 17.92 and 17.93 below compare a sum over X bonds with a sum over Y bonds through Lemma 17.91, which needs a C^1 field on the block that equals the Y kernel on the Y lattice. The Fourier formula of Lemma 17.32 evaluated off the lattice (the sharp-zone extension) is not admissible for this: the symbol does not vanish on the faces of the zone, so the extension and its gradient have algebraic tails of order $\eta'^3/|r-z|$ and $\eta'^2/|r-z|$ off the lattice, not summable over a torus in $d=3$. The field used is the cubic-Hermite cardinal interpolation of the lattice kernels: it is C^1, equals the lattice kernel on Λ_Y, has exponentially decaying value and gradient by the lattice bounds alone, and obeys the multiplier comparison of Lemma 17.32 at the X sites with an explicit constant.

Definition 17.84 (cubic-Hermite cardinal extension)**.** $S(u) := 3u^2 - 2u^3$, $\psi(s) := S(1-|s|)\mathbf{1}_{|s|<1} = (1-|s|)^2(1+2|s|)\mathbf{1}_{|s|<1}$, $\Psi(v) := \prod_\alpha \psi(v\cdot f_\alpha/\eta')$ (the frame of Y_k). For F on $\Lambda_Y = R^{-1}(\eta'\mathbb{Z}^3)$ put $(I_S F)(r) := \sum_{y\in\Lambda_Y} \Psi(r-y)F(y)$, $r\in\mathbb{R}^3$. The extensions are $E^S(r,z) := I_S[H^Y(\cdot,z)](r)$ for the level-r lift (scalar and photon), $E^S_j(r,z) := I_S[H^Y_j(\cdot,z)](r)$ for the level-j lift, $\tilde S^S_j(r,y) := \sum_{z,z'} E^S_j(r,z)C^Y_j(z,z')\overline{H^Y_j(y,z')} = I_S[S^Y_j(\cdot,y)](r)$ for $y\in\Lambda_Y$ ($S^Y_j = H^Y_j C^Y_j H^{Y*}_j$, Definition 17.28), likewise for the photon slices, and $E^S\varphi := I_S[H^Y\varphi]$ for a coarse field φ (componentwise in any frame: I_S is scalar and linear). The extension acts in the first argument only; second arguments stay on Λ_Y.

Lemma 17.85 (cell identity)**.** *Write the frame coordinates of r as $f_\alpha\cdot r = \eta'(n_\alpha+u_\alpha)$, $n\in\mathbb{Z}^3$, $u\in[0,1)^3$, with cell corners $y_\sigma := R^{-1}\eta'(n+\sigma)$, $\sigma\in\{0,1\}^3$, and weights $\omega_\sigma(u) := \prod_\alpha \omega_{\sigma_\alpha}(u_\alpha)$, $\omega_0 := 1-S$, $\omega_1 := S$ (so $\omega_\sigma\ge 0$, $\sum_\sigma\omega_\sigma = 1$). Then* (i) *$I_SF(r) = \sum_\sigma \omega_\sigma(u)F(y_\sigma)$;* (ii) *$\partial_{f_1} I_SF(r) = S'(u_1)\sum_{\sigma'\in\{0,1\}^2}\omega_{\sigma'}(u_2,u_3)\,(\partial_1 F)(y_{(0,\sigma')})$ with the lattice forward difference $\partial_1 F(y) := [F(y+\eta' f_1)-F(y)]/\eta'$, and likewise for f_2, f_3;* (iii) *$|I_SF(r)| \le \max_\sigma |F(y_\sigma)|$ and $|\partial_{f_\alpha} I_SF(r)| \le \frac{3}{2}\max|\partial_\alpha F|$ over the four cell edges parallel to f_α, so $|\nabla I_SF(r)| \le \frac{3\sqrt3}{2}\max_{\text{edges}}|\partial F|$;* (iv) *$I_SF\in C^1(\mathbb{R}^3)$ and $I_SF = F$ on Λ_Y.*

Proof. Only the corners with $|(r-y)\cdot f_\alpha| < \eta'$ for all α contribute, i.e. the y_σ; for them $\psi((r-y_\sigma)\cdot f_\alpha/\eta') = \psi(u_\alpha-\sigma_\alpha)$, which is $S(1-u_\alpha) = 1-S(u_\alpha)$ for $\sigma_\alpha = 0$ and $S(u_\alpha)$ for $\sigma_\alpha = 1$ ($S(u)+S(1-u)=1$): this is (i). Differentiating (i) in $r\cdot f_1 = \eta'(n_1+u_1)$ gives $\eta'^{-1}S'(u_1)\sum_{\sigma'}\omega_{\sigma'}[F(y_{(1,\sigma')}) - F(y_{(0,\sigma')})]$, which is (ii). (iii) follows from $\omega\ge 0$, $\sum\omega = 1$, $0\le S'(u) = 6u(1-u)\le\frac32$. (iv) $\psi'(s) = -6s(1-|s|)$ on $|s|<1$ vanishes at $s = 0,\pm1$, so $\psi\in C^1(\mathbb{R})$, and the sum is locally finite; $\psi(m) = \delta_{m0}$ for $m\in\mathbb{Z}$. □

Lemma 17.86 (decay of the extensions)**.** *The lattice inputs are* [13, (3.6), (3.7), (3.63)] *for the standard flow: $|H^Y(y,z)|, |\partial H^Y(y,z)| \le Ce^{-\delta_0|y-z|}$ ((3.7) with $f = Q^*\delta_z$, supported in the block of z, $\|f\|_\infty\le 1$; for the photon, "the same bounds for the vector field operators",* [13, p. 656]*); at level j, in the unit scale of level j, $|H^Y_j|, |\partial H^Y_j| \le Ce^{-\delta_0|\cdot|}$ and $|C^Y_j(z,z')| \le Ce^{-\delta_0|z-z'|}$. Then for $0<\delta<\delta_0$, uniformly in j, r, η', the torus and $m_j\ge 0$, for all $r\in\mathbb{R}^3$, z, and $y\in\Lambda_Y$, in the unit scale of level r:*

(a) *$|E^S(r,z)|$, $\frac{2}{3\sqrt3}|\nabla_r E^S(r,z)| \le Ce^{\sqrt3\delta_0\eta'}e^{-\delta_0|r-z|}$, and the same for E^S_j in the unit scale of level j;*

(b) *$|\tilde S^S_j(r,y)| \le C\ell_j^{-1}e^{-\delta|r-y|/\ell_j}$, $|\nabla_r\tilde S^S_j(r,y)|$, $|\partial_y\tilde S^S_j(r,y)| \le C\ell_j^{-2}e^{-\delta|r-y|/\ell_j}$ and $|\nabla_r\partial_y\tilde S^S_j(r,y)| \le C\ell_j^{-3}e^{-\delta|r-y|/\ell_j}$ (∂_y a lattice difference in the second argument);*

(c) *$E^S(\cdot,z)$, $\tilde S^S_j(\cdot,y)$, $E^S\varphi$ are C^1 and equal $H^Y(\cdot,z)$, $S^Y_j(\cdot,y)$, $H^Y\varphi$ on Λ_Y.*

Proof. (a) Lemma 17.85(iii) with $F = H^Y(\cdot,z)$ and $|y_\sigma - r| \le \sqrt3\eta'$. (b) In the unit scale of level j put $F(y') := S^Y_j(y',y)$, resp. $\partial_y S^Y_j(y',y)$ (∂_y acts on the second argument and commutes with I_S). With $S^Y_j = H^Y_j C^Y_j H^{Y*}_j$, F and its lattice gradient in y' are sums $\sum_{z,z'}$ of products of three kernels with at most one lattice difference each, so with $e^{-\delta_0(|y'-z|+|z-z'|+|z'-y|)} \le e^{-\delta|y'-y|}e^{-(\delta_0-\delta)(|y'-z|+|z-z'|)}$ they are $\le C^3K_\delta e^{-\delta|y'-y|}$, $K_\delta := (\sum_{v\in\mathbb{Z}^3} e^{-(\delta_0-\delta)(|v|-\sqrt3)})^2$. Lemma 17.85(iii) at the spacing $\eta'/\ell_j \le \frac15$ adds $e^{\sqrt3\delta/5}$ and the factor $\frac{3\sqrt3}{2}$ for the gradient. Rescale by [13, (2.20)]:

kernel ℓ_j^{-1}, each derivative ℓ_j^{-1}. (c) Lemma 17.85(iv). Only lattice inputs with at most one derivative occur; no same-argument second derivative of a block-spin kernel is used. □

Lemma 17.87 (multiplier form at the X sites)**.** *Unit scale of level r; $x \in \Gamma_1$, $s \in \mathbb{R}^3$, $|s| \le \sqrt{13}\eta'$. In the notation of Lemma 17.32,*

$$E^S(x+s,z) = (2\pi)^{-3}\int_{V_\eta} dq\, e^{\mathrm{i}q\cdot(x-z)} u_{r-1}(q) M_S^Y(q;s) A^Y(q), \qquad M_S^Y(q;s) := \sum_{l\in L_2^Y} \frac{w_l(q;s)\, u_Y(q+l)}{\Delta_Y'(q+l)}, \qquad w_l(q;s$$

$\chi(Q) := \prod_\alpha \hat\psi(\eta' Q\cdot f_\alpha)$, $\hat\psi(t) = \int \psi(\sigma) e^{-\mathrm{i}t\sigma} d\sigma = 12(2-2\cos t - t\sin t)/t^4$, $\Lambda_Y^* = \frac{2\pi}{\eta'} R^{-1}\mathbb{Z}^3$; *and on V_η: $|w_l| \le W^3 \le 1.20$ for $l \ne 0$, $|w_0(q;s) - e^{\mathrm{i}q\cdot s}| \le 0.106\,\eta|q|$, $|M_S^Y(q;s) - \Delta_R^{-1}| \le 207\,\eta|q|^{-1}$. Consequently, for $x' = x + s \in \Lambda_X$ in the tile of x, $|H^X(x',z) - E^S(x',z)| \le C_E^S J_1 \eta$ with $C_E^S := 380a + 0.2a(C_*/20 + 8.3a)$ (C_* of Lemma 17.32), and the same at level j in the unit scale of level j.*

Proof. Poisson summation over Λ_Y ($\hat\Psi(Q) = \eta'^3\chi(Q)$): $\sum_y \Psi(r-y)e^{\mathrm{i}Q\cdot y} = \sum_L \chi(Q+L)e^{\mathrm{i}(Q+L)\cdot r}$. Inserted in $H^Y(y,z) = (2\pi)^{-3}\int_{V_{\eta'}} e^{\mathrm{i}Q\cdot(y-z)} h(Q) dQ$ (h periodic under Λ_Y^*, $e^{\mathrm{i}L\cdot z} = 1$) this gives $E^S(r,z) = (2\pi)^{-3}\int_{\mathbb{R}^3} e^{\mathrm{i}Q\cdot(r-z)}\chi(Q)h(Q)dQ$; splitting $\mathbb{R}^3$ into the translates $V_\eta + l + L$ and using $e^{\mathrm{i}(l+L)\cdot x} = 1$ gives the formula. Bounds: $\psi \ge 0$ gives $|\hat\psi| \le \hat\psi(0) = 1$; $\hat\psi(2\pi n) = 0$ for $n \ne 0$; for $|\tau| \le \pi/5$ and $n \ne 0$, $|2 - 2\cos\tau - (2\pi n + \tau)\sin\tau| \le \tau^4/12 + 2\pi|n||\tau| \le 1.004 \cdot 2\pi|n||\tau|$ and $|2\pi n + \tau| \ge 0.9 \cdot 2\pi|n|$, so $|\hat\psi(2\pi n + \tau)| \le 0.074|\tau|/|n|^3$ and $T(\tau) := \sum_{n\ne 0}|\hat\psi(\tau + 2\pi n)| \le 2\zeta(3)\cdot 0.074|\tau| \le 0.18|\tau|$ (below the cruder $0.213|\tau|$ is used); $1 - \hat\psi(t) \le t^2/15$ on $[0,\pi]$ (the series $1 - t^2/15 + t^4/560 - \dots$); $W := \sup_t \sum_n |\hat\psi(t + 2\pi n)| \le 1.06$ (a one-dimensional bound: the sum is 1 at the integers and its oscillation on a period is controlled by the bound on T). For $l \ne 0$: $|w_l| \le \prod_\alpha \sum_n |\hat\psi(t_\alpha + 2\pi n)| \le W^3$, $t_\alpha := \eta'(q+l)\cdot f_\alpha$. For $l = 0$ ($|t_\alpha| \le \pi/5$ on V_η, $t := \eta' q$): $|w_0 - e^{\mathrm{i}q\cdot s}| \le (1 - \chi(q)) + [\prod_\alpha(1 + T(t_\alpha)) - 1] \le |t|^2/15 + (e^{0.213\sum_\alpha |t_\alpha|} - 1) \le (0.073 + 0.454)|t| \le 0.106\,\eta|q|$ (convexity on $\sum|t_\alpha| \le 3\pi/5$). Main term: $|w_0 u_Y - 1| \le 0.106\eta|q| + |q||s| + (1 - u_Y) \le (0.106 + 0.722 + 0.218)\eta|q|$, over $\Delta_Y' \ge \frac{4}{\pi^2}|q|^2$: $\le 2.59\eta|q|^{-1}$; $|\Delta_Y'^{-1} - \Delta_R^{-1}| \le 1.12\eta|q|^{-1}$ (proof of Lemma 17.32); the 124 aliases: $\le 124\, W^3\eta^2/4 \le 37.2\eta^2 \le 203\eta|q|^{-1}$ ($|q| \le \sqrt{3}\pi/\eta$). So $|M_S^Y - \Delta_R^{-1}| \le 207\eta|q|^{-1}$, and with $|M^X - \Delta_R^{-1}| \le 172.1\eta|q|^{-1}$ at the same offset s (proof of Lemma 17.32), $|M^X - M_S^Y| \le 380\eta|q|^{-1}$. Then the multiplier step of Lemma 17.32 verbatim: $|M^X A^X - M_S^Y A^Y| \le 380a\eta|q|^{-1} + 0.2a\eta^2 A^Y |M_S^Y| \le C_E^S \eta|q|^{-1}$; integrate against $|u_{r-1}|$. □

Corollary 17.88 (decaying forms at the X sites)**.** *Unit scale of level r; $x' \in \Lambda_X$ in the tile of $x \in \Gamma_1$, $x'' = \tau x'$; $y' \in \Lambda_X$, $y'' = \tau y'$.*

(a) *(lifts) $|H^X(x',z) - E^S(x',z)| \le C(C_E^S J_1\eta)^{1/2} e^{-\delta|x-z|/2}$ and $|H^Y(x'',z) - E^S(x',z)| \le C(375a\, J_1\eta)^{1/2} e^{-\delta|x-z|/2}$; the same for the photon lifts and for E_j^S at level j (unit scale of level j, $\eta \to \eta_j = 5^{1-j}$).*

(b) *(slices) for $2 \le j < r$: $|S_j^X(x',y') - \tilde S_j^S(x',y'')| \le C(\eta'/\ell_j)^{1/2}\ell_j^{-1} e^{-\delta|x'-y'|/\ell_j}$.*

(c) *(slices, the derivative at the lattice end) for $2 \le j < r$: $|\partial_{y''} S_j^Y(x'',y'') - \partial_{y''}\tilde S_j^S(x',y'')| \le C(\eta'/\ell_j)^{1/2}\ell_j^{-2} e^{-\delta|x'-y'|/\ell_j}$.*

For $j \in \{0,1\}$ each term is bounded separately by Lemma 17.86 and the per-flow bounds of Corollary 17.36(a); there is no gain ($\ell_j \le 5\eta'$).

Proof. (a) Lemma 17.87 gives the sup bound $C_E^S J_1\eta$; the second form is the same computation with $M^Y(q)$ at the cube offset $r_Y = x'' - x$ in place of M^X: $|w_0 u_Y - e^{\mathrm{i}q\cdot r_Y} u_Y| \le (0.106 + c_\tau/5)\eta|q|$ and $\sum_{l\ne 0}|w_l - e^{\mathrm{i}(q+l)\cdot r_Y}| \le 124(W^3 + 1)$, giving $\le 375a\, J_1\eta$. The decay is Lemma 17.86(a) for E^S and the separate bounds $|H^F(x^F,z)| \le Ce^{-\delta|x^F - z|}$ of Lemma 17.32; take the geometric mean ($|x' - x| \le \sqrt{13}\eta'$). (b) In the unit scale of level j,

$$S_j^X(x',y') - \tilde S_j^S(x',y'') = \sum_{z,z'} \big[(H_j^X - E_j^S)(x',z)\, C_j^X(z,z')\overline{H_j^X(y',z')} + E_j^S(x',z)(C_j^X - C_j^Y)(z,z')\overline{H_j^X(y',z')} + E_j^S(x'$$

(a), the bound $|C_j^X - C_j^Y| \le C\eta_j e^{-\delta|z-z'|/2}$ of the proof of Lemma 17.35, Lemma 17.86(a) and the decaying form of Lemma 17.32 at matched lattice points; sum the exponentials and rescale ($\eta_j = 5\eta'/\ell_j$). This is Lemma 17.35 with E_j^S in place of $H_j^Y \circ \tau$ in the first argument. (c) The difference is $\sum_{z,z'}(H_j^Y(x'',z) - E_j^S(x',z))C_j^Y(z,z')\overline{\partial H_j^Y(y'',z')}$: the second form of (a), [13, (3.6)], and [13, (3.7)] for the lattice gradient at y''; one more factor ℓ_j^{-1} on rescaling. (Alternatively, by Lemma 17.86(c) the difference is $\mathcal{A}(x'') - \mathcal{A}(x')$ for $\mathcal{A} := \partial_{y''}\tilde{S}_j^S(\cdot, y'')$, so it is $\le c_\tau\eta' \sup|\nabla\mathcal{A}| \le C(\eta'/\ell_j)\ell_j^{-2}e^{-\delta|x'-y'|/\ell_j}$ by Lemma 17.86(b).) □

Lemma 17.89 (the slice derivative at matched points)**.** *Let $\alpha \in (0,1)$ be the Hölder exponent of* [13, (3.8)] *(as in Theorem 17.39) and put*

$$\theta_\partial := \frac{\alpha}{2(2+\alpha)}, \qquad \text{so that} \qquad \theta_1 = \frac{\alpha^2}{4(1+\alpha)^2} \le \theta_\partial \le \tfrac{1}{4}.$$

For $2 \le j < r$, in the unit scale of level r, let $\nabla_y^X S_j^X(x',y') := \sum_\mu \partial_\mu^X S_j^X(x',y')\, e_\mu$ be the vector of forward lattice differences of the X slice in its lattice end y', and $\nabla_y^Y S_j^Y(x'',y'') := \sum_\nu \partial_\nu^Y S_j^Y(x'',y'')\, f_\nu$ the same for the Y slice along the Y frame $f_\nu = R^{-1}e_\nu$, both read as vectors in the common frame (Lemma 17.29(iii)). Then for matched (x',y'), $(x'',y'') = (\tau x', \tau y')$,

$$\big|\nabla_y^X S_j^X(x',y') - \nabla_y^Y S_j^Y(x'',y'')\big| \;\le\; C\Big(\frac{\eta'}{\ell_j}\Big)^{\theta_\partial}\ell_j^{-2}e^{-\delta|x'-y'|/\ell_j},$$

with $C = C(\alpha)$ uniform in j, r and the torus; the same for the photon slices. For $j \in \{0,1\}$ each gradient is bounded separately by $C\ell_j^{-2}e^{-\delta|x'-y'|/\ell_j}$ ([13, (3.63)] for Y, Lemma 17.81(a) for X) and $(\eta'/\ell_j)^{\theta_\partial} \ge 5^{-1/4}$: there is no gain, and the factor is paid by a degree loss, as in Corollary 17.36(a).

Proof. Write $D_j := S_j^X - S_j^Y \circ \tau$ as in Corollary 17.36, so that $|D_j(p',q')| \le C_D(\eta'/\ell_j)^{1/2}\ell_j^{-1}e^{-\delta|p'-q'|/\ell_j}$ for every pair of matched points (Lemma 17.35; $j \ge 2$). The per-flow inputs are the gradient bound $|\nabla_y S_j^F(x,y)| \le C\ell_j^{-2}e^{-\delta|x-y|/\ell_j}$ and the Hölder clause of the gradient in the lattice end,

$$\big|\nabla_y S_j^F(x,y) - \nabla_y S_j^F(x,q)\big| \le C'\ell_j^{-2}\Big(\frac{|y-q|}{\ell_j}\Big)^{\alpha'}e^{-\delta\min(|x-y|,|x-q|)/\ell_j} \qquad (|y-q| \le 2\ell_j), \qquad \alpha' := \alpha/2,$$

for both flows. For $F = Y$ this is [13, (3.63)] ([13, Prop. 3.7], from [13, Thm. 3.3, (3.8)] by scaling) with King's constant, and $\alpha' < \alpha$ costs nothing on $|y-q| \le 2\ell_j$. For $F = X$ it is Lemma 17.81(a) read in the lattice end ($S_j^X = H_j^X C_j^X H_j^{X*}$ is symmetric), whose factor $C_\alpha \log^{3\alpha}(2 + \ell_j/\eta')\,(|y-q|/\ell_j)^\alpha$ is $\le C'(|y-q|/\ell_j)^{\alpha'}$: with $t := |y-q|/\ell_j \in [\eta'/\ell_j, 2]$ one has $\log(2+\ell_j/\eta') \le \log(2+1/t)$, and $t^{\alpha/2}\log^{3\alpha}(2+1/t)$ is bounded on $(0,2]$ (the device of Remark 17.83).

Fix a unit vector e_μ of the X frame and a multiple $\rho = N\eta'$ of η' with $\eta' \le \rho \le \ell_j$, to be chosen, and put $p' := y' + \rho e_\mu \in \Lambda_X$, $p'' := \tau p'$. All points below lie within $2\ell_j$ of y', so the decay factors of the inputs differ from $e^{-\delta|x'-y'|/\ell_j}$ by constants, which are not written.

The X side. Telescoping the forward differences along the segment, $S_j^X(x',p') - S_j^X(x',y') = \sum_{i=0}^{N-1}\eta'\,\partial_\mu^X S_j^X(x', y'+i\eta' e_\mu)$, and replacing each difference by the one at y' with the Hölder clause,

$$\Big|\rho^{-1}\big[S_j^X(x',p') - S_j^X(x',y')\big] - e_\mu\cdot\nabla_y^X S_j^X(x',y')\Big| \le C\ell_j^{-2}\Big(\frac{\rho}{\ell_j}\Big)^{\alpha'}e^{-\delta|x'-y'|/\ell_j}.$$

The Y side. $p''-y''$ is a vector of the Y lattice, $p''-y'' = \sum_\nu n_\nu\eta' f_\nu$ with $n_\nu\eta' = (p''-y'')\cdot f_\nu$, and $p''-y'' = \rho e_\mu + w$ with $w = (\tau p' - p') - (\tau y' - y')$, $|w| \le 2c_\tau\eta'$ (Definition 17.28). Join y'' to p'' by the lattice path that takes the $|n_1|$ steps along $\pm f_1$ first, then those along $\pm f_2$, then those along $\pm f_3$; it has at most $\sqrt{3}(\rho + 2c_\tau\eta')/\eta' + 3$ steps, and every point of it is within $\rho + 2c_\tau\eta' + \eta'$ of y''. Telescoping along the path (a step along $-f_\nu$ from q is $-\eta'\,\partial_\nu^Y S_j^Y(x'', q - \eta' f_\nu)$) and

replacing each difference by the one at y'' with the Hölder clause,

$$S_j^Y(x'',p'')-S_j^Y(x'',y'')=(p''-y'')\cdot\nabla_y^Y S_j^Y(x'',y'')+\mathrm{Rem},\qquad |\mathrm{Rem}|\le C\rho\,\ell_j^{-2}\Big(\frac{\rho+2c_\tau\eta'}{\ell_j}\Big)^{\alpha'}e^{-\delta|x'-y'|/\ell_j},$$

and $(p''-y'')\cdot\nabla_y^Y S_j^Y=\rho\,e_\mu\cdot\nabla_y^Y S_j^Y+w\cdot\nabla_y^Y S_j^Y$ with $|w\cdot\nabla_y^Y S_j^Y(x'',y'')|\le 2c_\tau\eta'\,C\ell_j^{-2}e^{-\delta|x'-y'|/\ell_j}$. Hence, with $(a+b)^{\alpha'}\le a^{\alpha'}+b^{\alpha'}$,

$$\Big|\rho^{-1}\big[S_j^Y(x'',p'')-S_j^Y(x'',y'')\big]-e_\mu\cdot\nabla_y^Y S_j^Y(x'',y'')\Big|\le C\ell_j^{-2}\Big[\Big(\frac{\rho}{\ell_j}\Big)^{\alpha'}+\Big(\frac{\eta'}{\ell_j}\Big)^{\alpha'}+\frac{\eta'}{\rho}\Big]e^{-\delta|x'-y'|/\ell_j}.$$

The comparison. The pairs $(x',y'),(x'',y'')$ and $(x',p'),(x'',p'')$ are matched, so

$$\rho^{-1}\Big|\big[S_j^X(x',p')-S_j^X(x',y')\big]-\big[S_j^Y(x'',p'')-S_j^Y(x'',y'')\big]\Big|\le\rho^{-1}\big(|D_j(x',p')|+|D_j(x',y')|\big)\le C\Big(\frac{\eta'}{\ell_j}\Big)^{1/2}\frac{\ell_j}{\rho}\,\ell_j^{-2}e^{-\delta|x}$$

Adding the three displays, with $s:=\rho/\ell_j\in[\eta'/\ell_j,1]$,

$$\big|e_\mu\cdot\big(\nabla_y^X S_j^X(x',y')-\nabla_y^Y S_j^Y(x'',y'')\big)\big|\le C\ell_j^{-2}e^{-\delta|x'-y'|/\ell_j}\Big[\Big(\frac{\eta'}{\ell_j}\Big)^{1/2}\frac{1}{s}+s^{\alpha'}+\Big(\frac{\eta'}{\ell_j}\Big)^{\alpha'}+\frac{\eta'}{\ell_j}\,\frac{1}{s}\Big].$$

Choose s as the smallest multiple of η'/ℓ_j that is $\ge(\eta'/\ell_j)^{1/(2(1+\alpha'))}$; since $1/(2(1+\alpha'))<1$ it lies in $[\eta'/\ell_j,1]$ and is $\le 2(\eta'/\ell_j)^{1/(2(1+\alpha'))}$. The first two terms are then $\le C(\eta'/\ell_j)^{\alpha'/(2(1+\alpha'))}$, and so are the last two, because $\alpha'\ge\alpha'/(2(1+\alpha'))$ and $1-\frac{1}{2(1+\alpha')}=\frac{1+2\alpha'}{2(1+\alpha')}\ge\frac{\alpha'}{2(1+\alpha')}$. With $\alpha'=\alpha/2$, $\frac{\alpha'}{2(1+\alpha')}=\frac{\alpha}{2(2+\alpha)}=\theta_\partial$. Summing over the three e_μ gives the vector bound; the photon slices are the scalar ones times the identity. The inequalities $\theta_1\le\theta_\partial\le\frac14$ are $\alpha(2+\alpha)\le 2(1+\alpha)^2$ and $2\alpha\le 2+\alpha$. □

The difference elements at a non-zero gauge field. After the constant-background reduction of Lemma 17.40, applied to both flows on their infinite lattices with the torus lines as exact periodisations, every propagator that is compared between the flows is an $A=0$ line $G_r(\Lambda_F,0)=\sum_{j<r}S_j^F$, exactly the kernels of Proposition 17.31, lemmas 17.32 and 17.35, and corollary 17.36, with all vertices in the region F_v, and the background enters only through the explicit coarse fields $A_r(w_1)\cdots A_r(w_q)$ of [13, (3.55)] (the same for both flows), the constant-background phases [13, (3.46)], the contour vertices [13, (3.24)–(3.26)] with their contour lines [13, (3.27)], the external lifted fields $\tilde A^{(r),F}$, and the per-flow remainders, which are bounded by Proposition 17.72(a) at the regular fields of Lemma 17.40(R). The elements that are not $A=0$ lines are compared here.

Lemma 17.90 (the lift of a constant field; the phases at matched vertices)**.** *The lift* [13, (2.18)] *of a constant coarse field B to either flow is the constant fine field $\varrho_r B$, $\varrho_r:=a_r/(a_r+\mu_0^2\delta_r^2)\le 1$, the same number in both flows (*[13, (3.43)]*). For matched $v'\in\Lambda_X$, $v''=\tau v'$ and the common root $y\in\Gamma_r$,*

$$\big|U(\varrho_r B(\Gamma^X_{y,v'}))-U(\varrho_r B(\Gamma^Y_{y,v''}))\big|\le e_r|B|\,c_\tau\eta'\le e\,p_r\mu_0^{-1}c_\tau\,\delta_r^{-1/2}\eta'$$

for B in the small-field class.

Proof. $Q_F^*\mathbf{1}_{\Gamma_r}=\mathbf{1}_{\Lambda_F}$ for the frame-covariant photon average (Lemma 17.19(iii): $(\hat Q^*\psi)(x)=\sum_\alpha\psi_\alpha(y(x))f_\alpha$, which for the frame components $\psi_\alpha=f_\alpha\cdot B$ of a constant B is B), and $(-\Delta+\mu_0^2\delta_r^2)\mathbf{1}=\mu_0^2\delta_r^2\mathbf{1}$, so $(M+a_rQ_F^*Q_F)\mathbf{1}=(\mu_0^2\delta_r^2+a_r)\mathbf{1}$ and $a_rG_r^{F,\mathrm{ph}}Q_F^*B=\varrho_rB$ for both flows. Both phases are $e^{\mathrm{i}e_r\varrho_rB\cdot(v-y)}$ with $v=v'$, v'' (Lemma 17.74(iii)); $|v'-v''|\le c_\tau\eta'$, $\varrho_r\le 1$, $|B|\le p_r(\mu_0\delta_r)^{-1}$ ([13, (3.2)]), $e_r=e\delta_r^{1/2}$. □

Lemma 17.91 (contour-line comparison)**.** *Let $\mathcal A$ be a C^1 vector field on a neighbourhood of the level-r block $z+R^{-1}[-\frac12,\frac12]^3$ with $|\mathcal A|\le P_0$, $|\nabla\mathcal A|\le P_1$ there. Let $a_b^X:=\mathcal A(x_b)\cdot e_\mu$ for the Λ_X-bond $b=(x_b,x_b+\eta' e_\mu)$ and $a_{b''}^Y:=\mathcal A(x_{b'})\cdot f_\alpha$ for the Λ_Y-bond $b''=(x_{b''},x_{b''}+\eta' f_\alpha)$ (site sampling, King's identification of bond and site fields). Then for $x'\in B_r^X(z)$, $x''=\tau x'$,*

$$\big|a^X(\Gamma^{X,(r)}_{z,x'})-a^Y(\Gamma^{Y,(r)}_{z,x''})\big|\;\le\;10\,(P_0+P_1)\,\eta'.$$

The same holds for the partial contours $\Gamma^{(j+1)}_{x_{j+1},x}$ of [13, (2.19)]*, with P_0, P_1 taken on the level-$(j+1)$ block, the sliver area $\le 1.5\eta'\ell_{j+1}+33\eta'^2$ in the flux term and the lengths $\le 1.5\ell_{j+1}$ (Y), $\le 2.1\ell_{j+1}$ (X) in the Riemann term: the bound is $\le (7.1P_0 + (6.6\ell_{j+1} + 66\eta')P_1)\eta' \le C(P_0 + P_1\ell_{j+1})\eta'$.*

Proof. Riemann step. For a bond b of length η' sampled at its initial point, $|\eta'\,\mathcal{A}(x_b)\cdot\hat b - \int_b \mathcal{A}\cdot dl| \le \eta'\cdot\eta' P_1$. Summing over the $\le 2.1/\eta'$ bonds of Γ^X and the $\le 1.5/\eta'$ bonds of Γ^Y: $|a^X(\Gamma^X) - \int_{\Gamma^X}\mathcal{A}| + |a^Y(\Gamma^Y) - \int_{\Gamma^Y}\mathcal{A}| \le 3.6\,P_1\eta'$. *Stokes step.* $\int_{\Gamma^X}\mathcal{A} - \int_{\Gamma^Y}\mathcal{A} = \oint_{\partial\Sigma}\mathcal{A} - \int_{[x',x'']}\mathcal{A} = \int_\Sigma \operatorname{curl}\mathcal{A}\cdot dS - \int_{[x',x'']}\mathcal{A}$ with the surface $\Sigma = \Sigma(z,x')$ of Lemma 17.73(iii) (piecewise planar, so Stokes' theorem applies piece by piece and the interior edges cancel); $|\operatorname{curl}\mathcal{A}| \le 2P_1$, $\operatorname{area}(\Sigma) \le 1.5\eta' + 33\eta'^2 \le 3\eta'$ for $r \ge 2$, and $|[x',x'']| \le c_\tau\eta'$. Total $\le (3.6+6)P_1\eta' + 7.1P_0\eta' \le 10(P_0+P_1)\eta'$. For partial contours the same decomposition applies to the sub-polygon up to level $j+1$ (Lemma 17.73(i),(iii)). □

Lemma 17.92 (contour vertices at the lifted background)**.** *Let $A'^F = \vartheta\tilde A^{(r),F} + za_F$ be the reduced background of the flow F (Lemma 17.40). Then for matched (y,x'), (y,x''),*

$$\big|U(A'^X(\Gamma^{X,(r)}_{y,x'})) - U(A'^Y(\Gamma^{Y,(r)}_{y,x''}))\big| \;\le\; e_r\big|A'^X(\Gamma^X_{y,x'}) - A'^Y(\Gamma^Y_{y,x''})\big| \;\le\; C\,e_r\,p_r^2\,\eta^{1/2},$$

and hence the kernels of $F^X_{2,r}(A'^X,0)$ and of $F^Y_{2,r}(A'^Y,0)\circ\tau$ ([13, (3.51)], the contour vertices (3.24)–(3.26)) differ entrywise by $\le Ce_rp_r^2\eta^{1/2}$ against their own size $\le Ce_rp_r^2$: a relative gain $\eta^{1/2}$, and an absolute gain $\eta^{1/2}e\,p_r^2\delta_r^{1/2}$. The products (3.24)–(3.26)×(3.26) at the same y inherit the bound.

Proof. Let $\mathcal{A} := \vartheta\,E^S_{\rm ph}\tilde A_r + za^{\rm c}$, where $E^S_{\rm ph}\tilde A_r := I_S[H^{Y,{\rm ph}}\tilde A_r]$ is the cubic-Hermite extension of the photon lift of $\tilde A_r$ along the flow Y (Definition 17.84; componentwise in any frame) and $a^{\rm c}$ the smooth Cameron–Martin form (Lemma 17.21). $\mathcal{A}$ is C^1 (Lemma 17.85(iv)). By Lemma 17.85(iii) and Lemma 17.40(R), $|E^S_{\rm ph}\tilde A_r(r)| \le \max_\sigma|\tilde A^{(r),Y}(y_\sigma)| \le Cp_r(|r-x_0|+\sqrt{3}\eta')$ and $|\nabla E^S_{\rm ph}\tilde A_r(r)| \le \frac{3\sqrt3}{2}\max_{\rm edges}|\partial\tilde A^{(r),Y}| \le Cp_r$; with Lemma 17.21 for $a^{\rm c}$, $P_0,P_1 \le Cp_r^2$ on the vertex region (of diameter $\le Cp_r$). Then $a^Y_{b''} = \mathcal{A}(x_{b''})\cdot f_\alpha$ exactly (Lemma 17.86(c); Lemma 17.21 for a_Y), while

$$|a^X_b - \mathcal{A}(x_b)\cdot e_\mu| \le \vartheta\sum_z |H^{X,{\rm ph}}(x_b,z) - E^S_{\rm ph}(x_b,z)||\tilde A_r(z)| + |a_{X,\mu}(x_b) - a^{\rm c}\cdot e_\mu| \le C(C_E^S J_1\eta)^{1/2}p_r^2 + C\eta'^2$$

by Corollary 17.88(a) at the X site x_b (the decaying form; $|\tilde A_r(z)| \le p_r(|z-x_0|+1)$, the contributions from far away as in Lemma 17.40(R)) and Lemma 17.21. Summing the lift mismatch with weight η' over the $\le 2.1/\eta'$ bonds of Γ^X gives $\le C\eta^{1/2}p_r^2$, and Lemma 17.91 gives $\le 10(P_0+P_1)\eta' \le Cp_r^2\eta'$ for the rest. Multiply by e_r for the phases ($|U(a)-U(a')| \le e_r|a-a'|$), and by η'^3 per summand of (3.51). □

Lemma 17.93 (the contour lines [13, (3.27)])**.** *For $0 \le j < r$ let $S_j^{F,{\rm ph}}$ be the stage-j photon slice kernel (an $A=0$ kernel: the scalar slice S_j^F times the identity, frame-free), and let $\mathcal{L}_j^F(x;y) := \sum_{b_1\in\Gamma^{F,(j+1)}_{x_{j+1},x}}\eta'\,S_j^F(x_{b_1},y)\,\hat b_1 \in \mathbb{R}^3$ ($\hat b_1$ the direction of b_1, $y\in\Lambda_F$) be King's contour line (3.27) as a vector: its component along the direction of the photon bond $b = (y, y+\eta'\hat e)$ is $G_{(j)}(\Gamma^{(j+1)}_{x_{j+1},x}, b)$ of* [13, (3.27)]*. Lines are compared as vectors (Lemma 17.29(iii)). Then for matched (x',y'), $(x'',y'') = (\tau x',\tau y')$,*

$$\big|\mathcal{L}_j^X(x';y') - \mathcal{L}_j^Y(x'';y'')\big| \;\le\; C\Big(\frac{\eta'}{\ell_j}\Big)^\theta e^{-\delta\,\mathrm{dist}(\Gamma^{(j+1)},y')/\ell_j}, \qquad \theta = \tfrac12,$$

against the unweakened bound $Ce^{-\delta\,\mathrm{dist}\,/\ell_j}$ of each line: a slice-j line weakened by the degree θ in the sense of Theorem 17.39, with the prefactor η'^θ. With one lattice derivative at the lattice end y (the gradients of the two lines read as vectors in the common frame, Lemma 17.89) the same holds with one more factor ℓ_j^{-1} and, in place of $(\eta'/\ell_j)^{1/2}$, the gain $(\eta'/\ell_j)^{\theta_\partial}$ of Lemma 17.89, $\theta_\partial = \alpha/(2(2+\alpha)) \ge \theta_1$: a slice-$j$ line with one derivative, weakened by the degree θ_∂.

Proof. Keep the second argument of the Y kernel at the lattice point y'' throughout (no extension in the second argument is used) and write

$$\mathcal{L}_j^X(x';y')-\mathcal{L}_j^Y(x'';y'')=\sum_{b_1\in\Gamma^X}\eta'\big[S_j^X(x_{b_1},y')-\tilde S_j^S(x_{b_1},y'')\big]\hat b_1+\Big[\sum_{b_1\in\Gamma^X}\eta'\tilde S_j^S(x_{b_1},y'')\hat b_1-\sum_{b_1''\in\Gamma^Y}\eta' S_j^Y(x_{b_1''},y'')\hat b_1''\Big].$$

First sum ($2\le j<r$): Corollary 17.88(b) at the X sites x_{b_1}, per bond $\le C(\eta'/\ell_j)^{1/2}\ell_j^{-1}e^{-\delta|x_{b_1}-y'|/\ell_j}$; with weight η' over the $\le 10.5\ell_j/\eta'$ bonds of the partial X contour (Lemma 17.73(i)): $\le C(\eta'/\ell_j)^{1/2}e^{-\delta\,\mathrm{dist}(\Gamma^X,y')/\ell_j}$. *Bracket*: for a unit vector u, $u\cdot[\,\cdot\,]=a^X(\Gamma^X)-a^Y(\Gamma^Y)$ for the field $\mathcal{A}_u(r):=\tilde S_j^S(r,y'')\,u$, which is C^1 (Lemma 17.86(c)), whose Y samples are exactly $S_j^Y(x_{b''},y'')\,u\cdot f_\alpha$, and which has $P_0\le C\ell_j^{-1}e^{-\delta\,\mathrm{dist}(\Gamma,y'')/\ell_j}$, $P_1\le C\ell_j^{-2}e^{-\delta\,\mathrm{dist}(\Gamma,y'')/\ell_j}$ on the level-$(j+1)$ block (Lemma 17.86(b); the block's diameter $\sqrt{3}\ell_{j+1}$ and $|y'-y''|\le c_\tau\eta'$ change only the constant). Lemma 17.91 for partial contours gives $\le C(P_0+P_1\ell_{j+1})\eta'\le C(\eta'/\ell_j)e^{-\delta\,\mathrm{dist}\,/\ell_j}$ (flux, end segment and Riemann terms together: the kernel difference along the contour plus the flux of the slice kernel through the sliver, as Corollary 17.36(c) states); u over an orthonormal basis gives the vector bound. The matching $y'\leftrightarrow y''$ of the second argument sits inside Corollary 17.88(b). *The derivative version* ($2\le j<r$). With the lattice derivative on the lattice end, split the first sum at the matched Y point: $\sum_{b_1\in\Gamma^X}\eta'\big[\nabla_y^X S_j^X(x_{b_1},y')-\nabla_y^Y S_j^Y(\tau x_{b_1},y'')\big]\hat b_1+\sum_{b_1\in\Gamma^X}\eta'\big[\nabla_y^Y S_j^Y(\tau x_{b_1},y'')-\nabla_y^Y\tilde S_j^S(x_{b_1},y'')\big]\hat b_1$ (gradients as vectors, the matrix $[\,\cdot\,]\hat b_1$ contracted on the line's index). The first sum is Lemma 17.89 at the matched pairs (x_{b_1},y'), $(\tau x_{b_1},y'')$, per bond $\le C(\eta'/\ell_j)^{\theta_\partial}\ell_j^{-2}e^{-\delta|x_{b_1}-y'|/\ell_j}$, hence $\le C(\eta'/\ell_j)^{\theta_\partial}\ell_j^{-1}e^{-\delta\,\mathrm{dist}(\Gamma^X,y')/\ell_j}$ after the sum over the $\le 10.5\ell_j/\eta'$ bonds with weight η'; the second sum is Corollary 17.88(c) per bond, $\le C(\eta'/\ell_j)^{1/2}\ell_j^{-1}e^{-\delta\,\mathrm{dist}\,/\ell_j}$ after the same sum; and the bracket is Lemma 17.91 for the C^1 field $\partial_{y''}\tilde S_j^S(\cdot,y'')\,u$ (Lemma 17.86(b),(c): $P_0\le C\ell_j^{-2}e^{-\delta\,\mathrm{dist}\,/\ell_j}$, $P_1\le C\ell_j^{-3}e^{-\delta\,\mathrm{dist}\,/\ell_j}$), $\le C(\eta'/\ell_j)\ell_j^{-1}e^{-\delta\,\mathrm{dist}\,/\ell_j}$. The weakest gain is the first sum's, $(\eta'/\ell_j)^{\theta_\partial}$. $j\in\{0,1\}$: each line, also with the derivative, is bounded separately by $Ce^{-\delta\,\mathrm{dist}\,/\ell_j}$ (times ℓ_j^{-1} with the derivative) ([13, (3.64)] for Y; Lemma 17.81(a) summed along $\le 10.5\ell_j$ for X), and $(\eta'/\ell_j)^{1/2}\ge 5^{-1/2}$: there is no gain and the factor η'^{θ} is paid by a degree loss, exactly as in Corollary 17.36(a). □

Lemma 17.94 (double contour lines)**.** *For $0\le j<r$ and $x_1,x_2\in\Lambda_F$ let*

$$\mathcal{D}_j^F(x_1;x_2):=\sum_{b_1\in\Gamma^{F,(j+1)}_{x_1,j+1,x_1}}\ \sum_{b_2\in\Gamma^{F,(j+1)}_{x_2,j+1,x_2}}\eta'^2 S_j^F(x_{b_1},x_{b_2})\,\hat b_1\cdot\hat b_2,$$

the line joining two contour factors of [13, (3.26)] *(at one vertex with $m\ge 2$, at the two factors of a product (3.26)×(3.26), or at two vertices). Then $|\mathcal{D}_j^F(x_1;x_2)|\le C\ell_j e^{-\delta|x_{1,j+1}-x_{2,j+1}|/\ell_j}$ for both flows, and at matched points $|\mathcal{D}_j^X(x_1';x_2')-\mathcal{D}_j^Y(x_1'';x_2'')|\le C(\eta'/\ell_j)^{1/2}\ell_j e^{-\delta|x_{1,j+1}-x_{2,j+1}|/\ell_j}$. In Bałaban's degree $D(H)=\sum_l d_l+\sum_v a_v+3(|V(H)|-1)$ ([3, (2.1)–(2.2)], with the two summation points of a block product as vertices, as in Corollary 17.41) a double line has $d_{l_0}=+1$, and every connected all-scale-summed subgraph H containing it has $D(H)\ge 1$: the double line is never the binding constraint of the weakened-element bound.*

Proof. $\mathcal{D}_j^F(x_1;x_2)=\sum_{b_2}\eta'\hat b_2\cdot\mathcal{L}_j^F(x_1;x_{b_2})$. The size bound: lengths $\le 10.5\ell_j$ (X), $\le 7.5\ell_j$ (Y) (Lemma 17.73(i)), Lemma 17.81(a), [13, (3.63)]; since both contours lie in open rotated cubes of side $5\ell_j$ about their roots, $e^{-\delta|x_{b_1}-x_{b_2}|/\ell_j}\le e^{5\sqrt{3}\delta}e^{-\delta|x_{1,j+1}-x_{2,j+1}|/\ell_j}$. For $j\ge 2$ put $\mathcal{V}(r):=\sum_{b_1''}\eta'\,I_S[S_j^Y(x_{b_1''},\cdot)](r)\,\hat b_1''$ (extension in the second argument); then

$$\mathcal{D}^X-\mathcal{D}^Y=\sum_{b_2}\eta'\hat b_2\cdot[\mathcal{L}_j^X(x_1';x_{b_2})-\mathcal{L}_j^Y(x_1'';\tau x_{b_2})]+\sum_{b_2}\eta'\hat b_2\cdot[\mathcal{V}(\tau x_{b_2})-\mathcal{V}(x_{b_2})]+[a^X(\Gamma_2^X)-a^Y(\Gamma_2^Y)]_{\mathcal{A}=\mathcal{V}}$$

exactly (Lemma 17.29(i); $\tau x_{b_2}\in\Lambda_Y$); the three terms are bounded by Lemma 17.93 summed along Γ_2^X, by $c_\tau\eta'\sup|\nabla\mathcal{V}|$ with $|\nabla\mathcal{V}|\le C\ell_j^{-1}$ (Lemma 17.85(iii), Lemma 17.86(b)), and by Lemma 17.91 for partial contours with $P_0\le C$, $P_1\le C\ell_j^{-1}$. For $j\le 1$ use the size bounds and $(\eta'/\ell_j)^{1/2}\ge 5^{-1/2}$. Degree: the size $C\ell_j$ against ℓ_j^{-1} for a slice line ($d_l=-1$) and ℓ_j^0 for a

one-ended contour line ($d_l = 0$) gives $d_{l_0} = +1$ (Corollary 17.41: $d_l = -1 + \#(\text{contour ends})$). For a connected all-scale-summed H containing l_0: if $H \setminus l_0$ is connected (or l_0 is a self-loop), $D(H) = D(H \setminus l_0) + 1 \geq 1$, because $D \geq 0$ for every all-scale-summed subgraph that does not contain l_0 (the per-flow renormalised structure, Corollary 17.41); if l_0 is a bridge between H_1 and H_2, $D(H) = D(H_1) + D(H_2) + 1 + 3 \geq 4$. So $D(H) - \theta \geq \frac{1}{2}$ for $\theta = \frac{1}{2}$. □

Proposition 17.95 (every difference element at $A \neq 0$)**.** *In the setting of Proposition 17.72(b), after Lemma 17.40, order the elements of a renormalised family consistently across the two flows and telescope as in Proposition 17.38. Every difference element $e_i^X - e_i^Y$ is of one of the following types, with the stated bound:*

(i) *an $A = 0$ line: a fine slice S_j^F, $j < r$ (Corollary 17.36(a): $C_D \eta'^{\theta} \ell_j^{-1-\theta} e^{-\delta|x'-y'|/\ell_j}$, $\theta = \frac{1}{2}$), a coarse slice or Γ_n^F (Corollary 17.36(b): $C\eta^{1/2}\ell_j^{-1} e^{-\delta|x'-y'|/\ell_j}$), the coarse multiplier (Proposition 17.31: $0.2a\eta^2$ relative), on Λ_F;*

(ii) *a contour line (3.27): Lemma 17.93, gain $(\eta'/\ell_j)^{1/2}$; with one lattice derivative at its lattice end, gain $(\eta'/\ell_j)^{\theta_\partial}$, $\theta_\partial = \alpha/(2(2+\alpha))$ (Lemma 17.89);*

(iii) *a contour vertex (3.24)–(3.26): Lemma 17.92, entrywise $\leq C e_r p_r^2 \eta^{1/2}$, relative gain $\eta^{1/2}$;*

(iv) *an external lifted field $\tilde{A}^{(r),F}$, $H^F \phi_r$ or its covariant derivative: Lemmas 17.32 and 17.33, $\leq C\eta^{1/2}$ times the field norm;*

(v) *a constant-background phase (3.46): Lemma 17.90, $\leq e\, p_r \mu_0^{-1} c_\tau \delta_r^{-1/2} \eta'$;*

(vi) *a hopping vertex of $V_r(A', 0)$ or a vertex function (couplings, partition of unity, the Cameron–Martin form): Lemma 17.29(ii),(iii), unchanged;*

(vii) *a double contour line, joining two contour factors of (3.26): Lemma 17.94, size $\leq C\ell_j$, relative gain $(\eta'/\ell_j)^{1/2}$ on slice j.*

Each weakened element therefore obeys the bound of its type weakened by the degree $\theta = \frac{1}{2}$ (by $\theta_\partial \geq \theta_1$ for a contour line carrying a lattice derivative, which is all Theorem 17.39 asks) with a constant $\leq C\, p_r^2\, \delta_r^{-1/2}$ (the worst of (iii) and (v)); the per-flow remainders (Lemma 17.40(2)) contribute $(C e_r p_r^2)^{n+1} + e^{-\delta' p_r}$ per graph.

Proof. Lemma 17.40 lists what carries the background after the reduction; the compared lines are $A = 0$ lines, the contour lines, one-ended (3.27) and double (both ends on contours), are the only lines whose shape depends on the flow, and the only background-dependent vertices are the contour vertices (3.24)–(3.26) ([13, p. 658]: "The remaining vertices occur when the renormalization transformation of the scalar field is expanded about $A' = 0$") and the phases (3.46). The external coarse fields $A_r(w_i)$ of (3.55) are common. The bounds are the cited lemmas. □

Proof of Proposition 17.72(b). The difference elements $\Delta\Gamma$, ΔH, $\Delta(A^{-1})$ at regular A are, after the reduction, the $A = 0$ differences of Corollary 17.36 (types (i) and (iv) of Proposition 17.95), with the factor $\eta'^{1/2}$. The contour vertices at the reduced background differ entrywise by a relative $\eta^{1/2}$ against their size $C e_r p_r^2$ (Lemma 17.92); the constant-background phases at matched vertices differ by $\leq e\, p_r \mu_0^{-1} c_\tau \delta_r^{-1/2} \eta'$ (Lemma 17.90); the per-slice pieces carried by the contour vertex, with one end on a contour ([13, (3.27)]) or both ends on contours, obey the slice-j bounds [13, (3.63)–(3.65)] in both flows ((a) for the hybrid flow, [13, Prop. 3.7] for the standard flow) with the relative mismatch $(\eta'/\ell_j)^{1/2}$ (Lemmas 17.93 and 17.94), the one-ended lines also with one lattice derivative at the lattice end, there with the relative mismatch $(\eta'/\ell_j)^{\theta_\partial}$ of Lemma 17.89 (Lemma 17.93; a double line has no lattice end). A constant background is pure gauge on the infinite lattice and gives the same phase, up to the matching segment, for every contour (Lemma 17.90); what would break (b) are mis-paired contours or mis-rooted blocks, both excluded by Definition 17.15 and Convention 17.17(i). The gains are not uniform in the size of the constant part of the background: they carry the small-field bound [13, (3.2)], as King's own reduction does. □

Proposition 17.96 (King's upper bound for the rotated lattice)**.** *For Y_k on T and every real smooth periodic source $s = (zG, h)$, with the free-boundary convention of Definition 15.6 and the*

centred blocks of Convention 17.17,

$$\mathsf{Z}_{Y_k,T}(s) \le e^{C\delta_r^q|T|}\int \chi_r\, e^{-S_T^{(r)}} + e^{-p(\delta_r)^2+C|T|}\,\varepsilon^{-q},$$

$C = C(s,|T|)$, $q = q(s) < \infty$, *with* $q = 0$ *when* $h = 0$.

Proof. On $T_2 = R^{-1}D_J$, which is aligned in Y's frame, the passive rotation R (Lemma 15.9(b)) maps (Y_k, T_2) to (X_k, D_J) with its standard frame, so the statement is King's Theorem 3.1 [13, (3.4)], with the additive large-field term in the form in which King uses it ([13, (3.11)], [14, (4.7)]), on King's torus at block factor 5: this is Theorem F.2(iii) with Lemma 5.1, read with the torus-convention counterterms replaced by the free-boundary ones at a constant-shifted source (Proposition 17.121(b)) and with corner-rooted blocks replaced by centred ones (Convention 17.17(i)); King's constant depends on g, h and $|T|$ and is uniform under a shift of the scalar source by a bounded constant, which is the uniformity of [13, Thm. 3.1] in the mass over a compact mass set ([13, p. 655]). On $T_1 = D_J$ the frame torus of Y_k is RD_J, skew in Y's frame at every scale; this instance is Proposition 17.117. For a walled source ($\int h \neq 0$) the constant of the large-field term inherits [13, (2.23)], which fails as printed when $\int h \neq 0$ (Remark 16.1); what the inductive proof supplies is a bounded h-linear c-number per step, so the large-field term is read with the polynomial envelope factor ε^{-q}, which Proposition 17.134 absorbs. □

The cumulant bound for the step Gaussian. The fluctuation integral of every block-spin step of both flows has the form $\int \chi_b\, e^V d\mu_P$: μ_P is a centred Gaussian whose precision is coercive and has exponentially decaying off-diagonal entries, V is a real polynomial with exponentially decaying coefficients, and χ_b is a product over sites of indicators of a fixed convex body scaled by the threshold b. Theorem 17.112 below is the order-$\bar{n}$ cumulant expansion of such an integral, with an error proportional to the volume of the support of V and, in the lower bound, a normalisation term proportional to $|\Lambda|$. The printed cluster-expansion lemma of Benfatto and coauthors, which King cites for this step, is stated for the nearest-neighbour Markov field with a cutoff growing with the distance to the support of the interaction; its proof uses the Markov property and the nearest-neighbour formulas for conditional Gaussians in four places. The proof below keeps its architecture (cells with corridors, four shifted pavements, the cutoff moved from the corridors into the cells) and replaces those four uses by the decoupling and conditioning lemmas of this part, which hold for any coercive precision with exponentially decaying entries.

The setting. Let $\mathbb{L}$ be one of: the lattice $\mathbb{Z}^3$; a cubic torus $(\mathbb{Z}/N\mathbb{Z})^3$; a skew torus $\mathbb{Z}^3/mG_\pm$, $m \in \mathbb{N}$, where $G_+ := \langle (40,-30,0),(30,40,0),(0,0,50)\rangle$ and G_- is its mirror image under $x_2 \mapsto -x_2$. These are the frame period lattices of the skew tori of Proposition 17.117 in the units of a step lattice: $\Gamma' = R^{\mathrm{T}}(50MJ\,\mathbb{Z}^3) = MJ\,G_+$ for RD_J in Y's frame and $RP_J = MJ\,G_-$ for RD_J in X's frame, so $m = MJ$ divided by the spacing of the step, an integer because every block lattice contains P_J (Lemma 17.14(iii)). Distances are quotient Euclidean distances, $|x-y| := \min_g |\tilde{x}-\tilde{y}-g|$ over the period lattice (a metric), and $\operatorname{dist}(x,S) := \min_{y\in S}|x-y|$. $\Lambda \subset \mathbb{L}$ is finite, $z = (z_x)_{x\in\Lambda}$ with $z_x \in \mathbb{R}^n$, and a complex Hermitian operator on $\mathbb{C}^\Lambda$ is read in its real form (blocks $\left(\begin{smallmatrix}\mathrm{Re} & -\mathrm{Im}\\ \mathrm{Im} & \mathrm{Re}\end{smallmatrix}\right)$), which preserves the spectrum and the block operator norms. The hypotheses are:

(R1) $\mu_P := N(0,P^{-1})$ with P real symmetric on $(\mathbb{R}^n)^\Lambda$ and $P \ge \gamma_0 > 0$ (no upper bound is needed);

(R2) $\|P(x,y)\| \le c_P e^{-\delta_P|x-y|}$ for $x \neq y$ (block operator norms); the diagonal blocks are arbitrary;

(R3) $V = \sum_{Y\subset J} V_Y$ with $J \subset \Lambda$, where V_Y is a real polynomial in $(z_y)_{y\in Y}$ of degree at most D, without constant term, each of whose monomials involves every site of Y (its *support*); $\ell(Y)$ is the length of a shortest connected graph on a vertex set containing Y (edges between points of $\mathbb{L}$, of length their distance); $|V_Y|_1$ is the sum of the moduli of the coefficients; for $\theta \ge 0$ and for any polynomial $F = \sum_Y F_Y$ sorted by supports in this way,

$$\|F\|_\theta := \sup_{x\in\mathbb{L}} \sum_{Y\ni x} e^{\theta\ell(Y)}|F_Y|_1, \qquad \mathsf{A} := \|V\|_\varsigma \ \text{ for a fixed rate } \varsigma > 0, \qquad V_R := \sum_{Y\subset R} V_Y;$$

(R4) $\chi_\beta(z) := \prod_{x\in\Lambda} \mathbf{1}\{z_x \in \beta K\}$ with $K \subset \mathbb{R}^n$ a closed symmetric convex body, $\bar{B}(1) \subset K \subset \bar{B}(c_K)$: the Euclidean ball ($c_K = 1$) or Bałaban's two-component cutoff $K = \{|z^A| \le 1,\ |z^\phi| \le 1\}$, $c_K = \sqrt{2}$, which restricts the photon and the scalar fluctuation at each site separately.

$\kappa_\mu(F_1,\dots,F_k)$ denotes the joint cumulant under a law μ and $\kappa_\mu(F;k) := \kappa_\mu(F,\dots,F)$ the k-th cumulant, the truncated expectation $\langle F^k\rangle^T$. For $X \subset \Lambda$, P_{XX} is the principal block, $G_X := (P_{XX})^{-1}$ and $C := P^{-1} = G_\Lambda$; for $Y := \Lambda\setminus X$ the conditional law of z_X given z_Y under μ_P is $N(-G_X P_{XY} z_Y,\, G_X)$, by completing the square in the exponent. $\chi_\beta^X := \prod_{x\in X} \mathbf{1}\{z_x \in \beta K\}$.

Lemma 17.97 (lattice facts)**.** (i) *$x \mapsto |x - y_0|$ and $x \mapsto \operatorname{dist}(x,S)$ are 1-Lipschitz.* (ii) *For every $f \ge 0$ and $x \in \mathbb{L}$, $\sum_{y\in\mathbb{L}} f(|x-y|) \le \sum_{z\in\mathbb{Z}^3} f(|z|)$.* (iii) *For $x,y \in Y$, $|x-y| \le \ell(Y)$; a shortest connected graph for Y joins any two of its vertices by a path of length at most $\ell(Y)$; and $\ell(Y') \le \ell(Y)$ for $Y' \subset Y$.* (iv) *For $\theta \le \varsigma$ and $r \ge 0$ the part of V made of the monomials of length $\ell(Y) \ge r$ has $\|\cdot\|_\theta \le \mathsf{A} e^{-(\varsigma-\theta)r}$; in particular $\sum_{Y\ni x,\ \ell(Y)\ge r} |V_Y|_1 \le \mathsf{A} e^{-\varsigma r}$.*

Proof. (i) is the triangle inequality of the quotient metric. (ii) Fix a lift $\tilde{x}$ and for each y choose the lift $\tilde{y}$ nearest to $\tilde{x}$, so that $f(|x-y|) = f(|\tilde{x}-\tilde{y}|)$; distinct y have distinct lifts, so the sum is a sub-sum of $\sum_{\tilde{z}\in\mathbb{Z}^3} f(|\tilde{x}-\tilde{z}|)$. (iii) A path in the graph is no longer than the graph, and by the triangle inequality the distance of its endpoints is at most its length; a graph for Y is a graph for Y'. (iv) is the definition of the norm, since $e^{\theta\ell} = e^{\varsigma\ell}e^{-(\varsigma-\theta)\ell}$. □

Constants. Put $\tilde{c}(\theta) := \sum_{z\in\mathbb{Z}^3} e^{-\theta|z|}$, $\mathsf{K} := \sum_{z\in\mathbb{Z}^3} |z| e^{-\delta_P|z|/2}$, $c_d := \tilde{c}(\delta_P/2)$, and

(40)
$$\mu_1 := \min\Big\{\frac{\delta_P}{2}, \frac{\gamma_0}{2c_P\mathsf{K}}\Big\}, \quad g_0 := \frac{2}{\gamma_0}, \quad \mathsf{M}_1 := \frac{2c_P}{\gamma_0}\tilde{c}(\mu_1/2)^2, \quad c_{\mathrm{D}} := \frac{4c_P}{\gamma_0^2}\tilde{c}(\mu_1/2)^2, \quad \gamma_K := \min\Big\{\frac{1}{2}, \frac{1}{2c_K\mathsf{M}_1}\Big\},$$

By Lemma 17.97(ii) every sum over $\mathbb{L}$ of a function of the distance is bounded by the corresponding sum over $\mathbb{Z}^3$, so these constants serve on every $\mathbb{L}$. They depend on $(\gamma_0, c_P, \delta_P, c_K, \varsigma)$ only, and no lower bound on c_P is used anywhere ($c_P \to 0$ gives $\mu_1 = \delta_P/2$, $\mathsf{M}_1 \to 0$, $\gamma_K = \frac{1}{2}$).

Lemma 17.98 (Combes–Thomas bound)**.** *For $X \subset \Lambda$ and $r \ge 0$, $\|(P_{XX}+r)^{-1}(x,y)\| \le 2(\gamma_0+r)^{-1}e^{-\mu_1|x-y|}$. In particular $\|G_X(x,y)\| \le g_0 e^{-\mu_1|x-y|}$ for every X, and $\|C(x,y)\| \le g_0 e^{-\mu_1|x-y|}$.*

Proof. Fix $y_0 \in X$ and $f := |\cdot - y_0|$, and conjugate: $P^f := e^{\mu_1 f} P_{XX} e^{-\mu_1 f}$. Then $(P^f - P_{XX})(x,y) = P(x,y)\big(e^{\mu_1(f(x)-f(y))} - 1\big)$ vanishes on the diagonal and, by (R2), Lemma 17.97(i) and $e^t - 1 \le te^t$ for $t \ge 0$, has block norm at most $c_P e^{-\delta_P|x-y|}\mu_1|x-y|e^{\mu_1|x-y|} \le c_P\mu_1|x-y|e^{-\delta_P|x-y|/2}$, since $\mu_1 \le \delta_P/2$; by Schur's test and Lemma 17.97(ii) its row and column sums are at most $c_P\mu_1\mathsf{K} \le \gamma_0/2$, so $\|P^f - P_{XX}\| \le \gamma_0/2$. Hence $\operatorname{Re}\langle u, (P^f+r)u\rangle \ge (\gamma_0/2+r)\|u\|^2$ for every u ($P_{XX} \ge \gamma_0$ as a principal block), so $\|(P^f+r)^{-1}\| \le (\gamma_0/2+r)^{-1} \le 2(\gamma_0+r)^{-1}$, and $(P_{XX}+r)^{-1}(x,y_0) = e^{-\mu_1 f(x)}(P^f+r)^{-1}(x,y_0)e^{\mu_1 f(y_0)}$ has norm at most $2(\gamma_0+r)^{-1}e^{-\mu_1|x-y_0|}$. □

Lemma 17.99 (conditioning)**.** *Let $X \subset \Lambda$ and $Y := \Lambda \setminus X$. Then $0 < G_X \le C_{XX} \le \gamma_0^{-1}$, so every component variance of $N(\cdot, G_X)$ and of μ_P is at most γ_0^{-1}. The conditional mean $h := -G_X P_{XY} z_Y$ obeys*

$$|h(x)| \le \mathsf{M}_1 e^{-(\mu_1/2)\operatorname{dist}(x,Y)}\|z_Y\|_\infty \qquad (x \in X),$$

in particular $\|h\|_\infty \le \beta/2$ whenever $\|z_Y\|_\infty \le c_K\gamma_K\beta$. For $x,y \in X$,

$$\|(C - G_X)(x,y)\| \le c_{\mathrm{D}} \exp\Big(-\frac{\mu_1}{8}\big(\operatorname{dist}(x,Y) + \operatorname{dist}(y,Y)\big) - \frac{\mu_1}{4}|x-y|\Big).$$

Proof. $C_{XX} = (P_{XX} - P_{XY}P_{YY}^{-1}P_{YX})^{-1} \ge P_{XX}^{-1} = G_X$, because the subtracted operator is non-negative and inversion reverses the order of positive operators; and $C_{XX} \le \|C\| \le \gamma_0^{-1}$. For the mean, Lemma 17.98 and (R2) (the pairs $(u,v) \in X \times Y$ are off-diagonal) give $|h(x)| \le \sum_{u\in X, v\in Y} g_0 e^{-\mu_1|x-u|} c_P e^{-\delta_P|u-v|}|z_v|$; as $\delta_P \ge 2\mu_1$ and $|x-u|+|u-v| \ge |x-v| \ge \operatorname{dist}(x,Y)$, the exponent is at most $-\frac{\mu_1}{2}\operatorname{dist}(x,Y) - \frac{\mu_1}{2}(|x-u|+|u-v|)$, and the double sum of the last factor is at most $\tilde{c}(\mu_1/2)^2$ by Lemma 17.97(ii). The particular case is $\mathsf{M}_1 c_K \gamma_K \le \frac{1}{2}$. For the covariance difference, row X of $PC = \mathbf{1}$ reads $P_{XX}C_{XX} + P_{XY}C_{YX} = \mathbf{1}$, so $C_{XX} - G_X = -G_X P_{XY} C_{YX}$, with

kernel $-\sum_{u\in X,v\in Y} G_X(x,u)P(u,v)C(v,y)$, bounded by $g_0^2 c_P \sum_{u,v} e^{-\mu_1 L}$ with $L := |x-u|+|u-v|+|v-y|$ (using $\delta_P \geq \mu_1$). Since $L \geq |x-y|$, $L \geq |x-v| \geq \operatorname{dist}(x,Y)$ and $L \geq |v-y| \geq \operatorname{dist}(y,Y)$, write $e^{-\mu_1 L} = e^{-\mu_1 L/4} e^{-\mu_1 L/4} e^{-\mu_1 L/2} \leq e^{-\frac{\mu_1}{8}(\operatorname{dist}(x,Y)+\operatorname{dist}(y,Y))} e^{-\frac{\mu_1}{4}|x-y|} e^{-\frac{\mu_1}{2}(|x-u|+|u-v|)}$ and sum the last factor over u, v. □

Lemma 17.100 (decoupling across corridors)**.** *Let $\Lambda = \Gamma \sqcup D$ with $D = \bigsqcup_i D_i$ and $\operatorname{dist}(D_i, D_j) \geq w_s$ for $i \neq j$. Let $\mu_P(\cdot \,|\, z_\Gamma)$ be the conditional law of z_D given z_Γ and $\nu_{\mathrm{loc}}(\cdot \,|\, z_\Gamma) := \bigotimes_i N(-G_{D_i} P_{D_i\Gamma} z_\Gamma,\, G_{D_i})$ its decoupled version; put $\omega := c_P c_d e^{-\delta_P w_s/2}$ and suppose $\omega \leq \gamma_0/2$. On $\{\|z_D\|_\infty \leq \beta,\ \|z_\Gamma\|_\infty \leq \beta\}$,*

$$\Big|\log \frac{d\mu_P(\cdot \,|\, z_\Gamma)}{d\nu_{\mathrm{loc}}(\cdot \,|\, z_\Gamma)}\Big| \leq \omega_{\mathrm{tot}} := \omega|D|\Big[\frac{\beta^2}{2} + \frac{n}{\gamma_0} + \frac{(c_P \tilde{c}(\delta_P)\beta)^2}{2\gamma_0^2}\Big],$$

hence $e^{-\omega_{\mathrm{tot}}} \int F\, d\nu_{\mathrm{loc}} \leq \int F\, d\mu_P(\cdot \,|\, z_\Gamma) \leq e^{\omega_{\mathrm{tot}}} \int F\, d\nu_{\mathrm{loc}}$ for every $F \geq 0$ supported in that set. Each factor of ν_{loc} is a law of the type of Lemma 17.99, with $X = D_i$, $Y = \Lambda \setminus D_i$ and $z_{D_j} = 0$ for $j \neq i$.

Proof. Put $u := P_{D\Gamma} z_\Gamma$, $B := \bigoplus_i P_{D_i D_i}$ and $O := P_{DD} - B$ (the blocks $P_{D_i D_j}$, $i \neq j$). The two laws are $N(-P_{DD}^{-1}u, P_{DD}^{-1})$ and $N(-B^{-1}u, B^{-1})$, since $(B^{-1}u)_{D_i} = G_{D_i} P_{D_i\Gamma} z_\Gamma$. The logarithm of the density of $N(-A^{-1}u, A^{-1})$ is $\frac{1}{2}\log\det(A/2\pi) - \frac{1}{2}\langle z, Az\rangle - \langle z, u\rangle - \frac{1}{2}\langle u, A^{-1}u\rangle$; the linear terms agree for $A = P_{DD}$ and $A = B$, so the log-ratio is $\frac{1}{2}\log\det(P_{DD}B^{-1}) - \frac{1}{2}\langle z_D, Oz_D\rangle - \frac{1}{2}\langle u, (P_{DD}^{-1} - B^{-1})u\rangle$. Every entry of O is an off-diagonal entry of P between points at distance at least w_s, so by (R2) and Lemma 17.97(ii) its row and column sums are at most $c_P e^{-\delta_P w_s/2}\tilde{c}(\delta_P/2) = \omega$, and $\|O\| \leq \omega$. First term: $|\frac{1}{2}\langle z_D, Oz_D\rangle| \leq \frac{1}{2}\beta^2 \sum_{x\in D}\sum_y \|O(x,y)\| \leq \frac{1}{2}\beta^2|D|\omega$. Second term: $\det(P_{DD}B^{-1}) = \det(\mathbf{1} + B^{-1/2}OB^{-1/2})$; the symmetric operator $B^{-1/2}OB^{-1/2}$ has $n|D|$ eigenvalues, each of modulus at most $\omega/\gamma_0 \leq \frac{1}{2}$ ($B \geq \gamma_0$ as a direct sum of principal blocks), and $|\log(1+t)| \leq 2|t|$ for $|t| \leq \frac{1}{2}$; so the term is at most $n|D|\omega/\gamma_0$. Third term: $P_{DD}^{-1} - B^{-1} = -P_{DD}^{-1}OB^{-1}$ has norm at most ω/γ_0^2, and $|u_x| \leq \sum_{v\in\Gamma} c_P e^{-\delta_P|x-v|}\beta \leq c_P\tilde{c}(\delta_P)\beta$, so $\|u\|^2 \leq |D|(c_P\tilde{c}(\delta_P)\beta)^2$. The three bounds add up to ω_{tot}. □

Lemma 17.101 (tails)**.** *Let $\nu = N(h, G)$ on $(\mathbb{R}^n)^X$ with $\|h\|_\infty \leq \beta/2$ and $G \leq \gamma_0^{-1}$. Then $\nu(\exists x \in X :\ |z_x| > \beta) \leq \epsilon(X,\beta) := 2n|X|e^{-\gamma_0\beta^2/(8n)}$; since $\bar{B}(\beta) \subset \beta K$, also $\nu(1 - \chi_\beta^X) \leq \epsilon(X,\beta)$.*

Proof. $|z_x| > \beta$ forces $|\zeta_x| > \beta/2$ for $\zeta := z - h$, hence $|\zeta_x^a| > \beta/(2\sqrt{n})$ for some component a; $\zeta_x^a \sim N(0,\sigma^2)$ with $\sigma^2 \leq \gamma_0^{-1}$, $\Pr(|N(0,\sigma^2)| > t) \leq 2e^{-t^2/(2\sigma^2)}$, and a union bound over the $n|X|$ components gives the claim. □

Lemma 17.102 (the normalisation of the cutoff)**.** *Let μ be a centred Gaussian on $(\mathbb{R}^n)^\Lambda$ all of whose component variances are at most γ_0^{-1}, and let $2e^{-\gamma_0\beta^2/(2n)} \leq \frac{1}{2}$. Then $\mu(\chi_\beta) \geq \exp\big(-4n|\Lambda|e^{-\gamma_0\beta^2/(2n)}\big)$.*

Proof. $\{z_x \in \beta K\} \supset \{|z_x| \leq \beta\} \supset \bigcap_{a\leq n}\{|z_x^a| \leq \beta/\sqrt{n}\}$. By Šidák's inequality (Z. Šidák, J. Amer. Statist. Assoc. 62 (1967), 626–633; C. G. Khatri, Ann. Math. Statist. 38 (1967), 1853–1867): for a centred Gaussian vector $(\xi_1,\dots,\xi_m)$ and $t_1,\dots,t_m > 0$, $\Pr(\bigcap_i\{|\xi_i| \leq t_i\}) \geq \prod_i \Pr(|\xi_i| \leq t_i)$; this is the case of one slab against a product of slabs of the Gaussian correlation inequality, and it is applied here to the $n|\Lambda|$ components of the real centred vector z. Each factor is at least $1 - 2e^{-\gamma_0\beta^2/(2n)}$, and $\log(1-u) \geq -2u$ for $u \leq \frac{1}{2}$. □

Lemma 17.103 (cumulants of bounded variables)**.** *For any probability measure ν and real variables with $|X_j| \leq M_j$, $|\kappa_\nu(X_1,\dots,X_k)| \leq k!\,2^{k-1}\prod_j M_j$. If $|X| \leq M$, then $\big|\log\int e^X d\nu - \sum_{k=1}^{\bar{n}} \kappa_\nu(X;k)/k!\big| \leq 2^{\bar{n}} M^{\bar{n}+1} \leq (2M)^{\bar{n}+1}$.*

Proof. In the Möbius formula $\kappa_\nu(X_1,\dots,X_k) = \sum_\pi (-1)^{|\pi|-1}(|\pi|-1)!\prod_{B\in\pi}\mathbb{E}_\nu\prod_{j\in B} X_j$, over the set partitions π of $\{1,\dots,k\}$, each product of moments is at most $\prod_j M_j$, and $\sum_\pi(|\pi|-1)!$ is at most the number of ordered set partitions, which is at most $k!\,2^{k-1}$ (an ordered partition is a permutation cut into consecutive blocks). For the second claim, $f(s) := \log\int e^{sX} d\nu$ has

$f^{(m)}(s) = \kappa_{\nu_s}(X; m)$ with $\nu_s \propto e^{sX}\nu$ a probability measure under which still $|X| \le M$; Taylor's formula at $s = 0$ with the Lagrange remainder $f^{(\bar n+1)}(s_*)/(\bar n + 1)!$ and the first claim give the bound. □

Lemma 17.104 (the cutoff in the exponent)**.** *Let ν be a probability measure, χ an indicator with $\nu(1-\chi) \le \epsilon$, W measurable with $|W| \le M$, and $e^{2M}\epsilon \le \frac12$. Then* (i) $\int \chi e^W d\nu \ge e^{-2e^{2M}\epsilon} \int e^W d\nu$, *and* (ii) $\int \chi e^W d\nu \le \int e^{\chi W} d\nu \le e^{2e^M\epsilon} \int \chi e^W d\nu$.

Proof. (i) $\int (1-\chi)e^W \le e^M\epsilon \le e^{2M}\epsilon \int e^W$, because $\int e^W \ge e^{-M}$; and $1 - u \ge e^{-2u}$ for $u \le \frac12$. (ii) On $\{\chi = 0\}$, $e^{\chi W} = 1 \ge 0 = \chi e^W$, so $\int e^{\chi W} = \int \chi e^W + \nu(1-\chi)$; and $\int \chi e^W \ge e^{-M}(1-\epsilon) \ge e^{-M}/2$, so $\nu(1-\chi) \le 2\epsilon e^M \int \chi e^W$. □

The lemma is applied with $W = \Psi\chi$, for which $\chi e^W = \chi e^\Psi$.

Lemma 17.105 (connected Wick expansion)**.** *Let ζ be a centred Gaussian vector with covariance G and let $F_j = \zeta^{\alpha_j}$, $j = 1, \dots, k$, be monomials, each a multiset of legs. Then $\kappa(F_1, \dots, F_k) = \sum_\pi \prod_{\{u,v\}\in\pi} G(u, v)$, the sum over the pairings π of all legs that connect the k factors (the graph on $\{1, \dots, k\}$ with an edge for each pair joining legs of two different factors is connected; pairs inside a factor are allowed). For $N(h, G)$ the same holds after the shift $z = h + \zeta$, the constants dropping out of joint cumulants with $k \ge 2$.*

Proof. Isserlis' formula gives $\mathbb{E}\prod_{j\in B} F_j = \sum \prod G$ over all pairings of the legs of the factors in B. Grouping each pairing by the partition of B into the connected components of its factor graph gives $\mathbb{E}\prod_{j\in B} F_j = \sum_\sigma \prod_{b\in\sigma} \mathsf{C}(b)$, with $\mathsf{C}(b)$ the sum over the connected pairings of the legs of b; the moment–cumulant relation $\mathbb{E}\prod_{j\in B} F_j = \sum_\sigma \prod_{b\in\sigma} \kappa(b)$ determines the $\kappa(b)$ from the moments recursively, so $\kappa = \mathsf{C}$. □

Lemma 17.106 (anchored connected bound)**.** *Let $\nu = N(h, G)$ on $(\mathbb{R}^n)^X$ with $\|G(x, y)\| \le g_0 e^{-\theta|x-y|}$ and $\|h\|_\infty \le \beta$. Let $F_1, \dots, F_k$ ($k \ge 2$) be polynomials of degree at most D in the variables of X, sorted by supports as in (R3), with $\|F_j\|_{\theta'} \le a_j$ for some $\theta' \le \theta$, and let $S_1, S_2 \subset \mathbb{L}$ be such that every monomial of F_1 has support meeting S_1 and every monomial of F_2 has support inside S_2. Then*

$$|\kappa_\nu(F_1, \dots, F_k)| \le |S_1 \cap X|\, K_k(\theta')\, (1+\beta)^{kD} \prod_j a_j\, e^{-(\theta'/2)\,\mathrm{dist}(S_1,S_2)}, \qquad K_k(\theta') := (kD)^{kD/2} \max\{1, g_0\}^{kD/2}\, \tilde c(\theta'/$$

and the same bound without the distance factor when no S_2 is given (the anchored bound). For $k = 1$, $|\mathbb{E}_\nu F_1| \le |S_1 \cap X|\, D^{D/2} \max\{1, g_0\}^{D/2}(1 + \beta)^D a_1$. For centred ν the factor $(1 + \beta)^{kD}$ is absent.

Proof. (1) Shift $z = h + \zeta$: a monomial z^α with support Y becomes $\sum_{\alpha' \le \alpha} \binom{\alpha}{\alpha'} h^{\alpha-\alpha'} \zeta^{\alpha'}$, whose coefficients have modulus sum at most $\prod(1 + |h|)^\alpha \le (1 + \beta)^D$; the pieces are kept grouped under the original support Y (their sites lie in Y), so the shifted F_j' has $\|F_j'\|_{\theta'} \le (1 + \beta)^D a_j$, and the constant pieces drop out for $k \ge 2$. For $h = 0$ there is no shift and no factor. (2) By Lemma 17.105, $|\kappa_\nu(F_1, \dots, F_k)| \le \sum_{Y_1,\dots,Y_k} \sum_{\text{monomials}} \prod |\text{coefficient}| \sum_{\text{connected } \pi} \prod \|G\|$; the factors have at most kD legs in all, so for fixed monomials there are at most $(kD)^{kD/2}$ pairings. (3) For a connected π fix a spanning tree of its factor graph, rooted at factor 1, with $k - 1$ tree pairs. Sum the root over the sites $a \in S_1 \cap X$ and the supports $Y_1 \ni a$: at most $|S_1 \cap X|\, \|F_1'\|_{\theta'}$ times the rest. Each child factor j is entered through a tree pair (y, y') with y in the parent's support and $y' \in Y_j$, which gives $\sum_{y'} g_0 e^{-\theta|y-y'|} \sum_{Y_j \ni y'} |F'_{j,Y_j}|_1$; the pairs not in the tree are bounded by g_0 each. (4) Write $|F'_{j,Y}|_1 = \big(|F'_{j,Y}|_1 e^{\theta'\ell(Y)}\big) e^{-\theta'\ell(Y)}$ and $e^{-\theta|y-y'|} \le e^{-(\theta'/2)|y-y'|} e^{-(\theta'/2)|y-y'|}$. The tree path from factor 1 to factor 2 lifts to a path in $\mathbb{L}$ from $a \in S_1$ to a point of $Y_2 \subset S_2$, made of tree pairs and of paths inside shortest connecting graphs of the traversed supports Y_j, each of length at most $\ell(Y_j)$ (Lemma 17.97(iii)); hence $e^{-(\theta'/2)(\sum_j \ell(Y_j) + \sum_{\text{tree}} |y-y'|)} \le e^{-(\theta'/2)\,\mathrm{dist}(S_1,S_2)}$. The remaining halves give $\tilde c(\theta'/2)$ per tree pair (Lemma 17.97(ii)) and $\|F_j'\|_{\theta'}$ per factor. For $k = 1$ the same count applies without a tree. □

Lemma 17.107 (cumulants with and without the cutoff)**.** *Let $\nu = N(h, G)$ on $(\mathbb{R}^n)^X$ with $\|h\|_\infty \le \beta/2$, $G \le \gamma_0^{-1}$ and $\beta \ge 2$; put $\chi := \chi_\beta^X$ and $\epsilon := \nu(1-\chi)$. Let $F_1, \dots, F_k$ ($k \le \bar{n}$) be polynomials in the variables of X of degree at most D with $\|F_j\|_0 \le a_j$, and assume the moment condition*

$$2^q\,|X|\,(2n)^{1/2}\big(n^q(2q-1)!!\big)^{1/2}\gamma_0^{-q/2}\,e^{-\gamma_0\beta^2/(4n)} \le (2\beta)^q \qquad (1 \le q \le 2\bar{n}D). \tag{41}$$

Then $|\kappa_\nu(\chi F_1, \dots, \chi F_k) - \kappa_\nu(F_1, \dots, F_k)| \le k \cdot k! \cdot 4^k \prod_j (a_j|X|)\,(2\beta)^{kD}\,\epsilon^{1/2}$.

Proof. Let $Z := \max\{1, \max_{x\in X}|z_x|\}$ and $\zeta := z - h$. If $\max_x|\zeta_x| \le \beta$ then $Z \le 2\beta$, otherwise $Z \le 2\max_x|\zeta_x|$; so $\mathbb{E}Z^q \le (2\beta)^q + 2^q\sum_x \mathbb{E}\big[|\zeta_x|^q\mathbf{1}\{|\zeta_x| > \beta\}\big] \le 2(2\beta)^q$ by the Cauchy–Schwarz inequality, $\mathbb{E}|\zeta_x|^{2q} \le n^q(2q-1)!!\gamma_0^{-q}$ (Hölder's inequality over the n components and the Gaussian moments), $\Pr(|\zeta_x| > \beta) \le 2ne^{-\gamma_0\beta^2/(2n)}$ and (41). Since $|F_j| \le a_j|X|Z^D$, every product of moments obeys $|\mathbb{E}\prod_{j\in B}F_j| \le 2\prod_{j\in B}(a_j|X|)(2\beta)^{|B|D}$, and $|\mathbb{E}[(1-\chi)\prod_{j\in B}F_j]| \le \epsilon^{1/2}(\mathbb{E}\prod_{j\in B}F_j^2)^{1/2} \le \sqrt{2}\,\epsilon^{1/2}\prod_{j\in B}(a_j|X|)(2\beta)^{|B|D}$. In the Möbius formula ($\chi^{|B|} = \chi$ on each block) replace the blocks one at a time: each partition contributes at most $k\cdot\sqrt{2}\cdot 2^{k-1}\epsilon^{1/2}\prod_j(a_j|X|)(2\beta)^{kD}$, and $\sum_\pi(|\pi|-1)! \le k!\,2^{k-1}$. □

The moment condition (41) is monotone: once it holds for a pair $(|X|, \beta)$ it holds for smaller $|X|$ and larger β.

Lemma 17.108 (comparing two Gaussian laws)**.** *Let $\nu, \tilde{\nu}$ be Gaussian laws of $(z_x)_{x\in S}$, S finite, with means $h, \tilde{h}$ of sup norm at most $\beta/2$ and covariances $G, \tilde{G}$ all of whose blocks have norm at most γ_0^{-1}, and let $\sup_x|h(x) - \tilde{h}(x)| \le \eta_1$, $\sup_{x,y}\|(G-\tilde{G})(x,y)\| \le \eta_2$. For polynomials $F_1, \dots, F_k$ in the variables of S of degree at most D with $\|F_j\|_0 \le a_j$,*

$$|\kappa_\nu(F_1, \dots, F_k) - \kappa_{\tilde{\nu}}(F_1, \dots, F_k)| \le k \cdot k! \cdot 4^k\,(kD)^{kD+1}\,(1+\beta+\gamma_0^{-1})^{kD}\prod_j(a_j|S|)\,\max\{\eta_1, \eta_2\}.$$

Proof. A moment of $L \le kD$ coordinate variables is, by Isserlis' formula for a Gaussian with mean, a sum over at most L^L partial pairings of products of at most L factors (covariance entries or means), each of modulus at most $1+\beta+\gamma_0^{-1}$ under both laws; replacing the factors one at a time, two corresponding products differ by at most $L\max\{\eta_1,\eta_2\}(1+\beta+\gamma_0^{-1})^L$. The coefficient sums contribute $\prod_j(a_j|S|)$, and the Möbius formula with the telescoping of Lemma 17.107 gives the bound. □

Lemma 17.109 (pavements)**.** *Let $b \ge b_{\mathrm{geo}} := 576$, $q_b := \lceil b^2\rceil$, $w_b := \lceil b^{3/2}\rceil$ and $s_b := \lfloor q_b/4\rfloor$, so that $s_b \ge 5w_b + 2$, $q_b \ge 6w_b$ and $3q_b \le 4b^2$; put $v_b := (4b^2)^3 = 64\,b^6$. On each $\mathbb{L}$ of the setting there are four partitions $\mathcal{P}^i$, $i = 0,1,2,3$, of $\mathbb{L}$ into* cells $\square$*, with* levels $\lambda_i : \mathbb{L} \to [0,\infty]$*, such that*

(G1) *λ_i is 1-Lipschitz;*
(G2) *if x, y lie in different cells of $\mathcal{P}^i$, then $|x-y| \ge \lambda_i(x) + \lambda_i(y)$;*
(G3) *$\bigcap_{i=0}^3\{x :\ \lambda_i(x) < 2.5w_b + 1\} = \emptyset$;*
(G4) *every cell has at most v_b sites; in particular, if some $\mathcal{P}^i$ consists of a single cell, then $|\mathbb{L}| \le v_b$.*

Proof. Numbers. For $b \ge 576$: $s_b \ge b^2/4 - 1 \ge 5b^{3/2} + 7 \ge 5w_b + 2$ (the middle inequality is $\sqrt{b} \ge 20 + 32b^{-3/2}$, true for $\sqrt{b} \ge 24$), $q_b \ge b^2 \ge 6b^{3/2} + 6 \ge 6w_b$, and $3q_b \le 3b^2 + 3 \le 4b^2$.

Coordinates. Each $\mathbb{L}$ has three coordinate cycles $\mathbb{I}_a$, $a = 1,2,3$, with a well-defined map $x \mapsto x_a \in \mathbb{I}_a$ and $N_a := |\mathbb{I}_a| \in \mathbb{N}\cup\{\infty\}$: for $\mathbb{Z}^3$, $\mathbb{I}_a = \mathbb{Z}$; for $(\mathbb{Z}/N\mathbb{Z})^3$, $\mathbb{I}_a = \mathbb{Z}/N\mathbb{Z}$; for $\mathbb{Z}^3/mG_\pm$, $\mathbb{I}_a := \mathbb{Z}/N_a\mathbb{Z}$ with $N_a\mathbb{Z} := \pi_a(mG_\pm)$ the projection of the period lattice on the coordinate a, that is $N_1 = N_2 = 10m$ and $N_3 = 50m$ (the entries $40, \mp 30, 0$ and $\pm 30, 40, 0$ have greatest common divisor 10); the map is well defined because the period lattice $G := mG_\pm$ is contained in $G' := \bigoplus_a N_a\mathbb{Z}$. In every case $\mathrm{dist}_{\mathbb{I}_a}(x_a, y_a) \le |\tilde{x}_a - \tilde{y}_a| \le |\tilde{x} - \tilde{y}| = |x-y|$ for lifts realising the quotient distance. For the skew tori we record: $\det G = (40\cdot 40 + 30\cdot 30)\cdot 50\cdot m^3 = 125000\,m^3 = (50m)^3$, $[G' : G] = 125000/5000 = 25$, and $G \cap \{g_3 = 0\} = m\langle(40, \mp 30, 0), (\pm 30, 40, 0)\rangle$ has index $2500m^2$ in $\mathbb{Z}^2 \times \{0\}$.

Cuts. A coordinate a is *cut* if $N_a \geq q_b$ (always for $\mathbb{Z}^3$). For a cut coordinate choose a set $B_a^0 \subset \mathbb{I}_a + \frac{1}{2}$ of cuts all of whose (cyclic) gaps lie in $[q_b, 2q_b)$: for $\mathbb{I}_a = \mathbb{Z}$ take $q_b\mathbb{Z} + \frac{1}{2}$; on a cycle take $p := \lfloor N_a/q_b \rfloor \geq 1$ cuts with gaps $\lfloor N_a/p \rfloor$ or $\lceil N_a/p \rceil$, which lie in $[q_b, 2q_b - 1]$. Put $B_a^i := B_a^0 + is_b$, $i = 0, \dots, 3$. Cuts of B_a^i and of $B_a^{i'}$, $i \neq i'$, are at cyclic distance at least s_b: two shifts of the same cut differ by $|i - i'|s_b \in [s_b, 3s_b]$, and $N_a - 3s_b \geq q_b - 3s_b \geq s_b$; shifts of different cuts differ by at least $q_b - 3s_b \geq s_b$. For $x \in \mathbb{L}$ put $\lambda_i(x) := \min_{a \text{ cut}} \operatorname{dist}_{\mathbb{I}_a}(x_a, B_a^i)$, and $\lambda_i := +\infty$ if no coordinate is cut. The *cut set* C^i is the set of points ξ of the continuum torus ($\mathbb{R}^3$, $\mathbb{R}^3/N\mathbb{Z}^3$ or $\mathbb{R}^3/G$) with $\xi_a \in B_a^i$ for some cut coordinate a, and the cells of $\mathcal{P}^i$ are the intersections of $\mathbb{L}$ with the connected components of the complement of C^i; no site lies on a cut, so the cells partition $\mathbb{L}$.

(G1) is the Lipschitz property of $\operatorname{dist}_{\mathbb{I}_a}(\cdot, B_a^i)$ together with $\operatorname{dist}_{\mathbb{I}_a}(x_a, y_a) \leq |x - y|$. *(G3)*: if $\lambda_i(x) < 2.5w_b + 1$ for all four i, each minimum is attained at a cut coordinate, and since there are at most three, two indices $i \neq i'$ use the same coordinate a; then a cut of B_a^i and a cut of $B_a^{i'}$ are at cyclic distance less than $2(2.5w_b + 1) = 5w_b + 2 \leq s_b$, a contradiction. If no coordinate is cut the four sets are empty. *(G2)*: lift x, y to $\tilde{x}, \tilde{y} \in \mathbb{R}^3$ with $|\tilde{x} - \tilde{y}| = |x - y|$. The lift $\tilde{C}^i$ of the cut set is the union of the planes $\{\xi_a = c\}$ over the cut coordinates a and the lifts c of the cuts of B_a^i; it is G'-periodic, hence G-periodic, the components of $\mathbb{R}^3 \setminus \tilde{C}^i$ are the products of open gap intervals in the cut coordinates and of $\mathbb{R}$ in the others, and the projection maps them onto the components of the complement of C^i. If the segment $[\tilde{x}, \tilde{y}]$ met none of the planes it would lie in one component, and x, y would lie in one cell; so it crosses a plane $\{\xi_a = c\}$ with c between $\tilde{x}_a$ and $\tilde{y}_a$, and $|x - y| \geq |\tilde{x}_a - \tilde{y}_a| = |\tilde{x}_a - c| + |c - \tilde{y}_a| \geq \operatorname{dist}_{\mathbb{I}_a}(x_a, B_a^i) + \operatorname{dist}_{\mathbb{I}_a}(y_a, B_a^i) \geq \lambda_i(x) + \lambda_i(y)$. *(G4)*: a component Q of $\mathbb{R}^3 \setminus \tilde{C}^i$ projects onto its cell injectively modulo $G_Q := \{g \in G : \ Q + g = Q\} = \{g \in G : \ g_a = 0 \text{ for every cut } a\}$: if Q and $Q + g$ meet they coincide, $\tilde{C}^i$ being G-invariant. On $\mathbb{Z}^3$, and whenever all three coordinates are cut, $G_Q = 0$ and the cell is a box with sides at most $2q_b - 1$, hence has at most $(2q_b)^3$ sites. On a cubic torus the only other case is that no coordinate is cut, when $|\mathbb{L}| = N^3 < q_b^3$. On $\mathbb{Z}^3/mG_\pm$, where $N_3 = 5N_1$, two further cases occur: only the third coordinate is cut ($10m < q_b \leq 50m$), the cell is the product of $\mathbb{Z}^2/(G \cap \{g_3 = 0\})$ with an interval of fewer than $2q_b$ integers and has fewer than $2500m^2 \cdot 2q_b = 25\,(10m)^2 \cdot 2q_b < 50\,q_b^3$ sites; or no coordinate is cut ($50m < q_b$) and $|\mathbb{L}| = \det G = (50m)^3 < q_b^3$. With $q_b \leq b^2 + 1 \leq \frac{257}{256}b^2$ the three counts are below $8.1\,b^6$, $50.6\,b^6$ and $1.02\,b^6$, hence below $v_b = 64\,b^6$. □

Layers. For a cell $\square$ of $\mathcal{P}^i$ write $\lambda := \lambda_i$ and define, all intersected with $\square \cap \Lambda$:

$$\Gamma_1(\square) := \{\lambda < w_b\}, \quad \Gamma_2(\square) := \{w_b \leq \lambda < 2w_b\}, \quad \Gamma_3(\square) := \{1.5w_b \leq \lambda < 2w_b\}, \quad \square' := \{\lambda \geq 2w_b\}, \quad \Gamma_4(\square) :$$

$\operatorname{Deep}(\square) := \{\lambda \geq 2.5w_b\}$ and $D_\square := \Gamma_2(\square) \cup \square' = \{\lambda \geq w_b\}$. By (G1) and (G2), two points of one cell at levels $\lambda(x) \geq \lambda(y)$ are at distance at least $\lambda(x) - \lambda(y)$, and two points of different cells at distance at least $\lambda(x) + \lambda(y)$. Hence: the sets $D_\square$ of distinct cells are at distance at least $2w_b$; $\operatorname{dist}(D_\square, \Gamma_1(\tilde{\square})) \geq w_b$ and $\operatorname{dist}(\operatorname{Deep}(\square), \tilde{\square} \cap \{\lambda \geq 1.5w_b\}) \geq 4w_b$ for $\tilde{\square} \neq \square$; and $\operatorname{dist}(\operatorname{Deep}(\square), \Lambda \setminus \square') \geq \frac{1}{2}w_b$, $\operatorname{dist}(\square \cap \{\lambda \geq 1.5w_b\}, \Lambda \setminus D_\square) \geq \frac{1}{2}w_b$, $\operatorname{dist}(\operatorname{Deep}(\square), \Lambda \setminus (\square \cap \{\lambda \geq 1.5w_b\})) \geq w_b$, $\operatorname{dist}(\Gamma_4(\square), \square \cap \{\lambda < 1.5w_b\}) \geq \frac{1}{2}w_b$, $\operatorname{dist}(\square', \Gamma_2(\square) \setminus \Gamma_3(\square)) \geq \frac{1}{2}w_b$, $\operatorname{dist}(\Gamma_1(\square), \square') \geq w_b$.

One stage. Fix i, the pavement $\mathcal{P}^i$ with its layers, a set $J' \subset J$ (write $V' := V_{J'}$) and a threshold $\beta \in [b_{\min}, b_{\max}]$, where $b_{\min} := \gamma_K^4 b$ and $b_{\max} := \gamma_K^{-4} b$. Put $\Phi(\beta; R) := \int \chi_\beta e^{V_R} d\mu_P$, $\chi_\beta^\square := \chi_\beta^{D_\square}$, $\mathcal{B}' := \{\square \in \mathcal{P}^i : \ \square \cap J' \neq \emptyset\}$ (the *J-cells*), $N' := |\mathcal{B}'| \leq |J'|$, $D := \bigcup_{\square \in \mathcal{B}'} D_\square$, $\Gamma := \Lambda \setminus D$ and $J'' := J' \cap \{\lambda_i < 2.5w_b\}$. The monomials of V', all of which have support in J', are sorted as follows.

- V_Γ: support in Γ. For $\square \in \mathcal{B}'$, $\Psi_\square$: support in $D_\square$, or support in $\Gamma_1(\square) \cup \Gamma_2(\square)$ meeting both. $R_0 := V' - V_\Gamma - \sum_\square \Psi_\square$.
- Ψ_1: support in $\square \cap \{\lambda \geq 1.5w_b\}$ and meeting $\square'$. Ψ_2: support in $\Gamma_1(\square) \cup \Gamma_2(\square)$ and meeting $\Gamma_2(\square)$. $\Psi_3 := \Psi_\square - \Psi_1 - \Psi_2$: the monomials of $\Psi_\square$ meeting $\square'$ and $\Gamma_2(\square) \setminus \Gamma_3(\square)$.
- Ψ_1': the monomials of Ψ_1 with support in $\Gamma_3(\square) \cup \Gamma_4(\square)$ (they meet $\Gamma_4(\square)$). $\Psi_1'' := \Psi_1 - \Psi_1'$: the monomials of Ψ_1 meeting $\operatorname{Deep}(\square)$. $W_\square := \Psi_1' + \Psi_2$ and $R_1 := V_{J''} - V_\Gamma - \sum_\square W_\square$.

Ψ_2 depends on $z_{\Gamma_1(\square)}$ as well as on $z_{D_\square}$. Under the conditional laws below it is read as a polynomial in $z_{D_\square}$ whose coefficients depend on $z_{\Gamma_1(\square)}$: every such monomial keeps at least one variable in $D_\square$, its support shrinks to $Y\cap D_\square$ (so its length does not increase, Lemma 17.97(iii)), and on $\{\|z_\Gamma\|_\infty\le c_K\beta\}$ its coefficient sums grow by at most the factor $(c_K\beta)^D$. The other pieces involve $z_{D_\square}$ only.

Length facts. V_Γ, Ψ_1' and Ψ_2 consist of monomials of $V_{J''}$ (their supports lie in $J'\cap\{\lambda<w_b\}$, in $\Gamma_3\cup\Gamma_4$ and in $\{\lambda<2w_b\}$ respectively; a site of J' in a cell $\square$ with $\lambda\ge w_b$ lies in $D_\square$, since then $\square\in\mathcal{B}'$), and the three classes are disjoint, so R_1 is a sub-sum of the monomials of $V_{J''}$. Every monomial of R_0 meets some $D_\square$, $\square\in\mathcal{B}'$, and has $\ell\ge w_b$: if it lies in $\square$ it meets $\Gamma_1(\square)$ (its support is not inside $D_\square$) and $\square'$ (its support is not inside $\Gamma_1\cup\Gamma_2$); otherwise it meets a second cell, in $D_{\tilde\square}$ or in $\Gamma_1(\tilde\square)$. Every monomial of Ψ_3 has $\ell\ge\frac12 w_b$. Every monomial of R_1 meets $\Gamma_2(\square)\cup\Gamma_4(\square)$ for some $\square\in\mathcal{B}'$ and has $\ell\ge\frac12 w_b$: it meets some $D_\square$ inside $\{\lambda<2.5w_b\}$; if it lies in $\square$ and meets $\Gamma_4(\square)$, it is not inside $\Gamma_3\cup\Gamma_4$ (it is not in Ψ_1'), so it meets $\{\lambda<1.5w_b\}$; if it lies in $\square$ and misses $\Gamma_4(\square)$, its support is inside $\Gamma_1\cup\Gamma_2$ and meets Γ_2, so it belongs to Ψ_2, which is excluded; if it meets a second cell, $\ell\ge w_b$. On $\{\|z\|_\infty\le c_K\beta\}$, with $\beta\le b_{\max}$ and $|D_\square|\le v_b$, Lemma 17.97(iv) gives

$$|R_0|,\,|R_1|\le N'v_b\mathsf{A}(c_Kb_{\max})^De^{-\varsigma w_b/2},\qquad |\Psi_\square|,\,|W_\square|\le M:=\mathsf{A}v_b(c_Kb_{\max})^D,\qquad |\Psi_3|\le Me^{-\varsigma w_b/2},$$

and $M=\mathsf{m}_*\,\mathsf{A}b^{D+6}$ with $\mathsf{m}_*:=64\,c_K^D\gamma_K^{-4D}$, so $M\le\mathsf{m}_*$ under $\mathsf{A}b^{D+6}\le1$.

Thresholds. Put $\epsilon_\beta:=2nv_be^{-\gamma_0\beta^2/(8n)}$, with $b_{\min}$, $b_{\max}$ and m_* as above. The threshold of Theorem 17.112 is $b_*:=\max\{576,\,b_{\rm dec},\,b_{\rm tail},\,b_{\rm mc},\,b_{\rm gci}\}$, where $b_{\rm dec}$, $b_{\rm tail}$, $b_{\rm mc}$ and $b_{\rm gci}$ are the least numbers beyond which, respectively, $c_Pc_de^{-\delta_Pb^{3/2}}\le\gamma_0/2$; $2nv_be^{2\mathsf{m}_*}e^{-\gamma_0\gamma_K^8b^2/(8n)}\le\frac12$ and $\gamma_K^4b\ge2$; the moment condition (41) holds with $|X|=v_b$ and $\beta=b_{\min}$; and $2e^{-\gamma_0\gamma_K^8b^2/(2n)}\le\frac12$. Each of these inequalities, once true, stays true for larger b, and all of them depend only on $(n,\bar n,D,\gamma_0,c_P,\delta_P,c_K)$.

Lemma 17.110 (one stage)**.** *Let $b\ge b_*$ and $\beta\in[b_{\min},b_{\max}]$, with $\beta/\gamma_K\le b_{\max}$ in* (U)*. Then*

$$\text{(L)}\qquad \Phi(\beta;J')\ge\Phi(\gamma_K\beta;J'')\,\exp\Big\{\sum_{k=1}^{\bar n}\frac{1}{k!}\big[\kappa_{\mu_P}(V_{J'};k)-\kappa_{\mu_P}(V_{J''};k)\big]-N'E_*\Big\},$$

$$\text{(U)}\qquad \Phi(\beta;J')\le\Phi(\beta/\gamma_K;J'')\,\exp\Big\{\sum_{k=1}^{\bar n}\frac{1}{k!}\big[\kappa_{\mu_P}(V_{J'};k)-\kappa_{\mu_P}(V_{J''};k)\big]+N'E_*\Big\},$$

with the per-cell error $E_*=E_*(b,\mathsf{A}):=e_R+e_{\rm dec}+e_{\rm ext}+e_{\rm cum}+e_3+e_{\rm mix}+e_{\rm rep}+e_{\rm bk}$ defined in the proof and in Lemma 17.111. All its terms except $e_{\rm cum}=2(2M)^{\bar n+1}$ are increasing in A and carry one of the factors $e^{-\varsigma w_b/2}$, $e^{-\delta_Pw_b}$, $e^{-\bar\mu w_b/4}$, $e^{-\mu_1w_b/8}$, $\epsilon(v_b,b_{\min})^{1/2}$.

Proof. Recall $\epsilon_\beta=\epsilon(v_b,\beta)=2nv_be^{-\gamma_0\beta^2/(8n)}$, which decreases in β.

Lower bound. (L1) $V'=V_\Gamma+\sum_\square\Psi_\square+R_0$, so on $\operatorname{supp}\chi_\beta$, $e^{V'}\ge e^{-|R_0|_{\max}}e^{V_\Gamma+\sum_\square\Psi_\square}$. (L2) Shrink the cutoff on Γ: $\chi_\beta\ge\chi^\Gamma_{\gamma_K\beta}\chi^D_\beta$, as $\gamma_K\beta K\subset\beta K$. Disintegrate μ_P into its marginal on z_Γ and the conditional law of z_D. The z_D-integrand $\prod_\square\chi^\square_\beta e^{\Psi_\square}$ is non-negative and supported in $\{\|z_D\|_\infty\le c_K\beta\}$, and $\|z_\Gamma\|_\infty\le c_K\gamma_K\beta\le c_K\beta$ on $\operatorname{supp}\chi^\Gamma_{\gamma_K\beta}$; Lemma 17.100 with the pieces $D_\square$, $\square\in\mathcal{B}'$ (separation $2w_b$, so $\omega=c_Pc_de^{-\delta_Pw_b}\le\gamma_0/2$ because $b\ge b_*$), gives

$$\int\mu_P(dz_D\,|\,z_\Gamma)\prod_\square\chi^\square_\beta e^{\Psi_\square}\ge e^{-\omega_{\rm tot}}\prod_{\square\in\mathcal{B}'}\int\nu_\square(dz_{D_\square})\,\chi^\square_\beta e^{\Psi_\square},\qquad \nu_\square:=N(h_\square,G_{D_\square}),\quad h_\square:=-G_{D_\square}P_{D_\square\Gamma}z_\Gamma,$$

with $\omega_{\rm tot}\le N'v_bc_Pc_de^{-\delta_Pw_b}\big[(c_K\beta)^2/2+n/\gamma_0+(c_P\tilde c(\delta_P)c_K\beta)^2/(2\gamma_0^2)\big]$; only the J-cells are decoupled, which is why the error is proportional to N'. (L3) By Lemma 17.99, $\|h_\square\|_\infty\le\beta/2$ and $G_{D_\square}\le\gamma_0^{-1}$; by Lemma 17.101, $\nu_\square(1-\chi^\square_\beta)\le\epsilon_\beta$; by Lemma 17.104(i) with $W:=\Psi_\square\chi^\square_\beta$, $|W|\le M$ (and $e^{2M}\epsilon_\beta\le\frac12$ because $b\ge b_*$), $\int\nu_\square\chi^\square_\beta e^{\Psi_\square}\ge e^{-2e^{2M}\epsilon_\beta}\int\nu_\square e^W$. (L4) By Lemma 17.103, $\log\int\nu_\square e^W\ge\sum_{k\le\bar n}\kappa_{\nu_\square}(W;k)/k!-(2M)^{\bar n+1}$. (L5) Expand $W=\Psi_1'\chi+\Psi_1''\chi+\Psi_2\chi+\Psi_3\chi$ ($\chi=\chi^\square_\beta$) multilinearly: $\kappa_{\nu_\square}(W;k)$ is the sum over the at most 4^k words of

length k in these four letters of their joint cumulants. Words containing $\Psi_3\chi$: by Lemma 17.103 each is at most $k!\,2^{k-1}M^k e^{-\varsigma w_b/2}$, so they contribute at most $e_3 := \sum_{k\le\bar n}(8M)^k e^{-\varsigma w_b/2}$. (L6) Words with at least one $\Psi_1''\chi$ and one $\Psi_2\chi$ (so $k\ge 2$): Lemma 17.107 removes the cutoffs (with $X = D_\square$, $|X|\le v_b$, $a_j \le \mathsf{A}(c_K\beta)^D$, and the moment condition holding because $b \ge b_*$), then Lemma 17.106 with $S_1 := \mathrm{Deep}(\square)$ (met by every monomial of Ψ_1''), $S_2 := \square\cap\{\lambda < 2w_b\}$ (containing the supports of Ψ_2 read in $z_{D_\square}$), $\mathrm{dist}(S_1,S_2)\ge \frac12 w_b$, $\theta = \mu_1$ (Lemma 17.98) and $\theta' = \bar\mu$ ($\|\cdot\|_{\bar\mu}\le\|\cdot\|_\varsigma$) bounds each by $v_b K_k(\bar\mu)(1+\beta)^{kD}(\mathsf{A}(c_K\beta)^D)^k e^{-\bar\mu w_b/4}$; in total these words contribute at most

$$e_{\mathrm{mix}} := \sum_{k=2}^{\bar n}\frac{4^k}{k!}\Big[k\,k!\,4^k\big(\mathsf{A}(c_K b_{\max})^D v_b\big)^k (2b_{\max})^{kD}\epsilon_{b_{\min}}^{1/2} + v_b K_k(\bar\mu)(1+b_{\max})^{kD}\big(\mathsf{A}(c_K b_{\max})^D\big)^k e^{-\bar\mu w_b/4}\Big].$$

(L7) Words with at least one $\Psi_1''\chi$ and neither $\Psi_2\chi$ nor $\Psi_3\chi$ (all factors $\Psi_1'\chi$ or $\Psi_1''\chi$, with supports in $S := \square\cap\{\lambda\ge 1.5w_b\}$): Lemma 17.107 removes the cutoffs, and Lemma 17.108 compares the marginal of $\nu_\square$ on S (mean $h_\square|_S$, covariance $(G_{D_\square})_{SS}$) with that of μ_P (centred, covariance C_{SS}): since $\mathrm{dist}(S,\Lambda\setminus D_\square)\ge\frac12 w_b$, Lemma 17.99 gives $\eta_1\le \mathsf{M}_1 c_K\gamma_K\beta e^{-\mu_1 w_b/4}$ and $\eta_2\le c_{\mathrm D} e^{-\mu_1 w_b/8}$. This replaces these words by the z_Γ-independent quantity

$$c_\square := \sum_{k=1}^{\bar n}\frac{1}{k!}\big[\kappa_{\mu_P}(\Psi_1'+\Psi_1'';k) - \kappa_{\mu_P}(\Psi_1';k)\big]$$

with an error at most

$$e_{\mathrm{rep}} := \sum_{k=1}^{\bar n}\frac{2^k}{k!}\,k\,k!\,4^k\big(\mathsf{A}(c_K b_{\max})^D v_b\big)^k\Big[(2b_{\max})^{kD}\epsilon_{b_{\min}}^{1/2} + (kD)^{kD+1}(1+b_{\max}+\gamma_0^{-1})^{kD}\big(\mathsf{M}_1 c_K b_{\max}+c_{\mathrm D}\big)e^{-\mu_1 w_b/8}\Big].$$

What is left of $\sum_k \kappa_{\nu_\square}(W;k)/k!$ are the words in $\Psi_1'\chi$ and $\Psi_2\chi$ alone, that is $\sum_k\kappa_{\nu_\square}(W_\square\chi;k)/k!$. (L8) By Lemma 17.103 backwards and $e^{\chi W_\square}\ge\chi e^{W_\square}$, $\sum_k\kappa_{\nu_\square}(W_\square\chi;k)/k!\ge\log\int\nu_\square\chi_\beta^\square e^{W_\square} - (2M)^{\bar n+1}$. (L9) $\prod_\square\int\nu_\square\chi_\beta^\square e^{W_\square} = \int\nu_{\mathrm{loc}}\prod_\square\chi_\beta^\square e^{W_\square}\ge e^{-\omega_{\mathrm{tot}}}\int\mu_P(dz_D\,|\,z_\Gamma)\prod_\square\chi_\beta^\square e^{W_\square}$ by Lemma 17.100 backwards (the integrand is non-negative and supported in the required set). The $c_\square$ do not depend on z_Γ, so they come out of the z_Γ integral; with $\chi_\beta^D\ge\chi_{\gamma_K\beta}^D$ and $V_{J''} = V_\Gamma + \sum_\square W_\square + R_1$,

$$\int d\mu_P\,\chi_{\gamma_K\beta}^\Gamma\chi_\beta^D\,e^{V_\Gamma+\sum_\square W_\square}\ge e^{-|R_1|_{\max}}\int d\mu_P\,\chi_{\gamma_K\beta}\,e^{V_{J''}} = e^{-|R_1|_{\max}}\,\Phi(\gamma_K\beta;J'').$$

(L10) By Lemma 17.111, $\sum_{\square\in\mathcal{B}'}c_\square\ge\sum_{k\le\bar n}\frac{1}{k!}[\kappa_{\mu_P}(V_{J'};k)-\kappa_{\mu_P}(V_{J''};k)] - N' e_{\mathrm{bk}}$. Collecting the errors, with $e_R := 2v_b\mathsf{A}(c_K b_{\max})^D e^{-\varsigma w_b/2}$ (for R_0 and R_1), $e_{\mathrm{dec}} := 2v_b c_P c_d e^{-\delta_P w_b}\big[(c_K b_{\max})^2/2 + n/\gamma_0 + (c_P\tilde c(\delta_P)c_K b_{\max})^2/(2\gamma_0^2)\big]$ (two uses of Lemma 17.100), $e_{\mathrm{ext}} := 4e^{2M}\epsilon_{b_{\min}}$ and $e_{\mathrm{cum}} := 2(2M)^{\bar n+1}$, proves (L).

Upper bound. The same steps with the following changes. (U2) Enlarge the cutoff on D: $\chi_\beta\le\chi_\beta^\Gamma\chi_{\beta'}^D$ with $\beta' := \beta/\gamma_K$; the z_D-integrand is non-negative and supported in $\{\|z_D\|_\infty\le c_K\beta'\}$, and $\|z_\Gamma\|_\infty\le c_K\beta = c_K\gamma_K\beta'$, so Lemma 17.99 gives $\|h_\square\|_\infty\le\beta'/2$ and the upper half of Lemma 17.100 applies with the radius $c_K\beta'$. (U3) $\int\nu_\square\chi_{\beta'}^\square e^{\Psi_\square}\le\int\nu_\square e^{\Psi_\square\chi_{\beta'}^\square}$. (U4)–(U7) as (L4)–(L7) with β' in place of β, using the upper half of Lemma 17.103. (U8) $\sum_k\kappa_{\nu_\square}(W_\square\chi;k)/k!\le\log\int\nu_\square e^{W_\square\chi} + (2M)^{\bar n+1}\le\log\int\nu_\square\chi_{\beta'}^\square e^{W_\square} + 2e^M\epsilon_{\beta'} + (2M)^{\bar n+1}$ by Lemma 17.104(ii). (U9) The upper half of Lemma 17.100 backwards, then $\chi_\beta^\Gamma\chi_{\beta'}^D\le\chi_{\beta'}$ and $e^{V_\Gamma+\sum_\square W_\square}\le e^{|R_1|_{\max}}e^{V_{J''}}$. (U10) The upper half of Lemma 17.111. Since all thresholds used lie in $[b_{\min},b_{\max}]$, every error is again bounded by E_*. □

Lemma 17.111 (bookkeeping). $\big|\sum_{\square\in\mathcal{B}'}c_\square - \sum_{k=1}^{\bar n}\frac{1}{k!}\big[\kappa_{\mu_P}(V_{J'};k)-\kappa_{\mu_P}(V_{J''};k)\big]\big|\le N' e_{\mathrm{bk}}$ *with* $e_{\mathrm{bk}} := \sum_{k=1}^{\bar n}\frac{6^k}{k!}\,v_b K_k(\bar\mu/2)\,\mathsf{A}^k e^{-\bar\mu w_b/4}$.

Proof. Here the law is the centred μ_P, so Lemma 17.106 applies with the covariance decay $g_0 e^{-\mu_1|x-y|}$ of Lemma 17.98 and without the factor $(1+\beta)^{kD}$. (a) $V_{J'} = V_{J''} + \sum_{\square\in\mathcal{B}'}X_\square + R_2$ with $X_\square := \Psi_1''$ of the cell $\square$: a monomial of $V_{J'}$ not inside J'' has a site of J' at level at least $2.5w_b$, so it meets $\mathrm{Deep}(\square)$ for a J-cell $\square$; if its support lies in $\square\cap\{\lambda\ge 1.5w_b\}$ it belongs to Ψ_1

and not to Ψ_1', that is to $X_\square$; otherwise it has $\ell \geq \operatorname{dist}(\operatorname{Deep}(\square), \Lambda \setminus (\square \cap \{\lambda \geq 1.5 w_b\})) \geq w_b$ and is put into R_2. (b) Expand $\kappa_{\mu_P}(V_{J''} + X + R_2; k)$, $X := \sum_\square X_\square$, into words. Words with at least one R_2: Lemma 17.106 anchored at $S_1 := \bigcup_{\square \in \mathcal{B}'} \operatorname{Deep}(\square)$ (met by every monomial of R_2; $|S_1 \cap \Lambda| \leq N' v_b$) with $\theta' = \bar\mu/2$ and $\|R_2\|_{\bar\mu/2} \leq \mathsf{A} e^{-\bar\mu w_b/2}$ (Lemma 17.97(iv)), the other factors having $\|\cdot\|_{\bar\mu/2} \leq \mathsf{A}$: at most $N' v_b K_k(\bar\mu/2) \mathsf{A}^k e^{-\bar\mu w_b/2}$ each. Words with some X and no R_2: sort them by the cell $\square$ of the first X factor and write every other X factor as $X_\square + X_{\neq\square}$. Words containing an $X_{\neq\square}$: Lemma 17.106 with $S_1 = \operatorname{Deep}(\square)$, $S_2 := \bigcup_{\tilde\square \neq \square} (\tilde\square \cap \{\lambda \geq 1.5 w_b\})$, $\operatorname{dist}(S_1, S_2) \geq 4 w_b$ and $\theta' = \bar\mu$: at most $v_b K_k(\bar\mu) \mathsf{A}^k e^{-2\bar\mu w_b}$ each. Words with only $X_\square$ and $V_{J''}$ factors: for this cell write $V_{J''} = \Psi_1' + Y^{\mathrm{long}} + Y^{\mathrm{far}}$, where Y^{long} consists of the monomials of $V_{J''}$ meeting $\Gamma_4(\square)$ that are not in Ψ_1' (they are not inside $\Gamma_3 \cup \Gamma_4$, so they meet $\{\lambda < 1.5 w_b\}$ or another cell, and have $\ell \geq \frac12 w_b$) and Y^{far} of the rest (supports in $J'' \setminus \Gamma_4(\square)$, at distance at least $\frac12 w_b$ from $\operatorname{Deep}(\square)$). The words in $X_\square$ and Ψ_1' alone with at least one $X_\square$ add up to exactly $\kappa_{\mu_P}(\Psi_1' + \Psi_1''; k) - \kappa_{\mu_P}(\Psi_1'; k)$, which is $k!$ times the k-th term of $c_\square$. Words with a Y^{far}: Lemma 17.106 with $S_1 = \operatorname{Deep}(\square)$, $S_2 = J'' \setminus \Gamma_4(\square)$, $\theta' = \bar\mu$: at most $v_b K_k(\bar\mu) \mathsf{A}^k e^{-\bar\mu w_b/4}$ each. Words with a Y^{long} and no Y^{far}: anchored at $\operatorname{Deep}(\square)$ with $\theta' = \bar\mu/2$ and $\|Y^{\mathrm{long}}\|_{\bar\mu/2} \leq \mathsf{A} e^{-\bar\mu w_b/4}$: at most $v_b K_k(\bar\mu/2) \mathsf{A}^k e^{-\bar\mu w_b/4}$ each. Every slot of a word takes one of six roles (Ψ_1', Y^{far}, Y^{long}, $X_\square$, $X_{\neq\square}$, R_2), so there are at most 6^k words for each k; each word outside $c_\square$ is bounded, per cell, by $v_b K_k(\bar\mu/2) \mathsf{A}^k e^{-\bar\mu w_b/4}$ (as $K_k(\bar\mu) \leq K_k(\bar\mu/2)$ and all rates are at least $\bar\mu w_b/4$), and the words made of $V_{J''}$ alone add up to $\kappa_{\mu_P}(V_{J''}; k)$. □

Theorem 17.112 (the cumulant bound for the step Gaussian)**.** *In the setting* (R1)–(R4) *above (any $\mathbb{L}$ among $\mathbb{Z}^3$, the cubic tori and the skew tori, any finite $\Lambda \subset \mathbb{L}$, the Euclidean or the two-component cutoff) with b_* the threshold fixed before Lemma 17.110, there are $S < \infty$ and $\rho_3 > 0$, all three depending only on $(n, \bar n, D, \varsigma, \gamma_0, c_P, \delta_P, c_K)$, such that for $b \geq b_*$ and $\mathsf{A}\, b^{D+6} \leq 1$*

$$\log \int \chi_b\, e^V d\mu_P \leq \sum_{k=1}^{\bar n} \frac{\kappa_{\mu_P}(V; k)}{k!} + S|J|\big[(\mathsf{A}\, b^{D+6})^{\bar n+1} + e^{-\rho_3 b^{3/2}}\big],$$

$$\log \int \chi_b\, e^V d\mu_P \geq \sum_{k=1}^{\bar n} \frac{\kappa_{\mu_P}(V; k)}{k!} - S|J|\big[(\mathsf{A}\, b^{D+6})^{\bar n+1} + e^{-\rho_3 b^{3/2}}\big] - 4n|\Lambda|\, e^{-\gamma_0 \gamma_K^8 b^2/(2n)},$$

where $\kappa_{\mu_P}(V; k)$ are the truncated expectations of V under μ_P without cutoff and $\gamma_K = \min\{\frac12, (2c_K \mathsf{M}_1)^{-1}\}$ is the constant of (40)*. The constants do not depend on Λ, on J, on the torus, or on P and V otherwise, and no translation invariance is used. For the step Gaussians of both flows, at every step j below the stopping scale, on $T^{(j)}$, on a principal block of it or on a Neumann sub-domain that is a union of j-blocks, in the X-frame and in the Y-frame steps alike, the hypotheses hold with γ_0 Bałaban's coercivity constant* [1, (2.33)]*, δ_P the decay rate of the step precision divided by $\sqrt3$, and c_P a constant of (a, M).*

Proof. The hypotheses for the step Gaussians. The lower bound $\Delta^{(j)} + a5^{-2} Q^* Q \geq \gamma_0$ is [1, (2.33)] (for the hybrid flow by Proposition 17.72(a)), with constants independent of A, j and the domain; this is (R1), which needs no upper bound. The kernel bound (R2) is the exponential decay of $(M_j)^{-1}$ by Combes–Thomas conjugation (for the hybrid flow, the kernel bound of Proposition 17.72(a)) (whose first-order hopping part is anti-Hermitian and whose remainder has norm $\leq 3d\delta^2$ uniformly in the spacing), with $\operatorname{dist}(B(u), B(v)) \geq |u-v| - \sqrt3$, and the block term $a5^{-2} Q^* Q$ has range $\sqrt3 \cdot 5$; only the off-diagonal entries are constrained, so a diagonal term (such as the shell mass of the shell model) leaves the constants unchanged; Bałaban's bounds are stated in the sup metric, and $|x-y| \leq \sqrt3\, d^*(x, y)$ converts them to the quotient Euclidean distance at the rate divided by $\sqrt3$; principal blocks and Neumann sub-domains inherit (R1) and (R2) as sub-matrices. The interaction at step j is a real polynomial of fixed degree in the fluctuation variables (the analytic remainder vertices of order $\bar n + 1$ are taken out of the exponent on the support of the cutoff as $e^{\pm C \ell_j^{7-\beta} |\Lambda|}$, as [2, (2.31)] does), with $\mathsf{A} = \|V\|_\varsigma \leq s_0\, O(1) \ell_j^{\beta_0}$ by Corollary 17.41 at ς half of Bałaban's coefficient decay rate: a coefficient bound $O(1) \ell_j^{\beta_0} e^{-\delta_0 d(x_1, \dots, x_p)}$, with d the

length of the shortest graph connecting the points [1, (3.58)], gives $\|V\|_{\delta_0/2} \le s_0 O(1)\ell_j^{\beta_0}$ with $s_0 := \sum_{p\le D} p\, n^p\, \tilde c(\delta_0/(2p))^{p-1}$, because $d(x_1,\dots,x_p) \ge (p-1)^{-1}\sum_{i\ge 2}|x_i - x_1|$; this is (R3). The cutoff is Bałaban's, a product over sites of $\mathbf{1}\{|A'(x)| \le b\}\mathbf{1}\{|\phi'(x)| \le b\}$ [1, (3.43), (3.50)], which is (R4) with $c_K = \sqrt{2}$. Finally $\mathsf{A}b^{D+6} \le 1$ and $b \ge b_*$ hold at $b = p_1(\ell_j) = b_1(1+\log \ell_j^{-1})^{p_1}$ for ℓ_j below Bałaban's final spacing, since $\ell_j^{\beta_0}$ beats every power of $1+\log\ell_j^{-1}$; the exponent $p_1 \ge 1$ is a choice within Bałaban's freedom, under which $e^{-\rho_3 b^{3/2}}$ and $e^{-\gamma_0\gamma_K^8 b^2/(2n)}$ are $O(\ell_j^\kappa)$ for every κ.

The thresholds. The threshold b_* is the one fixed before Lemma 17.110; it depends only on $(n, \bar n, D, \gamma_0, c_P, \delta_P, c_K)$.

Small volume. If some pavement consists of a single cell, then $|\Lambda| \le v_b$ by (G4) and no pavement is needed: with $\nu = \mu_P$ (centred), $\epsilon := \mu_P(1-\chi_b) \le \epsilon_b$ by Lemma 17.101 and $M := \mathsf{A}|\Lambda|(c_K b)^D \le \mathsf{m}_*$, Lemma 17.104 moves the cutoff into the exponent at a cost of at most $2e^{2M}\epsilon$, Lemma 17.103 expands to order $\bar n$ at a cost of $(2M)^{\bar n+1}$, and Lemma 17.107 replaces $\kappa_{\mu_P}(V\chi_b;k)$ by $\kappa_{\mu_P}(V;k)$ at a cost of $k\,k!\,4^k(\mathsf{A}|\Lambda|)^k(2b)^{kD}\epsilon^{1/2}/k!$ for each k; the total is at most $E_* \le |J|E_*$ in both directions. (If $J = \emptyset$ then $V = 0$ and the statement reduces to $0 \ge \log\mu_P(\chi_b) \ge -4n|\Lambda|e^{-\gamma_0 b^2/(2n)}$, which is Lemma 17.102.)

Assembly. Otherwise iterate Lemma 17.110 over $i = 0,1,2,3$ with $J_0 := J$, $J_{i+1} := J_i \cap \{\lambda_i < 2.5 w_b\}$ (the set J'' of stage i), and the thresholds $\beta_i := \gamma_K^i b$ for the lower bound and $\beta_i := \gamma_K^{-i} b$ for the upper bound, all of which lie in $[b_{\min}, b_{\max}]$ together with β_3/γ_K. By (G3), $J_4 \subset J \cap \bigcap_i \{\lambda_i < 2.5 w_b\} = \emptyset$, so $V_{J_4} = 0$ and the cumulant increments of the four stages telescope to $\sum_{k\le\bar n}\kappa_{\mu_P}(V_J;k)/k!$; the numbers of J-cells add up to $\sum_i N_i \le 4|J|$. The terminal factor is $\Phi(\gamma_K^4 b;\emptyset) = \mu_P(\chi_{b_{\min}}) \ge \exp\big(-4n|\Lambda|e^{-\gamma_0\gamma_K^8 b^2/(2n)}\big)$ in the lower bound, by Lemma 17.102 (μ_P is centred with component variances at most $\|C\| \le \gamma_0^{-1}$), and $\Phi(\gamma_K^{-4}b;\emptyset) = \mu_P(\chi_{b_{\max}}) \le 1$ in the upper bound. Hence

$$\log\int \chi_b e^V d\mu_P - \sum_{k=1}^{\bar n}\frac{\kappa_{\mu_P}(V;k)}{k!} \le 4|J|E_*(b,\mathsf{A}), \qquad \log\int \chi_b e^V d\mu_P - \sum_{k=1}^{\bar n}\frac{\kappa_{\mu_P}(V;k)}{k!} \ge -4|J|E_*(b,\mathsf{A}) - 4n|\Lambda|e^-$$

The constants. $e_{\rm cum} = 2(2M)^{\bar n+1} = 2(2\mathsf{m}_*)^{\bar n+1}(\mathsf{A}b^{D+6})^{\bar n+1}$ exactly. Every other term of E_* is increasing in A, so under $\mathsf{A}b^{D+6} \le 1$ it is at most its value at $\mathsf{A} = b^{-D-6}$; call the sum of these values $Q(b)$. It is a finite sum of polynomials in b times one of $e^{-\varsigma w_b/2}$, $e^{-\delta_P w_b}$, $e^{-\bar\mu w_b/4}$, $e^{-\mu_1 w_b/8}$, $e^{-2\bar\mu w_b}$ and $\epsilon_{b_{\min}}^{1/2}$, each of which is at most a polynomial in b times $e^{-\bar\mu b^{3/2}/8}$ (since $w_b \ge b^{3/2}$, $\bar\mu \le \mu_1 \le \delta_P/2$ and $\bar\mu \le \varsigma$). With $\rho_3 := \bar\mu/16$ and

$$S := 4\max\Big\{2(2\mathsf{m}_*)^{\bar n+1},\ \sup_{b\ge b_*} e^{\rho_3 b^{3/2}}Q(b)\Big\} < \infty$$

one has $4|J|E_*(b,\mathsf{A}) \le S|J|\big[(\mathsf{A}b^{D+6})^{\bar n+1} + e^{-\rho_3 b^{3/2}}\big]$, which is the statement. S, b_*, ρ_3 and γ_K depend only on $(n, \bar n, D, \varsigma, \gamma_0, c_P, \delta_P, c_K)$; nothing depends on Λ, on J, on $\mathbb{L}$ or its period lattice, on P and V beyond the stated bounds, or on any translation invariance, of which none is used; no lower bound on c_P and no upper bound on P enters. □

Proposition 17.113 (the hybrid lower bound)**.** *For the hybrid flow on T and every real smooth periodic source s,*

$$\mathsf{Z}_{X_k,T}(s) \ge e^{-C(1+k)^p\delta_r^\sigma|T|}\int \chi_r\, e^{-\hat s_T^{(r)}}.$$

Proof. This is the analogue of [14, Prop. 4.1, (4.5)] for the hybrid flow (the print has "$+C$" in the exponent; in a lower bound it must be "$-C$"; the factor $(1+k)^p$ is that of Corollary 17.41 for the hybrid flow and is harmless, $(1+k)^p\delta_r^\sigma \to 0$ by Lemma 17.26). Its inputs are: the block-spin algebra of the tile average (Lemma 17.19, Lemma 17.16); the hybrid analogues of [1, Props. 2.1–2.3], the decay and regularity of G_j, $\Delta^{(j)}$, $C^{(j)}$ along the hybrid flow with the paired contours, with [14, Prop. 5.6] for the restricted inverses on unions of big blocks, which is Proposition 17.72(a) at every level ($j \ge 2$ by its proof, $j \in \{0,1\}$ by the same proofs at the tile and first Y-frame levels); the cumulant lower bound for the step Gaussian, Theorem 17.112,

whose extra lower-bound term $4n|\Lambda|e^{-\gamma_0\gamma_K^8 b^2/(2n)}$ at the threshold $b_j = p_1(t_j) = b_1(1+\ln t_j^{-1})^{p_1}$, $t_j = 5^j\varepsilon$, $p_1 \geq 1$ [1, (3.43), (3.50)], on $\Lambda = T_1^{(j)}$ with $|T_1^{(j)}| = |T|t_j^{-3}$, is $\leq C_\kappa t_j^\kappa |T_1^{(j)}|$ for every κ, of the form of Bałaban's own step error, and summed over $j < r$ is $\leq C'_\sigma \delta_r^\sigma |T|$, uniformly in T, k, r; and Bałaban's lower-bound argument [1, §§3–4] at block factor 5 (Theorem F.2) on the torus on which the steps are performed. The hybrid flow runs step 1 in X's frame and steps $2,\dots,r$ in Y's frame. On T_1, aligned for X and skew for Y, steps $2,\dots,r$ are steps on a torus skew in the frame of their blocks, and the lower-bound argument is the hybrid instance of Proposition 17.117, with the constant $C(1+k)^p$ of Corollary 17.41 for the hybrid flow (Bałaban's Neumann cube step being replaced by the constant-background reduction of Lemma 17.40). On T_2, skew for X and aligned for Y, only step 1 is skew: the fine axis lattice X_k on the torus with period lattice $R^{-1}P_J$; its blocks tile T_2 (Lemma 17.14(ii), Lemma 17.19(i)), its fluctuation covariance has mass of order ε^{-1} in physical units, so the torus images of its kernels are $O(e^{-c\,50MJ/\varepsilon})$, and the step lands on $Y_{k-1} \subset T_2$, aligned for every later step. Both instances pass through the bridge Proposition 17.121 (King adopts the free-boundary counterterm on [14, p. 330] and appeals to [1] for (4.5)) and are stated for the centred-block hybrid action (Convention 17.17). □

Proposition 17.114 (the polynomial lower bound for the rotated lattice)**.** *For Y_k on T and every source $s = (zG, h_0)$ with $h_0 = -n\psi$, $n \geq 0$: $\mathsf{Z}_{Y_k,T}(s) \geq e^{-C_1|T|}\varepsilon^{\,q(s)}$ for some $q(s) < \infty$.*

Proof. At $h = 0$ this is the lower half of [14, (4.2)] ("The results of [Ba 1–4] apply to both these models"), Bałaban's lower bound [1, (1.14)]: on T_2, aligned in Y's frame, by Theorems F.1 and F.2 after the passive rotation, with the bridge Proposition 17.121; on T_1 by the skew instance Proposition 17.117. The source zG is removed by Jensen's inequality: $\mathsf{Z}_{Y_k,T}(zG,h) = \mathsf{Z}_{Y_k,T}(0,h)\langle e^{zF(G)}\rangle_{(T,h),Y_k} \geq \mathsf{Z}_{Y_k,T}(0,h)\, e^{z\langle F(G)\rangle} = \mathsf{Z}_{Y_k,T}(0,h)$ for real z, since $\langle F(G)\rangle = 0$ under the charge conjugation of Definition 15.6. At the mass-increasing wall $h_0 = -n\psi$, $n > 0$, the two-sided ε-uniform bound [14, (4.2)] is false (Remark 16.1) and only the polynomial form is asked for; it is Theorem 17.116, at every King point, or alternatively Proposition 17.115 from the source-free instance and an upper bound on the one-point Wick function. □

Proposition 17.115 (Jensen: the wall instance from the source-free instance)**.** *Let $h_0 = -n\psi \leq 0$ be a wall. For every real t, $\Lambda \in \{X_k, Y_k\}$ and $T \in \{T_1, T_2\}$,*

$$\mathsf{Z}_{\Lambda,T}(tG,h_0) \;\geq\; \mathsf{Z}_{\Lambda,T}(0,0)\,\exp\Big[-n\sum_x \varepsilon^3\psi(x)\big\langle|\phi(x)|^2 - 2\mathsf{W}_\varepsilon\big\rangle_{T,0,\Lambda}\Big],$$

the expectation in the source-free periodic state. On Y_k the one-point function $w_k(T) := \langle|\phi(x)|^2 - 2\mathsf{W}_\varepsilon\rangle_{T,0,Y_k}$ is independent of x ($P_J \subset Y_k$, Lemma 17.14), so the right side is $\mathsf{Z}_{Y_k,T}(0,0)e^{-n\mathsf{S}_{Y_k}(\psi)w_k(T)}$, and an upper bound $w_k(T) \leq C(1+k)$ gives the polynomial form of Proposition 17.114 at the wall with $q(tG,h_0) = q(0,0) + nC\mathsf{S}(\psi)/\ln 5$.

Proof. $W_\Lambda(h_0) = \sum_x \varepsilon^3 h_0(x)|\phi(x)|^2 - 2\mathsf{W}_\varepsilon\mathsf{S}_\Lambda(h_0)$ gives $\mathsf{Z}(tG,h_0) = e^{-2\mathsf{W}_\varepsilon\mathsf{S}(h_0)}\mathsf{Z}(0,0)\langle e^{tF(G)+\sum\varepsilon^3 h_0|\phi|^2}\rangle_{T,0,\Lambda}$. Put $X := -tF(G) + n\sum_x \varepsilon^3\psi(x)|\phi(x)|^2$; then $\langle X\rangle = n\sum_x \varepsilon^3\psi(x)\langle|\phi(x)|^2\rangle$ because $\langle F(G)\rangle = 0$ by charge conjugation. Jensen's inequality $\langle e^{-X}\rangle \geq e^{-\langle X\rangle}$ and $-2\mathsf{W}_\varepsilon\mathsf{S}(h_0) - \langle X\rangle = -n\sum_x \varepsilon^3\psi(x)(\langle|\phi(x)|^2\rangle - 2\mathsf{W}_\varepsilon)$ give the bound. Only an upper bound on the one-point function is used, only on Y_k and only without sources; the mass-decreasing sign would need a lower bound, which Jensen's inequality does not give. □

Theorem 17.116 (floor shift: the polynomial lower bound at the wall at every King point)**.** *Let m^2 be any King point and h a real P_J-periodic function (it may depend on k) with $\|h\|_\infty \leq H$ uniformly in k. Write $a_s := s + \delta m^2(s;\varepsilon)$ for the bare scalar coefficient at the King point s and let M^2 be a King point with $\Delta_0(M^2) := \liminf_{\varepsilon\to 0}[a_M - a_m] > 2H$; such an M^2 exists for every coupling and every H, since $\Delta_0(M^2) \to +\infty$ as $M^2 \to \infty$ (the coefficients of $\partial_s\delta m^2$ are non-negative at order λ, $O(s^{-1/2})$ at order e^2 and $O(s^{-1})$ at the two-loop orders, appendix E.2). Then on $\Lambda \in \{Y_k, X_k\}$, $T \in \{T_1, T_2\}$,*

$$\ln \mathsf{Z}^{m^2}_{\Lambda,T}(0,h) \geq -C|T| - C(1+k)|T| - qk\ln 5, \qquad \textit{hence} \qquad \mathsf{Z}^{m^2}_{\Lambda,T}(tG,h) \geq e^{-C_1|T|}\varepsilon^{\,q'} \quad \textit{for every real } t.$$

This is the polynomial form of Proposition 17.114 at the wall $h_0 = -n\psi$ for every $n > 0$; the sign of h is used nowhere.

Proof. The inputs are: (i) the source-free lower bound [1, (1.14)] at the one fixed King point M^2 (Theorem F.1); (ii) the upper bound [2, (1.3)] at one fixed M_+^2 with $\liminf_{\varepsilon\to 0}[a_{M_+} - a_M] > 0$ (which exists by the same growth); on T_1 the skew instances of Proposition 17.117; (iii) the bounds on the coefficients of the one-point Wick function through order 13 in the couplings on a compact mass interval, from Reisz's power-counting theorem [20, Thm. 1] with the counterterm conditions [13, (3.33)] (the order-e^2 coefficient is computed in appendix E.1), and the boundedness of $Y := 1 + \partial_s \delta m^2$ on $[m^2, M_+^2]$ (appendix E.2); (iv) exact identities: the bare-mass form $\mathsf{Z}^s(T; 0, u\mathbf{1}) = e^{-P_s - 2c_s u|T|}\hat{Z}(a_s - 2u)$ with $\hat{Z}(\gamma)$ the convention-free integral at bare coefficient γ and $P_s = E_0(s) + E_1(s)$, the integrated counterterm identity for $P_M - P_m$, Jensen's inequality, the monotonicity of $\langle \sum_x \varepsilon^3 |\phi|^2 \rangle_\gamma$ in γ, translation invariance on Λ and charge conjugation. *Step 1 (floor shift).* $e^{\sum \varepsilon^3 h|\phi|^2} = e^{h_{\min} \sum \varepsilon^3 |\phi|^2} e^{\sum \varepsilon^3 (h - h_{\min})|\phi|^2}$ moves the wall into the bare coefficient, $\gamma' := a_m - 2h_{\min}$, and Jensen's inequality bounds the remaining factor below by $\exp[\sum \varepsilon^3 (h - h_{\min}) \langle |\phi|^2 \rangle_{\gamma'}] \geq 1$. *Step 2 (the higher King point).* $u := \frac{1}{2}(a_M - \gamma') = \frac{1}{2}(a_M - a_m + 2h_{\min})$ lies in $[0, C]$ for $\varepsilon \leq \varepsilon_0$, by $h_{\min} \geq -H$, $\Delta_0(M^2) > 2H$ and the finite upper limit of $a_M - a_m$; since $-\partial_\gamma \ln \hat{Z} = \frac{1}{2}\langle \sum \varepsilon^3 |\phi|^2 \rangle_\gamma$ is nonincreasing in γ, $\ln \hat{Z}(\gamma') \geq \ln \hat{Z}(a_M) + u|T|(2c_M + w_k(M))$, where input (i) bounds $\ln \hat{Z}(a_M) = \ln \mathsf{Z}^{M^2}(0, 0) + P_M$ below by $-C|T|$ plus the c-numbers. *Step 3 (c-numbers).* By the integrated counterterm identity the four ε^{-1}-divergent c-numbers collapse exactly to $2u(c_M - c_m)|T| = O(|T|)$; the remainder $|T| \int_{m^2}^{M^2} (c_m - c_s) Y \, ds$ is bounded below by $-C|T|$ from $0 \leq c_m - c_s \leq C$ and $|Y| \leq C$ (its sign is not needed), and the perturbative remainder is $\geq -C(1 + k)|T|$ by (iii). *Step 4.* $w_k(M) \geq -C(1 + k)$ is the mirror bound from (i)–(iii) at M^2, M_+^2: the same floor shift with the roles of the two points exchanged and the upper bound (ii) in place of the lower one. *Step 5.* Assemble; the source tG is removed as in Proposition 17.114. No stability bound at m^2 itself, no bound on $w_k(m^2)$ and no two-sided Wick bound is used. The statement is in the free-boundary convention; the c-number mismatch with the torus convention is $\Delta c_T = O(e^{-c\ell(T)})$ (Proposition 17.121(b)). □

The skew-torus instances. Call a torus $\mathbb{R}^3/\Pi$ *aligned* in the frame $(f_\alpha) = (Oe_\alpha)$ of a lattice $O(s\mathbb{Z}^3)$ if $\Pi = O(\ell_1\mathbb{Z} \times \ell_2\mathbb{Z} \times \ell_3\mathbb{Z})$ for some $\ell_\mu > 0$, that is, if the torus is a rectangular box with periodic boundary conditions whose sides are parallel to the frame vectors, and *skew* in that frame otherwise. [1, 2, 3] and [13] are written for a torus aligned in the frame of the blocks ([13, (2.21)]). $T_1 = D_J$ is aligned (a cube) in the frame of X_k and skew in the frame of Y_k; $T_2 = R^{-1}D_J$ is aligned for Y_k and skew for X_k. On T_1 every block-spin step performed in Y's frame, that is all r steps of the standard flow and steps $2, \dots, r$ of the hybrid flow, is a step on a torus skew in the frame of its blocks; on T_2 only the first hybrid step is.

Proposition 17.117 (the skew-torus instances)**.** *King's Theorem 3.1* [13, (3.4)] *(Bałaban's upper bound with the renormalisation of* [3]*) and Bałaban's lower-bound machinery* [1, §§2–4, (1.14)] *hold for the standard flow of Y_k on T_1, the torus RD_J in Y's frame, which is skew, with the constants of the aligned case times at most* 125 *in the linear part; and Bałaban's lower-bound argument holds for steps $2, \dots, r$ of the hybrid flow on T_1 with the constant $C(1 + k)^p$ of Corollary 17.41. In both, the torus enters* [1, 2, 3] *only through the embeddings of the block lattices, the external fields and the image terms of the periodised kernels; no torus symmetry and no torus Fourier analysis is used in the proof of* [3, Prop. 1]*. The statements hold under the provisos $J \in 5^{n_*}\mathbb{N}$ and $J \geq J_0(\delta_r)$ of Theorem 17.13.*

The proof consists of three transfer lemmas. Let $T' := RD_J$ be the skew torus in Y's frame with frame period lattice $\Gamma' := R^{\mathrm{T}}(50MJ\,\mathbb{Z}^3)$ (read in the frame coordinates), its cubic cover $\tilde{T} := \mathbb{R}^3/250MJ\,\mathbb{Z}^3$ (a cube aligned in Y's frame, side $250MJ \in M\mathbb{N}$), the covering $\pi : \tilde{T} \to T'$ with deck group $\mathcal{G} = \Gamma'/250MJ\,\mathbb{Z}^3$ of order 125, and the sup-systole $s_\infty(\Gamma') = 40MJ$ (Bałaban's metric is the sup metric, [1, p. 605]).

Lemma 17.118 (folding)**.** $250MJ\,R^{-1}\mathbb{Z}^3 = 50MJ\,(5R^{-1})\mathbb{Z}^3 \subset 50MJ\,\mathbb{Z}^3 = P_J$ *because $5R^{-1}$ is integral, with index $|\det 5R^{-1}| = 125$; so $\tilde{T}$ covers T' with a deck group of order* 125*. Every*

block lattice Y_{k-j} contains P_J (Lemma 17.14(iii)), so the deck translations commute with every block average and the Gaussian quadratic forms of the flow are deck-invariant; π preserves bonds, blocks, big blocks and the contours, $\mathrm{dist}_{\tilde T}(\tilde x, \pi^{-1}S) = \mathrm{dist}_{T'}(x, S)$ *in the quotient sup metric, and lifted operators intertwine:* $G_j(\pi^{-1}\Omega, A\circ\pi)(f\circ\pi) = (G_j(\Omega, A)f)\circ\pi$*, likewise D^η, the Hölder quotients, δG_j, $C^{(j)}$. Hence every kernel of a Y-frame flow on T' (G_j, $\Delta^{(j)}$, $C^{(j)}$, the slice kernels) is the* 125*-fold periodisation of the corresponding kernel on the aligned torus $\tilde T$, and* [1, Props. 2.1–2.3] *(=*[4]*) and* [13, Thm. 3.3] *hold on T' with the $\tilde T$ constants times at most* 125 *(fibre points are at least s_∞ apart, and the finite sum of exponentially decaying terms is absorbed).*

Proof. The covering step of the aligned case with $(\mathbb{Z}/3)^3$ replaced by $\mathcal{G}$: a non-wrapping parallelepiped lifts to disjoint parallelepipeds [1, p. 611], and positivity on T' is positivity on $\mathcal{G}$-invariant functions. The deck group commutes with the tile average as well ($P_J \subset Y_{k-1}$), so the lemma covers the hybrid steps. □

Lemma 17.119 (restriction principle)**.** *Call a functional Φ liftable of degree $a \in \{0, 1\}$ if $\Phi_{\tilde T}(\xi\circ\pi) = 125^a \Phi_{T'}(\xi)$ (local sums $a = 1$, sup-type $a = 0$, quadratic forms in Bałaban's operators on unions of big blocks $a = 1$, background fields $a = 0$). If an inequality $\Phi \le \Psi$ between liftable functionals of the same degree holds on $\tilde T$ for all configurations obeying block-local restrictions χ and all admissible regions, it holds on T' with the same constants. Moreover $\ell^2(T')$ is the $\mathcal{G}$-invariant subspace of $\ell^2(\tilde T)$ (inner product times* 125*) and lifted operators commute with $\mathcal{G}$, so $\|X_{T'}\| \le \|\tilde X\|$ and $\tilde X \ge 0 \Rightarrow X_{T'} \ge 0$. The lemma does not carry integrals over field configurations (the $\mathcal{G}$-invariant model on $\tilde T$ is a different measure); those are handled by Lemma 17.120.*

Proof. Lifted configurations obey the lifted restrictions; lifted regions $\pi^{-1}\Lambda$ are unions of big blocks with the same distance and corridor data (Lemma 17.118); quadratic forms scale by 125 by the intertwining; backgrounds are images of intertwined operators (the model is abelian). Evaluate the $\tilde T$ inequality at $\xi\circ\pi$ and divide by 125^a. □

Lemma 17.120 (type closure)**.** *With $\sigma_j := s_\infty(\Gamma')/(5^j\varepsilon)$, every step of* [1, §§2–4], [2, §§2–3], [3, §§1–3] *and* [13, §3] *is of one of six types and holds verbatim on T' with the aligned constants:* (I) *exact identities on a finite graph (Gaussian integration, composition of the block-spin maps, block normalisations, set decompositions);* (II) *pointwise bounds on the integrand (dropping non-negative terms, Neumann decoupling, indicator $\le$ exponential, rescalings);* (III) *operator inequalities valid on any finite graph (the diamagnetic inequality,* $\log\det(1+X) \le \mathrm{Tr}\, X$*,* $\mathrm{Tr}(X\mathbf{1}_\Lambda) \le \|X\|\,|\Lambda|$*, $-\Delta_A \ge 0$, also with Neumann restriction);* (IV) *decay and norm bounds for G_j, C_j, $\Delta^{(j)}$, H_j, δG_j on unions of large blocks, wrapping allowed (Lemmas 17.118 and 17.119);* (V) *local arguments on a set of sup-diameter D (gauge-outs, Stokes, local expansions, the contour conventions inside a block, "sets of sup-diameter $< \sigma/2$ do not wrap"), verbatim when $D < \sigma_j$: the sets are blocks, big blocks and corridors whose width in scale-j units grows only logarithmically in t_j^{-1}, and since the product of the spacing with the corridor width increases along the flow the binding case is the coarsest scale δ_r, which is the condition* $40MJ \ge \delta_r \max\{6\sqrt{3}\, r(\delta_r)+2,\, 2/\delta_0\}$*, i.e. $J \ge J_0(\delta_r)$;* (VI) *counts ($|T|$, $|\Lambda|$, $|\partial\Lambda|$, sums over scales, and the estimate* [2, (3.42)]*) in the form of Theorem F.2(d), whose box partition descends to T' under the same non-wrapping condition; the printed cube count* [2, (3.45)] *and the first inequality of (3.46) are false as printed, at every block factor, and are not used): they depend on the torus only through the block count. The large-field analysis (*[2, §3], [1, §4], [13, §3.2]*) uses (II)–(III) on large-field blocks, (I), (III), (IV) for the fluctuation integral with Neumann decoupling, (VI) for the sum over large-field configurations,* $\prod_j \sum_\Lambda e^{-p(t_j)|\Lambda|} \le e^{C|T|}$*, and (I)/(IV)/(V) with the perturbative estimates for the collar terms; the lower bound* [1, §4] *restricts to the whole-torus small-field region by Jensen's inequality and uses (I)–(IV) only. The torus enters the prints only through admissibility, the systole, the kernels and their images, counts, counterterms and Wick images, external-field periodisation and symmetries, and once through a coordinate-parity split of the bonds inside* [4, Prop. 3.1′] *(=*[2, (3.29)]*), a quadratic-form statement that folds from the cover (Lemma 17.119;*

on T' the periods are even at every scale, so it also holds verbatim); the one momentum-space formula of [3]*, (3.28) on p. 441, uses the cubic symmetry with an image-term defect $e^{-c\sigma}$.*

Proof. Each display of [1, Ch. 1–4], [2, pp. 556–594], [3, pp. 411–445] and [13, App.], read from the page images, is assigned to a type or to one of the entry points listed; types (I)–(III) are torus-free, (IV) is Lemmas 17.118 and 17.119, (V) is the non-wrapping condition, and (VI) is the block count. The one external lemma, the cumulant bound, is Theorem 17.112 on the skew tori. □

Proof of Proposition 17.117. The linear part is Lemma 17.118; the nonlinear part (the small- and large-field analysis) is Lemma 17.120 relative to the same bounds on aligned tori at block factor 5, which are Theorem F.2. The hybrid steps $2,\dots,r$ act on composite blocks with the hybrid kernels of Proposition 17.72(a); Bałaban's Neumann cube step is replaced by the constant-background reduction of Lemma 17.40 and the whole-torus form of [3, Prop. 1] for the hybrid flow is Corollary 17.41, whence the constant $C(1+k)^p$. The provisos: $J \in 5^{n_*}\mathbb{N}$ because the frame period lattice Γ' must lie on the 5-adic grid $M_b 5^{n_*}\mathbb{Z}^3$, n_* the number of large-field scales of the Y flow, fixed by the couplings; and the non-wrapping cap $J \geq J_0(\delta_r)$ of type (V). Both are harmless and cofinal: in the flow of this subsection the coarsest spacing is $\delta_r \to 0$ (Lemma 17.26), so no condition on J remains for large k. □

The counterterm bridge.

Proposition 17.121 (the counterterm and Wick-constant conventions)**.** [13] *and* [1, 2, 3] *are proved for the model in which the mass counterterm is defined by* [13, (3.33)] *with the sum over the torus and torus propagators in the graphs, the Wick subtraction by* [13, (2.8)] *with the torus propagator, the vacuum counterterm E_1 by* [13, (3.35)] *as vacuum graphs on the torus, and the source vertex by* [13, (3.21)] *with torus propagators. Definition 15.6 uses the free-boundary conventions of* [14, p. 330]*. For a torus T with shortest period $\ell(T)$ ($= 50MJ$ for T_1, T_2) and spacing s, write $\delta m^2_T(s)$, $E_{1,T}(s)$, $c_T(s)$ for the torus-convention quantities and*

$$\Delta_T(s) := \delta m^2(s) - \delta m^2_T(s), \qquad \Delta E_T(s) := E_1(s) - E_{1,T}(s), \qquad \Delta c_T(s) := \mathsf{W}_s - c_T(s) = -\sum_{\gamma \in P\setminus 0} C_s(\gamma).$$

(a) *With $N = 2$, $\mathsf{Z}_{\Lambda,T}(G,h) = e^{-\Delta E_T|T| - N\Delta c_T \mathsf{S}_\Lambda(h)} \int \prod dA \prod d\phi\, e^{-S^{\mathrm{tor}}_\Lambda - \frac{1}{2}\Delta_T \sum_x s^3|\phi(x)|^2 + F_\Lambda(G) + W^{\mathrm{tor}}_\Lambda(h)}$: the free-boundary model is the torus-convention model with the bare mass shifted by Δ_T and its counterterms frozen at m^2, times a c-number and a shift of the Wick constant. Both block-spin flows carry these c-numbers through unchanged, so the same identity holds between the effective actions at every step.*

(b) *$\Delta_T(s)$, $\Delta c_T(s)$ and $\Delta E_T(s)$ are bounded uniformly in $s \leq s_0$, and $|\Delta_T(s) - \Delta_T(0)| \leq Cs$. Put $\Delta_* := \sup_{s\leq s_0} |\Delta_T(s)|$ and $u(s) := -\Delta_T(s)/2$. The free-boundary effective polynomial of Definition 17.27 equals King's torus-convention polynomial at the constant-shifted source $h + u(s)\mathbf{1}$, minus a c-number, up to graphs that carry at least one u-vertex and have order ≥ 14 in the couplings, whose sum is bounded by $C\delta_r^\sigma|T|$ on $\operatorname{supp}\chi_r$; and Proposition 17.134, the only consumer of Propositions 17.96, 17.113 and 17.114, is c-number blind: multiplying each lattice's Z and small-field integral by an arbitrary positive constant leaves every step and the conclusion unchanged. Consequently the extensive bounds Propositions 17.96, 17.113 and 17.114, proved for the torus-convention model at the sources $(zG, h + u(s)\mathbf{1})$, hold for the model of Definition 15.6 as consumed; for Propositions 17.96 and 17.113 the constant-source shift needs King's and Bałaban's bounds with constants uniform over the family $\{h + u\mathbf{1} : |u| \leq \Delta_*/2\}$, which is the uniformity of* [13, Thm. 3.1] *in the scalar mass over a compact mass set, King's own clause "C depends on g, h and $|T|$"; for Proposition 17.114 the shift is removable by Jensen's inequality.*

(c) *The ratios are convention-free up to a shift of the source by a constant: for every Λ, G, h,*

$$\mathsf{Z}^{\mathrm{free}}_{\Lambda,T}(G,h) = \exp\big[-(\Delta E_T(s) + \Delta_T(s)c_T(s))|T| - 2\Delta c_T(s)\mathsf{S}_\Lambda(h)\big]\, \mathsf{Z}^{\mathrm{tor}}_{\Lambda,T}\big(G, h - \tfrac{1}{2}\Delta_T(s)\mathbf{1}\big),$$

with a prefactor independent of G; so the walled curvature ratio $\langle e^{tF(G)}\rangle_{\mathfrak{c},\Lambda}$ is the same number in both conventions at the shifted wall $h_0 - \Delta_T(\varepsilon_k)/2$.

Proof. (a) and (c) are exact algebra: the two actions differ by the counterterms $\frac{1}{2}\Delta_T \sum_x s^3|\phi|^2 + \Delta E_T|T|$ and the Wick squares by the c-number $-2\Delta c_T \mathsf{S}_\Lambda(h)$; a field-independent constant passes through every block-spin step (Lemma F.3). (b) For $\Delta c_T = -\sum_{\gamma\neq 0} C_s(\gamma)$ the bound is the exponential decay of the massive lattice propagator, $|C_s(\gamma)| \le Ce^{-c|\gamma|}$ uniformly in s, summed over the non-zero periods. For Δ_T, in each of the finitely many graphs of [13, (3.33)] (orders $a + 2b \le 4$) the torus-minus-free difference replaces one propagator by its image sum, which is smooth and removes the coincident-point singularity, so that the remaining singular factors are summable in three dimensions uniformly in s; this gives the boundedness, and the rate $|\Delta_T(s) - \Delta_T(0)| \le Cs$ follows at one loop from the explicit integrals and at the two-loop orders $(0,2)$, $(2,1)$, $(4,0)$ from a graph-by-graph computation of the thirteen two-loop topologies at order $(4,0)$, the degree-zero loops evaluated inner-first and symmetrised, using that the $e^2\lambda$ vertex-type loop vanishes by charge conjugation and the exchange composed with the photon self-energy vanishes by lattice transversality. For ΔE_T the same image-sum argument applies to the vacuum graphs of [13, (3.35)], with one vertex fixed. The statement on the effective polynomials: a constant shift of the scalar source enters King's expansion as a vertex $u\sum \eta^3|\phi|^2$ with a bounded coefficient, and the graphs with a u-vertex and order ≤ 13 in the couplings are exactly the terms by which the two polynomials differ beyond the c-number, so the remainder is a finite set of graphs of order ≥ 14 with running power $\ge 7 > 3$, bounded by $C\delta_r^\sigma|T|$. The c-number blindness of Proposition 17.134 is read off its proof. □

The existence of the walled axis-lattice limits.

Theorem 17.122 (the walled torus limit on the axis lattice)**.** *For every walled configuration $\mathfrak{c} = (J, \psi, n)$ of the class of Theorem 17.13 and every real smooth P_J-periodic 2-form G the limit*

$$\mathcal{W}_{\mathfrak{c}}(tG) := \lim_{k\to\infty} \frac{\mathsf{Z}_{X_k,T_1}(tG, h_0)}{\mathsf{Z}_{X_k,T_1}(0, h_0)} = \lim_{k\to\infty} \langle e^{tF(G)}\rangle_{\mathfrak{c},X_k}$$

exists for every real t; it is the same number in the free-boundary convention of Definition 15.6 and in King's torus convention at the shifted constant source $h_0 - \Delta_T(\varepsilon_k)/2$ (Proposition 17.121(c)), and it equals the limit of the torus-convention ratios $\mathsf{Z}^{\mathrm{tor}}_{X_k,T_1}(tG, h_\infty)/\mathsf{Z}^{\mathrm{tor}}_{X_k,T_1}(0, h_\infty)$ at the frozen wall $h_\infty := h_0 - \Delta_T(0)/2$. The convergence is locally uniform in $t \in \mathbb{C}$, the limit is entire, and all moments of $F(G)$ converge.

This is King's torus theorem [13, Thm. 2.1] with the wall as his scalar source, read on the normalised ratio, in the only form in which it is true at a wall: the convergence of King's effective action [13, Thm. 3.4] modulo a field-independent constant of its scalar-source sector, which cancels in the ratio, with the drift-free stability bound [13, (2.23)] replaced by a polynomial envelope. The proof is given in the following three groups of statements; it compares the walled ratio, which is King's torus-convention ratio at a wall shifted by a bounded constant that moves with k by $O(\varepsilon_k)$, with the torus-convention ratio at a frozen wall on the same lattice, proves the frozen family Cauchy by telescoping one step at a time, and bounds the effect of the constant shift of the source by a per-site bound with its volume factor against the rate of Proposition 17.121(b).

The proof of Theorem 17.122 occupies the rest of this paragraph, up to the subsubsection on the forward bound. It is a one-lattice statement: only the axis lattice $X_k = \Lambda_k$ on the torus $T := T_1 = D_J$ occurs, with King's standard block-spin flow at block factor 5 and King's own corner-rooted blocks (the centred blocks of Convention 17.17 play no role here), and neither the rotated lattice, nor the hybrid flow, nor Theorem 17.6 is used. Throughout, $\mathfrak{c} = (J, \psi, n)$ is in the class of Theorem 17.13, $h_0 := -n\psi$, G is a real smooth P_J-periodic 2-form sampled in the frame of X_k, $t \in \mathbb{R}$, and $r = r(k)$, $\delta_r = 5^r\varepsilon_k = 5^{-\lfloor \varrho k\rfloor}$ are the depth and the coarse spacing of Lemma 17.26. The argument compares two ratios on the same lattice: the walled curvature ratio, which by the bridge of Proposition 17.121 is King's torus-convention ratio at a wall that moves with k by a constant of order ε_k, and the torus-convention ratio at a frozen wall; the frozen family is Cauchy by King's renormalisation theorem read modulo the field-independent constant of its scalar-source sector, and the moving family stays close to the frozen one because a constant shift of the scalar source changes the effective action by a bounded amount per site, which the rate $|u_k| \le C\varepsilon_k$ of Proposition 17.121(b) beats.

Conventions, the bridge and the a priori bounds.

Lemma 17.123 (the moving and the frozen wall)**.** *Write* $\mathsf{Z}_k^{\mathrm{tor}}(t;h) := \mathsf{Z}_{X_k,T}^{\mathrm{tor}}(tG,h)$ *for the torus-convention functional of Proposition 17.121(c) and*

$$h_k := h_0 - \tfrac{1}{2}\Delta_T(\varepsilon_k)\mathbf{1}, \qquad h_\infty := h_0 - \tfrac{1}{2}\Delta_T(0)\mathbf{1}, \qquad u_k := h_k - h_\infty = \tfrac{1}{2}\big(\Delta_T(0) - \Delta_T(\varepsilon_k)\big),$$

a constant with $|u_k| \le C\varepsilon_k$*. Then for every* k *and every real* t

$$r_k(t) := \langle e^{tF(G)}\rangle_{\mathfrak{c},X_k} = \frac{\mathsf{Z}_{X_k,T}(tG,h_0)}{\mathsf{Z}_{X_k,T}(0,h_0)} = \frac{\mathsf{Z}_k^{\mathrm{tor}}(t;h_k)}{\mathsf{Z}_k^{\mathrm{tor}}(0;h_k)}, \qquad s_k(t) := \frac{\mathsf{Z}_k^{\mathrm{tor}}(t;h_\infty)}{\mathsf{Z}_k^{\mathrm{tor}}(0;h_\infty)}.$$

The family $\{h_k\}_k \cup \{h_\infty\}$ *is bounded in every* $C^N(T)$ *norm,* $\|h\|_{C^N} \le \|h_0\|_{C^N} + \Delta_*/2$*;* X_k/P_J *is King's lattice on his torus* [13, (2.21)]*, so the right side of the identity for* r_k *is literally the object of* [13, Thms. 3.1, 3.4] *with the scalar source* h_k*, and King's constants, which depend on the sources only through bounds on finitely many derivatives (*[13, pp. 664–665, 677]*: "the sources* g, h *are bounded by* C*", "their derivatives are uniformly bounded"), are uniform over the family. The physical mass* m^2 *is the same in both conventions and is never changed.*

Proof. The first identity is the walled-ratio display of Definition 15.7 at $g = 0$, with $-w = h_0$ (see also Lemma 15.9(c)): the wall is the scalar source h_0 up to a constant that cancels in the ratio. The second is Proposition 17.121(c) at $\Lambda = X_k$, whose prefactor does not depend on G and cancels in the ratio. The existence of $\Delta_T(0) = \lim_{s\to 0}\Delta_T(s)$ and $|u_k| \le C\varepsilon_k$ are the rate clause of Proposition 17.121(b), and $|\Delta_T(\varepsilon_k)|, |\Delta_T(0)| \le \Delta_*$ give the C^N bound, the shifts being constants. No sign condition on the scalar source enters King's bounds: $\lambda > 0$ makes every single-site measure finite for every real source. □

Lemma 17.124 (a priori bounds on the ratios)**.** *Let* $C_G := \sup_k q_{X_k}(G)/2 < \infty$ *(Lemma 17.22). For every* k*, every real smooth* P_J*-periodic* h *and every real* t*,*

$$1 \le \frac{\mathsf{Z}_k^{\mathrm{tor}}(t;h)}{\mathsf{Z}_k^{\mathrm{tor}}(0;h)} \le e^{t^2 q_{X_k}(G)/2} \le e^{C_G t^2}, \qquad \left|\frac{\mathsf{Z}_k^{\mathrm{tor}}(z;h)}{\mathsf{Z}_k^{\mathrm{tor}}(0;h)}\right| \le \frac{\mathsf{Z}_k^{\mathrm{tor}}(\operatorname{Re} z;h)}{\mathsf{Z}_k^{\mathrm{tor}}(0;h)} \le e^{C_G(\operatorname{Re} z)^2} \quad (z \in \mathbb{C}),$$

and the same in the free-boundary convention. Consequently, if a sequence of such ratios converges for every real t*, it converges locally uniformly on* $\mathbb{C}$ *to an entire function, and all moments of* $F(G)$ *converge to the Taylor coefficients of the limit.*

Proof. The torus-convention walled state is a BFS measure of King type in the sense of the paragraph after Definition 15.7: the torus counterterm and Wick constant only change the Gaussian factor of the single-site measures and a constant. By Lemma 2.8 with $F_{X_k}(G) = \langle A, \ell_{X_k}\rangle$ (Lemma 17.21), $\langle e^{tF(G)}\rangle \le e^{t^2 q_{X_k}(G)/2}$; by Jensen's inequality and $\langle F(G)\rangle = 0$ under the charge conjugation of Definition 15.6, $\langle e^{tF(G)}\rangle \ge 1$. For complex z, $|e^{zF}| = e^{(\operatorname{Re} z)F}$. The functions $z \mapsto \mathsf{Z}_k^{\mathrm{tor}}(z;h)/\mathsf{Z}_k^{\mathrm{tor}}(0;h)$ are entire (the integral converges absolutely for every complex z by the same Gaussian bound) and are bounded on every disc $|z| \le \mathsf{R}$ by $e^{C_G\mathsf{R}^2}$, uniformly in k; convergence on $\mathbb{R}$ then gives locally uniform convergence on $\mathbb{C}$ by Vitali's theorem [10], and Cauchy's estimates give the moments. □

King's flow on the axis lattice. For a lattice $X_{k'}$ and a depth r' with $5^{r'}\varepsilon_{k'} < 1$, King's standard flow [13, §3] on $X_{k'}/P_J$ after r' steps produces the unit lattice $T^{(r')}$ of physical spacing $\delta := 5^{r'}\varepsilon_{k'}$, the small-field indicator $\chi_{r'}$ [13, (3.2)], on whose support $|\phi_{r'}| \le p(\delta)(\lambda\delta)^{-1/4}$, and the effective action $S_{k'}^{(r')}(tG,h)$ of Definition 17.27, here in the torus-convention reading for which [13] is written: its Gaussian parts (quadratic forms, $\ln N_{r'}$, $\ln Z_{r'}(A^{(r')})$) are defined at the mass m^2 without sources [13, (2.13)–(2.16)], and the sources enter only through the curvature vertex [13, (3.20)] and the h-vertex [13, (3.21)] of the perturbative part $E'_{r'}$, expanded to King's order $\bar{n} = 13$ with a remainder. Put

$$I_{k'}^{(r')}(t;h) := \int \chi_{r'}\, e^{-S_{k'}^{(r')}(tG,h)}\, dA_{r'}\, d\phi_{r'}, \qquad \rho_{k'}^{(r')}(t;h) := \frac{I_{k'}^{(r')}(t;h)}{I_{k'}^{(r')}(0;h)},$$

and $I_k := I_k^{(r(k))}$, $\rho_k := \rho_k^{(r(k))}$ at the lattice's own depth. Two facts are used throughout: the unit lattice has $|T^{(r')}| = |T|\delta^{-3}$ sites ([13, (2.15)] in unit-lattice counting); and $5^r\varepsilon_k = 5^{r+1}\varepsilon_{k+1} = \delta_r$, so that X_k at depth r and X_{k+1} at depth $r+1$ live on the same unit lattice $T^{(r)}$ with the same χ_r and the same field bounds ([13, p. 656]: "since $L^k\varepsilon = L^{k+n}\varepsilon'$, the fields A_k, ϕ_k have the same bounds in each integral").

The two stability inputs at the walls.

Proposition 17.125 (King's upper bound on the axis lattice, at a wall)**.** *For h in the family of Lemma 17.123, every k and every admissible depth r', with $\delta = 5^{r'}\varepsilon_k$,*

$$e^{-C\delta^\sigma|T|}\, I_k^{(r')}(t;h) \le \mathsf{Z}_k^{\mathrm{tor}}(t;h) \le e^{C\delta^\sigma|T|}\, I_k^{(r')}(t;h) + B_k, \qquad B_k := e^{-p(\delta)^2 + C|T|}\varepsilon_k^{-q},$$

with $\sigma > 0$, $C = C(t, G, h_0, |T|)$ locally uniform in t, and $q = q(t, G, h_0) < \infty$, all independent of k, of r' and of h in the family.

Proof. This is [13, Thm. 3.1, (3.3)–(3.4)] for King's torus with the sources (tG, h), with the additive large-field term in the form in which King uses it ([13, (3.11)], [14, (4.7)]), at block factor 5 on the torus D_J aligned with King's blocks: Theorem F.2(iii) with Lemma 5.1, in the torus convention for which it is printed, so that no bridge is crossed. Two readings are made, the same two as in Proposition 17.96: the uniformity over the bounded source family (Lemma 17.123); and the envelope. The factor $e^{C|T|}$ of the large-field term is the ultraviolet stability bound [13, (2.23)], which fails as printed when $\int h \neq 0$ (Remark 16.1); what the inductive proof supplies at a wall is a bounded h-linear c-number per step, $e^{C|T|+C'(1+k)|\int h|} = e^{C|T|}\varepsilon_k^{-q}$. The loss is harmless against $e^{-p(\delta_r)^2}$, since $p(\delta_r)^2$ grows at least quadratically in k (Lemma 17.26) while $\ln \varepsilon_k^{-q} = qk \ln 5$ is linear. The sources enter the proof through the vertices (3.20)–(3.21) only, with constants depending on $\sup|G|$, $\sup|\nabla G|$, $\sup|h|$, $\sup|\nabla h|$ ([13, pp. 664–665, 677]). □

Proposition 17.126 (the polynomial lower bound at the wall, torus convention)**.** *For h in the family of Lemma 17.123 and every k: $\mathsf{Z}_k^{\mathrm{tor}}(0;h) \ge e^{-C_1|T|}\varepsilon_k^{q_1}$, with C_1, q_1 independent of k and of h in the family.*

Proof. This is Theorem 17.116 on $\Lambda = X_k$, $T = T_1$, for the bounded family ($\|h\|_\infty \le H := n\|\psi\|_\infty + \Delta_*/2$; the sign of h is not used there), read in the torus convention: the proof of Theorem 17.116 moves the wall into the bare scalar coefficient, compares with a higher King point by Jensen's inequality and the monotonicity of $\langle \sum_x \varepsilon^3 |\phi|^2 \rangle$ in that coefficient, and uses the source-free stability bounds [1, (1.14)], [2, (1.3)] at the higher points together with the integrated counterterm identity and the perturbative bounds on the one-point Wick function. Each of these steps is available for the torus-convention model with the same constants: [1, (1.14)] and [2, (1.3)] are printed for it (Theorem F.2(iii)); its bare coefficient, Wick constant and vacuum constant differ from the free-boundary ones by $\Delta_T(s)$, $\Delta c_T(s)$, $\Delta E_T(s)$, which are bounded uniformly in s (Proposition 17.121(b)) and shift the King points by bounded amounts only; and the perturbative coefficients of the torus one-point function differ from the infinite-lattice ones by image sums, exponentially small in $\ell(T)$. The bound reads $\ln \mathsf{Z}_k^{\mathrm{tor}}(0;h) \ge -C|T| - C(1+k)|T| - qk\ln 5$, which is the claim with $q_1 = q + C|T|/\ln 5$.

At the mass-increasing wall the same conclusion also follows from Jensen's inequality as in Proposition 17.115, in the torus convention: $\mathsf{Z}_k^{\mathrm{tor}}(0;h) \ge \mathsf{Z}_k^{\mathrm{tor}}(0;0)\exp[\mathsf{S}_{X_k}(h)\, w_k^{\mathrm{tor}}]$ with $w_k^{\mathrm{tor}} := \langle |\phi(x)|^2 \rangle_{0,0}^{\mathrm{tor}} - 2c_T(\varepsilon_k)$, independent of x by translation invariance ($P_J \subset X_k$), where $\mathsf{Z}_k^{\mathrm{tor}}(0;0) \ge e^{-C|T|}$ is [1, (1.14)] and $\mathsf{S}_{X_k}(h) = -n\mathsf{S}_{X_k}(\psi) - \frac{1}{2}\Delta_T|T| < 0$ by the wall condition of Theorem 17.13; together with an upper bound $w_k^{\mathrm{tor}} \le C(1+k)$ on the source-free one-point Wick function, which is not established here and not needed, this gives the claim with $q_1 = C|\mathsf{S}_{X_k}(h)|/\ln 5$. This is the only place in the proof where the sign of the wall would enter; the route through Theorem 17.116 does not use it. □

The ratios are small-field ratios up to $o(1)$.

Lemma 17.127 (the large-field term is negligible)**.** *Let h be in the family of Lemma 17.123, r' an admissible depth for X_k, $\delta = 5^{r'}\varepsilon_k$, $x := C\delta^\sigma|T|$ with the constant of Proposition 17.125, and $b_k(h) := 2B_k/\mathsf{Z}_k^{\mathrm{tor}}(0;h)$. Then*

$$b_k(h) \le 2e^{-p(\delta)^2+(C+C_1)|T|}\varepsilon_k^{-q-q_1},$$

and whenever $b_k \le \frac12$ and $2x \le \ln 2$:

(42)

$$e^{-2x}(1-b_k)\,\rho_k^{(r')}(t;h) \le \frac{\mathsf{Z}_k^{\mathrm{tor}}(t;h)}{\mathsf{Z}_k^{\mathrm{tor}}(0;h)} \le e^{2x}\rho_k^{(r')}(t;h)+b_k, \qquad \rho_k^{(r')}(t;h) \le 4e^{C_G t^2}, \qquad \Big|\frac{\mathsf{Z}_k^{\mathrm{tor}}(t;h)}{\mathsf{Z}_k^{\mathrm{tor}}(0;h)} - \rho_k^{(r')}(t;h)\Big| \le 4e^{C}$$

At the lattice's own depth $r = r(k)$, $x = C\delta_r^\sigma|T| \to 0$ and $b_k(h) \to 0$ as $k \to \infty$ (Lemma 17.26: $p(\delta_r)^2$ grows at least quadratically in k, the exponent of $\varepsilon_k^{-q-q_1}$ linearly), so $|r_k(t) - \rho_k(t;h_k)| \to 0$ and $|s_k(t) - \rho_k(t;h_\infty)| \to 0$ for every real t, locally uniformly in t.

Proof. Write $Z(t) := \mathsf{Z}_k^{\mathrm{tor}}(t;h)$, $I(t) := I_k^{(r')}(t;h)$, $\rho := \rho_k^{(r')}(t;h)$, $B := B_k$, $b := b_k(h)$. The bound on b is Propositions 17.125 and 17.126. By Proposition 17.125 at t and at 0: $Z(t) \le e^x I(t)+B$ and $Z(0) \ge e^{-x}I(0)$ give $Z(t)/Z(0) \le e^{2x}\rho+B/Z(0) \le e^{2x}\rho+b$; and $Z(t) \ge e^{-x}I(t) = e^{-x}\rho I(0)$ with $I(0) \ge e^{-x}(Z(0)-B)$ give $Z(t)/Z(0) \ge e^{-2x}\rho(1-B/Z(0)) \ge e^{-2x}(1-b)\rho$. Hence, by Lemma 17.124, $\rho \le e^{2x}(1-b)^{-1}Z(t)/Z(0) \le 2e^{2x}e^{C_G t^2} \le 4e^{C_G t^2}$. For the difference: $Z(t)/Z(0)-\rho \le (e^{2x}-1)\rho+b$ and $\rho - Z(t)/Z(0) \le \rho[(1-e^{-2x})+b] \le \rho[(e^{2x}-1)+b]$; the factor $(1+4e^{C_G t^2})$ on b covers both sides. The last statement is the case $r' = r(k)$, $h \in \{h_k, h_\infty\}$, with the identities of Lemma 17.123. □

Graphs, degrees and the census of the source subgraphs. King's perturbative part $P_k^{(r)}(tG,h)$ is a finite sum of connected renormalised graphs H [13, (3.29), (3.37)] with the vertices [13, (3.18)–(3.26)]; the number of graphs depends on $\bar n$ only, each graph is multilinear in its source insertions, and its value on $\operatorname{supp}\chi_r$ is $\sum_{\{y\},\{z\}} E^{(r)}(H;\{y\},\{z\})\times(\text{external fields at }\{y\})$, with a kernel that depends on the coarse background $A^{(r)}$ through its lines $G_r(A^{(r)})$ [13, (3.17)], so that a graph without external legs still depends on the background. Degrees are King's [13, (3.30)–(3.31)], as in Lemma 17.60, computed on the point-split graph (the two summation points of a block product as vertices, as in Corollary 17.41), with two conventions. The curvature vertex [13, (3.20)] is counted with $D_v = \frac52$, the lattice curl of its photon leg having been summed by parts onto the smooth G (exact on the torus, $|d_\eta^* G| \le C\|\nabla G\|_\infty$); with the curl charged to the vertex, $D_v = \frac32$ and the tree with two curvature vertices would have degree 0, so the summation by parts is necessary. The h-vertex [13, (3.21)] has $D_v = 2$ (Lemma 17.60). Every connected subgraph containing an h-vertex and a curvature vertex therefore has $D \ge 2+\frac52-3 = \frac32 > 0$, and the smallest vacuum graph with two curvature vertices and two current vertices has $D = 5$.

Each graph value is split at zero coarse fields, $E(H) = E(H)|_0 + [E(H)-E(H)|_0]$, where $|_0$ means $(A_r,\phi_r) = (0,0)$, and we put $E_N(H) := E(H)$ if H has an external scalar leg or a curvature vertex and $E_N(H) := E(H)-E(H)|_0$ otherwise: the field- or source-dependent part. The field-free, G-free values $E(H)|_0$ of the remaining graphs (h-linear or h-nonlinear vacuum graphs without curvature vertex, and the h-free vacuum energy) are numbers depending on k, r, h only; nothing is claimed about their convergence, and for the order-e^2 h-linear family it fails (Remark 16.1: the Wick constant drifts by $c_*\ln 5$ per step).

Lemma 17.128 (the census of the source subgraphs)**.** *Expand every graph in the coarse background as in [13, (3.43)–(3.55)] (Lemma 17.40(1) for one flow): $E(H)$ becomes a finite sum of graphs $\tilde H$ with lines at zero background, explicit background legs $\tilde A$ at block points, and remainder factors, and $E(H)|_0$ is the core $\tilde H_0$ without background legs. Then:*

(a) *Every $\tilde H$ entering $E_N(H)$ is connected and has an external leg (scalar, background or dG) or a curvature vertex.*

(b) *For such $\tilde H$, every connected ordered subgraph γ (lines with scale indices below r) has strictly positive King degree or vanishes identically. In detail: if γ contains a block point, $D(\gamma) \ge 1$ (≥ 2 with a source vertex); if γ contains an h-vertex, $D(\gamma) \ge \frac12$ unless γ is a vacuum graph*

with exactly one h-vertex and no curvature vertex (then $\gamma = \tilde{H}$, excluded by (a)) or γ is the h-loop through a single current vertex [13, (3.22)] *with its photon leg external ($D = 0$: the current tadpole of Lemma 17.60 with the h-insertion), which vanishes identically at zero background ($\operatorname{Tr} q = 0$); if γ contains a curvature vertex and no h-vertex, $D(\gamma) \geq \frac{1}{2}$ unless γ is the curvature vertex joined by a photon line to a current tadpole ($D = 0$), which vanishes identically at zero background for the same reason; if γ is source-free, $D(\gamma) > 0$ after Bałaban's resummed rewriting* [3, (3.5)]*, the input of Corollary 17.41.*

(c) *The rewriting of the source-free subdivergences inside a graph that carries source vertices (Bałaban's Taylor subtraction at the subdivergence with its counterterm partner,* [3, (3.5)], [13, (3.83)–(3.85)]*) does not lower the degree of a prefix component that contains a source vertex below the value counted in (b).*

(d) *The only connected subgraphs containing a source vertex with King degree ≤ 0 in the raw count are the field-free h-linear vacuum graphs of order ≤ 2 in the couplings, the family $\mathcal{T}_h$ of Lemma 17.60 (which have no external leg, hence occur only as whole graphs), and the two vanishing exceptions of (b).*

Consequently the minimum α_0 of the degrees of the non-vanishing ordered subgraphs over the finitely many graphs and orderings is strictly positive.

Proof. (a) The core $\tilde{H}_0$ is the same in $E(H)$ and $E(H)|_0$ and cancels in $E_N(H)$ when H has no external scalar leg and no curvature vertex; every other $\tilde{H}$ carries a background leg, an external dG line or a remainder factor (the remainders are bounded separately, Lemma 17.40(2)). (b) The degree arithmetic is that of Lemma 17.60: inserting an h-vertex into a scalar line adds 2; source-free connected subgraphs with an external leg have $D \geq -1$ except the current tadpole ($D = -2$), so a subgraph with an h-vertex and an external leg has $D \geq 1$ except the h-loop through one current vertex ($D = 0$); a vacuum subgraph with exactly one h-vertex has $D \leq 0$ only at order ≤ 2 in the couplings, the exhaustive enumeration of Lemma 17.60, and it is then the whole graph; with two or more h-vertices $D \geq 1$. For a curvature vertex, which attaches by a photon leg and contributes $\frac{5}{2}$, the same enumeration leaves the one exception named. Both exceptions vanish at zero background because the current one-point function of the loop vanishes, $\operatorname{Tr} q = 0$, exactly as for the current tadpole in Lemma 17.60. A component containing a block point has degree ≥ 1 by the point-split count of Corollary 17.41 (every component containing a block point has Bałaban's degree ≥ 1, [3, (2.1)–(2.2)]), and a source vertex adds at least 1 more. (c) Bałaban's Taylor subtraction at a source-free subdivergence S raises the degree of S by θ_S and lowers by θ_S the degree of the external line of S that receives the derivative; the construction, run on the whole graph with the source vertices counted as vertices, rewrites S only in orderings in which every line of S is finer than every external line of S. Let γ be a component of an ordered subgraph. If the line that receives the derivative lies in γ, then every line of S, being finer, lies in the same ordered subgraph, and S, connected to that line, lies in γ: the two changes cancel. Otherwise the degree of γ can only increase. By induction over the forest of subdivergences, finest first, no component falls below its value in (b); the source vertices, which are never inside a rewritten S, keep their contributions 2 and $\frac{5}{2}$. (d) is read off the same arithmetic. □

The restricted renormalisation theorem. King's renormalisation theorem [13, Thm. 3.5] asserts the per-site bound (3.38) and the convergence (3.39) of the graph kernels for all graphs. As printed it is false on the order-e^2 h-linear field-free family (Remark 16.1), and the convergence of the field-free, G-free constants is not needed here. What is used is the following restricted form.

Proposition 17.129 (the restricted renormalisation theorem)**.** *Let h be in the family of Lemma 17.123, t real, and let $\gamma_*, \beta_*, \sigma$ be the exponents of* [13, Thm. 3.4].

(i) (convergence) *For every graph H and every k, with $r = r(k)$, the field- or source-dependent parts obey, after the sum over the positions, for any $\gamma' < \gamma_*$, $\beta' > \frac{7}{2}$, $\sigma' < \sigma$,*

$$\sum_{\text{positions}} \left|E^{(r+1)}_{N,k+1}(H) - E^{(r)}_{N,k}(H)\right| \times |\text{external fields}| \leq C\left[5^{-\gamma' r}\delta_r^{-\beta'} + \delta_r^{\sigma'}\right]|T| \quad \text{on } \operatorname{supp}\chi_r,$$

the two kernels living on the common unit lattice $T^{(r)}$ of X_k at depth r and X_{k+1} at depth $r+1$; $C = C(t, G, h_0, |T|)$ is independent of k and locally uniform in t.

(ii) (per-site bound) *For every graph H with at least one h-vertex, other than the degree-0 members of the order-e^2 h-linear field-free family,*

$$|E_k^{(r)}(H; \{y\}, \{z\})| \le C(1+k-r)^{p_0}\, \delta_r^{\varrho_H}\, p(\delta_r)^{n_1}\, e^{-\delta\, \mathrm{dist}(\{y\},\{z\})}, \qquad \varrho_H \ge 0,$$

and for the degree-0 members of that family the same bound with the extra factor $(1+r)$, which is necessary there. For a graph at non-zero background the bound is read through the expansion of Lemma 17.128: the core at zero background falls under the second clause if H is in that family and under the first otherwise, and every other term has an external leg and falls under the first. C is linear in $\sup|h|$ for a distinguished h-vertex and depends on the other sources through sup norms; p_0, n_1 depend on $\bar{n}$ only.

Proof. King's proof of (3.39) [13, pp. 660–662] reduces it to the convergence of a single graph with one line replaced by the difference of the two flows' lines (3.56), and that to [13, Prop. 3.6], the one-flow estimate for a renormalised family with one element weakened by a small degree, which King asserts on p. 665 "by redoing the analysis of Sect. 3.4" and which is Corollary 17.41. Corollary 17.41 applies to a family exactly when every connected ordered subgraph of its non-vanishing graphs has strictly positive degree after Bałaban's rewriting: a zero-degree ultraviolet subgraph containing the replaced line would give $\sum_j (5^j\eta)^{-\gamma} \sim 5^{\gamma r}$ and cancel the gain $5^{-\gamma r}$. By Lemma 17.128 this holds, with the minimum $\alpha_0 > 0$, for every $\tilde{H}$ entering $E_N(H)$. The replaced line is a fine line of [13, (3.17)] with scale index below r; the lines with indices between r and k occur only on the counterterm lines of the δm^2 graphs, never at a source vertex, so the counterterm-attachment clause [13, (3.78)] and the one-vertex attachment are the source-free statements and are unchanged by the presence of the sources. The remaining steps are King's: the kernel bounds [13, Thm. 3.3, Props. 3.7–3.9] for the lines and the fine-field reconstructions, printed for the torus convention at block factor 5 (Theorem F.2); the sampling of the sources $x' \to x$ (p. 665), which costs $C\varepsilon_k\|\nabla h\|_\infty + C\varepsilon_k\|\nabla G\|_\infty \le C5^{-\gamma r}$; the tree-level h-term $\delta_r^2 \sum \eta^3 h|\phi^{(r)}|^2$, which has no internal line and is compared directly through the convergence of the fine-field reconstructions $a_r G_r Q_r^*$ [13, Thm. 3.3, §3.6]; the non-polynomial terms of (3.14) by [13, §3.6, (3.89)–(3.98), Prop. 3.10]; the Taylor remainders beyond order $\bar{n}$, $C\delta_r^\sigma|T|$ per flow (p. 665); and the position sum (3.40)–(3.42), which gives $|T|$ with one free position, the others tied by the exponential tree decay. The factor $(1+k-r)^{p_0}$ of Corollary 17.41 is a power of $\log \delta_r^{-1}$ and is absorbed by $\gamma' < \gamma_*$, $\sigma' < \sigma$, and the field norms $(\lambda\delta_r)^{-s/4} p(\delta_r)^n$ of the external legs by $\beta' > \frac{7}{2}$. This proves (i).

(ii) For a graph outside the degree-0 members of the order-e^2 h-linear field-free family, Lemma 17.128 gives strictly positive ordered degrees for every non-vanishing derived graph (the excluded vacuum case of the census is now the family itself), and Corollary 17.41 without weakening gives the per-site bound with the factor $(1+k-r)^{p_0}$, King's "possible addition of some power of $\ln(L^k\varepsilon)^{-1}$" (p. 666); the sup-norm dependence on the sources is King's (p. 664, 677), the distinguished vertex entering linearly. The two h-linear field-free members of order 0 and λ are bounded directly: the subtracted tadpole at zero coarse field is the infrared remainder $-N\sum \eta^3 \delta_r^2 h\, F_r(x,x)$ of the coincident propagator, bounded per site at unit scale, and the figure-eight minus its δm_1^2 partner leaves a bubble of positive ordered degree times an infrared tadpole line, as in step 3 of the proof of Corollary 17.41. For the degree-0 members of the order-e^2 family (the exchange and seagull tadpoles with their δm_2^2 partner) every proper ordered subgraph has $D > 0$ (a proper component with the h-vertex is not a vacuum graph, the h-loop vanishes, and the source-free components are the self-energy and the photon tadpole, renormalised by the δm_2^2 partner) and only the whole graph has $D = 0$: the Taylor identity for the self-energy with its δm_2^2 partner, Bałaban's general step at the whole graph with no final renormalisation (there is none to perform), as in King's worked case [13, (3.81)–(3.87)], makes every proper prefix component of degree $\ge \alpha$ and leaves the whole graph at degree 0, so in the multiscale sum every geometric series keeps its constant except the last, $\sum_{j \le r-1} (5^j\eta)^0 = r$: this is the factor $(1+r)$, the zero-degree logarithm of Corollary 17.41. It is necessary: the value of

the family at a constant source is u times the drift of Remark 16.1, of size r at fixed δ_r. The infrared counterterm lines are handled as in (i). □

The Cauchy estimate at the frozen wall.

Proposition 17.130 (King's Cauchy estimate modulo the field-free constant)**.** *Let $h = h_\infty$ and $t \in \mathbb{R}$, and define the field-free constant $a_k^{(r)}(h) := S_k^{(r)}(0,h)\big|_{A_r=\phi_r=0}$ (the whole value of the effective action at zero fields and zero curvature source) and $d_k := a_k^{(r)}(h) - a_{k+1}^{(r+1)}(h)$, a number independent of t and of G. Then on $\operatorname{supp}\chi_r$, with γ', β', σ' as in Proposition 17.129(i),*

$$\big|S_k^{(r)}(tG,h) - S_{k+1}^{(r+1)}(tG,h) - d_k\big| \le \eta_k |T|, \qquad \eta_k := C\big[5^{-\gamma' r}\delta_r^{-\beta'} + \delta_r^{\sigma'}\big],$$

C independent of k and locally uniform in t.

Proof. Both effective actions live on $T^{(r)}$ with the same χ_r. Their Gaussian parts converge by [13, Thm. 3.3] (the step King takes for (3.9) before the perturbative part), and their difference is a quadratic form in the fields plus a constant, the constant being part of d_k. In the perturbative parts, split every graph as $E(H) = E(H)|_0 + E_N(H)$ when H has no external scalar leg and no curvature vertex: the sum of the values $E(H)|_0$ over these graphs, at $G = 0$, is by definition the field-free constant $a_k^{(r)}(h)$ minus the Gaussian constant, and the graphs with a curvature vertex and no external leg are included in E_N and converge by Proposition 17.129(i). So $S_k^{(r)}(tG,h) - S_{k+1}^{(r+1)}(tG,h) - d_k$ is a sum of differences $E_N^{(r+1)}(H) - E_N^{(r)}(H)$ over the finitely many graphs, each bounded by Proposition 17.129(i), plus the two Taylor remainders, $\le C\delta_r^\sigma|T|$ each, plus the non-polynomial terms, included in Proposition 17.129(i). This is King's [13, (3.9)] without the field-free h-sector constant, which is exactly what is false in the printed form at $\int h \neq 0$ (Remark 16.1: the order-e^2 h-linear constant drifts by $c_* \ln 5 \int h$ per step, so $d_k \not\to 0$); d_k is t- and G-independent by construction and cancels between numerator and denominator of every ratio below. Only one step, $n = 1$, is used. □

The constant-source shift on one lattice.

Lemma 17.131 (the constant-source shift of the effective action)**.** *Let h, h' be in the family of Lemma 17.123 with $h - h' = u\mathbf{1}$ constant. Then on $\operatorname{supp}\chi_r$, for every real t,*

$$\big|S_k^{(r)}(tG,h) - S_k^{(r)}(tG,h')\big| \le C(1+r)\,|u|\,p(\delta_r)^{n_1}\delta_r^{-3}|T| + 2C\delta_r^\sigma|T|, \tag{43}$$

with $C = C(t, G, h_0, |T|)$ independent of k and u and locally uniform in t, n_1 the exponent of Proposition 17.129(ii) enlarged to absorb $(1+k-r)^{p_0}$, and σ the remainder exponent.

Proof. Where h enters. By [13, (3.14)–(3.17)] the Gaussian parts of $S_k^{(r)}$ are defined at the mass m^2 without sources, and the sources sit in E_r' only, through the vertices (3.20), (3.21); the h-vertex is linear in h with the explicit coupling $\delta_r^2 \le 1$. So $S_k^{(r)}(tG,h) - S_k^{(r)}(tG,h') = [P_k^{(r)}(h) - P_k^{(r)}(h')] + [R(h) - R(h')]$, P the Taylor polynomial of order $\bar n = 13$ and R the remainder. *The remainder.* $|R(h)|, |R(h')| \le C\delta_r^\sigma|T|$ on $\operatorname{supp}\chi_r$, uniformly over the source family ([13, p. 665]); this is the second term, and no derivative of R in u is taken. *The polynomial.* $P_k^{(r)}(h)$ is a finite sum of connected renormalised graphs with $m \ge 0$ h-vertices, each multilinear in its source insertions; by multilinearity $P(h) - P(h')$ is the sum over the graphs with $m \ge 1$ of $\sum_{i=1}^m [H$ with the i-th h-vertex carrying the constant u and the others carrying h or $h']$. The value of each such graph on $\operatorname{supp}\chi_r$ is $\sum_{\{y\},\{z\}} E_k^{(r)}(H;\{y\},\{z\}) \times$ (external fields at $\{y\}$). By Proposition 17.129(ii), $|E_k^{(r)}(H)| \le C(1+r)(1+k-r)^{p_0}\delta_r^{\varrho_H}p(\delta_r)^{n_1}e^{-\delta\,\mathrm{dist}}$ with $\varrho_H \ge 0$ and C linear in $|u|$ for the distinguished vertex (a constant source has zero derivatives), the factor $(1+r)$ being present only for the degree-0 members of the order-e^2 h-linear field-free family and harmless elsewhere; the external scalar fields obey $|\phi_r(y)| \le p(\delta_r)(\lambda\delta_r)^{-1/4}$ [13, (3.2)], with $\delta_r^{-s/4}$ absorbed by $\varrho_H \ge s/4$ for s external scalar legs, and the external photon legs are bounded likewise by (3.2), so the product of fields and kernel is $\le C|u|(1+r)(1+k-r)^{p_0}p(\delta_r)^{n_1+s}$ per position; this is the whole content of King's remark that the sources "may be bounded when

making a perturbative expansion" and "do not affect the leading order positivity of the action" (p. 677). The graph is connected and its kernel decays exponentially in the tree distance of all its vertices, so the sum over all positions is $\le C|T^{(r)}| = C|T|\delta_r^{-3}$: one free position, the others tied. Collecting the finitely many graphs, with $(1+k-r)^{p_0} \le C(1+\log\delta_r^{-1})^{p_0}$ absorbed by enlarging n_1, gives the first term of (43). □

The bound (43) keeps no smallness from the explicit δ_r^2 of the h-vertex, from the running couplings or from the positive degrees: it is a per-site bound $O((1+r)p^{n_1})$ times the site count $|T|\delta_r^{-3}$. The true size of the worst term, the tree-level term $u\delta_r^2\sum\eta^3|\phi^{(r)}|^2$, is $|u|p^2\lambda^{-1/2}\delta_r^{-3/2}|T|$ on $\operatorname{supp}\chi_r$: a constant scalar source is a bare-mass shift, and its effect is a running-mass term of the same order as the mass term already present, which does not tend to zero. The smallness comes from the rate below, not from the effective action.

Lemma 17.132 (the rate and the choice of ϱ)**.** *With $u_k = \frac12(\Delta_T(0)-\Delta_T(\varepsilon_k))$ of Lemma 17.123, $|u_k| \le C\varepsilon_k = C5^{-k}$ (Proposition 17.121(b)), and*

$$\eta'_k := C(1+r)|u_k|p(\delta_r)^{n_1}\delta_r^{-3} + 2C\delta_r^{\sigma} \le C(1+k)\,5^{-k+3\lfloor\varrho k\rfloor}\big(b_0(1+k\ln 5)\big)^{\mathrm{p}n_1} + 2C\,5^{-\sigma\lfloor\varrho k\rfloor} \longrightarrow 0,$$

since $\varrho < \frac19 < \frac13$ (Lemma 17.26): the polynomial factor $(1+k)^{1+\mathrm{p}n_1}$ is beaten by the geometric $5^{-(1-3\varrho)k}$. Hence, on $\operatorname{supp}\chi_r$, $|S_k^{(r)}(tG,h_k) - S_k^{(r)}(tG,h_\infty)| \le \eta'_k|T|$, and, exponentiating under the common χ_r (same lattice, same depth, same fields),

$$I_k(t;h_k) = e^{\theta_k(t)|T|}I_k(t;h_\infty), \quad |\theta_k(t)| \le \eta'_k, \qquad \text{so} \qquad \frac{\rho_k(t;h_k)}{\rho_k(t;h_\infty)} \in [e^{-2\eta'_k|T|}, e^{2\eta'_k|T|}], \qquad |\rho_k(t;h_k)-\rho_k(t;h_\infty)| \le$$

for k so large that $2\eta'_k|T| \le 1$, locally uniformly in t.

Proof. The first display is (43) with $u = u_k$, $\delta_r = 5^{-\lfloor\varrho k\rfloor}$, $1+r \le 1+k$ and $p(\delta_r) \le b_0(1+k\ln 5)^{\mathrm{p}}$; both terms tend to 0 exactly when $0 < \varrho < \frac13$, and every condition of Lemma 17.26 on ϱ is an upper bound, so its window $\varrho < \frac19$ needs no change. The second: $I_k(t;h_k) = \int \chi_r e^{-S(tG,h_\infty)}e^{-[S(tG,h_k)-S(tG,h_\infty)]}$ lies in $[e^{-\eta'_k|T|}, e^{\eta'_k|T|}]I_k(t;h_\infty)$, and the same at $t=0$; divide, and use $\rho_k(t;h_\infty) \le 4e^{C_G t^2}$ (Lemma 17.127) with $e^x - 1 \le 2x$ for $0 \le x \le 1$. □

The frozen family is Cauchy.

Lemma 17.133 (one step at the frozen wall)**.** *Fix $t \in \mathbb{R}$ and write $r = r(k)$. For $k \ge k_0$ (so that $\delta_r < 1$, $b_k, \tilde b_{k+1} \le \frac12$, $2C\delta_r^{\sigma}|T| \le \ln 2$ and $2\eta_k|T| \le 1$), with $\tilde\rho_{k+1}(t) := \rho_{k+1}^{(r+1)}(t;h_\infty)$ the small-field ratio of X_{k+1} at depth $r+1$ (which is $r(k+1)$ or $r(k+1)+1$),*

$$\epsilon(k) := |s_k(t) - s_{k+1}(t)| \le 16e^{C_G t^2}|T|\big(2C\delta_r^{\sigma} + \eta_k\big) + (1+4e^{C_G t^2})\big(b_k + \tilde b_{k+1}\big),$$

where $b_k \le 2e^{(C+C_1)|T|}e^{-p(\delta_r)^2}5^{(q+q_1)k}$ and $\tilde b_{k+1} \le 2e^{(C+C_1)|T|}e^{-p(\delta_r)^2}5^{(q+q_1)(k+1)}$. Consequently $\sum_k \epsilon(k) < \infty$, $(s_k(t))_k$ is Cauchy, and $W^{\mathrm{tor}}(t) := \lim_k s_k(t)$ exists, locally uniformly in t.

Proof. (B1) The Cauchy estimate at one step. By Proposition 17.130, on $\operatorname{supp}\chi_r$: $S_k^{(r)}(tG,h_\infty) = S_{k+1}^{(r+1)}(tG,h_\infty) + d_k + R_k(A_r,\phi_r;t)$ with $|R_k| \le \eta_k|T|$ and d_k independent of the fields and of t. Exponentiate and integrate against the common χ_r: $I_k^{(r)}(t;h_\infty) = e^{-d_k}I_{k+1}^{(r+1)}(t;h_\infty)e^{\theta|T|}$ with $|\theta| \le \eta_k$, and the same at $t=0$ with the same d_k. Dividing,

$$\rho_k(t;h_\infty) = \tilde\rho_{k+1}(t)\,e^{\theta'|T|}, \quad |\theta'| \le 2\eta_k, \qquad \text{so} \qquad |\rho_k(t;h_\infty) - \tilde\rho_{k+1}(t)| \le \tilde\rho_{k+1}(t)\big(e^{2\eta_k|T|}-1\big).$$

(B2) The large-field terms at the two depths. Lemma 17.127 uses only Proposition 17.125 at the given lattice and depth and Proposition 17.126 for that lattice, which is depth-free; both hold for X_k at depth r and for X_{k+1} at depth $r+1$ ($\delta_{r+1}(X_{k+1}) = \delta_r < 1$), with the stated bounds on b_k and $\tilde b_{k+1}$. Hence $\rho_k(t;h_\infty), \tilde\rho_{k+1}(t) \le 4e^{C_G t^2}$ and, by (42) at both lattices,

$$|s_k(t)-\rho_k(t;h_\infty)| \le 4e^{C_G t^2}(e^{2C\delta_r^{\sigma}|T|}-1)+(1+4e^{C_G t^2})b_k, \qquad |s_{k+1}(t)-\tilde\rho_{k+1}(t)| \le 4e^{C_G t^2}(e^{2C\delta_r^{\sigma}|T|}-1)+(1+4e^{C_G t^2}$$

(B3) One step. By (B1), (B2) and the triangle inequality,

$$\epsilon(k) \le 4e^{C_G t^2}\big[2(e^{2C\delta_r^{\sigma}|T|}-1) + (e^{2\eta_k|T|}-1)\big] + (1+4e^{C_G t^2})(b_k + \tilde b_{k+1}),$$

and $e^x - 1 \le 2x$ for $0 \le x \le 1$ gives the claim. *(B4) Summability.* With $\delta_r = 5^{-\lfloor \varrho k \rfloor} \le 5 \cdot 5^{-\varrho k}$ and $r = k - \lfloor \varrho k \rfloor \ge (1-\varrho)k$: $\delta_r^{\sigma} \le 5^{\sigma} 5^{-\sigma \varrho k}$; $5^{-\gamma' r} \delta_r^{-\beta'} \le 5^{\beta'} 5^{-[\gamma' - (\gamma'+\beta')\varrho]k}$, geometric once $\varrho < \gamma'/(\gamma'+\beta')$, which holds for $\gamma' < \gamma_*$ close to γ_* and $\beta' \in (\frac{7}{2}, 5)$, because the window $\varrho < \gamma/(\gamma+\beta)$ of Lemma 17.26, with $\gamma \le \gamma_*$ and $\beta \ge 5$, lies strictly inside $\varrho < \gamma_*/(\gamma_* + \beta')$; and $b_k + \tilde{b}_{k+1} \le 4e^{(C+C_1)|T|} 5^{q+q_1} \exp[-b_0^2 (1 + \lfloor \varrho k \rfloor \ln 5)^{2\mathrm{p}} + (q+q_1) k \ln 5]$, a quadratic-in-$k$ negative exponent against a linear positive one ($\mathrm{p} \ge 1$, Lemma 17.26), hence $\le C e^{-ck^2 + C'k}$. So $\sum_{k \ge k_0} \epsilon(k) < \infty$, and for $k_0 \le k < k'$, $|s_k(t) - s_{k'}(t)| \le \sum_{j=k}^{k'-1} \epsilon(j) \le \sum_{j \ge k} \epsilon(j) \to 0$. All constants are uniform for t in compacts. No uniformity in the number of steps is used or claimed: a sequence with summable consecutive differences is Cauchy. □

This lemma alone is King's torus theorem [13, Thm. 2.1] with the frozen wall h_∞ as his scalar source, read on the normalised ratio: the differences from King's argument [13, (3.10)–(3.13)] are that the drift-free stability bound $|\ln Z| \le C|T|$ [13, (2.23)], false at a wall, is replaced by the polynomial envelope of Propositions 17.125 and 17.126, that the field-free h-sector constant, which does not converge, is removed by the ratio, and that the uniformity in the number of steps is replaced by summation over single steps.

Assembly.

Proof of Theorem 17.122. Fix $t \in \mathbb{R}$. By Lemma 17.127 at the two walls and Lemma 17.132,

$$|r_k(t) - s_k(t)| \le |r_k(t) - \rho_k(t; h_k)| + |\rho_k(t; h_k) - \rho_k(t; h_\infty)| + |\rho_k(t; h_\infty) - s_k(t)| \longrightarrow 0,$$

and by Lemma 17.133 $s_k(t) \to W^{\mathrm{tor}}(t)$. Hence $\lim_k r_k(t) = \lim_k \langle e^{tF(G)} \rangle_{\mathfrak{c}, X_k}$ exists and equals $W^{\mathrm{tor}}(t)$, for every real t; by Lemma 17.123 this is the limit of the torus-convention ratios at the moving wall $h_k = h_0 - \Delta_T(\varepsilon_k)/2$, which coincide with $r_k(t)$ term by term, and of those at the frozen wall $h_\infty = h_0 - \Delta_T(0)/2$. By Lemma 17.124 the convergence is locally uniform on $\mathbb{C}$, the limit $\mathcal{W}_{\mathfrak{c}}(zG)$ is entire, and all moments of $F(G)$ converge. The limits at h_0 itself in the two conventions are not claimed equal, and nothing below uses that.

What the proof does not use. No value of the constants d_k; no Wick scheme; no convergence of $\mathsf{Z}_k^{\mathrm{tor}}(t; h)$ or $\mathsf{Z}_k^{\mathrm{tor}}(0; h)$ separately (both drift like $\varepsilon_k^{\pm q}$); no comparison of two lattices, no rotated lattice, no skew torus, no hybrid flow and no centred blocks; no change of mass; the bridge Proposition 17.121 only through the exact identity (c), the boundedness of Δ_T and the rate $|\Delta_T(s) - \Delta_T(0)| \le Cs$ of (b); and the sign of the wall nowhere, when the lower bound is taken from Theorem 17.116. The statement is at fixed $\mathfrak{c}$, G and t, with $\varepsilon_k \to 0$ along $\varepsilon_k = 5^{-k}$; nothing depends on the later infimum over configurations or on the wall-height index of the shell construction. □

The forward bound and the assembly.

Proposition 17.134 (the forward bound)**.** *For $T \in \{T_1, T_2\}$ and every real source $s = (zG, h)$ with h a wall $-n\psi$ or its transport,*

$$\mathsf{Z}_{Y_k, T}(s) \le (1 + \delta_k)\, e^{-c_k \mathsf{S}_{X_k}(h)}\, \mathsf{Z}_{X_k, T}(s), \qquad \delta_k = \delta_k(T, s) \to 0,$$

with the constant c_k of Lemma 17.61, independent of T and of the source, and $|c_k| \le C\varepsilon_k^{-q}$. No bound on c_k is used.

Proof. Write $I_Y := \int \chi_r e^{-S^{(r)}}$ and $I_X := \int \chi_r e^{-\hat{s}^{(r)}}$. By Proposition 17.96, $\mathsf{Z}_Y \le e^{C\delta_r^{\sigma}|T|} I_Y + B$ with $B := e^{-p(\delta_r)^2 + C|T|} \varepsilon^{-q}$. By Proposition 17.114 (at the wall, Theorem 17.116), $B \le b\, \mathsf{Z}_Y$ with $b := e^{-p(\delta_r)^2 + (C+C_1)|T|} \varepsilon^{-q-q'}$; since $p(\delta_r)^2$ grows at least quadratically in k (Lemma 17.26) while $\ln \varepsilon^{-q-q'} = (q+q')k \ln 5$ grows linearly, $b \to 0$, so $\mathsf{Z}_Y \le (1-b)^{-1} e^{C\delta_r^{\sigma}|T|} I_Y$: the large-field term is absorbed on the left, which is why no bound on c_k and no lower bound for the axis lattice is needed. On $\operatorname{supp} \chi_r$, Theorem 17.62 gives $e^{-S} \le e^{-\hat{s} - c_k \mathsf{S}_{X_k}(h) + \vartheta_k |T|}$, so $I_Y \le e^{-c_k \mathsf{S}_{X_k}(h) + \vartheta_k |T|} I_X$. By Proposition 17.113, $I_X \le e^{C(1+k)^p \delta_r^{\sigma} |T|} \mathsf{Z}_X$. So $\delta_k = (1-b)^{-1} e^{(2C(1+k)^p \delta_r^{\sigma} + \vartheta_k)|T|} - 1$, which tends to 0 by Lemma 17.26. Multiplying Z and the small-field integrals of either lattice by a positive constant changes nothing above, which is the c-number blindness used in Proposition 17.121(b). □

Proposition 17.135 (mass-increasing walls only)**.** *Let $\mathcal{S}^- := \{(tG, h_0) : t \in \mathbb{R}\}$ with $h_0 = -n\psi$, $n > 0$, a family not closed under $h \mapsto -h$, and assume the forward bounds of Proposition 17.134 on T_1 for $s \in \mathcal{S}^-$ and on T_2 for their S-images, with the common constant c_k. Then the conclusions* (a) *and* (c) *of Theorem 17.6 hold for the sources of $\mathcal{S}^-$, with $v = v' = 0$, $c = c' = c_k$ and $c_k \to 0$ (Proposition 17.71); the sources $(tG, +n\psi)$ do not occur.*

Proof. Part (a) of Theorem 17.6 uses only the two forward bounds, one source at a time: the upper bound is the forward bound on T_1, the lower bound the forward bound on T_2 for $S \cdot s$ pulled back by Lemma 17.4, with $\mathsf{S}_\Lambda(h) = \int h + O(\varepsilon^N)$ (Lemma 17.25) and $N > q$. Part (b) is the only place where the pair $(G, \pm h)$ enters, and its conclusion $c + c' \to 0$ is here supplied directly by $c = c' = c_k \to 0$; the interval of (a) being non-empty gives $2c_k \int h_0 \le o(1)$ with $\int h_0 < 0$, that is $\liminf c_k \ge 0$, consistently. Part (c) then follows as in that proof: both ends of the interval in (a) tend to 1. □

Proof of Theorem 17.13. Fix $\mathfrak{c} = (J, \psi, n)$ in the class and G, and let $\mathcal{S}^- = \{(tG, h_0) : t \in \mathbb{R}\}$, $h_0 = -n\psi$. *Step 1 (comparison of the effective actions).* Theorem 17.62 on T_1 for $s \in \mathcal{S}^-$ and on T_2 for $S \cdot s$ (Lemma 15.9(b) transports the own-frame sources; the S-image of a wall is a wall on T_2), with the common constant c_k of Lemma 17.61, which does not depend on the torus. Inputs: Theorem 17.39, Proposition 17.72, [13, Thm. 3.3], Convention 17.17(ii), Remark 17.24. *Step 2 (forward bounds).* Proposition 17.134 on T_1 and on T_2 for these sources. Inputs: Propositions 17.96, 17.113 and 17.114 on both tori, hence Theorem F.2, Proposition 17.121 and Convention 17.17 throughout; on T_1 every Y-frame step is a skew step (Proposition 17.117); on T_2 only the first hybrid step is skew, with image terms $O(e^{-c\,50MJ/\varepsilon})$. The instance of Proposition 17.114 at the wall is Theorem 17.116. *Step 3 (sandwich).* Proposition 17.135 with Proposition 17.71: conclusion (c) of Theorem 17.6 for the balanced pair $s_1 = (tG, h_0)$, $s_0 = (0, h_0)$, whose scalar sources have the same integral, gives

$$u_k(t) := \langle e^{tF(G)} \rangle_{\mathfrak{c}, Y_k} - \langle e^{tF(G)} \rangle_{\mathfrak{c}, X_k} \to 0 \qquad \text{for every real } t.$$

Step 4 (complex coefficient, moments). By Lemma 2.8 and Lemma 17.21, $1 \le \langle e^{tF} \rangle \le e^{t^2 q_\Lambda/2}$ with $q_\Lambda \to q$ on both lattices (Lemma 17.22), and $|\langle e^{zF} \rangle| \le \langle e^{(\operatorname{Re} z)F} \rangle$; so u_k is bounded on strips uniformly in k and tends to 0 on $\mathbb{R}$; Vitali's theorem [10] gives (i), and Cauchy's estimates the moments. *Step 5 (existence).* By Theorem 17.122 the axis family converges for real t; by the same argument it converges locally uniformly on $\mathbb{C}$ to an entire function $\mathcal{W}_\mathfrak{c}(zG)$, and by (i) so does the rotated family, to the same limit. This is (ii). *Step 6 (other generators).* Lemma 17.3(c): the statement for (R^{-1}, T_1) and for the other planes follows from the one for (R, T_1) by the passive reflections and permutations, which preserve X_k and T_1 and transport own-frame sources to own-frame sources, and the class of configurations is invariant under them. This is (iii). No reverse step ("switching around", [14, p. 337]), no comparison of two spacings, no Wick scheme and no value of any h-linear constant, no convergence of the rotated family on its own and no infinite-volume limit is used: the two-lattice content is Steps 1–4, and existence is the one-lattice question of Step 5. □

Completion of the proof of Theorem 17.1. By Lemma 17.3(c) it suffices to treat $R = R_{12}(\theta_0)$, and by Remark 18.6 the configurations to be treated are those of the class of Theorem 17.13. For such $\mathfrak{c}$ and G, Steps 1–2 of the proof of Theorem 17.13 give the bound (F) of Definition 17.5 on both tori for the sources (tG, h_0), in total form: $v_k(T_1) = v_k(T_2) = 0$ and $c_k(T_1) = c_k(T_2) = c_k$, with $|c_k| \le C\varepsilon_k^{-q}$ (Lemma 17.61), so that (F_0) holds for these sources, (V) and (C) hold trivially, and moreover $c_k \to 0$ (Proposition 17.71). Proposition 17.135 supplies conclusion (c) of Theorem 17.6 for the family $\mathcal{S}^-$ without the sources $(tG, +n\psi)$, and Corollary 17.7 then gives $\langle e^{zF(G)} \rangle_{\mathfrak{c}, R^{-1}\Lambda_k} - \langle e^{zF(G)} \rangle_{\mathfrak{c}, \Lambda_k} \to 0$ locally uniformly in z; with the existence of $\lim_k \langle e^{tF(G)} \rangle_{\mathfrak{c}, \Lambda_k}$ for real t (Theorem 17.122) its last clause gives (K_0) for $\mathfrak{c}$ and G. □

18. Walled tori, the closing estimate and rotation invariance

18.1. **Walled tori and the closing estimate.**

Definition 18.1 (admissible walls and walled values)**.** For a bounded set $X \subset \mathbb{R}^3$ let $\mathcal{A}(X)$ be the set of walled configurations $\mathfrak{c} = (J', w)$ of definition 15.6 with $N_1(X) \subset (-25MJ', 25MJ')^3$ and $w = 0$ on $N_1(X)$. The interior $(-25MJ', 25MJ')^3$ of the cell $\Gamma_{J'}$ is a fundamental domain of $D_{J'}$ for both lattices, $R^{-1}\Lambda_k$ descending to $D_{J'}$ for $R \in \mathcal{R}_0$ by lemma 15.3(b), and sources supported in X are periodised from it. If $X \subset Y$ then $\mathcal{A}(Y) \subset \mathcal{A}(X)$. Recall the walled ratio $T^a_{\mathfrak{c},\Lambda}(f,g) = e^{-a_k \mathsf{S}_\Lambda(g)} \langle \Theta(\mathrm{i}f, g)\rangle_{\mathfrak{c},\Lambda}$ of a lattice Λ of spacing ε_k descending to $D_{J'}$, defined for $(f,g) \in \mathcal{K}$ with $w = 0$ on $\operatorname{supp} g$ (lemma 15.9(c)). For $g \neq 0$ assume $(\mathrm{K})_a$ (Definition 17.2). For $(f,g) \in \mathcal{K}$ supported in X and $\mathfrak{c} \in \mathcal{A}(X)$ the periodised sources are smooth, and by lemma 17.3(a) the *walled value*

$$\mathcal{W}^a_{\mathfrak{c}}(f,g) := \lim_{k\to\infty} T^a_{\mathfrak{c},\Lambda_k}(f,g) = \lim_{k\to\infty} T^a_{\mathfrak{c},R^{-1}\Lambda_k}(f,g) \qquad (R \in \mathcal{R}_0)$$

exists in $(0,\infty)$. For $b > 0$ and $(f,g) \in \mathcal{K}$ supported in Q_b let $I^a_b(f,g) := \inf_{\mathfrak{c}\in\mathcal{A}(Q_b)} \mathcal{W}^a_{\mathfrak{c}}(f,g)$, which is nondecreasing in b, and

$$\mathcal{W}^{a,\mathrm{env}}(f,g) := \lim_{b\to\infty} I^a_b(f,g) \in (0,\infty].$$

At $g = 0$ these definitions rest on (K_0) alone (Theorem 17.1; lemma 17.3(a) at the complex coefficient i), and they do not depend on the scheme (lemma 15.9(c)): $\mathcal{W}_{\mathfrak{c}}(f,0) = \lim_k \langle e^{\mathrm{i}F_{\Lambda_k}(f)}\rangle_{\mathfrak{c},\Lambda_k} \in [-1,1]$, and we write $I_b(f,0)$, $\mathcal{W}^{\mathrm{env}}(f,0)$. King's printed state at a curvature source is written $Z^K_\infty(\mathrm{i}f,0)$ in this part, the second slot recording the absence of a scalar source; it is $Z^K_\infty(\mathrm{i}f)$ of theorem 7.6(b), and $Z^K_J(\mathrm{i}f)$ denotes the fixed-cube limit there.

Lemma 18.2 (the free direction)**.** (a) *Assume* $(\mathrm{W})_a$ *and* $(\mathrm{K})_a$*. If* $(f,g) \in \mathcal{K}$ *is supported at* ℓ^∞*-distance* > 1 *from* $\mathbb{R}^3 \setminus C_J$ *and* $\mathfrak{c} \in \mathcal{A}(\bar{C}_J)$*, then* $\mathcal{W}^a_{\mathfrak{c}}(f,g) \geq Z^a_J(\mathrm{i}f,g)$*, where* $Z^a_J = \lim_k Z^a_{k,J}$ *is the fixed-cube limit of theorem 16.6(i).*

(b) *Assume* $(\mathrm{W})_a$ *and* $(\mathrm{K})_a$*. Then* $\mathcal{W}^{a,\mathrm{env}} \geq Z^a_\infty$ *on* $\mathcal{K}$*.*

(c) *If the real 2-form* $f \in C^\infty_c$ *is supported at* ℓ^∞*-distance* > 1 *from* $\mathbb{R}^3 \setminus C_J$ *and* $\mathfrak{c} \in \mathcal{A}(\bar{C}_J)$*, then* $\mathcal{W}_{\mathfrak{c}}(f,0) \geq Z^K_J(\mathrm{i}f)$*.*

(d) $\mathcal{W}^{\mathrm{env}}(f,0) \geq Z^K_\infty(\mathrm{i}f,0) \geq 0$ *for every real 2-form* $f \in C^\infty_c$*.*

Proof. (a) Fix k. The set $X_J = C_J \cap \Lambda_k$ and its nearest neighbours lie in $N_1(\bar{C}_J)$, hence in the interior of the cell, and the wall vanishes there; so lemma 15.8 applied to the walled torus state $\langle\cdot\rangle_{\mathfrak{c},\Lambda_k}$ with $X = X_J$ has as its X-factor the comparison cube $\langle\cdot\rangle^{(a)}_{k,J}$ of definition 2.1, and gives $\langle\Theta(\mathrm{i}f,g)\rangle^{(a)}_{k,J} \leq \langle\Theta(\mathrm{i}f,g)\rangle_{\mathfrak{c},\Lambda_k}$; every plaquette and site at which (f,g) does not vanish lies in X_J by the support condition. The multiplier $e^{-a_k \mathsf{S}_{\Lambda_k}(g)}$ is the same positive constant on both sides (lemma 15.9(c), lemma 2.11(a)), so $Z^a_{k,J}(\mathrm{i}f,g) \leq T^a_{\mathfrak{c},\Lambda_k}(f,g)$. Let $k \to \infty$: the left side converges to $Z^a_J(\mathrm{i}f,g)$ by theorem 16.6(i), the right side to $\mathcal{W}^a_{\mathfrak{c}}(f,g)$ by definition 18.1.

(b) If $\bar{C}_J \subset Q_b$ then $\mathcal{A}(Q_b) \subset \mathcal{A}(\bar{C}_J)$, so $I^a_b(f,g) \geq Z^a_J(\mathrm{i}f,g)$ by (a) for every J at which the support condition of (a) holds; let $b \to \infty$ and then $J \to \infty$ (theorem 16.6(iii)).

(c) At $g = 0$ no multiplier enters. As in (a), lemma 15.8 gives $Z_{k,J}(\mathrm{i}f) = \langle e^{\mathrm{i}F(f)}\rangle^{(a)}_{k,J} \leq T_{\mathfrak{c},\Lambda_k}(f,0)$, and the first inequality of lemma 4.2, whose support condition (S) holds for X_J because f is supported at distance > 1 from $\mathbb{R}^3 \setminus C_J$, gives $Z^K_{k,J}(\mathrm{i}f) = \langle e^{\mathrm{i}F(f)}\rangle^K_{k,J} \leq Z_{k,J}(\mathrm{i}f)$. Hence $Z^K_{k,J}(\mathrm{i}f) \leq T_{\mathfrak{c},\Lambda_k}(f,0)$ at every k. Let $k \to \infty$: the left side converges to $Z^K_J(\mathrm{i}f)$ by theorem 7.6(b) under (E_C), the right side to $\mathcal{W}_{\mathfrak{c}}(f,0)$ by (K_0) (lemma 17.3(a)).

(d) Fix f and $J \geq J_0(f)$. For $b \geq MJ$ we have $\bar{C}_J = Q_{MJ} \subset Q_b$, so $\mathcal{A}(Q_b) \subset \mathcal{A}(\bar{C}_J)$ and $I_b(f,0) \geq Z^K_J(\mathrm{i}f)$ by (c); letting $b \to \infty$, $\mathcal{W}^{\mathrm{env}}(f,0) \geq Z^K_J(\mathrm{i}f)$ for every $J \geq J_0(f)$. By corollary 4.3 and theorem 7.6(b), $J \mapsto Z^K_J(\mathrm{i}f)$ is nondecreasing with values in $[0,1]$, and by corollary 4.4(ii)(β) it converges to $Z^K_\infty(\mathrm{i}f,0)$; let $J \to \infty$. □

Lemma 18.2(a),(c) compare a walled value with a cube of fixed size. They do not assert $\mathcal{W}^a_{\mathfrak{c}} \geq Z^a_\infty$ for an individual configuration, and this need not hold: for instance, at $e = \lambda = 0$, $f = 0$ and $g \geq 0$ small, as $n \to \infty$ the walled value of (J', ψ, n) decreases (lemma 2.7(c)) towards the value of the Dirichlet state on the set where $\psi = 0$, which is smaller than the infinite-volume value, as the Gaussian formula for $\langle e^{:\phi^2:(g)}\rangle$ shows.

Lemma 18.3 (independence of the orientation)**.** *Assume* $(\mathrm{K})_a$ *if* $g \neq 0$*. For* $R \in \mathcal{R}_0$ *and* $(f,g) \in \mathcal{K}$*,*

$$\lim_{L\to\infty} \inf_{\mathfrak{c}\in\mathcal{A}(R^{-1}Q_{ML})} \mathcal{W}^a_{\mathfrak{c}}(f,g) = \mathcal{W}^{a,\mathrm{env}}(f,g),$$

the infimum being defined as soon as $\operatorname{supp} f \cup \operatorname{supp} g \subset Q_{5ML/7}$*. At* $g = 0$ *the same holds, with no assumption, for* $\mathcal{W}_{\mathfrak{c}}(f,0)$ *and* $\mathcal{W}^{\mathrm{env}}(f,0)$*.*

Proof. By lemma 15.3(c), $Q_{5ML/7} \subset R^{-1}Q_{ML} \subset Q_{7ML/5}$, hence $\mathcal{A}(Q_{7ML/5}) \subset \mathcal{A}(R^{-1}Q_{ML}) \subset \mathcal{A}(Q_{5ML/7})$, and $I^a_{5ML/7} \le \inf_{\mathcal{A}(R^{-1}Q_{ML})} \mathcal{W}^a_{\mathfrak{c}} \le I^a_{7ML/5}$. Both bounds tend to $\mathcal{W}^{a,\mathrm{env}}$ as $L \to \infty$. □

Theorem 18.4 ((TIGHT_0), (TIGHT_2): the closing estimate)**.** (TIGHT_0)*:* $\mathcal{W}^{\mathrm{env}}(f,0) \le Z^K_\infty(\mathrm{i}f,0)$ *for every real* 2*-form* $f \in C^\infty_c$*.* (TIGHT_2)*: for every real* 2*-form* $f \in C^\infty_c$*,*

$$\lim_{b\to\infty} \sup_{\mathfrak{c}\in\mathcal{A}(Q_b)} \sigma^2_{\mathfrak{c}}(f) \ge S^K_2(f), \qquad \sigma^2_{\mathfrak{c}}(f) := \lim_k \langle F_{\Lambda_k}(f)^2\rangle_{\mathfrak{c},\Lambda_k}, \quad S^K_2(f) := S^K_2(f,f) = \lim_J \lim_k \langle F(f)^2\rangle^K_{k,J}.$$

(TIGHT_0) involves only (K_0), which defines its left side, and King's printed state, which exists by (E_C). It is proved in the heavy-shell sections: Theorem 19.10 derives it from the decoupling property (RES_0) of Definition 19.8, which Theorem 20.5 establishes at one configuration from the heavy-shell estimates (HS); (TIGHT_2) follows from (TIGHT_0) by Lemma 18.7(c). In the Wick sector the closing statement is a property of the scheme, not proved here (it is listed among the open problems, in the final section) and used only by the optional extension of the Wick leg (Theorem 23.7):

Definition 18.5 ($(\mathrm{TIGHT})_a$: the closing property, Wick sector)**.** Let a be W-admissible and assume $(\mathrm{K})_a$. $(\mathrm{TIGHT})_a$: $\mathcal{W}^{a,\mathrm{env}} \le Z^a_\infty$ on $\mathcal{K}$. Equivalently: for every $b > 0$, every $(f,g) \in \mathcal{K}$ supported in Q_b and every $\eta > 0$ there is $\mathfrak{c} \in \mathcal{A}(Q_b)$ with $\mathcal{W}^a_{\mathfrak{c}}(f,g) \le Z^a_\infty(\mathrm{i}f,g) + \eta$.

The equivalence in $(\mathrm{TIGHT})_a$ holds because I^a_b is nondecreasing in b. By lemma 18.2, $(\mathrm{TIGHT})_a$ is equivalent to $\mathcal{W}^{a,\mathrm{env}} = Z^a_\infty$ on $\mathcal{K}$, and (TIGHT_0) to $\mathcal{W}^{\mathrm{env}}(\cdot,0) = Z^K_\infty(\mathrm{i}\cdot,0)$: *King's printed state is the limit of the lower envelopes of walled tori, the continuum limit being taken first.* The limit defining $\sigma^2_{\mathfrak{c}}$ exists by (K_0), since $t \mapsto \langle e^{\mathrm{i}tF(f)}\rangle_{\mathfrak{c},\Lambda_k}$ converges locally uniformly as an entire function (lemma 17.3(a)) and second derivatives of locally uniformly convergent entire functions converge; S^K_2 is the two-point Schwinger function of theorem 7.6(c), with the lattice second moments converging to it. The supremum over $\mathcal{A}(Q_b)$ is nonincreasing in b, so the limit in (TIGHT_2) exists.

Remark 18.6 (the configuration class). The proofs below use (K_0) only for the walled configurations that enter them, and they use the infimum over $\mathcal{A}(R^{-1}Q_{ML})$ only through an upper bound by the walled value of one configuration supplied by (TIGHT_0). Hence (K_0) and (TIGHT_0) may be read on any common sub-class of walled configurations which is stable under raising the wall height and contains the single-level configuration produced by theorem 19.10: the infimum is then taken over that sub-class, (TIGHT_0) is read with the same sub-class, and lemma 18.3 holds on it, since its proof uses only the inclusions between shrinking families. The sub-class on which the proof of (K_0) along King's lines [14] supplies the comparison consists of tori whose side is a large multiple of a power of 5, of mass-increasing walls whose mean over the torus exceeds a fixed multiple of the torus volume, and of a source box and torus aligned with the blocks of the final scale. The configuration of theorem 19.10 lies in it: its wall is $(n+K)\psi$ with $n + K \ge 1$, mass-increasing; the condition on the wall mean is linear in the height and holds once K is large at a fixed torus, which remark 19.11(iii) allows; the torus and the alignment of the source box are choices made in theorem 20.5.

Lemma 18.7. (a) *If* a, a' *are* W*-admissible and* $(\mathrm{K})_a$ *holds, then* $(\mathrm{TIGHT})_a$ *and* $(\mathrm{TIGHT})_{a'}$ *are equivalent.*

(b) *Assume* $(\mathrm{W})_a$ *and* (E_C)*. Then* $(\mathrm{TIGHT})_a$ *implies* (TIGHT_0)*.*

(c) (TIGHT_0) *implies* (TIGHT_2)*; and* $\limsup_b \sup_{\mathcal{A}(Q_b)} \sigma^2_{\mathfrak{c}}(f) \le S^K_2(f)$ *always, so* (TIGHT_2) *is an equality.*

Proof. (a) By lemma 16.4(b), $a'_k - a_k \to \kappa_{a'a}$, and $T^{a'}_{\mathfrak{c},\Lambda} = e^{-(a'_k-a_k)\mathsf{S}_\Lambda(g)} T^a_{\mathfrak{c},\Lambda}$ by lemma 15.9(c); since $\mathsf{S}_{\Lambda_k}(g) \to \int g$, $\mathcal{W}^{a'}_{\mathfrak{c}} = e^{-\kappa_{a'a}\int g}\mathcal{W}^a_{\mathfrak{c}}$ with a factor independent of $\mathfrak{c}$. The same factor relates $Z^{a'}_\infty$ and Z^a_∞ (theorem 16.6(vi)).

(b) At $g = 0$ the walled values are scheme-free (lemma 15.9(c)), so the left sides of $(\mathrm{TIGHT})_a$ and (TIGHT_0) agree, and the right side of $(\mathrm{TIGHT})_a$ is $Z^a_\infty(\mathrm{i}f, 0) = \lim_J \lim_k Z_{k,J}(\mathrm{i}f)$, the iterated limit of the comparison family, which exists under $(\mathrm{W})_a$ (theorem 16.6(iii)) and equals $Z^K_\infty(\mathrm{i}f, 0)$ under (E_C) by corollary 4.4(ii)(γ).

(c) Fix f, b and $\eta' > 0$. All functions of t below are real and even. Since $\cos x \geq 1 - \frac{x^2}{2}$ and $\cos x \leq 1 - \frac{x^2}{2} + \frac{x^4}{24}$ at every lattice level, and the moments converge in the limits (for the walled torus by lemma 17.3(a), for King's cube by theorem 7.6(c)),

$$\mathcal{W}_{\mathfrak{c}}(tf, 0) \geq 1 - \tfrac{t^2}{2}\sigma^2_{\mathfrak{c}}(f), \qquad Z^K_\infty(\mathrm{i}tf, 0) \leq 1 - \tfrac{t^2}{2}S^K_2(f) + \tfrac{t^4}{24}M_4(f),$$

with $M_4(f) := \partial_t^4 Z^K_\infty(tf, 0)|_{t=0} = \lim_J \lim_k \langle F(f)^4\rangle^K_{k,J} < \infty$. By (TIGHT_0) applied to tf, for every $t > 0$ and $\eta > 0$ there is $\mathfrak{c} \in \mathcal{A}(Q_b)$ with $\mathcal{W}_{\mathfrak{c}}(tf, 0) \leq Z^K_\infty(\mathrm{i}tf, 0) + \eta$, hence $\sigma^2_{\mathfrak{c}}(f) \geq S^K_2(f) - \frac{t^2}{12}M_4(f) - \frac{2\eta}{t^2}$. Choosing t with $t^2 M_4(f)/12 < \eta'/2$ and then $\eta = t^2\eta'/4$ gives $\sup_{\mathcal{A}(Q_b)} \sigma^2_{\mathfrak{c}}(f) \geq S^K_2(f) - \eta'$.

For the second claim, let $\mathfrak{c} \in \mathcal{A}(\bar{C}_J)$ with f supported at distance > 1 from $\mathbb{R}^3 \setminus C_J$. By lemma 18.2(c) applied to tf, $\mathcal{W}_{\mathfrak{c}}(tf, 0) \geq Z^K_J(\mathrm{i}tf)$ for all real t, with equality $(= 1)$ at $t = 0$; both sides are even and smooth, so comparing second derivatives at $t = 0$ gives $\sigma^2_{\mathfrak{c}}(f) \leq v_J(f) := \lim_k \langle F(f)^2\rangle^K_{k,J}$, the limit existing by theorem 7.6(c). Hence $\sup_{\mathcal{A}(Q_b)} \sigma^2_{\mathfrak{c}}(f) \leq v_J(f)$ whenever $\bar{C}_J \subset Q_b$. By corollary 4.3, $\langle e^{\mathrm{i}tF(f)}\rangle^K_{k,J} \leq \langle e^{\mathrm{i}tF(f)}\rangle^K_{k,J+1}$ for all real t, with equality at $t = 0$; comparing second derivatives, the cube variances $\langle F(f)^2\rangle^K_{k,J}$ are nonincreasing in J at every k, hence so are the $v_J(f)$, and they converge to $S^K_2(f)$ (theorem 7.6(c)). Therefore $\lim_b \sup_{\mathcal{A}(Q_b)} \sigma^2_{\mathfrak{c}}(f) \leq \inf_J v_J(f) = S^K_2(f)$. □

Remark 18.8 (the role of the closing estimate). Apart from the closing estimate, the proofs of theorems 18.9 and 18.12 below use only: monotonicity of the cube functionals in J (on $\mathcal{K}$ and on the decreasing class), the comparison "cube $\leq$ walled torus" on $\mathcal{K}$ and "cube $\geq$ walled torus" on the decreasing class on each lattice, the equality of walled values on the two lattices, passive covariance, and the existence of the limits. All of these hold for the following family of Gaussian functionals, in which the analogue of King's state is not rotation invariant. Let $A = \mathrm{diag}(1, 2, 3)$, $\mu > 0$, and let C_I, C_R, C_U be the translation-invariant operators on $L^2(\mathbb{R}^3)$ with Fourier symbols $\hat{c}_I(k) = (\langle k, Ak\rangle + \mu^2)^{-1}$, $\hat{c}_R(k) = \hat{c}_I(Rk)$ and $\hat{c}_U = \min(\hat{c}_I, \hat{c}_R)$. Let the "cube of size J" on the lattice $X \in \{I, R\}$ have covariance $C_X + J^{-1}$, let every "walled torus" have covariance C_U on both lattices, and let a covariance C define the functionals $e^{-\frac{1}{2}\langle f, Cf\rangle}$ on the cone and $e^{\frac{1}{2}\langle f, Cf\rangle}$ on the decreasing class. Every relation listed reduces to $C_U \leq C_X + J^{-1}$ or to the monotonicity of $C_X + J^{-1}$ in J, and passive covariance holds since $\langle f, C_R f\rangle = \langle R\cdot f, C_I\, R\cdot f\rangle$. But $e^{-\frac{1}{2}\langle R\cdot f, C_I R\cdot f\rangle} \neq e^{-\frac{1}{2}\langle f, C_I f\rangle}$ in general, and the closing estimate fails since $C_U \neq C_I$. Hence the closing estimate is not a consequence of the other properties, and an input asserting "walled torus $\leq$ cube + small" is needed. (The same family shows that any placement of the two-lattice comparison on a finite box must also carry an input beyond the correlation inequalities: there the input is the two-lattice comparison itself, which fails for $C_R \neq C_I$.)

18.2. **Rotation invariance.**

Theorem 18.9 (invariance under the generators, Wick sector)**.** *Let a be* W*-admissible (*$(\mathrm{W})_a$*), and assume* $(\mathrm{K})_a$ *and* $(\mathrm{TIGHT})_a$*. Then for every $R \in \mathcal{R}_0$ and every $(f, g) \in \mathcal{K}$,*

$$Z^a_\infty(\mathrm{i}\, R\cdot f, R\cdot g) = Z^a_\infty(\mathrm{i}f, g),$$

and consequently the same holds for every $R \in G_0$.

Proof. Fix $R \in \mathcal{R}_0$ and $(f, g) \in \mathcal{K}$ supported in Q_ρ.

Step 0 (geometry). Take L with $ML \geq \frac{7}{5}\rho + 2$. By lemma 15.3(c), $R\cdot(f, g)$ is supported in $Q_{7\rho/5}$, at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_L$, and (f, g) is supported in $Q_{5ML/7} \subset R^{-1}Q_{ML}$. Let

$\mathfrak{c} \in \mathcal{A}(R^{-1}Q_{ML})$. Then $R^{-1}\Lambda_k$ descends to $D_{J'}$ (lemma 15.3(b)), and $B'_k := R^{-1}(C_L \cap \Lambda_k)$ is a set of sites of $R^{-1}\Lambda_k$ which, with its nearest neighbours, lies in the interior of the cell, and on which $w = 0$. Since $R \cdot (f, g)$ has distance > 1 from $\mathbb{R}^3 \setminus C_L$, every plaquette and site of $R^{-1}\Lambda_k$ at which (f, g) does not vanish lies in B'_k.

Step 1 (cutting on the rotated lattice). By lemma 15.8, applied to the walled torus state $\langle\cdot\rangle_{\mathfrak{c},R^{-1}\Lambda_k}$ with $X = B'_k$, whose X-factor is the Neumann–Dirichlet state on B'_k with the scalar alive on all of B'_k, and by the multiplier being the same constant on both sides (lemma 15.9(c), lemma 2.11(a)), at every k,

$$Z^a_{R^{-1}\Lambda_k,B'_k}(\mathrm{i}f, g) \le T^a_{\mathfrak{c},R^{-1}\Lambda_k}(f, g).$$

Step 2 (passive rotation). By lemma 15.9(b) with $O = R$, which maps $R^{-1}\Lambda_k$ with its frame $(R^{-1}e_\alpha)$ onto Λ_k with the frame (e_α) and B'_k onto X_L, $Z^a_{R^{-1}\Lambda_k,B'_k}(\mathrm{i}f, g) = Z^a_{k,L}(\mathrm{i}\, R \cdot f, R \cdot g)$.

Step 3 (continuum limits). As $k \to \infty$ the left side converges by theorem 16.6(i) (the sources $R \cdot (f, g)$ being at distance > 1 from $\mathbb{R}^3 \setminus C_L$) and the right side by $(\mathrm{K})_a$ (definition 18.1):

$$Z^a_L(\mathrm{i}\, R \cdot f, R \cdot g) \le \mathcal{W}^a_{\mathfrak{c}}(f, g) \qquad \text{for every } \mathfrak{c} \in \mathcal{A}(R^{-1}Q_{ML}).$$

Step 4 (infimum, volume limit, closing hypothesis). Take the infimum over $\mathfrak{c} \in \mathcal{A}(R^{-1}Q_{ML})$ and let $L \to \infty$. By theorem 16.6(iii), lemma 18.3 and $(\mathrm{TIGHT})_a$,

$$Z^a_\infty(\mathrm{i}\, R \cdot f, R \cdot g) = \lim_{L\to\infty} Z^a_L(\mathrm{i}\, R \cdot f, R \cdot g) \le \mathcal{W}^{a,\mathrm{env}}(f, g) \le Z^a_\infty(\mathrm{i}f, g).$$

Step 5 (the inverse). $R^{-1} \in \mathcal{R}_0$ (lemma 15.3(a)) and $R \cdot (f, g) \in \mathcal{K}$, so Steps 0–4 for $(R^{-1}, R \cdot (f, g))$ give $Z^a_\infty(\mathrm{i}f, g) \le Z^a_\infty(\mathrm{i}\, R \cdot f, R \cdot g)$. Since $\mathcal{K}$ is G_0-invariant and the transport is a left action, the claim extends to G_0. □

Remark 18.10 (the order of limits). The order of limits is: at fixed R, (f, g), L and $\mathfrak{c}$, first $k \to \infty$; then the infimum over $\mathfrak{c}$; then $L \to \infty$. The continuum limit is always taken first, at fixed volume and fixed wall, as in King's construction [14, p. 336], and the torus comparison is used only as an equality of continuum limits, never at a fixed lattice spacing.

Theorem 18.11 (closure: from the generators to $O(3)$, Wick sector)**.** *Let a be* W*-admissible* ($(\mathrm{W})_a$)*, and suppose* $Z^a_\infty(\mathrm{i}\, R \cdot f, R \cdot g) = Z^a_\infty(\mathrm{i}f, g)$ *for all* $(f, g) \in \mathcal{K}$ *and* $R \in \mathcal{R}_0$*. Then for every* $R \in O(3)$*, every* N*, all real* 2*-forms* f_j *and real* h_j *in* C^∞_c*, and all* $(z, w) \in \mathbb{C}^{2N}$*,*

$$Z^a_\infty\Big(\sum_j z_j\, R \cdot f_j, \sum_j w_j\, R \cdot h_j\Big) = Z^a_\infty\Big(\sum_j z_j f_j, \sum_j w_j h_j\Big).$$

Proof. Complex coefficients. The hypothesis passes to G_0, since $\mathcal{K}$ is invariant under $O(3)$ and $\mathcal{R}_0 = \mathcal{R}_0^{-1}$. For $R \in G_0$ and a finite family, split $h_j = h_j^+ - h_j^-$ as in the proof of theorem 16.6(iii). The difference of the two sides, as a function of $(z, w^+, w^-) \in \mathbb{C}^{3N}$, is entire by theorem 16.6(iii) and vanishes on $\Omega_+ = (\mathrm{i}\mathbb{R})^N \times (0, \infty)^{2N}$, where all sources lie in $\mathcal{K}$; Ω_+ has non-empty interior within a totally real subspace of maximal dimension, hence a set of uniqueness [10], and the difference vanishes identically.

Closure in $SO(3)$. At real coefficients, $\Phi^a(R \cdot s) = \Phi^a(s)$ for $s \in \mathcal{E}_\rho$ (the pairs of real sources supported in $\bar{B}_\rho$, theorem 16.6(iv)) and $R \in G_0$; note $R \cdot s \in \mathcal{E}_\rho$ since $R\bar{B}_\rho = \bar{B}_\rho$. The map $R \mapsto R \cdot s \in \mathcal{E}_\rho$ is continuous on $O(3)$, and Φ^a is continuous on $\mathcal{E}_\rho$ by theorem 16.6(iv). Hence $G := \{R \in O(3) : \Phi^a(R \cdot s) = \Phi^a(s) \text{ for all } \rho \text{ and } s \in \mathcal{E}_\rho\}$ is a closed subgroup of $O(3)$ containing G_0, so $G \supset SO(3)$ by lemma 15.3(d).

Reflection. Let $\sigma_3 = \mathrm{diag}(1, 1, -1)$ and $(f, g) \in \mathcal{K}$ supported at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_{J-1}$. Lemma 2.13 gives, at every k, $Z^a_{k,J-1}(\mathrm{i}f, g) \le Z^a_{k,J}(\mathrm{i}\, \sigma_3 \cdot f, \sigma_3 \cdot g) \le Z^a_{k,J+1}(\mathrm{i}f, g)$. The family $\sigma_3 \cdot (f, g)$ is supported at distance > 1 from $\mathbb{R}^3 \setminus C_J$, the region $\{\mathrm{dist}_\infty(\cdot, \mathbb{R}^3 \setminus C_{J-1}) > 1\}$ being σ_3-symmetric and contained in C_J; so all three terms converge as $k \to \infty$ by theorem 16.6(i), and then as $J \to \infty$ by theorem 16.6(iii): $Z^a_\infty(\mathrm{i}\, \sigma_3 \cdot f, \sigma_3 \cdot g) = Z^a_\infty(\mathrm{i}f, g)$ on $\mathcal{K}$. The first step, with σ_3 in place of R, gives $\sigma_3 \in G$, so $G = O(3) = SO(3) \cup \sigma_3 SO(3)$.

Conclusion. For $R \in O(3)$ both sides of the claim are entire in $(z, w) \in \mathbb{C}^{2N}$ (theorem 16.6(iii)) and agree on $\mathbb{R}^{2N}$, since $R \in G$; hence they agree on $\mathbb{C}^{2N}$. □

Theorem 18.12 (the curvature sector)**.** *King's printed state on curvature sources is $O(3)$-invariant: for every $R \in O(3)$, all real 2-forms $f_1, \dots, f_N \in C_c^\infty$ and all $z \in \mathbb{C}^N$,*

$$Z_\infty^K\Big(\sum_j z_j\, R\cdot f_j, 0\Big) = Z_\infty^K\Big(\sum_j z_j f_j, 0\Big),$$

and the curvature Schwinger functions S_n^K of theorem 7.6(c) are $O(3)$-covariant, F transforming as a 2-form. Moreover, for every Euclidean motion $x \mapsto Rx + b$, $Z_\infty^K(\sum_j z_j\, \tau_b R\cdot f_j, 0) = Z_\infty^K(\sum_j z_j f_j, 0)$.

Proof. The inputs are the existence theorem (E_C) (Theorem 3.2), the torus comparison (K_0) (Theorem 17.1) and the closing estimate (TIGHT_0) (Theorem 18.4), the last two on the class of remark 18.6.

Invariance under the generators at imaginary points. Fix $R \in \mathcal{R}_0$ and a real 2-form $f \in C_c^\infty$ supported in Q_ρ. At $g = 0$ the walled values are those of (K_0) (definition 18.1). Steps 0–2 of the proof of theorem 18.9 run unchanged with the source f alone: with L, $\mathfrak{c} \in \mathcal{A}(R^{-1}Q_{ML})$ and B_k' as there, lemma 15.8 on the rotated lattice and lemma 15.9(b) give, at every k,

$$Z_{k,L}(\mathrm{i}\,R\cdot f) = \langle e^{\mathrm{i}F(R\cdot f)}\rangle_{k,J}^{(a)}\big|_{J=L} = Z_{R^{-1}\Lambda_k, B_k'}(\mathrm{i}f) \le T_{\mathfrak{c}, R^{-1}\Lambda_k}(f,0),$$

the middle term being the functional of the comparison cube X_L. *Step 3.* The right side converges to $\mathcal{W}_{\mathfrak{c}}(f,0)$ by (K_0) (lemma 17.3(a)), so $\limsup_k Z_{k,L}(\mathrm{i}\,R\cdot f) \le \mathcal{W}_{\mathfrak{c}}(f,0)$; and since $R\cdot f$ is supported at distance > 1 from $\mathbb{R}^3 \setminus C_L$ (Step 0), corollary 4.4(ii)(γ) under (E_C) gives $Z_L^K(\mathrm{i}\,R\cdot f) \le \liminf_k Z_{k,L}(\mathrm{i}\,R\cdot f)$. Hence

$$Z_L^K(\mathrm{i}\,R\cdot f) \le \mathcal{W}_{\mathfrak{c}}(f,0) \qquad \text{for every } \mathfrak{c} \in \mathcal{A}(R^{-1}Q_{ML}) \text{ of the class.}$$

Step 4. Take the infimum over $\mathfrak{c}$ and let $L \to \infty$: the left side converges to $Z_\infty^K(\mathrm{i}\,R\cdot f, 0)$ by corollary 4.4(ii)(β), the infima converge to $\mathcal{W}^{\mathrm{env}}(f,0)$ by lemma 18.3 (on the class, remark 18.6), and (TIGHT_0) gives

$$Z_\infty^K(\mathrm{i}\,R\cdot f, 0) \le \mathcal{W}^{\mathrm{env}}(f,0) \le Z_\infty^K(\mathrm{i}f, 0).$$

Step 5. $R^{-1} \in \mathcal{R}_0$ and $R\cdot f$ is a real 2-form in C_c^∞, so the same for $(R^{-1}, R\cdot f)$ gives the reverse inequality: $Z_\infty^K(\mathrm{i}\,R\cdot u, 0) = Z_\infty^K(\mathrm{i}u, 0)$ for every real 2-form $u \in C_c^\infty$ and $R \in \mathcal{R}_0$, hence for every $R \in G_0$, the transport being a left action.

Closure in $SO(3)$. Fix a real 2-form $u \in C_c^\infty$. The map $R \mapsto R\cdot u$ is continuous from $O(3)$ to $L^2(\mathbb{R}^3;\Lambda^2)$: for $R \to R_0$ the 2-forms $R\cdot u$ converge to $R_0\cdot u$ uniformly, with supports in one compact set. By the Lipschitz bound of corollary 8.3, $|Z_\infty^K(\mathrm{i}R\cdot u) - Z_\infty^K(\mathrm{i}R_0\cdot u)| \le \sqrt{2}\,\|R\cdot u - R_0\cdot u\|_2$, so $R \mapsto Z_\infty^K(\mathrm{i}\,R\cdot u, 0)$ is continuous on $O(3)$. It is constant on G_0, which is dense in $SO(3)$ (lemma 15.3(d)); hence $Z_\infty^K(\mathrm{i}\,R\cdot u, 0) = Z_\infty^K(\mathrm{i}u, 0)$ for all $R \in SO(3)$.

The reflection. By theorem 8.5 with $O = \sigma_3 \in O_h$, $Z_\infty^K(\mathrm{i}\,\sigma_3\cdot u, 0) = Z_\infty^K(\mathrm{i}u, 0)$ for every real u. Since $O(3) = SO(3) \cup \sigma_3 SO(3)$ and $(\sigma_3 R)\cdot u = \sigma_3\cdot(R\cdot u)$, the identity at imaginary points holds for every $R \in O(3)$.

All complex coefficients. Fix $R \in O(3)$ and the family $f_1, \dots, f_N$. By theorem 7.6(b) both sides of the claim are entire functions of $z \in \mathbb{C}^N$. At $z = \mathrm{i}t$, $t \in \mathbb{R}^N$, we have $\sum_j \mathrm{i}t_j\, R\cdot f_j = \mathrm{i}\,R\cdot u_t$ with $u_t = \sum_j t_j f_j$, so the two sides agree on $(\mathrm{i}\mathbb{R})^N$, a totally real subspace of maximal dimension, hence a set of uniqueness [10]; they agree on $\mathbb{C}^N$. Differentiating at $z = 0$ gives $S_n^K(R\cdot f_1, \dots, R\cdot f_n) = S_n^K(f_1, \dots, f_n)$ for every n, the $O(3)$-covariance of the Schwinger functions.

Euclidean motions. By corollary 8.3, $Z_\infty^K(\sum_j z_j \tau_b R\cdot f_j, 0) = Z_\infty^K(\sum_j z_j R\cdot f_j, 0)$ for every $b \in \mathbb{R}^3$, and the right side equals $Z_\infty^K(\sum_j z_j f_j, 0)$ by the above. □

Corollary 18.13 (the curvature two-point function)**.** *The two-point Schwinger function $S_2^K(f_1, f_2) = \lim_J \lim_k \langle F(f_1)F(f_2)\rangle_{k,J}^K$ of theorem 7.6(c) satisfies $S_2^K(R\cdot f_1, R\cdot f_2) = S_2^K(f_1, f_2)$ for all $R \in O(3)$ and all real 2-forms $f_1, f_2 \in C_c^\infty$.*

Proof. Of the closing estimate only (TIGHT_2) is used. Let $R \in \mathcal{R}_0$ and f a real 2-form. Steps 0–3 of the proof of theorem 18.12, with the source tf, give $Z_L^K(\mathrm{i}t\,R\cdot f) \le \mathcal{W}_{\mathfrak{c}}(tf, 0)$ for all real t

and all $\mathfrak{c} \in \mathcal{A}(R^{-1}Q_{ML})$. Both sides are even and smooth in t and equal 1 at $t = 0$; comparing second derivatives, $v_L(R \cdot f) \geq \sigma^2_{\mathfrak{c}}(f)$, where $v_L(u) := \lim_k \langle F(u)^2\rangle^K_{k,J}|_{J=L}$ is the cube variance (theorem 7.6(c)). Since $\mathcal{A}(Q_{7ML/5}) \subset \mathcal{A}(R^{-1}Q_{ML})$ and $b \mapsto \sup_{\mathcal{A}(Q_b)} \sigma^2_{\mathfrak{c}}(f)$ is nonincreasing,

$$v_L(R \cdot f) \geq \sup_{\mathcal{A}(Q_{7ML/5})} \sigma^2_{\mathfrak{c}}(f) \geq \lim_{b\to\infty} \sup_{\mathcal{A}(Q_b)} \sigma^2_{\mathfrak{c}}(f) \geq S^K_2(f)$$

by (TIGHT$_2$). The cube variances converge to $S^K_2(R \cdot f)$ as $L \to \infty$ (theorem 7.6(c)), so $S^K_2(R \cdot f) \geq S^K_2(f)$, and the inverse generator gives equality; hence $S^K_2(u, u)$ is G_0-invariant.

The bound. By theorem 7.6(a), $\langle e^{tF(f)}\rangle^K_{k,J} \leq \exp(\frac{1}{2}t^2\mu_0^{-2}\|d^*_k f\|^2_{2,k})$ for real t, with equality at $t = 0$; comparing second derivatives, $\langle F(f)^2\rangle^K_{k,J} \leq \mu_0^{-2}\|d^*_k f\|^2_{2,k}$, and $\|d^*_k f\|_{2,k} \to \|d^* f\|_{L^2}$ by (1). Letting $k \to \infty$ and then $J \to \infty$, $S^K_2(f) \leq \mu_0^{-2}\|d^* f\|^2_{L^2}$; and $S^K_2(f) \geq 0$ since $Z^K_\infty(\mathrm{i}t f, 0) \leq 1 = Z^K_\infty(0, 0)$ (theorem 7.6(b)). (The continuum bound of theorem 7.6(b) gives the same at once.)

Closure. S^K_2 is a positive semidefinite symmetric bilinear form, so $|S^K_2(f, f) - S^K_2(f', f')| = |S^K_2(f - f', f + f')| \leq S^K_2(f - f')^{1/2} S^K_2(f + f')^{1/2}$, and with the bound, $R \mapsto S^K_2(R \cdot f, R \cdot f)$ is continuous on $O(3)$, because $d^*(R \cdot f) \to d^*(R_0 \cdot f)$ in L^2 as $R \to R_0$ (uniform convergence of the first derivatives on one compact set). Density of G_0 gives $SO(3)$; theorem 8.5 with $O = \sigma_3$, differentiated twice at $z = 0$, gives σ_3, hence $O(3)$; and polarisation gives the bilinear form. □

19. The heavy shell I: the photon switch, two-level walls and the dead shell

This section and the next prove the closing estimate (TIGHT$_0$) of Theorem 18.4. Here we prove the photon boundary switch (Lemma 19.1), define the two-level walls and the dead shell, and show that each of the two shell-decoupling properties (SH$_0$) and (RES$_0$) implies (TIGHT$_0$) (Theorems 19.6 and 19.10); the next section proves (RES$_0$) from the heavy-shell estimates.

Throughout this section $\Lambda = \Lambda_k$ is the axis lattice, $\varepsilon = \varepsilon_k \leq 1/\mu_0$, and

$$m_\varepsilon := \frac{2}{\varepsilon} \operatorname{arcsinh}\Big(\frac{\varepsilon\mu_0}{2}\Big);$$

since $t \mapsto \operatorname{arcsinh}(t)/t$ is decreasing on $(0, \infty)$ and $\varepsilon\mu_0/2 \leq \frac{1}{2}$, $m_\varepsilon \geq 0.96\,\mu_0$. Let $G_{\mathbb{Z}^3}$ be the Green's function of simple random walk on $\mathbb{Z}^3$ and

$$\mathsf{a} := \sup_{z\in\mathbb{Z}^3\setminus\{0\}} |z|_\infty\, G_{\mathbb{Z}^3}(0, z),$$

which is finite by the asymptotics of the Green's function of simple random walk in three dimensions [15, Chap. 4]. For a set $X \subset \mathbb{R}^3$ put $\mathrm{vol}_\varepsilon(X) := \varepsilon^3\#(X \cap \Lambda)$, and $\mathrm{vol}(X)$ for the Lebesgue measure. King cubes are $C_L = [-ML, ML)^3$ with $X_L = C_L \cap \Lambda_k$ (Definition 15.2); the comparison cube on X_L is $\langle\cdot\rangle^{(a)}_{k,J}$ with $J = L$ and its functional is $Z^a_{k,L}$ (Definitions 2.1 and 2.3).

19.1. **Photon boundary conditions across a matter-free collar.** The photon in Feynman form consists of three independent scalar Gaussian fields A_α, $\alpha = 1, 2, 3$, and on a finite graph $\mathcal{G}$ of sites of Λ each has the precision $K_\mathcal{G} = \varepsilon L_\mathcal{G} + \mu_0^2\varepsilon^3$, with $L_\mathcal{G}$ the graph Laplacian of $\mathcal{G}$ with unit conductances and $\langle u, v\rangle = \sum_x u(x)v(x)$; King's Neumann photon on a cube is the case of the induced subgraph [14, (2.5), p. 333], and the periodic photon the case of the torus graph.

Lemma 19.1 (photon boundary switch)**.** *Let $\mathsf{H} = Q_{b'}$ be a closed cube and C_L a King cube with $\mathrm{dist}_\infty(\mathsf{H}, \mathbb{R}^3 \setminus C_L) = D + 3$, where $D \geq \max(1, 3/\mu_0)$ and $\varepsilon \leq \min(1/\mu_0, 1/4)$. Let $\langle\cdot\rangle_\natural$ and $\langle\cdot\rangle_N$ be BFS measures with the same matter part: the sites of $\mathsf{H} \cap \Lambda$, BFS single-site measures (arbitrary; wall and killed sites included), and hopping terms only between neighbouring sites of H, coupled to the photon variables of the bonds of H as in King's action. Their photon parts are:*

- *(♮) the variables $A_\alpha(x)$ with $x, x + \varepsilon e_\alpha \in C_L$, with the law of the periodic photon of a torus whose cell contains $N_1(C_L)$, conditioned on $A_\beta(y) = 0$ for all other (β, y);*
- *(N) King's Neumann photon on $C_L \cap \Lambda$.*

Let $\Theta = e^{\mathrm{i}F(f)}e^{\sum_x \varepsilon^3 g(x) r_x^2 - c}$ *with* f *supported on plaquettes of* H, $g \ge 0$ *supported in* H, *and* $c \in \mathbb{R}$. *Then*

$$0 \le \langle\Theta\rangle_\natural - \langle\Theta\rangle_N \le \langle|\Theta|^2\rangle_N^{1/2}\big(e^{\tau}-1\big)^{1/2}, \qquad \tau \le C_{\mathrm{GB}}\,\mathsf{a}\,\mathrm{vol}_\varepsilon(N_2(\mathsf{H}))\,D^{-1}e^{-\frac{3}{2}m_\varepsilon D},$$

with an absolute constant C_{GB}.

Proof. Step 1 (marginals and order). Fix a component α. Let $X_\alpha := \{x \in C_L \cap \Lambda : x + \varepsilon e_\alpha \in C_L\}$; conditioning a Gaussian on the vanishing of some coordinates leaves the principal submatrix of the precision on the others, so in ($\natural$) the variables $A_\alpha(x)$, $x \in X_\alpha$, have precision $K_\natural = \varepsilon(6 - \mathrm{Adj}_{X_\alpha}) + \mu_0^2\varepsilon^3$, where Adj_{X_α} is the adjacency matrix of X_α. In (N) the variables are $A_\alpha(x)$, $x \in C_L \cap \Lambda$, with precision $K_N = \varepsilon L_{C_L} + \mu_0^2\varepsilon^3$. The principal submatrix of K_N on X_α is $\varepsilon(\deg_{C_L} - \mathrm{Adj}_{X_\alpha}) + \mu_0^2\varepsilon^3 \le K_\natural$, and the Schur complement of K_N on X_α is at most its principal submatrix; hence $C_\natural := K_\natural^{-1} \le C_N|_{X_\alpha}$, $C_N := K_N^{-1}$. Let $\hat X := \{(\alpha, x) : x, x + \varepsilon e_\alpha \in \mathsf{H}\}$. The integrands depend on A only through $A_{\hat X}$, whose laws are $N(0, C_\natural|_{\hat X})$ and $N(0, C_N|_{\hat X})$, with $C_\natural|_{\hat X} \le C_N|_{\hat X}$. Put $M := (C_\natural|_{\hat X})^{-1} - (C_N|_{\hat X})^{-1} \ge 0$. The $\natural$-measure is obtained from the N-measure, after integrating out the photon variables off $\hat X$, by adding $\frac12\langle A, M\rangle A$ to the action; writing $M = \sum_j \lambda_j v_j v_j^T$ with $\lambda_j \ge 0$, this adds $\frac{s}{2}\sum_j \lambda_j \langle v_j, A\rangle^2$ at $s = 1$, and by Lemma 2.7(a) along $s \in [0,1]$, $\langle\Theta\rangle_\natural \ge \langle\Theta\rangle_N$.

Step 2 (Cauchy–Schwarz). With $Q := \langle A, M\rangle A \ge 0$ and $\varsigma := e^{-Q/2}$, $\langle\Theta\rangle_\natural = \langle\Theta\varsigma\rangle_N / \langle\varsigma\rangle_N$, so

$$\langle\Theta\rangle_\natural - \langle\Theta\rangle_N = \frac{\langle(\Theta - \langle\Theta\rangle_N)(\varsigma - \langle\varsigma\rangle_N)\rangle_N}{\langle\varsigma\rangle_N} \le \langle|\Theta|^2\rangle_N^{1/2}\Big(\frac{\langle e^{-Q}\rangle_N}{\langle e^{-Q/2}\rangle_N^2} - 1\Big)^{1/2} \le \langle|\Theta|^2\rangle_N^{1/2}\big(e^{\langle Q\rangle_N} - 1\big)^{1/2},$$

by $\langle e^{-Q}\rangle_N \le 1$ and Jensen's inequality, $\langle e^{-Q/2}\rangle_N \ge e^{-\langle Q\rangle_N/2}$. With $M = \sum_j \lambda_j v_j v_j^T$ as in Step 1, Lemma 15.10(b) in the N-measure, whose Gaussian has covariance C_N, gives $\langle Q\rangle_N = \sum_j \lambda_j \langle\langle v_j, A\rangle^2\rangle_N \le \sum_j \lambda_j \langle v_j, C_N v_j\rangle = \mathrm{Tr}(M C_N|_{\hat X}) =: \tau$.

Step 3 (the trace). Let $\Delta := C_N|_{X_\alpha} - C_\natural \ge 0$, for each α. Then $M C_N|_{\hat X} = (C_\natural|_{\hat X})^{-1}\Delta_{\hat X}$, so $\tau = \mathbb{E}\langle Y, (C_\natural|_{\hat X})^{-1} Y\rangle$ with $Y \sim N(0, \Delta_{\hat X})$. Since the inverse of a principal block of the covariance is the minimum of the precision form over extensions, $\langle a, (C_\natural|_{\hat X})^{-1} a\rangle = \min\{\langle u, K_\natural u\rangle : u|_{\hat X} = a\}$. Let $\tilde Y \sim N(0, \Delta)$ on X_α and $\eta(x) := (1 - \mathrm{dist}_\infty(x, \mathsf{H}))_+$; then $\eta\tilde Y$ is an extension of $\tilde Y|_{\hat X}$, so $\tau \le \sum_\alpha \mathbb{E}\langle \eta\tilde Y, K_\natural \eta\tilde Y\rangle$. Let int be the set of $x \in X_\alpha$ whose six neighbours lie in X_α; it contains $N_2(\mathsf{H}) \cap \Lambda$. At $x \in \mathrm{int}$ the rows of $K_\natural$, K_N and of the infinite-lattice precision K_∞ coincide, so $(K_\infty \Delta)(x, w) = (K_N C_N)(x, w) - (K_\natural C_\natural)(x, w) = 0$ for all $w \in X_\alpha$, hence $\mathrm{Var}((K_\infty \tilde Y)(x)) = 0$ and $K_\infty \tilde Y = 0$ on int almost surely. For u with $K_\infty u = 0$ on $\mathrm{int} \supset \mathrm{supp}\,\eta$, summation by parts and the identity $(\eta_x u_x - \eta_y u_y)^2 - (u_x - u_y)(\eta_x^2 u_x - \eta_y^2 u_y) = (\eta_x - \eta_y)^2 u_x u_y$ give $\langle \eta u, K_\natural \eta u\rangle = \langle \eta u, K_\infty \eta u\rangle = \varepsilon \sum_{\langle xy\rangle} (\eta_x - \eta_y)^2 u_x u_y$. Since $|\eta_x - \eta_y| \le \varepsilon$ and $|\Delta(x,y)| \le \frac12(\Delta(x,x) + \Delta(y,y))$,

$$\tau \le \sum_\alpha \varepsilon^3 \sum_{\langle xy\rangle \subset N_2(\mathsf{H})} |\Delta(x,y)| \le 9\,\mathrm{vol}_\varepsilon(N_2(\mathsf{H})) \sup_{\alpha,\ x \in N_2(\mathsf{H})} \Delta(x,x).$$

Step 4 (decay of Δ*).* For $K = \mathcal{D} - \varepsilon \mathrm{Adj}_{\mathcal{G}}$ with $\mathcal{D}$ diagonal, $K^{-1}(x,y) = \sum_\omega \prod_{(u\to v)\in\omega} (\varepsilon/\mathcal{D}(u)) \cdot \mathcal{D}(y)^{-1}$, the sum over walks ω from x to y in $\mathcal{G}$. For $K_\natural$ and K_N the weights of steps from sites of int coincide. Decomposing walks from $x \in N_2(\mathsf{H})$ at their first visit to $\Sigma := X_\alpha \setminus \mathrm{int}$, $K^{-1}(x,y) = G^{\mathrm{int}}(x,y) + \sum_{z\in\Sigma} \mathsf{h}(x,z) K^{-1}(z,y)$ with G^{int} and the hitting kernel h the same for both precisions. Hence

$$\Delta(x,x) = \sum_{z\in\Sigma} \mathsf{h}(x,z)\big(K_N^{-1}(z,x) - K_\natural^{-1}(z,x)\big) \le \Big(\sum_{z\in\Sigma} \mathsf{h}(x,z)\Big) \sup_{z\in\Sigma} K_N^{-1}(z,x).$$

The first factor is the probability that simple random walk from x, killed with probability $\mu_0^2\varepsilon^2/(6 + \mu_0^2\varepsilon^2)$ per step, reaches Σ, which lies at ℓ^∞-distance at least D from x. With $\cosh\lambda = 1 + \mu_0^2\varepsilon^2/2$, that is $\lambda = \varepsilon m_\varepsilon$, $e^{\pm\lambda S_n^\beta}\mathbf{1}_{\{\text{alive at } n\}}$ is a martingale for each coordinate S^β of the walk; optional stopping and a union over the six half-spaces give $\sum_z \mathsf{h}(x,z) \le 6e^{-m_\varepsilon D}$. The same

argument, with the decomposition at the first exit from the ℓ^∞-ball of radius $|y|_\infty/2$ and the bound $K^{-1}_{\varepsilon\mathbb{Z}^3,\mu_0} \le K^{-1}_{\varepsilon\mathbb{Z}^3,0} = (6\varepsilon)^{-1} G_{\mathbb{Z}^3}(\cdot/\varepsilon)$ entrywise, gives the free bound

$$G_\varepsilon(y) := K^{-1}_{\varepsilon\mathbb{Z}^3}(0,y) \le 4\mathsf{a}\,|y|_\infty^{-1} e^{-m_\varepsilon |y|_\infty/2} \qquad (|y|_\infty \ge 4\varepsilon).$$

King's Neumann precision on the box $C_L \cap \Lambda$ has the Green's function $K_N^{-1}(z,x) = \sum_{x'} G_\varepsilon(z - x')$, the sum over the images x' of x under the reflections of $\varepsilon\mathbb{Z}^3$ in the planes half a lattice spacing outside the faces of the box (method of images; the mass term is uniform). Every image other than x lies outside C_L at ℓ^∞-distance at least D from every site of C_L, and so does x from $z \in \Sigma$. If $s \ge 2D$ is the side of C_L, at most $8(j+3)^3$ images satisfy $|z - x'|_\infty < D + (j+1)s$; since $m_\varepsilon D \ge 2$,

$$K_N^{-1}(z,x) \le \sum_{j\ge 0} 8(j+3)^3 \,\frac{4\mathsf{a}\, e^{-m_\varepsilon(D+js)/2}}{D+js} \le C_1\,\mathsf{a}\, D^{-1} e^{-m_\varepsilon D/2}$$

with an absolute constant C_1. Combining, $\Delta(x,x) \le 6C_1 \mathsf{a} D^{-1} e^{-\frac{3}{2} m_\varepsilon D}$, and Step 3 gives the claim with $C_{\mathrm{GB}} = 54 C_1$. □

The Gaussian measures in Lemma 19.1 are compared only on the bonds of H, where no live matter crosses the region in which the two photon boundary conditions differ; this is what makes the bound uniform in ε. A comparison in which the photon is pinned or cut on a surface that live matter crosses is not of this type: in formal perturbation theory, at one loop, the order-e^2 difference of two such states diverges logarithmically as $\varepsilon \to 0$, because the free-boundary mass counterterm compensates the photon self-energy of the matter only where the photon has its bulk covariance.

19.2. **Shell configurations, two-level walls, and shell-kill decoupling.**

Definition 19.2 (shell configurations and shell profiles)**.** Let $b > 0$. A *shell configuration* for Q_b is $\mathfrak{s} = (\mathfrak{c}, b', L)$, where $\mathfrak{c} = (J', \psi, n)$ with $\psi \in C^\infty(D_{J'}; [0,\infty))$ and $n \ge 0$ is a walled configuration in $\mathcal{A}(Q_b)$ (so $\psi = 0$ on $N_1(Q_b)$; Definitions 15.6 and 18.1), $\mathsf{H} := Q_{b'}$ with $b' \ge b + 3$, and C_L is a King cube with $\operatorname{dist}_\infty(\mathsf{H}, \mathbb{R}^3 \setminus C_L) = D(\mathfrak{s}) + 3$, $D(\mathfrak{s}) \ge \max(1, 3/\mu_0)$, and $N_1(\bar{C}_L) \subset (-25MJ', 25MJ')^3$. Its *shell* is $U := C_L \setminus \mathsf{H}$, and we require the *plateau condition* $\psi \ge 1$ on $N_1(U)$ (compatible with $\psi = 0$ on $N_1(Q_b)$, since $\operatorname{dist}_\infty(U, Q_b) \ge 3$). Put

$$\tau(\mathfrak{s}) := C_{\mathrm{GB}}\,\mathsf{a}\,\operatorname{vol}(N_3(\mathsf{H}))\, D(\mathfrak{s})^{-1} e^{-1.44\,\mu_0 D(\mathfrak{s})}, \tag{44}$$

with the constant C_{GB} of Lemma 19.1. A *shell profile* for $\mathfrak{s}$ is $\chi \in C^\infty(D_{J'}; [0,1])$ with $\chi = 1$ on U and $\operatorname{supp}\chi \subset N_1(U)$; then $\chi \le \psi$ automatically, and $\chi = 0$ on $N_1(Q_b)$. Its *dead set* is $D_\chi := \{\chi > 0\} \subset N_1(U)$; every site of C_L within ε of $\mathbb{R}^3 \setminus C_L$, and every site of U, lies in D_χ. We call $(\mathfrak{s}, \chi)$ a *two-level configuration* for Q_b.

Definition 19.3 (two-level states and the tail)**.** Let $(\mathfrak{s}, \chi)$ be a two-level configuration for Q_b, $(f,g) \in \mathcal{K}$ supported in Q_b, a a scheme, and k with $\varepsilon_k \le \min(1/\mu_0, 1/4)$. For $K \in [0,\infty)$ let

$$\Psi_k(n,K) := T^a_{(J',\, n\psi + K\chi),\Lambda_k}(f,g)$$

be the walled ratio (Definition 15.7) of the configuration with the *two-level wall* $n\psi + K\chi$, which lies in $\mathcal{A}(Q_b)$; and let

$$\Psi_k(n,\infty) := e^{-a_k \mathsf{S}_{\Lambda_k}(g)} \big\langle \Theta(\mathrm{i}f, g) \big\rangle$$

in the walled torus state of $\mathfrak{c} = (J', n\psi)$ with the matter killed on $D_\chi \cap \Lambda_k$ (Definition 15.6(d)), the *dead-shell state*. Put $\operatorname{tail}_k(K) := \Psi_k(n,K) - \Psi_k(n,\infty)$. At $g = 0$ the multiplier is 1 and nothing depends on the scheme (Lemma 15.9(b)).

Lemma 19.4 (elementary properties)**.** (a) *$K \mapsto \Psi_k(n,K)$ is nonincreasing on $[0,\infty)$, $\Psi_k(n,K) \ge \Psi_k(n,\infty)$, and $\Psi_k(n,K) \to \Psi_k(n,\infty)$ as $K \to \infty$. Hence $\operatorname{tail}_k(K) \ge 0$ is nonincreasing in K and tends to 0 as $K \to \infty$, at every fixed k.*

(b) *(two-level sandwich) For $n, K \ge 0$: $\Psi_k(n+K, 0) \le \Psi_k(n,K) \le \Psi_k(n,0)$.*

Proof. All states involved are BFS measures of King type (the paragraph after Definition 15.7), the observable $\Theta(\mathrm{i}f,g)$ is of class $\mathcal{I}$, and the multiplier $e^{-a_k \mathsf{S}_{\Lambda_k}(g)}$ is one positive constant for all of them (Lemma 2.11(a), Lemma 15.9(b)). (a) Increasing K to K' multiplies the single-site measures on D_χ by $e^{-(K'-K)\varepsilon^3\chi(x)r_x^2}$, which does not increase the expectation by Lemma 2.7(c). Killing $D_\chi \cap \Lambda_k$ is the limit $K \to \infty$ (Lemma 2.9 with the rates $\chi > 0$ on D_χ; $|\Theta(\mathrm{i}f,g)| \le e^{C\sum_x r_x^2}$), and it does not increase the expectation (Lemma 2.7(c), killing clause, the state being of King type). (b) The wall of $\Psi_k(n+K,0)$ exceeds that of $\Psi_k(n,K)$ by $K(\psi-\chi) \ge 0$, since $\chi \le \psi$; the wall of $\Psi_k(n,K)$ exceeds that of $\Psi_k(n,0)$ by $K\chi \ge 0$. Apply Lemma 2.7(c). □

Part (b) is the two-level sandwich (Lemma 19.7 below in the form used); it is the only place where $\chi \le \psi$ is needed. In words: raising the matter mass on the shell alone, by any amount, gives a state above the one in which the whole plateau is raised by the same amount, and below the one with the shell untouched.

Definition 19.5 ($(\mathrm{SH})_a$ and (SH_0): shell-kill decoupling)**.** $(\mathrm{SH})_a$ is the property: for every $b > 0$, every $(f,g) \in \mathcal{K}$ supported in Q_b and every $\eta > 0$ there is a two-level configuration $(\mathfrak{s},\chi)$ for Q_b with $\tau(\mathfrak{s}) \le \eta$ and

$$\liminf_{k\to\infty} \mathrm{tail}_k(0) = \liminf_{k\to\infty} \big[\Psi_k(n,0) - \Psi_k(n,\infty)\big] \le \eta.$$

(SH_0) is the same statement for $g = 0$; it does not depend on the scheme.

Since $\tau(\mathfrak{s})$ depends only on H and the width $D(\mathfrak{s})$, the condition $\tau(\mathfrak{s}) \le \eta$ only asks the shell to be wide enough, logarithmically in b' and $1/\eta$. The property involves one lattice, a single walled torus and a single dead set, a fixed wall height, and the lim inf over the lattice spacings; it involves no limit $n \to \infty$ and no comparison between lattices.

Theorem 19.6 (shell-kill decoupling implies the closing estimate)**.** *Assume* $(\mathrm{W})_a$ *and* $(\mathrm{K})_a$*. Then* $(\mathrm{SH})_a$ *implies* $(\mathrm{TIGHT})_a$*. In the curvature sector,* (SH_0) *implies* (TIGHT_0)*. More precisely, for every two-level configuration* $(\mathfrak{s},\chi)$ *for* Q_b*, every* $(f,g) \in \mathcal{K}$ *supported in* Q_b *and every* k *with* $\varepsilon_k \le \min(1/\mu_0, 1/4)$*,*

$$\Psi_k(n,\infty) \le Z^a_{k,L}(\mathrm{i}f,g) + Z^a_{k,L}(0,2g)^{1/2}\big(e^{\tau(\mathfrak{s})}-1\big)^{1/2}, \tag{45}$$

where $Z^a_{k,L}$ *is the functional of the comparison cube on* $X_L = C_L \cap \Lambda_k$*. Hence, under* $(\mathrm{W})_a$*,* $\limsup_k \Psi_k(n,\infty) \le Z^a_\infty(\mathrm{i}f,g) + Z^a_\infty(0,2g)^{1/2}(e^{\tau(\mathfrak{s})}-1)^{1/2}$*; and at* $g = 0$*, under* (E_C) *alone,* $\limsup_k \Psi_k(n,\infty) \le Z^K_\infty(\mathrm{i}f,0) + (e^{\tau(\mathfrak{s})}-1)^{1/2}$*.*

Proof. Fix k with $\varepsilon = \varepsilon_k \le \min(1/\mu_0, 1/4)$ and write $\Theta = \Theta(\mathrm{i}f,g)$. The multiplier $e^{-a_k \mathsf{S}_{\Lambda_k}(g)}$ is one positive constant for every state below (Lemma 2.11(a)); we compare the expectations of Θ and of $|\Theta|^2 = \Theta(0,2g)$ and restore it at the end. Let E be the set of sites of the torus $D_{J'}$ outside C_L.

Step 1 (pinning). In the dead-shell state, add $s|A_\beta(y)|^2$ to the action for every photon variable on a bond with an endpoint in E, and let $s \to \infty$. By Lemma 2.7(a) the expectation of Θ does not decrease, and in the limit the photon is the conditioned law ($\natural$) of Lemma 19.1, the periodic photon of $D_{J'}$ (whose cell contains $N_1(C_L)$ by Definition 19.2) conditioned on the vanishing of all variables except the $A_\alpha(x)$ with $x, x+\varepsilon e_\alpha \in C_L$. In this state the hopping terms within E do not involve A, and there are no hopping terms between E and live sites: every site of C_L adjacent to E lies in D_χ and is dead. So the matter on E is a factor independent of A and cancels (Lemma 2.7(e)). What remains is the $\natural$-measure of Lemma 19.1, with the matter on H (dead on $D_\chi \cap \mathsf{H}$), the wall $n\psi$ in its single-site measures, and Dirichlet conditions at $\partial\mathsf{H}$, since $U \subset D_\chi$ is dead. Hence $\Psi_k(n,\infty) \le e^{-a_k \mathsf{S}_{\Lambda_k}(g)}\langle\Theta\rangle_\natural$.

Step 2 (photon switch). By Lemma 19.1, $\langle\Theta\rangle_\natural \le \langle\Theta\rangle_N + \langle|\Theta|^2\rangle_N^{1/2}(e^\tau-1)^{1/2}$ with $|\Theta|^2 = \Theta(0,2g)$; here f is supported on plaquettes of H and g in H because $b' \ge b+3$, and $m_\varepsilon \ge 0.96\mu_0$ together with $\mathrm{vol}_\varepsilon(N_2(\mathsf{H})) \le \mathrm{vol}(N_3(\mathsf{H}))$ (for $\varepsilon \le 1$) give $\tau \le \tau(\mathfrak{s})$.

Step 3 (to the comparison cube). The N-measure is the limit $s \to \infty$ of the walled Neumann–Dirichlet state on X_L, with the wall $n\psi$ in its single-site measures, in which the single-site

measures on $D_\chi \cap X_L$ are multiplied by $e^{-s\varepsilon^3\chi(x)r_x^2}$ (Lemma 2.9; the matter alive in the N-measure is that of $X_L \setminus D_\chi \subset \mathsf{H}$, its Dirichlet terms toward the dead sites being the diagonal parts of the bonds into them). By Lemma 2.7(c) the expectations of $\Theta(\mathrm{i}f, g)$ and $\Theta(0, 2g)$ do not decrease as s decreases to 0, nor when the wall is then removed. The result is the comparison cube $\langle\cdot\rangle^{(a)}_{k,J}$ on X_L (Definition 2.1), in which both observables are the same functions of the shared variables, every site with $g \neq 0$ and every plaquette of f lying in H, at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_L$. Restoring the multiplier (Lemma 2.11(a); for $\Theta(0, 2g)$ it is its square), this is (45).

Step 4 (limits). Under $(\mathrm{W})_a$ the cube functionals converge, $Z^a_{k,L} \to Z^a_L$, by Theorem 16.6(i), and $Z^a_L \le Z^a_\infty$ on $\mathcal{K}$ by Theorem 16.6(iii), with $(0, 2g) \in \mathcal{K}$. This gives the lim sup bound. Now let $(\mathfrak{s}, \chi)$ be as in $(\mathrm{SH})_a$ for (f, g) and η. Then $T^a_{\mathfrak{c},\Lambda_k}(f, g) = \Psi_k(n, 0) = \mathrm{tail}_k(0) + \Psi_k(n, \infty)$, the left side converges to $\mathcal{W}^a_{\mathfrak{c}}(f, g)$ by $(\mathrm{K})_a$ (Lemma 17.3(a), Definition 18.1), and taking $\liminf_k$, with $\liminf_k(x_k + y_k) \le \liminf_k x_k + \limsup_k y_k$,

$$\mathcal{W}^a_{\mathfrak{c}}(f, g) \le \eta + Z^a_\infty(\mathrm{i}f, g) + Z^a_\infty(0, 2g)^{1/2}\big(e^\eta - 1\big)^{1/2}.$$

Since $\mathfrak{c} \in \mathcal{A}(Q_b)$ and η is arbitrary, $I^a_b(f, g) \le Z^a_\infty(\mathrm{i}f, g)$ for every b, which is $(\mathrm{TIGHT})_a$.

At $g = 0$ the multiplier is 1 and $\langle|\Theta|^2\rangle = 1$, so Step 3 gives $\Psi_k(n, \infty) \le Z_{k,L}(\mathrm{i}f) + (e^{\tau(\mathfrak{s})} - 1)^{1/2}$ with $Z_{k,L}(\mathrm{i}f) = \langle e^{\mathrm{i}F(f)}\rangle^{(a)}_{k,J}$ at $J = L$. Under (E_C), since f is supported at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_L$, Corollary 4.4(ii)($\gamma$) gives $\limsup_k Z_{k,L}(\mathrm{i}f) \le Z^K_{L+1}(\mathrm{i}f)$, and $Z^K_{L+1}(\mathrm{i}f) \le Z^K_\infty(\mathrm{i}f, 0)$ because $J \mapsto Z^K_J(\mathrm{i}f)$ is nondecreasing (Corollary 4.3, Theorem 7.6(b)) and converges to $Z^K_\infty(\mathrm{i}f, 0)$ (Corollary 4.4(ii)(β)). This is the lim sup bound at $g = 0$. With $(\mathfrak{s}, \chi)$ as in (SH_0), $T_{\mathfrak{c},\Lambda_k}(f, 0) = \Psi_k(n, 0) \to \mathcal{W}_{\mathfrak{c}}(f, 0)$ by (K_0) (Lemma 17.3(a)), and the same lim inf argument gives $\mathcal{W}_{\mathfrak{c}}(f, 0) \le \eta + Z^K_\infty(\mathrm{i}f, 0) + (e^\eta - 1)^{1/2}$, hence $I_b(f, 0) \le Z^K_\infty(\mathrm{i}f, 0)$ for every b, which is (TIGHT_0). □

The theorem is stated for the cubic shell $U = C_L \setminus \mathsf{H}$ of Definition 19.2, in which C_L is a King cube and H a cube inside it. For the configurations of the next section, whose shell is the exterior of a box in the torus, the three steps of the proof are carried out in Proposition 20.1 ((DS)).

19.3. **Heavy-shell decoupling.**

Lemma 19.7 (two-level sandwich)**.** *For every two-level configuration, every $(f, g) \in \mathcal{K}$ supported in Q_b, every k with $\varepsilon_k \le \min(1/\mu_0, 1/4)$ and $n, K \ge 0$,*

$$\Psi_k(n + K, 0) \;\le\; \Psi_k(n, K) \;=\; \Psi_k(n, \infty) + \mathrm{tail}_k(K) \;\le\; \Psi_k(n, 0), \qquad \mathrm{tail}_k(K) \ge 0.$$

Proof. Lemma 19.4. □

Definition 19.8 ($(\mathrm{RES})_a$ and (RES_0): heavy-shell decoupling at one configuration)**.** $(\mathrm{RES})_a$ is the property: for every $b > 0$, every $(f, g) \in \mathcal{K}$ supported in Q_b and every $\eta > 0$ there are a two-level configuration $(\mathfrak{s}, \chi)$ for Q_b with $\tau(\mathfrak{s}) \le \eta$ and a shell mass $K \in [0, \infty)$ such that

$$\liminf_{k\to\infty} \mathrm{tail}_k(K) = \liminf_{k\to\infty} \big[\Psi_k(n, K) - \Psi_k(n, \infty)\big] \le \eta.$$

(RES_0) is the same statement for $g = 0$; it does not depend on the scheme.

In words: in a walled torus, once the scalar mass squared on a thick shell of the heavy plateau has been raised by $2K$, removing the matter on the shell altogether changes the expectation of the observable by at most η in the continuum limit, for one geometry and one K, with no rate in K and no uniformity in anything. By Lemma 19.4(a) the bracket tends to 0 as $K \to \infty$ at every fixed k; the property says that the heavy-shell limit and the continuum limit may be interchanged at one configuration. The quantifiers are the weakest that Theorem 19.10 accepts: one configuration, one K (with $K = 0$ allowed, which is shell-kill decoupling, Definition 19.5), the lim inf along the spacings, and no uniformity in n, L, $D(\mathfrak{s})$ or the torus.

Proposition 19.9 ($(\mathrm{SH})_a$ implies $(\mathrm{RES})_a$)**.** *For every two-level configuration and every $K \ge 0$, $\mathrm{tail}_k(K) \le \mathrm{tail}_k(0)$. Hence $(\mathrm{SH})_a$ implies $(\mathrm{RES})_a$ with the same configurations and every K, and likewise (SH_0) implies (RES_0).*

Proof. Lemma 19.4(a). □

Theorem 19.10 ((RES)$_a$ implies (TIGHT)$_a$)**.** *Assume* (W)$_a$ *and* (K)$_a$*. Then* (RES)$_a$ *implies* (TIGHT)$_a$*. In the curvature sector,* (RES$_0$) *implies* (TIGHT$_0$)*. More precisely, if* $(\mathfrak{s},\chi)$ *and* K *are as in* (RES)$_a$ *for* (f,g) *and* η*, then the single-level configuration* $\mathfrak{c}' := (J',\psi,n+K) \in \mathcal{A}(Q_b)$ *satisfies*

$$\mathcal{W}^a_{\mathfrak{c}'}(f,g) \le Z^a_\infty(\mathrm{i}f,g) + Z^a_\infty(0,2g)^{1/2}\big(e^\eta-1\big)^{1/2} + \eta,$$

and at $g=0$*, with* $(\mathfrak{s},\chi)$ *and* K *as in* (RES$_0$)*,* $\mathcal{W}_{\mathfrak{c}'}(f,0) \le Z^K_\infty(\mathrm{i}f,0) + (e^\eta-1)^{1/2} + \eta$*.*

Proof. By Lemma 19.7, $T^a_{\mathfrak{c}',\Lambda_k}(f,g) = \Psi_k(n+K,0) \le \Psi_k(n,\infty) + \mathrm{tail}_k(K)$ at every k. The left side converges to $\mathcal{W}^a_{\mathfrak{c}'}(f,g)$ by (K)$_a$, since $\mathfrak{c}'$ is a walled configuration in $\mathcal{A}(Q_b)$ (Lemma 17.3(a), Definition 18.1); hence, using $\liminf_k(x_k+y_k) \le \limsup_k x_k + \liminf_k y_k$,

$$\begin{aligned}\mathcal{W}^a_{\mathfrak{c}'}(f,g) = \liminf_k \Psi_k(n+K,0) &\le \limsup_k \Psi_k(n,\infty) + \liminf_k \mathrm{tail}_k(K)\\ &\le Z^a_\infty(\mathrm{i}f,g) + Z^a_\infty(0,2g)^{1/2}\big(e^{\tau(\mathfrak{s})}-1\big)^{1/2} + \eta\end{aligned}$$

by Theorem 19.6 and (RES)$_a$. With $\tau(\mathfrak{s}) \le \eta$ this is the displayed bound; as η is arbitrary and $\mathfrak{c}' \in \mathcal{A}(Q_b)$, $I^a_b(f,g) \le Z^a_\infty(\mathrm{i}f,g)$ for every b, which is (TIGHT)$_a$. At $g=0$ the same argument runs with the $g=0$ clause of Theorem 19.6 under (E$_C$), the convergence of the left side given by (K$_0$), and $Z(0,0)=1$; it gives $I_b(f,0) \le Z^K_\infty(\mathrm{i}f,0)$ for every b, which is (TIGHT$_0$). □

Remark 19.11 (on the proof). (i) The two-level state is only an intermediate inequality; the configuration that enters (TIGHT)$_a$, respectively (TIGHT$_0$), is the single-level wall $(n+K)\psi$, and the only limit input is the existence of its walled value, which is part of (K)$_a$, respectively (K$_0$) (the existence of the torus limits on the axis lattice, Remark 17.11). No existence statement is needed for the two-level state or for the dead-shell state. (ii) The lattice statements of this subsection hold verbatim for the sharp profile $\chi = \mathbf{1}_U$ (then $D_\chi = U$ and $\chi \le \psi$ needs only $\psi \ge 1$ on U); smoothness of χ is used only in Theorem 20.8, where the two-level walled value must exist. (iii) Since $\mathrm{tail}_k(K)$ is nonincreasing in K, (RES)$_a$ for one K implies it for every larger K; so K may be taken as large as one likes, which is used to place $n+K$ in any prescribed window of wall heights (Remark 18.6).

20. The heavy shell II: the heavy-shell estimates and their continuum form

20.1. **The heavy-shell estimates.** In the curvature sector (RES$_0$) is proved at one configuration, from two estimates on King's walled torus model with the shell mass inside the matter covariance, the heavy-shell estimates (HS) of Theorem 20.2. The shell is now taken to be the whole exterior of the source box in the torus. Let $(\mathfrak{s},\chi)$ be a two-level configuration in the sense of Definitions 19.2 and 19.3, with the walled torus $T := D_{J'}$ of $\mathfrak{c} = (J',\psi,n)$ and the source box $\mathsf{H}' := \mathsf{H} = Q_{b'}$, in which the shell $C_L \setminus \mathsf{H}$ is replaced by

$$U := T \setminus \mathsf{H}',$$

so that $\partial U = \partial \mathsf{H}'$ is a single convex surface with no re-entrant edge (on a cubic shell $C_L \setminus \mathsf{H}$ the derivative clause of King's bound on the block propagators, [13, Thm. 3.3], fails at the re-entrant edges); the plateau condition $\psi \ge 1$ on $N_1(U)$ is kept, the profile is the sharp one, $\chi = \mathbf{1}_U$, which Remark 19.11(ii) allows for every lattice statement of the preceding section, so that the dead set is $D_\chi = U$, and $\tau(\mathfrak{s})$ is computed, as in Definition 19.2, from H and a King cube C_L with $\mathsf{H}' \subset C_L \subset T$ at ℓ^∞-distance $D(\mathfrak{s})+3$ from H'. For $\varkappa \in [0,\infty)$ let $\Psi_k(n,\varkappa)$ be the two-level expectation of Definition 19.3 with the shell mass $\varkappa\mathbf{1}_U$ added to the wall: by Definition 15.6(c) the wall $n\psi + \varkappa\mathbf{1}_U$ raises the scalar mass squared by $2n\psi$ on the plateau and by a further $2\varkappa$ on U, and $\Psi_k(n,K)$, $\Psi_k(n,\infty)$ of Definition 19.8 are the cases $\varkappa = K$ and $\varkappa = \infty$ (the matter killed on U). Let $Z_\varkappa(zf,0)$ be the corresponding partition function with the curvature source zf, so that $Z_\varkappa(zf,0)/Z_\varkappa(0,0) = \langle e^{zF_{\Lambda_k}(f)}\rangle_\varkappa$ is the expectation of the source weight in the walled torus state with the shell mass $\varkappa\mathbf{1}_U$. Here $Z_\varkappa(\,\cdot\,,0)$ is the unnormalised functional of this walled state in the convention of Definition 15.7, with the wall $n\psi$ and the shell mass $\varkappa\mathbf{1}_U$ in its single-site measures; it is not the normalised functional Z^a of the first sections, and Theorem 20.5 uses it

only through the ratios $Z_\varkappa(zf,0)/Z_\varkappa(0,0)$. Finally $n_* := 1$: the estimates of (HS) are read at plateau heights $n \geq n_*$, at which the mass squared on the plateau is raised by at least 2.

The first statement carries the three steps of the dead-shell chain, proved in the preceding section for the cubic shell $C_L \setminus \mathsf{H}$, to the exterior-shell geometry.

Proposition 20.1 ((DS): the dead-shell chain for the exterior shell)**.** *For the two-level configurations with the exterior shell $U = T \setminus \mathsf{H}'$ the three steps of the proof of Theorem 19.6 hold: pinning the photon outside a King cube $C_L \supset \mathsf{H}'$ all of whose sites outside H' are dead, the photon switch across the matter-free collar between H' and ∂C_L, and the passage to the comparison cube on $X_L = C_L \cap \Lambda_k$. Consequently the dead-shell bound (45) holds for them: at every k with ε_k small, $\Psi_k(n,\infty) \leq Z_{k,L}(\mathrm{i}f) + (e^{\tau(\mathfrak{s})} - 1)^{1/2}$ at $g = 0$.*

Proof of Proposition 20.1. Fix a two-level configuration with the exterior shell: the walled torus $T = D_{J'}$ of $\mathfrak{c} = (J', \psi, n)$, the source box $\mathsf{H}' = Q_{b'}$ with $b' \geq b + 3$, the sharp profile $\chi = \mathbf{1}_U$ with $U = T \setminus \mathsf{H}'$, so that the dead set is $D_\chi = U$, and a King cube C_L with $\mathsf{H}' \subset C_L \subset T$, $\mathrm{dist}_\infty(\mathsf{H}', \mathbb{R}^3 \setminus C_L) = D(\mathfrak{s}) + 3$, $D(\mathfrak{s}) \geq \max(1, 3/\mu_0)$ and $N_1(\bar{C}_L) \subset (-25MJ', 25MJ')^3$; let f be a real 2-form supported in Q_b, $g = 0$, and k with $\varepsilon = \varepsilon_k \leq \min(1/\mu_0, 1/4)$. Write $\Theta := e^{\mathrm{i}F_{\Lambda_k}(f)}$ and $E := T \setminus C_L$ for the set of sites of the torus outside C_L. Since $\mathrm{supp}\, f \subset Q_b$ and $b' \geq b + 3$, every plaquette of f lies in H' at ℓ^∞-distance > 1 from $\partial\mathsf{H}'$. The dead-shell state $\Psi_k(n,\infty) = \langle\Theta\rangle$ is the walled torus state of $\mathfrak{c}$ with the matter killed on $U \cap \Lambda_k$ (Definition 19.3): the single-site measure at every site of U is the point mass at $\phi = 0$, every hopping term with an endpoint in U is reduced to its diagonal part at the other endpoint, and the photon is the periodic photon of T with mass μ_0 (Definition 15.6(d)). The matter alive is that of $\mathsf{H}' \cap \Lambda_k$, with the wall $n\psi$ in its single-site measures; since $\psi = 0$ on $N_1(Q_b)$ and $\psi \geq 1$ on $N_1(U)$, the wall is a fixed non-negative function on H'. The two differences from the cubic shell of Definition 19.2 are that the dead region U now extends to the whole exterior of H' in T, so that no live matter sits outside H', and that $\partial U = \partial\mathsf{H}'$ is a single convex surface. Both only simplify the three steps.

Step 1 (pinning). For $s \geq 0$ add $s|A_\beta(y)|^2$ to the action for every photon variable $A_\beta(y)$ on a bond with an endpoint in E, that is for every (β, y) with $y \in E$ or $y + \varepsilon e_\beta \in E$. The resulting measures are BFS measures of King type (the paragraph after Definition 15.7: adding $s|\ell(A)|^2$ to the action is one of the admitted operations), so by Lemma 2.7(a) the expectation of Θ, which is of class $\mathcal{I}$, does not decrease in s; let $s \to \infty$. In the limit the pinned variables are fixed at 0, and by the Gaussian structure of the photon part the law of the remaining variables $A_\alpha(x)$, $x, x + \varepsilon e_\alpha \in C_L$, is the periodic photon of T conditioned on the vanishing of all the others: this is the law ($\natural$) of Lemma 19.1, whose cell contains $N_1(C_L)$ by hypothesis. In the pinned measure the matter outside H' is dead, so the hopping terms with both endpoints in U contribute nothing (their cosine parts vanish at $\phi = 0$ and their diagonal parts vanish as well), and a hopping term between a site of H' and a dead neighbour in U is its diagonal part $\frac{\varepsilon}{2} r_x^2$ at the live endpoint, a Dirichlet term independent of A. In particular no hopping term couples E to a live site, since every site of C_L adjacent to E lies in $C_L \setminus \mathsf{H}' \subset U$ and is dead, and the matter on E is a product of point masses, a factor 1. What remains is exactly the $\natural$-measure of Lemma 19.1: the matter on $\mathsf{H}' \cap \Lambda_k$ with BFS single-site measures (the wall $n\psi$ and the Dirichlet terms toward U included), hopping terms only between neighbouring sites of H', coupled to the photon variables of the bonds of H' as in King's action, and the photon law ($\natural$) on the bonds of C_L. Hence

$$\Psi_k(n,\infty) \;\leq\; \langle\Theta\rangle_\natural .$$

Step 2 (the photon switch across the collar). The collar $C_L \setminus \mathsf{H}' \subset U$ carries no live matter, and $\mathrm{dist}_\infty(\mathsf{H}', \mathbb{R}^3 \setminus C_L) = D(\mathfrak{s}) + 3$ with $D(\mathfrak{s}) \geq \max(1, 3/\mu_0)$ and $\varepsilon \leq \min(1/\mu_0, 1/4)$. So Lemma 19.1 applies to the pair of measures with the common matter part of Step 1 and the photon parts ($\natural$) and (N), the latter King's Neumann photon on $C_L \cap \Lambda_k$; its hypotheses on the observable hold with $c = 0$ and $g = 0$, f being supported on plaquettes of H'. It gives

$$\langle\Theta\rangle_\natural \;\leq\; \langle\Theta\rangle_N + \langle|\Theta|^2\rangle_N^{1/2} \big(e^\tau - 1\big)^{1/2} = \langle\Theta\rangle_N + \big(e^\tau - 1\big)^{1/2}, \qquad \tau \leq C_{\mathrm{GB}}\, \mathsf{a}\, \mathrm{vol}_\varepsilon(N_2(\mathsf{H}'))\, D(\mathfrak{s})^{-1} e^{-\frac{3}{2} m_\varepsilon D(\mathfrak{s})},$$

since $|\Theta|^2 = 1$ at $g = 0$. With $m_\varepsilon \geq 0.96\mu_0$ and $\mathrm{vol}_\varepsilon(N_2(\mathsf{H}')) \leq \mathrm{vol}(N_3(\mathsf{H}'))$ for $\varepsilon \leq 1$ this is $\tau \leq \tau(\mathfrak{s})$, the constant (44) computed from $\mathsf{H} = \mathsf{H}'$ and $D(\mathfrak{s})$, as in Step 2 of the proof of

Theorem 19.6. The geometry of the shell outside C_L plays no role here: the lemma sees only the matter on H' and the photon on C_L.

Step 3 (to the comparison cube). The N-measure of Step 2 is the Neumann–Dirichlet state on $X_L = C_L \cap \Lambda_k$ of Definition 15.6(a) with the wall $n\psi$ in its single-site measures and the matter killed on $(C_L \setminus \mathsf{H}') \cap \Lambda_k$: King's Neumann photon on C_L, Dirichlet terms for the matter of H' toward its dead neighbours and, for the sites of C_L adjacent to $\mathbb{R}^3 \setminus C_L$, toward the outside (these sites are dead, so the two readings of their Dirichlet terms coincide). By Lemma 2.9, applied with the set $(C_L \setminus \mathsf{H}') \cap \Lambda_k$ and the rate 1 on it, this state is the limit $s \to \infty$ of the walled Neumann–Dirichlet states on X_L in which the single-site measures on $(C_L \setminus \mathsf{H}') \cap \Lambda_k$ are multiplied by $e^{-s\varepsilon^3 r_x^2}$; $|\Theta| \le 1$ is of the form admitted there. By Lemma 2.7(c) the expectation of Θ does not decrease when s decreases to 0 (the single-site weights $e^{-s\varepsilon^3 r_x^2}$ are removed), nor when the wall factors $e^{-\varepsilon^3 n\psi(x) r_x^2}$, $x \in X_L$, are then removed (the same clause, the wall being a non-negative multiplier of the form e^{-sr^2} at each site). The result is the comparison cube $\langle\cdot\rangle_{k,J}^{(a)}$ on X_L (Definition 2.1): the Neumann photon on C_L, the scalar alive on all of X_L with the Dirichlet condition outside C_L, no wall. In it Θ is the same function of the shared variables, every plaquette of f lying in H' at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_L$. Hence $\langle\Theta\rangle_N \le \langle e^{\mathrm{i}F(f)}\rangle_{k,J}^{(a)}|_{J=L} = Z_{k,L}(\mathrm{i}f)$, and with Steps 1–2

$$\Psi_k(n,\infty) \;\le\; Z_{k,L}(\mathrm{i}f) + \big(e^{\tau(\mathfrak{s})} - 1\big)^{1/2},$$

which is (45) at $g = 0$ for the exterior-shell configuration. The three steps use the shape of U only through two facts, that every site of C_L adjacent to E is dead (Step 1) and that the collar between H' and ∂C_L carries no live matter (Step 2); both hold because $U \supset C_L \setminus \mathsf{H}'$. The two-level sandwich Lemma 19.7, which Theorem 19.10 applies to this configuration, needs $\chi \le \psi$ pointwise, that is $\psi \ge 1$ on $U = T \setminus \mathsf{H}'$; this is the plateau condition $\psi \ge 1$ on $N_1(U)$ of Definition 19.2, kept for the exterior shell, and it is compatible with $\psi = 0$ on $N_1(Q_b)$ because $\mathrm{dist}_\infty(U, Q_b) \ge 3$. □

With Proposition 20.1 in place, the proof of Theorem 19.10 runs verbatim for the exterior-shell configurations of Theorem 20.5: its only use of the shell geometry is the dead-shell bound (45), which the proposition supplies, and Lemma 19.7, which holds for the sharp profile by Remark 19.11(ii).

Theorem 20.2 ((HS): the heavy-shell estimates)**.** *Let $(\mathfrak{s}, \chi)$ be a two-level configuration with the exterior shell $U = T \setminus \mathsf{H}'$, with H' and T unions of blocks of the final scale of King's block-spin flow, with H' of side at least two final blocks and at distance at least three final blocks from its periodic images in every direction, and with every corner coordinate of H', counted from the origin in units of the final block, congruent to $M'/2$ modulo M', where the large-block integer is $M = 5M'$ with M' even (the grid-offset condition; King's tori satisfy the clauses on T, and in each direction every interval of M' final blocks contains an admissible corner coordinate) and plateau height $n \ge n_*$, let $g = 0$ and let f be a real 2-form in C_c^∞ supported in Q_b. There are $\eta_1 > 0$ and, for every stopping scale $\eta_0 \in 5^{-\mathbb{N}}$ with $\eta_0 \le \eta_1$: numbers $\mathfrak{a}(\eta_0), \mathfrak{b}(\eta_0) \in [0,\infty)$ with $\mathfrak{a}(\eta_0), \mathfrak{b}(\eta_0) \to 0$ as $\eta_0 \to 0$; a threshold $\varkappa_0(\eta_0) < \infty$; a spacing bound $\varepsilon_0(\eta_0) > 0$; a function $r(\,\cdot\,;\eta_0)$ on $[\varkappa_0,\infty)$ with $r(\varkappa;\eta_0) \to 0$ as $\varkappa \to \infty$; and, for every k with $\varepsilon_k \le \varepsilon_0(\eta_0)$, a measure space $(\tilde\chi, db)$ not depending on $\varkappa$ and measurable functions $\tilde S_\varkappa(\,\cdot\,;z)$ on $\tilde\chi$ for $\varkappa \ge \varkappa_0$ and $z \in [-1,1]$ (both may depend on k; the index is suppressed); such that, with*

$$I_\varkappa^{\mathrm{cap}}(z) := \int_{\tilde\chi} e^{-\tilde S_\varkappa(b;z)}\, db, \qquad \Delta_z(b) := \tilde S_{\varkappa'}(b;z) - \tilde S_\varkappa(b;z),$$

for every such k, all $\varkappa_0 \le \varkappa \le \varkappa' < \infty$ and all $z \in [-1,1]$:

(i) (the two-sided bound) $e^{-\mathfrak{a}}\, I_\varkappa^{\mathrm{cap}}(z) \;\le\; Z_\varkappa(zf,0) \;\le\; e^{\mathfrak{a}}\, I_\varkappa^{\mathrm{cap}}(z) + \mathfrak{b}\, Z_\varkappa(0,0)$;

(ii) (the $\varkappa$-Cauchy property) $\sup_{b,b' \in \tilde\chi} |\Delta_z(b) - \Delta_0(b')| \;\le\; 2\, r(\varkappa;\eta_0)$.

The shell model and the stopping-scale integrals. Fix the exterior-shell configuration of Theorem 20.2: the torus $T = D_{J'}$, the box $\mathsf{H}' = Q_{b'}$, both aligned with the blocks of the final scale as Theorem 20.2 requires (block alignment and grid offset), the plateau height $n \ge n_* = 1$ and

the shell $U = T \setminus \mathsf{H}'$. The *shell model* at shell mass $\varkappa \in [0,\infty)$ is King's walled torus model of Definition 15.6(c) with the wall $n\psi + \varkappa \mathbf{1}_U$: its matter operator is $-\Delta_A + m^2 + \delta m_\varepsilon^2 + 2n\psi + 2\varkappa\mathbf{1}_U$, and $\varkappa\mathbf{1}_U$ is kept inside the matter covariance of every block-spin step. King's block-spin flow [13, §3] is run on Λ_k, $\varepsilon = \varepsilon_k = 5^{-k}$, through r steps down to the *stopping scale* $\eta_0 := 5^{r-k} \in 5^{-\mathbb{N}}$, at which the plateau is massive in block units (no quantitative heaviness of the plateau is used; only $n \geq 1$ enters). The shell mass in block units at the stopping scale is $w_r := \varkappa\eta_0^2$; the *frozen regime* is $w_r \geq c_0$ for a fixed $c_0 \in (0, c_0^*]$, and the threshold of Theorem 20.2 is

$$\varkappa_0 = \varkappa_0(\eta_0) := \hat{c}_0/\eta_0^2, \qquad \hat{c}_0 := \max(c_0, 1).$$

The output of the flow is the effective action $S^{(r),\varkappa}(A_r, \phi_r; zf, 0)$ on the block fields of the coarse lattice $5^r\varepsilon\,\mathbb{Z}^3/P_{J'}$, in King's unit scale, with the curvature source zf carried as in [13, (3.20)], and the small-field indicator χ_r of [13, (3.2)]; the shell model carries in addition two tests at the stopping scale, a *near-shell test* τ_r on the block fields of U and a *first-layer test* λ_r on the block fields of the first light layer of H', both with thresholds that depend on w_r through increasing functions, and a weakened gradient family $\chi_r^\flat \geq \chi_r$ with no clause on the bonds inside U or crossing ∂U. The stopping-scale integrals are

$$I_\varkappa^\sharp(z) := \int (dA_r)(d\phi_r)\, \chi_r \tau_r \lambda_r\, e^{-S^{(r),\varkappa}}, \qquad I_\varkappa(z) := \int (dA_r)(d\phi_r)\, \chi_r^\flat \tau_r \lambda_r\, e^{-S^{(r),\varkappa}} \;\geq\; I_\varkappa^\sharp(z).$$

For $\varkappa \geq \varkappa_0$ the thresholds of τ_r, λ_r are dominated by the $\varkappa$-free caps $C_7 p_r$ on U and $C_8 p_r$ on the first layer ($p_r = p(\eta_0)$ King's small-field function), and the *capped* integral $I_\varkappa^{\mathrm{cap}}(z)$ of Theorem 20.2 is $I_\varkappa$ with τ_r, λ_r replaced by the indicators of these caps. Under the standing smallness $\eta_0 \leq 5/(\lambda C_7^4)$ the cap on U implies the block-sup clause of $\chi_r^\flat$ there, so the capped test set is a product: with $b := (A_r, \phi_r|_{\mathsf{H}'})$ the *exterior block data* and $\phi_U := \phi_r|_U$,

$$\tilde{\chi}^{\mathrm{cap}} := \{b :\ \chi_r^\flat \lambda_r^{\mathrm{cap}}(b) = 1\}, \qquad \tilde{S}_\varkappa^{\mathrm{cap}}(b; z) := -\ln \int \prod_{y \in U^{(r)}} d\phi_r(y)\, \mathbf{1}\{|\phi_r| \leq C_7 p_r \text{ on } U^{(r)}\}\, e^{-S^{(r),\varkappa}(A_r,\phi_r;zf,0)},$$

so that $I_\varkappa^{\mathrm{cap}}(z) = \int_{\tilde{\chi}^{\mathrm{cap}}} e^{-\tilde{S}_\varkappa^{\mathrm{cap}}(b;z)}\, db$ exactly: this is the measure space $(\tilde{\chi}, db)$ and the reduced small-field action of Theorem 20.2, and the test set $\tilde{\chi}^{\mathrm{cap}}$ is one and the same for every $\varkappa \geq \varkappa_0$, which is the threshold clause of that theorem. Every statement below is uniform in $\varepsilon \leq \varepsilon_0(\eta_0)$, in $\varkappa$ on the range stated, in $z \in [-1, 1]$, and holds for the model with the free-boundary counterterm δm_ε^2 of Definition 2.1: the torus-convention counterterm of [13] differs from it by a number $\Delta_T(\varepsilon)$ bounded uniformly in ε (Proposition 17.121(b) above), and every bound of the shell analysis holds with constants depending on $\sup_\varepsilon |\Delta_T(\varepsilon)|$ as well.

Proposition 20.3 (the two-sided bound with the shell inside)**.** *Let $L = 5$, the couplings e, λ, m, μ_0 in King's range, (H', T) block-aligned with the grid offset as in Theorem 20.2, $n \geq 1$, and f as in Theorem 20.2. There are $\eta_1 > 0$, $\sigma > 0$, $c > 0$ and, for every $\eta_0 \in 5^{-\mathbb{N}}$ with $\eta_0 \leq \eta_1$, a spacing bound $\varepsilon_0(\eta_0) > 0$ and a constant $C < \infty$ depending on $(\lambda, e, m, \mu_0, |T|, n, f, \mathsf{H}', c_0)$ but not on η_0, $\varkappa$ or ε, such that for all $\varepsilon \leq \varepsilon_0(\eta_0)$ and $z \in [-1, 1]$, with*

$$\mathfrak{a}(\eta_0) := C\eta_0^\sigma |T|, \qquad \mathfrak{b}(\eta_0) := e^{-c\,p(\eta_0)^2 + C|T|},$$

$$\begin{aligned}
Z_\varkappa(zf, 0) &\geq e^{-\mathfrak{a}}\, I_\varkappa^\sharp(z) && (\varkappa \in [0,\infty)),\\
Z_\varkappa(zf, 0) &\geq e^{-\mathfrak{a}}\, I_\varkappa^{\mathrm{cap}}(z) && (\varkappa \in [\varkappa_0,\infty)),\\
Z_\varkappa(zf, 0) &\leq e^{\mathfrak{a}}\, I_\varkappa(z) + \mathfrak{b}\, Z_\varkappa(0,0) \;\leq\; e^{\mathfrak{a}}\, I_\varkappa^{\mathrm{cap}}(z) + \mathfrak{b}\, Z_\varkappa(0,0) && (\varkappa \in [0,\infty)).
\end{aligned}$$

In particular $\mathfrak{a}, \mathfrak{b} \to 0$ as $\eta_0 \to 0$ at fixed $(n, |T|)$, uniformly in $\varkappa$ and $\varepsilon \leq \varepsilon_0(\eta_0)$, and clause (i) *of Theorem 20.2 holds for $\varkappa \geq \varkappa_0$ with the capped integral.*

Proof. The two lines are King's Theorem 3.1 [13, (3.3)–(3.4)], that is Bałaban's lower bound [1, §§3–4] and upper bound [2, §§2–3] with the renormalisation of [3], re-run for the shell model, at block factor 5 (Theorem F.2) and with the remainder of the upper line measured against $Z_\varkappa(0,0)$. The proof has four layers, each uniform in $\varkappa \in [0,\infty]$.

(1) Exact algebra. The block-spin transformation is a Gaussian identity valid for every positive matter covariance: with $\varkappa\mathbf{1}_U$ a non-negative diagonal operator, every object of [13, (2.13)–(2.17),

(3.14)] is defined, the composition law of the steps, the scale decomposition of the fluctuation covariance into slices, the minimiser identity for the background $\phi_\varkappa^{(j)} = a_j (M_j^\varkappa)^{-1} Q_j(A)^* \phi_j$ and the Schur-complement monotonicity in $\varkappa$ hold verbatim, and every scalar line of the flow is a leg of block data (the slice covariances are $\ell\, C^{(j),\varkappa} \ell^\dagger$ with legs ℓ). A two-line positivity argument inserts the near-shell test at every scale at no cost.

(2) Operator bounds. King's Theorem 3.3 [13, (3.5)–(3.8)] for the shell model in the forms consumed: exponential decay, sup and first-difference bounds for the step operators G_j, $\Delta^{(j)}$, $C^{(j)}$ everywhere, uniformly in $\varkappa$ (coercivity, since $\varkappa \mathbf{1}_U \geq 0$ adds to the quadratic form; ℓ^2 decay by conjugation, which $\varkappa \mathbf{1}_U$ commutes with; Loewner monotonicity for the unit-lattice kernels; Kato's inequality with the free Laplacian for the fine sup bounds); the derivative clause for the background of small-field data; Hölder bounds on the light side; and on the heavy side the amplitude bounds of a barrier argument (a constant supersolution of mass $\sqrt{w_j}$ in block units), with the first-difference bounds at the heavy points paired with the amplitude of the other leg of the pair. The pointwise difference bounds of the line kernels, $|D_A G(x,y)| \leq C|x-y|_\eta^{-2} e^{-c|x-y|}$ and, with one difference at each end of one line, $|D_A^x D_A^y G(x,y)| \leq C|x-y|_\eta^{-3} e^{-c|x-y|}$, uniformly in $\varkappa$ on both sides of ∂U, follow from the scale decomposition of (1), one difference per variable carried by its own leg, the leg bounds and a geometric sum over the scales. On the Neumann sub-domains of Bałaban's splitting the first-difference and Hölder clauses are used at points at distance at least Bałaban's localisation radius from the complement, where they hold; the convexity of $\partial U = \partial \mathsf{H}'$ enters here, since at a re-entrant edge the first-difference bound fails uniformly in the spacing.

(3) Perturbative bounds. Bałaban's renormalisation bound [3, Prop. 1, (1.33)] for the renormalised families of [13, §3.5] of order at most 13 with at least one external field, with the shell model's propagators and legs, in Bałaban's product form with a $\varkappa$-adapted norm on the scalar external fields, and the restricted form of King's bound [13, (3.38)] on the field-dependent families; the Taylor step at a divergent two-scalar-leg subgraph acts on the photon legs, whose norms do not see the shell mass, with the remainder controlled by the current two-point kernel bound $|\Pi_j^\varkappa(x,y)| \leq C|x-y|^{-4}$ from the difference bounds of (2); the two-photon-leg and mixed classes are combined by the Ward–Takahashi identities, which hold for the shell model since $\varkappa \mathbf{1}_U$ is gauge invariant, used for full step-j families and never for a fixed ordering of the lines; a mismatch vertex M (the difference of the shell-adapted and the bulk subtraction inside a renormalised pair) is a two-leg vertex of bounded coefficient. The block-data form of the bound, with King's legs $\phi_\varkappa^{(r)}$ and the small-field data $|\phi_r| \leq p$, $|\partial A_r| \leq p$, $|D_{A_r} \phi_r| \leq p$ of [13, (3.2)], is derived from the product form on the photon legs by a local gauge transformation that removes the constant part of the background before the field itself is bounded, King's data satisfying $|A_r| \leq p\,(\mu_0 \ell_{r-1})^{-1}$ but no uniform bound; the same reduction gives the light-side first moment of the photon two-leg kernel, summed against the background and its first differences, which the near-face analysis of [14, §5.2] transplanted to the shell model consumes. [proof of the bound, uniform in the shell mass, on the first-order term in the background field of the determinant of the step fluctuation Gaussian on the Neumann sub-domains of Bałaban's splitting, where the large-field analysis of the upper bound uses it: being completed]

(4) The two bounds. The lower bound is [1, §§3–4]: restriction to the small-field region by Jensen's inequality, the inductive hypothesis on King's Appendix class [13, (A.1)–(A.8)] for the shell model (with a background-gradient and a block-mismatch clause, which compose from step to step), the cumulant bound for the step Gaussian with $\varkappa$-free constants (Theorem 17.112), and the final step [1, (3.66)–(3.69)]; the restriction to the capped set at the stopping scale is a positivity step inside the class, which gives the second line for $\varkappa \geq \varkappa_0$. The upper bound is [2, §§2–3]: the large/small-field split at every scale, with the near-shell test paid at every scale and the first-layer test at the stopping scale, the sum over large-field configurations $\prod_j \sum_\Lambda e^{-p(\ell_j)|\Lambda|} \leq e^{C|T|}$, the propagator-domain comparisons of [2, Prop. 2.6] in the form Lemma 5.1 with the regrouping Lemma 5.2, and the continuation of the large-field terms past the stopping scale, which under block alignment runs the shell model at the scales below η_0; the remainder is $\mathfrak{b}\, Z_\varkappa(0,0)$ in normalisation-free form by the ultraviolet-stability gap, as in (2). The upper capped line

follows from $I_\varkappa \le I_\varkappa^{\mathrm{cap}}$ (the cap tests dominate τ_r, λ_r pointwise as indicators and the integrand is positive); the lower capped line is the positivity step inside the class just described, the capped test set lying in King's Appendix class for $\varkappa \ge \varkappa_0$. □

Proposition 20.4 (the $\varkappa$-Cauchy property of the reduced action)**.** *In the setting of Proposition 20.3 fix $\alpha_0 \in (0, \frac{1}{2}]$. There is $\eta_2 \le \eta_1$ such that for every $\eta_0 \in 5^{-\mathbb{N}}$ with $\eta_0 \le \eta_2$ there is $C_{\mathfrak{c}}(\eta_0) < \infty$, depending on $(T, \mathsf{H}', n, \lambda, e, m, \mu_0, \alpha_0, c_0, \eta_0)$ and on f but not on ε, $\varkappa$, z, such that for all $\varepsilon \le \varepsilon_0(\eta_0)$, all $\varkappa_0 \le \varkappa \le \varkappa' < \infty$ and all $z \in [-1, 1]$,*

$$\sup_{b,b' \in \tilde{\chi}^{\mathrm{cap}}} \left|\Delta_z(b) - \Delta_0(b')\right| \;\le\; 2R(\varkappa, \varkappa') \;\le\; 2\, r(\varkappa; \eta_0), \qquad R(\varkappa, \varkappa') := \int_{\varkappa\eta_0^2}^{\varkappa'\eta_0^2} \theta(w)\, dw + \tau(\varkappa),$$

$$\theta(w) := C_{\mathfrak{c}}\, w^{-1} w^{-(1-\alpha_0)/2} (1+\log w)^2, \qquad \tau(\varkappa) := C_{\mathfrak{c}}\, (\varkappa\eta_0^2)^{-\frac{1}{2}+\frac{\alpha_0}{4}}, \qquad r(K; \eta_0) := C'_{\mathfrak{c}}\, (K\eta_0^2)^{-(1-\alpha_0)/2} \big(1+\log(K$$

with $\Delta_z(b) = \tilde{S}_{\varkappa'}^{\mathrm{cap}}(b; z) - \tilde{S}_{\varkappa}^{\mathrm{cap}}(b; z)$ as in Theorem 20.2; $r(K; \eta_0) \to 0$ as $K \to \infty$ at fixed η_0. This is clause (ii) *of Theorem 20.2.*

Proof. The statement is derived from a bound on King's full explicit action in all block fields: there is a field-free function $\varkappa \mapsto c_\varkappa$ on $[\varkappa_0, \infty)$, independent of z, with

$$\sup_{(A_r, \phi_r) \in \mathrm{supp}(\chi_r^{\flat} \tau_r^{\mathrm{cap}} \lambda_r^{\mathrm{cap}})} \left|S^{(r),\varkappa} - S^{(r),\varkappa'} - (c_\varkappa - c_{\varkappa'})\right| \;\le\; R(\varkappa, \varkappa'),$$

and on the capped set, whose U-section is the same cube $\{|\phi_U| \le C_7 p_r\}$ for every exterior datum b and every $\varkappa$, the reduced actions inherit the bound with the same $c_\varkappa$ (the logarithm of an integral of e^{-S} over a fixed set moves by at most the sup of the change of S); taking the mixed difference removes $c_\varkappa$. The full-field bound is proved in derivative form: $\partial_\varkappa S^{(r),\varkappa}$ is King's sum of renormalised graphs with one insertion of $\partial_\varkappa$ of the matter operator, that is of $\mathbf{2}\mathbf{1}_U$ on a line, plus the $\varkappa$-derivative of the block-free part of the background determinant. For an insertion on a line with a scalar connection to the rest of the graph the heavy matter on U pays $w_j^{-1/2}$ per deep end at every frozen scale (the barrier bounds of layer (2) of the proof of Proposition 20.3, with the one-loop z^2-term and its positions in U treated in both regimes); for an insertion whose matter component through it is closed and connects only through photon lines, which do not see the shell mass, the members with the same full lines are grouped into one class, one gauge identity is applied to the class, and only then is it expanded into King's members with the background rebased for each, the base layer being a first-moment bound, uniform in $\varkappa$, for the two-photon kernel built from full lines. In the units of the stopping scale, with $\hat{\varkappa} = \varkappa\eta_0^2$, the sum over the members less its field-free part is at most $\theta(\hat{\varkappa})$ per unit of $\hat{\varkappa}$, integrable at infinity; the Hölder remainders of the Taylor step for lines used as legs and the lattice regime of the scale sum give the exponent $(1-\alpha_0)/2$. The block-free part of the determinant has the monotone tail $0 \le \Phi^{\mathrm{bf}}(\varkappa) - \Phi^{\mathrm{bf}}(\infty) \le \tau(\varkappa)$, with the bracket for the loops winding through U (whose contribution is positive even at zero background curvature) bounded, under block alignment, by $C\hat{\varkappa}^{-1/2}$. Integrating in $\varkappa$ gives $R(\varkappa, \varkappa')$, and the sum over the dyadic intervals of $\hat{\varkappa}$ gives $r(K; \eta_0)$. The perturbative inputs are those of layer (3) of the proof of Proposition 20.3, the restricted form of [13, (3.38)] being re-run for the shell vertex set; the block-data form is consumed at the step marked there. □

Proof of Theorem 20.2. Take η_1 of the theorem to be the η_2 of Proposition 20.4, $\mathfrak{a}, \mathfrak{b}$ and $\varepsilon_0(\eta_0)$ as in Proposition 20.3, $\varkappa_0 = \varkappa_0$, $r(\cdot; \eta_0)$ as in Proposition 20.4, and $(\tilde{\chi}, db) = (\tilde{\chi}^{\mathrm{cap}}, db)$ with $\tilde{S}_\varkappa = \tilde{S}_\varkappa^{\mathrm{cap}}$. Clause (i) is the second and third lines of Proposition 20.3 for $\varkappa \ge \varkappa_0$; clause (ii) is Proposition 20.4; the test set does not depend on $\varkappa$ on $[\varkappa_0, \infty)$, as stated. □

In words. King's block-spin flow [13, §3] is run for the model in which $\varkappa\mathbf{1}_U$ is part of the matter covariance at every scale, down to a stopping scale η_0 at which the plateau is massive in block units; $\varkappa_0(\eta_0)$, of the order of η_0^{-2}, is the mass above which the shell is heavy in block units at that scale. $I_\varkappa^{\mathrm{cap}}(z)$ is the stopping-scale integral of the flow over a capped set $\tilde{\chi}$ of block fields, the *test set*, and $\tilde{S}_\varkappa(b; z)$ is the reduced small-field action at the block field b, shell mass $\varkappa$ and source coefficient z, both for the model at the spacing ε_k: the flow from ε_k to η_0 depends

on ε_k, and so do the test set and the reduced action, whereas the constants $\mathfrak{a}, \mathfrak{b}, \varkappa_0, \varepsilon_0$ and the function r do not. Clause (i) is the two-sided bound of King's and Bałaban's type (small-field integral below, small-field integral plus a large-field remainder above) for the model with the shell mass inside the covariance, uniformly in $\varkappa \geq \varkappa_0$ and in the spacing; clause (ii) says that the dependence of the reduced action on the shell mass is, up to a quantity that tends to 0 as the shell gets heavier, the same at every block field and every source coefficient. That the test set is one and the same for all $\varkappa \geq \varkappa_0$ is where the threshold enters: no term from a change of representation of the stopping-scale integral arises between $\varkappa$ and $\varkappa'$, and (ii) compares two integrands on one set. The proof is Propositions 20.3 and 20.4; Remark 20.6 lists its inputs beyond the printed literature.

Theorem 20.5 (the heavy-shell estimate from the shell bounds)**.** *Let $g = 0$ and let $(\mathfrak{s}, \chi)$ be a two-level configuration with the exterior shell $U = T \setminus \mathsf{H}'$, plateau height $n \geq n_*$ and H', T satisfying the block-alignment and grid-offset clauses of Theorem 20.2, and let f be a real 2-form in C_c^∞ supported in Q_b. Then for every $\eta > 0$ there are a stopping scale η_0 and a shell mass $K < \infty$ with*

$$\limsup_{k \to \infty} \mathrm{tail}_k(K) \leq \eta;$$

in particular (RES_0) *holds for $(f, 0)$ and η at every such configuration with $\tau(\mathfrak{s}) \leq \eta$, with this K. Consequently the closing estimate* (TIGHT_0) *holds (Theorem 19.10, with Theorem 17.1 and proposition 20.1), with the single-level configuration $(J', \psi, n + K)$ in the class of Remark 18.6.*

Proof. Fix $(\mathfrak{s}, \chi)$, the source f and $g = 0$. For $\varkappa \in [0, \infty]$ and $z \in \mathbb{C}$ put $\Psi_\varkappa(z) := Z_\varkappa(zf, 0)/Z_\varkappa(0, 0) = \langle e^{zF_{\Lambda_k}(f)} \rangle_\varkappa$, where for $\varkappa = \infty$ the state is the walled torus state with the matter killed on $D_\chi = U$. The shell mass $\varkappa \mathbf{1}_U$ is the second level $\varkappa\chi$ of the wall, so by Definition 19.3, at $g = 0$, $\Psi_K(\mathrm{i}) = \Psi_k(n, K)$, $\Psi_\infty(\mathrm{i}) = \Psi_k(n, \infty)$ and $\mathrm{tail}_k(K) = \Psi_K(\mathrm{i}) - \Psi_\infty(\mathrm{i})$. For $\varkappa \geq \varkappa_0$ and $z \in [-1, 1]$ put $\mathcal{J}_\varkappa(z) := I_\varkappa^{\mathrm{cap}}(z)/I_\varkappa^{\mathrm{cap}}(0)$, with the test set and the reduced action of (HS) at the spacing ε_k under consideration. Every constant below is uniform in k with $\varepsilon_k \leq \varepsilon_0(\eta_0)$.

Step 0: the disc bound. At fixed spacing $z \mapsto \Psi_\varkappa(z)$ is entire, a finite-dimensional integral of $e^{zF(f)}$ against a measure with Gaussian photon tails. For $|z| \leq 2$, $|e^{zF(f)}| \leq e^{2F(f)} + e^{-2F(f)}$. Each state $\langle \cdot \rangle_\varkappa$, $\varkappa \in [0, \infty]$, is a BFS measure whose Gaussian is the periodic massive photon Gaussian of the torus, with covariance C; the wall, the shell mass and the killing act in the single-site measures only. So Lemma 2.8 with $s = \pm 2$ and v the vector with $\langle v, A \rangle = F_{\Lambda_k}(f)$ gives $\langle e^{\pm 2F(f)} \rangle_\varkappa \leq e^{2\langle v, Cv \rangle}$, hence

$$|\Psi_\varkappa(z)| \leq M_\circ := 2e^{2\langle v, Cv \rangle} \qquad (|z| \leq 2,\ \varkappa \in [0, \infty]).$$

Here $\langle v, Cv \rangle$, the variance of $F(f)$ under the free massive photon on the torus, is bounded uniformly in the spacing for smooth f: the periodic photon form is at least μ_0^2 times the ℓ^2 norm, so $\langle v, Cv \rangle \leq \mu_0^{-2} \|d_k^* f\|_{2,k}^2$, a Riemann sum for $\mu_0^{-2} \|d^* f\|_{L^2}^2$ (as in Corollary 17.7). At $g = 0$ no matter moment enters the disc bound. For real z, $\Psi_\varkappa(z) \to \Psi_\infty(z)$ as $\varkappa \to \infty$ at fixed spacing, by the substitution and dominated convergence of the proof of Lemma 2.9 applied with the set U, the rate $\chi \equiv 1$ on U and $\Theta = e^{zF(f)}$, which is of the form admitted there ($|\Theta| = e^{\ell(A)}$ with $\ell = zF(f)$ real linear), the factor $e^{zF(f)}$ being integrable against the photon Gaussian by the same bound.

Step 1: the sandwich. Let $\eta_0 \leq \eta_1$ be so small that $\mathfrak{b}(\eta_0) \leq \frac{1}{2}$, fix k with $\varepsilon_k \leq \varepsilon_0(\eta_0)$, and let $\varkappa \geq \varkappa_0$, $z \in [-1, 1]$; write $\mathfrak{a} = \mathfrak{a}(\eta_0)$, $\mathfrak{b} = \mathfrak{b}(\eta_0)$. By (HS)(i) at $z = 0$, $e^{-\mathfrak{a}}(1 - \mathfrak{b}) Z_\varkappa(0, 0) \leq I_\varkappa^{\mathrm{cap}}(0) \leq e^{\mathfrak{a}} Z_\varkappa(0, 0)$; by (HS)(i) at z and these two bounds,

$$e^{-2\mathfrak{a}}(1 - \mathfrak{b})\, \mathcal{J}_\varkappa(z) \ \leq\ \Psi_\varkappa(z) \ \leq\ e^{2\mathfrak{a}} \mathcal{J}_\varkappa(z) + \mathfrak{b}.$$

Hence $\mathcal{J}_\varkappa(z) \leq 2e^{2\mathfrak{a}} M_\circ =: \bar{\mathcal{J}}$ and $|\Psi_\varkappa(z) - \mathcal{J}_\varkappa(z)| \leq \bar{\mathcal{J}}(e^{2\mathfrak{a}} - e^{-2\mathfrak{a}} + \mathfrak{b}) + \mathfrak{b} =: m_1(\eta_0)$, and $m_1(\eta_0) \to 0$ as $\eta_0 \to 0$ because $\mathfrak{a}, \mathfrak{b} \to 0$.

Step 2: the $\varkappa$-Cauchy property. Let $\varkappa_0 \le \varkappa \le \varkappa' < \infty$ and $z \in [-1,1]$. Since at this k the test set $\tilde\chi$ is the same for $\varkappa$ and $\varkappa'$,

$$\frac{\mathcal{J}_\varkappa(z)}{\mathcal{J}_{\varkappa'}(z)} = \frac{\iint_{\tilde\chi\times\tilde\chi} e^{-\tilde S_\varkappa(b;z)-\tilde S_{\varkappa'}(b';0)}\,db\,db'}{\iint_{\tilde\chi\times\tilde\chi} e^{-\tilde S_{\varkappa'}(b;z)-\tilde S_\varkappa(b';0)}\,db\,db'},$$

and the two integrands differ pointwise by the factor $e^{\Delta_z(b)-\Delta_0(b')}$, which lies in $[e^{-2r}, e^{2r}]$ with $r := r(\varkappa;\eta_0)$ by (HS)(ii). So $\mathcal{J}_\varkappa(z)/\mathcal{J}_{\varkappa'}(z) \in [e^{-2r}, e^{2r}]$ and $|\mathcal{J}_\varkappa(z) - \mathcal{J}_{\varkappa'}(z)| \le \bar{\mathcal{J}}(e^{2r}-1)$. Put $\bar r(K;\eta_0) := \sup_{\varkappa\ge K} r(\varkappa;\eta_0)$, which tends to 0 as $K\to\infty$ at fixed η_0. With Step 1, for all $\varkappa' \ge \varkappa \ge K \ge \varkappa_0$,

$$|\Psi_\varkappa(z) - \Psi_{\varkappa'}(z)| \;\le\; m(K,\eta_0) := 2m_1(\eta_0) + \bar{\mathcal{J}}\big(e^{2\bar r(K;\eta_0)}-1\big), \qquad z\in[-1,1],$$

uniformly in k with $\varepsilon_k \le \varepsilon_0(\eta_0)$.

Step 3. Letting $\varkappa'\to\infty$ at fixed spacing (Step 0): $|\Psi_K(z) - \Psi_\infty(z)| \le m(K,\eta_0)$ for $z\in[-1,1]$ and $K \ge \varkappa_0$.

Step 4: two constants. $\Xi := \Psi_K - \Psi_\infty$ is entire, $|\Xi| \le 2M_\circ$ on $\{|z|\le 2\}$ and $|\Xi| \le m(K,\eta_0)$ on $[-1,1]$. If $m(K,\eta_0) = 0$, then Ξ vanishes on $[-1,1]$ and so identically. Otherwise $\log|\Xi|$ is subharmonic on the domain $\Omega := \{|z|<2\}\setminus[-1,1]$, bounded above by $\log(2M_\circ)$ there, with upper limit at most $\log m(K,\eta_0)$ at every point of the slit; by the two-constants theorem [19, §4.3] it is dominated on Ω by the harmonic function with boundary values $\log(2M_\circ)$ on $|z|=2$ and $\log m(K,\eta_0)$ on the slit. At the point $\mathrm{i}\in\Omega$,

$$|\Xi(\mathrm{i})| \le (2M_\circ)^{1-\omega}\, m(K,\eta_0)^\omega,$$

where $\omega\in(0,1)$ is the harmonic measure of the slit $[-1,1]$ at i in Ω, an absolute constant.

Step 5: the real point. The charge conjugation $(A,\vartheta)\mapsto(-A,-\vartheta)$ preserves the action, the wall, the shell mass and the measure (Definition 15.6), and reverses the sign of $F(f)$; so $\Psi_\varkappa(-z) = \Psi_\varkappa(z)$, and since $\Psi_\varkappa(\bar z) = \overline{\Psi_\varkappa(z)}$, the value $\Psi_\varkappa(\mathrm{i})$ is real. Thus $\Xi(\mathrm{i}) = \mathrm{tail}_k(K) \ge 0$ (Lemma 19.4(a)) and

$$0 \le \mathrm{tail}_k(K) \le (2M_\circ)^{1-\omega}\, m(K,\eta_0)^\omega \qquad \text{for every } \varepsilon_k \le \varepsilon_0(\eta_0).$$

Choice of the parameters. Given $\eta>0$: take $\delta>0$ with $(2M_\circ)^{1-\omega}\delta^\omega \le \eta$; then $\eta_0 \le \eta_1$ with $\mathfrak{b}(\eta_0) \le \frac12$ and $2m_1(\eta_0) \le \delta/2$, which is possible because $\mathfrak{a},\mathfrak{b}\to 0$; then $K \ge \varkappa_0(\eta_0)$ with $\bar{\mathcal{J}}(e^{2\bar r(K;\eta_0)}-1) \le \delta/2$, which is possible because $r(K;\eta_0)\to 0$ as $K\to\infty$ at fixed η_0; and only then $\varepsilon_k \le \varepsilon_0(\eta_0)$. Then $m(K,\eta_0)\le\delta$ and $\mathrm{tail}_k(K)\le\eta$ for all $\varepsilon_k \le \varepsilon_0(\eta_0)$, so $\limsup_k \mathrm{tail}_k(K) \le \eta$. This is the bracket of (RES_0) at $(\mathfrak{s},\chi)$ with this K, in the quantifier order of Definition 19.8: one configuration, one K, no rate, the limit along the spacings last; since $\tau(\mathfrak{s})$ depends only on H and the width $D(\mathfrak{s})$, a configuration with $\tau(\mathfrak{s})\le\eta$, with b' in the residue class of the grid-offset condition and T a King torus, so that H' and T satisfy the alignment clauses of Theorem 20.2 (every interval of length M' final blocks contains an admissible b'), is available for every b, f and η, and (RES_0) holds. Since tail_k is nonincreasing in K (Lemma 19.4(a)), K may be enlarged at will (Remark 19.11(iii)), which places $(J',\psi,n+K)$ in the class of Remark 18.6. The proof of Theorem 19.10 uses the dead-shell bound (45) at this configuration, which is (DS) (Proposition 20.1), and the existence of the walled value of $(J',\psi,n+K)$, which is part of (K_0) (Theorem 17.1), together with King's printed state $((\mathrm{E}_C))$; it then gives (TIGHT_0). □

Proof of Theorem 15.1. Theorem 20.5 gives, from the heavy-shell estimates (HS) (Theorem 20.2), the decoupling (RES_0) at one two-level configuration for each b, f and η, and its final clause, Theorem 19.10 with the torus comparison (K_0) (Theorem 17.1) and the dead-shell chain (DS) (Proposition 20.1), gives the closing estimate (TIGHT_0). With (E_C) (Theorem 3.2), (K_0) and (TIGHT_0), Theorem 18.12 gives the invariance of $Z^K_\infty(\cdot,0)$ under $O(3)$ for all complex coefficients, and Corollary 8.3 adds the translations. □

Remark 20.6 (the inputs of the proof of (HS)). The two clauses of (HS) are proved by a renormalisation-group analysis of King's walled torus model [13, 14] with the shell mass kept inside the matter covariance at every scale, run down to a stopping scale at which the plateau is massive in block units. Its ingredients beyond the printed literature, by content: Bałaban's

asserted block-factor-5 estimates re-run for the shell model; a frozen-scale sub-domain bound in scale-free form; the sub-domain form of the perturbative bound with a tangentially localised amplitude; pointwise first- and mixed second-difference bounds on the line kernels, on the torus and on the Neumann sub-domains used, uniformly in the shell mass and on both sides of ∂U; the light-side regularity estimates on sub-domains and a localisation lemma for the local inverses near the edges and corners of the source box; the tail of the block-free determinant factor with its bracket for the loops winding through U; heavy-kernel bounds for the matter operator with the shell mass; the $\varkappa$-derivative bound for clusters closed in matter, through a gauge identity for full lines; the restricted renormalisation theorem for the shell vertex set; a first-moment bound, uniform in $\varkappa$, for the two-photon kernel built from full lines; King's near-face analysis [14, §5.2] transplanted to the shell model; Bałaban's all-orders enumeration [3, §§2–3] with one inserted vertex; King's Appendix class [13, (A.1)–(A.8)] for the shell model. The counterterm offset between the torus and the free-boundary convention enters as a bounded constant. Of the printed literature it uses, besides the theorems as printed, the statements named in Remark 17.10 together with [2, Lemma 2.7, (2.113)]; the shell analysis shares with the torus comparison and with the existence theorem (section 5) Bałaban's same-scale estimates at block factor 5, their reading at a scalar wall with unwalled photon lines, and the orbit regrouping of the cut graph families. Several forms in which such shell estimates might be posed are false and are not used: a sub-domain clause with an absolute constant and a rate in the penetration depth, which fails when the heaviness threshold of the shell in block units is small (the scale-free form is used); the determinant tail without the winding bracket and with a right side proportional to the square of the background curvature, which fails on the torus (a constant background has zero curvature while the winding loops contribute a positive tail); a Yukawa form of the heavy-pair clause, which fails for pairs in the first layer; and a supremum bound for the full covariance next to ∂U in the lattice regime. Theorem 20.5 uses the determinant tail only through a bound on it that tends to 0 as $\varkappa \to \infty$, so a change of its exponent or a logarithmic factor leaves the theorem unchanged. A photon Ward identity for each fixed ordering of King's graphs would not supply the $\varkappa$-derivative bound, since that identity is false; the identities that hold are those for full lines and for Bałaban's resummed organisation, which keeps a convergent remainder, and in its place the bound rests on the first-moment bound for the two-photon kernel. (DS) (Proposition 20.1) carries the three steps of Theorem 19.6, whose proof is written for a shell between two aligned cubes, to the geometry with one convex face.

20.2. **The continuum form.** This subsection is not on the chain of Theorem 15.1: it records an equivalent form of $(\mathrm{RES})_a$, which is not used, and the existence statement (E_∞) introduced here is used by nothing. Both sides of the bracket in $(\mathrm{RES})_a$ have continuum limits, the first by $(\mathrm{K})_a$ and the second under one further existence statement; in those terms the property says that two continuum objects coincide up to η. Throughout, $(\mathfrak{s}, \chi)$ is a two-level configuration for Q_b in the sense of Definition 19.2 (smooth profile χ, shell $U = C_L \setminus \mathsf{H}$), and $(f, g) \in \mathcal{K}$ is supported in Q_b.

Proposition 20.7 (the dead-shell state between two cube states)**.** *Let $B_k := e^{-a_k \mathsf{S}_{\Lambda_k}(g)} \langle \Theta(\mathrm{i}f, g) \rangle$ in the Neumann–Dirichlet state on $C_L \cap \Lambda_k$ (Neumann photon on C_L, Dirichlet matter outside C_L) whose single-site measures are those of the dead-shell walled torus: the wall $n\psi$, and the matter killed on $D_\chi \cap C_L$. Then at every k with ε_k small*

$$B_k \;\le\; \Psi_k(n, \infty) \;\le\; B_k + Z^a_{k,L}(0, 2g)^{1/2} \big(e^{\tau(\mathfrak{s})} - 1\big)^{1/2}, \qquad B_k \le Z^a_{k,L}(\mathrm{i}f, g).$$

Proof. The lower bound is Lemma 15.8 applied to the dead-shell torus state and $X = C_L \cap \Lambda_k$ (X and its neighbours lie in one fundamental cell, as $N_1(\bar{C}_L) \subset (-25MJ', 25MJ')^3$): the only couplings between X and its complement are photon gradient terms, since every site of C_L adjacent to the complement is dead, and deleting them lowers the expectation and factorises the measure, whose X-factor is the state defining B_k. The upper bound is Steps 1–2 of the proof of Theorem 19.6, since the N-measure there is exactly the state defining B_k, and $e^{-2a_k \mathsf{S}_{\Lambda_k}(g)} \langle \Theta(0, 2g) \rangle_N \le Z^a_{k,L}(0, 2g)$ by its Step 3. The last inequality is that Step 3. □

Theorem 20.8 (the continuum form)**.** *Assume* $(\mathrm{K})_a$ *if* $g \neq 0$ *(at* $g = 0$ *the torus comparison is Theorem 17.1); for the sandwich in* (d) *at* $g \neq 0$ *assume also* $(\mathrm{W})_a$*.*

(a) *For every* $K \in [0,\infty)$ *the limit* $\Psi^0(n,K) := \lim_k \Psi_k(n,K)$ *exists, equals the walled value* $\mathcal{W}^a_{(J',n\psi+K\chi)}(f,g)$*, and is nonincreasing in* K*; so* $\Psi^0(n,\infty^-) := \lim_{K\to\infty} \Psi^0(n,K)$ *exists.*

(b) $\liminf_k \mathrm{tail}_k(K) = \Psi^0(n,K) - \limsup_k \Psi_k(n,\infty)$ *for every* K*, and* $\limsup_k \Psi_k(n,\infty) \le \Psi^0(n,\infty^-)$*.*

(c) $(\mathrm{RES})_a$ *holds at* $(\mathfrak{s},\chi)$ *with tolerance* η *for some* K *if and only if*

$$\Psi^0(n,\infty^-) = \lim_{K\to\infty} \lim_{k\to\infty} \Psi_k(n,K) \;\le\; \limsup_{k\to\infty} \Psi_k(n,\infty) + \eta.$$

(d) *If the* dead-shell torus value $\Psi^0(n,\infty) := \lim_k \Psi_k(n,\infty)$ *exists, then* (c) *reads* $\lim_{K\to\infty} \Psi^0(n,K) \le \Psi^0(n,\infty) + \eta$*, the reverse inequality* $\lim_K \Psi^0(n,K) \ge \Psi^0(n,\infty)$ *always holds, and*

$$B^0 \le \Psi^0(n,\infty) \le B^0 + Z^a_\infty(0,2g)^{1/2}\big(e^{\tau(\mathfrak{s})} - 1\big)^{1/2}$$

for every limit point B^0 *of* (B_k)*; in particular* $\Psi^0(n,\infty) \le Z^a_\infty(\mathrm{i}f,g) + Z^a_\infty(0,2g)^{1/2}(e^{\tau(\mathfrak{s})} - 1)^{1/2}$*. At* $g = 0$ *the sandwich reads* $B^0 \le \Psi^0(n,\infty) \le B^0 + (e^{\tau(\mathfrak{s})} - 1)^{1/2}$ *and* $\Psi^0(n,\infty) \le Z^K_\infty(\mathrm{i}f,0) + (e^{\tau(\mathfrak{s})} - 1)^{1/2}$*.*

Proof. (a) $(J', n\psi + K\chi)$ is a walled configuration in $\mathcal{A}(Q_b)$ (its wall is smooth and vanishes on $N_1(Q_b)$), so Lemma 17.3(a) gives the limit, under (K_0) at $g = 0$; monotonicity in K passes to the limit (Lemma 19.4(a)). (b) $\liminf_k(x_k - y_k) = \lim_k x_k - \limsup_k y_k$ when x_k converges; and $\Psi_k(n,\infty) \le \Psi_k(n,K)$ for every k and K. (c) By (b), $(\mathrm{RES})_a$ for some K says $\Psi^0(n,K) \le \limsup_k \Psi_k(n,\infty) + \eta$ for some K, which by monotonicity in K is the same as $\inf_K \Psi^0(n,K) \le \limsup_k \Psi_k(n,\infty) + \eta$. (d) The first two statements are (b), (c) with the limit in place of the lim sup; the sandwich is Proposition 20.7 in the limit along a subsequence on which $B_k \to B^0$. At $g \neq 0$, $Z^a_{k,L} \to Z^a_L \le Z^a_\infty$ on $\mathcal{K}$ (Theorem 16.6(i),(iii), under $(\mathrm{W})_a$), with $(0,2g) \in \mathcal{K}$. At $g = 0$, $Z^a_{k,L}(0,0) = 1$, and $B^0 \le \limsup_k Z_{k,L}(\mathrm{i}f) \le Z^K_{L+1}(\mathrm{i}f) \le Z^K_\infty(\mathrm{i}f,0)$ by Corollary 4.4(ii)(γ) and the monotonicity of $J \mapsto Z^K_J(\mathrm{i}f)$ (Theorem 7.6(b)); the support condition of that corollary holds since $\operatorname{supp} f \subset Q_b \subset \mathsf{H}$ and $\mathrm{dist}_\infty(\mathsf{H}, \mathbb{R}^3 \setminus C_L) \ge 4$. □

Definition 20.9 ((RES^0): the continuum form)**.** (RES^0) is the property: for every $b > 0$, $(f,g) \in \mathcal{K}$ supported in Q_b and $\eta > 0$ there is a two-level configuration $(\mathfrak{s},\chi)$ for Q_b with $\tau(\mathfrak{s}) \le \eta$ for which the dead-shell torus value $\Psi^0(n,\infty)$ exists and

$$\lim_{K\to\infty} \Psi^0(n,K) \;\le\; \Psi^0(n,\infty) + \eta.$$

By Theorem 20.8, (RES^0) implies $(\mathrm{RES})_a$, and $(\mathrm{RES})_a$ implies (RES^0) whenever the dead-shell torus values of the configurations in question exist; the two are the same statement about the interchange of the limits $K \to \infty$ and $k \to \infty$, and in (RES^0) no lattice spacing appears. The existence of $\Psi^0(n,K)$ is covered by $(\mathrm{K})_a$ because the two-level wall $n\psi + K\chi$ is a smooth source; with the sharp profile $\mathbf{1}_U$ it would not be. The existence of $\Psi^0(n,\infty)$, the continuum limit of a torus state with an interior region of Dirichlet matter, is the statement (E_∞). It would be a theorem of King's type; King does not print it: the Dirichlet region is bounded by lattice hyperplanes only when $\partial\mathsf{H}$ is, and it would have to come from the methods of [14] for Dirichlet matter on a cube, applied to a shell between two aligned cubes; the statement is not in [13, 14]. Neither (E_∞) nor (RES^0) is consumed by any result of this paper, and the lattice form $(\mathrm{RES})_a$ needs neither of them nor any limit beyond $(\mathrm{K})_a$.

20.3. **The fixed-spacing statement and the shape of a counterexample.**

Proposition 20.10 (fixed spacing)**.** *For every two-level configuration, every* $(f,g) \in \mathcal{K}$ *supported in* Q_b *and every* k*,* $\mathrm{tail}_k(K) \to 0$ *as* $K \to \infty$*. Consequently, if* $(\mathrm{RES})_a$ *fails for some* (f,g)*,* η*, then for every two-level configuration* $(\mathfrak{s},\chi)$ *for* Q_b *with* $\tau(\mathfrak{s}) \le \eta$ *and every* K*,* $\mathrm{tail}_k(K) > \eta$ *for all large* k*, while* $\mathrm{tail}_k(K) \le \eta$ *for* $K \ge K_*(k) < \infty$*; hence* $K_*(k) \to \infty$ *as* $k \to \infty$*.*

Proof. The first statement is Lemma 19.4(a). The second is its contrapositive form: the failure of $(\mathrm{RES})_a$ at $(\mathfrak{s}, \chi)$ means $\liminf_k \mathrm{tail}_k(K) > \eta$ for every K, so $\mathrm{tail}_k(K) > \eta$ for all large k; and $K_*(k) := \inf\{K : \mathrm{tail}_k(K) \le \eta\}$ is finite by the first statement. If $K_*(k) \le K_0$ along a subsequence, then $\mathrm{tail}_k(K_0) \le \eta$ along it, contradicting the previous sentence. □

So any failure of $(\mathrm{RES})_a$ would be carried by shell matter whose mass squared tends to infinity with k: a lattice-scale effect that disappears at every fixed cutoff.

20.4. **Why the wall, and why no soft argument.**

Proposition 20.11 (no common lattice hyperplanes)**.** *Let $R = R_{\mu\nu}(\pm\theta_0) \in \mathcal{R}_0$ and let ρ be the third index.*

(a) *There is no lattice $P \subset \mathbb{R}^3$ of rank 3 with $RP = P$.*
(b) *An affine plane is a lattice hyperplane of both Λ_k and $R^{-1}\Lambda_k$ (that is, of the form $\{x_\alpha = c\}$ and of the form $\{(R^{-1}e_\beta)\cdot x = c'\}$) only if it is of the form $\{x_\rho = c\}$.*
(c) *Consequently no bounded region of $\mathbb{R}^3$ has all its faces on lattice hyperplanes of both lattices; the only unbounded regions with that property are the slabs $\{c_1 \le x_\rho \le c_2\}$. The bounded geometries to which both lattices descend are the tori $\mathbb{R}^3/P$ with $P \subset \Lambda_k \cap R^{-1}\Lambda_k$, for instance $P = 50MJ\,\mathbb{Z}^3$; by* (a)*, no such P is R-invariant, so the rotated torus $R^{-1}D_J$ is never the torus D_J.*

Proof. (a) If $RP = P$, then in a $\mathbb{Z}$-basis of P the map R is an integer matrix, whose characteristic polynomial has integer coefficients; but the characteristic polynomial of R is $(x-1)(x^2 - \frac{8}{5}x + 1)$. (b) The normals are e_α and $R^{-1}e_\beta$; $e_\alpha = \pm R^{-1}e_\beta$ means $Re_\alpha = \pm e_\beta$, and for $\alpha \in \{\mu, \nu\}$ the vector Re_α has the components $\pm\frac{3}{5}, \pm\frac{4}{5}$ in the (μ,ν)-plane. (c) A bounded region with faces on hyperplanes needs face normals spanning $\mathbb{R}^3$; by (b) only e_ρ is available. Both lattices descend to $\mathbb{R}^3/P$ exactly when $P \subset \Lambda_k \cap R^{-1}\Lambda_k$, which contains $50MJ\,\mathbb{Z}^3$ by Lemma 15.3(b). □

The ultraviolet analysis of [13, 14] needs boundaries that are lattice hyperplanes of the lattice being analysed ([14, p. 326]: "These hold only when the boundary is a lattice hyperplane orthogonal to a lattice coordinate direction. Clearly the two lattices in Fig. 1 never share such a boundary. Our solution of this problem is to use periodic boundary conditions."). Proposition 20.11 makes this precise: the objects on which the two lattices can be compared by King's method are tori with smooth sources, and smooth walls are the only admissible way to localise a torus; a Dirichlet region cannot be aligned with both lattices. The correlation inequalities give exactly "cube $\le$ walled torus" on the cone and never the reverse (Lemma 15.8; Remark 18.8), so the comparison "walled torus $\le$ King's state $+\,\eta$" must be supplied by an estimate on King's measure; in the curvature sector that estimate is (HS), through Theorem 20.5.

Remark 20.12 (the content of $(\mathrm{RES})_a$). (i) At every fixed spacing $(\mathrm{RES})_a$ holds (Proposition 20.10); its content is uniformity in the spacing. (ii) It is a decoupling statement for a heavy, weakly coupled region: on the shell the dimensionless couplings are of order $\lambda/\sqrt{K}$ and $e^2/\sqrt{K}$, and the plateau mass n is free. The difficulty is not the shell but the comparison of two states that differ on the shell, seen through the interacting plateau and the light core, uniformly in the spacing. (iii) No correlation inequality bounds the tail from above: Lemma 2.7 gives signs and monotonicity, and the one comparison through a relative density that is uniform in ε is the Gaussian photon switch across a matter-free collar (Lemma 19.1); for the matter, the couplings across the boundary of the shell contribute to the logarithm of the relative density a term whose variance is of order $|\partial U|\varepsilon^{-2}$. (iv) In the curvature sector the estimate is obtained from (HS) in Theorem 20.5. The proof of (HS) is a renormalisation-group expansion of the walled model of [13, 14] with the shell mass kept inside the matter covariance at every scale, run down to a stopping scale at which the plateau is massive in block units, the dependence on the shell mass being controlled through the $\varkappa$-Cauchy property of the reduced small-field action on a test set that does not depend on $\varkappa$. No such expansion for the symmetric massive phase of the three-dimensional model is in the printed literature; Bałaban, Imbrie and Jaffe [5] develop a multiscale expansion of the high-frequency modes of the abelian Higgs model in

the Higgs regime and postpone the large-field and decoupling estimates. (HS) carries a window $n \geq n_*$ for the plateau height, which costs nothing for the theorem since n is free. (v) At $e = 0$ the photon is independent of the matter, and (RES_0) holds trivially ($\mathrm{tail}_k \equiv 0$ at $g = 0$), while $(\mathrm{RES})_a$ for $g > 0$ is a decoupling statement for the two-component φ^4_3 lattice model with a heavy shell. Nothing established at $e = 0$ has been carried to $e > 0$ in this problem, since King's mass counterterm at order e^2 is negative and of order e^2/ε, so that every comparison of the $e > 0$ model with an $e = 0$ model at the bare level is a comparison with a model deep in its broken phase.

Remark 20.13 (the chain of the rotation part). Every implication in the chain

$$(\mathrm{HS}) \xrightarrow{\text{Theorem 20.5}} (\mathrm{RES}_0) \xrightarrow{(\mathrm{E}_C),\ (\mathrm{K}_0),\ (\mathrm{DS});\ \text{Theorem 19.10}} (\mathrm{TIGHT}_0),$$

$$(\mathrm{K}_0) + (\mathrm{TIGHT}_0) \xrightarrow{\text{Theorem 18.12}} \text{invariance under } \mathcal{R}_0 \xrightarrow{\text{Theorem 18.12}} O(3) \xrightarrow{\text{Corollary 8.3}} \mathbb{R}^3 \rtimes O(3)$$

is proved in this part, with (E_C) (Theorem 3.2) wherever King's printed state is named (Lemma 18.2, Theorem 18.12); Theorem 15.1 is the composition. Its two ends, the heavy-shell estimates (HS) (Theorem 20.2) and the torus comparison (K_0) (Theorem 17.1), are renormalisation-group statements about King's walled torus model, proved in Propositions 20.3 and 20.4 and in section 17.3; (DS) (Proposition 20.1) is the dead-shell chain in the exterior-shell geometry.

The main theorem

21. The main theorem for King's printed state

We state the main theorem in full and prove it by assembling Sections 4–12. The object is King's printed state $Z^K_\infty(f, 0) = \lim_J \lim_k \langle e^{F(f)} \rangle^K_{k,J}$ on King's RG cube (definition 2.2), block factor 5, at a fixed photon mass $\mu_0 > 0$ and all positive couplings e, λ, m (lemma 22.1). Both sectors are proved on King's convention ND^K, on which the existence theorems theorems 3.2 and 3.5 are posed. In the Wick sector (Leg W), proved in its tube form, Form (B), the tube statements theorems 10.7 and 13.10 and lemma 13.11 are written on the comparison convention $\mathrm{ND}^{(a)}$ (definition 2.1) as templates; the two of them that the interlacing of lemma 4.2 does not transfer, the bound (31) and the fixed-cube limits on the tube for the K-family functionals $Z^{a,K}_{k,J}$, are proved directly on King's convention in theorem 21.3, after which the Wick sector follows on ND^K (proposition 21.4); the comparison family enters through its LIM states, which coincide with King's tube object (proposition 21.5). The torus inputs are proposition 3.8 and lemma 3.9, and the coupling condition is (T3) (Hypothesis 3.11), the coupling region of which contains the explicit regions of proposition 22.15; $\mathcal{K} = \{(\mathrm{i}f, g) : g \geq 0\}$ is the cone. Theorem 1.1 is the theorem below.

Theorem 21.1 (the axioms for King's printed state at fixed photon mass). *Let $e > 0$, $\lambda, m, \mu_0 > 0$, $\varepsilon_k = 5^{-k}$, $C_J = [-MJ, MJ)^3$ with King's block integer $M \geq 1$, and let $\mathrm{ND}^K_{k,J}$ be King's cube state.*

A. The object. *For every finite family $f = (f_1, \dots, f_N)$ of real 2-forms in $C^\infty_c(\mathbb{R}^3; \Lambda^2)$ the limit*

$$Z^K_\infty\Big(\sum_j z_j f_j, 0\Big) = \lim_{J\to\infty} \lim_{k\to\infty} \langle e^{\sum_j z_j F_{\Lambda_k}(f_j)} \rangle^K_{k,J}$$

exists locally uniformly in $z \in \mathbb{C}^N$, is entire, depends on the family only through the total source, is real and ≥ 1 at real z, lies in $[0, 1]$ at purely imaginary z, and satisfies $|Z^K_\infty(\sum_j z_j f_j, 0)| \leq e^{\|g(z)\|^2_2}$, $g(z) = \sum_j (\mathrm{Re}\, z_j) f_j$, with no dependence on μ_0, e, λ, m or M.

B. Leg C (the curvature field). *The Schwinger functions $S^K_n(f_1, \dots, f_n) = \partial_{z_1} \cdots \partial_{z_n} Z^K_\infty(\sum_j z_j f_j, 0)|_{z=0}$ satisfy:*

(i) OS0. *$|Z^K_\infty(\sum_j z_j f_j, 0)| \leq e^{\|g(z)\|^2_2}$ for all $z \in \mathbb{C}^N$, and also $\leq \exp(\frac{1}{2}\mu_0^{-2} \|\sum_j (\mathrm{Re}\, z_j) d^* f_j\|^2_2)$; the same two bounds, with the lattice norms $\|\cdot\|_{2,k}$ and d^*_k, hold at every (k, J) for the*

lattice functionals $\langle e^{\sum_j z_j F_{\Lambda_k}(f_j)}\rangle^K_{k,J}$. *For real test functions*

$$|S^K_n(f_1,\dots,f_n)| \le (\sqrt{2}\,e)^n (n!)^{1/2} \prod_{i=1}^n \|f_i\|_2 \le (e\pi/2)^n (n!)^{1/2} \prod_{i=1}^n |f_i|_4;$$

each S^K_n *extends to a continuous symmetric multilinear form on* $\mathcal{S}(\mathbb{R}^3;\Lambda^2_{\mathbb{C}})^n$, *a tempered distribution, the bound carrying the factor* 2^n *for complex test functions; so* E0″ *holds with* $s=4$, $\sigma_n = (e\pi)^n (n!)^{1/2}$ *and every* $\beta > \frac{1}{2}$.

(ii) OS1. *The* S^K_n *are invariant under all translations of* $\mathbb{R}^3$ *and under the cubic group* O_h *about every point,* F *transforming as a 2-form. Moreover* Z^K_∞ *is invariant under the Euclidean group* $\mathbb{R}^3 \rtimes O(3)$ *and the* S^K_n *are Euclidean covariant.*

(iii) OS2. *Reflection positivity about every coordinate plane* $\{x_\nu = c\}$, $\nu \in \{1,2,3\}$, $c \in \mathbb{R}$: *for finite families* $\zeta_1,\dots,\zeta_N$ *of real 2-form sources supported in* $\{x_\nu > c\}$ *with arbitrary complex coefficients,* $\sum_{i,j} \bar\gamma_i \gamma_j Z^K_\infty(\theta\zeta_i^* \sqcup \zeta_j, 0) \ge 0$ *for all* $\gamma \in \mathbb{C}^N$ *(the matrix form for exponentials), and* $S(\theta(P)P) \ge 0$ *for every polynomial* P *in the smeared curvature with sources supported in* $\{x_\nu > c\}$ *(the polynomial form). The same holds about every plane* $\{x\cdot n = c\}$, n *a unit vector.*

(iv) OS3. S^K_n *is real for real test functions, symmetric, and zero for odd* n.

(v) The measure. *For every even* $r \ge 14$ *there is exactly one Borel probability measure* $\mu^{K,H}$ *on* $\mathcal{H}^{-r}(\mathbb{R}^3;\Lambda^2)$ *whose moments on* C^∞_c *are the* S^K_n; *its characteristic functional is* $Z^K_\infty(\mathrm{i}\,\cdot, 0)$, *extended by continuity to* $\mathcal{S}$; *its image* μ^K_S *on* $H^{-r}_{\mathrm{loc}}(\mathbb{R}^3;\Lambda^2)$ *is the unique Borel probability measure there with this characteristic functional, and the unique one with these moments. For every real* $\chi \in C^\infty_c(\mathbb{R}^3)$ *and every box* Q *containing* $\operatorname{supp}\chi$ *in its interior, the laws of the localised lattice curvature fields* $\chi\Phi^K_{k,J}$ *under* $\mathrm{ND}^K_{k,J}$ *on* $H^{-r}_{\mathrm{per}}(Q;\Lambda^2)$ *converge weakly, along the full sequence* $k\to\infty$ *at every fixed* $J \ge J_0(\chi)$ *and then as* $J\to\infty$, *to the law of* $\chi\Phi$ *under* μ^K_S. *Nothing is asserted about the laws of the unlocalised lattice fields at fixed* J.

(vi) Weak OS4. μ^K_S *is* $\mathbb{R}^3 \rtimes O_h$*-invariant and is an integral* $\int \mu_\xi \, d\nu(\xi)$ *of translation-ergodic, reflection-positive,* θ_j*-invariant components which cluster in the mean and under diagonal separation, have a unique joint vacuum in each coordinate direction, and satisfy the factorial growth bound* $|S^\xi_N(u_1,\dots,u_N)| \le \sigma^\xi_N \prod_i \|u_i\|_{r+2}$, $\sigma^\xi_N \le \alpha_\beta(\xi)(N!)^\beta$ *for every* $\beta > 1$, *with component constants (no tube bound); there is an* O_h*-invariant decomposition into mixtures of* $k \in \{1,2,3,6\}$ *ergodic measures; and* ν*-almost every component satisfies* OS4 *in the weak form (ergodicity under* $\mathbb{R}^3$*). Moreover* ν*-almost every component is* $\mathbb{R}^3 \rtimes O(3)$*-invariant, reflection positive about every plane, and satisfies* OS4 *in each of its forms for every time direction, with* $\ker H_n = \mathbb{C}\Omega_n$ *in every direction* n; *its moments are real and symmetric. The vanishing of a component's odd moments is not claimed.*

C. Leg W, Form (B), on King's convention. *Let the couplings lie in the coupling region of Hypothesis 3.11, the set on which* (T3) *holds, which contains the explicit regions of proposition 22.15, and fix an admissible scheme* a *(definition 3.6;* $z = a^\sharp$ *the reference scheme). Let* $\sigma_+ = s_1$, $H = s_1/16$, $\kappa = C_U/s_1$, K_- *be the constants of* (30) *(*K_- *built from* B_{cell}*; for* $a \ne z$ *with* C_U *replaced by* $C_U + s_1 \sup_k |z_k - a_k|$*),* $\sigma_0 = H/4$ *and* $K' = \kappa\pi^2 + K_-$. *Then for every finite family* $\mathcal{F} = (f_1,\dots,f_n;h_1,\dots,h_p)$ *of real 2-forms and real scalars the tube object*

$$G^K_\infty(\zeta) = \lim_{J\to\infty}\lim_{k\to\infty} Z^{a,K}_{k,J}\Big(\sum_i z_i f_i, \sum_j w_j h_j\Big), \qquad \zeta = (z,w) \in \mathcal{D}_{\sigma_+,H} = \{\|k_+(\zeta)\|_\infty < \sigma_+/2,\ \|k_-(\zeta)\|_\infty < H/2$$

exists locally uniformly on the tube and is holomorphic there; on the Wick tube of definition 10.1 the limit $Z^{a,K}_{\infty,T} = \lim_J \lim_k Z^{a,K}_{k,J}$ *exists and equals* G^K_∞ *where both are defined. The joint Schwinger functions* $S_{n,p}(f_1,\dots,f_n;h_1,\dots,h_p) = \partial^{(n)}_z \partial^{(p)}_w G^K_\infty(0)$ *of the pair* (F, W^a) *satisfy:*

(i) OS0. $|G^K_\infty(\zeta)| \le \exp(\|g(\zeta)\|_2^2 + \kappa\|k_+(\zeta)\|_{L^1} + K_-|k_-(\zeta)|_w)$ *on* $\mathcal{D}_{\sigma_+,H}$, *and the same bound with lattice norms holds for* $Z^{a,K}_{k,J}$ *uniformly in* $k \ge 0$ *and* $J \ge J_1(R)$ *when the supports lie*

in Q_R; for real test functions

$$|S_{n,p}(f_1,\dots,f_n;h_1,\dots,h_p)| \le e^{K'\sigma_0}(\sqrt{2}\,e)^n (n!)^{1/2}(e/\sigma_0)^p\, p! \prod_{i=1}^{n}\|f_i\|_2 \prod_{j=1}^{p}|h_j|_w,$$

with the factor 2^{n+p} for complex test functions; the $S_{n,p}$ are tempered distributions and E0″ *holds with $s=4$ and every $\beta>1$. The constants σ_+, H, κ, K_- depend on (e,λ,m,μ_0,M,a), on the fixed auxiliary choices M_1, χ_0 and the amplitude $s_1\in(0,\sigma_1)$ of* (I2)*, and on nothing else; they are not uniform in μ_0.*

(ii) OS1. *$G^K_\infty(\tau_b R\cdot\zeta)=G^K_\infty(\zeta)$ on $\mathcal{D}_{\sigma_+,H}$ for every $b\in\mathbb{R}^3$ and $R\in O_h$, F transforming as a 2-form and W^a as a scalar (the transformed family has the same tube); hence the $S_{n,p}$ are invariant under translations and under O_h about every point. No Euclidean clause is asserted.*

(iii) OS2. *For finite families $\zeta_1,\dots,\zeta_N$ of real sources supported in $\{x_\nu>c\}$ with complex coefficients of amplitude $\mathsf{a}(\zeta_i)<\sigma_W:=\sigma_+/8$ (in particular all purely imaginary Wick coefficients with arbitrary complex curvature coefficients), $\sum_{i,j}\bar\gamma_i\gamma_j Z^{a,K}_{\infty,T}(\theta\zeta_i^*\sqcup\zeta_j)\ge 0$ about every coordinate plane; and the polynomial form $S(\theta(P)P)\ge 0$ for polynomials in the smeared F and W^a with sources in $\{x_\nu>c\}$.*

(iv) OS3. *The $S_{n,p}$ are symmetric multilinear in the f's and in the h's, real for real test functions, and zero for odd n.*

(v),(vi) The measure and weak OS4. *Clauses (v) and (vi) of Leg C hold for the joint field on $\mathcal{H}^{-r}(\mathbb{R}^3;\Lambda^2\oplus\mathbb{R})$ and $H^{-r}_{\mathrm{loc}}(\mathbb{R}^3;\Lambda^2\oplus\mathbb{R})$, every even $r\ge 14$, with moments $S_{n,p}$, characteristic functional $(u,v)\mapsto G^K_\infty(\mathrm{i}u,\mathrm{i}v)$ on real test functions (purely imaginary coefficients have $k(\zeta)=0$ and lie in the tube), the component growth bound with the constants of (i), and the local convergence of the laws of the scheme-shifted lattice fields $\chi(F_{\Lambda_k},W^a_{\Lambda_k})$ under $\mathrm{ND}^K_{k,J}$ along the full sequence $k\to\infty$ at every fixed $J\ge\max(J_0(\chi),J_1(R_\chi))$, then $J\to\infty$. No Euclidean clause is asserted. Two admissible schemes a,a' give objects differing by the shift $W\mapsto W-\kappa_{a'a}\int(\cdot)$ of the Wick component, $\kappa_{a'a}=\lim_k(a'_k-a_k)$; joint cumulants of order ≥ 2 are scheme-free.*

No bound at positive scalar sources beyond the cone $\sup q\le s_1$ of corollary 13.3 is used.

D. The comparison family. *Under the conditions of C, every increasing sequence of spacings has a continuum subsequence $\mathcal{K}$ for the Wick sector of the comparison family $\mathrm{ND}^{(a)}$ (definition 14.1), and for every such $\mathcal{K}$ and every family the LIM state $Z^{a,\mathcal{K}}_\infty=\lim_J\lim_{k\in\mathcal{K}}Z^a_{k,J}$ exists on $\mathcal{D}_{\sigma_+,H}$ and equals G^K_∞ (proposition 21.5). Hence clauses (i)–(vi) of C hold for the Wick-sector LIM states of the comparison family, which coincide with King's tube object and do not depend on $\mathcal{K}$; in (v) the laws of the localised lattice fields $\chi(F_{\Lambda_k},W^a_{\Lambda_k})$ under $\mathrm{ND}^{(a)}_{k,J}$ converge along $\mathcal{K}$ at every fixed J with $MJ>R(\chi)+\ell_c$, then as $J\to\infty$, to the law of the joint field under the measure of C(v).*

Proof. *A.* Theorem 7.6(b): by theorem 3.2 the two limits exist locally uniformly on $\mathbb{C}^N$ and are entire, $|Z^K_J|,|Z^K_\infty|\le e^{\|g(z)\|_2^2}$, $J\mapsto Z^K_J(\mathrm{i}t)$ is nondecreasing with values in $[0,1]$, and $Z^K_\infty(x)\ge 1$ at real x; realness at real z is the symmetry $(A,\vartheta)\mapsto(-A,-\vartheta)$ of every $\mathrm{ND}^K_{k,J}$, closed under limits. Each lattice functional depends on the family only through its total source, hence so do the limits. The bound is absolute (remark 7.5).

B. (i) is theorem 7.6: the finite-volume bound (a) with $\min\{\|g\|^2_{2,k},\frac12\mu_0^{-2}\|d^*_k g\|^2_{2,k}\}$ at every (k,J); the two limit bounds (b), the second being (b)'s second bound, carried from the lattice by $\|d^*_k g\|_{2,k}\to\|d^*g\|_2$, (1); the Schwinger bounds, the extension to $\mathcal{S}$, the factor 2^n and E0″ (c). (ii): translations are corollary 8.3 (the lattice sandwich lemma 8.2, the two limits, the identity theorem, density); the cubic group about every point is theorem 8.5; the Euclidean clause is theorem 8.7, that is theorem 18.12 (the rotation theorem theorem 3.13) with corollary 8.3. (iii): the matrix form about every coordinate plane and the polynomial form are lemma 10.8 with proposition 10.10; a plane $\{x\cdot n=c\}$ is the image of a coordinate plane under a Euclidean motion γ, and by (ii) $Z^K_\infty(\theta_\gamma\zeta_i^*\sqcup\zeta_j,0)=Z^K_\infty(\gamma^{-1}\cdot(\theta_\gamma\zeta_i^*\sqcup\zeta_j),0)$ is a Gram entry about the

coordinate plane for the sources $\gamma^{-1}\cdot\zeta_i$, supported in the corresponding coordinate half-space. (iv) is proposition 10.9 for $Z^K_\infty(\cdot,0)$ (through lemma 10.8; charge conjugation gives the vanishing for odd n). (v) is theorem 11.1(a), including the local convergence of the lattice laws along the full sequence, with remark 11.2 for the unlocalised fields; (vi) is theorem 11.1(b) and (c), whose $O(3)$ input is (ii)'s Euclidean clause; the disclaimer on odd moments is that of theorem 11.1(c).

C. Under (T3), proposition 12.6 gives (I2) with amplitude $s_1\in(0,\sigma_1)$ and constant C_U from the torus inputs proposition 3.8 and lemma 3.9; for $a\neq z$ the difference z_k-a_k converges by admissibility, so $\sup_k|z_k-a_k|<\infty$ and C_U is replaced as stated (lemma 14.5); (I3) is proposition 12.2, the correlation inequalities are lemma 2.7, and the cell constant $B^{(a)}_{\mathrm{cell}}$ is finite (lemma 14.4). *(i)*: theorem 21.3 gives the bound (46) for every $k\geq 0$ and $J\geq J_1(R)$, the fixed-cube limits G^K_J and the cube limit G^K_∞ locally uniformly on $\mathcal{D}_{\sigma_+,H}$, their holomorphy, the bound in the limit, the Schwinger bounds, the factor 2^{n+p} and E0″, with the stated dependence of the constants and no uniformity in μ_0. On the Wick tube $T_{\mathcal{F}'}$ of the enlarged family of definition 10.1, whose scalar members are nonnegative and whose points have $2\|k_1\|_\infty<\sigma_+/2$, the fixed-cube limits of $Z^{a,K}_{k,J}$ exist locally uniformly by theorem 6.17(i), whose domain $\{\sup_x 2k_1<s_W\}$, $s_W=s_1$, contains $T_{\mathcal{F}'}$, for the reference scheme and hence for a (definition 3.6); their cube limit $Z^{a,K}_{\infty,T}$ is proposition 10.5(a) run on King's convention (proposition 21.4(c)), and it equals G^K_∞ on $T_{\mathcal{F}'}\cap\mathcal{D}_{\sigma_+,H}$ as the same J-limit of the same fixed-cube limits. *(ii)*: proposition 21.4(b). *(iii)*: proposition 21.4(c), that is theorem 10.7 with proposition 10.5 run on King's convention, and the polynomial form proposition 10.10. *(iv)*: proposition 10.9 for $Z^{a,K}_{\infty,T}$ and G^K_∞ (proposition 21.4(c)). *(v),(vi)*: proposition 21.4(d). The Gram points of (v)'s characteristic functional have purely imaginary coefficients and lie in the tube, and the reflection positivity consumed there is (iii) at amplitude 0 (remark 21.6). *The scheme clause*: for admissible a,a', $Z^{a',K}_{k,J}(\zeta)=e^{-(a'_k-a_k)\sum_j w_j \mathsf{S}_{\Lambda_k}(h_j)}Z^{a,K}_{k,J}(\zeta)$ (definition 2.3), where $a'_k-a_k=(z_k-a_k)-(z_k-a'_k)\to\kappa_{a'a}$ and $\mathsf{S}_{\Lambda_k}(h_j)\to\int h_j$; the factor converges locally uniformly in ζ, so the fixed-cube limits and then the cube limits satisfy $G^{K,a'}_\infty(\zeta)=e^{-\kappa_{a'a}\sum_j w_j\int h_j}G^{K,a}_\infty(\zeta)$ on the tube, the shift $W\mapsto W-\kappa_{a'a}\int(\cdot)$ of the Wick component; a factor $e^{\ell(w)}$ with ℓ linear changes only the first cumulants in w, so the joint cumulants of order ≥ 2 are scheme-free.

D. Proposition 21.5; the clauses of C then hold for $Z^{a,\mathcal{K}}_\infty=G^K_\infty$, and the convergence of the laws of the localised comparison-cube fields along $\mathcal{K}$ is clause (v) of the Wick sector of theorem 14.9, whose hypotheses ($\mathrm{B}_{\mathrm{cell}}$) and (SCH) hold for every admissible scheme under (T3) (lemmas 14.4 and 14.5). □

Remark 21.2 (the Euclidean clause). The Euclidean clauses of B(ii), (iii), (vi) rest on the rotation theorem theorem 3.13, proved in the rotation part (theorem 15.1): the walled tori on the two lattices have the same continuum limits (theorem 17.1), King's printed state is the lower envelope of the walled tori (theorem 18.4), and that closing estimate comes from the heavy-shell estimates on King's block-spin flow (theorem 20.2); the Wick sector carries no Euclidean clause (theorem 23.7).

Theorem 21.3 (the Wick tube on King's convention)**.** *Let the couplings satisfy* (T3) *and let a be an admissible scheme. Let $\sigma_+=s_1$, H, κ be the constants of* (30) *and let K_- be the constant there, computed from the cell constant $B^{(a)}_{\mathrm{cell}}$ of the comparison convention (lemma 14.4), which is finite. Then for every finite family of real sources with supports in Q_R, every $k\geq 0$, every $J\geq J_1(R)$ and every $\zeta\in\mathcal{D}_{\sigma_+,H}$,*

$$|Z^{a,K}_{k,J}(\zeta)|\;\leq\;Z^{a,K}_{k,J}(\operatorname{Re}\zeta)\;\leq\;\exp\big(\|g(\zeta)\|^2_{2,k}+\kappa\|k_+(\zeta)\|_{1,k}+K_-|k_-(\zeta)|_w\big),\tag{46}$$

the bound (31) *for the K-family functionals with the constants of the comparison convention; and the fixed-cube limits $G^K_J(\zeta)=\lim_{k\to\infty}Z^{a,K}_{k,J}(\zeta)$ exist for every $\zeta\in\mathcal{D}_{\sigma_+,H}$, locally uniformly, so that theorem 13.10(i),(ii) hold on ND^K: G^K_J and $G^K_\infty=\lim_J G^K_J$ exist locally uniformly on $\mathcal{D}_{\sigma_+,H}$ and are holomorphic, the bound* (46) *passes to the limits, and the Schwinger bounds and* E0″ *of theorem 21.1C(i) hold for G^K_∞; moreover $G^K_\infty>0$ at the real points of $\mathcal{D}_{\sigma_+,H}$ and $G^K_J\leq G^K_\infty$ at its cone points.*

Proof. Throughout, $\mathcal{F} = (f_1, \dots, f_n; h_1, \dots, h_p)$ is a family of real sources with supports in Q_R, $J \geq J_1(R)$, so that $MJ \geq R + \ell_c \geq R + 10$: every source lies at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_J$, the support condition (S) holds for X_J, and $J \geq J_0(\mathcal{F})$. For $\zeta \in \mathcal{D}_{\sigma_+,H}$ write $g = g(\zeta)$ and $k = k(\zeta) = k_+ - k_-$ (definition 2.5); by lemma 2.6(E3) the sampled sources satisfy $0 \leq 2k_+ \leq s_1$ and $0 \leq 2k_- \leq H$ on Λ_k. The inputs are: (I2) with amplitude $s_1 \in (0, \sigma_1)$ and constant C_U, from (T3) and the torus inputs proposition 3.8 and lemma 3.9 through proposition 12.6, for $a \neq z$ with the constant $C_U + s_1 \sup_k |z_k - a_k|$, finite by definition 3.6 (lemma 14.5); (I3), which is proposition 12.2; (I4), which is lemma 2.7; and the cell constant $B^{(a)}_{\mathrm{cell}} < \infty$ of the comparison convention (lemma 14.4). Under these, corollary 13.3 and proposition 13.9 hold on the comparison convention and lemma 4.7(d) on King's.

Step 1 (modulus). Lemma 2.11(d) holds for every state of definition 2.1, killed sites included: $|Z^{a,K}_{k,J}(\zeta)| \leq Z^{a,K}_{k,J}(\operatorname{Re}\zeta)$. It therefore suffices to bound $Z^{a,K}_{k,J}$ at the real point $\operatorname{Re}\zeta$, where $Z^{a,K}_{k,J}(\operatorname{Re}\zeta) = N^{a,K}(g,k) := e^{-a_k \mathsf{S}(k)} \langle e^{F(g) + :\phi^2:(k)} \rangle^K_{k,J}$.

Step 2 (curvature off). Let $\langle\cdot\rangle_{K,k}$ be King's cube with each single-site measure at an alive site multiplied by $e^{\varepsilon^3 k(x) r_x^2}$; it is a BFS measure (the factor keeps $\int e^{\alpha r^2} d\rho_x < \infty$ because $\lambda > 0$, and the killed sites are unchanged), and $N^{a,K}(g,k) = N^{a,K}(0,k)\, \langle e^{F(g)} \rangle_{K,k}$. Its Gaussian is the Neumann photon Gaussian of X_J, the same for both conventions, with covariance C as in lemma 7.1. By (S) every plaquette with $g \neq 0$ has its corners in X_J, so $F(g) = \langle v, A \rangle_k$ with $v = d_k^* g$, and lemma 2.8 (with $s = 1$) and lemma 7.1 give $\langle e^{F(g)} \rangle_{K,k} \leq e^{\langle v, Cv \rangle_k / 2} \leq e^{\|g\|^2_{2,k}}$. This is the chain of theorem 7.6(a) in the presence of the real scalar source k. Hence $N^{a,K}(g,k) \leq e^{\|g\|^2_{2,k}} N^{a,K}(0,k)$.

Step 3 (the split). Lemma 2.11(f) at $f = 0$ with $k_1 = k_+$, $k_2 = k_-$: $N^{a,K}(0, k_+ - k_-) \leq N^{a,K}(0, 2k_+)^{1/2} N^{a,K}(0, -2k_-)^{1/2}$.

Step 4 (the positive leg on King's convention). Put $q := 2k_+$, a lattice function with $0 \leq q \leq s_1$ supported in $Q_R \cap \Lambda_k \subset \mathrm{alive}^K_J$. The observable $e^{\sum_x \varepsilon^3 q(x) r_x^2}$ is $\Theta_{0,q'}$ with $q' = \varepsilon^3 q \geq 0$, of class $\mathcal{I}$, and (S) holds for X_J; Step 1 of the proof of lemma 4.2 (the kill step on $\mathcal{I}$) gives $\langle \Theta_{0,q'} \rangle^K_{k,J} \leq \langle \Theta_{0,q'} \rangle^{(a)}_{k,J}$, and the multiplier $e^{-(a_k + 2\mathsf{W}_\varepsilon)\mathsf{S}(q)}$ is one positive constant for both states (lemma 2.11(a)). Hence

$$N^{a,K}_{k,C_J}(0,q) \;\leq\; N^{a}_{k,C_J}(0,q) \;\leq\; e^{\kappa \|q\|_{1,k}}$$

by corollary 13.3, whose hypotheses (I2), (I3) hold (this is the cone instance of remark 4.8, as in remark 13.12). So $\frac{1}{2} \ln N^{a,K}(0, 2k_+) \leq \kappa \|k_+\|_{1,k}$.

Step 5 (the negative leg on King's convention). Put $q := 2k_-$, a lattice function with $0 \leq q \leq H$ supported in Q_R, and $J \geq J_1(R)$. Lemma 4.7(d) gives $\ln N^{a,K}_{k,C_J}(0,-q) \leq K_- |q|_w$ with the printed constant K_- of (30), built from the cell constant $B^{(a)}_{\mathrm{cell}}$ of the comparison cell, which is finite by lemma 14.4: by lemma 4.7(c) every cell on which one of the cutoff sources $8\chi_s q$ does not vanish is a full cell inside C_J containing no killed site, so each factor of the wall insertion of lemma 4.7(a) that carries a source is a comparison-cell functional, to which lemmas 13.6 and 13.7 apply as printed, and the sum over cells is lemma 13.8; the scheme multiplier factorises over the cells (lemma 2.11(a)). The direction of the kill matters here and is the content of lemma 4.7(e). Hence $\frac{1}{2} \ln N^{a,K}(0, -2k_-) \leq \frac{1}{2} K_- |2k_-|_w = K_- |k_-|_w$.

Step 6 (assembly). Steps 1–5 give, for every $k \geq 0$, $J \geq J_1(R)$ and $\zeta \in \mathcal{D}_{\sigma_+,H}$,

$$|Z^{a,K}_{k,J}(\zeta)| \leq Z^{a,K}_{k,J}(\operatorname{Re}\zeta) \leq \exp\big(\|g(\zeta)\|^2_{2,k} + \kappa \|k_+(\zeta)\|_{1,k} + K_- |k_-(\zeta)|_w \big),$$

which is (46). The constants are those of (30), with $B^{(a)}_{\mathrm{cell}}$ in K_- and, for $a \neq z$, $C_U + s_1 \sup_k |z_k - a_k|$ in κ.

Step 7 (the fixed-cube limits). Each $Z^{a,K}_{k,J}$ is entire in $\zeta \in \mathbb{C}^{n+p}$: it is a finite-dimensional integral with Gaussian decay in A ($\mu_0 > 0$) and $e^{-\lambda \varepsilon^3 r^4}$ decay in r at every alive site, the killed sites carrying no variables, and differentiation under the integral sign is justified. On a compact $Q \subset \mathcal{D}_{\sigma_+,H}$ the family $\{Z^{a,K}_{k,J}\}_{k \geq 0}$ is uniformly bounded, by (46) and lemma 2.6(E1)–(E3): $\|g\|^2_{2,k} \leq C_0 |g|^2_4$, $\|k_+\|_{1,k} \leq C_0' |k_+|_w$ and $|k_\pm|_w \leq \sum_j |\operatorname{Re} w_j| |h_j|_w$, all continuous in

ζ. At every real point of $\mathcal{D}_{\sigma_+,H}$ the sequence converges: there $2\|k_+\|_\infty < \sigma_+ = s_1 = s_W$ and the negative part is unrestricted, so theorem 3.5, which is posed on King's convention (corollary 6.18(a)), gives the limit in $(0,\infty)$ for the reference scheme; for the admissible scheme a, $Z^{a,K}_{k,J} = e^{-(a_k-z_k)\sum_j w_j \mathsf{S}_k(h_j)} Z^{z,K}_{k,J}$ with $a_k - z_k$ convergent and $\mathsf{S}_k(h_j) \to \int h_j$ (definition 3.6, remark 6.19). The real points of $\mathcal{D}_{\sigma_+,H}$ form a non-empty open subset of $\mathbb{R}^{n+p}$ (definition 2.5), a set of uniqueness in the connected open $\mathcal{D}_{\sigma_+,H}$; by Montel's and Vitali's theorems (the fact (T3) stated before theorem 13.10),$Z^{a,K}_{k,J} \to G^K_J$ locally uniformly on $\mathcal{D}_{\sigma_+,H}$, G^K_J is holomorphic there, and (46) passes to the limit with $\|g\|_{2,k} \to \|g\|_2$, $\|k_+\|_{1,k} \to \|k_+\|_{L^1}$ (k_+ is continuous with compact support) and the sampled $|k_-|_w$ at most the continuum one (E3):

$$|G^K_J(\zeta)| \le \exp\big(\|g(\zeta)\|_2^2 + \kappa\|k_+(\zeta)\|_{L^1} + K_-|k_-(\zeta)|_w\big) \qquad (\zeta \in \mathcal{D}_{\sigma_+,H},\ J \ge J_1(R)), \tag{47}$$

a bound free of J. At real points $G^K_J > 0$ (theorem 3.5).

Step 8 (the cube limit on a sub-tube). Split $h_j = h_j^+ - h_j^-$ with $h_j^- := \|h_j\|_\infty \chi_j$, $\chi_j \in C_c^\infty$, $0 \le \chi_j \le 1$, $\chi_j = 1$ on $\operatorname{supp} h_j$, $\operatorname{supp}\chi_j \subset Q_{R+1/2}$, and $h_j^+ := h_j + h_j^- \ge 0$. For the split family $\mathcal{F}'' = (f; h^+, h^-)$ put $F^K_{k,J}(z, w^+, w^-) := Z^{a,K}_{k,J}(\sum_i z_i f_i, \sum_j (w_j^+ h_j^+ + w_j^- h_j^-))$ and $\mathcal{D}'' := \{(z, w^+, w^-) \in \mathbb{C}^{n+2p} : |\operatorname{Re} w_j^\pm| < \delta_j\}$ with $\sum_j \delta_j(\|h_j^+\|_\infty + \|h_j^-\|_\infty) < H/2$. On $\mathcal{D}''$ the scalar source k'' of the split family has $\|k''_\pm\|_\infty \le \|k''\|_\infty < H/2 \le \sigma_+/2$, so $\mathcal{D}''$ lies in the tube of $\mathcal{F}''$, whose supports lie in $Q_{R+1/2}$; Steps 1–7 for $\mathcal{F}''$ and $J \ge J_1(R+\frac12)$ give $F^K_{k,J} \to F^K_J$ locally uniformly on $\mathcal{D}''$ with the J-free bound (47) for $\mathcal{F}''$. On $\Omega''_+ := \mathcal{D}'' \cap ((\mathrm{i}\mathbb{R})^n \times (0,\infty)^{2p})$ the curvature coefficients are imaginary and the total scalar source $\sum_j (w_j^+ h_j^+ + w_j^- h_j^-)$ is nonnegative, so the observable is of class $\mathcal{I}$ and (S) holds for X_J ($MJ \ge R + \frac12 + \ell_c$); corollary 4.3 gives $F^K_{k,J} \le F^K_{k,J+1}$ at every k (the scheme multiplier is common, lemma 2.11(a)), and $k \to \infty$ gives $F^K_J \le F^K_{J+1}$ on Ω''_+. Being nondecreasing and bounded there, $J \mapsto F^K_J$ converges on Ω''_+, a non-empty open subset of the maximal totally real subspace $(\mathrm{i}\mathbb{R})^n \times \mathbb{R}^{2p}$; $\mathcal{D}''$ is convex, hence connected, and the Montel–Vitali fact (T3) stated before theorem 13.10 gives $F^K_J \to F^K_\infty$ locally uniformly on $\mathcal{D}''$. At $w_j^+ = w_j$, $w_j^- = -w_j$ the total scalar source is $\sum_j w_j(h_j^+ - h_j^-) = \sum_j w_j h_j$, so $F^K_{k,J}(z, w, -w) = Z^{a,K}_{k,J}(\zeta)$ exactly at every k; hence $G^K_J \to G^K_\infty$ locally uniformly on $\mathcal{D}' := \{\zeta : |\operatorname{Re} w_j| < \delta_j\}$, which lies in $\mathcal{D}_{\sigma_+,H}$ because there $\|k_\pm\|_\infty \le \|k\|_\infty < \sum_j \delta_j \|h_j\|_\infty < H/2$.

Step 9 (the whole tube). The family $\{G^K_J\}_{J \ge J_1(R+1/2)}$ is locally uniformly bounded on $\mathcal{D}_{\sigma_+,H}$ by (47); by Montel's theorem every subsequence has a further subsequence converging locally uniformly to a holomorphic function, any two such limits agree on the open set $\mathcal{D}'$ by Step 8, and $\mathcal{D}_{\sigma_+,H}$ is connected, so all subsequential limits coincide: $G^K_J \to G^K_\infty$ locally uniformly on $\mathcal{D}_{\sigma_+,H}$, G^K_∞ is holomorphic, and (47) passes to G^K_∞. The finitely many indices $J_1(R) \le J < J_1(R+\frac12)$ do not affect the limit.

Step 10 (positivity and the cone). For a real $\zeta \in \mathcal{D}_{\sigma_+,H}$ pick $\epsilon > 0$ with $-\epsilon\zeta \in \mathcal{D}_{\sigma_+,H}$. The function $\varphi_J(t) := \ln G^K_J(t\zeta)$ is finite and convex on $[-\epsilon, 1]$ (a pointwise limit of the convex functions $\ln Z^{a,K}_{k,J}(t\zeta)$, lemma 2.11(d), with positive limits at real points) and $\varphi_J(0) = 0$, so $\varphi_J(1) \ge -\varphi_J(-\epsilon)/\epsilon \ge -C_\epsilon/\epsilon$, C_ϵ the exponent of (47) at $-\epsilon\zeta$; hence $G^K_\infty(\zeta) \ge e^{-C_\epsilon/\epsilon} > 0$. At the cone points of $\mathcal{D}_{\sigma_+,H}$ ($z \in (\mathrm{i}\mathbb{R})^n$, $w \in \mathbb{R}^p$, $k(\zeta) \ge 0$), corollary 4.3 at every k and $k \to \infty$ give $G^K_J \le G^K_{J+1} \le G^K_\infty$.

Step 11 (the Schwinger bounds and E0″*).* Put $S^K_{n,p}(f; h) := \partial_z^{(n)} \partial_w^{(p)} G^K_\infty(0)$, $\sigma_0 = H/4$, $K' = \kappa\pi^2 + K_-$. Normalise $\|f_i\|_2 = 1$ and let $\rho_j := \sigma_0/(p|h_j|_w)$. On the polydisc $\{|w_j| \le \rho_j\} \times \mathbb{C}^n$ we have $\|k\|_\infty \le \sum_j |\operatorname{Re} w_j| \|h_j\|_\infty \le \sum_j \rho_j |h_j|_w = \sigma_0 < H/2 \le \sigma_+/2$ (lemma 2.6(E2)), so the polydisc lies in $\mathcal{D}_{\sigma_+,H}$, where G^K_∞ exists with the bound (47), whose exponent is at most $\|g(z)\|_2^2 + \kappa\pi^2|k_+|_w + K_-|k_-|_w \le \|g(z)\|_2^2 + K'|k|_w \le \|g(z)\|_2^2 + K'\sigma_0$ (E2). Cauchy's estimate for the mixed first derivative in the w's at $w = 0$, z fixed, gives $|\partial_w^{(p)} G^K_\infty(z,0)| \le \prod_j \rho_j^{-1} \sup|G^K_\infty| \le (p/\sigma_0)^p \prod_j |h_j|_w \, e^{K'\sigma_0} e^{\|g(z)\|_2^2}$, and $p^p \le e^p p!$ (E4). The function $z \mapsto \partial_w^{(p)} G^K_\infty(z,0)$ is entire ($\mathbb{C}^n \times \{w\} \subset \mathcal{D}_{\sigma_+,H}$ for $|\operatorname{Re} w_j|$ small) and bounded by $A\, e^{\|g(z)\|_2^2}$ with $A := e^{K'\sigma_0}(e/\sigma_0)^p p! \prod_j |h_j|_w$; on the polydisc of radius $r = (2n)^{-1/2}$ about 0, $\|g(z)\|_2 \le \sum_i |\operatorname{Re} z_i| \le nr$, so Cauchy's estimate

for the mixed first z-derivative gives $|S^K_{n,p}| \le A\, r^{-n} e^{n^2 r^2} = A\,(2n)^{n/2} e^{n/2} \le A\,(\sqrt{2}e)^n (n!)^{1/2}$ by (E4). Restoring the normalisation by multilinearity,

$$|S^K_{n,p}(f_1,\dots,f_n;h_1,\dots,h_p)| \le e^{K'\sigma_0}(\sqrt{2}e)^n (n!)^{1/2} (e/\sigma_0)^p\, p! \prod_i \|f_i\|_2 \prod_j |h_j|_w .$$

By (E2), $\|f_i\|_2 \le \frac{\pi}{2\sqrt{2}}|f_i|_4$ and $|h_j|_w \le |h_j|_4$, and $(n!)^{1/2} p! \le (n+p)!$, so the right side is at most $\sigma_{n+p} \prod_i |f_i|_4 \prod_j |h_j|_4$ with $\sigma_N = e^{K'\sigma_0} C_1^N N!$, $C_1 = 2\max(e\pi/2, e/\sigma_0)$; for complex test functions each argument is split into real and imaginary parts, which gives the factor $2^{n+p} \le 2^N$ absorbed in C_1. The $S^K_{n,p}$ are therefore continuous multilinear forms in the norms $|\cdot|_4$, extend to $\mathcal{S}(\mathbb{R}^3;\Lambda^2_{\mathbb{C}})^n \times \mathcal{S}(\mathbb{R}^3;\mathbb{C})^p$ and are tempered distributions, as in theorem 7.4(c), and $C_1^N/(N!)^{\beta-1} \to 0$ for every $\beta > 1$: this is E0″ with $s = 4$ and every $\beta > 1$, with the constants of theorem 21.1C(i). □

Proposition 21.4 (the Wick sector on King's convention)**.** *Let the couplings satisfy* (T3) *and let a be an admissible scheme. Then, with the constants of theorem 21.1C:*

(a) *$G^K_J := \lim_k Z^{a,K}_{k,J}$ exists locally uniformly on $\mathcal{D}_{\sigma_+,H}$ for every $J \ge J_1(R)$, $G^K_\infty := \lim_J G^K_J$ exists locally uniformly on $\mathcal{D}_{\sigma_+,H}$, both are holomorphic, $|G^K_\infty(\zeta)| \le \exp(\|g(\zeta)\|_2^2 + \kappa\|k_+(\zeta)\|_{L^1} + K_-|k_-(\zeta)|_w)$, and the Schwinger bounds and* E0″ *of theorem 21.1C(i) follow.*

(b) *$G^K_\infty(\tau_b R\cdot\zeta) = G^K_\infty(\zeta)$ on $\mathcal{D}_{\sigma_+,H}$ for all $b \in \mathbb{R}^3$, $R \in O_h$, W^a a scalar.*

(c) *For every family $\mathcal{F}'$ with nonnegative scalar members the Wick-tube object $Z^{a,K}_{\infty,T} := \lim_J \lim_k Z^{a,K}_{k,J}$ exists locally uniformly on $T_{\mathcal{F}'}$ and equals G^K_∞ on $T_{\mathcal{F}'} \cap \mathcal{D}_{\sigma_+,H}$; reflection positivity about every coordinate plane holds in the matrix form at amplitude $< \sigma_W$ and in the polynomial form, and* OS3*, for $Z^{a,K}_{\infty,T}$, G^K_∞ and their Schwinger functions.*

(d) *The measure and weak* OS4 *of theorem 21.1C(v),(vi) for the joint field under* ND^K*, with the local convergence of the laws of $\chi(F_{\Lambda_k}, W^a_{\Lambda_k})$ under $\mathrm{ND}^K_{k,J}$ along the full sequence $k \to \infty$ at every fixed $J \ge \max(J_0(\chi), J_1(R_\chi))$, then $J \to \infty$.*

Proof. The proofs of theorem 13.10(i), lemma 13.11, theorem 8.5 with scalar sources, theorem 10.7 with proposition 10.5, and theorem 11.1 in the Wick sector are run on ND^K with the replacements below; every other step of theirs is convention-free, since it sees the states only as members of the class of lemma 2.7 or concerns the limit object alone.

(a) is theorem 21.3, whose proof contains the fixed-cube limits (Step 7), the cube limit on the connected tube with the cube monotonicity of corollary 4.3 in place of the first inequality of lemma 2.10 (Steps 8–9), the positivity at real points and the cone monotonicity (Step 10), and the Schwinger bounds by Cauchy's estimates on a polydisc in the tube (Step 11).

(b) The steps of theorem 8.5 with scalar sources. Coordinate permutations are exact on ND^K for (f,h) by lemma 8.4. The reflection sandwich is lemma 4.5(b) at the cone points of the tube: for $(f,g) \in \mathcal{K}$ with $\|g\|_\infty < \sigma_+/2$ the point $(\mathrm{i},1)$ lies in $\mathcal{D}_{\sigma_+,H}$, so (a) supplies the three k-limits and the two J-limits of the two-step sandwich $C^K_{J\mp 2}$. Conjugation and group closure on the cone are unchanged. Off the cone, the identity theorem on the split family's tube of lemma 13.11, where (a) gives the limits, extends the identity from the cone points to the connected tube; so lemma 4.5 is needed on $\mathcal{K}$ only, and no class-$\mathcal{D}$ form of it enters (the proof splits $h = h^+ - h^-$ and never applies the sandwich to a negative scalar source). Translations are lemma 8.2, stated on $\mathcal{K}$ and applied at the cone points of the tube with (a) for the limits, then the continuity route of corollary 8.3 with the Wick-sector second-moment bound, which Cauchy's estimates give from (31) on ND^K; the translated family needs $J \ge \max(J_0+1, J_1(R+M+\frac{1}{2}))$. Invariance about every point and differentiation concern the limit object.

(c) Theorem 10.7 and proposition 10.5 on ND^K: in the sandwich of the J-limit the upper box is lemma 4.6 (the printed boxes C_J, C_{J+2n}) and the lower end carries the kill step of remark 4.8; the positive-leg bound corollary 13.3 holds on ND^K by remark 4.8 prepended to proposition 13.2 (remark 13.12); the cube monotonicity in both classes is corollary 4.3; the fixed-cube limits on $T_{\mathcal{F}'}$ are theorem 6.17(i), whose domain $\{\sup_x 2k_1 < s_W\}$, $s_W = s_1$, contains $T_{\mathcal{F}'}$ (for the reference scheme, hence for a by definition 3.6), and on $\mathcal{D}_{\sigma_+,H}$ they are (a); the

cube limit on $T_{\mathcal{F}'}$ is then Montel's theorem with the uniqueness set of definition 10.1(iii), as in proposition 10.5; the translations to the planes $c \notin D$ are (b). OS3 and the polynomial form use only the existence and holomorphy of the limit near 0 and charge conjugation (propositions 10.9 and 10.10).

(d) The steps of theorem 11.1 on $\mathcal{V} = \Lambda^2 \oplus \mathbb{R}$ with the scheme-shifted field: (31) on ND^K in the moment bounds, (a) for the limit functionals and the lattice laws, (b) and (c) for the invariance and the reflection positivity consumed by the abstract chain of [24]. The second moments of the lattice field under $\mathrm{ND}^K_{k,J}$ come from (31) on ND^K by Cauchy's estimates, so tightness and the identification of the limit law run as in the curvature sector, with the fixed-cube limits of (a) along the full sequence, for $J \geq \max(J_0(\chi), J_1(R_\chi))$; the remaining steps concern the limit measure. □

Proposition 21.5 (the comparison family on the Wick tube)**.** *Let the couplings satisfy* (T3) *and let a be an admissible scheme.*

(a) *Every increasing sequence of spacings has a continuum subsequence $\mathcal{K}$ for the Wick sector of the comparison family (definition 14.1).*

(b) *For every such $\mathcal{K}$, every family $\mathcal{F}$ of real sources with supports in Q_R and every $\zeta \in \mathcal{D}_{\sigma_+,H}$, the LIM state of definition 14.2 is King's tube object:*

$$Z^{a,\mathcal{K}}_\infty(\zeta) \;:=\; \lim_{J\to\infty} \lim_{m\to\infty} Z^a_{k_m,J}(\zeta) \;=\; G^K_\infty(\zeta) \qquad (\zeta \in \mathcal{D}_{\sigma_+,H}).$$

In particular the Wick-sector LIM state does not depend on $\mathcal{K}$, and the cube limit of every subsequential fixed-cube limit of the comparison family on the tube is G^K_∞.

Proof. (a) By lemmas 14.4 and 14.5 the scheme has the cell bound and the scheme bound, so lemma 14.7(b) applies.

(b) Fix $\mathcal{K} = (k_m)$ and $\mathcal{F}$. By lemma 14.8 and the Wick sector of lemma 14.7(b), $Z^{a,\mathcal{K}}_\infty$ exists locally uniformly on $\mathcal{D}_{\sigma_+,H}$ and is holomorphic there; by theorem 21.3 so is G^K_∞. Take the split family $\mathcal{F}'' = (f; h^+, h^-)$ of Step 8 of the proof of theorem 21.3, with supports in $Q_{R+1/2}$, the domain $\mathcal{D}''$ and the set $\Omega''_+ = \mathcal{D}'' \cap ((\mathrm{i}\mathbb{R})^n \times (0,\infty)^{2p})$ defined there; write $F^K_{k,J}$ and $F^{(a)}_{k,J}$ for the functionals of $\mathcal{F}''$ on King's and on the comparison cube. Let J be so large that $MJ > R + \frac{1}{2} + \ell_c$. At a point of Ω''_+ the curvature coefficients are purely imaginary and the total scalar source is nonnegative, so the observable is of class $\mathcal{I}$, and the support condition (S) holds for X_J; lemma 4.2 (with its clause for the scheme functionals) gives, at every k,

$$F^K_{k,J} \;\leq\; F^{(a)}_{k,J} \;\leq\; F^K_{k,J+1} \qquad \text{on } \Omega''_+.$$

Along $\mathcal{K}$ the two ends converge, by theorem 21.3 for the family $\mathcal{F}''$ (whose fixed-cube limits exist along the full sequence, hence along $\mathcal{K}$), to the fixed-cube limits F^K_J, F^K_{J+1} of Step 8 of that proof, and the middle term converges to the fixed-cube limit $F^{(a),\mathcal{K}}_J$ of the comparison family along $\mathcal{K}$ (definition 14.1 for $\mathcal{F}''$, whose tube contains $\mathcal{D}''$ and whose threshold is $MJ > R(\mathcal{F}'') + \ell_c$, $R(\mathcal{F}'') \leq R + \frac{1}{2}$). Letting $J \to \infty$, the two ends tend to F^K_∞ (Steps 8–9 of the proof of theorem 21.3) and the middle term to the LIM state $F^{(a),\mathcal{K}}_\infty$ of $\mathcal{F}''$ (lemma 14.8). Hence $F^{(a),\mathcal{K}}_\infty = F^K_\infty$ on Ω''_+, a non-empty open subset of the maximal totally real subspace $(\mathrm{i}\mathbb{R})^n \times \mathbb{R}^{2p}$, and both functions are holomorphic on the connected open set $\mathcal{D}''$; by the identity theorem (the fact (T2) stated before theorem 13.10) they agree on $\mathcal{D}''$. At $w^+_j = w_j$, $w^-_j = -w_j$ the lattice functionals of $\mathcal{F}''$ coincide exactly with those of $\mathcal{F}$ at ζ, on both conventions and at every k (equal total sources); so $Z^{a,\mathcal{K}}_\infty = G^K_\infty$ on $\mathcal{D}' = \{\zeta : |\operatorname{Re} w_j| < \delta_j\} \subset \mathcal{D}_{\sigma_+,H}$. Both are holomorphic on the connected $\mathcal{D}_{\sigma_+,H}$ and agree on the open set $\mathcal{D}'$, hence on $\mathcal{D}_{\sigma_+,H}$. The last clause is definition 14.2: every subsequential fixed-cube limit on the tube is the fixed-cube limit along some continuum subsequence. □

Remark 21.6 (the three amplitude constants). Three amplitudes occur in the Wick sector, all built from s_1: the matrix form of OS2 is proved at amplitude $\mathsf{a}(\zeta_i) < \sigma_W = \sigma_+/8$ on the Wick tube (theorem 10.7); the tube $\mathcal{D}_{\sigma_+,H}$ has half-widths $\sigma_+/2$ for the positive and $H/2 = \sigma_+/32$ for the negative part of the real Wick source, and the form of OS2 proved for LIM states

(theorem 14.9) takes $\mathsf{a}(\zeta_i) < H/2 < \sigma_W$, so that every Gram point lies in $\mathcal{D}_{\sigma_+,H}$, where those limits exist; and $\sigma_0 = H/4$ is the radius of the polydisc in Cauchy's estimates for E0″. The consumer of OS2 in clauses (v),(vi), the reflection positivity of exponentials $e^{\mathrm{i}\Phi(u)}$ with real u in the clustering theorems of [24], needs purely imaginary Wick coefficients only, that is amplitude 0 (remark 10.11).

Remark 21.7 (the photon mass and the couplings). The photon mass $\mu_0 > 0$ is fixed throughout theorem 21.1. No relation $\mu_0 \le m$ is used or needed: the one input whose printed proof in the torus paper [24] carries the restriction $\mu_0 \in (0, \mu_*]$, the lattice reflection positivity of King's tori, holds for all $\mu_0 > 0$, since that proof uses no bound on μ_0 and the paper's own lemma on the finite-volume photon covariance lifts the scope (proposition 12.2, lemma 22.1(d)). Reflection positivity of the cube limit needs the margin $R(\mu_0) = \pi\sqrt{3}/(2\mu_0)$ of (20) inside the upper half of the sandwich (lemma 9.10) and nowhere in the conclusion. The constants of Leg C are absolute; those of Leg W depend on μ_0 and are not uniform in it. The couplings range over all of $e > 0$, $\lambda, m, \mu_0 > 0$: their quantifier is the sources' (lemma 22.1(e), remark 22.16), the stopping scale of the lattice constructions behind theorems 3.2 and 3.5 and proposition 3.8 being chosen after the point and the datum, with the mass cap of [2, Prop. 3.1] one more one-sided cap on that scale (remark 22.2); so the theorem contains no sentence "e, λ sufficiently small", and the only coupling condition of the Wick sector is the one-sided sign clause (T3) (lemma 22.1(b)), proved on the explicit regions of proposition 22.15.

22. The coupling range

This section collects what the sources behind the inputs of theorem 21.1 say about the couplings and shows that the only coupling condition anywhere in this paper is the one-sided sign clause (T3) of the Wick leg, which proposition 22.15 proves on explicit regions.

King's range, units, two letters J. Let $e > 0$ and $\lambda, m, \mu_0 > 0$. Every constant of the cited constructions, including the number J_{K} of renormalisation steps after which the running couplings are small and the stopping scale $5^{-J_{\mathrm{K}}}$ (a member of the family $\{5^{-k}\}$), may depend on (e, λ, m, μ_0), on the large-block integer M and on the cube or torus; no numerical restriction is placed on the couplings themselves. That a stopping scale exists at every positive point is part of the cited constructions ([13, Thm. 3.1], [1, 2, 3, 4], and the existence and rotation parts of this paper), under the mass cap of [2, Prop. 3.1], a one-sided cap on the stopping scale (remark 22.2). This is what "King's range" means here: all positive (e, λ, m, μ_0), with a stopping scale chosen after them. The cube index J of C_J is ours; King's stopping index, the "fixed integer J" of [13, Thm. 3.1], is written J_{K}. Couplings are measured in the units in which the coarsest spacing is 1; in $d = 3$ all four of e^2, λ, m, μ_0 have the dimension of a mass, rescaling to the unit lattice at spacing ε replacing m^2, λ, e by $m^2\varepsilon^2, \lambda\varepsilon, e\varepsilon^{1/2}$ [13, Sect. 0] and μ_0^2 like m^2.

The relative bare coefficient. Let $\alpha_k^\infty(M) = M^2 + \delta m_k^2(M)$, $s_k^\infty(M_1) = \frac{1}{2}(\alpha_k^\infty(m) - \alpha_k^\infty(M_1))$ and $\sigma_1(M_1) = \liminf_k s_k^\infty(M_1)$ be as in lemma 3.9; the theorem is stated through σ_1 alone. Describing *where* (T3) fails uses one further statement about the continuum limit of this coefficient, which is of the same kind as lemma 3.9 (remark 3.10) and which we name rather than use tacitly:

(CT_∞) *For every $M > 0$ the limit*

$$\alpha_\infty(M) - \alpha_\infty(m) := \lim_{k\to\infty} \big[\alpha_k^\infty(M) - \alpha_k^\infty(m)\big]$$

exists and is a polynomial of degree ≤ 2 in (e^2, λ) with finite coefficients, differentiable in M^2 at $M = m$.

The divergent parts of the counterterm ($1/\varepsilon$ at one loop, $\ln(1/\varepsilon)$ at two loops) do not depend on the scalar mass and cancel in the difference. (CT_∞) is proved in theorem F.159(b); it is not a hypothesis of theorem 21.1 and enters only the description of the failure set. Under (CT_∞) put

$$s_\infty(M_1) := \tfrac{1}{2}\big(\alpha_\infty(m) - \alpha_\infty(M_1)\big), \qquad s_* := \sup_{M>0} s_\infty(M) \in [0, \infty],$$

$$E_W := \{s_* = 0\} = \{m \text{ is a global minimiser of } M \mapsto \alpha_\infty(M) \text{ on } (0,\infty)\}, \qquad E_W^{<} := \{m \text{ minimises } \alpha_\infty \text{ on } (0,$$

subsets of the coupling space ($s_* \geq 0$ because $M = m$ gives 0; the value $+\infty$ is harmless below). The two sign clauses are

(NE) m is not a global minimiser of α_∞ on $(0,\infty)$, i.e. $s_* > 0$, i.e. $(e,\lambda,m,\mu_0) \notin E_W$;

(NE$^<$) some $M_1 \in (0,m)$ has $\alpha_\infty(M_1) < \alpha_\infty(m)$, i.e. $(e,\lambda,m,\mu_0) \notin E_W^<$ (the one-sided form; it implies (NE)).

Throughout, $'$ denotes d/dM^2, N is the number of real components of the scalar field, and "one-loop order" means the orders e^2 and λ.

Lemma 22.1 (the coupling range)**.** *Let $e > 0$ and $\lambda, m, \mu_0 > 0$.*

(a) The curvature sector. *The existence theorem of the curvature sector, theorem 3.2 (the existence clause of* [14, Thm. 2.1] *at zero scalar source), carries no condition on e, λ, m, μ_0. The correlation inequalities it uses, lemma 2.7 (Theorem 6.2(1),(2) of* [6]*), have the standing hypotheses of a finite single-site measure, positive hopping coefficients and a positive photon covariance, which for King's action are $\lambda > 0$ and $\mu_0 > 0$; the margin $R(\mu_0) = \pi\sqrt{3}/(2\mu_0)$ of lemma 9.10 needs $\mu_0 > 0$ only; and no lemma proved in this paper for the curvature sector carries a condition on the couplings beyond those of its inputs. Hence the curvature sector of theorem 21.1 holds on all of King's range, with no exceptional set. The one proof-internal proviso in the Bałaban antecedents of theorem 3.2, the mass clause of* [2, Prop. 3.1]*, is a cap on the dimensionless scalar mass at the stopping scales and no condition on the bare couplings (remark 22.2). Of the bounds of the paper only the absolute bounds of theorem 7.6 are uniform in μ_0.*

(b) The Wick sector, Form (B). *The inputs of Form (B) beyond theorem 3.5, proposition 3.8, and lemma 3.9 and* (T3) *— proposition 12.2 and lemmas 2.7, 12.4, 12.9 and 12.10 and the tube statements theorems 10.7 and 13.10 and lemma 13.11, whose conditions on the Wick amplitudes are not conditions on the couplings — print no condition on the bare couplings beyond positivity ($\lambda > 0$, $\mu_0 > 0$); proposition 3.8 is used at the two masses m and M_1. The only coupling condition of Form (B) is therefore* (T3)*, which proposition 22.15 proves on explicit regions. Under* (CT_∞)*:*

(i) *$\sigma_1(M_1) = \lim_k s_k^\infty(M_1) = s_\infty(M_1)$ for every $M_1 > 0$. Hence* (T3) *holds if and only if* (NE$^<$)*: it fails exactly on $E_W^<$. If proposition 3.8 and lemma 3.9 are available also at auxiliary masses $M_1 > m$, then by proposition 12.11 the two-sided clause* (NE) *replaces* (T3)*, with failure set exactly E_W; the branch $M_1 > m$ is needed exactly on $E_W^< \setminus E_W$, for instance when $\alpha'_\infty(m) < 0$ and m minimises α_∞ on $(0,m]$.*

(ii) *To one-loop order, for every $M_1 > 0$,*

$$s_\infty(M_1) = \tfrac{1}{2}(m^2 - M_1^2) + \frac{\lambda(N+2)(m-M_1)}{2\pi} - e^2\,\frac{(m-M_1)\,(m\mu_0 + mM_1 + M_1\mu_0)}{8\pi(\mu_0+M_1)(m+\mu_0)},$$

so that

$$\alpha'_\infty(m) = 1 + \frac{\lambda(N+2)}{2\pi m} - e^2\,\frac{m+2\mu_0}{8\pi(m+\mu_0)^2}, \qquad \frac{m+2\mu_0}{8\pi(m+\mu_0)^2} \leq \frac{1}{4\pi(m+\mu_0)}, \tag{48}$$

and to one-loop order (T3) *holds if and only if*

$$e^2 < e_c^2(\lambda,m,\mu_0) := \max\Big\{\frac{8\pi(m+\mu_0)^2}{m+2\mu_0}\Big(1+\frac{\lambda(N+2)}{2\pi m}\Big),\ 4\pi(m+\mu_0)\Big(1+\frac{\lambda(N+2)}{\pi m}\Big)\Big\}, \tag{49}$$

the first entry being the larger if and only if $\lambda(N+2)\mu_0 \leq \pi m^2$. So the one-loop failure set of (T3) *is the half-line $e^2 \geq e_c^2$. The first entry is the zero $e_*^2 \geq 4\pi(m+\mu_0)$ of* (48)*; at $m = \mu_0 = 1$, $\lambda = 0$ it equals $32\pi/3 \approx 33.5$ (for orientation only). Nothing finer than one-loop order is claimed about $E_W^<$.*

(iii) *For the two-sided set, $E_W \subset \{\alpha'_\infty(m) = 0\}$; for each (λ,m,μ_0) at which the polynomial $e^2 \mapsto \alpha'_\infty(m)$, of degree ≤ 2, is not identically zero, this is at most two values of e^2. At $\mu_0 = m$ the window $0 < \lambda, e^2 < 0.1m$ of the torus paper* [24] *lies outside $E_W^<$, by (d).*

(c) The existence and rotation parts. *The existence theorems theorems 3.2 and 3.5 and the rotation theorem theorem 3.13 are proved for $e > 0$, $\lambda, m, \mu_0 > 0$ in King's range, and none*

carries a condition on the bare couplings beyond positivity: every smallness condition of [1, 2, 3, 4] *that they use (listed in the proof) is a one-sided cap on a stopping scale* $\beta \le 1$*, imposed last in the schedule of constants, and an admissible* β *exists at every positive point.*

(d) Comparison with the torus paper. *The window* $0 < \lambda < 0.1m$*,* $0 < e^2 < 0.1m$*,* $\mu_0 \in (0, m]$ *of* [24] *is a hypothesis of its torus theorem only and is not inherited by King's state. Nor is* $\mu_0 \le m$*: the one King-side input printed under it, the torus paper's reflection-positivity proposition behind proposition 12.2 (stated for* $\mu_0 \in (0, \mu_*]$*,* $\mu_* = m$*), is lifted to all* $\mu_0 > 0$ *by a proof that uses only site-independent coefficients,* $\lambda > 0$ *and even sides. At* $\mu_0 = m$ *the torus paper's certificate gives* $s_\infty(0.7m) \ge 0.1595\, m^2$ *on the whole window (identifying the two torus counterterm schemes at* $\mu_0 = m$*, which compute the same graphs), so* (T3) *holds there with* $M_1 = 0.7m$ *and an explicit margin; proposition 22.15 sharpens this and gives explicit regions at every* $\mu_0 \in (0, m]$ *(remark 22.3).*

(e) The quantifier. *For all* $e > 0$*,* $\lambda, m, \mu_0 > 0$ *and each datum there are an integer* J_{K} *(*[13, Thm. 3.1]*) and a stopping scale* $\beta \le 1$ *such that the running couplings* $(e\beta^{1/2}, \lambda\beta, m\beta, \mu_0\beta)$ *satisfy finitely many one-sided caps on* β *and on three of the four of them (*$\mu_0\beta$ *is uncapped in the sources); no condition is placed on the bare couplings, and the theorems consumed from* [13, 14] *carry no coupling hypothesis. A sentence "*e, λ *sufficiently small,* $m, \mu_0 > 0$ *fixed", read as a hypothesis on the bare couplings, is strictly stronger and is printed in none of the sources (remark 22.16).*

Proof. (a). Theorem 2.1 of [14] asserts, for g, h supported in C_J and all $z, w \in \mathbb{C}$, the existence of $\lim_K \langle \Theta(zg, wh)\rangle^{\varepsilon_K}_{N,D,C_J}$; no hypothesis on the couplings precedes it. Its proof rests on [13, Thm. 3.1] (the convergence of the effective action, referred there to [1, 2, 3, 4]), stated "for some fixed integer J", our J_{K}; neither it nor [13, Thms. 2.1, 3.4] carries a coupling hypothesis, the smallness in [13] being of the fields and of the stopping spacing. So the one proviso of [2, Prop. 3.1] is upstream of theorem 3.2. For King's action the single-site measure $e^{-\varepsilon^3[\frac{1}{2}\alpha r^2 + \lambda r^4]} r\, dr$ is finite iff $\lambda > 0$ (or $\lambda = 0$, $\alpha > 0$), the hopping weights are positive, and the Feynman-form photon covariance is positive iff $\mu_0 > 0$; the "$g > 0$" of Theorem 6.2 of [6] is the sign of the scalar source, and lemma 2.7 covers $g = 0$. The margin of lemma 9.10 concerns the photon Gaussians of the two boxes alone, so μ_0 is its only parameter (no fixed margin works at $\mu_0 = 0$). The lemmas of this paper see only the form of the action.

The torus inputs. Proposition 3.8 is Bałaban's stability theorem [1, (1.14)], [2, (1.3)], with constants independent of ε and of the torus, under $\lambda > 0$, $\mu_0^2 > 0$, $m^2 > 0$. Every printed smallness in [1, 2, 3, 4] is of one of three kinds. *Running charge:* the bounds hold "for $e(L^k\varepsilon)$ sufficiently small", $e(L^k\varepsilon) = e\,(L^k\varepsilon)^{(4-d)/2}$ [1, (2.23)]; [4] proves them for a unit-lattice model whose charge is this running charge (its theorem is stated there as Proposition 2.1 of [1]), so "e sufficiently small" there is a cap on $e\beta^{1/2}$ at the stopping scale β. *Scale:* the stopping scale ε_0 with $L^K\varepsilon \le \varepsilon_0 < L^{K+1}\varepsilon$ is "some fixed sufficiently small number" [3], and "$L^{k+1}\varepsilon \le 1$", "$L^k\varepsilon$ sufficiently small" [1, 2] are caps on it; Bałaban's units have $\lambda = 1$ (in $d = 3$: spacing $\lambda\varepsilon$, mass squared m^2/λ^2, charge $e\lambda^{-1/2}$, the running charge invariant), so his "ε small" is "$\lambda\varepsilon$ small" in ours, and his constants may depend on $e\lambda^{-1/2}$, m/λ, μ_0/λ: a dependence, not a condition. *Mass:* Proposition 3.1 of [2] carries the sentence "we assume $m^2 \le O(1)$ also"; this is the mass cap of that proposition (remark 22.2). Lemma 3.9 carries no coupling condition: the bound on the relative bare coefficient holds for every mass in a compact interval containing m, on either side of m, with a constant depending on the couplings (remark 3.10; the power-counting theorem of [20] that enters it has none). Lemma 12.10 needs a finite single-site integral ($\lambda > 0$) and a finite photon Gaussian on the torus ($\mu_0 > 0$; at $\mu_0 = 0$ the constant mode lies in the kernel); lemma 12.9 needs massive propagators; proposition 12.2 needs finite single-site moments and a positive Proca covariance and does not see the counterterm. The tube statements restrict Wick amplitudes only. So the only coupling condition of Form (B) is the sign clause of proposition 12.6, i.e. (T3).

(b)(i). By lemma 12.9, $s_k^\infty(M_1)$ is the fixed-k volume limit of the torus coefficient, the relative bare coefficient on Λ_k in the free convention; by (CT_∞) its limit $s_\infty(M_1)$ exists, hence $\sigma_1(M_1) = \liminf_k = \lim_k = s_\infty(M_1)$. Thus some $M_1 \in (0, m)$ has $\sigma_1(M_1) > 0$ iff ($\mathrm{NE}^<$),

and some $M_1 > 0$ has $\sigma_1(M_1) > 0$ iff (NE). Proposition 12.11 gives (I2) from proposition 3.8 and lemma 3.9 at an auxiliary mass on either side of m, so with those inputs (NE) suffices; on $E_W^< \setminus E_W$ every $M_1 < m$ fails and some $M_1 > m$ works.

(b)(ii). The closed form is the one-loop evaluation of King's counterterm [13, (3.32)–(3.33)] (tadpole and zero-momentum photon-exchange graphs; the $1/\varepsilon$ divergence cancels in the difference). It equals $\frac{1}{2}\int_{M_1^2}^{m^2} \alpha'_\infty(\mu)\, d\mu^2$ with $\alpha'_\infty(\mu) = 1 + \frac{\lambda(N+2)}{2\pi\mu} - e^2 \frac{\mu+2\mu_0}{8\pi(\mu+\mu_0)^2}$, an orientation-free definite integral, valid on either side of m; at $\mu = m$ this is (48), whose inequality is $m \geq 0$. For $M_1 \in (0, m)$ write $s_\infty(M_1) = \frac{m-M_1}{2}\beta(M_1)$ with

$$\beta(M_1) = (m + M_1) + c - e^2\, \frac{m\mu_0 + (m+\mu_0)M_1}{4\pi(m+\mu_0)(\mu_0+M_1)}, \qquad c := \frac{\lambda(N+2)}{\pi};$$

$s_\infty(M_1) > 0$ iff $\beta(M_1) > 0$ iff $e^2 < e_c^2(M_1) := \big[(m+M_1)+c\big]\frac{4\pi(m+\mu_0)(\mu_0+M_1)}{m\mu_0+(m+\mu_0)M_1}$. With $\mathcal{N}(M_1) := (m+M_1+c)(\mu_0+M_1)$ and $\mathcal{D}(M_1) := m\mu_0 + (m+\mu_0)M_1$, the derivative of $e_c^2(M_1)$ has the sign of $\mathcal{N}'\mathcal{D} - \mathcal{N}\mathcal{D}' = 2m\mu_0 M_1 + (m+\mu_0)M_1^2 - c\mu_0^2$, positive on $(0, m)$ when $c = 0$ and, when $c > 0$, negative then positive with one zero. So $e_c^2(\cdot)$ is increasing, or decreasing then increasing, on $(0, m)$, and its supremum over the open interval is the larger of the two end values, not attained: $M_1 \to m^-$ gives $\frac{8\pi(m+\mu_0)^2}{m+2\mu_0}(1 + \frac{c}{2m})$, the zero e_*^2 of (48), and $M_1 \to 0^+$ gives $4\pi(m+\mu_0)(1+\frac{c}{m})$. Hence to one-loop order (T3) holds iff $e^2 < e_c^2(\lambda, m, \mu_0)$ as in (49), and fails for every $e^2 \geq e_c^2$ (then $\beta \leq 0$ for every M_1); the first entry is the larger iff $(m+\mu_0)(2m+c) \geq (m+c)(m+2\mu_0)$, i.e. $c\mu_0 \leq m^2$.

(b)(iii). Under (CT_∞), $M^2 \mapsto \alpha_\infty(M)$ is differentiable at m, and a global minimum at the interior point m is a critical point: $E_W \subset \{\alpha'_\infty(m) = 0\}$. A polynomial of degree ≤ 2 in e^2 that is not identically zero has at most two zeros. The clause at $\mu_0 = m$ is remark 22.3.

(c). The existence and rotation parts fix $e > 0$, $\lambda, m, \mu_0 > 0$ in King's range and the block factor 5, and fix their constants after the couplings. In their schedule of constants, β is fixed last as the minimum of finitely many caps, each one-sided and monotone in β with a right side that is a positive finite function of the couplings and of the entries fixed before it; so an admissible β exists at every positive point. Bałaban's own thresholds and King's J_{K} are imposed on β after M is fixed.

(d). The torus paper states that its numerical window enters through its counterterm-gap input alone and that none of its other conditions is a condition on e or λ. Its reflection-positivity proposition for King's tori is stated for $\mu_0 \in (0, \mu_*]$, $\mu_* = m$, while the two lemmas on the periodic photon covariance that carry its proof are stated for every $\mu > 0$; the proof of proposition 12.2 uses only site-independent coefficients, $\lambda > 0$ and even sides, and so holds for every $\mu_0 > 0$. The certificate is remark 22.3.

(e). The quantifier is read off [13, Thm. 3.1] ("for some fixed integer J, for all $K \geq J$, and all $k \leq K - J$") and the stopping-scale caps listed above; the last sentence is remark 22.16. □

Remark 22.2 (the mass clause of Bałaban's upper bound). Proposition 3.1 of [2] asserts an inequality (3.26), summed over the scales $k = 0, \ldots, K$, with a constant $\gamma_0 = \gamma_0(d, a)$ and mass terms $(L^k\varepsilon)^d m^2 |\phi_k|^2$; rescaled to the unit lattice, its k-th term is the per-scale inequality (3.29) with the mass term $(L^k\varepsilon)^2 m^2 |\phi'_k|^2$, so $(L^k\varepsilon)^2 m^2$ is the dimensionless mass at scale k. The proposition carries the sentence "we assume $m^2 \leq O(1)$ also", m^2 being the bare scalar mass squared in Bałaban's units ($\lambda = 1$), i.e. m^2/λ^2 in ours. The clause under which this paper cites the proposition concerns the scales at which it is applied:

> *Inequality (3.29) of* [2] *holds, with $\gamma_0 = \gamma_0(d, a)$, at every scale $k \leq K$ at which $(L^k\varepsilon)^2 m^2 \leq c_m(d, a)$; hence (3.26) holds whenever this is so for every $k \leq K$.*

Here $c_m(d, a)$ is a constant depending on d and a only. The clause belongs to the statement of the proposition: the mass that enters its printed proof, referred to [4, Sect. 4], is the unit-lattice mass at scale k, the text of [2] makes no further use of the sentence, and the proposition is cited here under the clause. It is then one more one-sided cap on the stopping scale $\beta = \varepsilon_0$ ($(L^k\varepsilon)^2 m^2 \leq \beta^2 m^2$ at every scale used), of the kind every other smallness condition of [1, 2, 3, 4]

has; a further monotone cap on β leaves the existence of an admissible stopping scale unchanged, and nothing is imposed on the bare mass.

Read instead as an absolute bound on m^2/λ^2, the sentence would exclude small λ at fixed m, including all of the window $\lambda < 0.1m$ of [24] ($m^2/\lambda^2 > 100$), and would be an unprinted hypothesis of the theorem (1.3) of [2], whose only printed condition on m is $m^2 > 0$; that is not the reading under which the proposition is proved, and it is used nowhere. The clause sits upstream of proposition 3.8 at both masses, of theorems 3.2 and 3.5 (through [13, Thm. 3.1]) and of the steps of the rotation part that rest on [1, 2], as every cap on the stopping scale does; it is not a condition of this paper.

Remark 22.3 (the torus paper's window and its certificate at $\mu_0 = m$). (i) *The certificate.* The torus paper's coefficient $a_{K,T}(M) = M^2 + \delta m^2_{K,T}(\varepsilon; M)$ is the counterterm of [13, (3.32)–(3.33)] (orders $2 \le \alpha + 2\beta \le 4$, zero-momentum normalisation) computed on the periodic torus with the periodic scalar propagator of mass M and the periodic photon propagator of the reference mass μ_*, without block averaging; its theorem on this coefficient gives, for $m > 0$, $\mu_* = m$, $M_1 = 0.7m$, uniformly in $\varepsilon \le 1/m$ and in every periodic torus of side $\ge 96/m$, the explicit two-loop bound $a_{K,T}(m) - a_{K,T}(0.7m) \ge 0.319\, m^2$. King's torus-convention coefficient $\alpha^{\mathrm{tor}}_{k,T}(M)$ of definition 12.1 is the same polynomial over the same orders, as translation-invariant lattice graphs on the torus with the photon of mass μ_0 and the same normalisation, the counterterm graphs of [13] being defined with the full lattice propagators, without block averaging, as we read that paper. So at $\mu_0 = m = \mu_*$ the two schemes compute the same graphs and $a_{K,T}(M) = \alpha^{\mathrm{tor}}_{k,T}(M)$ (an identification of ours). Then $s_{k,T}(0.7m) = \frac{1}{2}(a_{K,T}(m) - a_{K,T}(0.7m)) \ge 0.1595\, m^2$ for all $\varepsilon_k \le 1/m$ and every T_J with $2MJ \ge 96/m$; lemma 12.9 ($J \to \infty$ at fixed k) gives $s^\infty_k(0.7m) \ge 0.1595\, m^2$ for every such k, hence $\sigma_1(0.7m) \ge 0.1595\, m^2 > 0$: at $\mu_0 = m$, (T3) holds on the whole window with $M_1 = 0.7m$ and an explicit margin, without (CT_∞) and without C_4; under (CT_∞) the window lies outside $E^<_W$.

(ii) *Away from* $\mu_0 = m$. The torus paper's counterterm is computed at the photon mass $\mu_* = m$, King's at μ_0; the two coefficients differ by a term of order $e^2|m - \mu_0|$ (the zero-momentum photon-exchange graph at the two photon masses), so the certificate does not transfer to $\mu_0 \ne m$. There the comparison is through one-loop order: by (49), $e_c^2 \ge 4\pi(m + \mu_0) > 0.1m$, so the one-loop form of (T3) holds on the torus paper's window at every $\mu_0 > 0$; beyond one loop, proposition 22.15 places the window inside the region where (T3) holds for every $\mu_0 \in [0.03m, m]$, and a box shrinking like $\sqrt{\mu_0}$ in e^2 for every $\mu_0 \in (0, m]$; the exact position of $E^<_W$ is not determined.

The explicit coupling region. The sign clause (T3) is proved below on an explicit region of the (λ, e^2)-plane at every photon mass $\mu_0 \in (0, m]$, with $N = 2$ and the auxiliary mass $M_1 = 0.7m$. The argument has two parts: the three low orders of King's counterterm, whose contributions to $s^\infty_k(M_1)$ are lattice integrals with exact limits (lemmas 22.4, 22.5 and 22.7 to 22.9), and the two orders $e^2\lambda$ and e^4, whose contributions are bounded from below, uniformly in k, through the two-loop majorants of the torus paper's Appendix G [24], read at the photon mass μ_0 in place of its reference mass, with the torus floor removed (lemmas 22.11 and 22.12), made uniform on intervals of photon masses (lemma 22.13) and, in a weaker closed form, uniform in the photon mass (lemma 22.14). The resulting lower bound σ_{low} on $\sigma_1(0.7m)$, (55), is explicit through table 1.

King's counterterm order by order. Throughout this subsection and the next, $\varepsilon = \varepsilon_k$, $\Lambda = \Lambda_k$, $\mathsf{B} = \mathsf{B}_\varepsilon := [-\pi/\varepsilon, \pi/\varepsilon)^3$ is the Brillouin zone with $\int_{\mathsf{B}} := \int_{\mathsf{B}} \frac{d^3p}{(2\pi)^3}$, $\hat p_\mu := \frac{2}{\varepsilon}\sin\frac{\varepsilon p_\mu}{2}$ and $\hat p^2 := \sum_\mu \hat p_\mu^2$ (periodic in p), and for a mass $M > 0$

$$G_M(p) := \frac{1}{\hat p^2 + M^2}, \qquad \hat D(p) := \frac{1}{\hat p^2 + \mu_0^2}, \qquad C_M(x) := \int_{\mathsf{B}} e^{\mathrm{i}p\cdot x} G_M(p), \qquad D(x) := \int_{\mathsf{B}} e^{\mathrm{i}p\cdot x}\hat D(p) \quad (x \in \Lambda)$$

are the free scalar and photon propagators of King's action on Λ (per real scalar component and per photon component; $C_M(0)$ is the free Wick constant at mass M, $C_m(0) = \mathsf{W}_\varepsilon$). N

is the number of real components of the scalar field. King's free-boundary counterterm [13, (3.32)–(3.33)], [14, p. 330] at scalar mass M is the polynomial

$$\delta m^2_\varepsilon(M) = \sum_{(\alpha,\beta)} e^\alpha \lambda^\beta \, \delta m^2_{(\alpha\beta)}(\varepsilon; M), \qquad (\alpha,\beta) \in \{(0,1),(2,0),(0,2),(2,1),(4,0)\}, \tag{50}$$

over the superficially divergent orders $2 \le \alpha + 2\beta \le 4$ with the odd powers of e dropped (they vanish: an odd power of e comes with an odd number of photon fields, whose centred Gaussian expectation is zero); the coefficient $\delta m^2_{(\alpha\beta)}(\varepsilon; M)$ is fixed recursively by the condition that the order-$e^\alpha\lambda^\beta$ part of the amputated one-particle-irreducible two-point function of ϕ at zero external momentum, computed on Λ with the propagators above and with all counterterm vertices of lower order inserted, be cancelled by the counterterm vertex of that order. In the sign convention in which a term of the expansion of e^{-V} that reduces to $+\frac12 \sum_{x,y} \varepsilon^6 \phi_a(x) \mathsf{K}(x-y) \phi_a(y)$ contributes $+\hat{\mathsf{K}}(p)$ to the self-energy $\Sigma(p)$, so that the inverse propagator per component is $\hat p^2 + M^2 - \Sigma(p)$ and the counterterm vertex $-\frac12 \delta m^2 \sum \varepsilon^3 |\phi|^2$ contributes $-\delta m^2$, the condition reads $\delta m^2_{(\alpha\beta)}(\varepsilon; M) = \Sigma^{(\alpha\beta)}(0; M)$, $\Sigma^{(\alpha\beta)}$ the order-$e^\alpha\lambda^\beta$ part of the sum of the one-particle-irreducible two-point graphs built from the vertices of King's action and from the counterterm vertices of lower order. Every such graph is a finite sum of absolutely convergent integrals over B^L, $L \le 2$ (all masses are $\ge \min(M, \mu_0) > 0$ and B is compact), so each coefficient is a finite real number, smooth in M (a rational function of M^2 without poles on $(0,\infty)$ under each integral). With $\alpha_k^\infty(M) = M^2 + \delta m^2_{\varepsilon_k}(M)$ as in lemma 3.9 and $M_1 \in (0, m)$:

Lemma 22.4 (structure)**.** *Put $S_{\alpha\beta,k} := \frac12 \big[\delta m^2_{(\alpha\beta)}(\varepsilon_k; m) - \delta m^2_{(\alpha\beta)}(\varepsilon_k; M_1)\big]$, numbers free of the couplings. Then*

$$s_k^\infty(M_1) = \tfrac12 (m^2 - M_1^2) + \lambda S_{01,k} + e^2 S_{20,k} + \lambda^2 S_{02,k} + e^2\lambda S_{21,k} + e^4 S_{40,k}. \tag{51}$$

In $d = 3$, $S_{01,k}$ and $S_{20,k}$ have the dimension of a mass and $S_{02,k}$, $S_{21,k}$, $S_{40,k}$ are dimensionless.

Proof. $s_k^\infty(M_1) = \frac12(\alpha_k^\infty(m) - \alpha_k^\infty(M_1))$ and (50). □

Lemma 22.5 (the three low orders, exact)**.** *For every ε and $M > 0$:*

(i) $\delta m^2_{(01)}(\varepsilon; M) = -4(N+2)\, C_M(0)$;

(ii) $\delta m^2_{(20)}(\varepsilon; M) = \int_{\mathsf{B}} \hat k^2 \hat D(k) G_M(k)\, dk - 3D(0) = -2D(0) - M^2 \int_{\mathsf{B}} \hat D G_M$;

(iii) $\delta m^2_{(02)}(\varepsilon; M) = 32(N+2) \sum_{x\in\Lambda} \varepsilon^3 C_M(x)^3$.

Consequently, with $J_\varepsilon := \int_{\mathsf{B}} G_m G_{M_1}$, $I_\varepsilon := \int_{\mathsf{B}} \hat k^2 \hat D(k) G_m(k) G_{M_1}(k)\, dk$ and $F_\varepsilon := \sum_{x\in\Lambda} \varepsilon^3 [C_{M_1}(x)^3 - C_m(x)^3]$,

$$S_{01,k} = 2(N{+}2)\big(C_{M_1}(0) - C_m(0)\big) = 2(N{+}2)(m^2 - M_1^2) J_{\varepsilon_k}, \qquad S_{20,k} = -\tfrac12 (m^2 - M_1^2) I_{\varepsilon_k}, \qquad S_{02,k} = -16(N{+}2 \tag{52}$$

Proof. Write $\phi = (\phi_1, \dots, \phi_N)$ with $\phi \cdot \psi = \sum_a \phi_a \psi_a$ and, for $N = 2$, $\phi \wedge \psi = \phi_1\psi_2 - \phi_2\psi_1 = J_{ab}\phi_a\psi_b$ with $J = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, $J^2 = -1$. With $y = x + \varepsilon e_\mu$ and $\theta = e\varepsilon A_\mu(x)$ the hopping term of King's action is $\frac12 \varepsilon |e^{\mathrm{i}\theta}\phi(y) - \phi(x)|^2 = \frac12 \varepsilon (|\phi(x)|^2 + |\phi(y)|^2) - \varepsilon \cos\theta\, \phi(x)\cdot\phi(y) + \varepsilon \sin\theta\, \phi(x) \wedge \phi(y)$; expanding $\cos\theta$ and $\sin\theta$ gives the free action $S_0 = \frac12 \sum_x \varepsilon^3 [\sum_\mu |\nabla_\mu \phi|^2 + M^2 |\phi|^2] + \frac12 \sum_x \varepsilon^3 [\sum_{\mu,\nu} (\nabla_\nu A_\mu)^2 + \mu_0^2 |A|^2]$, $\nabla_\mu f(x) = (f(x+\varepsilon e_\mu) - f(x))/\varepsilon$, whose covariances are $\langle \phi_a(x)\phi_b(y)\rangle_0 = \delta_{ab} C_M(x-y)$ and $\langle A_\mu(x) A_\nu(y)\rangle_0 = \delta_{\mu\nu} D(x-y)$, and the vertices

$$V_1 = e \sum_{x,\mu} \varepsilon^2 A_\mu(x)\, \phi(x) \wedge \phi(y), \qquad V_2 = \tfrac{e^2}{2} \sum_{x,\mu} \varepsilon^3 A_\mu(x)^2\, \phi(x)\cdot\phi(y), \qquad V_\lambda = \lambda \sum_x \varepsilon^3 (\phi\cdot\phi)^2, \qquad V_{\mathrm{ct}} = \tfrac12 \delta m^2 \sum_x$$

together with vertices of order e^3, e^4 and higher, which do not enter the three orders below. The graphs are those of $-V + \frac12 V^2 - \cdots$ with the normalisation fixed above; by $O(N)$ symmetry they are proportional to δ_{ab}.

(i) At order λ the only one-particle-irreducible two-point graph is the tadpole of $-V_\lambda$. In $(\phi\cdot\phi)^2 = \phi_a\phi_a\phi_b\phi_b$ one of the six pairs of fields is contracted: the pairs $(1,2)$ and $(3,4)$ each give $N\, C_M(0)\, \phi\cdot\phi$, the four mixed pairs each give $C_M(0)\, \phi\cdot\phi$, in all $2(N+2) C_M(0)\, \phi\cdot\phi$. So

$-V_\lambda$ contributes $-2(N+2)\lambda C_M(0)\sum_x \varepsilon^3\phi\cdot\phi = \frac12\sum \phi_a \mathsf{K}\phi_a$ with $\hat{\mathsf{K}} = -4(N+2)\lambda C_M(0)$, and $\delta m^2_{(01)} = \Sigma^{(01)}(0)/\lambda = -4(N+2)C_M(0)$.

(ii) At order e^2 there are two graphs. *Seagull:* contracting $A_\mu(x)^2$ to $D(0)$ in $-V_2$ gives $-\frac{e^2}{2}D(0)\sum_{x,\mu}\varepsilon^3\phi(x)\cdot\phi(x+\varepsilon e_\mu)$, i.e. $\Sigma_{\text{seag}}(p) = -e^2D(0)\sum_\mu\cos(\varepsilon p_\mu)$ after symmetrisation, and $\Sigma_{\text{seag}}(0) = -3e^2D(0)$. *Exchange:* in momentum space $V_1 = \sum_\mu\iiint \hat A_\mu(k)\hat\phi_i(p)\hat\phi_j(p')\,W^\mu_{ij}(p,p')\,(2\pi)^3\delta(k+p+p')$ with the symmetrised vertex factor $W^\mu_{ij}(p,p') = \mathrm{i}e\,J_{ij}\,e^{-\mathrm{i}\varepsilon k_\mu/2}\frac1\varepsilon\sin\frac{\varepsilon(p'-p)_\mu}{2}$ (from $\frac12 J_{ij}(e^{\mathrm{i}\varepsilon p'_\mu} - e^{\mathrm{i}\varepsilon p_\mu}) = J_{ij}e^{\mathrm{i}\varepsilon(p+p')_\mu/2}\,\mathrm{i}\sin\frac{\varepsilon(p'-p)_\mu}{2}$). In $\frac12 V_1^2$ contract the photon pair, $\langle\hat A_\mu\hat A_\nu\rangle = \delta_{\mu\nu}(2\pi)^3\delta(k+k')\hat D(k)$, and one scalar pair (four equivalent choices); with external pair $\hat\phi_i(p)\hat\phi_l(-p)$ the internal line carries $p+k$ and the two vertex factors multiply to $(\mathrm{i}e)^2(J^2)_{il}\frac{1}{\varepsilon^2}\sin^2\frac{\varepsilon(2p+k)_\mu}{2} = e^2\delta_{il}\frac{1}{\varepsilon^2}\sin^2\frac{\varepsilon(2p+k)_\mu}{2}$, the phases cancelling. Hence $\Sigma_{\text{exch}}(p) = e^2\int_{\mathsf B}\hat D(k)G_M(p+k)\sum_\mu\big(\frac2\varepsilon\sin\frac{\varepsilon(2p+k)_\mu}{2}\big)^2dk$ and $\Sigma_{\text{exch}}(0) = e^2\int_{\mathsf B}\hat D(k)\hat k^2G_M(k)\,dk$. So $\delta m^2_{(20)} = \int_{\mathsf B}\hat k^2\hat D G_M - 3D(0)$; the second form follows from $\hat k^2\hat D G_M = \hat D(1-M^2G_M)$ and $\int_{\mathsf B}\hat D = D(0)$.

(iii) At order λ^2 (and $e=0$) the one-particle-irreducible two-point graphs with two quartic vertices u,v are the sunset (external legs at u and at v, three lines u–v) and the tadpole with a tadpole inserted on its loop line (external legs at u, two lines u–v, one loop at v); the configurations with a single bridge are not one-particle irreducible and the rest are disconnected. The insertion graph is cancelled exactly by the graph with the counterterm vertex $-V^{(01)}_{\text{ct}}$ on the loop line, because the order-λ self-energy with its counterterm, $\Sigma^{(01)}(q) - \lambda\delta m^2_{(01)}$, vanishes for every q (the tadpole does not depend on the momentum). For the sunset, in $\frac12 V_\lambda^2 = \frac12\lambda^2\sum_{u,v}\varepsilon^6(\phi\cdot\phi)^2_u(\phi\cdot\phi)^2_v$ choose the external field at u (4 ways) and at v (4 ways) and pair the remaining triples in the 3! ways: two pairings give $N\delta_{ab}$ and four give δ_{ab}, in all $2(N+2)\delta_{ab}$; so $\frac12 V_\lambda^2$ contributes $\frac12\sum_{u,v}\varepsilon^6\phi_a(u)\mathsf K(u-v)\phi_a(v)$ with $\mathsf K(x) = 32(N+2)\lambda^2C_M(x)^3$, and $\delta m^2_{(02)} = \hat{\mathsf K}(0)/\lambda^2 = 32(N+2)\sum_x\varepsilon^3C_M(x)^3$.

The formulas (52) follow from (i)–(iii) and the resolvent identity $G_{M_1} - G_m = (m^2 - M_1^2)G_{M_1}G_m$; the M-independent tadpole $3D(0)$ and the term $-2D(0)$ cancel in $S_{20,k}$. □

At $N=2$ the three coefficients are the terms $-16\lambda C_T(M)$, $-2e^2D_T - e^2J_T(M)$ and $128\lambda^2S_T(M)$ of the exact two-loop form of the torus coefficient in the torus paper [24, App. G, Lemma G.1], read on Λ in place of the torus; the orders $e^2\lambda$ and e^4 are the terms $16e^2\lambda F_T(M)$ and $e^4Q_T(M)$ there. Every graph of these two orders is an absolutely convergent integral over $\mathsf B^2$ at fixed ε (massive propagators, bounded trigonometric vertex factors), so $S_{21,k}$ and $S_{40,k}$ are finite real numbers for every k; their bounds, uniform in k, are the subject of the next subsection.

Lemma 22.6 (four lattice facts)**.** (L1) *For $p\in\mathsf B$: $cp^2\le\hat p^2\le p^2$ with $c = 4/\pi^2$. Hence, for a mass $a>0$, $\bar G_a(p) := (p^2+a^2)^{-1} \le G_a(p) \le \bar G^c_a(p) := (cp^2+a^2)^{-1}$ on $\mathsf B$, and for $a,b>0$: $\int_{\mathsf B}\bar G^c_a\bar G^c_b \le \int_{\mathbb R^3}\bar G^c_a\bar G^c_b\frac{d^3p}{(2\pi)^3} = \frac{\pi^2}{32(a+b)}$, $\int_{\mathsf B}(\bar G^c_a)^2 \le \frac{\pi^2}{64a}$.* (L2) *$C_M(x) = \varepsilon^{-3}\int_0^\infty e^{-tM^2}(e^{t\Delta_\varepsilon})_{x,0}\,dt$ with $(e^{t\Delta_\varepsilon})_{x,y} = e^{-6t/\varepsilon^2}\sum_{n\ge0}\frac{(t/\varepsilon^2)^n}{n!}\#\{n\text{-step nearest-neighbour paths } y\to x\} > 0$. Hence $C_M(x) > 0$ for every $x\in\Lambda$, $M\mapsto C_M(x)$ is strictly decreasing, $\sum_x\varepsilon^3C_M(x) = M^{-2}$, and the resolvent identity $C_{M_1} - C_m = (m^2-M_1^2)\,C_{M_1} * C_m$ holds pointwise with a positive right side, $(f*g)(x) := \sum_y\varepsilon^3 f(x-y)g(y)$.* (L3) *With $\int := \int_{\mathbb R^3}\frac{d^3p}{(2\pi)^3}$ and $a,b,c>0$: $\int\frac{1}{(p^2+a^2)(p^2+b^2)} = \frac{1}{4\pi(a+b)}$, $\int\frac{1}{(p^2+a^2)^2} = \frac{1}{8\pi a}$, $\int\frac{1}{(p^2+a^2)(p^2+b^2)(p^2+c^2)} = \frac{1}{4\pi(a+b)(b+c)(c+a)}$.* (L4) *If h_ε on $\mathsf B_\varepsilon$ satisfies $|h_\varepsilon|\le\bar h\in L^1(\mathbb R^3)$ and $h_\varepsilon(p)\to h(p)$ for every p as $\varepsilon\to0$, then $\int_{\mathsf B_\varepsilon}h_\varepsilon\to\int_{\mathbb R^3}h$. As $\varepsilon\to0$, $\hat p^2\to p^2$ at every fixed p, so $G_M(p)\to\bar G_M(p)$ and $\hat D(p)\to\bar G_{\mu_0}(p)$.*

Proof. (L1) $|\sin x|\le|x|$ and $\sin x\ge 2x/\pi$ on $[0,\pi/2]$, at $x = \varepsilon|p_\mu|/2\le\pi/2$; the integrals follow from (L3) by the scaling $p\mapsto c^{-1/2}p$, $c^{-3/2} = \pi^3/8$. (L2) $(-\Delta_\varepsilon + M^2)^{-1} = \int_0^\infty e^{-tM^2}e^{t\Delta_\varepsilon}dt$ as matrices on Λ, with kernel $\varepsilon^3C_M(x-y)$; the path expansion is $e^{t\Delta_\varepsilon} = e^{-6t/\varepsilon^2}e^{(t/\varepsilon^2)\mathcal A}$ with $\mathcal A$ the adjacency matrix; $\sum_x(e^{t\Delta_\varepsilon})_{x,0} = 1$; the identity is $G_{M_1} - G_m = (m^2-M_1^2)G_{M_1}G_m$ in Fourier space. (L3) Radially, $\frac{1}{2\pi^2}\int_0^\infty\frac{p^2dp}{(p^2+a^2)(p^2+b^2)} = \frac{1}{2\pi^2}\cdot\frac{\pi}{2(a+b)}$, the second integral is the limit

$b \to a$, and the third follows by partial fractions from the first. (L4) Dominated convergence after extension by 0. □

Lemma 22.7 (order λ). *For every k:* (i) $S_{01,k} > 0$; (ii) $J_\varepsilon \le \frac{\pi^2}{32(m+M_1)}$, *hence* $S_{01,k} \le \frac{(N+2)\pi^2(m-M_1)}{16}$; (iii) $J_{\varepsilon_k} \to \frac{1}{4\pi(m+M_1)}$, *hence* $S_{01,k} \to \frac{(N+2)(m-M_1)}{2\pi}$ *as* $k \to \infty$.

Proof. (i) (52) and $G_m G_{M_1} > 0$. (ii) (L1). (iii) (L4) with $\bar h = \bar G^c_m \bar G^c_{M_1}$, and (L3). □

Lemma 22.8 (order e^2). *For every k:* (i) $S_{20,k} < 0$; (ii) $0 < I_\varepsilon \le \bar I := \frac{\pi^2}{32(m+M_1)}$, *a bound free of* μ_0 *and* k; (iii) $I_{\varepsilon_k} \to I_\infty(\mu_0) := \frac{m\mu_0+mM_1+M_1\mu_0}{4\pi(m+M_1)(\mu_0+M_1)(m+\mu_0)}$, *hence* $S_{20,k} \to -\frac12(m^2-M_1^2)I_\infty(\mu_0) = -\frac{(m-M_1)(m\mu_0+mM_1+M_1\mu_0)}{8\pi(\mu_0+M_1)(m+\mu_0)}$; (iv) $\mu_0 \mapsto I_\infty(\mu_0)$ *is strictly decreasing on* $(0,\infty)$, *with* $I_\infty(\mu_0) \uparrow I^* := \frac{1}{4\pi(m+M_1)}$ *as* $\mu_0 \downarrow 0$.

Proof. (i) (52). (ii) $\hat k^2 \hat D(k) \in [0,1)$ and (L1). (iii) (L4) with $\bar h = \bar G^c_m \bar G^c_{M_1}$; the limit is $\int \frac{k^2}{(k^2+\mu_0^2)(k^2+m^2)(k^2+M_1^2)} = \frac{1}{4\pi(m+M_1)} - \frac{\mu_0^2}{4\pi(\mu_0+m)(m+M_1)(M_1+\mu_0)}$ by (L3), and $(\mu_0+M_1)(m+\mu_0) - \mu_0^2 = m\mu_0 + mM_1 + M_1\mu_0$. (iv) With $\mathcal N(\mu) = (m+M_1)\mu + mM_1$ and $\mathcal D(\mu) = (\mu+M_1)(m+\mu)$, $\mathcal N'\mathcal D - \mathcal N\mathcal D' = -(m+M_1)\mu^2 - 2mM_1\mu < 0$. □

Lemma 22.9 (order λ^2). *For every k:* (i) $F_\varepsilon > 0$, *so* $S_{02,k} < 0$; (ii) $F_\varepsilon \le 3(m^2-M_1^2)\big(\int_{\mathsf B} G^2_{M_1}\big)^2 \le \frac{3\pi^4(m^2-M_1^2)}{4096M_1^2}$, *hence* $|S_{02,k}| \le \bar L := \frac{3(N+2)\pi^4(m^2-M_1^2)}{256M_1^2}$; (iii) $F_{\varepsilon_k} \to \frac{\ln(m/M_1)}{16\pi^2}$, *hence* $S_{02,k} \to -L(M_1) := -\frac{(N+2)\ln(m/M_1)}{\pi^2}$. *The same holds with any $M > 0$ in place of M_1:* $\lim_k[\delta m^2_{(02)}(\varepsilon_k;m) - \delta m^2_{(02)}(\varepsilon_k;M)] = -\frac{N+2}{\pi^2}\ln\frac{m^2}{M^2}$ *(each $\delta m^2_{(02)}(\varepsilon_k;M)$ alone diverges like $\ln(1/\varepsilon_k)$).*

Proof. (i) By (L2), $C_{M_1}(x) > C_m(x) > 0$ pointwise, and the sum converges absolutely ($C_M \in \ell^1 \cap \ell^\infty$).

(ii) Pointwise $C^3_{M_1} - C^3_m = (C_{M_1} - C_m)(C^2_{M_1} + C_{M_1}C_m + C^2_m) \le 3(C_{M_1} - C_m)C^2_{M_1}$, and by (L2) $C_{M_1} - C_m = (m^2 - M_1^2)C_{M_1} * C_m \le (m^2 - M_1^2)C_{M_1} * C_{M_1}$. For $f,g,h \in \ell^1(\Lambda)$, $\sum_x \varepsilon^3 f(x)g(x)h(x) = \int_{\mathsf B}\int_{\mathsf B} \hat f(p)\hat g(q)\hat h(-p-q)$ (Fourier inversion and Fubini, $\hat h$ periodic); here $\hat f = G^2_{M_1}$, $\hat g = \hat h = G_{M_1}$, all even. By $G(q)G(p+q) \le \frac12[G(q)^2 + G(p+q)^2]$ and the measure-preserving shift $q \mapsto p+q$ of $\mathsf B$,

$$\int_{\mathsf B}\int_{\mathsf B} G_{M_1}(p)^2 G_{M_1}(q)G_{M_1}(p+q) \le \Big(\int_{\mathsf B} G^2_{M_1}\Big)^2 \le \Big(\frac{\pi^2}{64M_1}\Big)^2$$

by (L1). Multiply by $3(m^2 - M_1^2)$, and by $16(N+2)$ for $\bar L$.

(iii) Write $G_1 := G_{M_1}$, $G_2 := G_m$ (at spacing ε) and $\Delta := G_1 - G_2 = (m^2 - M_1^2)G_1G_2 \in [0, (m^2 - M_1^2)G_1^2]$, so that, by the formula of (ii), $F_\varepsilon = \int_{\mathsf B}\int_{\mathsf B}[G_1G_1G_1 - G_2G_2G_2](p,q,p+q) = T_1 + T_2 + T_3$ with $T_1 = \iint \Delta(p)G_1(q)G_1(p+q)$, $T_2 = \iint G_2(p)\Delta(q)G_1(p+q)$, $T_3 = \iint G_2(p)G_2(q)\Delta(p+q)$, all ≥ 0 (telescoping). Consider T_3; the other two are the same argument after the measure-preserving change of variables $(p,q) \mapsto (q, -p-q \bmod \mathsf B)$ on $\mathsf B \times \mathsf B$. Split $\mathsf B \times \mathsf B = N_\varepsilon \cup U_\varepsilon$ with $U_\varepsilon := \{(p,q) : |p_\mu + q_\mu| > \pi/\varepsilon \text{ for some } \mu\}$, the set where $p+q$ is folded back into $\mathsf B$. *The folding set is negligible:* on U_ε, $\max(|p|,|q|) \ge \pi/(2\varepsilon)$, so by (L1) $\min(G_2(p), G_2(q)) \le (c\pi^2/(4\varepsilon^2))^{-1} = \varepsilon^2$; with $\Delta \le (m^2 - M_1^2)G_1G_2$, the substitution $r = p+q$ (mod $\mathsf B$) at fixed q, and $\int_{\mathsf B} G_2 \le \int_{|p|\le\sqrt3\pi/\varepsilon} \frac{d^3p}{(2\pi)^3}\frac{1}{cp^2} = \frac{\sqrt3\pi}{8\varepsilon}$ (L1),

$$\iint_{U_\varepsilon} G_2(p)G_2(q)\Delta(p+q) \le 2(m^2-M_1^2)\,\varepsilon^2 \int_{\mathsf B} G_2 \int_{\mathsf B} G_1G_2 \le 2(m^2-M_1^2)\,\varepsilon^2 \cdot \frac{\sqrt3\pi}{8\varepsilon} \cdot \frac{\pi^2}{32(m+M_1)} \xrightarrow[\varepsilon\to0]{} 0.$$

The rest converges: on N_ε the integrand, extended by 0, converges pointwise (every fixed (p,q) lies in N_ε for small ε, and (L4)) to $(m^2-M_1^2)\bar G_m(p)\bar G_m(q)\bar G_{M_1}(p+q)\bar G_m(p+q)$ and is dominated by $(m^2-M_1^2)\bar G^c_{M_1}(p)\bar G^c_{M_1}(q)\bar G^c_{M_1}(p+q)^2 \in L^1(\mathbb R^6)$ (substitute $r = p+q$ and use $\bar G^c(q)\bar G^c(r-q) \le \frac12[\bar G^c(q)^2 + \bar G^c(r-q)^2]$; the integral is at most $(\int(\bar G^c_{M_1})^2)^2 < \infty$). So $T_3 \to T_3^\infty := (m^2 - M_1^2)\iint_{\mathbb R^6} \bar G_m(p)\bar G_m(q)\bar G_{M_1}(p+q)\bar G_m(p+q)$, and likewise T_1, T_2. *Back to position space:* with

the continuum Green's function $\bar C_M(x) = \frac{e^{-M|x|}}{4\pi|x|}$ (Fourier transform $\bar G_M$) and $\hat h := \bar G_{M_1}\bar G_m \in L^1\cap L^2$, $h = \bar C_{M_1} * \bar C_m = (\bar C_{M_1} - \bar C_m)/(m^2 - M_1^2)$, a bounded continuous function in L^2: for fixed p, $\int \bar G_m(q)\hat h(p+q)\,dq = \widehat{(\bar C_m h)}(p)$ as a Plancherel identity for the L^2 functions $\bar G_m$ and $\hat h$ ($\bar C_m h \in L^1\cap L^2$, being $O(|x|^{-1})$ near 0 and exponentially small at infinity), and $\int \bar G_m(p)\widehat{(\bar C_m h)}(p)\,dp = \int d^3x\, \bar C_m(x)(\bar C_m h)(x)$ by Plancherel again ($\bar C_m \in L^2(\mathbb{R}^3)$). Hence $T_3^\infty = \int d^3x\, \bar C_m^2(\bar C_{M_1} - \bar C_m)$, $T_1^\infty = \int \bar C_{M_1}^2(\bar C_{M_1} - \bar C_m)$, $T_2^\infty = \int \bar C_m \bar C_{M_1}(\bar C_{M_1} - \bar C_m)$, each absolutely convergent (the integrands are $O(|x|^{-2})$ at 0), and

$$\lim_{\varepsilon\to 0} F_\varepsilon = \int_{\mathbb{R}^3} d^3x\,\big[\bar C_{M_1}(x)^3 - \bar C_m(x)^3\big] = \frac{4\pi}{(4\pi)^3}\int_0^\infty \frac{e^{-3M_1x} - e^{-3mx}}{x}\,dx = \frac{\ln(m/M_1)}{16\pi^2}$$

by Frullani's integral $\int_0^\infty (e^{-ax} - e^{-bx})\frac{dx}{x} = \ln\frac{b}{a}$. Multiplying by $-16(N+2)$ gives the limit of $S_{02,k}$. For an arbitrary $M > 0$ in place of M_1 the same proof applies (for $M > m$ with the roles of the two masses exchanged); the divergence of a single $\delta m^2_{(02)}(\varepsilon; M)$ is that of $\sum_x \varepsilon^3 C_M(x)^3$, whose terms near $x = 0$ are of size ε^{-3} on $O(1)$ sites, with the logarithm from the region $\varepsilon \ll |x| \ll 1/M$, where $C_M(x) \approx (4\pi|x|)^{-1}$. □

Remark 22.10 (the one-loop form and the $e = 0$ limit). Lemmas 22.7 to 22.9 give the limits $\lim_k S_{01,k} = \frac{(N+2)(m-M_1)}{2\pi}$, $\lim_k S_{20,k} = -\frac12(m^2 - M_1^2)I_\infty(\mu_0)$ and $\lim_k S_{02,k} = -L(M_1)$ for every $M_1 \in (0, m)$ and every $\mu_0 > 0$: inserted into (51) they are the one-loop closed form of lemma 22.1(b)(ii) and, at $e = 0$, the exact limit $\lim_k s_k^\infty(M_1) = \frac12(m^2 - M_1^2) + \frac{(N+2)\lambda(m-M_1)}{2\pi} - \frac{(N+2)\lambda^2}{\pi^2}\ln\frac{m}{M_1}$ of proposition 22.15(vi). The constants $\bar I$, $\bar L$ and the limits of $S_{01,k}$, $S_{02,k}$ do not involve the photon mass; of the three low orders only I_∞ depends on μ_0, and $I_\infty(\mu_0) \le I^*$ for every $\mu_0 > 0$ (lemma 22.8(iv)).

The two-loop orders at every photon mass. From here to the end of the proof of proposition 22.15 we use the units $m = 1$ (λ and e^2 in units of m, σ in units of m^2), $N = 2$ and $M_1 = 0.7$; "$\varepsilon_k \le 1$" means $\varepsilon_k \le 1/m$. The two-loop coefficients $S_{21,k}$, $S_{40,k}$ are bounded from below through the torus paper's Appendix G [24], whose statements are applied with their hypotheses matched here; that appendix is written on periodic tori T of side $\ell_T \ge \ell_0$ with $\ell_T/\varepsilon \in 2\mathbb{N}$, in units $m = 1$, and its statements carry the reference photon mass $\mu_* \in (0, 1]$ of its photon propagator $\hat D = (\hat k^2 + \mu_*^2)^{-1}$ as a parameter; we read them at $\mu_* = \mu_0$. Its objects are the following.

(G1) *The torus coefficient* $a_{K,T}(M)$ [24, Def. 2.2]: King's recursion (50) computed on T with the periodic propagators of T at scalar mass M and photon mass μ_*. Read at $\mu_* = \mu_0$, this is the bare coefficient $\alpha^{\mathrm{tor}}_{k,T}(M) = M^2 + \delta m^2_{k,T}(M)$ of King's torus convention (definition 12.1(ii)): both are the same recursion with the same propagators. Its exact two-loop form [24, Lemma G.1] is, at $N = 2$,

$$a_{K,T}(M) = M^2 - 16\lambda C_T(M) - 2e^2 D_T - e^2 J_T(M) + 128\lambda^2 S_T(M) + 16e^2\lambda F_T(M) + e^4 Q_T(M),$$

with $C_T(M) = \int_T \hat G_M$, $D_T = \int_T \hat D$, $J_T(M) = M^2\int_T \hat D\hat G_M$, $S_T(M) = \sum_{x\in T}\varepsilon^3 G_M(x)^3$, F_T the order-$e^2\lambda$ and Q_T the order-e^4 coefficient, the term $\frac34\varepsilon^2 D_T^2$ of Q_T being independent of M; $M \mapsto a_{K,T}(M)$ is smooth. (The three low orders are lemma 22.5 on the torus.)

(G2) *The one-loop majorants* [24, Lemma G.15]: for $\alpha, \beta \ge 0$ with $g := \alpha + \beta > \frac32$, $a, b > 0$, $\nu := \min(a, b)$ and $\ell \in (0, \infty]$,

$$\mathcal{I}_{\alpha,\beta}(a, b; \ell) := c_N C_3 + \frac{3\pi}{16\ell}C_2 + \frac{3}{4\ell^2}C_1 + \frac{a^{-2\alpha}b^{-2\beta}}{\ell^3}, \qquad c_N = \frac{\pi^3}{8}, \tag{53}$$

each $C_d = C_d(\alpha, \beta; a, b)$ the minimum of its admissible closed forms: for C_3 the ν-form $\Gamma(g - \frac32)\nu^{3-2g}/((4\pi)^{3/2}\Gamma(g))$, the Beta form in a, $\frac{1}{4\pi^2}B(c, \alpha - c)a^{2c-2\alpha}$ with $c = \frac{3-2\beta}{2}$ if $\beta < \frac32$, the same with $(a, \alpha) \leftrightarrow (b, \beta)$ if $\alpha < \frac32$, the exact values $\frac{1}{4\pi(a+b)}$ at $(\alpha, \beta) = (1, 1)$ and $\frac{1}{8\pi a(a+b)^2}$ at $(2, 1)$, and the Hölder form $\gamma_3(g)a^{(3-2g)\alpha/g}b^{(3-2g)\beta/g}$, $\gamma_3(g)a^{3-2g}$ the closed form of $\int(u + a^2)^{-g}$; for C_2 the ν-form $\nu^{2-2g}/(g-1)$, the Beta form $a^{2(1-g)}B(1-\beta, g-1)$

if $0<\beta<1$ and its symmetric form, and the Hölder form; for C_1 the ν-form $\sqrt{\pi}\Gamma(g-\frac12)\nu^{1-2g}/\Gamma(g)$, the Beta form $a^{1-2g}B(\frac12-\beta,g-\frac12)$ if $0<\beta<\frac12$ and its symmetric form, and the Hölder form. For every torus of side ℓ and every ε, $\int_T(\hat k^2+a^2)^{-\alpha}(\hat k^2+b^2)^{-\beta}\le \mathcal{I}_{\alpha,\beta}(a,b;\ell)$, on $\varepsilon\mathbb{Z}^3$ with $\ell=\infty$; $\mathcal{I}_{\alpha,\beta}$ is decreasing in ℓ, and at $(a,b)=(M,\mu)$ every admissible form is non-increasing in M. The majorants, all at $(M,\mu;\ell_0)$ with $\mu=\mu_*$, are $\bar\tau=\mathcal{I}_{2,0}$, $\bar\psi=\mathcal{I}_{1,1}$, $\bar\delta=\mathcal{I}_{0,2}$, $\bar\Theta=\mathcal{I}_{1,3/4}+\mathcal{I}_{3/4,1}$, $\bar J_s=\mathcal{I}_{3-s/2,0}$ and $\bar J'_s=\mathcal{I}_{2-s/2,1}$ for $s\in\{0,1,\frac32\}$, $\bar\Lambda_0=\mathcal{I}_{3/4,1}$, $\bar\Lambda_1=\mathcal{I}_{5/4,1}+\mathcal{I}_{2,1/4}$, $\bar\Lambda_2=\mathcal{I}_{1/4,2}+\mathcal{I}_{1,5/4}$: fourteen distinct pairs (α,β), all with $g\ge\frac74>\frac32$; and $\bar\omega:=w+\hat E(\mu,\ell_0)$ with $w=0.252731$ (used as $w\le0.25273101$), $\hat E:=\min\{\bar E,\tilde E\}$, where $\bar E(\mu,\ell)=\frac{7.37}{\ell}+\frac{3\pi}{4\ell}E_1(\frac{\mu^2\ell^2}{8})+\frac{12\pi\,\mathrm{erfc}(\mu\ell/(2\sqrt2))}{\mu\ell^2}+\frac{64e^{-\mu^2\ell^2/8}}{\mu^2\ell^3}$ [24, Lemma H.31] and $\tilde E\ge0$ [24, Lemma G.14]; for $\varepsilon\le1$ and $\ell_T\ge\ell_0$, $\varepsilon D_T\le\bar\omega$ [24, Lemma G.14(c)], and on $\varepsilon\mathbb{Z}^3$ ($\ell_0=\infty$) one takes $\bar\omega=w$ [24, Lemma G.24].

(G3) *The slope bounds* [24, Lemma G.2]: for $\varepsilon\le1$, $\ell_T\ge\ell_0$, $M>0$, with the majorants at $(M,\mu_*;\ell_0)$, $\bar\beta:=2M[2\bar\tau+8\bar\psi]$ and

$$j:=2M\min\{\bar\tau,\bar\psi\},\qquad \varphi_3:=4M\big[2\sqrt2\,\bar\Theta\bar J_{3/2}+|\mu_*^2-2M^2|\bar\psi\bar J_0+3\sqrt3\,\bar\omega\bar J_1\big]+\bar\beta\bar\tau,$$
$$b_4:=\varphi_A+28M\bar\delta\bar\tau+\varphi_{D2},\quad \varphi_A:=6M\big[2\sqrt2\,\bar\Theta\bar J'_{3/2}+|\mu_*^2-2M^2|\bar\psi\bar J'_0+3\sqrt3\,\bar\omega\bar J'_1\big]+\bar\beta\bar\psi,$$
$$\varphi_{D2}:=2M\bar\psi(2\bar\psi+4\bar\delta)+\min\{8M\bar\psi(6\bar\psi+2\bar\tau+4\bar\delta),\ 2^{5/2}M(2\bar\Lambda_1+4\bar\Lambda_2)\bar\Lambda_0\},$$

one has $0\le\partial_M J_T\le j(M)$, $0\le-128\,\partial_M S_T\le768M\bar\tau(M)^2$, $|\partial_M F_T|\le\varphi_3(M)$ and $\partial_M Q_T\ge-b_4(M)$.

(G4) *The two-loop gap* [24, Prop. G.3]: for $0<M_1<M_2$, $\mu_*\in(0,1]$, $\ell_0>0$, every $\varepsilon\le1$ and every torus of side $\ell_T\ge\ell_0$, $a_{K,T}(M_2)-a_{K,T}(M_1)\ge(M_2^2-M_1^2)+16\lambda h_1-e^2\bar j-\lambda^2\bar s-e^2\lambda\bar B_3-e^4\bar B_4^-$ with $h_1\ge0$, $\bar j=\int_{M_1}^{M_2}j$, $\bar s=\int768M\bar\tau^2$, $\bar B_3=16\int\varphi_3$, $\bar B_4^-=\int b_4$; its proof writes each order as $\int_{M_1}^{M_2}\partial_M(\cdot)$ and applies (G3) *order by order*: $-[J_T(M_2)-J_T(M_1)]\ge-\bar j$, $128[S_T(M_2)-S_T(M_1)]\ge-\bar s$, $16[F_T(M_2)-F_T(M_1)]\ge-\bar B_3$, $Q_T(M_2)-Q_T(M_1)\ge-\bar B_4^-$.

(G5) *The cell rule* (the proof of [24, Cor. G.4]): for a density $\phi\in\{j,768M\bar\tau^2,16\varphi_3,b_4\}$ and a cell $[a,b]\subset(0,\infty)$ let $\phi^{\mathrm{cell}}[a,b](\mu,\ell_0)$ be the number obtained from the formula for ϕ by evaluating every majorant at the left endpoint a, the prefactor M at the right endpoint b, the prefactor $|\mu^2-2M^2|$ at its maximum over $\{a,b\}$ and $\bar\omega$ as it is; since the majorants are non-negative and non-increasing in M and the formulas are sums, products and minima with non-negative coefficients, $\phi\le\phi^{\mathrm{cell}}[a,b]$ on $[a,b]$, and for a partition $M_1=M^{(0)}<\cdots<M^{(n)}=M_2$

$$\int_{M_1}^{M_2}\phi\,dM\ \le\ \Phi_n(\mu,\ell_0):=\sum_{i=0}^{n-1}\big(M^{(i+1)}-M^{(i)}\big)\,\phi^{\mathrm{cell}}[M^{(i)},M^{(i+1)}](\mu,\ell_0).\tag{54}$$

We write $\bar j_n,\bar s_n,\bar B_{3,n},\bar B_{4,n}^-$ for the four upper sums on the uniform partition of $[0.7,1]$ into n cells, and $\bar X_n$ for any of them; $n=400$ unless said otherwise.

Lemma 22.11 (the four orders on Λ_k at every photon mass)**.** *Let $\mu_0\in(0,1]$, $\ell_0>0$, $n\ge1$. For every k with $\varepsilon_k\le1$:*

$$2S_{20,k}\ge-\bar j_n(\mu_0,\ell_0),\qquad 2S_{02,k}\ge-\bar s_n,\qquad 2S_{21,k}\ge-\bar B_{3,n}(\mu_0,\ell_0),\qquad 2S_{40,k}\ge-\bar B_{4,n}^-(\mu_0,\ell_0).$$

Proof. Fix such a k, and let $T=T_J$ be King's tori (definition 12.1; side $2MJ$, $2MJ\cdot5^k$ sites per side, an even number), J so large that $2MJ\ge\ell_0$. By (G1) at $\mu_*=\mu_0$, the coefficients of the torus-convention bare coefficient $\alpha_{k,T}^{\mathrm{tor}}(M)$ in the five orders are the terms of the exact two-loop form, so that for $S^T_{\alpha\beta,k}:=\frac12[\delta m^2_{(\alpha\beta),T}(\varepsilon_k;1)-\delta m^2_{(\alpha\beta),T}(\varepsilon_k;0.7)]$,

$$2S^T_{20,k}=-[J_T(1)-J_T(0.7)],\quad 2S^T_{02,k}=128[S_T(1)-S_T(0.7)],\quad 2S^T_{21,k}=16[F_T(1)-F_T(0.7)],\quad 2S^T_{40,k}=Q_T(1)-$$

the M-independent terms $-2D_T$ and $\frac34\varepsilon^2D_T^2$ cancelling in the differences. The hypotheses of (G3) and (G4) are met: $\varepsilon_k\le1$, $\ell_T\ge\ell_0$, $\mu_*=\mu_0\in(0,1]$, $M\in[0.7,1]$, and the lemmas behind (G3) carry μ_* as a parameter in their statements ((G2): $a,b>0$ and $g>\frac32$ for every instance;

$\varepsilon \le 1$ and $\ell_T \ge \ell_0$ for $\bar\omega$; $\ell_T/\varepsilon \in \mathbb{N}$, $\ell_T/\varepsilon \ge 2$ for $\bar E$). Hence, by the order-by-order bounds of (G4) with the majorants at (μ_0, ℓ_0) and by the cell rule (54), $2S^T_{20,k} \ge -\bar j_n(\mu_0,\ell_0)$, $2S^T_{02,k} \ge -\bar s_n$, $2S^T_{21,k} \ge -\bar B_{3,n}(\mu_0,\ell_0)$, $2S^T_{40,k} \ge -\bar B^-_{4,n}(\mu_0,\ell_0)$, uniformly in J ($\bar s$ does not depend on μ: $\bar\tau = \mathcal{I}_{2,0}$ has $\beta = 0$). Now let $J \to \infty$ at fixed k. By lemma 12.9, $\alpha^{\mathrm{tor}}_{k,T_J}(M) \to \alpha^{\infty}_k(M)$ for $M \in \{0.7, 1\}$, and its proof (appendix F.3) establishes the convergence of the recursion (50) order by order: $\delta m^2_{(\alpha\beta),T_J}(\varepsilon_k; M) \to \delta m^2_{(\alpha\beta)}(\varepsilon_k; M)$ for each of the five orders (alternatively: a polynomial of bounded degree in (e^2, λ) that converges at every point of an open set converges coefficient by coefficient, and lemma 12.9 holds at every point of King's range). So $S^{T_J}_{\alpha\beta,k} \to S_{\alpha\beta,k}$, and closed inequalities pass to the limit. □

Lemma 22.12 (the torus floor can be removed)**.** *For every $\mu_0 \in (0, 1]$, $n \ge 1$ and every k with $\varepsilon_k \le 1$, the four inequalities of lemma 22.11 hold with $\bar X_n(\mu_0, \infty)$, the upper sums formed with the $\varepsilon\mathbb{Z}^3$ majorants $\mathcal{I}_{\alpha,\beta}(M, \mu_0; \infty) = c_N C_3(M, \mu_0)$ and with $\bar\omega = w$.*

Proof. Fix k, μ_0, n. By lemma 22.11, $2S_{\alpha\beta,k} \ge -\bar X_n(\mu_0, \ell_0)$ for *every* $\ell_0 > 0$: the torus floor is a free parameter of (G4), and the tori T_J of the volume limit eventually exceed any ℓ_0. It therefore suffices that $\bar X_n(\mu_0, \ell_0) \to \bar X_n(\mu_0, \infty)$ as $\ell_0 \to \infty$. Each $\bar X_n$ is a finite sum of cell values, and each cell value is a fixed finite expression (sums, products and minima, with the fixed prefactors) in the majorants $\mathcal{I}_{\alpha,\beta}(a, \mu_0; \ell_0)$ at the left endpoint a of the cell and in $\bar\omega(\mu_0, \ell_0)$, hence continuous in these. As $\ell_0 \to \infty$, $\mathcal{I}_{\alpha,\beta}(a, \mu_0; \ell_0) \to c_N C_3(a, \mu_0)$ by (53), since C_2 and C_1 are finite numbers that do not depend on ℓ_0; and $0 \le \hat E(\mu_0, \ell_0) \le \bar E(\mu_0, \ell_0) \to 0$, each of the four terms of $\bar E$ being a positive constant times ℓ_0^{-1}, $\ell_0^{-1} E_1(\mu_0^2\ell_0^2/8)$, $\ell_0^{-2}\mathrm{erfc}(\mu_0\ell_0/(2\sqrt{2}))$ or $\ell_0^{-3} e^{-\mu_0^2\ell_0^2/8}$, so $\bar\omega \to w$. □

Lemma 22.13 (monotonicity in the photon mass; interval bounds)**.** *Let $0 < \mu_{\mathrm{lo}} \le \mu_{\mathrm{hi}} \le 1$. Define the* interval upper sum $\bar X^{\mathrm{int}}_n([\mu_{\mathrm{lo}}, \mu_{\mathrm{hi}}])$ *as $\bar X_n(\cdot, \infty)$ with every majorant evaluated at $\mu = \mu_{\mathrm{lo}}$ and, on the cell $[a, b]$, the prefactor $|\mu^2 - 2M^2|$ replaced by $\max\{|\mu^2 - 2M^2| : \mu \in \{\mu_{\mathrm{lo}}, \mu_{\mathrm{hi}}\}, M \in \{a, b\}\}$. Then $\bar X_n(\mu, \infty) \le \bar X^{\mathrm{int}}_n([\mu_{\mathrm{lo}}, \mu_{\mathrm{hi}}])$ for every $\mu \in [\mu_{\mathrm{lo}}, \mu_{\mathrm{hi}}]$, so the inequalities of lemma 22.12 hold with $\bar X^{\mathrm{int}}_n([\mu_{\mathrm{lo}}, \mu_{\mathrm{hi}}])$ for all $\mu_0 \in [\mu_{\mathrm{lo}}, \mu_{\mathrm{hi}}]$ at once. Moreover, if $\mu_{\mathrm{hi}} \le 0.989949$, the interval upper sum equals the point upper sum $\bar X_n(\mu_{\mathrm{lo}}, \infty)$.*

Proof. A cell value is non-decreasing in each majorant and in the prefactor (sums, products and minima of non-negative quantities with non-negative coefficients). (i) Each admissible closed form of C_3 is non-increasing in $b = \mu$: the ν-form, since $\nu = \min(a, b)$ is non-decreasing in b and $3 - 2g < 0$; the Beta form in a is b-free; the Beta form in b carries $b^{2c'-2\beta}$ with $c' = \frac{3-2\alpha}{2} < \beta$, as $g > \frac{3}{2}$; the Hölder form carries $b^{(3-2g)\beta/g}$; the exact forms $\frac{1}{4\pi(a+b)}$ and $\frac{1}{8\pi a(a+b)^2}$ decrease in b. A minimum of non-increasing functions is non-increasing, so every $\mathcal{I}_{\alpha,\beta}(a, \mu; \infty)$ is non-increasing in μ, and $\bar\omega = w$ does not depend on μ. (ii) $\mu \mapsto |\mu^2 - 2M^2|$ is convex in μ^2 and $M \mapsto |\mu^2 - 2M^2|$ convex in M^2, so the maximum over the rectangle $[\mu_{\mathrm{lo}}, \mu_{\mathrm{hi}}] \times [a, b]$ is attained at a corner. Hence every ingredient of the cell value at μ is at most the corresponding ingredient of the interval cell value. For the last claim: on $[0.7, 1]$ one has $2M^2 \ge 0.98 > 0.989949^2$, so $|\mu^2 - 2M^2| = 2M^2 - \mu^2$ on the whole rectangle, maximal at (μ_{lo}, b), which is also the point prefactor at μ_{lo}. (For the interval $[0.98, 0.99]$ the corner $(0.99, 0.7)$ of the first cell gives $|0.9801 - 0.98| = 10^{-4}$, below the value $2 \cdot 0.70075^2 - 0.98^2 = 0.0217$ at $(0.98, 0.70075)$, so the equality holds there too; only the interval $[0.99, 1]$ differs, by 8.5786210 against the point value 8.5785880 of $\bar B_3$.) □

Lemma 22.14 (a closed form, uniform in the photon mass)**.** *For $M \in [0.7, 1]$ put $\bar\tau^*(M) := \mathcal{I}_{2,0}(M; \infty) = \frac{c_N}{8\pi M}$, $\bar\psi^*(M) := \frac{c_N}{4\pi M}$, and for $X \in \{\Theta, J_s, J'_s\}$ let $X^*(M)$ be c_N times the minimum over the μ-free admissible forms of C_3 only (the Beta forms in $a = M$ and, for $\beta = 0$, the Hölder form $\gamma_3(g)M^{3-2g}$; the ν-forms, the Beta forms in b, the exact form at $(2, 1)$ and the Hölder forms with $\beta > 0$ are dropped, and the exact form at $(1, 1)$ is replaced by its bound $\frac{1}{4\pi a}$, which is $\bar\psi^*$). Put $\bar\beta^* := 2M[2\bar\tau^* + 8\bar\psi^*]$,*

$$\varphi^*_3(M) := 4M\big[2\sqrt{2}\,\bar\Theta^* \bar J^*_{3/2} + 2M^2\bar\psi^* \bar J^*_0 + 3\sqrt{3}\,w\,\bar J^*_1\big] + \bar\beta^*\bar\tau^*, \qquad \varphi^*_A(M) := 6M\big[2\sqrt{2}\,\bar\Theta^* \bar J'^*_{3/2} + 2M^2\bar\psi^* \bar J'^*_0 + 3\sqrt{}$$

$$P(M) := \varphi^*_A + 52M\bar\psi^{*2} + 16M\bar\psi^*\bar\tau^*, \qquad Q(M) := \frac{c_N}{8\pi} M\big(28\bar\tau^* + 40\bar\psi^*\big).$$

Then $16\varphi_3(M;\mu,\infty) \le 16\varphi_3^*(M)$ *and* $b_4(M;\mu,\infty) \le P(M) + Q(M)/\mu$ *for all* $M \in [0.7,1]$ *and* $\mu \in (0,1]$. *Consequently, with the* 400*-cell upper sums* $B_3^* := \sum h\,(16\varphi_3^*)^{\mathrm{cell}}$, $P^* := \sum hP^{\mathrm{cell}}$, $Q^* := \sum hQ^{\mathrm{cell}}$ *(majorants at the left endpoint, the prefactors* M *and* $2M^2$ *at the right endpoint), for every* $\mu_0 \in (0,1]$ *and every* k *with* $\varepsilon_k \le 1$*:*

$$2S_{21,k} \ge -B_3^*, \qquad 2S_{40,k} \ge -\big(P^*+Q^*/\mu_0\big), \qquad B_3^* = 13.187549, \quad P^* = 4.7629000, \quad Q^* = 0.77120851.$$

Proof. Each starred majorant is an upper bound of the corresponding majorant at $(M,\mu;\infty)$ for every $\mu > 0$: a minimum over a subset of the admissible forms is at least the minimum over all of them, and $\mathcal{I}_{1,1}(M,\mu;\infty) \le c_N \frac{1}{4\pi(M+\mu)} \le \frac{c_N}{4\pi M}$. Further $\bar\delta(\mu) = \mathcal{I}_{0,2}(M,\mu;\infty) \le c_N \frac{1}{4\pi^2} B(\frac32,\frac12)\mu^{-1} = \frac{c_N}{8\pi\mu}$ (the Beta form in b, admissible since $\alpha = 0 < \frac32$ and $\beta = 2 > \frac32$); $|\mu^2 - 2M^2| \le 2M^2$, since $\mu^2 \le 1 \le 4M^2$; $\bar\omega = w$ (lemma 22.12); and in φ_{D2} the minimum is at most its first branch. Substituting these bounds into the formulas (G3), which are non-decreasing in every majorant, gives $16\varphi_3 \le 16\varphi_3^*$ and

$$b_4 \le \varphi_A^* + 28M\bar\tau^*\frac{c_N}{8\pi\mu} + 2M\bar\psi^*\Big(2\bar\psi^* + \frac{4c_N}{8\pi\mu}\Big) + 8M\bar\psi^*\Big(6\bar\psi^* + 2\bar\tau^* + \frac{4c_N}{8\pi\mu}\Big) = P(M) + \frac{Q(M)}{\mu}.$$

The starred majorants are non-increasing in M (powers $M^{2c-2\alpha}$ with $c < \alpha$, and $1/M$), so the cell rule (54) applies to $16\varphi_3^*$, P and Q with the prefactors M, $2M^2$ at the right endpoint. The two inequalities follow *cellwise* from lemma 22.12: for a cell $[a,b]$ the cell value of $16\varphi_3$ (resp. b_4) at (μ_0,∞) is a non-decreasing expression in the majorants at $(a,\mu_0;\infty)$, in $\bar\omega$ and in the prefactor, each of which is at most its starred counterpart at a (with $\bar\delta \le c_N/(8\pi\mu_0)$, $\max_{M\in\{a,b\}}|\mu_0^2 - 2M^2| \le 2b^2$, $\bar\omega = w$); so the cell value is at most the cell value of $16\varphi_3^*$ (resp. of $P + Q/\mu_0$). Summing over the 400 cells, $\bar B_{3,400}(\mu_0,\infty) \le B_3^*$ and $\bar B^-_{4,400}(\mu_0,\infty) \le P^* + Q^*/\mu_0$, and lemma 22.12 gives the two inequalities. The three numbers are the 400-cell sums, computed as described after table 1. (The slope bounds (G3) are not invoked with the starred majorants in place of the one-loop majorants: as printed they make only $\bar\tau$, $\bar\psi$ generic; the domination above is all that is used.) □

The numbers. Let $\hat\mu = \mu_0/m \in (0,1]$. The grid is $\{1, 0.99, 0.98, 0.95, 0.9, 0.85, \dots, 0.15, 0.1, 0.08, 0.06, 0.05, 0.04$ (35 points), and the *rows* of table 1 are the 34 intervals $[\mu_{\mathrm{lo}}, \mu_{\mathrm{hi}}]$ between consecutive grid points. For $\hat\mu \in [10^{-4}, 1]$ the *row of* $\hat\mu$ is the row containing $\hat\mu$ (at a grid point, where two rows contain $\hat\mu$, the row with $\mu_{\mathrm{lo}} = \hat\mu$), and $\bar B_3(\hat\mu)$, $\bar B_4^-(\hat\mu)$, $\bar j(\hat\mu)$ denote the interval upper sums $\bar B^{\mathrm{int}}_{3,400}$, $\bar B^{-,\mathrm{int}}_{4,400}$, $\bar j^{\mathrm{int}}_{400}$ of that row (lemma 22.13), and $\bar s := \bar s_{400} = 6.5231964$; for $\hat\mu < 10^{-4}$, below the table, we read $\bar B_3(\hat\mu) := B_3^*$, $\bar B_4^-(\hat\mu) := P^* + Q^*/\hat\mu$ and $\bar j(\hat\mu) := 0.092610085$ (lemma 22.14 for the first two; the last is the μ-free branch $\int 2M\bar\tau$ of $\bar j$, see the proof of proposition 22.15(i)). With $A := \frac12(1 - M_1^2) = 0.255$, $b := \frac{4(1-M_1)}{2\pi} = 0.1909859$, $L := \frac{4\ln(1/M_1)}{\pi^2} = 0.1445549$ (lemmas 22.7 to 22.9 at $N = 2$, $M_1 = 0.7$), $I_\infty(\hat\mu) = \frac{0.7+1.7\hat\mu}{4\pi\cdot 1.7\,(\hat\mu+0.7)(1+\hat\mu)}$ (so $I_\infty(1) = 0.0330425$) and $I^* = \frac{1}{4\pi\cdot 1.7} = 0.0468103$, put

$$\sigma_{\mathrm{low}}(\hat\lambda,\hat e^2;\hat\mu) := A\big(1-\hat e^2 I_\infty(\mu_{\mathrm{lo}})\big) + b\hat\lambda - L\hat\lambda^2 - \tfrac12\bar B_3(\hat\mu)\,\hat e^2\hat\lambda - \tfrac12\bar B_4^-(\hat\mu)\,\hat e^4 \qquad (\hat\mu \in [10^{-4},1]), \tag{55}$$

$$\sigma^*(\hat\lambda,\hat e^2;\hat\mu) := A\big(1-\hat e^2 I^*\big) + b\hat\lambda - L\hat\lambda^2 - \tfrac12 B_3^*\,\hat e^2\hat\lambda - \tfrac12\big(P^* + Q^*/\hat\mu\big)\hat e^4 \qquad (\hat\mu \in (0,1]), \tag{56}$$

and $\sigma_{\mathrm{low}}(\cdot;\hat\mu) := \sigma^*(\cdot;\hat\mu)$ for $\hat\mu < 10^{-4}$. The derived columns of the table are: $e^2_{\max}$, the positive root of $\sigma_{\mathrm{low}}(0,\cdot;\hat\mu)$, $e^2_{\max} = \big(-AI_\infty + \sqrt{A^2I_\infty^2 + 2A\bar B_4^-}\big)/\bar B_4^-$; the side c of the largest box $[0,c]^2$ on which $\sigma_{\mathrm{low}} \ge 0$; and the value $\sigma_{\mathrm{low}}(0.1,0.1;\hat\mu)$, where a negative entry means only that the bound does not place $[0,0.1]^2$ in the region.

How the entries were computed, and what is exact. The upper sums are the finite sums (54) over the 400 cells of the formulas (G2), (G3), with the interval rule of lemma 22.13; their validity as upper bounds is the inequality (54), which is exact once the finitely many closed forms are evaluated correctly. At $\ell_0 = \infty$ only the $c_N C_3$ term of (53) and $\bar\omega = w \le 0.25273101$ enter, so no series and no special function beyond Γ and B is involved. The evaluation was carried out in floating-point arithmetic with 50 significant digits; every final upper sum was multiplied

$[\mu_{\mathrm{lo}}, \mu_{\mathrm{hi}}]$	$\bar{j}$	$\bar{B}_3$	$\bar{B}_4^-$	$AI_\infty(\mu_{\mathrm{lo}})$	$e^2_{\max}(\lambda=0)$	c	$\sigma_{\mathrm{low}}(0.1,0.1)$
$[0.99, 1]$	0.085313936	8.5786210	1.5854190	0.008458	0.5618	0.2390	0.2209
$[0.98, 0.99]$	0.085757165	8.6036471	1.6023646	0.008491	0.5588	0.2385	0.2207
$[0.95, 0.98]$	0.087020288	8.6804035	1.6554211	0.008589	0.5498	0.2369	0.2201
$[0.9, 0.95]$	0.088887935	8.8139269	1.7519811	0.008757	0.5345	0.2341	0.2189
$[0.85, 0.9]$	0.090427040	8.9551127	1.8603008	0.008930	0.5188	0.2313	0.2176
$[0.8, 0.85]$	0.091595352	9.1048479	1.9826438	0.009108	0.5026	0.2283	0.2163
$[0.75, 0.8]$	0.092342924	9.2641791	2.1218953	0.009291	0.4858	0.2251	0.2147
$[0.7, 0.75]$	0.092610085	9.4343534	2.2812536	0.009480	0.4686	0.2217	0.2131
$[0.65, 0.7]$	0.092610085	9.6168755	2.4625429	0.009673	0.4511	0.2182	0.2112
$[0.6, 0.65]$	0.092610085	9.8135862	2.6698136	0.009871	0.4333	0.2145	0.2092
$[0.55, 0.6]$	0.092610085	10.026776	2.9084862	0.010073	0.4152	0.2105	0.2069
$[0.5, 0.55]$	0.092610085	10.259350	3.1856310	0.010279	0.3969	0.2063	0.2044
$[0.45, 0.5]$	0.092610085	10.515087	3.5155645	0.010488	0.3779	0.2017	0.2014
$[0.4, 0.45]$	0.092610085	10.799045	3.9183359	0.010697	0.3580	0.1967	0.1979
$[0.35, 0.4]$	0.092610085	11.118258	4.4224644	0.010906	0.3371	0.1912	0.1938
$[0.3, 0.35]$	0.092610085	11.477171	5.0733664	0.011111	0.3148	0.1850	0.1887
$[0.25, 0.3]$	0.092610085	11.851059	5.9437544	0.011309	0.2910	0.1781	0.1825
$[0.2, 0.25]$	0.092610085	12.193509	7.0632067	0.011495	0.2670	0.1709	0.1752
$[0.15, 0.2]$	0.092610085	12.433452	8.5440078	0.011662	0.2429	0.1635	0.1665
$[0.1, 0.15]$	0.092610085	12.659419	11.303445	0.011801	0.2113	0.1525	0.1516
$[0.08, 0.1]$	0.092610085	12.756284	13.316343	0.011846	0.1948	0.1460	0.1411
$[0.06, 0.08]$	0.092610085	12.857230	16.620716	0.011884	0.1744	0.1370	0.1240
$[0.05, 0.06]$	0.092610085	12.909321	19.239396	0.011899	0.1621	0.1310	0.1107
$[0.04, 0.05]$	0.092610085	12.962542	23.145186	0.011912	0.1479	0.1233	0.0909
$[0.03, 0.04]$	0.092610085	13.016932	29.623519	0.011923	0.1308	0.1132	0.0582
$[0.02, 0.03]$	0.092610085	13.072534	42.530531	0.011931	0.1092	0.0987	−0.0066
$[0.015, 0.02]$	0.092610085	13.100802	55.411533	0.011933	0.0957	0.0887	−0.0712
$[0.01, 0.015]$	0.092610085	13.129390	81.146532	0.011935	0.0791	0.0753	−0.2000
$[0.005, 0.01]$	0.092610085	13.158304	158.29597	0.011937	0.0566	0.0555	−0.5859
$[0.002, 0.005]$	0.092610085	13.175811	389.67594	0.011937	0.0361	0.0360	−1.7428
$[0.001, 0.002]$	0.092610085	13.181673	775.28604	0.011937	0.0256	0.0256	−3.6709
$[0.0005, 0.001]$	0.092610085	13.184609	1546.4975	0.011937	0.0181	0.0181	−7.5270
$[0.0002, 0.0005]$	0.092610085	13.186373	3860.1248	0.011937	0.0114	0.0114	−19.095
$[0.0001, 0.0002]$	0.092610085	13.186961	7716.1679	0.011937	0.0081	0.0081	−38.375

TABLE 1. The interval upper sums $\bar{j}$, $\bar{B}_3$, $\bar{B}_4^-$ of lemma 22.13 on the 400-cell partition of $[0.7, 1]$ at $\ell_0 = \infty$ (lemma 22.12), valid for every $\hat{\mu}$ in the row; $\bar{s} = 6.5231964$ on every row. The three upper sums are rounded upward to eight significant digits; the derived columns $e^2_{\max}$, c are rounded down and $\sigma_{\mathrm{low}}(0.1, 0.1)$ is rounded down. Units $m = 1$, $N = 2$, $M_1 = 0.7$.

by $1 + 10^{-30}$ and rounded upward to eight significant digits, the derived region numbers were computed from the rounded upper sums with coefficients rounded up and sizes rounded down, and a second run at 30 digits agrees with the first to all printed digits. The rounding of the elementary and Gamma functions at 50 digits is many orders below the safety factor and the final rounding; interval validity rests on lemma 22.13 and finitely many evaluations, not on interval arithmetic. The same evaluation at $\hat{\mu} = 1$ and $\ell_0 = 96$, with $\bar{\omega} = w + \min\{\bar{E}, \tilde{E}\}$ as in (G2), reproduces the four numbers 0.0886293, 7.179064, 9.330427, 1.703784 of [24, Cor. G.4] to every printed digit (it gives 0.0886292849, 7.17906354, 9.3304261, 1.70378338); this checks the transcription of the formulas and of the set of admissible forms against the torus paper's printed certificate, which exercises every form that is active at $\mu = 1$. Internal consistency of the table: $\bar{s}_{400} = 6.5231964$ at every $\hat{\mu}$ ($\bar{s}$ is μ-free); $\bar{j} = 0.092610085$ for all $\hat{\mu} \le 0.7$, the μ-free branch $\int 2M\bar{\tau}$ of $\bar{j}$ (analytically $\frac{c_N}{4\pi}(1-M_1) = 0.09253$ before the cell rounding); the interval values equal the point values at μ_{lo} except on $[0.99, 1]$ (lemma 22.13); $\bar{B}^-_{4,400}(\hat{\mu}, \infty) \le P^* + Q^*/\hat{\mu}$ at every grid point ($7716.17 \le 7716.85$ at $\hat{\mu} = 10^{-4}$, $1.5854 \le 5.534$ at $\hat{\mu} = 1$); and $\hat{\mu}\,\bar{B}_4^-(\hat{\mu}) \to 0.7716$ at $\hat{\mu} = 10^{-4}$ against $Q^* = 0.7712$. The closed form of lemma 22.14 is weaker than the table near $\hat{\mu} = 1$ (it uses $\bar{\psi}^*$, $|\mu^2 - 2M^2| \le 2M^2$ and the first branch of the minimum) and is within one per cent of it for $\hat{\mu} \le 10^{-2}$.

Proposition 22.15 (the explicit coupling region of (T3) at every photon mass)**.** *Let $N = 2$, $M_1 = 0.7m$ and $\mu_0 \in (0, m]$; write $\hat\lambda = \lambda/m$, $\hat e^2 = e^2/m$, $\hat\mu = \mu_0/m$. Let $\delta m_k^2(M)$ be King's counterterm (50) on Λ_k at the photon mass μ_0, $S_{\alpha\beta,k}$ the coefficients of lemma 22.4, $\sigma_1(M_1) = \liminf_k s_k^\infty(M_1)$ as in lemma 3.9, and let $\sigma_{\rm low}$, σ^*, $\bar j(\hat\mu)$, $\bar s$, $\bar B_3(\hat\mu)$, $\bar B_4^-(\hat\mu)$, A, b, L, I_∞, I^* be as defined before table 1, B_3^*, P^*, Q^* as in lemma 22.14, and $\lambda_c := \frac{b+\sqrt{b^2+4AL}}{2L} = 2.14398\ldots$.*

(i) (The five orders.) *For every k with $\varepsilon_k \le 1/m$, in units $m = 1$: $S_{01,k} > 0$; $2S_{20,k} \ge -\bar j(\hat\mu) \ge -0.092611$; $2S_{02,k} \ge -\bar s = -6.5231964$; $2S_{21,k} \ge -\bar B_3(\hat\mu)$ and $2S_{21,k} \ge -B_3^*$; $2S_{40,k} \ge -\bar B_4^-(\hat\mu)$ and $2S_{40,k} \ge -(P^* + Q^*/\hat\mu)$.*

(ii) (The lower bound.) $\sigma_1(0.7m) \ge m^2 \max\{\sigma_{\rm low}(\hat\lambda, \hat e^2; \hat\mu),\ \sigma^*(\hat\lambda, \hat e^2; \hat\mu)\}$.

(iii) (The region.) (T3) *holds on $R(\mu_0) := \{(\lambda, e^2) : \sigma_{\rm low}(\hat\lambda, \hat e^2; \hat\mu) > 0\}$, with $\sigma_1(0.7m) \ge m^2\sigma_{\rm low} > 0$ there. $R(\mu_0)$ contains the box $[0, c(\hat\mu)m]^2$ and the segment $\{\lambda = 0,\ e^2 < e^2_{\max}(\hat\mu)m\}$, with c and $e^2_{\max}$ from table 1 for $\hat\mu \in [10^{-4}, 1]$, and the segment $\{e = 0,\ \lambda < \lambda_c m\}$ for every $\hat\mu$. The map $\hat\mu \mapsto \sigma_{\rm low}(\hat\lambda, \hat e^2; \hat\mu)$ is non-decreasing (on $[10^{-4}, 1]$ a step function of $\hat\mu$, constant on each row of the table; below 10^{-4} the closed form σ^*), so $R(\mu_0) \supseteq R(\mu_0')$ for $\mu_0 \ge \mu_0'$. In particular $[0, 0.2m]^2 \subseteq R(\mu_0)$ for $\mu_0 \in [0.5m, m]$; $[0, 0.15m]^2 \subseteq R(\mu_0)$ for $\mu_0 \in [0.1m, m]$; $[0, 0.1m]^2 \subseteq R(\mu_0)$ for $\mu_0 \in [0.03m, m]$, with $\sigma_1(0.7m) \ge 0.0582\, m^2$ at $(\lambda, e^2) = (0.1m, 0.1m)$; $[0, 0.075m]^2 \subseteq R(\mu_0)$ for $\mu_0 \in [0.01m, m]$; and at $\mu_0 = m$, $R(m)$ contains the box $[0, 0.239m]^2$ and the segment $\{\lambda = 0,\ e^2 < 0.5618\, m\}$, and $\sigma_1(0.7m) \ge 0.2209\, m^2$ on $[0, 0.1m]^2$.*

(iv) (Uniform in the photon mass; closed form.) (T3) *holds whenever $\sigma^*(\hat\lambda, \hat e^2; \hat\mu) > 0$, with $\sigma_1(0.7m) \ge m^2\sigma^*$; in particular, for every $\mu_0 \in (0, m]$, on each of the boxes $\{\lambda \le \lambda_1 m,\ e^2 \le c_1\sqrt{\mu_0 m}\}$ with $(\lambda_1, c_1) \in \{(0, 0.3014), (0.05, 0.2533), (0.1, 0.2152), (0.2, 0.1616), (0.5, 0.0884), (1$*

(v) (Every k.) *For every k with $\varepsilon_k \le 1/m$: $s_k^\infty(0.7m) \ge m^2\big[A - \frac12\bar j(\hat\mu)\hat e^2 - \frac12\bar s\hat\lambda^2 - \frac12\bar B_3(\hat\mu)\hat e^2\hat\lambda - \frac12\bar B_4^-(\hat\mu)\hat e^4\big]$.*

(vi) ($e = 0$; exact.) *At $e = 0$, for every $M_1 \in (0, m)$, every $\mu_0 > 0$ and every $N \ge 1$: $\lim_k s_k^\infty(M_1) = \frac12(m^2 - M_1^2) + \frac{(N+2)\lambda(m - M_1)}{2\pi} - \frac{(N+2)\lambda^2}{\pi^2}\ln\frac{m}{M_1}$ exactly, and* (T3) *holds if and only if $\lambda < \frac{\pi}{4}\big(1 + \sqrt{1 + \frac{16}{N+2}}\big)m$, which is $\frac{\pi}{4}(1 + \sqrt5)\, m = 2.5416\ldots\, m$ for $N = 2$; at $M_1 = 0.7m$, $N = 2$, the exact window is $\lambda < \lambda_c m$, $\lambda_c = 2.14398\ldots$, in particular $\lambda \le 2.1439\, m$.*

(vii) (Two-sided.) *On each region of (iii), (iv) and (vi) the two-sided clause* (NE) *holds as well, with no use of proposition 12.11.*

Outside these regions the sign clause (T3) is not decided here. The region is explicit at every photon mass but not uniform in it: by (56), $e^2_{\max}(\hat\mu) \sim \sqrt{2A/Q^*}\sqrt{\hat\mu} = 0.81\sqrt{\hat\mu}$ as $\hat\mu \to 0$ (table 1: $\hat\mu\,\bar B_4^-(\hat\mu) \to 0.77$), and $\bigcap_{\mu_0 \in (0,m]} R(\mu_0) = \{e = 0,\ \lambda < \lambda_c m\}$, since for $e > 0$ the term $-\frac12(Q^*/\hat\mu)\hat e^4$ of σ^* dominates as $\hat\mu \to 0$. The $1/\hat\mu$ sits in the order-e^4 majorant alone: the low orders and B_3^* are uniform in the photon mass.

Proof of proposition 22.15. Units $m = 1$, $N = 2$, $M_1 = 0.7$; $\hat\mu = \mu_0 \in (0, 1]$.

(i). $S_{01,k} > 0$ is lemma 22.7(i). The four bounds with the interval upper sums are lemma 22.12 with lemma 22.13, applied to the row of $\hat\mu$ (for $\hat\mu \ge 10^{-4}$); the bounds with B_3^* and $P^* + Q^*/\hat\mu$ are lemma 22.14; for $\hat\mu < 10^{-4}$ the first bound is the μ-free branch of $\bar j$: $j \le 2M\bar\tau$ gives $\bar j_{400}(\hat\mu, \infty) \le \sum h\,(2M\bar\tau^*)^{\rm cell} = 0.092610085$ for every $\hat\mu$, by the cell rule (54). The numbers $\bar j \le 0.092610085 < 0.092611$ and $\bar s = 6.5231964$ are read off table 1. The interval value $\bar B_4^-(\hat\mu)$, taken at $\mu_{\rm lo}$, and the closed form $P^* + Q^*/\hat\mu$, taken at $\hat\mu$, are two separate bounds, neither of which dominates the other on a whole row.

(ii). By (51) and lemma 22.7(iii), lemma 22.8(iii), lemma 22.9(iii), the first four coefficients converge, so

$$\sigma_1(0.7) = \liminf_k s_k^\infty(0.7) = A + b\hat\lambda - \hat e^2 A I_\infty(\hat\mu) - L\hat\lambda^2 + \liminf_k\big(\hat e^2\hat\lambda S_{21,k} + \hat e^4 S_{40,k}\big),$$

and by (i), for all but finitely many k (those with $\varepsilon_k \le 1$), $\hat e^2\hat\lambda S_{21,k} + \hat e^4 S_{40,k} \ge -\frac12\bar B_3(\hat\mu)\hat e^2\hat\lambda - \frac12\bar B_4^-(\hat\mu)\hat e^4$; with $I_\infty(\hat\mu) \le I_\infty(\mu_{\rm lo})$ (lemma 22.8(iv)) this is $\sigma_1(0.7) \ge \sigma_{\rm low}$. The same argument

with the closed-form bounds of (i) and $I_\infty(\hat\mu) \le I^*$ gives $\sigma_1(0.7) \ge \sigma^*$. (Cellwise, the interval value of $16\varphi_3$ at $\mu_{\rm lo}$ is at most the starred cell value, so $\bar B_3(\hat\mu) \le B_3^*$ always holds; for b_4 the interval value carries $\bar\delta(\mu_{\rm lo}) \approx c_N/(8\pi\mu_{\rm lo})$ while the closed form carries $Q^*/\hat\mu$, so neither bound dominates the other on a whole row; hence the maximum in (ii).)

(iii). The first sentence is (ii) and the definition of (T3) (Hypothesis 3.11). $\sigma_{\rm low}$ is concave in $\hat\lambda$ at fixed $\hat e^2$ (the term $-L\hat\lambda^2$) and concave in $\hat e^2$ at fixed $\hat\lambda$ (the term $-\frac12\bar B_4^-\hat e^4$), so its minimum over a box $[0,c]^2$ is attained at a corner: the minimum over $\hat\lambda \in [0,c]$ of a concave function is attained at an endpoint, the two resulting functions of $\hat e^2$ are concave and so is their minimum, whose minimum over $[0,c]$ is again at an endpoint. The side $c(\hat\mu)$ of table 1 is the largest c for which the four corner values are ≥ 0, rounded down; $e^2_{\max}$ is the positive root of $\sigma_{\rm low}(0,\cdot) = A(1-\hat e^2 I_\infty) - \frac12\bar B_4^-\hat e^4$, rounded down; and at $e=0$, $\sigma_{\rm low} = A + b\hat\lambda - L\hat\lambda^2 > 0$ if and only if $\hat\lambda < \lambda_c = \frac{b+\sqrt{b^2+4AL}}{2L} = 2.14398\ldots$ (as in (vi)). *Monotonicity in $\hat\mu$:* on $[10^{-4},1]$, $\sigma_{\rm low}(\cdot;\hat\mu)$ is a step function of $\hat\mu$, built row by row from $I_\infty(\mu_{\rm lo})$ and the interval values $\bar B_3$, $\bar B_4^-$ of the row, which enter with negative coefficients; so it suffices that these three row values do not decrease from each row to the row below it. For every row but the top one the interval values are the point values at $\mu_{\rm lo}$ (lemma 22.13, last clause, and the remark at the end of its proof for the row $[0.98,0.99]$), the point values are non-increasing in μ (lemma 22.13(i), with the point prefactor $\max_{M\in\{a,b\}}|\mu^2-2M^2| = 2b^2-\mu^2$ non-increasing in μ for $\mu \le 0.989949$) and so is I_∞ (lemma 22.8(iv)); hence the values of a row, taken at its $\mu_{\rm lo}$, are at most those of the row below, whose left end is smaller. For the top row $[0.99,1]$ this is read off the table ($8.5786210 \le 8.6036471$, $1.5854190 \le 1.6023646$, $0.008458 \le 0.008491$); by inspection each of the three columns is non-decreasing down the table through all 34 rows. Across $\hat\mu = 10^{-4}$: σ^* is non-decreasing in $\hat\mu$, and $\sigma^*(\cdot;\hat\mu) \le \sigma_{\rm low}(\cdot;10^{-4})$ for $\hat\mu < 10^{-4}$ because $I^* \ge I_\infty(10^{-4})$, $B_3^* \ge \bar B_3(10^{-4}) = 13.186961$ and $P^* + Q^*/\hat\mu > 7716.84 > \bar B_4^-(10^{-4}) = 7716.1679$. At a grid point both rows containing it give valid bounds, and the convention selects the row with $\mu_{\rm lo} = \hat\mu$. Hence $\hat\mu \mapsto \sigma_{\rm low}(\hat\lambda,\hat e^2;\hat\mu)$ is non-decreasing and $R(\mu_0) \supseteq R(\mu_0')$ for $\mu_0 \ge \mu_0'$. *The listed boxes:* $c \ge 0.2063$ on the row $[0.5,0.55]$, $c \ge 0.1525$ on $[0.1,0.15]$, $c \ge 0.1132$ and $\sigma_{\rm low}(0.1,0.1) \ge 0.0582$ on $[0.03,0.04]$, $c \ge 0.0753$ on $[0.01,0.015]$ (table 1), together with the monotonicity; at $\hat\mu = 1$ the row $[0.99,1]$ gives $c \ge 0.2390$ and $e^2_{\max} \ge 0.5618$.

(iv). σ^* is concave in each variable separately, so again the corners decide. Let $(\hat\lambda,\hat e^2)$ lie in the box $\{\hat\lambda \le \lambda_1,\ \hat e^2 \le c_1\sqrt{\hat\mu}\}$ with $\hat\mu \le 1$. At a corner $(\hat\lambda, c_1\sqrt{\hat\mu})$, $\hat\lambda \in \{0,\lambda_1\}$,

$$\sigma^* = A - AI^*c_1\sqrt{\hat\mu} + b\hat\lambda - L\hat\lambda^2 - \tfrac12 B_3^* c_1\sqrt{\hat\mu}\,\hat\lambda - \tfrac12\big(P^*\hat\mu + Q^*\big)c_1^2 \;\ge\; A - AI^*c_1 + b\hat\lambda - L\hat\lambda^2 - \tfrac12 B_3^* c_1\hat\lambda - \tfrac12(P^*+Q^*)c_1^2 = \sigma^*$$

using $\sqrt{\hat\mu} \le 1$ and $\hat\mu \le 1$ in the negative terms; the corners with $e=0$ give $A>0$ and $A + b\lambda_1 - L\lambda_1^2 > 0$ ($\lambda_1 \le 1 < \lambda_c$). So it suffices that $\sigma^*(0,c_1;1) \ge 0$ and $\sigma^*(\lambda_1,c_1;1) \ge 0$ for each listed pair, which is how the c_1 were determined (bisection on c_1, rounded down); explicitly, $\sigma^*(0,c_1;1) = A(1-I^*c_1) - \frac12(P^*+Q^*)c_1^2 = 3.8\cdot10^{-5}$ at $c_1 = 0.3014$, and $\sigma^*(\lambda_1,c_1;1)$ equals $1.2\cdot10^{-4}$, $4.1\cdot10^{-5}$, $1.1\cdot10^{-4}$, $2.3\cdot10^{-4}$, $6.3\cdot10^{-4}$ at the five pairs with $\lambda_1 > 0$, each positive.

(v). (51), $\hat\lambda S_{01,k} \ge 0$ (lemma 22.7(i)) and (i).

(vi). At $e=0$ only the orders $(0,1)$ and $(0,2)$ of (50) are present, so by lemma 22.7(iii), lemma 22.9(iii) (for an arbitrary $M_1 \in (0,m)$, and now in general units) $\lim_k s_k^\infty(M_1) = g_\lambda(M_1) := (m-M_1)\big[\frac{m+M_1}{2} + \frac{(N+2)\lambda}{2\pi}\big] - \frac{(N+2)\lambda^2}{\pi^2}\ln\frac{m}{M_1}$ exactly, and $\sigma_1(M_1) = g_\lambda(M_1)$. As a function of $M_1 \in (0,m]$: $g_\lambda(m) = 0$, $g_\lambda'(M_1) = -M_1 - \frac{(N+2)\lambda}{2\pi} + \frac{(N+2)\lambda^2}{\pi^2 M_1}$, $g_\lambda''(M_1) = -1 - \frac{(N+2)\lambda^2}{\pi^2 M_1^2} < 0$, so g_λ is strictly concave. If $g_\lambda'(m) \ge 0$, concavity gives $g_\lambda(M_1) < g_\lambda(m) + g_\lambda'(m)(M_1-m) \le 0$ for every $M_1 < m$, and (T3) fails; if $g_\lambda'(m) < 0$, then $g_\lambda > 0$ on a left neighbourhood of m, and (T3) holds. Now $g_\lambda'(m) < 0$ if and only if $x = \lambda/m$ lies below the positive root of $\frac{N+2}{\pi^2}x^2 - \frac{N+2}{2\pi}x - 1$, which is $x_* = \frac{\pi}{4}\big(1 + \sqrt{1+\frac{16}{N+2}}\big)$, equal to $\frac{\pi}{4}(1+\sqrt5) = 2.5416\ldots$ for $N=2$. At a fixed M_1, $g_\lambda(M_1) > 0$ if and only if $L\lambda^2 - b\lambda - A < 0$ with A, b, L at M_1, i.e. $\lambda < \lambda_c(M_1)m := \frac{b+\sqrt{b^2+4AL}}{2L}m$; at $M_1 = 0.7m$, $\lambda_c = 2.14398\ldots$.

(vii). By theorem F.159(b) the relative limit $\alpha_\infty(M) - \alpha_\infty(m)$ exists for every $M > 0$ and $\sigma_1(M_1) = \lim_k s_k^\infty(M_1) = \frac12(\alpha_\infty(m) - \alpha_\infty(M_1))$. On each region of (iii), (iv), (vi), $\sigma_1(0.7m) > 0$

(resp. $\sigma_1(M_1) > 0$), so $\alpha_\infty(0.7m) < \alpha_\infty(m)$: m is not a minimiser of α_∞ on $(0, m]$, which is (NE$^<$), and (NE$^<$) implies (NE) (lemma 22.1(b)(i)). Proposition 12.11 is needed only on $E_W^< \setminus E_W$, which these regions avoid. □

Remark 22.16 (the quantifier over the couplings). The theorems of this paper print the sources' quantifier of lemma 22.1(e), in the units of the coarsest spacing, with the cube family C_J not rescaled and $\beta = 5^{-J_K}$. A sentence "e, λ sufficiently small, $m, \mu_0 > 0$ fixed" describes *where* the smallness lives (the running couplings, through Bałaban's ε_0), but as a hypothesis on the bare couplings it is strictly stronger than what the sources need. (i) All four couplings have mass dimension one, so its dimensionless content is a bound on e^2/m and λ/m, while the sources' hypothesis on the bare couplings is empty beyond positivity: the point $(e^2, \lambda, m, \mu_0) = (100, 100, 1, 1)$ is admitted by every source (with a large J_K) and excluded by the phrase in every system of units. (ii) In the units in which the stopping scale is 1, m and μ_0 are not fixed but rescaled to $m\beta$ and $\mu_0\beta$, of which $m\beta$ is capped (by the mass cap of [2, Prop. 3.1] and by a further cap of the schedule) and $\mu_0\beta$ is not: in those units the theorem reads "three couplings small, the fourth rescaled". (iii) Even at a fixed stopping scale β_0 the admissible set is not a product neighbourhood of $(0, 0)$ in (e, λ): one cap of the schedule of constants of the rotation part has the form $e \le \lambda^{1/4} c(\beta_0)$, a cusp that fails as $\lambda \to 0$ at fixed e. So the phrase names no set exactly, and the sources' form survives because β can be taken to 0.

23. Open problems

Throughout, K denotes King's printed state of theorem 21.1: the Neumann–Dirichlet cube limit on King's RG cube at block factor 5, fixed photon mass $\mu_0 > 0$, with an admissible scheme a in the Wick-square sector; S denotes a torus limit state of the authors' earlier paper [24] (block factor 2, torus counterterm scheme with reference photon mass m, $0 < \lambda, e^2 < 0.1m$, $\mu_0 \in (0, m]$). No identification between the two is proved. "Symmetric phase" is, for both, the expectation that the states lie above the surface predicted by one-loop perturbation theory; it is neither a hypothesis nor a conclusion of any theorem here, and theorem 21.1 holds on all of King's coupling range (lemma 22.1), the surface included. The problems below are extensions of the main theorem, not conditions on it.

23.1. **Relation to the torus states.** Three hypotheses bear on the relation of K to S: (FAM$_L$), King's cube construction exists on the block-factor-2 family in the torus counterterm scheme; (KI$_L$), the fixed-torus continuum limit exists in that family; (CE), a convergent unit-scale cluster expansion with sources on the cubic tori, their tilted copies and their cuts along King's cubes, uniform in the lattice spacing. Under (FAM$_L$) and a sign or decay hypothesis on the difference of the two counterterms, every torus state dominates the King state at a matched bare mass on the observables $e^{\mathrm{i}F(f)+:\phi^2:(k)}$, $k \ge 0$, a one-sided inequality carrying no information about rotations. Under (CE) at a point of a massive window ($\lambda m^2 \ge K_0 e^6$, $K_0 \ge 1/(18\pi^2)$, small e^2, λ) every torus limit state equals the cube limit of the same family and scheme in every order of limits; with (FAM$_L$) or (KI$_L$) it is unique, exponentially clustering and ergodic; with King's cube limits on both families it equals K at an explicitly shifted bare mass. Whether (CE) holds anywhere at $e > 0$ is open. Two obstacles are structural: the torus paper computes its mass counterterm with the photon propagator frozen at m while King's free-boundary counterterm uses the model's own μ_0, a difference of order $e^2|m - \mu_0|$, not small (remark 22.3); and King's construction needs block factor 5 for the rotation by $\arctan(3/4)$, while no lattice family admits both a dyadic structure and a rotation of infinite order. On the one-loop surface $\lambda m^2 = e^6/(72\pi^2)(1 + O(e^2/m))$ the two constructions are expected to select different phases (periodic tori the broken one, Dirichlet faces the symmetric one); this is a heuristic, not a theorem, and no statement of this paper uses the phase.

23.2. **Uniqueness and ergodicity of K.** Uniqueness of King's cube limit is part of the construction (monotonicity in the cube), but uniqueness of the infinite-volume state, in the sense of King's expected identity between cube limits and periodic-box limits, is not proved, and neither is ergodicity of K under translations: theorem 21.1 gives clustering only on the ergodic

components. The usual route, a cluster expansion with a mass gap, is (CE). This is distinct from the independence of a LIM state from its continuum subsequence, (UNIQ), which holds (proposition 14.6, from theorem 3.13 and from theorem 3.2; proposition 14.12, corollary 4.4). Cubic symmetry of K is a theorem (theorem 8.5): it needs only the uniqueness of the monotone cube limit and the reflection sandwich (remark 8.6).

23.3. **Non-triviality.** For the torus states the truncated correlation of $F(g)^2$ with the Wick square is non-positive, and strictly negative at small couplings under the two hypotheses of [24, Thm. 12.9]: the one-loop statement (C) on the continuum coefficient and a remainder bound on it, which that paper's uniform estimates do not provide. The analogous statements for King's cube, and a non-vanishing two-point function, are left to later work. For K nothing is proved here beyond the sign inequalities that hold on every lattice. Whether K is non-Gaussian, or even not decoupled, is open.

23.4. **The limit of vanishing photon mass.** Every constant of the Wick leg of theorem 21.1 depends on μ_0 and is not uniform in it (remark 21.7); the absolute bounds of the curvature leg are. The limit $\mu_0 \downarrow 0$, taken along a frozen family (counterterm and scheme frozen at a reference mass, so that the members are monotone in μ on the cone by the correlation inequalities), is left to later work: in the curvature sector it needs the existence theorem for every member and nothing else; in the Wick sector it needs a cell bound uniform in the photon mass (the analogue, along the family, of the cell bound ($\mathrm{B}_{\mathrm{cell}}$) of lemma 14.4, which at a fixed mass is a lemma; the clause "several models at once" of lemma 14.7 is formulated for this purpose), together with the torus inputs at the reference mass and the transfer of theorem 21.3 for the members on King's convention; the identification of the members with King's models at shifted scalar masses, masslessness proper (the vanishing of the photon mass in the limit theory) and the two-point function are further questions there. None of these enters the present paper.

23.5. **The optional extension: (W2) at arbitrary amplitudes, Form (A), and the Euclidean clause of the Wick sector.** Form (B) of theorem 21.1 lives on the tube $\mathcal{D}_{\sigma_+,H}$: the only positive-source bound it uses is the one proved on the cone $\sup q \le s_1 < \sigma_1(M_1)$ (corollary 13.3), and the only existence statement is theorem 3.5, whose positive scalar part is below the same threshold. Above the threshold, the existence of the fixed-cube limits and the uniform positive-source bound are the hypothesis (W2) below, which is open; the Euclidean clause of Form (A) and its King-convention form are the further hypotheses (R_W) and (FA^K).

Hypothesis 23.1 ((W2): the Wick sector at arbitrary amplitudes)**.** Let a be an admissible scheme. (a) For every finite family of real 2-forms and real functions, every $J \ge J_0(f,h)$ and *every* real coefficient vector (z,w), the limit $\lim_k Z^a_{k,J}(z,w)$ exists and lies in $(0,\infty)$. (b) For every $q \in C^\infty_c(\mathbb{R}^3)$ with $q \ge 0$,

$$\sup_{J \ge J_q} \lim_{k\to\infty} Z^a_{k,J}(0,q) < \infty,$$

where J_q is the least J with $\operatorname{supp} q$ at ℓ^∞-distance greater than 1 from $\mathbb{R}^3 \setminus C_J$.

Remark 23.2 (on (W2)). (i) Clause (a) is theorem 3.5 with the threshold on the positive scalar part removed. On the cone of amplitudes $\sup q \le s_1 < \sigma_1(M_1)$, clause (b) is proved in this paper in L^1 form (corollary 13.3, on the comparison convention; on ND^K in lim sup form by remark 4.8). Above the threshold both clauses are open; the exact obstruction is the free-energy bound (B^-)[u] at amplitude u above the threshold (Hypothesis 23.9 below): the ε-uniform stability of King's torus model with its bare coefficient shifted by $-2u$. (ii) No clause of theorem 21.1 uses (W2). The Wick result of the main theorem is Form (B) on the tube $\mathcal{D}_{\sigma_+,H}$, where only the proved cone instance enters, through theorem 13.10; (W2) is a hypothesis of Form (A) (theorem 23.7) and of (R_W). In the rotation part, (W2) for a scheme a is the W-admissibility $(\mathrm{W})_a$ of definition 16.3.

Hypothesis 23.3 ((R_W): rotation invariance of the Wick sector at arbitrary amplitudes)**.** For an admissible scheme a satisfying (W2), the full object $Z^a_\infty = \lim_J \lim_k Z^a_{k,J}$ of the comparison family (whose existence and entireness follow from (W2) as in lemma 10.2) is invariant under

every Euclidean motion for all complex coefficients, F transforming as a 2-form and W^a as a scalar.

Remark 23.4 (on (R_W)). (R_W) enters only in the Euclidean clause of Form (A) (theorem 23.7). Form (B) has no Euclidean clause and does not assume it. The rotation part proves the curvature sector; its Wick-sector theorems (theorems 18.9 and 18.11) give the $O(3)$ clause of (R_W) for a scheme a (its translations are theorem 23.7(ii)) under the torus comparison with scalar sources and the closing hypothesis for that scheme (definitions 17.2 and 18.5).

Hypothesis 23.5 ((FA^K): Form (A) on King's convention)**.** On King's convention ND^K, under (W2) read for ND^K: the limit functional $Z_\infty^{a,K}$ is entire in all coefficients; the lattice symmetries O_h about every point hold for the joint field at all amplitudes; reflection positivity holds at arbitrary complex coefficients; the measure exists off the tube; and the Euclidean clause of the Wick sector holds for the King-family object.

Remark 23.6 (on (FA^K)). (FA^K) is the King-convention counterpart of Form (A) as the comparison-convention results give it (theorem 23.7, last paragraph). The main theorem makes no use of it: its Wick leg on King's convention is Form (B) on the tube (theorem 21.3, proposition 21.4).

Theorem 23.7 (Form (A): the Wick leg at arbitrary amplitudes)**.** *Let the couplings satisfy* (T3)*, and let a be an admissible scheme satisfying* (W2) *for the comparison convention* $\mathrm{ND}^{(a)}$*. Then for every finite family $(f_1,\dots,f_n;h_1,\dots,h_p)$ of real 2-forms and real scalars the full object $Z_\infty^a(\zeta)=\lim_J\lim_k Z_{k,J}^a(\sum_i z_i f_i,\sum_j w_j h_j)$ exists for every $\zeta=(z,w)\in\mathbb{C}^{n+p}$, locally uniformly, Z_∞^a is entire, and $Z_\infty^a=G_\infty$ on $\mathcal{D}_{\sigma_+,H}$. Its Schwinger functions are those of Form (B), and:*

(i) OS0 *as in Form (B)(i): the tube bound on $\mathcal{D}_{\sigma_+,H}$ and the bounds on the $S_{n,p}$; no quantitative bound on Z_∞^a off the tube is asserted.*

(ii) OS1*: $Z_\infty^a(\tau_b R\cdot\zeta)=Z_\infty^a(\zeta)$ for every $\zeta\in\mathbb{C}^{n+p}$, $b\in\mathbb{R}^3$, $R\in O_h$ (F a 2-form, W^a a scalar).*

(iii) OS2 *in matrix form about every coordinate plane for exponentials with arbitrary complex coefficients (theorem 23.8), and in polynomial form (proposition 10.10).*

(iv) OS3 *as in Form (B)(iv).*

(v),(vi) *As in Form (B), the characteristic functional $(u,v)\mapsto Z_\infty^a(\mathrm{i}u,\mathrm{i}v)$ being the restriction of the entire Z_∞^a to the imaginary axes.*

Euclidean clause. *If in addition* (R_W) *holds for the comparison family, Z_∞^a is invariant under every Euclidean motion, its mixed Schwinger functions are $\mathbb{R}^3\rtimes O(3)$-covariant, and ν-almost every ergodic component of the measure of (v) is $\mathbb{R}^3\rtimes O(3)$-invariant and reflection positive about every plane (no vanishing of odd moments is asserted for a component). This is the only Euclidean clause of the Wick sector.*

On King's convention. *The same statements for the K-family object $Z_\infty^{a,K}$ under* (W2) *read for* ND^K *are the content of* (FA^K) *(Hypothesis 23.5); nothing is derived for it from the comparison-family statements, since* (W2) *on* ND^K *does not give* (W2) *for the comparison family (corollary 4.4(iv)).*

Indications. Existence at all complex coefficients, locally uniform convergence and entireness are lemma 10.2 under (W2) (version (A_∞) of lemma 10.3); $Z_\infty^a=G_\infty$ on $\mathcal{D}_{\sigma_+,H}$ as the same J-limit of the same Z_J^a (theorem 13.10). (i) is Form (B)(i). (ii): Form (B)(ii) is the identity on $\mathcal{D}_{\sigma_+,H}$, an open neighbourhood of the origin of $\mathbb{C}^{n+p}$, and both sides are entire. (iii) is theorem 23.8 with proposition 10.10; (iv) is proposition 10.9 for Z_∞^a; (v),(vi): the measure of Form (B) is built from G_∞ on the imaginary axes, which lie in $\mathcal{D}_{\sigma_+,H}$. The Euclidean clause is (R_W) itself; the component clause follows through the ergodic decomposition of Form (B)(vi) as in the curvature sector (theorem 11.1(c)). □

For a LIM state of a continuum subsequence (theorem 14.9) the corresponding Form (A) requires both clauses of (W2): the existence of the limits at every real coefficient vector and the uniform bound at every amplitude.

Theorem 23.8 (reflection positivity at arbitrary complex coefficients)**.** *Let a be an admissible scheme satisfying* (W2)*, let $\nu \in \{1,2,3\}$, $c \in \mathbb{R}$, and let $\zeta_1, \dots, \zeta_N$ be finite families of real sources $f_{il} \in C_c^\infty(\mathbb{R}^3; \Lambda^2)$, $h_{il} \in C_c^\infty(\mathbb{R}^3)$ supported in $\{x_\nu > c\}$, with arbitrary complex coefficients z_{il}, w_{il}. Then*

$$\sum_{i,j=1}^{N} \bar\gamma_i \gamma_j \, Z_\infty^a\big(\theta\zeta_i^* \sqcup \zeta_j\big) \;\geq\; 0 \qquad \text{for all } \gamma \in \mathbb{C}^N.$$

Refinement. *Let B_1, B'', the vertex sources q_v ($v \in \mathrm{vert}(B'')$) and q_ζ be those of definition 10.1 for the plane $\{x_\nu = c\}$. (a) If $c \in D$, the proof uses* (W2)*(b) only at the finitely many sources q_v, and it suffices to assume it at the single source $(1+\eta)q_\zeta$ for any one $\eta > 0$ (remark 10.4, one direction only); all other steps use* (W2)*(a) and the inequalities of lemma 2.7, and the tube machinery is not needed. (b) If $c \notin D$, with $c' \in D$ and $b := (c'-c)e_\nu$: the finitely many instances q_v and $\tau_b q_v$, $v \in \mathrm{vert}(B'')$, together with the tube inputs* (I2), (I3) *of theorem 10.7. Such a plane needs a non-lattice translation, which finitely many instances of* (W2)*(b) do not supply; it is carried by the translation invariance of the tube object (proposition 10.5(b)).*

Proof. Let $c \in D$ and let $\mathcal{F}$, Φ_k, ζ_{ij}^*, $\mathcal{T}_1$, $T_\mathcal{F}$, U be as in definition 10.1. By lemma 10.2, under (W2)(a) and (W2)(b) at the q_v, $\{Z_J^a\}_{J \geq J_0}$ is uniformly bounded on $\mathcal{T}_1$, $Z_J^a \to Z_\infty^a$ locally uniformly on $\mathcal{T}_1$, Z_∞^a is holomorphic there and $Z_J^a \leq Z_\infty^a$ on $U \cap \mathcal{T}_1$. By lemma 10.3(A), $\limsup_k \sup_{\mathcal{T}_1} |\Phi_k| \leq C$ with C independent of $J \geq J_0$, and at fixed J the Φ_k are locally bounded on $\mathbb{C}^m$ ((W2)(a) only). Montel's theorem gives a subsequence along which $\Phi_k \to \Phi^J$ locally uniformly, Φ^J entire; the matrices $(\Phi_k(\zeta_{ij}^*))_{ij}$ are positive semidefinite for large k (lemma 9.3(b)), hence so is their limit, and $\sup_{\mathcal{T}_1} |\Phi^J| \leq C$ uniformly in J. On U every lattice value is real and the scheme multipliers of the two states agree, so the $\mathcal{I}$ halves of lemmas 9.9 and 9.10 give $Z_{k,J}^a \leq \Phi_k \leq Z_{k,J+2n}^a$, hence $Z_J^a \leq \Phi^J \leq Z_{J+2n}^a$ on U. On $U \cap \mathcal{T}_1$, $Z_J^a \uparrow Z_\infty^a$ (lemma 10.2), so $\Phi^J \to Z_\infty^a$ there; by the uniform bound on $\mathcal{T}_1$, Montel's theorem and the uniqueness set $U \cap \mathcal{T}_1$ (definition 10.1), $\Phi^J \to Z_\infty^a$ locally uniformly on the connected $\mathcal{T}_1$, in particular at the ζ_{ij}^*, and positivity passes to the limit. The only instances of (W2)(b) used are the q_v of lemmas 10.2 and 10.3; remark 10.4 replaces them by $(1+\eta)q_\zeta$.

Let $c \notin D$, $c' \in D$, $b := (c'-c)e_\nu$. The $\tau_b\zeta_i$ are supported in $\{x_\nu > c'\}$ and $\theta_{c'}(\tau_b\zeta)^* = \tau_b(\theta_c\zeta^*)$, so the display at c follows from the display at c' for the translated families once Z_∞^a is translation invariant at the points in question. The enlarged family of the translated problem at c' is $\tau_b\mathcal{F}$, with the same coefficient space, matrix points, B_1, B'', and vertex sources $\tau_b q_v$; the argument above for $\tau_b\mathcal{F}$ gives the positivity of $(Z_\infty^a(\tau_b \cdot \zeta_{ij}^*))_{ij}$, and $\zeta \mapsto Z_\infty^a(\tau_b \cdot \zeta)$, $\zeta \mapsto Z_\infty^a(\zeta)$ are holomorphic on $\mathcal{T}_1$ by lemma 10.2. On the non-empty open set $\mathcal{T}_1 \cap T_\mathcal{F}$ both equal the tube limits $Z_{\infty,T}^a(\tau_b \cdot \zeta)$, $Z_{\infty,T}^a(\zeta)$, which agree by proposition 10.5(b); by the identity theorem they agree on $\mathcal{T}_1$, in particular at the ζ_{ij}^*. □

Hypothesis 23.9 ((B$^-$)[u]: stability of the shifted torus model)**.** Let $u > 0$. In the free convention and the scheme a,

$$\sup_{k \geq 0} \limsup_{J \to \infty} \alpha_{k,T_J}^a(u) < \infty, \qquad \alpha_{k,T}^a(u) = |T|^{-1} \ln N_{k,T}^a(0, u\mathbf{1}),$$

the free-energy density of the torus functional at the constant scalar source u; that is, (I2) of definition 12.5 at the amplitude $s_1 = u$.

For u below the threshold s_* of lemma 22.1, (B$^-$)[u] is proposition 12.6 (or proposition 12.11) at an auxiliary mass M_1 with $\sigma_1(M_1) > u$, so Form (B), whose constants are built from (I2) at one such amplitude, already contains every amplitude at which the positive-source bound is known. For $u \geq s_*$ the hypothesis is the exact obstruction. By lemma 12.7, exact at fixed (k,T), $\alpha_{k,T}^z(u)$ is, up to explicit c-numbers, $|T|^{-1}$ times the difference of the free energies of King's torus model with bare scalar coefficient $\alpha_{k,T}^{\mathrm{tor}}(m) - 2u$ and with $\alpha_{k,T}^{\mathrm{tor}}(m)$, both with the counterterms of the mass m: the second is proposition 3.8 at m, the first the ultraviolet stability of the shifted model. For $u < s_*$ the shifted coefficient is that of King's model at a positive mass M_1; for $u \geq s_*$ it lies, asymptotically in k, below $\inf_{M>0} \alpha_k^\infty(M)$, and no positive scalar mass

realises it. The printed proofs of [1, 2] expand about the Gaussian at the positive scalar mass m and use its mass term in the small-field bounds; no printed theorem covers a bare coefficient below every positive-mass value. Conversely $(\mathrm{B}^-)[u]$ is sufficient for (W2)(b): proposition 13.2 at $s_1 = u$, whose steps do not depend on the amplitude, gives $\ln N^a_{k,X}(0,q) \le (C_U/u)\|q\|_{1,k}$ for $0 \le q \le u$ on every Neumann–Dirichlet box, hence (W2)(b) in L^1 form on $\sup q \le u$. It is not asserted that (W2) implies $(\mathrm{B}^-)[u]$: the inequalities available bound a constant-source density from below by local sources, never from above, so $(\mathrm{B}^-)[u]$ is sufficient and not shown necessary. Also open: an L^1 bound for the negative part of the Wick source on cubes, as on the torus paper's tori (the tube bound (31) carries the weighted norm $|k_-|_w$).

23.6. **Rotation invariance of the Wick sector.** Theorem 21.1 gives invariance under $\mathbb{R}^3 \rtimes O(3)$ for the curvature sector; for the Wick sector it gives invariance under $\mathbb{R}^3 \rtimes O_h$ on the tube and no Euclidean clause. Euclidean invariance of the joint field is the hypothesis (R_W) of the optional extension (Hypothesis 23.3, theorem 23.7), and on King's convention (FA^K) (Hypothesis 23.5); whether the heavy-shell comparison of the rotation part extends to the Wick square, on the tube or at arbitrary amplitudes, is open. Cubic symmetry is a theorem in every case. The two statements the rotation part would need for the Wick sector are open: the torus comparison with scalar sources $(\mathrm{K})_a$ (definition 17.2) and the closing property $(\mathrm{TIGHT})_a$ (definition 18.5) for a W-admissible scheme; given both, theorems 18.9 and 18.11 give (R_W). In the curvature sector both are theorems (theorems 17.1 and 18.4).

23.7. **The coupling condition.** The sign clause (T3) is proved on the explicit regions of proposition 22.15 (at $\mu_0 = m$ on a region containing $0 \le \lambda, e^2 \le 0.1m$; at every $\mu_0 \in (0, m]$ on explicit boxes; exactly for $\lambda < \frac{\pi}{4}(1+\sqrt{5})\, m$ at $e = 0$), and the Wick leg is stated there. Outside those regions the exact position of the failure set $E_W^<$ is not determined; the two-sided clause that m is not a global minimiser of α_∞, with the auxiliary mass on either side of m (proposition 12.11, lemma 22.1(b)), would enlarge the region. The cell bound $(\mathrm{B}_{\mathrm{cell}})$, the scheme bound (SCH) and the independence of the subsequence (UNIQ) of the statements on continuum subsequences are proved (lemmas 14.4 and 14.5 and proposition 14.6, proposition 14.12); a LIM state of a scheme that is not admissible, or a LIM state along a family of photon masses, would require the first two to be re-established for it. The mass clause of [2, Prop. 3.1], a cap on the dimensionless scalar mass at the stopping scales where that proposition is used, belongs to the statement of the proposition, which is cited under it, and imposes nothing on the bare mass (remark 22.2).

Appendices

Appendix A. Proof of Proposition 5.4: the small-field representation on King's cube

This appendix proves proposition 5.4, including its clause (R_0). The proof is King's proof of [13, Thm. 3.1], which is Bałaban's lower and upper bounds [1, 2] with the modifications of [13, Appendix], run on King's RG cube C_J^K (definition 2.2) with King's block-spin flow of section 5.1. Two things change relative to the torus. First, the scalar field is killed on the Dirichlet planes of the cube and averaged with King's Dirichlet weight; every place where Bałaban's or King's arguments use that the block average is uniform ($QQ^* = 1$, P a projector, the constant is reproduced by the block average) is replaced by the corresponding fact for the weighted average (appendix A.1), and every place where the torus has no boundary acquires a Dirichlet face, at which the scalar background vanishes linearly (appendix A.3). Second, the photon has Neumann conditions; every photon object of the cube is the even restriction of the same object on a doubled torus, so every photon estimate of the torus analysis holds on the cube with its constant multiplied by at most 8 (appendix A.2). The large-field analysis near the Dirichlet faces uses King's face test [14, (5.41)] as a large-field family at every scale, paid by the Dirichlet-face positivity of [14, Prop. 5.4] in an elementary form valid for every gauge field (appendix A.5). The lower bound (appendix A.7) carries King's Appendix class with a background-gradient clause and a block-mismatch clause in place of King's Laplacian clause,

and the upper bound (appendix A.8) runs Bałaban's large-field decomposition on the cube; the assembly, the normalisation-free form of the remainder and the clause (R_0) are in appendix A.9.

Throughout, $d = 3$, $L = 5$, the couplings (m^2, μ_0, e, λ) with $m^2, \mu_0, \lambda > 0$ are fixed, as are $J \geq J_0$ and the source $s \in \{u_x, 0\}$ (so $h = 0$: the scalar source is treated in section 6). Constants C, c, δ change from line to line and depend only on the quantities listed where they are introduced; they never depend on the spacing index k, the step j, the field or the domain unless this is said.

A.1. **The cube in block units, and the weighted block average.**
Block units. Fix the spacing index k and a step j, $0 \leq j \leq k$, and write everything in *unit-j units*: the j-blocks (cubes of side $5^j\varepsilon_k = \eta$ in physical units) have side one, and the fine lattice has spacing $\vartheta := 5^{-j}$, with $1/\vartheta \in 5\mathbb{N}$ for $j \geq 1$. The number of blocks per side is $N := 2MJ/\eta \in \mathbb{N}$ (the side $2MJ$ of C_J is a multiple of every block side, section 5.1). After a translation the picture is: fine sites $\Lambda := \vartheta\mathbb{Z}^3$, the *killed set* $\mathsf{K} := \{x \in \Lambda \cap [0,N]^3 : \ x_\mu \in \{0, N\}$ for some $\mu\}$ (the two Dirichlet planes of C_J^K in each direction, $2MJ$ apart), the *live set* $X_{\text{live}} := \{x \in \Lambda : \ 0 < x_\mu < N \ \forall\mu\}$, which is alive$_J^K$ of definition 2.2 ($N/\vartheta - 1$ sites per side), the blocks $B(y) := (y + [0,1)^3) \cap \Lambda$, $y \in \mathbb{Z}^3$, $0 \leq y_\mu \leq N - 1$, and the *face layer* L_1 of blocks with some $y_\mu \in \{0, N-1\}$. The bottom Dirichlet plane $x_\mu = 0$ is the first site plane of the blocks with $y_\mu = 0$ (inside those blocks); the top plane $x_\mu = N$ is the first site plane of the blocks $y_\mu = N$, which lie outside the cube. We write ∂D for the union of the Dirichlet planes, $t(x) := \min_\mu \min(x_\mu, N - x_\mu)$ for the face distance of a live site ($t = \vartheta$ on the first live layer F_1, the live sites with a neighbour in K), and a *face bond* is a bond from a live site to a killed site. By definition 2.1, a face bond carries only the diagonal term $\frac{1}{2}\vartheta^{d-2}|\phi(x)|^2$ at its live end: it has no link variable, and $D_A\phi(b) = -\vartheta^{-1}\phi(x)$ on it whatever A. The photon is a site field on X_J with Neumann conditions on the bond planes $x_\mu = -\vartheta/2$, $N - \vartheta/2$ (definition 2.2).
The weight. King's Dirichlet weight [14, (3.8)–(3.9)], in the form in which the averaging blocks have their top layers on the Dirichlet planes (section 5.1), is the function

$$w(x) \ := \ W \prod_{\mu=1}^{3} \mathbf{1}\big[x_\mu/\vartheta \not\equiv 0 \ (\mathrm{mod}\ 5)\big], \qquad W := (5/4)^3 = 125/64,$$

on the fine sites of a block: zero on the three site planes of the block that lie in the residue class of the Dirichlet planes (the *zero-weight layers*), the constant W elsewhere. Since $1/\vartheta \equiv 0$ (mod 5), the block corners and all of K lie in the zero-weight class, and $\sum_{x\in B(y)} \vartheta^3 w(x) = 1$ for every block of the cube. The weight has period 5ϑ, so it acts at the finest step only: the coarse averages above scale one are uniform, $Q_{j+1} = Q\,Q_j$ with Q the uniform one-step average with $QQ^* = 1$, and the constants a_j of [13, (2.13)] are those of the uniform flow. The block average and its adjoint are

$$Q(A)\phi(y) \ := \sum_{x \in B(y) \cap X_{\text{live}}} \vartheta^3 \, w(x) \, U(A(\Gamma_{y,x})) \, \phi(x), \qquad P(A) := Q(A)^* Q(A),$$

with King's hierarchical contours $\Gamma_{y,x}$ [13, (2.10)–(2.12)], chosen coordinate-monotone between the block centre and x; such a path between two live sites of one block meets no killed site, since the killed sites of a block sit at the extreme value of a coordinate. The adjoint is taken for the measure ϑ^3 on fine sites and counting measure on blocks.

Definition A.1 (weight class)**.** A block weight is a function $w \geq 0$ on the fine sites of a unit block with (W1) $\sum_{x\in B} \vartheta^3 w(x) = 1$; (W2) $w \leq W$; (W3) $w = 0$ on every killed site of the block; (W4) the same w on every block up to translation. The uniform weight ($w \equiv 1$, $W = 1$) and King's weight above ($W = 125/64$) are in the class.

Lemma A.2 (the weighted average)**.** *For every weight in the class and every gauge field A, with $W_2 := \sum_{x\in B} \vartheta^3 w(x)^2 \in [1, W]$ (for King's weight $W_2 = W$):*
(Q1) $Q(0)\mathbf{1} = \mathbf{1}$ on every block, and $Q(A)$ maps the transported constant to the constant;
(Q2) $Q(A)Q(A)^* = W_2\,\mathbf{1}$;

(Q3) $P = W_2\Pi$ *with* Π *an orthogonal projector;* $0 \le P \le W_2$ *and* $\|1-P\| = \max(1, |1-W_2|) \le 1$ *since* $W_2 \le W < 2$*;*
(Q4) $|(Qu)(y)| \le W_2^{1/2} \|\mathbf{1}_{B(y)} u\|_2$ *and* $\|Q^* e_y\|_2 = W_2^{1/2}$*;*
(Q5) $|(Qu)(y)| \le \sup_{B(y)} |u|$*;*
(Q6) $|Q^* b|_\infty \le W |b|_\infty$*,* $\|P\|_{\infty\to\infty} \le W$*, and the Schur norm* $\sup_x \sum_{x'} \vartheta^3 |P(x,x')| \le W$*;*
(Q7) $\sum_x \vartheta^3 |(Pu)(x)| = \sum_y |(Qu)(y)|$*;*
(Q8) $P(x,x') = 0$ *unless* x, x' *lie in one block, and* P *vanishes on killed sites:* P *never couples a live site to a killed one, and every union of blocks is invariant;*
(Q9) under $\phi \mapsto e^{eq\lambda}\phi$*,* $A \mapsto A + d\lambda$*, the operators* $Q(A)$*,* $P(A)$ *transform covariantly.*

Proof. (Q1), (Q2): (W1) and the orthogonality of the transports U. (Q3): $P^2 = Q^*(QQ^*)Q = W_2 P$, and $1 - P$ has the eigenvalues 1 on $\Pi^\perp$ and $1 - W_2$ on Π. (Q4): Cauchy–Schwarz with the weight w and (W1); $\|Q^* e_y\|^2 = \sum \vartheta^3 w^2$. (Q5): $|Qu| \le \sum \vartheta^3 w |u| \le \sup |u|$ by (W1). (Q6): $(Q^* b)(x) = w(x) U^* b(y)$ and $(Pu)(x) = w(x) \sum_{x'} \vartheta^3 w(x') U\, u(x')$. (Q7): $|(Pu)(x)| = w(x) |(Qu)(y)|$ and (W1). (Q8): the definitions and (W3). (Q9): the contour transports transform with the endpoint gauge functions, and w is a scalar. □

Where a step of Bałaban's or King's torus analysis uses $QQ^* = 1$, $P^2 = P$, or that the block average of a constant is that constant, the cube version below uses the corresponding item of lemma A.2 and acquires a factor depending on W only. Three identities change in substance and are proved below: the background of a constant datum (lemma A.14), the block-mismatch identity (lemma A.21), and the minimiser equation with $QQ^* = W_2$ (used in lemma A.21).
The operators. In unit-j units let $V_j := m^2\eta^2 \le c_V$ be the mass term, A a real gauge field on the live bonds of X_{live} with $U(A_b) = \exp(\vartheta\, e_j q A_b)$, $e_j := e\eta^{1/2}$, and

$$M(\Omega, A) \;:=\; -\Delta_{A,\Omega}^{N,D} + V_j + a_j P(A) \qquad \text{on } \ell^2(\Omega \cap X_{\text{live}}),$$

for a *sub-domain* Ω (a union of j-blocks of the cube; $\Omega = X_{\text{live}}$ is the whole cube): the Laplacian keeps every bond with both ends in $\Omega \cap X_{\text{live}}$, adds ϑ^{-2} to the diagonal for each face bond (Dirichlet, no link), and on bonds from $\Omega \cap X_{\text{live}}$ to $X_{\text{live}} \setminus \Omega$ either drops them (Neumann cut) or adds the diagonal (Dirichlet cut). The *Neumann boundary* $\partial_N\Omega$ is the set of block midplanes between a block of Ω and a block of the cube not in Ω; the *depth* of a point is $\operatorname{depth}(x) := \operatorname{dist}_\infty(x, \partial_N\Omega)$ ($+\infty$ on the whole cube). The Dirichlet faces of the cube are part of the operator, never part of $\partial_N\Omega$: every block touching a Dirichlet plane contains that plane's killed sites (bottom) or ends one site before it (top), so no sub-domain has a Neumann face on a Dirichlet plane. The *leg* of block data b is $u := a_j M(\Omega, A)^{-1} Q(A)^* b$; the *background* of the block field ϕ_j is $\phi^{(j)} := a_j M^{-1} Q^* \phi_j$ [13, (2.14)–(2.16)]; the block form is $\Delta^{(j)} := a_j - a_j^2 Q M^{-1} Q^*$ and the fluctuation covariance $C^{(j)} := (\Delta^{(j)} + aL^{-2} Q_c^* Q_c)^{-1}$ with Q_c the uniform coarse average. A field A is *regular* on Ω at scale j if it obeys [13, (3.5)]: $|\partial_\mu A_\nu| \le K_j$ with $\theta_j := e_j K_j \le \theta_0$ small; King's small-field class gives $\theta_j \le C e_j p_j$, small for $\eta \le \eta_1$. Gauge covariance on the cube: for any real λ on the sites, $M(\Omega, A + d\lambda) = U_\lambda M(\Omega, A) U_\lambda^*$ with U_λ the multiplication by $e^{eq\lambda}$, $Q(A + d\lambda) = U_\lambda^c Q(A) U_\lambda^*$, and on a face bond $|D_{A+d\lambda}(U_\lambda u)| = |D_A u| = \vartheta^{-1}|u(x)|$; so every bound at a constant field $B = d(B \cdot x)$ on a box equals the bound at 0 for a datum of the same modulus (the contours lie in the block, the Dirichlet condition is multiplication-invariant, the weight is a scalar).
The small-field set and the face test. The small-field indicator χ_j of section 5.1 is [13, (3.2)] on the blocks of the cube, with the gradient clause $|D_{A_j}\phi_j(\beta)| \le p_{j-1}$ on every block bond β of the cube (all of which are cube-internal), times King's face test [14, (5.41)]

$$\lambda_j \;:=\; \prod_{y' \in L_1} \chi\big(|\phi_j(y')| \le \sigma_j\big), \qquad \sigma_j := \Big(\frac{2}{\theta_2 \gamma_D}\Big)^{1/2} p_j \;\le\; C_8\, p_j, \quad \gamma_D := \frac{a_j}{1 + a_j W^2},$$

with $\theta_2 \in (0,1)$ a fixed share and γ_D the Dirichlet-face positivity constant of lemma A.22; $p_j := p(\eta) = b_0(1 + \ln \eta^{-1})^p$ and $M_j := p_{j-1}(\lambda \ell_{j-1})^{-1/4}$ is the block-sup threshold of [13, (3.2)] ($\ell_j := 5^j \varepsilon_k$ the physical block side, $\lambda(\ell_j) = \lambda \ell_j$ the running coupling in $d = 3$). The constant C_8 depends on (a, W, θ_2, L) only.

A.2. **The photon sector: even restriction of the doubled torus.** Let $2X$ be the torus of period $2N$ in each direction (physical period $4MJ$), X_J embedded as $\{0,\dots,N-\vartheta\}^3$ after the translation above, and $\mathcal{W} \cong (\mathbb{Z}_2)^3$ the group generated by the reflections r_μ in the photon planes $x_\mu = -\vartheta/2$ (on $2X$ the reflection in $x_\mu = N - \vartheta/2$ is the same element). Since $4MJ \equiv 0$ modulo every block side, the blocks of X_J are blocks of the periodic tiling of $2X$, and $\mathcal{W}$ maps blocks to blocks. For a photon component A_μ (a site field) let E be the even extension to $2X$, and E_c the same on the coarse lattices.

Lemma A.3 (photon objects are even restrictions)**.** *For every photon object O of Bałaban's and King's construction built from the Neumann form $-\Delta^N_{X_J} + \mu_0^2\eta^2$ and the uniform block averages on X_J, and its counterpart O_{2X} on the periodic torus $2X$:*

(a) the propagators G_j, the block forms $\Delta^{(j)}$, the fluctuation covariances $C^{(j)}$, the background map $A_j \mapsto A^{(j)} = a_j G_j Q_j^* A_j$ and the slices satisfy $O_{2X}\mathsf{E} = \mathsf{E}\,O$;

(b) quadratic forms: $\langle \mathsf{E}A, \Delta^{(j)}_{2X}\mathsf{E}A\rangle = 8\langle A, \Delta^{(j)}A\rangle$, and likewise for every positive form above;

(c) for every coarse $\Lambda \subset X^{(j)}$ (resp. every fine union of blocks $\Omega \subset X_J$) the Dirichlet-conditioned compressed operators on Λ and the Neumann operators on Ω intertwine with E and the $\mathcal{W}$-invariant sets $\mathcal{W}\Lambda$, $\mathcal{W}\Omega$;

(d) if Bałaban's cut-off kernel ζ of [2, (2.44)] *is applied image-wise on $2X$ (the cube kernel being $K_X(x,y) := \sum_{g\in\mathcal{W}} \zeta(x,gy)K_{2X}(x,gy)$), the cut-off objects of the cube are even restrictions of those of $2X$ and the support and size properties of ζ hold for them with the torus constants;*

(e) every positivity bound of the photon sector ([2, (3.29), Prop. 3.1] *photon part, the last-step mass bound $G'_k \geq \gamma_1 \mu_0^2 \ell_k^2$ of* [2, (3.11)]*, the photon exponents of* [2, (3.26)]*) holds on X_J with the constant of $2X$.*

Proof. (a) The Neumann Laplacian on X_J is the periodic Laplacian of $2X$ on $\mathcal{W}$-even fields with reflection planes between sites: every plane-crossing bond of $2X$ joins a site to its mirror image, where an even field has zero difference, and every bond of X_J appears 8 times. Block averages commute with E (blocks go to blocks) and $Q^*_{2X}\mathsf{E}_c = \mathsf{E}Q^*_X$ (the adjoint of a $\mathcal{W}$-covariant uniform average is the uniform block extension). Hence $(M_{2X} + aQ^*Q)\mathsf{E} = \mathsf{E}(M_X + aQ^*Q)$; the torus operator commutes with $\mathcal{W}$, preserves the even subspace and is invertible on it, so $G_{2X}\mathsf{E} = \mathsf{E}G_X$, and $\Delta^{(j)}$, $C^{(j)}$, the background map and the slices follow from the algebra of [13, (2.13)–(2.17)], which is the same on both sides. (b) $\mathsf{E}^*\mathsf{E} = 8$ on X_J. (c) Compression to a $\mathcal{W}$-invariant set commutes with $\mathcal{W}$, and $P_{\mathcal{W}\Lambda}\mathsf{E} = \mathsf{E}P_\Lambda$; then (a) on the invariant subspace. For Neumann conditions on Ω, the bonds of $\mathcal{W}\Omega$ are the images of those of Ω plus plane-crossing bonds, on which even fields have zero difference. (d) $\zeta(x,gy)$ depends on $|x-gy|$ only; the image-wise kernel commutes with $\mathcal{W}$ by construction and restricts to the cube kernel by definition; the support and size statements are statements about $\zeta(x,\cdot)$ on $2X$, and $|x-gy| \geq |x-y|$ for $x,y \in X_J$ keeps every support statement. The cut is a device inside the definition of Bałaban's background, so the choice is free. (e) By (b): the cube form of an even field is $1/8$ of the torus form, and the right-hand sides (gradient sums, mass sums, block sums) scale by the same factor since plane-crossing gradients of even fields vanish. □

Lemma A.4 (regularity is preserved)**.** *(a) A block photon configuration A_j on X_J satisfies the photon clauses of* [13, (3.2)] *if and only if $\mathsf{E}_c A_j$ does on $2X$. (b) If A on X_J is regular in the sense of* [13, (3.5)]*, or obeys a Hölder bound with constant C, so does $\mathsf{E}A$ on $2X$ with the same constant.*

Proof. (a) Plane-crossing coarse gradients of $\mathsf{E}_c A_j$ vanish; the other clauses are copies. (b) $\mathsf{E}A = A \circ F$ with F the coordinatewise folding map $2X \to X_J$, which is 1-Lipschitz; lattice differences across a plane vanish, the others are copies. □

Lemma A.5 (Neumann photon lines are image sums)**.** *Every photon line of the cube (the fluctuation covariances and slices of lemma A.3(a) and the full covariance $(-\Delta^N + \mu_0^2\eta^2)^{-1}$) satisfies $K_X(x,y) = \sum_{g\in\mathcal{W}} K_{2X}(x,gy)$ for $x,y \in X_J$. Hence every line bound of the torus*

analysis (decay, pointwise and derivative clauses, Hölder clauses in either variable, moment forms) holds for K_X with the constant multiplied by 8 and the same rates.

Proof. Kernel form of lemma A.3(a): $(K_X f)(x) = (K_{2X}\mathsf{E}f)(x) = \sum_y \sum_g K_{2X}(x, gy) f(y)$. Each image term obeys the torus bound at the pair (x, gy), and $|x - gy| \geq |x - y|$; derivatives in x act on $K_{2X}(\cdot, gy)$, derivatives in y through the lattice isometry $y \mapsto gy$. □

Consequently every photon statement that the torus analysis takes from [1, 2, 3] holds on X_J: apply it on the torus $2X$ (whose period $4MJ$ is a multiple of every large-block side, and whose sides exceed the diameter of every box used) to the even data and restrict. For $x \in X_J$, $\mathrm{dist}_{2X}(x, \partial(\mathcal{W}\Omega)) \geq \mathrm{dist}(x, \partial\Omega \setminus \partial X_J)$, so the restricted-form clauses transfer, and at the faces of X_J the photon has no boundary at all in the doubled picture. The photon tadpole and exchange graphs of the counterterm that contain a Neumann line $C_N - C_{\mathrm{free}} = \sum_{g\neq 1} C_{\mathrm{free}}(\cdot, g\,\cdot)$ are bounded by the same image argument: with $\varrho(x)$ the distance of x to the nearest photon plane, $|x - gx| \geq 2\varrho(x)$ for $g \neq 1$ and $|C_{\mathrm{free}}(x, z)| \leq C \min(\vartheta^{-1}, |x - z|^{-1}) e^{-\mu_0|x-z|}$, so the photon-image part of the mass counterterm is $O\big((e^2 + e^4)(\min(\vartheta^{-1}, \varrho^{-1}) + 1)\big)$ at order ≤ 4, the same size as the scalar boundary remainders near the face, and it is absorbed in the zone-B bounds of proposition A.27 (it is the photon-image part of the boundary remainder families there).

A.3. **Operator bounds on the cube and its sub-domains.** The statements of this subsection are [13, Thm. 3.3] and Bałaban's propagator bounds [4, Sects. 1–2] for the cube operator $M(\Omega, A)$ of appendix A.1, on every sub-domain Ω (any union of j-blocks of the cube, meeting the Dirichlet planes or not, re-entrant edges allowed, Neumann or Dirichlet cuts on $\partial_N\Omega$), at every scale $1 \leq j \leq K_f$, uniformly in ϑ, J, k, the shape of Ω and the regular field A. At $j = 0$ (fixed spacing 5^{-1}) every clause holds with L-dependent constants by finite dimensionality. The constants depend on (d, L, a, W, c_V) and King's constants only.

Lemma A.6 (coercivity)**.** *There are $\gamma = \gamma(d, a, W) > 0$ and $\theta_0 > 0$ such that $M(\Omega, A) \geq \gamma$ for every Ω, $\vartheta \leq 1/5$, $V_j \geq 0$ and A regular with $\theta_j \leq \theta_0$.*

Proof. Dropping the bonds between distinct blocks lowers the form (each dropped term is ≥ 0; Dirichlet terms only add), and $P(A)$ is block diagonal (Q8), so $M \geq \bigoplus_B M_B$ with $M_B := -\Delta^N_{A,B} + a_j P_B(A)$ on $\ell^2(B \cap X_{\mathrm{live}})$, the killed sites of a face block entering as a Dirichlet term. At $A = 0$ write $\phi = c + \psi$ on B with ψ orthogonal to constants (after extending ϕ by 0 to the killed sites, which does not change $P\phi$ by (W3) and only raises the gradient form). The Neumann gap of the unit cube gives $\langle\psi, -\Delta^N\psi\rangle \geq g_P\|\psi\|^2$ with $g_P = 4\vartheta^{-2}\sin^2(\pi\vartheta/2) \geq 8$ for $\vartheta \leq \frac{1}{2}$; by (Q1), (Q4), $|Q\phi| = |c + Q\psi| \geq |c| - W^{1/2}\|\psi\|$. Hence $\langle\phi, M_B\phi\rangle \geq g_P\|\psi\|^2 + a(1 - L^{-2})(|c| - W^{1/2}\|\psi\|)_+^2 \geq \gamma_*(|c|^2 + \|\psi\|^2)$ with $\gamma_* = \gamma_*(a, W) > 0$ by the springs-in-series minimisation $\min_{s\geq 0}[W^{-1}s^2 + a'(F_0 - s)_+^2] = F_0^2/(a'^{-1} + W)$; explicitly $\gamma \geq (1 - L^{-2})\min\{a/2, 8/(1 + 2W)\}$. For regular A: on the simply connected block, $A = A(x_B) + A''$ with $A(x_B)$ constant, pure gauge, removed by a unitary gauge transformation that preserves both terms, and $|A''| \leq \sqrt{3}K_j$; then $\|D_{A''}\phi - D_0\phi\| \leq 2\theta_j\|\phi\|$ and $|Q(A'')\phi - Q(0)\phi| \leq 3\sqrt{3}\,W^{1/2}\theta_j\|\phi\|$ (contours of length ≤ 3), so $\langle\phi, M_B\phi\rangle \geq (\gamma_*/2 - C\theta_j)\|\phi\|^2$, which is the claim for $\theta_j \leq \theta_0$. □

Lemma A.7 (Combes–Thomas; unit-lattice kernels)**.** *(a) For every 1-Lipschitz g, $\|e^{\delta_0 g} M^{-1} e^{-\delta_0 g}\|_{2\to 2} \leq 2/\gamma$, hence $\|\mathbf{1}_X M^{-1}\mathbf{1}_Y\| \leq (2/\gamma)e^{-\delta_0\,\mathrm{dist}(X,Y)}$, with $\delta_0 = \delta_0(\gamma, a, W, d)$. (b) $|\Delta^{(j)}(\Omega, A)(u, v)| \leq Ce^{-\delta_0|u-v|}$. (c) If Ω is a union of $(j + 1)$-blocks, then $\gamma_0'' \leq \Delta^{(j)} + aL^{-2}Q_c^*Q_c \leq a_j + a$ on the blocks of Ω, with $\gamma_0'' = \gamma_0''(d, L, a, W) > 0$ for $A = 0$ and, at regular A with $\theta_j \leq \theta_0'$, half of it; consequently $|C^{(j)}(u, v)| \leq Ce^{-\delta_1|u-v|}$.*

Proof. (a) Conjugation multiplies the hopping entry (x, x') by $e^{\delta(g(x)-g(x'))}$, $|g(x)-g(x')| \leq \vartheta$; the first-order part is anti-Hermitian and drops from $\mathrm{Re}\langle\phi, M_\delta\phi\rangle$, the remainder has norm $\leq 3d\delta^2$; the block term changes by $\leq (e^{\delta\sqrt{3}} - 1)W \leq 2\sqrt{3}\,W\delta$ (its Schur norm, (Q6)); Dirichlet terms are diagonal and commute. So $\mathrm{Re}\,M_\delta \geq \gamma - 3d\delta^2 - 2\sqrt{3}Wa\delta \geq \gamma/2$ for $\delta \leq \delta_0$. (b) $\Delta^{(j)}(u, v) = a_j\delta_{uv} - a_j^2\langle Q^*e_u, M^{-1}Q^*e_v\rangle$, $\|Q^*e_u\| = W_2^{1/2}$ (Q4), and (a). (c) The upper bound is $\Delta^{(j)} \leq a_j$,

from the minimiser identity $\langle\psi,\Delta^{(j)}\psi\rangle = \min_\Phi[\langle\Phi,H\Phi\rangle + a_j\|\psi - Q\Phi\|^2]$, $H := -\Delta_A + V_j$ (take $\Phi = 0$). Lower bound: dropping the fine bonds between distinct $(j+1)$-blocks Y lowers H and hence $\Delta^{(j)}$ (monotonicity of the minimiser identity), and $Q_c^*Q_c$ is $(j+1)$-block diagonal, so it suffices to bound $F_Y(\psi) := \langle\psi,(\Delta_Y^{(j)} + aL^{-2}Q_c^*Q_c)\psi\rangle$ from below on the L^3 coarse sites of one $(j+1)$-block Y (Neumann on its faces, Dirichlet at killed sites). $Q_c^*Q_c$ is the projector onto constants on Y; write $\psi = c + \psi_\perp$. Since $Q\mathbf{1} = \mathbf{1}$ (Q1), $\|\psi - Q\Phi\|^2 \geq \|\psi_\perp - (Q\Phi)_\perp\|^2 \geq (\|\psi_\perp\| - \|Q(\Phi - \langle\Phi\rangle)\|)_+^2 \geq (\|\psi_\perp\| - (WC_Ps)^{1/2})_+^2$ when $\langle\Phi,-\Delta_Y^N\Phi\rangle \leq s$, by (Q4) and the Neumann Poincaré inequality on the live sites of Y (a cube of side L, possibly minus a killed face layer, whose Dirichlet bonds only add; constant $C_P(L)$ uniform in ϑ). Springs in series: $\Delta_Y^{(j)}(\psi) \geq \|\psi_\perp\|^2/(WC_P + 1/a_j)$, and the coarse term is $aL^{-2}|c|^2L^3$; so $F_Y \geq \min\{(WC_P + 1/a)^{-1}, aL^{-2}\}\|\psi\|^2$ at $A = 0$. At regular A gauge away the constant part on Y (diameter $\sqrt{3}L$) at cost $C(L)\theta_j$. The decay of $C^{(j)}$ then follows from the lower bound and (b) by Combes–Thomas for exponentially decaying unit-lattice kernels (the conjugation argument of (a) with the Schur bound $\|e^{tg}Xe^{-tg} - X\| \leq Ct\sum_{z\in\mathbb{Z}^3}|z|e^{-\delta_0|z|/2}$), which is [4, (5.6)–(5.7)]. □

Lemma A.8 (Kato, and sup bounds with decay)**.** *Let $u = M(\Omega,A)^{-1}f$. (a) $(-\Delta_{0,\Omega}^{N,D} + V_j)|u| \leq |f - a_jP(A)u|$ pointwise on $\Omega \cap X_{\rm live}$, for every A and $V_j \geq 0$ (the free Laplacian with the same cuts and the same Dirichlet terms). (b) For block data b on the blocks of Ω and the leg $u = a_jM^{-1}Q^*b$,*

$$|u(x)| \leq C\sum_v e^{-\delta_2\,\mathrm{dist}(x,B(v))}|b(v)| \leq C|b|_\infty, \qquad |(Qu)(v)| \leq C\sum_{v'} e^{-\delta_0|v-v'|}|b(v')|,$$

at every point of every Ω and every A (given lemma A.6). (c) The same with Q^*b replaced by any f with block-L^2 data $\|f\mathbf{1}_{B(v)}\|_2 \leq |b(v)|$; in particular $\|M^{-1}f\|_\infty \leq C\|f\|_\infty$ and $|M^{-1}f(x)| \leq Ce^{-\delta_2\,\mathrm{dist}(x,\mathrm{supp}\,f)/2}\|f\|_\infty$ for bounded f.

Proof. (a) At $u(x) = 0$ the left side is ≤ 0. For $u(x) \neq 0$, $\mathrm{Re}[\bar u(x)(u(x) - U_{xx'}u(x'))]/|u(x)| \geq |u(x)| - |u(x')|$ since $U_{xx'}$ is orthogonal; a face bond contributes $\vartheta^{-2}|u(x)|$ on both sides; sum over the kept bonds and use $(-\Delta_A + V_j)u = f - a_jPu$. (b), (c) Block averages first: $\|\mathbf{1}_{B(v)}u\|_2 \leq (2a/\gamma)\sum_{v'} e^{-\delta_0\,\mathrm{dist}}\|\mathbf{1}_{B(v')}f\|_2$ by lemma A.7(a), and $|(Qu)(v)| \leq W^{1/2}\|\mathbf{1}_{B(v)}u\|_2$ (Q4). Then with $M_1 := -\Delta_{0,\Omega}^{N,D} + V_j + a_j$ write $u = M_1^{-1}[f + a_j(1-P)u]$ in the sense of (a): $|u| \leq M_1^{-1}|f + a_j(1-P)u|$ by inverse positivity of the M-matrix M_1, with $\|\mathbf{1}_B(1-P)u\|_2 \leq \|\mathbf{1}_Bu\|_2$ by (Q3). The kernel of M_1^{-1} is dominated entrywise by that of the Neumann Yukawa operator $(-\Delta_{\Omega'}^N + a_j)^{-1}$ on the union Ω' of the blocks of Ω and of the blocks adjacent to its top face blocks (so that the killed sites adjacent to $\Omega \cap X_{\rm live}$ are sites of Ω'): M_1 is a principal compression of that M-matrix plus a non-negative diagonal, and for non-singular M-matrices the inverse of a principal compression is dominated entrywise by the compression of the inverse (Schur complement, $(M^{-1})_{SS} = (M_{SS} - M_{SR}M_{RR}^{-1}M_{RS})^{-1}$ with $M_{SR}M_{RR}^{-1}M_{RS} \geq 0$ entrywise). Lemma A.9 gives the block summability $\sup_x\sum_B e^{\delta_2'\,\mathrm{dist}(x,B)}\|K_a^N(x,\cdot)\|_{L^2(B)} \leq C$ of the Neumann Yukawa kernel on every union of blocks, uniformly in ϑ; combining the two decays through $\mathrm{dist}(x,B') \leq \mathrm{dist}(x,B) + \sqrt{3} + \mathrm{dist}(B,B')$ gives (b) and (c) with $\delta_2 \leq \delta_0$. For bounded f only the blocks meeting $\mathrm{supp}\,f$ carry data. □

Lemma A.9 (block summability of the Neumann Yukawa kernel)**.** *Let Ω' be a union of unit blocks of $\vartheta\mathbb{Z}^3$ (any shape) with Neumann conditions and $K_a^N := (-\Delta_{\Omega'}^N + a_j)^{-1}$, with kernel for the measure ϑ^3. There are $C_Y', \delta_2' > 0$ depending on a only such that, uniformly in $\vartheta \leq 1$ and Ω', $\sup_x\sum_B e^{\delta_2'\,\mathrm{dist}(x,B)}\|K_a^N(x,\cdot)\|_{L^2(B)} \leq C_Y'$ and $K_a^N(x,y) \leq C_Y'e^{-2\delta_2'|x-y|}$ for $|x-y| \geq \frac12$.*

Proof. $-\Delta_{\Omega'}^N$ generates the reflecting continuous-time simple random walk on Ω', so $K_a^N(x,y) = \int_0^\infty e^{-a_jt}p_t(x,y)\,dt$ with p_t symmetric and $\|K_a^N(x,\cdot)\|_2^2 = \int_0^\infty e^{-a_ju}u\,p_u(x,x)\,du$. *On-diagonal bound.* On one unit cube Q the discrete Nash inequality $\|f\|_{2,Q}^{2+4/d} \leq C_N(E_Q(f) + \|f\|_{2,Q}^2)\|f\|_{1,Q}^{4/d}$ holds with C_N independent of ϑ (the discretisations of one fixed convex domain); it globalises to Ω' with no geometric hypothesis: with $a_Q := \|f\|_{2,Q}^2$, $e_Q := E_Q(f) + a_Q$, $b_Q := \|f\|_{1,Q}$

the cube inequality reads $a_Q \le C_N^{d/(d+2)} e_Q^{d/(d+2)} b_Q^{4/(d+2)}$, and summing over the cubes with Hölder's inequality at the exponents $\frac{d+2}{d}, \frac{d+2}{2}$, then $\sum b_Q^2 \le (\sum b_Q)^2$ and $\sum_Q E_Q \le E_{\Omega'}$, gives the Nash inequality on $\ell^2(\Omega')$. By [26, Thm. 2.1] this is equivalent to $p_u(x,x) \le Cu^{-d/2}$ for $u \le 1$; trivially $p_u(x,x) \le \vartheta^{-d}$ and $p_u \le p_1$ for $u \ge 1$. In $d = 3$ the integral $\int_0^{\vartheta^2} \vartheta^{-3} u\, du + \int_{\vartheta^2}^1 Cu^{-1/2} du + \int_1^\infty e^{-a_j u} Cu\, du$ is bounded, which bounds the near blocks. *Far points*, $D := |x-y| \ge \frac{1}{2}$: for a 1-Lipschitz g the conjugated form satisfies $\mathrm{Re}\, e^{\theta g}(-\Delta^N_{\Omega'}) e^{-\theta g} \ge -\lambda(\theta)$, $\lambda(\theta) := 2d\vartheta^{-2}(\cosh(\theta\vartheta) - 1) \le d\theta^2 \cosh(\theta\vartheta)$ (bond by bond, each site in $\le 2d$ bonds), so $\|\mathbf{1}_F e^{t\Delta^N} \mathbf{1}_E\| \le \exp(-\theta D + d\theta^2 t \cosh(\theta\vartheta))$ for sets at distance D, which is $\le e^{-D^2/(5dt)}$ for $t \ge \vartheta D$ and $\le e^{-D/(8d\vartheta)}$ for $t \le \vartheta D$. For $t \le \vartheta D$ this gives $p_t(x,y) \le \vartheta^{-d} e^{-D/(8d\vartheta)}$ and $\int_0^{\vartheta D} e^{-a_j t} p_t\, dt \le Ce^{-D/(16d)}$. For $\vartheta D \le t \le 1$, Davies' method [27] (the twisted L^p energy inequalities for the conjugated semigroup, with the Nash inequality above and $\lambda_p(\theta) \le C_p \theta^2 \cosh(\theta\vartheta)$, give the $2 \to \infty$ bound and duality the $1 \to 2$ bound) gives $p_t(x,y) \le Ct^{-d/2} e^{-\theta D + C_0 dt\theta^2 \cosh(\theta\vartheta)}$, and $\theta = D/(2C_0 dt)$ gives $p_t \le Ct^{-d/2} e^{-cD^2/t}$, whose time integral is $\le Ce^{-cD^2/2}$. For $t \ge 1$ write $p_t(x,y) = \langle p_{1/2}(y,\cdot), e^{(t-1)\Delta^N} p_{1/2}(x,\cdot)\rangle$, split each $p_{1/2}$ into its part within $D/3$ of its base point and the rest; the main term is $\le Ce^{-D^2/(45dt)}$ by the Davies–Gaffney bound with $\|p_{1/2}(x,\cdot)\|_2^2 = p_1(x,x) \le C$, and the other terms are $\le Ce^{-cD^2}$; then $\int_1^\infty e^{-a_j t} e^{-c_1 D^2/t} dt \le (2C/a_j) e^{-\sqrt{2a_j c_1} D}$. Altogether $K_a^N(x,y) \le C'_Y e^{-2\delta'_2 D}$ with $2\delta'_2 = \min\{\frac{1}{48d}, \frac{c}{2}, \sqrt{\frac{3}{2} a c_1}\}$, and the block sum follows (at most 3^d near blocks; far blocks by the pointwise bound and unit volume). □

Lemma A.10 (linear vanishing at the Dirichlet face)**.** *For every Ω, every A (given lemma A.6), block data b on the blocks of Ω and every $x \in \Omega \cap X_{\mathrm{live}}$: $|u(x)| \le C_{\mathrm{DV}} |b|_\infty \min(1, t(x))$ for the leg u; the same for $u = M^{-1} f$ with bounded f ($C_{\mathrm{DV}} \|f\|_\infty$).*

Proof. Fix the face μ with $t_\mu(x) = t(x) \le 1$ (otherwise lemma A.8). Let $F_0 := a_j W(1+C)|b|_\infty$ bound $|a_j Q^*(b - Qu)|$ ((Q6), lemma A.8 for $|Qu|$) and S_0 the sup bound of lemma A.8. Put $\rho(s) := (S_0 + F_0)s - F_0 s^2/2$ on $[0, 1+\vartheta]$: increasing there, $\rho(0) = 0$, $\rho(1) \ge S_0$, $\rho \ge 0$, and $-\Delta_s \rho = F_0$ exactly (second difference of a quadratic). Comparison region $R :=$ the first-layer slab $\Omega \cap X_{\mathrm{live}} \cap \{0 < t_\mu \le 1\}$, with $\Phi(x) := \rho(t_\mu(x))$. On R: $(-\Delta^{N,D}_{0,\Omega} + V_j)\Phi \ge F_0 \ge$ the right-hand side of Kato (lemma A.8(a)), because tangential bonds and tangential Neumann faces see Φ constant, a Neumann face of Ω at the far side of the first-layer block drops a neighbour with larger Φ (which raises $-\Delta^N \Phi$), and on face bonds the killed neighbour carries $\Phi = \rho(0) = 0 = |u|$. The boundary of R inside $\Omega \cap X_{\mathrm{live}}$ consists of sites at $t_\mu \in \{1, 1+\vartheta\}$, where $|u| \le S_0 \le \Phi$. The operator $-\Delta^{N,D}_{0,R} + V_j$ with these data is a non-singular M-matrix ($V_j > 0$), so the maximum principle gives $|u| \le \Phi \le (S_0 + F_0) t_\mu$ on R. Other Dirichlet faces met by R carry $u = 0 \le \Phi$. For bounded f take $F_0 := \|f\|_\infty + a_j WC \|f\|_\infty$. □

No hypothesis on the faces of Ω is used: on the cube every sub-domain carries the Dirichlet condition of the cube at the Dirichlet planes, and the barrier sees the Neumann cuts only through the zero drop of a function constant along them.

Lemma A.11 (tangentially localised amplitude)**.** *There are $\delta_5 > 0$ and C such that for every Ω, every A, every block v of Ω, every face μ and every x in the first-layer slab of μ, the leg $l(\cdot, v)$ of the unit datum on v satisfies $|l(x,v)| \le C(\vartheta + t_\mu(x)) e^{-\delta_5 \operatorname{dist}(x, B(v))}$; consequently $|u(x)| \le C|b|_\infty (\vartheta + t(x)) e^{-\delta_5 \operatorname{dist}(x, \operatorname{supp} b)/2}$ there, and the same for bounded sources supported in a union of blocks.*

Proof. The barrier of lemma A.10 is multiplied by tangential weights: for each $\nu \ne \mu$ put $f_\nu(s) := \exp(-\delta|\tilde\psi(s) - \tilde\psi(v_\nu)|)$ with $\tilde\psi(s) := s + (2\pi)^{-1} \sin(2\pi(s + \frac{1}{2} + \frac{\vartheta}{2}))$, whose derivative vanishes exactly on the block midplanes $s \in \mathbb{Z} - \vartheta/2$, so that f_ν takes equal values at the two sites adjacent to any block midplane and every Neumann cut of Ω (which lies on a block midplane) drops a bond of zero difference. The tangential directions are intervals $[0, N]$ with killed ends, on which $u = 0 \le$ barrier, so no periodisation is needed. Term by term, $-\Delta_\nu f_\nu \ge -c_\delta f_\nu$ with $c_\delta = (2\pi\delta + 4\delta^2) e^{3\delta}$ (the kink at v_ν is concave). With $\Phi := \rho(t_\mu) \prod_{\nu \ne \mu} f_\nu$ and F_0, S_0

enlarged by a fixed factor, $(-\Delta^{N,D}+V_j)\Phi \geq \prod f_\nu[F_0 - 2(c_\delta + 8\delta)\sup\rho] \geq \frac{1}{2}F_0\prod f_\nu$ for $\delta \leq \delta_5$ small; the source $|a_jQ^*(\delta_v - Ql)| \leq F_0 e^{-\delta_2\,\mathrm{dist}(\cdot,B(v))}$ (lemma A.8) is dominated by $\frac{1}{2}F_0\prod f_\nu$ for $\delta \leq \delta_2/\sqrt{3}$, and so are the data at $t_\mu \in \{1, 1+\vartheta\}$. The maximum principle gives $|l| \leq \Phi$, and $\prod f_\nu \leq e^{2\delta/\pi}e^{-\delta\sum_\nu |x_\nu - v_\nu|}$ with $\mathrm{dist}(x, B(v)) \leq \sum_\nu |x_\nu - v_\nu| + (t_\mu(v)-1)_+ + 3$. For bounded sources decompose $f = \sum_v f\mathbf{1}_{B(v)}$, apply the barrier block by block and sum, halving the rate. □

Interior estimates. For regular A, $r \in [2\vartheta, 1]$ and a ball $B_{2r}(x_0)$ of sites on which $-\Delta_A u = g$ holds as a full-lattice equation, $\sup_{B_r}|D_Au| \leq C(r^{-1}\sup_{B_{2r}}|u| + r\sup_{B_{2r}}|g|)$, together with the α-Hölder bound of D_Au on B_r; this is [4, (2.31)–(2.33)], which gauges away the constant part of A on the ball and perturbs from the free lattice Green's function. On the cube the full-lattice equation holds at every live site whose neighbours are live or killed (extend u by 0 to K: a face bond carries no link), so the estimate applies to every ball containing no killed site and no site within one bond of $\partial_N\Omega$; the open ball $B_{t(x)}(x)$ contains no killed site.

Lemma A.12 (first differences with decay, up to the face)**.** *For every Ω and regular A, for the leg u of data b supported in a set S of blocks, and for $u = M^{-1}f$ with bounded f supported in S: $|D_Au(b')| \leq C\|b\|_\infty\, e^{-\delta_3\,\mathrm{dist}(b',S)}$ (resp. $C\|f\|_\infty$) at every bond b' of $\Omega \cap X_{\mathrm{live}}$ whose endpoints have depth $\geq \frac{1}{4}$, face bonds included; on the whole cube at every bond. The α-Hölder quotients of D_Au on the balls used in the proof obey the same bounds with the same decay.*

Proof. Let x be the base of b' and $D := \mathrm{dist}(x, S)$. For $\vartheta > \frac{1}{32}$ every clause follows from $|D_Au(b')| \leq 2\vartheta^{-1}\max_{x'\sim x}|u(x')|$ and lemma A.8; let $\vartheta \leq \frac{1}{32}$. (1) Face bonds: $|D_Au| = |u(x)|/\vartheta \leq C_{\mathrm{DV}}\|b\|$ by lemma A.10, with decay by lemma A.11. (2) $t(x) \leq 4\vartheta$: $|D_Au(b')| \leq 2\vartheta^{-1}\max_{x'\sim x}|u(x')| \leq Ce^{-\delta_5 D/2}$ by lemma A.11 (amplitude $\vartheta + t \leq 6\vartheta$). (3) $4\vartheta < t(x) \leq \frac{1}{8}$: $r := t(x)/2 \geq 2\vartheta$; $B_{2r}(x) = B_{t(x)}(x)$ has no killed site and, as $\mathrm{depth}(x) \geq \frac{1}{4} \geq t(x) + \frac{1}{8}$, no site within one bond of $\partial_N\Omega$; the source $g := -V_ju - a_jPu + a_jQ^*b$ (resp. $f - V_ju - a_jPu$) obeys $|g| \leq Ce^{-\delta_5(D-t)/2}$ on the ball, and $\sup_{B_t}|u| \leq Ct\,e^{-\delta_5(D-t)/2}$ by lemma A.11 (the sites of $B_t(x)$ have $t_\mu \leq 2t \leq \frac{1}{4}$); the interior estimate gives $|D_Au(b')| \leq C(r^{-1}t + r)e^{-\delta_5 D/2} \leq Ce^{-\delta_5 D/2}$ and $[D_Au]_{\alpha;B_r} \leq Ct^{-\alpha}e^{-\delta_5 D/2}$. (4) $t(x) > \frac{1}{8}$: $r := \frac{1}{16}$, $B_{1/8}(x)$ has no killed site and no site within one bond of $\partial_N\Omega$ ($\mathrm{depth} \geq \frac{1}{4} \geq \frac{1}{8} + 2\vartheta$); the interior estimate with lemma A.8 gives $|D_Au|, [D_Au]_\alpha \leq Ce^{-\delta_2(D-1/8)}$. Take $\delta_3 := \delta_5/4$. □

Corollary A.13 (external legs)**.** *With the amplitude profile $\rho_j(x) := \min\{1, \vartheta + t(x)\}$ and Bałaban's external-field norm [3, (1.32)] (sup with the amplitude, first differences, α_0-Hölder quotients of the first differences on unit balls), for base points of depth ≥ 2 on Ω (every base point on the whole cube): (a) $\|a_jM^{-1}Q^*b\| \leq C\|b\|_\infty$ for the leg of block data; (b) $\|G_{(j)}(\cdot, z)\| \leq C$ for the slice lines $G_{(j)} = G_j - G_{j-1}$ on unions of $(j+1)$-blocks, $j \geq 1$.*

Proof. (a) The sup clause with the amplitude $\rho_j^{1/2}$ from lemmas A.8 and A.10 by the geometric mean $\min(P, Q) \leq P^{1/2}Q^{1/2}$; the first-difference clause from lemma A.12; the Hölder clause: for $|x-y| \geq r(x)$ the quotient is $\leq 2\sup_{B_1(x)}|D_Au|r^{-\alpha_0}$ and $\rho^{1/2}r^{-\alpha_0} \leq C$ in each case of the proof of lemma A.12 ($\rho \leq 2t$, $r = t/2$, $\alpha_0 \leq \frac{1}{2}$; or $r = \frac{1}{8}$; or $\rho \leq 5\vartheta$, $r \asymp \vartheta$); for $|x-y| < r(x)$ the Hölder bounds of that proof give $\rho^{1/2}t^{-\alpha_0} \leq C$. (b) The slice identity $G_{(j)} = a_jG_jQ^*C^{(j)}QG_j$ of [13, (2.17)], which is algebra valid for every weight, with lemmas A.7 and A.8. □

A.4. **The face-localised difference and the background of small-field data.**

Lemma A.14 (reproduction of the constant)**.** *On the doubled torus $2X$ with the periodic extension of the weight (the killed sites of the cube being ordinary zero-weight sites there), and on the cube, $a_jG_jQ^*\mathbf{1} = \mathbf{1} - V_jG_j\mathbf{1}$ at $A = 0$, where on the cube $G_j := M(X_{\mathrm{live}}, 0)^{-1}$ and the identity holds up to the face term of lemma A.15; and $\|G_j\mathbf{1}\|_\infty \leq C$.*

Proof. $Q\mathbf{1} = \mathbf{1}$ (Q1), so $P\mathbf{1} = Q^*\mathbf{1}$. With $v := \mathbf{1} - V_jG_j\mathbf{1}$: $(-\Delta + V_j + a_jP)v = V_j + a_jP\mathbf{1} - V_j = a_jQ^*\mathbf{1}$ on the torus, where $-\Delta\mathbf{1} = 0$. On the cube $-\Delta^D\mathbf{1} = \mathsf{B}$, the face term of lemma A.15. The sup bound is lemma A.8(c). □

Lemma A.15 (the Dirichlet Poisson term decays)**.** *Let* $\mathsf{B}(x) := \vartheta^{-2}\#\{\text{killed neighbours of } x\}$ *(supported on* F_1*) and* $v_0 := M(X_{\rm live},0)^{-1}\mathsf{B}$*. Then* $|v_0(x)| \le Ce^{-(\delta_2/2)(t(x)-4)}$ *for every* x*, and* $|D_0 v_0(b')| \le Ce^{-(\delta_2/2)(t(b')-5)}$ *at every bond with* $t \ge 6$*.*

Proof. (1) $M(X_{\rm live},0)\mathbf{1} = \mathsf{B} + V_j + a_j Q^*\mathbf{1}$ on $X_{\rm live}$, so $h^0 := a_j M^{-1}Q^*\mathbf{1} = \mathbf{1} - M^{-1}V_j - v_0$ and v_0 is bounded by $1 + Cc_V + C$ (lemma A.8). (2) Let $\chi(x) := \prod_\mu \psi(x_\mu)\psi(N - x_\mu)$ with ψ smooth, 0 on $(-\infty, 3.25]$, 1 on $[3.75, \infty)$, constant on a ϑ-neighbourhood of every block plane. Then $M(\chi v_0) = [M,\chi]v_0$ (since $\mathsf{B} = 0$ where $\chi \ne 0$), supported on the blocks at face distance in $[3,4]$; the Caccioppoli inequality on a two-block neighbourhood (where $Mv_0 = 0$; the bond sum over $X_{\rm live}$ has no boundary term, the face bonds entering with $v_0 = 0$ at K) bounds $\|\mathbf{1}_B[M,\chi]v_0\|_2 \le C\sup|v_0| \le C$, and lemma A.8(c) with block-L^2 data gives $|\chi v_0(x)| \le C\sum_{B\in\text{shell}} e^{-\delta_2 \operatorname{dist}(x,B)} \le Ce^{-(\delta_2/2)(t(x)-4)}$. (3) The interior estimate on $B_{1/4}(x)$ at $t \ge 6$ with $g = -V_j v_0 - a_j P v_0$. □

Lemma A.16 (face-localised difference, whole cube, regular A)**.** *Let* $T := 2X$ *carry the periodic extension of the weight and the even extension* $A_T := \mathsf{E}A$ *of a regular field* A *(regular with the same constant by lemma A.4(b)), and* $M_T := -\Delta_{A_T} + V_j + a_j P^T$*. For bounded* F *supported in* $X_{\rm live}$ *put* $v := M_T^{-1}F$*,* $u := M(X_{\rm live}, A)^{-1}F$ *and* $w := v|_{X_{\rm live}} - u$*. Then for every site* x *and fine bond* b *of the cube,*

$$|w(x)| + |D_A w(b)| \;\le\; C\exp\big[-\delta_F\big(\operatorname{dist}(\cdot,\partial D) + \operatorname{dist}(\operatorname{supp}F,\partial D) + \operatorname{dist}(\cdot,\operatorname{supp}F)\big)\big]\,\|F\|_\infty,$$

with C, δ_F *depending on* (d, L, a, W, c_V) *and King's constants only.*

Proof. (1) For $x \in X_{\rm live}$, $(M_T v)(x)$ and $(M(X_{\rm live},A)v|_X)(x)$ differ only by the hopping to killed neighbours, since $P^T(x,\cdot)$ is supported in the block of x, whose weighted sites are live (Q8). So $(M(X_{\rm live},A)v|_X)(x) = F(x) + h(x)$ with $h(x) := \vartheta^{-2}\sum_{x'\in\mathsf{K},x'\sim x} U_{xx'}v(x')$ supported on F_1, and $w = M^{-1}h$. (2) Near the face, $\operatorname{dist}(x,\partial D) \le D_0 := 4 + 2\sqrt{3}$: $|w| \le |v| + |u|$ and $|D_A w| \le |D_A v| + |D_A u|$; on the torus $|v(x)| \le Ce^{-\delta\operatorname{dist}(x,\operatorname{supp}F)}\|F\|$ (lemma A.8(c) on T) and $|D_A v(b)| \le Ce^{-\delta\operatorname{dist}(b,\operatorname{supp}F)/2}\|F\|$ (interior estimate on unit balls); on the cube the same for u up to the face (lemma A.8(c), lemma A.12 in the bounded-source form). As $\operatorname{dist}(\operatorname{supp}F,\partial D) \le \operatorname{dist}(x,\operatorname{supp}F) + D_0$ near the face, the claim holds there with $\delta_F \le \delta/4$. (3) Away from the face: let χ be smooth, $\chi = 0$ on $\{\operatorname{dist}(\cdot,\partial D) \le 1 + \sqrt{3}\}$, $\chi = 1$ on $\{\operatorname{dist} \ge D_0 - 1\}$, $|\nabla\chi| \le 1$, $|\nabla^2\chi| \le C$. Then $\chi h = 0$ and $M(\chi w) = [M,\chi]w =: g$ with $g = [-\Delta_A,\chi]w + a_j[P,\chi]w$; the lattice Leibniz rule gives $|[-\Delta_A,\chi]w| \le C(|w| + |D_A w|)$ on the shell $S := \{\sqrt{3} \le \operatorname{dist}(\cdot,\partial D) \le D_0\}$, and $|[P,\chi]w(x)| \le \sum_{x'}\vartheta^3|P(x,x')||\chi(x) - \chi(x')||w(x')| \le 2\sqrt{3}\,W\sup_{B(x)}|w|$ by (Q6), (Q8), supported on the blocks meeting the shell. So g is bounded, supported in S' (the union of S with the blocks meeting it, at distance $\le D_0 + \sqrt{3}$ from ∂D and ≥ 1 from K), and by (2) $|g(z)| \le Ce^{-\delta\operatorname{dist}(z,\operatorname{supp}F)/2}\|F\|$. (4) By lemma A.8(c) in the decaying form, $|\chi w(x)| \le C\sup_{z\in S'} e^{-\delta_2|x-z|/2}|g(z)|$, hence for $\operatorname{dist}(x,\partial D) \ge D_0$: $|w(x)| \le C\sup_{z\in S'} e^{-\delta_2|x-z|/2-\delta\operatorname{dist}(z,\operatorname{supp}F)/2}\|F\|$; with $|x-z| \ge \operatorname{dist}(x,\partial D) - D_0 - \sqrt{3}$, $\operatorname{dist}(z,\operatorname{supp}F) \ge \operatorname{dist}(\operatorname{supp}F,\partial D) - D_0 - \sqrt{3}$ and $|x-z| + \operatorname{dist}(z,\operatorname{supp}F) \ge \operatorname{dist}(x,\operatorname{supp}F)$, averaging the exponents gives the claim for $|w|$ with $\delta_F := \min(\delta,\delta_2)/6$. (5) For $\operatorname{dist}(x,\partial D) \ge D_0$, w solves $-\Delta_A w = -V_j w - a_j P w$ on $B_{1/4}(x)$ ($h = 0$ there), so the interior estimate gives $|D_A w| \le CW\sup_{B_{1/2}(x)+\text{one block}}|w|$, which obeys the bound of (4) with D_0 replaced by $D_0 + 1$. □

The comparison operator is the torus operator with an arbitrary regular extension of A; the scalar is never reflected, so no regularity of a reflected field is needed. No flat-face condition is used: the cut-off sees only $\operatorname{dist}(\cdot,\partial D)$, so edges and corners of the cube are included.

Lemma A.17 (the Dirichlet image effect on a sub-domain, sup form)**.** *Let* Ω *be a sub-domain and* $\Omega^\sharp$ *a Neumann domain without Dirichlet face that contains* $\Omega \cap X_{\rm live}$ *and the killed sites adjacent to it, adds no bond from* $\Omega \cap X_{\rm live}$ *except the bonds to those sites, and carries a regular extension* $A^\sharp$ *of* A *(for instance* Ω *together with its mirror blocks across the Dirichlet planes it meets, with the even extension of every component of* A*). For data* b *on the blocks of* Ω *(zero on the added blocks) let* $u := a_j M(\Omega,A)^{-1}Q^*b$ *and* $u^\sharp := a_j M(\Omega^\sharp,A^\sharp)^{-1}Q^*b$*. Then for every*

$x \in \Omega \cap X_{\mathrm{live}}$, $|u(x)-u^\sharp(x)| \le C_U|b|_\infty e^{-(\delta_2/2)(t(x)-4)}$ with $C_U = C_U(d,a,W,\gamma)$, free of Bałaban's large-block size M, of ϑ, j, Ω and A; and $|D_A(u-u^\sharp)(b')| \le C|b|_\infty e^{-(\delta_2/2)(t(b')-5)}$ at bonds with $t \ge 6$ and depth ≥ 1.

Proof. $w := u - u^\sharp|_\Omega$ solves $M(\Omega)w = 0$ at every site of $\Omega \cap X_{\mathrm{live}}$ off F_1 (same bonds, same blocks; the killed sites have zero weight). The product cut-off of lemma A.15, the Caccioppoli inequality in the region $t \ge 2.25$, $\sup|w| \le |u| + |u^\sharp| \le C|b|$ (lemma A.8 on both domains) and lemma A.8(c) with block-L^2 data give the sup bound exactly as in lemma A.15; the derivative bound is the interior estimate. □

Lemma A.18 (transported constant up to the face)**.** *Let A be regular on X_{live}, b a fine bond with base x_0, $\lambda_0(z) := \sum_\mu A_\mu(x_0)z_\mu$, $A' := A - d\lambda_0$ (gauge equivalent to A on the simply connected cube, with $A' = 0$ on the bonds at x_0 and $|A'_{b''}| \le K_j|b''-x_0|_1$), and $h := a_j M(X_{\mathrm{live}},A')^{-1}Q(A')^*\mathbf{1}$. Then $|D_{A'}h(b)| \le C_H(V_j + \theta_j + e^{-\delta_h t(x_0)})$ with $\delta_h := \delta_2/2$ and $C_H = C_H(d,L,a,W,\gamma,c_V)$.*

Proof. If $t(x_0) < 6$ the claim is lemma A.12 (h is the leg of the datum $\mathbf{1}$). Let $t(x_0) \ge 6$. (1) At $A = 0$: by lemma A.15(1), $D_0h^0 = -D_0M_0^{-1}V_j - D_0v_0$, so $|D_0h^0(b)| \le CV_j + Ce^{-\delta_h(t-5)}$ (lemma A.12 for the bounded source V_j, lemma A.15). (2) $w := h - h^0 = M(A')^{-1}S$ with $S := a_j(Q(A')^* - Q(0)^*)\mathbf{1} - (M(A') - M(0))h^0$. Pointwise $|S(z)| \le C\theta_j(1+|z-x_0|)^2$: the block terms by $|U(A'(\Gamma)) - 1| \le e_j|A'||\Gamma| \le C\theta_j(1+|z-x_0|)$ and $|h^0| \le C$; the Laplacian terms by pairing the two bonds of each direction, $\vartheta^{-2}[(1-U_+)+(1-U_-)]h^0(z) + \vartheta^{-2}\sum_\pm(1-U_\pm)(h^0(z_\pm)-h^0(z))$, where the first bracket is $-\vartheta e_j q(a_+ - a_-) + O(\vartheta^2 e_j^2|A'|^2)$, so its ϑ^{-2}-multiple is $\le \theta_j + C\theta_j^2|z-x_0|^2$, and the second is $\le \sum e_j|A'||D_0h^0| \le C\theta_j|z-x_0|$ (lemma A.12 at $A=0$); at a first-layer site the normal pair is incomplete and the unpaired term is $\le \vartheta^{-1}e_j|A'| \cdot 2C_{\mathrm{DV}}\vartheta \le C\theta_j|z-x_0|$ by lemma A.10. (3) $|w(x)| \le C\theta_j(1+|x-x_0|)^2$ by lemma A.8(c), shell by shell around x_0. (4) The interior estimate for w on $B_{1/4}(x_0)$ (no killed site, $t \ge 6$) with $-\Delta_{A'}w = S - V_jw - a_jP(A')w$, bounded by $C\theta_j$ on $B_{1/2}(x_0)$, gives $|D_{A'}w(b)| \le C\theta_j$. (5) $D_{A'}h(b) = D_0h^0(b) + (D_{A'} - D_0)h^0(b) + D_{A'}w(b)$, and the middle term vanishes since $A' = 0$ on the bonds at x_0. □

Proposition A.19 (derivative clause for backgrounds of small-field data)**.** *Let A be regular on X_{live}, ϕ_j block data, b a fine bond with base x_0 in the block y_0, $r_j := R(1+\ln \ell_j^{-1})^\varrho$ Bałaban's region radius, and assume: (G) $|D_{A_j}\phi_j(\beta)| \le p_j$ on every block bond within block distance $2r_j$ of y_0; (S) $|\phi_j| \le M_j$; (F) King's face test $|\phi_j(y)| \le \sigma_j \le C_8p_j$ on the face blocks within $2r_j$ of y_0; and $M_j(V_j + \theta_j) \le p_j$, $M_je^{-\delta_* r_j} \le p_j$ with $\delta_* := \min(\delta_3,\delta_h)/2$ (both hold for $\eta \le \eta_1$ and $R \ge R_*$: $M_j\theta_j \le Ce\lambda^{-1/4}L^{1/4}(1+\ln L)^p p_j^2 \ell_j^{1/4}$). Then the background $\phi^{(j)} = a_jM(X_{\mathrm{live}},A)^{-1}Q(A)^*\phi_j$ satisfies $|D_A\phi^{(j)}(b)| \le C_{\mathrm{legs}}p_j$ at every fine bond of the cube, face bonds included, $C_{\mathrm{legs}} = C(1+C_8)$. On a sub-domain Ω, under the same hypotheses for the data on X_{live}, the domain background $\phi_\Omega^{(j)} := a_jM(\Omega,A)^{-1}Q^*(\phi_j|_\Omega)$ satisfies, at every bond of depth $\ge \frac12$,*

$$|D_A\phi_\Omega^{(j)}(b)| \;\le\; C_{\mathrm{legs}}p_j + CM_je^{-\delta'(\mathrm{depth}(b)-4)_+}, \qquad \delta' := \delta_3/2,$$

hence $\le (C_{\mathrm{legs}}+1)p_j$ at depth $\ge \rho_j := 4 + \delta'^{-1}\ln(CM_j/p_j)$, and $\rho_j \le r_j$ once $R \ge R'_*$.

Proof. Gauge to A' as in lemma A.18 (every quantity is gauge covariant). (1) For blocks y with $|y-y_0| \le r_j$ let U_y be the coarse transporter along the coordinate-ordered block path from y_0 to y (inside the convex cube); telescoping (G) gives $\phi_j(y) = U_yc + \psi(y)$, $c := \phi_j(y_0)$, $|\psi(y)| \le p_j|y-y_0|_1$; for $|y-y_0| > r_j$ put $\psi(y) := \phi_j(y)$, $|\psi| \le M_j$. In the gauge A', $|U_y - 1| \le C\theta_j(1+|y-y_0|)^2$. (2) The ψ-part: $|D_{A'}[a_jM^{-1}Q^*\psi](b)| \le \sum_y |D_{A'}l(b,y)||\psi(y)| \le C(p_j + M_je^{-\delta_3 r_j}) \le Cp_j$ by the decaying clause of lemma A.12; the transport correction $|c||D_{A'}[a_jM^{-1}Q^*((U-1)\mathbf{1}_{|y-y_0|\le r_j})](b)| \le CM_j\theta_j \le Cp_j$; the truncation $|c||D_{A'}[a_jM^{-1}Q^*\mathbf{1}_{|y-y_0|>r_j}](b)| \le CM_je^{-\delta_3 r_j} \le Cp_j$. (3) The constant part: $|c||D_{A'}h(b)| \le |c|C_H(V_j + \theta_j + e^{-\delta_h t(x_0)})$ by lemma A.18. (4) The cap: if $t(x_0) \le r_j$, transport along the normal block path from y_0 to the face layer (length $\le t(x_0)+1$) and (F) give $|c| \le \sigma_j + p_j(t(x_0)+1)$, so $|c|e^{-\delta_h t} \le C(1+C_8)p_j$; if $t > r_j$, $|c|e^{-\delta_h t} \le M_je^{-\delta_h r_j} \le p_j$; and $|c|C_H(V_j+\theta_j) \le Cp_j$. On Ω: $w := \phi_\Omega^{(j)} - \phi^{(j)}|_\Omega$ solves $M(\Omega)w = 0$

at depth ≥ 1 (same bonds and blocks there); with $\chi := \psi(\tilde{d}_N)$, $\tilde{d}_N$ a mollified depth constant near block planes, $M(\Omega)(\chi w) = [M, \chi]w =: F$, supported at depth in $[3.25, 3.75]$ plus one block, with $|F| \leq C(|w| + |D_A w| + \sup_{\text{block}} |w|) \leq CM_j$ (lemma A.8 for both backgrounds, lemma A.12 at depth ≥ 3); lemma A.12 for bounded sources gives $|D_A(\chi w)(b)| \leq CM_j e^{-\delta_3(\text{depth}(b)-4)}$, and $\chi = 1$ near b when $\text{depth}(b) \geq 4$; for $\frac{1}{2} \leq \text{depth}(b) < 4$ use lemma A.12 on both backgrounds. □

The face test is necessary: for the constant data $\phi_j \equiv c$ the background vanishes at the Dirichlet face and $|D_A \phi^{(j)}|$ is of order $|c|$ on the first block; this is why King imposes [14, (5.41)] at each step.

Lemma A.20 (face-localised background of a covariantly constant datum, regular A)**.** *Let Ω be a box of large blocks with Dirichlet conditions on its faces (the cube, or a sub-box), y_0 a block, $c \in \mathbb{C}$, T a tree of coordinate-ordered block paths from y_0, $\Theta_A(y) := \overline{U(A(\Gamma_{\mathsf{T}}(y_0 \to y)))}\, c$ the datum covariantly constant along T, and $u_A := a_j M(\Omega, A)^{-1} Q(A)^* \Theta_A$. There are $\delta_1 > 0$, C and $\theta_* > 0$ such that for A regular on Ω with $\theta_j \leq \theta_*$, every bond b of $\bar{\Omega}$ (face bonds included) and every block y: (a) $|D_{A,b} u_A| \leq |c|[CV_j + Ce^{-\delta_1 \operatorname{dist}(b, \partial D)} + C\theta_j(1 + |b - y_0|)^3]$; (b) $|\Theta_A(y) - (Q(A)u_A)(y)| \leq |c|[CV_j + Ce^{-\delta_1 \operatorname{dist}(B(y), \partial D)} + C\theta_j(1 + |y - y_0|)^3]$; (c) $\sup |u_A| \leq C|c|$.*

Proof. Step 1 ($A = 0$, $\Theta = c$). The Dirichlet operator of a box of large blocks with King's weight is the odd restriction of the free operator on $\vartheta\mathbb{Z}^3$ with the periodic weight: the odd extension of a function vanishes on the Dirichlet planes, the weight couples no live site across a plane, and the free operator preserves the odd subspace, on which it is unitarily $M(\Omega, 0)$ (coercivity gives uniqueness). So $M(\Omega, 0)^{-1} F = (G^{\text{free}} \mathsf{E}_{\text{odd}} F)|_{\bar{\Omega}}$ and $\mathsf{E}_{\text{odd}} Q^* c = Q^*(cs)$ with $s(y) = \pm 1$ the coarse odd extension. Write $s = 1 - (1 - s)$: by lemma A.14 on the free lattice, $a_j G^{\text{free}} Q^* \mathbf{1} = \mathbf{1} - V_j G^{\text{free}} \mathbf{1}$, so $u_0 = c[\mathbf{1} - V_j G^{\text{free}} \mathbf{1} - a_j G^{\text{free}} Q^*(1 - s)]$ on $\bar{\Omega}$; $(1 - s)$ is supported on the image boxes with $s = -1$, at distance $\geq \operatorname{dist}(b, \partial D)$ from b, and bounded by 2. With the weighted-ℓ^1 bounds $\|DG^{\text{free}}\|_{\delta_1}, \|G^{\text{free}}\|_{\delta_1} \leq C$ of the free weighted operator (lemmas A.7 and A.8 on $\vartheta\mathbb{Z}^3$), $|Du_0(b)| \leq |c|[V_j \|DG^{\text{free}}\|_0 + 2a_j W \|DG^{\text{free}}\|_{\delta_1} e^{-\delta_1 \operatorname{dist}(b, \partial D)}]$ at every bond, face bonds included (the intertwining holds there); the same computation with Q in place of D gives (b), and (c) is $\|G^{\text{free}}\|_0$. Step 2 (constant B): gauge covariance, $u_B = U_\lambda u_0(c')$ with $|c'| = |c|$, $\lambda = B \cdot x$; (a)–(c) hold with the θ term absent. Step 3 (regular A). Fix the bond b and let B be the constant field with the components of A at reference bonds through b; $A' := A - B$ obeys $e_j |A'(x)| \leq d\theta_j(1 + |x - b|_1)$ on the box (sum the first differences along a lattice path). With $w := u_A - u_B$ (both vanish on ∂D): $M_A w = F := a_j[Q(A)^* \Theta_A - Q(B)^* \Theta_B] + (M_B - M_A)u_B$; the hopping difference is a first-order operator with coefficients $\leq C\theta_j(1 + |x - b|)$ and the block part has row sums $\leq C\theta_j(1 + |x - b|)$, the contour transports differ by $\leq e_j \sum_\Gamma \vartheta |A'|$, and $|u_B|, |D_B u_B| \leq C|c|$ by Steps 1–2; so $|F(x)| \leq C\theta_j |c|(1 + |x - b| + |y_0 - b|)^3$. Split $F = \sum_n F_n$ by shells $n \leq |x - b| < n + 1$; lemma A.12 in the bounded-source form (face bonds included) gives $|D_{A,b} M_A^{-1} F_n| \leq Ce^{-\delta'(n-1)} \|F_n\|_\infty$ and the same for $|M_A^{-1} F_n|$ within one block of b; summing, $|D_{A,b} w| + \sup_{|x-b| \leq 2} |w| \leq C\theta_j |c|(1 + |y_0 - b|)^3$. Finally $D_{A,b} u_B = D_{B,b} u_B$ since A and B agree on b (on a face bond both are the gauge-free $\vartheta^{-1} |u_B|$), which gives (a); (b) from the four-term split $\Theta_A - Q(A)u_A = [\Theta_B - Q(B)u_B] + [\Theta_A - \Theta_B] - [(Q(A) - Q(B))u_B] - Q(A)w$, and (c) from the sup clause with $\|F\|_\infty = a_j W |c|$. □

The face rate δ_1 is the free rate of the weighted operator at $A = 0$ and does not depend on Bałaban's large-block size M; this is what allows the near-layer constant R_0 of appendix A.8 to be fixed before M.

Lemma A.21 (the block-mismatch identities)**.** *For every weight, domain, boundary condition and gauge field, with $\phi^{(j)} = a_j M^{-1} Q(A)^* \phi_j$: (a) $\phi_j - Q(A)\phi^{(j)} = a_j^{-1} \Delta^{(j)}(A)\phi_j$; (b) with the* uniform *transported average $Q^u(A)\psi(y) := \sum_{x \in B(y)} \vartheta^3 U(A(\Gamma_{y,x}))\psi(x)$, exactly $a_j(\phi_j - Q(A)\phi^{(j)})(y) = Q^u(A)[(-\Delta_A + V_j)\phi^{(j)}](y)$, whereas with $Q(A)$ in place of $Q^u(A)$ the left side acquires the factor W_2.*

Proof. (a) $Q\phi^{(j)} = a_j Q M^{-1} Q^* \phi_j$ and $\Delta^{(j)} = a_j - a_j^2 Q M^{-1} Q^*$. (b) The minimiser equation is $(-\Delta_A + V_j)\phi^{(j)} = a_j Q^*(\phi_j - Q\phi^{(j)}) =: a_j Q^* m$, and pointwise $(Q^* m)(x) = w(x) U(\Gamma_{y,x})^* m(y)$ for

$x \in B(y)$; applying $Q^u(A)$ gives $a_j m(y) \sum_x \vartheta^3 w(x) = a_j m(y)$ by (W1) and $UU^* = 1$; applying $Q(A)$ gives $a_j W_2 m(y)$. □

Identity (b) is the form in which the block-mismatch clause of the lower bound is read on the cube; the naive summation by parts of the *weighted* average $Q[-\Delta\phi](y)$ produces, besides the boundary flux of the block, interior terms $\sum_{\text{bonds}} \vartheta^{d-2}(w(x) - w(x'))(\phi(x) - \phi(x'))$ across the zero-weight layers, whose absolute bound $2dW \sup|D\phi|/(L\vartheta)$ diverges as $\vartheta \to 0$; the uniform transported average has no interior terms, and (a) needs no fine-level summation by parts at all.

A.5. **The Dirichlet-face positivity and the face family.**

Lemma A.22 (Dirichlet-face positivity)**.** *In unit-j units let $y' \in L_1$ be a face block, $B(y')$ its unit cube, and $\Delta^{(j)}$ the block form of the cube at any gauge field A. Then for every block field ϕ and $\theta \in (0,1)$,*

$$\langle \phi, \Delta^{(j)} \phi \rangle \;\geq\; (1-\theta)\langle \phi, \Delta^{(j)} \phi \rangle + \theta\, \gamma_D\, |\phi(y')|^2, \qquad \gamma_D = \frac{a_j}{1 + a_j W^2}.$$

The same holds for the composite multiscale form of [2, (3.23)–(3.25)] *restricted to the fine sites of $B^j(y')$, in the form*

$$\sum_{\text{bonds of } \overline{B}} \varepsilon^d |D_A \phi_0|^2 + \sum_{i<j} a\, \ell_{i+1}^{d-2} \sum_{B^{i+1}(y'') \subset B} |\phi_{i+1}(y'') - Q(A)\phi_i(y'')|^2 \;\geq\; (W + a_j^{-1})^{-1} \ell_j^{d-2} |\phi_j(y')|^2$$

(physical units; the first sum over the bonds inside $B = B^j(y')$ and the face bonds from B to the killed layer), for every gauge field.

Proof. By the minimiser identity, $\langle \phi, \Delta\phi \rangle = \min_\Phi F(\Phi)$, $F = \langle \Phi, (-\Delta_A^D + V_j)\Phi \rangle + a_j \|\phi - Q\Phi\|^2$. In θF keep only the gradient terms of bonds inside $B(y')$, the Dirichlet terms of the face bonds of $B(y')$ (each $\vartheta^{d-2}|\Phi(x)|^2$) and the block term of y'; all dropped terms are ≥ 0. With $g := |\Phi|$, unitary transport gives $|D_A \Phi(b)| \geq |g(b_+) - g(b_-)|$, and the face terms are $\vartheta^{d-2} g(x)^2$. Discrete Friedrichs inequality on the unit cube with the killed layer behind one face: on each line normal to that face, $g(x) = \sum_{s \leq m} [g(s) - g(s-1)]$ with $g(0) := 0$ at the killed site and $m \leq \vartheta^{-1}$, so $g(x)^2 \leq \vartheta^{-1} \sum_{\text{line}} |\partial g|^2$; summing $\vartheta^d g^2$ over the line and over all lines, $\sum_{B(y')} \vartheta^d g^2 \leq E(g)$, the kept energy. Then $|(Q\Phi)(y')| \leq \sum \vartheta^d w |\Phi| \leq W (\sum \vartheta^d g^2)^{1/2} \leq W E^{1/2}$. So $\theta F \geq \theta \min_{s \geq 0} [W^{-2} s^2 + a_j (|\phi(y')| - s)_+^2] = \theta \gamma_D |\phi(y')|^2$ (springs in series), and minimising over Φ gives the first claim. Composite form: with $s_i := (\ell_i^d \sum_B |\phi_i|^2)^{1/2}$, Friedrichs along the normal lines from the killed layer gives $s_0^2 \leq \ell_j^2 E$ with E the first sum; Cauchy–Schwarz with the weights gives $\|Q_1 \phi_0\| \leq W^{1/2} s_0$ at the weighted step and $\|Q\phi_i\| \leq s_i$ at the uniform steps $i \geq 1$; the reverse triangle inequality gives $s_j \leq W^{1/2} s_0 + \sum_i t_i$, $t_i := \|\phi_{i+1} - Q\phi_i\|$, and Cauchy–Schwarz with the weights $(W\ell_j^2, \ell_{i+1}^2/a)$ gives $s_j^2 \leq (W\ell_j^2 + \ell_j^2/a_j)(E + \sum_i a \ell_{i+1}^{-2} t_i^2)$, using $\sum_{i<j} \ell_{i+1}^2/a = \ell_j^2/a_j$. □

This is the extra term $\gamma_0 \sum_{x \in \bar{\Omega}^{(k)}} |\phi(x)|^2$ of [14, Prop. 5.4, (5.26)], valid for every gauge field (King's printed proof [14, (5.28)–(5.29)] expands around a constant field and needs A regular). King's constant [14, (5.38)–(5.40)] is consistent with it.

Definition A.23 (the face family)**.** At every scale $0 \leq i \leq K_f$ of the upper bound's large-field decomposition, $i = 0$ included, the *face family* is the set of events $\{|\phi_i(y')| > \sigma_i\}$ at $y' \in L_1^{(i)} \cap \Lambda_7^{(i-1)\prime}$ (with $\Lambda_7^{(-1)\prime} :=$ everything), where $L_1^{(i)}$ is the set of i-blocks containing a site of F_1 (for $i \geq 1$ the face blocks of both faces in every direction; for $i = 0$ the first live layer F_1 itself, where ϕ_0 is the fine field), and $\sigma_i := (2/(\theta_2 \gamma_D^{(i)}))^{1/2} p_i \leq C_8 p_i$ with $\gamma_D^{(0)} := 1$. At the stopping scale the same test is the indicator λ_j of appendix A.1 (it is [14, (5.41)] with $C = \sigma_j/p_j$).

At $i = 0$ the payment is the θ_2-share of the face-bond term itself: $\langle \phi_0, (-\Delta_A^D + V)\phi_0 \rangle \geq \sum_{\text{face bonds } b \ni x} |D_A \phi_0(b)|^2 = n_F(x) |\phi_0(x)|^2$ (unit-0 units; no link on a face bond), with $n_F(x) \in$

$\{1,2,3\}$ the number of face bonds at x. The scale-0 member must be stated on the first live layer (not as the set of boundary blocks, which at $i=0$ would be the killed plane on one face), because the inherited cap of lemma A.26 at $k=1$ needs a cap on the first live layer of both faces.

Lemma A.24 (payment, and the corrected near-face estimates)**.** *(a) A violation of the face test at y' at scale i is paid by the factor $e^{-p_i^2/4}$ from the θ_2-share of the scale-i form (lemma A.22), and in the composite form of* [2, Prop. 3.1] *by its composite clause; violating blocks of different scales are disjoint as fine sets by Bałaban's region rules, so the payments add. (b) For $x\in\Lambda_4^{(i)}$ with $\operatorname{dist}(x,\partial D)\le r(\ell_i)$, a coordinate block path from x to the face layer of length $\le r(\ell_i)$ lies in $\Lambda_3^{(i)}\subset\Lambda_0^{(i)}$ (since $\operatorname{dist}(\Lambda_4,\Lambda_3^c)>r$), so on the complement of the face family and of the gradient family $|\phi_i(x)|\le\sigma_i+r(\ell_i)p_{i-1}\le Cp_i^2$, and the fluctuation field obeys $|\phi_i'(x)|\le Cp_i^2$ there. (c) King's printed threshold "$\operatorname{dist}(x,\partial T^{(k)})<p(L^k\varepsilon)$" in* [14, (5.43)–(5.45)] *cannot be used: $p(\ell)=b_0(1+\ln\ell^{-1})^p$ with $2p\ge(d+1)\varrho$ ([2, (3.54)]), so $p(\ell)/r(\ell)\to\infty$ and a contour of length up to $p(\ell)$ leaves $\Lambda_0^{(i)}$, where* [14, (5.41)] *is imposed; the correct threshold is one corridor $r(\ell_i)$, with which the conclusions of (b) hold. (d) For x at distance $>r(\ell_i)$ from ∂D the Dirichlet image effect on the background is $\le Ce^{-\delta_1 r(\ell_i)}M_i\le p_i$ (lemmas A.17 and A.20; $\varrho>1$), so* [2, (2.113)] *holds there as printed. (e) Within distance R_0 of the face either a gradient violation or a face-family element is present on a shortest block path from a block of $R_s^{(i)}$ to the face layer, and at face blocks off the families the kept quartic is $\ge -C\lambda(\ell_i)p_i^8$ per block, an error $\le C\ell_i^\alpha$ per block; this is* [14, (5.46)–(5.49)] *in event form.*

Proof. (a) Lemma A.22 with $\theta=\theta_2$ and σ_i chosen so that $\frac12\theta_2\gamma_D\sigma_i^2/4=p_i^2/4$; the disjointness is the nesting of Bałaban's regions applied to a pointwise family (the face family is a condition on one block variable at one block of the index set, of the same type as $|\psi(y)-(Q\phi)(y)|>p$, so the rules (2.4)–(2.10) of [2] hold with it verbatim). (b) Triangle inequality along the path, the gradient clause $|D_{A_i}\phi_i|\le p_{i-1}$ on each of $\le r$ bonds and $|U|=1$; the fluctuation $\phi'=\phi_i-aL^{-2}C^{(i)}Q^*\phi_{i+1}$ has its second term bounded by $C\sup|\phi_{i+1}|$ on a neighbourhood (decay of $C^{(i)}$, lemma A.7), which by (b) at scale $i+1$ is $\le Cp^2$. (c) Arithmetic: $p(\ell)/r(\ell)=(b_0/R)(1+\ln\ell^{-1})^{p-\varrho}$ with $p-\varrho=4$ at Bałaban's values $p=6$, $\varrho=2$. (d) Lemma A.17 (sup form) times the block-sup bound M_i; $e^{-\delta R(1+\ln\ell^{-1})^2}M_i\le p(\ell)$ for $\ell\le\eta_1$. (e) Along a shortest cube-internal block path from $x\in R_s^{(i)}$ (where $|\phi_i(x)|\ge M_i$) to $L_1^{(i)}$, of length $\le dR_0<s_b$ inside $\Lambda_7^{(i-1)\prime}$, either a bond carries a gradient violation, or the end block y' has $|\phi_i(y')|\ge M_i-dR_0p_i\ge\frac12M_i\ge\sigma_i$ (for $\lambda(\ell_i)\le(2C_8)^{-4}$ and $dR_0\le\frac12\lambda(\ell_i)^{-1/4}$, conditions on η_1), a face-family element; the per-block error is $\lambda(\ell_i)(Cp_i^2)^4$ by (b). □

Lemma A.25 (Bałaban's per-scale lower bound at the weight)**.** *For a union Λ of k-blocks with Neumann cuts and, where Λ meets ∂D, the Dirichlet condition of the cube, and A regular on Λ with $\theta_k\le\theta_*$,*

$$\langle\phi,\Delta^{(k)}(\Lambda,A)\phi\rangle\ \ge\ \gamma_0^w\Big[\sum_{\text{block bonds }\langle y,y'\rangle\subset\Lambda}|U(A_k(\langle y,y'\rangle))\phi(y')-\phi(y)|^2+V_k\sum_y|\phi(y)|^2\Big]-C(\theta_k+e_kp_k)^2\sum_{y\in\Lambda}|\phi(y)|^2$$

with $\gamma_0^w>0$ depending on (d,a,W) only. On King's small-field set $|\phi|\le M_k'$ the last term is $\le Ce_k^2p_k^2M_k'^2|\Lambda|=O(\ell_k^{\kappa_0})|\Lambda|$; at $A=0$ it is absent. This is [2, (3.29)] *at the weight.*

Proof. (1) Dirichlet faces only increase the form: the Dirichlet form on the live sites is the Neumann form plus the face terms $\vartheta^{d-2}|\phi(x)|^2\ge0$, so $\Delta_D^{(k)}\ge\Delta_N^{(k)}$ by the monotonicity of the minimiser identity. (2) $\langle\phi,\Delta\phi\rangle=\min_\Phi[\langle\Phi,H\Phi\rangle+a_k\|\phi-Q\Phi\|^2]$. For a block bond $\langle y,y'\rangle$: $|\phi(y)-U\phi(y')|^2\le3(|\phi(y)-Q\Phi(y)|^2+|Q\Phi(y)-UQ\Phi(y')|^2+|\phi(y')-Q\Phi(y')|^2)$. On the two-block box $B(y)\cup B(y')$ (simply connected) gauge to the constant field and compare: the averages and covariant differences at A and at the constant field differ by $\le C\theta_k\|\Phi\|_{B\cup B'}$, and $\|\Phi\|_B^2\le C(E_B(\Phi)+|\phi(y)|^2+|\phi(y)-Q\Phi(y)|^2)$ (block Poincaré with the weight); the block-bond transport $U(A_k)$ differs from the fine one by Ce_kp_k (the photon block-versus-background step [13, (A.2)–(A.6)], which is κ-free and holds on the cube by lemmas A.3 and A.4). At $A=0$,

$|\operatorname{avg}_w g - \operatorname{avg}_{w'} g|^2 \le 4WC_P(d)E_{B\cup B'}(g)$ by Cauchy–Schwarz with the weights ($\sum \vartheta^d w^2 = W$ on a unit block) and the two-block Poincaré inequality. Mass: $V_k|\phi(y)|^2 \le 2V_k(W\|\Phi\|^2_{B(y)} + |\phi(y) - Q\Phi(y)|^2)$. Summing (each block term counted $\le 2d$ times, each fine bond in ≤ 2 boxes) and absorbing the θ_k-parts gives the claim with $\gamma_0^w = [C(d)\max(4WC_P, 2W, (6d+2)/a_k)]^{-1}(1 - O(\theta_k))$. □

Lemma A.26 (the inherited face cap)**.** *Let $k \ge 1$ and fix a term of the step-$(k-1)$ large-field decomposition (a family of events with its bad union S_{k-1}). On the support of its characteristic functions (Bałaban's six families on $\Lambda_7^{(k-2)\prime}$ and the face family of scale $k-1$; at $k-1=0$ on the first live layer of fine sites), every face k-block $y \in L_1^{(k)}$ with $B(y) \subset \Lambda_0^{(k-1)}$ satisfies, in unit-k normalisation,*

$$|\phi_k(y)| \;\le\; L^{1/2}(\sigma_{k-1} + Lp_{k-1}) \;\le\; C_{\mathrm{IC}}\, p_k, \qquad C_{\mathrm{IC}} := L^{1/2}(1+\ln L)^p(C_8 + L),$$

free of ε, k, K, J and of M. By the nesting of the regions the cap holds at every face block of $\Lambda_2^{(k-1)\prime}$, in particular at every face block of the propagator domain $B^k(\Lambda_2^{(k-1)\prime})$, of the ring $\Lambda_6^{(k-1)\prime} \setminus \Lambda_7^{(k-1)\prime}$ and of S_k.

Proof. Every step-$(k-1)$ test located in $B(y)$ is imposed and passes: the face test at the $(k-1)$-blocks of $B(y)$ in $L_1^{(k-1)}$ (their index set contains them since $\Lambda_0^{(k-1)} \subset \Lambda_7^{(k-2)\prime}$), the gradient test on the block bonds with both ends in $B(y)$ (a violating bond would put its endpoints in S_{k-1}), and the new-field test $|\phi_k(y) - (Q\phi_{k-1})(y)| \le p_{k-1}$ at y (y lies in the prime of $\Lambda_7^{(k-2)\prime}$ and $y \in P_s^{(k-1)}$ would give $B(y) \subset S_{k-1}$). Since $B(y)$ is a face block, some $(k-1)$-block $y'' \subset B(y)$ is a face block with $|\phi_{k-1}(y'')| \le \sigma_{k-1}$; the gradient test along a block path inside $B(y)$ (length $\le L$) gives $|\phi_{k-1}| \le \sigma_{k-1} + Lp_{k-1}$ on all of $B(y)$, hence $|(Q\phi_{k-1})(y)| \le \sigma_{k-1} + Lp_{k-1}$ (Q5), and the new-field test gives $|\phi_k(y)| \le \sigma_{k-1} + (L+1)p_{k-1}$ in unit-$(k-1)$ normalisation, i.e. the claim after the conversion factor $L^{1/2}$ ($d=3$) and $p_{k-1} \le (1+\ln L)^p p_k$. At $k-1=0$ the step-0 face member is the cap on F_1 of definition A.23. □

Without the cap the fine-field class of [2, (2.55)] fails at the collar cubes: for constant data of size M_k the background's face gradient is of order M_k, not p_k. The cap is inherited from the previous step exactly as (2.55) is carried from step to step, as a characteristic function of the block field.

A.6. **The perturbative input on the cube.** Bałaban's perturbative bounds [3, Prop. 1] are consumed, in the lower bound at every step and in the upper bound at every scale, in the field-dependent form: for a renormalised family $\mathcal{R}$ of order $\le \bar n = 13$ of the expansion of the effective action, built from the lines (slices $G_{(i)}$ and G_k) and legs ($\phi^{(k)}$, photon backgrounds) of the step, with external fields in the class of corollary A.13 and possibly source vertices, the sum over orderings, indices and positions of the values of its members, with one vertex x_0 held, is bounded by $C\ell_k^{\beta_0} p(\ell_k)^{n_1}(1+K-k)^{p_0} e^{-\delta \operatorname{dist}(V_*)}$ times the external-field norms, with V_* the set of vertices carrying external legs or sources and $\beta_0 = \frac14 - \beta$; the R-terms (orders $> \bar n$) are $O(1)\ell_k^{\kappa'}|\Lambda^{(k)}|$ with $\kappa' > d$. On the torus this is [3, Prop. 1] with the hypotheses on the external fields supplied by [2, Lemma 2.4].

Proposition A.27 (the perturbative bounds on the cube)**.** *On C_J^K with King's flow, at every scale $k \le K_f$ and every regular A: (a) on the whole cube, the bound above holds for every renormalised family with the cube's lines and legs, with the counterterm King's free-boundary δm^2 ([14, p. 330], field- and domain-independent), in the organisation of [14, p. 330]: for a held vertex x_0 at face distance $t(x_0) \ge 1$ (zone I) with the torus constants; for $t(x_0) < 1$ (zone B) with the additional factor $\Lambda(x_0) \le C(1+\ln_+(1/t(x_0)))^{14} t(x_0)^{-(1-\beta)} e^{-\delta_\Lambda t(x_0)}$, which is summable: $\sum_{x_0 \in \text{unit cube}} \vartheta^d \Lambda(x_0) \le C$; (b) the same on every sub-domain Ω of the cube (any union of k-blocks, meeting ∂D or not) for the families with the lines $G_{(i)}(\Omega, A)$, $G_k(\Omega, A)$ and the legs $\phi_\Omega^{(k)}$, and with one line replaced by a difference line $G_k(\Omega, A) - G_k(\Omega_2, A)$ of two Neumann domains as an external pair, provided every vertex cube lies at depth $\ge R_*$ in every propagator domain of*

the family; in the difference case the bound carries the extra factor $\exp[-\delta_c(\mathrm{dist}(\mathrm{cube}(v), X_{\mathrm{live}} \setminus \Omega_2) + \mathrm{dist}(\mathrm{cube}(v'), X_{\mathrm{live}} \setminus \Omega_2))]$ *for the two ends* v, v' *of the difference line; (c) the consequences* [1, (3.58)] *hold with these inputs: the coefficients of the effective potential* $V^{(k)}$ *are* $O(1)\ell_k^{\beta_0}$ *and the remainders* $O(1)\ell_k^{\kappa'}|\Lambda^{(k)}|$*, uniformly in* k*,* K*,* ϑ *and the field.*

Proof. (a) Zone I is the torus bound: at face distance ≥ 1 every line is, up to the face-localised difference of lemma A.16 (exponentially small in the distance to the face and absorbed in the decay factors), the torus line at the same field, and the photon lines are the image sums of lemma A.5; the class of the external legs is corollary A.13. Zone B is the content of the boundary-remainder bounds established in the proof of proposition 5.5 (the scalar boundary remainders at zero background and the photon face terms of [14, pp. 331–332], with the slice bounds $|D_x^{n_x} D_y^{n_y} G^{\Omega}_{(i)}(b, b')| \leq C\ell_i^{-1-n_x-n_y} e^{-\kappa|x-y|/(2\ell_i)}$ and the cumulative line bound $|D_x^{n_x} D_y^{n_y} G^{\Omega}_{\leq i_0}(b, b')| \leq C|x-y|_\vartheta^{-1-n_x-n_y} e^{-\kappa|x-y|/(4\ell_{i_0})}$ at zero background on the cube and on King's localisation boxes, face bonds included, which follow from the odd-image representation of Step 1 of lemma A.20 and the free weighted bounds; and, at regular A, from lemmas A.7, A.8 and A.12 through the gauge-away of the constant part on each box): there every line is written as its free counterpart plus a boundary remainder localised at the face, and the remainder vertices are of higher degree, so that the degree bookkeeping of [3, Sect. 2] is unchanged while the sum over the position of the held vertex produces $\Lambda(x_0)$. The photon-image part of the counterterm is the photon-image part of the same remainder families (appendix A.2). (b) is theorem A.41 below. (c) is [3, p. 421]: the coefficient bounds use of the external scalar field exactly the sup clause, the gradient clause (P-grad) and the block clause (P-block) of lemma A.31, at the current vertex [3, (1.8)] and at the first two kernel vertices [3, (1.18)], every other family using the sup clause alone; on the cube the current vertex sits on live bonds only (a face bond carries no link) and the kernel vertices at weighted sites only (the weight vanishes at killed sites), so the two clauses are needed only there. □

A.7. **The lower bound.** Following [1, Sect. 3] and [13, Appendix]: $p_i := p(\ell_i)$, $M_i := p_i\lambda(\ell_i)^{-1/4}$, $M_i' := p_{i-1}\lambda(\ell_{i-1})^{-1/4} = L^{1/4}(p_{i-1}/p_i)M_i$ the block-sup threshold of [13, (3.2)], $p_1(\ell) := b_1(1 + \ln \ell^{-1})^{p_1}$ the fluctuation cut-off with $p - 1 \geq p_1$.

Definition A.28 (the class carried through the induction)**.** In unit-i units, with $\phi^{(i)} := a_i M(X_{\mathrm{live}}, A^{(i)})^{-1} Q(A^{(i)}$ the class $(\mathrm{A.1})_C^{(i)}$ is defined by

$$\sup_{x \in X_{\mathrm{live}}} |\phi^{(i)}(x)| \leq C_1 M_i, \qquad \sup_y |\phi_i(y)| \leq C_1 M_i,$$

$$(\mathrm{G})_i : \sup_b |D_{A^{(i)}} \phi^{(i)}(b)| \leq C_1' p_i \text{ over all bonds of the cube, face bonds included,} \qquad (\mathrm{E})_i : \sup_y |\phi_i(y) - Q(A^{(i)})$$

together with the photon clause $|\partial A^{(i)}|_\infty \leq C_\Gamma' p_i$. On a face bond $(\mathrm{G})_i$ reads $\vartheta^{-1}|\phi^{(i)}(x)| \leq C_1' p_i$ at the adjacent live site: a fine-level linear vanishing with slope $O(p_i)$.

The first two clauses are King's (A.1) with its Laplacian clause dropped; (G) and (E) are the two consequences of that clause which the lower bound consumes (lemma A.31). On the cube the Laplacian clause itself holds as a consequence ($(-\Delta_A^D + V_i)\phi^{(i)} = a_i Q^*(\phi_i - Q\phi^{(i)})$ gives $|\Delta_A^D \phi^{(i)}| \leq V_i C_1 M_i + a_i W C_1'' p_i$), but it is not needed.

Lemma A.29 (compatibility, coefficient one)**.** *Let* $\psi \in (\mathrm{A.1})_C^{(i+1)}$*,* $|\phi_i'|_\infty \leq p_1(\ell_i)$ *and* $\phi_i = aL^{-2}C^{(i)}Q_c^*\psi + \phi_i'$ *at the gauge field of Bałaban's step* ($A^{(i)} = A^{(i+1)} + A'^{(i)}$)*. Then* $\phi_i \in (\mathrm{A.1})_C^{(i)}$ *provided* $L^{3/4} \geq 3$*,* $C_1 \geq 3\max(C_A, 1)\max(1, b_1/b_0)$*,* $C_1' \geq 3C_D \max(1, b_1/b_0)$*,* $C_1'' \geq \max\{2(1 + C_{Q4})\max(1, b_1/b_0), C_1'\}$*, and the three smallness conditions on* $\varepsilon_0^{\mathrm{Ba}}$ *for the gauge-field remainders (below), where* C_A*,* C_D *are the constants of lemma A.8(b) and lemma A.12 for the leg of block data* ($C_A, C_D \leq CaW$) *and* $C_{Q4} \leq C_A$ *the constant in* $|Q(A)u|_\infty \leq C_{Q4}|b|_\infty$*.*

Proof. (1) *Exact composition.* By the composition law of the kernels and the minimiser identity (the minimiser of the composed functional is the minimiser of the one-step functional with

the composed average, the uniform coarse step having $Q_cQ_c^*=1$), $\phi^{(i)}(\phi_i)=\Phi_\psi+u'$ with $\Phi_\psi:=\phi^{(i+1)}(\psi)$ and $u':=a_iM^{-1}Q^*\phi_i'$, $|u'|_\infty\le C_A|\phi_i'|_\infty$. (2) *Units.* A field in unit-$(i+1)$ units is $L^{1/2}$ times the field in unit-i units and a length is L times, so a fine-bond difference quotient converts by $L^{-3/2}$ and a block average by $L^{-1/2}$; $p_{i+1}\le p_i$, $\lambda(\ell_{i+1})=L\lambda(\ell_i)$. (3) *Sup clauses.* $|\phi^{(i)}|_\infty\le|\Phi_\psi|_\infty+C_A|\phi_i'|_\infty\le C_1L^{-3/4}M_i+C_Ap_1(\ell_i)\le\frac13C_1M_i+\frac13C_1M_i$, the third share being left for the gauge-field remainder. Block clause: by the blockwise push-down identity $aL^{-2}C^{(i)}Q_c^*\psi=Q\Phi_\psi+\kappa'Q_c^*(\psi-Q_{i+1}\Phi_\psi)$ with $\kappa'=aL^{-2}/(a_i+aL^{-2})\le L^{-2}$ (minimise the composed functional over ϕ_i block by block: on each $(i+1)$-block the minimiser is $Q\Phi+\delta$ with δ constant on the block, and the one-dimensional minimisation gives $\delta=\kappa'(\psi-Q_{i+1}\Phi)$; this uses only that the coarse step is uniform), and $|Q\Phi|\le\sup|\Phi|$ (Q5), $|Q_c^*c|=|c|$: $|\phi_i|_\infty\le(1+2\kappa')C_1L^{-3/4}M_i+p_1(\ell_i)\le C_1M_i$. (4) $(\mathrm{G})_i$ at live bonds: $|D\Phi_\psi(b)|\le L^{-3/2}C_1'p_{i+1}\le\frac13C_1'p_i$ by $(\mathrm{G})_{i+1}$ at the same fine bond, and $|Du'(b)|\le C_D|\phi_i'|_\infty\le\frac13C_1'p_i$ by lemma A.12; the third share is the gauge-field remainder. At face bonds the same three terms, each a face-bond derivative: $D\Phi_\psi$ by $(\mathrm{G})_{i+1}$ on the same face bond, Du' by the face-bond clause of lemma A.12, and the remainder likewise; the difference $D_{B+A'}\Phi_\psi-D_B\Phi_\psi$ vanishes on a face bond (no link). (5) $(\mathrm{E})_i$: by the push-down identity and the exact composition,

$$\phi_i-Q(A)\phi^{(i)}(\phi_i)=\kappa'Q_c^*\big(\psi-Q_{i+1}\Phi_\psi\big)+\big(\phi_i'-Q(A)u'\big),$$

and at every block $|\cdot|\le\kappa'L^{-1/2}C_1''p_{i+1}+(1+C_{Q4})p_1(\ell_i)\le L^{-5/2}C_1''p_i+\frac12C_1''p_i\le C_1''p_i$. (6) *The gauge-field remainder.* The composition holds at a common gauge field; in Bałaban's step the push-down is taken at $B:=A^{(i+1)}$ while $\phi^{(i)}$ is built with $B+A'$. The background of ψ at $B+A'$ differs from Φ_ψ by δ solving $M(B+A')\delta=f$, $f=a_i[Q(B+A')^*-Q(B)^*]\omega_B-[M(B+A')-M(B)]\Phi_\psi$ with ω_B the pushed-down datum, $|\omega_B|_\infty\le C_1M_i$; expanding term by term, f is a bounded function with $|f|\le Ce_ip_1(\ell_i)(C_1M_i+C_1'p_i)$ on Bałaban's fluctuation class ($|A'|,|\partial A'|\le C_\gamma p_1(\ell_i)$ by [13, (2.18)] with [13, Thm. 3.3] for the vector operators and the cut-off [1, (3.50)]; on the cube by lemmas A.3 and A.4), involving only live bonds and contours inside blocks; by lemma A.8(c) and lemma A.12 (bounded source), $|\delta|_\infty+\sup_b|D_A\delta(b)|\le C_{\mathrm{rem}}e\,p_1(\ell_i)(C_1\lambda^{-1/4}\ell_i^{1/4}+C_1'\ell_i^{1/2})p_i$, which is $\le\frac13C_1'p_i$ and $\le\frac13C_1M_i$ under $C_{\mathrm{rem}}e\,C_1\lambda^{-1/4}\beta^{1/4}p_1(\beta)\le\frac16C_1'$, $C_{\mathrm{rem}}e\,\beta^{1/2}p_1(\beta)\le\frac16$ and $C_{\mathrm{rem}}e\,p_1(\beta)(\lambda^{-1/4}\beta^{1/4}+(C_1'/C_1)\beta^{1/2})\le\frac13$ at $\beta=\varepsilon_0^{\mathrm{Ba}}$ (monotone in β on $(0,e^{1-4p_1}]$, so valid at every $\ell_i\le\varepsilon_0^{\mathrm{Ba}}$); the block-averaged form of the remainder enters $(\mathrm{E})_i$ with the sup clause of lemma A.8(c) and is absorbed in the room $\frac12-L^{-5/2}\ge\frac13$ (here $C_1''\ge C_1'$ is used). The scale-$(i+1)$ class enters with coefficient one; no constant multiplies p_{i+1}, which is what [1, (3.53)–(3.54)] requires, and the conditions (3.54) are unchanged. □

Corollary A.30 (King's face test as a consequence at every scale of the lower bound)**.** *On* $(\mathrm{A}.1)_C^{(i+1)}$*, at every face block* y *of scale* i*, the push-down* $b_i:=aL^{-2}C^{(i)}Q_c^*\psi$ *obeys* $|b_i(y)|\le(1+\vartheta)L^{-3/2}C_1'p_i+L^{-5/2}C_1''p_i$*, hence* $|\phi_i(y)|\le$ *that* $+p_1(\ell_i)$ *on the induction set.*

Proof. $b_i=Q\Phi_\psi+\kappa'Q_c^*(\psi-Q_{i+1}\Phi_\psi)$. Φ_ψ vanishes on the killed layer; along a normal path of length $\le1+\vartheta$ from the killed site to any point of $B(y)$ every bond is a live or face bond, so $|\Phi_\psi|\le(1+\vartheta)\sup|D\Phi_\psi|\le(1+\vartheta)L^{-3/2}C_1'p_i$ by $(\mathrm{G})_{i+1}$ including its face part; Q is an average; the second term is $(\mathrm{E})_{i+1}$ with $\kappa'L^{-1/2}\le L^{-5/2}$. □

This is a consequence, not an inserted test: no block-field test is inserted at an intermediate scale of the lower bound (such a test would not be implied by the previous scale, the push-down norm having no numerical bound below one); King's remark that there is "no change in the proof of the lower bound" [14, p. 343] is correct in this sense once King's Laplacian clause is replaced by (G), (E).

Lemma A.31 (the two properties consumed by the $n+n'=1$ vertices)**.** *On* $(\mathrm{A}.1)_C^{(i)}$*, at every block* y*, every live* $x\in B(y)$ *and every live bond* b*: (P-grad)* $|D_{A^{(i)}}\phi^{(i)}(b)|\le C_1'p_i$*; (P-block)* $|\phi^{(i)}(x)-U(A^{(i)}(\Gamma_{x,y}))\phi_i(y)|\le C_Pp_i$ *with* $C_P:=C_1''+2dC_1'$.

Proof. (P-grad) is $(\mathrm{G})_i$. (P-block): $Q\phi^{(i)}(y) = \sum_{x'} \omega(x') U(\Gamma_{y,x'}) \phi^{(i)}(x')$ with $\omega := \vartheta^3 w \geq 0$, $\sum \omega = 1$; so $\phi^{(i)}(x) - U(\Gamma_{x,y})\phi_i(y) = U(\Gamma_{x,y})[Q\phi^{(i)}(y) - \phi_i(y)] + \sum_{x'} \omega(x')[\phi^{(i)}(x) - U(\Gamma_{x,y})U(\Gamma_{y,x'})\phi^{(i)}(x')]$; the first term is $(\mathrm{E})_i$, and each term of the sum is the covariant difference along the path $\Gamma_{y,x'}$ followed by $\Gamma_{x,y}$, of length $\leq 2d$, the exact telescoping sum of $\vartheta D_A \phi^{(i)}$ over its bonds (live sites only, by the choice of contours), hence $\leq 2dC_1' p_i$ by $(\mathrm{G})_i$. □

Lemma A.32 (final restriction at the stopping scale)**.** *Let $\chi_k \lambda_k$ be the small-field indicator of appendix A.1 at the stopping scale (King's* [13, (3.2)] *with the gradient clause on every block bond of the cube, times the face test). Then* $\mathrm{supp}(\chi_k \lambda_k) \subset (\mathrm{A}.1)_C^{(k)}$ *once $C_1 \geq \max(C_A, 1) L^{1/4}(1 + \ln L)^p$, $C_1' \geq C_G(1 + \ln L)^p$, $C_1'' \geq C_E(1 + \ln L)^p$, $V_k M_k' \leq p_k$ and $e_k M_k \leq 1$ (conditions on η_1), with C_G, C_E depending on $(C_8, \delta_1, \delta', C_A, L, a, d)$ only. At the final scale K_f the set $\{|\phi_{K_f}(y)| \leq 1 \ \forall y\}$ lies in $(\mathrm{A}.1)_C^{(K_f)}$.*

Proof. Facts on the final set: (f1) $|\phi_k(y')| \leq C_8 p_k$ on $L_1^{(k)}$; (f2) $|D_{A_k}\phi_k(\beta)| \leq p_{k-1}$ on every block bond; (f3) $|\phi_k| \leq M_k'$; hence (f5) $|\phi_k(y)| \leq C_9(1 + \mathrm{dist}(y, L_1^{(k)}))p_{k-1}$, $C_9 := \max(C_8, 1)$ (the cube is a box: every block is joined to the face layer by a cube-internal block path of that length), and (f6) $|\phi_k(y) - U(A_k(\Gamma_{y,y_0}^{\mathsf{T}}))\phi_k(y_0)| \leq |y - y_0|_1 p_{k-1}$ along a tree T of block paths. Sup clauses: (f3) and lemma A.8(b), with $M_k' \leq L^{1/4}(1 + \ln L)^p M_k$. $(\mathrm{G})_k$ at a bond b (live or face) in the block y_0: write $\phi_k = \phi_k(y_0)\Theta^{\mathrm{blk}} + (\phi_k - \phi_k(y_0)\Theta^{\mathrm{blk}})$ with Θ^{blk} the unit tree datum transported by A_k. Constant part: lemma A.20(a) with $|b - y_0| \leq 1$, the tree datum transported with the block field A_k differing from Θ_A by $\leq C e_k p_k |y - y_0||c|$ (the photon block-versus-background step [13, (A.2)–(A.6)], on the cube by lemmas A.3 and A.4): $\leq |\phi_k(y_0)|[CV_k + Ce^{-\delta_1 \mathrm{dist}(b,\partial D)} + C\theta_k + Ce_k p_k] \leq C[V_k M_k' + C_9 \sup_t (1+t) e^{-\delta_1(t-1)} p_{k-1} + (\theta_k + e_k p_k) M_k'] \leq C' p_k$ by (f3), (f5), $\mathrm{dist}(b, \partial D) \geq \mathrm{dist}(y_0, L_1^{(k)}) - 1$, $V_k M_k' \leq p_k$, $\theta_k \leq C e_k p_k$ and $e_k p_k M_k' \leq C_1 p_k$. Varying part: $\sum_y |D_{A,b} l(b,y)||\phi_k(y) - \phi_k(y_0)\Theta^{\mathrm{blk}}(y)| \leq \sum_y C e^{-\delta'|b-y|}|y - y_0|_1 p_{k-1} \leq C p_{k-1}$ by lemma A.12 in kernel form and (f6). $(\mathrm{E})_k$: by lemma A.21(a), $|\phi_k(y) - Q\phi^{(k)}(y)| = a_k^{-1}|\Delta^{(k)}\phi_k(y)|$; with the tree rooted at y, $\Delta^{(k)}\phi_k(y) = \sum_{y'} \Delta^{(k)}(y,y')(\phi_k(y') - \phi_k(y)\Theta^{\mathrm{blk}}(y')) + \phi_k(y)(\Delta^{(k)}\Theta^{\mathrm{blk}})(y)$; the kernel decays, $|\Delta^{(k)}(y,y')| \leq Ce^{-\delta'|y-y'|}$ (lemma A.7(b)), so the first sum is $\leq C p_{k-1}$ by (f6), and the second is $a_k|\phi_k(y)||\Theta^{\mathrm{blk}}(y) - Qu(y)| \leq a_k|\phi_k(y)|[CV_k + Ce^{-\delta_1 \mathrm{dist}(B(y),\partial D)} + C(\theta_k + e_k p_k)] \leq C p_k$ by lemma A.20(b), (f3), (f5). At the final scale the data are bounded by 1 and $(\mathrm{G})_{K_f}$, $(\mathrm{E})_{K_f}$ follow from lemmas A.8 and A.12(b) with $|b|_\infty \leq 1$, since $p_{K_f} \geq b_0 \geq 1$. □

The face test is what places the face blocks in the class: without it, constant data of size M_k' violate $(\mathrm{G})_k$ near the face by the factor M_k'/p_k (lemma A.20(a) is sharp at the face).

The steps of the lower bound. We follow [1, Sect. 3] with King's modifications [13, Appendix, p. 676]; E_1 is any field-independent constant (lemma A.44).

B1 (restriction to small fields; [1, (3.6)]**):**
The integrand of $Z_k(s)$ is positive, so inserting any indicator at any step gives a lower bound. We insert, at each step $i < k$, the indicator of the class $(\mathrm{A}.1)_C^{(i)}$ (a class on the backgrounds, the block sup and the block mismatch, not King's six block-field tests), and at the stopping scale k the indicator $\chi_k \lambda_k$, a subset of $(\mathrm{A}.1)_C^{(k)}$ by lemma A.32. Nothing is paid in a lower bound.

B2 (inductive hypothesis; [1, (3.26)–(3.32)]**):**
After i steps, $Z_k \geq e^{-a_i} \int \chi_{(\mathrm{A}.1)_C^{(i)}} e^{-S_k^{(i)}}$ with $a_i := \sum_{i'<i} C \ell_{i'}^{\kappa'} |X^{(i')}|$ and $S_k^{(i)}$ King's effective action [13, (3.14)] built on the cube; the identities for the normalisations and the representation of the fields after i steps are [13, (2.13)–(2.17)], exact for every weight and boundary condition (the block-spin RG is a Gaussian identity for any positive covariance; the composition law uses only that the coarse steps are uniform).

B3 (the fluctuation step; [1, (3.37)–(3.49), Props. 3.1–3.2]**):**
The Gaussian fluctuation integral is exact; the interaction is expanded to order $\bar{n} = 13$ [13, Appendix] with Cauchy estimates in the complexified couplings; the R-terms and the coefficient bounds are proposition A.27(c) under the hypotheses supplied by definition A.28

and lemma A.31; the one-loop determinants are bounded below by $\ln(1+x) \le x - x^2/2 + \dots$ (n odd) for the bounded operators $C^{(i)}$ of lemma A.7(c).

B4 (cut-off removal and the consistency conditions; [1, (3.50)–(3.55)]): The cumulant bound $\int \chi e^V d\mu \ge \exp[\sum_{n\le\bar n} E_n^T(V)/n! - R]$ with cut-off-free cumulants and $R = O(1)\ell_i^{\kappa'}|X_1^{(i)}|$, $\kappa' > d$, for a Gaussian measure with exponentially decaying precision (lemmas A.6 and A.7) and a real polynomial interaction with Bałaban's two-component cut-off: this is the lemma behind [1, Prop. 3.2], with its interaction input proposition A.27(c); the compatibility inequality $\chi_{(\mathrm{A}.1)_C^{(i+1)}}(\psi)\chi(|\phi_i'| \le p_1(\ell_i)) \le \chi_{(\mathrm{A}.1)_C^{(i)}}(\phi_i)$ is lemma A.29, with coefficient one on the scale-$(i+1)$ class, the rooms $L^{3/4}$ (sup clauses), $L^{3/2}$ ((G)), $L^{5/2}$ ((E)), and the conditions [1, (3.54)] unchanged.

B5 (King's Appendix: [13, (3.2)] in place of [1, (3.27)–(3.29)]): King's Laplacian clause is replaced by $(\mathrm{G})_i$, $(\mathrm{E})_i$, which compose with coefficient one (lemma A.29) and supply through lemma A.31 exactly the two properties of the external scalar field that [3, p. 421] draws from [2, Lemma 2.4] for his class; what [3, Prop. 1] uses of the external legs otherwise is their norm, corollary A.13(a). The final set $\chi_k\lambda_k$ is in the class by lemma A.32. The photon background's clauses [13, (A.2)–(A.6)] are unchanged on the cube (appendix A.2).

B6 (the final step at $\ell_f = \ell_{K_f}$; [1, (3.66)–(3.69)]): Expand in the small $A^{(K_f)}$ (proposition A.27(c)); $|V^{(K_f)}| \le O(1)\ell_f^{\beta_0}|X^{(K_f)}|$; the finite-dimensional integral over $\{|\phi_{K_f}(y)| \le 1, |A_{K_f}| \le 1\} \subset (\mathrm{A}.1)_C^{(K_f)}$ (lemma A.32) of $\exp[-\frac12\langle A_{K_f}, \Delta^{(K_f)}A_{K_f}\rangle - \frac12\langle\phi_{K_f}, \Delta^{(K_f)}\phi_{K_f}\rangle]$ is $\ge \exp(-O(1)|X^{(K_f)}|)$, $|X^{(K_f)}| = \ell_f^{-d}|C_J|$, using only the upper bounds $\Delta^{(K_f)} \le a_{K_f}\ell_f^{-2}$ on the integration region; and the normalisation identity [1, (3.69)] gives $Z_{K_f}Z_{K_f}(0)e^{-E_0} \ge \exp[-\frac12 a_{K_f}\ell_f^{d-2}(\operatorname{Tr} P_{K_f}G + \operatorname{Tr} P_{K_f}G^\gamma)]$ with $\operatorname{Tr} P_{K_f}G \le W_2 \operatorname{Tr}(\Pi G) \le W_2 N|X^{(K_f)}|\|G\|$ (Q3). Hence $Z_k(0) \ge e^{-E_-|C_J|}$ in the normalisation in which E_1 is the domain-adapted constant of [13, (3.35)] on C_J, with $E_- = O(\ell_f^{-d})$ depending on $\varepsilon_0^{\mathrm{Ba}}$ and King's data only.

B7 (the source): $F(u_x)$ is King's plaquette source [13, (2.7)], whose vertices [13, (3.20)] sit at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_J$ (zone I), so the face enters them only through propagators (proposition A.27).

Proposition A.33 (lower line)**.** *For $\eta \le \eta_1$ (so that [1, (3.54)] holds at all scales $\le k$ and $j \le k - J_*$) and $j \ge J_*$: $Z_k(s) \ge e^{-a} I_k^{(j)}(s)$ with $a = \sum_{i<j} C\ell_i^{\kappa'}|X^{(i)}| = C\eta^\sigma|C_J|$, $\sigma = \kappa' - d > 0$; and $Z_k(0) \ge e^{-E_-|C_J|}$ in the normalisation of B6.*

Proof. B1–B7. The sum $\sum_{i<j}\ell_i^{\kappa'}|X^{(i)}| = |C_J|\sum_{i<j}\ell_i^{\kappa'-d}$ is geometric and dominated by $i = j-1$, giving $a = C\eta^\sigma|C_J|$ with C independent of j and k (proposition A.27, lemma A.29, whose constants are free of k, j, ϑ); the final restriction is a positivity step inside the class (lemma A.32), so the threshold σ_k never enters a constant. □

A.8. **The upper bound.** We follow [2, Sects. 2–3] and [14, Sect. 5.2]. Bałaban's six large-field families (the tests on $|B - QA|$, $|\partial A|$, $|A|$, $|\psi - Q(A)\phi|$, $|D_A\phi|$, $|\phi|$ of [2, (2.2)]) are used at every scale $i \le K_f$, with the face family of definition A.23 as a seventh family at every scale, $i = 0$ included; at the stopping scale j the small-field term carries $\chi_j\lambda_j$. The large-field decomposition, its regions $\Lambda_0^{(i)} \supset \Lambda_1^{(i)} \supset \cdots \supset \Lambda_7^{(i)}$ (layers of thickness $r(\ell_i)$), the nesting rules and the inductive representation [2, (2.43)–(2.52)] depend only on the set of event blocks and the number of event types, so they hold verbatim with the face family, which is a pointwise condition on one block variable at one block of the index set $L_1^{(i)} \cap \Lambda_7^{(i-1)\prime}$. Four steps of Bałaban's printed proofs do not hold as printed at block factor 5 and are used in a replaced form: the cube step of [3, p. 433], the cut-off $\zeta^{(k)}$ of [2, (2.44)], the boxes in the proof of [2, Lemma 2.4], and the cube partition [2, (3.43)–(3.45)], the last replaced by a product partition into boxes of large blocks with sides in $[r + s_b, 2r + 3s_b]$, which has every property used in [2, Sect. 3C]; the replaced objects are defined from $(L, M, R, \ell_i, \bar n)$ alone and do not depend on the matter boundary condition, and

the replacements change none of the printed statements. Their block-factor-5 form, with proofs, follows.

The four replaced steps at block factor 5. Scale-i units (the block lattice of scale i has spacing 1); $u_i := 1 + \ln \ell_i^{-1}$, so $u_{i-1} = u_i + \ln L$, $r_i := r(\ell_i) = R u_i^{\varrho}$, $p_i = p(\ell_i) = b_0 u_i^{p}$ with $\varrho = 2$, $p = 6$; $s_b = LM$ is the large-block side; $\rho_i := (r(\ell_{i-1}) - s_b)/L$ is one previous-scale corridor in scale-i units ([2, p. 578]); X' denotes the blocks of X read as points of the next lattice [1, (1.16)]; $\Omega := B^i(\Lambda_2^{(i-1)\prime})$ is the propagator domain and $\Omega_1 := B^i(\Lambda_7^{(i-1)\prime})$ the vertex set of [3, (1.1)–(1.5)]; δ_0, c_{Ba} and the depth R_{Ba} are the constants of [1, Props. 2.1–2.3]. Everything below is a statement about Bałaban's regions, cut-offs, boxes and partitions, which are built from $(L, M, R, \ell_i, \bar n)$ and the set of event blocks alone; nothing depends on the matter boundary condition, so each statement holds on the cube exactly as on the torus.

Lemma A.34 (the supply of depth, and the criterion)**.** *(a)* $\operatorname{dist}(\Lambda_{m+s}^{(i-1)}, (\Lambda_m^{(i-1)})^c) > s\, r(\ell_{i-1}) - (s-1)$ *for* $s \ge 1$ *(scale-*$(i-1)$ *units), and in scale-*i *units* $\operatorname{dist}(B^i(\Lambda_{m+n}^{(i-1)\prime}), B^i(\Lambda_m^{(i-1)\prime})^c) \ge n\, r(\ell_{i-1})/L - 2$ *for* $n \le L+1$. *The hypothesis of* [3, Prop. 1] *asks depth* $> 2r_i$ *of* $\operatorname{supp}\tilde A$ *in* Ω*; at its use in* [2]*, where the domain is* $B^i(\Lambda_0^{(i)})$*, the regions supply* $\operatorname{dist}(\operatorname{supp}\tilde A, \Omega^c) > 2r_i - 2 - M_\theta - 2\eta$ *(the slice of* θ_i *and the fine blocks), which is all that the proof consumes. (b) Let* $\alpha, \beta > 0$, $c \ge 0$. *If* $\alpha L \le \beta$*, then* $\beta\rho_i - \alpha r_i \ge \alpha R\varrho \ln L\, u_i^{\varrho-1} - \beta s_b/L$ *for every* i*, so* $\beta\rho_i \ge \alpha r_i + c$ *at every scale once* $R \ge (c + \beta s_b/L)/(\alpha\varrho \ln L)$*. If* $\alpha L > \beta$*, then* $\beta\rho_i < \alpha r_i$ *for every* $\ell_i < \exp\big(1 - \ln L/((\alpha L/\beta)^{1/\varrho} - 1)\big)$*, whatever* R *and* c *are. If an object of size* θr_{i+1} *at scale* $i+1$ *must sit at depth* βr_i*, the containment holds at every scale iff* $L\theta \le \beta$*, and for* $L\theta > \beta$ *it fails for every* $\ell_i < \exp\big(1 - \ln L/(1 - (\beta/(L\theta))^{1/\varrho})\big)$ *(now* $r_{i+1}/r_i = (1 - \ln L/u_i)^{\varrho}$ *enters). (c) Where only exponential smallness across scales is used,* $e^{-\delta\beta\rho_i} \le C e^{-(\delta\beta/L) r_i}$*, which is* $O(\ell_i^{\kappa})$ *for every* κ *since* $\varrho > 1$.

Proof. (a) By [2, (2.8)] $(\Lambda_{m+1})^c$ contains every large block within r of $(\Lambda_m)^c$, so a point of Λ_{m+1} is at distance $> r$ from $(\Lambda_m)^c$; along a lattice geodesic from $x \in \Lambda_{m+s}$ to $(\Lambda_m)^c$ each further layer adds $> r - 1$. For the primed sets, points of $B^i(y)$, $B^i(y')$ with $y \in \Lambda_{m+n}$, $y' \in \Lambda_m^c$ differ coordinatewise by $(y - y') + \tau$, $|\tau| < L$, so $|x - x'|_\infty > n\, r(\ell_{i-1}) - (n-1) - L$, and dividing by L gives the claim for $n \le L+1$. (b) If $\beta/L \ge \alpha$ then $\beta\rho_i - \alpha r_i \ge \alpha(r_{i-1} - r_i) - \beta s_b/L$ and $r_{i-1} - r_i = R[(u_i + \ln L)^{\varrho} - u_i^{\varrho}] \ge R\varrho \ln L\, u_i^{\varrho-1}$. If $\alpha L > \beta$, $\beta\rho_i \le (\beta/L) r_{i-1}$ and $r_{i-1}/r_i = (1 + \ln L/u_i)^{\varrho} < \alpha L/\beta$ exactly when $u_i > \ln L/((\alpha L/\beta)^{1/\varrho} - 1)$. (c) $e^{-\delta\beta\rho_i} \le e^{\delta\beta s_b/L} e^{-(\delta\beta/L) r_{i-1}}$ and $e^{-cR(1+\ln \ell^{-1})^{\varrho}} \le \ell^{\kappa}$ for small ℓ because $\varrho > 1$. □

Every place where L enters [1, 2, 3] and [13, Thm. 3.1] is a constant, a rate, a convergent sum, a condition on $(b_0, R, M, \varepsilon_0^{\mathrm{Ba}})$, a divisibility, or a *cross-scale containment* $\beta\rho_i \ge \alpha r_i + c$, and by lemma A.34(b) a containment holds at every scale iff $\alpha L \le \beta$. At $L = 5$ exactly four printed steps fail this test (theorem F.2): the cube step of [3, p. 433] ($\alpha = \frac{3}{2}$, $\beta = 5$), the regularity of the vector background [2, Lemma 2.3] ($\alpha = 1$, $\beta = 3$), the boxes of [2, Lemma 2.4] ($\alpha = 4$, $\beta = 5$; this one fails at every $L \ge 2$), and the split [2, (2.89)] ($L\theta = L$, $\beta = 2$); the partition [2, (3.43)–(3.45)] is a divisibility entry which fails on some lattices at $L = 5$. These are gaps of the printed proofs, not counterexamples to the theorems, whose statements never mention the objects concerned.

(a) The cube step. On [3, p. 433] the localisation cubes of a renormalised class, which lie in a cube of side $\ell_{\bar n}$ depending on $\bar n$ only [3, p. 432], are enclosed in concentric cubes $\square_1 \subset \square$ of sides r_i, $3r_i$, sums of large blocks; $G_i(\Omega, \tilde B)$ is written $G_i(\square, \tilde B) + \delta G_i(\Omega, \square, \tilde B)$ with the difference treated as an external field, the regular field $\tilde B$ is replaced on $\square$ by its constant part B_0, B_0 is gauged away, and $G_i(\square, 0)$ is replaced by $G_i(0)$. The step presupposes $\square \subset \Omega$ (resp. $\square \subset \Omega_2$): the Neumann operator $G_i(\square, \tilde B)$ is built from the bonds of $\square$, where $\tilde B$ lives only if $\square \subset \Omega$, and [1, (2.26)] is stated for nested domains. Every vertex of every class lies within sup-distance 1 of $\Omega_1 \cup \operatorname{supp}\tilde A$ (the vertices [3, (1.6)–(1.7)] are summed over Ω_1; those of [3, (1.8)–(1.11)] carry g_i, $\operatorname{supp} g_i \subset \Omega_1$, or $\tilde A$; the pair vertices [3, (1.12)–(1.15)] carry A' or $\tilde A$ along a contour inside one

block). For an old vertex x_v of the class the containment reads

$$\operatorname{dist}(x_v, \Omega^c) > \tfrac{3}{2} r_\square + 2s_b + \ell_{\bar n} \implies \square \subset \Omega \text{ and } \operatorname{dist}(\square_1, \square^c) \geq r_\square \tag{57}$$

(cubes rounded outward to large blocks; triangle inequality). With the printed $r_\square = r_i$ and a vertex anchored at Ω_1, whose depth is $\geq 5r(\ell_{i-1})/L - 2$ by lemma A.34(a), this is a containment with $\alpha = \frac{3}{2}$, $\beta = 5$: it holds at every scale iff $L \leq 3$, and at $L = 5$ it fails for all small ℓ_i; the failing vertices (λ- and δm^2-vertices summed over Ω_1, g_i-localised photon legs near $\partial\Omega_1$) carry no $\tilde A$-leg, so no hypothesis on $\operatorname{supp} \tilde A$ helps.

Proposition A.35 (the cube step with a constant cube)**.** *Let $\ell^\sharp := 3\ell_{\bar n} + 4N_B$, where $N_B = n_1 + n_2 - 1$ is the number of insertions per line in the expansion of* [3, p. 433] *to order $\bar n$ (a function of $\bar n$ only), let $r^\flat$ be $\max\{3\ell^\sharp + R_{\mathrm{Ba}} + 12,\ 3R_{\mathrm{Ba}} + 15,\ 107\}$ rounded up to a multiple of s_b, and $D^\flat := \frac{3}{2} r^\flat + 2s_b + \ell^\sharp$. Run the step with concentric cubes $\square_1, \square$ of sides $r^\flat$, $3r^\flat$ centred at the component cube. If every old vertex of the class has depth $> D^\flat$ in Ω (resp. Ω_2), the proof of* [3, pp. 433–445] *goes through and gives* [3, (1.33)] *with a constant of the printed form, depending on $(\alpha_0, \bar n, d, a, M)$ only. If $R \geq R^\flat := D^\flat + 3$ and $R \geq M_\theta + 2$, then at $L = 5$ every class occurring in the use of* [3, Prop. 1] *in* [1, 2] *has its old vertices at depth $> D^\flat$, so* [3, Prop. 1] *holds at $L = 5$ in its printed form.*

Proof. The side $r_\square$ enters [3, pp. 433–434] in five places. (1) *Containment:* (57) with $r_\square = r^\flat$. (2) *Old-vertex classes lie in $\square_1$:* the expansion in $\tilde B'$ uses [1, (3.14), (3.44), (3.45)] to order $\bar n$; only lines between localisations whose boxes meet stay internal [3, p. 432], so along an expanded line a new vertex advances at most 2, and the new vertices lie within $2N_B$ of the class-component cube; the boxes of all graphs of a connected component lie in a grid cube of side $3\ell_{\bar n}$ concentric with one member's cube ("localizations ... in agreement", [3, p. 432]); so the class lies in a cube of side $\ell^\sharp$, and $3\ell^\sharp \leq r^\flat$. (3) *The two δG external fields are $O(1)$:* $\delta G_i(\Omega, \square, \tilde B)$ is evaluated at points of $\square_1$, at distance $\geq r^\flat \geq R_{\mathrm{Ba}} + 12$ from $\square^c$, with sources in $\square_1$, so [1, (2.26)] for the pair $(\square, \Omega)$ gives $\|h\, \delta G_i\, h'\|_{1,\alpha} \leq c_{\mathrm{Ba}} e^{-\delta_0 r^\flat}$; for $\delta G_i(\square, \eta\mathbb{Z}^d, 0)$ take a torus $T' \supset \square$ with side a multiple of $2s_b$ and at least $\max\{3s_b, 3r^\flat + 2s_b + 2\}$: at zero background $G_i(T', 0)$ is the periodisation of $G_i(\eta\mathbb{Z}^d, 0)$, so $\delta G_i(\square, \eta\mathbb{Z}^d, 0) = \delta G_i(\square, T', 0) + \sum_{n \neq 0} G_i(\eta\mathbb{Z}^d, 0)(\cdot, \cdot + n\ell')$, the first term obeying [1, (2.26)] on T' with constants independent of ℓ' (the constants of [4] depend on d, a, M, L only) and the second vanishing as $\ell' \to \infty$ by [1, (2.25)]. Both are external fields of dimension 0 [3, (2.1), p. 422], so an $O(1)$ bound is all that is used. (4) *The constant field:* for $x \in \square$ and c the centre, $|\tilde B(x) - \tilde B(c)| \leq d(\frac{3}{2} r^\flat + s_b) \sup_\square |\partial^\eta \tilde B| = O(r^\flat p_i) = O(p_i)$ with $B_0 := \tilde B(c)$, in place of the printed $O(r_i p_i) \leq O(p_i^2)$; every use of this bound is an upper bound monotone in the size of $\tilde B'$ (vertices weighted by $e(\ell_i)|\tilde B'|$, the remainder [3, (1.16)] by $(e(\ell_i)^{1-\alpha})^{n+n'}$, the norm comparison between the norms [3, (1.32)] at B_0 and at $\tilde B$ costing $(1 + O(p_i))^{\#\mathrm{legs}}$, absorbed by the slack $\ell_i^{(4-d)/4-\beta}$ of the applications [3, p. 421]); so a smaller cube only helps. (5) *Gauge-out of B_0:* $B_0 = \partial^\eta \lambda_0$ with $\lambda_0(x) = B_0 \cdot (x - c)$ on $\square$; the Neumann covariant Laplacian on the bonds of $\square$ and the averaging Q_i (contours inside blocks contained in $\square$) transform covariantly, $G_i(\square, B_0) = U(\lambda_0) G_i(\square, 0) U(\lambda_0)^*$, and every class is invariant when the external scalar fields are transformed simultaneously [2, (2.69)–(2.74)]; exact as long as $\square$ does not wrap around the torus (no condition on the cube). (6) *Ω-freeness after the split:* every old vertex lies in $\square_1 \subset \square \subset \Omega$ and every new vertex in $\square$, so the characteristic function of Ω is 1 wherever a vertex or a line end occurs; every object depending on Ω or on the cube ($\delta G_i(\Omega, \Omega_2, \tilde B)$, [3, (1.16)], the two δG's, the gauge factors) has been moved into external fields, used on [3, pp. 434–445] only through the norms [3, (3.3)] at points of $\square_1$; the Ω_1-dependent objects g_i, 1_{Ω_1}, $\delta m^2(e', g_i, \lambda', \Omega_1, x)$ stay in the vertices, fixed before the step and entering only identities between a graph and its counterterm with the same restriction; on [3, pp. 434–445] every propagator is $G_i(0)$, a slice, G^ξ, C^ξ or Π, every sum runs over $\eta\mathbb{Z}^d$, $\xi\mathbb{Z}^d$, the cubes $\Delta(v)$, Ω_1 or the torus identified with $\eta\mathbb{Z}^d$, Taylor's formula [3, (3.10)] is applied to legs and to g, g', and the λ-vertices are matched against counterterms with the same localisation [3, (3.19), pp. 444–445]: none of this refers to the cube side or to $\operatorname{dist}(\cdot, \Omega^c)$. (7) *Chains without old vertices* [3, p. 433]: a class with no old vertex is a path $v_1 – \cdots – v_m$ of new vertices of one product

$(G_0V)^mG_0$, $G_0 := G_i(\square, B_0)$, with the cut lines at its ends as external scalar fields; the printed route through [3, Prop. 2.1] is not used (its summation by parts [3, (2.8)] is printed on the torus and on the Neumann box produces a face term where the $\tilde B'$-localisation does not vanish); instead, with $\|\psi\|_E^2 := \langle\psi, G_0^{-1}\psi\rangle$, $G_0^{-1} \geq D_{B_0}^* D_{B_0}$ and $G_0^{-1} \geq \gamma_\square$ (constant field, gauged out), each vertex of [3, (1.8)–(1.11), (1.13)–(1.15)] is a bounded form in $\|\cdot\|_E$: the form of [3, (1.8)] is $\kappa \sum_{b\subset\square} \eta^d (D_{B_0}\phi)(b)\cdot M(b)\chi(b_-)$ with $|M(b)| \leq C(\sup|\tilde B'|)^{n'}$, bounded through $\|D_{B_0}\phi\| \leq \|\phi\|_E$ and $\|\chi\| \leq \gamma_\square^{-1/2}\|\chi\|_E$; [3, (1.9), (1.11)] likewise, with $|R_{\bar n+1}| \leq 1$; [3, (1.10)] carries no derivative and uses $\gamma_\square^{-1/2}$ on both legs; a block pair [3, (1.13)–(1.14)] uses $|\tilde B'(\Gamma)| \leq 2d \sup|\tilde B'|$ on its unit-block contours and $\sum_{x\in B(y)} \eta^d|\phi(x)| \leq \|\phi\|$; every form norm is $\leq C(e(\ell_i)\sup|\tilde B'|)^{n'_v}$. Hence $E(H) = \langle G_0^{1/2}K_1^*\Phi_1, \hat K_2 \cdots \hat K_{m-1} G_0^{1/2} K_m \Phi_m\rangle$, $\hat K_v := G_0^{1/2}K_vG_0^{1/2}$, is bounded by the product of the form norms times the end norms $\|h\Phi\|_E$, where $h = 1$ on the 2η-neighbourhood of the end cube, $|\partial^\eta h| \leq c_h$, and $\|h\Phi\|_E^2 = \|D_{B_0}(h\Phi)\|^2 + m^2\ell_i^2\|h\Phi\|^2 + a_i\|Q_i(B_0)h\Phi\|^2 \leq C(1+c_h)^2\|h\Phi\|_{1,0}^2$ by the Leibniz rule on a cube of side ≤ 4; so $\|1_{\square^{\eta}(v)}\Phi\| + \|1_{\square^{\eta}(v)}D_{B_0}\Phi\| \leq C\|h\Phi\|_{1,0}$ is all that is used of an end field. The cut lines are bounded pointwise on $\square$, with their first differences in each slot, at pairs of boxes at distance $\geq 1 - 2\eta \geq \frac35$, by lemma A.36: $|D^a_{B_0,x}D^b_{B_0,x'}G_0(x,x')| \leq Ce^{-\delta|x-x'|}$ for $|x-x'| \geq \frac35$ and $a, b \in \{0,1\}$, at every pair of points of $\square$, its faces included, with no depth condition; and a cut line K contracted between two functionals A, B bounded by the sup-and-difference norms of the two end regions obeys $|A(x \mapsto B(K(x,\cdot)))| \leq \|A\|\|B\| \max_{a,b\in\{0,1\}} \sup|D_x^a D_{x'}^b K(x,x')|$, because $x \mapsto B(K(x,\cdot))$ is linear in K and its differences are $B(D_xK(x,\cdot))$ bond by bond, so no hypothesis dist ≥ 1 on the two supports is needed and distance exactly 1 is covered. A cut line between a chain end and an old-vertex class is an external scalar field of that class, read through its norms [3, (3.3)] after localisation to the η-neighbourhood of the class's cube by a function h' as in lemma A.36(c), the chain slot anywhere in $\square$: by the same linearity, with B the class functional, bounded by Bałaban's norm [3, (1.32)] of its external field, $|A(x \mapsto B(h'K(x,\cdot)))| \leq \|A\|\|B\| \max_{a\in\{0,1\}} \sup_x \|h' D_x^a K(x,\cdot)\|$ with the norm [3, (1.32)] in the last factor, and lemma A.36(c) bounds that factor by $C(1+c_h)^2$ at every x of the chain-end region, the faces of $\square$ included. The vertex forms are sums over the bonds of $\square$, so vertices at the faces need no separate treatment. (8) *The small factors of* [2] *survive:* the factors $O(e^{-\delta_1 r_i})$ that [3, p. 422] produces at domain changes come from $\exp[-\delta_0 d(\{\square(v)\})]$ in [3, (1.33)], i.e. from the lines removed before the cube step, whose bound depends on Ω only; the two-domain inputs enter as external fields, linearly. So the only place where $r_\square = r_i$ was used is (1), where $r^\flat$ serves by hypothesis; the constant acquires the factors of (3) and (4), both of the printed kind. For the last claim: a vertex anchored at Ω_1 has depth $\geq 5r(\ell_{i-1})/5 - 3 \geq r_i - 3 \geq R - 3 \geq D^\flat$ (lemma A.34(a), $r(\ell_{i-1}) \geq r_i \geq R$); a vertex anchored at $\operatorname{supp}\tilde A$ has depth $> 2r_i - 2 - M_\theta - 2\eta \geq 2R - 2.4 - M_\theta > D^\flat$ (lemma A.34(a)); the other steps of the proof need depth at most $R_{\mathrm{Ba}} + 13 < D^\flat$ at old vertices or within $2\sqrt3 + 1$ of $\operatorname{supp}\tilde A$. $\square$

Lemma A.36 (two-slot bounds of the box propagator at a constant field)**.** *Let $i \geq 1$, $\eta := L^{-i}$ the fine spacing in scale-i units, $\square$ a box of large blocks (a product of intervals, each a union of large blocks) that does not wrap around the torus, with Neumann conditions on its faces, B_0 a constant field, and $G_0 := G_i(\square, B_0)$ the propagator of step (7) of the proof of proposition A.35, $G_0^{-1} = D_{B_0}^* D_{B_0} + m^2\ell_i^2 + a_iQ_i(B_0)^*Q_i(B_0)$ on $\ell^2(\square \cap \eta\mathbb{Z}^d)$ with the uniform block average Q_i. Kernels are taken for the measure η^d, and $D^a_{B_0,x}$, $a \in \{0,1\}$, denotes the identity or a covariant forward difference on a bond of $\square$ at x.* (a) *There are $C, \delta > 0$ depending on (d, L, a) and the mass range only, free of i, η, $\square$, B_0 and the positions, such that for all $x, x' \in \square$ and $a, b \in \{0,1\}$ ($d \geq 3$; here $d = 3$)*

$$|D^a_{B_0,x}D^b_{B_0,x'}G_0(x,x')| \;\leq\; C\,(\eta+|x-x'|)^{-(d-2+a+b)}\,e^{-\delta|x-x'|}, \qquad \text{so} \qquad |D^a_{B_0,x}D^b_{B_0,x'}G_0(x,x')| \leq 5C\,e^{-\delta|x-x'|}$$

(b) *No depth condition enters: the bound holds at every pair of points of $\square$, the bonds of its faces included.*

(c) One slot in Bałaban's norm. *Let Δ, Δ' be two cubes of $\square$ at distance ≥ 1, let h' be a localisation function on $\square$, $0 \leq h' \leq 1$, equal to 1 on the η-neighbourhood of Δ' and vanishing at*

distance $\geq \frac{1}{2}$ *from* Δ'*, with* $|\partial^\eta h'| \leq c_h$ *and* $|\partial^\eta \partial^\eta h'| \leq c_h$*, and let* x *lie within* η *of* Δ*. Then for* $a \in \{0,1\}$ *the field* $\Psi := h'(\cdot)\, D^a_{B_0,x} G_0(x,\cdot)$ *satisfies the three clauses of Bałaban's external-field norm* [3, (1.32)] *(sup, first differences,* α_0*-Hölder quotients of the first differences on unit balls) on all of* $\square$*, its faces included, with the bound* $C(1+c_h)^2$*,* C *of the same kind as above.*

Proof. (1) *Gauge-out.* With $\lambda_0(x) := B_0 \cdot (x - c)$ on the non-wrapping $\square$, $G_i(\square, B_0) = U(\lambda_0) G_i(\square, 0) U(\lambda_0)^*$ and $D_{B_0}(U(\lambda_0)\phi) = U(\lambda_0) D\phi$ bond by bond (step (5) of the proof of proposition A.35; $|U| = 1$), so every modulus in the claim equals the one at $B_0 = 0$. Let $B_0 = 0$.

(2) *Even images.* Let $\square = \prod_\mu [\alpha_\mu, \beta_\mu)$ and let $T_\square$ be the torus of periods $2(\beta_\mu - \alpha_\mu)$, each a multiple of $2s_b$, so that the large blocks of $\square$ tile $T_\square$ periodically; let $\mathcal{W}_\square \cong (\mathbb{Z}_2)^d$ be the group generated by the reflections in the face planes $x_\mu = \alpha_\mu - \eta/2$ of $\square$ (planes between sites on block boundaries; on $T_\square$ the reflection in $x_\mu = \beta_\mu - \eta/2$ is the same element). They map sites to sites, bonds to bonds and blocks to blocks, the uniform average commutes with them, and the mass term is constant; every bond of $T_\square$ that crosses a face plane joins a site to its mirror image, where a $\mathcal{W}_\square$-even field has zero difference, and the remaining bonds of $T_\square$ inside $\square$ are the bonds of $\square$. Hence, exactly as in lemma A.3(a) and lemma A.5, the Neumann operator on $\square$ is the even restriction of the periodic one, $G_i(T_\square, 0)\mathsf{E} = \mathsf{E}\, G_i(\square, 0)$ with E the even extension, and

$$G_i(\square, 0)(x, x') = \sum_{g \in \mathcal{W}_\square} G_i(T_\square, 0)(x, gx') \qquad (x, x' \in \square),$$

a difference in x' along a bond of $\square$ acting on the image terms along the reflected bonds. Every image satisfies $\mathrm{dist}_{T_\square}(x, gx') \geq |x - x'|$: coordinatewise, for $x_\mu, x'_\mu \in [\alpha_\mu, \beta_\mu - \eta]$ and r the reflection in $\beta_\mu - \eta/2$, $rx'_\mu - x_\mu = |x_\mu - x'_\mu| + 2(\beta_\mu - \frac{\eta}{2} - \max(x_\mu, x'_\mu)) \geq |x_\mu - x'_\mu|$, and the complementary distance around the period is $x_\mu + x'_\mu - 2\alpha_\mu + \eta \geq |x_\mu - x'_\mu|$.

(3) *Periodisation.* At zero field $G_i(T_\square, 0)(x, y) = \sum_{n \in \mathbb{Z}^d} G_i(\eta\mathbb{Z}^d, 0)(x, y + 2\ell n)$, $\ell := (\beta_\mu - \alpha_\mu)_\mu$ (the periodic operator is the periodisation of the free-lattice one, as in step (3) of the proof of proposition A.35), every term having $|x - y - 2\ell n| \geq \mathrm{dist}_{T_\square}(x, y)$, and at most $(2+k)^d$ values of n have $|x - y - 2\ell n| < k + 1$, every period being ≥ 1. So, with the free-lattice bound (4) below at the rate δ' and $\rho := |x - x'|$,

$$|D^a_x D^b_{x'} G_i(\square, 0)(x, x')| \;\leq\; 2^d C' (\eta + \rho)^{-(d-2+a+b)} \sum_{k \geq \lfloor \rho \rfloor} (2+k)^d e^{-\delta' k} \;\leq\; C(\eta+\rho)^{-(d-2+a+b)} e^{-\delta'\rho/2}.$$

(4) *The free lattice.* On $\eta\mathbb{Z}^d$ at zero field, $G_i(\eta\mathbb{Z}^d, 0) = G_1 + \sum_{j=1}^{i-1} G_{(j)}$ with the slices $G_{(j)} := G_{j+1} - G_j = a_j G_j Q^* C^{(j)} Q G_j$ (the Schur identity behind corollary A.13(b) and [13, (2.17)], with $C^{(j)}$ as in lemma A.7(c)), so that the sum telescopes to G_i. In unit-j units (j-blocks of side one, spacing L^{-j}) let $\ell_j(\cdot, u) := a_j G_j Q^* e_u$ be the leg of the unit datum on the j-block u; since G_j is real symmetric at zero field, $G_{(j)}(y, y') = a_j^{-1} \sum_{u,v} \ell_j(y, u) C^{(j)}(u, v) \ell_j(y', v)$, each variable sitting in its own leg, so

$$D^a_y D^b_{y'} G_{(j)}(y, y') = a_j^{-1} \sum_{u,v} (D^a \ell_j)(y, u)\, C^{(j)}(u, v)\, (D^b \ell_j)(y', v).$$

The legs obey $|D^a \ell_j(y, u)| \leq C e^{-\delta_2 \,\mathrm{dist}(y, B(u))}$ at every site and every bond of the free lattice, $a \in \{0, 1\}$ (the leg bounds [13, (3.64)], [3, (2.12)], at the uniform average $f = 1$ in proposition B.9; on the free lattice every bond is interior, so they are also lemma A.8(b) and the interior case (4) of the proof of lemma A.12), and $|C^{(j)}(u, v)| \leq C e^{-\delta_1 |u - v|}$ ([13, Thm. 3.3, (3.6)]; lemma A.7(c)). With $|y - y'| \leq \mathrm{dist}(y, B(u)) + |u - v| + \mathrm{dist}(y', B(v)) + \sqrt{d}$ and two geometric sums over $u, v \in \mathbb{Z}^d$, $|D^a_y D^b_{y'} G_{(j)}(y, y')| \leq C_S e^{-\delta_S |y - y'|}$ at every pair of points, $\delta_S := \frac{1}{2} \min\{\delta_1, \delta_2\}$. In scale-$i$ units the slice j has the kernel $s_j^{2-d} G_{(j)}(x/s_j, x'/s_j)$ with $s_j := L^{j-i}$, and a scale-i difference is s_j^{-1} times a unit-j one, so slice j contributes at most $C_S s_j^{-(d-2+a+b)} e^{-\delta_S |x - x'|/s_j}$; G_1, read in unit-1 units at the fixed spacing L^{-1}, obeys the same bound with $s := L\eta$ and a constant depending on L (its kernel is $L^d \langle e_y, G_1 e_{y'} \rangle$, bounded with the decay of lemma A.7(a), and a difference at spacing L^{-1} costs a factor $2L$). Finally, for $p := d - 2 + a + b > 0$ and s ranging over the powers of L

in $[L\eta, L^{-1}]$: $e^{-\delta_S\rho/s} \le e^{-\delta_S\rho/2}e^{-\delta_S\rho/(2s)}$ as $s \le 1$; the terms with $s \ge \max(\rho,\eta)$ sum to at most $(1-L^{-p})^{-1}\max(\rho,\eta)^{-p}$, and the terms with $s < \rho$ are $\rho^{-p}(\rho/s)^p e^{-\delta_S\rho/(2s)}$ with ρ/s running over powers of L, summable to at most $C_p\rho^{-p}$. Hence $|D^a_x D^b_{x'} G_i(\eta\mathbb{Z}^d,0)(x,x')| \le C'(\eta+\rho)^{-p}e^{-\delta'\rho}$ with $\delta' := \delta_S/2$, which is (4), and (3) gives the claim with $\delta := \delta'/2$. The last clause of the statement is $(\eta+\rho)^{-p} \le (5/3)^3 < 5$ for $\rho \ge \frac{3}{5}$.

(5) *The clause (c).* By (1) every modulus is the one at $B_0 = 0$. The reflections of (2) and the translations of (3) are isometries of the lattice mapping bonds to bonds, so each clause of $x' \mapsto D^a_x G_i(\square,0)(x,x')$ at a pair $x', y' \in \square$ is bounded by the sum over the images of the same clause of $x' \mapsto D^a_x G_i(\eta\mathbb{Z}^d,0)(x,x')$ at the pair $(gx'+2\ell n, gy'+2\ell n)$, whose points are at least as far from x as x', y' are, with the count of (3). Write $\Psi_0(x') := D^a_x G_i(\eta\mathbb{Z}^d,0)(x,x')$ and $\rho' := \min\{|x'-x|, |y'-x|\}$. *Claim:* on the free lattice at zero field the sup and first-difference clauses of Ψ_0 at x', and the Hölder quotient at a pair x', y' with $|x'-y'| \le 1$, are $\le Ce^{-\delta\rho'}$ whenever $\rho' \ge \frac{1}{20}$. Indeed, by the factorisation of (4), in unit-j units $D^a_x G_{(j)}(x,\cdot) = \sum_v \ell_j(\cdot,v)\, a_j^{-1}\beta_j(v)$ is the leg of the block data $a_j^{-1}\beta_j$, $\beta_j(v) := \sum_u (D^a\ell_j)(x,u)C^{(j)}(u,v)$, with $|\beta_j(v)| \le Ce^{-\delta_S \operatorname{dist}(x,B(v))}$ by the leg and $C^{(j)}$ bounds of (4) and one geometric sum. Split the data into the blocks with $\operatorname{dist}(x,B(v)) \le \rho'_j/2$, $\rho'_j := \rho'/s_j$ the distance in unit-j units, and the rest. The leg of bounded block data obeys the sup bound of lemma A.8(b), the first-difference bound of lemma A.12 and its Hölder bound on balls of radius $\frac{1}{16}$ with decay $e^{-\delta_3 \operatorname{dist}(\cdot,S)}$ from the support S of the data (the interior case, every bond of the free lattice being interior; printed: [13, (3.63)–(3.65)], [3, (2.12)]); the far part has data of size $\le Ce^{-\delta_S\rho'_j/2}$, and the near part has its support within $\rho'_j/2$ of x, so at distance $\ge \rho'_j/2$ from x' and y'; a pair at unit-j distance $> \frac{1}{16}$ is bounded by $2\cdot 16^{\alpha_0}$ times the first differences. So in unit-j units every clause of slice j is $\le Ce^{-\delta_6\rho'_j}$, $\delta_6 := \frac{1}{4}\min\{\delta_S,\delta_3\}$. In scale-$i$ units the sup clause of slice j scales as $s_j^{-(d-2+a)}$, the first differences as $s_j^{-(d-1+a)}$ and the Hölder quotients as $s_j^{-(d-1+a+\alpha_0)}$ (a scale-i pair at distance $t \le 1$ is a unit-j pair at distance t/s_j; for $t > s_j/16$ use two first differences, $2\cdot 16^{\alpha_0}s_j^{-(d-1+a)}t^{-\alpha_0} \le 2\cdot 16^{\alpha_0}s_j^{-(d-1+a+\alpha_0)}$); the slice G_1 at the fixed spacing contributes a bounded multiple of its first differences to the Hölder clause, every pair being at distance $\ge \eta$. The sum over s is the geometric sum of (4) with $d-1+a+\alpha_0$ in place of p, giving $\le C(\eta+\rho')^{-(d-1+a+\alpha_0)}e^{-\delta'\rho'} \le Ce^{-\delta'\rho'}$ for $\rho' \ge \frac{1}{20}$; this is the claim, and with the image count it holds for $x' \mapsto D^a_x G_i(\square,0)(x,x')$ at pairs of $\square$ with $\rho' \ge \frac{1}{20}$. *Localisation.* $\Psi = h'\Psi_\square$, $\Psi_\square := D^a_x G_i(\square,0)(x,\cdot)$, and $D(h'\Psi_\square) = h'(\cdot+\eta e)D\Psi_\square + (\partial^\eta h')\Psi_\square$ bond by bond. Both Ψ and $D\Psi$ vanish outside $S' := \{\operatorname{dist}(\cdot,\Delta') \le \frac{1}{2}+\eta\}$, whose points are at distance $\ge \frac{1}{2} - 2\eta \ge \frac{1}{10}$ from x ($\eta \le \frac{1}{5}$, x within η of Δ, $\operatorname{dist}(\Delta,\Delta') \ge 1$). Hence the sup and first-difference clauses of Ψ are those of $\Psi_\square$ on S' times $1+c_h$. For a Hölder pair x', y' with both points at distance $\ge \frac{1}{20}$ from x, the Leibniz rule and the bounds $|h'(x')-h'(y')| \le dc_h|x'-y'|$, $|\partial^\eta h'(x') - \partial^\eta h'(y')| \le dc_h|x'-y'|$ give the quotient $\le C(1+c_h)^2$ from the three clauses of $\Psi_\square$; a pair with one point y' within $\frac{1}{20}$ of x has $D\Psi(y') = 0$ and $|x'-y'| \ge \frac{1}{10} - \frac{1}{20} = \frac{1}{20}$ whenever $D\Psi(x') \ne 0$, so its quotient is $\le 20^{\alpha_0}$ times a first difference. This is (c). □

On the cube the same device, a constant box Q_x of large blocks around each vertex cube at depth $\ge R_*$, is step (1) of the proof of theorem A.41; the depth supply is lemma A.34(a) there too.

(b) The cut-off $\zeta^{(i)}$. In [2, (2.44)] $A^{(i)} = a_i\ell_i^{-2}\zeta^{(i)}G_iQ_i^*A$ with $\operatorname{supp}\zeta^{(i)}(\cdot,y) \subset \{|x-y| < r_i - 2M\}$, $\zeta^{(i)}(x,y) = 1$ for $|x-y| \le \frac{1}{2}r_i$. The printed proof of [2, (2.60)], $\partial^\eta A^{(i)} = O(p_i)$ on Ω, bounds the first two terms of [2, (2.61)] by the restrictions [2, (2.55)] at every coarse point y' with $\zeta^{(i)}(x,y') \ne 0$; [2, (2.55)] is available on $\Lambda_{-1}^{(i-1)'}$, which contains Ω at depth $\ge 3r(\ell_{i-1})/L - 2$ (lemma A.34(a), three layers). The containment $3r(\ell_{i-1})/L - 2 \ge r_i - 2M$ has $\alpha = 1$, $\beta = 3$: it holds at every scale iff $L \le 3$. The gap is real: by [2, (2.9)] the restrictions survive the partial resummation only on Λ'_{-1}, Λ^*_{-1}, Λ_{-1}, so at a large block which stayed large at every finer scale $A(y')$ is unrestricted and the contribution $a_i\zeta^{(i)}(x,y')(G_iQ_i^*)(x,y')A(y')$ has no bound; hence [2, (2.60)] as printed is not established in a collar of width $\approx r_i - 3r(\ell_{i-1})/L$ along $\partial\Omega$, while [2,

p. 571] uses it on all of Ω and [3, Prop. 1] assumes it on [3, p. 412]. The same cut-off enters the split [2, (2.89)] at the next scale.

Proposition A.37 (the shrunk cut-off)**.** *Replace* $\zeta^{(i)}$ *in* [2, (2.44)]*, at every scale, by* $\zeta_\theta^{(i)}$ *with* $\operatorname{supp}\zeta_\theta^{(i)}(\cdot,y)\subset\{|x-y|<\theta r_i-2M\}$, $\zeta_\theta^{(i)}(x,y)=1$ *for* $|x-y|\le\frac12\theta r_i$, $|\partial^\eta\zeta_\theta|\le1$, $|\partial^\eta\partial^\eta\zeta_\theta|\le2$ *(a* $C^{1,1}$ *radial profile), with* $\theta=\theta_5:=\frac25$*; the first-step cut-off of* [2, p. 561] *already has range* $r/(2L)$ *in the units of the next scale, so this returns to the first-step convention. Let* $R\ge\max\{10(M+1),M_\theta+2,(M_\theta+7)/11.6\}$*, where* θ_{i+1} *equals* 1 *on* $B^i(\Lambda_2^{(i)})$ *and* 0 *at distance* $\ge M_\theta$ *from it (*$M_\theta<M$ *by* [2, p. 567]*, where* θ_{i+1} *"varies smoothly" on a slice of thickness* $<M$*; we take* $M_\theta=2M/3$ *and a* $C^{1,1}$ *profile). Then at* $L=5$*:*

(Z1) *for* $y\in\Lambda_2^{(i-1)\prime}$*,* $x\in B^i(y)$*, the* ζ_θ*-ball* $\{|x-y'|<\theta r_i-2M\}$ *lies in* $\Lambda_{-1}^{(i-1)\prime}$*, where* [2, (2.55)] *holds;*

(Z2) [2, (2.59)–(2.60), (2.64)] *hold with* $O(p_i)$ *replaced by* $O_L(p_i)$*, the constant containing* $p_{i-1}/p_i\le(1+\ln L)^p$*;*

(Z3) *the terms carrying* $1-\zeta_\theta^{(i)}$ *in* [2, (2.61), (2.64)]*, the terms carrying* $1-\zeta_\theta^{(i)}$*,* $1-\zeta_\theta^{(i+1)}$ *in* [2, (2.88)–(2.90)]*, and the second term of* [2, (2.89)] *(the one with* $\delta C_{\Lambda_0}^{(i)}$*) are* $O(\ell_i^\kappa)$ *for every* κ*, with their derivatives and the* α_0*-Hölder quotients of their derivatives;*

(Z4) *the transition width* $\frac12\theta r_i-2M\ge2$ *required by the* $C^{1,1}$ *radial profile with* $|\partial^\eta\zeta_\theta|\le1$ *and* $|\partial^\eta\partial^\eta\zeta_\theta|\le2$ *holds;*

(Z5) *the split* [2, (2.89)] *holds* as printed*, with the full* B *in its fifth term, and the* $\zeta_\theta^{(l)}$*-balls around* $\operatorname{supp}\theta_l$ *lie in* $\Lambda_{-1}^{(l-1)\prime}$ *at every scale* l*.*

No ζ *obeying the printed* [2, (2.44)] *makes* [2, (2.89)] *exact at* $L=5$ *for small* ℓ_i *(the minimal printed support* $\frac12 r$ *gives* $L\theta=\frac52>2$*), so the shrunk cut-off is necessary at* $L=5$ *if* [2, (2.89)] *is kept in Bałaban's form.*

Proof. (Z1): $\theta r_i\le3r(\ell_{i-1})/L-2+2M$ is lemma A.34(b) with $\alpha=\frac25$, $\beta=3$, $\alpha L=2\le3$, margin $\ge r(\ell_{i-1})/5-2\ge0$ for $R\ge10$. (Z2): in [2, (2.61)] with ζ_θ the first term sums over y' in the ζ_θ-ball, where by (Z1) $|A(y')-A(y)|\le c_1p_{i-1}L|y'-y|$ (a path of $\le L|y'-y|$ bonds with $|\partial A|\le c_1p_{i-1}$) and [2, Prop. 2.2] gives the kernel bound $c_{\mathrm{Ba}}e^{-\delta_0|x-y'|}$, so the term is $O(p_{i-1})=O_L(p_i)$; the third term is the exact [2, (2.62)–(2.63)] ($G_iQ_i^*1$ is constant); the second is a tail of (Z3); [2, (2.64)] is the same for the derivative, with $\partial^\eta(G_iQ_i^*1)=0$. (Z3): the $(1-\zeta_\theta)$ term of [2, (2.61)] multiplies the single constant $A(y)$, $|A(y)|\le c_1p_{i-1}/(\mu_0\ell_{i-1})$, with a kernel supported at $|x-y'|\ge\frac12\theta r_i$ and bounded by $c_{\mathrm{Ba}}e^{-\delta_0|x-y'|}$, so it is $\le Cr_i^de^{-\delta_0\theta r_i/2}p_{i-1}/(\mu_0\ell_{i-1})\le\ell_i^\kappa$ for small ℓ_i (lemma A.34(c)); in [2, (2.89)] the third and fourth terms carry $1-\zeta_\theta^{(i)}$, $1-\zeta_\theta^{(i+1)}$ acting on $\Lambda_0^{(i)\prime}B$ with $|B|\le c_1p/(\mu_0\ell)$ there, and the same bound gives them, their derivatives and Hölder quotients (kernel by kernel: [1, (2.34), (2.36)] for $C^{(i)}$, δC_{Λ_0}, the composition identity of (Z5) for $G_{i+1}^\eta Q_{i+1}^*$, the $C^{1,1}$ profile at scales i, $i+1$); the second term $T_2(x)=a_iaL^{-2}\theta_{i+1}(x)\sum_y\zeta^{(i)}(x,y)(G_iQ_i^*)(x,y)(\delta C_{\Lambda_0}^{(i)}Q^*\Lambda_0^{(i)\prime}B)(y)$ has, for $x\in\operatorname{supp}\theta_{i+1}$ and $\zeta(x,y)\ne0$, $\operatorname{dist}(y,\Lambda_0^{(i)c})>r_i-M_\theta-2$ (two layers of [2, (2.8)] minus the range of ζ), and $\delta C_{\Lambda_0}(y,y')$ decays exponentially in $|y-y'|$ and in the distances of y, y' to $\Lambda_0^{(i)c}$ [1, (2.35)], so $|T_2|+|\partial^\eta T_2|\le Cr_i^de^{-\delta'(r_i-M_\theta-2)}p/(\mu_0\ell)\le C_\kappa\ell_i^\kappa$. (Z4): $\frac12\theta r_i-2M\ge2$ iff $r_i\ge(4+4M)/\theta=10(M+1)$, and $r_i\ge R$. (Z5): on the unit lattice of scale i, with Q the one-step average, $P_i=Q_i^*Q_i$, $P_{i+1}=Q_i^*Q^*QQ_i$, $G_i=(H+a_iP_i)^{-1}$, $G_{i+1}^\eta=(H+a_{i+1}L^{-2}P_{i+1})^{-1}$, $\Delta^{(i)}=a_i-a_i^2Q_iG_iQ_i^*$, $C^{(i)}=(aL^{-2}Q^*Q+\Delta^{(i)})^{-1}$ and $a_{i+1}=aa_i/(a_i+aL^{-2})$ [1, (2.13), (2.21), (2.31)], one has the composition identity $a_iaL^{-2}G_iQ_i^*C^{(i)}Q^*=a_{i+1}L^{-2}G_{i+1}^\eta Q_{i+1}^*$: apply $H+a_{i+1}L^{-2}P_{i+1}$ to the left side, use $HG_i=1-a_iP_iG_i$, $1-a_iQ_iG_iQ_i^*=\Delta^{(i)}/a_i$, $\Delta^{(i)}C^{(i)}=1-aL^{-2}Q^*QC^{(i)}$ and $Q^*QQ^*=Q^*$; the coefficient of $Q^*QC^{(i)}Q^*$ vanishes iff $a_{i+1}(a_i+aL^{-2})=aa_i$, which is [1, (2.13)], and the remaining coefficient is $a_{i+1}L^{-2}$ by the same relation (this is [1, (2.41)] in adjoint, rescaled form). By [2, (2.80)] and the Dirichlet condition [1, (2.32)], $C_{\Lambda_0}^{(i)}Q^*B=C_{\Lambda_0}^{(i)}Q^*\Lambda_0^{(i)\prime}B$; writing $C_{\Lambda_0}=C^{(i)}+\delta C_{\Lambda_0}$ and $\zeta=1-(1-\zeta)$

and using the identity, $\theta_{i+1}A^{(i)}$ equals the five terms of [2, (2.89)] with $\Lambda_0^{(i)\prime}B$ in the fifth term, identically in A', B; the printed form differs by the defect $a_{i+1}L^{-2}\theta_{i+1}\zeta_\theta^{(i+1)}G_{i+1}^{\eta}Q_{i+1}^{*}\Lambda_0^{(i)\prime c}B$, which vanishes when $\zeta_\theta^{(i+1)}(x,y)=0$ for all x with $\theta_{i+1}(x)\neq 0$ and $y\notin\Lambda_0^{(i)\prime}$, i.e. when $L(\theta r_{i+1}-2M)\le 2r_i-M_\theta-2-L$ (such x lie within $M_\theta+1$ of $\Lambda_2^{(i)}$, such y within L of $\Lambda_0^{(i)c}$, and $\operatorname{dist}(\Lambda_2^{(i)},\Lambda_0^{(i)c})>2r_i-1$). With $L\theta_5=2$ this reads $2(r_i-r_{i+1})\ge M_\theta+7-10M$, and $r_i-r_{i+1}=R\ln 5\,(2u_{i+1}+\ln 5)\ge 5.81R$. The balls around $\operatorname{supp}\theta_l$ reach at most $2r_l-10M+M_\theta+1$ beyond $\Lambda_2^{(l-1)}$ (scale-$(l-1)$ units), while $\Lambda_{-1}^{(l-1)}$ contains the $(3r(\ell_{l-1})-2)$-neighbourhood of $\Lambda_2^{(l-1)}$ (lemma A.34(a)): $2r_l+M_\theta+8\le 3r(\ell_{l-1})+10M$ holds since $r_l\le r(\ell_{l-1})$ and $r(\ell_{l-1})\ge R\ge M_\theta+2$. For the printed $\theta=1$ the defect condition has $L\theta=L$, $\beta=2$, so it holds at every scale iff $L\le 2$; on the reading that absorbs the defect into $\tilde B^{(i)}$, the $\zeta^{(i+1)}$-balls must stay inside $\Lambda_{-1}^{(i)\prime}$, $\beta=3$, hence $L\le 3$. □

The other uses of [2, Lemma 2.3] are covered by (Z2): [2, (2.68)] on the box of (c) below; [2, (2.95)–(2.98)] at each scale l together with (Z5); the regularity of $\tilde B$ needed by [3, Prop. 1] at the background $\theta_iA^{(i)}$ on Ω, where $\theta_i=1$; and the first-step use [2, (2.25)], whose cut-off has $\theta=1/(2L)<3/L$. On the consumer side, the all-small-field term of [13, (3.4)] is affected only through the cut background, and the difference between the actions at $A^{(j)}$ and at A_θ is $O(\eta^{s'})|X^{(j)}|$ on $\operatorname{supp}\chi_j$, as used in the proof of proposition A.43.

(c) The boxes of [2, Lemma 2.4]. The printed proof [2, p. 572] takes $\square_1,\square_2$ as the sums of large blocks of $\Lambda_7^{(i-1)\prime}$ within $2r_i$, $4r_i$ of y and $\square=B^i(\square_2)$. For $y\in\Lambda_7$ at distance $O(1)$ from Λ_7^c the step [2, (2.67)] is not justified: the dropped source $\Lambda_6\phi-\square_1\phi$ contains ϕ on $\Lambda_6\setminus\Lambda_7$ at distance $O(1)$ from x, where $a_iG_iQ_i^*$ is $O(1)$, so this error is $O(\Upsilon_i)$, $\Upsilon_i:=c_1(\lambda\ell_{i-1})^{-1/4}p_{i-1}$, not $O(\ell_i^\kappa)$; the domain change $\Omega\to\square$ happens at distance $O(1)$ from x, so [1, (2.26)] gives no smallness; and x may lie within R_{Ba} of $\partial\square$, where [1, (2.24)–(2.25)] need a parallelepiped with a constant field near its boundary, while $\Lambda_7\cap(\text{cube})$ has re-entrant edges. An untruncated box of radius $4r_i$ inside Ω would need $4r_i\le 5\rho_i$, which fails for small ℓ_i at every $L\ge 2$ ($\alpha=4$, $\beta=5$). The only consumer is the quartic convergence factor [2, (2.78)–(2.79)], a sum over $y\in\Lambda_7^{(i-1)\prime}\cap\Lambda_7^{(i)c}$, which contains every point of Λ_7 within $2r_i$ of Λ_7^c: the edge points are exactly the ones used.

Lemma A.38 ([2, Lemma 2.4] with a complete proof, every L)**.** *Assume* [2, (2.55)]*, and:* $5(r(\varepsilon_0^{\mathrm{Ba}})-s_b)/L\ge R_{\mathrm{Ba}}+c_d+2s_b$, $4r(\varepsilon_0^{\mathrm{Ba}})\ge R_{\mathrm{Ba}}+c_d$, $R\ge 10s_b+10$, *with* $c_d:=\sqrt{d}\,s_b$; $\varepsilon_0^{\mathrm{Ba}}$ *small enough that* $r(\ell_{i-1})/r(\ell_i)\le 2L$ *for all* i *(at* $L=5$, $\varrho=2$ *this is* $(1+\ln 5)^2\le 10$, *true for all* $\varepsilon_0^{\mathrm{Ba}}\le 1$*); the mass condition* $\Upsilon_i m^2\ell_i^2(G_i(\square,0)Q_i^*1)(x)=O(p_i)$ *(Bałaban's own use of* [2, (2.75)]*); and* $Ca_i\Upsilon_i(1+r_i)^{d-1}\exp(-\delta_0[(r(\ell_{i-1})-s_b)/L-c_d])\le\ell_i^\kappa$, *which holds for all small* $\varepsilon_0^{\mathrm{Ba}}$ *at every* $R>0$ *because* $\varrho>1$. *Then* [2, (2.65)–(2.66)] *hold for every* $y\in\Lambda_7^{(i-1)\prime}$, $x\in B^i(y)$, $b\subset B^i(\Lambda_7^{(i-1)\prime})$, *with constants independent of* ε, i, *the lattice and the position of* y *relative to* $\partial\Lambda_7$.

Proof. Put $s_i:=\min\{4r_i,5\rho_i-2s_b\}$, let $\square_2$ be the cube of large blocks of sup-radius s_i around the large block of y, $\square:=B^i(\square_2)$, and $\square_1\phi:=1_{Q(y,s_i/2)}\Lambda_6\phi$ with $Q(y,s)$ the sup-ball. Since $\operatorname{dist}(\Lambda_7,\Lambda_2^c)\ge 5\rho_i$ (lemma A.34(a)), $\square\subset\Omega$; $\square$ is a rectangular parallelepiped and a union of large blocks, and $s_i-c_d\ge R_{\mathrm{Ba}}$ at every i by the hypotheses. (1) *Source cut:* $\Lambda_6\phi-\square_1\phi$ is supported at distance $\ge s_i/2-c_d$ from x, and $\operatorname{dist}(x,\Omega^c)\ge 5\rho_i-c_d\ge R_{\mathrm{Ba}}$, so [1, (2.25)] on Ω with the sup bound Υ_i on ϕ gives $|a_i(G_i(\Omega,A^{(i)})Q_i^*(\Lambda_6\phi-\square_1\phi))(x)|\le Ca_ie^{-\delta_0(s_i/2-c_d)}\Upsilon_i$. (2) *Domain change:* $G_i(\square,A)=G_i(\Omega,A)+\delta G_i(\square,\Omega,A)$, and by [1, (2.26)] the second term applied to $Q_i^*\square_1\phi$ at x carries $e^{-\delta_0(3s_i/2-2c_d)}$; the same for D^η by [2, (2.58)]. Hence $\phi^{(i)}(x)=a_i(G_i(\square,A^{(i)})Q_i^*(A^{(i)})\square_1\phi)(x)+O(a_i\Upsilon_ie^{-\delta_0(s_i/2-c_d)})$, and likewise for the covariant derivative: this is [2, (2.67)] with the boxes chosen above. (3) *Constant background:* $A_0:=A^{(i)}(y)$; joining points of the convex box by straight lattice paths and using [2, (2.60)] (proposition A.37(Z2)) on $B^i(\Lambda_2^{(i-1)\prime})\supset\square$, $|A^{(i)}(z)-A_0|\le O(p)\sqrt{d}\,s_i\le O(pr_i)$ on $\square$; the expansion [1, (3.44)] and [1, Prop. 2.2] on the parallelepiped $\square$ replace $A^{(i)}$ by A_0 with error $O(\ell_i^{\kappa_0})$, as in [2, (2.68)]. (4) *Gauge-out:* $A_0=\partial^\eta\lambda_0$ with $\lambda_0(z)=A_0\cdot(z-y)$ on $\square$, so $U(A_0(\Gamma_{z,y}))=$

$e^{ieA_0\cdot(z-y)}$ for every contour inside $\square$, $G_i(\square,\partial\lambda_0) = U(\lambda_0)G_i(\square,0)U(\lambda_0)^*$ and $Q_i(\partial\lambda_0) = U(\lambda_0)_{\text{coarse}}Q_i(0)U(\lambda_0)^*$: [2, (2.69)–(2.74)]; legitimate because $\square$, of diameter $\le 8r_i+s_b$, does not wrap around (on the cube, no condition). Write $\phi' := U(\lambda_0)^{-1}\phi$. (5) *Reproduction of constants:* by [2, (2.75)], $a_iG_i(\square,0)Q_i^*1 = 1 - m^2\ell_i^2G_i(\square,0)Q_i^*1$, hence $a_i(G_i(\square,0)Q_i^*\square_1\phi')(x) - \phi'(y) = a_i(G_i(\square,0)Q_i^*[\square_1(\phi'-\phi'(y))])(x) - \phi'(y)a_i(G_i(\square,0)Q_i^*1_{\square\setminus\square_1})(x) - \phi'(y)m^2\ell_i^2(G_i(\square,0)Q_i^*1)(x)$. The first term is $O(p)$: on the convex cube $Q(y,s_i/2)\subset\Lambda_{-1}$ the restriction $|D\phi|\le c_1p$ gives $|\phi'(z)-\phi'(y)|\le O(p)|z-y|$ and the kernel decays exponentially. The second: $\square\setminus\square_1 \subset (Q(y,s_i)\setminus Q(y,s_i/2))\cup(Q(y,s_i/2)\setminus\Lambda_6)$ lies at distance $\ge \min\{s_i/2,\rho_i\}-c_d$ from x, so it is $\le c_{\text{Ba}}e^{-\delta_0(\min\{s_i/2,\rho_i\}-c_d)}\Upsilon_i$ by [1, (2.25)] on the parallelepiped. The third is the mass-defect term of the printed proof, $O(p_i)$ under the mass condition. (6) *Derivatives:* a bond $b\subset B^i(y)$ lies at distance $\ge s_i-c_d\ge R_{\text{Ba}}$ inside $\square$ and $\ge 5\rho_i-c_d$ inside Ω, so the printed derivative bounds apply at b and (1)–(5) give [2, (2.66)] in the form [2, (2.77)]; no re-entrant edge is approached. *Errors:* $s_i/2\ge\rho_i$ iff $\rho_i\le 2r_i$ and $3\rho_i\ge 2s_b$, the first by $r(\ell_{i-1})/r_i\le 2L$, the second by $R\ge 10s_b+10$; so every error is $\le Ca_i\Upsilon_i(1+r_i)^{d-1}e^{-\delta_0(\rho_i-c_d)}$, which is $\le \ell_i^\kappa$ by the last hypothesis; the rest of the printed proof, [2, (2.69)–(2.77)], runs on $\square$. Nothing depends on L except through the named constants. □

At $L=5$ the hypotheses read $r(\varepsilon_0^{\text{Ba}})\ge R_{\text{Ba}}+(3+\sqrt{3})s_b$, $r(\ell_{i-1})/r_i\le 6.81\le 10$, $R\ge 10s_b+10$: lower bounds on R and caps on $\varepsilon_0^{\text{Ba}}$, none restricting L. On the cube the same device (a box of large blocks inside the propagator domain about the collar cubes, with the cap inherited from the previous step) is the King box of D8.

(d) The partition [2, (3.43)–(3.45)]. The printed "regular partition into a lattice of cubes, each a sum of large blocks, with side in $(r,2r)$" needs, per coordinate, a divisor c of the number N of large blocks with $r/s_b < c < 2r/s_b$; at $L=2$ a power of 2 always works, while at $L=5$ the divisors of $N_i = 2MJ/(\ell_is_b)$, a number of the form $2\cdot 5^bJ$, have gaps of ratio up to $\frac{5}{2}$, so the window can contain no divisor, and with integer sides and even N the printed partition fails exactly when r_i/s_b is an integer. The partition is used in [2, Sect. 3C] only through four properties: the covering [2, (3.43)]; the neighbour count 3^d in [2, (3.44)–(3.45)]; the volume bounds $(2r)^d$ in [2, (3.48)–(3.49)]; and the corridor step of [2, p. 593].

Lemma A.39 (the product partition and the cube count)**.** *Let $q_i := \lceil (r_i+s_b)/s_b\rceil$ and let every coordinate carry $N\ge q_i$ large blocks (on King's cube, $N = 2MJ/(\ell_is_b)$ large blocks per side in scale-i units, which exceeds q_i for $J\ge J_0$; on a torus, N per period). Cut each coordinate interval into consecutive arcs of between q_i and $2q_i-1$ large blocks (write $N = nq_i+\varrho'$, $0\le\varrho'<q_i$, and give the ϱ' extra blocks to the n arcs, at most q_i-1 each), and let $\mathcal{Q}_i$ be the product partition into boxes. Call $Q\sim Q'$ if their arc indices differ by at most one in every coordinate. Then:* (Q1) *every box is a union of large blocks;* (Q2) *every box is a product of intervals of at most $2r_i+3s_b\le(2+3s_b/R)r_i$ points;* (Q3a) *every box has at most 3^d boxes $Q'\sim Q$;* (Q3b) *if $\operatorname{dist}(z,x)\le r_i+s_b$ then $Q(z)\sim Q(x)$. Consequently, with $I_i := \{z : \operatorname{dist}(z,(\Lambda_7^{(i-1)\prime})^c)\ge s_b\}$ and $\mathcal{C}_i := \{Q\in\mathcal{Q}_i : Q\cap\Lambda_0^{(i)c}\cap I_i\ne\emptyset\}$, every $Q\in\mathcal{C}_i$ satisfies $Q\sim Q(x_e)$ for the charge point x_e of some element e of the family F_i of large-field events at scale i, hence $|\mathcal{C}_i|\le 3^d|F_i|$; the volume bound $(2r)^d$ becomes $(2r_i+3s_b)^d$ and the corridor bound $22r$ becomes $22r_i+3s_b$, both $O(1)r_i^d$.*

Proof. (Q1) Arcs are unions of large-block intervals. (Q2) An arc has at most $(2q_i-1)s_b < 2(r_i+s_b)+s_b$ points, and $s_b\le(s_b/R)r_i$. (Q3a) The boxes $\sim Q$ are the 3^d neighbours of Q in the product structure. (Q3b) Every arc has $\ge q_is_b\ge r_i+s_b$ points, so if $|z-x|_\infty\le r_i+s_b$ the two arcs in each coordinate are equal or consecutive (otherwise a whole arc lies strictly between and the coordinate distance is $\ge q_is_b+1$). For the count: let $Q\in\mathcal{C}_i$ and $z\in Q\cap\Lambda_0^{(i)c}\cap I_i$; its large block β lies in Q by (Q1) and, having sup-diameter $<s_b$, in $\Lambda_7^{(i-1)\prime}$; since $z\notin\Lambda_0^{(i)}$, β is not one of the blocks of [2, p. 570], so $\operatorname{dist}(\beta,S_i)\le r_i$ where S_i is the point set of F_i; take $z'\in\beta$, $y\in S_i$ with $\operatorname{dist}(z',y)\le r_i$, y a point of an element e with $\operatorname{dist}(y,x_e)\le L-1<s_b$; then $\operatorname{dist}(z',x_e)\le r_i+s_b$ and $Q = Q(z')\sim Q(x_e)$ by (Q3b); count with (Q3a). The printed [2, (3.45)] and the first inequality of [2, (3.46)] do not hold as printed (a large block straddling $\partial\Lambda_7^{(i-1)\prime}$ lies in $\Lambda_0^{(i)c}$ with F_i possibly empty); the count above, on the interior I_i, becomes the count of D14

over the paid elements through lemma A.40; in D14 the boxes of $\mathcal{C}_{i'}$ for $i' < i$ are carried along by the corridor construction of [2, (3.48)–(3.50)] with the thickness $7(r_{i-1} + s_b) + Ls_b$ inside Bałaban's $(9r, 10r)$ once $(7 + L)s_b < 2r_{i-1}$. □

Lemma A.40 (the count over the paid elements)**.** *Let the boxes $\mathcal{Q}_i$, the relation $\sim$, the arcs of q_i to $2q_i - 1$ large blocks, the set $\mathcal{C}_i$ and the family F_i be those of lemma A.39, with the charge point x_e of an element e as in its proof (every point of e lies within* $\operatorname{dist} \le L-1$ *of x_e). Let $F_i^{\mathrm{paid}} \subset F_i$, and suppose that every $e \in F_i \setminus F_i^{\mathrm{paid}}$ has a* witness $w(e) \in F_i^{\mathrm{paid}}$ *with* $\operatorname{dist}(x_e, x_{w(e)}) \le r_i + s_b$. *Then*

$$|\mathcal{C}_i| \;\le\; 5^d\, |F_i^{\mathrm{paid}}|.$$

In D14 the unpaid elements are the near R_s elements, the blocks $y \in R_s^{(i)}$ within ℓ^∞-distance R_0 of the face layer, and F_i^{paid} consists of Bałaban's six families, R_s restricted to its far elements, and the face family; the witness of D7(b) (lemma A.24(e)) has a point within distance $dR_0 + 1$ of a point of e, so $\operatorname{dist}(x_e, x_{w(e)}) \le dR_0 + 1 + 2(L-1) \le s_b + L - 2 < r_i + s_b$ *by $s_b \ge dR_0 + L + 1$ (D7(c)) and $r_i \ge R > L$, and the hypothesis holds.*

Proof. Let $Q \in \mathcal{C}_i$. The proof of lemma A.39 gives $z' \in Q$ and $e \in F_i$ with $\operatorname{dist}(z', x_e) \le r_i + s_b$. Put $e' := e$ if $e \in F_i^{\mathrm{paid}}$ and $e' := w(e)$ otherwise; in both cases $e' \in F_i^{\mathrm{paid}}$ and $\operatorname{dist}(z', x_{e'}) \le 2(r_i + s_b) \le 2q_i s_b$, since $q_i s_b \ge r_i + s_b$. In each coordinate the arc indices of $Q = Q(z')$ and of $Q(x_{e'})$ differ by at most 2: if they differed by 3 or more, two whole arcs, of at least $q_i s_b$ points each, would lie strictly between the two coordinates, whose distance would then be $\ge 2q_i s_b + 1$. A box has at most 5^d boxes whose arc indices differ from its own by at most 2 in every coordinate (fewer on a torus with fewer than five arcs in some coordinate). So $Q \mapsto e'$ maps $\mathcal{C}_i$ into F_i^{paid} with fibres of at most 5^d boxes, which is the claim. For the application: a near R_s element e at the block y has $\operatorname{dist}(y, x_e) \le L - 1$; lemma A.24(e) gives, on a block path of length $\le dR_0$ from y to the face layer, a bond of the gradient family or a block of the face family, that is an element $w(e)$ of a paid family one of whose points lies within distance $dR_0 + 1$ of y and within $L - 1$ of $x_{w(e)}$; the triangle inequality gives $\operatorname{dist}(x_e, x_{w(e)}) \le dR_0 + 2L - 1$, and $dR_0 + 2L - 1 \le s_b + L - 2$ is $s_b \ge dR_0 + L + 1$. □

The replaced objects of (a)–(d), $r^\flat$, $D^\flat$, $\zeta_\theta^{(i)}$, the box $\square$ of lemma A.38 and the partition $\mathcal{Q}_i$, are built from $(L, M, R, \ell_i, \bar{n})$ and the set of event blocks alone, so they are the same on the cube as on the torus (theorem F.2); they are auxiliary objects of the proofs, not of the statements, and under (b) the all-small-field action changes by $O(\ell_i^\kappa)|X^{(i)}|$, which the statements absorb. The parameter conditions they add are lower bounds on R ($R \ge R^\flat$; $R \ge 10(M+1)$, $R \ge M_\theta + 2$; $R \ge 10s_b + 10$; $R \ge 6L\sqrt{d}$ for the covering of D14) and caps on $\varepsilon_0^{\mathrm{Ba}}$, all fixed in Bałaban's order of choices (d, a, $\bar{n}$, α_0, L, M, R, ϱ, p, b_0, then $\varepsilon_0^{\mathrm{Ba}}$), so the order is acyclic.

Theorem A.41 (the perturbative bounds on sub-domains)**.** *Fix a scale $k \le K_f$, a union Ω of k-blocks of the cube (any shape, meeting ∂D or not), A regular on Ω with $\theta_k \le \theta_0$, and a renormalised family $\mathcal{R}^\Omega$ of order ≤ 13 of Bałaban's expansion with the lines $G_{(i)}(\Omega, A)$, $G_k(\Omega, A)$, the legs $\phi_\Omega^{(k)}$, photon legs of the class of* [3, p. 412]*, possibly source vertices, possibly one difference line $\delta G_k(\Omega, \Omega_2; A)$ of two Neumann domains as an external pair, with at least one external field or source vertex, and all vertex cubes at depth $\ge R_* := 48$ in every propagator domain. Then for one held vertex x_0 (or none),*

$$\sum_{\text{orderings, indices, positions}} |E^{(k)}(G)| \;\le\; C\ell_k^{\beta_0} p(\ell_k)^{n_1} (1 + K - k)^{p_0} e^{-\delta \operatorname{dist}(V_*)} \Lambda(x_0)\, \Pi_b,$$

with $\Lambda(x_0)$ as in proposition A.27(a), $\Pi_b = 1$ without a difference line and $\Pi_b = \exp[-\delta_c(\operatorname{dist}(\mathrm{cube}(v), X_{\mathrm{live}} \setminus \Omega_2) + \operatorname{dist}(\mathrm{cube}(v'), X_{\mathrm{live}} \setminus \Omega_2))]$ for the ends v, v' of the difference line; the constants do not depend on k, K, ϑ, J, Ω, A or x_0. The same holds for a finite set of such difference lines and difference legs, each contributing its own factor, with the constant multiplied by $C_1^{|S|}$.

Proof. The proof of proposition A.27(a) is run on $\mathcal{R}^\Omega$ with the following changes. (1) *Localisation.* Every line and leg of the family is localised on a Dirichlet box Q_x of large blocks around its

vertex cube, inside Ω at depth $\geq R_*$: the difference between the operator on Ω and the operator on Q_x is a cut-off and commutator argument of the type of lemma A.16 (a smooth cut-off at the Neumann faces of Q_x, the commutator bounded by lemma A.12, the result bounded by lemma A.8(c)), exponentially small in the depth, and the box-vs-whole-cube difference near ∂D is lemma A.17; so every line at a vertex of depth $\geq R_*$ is the line of the box up to $Ce^{-\delta R_*}$ times the box line, with the Combes–Thomas rate, free of Bałaban's M. (2) *Zero-background identities on the box.* The five identities used in the organisation of [14, p. 330] (the free counterterm, the bulk rewriting of the renormalised families, the constant-datum identity lemma A.14, the slice identity, and the Ward-type cancellation of the zeroth photon moment, which on a box is exact since a constant gauge field is pure gauge there) hold on the Dirichlet box by the odd-image representation of Step 1 of lemma A.20, and at a regular background by gauging away the constant part of A on the box (gauge covariance, appendix A.1) and expanding to first order in the remainder, whose coefficients are $O(\theta_k(1+|x-x_0|))$ (lemma A.20, Step 3); the first-order photon tadpole at a background is bounded by the same expansion. (3) *Remainder vertices.* Each boundary-remainder vertex is a layered vertex of the class of proposition A.27(a) conjugated by the gauge transformation of (2), plus an ordinary vertex of degree raised by $D'+2 \geq 1$, plus an any-degree term with the small factor of (1); conjugation does not change absolute values, so the degree bookkeeping of [3, Sect. 2] is unchanged and the extra vertices only raise the containing stages. (4) *Lines near the face, difference lines.* The decay along paths of the lines on Ω is lemmas A.7 and A.8; the Hölder clause of a difference line near ∂D comes from the proof of lemma A.12, cases (3)–(4), applied to $w = M(\Omega_2)^{-1}g$ with g on the cut shell. The difference line $\delta G_k(\Omega,\Omega_2) = G_k(\Omega) - G_k(\Omega_2)$ is $G_k(\Omega)YG_k(\Omega_2)$ with Y the bond terms of the cut, so its kernel between two points carries the factor $e^{-\delta_c(\mathrm{dist}(\cdot,X_{\mathrm{live}}\setminus\Omega_2)+\mathrm{dist}(\cdot,X_{\mathrm{live}}\setminus\Omega_2))}$ by Combes–Thomas on both sides, which is Π_b. (5) *Telescoping.* A family some of whose lines are written $o = o' - \Delta o$ is the finite sum $\sum_S \mathcal{R}^{(S)}$ over the sets S of decomposed objects, $\mathcal{R}^{(S)}$ carrying $\prod_S(-\Delta o)$ and $\prod_{O\setminus S} o'$; $\mathcal{R}^{(\emptyset)}$ is the family with every object replaced by its counterpart on Ω, with the same counterterms, and for $S \neq \emptyset$, $\mathcal{R}^{(S)}$ is the renormalised family of the cut graphs $G \setminus S$, with every subtraction of the divergent subgraphs of $G \setminus S$ and the same counterterm values, the divergent subgraphs containing a line of S carrying no subtraction: this is the regrouping lemma 5.2, and no subtraction is lost. Each $\mathcal{R}^{(S)}$ is then a family with $|S|$ admissible external pairs, bounded by (1)–(4) with the factor $\Pi_S = \prod_{\sigma\in S} C_\sigma e^{-\delta_\sigma(\cdots)}$; the constant per pair is $C_1 = C_1(d,L,\bar{n})$. (6) *Depth.* At every consumer (D5, D6, D7, D10 below) the vertex cubes lie in a Λ_7-level set and the propagator domains are Λ_2-level, five nested layers apart ([3, p. 411], [2, (2.53), (2.56), Prop. 2.6]), so their depth in every propagator domain is $\geq \lfloor R \rfloor - 1 \geq 51 > R_*$ once $R \geq \max(52, \lceil R_{\mathrm{Ba}} \rceil + 1)$, where R_{Ba} is Bałaban's depth constant of [4, (2.19)]. □

Proposition A.42 (changes of propagator domain)**.** *At every scale $k \leq K_f$ of the upper bound, the domain-change statements of* [2, (2.18)–(2.50), (2.108)–(2.115)] *hold on the cube with the weighted flow: (i) the comparison of a Dirichlet-conditioned covariance C_Λ with the covariance of the Neumann domain $B^k(\Lambda_2^{(k)})$, $C_\Lambda - C|_{\Lambda\Lambda} = C_\Lambda X_{\Lambda\Lambda^c} C_{\Lambda^c\Lambda}$, is $O(M_{k+1}e^{-cr(\ell_k)})$ in the external-field norm; (ii) the scalar translation* [2, (2.23)] *and its replacement* [2, (2.25)] *differ by $O(M_1 e^{-\delta r(\varepsilon)}) = O(\varepsilon^{\kappa'})$ for every κ'; (iii) the change* [2, (2.108)] *of the block operators $\Delta^{(k)}$ between $B^k(\Lambda_2^{(k-1)\prime})$, $B^k(\Lambda_2^{(k-1)\prime} \cap \Lambda_5^{(k)c})$ and $B^k(\Lambda_2^{(k)})$ obeys* [2, (2.109)], *$|h_k(x,x')| \leq Ce^{-c\min(r(\ell_k),4r(\ell_{k-1})/L)}e^{-\delta|x-x'|}$; (iv) the Neumann cuts along $\partial\Lambda_6$, $\partial\Lambda_7$ inside the Dirichlet-restricted covariances* [2, (2.38)–(2.40)] *cost $O(\varepsilon^{\kappa'})|\partial\Lambda_5|$, resp. $O(\varepsilon^{\kappa'})|\Lambda_5|$; (v) the exact recursion* [1, (2.41)] *on the Neumann domain $B^k(\Lambda_2^{(k)})$ holds (it uses only the uniform coarse step); (vi) the restricted fluctuation covariance $C^{(k)}_{\Lambda_5^{(k)}}$ of* [2, (2.115)] *and the Neumann hole of* [2, (2.46)–(2.47)] *cost $O(\ell_{k-1}^{\kappa'})|\Lambda_6^{(k-1)}|$; (vii) the photon statements are those of the torus by lemma A.3(c). The constants depend on (d,L,a,W,c_V) and King's and Bałaban's constants only, not on J, K, k, ε, the domains (any unions of blocks in the classes where the objects are defined, the face met or not, re-entrant edges allowed) or the field.*

Proof. Each statement is an identity of resolvents (the bond cut $C_\Lambda - C'_\Lambda = -C_\Lambda Y_{\Lambda\Lambda} C'_\Lambda$, the restriction identity above, the Schur complement for the hole) followed by the Combes–Thomas decay of lemma A.7(a) on each factor and the coercivity lemma A.6 on every domain (which is what replaces Bałaban's γ'_0; the step-0 lower bound for every field is the Neumann gap with the weight, as in the proof of lemma A.6), the sup bounds of lemma A.8 and the legs' norms of corollary A.13 on the domains involved; the changes relative to the torus are constants only (the factor W from (Q5), (Q6) wherever a block average is bounded, and γ''_0 of lemma A.7(c) in place of Bałaban's γ'_0). The distances $r(\ell_k)$, $4r(\ell_{k-1})/L$ between the sets compared are Bałaban's region geometry, unchanged. The photon part is lemma A.3(c). □

The steps of the upper bound.

D1 (large/small split; [2, (2.2)–(2.3)]):
At each step the fields B, A, ψ, ϕ are tested; each test is tied to a positive term of the action and a violation yields $e^{-c_0^{\mathrm{Ba}} p(\ell_i)^2}$; with the Dirichlet condition the matter Gaussian only gains the positive face terms, which lower the integrand, and every printed large-field factor is monotone in them.

D2 (the face family):
At every scale $0 \le i \le K_f$ the face test with threshold σ_i is added to the six families (definition A.23); a violation is paid $e^{-p_i^2/4}$ by lemma A.24(a) from the θ_2-share of the scale-i form, the $(1-\theta_2)$-share remaining for D12. The family is forced: without it the background of face data of size M_i has a face gradient of order M_i (proposition A.19, remark), which violates the fine-field class [2, (2.55)].

D3 (region geometry; [2, (2.4)–(2.10), (2.43)]):
As said above, verbatim with seven families. The pair $B^i(y) \cup B^i(y')$ of lemma A.22's payment is one block here, so no clean-pair condition arises; tested blocks are clean (no Neumann cut through $B^i(y')$) by the nesting rules.

D4 (scalar large fields; [2, (2.11)–(2.13)]):
$\lambda(\varepsilon)|\phi|^4 + \frac{1}{2}\delta m^2 \varepsilon^2 |\phi|^2 \ge -O(\varepsilon^{\beta_0})$ and $\ge \frac{1}{2}\lambda(\varepsilon)|\phi|^4$ for $|\phi|^2 \ge O(1 + \ln \varepsilon^{-1})$: unchanged, δm^2 being King's free-boundary counterterm, a vertex, weight-independent.

D5 (translation and splitting; [2, (2.18)–(2.19)]):
The photon translation with the Dirichlet-conditioned $C^{(0)}_{\Lambda_0}$ is lemma A.3(c); the scalar analogue uses the operators on sub-regions with lemmas A.6 to A.8, A.10 and A.11 at every point and lemma A.12 and corollary A.13 in the restricted form (depth $\ge R_{\mathrm{Ba}}$, which is where Bałaban's splitting evaluates them: his vertex sets are Λ_7-level and his propagator domains Λ_2-level), theorem A.41 for the perturbative terms on these sub-domains and proposition A.42(i), (ii), (iv), (vii) for the changes of domain and of boundary condition. No face hypothesis on the splitting domains is needed on the cube.

D6 (expansion to order $\bar{n}$, cumulants, removal of the interaction from Λ_7^c; [2, (2.22)–(2.28), Prop. 2.6]):
Proposition A.27(c) and theorem A.41 with the difference lines of the pair of propagator domains of [2, Prop. 2.6] as external pairs (the telescoped families are finite signed sums of members with $s \ge 1$ difference objects, each admissible with the gain distance $\mathrm{dist}(\cdot, X_{\mathrm{live}} \setminus \Omega_2)$, so the extra factor is $e^{-5\delta_c r(\ell_k)} = O(\ell_k^{\kappa})$ at the margin $> 5r(\ell_k)$ of Bałaban's Proposition 2.6: this is the interaction-term bound lemma 5.1 on the cube, with the printed $e^{-5\delta_0 r(\ell_k)}$ replaced by $e^{-5\delta_c r(\ell_k)}$); the replacements [2, (2.108)–(2.115)] by proposition A.42(iii), (v), (vi).

D7 (the collar quartic; [2, Prop. 2.1, (2.57), (2.79)]):
Theorem A.41 on the propagator domain of [2, (2.56)] keeps $-\lambda(\ell_i) \sum_{x \in B^i(\mathrm{collar})} \vartheta^d |\phi^{(i)}(x)|^4$ exactly. The step [2, (2.79)] rests on [2, (2.65)], i.e. on $|\phi^{(i)}(x)| \ge \frac{1}{2}|\phi_i(y)|$ for $x \in B^i(y)$, $y \in R_s^{(i)}$. (b) *The near layers:* for $y \in R_s^{(i)}$ within ℓ^∞-distance R_0 of the face layer, lemma A.24(e) supplies a witness (a gradient violation or a face-family element) within distance $dR_0 + 1$, paid by its own family. (c) *All other blocks:* for y at distance $> R_0$ from ∂D, the background on $B^i(y)$ differs from the bulk background by the image effect, bounded by $C' e^{-\delta_1(R_0-2)} M_i \le \frac{1}{4}|\phi_i(y)|$ on the box $\Omega_b := X_{\mathrm{live}} \cap$(the ℓ^∞ block ball of radius $\lfloor \mathrm{depth}(x)/2 \rfloor$ about y) (lemma A.20,

Step 1's box-versus-free factor at $A = 0$ and Step 3 at regular A) plus the box-versus-domain difference (a cut-off and commutator argument at the Neumann faces of Ω_b, as in theorem A.41(1), of size $CM_i e^{-(\delta_2/2)(\mathrm{depth}(x)/2-8)} \le p_i$ at the D7 points, which lie several r-layers inside the propagator domain), with R_0 the smallest integer ≥ 3 with $C' e^{-\delta_1(R_0-2)} \le \frac{1}{4}$; R_0 depends on (d, L, a, W, c_V) only and is fixed before Bałaban's M ($s_b = LM \ge dR_0 + L + 1$). Under the caps $(C_B + 3)\lambda(\ell_i)^{1/4} \le \frac{1}{8}$, $C'(V_i + \theta_i) \le \frac{1}{8}$, this gives $|\phi^{(i)}(x)| \ge \frac{1}{2}|\phi_i(y)|$, hence [2, (2.79)] with $\frac{1}{2} \to \frac{1}{16}$, and the exponent of R_s in [2, (3.41)] changes only in its constant.

D8 (the backgrounds in the regular and fine-field classes; [2, Lemmas 2.3–2.4, Prop. 2.2]): $A^{(i)}$: lemma A.3(a), (d) and lemma A.4. $\phi^{(i)}$: [2, Prop. 2.2] is lemma A.12 (every bond, face bonds included); the fine-field class [2, (2.55)], $|D_{\tilde{A}}\phi^{(i)}(b)| \le c_1' p_{i-1}$ on $B^i(\Lambda_7^{(i)})$, is proposition A.19 with the cap $|\phi_i(y_0)| \le \sigma_i + \mathrm{dist}(y_0, L_1^{(i)})p_i$ for dist $\le r_i$ (transport along a block path to the face layer inside $\Lambda_7^{(i-1)\prime} \setminus S_i$, since $\mathrm{dist}(\Lambda_7^{(i)}, X_i) > 7r_i$) and $e^{-\delta_1 r_i} M_i \le p_i$ for dist $> r_i$; the product $(\sigma_i + tp_i)e^{-\delta_1 t} \le Cp_i$. At the collar cubes $\square \subset B^i(\Lambda_7^{(i-1)\prime})$ of [2, Prop. 2.1], where the data of [2, (2.56)] are cut one previous-scale layer away and the scale-i tests are known only off the bad set, the cap is inherited from step $i-1$ (lemma A.26), inserted as a characteristic function of ϕ_i as [2, (2.55)] is carried to step i; with it, for every fine bond b within 1 of $\square$: $|D_A \phi_\Omega^{(i)}(b)| \le C_G p_i$ and $|\phi_i(y) - (Q\phi_\Omega^{(i)})(y)| \le C_K p_i$ at the block y of $\square$, by proposition A.19 on the King box of radius $\lfloor m_i/2 \rfloor$ about $\square$ inside the propagator domain ($m_i := r(\ell_{i-1})/L \ge 70$), the truncation entering as the tail $Cc_1 M_i e^{-\delta_c r(\ell_{i-1})/L} \le p_i$ and the box-versus-domain difference as in D7(c); the block clause from lemma A.21. Every use of [2, (2.55)] in [2] is polynomial in p and paid by factors e^{-cp^2}.

D9 (fluctuation fields small; [2, Lemmas 2.5, 2.7]): $|A'| \le O(1)p$ [2, (2.86)] by lemma A.3; $|\phi'| \le O(1)p$ [2, (2.114)] from the block tests and the covariance bounds (lemma A.7(c), whose lower bound at the weight is lemma A.25 plus the coarse block Poincaré inequality); near the face lemma A.24(b)–(d) with the corrected threshold $r(\ell_i)$ (the contour of [14, (5.44)] must stay where the gradient test holds, which a path of length $\le r_i$ from $x \in \Lambda_4^{(i)}$ does, since $\mathrm{dist}(\Lambda_4^{(i)}, \Lambda_0^{(i)c}) \ge 4r_i$). Lemma A.24(d) supplies [2, (2.113)] at distance $> r_i$ from the face.

D10 (the Z-factors and the tail removal; [2, (2.92)–(2.109)]): $\det(C^{(i)})^{\pm 1/2}$ with $C^{(i)}$ bounded (lemma A.7); the removal of the tail field $\tilde{B}^{(i)}$ of [2, (2.89)–(2.90)] from the Z-factors, the lines and legs and the quadratic form costs $C\ell_i^{\kappa'}|\Lambda^{(i)}|$: the matter form by the envelope argument at the regular backgrounds, the determinant by the exact block split $\ln\det M = \ln\det(-\Delta_A + V) + \ln\det(1 + a_i Q(-\Delta_A + V)^{-1}Q^*)$ with the coarse determinant bounded in summed form and the block-free one by its closed-loop representation at the two regular backgrounds, and the perturbative part by theorem A.41 with the tail as an external photon field; on sub-domain histories the tail lies at distance $> r(\ell_i)/2$ from every place where the domain operators differ from the cube operators, so the domain remainders are $O(\ell_i^\kappa)e^{-cr(\ell_i)}$ times the volume.

D11 (the last step; [2, (3.11)–(3.12)]): $G_k' \ge \gamma_1 \mu_0^2 \ell_k^2$ (photon, lemma A.3(e)); $G_k'' \ge \gamma_1 m^2 \ell_k^2$ on $\Lambda_5^{(k)}$ from $V \ge m^2$ and the ℓ^2 bounds of lemma A.7, the Dirichlet terms only adding; $\|H_k'\|, \|H_k''\| \le O(1)e^{-\delta_1 r(\ell_k)}$ requires only ℓ^2 bounds (lemma A.7(a)).

D12 (positivity; [2, Prop. 3.1, (3.26)]): For every configuration in the support of the characteristic functions of [2, (3.21)] (the fine-field class on each collar), in the multiscale form [2, (3.23)–(3.25)]: [2, (3.26)] on the $(1-\theta_2)$-share follows from the Neumann separation [2, (3.27)–(3.28)] (printed algebra, weight-free) and lemma A.25 (the per-scale lower bound [2, (3.29)] at the weight, whose ϕ-independent error $O(\ell_k^{\kappa_0})|\Lambda_k|$ is carried on that support exactly as in print), with the Dirichlet faces only adding; the photon exponents are unchanged (lemma A.3(e)); the face family is paid on the θ_2-share by the composite clause of lemma A.22, on blocks of the index set, and payments of different scales are on disjoint fine sets and add. The convergence factors [2, (3.31)–(3.34)]

follow with $(1-\theta_2)$ on the two matter positivity exponents, the photon exponents unchanged, plus $\exp(-\frac{1}{4}p_i^2|\text{face violations at scale } i|)$ at every scale.

D13 (integration of the block fields; [2, (3.32)–(3.38)]): The normalisation identities are the minimiser identity; the exceptional integrals over $|\phi_{i+1}(y)-(Q\phi_i)(y)|>p$ as printed; after the rescaling, [2, (3.37)] is $\int d\phi_0 \exp(-\frac{1}{2}\langle\phi_0,(-\Delta^D_{\tilde A}+V)\phi_0\rangle)$ with the Dirichlet Laplacian and $V\geq m^2>0$, and the diamagnetic inequality of [6, Sect. 3.1, (3.4)–(3.7)] in its site-diagonal form (the resolvent Neumann series with the site-diagonal $\vartheta d_x+\vartheta^3V(x)+$ the face terms, every walk weight non-negative and $\|U(\tilde A)\|=1$, so $|(-\Delta^D_{\tilde A}+V)^{-1}(x,y)|\leq(-\Delta^D+V)^{-1}(x,y)$, and the interpolation of [6, Sect. 3.3, (3.16)–(3.17)]) gives $\leq\int\exp(-\frac{1}{2}\langle\phi,(-\Delta^D+V)\phi\rangle)=e^{E_{0,s}}$.

D14 (combinatorics; [2, Sect. 3C, (3.51)–(3.54)]): Purely combinatorial: $|\Lambda_0^{(i)c}|\leq O(1)r(\ell_i)^d(|\mathcal{C}_0|+\cdots+|\mathcal{C}_i|)$ and $-\sum_i(c_0'p_i^2-O(1)r(\ell_i)^{d+1})|\mathcal{C}_i|\leq 0$ if $2p>(d+1)\varrho$ (here $12>8$) and b_0 is large, with the product partition into boxes of large blocks in place of [2, (3.43)–(3.45)] and the cube count $|\mathcal{C}_i|\leq 5^d|F_i^{\text{paid}}|$ of lemma A.40, the paid part of the family comprising Bałaban's six families (R_s restricted to its far elements) and the face family (the near R_s elements are charged to their witnesses of D7(b) and not counted); the constant becomes $c_0':=5^{-d}\min\{\frac{1}{8}(1-\theta_2)a,\frac{1}{4}(1-\theta_2)\gamma_0^w,\frac{1}{4},\frac{1}{16}\}$, the region entropy $2^{7|\Lambda_0^{(i)c}|}$ is absorbed in the $O(1)(1+\ln\ell_i^{-1})|\Lambda_0^{(i)c}|$ of [2, (3.47)]. We use half of each payment for [2, (3.54)] and keep $e^{-\frac{1}{2}c_0'p_i^2|\mathcal{C}_i|}$.

D15 (final step; [2, (2.116)–(2.118)]): $P^{(K_f)}\leq O(\ell_f^{\beta_0})|\Lambda_7^{(K_f)}|$, $\sum_{i\leq K_f}O(\ell_i^{\kappa'})|X^{(i)}|\leq O(1)|C_J|$ (geometric, $\kappa'>d$), and the final integral with the region entropy gives $\exp(O(1)|X^{(K_f)}|)$, $|X^{(K_f)}|=\ell_f^{-d}|C_J|$: proposition A.27(c).

Proposition A.43 (upper line)**.** *Under the hypotheses of proposition A.33, $Z_k(s)\leq e^aI_k^{(j)}(s)+e^{-\frac{1}{2}c_0'p(\eta)^2+E_+|C_J|}$, with a as there, c_0' the constant of D14 and E_+ depending on $\varepsilon_0^{\text{Ba}}$ and King's data only.*

Proof. In the inductive formula [2, (2.43)] at scale j (seven families at every scale $i\leq j$, the gradient family at scale j imposed on every block bond of the cube), the term with no large-field block at any scale $i\leq j$ is $\int\chi_j\lambda_j\,e^{-S_k^{(j)}[A_\theta]+\sum_{i<j}O(1)\ell_i^{\kappa'}|X^{(i)}|}$ (D1–D10), with Bałaban's cut background $A_\theta=a_j\ell_j^{-2}\zeta_\theta^{(j)}G_jQ^*A_j$ in every scalar expression [2, (2.44), (2.50)]; the difference $|S_k^{(j)}[A^{(j)}]-S_k^{(j)}[A_\theta]|\leq C\eta^{s'}|X^{(j)}|$ on $\operatorname{supp}\chi_j$ (the matter form by the envelope argument between two regular backgrounds; the determinant by the block split of D10 at the two backgrounds; the perturbative part by proposition A.27 with the background-type photon field $A^{(j)}-A_\theta$), so the term is $\leq e^aI_k^{(j)}(s)$ after enlarging C in a. Every other term has a large-field block at some scale $i\leq j$ and carries the kept factor $e^{-\frac{1}{2}c_0'p_i^2}\leq e^{-\frac{1}{2}c_0'p(\eta)^2}$ (p is decreasing in the scale). Continue the RG for these terms to the final scale $\ell_f\asymp\varepsilon_0^{\text{Ba}}$ (D11–D15, with the face family imposed and paid at each step, D2); the sum over all admissible histories with the remaining half of the payments and [2, (3.54)] is $\leq e^{O(1)|X^{(K_f)}|}=e^{E_+|C_J|}$, $E_+=O(\ell_f^{-d})$ ([2, (2.117), (3.42)–(3.47)] with (3.54)). The constants are those of D1–D15: independent of k, j and ϑ, each step's constants being so (the payments $e^{-p_i^2/4}$ are exact for every scale by the choice of σ_i; at every scale the lattice regime is the only regime on the cube, and lemmas A.10 and A.12 cover it uniformly in $\vartheta\leq 1$). □

A.9. **Assembly, the normalisation-free form, and the clause** (R_0)**.**

Lemma A.44 (field-independent constants cancel)**.** *Let $Z'(s)$, $I'(s)$ be $Z_k(s)$ and $I_k^{(j)}(s)$ computed with the effective action shifted by a field- and source-independent constant c (a different E_1, a different normalisation of the measure, the domain-adapted E_0, E_1 of [13, (3.35)]). If $e^{-a}I'\leq Z'\leq e^aI'+bZ'(0)$, then $e^{-a}I_k^{(j)}\leq Z_k\leq e^aI_k^{(j)}+bZ_k(0)$ with the same a,b.*

Proof. $Z' = e^{-c}Z_k$ and $I' = e^{-c}I_k^{(j)}$ for every s; divide. □

Proof of proposition 5.4. Fix $\eta_1 \le \min(5^{-J_*}, \varepsilon_0^{\mathrm{Ba}})$ and smaller where the conditions on η_1 collected above require it (lemmas A.29 and A.32 and proposition A.19, D7, D14), so that [1, (3.54)] holds at every scale $\le j$ for $b_0 \ge b_0^*$; the condition $j \ge J_*$ is Bałaban's $\varepsilon \le \varepsilon_0(\eta) = \eta 5^{-J_*}$ and $j \le k - J_*$ is King's $\eta \le 5^{-J_*}$, so that $j \le K_f \le k$. Propositions A.33 and A.43 give, in the normalisation of B6,

$$e^{-a}I_k^{(j)}(s) \le Z_k(s) \le e^{a}I_k^{(j)}(s) + e^{-\frac{1}{2}c_0' p(\eta)^2 + E_+|C_J|}, \qquad Z_k(0) \ge e^{-E_-|C_J|},$$

hence the remainder is $\le e^{-\frac{1}{2}c_0' p(\eta)^2 + (E_+ + E_-)|C_J|} Z_k(0)$. By lemma A.44 the same holds for Z_k and $I_k^{(j)}$ in the normalisation of section 5.1. This is (2) with $C\eta^\sigma$ in place of $C\eta^\sigma|C_J|$ (the volume $|C_J| = (2MJ)^3$ is absorbed in C, which may depend on J), $c = \frac{1}{2}c_0'$ and $\rho(\eta) = e^{-cp(\eta)^2 + C}$ with $C \ge (E_+ + E_-)|C_J|$. The constants are independent of k and j: of j (i.e. of $\eta \le \eta_1$) because every step's constants are, and of k (i.e. of ϑ) because no step depends on the spacing except through the lattice regime, which lemmas A.10 to A.12 cover with the same constants; the dependence on x enters through the source vertices only (B7, proposition A.27). The clause (R_0) is theorem A.45 below. □

Theorem A.45 (the remainder is the large-field part of Bałaban's decomposition)**.** *In the setting of proposition 5.4, let $P := \prod_{i \le j} P^{(i)}$, where $P^{(i)}$ is the product, over all points, bonds and blocks of the scale-i index sets of the all-small history (the whole cube's lattices at every scale), of the small-field indicators of the families of scale i (Bałaban's six families, the face family, and at $i = j$ King's χ_j and λ_j). Then: (a) P is a function on the joint variable space of Bałaban's block-spin representation with values in $\{0,1\}$, the same for every source s; (b) $\int P\, d\mu_{k,s}$ is the exact contribution of the all-small history at scale j, and $\int P\, d\mu_{k,0} \le e^{C\eta^\sigma} I_k^{(j)}(0)$; (c) $\int (1-P)\, d\mu_{k,0} \le e^{-\frac{1}{2}c_0' p(\eta)^2 + E_+|C_J|} \le \rho(\eta) Z_k(0)$.*

Proof. The joint measure. Each block-spin step inserts $1 = N_i^{-1} \int dB\, d\psi \exp(-\frac{a}{2}|B - QA|^2 - \frac{a}{2}|\psi - Q(A)\phi|^2)$ [2, (2.1)], with N_i field-independent; so $Z_k(s) = N^{-1} \int d\mu_{k,s}$ with $d\mu_{k,s} := e^{-S} e^{X(s)} \prod_i [\text{kernels}]\, d(\text{all variables})$ a positive measure, and N independent of s. *The tested quantities are source-free:* they are fields, block fields and fluctuation fields (differences between a field and a background, [13, (3.16)]), and backgrounds are linear maps of block fields built from the source-free quadratic forms; the source enters only the vertices [13, (3.20)] and the factor $e^{X(s)}$. This gives (a). *Decomposition at one scale.* Expanding $\prod[\chi(\text{small}) + \chi(\text{large})]$ over the points, bonds and blocks of scale i gives [2, (2.4)–(2.6)]: $1 =$ a sum over large-field tuples of products of indicators; the regrouping by Λ_0 and the partial resummation [2, (2.9)–(2.10)] replace, for fixed $\Lambda_0 \ne X_{\mathrm{live}}$, the indicator of "the true tuple generates Λ_0" by a sum over minimal admissible tuples of non-negative indicator products, which is $\ge$ the indicator it replaces; for $\Lambda_0 = X_{\mathrm{live}}$ the only admissible tuple is the empty one and its term is exactly the product of all small-field indicators of that scale. *Induction over scales.* In each history the next scale's decomposition is inserted where the RG continues; inside a term the translations are changes of variables with unit Jacobian, the Gaussian integrations are exact, and the removal of interactions from large-field regions and the large-field estimates [2, (2.11)–(2.14)] are termwise upper bounds of a positive integrand; this is why [2, (2.43)] is an inequality bounding Z_k itself. *Telescoping*, exact and pointwise, with $P_i := \prod_{i' \le i} P^{(i')}$, $P_{-1} := 1$: $1 - P = \sum_{i=0}^{j} P_{i-1}(1 - P^{(i)})$, the i-th summand being the indicator of "all tests of scales $< i$ pass everywhere, some test of scale i fails"; by the identity-or-over-count at each scale, $P_{i-1}(1 - P^{(i)}) \le$ the sum of the weights w_H of the histories H whose first large-field block is at scale i, and summing over i, $1 - P \le \sum_{H \ne H_0} w_H$, H_0 the all-small history (weight P). Integrate against the positive measure $d\mu_{k,0}$: $N^{-1} \int (1 - P) d\mu_{k,0} \le \sum_{H \ne H_0} [\text{the (2.43)-term of } H \text{ continued to the final scale}]$, each history's exact contribution being bounded by its term because every operation after the insertion of the indicators is an identity or a termwise upper bound. The proof of proposition A.43 bounds exactly this sum by $e^{-\frac{1}{2}c_0' p(\eta)^2 + E_+|C_J|}$, and $Z_k(0) \ge e^{-E_-|C_J|}$ in the same normalisation (B6),

which gives (c). (b): the all-small history's exact contribution is $\int P\, d\mu_{k,s}$, which the proof of proposition A.43 bounds by $e^a I_k^{(j)}(s)$. The normalisation-free form follows by lemma A.44. Positivity of $(1-P)\, d\mu_{k,0}$ holds because $P \le 1$; whether [2, (2.9)] is an identity or an over-count does not matter, only $1 - P \le \sum_{H \neq H_0} w_H$ is used. □

Appendix B. Proof of proposition 5.5

This appendix proves proposition 5.5. Throughout, k is the spacing index, j the number of renormalisation steps, $\eta = 5^{j-k}$ the block scale, and the two members of a diagonal pair are the k-model after j steps and the $(k+1)$-model after $j+1$ steps (section 5.1); quantities of the second member carry a prime. The scalar source vanishes, $s = u$; by remark 5.8(i) nothing below concerns the vacuum constant $v_{k,j}$, and "field-free, source-free" components are dropped wherever they occur.

B.1. Units, operators, and the decomposition of the effective action.

Units. Fix a depth $0 \le i \le j$ and work in *unit units of depth* i: the lattice X_i of the i-th block variables has spacing 1, the physical length unit is $l_i := 5^i \varepsilon_k$ ($l_j = \eta$), the fine lattice has spacing 5^{-i}, $e_i := e l_i^{1/2}$, $m_i^2 := m^2 l_i^2$, $\mu_i^2 := \mu_0^2 l_i^2$, and the rescaled fine field is $A_i(x) := l_i^{1/2} A(l_i x)$, so that a unit charge sees $U(A_b) = \exp(e_i 5^{-i} q A_{i,b})$, q the real antisymmetric generator with $q^2 = -1$. At depth j we write $\rho_i := 5^{i-j}$ for the scale of the slice of index $i \le j$ ($\rho_j = 1$). The $(k+1)$-model at depth $i+1$ lives on the same lattice X_i with fine spacing 5^{-i-1}. All statements hold at every point (e, λ, m^2, μ_0) of King's range; constants depend on it, on J, on King's b_0, p and on the order $\bar{n} = 13$ of his expansion, and on nothing else unless said.

Operators. ([13, (2.10)–(2.17)], [14, (3.2)]). On the cube C_J^K the scalar has the Dirichlet condition of definition 2.2 and the bonds to the Dirichlet planes are kept; $H_A := D_A^* D_A$ is the covariant Laplacian, $M_0(A) := H_A + m_i^2$. $Q_i(A)$ averages the fine field over the i-th blocks along King's hierarchical staircases $\Gamma_{y,x}^{(i)}$ (of length at most 3 in unit units, inside the block) with weights $w(x) = 5^{-3i} f(x)$, $f = 1$ above the finest step and $f = f_D$ at the finest step, $\sum_x w = 1$ and density at most $\varrho := (5/4)^3$. Then

$$M(A) := M_0(A) + a_i Q_i(A)^* Q_i(A), \quad G_i(A) := M(A)^{-1}, \quad H_i(A) := a_i G_i Q_i^*, \quad \Delta^{(i)}(A) := a_i - a_i Q_i H_i(A),$$

and with Q the plain one-step average $X_i \to X_{i+1}$, $K_i(A) := \Delta^{(i)}(A) + a 5^{-2} Q(A)^* Q(A) = C^{(i)}(A)^{-1}$. The photon operators are the same with $f = 1$, $m \to \mu_0$, the Neumann condition of definition 2.2, and blocks that tile C_J; the background is $A^{(j)} := H_j^{\mathrm{ph}} A_j$. On the support of χ_j (King's [13, (3.2)] with the face factor [14, (5.41)]) $|A_j|, |dA_j|, |\phi_j|$ are bounded by powers of $p = p(\eta)$, and $\varpi_i := e_i \sup |dA_i^{(j)}|$ is the smallness parameter.

The decomposition. By [13, (3.90)] the effective action after j steps is

$$S_k^{(j)} = \tfrac{1}{2}\langle A_j, \Delta_{\mathrm{ph}}^{(j)} A_j\rangle + \ln N_j + \tfrac{1}{2}\langle \phi_j, \Delta^{(j)}(A^{(j)})\phi_j\rangle + \ln N_j(0) - \ln \frac{Z_j(A^{(j)})}{Z_j(0)} + P^{(j),1}, \tag{58}$$

where $Z_j(A)$ is the normalised scalar Gaussian integral [13, (2.15)] and $P^{(j),1}$ the sum of King's graphs [13, Sect. 3.3] with the source vertex [13, (3.20)] and the free-boundary mass counterterm of [14, p. 330]. The constants $\ln N_j$, $\ln N_j(0)$ and the field- and source-free part of $P^{(j),1}$ belong to $v_{k,j}$. The remaining pieces, each compared across the diagonal pair on $\operatorname{supp}\chi_j$, are:

- (B1) the photon quadratic form, $\tfrac{1}{2}\langle A_j, (\Delta_{\mathrm{ph}}^{(j)} - \Delta_{\mathrm{ph}}^{\prime(j+1)}) A_j\rangle$;
- (B2) the scalar quadratic form $\tfrac{1}{2}\langle \phi_j, \Delta^{(j)}(A^{(j)})\phi_j\rangle$;
- (B3) the background determinant $-\ln[Z_j(A^{(j)})/Z_j(0)]$;
- (B4) the graphs of $P^{(j),1}$ containing a curvature vertex;
- (B5) the source-free, field-dependent graphs of $P^{(j),1}$ localised away from the faces (zone I, appendix B.5);
- (B5′) the same, localised at the faces (zone B);
- (B6) the members of (B3)–(B5′) containing King's block vertices [13, (3.25)–(3.26)].

Proposition 5.5 follows from proposition B.2, corollaries B.8 and B.19, and theorems B.37, B.51 and B.58 and the position sum lemma B.42, assembled in appendix B.10.

B.2. **The photon quadratic form.** Let Ω be a cube whose faces lie between lattice sites on block planes at every scale $\le j$ (this is C_J: its side $2MJ$ is an integer), 2Ω the doubled torus, W the group generated by the reflections of 2Ω in the faces of Ω, and E_f, E_c the even extensions of fine and coarse fields from Ω to 2Ω. King's photon action [14, (2.5)] is componentwise, and the Neumann condition of definition 2.2 keeps the gradient pairs inside Ω: each component is a scalar field with the Neumann (induced-subgraph) Laplacian M_Ω.

Lemma B.1 (even restriction)**.** *Let the photon blocks tile Ω. Then $\Delta^{(i)}_{2\Omega}E_c = E_c\Delta^{(i)}_\Omega$, $G_{2\Omega}E_f = E_fG_\Omega$ and $H_{2\Omega}E_c = E_fH_\Omega$ (photon operators at $A=0$). Hence $\langle A, \Delta^{(i)}_\Omega A\rangle_\Omega = 2^{-3}\langle E_cA, \Delta^{(i)}_{2\Omega}E_cA\rangle_{2\Omega}$, and every bound on the torus photon kernels of* [13, Thm. 3.3, Props. 3.8, 3.10] *on 2Ω holds verbatim for the Neumann photon on Ω. Moreover $H_\Omega 1 = c_i 1$ with $c_i = a_i/(a_i+\mu_i^2)$.*

Proof. The Neumann Laplacian on Ω is the restriction of the torus Laplacian on 2Ω to W-even fields (the reflection planes lie between sites), so $M_{2\Omega}E_f = E_fM_\Omega$. Since W maps blocks to blocks, $Q_{2\Omega}E_f = E_cQ_\Omega$ and $Q^*_{2\Omega}E_c = E_fQ^*_\Omega$ (the adjoint of a W-covariant uniform block average is the uniform block extension, which maps even fields to even fields). Hence $(M_{2\Omega} + aQ^*Q)E_f = E_f(M_\Omega + aQ^*_\Omega Q_\Omega)$; inverting on the invariant even subspace gives $G_{2\Omega}E_f = E_fG_\Omega$, then $H_{2\Omega}E_c = aG_{2\Omega}Q^*E_c = aG_{2\Omega}E_fQ^*_\Omega = E_fH_\Omega$ and $\Delta_{2\Omega}E_c = aE_c - aQE_fH_\Omega = E_c(a - aQ_\Omega H_\Omega)$. The quadratic-form identity follows from $E_c^*E_c = 2^3$ on even fields. Constants are Neumann-harmonic and $Q_\Omega 1 = 1$, which gives $H_\Omega 1 = c_i 1$. If the blocks do not tile Ω the identity fails at the faces; the tiling is part of the definition of the flow (section 5.1), and the state does not depend on the flow. □

Proposition B.2 ((B1))**.** *On* $\operatorname{supp}\chi_j$, $\left|\langle A_j, (\Delta^{(j)}_{\mathrm{ph}} - \Delta'^{(j+1)}_{\mathrm{ph}})A_j\rangle\right| \le C5^{-2j}p^2(\mu_0\eta)^{-2}\eta^{-3}|C_J|$.

Proof. [13, Prop. 3.10] bounds the difference of the Fourier symbols of $\Delta^{(j)}$ and $\Delta'^{(j+1)}$ on any torus that is a union of blocks at every scale by $C5^{-2j}$ relative to $\Delta^{(j)}(p) \le a_j$; its proof [13, Sect. 4, Lemma 4.2] is the $f=1$ case of proposition B.10(a) below. Apply it on $2C_J$, take the supremum over the momenta of $2C_J$, and use lemma B.1 with $\|E_cA_j\|^2 = 2^3\|A_j\|^2$ and $|A_j| \le Cp(\mu_0\eta)^{-1}$ on $\operatorname{supp}\chi_j$ [13, (3.2)]; the number of coarse sites is $\eta^{-3}|C_J|$. This is [13, (3.92)] with $|T|$ replaced by $|C_J|$. □

B.3. **The scalar quadratic form.**

Lemma B.3 (the form is a two-leg tree)**.** *For every bond field A and every ϕ_j on X_j,*

$$\langle\phi_j, \Delta^{(j)}(A)\phi_j\rangle = a_j\|\phi_j\|^2 - a_j\langle\phi_j, Q_j(A)\phi^{(j)}\rangle, \qquad \phi^{(j)} := H_j(A)\phi_j = a_jG_j(A)Q_j(A)^*\phi_j.$$

Proof. $\Delta = a_j - a_j^2QGQ^* = a_j - a_jQ(a_jGQ^*)$. □

The second term is King's vertex [13, (3.24)×(3.25)] at $\phi'=0$ with no photon leg, and its external line is $\phi^{(j)}$, [13, (3.28)]. The object has no internal line and no loop; its expansion in the background (proposition B.7) consists of chains, every subgraph of which has non-negative degree, so it needs no counterterm and no face remainder. This is the content of [13, p. 669], "a graph with two external scalar fields".

Lemma B.4 (the photon background)**.** *On* $\operatorname{supp}\chi_j$*: (i) $|A^{(j)}|_\infty \le C_Hp(\mu_0\eta)^{-1}$; (ii) $|dA^{(j)}|_\infty \le C_Fp$ and $[dA^{(j)}]_\alpha \le C_Fp$, hence $|F(A^{(j)})|_\infty \le 2C_Fp$; (iii) $A'^{(j+1)}$ and $A^{(j)}$ agree at the common fine sites up to $C5^{-\gamma j}p(\mu_0\eta)^{-1}$ in sup norm and $C5^{-\gamma j}p$ in the first difference. In the units of depth $i \le j$, $\varpi_i \le (l_i/\eta)^2\varpi_j$ and $\varpi_j \le C_Fe\eta^{1/2}p$.*

Proof. (i) is [13, (3.7)] for H^{ph} with $f = Q^*A_j$. (ii): $H1 = c_j1$ (lemma B.1), so $dHf(x) = dH(f - f(z_x))(x)$; the kernel of dH decays like $Ce^{-\delta|x-z|}$ by [13, (3.7)], $|f(z) - f(z_x)| \le |dA_j|_\infty|z - z_x|_1$ with $|dA_j| \le p$ by [13, (3.2)]; the Hölder part follows from [13, (3.8)] in the same way. (iii) is [13, (3.71)] applied to A_j minus a constant, with the constant identity. On C_J use lemma B.1; across a reflection plane the even extension has zero difference, so $|dE_cA_j| = |dA_j|$. The scaling of ϖ is the definition. □

Lemma B.5 (block coercivity)**.** *Let X be a lattice box with any Neumann or Dirichlet scalar condition, or a torus, tiled by unit blocks Δ, with probability weights of density $\le \varrho$ and staircases of length $\le C_\Gamma$ inside the blocks. Put $C_P := \frac{1}{4}$, $c_0 := [\max\{(2+4(\varrho+1)^2)C_P, 4/a\}]^{-1}$ and $\nu_0 := (c_0/(16(3+aC_\Gamma^2\varrho)))^{1/2}$. If $e_i|F(C)|_\infty \le \nu_0$, then $D_C^*D_C + aQ(C)^*Q(C) \ge c_0/4$; hence $M(C) \ge c_0/8$ whenever $|m_i^2| \le c_0/8$.*

Proof. Drop the bonds between blocks and the Dirichlet boundary bonds (both non-negative): the form is $\ge \sum_\Delta q_\Delta$ with $q_\Delta(\psi) = \sum_{b\in\Delta} 5^{-i}|U(C_b)\psi(b_+) - \psi(b_-)|^2 + a|Q(C)\psi(y_\Delta)|^2$. *Free case.* With $\bar\psi$ the plain block mean and Q^w the weighted mean, $|Q^w\psi - \bar\psi| \le (\varrho+1)\|\psi - \bar\psi\|$, and the discrete Neumann Poincaré inequality on a box of side ≤ 1 has constant $\le C_P$ ($\lambda_2 = 4n^2\sin^2(\pi/2n) \ge 4$); hence $\|\psi\|^2 \le (2+4(\varrho+1)^2)C_P\|d\psi\|^2 + 4|Q^w\psi|^2$, i.e. $q_\Delta^0 \ge c_0\|\psi\|^2$. *Gauge.* q_Δ is gauge invariant. In the comb gauge on the spanning tree of Δ rooted at y_Δ every non-tree bond carries the holonomy of its fundamental cycle, which bounds a surface of at most $2\cdot 5^i$ plaquettes, so $|U(\tilde C_b) - 1| \le 2e_i5^{-i}|F|$, and a staircase picks up at most $2C_\Gamma e_i|F|$. Hence $\|D_C\psi\|^2 \ge \frac{1}{2}\|d\tilde\psi\|^2 - 12e_i^2|F|^2\|\psi\|^2$ and $|Q\psi|^2 \ge \frac{1}{2}|Q^w\tilde\psi|^2 - 4C_\Gamma^2\varrho e_i^2|F|^2\|\psi\|^2$, so $q_\Delta \ge (c_0/2 - 4(3+aC_\Gamma^2\varrho)e_i^2|F|^2)\|\psi\|^2 \ge \frac{c_0}{4}\|\psi\|^2$. □

Lemma B.6 (one interpolation step)**.** *Let $C = B + A'$, $V := M(B) - M(C)$, $\epsilon_1 := \sqrt{3}e_i|A'|_\infty$, $\epsilon_2 := C_\Gamma\varrho^{1/2}e_i|A'|_\infty$, $c' := c_0/8$ and $D := G(C)^{1/2}VG(C)^{1/2}$. Then $\|D\| \le \theta := 2\sqrt{2}(\epsilon_1 + \sqrt{a}\epsilon_2)c'^{-1/2} + (\epsilon_1^2 + a\epsilon_2^2)c'^{-1}$.*

Proof. With $E := D_B - D_C$ and $F_Q := Q(B) - Q(C)$: $V = D_C^*E + E^*D_C + E^*E + a(Q_C^*F_Q + F_Q^*Q_C + F_Q^*F_Q)$, $\|E\| \le \epsilon_1$ (since $|1 - U(A'_b)| \le e_i5^{-i}|A'_b|$) and $\|F_Q\| \le \epsilon_2$ (staircase phases and Cauchy–Schwarz). For $\psi = G^{1/2}\varphi$, $\|D_C\psi\|^2 + a\|Q_C\psi\|^2 = \langle\psi, (M - m_i^2)\psi\rangle \le 2\|\varphi\|^2$ and $\|\psi\|^2 \le \|\varphi\|^2/c'$; insert into $|\langle\psi, V\psi\rangle|$. □

Proposition B.7 (convergence of the scalar effective operator)**.** *On $\operatorname{supp}\chi_j$, for $1 \le i \le j$, with $\Delta_k^{(i)}(A_k^{(j)})$ and $\Delta_{k+1}^{\prime(i+1)}(A_{k+1}^{\prime(j+1)})$ on the common lattice X_i, $\gamma' := \min(\gamma, \alpha)$ and every $\varsigma > 0$,*

$$\delta_\Delta(i) := \|\Delta_k^{(i)}(A) - \Delta_{k+1}^{\prime(i+1)}(A')\|_{\ell^2(X_i)} + \|\Delta_k^{(i)}(0) - \Delta_{k+1}^{\prime(i+1)}(0)\|_{\ell^2(X_i)} \le C_\varsigma p^c(\mu_0\eta)^{-1}(5^{-\gamma' i} + \eta^\varsigma).$$

The same bound holds for the s-derivatives of order ≤ 2 of $K_i(s\tilde A)$ in the gauge of lemma B.17, with weights $e^{-\delta'|\cdot - x|}$.

Proof. Inputs. The zero-background line bounds of the f_D flow on C_J^K and on boxes with faces on block planes: the slice bounds [13, (3.63)–(3.65)] with decay and Hölder clauses, the slice convergence [13, (3.73)–(3.75)] and the external-line convergence [13, (3.71), (3.74)] at $n = 1$. On the free lattice they are proposition B.9; on the cube they follow by the odd image sum of proposition B.10(c) (the image terms obey the free bounds with decay in $|x - gy|$). *Step 1 (Schur).* It suffices to show $|\delta\Delta(y, y')| \le C\theta_i e^{-\delta|y-y'|}$ with θ_i the right side. *Step 2 (gauge).* Take $B := A_k(x_0)$ at a common coarse site x_0 next to y, the same in both models, and $\lambda(z) := B\cdot(z - x_0)$; on C_J this is a global gauge transformation, under which the Dirichlet condition and the weights are invariant. Then $\Delta(A)(y, y') = e^{eq\lambda(y)}\Delta(\tilde A)(y, y')e^{-eq\lambda(y')}$ with $\tilde A := A - B$, $|\tilde A(z)| \le C_F p|z - x_0|$ (lemma B.4(ii), [13, (3.45)]); the phases of the two models agree up to $e_i|B - B'||y - y'| \le C5^{-\gamma j}p^2(\mu_0\eta)^{-1}$ by lemma B.4(iii). *Step 3 (expansion).* Taylor-expand $\Delta(s\tilde A)(y, y')$ at $s = 0$ to order n_T, using the exact algebra [13, (3.47)–(3.53)] at $B = 0$. The coefficient of s^m is a finite sum of zero-background chains $Q^{(e_0)}GV_1G\cdots V_rGQ^{(e_r)*}$, V_i the vertices of [13, (3.53)] with $\tilde A$-legs, $Q^{(e)} \in \{Q(0), F_2(\tilde A, 0)\}$. The remainder has background lines $G(s\tilde A)$, whose Combes–Thomas decay follows from lemma B.5 (M is finite-range plus a form-small perturbation) and whose vertex sizes follow from lemma B.6; it is $\le C^{n_T}(C\varpi_i p)^{n_T+1}e^{-\delta|y-y'|} \le C\eta^\varsigma$ for $n_T \ge 2\varsigma + 2$, separately in each model. *Step 4 (chains).* Compare one chain across the two models at the common fine sites ($\varepsilon_k\mathbb{Z}^3 \subset \varepsilon_{k+1}\mathbb{Z}^3$) with the discrepancy $d_1(u', u) := \sup|u' - u| + \sup|d'u' - du|$, propagated along the chain: G maps $|\cdot|_\infty$ to $C^{1,\alpha}$; a hopping vertex maps $C^{1,\alpha}$ to C^α with size $C\varpi_i$ and two-scale discrepancy $C\varpi_i[5^{-\alpha i}|u'|_{1,\alpha} + d_1(u', u) + d_1(\tilde A', \tilde A)]$ (because $|d^hv - d^{h/5}v| \le h^\alpha[dv]_\alpha$); lines are compared by [13, (3.73)–(3.75)]; a Q^*-type output is always followed by a line, $GQ^* = a^{-1}H$, and [13, (3.71)] applies, and for F_2^* outputs the staircase-integrated lines [13,

(3.74)]; at the end vertex $Q(\tilde A)$ the staircases nest, $\Gamma'^{(i+1)}_{y,x'} = \Gamma^{(i)}_{y,x} \cup \Gamma_{x,x'}$ (the $(k+1)$-model's i-blocks are the k-model's $(i-1)$-blocks), so $|U(\tilde A'(\Gamma'))-U(\tilde A(\Gamma))| \le e_i(C_\Gamma \sup|\tilde A'-\tilde A|+3\cdot 5^{-i}|\tilde A'|)$; the weights satisfy $\sum_{x'\in B^1(x)} w'(x') = 5^{-3i}$ against $w(x) = 5^{-3i} f_D(x)$, and the resulting f_D-weighted minus plain average over a finest block is $\le C5^{-\alpha i}$ since the integrand is Lipschitz there. After at most n_T steps the chain discrepancy is $\le C_{n_T}(5^{-\gamma i} + 5^{-\alpha i} + d_1(\tilde A', \tilde A))\varpi_i^r p^c e^{-\delta|y-y'|}$. No subgraph degree is used: D^* always acts on a $C^{1,\alpha}$ function, so doubly differentiated lines never occur. *Step 5.* Sum the finitely many chains and add Steps 2–3. □

Corollary B.8 ((B2)). *On* $\operatorname{supp}\chi_j$, $|\mathrm{B2}-\mathrm{B2}'| \le \frac12 \delta_\Delta(j)\|\phi_j\|^2 \le Cp^2\lambda^{-1/2}\eta^{-7/2}|C_J|\,\delta_\Delta(j)$, *with* $\delta_\Delta(j) \le C_\varsigma p^c(\mu_0\eta)^{-1}(5^{-\gamma' j} + \eta^\varsigma)$. *In particular (B2) obeys the bound* (3).

Proof. Proposition B.7 at $i = j$ and the bound $|\phi_j(y)| \le Cp[\lambda\eta]^{-1/4}$ of [13, (3.2)]. □

The zero-background symbol. The following statement, used above through the image sum, is the Dirichlet-weight form of [13, Lemma 4.2, Prop. 3.10] and [14, App. A, (A.10)–(A.11)].

Proposition B.9 (the free f_D flow). *On* $5^{-j}\mathbb{Z}^3$ *at* $A = 0$, *for* $f \in \{f_D, 1\}$, *uniformly in* $j \ge 1$ *and in the mass* $m^2\eta^2 \in [0, m^2]$: *the weighted* ℓ^1 *bounds of* [13, Thm. 3.3, (3.6)–(3.8)] *for* G_j, DG_j *and their Hölder quotients; the slice bounds* [13, Prop. 3.7, (3.62)–(3.65)], *including the slices of index* < 0 *of the finer model; and the one-step comparison bounds* [13, Props. 3.8–3.9, (3.71), (3.73)–(3.75)] *in all their lines.*

Proof. This is King's Fourier argument [13, Sect. 4] with the averaging weight f, as [14, App. A] prescribes. The symbol of $Q_j^*Q_j$ at a coarse momentum z is $\sum_l |\hat f_j(z+l)|^2$ over the 5^{3j} aliases l, where $\hat f_j(q) = \prod_\mu \sigma_j(q_\mu)$ factorises, σ_j being the product over the levels of the one-step symbols; for f_D the finest factor has centre of mass at the geometric centre of the block, so $\hat f_j(q) = e^{-iq\cdot(\frac12,\frac12,\frac12)}R_j(q)$ with R_j real and even. The alias majorant $|\hat f_j(z+l)| \le C\prod_\mu \min(1, |z_\mu|/|(z+l)_\mu|)$ (with the modulus $\frac14$ at the alias centres for f_D, in place of the $f = 1$ value 0) makes $\sum_l |\hat f_j(z+l)|^2 \hat h(z+l)^{-1}$ converge uniformly, $\hat h$ the symbol of $-\Delta + m_j^2$, and $\Delta^{(j)}(z) = (a_j^{-1} + \sum_l |\hat f_j(z+l)|^2(\hat h(z+l) + a_j\sum_{l'}|\hat f_j(z+l')|^2)^{-1})^{-1}$ by [13, (4.2)]; the symbols extend analytically to a strip $|\operatorname{Im} z| \le \kappa$ and are bounded below there, which gives the exponential decay of G_j, $\Delta^{(j)}$ and $C^{(j)}$ (a Combes–Thomas argument on the unit lattice for the operator form). The slices are the factorisation [13, (2.17)], $G_{(i)} = a_i^2\rho_i^{-4}(G_iQ_i^*)C^{(i)}(Q_iG_i)$, whose outer factors are the profiles H_i bounded in $C^{1,\alpha}$ at scale ρ_i by the same symbol bounds; this gives (3.62)–(3.65). The one-step comparison: $\hat f'_{j+1} = \hat f_j \cdot \sigma_{\text{fine}}$ and $a_{j+1}^{-1} = a^{-1} + a_j^{-1}5^{-2}$, so $|\Delta^{(j)}(z) - \Delta'^{(j+1)}(z)| \le C5^{-2j}\Delta^{(j)}(z)$ for real z (the two symbols differ by one more alias level, whose relative size is $O(5^{-2j})$ because the centres of mass coincide for f_D and move by half a fine spacing for $f = 1$), and $|\Delta^{(j)}(y,y') - \Delta'^{(j+1)}(y,y')| \le C5^{-\gamma' j}e^{-\kappa|y-y'|}$ for $\gamma' < 2$ by a contour shift; (3.71), (3.73)–(3.75) follow factor by factor from the slice factorisation, as in [13, Sect. 4]. □

Proposition B.10 (the Dirichlet symbol on the cube). *(a)* $|\Delta^{(j)}(z) - \Delta'^{(j+1)}(z)| \le C5^{-2j}\Delta^{(j)}(z)$ *for real* z, *hence* $\|\Delta^{(j)} - \Delta'^{(j+1)}\|_{\ell^2(\mathbb{Z}^3)} \le Ca5^{-2j}$. *(b)* $|\Delta^{(j)}(y,y') - \Delta'^{(j+1)}(y,y')| \le C5^{-\gamma' j}e^{-\kappa|y-y'|_1}$ *for every* $\gamma' < 2$. *(c) Let* Ω *be a box with faces on block planes whose Dirichlet planes carry the zeros of* f_D *(in particular* C_J^K*), and* $\Delta^{(j)}(\Omega) := a_j - a_j^2Q_jG_j(\Omega)Q_j^*$ *at* $A = 0$. *Then* $\Delta^{(j)}(\Omega)(y,y') = \sum_{g\in\mathcal{G}} \operatorname{sgn}(g)\Delta^{(j)}(y, gy')$ *over the group* $\mathcal{G}$ *generated by the reflections in the Dirichlet planes,* $\|\Delta^{(j)}(\Omega) - \Delta'^{(j+1)}(\Omega)\| \le Ca5^{-2j}$, *and the kernel bound (b) holds with* $\kappa/2$. *Consequently, on* $\operatorname{supp}\chi_j$ *the zero-background part of (B2) differs across the pair by at most* $Ca5^{-2j}p^2\lambda^{-1/2}\eta^{-7/2}|C_J|$.

Proof. (a), (b) are in the proof of proposition B.9. (c): the Dirichlet Laplacian on Ω is the restriction of the free one to $\mathcal{G}$-odd fields, and the f_D blocks are mapped to blocks by $\mathcal{G}$ (the Dirichlet planes carry the weight's zeros, which is what the choice of f_D achieves [14, (3.8)–(3.9)]), so $E_c\Delta^{(j)}(\Omega) = \Delta^{(j)}E_c$ with E_c the coarse odd extension, exactly as in lemma B.1 with signs; this is the image sum, absolutely convergent by (b). For the operator norm periodise to

the doubled box: $\Delta^{(j)}(\Omega)$ is the restriction of the torus operator to odd fields, $\langle\psi,\Delta^{(j)}(\Omega)\psi\rangle = 2^{-3}\langle E_c\psi,\Delta^{(j)}_{2\Omega}E_c\psi\rangle$ with $\|E_c\psi\|^2 = 2^3\|\psi\|^2$, and the difference has norm $\le \sup_z|\Delta^{(j)}(z) - \Delta'^{(j+1)}(z)|$. The last claim is corollary B.8's arithmetic with δ_Δ replaced by $Ca5^{-2j}$. □

B.4. **The background determinant.** Write $A := A^{(j)}$ and, with $R := 1 + \lfloor(\mu_0\eta)^{-1}\rfloor$, $C_r := (r/R)A$, $B_r := C_{r-1}$, $V_r := M(B_r) - M(C_r)$, $D_r := G(C_r)^{1/2}V_rG(C_r)^{1/2}$ $(1 \le r \le R)$, King's interpolation [13, (3.93)]. Determinants are real determinants on $\mathbb{R}^N$-valued fields.

Lemma B.11 (the identities [13], (3.94)–(3.95))**.** *On every finite domain,* $\ln[Z_j(A)/Z_j(0)] = \frac12\ln\det[G_j(A)G_j(0)^{-1}] = \frac12\sum_{r=1}^R\ln\det(I+D_r)$.

Proof. The normalising prefactor of [13, (2.15)] does not depend on A, so the ratio of Gaussian integrals is $\det(G(A)/G(0))^{1/2}$; $G(C)G(B)^{-1} = I+G(C)V$ and $\det(I+XY) = \det(I+YX)$. □

Lemma B.12 (the interpolation steps are small)**.** *There is $\eta_0(e,\mu_0,m,\lambda)$ such that for $\eta\le\eta_0$, on* $\operatorname{supp}\chi_j$, $\|D_r\| \le \theta \le Ce\eta^{1/2}p \le \frac12$ *for every r. For symmetric D with $\|D\|\le\theta<1$ and $n\ge1$,*

$$\Big|\operatorname{Tr}\ln(I+D) - \sum_{q=1}^{n}\frac{(-1)^{q-1}}{q}\operatorname{Tr}D^q\Big| \le \frac{\operatorname{Tr}|D|^{n+1}}{(n+1)(1-\theta)}.$$

Proof. $|A'| = |A|/R \le C_H5p$ by lemma B.4(i) and $|F(C_r)| \le |F(A)| \le 2C_Fp$ by (ii), so lemma B.5 applies for $\eta\le\eta_0$ and lemma B.6 gives $\|D_r\|\le\theta$. The series bound: diagonalise; for $|x|\le\theta$, $|\ln(1+x) - S_n(x)| \le \sum_{q>n}|x|^q/q$. (The one-sided inequality $\ln(I+D)\ge S_n(D)$ for even n printed in [13, (3.96)] fails when D has a negative eigenvalue, e.g. $\ln\frac12 < S_2(-\frac12)$; the two-sided form above is what is used.) □

Lemma B.13 (the remainder of the series)**.** *Let* $\mathfrak{q} > 3$ *(King uses $\mathfrak{q} = 2\bar n+1 = 27$). Under lemma B.12, with ϵ_1,ϵ_2 as in lemma B.6,*

$$\operatorname{Tr}|D_r|^{\mathfrak{q}} \le C_{\mathfrak{q}}\big[(\epsilon_1+\sqrt{a}\epsilon_2)(1+\epsilon_1+\sqrt{a}\epsilon_2)\big]^{\mathfrak{q}}N|X_j|$$

uniformly in the fine spacing, $|X_j|$ the number of unit sites; hence, for both members of the pair, $\sum_{r\le R}|\mathrm{Rem}_r| \le CR(Ce\eta^{1/2}p)^{27}N\eta^{-3}|C_J| \le C\eta^{\sigma+1}|C_J|$ *for every $\sigma\le8$. This is the remainder line of* [13, (3.98)]*, on the cube and on the torus.*

Proof. (1) Schatten–Hölder, $\|XYZ\|_{S_{\mathfrak{q}}} \le \|X\|\|Y\|\|Z\|_{S_{\mathfrak{q}}}$ [22, Sect. VI.6], with the decomposition of V in lemma B.6 and $\|G^{1/2}D_C^*\| \le \sqrt2$, $\|\sqrt{a}Q_CG^{1/2}\| \le \sqrt2$, $\|G^{1/2}\| \le c'^{-1/2}$, gives $\|D\|_{S_{\mathfrak{q}}} \le C(\epsilon_1+\sqrt{a}\epsilon_2)(1+\epsilon_1+\sqrt{a}\epsilon_2)\|G^{1/2}\|_{S_{\mathfrak{q}}}$ and $\|G^{1/2}\|^{\mathfrak{q}}_{S_{\mathfrak{q}}} = \operatorname{Tr}G^{\mathfrak{q}/2}$. (2) $M(C)\ge c'$ and $M(C)\ge H_C + m_i^2$ give $G \le 2(H_C+c'/2)^{-1}$, so $\operatorname{Tr}G^s \le 2^s\operatorname{Tr}(H_C+c'/2)^{-s}$ by operator monotonicity. (3) Diamagnetic bound: the Trotter product with entrywise domination (the diagonal parts of H_C and H_0 agree, also with the Dirichlet bonds) gives $\|e^{-tH_C}(x,x)\| \le e^{-tH_0}(x,x)$ [6], hence $\operatorname{Tr}(H_C+c)^{-s} \le N\operatorname{Tr}(H_0+c)^{-s}$. (4) The eigenvalues of H_0 on a box of sides $l_\mu\ge1$ at spacing h are $\ge\sum_\mu 4n_\mu^2/l_\mu^2$, so $\operatorname{Tr}(H_0+c)^{-s} \le C_{s,c}\prod_\mu l_\mu = C|X_j|$ for $s>3/2$, uniformly in h. (5) Take $s = \mathfrak{q}/2$ and apply lemma B.12 with $n=26$. □

Remark B.14 (the middle line of [13], (3.98)). The per-order bound printed there, $q^{-1}|\operatorname{Tr}D_r^q - \operatorname{Tr}D_r'^q| \le C(5^{-\gamma j}\eta^{-\beta}+\eta^{\sigma+1})|T|$, does not hold for $q=2$. For a constant A_j on the torus (which lies in $\operatorname{supp}\chi_j$) and a fixed block scale, a Fourier computation on each eigenplane of q gives $-\frac12\operatorname{Tr}D_r^2 = -\frac12c_*\delta^2|X_j|5^j + O(|X_j|)$, $\delta = e_j|A'|$, $c_* = \frac{4}{(2\pi)^3}\int_{[-\pi,\pi]^3}\sin^2u_1(\sum_\mu 4\sin^2\frac{u_\mu}{2})^{-2}d^3u = 0.1712\ldots$ (the resolvent identity $G = G_m - G_maQ^*QG$ shows that the block part, of rank $\le 3N|X_j|$, contributes $O(|X_j|)$), so the difference across the pair grows like 5^j; the $q=1$ term diverges with the opposite sign. Only the sum over q converges. The argument below therefore does not expand in q at a fixed background; it uses the scale decomposition of the determinant, in which each scale contributes $O(e_i^2|dA_i|^2)$ per site without any cancellation between orders.

Lemma B.15 (two exact splittings)**.** *(i) If* $M_0(A) > 0$*, then* $\ln\det[G_j(A)G_j(0)^{-1}] = \ln\det{}_{X_j}[\Delta^{(j)}(A)\Delta^{(j)}(0)^{-1}]$ $[\ln\det M_0(A) - \ln\det M_0(0)]$*. (ii) With* $\Delta^{(0)} := M_0$ *and* $C^{(i)} := (\Delta^{(i)} + al_{i+1}^{-2}Q^*Q)^{-1}$ *in physical units,*

$$\ln\det G_j(A) - \ln\det G_j(0) = \sum_{i=0}^{j-1}\big[\ln\det C^{(i)}(A) - \ln\det C^{(i)}(0)\big],$$

and the Schur complement $al_{i+1}^{-2} - a^2 l_{i+1}^{-4} QC^{(i)}Q^*$ *equals King's* $\Delta^{(i+1)}$ *of* [13, (2.14)]*, with* $a_{i+1}^{-1} = a^{-1} + a_i^{-1}5^{-2}$.

Proof. (i) $\det M = \det M_0 \det(I_c + aQM_0^{-1}Q^*)$, and Woodbury gives $(I_c + aQM_0^{-1}Q^*)^{-1} = I - aQGQ^* = \Delta/a$. (ii) Sylvester and Woodbury at each step give $\det C^{(i)} = \det\Delta_{\mathrm{rec}}^{(i+1)}(al_{i+1}^{-2})^{-\dim}/\det\Delta^{(i)}$, which telescopes to $\det\Delta_{\mathrm{rec}}^{(j)}/\det M_0$ up to an A-free constant; compare with (i). Integrating out ϕ_i in the composition of the Gaussian smearings with variances $a_i^{-1}l_i^2$ and $a^{-1}l_{i+1}^2$ gives the variance $a_{i+1}^{-1}l_{i+1}^2$, because $QQ^* = I$ for the plain one-step average and the averages compose, $QQ_i = Q_{i+1}$, the staircases being hierarchical; f_D enters only Q_1, so the steps $i \geq 1$ are plain [14, (3.2)]. □

Lemma B.16 (coarse coercivity and decay at every scale)**.** *In the units of depth* i*, if* $e_i|F_i|\cdot 25 \leq \nu_0$ *and* $|m_i^2| \leq c_0/2$ *(constants of lemma B.5 at block side 5), then* $c_K \leq K_i(A) \leq a_i + a5^{-2}$ *with* $c_K > 0$ *independent of* i *and of the fine spacing. Moreover (Combes–Thomas)* $\|e^w G_i e^{-w}\| \leq 2/c'$ *and* $\|e^w(K_i + t)^{-1}e^{-w}\| \leq 2/(c_K + t)$ *for* $|\nabla w| \leq \delta_{CT}$*, uniformly in* $t \geq 0$.

Proof. $\langle\psi, \Delta^{(i)}\psi\rangle = \inf_\varphi[\langle\varphi, M_0\varphi\rangle + a_i\|\psi - Q_i\varphi\|^2]$; with the minimiser φ and $r := \psi - Q_i\varphi$, $\langle\psi, K_i\psi\rangle = \langle\varphi, M_0\varphi\rangle + a_i\|r\|^2 + a5^{-2}\|Q\psi\|^2$. Lemma B.5 for the $(i+1)$-step average $Q_{i+1} = QQ_i$ at block side 5 gives $c_0\|\varphi\|^2 \leq \|D\varphi\|^2 + a'\|Q_{i+1}\varphi\|^2$; with $\|Q_{i+1}\varphi\|^2 \leq 2\|Q\psi\|^2 + 2\|r\|^2$, $\|D\varphi\|^2 \leq \langle\varphi, M_0\varphi\rangle + |m_i^2|\|\varphi\|^2$ and $\|\psi\|^2 \leq 2\varrho\|\varphi\|^2 + 2\|r\|^2$ this gives $\|\psi\|^2 \leq C_K\langle\psi, K_i\psi\rangle$; the upper bound is $\|Q\| \leq 1$. Combes–Thomas: M has range ≤ 1, and $e^wMe^{-w} - M$ is form-small, $|\langle\psi, B\psi\rangle| \leq C\delta_{CT}(\|\psi\|\|D\psi\| + \|\psi\|^2)$; combine with the lower bounds. □

Lemma B.17 (per-site bound in a local gauge)**.** *For* $x \in X_i$ *let* $g_x(A) := \mathrm{Tr}_N[\ln K_i(A)](x,x)$*, so that* $\ln\det C^{(i)} = -\sum_x g_x$*. Then* $|g_x(A) - g_x(0)| \leq CN(e_i|dA_i|_{\mathrm{loc},x})^2$*, where* $|\cdot|_{\mathrm{loc},x}$ *is the supremum weighted by* $e^{-\delta'|\cdot - x|}$.

Proof. (1) Gauge covariance of K_i (all staircases transform at their endpoints) gives $g_x(A) = g_x(\tilde A_x)$ with $\tilde A_x := A - A(x)$, a global gauge transformation on C_J, and $|\tilde A_x(z)| \leq |dA|_\infty|z - x|_1$. (2) Let P be orthogonal with $PqP^{-1} = -q$. Then $K_i(-A) = PK_i(A)P^{-1}$, so $g_x(s\tilde A)$ is even in s and $\partial_s g_x(0) = 0$. (3) $g_x(\tilde A) - g_x(0) = \int_0^1(1-s)\partial_s^2 g_x(s\tilde A)\,ds$ with $\partial_s^2 \ln K = \int_0^\infty[RK''R - 2RK'RK'R]\,dt$, $R = (K+t)^{-1}$. (4) With $w := \delta_{CT}|\cdot - x|$, $|[RK''R](x,x)| \leq \|e^wRe^{-w}\|^2\|e^{-w}K''e^{-w}\|$; K', K'' are fine-level chains with lines $G_i(s\tilde A)$ (lemma B.16) and vertices linear or quadratic in $\tilde A$, whose growing coefficients $|\tilde A_x(z)| \leq |dA||z - x|$ are absorbed by e^{-w} ($\sum_z e^{-2\delta_{CT}|z-x|}|z-x|^2 \leq C$), so $\|e^{-w}K''e^{-w}\| + \|e^{-w}K'e^{-w}\|^2 \leq C(e_i|dA|_{\mathrm{loc},x})^2$. (5) The t-integral converges like $\int(c_K + t)^{-2}dt$. □

Theorem B.18 (the background determinant converges)**.** *On* $\mathrm{supp}\,\chi_j$*, with* $F_{\mathrm{ph}} := \sup|dA^{(j)}|$ *in physical units (*$\leq Cp\eta^{-3/2}$ *by lemma B.4):*

(i) $|\ln[Z_j(A)/Z_j(0)]| \leq CNe^2F_{\mathrm{ph}}^2|C_J|\sum_{i<j}l_i \leq CNe^2p^2\eta^{-2}|C_J|$;

(ii) $\big|\ln[Z_j(A)/Z_j(0)] - \ln[Z'_{j+1}(A')/Z'_{j+1}(0)]\big| \leq CNe^2F_{\mathrm{ph}}^2|C_J|\big[\sum_{i=1}^{j-1}l_i\delta_{\mathrm{conv}}(i) + 3\varepsilon_k + \eta^\varsigma\big]$ *with* $\delta_{\mathrm{conv}}(i) \leq Cp^c(\mu_0\eta)^{-1}(5^{-\gamma' i} + \eta^\varsigma)$ *from proposition B.7; hence the left side is* $\leq CNe^2p^{c+2}(\mu_0\eta)^{-1}\eta^{-2}(5^{-\gamma' j} + 5^{-j} + \eta^\varsigma)|C_J|$.

Proof. (1) Lemma B.15(ii) writes both members as sums over scales. (2) Pair the k-model's step i with the $(k+1)$-model's step $i+1$ (same physical scale, same coarse lattice X_i); for $i \geq 1$ both one-step averages are plain. (3) The unmatched steps, the k-model's step 0 (weight f_D) against the $(k+1)$-model's step 1 (plain), and the $(k+1)$-model's finest step 0, are bounded by lemma B.17 alone, each by $CNe^2F_{\mathrm{ph}}^2|C_J|\varepsilon_k$. (4) Lemma B.17 per site, times $\#X_i = |C_J|l_i^{-3}$ and $(e_i|dA_i|)^2 =$

$e^2 l_i^4 F_{\mathrm{ph}}^2$, gives (i). (5) For (ii) compare per site: with the expansion of proposition B.7, Step 3, in s (order n_T, remainder bounded separately), the difference of $\partial_s^2 g_x$ between the two members is $\le CN(e_i|dA_i|)^2\delta_{\mathrm{conv}}(i)$; integrate in s and sum over x and i, using $\sum_i l_i 5^{-\gamma' i} \le C\eta 5^{-\gamma' j}$. □

No Ward identity, renormalisation or expansion in orders enters; lemma B.17 is uniform up to the Dirichlet faces, since the Dirichlet condition only adds non-negative bonds, so no face remainder arises for (B3).

Corollary B.19 ((B3)). *On* $\operatorname{supp}\chi_j$, $|\mathrm{B3}-\mathrm{B3}'|$ *obeys the bound of theorem B.18(ii), which is of the form (3).*

B.5. **The graphs: localisation, zones, and the reduction away from the faces.** From here on everything is in the unit units of depth j: the live sites are $T_\eta := X_j$, the fine spacing is 5^{-j}, F denotes the scalar Dirichlet planes of ∂T_η, and $t(x) := \operatorname{dist}(x,F) \ge 5^{-j}$ for a live site. King localises every graph of $P^{(j),1}$ [13, p. 661]: $\square$ is the smallest union of large blocks (side M) containing its vertices, $\square'' \subset \square'$ the smallest boxes containing the large blocks within p, resp. $2p$, of $\square$, truncated at ∂T_η; $\Omega := \square' \cap T_\eta$ is a rectangular box with faces on block planes at every level (this is where the integer side $2MJ$ enters), and the internal lines are the slices $G_{(i)}(\square',0)$ of the f_D flow with the Dirichlet condition on all of $\partial\Omega$, the photon slices with the Neumann condition, and the lines $\delta G_j(\square', A^{(j)})$, external in King's sense. A localisation is in *zone I* if $\square' \cap F = \emptyset$ and in *zone B* otherwise; the faces of Ω not on F lie at distance $\ge 2p$ from $\square$. Lines obey King's slice bounds [13, (3.63)–(3.65)] on Ω up to and including the faces (proposition B.10(c) for the scalar, lemma B.1 for the photon, both image sums of free kernels), and the external objects obey [14, (5.2)–(5.3)]: this is King's Prop. 5.1 on boxes whose faces lie on block planes, which follows from the image sums and a bootstrap on the sup of $|u| + |D_A u|$ with the commutator localisation of [14, Lemma 5.3], with no a priori sup bound. The range of j is $0 \le j \le k - J_0$, $5^{-J_0} \le \delta_*$, with $\delta_*^{-1} > 3p(\delta_*) + 4M + 1$.

Lemma B.20 (zones). *(i) Every localisation with* $R := \operatorname{dist}(\{y\},\{z\}) \le p$ *that contains a source vertex lies in zone I. (ii) Every zone-B localisation containing a source vertex has* $R \ge \eta^{-1} - 2p - 4M \ge \eta^{-1}/2$. *(iii) Every source vertex* s *has* $t(s) \ge \eta^{-1}/2 \ge 1$ *and* $\operatorname{dist}(s, \partial\square' \setminus F) \ge 2p - \sqrt{3}$.

Proof. The sources are supported at continuum ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_J$, i.e. $\ge \eta^{-1}$ unit lengths from ∂T_η; after the summation by parts of lemma B.22 the summed vertex sits on bonds at distance $\le 1-\varepsilon$ from nothing (its weight vanishes within $1-\varepsilon$ of the faces). In (i) every vertex is within p of the source vertex, $\square$ within $p+M$, $\square'$ within $3p+3M < \eta^{-1}$. (ii), (iii) are the same arithmetic. □

Lemma B.21 (zone I is the free f_D flow). *Let* $\tilde H$ *be a graph localised in zone I and* $\tilde H^{\mathrm{free}}$ *the same graph with every internal scalar slice replaced by the free* f_D *slice on* $5^{-j}\mathbb{Z}^3$ *and every internal photon slice by its free slice. Then* $E(\tilde H) = E(\tilde H^{\mathrm{free}}) + \mathcal{R}$, *where every term of* $\mathcal{R}$ *carries one image line of size* $\le C\rho_i^{-1} e^{-4\delta_0 p/\rho_i}$ *(with derivatives and Hölder quotients), so that an image line has any prescribed degree and* $|\mathcal{R}| \le Ce^{-\delta_0 p/2} \times$ *(the bound of* $\tilde H^{\mathrm{free}}$ *with that line's degree raised). Since* $p(\eta) = b_0(1+\ln\eta^{-1})^p$ *with* $p > 1$, $e^{-\delta_0 p/2} \le C\eta^{3+\sigma+1/2}$. *The free-boundary counterterm of* [14, p. 330] *renormalises* $\tilde H^{\mathrm{free}}$ *exactly, because the full free covariance does not depend on the averaging weight.*

Proof. $G_{(i)}(\square',0) - G_{(i)}(0)$ is the signed sum over the non-identity elements g of the reflection group of Ω of free slices $G_{(i)}(x, gy)$ (proposition B.10(c)); for $x, y \in \square$ and $\operatorname{dist}(\square, \partial\square') \ge 2p$, $|x - gy| \ge 4p$, and the free slice bounds give the size. The photon slices are the even image sums of lemma B.1, and in zone I $\operatorname{dist}(\square, \partial T_\eta) \ge 2p$ as well. The lines $\delta G_j(\square', A)$ carry $e^{-\delta_0 \operatorname{dist}(x,\partial\square')}$ by [14, (5.3)], also $O(e^{-\delta_0 p})$. Expanding each line as free plus image gives the decomposition. □

The curvature vertex. Let f be a real 2-form of the family, g^ε its centre sampling, and $F(p)$ the plaquette curvature.

Lemma B.22 (the summed curvature vertex). *(a) For every bond field* A *on* C_J, $\sum_p g^\varepsilon(p)F(p) = \sum_b w(b)A_b$ *exactly, with* $w(y,\mu) = \sum_{\nu>\mu} \varepsilon^{-1}[g(p_{\mu\nu}(y)) - g(p_{\mu\nu}(y - \varepsilon e_\nu))] + \sum_{\nu<\mu} \varepsilon^{-1}[g(p_{\nu\mu}(y -$

$\varepsilon e_\nu)) - g(p_{\nu\mu}(y))]$, *no boundary term, and* $w = 0$ *on bonds within* $1-\varepsilon$ *of* ∂C_J. *(b)* $d^*w = 0$, *and* $\sum_b w(b)B_b = 0$ *for constant* B. *(c)* $w(b) = W^\varepsilon_\mu(m_b)$ *with* $W^\varepsilon_\mu(x) = \sum_{\nu\neq\mu}\varepsilon^{-1}[f_{\mu\nu}(x + \frac{\varepsilon}{2}e_\nu) - f_{\mu\nu}(x - \frac{\varepsilon}{2}e_\nu)]$, *the restriction to bond midpoints of a smooth function with* $\|\partial^\alpha W^\varepsilon\|_\infty \le 2\max\|f_{\mu\nu}\|_{C^{|\alpha|+1}}$ *uniformly in* ε; *so* w *has sup and lattice* $(1,\alpha)$*-norms bounded by* $C\max\|f_{\mu\nu}\|_{C^{2,\alpha}}$. *(d) For bonds* b' *at spacing* ε_{k+1} *and* b *at spacing* ε_k *in the same direction with* $|m_{b'} - m_b| \le \sqrt{3}\varepsilon_k$, $|w_{\varepsilon_{k+1}}(b') - w_{\varepsilon_k}(b)| \le (2\sqrt{3}+1)\varepsilon_k \max\|f_{\mu\nu}\|_{C^2}$.

Proof. (a) is a finite rearrangement: each A_b collects the plaquettes containing b with their orientation signs; a plaquette containing a bond of C_J but not lying in C_J has its centre within $\varepsilon/2$ of ∂C_J, where $f = 0$. The Neumann condition never enters. (b) $d^*d^* = 0$ on plaquette functions, $F(B) = 0$. (c) reads off the centres, $f_{\nu\mu} = -f_{\mu\nu}$, and W^ε_μ is a segment average of $\sum_\nu \partial_\nu f_{\mu\nu}$. (d) is (c) and the mean value theorem. □

In King's step-j vertex [13, (3.20)] this gives $\tau\eta^{3/2}\sum_b 5^{-3j}w_\eta(b)(A'_b + A^{(j)}_b)$: one undifferentiated photon end, no explicit power of the spacing, vertex weight $u = \frac{5}{2}$ in the degree count of [13, (3.30)]. By (b) the vertex is invariant under the gauge transformation [13, (3.46)] and the constant shift [13, (3.43)–(3.45)] drops out of it. Graphs with an odd number of curvature vertices and no external field vanish at $A_j = \phi_j = 0$ by the symmetry $(A,\phi) \mapsto (-A,\bar\phi)$.
The degree count on the free f_D flow. Bałaban's renormalised degree count [3, Sect. 3, (3.5)], in the form King uses on the torus [13, Sect. 3.5, (3.77)–(3.78)]: after the rewriting of the divergent subgraphs (Taylor transport of legs, Hölder splits, resummation over orderings up to the lowest external index j_0, and evaluation of the resummed coefficients c_S on the free lattice, where they vanish by the counterterm, by sign pairing, by translation invariance or parity, or by the Ward identity of the vacuum polarisation), every ordered ultraviolet stage of a source-free renormalised family has positive degree, and the infrared counterterm lines obey [13, (3.78)]. We call this (DEG). It is printed for the $f = 1$ flow; the lines of the zone-I graphs are those of the f_D flow, and the transfer is the following.

Lemma B.23 (the two flows differ by a top-scale kernel)**.** *Let* $\Delta G_i := G_i^{f_D} - G_i^1$ *on* $5^{-i}\mathbb{Z}^3$ *at* $A = 0$, *in the units of depth* i. *For* $a,b \in \{0,1\}$ *and every* $\gamma < 1$, $|\partial_x^a\partial_y^b \Delta G_i(x,y)| \le C_\gamma 5^{-\gamma i}e^{-\kappa'|x-y|}$, *and* $\partial_x\Delta G_i(\cdot,y)$ *is* α'*-Hölder with constant* $C_{\alpha'}5^{-si}$, $s = (1-\alpha')/(1+\alpha')$, *times the same decay. In the units of depth* j *the level-*i *kernel is the rescaled one, so* ΔG_i *obeys the slice bounds* [13, (3.63)–(3.65)] *at the index* i *times* $5^{-\gamma i}$*: a line with no short-distance singularity.*

Proof. Both operators have the same $C^{-1} = -\Delta + m_i^2$, so $\Delta G_i = G^{f_D}RG^1$ with $R(u,u') = a_i[1 - w(u)w(u')]\mathbf{1}[B(u) = B(u')]$. Write $1 - w(u)w(u') = (1 - w(u)) + w(u)(1 - w(u'))$; using $\sum_{u\in B(y)} 5^{-3i}G^{f_D}(x,u)w(u) = H^{f_D}(x,y)/a_i$ one gets $\Delta G_i(x,x') = a_i\sum_y D_0(x,y)g_1(y,x') + \sum_y H^{f_D}(x,y)g_2(y,x')$ with $D_0(x,y) := \sum_{u\in B(y)} 5^{-3i}(1-w(u))G^{f_D}(x,u)$, $g_1 := \sum_{u'\in B(y)} 5^{-3i}G^1(u',x')$, $g_2 := \sum_{u'\in B(y)} 5^{-3i}(1-w(u'))G^1(u',x')$. *Dipole lemma:* every unit block is a union of finest cells c of side 5^{1-i} on each of which $\sum_{u\in c}(1-w(u)) = 0$ (since $\sum_{u\in c} w = \prod_\mu \sum_{r=0}^4 w_1(r) = 5^3 = |c|$), so $\sum_{u\in B} 5^{-3i}(1-w)h(u) = \sum_c \sum_{u\in c} 5^{-3i}(1-w(u))[h(u) - h(u_c)]$ is $\le C\sum_c |c|\operatorname{osc}_c h$, and with $|\partial h| \le C(|x-v|+5^{-i})^{-\mathsf{p}}$ this is $\le C5^{-i}$ for $\mathsf{p} = 2$ and $\le C5^{-i}(1+i)$ for $\mathsf{p} = 3$. The free kernel bounds $|\partial^a G| \le C(|x-u|+5^{-i})^{-1-a}e^{-\kappa|x-u|}$ (proposition B.9) then give $|D_0|, |g_2| \le C5^{-i}e^{-\kappa\,\mathrm{dist}}$, $|\partial D_0|, |\partial g_2| \le C5^{-i}(1+i)e^{-\kappa\,\mathrm{dist}}$, $|g_1|, |\partial g_1| \le Ce^{-\kappa\,\mathrm{dist}}$, $|\partial^a H^{f_D}| \le Ce^{-\kappa|x-y|}$; every product contains one dipole factor and the y-sums converge. For the Hölder clause interpolate the cell oscillation of $h(u) = \partial_x G(x,u) - \partial_x G(\tilde x,u)$ between the Hölder bound $C|x-\tilde x|^\alpha r_c^{-2-\alpha}$ and $C5^{1-i}r_c^{-3}$ with $\alpha = (1+\alpha')/2$. □

Theorem B.24 ((DEG) on the free f_D flow)**.** *Every source-free renormalised prefix component* S *of the free* f_D *flow, at zero background, admits the rewriting of* [3, Sect. 3] *with: (i) every resummed coefficient that vanishes exactly for* $f = 1$ *vanishes exactly for* f_D*; (ii)* $c_a^{f_D} = c_a^1 + c_a(M_S)$ *with* $|c_a^1| \le C\rho_{j_0}^{D(S)+|a|}$ *the* $f = 1$ *moment and* $|c_a(M_S)| \le C_\gamma 5^{-\gamma i_0}\rho_{j_0}^{D(S)+|a|}$, i_0 *the depth of* j_0*; (iii) the same remainder shapes as for* $f = 1$. *Hence* (DEG) *holds for the* f_D *flow, with the degree of every stage unchanged, and the infrared half* [13, (3.78)] *holds with the mass-counterterm blob* U^{f_D} *differing from* U^1 *by* $C(\varepsilon_k/\eta)^\gamma\eta^{1-\varsigma}$ *in the units of* [13, (3.78)]. *The* $f = 1$

input of (ii) on the rows whose coefficient vanishes by the zero-momentum gauge covariance (the Ward rows) is theorems B.30 and B.35 of appendix B.6; on every other row it is the member-wise identity named there.

Proof. Resum the lines of S to $G_{j_0}^{f_D}$, write $G^{f_D} = G^1 + \Delta G$ on every line and expand multilinearly: $F_S^{f_D} = F_S^1 + M_S$, where F_S^1 is the inside family of the same class with $f = 1$ lines (the members are the same graphs, since every vertex of a divergent class is weight-free; the counterterms δm^2, E_1 have the same value in both flows; a counterterm piece all of whose lines are G^1 is the $f = 1$ decomposition of the same counterterm), and M_S is the sum over the members and the non-empty sets T of lines carrying ΔG. (i): the exact-zero mechanisms (counterterm per scale assignment, pairing, parity, telescoping of a summable kernel) are identities for every charge-diagonal line kernel when partners carry the same lines, so they hold for the f_D lines directly. (ii): in a term of M_S, treat each ΔG line as one line of the fixed index j_0; by lemma B.23 it obeys the index-j_0 slice bounds times $5^{-\gamma i_0}$. In any ordering the first ΔG line forces the index j_0 on every later line, so the free index sums run over G^1 slices only, and the prefixes that contain a ΔG line enter no geometric sum. Each prefix component γ with $D(\gamma) \le 0$ before the first ΔG line is a source-free, block-free $f = 1$ subgraph whose complete inside family (partners included) is present at fixed outside, because replacing γ by another member of its class does not touch T; renormalise it by the $f = 1$ rewriting, finest first, with the ΔG ends as legs at scale ρ_{j_0} (Lipschitz, with α'-Hölder first difference, which is what the transport and Taylor steps use), debits landing on a ΔG line paid at ρ_{j_0}. The doubly differentiated single line between two current vertices whose other ends carry ΔG is rewritten by the free-lattice identity $\sum D_x D_y G^1(\cdot) = 0$ against a constant, whose transport remainder carries $|\Delta G(\cdot, v_2) - \Delta G(\cdot, v_1)| \le C|v_1 - v_2| 5^{-\gamma i_0} \rho_{j_0}^{-2}$. After this every prefix with a non-trivial index sum has positive degree, the lines obey their slice bounds, the moment factors $|x_i - x|^{a_i}$ are distributed along paths of lines (costing $\rho_{j_0}^{a_i}$ on a ΔG line), and the multiscale bound (theorem B.40(a) with the root held) gives $|c_a(M_S)| \le C \rho_{j_0}^{D(S)+|a|} 5^{-\gamma i_0 |T|}$. (iii) is the same. Infrared half: $U^{f_D} = U^1 -$ (terms with at least one ΔG_j soft line), and each such term is a skeleton term of the $f = 1$ argument in which the soft line obeys the soft-line bounds with the constant $C(\varepsilon_k/\eta)^\gamma$ (lemma B.23 in physical units), which gives the stated difference. □

B.6. **The Ward rows of the degree count.** The source-free, block-free classes of King's expansion with $D \le 0$ are, by order and external profile ([13, (3.30)–(3.31)]: $D = \frac{o}{2} - 3 + \frac{1}{2}(E_S + E_A) + \frac{3}{2} E_{S_D}$ for a member of order o with E_S undifferentiated scalar, E_{S_D} differentiated scalar and E_A photon legs), the vacuum classes, the photon one- and three-point classes, the scalar two-point classes with and without one photon leg, the photon two-point classes of orders e^2, e^4, $e^2\lambda$, and the tree [3, (2.4)]. The resummed coefficient c_S of such a class vanishes, or is coarse-dominated, by one of six mechanisms: the counterterm matched per scale assignment [13, (3.33)–(3.35)]; charge parity; a derivative at the end of a line; the locality of the tadpoles; a reflection; and the zero-momentum gauge covariance of the full fixed-order two-photon function and of the photon insertions into the exchange graph Σ_1. The rows of the last kind, the photon two-point classes (all 24 configurations, with either photon leg an A'-line or an $\tilde{A}$ field) and the scalar two-point classes with one photon inserted into Σ_1 at a vertex or on its scalar line, or into the λ-tadpole loop, are the *Ward rows*. They differ from the others in two respects. Their identity holds for the *whole inside family* with all internal lines resummed and replaced by one kernel, not member by member or at a fixed index assignment; so at a fixed localisation tuple the localisation weights must be common to all members of the family, which under King's rule of one partition of unity per internal vertex [13, (3.36)] they are not (at order e^2 the bubble has two internal vertices and the J_2-loop one, so at a fixed tuple the bubble has no partner, and its weighted zeroth moment grows like ξ^{-1}). And in the $R > p$ branch [13, (3.38)] King cuts the lines longer than $3\sqrt{d}$ by vertex positions [13, p. 666], which the identities sum over. Both are met by a reading of King's localisation and cut on data fixed by the tuple alone; this subsection gives it, on the $f = 1$ flow, and proves the coarse domination $|c_a| \le C \rho_{j_0}^{D(S)+|a|}$ of theorem B.24(ii) on

the Ward rows. Throughout, S is a prefix component of an ordering with $D(S) \le 0$, no source, block or h-vertex, legs Φ_e (external fields or lines of index $\ge j_0$ evaluated as fields) of indices j_e, $j_0 := \min_e j_e$; its *inside family* F_S is the set of all connected block-free subgraphs of the same order and profile occurring in place of S with the same legs at the same outside points, the δm^2-vertex and δm^2-inserted graphs included [3, p. 431]; the *admitted set* A_S is all internal indices $\le j_0$; and for a reference vertex v_0 (the vertex of the coarsest leg) and a multi-index a on the other legs the moments are $c_a(x) := \sum_{\text{internal positions}} 5^{-3j(V-1)} K_{F_S}(x; x_2, \dots) \prod_{e\ge 2} (x_e - x)^{a_e}/a!$, x the position of v_0, internal localisations summed.

Lemma B.25 (resummation)**.** *For fixed outside indices, the sum over all orderings of the lines of S and all index assignments in A_S equals the expression with every line of S and of every member of F_S (the internal lines of δm^2 units included) replaced by the resummed line $G_{j_0} = \sum_{i\le j_0} G_{(i)}$ (resp. $G^{\mathrm{ph}}_{j_0}$). Rescaled by ρ_{j_0} to the lattice $\xi\mathbb{Z}^3$, $\xi := 5^{-j_0}$, let $C^\xi := (-\Delta + 1)^{-1}$ (scalar $\otimes 1$; $\delta_{\mu\nu}$ for the photon) and let F^C_S be the family with every line, counterterm-internal lines included, replaced by C^ξ on $\xi\mathbb{Z}^3$. Then for $|a| \le n := -D(S)$*

$$c_a(F_S) = \rho_{j_0}^{D(S)+|a|} \Big[c_a(F^C_S) + \sum_{\emptyset \ne T} c_a(F_{S,T}) \Big], \qquad |c_a(F_{S,T})| \le C \text{ uniformly in } \xi, x, j_0,$$

where $F_{S,T}$ has the lines of T replaced by $\Delta := G_{j_0} - C^\xi$ (rescaled) or by the periodisation tail on a torus, and the others by C^ξ.

Proof. The first statement is [13, (3.57)–(3.58)] and multilinearity: King's tie rule is replaced by a simultaneous S-first tie-break, which changes nothing in the total. For the second, the resolvent identity $G = C^\xi + G(1 - \tilde m^2 - \tilde a P)C^\xi$ [3, p. 437] ($P = Q^*Q$, $\tilde m^2 = m^2 l^2_{j_0}$, $\tilde a = a_{j_0}$) gives $\Delta = G(1 - \tilde m^2 - \tilde a P)C^\xi$; with $|G(x,y)| \le C|x-y|^{-1}_\xi e^{-\delta|x-y|}$, $|DG| \le C|x-y|^{-2}_\xi e^{-\delta|x-y|}$, $|D_x D_y G| \le C|x-y|^{-3}_\xi e^{-\delta|x-y|}$ (proposition B.9; [13, Thm. 3.3]), the same for C^ξ, and P bounded with unit-range kernel, convolution in $d = 3$ gives $|\Delta| \le C$, $|D\Delta| \le C(1 + \ln_+(1/|x-y|_\xi))$, $|D_x D_y \Delta| \le C|x-y|^{-1}_\xi$, all times $e^{-\delta'|x-y|}$: a Δ-line is a unit-scale kernel, undifferentiated bounded and Lipschitz below scale 1, differentiated two orders better than the G-line (degree $+2$ up to a logarithm absorbed by an ϵ-loss). The replacement is multilinear, the δm^2 partners' internal lines being expanded by the same rule, so every all-C subgraph of $F_{S,T}$ is renormalised inside $F_{S,T}$ by its own partners. On a torus $C^\xi_T = \sum_\gamma C^\xi(\cdot + \gamma)$ and the $\gamma \ne 0$ tail is smooth and $O(e^{-\delta|\gamma|})$, one more Δ-type line. Decompose the C^ξ lines into scales (C^ξ is the $A = 0$, mass-1 instance of [13, (2.13)–(2.17)] on $\xi\mathbb{Z}^3$ with the bounds [13, (3.63)–(3.65)]) and order all lines: a prefix component of an output term is either all-C, hence a proper subgraph of a member of F_S (lower order, or the same order and more legs), whose class is rewritten by the same statement for the C^ξ line system (induction on order, then on the number of legs, which is well founded since $E \le 4$ for every divergent class); or it contains a Δ-line: undifferentiated, it sits at the top scale and enters no fine-scale sum; differentiated, its degree is $\ge D_{\mathrm{plain}} + 2 - \epsilon \ge 1 - \epsilon > 0$, because every non-vanishing non-vacuum source-free connected subgraph has $D \ge -1$ (the $D = -2$ classes are the current tadpole, which vanishes by charge parity, and vacuum graphs, which are not prefix components of a non-vacuum S). So every ordered subgraph of every output term has positive degree, the multiscale bound theorem B.40(a) applies, and the final scale sum is coarse-dominated. □

Lemma B.26 (zero-momentum gauge covariance)**.** *Let the lines be gauge covariant in a background B, $C_B(x,y) \mapsto g(x) C_B(x,y) g(y)^T$ under $B \mapsto B + \nabla\chi$, $g = e^{eq\chi}$, with King's vertices [13, (3.18)–(3.23)] at background B, and on $\xi\mathbb{Z}^3$ let B be constant, $\chi_B(x) = B \cdot x$. For a labelled contraction G with two external scalar fields at x, x_2, $K^G_B(x, x_2) = g(x) K^G_0(x, x_2) g(x_2)^T$, and for G with photon legs only $F^G_B = F^G_0$. Differentiating at $B = 0$ in direction μ:* (a) *with $W(A) := \ln Z(A) - \ln Z(0)$, $Z(A) = \int d\mu(A_{\mathrm{int}}) \int d\mu_{C_0}(\phi) e^{V(\phi, A + A_{\mathrm{int}})}$, $W(A+B) = W(A)$ for constant B on $\xi\mathbb{Z}^3$, hence for the full fixed-order connected two-photon function $\Pi = \partial^2 W/\partial A \partial A|_0$, all its labelled contractions and δm^2-inserted ones included, $\sum_{b_2 \in \mathrm{bonds}_\nu} \Pi_{\mu\nu}(b_1, b_2) = \partial_{A_{b_1}} \partial_{B_\nu} W(A +$*

$B)|_0 = 0$; (b) *for every two-scalar-leg parent G with scalar charge structure,* $\sum_{b\in\mathrm{bonds}_\mu} K^{(G+\mu)}(x,x_2;b) = eq(x-x_2)_\mu K_0^G(x,x_2)$, *so the zeroth moment of the insertion family of G equals $-eq$ times the first moment of G. The insertions are: on a scalar line $-C(\partial_B(-\Delta_B))C$, the vertex J_1; at a vertex $J_m(B)$ through $U(B_b)$ a J_{m+1}-type vertex and through D_B a Q_{m+1}-type vertex; none on photon lines, λ or M.*

Proof. (a) $Z(A+\nabla\chi) = Z(A)$ by the change of variables $\phi \mapsto e^{eq\chi}\phi$, a constant B is $\nabla(B\cdot x)$ on $\xi\mathbb{Z}^3$, and the order-by-order expansion of W in e commutes with the shift because W depends on eA only. (b) Per contraction the value is a contraction of covariant tensors with covariant propagators, the internal phases cancel, $\partial_{B_\mu} g = eqx_\mu g$, and K_0^G commutes with q; the insertion list is Leibniz on C_B and on $J_m(B)$ [13, (3.22)] (Q_m carries no U and is B-independent). □

The identity is used only for F_S^C of lemma B.25, whose lines are full resummed lines replaced by C^ξ (the total-field form); no identity is applied to a fixed ordering or a fixed index assignment. On a torus (a), (b) hold up to the holonomy response $O(e^{-\delta|T|_\xi})$, one more Δ-type term.

Lemma B.27 (reflection)**.** *Let g_s, c be the even scalar and photon kernels C^ξ on $\xi\mathbb{Z}^3$.* (a) *The exchange kernel Σ_1 with both S_D legs on the scalar line and $\mu = \nu$ is $k(y) = -e^2\sum_\mu(\Delta_\mu g_s)(y)c(y)$, $y = x - x_2$, which is even, so its first moment vanishes.* (b) *For the order-e^2 two-photon family (J_1J_1 bubble, J_2 loop, Q_2 loop) with zeroth moment $M^{(0)} = 0$, every first moment $M^{(1)}_{\mu\nu\sigma} := \sum_{b_2}\Pi_{\mu\nu}(b_1,b_2)(b_2-b_1)_\sigma$ vanishes.*

Proof. (a) With $b = (x-\xi e_\mu, x)$, $b' = (x_2 - \xi e_\mu, x_2)$, $\partial_b\partial_{b'} g_s(x-x_2) = -\Delta_\mu g_s(x-x_2)$ and the photon line is $c(x-x_2)$. (b) Under the reflection R_σ the bond (z,μ) goes to $(Rz - \xi e_\sigma\delta_{\mu\sigma}, \mu)$ and $A_\mu \mapsto s_\mu A_\mu$ ($s_\sigma = -1$, else $+1$); the family is the expansion of a reflection-invariant action, so $\Pi_{\mu\nu}(b_1,b_2) = s_\mu s_\nu \Pi_{\mu\nu}(R_\mu b_1, R_\nu b_2)$, hence $M^{(1)}_{\mu\nu\sigma} = s_\mu s_\nu[-M^{(1)}_{\mu\nu\sigma} - \xi(\delta_{\nu\sigma} - \delta_{\mu\sigma})M^{(0)}_{\mu\nu}]$ under R_σ and $= s'_\mu s'_\nu M^{(1)}_{\mu\nu\sigma}$ under R_τ, $\tau \neq \sigma$. With $M^{(0)} = 0$: if $s_\mu s_\nu = +1$ under R_σ then $M^{(1)} = 0$; otherwise exactly one of μ,ν is σ, say $\mu = \sigma \neq \nu$, and R_ν gives $M^{(1)} = -M^{(1)}$. □

The localisation by centres. King localises every graph of $P^{(j),1}$ by inserting $1 = \sum_z \chi_z$ at each internal vertex [13, (3.36)] and summing over the tuple $(\{y_i\},\{z\})$ [13, (3.37)]. We attach the partitions of unity to *centres* instead: (i) one centre to every A' field of a J_m or Q_m vertex (m centres at J_m [13, (3.22)], m at Q_m [13, (3.23)]); (ii) one centre to every other vertex (λ [13, (3.18)], block vertices [13, (3.24)–(3.26)], h-, g- and source vertices). For each centre c insert $1 = \sum_{z_c}\chi_{z_c}(p_c)$, p_c the position of the vertex carrying c; the tuple of a graph is $\tau = (\{y_i\},(z_c)_c)$; □ is the smallest cube of large blocks containing every y_i and every z_c (the centres of the localised counterterm graphs included), and □′, □″, x_0, $B = A_j(x_0)$, $\tilde{A}$ and $R := \mathrm{dist}(\{y_i\},\{z_c\})$ are as in [13, pp. 660–662]. King's index rules [13, p. 667] are kept: a δm^2 graph is localised only if all its integers are $< j$, and an E_1 piece with $m_1 < m$ gets one centre at one vertex connected to $l(m)$. King's printed rule is this rule with one centre per vertex in place of (i).

Lemma B.28 (King's localisation argument with centres)**.** (a) *The only properties of* [13, (3.36)] *used in* [13, Sects. 3.4–3.5] *are: $\sum_z \chi_z = 1$ per inserted factor; $0 \le \chi \le 1$ with support of size $O(1)$ and bounded derivatives; every localised vertex lies within $O(1)$ of a centre in □; and the number of centres is bounded by a function of $\bar{n}$ and the order. The rule by centres has all of them, so* [13, Thm. 3.5, Prop. 3.6, (3.40)–(3.42), (3.43)–(3.57), (3.68), (3.85)] *and the step comparison of* [13, p. 665] *hold verbatim, with constants depending on $\bar{n}$ and d (the centres of one vertex lie within $\frac{4}{3}\sqrt{d}$ of each other, so the tree distance over centres exceeds that over vertices by at most $C(\bar{n})$).* (b) Centre count. *For $S' \in F_S$ with e-order α, λ-order β and $E_{\tilde{A}}$ legs that are $\tilde{A}$ fields, the number of centres is $N(S') = \alpha - E_{\tilde{A}} + \beta$; it is the same for every member of F_S.* (c) *Consequently every graph of a renormalisation orbit carries the same tuple, under a fixed bijection of the centres of S and S', hence the same □, □′, x_0, B, $\tilde{A}$ and the same branch of* [13, (3.38)/(3.39)]*; the same along any sequence of replacements, nested units included.* (d) *In the order-e^2 photon two-point family every centre sits at the S-end of an A'-line leg: no member has an internal photon line or a λ vertex, and two $\tilde{A}$ legs are impossible.*

Proof. (a) Each inserted factor sums to 1, so [13, (3.37)] is an identity for any number of centres; the uses on [13, pp. 659–668] are: the definition of the tree distance and the decay $\exp[-\delta\,\mathrm{dist}]$ (co-located centres cost a bounded factor); the dichotomy and $\sum_{y,z} e^{-\frac{1}{2}\delta R} \le C\eta^{-3}|C_J|$; the cubes; $B = A_j(x_0)$ on $\square'$; the gauge transformation and θ; $|x_0 - z_c| \le Cp$ by the tree decay; [13, (3.68)], the weight of a vertex being a product of χ's in $[0,1]$ supported in a unit cube; the root sum by $|\square'|$; the replacement of $\chi(x')$ by $\chi(x)$ in the step comparison, since products of χ's have bounded derivatives; δG bounds "since $x, y \in \square$", every vertex lying within $\frac{2}{3}\sqrt{d}$ of each of its centres; the counterterm localisation with the index rules; and the fold [13, (3.85)] of a leg's χ into the leg. (b) Each A' field at a J- or Q-vertex carries one factor e_j [13, (3.22)–(3.23)]; each $\tilde{A}$ carries one e_j [13, (3.47)–(3.53)] and is never contracted; each λ vertex carries one λ_j. So $\alpha = \sum_v (a_v + t_v)$ with a_v the A' fields and t_v the $\tilde{A}$ legs at v, $E_{\tilde{A}} = \sum_v t_v$, $\beta = |\Lambda(S')|$, and $N = \sum_v a_v + |\Lambda(S')|$. F_S fixes (α, β) and the legs; every identity used acts within one (α, β), since gauge invariance holds for every value of e and λ and δm^2, E_1 are sums of graphs of definite order [13, (3.33)–(3.35)]. (c) The reduction is graph-local and the listed objects are functions of the tuple. (d) $\alpha = 2$, $\beta = 0$, $E_{A'} + E_{\tilde{A}} = 2$, so there are exactly $E_{A'}$ A' fields, the S-ends of the A'-line legs; a J/Q vertex needs an A' field, and a subgraph of $\tilde{A}$-only vertices is the tree [3, (2.4)], not a two-photon graph. □

Lemma B.29 (frozen weights)**.** *Let S be a Ward-row unit at a fixed tuple, x its reference point, and for a member S' let $W_{S'} = \prod_{c\in\mathrm{legC}} \chi_{z_c}(p_c) \prod_{c\in\mathrm{intC}} \chi_{z_c}(p_c)$, where* legC *are the centres at the S-ends of the A'-line legs (common to all members) and* intC *the remaining $N - E_{A'}$ centres (common by lemma B.28(b), identified by the fixed bijection). Put $\tilde{\Phi}_\ell := \chi_{z_\ell}\Phi_\ell$ for each A'-line leg, $\tilde{\Phi} := \Phi$ for the other legs, and $W^\circ := \prod_{c\in\mathrm{intC}} \chi_{z_c}(x)$, a member-independent number. Then*

$$\sum_{S'\in F_S} \sum_{\mathrm{pos}} K_{S'} W_{S'} \prod_\ell \Phi_\ell = W^\circ \sum_{S'} \sum_{\mathrm{pos}} K_{S'} \prod_\ell \tilde{\Phi}_\ell + \sum_{S'} \sum_{\mathrm{pos}} K_{S'} \Big[\prod_{c\in\mathrm{intC}} \chi_{z_c}(p_c) - W^\circ \Big] \prod_\ell \tilde{\Phi}_\ell .$$

*The first term is the unweighted family sum against the legs $\tilde{\Phi}$ (unit-scale smooth factors times legs, Hölder or $C^{1,\alpha}$ at the leg's own scale, [13, (3.85), p. 665]), times a constant. The second term is a sum of remainder terms with a credit ρ_{j_l} on a line l of S and a debit on a unit-scale weight (cost $\rho_j^{-1} = 1$), so $D'(S) = D(S) + 1$ and $D'(\gamma) \ge D(\gamma)$ for $\gamma \subsetneq S$; it is absent for the order-e^2 photon two-point class (*intC $= \emptyset$ *by lemma B.28(d)), the only Ward row with $D = -1$, where the $n = 1$ Taylor step acts on $\tilde{\Phi}$, King's fold* [13, (3.85)].

Proof. Add and subtract; for products of $[0,1]$-valued functions $|\prod a_i - \prod b_i| \le \sum |a_i - b_i|$, so $|\prod \chi(p_c) - W^\circ| \le \sup|\nabla\chi| \sum_c |p_c - x| \le C \sum_{l\in P} |\mathrm{step}_l|$ along a path P of lines of S' from x to the vertices carrying intC; each step is absorbed by part of its line's decay at the price ρ_{j_l} [3, (3.13)–(3.14), (3.19)]. Every Ward row other than the order-e^2 photon class has $D(S) = 0$, hence $D'(S) = 1 > 0$. □

Theorem B.30 (the Ward rows, $R \le p$)**.** *Let the localisation be by centres, and let S be a Ward-row prefix component at a fixed tuple in the branch $R \le p$ of* [13, (3.39)]*, after King's reduction* [13, (3.43)–(3.55)] *with the gauge $B = A_j(x_0)$ of the cube. Then the sum over F_S and A_S is a sum of terms of three shapes:* (R0) *zero;* (R1) *local terms $\sum_x 5^{-3j} c_a(x) \prod \partial^a \tilde{\Phi}(x)$, $|a| \le n = -D(S)$, with $|c_a(x)| \le C\rho_{j_0}^{D(S)+|a|}$ uniformly in x, j_0, k, η and the tuple, times W° and the frozen $\tilde{A}(x)$-legs;* (R2) *remainders with a credit ρ^θ on a line of S and a debit of order $\le \theta$ on one leg of S or on a unit-scale weight, $D'(S) > 0$ and $D'(\gamma) \ge D(\gamma)$ for every prefix component $\gamma \subsetneq S$. The constant depends on $\bar{n}$, d, L, α only.*

Proof. (1) By lemma B.28(c) every member of F_S, and every member of every family met in the nesting, is reduced with the same $\square$, $\square'$, x_0, B, $\tilde{A}$ and lies in the same case below. (2) Insert at every $\tilde{A}$-vertex v the identity $1 = \zeta(v) + (1 - \zeta(v))$, $\zeta(v) := \psi(\mathrm{dist}(v, \square)/p)$, $\psi \in C^\infty$, $\psi = 1$ on $[0,1]$, 0 on $[\frac{3}{2}, \infty)$, $|\psi'| \le 4$. An $\tilde{A}$-vertex, an insertion vertex of the expansion [13, (3.52)–(3.53)] of one propagator in $\tilde{A}$, has exactly two scalar line-ends, both on the chain of

that propagator, and external $\tilde{A}$-legs only; so a connected subgraph all of whose vertices are $\tilde{A}$-vertices is a path of r vertices inside one chain, with $E_S + E_{S_D} = 2$, $E_A \geq r$ and order $\geq r$, hence $D \geq \frac{r}{2} - 3 + \frac{1}{2}(2 - E_{S_D} + r) + \frac{3}{2}E_{S_D} = r - 2 + E_{S_D}$ by the degree formula of appendix B.6, and $D \leq 0$ only for $r = 2$, $E_{S_D} = 0$, $E_A = 2$: the tree [3, (2.4)], which is not a Ward row (its member-wise identity at every tuple, the cut of its position sums at $\partial\square'$ included, is proposition B.32 below). So S contains a vertex u of the original graph, localised by its centre within $\sqrt{d}$ of $\square$. A term in which some $\tilde{A}$-vertex v carries $1 - \zeta$ has $\operatorname{dist}(v, \square) \geq p$, hence $|u - v| \geq p - \sqrt{d}$; a term with $\operatorname{dist}(x, \square) > p/2$ has $|x - u| \geq p/2 - \sqrt{d}$; a term with a line $\delta G_{(i)}(\square')$ has either both ends of that line at distance $\geq p/2$ from $\partial\square'$, or an end w with $\operatorname{dist}(w, \square) \geq \frac{3}{2}p$, hence $|w - u| \geq \frac{3}{2}p - \sqrt{d}$. Two vertices of S at distance $\geq t$ are joined by a path of at most $V - 1$ lines of S, $V \leq C(\bar{n})$ its number of vertices, so some line ℓ of S, of index i, has length $\geq t/V$, and $e^{-\delta_0|\ell|/\rho_i} \leq C_\theta (V\rho_i/(\delta_0 t))^\theta e^{-\delta_0|\ell|/(2\rho_i)}$ for every $\theta \geq 0$ (the configuration space is split by the first line of the path of length $\geq t/V$; the split is sharp but harmless, since no identity is used on these terms); a $\delta G_{(i)}(\square')$ line with both ends at distance $\geq p/2$ from $\partial\square'$ carries $e^{-\delta_1 p/(2\rho_i)}$ by lemma B.21. In every case the line supplies a credit of any order, $\leq C_\theta(\rho_i/p)^\theta$ with $\theta = 3 > -D(S)$, on a line of S with no debit and no identity (proposition B.9, lemma B.21); these terms are of shape (R2), bounded with absolute values, and S needs no rewriting. (3) In the remaining *bulk* case ($\operatorname{dist}(x, \square) \leq p/2$, every line a free slice, every $\tilde{A}$-vertex carrying ζ, so $\zeta(x) = 1$ and all position sums are sums over the whole lattice) lemma B.29 turns the weighted family into W° times the unweighted family against $\tilde{\Phi}$ plus (R2) weight units, and the unweighted family is the zero-background free-lattice object of lemma B.25: resum and replace by C^ξ; for the photon two-point classes F_S^C is the full fixed-order connected two-photon function of the C^ξ theory, all ways of filling the inside and the δm^2 units included [3, p. 431], so $c_0(F_S^C) = 0$ by lemma B.26(a), and at order e^2 also $c_1(F_S^C) = 0$ by lemma B.27(b); for the insertion classes the inside families are exactly the insertion families of the order-2 parents Σ_1, the λ-tadpole and the Q_2-tadpole, so $c_0(F_S^C) = -eq \times (\text{first moment of the parent})^C = 0$ by lemma B.26(b), lemma B.27(a) and the locality of the tadpoles; lemma B.25 bounds the remaining terms by $C\rho_{j_0}^{D+|a|}$. The transport and Taylor steps [3, (3.10), (3.34)], with the reference vertex the vertex of the coarsest leg and the Taylor leg the finer one, then give (R1) and (R2) with the degree arithmetic of [3, Sect. 3]: each remainder factor $|x_e - x|^\theta$ is bounded along a path of lines of S at the price $\rho_{j_l}^\theta$, the Hölder quotient of the leg costs $\rho_{j_e}^{-\theta}$ [13, (3.65), (3.85)], $D'(S) = D(S) + \theta > 0$, and a proper prefix component does not contain the debited leg. (4) Sub-units are rewritten first, finest first, at the same tuple; each rewriting is an identity on its own family, so the moments of S are those of its original weighted family kernel, to which (3) applies; a sub-unit's frozen weights are frozen again at x, since products of $[0, 1]$-valued unit-scale factors compose; each leg receives at most one unit's debit, of order $\leq 1 + \alpha$; a weight debit is never a leg debit; and one gauge, $B = A_j(x_0)$ per cube, is used throughout. □

The tree near the face of the reduction cube. Step (2) of the proof of theorem B.30 shows that the only source-free connected subgraph with $D \leq 0$ all of whose vertices are $\tilde{A}$-vertices is the tree T of [3, (2.4)]: two insertion vertices of [13, (3.52)–(3.53)] of J_1 type joined by a doubly differentiated line, $D(T) = 0$, no partner, inside family $\{T\}$. Its member-wise identity, a derivative at the end of a line (appendix B.6), is a sum over the whole lattice; at a fixed tuple the positions of insertion vertices are summed over the reduction cube $\square'$ with a sharp cut ([13, p. 668]: only through the expansion [13, (3.52)] can vertices approach $\partial\square'$), so near $\partial\square'$ the identity leaves a face term. The lines are decomposed scale by scale on $\square'$, $G_{(i)}(\square', 0) = G_{(i)}(0) + \delta G_{(i)}(\square')$, the difference kept as an internal line of index i (as in step (2) of the proof of theorem B.30); we write $F_i(v) := \exp[-\delta_1 \operatorname{dist}(v, \partial\square')/\rho_i]$ with $\delta_1 := \delta_0/2$, δ_0 the decay rate of the free slices [13, (3.63)].

Lemma B.31 (face factors)**.** (a) *For $u, v \in \square'$ the line $\delta G_{(i)}(\square')(u, v)$ obeys the free slice bounds* [13, (3.63)–(3.65)] *(sup, first differences, Hölder quotients of the first differences) at index i with half the rate, times $F_i(u)F_i(v)$ (for a Hölder pair $u, \tilde{u}$, F_i taken at the point of the pair nearer*

to $\partial\square'$). (b) *Let a term of the ordered bound of theorem B.40(a) carry the factor $F_i(v)$ at a vertex v of the tree T, i the index of its line ℓ. Then the term is an* (R2) *unit in the sense of theorem B.30: a credit ρ_i on ℓ and a debit $\rho_{i'}^{-1}$ on one leg l' of T, of index $i' \geq i$, or no debit when the surviving vertex of T is the root of the graph; $D'(T) = 1$, and $D'(\gamma) \geq D(\gamma)$ for every prefix component $\gamma \supsetneq T$.*

Proof. (a) The boundary clause of [13, Thm. 3.3] gives $\delta G_j(\square')$ and $\delta C^{(j)}(\square')$ their bounds with the extra factor $e^{-\delta_0 \operatorname{dist}(\{u,v\},\partial\square')/\rho_j}$ (the clause used for the lines $\delta G_j(\square', A)$ in the proof of lemma B.21 through [14, (5.3)]). Writing each factor of the slice $G_{(i)} = a_i G_i Q^* C^{(i)} Q G_i$ on $\square'$ as its free-lattice part plus its δ-part and telescoping, every term of $\delta G_{(i)}(\square')$ carries one such factor, and the decay $e^{-\delta_0|\cdot|/\rho_i}$ of the factors (the δ-factor's own included) moves it to both u and v at half the rate ($\min + \text{distance} \geq \max \geq \frac{1}{2}(\operatorname{dist}(u,\partial\square') + \operatorname{dist}(v,\partial\square'))$), which is $F_i(u)F_i(v)$ with $\delta_1 = \delta_0/2$ against the free bounds at the rate $\delta_0/2$; the Hölder clause follows from the sup and first-difference clauses of the factors as [13, Prop. 3.7] follows from [13, Thm. 3.3], the boundary factor of the differentiated factor taken at the point of the pair nearer to $\partial\square'$, a pair at distance $> \rho_i$ being bounded by two first differences. On the f_D flow the same follows from the image sum $\delta G_{(i)}(\square')(u,v) = \sum_{g\neq 1} \operatorname{sgn}(g)\, G_{(i)}(0)(u, gv)$ of proposition B.10(c), since $|u - gv| \geq \max\{|u-v|, \operatorname{dist}(u,\partial\square') + \operatorname{dist}(v,\partial\square')\}$ for every $g \neq 1$ and the images at distance $\leq t$ number $O((1+t/p)^d)$. (b) In the ordered bound the line $\ell = (v, v')$ is shrunk by summing one endpoint against its kernel; $\operatorname{dist}(\cdot,\partial\square')$ is 1-Lipschitz, so $F_i(v) \leq e^{\delta_1|v-v'|/\rho_i} F_i(v')$ moves the factor to whichever endpoint survives, at the cost of half of the decay of ℓ. The surviving vertex x of T is then summed against a leg l' of T, a line of index $i' \geq i$ with the kernel profile $|K_{l'}(u-x)| \leq C\rho_{i'}^{-c} e^{-\delta_0|u-x|/\rho_{i'}}$ of [13, (3.63)–(3.65)] (a cut-line leg, lemma B.34(b), has the unit-scale profile, $i' = j$), or it is the root. Since $F_i(x) \leq \sum_\pi e^{-\delta_1 \operatorname{dist}(x,\pi)/\rho_i}$ over the $2d$ face planes of $\square'$, each term depending on one coordinate only, summing that coordinate first gives $\sum_{x_\mu} 5^{-j} e^{-\delta_1|x_\mu - \pi_\mu|/\rho_i} \leq C\rho_i$ (as $5^{-j} \leq \rho_i$), while the sum of $|K_{l'}|$ over the other $d-1$ coordinates is at most $C\rho_{i'}^{-c+d-1}$; so

$$\sum_x 5^{-3j} |K_{l'}(u-x)|\, F_i(x) \;\leq\; C\,\frac{\rho_i}{\rho_{i'}}\,\rho_{i'}^{d-c},$$

the bound $C\rho_{i'}^{d-c}$ of that step of the induction times $\rho_i/\rho_{i'}$. At the root, where the final position sum is paid by $|\square'|$ ([13, p. 664]), $\sum_{x\in\square'} 5^{-3j} F_i(x) \leq C\rho_i \cdot (\text{area of } \partial\square') \leq C(\rho_i/p)\,|\square'|$, the side of $\square'$ being $\geq 4p$. In the telescoped product of the ordered bound (lemma B.38(ii)) the factor $\rho_i/\rho_{i'}$ raises by one the exponent of exactly those prefixes that contain ℓ and not l', so $D'(T) = D(T) + 1 = 1$ and $D'(\gamma) \in \{D(\gamma), D(\gamma)+1\}$ for $\gamma \supsetneq T$; at the root the credit has no debit. □

Proposition B.32 (the tree at every tuple)**.** *Let T be the tree* [3, (2.4)] *as a prefix component at a fixed tuple, in either branch (with $\square'_K$ in place of $\square'$ in the branch $R > p$), with vertices v_1, the reference point x, and v_2, both summed over $\square'$, its line ℓ of index i, and the products $W(v_1)$, $W'(v_2)$ of the factors at its vertices (an $\tilde{A}$ field, the weight ζ or $1-\zeta$ of step (2) of the proof of theorem B.30, and one chain-end line of index $\geq i$, a cut line in the branch $R > p$). Then $\sum_{v_1,v_2\in\square'} \ell(v_1,v_2) W(v_1) W'(v_2)$ is a sum of terms of the shapes* (R0)–(R2) *of theorem B.30, with $D'(T) \geq \min\{\alpha, 1\} > 0$ and $D'(\gamma) \geq D(\gamma)$ for $\gamma \supsetneq T$. In particular the member-wise identity of the tree holds at every tuple, near $\partial\square'$ included, which is what corollary B.36(a) uses for it.*

Proof. If $\ell = D_{b_1} D_{b_2} G_{(i)}(0)$, write

$$\sum_{v_1,v_2\in\square'} \ell\, W\, W' = \sum_{v_1\in\square'} W(v_1) \sum_{v_2\in\square'} \ell(v_1,v_2)\big[W'(v_2) - W'(v_1)\big] + \sum_{v_1\in\square'} W(v_1) W'(v_1) \Big[\sum_{v_2} \ell(v_1,v_2) - \sum_{v_2\notin\square'} \ell(v_1,v_2)\Big]$$

The full sum over v_2 vanishes exactly, a lattice difference summed over the whole lattice (or the torus) being zero, the derivative at the end of a line: (R0). The first term is the transport step of [3, (3.10)] with $n = 0$: the factor $|v_1 - v_2|^\alpha$ is absorbed by the decay of ℓ at the price ρ_i^α, a credit on ℓ, and the α-Hölder quotient of W' costs $\rho_{i'}^{-\alpha}$ on the chain-end line at v_2, which is

Hölder at its own scale up to $\partial\square'$ (the free part by [13, (3.65)], the δG part by lemma B.31(a), a cut line by lemma B.34(b)), and $O(1)$ on the smooth factors $\tilde{A}$ ([13, (3.45)]) and ζ ($|\nabla\zeta| \le 4/p$): (R2) with $D'(T) = \alpha$. In the cut term $v_1 \in \square'$ and $v_2 \notin \square'$, so $|v_1 - v_2| \ge \operatorname{dist}(v_1, \partial\square')$ and half of the decay of ℓ gives $F_i(v_1)$; the sum over v_2 is then bounded by the kernel integral of ℓ, as at every step of the ordered bound, and lemma B.31(b) makes the term an (R2) unit with credit 1. If $\ell = \delta G_{(i)}(\square')$, lemma B.31(a) gives $F_i(v_1)$ (the factor $F_i(v_2)$ is dropped) and lemma B.31(b) applies; no identity is used. Since T has no partner and its inside family is $\{T\}$, the rewriting is an identity term by term, and the nesting of step (4) of the proof of theorem B.30 is unchanged. Away from $\partial\square'$ and on the torus this is Bałaban's treatment of [3, (2.4)] in [3, Prop. 2.1]; the cut term is the face term that the cut of the position sums produces. □

The cut by clusters. In the branch $R > p$ King proves [13, (3.38)] by treating every propagator longer than $3\sqrt{d}$ as an external line, so that the graph factorises into localised subgraphs, each reduced with its own cube [13, p. 666]. That cut is by vertex positions: at a fixed tuple the partition into subgraphs, hence each subgraph's $\square$, x_0, B and $\tilde{A}$, varies with the positions that the identities of appendix B.6 sum over, within one member as well as across members. We cut by the centres instead. Units of depth j, $d = 3$; $c_0 := \frac{13}{3}\sqrt{d}$ and $R_0 := c_0 - \frac{4}{3}\sqrt{d} = 3\sqrt{d}$, King's threshold. Fix a graph H of $P^{(j),1}$ localised by centres and a tuple τ with point set $P(\tau)$ (the y_i and the z_c).

(C1) *Clusters.* Let Γ_τ be the graph on $P(\tau)$ with an edge between two points at Euclidean distance $\le c_0$; the clusters K are the components of Γ_τ, with point sets P_K.

(C2) *Vertices.* A vertex with centres is assigned to the cluster of its centres (well defined on the support of the weights, (a′) below); a block or y-point carries its own point of $P(\tau)$. An unlocalised vertex is either the attachment vertex of a δm^2 point vertex whose blob carries an index $\ge j$ ([13, p. 667]: no partition of unity is introduced in that graph) or the vertex of a one-vertex graph; the former has exactly two scalar legs [13, (3.19)], and each maximal path through such vertices is read as one *dressed line* between localised ends; one-vertex graphs have no internal line, no divergent subgraph and no cube, and are bounded by the decay of their external lines; an E_1 piece localised at one vertex forms one cluster.

(C3) *Cut.* A line (scalar, photon, external line $a_j G_j Q_j^*$ to a y-point, fine δm^2-blob line or dressed line) is *cut* iff its end-vertices lie in different clusters. A cut line is kept whole, with no scale decomposition, and treated as an external line of both clusters; the *cluster graph* H_K is H restricted to the vertices of K, with its uncut lines internal and its cut lines as external fields of index j.

(C4) *Reduction and families.* Each H_K is reduced by [13, (3.43)–(3.55)] with its own cube: $\square_K$ the smallest cube of large blocks containing P_K, $\square'_K$, $\square''_K$ its $2p$- and p-neighbourhoods, x_0^K the centre of $\square_K$, $B_K = A_j(x_0^K)$, $\tilde{A}_K = A^{(j)} - B_K$ on $\square'_K$; cut lines keep the background $A^{(j)}$ and receive King's unitary factors [13, (3.46)] at their ends, as [13, p. 661] prescribes for a vertex connected to an external line. Renormalisation families are formed inside each H_K [13, p. 666].

The only change from [13, p. 666] is the cut criterion: King decides by vertex positions, (C3) by the clusters of the centres, which are summation variables of [13, (3.36)] fixed in τ.

Lemma B.33 (the cluster lemma)**.** *Let S be a prefix component of an ordering of a cluster graph H_K with inside family F_S.* (a′) *On the support of the localisation weights the centres of one vertex lie in one cluster.* (a) *For $S' \in F_S$ and $H' := (H \setminus S) \cup S'$: H' carries the same tuple under the bijection of lemma B.28(c), hence the same clusters, cubes $\square_K, \square'_K, \square''_K$, points x_0^K, gauges B_K, fields $\tilde{A}_K$ and the same branch; every vertex and line of S' lies in K; the legs of S' are the legs of S, each with the same cut/uncut status; every line of H' outside S' has its status in H. The partition depends on no vertex position.* (b) *On the supports a cut line u–v has $|u - v| \ge R_0 = 3\sqrt{d}$, and a cut external line from v to the block of a y-point has length $> R_0$.* (c) $\operatorname{diam} P_K \le c_0(|P_K| - 1)$, *every vertex of K lies within $\frac{2}{3}\sqrt{d}$ of P_K, and $|\square'_K| \le C(p + c_0 N + M)^d \le C' p^d$, $N := |P(\tau)| \le C(\bar{n})$.* (d) *The quotient graph (clusters as nodes, cut*

lines as edges) is connected, and for every spanning tree T of it, $\operatorname{dist}(P(\tau)) \le \sum_{\ell\in T}|\ell| + C(N)$, *$|\ell|$ the distance between the ends of ℓ.*

Proof. (a′) Two centres c, c' of a vertex v have $v \in \operatorname{supp}\chi_{z_c} \cap \operatorname{supp}\chi_{z_{c'}}$, so $|z_c - z_{c'}| \le \frac{4}{3}\sqrt{d} < c_0$ and they are adjacent in Γ_τ. (a) The clusters are a function of $P(\tau)$, which is common by lemma B.28(c), and the cluster data are functions of P_K; the centres of S' are those of S as points, all in K because S is connected through uncut lines of H_K, so by (a′) every vertex of S' is in K and every line of S' joins two vertices of K; a leg of S' joins a vertex of S' to the same outside end as the corresponding leg of S, in some cluster K_w, and its status is "cut iff $K_w \neq K$" in both; lines outside S' are untouched; no step uses a position. (b) Let z, z' be centres of u, v in different clusters: $|z - z'| > c_0$ and $|u - z|, |v - z'| \le \frac{2}{3}\sqrt{d}$. (c) A path in Γ_τ has steps $\le c_0$, and $p \ge 1$. (d) Join the points inside each P_K by a tree with steps $\le c_0$ and the clusters along T through the centres of the endpoints of each ℓ, at the cost $|\ell| + \frac{4}{3}\sqrt{d}$ per edge. □

So both of King's phrases, "treat it as an external line" and "the graph then factorizes into a product of localised subgraphs" [13, p. 666], become exact, with the same factorisation for every member and every position of every family.

Lemma B.34 (regularised cut lines)**.** *Fix a radial $\bar\psi \in C^\infty(\mathbb{R}^d)$, $0 \le \bar\psi \le 1$, $\bar\psi(w) = 0$ for $|w| \le \frac{1}{2}$, $\bar\psi(w) = 1$ for $|w| \ge 1$, and for a cut line ℓ between $u \in K$ and $v \in K'$ put $\ell^\flat(u,v) := \bar\psi(u - v)\ell(u,v)$ (for an external line to a y-point, $\bar\psi(u - y)$), and $\ell^{\flat,\mathrm{ph}} := U(B_K(\Gamma_{x_0^K,u}))^{-1}\ell^\flat U(B_{K'}(\Gamma_{x_0^{K'},v}))$ for the phase-dressed form of (C4).* (a) *Replacing every cut line by $\ell^\flat$ leaves $E^{(j)}(H;\tau)$ unchanged, term by term: for every member of every family, every ordering and every position configuration, each evaluation pair (u', v') at which a cut line enters, the lattice differences of the vertex functions at its ends included, has $|u' - v'| \ge R_0 - 4\cdot 5^{-j} > 1$, where $\bar\psi = 1$.* (b) *$\ell^{\flat,\mathrm{ph}}$ is $C^{1,\alpha} \otimes C^{1,\alpha}$ at unit scale in both arguments with decay $e^{-\delta_1|u-v|}$, at all positions and with no distance condition, the constant depending on $\bar n$, d, L, α only, times $(1 + k - j)^5$ for a dressed line (the size of the coarse δm^2 blob); this is* [3, (3.1)] *in the form of a cut line with the endpoint localisations.*

Proof. (a) The ends of a cut line are localised in the original sum: (C2) assigns to a cluster localised vertices, whose weights are present, block and y-points, which are points of $P(\tau)$, the vertices of one-centre E_1 pieces, none of whose lines is cut, and unlocalised coarse δm^2 vertices, which lie strictly inside dressed lines and are never an end; so lemma B.33(b) applies at every evaluation pair up to the lattice differences, each of which moves an end by at most 5^{-j}. (b) For $|u - v| \ge \frac{1}{2}$ the slice bounds [13, Thm. 3.3, Props. 3.7–3.9] (proposition B.9, proposition B.10) of the whole line, its derivatives and its mixed second differences are those of a unit-scale kernel with the stated decay, and the phases [13, (3.46)] and $\bar\psi$ are smooth at unit scale; for $|u - v| < \frac{1}{2}$ the kernel vanishes. □

Theorem B.35 (the Ward rows at every tuple)**.** *Let the localisation be by centres and the cut by clusters, with every cut line replaced by $\ell^\flat$, and let S be a Ward-row prefix component of an ordering of a cluster graph H_K at a fixed tuple τ (in particular in the branch $R > p$), with A'-line legs and scalar legs that may be cut lines. Then the sum over F_S and A_S is a sum of terms of the shapes of theorem B.30 with $\square \to \square_K$:* (R0) *zero;* (R1) $\sum_x 5^{-3j} c_a(x) \prod \partial^a \tilde\Phi(x)$, $|a| \le -D(S)$, $|c_a| \le C\rho_{j_0}^{D(S)+|a|}$, *times W° and the frozen $\tilde A_K(x)$-legs, a cut-line leg counting as index j;* (R2) *a credit ρ^θ on a line of S with a debit of order $\le \theta$ on one leg (a cut-line leg costs* 1*, by lemma B.34(b)) or on a unit-scale weight, $D'(S) > 0$, $D'(\gamma) \ge D(\gamma)$ for $\gamma \subsetneq S$. C is uniform in x, j_0, k, η and τ and depends on $\bar n$, d, L, α only, up to the dressed-line factor $(1 + k - j)^5$ of lemma B.34(b), absorbed in $(1 + k)^{p_0}$ of corollary B.36(b); the decay budget is $\bar n + 4$ draws per line, and every evaluation point of a cut-line end is joined to a localised anchor of its own cluster by at most $C(\bar n)$ lines of rewritten units.*

Proof. The proof of theorem B.30 is run inside H_K with these changes. By lemma B.33(a) every member of every family met in the nesting is reduced with the same cluster data, and cut lines never lie inside a unit: no ordered subgraph contains a cut line (it carries index j and is external

to both clusters), so a loop through a cut line is never a divergent unit. The frozen-weight identity of lemma B.29 holds per cluster, the legs that are cut lines entering through $\ell^{\flat,\mathrm{ph}}$, which by lemma B.34(b) is a $C^{1,\alpha}$ leg at unit scale, so its debit in the Taylor and transport steps costs 1; the identities of lemmas B.26 and B.27 act only on F_S^C after lemma B.25, inside one cluster, and the cut lines are legs that enter only through the transport and Taylor steps after the identity, so the cut adds no identity and truncates no member line. The cut-line factor of an output term is bounded by $Ce^{-\delta_2 d(K,K')}$ times transfer factors on unit lines: each evaluation point of a cut-line end is joined to a localised anchor by at most $C(\bar{n})$ lines of its own cluster's rewritten units, the transfer takes one more decay draw per line, and the bound factorises over clusters. Nesting is as in theorem B.30(4), with $\Box_K$ in place of $\Box$. □

Corollary B.36 (the degree count on every row; the bound in the branch $R > p$)**.** (a) *With the localisation by centres and the cut by clusters, every source-free class with $D \le 0$ obeys* (DEG) *at every tuple: the non-Ward rows by their member-wise identities (charge parity, the derivative at a line end, reflection, the bounded η-suppressed tadpole) or by the counterterm matching between partners of identical vertex structure, whose centre multisets coincide by lemma B.28(b) (the only non-Ward $D = -1$ class with a Taylor step is Σ_1, whose two centres sit at its two vertices, the attachment points of its scalar legs, so its non-reference centre is folded into the finer leg,* [13, (3.85)]*, exactly as under King's rule); the Ward rows by theorems B.30 and B.35.* (b) *For every graph whose divergent units are such classes,* $|E^{(\theta)}(H_{\mathrm{ren}};\tau)| \le C\eta^{\rho_0}p^{n_1}(1+k)^{p_0}\exp[-\delta\,\mathrm{dist}(P(\tau))]$*, $\theta \in \{k, k+n\}$, at every tuple, which is* [13, (3.38)] *in the branch $R > p$; with* [13, (3.39)] *from theorem B.30 in the branch $R \le p$ this gives King's per-graph estimate* [13, (3.42)]*.* (c) *At tuples with at least two clusters every connected component of every cluster graph carries a cut-line leg, so the vacuum members and the component-wise cancellation of* [13, p. 667] *do not arise there, and $E(H;A^{(j)};\tau)$ and $E(H;0;\tau)$ are bounded separately; a closed scalar loop of internal lines of one cluster graph none of whose vertices carries another charged field vanishes when the total q-power on it is odd (no phase sits on the loop), which is the per-cluster form of charge parity.*

Proof. (a) is the list of mechanisms of appendix B.6 with the two theorems. (b) Per cluster, theorem B.35 and (a) give the rewritten form with every stage of positive degree; the multiscale bound theorem B.40(a) with the root held, the cut-line factors of lemma B.34 and lemma B.33(d) (the tree distance of the tuple is controlled by the cut lines of a spanning tree) give the exponential in $\mathrm{dist}(P(\tau))$, and the volume factors $|\Box'_K| \le C'p^d$ of lemma B.33(c) give p^{n_1}. (c) A cluster graph is connected to the rest of H through cut lines only, and H is connected; on a closed loop with no charged external field the internal phases cancel and $\mathrm{Tr}\, q^{2r+1} = 0$. □

For theorem B.24 this supplies the $f = 1$ input of (ii) on the Ward rows, $|c_a^1| \le C\rho_{j_0}^{D(S)+|a|}$, at every tuple of either branch; the transfer to the f_D flow is (i)–(iii) there, the exact-zero and Ward mechanisms being identities for every charge-diagonal line kernel when partners carry the same lines, and lemma B.28(b) holding for the f_D members verbatim since the weights do not enter the centre count.

Theorem B.37 ((B4), (B5): the curvature classes and the source-free classes in zone I)**.** *Let H be a graph of King's expansion at step j on C_J^K at $h = 0$ that is field-dependent, or contains a curvature vertex, and let it be localised in zone I. Then* [13, (3.39)] *holds for it, and* [13, (3.38)] *with the extra factor $(1+k-j)^{p_0}$; summed over positions by lemma B.42, the zone-I contribution to $|S_{N,k}^{(j)} - S_{N,k+1}^{\prime(j+1)}|$ is $\le C[5^{-\gamma' j}\eta^{-\beta'} + \eta^{\sigma'}]|C_J|$, with C depending on the curvature sources through $\max\|f_{\mu\nu}\|_{C^{2,\alpha}}$ only.*

Proof. By lemma B.21 it suffices to treat $\tilde{H}^{\mathrm{free}}$ on the free f_D flow, with the error of the form of [13, (3.39)]. For the free flow run King's proof of [13, Thm. 3.5] and of the step bound [13, Prop. 3.6]: the census of ordered subgraph components (every component containing a source vertex has degree $\ge \frac{1}{2}$, except the field-independent, source-linear family, which is absent at $h = 0$ and vanishes for the curvature source at $A_j = \phi_j = 0$ by charge parity, and the single-curvature-vertex tadpole, which vanishes by $\mathrm{Tr}\, q = 0$); the ordered multiscale bound (theorem B.40(a)

with the degree count (DEG) of theorem B.24 for the source-free components); the one-factor-at-a-time replacement of [13, pp. 663–665] with the line comparisons of proposition B.9 and, for the curvature vertex, lemma B.22(d), which carries the gain $\varepsilon_k \le 5^{-\gamma j}$ with a bounded summed vertex; and the identities [13, (3.43)–(3.55)], under which the curvature vertex is invariant (lemma B.22(b)). The curvature source enters the proof only through the sup and $(1,\alpha)$-norms of w (lemma B.22(c)), uniformly in ε. The free-boundary counterterm renormalises the free graphs exactly (lemma B.21). □

B.7. **Block vertices and the split degree.** King's block products [13, (3.24)×(3.26), (3.25)×(3.26), (3.26)×(3.26)] carry one or two summation points x, x' in the block $B^j(y)$ and photon fields integrated along the staircases $\Gamma^{(j)}_{y,x}$. Following [3, (1.13)–(1.15), (2.1)–(2.2), (2.12)–(2.14)] we read a graph $\tilde H$ with block products as the *split graph* G^{sp}: the points of the block products are vertices with measure 5^{-3j} and no explicit power of the spacing; a photon field on a staircase, contracted at slice i, enters through the staircase-integrated kernel $g^{(i)}(x,\cdot) := G_{(i)}(\Gamma^{(i+1)}_{x_{i+1},x},\cdot)$, read as a scale-summed line of dimension $d_l := 2-3-\#\mathrm{der}+c_l$, $c_l \in \{0,1,2\}$ the number of its staircase ends; the block kernel $\kappa_y(x,x') := \mathbf{1}[x,x' \in B^j(y)]$ (times $-a_j$ or $-a_j/2$) is a line of the single index j and dimension 0. The *split degree* $D^{\mathrm{sp}}(c) := \sum_{v\in c} w_v - 3$ of a connected subgraph c counts, at each vertex, 3 plus the explicit powers of the spacing, minus $\frac12$ per undifferentiated and $\frac32$ per differentiated end of a line of c, plus 1 per photon field of a point whose line lies in c; it is Bałaban's degree [3, (2.1)–(2.2)]. (King's [13, (3.30)] credits "$+m$" at a block vertex for all its m photon fields; on subgraphs only the fields whose lines lie in the subgraph may be credited, as Bałaban's text says.)

Lemma B.38 (dictionary and telescoping)**.** *(i) For a connected all-scale-summed subgraph H of G^{sp} with components $c_1,\dots,c_r$ after removing the κ-lines, the degree of the multiscale lemma, $D_{MS}(H) := \sum_{l\in H} d_l + \sum_{v\in H} a_v + 3(|V(H)|-1)$, equals $\sum_i D^{\mathrm{sp}}(c_i) + 3(r-1)$. (ii) For an ordering $l(1),\dots,l(m)$ with indices $i_1 \le \cdots \le i_m$,*

$$\prod_l \rho_{i_l}^{d_l} \prod_v \rho_{i(v)}^{a_v} \prod_{\text{tree lines}} \rho_{i_l}^{3} = \prod_{t=1}^{m} \rho_{i_t}^{D(H_t)-D(H_{t-1})} = \Big[\prod_{t<m} \Big(\frac{\rho_{i_t}}{\rho_{i_{t+1}}}\Big)^{D(H_t)}\Big] \rho_{i_m}^{D(H_m)},$$

$i(v)$ the index of the first line at v. So a factor attached to a line enters exactly the stages containing that line, and a weight attached to a vertex enters from its first line on.

Proof. (i) Each end of a c-line at v contributes $-\frac12$ or $-\frac32$ to w_v, so a line contributes $2-3-\#$der; a staircase end contributes $+1$ at its point, booked on the line as c_l; sum, and add the κ-lines ($d_\kappa = 0$), which join the r components. (ii) Adding $l(t)$ changes D by $d_{l(t)}$, plus a_v for each new endpoint, plus 3 if $l(t)$ is a tree line. □

Lemma B.39 (line bounds for the split graph)**.** *(i) $|g^{(i)}(x,b)| \le C\rho_i e^{-\delta_0|x-b|/\rho_i}$ (one staircase end; [13, (3.64)], [3, (2.12)]); directly from [13, (3.63)], since $\Gamma^{(i+1)}_{x_{i+1},x}$ has at most $15\cdot 5^j\rho_i$ bonds each weighted by 5^{-j} and lies within $5\sqrt{3}\rho_i$ of x. (ii) Two staircase ends: $C\rho_i^2$. (iii) Derivatives and Hölder differences in the non-staircase variable: [13, (3.63), (3.65)] summed over the staircase. (iv) The finer model's slices $-1 \le i$: the same. (v) Comparison across the pair: [13, (3.74)] for the staircase line; its Hölder form from (3.74) and (3.75), the finer staircases subdividing the coarser ones except for the initial piece in the fine block, which costs $C5^{-j}\rho_i^{-1-\alpha} \le C5^{-\gamma j}\rho_i^{-\alpha-\gamma}$. (vi) $|\kappa^{(i)}(x,x')| \le e^{\delta\sqrt{3}}\delta_{ij}e^{-\delta|x-x'|}$. (vii) On C_J^K, (i)–(iii) hold with the cube slice bounds (proposition B.10(c), lemma B.1); the staircases lie inside the blocks, hence inside C_J.*

Theorem B.40 (the split degree governs the multiscale bound)**.** *Let $\mathcal{R}$ be a renormalised family of King's expansion of order $\le \bar n$, written as a finite sum of graphs G with the source vertices admitted, such that for each admitted ordering of the lines of G^{sp} every component of every ultraviolet prefix has current degree $D' > 0$ after the rewriting ((*DEG*) in the split count), the infrared lines obey [13, (3.78)], and the lines obey lemma B.39. Let α_0 be the minimum of these degrees. Then, on the torus and on C_J^K:*

(a) *the sum over orderings, compatible indices and positions of* $|E^{(j)}(G)|$ *is* $\le C\eta^{\rho_0}p^{n_1}(1+k)^{p_0}\exp[-\delta\,\mathrm{dist}(\textit{external vertices})]$, *and the same holds with one element weakened by* $\theta<\alpha_0$ *(a line with* $c_l\to c_l\rho_{i_l}^{-\theta}$*, a vertex with an explicit power* $a_v\ge\theta$*, or an external field);*
(b) $|E'^{(j+1)}(G)-E^{(j)}(G)|\le C\eta^{\rho_0'}5^{-\gamma j}\exp[-\delta\,\mathrm{dist}]$ *for* $0<\gamma<\alpha_0$*;*
(c) *every rewritten subgraph is source-free, point-free and a component of a prefix of* G^{sp}*, and* $D'(\gamma)\ge D^{\mathrm{sp}}(\gamma)$ *for every component* γ *of every prefix;*
(d) *the exponent of every stage of the ordered bound is* $D'(H_t)$ *in the split count; no per-vertex credit for staircase fields enters.*

Proof. (a) Apply the multiscale lemma of [13, (3.66)–(3.70)] (a line-by-line induction over the ordering, each step shrinking one line and summing one endpoint, the index sums geometric by the positive degree) to the connected graph G^{sp} with the lines and bounds of lemma B.39. Its degree is D^{sp} by lemma B.38(i); a subgraph containing a κ-line has $D_{MS}=\sum_i D^{\mathrm{sp}}(c_i)+3(r-1)$ with each c_i containing a point, hence ≥ 1 by lemma B.41. Infrared stage: points lie in no counterterm unit, so no infrared line ends at a point; κ has index $j\le$ every infrared index, and shrinking it (put it first among the ties) transfers only external lines at x' to x and gives $\sum_{x'}5^{-3j}\kappa\le 1$, shifting every later prefix degree by the same constant; so the infrared differences of [13, (3.80)] are unchanged. Fixed lines, remainders and the position sum as in [13, pp. 663–664]; the weakened element by the same induction with $D'-\theta\ge\alpha_0-\theta>0$. (b) [13, pp. 663–665] verbatim on G^{sp}: the finer slices give $5^{-\gamma j}$ through the finest sum; otherwise replace one factor at a time (external lines by (3.71), sources by sampling, the phase by (3.72), plain lines by (3.73), staircase lines by (3.74), Hölder lines by (3.75) and lemma B.39(v)), each error being the graph with one element weakened by γ; then replace the point sums, $\sum_{x'\in B^1(x)}5^{-3(j+1)}=5^{-3j}$ exactly, and $\kappa'=\kappa$ exactly since the unit blocks of both members coincide. (c) A rewritten subgraph is divergent, so by lemma B.41 it contains no point and is one of Bałaban's divergent classes, which contain no block vertex [3, pp. 428–431]; debits land on legs of S, and if a leg lies in γ then $S\subset\gamma$, so the net change per component is ≥ 0. (d) is lemma B.38(ii). On C_J^K use lemma B.39(vii) and bound position sums over C_J by sums over $\mathbb{Z}^3$. □

Lemma B.41 (block-vertex components are convergent)**.** *Let* γ *be a connected source-free component of King's expansion at zero background, on* C_J^K *or on the torus, containing a point of a block product and properly contained in a connected graph. Then* $D^{\mathrm{sp}}(\gamma)\ge 1$*, or* γ *vanishes identically;* $D^{\mathrm{sp}}(\gamma)\ge 2$ *if* γ *contains a second point or a source vertex.*

Proof. A proper connected subgraph has $E_\gamma\ge 1$ external legs. Every King vertex has an even number of scalar legs and each internal scalar line uses two ends, so E_γ has the parity of the number of external photon legs. With $D(\gamma)=\sum_v u_v-3+E_\gamma/2$ [13, (3.30)–(3.31)], a point with a photon field whose line lies in γ has $w\ge\frac{5}{2}$ and every other vertex $w\ge 0$: if $E_\gamma\ge 2$, $D^{\mathrm{sp}}\ge\frac{5}{2}-3+1=\frac{1}{2}$, hence ≥ 1 by integrality (D^{sp} is an integer, since a_v and the number of differentiated internal ends are integers and $\sum_v n_v=2\#\mathrm{lines}+E_\gamma$). If $E_\gamma=1$ the leg is a photon; at zero background every scalar line is component-diagonal and every vertex carries q^{m_v}, so each closed charged loop carries $\mathrm{Tr}\,q^{n_c}$ with $\sum_c n_c$ odd, and $\mathrm{Tr}\,q^{\mathrm{odd}}=0$. A second point or a source vertex adds $\ge\frac{3}{2}$ (curvature) or 2 (point, scalar source). □

Lemma B.42 (the position sum)**.** *Let a family of localised graphs on* C_J^K *satisfy, for each localisation, the root-summed bound* $\sum_{\textit{other positions}}|E^{(j)}(H;\{y\},\{z\})|e^{\delta'\,\mathrm{dist}}\le C\eta^{\rho_0}p^{n_1}(1+k)^{p_0}\Lambda(y_1)$ *with the root* y_1 *fixed and a root weight* Λ *with* $\sum_{y_1\in Q}5^{-3j}\Lambda(y_1)\le C$ *for every unit cube* Q*. Then the sum over all localisations of the field- and source-dependent part is* $\le C\eta^{\rho_0}p^{n_1}(1+k)^{p_0}\eta^{-3}|C_J|$ *for the* $R\le p$ *branch, and* $\le C\eta^{3+\sigma}\eta^{-3}|C_J|$ *for the* $R>p$ *branch, where the factor* $e^{-(\delta-\delta')p}$ *is* $\le C\eta^{3+\sigma}$ *because* $p(\eta)$ *grows faster than every logarithm. On the diagonal sequence* $j=j(k)$ *of theorem 5.7 the powers* $p^{n_1}(1+k)^{p_0}$ *are* $\le C_\epsilon 5^{\epsilon j}$*, so a bound* $5^{-\gamma j}\times$*(this) is* $\le C5^{-\gamma' j}\eta^{-\beta'}|C_J|$ *for any* $\gamma'<\gamma$ *and* $\beta':=\max(3-\rho_0,0)$.

Proof. This is [13, (3.40)–(3.42)] with the position sum over $\eta^{-3}|C_J|$ unit sites and the root weight summed by its hypothesis; the counterterm units' vertices off T_η are internal to a constant and do not enter. □

B.8. **Zone B: the scalar face remainders.** In zone B the images of the lines across the faces on F are at distance $2t(x)$, not $\geq 4p$, so lemma B.21 fails, and the free-boundary counterterm renormalises the Dirichlet self-energies only up to remainders [14, pp. 331–332]. We write the cube rewriting as the free one plus remainders, and bound the remainders by face-layer weights.

Lemma B.43 (the rewriting steps are identities on the cube)**.** *(a) Taylor transport along axis paths between live sites of Ω, Hölder splits, the resolvent splitting $G = C^\xi + G(C^{\xi,-1} - G^{-1})C^\xi$ with $C^\xi := (-\Delta^\Omega + \xi^{-2})^{-1}$, rescaling and resummation over orderings are pointwise identities on Ω. (b) On a bond with one end on a Dirichlet plane the hopping term is $|\phi(b_\pm)|^2$, independent of A_b; the vertices on such bonds sum to zero and are dropped, so every vertex sits on bonds with both ends live. (c) For f vanishing off the live sites and w supported on live bonds, $\sum_b (\nabla_b f) w(b) = -\sum_x f(x)(\nabla^* w)(x)$ exactly, with no boundary term: this is* [3, (2.8)] *and* [14, (3.16)] *on the cube.*

Proof. Finite sums on $5^{-j}\mathbb{Z}^3$; (b) uses $|U(A_b)\phi(b_+) - 0|^2 = |\phi(b_+)|^2$. □

Definition B.44 (cube rewriting, remainder vertices)**.** Apply the steps of the free rewriting to a zone-B family with cube lines and live-site positions. At each evaluation of a resummed coefficient $c_S(x)$ (a *C-site* S, rooted at x) write $c_S^C(x) = c_S^F(x) + \Delta_S(x)$, c_S^F the value with free f_D lines and positions over $5^{-j}\mathbb{Z}^3$; the *remainder* $\Delta_S(x)$ is attached at x as a generalised *remainder vertex* v_S with the legs of the evaluated term, and every graph carrying a remainder is *frozen*: no later step acts on a stage containing a remainder vertex. The *moment degree* $D'(S)$ is $D(S)$ for a zeroth moment and $D(S) + 1$ for a first moment.

Lemma B.45 (comparison decomposition)**.** *Every zone-B renormalised family $\mathcal{R}$ of order $\leq \bar{n}$ on C_J^K is a finite sum $\sum G + \sum G'$ of* bulk-type *graphs (the free rewriting's outputs with cube lines and live positions, free coefficients at the C-sites; they obey* (DEG) *of theorem B.24, degrees being combinatorial, and their lines obey the cube slice bounds) and* remainder *graphs (each containing a remainder vertex, with every stage containing it unrewritten). The free-boundary counterterm units are the partners of the bulk-type part exactly, as in zone I.*

Proof. Induction along the rewriting: lemma B.43, the identity $c^C = c^F + \Delta$, and the fact that an inner remainder is never inside a collapsed coefficient (its graph is frozen before the outer evaluation). □

Theorem B.46 (census of the scalar remainders)**.** *Let S be a C-site of a zone-B family other than the vacuum polarisation sites of appendix B.9 and the field-free vacuum components. Then (i) S contains no block point and no source vertex; (ii) $D'(S) \in \{-1, 0\}$, S being the zeroth moment of a scalar self-energy of order 2 or 4, the first moment of the order-2 one-photon exchange, a one-leg-differentiated two-point class, or a $\phi\phi A$ class; (iii) $D'(S) = -1$ iff S is the zeroth moment of an order-2 scalar two-point class with both legs undifferentiated (the one-photon exchange after* [14, (3.16)]*, the λ-tadpole, the seagull tadpole); then v_S has exactly two scalar legs, both undifferentiated at v_S; (iv) every connected non-vacuum source-free stage that is not charge-odd has $D \geq -1$. Moreover, every connected stage H of a non-vanishing member that contains a line at a scalar remainder vertex and has $E_H \geq 1$ satisfies $D(H) \geq 0$, computed with $u_{v_S} = D'(S) + 3 - E_S/2 - \epsilon_v$ ($\epsilon_v = 1$ if a first-moment derivative sits on a leg internal to H), and $D(H) \geq 1$ if H contains a block point.*

Proof. (i) A block point gives $D^{\mathrm{sp}} \geq 1$ (lemma B.41), so no evaluation acts there; sources by lemma B.20. (ii) is Bałaban's list of evaluations [3, pp. 429–445] for the point-free vertex set, which is King's [13, (3.18), (3.19), (3.22), (3.23)], with the degrees from [13, (3.30)–(3.31)]: for two scalar legs $D = \sum u - 2$, the one-photon exchange (two current vertices with internal differentiated ends, $u = \frac{1}{2}$ each) and the tadpoles ($u = 1$) have $D = -1$, order 4 has $D = 0$, order ≥ 6 is convergent; a first moment adds 1; a derivative on an external leg does not count,

so the one-leg-differentiated classes have $D = 0$; for $\phi\phi A$ legs $D = \sum u - \frac{3}{2} \geq 0$, and D is an integer (the parity count below), so the next value is 1. (iii) by (ii): the current vertex's derivative is on the internal line, the transported leg is undifferentiated. (iv) $E \geq 5$: $D > 0$ [3, (2.17)]; $E = 4$: $D \geq 0$; $E = 3$: $D \geq 0$; $E = 2$: $D \geq -1$; a scalar one-leg graph needs a point, a photon one-leg graph is charge-odd. *The stage bound.* v_S carries $a_v = D'(S)$ (lemma B.48 books $\rho_{j_0}^{D'}$ at $j(v) = j_0$), $n_v = E_S$ legs and $\delta_v = \epsilon_v$, so u_v is as stated and $n_v \equiv 2u_v \pmod 2$: $D(H)$ is an integer. For a zeroth moment $u_v = \sum_{w\in S} u_w$, so $D(H) = D(\hat{H})$ with v re-expanded; a first moment adds $1 - \epsilon_v \geq 0$. Units met by H through a counterterm line are replaced by point vertices of $u = 2$, each whole order-2 unit costing ≥ -1 and every other ≥ 0; with k_1, k_2 their numbers, $D(H) \geq D'(S) - E_S/2 - \epsilon_v + k_1 + 2k_2 + \sum_{\text{std}} u + E_{H'}/2$. If $k_2 \geq 1$, $D(H) \geq 0$; if $k_2 = 0 < k_1$, $E_{H'} = E_H \geq 1$ gives $D(H) \geq -\frac{1}{2}$, hence ≥ 0. If $k_1 = k_2 = 0$ put $X := \sum_{w\neq v} u_w + E_H/2$, so $D(H) = D'(S) - E_S/2 - \epsilon_v + X$ with $X - E_S/2 \in \mathbb{Z}$. For $E_S = 2$: $X \geq 1$ (a self-loop at v alone leaves $E_H = 0$); $X = 1$ forces $E_H = 1$ and $H = \{v, w\}$ with w a current vertex whose two scalar ends close both legs of v, a scalar cycle v–w–v carrying $\mathrm{Tr}(M_v q) = 0$ ($M_v \propto 1$), so the member vanishes; hence $X \geq 2$ and $D(H) \geq D'(S) - \epsilon_v + 1 \geq 0$. For $E_S = 3$: $X \in \frac{1}{2} + \mathbb{Z}$, $X = \frac{1}{2}$ forces a self-loop closing the two scalar legs of v, whose cycle carries $\mathrm{Tr}\, q^{\text{odd}} = 0$; hence $X \geq \frac{3}{2}$ and $D(H) \geq D'(S) = 0$. Block points: $D(H) \geq D(\hat{H}) \geq 1$ by lemma B.41. □

Lemma B.47 (geometry of a remainder)**.** *Let a configuration of a C-site S rooted at a live site x contain a boundary element: an image line (u, gv), $g \neq 1$ in the reflection group of Ω, or a free-side internal position y that is not live. Then the lengths $l_l := |u_l - g_l v_l|$ of the lines on a path from x to the element sum to $\geq t(x)/2$; if g involves a face of Ω not on F, then $l_l \geq 2p$.*

Proof. $|u - gv| \geq \mathrm{dist}(u, \partial\Omega) + \mathrm{dist}(v, \partial\Omega)$ and $|a - b| \leq |a - gb|$ for every path line, so $t(x) \leq |x - u| + t(u) \leq \sum_{\text{path}} l + l_l$, up to the photon half-offset $t_N(u) \geq t(u)/2$; an outside vertex gives $t(x) \leq |x - y|$; the vertices of S lie in □, at distance $\geq 2p$ from the faces not on F. □

Lemma B.48 (layer weight of a remainder vertex)**.** *Let S be a scalar C-site, j_0 the lowest index of its legs, $\rho := \rho_{j_0}$, $D' := D'(S)$, x live, $t := t(x)$. Then*

$$|\Delta_S(x)| \leq C\rho^{D'}\big[w_{D'}(t/\rho) + e^{-\delta p/\rho}\big], \qquad w_{-1}(s) := s^{-1}e^{-\delta s}, \quad w_0(s) := (1 + \ln_+(1/s))e^{-\delta s},$$

and across the pair $|\Delta'_S(x) - \Delta_S(x)| \leq C5^{-\gamma j}\rho^{D'-\gamma}[(\rho/t)^\gamma w_{D'}(t/\rho) + e^{-\delta p/\rho}]$ for $0 < \gamma < \alpha_0$. C, δ depend on $\bar{n}$, α, α_0 and the line constants only.

Proof. Expand the internal lines of S into slices of index $\leq j_0$ and fix an ordering; expand Δ_S multilinearly, each term having a boundary element (lemma B.45); image lines are free kernels at (u, gv) with $|K^{\text{free}}(u, gv)| \leq C\rho_i^{d_l} e^{-\delta_1 l_l/\rho_i}$, and the image sums converge since the sides of Ω are $\geq 1 \geq \rho$. Run the multiscale induction (theorem B.40(a)) with the root held, splitting each line's kept half-exponential into quarters (external distance, transfer, image count, lemma B.47); since $\rho_{i_l} \leq \rho_{i_m}$ for the top internal index i_m, lemma B.47 gives $\prod_l e^{-(\delta_1/8)l_l/\rho_{i_l}} \leq e^{-(\delta_1/16)t/\rho_{i_m}}$, or $e^{-\delta_1 p/(4\rho_{i_m})}$ for an off-F image. The proper stages of S have $D \geq \alpha_0 > 0$ ((DEG): "the only divergent graphs can be the whole graphs", [3, p. 445]), so their index sums are geometric, and the telescoped product (lemma B.38(ii)) leaves $\rho_{i_m}^{D'}$ (a first-moment factor $|x' - x|$, distributed along a chain with $|a-b| \leq |a-gb|$, adds +1). Hence $|\Delta_S(x)| \leq C\sum_{i_m \leq j_0} \rho_{i_m}^{D'} e^{-\delta' t/\rho_{i_m}} + (\text{off-}F)$, which is $\leq Ce^{-\delta t/\rho}(1 + \ln_+(\rho/t))$ for $D' = 0$ and $\leq Ct^{-1}e^{-\delta t/\rho}$ for $D' = -1$; the off-F part is $\leq C_N e^{-\delta p/2}$ for every N. The pair difference: replace one line at a time by its difference ([13, Props. 3.8–3.9] on Ω, via proposition B.10(c)), which weakens it by $\rho_{i_l}^{-\gamma}$ with the prefactor $5^{-\gamma j}$; the finer slices $i < 0$ give $5^{-\gamma j}$ through the finest stage ($D(H_1) \geq \alpha_0 > \gamma$), a top-only finer line (the λ-tadpole) giving $\sum_{i<0} \rho_i^{-1} e^{-\delta' t/\rho_i} \leq Ct^{-1}(5^{-j}/t)^\gamma$; the point-sum replacement is exact since both members have the same planes. □

Lemma B.49 (the borrowed Hölder factor)**.** *Let x be live, $t = t(x)$, $0 < \alpha < 1$, $m_\alpha(x; \rho) := \min(1, t/\rho)^\alpha$. (i) For a cube slice with any far-end decoration O_z of order $\mathfrak{q} \in \{0, 1, 1 + \alpha'\}$: $|O_z G^\Omega_{(i)}(x, z)| \leq C\rho_i^{-1-\mathfrak{q}} m_\alpha(x; \rho_i) e^{-\delta|x-z|/\rho_i}$. (ii) $|\phi^{(j)}(x)| \leq Cm_1(x; 1)\|\phi_j\|_\infty$ and $|a_j G_j Q_j^*(x, y)| \leq$*

$Cm_\alpha(x;1)e^{-\delta|x-y|}$. *(iii)* $|\delta G_j(\square', A^{(j)})(x,y)| \le Cm_\alpha(x;1)e^{-\delta_0 \operatorname{dist}(\{x,y\},\partial\Omega\setminus F)}$. *(iv) If the line also carries a pair difference (bound* $5^{-\gamma_1 j}\rho^{-\gamma_1}\times$ *its bound), its difference obeys the difference bound with* $\gamma = \gamma_1/2$ *times* $m_{\alpha/2}$.

Proof. (i) The mixed clause: each slice factorises as $G_{(i)} = a_i^2\rho_i^{-4}(G_iQ_i^*)C^{(i)}(Q_iG_i)$ [13, (2.17), (3.60)] with box operators on Ω, so the x-dependence sits in the left factor and the z-dependence in the right one, and $|O_zG_{(i)}(x,z) - O_zG_{(i)}(x',z)| \le C|x-x'|^\alpha\rho_i^{-1-\alpha-\mathfrak{q}}e^{-\delta \operatorname{dist}(\{x,x'\},z)/\rho_i}$ by the Hölder clause of the left factor and the decorated clause of the right one; with $x' = x_P$ the foot of x on the nearest Dirichlet plane, where $G_{(i)}(x_P,\cdot) = 0$, this is the claim for $t \le \rho_i$, and for $t > \rho_i$ use the single-point bound. This is King's [14, (3.19)] for every leg type. (ii) $\phi^{(j)}$ vanishes on the Dirichlet planes and $|D_A\phi^{(j)}| \le C\|\phi_j\|$ over face bonds [14, (5.2)]; telescope along the $t5^j$ bonds from x to x_P. (iii) Both kernels vanish on $F\cap\partial\Omega$, and [14, (5.3)] gives the derivative and Hölder clauses of the difference with decay from the faces not on F. (iv) $\min(a,b) \le a^{1/2}b^{1/2}$, both members' slices vanishing on the same planes. □

Lemma B.50 (charging rule and the weight after borrowing)**.** *In a non-vanishing member of a remainder graph one can choose for each* $D' = -1$ *remainder vertex* v *one of its two scalar leg ends at* v *so that every line is charged at most once. Fix* $0 < \beta < \alpha/2$ *and move the factor* $m_\beta(v;\rho_l)$ *of the charged leg* l *(lemma B.49) from the line to* v_S*. Then the line keeps its standard bound, and* v_S *carries* $\omega'_S(x) := \rho|\Delta_S(x)|m_\beta(x;\rho_l) \le C[(\rho/t)^{1-\beta}e^{-\delta t/\rho} + e^{-\delta p/(2\rho)}]$*, a weight of class* $\mathcal{L}(\rho;0,1-\beta)$ *(definition B.53) plus an off-F term of any degree; across the pair the difference is* $5^{-\gamma j}\rho^{-\gamma}$ *times a weight of class* $\mathcal{L}(\rho;0,1-\beta+\gamma)$*,* $0 < \gamma < \beta$.

Proof. Let $\mathcal{R}$ be the multigraph on the $D' = -1$ remainder vertices with the member lines joining two of them as edges. Each node has degree ≤ 2, so $\mathcal{R}$ is a union of paths and cycles; a cycle uses both legs of each node and is a whole component with $E = 0$, a field-free vacuum component, which is dropped. Orient each path and charge each node to its left leg end; the leftmost node's left leg is a line to a vertex outside $\mathcal{R}$ or an external field, and each edge is charged once, at its right end. The weight: $m_\beta(x;\rho_l) \le m_\beta(x;\rho)$ for $\rho_l \ge \rho$, and lemma B.48. Without the borrowing the $D' = -1$ layer sum diverges like $\ln(\rho5^j)$ per unit face area; with any $\beta > 0$ it is bounded. This is King's "extra degree of convergence ... borrowed from a line external to H" [14, p. 332], with the rule that no line gives up two factors. □

Theorem B.51 ((B5′), scalar part)**.** *Let* $\mathcal{R}$ *be a renormalised family of order* $\le \bar n$ *of King's expansion of* $P^{(j),1}$ *on* C_J^K *at zero background, localised in zone B, with at least one external field or at least one source vertex, rewritten by definition B.44 (its vacuum-polarisation prefix components by theorem B.58). Fix* $0 < \alpha < 1$*,* $0 < \beta < \alpha/2$ *and* $0 < \gamma < \min(\alpha_0^B, \beta)$*,* α_0^B *the minimum positive stage degree over the finitely many graphs and orderings. Then:*

- *(a) with one vertex* x_0 *held (or none), the sum over orderings, compatible indices and positions of* $|E^{(j)}(G)|$ *over the members of* $\mathcal{R}$ *is* $\le C\eta^{\rho_0}p^{n_1}(1+k)^{p_0}\exp[-\delta \operatorname{dist}(V_*)]\Lambda(x_0)$*,* V_* *the vertices with external legs, the held vertex and the source vertices, with* $\Lambda(x_0) \le C(1+\ln_+(1/t(x_0)))^{\bar n+1}t(x_0)^{-(1-\beta)}e^{-\delta_\Lambda t(x_0)}$*,* $\Lambda = 1$ *if no vertex is held,* $\Lambda(x_0) \le C$ *if* $t(x_0) \ge 1$ *(in particular at every source vertex), and* $\sum_{x_0\in Q} 5^{-3j}\Lambda(x_0) \le C$ *for every unit cube* Q*;*
- *(b) across the pair,* $|E'^{(j+1)} - E^{(j)}| \le C5^{-\gamma j}\times$*(the bound (a) with* $t^{-(1-\beta)}$ *replaced by* $t^{-(1-\beta+\gamma)}$*);*
- *(c) summed over the zone-B localisations (lemma B.42), the zone-B part of the source-free field-dependent effective-action difference is* $\le C5^{-\gamma j}\eta^{\rho_0}p^{n_1}(1+k)^{p_0}|\partial T^{(j)}| \le C'5^{-\gamma' j}\eta^{-\beta'}|C_J|$ *on the diagonal sequence, for every* $\gamma' < \gamma$ *and* $\beta' = \max(2-\rho_0, 0)$*; for source hosts the position sum is the host's, with the root at the source and the factor* $e^{-\delta R/2}$*,* $R \ge \eta^{-1}/2$.

The constants depend on the point of King's range, $\bar n$*,* α,β,γ *and the line constants, not on* j,k,η,J,x_0.

Proof. Lemma B.45 splits $\mathcal{R}$ into bulk-type and remainder graphs. *Bulk-type graphs* are generalised graphs with (DEG) (theorem B.24; with sources, the host's own census as in theorem B.37), lines obeying the cube slice bounds, vertices bounded by sup norms (sources by

$\|w\|_\infty$, lemma B.22(c)); theorem B.40(a) holds on C_J^K verbatim (the induction uses no periodicity), so (a) holds with $\Lambda \equiv 1$, and (b) is theorem B.40(b) with the cube line differences. *Remainder graphs* satisfy the hypotheses of theorem B.57: lines obey the standard bounds (a charged leg keeps its bound, lemma B.50); the layered vertices are the $D' = 0$ remainder vertices, of class $\mathcal{L}(\rho_{j_0}; 1, 0)$ (lemma B.48), the $D' = -1$ ones after borrowing, of class $\mathcal{L}(\rho_{j_0}; 0, 1-\beta)$ (lemma B.50), and the vacuum-polarisation face pieces (theorem B.58(b)), each of width $\rho_v \le \rho_{j(v)}$ since $j(v) = j_0$, with off-F terms of any prescribed degree. (DEG^B): every connected component of every admitted stage has $D \ge 0$, and $D = 0$ only with a layered vertex: stages containing a scalar remainder vertex by theorem B.46; stages containing a vacuum-polarisation piece by lemma B.54; stages containing a block point by lemma B.41; stages containing a source vertex and a layered vertex (*mixed*) by lemma B.52; all other stages are bulk-type stages of the frozen graph with $D > 0$ by (DEG). Theorem B.57(i) gives (a), with $\theta_{\max} = 1 - \beta < 1$ so that the root weight is summable, and $\Lambda(x_0) \le C$ at $t(x_0) \ge 1$. Theorem B.57(ii) gives (b): its pair hypotheses are the line differences (proposition B.10(c), lemma B.1), the coefficient differences (lemma B.48, theorem B.58(c)), the layered weights in rate form (lemma B.50, class $\theta + \gamma < 1$), the sources by sampling and lemma B.22(d), and $\gamma < \min(\alpha_0^B, \beta)$. The infrared stage is unchanged: infrared lines lie inside free counterterm units, which contain no remainder, no source and no layered vertex. Field-independent members with a source vertex (curvature classes with $n_g \ge 2$ even): $\mathcal{R}$ of lemma B.50 has no cycle there, because a cycle would be a whole component made of remainder vertices, and the member is connected and contains a source vertex, which is not a remainder vertex. (c): theorem B.58(d) verbatim, with $\sum_{x_0} 5^{-3j}\Lambda(x_0) \le C$ per unit face cell (lemma B.56(b) at width 1) and $|\partial T^{(j)}| \le C\eta^{-2}|C_J|$. □

Lemma B.52 (stages containing a source vertex and a layered vertex)**.** *Let H be a connected component of an ultraviolet stage of a non-vanishing remainder member, containing a source vertex s and a layered vertex v. Then (a) in the induction of theorem B.57, reserving half of every ultraviolet line's decay and half of every layered weight's exponential, the reserved factors give $\le C_N \rho_{i_{t_*}}^N$ for every N at the first step t_* at which such a component appears, raising the exponent of every later ultraviolet step by N; (b) $D(H) \ge 0$; (c) no C-site contains a source vertex.*

Proof. (a) A path P of lines of the stage joins s to x_v, each of index $\le i_{t_*}$; successive vertices on P are attached within one fine spacing, so $\sum_{l\in P}|l| \ge |s - x_v| - \bar{n}5^{-j}$, and with $t(s) \ge \eta^{-1}/2 \ge 4$ (lemma B.20(iii)) and t 1-Lipschitz, $\sum_P |l| + t(x_v) \ge t(s) - \bar{n}5^{-j} \ge 1$; also $\rho_v \le \rho_{i_{t_*}}$. Hence the reserved factors are $\le \exp[-c_1/\rho_{i_{t_*}}] \le C_N \rho_{i_{t_*}}^N$, and $\rho_{i_{t_*}}^N \le \prod_{t\ge t_*}(\rho_{i_t}/\rho_{i_{t+1}})^N$. (b) Re-expand the generalised vertices of H: $D(H) \ge D^{\mathrm{sp}}(\tilde{H})$ (zeroth moments give equality, first moments and the bulk rewriting never lower a containing degree, theorem B.40(c)); replace the units met through a counterterm line by points of $u = 2$ as in theorem B.46. If some unit is met, $D^{\mathrm{sp}} \ge u_s - 2$; otherwise $\tilde{H}$ contains s and the vertex set of the layered vertex's class, whose standard vertices give $\sum u \ge 1$, so again $D^{\mathrm{sp}} \ge u_s - 2$. With $u_s = 2$ (scalar source) or $\frac{5}{2}$ (summed curvature vertex) this is ≥ 0. (c) A connected subgraph with a source vertex has $D \ge \frac{1}{2}$ except the field-independent source-linear family (absent at $h = 0$; for the curvature source it vanishes by charge parity) and the single-vertex tadpole ($\mathrm{Tr}\, q = 0$): theorem B.37's census. □

B.9. **Zone B: the vacuum-polarisation face pieces and the ordered bound with layered vertices.**

Definition B.53 (layered weights)**.** A weight ω on Ω is *layered* of class $\mathcal{L}(\rho; m, \theta)$ ($5^{-j} \le \rho \le 1$, $m \le \bar{n}$, $0 \le \theta < 1$, $\delta_b > 0$) if $|\omega(x)| \le (1 + \ln_+(\rho/t(x)))^m (\rho/t(x))^\theta e^{-\delta_b t(x)/\rho}$ for every site. A vertex is *layered* if its vertex function is a bounded function times $\omega_v(x_v)$ with $\omega_v \in \mathcal{L}(\rho_v; m, \theta)$ and $\rho_v \le \rho_{j(v)}$, $j(v)$ the lowest index of the lines at v.

Lemma B.54 (integrality and the photon two-point census)**.** *(i) For a connected γ, $D(\gamma) = \sum_v u_v - 3 + E_\gamma/2$ is an integer. (ii) Let γ be connected, source-free, with exactly two external legs,*

both photon. Then $D(\gamma) \geq -1$, with equality iff γ is the one-loop bubble (two current vertices joined by two scalar lines) or a tadpole (one seagull vertex with its scalar ends joined); all others have $D \geq 0$. (iii) Let m be a non-vanishing member, S' a connected photon two-point prefix component and H a connected component of a stage with $S' \subsetneq H$, $E_H \geq 1$. Then $D(H) \geq 0$, and $D(H) \geq D(S') + 1$ if $D(S') = -1$.

Proof. (i) a_v and the number of differentiated internal ends are integers, so $n_v \equiv 2u_v \pmod 2$, and $\sum_v n_v = 2\#\text{lines} + E_\gamma$. (ii) $D = \sum u - 2$; the two photon ends sit at vertices of total $u \geq 1$, with equality only for the bubble and the tadpoles (block vertices have $u \geq \frac{5}{2}$, and a $u = 0$ merged counterterm vertex drags a $u \geq \frac{1}{2}$ vertex of its unit into γ). (iii) $\sum_{v \in S'} u_v = D(S') + 2$; H contains a leg of S' and its other end w, $u_w \geq \frac{1}{2}$, and $D(H) = D(S') - 1 + (\Sigma' + E_H/2)$ with $\Sigma' := \sum_{v \in H \setminus S'} u_v$, $\Sigma' + E_H/2$ an integer ≥ 1. If $D(S') = -1$ and $D(H) < 0$ then $\Sigma' = \frac{1}{2}$, $E_H = 1$, $H \setminus S' = \{w\}$ a current vertex whose differentiated end closes a loop at w with one photon vertex, $\mathrm{Tr}[V_b^{\mathrm{hop}} G_{(i)}] = i(G_{(i)}(x, x + e_\mu) - G_{(i)}(x + e_\mu, x)) = 0$: excluded. □

By (ii) a first-moment extraction is needed only for the one-loop bubble, so the face piece below exists only there; the local part of the vacuum polarisation (the zeroth moment of the part with lines C^ξ only) vanishes identically on Ω, faces included, by gauge invariance of the scalar functional under constant shifts of the photon field on the simply connected cube (an identity for the full kernel, not a per-ordering one), so no $D = -1$ photon piece carries a face weight.

Lemma B.55 (the face piece)**.** *Let $\Pi^\xi(b, b')$ be the one-loop bubble with two lines $C_D^\xi := (-\Delta_D^\Omega + \rho^{-2})^{-1}$ minus the seagull, and $M(x) := \sum_{b'} \Pi^\xi(b, b')(x' - x)_s$ its first moment at $b = (x, \mu)$. Then $|M(x)| \leq C(1 + \ln_+(\rho/t(x)))e^{-\delta t(x)/\rho} + Ce^{-\delta p/\rho}$, uniformly in $5^{-j} \leq \rho$.*

Proof. $C_D^\xi(x, y) = \sum_g \mathrm{sgn}(g) C^\xi(x, gy)$ over the reflection group of Ω. The term with both identity images and x' summed over $5^{-j}\mathbb{Z}^3$ is the free first moment, which vanishes by the symmetries $C^\xi(-y) = C^\xi(y)$ and $\mu \leftrightarrow \mu'$ [3, (3.27)–(3.28)]. With $|\nabla^a C^\xi(x)| \leq C_a(|x| + 5^{-j})^{-1-a} e^{-c|x|/\rho}$ [15, Thm. 4.3.1, Cor. 4.3.3], each bubble term carries two lattice differences and $|\Pi_{\mathrm{free}}(x, x')| \leq C(|x - x'| + 5^{-j})^{-4} e^{-c|x-x'|/\rho}$. The free term over x' outside Ω has $|x' - x| \geq t - 5^{-j}$ and contributes $\leq C(1 + \ln_+(\rho/t))e^{-ct/\rho}$. A term with an image line: for a reflection g in a face plane, $|x - gx'| \geq \max(|x - x'|, t)$; a term whose image line carries q of the two differences is $\leq C|x - x'|^{-1-a} \max(|x - x'|, t)^{-1-\mathsf{q}} e^{-c \max/\rho}$, $a + \mathsf{q} = 2$, and its first moment is $\leq C[t^{-1-\mathsf{q}} \int_0^t r^{2-a} dr + \int_t^\infty r^{1-a-\mathsf{q}} e^{-cr/2\rho} dr] e^{-ct/2\rho} \leq C[1 + \ln_+(\rho/t)] e^{-ct/2\rho}$. Terms with two image lines or composed reflections decay faster; images across the faces of $\square'$ not on F give $e^{-\delta p/\rho}$. □

Lemma B.56 (layer transfer)**.** *Let $\omega \in \mathcal{L}(\rho; m, \theta)$, $\rho \leq \rho' \leq 1$, $\delta > 0$, $\delta_2 := \min(\delta, \delta_b)/2$. Then (a) $\sup_y e^{\delta_2 t(y)/\rho'} \rho'^{-3} \sum_x 5^{-3j} e^{-\delta|x-y|/\rho'} |\omega(x)| \leq C\rho/\rho'$; (b) $\sum_{x \in Q} 5^{-3j} |\omega(x)| \leq Cs^2\rho$ for every cube Q of side $s \geq \rho$.*

Proof. $t(x) \geq t(y) - |x - y|$ gives $\delta|x - y|/2\rho' + \delta_b t(x)/2\rho \geq 2\delta_2 t(y)/\rho'$, which produces $e^{-\delta_2 t(y)/\rho'}$ from half of each exponential. For the rest, $t(x) = \min_P \mathrm{dist}(x, P)$ over the (at most six) face planes, the profile $f(s) := (1 + \ln_+(\rho/s))^m (\rho/s)^\theta e^{-\delta_b s/2\rho}$ is decreasing, and for one plane the sum factorises: $\sum_{x_\parallel} 5^{-2j} e^{-\delta|x_\parallel - y_\parallel|/2\rho'} \leq C\rho'^2$ and $\sum_{n \geq 1} 5^{-j} f(n5^{-j}) \leq C\rho$ because $\int_0^\infty (1 + \ln_+(1/u))^m u^{-\theta} e^{-\delta_b u/2} du < \infty$ for $\theta < 1$. (b) is the same with the cube's cross-section. □

Theorem B.57 (the ordered bound with layered vertices)**.** *Let G be a generalised graph of the cube expansion with its divergent sub-stages rewritten, every line obeying the slice bounds on Ω up to F, vertex functions bounded by sup norms except finitely many layered vertices $v_1, \ldots, v_r$ with ultraviolet lines, and assume* (DEG^B)*: for every admitted ordering and every stage, every connected component H has $D(H) \geq 0$, with $D(H) = 0$ only if H contains a layered vertex; let $\alpha_0^B > 0$ be the minimum positive degree. (i) With one vertex x_0 held (or none), the sum over orderings, compatible indices and positions of $|E^{(j)}(G)|$ is $\leq C\eta^{\rho_0} p^{n_1} (1 + k)^{p_0} \exp[-\delta\, \mathrm{dist}(\textit{external})] \Lambda(x_0)$ with $\Lambda(x_0) \leq C(1 + \ln_+(1/t(x_0)))^{\bar{n}+1} t(x_0)^{-\theta_{\max}} e^{-\delta_\Lambda t(x_0)}$, $\Lambda \leq$*

C if $t(x_0) \geq 1$, $\sum_{x_0 \in Q} 5^{-3j}\Lambda(x_0) \leq C$ per unit cube, $\Lambda = 1$ if no vertex is held. (ii) Let $0 < \gamma < \min(\alpha_0^B, 1-\theta_{\max})$. If every line, vertex function and layered weight obeys a pair bound $|F' - F| \leq 5^{-\gamma j}\rho_F^{-\gamma} \times$(the bound of F), a layered weight's difference lying in $\mathcal{L}(\rho_v; m, \theta_v + \gamma)$, then $|E'^{(j+1)}(G) - E^{(j)}(G)| \leq C5^{-\gamma j} \times$(the bound (i)).

Proof. (i) Run the multiscale induction of theorem B.40(a) with one change: when the line $l(t)$ being shrunk has an endpoint composite carrying a layer weight of width $\rho_w \leq \rho_{i_t}$ and its other endpoint is not held, sum the layered endpoint, splitting the retained half-exponential of $l(t)$ into three parts (the sum, the transfer of the summed endpoint's other half-exponentials, the layer transfer) and using lemma B.56(a) in place of the plain sum: the factor is $C\rho_{i_t}^3(\rho_w/\rho_{i_t})$, and the surviving vertex carries a layer weight of width ρ_{i_t}. If both endpoints are layered, sum one and keep the narrower weight; if the other endpoint is x_0, sum nothing new and move the weight to x_0. The gains of a layered composite created at step t_1 and summed at $t_2 < \cdots < t_q$ multiply to $\leq \prod_{t=t_1}^{t_q-1}(\rho_{i_t}/\rho_{i_{t+1}})$, i.e. $+1$ in the exponent at every step in $[t_1, t_q)$ (an exact rewriting, each ratio ≤ 1); if the composite is never merged into x_0, the last step is the volume sum, lemma B.56(b). At every step the exponent is $D(H_t)$ plus the number of active layered composites, which by (DEG^B) is $\geq \min(\alpha_0^B, 1) > 0$, except at steps where a $D = 0$ component contains x_0 and an inactive layered composite: there the layer factor $e^{-\delta' t(x_0)/\rho_*}$ at x_0 forces $\rho_* \geq ct(x_0)$ up to a summable tail, the number of admissible index tuples for the at most $\bar{n}$ such steps is $\leq C(1 + \log_5(1/t(x_0)))_+^{\bar{n}}$, the sum over the merge index gives one more logarithm and $e^{-\delta_\Lambda t(x_0)}$, and a θ-weight adds $t(x_0)^{-\theta}$: this is $\Lambda(x_0)$. The infrared stage is unchanged, no infrared slice meeting a layered vertex. (ii) theorem B.40(b) with (i) in place of the plain bound: the finer slices carry $\rho_i^D = \rho_i^{D-\gamma}\rho_i^\gamma$ with $\rho_i^\gamma \leq 5^{-\gamma j}$ for $i \leq 0$, the exponents $D - \gamma$ and $D + 1 - \gamma$ staying positive; the replacement telescoping weakens one element by γ with the prefactor $5^{-\gamma j}$, a weakened layered weight being of class $\theta_v + \gamma < 1$, still admissible in lemma B.56. □

Theorem B.58 ((B5′), photon part)**.** *Let a member family of the cube expansion be localised in zone B, every photon two-point prefix component S' rewritten by the zeroth- and first-moment extraction of* [3, (3.26)–(3.31)] *with the lines $G_{j_0} = C_D^\xi + G_{j_0}(1 - m_{j_0}^2 - a_{j_0}P_{j_0})C_D^\xi$, and every other divergent source-free stage rewritten as in definition B.44. Inside each collapsed piece write every cube line as free line plus image part and every internal sum over Ω as the sum over $5^{-j}\mathbb{Z}^3$ minus the sum outside, and rewrite the free piece by the free rules. Then: (a) the cube is compared with the free lattice only inside the collapsed pieces and at the two cancellation sites of the bulk structure (the free counterterm against the scalar local parts, and the reflection symmetry for the first moment), piece by piece: the zeroth moment of the C_D^ξ-part vanishes on both sides; the residual $\Delta c^{S'}$; the face piece M; and the remaining differences. (b) Each difference piece is a layered vertex at stage $j_0(S')$: $M \in \mathcal{L}(\rho_{j_0}; 1, 0)$ (lemma B.55), one leg differentiated, of the degree of the bulk first-moment vertex; $\Delta c^{S'}$, with $\sup \leq C\rho_{j_0}^{-1}e^{-\delta t/\rho_{j_0}}$ (one line replaced by its image part, $\sum_w 5^{-3j}|\nabla G_{j_0}^{\mathrm{img}}(x,w)| \leq C\rho_{j_0}e^{-\delta' t/\rho_{j_0}}$ since $|x - gw| \geq \max(|x-w|, t)$), of class $\mathcal{L}(\rho_{j_0}; 0, 0)$ and degree $D(S')$; the remaining pieces, ultraviolet-finite at j_0 with an image line, of class $\mathcal{L}(\rho_{j_0}; 0, 0)$ by the per-scale sum of* [14, (3.17)–(3.18)] *in the $D > 0$ case, with degrees $\geq D(S') + 1 + \alpha$ (Taylor remainders) and ≥ 1 (block members, lemma B.41). No piece has weight t^{-1}. (c)* (DEG^B) *holds for every difference graph (lemma B.54(iii), the rewriting never lowering the degree of a containing stage), so theorem B.57 applies: the pair bound with $5^{-\gamma j}$, the face piece's difference being of class $\mathcal{L}(\rho_{j_0}; 1, \gamma)$ times $5^{-\gamma j}\rho_{j_0}^{-\gamma}$ by the proof of lemma B.55 with one line replaced by its difference. (d) In a host rooted at an interior source vertex ($t(x_0) \geq \eta^{-1}/2$), $\Lambda(x_0) \leq C$: the zone-B photon pieces cost a constant. For the source-free field-dependent classes, root-summed, $\sum_{x_0} 5^{-3j}\Lambda(x_0) \leq C$ per unit face cell (lemma B.56(b)), the pair difference carries $5^{-\gamma j}$, and the zone-B total is $\leq C|\partial T^{(j)}| \leq C\eta|T^{(j)}|$ relative to the bulk form.*

Proof. (a) The image representations are exact and the rewriting rules are identities on each side; the difference of two rewritten pieces is the sum of the differences of their parts. (b) lemma B.55, the one-loop computation with one image line, and the per-scale sum with lemma B.54(ii)

(no $D=-1$ piece other than the one-loop class, whose C_D^{ξ}-zeroth moment is exactly zero). (c) lemma B.54(iii) and the degree bookkeeping of the pieces. (d) theorem B.57(i), (ii) and lemma B.56(b). □

B.10. **Assembly.**

Proof of proposition 5.5. Fix $J \geq J_0$, f, x, $s \in \{u_x, 0\}$, and $k \geq k_0$, $J_* \leq j \leq k - J_*$ with $\eta \leq \eta_1$, where k_0, η_1 are such that $j \leq k - J_0$, $\eta \leq \min(\eta_0, \delta_*)$ (lemma B.12, lemma B.20) and $J_* \geq 1$. On $\operatorname{supp} \chi_j$ write $S_{N,k}^{(j)} - S_{N,k+1}^{\prime(j+1)}$ by (58) for both members; the constants $\ln N_j$, $\ln N_j(0)$, $\ln N'_{j+1}$, $\ln N'_{j+1}(0)$ and the field- and source-free parts of $P^{(j),1}$, $P^{\prime(j+1),1}$ are in $v_{k,j}$, $v_{k+1,j+1}$ and drop out. The remaining difference is the sum of (B1)–(B5′), with (B6) inside (B3)–(B5′):

- (B1) $\leq C5^{-2j}p^2(\mu_0\eta)^{-2}\eta^{-3}|C_J|$ by proposition B.2;
- (B2) $\leq C_\varsigma p^{c+2}\lambda^{-1/2}(\mu_0\eta)^{-1}\eta^{-7/2}(5^{-\gamma' j}+\eta^\varsigma)|C_J|$ by corollary B.8 (its zero-background part also by proposition B.10(c));
- (B3) $\leq CNe^2p^{c+2}(\mu_0\eta)^{-1}\eta^{-2}(5^{-\gamma' j}+5^{-j}+\eta^\varsigma)|C_J|$ by corollary B.19;
- (B4) and (B5): the curvature classes and the source-free field-dependent classes; their zone-I localisations by theorem B.37 (through lemma B.21, lemma B.22, theorem B.24, theorem B.40), their zone-B tails (the $R > p$ branch of a host rooted at the interior source vertex, lemma B.20(ii)) by theorem B.51(a) and theorem B.58(d) at $\Lambda \leq C$ with the factor $e^{-\delta R/2}$, and the position sum by lemma B.42: together $\leq C[5^{-\gamma' j}\eta^{-\beta'}+\eta^{\sigma'}]|C_J|$;
- (B5′): the source-free field-dependent classes localised in zone B, with the rate, by theorem B.51(b),(c) and theorem B.58(c),(d) (mixed members with scalar remainders and photon face pieces included): $\leq C5^{-\gamma' j}\eta^{-\beta'}|C_J|$;
- (B6): the members with block vertices are bounded inside each of the above through lemma B.41 and theorem B.40, which supply the degree and the ordered bound in the split count; their pair bound is theorem B.40(b).

Every piece is of the form $C(5^{-\gamma j}\eta^{-\beta}+\eta^\sigma)$ for some $\gamma \in (0,1)$, $\beta, \sigma > 0$ (the factor $|C_J|$ and the powers of $p = p(\eta)$, $(1+k)$ are absorbed on the range $j \leq k$, $\eta = 5^{j-k}$, into $\eta^{-\beta}$ by an arbitrarily small loss in β, and $5^{-\gamma' j}$ with $\gamma' \geq 1$ is weakened to $\gamma < 1$). Take the smallest γ, the largest β and the smallest σ among the pieces, and the sum of the constants as C_3. The constants depend on the point of King's range, J (through $|C_J|$ and J_0), f (through $\max\|f_{\mu\nu}\|_{C^{2,\alpha}}$ and β_f) and x (through $|x|$ in the source vertex), and on nothing else. This is (3). □

Appendix C. Proof of proposition 6.6: the small-field lines at a non-negative scalar source

This appendix proves proposition 6.6: on King's cube C_J^K, at a real source $s = (u, h)$ with $h \geq 0$, the lower line $Z_k(s) \geq e^{-C\eta^\sigma} I_k^{(j)}(s)$ and the small-field half of the upper line, $\int P\, d\mu_{k,s} \leq e^{C\eta^\sigma} I_k^{(j)}(s)$, hold on both members of every diagonal pair, uniformly on compact source families. The large-field remainder at $s \neq 0$ is not part of the statement; it is supplied where needed by lemma 6.12.

The proof is the insertion of the scalar source into the small-field chain of the proof of proposition 5.4, that is, into Bałaban's inductive lower bound [1, Sect. 3] and into the term of his upper bound [2, (2.43)] in which no large-field block occurs at any scale, both run on C_J^K with King's flow of section 5.1 and with the curvature source u already present. The point is that the scalar source enters King's effective action through one vertex only, [13, (3.21)], whose coupling carries two powers of the running scale where the other vertices carry at most one and a half; every step of the chain is either blind to it, or a bound on the interaction families which that margin leaves unchanged, or a piece of bookkeeping that is written out below. The notation is that of sections 5.1 and 6.1.

C.1. **Scales, units, and the source vertex.** Fix the couplings, $J \geq J_0$ and a spacing $\varepsilon_k = 5^{-k}$. At step i of King's flow ($0 \leq i \leq j$) the block side is $\ell_i := 5^i\varepsilon_k$; the blocks of side ℓ_i tiling X_J form the lattice $\Lambda^{(i)}$, $|\Lambda^{(i)}| = \ell_i^{-3}|C_J|$; and $p_i := p(\ell_i) = b_0(1+\ln \ell_i^{-1})^p$ is Bałaban's threshold at

that scale, with $p = 6$ as in [1, (3.54)]. In *unit-i units* lengths are measured in ℓ_i, the fine lattice has spacing 5^{-i} and $\sum_{x\in B} 5^{-3i} = 1$ over the fine sites of a block B. At the stopping scale $i = j$ one has $\ell_j = \eta$; on a diagonal pair $\eta = \eta_k$.

The sources are supported at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_J$, so in unit-$i$ units every source vertex lies at distance $> 1/\ell_i$ from the Dirichlet faces. We write $\mathsf{S}(s) = \sum_i |x_i| \max_{\mu\nu} \|f_{i,\mu\nu}\|_{C^{2,\alpha}} + \|h\|_\infty$ as in section 6.3 and fix S_0 with $\mathsf{S}(s) \le S_0$.

The source vertex. In King's generating function [13, (3.17)] the scalar source appears as the vertex [13, (3.21)], which at step i reads, in unit-i units,

$$\theta\, \ell_i^2 \sum_x 5^{-3i}\, h(x) \sum_{a=1}^{N} \Big[\big(\phi'_a(x) + \phi_a^{(i)}(x)\big)^2 - C^{(i)}(x,x) \Big], \tag{59}$$

with θ the amplitude parameter of the vertex ($\theta = 1$ in the action), ϕ' the fluctuation field, $\phi^{(i)}$ the background (external) scalar field of the step, and $C^{(i)}(x,x)$ King's Wick constant, the diagonal of the free scalar covariance at the running scale ($(-\Delta^{\eta,\Omega} + m^2(5^i\varepsilon_k)^2)^{-1}(x,x)$ in King's notation; at $i = 0$ it is the free Wick constant $\mathsf{W}_{\varepsilon_k}$ of definition 2.3). Its *coupling power* is 2: the other vertices of King's list [13, (3.18)–(3.26)] carry $\lambda\ell_i$ (the quartic), $(e\ell_i^{1/2})^m$, $m \ge 1$ (the current vertices), $e^2\ell_i$ (the seagull), $\tau\ell_i^{3/2}$ (the curvature source [13, (3.20)]) and $O(\ell_i)$ (the mass counterterm, by [3, (1.29)]). The vertex (59) carries no photon field, is gauge invariant and charge neutral, and has at most two scalar legs.

C.2. **Where the sources sit.**

Lemma C.1 (where the sources sit)**.** *Let $S_k^{(i)}(\cdot; s)$ be King's effective action after i steps on C_J^K at the source $s = (u, h)$, and $P_k^{(i)}(\cdot; s)$ its interaction part* [13, (3.17)].

(a) *For every i and s, $S_k^{(i)}(\cdot; s) - S_k^{(i)}(\cdot; 0) = P_k^{(i)}(\cdot; s) - P_k^{(i)}(\cdot; 0)$: the Gaussian parts $\frac12\langle A_i, \Delta^{(i)} A_i\rangle$, $\frac12\langle \phi_i, \Delta^{(i)}(A^{(i)})\phi_i\rangle$, the normalisation constants $\ln Z_i$, $\ln Z_i(A^{(i)})$ and the vacuum counterterms E_0, E_1 are source-free.*

(b) *The indicator χ_i (King's* [13, (3.2)] *times the face factor* [14, (5.41)]*), every test of Bałaban's decomposition of unity at every scale* [2, (2.2)–(2.10)]*, and the small-field classes inserted at the intermediate scales of the lower bound* [1, Sect. 3] *are source-free functions of the block and fluctuation fields. Hence the product P of the small-field factors, the admissible histories of large-field blocks and the region geometry are source-free.*

(c) *Every renormalisation-group operator (the averaging operators, the block covariances $\Delta^{(i)}$, $C^{(i)}$, the propagators G_i, the backgrounds, the sub-domain operators and the difference lines between propagator domains), every object of the photon sector, every positivity statement and every Gaussian integration of the chain is source-free.*

(d) *At step 0 the scalar source enters the bare interaction as $-\varepsilon_k^2 h(x)\big[|\phi(x)|^2 - N\, c_W(x)\big]$ in the exponent, in unit-ε_k units ($\varepsilon_k^3 h|\phi_{\rm phys}|^2 = \varepsilon_k^2 h|\phi_{\rm unit}|^2$ with $\phi_{\rm phys} = \varepsilon_k^{-1/2}\phi_{\rm unit}$; $c_W := \varepsilon_k \mathsf{W}_{\varepsilon_k}$ is a unit-lattice Wick constant, $0 < c_W \le C_W$).*

(e) *Consequently a step of the chain is touched by the scalar source only if it bounds, expands or removes $P_k^{(i)}$ or the bare interaction, or assembles such bounds. Every other step is h-blind.*

Proof. (a) King's [13, (3.14)] reads $S^{(k),1}(T^{(k)}, A_k, \phi_k, g, h) = \frac12\langle A_k, \Delta^{(k)} A_k\rangle - \ln Z_k + \frac12\langle \phi_k, \Delta^{(k)}(A^{(k)})\phi_k\rangle - \ln Z_k(A^{(k)}) + E_0 + P^{(k),1}(T^{(k)}, A_k, \phi_k, g, h)$, with E_0 given by [13, (3.15)], a free Gaussian integral; the sources g, h occur in $P^{(k),1}$ alone. By [13, (3.17)], $P^{(k),1}$ is a sum of derivatives with respect to the parameters $e', \lambda', \tau, \theta$ of $E'_k = -\ln \int \prod_j d\mu_{C(j),L^j\eta}(A'_j)\, d\mu_{G_k(A^{(k)})}(\phi')\, e^{V^{(0),\eta}}$, whose fluctuation measures are source-free Gaussians, and in the vertex list [13, (3.18)–(3.26)] of $V^{(0),\eta}$ the curvature source occurs only in (3.20) and the scalar source only in (3.21). The same holds for E_1, King's [13, (3.35)] on the cube, which is built from the source-free part of the same expansion. (b) [13, (3.2)] and [14, (5.41)] are inequalities on (A_i, ϕ_i); Bałaban's tests [2, (2.2)–(2.10)] and the classes of [1, Sect. 3] (King's [13, (A.1)]) are inequalities on the fields. (c) The operators are built from the Gaussian forms and the averaging operators, which carry

no source; the photon sector does not see h ((59) has no photon field); the positivity statements and Gaussian integrations are statements about the source-free kernels. (d) [13, (3.21)] at $k = 0$ is the bare source with King's Wick constant; Bałaban's bare interaction [2, p. 559] is $\lambda(\varepsilon)|\phi|^4 + \frac{1}{2}\delta m^2\,\varepsilon^2|\phi|^2$, to which the source adds the displayed term. (e) is the summary of (a)–(d). □

C.3. **Power counting with the source vertex.** In the power counting of [13, Sect. 3] and [3, Sects. 1–3] a vertex v whose coupling carries $\ell_i^{u_v}$ in unit-i units contributes u_v, and a connected graph γ with E_γ external legs (scalar or photon) has degree

$$D(\gamma) \;:=\; \sum_{v\in\gamma} u_v - 3 + \tfrac{1}{2}E_\gamma; \tag{60}$$

the subgraphs that are renormalised (the divergent ones) are those with $D(\gamma) \le 0$. The quartic has $u = 1$, the current vertex of order m has $u = m/2$, the seagull $u = 1$, the mass counterterm $u \ge 1$, the curvature vertex $u = 3/2$, the remainder vertices of order $\bar{n}+1 = 14$ have $u \ge 7$, and the source vertex (59) has $u = 2$.

Lemma C.2 (the divergent graphs containing a source vertex)**.** *Let γ be a connected graph with at least one line.*

(i) *If γ contains exactly one vertex (59) and no curvature vertex, then $D(\gamma) \le 0$ only in the following cases:* (T) *$E_\gamma = 0$, the graph is a field-free h-linear vacuum graph whose other vertices have total coupling power ≤ 1, that is, of order $(\alpha,\beta) \in \{(0,0),(0,1),(2,0)\}$ in (e^2,λ), with the mass counterterm counted as its partner;* (F) *$\gamma = \gamma_F$, the graph with the source vertex at x, one current vertex of order $m = 1$ at y, the two scalar lines joining x and y, and the photon leg of y external; $D(\gamma_F) = 0$, and its amplitude vanishes identically.*

(ii) *If γ contains two or more vertices (59), or one of them and a curvature vertex, then $D(\gamma) \ge \frac{1}{2}$.*

Consequently no subgraph containing a source vertex and an external leg is renormalised, and the renormalisation of the h-free divergent subgraphs inside a family with a source vertex is the one at $h = 0$.

Proof. (i) With one source vertex, $D(\gamma) = \sum_{v\neq\text{source}} u_v - 1 + \frac{1}{2}E_\gamma$, so $D \le 0$ forces $\sum_{v\neq\text{source}} u_v + \frac{1}{2}E_\gamma \le 1$. If $E_\gamma = 0$ the other vertices have total coupling power ≤ 1: none (the Wick tadpole, order $(0,0)$), one quartic or one mass counterterm (order $(0,1)$), or two current vertices of order 1, or one seagull, or one current vertex of order 2 (order $(2,0)$); a single current vertex of order 1 is excluded at $E_\gamma = 0$, its photon leg having no partner. If $E_\gamma = 1$ the other vertices have total power $\le \frac{1}{2}$, so there is exactly one current vertex of order 1; the external leg is its photon leg (a lone scalar leg cannot be external when the source vertex's two legs are both to be contracted and the current vertex's two scalar legs are the only other ones), both scalar legs of each vertex are contracted into lines, and the graph is γ_F. If $E_\gamma = 2$ there is no other vertex and no line, which is excluded. If $E_\gamma \ge 3$ then $D \ge \frac{1}{2}$. The amplitude of γ_F is the scalar loop with one insertion of $|\phi|^2$ and one insertion of the current $e\,A\cdot(\phi, q\,\partial\phi)$, whose component trace is $\operatorname{Tr} q$ with q the antisymmetric charge generator of the $U(1)$ action on $\mathbb{R}^N$; $\operatorname{Tr} q = 0$, so γ_F vanishes. (ii) Two source vertices give $\sum_v u_v \ge 4$, hence $D \ge 1$; a source vertex and a curvature vertex give $\sum_v u_v \ge \frac{7}{2}$, hence $D \ge \frac{1}{2}$. The consequence: by (i), (ii), the only divergent subgraphs with a source vertex are vacuum graphs ($E = 0$), which are not Taylor-subtracted at external legs, and γ_F, which is zero; the divergent subgraphs with external legs are therefore source-free, and they are renormalised as at $h = 0$. □

The three field-free h-linear families of case (T) are exactly the members excluded from the class $\mathcal{N}_h$ of lemma 6.1. In the small-field chain they are never bounded: they are field-independent constants of the effective action, carried exactly (appendix C.5, step B7).

C.4. **The margin.** The graph bound used throughout the chain is Bałaban's bound in product form [3, Prop. 1, (1.33)], King's [13, Thm. 3.5, (3.38)] on the cube: in unit-i units, on Bałaban's small-field set, every renormalised connected family of $P_k^{(i)}$ with at least one external field has a coefficient bound, per block, $O(1)\,\ell_i^{\rho}\,p_i^{n_1}e^{-\delta_0 d}$, where d is the tree length of the vertex positions and the power ρ is the sum over the vertices of the coupling power u_v, less $\frac14$ for every external scalar leg: on Bałaban's intermediate field set the external scalar field is bounded by $|\psi(y)| \le 2\lambda(\varepsilon)^{-1/4}p(\varepsilon)$ [2, p. 559], that is by $C\,p_i\,\ell_i^{-1/4}$ in unit-i units, while lines and photon legs cost no power. Writing $\epsilon_v := u_v - \frac14\,(\text{number of external scalar legs of } v)$, the power is $\rho = \sum_v \epsilon_v$, and $\epsilon_v \ge 0$ for every vertex of the list [3, (1.6)–(1.15)]. For the families with a source vertex this bound is the cube form of the (3.38) bound of lemma 6.1; the vertex (59) is admitted into the list as an ordinary local two-leg vertex with coefficient bounded by $\|h\|_\infty \le S_0$, since by lemma C.2 it creates no new divergent subgraph with external legs, so that the renormalisation structure of the families, and with it the proof of the product-form bound, is the one at $h = 0$.

Lemma C.3 (the margin of the source families)**.** *In unit-i units, on Bałaban's step-i intermediate field set, every renormalised connected family of $P_k^{(i)}(\cdot\,;s)$ with at least one source vertex (59) and at least one external field (scalar, block or photon) has a coefficient bound, per block,*

$$C(S_0)\,\ell_i^{3/2}\,p_i^{N'}\,(1+\log_5\ell_i^{-1})^{p_0} \;\le\; C'(S_0)\,\ell_i^{5/4},$$

with $C'(S_0)$ independent of i, k and of s with $\mathsf{S}(s) \le S_0$. The same holds for the families with a curvature vertex and no source vertex.

Proof. The source vertex has coupling $\theta\ell_i^2 h$ and at most two scalar legs, so $\epsilon_v \ge 2 - \frac12 = \frac32$; the curvature vertex [13, (3.20)] has $\epsilon_v = \frac32$ (no scalar leg); every other vertex has $\epsilon_v \ge 0$. Hence $\sum_v \epsilon_v \ge \frac32$ for every family containing a source or a curvature vertex. By lemma C.2 no Taylor subtraction sits at a subgraph containing the source vertex, so the product-form bound applies with (59) as a local vertex of sup norm $\le S_0$; the intermediate set enlarges the leg bounds of King's set [13, (3.2)] by a fixed factor, which is absorbed into $C(S_0)$; and $p_i^{N'}(1+\log_5\ell_i^{-1})^{p_0} \le C\ell_i^{-1/4}$. The constants depend on s only through $\|h\|_\infty$ and $\max\|f_{i,\mu\nu}\|_{C^{2,\alpha}}$, hence through S_0. Near the Dirichlet faces: the source vertices lie at unit-i distance $> 1/\ell_i$ from $\mathbb{R}^3 \setminus C_J$, so a family with a source vertex and a vertex within the face zone has tree length $d > 1/\ell_i$ and carries the additional factor $e^{-\delta_0 d} \le e^{-\delta_0/(2\ell_i)} \le \ell_i^{10}$ beyond the face bounds of the proof of proposition 5.4; it enters every estimate through this tail only. □

The consequence used below is that the uniform exponent $\beta_0 = \frac14 - \beta$ of the coefficient bounds [1, Prop. 3.2] and the summability condition $(\bar n+1)\beta_0 > 3$ of the cumulant step hold at s unchanged, since $\frac54 \ge \beta_0$; and that the remainder terms of order $> \bar n$ and the terms of total power > 3 in ℓ_i [1, Prop. 3.1] keep their bound $O_{S_0}(\ell_i^{\kappa'})|\Lambda^{(i)}|$ with the same $\kappa' > 3$.

C.5. **The two small-field lines.**

Theorem C.4 (the small-field lines at $h \ge 0$)**.** *Fix $J \ge J_0$, $\alpha \in (0, \frac12)$ and $S_0 > 0$. There are k_0, $C(S_0)$ and $\sigma = \kappa' - 3 > 0$, depending on the couplings, J, α, S_0 and on nothing else, such that for every real source $s = (u, h)$ with $h \ge 0$ and $\mathsf{S}(s) \le S_0$, every $k \ge k_0$ and both members $(k', j') \in \{(k, j(k)), (k+1, j(k)+1)\}$ of the diagonal pair, with $b := C(S_0)|C_J|\,\eta_k^{\sigma}$,*

$$Z_{k'}(s) \;\ge\; e^{-b}\,I_{k'}^{(j')}(s) \qquad \text{and} \qquad \int P\,d\mu_{k',s} \;\le\; e^{b}\,I_{k'}^{(j')}(s).$$

Since $C(S_0)|C_J|$ depends on J and S_0 only, this is (11), and proposition 6.6 follows.

Proof. We fix one member, write k for its spacing index and j for its number of steps, and run the small-field chain of the proof of proposition 5.4 at the source s instead of $(u, 0)$. By lemma C.1(e) only the steps that bound, expand or remove the interaction, or assemble such bounds, see the change; we go through them.

The lower line is Bałaban's inductive lower bound [1, Sect. 3], in the following seven steps.

B1 (restriction to small fields, [1, (3.6)]*).* The integrand $e^{-S}e^{F(u)+:\phi^2:(h)}$ of $Z_k(s)$ is positive, the sources being real. Inserting the source-free indicators (lemma C.1(b)) of the small-field classes at the scales $i<j$ and of χ_j at the stopping scale gives a lower bound. Nothing is paid for in a lower bound.

B2 (the inductive hypothesis, [1, (3.26)–(3.32)]*).* After i steps, $Z_k(s) \geq e^{-a_i}\int \chi_{(i)} e^{-S_k^{(i)}(\cdot;s)}$ with $a_i := \sum_{i'<i} C(S_0)\ell_{i'}^{\kappa'}|\Lambda^{(i')}|$, where $S_k^{(i)}(\cdot;s)$ is King's effective action whose vertex list contains [13, (3.20)–(3.21)] from the start, and $\chi_{(i)}$ is the inserted class indicator. The identities for the normalisation constants and for the representation of the fields after i steps are source-free (lemma C.1(c)).

B3 (the fluctuation step, [1, (3.37)–(3.49), Props. 3.1–3.2]*).* The Gaussian fluctuation integral is exact and source-free. The interaction, now with the source families, is expanded to order $\bar{n}=13$ with Cauchy estimates in the complexified parameters $(e',\lambda',\tau,\theta)$ of [13, (3.17)]; θ is the amplitude of (59), and on $|\theta|\leq 2$ the coupling of (59) is at most $2\ell_i^2 S_0$, so lemma C.3 applies with S_0 replaced by $2S_0$. The remainder terms and the terms of total power >3 are bounded by $O_{S_0}(\ell_i^{\kappa'})|\Lambda^{(i)}|$, $\kappa'>3$, and the coefficients of the polynomial part by $O_{S_0}(\ell_i^{\beta_0})$ per block, as at $h=0$: by lemma C.3 every family with a source vertex and an external field has coefficient $O(\ell_i^{5/4})$, within both bounds. The one-loop determinants are source-free.

B4 (cutoff removal and the consistency conditions, [1, (3.50)–(3.55)]*).* The cumulant bound used at this step in the proof of proposition 5.4 is blind to the weights and to the vertex list: it needs only that the interaction be a real polynomial in the fluctuation variables without constant term, with coefficients bounded as in B3. At the source s the fluctuation-independent parts of the expanded interaction, namely the background part $\theta\ell_i^2\sum 5^{-3i}h|\phi^{(i)}|^2$ of (59), the Wick member, and every field-free family, are pulled out of the exponent exactly, into $S_k^{(i+1)}(\cdot;s)$; what remains is a polynomial in the fluctuation without constant term, with coefficients as in B3. The compatibility inequality [1, (3.53)] between the classes of consecutive scales is a statement on fields (lemma C.1(b)), and the conditions [1, (3.54)] on p are unchanged.

B5 (the classes). The background classes inserted at the intermediate scales are source-free.

B6 (the final step, [1, (3.66)–(3.69)]*).* It gives the lower bound $Z_k(0)\geq e^{-E_-|C_J|}$ on the partition function at zero source; it is used in the proof of proposition 5.4 at $s=0$ only and is not needed here.

B7 (the sources). The curvature source is King's plaquette source [14, (2.7)]. The source vertex (59) is switched on. Its field-independent part, $-\theta\ell_i^2\sum_x 5^{-3i}h(x)\,N\,C^{(i)}(x,x)$ together with every field-free family it generates, is a c-number of $S_k^{(i)}(\cdot;s)$ at every step. A field- and fluctuation-independent constant in the action multiplies the integral and every intermediate integrand by the same factor; so in the small-field half it is common to $Z_k(s)$ and to $I_k^{(j)}(s)$, with King's Wick constant inside $S_k^{(j)}(\cdot;s)$ on both sides. No bound on these constants is needed, and none is used.

Assembly ([1, Sect. 3, end]*).* After j steps, $Z_k(s)\geq e^{-a}\int\chi_j e^{-S_k^{(j)}(\cdot;s)} = e^{-a}I_k^{(j)}(s)$ with

$$a=\sum_{i<j} C(S_0)\,\ell_i^{\kappa'}|\Lambda^{(i)}| = C(S_0)|C_J|\sum_{i<j}\ell_i^{\kappa'-3} \;\leq\; C(S_0)|C_J|\,\eta^{\sigma}, \qquad \sigma=\kappa'-3>0,$$

the sum being geometric and dominated by $i=j-1$, $\ell_{j-1}=\eta/5$; the constant does not depend on k or j. The restriction to χ_j at the last step is a positivity step inside the class, so the thresholds of χ_j never enter a constant.

The upper line's small-field part. In Bałaban's inductive formula [2, (2.43)] at the scale j, with the decomposition of unity of [2, (2.2)–(2.10)] inserted at every scale $i\leq j$ (source-free, lemma C.1(b)), $Z_k(s)$ is a sum over histories of large-field blocks, and $\int P\,d\mu_{k,s}$ is the term of the history with no large-field block at any scale $i\leq j$. Along this history the steps of the upper bound (the restriction to the small-field set, the translation to the background, the expansion of the interaction to order $\bar{n}$ with its cumulants, the bounds on the background and fluctuation fields, the Z-factors and the Gaussian integrations, [2, Sect. 2]) are carried out with the source families present and nothing is removed. Of these steps only the expansion of the interaction

sees the source; it gives, by lemma C.3 exactly as in B3, the same remainder bound $O_{S_0}(\ell_i^{\kappa'})|\Lambda^{(i)}|$ at every scale $i < j$, and the field-free constants are carried exactly as in B7. The replacement of Bałaban's cut photon background by the full one in the final integrand is an operation in the photon sector, source-free, done as in the proof of proposition 5.4. Hence

$$\int P\, d\mu_{k,s} \;\le\; \int \chi_j\, e^{-S_k^{(j)}(\cdot;s)+\sum_{i<j} C(S_0)\ell_i^{\kappa'}|\Lambda^{(i)}|} \;=\; e^a\, I_k^{(j)}(s),$$

with the same a as in the lower line; the last step uses that χ_j is the indicator of $I_k^{(j)}$ and that the integrand is positive, the exponent being real.

Both members. Both members of a diagonal pair have the stopping scale η_k, the same unit lattice and the same indicator $\chi_{j(k)}$ (section 5.1); every constant above depends on (k, j) only through η, so one $b = C(S_0)|C_J|\eta_k^\sigma$ serves both members, and $I_k^{(j)}(s)$ is the same kind of integral on both.

Uniformity on compact families. The source enters every constant only through $\|h\|_\infty$, by way of the coupling of (59) (no Taylor subtraction sits at a source subgraph, lemma C.2, so no derivative of h is ever taken), and through $\max_{\mu\nu}\|f_{i,\mu\nu}\|_{C^{2,\alpha}}$ as in the proof of proposition 5.4. The constants are non-decreasing functions of S_0; so the statement is uniform on $\{\mathsf{S}(s) \le S_0\}$. The sign $h \ge 0$ is not used in this proof; it is the hypothesis under which lemma 6.12 supplies the large-field remainder, and it is kept in the statement because the proposition is consumed only there.

Every point of King's range. Each step holds at every point (e, λ, m, μ_0) of King's range with point-dependent constants, in particular at the mass M_2 of section 6.1 with the same e, λ, μ_0. □

C.6. **The insertion, step by step.** For completeness we list how each step of the small-field chain of the proof of proposition 5.4 is affected by the scalar source; the marks are: **blind**, the step is source-free by lemma C.1; **family**, the step is a bound on the families of the interaction, read at s through lemma 6.1 and lemma C.3; **rewritten**, the step's bookkeeping changes at s and is written out in appendix C.5.

step	mark	reason
the operator layer: block covariances, propagators and their decay, sub-domain operators, difference lines, Green's function regularity [4]	blind	h is a vertex; it enters no Gaussian form
the graph bound in product form [3, Prop. 1], King's [13, (3.38)], on the cube	family	source families by lemma 6.1; the margin by lemma C.3; the face zone through tails only
the Taylor step at a divergent two-leg subgraph	family	no subtraction at a source subgraph (lemma C.2; γ_F vanishes)
the Ward identities for the renormalised classes	family	(59) is gauge invariant; the identities are unchanged
the bounds on propagator sub-domains and the domain-change estimate [2, Prop. 2.6], lemma 5.1	family	the same reading with (59) in the list
[1, Props. 3.1–3.2]: remainder terms and coefficient bounds	family	at s with lemma C.3; exponents unchanged
the uniform exponent β_0	family	lemma C.3: $\epsilon_v \geq \frac{3}{2}$ at the source vertex
the small-field tests, the classes, their compatibility, the external-field properties	blind	functions of the fields
B1 positivity	rewritten	real sources
B2 inductive hypothesis	rewritten	$S_k^{(i)}(\cdot; s)$
B3 fluctuation step	family	Cauchy estimates also in θ
B4 cutoff removal	family	constants pulled out exactly
B5 classes	blind	
B6 final step	blind	consumed at $s = 0$ only
B7 sources	rewritten	the source vertex switched on
lower-line assembly	rewritten	the lower line at s
positivity lemmas, the decomposition of unity, the region geometry	blind	
the restriction to small fields and the translation to the background in the upper bound [2, (2.11)–(2.14), (2.43)]	rewritten	the bare source re-expressed after the translation, exactly
the expansion of the interaction in the upper bound	rewritten	with the source families; nothing removed on the small-field history
backgrounds, fluctuation bounds, Z-factors, last-step masses, positivity, diamagnetic integration	blind	
the per-block costs of the final scale and the final-scale bound	blind	consumed in the large-field remainder only
the upper-line assembly	rewritten	the small-field part at s

Also outside the enumerated steps: the basic positivity lemma, the composition identities of the flow and the Schur complements are blind; the definition of the effective action carries the vertex list with (59) switched on; the c-number lemma (a field- and source-independent constant multiplies every integrand by the same factor) is used at B7 with King's Wick constant in place of a scheme constant; the face lemmas of the cube (the Dirichlet mismatch of the counterterm, the face terms of the photon sector) are blind, carrying no source vertex.

Appendix D. The one-step bound with the scheme c-number: proof of proposition 6.7

This appendix proves proposition 6.7. The notation is that of sections 5.1 and 6.1: k is the spacing index, j counts renormalisation steps, $\eta = 5^{j-k}$ is the block scale, the diagonal pair at k consists of the k-model after $j(k)$ steps and the $(k+1)$-model after $j(k)+1$ steps (both with block scale $\eta_k = 5^{j(k)-k}$, the same unit lattice and the same indicator $\chi_{j(k)}$), $S_{N,k}^{(j)}(\cdot; s) = S_k^{(j)}(\cdot; s) - v_{k,j}$ is the normalised action, $\mathsf{S}_k(h) = \sum_{x \in \Lambda_k \cap C_J} \varepsilon_k^3 h(x)$ and $c_k(h) = z_k \mathsf{S}_k(h)$. The cube is King's RG cube C_J^K (definition 2.2), $J \geq J_0$ is fixed, and every source is supported at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_J$.

The proof has three parts. First, on the support of χ_j the normalised action at the source (u,h) is split exactly into the normalised action at $(u,0)$, the members of the family $\mathcal{N}_h$ of lemma 6.1, and the h-linear vacuum constant $\mathcal{V}_k^{(j)}(h)$ of the three lowest orders (appendix D.1). Second, the one-step differences of the first two pieces are bounded by proposition 5.5 and lemma 6.1, uniformly on compact families by a polarisation argument (appendix D.2). Third, $\mathcal{V}_k^{(j)}(h) + c_k(h)$ is a local functional of h: it equals $\bar\beta_k^{(j)} \int h$, with $\bar\beta_k^{(j)}$ the block mean of an infinite-lattice one-point density, up to errors that are super-exponentially small in $1/\eta$, and $\bar\beta_k^{(j)}$ converges along the diagonal pairs at a power rate, the *bulk sequence* (appendix D.3). The three pieces are assembled in appendix D.4.

D.1. **The split of the effective action.** On the support of χ_j King's effective action after j steps is [13, (3.14)] (cube form [14, (3.12)])

$$S_k^{(j)}(A_j,\phi_j;s) = \tfrac12\langle A_j,\Delta^{(j)}A_j\rangle - \ln Z_j + \tfrac12\langle\phi_j,\Delta^{(j)}(A^{(j)})\phi_j\rangle - \ln Z_j(A^{(j)}) + E_0 + P^{(j),1}(A_j,\phi_j;s),$$

where $P^{(j),1}$ is the sum of the connected renormalised graphs H: the Taylor coefficients of King's E_j' [13, (3.17)] in the parameters $(e',\lambda',\tau,\theta)$ up to order $\bar n = 13$. The sources enter E_j' only through the vertices [13, (3.20)] (the curvature source, parameter τ, linear in u) and [13, (3.21)] (the scalar source, parameter θ, linear in h). For a graph H let $n_u(H)$ and $n_h(H)$ be its numbers of curvature and of scalar source vertices, $(\alpha(H),\beta(H))$ its order in (e^2,λ), $E(H;A_j,\phi_j;s)$ its value and $E_N(H;\cdot;s) := E(H;\cdot;s) - E(H;0,0;s)$ its field-dependent part. At every field configuration $E(H;\cdot;(u,h))$ is a homogeneous polynomial of degree $n_u(H)$ in u and of degree $n_h(H)$ in h, and $n_u(H)+n_h(H)\le\bar n$. The number of graphs is finite and depends on $\bar n$ only.

Definition D.1 (the three families)**.** Put $P_0 := \{(0,0),(0,1),(2,0)\}$ and

$$\mathcal{T}_h := \{H:\ n_h(H)=1,\ n_u(H)=0,\ (\alpha(H),\beta(H))\in P_0\},\qquad \mathcal{V}_k^{(j)}(h) := \sum_{H\in\mathcal{T}_h} E(H;0,0;(0,h)),$$

the *h-linear vacuum constant of the three lowest orders*; it is linear in h and does not depend on u. The family $\mathcal{N}_h$ consists of the field-dependent parts $E_N(H;\cdot;(u,h))$ of all graphs with $n_h(H)\ge1$ and of the field-independent values $E(H;0,0;(u,h))$ of all graphs with $n_h(H)\ge1$ and $H\notin\mathcal{T}_h$. This is the family of lemma 6.1; the excepted graphs are exactly those of $\mathcal{T}_h$, which carry no curvature vertex. In particular the h-linear graphs of the orders P_0 which do carry curvature vertices (for instance the order-e^2 graph in which the seagull vertex joins two curvature legs to the scalar loop through the vertex [13, (3.21)]) belong to $\mathcal{N}_h$.

Lemma D.2 (the split)**.** *On the support of χ_j, for every real source (u,h),*

$$S_{N,k}^{(j)}(\cdot;(u,h)) \;=\; S_{N,k}^{(j)}(\cdot;(u,0)) \;+\; \sum_{T\in\mathcal{N}_h} T \;+\; \mathcal{V}_k^{(j)}(h). \tag{61}$$

Proof. The quadratic forms, $\ln Z_j$, $\ln Z_j(A^{(j)})$, E_0 and the vacuum counterterm E_1 inside $V^{(0)}$ are source-free: the sources enter E_j' only through the vertices [13, (3.20)–(3.21)], so they appear in $S_k^{(j)}$ only through $P^{(j),1}$. A graph with $n_h(H)=0$ takes the same value at (u,h) and at $(u,0)$; a graph with $n_h(H)\ge1$ vanishes at $h=0$. Hence

$$S_k^{(j)}(\cdot;(u,h)) - S_k^{(j)}(\cdot;(u,0)) = \sum_{n_h(H)\ge1} E(H;\cdot;(u,h)) = \sum_{n_h(H)\ge1} E_N(H;\cdot;(u,h)) + \sum_{\substack{n_h(H)\ge1\\ H\notin\mathcal{T}_h}} E(H;0,0;(u,h)) + \sum_{H\in\mathcal{T}_h}$$

and for $H\in\mathcal{T}_h$ one has $n_u(H)=0$, so $E(H;0,0;(u,h)) = E(H;0,0;(0,h))$. Subtracting the source-free constant $v_{k,j}$ from both actions gives (61). □

D.2. **The field-dependent and the h-nonlinear parts.** For a diagonal pair write $D[T] := T|_{(k,j(k))} - T|_{(k+1,j(k)+1)}$ for the one-step difference of a member T of the expansion, a function on the support of $\chi_{j(k)}$ (the two members live on the same unit lattice).

Lemma D.3 (rows at a fixed source)**.** *Fix J, a family $f_1,\dots,f_n$, $x\in\mathbb{R}^n$ and a real h; let $s=(u_x,h)$. There is k_0 such that for $k\ge k_0$ (so that $J_*\le j(k)$ and $\eta_k\le\eta_1$), with $j=j(k)$, on the support of χ_j:*

(a) $\big|S^{(j)}_{N,k}(\cdot;(u_x,0)) - S^{(j+1)}_{N,k+1}(\cdot;(u_x,0))\big| \le C_3\big(5^{-\gamma j}\eta_k^{-\beta}+\eta_k^{\sigma}\big)$, *with the constants of proposition 5.5 at $s=u_x$;*

(b) $\big|\sum_{T\in\mathcal{N}_h} D[T]\big| \le C_h\big(5^{-\gamma' j}\eta_k^{-\beta'}+\eta_k^{\sigma'}\big)|C_J|$, *with C_h the constant of lemma 6.1 at the source s times the number of members of $\mathcal{N}_h$, and γ',β',σ' as there.*

Proof. (a) is (3) at $s=u_x$, applied with $j=j(k)$; both members of the pair are admissible since $j(k)+1\le(k+1)-J_*$. (b) is the third clause of lemma 6.1 for each member of $\mathcal{N}_h$, which bounds $D[T]$ on the support of χ_j by $C[5^{-\gamma' j}\eta^{-\beta'}+\eta^{\sigma'}]|C_J|$, summed over the finitely many members. □

The proposition asks for bounds uniform on a compact family of sources. The families to which it is applied in section 6 are of the form $\Sigma=\{(u_x,h_y):(x,y)\in Y\}$ with $h_y:=\sum_j y_j h_j$, $Y\subset\mathbb{R}^{n+p}$ compact, for fixed $f_1,\dots,f_n$ and $h_1,\dots,h_p$; we prove the uniformity for such families.

Lemma D.4 (uniformity on compact families)**.** *Let Σ be as above and let $W:=\{(f_i,0)\}_{i\le n}\cup\{(0,h_j)\}_{j\le p}$. Let $\mathcal{S}$ be the finite set of sources $\sum_{i=1}^{d}\epsilon_i w_i$ with $d\le\bar n$, $w_i\in W$ and $\epsilon_i\in\{\pm1\}$. There is $C(Y)<\infty$ such that for every function T of the source which is, at every field configuration, a homogeneous polynomial of degree $d\le\bar n$ in the source $w=(u,h)$,*

$$\sup_{(x,y)\in Y}\big|T(u_x,h_y)\big| \;\le\; C(Y)\max_{s'\in\mathcal{S}}\big|T(s')\big|$$

pointwise in the fields. This applies to every $E(H)$, $E_N(H)$, to every member of $\mathcal{N}_h$ and to their one-step differences $D[\cdot]$.

Proof. Let M be the symmetric d-linear form with $T(w)=M(w,\dots,w)$. By the polarisation identity, $M(w_1,\dots,w_d)=(2^d d!)^{-1}\sum_{\epsilon\in\{\pm1\}^d}\epsilon_1\cdots\epsilon_d\,T(\sum_i\epsilon_i w_i)$. Expanding $(u_x,h_y)=\sum_i x_i(f_i,0)+\sum_j y_j(0,h_j)$ by multilinearity, $T(u_x,h_y)$ is a sum of at most $(n+p)^d$ monomials in (x,y) of degree d, each multiplied by a value $M(w_{a_1},\dots,w_{a_d})$ with $w_{a_i}\in W$, which by polarisation is bounded by $\max_{s'\in\mathcal{S}}|T(s')|$. The monomials are bounded on the compact set Y. Each graph's value is such a polynomial (appendix D.1), and so is $D[\cdot]$ of it, since both members of a diagonal pair are evaluated at the same source. □

Corollary D.5 (rows K0 and SRC on a compact family)**.** *For Σ as above there are k_0 and constants C^{K0}, C^{SRC}, depending on the couplings, J, f, $h_1,\dots,h_p$ and Y only, such that for $k\ge k_0$, $j=j(k)$, every $\sigma=(u_x,h_y)\in\Sigma\cup\{0\}$ and every point of the support of χ_j,*

$$\big|S^{(j)}_{N,k}(\cdot;(u_x,0)) - S^{(j+1)}_{N,k+1}(\cdot;(u_x,0))\big| \le \theta_k^{\mathrm{K0}} := C^{\mathrm{K0}}\big(5^{-\gamma j}\eta_k^{-\beta}+\eta_k^{\sigma}\big), \tag{62}$$

$$\Big|\sum_{T\in\mathcal{N}_{h_y}} D[T]\Big| \le \theta_k^{\mathrm{SRC}} := C^{\mathrm{SRC}}\big(5^{-\gamma' j}\eta_k^{-\beta'}+\eta_k^{\sigma'}\big). \tag{63}$$

Proof. Write $S^{(j)}_{N,k}(\cdot;(u_x,0)) = S^{(j)}_{N,k}(\cdot;0)+\sum_{n_u(H)\ge1,\,n_h(H)=0}E(H;\cdot;(u_x,0))$ (as in the proof of lemma D.2, with the roles of u and h exchanged). The first term's one-step difference is (3) at $s=0$. The second is a polynomial in x without constant term, and its one-step difference is bounded, by lemma D.4, by $C(Y)$ times the maximum over the finitely many curvature sources $s'=(u_{x'},0)\in\mathcal{S}$ of $|D[S_N(\cdot;(u_{x'},0))]|+|D[S_N(\cdot;0)]|$, each bounded by lemma D.3(a) at x'. This gives (62) with C^{K0} the largest of the finitely many constants. Likewise $\sum_{T\in\mathcal{N}_{h_y}}D[T]$ is, at each field configuration, a polynomial in (x,y) (every member has $n_h\ge1$, so it has no y-free term), and lemma D.4 bounds it by $C(Y)$ times the maximum over $s'\in\mathcal{S}$ of $|\sum_{T\in\mathcal{N}_{h'}}D[T](s')|$,

each bounded by lemma D.3(b) at the source s'; $|C_J|$ is absorbed into C^{SRC}. The constants k_0 are the largest of the finitely many. □

D.3. The h-linear constant of the three lowest orders.

D.3.1. *The constant as a fluctuation one-point function.*

Lemma D.6 (the vacuum constant is a one-point function)**.** *Let $\langle\cdot\rangle^{\mathrm{fl}}_{k,j}$ be King's fluctuation measure* [13, (3.17)] *on C^K_J at zero block fields $A_j = \phi_j = 0$: the Gaussian measure of the fluctuation fields with the covariances $G_j(0)$ and $G^{\mathrm{ph}}_{(i)}$ of the j steps, weighted by $e^{V^{(0)}}$ with the vertices* [13, (3.18)–(3.26)] *at zero source, and let $[\,\cdot\,]_{P_0}$ denote the sum of the Taylor coefficients at the orders P_0 in (e^2,λ). Then*

$$\mathcal{V}^{(j)}_k(h) = \sum_{x\in\Lambda_k\cap C_J} \varepsilon_k^3 h(x)\, v^{(j)}_k(x), \qquad v^{(j)}_k(x) = -\Big[\big\langle r_x^2 - 2\mathsf{W}_{\varepsilon_k}\big\rangle^{\mathrm{fl}}_{k,j}\Big]_{P_0}. \tag{64}$$

Proof. At $A_j = \phi_j = 0$ the background fields vanish, the quadratic forms and the normalisation constants in [13, (3.14)] do not depend on θ, and $P^{(j),1}(0,0;(0,\theta h))$ is the Taylor polynomial of $E'_j = -\ln\int\prod d\mu\, e^{V^{(0)}}$. The θ-derivative of $V^{(0)}$ at $\phi^{(j)} = 0$ is the vertex [13, (3.21)], which in unscaled units is $\sum_x \varepsilon_k^3 h(x)(r_x^2 - 2\mathsf{W}_{\varepsilon_k})$. Differentiating under the finite-dimensional integral, $\partial_\theta E'_j|_{\theta=0} = -\langle\partial_\theta V^{(0)}\rangle^{\mathrm{fl}}_{k,j}$; the graphs of $\mathcal{T}_h$ are the Taylor coefficients of this derivative at the orders P_0 (and at $\tau = 0$). □

D.3.2. *The bulk density.*

Definition D.7 (King's block spin at zero field on the infinite lattice; the bulk density)**.** On the infinite lattice $\Lambda_k = \varepsilon_k\mathbb{Z}^3$ run King's block-spin averaging at $A = 0$ with block factor 5: the uniform photon average, and for the scalar the weighted average [14, (3.1)–(3.2)] with the Dirichlet weight f_D of [14, (3.8)] at the finest step and the uniform weight above it (as on the cube, section 5.1). Here f_D is $5\varepsilon_k$-periodic, equal to the constant $(5/|S|)^3$ on the sites of each $5\varepsilon_k$-block whose coordinates relative to the block's corner lie in $S\varepsilon_k$, where $S \subset \{0,\dots,4\}$ is a set of $|S| \in \{3,4\}$ consecutive integers, and zero on the remaining layer(s); its block averages are 1 and $\|f_D\|_\infty \le (5/3)^3$. (The printed weight and the corrected weight of section 5.1 are both of this form; nothing below depends on which.) Let Q_j be the composite average onto the blocks of side $\eta = 5^j\varepsilon_k$, Q_j^* its adjoint for the inner products $\sum\varepsilon_k^3$, $\sum\eta^3$. The fluctuation covariance after j steps is

$$G_j := \big(-\Delta_{\varepsilon_k} + m^2 + \mathfrak{a}_j\eta^{-2}Q_j^*Q_j\big)^{-1}, \qquad \mathfrak{a}_j = \mathfrak{a}(1-5^{-2})(1-5^{-2j})^{-1}$$

[13, (2.13)], [14, (3.5)], and the free propagator splits as $C := (-\Delta_{\varepsilon_k} + m^2)^{-1} = G_j + \Gamma_j$; the photon is the same with m^2 replaced by μ_0^2, the uniform weight and $A = 0$: $D = D_j + \Gamma^{\mathrm{ph}}_j$ with $D_j = \sum_{i<j} G^{\mathrm{ph}}_{(i)}$ [13, (2.17)]. Let $v^{(j)}_{k,\infty}(x)$ be the density of lemma D.6 for this flow on Λ_k (the same finitely many graphs, with the free lines G_j, $G^{\mathrm{ph}}_{(i)}$ on Λ_k and the vertices summed over Λ_k), and

$$\beta^{(j)}_k(x) := v^{(j)}_{k,\infty}(x) + z_k, \qquad \bar\beta^{(j)}_k := \eta^{-3}\sum_{x\in\Lambda_k\cap[0,\eta)^3} \varepsilon_k^3\,\beta^{(j)}_k(x).$$

$\beta^{(j)}_k$ is $\eta\mathbb{Z}^3$-periodic.

D.3.3. *Decay of the fluctuation covariance at zero field.*

Lemma D.8 (decay at $A = 0$)**.** *Let $f \ge 0$ be a weight on $\varepsilon\mathbb{Z}^3$ with $\|f\|_\infty < \infty$ and block averages $\eta^{-3}\sum_{x\in B(y)}\varepsilon^3 f(x) = 1$ over blocks $B(y)$ of side η; let $(Q_f\phi)(y) := \eta^{-3}\sum_{x\in B(y)}\varepsilon^3 f(x)\phi(x)$, so that $Q_f^*Q_f = M_f\mathsf{P}M_f$ with P the uniform block average and M_f multiplication by f, and*

$H := -\Delta_\varepsilon + m^2 + b\eta^{-2}Q_f^*Q_f$ with $b \in [b_-, b_+] \subset (0,\infty)$. There are $c_0, \alpha_0 > 0$, depending on $b_\pm$ and $\|f\|_\infty$ only, such that $H \geq c_0\eta^{-2}$ and, for every unit vector w and $|\alpha| \leq \alpha_0$, $\|e^{\alpha w\cdot x/\eta}H^{-1}e^{-\alpha w\cdot x/\eta}\| \leq 2c_0^{-1}\eta^2$. Consequently the kernel with respect to ε^3 obeys

$$|H^{-1}(x,y)| \leq 2c_0^{-1}\eta^2\varepsilon^{-3}e^{-\alpha_0|x-y|_\infty/\eta}.$$

(King's weights: $f \equiv 1$ on the steps above the finest one, $f = f_D$ at the finest step.)

Proof. Gap. On a block B write $\bar\phi$ for the uniform mean and $\ell(\phi) := \langle f\phi\rangle_B = (Q_f\phi)(y)$, so $\ell(1) = 1$ and $|\ell(\psi)| \leq \|f\|_\infty\langle|\psi|\rangle_B$. Then $\bar\phi = \ell(\phi) - \ell(\phi - \bar\phi)$, and the discrete Poincaré inequality on a cube of side η and mesh ε (uniform in ε/η) gives $\langle|\phi|^2\rangle_B \leq 2(1+\|f\|_\infty)^2C_P\eta^2\langle|\nabla\phi|^2\rangle_B + 2|\ell(\phi)|^2$. Sum over blocks: $-\Delta_\varepsilon$ dominates the sum of the block Laplacians, and $\sum_y \eta^3|\ell(\phi)|^2 = \langle\phi, Q_f^*Q_f\phi\rangle$. Hence $\|\phi\|^2 \leq C\max(1,b^{-1})\eta^2\langle\phi, H\phi\rangle$. *Combes–Thomas argument* [25]. With $\Phi = \alpha w\cdot x/\eta$, $e^\Phi(-\Delta_\varepsilon)e^{-\Phi} = -\Delta_\varepsilon + R$ with $\mathrm{Re}\langle\psi, R\psi\rangle \geq -\sum_\mu 2\varepsilon^{-2}(\cosh(\alpha\varepsilon/\eta) - 1)\|\psi\|^2 \geq -C\alpha^2\eta^{-2}\|\psi\|^2$ (the antisymmetric part of the conjugated hopping is skew), and $\|e^\Phi M_f\mathsf{P}M_fe^{-\Phi} - M_f\mathsf{P}M_f\| \leq \|f\|_\infty^2(e^{\sqrt{3}\alpha}-1)$ since P has range one block. For $\alpha \leq \alpha_0$ small, $\mathrm{Re}\, e^\Phi He^{-\Phi} \geq \frac{c_0}{2}\eta^{-2}$. The kernel bound follows from $|K(x,y)| = \varepsilon^{-6}|\langle\mathbf{1}_x, K\mathbf{1}_y\rangle|$ and $\|\mathbf{1}_x\|^2 = \varepsilon^3$. □

D.3.4. *Locality of the vacuum constant.*

Lemma D.9 (locality)**.** *For every real h supported at ℓ^∞-distance > 1 from $\mathbb{R}^3 \setminus C_J$ and $0 \leq j \leq k - J_0$,*

$$\left|\mathcal{V}_k^{(j)}(h) + z_k\mathsf{S}_k(h) - \sum_{x\in\Lambda_k}\varepsilon_k^3h(x)\,\beta_k^{(j)}(x)\right| \;\leq\; C\,\|h\|_{L^1}\,\varepsilon_k^{-\nu}\,e^{-c/\eta}, \tag{65}$$

with $C, \nu, c > 0$ depending on the couplings and J only. Along the diagonal pairs the right side is $\leq C\|h\|_{L^1}5^{\nu k}e^{-c5^{\alpha k-1}}$ for both members, which decays faster than any geometric sequence.

Proof. By lemma D.6, $v_k^{(j)}(x)$ is a finite sum of Feynman graphs of King's fluctuation measure with one vertex [13, (3.21)] at x and lines $G_j(C_J^K)$, $G_{(i)}^{\mathrm{ph}}(C_J^K)$ (and their differences and contour sums [13, (3.27)] for the averaging vertices), the other vertices summed over C_J^K; the counterterm $\delta m^2{}_{\varepsilon_k}$ and the Wick constant $\mathsf{W}_{\varepsilon_k}$ are the free-boundary numbers, hence the same on C_J^K and on $\varepsilon_k\mathbb{Z}^3$. *Image representations.* The region operators are signed sums of free ones over a reflection group: [14, (3.7)] for the photon (Neumann) and [14, (3.9)] for the scalar (Dirichlet, with the weight f_D; "if we had taken $f = 1$, the representation (3.9) would not hold"), with every non-identity image at distance $\geq \mathrm{dist}(x, \partial C_J^K) + \mathrm{dist}(y, \partial C_J^K)$ from the pair (x,y); the resulting locality bound is [14, Prop. 5.1, (5.3)]. The same holds for every slice and for the lattice differences. *Decay.* Every free fine line obeys lemma D.8: $|G_j(u,w)| \leq C\varepsilon_k^{-3}\eta^2e^{-\alpha_0|u-w|/\eta}$, the same for the slices $G_{(i)}$ with η replaced by $5^{i+1}\varepsilon_k \leq \eta$, and for lattice differences with an extra factor ε_k^{-1} each. *Graph bound.* Expand every region line as the free line plus its non-identity images, and every vertex sum over C_J^K as the sum over $\varepsilon_k\mathbb{Z}^3$ minus the sum over the complement. Every term other than the infinite-lattice graph contains an image line (u, gw), with $|u - gw| \geq \mathrm{dist}(u, \partial C_J^K) + \mathrm{dist}(w, \partial C_J^K)$, or a vertex outside C_J^K, and the path of fine lines from x to it has length $\geq \mathrm{dist}(x, \partial C_J^K) - \mathrm{dist}(u, \partial C_J^K)$, respectively $\geq \mathrm{dist}(x, \partial C_J^K)$; in both cases the total length along the path is $\geq \mathrm{dist}(x, \partial C_J^K) \geq 1$. The infinite-lattice graphs give $\sum_x \varepsilon_k^3h(x)v_{k,\infty}^{(j)}(x)$, and $z_k\mathsf{S}_k(h)$ adds z_k per site, which gives $\sum_x \varepsilon_k^3h(x)\beta_k^{(j)}(x)$ (the sums over $\Lambda_k \cap C_J$ and over Λ_k agree, h vanishing outside C_J). In every other term the product of the exponential factors along the path is at most $e^{-\alpha_0/(2\eta)}$ times the remaining half of each factor, which makes the vertex sums converge. The singular and ε_k-dependent prefactors (propagator sup norms, differences, $|\delta m^2{}_{\varepsilon_k}| \leq C\varepsilon_k^{-1}$, $|C_J|$, the finitely many graphs of order $\leq e^2$) are at most $C\varepsilon_k^{-\nu}$. Along the diagonal pairs $\eta_k^{-1} = 5^{k-j(k)} \geq 5^{\alpha k-1}$, so $\varepsilon_k^{-\nu}e^{-c/\eta_k} \leq 5^{\nu k}e^{-c5^{\alpha k-1}}$; the second member has $\varepsilon_{k+1} = \varepsilon_k/5$ and the same η_k. □

D.3.5. *Aliasing.*

Lemma D.10 (aliasing)**.** *Let β be an $\eta\mathbb{Z}^3$-periodic function on Λ_k with block mean $\bar\beta := \eta^{-3}\sum_{x\in\Lambda_k\cap[0,\eta)^3}\varepsilon_k^3\beta(x)$, and $h\in C_c^\infty(\mathbb{R}^3)$. Then for every integer $r\ge 2$*

$$\Big|\sum_{x\in\Lambda_k}\varepsilon_k^3 h(x)\beta(x)-\bar\beta\,\mathsf{S}_k(h)\Big|\le C_r\|\beta\|_\infty\|\Delta^r h\|_{L^1}\,\eta^{2r},\qquad \big|\mathsf{S}_k(h)-\textstyle\int h\big|\le C_r\|\Delta^r h\|_{L^1}\,\varepsilon_k^{2r}. \tag{66}$$

Proof. Let $\Gamma^*:=\frac{2\pi}{\eta}\mathbb{Z}^3$ and $\Lambda^*:=\frac{2\pi}{\varepsilon_k}\mathbb{Z}^3\subset\Gamma^*$ be the dual lattices. The periodic function β has the finite Fourier expansion $\beta(x)=\sum_{\kappa\in\Gamma^*/\Lambda^*}\hat\beta_\kappa e^{i\kappa\cdot x}$ with $|\hat\beta_\kappa|\le\|\beta\|_\infty$ and $\hat\beta_0=\bar\beta$. For $\hat h(p):=\int h(x)e^{-ip\cdot x}dx$ (a Schwartz function) the Poisson summation formula on Λ_k gives $\sum_{x\in\Lambda_k}\varepsilon_k^3 h(x)e^{i\kappa\cdot x}=\sum_{l\in\Lambda^*}\hat h(l-\kappa)$. For $\kappa=0$ this is $\mathsf{S}_k(h)=\int h+\sum_{l\in\Lambda^*\setminus 0}\hat h(l)$; since $|\hat h(p)|\le\|\Delta^r h\|_{L^1}|p|^{-2r}$ and $\sum_{l\in\Lambda^*\setminus0}|l|^{-2r}=(\varepsilon_k/2\pi)^{2r}\sum_{n\in\mathbb{Z}^3\setminus0}|n|^{-2r}$, finite for $2r>3$, the second bound follows. For $\kappa\neq0$ the points $l-\kappa$, $l\in\Lambda^*$, $\kappa\in(\Gamma^*/\Lambda^*)\setminus0$, run exactly once through $\Gamma^*\setminus\Lambda^*\subset\Gamma^*\setminus0$, all of modulus $\ge 2\pi/\eta$; hence $\sum_{\kappa\neq0}|\hat\beta_\kappa||\sum_l\hat h(l-\kappa)|\le\|\beta\|_\infty\|\Delta^r h\|_{L^1}\sum_{p\in\Gamma^*\setminus0}|p|^{-2r}=C_r\|\beta\|_\infty\|\Delta^r h\|_{L^1}\eta^{2r}$. □

D.3.6. *The free lattice propagator at a rate.*

Lemma D.11 (the free lattice propagator approaches the continuum one)**.** *For $t>0$ let $C_t(\cdot;m')$ be the kernel of $(-\Delta_t+m'^2)^{-1}$ on $t\mathbb{Z}^3$ with respect to $t^3\sum$, and $C_0(r;m')=(4\pi|r|)^{-1}e^{-m'|r|}$ the continuum kernel. There is an absolute constant C_* such that for all $t>0$, all $m'\ge0$ and all $r\in t\mathbb{Z}^3\setminus0$,*

$$|C_t(r;m')-C_0(r;m')|\le C_*\,t^2|r|^{-3}.$$

Proof. Scaling and heat kernels. $-\Delta_t+m'^2=t^{-2}(-\Delta_1+M^2)$ with $M=m't$, so for $r=tn$, $n\in\mathbb{Z}^3\setminus0$,

$$C_t(tn;m')=t^{-1}\int_0^\infty e^{-M^2\tau}P_\tau(n)\,d\tau,\qquad C_0(tn;m')=t^{-1}\int_0^\infty e^{-M^2\tau}G_\tau(n)\,d\tau,$$

where $P_\tau(n)=\prod_\mu q_\tau(n_\mu)$ with $q_\tau(\ell)=e^{-2\tau}I_\ell(2\tau)=(2\pi)^{-1}\int_{-\pi}^{\pi}e^{i\ell\vartheta-2\tau(1-\cos\vartheta)}d\vartheta$ is the heat kernel of Δ_1 on $\mathbb{Z}^3$ and $G_\tau(n)=(4\pi\tau)^{-3/2}e^{-|n|^2/4\tau}$ the continuum one. Since $0<e^{-M^2\tau}\le1$, $|C_t-C_0|\le t^{-1}\int_0^\infty|P_\tau(n)-G_\tau(n)|d\tau$, and it suffices to show $\int_0^\infty|P_\tau(n)-G_\tau(n)|d\tau\le C|n|^{-3}$: the mass enters only as a positive weight ≤1, so the bound is uniform in $m'\ge0$.

One-dimensional local limit theorem with a rate. Let $g_\tau(\ell)=(4\pi\tau)^{-1/2}e^{-\ell^2/4\tau}$. We claim that for $\tau\ge1$ and $\ell\in\mathbb{Z}$, $|q_\tau(\ell)-g_\tau(\ell)|\le C\tau^{-3/2}e^{-c\min(\ell^2/\tau,|\ell|)}$, and that $|q_\tau(\ell)|,|g_\tau(\ell)|\le C\tau^{-1/2}e^{-c\min(\ell^2/\tau,|\ell|)}$ for all $\tau>0$, and moreover $|q_\tau(\ell)|\le C\min(1,\tau^{-1/2})e^{-c\min(\ell^2/\tau,|\ell|)}$. The second bound: for g it is elementary; for q shift the contour of the ϑ-integral to $\operatorname{Im}\vartheta=\xi$ with $\xi:=\operatorname{sgn}(\ell)\min(|\ell|/4\tau,1)$ (the integrand is entire and 2π-periodic, so the vertical sides cancel), which produces the factor $e^{-\ell\xi+2\tau(\cosh\xi-1)}\le e^{-|\ell\xi|/2}\le e^{-c\min(\ell^2/\tau,|\ell|)}$, while $\operatorname{Re}[2\tau(1-\cos\vartheta)\cosh\xi]\ge c\tau\vartheta^2$ on $|\vartheta|\le\pi$ gives the factor $C\tau^{-1/2}$; the shifted ϑ-integrand has modulus ≤1, which gives the factor 1 in the $\min(1,\tau^{-1/2})$ form. For the first bound, in the region $|\ell|\le\tau^{3/4}$ shift both integrals to $\operatorname{Im}z=\xi:=\ell/2\tau$ (so $|\xi|\le\frac12\tau^{-1/4}$); the Gaussian's integral over $|\vartheta|>\pi$ contributes $\le C\tau^{-1}e^{-\ell^2/4\tau-\pi^2\tau}$; on $|\vartheta|\le\pi$ write $\varphi(z)=2(1-\cos z)$, $\psi(z)=z^2$, use $e^{-\tau\varphi}-e^{-\tau\psi}=-\tau(\varphi-\psi)\int_0^1e^{-\tau(s\varphi+(1-s)\psi)}ds$ with $|\varphi-\psi|\le C|z|^4$ on the strip, $\operatorname{Re}\psi=\vartheta^2-\xi^2$ and $\operatorname{Re}\varphi\ge\frac{4}{\pi^2}\vartheta^2-\xi^2-C\xi^4$; the common prefactor is $e^{-\ell\xi+\tau\xi^2+C\tau\xi^4}=e^{-\ell^2/4\tau}O(1)$ since $\ell^4/\tau^3\le1$ there, and the remaining integral $\tau\int(\vartheta^2+\xi^2)^2e^{-4\tau\vartheta^2/\pi^2}d\vartheta\le C\tau^{-3/2}(1+(\ell^2/\tau)^2)$; altogether $|q-g|\le C\tau^{-3/2}e^{-\ell^2/8\tau}$. In the region $|\ell|>\tau^{3/4}$ one has $\min(\ell^2/\tau,|\ell|)\ge\tau^{1/2}$, and the second bound gives $|q|,|g|\le C\tau^{-1/2}e^{-c\min}\le C\tau^{-3/2}\sup_\tau(\tau e^{-c\tau^{1/2}/2})e^{-c\min/2}$.

Three dimensions. Telescope $P_\tau-G_\tau=\sum_\mu\big(\prod_{\nu<\mu}g_\tau(n_\nu)\big)(q_\tau(n_\mu)-g_\tau(n_\mu))\big(\prod_{\nu>\mu}q_\tau(n_\nu)\big)$ and let $\varrho=|n|_\infty\ge1$. For $\tau\ge1$ every factor carries its own decay, so $|P_\tau-G_\tau|\le C\tau^{-5/2}e^{-c\min(\varrho^2/\tau,\varrho)}$; $\int_1^\varrho$ gives $\le Ce^{-c\varrho}$ and $\int_\varrho^\infty\tau^{-5/2}e^{-c\varrho^2/\tau}d\tau\le\Gamma(\frac32)(c\varrho^2)^{-3/2}$. For $\tau\le1$,

$|P_\tau| \le Ce^{-c\varrho}$ by the second bound in its $\min(1,\tau^{-1/2})$ form (for $\tau \le 1$ and $\ell \ne 0$, $\min(\ell^2/\tau, |\ell|) = |\ell|$), and $G_\tau(n) \le Ce^{-\varrho^2/8}$. Hence $\int_0^\infty |P_\tau - G_\tau| d\tau \le C|n|^{-3}$. □

At $m' = 0$ the lemma is the Green's function asymptotics of simple random walk [15, Thm. 4.3.1], $C_t(r;0) = t^{-1}g(r/t)$ with $g(n) = (4\pi|n|)^{-1} + O(|n|^{-3})$; the proof above is that argument with the mass carried as a weight. Since $|r| \ge t$, the lemma gives $|C_t(r;m') - C_0(r;m')| \le C_* t^\gamma |r|^{-1-\gamma}$ for every $\gamma \in (0,2]$.

D.3.7. *The densities at the three lowest orders.*

Lemma D.12 (the bulk density at $A = 0$)**.** *With the notation of definition D.7 on $\varepsilon_k\mathbb{Z}^3$, and $c_\lambda > 0$ the combinatorial tadpole constant of King's order-λ counterterm ($\delta m^{2(\lambda)} = -c_\lambda \lambda \mathsf{W}_{\varepsilon_k}$, [13, (3.33)]), the three orders of $\beta_k^{(j)} = \beta_{(0,0)} + \beta_{(0,1)} + \beta_{(2,0)}$ are*

$$\beta_{(0,0)}(x) = 2\,\Gamma_j(x,x), \qquad \beta_{(0,1)}(x) = -2c_\lambda\lambda \sum_y \varepsilon_k^3 G_j(x,y)^2\,\Gamma_j(y,y),$$

$$\beta_{(2,0)}(x) = e^2\big[\mathfrak{g}_x(C,C,C;D) - \mathfrak{g}_x(G_j,G_j,G_j;D_j)\big] - \mathfrak{a}_x,$$

where $\mathfrak{g}_x(G_1,G_2,G_3;D')$ is the coefficient of e^2 in $\langle r_x^2\rangle$ of the Gaussian matter model with outer lines G_1, G_3, internal scalar line G_2, photon covariance D', King's bond coupling (exchange and seagull) and the free counterterm $\delta m^{2(e^2)}$; $e^2\mathfrak{g}_x(C,C,C;D) = z_k$ (the order-e^2 coefficient of $\langle|\phi(0)|^2\rangle$ on $\varepsilon_k\mathbb{Z}^3$ with the order-e^2 counterterm inserted, definition 2.3); and $\mathfrak{a}_x$ is the order-e^2 contribution of the A-dependence of King's averaging, the vertices [13, (3.24)–(3.26)] with the contour convention [13, (2.19), (3.27)], to $\langle r_x^2\rangle^{\mathrm{fl}}$.

Proof. Order 1: $\langle r_x^2\rangle^{\mathrm{fl}}_{(0,0)} = 2G_j(x,x)$, so $v_{(0,0)} = -2(G_j(x,x) - \mathsf{W}_{\varepsilon_k}) = 2\Gamma_j(x,x)$, and z_k has no order-1 part. Order λ: first-order perturbation theory with the vertices [13, (3.18)–(3.19)] at $\phi^{(j)} = 0$ gives $\langle r_x^2\rangle^{\mathrm{fl}}_{(0,1)} = -2c_\lambda\lambda\sum_y \varepsilon_k^3 G_j(x,y)^2(G_j(y,y) - \mathsf{W}_{\varepsilon_k})$, and $G_j(y,y) - \mathsf{W}_{\varepsilon_k} = -\Gamma_j(y,y)$; z_k has no order-λ part either. Order e^2: the θ-linear vacuum graphs of order e^2 built from the original vertices are the exchange graph, the seagull graph and the δm^2 insertion with the lines G_j, D_j; the photon tadpole $\langle J\rangle$ vanishes by charge conjugation; the remaining graphs, with the averaging vertices, form $\mathfrak{a}$. Adding $z_k = e^2\mathfrak{g}_x(C,C,C;D)$ gives the formula. □

D.3.8. *Order one.*

Lemma D.13 (the bulk sequence at order 1)**.** *There is C, depending on $\mathfrak{a}$ and on an upper bound m_0 for $m\eta$, such that for all k and $1 \le j \le k-1$,*

$$\|\beta_{(0,0)}\|_\infty \le C\eta^{-1}, \qquad \big|\bar\beta^{(j)}_{(0,0),k} - \bar\beta^{(j+1)}_{(0,0),k+1}\big| \le C\eta^{-1}5^{-j/2}.$$

Reduction to unit block scale. Put $t := \varepsilon_k/\eta = 5^{-j}$, $m' := m\eta$, $b := \mathfrak{a}_j$. The map $x \mapsto x/\eta$ carries $-\Delta_{\varepsilon_k} + m^2 + \mathfrak{a}_j\eta^{-2}Q_j^*Q_j$ to $\eta^{-2}(-\Delta_t + m'^2 + bQ^*Q)$ on $\Lambda_t := t\mathbb{Z}^3$ with unit blocks; kernels with respect to the lattice volume element scale as $\Gamma_j(x,x') = \eta^{-1}\Gamma_t(x/\eta, x'/\eta)$. Hence $\bar\beta_{(0,0)} = \eta^{-1}F(t;m',b)$ with $F(t;m',b) := 2t^3\sum_{x\in\Lambda_t\cap[0,1)^3}\Gamma_t(x,x)$, and the diagonal pair changes $(t,b) \to (t/5, \mathfrak{a}_{j+1})$ at fixed m', with $|\mathfrak{a}_j - \mathfrak{a}_{j+1}| \le C5^{-2j}$. It suffices to prove, uniformly in $m' \in (0, m_0]$ and $b \in [\mathfrak{a}(1-5^{-2}), \mathfrak{a}]$:

(i) $0 \le \Gamma_t(x,x) \le C$; (ii) $|\Gamma_t(x,x') - \Gamma_0(x,x')| \le Ct^{1/2}$ for $x, x' \in \Lambda_t$; (iii) $|\nabla_x\Gamma_0(x,x)| \le C$; (iv) $|\partial_b\Gamma_t(x,x')| \le C$,

where Γ_0 is the continuum block-spin coarse covariance defined below. Then $|F(t) - F(0)| \le 2\sup|\Gamma_t - \Gamma_0| + 2|t^3\sum\Gamma_0(x,x) - \int_{[0,1)^3}\Gamma_0(u,u)du| \le Ct^{1/2}$ and $|F(t;b) - F(t;b')| \le C|b-b'|$, which gives the lemma.

Proof of lemma D.13. Fourier representation. For $t > 0$ let $\mathrm{BZ}_t = [-\pi/t, \pi/t)^3$, $\omega_t(p) = \sum_\mu 4t^{-2}\sin^2(tp_\mu/2)$, $\hat C_t(p) = (\omega_t(p) + m'^2)^{-1}$; for $t = 0$, $\mathrm{BZ}_0 = \mathbb{R}^3$, $\omega_0 = |p|^2$. Write $p = P + 2\pi n$ with $P \in \mathbb{T} := [-\pi,\pi)^3$ and $n \in N_t$, the set of t^{-3} aliases with $p \in \mathrm{BZ}_t$ ($N_0 = \mathbb{Z}^3$). The averaging symbol is $s_t^f(p) := t^3\sum_{x\in\Lambda_t\cap[0,1)^3} f(x)e^{ip\cdot x}$; for the uniform weight

$s_t(p) = t^3 \sum_{i\in\{0,\dots,t^{-1}-1\}^3} e^{itp\cdot i}$, $s_0(p) = \int_{[0,1)^3} e^{iu\cdot p}du$. With $(Q\phi)(y) = \sum_{x\in B(y)} t^3 f(x)\phi(x)$ and $Q^*v(x) = f(x)v(y(x))$ one has $QQ^* = 1$ and $(QC_tQ^*)\hat{}(P) = W_t(P) := \sum_{n\in N_t} |s_t^f(p)|^2 \hat{C}_t(p)$. Woodbury's identity gives $\Gamma_t = C_tQ^*(b^{-1} + QC_tQ^*)^{-1}QC_t$, hence, by Parseval on $\mathbb{Z}^3$,

$$\Gamma_t(x,x') = \int_{\mathbb{T}} \frac{\overline{\hat{v}_x^t(P)}\,\hat{v}_{x'}^t(P)}{b^{-1}+W_t(P)} \tfrac{dP}{(2\pi)^3}, \qquad \hat{v}_x^t(P) := \sum_{n\in N_t} s_t^f(p)\hat{C}_t(p)e^{-ip\cdot x},$$

and Γ_0 is *defined* by the same formula at $t = 0$ with s_0 (the continuum block spin). All sums below converge absolutely.

Elementary bounds, uniform weight ($t \in [0,1]$, $P \in \mathbb{T}$, $n \in N_t$, $p = P + 2\pi n$). (E1) $|s_t(p)| = \prod_\mu |t\sin(p_\mu/2)/\sin(tp_\mu/2)| \le \sigma(p) := \prod_\mu \min(1, \pi|\sin(p_\mu/2)|/|p_\mu|)$, using $|\sin(tp_\mu/2)| \ge t|p_\mu|/\pi$ on BZ_t (and $|s_0| \le \sigma$). If $n_\mu \ne 0$ then $|p_\mu| \ge \pi|n_\mu|$, so $\sigma(p) \le \rho(n) := \prod_{\mu:n_\mu\ne0} \frac{\pi}{2|n_\mu|}$; for $n = 0$, $|s_t(P)| \ge (2/\pi)^3$. (E2) $\frac{4}{\pi^2}|p|^2 \le \omega_t(p) \le |p|^2$, so $(|p|^2+m'^2)^{-1} \le \hat{C}_t(p) \le \frac{\pi^2}{4}(|p|^2+m'^2)^{-1}$, and $\hat{C}_t(p) \le \frac{3}{4}|n|^{-2}$ for $n \ne 0$. (E3) For $\kappa < 2$, $S_\kappa := \sum_{n\ne0}\rho(n)|n|^{-2+\kappa} < \infty$. (E4) For $p \in \mathrm{BZ}_t$, $s_t(p) = s_0(p)\prod_\mu \mathrm{r}(itp_\mu)$ with $\mathrm{r}(z) = z/(e^z-1)$, $|\mathrm{r}(i\vartheta)| \le \pi/2$ and $|\mathrm{r}(i\vartheta)-1| \le C|\vartheta|$ for $|\vartheta| \le \pi$, so $|s_t(p) - s_0(p)| \le C\min(1, t|p|)\sigma(p)$; and $|\omega_0 - \omega_t| \le t^2|p|^4/12$ gives $|\hat{C}_t - \hat{C}_0| \le C\min(t^2, (|p|^2+m'^2)^{-1}) \le Ct(|p|^2+m'^2)^{-1/2}$.

(i). By (E1)–(E3), $|\hat{v}_x^t(P)| \le \frac{\pi^2}{4}(P^2+m'^2)^{-1} + \frac{3}{4}S_0 \le C(P^2+m'^2)^{-1}$, and $W_t(P) \ge |s_t(P)|^2\hat{C}_t(P) \ge (2/\pi)^6(P^2+m'^2)^{-1}$, so $(b^{-1}+W_t)^{-1} \le (\pi/2)^6(P^2+m'^2)$. The integrand of $\Gamma_t(x,x)$ is ≥ 0 and $\le C(P^2+m'^2)^{-1}$, integrable on $\mathbb{T}$ uniformly in m'.

(ii). Put $a_t(n) := s_t(p)\hat{C}_t(p)$ for $n \in N_t$ and $a_t(n) := 0$ otherwise. For $n \in N_t$, by (E4), $|a_t(n) - a_0(n)| \le Ct\sigma(p)(|p|^2+m'^2)^{-1/2}$, which is $\le Ct(P^2+m'^2)^{-1/2}$ for $n = 0$ and $\le Ct\rho(n)|n|^{-1}$ for $n \ne 0$. For $n \notin N_t$, $|n| \ge (3t)^{-1}$ and $|a_0(n)| \le C\rho(n)|n|^{-2} \le 3Ct\rho(n)|n|^{-1}$. The phases are the same for t and 0. Hence, with (E3) at $\kappa = 1$, $|\hat{v}_x^t - \hat{v}_x^0| \le Ct(P^2+m'^2)^{-1/2}$ and $|W_t - W_0| \le Ct(P^2+m'^2)^{-1/2}$, and

$$|\Gamma_t(x,x') - \Gamma_0(x,x')| \le \int_{\mathbb{T}} \Big[\frac{|\hat{v}_x^t - \hat{v}_x^0||\hat{v}_{x'}^t| + |\hat{v}_x^0||\hat{v}_{x'}^t - \hat{v}_{x'}^0|}{b^{-1}+W_t} + \frac{|\hat{v}_x^0||\hat{v}_{x'}^0||W_t - W_0|}{(b^{-1}+W_t)(b^{-1}+W_0)}\Big] \tfrac{dP}{(2\pi)^3} \le Ct\int_{\mathbb{T}} (P^2+m'^2)^{-1/2}$$

(iii). $|\nabla_x\hat{v}_x^0(P)| \le \sum_n |p||a_0(n)| \le C(P^2+m'^2)^{-1/2} + CS_1$, so $|\nabla_x\Gamma_0(x,x)| \le 2\int |\nabla\hat{v}_x^0||\hat{v}_x^0|(b^{-1}+W_0)^{-1} \le C$. *(iv).* $\partial_b(b^{-1}+W)^{-1} = b^{-2}(b^{-1}+W)^{-2} \le b^{-2}(\pi/2)^{12}(P^2+m'^2)^2$, so $|\partial_b\Gamma_t(x,x')| \le C$.

The Dirichlet weight. f_D acts at the finest step only (definition D.7). At unit block scale, splitting $x = 5ti + tm$ with $m \in \{0,\dots,4\}^3$ gives $s_t^f(p) = s_{5t}(p)\,\varphi(tp)$, $\varphi(\vartheta) = \prod_\mu \varphi_1(\vartheta_\mu)$, $\varphi_1(\vartheta) = |S|^{-1}\sum_{m\in S} e^{im\vartheta}$, with s_{5t} the uniform symbol at spacing $5t$; $|\varphi_1| \le 1$, $|\varphi_1(\vartheta) - 1| \le C|\vartheta|$, and $|\varphi_1(\vartheta)| \ge c_S > 0$ for $|\vartheta| \le \pi/5$ (its modulus is $|\sin(|S|\vartheta/2)|/(|S||\sin(\vartheta/2)|)$, whose first zero is at $2\pi/|S| \ge \pi/2$). Repeat the proof with s_t^f in place of s_t; only (E1) and (E4) change. Write $p \in \mathrm{BZ}_t$ as $p = q + 2\pi\ell/(5t)$ with $q \in \mathrm{BZ}_{5t}$, $\ell \in \{-2,\dots,2\}^3$; s_{5t} is $2\pi/(5t)$-periodic, and $q = P + 2\pi n'$ with $n' \in \mathbb{Z}^3$ (as $1/(5t) = 5^{j-1} \in \mathbb{N}$). For $\ell = 0$: $|s_t^f(p) - s_0(p)| \le |s_{5t} - s_0||\varphi| + |s_0||\varphi(tp) - 1| \le C\min(1, t|p|)\sigma(p)$ by (E4) at spacing $5t$, so (ii) goes through verbatim. For $\ell \ne 0$: $|s_t^f(p)| \le |s_{5t}(q)| \le \sigma(q)$ by (E1) at spacing $5t$, and $|p_\mu| \ge \pi/(5t)$ for some μ, so $\hat{C}_t(p) \le 25t^2/4$; summing $\sigma(q)$ over $|n'_\mu| \le 1/(10t)$ gives $\le C(1+\ln t^{-1})^3$. So these aliases contribute $\le Ct^2(1+\ln t^{-1})^3$ to $\hat{v}_x^{f,t}$ and to W_t^f, while the corresponding $a_0(n)$ lie in the tail $|n| \ge c/t$. The lower bound becomes $W_t^f(P) \ge (2/\pi)^6 c_S^6(P^2+m'^2)^{-1}$ (for $j \ge 1$, $|tP_\mu| \le \pi/5$). Hence (i)–(iv) hold for f_D with Ct replaced by $Ct(1+\ln t^{-1})^3 \le C't^{1/2}$, with the same continuum limit Γ_0.

Conclusion. $\|\beta_{(0,0)}\|_\infty = 2\eta^{-1}\sup_x \Gamma_t(x,x) \le C\eta^{-1}$ by (i), and $|\bar{\beta}^{(j)}_{(0,0),k} - \bar{\beta}^{(j+1)}_{(0,0),k+1}| = \eta^{-1}|F(t;m',\mathfrak{a}_j) - F(t/5;m',\mathfrak{a}_{j+1})| \le \eta^{-1}\big(|F(t) - F(0)| + |F(0;\mathfrak{a}_j) - F(0;\mathfrak{a}_{j+1})| + |F(t/5) - F(0)|\big) \le C\eta^{-1}(t^{1/2} + 5^{-2j})$. □

D.3.9. *Order λ.*

Lemma D.14 (the bulk sequence at order λ)**.** *For every $\gamma'' \in (0, \frac{1}{2})$ there is C such that for all k and $1 \le j \le k-1$,*

$$\|\beta_{(0,1)}\|_\infty \le C\lambda(1+j)^3, \qquad \big|\bar\beta^{(j)}_{(0,1),k} - \bar\beta^{(j+1)}_{(0,1),k+1}\big| \le C\lambda\big(5^{-\gamma'' j} + 5^{-2j}\big).$$

Proof. At unit block scale (the reduction of lemma D.13, with $t = 5^{-j}$) $\beta_{(0,1)}(x) = -2c_\lambda \lambda t^3 \sum_y G_t(x,y)^2 \Gamma_t(y,y)$, a dimensionless sum. *Decay.* $G_t = C_t - \Gamma_t$; by lemma D.11 and $C_t(x,x) \le Ct^{-1}$, $|C_t(x,y)| \le C(|x-y|+t)^{-1}$, and $|\Gamma_t(x,y)| \le \Gamma_t(x,x)^{1/2}\Gamma_t(y,y)^{1/2} \le C$ by lemma D.13(i); so $|G_t(x,y)| \le C(|x-y|+t)^{-1} + C$. Lemma D.8 at unit block scale gives $|G_t(x,y)| \le Ct^{-3}e^{-\alpha_0|x-y|_\infty} \le Ct^{-3}e^{-\alpha_0|x-y|/\sqrt{3}}$; use the first bound for $|x-y| \le R_t := 6\sqrt{3}\,\alpha_0^{-1}\ln t^{-1}$ and the second beyond. With lemma D.13(i) this gives the sup bound: $t^3\sum_{|x-y|\le R_t}\big(|x-y|_t^{-1}+1\big)^2 \le C(1+R_t)^3$ and the part $|x-y| > R_t$ is $\le Ct^3$ (as below), so $\|\beta_{(0,1)}\|_\infty \le C\lambda(1+\ln t^{-1})^3 \le C\lambda(1+j)^3$. *Increment.* For $x \ne y$, lemma D.11 and lemma D.13(ii) give $|G_t(x,y) - G_0(x,y)| \le Ct^2|x-y|^{-3} + Ct^{1/2}$, where $G_0 = C_0 - \Gamma_0$. Hence

$$t^3\sum_y |G_t^2 - G_0^2|(x,y)\,\Gamma_t(y,y) \le Ct^3 \sum_{0<|r|\le R_t}\big(t^2|r|^{-3} + t^{1/2}\big)\big(|r|^{-1}+1\big) + Ct^3 G_t(x,x)^2 + (\text{the part } |r| > R_t) \le Ct^{1/2}(1$$

the part $|r| > R_t$ being $\le Ct^{-6}e^{-2\alpha_0 R_t/\sqrt{3}}\sum_{|r|>R_t} t^3 e^{-2\alpha_0(|r|-R_t)/\sqrt{3}} \le Ct^{-6}t^{12}(1+R_t)^2 \le Ct^3$; further $|\Gamma_t(y,y) - \Gamma_0(y,y)| \le Ct^{1/2}$; and the Riemann sum of the integrable function $G_0(x,\cdot)^2\Gamma_0(\cdot,\cdot)$ (continuum bounds $|\nabla^i G_0(x,y)| \le C|x-y|^{-1-i}e^{-\delta|x-y|}$, $i \le 1$, and lemma D.13(iii)) differs from its integral by $\le Ct^{1/2}$. The cell average in x and the dependence on $b = \mathfrak{a}_j$ are handled as in lemma D.13; $Ct^{1/2}(1+\ln t^{-1})^3 \le C5^{-\gamma'' j}$. □

D.3.10. *Order e^2.*

Lemma D.15 (the bulk sequence at order e^2)**.** *For every $\vartheta \in (0, \frac{1}{2})$ there are C, C_ϑ and $q \ge 0$, depending on ϑ, $\mathfrak{a}$, e, m, μ_0 and an upper bound m_0 for $\max(m,\mu_0)\eta$, such that for all k and $1 \le j \le k - J_0$,*

$$\|\beta_{(2,0)}\|_\infty \le Ce^2\,\eta^{-1}(1+j)^q, \qquad \big|\bar\beta^{(j)}_{(2,0),k} - \bar\beta^{(j+1)}_{(2,0),k+1}\big| \le C_\vartheta e^2\,5^{-\vartheta j}\eta^{-1}.$$

Both bounds hold for the original-vertex part $e^2[\mathfrak{g}_x(C,C,C;D) - \mathfrak{g}_x(G_j,G_j,G_j;D_j)]$ and for the averaging part $\mathfrak{a}_x$ of lemma D.12 separately; for the first, the increment bound holds for every $\vartheta \in (0,1)$.

The proof of lemma D.15 occupies the rest of this subsection. The two parts of $\beta_{(2,0)}$ are treated separately: the original-vertex part $e^2[\mathfrak{g}_x(C,C,C;D) - \mathfrak{g}_x(G_j,G_j,G_j;D_j)]$ in §§B2.1–B2.6 and the averaging part $\mathfrak{a}_x$ in §§B2.7–B2.10, the last of which also adds the two. Throughout, $L := 5$ is the block factor.

B2.1 Unit block scale, the order-e^2 density, the sandwich. Normalisation. As in the reduction before lemma D.13: $t := \varepsilon_k/\eta = 5^{-j}$, $\tau \in \{t, t/5\}$ is the spacing of one member of the diagonal pair at unit block scale, $\Lambda_\tau := \tau\mathbb{Z}^3$, $m' := m\eta$, $\mu' := \mu_0\eta$, both in $(0, m_0]$, and $b := \mathfrak{a}_j \in [\mathfrak{a}(1-5^{-2}), \mathfrak{a}]$ with $|\mathfrak{a}_j - \mathfrak{a}_{j+1}| \le C5^{-2j} = Ct^2$. Kernels are taken with respect to the counting measure $\tau^3\sum$. $C_\tau := (-\Delta_\tau + m'^2)^{-1}$; Q is the block average over the unit cubes (for the weight f_D, $Q_f = QM_f$ with the weight on the 5τ-blocks, as in the proof of lemma D.13), $P := Q^*Q$; $\Gamma_\tau := C_\tau Q^*(b^{-1} + QC_\tau Q^*)^{-1}QC_\tau$ is the coarse line and $G_\tau := C_\tau - \Gamma_\tau = (-\Delta_\tau + m'^2 + bP)^{-1}$ the fluctuation (outer) line (Woodbury). The photon is componentwise the same with $m' \to \mu'$, the uniform block average and the same b [13, (2.6), (2.10), (2.16)–(2.17)]: $D_\tau := (-\Delta_\tau + \mu'^2)^{-1}$, $\Gamma^{\mathrm{ph}}_\tau$, $G^{\mathrm{ph}}_\tau := D_\tau - \Gamma^{\mathrm{ph}}_\tau$; for f_D the photon averaging is unchanged [14, p. 328]. The diagonal pair keeps η, m', μ' and the unit cell fixed and changes $(\tau, b, \text{weight}) = (t, \mathfrak{a}_j, f \text{ on } 5t\text{-blocks}) \to (t/5, \mathfrak{a}_{j+1}, f \text{ on } t\text{-blocks})$. Scaling: $\Gamma_j(x,x') = \eta^{-1}\Gamma_t(x/\eta, x'/\eta)$, and e^2 has the dimension of a mass, so the order-e^2 density is e^2 times a dimensionless unit-scale quantity; all bounds below are uniform in $m', \mu' \in (0, m_0]$, b and $\tau \in (0,1]$ unless said otherwise, and $\eta \le 1$ is absorbed into constants.

Lemma D.16 (the order-e^2 density)**.** *At $\lambda = 0$ and zero block fields the matter is Gaussian given A with quadratic form $M_A = -\Delta_A + m'^2 + \delta$, $(\Delta_A\phi)(u) = \tau^{-2}\sum_\mu [e^{ie'\tau A_\mu(u)}\phi(u+\tau e_\mu) + e^{-ie'\tau A_\mu(u-\tau e_\mu)}\phi(u-\tau e_\mu) - 2\phi(u)]$, where δ is King's free order-e^2 counterterm* [13, (3.33)]*, the same number in the full and in the fluctuation measure* [14, p. 330]*. Writing $M_A = M_0 + e'M_1 + e'^2(M_2+\delta) + O(e'^3)$,*

$$(M_1\phi)(u) = -i\tau^{-1}\sum_\mu [A_\mu(u)\phi(u+\tau e_\mu) - A_\mu(u-\tau e_\mu)\phi(u-\tau e_\mu)], \quad (M_2\phi)(u) = \tfrac{1}{2}\sum_\mu [A_\mu(u)^2\phi(u+\tau e_\mu) + A_\mu(u-$$

With matter covariance G and photon covariance D', the coefficient of e'^2 in $\langle r_x^2\rangle = 2\mathbb{E}_A[M_A^{-1}(x,x)]$ is $-2(G\,\Pi[G,D']\,G)(x,x)$ with

$$\Pi[G,D'] := \mathbb{E}_{D'}[M_2] + \delta - \mathbb{E}_{D'}[M_1 G M_1] \qquad (\textit{seagull} + \textit{counterterm} - \textit{exchange});$$

explicitly $\mathbb{E}[M_2](u,v) = \frac{1}{2}\tau^{-3}\sum_\mu [D'(u,u)\mathbf{1}_{v=u+\tau e_\mu} + D'(u-\tau e_\mu, u-\tau e_\mu)\mathbf{1}_{v=u-\tau e_\mu}]$ and $\mathbb{E}[M_1 G M_1](u,v) = -\tau^{-2}\sum_\mu\sum_{\sigma,\sigma'\in\{0,1\}}(-1)^{\sigma+\sigma'} D'(u-\sigma\tau e_\mu, v-(1-\sigma')\tau e_\mu)\, G(u+(1-2\sigma)\tau e_\mu, v-(1-2\sigma')\tau e_\mu)$. Hence, at unit block scale,

(67)

$$\beta_{(2,0)}(x) + \mathfrak{a}_x = -2e^2\big[(C_\tau\Pi_C C_\tau)(x,x) - (G_\tau\Pi_G G_\tau)(x,x)\big], \qquad \Pi_C := \Pi[C_\tau, D_\tau], \quad \Pi_G := \Pi[G_\tau, G_\tau^{\mathrm{ph}}];$$

Π_C is translation invariant. (Overall signs and the factor 2 play no role in the bounds and are not tracked.)

Proof. This is the Gaussian algebra of lemma D.12 and lemma E.1: $(M_A)^{-1} = M_0^{-1} - e'M_0^{-1}M_1M_0^{-1} + e'^2[M_0^{-1}M_1M_0^{-1}M_1M_0^{-1} - M_0^{-1}(M_2+\delta)M_0^{-1}] + O(e'^3)$; the determinant does not contribute at order e'^2 since $\operatorname{Tr} M_0^{-1}M_1 = 0$, and $\mathbb{E}_{D'}[M_1] = 0$. The kernels follow from $\mathbb{E}[A_\mu(u)A_\nu(v)] = \delta_{\mu\nu}D'(u,v)$. In the full model $G = C_\tau$, $D' = D_\tau$; in King's fluctuation measure at zero block fields $G = G_\tau$, $D' = G_\tau^{\mathrm{ph}}$ (definition D.7); the difference of the two is the original-vertex part plus the averaging part, by lemma D.12. □

Lemma D.17 (sandwich)**.** *With $\Gamma := \Gamma_\tau = C_\tau - G_\tau$ (indices dropped),*

(68)

$$C\Pi_C C - G\Pi_G G = \underbrace{\Gamma\Pi_C C + C\Pi_C\Gamma}_{\text{(I)}}\ \underbrace{-\Gamma\Pi_C\Gamma}_{\text{(II)}} + \underbrace{G(\Pi_C - \Pi_G)G}_{\text{(III)}}, \qquad \Pi_C - \Pi_G = \Sigma^{\mathrm{sg}}[\Gamma^{\mathrm{ph}}] - \Sigma^{\mathrm{ex}}[\Gamma, D] - \Sigma^{\mathrm{ex}}[G, \Gamma^{\mathrm{ph}}$$

where $\Sigma^{\mathrm{ex}}[G,D'] := \mathbb{E}_{D'}[M_1GM_1]$ and $\Sigma^{\mathrm{sg}}[D'] := \mathbb{E}_{D'}[M_2]$. The counterterm δ cancels in (III)*; only* (I), (II) *contain the renormalised translation-invariant self-energy Π_C.*

Proof. Expand $G\Pi_C G = (C-\Gamma)\Pi_C(C-\Gamma)$ and add $G(\Pi_C - \Pi_G)G$. The second identity is the bilinearity of $\mathbb{E}_{D'}[M_1GM_1]$ in (G,D') and the linearity of $\mathbb{E}_{D'}[M_2]$ in D', with $\Gamma = C - G$, $\Gamma^{\mathrm{ph}} = D - G^{\mathrm{ph}}$. □

Lemma D.18 (derivative placement)**.** *The current at the bond (u,μ) is $i[\bar\phi(u)\phi(u+\tau e_\mu) - \bar\phi(u+\tau e_\mu)\phi(u)] = i\tau[\bar\phi\nabla_\mu\phi - (\nabla_\mu\bar\phi)\phi](u)$, ∇_μ the forward difference. Hence in every Wick contraction of an exchange term, at each of the two current vertices exactly one of the two matter lines meeting there carries one forward difference, and every term of $(G\Sigma^{\mathrm{ex}}[\cdot,\cdot]G)(x,x)$ is a triangle*

$$\sum_{u,v\in\Lambda_\tau}\tau^6\, k_1(x,u)\, k_2(u,v)\, k_3(v,x), \qquad k_1,k_3 \in \{G, \nabla G\}, \quad k_2 = (\textit{inner matter line})\times(\textit{photon line}),$$

with arguments shifted by at most one lattice unit. When both differences fall on the inner matter line ($\nabla_u\nabla_v\Gamma(u,v)$ or $\nabla_u\nabla_v G(u,v)$), one summation by parts in u (all sums are finite at fixed τ and absolutely convergent; Λ_τ has no boundary),

$$\sum_u G(x,u)k(u,v)\nabla_u F(u,v) = -\sum_u \big[(\nabla_u^* G)(x,u)k(u,v) + G(x,u-\tau e_\mu)(\nabla_u^* k)(u,v)\big]F(u,v),$$

k the photon line, moves one of them. Afterwards every line carries at most one difference. With $w_s(r) := (|r|+\tau)^{-s}$ the exponent bookkeeping is $G \mapsto 1$, $\nabla G \mapsto 2$, $D \mapsto 1$, $\nabla D \mapsto 2$, $\Gamma, \nabla\Gamma, \Gamma^{\mathrm{ph}}, \nabla\Gamma^{\mathrm{ph}} \mapsto 0$ (bounded, lemma D.23 below). Every triangle has slots (a,b,c) with $a,c \le 2$, $b \le 2$, $a+b+c \le 5$; the seagull term $G\Sigma^{\mathrm{sg}}[\Gamma^{\mathrm{ph}}]G$ is a two-point sum $\sum_u \tau^3 G(x,u)\Gamma^{\mathrm{ph}}(u,u)G(u\pm\tau e_\mu, x)$ with slots $(1,1)$ and no difference.

Proof. The current identity is algebra. The slot list: the inner line k_2 is one of $\Gamma\cdot D$, $\Gamma\cdot\nabla D$, $\nabla\Gamma\cdot D$, $G_{\mathrm{in}}\cdot\Gamma^{\mathrm{ph}}$, $\nabla G_{\mathrm{in}}\cdot\Gamma^{\mathrm{ph}}$, $G_{\mathrm{in}}\cdot\nabla\Gamma^{\mathrm{ph}}$ (slot $b\in\{1,2\}$), and k_1,k_3 have slots in $\{1,2\}$; e.g. $\nabla G\,|\,\Gamma D\,|\,\nabla G$ is $(2,1,2)$, $\nabla G\,|\,\nabla G_{\mathrm{in}}\Gamma^{\mathrm{ph}}\,|\,G$ is $(2,2,1)$, $G\,|\,\nabla^*D\,\nabla\Gamma\,|\,G$ is $(1,2,1)$. The summation by parts is needed because $\nabla\nabla\Gamma_\tau$ is bounded only by $C(1+\ln\tau^{-1})^3$ (the alias sum $\sum_{N_\tau}\rho(n)$, (E1)–(E3) in the proof of lemma D.13). □

B2.2 The renormalised one-loop self-energy. Let $\hat\Pi_\tau(p):=\sum_r\tau^3\Pi_C(r)e^{-ip\cdot r}$ be the symbol of Π_C (BZ_τ-periodic). By lemma E.1 (whose Π equals $-\frac12\Pi_C$ in the normalisation of lemma D.16, since its K is $\frac12M_0$ and its M is $\frac12(M_A-M_0)$) and the scaling of its part (ii), in the notation of appendix E.1 (unit lattice, masses $a=\tau m'$, $b=\tau\mu'$),

(69)
$$\hat\Pi_\tau(p)=\tau^{-1}\bar\pi_{\tau m',\tau\mu'}(\tau p),\qquad \hat\Pi_0(p):=-2\Big[|p|^2B^{\mathrm{c}}_{m',\mu'}(p)+\tfrac12(\mu'^2-2m'^2)\big(B^{\mathrm{c}}_{m',\mu'}(p)-B^{\mathrm{c}}_{m',\mu'}(0)\big)\Big],$$

the second being -2 times the continuum renormalised one-loop self-energy ($B^{\mathrm{c}}_{ta,tb}(tq)=t^{-1}B^{\mathrm{c}}_{a,b}(q)$, lemma E.5(i), so $\hat\Pi_0(p)=-2\tau^{-1}[|q|^2B^{\mathrm{c}}_{a,b}(q)+\frac12(b^2-2a^2)(B^{\mathrm{c}}_{a,b}(q)-B^{\mathrm{c}}_{a,b}(0))]$ at $q=\tau p$).

Lemma D.19 (self-energy bounds)**.** *Uniformly in $\tau\in(0,1]$ and $m',\mu'\in(0,m_0]$, with C depending on m_0 only:* (Π-0) $\hat\Pi_\tau(0)=0=\hat\Pi_0(0)$*;* (Π-a) $|\hat\Pi_\tau(p)|\le C(|p|+m_0)$ *for $p\in\mathrm{BZ}_\tau$ and $|\hat\Pi_0(p)|\le C(|p|+m_0)$ on $\mathbb{R}^3$;* (Π-b) $|\hat\Pi_\tau(p)-\hat\Pi_0(p)|\le C[\tau^2|p|^3+\tau|p|^2+\tau m_0^2]\le C\tau(1+|p|)(|p|+m_0)$ *for $p\in\mathrm{BZ}_\tau$.*

Proof. By lemma E.3, at $q=\tau p\in\mathsf{B}$, $-\frac12\bar\pi_{a,b}(q)=\hat q^2B_{a,b}(q)+\frac12(b^2-2a^2)(B_{a,b}(q)-B_{a,b}(0))+\rho_{a,b}(q)$, and by (69) $\hat\Pi_\tau(p)-\hat\Pi_0(p)=-2\tau^{-1}\big\{-\frac12\bar\pi_{a,b}(q)-[|q|^2B^{\mathrm{c}}+\frac12(b^2-2a^2)(B^{\mathrm{c}}(q)-B^{\mathrm{c}}(0))]\big\}$, where

$$-\tfrac12\bar\pi_{a,b}(q)-\big[|q|^2B^{\mathrm{c}}+\tfrac12(b^2-2a^2)(B^{\mathrm{c}}(q)-B^{\mathrm{c}}(0))\big]=(\hat q^2-|q|^2)B^{\mathrm{c}}(q)+\hat q^2(B-B^{\mathrm{c}})(q)+\tfrac12(b^2-2a^2)\big[(B-B^{\mathrm{c}})(q)-(B-$$

By lemma E.4(c), $|\hat q^2-|q|^2|\le|q|^4/12$, and $B^{\mathrm{c}}(q)\le1/(8|q|)$ (lemma E.5(i)), so the first term is $\le|q|^3/96$; $|B-B^{\mathrm{c}}|\le C_0$ on B (lemma E.5(ii)) bounds the second by $C_0|q|^2$ and the third by $2C_0(a^2+b^2)$; $|\rho|\le\frac34d_0\hat q^2\le\frac34d_0|q|^2$ (lemma E.6). Multiplying by $2\tau^{-1}$ and substituting $q=\tau p$, $a^2+b^2=\tau^2(m'^2+\mu'^2)\le2\tau^2m_0^2$ gives the first form of (Π-b); on BZ_τ, $\tau|p|\le\sqrt3\pi$ gives the second. For (Π-a): $0<B^{\mathrm{c}}(q)\le B^{\mathrm{c}}(0)=(4\pi(a+b))^{-1}$ and $\frac12|b^2-2a^2|\le(a+b)^2$, so $|\hat\Pi_0(p)|\le2\tau^{-1}[|q|/8+(a+b)/(4\pi)]=|p|/4+(m'+\mu')/(2\pi)$; then $|\hat\Pi_\tau|\le|\hat\Pi_0|+C\tau(1+|p|)(|p|+m_0)\le C(|p|+m_0)$ on BZ_τ. (Π-0): $\hat q=0$ at $q=0$, and $\rho(0)=0$; $\hat\Pi_\tau(0)=0$ is the statement that King's counterterm [13, (3.33)] makes the order-e^2 one-particle-irreducible two-point function vanish at zero momentum on the infinite lattice (lemma E.1). □

B2.3 The alias sums with the Dirichlet weight. We use the Fourier representation from the proof of lemma D.13: for $x,x'\in\Lambda_\tau$,

$$\Gamma_\tau(x,x')=\int_{\mathbb{T}}\overline{\hat v^\tau_x(P)}\,\hat v^\tau_{x'}(P)\,\hat M_\tau(P)\,\tfrac{dP}{(2\pi)^3},\qquad \hat M_\tau:=(b^{-1}+W_\tau)^{-1},\quad a_\tau(n):=s_\tau(p)\hat C_\tau(p),$$

with $p=P+2\pi n$, $n\in N_\tau$, and (E1)–(E4) as there; for the weight f_D, $s^f_\tau=s_{5\tau}(p)\varphi(\tau p)$, a^f_τ, $\hat v^{f,\tau}_x$, W^f_τ, $\hat M^f_\tau$ are the same expressions with s^f_τ in place of s_τ (the Poisson-summation derivation uses only that f_τ is 1-periodic, so $Q_fC_\tau Q_f^*$ commutes with $\mathbb{Z}^3$-translations; $Q_fQ_f^*\ne1$ is not used). The continuum object is the same Γ_0 for both weights. Consequences recorded in that proof: $|\hat v^\tau_x|\le C(P^2+m'^2)^{-1}$, $\hat M_\tau\le\min(b,(\pi/2)^6(P^2+m'^2))$, $|\hat v^\tau_x-\hat v^0_x|,|W_\tau-W_0|\le C\tau(P^2+m'^2)^{-1/2}$, hence $|\hat M_\tau-\hat M_0|\le\hat M_\tau\hat M_0|W_\tau-W_0|\le C\tau(P^2+m'^2)^{3/2}$; for $n\ne0$, $c|n|\le|p|\le C|n|$; all uniform in $m'\in(0,m_0]$.

Lemma D.20 (alias sums)**.** *For $\tau\le1$, $P\in\mathbb{T}$ and $\kappa\in[0,2)$, uniform weight:*

$$\sum_{n\in N_\tau}|p|^\kappa|a_\tau(n)|\le C(P^2+m'^2)^{(\kappa-2)/2}+CS_\kappa\le C(P^2+m'^2)^{(\kappa-2)/2}.\tag{70}$$

For the weight f_D and $\tau=5^{-j}\le\frac15$, write $p=q+2\pi\ell/(5\tau)$ with $q\in\mathrm{BZ}_{5\tau}$, $\ell\in\{-2,\dots,2\}^3$, $q=P+2\pi n'$, $n'\in N_{5\tau}$ (a bijection $n\leftrightarrow(n',\ell)$); $\ell=0$ are the uniform aliases, $\ell\ne0$ the D-aliases ($|p|_\infty\ge\pi/(5\tau)$, so $\hat C_\tau(p)\le25\tau^2/4$, and $|p|\le\sqrt3\pi/\tau$). Then

(A1) $\sum_{\ell\neq 0}|p|^\kappa|a_\tau^f(n)|\le C\tau^{2-\kappa}(1+\ln\tau^{-1})^3$;
(A2) $\sum_{n\in N_\tau}|p|^\kappa|a_\tau^f(n)|\le C(P^2+m'^2)^{(\kappa-2)/2}+CS_\kappa+C\tau^{2-\kappa}(1+\ln\tau^{-1})^3\le C(P^2+m'^2)^{(\kappa-2)/2}$;
(A3) $W_\tau^f(P)\ge(2/\pi)^6c_S^6(P^2+m'^2)^{-1}$, *hence* $\hat M_\tau^f\le\min(b,C(P^2+m'^2))$; $|\hat v_x^{f,\tau}|\le C(P^2+m'^2)^{-1}$; $W_\tau^f\le C(P^2+m'^2)^{-1}$;
(A4) $|\hat v_x^{f,\tau}-\hat v_x^0|+|W_\tau^f-W_0|\le C\tau(P^2+m'^2)^{-1/2}$; $|\hat M_\tau^f-\hat M_0|\le C\tau(P^2+m'^2)^{3/2}$; $|\nabla^{(\tau)}\hat v_x^{f,\tau}-\partial\hat v_x^0|\le C\tau(1+\ln\tau^{-1})^3$, *where* $\nabla^{(\tau)}$ *multiplies the n-th term by* $d_\tau(p_\mu):=(e^{-i\tau p_\mu}-1)/\tau$ *and* ∂ *by* $-ip_\mu$.

Proof. (70): the $n=0$ term is $\le|P|^\kappa\frac{\pi^2}{4}(P^2+m'^2)^{-1}$ by (E1)–(E2), and the aliases give $\sum_{n\neq0}\rho(n)C|n|^\kappa|n|^{-2}=CS_\kappa$ (E3). (A1): each D-alias term is $\le(\sqrt3\pi/\tau)^\kappa(25\tau^2/4)\sigma(q)$ by (E1) at spacing 5τ ($|s_\tau^f(p)|\le|s_{5\tau}(q)|\le\sigma(q)$, as in the Dirichlet-weight paragraph of lemma D.13), and $\sum_{\ell\neq0}\sum_{n'\in N_{5\tau}}\sigma(q)\le124\big[1+\sum_{\emptyset\neq A\subset\{1,2,3\}}\prod_{\mu\in A}\sum_{1\le|n'_\mu|\le1/(10\tau)}\frac{\pi}{2|n'_\mu|}\big]\le C(1+\ln\tau^{-1})^3$. (A2): the $\ell=0$ terms are bounded as in (70) ($|s^f|\le\sigma(p)\le\rho(n)$); add (A1), and $\tau^{2-\kappa}(1+\ln\tau^{-1})^3\le C_\kappa$. (A3): the $n=0$ term of W^f with $|s_\tau^f(P)|\ge(2/\pi)^3c_S^3$ (valid for $|\tau P_\mu|\le\pi/5$, i.e. $\tau\le\frac15$) and (E2); the upper bounds from (A2) at $\kappa=0$. (A4): telescope $a_\tau^f(n)-a_0(n)=(s_\tau^f-s_0)\hat C_\tau+s_0(\hat C_\tau-\hat C_0)$. Uniform aliases: $|s_\tau^f(p)-s_0(p)|\le|s_{5\tau}(p)-s_0(p)||\varphi|+|s_0(p)||\varphi(\tau p)-1|\le C\min(1,\tau|p|)\sigma(p)$ ((E4) at spacing 5τ and $|\varphi_1(\vartheta)-1|\le C|\vartheta|$), and (E4) for $\hat C$, give $|a_\tau^f(n)-a_0(n)|\le C\tau\sigma(p)(|p|^2+m'^2)^{-1/2}$, summed $\le C\tau(P^2+m'^2)^{-1/2}+C\tau S_1$, as for the uniform weight. D-aliases: $\sum|a_\tau^f(n)|\le C\tau^2(1+\ln\tau^{-1})^3$ by (A1) at $\kappa=0$, and their continuum partners $a_0(n)$ with $|n|\ge1/(10\tau)$: $\sum_{|n|\ge N}\rho(n)|n|^{-2}\le CN^{-2}(1+\ln N)^2$. Continuum-only aliases $n\notin N_\tau$ ($|n|\ge(3\tau)^{-1}$): $\le C\tau$ as in lemma D.13(ii). All are $\le C\tau\le C\tau(P^2+m'^2)^{-1/2}$ on $\mathbb{T}$. For W^f-W_0 there is one more factor $|s|\le1$; $|\hat M_\tau^f-\hat M_0|\le\hat M_\tau^f\hat M_0|W_\tau^f-W_0|$. Derivative clause: $|d_\tau(\vartheta)|\le|\vartheta|$, $|d_\tau(\vartheta)+i\vartheta|\le\tau\vartheta^2/2$; so $|\nabla^{(\tau)}\hat v_x^{f,\tau}-\partial\hat v_x^0|\le\sum_{n\in N_\tau}[|a_\tau^f-a_0||p|+|a_0|\tau p^2/2]+\sum_{n\notin N_\tau}|a_0||p|$; the uniform aliases give $C\tau\sum_{n\in N_\tau}\sigma(p)[|p|(|p|^2+m'^2)^{-1/2}+p^2(|p|^2+m'^2)^{-1}]\le C\tau(1+\sum_{N_\tau\setminus0}\rho(n))\le C\tau(1+\ln\tau^{-1})^3$, the D-aliases $C\tau(1+\ln\tau^{-1})^3$ by (A1) at $\kappa=1$, and $\sum_{|n|\ge N}\rho(n)|n|^{-1}\le CN^{-1}(1+\ln N)^2$ the rest. □

B2.4 Rows (I) *and* (II)*: the translation-invariant part with rate.*

Lemma D.21 (cell averages)**.** *Let* $\langle F\rangle_{\text{cell}}:=\tau^3\sum_{x\in\Lambda_\tau\cap[0,1)^3}F(x,x)$. *With* $Z_\tau(P):=\sum_{n\in N_\tau}|s_\tau|^2\hat C_\tau^3\hat\Pi_\tau(p)$, $W_{2,\tau}:=\sum_n|s_\tau|^2\hat C_\tau^2$, $Y_\tau:=\sum_n|s_\tau|^2\hat C_\tau^2\hat\Pi_\tau(p)$,

$$\langle(\mathrm{I})\rangle_{\text{cell}}=2\int_{\mathbb{T}}\hat M_\tau Z_\tau\,\tfrac{dP}{(2\pi)^3},\qquad\langle(\mathrm{II})\rangle_{\text{cell}}=-\int_{\mathbb{T}}\hat M_\tau^2W_{2,\tau}Y_\tau\,\tfrac{dP}{(2\pi)^3},$$

and the same with s_τ^f, $\hat M_\tau^f$ *for the weight* f_D. *The continuum* ($\tau=0$) *expressions are the same with* $N_0=\mathbb{Z}^3$, s_0, $\hat C_0$, $\hat\Pi_0$.

Proof. $(\Gamma\Pi_C C)(x,x)=\sum_{y,z}\tau^6\Gamma(x,y)\Pi_C(y-z)C(z-x)=\int_{\mathbb{T}}\hat M_\tau\overline{\hat v_x(P)}\sum_n a_\tau(n)\hat C_\tau(p)\hat\Pi_\tau(p)e^{-ip\cdot x}\,\frac{dP}{(2\pi)^3}$ (sum the translation-invariant lines in Fourier). Averaging over $x\in\Lambda_\tau\cap[0,1)^3$ with weight τ^3 kills the off-diagonal aliases, $\tau^3\sum_x e^{i(p-p')\cdot x}=\tau^3\sum_x e^{2\pi i(n-n')\cdot x}=\delta_{nn'}$ ($n,n'\in N_\tau$). For (II): $(\Gamma\Pi_C\Gamma)(x,x)=\iint\hat M(P)\hat M(P')\overline{\hat v_x(P)}[\sum_{y,z}\tau^6\hat v_y(P)\Pi_C(y-z)\overline{\hat v_z(P')}]\hat v_x(P')$; the bracket is $\sum_{n,n'}a_\tau(n)\overline{a_\tau(n')}\hat\Pi_\tau(p)\tau^3\sum_z e^{-i(p-p')\cdot z}$, and $\sum_{z\in\Lambda_\tau}\tau^3e^{-i(p-p')\cdot z}$ with $p,p'\in\mathrm{BZ}_\tau$ forces $P=P'$ (as a δ on $\mathbb{T}$) and $n=n'$; the cell average of $|\hat v_x|^2$ then gives $W_{2,\tau}$. The weight enters s only, and the Poisson step is weight-free. □

Theorem D.22 (rows (I)+(II))**.** *For both weights, uniformly in* $b\in[\mathfrak{a}(1-5^{-2}),\mathfrak{a}]$ *and* $m',\mu'\in(0,m_0]$ *(for* f_D*:* $\tau=5^{-j}\le\frac15$*),*

$$\big|\langle(\mathrm{I})+(\mathrm{II})\rangle_{\text{cell},\tau}-\langle(\mathrm{I})+(\mathrm{II})\rangle_{\text{cell},0}\big|\le C\tau(1+m'^{-1}),\qquad\big|\langle(\mathrm{I})+(\mathrm{II})\rangle_{\text{cell},0}(b)-\langle(\mathrm{I})+(\mathrm{II})\rangle_{\text{cell},0}(b')\big|\le C|b-b'|.$$

Hence the diagonal increment obeys $\big|[(\mathrm{I})+(\mathrm{II})]_{t/5}(\mathfrak{a}_{j+1})-[(\mathrm{I})+(\mathrm{II})]_t(\mathfrak{a}_j)\big|\le Ct(1+m'^{-1})+Ct^2\le C't(1+m'^{-1})$.

Proof. (i) Majorants. $n=0$ ($p=P\in\mathbb{T}$): $|s|\le 1$, $\hat C\le C(P^2+m'^2)^{-1}$, $|\hat\Pi|\le C(|P|+m_0)$ (Π-a), $\hat M\le C(P^2+m'^2)$ ((A3) for f_D). So $|\hat M Z^{(0)}|\le C(|P|+m_0)(P^2+m'^2)^{-2}$ and $\int_{\mathbb{T}}\le C\int_0^{\sqrt{3}\pi}(\varrho+m_0)\varrho^2(\varrho^2+m'^2)^{-2}d\varrho\le C(1+\ln(1/m')+m_0/m')\le C(1+m'^{-1})$. Uniform aliases $n\neq 0$: $|s|^2\le\rho(n)^2$, $\hat C\le\frac34|n|^{-2}$, $|\hat\Pi|\le C|n|$, so $|\hat M Z^{(n)}|\le C\rho(n)^2|n|^{-5}$ and $\sum_{n\neq0}\le CS_0$. D-aliases (f_D): $|\hat M^f Z^{f,(n)}|\le C\sigma(q)^2(25\tau^2/4)^3C|p|\le C\tau^5\sigma(q)^2$, summed $\le C\tau^5(1+\ln\tau^{-1})^3$ by the count in (A1); likewise W_2^f ($\le C\tau^4(1+\ln\tau^{-1})^3$) and Y^f ($\le C\tau^3(1+\ln\tau^{-1})^3$). *(ii) Differences at a common $n\in N_\tau$.* Telescope the product $\hat M|s|^2\hat C^3\hat\Pi$ one factor at a time; each factor's difference is $\le C\tau(1+|p|)$ times its majorant: $|s_\tau-s_0|\le C\tau|p|\sigma$ (E4; for f_D and $\ell=0$ the bound of the proof of (A4)), $|\hat C_\tau-\hat C_0|\le C\tau(|p|^2+m'^2)^{-1/2}=C\tau(|p|^2+m'^2)^{1/2}(|p|^2+m'^2)^{-1}$ (E4), (Π-b), and $|\hat M_\tau-\hat M_0|\le C\tau(P^2+m'^2)^{3/2}=C\tau(P^2+m'^2)^{1/2}(P^2+m'^2)$. For $n=0$ this gives $\int_{\mathbb{T}}C\tau(1+|P|)(|P|+m_0)(P^2+m'^2)^{-2}\le C\tau(1+m'^{-1})$; for $n\neq0$, $\sum C\tau(1+|n|)\rho(n)^2|n|^{-5}\le C\tau$. D-aliases: bound the τ-term by its majorant of (i) ($O(\tau^3(1+\ln\tau^{-1})^3)$ at worst) and its continuum partner by the tail $|n|\ge 1/(10\tau)$: $|\hat M_0 Z_0^{(n)}|\le C\rho(n)|n|^{-5}$, $\sum_{|n|\ge1/(10\tau)}\le C\tau^2$. *(iii) Continuum-only aliases $n\notin N_\tau$ ($|n|\ge(3\tau)^{-1}$):* $|\hat M_0 Z_0^{(n)}|\le C\rho(n)|n|^{-5}$, $\sum_{|n|\ge1/(3\tau)}\le C\tau^2$. *(iv)* (II). The same bookkeeping for $\hat M^2W_2Y$ with the majorants $\hat M^2\le C(P^2+m'^2)^2$, $W_2^{(0)}\le C(P^2+m'^2)^{-2}$, $Y^{(0)}\le C(|P|+m_0)(P^2+m'^2)^{-2}$, alias parts $\le CS_0$ (and the D-alias bounds of (i)); every product of majorants is $\le C[(|P|+m_0)(P^2+m'^2)^{-2}+1]$, integrable with $\int\le C(1+m'^{-1})$, and each factor's difference again costs $C\tau(1+|p|)$. *(v) b-dependence at $\tau=0$.* $\partial_b\hat M=b^{-2}\hat M^2\le C(P^2+m'^2)^2$, so $|\partial_b\langle(\mathrm{I})\rangle_0|\le C\int(P^2+m'^2)^2(|P|+m_0)(P^2+m'^2)^{-3}\le C$ and $|\partial_b\langle(\mathrm{II})\rangle_0|\le C\int(P^2+m'^2)^3(|P|+m_0)(P^2+m'^2)^{-4}\le C$ (no infrared loss). The continuum expressions are weight-free (same Γ_0). *(vi) Diagonal increment.* Insert $\langle\cdot\rangle_0(\mathfrak{a}_{j+1})$ and $\langle\cdot\rangle_0(\mathfrak{a}_j)$; $|\mathfrak{a}_j-\mathfrak{a}_{j+1}|\le Ct^2$. Each member is compared with the continuum with its own weight and spacing (t or $t/5$). □

B2.5 The lines of row (III).

Lemma D.23 (coarse lines at spacing τ)**.** *Let $\theta\in(0,1)$. There is $C=C(\theta,\mathfrak{a},m_0)$ such that for all $\tau\in(0,1]$ ($\tau\le\frac15$ for f_D), $m'\in(0,m_0]$, $b\in[\mathfrak{a}(1-5^{-2}),\mathfrak{a}]$, $x,x',h,h'\in\Lambda_\tau$ and every forward difference $\nabla=\nabla_\mu^{(\tau)}$ acting on x or on x':*

(Γ1) $0\le\Gamma_\tau(x,x)\le C$, $|\Gamma_\tau(x,x')|\le C$;
(Γ2) $|\nabla\Gamma_\tau(x,x')|\le C$;
(Γ3) $|\nabla\Gamma_\tau(x+h,x'+h')-\nabla\Gamma_\tau(x,x')|\le C(|h|+|h'|)^\theta$;
(Γ4) *for $x,x'\in\Lambda_t$ ($t=5^{-j}$, both members of the diagonal pair, each with its own weight):* $|\Gamma_{t/5}(x,x')-\Gamma_t(x,x')|\le Ct$ *and* $|\nabla^{(t/5)}\Gamma_{t/5}(x,x')-\nabla^{(t)}\Gamma_t(x,x')|\le Ct(1+\ln t^{-1})^3$;
(Γ5) $|\partial_b\Gamma_\tau(x,x')|\le C$, $|\partial_b\nabla\Gamma_\tau(x,x')|\le C$.

The same holds for the photon coarse line $\Gamma_\tau^{\mathrm{ph}}$ with $m'\to\mu'$ (uniform weight).

Proof. In the Fourier representation a forward difference in x multiplies the n-th term of $\hat v_x^\tau$ by $d_\tau(p_\mu)$, $|d_\tau(\vartheta)|\le|\vartheta|$, and a displacement $x\to x+h$ by $e^{-ip\cdot h}$, $|e^{-ip\cdot h}-1|\le\min(2,|p||h|)\le 2^{1-\theta}|p|^\theta|h|^\theta$. Write $(*)_\kappa$ for (70), resp. (A2) for f_D, and use $\hat M\le C(P^2+m'^2)$ ((A3) for f_D). (Γ1) is lemma D.13(i), including its Dirichlet-weight paragraph. (Γ2): $|\nabla\hat v_x|\le(*)_1$, $|\hat v_{x'}|\le(*)_0$: the integrand is $\le C(P^2+m'^2)^{-1/2}$, integrable on $\mathbb{T}$ uniformly in m' (also at $m'=0$). (Γ3): displacing x: $|\sum_n a_\tau d_\tau(p_\mu)(e^{-ip\cdot h}-1)e^{-ip\cdot x}|\le C|h|^\theta(*)_{1+\theta}\le C|h|^\theta(P^2+m'^2)^{(\theta-1)/2}$, times $|\hat v_{x'}|\hat M\le C$; displacing x': $|\hat v_{x'+h'}-\hat v_{x'}|\le C|h'|^\theta(*)_\theta\le C|h'|^\theta(P^2+m'^2)^{(\theta-2)/2}$, times $|\nabla\hat v_x|\hat M\le C(P^2+m'^2)^{1/2}$; in both cases the integrand is $C|h|^\theta(P^2+m'^2)^{(\theta-1)/2}$, integrable since $(\theta-1)/2>-3/2$. For f_D the D-alias part of $(*)_{1+\theta}$ is $C\tau^{1-\theta}(1+\ln\tau^{-1})^3\le C_\theta$ (A2), so the constant stays τ-uniform. ($\theta=1$ fails only because $S_2=\infty$.) (Γ4), first clause: lemma D.13(ii) at both spacings, each member compared with the same Γ_0 ($\Lambda_t\subset\Lambda_{t/5}$; for f_D by (A4), each with its own weight): $\le Ct/5+Ct$. Second clause: compare both with $\partial\Gamma_0(x,x'):=\int\overline{\sum_n a_0(n)(-ip_\mu)e^{-ip\cdot x}}\,\hat v^0_{x'}\hat M_0\frac{dP}{(2\pi)^3}$. Telescoping $\nabla\hat v_x^\tau\,\hat v_{x'}^\tau\hat M_\tau-\partial\hat v_x^0\,\hat v_{x'}^0\hat M_0$, the first term is $|\nabla\hat v_x^\tau-\partial\hat v_x^0|\,|\hat v_{x'}^\tau|\hat M_\tau\le C\tau(1+\ln\tau^{-1})^3$ (the derivative clause of (A4); for the uniform

weight the same computation without D-aliases), and the other two, $|\partial\hat{v}^0_x||\hat{v}^\tau_{x'} - \hat{v}^0_{x'}|\hat{M}_\tau$ and $|\partial\hat{v}^0_x||\hat{v}^0_{x'}||\hat{M}_\tau - \hat{M}_0|$, are $\le C\tau(P^2+m'^2)^{-1/2}\cdot C$ by §B2.3; integrate over $\mathbb{T}$ and apply at $\tau = t$ and $\tau = t/5$. (Γ5): $\partial_b\hat{M} = b^{-2}\hat{M}^2 \le C(P^2+m'^2)^2$, so $|\partial_b\Gamma_\tau| \le C\int(P^2+m'^2)^{-2+2}\frac{dP}{(2\pi)^3} \le C$ and $|\partial_b\nabla\Gamma_\tau| \le C\int(P^2+m'^2)^{-1/2-1+2}\frac{dP}{(2\pi)^3} \le C$. Photon: the same formulas with $m' \to \mu'$, the same Q and b [13, (2.16)–(2.17)]. □

Lemma D.24 (the free lattice propagator with differences)**.** *For forward lattice differences ∇^α_t of order $|\alpha| \le 2$ there is C such that for all $t > 0$, $m' \ge 0$ and $r \in t\mathbb{Z}^3 \setminus 0$,*

$$|\nabla^\alpha_t C_t(r;m') - \partial^\alpha C_0(r;m')| \le C\,t\,|r|^{-2-|\alpha|}.$$

The same holds for the photon kernel $D_t(\cdot;\mu')$.

Proof. In the notation of the proof of lemma D.11, for $r = tn$: $\nabla^\alpha_t C_t(tn) = t^{-1-|\alpha|}\int_0^\infty e^{-M^2\tau}(\nabla^\alpha_1 P_\tau)(n)\,d\tau$ and $\partial^\alpha C_0(tn) = t^{-1-|\alpha|}\int_0^\infty e^{-M^2\tau}(\partial^\alpha G_\tau)(n)\,d\tau$, with ∇_1 the unit-lattice difference, so it suffices to show $\int_0^\infty |\nabla^\alpha_1 P_\tau - \partial^\alpha G_\tau|(n)\,d\tau \le C|n|^{-2-|\alpha|}$ ($t^{-1-|\alpha|}|n|^{-2-|\alpha|} = t|r|^{-2-|\alpha|}$). *One dimension.* $\nabla^j q_\tau(\ell) = (2\pi)^{-1}\int_{-\pi}^{\pi} e^{i\ell\vartheta}(e^{i\vartheta}-1)^j e^{-2\tau(1-\cos\vartheta)}d\vartheta$ and $\partial^j g_\tau(\ell) = (2\pi)^{-1}\int_{\mathbb{R}} e^{i\ell\vartheta}(i\vartheta)^j e^{-\tau\vartheta^2}d\vartheta$. Repeat the one-dimensional argument of lemma D.11 with these insertions: on the shifted contour $|e^{iz}-1| \le C|z|$ and $|(e^{iz}-1)^j - (iz)^j| \le C|z|^{j+1}$. The term $(iz)^j(e^{-\tau\varphi} - e^{-\tau\psi})$ carries one more factor $|z|^j \le C(\vartheta^2+\xi^2)^{j/2}$, which turns $\tau^{-3/2}$ into $\tau^{-(3+j)/2}$; the term $[(e^{iz}-1)^j - (iz)^j]e^{-\tau\varphi}$ is $O(|z|^{j+1})$ and gives $\tau^{-(2+j)/2}$. Hence, for $\tau \ge 1$, $|\nabla^j q_\tau(\ell) - \partial^j g_\tau(\ell)| \le C\tau^{-(2+j)/2}e^{-c\min(\ell^2/\tau,|\ell|)}$, and separately $|\nabla^j q_\tau(\ell)|, |\partial^j g_\tau(\ell)| \le C\tau^{-(1+j)/2}e^{-c\min(\ell^2/\tau,|\ell|)}$ for all $\tau > 0$ (the same contour shift). *Three dimensions.* $\nabla^\alpha_1 P_\tau(n) = \prod_\mu \nabla^{\alpha_\mu} q_\tau(n_\mu)$; telescope as in lemma D.11 with $\varrho = |n|_\infty \ge 1$. For $\tau \ge 1$ the integrand is $\le C\tau^{-(4+|\alpha|)/2}e^{-c\min(\varrho^2/\tau,\varrho)}$ (one differenced factor at $\tau^{-(2+j)/2}$, the two others at their own decay); $\int_1^\varrho \le Ce^{-c\varrho}$ and $\int_\varrho^\infty \tau^{-(4+|\alpha|)/2}e^{-c\varrho^2/\tau}d\tau \le C\varrho^{-2-|\alpha|}$. For $\tau \le 1$ write each forward difference as a sum of at most $2^{|\alpha|}$ shifted values: $|\nabla^\alpha_1 P_\tau(n)| \le 2^{|\alpha|}\max_{|n'-n|_\infty\le 2}|P_\tau(n')| \le Ce^{-c(\varrho-2)}$, since the contour shift at $\xi = \mathrm{sgn}(\ell)$ gives $|q_\tau(\ell)| \le e^{-|\ell|+2\tau(\cosh 1-1)} \le Ce^{-|\ell|}$ for $\tau \le 1$ (the shifted ϑ-integrand has modulus ≤ 1); and $|\partial^\alpha G_\tau(n)| \le C\tau^{-(3+|\alpha|)/2}e^{-\varrho^2/4\tau}$, integrable with $\int_0^1 \le Ce^{-\varrho^2/8}$. The finitely many n with $\varrho \le 3$ are covered by $\int_0^\infty(1+\tau)^{-(3+|\alpha|)/2}d\tau < \infty$ for the lattice term (bounded for $\tau \le 1$ by the same shifted-value argument, $\le C\tau^{-(3+|\alpha|)/2}$ for $\tau \ge 1$ by the separate bounds) and by the Gaussian factor for the continuum term. The mass enters only through $0 < e^{-M^2\tau} \le 1$. □

Lemma D.25 (free and outer lines)**.** *Let $w_s(r) := (|r|+\tau)^{-s}$ on Λ_τ. Uniformly in $\tau \in (0,1]$ and $m', \mu' \ge 0$:*

(T1) *$|\nabla^\alpha C_\tau(r)| \le Cw_{1+|\alpha|}(r)$ for all $r \in \Lambda_\tau$ and $|\alpha| \le 2$; the same for D_τ.*

(T2) *For $r \in \Lambda_t \setminus 0$: $|C_{t/5}(r) - C_t(r)| \le Ct^2|r|^{-3} \le Ctw_2(r)$ and $|\nabla^{(t/5)}C_{t/5}(r) - \nabla^{(t)}C_t(r)| \le Ct|r|^{-3} \le Ctw_3(r)$.*

(T3) *For $r, r' \in \Lambda_\tau$ with $|r| \ge 8t$, $|r'-r| \le 2\sqrt{3}t$, $\tau \le t$ and $|\alpha| \le 1$: $|\nabla^\alpha C_\tau(r') - \nabla^\alpha C_\tau(r)| \le Ctw_{2+|\alpha|}(r)$.*

For the outer line $G_\tau = C_\tau - \Gamma_\tau$ (both weights; $b \in [\mathfrak{a}(1-5^{-2}), \mathfrak{a}]$) and $i \le 1$:

(O1) *$|\nabla^i G_\tau(x,y)| \le Cw_{1+i}(x-y)$ for $|x-y| \le 1$ and $|\nabla^i G_\tau(x,y)| \le C$ for all x, y; hence on a ball $|x-y| \le R$, $|\nabla^i G_\tau(x,y)| \le C(R+1)^{1+i}w_{1+i}(x-y)$.*

(O2) *With $c := \alpha_0/(2\sqrt{3})$ and $R_\tau := (16\sqrt{3}/\alpha_0)\ln\tau^{-1} + 1$: $|\nabla^i G_\tau(x,y)| \le C\tau^4 e^{-c|x-y|}$ for $|x-y| \ge R_\tau$, where $\alpha_0 = \alpha_0(\mathfrak{a}, \|f\|_\infty) > 0$ is the constant of lemma D.8, independent of τ, m' and b.*

The photon lines D_τ, G^{ph}_τ obey the same bounds with μ'.

Proof. (T1): by lemma D.24, $|\nabla^\alpha C_\tau(r) - \partial^\alpha C_0(r)| \le C\tau|r|^{-2-|\alpha|} \le C|r|^{-1-|\alpha|}$ for $r \ne 0$ ($\tau \le |r|$), and $|\partial^\alpha C_0(r)| \le C|r|^{-1-|\alpha|}$ for the Yukawa kernel $C_0 = e^{-m'|r|}/(4\pi|r|)$, uniformly in $m' \ge 0$; at $r = 0$, $C_\tau(0) = \int_{\mathrm{BZ}_\tau}\hat{C}_\tau\frac{dP}{(2\pi)^3} \le C\tau^{-1}$ and $|\nabla^\alpha C_\tau| \le 2^{|\alpha|}\tau^{-|\alpha|}C_\tau(0)$ ($|C_\tau(r)| \le C_\tau(0)$ by positive definiteness). Together $|\nabla^\alpha C_\tau(r)| \le C(|r|+\tau)^{-1-|\alpha|}$. (T2): lemma D.11 at both

spacings, $|C_{t/5}-C_0|+|C_0-C_t|\le 2C_*t^2|r|^{-3}$, and $t^2|r|^{-3}\le 4tw_2(r)$ since $|r|\ge t$; lemma D.24 likewise for the differences (the coarse forward difference of $C_{t/5}$ at spacing $t/5$ is taken at the points of Λ_t with step $t/5$, as the statement says). (T3): a lattice path in Λ_τ from r to r' has $\le 3\cdot 2\sqrt{3}t/\tau$ steps; along it $|r''|\ge |r|-2\sqrt{3}t\ge |r|/2$; each step changes $\nabla^\alpha C_\tau$ by $\le \tau\sup|\nabla^{\alpha+e_\nu}C_\tau(r'')|\le C\tau|r|^{-2-|\alpha|}2^{2+|\alpha|}$ by (T1) at order $|\alpha|+1\le 2$; the sum is $\le Ctw_{2+|\alpha|}(r)$. (O1): (T1) and (Γ1), (Γ2) of lemma D.23; on $|r|\le R$, $C\le C(R+1)^s w_s(r)$. (O2): lemma D.8 at unit block scale ($\eta=1$, $H=-\Delta_\tau+m'^2+bM_f\mathsf{P}M_f$, $b\in[\mathfrak{a}(1-5^{-2}),\mathfrak{a}]$, block averages of f equal to 1): $|G_\tau(x,y)|\le 2c_0^{-1}\tau^{-3}e^{-\alpha_0|x-y|_\infty}$, so $|\nabla G_\tau|\le C\tau^{-4}e^{-\alpha_0(|x-y|_\infty-\tau)}$; with $|r|_\infty\ge |r|/\sqrt{3}$, $e^{-\alpha_0|r|/\sqrt{3}}=e^{-c|r|}e^{-c|r|}\le e^{-cR_\tau}e^{-c|r|}\le \tau^8 e^{-c|r|}$ for $|r|\ge R_\tau$. Photon: (T1)–(T3) with μ' (lemmas D.11 and D.24 hold for every mass), (O1)–(O2) with the uniform weight. □

Lemma D.26 (power counting on confined triangles)**.** *Let $\tau\in(0,1]$, $R\ge 2$, $x\in\Lambda_\tau$, $a,b,c\in[0,3]$ with $\Sigma:=a+b+c\le 6$, and $E_R:=\{(u,v)\in\Lambda_\tau^2:|x-u|\le R,|x-v|\le R\}$. Then, with C depending only on an upper bound for Σ,*

$$\tau^6\sum_{(u,v)\in E_R} w_a(x-u)w_b(u-v)w_c(v-x)\le C(1+\ln(R/\tau))^4R^{6-\Sigma}. \tag{71}$$

Two-point version: $\tau^3\sum_{|x-u|\le R}w_a(x-u)w_c(u-x)\le C(1+\ln(R/\tau))R^{(3-a-c)_+}\tau^{-(a+c-3)_+}$. Tail version: for $s\le 3$ and $\lambda>0$, $\tau^3\sum_u e^{-\lambda|x-u|}w_s(x-u)\le C_\lambda(1+\ln\tau^{-1})$.

Proof. Coincident terms. $u=x$: $\tau^3 w_a(0)=\tau^{3-a}$, and the v-sum is $\tau^3\sum_{|v-x|\le R}w_{b+c}(v-x)\le C(1+\ln(R/\tau))R^{(3-b-c)_+}\tau^{-(b+c-3)_+}$ (shells); the product is $\le C(1+\ln(R/\tau))R^{6-\Sigma}$ (if $b+c\ge 3$ the τ-exponent is $6-\Sigma\ge 0$; if $b+c<3$, $\tau^{3-a}R^{3-b-c}\le R^{6-\Sigma}$). $v=x$ and $u=v$: the same. *Non-coincident terms.* For $r\in\Lambda_\tau\setminus 0$ and r' in the sub-cube Q'_r of side $\tau/4$ centred at r, $|r'-r|\le\sqrt{3}\tau/8$, so $|r'|\le 2(|r|+\tau)$ and $\tau^3 w_s(r)\le 64\cdot 2^s\int_{Q'_r}|r'|^{-s}dr'$. Applied to the three factors with $u'\in Q'_u$, $v'\in Q'_v$ (so $|(u'-v')-(u-v)|\le\sqrt{3}\tau/4$ and $|u'-v'|\le 2(|u-v|+\tau)$): for distinct lattice points the sub-cubes are disjoint and every pairwise distance among x,u',v' is $\ge\tau-\sqrt{3}\tau/4\ge\tau/2$, whence

$$\tau^6\sum_{E_R,\ \text{non-coinc.}}\prod w\le 64^2\,2^\Sigma\iint_\Omega|x-u'|^{-a}|u'-v'|^{-b}|v'-x|^{-c}\,du'dv',\quad \Omega:=\{|x-u'|,|x-v'|\le R+1,\ \text{all pairwise}$$

Dyadic shells in $\varrho:=\max(|x-u'|,|x-v'|)\in[\tau/2,R+1]$: scaling $(u'-x,v'-x)=2^i(u'',v'')$ on the shell $2^i\le\varrho<2^{i+1}$ gives $2^{i(6-\Sigma)}J_i$, $J_i:=\int_{1\le\varrho''<2,\ \text{distances}\ge\delta_i:=\tau/2^{i+1}}|u''|^{-a}|u''-v''|^{-b}|v''|^{-c}$. On the shell one of $|u''|,|v''|$ is ≥ 1, say $|u''|\ge 1$, so $|u''|^{-a}\le 1$; the remaining $\int_{|v''|\le 2}|v''|^{-c}|u''-v''|^{-b}dv''$ with $b,c\le 3$ and the two singular points at distance ≥ 1 is $\le C(1+\ln(2^i/\tau))^2$ (a slot equal to 3 costs one logarithm, a slot <3 none), and the u''-integral over $1\le|u''|\le 2$ is bounded. So $J_i\le C(1+\ln(R/\tau))^2$ and $\sum_{i:\tau/4\le 2^i\le R+1}2^{i(6-\Sigma)}\le C(1+\ln(R/\tau))R^{6-\Sigma}$ ($\Sigma\le 6$). Total $\le C(1+\ln(R/\tau))^4R^{6-\Sigma}$. Two-point and tail versions: shells in one variable. □

Lemma D.27 (one inner sum)**.** *For $b,c\in[0,3]$ and $u,x\in\Lambda_\tau$: $\tau^3\sum_{v\in\Lambda_\tau,|v-x|\le R}w_b(u-v)w_c(v-x)\le C(1+\ln(R/\tau))\cdot\{w_{b+c-3}(u-x)$ if $b+c>3$; 1 if $b+c=3$; R^{3-b-c} if $b+c<3\}$.*

Proof. Split into $|v-x|\le|u-x|/2$ (there $w_b(u-v)\le 2^b w_b(u-x)$), $|v-u|\le|u-x|/2$ (there $w_c(v-x)\le 2^c w_c(u-x)$) and the rest (there $|v-u|\ge|u-x|/2$ and $|v-x|\ge|u-x|/2$, so $w_bw_c\le Cw_{b+c}(v-x)$ up to the roles), and sum in shells. □

B2.6 Row (III) *with rate, the sup bound, and the original-vertex part.*

Theorem D.28 (row (III))**.** *Let $\mathcal{T}_\tau(b):=\tau^3\sum_{x\in\Lambda_\tau\cap[0,1)^3}\big(G(\Pi_C-\Pi_G)G\big)_\tau(x,x)$ (all lines at spacing τ with block coupling b and the weight of that member). For every $\theta<1$ there is C_θ, depending on θ, $\mathfrak{a}$, m_0 and $\|f\|_\infty$ only, such that for all $t=5^{-j}\le 1$ ($\le\frac15$ for f_D) and $m',\mu'\in(0,m_0]$:*

$$\text{(III-sup)}\quad \sup_x\big|\big(G(\Pi_C-\Pi_G)G\big)_t(x,x)\big|\le C(1+\ln t^{-1})^{14};\qquad \text{(III-inc)}\quad \big|\mathcal{T}_{t/5}(\mathfrak{a}_{j+1})-\mathcal{T}_t(\mathfrak{a}_j)\big|\le C_\theta t^\theta.$$

Proof. By lemmas D.17 and D.18, $(G(\Pi_C-\Pi_G)G)(x,x)$ is a finite sum (independent of τ) of triangle terms $T_\tau(x)=\tau^6\sum_{u,v}k_1(x,u)k_2(u,v)k_3(v,x)$ and of two-point (seagull) terms, with $k_1,k_3\in\{G_\tau,\nabla G_\tau\}$ (slots $a,c\in\{1,2\}$), $k_2\in\{\Gamma D,\Gamma\nabla D,\nabla\Gamma D,G_{\rm in}\Gamma^{\rm ph},\nabla G_{\rm in}\Gamma^{\rm ph},G_{\rm in}\nabla\Gamma^{\rm ph}\}$ (slot $b\in\{1,2\}$), $a+b+c\le 5$, arguments shifted by at most one lattice unit. Put $R:=R_{t/5}+5\le C(1+\ln t^{-1})$ and split each outer line $k_i=k_i^{\rm in}+k_i^{\rm out}$, $k^{\rm in}:=k\mathbf{1}_{|r|\le R}$. *Sup bounds* (L1): $|k_1|\le C(R+1)^a w_a$, $|k_3|\le C(R+1)^c w_c$ on $|r|\le R$ and $\le C\tau^4e^{-c|r|}$ on $|r|\ge R-5$ (O1), (O2); $|k_2|\le Cw_b$ ((Γ1), (Γ2), (T1), (O1)), with C replaced by $C(2R+1)^b$ for an inner $G_{\rm in}$, $\nabla G_{\rm in}$ on the confined region $|u-v|\le 2R$ (a polylogarithm, absorbed below); where an outer line is "out", such an inner line is bounded by $Cw_b(u-v)$ on $|u-v|\le 1$ and by C beyond (slot 0). The weight enters only through lemmas D.23 and D.25; the kernels are evaluated, never their weights.

Step 0 (sup bound). Core (both outer lines "in"): lemma D.26 with $\Sigma\le 5$ gives $\le C(R+1)^{a+c}(2R+1)^b(1+\ln(R/\tau))^4R^{6-\Sigma}\le C(1+\ln t^{-1})^{13}$. Terms with $k_1^{\rm out}$: $\le C\tau^4\sum_u\tau^3e^{-c|x-u|}[\tau^3\sum_v w_b(u-v)|k_3(v,x)|]$; by lemma D.27 (the v-sum over the ball $|v-x|\le R$) and the $k_3^{\rm out}$ tail, the bracket is $\le C(R+1)^{c+2}(1+\ln\tau^{-1})(1+w_{(b+c-3)_+}(u-x))+C\tau^4(1+\ln\tau^{-1})$ (the case $b+c<3$ of lemma D.27, with slot 0 for an inner $G_{\rm in}$ beyond $|u-v|\le 1$, gives $R^{3-b-c}\le R^2$), and the tail version of lemma D.26 gives $\le C\tau^4(1+\ln\tau^{-1})^q$. Same for $k_3^{\rm out}$. Seagull terms: the two-point version of lemma D.26. Hence (III-sup) with exponent 14, uniformly in m',μ': no line needs mass decay, the outer lines decaying at the m'-independent rate c.

Step 1 (decomposition of the increment). With the cells $c(x):=x+(t/5)\{0,\dots,4\}^3$ (they tile $\Lambda_{t/5}$, and the fine unit cell is the disjoint union of the $c(x)$, $x\in\Lambda_t\cap[0,1)^3$), for each triangle term

$$\mathcal{T}_{t/5}(\mathfrak{a}_j)-\mathcal{T}_t(\mathfrak{a}_j)=\mathcal{D}+\mathcal{C},\quad \mathcal{D}:=t^9\sum_{x,u,v\in\Lambda_t}\Big[5^{-9}\sum_{x'\in c(x),u'\in c(u),v'\in c(v)}F_{t/5}(x',u',v')-F_{t/5}(x,u,v)\Big],\quad \mathcal{C}:=t^9\sum_{x,u,v\in\Lambda_t}[F_t$$

$F_\tau(x,u,v):=k_1k_2k_3$ at spacing τ (x restricted to the unit cell). Near-diagonal set $\mathcal{N}:=\{\min(|x-u|,|u-v|,|v-x|)<8t\}$. (For f_D the fine member's weight lives on the t-blocks, which are exactly the cells $c(x)$, the coarse member's on the $5t$-blocks.)

Step 2 ($\mathcal{N}$). Each of the four sums restricted to $\mathcal{N}$ (fine triples whose coarse parent is in $\mathcal{N}$ lie in $\mathcal{N}':=\{\min<8t+2\sqrt{3}t\}$) is bounded by (L1) alone. Case $|x-u|<8t$: $\le 17^3$ values of u (resp. $(17\cdot5)^3$ fine ones), $|F|\le C(R+1)^{a+c}\tau^{-a}w_b(u-v)w_c(v-x)$; lemma D.27 ($R$-confinement by k_3, or its tail) gives $\tau^3\sum_v\le C(1+\ln\tau^{-1})R^3t^{-(b+c-3)_+}$, so the contribution is $\le Ct^3t^{-a}t^{-(b+c-3)_+}(1+\ln t^{-1})^8\le Ct(1+\ln t^{-1})^8$, since $a\le 2$ and $a+b+c\le 5$ give $3-a-(b+c-3)_+\ge\min(3-a,6-\Sigma)\ge 1$. Case $|v-x|<8t$: symmetric. Case $|u-v|<8t$: $\le 17^3$ values of v per u, $|F|\le C(R+1)^{a+c}w_a(x-u)t^{-b}w_c(v-x)$ with $w_c(v-x)\le 2^cw_c(u-x)$ if $|u-x|\ge 16t$ and $\le t^{-c}$ otherwise: $\le Ct^{3-b}(R+1)^{a+c}[t^3\sum_{|x-u|\le R}w_{a+c}(x-u)+t^3\cdot 33^3t^{-a-c}]\le Ct(1+\ln t^{-1})^8$ ($b\le 2$, $a+c\le 4$, $\Sigma\le 5$). Two-point terms: the same with one fewer sum.

Step 3 ($\mathcal{C}$ off $\mathcal{N}$: same points, two spacings). $F_{t/5}-F_t=\sum_\ell(k_\ell^{t/5}-k_\ell^t)\prod_{\rm others}$ (mixed products, every factor bounded by (L1) at its own spacing). Line differences at common points $x,u,v\in\Lambda_t$ with all distances $\ge 8t$: translation-invariant factors $C,\nabla C,D,\nabla D$: $\le Ctw_{s+1}$ by (T2), slot $s\to s+1$; coarse factors $\Gamma,\Gamma^{\rm ph}$: $\le Ct$, $\nabla\Gamma,\nabla\Gamma^{\rm ph}$: $\le Ct(1+\ln t^{-1})^3$ by (Γ4), slots unchanged (for f_D, each member with its own weight, (Γ4) as stated); outer lines $G=C-\Gamma$, ∇G: the sum of a translation-invariant and a coarse difference, hence $\le Ctw_{s+1}+Ct(1+\ln t^{-1})^3\le Ct(1+\ln t^{-1})^3(R+1)^{s+1}w_{s+1}$ on $|r|\le R$, and $\le 2C\tau^4e^{-c|r|}$ on $|r|\ge R$ (both members are tails there, (O2), with the same α_0 for both spacings). An argument shifted by one lattice unit at the two spacings (te_μ versus $(t/5)e_\mu$) is a displacement $\le t$ and falls under Step 4. For a differenced inner line the two outer lines still confine u,v to R-balls (or are tails); for a differenced outer line its "in" part is confined by definition and its "out" part carries $e^{-c|r|}$: no mass decay is used. lemma D.26 with $\Sigma\le 6$, $R\le C(1+\ln t^{-1})$, and the tail version: $|\mathcal{C}\text{ off }\mathcal{N}|\le Ct(1+\ln t^{-1})^q$.

Step 4 ($\mathcal{D}$ off $\mathcal{N}$: displaced arguments, one spacing $\tau=t/5$). For $(x',u',v')\in c(x)\times c(u)\times c(v)$ every pairwise displacement is $\le 2\sqrt{3}t\le|r|/4$ ($|r|\ge 8t$), so $|r'|\ge|r|/2$ and (L1) at the displaced arguments is $\le 2^s$(L1) at (x,u,v); R-confinement holds with $R\pm 4$. Telescope over the lines:

translation-invariant factors by (T3): $\le Ctw_{s+1}(r)$; coarse factors $\Gamma, \Gamma^{\mathrm{ph}}$ by (Γ2) along lattice paths: $\le Ct$; $\nabla\Gamma, \nabla\Gamma^{\mathrm{ph}}$ by (Γ3): $\le Ct^{\theta}$ (slots unchanged; for f_D the finest-scale modulation of $\nabla\Gamma$ is inside the constant of (Γ3)); outer lines: translation-invariant plus coarse parts as above, tails by (O2). lemma D.26 ($\Sigma \le 6$): $|\mathcal{D}\text{ off }\mathcal{N}| \le C_\theta t^{\theta}(1+\ln t^{-1})^{q}$. Two-point terms: the same.

Step 5 (b). $|\mathcal{T}_{t/5}(\mathfrak{a}_{j+1}) - \mathcal{T}_{t/5}(\mathfrak{a}_j)|$: telescope over the lines; translation-invariant lines do not depend on b. Inner coarse factors ($\Gamma, \Gamma^{\mathrm{ph}}$, and G_{in} through Γ): (Γ5) gives $\le C|\mathfrak{a}_j - \mathfrak{a}_{j+1}| \le Ct^2$, confined by the two outer lines as in Steps 3–4. A differenced outer line: the same in/out dichotomy. lemma D.8 is uniform in $b \in [\mathfrak{a}(1-5^{-2}), \mathfrak{a}]$, so for $|r| \ge R$ both members obey (O2), $|k(\mathfrak{a}_{j+1})(r) - k(\mathfrak{a}_j)(r)| \le 2C\tau^4 e^{-c|r|}$; for $|r| \le R$ use (Γ5): $\le Ct^2(R+1)^s w_s(r)$, s its slot. lemma D.26 and its tail version give $\le Ct^2(1+\ln t^{-1})^{q}$.

Summing Steps 2–5 and absorbing the logarithms into t^{θ} ($\theta < 1$ arbitrary) gives (III-inc). Summation by parts (lemma D.18) was on the infinite lattice, so there is no boundary term, and all sums are absolutely convergent at fixed τ (outer lines decay exponentially by (O2), inner lines are bounded by w_1 at worst). □

Proposition D.29 (the sup bound at order e^2)**.** *For both weights, for all k and $1 \le j \le k$, with C independent of k and j (depending on m, $\mathfrak{a}$, m_0, $\|f\|_\infty$):*

$$\big\|e^2[\mathfrak{g}_x(C,C,C;D) - \mathfrak{g}_x(G_j,G_j,G_j;D_j)]\big\|_\infty = \|\beta_{(2,0)} + \mathfrak{a}\|_\infty \le Ce^2\big[(1+(m\eta)^{-1}) + (1+j\ln 5)^{14}\big] \le Ce^2\eta^{-1}(1+j)^{14}.$$

Proof. Pointwise (no cell average), at unit block scale with $\tau = t = 5^{-j}$: $(\Gamma\Pi_C C)(x,x) = \int_{\mathbb{T}} \hat{M}\,\overline{\hat{v}_x}\,\hat{u}_x \frac{dP}{(2\pi)^3}$ with $\hat{u}_x := \sum_{n\in N_\tau} s_\tau \hat{C}_\tau^2 \hat{\Pi}_\tau(p) e^{-ip\cdot x}$; by (E1)–(E3) and ($\Pi$-a), $|\hat{u}_x| \le C(|P| + m_0)(P^2+m'^2)^{-2} + C\sum_{n\ne 0}\rho(n)|n|^{-3} \le C(|P|+m_0)(P^2+m'^2)^{-2}$ (for f_D the D-aliases add $\sum_{\ell\ne 0}\sigma(q)(25\tau^2/4)^2 C\sqrt{3}\pi/\tau \le C\tau^3(1+\ln\tau^{-1})^3$), and $|\hat{v}_x|\hat{M} \le C$ ((A3) for f_D); so $|(\Gamma\Pi_C C)(x,x)| \le C\int_{\mathbb{T}}(|P|+m_0)(P^2+m'^2)^{-2}\frac{dP}{(2\pi)^3} \le C(1+\ln(1/m') + m_0/m') \le C(1+m'^{-1})$ (this is the infrared logarithm of the order-e^2 Wick constant). $(C\Pi_C\Gamma)(x,x)$: the same. $(\Gamma\Pi_C\Gamma)(x,x) = \int \hat{M}^2|\hat{v}_x|^2 Y_\tau \frac{dP}{(2\pi)^3} \le C\int (P^2+m'^2)^2(P^2+m'^2)^{-2}(|P|+m_0)(P^2+m'^2)^{-2}\frac{dP}{(2\pi)^3} \le C(1+m'^{-1})$. (III): (III-sup). By (67)–(68) the original-vertex density is $-2e^2[(\mathrm{I})+(\mathrm{II})+(\mathrm{III})]$ at unit block scale; with $m' = m\eta$, $1+m'^{-1} = 1+(m\eta)^{-1}$ and $(1+\ln t^{-1})^{14} = (1+j\ln 5)^{14}$. The last inequality uses $\eta \le 1$ and $(m\eta)^{-1} \le m^{-1}\eta^{-1}$. □

Theorem D.30 (the original-vertex part)**.** *Let $\theta \in (0,1)$. There are C, C_θ (depending on θ, $\mathfrak{a}$, $m_0 \ge \max(m,\mu_0)\eta$, e, m and $\|f\|_\infty$) such that for all k and $1 \le j \le k$ ($t = 5^{-j}$, $\eta = 5^j\varepsilon_k \le 1$), for both weights,*

$$\|\beta_{(2,0)} + \mathfrak{a}\|_\infty \le Ce^2\big[(1+(m\eta)^{-1}) + (1+j\ln 5)^{14}\big] \le Ce^2\eta^{-1}(1+j)^{14},$$

$$\big|(\bar{\beta}_{(2,0)} + \bar{\mathfrak{a}})_k^{(j)} - (\bar{\beta}_{(2,0)} + \bar{\mathfrak{a}})_{k+1}^{(j+1)}\big| \le C_\theta e^2\big[t(1+(m\eta)^{-1}) + t^{\theta}\big] \le C_\theta e^2\, 5^{-\theta j}\eta^{-1},$$

where $\bar{\mathfrak{a}}$ is the block mean of $\mathfrak{a}_x$. This is the statement of lemma D.15 for the original-vertex part, with $\vartheta = \theta$ arbitrary in $(0,1)$ and $q = 14$.

Proof. By (67)–(68), $\beta_{(2,0)} + \mathfrak{a} = -2e^2[(\mathrm{I})+(\mathrm{II})+(\mathrm{III})]$ at unit block scale, and the block mean $\eta^{-3}\sum_{x\in\Lambda_k\cap[0,\eta)^3}\varepsilon_k^3(\cdot)$ is the cell average $\langle\cdot\rangle_{\mathrm{cell}}$ of lemma D.21 at unit block scale. The diagonal pair $(k,j) \to (k+1, j+1)$ keeps η, m', μ' and the unit cell fixed and changes $(\tau, b, \text{weight}) = (t, \mathfrak{a}_j, f \text{ on } 5t\text{-blocks}) \to (t/5, \mathfrak{a}_{j+1}, f \text{ on } t\text{-blocks})$ (the reduction of lemma D.13 and its Dirichlet-weight paragraph). Theorem D.22: $|[(\mathrm{I})+(\mathrm{II})]_{t/5}(\mathfrak{a}_{j+1}) - [(\mathrm{I})+(\mathrm{II})]_t(\mathfrak{a}_j)| \le Ct(1+m'^{-1})$. Theorem D.28: $|\mathcal{T}_{t/5}(\mathfrak{a}_{j+1}) - \mathcal{T}_t(\mathfrak{a}_j)| \le C_\theta t^{\theta}$. Sup bound: proposition D.29. With $m' = m\eta$: $t(1+m'^{-1}) \le (1+m^{-1})5^{-j}\eta^{-1}$ and $t^{\theta} = 5^{-\theta j} \le 5^{-\theta j}\eta^{-1}$. For f_D the restriction $j \ge 1$ ($\tau \le \frac{1}{5}$ for both members) is the one in the statement of lemma D.15. □

B2.7 The averaging vertices: structure and the photon slices. $\mathfrak{a}_x$ is the order-e^2 contribution to $\langle r_x^2\rangle^{\mathrm{fl}}$ of the A-dependence of King's block averaging, the vertices [13, (3.24)–(3.26)] with the contour convention [13, (2.19), (3.27)]. We work at unit block scale ($t = 5^{-j}$, Λ_t, $m', \mu' \in (0, m_0]$, $b = \mathfrak{a}_j$) and write $\kappa_0 := \alpha_0/\sqrt{3}$, so that lemma D.8 at unit block scale reads $|G_\tau(x,y)| \le C\tau^{-3}e^{-\kappa_0|x-y|}$ and, at block side ℓ, $|D_i(u,v)| \le C\ell^2\tau^{-3}e^{-\kappa_0|u-v|/\ell}$.

Photon slices. For $0 \le i \le j$ let $\ell_i := 5^i t$, P_i the uniform block average onto the blocks of side ℓ_i (corner blocks [13, (2.12)]), $D_i := (-\Delta_t + \mu'^2 + \mathfrak{a}_i \ell_i^{-2} P_i)^{-1}$ ($D_0 := 0$), and $S_i := D_{i+1} - D_i$ for $i < j$. By [13, (2.17)–(2.18)] the photon fluctuation field after j steps is $A' = \sum_{i<j} A'_i$ with independent Gaussian slices, $\langle A'_{i,\mu}(u) A'_{i',\nu}(v)\rangle = \delta_{ii'}\delta_{\mu\nu} S_i(u,v)$ (definition D.7: $D_j = \sum_{i<j} G^{\mathrm{ph}}_{(i)}$).

Contours. By [13, (2.19)], $A(\Gamma_{y,x}) = \sum_{i<j} A'_i(\Gamma^{(i+1)}_{x_{i+1},x})$, $A(\Gamma) := \sum_{\mathsf{b}\in\Gamma} t A_{\mathsf{b}}$, where x_{i+1} is the corner of the ℓ_{i+1}-block of x and $\Gamma^{(i+1)}_{x_{i+1},x}$ is a staircase of $\le 3\sum_{i'\le i} 5^{i'+1}$ fine bonds, hence of length $\le C_L \ell_i$, $C_L := 3L^2/(L-1)$: slice i sees only the sub-contour inside its own ℓ_{i+1}-block.

Vertices. With $U(\vartheta) = 1 + \vartheta J - \vartheta^2/2 + O(\vartheta^3)$ (J the complex structure on the two real components) and $V_{\mathrm{av}} = -\frac{b}{2}\sum_y [|Q(A)\phi|^2 - |Q\phi|^2]$, $(Q(A)\phi)(y) = \sum_{x\in B(y)} t^3 U(e' A(\Gamma_{y,x}))\phi(x)$ [13, (3.24)–(3.26)] at zero block fields:

$$V_1 = -be' \sum_y \sum_{x\in B(y)} t^3 A(\Gamma_{y,x})\, \langle Q\phi(y), J\phi(x)\rangle, \qquad V_2 = \tfrac{be'^2}{2} \sum_{y,x} t^3 A(\Gamma_{y,x})^2 \langle Q\phi(y), \phi(x)\rangle - \tfrac{be'^2}{2} \sum_y \Big| \sum_{x\in B(y)} t^3 A(\Gamma$$

$$V_h = e' t^3 \sum_{\mathsf{b}} A_{\mathsf{b}} \langle \nabla\phi(\mathsf{b}), J\phi(\mathsf{b}_+)\rangle = -e' t^2 \sum_{\mathsf{b}} A_{\mathsf{b}} \langle \phi(\mathsf{b}_-), J\phi(\mathsf{b}_+)\rangle,$$

the last being the hopping vertex [13, (3.22)] at $m = 1$ (one difference quotient) at unit block scale ($\langle \phi(\mathsf{b}_+), J\phi(\mathsf{b}_+)\rangle = 0$; its sign plays no role). At order e'^2 of $\langle |\phi(x_0)|^2\rangle^{\mathrm{fl}}$, since $\langle A'\rangle = 0$ kills the first order and everything without an averaging vertex is the original-vertex part, $\mathfrak{a}_{x_0}$ consists of the four classes of connected correlations

$$\mathrm{E1} + \mathrm{E2} = \langle |\phi(x_0)|^2; V_2\rangle, \qquad \mathrm{E3} = \tfrac{1}{2}\langle |\phi(x_0)|^2; V_1; V_1\rangle, \qquad \mathrm{E4} = \langle |\phi(x_0)|^2; V_1; V_h\rangle.$$

Their photon kernels are $\mathcal{V}(x) := \sum_i \mathrm{Var}(A'_i(\Gamma^{(i+1)}_{x_{i+1},x}))$, $\mathcal{K}(x,x') := \sum_i \langle A'_i(\Gamma(x)) A'_i(\Gamma(x'))\rangle$ and $\mathcal{K}_{\mathsf{b}}(x) := \sum_i \mathcal{K}_{i,\mathsf{b}}(x)$, $\mathcal{K}_{i,\mathsf{b}}(x) := \sum_{\mathsf{b}'\in\Gamma_i(x),\ \mathsf{b}'\parallel\mathsf{b}} t S_i(\mathsf{b}', \mathsf{b})$; the scalar lines are $G = G_t$, (QG), (QGQ^*). Each class is a two- or three-line cycle through x_0 (E1/E2: two lines from x_0 to legs in one block; E3/E4: x_0–v_1–v_2–x_0), and E4 carries one differentiated line: the J-antisymmetry at b gives $2(QG)(y,x_0)[G(x_0,\mathsf{b}_-)G(x,\mathsf{b}_+) - G(x_0,\mathsf{b}_+)G(x,\mathsf{b}_-)]$, a forward difference at b on one of the two lines.

Lemma D.31 (photon slices)**.** (S0) *S_i is the slice covariance of* [13, (2.17)]*: the joint Gaussian of (ϕ,ψ_i) there has marginal ϕ-covariance D_{i+1}, because $\ell_{i+1}^2/\mathfrak{a}_{i+1} = \ell_i^2/\mathfrak{a}_i + \ell_{i+1}^2/\mathfrak{a}$ (algebra from* [13, (2.13)]*), and by the law of total covariance $D_{i+1} = \mathrm{Cov}(\phi|\psi) + \mathrm{Cov}(\mathbb{E}[\phi|\psi]) = D_i + \mathfrak{a}_i^2 \ell_i^{-4} D_i Q_i^* C^{(i)} Q_i D_i$, the term of* [13, (2.17)].

(S1) $0 \le S_i \le D - D_i$*: $P_{i+1} = Q_i^* Q'^* Q' Q_i \le P_i$ ($Q'Q'^* = 1$) and $\mathfrak{a}_i \ell_i^{-2} \ge \mathfrak{a}_{i+1}\ell_{i+1}^{-2}$, so $D_i \le D_{i+1} \le D$.*

(S2) $|S_i(u,v)| \le C\ell_i^{-1}$*: $S_i(u,u) \le (D-D_i)(u,u)$ by (S1), Cauchy–Schwarz, and $(D-D_i)(u,u) = \ell_i^{-1}\Gamma_{t/\ell_i}(u/\ell_i, u/\ell_i) \le C\ell_i^{-1}$ by lemma D.13(i) at relative spacing $t/\ell_i = 5^{-i}$, mass $\mu'\ell_i \le m_0$, coupling $\mathfrak{a}_i$; for $i = 0$, $D(u,u) \le C/t$. Uniform in μ'.*

(S3) $|\nabla_u S_i(u,v)| \le C\ell_i^{-2}$*: (T2) rescaled for $D - D_i$ and $D - D_{i+1}$ ($i \ge 1$), (T1) for $i = 0$.*

(S4) $|S_i(u,v)| \le C\ell_i^{-1} \min\big(1, e^{(\kappa_0/2)(\rho_i - |u-v|)/\ell_{i+1}}\big)$ *with $\rho_i := (6/\kappa_0)\ell_{i+1}\ln(\ell_{i+1}/t) \le C\ell_i(i+1)$: lemma D.8 for D_i and D_{i+1} at block side $\le \ell_{i+1}$, half the exponent paying the prefactor $\ell_{i+1}^2 t^{-3}$ beyond ρ_i.*

(S5) *For $u,v \in \Lambda_t \subset \Lambda_{t/5}$ and $i \ge 1$: $|S_i^{(t)}(u,v) - S_{i+1}^{(t/5)}(u,v)| \le Ct\ell_i^{-2}$: the coarse slice i and the fine slice $i+1$ have the same block sides ℓ_i, ℓ_{i+1}; compare each of the four coarse lines with the continuum one at its block side by lemma D.13(ii) rescaled ($\ell^{-1}\cdot C(t/\ell)$), and the coupling change $\mathfrak{a}_i \to \mathfrak{a}_{i+1}$ by lemma D.13(iv) rescaled ($C\ell_i^{-1}5^{-2i} = Ct^2\ell_i^{-3} \le Ct\ell_i^{-2}$). Slice 0 of both members and the extra finest slice of the finer member are not matched; each contributes $O(t\cdot\mathrm{polylog})$ (E4: $O(t^2\cdot\mathrm{polylog})$) by lemma D.32(i),(ii) below, since $\ell_0 = t$ for both.*

Lemma D.32 (contour kernels)**.** (i) $\mathrm{Var}(A'_i(\Gamma^{(i+1)})) \le (C_L\ell_i)^2 C\ell_i^{-1} = C\ell_i$*; $\mathcal{V} \le C\sum_{i<j}\ell_i \le C$; $|\mathcal{K}_i| \le C\ell_i$ and $|\mathcal{K}| \le C$ (Cauchy–Schwarz).*

(ii) $|\mathcal{K}_{i,\mathsf{b}}(x)| \le C$ *($\le C_L\ell_i/t$ bonds of weight t, (S2)); up to the tail of (S4) it is supported in* $|\mathsf{b}-x| \le r_i := C_L\ell_i + \rho_i \le C\ell_i(i+1)$.

(iii) *For $x \in \Lambda_t$ (corner blocks, $x_1 = x$) the finer member's sub-contours coincide with the coarse ones as point sets and its finest slice sees the empty contour; for x' in the coarse cell of x every slice gains $\Gamma_{x,x'}$ ($\le 3(L-1)$ fine bonds, length $\le 3t$).*

(iv) *Replacing the fine bond sum along a contour by the coarse one in the same kernel costs $\le 2(C_L\ell_i)^2 C\ell_i^{-2} t = Ct$ per slice ((S3)).*

(v) *In-cell oscillation of the finer member's kernels: $|\mathcal{K}'(x',\cdot) - \mathcal{K}'(x,\cdot)| \le Ct(j+1)$; for $\mathcal{K}'_{\mathsf{b}}$ the E4 cost is $\le Ctj^2$.*

Proof. (i): the variance of A'_i along a contour of $\le C_L\ell_i/t$ bonds of length t is bounded by $(\text{length})^2 \sup|S_i|$, (S2); $\sum_{i<j}\ell_i = t(5^j-1)/4 \le \frac{1}{4}$. (ii): (S2) and (S4), the bonds of $\Gamma_i(x)$ lying within $C_L\ell_i$ of x. (iii) is the corner-block convention [13, (2.12), (2.19)]. (iv): (S3), the Riemann error of a bond sum with step t over a path of length $C_L\ell_i$ against a kernel with gradient $\le C\ell_i^{-2}$. (v): (iii) and (S2), (S3), summed over the $\le j+1$ slices. □

Lemma D.33 (scalar lines)**.** *Write $|r|_t := |r| + t$ and $R_t := (6/\kappa_0)\ln t^{-1} = 6\sqrt{3}\,\alpha_0^{-1}\ln t^{-1}$, the radius used in the proof of lemma D.14 (in §§B2.7–B2.10 R_t denotes this radius, not the R_τ of (O2) in lemma D.25); $G = C_t - \Gamma_t$.*

(G1) $|C_t(r)| \le C|r|_t^{-1}$, $|\nabla C_t| \le C|r|_t^{-2}$, $|\nabla\nabla C_t| \le C|r|_t^{-3}$ *(T1); $|C_t - C_{t/5}|(r) \le Ct^2|r|^{-3}$ and $|\nabla^{(t)}C_t - \nabla^{(t/5)}C_{t/5}|(r) \le Ct|r|^{-3}$ for $r \in \Lambda_t \setminus 0$ (lemmas D.11 and D.24 at both spacings against C_0, ∂C_0).*

(G2) $|\Gamma_t|, |\nabla\Gamma_t| \le C$; $|\Gamma_t - \Gamma_{t/5}| \le Ct + C5^{-2j}$ *((Γ4) and (Γ5) for the change of b); $|\partial_b\Gamma_t| \le C$; $|\nabla\nabla\Gamma_\tau(u,v)| \le C\sum_{n\in N_\tau}|p|^2\sigma\hat{C}_\tau \le C\sum_{|n|\le C/\tau}\rho(n) \le C(1+\ln\tau^{-1})^3$ (uniform in m').*

(G3) $|G(x,y)| \le Ct^{-3}e^{-\kappa_0|x-y|}$ *(lemma D.8); hence $|G| \le C|r|_t^{-1} + C$ on $|r| \le R_t$ and $\le Ct^3$ beyond; $|(QG)|, |(QGQ^*)| \le C$ with the same tail; $t^3G(x,x) \le Ct^2$.*

(G4) *Gradient line: $|\nabla G| \le C|r|_t^{-2} + C$; $|\nabla_{\mathsf{b}}(QG)(y,\mathsf{b})| \le C$ uniformly in y.*

Proof. (G1), (G2) restate lemmas D.11 and D.23 to D.25; the second-difference bound on Γ_τ follows from the Fourier representation with two factors $|d_\tau(p_\mu)| \le |p|$ and (E1)–(E3). (G3), (G4): (G1), (G2) and lemma D.8; $(QG)(y,\cdot)$ is a block average of G. □

Lemma D.34 (tails of the contour kernels)**.** *Fix $i \ge 1$, $r := |\mathsf{b}-x|$, $\lambda_i := 2\ell_{i+1}/\kappa_0$, $M := (2/\kappa_0)\ln t^{-1}$ and $\varrho_i := r_i + M\ell_{i+1}$; for x, b coarse let $\Delta\mathcal{K}_{i,\mathsf{b}}(x) := \mathcal{K}^{(t)}_{i,\mathsf{b}}(x) - \mathcal{K}^{(t/5)}_{i+1,\mathsf{b}}(x)$ (coarse slice i matched with fine slice $i+1$).*

(T1) $|\Delta\mathcal{K}_{i,\mathsf{b}}(x)| \le Ct/\ell_i$ *for every r ((S5) and lemma D.32(iv): $\le (C_L\ell_i/t)\cdot t\cdot C t\ell_i^{-2}$).*

(T2) $|\mathcal{K}^{(t)}_{i,\mathsf{b}}(x)| \le Ce^{-(r-r_i)/\lambda_i}$ *and $|\mathcal{K}^{(t/5)}_{i+1,\mathsf{b}}(x)| \le CL^3e^{-(r-r_i)/\lambda_i}$ for $r > r_i$, hence $|\Delta\mathcal{K}_{i,\mathsf{b}}(x)| \le CL^3e^{-(r-r_i)/\lambda_i}$.*

(T3) *(E4 cost, both Wick families.) Let $\kappa \ge 0$ be radial and non-increasing with $\kappa(r) \ge |\Delta\mathcal{K}_{i,\mathsf{b}}(x)|\mathbf{1}_{r>r_i}$, e.g. $\kappa_i(r) := \min(Ct/\ell_i, CL^3e^{-(r-r_i)_+/\lambda_i})$. Then the E4 term with the kernel $\Delta\mathcal{K}_{i,\mathsf{b}}\mathbf{1}_{r>r_i}$ is $\le Ct(1+j)^3R_t^3$ per slice, and the same holds for the kernel $|\mathcal{K}^{(\tau)}_{i,\mathsf{b}}|\mathbf{1}_{r>r_i}$ of a single member with Ct replaced by $C\ell_{i+1}$.*

Proof. (T2): bound each member separately. Coarse member: the contour $\Gamma^{(i+1)}_{x_{i+1},x}$ ends at x and has length $\le C_L\ell_i$, so each of its bonds b' has $|\mathsf{b}'-\mathsf{b}| \ge r - C_L\ell_i$, and (S4) gives $|S_i(\mathsf{b}',\mathsf{b})| \le C\ell_i^{-1}e^{(\kappa_0/2)(\rho_i+C_L\ell_i-r)/\ell_{i+1}} = C\ell_i^{-1}e^{-(r-r_i)/\lambda_i}$; $\le C_L\ell_i/t$ bonds of weight t give $\le Ce^{-(r-r_i)/\lambda_i}$. Finer member (slice $i+1$ of $\Lambda_{t/5}$): same scale ℓ_i, same block side ℓ_{i+1}, contour length $3(t/5)\sum_{i'\le i+1}5^{i'+1} \le C_L\ell_i$; its (S4) shift is $(6/\kappa_0)\ell_{i+1}\ln(\ell_{i+1}/(t/5)) = \rho_i + (6/\kappa_0)\ell_{i+1}\ln 5$, which multiplies the bound by $e^{3\ln 5} = L^3$. (T3): E4 contracts $|\phi(x_0)|^2$ against $V_1 \ni \langle Q\phi(y), J\phi(x)\rangle$ and $V_h \ni \langle\phi(\mathsf{b}_-), J\phi(\mathsf{b}_+)\rangle$ (self-loops vanish, $\operatorname{Tr} J = 0$; x_0 joined to both fields of one vertex leaves a self-loop on the other), so there are two Wick families, both carrying the kernel $\mathcal{K}_{i,\mathsf{b}}(x)$: (i) x_0–$Q\phi(y)$, x–b, and (ii) x_0–$\phi(x)$, $Q\phi(y)$–b. *Family (i).* By (G3)–(G4), uniformly in b, for both placements of the difference (undifferentiated or gradient line from b to x),

$\sum_x t^3(|G(x,\mathsf{b}_+)|+|\nabla G(x,\mathsf{b})|)\kappa(|x-\mathsf{b}|)\le C\int(r^{-2}+r^{-1}+1)\kappa(r)d^3r+Ct\sup\kappa$ (the $O(1)$ points with $|x-\mathsf{b}|\le 2t$ cost $\le t^3Ct^{-2}$ each). The outer factors: $\sum_{\mathsf{b}} t^3(|\nabla G(\mathsf{b},x_0)|+|G(x_0,\mathsf{b})|)\le CR_t^3$ (beyond R_t a difference quotient is $\le 2t^{-1}Ct^{-3}e^{-\kappa_0(|r|-t)}$, total $\le Ct^2R_t^2$) and $|(QG)(y,x_0)|\le C$. With $r_i\le C\ell_i(i+1)$, $M\ell_{i+1}=(2L\ln L/\kappa_0)\ell_i j$ and $\ell_i\le 1$: $\varrho_i\le C\ell_i(1+j)$ and $\lambda_i\le 2L/\kappa_0$. Shell $r_i<r\le\varrho_i$: $\kappa\le Ct/\ell_i$ by (T1), $\int_{r\le\varrho}(r^{-2}+r^{-1}+1)d^3r\le C\varrho(1+\varrho)^2$, cost $\le C(t/\ell_i)\ell_i(1+j)(1+j)^2=Ct(1+j)^3$. Region $r>\varrho_i$: by (T2), $\kappa\le CL^3e^{-(\kappa_0/2)M}e^{-(r-\varrho_i)/\lambda_i}=CL^3te^{-(r-\varrho_i)/\lambda_i}$, and $\int_{r>\varrho}(r^{-2}+r^{-1}+1)e^{-(r-\varrho)/\lambda}d^3r\le C\lambda(1+\varrho+\lambda)^2\le C(1+j)^2$: cost $\le Ct(1+j)^2$. Near-diagonal: $Ct\sup\kappa\le Ct$. *Family (ii)* (lines $G(x,x_0)$, $G(x_0,\mathsf{b}_\mp)$, $(QG)(y,\mathsf{b}_\pm)$, y the block of x): the J-antisymmetry at b puts one forward difference at b on $G(x_0,\cdot)$ or on $(QG)(y,\cdot)$, and $|(QG)(y,\mathsf{b})|,|\nabla_{\mathsf{b}}(QG)(y,\mathsf{b})|\le C$ (G3)–(G4). The line from x now runs to x_0. Split the x-sum: on $|x-x_0|\ge|x-\mathsf{b}|$, $|G(x,x_0)|\le C|x-\mathsf{b}|_t^{-1}+C$ ($|\cdot|_t^{-1}$ non-increasing); on $|x-x_0|<|x-\mathsf{b}|$, $\kappa(|x-\mathsf{b}|)\le\kappa(|x-x_0|)$. Hence, uniformly in b and x_0, $\sum_x t^3|G(x,x_0)|\kappa(|x-\mathsf{b}|)\le 2\sup_z\sum_x t^3(C|x-z|_t^{-1}+C)\kappa(|x-z|)\le C\int(r^{-1}+1)\kappa(r)d^3r+Ct\sup\kappa$, which for $\kappa=\kappa_i$ is dominated by the three regions above ($r^{-1}+1\le r^{-2}+r^{-1}+1$; the first region's integral already runs over the whole ball $r\le\varrho_i$). The outer factor $\sum_{\mathsf{b}} t^3(|\nabla G(\mathsf{b},x_0)|+|G(x_0,\mathsf{b})|)\le CR_t^3$ is family (i)'s. Same per-slice cost $Ct(1+j)^3R_t^3$. For a single member, (T2) alone gives $\kappa\le Ce^{-(r-r_i)/\lambda_i}$ on $r>r_i$ and $\int_{r>r_i}(r^{-2}+r^{-1}+1)e^{-(r-r_i)/\lambda_i}d^3r\le C\lambda_i(1+r_i+\lambda_i)^2\le C\ell_{i+1}(1+j)^2$. The same holds for any radial non-increasing κ. □

B2.8 The sup bound for the averaging part.

Proposition D.35 (the averaging part, sup bound)**.** *For $f=1$, all k and $1\le j\le k-J_0$, there are q and C (uniform in $m',\mu'\in(0,m_0]$, b, k, j) with $\sup_x|\mathfrak{a}_x|\le Ce^2(1+\ln t^{-1})^q$ at unit block scale, i.e. $\|\mathfrak{a}\|_\infty\le Ce^2\eta^{-1}(1+j)^q$.*

Proof. Bound each class by |photon kernel| × |lines|, with the vertex sums restricted to $|\cdot-x_0|\le R_t$; the tails cost $\le Ct^3R_t^q$ by (G3). *E1:* $2be'^2\sum_x t^3\mathcal{V}(x)(QG)(y,x_0)G(x,x_0)\le C\sum_{|x-x_0|\le R_t}t^3(|x-x_0|_t^{-1}+1)\le C(1+R_t)^3$ ($\mathcal{V}\le C$, lemma D.32(i), $|(QG)|\le C$). *E2, E3:* $|\mathcal{K}|\le C$ and $\sum t^3t^3(|x_0-u|_t^{-1}+1)(|u-v|_t^{-1}+1)(|v-x_0|_t^{-1}+1)$ over the R_t-balls $\le C(1+R_t)^6$ (local integrability in three dimensions; coincident points cost Ct^2). *E4:* per slice i, with lemma D.32(ii) in its support form, $\sum_{\mathsf{b}} t^3|\nabla G(\mathsf{b},x_0)|\sum_{|x-\mathsf{b}|\le r_i}t^3|G(x,\mathsf{b}_+)|\,|(QG)(y,x_0)|\le Cr_i^2(1+r_i)(1+R_t)^3$ (the ball sum of $|G(x,\mathsf{b}_+)|\le C|x-\mathsf{b}|_t^{-1}+C$ over $|x-\mathsf{b}|\le r_i$ is $\le C(r_i^2+r_i^3)$), the tail $|\mathsf{b}-x|>r_i$ costs $\le C\ell_{i+1}(1+j)^2R_t^3$ (lemma D.34(T3), single member), summable over i, and the other placements of the difference and the second Wick family are identical ($|\nabla_{\mathsf{b}}(QG)(y,\mathsf{b})|\le C$). Summing over $i<j$, $\sum_i r_i^2(1+r_i)\le C(1+j)^3$ (dominated by $i\approx j$). Hence $\sup_x|\mathfrak{a}_x|\le Ce^2(1+\ln t^{-1})^q$ at unit block scale, with $e'^2=e^2\eta$ and the factor η^{-1} of the density; q is not pinned, and only its existence is used. □

B2.9 The averaging part along the diagonal pairs.

Lemma D.36 (gradient of the coarse-line pair difference)**.** *Let $g:=(\Gamma_t-\Gamma_{t/5})|_{\Lambda_t\times\Lambda_t}$ (both lines with the same b). Then*

$$|\nabla^{(t)}\Gamma_t-\nabla^{(t)}\Gamma_{t/5}|\le Ct^{1/2}(1+\ln t^{-1})^3,\qquad |\nabla^{(t)}\Gamma_{t/5}-\nabla^{(t/5)}\Gamma_{t/5}|\le Ct(1+\ln t^{-1})^3\quad\text{on }\Lambda_t\times\Lambda_t.$$

Proof. $A:=\sup|g|\le Ct$ by (Γ4), and $B:=\sup|\nabla^{(t)}\nabla^{(t)}g|\le C(1+\ln t^{-1})^3$ by (G2) (coarse second differences of $\Gamma_{t/5}$ are averages of fine ones). Discrete Landau inequality: $g(u+Nte_\mu)-g(u)=Nt\,\nabla_\mu^{(t)}g(u)+t^2\sum_{i<N}\sum_{i'<i}\nabla\nabla g$, so $|\nabla^{(t)}g|\le 2A/(Nt)+NtB/2$; take $N=\lfloor t^{-1/2}\rfloor$. The second bound: a coarse difference quotient of $\Gamma_{t/5}$ is the average of five fine ones along the step, and consecutive fine ones differ by $\le(t/5)B$. □

Proposition D.37 (the averaging part, increment)**.** *For $f=1$, all k and $1\le j\le k-J_0$, with q as in proposition D.35 (enlarged if necessary) and C uniform in $m',\mu'\in(0,m_0]$:*

$$\big|\bar{\mathfrak{a}}_k^{(j)}-\bar{\mathfrak{a}}_{k+1}^{(j+1)}\big|\le Ce^2\eta^{-1}\big[t^{1/2}(1+j)^q+5^{-2j}\big].$$

Proof. Both members are evaluated at unit block scale on the same unit cell; the coarse member has spacing t, coupling $\mathfrak{a}_j$ and slices $i < j$, the finer one spacing $t/5$, coupling $\mathfrak{a}_{j+1}$ and slices $i' \le j$. Every term keeps a cycle through x_0 with unit-scale-decaying lines, so all vertex sums converge uniformly in m', μ' (the mass-free decay (G3)). Telescope one factor at a time in four steps.

(1) Unmatched slices (slice 0 of the coarser member, the two finest slices of the finer member): $O(t \cdot \text{polylog})$ by (S5) and lemma D.32(i),(ii) ($\ell_0 = t$ in both members).

(2) Common slices $i \ge 1$ at coarse vertex points. (S5) and lemma D.32(iv) give Ct per slice on $\mathcal{V}$ and $\mathcal{K}$ (total Ctj). On $\mathcal{K}_{\mathsf{b}}$: per slice $|\Delta\mathcal{K}_{i,\mathsf{b}}| \le Ct/\ell_i$ (lemma D.34(T1)), which is not small for small i; in the region $|\mathsf{b} - x| \le r_i$ the E4 cost is $(t/\ell_i)\, r_i^2(1 + r_i)R_t^3$ (the ball sum of $|G(x, \mathsf{b}_+)| \le C|x - \mathsf{b}|_t^{-1} + C$ over $|x - \mathsf{b}| \le r_i$ is $\le C(r_i^2 + r_i^3)$), and $\sum_i (t/\ell_i) r_i^2(1 + r_i)R_t^3 \le Ct \sum_i \ell_i (i + 1)^3 R_t^3 \le Ct(1 + j)^3 R_t^3$ ($r_i \le C\ell_i(i + 1)$); in the shell $|\mathsf{b} - x| > r_i$ the pointwise bound carries no decay and would only give $\sum_i t/\ell_i = O(1)$, so there lemma D.34(T3) is used, with the per-slice cost $Ct(1 + j)^3 R_t^3$ for both Wick families: the shell part is $\le Ct(1 + j)^4 R_t^3$.

(3) Lines at coarse points. C-part: $|C_t - C_{t/5}| \le Ct^2|r|^{-3}$ and $|\nabla^{(t)} C_t - \nabla^{(t/5)} C_{t/5}| \le Ct|r|^{-3}$ (G1), summable against one more line at the cost of one logarithm; Γ-part: $|\Gamma_t - \Gamma_{t/5}| \le Ct + C5^{-2j}$ (G2); gradient-Γ part: lemma D.36, $\le Ct^{1/2}(1 + \ln t^{-1})^3$. This is the only step with rate $t^{1/2}$.

(4) Fine vertex points → coarse. Kernel oscillation by lemma D.32(v) ($\le Ctj^2$ for $\mathcal{K}'_{\mathsf{b}}$ in the region $|\mathsf{b} - x| \le r_i$); line oscillation $t(|r|_t^{-2} + 1)$; gradient-line oscillation $Ct|r|_t^{-3} + Ct(1 + \ln t^{-1})^3$ (G1), (G2); near-diagonal points ($|r| < 2t$) bounded directly, $O(t)$. Shell: by lemma D.32(iii) the in-cell difference $D_{i'}(x', \mathsf{b}) := \mathcal{K}'_{i',\mathsf{b}}(x') - \mathcal{K}'_{i',\mathsf{b}}(x)$ (x' in the coarse cell of x) is the sum of $(t/5)S'_{i'}(\mathsf{b}', \mathsf{b})$ over the $\le 3(L - 1)$ fine bonds b' of $\Gamma_{x,x'}$, all within $3t$ of x. With $\ell' := 5^{i'} t/5$ the scale of fine slice i' ($0 \le i' \le j$), $\lambda' := 2L\ell'/\kappa_0$, $\rho' := (6/\kappa_0)L\ell' \ln(L^2\ell'/t) \le C\ell'(1 + j)$ and $r' := \rho' + 3t$, (S2) and (S4) of the finer member give $|D_{i'}(x', \mathsf{b})| \le C(t/\ell') \min(1, e^{-(r - r')/\lambda'})$, $r = |\mathsf{b} - x|$. The bond move ($\mathsf{b} \to$ the parallel bond based at the coarse point of its cell, $|\mathsf{b} - \mathsf{b}_c| \le 3t$) obeys the same form $\min(Ct/\ell', CL^3 e^{-(r - r')/\lambda'})$: pointwise by (S3) ($|\nabla S'_{i'}| \le C\ell'^{-2}$, $\le C_L \ell' L/t$ bonds of weight $t/5$, step $\le 3t$), tail by (S4) applied to each of the two terms. lemma D.34(T3) for both Wick families ($t/\ell' \le L$; for the bond move with the split at $r' + ML\ell'$) gives per slice $\le Ct(1 + j)^3$, over the $\le j + 1$ fine slices $\le Ct(1 + j)^4$, times the outer factor CR_t^3.

Coupling change. $|\mathfrak{a}_j - \mathfrak{a}_{j+1}| \le C5^{-2j}$ with (G2) ($|\partial_b \Gamma| \le C$) and (S5): $C5^{-2j}$.

Summing (1)–(4): $|\bar{\mathfrak{a}}_k^{(j)} - \bar{\mathfrak{a}}_{k+1}^{(j+1)}| \le Ce^2\eta^{-1}[t^{1/2}(1 + j)^q + 5^{-2j}]$, the factors $e^2\eta^{-1}$ as in proposition D.35. □

B2.10 The averaging part with the Dirichlet weight, and the conclusion.

Proposition D.38 (the averaging part, weight f_D)**.** *Propositions D.35 and D.37 hold for the weight f_D of definition D.7, with constants depending in addition on $\|f_D\|_\infty$.*

Proof. Setting. The photon kernels are unchanged (the photon averaging is unchanged, [14, p. 328], so [13, (2.17)] and lemma D.31 hold verbatim). The vertex is $V_{\text{av}}^f = -\frac{b}{2} \sum_y [|Q^f(A)\phi|^2 - |Q^f\phi|^2]$ with $(Q^f(A)\phi)(y) = \sum_{x \in B(y)} t^3 f(x) U(e' A(\Gamma_{y,x}))\phi(x)$ [14, (3.2)], the weight multiplying the fine field at every slice; $Q_{j+1}^f = Q_1 Q_j^f$, $P_{i+1}^f \le P_i^f$ as before; the weight contributes bounded factors $\le \|f_D\|_\infty^2$. *Lines.* $G^f = C_t - \Gamma_t^f$ with the Dirichlet-weight clause of lemma D.13 (the same Γ_0, lower bound (A3) for $j \ge 1$), lemma D.8 with f_D, and the gradient and second-difference bounds of lemma D.33 with the D-aliases contributing $\le |p|^i\, Ct^2(1 + \ln t^{-1})^3 \le C \cdot \text{polylog}$ for $i \le 2$ ($|p| \le C/t$, $\hat{C}_t \le 25t^2/4$, (A1)). With these inputs the proofs of propositions D.35 and D.37 go through verbatim, except for the weight mismatch of the pair: $f^{(t)}$ is $5t$-periodic with $5t$-block average 1, $f^{(t/5)}$ is t-periodic with t-cell average 1 [14, (3.1)]. *Weight normalisation.* For an integrand F' on the finer lattice and F on the coarser one,

$$\sum_{x'} (t/5)^3 f'(x') F'(x') = \sum_{x \in \Lambda_t} t^3 F'(x) + \sum_x t^3 5^{-3} \sum_{x' \in c(x)} f'(x')[F'(x') - F'(x)],$$

$$\sum_x t^3 f(x)F(x) = \sum_x t^3 F(x) + \sum_{\text{blocks } B} \sum_{x\in B} t^3 (f(x)-1)[F(x)-F(x_B)].$$

Both remainders are $\le \|f\|_\infty$ times the in-cell, resp. in-block, oscillation of the integrand, which is step (4) of the proof of proposition D.37 over $5t$-blocks: $O(t\cdot \text{polylog})$ (near-diagonal points $\le Ct^2$). The same reduction applies to the $(Q^f G)$ lines. Hence the increment bound holds for f_D with the same rate $t^{1/2}$. □

Proof of lemma D.15. By lemma D.12, $\beta_{(2,0)} = (\beta_{(2,0)} + \mathfrak{a}) - \mathfrak{a}$, the first bracket being the original-vertex part. For it, theorem D.30 gives the sup bound $Ce^2\eta^{-1}(1+j)^{14}$ and the increment bound $C_\vartheta e^2 5^{-\vartheta j}\eta^{-1}$ for every $\vartheta \in (0,1)$, for both weights, for $1 \le j \le k$. For the averaging part, propositions D.35, D.37 and D.38 give $\|\mathfrak{a}\|_\infty \le Ce^2\eta^{-1}(1+j)^q$ and $|\bar{\mathfrak{a}}_k^{(j)} - \bar{\mathfrak{a}}_{k+1}^{(j+1)}| \le Ce^2\eta^{-1}[5^{-j/2}(1+j)^q + 5^{-2j}] \le C_\vartheta e^2 5^{-\vartheta j}\eta^{-1}$ for every $\vartheta \in (0,\frac12)$, for both weights, for $1 \le j \le k - J_0$. Adding, with q the larger of 14 and the exponent of the averaging part, gives both bounds of lemma D.15 with the constants depending on ϑ, $\mathfrak{a}$, e, m, μ_0 and m_0 (and on $\|f_D\|_\infty \le (5/3)^3$). □

D.3.11. *The bulk sequence.*

Lemma D.39 (the bulk sequence)**.** *Fix $\gamma_b \in (0,\frac12)$ and put $\beta_b := 1$. There is C such that for all k and $1 \le j \le k - J_0$,*

$$|\bar\beta_k^{(j)} - \bar\beta_{k+1}^{(j+1)}| \;\le\; C\,5^{-\gamma_b j}\eta^{-\beta_b}, \tag{72}$$

and along the diagonal pairs, for $k \ge k_0$ and $j = j(k)$,

$$\|\beta_k^{(j)}\|_\infty \;\le\; C\,\eta_k^{-2}. \tag{73}$$

Proof. Add the bounds of lemmas D.13 to D.15 with $\gamma'' = \vartheta = \gamma_b$: the increments are $\le C\eta^{-1}(5^{-j/2} + 5^{-\gamma_b j} + 5^{-2j}) \le C5^{-\gamma_b j}\eta^{-1}$ (as $\eta \le 1$). For the sup bound along the diagonal pairs, $j(k) \le k$ and $\eta_k^{-1} = 5^{k-j(k)} \ge 5^{\alpha k - 1}$, so $(1+j)^3 \le (1+k)^3 \le C_\alpha \eta_k^{-1}$ and $(1+j)^q \le (1+k)^q \le C_\alpha\eta_k^{-1}$, and $\|\beta_k^{(j)}\|_\infty \le C\eta_k^{-1} + C\lambda(1+k)^3 + Ce^2\eta_k^{-1}(1+k)^q \le C\eta_k^{-2}$. □

D.4. **Proof of proposition 6.7.** Let (γ,β,σ) be the exponents of proposition 5.5, (γ',β',σ') those of lemma 6.1, and (γ_b,β_b) those of lemma D.39. The diagonal pairs are taken with

$$0 < \alpha < \min\Big\{\tfrac12,\ \frac{\gamma}{\gamma+\beta},\ \frac{\gamma'}{\gamma'+\beta'},\ \frac{\gamma_b}{\gamma_b+\beta_b}\Big\}; \tag{74}$$

the Cauchy arguments of theorems 5.7 and 6.14 hold for every $\alpha \in (0,\frac12)$ with $\alpha < \gamma/(\gamma+\beta)$, so the two further restrictions cost nothing, and one value of α is fixed once for the whole existence part. Let $\Sigma = \{(u_x, h_y) : (x,y) \in Y\}$ be the compact family, as in appendix D.2.

Step 1 (the split). Fix $k \ge k_0$, $j := j(k)$ and $\sigma = (u_x, h_y) \in \Sigma$, and write $h := h_y$. Both members of the diagonal pair live on the same unit lattice with the same indicator χ_j, and lemma D.2 applies to each. Hence, on the support of χ_j,

$$S_{N,k}^{(j)}(\cdot;\sigma) + c_k(h) - S_{N,k+1}^{(j+1)}(\cdot;\sigma) - c_{k+1}(h) = \big[S_{N,k}^{(j)}(\cdot;(u_x,0)) - S_{N,k+1}^{(j+1)}(\cdot;(u_x,0))\big] + \sum_{T\in\mathcal{N}_h} D[T] + \Big(\big[\mathcal{V}_k^{(j)}(h) + c_k(h)\big] - \big[\mathcal{V}_{k+1}^{(j+1)}(h) + c_{k+1}(h)\big]\Big).$$

At $\sigma = 0$ the last two terms are absent and the left side is the left side of (3) at $s = 0$, $j = j(k)$.

Step 2 (rows K0 and SRC). By corollary D.5 the first two terms are bounded by $\theta_k^{\mathrm{K0}} + \theta_k^{\mathrm{SRC}}$, uniformly on $\Sigma \cup \{0\}$.

Step 3 (row LOW). The third term is field-independent and linear in h, and it vanishes at $h = 0$. By lemma D.9 and lemma D.10 with $r = 2$,

$$\mathcal{V}_k^{(j)}(h) + c_k(h) = \bar\beta_k^{(j)} \int h + E_k(h), \qquad |E_k(h)| \le C\|h\|_{L^1}\varepsilon_k^{-\nu}e^{-c/\eta_k} + C\|\beta_k^{(j)}\|_\infty \|\Delta^2 h\|_{L^1}\big(\eta_k^4 + \varepsilon_k^4\big),$$

where $c_k(h) = z_k \mathsf{S}_k(h)$ was used in the form of (65), and $|\bar\beta_k^{(j)}| \le \|\beta_k^{(j)}\|_\infty$. The same holds for the $(k+1)$-model after $j+1$ steps, with $\varepsilon_{k+1} = \varepsilon_k/5$ and the same η_k. By (72) and (73), the third term is bounded by

$$\theta_k^{\mathrm{LOW}}(h) := C\,5^{-\gamma_b j}\eta_k^{-\beta_b}\Big|\int h\Big| + C\|h\|_{L^1}5^{\nu k}e^{-c5^{\alpha k-1}} + C\|\Delta^2 h\|_{L^1}\big(\eta_k^2 + \eta_k^{-2}\varepsilon_k^4\big).$$

Its coefficients are linear in $(|\int h|, \|h\|_{L^1}, \|\Delta^2 h\|_{L^1})$, which are bounded on $\{h_y : y \in Y\}$; so $\theta_k^{\mathrm{LOW}}(h_y) \le \theta_k^{\mathrm{LOW}}$ with θ_k^{LOW} independent of $\sigma \in \Sigma$, and $\theta_k^{\mathrm{LOW}}(0) = 0$.

Step 4 (geometric decay). Along the diagonal pairs $\eta_k \le 5^{-\alpha k}$, $j(k) \ge (1-\alpha)k - 1$ and $k - j(k) \le \alpha k + 1$ (Step 6 of the proof of theorem 5.7). Hence η_k^σ, $\eta_k^{\sigma'}$, η_k^2 and $\eta_k^{-2}\varepsilon_k^4 \le 5^{2(\alpha k+1)-4k}$ decay geometrically; $5^{\nu k}e^{-c5^{\alpha k-1}}$ decays faster than any geometric sequence; and for $(\gamma_*, \beta_*) \in \{(\gamma,\beta), (\gamma',\beta'), (\gamma_b,\beta_b)\}$,

$$5^{-\gamma_* j(k)}\eta_k^{-\beta_*} \le 5^{\gamma_*+\beta_*}\,5^{-(\gamma_*(1-\alpha)-\beta_*\alpha)k},$$

which decays geometrically because $\alpha < \gamma_*/(\gamma_*+\beta_*)$ by (74). Therefore

$$\theta_k := \theta_k^{\mathrm{K0}} + \theta_k^{\mathrm{SRC}} + \theta_k^{\mathrm{LOW}}$$

decays geometrically, does not depend on $\sigma \in \Sigma$, and by Steps 1–3 the left side of (12) is bounded on the support of $\chi_{j(k)}$ by $\theta_k(\sigma) := \theta_k^{\mathrm{K0}} + \theta_k^{\mathrm{SRC}} + \theta_k^{\mathrm{LOW}}(h_\sigma) \le \theta_k$ for $\sigma \in \Sigma$, and by $\theta_k(0) := \theta_k^{\mathrm{K0}}$ at $\sigma = 0$. This is proposition 6.7. □

Remark D.40. The proof uses no convergence and no bound of the vacuum constants $v_{k,j}$, no bound on $Z_k(s)$, and no property of the scheme z beyond $|z_k| \le C(1+k)$ (lemma 6.8(a)) and the identity $\beta_k^{(j)} = v_{k,\infty}^{(j)} + z_k$ of definition D.7: the c-number $c_k(h) = z_k\mathsf{S}_k(h)$ is the one for which $\mathcal{V}_k^{(j)}(h) + c_k(h)$ is, up to the errors of lemmas D.9 and D.10, the block mean of the bulk density times $\int h$, and the bulk density is the difference between the low-order one-point function of the renormalised Wick square in the full lattice model and in King's fluctuation measure.

Appendix E. The c-numbers and the growth of the bare coefficient with the mass

This appendix proves lemma 6.8 and proposition 6.9. Both are statements about lattice Feynman integrals on the infinite lattice $\varepsilon\mathbb{Z}^3$: the first about the order-e^2 coefficient X_ε of definition 2.3, which defines the reference scheme $z_k = e^2 X_{\varepsilon_k}$, the second about King's free-boundary mass counterterm $\delta m^2{}_\varepsilon(M)$ [13, (3.32)–(3.33)], [14, p. 330] at its five orders in (e^2, λ). Appendix E.1 gives the exact asymptotics of X_ε (theorem E.7), from which lemma 6.8 follows; appendix E.2 computes the continuum limit of $\delta m^2{}_\varepsilon(M) - \delta m^2{}_\varepsilon(m)$ in closed form (theorem E.26) and derives from it the lower bound of proposition 6.9 (theorem E.28). Throughout, $N = 2$ (two real scalar components, $|\phi|^2 = r^2$), the quartic term is $\lambda|\phi|^4$, and the photon is in Feynman form with mass μ_0 [13, (2.4)], [14, (2.5)].

E.1. The order-e^2 Wick constant.

Conventions on the unit lattice. On $\mathbb{Z}^3$ put $\mathsf{B} := [-\pi,\pi]^3$, $\int_k := \int_{\mathsf{B}} \frac{d^3k}{(2\pi)^3}$, $\hat q_\alpha^2 := 4\sin^2(q_\alpha/2)$, $\hat q^2 := \sum_\alpha \hat q_\alpha^2$, and for $a, b > 0$

$$\hat G_a(q) := (\hat q^2 + a^2)^{-1}, \qquad D_b(k) := (\hat k^2 + b^2)^{-1},$$

$$\hat\Pi_{a,b}(p) := \int_k D_b(k) \sum_\alpha \Big[\cos p_\alpha - 4\sin^2\tfrac{2p_\alpha + k_\alpha}{2}\,\hat G_a(p+k)\Big],$$

$$\bar\pi_{a,b}(p) := \hat\Pi_{a,b}(p) - \hat\Pi_{a,b}(0), \qquad X_1(a,b) := -2\int_p \hat G_a(p)^2\,\bar\pi_{a,b}(p).$$

Lemma E.1 (the formula for X_ε)**.** (i) *On $\mathbb{Z}^3$ with King's action at unit spacing, scalar mass a, photon mass b and charge e, the coefficient of e^2 in $\langle|\phi(0)|^2\rangle$ at $\lambda = 0$, with the order-e^2 mass counterterm fixed by* [13, (3.33)]*, is $X_1(a,b)$.* (ii) *$X_\varepsilon(m;\mu) = X_1(\varepsilon m, \varepsilon\mu)$ for the quantity X_ε of definition 2.3 at scalar mass m and photon mass μ. In particular $z_k = e^2 X_1(\varepsilon_k m, \varepsilon_k\mu_0)$.*

Proof. (i) Work on $\mathbb{Z}^3/N\mathbb{Z}^3$ and let $N\to\infty$ at the end (all integrands below are smooth and periodic for $a,b>0$, so the Riemann sums converge). King's matter action at unit spacing is $\frac12\sum_{x,\alpha}|e^{\mathrm{i}eA_\alpha(x)}\phi(x+e_\alpha)-\phi(x)|^2+\frac12(a^2+\delta m^2)\sum_x|\phi|^2$ [14, (2.3), (2.5)], i.e. $\bar\phi(K+M(A))\phi$ with $K=\frac12(-\Delta+a^2)$ and $M(A)(x,x+e_\alpha)=-\frac12(e^{\mathrm{i}eA_\alpha(x)}-1)$, $M(A)(x+e_\alpha,x)=-\frac12(e^{-\mathrm{i}eA_\alpha(x)}-1)$. The photon is Gaussian with $\langle A_\alpha A_\beta\rangle=\delta_{\alpha\beta}(-\Delta+b^2)^{-1}$ (Feynman form). At fixed A the ϕ-integral gives $\langle\phi(0)\bar\phi(0)\rangle_A=(K+M)^{-1}(0,0)$ and the weight $\det(K+M)^{-1}$. Write $M=eM_1+e^2M_2+O(e^3)$. Then $(K+M)^{-1}=K^{-1}-K^{-1}MK^{-1}+K^{-1}MK^{-1}MK^{-1}-\dots$, and $\operatorname{Tr}K^{-1}M_1=\sum_{x,\alpha}K^{-1}(e_\alpha)(-\frac{\mathrm{i}}{2}A_\alpha(x)+\frac{\mathrm{i}}{2}A_\alpha(x))=0$ identically, so the determinant does not contribute at order e^2. Hence the order-e^2 part of $\langle\phi\bar\phi\rangle$ is $K^{-1}\Pi K^{-1}$ with $\Pi=-\langle M_2\rangle+\langle M_1K^{-1}M_1\rangle-\frac12\delta m^2{}_{e^2}$. In momentum space the vertex of M_1 between $\phi(p)$ and $\bar\phi(q)$ is $e^{-\mathrm{i}k_\alpha/2}\sin\frac{(p+q)_\alpha}{2}$ ($k=q-p$), and $\langle A_\alpha^2\rangle=\int_k D_b$; so $\Pi(p)=2\int_k D_b(k)\hat G_a(p-k)\sum_\alpha\sin^2\frac{(2p-k)_\alpha}{2}-\frac12\int_k D_b\sum_\alpha\cos p_\alpha-\frac12\delta m^2{}_{e^2}$, which is $-\frac12\hat\Pi_{a,b}(p)-\frac12\delta m^2{}_{e^2}$ (substitute $k\mapsto-k$ in $\hat\Pi$). By [13, (3.33)] the full order-e^2 one-particle-irreducible two-point function vanishes at zero momentum, so $\Pi(p)=-\frac12\bar\pi_{a,b}(p)$. With $K^{-1}=2\hat G_a$, $\langle|\phi(0)|^2\rangle_{e^2}=\int_p 4\hat G_a^2\Pi=-2\int_p\hat G_a^2\bar\pi_{a,b}=X_1(a,b)$.

(ii) Substitute $p=p'/\varepsilon$, $k=k'/\varepsilon$ in the same expression on $\varepsilon\mathbb{Z}^3$: $\hat p_\varepsilon=\varepsilon^{-1}\hat p'$, $\hat G_\varepsilon=\varepsilon^2\hat G_{\varepsilon m}$, $D_\varepsilon=\varepsilon^2D_{\varepsilon\mu}$, the vertex scales as ε^{-1}, so $\hat\Pi_\varepsilon(p)=\varepsilon^{-1}\hat\Pi_{\varepsilon m,\varepsilon\mu}(p')$ and $X_\varepsilon=\varepsilon^{-3}\varepsilon^4\varepsilon^{-1}X_1$. □

Lemma E.2 (a lattice identity)**.** *For real u,w, with $\hat x^2:=4\sin^2(x/2)$:* $4\sin^2\frac{u+w}{2}=2\hat u^2+2\hat w^2-\widehat{(u-w)}^2-\hat u^2\hat w^2$.

Proof. With $\hat x^2=2-2\cos x$ the right side is $2+2\cos(u-w)-4\cos u\cos w=2-2\cos u\cos w+2\sin u\sin w=2-2\cos(u+w)$. □

Lemma E.3 (exact decomposition of the order-e^2 self-energy)**.** *For $a,b>0$ and $p\in\mathbb{R}^3$ put*

$$B_{a,b}(p):=\int_k D_b(k)\hat G_a(p-k),\qquad J^\alpha_{a,b}(p):=\int_k D_b(k)\hat G_a(p-k)\,\widehat{(p-k)}_\alpha^2,\qquad d_b:=\int_k D_b(k).$$

Then

$$-\tfrac12\bar\pi_{a,b}(p)=\hat p^2B_{a,b}(p)+\tfrac12(b^2-2a^2)\big(B_{a,b}(p)-B_{a,b}(0)\big)+\rho_{a,b}(p),\qquad \rho_{a,b}(p):=-\tfrac12\sum_\alpha\hat p_\alpha^2J^\alpha_{a,b}(p)+\tfrac14d_b\,\hat p^2,$$

and therefore

$$X_1(a,b)=4\int_p\hat G_a(p)^2\Big[\hat p^2B_{a,b}(p)+\tfrac12(b^2-2a^2)\big(B_{a,b}(p)-B_{a,b}(0)\big)+\rho_{a,b}(p)\Big].$$

Proof. Substitute $k\mapsto-k$ in $\hat\Pi$ (D_b is even and the integrand is periodic): $\hat\Pi(p)=d_b\sum_\alpha\cos p_\alpha-\int_k D_b(k)\hat G_a(p-k)\sum_\alpha4\sin^2\frac{(2p-k)_\alpha}{2}$. Lemma E.2 with $u=p_\alpha$, $w=(p-k)_\alpha$ ($u+w=(2p-k)_\alpha$, $u-w=k_\alpha$), summed over α:

$$\textstyle\sum_\alpha4\sin^2\frac{(2p-k)_\alpha}{2}=2\hat p^2+2\widehat{(p-k)}^2-\hat k^2-\sum_\alpha\hat p_\alpha^2\widehat{(p-k)}_\alpha^2=2\hat p^2-2a^2+b^2+2\hat G_a(p-k)^{-1}-D_b(k)^{-1}-\sum_\alpha\hat p_\alpha^2\widehat{(p-k)}_\alpha^2$$

Multiply by $D_b(k)\hat G_a(p-k)$ and integrate; $\int_k\hat G_a(p-k)=\int_k\hat G_a=:g_a$ by periodicity: $\int_k D_b\hat G_a(p-\cdot)\sum_\alpha4\sin^2\frac{(2p-k)_\alpha}{2}=(2\hat p^2-2a^2+b^2)B(p)+2d_b-g_a-\sum_\alpha\hat p_\alpha^2J^\alpha(p)$. With $\sum_\alpha\cos p_\alpha=3-\frac12\hat p^2$: $\hat\Pi(p)=d_b-\frac12d_b\hat p^2-(2\hat p^2-2a^2+b^2)B(p)+g_a+\sum_\alpha\hat p_\alpha^2J^\alpha(p)$. Subtract the value at $p=0$, where $\hat p=0$. □

The same algebra with $(2p-k)^2=2p^2+2(p-k)^2-k^2$ gives the continuum function $\pi^{\mathrm c}(q)=(2q^2-2m^2+\mu^2)B^{\mathrm c}(q)-(\mu^2-2m^2)B^{\mathrm c}(0)$; lemma E.3 is its exact lattice form, the only lattice-specific terms being ρ (of degree 2 in $\hat p$) and the difference between the lattice and the continuum bubbles.

Lemma E.4 (elementary)**.** (a) *For $|x|\le\pi$: $\frac{4}{\pi^2}x^2\le4\sin^2\frac{x}{2}\le x^2$ and $0\le x^2-4\sin^2\frac{x}{2}\le x^4/12$. For all real x: $4\sin^2\frac{x}{2}\ge\frac{4}{\pi^2}\operatorname{dist}(x,2\pi\mathbb{Z})^2$.* (b) *For $p\in\mathsf B$, $\operatorname{dist}(p,2\pi\mathbb{Z}^3)=|p|$.* (c) *Hence for $q\in\mathsf B$: $\frac{4}{\pi^2}|q|^2\le\hat q^2\le|q|^2$, $0\le|q|^2-\hat q^2\le|q|^4/12$; and $\hat q^2\ge\frac{4}{\pi^2}\operatorname{dist}(q,2\pi\mathbb{Z}^3)^2$ for all $q\in\mathbb{R}^3$.*

Proof. (a) $\sin y \geq \frac{2}{\pi}y$ on $[0,\frac{\pi}{2}]$ (concavity); $1-\frac{x^2}{2} \leq \cos x \leq 1-\frac{x^2}{2}+\frac{x^4}{24}$ gives $x^2-\frac{x^4}{12} \leq 2-2\cos x \leq x^2$; the last claim by periodicity. (b) $|p_\alpha - 2\pi n_\alpha| \geq |p_\alpha|$ if $|p_\alpha| \leq \pi$. (c) Sum over α; $\sum_\alpha q_\alpha^4 \leq |q|^4$. □

Lemma E.5 (bubble comparison)**.** *Let* $B^{\mathrm{c}}_{a,b}(p) := \int_{\mathbb{R}^3} \frac{d^3k}{(2\pi)^3} \frac{1}{(|k|^2+b^2)(|p-k|^2+a^2)}$. *Then:*

(i) $B^{\mathrm{c}}_{a,b}(p) = \dfrac{\arctan(|p|/(a+b))}{4\pi|p|}$, $B^{\mathrm{c}}_{a,b}(0) = \dfrac{1}{4\pi(a+b)}$; $|p| \mapsto B^{\mathrm{c}}$ *is non-increasing,* $0 < B^{\mathrm{c}}_{a,b}(p) \leq \min\{\frac{1}{4\pi(a+b)}, \frac{1}{8|p|}\}$, *and* $B^{\mathrm{c}}_{ta,tb}(tq) = t^{-1}B^{\mathrm{c}}_{a,b}(q)$.

(ii) *For all* $a,b>0$ *and* $p \in \mathsf{B}$: $|B_{a,b}(p) - B^{\mathrm{c}}_{a,b}(p)| \leq C_0 := 2\cdot 10^3$.

(iii) *For* $p \in \mathsf{B}\setminus\{0\}$ *and fixed* $a,b>0$, *as* $t \downarrow 0$: $B_{ta,tb}(p) \uparrow B_{0,0}(p) := \int_k \hat{k}^{-2}\widehat{(p-k)}^{-2} < \infty$ *and* $B^{\mathrm{c}}_{ta,tb}(p) \to \frac{1}{8|p|}$.

Proof. (i) $\int_{\mathbb{R}^3} \frac{e^{\mathrm{i}k\cdot x}d^3k}{(2\pi)^3(|k|^2+b^2)} = \frac{e^{-b|x|}}{4\pi|x|}$, so $B^{\mathrm{c}}(p) = \int d^3x\, e^{-\mathrm{i}p\cdot x}\frac{e^{-(a+b)|x|}}{16\pi^2|x|^2} = \frac{1}{4\pi}\int_0^\infty e^{-(a+b)r}\frac{\sin(|p|r)}{|p|r}dr = \frac{\arctan(|p|/(a+b))}{4\pi|p|}$. $y \mapsto \arctan(y)/y$ is decreasing and ≤ 1, and $\arctan \leq \frac{\pi}{2}$. Homogeneity is read off.

(ii) *Case* $|p| \geq \frac{1}{2}$. $B^{\mathrm{c}} \leq \frac{1}{4}$ and $0 < B_{a,b}(p) \leq B_{0,0}(p)$. Split B into $V_1 = \{|k| < \frac{1}{4}\}$, $V_2 = \{\mathrm{dist}(k-p, 2\pi\mathbb{Z}^3) < \frac{1}{4}\}$ (disjoint from V_1, because $\mathrm{dist}(p,2\pi\mathbb{Z}^3) = |p| \geq \frac{1}{2}$ by lemma E.4(b)) and the rest. On V_1, $\mathrm{dist}(p-k,2\pi\mathbb{Z}^3) \geq \frac{1}{4}$, so $\widehat{(p-k)}^{-2} \leq 4\pi^2$ (lemma E.4(c)), and $\int_{V_1}\hat{k}^{-2} \leq \frac{\pi^2}{4}\int_{|k|<1/4}\frac{d^3k}{(2\pi)^3|k|^2} = \frac{1}{32}$. On V_2, $\mathrm{dist}(k,2\pi\mathbb{Z}^3) \geq \frac{1}{4}$, so $\hat{k}^{-2} \leq 4\pi^2$, and by periodicity $\int_{V_2}\widehat{(p-k)}^{-2} \leq \frac{1}{32}$. On the rest both factors are $\leq 4\pi^2$. So $B_{0,0}(p) \leq \frac{\pi^2}{4} + 16\pi^4 < 1.6\cdot 10^3$.

Case $|p| < \frac{1}{2}$. Let $U = \{|k| < 2\} \subset \mathsf{B}$. For $k \in U$, k and $p-k$ lie in the ball of radius $\frac{5}{2} < \pi$, which is inside B. By lemma E.4(c), $0 \leq D_b(k) - (|k|^2+b^2)^{-1} = \frac{|k|^2-\hat{k}^2}{(\hat{k}^2+b^2)(|k|^2+b^2)} \leq \frac{|k|^4/12}{(4/\pi^2)|k|^4} = \frac{\pi^2}{48}$, and the same for $\hat{G}_a(p-k)$. So on U, $|D_b(k)\hat{G}_a(p-k) - (|k|^2+b^2)^{-1}(|p-k|^2+a^2)^{-1}| \leq \frac{\pi^2}{48}\big(\frac{\pi^2}{4}|p-k|^{-2} + |k|^{-2}\big)$, whose integral over U is $\leq \frac{\pi^2}{48}\big(\frac{\pi^2}{4}\cdot\frac{5}{4\pi^2} + \frac{1}{\pi^2}\big) < 0.09$. On $\mathsf{B}\setminus U$: $\hat{k}^2 \geq \frac{16}{\pi^2}$, and $\mathrm{dist}(p-k,2\pi\mathbb{Z}^3) \geq \frac{3}{2}$ (for $n=0$, $|p-k| \geq \frac{3}{2}$; for $n \neq 0$ some $|p_\alpha - k_\alpha - 2\pi n_\alpha| \geq 2\pi - \frac{1}{2} - \pi$), so the lattice integrand is $\leq \frac{\pi^2}{16}\cdot\frac{\pi^2}{9}$ and its integral is < 0.7. On $\mathbb{R}^3\setminus U$: $|p-k| \geq \frac{3}{4}|k|$, so the continuum integrand is $\leq \frac{16}{9}|k|^{-4}$, with integral $\frac{4}{9\pi^2} < 0.05$. Total < 1.

(iii) The integrand increases as $t \downarrow 0$ to $\hat{k}^{-2}\widehat{(p-k)}^{-2}$, which is integrable for $p \notin 2\pi\mathbb{Z}^3$ by the splitting of the first case with radius $|p|/4$; monotone convergence. The continuum limit is read off from (i). □

Lemma E.6 (d and J)**.** $d_b \leq d_0 := \int_k \hat{k}^{-2} \leq \frac{\pi^2}{4}\int_{\mathsf{B}}\frac{d^3k}{(2\pi)^3|k|^2} < \infty$, $0 \leq J^\alpha_{a,b}(p) \leq d_b$, *hence* $|\rho_{a,b}(p)| \leq \frac{3}{4}d_0\,\hat{p}^2$. *As* $t \downarrow 0$, $d_{tb} \uparrow d_0$ *and* $J^\alpha_{ta,tb}(p) \to J^\alpha_{0,0}(p) := \int_k \hat{k}^{-2}\widehat{(p-k)}_\alpha^2\widehat{(p-k)}^{-2}$.

Proof. $\hat{G}_a(q)\hat{q}_\alpha^2 \leq 1$; dominated convergence with the integrable majorant $\hat{k}^{-2}$ (the integrand converges off the null set $k \in p + 2\pi\mathbb{Z}^3$). □

Theorem E.7 (asymptotics of the order-e^2 Wick constant)**.** *Fix* $m,\mu>0$. *As* $t \downarrow 0$,

$$X_1(tm,t\mu) = \frac{1}{4\pi^2}\ln\frac{1}{t} + \kappa(m,\mu) + o(1), \qquad \kappa(m,\mu) = \kappa_{\mathrm{lat}} + \kappa_{\mathrm{c}}(m,\mu),$$

where, with $r_0(p) := B_{0,0}(p) - \frac{1}{8|p|}$ *and* $\rho_0(p) := -\frac{1}{2}\sum_\alpha \hat{p}_\alpha^2 J^\alpha_{0,0}(p) + \frac{1}{4}d_0\hat{p}^2$,

$$\kappa_{\mathrm{lat}} := 4\int_p \frac{r_0(p)}{\hat{p}^2} + 4\int_p\frac{\rho_0(p)}{\hat{p}^4} + 4\int_p\Big(\frac{1}{\hat{p}^2} - \frac{1}{|p|^2}\Big)\frac{1}{8|p|} + \frac{\ln\pi}{4\pi^2} + \frac{1}{16\pi^3}\int_{\mathsf{B}\setminus\{|p|<\pi\}}\frac{d^3p}{|p|^3},$$

$$\kappa_{\mathrm{c}}(m,\mu) := -\frac{\ln m + \frac{1}{2}}{4\pi^2} + \frac{1}{2\pi^3}\int_{\mathbb{R}^3}\frac{u_{m,\mu}(q)}{(q^2+m^2)^2}\,d^3q, \qquad u_{m,\mu}(q) := q^2 B^{\mathrm{c}}_{m,\mu}(q) + \tfrac{1}{2}(\mu^2-2m^2)\big(B^{\mathrm{c}}_{m,\mu}(q) - B^{\mathrm{c}}_{m,\mu}(0$$

All integrals converge absolutely; κ_{lat} *does not depend on* (m,μ), *and* $\kappa_{\mathrm{c}}(m,\mu) = -\frac{\ln m}{4\pi^2} + \kappa_{\mathrm{c}}(1,\mu/m)$.

Proof. Put $a = tm$, $b = t\mu$, $t \in (0,1]$, $m_* := \max(m,\mu)$, $r := B - B^{\mathrm{c}}$ ($|r| \le C_0$ on B, lemma E.5(ii)), and $G^{\mathrm{c}}_a(p) := (|p|^2 + a^2)^{-1}$. Two integrals are used repeatedly: $\int_p \hat p^{-2} < \infty$, and

$$\int_p \hat G_a^2 \le \int_{\mathbb{R}^3} \frac{d^3p}{(2\pi)^3((4/\pi^2)p^2 + a^2)^2} = \frac{\pi^2}{64a}.$$

By lemma E.3, $X_1 = M(t) + E_1 + E_2 + E_3$ with

$$M(t) := 4\int_p \hat G_a^2\big[\hat p^2 B^{\mathrm{c}} + \tfrac12(b^2-2a^2)(B^{\mathrm{c}}(p) - B^{\mathrm{c}}(0))\big], \qquad E_1 := 4\int_p \hat G_a^2 \hat p^2 r,$$
$$E_2 := 2(b^2-2a^2)\int_p \hat G_a^2 (r(p) - r(0)), \qquad E_3 := 4\int_p \hat G_a^2 \rho.$$

Step 1. $|4\hat G_a^2 \hat p^2 r| \le 4C_0 \hat p^{-2}$ and $|4\hat G_a^2\rho| \le 3d_0\hat p^{-2}$; the integrands converge pointwise on $\mathsf{B}\setminus\{0\}$ (lemma E.5(iii), lemma E.6; $\hat G_{ta}(p) \to \hat p^{-2}$). By dominated convergence $E_1 \to 4\int_p r_0 \hat p^{-2}$ and $E_3 \to 4\int_p \rho_0 \hat p^{-4}$.

Step 2. $|E_2| \le 6m_*^2 t^2 \cdot 2C_0 \cdot \frac{\pi^2}{64tm} = O(t)$.

Step 3. Write $M(t) = M^{\mathrm{c}}(t) + E_4 + E_5$, where M^{c} is M with $\hat G_a^2\hat p^2$ replaced by $(G^{\mathrm{c}}_a)^2|p|^2$ and $\hat G_a^2$ by $(G^{\mathrm{c}}_a)^2$ (still integrated over the cube B), $E_4 := 4\int_p \big(\hat G_a^2\hat p^2 - (G^{\mathrm{c}}_a)^2|p|^2\big)B^{\mathrm{c}}$, and $E_5 := 2(b^2-2a^2)\int_p \big(\hat G_a^2 - (G^{\mathrm{c}}_a)^2\big)(B^{\mathrm{c}}(p) - B^{\mathrm{c}}(0))$. For $f(x) = x(x+a^2)^{-2}$, $|f'(x)| \le (x+a^2)^{-2}$; both $\hat p^2$ and $|p|^2$ lie in $[\frac{4}{\pi^2}|p|^2, |p|^2]$ and differ by $\le |p|^4/12$, so $|\hat G_a^2\hat p^2 - (G^{\mathrm{c}}_a)^2|p|^2| \le \frac{\pi^4}{192}$. With $B^{\mathrm{c}} \le \frac{1}{8|p|}$ the integrand of E_4 has an integrable majorant, and it converges pointwise; so $E_4 \to 4\int_p (\hat p^{-2} - |p|^{-2})\frac{1}{8|p|}$. For $g(x) = (x+a^2)^{-2}$, $|g'(x)| \le 2(x+a^2)^{-3}$, hence $|\hat G_a^2 - (G^{\mathrm{c}}_a)^2| \le \frac{\pi^4}{96}\big(\frac{4}{\pi^2}|p|^2 + a^2\big)^{-1}$; with $|B^{\mathrm{c}}(p) - B^{\mathrm{c}}(0)| \le \frac{1}{4\pi t(m+\mu)}$, $|E_5| \le 6m_*^2t^2 \cdot \frac{\pi^4}{96}\cdot\frac{1}{4\pi t(m+\mu)}\cdot\frac{\pi^2}{4}\int_p |p|^{-2} = O(t)$.

Step 4: $M^{\mathrm{c}}(t)$. Fix $\varrho \in (0,1]$ and split B into the ball $\{|p| < \varrho\}$ and the rest. On the rest the first part of the integrand, $4(G^{\mathrm{c}}_a)^2|p|^2B^{\mathrm{c}}$, is $\le \frac{1}{2|p|^3}$ and converges to $\frac{1}{2|p|^3}$; the second part is $\le |p|^{-4}\cdot 3m_*^2t^2\cdot\frac{1}{4\pi t(m+\mu)} \to 0$ uniformly on $|p| \ge \varrho$; so the outer part tends to $\int_{\mathsf{B}\setminus\{|p|<\varrho\}} \frac{d^3p}{(2\pi)^3\,2|p|^3} = \frac{\ln(\pi/\varrho)}{4\pi^2} + \frac{1}{16\pi^3}\int_{\mathsf{B}\setminus\{|p|<\pi\}}\frac{d^3p}{|p|^3}$. On the ball substitute $p = tq$; by the homogeneity in lemma E.5(i) and $G^{\mathrm{c}}_{ta}(tq) = t^{-2}G^{\mathrm{c}}_a(q)$, the inner part equals exactly

$$\frac{4}{(2\pi)^3}\int_{|q|<\varrho/t} \frac{q^2 B^{\mathrm{c}}_{m,\mu}(q) + \frac12(\mu^2-2m^2)(B^{\mathrm{c}}_{m,\mu}(q) - B^{\mathrm{c}}_{m,\mu}(0))}{(q^2+m^2)^2}\,d^3q = \frac{1}{2\pi^3}\int_{|q|<\varrho/t}\frac{|q|/8 + u_{m,\mu}(q)}{(q^2+m^2)^2}\,d^3q.$$

$u_{m,\mu}$ is continuous and bounded on $\mathbb{R}^3$ ($u(0) = 0$; $u(q) \to -\frac{m+\mu}{4\pi} - \frac{\mu^2-2m^2}{8\pi(m+\mu)}$ as $|q| \to \infty$, from $\arctan y = \frac{\pi}{2} - \frac{1}{y} + O(y^{-3})$), so its part converges to $\frac{1}{2\pi^3}\int_{\mathbb{R}^3} u(q^2+m^2)^{-2}$. The $|q|/8$ part is $\frac{1}{4\pi^2}\int_0^Q \frac{q^3dq}{(q^2+m^2)^2} = \frac{1}{8\pi^2}\big[\ln\frac{Q^2+m^2}{m^2} + \frac{m^2}{Q^2+m^2} - 1\big] = \frac{1}{4\pi^2}\big(\ln\frac{Q}{m} - \frac12\big) + o(1)$, $Q = \varrho/t$. Adding inner and outer parts, $\ln\varrho$ cancels and $M^{\mathrm{c}}(t) = \frac{1}{4\pi^2}\ln\frac1t + \kappa_{\mathrm{c}}(m,\mu) + \frac{\ln\pi}{4\pi^2} + \frac{1}{16\pi^3}\int_{\mathsf{B}\setminus\{|p|<\pi\}}|p|^{-3}d^3p + o(1)$.

Collecting Steps 1–4 gives the claim. Absolute convergence of the integrals in κ_{lat} is the majorant of Steps 1 and 3. The last statement is the substitution $q = mq'$ and the degree-one homogeneity of u in (q,m,μ). □

Proof of lemma 6.8. (a) By lemma E.1(ii) and theorem E.7 with $t = \varepsilon_k = 5^{-k}$, $z_k = e^2X_1(\varepsilon_k m, \varepsilon_k\mu_0) = \frac{e^2\ln 5}{4\pi^2}\,k + e^2\kappa(m,\mu_0) + o(1)$, so $|z_k| \le C(1+k)$ with C depending on the couplings only. (Also $z_k \to +\infty$, and $z_k - z_{k-1} \to \frac{e^2}{4\pi^2}\ln 5$.)

(b) Fix $M > 0$ and write $\mathsf{W}_\varepsilon(M) = \int_{[-\pi/\varepsilon,\pi/\varepsilon)^3}\frac{d^3q}{(2\pi)^3}\,(\hat q^2 + M^2)^{-1}$, $\hat q^2 = \sum_\alpha \frac{4}{\varepsilon^2}\sin^2\frac{\varepsilon q_\alpha}{2}$. Then $\mathsf{W}_\varepsilon(m) - \mathsf{W}_\varepsilon(M) = (M^2-m^2)\int\frac{d^3q}{(2\pi)^3}\frac{1}{(\hat q^2+m^2)(\hat q^2+M^2)}$; by lemma E.4(c) (scaled to $\varepsilon\mathbb{Z}^3$) the integrand is bounded by $\frac{\pi^4}{16}\big((q^2+m^2)(q^2+M^2)\big)^{-1}$, integrable on $\mathbb{R}^3$, and converges pointwise, so the Wick difference converges to $(M^2-m^2)\frac{1}{4\pi(m+M)} = \frac{M-m}{4\pi}$ and is bounded in k. For the scheme difference, theorem E.7 at the masses (m,μ_0) and (M,μ_0) with the same $t = \varepsilon_k$

gives $z_k(m) - z_k(M) = e^2\big[X_1(\varepsilon_k m, \varepsilon_k\mu_0) - X_1(\varepsilon_k M, \varepsilon_k\mu_0)\big] \to e^2\big[\kappa_{\rm c}(m,\mu_0) - \kappa_{\rm c}(M,\mu_0)\big]$: the logarithms and $\kappa_{\rm lat}$ cancel. Hence $\Delta_k(M)$ converges, in particular $\sup_k |\Delta_k(M)| < \infty$. □

E.2. **King's bare coefficient and its growth with the mass.**
The object. King's bare coefficient at scalar mass M and spacing ε is $\alpha_\varepsilon(M) = M^2 + \delta m^2{}_\varepsilon(M)$, where $\delta m^2{}_\varepsilon(M)$ is defined by [13, (3.32)–(3.33)] with the propagators of the infinite lattice $\varepsilon\mathbb{Z}^3$ [14, p. 330]: it is the finite sum

$$\delta m^2{}_\varepsilon(M) = \lambda\, d_{01} + e^2 d_{20} + e^4 d_{40} + \lambda e^2 d_{21} + \lambda^2 d_{02},$$

over the orders $2 \le \alpha + 2\beta \le 4$ in $e^\alpha\lambda^\beta$, each coefficient $d_{\alpha\beta} = d_{\alpha\beta}(M^2;\varepsilon)$ being the zero-momentum value of the one-particle-irreducible two-point function of its order with all lower-order counterterms inserted; the odd powers of e vanish, since the action is invariant under $(A,\phi) \mapsto (-A, R\phi)$ with R a reflection of the two components. There is no photon and no wave-function counterterm. With $\alpha_k(M) := \alpha_{\varepsilon_k}(M)$, the quantities of section 6.1 are $u_k(M) = \frac12(\alpha_k(M) - \alpha_k(m))$ and $s_-(M) = \liminf_k u_k(M)$.
Lattice formulas. On $\varepsilon\mathbb{Z}^3$ put $\mathsf{B}_\varepsilon := [-\pi/\varepsilon, \pi/\varepsilon)^3$, $\langle f\rangle := \int_{\mathsf{B}_\varepsilon} f\,\frac{d^3q}{(2\pi)^3}$, $\langle\langle f\rangle\rangle$ for the double integral, $\hat q^2 := \sum_\alpha \frac{4}{\varepsilon^2}\sin^2\frac{\varepsilon q_\alpha}{2}$, $s := M^2$, $C(q) := (\hat q^2 + s)^{-1}$ and $D(k) := (\hat k^2 + \mu_0^2)^{-1}$. Evaluating King's graphs [13, (3.34)] with the Feynman rules of his action (the current vertex of lemma E.1, the seagull, the quartic vertex $\lambda|\phi|^4$) gives, with $y_{\alpha\beta} := \partial_s d_{\alpha\beta}$:

(i) $d_{01} = -16\langle C\rangle$ (the quartic tadpole), $y_{01} = 16\langle C^2\rangle > 0$;
(ii) $d_{20} = \langle \hat q^2 DC\rangle - 3\langle D\rangle$ (photon exchange and seagull tadpole), $y_{20} = -\langle \hat q^2 DC^2\rangle < 0$;
(iii) $d_{02} = 128\langle\langle C(q)C(r)C(q+r)\rangle\rangle$ (the sunset; the tadpole-on-tadpole cancels identically against the inserted d_{01}), $y_{02} = -384\langle\langle C(q)^2C(r)C(q+r)\rangle\rangle < 0$;
(iv) $d_{21} = -8X_\varepsilon(M;\mu_0) = -16\langle C^2\sigma_\varepsilon\rangle$, where $\sigma_\varepsilon := \Sigma^{(20)}_{\rm ren}/e^2$ is the renormalised order-e^2 self-energy per real component ($-\bar\pi$ of lemma E.1 scaled to $\varepsilon\mathbb{Z}^3$, twice the function $-\frac12\bar\pi$ of lemma E.3; cf. (80)): the quartic tadpole with the order-e^2 self-energy inserted on its line is the only non-vanishing graph of this order, the tadpole-on-tadpole being cancelled by the inserted d_{01} and the two remaining classes vanishing identically at zero external momentum by the oddness of the current vertex;
(v) d_{40} is the sum over the twelve two-loop topologies of order e^4 (listed in the proof of lemma E.9(v) and, by continuum class, in lemma E.25 below), with the order-e^2 counterterm inserted on the exchange line.

Each $d_{\alpha\beta}(s;\varepsilon)$ is a smooth function of $s > 0$ at fixed ε (an integral of a smooth integrand over a compact zone), and the *exact identity*

$$\alpha_\varepsilon(M) - \alpha_\varepsilon(m) = (M^2 - m^2) + \sum_{\alpha\beta} e^\alpha\lambda^\beta \int_{m^2}^{M^2} y_{\alpha\beta}(s;\varepsilon)\,ds \tag{75}$$

holds for all $M, m > 0$. The formula (i), (ii) at one loop is the one behind (48).

Lemma E.8 (one-loop and sunset bounds on $\varepsilon\mathbb{Z}^3$)**.** *For every $\varepsilon > 0$ and $s > 0$:* (i) $0 < y_{01}(s) \le \dfrac{\pi^2}{4\sqrt{s}}$; (ii) $-\dfrac{\pi^2}{32(\sqrt{s}+\mu_0)} \le y_{20}(s) < 0$; (iii) $-\dfrac{81\pi^4}{32\,s} \le y_{02}(s) < 0$.

Proof. By lemma E.4(c) scaled to $\varepsilon\mathbb{Z}^3$, $\hat q^2 \ge \frac{4}{\pi^2}|q|^2$ on B_ε, so $C(q) \le \frac{\pi^2}{4}\tilde G(q)$ with $\tilde G(q) := (|q|^2 + \tilde a^2)^{-1}$, $\tilde a := \frac{\pi}{2}\sqrt{s}$, and $D(k) \le \frac{\pi^2}{4}(|k|^2 + \tilde b^2)^{-1}$, $\tilde b := \frac{\pi}{2}\mu_0$; moreover $C(q+r) \le \frac{\pi^2}{4}\tilde G(q+r-P)$ for the $P \in \frac{2\pi}{\varepsilon}\mathbb{Z}^3$ with $q + r - P \in \mathsf{B}_\varepsilon$, and for $q, r \in \mathsf{B}_\varepsilon$ there are 27 candidates P. (i) $16\langle C^2\rangle \le 16\cdot\frac{\pi^4}{16}\int_{\mathbb{R}^3}\frac{d^3q}{(2\pi)^3}\tilde G^2 = \pi^4\cdot\frac{1}{8\pi\tilde a} = \frac{\pi^2}{4\sqrt{s}}$. (ii) $\hat q^2 C \le 1$ gives $|y_{20}| \le \langle DC\rangle \le \frac{\pi^4}{16}\int_{\mathbb{R}^3}\frac{d^3k}{(2\pi)^3}\frac{1}{(|k|^2+\tilde b^2)(|k|^2+\tilde a^2)} = \frac{\pi^4}{16}\cdot\frac{1}{4\pi(\tilde a+\tilde b)} = \frac{\pi^2}{32(\sqrt{s}+\mu_0)}$ (lemma E.5(i)); strictness, since every summand is positive. (iii) $\langle\langle C(q)^2C(r)C(q+r)\rangle\rangle \le (\frac{\pi^2}{4})^4\int\tilde G(q)^2\Big[\sum_P\int_{\mathbb{R}^3}\tilde G(r)\tilde G(q+r-P)\,\frac{d^3r}{(2\pi)^3}\Big]\frac{d^3q}{(2\pi)^3} \le (\frac{\pi^2}{4})^4\cdot\frac{1}{8\pi\tilde a}\cdot\frac{27}{8\pi\tilde a} = \frac{27\pi^4}{4096\,s}$, by lemma E.5(i) ($B^{\rm c}_{\tilde a,\tilde a} \le \frac{1}{8\pi\tilde a}$). Multiply by 384. □

Lemma E.9 (two-loop lattice bounds on $\varepsilon\mathbb{Z}^3$)**.** (iv) *For every $\varepsilon > 0$ and $s > 0$, $|y_{21}(s;\varepsilon)| \le C_{21}(s)/s + C'_{21}\,\varepsilon/\sqrt{s}$, with*

$$C_{21}(s) := \frac{\pi^2}{4}\Big[\frac{135}{8}\pi^2 + \frac{135\pi^4}{128}\Big(2 + \frac{\mu_0^2}{s}\Big) + \frac{\pi^2}{16}\Big] \quad (\textit{non-increasing in } s), \qquad C'_{21} := \frac{3\sqrt{3}\,\pi^3}{32}.$$

(v) *There are $a_0 \in (0,1]$ and $C_{40} < \infty$, independent of μ_0, ε and s, such that $|y_{40}(s;\varepsilon)| \le C_{40}/s$ whenever $\varepsilon \le a_0/\sqrt{s}$.*

Proof of lemma E.9(iv). The lattice is $\varepsilon\mathbb{Z}^3$ with the conventions of the paragraph "Lattice formulas": $s = M^2$, $C(q) = (\hat q^2 + s)^{-1}$, $D(k) = (\hat k^2 + \mu_0^2)^{-1}$, $\langle\cdot\rangle$ the integral over B_ε, $\bar{\mathsf{B}}_\varepsilon$ the closed zone, $\hat x_\mu := \frac{2}{\varepsilon}\sin\frac{\varepsilon x_\mu}{2}$. Every integrand below is smooth and $\frac{2\pi}{\varepsilon}\mathbb{Z}^3$-periodic in each integration variable, so $k \mapsto p - k$ is a permitted substitution for every real p. Put

$$B_s(p) := \langle D(k)C(p-k)\rangle_k, \qquad \Xi_s(p) := \langle D(k)C(p-k)^2\rangle_k = -\partial_s B_s(p), \qquad J_s^\alpha(p) := \langle D(k)C(p-k)\widehat{(p-k)}_\alpha^{\,2}\rangle$$

$$s_0 := \langle D\rangle, \qquad b_s := \langle C^2\rangle, \qquad F_s(p) := \langle\widehat{(p+r)}^{\,2}D(p-r)C(r)\rangle_r, \qquad G_s(p) := \partial_s F_s(p) = -\langle\widehat{(p+r)}^{\,2}D(p-r)C$$

F_s is the photon-exchange graph and $s_0\sum_\mu\cos\varepsilon p_\mu$ the seagull tadpole at external momentum p: the function $\hat\Pi$ of lemma E.1, at spacing ε and after the substitution $r = p+k$, is $s_0\sum_\mu\cos\varepsilon p_\mu - F_s(p)$. Hence the self-energy per real component is

$$\sigma_\varepsilon(p) = -\bar\pi_\varepsilon(p) = F_s(p) - F_s(0) + s_0\sum_\mu(1-\cos\varepsilon p_\mu) = F_s(p) - F_s(0) + \tfrac12\varepsilon^2 s_0\,\hat p^2, \tag{76}$$

and since s_0 does not depend on s, $\partial_s\sigma_\varepsilon(q) = G_s(q) - G_s(0)$. Differentiating (iv) of the lattice formulas,

$$y_{21} = \partial_s d_{21} = -16\big\langle -2C^3\sigma_\varepsilon + C^2\,\partial_s\sigma_\varepsilon\big\rangle. \tag{77}$$

Lemma E.10 (elementary lattice inequalities, continued)**.** *Let $u, v \in \mathbb{R}^3$, $\tilde a := \frac{\pi}{2}\sqrt{s}$, $\tilde b := \frac{\pi}{2}\mu_0$, $\tilde G_a(w) := (|w|^2 + a^2)^{-1}$ on $\mathbb{R}^3$.*

(a) $|\hat u_\mu - \hat v_\mu| \le |u_\mu - v_\mu|$, $|\widehat{(u+v)}_\mu| \le |\hat u_\mu| + |\hat v_\mu|$, $|\widehat{(2v)}_\mu| \le 2|\hat v_\mu|$; *hence* $\widehat{(u+v)}^{\,2} \le 2\,\widehat{(u-v)}^{\,2} + 8\hat v^2$.

(b) *With $E(v) := \hat v^2$: $\partial_\mu E = \frac{2}{\varepsilon}\sin\varepsilon v_\mu$, $|\partial_\mu E| \le 2|\hat v_\mu|$, $\partial_\mu\partial_\nu E = 2\cos(\varepsilon v_\mu)\delta_{\mu\nu}$; and $|w\cdot \operatorname{Hess} C(v)\,w| \le 10\,C(v)^2|w|^2$ for all $w \in \mathbb{R}^3$.*

(c) *For $p \in \bar{\mathsf{B}}_\varepsilon$ and $k \in \mathsf{B}_\varepsilon$: $C(p-k) \le \frac{\pi^2}{4}\sum_{n\in\{-1,0,1\}^3}\tilde G_{\tilde a}\big(p-k-\frac{2\pi}{\varepsilon}n\big)$ and $C(p-k)^2 \le \big(\frac{\pi^2}{4}\big)^2\sum_n\tilde G_{\tilde a}\big(p-k-\frac{2\pi}{\varepsilon}n\big)^2$; for $k \in \mathsf{B}_\varepsilon$, $C(k) \le \frac{\pi^2}{4}\tilde G_{\tilde a}(k)$ and $D(k) \le \frac{\pi^2}{4}\tilde G_{\tilde b}(k)$.*

(d) *$s_0 \le d_0/\varepsilon$ with $d_0 = \int_k \hat k^{-2} \le \frac{\sqrt3\,\pi}{8}$ (d_0 as in lemma E.6); $0 \le J_s^\alpha(p) \le s_0$.*

(e) $\int_{\mathbb{R}^3}\tilde G_b(k)\tilde G_a(P-k)^2\frac{d^3k}{(2\pi)^3} = \dfrac{1}{8\pi a\,((a+b)^2+|P|^2)} \le \dfrac{1}{8\pi a(a+b)^2}$.

Proof. (a) The derivative of $x \mapsto \frac{2}{\varepsilon}\sin\frac{\varepsilon x}{2}$ is $\cos\frac{\varepsilon x}{2}$, of modulus ≤ 1; the addition theorem gives the second claim and $|\sin 2y| \le 2|\sin y|$ the third. Then $|\widehat{(u+v)}_\mu| = |\widehat{((u-v)+2v)}_\mu| \le |\widehat{(u-v)}_\mu| + 2|\hat v_\mu|$ and $(x+y)^2 \le 2x^2+2y^2$; sum over μ. (b) $|\frac{2}{\varepsilon}\sin\varepsilon v_\mu| = \frac{4}{\varepsilon}|\sin\frac{\varepsilon v_\mu}{2}\cos\frac{\varepsilon v_\mu}{2}| \le 2|\hat v_\mu|$. With $\partial_\mu C = -C^2\partial_\mu E$: $w\cdot\operatorname{Hess} C\,w = 2C^3(w\cdot\nabla E)^2 - 2C^2\sum_\mu\cos(\varepsilon v_\mu)w_\mu^2$, and $|w\cdot\nabla E| \le \sum_\mu 2|w_\mu||\hat v_\mu| \le 2|w||\hat v|$, so the modulus is $\le (8C^3\hat v^2 + 2C^2)|w|^2 \le 10C^2|w|^2$, since $C\hat v^2 \le 1$. (c) The components of $p-k$ lie in $[-\frac{2\pi}{\varepsilon},\frac{2\pi}{\varepsilon}]$, so a nearest point $\bar n$ of $\frac{2\pi}{\varepsilon}\mathbb{Z}^3$ lies in $\frac{2\pi}{\varepsilon}\{-1,0,1\}^3$. By lemma E.4(c) at spacing ε, $\widehat{(p-k)}^{\,2} \ge \frac{4}{\pi^2}|p-k-\bar n|^2$, so $C(p-k) \le (\frac{4}{\pi^2}|p-k-\bar n|^2 + s)^{-1} = \frac{\pi^2}{4}\tilde G_{\tilde a}(p-k-\bar n)$; a maximum of positive terms is at most their sum. For $k \in \mathsf{B}_\varepsilon$, $\bar n = 0$. (d) $k = k'/\varepsilon$ gives $\langle\hat k^{-2}\rangle = d_0/\varepsilon$, and $d_0 \le \frac{\pi^2}{4}\int_{|k|\le\sqrt3\pi}\frac{d^3k}{(2\pi)^3|k|^2} = \frac{\pi^2}{4}\cdot\frac{\sqrt3}{2\pi}$. $J_s^\alpha \le s_0$ because $C(v)\hat v_\alpha^2 \le C(v)\hat v^2 \le 1$. (e) By lemma E.5(i) the integral with $\tilde G_a$ in place of $\tilde G_a^2$ is $\frac{\arctan(|P|/(a+b))}{4\pi|P|}$; apply $-\partial/\partial(a^2) = -(2a)^{-1}\partial_a$ and $\partial_a\arctan\frac{x}{a+b} = -\frac{x}{(a+b)^2+x^2}$. □

Lemma E.11 (the decomposition on $\varepsilon\mathbb{Z}^3$)**.** *For every $p \in \mathbb{R}^3$,*

$$F_s(p) - F_s(0) = 2\hat{p}^2 B_s(p) + (\mu_0^2 - 2s)\big(B_s(p) - B_s(0)\big) - \varepsilon^2 \sum_\alpha \hat{p}_\alpha^2 J_s^\alpha(p),$$

hence $\sigma_\varepsilon(p) = 2\hat{p}^2 B_s(p) + (\mu_0^2 - 2s)(B_s(p) - B_s(0)) + 2\rho_\varepsilon(p)$ with $\rho_\varepsilon(p) := -\frac{1}{2}\varepsilon^2 \sum_\alpha \hat{p}_\alpha^2 J_s^\alpha(p) + \frac{1}{4}\varepsilon^2 s_0 \hat{p}^2$. (This is lemma E.3 at spacing ε.)

Proof. Lemma E.2 with $u = \varepsilon x$, $w = \varepsilon y$, divided by ε^2, reads $\widehat{(x+y)}^2 = 2\hat{x}^2 + 2\hat{y}^2 - \varepsilon^2 \hat{x}^2 \hat{y}^2 - \widehat{(x-y)}^2$ componentwise. With $x = p_\alpha$, $y = (p-k)_\alpha$, summed over α: $\widehat{(2p-k)}^2 = 2\hat{p}^2 + 2\widehat{(p-k)}^2 - \varepsilon^2 \sum_\alpha \hat{p}_\alpha^2 \widehat{(p-k)}_\alpha^2 - \hat{k}^2$. Substitute $r = p-k$ in F_s and insert $\widehat{(p-k)}^2 = C(p-k)^{-1} - s$, $\hat{k}^2 = D(k)^{-1} - \mu_0^2$:

$$F_s(p) = \big\langle D(k)C(p-k)\big[2\hat{p}^2 - 2s + \mu_0^2\big]\big\rangle_k + 2s_0 - \langle C\rangle - \varepsilon^2 \sum_\alpha \hat{p}_\alpha^2 J_s^\alpha(p),$$

using $\langle C(p-k)\rangle_k = \langle C\rangle$. Subtract the value at $p = 0$, where $\hat{p} = 0$: the linearly divergent constants $2s_0 - \langle C\rangle$ cancel exactly. The second form follows from (76). □

Lemma E.12 (one-loop blocks with explicit s)**.** *For every $\varepsilon > 0$ and $s > 0$:*

(P1) $b_s \le \dfrac{\pi^2}{64\sqrt{s}}$;

(P2) $B_s(0) \le \dfrac{\pi^2}{32(\sqrt{s} + \mu_0)}$ *and* $\sup_{p \in \bar{\mathsf{B}}_\varepsilon} B_s(p) \le \dfrac{27\pi^2}{32(\sqrt{s} + \mu_0)}$;

(P3) $\sup_{p \in \bar{\mathsf{B}}_\varepsilon} \Xi_s(p) \le \dfrac{27\pi^2}{64\, s^{3/2}}$;

(P4) $|G_s(p)| \le 2b_s + 8B_s(p)$ *for $p \in \bar{\mathsf{B}}_\varepsilon$; hence $|\partial_s \sigma_\varepsilon(q)| \le 4b_s + 16 \sup_{\bar{\mathsf{B}}_\varepsilon} B_s$ for $q \in \mathsf{B}_\varepsilon$;*

(P5) $|\sigma_\varepsilon(p)| \le 2\hat{p}^2 B_s(p) + 5(\mu_0^2 + 2s)\,|p|^2 \sup_{\bar{\mathsf{B}}_\varepsilon} \Xi_s + \frac{3}{2}\varepsilon^2 s_0 \hat{p}^2$ *for $p \in \mathsf{B}_\varepsilon$.*

Proof. (P1) By lemma E.10(c), $b_s \le (\frac{\pi^2}{4})^2 \int_{\mathbb{R}^3} \tilde{G}_{\tilde{a}}^2 \frac{d^3q}{(2\pi)^3} = \frac{\pi^4}{16} \cdot \frac{1}{8\pi\tilde{a}} = \frac{\pi^2}{64\sqrt{s}}$. (P2) At $p = 0$ both factors are at $k \in \mathsf{B}_\varepsilon$: $B_s(0) \le (\frac{\pi^2}{4})^2 \int \tilde{G}_{\tilde{b}} \tilde{G}_{\tilde{a}} = \frac{\pi^4}{16} \cdot \frac{1}{4\pi(\tilde{a}+\tilde{b})}$ by lemma E.5(i). At general $p \in \bar{\mathsf{B}}_\varepsilon$, lemma E.10(c) supplies 27 images, each integral being $\le \frac{1}{4\pi(\tilde{a}+\tilde{b})}$. (P3) Lemma E.10(c), (e): $\Xi_s(p) \le (\frac{\pi^2}{4})^3 \sum_n \int \tilde{G}_{\tilde{b}}(k) \tilde{G}_{\tilde{a}}(p - k - \frac{2\pi}{\varepsilon} n)^2 \frac{d^3k}{(2\pi)^3} \le \frac{27(\pi^2/4)^3}{8\pi\tilde{a}(\tilde{a}+\tilde{b})^2} \le \frac{27\pi^6/64}{8\pi(\pi/2)^3 s^{3/2}} = \frac{27\pi^2}{64\, s^{3/2}}$. (P4) Lemma E.10(a) with $u = p$, $v = r$: $\widehat{(p+r)}^2 D(p-r)C(r)^2 \le \big[2\widehat{(p-r)}^2 D(p-r) + 8\hat{r}^2 D(p-r)\big]C(r)^2 \le 2C(r)^2 + 8D(p-r)C(r)$, by $\hat{x}^2(\hat{x}^2 + \mu_0^2)^{-1} \le 1$ and $\hat{r}^2 C(r) \le 1$. Integrate over r; $\langle D(p-r)C(r)\rangle_r = B_s(p)$. The second claim is $\partial_s \sigma_\varepsilon = G_s(q) - G_s(0)$. (P5) In lemma E.11, $\varepsilon^2 \sum_\alpha \hat{p}_\alpha^2 J_s^\alpha \le \varepsilon^2 \hat{p}^2 s_0$ by lemma E.10(d), and the seagull term is $\frac{1}{2}\varepsilon^2 s_0 \hat{p}^2$. For the middle term: B_s is smooth on $\mathbb{R}^3$ and even ($k \mapsto -k$; D, C even), so $\nabla B_s(0) = 0$ and $B_s(p) - B_s(0) = \int_0^1 (1-t)\,(p\cdot\nabla)^2 B_s(tp)\,dt$ with $(p\cdot\nabla)^2 B_s(tp) = \langle D(k)\, p \cdot \mathrm{Hess}\, C(tp-k)\, p\rangle_k$. By lemma E.10(b), $|(p\cdot\nabla)^2 B_s(tp)| \le 10|p|^2 \Xi_s(tp)$ with $tp \in \bar{\mathsf{B}}_\varepsilon$, and $\int_0^1 (1-t)dt = \frac{1}{2}$. Finally $|\mu_0^2 - 2s| \le \mu_0^2 + 2s$. □

Proof of (iv). By (77), $|y_{21}| \le 16\big[2\langle C^3|\sigma_\varepsilon|\rangle + \langle C^2|\partial_s\sigma_\varepsilon|\rangle\big]$. For $q \in \mathsf{B}_\varepsilon$, $C^3\hat{q}^2 \le C^2$ and $C^3|q|^2 \le \frac{\pi^2}{4} C^3 \hat{q}^2 \le \frac{\pi^2}{4} C^2$ (lemma E.4(c)). So lemma E.12(P5) gives $2\langle C^3|\sigma_\varepsilon|\rangle \le b_s\big[4 \sup B_s + \frac{5\pi^2}{2}(\mu_0^2 + 2s) \sup \Xi_s + 3\varepsilon^2 s_0\big]$, and (P4) gives $\langle C^2|\partial_s\sigma_\varepsilon|\rangle \le b_s\big[4b_s + 16 \sup B_s\big]$, the suprema over $\bar{\mathsf{B}}_\varepsilon$. Hence

$$|y_{21}| \le 16\, b_s \Big[20 \sup B_s + \frac{5\pi^2}{2}(\mu_0^2 + 2s) \sup \Xi_s + 4b_s + 3\varepsilon^2 s_0\Big].$$

Insert (P1)–(P3) and lemma E.10(d), dropping $\mu_0 \ge 0$ from the denominators: $16 b_s \le \frac{\pi^2}{4\sqrt{s}}$; $20 \sup B_s \le \frac{135}{8}\frac{\pi^2}{\sqrt{s}}$; $\frac{5\pi^2}{2}(\mu_0^2 + 2s) \cdot \frac{27\pi^2}{64 s^{3/2}} = \frac{135\pi^4}{128}\big(2 + \frac{\mu_0^2}{s}\big)\frac{1}{\sqrt{s}}$; $4b_s \le \frac{\pi^2}{16\sqrt{s}}$; $3\varepsilon^2 s_0 \le \frac{3\sqrt{3}\,\pi}{8}\varepsilon$. Collecting,

$$|y_{21}(s;\varepsilon)| \le \frac{\pi^2}{4}\Big[\frac{135}{8}\pi^2 + \frac{135\pi^4}{128}\Big(2 + \frac{\mu_0^2}{s}\Big) + \frac{\pi^2}{16}\Big]\frac{1}{s} + \frac{\pi^2}{4} \cdot \frac{3\sqrt{3}\,\pi}{8} \cdot \frac{\varepsilon}{\sqrt{s}} = \frac{C_{21}(s)}{s} + C'_{21}\frac{\varepsilon}{\sqrt{s}},$$

and C_{21} is non-increasing in s because μ_0^2/s is. The only coefficient that grows with s, $|\mu_0^2 - 2s|$, has been paired with $B_s(p) - B_s(0) = O(|p|^2 s^{-3/2})$, the Taylor step being taken on the scalar line. □

Proof of lemma E.9(v). The proof rescales to unit scalar mass, writes King's coefficient as a forest-subtracted sum over the twelve graphs of order e^4, and bounds that sum uniformly in the rescaled photon mass. The power-counting input of [20, 21] enters at one mass configuration only (scalar and photon mass 1); the light-photon limit is handled by elementary bounds.

Lemma E.13 (exact scaling)**.** *Write $d_{\alpha\beta}(s;\mu_0;\varepsilon)$ for the coefficient at renormalised scalar mass squared s, photon mass μ_0 and spacing ε, and $n := \alpha/2 + \beta$. For every $t > 0$, $d_{\alpha\beta}(s;\mu_0;\varepsilon) = t^{2-n}\, d_{\alpha\beta}(s/t^2;\mu_0/t;t\varepsilon)$. Consequently*

$$y_{40}(s;\mu_0;\varepsilon) = s^{-1}\, Y\big(\mu_0/\sqrt{s},\ \varepsilon\sqrt{s}\big), \qquad Y(\nu,a) := y_{40}(1;\nu;a).$$

Proof. Consider a connected graph with L loops, I lines and vertices v of lattice degree d_v (1 for the current vertex, 0 for the seagull and the quartic vertex, $2 - n_v$ for the n_v-photon vertex of the hopping term), at zero external momentum. Substitute $q = tq'$ in every loop momentum: $q \in \mathsf{B}_\varepsilon$ iff $q' \in \mathsf{B}_{t\varepsilon}$, $d^3q = t^3 d^3q'$, and $\frac{4}{\varepsilon^2}\sin^2\frac{\varepsilon t q'_\mu}{2} = t^2 \cdot \frac{4}{(t\varepsilon)^2}\sin^2\frac{(t\varepsilon)q'_\mu}{2}$, so $\hat q^2$ at spacing ε equals t^2 times $\hat{q'}^2$ at spacing $t\varepsilon$. Each propagator therefore gives t^{-2}: $(\hat q^2 + s)^{-1} = t^{-2}(\hat{q'}^2 + s/t^2)^{-1}$ and $(\hat k^2 + \mu_0^2)^{-1} = t^{-2}(\hat{k'}^2 + \mu_0^2/t^2)^{-1}$. A vertex of the hopping term is $\varepsilon^{n_v-2}w(\varepsilon q)$ with w a trigonometric polynomial, and $\varepsilon^{n_v-2}w(\varepsilon q) = t^{2-n_v}(t\varepsilon)^{n_v-2}w\big((t\varepsilon)q'\big)$, i.e. t^{d_v} times the same vertex at spacing $t\varepsilon$; the quartic vertex is a constant. Hence the amplitude obeys $A(s,\mu_0,\varepsilon) = t^{3L-2I+\sum_v d_v} A(s/t^2,\mu_0/t,t\varepsilon)$, and for a two-point graph of order n the exponent is the superficial degree $3 - \frac{E}{2} - n = 2 - n$ ($E = 2$). A counterterm vertex $d_{\alpha'\beta'}$ of lower order n' inserted on a line contributes, by induction on the order, the factor $t^{2-n'}$ and one more propagator t^{-2}, which leaves the exponent $2 - n$ unchanged. $d_{\alpha\beta}$ is the zero-momentum value of such a sum [13, (3.32)–(3.33)], so the identity holds. Differentiate in s at fixed t and put $t = \sqrt{s}$: $y_{\alpha\beta}(s;\mu_0;\varepsilon) = s^{-n/2} y_{\alpha\beta}(1;\mu_0/\sqrt{s};\varepsilon\sqrt{s})$, and $n = 2$ at $(4,0)$. □

Conventions at unit scalar mass. For the rest of the proof the spacing is $a \in (0,1]$, $\mathsf{B}_a := [-\pi/a, \pi/a)^3$, $\langle f\rangle_a := \int_{\mathsf{B}_a} f\, \frac{d^3k}{(2\pi)^3}$, $\langle\langle\cdot\rangle\rangle_a$ for two loop momenta; $\hat e(k) := \sum_\mu \frac{4}{a^2}\sin^2\frac{ak_\mu}{2}$, so that $\frac{4}{\pi^2}|k|^2 \le \hat e(k) \le |k|^2$ on B_a (lemma E.4(c)) and $\hat e \le 12/a^2$; $\chi(k) := \hat e(k) + 1$, $C(q) := \chi(q)^{-1}$ (scalar mass 1), $D_\nu(k) := (\hat e(k) + \nu^2)^{-1}$ (photon mass $\nu > 0$; the propagator is diagonal in the photon indices), $D_0 := 1/\hat e$. In the per-component normalisation of (i)–(v) the current vertex carries $\tilde V_\mu(p,p') := -\frac{2}{a}\sin\frac{a(p+p')_\mu}{2}$ and the phase $e^{-\mathrm{i}ak_\mu/2}$, $k = p' - p$ the incoming photon momentum; the seagull is $e^{-\mathrm{i}ak_\mu/2}\cos\frac{a(p+p')_\mu}{2}\,\delta_{\mu\nu}$; the three- and four-photon vertices V_3, V_4 of the hopping term are a and a^2 times trigonometric polynomials of $a(p+p')$ and ak (lemma E.1). Numerical factors of the Feynman rules play no role and are suppressed. Soft and hard momenta: $Q := [-\frac12,\frac12]^3 \subset \mathsf{B}_a$, $Q_s := \{|k| \le \frac12\}$, hard $:= \mathsf{B}_a \setminus Q_s$. For hard k, $\hat e(k) \ge \pi^{-2}$, so $D_\nu \le (1+\pi^2)D_1$; for every k, $D_\nu \le D_0 \le \frac{\pi^2}{4}|k|^{-2}$ on B_a. In particular, for every $\nu > 0$ and $a \le 1$,

(78)

$$\sigma_1 := \langle \mathbf{1}_{Q_s} D_\nu\rangle_a \le \tfrac{1}{16}, \qquad \langle \mathbf{1}_{Q_s} D_\nu^2 |k|^2\rangle_a \le \tfrac{\pi^2}{64}, \qquad \langle C^2\rangle_a \le \tfrac{\pi^2}{64}, \qquad \langle \chi^{-3/2}\rangle_a \le C\big(1 + \ln\tfrac{1}{a}\big),$$

by $\frac{\pi^2}{4}\int_{|k|\le 1/2}\frac{d^3k}{(2\pi)^3|k|^2} = \frac{1}{16}$, $(\frac{\pi^2}{4})^2\int_{|k|\le1/2}\frac{d^3k}{(2\pi)^3|k|^2} = \frac{\pi^2}{64}$, $(\frac{\pi^2}{4})^2\int_{\mathbb{R}^3}\frac{d^3k}{(2\pi)^3(|k|^2+\pi^2/4)^2} = \frac{\pi^2}{64}$ and $(\frac{\pi^2}{4})^{3/2}\int_{|k|\le\sqrt3\pi/a}\frac{d^3k}{(2\pi)^3(|k|^2+\pi^2/4)^{3/2}}$. Throughout, C denotes a constant depending only on a_0 (fixed in lemma E.16), never on ν, a or s; it may change from line to line.

Subtraction operators, degrees, chords. For a one-particle-irreducible subgraph γ of Γ with external momenta q^γ and internal momenta k^γ, the subtraction operator of order $\delta \in \{0,1\}$ acts on the integrand $F_\gamma(k^\gamma;q^\gamma)$ at fixed k^γ [21, Def. 3.1]:

$$t^0_\gamma F_\gamma = F_\gamma(k^\gamma;0), \qquad t^1_\gamma F_\gamma = F_\gamma(k^\gamma;0) + \sum_\rho P_\rho(q^\gamma)\,\partial_{q_\rho} F_\gamma(k^\gamma;0), \qquad P_\rho(q) = a^{-1}p_\rho(aq),$$

with p_ρ smooth and 2π-periodic, $p_\rho(x) = x_\rho + O(|x|^2)$ ([21, Def. 3.1, (3-2), and App. A]; the same P_ρ is used throughout). These operators preserve periodicity and act as the Taylor operators do on the degrees. The degree of γ is $\omega(\gamma) = 3L(\gamma) - 2I(\gamma) + \sum_{v\in\gamma} d_v = 3 - \frac{E(\gamma)}{2} - n(\gamma)$ (E external legs, n the order); it is the UV-divergence degree of [21, (2-28)–(2-30)] in three dimensions, computed with the lattice degrees of [20, Defs. 2.2–2.3], the lattice degree of every vertex function being its continuum dimension. The lines of Γ are ordered with the photon lines first, and the chord set of every connected subgraph is the one determined by this order through the rule of [21, (2-24)]: the chords of Γ carry the loop momenta, all other line momenta follow by conservation, the chords of a subgraph γ are chords of Γ, and the internal momenta of γ are its chord momenta, which carry no external momentum of γ [21, (2-16)].

Lemma E.14 (the forest form of King's coefficient)**.** *On $\varepsilon\mathbb{Z}^3$, at any spacing and masses, with signed combinatorial weights c_Γ whose values play no role below,*

$$d_{40} = \sum_\Gamma c_\Gamma \,\langle\langle R' F_\Gamma\rangle\rangle, \qquad y_{40} = \partial_s d_{40} = -\sum_\Gamma \sum_l c_\Gamma \,\langle\langle R' F_{\Gamma,\mathrm{dbl}(l)}\rangle\rangle,$$

where Γ runs over the twelve one-particle-irreducible two-point graphs of order e^4 at zero external momentum (listed next), F_Γ is the integrand of the Feynman rules without counterterms, $R' := 1 - \sum_\gamma t_\gamma^{\omega(\gamma)}$ with γ running over the proper one-particle-irreducible subgraphs with $\omega(\gamma) \geq 0$, l runs over the scalar lines of Γ, and $F_{\Gamma,\mathrm{dbl}(l)}$ is F_Γ with the propagator of l squared.

Proof. King's scheme [13, (3.32)–(3.33)] has scalar mass counterterms at the orders $2 \leq \alpha+2\beta \leq 4$, each fixed by the vanishing at zero momentum of the one-particle-irreducible two-point function of its order with the lower-order counterterms inserted, and no photon or vertex counterterm. Compare it with the scheme in which every one-particle-irreducible type of non-negative degree receives the counterterm $-t^\omega$ of its zero-momentum Taylor polynomial, computed recursively in the order; its amplitudes are the zone integrals of the forest-subtracted integrands [21, Sect. 4], and since the forests of proper divergent subgraphs have at most one element here (property (iv) below), the forest operator is R'. The proper divergent subgraphs of the graphs of order e^4 are of three types: (a) the one-loop scalar self-energy of order e^2 (exchange graph, seagull tadpole; $\omega = 1$); (c) the one-loop vacuum polarisation of order e^2 (bubble, seagull loop; $\omega = 1$); (d) the one-loop $\phi\bar\phi A$ vertex of order e^3 (triangle, two swordfish graphs, A^3-tadpole; $\omega = 0$). The two schemes agree on $\varepsilon\mathbb{Z}^3$: (a) the order-1 part $t^1_\gamma - t^0_\gamma$ of the exchange integrand $f(q;p) = \widehat{(p+q)}^{\,2} D(p-q)C(q)$ is $\sum_\rho P_\rho(p)\partial_{p_\rho} f(q;0)$, which is odd in the loop momentum q (D, C even), while the seagull-tadpole integrand $D(k)\cos(ap_\mu)$ has zero gradient at $p = 0$; so the order-1 term integrates to zero identically in the remaining momenta, and the order-0 term is King's d_{20}-insertion. (c) The one-loop polarisation $\Pi^{(20)}_{\mu\nu}(k)$ (bubble plus seagull loop) satisfies $\Pi^{(20)}_{\mu\nu}(0) = \pm\langle\partial_\nu\big[\frac{2}{a}\sin(aq_\mu)\,C(q)\big]\rangle_a = 0$ for all μ,ν, the integral of a derivative of a periodic function (on the diagonal the seagull loop is the derivative of the current loop; off the diagonal the bubble alone is a derivative; this is the lattice Ward identity used after (81)), and $\nabla_k\Pi^{(20)}(0) = 0$: writing $\Pi^{(20)}_{\mu\nu}(k) = e^{-\mathrm{i}a(k_\mu-k_\nu)/2}T_{\mu\nu}(k)$, T is even in k (substitute $q \mapsto -q$), so $\nabla T(0) = 0$ and $\nabla_k\Pi^{(20)}(0) = -\frac{\mathrm{i}a}{2}(e_\mu - e_\nu)\Pi^{(20)}_{\mu\nu}(0) = 0$. Hence $t^1_k\Pi^{(20)} = 0$: the photon counterterm of the forest scheme vanishes, as King's does. (d) The vertex counterterm is the value at zero momenta of the sum of the one-loop vertex graphs, and each vanishes by $q \mapsto -q$: in the triangle $\tilde V_\mu(q,q) = -\frac{2}{a}\sin(aq_\mu)$ is odd, the other factors are even and the two phases of the internal photon cancel; the two swordfish graphs sum to $\langle 2\mathrm{i}\sin(aq_\mu)D(q)C(q)\rangle_a = 0$; the A^3-tadpole is proportional to $\sin\frac{a(p+p')_\mu}{2} = 0$ at $p = p' = 0$. So neither scheme has a vertex counterterm. Therefore the zero-momentum value of the order-e^4 two-point function with King's insertions is $\sum_\Gamma c_\Gamma\langle\langle R'F_\Gamma\rangle\rangle$. Differentiating in s: the operators t_γ act on momenta at fixed masses, so ∂_s commutes with R', and $\partial_s C = -C^2$ doubles one scalar line at a time. □

The twelve graphs. They are the one-particle-irreducible two-point placements of the vertex sets V_1^4, $V_2V_1^2$, V_2^2, V_3V_1, V_4; every other placement contains a bridge. With the names of the table in lemma E.25:

	vertices	graph	proper subgraphs with $\omega \geq 0$ (type; photon lines inside)	cl
Γ_0	V_1^4	rainbow	inner exchange (a; 1)	A
Γ_1	V_1^4	exchange, bubble inserted	bubble (c; 0)	B
Γ_2	V_1^4	crossed	two triangles (d; 1), overlapping	A
Γ_3	$V_2V_1^2$	exchange, seagull tadpole on the inner line	seagull tadpole (a; 1)	A
Γ_4	$V_2V_1^2$	seagull tadpole, bubble inserted	bubble (c; 0)	B
Γ_5	$V_2V_1^2$	seagull at an end (two mirror graphs)	swordfish vertex (d; 1)	A
Γ_6	$V_2V_1^2$	exchange, seagull loop inserted	seagull loop (c; 0)	B
Γ_7	$V_2V_1^2$	seagull in the middle	two swordfish vertices (d; 1), sharing V_2	A
Γ_8	V_3V_1	V_3-tadpole vertex (two orientations)	A^3-tadpole (d; 1), no scalar line	A
Γ_9	V_2^2	seagull tadpole, seagull loop inserted	seagull loop (c; 0)	B
Γ_{10}	V_2^2	melon	none	A
Γ_{11}	V_4	V_4 double tadpole	none	n

By inspection of the twelve graphs: (i) $\omega(\Gamma) = 0$, and for every scalar line l, $\omega_{\mathrm{dbl}}(\Gamma) = -2$, where $\omega_{\mathrm{dbl}}(\gamma) := \omega(\gamma) - 2 \cdot \mathbf{1}[l \in \gamma]$ is the degree of the doubled graph (the doubled line is a two-point insertion of degree 0); (ii) every proper one-particle-irreducible subgraph with $\omega \geq 0$ is one-loop, of type (a), (c) or (d), and contains at most one photon line, which, the photon lines being first in the order, is that subgraph's own chord; the subgraphs of type (c) contain no photon line; (iii) no subgraph containing l has $\omega_{\mathrm{dbl}} \geq 0$; (iv) two proper divergent subgraphs of the same Γ always share a line or a vertex (the triangles of Γ_2 overlap, the swordfish vertices of Γ_7 share the seagull), so the forests of proper subgraphs in the sense of [21, Sect. 2] (members nested or with no common lines and vertices) have at most one element; (v) in class A both photon lines are chords of Γ, with independent loop momenta k_1, k_2, and the scalar lines carry integer combinations of them; in class B one photon line is a chord with momentum k, the other photon line carries $\pm k$, and the second chord is a scalar line of the closed scalar loop, with momentum q.

By (iv), $R' = 1 - \sum_\gamma t_\gamma$ splits according to whether γ contains the doubled line: for every Γ and scalar line l, pointwise,

$$R' F_{\Gamma,\mathrm{dbl}(l)} = R_{\mathrm{dbl}} F_{\Gamma,\mathrm{dbl}(l)} - \sum_{\gamma \ni l,\ \omega(\gamma)\geq 0} \left(t_\gamma^{\omega(\gamma)} F_{\gamma,\mathrm{dbl}(l)}\right) F_{\bar\Gamma_\gamma}, \qquad R_{\mathrm{dbl}} := 1 - \sum_{\gamma \not\ni l,\ \omega(\gamma)\geq 0} t_\gamma^{\omega(\gamma)}, \tag{79}$$

where $F_{\bar\Gamma_\gamma}$ is the integrand of the lines and vertices of Γ outside γ. R_{dbl} is the forest formula of the doubled graph with the labels ω_{dbl} on every one-particle-irreducible subgraph: by (iii) the subgraphs containing l, and Γ itself, have negative label and are not subtracted. The subtracted terms ("removed terms") are polynomials of degree $\leq \omega(\gamma)$ in $P(q^\gamma)$ whose coefficients are integrals over k^γ.

Lemma E.15 (photon factorisation)**.** *Fix Γ, a scalar line l and the photon mass ν, and write $F^{(\nu)}$ for the integrand at that photon mass. In class A, $R_{\mathrm{dbl}} F^{(\nu)}_{\Gamma,\mathrm{dbl}(l)}(k_1,k_2) = D_\nu(k_1) D_\nu(k_2)\, R_{\Gamma,l}(k_1,k_2)$, and each removed term of (79) is $D_\nu(k^\gamma) D_\nu(k^{\bar\Gamma})$ times a ν-free function. In class B, $R_{\mathrm{dbl}} F^{(\nu)}_{\Gamma,\mathrm{dbl}(l)}(k,q) = D_\nu(k)^2 R_{\Gamma,l}(k,q)$, and the same holds for $R' F^{(\nu)}_{\mathrm{dbl}}$ and for the removed term. The functions $R_{\Gamma,l}$ and the cofactors do not depend on ν.*

Proof. Every factor of F_Γ other than the two photon propagators is ν-free: the vertex functions are mass-free trigonometric polynomials and the scalar lines have mass 1. A subtraction operator t_γ Taylor-expands the integrand of γ in γ's external momenta at fixed internal momenta, and γ's internal momentum is its chord momentum, which carries no external momentum of γ [21, (2-16), (2-24)]. By (ii) every subtracted γ contains either exactly one photon line, which is γ's own chord, so that its propagator is a function of γ's internal momentum alone and is untouched by t_γ, or no photon line (type (c)), its photon legs being external to γ, so that the attached photon propagators belong to $\bar\Gamma_\gamma$ and are cofactors. Hence no subtraction operator acts on a photon propagator, and the photon propagators factor out of every term of (79) with the momenta of (v). □

Lemma E.16 (the fixed-mass bound)**.** *There are $a_0 \in (0,1]$ and $K < \infty$ such that, at scalar and photon mass 1, for every Γ and every scalar line l of Γ,*

$$\sum_{|\beta|\le 6} \sup_{0<a\le a_0} \big\langle\big\langle \big|\partial^\beta R_{\mathrm{dbl}} F^{(1)}_{\Gamma,\mathrm{dbl}(l)}\big|\big\rangle\big\rangle_a \le K,$$

∂^β running over the derivatives in the two loop momenta.

Proof. $R_{\mathrm{dbl}}F^{(1)}$ is a forest-subtracted lattice Feynman integrand in the sense of [21]: the propagators are of the form $(\hat e(l)+\mu^2)^{-1}$ with $\mu = 1$, the numerators are trigonometric polynomials of the class of [20, Defs. 2.1–2.5] whose lattice degrees equal their continuum dimensions, the doubled line is a two-point insertion of degree 0, and the labels $\delta(\gamma) = \omega_{\mathrm{dbl}}(\gamma)$ are the degrees of the doubled graph itself, so the conditions (4-3) and (4-4) of [21] hold (remark after (4-4) there), with $t^\delta = 0$ for $\delta < 0$. The chord assignment obeys [21, (2-24)], and the line momenta are natural in the sense of [20, Def. 3.1]: every subtracted subgraph is one-loop, so its internal momentum is, up to sign, the momentum of one of its lines, and the set of line momenta of a connected graph without cut lines is natural ([20], remark after Def. 3.1). By the renormalization theorem of [21, Sects. 4–5], whose proof applies verbatim in three dimensions (the dimension enters only through the degree count; cf. the remark before Def. 2.5 of [20]), the integrand $R_{\mathrm{dbl}}F^{(1)}$ is $\frac{2\pi}{a}$-periodic, lies in the class of [20] and has negative degree on every Zimmermann subspace, the overall degree being $\omega_{\mathrm{dbl}}(\Gamma) = -2$; by [20, Lemmas 2.1–2.2] the derivatives ∂^β stay in the class and do not raise any of these degrees. The three-dimensional form of the renormalization theorem is read off its proof as follows. The dimension d enters that proof at three places only: through the term $d\,m(\gamma)$, $m(\gamma)$ the number of loops, in the divergence degree [21, (2-28), (2-31)]; through the quantities $M_U(\gamma) = d\sum m$ of [21, (5-10)] and the bound $M_U(\Gamma) \ge dh$ of [21, App. C], which is d times an inequality between loop counts and ranks of the coefficient matrices of the loop-momentum parametrisation of the Zimmermann subspace, the same for every d; and through the target inequality [21, (5-5)], $dh + \mathrm{degr}_t R_\Gamma < 0$. The degree bounds (5-11)–(5-13) of [21, Lemma 5.2] are proved by induction over the forest from the lattice degrees of the single propagators and vertex functions, which do not involve the dimension, from the subtraction degrees and from the loop counts, which add under the reduction of a subgraph, the same way as the degrees of (2-28) do; so with the degrees counted in d dimensions and labels satisfying (4-3) and (4-4) in those degrees, every line of the induction holds for every d, and its conclusion in $d = 3$, $3h + \mathrm{degr}_t R_\Gamma < 0$ on every Zimmermann subspace, is the hypothesis (3.9) of the power counting theorem of [20]. Everything else in the proof (the subtraction operators and their degree calculus, the forest algebra, the periodicity of R_Γ) treats the momenta as lists of real variables and does not depend on the dimension. The power counting theorem [20, Thm. 1] applies to each $\partial^\beta R_{\mathrm{dbl}}F^{(1)}$, and its proof bounds the integrand on every sector by non-negative majorants whose integrals over the zone are bounded uniformly in $a \le a_0$ when all degrees are negative ([20, Sect. 4 and Thm. 2]), which is the uniform absolute bound claimed; a_0 and K depend on the masses, here $(1,1)$, and on the finitely many integrands (Γ, l, β). □

Lemma E.17 (derivative bounds at unit mass)**.** *Let $a \le 1$ and $q, k \in \mathbb{R}^3$.*

(i) $\big|\sqrt{\hat e(q+k)} - \sqrt{\hat e(q)}\big| \le |k|$*; hence $C(q+k) \le 2C(q)$ and $\chi(q+k) \le 2\chi(q)$ for $|k| \le \frac12$.*

(ii) $a^2 \le 13/\chi(q)$.

(iii) $|\partial_\mu \hat e| \le 2\sqrt{\hat e}$*, and every derivative of $\hat e$ of order ≥ 2 is bounded by 2. Consequently, for $|k| \le \frac12$ and all multi-indices b, β: $|\partial_q^b \partial_k^\beta C(q+k)| \le C_{b,\beta}\,\chi(q)^{-1-(|b|+|\beta|)/2}$, $|\partial_q^b \partial_k^\beta \tilde V_\mu(q, q+k)| \le C_{b,\beta}\,\chi(q)^{(1-|b|-|\beta|)/2}$, and the derivatives of $\cos\frac{a(2q+k)_\mu}{2}$ and of the phases $e^{\pm \mathrm{i}ak_\mu/2}$ are bounded by $C_{b,\beta}\,\chi(q)^{-(|b|+|\beta|)/2}$: each derivative gains a factor $\chi(q)^{-1/2}$ relative to the bound of the undifferentiated factor ($C(q+k) \le 2\chi(q)^{-1}$, $|\tilde V_\mu(q,q+k)| \le 2\sqrt{\hat e(q)} + |k| \le 3\chi(q)^{1/2}$, $|\cos|, |e^{\mathrm{i}\cdot}| \le 1$).*

(iv) $|P_\rho(k)| \le C\sqrt{\hat e(k)}$ *for all k, and $|P_\rho(k) - k_\rho| \le Ca|k|^2$ for $|k| \le \frac12$.*

Proof. (i) With $v_\mu(q) := \frac{2}{a}\sin\frac{aq_\mu}{2}$, $\sqrt{\hat e} = |v|$ and $|v_\mu(q+k) - v_\mu(q)| \le |k_\mu|$ by $|\sin x - \sin y| \le |x-y|$; the triangle inequality gives the first claim. If $x := \sqrt{\hat e(q)} \ge \frac12$, then $\hat e(q+k) + 1 \ge (x -$

$\frac{1}{2})^2+1 \geq \frac{x^2+1}{2}$ (equivalent to $(x-1)^2+\frac{1}{2} \geq 0$); if $x < \frac{1}{2}$, then $\hat{e}(q+k)+1 \geq 1 \geq \frac{x^2+1}{2}$. Also $\hat{e}(q+k)+1 \leq (x+\frac{1}{2})^2+1 \leq 2x^2+2$. (ii) $\hat{e} \leq 12/a^2$ and $a \leq 1$. (iii) $\partial_\mu \hat{e} = \frac{2}{a}\sin(aq_\mu) = \frac{4}{a}\sin\frac{aq_\mu}{2}\cos\frac{aq_\mu}{2}$, of modulus $\leq 2|v_\mu|$; $\partial_\mu\partial_\nu\hat{e} = 2\cos(aq_\mu)\delta_{\mu\nu}$, and each further derivative multiplies by $a \leq 1$. A derivative of $C = \chi^{-1}$ produces $C^2\partial\hat{e}$ with $|\partial\hat{e}| \leq 2\chi^{1/2}$, a gain of $2\chi^{-1/2}$; a derivative of a factor $\partial\hat{e}$ gives a factor $\leq 2 \leq 2\chi^{1/2}\chi^{-1/2}$. The j-th derivative of $\tilde{V}_\mu(q,q+k) = -\frac{2}{a}\sin\frac{a(2q+k)_\mu}{2}$ is bounded by $2a^{j-1} \leq 2\cdot 13^{(j-1)/2}\chi(q)^{-(j-1)/2}$ for $j \geq 1$ by (ii), and that of $\cos\frac{a(2q+k)_\mu}{2}$ or $e^{\pm iak_\mu/2}$ by $a^j \leq 13^{j/2}\chi(q)^{-j/2}$. For the factor $C(q+k)$ and its derivatives use $\chi(q+k) \geq \frac{1}{2}\chi(q)$ from (i). Induction on $|b|+|\beta|$ gives the stated bounds. (iv) Applying [21, (3-2)] to $F \equiv q_\rho$ gives $P_\rho(0) = 0$ and $\nabla P_\rho(0) = e_\rho$, so $p_\rho(x) = x_\rho + O(|x|^2)$, and by periodicity p_ρ vanishes on $2\pi\mathbb{Z}^3$. Hence $|p_\rho(x)| \leq \|\nabla p_\rho\|_\infty \operatorname{dist}(x, 2\pi\mathbb{Z}^3) \leq \|\nabla p_\rho\|_\infty \frac{\pi}{2}\big(4\sum_\mu \sin^2\frac{x_\mu}{2}\big)^{1/2}$ by lemma E.4(c), and $|P_\rho(k)| = a^{-1}|p_\rho(ak)| \leq C\sqrt{\hat{e}(k)}$; and $|P_\rho(k) - k_\rho| = a^{-1}|p_\rho(ak) - ak_\rho| \leq a^{-1}C|ak|^2 = Ca|k|^2$ for $|ak| \leq \frac{1}{2}$. □

Lemma E.18 (box inequality)**.** (i) *For $g \in C^d$ on a cube $Q_d \subset \mathbb{R}^d$ of unit side,* $\sup_{Q_d}|g| \leq \sum_{S\subset\{1,\dots,d\}}\int_{Q_d}|\partial_S g|$, $\partial_S := \prod_{i\in S}\partial_i$ *($\partial_\emptyset g = g$).* (ii) *For G smooth and $\frac{2\pi}{a}$-periodic in $(k_1,k_2) \in \mathbb{R}^6$,*

$$\sup_{k_2\in Q}\big\langle|G(\cdot,k_2)|\big\rangle_{a,k_1} \leq \sum_{S\subset\{1,2,3\}}\int_Q \big\langle|\partial_S^{k_2}G(\cdot,k_2)|\big\rangle_{a,k_1}\,dk_2, \qquad \sup_{Q^2}|G| \leq \sum_{S\subset\{1,\dots,6\}}\int_{Q^2}|\partial_S G|.$$

Proof. (i) For $x, y \in Q_d$ the fundamental theorem of calculus, applied coordinate by coordinate, gives $g(x) = \sum_S \int_{[y_S,x_S]}(\partial_S g)(t_S, y_{S^c})\,dt_S$, so $|g(x)| \leq \sum_S \int_{[0,1]^{|S|}}|(\partial_S g)(t_S,y_{S^c})|\,dt_S$ for every y; average over $y \in Q_d$. (ii) Apply (i) in k_2 at fixed k_1 and integrate over k_1 (Fubini), respectively apply (i) in the six variables. □

Applied to $G = \chi(k_2)R_{\mathrm{dbl}}F^{(1)}$, respectively $G = \chi(k_1)\chi(k_2)R_{\mathrm{dbl}}F^{(1)}$: on Q, $\chi \leq \frac{7}{4}$, $|\partial_i\chi| = |\frac{2}{a}\sin(ak_i)| \leq 2|k_i| \leq 1$, and the mixed derivatives of χ vanish ($\hat{e}$ is a sum of functions of one coordinate each), so by the Leibniz rule both right sides of (ii) are $\leq CK$, with K of lemma E.16 (derivatives of order ≤ 3, respectively ≤ 6) and a numerical C.

Lemma E.19 (the polarisation remainder)**.** *Let $F_b(q;k) := e^{-ia(k_\mu-k_\nu)/2}\tilde{V}_\mu(q,q+k)\tilde{V}_\nu(q+k,q)C(q)C(q+k)$ be the bubble integrand at unit mass with photon indices μ,ν, or either of its doubled versions ($C(q) \to C(q)^2$ or $C(q+k) \to C(q+k)^2$). For $a \leq 1$ and $|k| \leq \frac{1}{2}$,*

$$\big|\big\langle(1-t_k^1)F_b(\cdot;k)\big\rangle_a\big| \leq C_{\mathrm{VP}}|k|^2,$$

and the left side vanishes at $k = 0$. Moreover $\Pi_{\mathrm{dbl}}(0) := \langle F_{b,\mathrm{dbl}}(\cdot;0)\rangle_a$ obeys $|\Pi_{\mathrm{dbl}}(0)| \leq 4\langle C^2\rangle_a \leq \frac{\pi^2}{16}$ and $\nabla_k\langle F_{b,\mathrm{dbl}}(\cdot;k)\rangle_a\big|_{k=0} = -\frac{ia}{2}(e_\mu - e_\nu)\Pi_{\mathrm{dbl}}(0)$, of modulus $\leq a|\Pi_{\mathrm{dbl}}(0)|$.

Proof. For fixed q, $(1-t_k^1)F_b(q;k) = \big[F_b(q;k) - F_b(q;0) - k\cdot\nabla_k F_b(q;0)\big] + \big(k - P(k)\big)\cdot\nabla_k F_b(q;0)$. The bracket equals $\int_0^1(1-\theta)\,k^{\mathsf{T}}\nabla_k^2 F_b(q;\theta k)\,k\,d\theta$, and by lemma E.17(iii), $|\partial_k^\beta F_b(q;k')| \leq C\chi(q)^{-1-|\beta|/2}$ for $|k'| \leq \frac{1}{2}$ (two vertex factors of size $\chi^{1/2}$, two propagators χ^{-1}, each derivative gaining $\chi^{-1/2}$; the doubled versions are smaller by a factor $2\chi^{-1}$). Hence $\langle|\nabla_k^2 F_b(\cdot;\theta k)|\rangle_a \leq C\langle\chi^{-2}\rangle_a \leq C$ by (78), and the bracket contributes $\leq C|k|^2$. For the second term, $|k - P(k)| \leq Ca|k|^2$ by lemma E.17(iv) and $\langle|\nabla_k F_b(\cdot;0)|\rangle_a \leq C\langle\chi^{-3/2}\rangle_a \leq C(1+\ln\frac{1}{a})$ by (78); and $a(1+\ln\frac{1}{a}) \leq 1$ for $a \leq 1$. At $k = 0$ the remainder vanishes identically in q, since $P(0) = 0$. For the last claims: at $k = 0$ the integrand is $\tilde{V}_\mu(q,q)\tilde{V}_\nu(q,q)C(q)^3$ with $|\tilde{V}_\mu(q,q)| = \frac{2}{a}|\sin(aq_\mu)| \leq 2\sqrt{\hat{e}(q)}$, so $|\Pi_{\mathrm{dbl}}(0)| \leq 4\langle\hat{e}C^3\rangle_a \leq 4\langle C^2\rangle_a$; and $\langle F_{b,\mathrm{dbl}}(\cdot;k)\rangle_a = e^{-ia(k_\mu-k_\nu)/2}T_{\mathrm{dbl}}(k)$ with T_{dbl} even in k (the substitution $q \mapsto -q$ maps the integrand at k to the integrand at $-k$, the doubled line to itself), so $\nabla T_{\mathrm{dbl}}(0) = 0$. □

Proposition E.20 (the unit-mass bound, uniform in the photon mass)**.** *With a_0 of lemma E.16 there is $C_U < \infty$, depending on a_0 only, such that $|Y(\nu,a)| \leq C_U$ for all $\nu > 0$ and $a \in (0,a_0]$.*

Proof. By lemma E.14 at $s = 1$, $Y(\nu,a)$ is a finite sum of terms $c_\Gamma\langle\langle R'F^{(\nu)}_{\Gamma,\mathrm{dbl}(l)}\rangle\rangle_a$. Each is bounded by a constant of the stated kind.

(0) Vanishing terms. Γ_{11} has no scalar line. For Γ_6 and Γ_9 the polarisation subgraph is the seagull loop; the seagull vertex depends on its two photon momenta only through their sum, which is 0 at zero external momentum, so the loop integrand $F_{\mathrm{VP}}(q;k)$ does not depend on k and $(1-t^1_k)F_{\mathrm{VP}} \equiv 0$. Hence $R'F_{\mathrm{dbl}} \equiv 0$ for every l: if l lies on the path, $R' = R_{\mathrm{dbl}}$ subtracts the loop; if l lies in the loop, $R'F_{\mathrm{dbl}} = (1-t^1_k)F_{\mathrm{dbl}} = 0$.

(1) Class A ($\Gamma_0, \Gamma_2, \Gamma_3, \Gamma_5, \Gamma_7, \Gamma_8, \Gamma_{10}$), the R_{dbl} part. By lemma E.15, $|R_{\mathrm{dbl}}F^{(\nu)}| = D_\nu(k_1)D_\nu(k_2)|R_{\Gamma,l}(k_1,k_2)$ $D_\nu(k_1)\chi(k_1)\,D_\nu(k_2)\chi(k_2)\,|R_{\mathrm{dbl}}F^{(1)}|$. Split B^2_a according to whether k_1, k_2 are hard or soft. On hard momenta $D_\nu\chi \le 1+\pi^2$. Hence

$$\langle\langle \mathbf{1}_{hh}|R_{\mathrm{dbl}}F^{(\nu)}|\rangle\rangle_a \le (1+\pi^2)^2\langle\langle|R_{\mathrm{dbl}}F^{(1)}|\rangle\rangle_a \le (1+\pi^2)^2K,$$
$$\langle\langle \mathbf{1}_{hs}|R_{\mathrm{dbl}}F^{(\nu)}|\rangle\rangle_a \le (1+\pi^2)\,\sigma_1 \sup_{k_2\in Q}\big\langle|\chi(k_2)R_{\mathrm{dbl}}F^{(1)}(\cdot,k_2)|\big\rangle_{a,k_1} \le (1+\pi^2)\sigma_1 CK,$$

by lemma E.18(ii) and lemma E.16, and the same for sh with $k_1 \leftrightarrow k_2$; and $\langle\langle \mathbf{1}_{ss}|R_{\mathrm{dbl}}F^{(\nu)}|\rangle\rangle_a \le \sigma_1^2 \sup_{Q^2}|\chi(k_1)\chi(k_2)R_{\mathrm{dbl}}F^{(1)}| \le \sigma_1^2 CK$. With $\sigma_1 \le \frac{1}{16}$ from (78), $|\langle\langle R_{\mathrm{dbl}}F^{(\nu)}\rangle\rangle_a| \le C$.

(2) Class A, the removed terms of (79). Type (d) (Γ_2, Γ_5, Γ_7): the coefficient $\langle F_{\gamma,\mathrm{dbl}}(\cdot;0)\rangle_a$ is the one-loop vertex subgraph with the line l doubled at zero external momenta, and it vanishes by $q \mapsto -q$: for the triangle the integrand is $\sum_\alpha \tilde V_\alpha(0,q)\tilde V_\mu(q,q)\tilde V_\alpha(q,0)\,D_\nu(q)C(q)^3$ up to the two phases of the internal photon, which cancel, with $\tilde V_\mu(q,q) = -\frac{2}{a}\sin(aq_\mu)$ odd and the other factors even; for the swordfish vertex (the seagull with one external photon leg, its other leg joined by the internal photon to a current vertex) the integrand is $(1+e^{-\mathrm{i}aq_\mu})(e^{\mathrm{i}aq_\mu}-1)D_\nu(q)C(q)^2 = 2\mathrm{i}\sin(aq_\mu)D_\nu(q)C(q)^2$, odd. ($\Gamma_8$ has no removed term, its A^3-tadpole containing no scalar line; the subtraction t^0 of that tadpole is 0 anyway, as $V_3 \propto \sin\frac{a(p+p')_\mu}{2}$.) Type (a) ($\Gamma_0$ with l the inner line): $t^1_\gamma F_{\gamma,\mathrm{dbl}} = F_{\gamma,\mathrm{dbl}}(k_2;0) + P(k_1)\cdot\nabla F_{\gamma,\mathrm{dbl}}(k_2;0)$, and the order-1 term is odd in the inner loop momentum k_2 (as in the proof of lemma E.14(a)), so the removed term's double integral is $N_\gamma\, I_{\bar\Gamma}$ with

$$N_\gamma = \big\langle D_\nu(k)\,\hat e(k)\,C(k)^2\big\rangle_a, \qquad I_{\bar\Gamma} = \big\langle D_\nu(k)\,\hat e(k)\,C(k)^2\big\rangle_a,$$

the inner exchange graph with its scalar line doubled at zero external momentum, and the outer exchange graph with the contracted subgraph as a unit vertex between its two scalar lines of equal momentum; here $\sum_\mu \tilde V_\mu(0,-k)\tilde V_\mu(-k,0) = \hat e(k)$. Since $D_\nu\hat e \le 1$, both are $\le \langle C^2\rangle_a \le \frac{\pi^2}{64}$.

(3) Class B (Γ_1, Γ_4). One photon chord k, one polarisation chord q. By lemma E.15 every term is $D_\nu(k)^2$ times a ν-free function, which is the path factor times the subtracted bubble: the path factor is $\tilde V_\mu(0,-k)\tilde V_\nu(-k,0)C(k)$ for Γ_1 (with $C(k)^2$ when l is the path line), bounded entrywise by $\hat e(k)C(k) \le 1$, and a constant for Γ_4 (the seagull at zero scalar momenta and zero total photon momentum). *Soft k:* use $R'F_{\mathrm{dbl}} = D_\nu(k)^2\,\mathrm{path}(k)\,(1-t^1_k)F_{b(,\mathrm{dbl})}(q;k)$ itself (R' subtracts the bubble by definition; for l on the path, $R' = R_{\mathrm{dbl}}$). By lemma E.19 its q-integral is bounded by $C_{\mathrm{VP}}|k|^2|\mathrm{path}(k)|$, and $\langle \mathbf{1}_{Q_s}D_\nu(k)^2|k|^2\rangle_a \le \frac{\pi^2}{64}$ by (78), uniformly in ν. *Hard k:* $R'F_{\mathrm{dbl}} = R_{\mathrm{dbl}}F_{\mathrm{dbl}}$ minus the removed term, which is present only when l lies in the bubble. The R_{dbl} part is $\le (1+\pi^2)^2\langle\langle|R_{\mathrm{dbl}}F^{(1)}|\rangle\rangle_a \le (1+\pi^2)^2K$. The removed part is

$$\big\langle \mathbf{1}_{\mathrm{hard}}\,D_\nu(k)^2\,|\mathrm{path}(k)|\,\big|\Pi_{\mathrm{dbl}}(0)+P(k)\cdot\nabla_k\langle F_{b,\mathrm{dbl}}(\cdot;k)\rangle_a\big|_{k=0}\big|\big\rangle_{a,k} \le (1+\pi^2)^2\,|\Pi_{\mathrm{dbl}}(0)|\,\big\langle D_1^2\big(1+Ca\sqrt{\hat e}\big)\big\rangle_a \le C\Big(1+$$

by lemma E.19, lemma E.17(iv) and (78) ($D_1^2\sqrt{\hat e} \le \chi^{-3/2}$). ($\Gamma_1$ vanishes identically on $\varepsilon\mathbb{Z}^3$ by the transversality noted in lemma E.25(b); the bound covers it regardless.)

Summing (0)–(3) over the finitely many pairs (Γ, l) gives $|Y(\nu,a)| \le C_U$, where C_U collects K and numerical constants, and depends on a_0 only. □

Proof of (v). By lemma E.13, $y_{40}(s;\varepsilon) = s^{-1}Y(\mu_0/\sqrt{s}, \varepsilon\sqrt{s})$. For $\varepsilon \le a_0/\sqrt{s}$ the second argument lies in $(0,a_0]$, and proposition E.20 gives $|y_{40}(s;\varepsilon)| \le C_U/s$ for every $\mu_0 > 0$. So (v) holds with $C_{40} := C_U$, which depends neither on μ_0 nor on ε nor on s. In scaled form, $s\,y_{40}$ is bounded by a function of $\mu_0/\sqrt{s}$ and $\varepsilon\sqrt{s}$ that stays bounded as $\mu_0/\sqrt{s} \to 0$: no $\ln(\sqrt{s}/\mu_0)$ occurs. □

The mechanism of (iv) is the one of lemma E.3: the only coefficient that grows with s, $|\mu_0^2 - 2s|$, multiplies $B_s(p) - B_s(0) = O(|p|^2 s^{-3/2})$, the Taylor step being taken on the scalar

line; the linearly divergent constants cancel exactly in the decomposition (lemma E.11) before any bound. In (v), $y_{40}(s;\varepsilon) = s^{-1}Y(\mu_0/\sqrt{s}, \varepsilon\sqrt{s})$ by scaling (lemma E.13), and Y is bounded uniformly in its first argument and in the rescaled spacing $a = \varepsilon\sqrt{s} \le a_0$ (proposition E.20); the power-counting input enters at one mass configuration only, and the constants C_{40}, a_0 are those of lemma E.16.

Theorem E.21 (k-uniform lower bound)**.** *Let $e, \lambda \ge 0$, $\mu_0, m > 0$, $M \ge m$ and $\varepsilon_k \le a_0/M$. Then*

$$2u_k(M) \;\ge\; (M^2-m^2) - \frac{\pi^2}{16}e^2(M-m) - 2\Big[C_{40}e^4 + C_{21}(m^2)\lambda e^2 + \frac{81\pi^4}{32}\lambda^2\Big]\ln\frac{M}{m} - 2C'_{21}\lambda e^2\varepsilon_k(M-m).$$

Consequently $s_-(M) \ge \frac{1}{2}\Phi_A(M)$ with $\Phi_A(M) := (M^2 - m^2) - \frac{\pi^2}{16}e^2(M-m) - 2[C_{40}e^4 + C_{21}(m^2)\lambda e^2 + \frac{81\pi^4}{32}\lambda^2]\ln\frac{M}{m} \to \infty$ as $M \to \infty$; this is (NEG_∞)*, and $s_-(M_2) > 0$ at every M_2 with $\Phi_A(M_2) > 0$.*

Proof. Insert lemmas E.8 and E.9 into (75): $y_{01} > 0$ drops the λ term; $\int_{m^2}^{M^2} |y_{20}|\,ds \le \int \frac{\pi^2}{32\sqrt{s}}\,ds = \frac{\pi^2}{16}(M-m)$; $\int |y_{02}| \le \frac{81\pi^4}{32}\ln\frac{M^2}{m^2}$; $\int |y_{21}| \le C_{21}(m^2)\ln\frac{M^2}{m^2} + 2C'_{21}\varepsilon(M-m)$, since $C_{21}(s) \le C_{21}(m^2)$ for $s \ge m^2$; $\int |y_{40}| \le C_{40}\ln\frac{M^2}{m^2}$, since $\varepsilon \le a_0/M \le a_0/\sqrt{s}$ on $[m^2, M^2]$. Finally $\ln\frac{M^2}{m^2} = 2\ln\frac{M}{m}$, and the lim inf drops the ε_k term. □

The constants of theorem E.21 are crude, and C_{40}, a_0 are not explicit. The exact continuum limit, computed next, gives explicit and essentially sharp constants.

Continuum conventions. $\int_k := \int_{\mathbb{R}^3}\frac{d^3k}{(2\pi)^3}$. Complex field $\phi_c = (\phi_1 + \mathrm{i}\phi_2)/\sqrt{2}$, so that $\frac{1}{2}|D_A\phi|^2 = |D_A\phi_c|^2$ and $\lambda|\phi|^4 = 4\lambda|\phi_c|^4$. Lines $G(q) = (q^2+M^2)^{-1}$, $D(k) = (k^2+\mu^2)^{-1}\delta_{\mu\nu}$ with $\mu := \mu_0$. Vertices (coefficients in the action): current $-e(p+p')_\mu$ (p in, p' out); seagull $2e^2\delta_{\mu\nu}$; quartic 16λ; counterterm δm^2. Σ is the one-particle-irreducible self-energy, and King's rule is $\delta m^2{}_{(\alpha\beta)} = \Sigma^{(\alpha\beta)}_{\mathrm{int}}(0)$. One-loop integrals: $I := \int_k DG = \frac{1}{4\pi(M+\mu)}$; $J := \int_k Dq^2G^2 = \frac{M+2\mu}{8\pi(M+\mu)^2}$; $\int G^2 = \frac{1}{8\pi M}$; $B^{\mathrm{c}}_{ab}(q) = \int_k G_a(k)G_b(q+k) = \frac{\arctan(|q|/(a+b))}{4\pi|q|}$ (lemma E.5(i)). Two-loop vacuum integrals with lines $q, k, q+k$ of masses a, b, c and $\mathsf{s} := a+b+c$: $V(a,b,c) := \iint G_a(q)G_b(k)G_c(q+k)$, of which only differences are used, $V(a,b,c) - V(a',b',c') = -\frac{1}{16\pi^2}\ln\frac{\mathsf{s}}{\mathsf{s}'}$ (absolutely convergent in momentum space, and equal to the position-space Frullani integral $\frac{1}{16\pi^2}\int_0^\infty (e^{-\mathsf{s}r} - e^{-\mathsf{s}'r})\frac{dr}{r}$ by Plancherel, since $\tilde{G}_a(x) = \frac{e^{-a|x|}}{4\pi|x|}$); and $V_2(a;b,c) := \iint G_a(q)^2G_b(k)G_c(q+k) = \frac{1}{32\pi^2 a\,\mathsf{s}}$, because $\widetilde{(G_a^2)}(x) = \frac{e^{-a|x|}}{8\pi a}$. Write $V(\mathsf{s})$ for V with mass sum s; only $V_1 := V(2\mu+M)$ and $V_2 := V(\mu+2M)$ occur. Two one-loop functions enter:

$$\sigma(q) := \Sigma^{(20)}_{\mathrm{ren}}(q)/e^2 = 2q^2B^{\mathrm{c}}_{M\mu}(q) + (\mu^2 - 2M^2)\big(B^{\mathrm{c}}_{M\mu}(q) - B^{\mathrm{c}}_{M\mu}(0)\big) \;=\; 2u_{M,\mu}(q) + \tfrac{|q|}{4}, \tag{80}$$

$$\Pi_{\mu\mu}(k) = -e^2\big[k^2B^{\mathrm{c}}_{MM}(k) + 4M^2\big(B^{\mathrm{c}}_{MM}(k) - B^{\mathrm{c}}_{MM}(0)\big)\big], \tag{81}$$

the continuum renormalised order-e^2 self-energy (the continuum analogue after lemma E.3, with u as in theorem E.7) and the trace of the transverse one-loop vacuum polarisation. For (81): the current loop is $e^2\int(2l+k)^2GG' = e^2[4T - (k^2+4M^2)B^{\mathrm{c}}_{MM}]$ with $T := \int G$, by $(2l+k)^2 = 2G^{-1} + 2G'^{-1} - k^2 - 4M^2$; the seagull loop is $-6e^2T$, k-independent. On the lattice the full one-loop polarisation vanishes at $k = 0$ exactly (the lattice Ward identity of King's gauge-invariant action), so the continuum limit is the subtracted form $\Pi^{\mathrm{cur}}(k) - \Pi^{\mathrm{cur}}(0)$; since $4\int l^2G^2 = 4T - 4M^2B^{\mathrm{c}}_{MM}(0)$, T cancels identically and (81) follows, with no regularisation. Check: $\Pi_{\mu\mu}(k) = -\frac{e^2k^2}{12\pi M} + O(k^4)$.

Lemma E.22 (the one-loop limits)**.** *For all $M, m > 0$, as $\varepsilon \to 0$: $d_{01}(M) - d_{01}(m) \to F_{01}(M) - F_{01}(m)$ and $d_{20}(M) - d_{20}(m) \to F_{20}(M) - F_{20}(m)$, where*

$$F_{01}(M) = \frac{4M}{\pi}, \qquad F_{20}(M) = -\frac{M^2}{4\pi(M+\mu)}, \qquad D_{20} := F_{20}(M) - F_{20}(m) = -\frac{(M-m)(Mm+M\mu+m\mu)}{4\pi(M+\mu)(m+\mu)} \in$$

Proof. $d_{01}(M)-d_{01}(m) = 16(M^2-m^2)\langle C_M C_m\rangle$ and $d_{20}(M)-d_{20}(m) = -(M^2-m^2)\langle \hat{q}^2 D C_M C_m\rangle$. On B_ε, $\hat{q}^2 \geq \frac{4}{\pi^2}q^2$ and $\hat{q}^2 D \leq 1$, so both integrands are dominated by $(\frac{\pi^2}{4})^2\big((q^2+M^2)(q^2+m^2)\big)^{-1}$, integrable on $\mathbb{R}^3$, and converge pointwise. Dominated convergence: $16(M^2-m^2)\int G_M G_m = \frac{16(M-m)}{4\pi}$, and $(M^2-m^2)\int k^2 D G_M G_m = \frac{(M-m)(Mm+M\mu+m\mu)}{4\pi(M+\mu)(m+\mu)}$ via $\frac{k^2}{(k^2+a^2)(k^2+b^2)} = \frac{1}{b^2-a^2}\big[\frac{b^2}{k^2+b^2} - \frac{a^2}{k^2+a^2}\big]$, which is $F_{20}(M)-F_{20}(m)$. The bracket follows from $D_{20} = -\frac{M-m}{4\pi}\big(1-\frac{\mu^2}{(M+\mu)(m+\mu)}\big)$. □

Lemma E.23 (the sunset)**.** *For all $M, m > 0$, $d_{02}(M) - d_{02}(m) \to D_{02} := F_{02}(M) - F_{02}(m)$ with $F_{02}(M) = -\frac{8}{\pi^2}\ln M$; the M^2-derivative is $y_{02}^\infty(s) = -\frac{4}{\pi^2 s}$.*

Proof. For each $s > 0$, $y_{02}(s;\varepsilon) \to -384\iint G(q)^2 G(r) G(q+r) = -384\, V_2(M;M,M) = -\frac{384}{96\pi^2 M^2} = -\frac{4}{\pi^2 s}$: the lattice integrand, extended by zero outside B_ε^2, is dominated by the integrable function $(\frac{\pi^2}{4})^4 \sum_P \tilde{G}(q)^2\tilde{G}(r)\tilde{G}(q+r-P)$ of the proof of lemma E.8(iii) and converges pointwise (for fixed (q,r) eventually $q+r \in \mathsf{B}_\varepsilon$, so $P=0$). By lemma E.8(iii), $|y_{02}(s;\varepsilon)| \leq \frac{81\pi^4}{32 s}$ uniformly in ε, so dominated convergence in s on $[m^2, M^2]$ (or $[M^2, m^2]$) in (75) gives $d_{02}(M) - d_{02}(m) \to \int_{m^2}^{M^2}(-\frac{4}{\pi^2 s})\, ds = -\frac{8}{\pi^2}\ln\frac{M}{m}$. □

Lemma E.24 (the mixed order)**.** *For all $M, m > 0$, $d_{21}(M) - d_{21}(m) \to D_{21} := F_{21}(M) - F_{21}(m)$ with*

$$F_{21}(M) = \frac{2}{\pi^2}\ln(2M+\mu) + \frac{\mu}{2\pi^2(M+\mu)}, \qquad y_{21}^\infty(s) = \frac{dF_{21}}{dM^2} = \frac{8M^2+14M\mu+7\mu^2}{4\pi^2 M(M+\mu)^2(2M+\mu)} > 0.$$

In particular $D_{21} \geq 0$ for $M \geq m$.

Proof. Limit. $d_{21} = -8X_\varepsilon(M;\mu)$, and theorem E.7 with $t = \varepsilon$ at the masses (M,μ) and (m,μ) gives $d_{21}(M) - d_{21}(m) \to -8\big[\kappa_{\mathrm{c}}(M,\mu) - \kappa_{\mathrm{c}}(m,\mu)\big]$, the $\ln\frac{1}{\varepsilon}$ terms and κ_{lat} cancelling. No input beyond theorem E.7 is used. *Value.* By (80), $u_{M,\mu} = \frac{1}{2}\sigma - \frac{|q|}{8}$, so $\kappa_{\mathrm{c}}(M,\mu) = 2\int_q G^2\sigma - \frac{1}{2\pi^3}\int \frac{|q|/8}{(q^2+M^2)^2} d^3q - \frac{\ln M+\frac{1}{2}}{4\pi^2}$ in the sense of M^2-derivatives, each of which is absolutely convergent; the derivative of the last two terms is $\frac{1}{8\pi^2 M^2} - \frac{1}{8\pi^2 M^2} = 0$. So, up to an M-independent constant, $-8\kappa_{\mathrm{c}} = -16\int G^2\sigma$. Partial fractions in q^2, with $q^2 G^2 = G - M^2 G^2$: $\int G^2\sigma = 2\big[V(M,\mu,M) - M^2 V_2(M;\mu,M)\big] + (\mu^2 - 2M^2)\big[V_2(M;\mu,M) - \frac{B^{\mathrm{c}}_{M\mu}(0)}{8\pi M}\big]$, with $V(M,\mu,M) = V_2$ and $V_2(M;\mu,M) = \frac{1}{32\pi^2 M(2M+\mu)}$. Hence

$$-16\int G^2\sigma = -32V_2 + \frac{1}{2\pi^2}\Big[\frac{4M^2-\mu^2}{M(2M+\mu)} - \frac{2M^2-\mu^2}{M(M+\mu)}\Big] = \frac{2}{\pi^2}\ln(2M+\mu) + \frac{\mu}{2\pi^2(M+\mu)} + \text{const},$$

since the bracket equals $\frac{\mu}{M+\mu}$. Each identity holds between the M^2-derivatives, which are absolutely convergent (Fubini). Differentiating F_{21} gives y_{21}^∞; its numerator is positive. □

Lemma E.25 (the order e^4)**.** *For all $M, m > 0$, $d_{40}(M) - d_{40}(m) \to D_{40} := F_{40}(M) - F_{40}(m)$, where*

$$F_{40}(M) = -\frac{1}{4\pi^2}\ln M + \Phi_{40}\Big(\frac{\mu}{M}\Big),$$

$$\Phi_{40}(x) = \frac{-7+4x+8x^2-3x^3-3x^4+x^5}{32\pi^2(1+x)(1-x^2)^2} + \frac{(5-2x^2)\ln(2+x) - (9-10x^2+4x^4)\ln(1+2x)}{16\pi^2(1-x^2)^2},$$

continuous on $[0,\infty)$ (also at $x=1$), with $\Phi_{40}(0) = \frac{10\ln 2-7}{32\pi^2}$ and $\Phi_{40}(x) = -\frac{1}{4\pi^2}\ln x + O(1)$ as $x \to \infty$. The M^2-derivative is $y_{40}^\infty = -\frac{1}{8\pi^2 M^2} - \frac{\mu}{2M^3}\Phi'_{40}(\frac{\mu}{M})$.

Proof. (a) Existence of the limit. Fix $s > 0$. The integrand of $y_{40}(s;\varepsilon)$ is a sum of two-loop lattice Feynman integrands of the class treated in [20, 21]: standard lattice propagators with positive masses, trigonometric-polynomial numerators from the vertices of King's action (the current vertex V_1, the seagull V_2, and the three- and four-photon vertices V_3, V_4 of the lattice hopping term), and the insertion of d_{20} on the exchange line, which is exactly the zero-momentum Taylor subtraction of the order-e^2 two-point subgraph. The remaining divergent

subgraphs are the logarithmically divergent one-loop vertex subgraphs (triangle and swordfish); their zero-momentum Taylor terms vanish identically, each being odd under the reflection of its loop momentum by the oddness of the current vertex, with the two phases of the internal photon cancelling, so these subtractions may be inserted without changing the integrand. The integrand is then the subtracted integrand of the forest formula of [21], whose lattice degree of divergence, after the derivative in s, is negative on every Zimmermann subspace (the degree count of [20] in three dimensions, to which it extends as remarked there), the loop momenta being routed along the photon chords so that the set of subtracted subgraphs is natural in the sense of [20]. The power-counting theorem of [20] gives the convergence of $y_{40}(s;\varepsilon)$ to the integral over $\mathbb{R}^6$ of the pointwise limit of the subtracted integrand. In that limit the lattice-only structures vanish: V_3 and V_4 carry ε and ε^2 relative to V_1 (their photon tadpoles are $O(\varepsilon^{-1})$, so the dressed vertices are still $O(\varepsilon)$), the cosine factors of the seagull tend to 1, and the phases tend to 1. So $y_{40}^{\infty}(s)$ is the s-derivative of King's continuum graph classes with the subtractions above, and by lemma E.9(v), for $\varepsilon \le a_0/\max(M,m)$, $|y_{40}(s;\varepsilon)| \le C_{40}/s$ on the interval between m^2 and M^2, so dominated convergence in (75) gives $d_{40}(M) - d_{40}(m) \to \int_{m^2}^{M^2} y_{40}^{\infty}(s)\,ds$.

(b) The classes. The twelve two-loop one-particle-irreducible two-point topologies of order e^4 fall into the continuum classes of the following table (values per e^4, up to M-independent constants; $k_3 = k_1 + k_2$, $G_i = G(k_i)$, $D_i = D(k_i)$):

topology	class	value
rainbow, and seagull tadpole on the inner line, d_{20} inserted	R	$\int D(q)q^2G(q)^2\sigma(q)$
crossed	X	$V_1 - M^2I^2$
seagull at an end, both photons to the currents (two mirror graphs)	SV_2	$-2[(M^2-\mu^2)I^2 + 3V_1 + (\mu^2 - 4M$
two-photon seagull sunset (melon)	Me	$6V_1$
seagull tadpole with the one-loop polarisation inserted	SS	$V_2 - \frac{1}{32\pi^2}$
exchange with the polarisation inserted		0
seagull in the middle		0
V_3-tadpole vertex		0
V_4 double tadpole		M-independent

Here $W_2 := \iint D_1D_2G_2G_3 = \frac{V_1-V_2}{M^2-\mu^2}$, by $DG = \frac{D-G}{M^2-\mu^2}$. The last four rows: the polarisation insertions on the exchange line vanish identically on $\varepsilon\mathbb{Z}^3$, because the lattice current vertex at zero scalar momentum is $-\mathrm{i}\hat{k}'_\mu(k)$ and $\hat{k}'_\mu\Pi_{\mu\nu}(k) = 0$ (the lattice Ward identity; in the continuum $k_\mu\Pi_{\mu\nu} = 0$). The seagull in the middle factorises, at zero external momentum, into $(\int D(k)k_\mu G(k))^2$, a product of two odd integrals, so 0 in the continuum, and is $O(\varepsilon)$ on the lattice; the V_3 class is $O(\varepsilon)$ as said in (a); the V_4 double tadpole has no scalar line, so its s-derivative vanishes. The inner seagull tadpole of the second graph of the first row is momentum dependent on the lattice only at $O(\varepsilon q^2)$; its sum with the rainbow and the counterterm is the lattice renormalised self-energy, whose continuum limit is σ.

(c) The class values (continuum, zero external momentum; signs and combinatorics from the Feynman rules above). $\mathrm{X} = e^4\iint D_1D_2G_1G_2G_3\,[k_1\cdot(k_1+2k_2)][k_2\cdot(2k_1+k_2)]$. The exact identities $k_1\cdot(k_1+2k_2) = k_3^2 - k_2^2 = G_3^{-1} - G_2^{-1}$ and $k_2\cdot(2k_1+k_2) = G_3^{-1} - G_1^{-1}$ turn the integrand into $D_1D_2[G_1G_2G_3^{-1} - G_1 - G_2 + G_3]$ with $G_3^{-1} = k_1^2 + k_2^2 + 2k_1\cdot k_2 + M^2$; the products factorise, the $k_1\cdot k_2$ term is odd, and with $\int Dk^2G = T_\mu - M^2I$ (pointwise $DGk^2 = D - M^2DG$, $T_\mu := \int D$) the T_μ terms cancel identically: $\mathrm{X}/e^4 = 2I(T_\mu - M^2I) + M^2I^2 - 2IT_\mu + V_1 = V_1 - M^2I^2$. SV_2 (both mirror graphs): $2\mathrm{SV} = -4e^4\iint D_1D_2G_2G_3\,(k_1+2k_2)\cdot k_2$; with $(k_1+2k_2)\cdot k_2 = \frac{1}{2}(G_3^{-1} + 3G_2^{-1} - D_1^{-1} - 4M^2 + \mu^2)$ the four terms give $T_\mu I$, $3V_1$, $-IT_M$ ($T_M := \int G$) and $(\mu^2 - 4M^2)W_2$, and $T_\mu - T_M = (M^2-\mu^2)I$. $\mathrm{Me} = \frac{1}{2}(2e^2)^2\delta_{\mu\nu}\delta_{\mu\nu}\iint DDG = 6e^4V_1$; its M^2-derivative is $-6e^4\iint D_1D_2G_3^2 = -\frac{6e^4}{32\pi^2M(M+2\mu)}$. $\mathrm{SS} = -e^2\int D(k)^2\Pi_{\mu\mu}(k) = e^4\int D^2[k^2B^{\mathrm{c}}_{MM} + 4M^2(B^{\mathrm{c}}_{MM} - B^{\mathrm{c}}_{MM}(0))]$; with $k^2 = D^{-1} - \mu^2$ and $\int D^2 = \frac{1}{8\pi\mu}$: $\mathrm{SS}/e^4 = V_2 - \mu^2V_2(\mu;M,M) + 4M^2\big[V_2(\mu;M,M) - \frac{1}{8\pi M}\cdot\frac{1}{8\pi\mu}\big] = V_2 + \frac{2M-\mu}{32\pi^2\mu} - \frac{M}{16\pi^2\mu} = V_2 - \frac{1}{32\pi^2}$; the $1/\mu$ terms cancel. $\mathrm{R} = e^4\int Dq^2G^2\sigma = e^4\big[2\int Dq^4G^2B^{\mathrm{c}} + (\mu^2 - 2M^2)\int Dq^2G^2(B^{\mathrm{c}} - B^{\mathrm{c}}(0))\big]$, $B^{\mathrm{c}} = B^{\mathrm{c}}_{M\mu}$. Partial

fractions: $\frac{x^2}{(x+\mu^2)(x+M^2)^2} = \frac{a}{x+\mu^2} + \frac{1-a}{x+M^2} + \frac{c}{(x+M^2)^2}$ with $a = \frac{\mu^4}{(M^2-\mu^2)^2}$, $c = -\frac{M^4}{M^2-\mu^2}$, and $\frac{x}{(x+\mu^2)(x+M^2)^2} = \frac{A}{x+\mu^2} - \frac{A}{x+M^2} + \frac{C'}{(x+M^2)^2}$ with $A = -\frac{\mu^2}{(M^2-\mu^2)^2}$, $C' = \frac{M^2}{M^2-\mu^2}$. So

$$\mathrm{R}/e^4 = 2\big[aV_1+(1-a)V_2+c\,V_{2M}\big]+(\mu^2-2M^2)\big[A(V_1-V_2)+C'V_{2M}-JI\big], \qquad V_{2M} := V_2(M;\mu,M) = \frac{1}{32\pi^2M(2M}$$

No linear divergence is cancelled between the pieces: σ is convergent.

(d) Why these manipulations give the subtracted limit of (a). R, SS and Me use only the convergent functions σ and Π, partial fractions and V-differences; their M^2-derivatives are absolutely convergent integrals, and the identities hold between them. X and SV_2 contain a logarithmically divergent vertex subgraph. For these use the product cutoff $|k_1|, |k_2| < \Lambda$ in the photon-chord routing. Every subtracted subgraph's loop momentum is then a cutoff variable, and its zero-momentum Taylor term is odd in that variable, so it integrates to zero on the symmetric ball; hence the product-cutoff integral of the unsubtracted class equals that of its subtracted integrand, which converges to the subtracted value by dominated convergence. The reduction of X uses no shift of integration variables and is exact at every Λ. The term $\iint D_2G_2G(k_1+k_2)$ of SV_2 needs one shift of the linearly divergent k_1-subintegral; the surface term $E_\Lambda = \int_{|k_2|<\Lambda} D_2G_2\,\delta_\Lambda(k_2)$, $\delta_\Lambda(k_2) := \int_{|k_1|<\Lambda}[G(k_1)-G(k_1+k_2)]$, has $dE_\Lambda/dM^2 \to 0$, since $|\delta_\Lambda(k)| \le C|k|^2/\Lambda$ for $|k| \le \Lambda/2$ and $\le C\Lambda$ otherwise, and the M^2-derivative brings one more factor G. Hence the M^2-derivatives of X and SV_2 are as displayed.

(e) The sum. $F_{40} = \mathrm{X}+\mathrm{Me}+\mathrm{SV}_2+\mathrm{SS}+\mathrm{R}$. The coefficients of the logarithmically divergent V's are $1+6-6+1+2=4$, so F_{40} carries $\frac{4}{16\pi^2}\ln\frac{1}{\varepsilon} = \frac{1}{4\pi^2}\ln\frac{1}{\varepsilon}$, the universal two-loop coefficient of order e^4; the M-dependence is finite. Substituting the closed forms of V_1-V_2, W_2, I, J, V_{2M} and $V_2(\mu;M,M)$ and simplifying gives $F_{40}(M,\mu) = -\frac{1}{4\pi^2}\ln M + \Phi_{40}(\mu/M)$ as an exact identity between elementary functions; Φ_{40} is continuous at $x=1$ (the apparent poles of the two terms cancel), $\Phi_{40}(0) = \frac{10\ln 2-7}{32\pi^2}$, and $\Phi_{40}(x) = -\frac{1}{4\pi^2}\ln x + O(1)$ as $x\to\infty$ (decoupling). At $\mu\to 0$, $F_{40}(M)-F_{40}(m) \to -\frac{1}{4\pi^2}\ln\frac{M}{m}$ exactly: the massless-photon limit is infrared finite. The derivative formula is $\frac{d}{dM^2}F_{40} = \frac{1}{2M}\frac{d}{dM}F_{40}$. □

Theorem E.26 (the exact continuum limit of the bare coefficient)**.** *For all $M, m > 0$ the limit $u_\infty(M) := \lim_k u_k(M)$ exists, and*

$$2u_\infty(M) = (M^2-m^2)+\lambda D_{01}+e^2D_{20}+e^4D_{40}+\lambda e^2D_{21}+\lambda^2D_{02}, \qquad D_{\alpha\beta} := F_{\alpha\beta}(M)-F_{\alpha\beta}(m),$$

with F_{01}, F_{20} of lemma E.22, F_{02} of lemma E.23, F_{21} of lemma E.24 and F_{40} of lemma E.25. The M^2-derivatives are

$$y_{01} = \frac{2}{\pi M}, \quad y_{20} = -\frac{M+2\mu}{8\pi(M+\mu)^2}, \quad y_{02} = -\frac{4}{\pi^2M^2}, \quad y_{21} = \frac{8M^2+14M\mu+7\mu^2}{4\pi^2M(M+\mu)^2(2M+\mu)} > 0, \quad y_{40} = -\frac{1}{8\pi^2M^2}$$

and for large M, $2u_\infty(M) = M^2 + \big(\frac{4\lambda}{\pi} - \frac{e^2}{4\pi}\big)M - \big(\frac{e^4}{4\pi^2} - \frac{2\lambda e^2}{\pi^2} + \frac{8\lambda^2}{\pi^2}\big)\ln M + O(1)$; the coefficient of $\ln M$ is minus the coefficient of $\ln\frac{1}{\varepsilon}$ in the two-loop counterterm.

Proof. Lemmas E.22 to E.25 in (75) at $\varepsilon = \varepsilon_k$. □

Lemma E.27 (monotonicity of Φ_{40})**.** *$\Phi_{40}'(x) < 0$ for every $x \ge 0$. Consequently, for $M \ge m$, $D_{40} \ge -\frac{1}{4\pi^2}\ln\frac{M}{m}$; and $y_{40} > -\frac{1}{8\pi^2M^2}$ for all $M, \mu > 0$.*

Proof. (i) *Derivative.* Differentiating the closed form,

$$\Phi_{40}'(x) = \frac{N(x)}{32\pi^2(1-x^2)^3(1+x)^2(2+x)(1+2x)}, \qquad N(x) = (x^2-1)Q(x) + P(x)\ln\frac{1+2x}{2+x},$$

with $Q(x) = 16x^7+56x^6+24x^5-100x^4-83x^3+82x^2+109x+40$ and $P(x) = 8x(x-2)(x+1)^2(x+2)^2(2x+1)$. The denominator has the sign of $1-x$, so $\Phi_{40}' < 0$ iff $\operatorname{sign} N(x) = \operatorname{sign}(x-1)$. (ii) *Substitution* $x = \frac{1+t}{1-t}$, $t \in [-1,1)$. Then $\ln\frac{1+2x}{2+x} = \ln\frac{3+t}{3-t} = 2\operatorname{artanh}\frac{t}{3}$, and $\tilde N(t) := N(x(t))(1-t)^9 = A(t) + 2B(t)\operatorname{artanh}\frac{t}{3}$ with

$$A(t) = 576t-2496t^2+13264t^3-3056t^4+224t^5-544t^6+272t^7-48t^8, \qquad B(t) = 32(t-3)^2(t-1)^2(t+1)(t+3)(3t-1$$

Since $(1-t)^9 > 0$ and $\operatorname{sign}(x-1) = \operatorname{sign} t$, the claim is $\tilde N(t)/t^3 > 0$ on $[-1,1]$. (iii) *Exact series.* With $2\operatorname{artanh}\frac{t}{3} = \sum_{j\ge 0}\frac{2t^{2j+1}}{3^{2j+1}(2j+1)}$ truncated at $j=30$, the coefficients of t^0, t^1, t^2 of $\tilde N$ vanish exactly, and $\tilde N(t)/t^3 = S(t) + E(t)$, where S is a polynomial with constant term $c_0 = \frac{34160}{3}$ and $\sum_{k\ge 1}|c_k| \le 9103.93$ (exact rational arithmetic), and the tail obeys $|E(t)| \le 2\max_{|t|\le 1}|B| \cdot \frac{(1/3)^{63}}{63\cdot(8/9)} \le 4.9\cdot 10^{-28}$, with $\max|B| \le 15808$ (the sum of the absolute values of its coefficients). Hence $\tilde N(t)/t^3 \ge c_0 - 9103.93 - 4.9\cdot 10^{-28} > 2282 > 0$ on $[-1,1]$. (iv) *Uniform formula.* With $x^2 - 1 = \frac{4t}{(1-t)^2}$, $(x^2-1)^3 = \frac{64t^3}{(1-t)^6}$, so for $t \ne 0$

$$\Phi'_{40}(x(t)) = -\frac{\tilde N(t)/t^3}{2048\pi^2(1-t)^3(1+x)^2(2+x)(1+2x)},$$

and at $t = 0$ ($x = 1$) by continuity $\Phi'_{40}(1) = -\frac{c_0}{2048\pi^2\cdot 36} \approx -0.0157$. The denominator is positive on $[-1,1)$ and $\tilde N/t^3 > 2282$. *Consequences.* For $M \ge m$, $\mu/M \le \mu/m$, so $\Phi_{40}(\mu/M) \ge \Phi_{40}(\mu/m)$, which is $D_{40} \ge -\frac{1}{4\pi^2}\ln\frac{M}{m}$; and $y_{40} > -\frac{1}{8\pi^2 M^2}$ by the derivative formula of lemma E.25. □

Theorem E.28 (the explicit lower bound)**.** *For every $M \ge m$ and every $\mu_0 > 0$,*

$$2u_\infty(M) \;\ge\; (M-m)\Big(M + m + \frac{4\lambda}{\pi} - \frac{e^2}{4\pi}\Big) - \Big(\frac{e^4}{4\pi^2} + \frac{8\lambda^2}{\pi^2}\Big)\ln\frac{M}{m}.$$

With $M_2 := \max\{m,\ \frac{e^2}{4\pi} + (\frac{e^4}{4\pi^2} + \frac{8\lambda^2}{\pi^2})/m\} + 1$: $s_-(M_2) = u_\infty(M_2) \ge \frac12$, and $s_-(M) \ge \frac12(M-m)(M-M_2+1) \to \infty$ for $M \ge M_2$.

Proof. Term by term in theorem E.26, for $M \ge m$: $\lambda D_{01} = \frac{4\lambda}{\pi}(M-m)$; $e^2 D_{20} \ge -\frac{e^2}{4\pi}(M-m)$ (lemma E.22); $e^4 D_{40} \ge -\frac{e^4}{4\pi^2}\ln\frac{M}{m}$ (lemma E.27); $\lambda e^2 D_{21} \ge 0$ (lemma E.24); $\lambda^2 D_{02} = -\frac{8\lambda^2}{\pi^2}\ln\frac{M}{m}$. This is the bound. With $\ln\frac{M}{m} \le \frac{M-m}{m}$, $2u_\infty(M) \ge (M-m)\big[M + m + \frac{4\lambda}{\pi} - \frac{e^2}{4\pi} - (\frac{e^4}{4\pi^2} + \frac{8\lambda^2}{\pi^2})/m\big] \ge (M-m)(M-M_2+1)$ for $M \ge M_2$; at $M = M_2$ this is $\ge M_2 - m \ge 1$. Finally $s_-(M) = u_\infty(M)$, since the limit exists. □

The "+1" and the "$\frac12$" are in the mass unit of m, e^2, λ; replacing $+1$ by $+\delta$ gives $u_\infty(M_2(\delta)) \ge \delta^2/2$ by the same proof.

Proof of proposition 6.9. Theorem E.26 gives the existence of $u_\infty(M)$ for every $M > 0$, theorem E.28 the displayed inequality, the divergence $s_-(M) = u_\infty(M) \to \infty$ and the value $s_-(M_2) \ge \frac12$ at the stated M_2. The statement holds at every point of King's range, $e, \lambda \ge 0$ and $m, \mu_0 > 0$, with no exceptional set; theorem E.21 gives (NEG_∞) independently of the closed forms. In particular m is never a global maximiser of $M \mapsto \lim_k[\alpha_k(M) - \alpha_k(m)]$. At strong coupling $u_\infty(M)$ can be negative for M up to $O(\frac{e^2}{4\pi} + \frac{e^4}{4\pi^2 m})$ and is positive beyond. □

Remark E.29. Theorem E.26 holds for every $M > 0$, on either side of m. Hence for every auxiliary mass $M_1 > 0$ the quantity $\sigma_1(M_1) = \liminf_k \frac12(\alpha_k(m) - \alpha_k(M_1))$ of lemma 3.9 is the limit $-u_\infty(M_1)$, finite and explicit; its one-loop part is the formula of lemma 22.1(b)(ii), and its two-loop part is $-\frac12[e^4 D_{40} + \lambda e^2 D_{21} + \lambda^2 D_{02}]$ with the closed forms above. The threshold $s_* = \frac12\sup_M(\alpha_\infty(m) - \alpha_\infty(M))$ of (R_W) is thereby explicit as well.

Appendix F. Bałaban's torus estimates at block factor 5

This appendix proves the three statements about King's torus model that the Wick leg takes from the torus analysis: the stability bounds proposition 3.8 in King's normalisation at the two masses m and M_1; the c-number lemma lemma 3.9; and the fixed-spacing volume limit lemma 12.9, together with the boundedness clause used in lemma 12.10. The first is Bałaban's theorem, [1, (1.14)] and [2, (1.3)], read at block factor 5 and in King's convention (appendix F.2); the other two are statements about finite lattice Feynman integrals, proved from Reisz's power counting (appendices F.3 and F.4).

Throughout, k is the spacing index ($\varepsilon = \varepsilon_k = 5^{-k}$), T a King torus of side $\ell \in M\mathbb{N}$ (definition 12.1), $N = 2$ the number of real scalar components, and $L = 5$ wherever Bałaban's

letter is kept. The dual lattice of T is $T^* = (2\pi/\ell)\mathbb{Z}^3$ modulo $(2\pi/\varepsilon)\mathbb{Z}^3$, $\mathsf{B} = [-\pi/\varepsilon, \pi/\varepsilon)^3$ is the Brillouin zone, and for a function F on B^{L_0} (L_0 loop momenta) the torus Riemann sum is $\mathcal{R}_T[F] := (2\pi/\ell)^{3L_0} \sum_{p\in(T^*)^{L_0}} F(p)$, so that $\mathcal{R}_T[F] \to \int_{\mathsf{B}^{L_0}} F\,dp$ is the passage from the torus to $\varepsilon\mathbb{Z}^3$ at fixed ε.

F.1. **King's torus model and Bałaban's.** King's torus action at scalar mass M [13, (2.4)–(2.9), (3.32)–(3.35)] is the action of definition 12.1(ii): the Feynman-form photon with mass μ_0, the hopping terms, the single-site weights with bare coefficient $\alpha^{\mathrm{tor}}_{k,T}(M) = M^2 + \delta m^2{}_{k,T}(M)$, and the two field-independent constants $E_0(M)$ and $E_1(M)$. Here

(i) $\delta m^2{}_{k,T}(M) = \sum_{2\le a+2b\le 4} e^a\lambda^b\,\delta m^2{}_{(ab),T}(M)$ is fixed recursively by [13, (3.33)] on $\Omega = T$: the one-particle-irreducible scalar two-point function at zero momentum, computed with the periodic propagators of T at masses M and μ_0, vanishes order by order in (e,λ);

(ii) $E_0(M) = \ln\int e^{-S_{\mathrm{free}}}$ is the Gaussian normalisation [13, (2.6)], so that $E_0(M)/|T|$ is a Riemann sum $\mathcal{R}_T[\cdot]$ of $\frac{N}{2}\ln(\hat p^2 + M^2)$ plus the M-independent photon term;

(iii) $E_1(M) = \sum_{1\le a+b\le 13} \frac{e^a\lambda^b}{a!\,b!}\,\partial^{a+b}_{e,\lambda} \ln\int e^{-S+E_1}\big|_0$ is the Taylor part of $\ln Z$ through order $\bar n = 13$ [13, (3.35)], with the counterterm vertices $\delta m^2{}_{(ab),T}$ kept as numbers: a finite sum of connected vacuum graphs on T;

(iv) $c_{k,T}(M) = \mathcal{R}_T[(\hat p^2 + M^2)^{-1}]$ is the periodic tadpole, King's Wick constant.

$Z^{(M)}_{k,T} = \int e^{-S}$ is the source-free partition function; with the constants inside S, $\ln Z^{(M)}_{k,T} = \mathcal{L}(M) - E_1(M)$ where $\mathcal{L}(M) := \ln\langle e^{-V_M}\rangle_{\mu_M}$, μ_M the normalised Gaussian measure of the free fields at masses (M,μ_0) and V_M the interaction (quartic term, hopping couplings, counterterm $\frac12\delta m^2{}_{k,T}(M)\sum_x \varepsilon^3|\phi_x|^2$).

Bałaban's torus model [1, (1.1)–(1.13)] is the same action: his E_0 is the Gaussian normalisation and his E_1 [1, (1.13)] is the formula (iii), the same graphs with the same counterterm vertices [3, (1.24)]; his tori [1, (1.2)] have sides which are multiples of the large-block side, and every King torus is one of them (its side ℓ is a multiple of M, and $\ell\,5^k$ is a multiple of 5^j for every $j \le k$, so the blocks of every scale tile T). Thus $Z^{(M)}_{k,T}$ is exactly the quantity bounded by Bałaban's theorems at scalar mass M, and proposition 3.8 is a reading of those theorems: no change of convention is involved. Two things do have to be established: that the theorems hold at block factor 5, the factor of King's flow, and that the vacuum c-number which Bałaban's proof carries without a printed bound is extensive. Both are done in appendix F.2.

F.2. **The stability bounds at block factor 5.**

Theorem F.1 (torus stability at a fixed scalar mass)**.** *Let $e > 0$, $\lambda, \mu_0 > 0$ and $M > 0$. There are a stopping scale $\varepsilon_0 = \varepsilon_0(e,\lambda,M,\mu_0) \in (0,1]$ and constants $E_\pm(M) < \infty$ such that for every k with $\varepsilon_k \le \varepsilon_0$ and every King torus T,*

$$e^{-E_-(M)|T|} \;\le\; Z^{(M)}_{k,T} \;\le\; e^{E_+(M)|T|}.$$

In particular proposition 3.8 holds at $M = m$ and $M = M_1$, with K_1 the first k with $\varepsilon_k \le \min(\varepsilon_0(m), \varepsilon_0(M_1))$ and $J_*(k) = 1$.

The theorem is [1, (1.14)] (lower bound) and [2, (1.3)] (upper bound), "with constants E_-, E_+ independent of ε, T_ε", under $\lambda > 0$, $\mu_0^2 > 0$, $M^2 > 0$ and no smallness of e (the smallness conditions of [1, 2, 3, 4] are caps on the stopping scale, lemma 22.1 and remark 22.2). Its proof is Bałaban's, in three papers and about a hundred and ten printed pages, at a block factor L which the papers keep general and fix large. We record what has to be added for the reading used here, in four steps: the block factor, the vacuum c-number, the unprinted steps of the printed proof, and the mass.

Step 1: the block factor. Every place where the value of L enters the arguments of [1, 2, 3] and of [13, Thm. 3.1] (constants, rates, convergent sums, parameter conditions, divisibility and containment statements) is one of finitely many *L-entries*; the statement that all of them are finite, satisfiable and explicitly listed at $L = 5$ is written $[\text{B-}L]_5$. The statement that every estimate asserted without a printed argument in [1, 2, 3], and every printed argument, holds

at block factor L as asserted, with per-step constants independent of ε, of the torus and of the step, and independent of ε_0 except through the final-step constants of [1, (3.67)–(3.68)], [2, (2.116)–(2.118)], is written [B-base]$_5$.

Theorem F.2 (the block-factor-5 reading)**.** *Let $d = 3$, $L = 5$, and let $\bar{n}$ be Bałaban's expansion order ($\bar{n} \geq 6$ for the lower bound* [1, p. 605]*, $\bar{n} > 12$ for the upper bound* [2, p. 563]*). Make the following four replacements in the printed proofs:*

(a) *in the cube step of* [3, p. 433]*, the cubes of sides $r(L^k\varepsilon)$, $3r(L^k\varepsilon)$ are replaced by concentric cubes of constant sides $r^\flat$, $3r^\flat$, with $r^\flat$ a multiple of the large-block side bounded below by a constant depending on $\bar{n}$ and on Bałaban's R_0 only;*

(b) *in* [2, (2.44), p. 566]*, the cut-off $\zeta^{(k)}$ has support radius $\frac{2}{5}r(L^k\varepsilon) - 2M$ and equals 1 for $|x - y| \leq \frac{1}{5}r(L^k\varepsilon)$;*

(c) *in* [2, Lemma 2.4, p. 572]*, the two auxiliary boxes are replaced by the box of sup-radius $\min\{4r(t_k), 5\rho_k - 2M_b\}$ around the large block of y, with the source cut to the half-box;*

(d) *in* [2, (3.43)–(3.45), p. 592]*, the "regular partition into a lattice of cubes with side in $(r, 2r)$" is replaced by a product partition into boxes whose sides are integer numbers of large blocks in $(r, 2r + M_b]$.*

Impose Bałaban's parameter conditions (lower bounds on R and b_0, upper bounds on ε_0) with their $L = 5$ values. Then: (i) [B-L]$_5$ *holds;* (ii) *the replacements change no statement of* [1, 2, 3]*, and the four theorems* [1, (1.14)]*,* [2, (1.3)]*,* [3, Prop. 1]*,* [13, Thm. 3.1] *keep their printed form;* (iii) *given* [B-*base*]$_5$*, these four theorems hold at $L = 5$ on every torus aligned with the blocks, with constants depending on $(d, a, L, M_b, R, b_0, \varepsilon_0)$ only.*

Proof. (i) Sorting the L-entries by type, an entry that is a constant, a rate, a convergent sum or a condition on $(b_0, R, M, \varepsilon_0)$ satisfiable in Bałaban's order of choices is finite at every fixed L; the only entries that can fail at a fixed L as $\varepsilon \to 0$ are the four *cross-scale containments*: the cube step of [3, p. 433] (a cube of side $r(L^k\varepsilon)$ must contain the localisation cube of a renormalised class after the expansion, whose side depends on $\bar{n}$ only), the regularity of the vector background [2, Lemma 2.3] (the support of $\zeta^{(k)}$ must stay inside the region where the background is regular), the scalar background at the edge of Λ_7 [2, Lemma 2.4], and the split [2, (2.89), p. 575]. Each holds at $L = 5$ under its replacement (a)–(d): (a) makes the containment a statement about two fixed cubes; (b) gives $\zeta^{(k)}$ the room $\frac{2}{5}r - 2M$ inside a region of radius r; (c) and (d) replace shapes chosen for large L by shapes that exist at $L = 5$. All other entries are evaluated at $L = 5$ and are finite. (ii) The replaced objects are auxiliary cubes, cut-offs, boxes and partitions of the proofs, not of the statements; under (b) the small-field actions on the whole torus change by $O(t^\kappa)|T|$, which the statements absorb. (iii) The lower bound [1, (1.14)] is [1, Sect. 3] with its Propositions 2.1–2.3 (Bałaban's Green's function estimates, proved in [4]), Propositions 3.1–3.2 (the perturbative bounds, from [3, Prop. 1] at $L = 5$), the cumulant bound and the final step [1, (3.66)–(3.69)], at the $L = 5$ constants; the upper bound [2, (1.3)] is [2, Sects. 2–3] with the replacements; [13, Thm. 3.1] is [1, (3.26)] and [2, (2.43), (2.117)] at the step k with King's sources and field bounds, the additive large-field term of [13, (3.4)] being the sum over the large-field sequences bounded in [2, Sect. 3] by $\exp(-p(L^k\varepsilon)^2 + C|T|)$. □

Step 2: the vacuum c-number. In [3, (1.26)–(1.30), p. 418] the counterterm E_1 is split as $E_1 = E_K + E_{\text{fin}}$, where E_K has the same graphs with the block-spin propagators G_K, G_K^v of the final scale in place of the free ones and E_{fin} is called "only a finite error"; E_{fin} is carried through the induction as a global constant, and the final bounds $\exp(\pm E_\pm|T_\varepsilon|)$ absorb it. What this consumes, and all that it consumes, is an extensive bound on E_{fin}.

Lemma F.3 (c-number factorisation)**.** *Let $c(e, \lambda)$ be a polynomial in the couplings which depends neither on the fields nor on the sources nor on the block-spin history, and replace E by $E + c$ in the action. Then $Z[E + c] = e^{-c}Z[E]$ for every source; every intermediate integrand of the block-spin construction is multiplied by e^{-c}; and every step polynomial shifts by the same constant, $P^{(k)}[E + c] = P^{(k)}[E] - c$ in the convention of* [2, (2.43)]*. In particular $Z[E_0 + E_1] = e^{-E_{\text{fin}}}Z[E_0 + E_K]$, and no step polynomial of the chain run with $E_0 + E_K$ contains E_{fin}.*

Proof. A constant in the exponent factors out of every integral. Every operation of the construction is linear in the integrand or an exact identity: the block-spin transformations [1, (2.18)], the translations [2, (2.80)], the characteristic-function decompositions [2, (2.15), (2.43)] and the cumulant expansions; the characteristic functions restrict the fields, never the action. So the factor e^{-c} passes through every step unchanged. At the level of [3, (1.4)–(1.5)], $E_k[E+c] = E_k[E]+c$, and the Taylor part of a field-free polynomial in the couplings is that polynomial; by [3, (1.30)] the replacement $E_1 \to E_K = E_1 - E_{\text{fin}}$ removes exactly E_{fin} from $E_1(e, g_k, \lambda, \Omega_1)$ at every k. □

Theorem F.4 (extensivity of the vacuum c-number)**.** *On every torus aligned with the blocks, for every order (a,b) with $1 \le a+b \le \bar{n}$, the coefficient $E_{\text{fin}}^{(a,b)}$ of $e^a\lambda^b$ in E_{fin} satisfies $|E_{\text{fin}}^{(a,b)}| \le C_{ab}(1+S_K)^{N_{ab}}|T^{(K)}|$, where $|T^{(K)}|$ is the number of blocks of the final scale $l_K = L^K\varepsilon \in (\varepsilon_0/L, \varepsilon_0]$, $S_K := \max\{(Ml_K)^{-2}, (\mu_0 l_K)^{-2}\} + a_K^{-1} \le C\varepsilon_0^{-2}$, and C_{ab}, N_{ab} are free of ε, ε_0, K and the torus. Consequently $|E_{\text{fin}}| \le C_F(\varepsilon_0)\,|T|$ with $C_F(\varepsilon_0)$ independent of ε and of the torus.*

The mechanism is an exact identity for the difference of the free and the block-spin propagators, which makes every term of E_{fin} a sum over clusters glued by a nonnegative kernel with explicit row sums.

Lemma F.5 (block-spin factorisation and positivity)**.** *At zero background let $C_0 = (-\Delta+M^2)^{-1}$ on $\varepsilon\mathbb{Z}^3$ restricted to the torus, Q_K the block average over the blocks of side l_K, $P_K = Q_K^*Q_K$, $G_K = (-\Delta + M^2 + bP_K)^{-1}$ with $b = a_K l_K^{-2}$, $H_K := bG_KQ_K^*$ and $\Gamma_K := Q_KC_0Q_K^* + b^{-1}$ on block functions. Then:* (a) $C_0 - G_K = H_K\Gamma_K H_K^*$*;* (b) Γ_K *is the inverse of the block-spin form $\langle\psi, \Delta^{(K)}\psi\rangle = b\langle\psi,\psi\rangle - b^2\langle\psi, Q_KG_KQ_K^*\psi\rangle$ of* [1, (2.21)]*, i.e. the covariance of Bałaban's block spin at scale K, and H_K is his background map;* (c) *the kernel of Γ_K with respect to the block measure l_K^3 is entrywise ≥ 0 and has row sums $\sum_{B'} l_K^3\Gamma_K(B,B') = M^{-2} + b^{-1}$ for every B; in scale-K units, $\hat{\Gamma} := l_K\Gamma_K \ge 0$ has row sums $(Ml_K)^{-2} + a_K^{-1} \le S_K$, hence $\sup_{B,B'}\hat{\Gamma}(B,B') \le S_K$;* (d) *the same holds for the photon with μ_0 in place of M.*

Proof. $Q_KQ_K^* = 1$, so P_K is the orthogonal projection of [1, (2.20)]. The resolvent identity $G_K^{-1} = C_0^{-1} + bP_K$ gives $C_0 - G_K = G_KbP_KC_0 = C_0bP_KG_K$. (a) $H_K\Gamma_KH_K^* = b^2G_KP_KC_0P_KG_K + bG_KP_KG_K$, while $C_0bP_KG_K = [G_K + (C_0-G_K)]bP_KG_K = bG_KP_KG_K + (G_KbP_KC_0)bP_KG_K$; the two agree. (b) $(b - b^2Q_KG_KQ_K^*)(b^{-1} + Q_KC_0Q_K^*) = 1 + bQ_K(C_0 - G_K - bG_KP_KC_0)Q_K^* = 1$. (c) $-\Delta + M^2 = (6\varepsilon^{-2} + M^2)(1-W)$ with W entrywise ≥ 0 and row sums $6\varepsilon^{-2}/(6\varepsilon^{-2}+M^2) < 1$, so $C_0 = (6\varepsilon^{-2}+M^2)^{-1}\sum_{n\ge 0} W^n$ has a nonnegative kernel, and so have Q_K, Q_K^*; $C_0\mathbf{1} = M^{-2}\mathbf{1}$ and $Q_K\mathbf{1} = \mathbf{1} = Q_K^*\mathbf{1}$ give the row sums. The rescaling to scale-K units is [1, (2.22)]. (d) The photon covariance in Feynman form has a nonnegative kernel and $C\mathbf{1} = \mu_0^{-2}\mathbf{1}$. All identities are exact on every torus the blocks tile. □

Lemma F.6 (gluing by positivity and row sums)**.** *Let a term of E_{fin} be written, through lemma F.5(a), as a sum over clusters of vertices (the pieces of a vacuum graph joined by free lines) connected by lines carrying the kernel $\hat{\Gamma}$, with $|\Lambda|$ such lines. If each cluster c with $n_l(c)$ legs is bounded, uniformly in its position, by K_c times a tree decay $e^{-\delta_0 d(\cdot)}$ in its vertices and legs, then the term is at most $c_3|T^{(K)}|\prod_c \left(K_cC_c(C_HC_B)^{n_l(c)}\right) S_K^{|\Lambda|}$, with C_c, C_B the sums of the tree decays over the positions of one cluster and of one block, which do not depend on ε, ε_0, K or the torus.*

Proof. Choose a spanning tree of the cluster multigraph. Place one vertex of the root cluster ($c_3|T^{(K)}|$ choices), sum its other vertices and the blocks of its legs with the tree decays (factors C_c, C_B). Along a tree edge from a processed cluster to a new one, the block at one end is fixed and the other end is summed against the row of $\hat{\Gamma}$, a factor $\le S_K$ by lemma F.5(c); then the new cluster is placed as before. A non-tree edge has both ends fixed and contributes $\sup\hat{\Gamma} \le S_K$, since nonnegative entries are bounded by their row sums. The counting of blocks and cubes at a given torus distance is dominated by the count in $\mathbb{Z}^3$ (distinct objects have distinct lifts at the same distance), so C_c, C_B are torus-free. No decay of $C_0 - G_K$ is used: the infrared is controlled by the exact row sums, and this is the only place where ε_0 enters. □

Proof of theorem F.4. The vertices of the vacuum classes are the quartic, hopping and counterterm vertices, the latter at the orders $a + 2b \in \{2, 4\}$ only [3, (1.23)]. Expand each class in the free lines $C_0 = G_K + H_K \Gamma_K H_K^*$ (and the photon analogue); the terms with no Γ-line are the graphs of E_K and cancel in $E_{\text{fin}} = E_1 - E_K$; each remaining term is a cluster expansion of the type of lemma F.6, the clusters being bounded with their tree decay by Bałaban's perturbative estimates [3, Prop. 1] on the block-spin lines G_K and the regularity of the background fields $H_K\psi$ [1, Prop. 2.1, (2.24)–(2.25)], and the counterterm coefficients of the two-point pieces at orders 2 and 4 bounded by the same argument applied to the two-point classes rooted at a point ($c_3|T^{(K)}|$ replaced by 1). Lemma F.6 gives the bound; $|T^{(K)}| = |T| l_K^{-3}$ and $l_K > \varepsilon_0/L$. □

With lemma F.3 and theorem F.4 the printed chain is consistent: the inductive hypothesis [2, (2.43)] and the final bound [2, (2.117)] are run for $Z[E_0 + E_K]$, whose step polynomials contain no E_{fin}, and $Z_{k,T}^{(M)} = e^{-E_{\text{fin}}} Z[E_0 + E_K]$ differs from it by a factor $e^{O(1)|T|}$ with an ε- and torus-free constant, which is exactly what the statements [1, (1.14)], [2, (1.3)] allow (their constants may depend on ε_0). The lower bound follows the same way from the chain of [1, Sect. 3], at whose final step the c-number is the normalisation constant alone.

Step 3: the unprinted steps. The statement $[\text{B-base}]_5$ of theorem F.2(iii) is the list of the estimates which [1, 2, 3] assert without a printed argument, in four groups: (A) the Green's function estimates of [4] and [1, Sect. 2] (the transfer of [4, Cor. 2.3] to the propagators with mixed block pins, the domain-difference clause [4, (1.20)], the lower bound [4, (1.15)], the j-uniform analyticity neighbourhood of [4, Lemma 2.4], [4, (1.8)] on a general region, [1, Prop. 2.2]); (B) the steps of the lower bound ([1, Props. 3.1–3.2, (3.20)–(3.21), (3.34)–(3.36), (3.66)–(3.69)]); (C) the steps of the upper bound ([2, (2.11)–(2.14), Props. 2.1–2.2, 2.6, Lemma 2.7, (2.78), Sect. 3A, the k-uniformity of the per-step constants]); (D) [3, Prop. 1] with its twenty sketched steps, the counterterm moves of [3, p. 422] and the split [3, (1.26)–(1.31)]. Of these, group (A) is proved in full: [1, Prop. 2.3] (restated on [4, p. 574]) is an implication of the printed statements [4, (1.15), (1.16), (1.18), (1.20), (5.5), Cor. 2.3] read at source (with two harmless print findings: the equation numbers of [4, Sect. 3] are shifted by four against the journal numbering, and the bracket of [4, (3.9)] has the wrong sign, which the second resolvent identity corrects, only norms of the bracket being used), and the j-uniformity of [4, Lemma 2.4] is the following theorem.

Theorem F.7 ([4, Lemma 2.4] with constants independent of the scale)**.** *Let $d \geq 1$, $L \geq 3$ odd, $0 < a_- \leq a_+$. There are $\delta_* > 0$, c_0 and, for each $\alpha \in [0, 1)$, $c_1(\alpha)$, depending only on d, a_-, a_+, α, such that* [4, (2.35)–(2.36)] *hold for every $j \geq 0$, every $m_j^2 \geq 0$, every $a_j \in [a_-, a_+]$ and every rectangular parallelepiped (or periodic box) that is a union of unit blocks, with $\delta_0 = \delta_*/2$ in the ℓ^1 distance;* [4, (2.37)] *holds with constants depending on L but not on j.*

Proof. The operator is $-\Delta^\xi + m^2 + a Q_j^* Q_j$ on $\xi\mathbb{Z}^d$, $\xi = L^{-j}$, with Q_j the average over the unit blocks. Its Fourier symbol on the unit-lattice momenta $p' \in [-\pi, \pi]^d$ is $E(p') = \Delta^\xi(p')(1 + aS(p'))$ with $S(p') = \sum_l w(p' + l)/\Delta^\xi(p' + l)$, the sum over the N^d aliases l of p' ($N = L^j$) and $w(p) = \prod_\mu g(p_\mu)^2$, $g(t) = \xi \sin(t/2)/\sin(\xi t/2)$ [4, (2.44)–(2.48)]. On the strip $|\operatorname{Im} p'|_\infty \leq \delta$: each alias term is bounded by elementary estimates on $|g|$ and on $|\Delta^\xi|$ from below, uniformly in j; the denominator E is bounded below on the strip by a constant times $|\Delta^\xi(p')|$ at $\delta \leq \delta_*$, since the alias sum S is positive on the real section and its imaginary part is controlled by δ times a j-free constant; so the kernel $(G_j Q_j^*)(x, y)$ is the Fourier integral of a function analytic on the strip with j-free bounds, and shifting the contour by δ_* gives the exponential decay [4, (2.35)] with c_0, δ_* free of j; the first-difference part of [4, (2.35)] is the same integral with one factor $\hat\partial_\mu(p) = (e^{i\xi p_\mu} - 1)/\xi$, and the Hölder quotient [4, (2.36)] carries in addition $(e^{ip\cdot v'} - 1)/|v'|^\alpha$, $v' = x - x'$; both are bounded on the strip uniformly in ξ. Neumann faces and periodic boxes are reduced to the full lattice by reflection and periodisation, which preserve the block structure since the box is a union of blocks. [4, (2.37)] is the transfer to the mixed-pin operator and inherits the L-dependence of that operator's constants. □

Of groups (B)–(D), the following are proved from the printed text or by exact computation: [1, (3.20)–(3.21)], [1, (3.66)–(3.69)]; [2, (2.11)–(2.13)] given $|\delta m^2| \le c\varepsilon^{-1}$ (supplied by theorem F.85); [2, Prop. 2.2]; [2, Lemma 2.7] given the path estimate of [2, p. 560] at scale k; the replacement $A'' \to A'$ of [2, pp. 561–562] with error $O(\varepsilon^{\kappa})|\Lambda_7|$ for every κ; [2, (2.78)]; [2, (3.29)] through [4, Prop. 3.1′]; [3, p. 438]; and the interaction-term half of [2, Prop. 2.6] at a nonzero background, stated as (BAL_5)[int] (lemma 5.1) and proved through classes complete at every background. The remaining pieces are proved in Steps 3(B)–3(F) below: the perturbative form of the effective action [1, (3.34)–(3.36)] (Step 3(B), an algebraic identity between formal series); the counterterm moves of [3, pp. 411–422], i.e. the vacuum part of the effective action measured against the localised counterterms and its behaviour under the domain, localisation and counterterm changes of a step (Steps $3(\mathrm{D}_1)$–$3(\mathrm{D}_2)$); the large-field chain of [2, Sect. 3A] with the k-uniformity of the unqualified $O(1)$ lines of [2] and the ε_0-freedom of the per-step constants (Steps $3(\mathrm{C}_1)$–$3(\mathrm{C}_2)$); and the sketched steps of [3, Prop. 1] itself, whose regrouping step is $(\mathrm{REN}_{\mathrm{orb}})$ (lemma 5.2) and whose renormalisation cancellation is the induction of [3, pp. 444–445] (Steps $3(\mathrm{D}_3)$–$3(\mathrm{D}_4)$); the sizes of the mass counterterm, $|\delta m^2| \le c_m\varepsilon^{-1}$, $\delta m^2_{K,k}(L^k\varepsilon)^2 = O(L^k\varepsilon)$ and the finite part $\delta m^2_{\mathrm{fin}}$ in the form in which it is consumed (Step 3(E), theorems F.85, F.93 and F.99), together with the vacuum counterterm pieces $E_{K,k}$, [2, (2.14)] for the constant the chain carries, and [1, Props. 3.1–3.2], [2, p. 579] given Proposition 1 (theorem F.103, corollary F.104, and proposition F.105); and the remaining asserted estimates of [2, Sects. 2–3]: [2, p. 568, (2.47)] (proposition F.108), the quadratic-form step [2, (2.108)–(2.109)] (lemmas F.111 and F.112), the field-dependent part of [2, Prop. 2.1, (2.57)] (theorem F.128), the passage from [2, (3.5)] at $k = 0$ to [2, (3.21)] with the payment of [2, Sect. 3C] (lemmas F.114 and F.116), the region sum [2, (3.42)] at $L = 5$ with the counterterm moves paid in it (theorem F.125 and proposition F.126), and the g_k-localisation of $Q^{(k)}$ [2, p. 579] (proposition F.127) (Step 3(F)). Every constant in these steps is explicit and free of ε, the step and the torus, and free of ε_0 except through the caps on ε_0 and the final-step constants that $[\text{B-base}]_5$ allows (the bound on $\delta m^2_{\mathrm{fin}}$ at fixed ε_0, corollary F.100, enters only through the ε_0-free form of theorem F.99); so $[\text{B-base}]_5$ holds and theorem F.2(iii) applies.

Notation for Steps 3(B)–3(D). Throughout the rest of Step 3, $d = 3$, $L = 5$, k is the step, $t_k := L^k\varepsilon \le \varepsilon_0 \le 1$ is Bałaban's scale parameter (written t when k is fixed), $\eta := L^{-k}$, and all scale-k operators act on the unit lattice $T_1^{(k)}$ of scale k with the fine lattice $\eta\mathbb{Z}^3$ inside each unit cell. The rescaled couplings are $e_k := e\,t_k^{1/2}$, $\lambda_k := \lambda t_k$; $p(t) := b_0(1+\ln t^{-1})^{\bar p}$ and $r(t) := R(1+\ln t^{-1})^{\varrho}$, $\varrho > 1$, are the field bound and the layer width of [2, (2.7), (2.16)–(2.17)]; $\Lambda_i^{(k)}$, $i = -1,\dots,8$, are the regions of [2, (2.8)] at scale k (unions of large blocks, $\Lambda_{i+1}^{(k)c}$ the union of the large blocks within distance $r(t_k)$ of $\Lambda_i^{(k)c}$), and a primed region $\Lambda_i^{(k-1)'}$ is the scale-$(k-1)$ region read on $T_1^{(k)}$. M is the side of Bałaban's large blocks ([1, p. 607]: the block partition with L replaced by ML; written M_b in theorem F.2), not the scalar mass of theorem F.1, which is m or M_1 here. The expansion order is $\bar n$; a_k is the block-spin coefficient of [1, (2.13)], with $a_{k+1} = aa_k/(aL^{-2}+a_k)$, $a_1 = a$, so that $a_* := a(1-L^{-2}) \le a_k \le a$ for all k. $Q_k(A)$ is the covariant k-step block average, $P_k = Q_k^*Q_k$, $G_k(\Omega, A) = (-\Delta^{\eta,N}_{A,\Omega} + m_k^2 + a_kP_k(A))^{-1}$ the scalar propagator on a union Ω of blocks with Neumann conditions, $\Delta^{(k)}(\Omega, A) = a_k - a_k^2Q_kG_k(\Omega, A)Q_k^*$ the block-spin operator, $C_\Lambda^{(k)} = ((\Delta^{(k)} + aL^{-2}P)|_\Lambda)^{-1}$ the fluctuation covariance, as in [1, Sect. 2] and [4, Sect. 1]. For a kernel or external field, $\|\cdot\|_{1,\alpha_0}$ is the norm [3, (1.32)] (sup, one covariant derivative, its Hölder quotient of exponent α_0), in each variable for kernels of two variables. "Proposition 1" is [3, Prop. 1, (1.33)]: for a localised renormalised class G_{ren} with localisation $\Lambda = \{\square(v)\}$, $|E(G_{\mathrm{ren}}, \Lambda, \Phi)| \le O(1)e_k^{d_v}\lambda_k^{d_s}e^{-\delta_0 d(\Lambda)}\prod\|h\Phi\|_{1,\alpha_0}$, $d(\Lambda)$ the length of a shortest tree through the cubes of Λ; its proof at $L = 5$ is Step 3(D) below.

Step 3(B): the perturbative form of the effective action, [1, (3.34)–(3.36)]. [1, (3.36)] asserts, "by close inspection", that after k steps the polynomial part of the effective action is $S^{(k)} = \frac{1}{2}\langle A, \Delta^{(k)}A\rangle +$ the truncation to order $\bar n$ of the formal series $\mathfrak{E}_k$ of [1, (3.35)], built with the

contour convention [1, (3.34)]. This is an algebraic identity between formal power series, proved here. The rescalings [1, (1.22)] are changes of variables with field-independent Jacobians and change nothing below except additive constants.

Definition F.8 (grading, truncation, re-weighting)**.** (a) In [1, (3.35)] replace e' by a formal variable s wherever it occurs (on every fluctuation field, and in $\delta m^2(e',\lambda')$, $E(e',\lambda')$), and read [1, (3.34)] with j from 0 (which is [3, (1.3)]). Write $\mathfrak{E}_k(s,\lambda';A,\phi) := E_k(s,\lambda',eA^{(k),\varepsilon},\phi)$, a formal power series in (s,λ') whose coefficients are smooth functions of the block fields; the degree of a monomial is the power of s plus the power of λ'. (b) $T_{\bar n}$ truncates to degree $\le \bar n$ and "ev" evaluates at $(s,\lambda')=(e,\lambda)$; in this notation [1, (3.36)] reads $S^{(k)} = \frac12\langle A,\Delta^{(k)}A\rangle + \operatorname{ev} T_{\bar n}\mathfrak{E}_k$. (c) In step $k+1$, $A = A' + aL^{-2}C^{(k)}Q^*B$ [1, (3.41)] and $A^{(k),\varepsilon} = A'^{(k),\varepsilon} + A^{(k+1),\varepsilon}$ [1, (3.42)]; for a coefficient function f of the background $eA^{(k),\varepsilon}$, $\iota_k f$ is the Taylor series of $\tau \mapsto f(\tau A'^{(k),\varepsilon} + eA^{(k+1),\varepsilon})$ at $\tau=0$ with τ then replaced by s, so that each power of the new fluctuation carries one power of s (Bałaban's order count: every A'- or $\tilde A$-leg in [3, (1.8)–(1.15)] carries one factor $e(L^k\varepsilon)$).

Lemma F.9 (truncation commutes with a Gaussian integral)**.** *Let $d\mu \propto e^{-F_0}dX$ be a Gaussian law on a finite-dimensional space (F_0 a positive quadratic form), and let W be a formal series of degree ≥ 1 whose coefficients are polynomials in X. Then $-\log\langle e^{-W}\rangle_\mu$ is a well-defined formal series; its degree-n part is a finite combination of Gaussian moments of the parts $W_1,\dots,W_n$ of W of degrees $1,\dots,n$; hence $T_{\bar n}(-\log\langle e^{-W}\rangle_\mu) = T_{\bar n}(-\log\langle e^{-T_{\bar n}W}\rangle_\mu)$.*

Proof. $\langle e^{-W}\rangle = \sum_{m\ge0}(-1)^m\langle W^m\rangle/m!$; the degree-$n$ part of W^m vanishes for $m>n$ and for $m\le n$ involves only $W_1,\dots,W_{n-m+1}$. So $\langle e^{-W}\rangle = 1+u$ with u of degree ≥1, and the degree-n part of $-\log(1+u) = \sum_{m\ge1}(-1)^m u^m/m$ involves only the parts of u of degree $\le n$. Each coefficient is a Gaussian moment of a polynomial, hence finite. □

This is the cumulant expansion: Bałaban's "terms of the order $\le\bar n$" of an integrated action are exactly $T_{\bar n}$ of this series.

Lemma F.10 (composition of the scalar kernels)**.** *Let $Q\colon \mathbb{R}^n\to\mathbb{R}^m$ be linear with $QQ^* = c\,\mathbf{1}_m$ ($c>0$), and $\alpha,\beta>0$. Then for all $\psi\in\mathbb{R}^m$, $\chi\in\mathbb{R}^n$,*

$$\int_{\mathbb{R}^n} d\phi\, \exp\left(-\tfrac12\alpha|\psi-Q\phi|^2-\tfrac12\beta|\phi-\chi|^2\right) = (2\pi/\beta)^{n/2}(1+\alpha c/\beta)^{-m/2}\exp\left(-\tfrac12\gamma|\psi-Q\chi|^2\right), \qquad \gamma := \frac{\alpha\beta}{\beta+\alpha c}.$$

Proof. Write $\phi = \chi + \beta^{-1/2}\xi$ with ξ standard Gaussian; then $Q\phi = Q\chi + Z$ with $Z\sim N(0,(c/\beta)\mathbf{1}_m)$, and for $Z\sim N(0,\sigma^2\mathbf{1}_m)$, $\mathbb{E}\exp(-\frac12\alpha|v+Z|^2) = (1+\alpha\sigma^2)^{-m/2}\exp(-\frac12\frac{\alpha}{1+\alpha\sigma^2}|v|^2)$. Take $v = Q\chi-\psi$, $\sigma^2 = c/\beta$. □

The covariant one-step average $(Q(\beta)\phi)(z) = L^{-d}\sum_{y\in B(z)} U(\beta(\Gamma_{z,y}))\phi(y)$ has $Q(\beta)Q(\beta)^* = L^{-d}\mathbf{1}$ for every background β (the U are orthogonal, the blocks disjoint). With $\alpha = aL^{d-2}$ [1, (3.38)], $\beta = a_k$ and $c = L^{-d}$ the lemma gives $\gamma = L^{d-2}aa_k/(a_k+aL^{-2}) = L^{d-2}a_{k+1}$, the recursion [1, (2.13)]; the factor L^{d-2} is removed by the rescaling of ψ, and the prefactor depends neither on the background nor on ψ,χ. At a fixed external field this is [1, (2.10)–(2.16)]; the next two lemmas extend it to level-dependent fields.

Lemma F.11 (composite averages)**.** *Let β be a fine field and Φ assign a phase to each composite contour $\Gamma^{(k)}_{y,x}$. Then $(Q(\beta)Q_k(\Phi)\phi')(z) = (Q_{k+1}(\Phi')\phi')(z)$ with $\Phi'(\Gamma^{(k+1)}_{z,x}) := \beta(\Gamma_{z,x_k})+\Phi(\Gamma^{(k)}_{x_k,x})$.*

Proof. $U(a)U(b) = U(a+b)$ since all U commute; $B(z)\circ B^k(y) = B^{k+1}(z)$ and $\Gamma^{(k+1)}_{z,x} = \Gamma_{z,x_k}\cup\Gamma^{(k)}_{x_k,x}$ by [1, (2.2)]; the weights $L^{-d}\eta^d$ are those of the $(k+1)$-average. □

Lemma F.12 (the contour convention is the iterate, and j starts at 0)**.** *For $k\ge1$ put $\Phi_k(s)(\Gamma^{(k)}_{y,x}) := eA^{(k),\varepsilon}(\Gamma^{(k)}_{y,x}) + s\sum_{j=0}^{k-1}A'^{(j),\varepsilon}(\Gamma^{(j+1)}_{x_{j+1},x})$, which is [1, (3.34)] with j from 0 and the fluctuations weighted by s.* (a) *With $\beta := \iota_k(eA^{(k),\varepsilon}) = sA'^{(k),\varepsilon} + eA^{(k+1),\varepsilon}$ on the first-step contour Γ_{z,x_k} prescribed by* [1, (3.37)], *$\beta(\Gamma_{z,x_k}) + (\iota_k\Phi_k(s))(\Gamma^{(k)}_{x_k,x}) = \Phi_{k+1}(s)(\Gamma^{(k+1)}_{z,x})$.* (b) *At $k=0$ the first kernel carries $U(eA(\Gamma_{z,x}))$ with the full fine field $A = A'_0 + A^{(1),\varepsilon}$* [1, (3.15), p. 615],

and $\iota_0(eA(\Gamma_{z,x})) = sA'_0(\Gamma^{(1)}_{x_1,x}) + eA^{(1),\varepsilon}(\Gamma^{(1)}_{z,x}) = \Phi_1(s)(\Gamma^{(1)}_{z,x})$*: its s-term is the* $j = 0$ *term.* (c) *Hence the printed range* $j = 1, \dots, k-1$ *of* [1, (3.34)] *is not what the procedure produces: at* $k = 1$ *it would drop* $sA'_0(\Gamma^{(1)})$ *and with it every contour vertex* [3, (1.14)] *with* A'_0*-legs, which the first step does produce* [1, (3.15)–(3.16)]*; with* $j = 0, \dots, k-1$ *the convention is the iterate.*

Proof. (a) By [1, (2.2)], $\Gamma^{(k+1)}_{z,x} = \Gamma_{z,x_k} \cup \Gamma^{(k)}_{x_k,x}$ and $\Gamma^{(k+1)}_{x_{k+1},x} = \Gamma^{(k+1)}_{z,x}$ since $x_{k+1} = z$; by [1, (2.3)] line integrals add over unions of contours, so $\beta(\Gamma_{z,x_k}) + sA'^{(k)}(\Gamma^{(k)}_{x_k,x}) + eA^{(k+1)}(\Gamma^{(k)}_{x_k,x}) = sA'^{(k)}(\Gamma^{(k+1)}_{x_{k+1},x}) + eA^{(k+1)}(\Gamma^{(k+1)}_{z,x})$, and the terms with $j < k$ are unchanged because the points x_j, $j \le k$, of x are the same at both levels. (b) and (c) are read off the definitions. □

Lemma F.13 (the Gaussian chain on a domain; [1, (3.41), (3.42)] and [3, (1.27)])**.** *Let* $H > 0$ *be the covariant Neumann operator* $-\Delta^{\eta}_{B,\Omega} + m^2$ *on a union* Ω *of blocks at a fixed background* B*, or the vector operator at* $B = 0$ *componentwise; let* R_k *be the* k*-step composite average onto* $T_1^{(k)} \cap \Omega$*,* Q *the one-step average with* $QQ^* = c\,\mathbf{1}$*, and* $\beta_k, \alpha > 0$*. Consider the Gaussian density of* (ϕ', ϕ, ψ) *proportional to* $\exp(-\frac{1}{2}\langle\phi', H\phi'\rangle - \frac{1}{2}\beta_k|\phi - R_k\phi'|^2 - \frac{1}{2}\alpha|\psi - Q\phi|^2)$*, and put* $G_k := (H + \beta_k R_k^* R_k)^{-1}$*,* $\Delta^{(k)} := \beta_k - \beta_k^2 R_k G_k R_k^*$*,* $C^{(k)} := (\Delta^{(k)} + \alpha Q^* Q)^{-1}$*,* $\gamma := \alpha\beta_k/(\beta_k + \alpha c)$*,* $G_{k+1} := (H + \gamma(QR_k)^*(QR_k))^{-1}$*. Then:* (a) *given* ϕ*,* $\phi' \sim N(\beta_k G_k R_k^* \phi, G_k)$*, and the marginal density of* ϕ *is* $\propto \exp(-\frac{1}{2}\langle\phi, \Delta^{(k)}\phi\rangle)$*;* (b) *given* ψ*,* $\phi \sim N(\alpha C^{(k)} Q^* \psi, C^{(k)})$*, and* $\mathbb{E}[\phi' \mid \psi] = \beta_k G_k R_k^* \alpha C^{(k)} Q^* \psi = \gamma G_{k+1}(QR_k)^* \psi$*;* (c) $\mathrm{Cov}(\phi' \mid \psi) = G_{k+1} = G_k + \beta_k^2 G_k R_k^* C^{(k)} R_k G_k$*.*

(a) is the definition of $A^{(k)}$, $\phi^{(k)}$ and $\Delta^{(k)}$ [1, (3.29), (2.21)]; (b) is [1, (3.41)] (the conditional fluctuation $A' = A - \mathbb{E}[A \mid B]$ is independent of B with covariance $C^{(k)}$) together with the second equality of [1, (3.42)], $a_k G_k Q_k^* \, a L^{-2} C^{(k)} Q^* B = a_{k+1} L^{-2} G^{\eta}_{k+1} Q^*_{k+1} B$; (c) is, on the torus at $B = 0$, one step of [3, (1.27)], and on Ω the same identity with Neumann lines.

Proof. (a): complete the square in ϕ'. (b): by (a) the (ϕ, ψ)-density is $\propto \exp(-\frac{1}{2}\langle\phi, \Delta^{(k)}\phi\rangle - \frac{1}{2}\alpha|\psi - Q\phi|^2)$; complete the square in ϕ. The full density factorises as $f(\phi', \phi)g(\phi, \psi)$, so ϕ' and ψ are conditionally independent given ϕ and $\mathbb{E}[\phi' \mid \psi] = \mathbb{E}[\mathbb{E}[\phi' \mid \phi] \mid \psi] = \beta_k G_k R_k^* \mathbb{E}[\phi \mid \psi]$. Integrating ϕ first (lemma F.10 with $\chi = R_k\phi'$), the (ϕ', ψ)-density is $\propto \exp(-\frac{1}{2}\langle\phi', H\phi'\rangle - \frac{1}{2}\gamma|\psi - QR_k\phi'|^2)$, whence $\mathbb{E}[\phi' \mid \psi] = \gamma G_{k+1}(QR_k)^*\psi$ and $\mathrm{Cov}(\phi' \mid \psi) = G_{k+1}$. (c): the law of total covariance, $\mathrm{Cov}(\phi' \mid \psi) = \mathbb{E}[\mathrm{Cov}(\phi' \mid \phi, \psi) \mid \psi] + \mathrm{Cov}(\mathbb{E}[\phi' \mid \phi, \psi] \mid \psi) = G_k + \beta_k^2 G_k R_k^* C^{(k)} R_k G_k$. □

Lemma F.14 (diagonal truncation)**.** *Let* $H(s, \tau, \lambda')$ *be a formal series and* $h(s, \lambda') := H(s, s, \lambda')$*. Then* $T_{\bar n} h$ *at* $(s, \lambda') = (e, \lambda)$ *equals* $T^{\mathrm{tot}}_{\bar n} H$ *at* $(s, \tau, \lambda') = (e, e, \lambda)$*, where* $T^{\mathrm{tot}}_{\bar n}$ *truncates the total degree.*

Proof. The coefficient of $s^m \lambda'^{\beta}$ in h is $\sum_{\alpha+\gamma=m} H_{\alpha\gamma\beta}$. □

Theorem F.15 ([1, (3.36)] at every k)**.** *Let* $S^{(0)} := S^{\varepsilon}$ [3, (1.20)]*, and for* $k \ge 0$ *let* $S^{(k+1)}_{\mathrm{pol}}$ *be the polynomial part, of order* $\le \bar n$*, that the step* $k+1$ *of* [1, Sect. 3] *produces from* $S^{(k)}_{\mathrm{pol}}$*: the kernels of* [1, (3.37)] *(gauge* $T_{a,L}$*, scalar* $T_{a,L,A^{(k),\varepsilon}}$*), the translation* [1, (3.41)]*, the expansion of all coefficients in the new fluctuation* [1, (3.44)]*, the Gaussian integral and the cumulant expansion keeping the terms of order* $\le \bar n$ *(the restrictions* χ *and the terms of order* $> \bar n$ *are split off; they are the error terms of* [1, (3.26)] *and are not part of this statement). Then for every* $k \ge 0$*,*

$$S^{(k)}_{\mathrm{pol}} = \tfrac{1}{2}\langle A, \Delta^{(k)} A\rangle + \mathrm{ev}\, T_{\bar n}\mathfrak{E}_k + c_k,$$

where $\mathfrak{E}_k$ *is* [1, (3.35)] *with* [1, (3.34)] *read from* $j = 0$*, and* c_k *is a constant independent of the fields and of the couplings (a normalisation, entering* E_0 *only). With the printed range* $j \ge 1$ *the identity is false at* $k = 1$ *(lemma F.12(c)).*

Proof. Induction on k. *Base* $k = 0$. $\mathfrak{E}_0$ contains no fluctuation: $\mathfrak{E}_0 = \frac{1}{2}\langle\phi, (-\Delta_{eA} + m^2)\phi\rangle + \sum \varepsilon^d(\lambda'|\phi|^4 + \frac{1}{2}\delta m^2(s, \lambda')|\phi|^2) + E(s, \lambda') + \mathrm{const}$, and $\frac{1}{2}\langle A, \Delta^{(0)} A\rangle = \frac{1}{2}\langle A, (-\Delta + \mu_0^2)A\rangle$; $T_{\bar n}$ acts as the identity, since δm^2 has orders $2 \le \alpha + 2\beta \le 4 \le \bar n$ [3, p. 417] and E has orders $\le \bar n$. So $S^{(0)} = \frac{1}{2}\langle A, \Delta^{(0)} A\rangle + \mathrm{ev}\, T_{\bar n}\mathfrak{E}_0$. At $k = 0$ the inner kernel is absent ($Q_0 = 1$): the gauge

part of the step is lemma F.13(b) with $H = \Delta^{(0)}$, and the scalar part is [1, (3.35)] at $k = 1$ by definition, with $a_1 = a$ and the $j = 0$ contour term of lemma F.12(b).

Step $k \to k+1$. Define

$$\mathrm{Step}_k F(B,\psi) := -\log\Big[N_k \int dA \int d\phi\, \exp\big(-\tfrac{1}{2}aL^{d-2}|B-QA|^2 - \tfrac{1}{2}aL^{d-2}|\psi - Q(\iota_k e A^{(k),\varepsilon})\phi|^2\big) e^{-\iota_k F(A,\phi)}\Big],$$

followed by the rescaling [1, (1.22)].

(i) *Exact identity* for $F = \frac{1}{2}\langle A, \Delta^{(k)} A\rangle + \mathfrak{E}_k$, degree by degree (Fubini for Gaussian moments of polynomials). Gauge part: by lemma F.13(b), $\int dA \exp(-\frac{1}{2}aL^{d-2}|B-QA|^2 - \frac{1}{2}\langle A, \Delta^{(k)}A\rangle) f(A^{(k),\varepsilon}) = N \exp(-\frac{1}{2}\langle B, \Delta^{(k+1)} B\rangle) \int d\mu_{C^{(k)}}(A'_k) f(A'^{(k),\varepsilon} + A^{(k+1),\varepsilon})$, and ι_k puts the factor s on $A'^{(k),\varepsilon}$. Scalar part: write $e^{-\mathfrak{E}_k}$ as the nested integral [1, (3.35)]; the ϕ-integral is $\int d\phi \exp(-\frac{1}{2}\alpha|\psi - Q(\beta)\phi|^2 - \frac{1}{2}a_k|\phi - Q_k(\Phi_k)\phi'|^2) = N' \exp(-\frac{1}{2}a_{k+1}|\psi - Q(\beta)Q_k(\Phi_k)\phi'|^2)$ by lemma F.10, and $Q(\beta)Q_k(\iota_k\Phi_k) = Q_{k+1}(\Phi_{k+1})$ by lemmas F.11 and F.12(a). Covariant Laplacian: on every bond b, $(\iota_k\Phi_k)_b = sA'^{(k)}_b + eA^{(k+1)}_b + s\sum_{j<k} A'^{(j)}_b = (\Phi_{k+1})_b$. The λ'-terms and $\delta m^2(s,\lambda')$, $E(s,\lambda')$ are unchanged. Hence $\mathrm{Step}_k(\frac{1}{2}\langle A, \Delta^{(k)}A\rangle + \mathfrak{E}_k) = \frac{1}{2}\langle B, \Delta^{(k+1)}B\rangle + \mathfrak{E}_{k+1} + c'_k$.

(ii) *Truncation.* The degree-0 part of the integrand of Step_k is Gaussian in $X = (A'_k, \phi)$: its gauge part is given by (i), its scalar part is $\frac{1}{2}\langle\phi, \Delta^{(k)}(eA^{(k+1),\varepsilon})\phi\rangle + \frac{1}{2}\alpha|\psi - Q(eA^{(k+1),\varepsilon})\phi|^2$. Since ι_k only raises degrees, $T_{\bar n}\iota_k T_{\bar n} = T_{\bar n}\iota_k$, and lemma F.9 gives $T_{\bar n}\mathrm{Step}_k(T_{\bar n}F) = T_{\bar n}\mathrm{Step}_k(F)$.

(iii) *Evaluation.* Bałaban's step expands in the new fluctuation at the true coupling e and keeps total order $\le \bar n$; by lemma F.14 with τ the weight of the new fluctuation (ι_k is the diagonal $\tau = s$), his truncation is $\mathrm{ev}\, T_{\bar n}$ of the diagonal. Hence $S^{(k+1)}_{\mathrm{pol}} = \mathrm{ev}\, T_{\bar n}\mathrm{Step}_k(S^{(k)}_{\mathrm{pol}}) = \frac{1}{2}\langle B, \Delta^{(k+1)}B\rangle + \mathrm{ev}\, T_{\bar n}\mathfrak{E}_{k+1} + c_{k+1}$. □

The theorem is the polynomial-part identity that "close inspection" asserts, an implication of the printed definitions [1, (2.1)–(2.3), (3.29), (3.33), (3.35), (3.37)], [4, (1.2)] and of [1, (3.34)] read with j from 0, i.e. of the defining formula [3, (1.3)]. It does not cover the analytic content of a step (the restrictions, the order-$> \bar n$ remainders, the error terms of [1, (3.26)]): that is [1, Props. 3.1–3.2], which follow from Proposition 1 (Step 3(D)).

Step 3(D_1): the counterterm moves of [3, pp. 411–422]*, exact algebra.* Two statements of [3] on the vacuum part of the effective action carry no printed argument: that the field-independent part generated after k steps on the domains of the step, measured against the localised counterterms $E_k(\cdot, \Omega_1)$ of [3, (1.30)], is $O(t_k)$ per unit block ("obvious consequences of Proposition 1", [3, pp. 421–422]), and that the domain and localisation changes of a step, with the counterterm move of [3, p. 422], cost only collar terms ("even simpler and we omit them"). We write them as the statements $(\mathrm{VAC}^\flat)_k$ and $(\mathrm{DOM})_k$ (theorems F.28 and F.31) and prove them from Proposition 1. This part sets up the objects and their exact identities; the bounds are in Step 3(D_2).

Definition F.16 (datum at scale k)**.** A datum is $\mathfrak{D} = (\Omega, \Omega_1, g, c)$: Ω a union of big blocks of $T^{(k)}_1$ (the domain hypothesis of [1, Props. 2.1–2.3], [3, p. 411]), on whose η-points the fine scalar field lives with the Neumann covariant Laplacian; $\Omega_1 \subset \Omega$ a union of unit blocks; g smooth with $\mathrm{supp}\, g \subset \Omega_1$, $g = 1$ where $\mathrm{dist}(\cdot, \partial\Omega_1) \ge M$ and $\|g\|_{C^2} \le C_g$ ([3, p. 412] has $C_g = 1$; a fixed C_g is allowed, as for the partitions of unity of [3, p. 420]); $c(x)$ the coefficient of the vertices [3, (1.7)], namely [3, (1.29)] built with g at the scale of $\mathfrak{D}$, $c = \delta m^2_{\mathrm{fin}} t_k^2 + \delta m^2_{K,k} t_k^2 + \delta m^2_k(se(t_k)g, \lambda'\lambda(t_k), \Omega_1, x)$. The depth is $\rho(\mathfrak{D}) := \mathrm{dist}(\Omega_1, \Omega^c)$ in scale-k units. Bałaban's step-k datum is $\mathfrak{D}_k := (B^k(\Lambda_2^{(k-1)'}), B^k(\Lambda_7^{(k-1)'}), g_k, c_k)$; torus data are $\mathfrak{D}^T_{W,g} := (T, W, g, c_{W,g})$, and the plain torus datum $\mathfrak{D}^T$ has $W = T$, $g = 1$, $c = \delta m^2$.

Definition F.17 (the vacuum functionals of a datum)**.** $\Xi_k[\mathfrak{D}](A,\phi)$ is the nested integral inside the logarithm of [3, (1.4)], with (Ω, Ω_1, g_k) and the coefficient replaced by those of $\mathfrak{D}$, e' replaced by s, and the counterterm factor e^{-E_1} omitted (the vector covariances are the torus $C^{(j)}$ of [3, (1.4)]). $\mathrm{Gen}_k[\mathfrak{D}] := T_{[1,\bar n]} \log[\Xi_k[\mathfrak{D}](0,0)/\Xi_k[\mathfrak{D}](0,0)|_{s=\lambda'=0}]$ (the generated vacuum constants);

$F_k[\mathfrak{D}](A,\phi) := T_{[0,\bar n]} \log[\Xi_k[\mathfrak{D}](A,\phi)/\Xi_k[\mathfrak{D}](0,0)]$ (the field-dependent part, including its degree-0 Gaussian part); $\mathrm{Ct}_k[\mathfrak{D}] := E_k(se(t_k)g, \lambda'\lambda(t_k), \Omega_1)$ of [3, (1.30)], under Convention F.18; $\mathcal{R}_k[\mathfrak{D}] := \mathrm{Gen}_k[\mathfrak{D}] - \mathrm{Ct}_k[\mathfrak{D}]$.

By [3, (1.4)], $E_k(\cdot,\Omega,0,0) = E_1(e', g_k, \lambda', \Omega_1) - \log \Xi_k[\mathfrak{D}_k](0,0)$, and by [3, (1.30)], $E_1(e', g_k, \lambda', \Omega_1) = E_{\mathrm{fin}} + E_{K,k} + \mathrm{Ct}_k[\mathfrak{D}_k]$; so the field-independent part of the polynomial $\mathcal{P}^{(k)}(\Omega_1, \cdot)$ of [3, (1.5)], in the logarithmic convention, is $\mathcal{R}_k[\mathfrak{D}_k] - E_{\mathrm{fin}} - E_{K,k}$, with $E_{\mathrm{fin}} + E_{K,k}$ a global front factor.

Convention F.18 (how the localised counterterms are built)**.** (a) Every vertex [3, (1.7)] carries the coefficient c of [3, (1.29)] as a number, not opened, both in the expansion of [3, (1.4)] and in the graphs of Ct_k. (b) $\mathrm{Ct}_k[\mathfrak{D}]$ is the sum of the bare vacuum graphs (vertices [3, (1.6), (1.7), (1.8), (1.10)] with $n' = 0$, $\tilde B = 0$) of orders $1 \le \alpha + \beta \le \bar n$, with every line of index $< k$: scalar lines $G_k(T, 0)$ (the rescaled $G^{\varepsilon}_k(0)$ of [3, (1.25)–(1.28)]) and vector lines $G^{\mathrm{vec}}_k(T)$, the vertices localised exactly as the generated ones of [3, (1.4)]: [3, (1.6)–(1.7)] summed over Ω_1, g on each vector leg. (c) In $\delta m^2_k(se(t_k)g, \dots)$ the factor g sits on the vector legs of the coefficient's graphs, so a divergent self-energy subgraph of a generated graph and its [3, (1.7)] partner ([3, p. 420]: "the same graph G but with both external legs localized in x") carry g on corresponding legs.

(b) and (c) are printed: [3, (1.31), p. 419] displays the localised second-order photon counterterm with torus lines and g_k on both vector legs, and [3, p. 419] reads "localize the vertices (1.6) and (1.7) restricting summations to the set Ω_1. The other vertices are localized by functions g_k multiplying each leg of the vector field". (a) is forced: if the [3, (1.7)] vertices of Ct carried the full torus δm^2, Ct would contain on $\{g < 1\}$ self-energy subgraphs with legs gg' paired with an unlocalised coefficient, an uncancelled $O(e(t)^2\eta^{-1})$ per site; if the coefficients were opened into lines of index $< k$, Gen − Ct would contain $(\delta m^2_{\mathrm{fin}} t^2 + \delta m^2_{K,k} t^2) G_k(x,x) = O(t\eta^{-1})$ per site. On the torus datum, $E_{\mathrm{fin}} + E_{K,k} + \mathrm{Ct}_k[\mathfrak{D}^T] = E_1$. (Bałaban remarks on [3, p. 418] that it "is sufficient" to keep the orders $2 \le \alpha + 2\beta \le 6$ in E_1; under that restricted E_1 the convergent bare graphs of higher order are extra standalone renormalised families in $\mathcal{R}_k$, bounded by lemma F.27(a) below, and nothing else changes.)

Definition F.19 (graphs, families, localisations)**.** Expanding $\log \Xi_k[\mathfrak{D}](0,0)$ in (s, λ') gives connected vacuum graphs [3, pp. 414–415], and $\mathrm{Gen}_k = \mathrm{Gen}^{\mathrm{bare}}_k + \mathfrak{B}_k$: bare graphs (all vertices from [3, (1.6), (1.7), (1.8), (1.10)], from the expansion of $-\Delta^{\eta}_{sgA'}$ and of V) and contour graphs (at least one product [3, (1.18)] of the form (1.13)·(1.14) or (1.14)·(1.14), from the expansion of $a_k|Q_k(sgA')\phi'|^2 - a_k|Q_k(0)\phi'|^2$; [3, (1.12)] does not occur since the block field is 0). Scalar lines are $G_k(\Omega, 0)$ (at $\phi = 0$ the Gaussian part is $-\frac12\langle\phi', (-\Delta^{\eta} + m^2 t_k^2 + a_k P_k(0))\phi'\rangle$ on Ω); vector lines are $G^{\mathrm{vec}}_k(T)$, the covariance of A' [3, p. 414]. Localisation follows [3, p. 420], the families are the G_{ren} of [3, p. 420], $X(\Lambda)$ is the sum of the terms of X with localisation $\Lambda = \{\square(v)\}$, and $d(\Lambda)$ the tree length of [3, (1.33)].

Lemma F.20 (the generated bare part and the counterterm differ only in their lines)**.** *Under Convention F.18,* $\mathrm{Gen}^{\mathrm{bare}}_k[\mathfrak{D}]$ *and* $\mathrm{Ct}_k[\mathfrak{D}]$ *are sums over the same graphs, with the same localisations and vertex coefficients, with lines* $(G_k(\Omega,0), G^{\mathrm{vec}}_k(T))$ *and* $(G_k(T,0), G^{\mathrm{vec}}_k(T))$ *respectively. Hence* $\mathcal{R}_k[\mathfrak{D}] = \mathfrak{B}_k[\mathfrak{D}] + \Delta_k[\mathfrak{D}]$ *with*

$$\Delta_k[\mathfrak{D}] := \sum_G \sum_{\ell} G\big[G_k(T,0) \text{ on the scalar lines before } \ell;\ \delta_k \text{ on } \ell;\ G_k(\Omega,0) \text{ after } \ell\big], \qquad \delta_k := G_k(\Omega,0) - G_k(T,$$

the scalar lines of each G ordered so that the lines of each closed scalar loop are consecutive.

Proof. At zero block fields the bare vertex functions of [3, (1.4)] come from the Taylor expansion of $U(sgA'_b) = \exp(qs\eta g(b)A'_b)$ inside $-\Delta^{\eta}_{sgA'}$ and from V: they are [3, (1.8), (1.10)] with $n' = 0$, $\tilde B = 0$, together with [3, (1.6), (1.7)]; the bare vacuum graphs of S^{ε} [3, p. 416] come from the same expansion of the lattice covariant Laplacian, rescaled, and Convention F.18(a), (b) give them the same localisation and coefficients. The generated scalar lines are $G_k(\Omega, 0)$; the counterterm lines have index $< k$, and by [3, (1.27)] the index-$< k$ part of $G^{\varepsilon}_K(0)$ is $G^{\varepsilon}_k(0)$, which rescales to $G_k(T, 0)$; the vector lines agree. Multilinearity in the scalar lines, $\prod_\ell a_\ell - \prod_\ell b_\ell =$

$\sum_\ell(\prod_{\ell'<\ell} b_{\ell'})(a_\ell - b_\ell)(\prod_{\ell'>\ell} a_{\ell'})$, gives Δ_k. S^ε has no block kernel, so no vertex [3, (1.13)–(1.15)] occurs in Ct_k, and the contour graphs, which have no counterterm, stay in $\mathfrak{B}_k$. □

The single terms of Δ_k are never bounded one by one: the vacuum graph made of one [3, (1.7)] vertex and one scalar loop gives $c(x)\delta_k(x,x)$ with $c \sim e(t)^2\eta^{-1}$, cancelled only together with the graph carrying the self-energy S with δ_k on the closing line, $[S-c]\delta_k = [S-S(T)]\delta_k + [S(T)-c]\delta_k$. So Δ_k, and every difference below, is used only through renormalised families whose members carry the telescoping order induced from G on G/S (lemma F.23); with the ordering of lemma F.20 every closed scalar loop not containing ℓ carries uniform full lines, all $G_k(T,0)$ or all $G_k(\Omega,0)$, and no closed loop is ever decomposed by fixed scales or orderings. In the measure convention of [3, (2.1)] a lone tadpole has $D=-1$ and Proposition 1 never bounds it alone.

Lemma F.21 (exact vacuum staging on a fixed datum)**.** *Fix a datum $\mathfrak{D}$ at scale k and let $\mathfrak{D}\uparrow$ be the same datum read at scale $k+1$ (same fine domains, vertex functions and coefficient c). Let $d\mu_k$ be the Gaussian law of the step-k fluctuations at zero $(k+1)$-block fields with the composing covariances $C^{(k)}$ (vector, torus) and $C^{(k)}(\Omega) := (\Delta^{(k)}(\Omega,0)+\alpha Q^*Q)^{-1}$ (scalar, lemma F.13 with H the Neumann operator on Ω, $B=0$). For a datum with localisation g let $\iota_k^g f$ be the Taylor series of $\tau \mapsto f(\tau g A'^{(k),\varepsilon} + eA^{(k+1),\varepsilon})$ at $\tau=0$ with $\tau \to s$ ($\iota_k^g = \iota_k$ on the torus; lemmas F.11 and F.12(a) hold verbatim for it, line integrals of gA' being additive). Put*

$$\mathrm{out}_k[\mathfrak{D}] := T_{[1,\bar n]} \log \int d\mu_k \exp\big(\iota_k^g F_k[\mathfrak{D}] - (\iota_k^g F_k)_0 + \mathcal{K}_k\big), \quad (\iota_k^g F_k)_0 := F_k|_{s=\lambda'=0,A=0} = -\tfrac12\langle\phi,\Delta^{(k)}(\Omega,0)\phi\rangle, \quad \mathcal{K}_k$$

the vacuum constant generated by step $k+1$. Then: (a) $\mathrm{Gen}_{k+1}[\mathfrak{D}\uparrow] = \mathrm{Gen}_k[\mathfrak{D}] + \mathrm{out}_k[\mathfrak{D}]$*;* (b) *the lines of $\mathrm{Gen}_{k+1}[\mathfrak{D}\uparrow]$ are $G_{k+1}(\Omega,0) = G_k(\Omega,0) + \beta_k^2 G_k(\Omega,0)R_k^* C^{(k)}(\Omega)R_k G_k(\Omega,0)$ and $G_{k+1}^{\mathrm{vec}} = G_k^{\mathrm{vec}} + (\text{lifted } C^{(k)})$, so $\mathrm{out}_k[\mathfrak{D}]$ is the part of $\mathrm{Gen}_{k+1}[\mathfrak{D}\uparrow]$ with at least one line of index k;* (c) *with $\mathcal{V}_k[\mathfrak{D}] := \mathrm{Ct}_{k+1}[\mathfrak{D}\uparrow] - \mathrm{Ct}_k[\mathfrak{D}]$ (the bare vacuum graphs with all lines of index $\le k$ and at least one of index k, localised and coefficiented as in $\mathfrak{D}$), exactly*

$$\mathcal{R}_{k+1}[\mathfrak{D}\uparrow] = \mathcal{R}_k[\mathfrak{D}] + \mathrm{out}_k[\mathfrak{D}] - \mathcal{V}_k[\mathfrak{D}].$$

Proof. Fubini on the nested integral: the $(k+1)$-step composite kernel on Ω factors into the step-$(k+1)$ kernels and the k-step composite (lemmas F.10 to F.12(a), all valid on Ω), the gauge part being lemma F.13(b). So $\Xi_{k+1}[\mathfrak{D}\uparrow](0,0) = N\int d\mu_k \exp(\iota_k^g F_k[\mathfrak{D}] - (\iota_k^g F_k)_0 + \mathcal{K}_k)\,\Xi_k[\mathfrak{D}](0,0)$ up to terms of degree $> \bar n$: only the zero-background degree-0 part $-\frac12\langle\phi,\Delta^{(k)}(\Omega,0)\phi\rangle$ and the term $\frac12\alpha|Q(0)\phi|^2$ combine with the step kernels into the Gaussian weight of $d\mu_k$, and every other term of $\iota_k^g F_k$, and $\mathcal{K}_k$, has degree ≥ 1 and is kept. (The degree-0 part of F_k depends on the background A through the scalar fluctuation determinants, Bałaban's $Z^{(j)}$ factors [2, (2.52), (2.92), (2.105)], and through $-\frac12\langle\phi,\Delta^{(k)}(A)\phi\rangle$; ι_k^g turns that dependence into terms of degree ≥ 1, which belong to out_k, as on [2, p. 576]: "in the density $\varrho'^{(k)}$ only the factors $Z^{(j)}$ depend on this field". $\mathcal{K}_k$ is the fluctuation dependence of the new scalar kernel at $\psi=0$; it gives the [3, (1.12)]-type contour vertices on the new outer segment Γ_{z,x_k}, and without it (a) is false.) Take logarithms; constants pass through, and lemma F.9 truncates: this is (a). (b) is lemma F.13(c), with H the Neumann operator on Ω for the scalar and on the torus for the vector; expanding multilinearly in the lines identifies out_k with the terms having at least one index-k line. (c): $\mathcal{R}_{k+1} - \mathcal{R}_k = (\mathrm{Gen}_{k+1} - \mathrm{Gen}_k) - (\mathrm{Ct}_{k+1} - \mathrm{Ct}_k)$, the coefficient c being the same in Gen and Ct and held fixed between k and $k+1$. □

On the torus datum $\Delta = 0$ and (c) reads $\mathrm{out}_k = \mathcal{V}_k + \mathfrak{B}_{k+1} - \mathfrak{B}_k$; on a general datum (c) holds with the Δ terms, and a term of $\mathcal{R}_k$ is never bounded in absolute value except where a datum changes.

Lemma F.22 (comparison of data)**.** *Let $\mathfrak{D},\mathfrak{D}'$ be data at the same scale.* (a) *(Propagator domain.) If $\mathfrak{D}'$ is $\mathfrak{D}$ with Ω replaced by an admissible $\Omega' \subset \Omega$ and the same vertex datum, then $\mathcal{R}_k[\mathfrak{D}] - \mathcal{R}_k[\mathfrak{D}']$ is a sum of families of the two forms of lemma F.20 (contour or Δ), each with one designated line $\delta G_k(\Omega,\Omega',0)$ (in Bałaban's list, [3, p. 414]); likewise $F_k[\mathfrak{D}] - F_k[\mathfrak{D}']$ is a sum of families with one designated $\delta G_k(\Omega,\Omega',\cdot)$ line or one designated δ in the*

external-field map $a_k G_k(\cdot) Q_k^*$ [3, p. 415]. (b) *(Vertex datum.) If* $\mathfrak{D}'$ *is* $\mathfrak{D}$ *with* (Ω_1, g, c) *replaced by* (Ω_1', g', c')*, then* $\mathcal{R}_k[\mathfrak{D}] - \mathcal{R}_k[\mathfrak{D}']$ *(likewise* out_k*,* F_k*) is a sum of families of the forms of lemma F.20 with one designated vertex of one of three kinds: (b1) a factor* $(g - g')$ *on one vector leg, with the same change on the corresponding leg of the* [3, (1.7)] *partner's coefficient graph (Convention F.18(c)); (b2) a* [3, (1.6)] *or* [3, (1.7)] *vertex in* $\Omega_1 \triangle \Omega_1'$*; (b3) a* [3, (1.7)] *vertex carrying a change of the g-independent finite part of c (*$\delta m^2_{\mathrm{fin}}$*,* $\delta m^2_{K,\cdot}$*, or a coefficient graph with an index-k line, all regular;* $|\mathit{change}| \le C_T t_k (1 + \ln t_k^{-1})^5$ *per site by theorem F.99, and* $\delta m^2_{K,k}(L^k\varepsilon)^2 = O(L^k\varepsilon)$ *by theorem F.93, as asserted on* [3, p. 418]*). The designated factor is supported in* $Z(\mathfrak{D}, \mathfrak{D}') := (\Omega_1 \triangle \Omega_1') \cup \mathrm{supp}(g - g') \cup \mathrm{supp}(c - c')$ *up to exponentially decaying tails of the regular coefficient graphs.* (c) *(Step covariances.) For two choices* C, C' *of the step-k covariances,* $\mathrm{out}_k[\mathfrak{D}; C] - \mathrm{out}_k[\mathfrak{D}; C']$ *is a sum of step-k cumulant families with one designated contraction by* $\delta C := C - C'$*, lifted by* $\beta_k G_k R_k^*$ *(lemma F.13(b)).*

Proof. Telescope multilinearly: in the lines for (a), in the vertex multipliers for (b), in the contraction kernels for (c). In (b) the telescoping is symmetric, $g' = g + (g' - g)$ and $\mathbf{1}_{\Omega_1'} = \mathbf{1}_{\Omega_1} + (\mathbf{1}_{\Omega_1'} - \mathbf{1}_{\Omega_1})$ on all legs and sites at once, applied identically to Gen, Ct and the coefficient graphs; every term then carries at least one factor $(g' - g)$ or a site difference, the members of a renormalised pair carry the same set of changed legs and sites (Convention F.18(c)), and the number of terms is $\le 2^{\text{legs+sites}} \le C(\bar{n})$. (A lone [3, (1.7)] vertex is divergent and the [3, (1.6)] tadpole needs its partner, so the kinds are not bounded vertex by vertex.) By Convention F.18 Gen and Ct carry the same vertices, so each difference term again has the form $\mathfrak{B} + \Delta$ of lemma F.20: no bare vacuum graph with an uncancelled overall divergence survives; by (c) of the convention the change on a self-energy subgraph and on its partner are made on corresponding legs, so renormalised families stay renormalised. A coefficient change splits into the g-localised part of [3, (1.29)], $\delta m^2_k(\cdot g)$, which changes with g and is paired with the generated subgraphs (kind (b1)), and the g-independent finite parts, which change only with the scale index ($\delta m^2_{K,k} \to \delta m^2_{K,k+1}$), the moved piece being an index-k slice, a regular graph (kind (b3)). Here $(g' - g)$ is itself a multiplier of the class (C^2-norm $\le 2C_g$) supported in Z. □

Lemma F.23 (grouped families)**.** *Let* X *be any of* $\Delta_k[\mathfrak{D}]$*,* $\mathcal{R}_k[\mathfrak{D}] - \mathcal{R}_k[\mathfrak{D}']$*,* $\mathrm{out}_k[\mathfrak{D}; C] - \mathrm{out}_k[\mathfrak{D}; C']$*, written with the line telescoping of lemma F.20 (order induced on* G/S *from* G*), the symmetric expansion of lemma F.22(b) for vertex multipliers and sites, and the multilinear expansion in contraction kernels. Then* X *is a finite sum, of at most* $C(\bar{n})$ *terms, of renormalised families in the sense of* [3, p. 420] *(each divergent two-scalar-leg subgraph* S *accompanied by* G/S *with the* [3, (1.7)] *vertex and the same changed lines, legs and sites outside* S*), each with exactly the designated object of lemma F.27(b)–(d).*

Proof. A term with δ_k on a line of S has S convergent (δ_k is regular on the diagonal); a term with δ_k off S pairs with the G/S term with δ_k on the corresponding line, and $S - c_S = [S - S(T)] + [S(T) - c_S]$, both convergent ($c_S$ uses torus lines, Convention F.18(b), [3, (1.31)]). The vertex expansion is symmetric, so the subsets of changed legs and sites of S and of its partner coincide. Contraction kernels are lifted index-k kernels (regular) or their deep differences. Polarisation subgraphs keep bubble and seagull with the same multipliers on corresponding legs, and the local cancellation survives C^2 multipliers, the subtraction $\sum_{x'} \Pi(x, x')[m(x') - m(x) - \nabla m(x) \cdot (x' - x)]$ being UV-finite in $d = 3$. The grouping is exactly [3, p. 414]'s line list $\{G(\Omega), G(\Omega_2), \delta G\}$. □

Lemma F.24 (geometry)**.** (a) $\mathrm{dist}(\Lambda_{i+1}, \Lambda_i^c) > r$ *for every* i *(the definition* [2, (2.8)]*, with* $r = r(t_k)$ *at scale* k*).* (b) *At scale* k*, with* c_d *a dimensional constant:* $\mathrm{dist}(B^k(\Lambda_7^{(k)}), B^k(\Lambda_5^{(k)})^c) > 2r(t_k) - c_d$*,* $\mathrm{dist}(B^k(\Lambda_7^{(k)}), B^k(\Lambda_2^{(k)})^c) > 5r(t_k) - c_d$*, and* $\mathrm{dist}(B^k(\Lambda_7^{(k)}), B^k(\Lambda_7^{(k-1)'})^c) > 7r(t_k) - c_d$*, because* $\Lambda_7^{(k)} \subset \Lambda_0^{(k)} \subset \Lambda_7^{(k-1)'}$ [2, p. 570]. (c) $\rho(\mathfrak{D}_k) \ge (5/L) r(t_{k-1}) - c_d \ge r(t_k) - c_d$ *at* $L = 5$ *(the primed sets are the scale-*$(k-1)$ *sets read on* $T_1^{(k)}$*, distances divided by* L*, inclusions kept).* (d) *If* $7r(t_k) - c_d \ge M$ *then* $g_k \equiv 1$ *on* $B^k(\Lambda_7^{(k)})$*, and the coefficient* [3, (1.29)] *there differs from the full* δm^2 *only by tails* $e^{-\delta(7r - M)}$*.* (e) *Put* $\mathrm{Col}_k := \Lambda_7^{(k-1)'} \cap \Lambda_7^{(k)c}$ *and let* $D < 7r(t_k) - c_d$*; every scale-k block* $b \subset \Lambda_7^{(k)}$ *with* $\mathrm{dist}(b, \Lambda_7^{(k)c}) \le D$ *lies within* $D + 1$ *of a block of* Col_k*, hence for*

$\delta>0$, $\sum_{b\subset\Lambda_7^{(k)}} e^{-\delta\,\mathrm{dist}(b,\Lambda_7^{(k)c})}\le C(\delta)|\mathrm{Col}_k|_k+e^{-\delta(7r(t_k)-c_d)}N_k$ *with* $C(\delta):=\sum_{D\ge0}(2D+3)^3e^{-\delta D}$ *and* $N_k:=|\Lambda_7^{(k)}|_k$. (f) *Any zone Z of blocks within distance $D_Z<7r(t_k)-c_d$ of $\partial B^k(\Lambda_7^{(k)})$, on either side, has $|Z|_k\le(1+(2D_Z+3)^3)|\mathrm{Col}_k|_k$.*

Proof. (a): by [2, (2.8)], Λ_{i+1}^c is the union of the large blocks at distance $\le r$ from Λ_i^c. (b): iterate (a) along a shortest path; each crossing of a shell costs at least r up to the block size. (c), (d): from (b) and the definition of g_k. (e): the nearest point of $\Lambda_7^{(k)c}$ to b is within D, and since $(\Lambda_7^{(k-1)'})^c$ is farther than $7r-c_d>D$ that point lies in Col_k; the map $b\mapsto$ (a collar block within $D+1$) has fibres of size $\le(2D+3)^3$; sum over D. (f): the blocks of Z outside $\Lambda_7^{(k)}$ lie in Col_k (at most $|\mathrm{Col}_k|_k$ of them), those inside are covered by (e). □

Step 3(D_2): the counterterm moves of [3, pp. 411–422]*, bounds.*
Inputs. (I1) Proposition 1 in the scope Bałaban states for it: (I1a) the expressions of the expansion [3, pp. 414–415] (vertices [3, (1.6)–(1.15)] and products [3, (1.18)]; propagators $G_k(\Omega,\tilde B)$, $G_k(\Omega_2,\tilde B)$ and the vector G_k on any admissible $\Omega\supset\Omega_2$, the torus included; vacuum expressions included, since vacuum contour graphs are among "an arbitrary expression in the expansion", [3, p. 414], with $D(G)\ge2$); (I1b) $\delta G_k(\Omega,\Omega_2,\tilde B)$ and [3, (1.16)] treated as external fields [3, pp. 414, 419]; (I1c) external fields that are functions of many variables, in particular "expressions obtained by an integration and a cumulant expansion in kth step, where some external legs are contracted by propagators $C^{(k)}_{\Lambda_5^{(k)}}$" [3, p. 419]; (I1d) the localisation of the vector legs by smooth partitions of unity [3, p. 420], i.e. smooth multipliers on single legs from a fixed C^2-bounded class, with the $O(1)$ uniform in them, in the domains and in $\tilde B$ [3, p. 421]. (I2) Norms of the external fields: (I2a) δG at depth, from [4, Thm., (1.10)–(1.12)] in its restricted form made two-sided (lemma F.148): for cubes $\square,\square'$ at depths $\rho_\square,\rho_{\square'}\ge R_0$ from Ω'^c, $\|h_\square\delta G_k(\Omega,\Omega')h_{\square'}\|_{1,\alpha_0}\le Ce^{-(\delta_0/4)(\rho_\square+\rho_{\square'}+\mathrm{dist}(\square,\square'))}$; (I2b) block-level covariance differences (C_Λ against $C^{(k)}$, lemma F.147; domain differences of $\Delta^{(k)}$ and C_Λ, lemma F.149), lifted by $a_kG_kQ_k^*$ whose C^{1,α_0} regularity is [4, Thm., (1.9)–(1.10)] and [2, Prop. 2.2, (2.58)], vector versions at $B=0$ componentwise; (I2c) the lifted index-k kernels $\beta_kG_kR_k^*C^{(k)}R_kG_k\beta_k$, with the same inputs and the decay of $C^{(k)}$ (lemma F.145(c)). (I3) The size of the finite mass vertices: in scale-k units the non-renormalising coefficient $c_k=t_k^2(\delta m^2_{\mathrm{fin}}+\delta m^2_{K,k})$ of the [3, (1.7)] vertex obeys $|c_k|\le C_Tt_k(1+\ln t_k^{-1})^5$ with C_T free of ε_0 (theorem F.99; asserted on [3, p. 418] as $\delta m^2_{K,k}t_k^2=O(t_k)$, $\delta m^2_{\mathrm{fin}}=O(1)$, the first of which is theorem F.93), entering only through Proposition 1's order count ([3, p. 421]: "the vertices (1.7) give also a positive power of $L^k\varepsilon$") and in kind (b3) of lemma F.22; every bound of Steps 3(D_1)–3(D_2) stated with a factor t_k from such a vertex holds with $t_k(1+\ln t_k^{-1})^5$ in its place, which its consumers absorb (lemma F.101). Geometry: [2, (2.7)–(2.8)] and the bound $e^{-\delta c_*R(1+\log t^{-1})^\varrho}\le C_\kappa t^\kappa$ for every κ, uniformly in $t\in(0,1]$ ($\varrho>1$; lemma F.153).

Lemma F.25 (gluing two families along contractions)**.** *Let F_1,F_2 be families obeying* [3, (1.33)] *with multi-variable external fields on their legs, $|E(F_i,\Phi)|\le P_ie^{-\delta_0d_i}\|h\Phi\|_{1,\alpha_0}$, and let $K_1,\dots,K_r$ be two-variable kernels joining a leg of F_1 (cube $\square_j$) to a leg of F_2 (cube $\square'_j$), with $\|h_{\square_j}K_jh_{\square'_j}\|_{(1,\alpha_0)\otimes(1,\alpha_0)}\le\kappa_je^{-\delta'\,\mathrm{dist}(\square_j,\square'_j)}$ (the C^{1,α_0}-norm in each variable uniformly in the other, with the mixed derivative and Hölder terms). Then $|E(F_1\circ_KF_2,\Phi_{\mathrm{rest}})|\le P_1P_2\prod_j\kappa_j\,e^{-\delta_0(d_1+d_2)}e^{-\delta'\sum_j\mathrm{dist}(\square_j,\square'_j)}\|h\Phi_{\mathrm{rest}}\|$.*

Proof. $E(F_1\circ_KF_2)=E(F_1,\Psi\otimes\Phi_1)$ with $\Psi(x_1,\dots,x_r):=E(F_2,K_1(x_1,\cdot)\otimes\cdots\otimes K_r(x_r,\cdot)\otimes\Phi_2)$. Ψ is linear in each $K_j(x_j,\cdot)$; its sup, derivative and Hölder quotient in x_j are $E(F_2,\cdot)$ applied to $K_j(x_j,\cdot)$, $\partial_{x_j}K_j(x_j,\cdot)$ and the Hölder quotients of K_j in x_j, each bounded by $P_2e^{-\delta_0d_2}$ times the mixed norm. So $\|h\Psi\|_{1,\alpha_0}\le P_2e^{-\delta_0d_2}\prod_j\kappa_je^{-\delta'\,\mathrm{dist}}$; apply [3, (1.33)] to F_1. For $m\ge3$ families (a cumulant κ_m) induct, joining F_m to the group already joined. □

Proposition 1's many-variable norm ([3, p. 420]: [3, (1.32)] "extends in a natural way") is C^{1,α_0} per variable; mixed derivatives enter only through the joins of lemma F.25, where the

lifted kernels and their deep differences factorise with one derivative on each lifting factor; δG lines never serve as joining kernels, and self-contractions inside one family are many-variable external fields of that family (I1c).

Lemma F.26 (localisation sums)**.** *Let $N \le \bar N$ be the number of indices and $\delta > 0$. Then $\sum_{\Lambda:\square_{\iota_0}=\square_0} e^{-\delta d(\Lambda)} \le S_N(\delta) := N^{N-2}(e^{\delta}\sigma(\delta))^{N-1}$, uniformly in the torus, with $\sigma(\delta) := \sup_z \sum_{\square} e^{-\delta \operatorname{dist}(\square,z)}$. Consequently, with $S := \max_{N\le\bar N} S_N(\delta)$: for a set Z of blocks, $\sum_{\Lambda:\text{designated cube in } Z} Ct_k e^{-\delta d(\Lambda)} \le C\bar N S t_k |Z|_k$, and $\sum_{\Lambda:\text{anchor in the vertex domain}} Ct_k e^{-\delta\rho} e^{-\delta d(\Lambda)} \le C\bar N S t_k e^{-\delta\rho} N_k$.*

Proof. $e^{-\delta d} \le \sum_{\text{trees}} e^{-\delta L_T}$; there are N^{N-2} trees (Cayley); sum the cubes from the leaves inward, each edge contributing $\le e^{\delta}\sigma(\delta)$. □

Lemma F.27 (bounds per family and per localisation, from Proposition 1)**.** *There are $C_P < \infty$ and $\delta > 0$, depending only on $(d, N, \|q\|, a, L, M, \bar n, \alpha_0, R, C_g)$ and on the constants of (I1)–(I3), not on ε, k, the torus or the data, such that for every localisation Λ:* (a) *every grouped contour family ($\mathfrak B$-type, lemma F.23) on any datum obeys $|\cdot(\Lambda)| \le C_P t_k e^{-\delta d(\Lambda)}$;* (b) *a grouped Δ-type family, or any grouped family with a designated δG line or a designated δ in the external-field map, whose designated object sits at or joins cubes at depth $\ge \rho \ge R_0+2$ from the boundary where the two propagators differ, obeys $|\cdot(\Lambda)| \le C_P t_k e^{-\delta\rho} e^{-\delta d(\Lambda)}$;* (c) *every grouped step-k cumulant family (a term of out_k on any datum) obeys $|\cdot(\Lambda)| \le C_P t_k e^{-\delta d(\Lambda)}$, and with a designated δC contraction at depth $\ge \rho$ it obeys $|\cdot(\Lambda)| \le C_P t_k e^{-\delta\rho} e^{-\delta d(\Lambda)}$;* (d) *the families of (a)–(c) with one designated vertex of kind (b1), (b2) or (b3) of lemma F.22 obey the same bounds, and their Λ contains a cube that meets the zone Z.*

Proof. Scope. Each family lies in the scope of (I1): (a) is (I1a) without external field; (b) is (I1b) with the designated δG as a two-variable external field; (c) is (I1c), with lemma F.25 when a cumulant joins several families; (d) is (I1d): for (b1) the designated leg carries the multiplier $(g'-g)$, in the class and supported in Z; for (b2) the scalar vertices are localised by unit cubes anyway [3, p. 420]; for (b3) the vertex is a finite mass vertex with coefficient $\le Ct_k$ (I3), in Proposition 1's class inside the families of lemma F.22 (its loop line is δ_k, a lifted index-k line, or not a pure full-line loop), never alone. (A lone out-term is not bounded either: a figure-eight with one full and one index-k line is $\sim \lambda t \eta^{-1}$; the pairs in out_k are inherited from the renormalised vacuum part of F_k.) *Order.* Every vacuum family has order $o := d_v/2 + d_s \ge 1$: all A'-legs are contracted, so d_v is even, and if $d_v = 0$ the family contains a [3, (1.6)] vertex or a [3, (1.7)] vertex of positive order; so $e(t_k)^{d_v}\lambda(t_k)^{d_s} = e^{d_v}\lambda^{d_s} t_k^{o} \le Ct_k$ for $t_k \le 1$; a step-k cumulant family is a join of families of order ≥ 1. *External fields.* In (b) and (c), (I2a)–(I2c) bound the designated objects by $Ce^{-(\delta'/4)(\rho_\square+\rho_{\square'}+\operatorname{dist}(\square,\square'))}$ with $\rho_\square, \rho_{\square'} \ge \rho$, so $e^{-\delta'(\rho_\square+\rho_{\square'})/4} \le e^{-\delta'\rho/2}$ and $e^{-\delta' \operatorname{dist}(\square,\square')/4}$ supplies the tree link between the two cubes; the other contractions in (c) are lifted index-k kernels with $\|h\cdot h'\| \le Ce^{-\delta' \operatorname{dist}}$ (I2c). *Assembly.* [3, (1.33)] gives $O(1)Ct_k e^{-\delta_0 d}\prod(\text{norms})$; the tree length over all cubes is at most the sum of the constituent tree lengths and the link lengths. Take $\delta := \frac12 \min(\delta_0, \delta'/4)$ and $C_P :=$ (number of families of order $\le \bar n$) $\cdot\, O(1) \cdot C$. □

Theorem F.28 ($(\mathrm{VAC}^\flat)_k$, $0 \le k \le K$)**.** *Let $\mathfrak D$ be a datum at scale k with $\rho(\mathfrak D) \ge R_0 + 3$. Then for every localisation Λ, $|\mathcal R_k[\mathfrak D](\Lambda)| \le C_V t_k e^{-\delta d(\Lambda)}$ with $C_V := 2N_{\mathrm{fam}}C_P$, N_{fam} the number of families of order $\le \bar n$; consequently $|\mathcal R_k[\mathfrak D]| \le C_V \bar N S t_k |\Omega_1|_k$. At $\mathfrak D = \mathfrak D_K$ this gives $|\mathcal R_K| \le C_V \bar N S t_K |T|_K = C_V \bar N S t_K^{-2}|T| \le C_V \bar N S L^2 \varepsilon_0^{-2}|T|$, since $t_K \in (\varepsilon_0/L, \varepsilon_0]$; at $\mathfrak D = \mathfrak D_k$ the hypothesis holds, $\rho(\mathfrak D_k) \ge r(t_k) - c_d \ge R - c_d \ge R_0 + 3$ (lemma F.24(c) at $L = 5$), given $R \ge R_0 + 3 + c_d$ (admissible: R is chosen after M, [2, (2.7)]).*

Proof. $\mathcal R_k = \mathfrak B_k + \Delta_k$ (lemma F.20). $\mathfrak B_k$ is bounded by lemma F.27(a). Δ_k is bounded by lemma F.27(b) through its grouped families (lemma F.23): the designated $\delta_k = G_k(\Omega, 0) - G_k(T, 0)$ sits between vertices of Ω_1 plus one block, at depth $\ge \rho(\mathfrak D) - 1 \ge R_0 + 2$, as (b) requires. Sum over localisations with lemma F.26. □

Corollary F.29 (the vacuum clause of [2, Prop. 2.1, (2.57)] at zero background)**.** *Let $\mathfrak D' := (\Omega^{(k)}, B^k(\Lambda_7^{(k)}), g', c')$ with g' admissible for $B^k(\Lambda_7^{(k)})$ and c' its [3, (1.29)]. Then $\mathcal R_k[\mathfrak D_k] - \mathcal R_k[\mathfrak D']$*

is a sum of families with a designated vertex in $Z_{\mathrm{col}} := B^k(\mathrm{Col}_k) \cup \{g' < 1\}$*, and* $|\mathcal{R}_k[\mathfrak{D}_k] - \mathcal{R}_k[\mathfrak{D}']| \le Ct_k|\mathrm{Col}_k|_k$ *with* $C := C_P\bar{N}SC(\delta)(1+(2M+5)^3)$.

At $A^{(k)} = 0$, $\phi = 0$ the quartic term of [2, (2.57)] vanishes and E_{fin}, $E_{K,k}$ are unchanged, so this is the c-number clause of [2, (2.57)], with $O(t_k) \le O(t_k^{\kappa_0})$ per collar block, under Convention F.18 (the mass counterterm enters as a vertex coefficient, and the replacement of propagators acts on lines only).

Proof. Apply lemma F.22(b): the vertex domains differ by $\Omega_1 \triangle \Omega_1' = B^k(\mathrm{Col}_k)$; on $B^k(\Lambda_7^{(k)})$, $g_k = 1$ (lemma F.24(d)), so $\mathrm{supp}(g_k - g') \subset B^k(\mathrm{Col}_k) \cup \{g' < 1\}$ and $c_k - c'$ is supported there up to tails $e^{-\delta(7r-M)}$. Apply lemma F.27(d), then lemma F.24(f) with $D_Z = M+1$, then lemma F.26; the coefficient tails are summed by $\sum_x e^{-\delta\,\mathrm{dist}(x,Z)} \le C(\delta)|Z|$. □

Lemma F.30 (the new fluctuation at step $k+1$, as [2, pp. 575–581] treats it)**.** *At step* $k+1$, [2] *writes* $\theta_{k+1}A^{(k)} = \theta_{k+1}A'^{(k)} + \tilde{B}^{(k)} + \theta_{k+1}B^{(k+1)}$ [2, (2.88)–(2.89)], θ_{k+1} *a smooth cutoff of* $B^k(\Lambda_2^{(k)})$ *changing in a slice of thickness* M*, keeps* $(1-\theta_{k+1})A^{(k)}$ *in the background* $\tilde{B}$ [2, p. 579]*; expands the* $Z^{(j)}$ *factors* [2, (2.92)], $j < k$*, on their domains* $B^j(\Lambda_2^{(j)})$ *with restrictions* $\Lambda_5^{(j)}$*, moves them to* $B^k(\Lambda_0^{(k)})$*, composes them by* [2, (2.105)] *into the polynomial* $Q^{(k)}$ [2, (2.106)]*; drops the terms with vertices localised in* $B^k(\Lambda_7^{(k)c})$ *and relocalises with a smooth* $g^\sharp$ *(*$\mathrm{supp}\, g^\sharp \subset \Lambda_7^{(k)}$*,* $g^\sharp = 1$ *on* $\Lambda_8^{(k)}$*)* [2, p. 579]*; replaces the lines by* $G_k(B^k(\Lambda_2^{(k)}), B^{(k+1)})$ [2, Prop. 2.6, (2.107)]*; and integrates* A', ϕ' *conditionally on* $\Lambda_5^{(k)c}$ *with* $C^{(k)}_{\Lambda_5^{(k)}}$ [2, (2.115)]*. At zero* $(k+1)$*-fields (so* $\tilde{B}^{(k)} = 0 = \theta_{k+1}B^{(k+1)}$*) write* out_k^θ *for the vacuum generated before the drop, and* $\mathrm{out}_k^\theta = \mathrm{out}_k^\sharp + D_k$*, where* D_k *collects the terms with at least one new leg carrying* $(1-g^\sharp)\theta_{k+1}$ *and* $\mathrm{out}_k^\sharp$ *is kept in the action; the legs of the new-kernel term* $\mathcal{K}_k$ *of lemma F.21 count as new legs. Then, with* out_k *of lemma F.21 on* $\mathfrak{D}' = (\Omega^{(k)}, W, g', c')$*,* $W := B^k(\Lambda_7^{(k)})$*,*

$$\mathrm{out}_k^\sharp = \mathrm{out}_k[\mathfrak{D}'; C_{\mathrm{act}}] + X_0 + X_3,$$

with $|X_0| \le Ct_k e^{-\delta(5r(t_k)-M-c_d)}N_k$ *(the domain moves and the conditioned background* $\tilde{B} = \sum_{l\le k}(1-\theta_{l+1})\theta_l A^{(l)}$ *at zero new fields, every slice of which lies in* $B^{l-1}(\Lambda_2^{(l-1)}) \setminus B^l(\Lambda_2^{(l)}) \subset B^k(\Lambda_2^{(k)})^c$*, at distance* $\ge 5r(t_k) - M - c_d$ *from* W*),* $|X_3| \le Ct_k(2D+5)^3|\mathrm{Col}_k|_k$ *with* $D := \max(M, D^\sharp)$*,* $D^\sharp$ *the transition width of* $g^\sharp$ *(families with one new leg carrying* $(g^\sharp - g')$*, supported within* D *of* ∂W*), and* $|D_k| \le Ct_k(2+(2D+3)^3)|\mathrm{Col}_k|_k$ *(the support of* $(1-g^\sharp)\theta_{k+1}$ *lies in* $B^k(\Lambda_2^{(k)} \setminus \Lambda_7^{(k)}) \cup (\theta\text{-slice}) \cup \{g^\sharp < 1\}$*, the first two in* $B^k(\mathrm{Col}_k)$ *since* $\Lambda_2^{(k)} \subset \Lambda_0^{(k)} \subset \Lambda_7^{(k-1)'}$ *and* $\mathrm{dist}(\Lambda_2^{(k)}, \Lambda_7^{(k-1)'c}) > 2r(t_k) > M$*). In all three,* $t_k(2D+5)^3 \le C't_k^{\kappa_0}$*,* $\kappa_0 = 1/5$*, even for* $D \sim r(t_k)$.

Proof. All three are differences of grouped families (lemma F.23) with one designated object: a deep domain or background change (X_0: the Combes–Thomas bound for the background resolvent difference, (I2a) for domains, (I2b) for the $\Lambda_5^{(j)}$ restrictions; W is at depth $\ge 5r(t_k) - c_d$ in $B^k(\Lambda_2^{(k)})$ and nested inside every $B^j(\Lambda_2^{(j)})$, $\Lambda_5^{(j)}$), or a smooth multiplier $(g^\sharp - g')$ or $(1-g^\sharp)\theta_{k+1}$ on one new leg (C^2-bounded, (I1d)). Bound each by lemma F.27 and sum with lemmas F.24 and F.26(f); $\theta_k = 1 = \theta_{k+1}$ on $\mathrm{supp}\, g^\sharp$, so $g^\sharp\theta_k\theta_{k+1} = g^\sharp$. The last claim is $t(1+\log t^{-1})^{3\varrho} \le Ct^{1/5}$. □

Theorem F.31 ($(\mathrm{DOM})_k$, $0 \le k < K$)**.** *Bałaban's step* $k+1$ *([3, p. 412], [2, pp. 569–582]) consists of: (O1) the collar removal* [2, (2.57)]*, down to a vertex datum* $(B^k(\Lambda_7^{(k)}), g', c')$*; (O2) the translation and the expansion in the new fluctuation; (O3) the integration of the step-k fluctuations with the covariances* $C^{(k)}_{\Lambda_5^{(k)}}(B^k(\Lambda_2^{(k)}), 0)$ *(scalar) and* $C^{(k)}_{\Lambda_5^{(k)}}$ *(vector) of* [3, p. 419]*, and the cumulant expansion; (O4) the counterterm move* $k \to k+1$ *of* [3, p. 422]*,* $E_{K,k} = E_{K,k+1} + \mathcal{V}_k^{\mathrm{glob}}$ *and* $\delta m^2_{K,k} \to \delta m^2_{K,k+1}$ *with the index-k slice moved into* δm^2_{k+1}*, followed by*

localisation; (O5) the rescaling to scale $k+1$ and the new datum $\mathfrak{D}_{k+1}$. Let $\mathrm{Vac}^{\mathrm{act}}_{k+1}$ be the field-independent part of degree $1..\bar{n}$ of the resulting polynomial, in the logarithmic convention with the front factor $e^{-E_{\mathrm{fin}}-E_{K,k+1}}$, the removed collar constant $\mathcal{R}_k[\mathfrak{D}_k]-\mathcal{R}_k[\mathfrak{D}']$ (corollary F.29) and the term D_k of lemma F.30 being dropped and booked by their absolute values. Then

$$\mathrm{Vac}^{\mathrm{act}}_{k+1} = \mathcal{R}_{k+1}[\mathfrak{D}_{k+1}] + \mathcal{E}'_k + \mathcal{O}_k,$$

where $\mathcal{E}'_k$, $\mathcal{O}_k$ are finite sums of localised vacuum families with $|\mathcal{E}'_k| \le Ct_k(2D+5)^3|\mathrm{Col}_k|_k + Ct_k e^{-\delta c_* r(t_k)}N_k \le C' t_k^{\kappa_0}|\mathrm{Col}_k|_k + C' t_k^{16/5} N_k$ (D as in lemma F.30, $\kappa_0 = 1/5$), $|\mathcal{O}_k| \le Ct_k|\Lambda_7^{(k)c}|_k$, every family of $\mathcal{O}_k$ having a designated vertex in $\Lambda_7^{(k)c}$; the constants do not depend on ε, k, the torus or the regions. Equivalently, the field-independent output of step k, $\mathrm{out}^{\mathrm{act}}_k - \mathcal{V}^{\mathrm{glob}}_k$, equals $\mathcal{R}_{k+1}[\mathfrak{D}_{k+1}] - \mathcal{R}_k[\mathfrak{D}'] + \mathcal{E}'_k + \mathcal{O}_k$.

Proof. Put $W := B^k(\Lambda_7^{(k)})$, $\Omega^\natural := B^k(\Lambda_2^{(k)})$, $\mathfrak{D}' := (\Omega^{(k)}, W, g', c')$, $\mathfrak{D}^\natural := (\Omega^\natural, W, g', c')$; $\Omega^\natural \subset \Omega^{(k)}$ by the nesting $\Lambda_2^{(k)} \subset \Lambda_0^{(k)} \subset \Lambda_7^{(k-1)'} \subset \Lambda_2^{(k-1)'}$ [2, p. 570].

1. After (O1) the vacuum part is $\mathcal{R}_k[\mathfrak{D}']$, the front factor unchanged; the dropped constant is bounded by corollary F.29.

2. (Input domain.) By lemma F.22(a), $\mathcal{R}_k[\mathfrak{D}'] = \mathcal{R}_k[\mathfrak{D}^\natural] + X_1$ and $F_k[\mathfrak{D}'] = F_k[\mathfrak{D}^\natural] +$ (δ-families), the designated object $\delta G_k(\Omega^{(k)}, \Omega^\natural)$ at depth $\ge 5r(t_k) - c_d$ (lemma F.24(b)).

3. (Step covariances.) $\mathrm{out}^{\mathrm{act}}_k := \mathrm{out}^\sharp_k = \mathrm{out}_k[\mathfrak{D}'; C_{\mathrm{act}}] + X_0 + X_3$ (lemma F.30), and $\mathrm{out}_k[\mathfrak{D}'; C_{\mathrm{act}}] = \mathrm{out}_k[\mathfrak{D}^\natural; C_{\mathrm{comp}}] + X_2$ by lemma F.22(a) for the input F_k (step 2, then integrated) and lemma F.22(c) for $C_{\mathrm{act}} - C_{\mathrm{comp}}$: scalar $C^{(k)}_{\Lambda_5^{(k)}}(\Omega^\natural, 0) - C^{(k)}(\Omega^\natural, 0)$, vector $C^{(k)}_{\Lambda_5^{(k)}} - C^{(k)}$ (I2b), every designated object at depth $\ge 2r(t_k) - c_d$ from $\Lambda_5^{(k)c}$. By lemma F.27(b), (c) and lemma F.26, $|X_1| + |X_2| \le Ct_k e^{-\delta(2r(t_k)-c_d)}N_k$.

4. (Exact staging.) lemma F.21(c) on $\mathfrak{D}^\natural$: $\mathcal{R}_k[\mathfrak{D}^\natural] + \mathrm{out}_k[\mathfrak{D}^\natural; C_{\mathrm{comp}}] - \mathcal{V}_k[\mathfrak{D}^\natural] = \mathcal{R}_{k+1}[\mathfrak{D}^\natural\uparrow]$.

5. (The counterterm move.) The front factor splits, $e^{-E_{K,k}} = e^{-E_{K,k+1}} e^{-\mathcal{V}^{\mathrm{glob}}_k}$ with $\mathcal{V}^{\mathrm{glob}}_k = \mathcal{V}_k[\mathfrak{D}^T]$ (largest line index exactly k, full coefficient δm^2). Ct does not depend on the propagator domain, so $\mathcal{V}_k[\mathfrak{D}^\natural] = \mathcal{V}_k[\mathfrak{D}^T_{W,g'}]$. lemma F.21(c) on the two torus data (where $\Delta = 0$) gives $\mathcal{V}_k[\mathfrak{D}^T] = \mathrm{out}_k[\mathfrak{D}^T] - \mathfrak{B}_{k+1}[\mathfrak{D}^T\uparrow] + \mathfrak{B}_k[\mathfrak{D}^T]$ and the same for $\mathfrak{D}^T_{W,g'}$, hence

$$Y_k := \mathcal{V}^{\mathrm{glob}}_k - \mathcal{V}_k[\mathfrak{D}^\natural] = \big[\mathrm{out}_k[\mathfrak{D}^T] - \mathrm{out}_k[\mathfrak{D}^T_{W,g'}]\big] - \big[\mathfrak{B}_{k+1}[\mathfrak{D}^T\uparrow] - \mathfrak{B}_{k+1}[\mathfrak{D}^T_{W,g'}\uparrow]\big] + \big[\mathfrak{B}_k[\mathfrak{D}^T] - \mathfrak{B}_k[\mathfrak{D}^T_{W,g'}]\big].$$

By lemma F.22(b) each bracket is a sum of families with a designated vertex in $Z^T := W^c \cup \{g' < 1\}$ or in the zone where the coefficient changes; so Y_k needs no bound on index-k slices alone, only Proposition 1 for out-type and contour families. Sort by the cube of the designated vertex: families with that cube in $\Lambda_7^{(k)c}$ form $-\mathcal{O}_k$, families with it in W (then within $M+1$ of ∂W) go into $\mathcal{E}_k$. By lemma F.27(a), (c), (d), lemma F.26 and lemma F.24(f): $|\mathcal{O}_k| \le Ct_k|\Lambda_7^{(k)c}|_k$ and $|Y_k + \mathcal{O}_k| \le Ct_k(1 + (2M+5)^3)|\mathrm{Col}_k|_k$ ($\mathfrak{B}_{k+1}$ lives at scale $k+1$, which changes the constants by at most L^3).

6. (The new datum.) Compare $\mathfrak{D}_{k+1}$ with $\mathfrak{D}^\natural\uparrow$. The propagator domain changes from $\Omega^\natural$ to $B^{k+1}(\Lambda_2^{(k)'})$, a deep change (lemma F.22(a), depth $\ge 5r - c_d$). The vertex datum changes from (W, g', c') to $(B^{k+1}(\Lambda_7^{(k)'}), g_{k+1}, c_{k+1})$, including the coefficient move of [3, (1.28)–(1.29)], $c' - c_{k+1} = [\delta m^2_{[k]}(1) - \delta m^2_{[k]}(g_{k+1})] + [\delta m^2_k(g') - \delta m^2_k(g_{k+1})]$ with $\delta m^2_{[k]}$ the index-k slice (kinds (b3) and (b1)), all supported within $D_6 := M + LM + 2$ of ∂W up to exponential tails. By lemma F.22(b), lemma F.27(d) and lemma F.24(f), $|\mathcal{R}_{k+1}[\mathfrak{D}^\natural\uparrow] - \mathcal{R}_{k+1}[\mathfrak{D}_{k+1}]| \le Ct_k(1 + (2D_6+3)^3)|\mathrm{Col}_k|_k + Ct_k e^{-\delta(5r-c_d)}N_k$. (If the new legs carry $g^\sharp$, start from the mixed datum, g' on the old legs and $g^\sharp$ on the new; the difference is X_3.) The rescaling is exact.

7. (Assembly.) $\mathrm{Vac}^{\mathrm{act}}_{k+1} = \mathcal{R}_k[\mathfrak{D}'] + \mathrm{out}^{\mathrm{act}}_k - \mathcal{V}^{\mathrm{glob}}_k = \mathcal{R}_k[\mathfrak{D}^\natural] + X_1 + \mathrm{out}_k[\mathfrak{D}^\natural; C_{\mathrm{comp}}] + X_2 + X_0 + X_3 - \mathcal{V}_k[\mathfrak{D}^\natural] - Y_k = \mathcal{R}_{k+1}[\mathfrak{D}^\natural\uparrow] + X_1 + X_2 + X_0 + X_3 - Y_k = \mathcal{R}_{k+1}[\mathfrak{D}_{k+1}] + \mathcal{E}'_k + \mathcal{O}_k$, with $\mathcal{E}_k := X_1 + X_2 - (Y_k + \mathcal{O}_k) + (\mathcal{R}_{k+1}[\mathfrak{D}^\natural\uparrow] - \mathcal{R}_{k+1}[\mathfrak{D}_{k+1}])$ and $\mathcal{E}'_k := \mathcal{E}_k + X_0 + X_3$.

8. (The exponents.) $t_k \le t_k^{\kappa_0}$, and $e^{-\delta c_* R(1+\log t^{-1})^{\varrho}} \le C_\kappa t^\kappa$ for every κ gives $te^{-\delta c_* r(t)} \le Ct^{16/5}$, likewise for X_0's factor. (C_κ is free of ε_0 but can be very large; it is absorbed in the final constants $E_\pm$.) □

No term of $\mathcal{R}_k$ is bounded in absolute value except its collar part (corollary F.29) and differences with a designated object deep inside or in a zone; the restrictions χ, the non-polynomial remainders and the degree-0 normalisations are Step 3(C). The field-dependent analogue ([3, p. 412]: [3, (1.5)] after $k+1$ steps on the new domains) follows from lemmas F.21 and F.22 in the same way, with Proposition 1 under the weaker restrictions on new scalar fields [3, p. 421]; its interaction-term half at a nonzero background is (BAL$_5$)[int] (lemma 5.1).
The bad localisations of [3, p. 422]. "Passing from k to $k+1$ we have to change ... the terms in $\delta m^2_{K,k}(x)(L^k\varepsilon)^2$ and $E_{K,k}$ for which the operators in the decomposition (1.27) have the indices $j \le k$ and at least one of them is equal to k ... then localize them properly, which means that we have to estimate the terms with bad localizations"; the moved line of index k "is a regular function of both variables and we can treat it as an external field". Write $S_{(k)}$ for the slice of index k of [3, (1.27)], $G_{k+1} = G_k + S_{(k)}$.

Lemma F.32 (the transfer identity)**.** *On the torus at zero background, $\delta m^2_{K,k} - \delta m^2_{K,k+1} = \delta m^2_{k+1} - \delta m^2_k =: m_k$, and m_k is the sum of the graphs of [3, (1.23)] whose lines are all G_k or $S_{(k)}$ with at least one $S_{(k)}$ line; the same holds for $E_{K,k} - E_{K,k+1}$.*

Proof. By [3, (1.27)], $G_{K,k} = S_{(k)} + G_{K,k+1}$ and $G_{k+1} = G_k + S_{(k)}$, for scalar and vector lines alike. Expand each graph of δm^2_K multilinearly with every line written $G_k + S_{(k)} + G_{K,k+1}$: $\delta m^2_{K,k}$ collects the terms with at least one line in $\{S_{(k)}, G_{K,k+1}\}$, $\delta m^2_{K,k+1}$ those with at least one line in $G_{K,k+1}$, and their difference is the terms with all lines in $\{G_k, S_{(k)}\}$ and at least one $S_{(k)}$, i.e. $\delta m^2_{k+1} - \delta m^2_k$. The counterterm vertices inside these graphs are vertices, not lines (Convention F.18(a)), and are not expanded. □

Lemma F.33 (the bad-localisation terms of the moved mass counterterm)**.** *At step $k+1$ let the moved terms m_k, rescaled to the η-lattice, be localised by the rules of Convention F.18 with $\Omega_1^+ := B^k(\Lambda_7^{(k)})$ and g^+ (*$\operatorname{supp} g^+ \subset \Omega_1^+$, *$g^+ = 1$ at distance $\ge LM$ from $\partial\Omega_1^+$ in scale-k units, [3, p. 412]); assume $R \ge 5M+6$, so that $r - 1 > LM + 4$. Put* Bad $:= (\Omega_1^+)^c \cup \{z \in \Omega_1^+ : \operatorname{dist}(z, \partial\Omega_1^+) < LM+2\}$ *and $b_k(x) := m_k^{\mathrm{glob}}(x) - m_k^{\mathrm{loc}}(x)$ for $x \in \Omega_1^+$, the coefficient change of the [3, (1.7)] vertex caused by the move. Then:* (a) *any localised family of the step-$(k+1)$ expansion containing one [3, (1.7)] vertex with coefficient $b_k(x)t^2$ at x is a finite sum of Proposition 1 families, each with one $S_{(k)}$ line treated as an external field and at least one vertex localised in* Bad*, and $|E| \le O(1)e_k^{d_v}\lambda_k^{d_s}\prod$(external norms)$\, C_K e^{-(\delta'/2)(\operatorname{dist}(\square(x),\mathrm{Bad})-2)} e^{-(\delta'/2)d(\{\square\})}$, $\delta' := \min(\delta_0, \delta_K)$, C_K, δ_K the constants of the slice kernel;* (b) *the standalone vertex with both legs external satisfies $\sum_{x\in\Omega_1^+} \eta^3 |b_k(x)t^2||\Phi(x)|^2 \le Ct^{1/2}p^2\lambda^{-1/2}[|\Lambda_7^{(k)} \setminus \Lambda_8^{(k)}| + e^{-(\delta'/2)(r-LM-6)}|\Lambda_7^{(k)}|]$, and every family with $W \ge 2$ containing the b_k vertex obeys the same bound with $t^{1/2-\beta}$;* (c) *in Bałaban's form: for $x \in B^k(\Lambda_7^{(k)} \setminus \Lambda_8^{(k)})$ the size is a power of the rescaled couplings ($O(e_k^2 + \lambda_k)$ for the coefficient), for $x \in B^k(\Lambda_8^{(k)})$ it carries in addition $e^{-(\delta'/2)(r-LM-6)}$, the printed $e^{-\delta_1 r}$ with $\delta_1 = \delta'/4$ once $r \ge 2(LM+6)$.*

Proof. m_k^{loc} is the same sum of graphs as m_k^{glob} with weight $\mathbf{1}_{\Omega_1^+}$ on the [3, (1.6)–(1.7)] vertices and g^+ on the vector legs; telescoping the weights gives terms in which at least one internal vertex z carries $1 - \mathbf{1}_{\Omega_1^+}(z)$ or $1 - g^+ \ne 0$, so $z \in$ Bad. Each graph of m_k has an $S_{(k)}$ line (lemma F.32); cut one in a fixed order and regroup into the renormalised families of the cut graph (as in theorem F.57, the slice line playing the role of a cut line; the $S_{(k)}$ line belongs to no divergent subgraph of the cut graph); the slice kernel is regular in both variables with $|S_{(k)}| \le C_K e^{-\delta_K \operatorname{dist}}$ in the two-variable norm, the printed "regular function of both variables". Apply Proposition 1 to the resulting families, joining the components of a disconnected cut graph through the slice decay (lemma F.25). The tree contains $\square(x)$ and a cube in Bad, so $d(\{\square\}) \ge \operatorname{dist}(\square(x), \mathrm{Bad}) - 2$, and

half of the exponent is kept for the localisation sum. The order is ≥ 2, giving $O(e_k^2+\lambda_k)=O(t)$; with $|\Phi|^2 \leq C_\Phi^2 p^2 \lambda_k^{-1/2}$ the standalone vertex is $O(t^{1/2}p^2\lambda^{-1/2})$ per cube times the tree factor. Every cube within $LM+3$ of $\partial\Omega_1^+$ lies in $B^k(\Lambda_7^{(k)}\setminus\Lambda_8^{(k)})$ since $\operatorname{dist}(\Lambda_8,\Lambda_7^c)\geq r-1>LM+4$; for cubes in $B^k(\Lambda_8^{(k)})$, $\operatorname{dist}(\square(x),\mathrm{Bad})\geq r-1-LM-3$. Count the cubes and use $p(t)^n e^{-\delta r(t)}\leq t^\kappa$ for the exponential part. □

Lemma F.34 (the vacuum counterterm moves; "even simpler"). *$e_k := [E_{k+1}-E_k]^{\mathrm{glob}} - [E_{k+1}-E_k]^{\mathrm{loc}}$ (localised with Ω_1^+, g^+) is a c-number with $|e_k| \leq Ct|T_1^{(k)}\setminus\Lambda_8^{(k)}| \leq Ct(1+(17(r+M))^d)|\Lambda_0^{(k)c}|$; if $\Lambda_0^{(k)c}=\emptyset$ then $\mathrm{Bad}=\emptyset$ and $e_k=0$.*

Proof. As in lemma F.33 without external fields: Proposition 1 for vacuum families after cutting $S_{(k)}$ gives $O(t)$ per localisation (order ≥ 2). Root at a cube in Bad, which lies in $T_1^{(k)}\setminus\Lambda_8^{(k)}$ up to the $LM+2$ layer, and sum. By [2, (2.8)] every block of $\Lambda_0\setminus\Lambda_8$ is within $8(r+M)$ of Λ_0^c (each layer is r plus one large block of side M), and at most $(16(r+M)+1)^d \leq (17(r+M))^d$ unit blocks lie within that distance of a given block of Λ_0^c. □

Both moves are bounded by $Ct^{1/2-\beta}(r+M)^d|\Lambda_0^{(k)c}|+C_\kappa t^\kappa|\Lambda_7^{(k)}|$, against which the large-field ledger of [2, Sect. 3C] pays $(d+1)(1+\ln t^{-1})$ per unit of $|\Lambda_0^{(k)c}|$; that ledger is [2, (3.42)], part of the $L=5$ constants of theorem F.2.

Step 3(C_1): the large-field chain of [2, Sect. 3A]. [2, Sect. 3A] inverts the renormalisation transformations in the large-field region: the inductive hypothesis [2, (3.5)] at level k is reduced to the same statement at level $k-1$ at the cost of a factor $\exp(O(1)(L^k\varepsilon)^{\kappa_0}|T_\varepsilon|)$, through the Gaussian integral [2, (3.9)] with the perturbing kernels H_k', H_k'' [2, (3.8), (3.12)], the linear terms F_k', F_k'' [2, (3.10)], the lower bounds [2, (3.11)] and the Gaussian comparison [2, (3.17)–(3.20)]. What is consumed is that each inverse step costs $\exp(C_\kappa t_k^\kappa|T_1^{(k)}|)$ with C_κ independent of k, ε, ε_0 and the torus. We prove this with explicit constants. Besides the notation of Step 3(B), $M_v := c_A p(t)/(\mu_0 t)$ and $M_s := c_s p(t)(\lambda t)^{-1/4}$ are the small-field bounds [2, (2.16)–(2.17)] at scale k ($c_A := 2+2Ld\mu_0$, $c_s := 2+2Ld$, under the cap $\lambda t\leq 1$), $M(t) := c_M t^{-1}p(t) \geq \max(M_v, M_s)$ with $c_M := \max(c_A/\mu_0, c_s\lambda^{-1/4})$; scale-$(k-1)$ collar fields rescaled to scale k are bounded by $c_L M_v(t)$ with $c_L := L^{3/2}(1+\ln L)^{\bar p}$; $S_\delta := \sup_x \sum_z e^{-\delta|x-z|}$; the separation of a primed scale-$(k-1)$ layer from the next is $D_k := (r(t)-c_b)/L \geq r(t)/(2L)$ for $R\geq 2c_b$, c_b the large-block diameter. Two sup bounds replace every "for ε_0 small" by a constant: for $\kappa\geq 0$, $\delta>0$, $a>0$, $m\geq 0$,

$$E(\kappa,\delta) := \sup_{0<t\leq 1} t^{-\kappa}e^{-\delta r(t)} < \infty, \qquad P(a,m) := \sup_{0<t\leq 1} t^a p(t)^m < \infty,$$

(with $t=e^{1-u}$, $u\geq 1$: $\exp(\kappa(u-1)-\delta R u^\varrho)$ is bounded since $\varrho>1$, and $u^{m\bar p}e^{-a(u-1)}$ is bounded), depending only on $(\kappa,\delta,R,\varrho)$ resp. $(a,m,b_0,\bar p)$. Three kernel facts are used throughout (sup-distance, $d(\cdot)$ a 1-Lipschitz distance function): (K-a) if $|K(x,x')| \leq ce^{-\delta(|x-x'|+d(x)+d(x'))}$ and $d\geq D$ on Y, then $|K| \leq ce^{-2\delta D}e^{-\delta|x-x'|}$ on $Y\times Y$; (K-b) if $|K_0(x,x')| \leq ce^{-\delta|x-x'|}$ for $x\in Y$, $x'\in Z$, $\operatorname{dist}(Y,Z)\geq D$, then $|K_0| \leq ce^{-\delta D/2}e^{-\delta|x-x'|/2}$; (K-c) if $|K_1(x,z)| \leq c_1 e^{-\delta|x-z|}$, $|K_3(z',x')| \leq c_3 e^{-\delta|z'-x'|}$ and $|K_2(z,z')| \leq c_2 e^{-\delta(|z-z'|+d(z)+d(z'))}$, then $|(K_1K_2K_3)(x,x')| \leq c_1c_2c_3S_{\delta/4}^2 e^{-(\delta/4)(|x-x'|+d(x)+d(x'))}$ (with $E := |x-z|+|z-z'|+|z'-x'|+d(z)+d(z')$, the triangle inequality and the Lipschitz property give $|x-x'|+d(x)+d(x')\leq 2E$, so the exponent $\frac{\delta}{4}(|x-x'|+d(x)+d(x'))+\frac{\delta}{4}(|x-z|+|z'-x'|)\leq\delta E$; sum over z,z'); telescoping a product of n such kernels gives the decay of (K-c) with rate $\delta/4^{n-1}$.

Lemma F.35 (the kernel types of the inverse step). *At scale k let Y be a set of unit-lattice points and $D := D_k$. Each of the following kernels $K(x,x')$ obeys $|K(x,x')| \leq C_H e^{-\delta_H r(t)}e^{-\delta_H|x-x'|}$, with $\delta_H := \min(\delta_7, \delta_7^S/4, \theta_0/4, \theta_0(\gamma_L,L)/6, \theta_W)/(16L)$ (the rates of* [1, Prop. 2.2, (2.27), (2.29)]*, of the domain differences of C_Λ, of the Combes–Thomas bound and of the conditioning-region difference: proposition F.150, lemma F.149(b), lemma F.144 and lemma F.147) and C_H explicit, all free of k, ε, ε_0 and the torus: (a)* domain change*: $K = O(\Omega_1) - O(\Omega_2)$, $\Omega_1\subset\Omega_2$ unions*

of big blocks, $O \in \{\Delta^{(k)}(\cdot,\tilde A), C_\Lambda(\cdot,\tilde A)\}$ at a regular background, with $\operatorname{dist}(Y,\Omega_1^{(k)c}) \ge D$*; (a′)* conditioning-region change*: $K = \Delta^{(k)}_{\Lambda_5^{(k-1)}} - \Delta^{(k)} = (aL^{-2})^2 Q(C - C_{\Lambda_5})Q^*$ (from* [2, (2.46)]*, $\Delta^{(k)}_{\Lambda_5} = aL^{-2} - (aL^{-2})^2 Q C^{(k-1)}_{\Lambda_5} Q^*$), on Y with* $\operatorname{dist}(Y,\Omega^{(k-1)}\setminus\Lambda_5^{(k-1)}) \ge D$*; this Λ-change, not covered by (a), is the only change in the photon form of* [2, (2.47), p. 568] *not already of types (a)–(d); (b)* products*: $O_1\cdots O_n - O_1'\cdots O_n'$ ($n \le 3$) with factors among $\Delta^{(k)}$, C_Λ, $a_k G_k Q_k^*$ and adjoints, each pair O_i, O_i' of type (a) or equal; (c)* cut-off*: $K = \mathbf{1}_Y K_0 \mathbf{1}_Z$ with $|K_0(x,x')| \le ce^{-\delta|x-x'|}$ and* $\operatorname{dist}(Y,Z) \ge D$*; (d)* background change*: $K = O(\tilde A + \tilde B^{(k)}) - O(\tilde A)$ with $|\tilde B^{(k)}| \le C_\kappa t^\kappa$* [2, (2.90)]*, where $e^{-\delta_H r(t)}$ is replaced by $Ct^{\kappa+1/2}$ for every κ (not used in Sect. 3A, which has no background change; used in Step 3(C_2)).*

Proof. (a): [1, (2.29)] (proposition F.150) gives $c_7 e^{-\delta_7(|x-x'|+d(x)+d(x'))}$ with $d = \operatorname{dist}(\cdot,\Omega_1^{(k)c}) \ge D$ on Y, and the domain difference of C_Λ the same with $\delta_7^S/4$ (lemma F.149(b)); then (K-a). (a′): the difference $C - C_{\Lambda_5}$ at scale $k-1$ decays at rate $\theta_0(\gamma_L, L)/6$ (lemma F.147) in $|y-y'| + \operatorname{dist}(y,\Omega^{(k-1)}\setminus\Lambda_5^{(k-1)}) + \operatorname{dist}(y',\text{same})$, and on $\Lambda_5^{(k)}$ that distance is at least two layers; then the coarse transfer QKQ^* and (K-a). (b): (K-c) and its telescoping, rate $\ge \delta/16$ for $n \le 3$. (c): (K-b). In (a)–(c), $e^{-\delta D} \le e^{\delta c_b/L} e^{-\delta r(t)/L}$. (d): the analytic family of block operators in the background $\tilde A + \tau\tilde B^{(k)}$ (Step 3(C_2), lemma F.48) with $w = \|q\| e_k C_\kappa t^\kappa$. □

Lemma F.36 ([2, (3.8), (3.10)])**.** (i) *H_k of* [2, p. 584] *("we have applied formula (2.108) together with the remark following it"): by* [2, (2.108)] *read with the background $\tilde A^{(k)}$, the difference between $\langle\phi, \Delta^{(k)}(B^k(\Lambda_2^{(k-1)'}), \tilde A^{(k)})\phi\rangle$ on $\Lambda_6^{(k-1)'}$ and its three localised pieces is the sum of type-(a) kernels on $(\Lambda_6^{(k-1)'}\cap\Lambda_3^{(k)c})^2$ and on $(\Lambda_6^{(k-1)'}\cap\Lambda_3^{(k)c})\times(\Lambda_3^{(k)}\cap\Lambda_4^{(k)c})$ ($\Omega_1 = B^k(\Lambda_2^{(k-1)'}\cap\Lambda_5^{(k)c})$), a type-(c) kernel on $(\Lambda_6^{(k-1)'}\cap\Lambda_3^{(k)c})\times\Lambda_4^{(k)}$ (distance $\ge r - c_b$), and a type-(a) kernel on $(\Lambda_3^{(k)})^2$ ($\Omega_1 = B^k(\Lambda_2^{(k)}) \subset \Omega_2 = B^k(\Lambda_2^{(k-1)'})$); no background change occurs, since by* [2, (3.4)] *$\tilde A^{(k)} - \tilde A^{(k+1)} = (1-\theta_{k+1})\theta_k A^{(k)}$ vanishes on $B^k(\Lambda_2^{(k)})$. Hence* [2, (3.8)] *holds for h_k with $\delta_1 = \delta_0 = \delta_H$ and $O(1) = C_H' := 4C_H$.* (ii) *H_k', H_k'', F_k', F_k'' (*[2, p. 585]*: "terms associated with the changes in (2.47) and the discussion following, and the term H_k"): by* [2, (2.47), p. 568] *the changes are the domain splits of $\Delta^{(k)}_{\Lambda_5^{(k-1)}}(B^{k-1}(\Lambda_2^{(k-1)}), A^{(k)})$ (type (a)), the replacement of $\Delta^{(k)}_{\Lambda_5^{(k-1)}}$ by $\Delta^{(k)}$ on $\Lambda_6^{(k-1)'}$ in the third line of* [2, (2.47)] *for the scalar and the photon form (type (a′)), the restriction of the third and ninth terms of* [2, (2.46)] *to $\Lambda_5^{(k-1)'}\cap\Lambda_6^{(k-1)'c}$ (type (c)), and the Neumann conditions on $\partial B^{k-1}(\Lambda_2^{(k-1)}\cap\Lambda_7^{(k-1)c})$ in the seventh to ninth terms, whose operators are the products $\Delta C_{\Lambda_5}\Delta$ and $\Delta C_{\Lambda_5} Q^*$ (type (b)); so H_k', H_k'' obey* [2, (3.8)] *with a fixed multiple of C_H.* (iii) *F_k' (resp. F_k'') is the sum of the type-(c) and type-(a)/(b) remainders of the linear terms and of $\sum_{x'\notin\Lambda_5^{(k)}} h'(x,x')v(x')$ with v the collar fields, $|v| \le c_L M(t)$; hence $|F_k'(x)| \le C_H S_{\delta_H} e^{-\delta_H r(t)} c_L M(t) \le C_F e^{-\delta_H r(t)/2}$ with $C_F := c_L C_H S_{\delta_H} c_M \sup_t t^{-1} p(t) e^{-\delta_H r(t)/2} < \infty$, and the same for F_k''.*

Proof. Each item is the cited type with the cited separations (lemma F.35); the separations are those of the layers. Bałaban defines H_k', H_k'', F_k', F_k'' by reference ([2, p. 568]: "Terms omitted are described more precisely later"); the list of changes is his, and each is of one of the types. □

Lemma F.37 ([2, (3.8)] implies [2, (3.12)] for every κ)**.** *If $|h(x,x')| \le ce^{-\delta_1 r(t)} e^{-\delta_0|x-x'|}$ on Y, then the operator H on $\ell^2(Y)$ has $\|H\| \le cS_{\delta_0} e^{-\delta_1 r(t)} \le cS_{\delta_0} E(\kappa,\delta_1) t^\kappa$ for every κ.*

Proof. Schur test, then the definition of $E(\kappa,\delta_1)$. □

Lemma F.38 ([2, (3.11)] with $\gamma_1 = \frac12$, without field restriction or regularity of $\tilde A$)**.** *Let $t \le \tau_1 := \min\{1, a_*^{1/2}/\mu_0, a_*^{1/2}/m\}$. Then $G_k' \ge \frac12\mu_0^2 t^2$ and $G_k'' \ge \frac12 m^2 t^2$ on $\ell^2(\Lambda_5^{(k)})$.*

Proof. G_k' (resp. G_k'') is the quadratic part in $A_k|_{\Lambda_5}$ (resp. $\phi_k|_{\Lambda_5}$) of the first four terms of [2, (3.9)]. The block-spin terms $\frac12\sum_l a_{l-k}(L^{l-k})^{d-2}\sum|A_l - Q_{l-k}A_k|^2$ (resp. with $Q_{l-k}(\tilde A^{(k+1)})$) contribute

nonnegative forms. The forms $\langle \Lambda_5 A_k, \Delta^{(k)}_{\Lambda_5^{(k-1)}} \Lambda_5 A_k\rangle$ and $\langle \Lambda_5\phi_k, \Delta^{(k)}_{\Lambda_5^{(k-1)}}(B^k(\Lambda_2^{(k-1)'}), \tilde A^{(k)})\Lambda_5\phi_k\rangle$ are compressions of conditional forms: the subscript records the conditional integration of [2, (2.46)] over the intermediate field on $\Lambda_5^{(k-1)}$ with its collar values as separate variables, so their quadratic part is the partial elimination of that field in the two-level form (block-spin term of the step) $+\langle A', \Delta^{(k-1)}A'\rangle$, whose full elimination is $\Delta^{(k)}$ by the composition identity [1, (2.40)–(2.41)]. For $H \ge 0$, $m_0^2 \ge 0$, $a > 0$ and $\|Q\| \le 1$, the Schur complement $\Delta := a - a^2Q(H + m_0^2 + aQ^*Q)^{-1}Q^*$ satisfies $\langle\psi, \Delta\psi\rangle = \min_\phi[a\|\psi - Q\phi\|^2 + \langle\phi, (H + m_0^2)\phi\rangle] \ge \min_u[a\|\psi - u\|^2 + m_0^2\|u\|^2] = \frac{am_0^2}{a+m_0^2}\|\psi\|^2$; a partial minimisation over a smaller set is larger, and a compression of an operator $\ge c$ is $\ge c$ on its range. The fundamental operators carry the masses $\mu_0^2t^2$ (photon, $-\Delta^\eta + \mu_0^2(L^k\varepsilon)^2$) and m^2t^2 (scalar, $-\Delta^{\eta,N}_{A^{(k)}} + m^2(L^k\varepsilon)^2$, [2, p. 585]), and $\|Q_k(A)\| \le 1$ for every real background. So each form is $\ge a_k m_0^2t^2/(a_k + m_0^2t^2) \ge \frac12 m_0^2t^2$, $m_0 \in \{\mu_0, m\}$, since $m_0^2t^2 \le a_* \le a_k$. □

This answers the "corollary of Proposition 3.1" of [2, p. 586]: no restriction on the fields and no regularity of $\tilde A^{(k)}$ is needed, only $-\Delta^{\eta,N}_{A,\Omega} \ge 0$ for every real background.

Lemma F.39 ([2, (3.15), (3.16)])**.** *Let a field v_l on the collar $\Lambda_{-1}^{(l-1)'} \cap \Lambda_5^{(l)c}$ at scale $l \ge k+1$ obey $|v_l| \le M_v(t_l)$ (vector) or $M_s(t_l)$ (scalar), and let $v_l^{(k)} := L^{-(l-k)/2}v_l(L^{-(l-k)}\cdot)$ be its image in scale-k units. Then $|A_l^{(k)}| \le L^{-3(l-k)/2}M_v(t)$, $|\phi_l^{(k)}| \le L^{-3(l-k)/4}M_s(t)$ (Bałaban's $L^{-(l-k)d/2}$, $L^{-(l-k)d/4}$), and the fields Φ', Φ'' of* [2, (3.13)–(3.14)] *satisfy $|\Phi'(x)| \le aM_v(t)/(L^{7/2}-1)$, $|\Phi''(x)| \le aM_s(t)/(L^{11/4}-1)$ on $\Lambda_5^{(k)}$, both $\le aM(t) \le ac_MP(1,1)t^{-2}$.*

Proof. $t_l = L^{l-k}t$ and $p(t_l) \le p(t)$, so $L^{-(l-k)/2}c_Ap(t_l)/(\mu_0 t_l) \le L^{-3(l-k)/2}M_v(t)$, and likewise for the scalar. $|Q^*_{l-k}(\tilde A)\phi_l| = |\phi_l|$ (orthogonal transport) and $a_{l-k} \le a$, so $|\Phi'(x)| \le \sum_{l>k} aL^{-2(l-k)}L^{-3(l-k)/2}M_v(t)$, a geometric series; the same for Φ''. Finally $M(t) \le c_MP(1,1)t^{-2}$. □

Lemma F.40 (the linear coefficient)**.** *Let J_k' (J_k'') be the linear coefficient of $A_k|_{\Lambda_5}$ ($\phi_k|_{\Lambda_5}$) in* [2, (3.9)] *without F_k' (F_k''). Then $|J_k'(x)|, |J_k''(x)| \le C_JM(t)$ on $\Lambda_5^{(k)}$, hence $\|J\| \le C_JM(t)|\Lambda_5^{(k)}|^{1/2}$, with $C_J := a + c_{\mathrm{coll}} + c_Lc_f$: c_{coll} the $\ell^\infty \to \ell^\infty$ norm of the kernel of the conditional operator $\Delta^{(k)}_{\Lambda_5^{(k-1)}}$ from the collar $\Lambda_5^{(k-1)'} \cap \Lambda_5^{(k)c}$ to $\Lambda_5^{(k)}$ (its decay is that of $aL^{-2} - (aL^{-2})^2QC_{\Lambda_5}Q^*$ with $|C_{\Lambda_5}(y,y')| \le c_2e^{-\theta_0|y-y'|}$), c_f that of the kernels of the third and ninth terms of* [2, (2.46)] *(products $\Delta^{(k-1)}C^{(k-1)}_{\Lambda_5}Q^*$ of decaying kernels, $c_f \le aL^{-2}c_7c_2S_{\delta_7}S_{\theta_0}$).*

Proof. Φ' by lemma F.39; the collar term by the conditional operator's decay and $|A_k| \le M_v$ on the collar; f_k' is a sum of such kernels applied to scale-$(k-1)$ collar fields bounded by $c_LM(t)$. □

This replaces Bałaban's $O((L^k\varepsilon)^{-2})|\Lambda_5^{(k)}|^{1/2}$ [2, p. 587] by $C_Jc_Mt^{-1}p(t)|\Lambda_5|^{1/2}$.

Lemma F.41 (Gaussian comparison)**.** *Let V be a real Euclidean space of dimension n, G, H symmetric on V with $G \ge g > 0$ and $\|H\| \le h \le g/2$, and $J, F \in V$. Then*

$$\int_V e^{-\frac12\langle x,(G-H)x\rangle + \langle J+F,x\rangle}dx \le \int_V e^{-\frac12\langle x,Gx\rangle+\langle J,x\rangle}dx \cdot \exp\Big[\frac{nh}{g} + \frac{h\|J\|^2}{g^2} + \frac{2\|F\|\|J\|}{g} + \frac{\|F\|^2}{g}\Big].$$

Proof. Both integrals are Gaussian, $\int e^{-\frac12\langle x,Mx\rangle+\langle b,x\rangle} = (2\pi)^{n/2}(\det M)^{-1/2}e^{\frac12\langle b,M^{-1}b\rangle}$ for $M > 0$ (here $G - H \ge g/2$). Determinant ([2, (3.18)–(3.20)]): $\det(G-H) = \det G\det(I-X)$, $X := G^{-1/2}HG^{-1/2}$, $\|X\| \le h/g \le \frac12$; for an eigenvalue λ of X, $-\log(1-\lambda) \le 2|\lambda|$, so $-\frac12\log\det(I-X) \le \sum|\lambda_i| \le nh/g$. Exponent: $(G-H)^{-1} - G^{-1} = G^{-1}H(G-H)^{-1}$ has norm $\le 2h/g^2$ and $\|(G-H)^{-1}\| \le 2/g$, so $\frac12\langle J+F, (G-H)^{-1}(J+F)\rangle - \frac12\langle J, G^{-1}J\rangle \le h\|J\|^2/g^2 + 2\|F\|\|J\|/g + \|F\|^2/g$. □

Corollary F.42 ([2, (3.9)] against the unperturbed integral)**.** *Let $t \le \tau_1$ and assume $C'_H S_{\delta_H} e^{-\delta_H R} \le \mu_{\mathrm{lo}}^2/4$ and $\delta_H R \ge 2$, $\mu_{\mathrm{lo}}^2 := \min(\mu_0^2, m^2)$ (a condition on R, chosen before ε_0). Then for every $\kappa > 0$, [2, (3.9)] is at most [2, (3.9)] at $H'_k = H''_k = 0$, $F'_k = F''_k = 0$, times $\exp(C_\kappa t^\kappa |\Lambda_5^{(k)}|)$, with*

$$C_\kappa := \sup_{0<t\le 1} t^{-\kappa}\Big[\frac{2(d+N)h(t)}{\mu_{\mathrm{lo}}^2 t^2}+\frac{4h(t)C_J^2 M(t)^2}{\mu_{\mathrm{lo}}^4 t^4}+\frac{4(d+N)^{1/2}C_F e^{-\delta_H r(t)/2}C_J M(t)}{\mu_{\mathrm{lo}}^2 t^2}+\frac{2(d+N)C_F^2 e^{-\delta_H r(t)}}{\mu_{\mathrm{lo}}^2 t^2}\Big], \quad h(t$$

finite since every term is $e^{-\delta_H r(t)/2}$ times a power of $t^{-1}p(t)$.

Proof. $V := \ell^2(\Lambda_5^{(k)})^d \oplus \ell^2(\Lambda_5^{(k)})^N$, $n = (d+N)|\Lambda_5^{(k)}|$; $G := G' \oplus G'' \ge g := \frac{1}{2}\mu_{\mathrm{lo}}^2 t^2$ (lemma F.38); $H := H' \oplus H''$ with $h = h(t)$ (lemmas F.36 and F.37), and $h \le g/2$ for all $t \le 1$ by the cap ($e^{-\delta_H r(t)} \le e^{-\delta_H R} t^2$ when $\delta_H R \ge 2$); $\|J\| \le C_J M(t)|\Lambda_5|^{1/2}$ (lemma F.40); $\|F\| \le (d+N)^{1/2} C_F e^{-\delta_H r/2}|\Lambda_5|^{1/2}$ (lemma F.36(iii)). lemma F.41, with the scalar background treated as an independent external field ([2, p. 568]), gives the bound. □

Lemma F.43 (the collar forms, [2, p. 584])**.** *On the collar $\Lambda_5^{(k-1)'} \cap \Lambda_5^{(k)c}$, where $|A_k| \le M_v$, $|\phi_k| \le M_s$, every form $\frac{1}{2}\langle v, Kv\rangle$ with K of the types of lemma F.35 obeys $|\frac{1}{2}\langle v, Kv\rangle| \le \frac{1}{2}C'_H S_{\delta_H} e^{-\delta_H r(t)} M(t)^2 |\mathrm{collar}| \le C'_\kappa t^\kappa |\Lambda_5^{(k-1)'} \cap \Lambda_5^{(k)c}|$ with $C'_\kappa := \frac{1}{2}C'_H S_{\delta_H} c_L^2 c_M^2 \sup_t t^{-\kappa-2} p(t)^2 e^{-\delta_H r(t)} < \infty$ (Schur test).*

Proposition F.44 (the inductive step [2, (3.5)]$_k \Rightarrow$[2, (3.5)]$_{k-1}$)**.** *Assume [2, (3.5)] at level k. Then it holds at level $k-1$ with the additional factor $\exp(C_{\mathrm{LF},\kappa} t_k^\kappa |T_1^{(k)}|)$, $C_{\mathrm{LF},\kappa} := C_\kappa + 3C'_\kappa$: the T-transformations are integral operators with nonnegative kernels, so inequalities inside the nested brackets are preserved; the transformations not depending on $A_k, \phi_k|_{\Lambda_5^{(k)}}$ are moved out (Fubini–Tonelli); the forms of [2, (3.5)] at level k are merged by [2, (2.108)] into one form plus H_k (lemma F.36(i)), the forms evaluated on the collar cost lemma F.43, the remaining integral is [2, (3.9)] (lemma F.36(ii)) and corollary F.42 compares it with the $H = F = 0$ integral; the latter is, by the definitions [2, (2.45)–(2.50)] read backwards, the level-k factor of $\varrho''^{(k)}$ [2, (2.46)], and [2, (2.48)] replaces ϱ'' by ϱ (for $k < K$ its interior part is of H'-type and sits inside corollary F.42, its collar part is a lemma F.43 form; at $k = K$, where [2, p. 583] restricts B, ψ everywhere, the whole of [2, (2.48)] is a sum of lemma F.43 forms, which is the base case [2, (3.1)]). The sum over levels of the added factors is $\sum_l C_{\mathrm{LF},\kappa} t_l^\kappa |T_1^{(l)}| = \sum_l C_{\mathrm{LF},\kappa} t_l^{\kappa-d} |T_\varepsilon|$, Bałaban's $\exp(\sum_l O(1)(L^l\varepsilon)^{\kappa_0}|T_\varepsilon|)$ with $\kappa_0 = \kappa - d$ and $O(1) = C_{\mathrm{LF},\kappa}$ independent of l; the induction is finite (K steps).*

Every constant above is explicit and free of k, ε, ε_0 and the torus; the caps used are $t \le \tau_1$ and the condition on R of corollary F.42, neither involving ε_0.

The k-uniformity of the unqualified $O(1)$ lines of [2]. Eight lines of [2] carry an $O(1)$ without a stated dependence: [2, (2.61), (2.86), (2.91), (2.94), (2.95), (2.103), (2.117), (2.118)]. [2, (2.61)] and [2, (2.95)] inherit the $L = 5$ constants of [2, Lemmas 2.3, 2.4] (theorem F.2); [2, (2.118)] is the geometric sum of proposition F.44; [2, (2.117)]'s constants are c_5 (from [4, Prop. 3.1′]) and $c_7 = C_{2.57} + C_{2.78}$, with $C_{2.57}$ from corollary F.29 and its field-dependent analogue and $C_{2.78}$ arithmetic; the remaining three are the following lemmas.

Lemma F.45 ([2, (2.86)] with an explicit k-free constant)**.** *At scale k let $\Lambda := \Lambda_0^{(k)}$, B the new block photon field, A the scale-k photon field, and assume [2, (2.17)] at scale k on Λ'_{-1}: $|A(x) - (Q^*B)(x)| \le 2Ldp$, $|B(y) - B(y')| \le 3Ldp$ for neighbouring L-blocks, $|B(y)| \le 2p/(\mu_0 t)$. Then for $x \in \Lambda_1^{(k)}$, $|A'(x)| := |A(x) - aL^{-2}(C_\Lambda^{(k)} Q^* B)(x)| \le C_{2.86}\, p$ with*

$$C_{2.86} := 2Ld + c_2 S_{\theta_0}\Big[3dc_7 \sum_n e^{-\delta_7|n|}(|n|_1 + dL) + 2\mu_0 + \frac{4c_7 S_{\delta_7/2} E(1, \delta_7/4)}{\mu_0}\Big] + \frac{2c_2 c_7 S_{\delta_7/2} S_{\theta''} e^{\theta'' c_b} E(1, \theta'')}{\mu_0},$$

$\theta'' := \min(\theta_0, \delta_7/2)/2$, *assuming $R \ge 4(c_b + dL)$.*

Proof. Photon operators are at zero background on the torus [1, p. 608], componentwise. $f := \Lambda Q^* B$ lies in the range of $\Lambda P \Lambda$, so $aL^{-2} C_\Lambda f - f = -C_\Lambda \Lambda \Delta \Lambda f$ with $\Delta := \Delta^{(k)}$, and since $f(x) = (Q^*B)(x)$ on Λ, $A'(x) = [A(x) - (Q^*B)(x)] + (C_\Lambda \Lambda \Delta \Lambda f)(x)$. $\Delta \mathbf{1} = \delta_\mu \mathbf{1}$ with $\delta_\mu = a_k \mu_0^2 t^2/(a_k + \mu_0^2 t^2) \le \mu_0^2 t^2$, so for $z \in \Lambda$, with $y(z)$ the L-block of z, $(\Delta \Lambda f)(z) = \sum_{z' \in \Lambda} \Delta(z,z')[B(y(z')) - B(y(z))] + B(y(z))[\delta_\mu - \sum_{z' \notin \Lambda} \Delta(z,z')]$. First sum: $|\Delta(z,z')| \le c_7 e^{-\delta_7 |z-z'|}$ ([1, (2.27)], proposition F.150); if $|z-z'| \le r/2$ the axis-ordered chain of L-blocks from $y(z)$ to $y(z')$ stays in Λ_{-1} and [2, (2.17)] gives $|B(y(z')) - B(y(z))| \le 3dp(|z-z'|_1 + dL)$; if $|z-z'| > r/2$ use $|B| \le 2p/(\mu_0 t)$ and $e^{-\delta_7|n|} \le e^{-\delta_7 r/4} e^{-\delta_7 |n|/2}$, the tail being $\le 4c_7 S_{\delta_7/2} e^{-\delta_7 r/4} p/(\mu_0 t)$. Second term: $|B|\delta_\mu \le 2\mu_0 t p \le 2\mu_0 p$, and $|B| \sum_{z' \notin \Lambda} |\Delta(z,z')| \le (2p/(\mu_0 t)) c_7 S_{\delta_7/2} e^{-(\delta_7/2)\operatorname{dist}(z,\Lambda^c)}$. Apply $|C_\Lambda(x,z)| \le c_2 e^{-\theta_0 |x-z|}$ (lemma F.145(c)): the bounded parts give $c_2 S_{\theta_0}(\dots) p$, the boundary part gives $2c_2 c_7 S_{\delta_7/2} S_{\theta''} e^{-\theta'' \operatorname{dist}(x,\Lambda^c)} p/(\mu_0 t)$ with $\operatorname{dist}(x,\Lambda^c) \ge r - c_b$ for $x \in \Lambda_1$, and $E(1,\theta'')$ absorbs $t^{-1} e^{-\theta'' r}$. □

Lemma F.46 ([2, (2.91)], value clause)**.** *With $A'^{(k)} := a_k \zeta^{(k)} G_k Q_k^* A'$ (the first term of [2, (2.89)]), $|\zeta^{(k)}| \le 1$, $\operatorname{supp} \zeta^{(k)}(\cdot, y) \subset \{|x-y| < r(t) - 2M\}$ (theorem F.2(b)), and $|(G_k Q_k^*)(x,y)| \le C_1 e^{-\delta_* |x-y|_1}$ uniformly in k (theorem F.7): for x within M of $B^k(\Lambda_2^{(k)})$, $|\theta_{k+1} A'^{(k)}(x)| \le C_{2.91}\, p$ with $C_{2.91} := aC_1 S_*(C_{2.86} + C')$, $S_* := \sum_n e^{-\delta_* |n|_1/2}$, $C' := e^{\delta_*(c_b + M + 1)/2} S_*(c_A + 2aL^{-2} c_2 S_{\theta_0}) E(1, \delta_*/2)/\mu_0$.*

Proof. Split the y-sum: $y \in \Lambda_1^{(k)}$ uses lemma F.45; $y \in \Lambda_0^{(k)} \setminus \Lambda_1^{(k)}$, the only other y reached since $2(r - c_b) > r - 2M + M$, has $|A'(y)| \le M_v + 2aL^{-2} c_2 S_{\theta_0} p/(\mu_0 t)$ and $|x-y| \ge r - c_b - M - 1$, which gives the $E(1, \delta_*/2)$ term. □

The derivative clause of [2, (2.91)] is [2, Lemma 2.3, (2.60)] at $L = 5$ (theorem F.2).

Lemma F.47 ([2, (2.103), (2.94)])**.** (i) *Let n be odd and $I := [1 - \gamma_1/\gamma_0, 1 - \gamma_0/\gamma_1]$ with $\gamma_0 := \gamma_X$, $\gamma_1 := a + aL^{-2}$ (the two-sided bound $\gamma_X \le X_\Lambda \le a_k + aL^{-2}$ of lemma F.145(b), so I does not depend on j,k). Then $-\frac{1}{2}\ln(1-\lambda) \le \sum_{l \le n} \lambda^l/(2l) + c_n \lambda^{n+1}$ on I with $c_n := \max(0, \max_I g)$, $g(\lambda) := [-\frac{1}{2}\ln(1-\lambda) - \sum_{l \le n} \lambda^l/(2l)]/\lambda^{n+1}$, $g(0) := 1/(2(n+1))$: g is continuous on the compact I ($1 - \lambda \ge \gamma_0/\gamma_1 > 0$) and $\lambda^{n+1} \ge 0$.* (ii) *$O(1)(L^j \eta)^{d/2} p(L^k \varepsilon) \le O(1) p(L^j \varepsilon)$ for $j \le k$ with the same $O(1)$: $(L^j \eta)^{d/2} = L^{(j-k)d/2} \le 1$ and $p(t_k) \le p(t_j)$.*

Step 3(C_2): the $Z^{(j)}$ expansions of [2, p. 578]. At step $k+1$ the fluctuation determinants $Z^{(j)}$, $j < k$, of [2, (2.92)] are expanded in the new field. With $\tilde{B}$ the regular part of the configuration [2, (2.93)–(2.94)], $A_1 := \theta_{k+1} A'^{(k)}$ and $A_2 := \tilde{B}^{(k)}$, both rescaled to scale j, and $F_j(Y) := \ln Z^{(j)}(Y) = -\frac{1}{2} \ln \det X^{(j)}(Y) + \text{const}$, $X^{(j)}(Y) := \Lambda(\Delta^{(j)}(\Omega, Y) + aL^{-2} P(Y))\Lambda$ on $\ell^2(\Lambda; \mathbb{R}^N)$, $\Omega := B^j(\Lambda_2^{(j)})$, $\Lambda := \Lambda_5^{(j)}$ [2, (2.92), (2.99)], exactly

$$F_j(\tilde{B}+A_1+A_2) = F_j(\tilde{B}) + P_{\tilde{B}}[A_1] + R_j + D_j, \qquad R_j := F_j(\tilde{B}+A_1) - F_j(\tilde{B}) - P_{\tilde{B}}[A_1], \quad D_j := F_j(\tilde{B}+A_1+A_2) - F_j$$

$P_{\tilde{B}}[A_1]$ the order-$\bar{n}$ Taylor polynomial [2, (2.104)]. R_j (the order-$(\bar{n}+1)$ trace term) is bounded by the Cauchy estimate on the analytic family of lemma F.48 below: $\sum_{j<k} |R_j| \le C_\kappa t_k^\kappa |\Lambda_2^{(k)}|/(L^{\bar{n}+1-d} -$ 1) for $\kappa < (\bar{n}+1)/2$. The terms D_j "containing $\tilde{B}^{(k)}$", which [2, p. 578] bounds by $O((L^j\varepsilon)^\kappa)|B^{k-j}(\Lambda_2^{(k)})|$ without argument, are the content of this part: a per-site count at scale j gives only $C t_k^{\kappa+1/2} L^{(k-j)(d-1)} |\Lambda_2^{(k)}|$, which is not summable over $j < k$. The mechanism is that A_2 is smooth at scale k, so at scale j it is almost pure gauge on every region of size $O(1)$ (its curvature is $L^{-2(k-j)}$ times smaller than at scale k, its size only $L^{-(k-j)}$ times smaller), and so is $B := \tilde{B} + A_1$; $Z^{(j)}$ is exactly gauge invariant and even under charge conjugation, so after a local gauge fixing the first variation of each local piece of $\ln Z^{(j)}$ is bilinear in the two curvatures, $L^{-2(k-j)} \times L^{-2(k-j)}$ per scale-j unit cube against $L^{3(k-j)}$ cubes per scale-k cube, leaving $L^{-(k-j)}$, which sums. No identity on a single graph slot is used: the local pieces are exact gauge-invariant functions.

Setting. Fix a scale j, Ω a union of unit blocks of the fine lattice $\xi\mathbb{Z}^3$, $\xi = L^{-j}$ (a box or a torus, Neumann bonds), Λ a union of L-blocks in $\Omega^{(j)}$, $e_j := e t_j^{1/2}$, $U(z) := \exp(q e_j \xi z)$ for $z \in \mathbb{C}$ (q real antisymmetric, so U is entire, $U(z)^T = U(-z) = U(z)^{-1}$, orthogonal for real z).

For a real bond field B on Ω put $X(B) := X_\Lambda(\Omega, B)$ as above and $F(B) := -\frac{1}{2}\ln\det X(B)$. For a complex bond field Z let $\hat w_\Delta(Z) := \|q\|e_j \sup\{|Z_b| : b_- \in \Delta\}$ for a unit cube Δ and $w(Z) := \sup_\Delta \hat w_\Delta(Z)$; for a real B let $f(B) := \|q\|e_j \sup|dB|$ be its curvature amplitude. Hypotheses: (Cur) $f(B) \le c_{\rm cur}$, with $c_{\rm cur} > 0$ (depending on $d, \|q\|, a, L$) such that the defects $2c_p\varepsilon_g + d\kappa^2$ of the cube coercivity (lemma F.142) are at most half of $\min\{2\ell^{-2}, c_p\}$ for $\ell \in \{1, L\}$; lemma F.142 uses the field only through $|F|$, so under (Cur) it and lemmas F.143 to F.145 and the first bound of proposition F.150 hold for B with no other condition, in particular $\gamma_X \le X(B) \le \gamma_1 := a + aL^{-2}$ and $|\Delta^{(j)}(\Omega, B; y, y')| \le c_7 e^{-\delta_7|y-y'|}$ [1, Prop. 2.2, (2.27)], hence $|X(B)(y,y')| \le c_X e^{-\delta_X|y-y'|}$ with $c_X := c_7 + aL^{-2}e^{\delta_7 L}$, $\delta_X := \delta_7$. All constants below depend only on $(d, N, \|q\|, a, L, M)$ and the constants of theorem F.151, never on $j, k, \varepsilon, \varepsilon_0, \Omega, \Lambda$, the torus or B.

Lemma F.48 (the analytic family of block operators)**.** *In the setting above let B be real with $T_0 := T(B) := D_B^T D_B + m_j^2 + a_j Q_j(B)^T Q_j(B) \ge \gamma > 0$ on Ω, let A be a real bond field, $s_A := \sup_b |A_b|$, $w := \|q\|e_j s_A$, S_A the set of unit cubes containing an endpoint of a bond with $A_b \ne 0$, and for $\tau \in \mathbb{C}$ let $T(\tau)$, $G(\tau) := T(\tau)^{-1}$, $X(\tau) := a_j - a_j^2 Q_\tau G(\tau) Q_\tau^T + aL^{-2}P(\tau)$ be the operators in the background $B + \tau A$. Put $m_1 := 5\sqrt{d}\gamma^{-1/2} + 6a_+ d\gamma^{-1}$, $c_\rho := (72 m_1 dL)^{-1}$, $\theta_W := \theta_0/4$ (θ_0 the Combes–Thomas rate of group (A) at $(\gamma, 1)$). For $|\tau|w \le c_\rho$:* (i) *$T(\tau)$ is invertible, $G(\tau)$ analytic in τ, and $\|T_0^{-1/2}(T(\tau) - T_0)T_0^{-1/2}\| \le m_1|\tau|w \le 1/72$;* (ii) *$G(\tau)$ obeys the Combes–Thomas bounds of group (A) (plain, one derivative on either side, two derivatives) with rate θ_W and constant $c_{\rm CT} := 12\max(1, \gamma^{-1}, \gamma^{-1/2})e^{\theta_W}$;* (iii) *$\tilde W(\tau) := X(\tau) - X(0)$ has $|\tilde W(\tau)(y,y')| \le C_W|\tau|w \sum_{\Delta\in S_A} e^{-\theta_W(\operatorname{dist}(y,\Delta)+\operatorname{dist}(\Delta,y'))}$ with $C_W = C_W(d, N, a, a_+, L, \gamma, \theta_0)$;* (iv) *hence $\|\tilde W(\tau)\| \le C_W S^2|\tau|w$ and $\sum_{y,y'} |\tilde W(\tau)(y,y')| \le C_W S^2|\tau|w|S_A|$, $S := \sup_z \sum_y e^{-\theta_W \operatorname{dist}(y,z)}$. The same holds for every compression $\Lambda X(\tau)\Lambda$ and for $\Delta^{(j)}(\tau)$. The statement holds with τA replaced by any complex bond field Z with $w(Z) \le c_\rho$.*

Proof. $U(B_b + \tau A_b) = U(B_b)U(\tau A_b)$, $\|U(B_b)\| = 1$ and $\|e^Z - 1\| \le \|Z\|e^{\|Z\|}$; since $|\tau|wdL \le 1/72 < \ln 2$, $\|U(\tau A_b) - 1\| \le 2\xi|\tau|w$ and $\|U(\tau A(\Gamma)) - 1\| \le 2\operatorname{len}(\Gamma)|\tau|w$. Hence $\|D_\tau - D_0\| \le 2\sqrt{d}|\tau|w$ (each site is the b_+ of at most d bonds), $\|Q_\tau - Q_0\| \le 2d|\tau|w$, $\|Q_\tau\| \le 2$, $\|Q(\tau) - Q(0)\| \le 2dL|\tau|w$. (i): $T(\tau) - T_0 = (D_\tau - D_0)^T D_\tau + D_0^T(D_\tau - D_0) + a_j[(Q_\tau - Q_0)^T Q_\tau + Q_0^T(Q_\tau - Q_0)]$; with $\|D_0 T_0^{-1/2}\| \le 1$ and $\|T_0^{-1/2}\| \le \gamma^{-1/2}$ the three groups contribute at most $m_1|\tau|w$, so $T(\tau) = T_0^{1/2}(1 + K(\tau))T_0^{1/2}$ with $\|K\| \le 1/72$. (ii): conjugate by $e^{\theta\rho_f}$ as in the Combes–Thomas argument; the $\tau = 0$ part contributes $\le 1/8$ at $\theta \le \theta_0/4$, the τ-part $V_\theta(\tau) := e^{\theta\rho_f}(T(\tau) - T_0)e^{-\theta\rho_f}$ has every factor connecting points at distance ≤ 1, so each norm grows by at most $e^\theta \le e$, and $\|e^{\theta\rho} D_\tau e^{-\theta\rho} T_0^{-1/2}\| \le 3$, giving $\le 9m_1|\tau|w \le 1/8$; so $\|K_\theta(\tau)\| \le 1/4$ and the three bounds follow with $\|T_\theta(\tau)^{-1}\| \le \frac{4}{3}\gamma^{-1}$. (iii): $\tilde W(\tau) = -a_j^2[(Q_\tau - Q_0)G_\tau Q_\tau^T + Q_0(G_\tau - G_0)Q_\tau^T + Q_0 G_0(Q_\tau - Q_0)^T] + aL^{-2}[P(\tau) - P(0)]$ with $G_\tau - G_0 = -G_\tau(T(\tau) - T_0)G_0$. $(Q_\tau - Q_0)^T\delta_y$ vanishes unless $B(y) \in S_A$, then is supported in $B(y)$ with norm $\le 2d|\tau|w$, and $Q_\tau^T\delta_{y'}$ is supported in $B(y')$ with norm ≤ 2; by (ii) the entry is $\le 4dc_{\rm CT}e^{\theta_W}|\tau|we^{-\theta_W \operatorname{dist}(B(y),y')}$, and the third term is symmetric. $T(\tau) - T_0 = \sum_{\Delta\in S_A} V_\Delta$, V_Δ collecting the bonds with $b_- \in \Delta$ and the pins of the block Δ; for one Δ, e.g. $\langle g, G_\tau(D_\tau - D_0)^T \mathbf{1}_\Delta D_\tau G_0 g'\rangle = \langle \mathbf{1}_\Delta(D_\tau - D_0)G_\tau g, \mathbf{1}_\Delta D_\tau G_0 g'\rangle$, bounded by $2\sqrt{d}|\tau|w\|\mathbf{1}_{\Delta^+}G_\tau g\|\|\mathbf{1}_\Delta D_\tau G_0 g'\|$ with Δ^+ the union of the $\le 3^d$ unit cubes touching Δ, and the localised L^2 forms of (ii) bound the two factors by $3^d c_{\rm CT}e^{\theta_W}e^{-\theta_W \operatorname{dist}(\Delta,B(y))}\|g\|$ and $c_{\rm CT}e^{-\theta_W \operatorname{dist}(\Delta,B(y'))}\|g'\|$; with $g = Q_0^T\delta_y$, $g' = Q_\tau^T\delta_{y'}$ this is the Δ-term. $P(\tau) - P(0)$ is block-diagonal over L-blocks, vanishes on L-blocks avoiding S_A, with entries $\le 5dL|\tau|w \le 5dLe^{2\theta_W L}|\tau|we^{-\theta_W(\operatorname{dist}(y,\Delta)+\operatorname{dist}(\Delta,y'))}$ for a Δ in the same L-block. Summing, $C_W := a_+^2[8dc_{\rm CT}e^{2\theta_W} + 4\cdot 3^d c_{\rm CT}^2 e^{\theta_W}(4\sqrt{d} + 4a_+ d)] + 5adL^{-1}e^{2\theta_W L}$. (iv): Schur test and the sum of (iii). For a complex Z the same entry bounds hold since U is entire. □

Lemma F.49 (two kernel tools)**.** (i) *Let B be real with (Cur) and Z, Z' complex with $w(Z), w(Z') \le c_\rho$. Then $|(X(B+Z) - X(B+Z'))(y,y')| \le C'_W \sum_\Delta \hat w_\Delta(Z - Z')e^{-\theta_W(d(y,\Delta)+d(\Delta,y'))}$, $C'_W :=$*

$3^d e^{2\theta_W} C_W$. (ii) *Put* $K_X := c_X \sum_{n\in\mathbb{Z}^3} |n| e^{-\delta_X|n|/2}$, $S'' := e^2 \sup_y \sum_\Delta e^{-(\theta_W/2)d(y,\Delta)}$, $\theta_2 := \min\{\delta_X/2, \gamma_X/(8K_X), \theta_W/2, 1\}$ *and* $c_* := \min\{c_\rho, \gamma_X/(8C'_W S''^2)\}$. *For real* B *with (Cur), complex* Z *with* $w(Z) \le c_*$, *every* $s \ge 0$ *and every* 1*-Lipschitz* ρ *on the unit lattice,* $\|e^{\theta_2\rho}(s + X(B + Z))^{-1} e^{-\theta_2\rho}\| \le 2/(s + \gamma_X)$, *hence* $|(s + X(B + Z))^{-1}(y, y')| \le \frac{2}{s+\gamma_X} e^{-\theta_2|y-y'|}$.

Proof. (i) Assign each bond to the unit cube of its b_-, $Z - Z' = \sum_\Delta Y_\Delta$, order the cubes and set $Z_i := Z' + \sum_{i'\le i} Y_{\Delta_{i'}}$; Z_i agrees with Z or Z' bond by bond, so $w(Z_i) \le c_\rho$. Telescope $X(B+Z) - X(B+Z') = \sum_i [X(B+Z_i) - X(B+Z_{i-1})]$; the i-th term is $X(B+Z_{i-1}+\tau Y_{\Delta_i}) - X(B+Z_{i-1})$ at $\tau = 1$, bounded by lemma F.48(iii) at the complex base $B + Z_{i-1} + \tau Y_{\Delta_i}$, integrated in τ; only the bonds of Y_{Δ_i} change, with amplitude $\le \hat w_{\Delta_i}(Z - Z')$, and the cubes containing their endpoints are the $\le 3^d$ neighbours of Δ_i, costing e^{θ_W} each. (ii) Schur test with $|e^{\theta|n|} - 1| \le \theta|n|e^{\theta|n|}$: $\|e^{\theta\rho} X(B) e^{-\theta\rho} - X(B)\| \le \theta K_X \le \gamma_X/8$; by (i) with $Z' = 0$ and a Schur test, $\|e^{\theta\rho}(X(B + Z) - X(B))e^{-\theta\rho}\| \le C'_W S''^2 w(Z) \le \gamma_X/8$; $s + X(B) \ge s + \gamma_X$ is self-adjoint, so $e^{\theta\rho}(s + X(B + Z))e^{-\theta\rho} = (s + X(B))^{1/2}(1 + K)(s + X(B))^{1/2}$ with $\|K\| \le \frac12$. Entries: take $\rho := d(\cdot, y')$. □

Lemma F.50 (gauge-invariant local densities of $\ln Z^{(j)}$)**.** *For* B *real with (Cur) and* Z *complex with* $w(Z) \le c_*$ *define, for* $y \in \Lambda$,

$$Q_y(B + Z) := -\tfrac12 \int_0^\infty \mathrm{Tr}_N \left[(1 + s)^{-1}\delta_{yy} - ((s + X(B + Z))^{-1})(y, y)\right] ds.$$

Then: (a) $F(B) = \sum_{y\in\Lambda} Q_y(B)$, *and* $\sum_y Q_y(B + Z) = -\frac12 \ln\det X(B + Z)$ *with the branch continuous from* $Z = 0$*;* (b) *(gauge invariance)* $Q_y(B + d\lambda) = Q_y(B)$ *for every real* λ *on the fine sites of* Ω, $(d\lambda)_b := \xi^{-1}(\lambda(b_+) - \lambda(b_-))$*;* (c) *(C-parity)* $Q_y(-B) = Q_y(B)$*;* (d) *(single locality) for* $w(Z), w(Z') \le c_*$, $Q_y(B + Z)$ *is analytic and* $|Q_y(B + Z) - Q_y(B + Z')| \le C_1 \sum_\Delta \hat w_\Delta(Z - Z') e^{-\theta_2 d(y,\Delta)}$, $C_1 := 4NC'_W S^2_{\theta_2}/\gamma_X$, $S_\theta := \sup_z \sum_y e^{-\theta|y-z|}$*;* (e) *(multi-point locality) for real* $u_1, \dots, u_n$, $n \le \bar n + 1$, $|D^n Q_y(B)[u_1, \dots, u_n]| \le C_{(n)} \sum_{\Delta_1,\dots,\Delta_n} \prod_i \hat w_{\Delta_i}(u_i) e^{-(\theta_2/n)\sum_i d(y,\Delta_i)}$, $C_{(n)} := 2^{n-1} n^n 3^d e^{\theta_2} C_1 c_*^{1-n}$*;* (f) $DQ_y(0) = 0$, *and for real* Y *with* sY *under (Cur) for* $s \in [0, 1]$ *and real* u, $DQ_y(Y)[u] = \int_0^1 D^2 Q_y(sY)[Y, u] ds$.

Proof. (a) For a positive symmetric matrix, $\ln X = \int_0^\infty [(1 + s)^{-1} - (s + X)^{-1}] ds$ (spectral calculus; the integrand is $O(s^{-2})$); take the trace over $\Lambda \otimes \mathbb{R}^N$, which is $\sum_y \mathrm{Tr}_N$ of the diagonal blocks; for complex Z both sides are analytic on $w(Z) \le c_*$ by lemma F.49(ii) and agree on real Z. (b) Under $B \to B + d\lambda$, $U(B_b) \to g(b_+)U(B_b)g(b_-)^{-1}$ with $g(x) := \exp(q e_j \lambda(x))$ orthogonal, all commuting; hence $D_B \to g D_B g^{-1}$, $Q_j(B + d\lambda) = g_{\mathrm{unit}} Q_j(B) g_{\mathrm{fine}}^{-1}$ (the contours start at the unit point), $P(B + d\lambda) = g_{\mathrm{unit}} P(B) g_{\mathrm{unit}}^{-1}$, $G_j \to g_{\mathrm{fine}} G_j g_{\mathrm{fine}}^{-1}$, so $X(B + d\lambda) = g_{\mathrm{unit}} X(B) g_{\mathrm{unit}}^{-1}$ with g_{unit} block-diagonal over y, commuting with the compression to Λ; therefore $((s + X(B + d\lambda))^{-1})(y, y) = g(y)((s + X(B))^{-1})(y, y)g(y)^{-1}$ and Tr_N is invariant. (c) Write $q = O(\oplus_i \theta_i J \oplus 0)O^T$ and $c := O(\oplus_i \sigma_z \oplus 1)O^T$, orthogonal with $cqc^{-1} = -q$; then $U(-B_b) = cU(B_b)c^{-1}$ and $X(-B) = (1 \otimes c)X(B)(1 \otimes c)^{-1}$ (for $N = 2$, $q = J$, $c = \sigma_z$: complex conjugation). (d) $(s+X')^{-1} - (s+X)^{-1} = (s+X')^{-1}(X - X')(s+X)^{-1}$ with $X := X(B+Z)$, $X' := X(B+Z')$; bound the (y, y) block by lemma F.49(ii) for both resolvents and (i) for $X - X'$: $N(2/(s + \gamma_X))^2 C'_W \sum_\Delta \hat w_\Delta \sum_{z,z'} e^{-\theta_2|y-z|} e^{-\theta_W(d(z,\Delta)+d(\Delta,z'))} e^{-\theta_2|z'-y|}$; since $\theta_2 \le \theta_W/2$ and $|y - z| + d(z, \Delta) \ge d(y, \Delta)$ the double sum is $\le S^2_{\theta_2} e^{-\theta_2 d(y,\Delta)}$, and $\int_0^\infty 4(s+\gamma_X)^{-2} ds = 4/\gamma_X$; analyticity by dominated convergence. (e) Decompose each $u_i = \sum_\Delta u_{i,\Delta}$; fix $(\Delta_1, \dots, \Delta_n)$, $\alpha_i := \hat w_{\Delta_i}(u_i) > 0$, and put $g(\tau) := \sum_{S\subset\{1..n\}} (-1)^{n-|S|} Q_y(B + \sum_{i\in S} \tau_i u_{i,\Delta_i})$ on $|\tau_i| \le \rho_i := c_*/(n\alpha_i)$, where $w(\sum \tau_i u_{i,\Delta_i}) \le c_*$. For each i, pairing S with $S \cup \{i\}$ gives differences with $Z - Z' = \tau_i u_{i,\Delta_i}$, each $\le C_1 3^d e^{\theta_2}(c_*/n) e^{-\theta_2 d(y,\Delta_i)}$ by (d); so $|g| \le 2^{n-1} 3^d e^{\theta_2} C_1 (c_*/n) \min_i e^{-\theta_2 d(y,\Delta_i)} \le \dots e^{-(\theta_2/n)\sum_i d(y,\Delta_i)}$. The mixed derivative $\partial_1 \cdots \partial_n g(0) = D^n Q_y(B)[u_{1,\Delta_1}, \dots, u_{n,\Delta_n}]$; Cauchy on the polydisc divides by $\prod \rho_i$; sum over the Δ_i. (f) By (c), $DQ_y(0) = -DQ_y(0)$; then $\frac{d}{ds} DQ_y(sY)[u] = D^2 Q_y(sY)[Y, u]$, each sY satisfying (Cur). □

Lemma F.51 (curvature form of a background change at one density)**.** *Let* B, A *be real bond fields on* Ω *such that* $B + \sigma A$ *satisfies (Cur) for* $\sigma \in [0, 1]$, $y \in \Lambda$, $R \ge 2$ *with* $C_R(y) := \{x : |x -$

$y|_\infty \le R\} \subset \Omega$. *Put* $g_B := \|q\|e_j \sup\{|(dB)(p)| : p \subset C_R(y)\}$, g_A *likewise,* $w_B := \|q\|e_j \sup_\Omega |B|$, w_A *likewise. Then*

$$|Q_y(B+A)-Q_y(B)| \le C_2\big[(g_B+g_A)g_A+e^{-\theta_2 R/8}(w_B+w_A+Rg_B+Rg_A)(w_A+Rg_A)\big], \qquad C_2 := 64C_{(2)}S^2_{\theta_2/4}.$$

Proof. Local axial gauge. For $x \in C_R(y)$ let $\gamma_{y,x}$ be the axis-ordered fine path from y to x (inside the box) and $\lambda_B(x) := \sum_{b\in\gamma_{y,x}} \xi B_b$; extend to Ω by $\lambda_B(x) := \lambda_B(\pi x)$, π the coordinatewise clamp onto $C_R(y)$. λ_B is linear in B; put $\beta := B - d\lambda_B$, $\alpha := A - d\lambda_A$, so $(B+\sigma A) - d\lambda_{B+\sigma A} = \beta+\sigma\alpha$. *Inside bounds* ($b \subset C_R(y)$): $\xi\beta_b$ is the sum of $\xi^2(dB)(p)$ over the $\le |b-y|_1/\xi+1$ plaquettes of a strip inside $C_R(y)$, so $|\beta_b| \le (|b-y|_1+\xi)\sup_{C_R(y)}|dB|$, likewise for α. *Outside bounds*: π maps the ends of a bond to one point or to the ends of a bond $b' \subset C_R(y)$, so $|(d\lambda_B)(b)| \le |B_{b'}|+|\beta_{b'}| \le \sup|B| + (3R+1)\sup_{C_R}|dB|$ and $|\beta_b| \le 2\sup|B| + (3R+1)\sup_{C_R}|dB|$ for every b, likewise for α. *Gauge invariance and C-parity*: by lemma F.50(b), $Q_y(B+A) - Q_y(B) = Q_y(\beta+\alpha) - Q_y(\beta) = \int_0^1 DQ_y(\beta+\sigma\alpha)[\alpha]d\sigma$, and by (f) this is $\iint D^2Q_y(s(\beta+\sigma\alpha))[\beta+\sigma\alpha,\alpha]\,ds\,d\sigma$, every background real with (Cur). *Estimate*: apply (e) with $n=2$; call Δ inside if $\Delta \subset C_{R-2}(y)$ ($d(y,\Delta) \le R$), otherwise $d(y,\Delta) \ge R-3$. Inside, $\hat w_\Delta(\beta+\sigma\alpha) \le (3d(y,\Delta)+5)(g_B+g_A)$ and $\hat w_\Delta(\alpha) \le (3d(y,\Delta)+5)g_A$; outside, $\hat w_\Delta(\beta+\sigma\alpha) \le 2(w_B+w_A)+(3R+1)(g_B+g_A)$ and $\hat w_\Delta(\alpha) \le 2w_A + (3R+1)g_A$. $\sum_\Delta (3d+5)e^{-(\theta_2/2)d} \le 8S_{\theta_2/4}$ and $\sum_{\text{outside}} e^{-(\theta_2/2)d} \le S_{\theta_2/4}e^{-\theta_2(R-3)/4}$; expand the product of the two Δ-sums. □

Lemma F.52 (Bałaban's configuration near $\operatorname{supp}\theta_{k+1}$; [2, pp. 567, 575–577])**.** *Let* $0 \le j < k$ *and* $N_k := B^{k-1}(\Lambda_2^{(k-1)})$ *(rescaled to scale* j*), where* $\theta_k = 1$. (a) *On* N_k *the configuration* [2, (2.93)] *equals* $A^{(k)}$*, hence* $B := \tilde B + A_1 = A^{(k)} - \tilde B^{(k)}$ *on* N_k. (b) $\operatorname{dist}_j(\operatorname{supp}\theta_{k+1}, N_k^c \cup \Omega^c) \ge D_{kj} := L^{k-j}(r(t_k) - c_b - M)$ *in scale-*j *units.*

Proof. (a) $\theta_{l+1} = 1$ on $B^l(\Lambda_2^{(l)})$ [2, p. 567], and $B^{k-1}(\Lambda_2^{(k-1)}) \subset B^l(\Lambda_2^{(l)})$ for $l \le k-1$ by $\Lambda_2^{(i+1)} \subset \Lambda_0^{(i+1)} \subset \Lambda_7^{(i)'} \subset \Lambda_2^{(i)'}$ [2, pp. 566, 570]; so every term $(1-\theta_{l+1})\theta_l A^{(l)}$, $l \le k-1$, of [2, (2.93)] vanishes on N_k, where $\theta_k = 1$; by [2, (2.89)], $\theta_{k+1}B^{(k+1)} + \theta_{k+1}A'^{(k)} + \tilde B^{(k)} = \theta_{k+1}A^{(k)}$, so the configuration is $(1-\theta_{k+1})A^{(k)} + \theta_{k+1}A^{(k)} = A^{(k)}$. (b) $\operatorname{supp}\theta_{k+1}$ lies in the M-neighbourhood of $B^k(\Lambda_2^{(k)})$; $\Lambda_2^{(k)} \subset \Lambda_7^{(k-1)'}$, which lies five layers of width $\ge r(t_{k-1}) - c_b$ (scale $k-1$) inside $\Lambda_2^{(k-1)'}$, i.e. $\ge r(t_k) - c_b$ in scale-k units at $L=5$; multiply by L^{k-j}. □

The inputs read from [2]: (R1) $|\partial^\eta A^{(k)}| \le C_{Z2}p(t_k)$ on $B^k(\Lambda_2^{(k-1)'}) \supset N_k$ [2, (2.60)]; (R2) $|\tilde B^{(k)}| \le C_\kappa t_k^\kappa$ and $|\partial^\eta \tilde B^{(k)}| \le C'_\kappa t_k^\kappa$ for every κ [2, (2.90)]; (R3) (Cur) at scale j on Ω for $\tilde B + A_1 + \sigma A_2$, $\sigma \in [0,1]$: by [2, (2.94), (2.98)] the curvature amplitude is $\le 4\|q\|C_{98}et_j^{1/2}p(t_j)$, which is $\le c_{\text{cur}}$ under the cap $\|q\|C_{98}et^{1/2}p(t) \le c_{\text{cur}}/4$ for $t \le \varepsilon_0$; (R4) $w_B \le C_w t_j^{-1/2}p(t_j)$ on Ω, from the restrictions [2, (2.55)] at each scale $l \in (j, k+1]$ with [2, (2.59)] and the rescaling $e_j A^{(l),(j)} = L^{-(l-j)}e_l A^{(l)}$.

Theorem F.53 (the terms containing $\tilde B^{(k)}$ are summable over $j<k$)**.** *Let* $1 \le k \le K-1$ *and* $j < k$*. Then for every* κ*,*

$$|D_j| \le C_G(\kappa)(1+(k-j))^{4+\bar p}L^{-(k-j)}t_k^{1+\kappa}p(t_k)|\Lambda_2^{(k)}|, \quad \text{hence} \quad \sum_{j<k}|D_j| \le C'_G(\kappa)t_k^\kappa|\Lambda_2^{(k)}| \ \text{ and } \ |D_j| \le C''_G(\kappa,\beta)t_k^\kappa|$$

for every $\beta > 0$*, with* C_G, C'_G, C''_G *depending only on* $(d, N, \|q\|, a, L, M, R, \varrho, \bar p, b_0, e, \mu_0, c_1, C_{Z2}, C_\kappa, C'_\kappa, C_{98}, C_w)$ *and the constants of lemma F.50, free of* $j, k, \varepsilon, \varepsilon_0$*, the torus and the history.*

Proof. Fix $j < k$, $A := A_2$, $B := \tilde B + A_1$ at scale j, and $R := R_{kj} := 4 + (32/\theta_2)(k-j)\ln L$; require $R \ge R_*(\theta_2, L, c_b, M)$, fixed before ε_0, so that $2R_{kj}+3 \le D_{kj}$ for all $k-j \ge 1$ (possible since $D_{kj} \ge L^{k-j}R/2$ grows faster than R_{kj}). *Count*: $\operatorname{supp}A \subset \operatorname{supp}\theta_{k+1}$, and the set S_A of scale-j unit cubes meeting it has $|S_A| \le (2M+1)^3L^{3(k-j)}|\Lambda_2^{(k)}|$. *Class I*, y with $d(y,S_A) \le R$: by lemma F.52(b), $C_R(y) \subset \Omega \cap N_k$, and by (a) $B = A^{(k)} - \tilde B^{(k)}$ there, so by (R1), (R2) and the rescaling, $g_B \le 2\|q\|eL^{-2(k-j)}t_k^{1/2}(C_{Z2}p(t_k) + C'_\kappa t_k^\kappa)$, $g_A \le 2\|q\|eL^{-2(k-j)}t_k^{1/2}C'_\kappa t_k^\kappa$, $w_A \le \|q\|eL^{-(k-j)}t_k^{1/2}C_\kappa t_k^\kappa$, and $w_B \le C_w t_j^{-1/2}p(t_j) = C_w L^{(k-j)/2}t_k^{-1/2}p(t_j)$ with $p(t_j) \le (1+(k-$

$j)\ln L)^{\bar p}p(t_k)$ (R4). lemma F.51 (hypotheses by (R3)): the main term is $\le CL^{-4(k-j)}t_k^{1+\kappa}p(t_k)$; the other is $\le Ce^{-\theta_2 R/8}R^2L^{(k-j)/2}L^{-(k-j)}(1+(k-j)\ln L)^{\bar p}t_k^{\kappa}p(t_k) \le CR^2(1+(k-j)\ln L)^{\bar p}L^{-4.5(k-j)}t_k^{\kappa}p(t_k)$, since $e^{-\theta_2 R/8}\le L^{-4(k-j)}$. The number of class-I y is $\le (2R+3)^3|S_A|$. *Class II*, $d(y,S_A)>R$: by lemma F.50(d) with $Z=A$, $Z'=0$ ($w(A)\le c_*$ for $t_k\le\varepsilon_0$ by (R2), a cap), $|Q_y(B+A)-Q_y(B)|\le C_1 w_A\sum_{\Delta\in S_A}e^{-\theta_2 d(y,\Delta)}$, which summed over class II is $\le C_1 w_A|S_A|S_{\theta_2/2}e^{-\theta_2 R/2}\le CL^{3(k-j)}L^{-(k-j)}L^{-16(k-j)}t_k^{1/2+\kappa}(2M+1)^3|\Lambda_2^{(k)}|$. *Sum*: by lemma F.50(a), $D_j=\sum_{y\in\Lambda}[Q_y(B+A)-Q_y(B)]$; class I gives $(2R+3)^3(2M+1)^3L^{3(k-j)}|\Lambda_2^{(k)}|\times[CL^{-4(k-j)}t_k^{1+\kappa}p+CR^2(1+(k-j)\ln L)^{\bar p}L^{-4.5(k-j)}t_k^{\kappa}p]$, whose first part is $\le C(1+(k-j))^3L^{-(k-j)}t_k^{1+\kappa}p|\Lambda_2^{(k)}|$ and whose second part is smaller by $L^{-1.5(k-j)}$ up to polynomials at the cost of one power of t_k (take κ one larger in (R2)); class II is smaller still. *Sums over j*: $\sum_{m\ge1}(1+m)^{4+\bar p}L^{-m}<\infty$ and $t^{1+\kappa}p(t)\le P(1,1)t^{\kappa}$; for the per-$j$ form absorb $(1+m)^{4+\bar p}$ into $L^{\beta m}$. □

Per scale-j cube the bound is $\sim L^{-4(k-j)}t_k^{1+\kappa}p=t_j^4t_k^{\kappa-3}p$, i.e. $O(t_j^{4-\beta})$ for $\kappa\ge3$: the printed claim "$O((L^j\varepsilon)^{\kappa})|B^{k-j}(\Lambda_2^{(k)})|$ with $\kappa>d$, $\kappa_0>0$" holds with $\kappa=4-\beta$, $\kappa_0=1-\beta$, the double-curvature count.

Corollary F.54 (the $Z^{(j)}$ expansions, consumed form)**.** *For $1\le k\le K-1$, $\sum_{j<k}|F_j(\tilde B+A_1+A_2)-F_j(\tilde B)-P_{\tilde B}[A_1]|\le(C_\kappa/(L^{\bar n+1-d}-1)+C'_G(\kappa))t_k^{\kappa}|\Lambda_2^{(k)}|$ for every $\kappa<(\bar n+1)/2$, with per-j form $O(t_k^{\kappa})|\Lambda_2^{(k)}|L^{(j-k)\kappa_0}$, $\kappa_0=\min(\bar n+1-d,1-\beta)>0$. This covers the three categories of the sentence of* [2, p. 578]*: the order-$(\bar n+1)$ trace term, the R-vertices and the orders $>\bar n$ (R_j), and the terms containing $\tilde B^{(k)}$ (D_j);* [2, (2.104)] *is unchanged.*

Every gauge step above is the exact invariance of a full object, Q_y or F_j, under a gauge transformation of the background on all of Ω; the local axial gauge of lemma F.51 is used only to bound fields, never to split an identity, and no Ward identity on a photon line enters.

The per-step large-field constants are free of ε_0. The order of constants in [2, Sect. 3] requires that each per-step constant of the large-field chain be fixed before ε_0, so that the final caps on ε_0 (lemma 22.1) are consistent. A constant can depend on ε_0 only through a cap "for $e(t)$ sufficiently small" or through a bound stated at the final scale; in the first case the constant is the sup over $t\in(0,1]$ of an explicit function, finite by $E(\kappa,\delta)$, $P(a,m)<\infty$, in the second case it is replaced by the scale-local bound. The constants of the chain are: $c_{3.29}$ of [2, (3.29)], from [4, Prop. 3.1′, Sect. 4], whose remainder $O(1)e^2p^2(e)(|\phi(x)|^2+|\phi(x')|^2)$ in [4, (4.7)] is built from [4, Lemma 2.1, (4.10), (4.13), (4.22)] with γ_0'' depending "on d and a only", so that $c_{3.29}(k)=C_{\mathrm{Ba4}}e^{2-\alpha}C_\phi^2\lambda^{-1/2}t_k^{(1-\alpha)/2}p_k^2$ is scale-local; $C_{2.57}$ of [2, (2.57)], which depends on $(d,L,M,\alpha_0,\bar n,a,c_1,b_0,\bar p,R,\varrho,e,\lambda,N,\beta)$ and on the $O(1)$ of Proposition 1, with ε_0 entering only through the caps $p(t)^n\lambda(t)^{-m}e^{-\delta r(t)}\le t^{\kappa}$ (Bałaban's order of choices, [3, p. 421]); the counterterm constants, $|\delta m^2|\le c_m\varepsilon^{-1}$ (per-graph power counting, no ε_0) and the extensive bounds on the vacuum energy and on $\delta m^2_{K,k}$ (Step 2 and theorems F.28 and F.31), whose constants are built from renormalised power counting; and the Z-factor constant C'_G of theorem F.53, whose caps are preliminary. So every per-step constant is free of ε_0, and the order statement of theorem F.2 holds.

Step 3(D_3): [3, Prop. 1] *at $L=5$, the regrouping after the cut.* The proof of [3, Prop. 1, (1.33)] on [3, pp. 420–445] consists of twenty sketched steps and seven further sketched places. Most of them are direct computations on the printed definitions (the slices [3, (2.6), (2.10)–(2.12)], the summation by parts [3, (2.8)], the induction [3, (2.16)], the generalised graphs of [3, Prop. 2.2], the estimate [3, (2.17)], the Ward–Takahashi identities of [3, pp. 430–431] for full lines, the estimate [3, (3.1)], the cube steps of [3, p. 433], the norms [3, (1.32)] at B_0 and at $\tilde B$), whose written form is a transcription with the $L=5$ constants of theorem F.2, given in appendix H. Two steps need more than a transcription: the regrouping into renormalised classes after the separated lines are cut ([3, p. 432]: "if we remove a line in any graph of the class, then for the obtained graph we can find in this class all graphs necessary to form a new renormalized class"),

and the renormalisation of the divergent classes ([3, pp. 434–445]). They are proved here and in Step 3(D_4); the conventions are Bałaban's, made explicit.
Conventions. (CV) Degrees are computed in Bałaban's count: each of [3, (1.13), (1.14), (1.15)] is a vertex with its own degree ([3, p. 423 (v), p. 414]); the two factors of one product [3, (1.18)] share the block point y and are joined in the bound function by a contact, a top-scale line of dimension 0 whose kernel is the indicator that both summation points lie in $B^k(y)$. (Loc) Label boxes are closed grid boxes of side ≤ 2 on the block-corner grid ([3, p. 420]: "for each leg of the vector field a smooth partition of unity ... a support of each function is contained in a cube with sides of length 2"); a label is attached to each leg of a vector field (the support box of its partition function $H_y \subset y + [-1,1]^d$), to each [3, (1.6)]/[3, (1.7)] vertex (its unit cube) and to each block factor ($B^k(y)$), and every vertex lies in the intersection of the boxes of its labels. The number of label boxes of a graph is $V_{\rm lab} \le (2\bar n+1)V_{\rm Bal}$, $V_{\rm Bal} \le 2\bar n(4N_A+5)$, constants of $\bar n$ only; put $l_N := 2V_{\rm lab}(\bar n)$. (Agr) The counterterm graph $T_\gamma G$ of a two-scalar-leg subgraph γ is G with γ's two external legs attached at one point x [3, p. 417], x the leg point of the anchor vertex v_a of γ; γ's internal vertices keep their labels ([3, p. 420], the localisation is performed "for the terms of $\delta m_k^2(\dots)$, $E_k(\dots)$ also"), so G and $T_\gamma G$ have the same label multiset (Convention F.18(c)). (R-W′) Two graphs are "in agreement" [3, p. 432] if they carry the same label multiset; a *per-label class* of a given type and order is the set of all graphs of that type and order with a given label multiset, equivalently (with the partition functions written as multipliers $t_y H_y$ on the vector legs and $t_Q \mathbf{1}_Q$ on the [3, (1.6)]/[3, (1.7)] vertices) the coefficient of one monomial $\prod t$ in the expansion of [3, (1.5)]; this is how [3, p. 444] evaluates "a product of the values of all the localization functions at the point x'" and then takes the graph "with the summations unrestricted". (H_A) For an external field with a leg that can sit at several vertices of a class (the counterterm merge moves it to the anchor, leg exchanges move it too), the localisation h of [3, p. 420] ("$h = 1$ on box(v) and $h = 0$ outside some neighborhood of box(v)") is taken $= 1$ on the label boxes of every vertex at which the leg sits in some term of the class and 0 outside their $\frac14$-neighbourhood; for moves inside a cluster this is a neighbourhood of size $\le l_N$, and for moves across clusters the extra box is paid by lemma F.59(a). It costs nothing to the consumers, which sum over localisations [3, p. 421] or use fields localised at one point.

Definition F.55 (clusters; the cluster cut rule)**.** Let τ be a term of a per-label class and $\mathcal B$ its multiset of label boxes, common to the class. The *clusters* are the connected components of the graph on $\mathcal B$ in which two boxes are joined when they intersect; U_K is the union of the boxes of the cluster K. Each end of a line of τ lies at a vertex, all of whose labels contain it, so they intersect and lie in one cluster K(end) (for a counterterm vertex, the cluster of v_a). The *cut set* $S(\tau)$ is the set of lines whose two ends lie in different clusters. A cut line becomes the external field $h_K G h_{K'}$ (or its vector analogue), with h_K smooth, $h_K = 1$ on the η-neighbourhood of U_K, $\operatorname{supp} h_K$ in the $\frac14$-neighbourhood of U_K and $|\partial^\eta h_K|, \operatorname{Lip}(\partial^\eta h_K) \le c(d)$ (e.g. $h_K = 1 - \prod_Q (1-\varphi_Q)$ over the unit grid cubes Q of U_K, φ_Q a fixed bump). This is Bałaban's threshold 1 [3, p. 432], applied to clusters instead of single lines.

Lemma F.56 (properties of the cut rule)**.** (a) *If $K \ne K'$ then* $\operatorname{dist}_\infty(U_K, U_{K'}) \ge 1$*, and points of* $\operatorname{supp} h_K$ *and* $\operatorname{supp} h_{K'}$ *are at distance* $\ge \frac12$. (b) $\operatorname{diam}_\infty U_K \le 2|K| \le l_N$. (c) *Every kept line joins two vertices of one cluster, so every kept-connected component lies in one U_K.* (d) *Let γ be a subgraph all of whose label boxes lie in one cluster K; any operation that changes only the vertices of γ at which lines are attached (the counterterm merge of* [3, p. 417]*, the exchange of two external legs of γ), or replaces γ by another graph with the same labels, leaves S unchanged.* (e) *A connected subgraph whose vertices lie in at least two clusters contains a cut line and is disconnected by S, each piece lying in one cluster.* (f) *A subgraph whose vertices lie in one cluster has no cut line.*

Proof. (a) Two grid boxes either intersect or are at sup-distance ≥ 1; boxes of different clusters do not intersect; the collars add $\le \frac14$ on each side. (b) Two boxes of K are joined by a chain of at most $|K|-1$ intersecting boxes of side ≤ 2. (c) is the definition. (d) The moved ends stay at vertices of γ, whose labels are in K, and the other ends do not move; a replacement with the

same label multiset does not change $\mathcal{B}$, hence not the clusters. (e) Kept lines are intra-cluster, so a kept path cannot leave a cluster. (f) is immediate. □

Let G be a labelled graph and $\mathcal{C} = R(G)$ its class: the forest sum over $\mathcal{D}$, the partnered subgraphs of G (the one-particle-irreducible two-scalar-leg subgraphs with mass counterterms, [3, p. 430], and the vacuum components with E_1, [3, p. 432]), closed under leg exchange of 2ϕ and $\phi\phi A$ subgraphs and under per-label completion of $2A$ and $\phi\phi A$ subgraphs. Split $\mathcal{D} = \mathcal{D}_0 + \mathcal{D}_1$, $\mathcal{D}_0 := \{\gamma : \text{all label boxes of } \gamma \text{ in one cluster}\}$. For a forest $F = F_1 + F_0$ ($F_1 \subset \mathcal{D}_1$, $F_0 \subset \mathcal{D}_0(F_1)$, the elements of $\mathcal{D}_0$ nested in or disjoint from every element of F_1), the term $\tau_F := \prod_{\gamma\in F}(-T_\gamma)G$ is cut by its own cut rule.

Theorem F.57 (the regrouping after the cut)**.** (i) *For fixed F_1 the cut set $S(\tau_{F_1+F_0})$ is the same for every F_0; call it $S(F_1)$.* (ii) $\sum_F E(\tau_F) = \sum_{F_1} R_{\mathcal{D}_0(F_1)}[E(\mathrm{Cut}_{S(F_1)}Y(F_1))]$, $Y(F_1) := \prod_{\gamma\in F_1}(-T_\gamma)G$, *each F_0-operation acting at its place (inside a counterterm coefficient when nested in an element of F_1).* (iii) *Every element of $\mathcal{D}_1$ is disconnected by $S(\emptyset)$.* (iv) *Every connected divergent subgraph of $\mathrm{Cut}\,Y(F_1)$ (kept lines only) is an element of $\mathcal{D}_0(F_1)$, renormalised by $R_{\mathcal{D}_0(F_1)}$; or a $2A$ or $\phi\phi A$ subgraph inside one cluster whose per-label partners are terms of the same block (theorem F.62); or a* [3, (2.4)] *subgraph (no partner; treated by the summation by parts* [3, (2.8)]*); or a subgraph containing a counterterm vertex of F_1, which by lemma F.59 is convergent unless it is a vacuum graph or has one external vector leg (Bałaban's exception list,* [3, p. 432]*). So each block is a renormalised class of its cut graph (*($\mathrm{REN}_{\mathrm{orb}}$)*, lemma 5.2, is the same regrouping with the slice decomposition in place of the cut).*

Proof. (i) The F_0-operations move ends inside clusters (lemma F.56(d)). An element $\gamma_0 \in \mathcal{D}_0(F_1)$ contains no line whose end is moved by an element $\gamma \in F_1$: such a line is external to γ and attached at a vertex of γ, so γ_0 would share a vertex with γ, hence be nested with it by compatibility; γ_0 inside γ contains no external line of γ, and γ_0 containing γ would contain γ's cut line, against $\gamma_0 \in \mathcal{D}_0$. For vacuum elements of F_1 nothing moves: the vacuum graph and its counterterm graph (same graph, same labels, Convention F.18(b)) are cut identically. (ii) Every forest of $\mathcal{D}$ splits uniquely as $F_1 + F_0$ with F_1 a forest of $\mathcal{D}_1$ and F_0 a forest of $\mathcal{D}_0(F_1)$, and conversely; by (i) the cut is the same for all F_0 at fixed F_1, and Cut commutes with the F_0-operations because they act inside clusters and the elements of $\mathcal{D}_0$ have no cut line (lemma F.56(f)). (iii) lemma F.56(e). (iv) A divergent connected kept subgraph H of $\mathrm{Cut}\,Y(F_1)$ without a counterterm vertex of F_1 lies in one cluster (lemma F.56(c)) and avoids the internal vertices of the elements of F_1 (these sit inside the counterterm coefficients). If H is 2ϕ it belongs to $\mathcal{D}$, lies in one cluster and is disjoint from F_1, so $H \in \mathcal{D}_0(F_1)$; if H is $2A$ or $\phi\phi A$, theorem F.62; [3, (2.4)] needs no partner; subgraphs through a counterterm vertex of F_1 are lemma F.59. □

Corollary F.58 (the cut kernels)**.** *In every term of every block, every cut line joins two different clusters, so its kernel is evaluated on* $\mathrm{supp}\,h_K \times \mathrm{supp}\,h_{K'}$*, at distance* $\geq \frac{1}{2}$*. The estimate* [3, (3.1)] *holds with $h_K, h_{K'}$ in place of h, h': $\|h_K G_k(\Omega, \tilde{B}) h_{K'}\|_{1,\alpha} \leq Ce^{-(\delta_1/2)(\mathrm{dist}(U_K,U_{K'})-1/2)}$, its proof using only the slice bounds at the points of the supports, the separation, and $|\partial h|, \mathrm{Lip}(\partial h) \leq c_h$; the points used lie within $\frac{1}{4}+\eta$ of vertex label boxes, at the depth the hypotheses supply ($R \geq R_0 + 5$). The tree factor of* [3, (1.33)] *satisfies $d(\{\square(v)\}) \leq \sum_{\text{spanning tree of cut kernels}} \mathrm{dist}(U_K, U_{K'}) + \sum_K \mathrm{diam}\,U_K$, and the second sum is $\leq l_N^2$, a constant.*

With the printed per-line threshold $\mathrm{dist}(\square(v), \square(v')) \geq 1$ and the counterterm leg move of [3, p. 417], a line from the moved leg of γ to a vertex w could be cut in G and short in $T_\gamma G$, or conversely, and then either the cut kernel is not $O(1)$ or the cut does not commute with T_γ; the cluster rule removes this (the example of three boxes $[0,1], [1,2], [2,3]$ on one axis is one cluster, and the line is kept in G and in $T_\gamma G$), with Bałaban's threshold 1 and no larger cut threshold.

Lemma F.59 (disconnected counterterms are finite vertices)**.** *Let $\gamma \in \mathcal{D}_1$ (so disconnected by S), with anchor v_a, and let $c(x)$ be the coefficient of T_γ in $Y(F_1)$ with γ's labels fixed: $c(x) = \sum_{x'}$ of the amputated kernel of γ, renormalised by the inner forests, with its lines of $S(F_1)$ replaced by cut kernels. Assume* [3, (1.33)] *for all classes with fewer vertices than G (the induction on the number of vertices that the classes already use,* [3, p. 431]*: "our definition of*

the classes G_{ren} is in fact inductive"). Then (a) $|c(x)| \le C e_k^{d_v(\gamma)} \lambda_k^{d_s(\gamma)} \exp(-\delta' d(\text{labels of } \gamma))$ *uniformly in $x, k, \varepsilon, \eta, \Omega, \Omega_1, \Omega_2, \tilde{B}$ and the torus;* (b) *T_γ is therefore a* [3, (1.7)] *vertex with a finite counterterm, of degree $D(v) = 2$ (*[3, p. 423 (ii)]*); every connected subgraph containing it has $D \ge 1$ if $E \ge 2$ and $D \ge 0$ if $E = 1$, with equality only for the tadpole graph, and vacuum graphs are treated by the vacuum rule of* [3, p. 432]*;* (c) *c is an even polynomial in q, so the tadpole with this vertex vanishes after the gauge-out of B_0* [3, p. 433].

Proof. (a) The pieces of $\gamma|_{S(F_1)}$ lie in single clusters, each with fewer vertices than G since γ has at least two pieces; each piece with its inner renormalisation is a class of lower order whose external fields are the cut kernels, two-variable external fields in the sense of [3, p. 419]; [3, (1.33)] for each piece, the product over pieces, and corollary F.58 for the cut kernels give (a), the decay coming from the cut kernels between the clusters of γ. (b) A [3, (1.7)] vertex with a bounded coefficient has $D(v) = d - (d-2) = 2$, and the degree count of [3, (2.1)–(2.3)] applies with $\kappa = 2$ at that vertex. (c) γ has no external vector leg; every closed scalar index loop of γ carries an even number of vector legs or vanishes ($\operatorname{Tr} q^{\mathrm{odd}} = 0$), and the scalar path between γ's two legs then carries an even number as well, since the total number of vector-leg ends of γ is twice its number of photon lines. □

Lemma F.60 (per-label gauge invariance)**.** *Let Y be a periodic box or $\xi\mathbb{Z}^d$, X any bond field built from A' and external fields (e.g. $X = \sum_y t_y H_y e g A' + \tilde{A}$-type legs with their multipliers), F any polynomial in the $|\phi(x)|^2$ with x-dependent coefficients (e.g. $\sum_Q t_Q \lambda \mathbf{1}_Q |\phi|^4$, the* [3, (1.7)] *vertices with any coefficients, the restriction to Ω_1), and $Z(A;t) := \int d\mu(A') \int d\phi \exp[-\frac{1}{2}\langle \phi, (-\Delta_{A+X} + m^2)\phi\rangle - F(\phi)]$ with any Gaussian measure $d\mu(A')$. Then $Z(A + d\lambda; t) = Z(A;t)$ for every periodic (on $\xi\mathbb{Z}^d$: finitely supported) λ and every t. Consequently every t-monomial coefficient of every A-Taylor coefficient of $\log Z$ satisfies the Ward–Takahashi identities of* [3, pp. 430–431]*; in particular the two-vector-leg kernel Π^{lab} of a per-label class obeys $\sum_{y'} \Pi^{\mathrm{lab}}_{\mu\nu}(y, y') = 0$.*

Proof. $\phi = V_\lambda \psi$: the covariant Laplacian is covariant for the total field, $-\Delta_{A+d\lambda+X} = V_\lambda(-\Delta_{A+X})V_\lambda^*$ for every X, because only the total field enters U; F is invariant, being a function of the $|\phi(x)|^2$; the A'-integral is untouched since the transformation acts on A only. The identities follow coefficient by coefficient in t, and by (R-W′) the per-label class is one t-monomial. □

The identity is for complete per-label classes with full propagator lines, before any scale decomposition; it is never applied to a single ordering or scale assignment. In Step 3(D_4) it is applied only to the all-C^ξ part of a regrouped local coefficient (full lines C^ξ on $\xi\mathbb{Z}^3$), the remainders being bounded explicitly.

Lemma F.61 (multipliers to the anchor; [3, p. 444, (3.36)])**.** *For a node with anchor $\hat{x}$ and a smooth multiplier H (partition function, g_k) at a vertex x_i of the node, $H(x_i) = H(\hat{x}) + [H(x_i) - H(\hat{x})]$ with $|H(x_i) - H(\hat{x})| \le c|x_i - \hat{x}|$; the second term carries a factor $|x_i - \hat{x}|$ on a path inside the node, so its degree gains* 1*. After all smooth multipliers of the node (including those of its external legs) are moved to $\hat{x}$, the remaining sums over the node's internal positions are unrestricted, except for sharp indicators (lemma F.67).*

Theorem F.62 (the Ward–Takahashi and $\phi\phi A$ classes at fixed localisation)**.** *Let H be a $2A$ or $\phi\phi A$ subgraph of a term of a per-label class.* (a) *If the label boxes of H lie in one cluster, every per-label partner $\tilde{H}$ (same type, order, label multiset) lies entirely in that cluster and has no cut line, and so does every leg-exchange partner (lemma F.56(d), (f)); the per-label completion is therefore a sum of terms of one block, and at the renormalisation step the local coefficient of the completed node is the per-label class sum, to which lemma F.60 applies ($2A$ zeroth moment, after lemma F.61 has moved the multipliers of H's external legs to the anchor), as do lemmas F.61 and F.67 with the reflection identities of* [3, (3.30)] *($\phi\phi A$ zeroth moment, $2A$ first moment).* (b) *If they span at least two clusters, every partner is disconnected by S (lemma F.56(e)) and no identity is needed.*

Proof. lemma F.56. □

At fixed vertex localisations of all vertices of H the zeroth-moment sum would be restricted to boxes and would not vanish (boundary layers of width $L^{j_0}\eta$); Bałaban localises legs, not vertices [3, p. 420], and moves the localisation functions to one point before the identity is used [3, p. 444], and with (R-W′) the identity holds label by label.

Consequences for the neighbouring steps: under the cluster rule the kept components lie in clusters (lemma F.56(c)), and the class-component cube of [3, p. 432] has side $\le l_N$, common to the class; kept lines may join boxes at distance up to l_N, so [3, (3.2)] for components carries the constant $e^{\delta l_N}$. The cut on the cube after the $\tilde{B}'$-expansion [3, p. 433] is the cluster rule on the label boxes of the expanded class; every component with an old vertex lies in the cluster of an old vertex, of diameter $\le 2V^\sharp_{\mathrm{lab}}$ ($V^\sharp_{\mathrm{lab}}$ the label count after the expansion), so the side of the localisation cube of a connected renormalised class after the expansion is $l^\sharp := l_N + 4V^\sharp_{\mathrm{lab}}(\bar{n})$, which fixes $r^\flat$ in theorem F.2(a) through $r^\flat \ge 3l^\sharp + R_0 + 12$. Chain components lie in one cluster, so their exponential factor is $\ge e^{-2\delta V^\sharp_{\mathrm{lab}}}$.

Step 3(D_4): the renormalisation of the divergent classes, [3, pp. 434–445]. After the cube steps of [3, p. 433] a class component has scalar lines $G_k(0)$ on $\eta\mathbb{Z}^d$, sliced by [3, (2.6)], photon lines on the torus, vertices in the cube's inner part, all other objects external, and per-label full classes. Orderings are the lexicographic ones of the slice indices; for an ordering $\tilde{l}$ the components $G_i^{(\alpha)}$ along it form a laminar family. A *dangerous node* is a component with $D \le 0$; by the degree formula [3, (2.1)–(2.3)] it is one of: a [3, (2.4)] subgraph ($D = 0$); 2ϕ ($D \in \{-1, 0\}$, $D = -1$ only at one loop, [3, (3.6), (3.8)]); $2A$ ($D \in \{-1, 0\}$, $D = -1$ only at one loop, [3, (3.25)]); $\phi\phi A$ ($D = 0$, one loop: $D \ge V/2 + E/2 - 3$ with $E = 3$ forces $V \le 3$ and one loop); a vacuum component, cancelled by the vacuum counterterms [3, (1.24)] with the same labels and lines, up to the torus-versus-$\eta\mathbb{Z}^d$ image terms. Subgraphs with no external scalar leg and an odd number of external vector legs vanish as factors at $B = 0$ (their scalar index loops close inside them, $\operatorname{Tr} q^{\mathrm{odd}} = 0$). A one-particle-reducible 2ϕ node is not partnered ([3, p. 430]: "We consider one-particle-irreducible graphs only"); since $D(P_1 + \mathrm{line} + P_2) = D(P_1) + D(P_2) + 2$, a dangerous reducible 2ϕ chain consists of one-loop $D = -1$ parts, and after these are processed the chain is no longer dangerous. External vector fields inside vertices ($\tilde{A}$, and $\tilde{B}'$ after the cube step) count as external vector legs for the type of a node (an $\tilde{A}$-leg adds $\kappa = n' = 1$ where an external leg adds 1 to E), so the partnered 2ϕ nodes are exactly the 2ϕ nodes without such legs. A node S is *useful* along $\tilde{l}$ if its internal lines precede all its external lines, i.e. if it is a component. *Processing* a useful dangerous node S with an anchor $\hat{x}$ (a vertex of S): [3, (2.4)] subgraphs are handled inside the smallest dangerous non-[3, (2.4)] node containing them (lemma F.65), or alone by [3, (2.8)] at either vertex; 2ϕ: the pairing of G with its counterterm member, anchored at the vertex fixed by the counterterm convention (Agr), the exact identity [3, (3.19)] for $D = 0$ and [3, (3.11)] for $D = -1$, which leaves a first-order local vertex with the regrouped coefficient and a remainder of degree $1 + \alpha$; $2A$, $\phi\phi A$: Taylor expansion of the external fields at $\hat{x}$ (zeroth order for $D = 0$, first order for $D = -1$), with lemma F.61 for the multipliers and lemma F.67 for sharp indicators, the local terms carrying the regrouped coefficients of theorem F.69 and the remainders gaining α (resp. $1 + \alpha$) on a path inside S. Non-[3, (2.4)] inner nodes are processed before outer ones ([3, p. 445]: "assuming that it holds for all its subgraphs"), and the Taylor split of an outer node is applied to all terms produced by the inner processing (lemma F.64).

Lemma F.63 (exactness and degree preservation)**.** (a) *Each processing step is an identity, at fixed external positions, of the sum over the orderings in which S is useful: the Taylor formulas,* [3, (3.19), (3.26), (3.34), (2.8)] *and the regrouping into classes are identities, and the counterterm pairing is an identity of class sums (theorem F.57). Hence the local coefficient of S, summed over S's internal orderings, has the value of the regrouped* unprocessed *coefficient; more precisely, the processed expression evaluated at external fields of S that are polynomials of degree ≤ 1 in the positions equals the unprocessed one, because every identity holds for every external field.* (b) *Processing a useful node S changes degrees only for subgraphs of S containing a line of the Taylor/Hölder path ($+\alpha$, or $+1+\alpha$) and for subgraphs containing an external line of S that carries the Hölder quotient ($-\alpha$, or $-1-\alpha$); every component along $\tilde{l}$ that contains such an*

external line contains S, so its net change is 0*. In the local term S becomes a vertex v_S at $\hat{x}$ with vertex power $D(S)$ at its vertex scale $s(v_S) = L^{j_0}\eta$ (j_0 the lowest external index), and every component containing v_S has the degree of the original component with S, which is the convention* [3, (2.3)]*.*

Proof. (a) as listed; the inner processings are identities on complete groups of inner orderings, which are sub-orderings of S's internal orderings. (b) A later component that contains a vertex of S contains all of S, since S is a component and components grow by merging along $\tilde{l}$; an earlier component contains no external line of S, these having higher index. □

Lemma F.64 (inner processing acting on an outer node's external fields; [3, p. 445])**.** *Let S' be a dangerous non-*[3, (2.4)] *node strictly inside a useful dangerous node S along $\tilde{l}$, processed first with any anchor. In each term it produces, an external field F of S enters either unchanged or through a combination $\Delta[F]$ that vanishes for constant F (a difference $F(x_i) - F(x_j)$, a Taylor remainder $F(x_i) - F(x_j) - (x_i - x_j)\cdot\partial F(x_j)$, or a derivative $\partial F(x_i)$); call the latter Δ-terms. Then:* (a) *S's degree in a Δ-term is $\geq D(S) + \alpha$: the gain of S''s processing lies inside S and the cost on S's external line, which is not counted in $D(S)$; if $D(S) = 0$, S is not dangerous in a Δ-term, and if $D(S) = -1$, S is one-loop and contains no non-*[3, (2.4)] *dangerous node.* (b) *Apply S's Taylor split (for 2ϕ, its counterterm pairing) to every term, Δ-terms included: in a Δ-term the local part is Δ evaluated at a constant field, i.e.* 0*; for 2ϕ the corresponding term of the counterterm member vanishes, since in $T_S(\cdot)$ the external fields of S sit at $\hat{x}$ and Δ of them is* 0*; the remainder of a Δ-term is the Δ-term itself.* (c) *The local part of S, summed over all terms, is the processed expression evaluated at constant external fields of S, which by lemma F.63(a) equals the regrouped unprocessed coefficient; so theorem F.69 bounds it, and the identities of lemma F.60 and* [3, (3.30)] *apply to it.*

Proof. (a) is lemma F.63(b). (b) The external fields of S are evaluated at $\hat{x}$ by the zeroth-order split, and every Δ annihilates constants; the counterterm member's statement is the definition of the counterterm graph [3, p. 417]. (c) lemma F.63(a) at constant fields. □

So no anchor rule is needed: inner processings may act on outer fields; when they do, the outer node becomes convergent in those terms and its local coefficient is unchanged. Bałaban's choice (derivatives never on G_0's legs) is one admissible choice among several. The anchors of $2A$ and $\phi\phi A$ nodes are free; the anchor of a 2ϕ node is the vertex at which the counterterm graph of [3, p. 417] merges the legs, the same in G and in $T_S G$.

Lemma F.65 (the [3, (2.4)] subgraphs inside a node; [3, p. 443])**.** *Let S be a useful dangerous node of type 2ϕ, $2A$ or $\phi\phi A$ with anchor $\hat{x}$, and $\Sigma_1, \dots, \Sigma_r$ the fresh* [3, (2.4)] *components inside S along $\tilde{l}$ (vertex-disjoint, since a* [3, (1.8)] *vertex has one derivative leg). There is an order of transport and a choice of Hölder paths such that in every term of the processed expression each Σ_i either carries a Taylor/Hölder factor on its line, or has a vertex that carries no field external to S, at which* [3, (2.8)] *is applied with no derivative on an external field of S.*

Proof. Write $\Sigma_i = \{a_i, b_i\}$, with a_i the anchor if the anchor is a vertex of Σ_i. Step 1: for each i move every field external to S that sits at b_i onto a_i, to the Taylor order of S: $F(x_{b_i}) = F(x_{a_i})[+(x_{b_i} - x_{a_i})\cdot\partial F(x_{a_i})]+$ remainder, the remainder carrying $|x_{b_i} - x_{a_i}|^{\alpha}$ (resp. $^{1+\alpha}$) on Σ_i's line, which makes Σ_i's degree $\geq \alpha > 0$; the Σ_i are vertex-disjoint, so these steps commute, and Δ-fields are moved the same way. Step 2: transport all fields to $\hat{x}$ along any paths, as [3, (3.19), (3.26), (3.34)] require (for 2ϕ the two-step splitting $\phi_b(x') - \phi_b(x) = [\phi_b(x') - \phi_b(y)] + [\phi_b(y) - \phi_b(x)]$, y the point of a_i). In every term produced by step 2, b_i carries no field external to S: its fields were moved to a_i in step 1, step 2 moves fields only from a_i and other vertices, and fields at $\hat{x}$ are constants in x_{b_i}. [3, (2.8)] at b_i moves the derivative from Σ_i's line onto b_i's vector leg, onto g, or onto b_i's other scalar leg, all internal to S or carrying a constant field; the transformed component gets degree 1 and later components through b_i contain Σ_i's line. If Σ_i joins the two leg vertices of S (the type [3, (3.8)], or the one-loop $2A$ bubble), step 1 is S's own first-order Taylor expansion along Σ_i's line. Inside a regrouped local coefficient there are no external fields at all (they are factored out at $\hat{x}$), so the integration by parts is free there. □

Lemma F.66 (useless counterterms; [3, p. 438])**.** *Let γ be a 2ϕ subgraph with external lines l_a, l_b, fix the indices of all lines outside γ, and let j_* be the lexicographically smaller of the indices of l_a, l_b, $s_* := L^{j_*}\eta$. Let $c_{\mathrm{use}}(x; j_*)$ be the counterterm coefficient of T_γ restricted to the internal index assignments in which γ is useless (maximal internal index $\geq j_*$), with γ's inner divergences processed. Then $|c_{\mathrm{use}}(x; j_*)| \leq C s_*^{D(\gamma)}$ for $D(\gamma) = -1$ and $\leq C(1 + \log_L(1/s_*)) \leq C_{\alpha'} s_*^{-\alpha'}$ for $D(\gamma) = 0$; consequently, in assignments where γ is useless, $T_\gamma G$ is bounded by the bound function of G/γ (γ contracted to a vertex v_γ at the anchor) with vertex power $D(\gamma) - \alpha'$ at $s(v_\gamma) = s_*$.*

Proof. For a fixed top internal index $j_{\mathrm{top}} \geq j_*$ the bound function estimate of [3, (2.16)], with no sign condition on the degree and the anchor's cube fixed, bounds the internal sum by $C s_{j_{\mathrm{top}}}^{D(\gamma)}$; it needs positive degrees for γ's components along its orderings, which the inner processing supplies (induction on lines). Sum over $j_{\mathrm{top}} \in [j_*, k-1]$: for $D = -1$, $\sum_j s_j^{-1} \leq s_*^{-1}/(1 - L^{-1})$; for $D = 0$ the sum has $k - j_* = \log_L(1/s_*)$ terms. The lines of G/γ at v_γ are l_a, l_b, so $j(v_\gamma) = j_*$ and $s(v_\gamma) = s_*$, exactly the vertex scale of the bound function, and vertex powers that are powers of the vertex scale are allowed with any real exponent. In the useful assignments $T_\gamma G$ is paired with G by [3, (3.19)]/[3, (3.11)] at the same indices (lemma F.63(a)); in the useless ones G has no γ-divergence, since γ is not a component. The loss $\alpha' := \alpha/2$ is below the gains α of the Hölder remainders and below the half-integer gaps (degrees lie in $\frac{1}{2}\mathbb{Z}$). □

Lemma F.67 (sharp indicators, boundary layers)**.** *Let S be a dangerous node with $D(S) \leq 0$ whose local coefficient contains a sharp indicator $\mathbf{1}_F(x_b)$ at an internal vertex b, F the unit cube of a [3, (1.6)]/[3, (1.7)] vertex or Ω_1. Write $c(\hat{x}) = \mathbf{1}_F(\hat{x}) c^{\mathrm{unr}}(\hat{x}) + d(\hat{x})$ with c^{unr} the coefficient with $\mathbf{1}_F$ evaluated at $\hat{x}$. Then $|d(\hat{x})| \leq C s_S^{D(S)} \exp(-\delta' \operatorname{dist}(\hat{x}, \partial F)/s_S)$, s_S the top internal scale of S; summed over the cube Δ of side $s(v_S)$ of the local vertex, $\sum_{\hat{x}\in\Delta} \eta^d |d| \leq C s_S^{D(S)} (s_S/s(v_S)) s(v_S)^d$; and after summing over $s_S \leq s(v_S)$ the defect is bounded like a vertex of power $D(S)$ at $s(v_S)$, with no loss for $D(S) = 0$ and better than [3, (2.3)] for $D(S) = -1$.*

Proof. $|\mathbf{1}_F(x_b) - \mathbf{1}_F(\hat{x})| \leq \mathbf{1}\{\operatorname{dist}(\hat{x}, \partial F) \leq |x_b - \hat{x}|\}$, the internal lines of S at scales $\leq s_S$ decay as $\exp(-\delta |x_b - \hat{x}|/s_S)$, and the bound function at fixed top scale gives the size; only the points of Δ within $O(s_S)$ of ∂F contribute, a fraction $\leq C s_S/s(v_S)$ of Δ; then a geometric sum. □

Lemma F.68 (position-space slices and remainder lines)**.** (a) *Let K be a kernel on $\xi\mathbb{Z}^3$ with $|\Delta_T D_S K(y, y')| \leq C_K (\xi + \rho)^{-\omega - |S| - \alpha|T|} e^{-\delta\rho}$, $\rho := |y - y'|$, for the derivative and Hölder operations used. With χ smooth, $= 1$ on $[0, 1]$, 0 on $[L, \infty)$, $s_j = L^j \xi$, $\chi_0 := \chi(\rho/\xi)$, $\chi_j := \chi(\rho/s_j) - \chi(\rho/s_{j-1})$ and the top piece $1 - \sum_{j<J} \chi_j$ at $s_J = 1$, the slices $K_j := K\chi_j$ obey $|\Delta_T D_S K_j| \leq C' s_j^{-\omega - |S| - \alpha|T|} \exp(-\delta'\rho/s_j)$, so K is a generalised line of the bound function with the derivatives acting on the sliced line.* (b) *The remainder kernels $R = \breve{G} - C^\xi$ of the cube step ($|R| \leq C$, $|\partial R| \leq C(1 + |\log(\xi + \rho)|)$, $|\partial R \partial^*| \leq C(\xi + \rho)^{-1}$) have slices with bounds $s_j^{-(d-2)+1-|S|-\alpha|T|}$, one better than C^ξ-slices at every scale; for $|S| \geq 1$ the logarithm is dominated by the terms where the derivative falls on χ_j.* (c) *For an inner node at scale $s_{j'}$ inside a coefficient on $\xi\mathbb{Z}^3$, the cumulative $C^\xi_{<j'} = C^\xi \chi_{<j'}$ splits as $C^{(j')} + R'_{j'}$ with $C^{(j')} := (-\Delta^\xi + s_{j'}^{-2})^{-1}$, and after rescaling by $s_{j'}$, $R'_{j'}$ obeys the bounds of (b): $C^\xi - C^{(j')} = (s_{j'}^{-2} - 1) C^\xi * C^{(j')}$ by the resolvent identity, and the cutoff part is supported at $\rho \sim s_{j'}$.*

Proof. (a) On $\operatorname{supp}\chi_j$, ρ is of order s_j and $|\partial^k \chi_j| \leq C s_j^{-k}$; Leibniz. (b) Insert the bounds on R into (a). (c) Resolvent identity and the convolution bound of two decaying kernels. □

Theorem F.69 (bounds on the regrouped local coefficients)**.** *For every useful dangerous node S of type $2A$ or $\phi\phi A$ the regrouped local coefficients satisfy: $2A$: $|c^{(0)}(\hat{x})| \leq C s(v_S)^{D(S)}$ (zeroth moment), and for $D(S) = -1$ also $|c^{(1)}_\mu| \leq C$ (first moment); $\phi\phi A$: $|c^{(0)}| \leq C$; with C independent of η, k, ε, the labels, Ω, $\tilde{B}$ and the torus. For 2ϕ nodes the zeroth order cancels exactly against the counterterm member, and the $D = -1$ first moment has the bounded regrouped coefficient of [3, (3.11)].*

Proof. Induction on the number of lines of the node. *The value.* By lemma F.63(a), c is the regrouped value; rescaled to $\xi = L^{-j_0}$ it is $s(v_S)^{D(S)}$ times the same expression on $\xi\mathbb{Z}^3$, with lines $\breve{G}_l = C^\xi + R_l$; smooth multipliers are moved to $\hat{x}$ (lemma F.61, gain 1), sharp ones are treated by lemma F.67. *The all-C^ξ part* is the per-label full class with full lines C^ξ, internal sums unrestricted on $\xi\mathbb{Z}^3$, and every counterterm member re-expanded with its own lines ([3, p. 445]: "the same transformations in mass renormalization counterterms"), inner counterterm coefficients being fixed numbers, which lemma F.60 allows: the $2A$ zeroth moment is 0 by lemma F.60; the $2A$ first moment and the $\phi\phi A$ zeroth moment are 0 by the reflection identities of [3, (3.30)], applicable because the coefficients are now constant; the torus-versus-$\xi\mathbb{Z}^3$ identification is the image-term argument of [3, pp. 418–419]. *Terms with at least one R-line, $D(S) = -1$:* these are one-loop nodes, bounded by the cancellation $\sum_{y'}(\partial C\partial^*)(y' - y) = 0$ for the term $R \cdot \partial C\partial^*$, which uses only the bounds on R and the forward-difference structure of the other line, so it applies also to R'-lines (lemma F.68(c)) and to slices on which the derivative acts. *Terms with at least one R-line, $D(S) = 0$:* re-slice every line (lemma F.68(a), (b)) and apply the bound function estimate [3, (2.16)] to the coefficient (anchor fixed, external fields factored out). Along each ordering: a component containing an R-slice has degree $\geq D(H) + 1$, which is $\geq \frac{1}{2}$ unless $D(H) = -1$, and a $D(H) = -1$ component is a one-loop 2ϕ or $2A$ node with an R-line, processed by the previous case ($2A$) or by its counterterm member with the same R-line (2ϕ, exact; the first moment bounded as for [3, (3.11)]); a component without an R-slice has its original degree, and the dangerous ones are inner nodes, processed by the induction hypothesis (local coefficients bounded, remainders gaining α) and by lemma F.65; the whole coefficient has degree $\geq D(S) + 1 = 1$; inner 2ϕ nodes whose counterterm assignments are useless relative to the inner scale are bounded by lemma F.66 (loss α'). So every component has degree $\geq \min\{\frac{1}{2}, \alpha\} - \alpha' > 0$, and [3, (2.16)] bounds the sum uniformly in ξ. *$\phi\phi A$ remainder terms*: one-loop, so the inner components are trees, the only dangerous ones [3, (2.4)], handled by integration by parts inside the coefficient; the previous case applies with no inner nodes. □

Theorem F.70 ([3, (3.5)] for every class)**.** *For every class component after the cube steps, $\sum_G \sum_{\tilde{l}} \sum_{j \in J(\tilde{l})} E(G(j))$ is a finite sum of terms each of which is either a generalised bound function in which every component along every ordering of its support has degree $\geq \theta := \alpha/2$, or is cancelled exactly (vacuum components, 2ϕ zeroth orders). Hence* [3, (3.3)] *holds by the bound function estimate* [3, (2.16)].

Proof. Fix an ordering, i.e. an index assignment class $J(\tilde{l})$, and process the dangerous nodes along $\tilde{l}$ innermost first. *Order and anchors*: non-[3, (2.4)] nodes innermost first with any anchors (2ϕ: the counterterm anchor); each outer node's split applied to all terms produced by the inner processing, Δ-terms included (lemma F.64); [3, (2.4)] components treated inside the smallest dangerous non-[3, (2.4)] node containing them (lemma F.65), or alone by [3, (2.8)]. *Local terms*: 2ϕ, the zeroth order cancels exactly against the counterterm member at the same indices in the useful assignments, and the $D = -1$ first moment has its bounded coefficient; $2A$ and $\phi\phi A$, the local vertex v_S has coefficient bounded by theorem F.69, its value being the regrouped one (lemma F.63(a)); for $2A$ with $D = -1$ this vertex is the non-vanishing zero-momentum remainder vertex of [3, (3.30)], of power -1 at $s(v_S)$, i.e. degree $2 + D(S) = 1$ in the convention [3, (2.3)]. *Remainder terms* gain α or $1 + \alpha$ inside S and lose the same on S's external lines (lemma F.63(b)); where an inner processing acts on an outer node's external field, the outer node gains at least α and has no local part in that term (lemma F.64); so after all nodes are processed every component that was dangerous has degree $\geq \alpha$, and every other component keeps its degree, which is $\geq \frac{1}{2}$ since degrees lie in $\frac{1}{2}\mathbb{Z}$. *Useless counterterm assignments* are bounded by lemma F.66 at a loss $\alpha' = \alpha/2$. *Remainder vertices inside larger graphs*: a remainder vertex of a $2A$ node S has power $D(S)$ at $s(v_S)$; components containing it have the degree of the same component with S restored (lemma F.63(b)); if such a component is itself dangerous it is a node of the next level and is processed there, its local coefficient being the regrouped value with S unprocessed (lemma F.63(a)), so the remainder vertex never needs a bound beyond its power. *Induction*: this is Bałaban's "for each ordering ... the last divergent subgraph ... apply (3.5)

... then apply (3.19), (3.31) or (3.34) to G_0 itself" [3, pp. 444–445], an induction on the number of lines, theorem F.69 being applied inside it to coefficients of strictly smaller nodes. *Vacuum and exception classes*: vacuum components cancel against [3, (1.24)] up to image terms, and $1A/3A$ factors vanish at $B = 0$. □

[3, Prop. 1, (1.33)] at $L = 5$. With the steps above, every one of the twenty sketched steps and seven further places of [3, pp. 420–445] is a written proof: the external-field readings of [3, (1.16)] and of $\delta G_k(\Omega, \Omega_2, \tilde{B})$ [3, pp. 414, 424] (the latter at depth, as (I2a) of Step 3(D$_2$)), the slices and the summation by parts [3, (2.6)–(2.12)], the induction [3, (2.16)], [3, Prop. 2.2, (2.17), Cor. 2.3], the Ward–Takahashi identities for full lines and per label (lemma F.60), [3, (3.1)] in the cluster form (corollary F.58), the regrouping (theorem F.57), the component cube of side l_N, the cube split with $l^\sharp$ (theorem F.2(a)), the constant field B_0 with the k-uniform constant of Step 3(C), the $\tilde{B}'$-expansion and its classes, the chains without old vertex (by an L^2 chain bound), gauge covariance and the gauge-out of B_0, $\delta G_k(\square, \eta\mathbb{Z}^d, 0)$, the norms [3, (1.32)] at B_0 and at $\tilde{B}$ (the constant carrying $(1+\operatorname{diam}\operatorname{supp} f)^{1-\alpha}$), and the renormalisation (theorem F.70). The inputs outside the proposition are the propagator estimates of group (A) (theorem F.151) and theorem F.7; the parameter rows, all fixed after L, $\bar{n}$, R_0 and before ε_0, are $R \geq R_0 + 5$ (the depth of [3, (3.1)]), $R \geq R_{\text{flat}} + R_{\text{Ba}} + 2$ (R_{flat} the depth parameter of the cube step, $R_{\text{Ba}} \geq R_0$ the depth of the Hölder clause of the background, R_0 the constant defined in [4, (2.19)]; the Hölder depth of the cube step), $l^\sharp = l_N + 4V^\sharp_{\text{lab}}(\bar{n})$ in theorem F.2(a), torus side $\geq \max\{6\sqrt{3}\, r(t_k) + 2, 8/\delta_Z\}$, the cap $e_k(\sigma_0 + \sigma_1)(1 + \operatorname{diam}\square) \leq 1$, and $\alpha_0 < \frac{1}{2}$ for the external-field bound of the kernel [3, (1.16)] of order $m = n + n' \in \{4, 5\}$, any $\alpha_0 < 1$ for $m \geq 6$ (at the uses $m \geq \bar{n} + 1 \geq 14$). The constants are independent of ε, k, Ω, Ω_1, Ω_2, $\tilde{B}$ and the torus, as [3, (1.33)] asserts; they depend on L, which is fixed at 5. The torus uniformity of the $O(1)$ (not in Bałaban's printed list, [3, p. 421]) is the torus side of the propagator estimates of group (A): theorem F.7 contributes the periodic-box clause of [4, Lemma 2.4], lemma F.154 gives [1, Prop. 2.1] on a torus tiled by the large blocks with the constants of [4], and the remaining estimates of theorem F.151 hold on the torus with the same constants, their proofs being local. With this, [B-base]$_5$ of theorem F.2(iii) holds on every torus aligned with the blocks, with the single exception named next.

The walled reading. The rotation part (section 15) applies the bound [3, (1.33)] once more, on tori carrying the smooth mass wall, to the two-point classes of the cut expansion of the finite order-4 mass counterterm and to the line-derivative families of the wall vertex: the scalar lines are $G^v_s(T_\eta, 0)$ with a site potential $0 \leq v \leq \bar{v}$ (in particular $v = (m^2 + n\psi(t_s\cdot))t_s^2$), the photon lines are the unwalled torus propagator, the background is zero, the counterterms are the model's own, and in the two-point settings the root vertex is pinned. These classes are not per-label full classes: they are closed under the counterterm, two-vector-leg and bridgeless leg-exchange moves of definition G.1, but the exchange at a bridged $\phi\phi A$ subgraph carrying an external leg leaves the family. The statement needed, theorem F.74 below, is [3, (1.33)] for such classes with walled scalar lines, with constants polynomial in $\bar{v}$ and free of ε, ε_0, the step and the torus. Its proof is the proof above run with walled lines, and it differs from the bulk at exactly two places: the zero-field kernel bounds of group (A) are replaced by their walled form (lemma F.72), and at the $\phi\phi A$ nodes the reflection identity over the full class is replaced by the clauses of theorem F.74(iii), in which the exact identities of the bulk torus for the all-C^ξ parts (the equality of the two leg-exchanged pinned coefficients of a bridgeless node, the vanishing of a bridged node's coefficient) are kept and the parts carrying the wall remainder R^v become bounds, $|c_{S,1}(x) - c_{S,2}(x)| \leq C^v_7$ for the bridgeless pair and $\leq C^v_{\text{BR}}$ for a bridged node paired with the counterterm member of its inner two-leg part.

The walled model. Fix a step s, $1 \leq s \leq K$, and work in scale-s units: $\eta = L^{-s}$, $t_s := L^s\varepsilon$, $e_s := e(L^s\varepsilon)$ and $\lambda_s := \lambda(L^s\varepsilon)$ the running couplings of [3, (1.33)], and T_η the torus, aligned with the blocks, with all sides ≥ 1. Let v be a real function on the sites of T_η with

$$0 \leq v(x) \leq \bar{v} \qquad \text{for every } x,$$

and nothing else assumed about v: no smoothness, no constancy on any set, no symmetry. The walled scalar operator at zero background is

$$H^v := -\Delta^{\eta}_{T_\eta} + v + a_s P_s, \qquad G^v_s(T_\eta, 0) := (H^v)^{-1},$$

with $a_s P_s$ the block term of the step-s scalar propagator of [1]; at the constant $v = m^2 t_s^2$ this is the bulk line $G_s(T_\eta, 0)$ of Step 3(D). The wall of the rotation part is $v = (m^2 + n\psi(t_s\,\cdot))t_s^2$ with a profile $0 \le \psi \le 1$, so that $\bar v = (m^2+n)t_s^2 \le m^2 + n$ once $t_s \le 1$; the one-parameter family $v_\theta = (m^2 + \theta n\psi(t_s\,\cdot))t_s^2$, $\theta \in [0,1]$, which interpolates between the bulk and the wall, satisfies the same bound. The *walled model* is the perturbative scheme of [3, Sect. 1] in which every scalar line $G_s(T_\eta, 0)$, every slice [3, (2.6)] and every zero-field object built from them is replaced by its v-analogue, while the vector lines are unchanged (the torus vector propagator carries no wall), the background is zero, and the counterterms are the model's own: a divergent two-scalar-leg subgraph γ is renormalised by the vertex [3, (1.7)] with the coefficient $\delta m^2_{s,\gamma}[v](x)$ given by Bałaban's rule [3, p. 417] applied to the walled graph (γ with both external legs at x and its other leg point summed over the torus, with the same lines and the same labels, Convention F.18, (Agr)), and the vacuum counterterm is $E_s[v]$, i.e. [3, (1.24)] with walled lines. These counterterms depend on x, exactly as Bałaban's own do ([3, p. 418]: the propagators "are not translation invariant, so we have the dependence on x"), and the rule never uses translation invariance. Finite [3, (1.7)] vertices with bounded x-dependent coefficients $c_v(x)$ are admitted as well; in the rotation part these are the non-renormalising parts of the inner mass counterterms and the *wall vertex*, a two-scalar-leg vertex at z with coefficient $-n\psi(t_s z)t_s^2$, which arises when a line of the bulk model is differentiated with respect to the wall height. The lines of a graph of the walled model are the full lines $G^v_s(T_\eta, 0)$; the slices, orderings and regrouped lines of Step 3(D_4) are internal to the bounds.

Kernel notation. For a kernel $K(x_1, x_2)$ on a lattice of spacing $\xi \in (0,1]$ and $S, T \subset \{1,2\}$, D_S applies one forward difference ∂^ξ_μ in the variable x_i for each $i \in S$, and Δ_T applies one Hölder quotient $|x_i - \tilde x_i|^{-\alpha}[K(\dots, x_i, \dots) - K(\dots, \tilde x_i, \dots)]$ for each $i \in T$ (in a fixed order; $\alpha \in (0,1)$ is the Hölder exponent of Step 3(D)); $\rho := |x_1 - x_2|$. Put $h_p(r) := r^{-p}$ for $p > 0$, $h_0(r) := 1 + |\log r|$ and $h_p(r) := 1$ for $p < 0$. In this notation the bounds of lemma F.68(b) on a remainder kernel R read

$$|\Delta_T D_S R(x_1, x_2)| \le C\, h_{|S| + \alpha|T| - 1}(\xi + \rho)\, e^{-\delta\rho} \prod_{i \in T} |x_i - \tilde x_i|^\alpha, \qquad |S|, |T| \le 2, \tag{82}$$

and the zero-field, zero-mass form of the group-(A) estimates ([4, Lemma 2.4, (2.35)–(2.36)], theorem F.7), extended to a difference in each variable by the same contour-shift argument, one factor $(e^{i\xi p_\mu} - 1)/\xi$ per difference, for the pinned free operator $T := -\Delta^\xi_X + bP_Y$ (X a union of unit blocks with Neumann conditions, a periodic box or $\xi\mathbb{Z}^3$; Y the unit blocks, or the L-blocks when X is a union of L-blocks; P_Y the block-average projection; $b \in [b_-, b_+] \subset (0, \infty)$) reads

$$|\Delta_T D_S T^{-1}(x_1, x_2)| \le C_1 (\xi + \rho)^{-p(S,T)} e^{-\delta\rho} \prod_{i \in T} |x_i - \tilde x_i|^\alpha, \qquad p(S,T) := 1 + |S| + \alpha|T|, \tag{83}$$

at every pair of points of X, with $T \ge \gamma' > 0$ and C_1, δ, γ' depending only on $d = 3$, L, $b_\pm$ and α. (A quotient over two points at distance > 1 is bounded by the two values, so (83) is needed only at $|x_i - \tilde x_i| \le 1$; the same convention applies to every bound below.)

Lemma F.71 (lattice convolution)**.** *Let $0 \le p_1, p_2 < 3$ and $\delta > 0$. Then for all $x, y \in \xi\mathbb{Z}^3$ and $\xi \in (0,1]$,*

$$\sum_{z \in \xi\mathbb{Z}^3} \xi^3 (\xi + |x - z|)^{-p_1} e^{-\delta|x-z|} (\xi + |z - y|)^{-p_2} e^{-\delta|z-y|} \;\le\; C_K\, h_{p_1+p_2-3}(\xi + |x - y|)\, e^{-\delta|x-y|/2},$$

with $C_K = C_K(p_1, p_2, \delta)$. The same holds on a torus or a Neumann domain, the sum running over its sites.

Proof. Put $r := |x - y|$. Since $|x - z| + |z - y| \ge r$, the exponentials are $\le e^{-\delta r/2} e^{-\delta(|x-z| + |z-y|)/2}$. On $\{|x - z| \le r/2\}$ one has $\xi + |z - y| \ge (\xi + r)/2$, and $\sum_{|x-z| \le r/2} \xi^3 (\xi + |x - z|)^{-p_1} \le$

$C(\xi+r)^{3-p_1}$ because $p_1<3$; the region $\{|z-y|\le r/2\}$ is symmetric; on the rest both factors are $\le C(\xi+|z-x|)^{-(p_1+p_2)}$ up to the exponentials, and $\sum_{|z|\ge r/2}\xi^3(\xi+|z|)^{-(p_1+p_2)}e^{-\delta|z|/2}$ is $\le C(\xi+r)^{3-p_1-p_2}$ if $p_1+p_2>3$, $\le C(1+|\log(\xi+r)|)$ if $p_1+p_2=3$ and $\le C$ otherwise. For $\xi+r\ge 1$ the exponential factor dominates. On a torus or a Neumann domain the sites are a subset of $\xi\mathbb{Z}^3$ with the quotient (resp. restricted) distance, and the same sums bound the result. □

Lemma F.72 (walled zero-field kernels)**.** *Let X, Y, b and $T=-\Delta_X^\xi+bP_Y$ be as in (83), let $w\colon X\to[0,\bar v]$ and $T_w:=T+w$. Then:*

(a) (algebra) *$T_w\ge T\ge\gamma'$. The function w commutes with every multiplication operator ($e^{\beta\rho}$ for a Lipschitz ρ, cutoffs, the gauge factors V_λ) and cancels in every difference of two operators carrying the same w; hence the gauge covariance $H^w(A^\lambda)=V_\lambda H^w(A)V_\lambda^*$, the commutator identities with cutoffs, the slice identity* [3, (2.6)] *and its rescaling hold for the walled operator verbatim, the rescaled potential at scale s_j being $s_j^2w(s_j\,\cdot)\in[0,\bar v s_j^2]\subset[0,\bar v]$, and the slices of the walled propagator are $G^{v,\eta}_{(j)}=a_j^2s_j^{-4}G_j^vQ_j^*C^{(j),v}Q_jG_j^v$ with $G_j^v:=(-\Delta+v+a_js_j^{-2}P_j)^{-1}$ and $C^{(j),v}$ the block operator built from G_j^v. Every walled zero-field scalar kernel is a scalar kernel times $\mathbf{1}_N$, because w is a scalar.*

(b) (the walled kernel bound) *T_w^{-1} satisfies (83) at every pair of points, with C_1 replaced by $C_1^w:=C(d,L,\alpha,b_+,\gamma',\delta,C_1)(1+\bar v)^2$ and δ replaced by $\delta^w:=\min\{\delta/4,\beta'/\sqrt{3}\}/4$, where $\beta'\le1$ is the Combes–Thomas rate of lemma G.7 for T_w (it depends only on γ', b_+ and L, not on w).*

(c) (block sums) *For $|S|,|T|\le1$ and every block $B(y)$: $|\Delta_T D_S(T_w^{-1}\mathbf{1}_{B(y)})(x)|\le c^we^{-\delta^w|x-y|}\prod_{i\in T}|x-\tilde x|^\alpha$.*

(d) (the remainder) *On $\xi\mathbb{Z}^3$ or a periodic box let $C^\xi:=(-\Delta^\xi+1)^{-1}$ be the free unit-mass kernel of Step 3(D_4) and $R^w:=T_w^{-1}-C^\xi$. Then R^w satisfies (82) at every pair of points with $C=C_R(1+\bar v)^3$ and $\delta=\delta_R$, C_R and δ_R depending only on $d,L,\alpha,b_\pm,\gamma'$ and the constants of (83). In particular R^w is a remainder line in the sense of lemma F.68(b), with constant $C_R(1+\bar v)^3$, and so are its slices and the kernels of lemma F.68(c) built from it.*

(e) (slices, cube kernels) *The slice bounds* [3, (2.10)–(2.12)] *and the kernel bounds of the cube step of Step 3(D) hold for the walled slices $G^{v,\eta}_{(j)}$, including the two-ended entries and the mixed second difference, with constants depending polynomially on $\bar v$ and otherwise as in the bulk; the Combes–Thomas bound of lemma G.7 holds for H^v with the v-free constants.*

(f) (periodisation) *$G_s^v(T_\eta,0)(x,x')=\sum_n G_s^{v_{\mathrm{per}}}(\eta\mathbb{Z}^3,0)(\tilde x,\tilde x'+n)$, the sum over the period lattice of T_η, $\tilde x,\tilde x'$ lifts of x,x' and v_{per} the periodic extension of v; the sum converges absolutely, and the same identity holds slice by slice.*

Proof. (a) w is a real scalar multiplication operator: it is self-adjoint and ≥0, it commutes with every other multiplication operator, and it cancels in $H^w(A)-H^w(A')$ and in $[H^w,\chi]$. The slice identity is the resolvent algebra of $-\Delta+w$ with the pin terms, which uses no property of the zeroth-order operator beyond coercivity; rescaling by s_j multiplies a potential by s_j^2 and composes it with the dilation. Since v acts as $v\mathbf{1}_N$, the inverse of H^v at zero field is a scalar kernel times $\mathbf{1}_N$.

(b) *Step 1 (Combes–Thomas).* $T_w=D^*D+w+bP_Y$ is of the form of lemma G.7 with $V=w\ge0$, Π_B the block-average projections and $T_w\ge\gamma'$; so $\|e^{\beta\rho}T_w^{-1}e^{-\beta\rho}\|\le2/\gamma'$ for every 1-Lipschitz ρ and $0\le\beta\le\beta'$, ρ being $\sqrt{3}$-Lipschitz for the sup-metric, so that $\beta\le\beta'/\sqrt{3}$ is imposed, with β' depending only on γ', b_+ and the block size. *Step 2 (identity).* $T_w^{-1}=T^{-1}-T^{-1}wT^{-1}+T^{-1}wT_w^{-1}wT^{-1}$, which combines $T_w^{-1}=T^{-1}-T^{-1}wT_w^{-1}$ with $T_w^{-1}=T^{-1}-T_w^{-1}wT^{-1}$. *Step 3 (no operations).* Put $r:=|x-y|$. The first term is $\le C_1(\xi+r)^{-1}e^{-\delta r}$ by (83). The second is $\le\bar vC_1^2C_Kh_{-1}(\xi+r)e^{-\delta r/2}=\bar vC_1^2C_Ke^{-\delta r/2}$ by lemma F.71 with $p_1=p_2=1$. For the third write $a(z):=T^{-1}(x,z)w(z)$, $b(z'):=w(z')T^{-1}(z',y)$ and $\rho:=|\cdot-x|$; then $|\langle a,T_w^{-1}b\rangle|\le\|e^{\beta\rho}a\|_2\,(2/\gamma')\,\|e^{-\beta\rho}b\|_2$, and since $|z'-x|\ge r-|z'-y|$, $\|e^{-\beta\rho}b\|_2\le e^{-\beta r}\|e^{\beta|\cdot-y|}b\|_2$; both weighted norms are $\le\bar vC_1J$ with $J^2:=\sup_\xi\sum_z\xi^3(\xi+|z|)^{-2}e^{-(\delta-2\beta)|z|}<\infty$ (here $d=3$ and $\beta\le\delta/4$). Hence $|T_w^{-1}(x,y)|\le C^{(1)}(1+\bar v)^2(\xi+r)^{-1}e^{-\delta' r}$ with $\delta':=\min\{\delta/2,\beta\}$ (for

$\xi + r \le 1$ the constant terms are $\le (\xi + r)^{-1}$, for $\xi + r \ge 1$ the exponential absorbs the power). *Step 4 (general S, T).* Apply $\Delta_T D_S$ to the identity of step 2. The operations at x_1 act only on the first factor $T^{-1}(x_1, \cdot)$ and those at x_2 only on the last factor $T^{-1}(\cdot, x_2)$; w and the middle factor T_w^{-1} are never differentiated. Put $p_i := 1 + [i \in S] + \alpha[i \in T] \in [1, 2 + \alpha]$, so that $p(S, T) = p_1 + p_2 - 1$. The first term is (83). The second is, by (83) for the two factors and lemma F.71, $\le 2^{|T|} \bar{v} C_1^2 C_K h_{p_1+p_2-3}(\xi + \rho) e^{-\delta\rho/2} \prod |x_i - \tilde{x}_i|^\alpha$. For the third insert the bound of step 3 for the middle factor and apply lemma F.71 twice: the convolution of $(\xi + |x_1 - z|)^{-p_1}$ with $(\xi + |z - z'|)^{-1}$ gives $h_{p_1-2}(\xi + |x_1 - z'|)$, which is $\le C_\alpha (\xi + |x_1 - z'|)^{-q}$ with $q := \max\{p_1 - 2, 0\} + \alpha/2 < 3$ (this covers the logarithm at $p_1 = 2$), and the second convolution with $(\xi + |z' - x_2|)^{-p_2}$ gives $h_{q+p_2-3}(\xi + \rho)$. Since $p_1 + p_2 - 3 < p(S, T)$ and $q + p_2 - 3 < p(S, T)$ with $p(S, T) \ge 1$, both h-factors are $\le C(\xi + \rho)^{-p(S,T)}$ for $\xi + \rho \le 1$, and for $\xi + \rho \ge 1$ the exponentials dominate. Each application of lemma F.71 halves the rate: the second term decays at the rate $\delta/2$ and the third at $\delta'/4$. Taking $\beta := \min\{\delta/4, \beta'/\sqrt{3}\}$, so that $\delta' = \beta$, and collecting the constants gives $C_1^w = C(1 + \bar{v})^2$ and the rate $\delta^w = \min\{\delta/2, \beta/4\} = \beta/4$.

(c) Sum (b) over $x' \in B(y)$: since $p(S, T) \le 2 + \alpha < 3$, $\sum_{x' \in B(y)} \xi^3 (\xi + |x - x'|)^{-p} e^{-\delta^w |x - x'|} \le C(p) e^{-\delta^w \operatorname{dist}(x, B(y))}$.

(d) Since $(C^\xi)^{-1} = -\Delta^\xi + 1$ and T_w^{-1} exists, $R^w = T_w^{-1}\big((C^\xi)^{-1} - T_w\big) C^\xi = T_w^{-1}(1 - w) C^\xi - b\, T_w^{-1} P_Y C^\xi$. This is the identity [3, p. 437] ($G^\xi(0) = C^\xi + G^\xi(0)(1 - m^2 - aP) C^\xi$) with the constant $1 - m^2$ replaced by the function $1 - w(z)$, which is never differentiated: the operations at x_1 fall on $T_w^{-1}(x_1, \cdot)$, those at x_2 on $C^\xi(\cdot, x_2)$. In the first term, (b) for T_w^{-1}, the free bound (83) for C^ξ (it is the case $b = 0$, $m^2 = 1$ of the group-(A) estimates), $|1 - w| \le 1 + \bar{v}$ and lemma F.71 give $C(1 + \bar{v})^3 h_{p_1+p_2-3}(\xi + \rho) e^{-\delta_R \rho} \prod |x_i - \tilde{x}_i|^\alpha$ with $p_1 + p_2 - 3 = |S| + \alpha|T| - 1$. The second term is $b \sum_Y |Y|^{-1} (T_w^{-1} \mathbf{1}_Y)(x_1) (\mathbf{1}_Y^{\mathsf{T}} C^\xi)(x_2)$, and (c) for $T_w^{-1} \mathbf{1}_Y$ together with the block-sum bound for $C^\xi \mathbf{1}_Y$ give $C e^{-\delta_R \rho} \prod |x_i - \tilde{x}_i|^\alpha$ after the sum over Y, which is $\le h_{|S|+\alpha|T|-1}(\xi + \rho)$ for $\xi + \rho \le 1$. The slices of R^w and the kernels of lemma F.68(c) are obtained from these bounds exactly as in lemma F.68, which uses only the bounds (82).

(e) The slice $G_{(j)}^{v,\eta}$ is the composition of (a) with G_j^v bounded by (b) at scale j (unit-block units after rescaling, with the potential $s_j^2 v(s_j \cdot) \in [0, \bar{v}]$) and with the block-operator kernel $C^{(j),v}$, whose exponential bound uses only the coercivity $H^v \ge H^0 \ge \gamma_0$ and the Combes–Thomas argument of lemma G.7, so its constants are v-free; the bulk proof of [3, (2.10)–(2.12)] in Step 3(D) is a composition of these three bounds and of lattice sums, and the mixed second difference and the two-ended entries are the cases $|S| = 2$ of (b). The kernel bounds of the cube step are sums of slice bounds over the scales. The Combes–Thomas bound is lemma G.7 with $V = v$.

(f) The right side is well defined because the image sum converges absolutely by (b) on $\eta\mathbb{Z}^3$ with the periodic potential v_{per} (which lies in $[0, \bar{v}]$); it is periodic, and it solves the torus equation because the $\eta\mathbb{Z}^3$ operator with v_{per} commutes with the period translations; the torus inverse is unique. The slice identity (a) is compatible with the periodisation term by term. □

Classes closed under the moves. Recall the moves (P), (P′), (W), (X) and the classes of definition G.1, and the cut terms and orbits of definition G.2. A *decorated graph* of the walled model is a graph of the model (lines $G_s^v(T_\eta, 0)$ and the torus vector propagator, vertices [3, (1.6)–(1.15)] with their labels (Loc), counterterm vertices with the model's own coefficients, finite [3, (1.7)] vertices with bounded coefficients) whose external legs carry external fields which may be kernels of several variables [3, p. 419], in particular the two ends of a cut line; these legs are the *slots* of the graph. A finite set $\mathcal{C}$ of decorated graphs with a common slot structure (the same slots, with the same kinds, differentiations, labels and fields), a common order, a common label multiset and a common multiset of finite-vertex coefficients is *closed under the moves* if every move of the set M_0, consisting of (P) and (P′) at a divergent two-scalar-leg subgraph of a member, (W) at a divergent two-vector-leg subgraph, and (X) at a *bridgeless* divergent $\phi\phi A$ subgraph (at the member itself included, when the member is of one of these types), gives a member. The set M_0 is a subset of the moves of definition G.1 and has the properties (A1)–(A5) of that definition, which are properties of the individual moves; so lemma G.3 applies with $M = M_0$. Per-label

full classes are closed under the moves (definition G.1), and so is every class $\mathrm{Cl}_M(K)$, $M \supset M_0$, of a connected component K of a cut graph in lemma G.3(c), with the cut ends as slots. The restriction to bridgeless subgraphs in (X) is forced by the two-point settings: the exchange at a bridged divergent $\phi\phi A$ subgraph carrying an external leg produces a one-particle-reducible graph, which is not a term of a family of [3, (1.23)] graphs (theorem F.74(iii)(c) shows that no partner is needed there). In a *two-point setting* the members have two external scalar legs, undifferentiated, one of them, the *root*, at a vertex pinned at the point x, the other at a vertex that is summed; the coefficient is a function of x, and the bounds below hold uniformly in x. A $\phi\phi A$ subgraph S is *bridgeless* if no internal line of S disconnects it (a self-loop is not a bridge), and *bridged* otherwise.

The two families of the rotation part. (a) *The cut expansion of the finite order-4 mass counterterm.* The order-4 two-point coefficient of the walled model is computed once with the ε-level lines C_0^v, C^γ and once with the step-K lines G_K^v, G_K^γ; the difference, with the scalar lines decomposed as $C_0^v = G_K^v + R_K^v$ and the vector lines as $C^\gamma = G_K^\gamma + R_K^\gamma$, is the sum (84) of definition G.2 over the cut terms with at least one cut line, the cut ends carrying the kernels R_K^v, R_K^γ. By lemma G.3(c),(d) each orbit is a product of classes $\mathrm{Cl}_M(K)$ of the connected components K of its cut graph, each closed under the moves, with the cut ends as slots; the component carrying the root is a two-point class with the root pinned at x, the others are vacuum components with slots. (b) *The line-derivative family of the wall vertex.* Let $\mathcal{F}^{(4)}$ be the set of two-point graphs of [3, (1.23)] of coupling order 4: one-particle-irreducible, two undifferentiated external scalar legs, the root at x, the inner [3, (1.7)] vertices carrying the model's own step-s counterterm of their order and their non-renormalising parts as finite vertices. For $\Gamma \in \mathcal{F}^{(4)}$ and a scalar line l of Γ let $\Gamma_{l\to v}$ be Γ with l subdivided by a two-scalar-leg vertex v at a point z with a bounded coefficient $c(z)$ (the wall vertex), and

$$(\Gamma_{l\to v})_{\mathrm{ren}} \;:=\; \sum_{F} \prod_{\gamma\in F} (-T_\gamma)\, \Gamma_{l\to v},$$

the sum over the forests F of divergent two-scalar-leg subgraphs of Γ avoiding l (i.e. not containing l), T_γ the counterterm operation of (Agr). The family is $\sum_{\Gamma\in\mathcal{F}^{(4)}} \sum_l (\Gamma_{l\to v})_{\mathrm{ren}}$, with no cut line.

Lemma F.73 (the line-derivative family)**.** (a) *Every connected subgraph of $\Gamma_{l\to v}$ that contains v has degree $D \geq 1$ or vanishes. Hence the divergent subgraphs of $\Gamma_{l\to v}$ are exactly the divergent subgraphs of Γ avoiding l, and $(\Gamma_{l\to v})_{\mathrm{ren}}$ is the renormalised forest sum of $\Gamma_{l\to v}$.* (b) *A forest of divergent two-scalar-leg subgraphs of $\Gamma \in \mathcal{F}^{(4)}$ avoiding l has at most one element.* (c) *Group the terms $T_F\Gamma_{l\to v} := \prod_{\gamma\in F}(-T_\gamma)\Gamma_{l\to v}$ ($\Gamma \in \mathcal{F}^{(4)}$, l a scalar line of Γ, F as above) by label multiset and by the multiset of finite-vertex coefficients, the wall vertex included. Each group is a two-point class closed under the moves, with the root pinned at x.*

Proof. (a) The vertex v is a [3, (1.7)] vertex with a bounded coefficient, so it has $D(v) = 2$ and every connected subgraph H containing it has $D(H) \geq 1$ if H has at least two slots (lemma F.59(b), [3, Cor. 2.3]). A connected subgraph with exactly one slot vanishes: with a scalar slot by the symmetry $\phi \to -\phi$ of every vertex (its amputated kernel K_a with one open scalar index satisfies $K_a = -K_a$), with a vector slot at zero background because its scalar index loops close inside it and $\mathrm{Tr}\, q^{\mathrm{odd}} = 0$ (Step 3(D$_4$)); and a connected subgraph of $\Gamma_{l\to v}$ with no slot is $\Gamma_{l\to v}$ itself, which has two. The divergent subgraphs of $\Gamma_{l\to v}$ therefore avoid v, i.e. are subgraphs of Γ not containing l, with the same vertices, lines, slots and degree (lemma G.3(a)); conversely a divergent subgraph of Γ avoiding l is a subgraph of $\Gamma_{l\to v}$ with the same data.

(b) A proper divergent two-scalar-leg subgraph of Γ has coupling order 2 ([3, (1.23)]; the order-4 one is Γ itself, which contains l). Two such subgraphs cannot be nested, and two disjoint ones γ, γ' would exhaust the vertices of Γ: each has exactly two slots, the two external legs of Γ are among the four slots, so at most one line joins γ to γ'; if exactly one does it is a bridge of Γ, and if none does Γ is disconnected; both contradict $\Gamma \in \mathcal{F}^{(4)}$.

(c) Fix a group. *(P), (P′).* A divergent two-scalar-leg subgraph γ of a member $T_F\Gamma_{l\to v}$ avoids v and l by (a); the move replaces the term with forest F by the term with forest $F \cup \{\gamma\}$ (or

$F \setminus \{\gamma\}$), a forest avoiding l by (b), of the same Γ and l; the counterterm vertex carries γ's labels (Agr) and the model's own coefficient, so the labels and the finite-vertex coefficients of the group are kept. *(W).* A divergent two-vector-leg subgraph H of a member avoids v and l by (a). Replacing H by a graph H' of its per-label two-vector-leg class (same two vector slots, same order, same label multiset, its counterterm members included) changes nothing outside H: the line l with the vertex v on it and the two external scalar legs are untouched, the order and the labels are kept, and the result is one-particle-irreducible: since $T_F\Gamma_{l\to v}$ is one-particle-irreducible, the two attachment points of H are connected outside H (else H's vector line to the rest would be a bridge), so an internal bridge of H' separating its two slots is not a bridge of the new graph, and an internal bridge of H' with a slot-free side bounds a one-slot subgraph, which vanishes by (a). Smoothing v back into a line gives a graph of $\mathcal{F}^{(4)}$ with the counterterm vertices of F inside or outside H as the case may be (a forest element inside H is a counterterm member of H's class), so the result is a member of the group. *(X).* The exchange at a bridgeless $\phi\phi A$ subgraph P moves only the attachments at P's two scalar slots and no vertex: the line l keeps its ends up to this re-attachment and v stays on it, the external legs stay undifferentiated (both slots of P carry no difference, since P is divergent), and the result is one-particle-irreducible: for a line $\ell \notin P$, P is connected in both graphs minus ℓ with the same multiset of outer attachment points, and for $\ell \in P$, $P \setminus \ell$ is connected because P is bridgeless. The forest element, if any, is nested with P or disjoint from it (a divergent two-scalar-leg subgraph crossing a bridgeless order-3 $\phi\phi A$ subgraph would meet it in a single one-vector-leg vertex whose two internal lines of P would be its two scalar slots, leaving its vector line as a bridge of the forest element); a forest element sharing a vertex with P but no line (the tadpole at the [3, (1.6)] vertex of the double-line shape) is untouched, since (X) moves attachments only; so F is carried along. *Bridged $\phi\phi A$ subgraphs.* No move of M_0 acts there. (The exchange at a bridged divergent $\phi\phi A$ subgraph $\{u\} \cup S_2$ of a member, with the external leg of the member at u, would move that leg to S_2, which then hangs on the bridge alone: the result is one-particle-reducible and is not a term of the family. Such a member exists: u, u' one-vector-leg vertices [3, (1.8)] joined by a vector line, w a [3, (1.6)] vertex with a self-loop, scalar lines u–w and w–u', the root at u and the wall vertex on w–u'; the node $\{u, w\}$ is a dangerous bridged $\phi\phi A$ node of degree 0.) Finally every member has its root pinned at x, since no move moves a vertex. □

Theorem F.74 (the walled form of [3, Prop. 1] for classes closed under the moves)**.** *In the walled model at a step $s \le K$ (potential $0 \le v \le \bar{v}$ on the scalar lines only, vector lines unwalled, zero background, the model's own counterterms, finite [3, (1.7)] vertices with bounded coefficients, torus aligned with the blocks with sides ≥ 1) let $\mathcal{C}$ be a class closed under the moves, of order $\le \bar{n}$, with slot fields $\Phi_{\rm ext}$, $A_{\rm ext}$ and finite-vertex coefficients c_v. Then:*

(i) (regrouping) *Lemma G.3 (with $M = M_0$ in the two-point settings, under which the families of [3, (1.23)] graphs are closed) and theorem F.57 hold verbatim with walled lines; in particular the cut expansion of the finite order-4 mass counterterm is a finite sum over orbits, each orbit the contraction through the kernels R^v_K, R^γ_K of per-component classes closed under the moves, with the invariants of lemma G.3(b),(f); and the line-derivative family of the wall vertex is a finite disjoint union of two-point classes closed under the moves (lemma F.73).*

(ii) (the bound) *With $e_s = e(L^s\varepsilon)$ and $\lambda_s = \lambda(L^s\varepsilon)$,*

$$\big|E^v(\mathcal{C}; \{\square(v)\}, \Phi_{\rm ext}, A_{\rm ext})\big| \;\le\; O(1)^v\, e_s^{d_v} \lambda_s^{d_s} \prod_{v \text{ finite}} \sup_z |c_v(z)|\; e^{-\delta_0^v d(\{\square(v)\})} \prod \|h\Phi_{\rm ext}\|_{1,\alpha_0} \prod \|h' A_{\rm ext}\|_{1,\alpha_0},$$

the form [3, (1.33)] (E^v the class sum of lemma G.3(d) with walled lines): d_v, d_s count the couplings of the action vertices, a counterterm vertex counting as the exact counterterm of the order it replaces; the localisations h, h' are those of (H_A); the slot fields are kernels of several variables, each measured by its norm [3, (1.32)] in the variable at the slot; in a two-point setting the left side is the coefficient at the pinned root x, and the bound is uniform in x. The constants $O(1)^v$ and δ_0^v depend only on the order, on the parameters of Step 3(D) ($d = 3$, $L = 5$, a, N, $\|q\|$, $\bar{n}$, α_0, m, μ_0, M, R) and polynomially on $\bar{v}$; they

do not depend on ε*, on* $\varepsilon_0 \in (0, \varepsilon_0^*]$ *below the caps of Step 3, on* s*,* K*, the labels, the torus side,* x*, or on* v *beyond* $\bar{v}$*.*

(iii) (the $\phi\phi A$ node clauses) *Let* S *be a dangerous* $\phi\phi A$ *node of a member* G *of* $\mathcal{C}$ *along an ordering, and let* c *be its regrouped zeroth-order local coefficient in the form of theorem F.69: rescaled to* $\xi\mathbb{Z}^3$*, lines* $\breve{G}_l = C^\xi + R_l$ *with* $R_l = R_l^v$ *of lemma F.72(d) on the scalar lines and the bulk remainder on the vector lines, multipliers at the anchor, and the decomposition* $c = c^C + c^R$ *into the all-*C^ξ *part and the terms with at least one* R*-line.*

(a) S *bridgeless,* G^{sw} *the member obtained by* (X) *at* S*, a common anchor in* S*:* $c(S \text{ in } G) + c(S \text{ in } G^{\mathrm{sw}}) = 0$ *exactly, for any line kernels.*

(b) S *bridgeless with its two scalar slots at distinct vertices* $u_1 \neq u_2$*, the pinned root at* u_1 *in* G *and at* u_2 *in* G^{sw}*, anchors at the root;* $c_{S,1}(x)$*,* $c_{S,2}(x)$ *the two coefficients:* $c^C_{S,1}(x) = c^C_{S,2}(x)$*, and* $|c_{S,1}(x) - c_{S,2}(x)| \leq C_7^v$*.*

(c) S *bridged. Then* $S = \{u\} \cup S_2$*, where* u *is a one-vector-leg vertex* [3, (1.8)] *carrying one scalar slot and the vector slot of* S*, its difference acting on the bridge* $l_0 = (u, w)$*,* $w \in S_2$*, and* S_2 *is a two-scalar-leg subgraph of order* 2 *(the exchange bubble, the* [3, (1.6)] *tadpole, a seagull with a vector loop, or a* [3, (1.7)] *vertex); these are the shapes* [3, (2.21b)–(g)]*. If* u *is summed, then with the anchor in* S_2*,* $c = 0$ *for every member and any line kernels. If* u *carries the pinned root (anchor* u *at* x*), then* $c^C = 0$ *for every member, and the* R*-part of the pair* (S *with* S_2) + (S *with* T_{S_2})*, both members of* $\mathcal{C}$ *by* (P)*, is* $\leq C^v_{\mathrm{BR}}$ *in absolute value; the* R*-part of a single member is in general not bounded uniformly in* ξ*.*

C_7^v *and* C^v_{BR} *are the* $\phi\phi A$ *constant of theorem F.69 times a polynomial in* $\bar{v}$*, free of* ξ*,* j_0*,* s*,* ε*, the labels, the torus and* x*.*

The rotation part consumes (ii) and (iii) at the step $s = K$ for the per-component classes of the cut expansion in (i), and at every step $s = a \leq K$ for the line-derivative family, both with v in the family v_θ and at coupling order ≤ 4. The proof of (ii) is by induction on the number of vertices of the members, as the proof of [3, (1.33)] is (lemma F.59); the induction runs over classes closed under the moves, the completed pieces of a cut partnered subgraph being such classes of lower order.

Proof of (i). Lemma G.3 uses the lines only through the properties (A1)–(A5) of the moves, the multilinearity of the expansion (84) and the symmetry of the decomposed kernels; $R_K^v = C_0^v - G_K^v$ is a difference of two symmetric resolvents, hence symmetric. In the two-point setting the family of [3, (1.23)] graphs is closed under M_0: (P) contracts or expands a one-particle-irreducible two-leg subgraph (a bridge of the new graph would be one of the old), and (W), (X) at bridgeless subgraphs keep one-particle-irreducibility and the undifferentiated external legs by the arguments given in the proof of lemma F.73(c). Theorem F.57 and lemma F.56 are statements about labels and the cut rule. Degrees count lines (CV). So (i) is the bulk statement, read with walled lines, and the second half is lemma F.73(c). □

Proof of theorem F.74(iii). *The shapes.* A dangerous $\phi\phi A$ node has order 3 and one loop (Step 3(D_4)), and a node containing a finite [3, (1.7)] vertex is not dangerous (lemma F.59(b)); by [3, p. 429] its vertices are of the types [3, (1.6)–(1.8), (1.10)] or divergent-order counterterm vertices. With an odd number of vector legs the vertex multisets are three one-vector-leg vertices [3, (1.8)]; one one-vector-leg vertex with one two-vector-leg vertex of type [3, (1.8)] or [3, (1.10)]; one three-vector-leg vertex [3, (1.8)]; or one one-vector-leg vertex with a [3, (1.6)] vertex or with an order-2 counterterm vertex. The line count gives one loop in each case, so the internal lines form one cycle through all vertices (a triangle or a double line) in the bridgeless case, and a bridge otherwise. The single three-vector-leg vertex is excluded (a vector self-loop leaves a differentiated scalar slot, a scalar self-loop leaves three vector slots). The bridgeless shapes are: *T*, three one-vector-leg vertices in a triangle, the apex carrying the vector slot and the two base vertices the scalar slots, which are their undifferentiated legs, the base vertices' differences acting on their lines to the apex, and the apex's difference on one of its two lines; *B8*, resp. *B10*, a one-vector-leg vertex u and a two-vector-leg vertex w of type [3, (1.8)], resp.

[3, (1.10)], joined by a vector line and a scalar line, the slots being u's undifferentiated scalar leg, w's undifferentiated scalar leg and w's second vector leg; BL, a one-vector-leg vertex u and a [3, (1.6)] vertex w joined by two scalar lines ending at legs of different index pairs of w, the slots being the two remaining legs of w and u's vector leg. In each shape every scalar slot is an undifferentiated leg and the vector slot carries no difference, as the degree $D = 0$ requires. The bridged shapes are [3, (2.21a)–(g)]; [3, (2.21a)], a vector bridge whose other side is a one-vector-leg vertex with a scalar loop, vanishes identically, and in the remaining shapes: a side of the bridge with no slot would be a one-slot subgraph, which vanishes (lemma F.73(a)); a vector bridge is excluded: the side carrying the vector slot has two vector legs (the slot and the bridge), hence order ≥ 2, so the other side has order 1 and is a single one-vector-leg vertex with no loop, whose two scalar legs are then both external, i.e. both scalar slots lie on it, and its difference acts on a slot, against $D = 0$; so the bridge l_0 is scalar, each side carries one scalar slot, the side with the vector slot has order ≥ 1 and the other, a two-scalar-leg subgraph, order ≥ 2, whence orders $1 + 2$: the first side is a single one-vector-leg vertex u whose difference, not allowed on its scalar slot or its vector leg, acts on l_0, and S_2 is one of the order-2 two-scalar-leg graphs, none of which has a bridge. This is the structure asserted in (c), and it is what Bałaban's arrows in [3, (2.21b)–(g)] show.

(a): the common-anchor cancellation. Let $\sigma_i \in u_i$ be the two scalar slots of S, on outer lines or external fields F_i. G^{sw} differs from G only in the attachments at σ_1, σ_2, so S has the same vertices, internal lines, differences, vertex localisations and anchor $\hat{x}$ in both, and the external lines of S keep their indices, so that usefulness, the regrouped lines of the internal and external slices (which depend on the external indices only) and the transport of the multipliers to $\hat{x}$ (lemma F.61) agree. Hence the regrouped position kernel $K_S(y_1, y_2; \dots)$ of S, including its R-lines and the walled kernels, is the same function in both members, whatever the line kernels are. Its index structure is a matrix M_S on the open index path from σ_1 to σ_2: every closed index cycle inside S carries an even power of q (else $\operatorname{Tr} q^{\mathrm{odd}} = 0$ kills the term), and the total number of vector-leg ends at the vertices of S is $2(\text{internal vector lines}) + 1$, odd, so the open path carries q^r with r odd times even polynomials in q, and $M_S^{\mathsf{T}} = -M_S$. At zeroth order the external fields are constants at $\hat{x}$ and the two local terms are $c(\hat{x})F_1(\hat{x})^{\mathsf{T}} M_S F_2(\hat{x})$ and $c(\hat{x})F_2(\hat{x})^{\mathsf{T}} M_S F_1(\hat{x}) = -c(\hat{x})F_1(\hat{x})^{\mathsf{T}} M_S F_2(\hat{x})$, with the same $c(\hat{x})$; they cancel. Walled lines change nothing: they are scalar kernels times $\mathbf{1}_N$ (lemma F.72(a)), so M_S is unchanged; for slot fields which are kernels of several variables (the ends of R_K^v, R_K^γ), the zeroth-order term at $\hat{x}$ is their value at $\hat{x}$ in the slot variable, controlled by the norm [3, (1.32)]. The remainders of the Taylor expansion are terms of the processing of Step $3(\mathrm{D}_4)$, bounded term by term. This is Bałaban's reason for [3, (2.18)], "the same expressions but with the opposite signs" [3, p. 444], with the two members anchored at one vertex.

(b): the pinned root. Here $u_1 \neq u_2$ (the shapes T, B8, B10; in BL both scalar slots sit at w, so $u_1 = u_2$ and (a) applies with the anchor at w). The local terms are $c_{S,1}(x)\Phi$ and $-c_{S,2}(x)\Phi$, with $c_{S,i}$ the regrouped coefficient with u_i at x and the anchor at u_i (the sign from (a)). *Split.* By the steps of theorem F.69: rescale to $\xi\mathbb{Z}^3$ ($\xi = L^{-j_0}$, j_0 the lowest external index), where by lemma F.72(f) and the rescaling of lemma F.72(a) every regrouped scalar line is $\breve{G}_l^v = C^\xi + R_l^v$ with a potential in $[0, \bar{v}]$ and R_l^v as in lemma F.72(d), and every vector line is $C^\xi + R_l$ with the bulk remainder; move the multipliers to the anchor (lemma F.61); expand multilinearly, $c_{S,i} = c_{S,i}^C + c_{S,i}^R$. The shapes T, B8, B10 contain no [3, (1.6)] or [3, (1.7)] vertex, so no sharp indicator occurs and the internal sums are unrestricted on $\xi\mathbb{Z}^3$; and they contain no divergent proper subgraph (a two-scalar-leg or two-vector-leg subgraph inside a one-loop triangle or double line would need a bridge), so no inner counterterm occurs. *The all-C^ξ parts.* $c_{S,i}^C(x)$ is the same expression as in the bulk model: the same graph, vertex tensors and differences, the multipliers at the anchor, lines C^ξ, unrestricted sums. It is the zero-momentum value of one translation-invariant kernel on $\xi\mathbb{Z}^3$ with u_i pinned: with $K_S^C(y_{u_1}, y_{u_2}, \dots)$ the all-C^ξ kernel, $c_{S,1}^C(x) = \sum K_S^C(x, y_{u_2}, \dots) = \sum K_S^C(0, y, \dots)$ and $c_{S,2}^C(x) = \sum K_S^C(y_{u_1}, x, \dots) = \sum K_S^C(0, -y, \dots)$, both the total sum of $K_S^C(0, \cdot, \dots)$ over $\xi\mathbb{Z}^3$ (absolutely convergent at fixed ξ), so $c_{S,1}^C(x) = c_{S,2}^C(x)$, independent of x. The wall does not enter this term. *The R-parts.* Each

term of $c^R_{S,i}$ has at least one R-line. Lemma F.72(d) gives the bounds (82) for R^v_l at every pair of points, with constant $C_R(1+\bar v)^3$, so R^v_l is a remainder line of lemma F.68(b), and the step of theorem F.69 for "terms with at least one R-line, $D(S)=0$" applies: the node is one-loop, its inner components are trees, the only dangerous ones are the [3, (2.4)] subgraphs, treated by the summation by parts inside the coefficient (lemma F.65), every component after re-slicing has positive degree, and the bound function estimate [3, (2.16)] bounds each $c^R_{S,i}$ by $C(1+\bar v)^p$ uniformly in ξ, j_0, s, ε, the torus and x (the kernel bounds are pointwise, and the anchor is fixed at x). The exponent p counts the R-lines and the factors of $\bar v$ in their bounds, at most three lines here. *Conclusion.* $|c_{S,1}(x)-c_{S,2}(x)| = |c^R_{S,1}(x)-c^R_{S,2}(x)| \le 2C(1+\bar v)^p =: C^v_7$. This is the form of the $\phi\phi A$ clause of theorem F.69, $|c^{(0)}|\le C$, for the pair.

(c): bridged nodes. Write $\Sigma_2(w,w')$ for the amputated kernel of S_2 between its leg point w on the bridge and its other leg point w', with S_2's lines (walled or not, at the regrouped scale), and $F(y) := (\partial^\xi_\mu \breve G_{l_0})(x_u,y)$ for the bridge factor, u's difference acting on the bridge.

u summed. Choose the anchor $\hat x$ in S_2 (anchors of $\phi\phi A$ nodes are free, Step 3(D$_4$)). At zeroth order every external field of S is a constant at $\hat x$, and so are u's multipliers, which are smooth (lemma F.61; a one-vector-leg vertex carries no sharp indicator). The dependence of the regrouped kernel on x_u is then the single factor F, and $\sum_{x_u}\xi^3(\partial^\xi_\mu \breve G_{l_0})(x_u,y)=0$ for every summable kernel: a forward difference in the summed variable telescopes, on $\xi\mathbb{Z}^3$ and on the torus. So $c=0$ for every member and any line kernels, before any C^ξ/R split, and the same holds for the member with S_2 replaced by its counterterm vertex. No translation invariance of the lines is used.

u pinned. The anchor is u, at x. *All-C^ξ part.* On $\xi\mathbb{Z}^3$ with lines C^ξ and unrestricted sums, $\sum_{w'}\Sigma^C_2(w,w')$ is a constant g by translation invariance, and $\sum_w \xi^3(\partial^\xi_\mu C^\xi)(x,w)\,g = g\,\partial^\xi_\mu(C^\xi\mathbf{1})(x)=0$ since $C^\xi\mathbf{1}\equiv 1$. The counterterm member is the same computation with Σ_2 replaced by its own row sum. A sharp indicator inside S_2 occurs only at the [3, (1.6)] tadpole, which cancels exactly against its counterterm member label by label ((Agr); the counterterm carries the same unit-cube label), so the pair has no indicator defect; the exchange bubble and the seagull loop carry only the smooth labels of their vector legs. So $c^C=0$ per member (and per pair). This is Bałaban's "by translation invariance it can be written as a derivative of a constant" [3, p. 444] for [3, (2.21c)], and the exact cancellations he asserts for [3, (2.21b), (2.21d), (2.21f), (2.21g)], applied to the all-C^ξ parts. *Why the R-part is not bounded per member.* The term in which the bridge carries $R^v_{l_0}$ and S_2 carries lines C^ξ equals $g\sum_w \xi^3(\partial^\xi_\mu R^v_{l_0})(x,w) = g\,\partial^\xi_\mu(R^v\mathbf{1})(x)$, where $g\sim\xi^{-1}$ (the node S_2 has $D(S_2)=-1$, and the R-line is not inside it, so it gains nothing for S_2). In the bulk $R\mathbf{1}$ is constant ($-\Delta+m^2+aP$ maps constants to constants), and the term vanishes; under a wall $\breve G^v\mathbf{1} = \mathbf{1}-\breve G^v v$ is not constant, so the term is of order ξ^{-1} and is not bounded uniformly in ξ. *The pair.* Both members are in $\mathcal{C}$ by (P). Let f be the external field at S_2's far slot w' (constant at the anchor at zeroth order, Taylor-expanded at first order), and take the anchor convention (Agr) for S_2 with the anchor at the bridge end w, so that the counterterm member carries $\delta m^2_{S_2}(w)=\sum_{w'}\xi^3\Sigma_2(w,w')$, the row sum of the same kernel (the model's own counterterm, [3, p. 417]). Then

$$\sum_{w,w'}\xi^6F(w)\Sigma_2(w,w')f(w') - \sum_w \xi^3F(w)\,\delta m^2_{S_2}(w)f(w) \;=\; \sum_{w,w'}\xi^6F(w)\Sigma_2(w,w')\big[f(w')-f(w)\big],$$

an identity for any kernels, with no symmetry used. Its zeroth-order term (f constant) is 0 exactly, including the terms in which an R-line sits inside S_2 (these have $D(S_2)+1=0$ and are paired with the counterterm member carrying the same R-line, as in the step of theorem F.69 for $D(H)=-1$ components). The first-order term is $\sum_w\xi^3F(w)c_\mu(w)\,\partial_\mu f$ with $c_\mu(w) := \sum_{w'}\xi^3\Sigma_2(w,w')(w'-w)_\mu$ the first moment of S_2; its all-C^ξ part is bounded by [3, (3.16)] (the cancellation $\sum_{y'}(\partial C^\xi\partial^*)(y'-y)=0$ of theorem F.69), and its R^v-part by $|\partial R^v\partial^*|\le C_R(1+\bar v)^3(\xi+\rho)^{-1}e^{-\delta_R\rho}$ (lemma F.72(d)) against the other line of the bubble, a sum convergent in $d=3$; so $|c_\mu(w)|\le C(1+\bar v)^3$, and $\sum_w\xi^3|F(w)|\le C(1+\bar v)^3$ by the bounds on ∂C^ξ and ∂R^v. The Hölder remainder $f(w')-f(w)-(w'-w)\cdot\partial f(w)$ gives the node S a differentiated

external leg, after which every component has positive degree and [3, (2.16)] bounds the sum. If the bridge attaches at S_2's other leg and the anchor convention puts the counterterm at the far leg, the same identity holds with f and F exchanged, by the symmetry $\Sigma_2(w,w') = \Sigma_2(w',w)$ of the real symmetric scalar kernels at zero field (lemma F.72(a)). Altogether the R-part of the pair is $\le C(1+\bar v)^p =: C^v_{\mathrm{BR}}$, uniformly in ξ, j_0, s, ε, the labels, the torus and x. The bridged shapes without a counterterm member ([3, (2.21a), (2.21c), (2.21e)]: the one whose expression vanishes identically [3, p. 442], and those in which S_2 carries a difference on a leg or an extra power of η) need none: for them the all-C^ξ part vanishes as above, and the R-part is convergent per member, since the difference on S_2's leg, or the extra power of η, makes $D(S_2) \ge 0$ before the gain of the R-line. □

Remark F.75 (reducible two-scalar-leg chains). A one-particle-reducible two-scalar-leg chain of one-loop $D = -1$ parts is renormalised by the counterterms of its parts (Step 3(D_4)). The identity behind it, $T_{P_2}(b',c) = \delta_{b'c}\sum_{c'} K_{P_2}(b',c')$ with the same lines, indices and labels, holds for any kernels because the counterterm is computed with the model's own lines; so it holds verbatim in the walled model. The alternative route through translation invariance is not used.

Proof of theorem F.74(ii). The proof of [3, (1.33)] at $L = 5$ for the bulk model, Steps 3(D_3)–3(D_4) with the twenty sketched steps listed after theorem F.70, is run for the walled model and for a class $\mathcal{C}$ closed under the moves. It is organised as follows: first, where the scalar lines enter and what replaces each entry; second, where the class enters and what supplies each partner; third, the assembly and the induction.

Step 1: the scalar lines. At zero background the scalar lines of every graph, slice, counterterm and cut kernel are zero-field kernels, and the proof uses them in exactly four ways. *(1) Kernel bounds*: the group-(A) estimates (the slice bounds [3, (2.6), (2.10)–(2.12)], the kernel bounds of the cube step, the two-domain differences $\delta G_s(\square, \eta\mathbb{Z}^3, 0)$, the cut-kernel estimate [3, (3.1)] in the cluster form of corollary F.58, the Combes–Thomas bounds of the chains), each a composition of the pointwise bounds on the zero-field kernels with lattice sums; for walled lines these are lemma F.72(b),(c),(e), with constants polynomial in $\bar v$. The two-domain difference is bounded by the same resolvent identity as in the bulk, with lemma F.72(b) on each domain, since at zero field the bounds hold at every point of a Neumann domain and of the torus. *(2) Algebra*: the slice identity, its rescaling, the gauge covariance used in the gauge-out of B_0, the cancellation of the operator in the differences of the $\tilde B'$-expansion and of the domain changes; for walled lines these are lemma F.72(a), and at zero background the $\tilde B'$-expansion, the constant field B_0 and its gauge-out are void. *(3) The regrouped lines of the renormalisation*: in Step 3(D_4) the cumulative regrouped line of a node is, by the slice identity, the inverse of the operator pinned at the lowest external scale, which after rescaling to $\xi\mathbb{Z}^3$ is $\breve G_l = C^\xi + R_l$ (theorem F.69, lemma F.68); for walled lines the rescaled operator carries the potential $s^2 v_{\mathrm{per}}(s\,\cdot) \in [0,\bar v]$ (lemma F.72(a),(f)), so $\breve G^v_l = C^\xi + R^v_l$ with the *same* C^ξ and with R^v_l a remainder line of lemma F.68(b) with constant $C_R(1+\bar v)^3$ (lemma F.72(d)); hence every step of theorem F.69 that concerns terms with at least one R-line (the one-loop $D = -1$ cancellation, which uses only the bounds on the R-line and the forward-difference structure of the C^ξ-line; the re-slicing of lemma F.68; the bound function estimate [3, (2.16)]) runs verbatim, and every all-C^ξ expression, with the inner counterterm members re-expanded with their own lines, is literally the same expression as in the bulk model [3, pp. 437, 441]. *(4) The counterterm values*: the model's own, by definition (the walled model). The torus-versus-$\xi\mathbb{Z}^3$ identification of [3, pp. 418–419] is lemma F.72(f), with the image terms bounded, as in the bulk, by [3, (1.33)] for the classes of lower order into which they decompose (the induction below). The sharp-indicator lemma lemma F.67 uses only the decay of the internal lines, supplied by lemma F.72(b); the useless-counterterm lemma lemma F.66 and the [3, (2.4)] lemma lemma F.65 are kernel-free given the bounds; lemmas F.63 and F.64 are identities. Subgraphs with no scalar slot and an odd number of vector slots vanish at $B = 0$ also with walled lines, which are scalar kernels times $\mathbf{1}_N$ (lemma F.72(a)). A finite [3, (1.7)] vertex with an x-dependent coefficient never sits inside a dangerous node (lemma F.59(b), [3, Cor. 2.3]), so such coefficients enter only through their suprema, as the statement counts them.

Step 2: the class. The class enters the bulk proof only at the dangerous nodes, as the sum over a node's partners with the rest of the graph fixed (lemma F.63(a): the counterterm pairing is an identity of class sums; theorem F.69: the local coefficient is the class sum at the node; theorem F.62), at the cut counterterms of partnered subgraphs (lemma F.59, which needs [3, (1.33)] for the completed pieces of lower order), at the class-component cube (one label multiset), and in the regrouping after the cut (theorem F.57, kernel-free, (i)). Node by node, with walled lines: *two-scalar-leg nodes* are paired with their counterterm member (P), the exact identities [3, (3.19)] ($D = 0$) and [3, (3.11)] ($D = -1$) against the model's own counterterm with the same lines, the same localisation and (Agr); the $D = -1$ first moment has its all-C^ξ part bounded by [3, (3.16)] and its R^v-part by $|\partial R^v \partial^*| \le C(1+\bar v)^3(\xi+\rho)^{-1}e^{-\delta\rho}$, so $|c_\mu| \le C(1+\bar v)^3$; reducible chains by remark F.75. *Two-vector-leg nodes*: the per-label two-vector-leg class at the node, which (W) supplies; the zeroth moment of its all-C^ξ part vanishes by lemma F.60, applied as in theorem F.69 to the full lines C^ξ of the class sum (the lemma allows any x-dependent $|\phi|^2$ coefficients, so the inner counterterm and finite-vertex coefficients, and the potential itself, are admissible coefficients; lemma F.60 is thus verbatim in the walled model), the first moment of its all-C^ξ part by the reflection identity [3, (3.30)] on C^ξ, which is model-free; the R^v-parts by the R-line steps of theorem F.69, with the zero-momentum remainder vertex kept. *$\phi\phi A$ nodes*: in the bulk, theorem F.69 bounds the zeroth moment by the reflection identity [3, (3.37)] over the per-label full $\phi\phi A$ class; in a class closed under the moves this is replaced by (iii): at a bridgeless node the (X)-partner is in $\mathcal{C}$ and (iii)(a) cancels the pair exactly at a common anchor, or, when the node carries the pinned root at one of its two slot vertices, (iii)(b) bounds the pair by C_7^v; at a bridged node no partner is needed, (iii)(c), the counterterm member (P) of S_2 being in $\mathcal{C}$; and the pair bounds are of the form $|c^{(0)}| \le C$ that theorem F.70 consumes for a $\phi\phi A$ local vertex. [3, (2.4)] *subgraphs*: summation by parts, lemma F.65, kernel-free. *Vacuum components*: cancelled by the model's own $E_s[v]$ with the same labels and lines, up to image terms (Step 1). *One- and three-vector-leg subgraphs*: vanish (Step 1). The cut counterterms of partnered subgraphs (lemma F.59) are classes of lower order of the completed pieces, which are closed under the moves (a move at a divergent subgraph of a piece is a move of the uncut member, and the completion of a partnered subgraph is by the moves); and the class-component cube depends on the label multiset only, common to $\mathcal{C}$. Per-label full classes are closed under the moves, so the bulk statement for them is the special case $\mathcal{C}$ = full class; and every partner that the walled node mechanisms use is in $\mathcal{C}$ by the closure, while every other step is a bound per term.

Step 3: assembly and induction. With Steps 1 and 2, theorems F.69 and F.70 hold for $\mathcal{C}$ with walled lines, with the $\phi\phi A$ clause in the pair form of (iii); the induction on the number of lines inside theorem F.69 and on the number of vertices in lemma F.59 runs over classes closed under the moves of lower order (the completed pieces of a cut partnered subgraph, and the pieces into which an image term decomposes), to which the statement applies by the induction hypothesis; the base is the one-vertex graphs. The bound function estimate [3, (2.16)] then gives [3, (3.3)] for every class component, and the assembly of [3, pp. 420–445] listed after theorem F.70, with the cluster cut and the cube steps, gives [3, (1.33)] in the stated form, with d_v, d_s counting the action couplings, the counterterm vertices as exact counterterms, and the finite vertices through their suprema (they have $D(v) = 2$ whatever their coefficient, lemma F.59(b)); the localisation of an external field whose leg moves under the moves is (H_A); the slot fields of several variables are measured by [3, (1.32)] in the slot variable, as the two-variable external fields of [3, p. 419]; and in a two-point setting the root vertex is fixed at x throughout, every kernel bound used being pointwise, so the bound is uniform in x.

Constants. The walled inputs carry $(1+\bar v)^2$ (lemma F.72(b)), $(1+\bar v)^3$ (lemma F.72(d)) and the constants of lemma F.72(e), polynomial in $\bar v$; a graph of order $\le \bar n$ has a number of lines bounded by $\bar n$, so $O(1)^v$ is a polynomial in $\bar v$ of degree bounded in terms of $\bar n$, and δ_0^v is a fixed fraction of δ^w, δ_R and the bulk rates. The parameter rows of Step 3(D) are kept, with R_0 replaced by $R_0^v := R_0 + 10$ in every depth row for a scalar propagator bound and the torus side $\ge \max\{6\sqrt{3}\, r(t_s) + 2, 8/\delta_0^v\}$, all lower bounds on R, $r^\flat$ and the torus side or caps on ε_0 of

Bałaban's kind, fixed after R_0 and before ε_0; nothing in the order of the choices changes. The constants are therefore independent of ε, of $\varepsilon_0 \in (0, \varepsilon_0^*]$ below the caps, of s, K, the labels, the torus side, x, and of v beyond $\bar{v}$, and they depend on $L = 5$, on the parameters of Step 3(D) and polynomially on $\bar{v}$. □

Remark F.76 (non-circularity of the uses in the rotation part). Theorem F.74(ii) is a bound in which the suprema of the finite-vertex coefficients are parameters. Its uses in the rotation part at coupling order 4 carry, as finite vertices, only the non-renormalising parts of the order-2 inner counterterms, whose sizes come from the order-2 statement, and the wall vertex, whose coefficient is $\le nt_s^2$; the sizes of the order-4 finite vertices are then *consequences* of (ii) at order 4, not inputs to it. The induction of the proof runs on the number of vertices inside the theorem.

Remark F.77 (a bounded wall is not a boundary). Theorem F.74 is stated for a bounded site potential on the scalar lines with the vector lines unwalled, on a torus. It does not cover scalar or vector lines with Dirichlet or Neumann boundary conditions on a box, and the proof of (iii)(b) does not extend to them: a boundary changes the kernel by image terms $C^\xi(x - y^*)$ which, for x and y in the boundary layer, are as singular as C^ξ itself and so are not remainder lines in the sense of (82); with a Neumann vector line the odd part of the pinned pair is left uncancelled at every scale, and nothing bounds $|c_{S,1} - c_{S,2}|$ uniformly in ξ in the layer. A bounded potential changes the kernel by R^v, which satisfies (82) at *every* pair of points (lemma F.72(d); the multiplier $1 - w$ is never differentiated, and no smoothness of v is used), so there is no boundary layer, and the vector line is the translation-invariant torus propagator. This is why the rotation part compares King's cubes with walled tori and never applies the renormalisation theorem on a cube.

Step 3(E): the sizes of the mass counterterm. Three statements about the mass counterterm δm^2 are asserted in [2, 3] without proof and are consumed by every step of the chain: $\delta m^2 = O(\varepsilon^{-1})$ [2, p. 559], and the two finite-part statements of [3, p. 418], $\delta m^2_{K,k}(x) = O((L^k\varepsilon)^{-1})$ and "δm^2_{fin} convergent as $\varepsilon \to 0$" ("Future considerations will imply"), which enter the proof of Proposition 1 only through the size of the non-renormalising part of the [3, (1.7)] vertex [3, p. 421]. This step proves the first in the form stated, the second in the form stated, and the third in the form in which it is consumed: in scale-k units the non-renormalising coefficient $c_k := t_k^2(\delta m^2_{\text{fin}} + \delta m^2_{K,k})$ is $O(t_k(1 + \ln t_k^{-1})^5)$ with a constant free of ε_0 (theorem F.99), which every consumer absorbs (lemma F.101); δm^2_{fin} itself is bounded by a constant depending on ε_0 (corollary F.100).

The per-graph bound $\delta m^2 = O(\varepsilon^{-1})$. Setting [1, pp. 605–606], [3, pp. 413, 416–417]. $d = 3$, $T_\varepsilon := \prod_\mu (\varepsilon\mathbb{Z}/\ell_\mu\mathbb{Z})$ with all sides $\ell_\mu \ge 1$, $\ell_\mu/\varepsilon \in \mathbb{N}$ (write ℓ_T^{-3} for $\prod_\mu \ell_\mu^{-1}$); functions carry the ε^3-weighted product and $\hat{f}(p) := \sum_x \varepsilon^3 f(x) e^{-ipx}$ for $p \in T^* := \prod_\mu (2\pi/\ell_\mu)\mathbb{Z} \cap B_\varepsilon$, $B_\varepsilon := [-\pi/\varepsilon, \pi/\varepsilon)^3$. For $p \in \mathbb{R}^3$, $[p] \in B_\varepsilon$ is its representative modulo $(2\pi/\varepsilon)\mathbb{Z}^3$. $(\partial_\mu f)(x) := \varepsilon^{-1}(f(x + \varepsilon e_\mu) - f(x))$ has symbol $\hat{\partial}_\mu(p) = \varepsilon^{-1}(e^{i\varepsilon p_\mu} - 1)$, and $\hat{\Delta}(p) := \sum_\mu 4\varepsilon^{-2}\sin^2(\varepsilon p_\mu/2) = \sum_\mu |\hat{\partial}_\mu(p)|^2$. The free covariances are $C_0 := (-\Delta + m^2)^{-1}$ (scalar, $m > 0$) and $C := (-\Delta + \mu_0^2)^{-1}$ (vector, Feynman gauge, componentwise [1, (1.11)]). By [3, (1.20)–(1.21)], $\delta m^2 = \sum_{2 \le \alpha + 2\beta \le 4} e^\alpha \lambda^\beta \delta m^2_{(\alpha,\beta)}$ is fixed graph by graph: $\delta m^2_G := \sum_{x' \in T_\varepsilon} \varepsilon^3 \Sigma^\varepsilon_G(x - x')$, Σ^ε_G the amputated one-particle-irreducible two-point graph G (no difference on its external legs) with the vertices of [3, (1.6)–(1.8), (1.10)] at $\eta = \varepsilon$, $L^k\varepsilon = 1$, $\tilde{B} = 0$, $g_k = 1$ [3, (1.22)–(1.23)]; δm^2 is an $N \times N$ matrix polynomial in q^2, treated as a number. The vertices, the Taylor terms of $\frac{1}{2}|D_A\phi|^2 - \frac{1}{2}|\partial\phi|^2 = -\varepsilon^{-2}\phi(b_-) \cdot (U(A_b) - 1)\phi(b_+)$, are: (V_λ) $-\lambda \sum_x \varepsilon^3 |\phi|^4$, order 2; (V_δ) $-\frac{1}{2}\sum_x \varepsilon^3 \delta m_1^2 |\phi|^2$ with δm_1^2 the order-2 part of δm^2, order 2; (V_n^8) $e^n(-1)^n\varepsilon^{n-1}(n!)^{-1}\sum_b \varepsilon^3[(\partial\phi)(b) \cdot q^n\phi(b_-)](A_b)^n$, $n \ge 1$, order n; (V_n^{10}) $e^n\varepsilon^{n-2}(n!)^{-1}\sum_b \varepsilon^3[\phi(b_-) \cdot q^n\phi(b_-)](A_b)^n$, $n \ge 2$ even (for odd n, q^n is antisymmetric and the term vanishes), order n. The order of G is $\omega_G := \sum_v \mathrm{ord}(v) = \alpha + 2\beta$. The ε-exponent of a vertex is $s_v := 0$ (V_λ), $n - 1$ (V_n^8), $n - 2$ (V_n^{10}), and $s_v := -1$ for V_δ, the last used only after $|\delta m_1^2| \le C\varepsilon^{-1}$ is known; $s(G) := \sum_v s_v$.

Lemma F.78 (symbols). $\frac{4}{\pi^2}|[p]|^2 \le \hat{\Delta}(p) \le |[p]|^2$, $|\hat{\partial}_\mu(p)| \le |[p]|$, *and* $\hat{C}_0(p) = (\hat{\Delta}(p) + m^2)^{-1} \le \frac{\pi^2}{4}(|[p]|^2 + m^2)^{-1}$, *the same for* $\hat{C}$ *with* μ_0.

Proof. All three are periodic, so take $p = [p]$; use $|\sin t| \geq 2|t|/\pi$ for $|t| \leq \pi/2$ and $|\sin t| \leq |t|$; for the last claim, $\frac{4}{\pi^2}|p|^2 + m^2 \geq \frac{4}{\pi^2}(|p|^2 + m^2)$. □

Lemma F.79 (Riemann domination)**.** *Let $F\colon B_\varepsilon^n \to [0,\infty)$ satisfy $F(k) \leq C_1 F(k')$ whenever $|k_{i,\mu} - k'_{i,\mu}| \leq \pi/\ell_\mu$ for all i,μ. Then $C_1^{-1}(2\pi)^{-3n}\int_{B_\varepsilon^n} F \leq \ell_T^{-3n}\sum_{k\in(T^*)^n} F(k) \leq C_1(2\pi)^{-3n}\int_{B_\varepsilon^n} F$. For $F = \prod_c g_c(q_c(k))$ with $g_c(q) = (|[q]|^2 + m_c^2)^{-\beta_c}$, $\beta_c \geq 0$, each q_c one of $\pm k_i$, $\pm(k_1+k_2)$, the hypothesis holds with $C_1 = \prod_c (2 + 242/m_c^2)^{\beta_c}$.*

Proof. The boxes of sides $2\pi/\ell_\mu$ centred at the points of T^* tile B_ε, and $\ell_T^{-3} = (2\pi)^{-3}|\mathrm{box}|$; compare F at the centre with F on the cube. For the product form, $|[q_c(k)] - [q_c(k')]| \leq 2\sqrt{3}\pi/\min_\mu \ell_\mu \leq 11$, and for $|\delta| \leq 11$, $(|q'|+|\delta|)^2 + m^2 \leq 2|q'|^2 + 242 + m^2 \leq (2+242/m^2)(|q'|^2 + m^2)$. □

Lemma F.80 (radial integrals)**.** *For $\beta \in \mathbb{R}$, $m > 0$, $0 < \varepsilon \leq 1$, $r > 0$ let $I(\beta) := \int_{B_\varepsilon}(|p|^2 + m^2)^{-\beta}dp$ and $H_\beta(r) := \int_{B_\varepsilon \cap \{|p|\leq r\}}(|p|^2+m^2)^{-\beta}dp$. With $C = C(\beta, m)$: if $\beta < \frac{3}{2}$, $I(\beta) \leq C\varepsilon^{2\beta-3}$ and $H_\beta(r) \leq C(1+r)^{3-2\beta}$; if $\beta = \frac{3}{2}$, $I \leq C(1 + \ln\varepsilon^{-1})$ and $H \leq C(1+\ln(1+r))$; if $\beta > \frac{3}{2}$, $I, H \leq C$.*

Proof. $B_\varepsilon \subset \{|p| \leq \sqrt{3}\pi/\varepsilon\}$; in polar coordinates $(s^2+m^2)^{-\beta} \leq \max(m^{-2\beta}, (1+m^2)^{-\beta})$ on $s \leq 1$ and $\leq Cs^{-2\beta}$ on $s \geq 1$; $\int_1^{\max(\rho,1)} s^{2-2\beta}ds$ gives the three cases. □

Lemma F.81 (structure and degree)**.** *Let G be a graph of δm^2 of order $\omega_G \leq 4$. Then:* (a) *the loop number is $\mathcal{L} := I - V + 1 = (\omega_G - \omega_{\mathrm{ins}})/2$, ω_{ins} the total order of the V_δ vertices; so $\mathcal{L} \leq 2$, and a V_δ vertex occurs only in one-loop graphs of order 4, exactly one, of order 2.* (b) *The internal-line graph G' is connected and bridgeless; for $\mathcal{L} = 1$ it is a cycle, for $\mathcal{L} = 2$ a theta (two vertices of degree 3 joined by three internally disjoint chains) or a figure-eight (one vertex of degree 4 on two cycles); all other vertices have degree 2 in G'.* (c) *At zero external momentum all lines of a chain (or cycle) carry one momentum q_c (k_1, k_2, k_1+k_2 for a theta; k_1, k_2 for a figure-eight; k_1 for a cycle), and $|\delta m_G^2| \leq C_G\varepsilon^{s(G)}\ell_T^{-3\mathcal{L}}\sum_{k\in(T^*)^{\mathcal{L}}}\prod_c g_c(q_c)$ with $g_c(q) = (|[q]|^2 + m_c^2)^{-\beta_c}$, $\beta_c := \sum_{l\in c}(1 - d_l/2) \geq 0$, $d_l \in \{0,1,2\}$ the number of differences acting on the line l, $m_c \in \{m, \mu_0\}$ on a chain of one line type and $m_c := \min(m,\mu_0)$ on a mixed chain.* (d) *With $\omega(G) := 3\mathcal{L} - 2\sum_l(1 - d_l/2) - s(G)$ and, for a cycle H of G', $\omega(H) := 3 - 2\sum_{l\in H}(1-d_l/2) - s(H)$: $\omega(G) = 2 - \omega_G/2$, and every cycle $H \neq G'$ of a two-loop G' has $\omega(H) \leq 1$, with equality only if H is a two-point subgraph of order 2.*

Proof. (a) $\sum_v h_v = 2I + 2$ (h_v the number of legs), so $\mathcal{L} = \sum_v(h_v/2 - 1)$, and $h_v/2 - 1 = \mathrm{ord}(v)/2$ for V_λ ($h = 4$), V_n^8, V_n^{10} ($h = n+2$) and 0 for V_δ ($h = 2$). (b) G connected with external legs at vertices gives G' connected; 1PI gives no bridge, so no vertex of degree ≤ 1; $\sum_v(\deg_{G'} v - 2) = 2(\mathcal{L} - 1)$. For $\mathcal{L} = 2$ with two vertices u, w of degree 3, a path leaving u and returning to u would make the connection to w a bridge, so all three paths end at w. (c) Momentum conservation at the vertices with zero external momenta; per line $|\hat{C}(q)||\hat{\partial}(q)|^{d_l} \leq \frac{\pi^2}{4}(|[q]|^2 + m_c^2)^{-1+d_l/2}$ by lemma F.78; vertex phases have modulus 1, $\|q\| \leq 1$. (d) With $\mathcal{L} = I - V + 1$ and $2I = \sum h_v - 2$, $\omega(G) = \sum_v(h_v/2 - 3 - s_v + d_v) + 2$, $d_v = 1$ for V^8 vertices (the difference acts on an internal line) and 0 otherwise; the bracket is $-\mathrm{ord}(v)/2$ for every type (-1, -1, $(n+2)/2-3-(n-1)+1 = -n/2$, $(n+2)/2-3-(n-2) = -n/2$). For a cycle H (as many lines as vertices), $\omega(H) = 3 - \mathrm{ord}(H)/2 - E_H/2 - \#\{V^8$ vertices of H whose difference acts off $H\}$, $E_H := \sum_{v\in H}(h_v - 2)$: the bracket per vertex is $-1 - e_v/2$ (V_λ, $e_v = 2$), $-n/2 - e_v/2 - (1 - d_{v,H})$ (V_n^8), $-n/2 - e_v/2$ (V_n^{10}), $-1 - e_v/2$ (V_δ). For a proper cycle of a theta or a figure-eight, $E_H \geq 2$ (the third chain, resp. the other cycle, leaves H). $\mathrm{ord}(H) \geq 2$: a cycle through ≥ 2 vertices has order ≥ 2; a self-loop can sit only at the junction of a figure-eight, which has ≥ 4 internal legs, so it is V_λ, V_n^8 or V_n^{10} with $n \geq 2$. Hence $\omega(H) \leq 1$, with equality iff $\mathrm{ord}(H) = 2$, $E_H = 2$ and no difference leaves H. □

Lemma F.82 (one loop)**.** *If $\mathcal{L}(G) = 1$, then $|\delta m_G^2| \leq C\varepsilon^{s(G)}I(\beta)$ with β the β_c of the cycle, and $|\delta m_G^2| \leq C\varepsilon^{-1}$.*

Proof. The first bound is lemmas F.79 and F.81(c). Then $\omega(G) = 3 - 2\beta - s$ and lemma F.80. Order 2 (no V_δ, $s \geq 0$, $\omega = 1$): $s = 0$ gives $\beta = 1$, $I \leq C\varepsilon^{-1}$; $s \geq 1$ gives $\beta = 1 - s/2 < \frac{3}{2}$ and $\varepsilon^s I \leq C\varepsilon^{s+2\beta-3} = C\varepsilon^{-1}$. Order 4 with one V_δ ($s = s' - 1$, $s' \geq 0$ the sum over the other vertices, $\omega = 0$, $3 - 2\beta = s' - 1$): $s' = 0$ gives $\beta = 2$ and $C\varepsilon^{-1}$; $s' = 1$ gives $\beta = \frac{3}{2}$ and $C(1 + \ln \varepsilon^{-1})$; $s' \geq 2$ gives $\varepsilon^{s'-1}\varepsilon^{2\beta-3} = 1$. □

Lemma F.83 (theta)**.** *If G' is a theta with chains $c = 1, 2, 3$, then $|\delta m^2_G| \leq C\varepsilon^{-1}$.*

Proof. By lemmas F.79 and F.81(c), $|\delta m^2_G| \leq C\varepsilon^s T$ with $T := \iint_{B^2_\varepsilon} g_1(k) g_2(l) g_3(k + l)\, dk\, dl$; at order 4 with $\mathcal{L} = 2$ there is no V_δ, so $s \geq 0$. (1) For $a \in \{1, 2, 3\}$ let S_a be the set where $|[q_a]|$ is the smallest of $|[k]|, |[l]|, |[k + l]|$, b, c the other indices and q_* the one of q_b, q_c of smaller norm; each g decreases in the norm, so on S_a the integrand is $\leq g_a(q_a) g_b(|q_*|) g_c(|q_*|)$ with $|[q_a]| \leq |[q_*]|$. (2) Any two of $k, l, k + l$ are coordinates with unit Jacobian, so $\iint_{S_a} \leq 2 \int_{B_\varepsilon} g_b(q) g_c(q) H_{\beta_a}(|q|) dq$. (3) Put $\omega^0 := 6 - 2(\beta_1 + \beta_2 + \beta_3)$, $\omega^0_{bc} := 3 - 2(\beta_b + \beta_c)$; then $\omega(G) = \omega^0 - s = 0$ and $\omega(H_{bc}) = \omega^0_{bc} - s(H_{bc}) \leq 1$ by lemma F.81(d), with $s(H_{bc}) \leq s$. (4) $\beta_a < \frac{3}{2}$: $H \leq C(1 + |q|)^{3-2\beta_a} \leq C(|q|^2 + m^2)^{(3-2\beta_a)/2}$, so the sector is $\leq C\, I(\beta')$ with $3 - 2\beta' = \omega^0 = s$, and ε^s times it is $\leq C(1 + \ln \varepsilon^{-1})$. (5) $\beta_a = \frac{3}{2}$: $H \leq C(1 + \ln \varepsilon^{-1})$ and $\omega^0_{bc} = \omega^0 = s$, so the sector is $\leq C(1 + \ln \varepsilon^{-1})^2$. (6) $\beta_a > \frac{3}{2}$: $H \leq C$ and $\varepsilon^s I(\beta_b + \beta_c) \leq C \max(\varepsilon^{s-\omega^0_{bc}}, 1 + \ln \varepsilon^{-1})$ with $\varepsilon^{s-\omega^0_{bc}} \leq \varepsilon^{s(H_{bc})-\omega^0_{bc}} = \varepsilon^{-\omega(H_{bc})} \leq \varepsilon^{-1}$. Sum the six pieces. □

Lemma F.84 (figure-eight)**.** *If G' is a figure-eight with cycles $1, 2$, then $|\delta m^2_G| \leq C\varepsilon^s I(\beta_1) I(\beta_2) \leq C\varepsilon^{-1}$.*

Proof. The cycles carry independent momenta; $\omega(G) = 6 - 2\beta_1 - 2\beta_2 - s = 0$ and $\omega(H_i) \leq 1$. Both $\beta_i < \frac{3}{2}$: $\varepsilon^{s+2\beta_1-3+2\beta_2-3} = \varepsilon^{-\omega(G)} = 1$. $\beta_1 < \frac{3}{2} = \beta_2$: $\varepsilon^{-\omega(G)}(1 + \ln \varepsilon^{-1})$. $\beta_1 < \frac{3}{2} < \beta_2$: $\varepsilon^{s-(3-2\beta_1)} \leq \varepsilon^{-\omega(H_1)} \leq \varepsilon^{-1}$, using $s \geq s(H_1)$. Both $\geq \frac{3}{2}$: $\leq C(1 + \ln \varepsilon^{-1})^2$. □

Theorem F.85 (the mass counterterm per graph)**.** *For every graph G of δm^2 ([3, (1.22)–(1.23)]: all graphs of order $\alpha + 2\beta \leq 4$, $d = 3$), $|\delta m^2_G| \leq C_G \varepsilon^{-1}$ with $C_G = C_G(m, \mu_0, N)$, uniformly in $0 < \varepsilon \leq 1$ and in the torus (all sides ≥ 1). Hence $|\delta m^2| \leq c_m \varepsilon^{-1}$ with $c_m := \sum_G e^\alpha \lambda^\beta C_G$, a finite sum; c_m does not depend on ε_0.*

Proof. Order 2: by lemma F.81(a) these are one-loop graphs without V_δ, so lemma F.82 gives $|\delta m^2_1| \leq C\varepsilon^{-1}$, which justifies $s_v = -1$ for the V_δ vertices at order 4. Order 4: by lemma F.81 each graph is a cycle with one V_δ (lemma F.82), a theta (lemma F.83) or a figure-eight (lemma F.84). At order ≤ 4, $n \leq 4$ and $V \leq 4$, so there are finitely many graphs. □

The proof uses the momentum-space form: in x-space the photon–scalar bubble line $\partial C_0 \partial^*$ has a logarithmically divergent ℓ^1 norm, while its symbol $|\hat{\partial}|^2 \hat{C}_0$ is bounded (lemma F.78). Not every order-4 graph is $O(\ln \varepsilon^{-1})$: graphs with an order-2 two-point subgraph (the last graph of [3, (1.22)], the tadpole on a tadpole of [3, (1.23)]) are of size ε^{-1}. The third graph of [3, (1.22)] is lemma F.84 at the junction V_4^{10}: $\varepsilon^2 C_0(0)^2 = O(1)$, Bałaban's "except the third".

Corollary F.86 ([2, (2.11)–(2.13)])**.** *The bounds [2, (2.11)–(2.13)] on the counterterm vertices hold for Bałaban's torus δm^2 with constants free of ε, ε_0 and the torus.*

Proof. They are stated in Step 3 as consequences of $|\delta m^2| \leq c\varepsilon^{-1}$ by exact computation; theorem F.85 supplies the hypothesis. □

The block kernels at zero background. Setting [3, (1.25)–(1.28), p. 418]. On a torus T_ε aligned with the blocks (all sides ≥ 1), at $A = 0$ and in ε-units, put $l_j := L^j \varepsilon = t_j$, $G_j := (C_0^{-1} + a_j l_j^{-2} P_j)^{-1}$ for $1 \leq j \leq K$, with P_j the orthogonal projection onto the functions constant on the blocks of side l_j (so G_K is the operator of lemma F.5) and $G_0 := 0$; $S_j := G_{j+1} - G_j$ (the slice of index j; $S_0 = G_1$) and $R_j := C_0 - G_j$. Then $G_K = G_k + G_{K,k}$ with $G_{K,k} = \sum_{j=k}^{K-1} S_j$, which is [3, (1.27)], and R_K is the "operator with a regular kernel" of [3, (1.25)]: $R_K = G_K(a_K l_K^{-2} P_K) C_0 = C_0 (a_K l_K^{-2} P_K) G_K$. The vector kernels G^v_j, S^v_j, R^v_j are the same with μ_0 in place of m, componentwise (Feynman gauge; [1, p. 608]: at $A = 0$ the vector

transformations are the scalar ones with $N = d$). δm_K^2, E_K, δm_k^2, E_k are "the same expressions" [3, (1.26), (1.28)] with G_K, resp. G_k, in place of C_0 and C: the [3, (1.7)] vertices inside them keep their numerical coefficients δm_1^2, $\delta m_{(4)}^2$, and only the lines are replaced (Convention F.18; in [3, (1.23)] δm_1^2 is a symbol for a number). Write $\delta m^2[G, G^v]$, $E[G, G^v]$ for the class sums [3, (1.23)–(1.24)] with the lines (G, G^v), so that $\delta m^2 = \delta m^2[C_0, C]$, $\delta m_K^2 = \delta m^2[G_K, G_K^v]$, $\delta m_{\mathrm{fin}}^2 = \delta m^2 - \delta m_K^2$, $\delta m_{K,k}^2 = \delta m_K^2 - \delta m_k^2$, and likewise for E. Superscripts (2), (4) denote the parts of order $\alpha + 2\beta = 2$, 4 (the orders (3, 0), (1, 1) vanish, α odd).

Lemma F.87 (operator monotonicity)**.** *For $1 \le k \le K$ and every x:* (a) $C_{M_k} \le G_k \le C_0$ *as operators, $C_M := (-\Delta + m^2 + M^2)^{-1}$, $M_k^2 := a l_k^{-2}$;* (b) $0 \le R_k(x,x) \le c_R l_k^{-1}$ *and $|G_{K,k}(x,x)| \le c_R l_k^{-1}$, $c_R := C(m) a^{1/2}$;* (c) *the same for the vector kernels.*

Proof. (a) $0 \le a_k l_k^{-2} P_k \le M_k^2$ and inversion is operator-monotone decreasing. Diagonal entries are monotone ($K(x,x) = \langle e_x, K e_x\rangle$, $e_x := \varepsilon^{-3}\mathbf{1}_x$), so $C_{M_k}(0) \le G_k(x,x) \le C_0(0)$ and $0 \le R_k(x,x) \le C_0(0) - C_{M_k}(0) = \ell_T^{-3} \sum_p M_k^2 \hat{C}_0(p)(\hat{\Delta}(p) + m^2 + M_k^2)^{-1} \le C M_k^2 \int_{\mathbb{R}^3} ((p^2+m^2)(p^2 + M_k^2))^{-1} dp \le C' M_k$ (lemma F.79). (b) $G_{K,k}(x,x) = G_K(x,x) - G_k(x,x)$ lies between $C_{M_K}(0) - C_0(0) \ge -c_R l_K^{-1}$ and $C_0(0) - C_{M_k}(0)$, and $l_K \ge l_k$. □

Lemma F.88 (the slice ladder)**.** *Let $\sigma, \sigma' \in \{0,1\}$ count forward differences in the first and second variable, $y := x - x'$ (torus distance).* (a) $G_k = \sum_{j<k} S_j$ *and* $R_k = R_K + \sum_{j=k}^{K-1} S_j$. (b) *For $0 \le j < K$, $|\partial^\sigma \partial'^{\sigma'} S_j(x,x')| \le C_K l_j^{-1-\sigma-\sigma'} e^{-\delta_K |y|/l_j}$, with C_K, δ_K free of j, ε, K and the torus.* (c) $|\partial^\sigma \partial'^{\sigma'} G_k(x,x')| \le C(\varepsilon + |y|)^{-1-\sigma-\sigma'} e^{-\delta_K |y|/(2l_k)}$, *and* $|\partial^\sigma \partial'^{\sigma'} C_0(x,x')| \le C(\varepsilon + |y|)^{-1-\sigma-\sigma'} e^{-2c_*|y|}$ *with $c_* = c_*(m) > 0$ (several differences in one variable allowed, $\sigma + \sigma' \le 3$), on every torus with sides ≥ 1.* (d) *The same for the vector kernels.*

Proof. (a) is telescoping. (b) For $j \ge 1$ the slice identity [3, (1.27)]: $S_j = a_j^2 l_j^{-4} G_j Q_j^* C^{(j)} Q_j G_j$, with Q_j the block average and $C^{(j)} = ((\Delta^{(j)} + aL^{-2}P)|_{T^{(j)}})^{-1}$ the step-j fluctuation covariance. In scale-j units the legs $G_j Q_j^* \delta_{\hat{y}}$, their differences and Hölder quotients (over separations ≤ 2) obey $c_0 e^{-\delta_0 \operatorname{dist}(\hat{x}, B_j(\hat{y}))}$, with $c_1(\alpha) e^{2\delta_0}$ in place of c_0 for the Hölder quotients of exponent α (the printed bound decays in $\operatorname{dist}(\{\hat{x}, \hat{x}'\}, \hat{y}) \ge \operatorname{dist}(\hat{x}, B_j(\hat{y})) - 2$), by [4, Lemma 2.4, (2.35)–(2.36)] on the periodic box $T^{(j)}$ (a union of unit blocks, the torus being aligned with the blocks), in the form of theorem F.7: the constants are free of j and of the torus, and no depth condition arises (this is [1, Prop. 2.1, (2.24)–(2.25)] at $A = 0$ with the source $Q_j^* \delta_{\hat{y}}$, $\|Q_j^* \delta_{\hat{y}}\|_\infty = 1$); and $|C^{(j)}(\hat{y}, \hat{y}')| \le c_2 e^{-\theta_0 |\hat{y} - \hat{y}'|}$ by [4, (2.37)] with the j-free constants of theorem F.7, i.e. lemma F.145(c) at $A = 0$ with $\Lambda = T^{(j)}$, whose proof uses, of the geometry in (H1), only that $T^{(j)}$ is a union of L-blocks (the blocks of scale $j + 1 \le K$); keep half of each exponent for the two block sums (lemma F.26) and use $|\hat{x} - \hat{x}'| \le \operatorname{dist}(\hat{x}, B(\hat{y})) + |\hat{y} - \hat{y}'| + \operatorname{dist}(\hat{x}', B(\hat{y}')) + 2$; rescaling to ε-units gives l_j^{-1} for the kernel and l_j^{-1} per difference, and $\delta_K := \min(\delta_0, \theta_0)/2$. For $j = 0$, $S_0 = G_1$ in scale-1 units has the fixed spacing $1/L$: the ℓ^2 decay of G_1 at rate θ_0 (the Combes–Thomas bound of group (A), lemma F.144 for $T = G_1^{-1} \ge \gamma_\emptyset$ by lemma F.143, at the fixed spacing $1/L$ and with no depth restriction) gives the pointwise bounds with a constant depending on L, and the rescaling is as before. (c) Sum (b) over $j < k$: the slices with $l_j \ge |y|$ form a geometric sum $\le C \max(\varepsilon, |y|)^{-1-\sigma-\sigma'}$, and those with $l_j < |y|$ give $l_j^{-a} e^{-\delta_K |y|/l_j} \le C_a |y|^{-a} (l_j/|y|)$, again geometric; split $e^{-\delta_K |y|/l_j} \le e^{-\delta_K |y|/(2l_j)} e^{-\delta_K |y|/(2l_k)}$ for $j < k$. For C_0: on $\varepsilon\mathbb{Z}^3$ shift the contour of the Fourier integral of $(\hat{\Delta}(p) + m^2)^{-1}$ by $i\tau$, $\tau := m/4$, in the direction of the largest coordinate of y (for $\varepsilon\tau \le 1$, $\operatorname{Re} \hat{\Delta}(p + i\tau e) \ge \hat{\Delta}(p) - 1.1\tau^2$), together with the short-distance bound $\int_{B_\varepsilon} |p|^{\sigma+\sigma'} (|p|^2 + m^2)^{-1} dp \le C \varepsilon^{-1-\sigma-\sigma'}$; the torus kernel is the periodisation, and the images at distance $\ge \ell_T/2 \ge \frac{1}{2}$ are summable by the mass. (d) is the same with μ_0. □

Lemma F.89 (the remainder)**.** *On every torus aligned with the blocks with all sides ≥ 1, for $1 \le a \le K$ and $\sigma, \sigma' \in \{0,1\}$: $|\partial^\sigma \partial'^{\sigma'} R_a(x,x')| \le C_R \max(l_a, |y|)^{-1-\sigma-\sigma'} e^{-c_*|y|}$, $C_R = C(m, L, a, C_K, \delta_K)$, free of ε, a, K and the torus; the same for R_a^v.*

Proof. Case $|y| \geq l_a$: $R_a = C_0 - G_a$, and lemma F.88(c) bounds both kernels by $C(\varepsilon+|y|)^{-1-\sigma-\sigma'}$ times $e^{-2c_*|y|}$, resp. $e^{-\delta_K|y|/(2l_a)} \leq e^{-\delta_K|y|/2}$ ($l_a \leq 1$); take $c_* \leq \delta_K/4$. Case $|y| < l_a$: $R_a = G_a(a_a l_a^{-2} P_a)C_0$ [3, (1.25)], whose kernel is $a_a l_a^{-5} \sum_B (\int_B \partial^\sigma G_a(x,\cdot))(\int_B C_0 \partial'^{\sigma'*}(\cdot,x'))$, the sum over the blocks B of side l_a; by [4, (2.35)] at scale a in the form of theorem F.7 ($\mathbf{1}_B = Q_a^* \delta_{\hat{y}}$ in scale-a units; the torus is a periodic box that is a union of the blocks of side l_a), $|\int_B \partial^\sigma G_a(x,\cdot)| = |(\partial^\sigma G_a \mathbf{1}_B)(x)| \leq C l_a^{2-\sigma} e^{-\delta_0 \operatorname{dist}(x,B)/(4l_a)}$, by lemma F.88(c) $\int_B |C_0 \partial'^{\sigma'*}(\cdot,x')| \leq C l_a^{2-\sigma'}$, and $\sum_B e^{-\delta_0 \operatorname{dist}(x,B)/(4l_a)} \leq C$; so the kernel is $\leq C l_a^{-1-\sigma-\sigma'}$, and $e^{-c_*|y|} \geq e^{-c_*}$. □

Proposition F.90 (the order-2 infrared remainders)**.** *For $a, b \in \{1, \dots, K\}$ with $|a-b| \leq 1$ let $\Delta_{a,b}(x) := \delta m_1^2 - \delta m_1^2[G_a, G_b^v](x)$. Then $|\Delta_{a,b}(x)| \leq C_D l_{\min(a,b)}^{-1}$, $C_D = C(N, m, \mu_0, e, \lambda)$, uniformly in x, ε and the torus. In particular $\delta m^{2\,(2)}_{\text{fin}} = \Delta_{K,K} = O(l_K^{-1})$ and $\delta m^{2\,(2)}_{K,k} = \Delta_{k,k} - \Delta_{K,K} = O(l_k^{-1})$.*

Proof. By [3, (1.23)], $\delta m_1^2[G, G^v](x) = -4(N+2)\lambda G(x,x) + e^2 dq^2 G^v(x,x) + b(G, G^v)(x)$ with $b(G, G^v)(x) := -e^2 q^2 \sum_{x'} \varepsilon^3 \sum_\mu (\partial_\mu G \partial'^*_\mu)(x,x') G^v(x,x')$. (1) Tadpoles: $-4(N+2)\lambda R_a(x,x)$ and $e^2 dq^2 R_b^v(x,x)$ are bounded by lemma F.87. (2) $b(C_0, C) - b(G_a, G_b^v) = b(R_a, C) + b(G_a, R_b^v)$ by bilinearity. (3) $b(R_a, C)$: by lemma F.89 $|\partial\partial' R_a| \leq C \max(l_a, |y|)^{-3}$ and $|C(x,x')| \leq C(\varepsilon+|y|)^{-1}$ (lemma F.88(c), no additive constant), so $\sum_{x'} \varepsilon^3 |\cdot| \leq C[l_a^{-3} \int_{|y|\leq l_a} (\varepsilon+|y|)^{-1} dy + \int_{|y|\geq l_a} |y|^{-4} dy] \leq C l_a^{-1}$. (4) $b(G_a, R_b^v)$, zero sum: $\sum_{x'} \partial'^*_{x'} F(x') = 0$ on the torus, so $R_b^v(x,x')$ may be replaced by $R_b^v(x,x') - R_b^v(x,x)$; integrating $|\partial' R_b^v| \leq C \max(l_b, s)^{-2}$ along a lattice path gives $|R_b^v(x,x') - R_b^v(x,x)| \leq C \min(|y| l_b^{-2}, l_b^{-1})$. For $|y| \leq l_b$, $|\partial\partial' G_a| \leq C(\varepsilon + |y|)^{-3}$ gives $C l_b^{-2} \int_{|y|\leq l_b} |y|^{-2} dy \leq C l_b^{-1}$; for $|y| > l_b$, $|\partial\partial' G_a| \leq C(\varepsilon + |y|)^{-3} e^{-\delta_K|y|/(2l_a)}$ gives $C l_b^{-1} \int_{|y|>l_b} |y|^{-3} e^{-\delta_K|y|/(2l_a)} dy \leq C l_b^{-1}(1 + \ln_+(l_a/l_b)) \leq C(1 + \ln L) l_{\min(a,b)}^{-1}$. (Without the zero-sum step the bound would carry $\ln(l_b/\varepsilon)$.) □

The multi-cut expansion and the cut-line mechanism. Let $F[G]$ denote a class sum of [3, (1.23)] (two-point, rooted at x) or [3, (1.24)] (vacuum) with all lines in a kernel pair $G = (G, G^v)$, written as Proposition 1 writes it: the sum over the forest families $\{G_\mathcal{F}\}$ of its graphs, $\mathcal{F}$ a forest of divergent subgraphs, each maximal member contracted to its counterterm vertex, where a subgraph is a set of lines with the vertices it touches and its degree [3, (2.1)] counts only its own lines, and the coefficients of the counterterm vertices are fixed numbers (Convention F.18).

Lemma F.91 (the multi-cut expansion)**.** *Let $G = H + D$ be kernel pairs.* (a) $F[H+D] - F[H] = \sum_\Gamma \sum_{\emptyset \neq \Lambda \subset \text{lines}(\Gamma)} (\Gamma \setminus \Lambda)_{\text{ren}}[H; D \text{ on } \Lambda]$, *where $(\Gamma\setminus\Lambda)_{\text{ren}} := \{G_\mathcal{F} : \mathcal{F}$ a forest of divergent subgraphs of Γ none of which contains a line of $\Lambda\}$, every line not in Λ carries H, and every line of Λ is cut and carries D as a two-variable external field.* (b) $(\Gamma\setminus\Lambda)_{\text{ren}}$ *is exactly the forest family of the cut graph $\Gamma\setminus\Lambda$, and it factorises over the connected components (clusters) of $\Gamma\setminus\Lambda$.* (c) *With $H = G_{j+1}$, $D = -S_j$: $m_j := F[G_{j+1}] - F[G_j] = \sum_\Gamma \sum_{\Lambda\neq\emptyset} (-1)^{|\Lambda|+1} (\Gamma \setminus \Lambda)_{\text{ren}}[G_{j+1}; S_j \text{ on } \Lambda]$; every uncut line carries G_{j+1}.*

Proof. In $G_\mathcal{F}$ the lines inside members of $\mathcal{F}$ are contracted, so the remaining lines are those of $\Gamma \setminus \bigcup \mathcal{F}$; expanding each as $H + D$ gives $\sum_{\Lambda \subset \text{lines}(G_\mathcal{F})} G_\mathcal{F}[H \text{ off } \Lambda, D \text{ on } \Lambda]$, the term $\Lambda = \emptyset$ being $G_\mathcal{F}[H]$; counterterm vertices are vertices, not lines, and are not expanded. For fixed Λ, the forests that occur are those disjoint from Λ. (b) For a subgraph containing no line of Λ the degree counts only its own lines, so it is the same in Γ and in $\Gamma \setminus \Lambda$; hence the divergent subgraphs of $\Gamma \setminus \Lambda$ are exactly those of Γ avoiding Λ. A divergent subgraph is connected, so it lies in one cluster; forests and values factorise over clusters, which are joined only through the D-fields on Λ. (c) is (a) with $G_j = G_{j+1} - S_j$. □

Proposition 1 is multilinear in the coefficients of the [3, (1.7)] vertices: a [3, (1.7)] vertex of order $\operatorname{ord}(v) = \alpha + 2\beta$ whose coefficient is a fixed number c_v (in the units of the step, couplings included) enters the bound [3, (1.33)] through $\sup |c_v|\, t^{-\operatorname{ord}(v)/2}$ in place of the factor $t^{\operatorname{ord}(v)/2}$ that the coupling count $e_k^{d_v} \lambda_k^{d_s}$ assigns to it ([3, p. 421]: "the vertices (1.7) give also a positive power of $L^k \varepsilon$ depending on a term in (1.29)"). So a localised family of order o whose [3, (1.7)] vertices

have coefficients c_v is bounded by $O(1)t^{o/2}\prod_v(\sup|c_v|t^{-\mathrm{ord}(v)/2})e^{-\delta_0 d(\Lambda)}\prod\|h\Phi\|_{1,\alpha_0}$. This is the form in which Proposition 1 is used below and in Steps 3(D_1)–3(D_2).

Lemma F.92 (the cut-line mechanism)**.** *Fix $0 \le j < K$ and a class F of graphs of order o of* [3, (1.23)] *or* [3, (1.24)]*, and let $m_j = F[G_{j+1}] - F[G_j]$.* (a) *Each term of lemma F.91(c) is a product of clusters, renormalised families at step $j+1$ with all lines G_{j+1}, joined through the cut lines, which are two-variable external fields: in scale-$(j+1)$ units $|\partial^\sigma\partial'^{\sigma'}S_j(x,x')| \le C_K L^{1+\sigma+\sigma'}e^{-\delta_K L|x-x'|}$, and the two-variable norm $\|\cdot\|_{1,\alpha_0}$ of hS_jh' for localisations at boxes at distance r is $\le C_K L^{2+\alpha_0}e^{-\delta_K Lr}$ (lemma F.88(b); for $j=0$ its L-dependent constant).* (b) *The only* [3, (1.7)] *vertices in a two-point class of order ≤ 4 or a vacuum class are δm_1^2 and $\delta m_{(4)}^2$ vertices; in a cluster with lines G_{j+1} the non-renormalising part of a δm_1^2 vertex is $\Delta_{j+1,j+1} = O(l_{j+1}^{-1})$ (proposition F.90), so in scale-$(j+1)$ units its coefficient is $O(t_{j+1})$: it enters Proposition 1 with its printed power $t^{\mathrm{ord}/2} = t$.* (c) *Proposition 1 per cluster, the clusters joined through the cut fields as in lemma F.25, bounds each localised term by $O(1)C_K^{|\Lambda|}L^{c|\Lambda|}t_{j+1}^{o/2-\sum_v(\mathrm{ord}(v)/2-1)^+}e^{-\delta''d(\{\square\})}$, the sum over the counterterm vertices v of the class (a $\delta m_{(4)}^2$ vertex carries $t\,q(t)$, $q(t) = (1+\ln t^{-1})^5$, i.e. $t^{1-\beta}$ for every $\beta > 0$, not t^2: theorem F.99).* (d) *Summing over localisations (lemma F.26), with the root cube fixed: for a two-point class without $\delta m_{(4)}^2$ vertices, $|m_j(x)| \le Cl_j^{o/2-2}$; for a vacuum class, $|m_j| \le Ct_j^{o/2}|T^{(j)}|$; with the exponent of* (c)*, times $q(t_{j+1})$ per $\delta m_{(4)}^2$ vertex, when $\delta m_{(4)}^2$ vertices occur. The constants are free of ε, ε_0, j, K and the torus.*

Proof. (a) is lemma F.91(c) and the rescaling of lemma F.88(b) to scale $j+1$, where S_j has side $l_j/l_{j+1} = 1/L$. (b) In a two-point class of order ≤ 4 the [3, (1.7)] vertices have order ≤ 2 (a $\delta m_{(4)}^2$ vertex alone is the whole order); in a cluster with lines G_{j+1}, every δm_1^2 vertex is paired with the order-2 two-point subgraphs of the same class with lines G_{j+1}, and the pair sums to $\delta m_1^2 - \delta m_1^2[G_{j+1}, G_{j+1}^v] = \Delta_{j+1,j+1}$ at that site, plus renormalised two-point insertions, which Proposition 1 bounds. (c) Proposition 1 with the external fields of (a), in the multilinear form above; the cut fields tie the clusters, and the tree of the whole term has length at most the sum of the cluster trees and the cut-line distances. (d) In scale-$(j+1)$ units the two-point coefficient is $l_{j+1}^2 m_j(x)$, bounded by $Ct_{j+1}^{o/2}$ after the sum over localisations with the root fixed; for a vacuum class the root ranges over the $|T^{(j+1)}|$ boxes. The factor $t_{j+1}^{(\cdot)} = L^{(\cdot)}t_j^{(\cdot)}$ changes constants only; the number of terms is $\le 2^I$, I the number of lines. □

Theorem F.93 ($\delta m_{K,k}^2 = O((L^k\varepsilon)^{-1})$, [3, (1.28)])**.** *For $0 \le k < K$ and every x:* (a) $\delta m_{K,k}^2 = \sum_{j=k}^{K-1} m_j$ *with $m_j := \delta m_{j+1}^2 - \delta m_j^2$, the graphs of* [3, (1.23)] *with all lines in $\{G_j, S_j\}$ and at least one S_j line.* (b) *Order 2: for $j \ge 1$, $m_j^{(2)} = \Delta_{j,j} - \Delta_{j+1,j+1}$, so $|m_j^{(2)}| \le 2C_D l_j^{-1}$; for $j = 0$, $m_0^{(2)} = \delta m_1^2 - \Delta_{1,1}$ and $|\delta m_1^2| \le c_m l_0^{-1}$ (theorem F.85).* (c) *Order 4: $|m_j^{(4)}(x)| \le C_4$ (lemma F.92 with $o = 4$, $l_j^{o/2-2} = 1$; the only* [3, (1.7)] *vertices are δm_1^2 vertices).* (d) *Hence $|\delta m_{K,k}^2(x)| \le \sum_{j\ge k}(2C_D l_j^{-1} + C_4) \le Cl_k^{-1} + C_4(K-k) \le C'l_k^{-1}$, since $K - k \le \log_L(1/l_k) \le l_k^{-1}$ ($l_K \le \varepsilon_0 \le 1$); so $\delta m_{K,k}^2(x)(L^k\varepsilon)^2 = O(L^k\varepsilon)$, exactly as* [3, p. 418] *asserts, with a constant free of ε, ε_0, k, K and the torus.* (e) *$\delta m^{2\,(2)}_{\mathrm{fin}} = \Delta_{K,K} = O(l_K^{-1})$. The order-$(0,2)$ part of $\delta m_{K,k}^2$ obeys $|\cdot| \le C\lambda^2(1+\ln l_k^{-1})$ without Proposition 1.*

Proof. (a) is telescoping under Convention F.18; (b) and (e) are proposition F.90; (c) is lemma F.92(d); (d) is the sum. For the last clause: the order-λ^2 graphs are the sunset, the tadpole on a tadpole and the tadpole with a $\delta m_{(0,1)}^2$ vertex, the last two forming the class "λ-tadpole whose loop line carries the blob $4(N+2)\lambda(C_0(0) - G(x',x'))$"; the sunset part is $16(2N+4)\lambda^2\sum_{x'}\varepsilon^3[G_K^3 - G_k^3](x,x')$, bounded by lemma F.88 through $|G_{K,k}(x,x')| \le \sum_{j=k}^{K-1} Cl_j^{-1}e^{-\delta_K|y|/l_j}$ and $|G_K|, |G_k| \le C(\varepsilon+|y|)^{-1}$ by $\sum_{j=k}^{K-1} Cl_j^{-1}\int(\varepsilon+|y|)^{-2}e^{-\delta_K|y|/l_j}dy \le C(K-k)$; the tadpole class is $C_N\lambda^2\sum_{x'}\varepsilon^3[(G_K^2 -$

$G_k^2)R_K - G_k^2 G_{K,k}(x',x')]$, with $|R_K| \le c_R l_K^{-1}$, $|G_{K,k}(x',x')| \le c_R l_k^{-1}$ (lemma F.87), bounded by $C l_K^{-1} \sum_{j<K} l_j^{-1} l_j^2 + C l_k^{-1} l_k \le C$. □

The finite part $\delta m^2_{\mathrm{fin}}$. By lemma F.91(a) with $H = (G_K, G_K^v)$ and $D = (R_K, R_K^v)$, $\delta m^2_{\mathrm{fin}} = \delta m^2[C_0, C] - \delta m^2[G_K, G_K^v]$ is a finite sum of terms in which every cluster is a renormalised family at step K with lines G_K and every cut line is an R_K line. The order-2 part is theorem F.93(e). The order-4 part is bounded in theorem F.98 below, up to a polylogarithm; the proof is Proposition 1 at step K per cluster, with the clusters glued along the R_K lines by an infrared power count.

Lemma F.94 (the remainder in scale-K units)**.** *Let $\alpha_0 \in (0,1)$ and $t := t_K = l_K$.* (a) *For $\sigma, \sigma' \in \{0,1\}$, $|\partial^\sigma \partial'^{\sigma'} R_K(x,x')| \le C_R \max(l_K, |y|)^{-1-\sigma-\sigma'} e^{-c_*|y|}$ (lemma F.89), and the Hölder quotients of exponent α_0 (in units of l_K, over separations $\le 2l_K$) of $\partial^\sigma \partial'^{\sigma'} R_K$, $\sigma + \sigma' \le 2$, in either variable obey the same bound with $\max(l_K, |y|)^{-1-\sigma-\sigma'}$ (so the mixed quotient, $\sigma = \sigma' = 1$, carries $\max(l_K, |y|)^{-3} e^{-c_*|y|}$).* (b) *In scale-$K$ units, $\hat R := l_K R_K(l_K \cdot, l_K \cdot)$: for localisations h, h' at boxes of side ≤ 2 at distance r, $\|(h \otimes h')\partial^\sigma \partial'^{\sigma'} \hat R\|_{0,\alpha_0} \le C_R \max(1,r)^{-1-\sigma-\sigma'} e^{-c_* t r}$, and the two-variable norm $\|(h \otimes h')\hat R\|_{1,\alpha_0}$ is $\le C_R \max(1,r)^{-1} e^{-c_* t r}$.* (c) *The same for R_K^v. The constants are free of ε, K and the torus.*

Proof. (a) The sup and first-difference bounds are lemma F.89. For the Hölder quotients use $R_K = G_K(a_K l_K^{-2} P_K) C_0$ [3, (1.25)]: with the block average, $R_K(x,x') = a_K l_K^{-5} \sum_B (G_K \mathbf{1}_B)(x) u_B(x')$, $u_B(x') := \sum_{z \in B} \varepsilon^3 C_0(z,x')$, the sum over the blocks of side l_K. Since $G_K \mathbf{1} = (m^2 + a_K l_K^{-2})^{-1} \mathbf{1}$ on the torus, $\sum_B \partial_x (G_K \mathbf{1}_B)(x) = 0$ and $\partial_x R_K(x,x') = a_K l_K^{-5} \sum_B \partial_x (G_K \mathbf{1}_B)(x)[u_B(x') - u_{B_x}(x')]$, B_x the block of x. By [4, Lemma 2.4, (2.35)–(2.36)] at scale K in the form of theorem F.7 ($\mathbf{1}_B = Q_K^* \delta_{\hat y}$ in scale-K units; the torus is a periodic box that is a union of the blocks of side l_K, so no depth condition arises and the constants are free of the torus) rescaled to ε-units, $|G_K \mathbf{1}_B|(x) \le c_0 l_K^2 e^{-\delta_0 \operatorname{dist}(x,B)/l_K}$, $|\partial_x G_K \mathbf{1}_B| \le c_0 l_K e^{-\delta_0 \operatorname{dist}(x,B)/l_K}$, and the Hölder quotient of $\partial_x G_K \mathbf{1}_B$ (in units of l_K) obeys the same bound with $c_1(\alpha_0) e^{2\delta_0}$ in place of c_0, $c_1(\alpha_0)$ the constant of [4, (2.36)] (which decays in $\operatorname{dist}(\{\hat x, \hat x'\}, \hat y) \ge \operatorname{dist}(\hat x, B) - 2$ for separations $\le 2l_K$). By lemma F.88(c) along a path from B_x to B, for $|y| \ge 4l_K + 2\operatorname{dist}(B, B_x)$: $|u_B(x') - u_{B_x}(x')| \le C l_K^3 (\operatorname{dist}(B, B_x) + l_K) |y|^{-2} e^{-c_*|y|}$, each difference in x' adding $|y|^{-1}$; for the remaining B ($\operatorname{dist}(B, B_x) \ge |y|/4 - l_K$) bound the two terms separately by $|u_B| \le C l_K^3 \max(l_K, \operatorname{dist}(B, x'))^{-1}$. The sum over B converges by $e^{-\delta_0 \operatorname{dist}/l_K}$, which absorbs the polynomial factors and makes the remaining-B part $\le C e^{-\delta_0 |y|/(8 l_K)}$. This gives (a) for the Hölder quotient of $\partial_x R_K$ at $|y| \ge 4l_K$; for $|y| < 4l_K$ the same sum with $|u_B| \le C l_K^2$ gives $C l_K^{-2-\sigma'}$. The x'-variable is symmetric ($R_K = C_0 a_K l_K^{-2} P_K G_K$). The mixed quotient (the Hölder quotient of $\partial_x \partial_{x'} R_K$ in either variable) is the same sum with one more difference: lemma F.88(c) at $\sigma + \sigma' \le 3$ for the differences of u_B, and [4, (2.36)] (in the form of theorem F.7) for $\partial_x G_K \mathbf{1}_B$. (b) is (a) rescaled: a two-point kernel scales by l_K, each difference by l_K, and $e^{-c_*|y|} = e^{-c_* t r}$; the norm $\|\cdot\|_{1,\alpha_0}$ is the maximum of the sup, the first differences and their Hölder quotients. □

Lemma F.95 (derivative credit)**.** *Let a localised family at step K (zero background, torus) have an external leg l at a vertex v of type* [3, (1.8)]*, with the difference of v acting on l, and let f_l be the external field on l. Then* [3, (1.33)] *holds with $\|h f_l\|_{1,\alpha_0}$ replaced by $C_{\mathrm{DC}} \|\hat h\, \partial f_l\|_{0,\alpha_0}$, $\hat h$ a localisation equal to 1 on the 1-neighbourhood of $\operatorname{supp} h$, C_{DC} depending on (d, α_0) and the localisation family only; the same in each variable of a two-variable external field.*

Proof. Vertices are localised by weights on their summation variables (unit-cube indicators for [3, (1.6)–(1.7)], smooth partitions of unity on the vector legs for the others, [3, p. 420]), so f_l enters only through $(\partial f_l)(b)$ at the bonds of the localisation of v, weighted by the partition, and the value is unchanged under $f_l \to f_l - f_l(z_0)$, $z_0 \in \operatorname{supp} h$ (at zero background constants have zero difference). Apply [3, (1.33)] to $f_l - f_l(z_0)$: on $\operatorname{supp} h$, $|f_l(z) - f_l(z_0)| \le 3d \sup |\partial f_l|$ along an axis path, and $\|h(f_l - f_l(z_0))\|_{1,\alpha_0} \le C(\sup_{\operatorname{supp} \hat h} |f_l - f_l(z_0)| + \|\hat h\, \partial f_l\|_{0,\alpha_0})$. □

Lemma F.96 (infrared power counting for the gluing)**.** *Let Q be a finite connected multigraph (self-edges allowed) with a root node c_0; each non-root node c has a position $z_c \in \mathbb{Z}^3$, $z_{c_0} := 0$ (or positions on a torus, with the torus distance in the weights, torus balls in the count and the torus triangle inequality; every step below holds verbatim). A non-self edge e between c, c' has weight $w_e(z_c - z_{c'})$ with $0 \le w_e(z) \le \max(1,|z|)^{-a_e} e^{-\mu|z|}$, $a_e \ge 0$, $0 < \mu \le \frac{1}{2}$; self-edges have weight ≤ 1. For a partition P of the nodes into groups each inducing a connected subgraph put $\mathrm{IR}(P) := 3(|P|-1) - \sum_{e \text{ between groups}} a_e$ and $D := \max_P \mathrm{IR}(P) \ge 0$. Then $\sum_{(z_c)} \prod_e w_e \le C_Q (1+\log_2 \mu^{-1})^{|E|} \mu^{-D}$, C_Q depending on Q and (a_e) only.*

Proof. For a non-self edge put $j_e := 0$ if $|z_e| \le 1$ and $j_e := j$ if $2^{j-1} < |z_e| \le 2^j$; $J := \lceil \log_2 \mu^{-1} \rceil$, $\hat{j}_e := \min(j_e, J)$. Kruskal's rule with the edges in non-decreasing j_e gives a spanning tree τ; the configuration is fixed by the tree-edge vectors, so the number of configurations with a given scale vector is $\le \prod_{e\in\tau} C 2^{3j_e}$. Weights: $w_e \le 2^{a_e} 2^{-a_e j_e} e^{-\mu 2^{j_e-1}[j_e \ge 1]}$. For $e \in \tau$ with $j_e > J$: $2^{(3-a_e)j_e} e^{-\mu 2^{j_e-1}} \le 2^{(3-a_e)\hat{j}_e} g(j_e - J)$ with $g(n) := 2^{3n}\exp(-2^{n-2})$ summable; for $e \notin \tau$, $2^{-a_e j_e} \le 2^{-a_e \hat{j}_e}$. Telescoping: $\sum_{e\in\tau} 3\hat{j}_e - \sum_e a_e \hat{j}_e = \sum_{i=1}^{J} [3\#\{e \in \tau : j_e \ge i\} - \sum_{e: j_e \ge i} a_e]$; for level i let P_i be the partition into the components of the edges with $j_e < i$; by Kruskal's rule $\#\{e \in \tau : j_e \ge i\} = |P_i| - 1$, every edge between groups of P_i has $j_e \ge i$, and the other edges with $j_e \ge i$ only lower the bracket; so the bracket is $\le \mathrm{IR}(P_i) \le D$ and the total is $\le JD$, i.e. $2^{JD} \le (2/\mu)^D$. A rejected edge e closes a cycle of tree edges f with $j_f \le j_e$ (Kruskal's rule), and the triangle inequality along that path gives $2^{j_e - 1} < |z_e| \le |Q| 2^{\max_f j_f}$, so $\max_f j_f \le j_e \le \max_f j_f + \log_2 |Q| + 1$; hence each tree scale vector extends to at most $(\log_2 |Q| + 2)^{|E|-|\tau|}$ full scale vectors, there are $\le (J+1)^{|\tau|}$ truncated tree scale vectors, and the excess sums converge; the stated exponent $|E| \ge |\tau|$ is an upper bound. □

Lemma F.97 (degrees)**.** *For a connected subgraph g of a graph of Step 3(E) (vertex set and all lines among them) let $E(g)$ be the number of its legs not on its lines, $x(g)$ the number of its V^8 vertices whose difference acts on such a leg, and $\omega(g) := 3\mathcal{L}(g) - 2\sum_{l\in g}(1 - d_l/2) - \sum_{v\in g} s_v$ with $s(V_\delta) := -1$.* (a) *$\omega(g) = 3 - E(g)/2 - \mathrm{ord}(g)/2 - x(g)$, an integer.* (b) *(Gluing identity.) Let Γ be a graph of $\delta m^2_{(4)}$ at zero external momentum, rooted at x, Λ a set of its lines, the clusters the components of $\Gamma \setminus \Lambda$; give an R_K line e the exponent $a_e := 1 + d_e$, $d_e \in \{0,1,2\}$ the number of* [3, (1.8)] *differences acting on its ends. For a partition P of the clusters into groups connected through Λ, with $g_1, \dots, g_n$ the subgraphs induced by the groups, $\mathrm{IR}(P) = \sum_i \omega(g_i) - \omega(\Gamma)$.* (c) *(Subgraph degrees.) If Γ is a graph of $\delta m^2_{(4)}$ (two-point, order 4, one-particle irreducible, no difference on its external legs), then $\omega(\Gamma) = 0$, every connected subgraph $g \ne \Gamma$ has $\omega(g) \le 1$, with $\omega(g) = 1$ only if g is blob-shaped ($\mathrm{ord}(g) = 2$, $E(g) = 2$, $x(g) = 0$) and $\omega(g) \le 0$ otherwise, and no two vertex-disjoint connected subgraphs of Γ are both blob-shaped.*

Proof. (a) With $\mathcal{L} = I - V + 1$ and $2I = \sum_v h_v - E$: $\omega = \sum_v (h_v/2 - 3 - s_v + d_{v,g}) - E/2 + 3$, $d_{v,g} \in \{0,1\}$ indicating a V^8 difference acting on a line of g; the bracket is $-\mathrm{ord}(v)/2$ for V_λ, V^{10}_n, V_δ, and $-n/2 - (1 - d_{v,g})$ for V^8_n (lemma F.81(d)). Parity: $h_v \equiv \mathrm{ord}(v) \pmod 2$ for every type, so $E(g) \equiv \mathrm{ord}(g)$ and $\omega(g) \in \mathbb{Z}$. (b) Lines between groups are R_K lines; $\mathcal{L}(\Gamma) = \sum_i \mathcal{L}(g_i) + (E_{\mathrm{inter}} - n + 1)$, so $\omega(\Gamma) = \sum_i \omega(g_i) + 3(E_{\mathrm{inter}} - n + 1) - \sum_{\mathrm{inter}}(2 - d_e) = \sum_i \omega(g_i) - [3(n-1) - \sum_{\mathrm{inter}}(1 + d_e)]$. (c) $\omega(\Gamma) = 3 - 1 - 2 - 0 = 0$. For a proper connected g: $E(g) \ge 2$ ($E = 0$ forces $g = \Gamma$; $E = 1$ makes the single leg a bridge or disconnects Γ). The only order-1 vertex is V^8_1, whose single-vertex subgraph has $E = 3$ and $x = 1$. So $E + \mathrm{ord} + 2x \ge 4$ with equality only for $(E, \mathrm{ord}, x) = (2, 2, 0)$: $\omega = 3 - (E + \mathrm{ord} + 2x)/2 \le 1$, with equality exactly for blob-shaped g and $\omega \le 0$ otherwise (integrality). Two vertex-disjoint blob-shaped g_1, g_2 carry all of the order, hence all vertices, and each has exactly two external legs; the two external legs of Γ are among these four: if both sit on g_1, the two legs of g_2 must join g_1, which has none left; if one on each, the remaining two form a bridge, against one-particle irreducibility; if neither, Γ has no external leg. □

Theorem F.98 (the order-4 finite part)**.** *For every x and every torus aligned with the blocks with sides ≥ 1, $|\delta m^{2(4)}_{\mathrm{fin}}(x)| \le C_{\mathrm{DMf}} l_K^{-1} (1 + \ln l_K^{-1})^5$, with C_{DMf} free of ε, ε_0, K and the torus.*

Proof. (1) By lemma F.91(a), $\delta m^{2\,(4)}_{\rm fin}(x) = \sum_\Gamma \sum_{\Lambda\neq\emptyset}(\Gamma\setminus\Lambda)_{\rm ren}[G_K; R_K \text{ on } \Lambda](x)$, a finite sum (at most 4 vertices and 5 lines per Γ). (2) Each cluster is a renormalised family at step K with lines G_K; its [3, (1.7)] vertices are δm^2_1 vertices (no order-4 counterterm occurs in $\delta m^2_{(4)}$), whose non-renormalising part is $\Delta_{K,K} = O(l_K^{-1})$ (proposition F.90), of the printed size; in scale-K units Proposition 1 gives per localised cluster family $O(1)t^{o_c/2}e^{-\delta' d(\{\square\})}$ times the norms of its external fields, $\sum_c o_c = 4$. (3) The external fields are the ends of the R_K lines; an end at which a [3, (1.8)] difference acts is measured by the differenced kernel (lemma F.95); by lemma F.94(b) a line e between boxes at distance r contributes $\le C_R \max(1,r)^{-a_e}e^{-c_* t r}$, $a_e = 1+d_e$, and a line with both ends in one cluster $\le C_R$; clusters joined by R_K lines are joined as in lemma F.25. (4) For each cluster sum its boxes around a root box with half of its tree decay (lemma F.26); the distance of two attached boxes differs from that of the roots by at most $d_c + d_{c'}$, and $(1+d_c+d_{c'})^{a_e}e^{c_* t(d_c+d_{c'})}$ is absorbed by the other half ($c_* t \le \delta'/8$). (5) Lemma F.96 with $\mu := c_* t$ bounds the sum over the cluster roots by $C(1+\log t^{-1})^{|\Lambda|}t^{-D}$, and by lemma F.97(b),(c), $\mathrm{IR}(P) = \sum_i \omega(g_i) - \omega(\Gamma) \le 1$ for every partition (at most one blob-shaped group, every other group ≤ 0), so $D \le 1$. (6) In scale-K units the coefficient is $\delta m^2(x) l_K^2$, so $|\delta m^{2\,(4)}_{\rm fin}(x)| l_K^2 \le C t^2 t^{-1}(1+\log t^{-1})^5$ with $t = l_K$. □

The logarithm is the count of scale vectors in lemma F.96; the statement without it is not claimed, and nothing below needs it.

The non-renormalising vertex at every step. For $0 \le j \le K$ put $c_j(x) := l_j^2(\delta m^2_{\rm fin}(x) + \delta m^2_{K,j}(x))$, $\delta m^2_{K,K} := 0$: the coefficient, in scale-j units, of the part of the step-j vertex [3, (1.7), (1.29)] that is not absorbed into renormalised pairs ($c_0 = \varepsilon^2 \delta m^2$, since $\delta m^2_0 = 0$); the renormalising piece $\delta m^2_j(e_j, \lambda_j, \Omega_1, x)$ of [3, (1.29)] is of size l_j^2/ε-type by theorem F.85 rescaled and is cancelled inside the renormalised classes. Let $q(t) := (1+\ln t^{-1})^5$ and $Q_\beta := \sup_{0<t\le 1} t^\beta q(t) = (5/\beta)^5 e^{\beta-5}$ for $\beta \in (0,5]$.

Theorem F.99 (the vertex tolerance)**.** *Let $C_T := 2\max(C_D, c_m) + 2C_4/\ln L + C_{\rm DMf}$. For every $0 \le j \le K$ and x:* (a) $|c_j(x)| \le C_T l_j q(l_j)$*;* (b) *for $\beta \in (0,5]$, $|c_j(x)| \le C_T Q_\beta l_j^{1-\beta}$;* (c) $|c_j(x)| \le C_T Q_1$*;* (d) $|c_j(x)| \le C_T(1+\ln(L/\varepsilon_0))^5 l_j$*;* (e) *only the order-4 finite piece carries q:* $|c_j^{(2)}| \le \max(C_D, c_m) l_j$, $|l_j^2 \delta m^{2\,(4)}_{K,j}| \le (C_4/\ln L) l_j^2 (1+\ln l_j^{-1}) \le C l_j$, $|l_j^2 \delta m^{2\,(4)}_{\rm fin}| \le C_{\rm DMf} l_j q(l_j)$. *$C_T$ is free of ε, ε_0, j, K and the torus.*

Proof. Split $c_j = c_j^{(2)} + c_j^{(4)}$ by order. Order 2: for $j \ge 1$, $\delta m^{2\,(2)}_{\rm fin} + \delta m^{2\,(2)}_{K,j} = \Delta_{K,K} + (\Delta_{j,j} - \Delta_{K,K}) = \Delta_{j,j}$ (theorem F.93(b),(e)) and $|\Delta_{j,j}| \le C_D l_j^{-1}$; for $j = 0$ it is δm^2_1 with $|\delta m^2_1| \le c_m l_0^{-1}$; so $|c_j^{(2)}| \le \max(C_D, c_m) l_j$. Order 4, (K,j) piece: $|\delta m^{2\,(4)}_{K,j}| \le C_4(K-j)$ (theorem F.93(c)) and $K - j \le \log_L(1/l_j) \le (1+\ln l_j^{-1})/\ln L$, so $l_j^2 |\delta m^{2\,(4)}_{K,j}| \le (C_4/\ln L) l_j^2 (1+\ln l_j^{-1}) \le (C_4/\ln L) l_j$, as $t(1+\ln t^{-1}) \le 1$ on $(0,1]$. Order 4, finite piece: $f(t) := t^{-1}q(t)$ is decreasing on $(0,1]$ ($f'/f = -t^{-1}(1+5/(1+\ln t^{-1})) < 0$), and $l_j \le l_K$, so theorem F.98 gives $l_j^2 |\delta m^{2\,(4)}_{\rm fin}| \le C_{\rm DMf} l_j^2 f(l_K) \le C_{\rm DMf} l_j^2 f(l_j) = C_{\rm DMf} l_j q(l_j)$. Sum, with $q \ge 1$: (a). (b), (c) are $q(t) \le Q_\beta t^{-\beta}$, $tq(t) \le Q_1$. (d): $l_j^2 f(l_K) = l_j (l_j/l_K)(1+\ln l_K^{-1})^5 \le l_j (1+\ln(L/\varepsilon_0))^5$, since $l_j \le l_K$ and $l_K > \varepsilon_0/L$; the other parts are $\le C l_j$ already. □

Corollary F.100 ($\delta m^2_{\rm fin}$ at fixed ε_0)**.** $|\delta m^2_{\rm fin}(x)| \le C_T L \varepsilon_0^{-1}(1+\ln(L/\varepsilon_0))^5$*, uniformly in ε, x and the torus: Bałaban's "convergent as $\varepsilon \to 0$" [3, (1.26)], with a constant depending on ε_0.*

Proof. Theorem F.99(d) at $j = K$ ($c_K = l_K^2 \delta m^2_{\rm fin}$) and $l_K > \varepsilon_0/L$. □

Lemma F.101 (family tolerance)**.** *Let F be a localised renormalised family at a step $j \le K$, in any application of Proposition 1 of Steps 3(B)–3(D), and let $n_7(F) \le \bar n/2$ be the number of its non-renormalising [3, (1.7)] vertices. If a bound B_F on F was obtained with $|c_j| \le C l_j$ at each such vertex, then theorem F.99 gives the bound $(C_T/C)^{n_7} q(l_j)^{n_7} B_F$, with constants free of ε_0. Consequently:* (i) *a bound $O(1)t^\kappa p(t)^N e^{-\delta d}$ becomes $O(1)t^\kappa p(t)^N q(t)^{\bar n/2} e^{-\delta d}$, the same*

polylogarithmic class ($p(t)^N q(t)^n \le C_{N,n,\beta} t^{-\beta}$); (ii) *a conclusion "$C_\beta t^{\kappa-\beta}$ for every $\beta > 0$" holds with the same κ and a new ε_0-free C_β;* (iii) *a bound that used only $|c_j| \le C$ holds with $C_T Q_1$;* (iv) *a geometric step sum $\sum_{i=k}^{K-1} t_i^a q(t_i)^n$ is $\le (1-L^a)^{-1} t_k^a q(t_k)^n$ for $a < 0$ and $\le (1-L^{-a/2})^{-1} Q^n_{a/(2n)}$ for $a > 0$, $n \ge 1$. Every consumer of the counterterm sizes in Steps 3(B)–3(D) uses the* [3, (1.7)] *vertex only as a positive power of t (*[3, p. 421]*; the exponents $\kappa_0 > 0$, $\frac12 - \beta$, $\frac14 - \beta$ of* [1, (3.58)], [2, (2.57), (2.116)–(2.117)]*), so each tolerates the form* (a) *of theorem F.99; where such a bound was written with an exact factor t it holds with $tq(t)$.*

Proof. Proposition 1 is multilinear in the coefficients c_j (Step 3(E), before lemma F.92); replace $\sup|c_j| \le C l_j$ by $C_T l_j q(l_j)$ at each of the n_7 vertices; $n_7 \le \bar n/2$ since each [3, (1.7)] vertex has order ≥ 2 and the family has order $\le \bar n$. (i), (ii): $p(t) = b_0(1 + \ln t^{-1})^{\bar p}$, so $p^N q^n$ is a power of $1 + \ln t^{-1}$, bounded by $C_\beta t^{-\beta}$. (iii) is theorem F.99(c). (iv): for $a < 0$, $q(t_i) \le q(t_k)$ for $i \ge k$ and the series is geometric; for $a > 0$, $t^{a/2} q(t)^n \le Q^n_{a/(2n)}$ and $\sum_i t_i^{a/2} \le (1 - L^{-a/2})^{-1}$. □

The vacuum counterterm pieces. The vacuum counterterm E_1 is split by [3, (1.26), (1.28), (1.30)] as $E_1 = E_{\text{fin}} + E_{K,k} + E_k$, $E_K = E[G_K, G_K^v]$, $E_k = E[G_k, G_k^v]$ (class sums [3, (1.24)] with the counterterm coefficients kept as numbers, Convention F.18), $E_{K,k} := E_K - E_k$ and $E_{\text{fin}} := E_1 - E_K$. E_{fin} is a global constant, factored out of the chain by lemma F.3 and bounded by theorem F.4; the pieces $E_{K,k}$ are what the step-k action carries, and the following statements bound them with constants free of ε_0.

Lemma F.102 (the c-number of the step-k polynomial)**.** *At zero background on the torus, the field-independent part of the step-k polynomial* [3, (1.4)–(1.5)] *with vertex domain Ω_1 is $c_k(\Omega_1) = -(E_{\text{fin}} + E_{K,k}) + a_k(\Omega_1)$, where $a_k(\Omega_1)$ is the sum of the connected vacuum families containing at least one block-spin pair vertex* [3, (1.12)–(1.14)]*. On the chain run with $E_0 + E_K$ (lemma F.3) it is $-E_{K,k} + a_k(\Omega_1)$.*

Proof. By [3, (1.4)] the vacuum graphs of $-\log$ of the fluctuation integral at zero background without pair vertices have the vertices of the action localised in Ω_1 ([3, p. 419]: the vertices [3, (1.6)–(1.7)] by $\sum_{x\in\Omega_1}$, the others by g_k, and "we perform this localization for the terms of δm_k^2, E_k also" [3, p. 420]), the full δm^2 of [3, (1.29)] in the vertex [3, (1.7)], and lines $G_k(T, 0)$; order by order these are the graphs of $E_k(e' e_k g_k, \lambda' \lambda_k, \Omega_1)$ ([3, p. 418]: "the same form, only the propagators are replaced by $G_k^\varepsilon(0)$, G_k^ε"), while E_{fin} and $E_{K,k}$ carry no Ω_1 [3, (1.30), p. 419]. So the Ω_1-dependent vacuum part cancels against $-E_k(\Omega_1)$ in [3, (1.5)], and the pair-vertex families remain. □

Theorem F.103 (the vacuum pieces at step k)**.** *For $0 \le k < K$, on every torus aligned with the blocks with sides ≥ 1:* (a) *the pair-vertex vacuum families $a_k(\Omega_1)$ are $O(t_k)$ per localisation cube;* (b) *$|E_{K,k}| \le C_{\text{VK}} t_k |T^{(k)}| = C_{\text{VK}} \ell_T^3 l_k^{-2}$ for the classes without $\delta m^2_{(4)}$ vertex, and $\le C_\beta t_k^{1-\beta} |T^{(k)}|$ for every $\beta > 0$ for the classes with one;* (c) *at order λ, without Proposition 1: $E^{(0,1)}_{K,k} = \lambda N(N+2) \sum_x \varepsilon^3 G_{K,k}(x,x)[R_K(x,x) + R_k(x,x)]$, so $|E^{(0,1)}_{K,k}| \le 2\lambda N(N+2) c_R^2 \ell_T^3 l_k^{-2}$, and $0 \le E^{(0,1)}_{\text{fin}} = \lambda N(N+2) \sum_x \varepsilon^3 R_K(x,x)^2 \le \lambda N(N+2) c_R^2 \ell_T^3 l_K^{-2}$. The constants are free of ε, ε_0, k, K and the torus.*

Proof. (a) Proposition 1 with no external fields gives $O(1) e_k^{d_v} \lambda_k^{d_s} e^{-\delta_0 d}$ per localised family; a pair vertex contains a [3, (1.14)] member, of order ≥ 1 in e, and all A'-legs are contracted in pairs, so $d_v \ge 2$ and the family carries t_k (as in the proof of lemma F.27); sum over localisations with the root fixed (lemma F.26). (b) $E_{K,k} = \sum_{j=k}^{K-1} \epsilon_j$ with $\epsilon_j := E[G_{j+1}] - E[G_j]$, and lemma F.92(d) for vacuum classes of order o gives $|\epsilon_j^{(o)}| \le C t_j^{o/2} |T^{(j)}| = C \ell_T^3 l_j^{o/2-3}$; summed over $j \ge k$: $o = 2$ gives $C\ell_T^3 l_k^{-2}$, $o = 4$ gives $C\ell_T^3 l_k^{-1}$, $o = 6$ gives $C\ell_T^3 (K - k)$, $o \ge 8$ gives $C\ell_T^3$, all $\le C\ell_T^3 l_k^{-2}$ since $l_k^2 (K - k) \le 1$. The [3, (1.7)] vertices of the vacuum families have orders 2 or 4; the non-renormalising part of a δm_1^2 vertex at step $j+1$ is $\Delta_{j+1,j+1} = O(l_{j+1}^{-1})$ (proposition F.90), of the printed size, and that of a $\delta m^2_{(4)}$ vertex is $\sum_{i>j} m_i^{(4)} + \delta m^{2\,(4)}_{\text{fin}}$, whose coefficient in scale-$(j+1)$ units is $\le C_T l_{j+1} q(l_{j+1})$ (theorem F.99(e)), so such a vertex carries $t_j^{1-\beta}$ in place of t_j^2

and the classes containing one obey $C_\beta t_j^{o/2-1-\beta}|T^{(j)}|$; summed over $j \geq k$ with $o \geq 4$ this is $\leq C_\beta \ell_T^3 l_k^{-2-\beta} = C_\beta t_k^{-\beta} t_k |T^{(k)}|$. (c) $E^{(0,1)}[G] = -\lambda N(N+2)\sum_x \varepsilon^3 G(x,x)[G(x,x) - 2C_0(0)]$ (the figure-eight and the cycle with the $\delta m^2_{(0,1)} = -4(N+2)C_0(0)$ vertex); subtracting at G_K and G_k, resp. at C_0 and G_K, gives the two displayed identities, and lemma F.87 the bounds. □

Corollary F.104 ([2, (2.14)] for the constant the chain carries)**.** *On the chain run with $E_0 + E_K$ the vacuum constant at step 0 is $E_K = E_{K,0}$ ($G_0 = 0$), and $|E_K| \leq C\varepsilon|T_\varepsilon|$ for the classes without $\delta m^2_{(4)}$ vertex, $\leq C_\beta \varepsilon^{1-\beta}|T_\varepsilon|$ for the others, with constants free of ε, ε_0 and the torus. Read as a density, $E_K(X) := (E_K/|T_\varepsilon|)|X|$: $E_K = E_K(\Lambda_7) + E'_K(\Lambda_7^c)$ with $|E'_K(\Lambda_7^c)| \leq C_\beta \varepsilon^{1-\beta}|\Lambda_7^c|$, which is* [2, (2.14)] *with $\kappa_0 = 1-\beta$ for every $\beta > 0$ ($\kappa_0 = 1$ for the classes without $\delta m^2_{(4)}$ vertex), for the constant to which the chain applies it.*

Proof. Theorem F.103(b) at $k = 0$, $t_0|T^{(0)}| = \varepsilon|T_\varepsilon|$; lemma F.3 for the booking. □

This bound holds for the class sums only. The vacuum graph made of a scalar ring through three λ-vertices, each with one scalar self-loop, is a term of E_1 of order λ^3 whose value per ε-site is $\geq c_N \lambda^3 m^{-3}$ for small ε (its three insertions are $4(N+2)\lambda C_0(0) \sim \varepsilon^{-1}$ each and $\ell_T^{-3}\sum_p \hat{C}_0(p)^3 \geq c\, m^{-3}$); it cancels within the order-λ^3 coefficient against the rings with $\delta m^2_{(0,1)}$ insertions.

Proposition F.105 ([1, Props. 3.1–3.2] and [2, p. 579])**.** *Given Proposition 1 at $L = 5$ (Steps 3(D_3)–3(D_4)) with the vertex sizes of theorem F.99:* (i) *(*[1, Prop. 3.1]*) at step k of the lower bound, the sum of the R-terms (the remainders of the Taylor expansion in the background,* [3, (1.9), (1.11), (1.15)]*) and the sum of the field-dependent terms at whose vertices the powers of $L^k\varepsilon$ add up to more than d are $O(1)(L^k\varepsilon)^\kappa |T_1^{(k)}|$ with $\kappa > d$; the c-number part of the step-k expansion is $-E_{K,k} + a_k(\Omega_1)$, bounded by theorem F.103, and the total c-number the chain collects is $E_K = O(\varepsilon|T_\varepsilon|)$, which the final bound absorbs.* (ii) *(*[1, Prop. 3.2, (3.57)–(3.58)]*) the coefficients $v^{(k)}_{j;\mu}(B^{(k+1)}, \psi; x, y)$ of $V^{(k)}$ with $n + m \geq 1$ obey $|v^{(k)}| \leq O(1)(L^k\varepsilon)^{\kappa_0} \exp(-\delta_0 d(x_1, \dots, y_m))$ with $\kappa_0 > 0$ independent of k; the coefficient with $n = m = 0$ is a function of the new fields $(B^{(k+1)}, \psi)$, whose field-dependent graphs obey the same bound through their external legs, and whose pure c-number part is that of* (i)*: $O(t_k^{1-\beta})$ per localisation cube.* (iii) *(*[2, p. 579]*) at step k of the upper bound, on $\Omega = B^k(\Lambda_0^{(k)})$ with the background $\tilde{B}$ regular by* [2, (2.98)]*, the terms with R-vertices and the terms of order $> \bar{n}$ are $O((L^k\varepsilon)^\kappa)|\Lambda_2^{(k)}|$, the terms containing $\tilde{B}^{(k)}$ are $O(t_k^\kappa)|\Lambda_2^{(k)}|$ for every κ by corollary F.54, and the field-dependent terms of $Q^{(k)}$ localised in $B^k(\Lambda_7^{(k)c})$ are bounded by proposition F.127, the corresponding terms of $P^{(k)}$ being the subject of remark F.136; the c-number part is as in* (i).

Proof. (i) Every R-term carries $e(L^k\varepsilon)^{\bar{n}+1}$ with $\bar{n}+1$ background legs, and Proposition 1 with the field restrictions of [1, Sect. 3] bounds it; a field-dependent term with more than d powers of $L^k\varepsilon$ at its vertices is bounded by Proposition 1 with that power; the exponent count $\kappa > d$ is arithmetic on the printed orders and needs $\bar{n} \geq 6$ (theorem F.2). The c-number part is lemma F.102 with lemma F.3, bounded by theorem F.103(a),(b); the total is corollary F.104. For a c-number vacuum class of order > 6 no graph-by-graph bound of the printed kind holds (a ring of n step-k tadpole insertions $\lambda_k G_k(x,x) \sim \lambda t_k/\eta$ has size $\sim (\lambda t_k/\eta)^n$ per unit block), which is why the class sums of theorem F.103 are used. (ii) For $n+m \geq 1$ this is Proposition 1's localised bound at fixed external fields, with κ_0 its uniform exponent (lemma F.101 for the [3, (1.7)] vertices); the coefficient with $n = m = 0$ is free of the fluctuation fields but depends on $(B^{(k+1)}, \psi)$, its graphs with external legs fall under Proposition 1, and its pure c-number part is read per localisation cube as the density of $-E_{K,k} + a_k(\Omega_1)$. (iii) is (i)'s argument on $\Omega = B^k(\Lambda_0^{(k)})$: Proposition 1 is stated for a general Ω with a regular background [3, p. 412], and the depth and Hölder clauses of the background are those of [2, (2.94)–(2.98)]; the $\tilde{B}^{(k)}$ terms are corollary F.54; the $Q^{(k)}$ terms are proposition F.127 and the $P^{(k)}$ terms remark F.136. □

Step 3(F): the remaining asserted estimates of [2]. The estimates of [2, Sect. 2] asserted at the conditional integral [2, p. 568], at the quadratic-form step [2, (2.108)–(2.109)], at the collar [2, Prop. 2.1, (2.57)] (its field-dependent part), at the passage from [2, (3.5)] to [2, (3.21)], and the region-entropy estimate [2, (3.42)] are proved in this step, at block factor 5.

The conditional integral: [2, p. 568, (2.46)–(2.47)]. *Setting.* The step from scale $k-1$ to scale k, in scale-$(k-1)$ units, $t := L^{k-1}\varepsilon$; $\Lambda_i := \Lambda_i^{(k-1)}$, $\Omega := B^{k-1}(\Lambda_2)$, Λ' the set of coarse points (the L-blocks of the unit lattice) whose block lies in Λ; φ, A the new fields on coarse points, φ', A' the old fields on unit points; A as an operator argument is the background $A^{(k),\varepsilon}$. The exponential of [2, (2.46)] has ten terms: T_1, T_2 (old-field vector forms on $\Lambda_6^{(k-2)'} \cap \Lambda_5^c$ with $\Delta^{(k-1)}$ and $\Delta^{(k-1)} C_{\Lambda_5}^{(k-1)} \Delta^{(k-1)}$), $T_3 = -a(L^k\varepsilon)^{-2}\langle(\Lambda_6^{(k-2)'} \cap \Lambda_5^c)A', \Delta^{(k-1)} C_{\Lambda_5}^{(k-1)} Q^* A\rangle$, $T_4 = -\frac{1}{2}\langle A, \Delta_{\Lambda_5}^{(k),L^k\varepsilon} A\rangle$, T_5, T_6 (old-field forms on $\Lambda_6^{(k-2)'} \cap \Lambda_3^c$, not touched on p. 568), $T_7 = -\frac{1}{2}\langle(\Lambda_3 \cap \Lambda_5^c)\varphi', \Delta^{(k-1)}(\Omega, A)(\Lambda_3 \cap \Lambda_5^c)\varphi'\rangle$, T_8 the same with $\Delta C_{\Lambda_5}\Delta$, $T_9 = -a(L^k\varepsilon)^{-2}\langle(\Lambda_3 \cap \Lambda_5^c)\varphi', \Delta^{(k-1)} C_{\Lambda_5}^{(k-1)} Q^*(A)\varphi\rangle$, $T_{10} = -\frac{1}{2}\langle\varphi, \Delta_{\Lambda_5}^{(k),L^k\varepsilon}(\Omega, A)\varphi\rangle$. The coarse forms are [2, (2.27)]: $\langle\psi, \Delta_\Lambda^{(k)}\psi\rangle = aL^{d-2}\sum_{y\in\Lambda'}|\psi(y)|^2 - a^2L^{-4}\langle\psi, QC_\Lambda^{(k-1)}Q^*\psi\rangle$. [2, p. 568] asserts: (2.47), the localisation of T_{10} to the three forms on $\Lambda_6'^c$, $\Lambda_6' \cap \Lambda_7'^c$ and Λ_6' with the operators on $B^{k-1}(\Lambda_2 \cap \Lambda_7^c)$ ("Neumann boundary conditions") and on Ω, up to $O((L^{k-1}\varepsilon)^\kappa)|\Lambda_6|$; the same for T_4 "except that Neumann boundary conditions are not introduced" (vector forms on the torus); the restriction of the linear terms T_3, T_9 to $\Lambda_5' \cap \Lambda_6'^c$; and the Neumann cut in T_7–T_9. Readings: $S := \partial B^{k-1}(\Lambda_7)$, $\Omega_1 := B^{k-1}(\Lambda_2 \cap \Lambda_7^c)$, $\Omega_{\rm cut}$ is Ω with the bonds crossing S removed, so $\Omega_{\rm cut} = \Omega_1 \sqcup B^{k-1}(\Lambda_7)$; "Neumann conditions on $\partial B^{k-1}(\Lambda_2 \cap \Lambda_7^c)$" is $\Omega \to \Omega_{\rm cut}$, and the printed $\Delta_{\Lambda_5}(B^{k-1}(\Lambda_2 \cap \Lambda_7^c), A)$ is the Ω_1-block of $\Delta_{\Lambda_5}(\Omega_{\rm cut}, A)$ (lemma F.107(d); read literally the printed operator is undefined, since [2, (2.27)] needs $\Lambda \subset \Omega^{(k-1)}$); the third line of (2.47) reads $B^{k-1}(\Lambda_2)$; the vector forms are the instance $A = 0$, μ_0 for m, componentwise. Inputs: (G) the geometry of [2, (2.7)–(2.8)]: with $d_S(z) := \operatorname{dist}(B^{k-1}(z), S)$, $z \notin \Lambda_6$ gives $d_S(z) \geq r - c_d - 1$, $z \in \Lambda_3 \cap \Lambda_5^c$ gives $d_S(z) \geq 2(r - c_d) - 1$, and $\operatorname{dist}(\Lambda_6, \Lambda_5^c)$, $\operatorname{dist}(\Lambda_6^c, \Lambda_7) \geq r - c_d$, $r = r(t)$; (FR) the field restrictions [2, (2.16)–(2.17)] at scale $k-1$ [2, p. 560]: $|\varphi| \leq M_\varphi := cp(t)\lambda(t)^{-1/4}$, $|A| \leq M_A := cp(t)t^{-1}$ on Λ_5', the old fields the same on $\Lambda_3 \cap \Lambda_5^c$, and the old A' on $\Lambda_6^{(k-2)'} \cap \Lambda_5^c$ by the printed factor $\chi_{k-1,\Lambda_5^{(k-1)c}}$ of [2, (2.45)–(2.46)]; $M := \max(M_\varphi, M_A, 1) \leq Ct^{-2}$; (RG) $A^{(k),\varepsilon}$ regular on Ω ([2, p. 562, (2.98)]) and $e \leq e_0$; (A) the propagator estimates of group (A), i.e. Step 3(A) at the step index $k-1$, under its hypotheses (Ω a union of big blocks, every pin set $\Lambda \subset \Omega^{(k-1)}$ a union of L-blocks (the proofs of Step 3(A) use of (H1) only this), A regular on Ω by (RG), $e_{k-1} \leq e_1$): for every pin set Λ the operator $T_\Lambda(\Omega) := -\Delta_{A,\Omega}^{\eta,N} + m_{k-1}^2 + a_{k-1}P_{k-1}(\Omega \setminus B\Lambda) + a_kL^{-2}P_k(B\Lambda)$ is $\geq \gamma_L$ (lemma F.143, which is [4, (1.8)] on a general region); the Combes–Thomas bound $\|\mathbf{1}_Q T^{-1} f\|, \|\mathbf{1}_Q D_A T^{-1} f\| \leq Ce^{-\theta_0 \operatorname{dist}(Q, \operatorname{supp} f)}\|f\|$ for unit cubes Q and $T = D_A^* D_A + V + \sum_B c_B \Pi_B \geq \gamma$ ($V \geq 0$, Π_B projections on disjoint sets of diameter $\leq \ell_*$, $0 \leq c_B \leq a$), $\theta_0 = \theta_0(\gamma, \ell_*)$ (lemma F.144(i)–(ii), paired with the functions, resp. bond functions, supported in Q; $C := \max\{2\gamma^{-1}, (2\gamma^{-1/2} + 6\sqrt{d}\,\theta_0\gamma^{-1})e^{\theta_0}\}$; this contains the G_k clause of [4, Cor. 2.3]); the decay $|\Delta^{(k-1)}(\Omega, A; z, z')| \leq c_7 e^{-\delta_7|z-z'|}$ (the first bound of proposition F.150, which is [1, (2.27)]; its proof gives the constant $a + 2a^2\gamma_\emptyset^{-1}e^{\theta_0(\gamma_\emptyset,1)}$ and the rate $\theta_0(\gamma_\emptyset, 1)$, both depending on $(d, N, \|q\|, a, L)$ only) and $|C_\Lambda(y, y')| \leq c_2 e^{-\theta_0|y-y'|}$ (lemma F.145(c), which is [4, (1.16)]); the block inversion $X_\Lambda^{-1} = W^* G_{k-1}(\Omega, \Lambda, A)W + a_{k-1}^{-1} - (a_kL^{-2}/a_{k-1}^2)P_\Lambda$, $X_\Lambda := (\Delta^{(k-1)} + aL^{-2}P)|_\Lambda$, $W := Q_{k-1}^*\Lambda(1 - (a_k/a_{k-1})L^{-2}P_\Lambda)$ (lemma F.145(a), which is [4, (3.8)]); and the restriction difference $|C_\Lambda(y, y') - C^{(k-1)}(\Omega, A; y, y')| \leq c_4 \exp[-\frac{\theta_0}{6}(|y - y'| + \operatorname{dist}(y, \Omega^{(k-1)} \setminus \Lambda) + \operatorname{dist}(y', \Omega^{(k-1)} \setminus \Lambda))]$ (lemma F.147, which is [4, (1.18)]). On a torus these hold with the same constants (lemma F.154).

Lemma F.106 (Neumann cut)**.** *Let Ω be a union of big blocks with A regular on Ω, $Y \subset \Omega^{(k-1)}$ a union of large blocks, $S := \partial B^{k-1}(Y)$, $\Omega_{\rm cut}$ as above, $\Lambda \subset \Omega^{(k-1)}$ a pin set, $T := T_\Lambda(\Omega)$ and $T_{\rm cut} := T_\Lambda(\Omega_{\rm cut})$ (the Neumann Laplacian of $\Omega_{\rm cut}$, same pins). Then $T_{\rm cut} \geq \gamma_L$, and for*

$f, f' \in \ell^2(\Omega)$ supported in sets of diameter $\le s$, both at distance $\ge R \ge 4$ from S: $|\langle f, (T_{\rm cut}^{-1} - T^{-1})f'\rangle| \le c_F(s)\|f\|\|f'\| \exp[-\frac{\theta_0}{4}(R + \operatorname{dist}(\operatorname{supp} f, \operatorname{supp} f'))]$, c_F depending on (d, L, a, γ_L, s) only. (For $R < 4$ the bound follows from the Combes–Thomas bound applied to each resolvent.)

Proof. (1) The proof of lemma F.143 drops all bonds between cells, and the cut bonds join cells since Y is a union of large blocks, so $T_{\rm cut} \ge \gamma_L$; lemma F.144 holds for $T_{\rm cut}$ with the bond set of $\Omega_{\rm cut}$ (its proof uses only that each site is b_+ of at most d bonds). Equivalently: $\Omega_{\rm cut} = (\Omega \setminus B^{k-1}(Y)) \sqcup B^{k-1}(Y)$ is a disjoint union of two unions of big blocks, $T_{\rm cut} = T_{\Lambda\setminus Y}(\Omega \setminus B^{k-1}(Y)) \oplus T_{\Lambda\cap Y}(B^{k-1}(Y))$, and the two lemmas apply to each summand. (2) Let $\zeta(x) := \min\{1, \max\{0, \frac{4}{R}\operatorname{dist}(x, S) - 1\}\}$: $\zeta = 0$ where $\operatorname{dist}(\cdot, S) \le R/4$, $\zeta = 1$ where $\operatorname{dist}(\cdot, S) \ge R/2$, $|\zeta(x) - \zeta(x')| \le \frac{4}{R}|x - x'|$; so $\zeta = 1$ on $\operatorname{supp} f \cup \operatorname{supp} f'$ and ζ vanishes at both ends of every cut bond (they lie within $\eta \le 1 \le R/4$ of S). (3) Put $u := T^{-1}f'$; $T(\zeta u) = f' + [T, \zeta]u$. With $T - T_{\rm cut} = D_J^* D_J$, D_J the covariant difference on the cut bonds, $D_J(\zeta u) = 0$, so $T_{\rm cut}(\zeta u) = T(\zeta u)$ and $\zeta u = T_{\rm cut}^{-1}(f' + [T, \zeta]u)$; pairing with f: $\langle f, T_{\rm cut}^{-1} f'\rangle - \langle f, T^{-1} f'\rangle = -\langle f, T_{\rm cut}^{-1}[T, \zeta]T^{-1}f'\rangle$. (4) With ζ acting on bond functions by $\zeta(b_-)$, $D_A(\zeta\phi) = \zeta D_A\phi + K\phi$, $(K\phi)(b) := \eta^{-1}(\zeta(b_+) - \zeta(b_-))U(A_b)\phi(b_+)$, $\|K\| \le 4\sqrt{d}/R$; $[D_A^* D_A, \zeta] = D_A^* K - K^* D_A$; the potential commutes with ζ; $\|[\Pi_B, \zeta]\| \le 2\operatorname{osc}_B \zeta \le 8\sqrt{d}L/R$. Every term of $[T, \zeta]$ is supported in both variables in the layer $\mathcal{L} := \{R/4 - c \le \operatorname{dist}(\cdot, S) \le R/2 + c\}$, $c = \sqrt{d}L$, and the bonds where $K \ne 0$ are not cut bonds. (5) Cover $\mathcal{L}$ by unit cubes Q; the Combes–Thomas bounds for $T_{\rm cut}$ and T give $\|\mathbf{1}_Q D_{A,\rm cut} T_{\rm cut}^{-1} f\|, \|\mathbf{1}_Q T_{\rm cut}^{-1} f\| \le Ce^{-\theta_0 \operatorname{dist}(Q, \operatorname{supp} f)}\|f\|$ and the same for $T^{-1}f'$; writing the pairing as $\langle D_{A,\rm cut} T_{\rm cut}^{-1} f, KT^{-1}f'\rangle - \langle KT_{\rm cut}^{-1} f, D_A T^{-1} f'\rangle + \sum_B c_B \langle T_{\rm cut}^{-1} f, [\Pi_B, \zeta]T^{-1}f'\rangle$, each term couples neighbouring cubes only, so its modulus is $\le CR^{-1}\sum_{Q\subset\mathcal{L}} e^{-\theta_0(\operatorname{dist}(Q, \operatorname{supp} f) + \operatorname{dist}(Q, \operatorname{supp} f'))}\|f\|\|f'\|$. (6) For $Q \subset \mathcal{L}$ both distances are $\ge R/2 - 2c$ and their sum is $\ge \operatorname{dist}(\operatorname{supp} f, \operatorname{supp} f') - \sqrt{d}$; so $\theta_0(a + a') \ge \frac{\theta_0}{4}(R - 4c) + \frac{\theta_0}{4}(\operatorname{dist}(\operatorname{supp} f, \operatorname{supp} f') - \sqrt{d}) + \frac{\theta_0}{2}a$, and $\sum_Q e^{-(\theta_0/2)\operatorname{dist}(Q, \operatorname{supp} f)} \le C(1 + s)^d$ uniformly in the torus. □

Lemma F.107 (the operators of [2, (2.27), (2.46)] under the cut)**.** *Take $Y = \Lambda_7$, $\Omega = B^{k-1}(\Lambda_2)$. Write $\Delta := \Delta^{(k-1)}(\Omega, A)$, $\Delta^{\rm cut} := \Delta^{(k-1)}(\Omega_{\rm cut}, A)$; C_Λ, $C_\Lambda^{\rm cut}$ for $\Lambda \subset \Lambda_2$; $\Delta_\Lambda^{(k)}$, $\Delta_\Lambda^{(k),\rm cut}$ the coarse forms built from them, and $\Delta^{(k)} := \Delta_{\Lambda_2}^{(k)}$. There are $c, \theta_* > 0$, depending on $(d, N, \|q\|, a, L, \gamma_L)$ only, such that for unit points z, z' and coarse points y, y':* (a) *$|\Delta|, |\Delta^{\rm cut}|(z, z') \le ce^{-\theta_*|z-z'|}$, the same for C_Λ, $C_\Lambda^{\rm cut}$ and for the coarse forms;* (b) *$|\Delta - \Delta^{\rm cut}|(z, z') \le ce^{-\theta_*(d_S(z) + d_S(z') + |z-z'|)}$, the same for $C_\Lambda - C_\Lambda^{\rm cut}$ and $\Delta_\Lambda^{(k)} - \Delta_\Lambda^{(k),\rm cut}$ (with $d_S(y) := \operatorname{dist}(B^{k-1}B(y), S)$);* (c) *for $y, y' \in \Lambda'$, $|\Delta_\Lambda^{(k)} - \Delta^{(k)}|(y, y') \le ce^{-\theta_*(|y-y'| + \operatorname{dist}(y, \Lambda_2\setminus\Lambda) + \operatorname{dist}(y', \Lambda_2\setminus\Lambda))}$;* (d) *block structure: $G_{k-1}(\Omega_{\rm cut}, \Lambda, A) = G_{k-1}(\Omega_1, \Lambda\cap\Lambda_7^c, A) \oplus G_{k-1}(B(\Lambda_7), \Lambda\cap\Lambda_7, A)$, and likewise $\Delta^{\rm cut}$, $C_\Lambda^{\rm cut}$, $\Delta_\Lambda^{(k),\rm cut}$ split into their Ω_1- and $B(\Lambda_7)$-blocks;* (e) *for products $K_1K_2K_3$ of up to three kernels of* (a) *(and Q^*, block-local), $|K_1K_2K_3 - K_1^{\rm cut}K_2^{\rm cut}K_3^{\rm cut}|(z, z') \le c'e^{-(\theta_*/3)(d_S(z) + d_S(z') + |z-z'|)}$.*

Proof. (a) The first bound of proposition F.150 ([1, (2.27)]) for Δ and lemma F.145(c) ([4, (1.16)]) for C_Λ: on Ω directly, and on $\Omega_{\rm cut}$ because their proofs use only lemmas F.143 and F.144 and the algebra of lemma F.145(a), all valid on $\Omega_{\rm cut}$ by lemma F.106(1) (equivalently, by (d), whose proof is algebraic: each operator on $\Omega_{\rm cut}$ is the direct sum of its blocks on the two unions of big blocks Ω_1 and $B^{k-1}(\Lambda_7)$); for the coarse kernels, $|(QKQ^*)(y, y')| \le L^d \max_{z\in B(y), z'\in B(y')}|K(z, z')|$ and the diagonal term of [2, (2.27)] is local. (b) For Δ: $(\Delta - \Delta^{\rm cut})(z, z') = a_{k-1}^2\langle q(z), (G_{k-1}(\Omega_{\rm cut}, A) - G_{k-1}(\Omega, A))q(z')\rangle$, $q(z) := Q_{k-1}^*\delta_z$ of diameter $\le \sqrt{d}$ and norm 1; lemma F.106 with $\Lambda = \emptyset$ and $R = \min(d_S(z), d_S(z'))$, and $\min(d_S(z), d_S(z')) + |z - z'| \ge \frac{1}{2}(d_S(z) + d_S(z') + |z - z'|) - c''$. For C_Λ: by the block inversion on Ω and on $\Omega_{\rm cut}$ (algebra, needing only $T_\Lambda > 0$), W depends only on the pins, so $(C_\Lambda - C_\Lambda^{\rm cut})(y, y') = \langle W\delta_y, (G_{k-1}(\Omega, \Lambda) - G_{k-1}(\Omega_{\rm cut}, \Lambda))W\delta_{y'}\rangle$ with $W\delta_y$ supported in the L-block of y, norm ≤ 1; lemma F.106 with $s = \sqrt{d}L$. The coarse forms follow. (c) lemma F.147 ([4, (1.18)]; here $\Omega^{(k-1)} = \Lambda_2$, so $C^{(k-1)}(\Omega, A) = C_{\Lambda_2}$ and the distances are to $\Lambda_2 \setminus \Lambda$), then the coarse step. (d) The Neumann Laplacian of $\Omega_{\rm cut}$ is the direct sum over Ω_1 and $B(\Lambda_7)$; the pins are block-local and every block lies in one part,

since Λ_7 is a union of large blocks; so $T_\Lambda(\Omega_{\rm cut})$ is block diagonal, $\Delta^{\rm cut}(z,z')$ vanishes unless $B(z), B(z')$ lie in the same part, $X_\Lambda^{\rm cut}$ is block diagonal and so is its inverse. (e) Telescope $K_1K_2K_3 - K_1^{\rm cut}K_2^{\rm cut}K_3^{\rm cut} = (K_1 - K_1^{\rm cut})K_2K_3 + K_1^{\rm cut}(K_2 - K_2^{\rm cut})K_3 + K_1^{\rm cut}K_2^{\rm cut}(K_3 - K_3^{\rm cut})$; in the first term, with $X := d_S(z) + d_S(w) + |z-w|$, $Y := |w-w'|$, $Z := |w'-z'|$ and the target $T := d_S(z) + d_S(z') + |z-z'|$, the 1-Lipschitz property of d_S gives $T \le X + 2Y + 2Z$, so $\theta_*(X+Y+Z) \ge \frac{\theta_*}{3}T + \frac{\theta_*}{3}(Y+Z)$, and $e^{-(\theta_*/3)(Y+Z)}$ pays the sums over w, w' uniformly in the torus; the other terms are the same with the roles exchanged. (A target rate $\theta_*/2$ is false: nothing would be left for the w'-sum.) □

Proposition F.108 ([2, p. 568])**.** *Assume (G), (FR), (RG), (A). For every $\kappa > 0$ there is $t_0(\kappa) > 0$ such that for $t = L^{k-1}\varepsilon \le t_0$ each of the following replacements changes the exponential of* [2, (2.46)] *by at most $Ct^\kappa|\Lambda_6^{(k-1)}|$, C independent of k, ε, the torus and the fields:* (i) $T_{10} \to$ *the right side of* [2, (2.47)] *without its O-term;* (ii) $T_4 \to$ *its analogue with A in place of φ, every vector form the torus form;* (iii) $T_3 \to T_3$ *with A replaced by $(\Lambda_5' \cap \Lambda_6'^c)A$;* (iv) $T_9 \to T_9$ *with φ replaced by $(\Lambda_5' \cap \Lambda_6'^c)\varphi$ and $\Delta^{(k-1)}$, $C_{\Lambda_5}^{(k-1)}$ taken on $\Omega_{\rm cut}$, i.e. (by lemma F.107(d)) $\Delta^{(k-1)}(\Omega_1)$ and $C_{\Lambda_5\cap\Lambda_7^c}^{(k-1)}(\Omega_1)$;* (v) $T_7, T_8 \to$ *the same with every operator on $\Omega_{\rm cut}$. In particular $\kappa > d$ is allowed, as* [2, (2.48)] *requires.*

Proof. θ_* from lemma F.107; all sums are over unit or coarse points with torus-free constants. (1) Counting: for $z \notin \Lambda_6$, $d_S(z) \ge r - c$ by (G), and every unit cube meeting S is adjacent to a cube of Λ_7, so for any set X with $d_S \ge r - c$ on X, $\sum_{z\in X} e^{-\theta d_S(z)} \le e^{-\theta(r-c)/2}\sum_{Q\cap S\neq\emptyset}\sum_z e^{-\theta\,\mathrm{dist}(z,Q)/2} \le C_\theta e^{-\theta r/2}|\Lambda_7|$. (2) (i): write $\varphi = \varphi_o + \varphi_m + \varphi_i$ supported in $\Lambda_5' \setminus \Lambda_6'$, $\Lambda_6' \setminus \Lambda_7'$, Λ_7'; with $\Delta_5 := \Delta_{\Lambda_5}^{(k)}(\Omega)$ (symmetric), $\langle\varphi, \Delta_5\varphi\rangle = \langle\varphi_o, \Delta_5\varphi_o\rangle + 2\langle\varphi_o, \Delta_5\varphi_m\rangle + 2\langle\varphi_o, \Delta_5\varphi_i\rangle + \langle\varphi_m+\varphi_i, \Delta_5(\varphi_m+\varphi_i)\rangle$; by lemma F.107(d) the first two terms of (2.47) are $\frac12\langle\varphi_o, \Delta_{\Lambda_5}^{(k),\rm cut}\varphi_o\rangle + \langle\varphi_o, \Delta_{\Lambda_5}^{(k),\rm cut}\varphi_m\rangle$ and the third is $\frac12\langle\varphi_m+\varphi_i, \Delta^{(k)}(\varphi_m+\varphi_i)\rangle$. The difference is $E_1 + E_2 + E_3$: $E_1 := \frac12\langle\varphi_o, (\Delta_5 - \Delta_{\Lambda_5}^{(k),\rm cut})\varphi_o\rangle + \langle\varphi_o, (\Delta_5 - \Delta_{\Lambda_5}^{(k),\rm cut})\varphi_m\rangle$, by lemma F.107(b) and (1) $|E_1| \le cM^2\sum_{y\in\mathrm{supp}\,\varphi_o} e^{-\theta_* d_S(y)}\sum_{y'} e^{-\theta_*|y-y'|} \le CM^2e^{-\theta_* r/2}|\Lambda_7|$; $E_2 := \langle\varphi_o, \Delta_5\varphi_i\rangle$, with $|y-y'| \ge r-c$ between $\mathrm{supp}\,\varphi_o$ and Λ_7' (G), $|E_2| \le CM^2e^{-\theta_* r/2}|\Lambda_7|$; $E_3 := \frac12\langle\varphi_m+\varphi_i, (\Delta_5 - \Delta^{(k)})(\varphi_m+\varphi_i)\rangle$, by lemma F.107(c) with $\mathrm{dist}(y, \Lambda_2\setminus\Lambda_5) \ge r-c$ on Λ_6, $|E_3| \le CM^2e^{-\theta_* r}|\Lambda_6|$. (3) (ii): as (i) with E_1 absent (no cut) and the torus operators; E_3 by lemma F.147 ([4, (1.18)]) on the torus, at $A = 0$ with $\Omega = T_\varepsilon$ and $\Lambda = \Lambda_5$ (lemma F.154). (4) (iii): the error is $a(L^k\varepsilon)^{-2}\langle A'_{\rm out}, \Delta C_{\Lambda_5}Q^*(\Lambda_6'A)\rangle$ with $A'_{\rm out}$ supported in Λ_5^c; by lemma F.107(a) $|(\Delta C_{\Lambda_5}Q^*)(x,y')| \le ce^{-\theta_*|x-y'|/2}$ and $|x-y'| \ge r-c$ (G), so the error is $\le Ct^{-2}M^2e^{-\theta_* r/4}|\Lambda_6|$. (5) (iv): restricting φ is as (4); replacing Ω by $\Omega_{\rm cut}$ in Δ, C_{Λ_5} changes the kernel at (x,y') by $\le ce^{-(\theta_*/3)(d_S(x)+d_S(y')+|x-y'|)}$ (lemma F.107(e)) with $d_S(x) \ge 2(r-c) - 1$ on $\Lambda_3 \cap \Lambda_5^c$, hence by (1) $\le Ct^{-2}M^2e^{-\theta_* r/6}|\Lambda_7|$; then lemma F.107(d): φ' is supported off $B(\Lambda_7)$ and $Q^*\varphi$ lies in $\Lambda_5\setminus\Lambda_6 \subset \Lambda_7^c$, so only the Ω_1-blocks act. (6) (v): as (5) with both outer points in $\Lambda_3\cap\Lambda_5^c$. (7) Every error is $\le Ct^{-4}M^2e^{-\theta_* r(t)/6}(|\Lambda_6| + |\Lambda_7|) \le C't^{-8}e^{-\theta_* r(t)/6}|\Lambda_6|$ ($M \le Ct^{-2}$, $\Lambda_7 \subset \Lambda_6$), and $e^{-\theta_* r(t)/6} \le t^{\kappa+9}$ for $t \le t_0(\kappa)$ since $\varrho > 1$ (the bound of Step 3(D_2), Geometry). □

The proof uses neither the Theorem of [4] nor the two-sided depth bounds; "Proposition I.2.1" of the print is not needed for these terms. The terms omitted at [2, p. 568] ("described more precisely later, in (2.109)") are the quadratic-form step treated next.

The quadratic-form step [2, (2.108)–(2.109)]. *Setting.* Step k in scale-k units, $t := t_k$, $\eta := L^{-k}$; $\Omega_A := B^k(\Lambda_2^{(k-1)'})$ (the propagator domain of [2, (2.53), (2.56)]), $\Omega_B := B^k(\Lambda_2^{(k-1)'} \cap \Lambda_5^{(k)c})$ and $\Omega_C := B^k(\Lambda_2^{(k)})$ (the domains of [2, (2.108)]), all unions of big blocks; $\tilde B := A^{(k)}$ on Ω_A. On Ω_C, $\tilde B = B^{(k+1),\eta}$: $\theta_{k+1} = 1$ on $B^k(\Lambda_2^{(k)})$ [2, pp. 567, 576], $\Omega_C \subset B^k(\Lambda_0^{(k)})$, and by [2, p. 579] the configuration there is $(1-\theta_{k+1})A^{(k)} + \theta_{k+1}B^{(k+1),\eta}$; the operators $G_k(\Omega_C,\cdot)$, $\Delta^{(k)}(\Omega_C,\cdot)$, $C_\Lambda^{(k)}(\Omega_C,\cdot)$ depend on the field only through the bonds of Ω_C [4, (1.3)–(1.4)]. So the replacement of [2, Prop. 2.6] is the pure domain change $\Omega_A \to \Omega_C$: its interaction-term half is theorem G.15, and its quadratic-form half is the following. Layer distances [2, (2.8)]:

$\operatorname{dist}(\Lambda_{i+j}^{(k)}, \Lambda_i^{(k)c}) \geq j(r-1)$ and $\operatorname{dist}(\Lambda_7^{(k-1)'}, \Lambda_2^{(k-1)'c}) \geq r-1$ ($L=5$). Standing lower bound: $R \geq R_0+3$, so that $r-2 \geq R_0+1$ (R may be enlarged, [2, (2.7)]).

Lemma F.109 (discrete trace inequality)**.** *Let u be a function on a box Y of $\eta\mathbb{Z}^d$ with all its bonds, z a site, e a lattice direction and $S_z := \{z + m\eta e : 0 \leq m < \eta^{-1}\} \subset Y$. Then $|u(z)|^2 \leq 2\sum_{v\in S_z} \eta|u(v)|^2 + 2\sum_{b'\subset S_z} \eta|(D_A u)(b')|^2$.*

Proof. $|U|=1$ gives $||u(b'_+)| - |u(b'_-)|| \leq \eta|D_A u(b')|$, so $|u(z)| \leq |u(v)| + \sum_{b'\subset[z,v]} \eta|D_A u(b')|$ for every $v \in S_z$; average over v with weight η (total weight 1), apply Cauchy–Schwarz and square. □

Lemma F.110 (domain difference of $\Delta^{(k)}$, decaying from the new boundary)**.** *Let $\Omega' \subset \Omega$ be unions of big blocks, A regular on Ω, Σ the set of bonds of Ω with exactly one end in Ω', and $\operatorname{dist}(\Sigma, \Omega^c) \geq R_0+1$. Then for $y, y' \in \Omega'^{(k)}$: $|\Delta^{(k)}(\Omega', A; y, y') - \Delta^{(k)}(\Omega, A; y, y')| \leq c_Q \exp[-\delta_Q(|y-y'| + \operatorname{dist}(B(y), \Sigma) + \operatorname{dist}(B(y'), \Sigma))]$, $\delta_Q := \min(\delta_0, \theta_0)/4$, c_Q depending on $(c_0, \delta_0, \gamma_\emptyset, \theta_0, a, d)$. (This is stronger than [1, (2.29)], which decays from all of Ω'^c.)*

Proof. (1) With $H_\Omega := -\Delta_{A,\Omega}^{\eta;N} + m^2t^2 + a_k P_k(A)$: $H_\Omega = (H_{\Omega'} \oplus H_{\Omega\setminus\Omega'}) + D_\Sigma^* D_\Sigma$, D_Σ the covariant difference on the bonds of Σ (P_k and the mass are block diagonal and both pieces are unions of blocks). For f' on Ω', $u' := G_k(\Omega')f'$ and $u^- := u' \oplus 0$: $H_\Omega u^- = f' + D_\Sigma^* D_\Sigma u^-$, hence $(G_k(\Omega)f')|_{\Omega'} - u' = -(G_k(\Omega) D_\Sigma^* D_\Sigma u^-)|_{\Omega'}$. (2) With $q(y) := Q_k^*(A)\delta_y$: $\delta\Delta(y,y') = a_k^2\langle q(y), (G_k(\Omega) - G_k(\Omega'))q(y')\rangle = -a_k^2\langle D_\Sigma G_k(\Omega)q(y), D_\Sigma u^-\rangle$, $u' = G_k(\Omega')q(y')$. (3) For $b = (z,w) \in \Sigma$ with $z \in \Omega'$, $|(D_\Sigma u^-)(b)| = \eta^{-1}|u'(z)|$. (4) By [1, (2.25)] at b, the restricted form of the theorem of [4], on the torus by lemma F.154 (its hypotheses hold: Ω is a union of big blocks, A is regular on Ω, $\|q(y)\|_\infty = 1$, and $\operatorname{dist}(b, \Omega^c) \geq R_0$), $|D_b G_k(\Omega)q(y)| \leq c_0 e^{-\delta_0 \operatorname{dist}(b, B(y))}$. (5) For a unit block Y of Ω' let Σ_Y be the bonds of Σ with $z_b \in Y$ ($\#\Sigma_Y \leq 2d\eta^{-(d-1)}$); each z_b lies on a face of Y, the segments away from w stay in Y and those from one face partition Y, so lemma F.109 over the $\leq 2d$ faces gives $\sum_{b\in\Sigma_Y} \eta^{d-1}|u'(z_b)|^2 \leq 4d(\|\mathbf{1}_Y u'\|^2 + \|\mathbf{1}_Y D u'\|^2)$. (6) Lemma F.144 on Ω' with $T = H_{\Omega'} \geq \gamma_\emptyset$ (lemma F.143 with $\Lambda = \emptyset$; Ω' is a union of big blocks and A is regular on it) gives $\|\mathbf{1}_Y u'\| \leq 2\gamma_\emptyset^{-1} e^{-\theta_0 \operatorname{dist}(Y, B(y'))}$ and $\|\mathbf{1}_Y D u'\| \leq C e^{-\theta_0 \operatorname{dist}(Y, B(y'))}$, no depth restriction. (7) Cauchy–Schwarz on Σ_Y with weights η^{d-1}: $|\delta\Delta| \leq a^2 \sum_Y \sum_{\Sigma_Y} \eta^d |D_b G q(y)| \eta^{-1}|u'(z_b)| \leq C\sum_Y e^{-\delta_0 \operatorname{dist}(Y,B(y)) - \theta_0 \operatorname{dist}(Y,B(y'))}$ over the blocks Y adjacent to Σ. (8) For such Y, $\operatorname{dist}(Y, B(y)) + \operatorname{dist}(Y, B(y')) \geq \frac{1}{2}(\operatorname{dist}(B(y), B(y')) + \operatorname{dist}(B(y), \Sigma) + \operatorname{dist}(B(y'), \Sigma)) - 2$; keep half of the exponent for the sum over Y and use $|y-y'| \leq \operatorname{dist}(B(y), B(y')) + 1$. □

Lemma F.111 ([2, (2.108)] is an identity)**.** *For φ on $\Lambda_6^{(k-1)'}$ put $\varphi_o := (\Lambda_6^{(k-1)'} \cap \Lambda_3^{(k)c})\varphi$, $\varphi_i := \Lambda_3^{(k)}\varphi$, $\varphi_e := (\Lambda_3^{(k)} \cap \Lambda_4^{(k)c})\varphi$, $\varphi_4 := \Lambda_4^{(k)}\varphi$ ($\varphi_i = \varphi_e + \varphi_4$), $\Delta_A := \Delta^{(k)}(\Omega_A, \tilde{B})$, $\Delta_B := \Delta^{(k)}(\Omega_B, \tilde{B})$, $\Delta_C := \Delta^{(k)}(\Omega_C, B^{(k+1),\eta}) = \Delta^{(k)}(\Omega_C, \tilde{B})$. Then $\frac{1}{2}\langle \Lambda_6'\varphi, \Delta_A \Lambda_6'\varphi\rangle = \frac{1}{2}\langle\varphi_o, \Delta_B\varphi_o\rangle + \langle\varphi_o, \Delta_B\varphi_e\rangle + \frac{1}{2}\langle\varphi_i, \Delta_C\varphi_i\rangle + \frac{1}{2}\langle\Lambda_6'\varphi, H_k\Lambda_6'\varphi\rangle$ with $\frac{1}{2}\langle\varphi, H_k\varphi\rangle := \frac{1}{2}\langle\varphi_o, (\Delta_A - \Delta_B)\varphi_o\rangle + \langle\varphi_o, (\Delta_A - \Delta_B)\varphi_e\rangle + \langle\varphi_o, \Delta_A\varphi_4\rangle + \frac{1}{2}\langle\varphi_i, (\Delta_A - \Delta_C)\varphi_i\rangle$. With the block-spin term $\frac{1}{2}aL^{d-2}\sum|\psi - Q(\theta_k\tilde{B})\varphi|^2$, unchanged on both sides, this is [2, (2.108)] with the factor $\frac{1}{2}$ of its fourth term restored.*

Proof. $\Lambda_3^{(k)} \subset \Lambda_0^{(k)} \subset \Lambda_7^{(k-1)'} \subset \Lambda_6^{(k-1)'}$, so $\Lambda_6'\varphi = \varphi_o + \varphi_i$; the Δ's are real symmetric ($U \in SO(N)$, q real antisymmetric); expand and subtract. All forms are defined: $B^k(\Lambda_6' \cap \Lambda_3^c)$ and $B^k(\Lambda_3 \cap \Lambda_4^c)$ lie in Ω_B, $B^k(\Lambda_3)$ in Ω_C. □

Lemma F.112 ([2, (2.109)])**.** *For $y, y' \in \Lambda_6^{(k-1)'}$ the kernel of H_k obeys $|h_k(y,y')| \leq C_H e^{-\delta_H(r-2)} e^{-\delta_H|y-y'|}$, $\delta_H := \min(\delta_Q, \delta_7/2)/2$ (δ_7 the rate of proposition F.150, the form of [1, (2.27)] proved in Step 3(A)); and under [2, (2.55)], for $t \leq t_0(\kappa)$, $\frac{1}{2}|\langle\Lambda_6'\varphi, H_k\Lambda_6'\varphi\rangle| \leq Cc_1^2 p(t/L)^2 \lambda(t/L)^{-1/2} e^{-(\delta_H/2)(r-2)}|\Lambda_3^{(k)}| \leq C_\kappa t^\kappa |\Lambda_3^{(k)}|$, the printed error of [2, (2.108)].*

Proof. Four pieces. (oo), (oe): lemma F.110 with $(\Omega, \Omega') = (\Omega_A, \Omega_B)$; $\Omega_A \setminus \Omega_B = B^k(\Lambda_5^{(k)})$, so Σ consists of the bonds crossing $\partial B^k(\Lambda_5^{(k)})$, with $\operatorname{dist}(\Sigma, \Omega_A^c) \geq \operatorname{dist}(\Lambda_5^{(k)}, \Lambda_2^{(k-1)'c}) - 1 \geq r-2 \geq$

R_0+1; for $y\in\Lambda_3^c$, $\operatorname{dist}(B(y),\Sigma)\ge 2(r-1)-2$, for $y'\in\Lambda_3\cap\Lambda_4^c$, $\operatorname{dist}(B(y'),\Sigma)\ge r-2$; so $|(\Delta_A-\Delta_B)(y,y')|\le c_Q e^{-(\delta_Q/2)(r-2)}e^{-(\delta_Q/2)(|y-y'|+\operatorname{dist}(B(y),\Sigma)+\operatorname{dist}(B(y'),\Sigma))}$. (o4): for $y\in\Lambda_3^c$, $y'\in\Lambda_4$, $|y-y'|\ge r-1$, and proposition F.150 ([1, (2.27)], under the hypotheses of lemma F.110 on Ω_A) gives $|\Delta_A(y,y')|\le c_7 e^{-(\delta_7/2)(r-1)}e^{-(\delta_7/2)|y-y'|}$. (ii): lemma F.110 with (Ω_A,Ω_C), Σ the bonds crossing $\partial\Omega_C$, $\operatorname{dist}(\Sigma,\Omega_A^c)\ge r-2$, and $\operatorname{dist}(B(y),\Sigma)\ge r-2$ for $y\in\Lambda_3$. Assembling the kernels gives the kernel bound. Error: $\sup|\varphi|\le c_1 p(t/L)\lambda(t/L)^{-1/4}$ on $\Lambda_6'\subset\Lambda_{-1}'$ by [2, (2.55)]; the (oo) sum over $y,y'\in\Lambda_3^c$ is $\le Ce^{-(\delta_Q/2)(r-2)}\sum_y e^{-(\delta_Q/4)\operatorname{dist}(B(y),\Sigma)}\le Ce^{-(\delta_Q/2)(r-2)}|\Lambda_5^{(k)}|$ (at most $|\Lambda_5|$ unit blocks are adjacent to Σ from inside); in (oe), (o4), (ii) the second index runs over Λ_3 or Λ_4, giving $\le C|\Lambda_3^{(k)}|$ times the exponential; then $|\Lambda_5|\le|\Lambda_3|$, $p(t/L)\le(1+\ln L)^{\bar p}p(t)$, $\lambda(t/L)=\lambda_k/L$, and $p^2\lambda^{-1/2}e^{-\delta r}\le t^\kappa$ for $t\le t_0(\kappa)$. □

With [1, (2.29)] in place of lemma F.110 the error would be $O(t^\kappa)|\Lambda_6^{(k-1)'}|$, still summable at $\kappa>d$ but not the printed $|\Lambda_3^{(k)}|$.

From [2, (3.5)] *at* $k=0$ *to* [2, (3.21)]*, and the payment of* [2, Sect. 3C].

Lemma F.113 (monotone nesting)**.** *Every renormalisation transformation* $T^t_{a,L}$, $T^t_{a,L,\tilde A}$ *of* [2, Sect. 3] *is* $F\mapsto\int t(\cdot,F\text{-arguments})F$ *with a kernel* $t\ge 0$ *normalised by* [2, (3.32)]*; hence* $F\le F'$ *pointwise implies* $T[F]\le T[F']$*, and nested brackets preserve inequalities.*

Lemma F.114 ([2, (3.21)])**.** [2, (3.5)] *holds at* $k=0$ *(proposition F.44, the finite downward induction from the base case). At* $k=0$ *the innermost bracket of* [2, (3.5)] *is the bare Gaussian density* $\exp(-\frac12\langle A_0,(-\Delta^\varepsilon+\mu_0^2)A_0\rangle)\exp(-\frac12\langle\phi_0,(-\Delta^\varepsilon_{\tilde A^\varepsilon}+m^2)\phi_0\rangle)$ *with the bare characteristic functions* [2, (2.34)–(2.42), (2.47)–(2.49)]*; each characteristic function factorises into a vector condition and a scalar condition with the vector configuration as parameter,* ζ *placed inside the scalar bracket; integrating first over the scalar fields at fixed vector fields (Tonelli) and lemma F.113 give* [2, (3.21)]*, whose last factor is* $\exp(\sum_{k=1}^K C_\kappa t_k^{\kappa_0}|T_\varepsilon|)\le\exp(C_\kappa\varepsilon_0^{\kappa_0}(1-L^{-\kappa_0})^{-1}|T_\varepsilon|)$*, i.e.* [2, (2.118)].

Lemma F.115 (the residue of [2, Sect. 3B])**.** *In* [2, (3.30)–(3.40)] *the only factors left besides* $\prod\zeta''$ *and* $\exp(E_0)$ *are: from* [2, (3.31)], $\exp(c_{3.29}t_k^{\kappa_0}|\Lambda_k|)$ *with* $c_{3.29}$ *the remainder constant of* [2, Prop. 3.1, (3.29)] *(theorem F.2, the* ε_0*-freedom paragraph of Step 3(*C_2*)); from* [2, (3.33)], $2^{N/2}\exp(-\frac14 ap(t_k)^2)\le\exp(-\frac18 ap(t_k)^2)$ *per exceptional block, valid once* $b_0^2\ge 4N\ln 2/a$ *(a condition on* b_0*, fixed before* ε_0*); from* [2, (3.36), (3.40)]*, the Jacobians* $\exp(\frac N2\ln 2\sum_k|\Lambda_k|)$ *and* $\exp(\frac d2\ln 2\sum_k|\Lambda_k|)$*; and from* [2, (3.38)] *the diamagnetic inequality and the normalisations* [2, (3.32)] *(equalities). Here* $\Lambda_k:=\Lambda_5^{(k-1)'}\cap\Lambda_5^{(k)c}$*, counted in points of the scale-k unit lattice. So the* $O(1)$ *of* [2, (3.39)–(3.40)] *is* $c_5:=\frac{N+d}{2}\ln 2+c_{3.29}\varepsilon_0^{\kappa_0}\le\frac{N+d}{2}\ln 2+c_{3.29}$*, and the* $O(1)$ *in front of* $(L^k\varepsilon)^{\kappa_0}|\Lambda_7^{(k-1)'}\cap\Lambda_7^{(k)c}|$ *in* [2, (2.117), (3.42)] *is* $c_7:=C_{2.57}+C_{2.78}$ *(Step 3(*C_2*), the* $O(1)$ *lines).* [2, (3.40)] *prints* $\sum_{k=1}^K|\Lambda_k|$ *where* [2, (3.36), (3.42)] *print* $\sum_{k=0}^K$*; the constants use* $\sum_{k=0}^K$*.*

Lemma F.116 (the payment)**.** *The residue* $\exp(c_5\sum_k|\Lambda_k|)$ *of lemma F.115 is exactly the second factor of the left side of* [2, (3.42)] *with* $O(1)=c_5$*, and the first factor carries* $O(1)=c_7$*. So* [2, (2.117)]*, hence the upper bound, follows from* [2, (3.42)] *with the constants* (c_5,c_7)*, which are free of* ε*,* K*, the torus, the history and* ε_0*;* [2, (3.42)] *itself is theorem F.125 below, which imposes its own cap on* ε_0 *after* c_5*,* c_7 *are fixed.*

Proof of lemmas F.113 to F.116. Lemma F.113: positivity of the Gaussian kernels. Lemma F.114: as stated, the $k=0$ bracket being the bare density by the definitions of the first step ([2, p. 588]: "using inequality (3.5) for $k=0$"), and $t_k=L^{k-K}t_K\le L^{k-K}\varepsilon_0$. Lemma F.115: arithmetic on the printed steps, with $p(t)\ge b_0$ for the b_0-condition. Lemma F.116: $|\Lambda_k|=|\Lambda_5^{(k-1)'}\cap\Lambda_5^{(k)c}|$ by definition ([2, p. 592]: "$|\cdot|$ means the number of elements"), an identity of terms. □

The region sum [2, (3.42), Sect. 3C] *at block factor* 5. *Setting.* $d=3$, $L=5$, $M\geq 1$, $s_b:=LM$ the side of a large block in lattice units of $T^{(k)}$ [1, p. 607]; $R>R_0$, $\theta:=s_b/R\leq\frac17$; $\varrho\geq 1$ and $\bar p$ with $2\bar p>\varrho d+2$ (Bałaban's $\varrho=2$, $\bar p=6$: exponent $2\bar p-2-\varrho d=4$); $\varepsilon_0\leq e^{-1}$; $\hat l_k:=1+\ln t_k^{-1}$, $r_k:=r(t_k)=R\hat l_k^{\varrho}$, $p_k:=p(t_k)=b_0\hat l_k^{\bar p}$. The metric is the torus sup-metric [1, (1.3)]; $N_w(X)$ is the closed w-neighbourhood; boxes are images of products of integer intervals. The torus $T^{(k)}=\mathbb{Z}^d/\Gamma_k$, $\Gamma_k=L\Gamma_{k+1}$, is aligned with the blocks: $\Gamma_k=\bigoplus_\mu P_{k,\mu}\mathbb{Z}$ with $s_b\mid P_{k,\mu}$ for $k\leq K-1$ (a condition on the torus: its sides are multiples of the side of a large block at scale $K-1$, hence at every scale up to $K-1$; cf. definition 12.1, whose sides lie in $M\mathbb{N}$, and definition G.17); so the large blocks and the L-blocks tile $T^{(k)}$ and the prime $X\mapsto X'$ of [1, (1.16)] is defined. Families and regions are as printed: at scale k a family $F_k=(P_v,Q_v,R_v,P_s,Q_s,R_s)$ inside $\Lambda_7^{(k-1)'}$ ($\Lambda_7^{(-1)'}:=T^{(0)}$); each element e has a point set $\mathrm{pts}(e)$ (a P-element its L-block; a bond or plaquette its ends or corners; an R-element its point) and a charge point $x_e\in\mathrm{pts}(e)$ (the lowest corner), with $\mathrm{pts}(e)\subset N_{L-1}(x_e)$; $S_k:=\bigcup_e\mathrm{pts}(e)$; $\Lambda_0^{(k)}$ is the union of the large blocks of $\Lambda_7^{(k-1)'}$ at distance $>r_k$ from S_k [2, p. 570], $\Lambda_i^{(k)}$ by [2, (2.8)], "admissible" as on [2, p. 566]. Write $X_k:=\Lambda_0^{(k)c}$, $\mathrm{Col}_k:=\Lambda_7^{(k-1)'}\cap\Lambda_7^{(k)c}$, $\mathrm{Sh}_k:=\Lambda_5^{(k-1)'}\cap\Lambda_5^{(k)c}$, $N_k:=|T^{(k)}|$, $I_k:=\{z\in T^{(k)}:\mathrm{dist}(z,T^{(k)}\setminus\Lambda_7^{(k-1)'})\geq s_b\}$ ($I_0=T^{(0)}$). Constants: $c_{\mathrm{box}}:=\min\{a3^{-d}/8,\gamma_0 3^{-d}/4,3^{-d}\}$ (Bałaban's c_0 of [2, p. 593]); $c_5,c_7\geq 0$, $\kappa_0>0$ the $O(1)$'s and the exponent of [2, (3.42)] (lemma F.115); $n_d:=4+2\max\{d,d(d-1)/2\}$ ($n_3=10$); $\nu_L:=14+(16+2L)\theta$, $\nu_L':=\nu_L+2+3\theta$, $S_L:=\sum_{m\geq 0}L^{-m}(1+m\ln L)^{\varrho}$, $C_D^0:=\nu_L'(S_L-1)$, $C_D:=C_D^0+14(1+\theta)$, $C_1:=n_d\ln 2+d+1+c_5+c_7$, $\Theta_L:=C_1C_D^dR^d/(c_{\mathrm{box}}b_0^2\ln L)$, $C_{\mathrm{fin}}:=L^d(c_7\varepsilon_0^{\kappa_0}+c_5)+e^{-(d+1)}\varepsilon_0^{d+1}/(L-1)$. At $L=5$, $\varrho=2$, $\theta=\frac17$: $S_5=3.470$, $C_D^0=49.76$, $C_D=65.76$.

The printed [2, Sect. 3C] is not followed: its partition [2, (3.43)] (cubes with side in $(r,2r)$) need not exist at $L=5$, and the count [2, (3.45)] with the first inequality of [2, (3.46)] is false as printed: at $k=1$, with one R_s-element at a point x_* of $T^{(0)}$ and $F_1=\emptyset$, $\Lambda_7^{(0)c}$ is a cube of large blocks whose prime is a union of scale-1 large blocks only if its corners are divisible by L, which fails for $L-1$ of every L positions of x_*; then a scale-1 large block straddling the face lies in $\Lambda_0^{(1)c}$ and meets $\Lambda_7^{(0)'}$, so the printed C_1 is non-empty while $3^d(|P_v^{(1)}|+\dots)=0$, and the first inequality of [2, (3.46)] reads $1\leq\exp(-c_0p^2|C_1|)<1$. The count below is on the interior I_k (lemma F.120), and the straddling layer is covered by the inherited boxes (lemma F.124).

Lemma F.117 (consequences of the printed rules)**.** *For every admissible sequence and $k\leq K-1$:* (a) *every point of every element of F_k lies in X_k;* (b) $\Lambda_i^{(k)c}\subset\Lambda_{i+1}^{(k)c}$ *and* $\Lambda_i^{(k)c}\subset N_{i(r_k+s_b-1)}(X_k)$, $i=0,\dots,8$; (c) *if $z\in X_k\cap I_k$, the large block $\beta\ni z$ lies in $\Lambda_7^{(k-1)'}$ and* $\mathrm{dist}(\beta,S_k)\leq r_k$; (d) $T^{(k)}\setminus\Lambda_7^{(k-1)'}=(\Lambda_7^{(k-1)c})'\subset X_k$; (e) $X_k,\mathrm{Col}_k,\mathrm{Sh}_k\subset\Lambda_7^{(k)c}$; (f) *if $X_i=\emptyset$ for all $i<j$ and $X_j\neq\emptyset$, then $\Lambda_7^{(j-1)'}=T^{(j)}$ and $X_j\cap I_j=X_j$.*

Proof. (a) The large block β of a point of $\mathrm{pts}(e)$ has $\mathrm{dist}(\beta,S_k)=0\leq r_k$, so it is not one of the blocks of [2, p. 570]; the large blocks tile $T^{(k)}$ and $\Lambda_0^{(k)}$ is a union of them. (b) [2, (2.8)] adds the large blocks at distance $\leq r_k$ (blocks of Λ_i^c included); a point of such a block is within $(s_b-1)+r_k$ of Λ_i^c; induct. (c) β has sup-diameter $s_b-1<s_b$, so $\beta\subset\Lambda_7^{(k-1)'}$, and being a candidate not chosen, $\mathrm{dist}(\beta,S_k)\leq r_k$. (d) $\Lambda_7^{(k-1)}$ is a union of L-blocks, so $(X^c)'=T^{(k)}\setminus X'$; $\Lambda_0^{(k)}\subset\Lambda_7^{(k-1)'}$. (e) by (b). (f) $X_i=\emptyset$ gives $\Lambda_7^{(i)c}=\emptyset$ by (b), so $\Lambda_7^{(i)'}=T^{(i+1)}$. □

Lemma F.118 (number of families)**.** *For an admissible $\Lambda_0^{(k)}$ the number of families F_k admissible for it is $\leq 2^{n_d|X_k|}$.*

Proof. By lemma F.117(a) every element has all its points in X_k; charge each to its charge point; a point is the charge point of at most one P_v-, one P_s-, one R_v-, one R_s-element and of at most $\max\{d,d(d-1)/2\}$ bonds or plaquettes of each Q-type; a family is a subset of these $\leq n_d|X_k|$ candidates. □

Lemma F.119 (the box partition)**.** *Let $q_k:=\lceil(r_k+s_b)/s_b\rceil$. For each $k\leq K-1$ there is a partition $\mathcal{Q}_k$ of $T^{(k)}$ into boxes and a symmetric relation $Q\sim Q'$ (equal or adjacent) such that:*

(Q1) every box is a union of large blocks; (Q2) every box is the image of a product of d integer intervals with at most $2r_k + 3s_b \le (2+3\theta)r_k$ points each; (Q3a) every box has at most 3^d boxes $Q' \sim Q$; (Q3b) if $\operatorname{dist}(z,x) \le r_k + s_b$ then $Q(z) \sim Q(x)$.

Proof. In each direction μ let $m_\mu := P_{k,\mu}/s_b$. If $m_\mu \ge q_k$, cut $\mathbb{Z}/P_{k,\mu}\mathbb{Z}$ into consecutive arcs, each a union of between q_k and $2q_k - 1$ consecutive large-block intervals $[js_b, (j+1)s_b)$ (write $m_\mu = nq_k + \rho'$, $0 \le \rho' < q_k$, and give the ρ' extra blocks to the n arcs, at most $q_k - 1$ each); if $m_\mu < q_k$ use one arc. The product of the lifted arc partitions is Γ_k-invariant, so its images form a partition of $T^{(k)}$; $Q \sim Q'$ if some lifts have arc indices differing by at most one in every coordinate. (Q1): arcs are unions of large-block intervals. (Q2): an arc has at most $(2q_k - 1)s_b < 2(r_k + s_b) + s_b$ points, or $P_{k,\mu} < q_k s_b \le r_k + 2s_b$ in the one-arc case; $s_b \le \theta r_k$. (Q3a): the boxes $\sim Q$ are the images of the 3^d neighbours of one lift. (Q3b): take lifts with $|\tilde z - \tilde x|_\infty = \operatorname{dist}(z,x) \le r_k + s_b$; in a direction with $m_\mu \ge q_k$ every arc has $\ge q_k s_b \ge r_k + s_b$ points, so the two arcs are equal or consecutive; in a one-arc direction the indices agree. □

Lemma F.120 (the cube count on the interior)**.** *Put $\mathcal{C}_k := \{Q \in \mathcal{Q}_k : Q \cap X_k \cap I_k \ne \emptyset\}$. For every family F_k admissible for $\Lambda_0^{(k)}$, every $Q \in \mathcal{C}_k$ satisfies $Q \sim Q(x_e)$ for some element e of F_k; hence $|\mathcal{C}_k| \le 3^d |F_k|$, $|F_k| := |P_v^{(k)}| + \cdots + |R_s^{(k)}|$. At $k = 0$ this is* [2, (3.44)].

Proof. Let $z \in Q \cap X_k \cap I_k$; its large block β lies in Q (Q1); by lemma F.117(c) there are $z' \in \beta$, $y \in S_k$ with $\operatorname{dist}(z', y) \le r_k$, $y \in \operatorname{pts}(e)$ with $\operatorname{dist}(y, x_e) \le L - 1 < s_b$; so $\operatorname{dist}(z', x_e) \le r_k + s_b$ and $Q = Q(z') \sim Q(x_e)$ by (Q3b); count with (Q3a). □

Lemma F.121 (the bound on ζ'')**.** $\zeta''_{\Lambda_0^{(k)}} \le 2^{n_d |X_k|} \exp(-c_{\mathrm{box}} p_k^2 |\mathcal{C}_k|)$.

Proof. The six coefficients $a/8, \gamma_0/4, \gamma_0/4, a/8, \gamma_0/4, 1$ of [2, (3.41)] are each $\ge 3^d c_{\mathrm{box}}$, so each term of [2, (3.41)] is $\le \exp(-3^d c_{\mathrm{box}} p_k^2 |F_k|) \le \exp(-c_{\mathrm{box}} p_k^2 |\mathcal{C}_k|)$ by lemma F.120 ($\mathcal{C}_k$ depends on the region sequence only); the number of terms is lemma F.118. □

Lemma F.122 (weighted region entropy)**.** *For $\theta' \ge 0$, $\sum_X e^{-\theta'|X|} \le \exp(N_k e^{-\theta'})$, the sum over all unions X of large blocks of $T^{(k)}$. With $\theta_k := (d+1)\hat l_k$: $\sum_{k<K} N_k e^{-\theta_k} \le e^{-(d+1)} \varepsilon_0^{d+1} |T^{(K)}|/(L-1)$.*

Proof. The large blocks partition $T^{(k)}$ into N_k/s_b^d blocks, so the sum is $(1 + e^{-\theta' s_b^d})^{N_k/s_b^d} \le \exp(N_k s_b^{-d} e^{-\theta' s_b^d}) \le \exp(N_k e^{-\theta'})$. Next $e^{-\theta_k} = e^{-(d+1)} t_k^{d+1}$ and $N_k t_k^d = |T^{(K)}| t_K^d$, so $\sum_{k<K} N_k e^{-\theta_k} = e^{-(d+1)} |T^{(K)}| t_K^d \sum_{k<K} t_k$ and $\sum_{k<K} t_k = t_K/(L-1) \le \varepsilon_0/(L-1)$. (A k-independent price would give $\sum_k N_k e^{-\theta'} \sim \varepsilon^{-d}|T|$: the logarithm in [2, (3.47)] is exactly the price $\hat l_k$.) □

Proposition F.123 (from the sum to the supremum)**.** *Let $\mathcal{F}(\Lambda)$ be the summand of* [2, (3.42)] *for the admissible sequence $\Lambda = (\Lambda_0^{(0)}, \dots, \Lambda_0^{(K-1)})$ and $\Xi(\Lambda) := \sum_{k<K} [(n_d \ln 2 + \theta_k)|X_k| + c_7 t_k^{\kappa_0} |\mathrm{Col}_k| + c_5 |\mathrm{Sh}_k|]$. For every set $\mathcal{A}$ of admissible sequences, $\sum_{\Lambda \in \mathcal{A}} \mathcal{F}(\Lambda) \le \exp(C_{\mathrm{fin}} |T^{(K)}|) \sup_{\Lambda \in \mathcal{A}} \exp(-\sum_{k<} \Xi(\Lambda))$.*

Proof. The $k = K$ terms of the two exponents of [2, (3.42)] are $\le (c_7 t_K^{\kappa_0} + c_5) L^d |T^{(K)}|$. By lemma F.121, $\mathcal{F}(\Lambda) \le e^{L^d(\dots)|T^{(K)}|} \Psi(\Lambda) e^{-\sum_{k<K} \theta_k |X_k|}$ with $\Psi(\Lambda) := \exp(-\sum_{k<K} c_{\mathrm{box}} p_k^2 |\mathcal{C}_k| + \Xi(\Lambda))$; bound Ψ by its supremum; $\Lambda \mapsto (X_0, \dots, X_{K-1})$ is injective and every X_k is a union of large blocks, so $\sum_{\Lambda \in \mathcal{A}} e^{-\sum \theta_k |X_k|} \le \prod_{k<K} \sum_X e^{-\theta_k |X|} \le \exp(\sum_{k<K} N_k e^{-\theta_k})$ by lemma F.122; collect into C_{fin}. □

Lemma F.124 (covering families; [2, (3.48)–(3.52)] repaired)**.** *Set $\mathcal{D}_0 := \mathcal{C}_0$ and, for $1 \le k \le K-1$, $\mathcal{D}_k := \{P_k(Q) : Q \in \mathcal{D}_{k-1}\} \cup \mathcal{C}_k$, $P_k(Q)$ the prime of the union of the L-blocks of $T^{(k-1)}$ meeting $N_{w_{k-1}}(Q)$, $w_{k-1} := 7(r_{k-1} + s_b) + L s_b$ (constructed on lifts; a box). Then for $k \le K-1$:* (i) *$X_k \subset \bigcup \mathcal{D}_k$;* (ii) *$|\mathcal{D}_k| \le \sum_{j \le k} |\mathcal{C}_j|$;* (iii) *every box of $\mathcal{D}_k$ has side $\le C_D^0 r_k$;* (iv) *$|\Lambda_i^{(k)c}| \le (C_D r_k)^d \sum_{j \le k} |\mathcal{C}_j|$ for $i \le 7$, and $|\Lambda_8^{(k)c}| \le (C_D^{(8)} r_k)^d \sum_{j \le k} |\mathcal{C}_j|$ with $C_D^{(8)} := C_D^0 + 16(1+\theta)$.*

Proof. (ii) is immediate. (i) Induction on k; at $k = 0$, $I_0 = T^{(0)}$. Let $z \in X_k$. If $z \in I_k$, $z \in Q(z) \in \mathcal{C}_k$. Otherwise $\operatorname{dist}(z, T^{(k)} \setminus \Lambda_7^{(k-1)'}) \le s_b - 1$, so by lemma F.117(d) there is z_1 with

$B(z_1) \subset \Lambda_7^{(k-1)c}$ and $\operatorname{dist}(z, z_1) \le s_b - 1$; by lemma F.117(b) at scale $k-1$ and the induction hypothesis every point u of $B(z_1)$ lies in $N_{7(r_{k-1}+s_b)}(Q)$ for some $Q \in \mathcal{D}_{k-1}$; lifting, a point of $B(z)$ is within $L(s_b-1)$ of u, so $B(z)$ meets $N_{w_{k-1}}(Q)$ and $z \in P_k(Q)$. (iii) A box born at scale j has side $s_j \le (2+3\theta)r_j$ (Q2); the union of the L-blocks meeting an interval of n points has at most $n + 2(L-1)$ points, so $s_{i+1} \le (s_i + 2w_i + 2L - 2)/L$, and $2w_i + 2L - 2 \le 14r_i + (16+2L)s_b \le \nu_L r_i$; iterating from $j < k$, $s_k \le L^{-(k-j)}s_j + \nu_L \sum_{i=j}^{k-1} L^{-(k-i)} r_i \le \nu'_L \sum_{i=j}^{k-1} L^{-(k-i)} r_i$, and $r_i \le (1+(k-i)\ln L)^{\varrho} r_k$ gives $s_k \le \nu'_L (S_L - 1) r_k$; at $k = j$, $2+3\theta \le C_D^0$. (iv) By lemma F.117(b) and (i), $\Lambda_i^{(k)c} \subset \bigcup_{Q\in\mathcal{D}_k} N_{i(r_k+s_b)}(Q)$, each of at most $(C_D^0 r_k + 2i(r_k+s_b))^d$ points; use (ii), with $i \le 7$, resp. $i = 8$. □

Theorem F.125 ([2, (3.42)] at $L = 5$)**.** *Let c_5, c_7, κ_0 be as in lemma F.115, let $c_8 \ge 0$ be free of k, ε, K, the torus and the history, and add to the left side of* [2, (3.42)] *the factor $\exp(\sum_{k<K} c_8 |\Lambda_8^{(k)c}|)$ (the counterterm moves, proposition F.126). Put $C_1^{(8)} := C_1 + c_8$, $\Theta_L^{(8)} := C_1^{(8)} (C_D^{(8)})^d R^d / (c_{\mathrm{box}} b_0^2 \ln L)$ and assume*

$$\text{(E.ent)} \qquad (1 + \ln(L/\varepsilon_0))^{2\bar{p}-2-\varrho d} \;\ge\; \Theta_L^{(8)}.$$

(a) *The (augmented) left side of* [2, (3.42)] *is $\le \exp(C_{\mathrm{fin}} |T^{(K)}|)$.* (b) *If (E.ent) holds with $2\Theta_L^{(8)}$, the part of the sum over the admissible sequences with some $X_k \ne \emptyset$ is $\le \exp(-c_{\mathrm{box}} p(t_{K-1})^2/2) \exp(C_{\mathrm{fin}} |T^{(K)}|)$.* (c) *Both bounds are uniform in ε, K, the torus and the history; (E.ent) is a cap on ε_0 fixed after d, a, L, M, R, b_0 and after c_5, c_7, c_8, in Bałaban's order of choices. With $c_8 = 0$ this is* [2, (3.42)] *as printed.*

Proof. (a) By proposition F.123 (with Ξ augmented by $\sum_{k<K} c_8 |\Lambda_8^{(k)c}|$, which is carried through its proof unchanged) it suffices that $\Xi(\Lambda) \le \sum_{j<K} c_{\mathrm{box}} p_j^2 |\mathcal{C}_j|$. By lemma F.117(e), $t_k^{\kappa_0} \le 1 \le \hat{l}_k$ and $\Lambda_7^{(k)c} \subset \Lambda_8^{(k)c}$, every coefficient in Ξ is $\le C_1^{(8)} \hat{l}_k$, so by lemma F.124(iv) and an exchange of sums, $\Xi \le C_1^{(8)} \sum_{k<K} \hat{l}_k |\Lambda_8^{(k)c}| \le C_1^{(8)} (C_D^{(8)})^d R^d \sum_{j<K} |\mathcal{C}_j| \sum_{k=j}^{K-1} \hat{l}_k^{1+\varrho d}$. $\hat{l}_k$ decreases in k and $K - j \le \log_L(\varepsilon_0/t_j) \le \hat{l}_j / \ln L$, so the inner sum is $\le \hat{l}_j^{2+\varrho d} / \ln L$ and $\Xi \le c_{\mathrm{box}} b_0^2 \Theta_L^{(8)} \sum_{j<K} \hat{l}_j^{2+\varrho d} |\mathcal{C}_j|$. For $j \le K-1$, $t_j \le \varepsilon_0/L$, so $\hat{l}_j \ge 1 + \ln(L/\varepsilon_0)$ and (E.ent) gives $\Theta_L^{(8)} \le \hat{l}_j^{2\bar{p}-2-\varrho d}$; hence $\Xi \le \sum_j c_{\mathrm{box}} b_0^2 \hat{l}_j^{2\bar{p}} |\mathcal{C}_j| = \sum_j c_{\mathrm{box}} p_j^2 |\mathcal{C}_j|$. (b) With $2\Theta_L^{(8)}$ the same chain gives $\Xi \le \frac{1}{2} \sum_j c_{\mathrm{box}} p_j^2 |\mathcal{C}_j|$; for a non-trivial sequence let j be the first index with $X_j \ne \emptyset$; by lemma F.117(f), $X_j \cap I_j = X_j$, so $|\mathcal{C}_j| \ge 1$, and $p_j \ge p_{K-1}$. (c) No step used a bound on L, ε, K, the torus or the history beyond the setting. □

Proposition F.126 (the counterterm moves in the ledger)**.** *Add at each step $k \le K-1$ the bounds of lemmas F.33 and F.34 to the exponent that* [2, (2.43), (2.115)] *carries into* [2, (2.117), (3.42)]*. Then the left side of* [2, (3.42)] *is bounded by theorem F.125(a), and its non-trivial part by* (b)*, with $c_8 := Q_1(C_{\mathrm{D22}} + C_{\mathrm{D23}})$, C_{D22}, C_{D23} the constants of those lemmas and Q_1 absorbing the factor $q(t_k)$ of a $\delta m^2_{(4)}$ vertex (lemma F.101), and the t^κ parts in the bulk sum* [2, (2.118)]*: $C_\kappa \sum_k t_k^\kappa |T^{(k)}| \le C_\kappa \varepsilon_0^{\kappa-d} (1 - L^{d-\kappa})^{-1} |T|$ for $\kappa > d$.*

Proof. $\Lambda_7^{(k)} \setminus \Lambda_8^{(k)} \subset \Lambda_8^{(k)c}$ and $T_1^{(k)} \setminus \Lambda_8^{(k)} = \Lambda_8^{(k)c}$, and $t_k^{1/2-\beta} \le 1$, $t_k q(t_k) \le Q_1$, so the moves are $\le c_8 |\Lambda_8^{(k)c}| + C_\kappa t_k^\kappa |T^{(k)}|$: the first part is a term of the augmented Ξ, the second a bulk term; they enter [2, (2.117)] with the same sign and in the same place as the collar terms, and lemma F.113 carries them through [2, Sect. 3A]. □

The g_k-localisation of $Q^{(k)}$ [2, p. 579].

Proposition F.127 (the terms of $Q^{(k)}$ removed by g_k)**.** *Let $1 \le k \le K-1$, $\Omega_c := B^k(\Lambda_0^{(k)})$, $\Phi := \theta_{k+1} A'^{(k)}$, and $Q^{(k)}[Y] := \sum_{j<k} \sum_{n=1}^{\bar{n}} \frac{1}{n!} \frac{d^n}{d\tau^n} F_j(\Omega_c; \tilde{B} + \tau Y)|_{\tau=0}$* [2, (2.105)–(2.106)]*, in the setting of theorem F.53. Let $c_g := \sup |\partial^\eta g_k|$ ($\operatorname{supp} g_k \subset \Lambda_7^{(k)}$, $g_k = 1$ on $\Lambda_8^{(k)}$). Then $|Q^{(k)}[\Phi] - Q^{(k)}[g_k \Phi]| \le F_H(t_k)(2M+1)^3 |\Lambda_8^{(k)c}|$ with $F_H(t) := C_H t p(t)^2 (1 + c_g)^2$, bounded on*

$(0,1]$ *and free of* ε, k, ε_0, *the torus and the history. The other two replacements of* [2, (2.107)] *(the background* $\tilde B \to B^{(k+1),\eta}$ *and the domain* $\Omega_c \to B^k(\Lambda_2^{(k)})$*) are* $O(t^\kappa)|\Lambda_7^{(k)}|$ *for every* κ.

Proof. Fix j and y; put $\Phi_1 := g_k\Phi$, $\Phi_2 := (1-g_k)\Phi$, $\operatorname{supp}\Phi_2 \subset \operatorname{supp}\theta_{k+1} \setminus B^k(\Lambda_8^{(k)})$. For each n, $D^nQ_y(\tilde B)[\Phi^n] - D^nQ_y(\tilde B)[\Phi_1^n]$ is a sum of n terms with at least one slot Φ_2 (multilinearity). Gauge-fix at y as in lemma F.51 (Q_y is gauge invariant, lemma F.50(b), and the axial map is linear). For $n=1$ use lemma F.50(f), which gives $D^2Q_y(s\beta)[\beta,\alpha_{\Phi_2}]$, bilinear in the curvature amplitudes of $\tilde B$ and Φ_2; for $n\ge 2$ use lemma F.50(e): each term is bounded by products of n local amplitudes, one of them that of Φ_2, plus terms suppressed by $e^{-\theta_2 R/(4n)}$ with $R := R_{kj}$ as in theorem F.53. Curvatures in scale-k units: $|\partial\Phi| \le Cp$ (the derivative clause of [2, (2.91)], Step 3(C_1): $\partial\Phi = (\partial\theta_{k+1})A'^{(k)} + \theta_{k+1}\partial A'^{(k)}$ with $|A'^{(k)}| \le C_{2.91}p$ by lemma F.46, $|\partial\theta_{k+1}| \le C$, and $\partial A'^{(k)} = \partial A^{(k)} - \partial B^{(k+1)} - \partial\tilde B^{(k)}$ by [2, (2.89)], bounded by (R1), [2, (2.98)] and (R2) of Step 3(C_2)), $|\partial\Phi_2| \le |\partial\Phi| + 2c_g|\Phi| \le C(1+c_g)p$, and on N_k, $\tilde B = A^{(k)} - \Phi - \tilde B^{(k)}$ (lemma F.52(a)) so $|\partial\tilde B| \le Cp$ by (R1), (R2); at scale j all three amplitudes are $\le C(1+c_g)L^{-2(k-j)}t_k^{1/2}p$. Per y the main terms are $\le C(1+c_g)^2L^{-4(k-j)}t_kp^2$, and the remaining terms are as in the proof of theorem F.53; the y within R of $\operatorname{supp}\Phi_2$ number $\le (2R+3)^3(2M+1)^3L^{3(k-j)}|\Lambda_8^{(k)c}|$; sum over y and over $j<k$ as there. The other two replacements act at distance $\ge 5L^{k-j}r_k$ (scale-j units) from $\operatorname{supp} g_k\Phi$, and lemma F.50(e) with one slot carrying the background or domain change gives $\le Ce^{-\theta_2 L^{k-j}r_k/(2\bar n)}$ per y (for the domain change, compare F_j on the two domains through the common sub-box; the difference is localised at the domain boundary by [1, (2.29)] and lemma F.50(d)). □

The collar: the field-dependent part of [2, Prop. 2.1, (2.57)]. *Setting.* Step k in scale-k units ($t := t_k$, $\eta := L^{-k}$, e_k, λ_k, $p = p(t)$, $r = r(t)$); $\Omega_A := B^k(\Lambda_2^{(k-1)'})$, $\Omega_1 := B^k(\Lambda_7^{(k-1)'})$, $\Omega_1' := B^k(\Lambda_7^{(k)})$, $\mathrm{Col} := \Omega_1 \setminus \Omega_1'$, $|\mathrm{Col}| = |\Lambda_7^{(k-1)'} \cap \Lambda_7^{(k)c}|$ unit cubes; $\tilde B := A^{(k)}$, which equals $\theta_kA^{(k)}$ on Ω_A; $\Phi := \phi^{(k)} = a_kG_k(\Omega_A,\tilde B)Q_k^*(\tilde B)\Lambda_6^{(k-1)'}\varphi$ [2, (2.56)]; $c_k(\Omega')$ the field-independent part of $P^{(k)}(\Omega',\tilde B,\varphi)$. Under [2, (2.55)], [2, Prop. 2.1, (2.57)] asserts $P^{(k)}(\Lambda_7^{(k-1)'},\theta_kA^{(k)},\varphi) = -\lambda_k\sum_{x\in B^k(\mathrm{Col})}\eta^d|\Phi(x)|^4 + P^{(k)}(\Lambda_7^{(k)},\theta_kA^{(k)},\varphi) + O((L^k\varepsilon)^{\kappa_0})|\mathrm{Col}|$, "a corollary of the analysis of the perturbation expansions" (the sketch is [3, p. 421]). Inputs: Proposition 1 with the vertex sizes of theorem F.99; [1, (2.24)–(2.25)] at points of Ω_1 (depth $\ge r-2 \ge R_0$); [2, (2.55), (2.60)] and the Hölder clause of [3, p. 412] for $\tilde B$; [2, Lemma 2.4, (2.65)–(2.66)] at every $y \in \Lambda_7^{(k-1)'}$ (theorem F.2(c)); the slice kernels of lemma F.88 at $A=0$; the layer distances $\operatorname{dist}(\Lambda_{i+j}^{(k)},\Lambda_i^{(k)c}) \ge j(r-1)$, $\operatorname{dist}(\Omega_1,\Omega_A^c) \ge r-2$.

Theorem F.128 (the collar quartic and its remainder)**.** $P^{(k)}(\Omega_1) - P^{(k)}(\Omega_1') = -\lambda_k\sum_{x\in\mathrm{Col}}\eta^3|\Phi(x)|^4 + R_{\mathrm{field}} + [c_k(\Omega_1) - c_k(\Omega_1')]$ *with* $|R_{\mathrm{field}}| \le C_\beta t^{1/2-\beta}|\mathrm{Col}|$ *for every* $\beta>0$ *and* $|c_k(\Omega_1) - c_k(\Omega_1')| \le Ct(1+\ln t^{-1})^5|\mathrm{Col}| \le C_\beta t^{1-\beta}|\mathrm{Col}|$, *under the caps of Step 3(*D_2*) (Geometry). So* [2, (2.57)] *holds with* $\kappa_0 = \frac12 - \beta$, *which implies the printed* $\frac{4-d}{4} - \beta = \frac14 - \beta$. *The constants depend on* $(d, L, M, \alpha_0, \bar n, a, c_1, b_0, \bar p, R, \varrho, e, \lambda, N, \beta)$, *on the* $O(1)$ *of Proposition 1 and on* c_0, δ_0, R_0 *of* [1, (2.24)]*; they are free of* ε, k, ε_0, *the sets and* φ.

The proof is lemmas F.129 to F.135.

Lemma F.129 (how Ω_1 enters)**.** *In* [3, (1.4)–(1.5)], Ω_1 *enters only through:* (i) $\mathbf{1}_{\Omega_1}$ *in the sums of the vertices* [3, (1.6)–(1.7)]; (ii) g_k *(*$\operatorname{supp} g_k \subset \Omega_1$, $g_k = 1$ *at distance* $\ge M$ *from* $\partial\Omega_1$, *derivatives up to order* 2 *bounded by* 1*) on every* A'*-leg, through* $e'g_kA' + A^{(k)}$, *the* A'*-legs of the pair vertices* [3, (1.14)] *included;* (iii) $\delta m_k^2(e'e_kg_k, \lambda'\lambda_k, \Omega_1, x)$ *and* $E_k(\dots,\Omega_1)$ *of* [3, (1.29)–(1.30)], *sums of graphs whose internal vertices are localised by* (i)–(ii) *(*[3, pp. 419–420]*: "We localize the vertices (1.6) and (1.7) restricting summations to the set* Ω_1. *The other vertices are localized by functions* g_k*"; "We perform this localization for the terms of* δm_k^2, E_k *also"). The Gaussian measures, the propagators* $G_k(\Omega_A,\tilde B)$ *and the external fields* Φ, φ *are* Ω_1*-free.*

Lemma F.130 (marking). *Put* $D_{\text{good}} := \{x \in \Omega_1' : \operatorname{dist}(x, \partial\Omega_1') \geq M+2\}$ *and* $\widetilde{\text{Col}} := \Omega_1 \setminus D_{\text{good}}$, *and localise every graph in unit cubes* [3, p. 420]. (a) *A localised family all of whose cubes (counterterm-internal vertices included) lie in* D_{good} *has the same value for* Ω_1 *and* Ω_1'. (b) $P(\Omega_1) - P(\Omega_1')$ *is the sum over the localised families with a cube meeting* $\widetilde{\text{Col}}$ *of* $E(\cdot;\Omega_1) - E(\cdot;\Omega_1')$. (c) $|\widetilde{\text{Col}}| \leq (1 + (2M+7)^d)|\text{Col}|$.

Proof. $\operatorname{dist}(\Omega_1', \Omega_1^c) \geq 7(r-1) - 1 > M + 3$, so $\partial\Omega_1'$ faces only Col. On D_{good} (with the side-2 supports of the leg partitions) $\mathbf{1}_{\Omega_1} = \mathbf{1}_{\Omega_1'} = 1$ and $g^{\Omega_1} = g^{\Omega_1'} = 1$; lemma F.129 for the counterterms. (c): every unit block of $\Omega_1' \setminus D_{\text{good}}$ is within $M+3$ of a block of Col, and each block of Col is so near to at most $(2M+7)^d$ of them. □

Lemma F.131 (external fields). *For localisations* h *at cubes in* Ω_1*:* $\|h\Phi\|_{1,\alpha_0} \leq C_\Phi p \lambda_k^{-1/4}$, *and the block fields* [3, (1.12)] *obey* $|\varphi(y)| \leq C_\Phi p \lambda_k^{-1/4}$ *for* $y \in \Lambda_7^{(k-1)'}$, *with* $C_\Phi := 3C_h a c_0 c_1 L^{1/4}(1 + \ln L)^{\bar{p}}$ *(*C_h *the localisation constant,* 3 *for the three parts of* [3, (1.32)]*). There are no external vector fields (*$n' = 0$ [3, p. 414]*; every* A'*-leg is contracted).*

Proof. $\Phi = a_k G_k(\Omega_A, \tilde{B}) f$ with $f := Q_k^*(\tilde{B})\Lambda_6'\varphi$, $\|f\|_\infty \leq \sup_{\Lambda_6'} |\varphi| \leq c_1 p(t/L)\lambda(t/L)^{-1/4}$ by [2, (2.55)] ($\Lambda_6' \subset \Lambda_{-1}'$); $p(t/L) \leq (1 + \ln L)^{\bar{p}} p(t)$, $\lambda(t/L) = \lambda_k/L$; apply [1, (2.24)–(2.25)] at points of Ω_1, at depth $\geq r - 2 \geq R_0$ in Ω_A, $\tilde{B}$ regular by [2, (2.60)]. □

Lemma F.132 (census and parity). *The expressions of* $P^{(k)}$ *are the vertices* [3, (1.6), (1.7), (1.8), (1.10)] *and the pairs from* [3, (1.12)–(1.14)]*, with* $n' = 0$ *(*[3, p. 414]*); a pair is the scalar product of two of* [3, (1.12)–(1.14)]*, at least one of the form* [3, (1.14)]*, with factor* $-a_k$ *or* $-\frac{1}{2}a_k$*, and* [3, (1.12)] *is an external field; the R-vertices do not occur (*[3, (1.5)] *has order* $\leq \bar{n}$*). Let* $s(v)$ *be the number of scalar legs of* v *(*[3, (1.12)] *counted) and* $X(v) := 2d_v(v) + 4d_s(v) - s(v)$*:* $X = 0$ *for* [3, (1.6)]*;* $X \geq 2$ *for* [3, (1.7)] *(coefficient of order* ≥ 2*);* $X = 2n - 2$ *for* [3, (1.8)] *and* [3, (1.10)] *at order* n*;* $X = 2m - 2$ *for a pair,* $m \geq 1$ *the total order of its* [3, (1.14)] *members. For a connected graph* G *with* l_s *internal scalar lines and* n_{ext} *external scalar legs,* $W(G) := 2d_v + 4d_s - n_{\text{ext}} = \sum_v X(v) + 2l_s$ *is even and* ≥ 0*, and* $W = 0$ *exactly for the lone* [3, (1.6)] *vertex with four external legs and for two order-*1 *objects (*[3, (1.8)] *with* $n = 1$*, or a pair* [3, (1.12)]·[3, (1.14)]$_1$, [3, (1.13)]·[3, (1.14)]$_1$ *with both scalar legs external) joined by one vector line.*

Proof. $\sum_v s(v) = n_{\text{ext}} + 2l_s$. $W = 0$ forces $l_s = 0$ and $X(v) = 0$ for every v; the $X = 0$ objects carry 0 or 1 A'-leg, all A'-legs are contracted and $P^{(k)}$ is connected, and a connected graph with vector lines only and at most one A'-leg per vertex has at most two vertices. □

Lemma F.133 (the families with $W \geq 2$). *For a localised family with* $W(G) \geq 2$*, at* $\Omega' \in \{\Omega_1, \Omega_1'\}$*:* $|E(G_{\text{ren}}, \{\square\}; \Omega')| \leq O(1) e^{d_v} \lambda^{d_s - n_{\text{ext}}/4} C_\Phi^{n_{\text{ext}}} p^{n_{\text{ext}}} t^{W/4} e^{-\delta_0 d(\{\square\})}$*; summed over the marked families of lemma F.130, this part of* R_{field} *is* $\leq C_\beta t^{1/2-\beta}|\text{Col}|$.

Proof. Proposition 1 with the external scalar fields of lemma F.131 and no external vector field; the background satisfies the hypotheses of [3, p. 412] ($|\partial\tilde{B}| \leq O(p)$ by [2, (2.60)], its Hölder clause), there is no $\tilde{A}$, and its $O(1)$ is independent of ε, k, the domains and $\tilde{B}$ [3, p. 421]. In $d = 3$, $e_k^{d_v} \lambda_k^{d_s} (p\lambda_k^{-1/4})^{n_{\text{ext}}} = e^{d_v} \lambda^{d_s - n_{\text{ext}}/4} p^{n_{\text{ext}}} t^{W/4}$, and $W \geq 2$ gives $t^{1/2}$; $n_{\text{ext}} \leq C(\bar{n})$, so $p^{n_{\text{ext}}} \leq C_\beta t^{-\beta}$; the [3, (1.7)] vertices with the finite parts δm^2_{fin}, $\delta m^2_{K,k}$ enter through theorem F.99 and lemma F.101. Root the tree at a marked cube ($\leq |\widetilde{\text{Col}}|$ choices, including the localised internal vertices of δm_k^2, E_k, lemma F.129(iii)) and sum the rest by lemma F.26; there are finitely many families of order $\leq \bar{n}$. □

Lemma F.134 (the terms with $W = 0$). (a) *The lone* [3, (1.6)] *vertex contributes to* $P(\Omega_1) - P(\Omega_1')$ *exactly* $-\lambda_k \sum_{x \in \text{Col}} \eta^3 |\Phi(x)|^4$ *(its sum is sharp over* Ω_1*,* [3, p. 413]*).* (b) *The single-line terms contribute* $\leq C_{\text{SL}} e^2 \lambda^{-1/2} p^4 t^{1/2}$ *per marked cube, hence* $\leq C t^{1/2-\beta}|\text{Col}|$ *in all.*

Proof of (b). The order-1 objects (both scalar legs external) are $V_8 = -e_k \sum_b \eta^3 [(D_{\tilde{B}}\Phi)(b) \cdot q\Phi(b_-)] g_k A_b'$ from [3, (1.8)]; $P_{12}(y) = a_k e_k \varphi(y) \cdot \sum_{x \in B^k(y)} \eta^3 A'(\Gamma_{y,x}^{(k)}) q U(\tilde{B}(\Gamma_{y,x}^{(k)}))\Phi(x)$ from

[3, (1.12)]·[3, (1.14)]$_1$; and $P_{13}(y)$, which is P_{12} with $\varphi(y)$ replaced by $-(Q_k(\tilde B)\Phi)(y)$. The weight g_k forces $y \in \Lambda_7^{(k-1)'}$. By [2, Lemma 2.4, (2.65)] at every $y \in \Lambda_7^{(k-1)'}$, $\Phi(x) = U(\tilde B(\Gamma^{(k)}_{x,y}))\varphi(y)+\mathrm{e}(x)$ with $|\mathrm{e}(x)| \le c_{65}p$; $U(\tilde B(\Gamma_{y,x}))U(\tilde B(\Gamma_{x,y})) = 1$ and $\varphi(y)\cdot q\varphi(y) = 0$ (q real antisymmetric), so the leading part of P_{12} vanishes and its coefficient is $\le ae_kC_\Phi p\lambda_k^{-1/4}c_{65}p$; the same for P_{13}, since $(Q_k(\tilde B)\Phi)(y) = \varphi(y) + O(p)$; for V_8, [2, (2.66)] gives $|D_{\tilde B}\Phi| \le c_{66}p$ on $B^k(\Lambda_7^{(k-1)'})$, so its coefficient is $\le e_kc_{66}pC_\Phi p\lambda_k^{-1/4}$. The vector line: $A' = \sum_j A'^{(j),\eta}$ with independent Gaussian A'_j of covariance $C^{(j)}$ [3, p. 414], and $A'(\Gamma^{(k)}_{y,x}) = \sum_j A'^{(j),\eta}(\Gamma^{(j+1)}_{x_{j+1},x})$ [3, (1.3), p. 412]; its covariance is slice-diagonal, $G_1^v + \sum_{j\ge1} S_j^v$ in scale-k units, with $|S_j^v(b,b')| \le Cl_j^{2-d}e^{-\delta_K|b-b'|/l_j}$, $l_j := L^{j-k}$ (lemma F.88(b),(d) rescaled; the $j=0$ term with its L-dependent constant), so $\sum_{b'}\eta^3|S_j^v(b,b')| \le Cl_j^2$, and the slice-$j$ contour $\Gamma^{(j+1)}$ has η-length $\le 1.25dLl_j$. The three pairings per marked cube: (V_8,V_8): $\le C(e_kp^2\lambda_k^{-1/4})^2\sum_j l_j^2 \le Ce^2\lambda^{-1/2}p^4t^{1/2}$; (V_8,P): $\sum_j(1.25dLl_j)Cl_j^2 \le C$, the same order; (P,P'): per pair of blocks $\sum_j(1.25dLl_j)^2Cl_j^{2-d}e^{-\delta_K\,\mathrm{dist}\,/l_j} = \sum_j C'l_je^{-\cdots}$ ($d=3$), and the sum over the second block with the decay is $\le C$ (the slice-j field is integrated only over a contour of length $O(l_j)$: no logarithm; the P-side bonds are summed through the η^3-weighted $x \in B^k(y')$, $x \mapsto b'$ being $O(1)$-to-one). Sum the marked cubes over $\widetilde{\mathrm{Col}}$. □

Lemma F.135 (the c-number clause)**.** (i) *At zero background on the torus, $c_k(\Omega_1) = -(E_{\mathrm{fin}} + E_{K,k}) + a_k(\Omega_1)$ (lemma F.102), so the collar difference within the vertex set of E_1 is exactly 0 at every order $\le \bar n$.* (ii) *At $(\Omega_A,\tilde B)$: $|c_k(\Omega_1) - c_k(\Omega_1')| \le Ct(1+\ln t^{-1})^5|\mathrm{Col}| \le C_\beta t^{1-\beta}|\mathrm{Col}|$.*

Proof. (ii) By lemma F.130 the difference is a sum over localised vacuum families with a cube meeting $\widetilde{\mathrm{Col}}$ (the vacuum graphs with their E-partners, [3, pp. 420, 422], and the pair-vertex vacuum graphs). Proposition 1 with no external fields gives $O(1)e_k^{d_v}\lambda_k^{d_s}e^{-\delta_0 d}$; a vacuum graph has $d_v \ge 2$ or $d_s \ge 1$ or contains a [3, (1.7)] vertex of order ≥ 2 (every expression of $P^{(k)}$ has order ≥ 1 in e unless it is [3, (1.6)] or [3, (1.7)]; each A'-leg carries one power of e; A'-legs are contracted in pairs), so $W = 2d_v + 4d_s \ge 4$ and $t^{W/4} \le t$, times $q(t) = (1+\ln t^{-1})^5$ per $\delta m^2_{(4)}$ vertex for the finite counterterm parts (theorem F.99, lemma F.101); no external field, so no power of p. Sum by lemma F.26 over roots in $\widetilde{\mathrm{Col}}$. At zero background this is corollary F.29. □

Proof of theorem F.128. Lemma F.130(b) splits the difference into the marked families; those with $W \ge 2$ are lemma F.133, those with $W = 0$ are lemma F.134, and the field-independent ones are lemma F.135. ε_0 enters only through the caps of Step 3(D$_2$) and the "$e(t)$ sufficiently small" of [1, (2.24)] and of Proposition 1; the torus uniformity of Proposition 1's $O(1)$ is the torus side of the propagator estimates of group (A). □

Remark F.136. The second sentence of [2, p. 579] bounds the terms localised in $B^k(\Lambda_7^{(k)c})$ by $O((L^k\varepsilon)^{\kappa_0})|\Lambda_2^{(k)} \cap \Lambda_7^{(k)c}|$: for $Q^{(k)}$ this is proposition F.127; for the terms of $P^{(k)}(\Lambda_7^{(k)}, \theta_kA^{(k)}, \varphi)$ whose vector leg is gated by g_k there it is proposition F.139 below, with $\kappa_0 = \frac12 - \beta$ and the part of the gate inside $\Lambda_7^{(k)} \setminus \Lambda_8^{(k)}$ booked as in corollary F.140; the field-dependent part of the final-step bound [2, (2.116)] is proposition F.141. Both run the mechanism of theorem F.128: the gated leg, resp. the whole final domain, takes the place of the collar vertex.

The g_k-gated terms of $P^{(k)}$ and the final step [2, p. 579, (2.116)]. *Setting.* Step k, $1 \le k \le K-1$, in scale-k units, with the Setting of theorem F.128 (Ω_A, Ω_1, $\Omega_1' = B^k(\Lambda_7^{(k)})$, $\mathrm{Col} = \mathrm{Col}_k$, $\Phi = \phi^{(k)}$, $p = p(t_k)$, $r = r(t_k)$). At step $k+1$ [2] writes $\theta_{k+1}A^{(k)} = \theta_{k+1}A'^{(k)} + \tilde B^{(k)} + \theta_{k+1}B^{(k+1),\eta}$ [2, (2.88)–(2.89)], with $A'^{(k)} = a_k\zeta^{(k)}G_kQ_k^*A'$ (lemma F.46), θ_{k+1} the smooth cutoff of $B^k(\Lambda_2^{(k)})$ changing in a slice of thickness $< M$ [2, p. 567], and every term of $\tilde B^{(k)}$ carrying the factor θ_{k+1}. Put

$$\tilde B := (1-\theta_{k+1})\theta_kA^{(k)} + \theta_{k+1}B^{(k+1),\eta}, \qquad a := \theta_{k+1}A'^{(k)} + \tilde B^{(k)}, \qquad \text{so that} \qquad \theta_kA^{(k)} = \tilde B + a,$$

since $\theta_k = 1$ on $\operatorname{supp}\theta_{k+1} \subset B^k(\Lambda_7^{(k-1)'}) \subset B^{k-1}(\Lambda_2^{(k-1)})$ (lemma F.52); $\tilde{B}$ is the background of [2, p. 582], regular by [2, (2.94)–(2.98)], and $\operatorname{supp} a \subset \operatorname{supp}\theta_{k+1}$, which lies within M of $B^k(\Lambda_2^{(k)})$, hence in $B^k(\Lambda_1^{(k)})$ ($\operatorname{dist}(\Lambda_2^{(k)}, \Lambda_1^{(k)c}) \geq r - 1 > M + 1$ under the caps of Step 3(D$_2$)). The gate is g_k ($\operatorname{supp} g_k \subset \Lambda_7^{(k)}$, $g_k = 1$ on $\Lambda_8^{(k)}$, $|\partial^\eta g_k| \leq c_g$, as in proposition F.127). The cutoffs are taken, as the printed "smooth" allows, from fixed classes: θ_{k+1} a fixed smooth function of the mollified distance to $B^k(\Lambda_2^{(k)})$, so that $|\partial^\eta \theta_{k+1}| \leq c_\theta$ and $|\partial^\eta \partial^\eta \theta_{k+1}| \leq c_{\theta 2}$ uniformly in k and η (a smooth function of the lattice sup-distance would have second differences of order η^{-1} along the edges of the slice); and $\zeta^{(k)}$ with $|\zeta^{(k)}| \leq 1$, $|\partial_x^\eta \zeta^{(k)}| \leq 1$, $|\partial_x^\eta \partial_x^\eta \zeta^{(k)}| \leq 2$ (the cut-off of theorem F.2(b) is so chosen: a fixed smooth profile of the mollified distance to y, as for θ_{k+1}, over its transition region of width $\frac{1}{5}r - 2M$, which is ≥ 2 for $r \geq 10M + 10$, a cap on ε_0 free of k). Put $\delta' := \min(\delta_*, \delta_0)$, $S'_* := \sum_{n \in \mathbb{Z}^3} e^{-\delta'|n|/2}$, and $C'(\delta')$ for the constant C' of lemma F.46 with δ' in place of δ_*.

Lemma F.137 (the field a)**.** *On* $\operatorname{supp}\theta_{k+1}$*, for* $t_k \leq t_0$*, where* $t_0 \in (0, 1]$ *is fixed by* $C_\kappa t_0^\kappa \leq C_{2.91} b_0$*,* $C'_\kappa t_0^\kappa \leq b_0$ *and* $C_\kappa^H t_0^\kappa \leq b_0$ *(*C_κ*,* C'_κ *the constants of (R2) of Step 3(*C_2*),* C_κ^H *that of (c) below; a cap on* ε_0*, free of* k*;* $p(t) \geq b_0$*):* (a) $|a| \leq C_{2.91} p + C_\kappa t_k^\kappa \leq 2C_{2.91} p$*;* (b) *for every fine bond* $b = (x, \mu)$ *with* x *or* $x + \eta e_\mu$ *in* $\operatorname{supp}\theta_{k+1}$*,* $|\partial_\mu^\eta a(x)| \leq C'_{2.91} p + C'_\kappa t_k^\kappa \leq (C'_{2.91} + 1)p$*, with*

$$C'_{2.91} := c_\theta C_{2.91} + a(C_1 e^{\delta_*} + c_0 e^{\delta_0}) S'_* (C_{2.86} + C'(\delta'));$$

(c) *for parallel bonds* b, b' *within distance* M *of* $\operatorname{supp}\theta_{k+1}$ *with* $|b - b'| \leq 1$*,* $|\partial^\eta a(b) - \partial^\eta a(b')| \leq C_{2.91}^H p\, |b - b'|^{\alpha_0}$*. The constants are free of* k*,* ε*,* ε_0*, the torus and the history. So* a *satisfies the hypotheses on an external vector field of* [3, p. 412] *with* $\|\cdot\|_{1,\alpha_0}$*-norm* $\leq C_a p$*,* $C_a := 2C_{2.91} + C'_{2.91} + 1 + C_{2.91}^H$.

Proof. (a) is lemma F.46 for $\theta_{k+1} A'^{(k)}$ and (R2) of Step 3(C$_2$) for $\tilde{B}^{(k)}$ [2, (2.90)]; $C_\kappa t^\kappa \leq C_{2.91} p$ for $t \leq t_0$.

(b) *Lattice distances.* Fine points are at mutual distances in $\eta\mathbb{Z}$, and the slice has thickness $< M$, so $\operatorname{supp}\theta_{k+1}$ lies within $M - \eta$ of $B^k(\Lambda_2^{(k)})$ and both x and $x + \eta e_\mu$ lie within M of it; lemma F.46 and its proof apply at both points (the proof bounds $A'^{(k)}$ itself there, without the factor θ_{k+1}). *Leibniz rule.* $\partial_\mu(\theta_{k+1} A'^{(k)})(x) = (\partial_\mu \theta_{k+1})(x) A'^{(k)}(x + \eta e_\mu) + \theta_{k+1}(x)(\partial_\mu A'^{(k)})(x)$. The first term is $\leq c_\theta C_{2.91} p$. In the second,

$$\partial_\mu A'^{(k)}(x) = a_k \sum_y \left[(\partial_{\mu,x} \zeta^{(k)})(x, y)(G_k Q_k^*)(x + \eta e_\mu, y) + \zeta^{(k)}(x, y)(\partial_\mu G_k Q_k^*)(x, y)\right] A'(y);$$

both kernels are $\leq (C_1 e^{\delta_*} + c_0 e^{\delta_0}) e^{-\delta'|x-y|}$, by $|\partial\zeta| \leq 1$, theorem F.7 ($|\cdot|_1 \geq |\cdot|$ and one bond shift) and the derivative clause of [2, Prop. 2.2, (2.58)] ([4, Thm., (1.9)–(1.10)], at zero background on the torus, where the depth restriction is void, lemma F.154); both vanish unless $|x - y| < r - 2M + \eta$, and such y lie in $\Lambda_0^{(k)}$ by the layer count of lemma F.46 ($2(r - c_b) > r - 2M + M + 1$). Split the y-sum as there: on $\Lambda_1^{(k)}$, lemma F.45 gives $\leq (\dots) S'_* C_{2.86} p$; on $\Lambda_0^{(k)} \setminus \Lambda_1^{(k)}$, $|x - y| \geq r - c_b - M - 1$ and the factor $e^{-\delta'|x-y|}$ with $E(1, \delta'/2)$ turn $p/(\mu_0 t)$ into $C'(\delta')p$. With $|\theta_{k+1}| \leq 1$ and $a_k \leq a$ the second term is $\leq a(C_1 e^{\delta_*} + c_0 e^{\delta_0}) S'_* (C_{2.86} + C'(\delta'))p$. For $\tilde{B}^{(k)}$ the derivative bound is (R2).

(c) Write $\partial(\theta_{k+1} A'^{(k)}) = (\partial\theta_{k+1})\,\mathsf{S} A'^{(k)} + \theta_{k+1} \partial A'^{(k)}$ (S the bond shift) and telescope the difference at b, b' term by term. The factors $\partial\theta_{k+1}$, θ_{k+1} and the kernel factor $\partial\zeta^{(k)}$ are Lipschitz on the η-lattice with constants $c_{\theta 2}$, c_θ and 2; at separation $|b - b'| \geq \eta$ a Lipschitz bound is $\leq$ the constant times $|b - b'|^{\alpha_0}$, since $|b - b'| \leq 1$; their cofactors ($A'^{(k)}$, $\partial A'^{(k)}$ and the kernel sums) are bounded by lemma F.46 and by (b), the kernel $(G_k Q_k^*)(\cdot + \eta e_\mu, y)$ of the $\partial\zeta^{(k)}$ term being itself Lipschitz in x with constant $c_0 e^{\delta_0} e^{-\delta_0 \operatorname{dist}(\cdot, y)}$ by the derivative clause of [2, (2.58)] (a difference along a lattice path of length $|b - b'| \leq 1$), summed over y as in (b). The θ-factors are telescoped first, so each cofactor is either evaluated at a bond of $\operatorname{supp}\theta_{k+1}$ or is a difference of two kernel sums of the type of (b) at points within $M + 1$ of $B^k(\Lambda_2^{(k)})$; the bounds of (b) hold there, since the kernel bounds are global on the torus and the layer

count with $M+1$ in place of M keeps the y-range in $\Lambda_0^{(k)}$. The term $\zeta^{(k)}(\partial G_k Q_k^*)A'$: its variation in b is the Hölder clause of [1, Prop. 2.1, (2.24)] ([4, Thm., (1.10)]) for $\partial G_k Q_k^* \delta_y$, $|(\partial G_k Q_k^*)(b,y) - (\partial G_k Q_k^*)(b',y)| \le c_0 |b-b'|^{\alpha_0} e^{-\delta_0 \min(\mathrm{dist}(b,y),\mathrm{dist}(b',y))}$; sum over y as in (b). For $\tilde{B}^{(k)}$: by [2, (2.89)] each of its three terms is θ_{k+1} times a kernel sum of the same type ($\zeta^{(k)} G_k Q_k^*$, its tail $(1-\zeta^{(k)})G_k Q_k^*$, or $\zeta^{(k)} G_k Q_k^*$ composed with the covariance difference of lemma F.147) applied to a block field bounded by $Cp/(\mu_0 t)$ on $\Lambda_0^{(k)'}$ (the restrictions [2, (2.17), (2.55)]); the same computation bounds its Hölder quotient, and the separation $\ge r - 2c_b - M$ between the evaluation points in $\operatorname{supp}\theta_{k+1}$ and the support of the field, through $E(1,\delta/2)$, gives $C_\kappa^H t_k^\kappa$ in place of p, as in (R2). □

Lemma F.138 (the gate)**.** *Let* Bad_k *be the set of scale-k unit cubes meeting* $\operatorname{supp}((1-g_k)a)$. *Then* $\mathrm{Bad}_k \subset N_M(B^k(\Lambda_2^{(k)})) \setminus B^k(\Lambda_8^{(k)}) \subset \Omega_1$, *every cube of* Bad_k *lies within M of a cube of* $\Lambda_2^{(k)} \cap \Lambda_8^{(k)c}$, *and*

$$|\mathrm{Bad}_k| \le (2M+1)^3 |\Lambda_2^{(k)} \cap \Lambda_8^{(k)c}| \le (2M+1)^3 \big(|\Lambda_2^{(k)} \cap \Lambda_7^{(k)c}| + |\Lambda_7^{(k)} \setminus \Lambda_8^{(k)}|\big), \qquad \Lambda_2^{(k)} \cap \Lambda_7^{(k)c} \subset \mathrm{Col}_k.$$

Proof. $(1-g_k)a$ vanishes outside $\operatorname{supp}\theta_{k+1} \subset N_M(B^k(\Lambda_2^{(k)}))$ and on $B^k(\Lambda_8^{(k)})$, where $g_k = 1$. Let $y \in \mathrm{Bad}_k$ and let x be a cube of $B^k(\Lambda_2^{(k)})$ within M of y. If $y \in \Lambda_2^{(k)}$, then $y \in \Lambda_2^{(k)} \cap \Lambda_8^{(k)c}$ itself. If $y \notin \Lambda_2^{(k)}$, then $\mathrm{dist}(x, \Lambda_2^{(k)c}) \le M+1 < 6(r-1) \le \mathrm{dist}(\Lambda_8^{(k)}, \Lambda_2^{(k)c})$, so $x \notin \Lambda_8^{(k)}$ and $x \in \Lambda_2^{(k)} \cap \Lambda_8^{(k)c}$. A cube of $\Lambda_2^{(k)} \cap \Lambda_8^{(k)c}$ has at most $(2M+1)^3$ cubes within M of it. The inclusions: $\Lambda_8^{(k)} \subset \Lambda_7^{(k)}$, and $\Lambda_2^{(k)} \subset \Lambda_0^{(k)} \subset \Lambda_7^{(k-1)'}$ [2, pp. 566, 570], so $\Lambda_2^{(k)} \cap \Lambda_7^{(k)c} \subset \Lambda_7^{(k-1)'} \cap \Lambda_7^{(k)c} = \mathrm{Col}_k$ and $N_M(B^k(\Lambda_2^{(k)})) \subset B^k(\Lambda_1^{(k)}) \subset \Omega_1$. □

In the Setting above, expand the step-k polynomial at the background $\theta_k A^{(k)} = \tilde{B} + a$ in powers of the new field a [1, (3.15), (3.44)], [2, p. 579]:

$$P^{(k)}(\Omega_1', \tilde{B} + a, \varphi) = \sum_{n_a=0}^{\bar{n}} P_{n_a}^{(k)}[a^{n_a}] + R_a^{(k)},$$

where $P_{n_a}^{(k)}[a^{n_a}]$ is the sum of the localised renormalised families of $P^{(k)}(\Omega_1', \tilde{B}, \varphi)$ carrying n_a external vector legs a: the vertices [3, (1.8)–(1.11)] with $n' \ge 1$, and the a-insertions on the scalar lines $G_k(\Omega_A, \tilde{B}+a)$ and on the external fields $\Phi(\tilde{B}+a)$ of [2, (2.56)]; $R_a^{(k)}$ collects the R-vertices and the orders $> \bar{n}$ in a. On p. 579 the new legs are localised by g_k. Splitting every leg as $a = g_k a + (1-g_k)a$ (multilinearity), put

$$T_{\mathrm{gate}}^{(k)} := \sum_{n_a=1}^{\bar{n}} \big(P_{n_a}^{(k)}[a^{n_a}] - P_{n_a}^{(k)}[(g_k a)^{n_a}]\big),$$

the sum of the terms with at least one leg carrying $(1-g_k)a$, i.e. with a vector leg localised where $g_k \ne 1$: in $B^k(\Lambda_7^{(k)c})$ or in $B^k(\Lambda_7^{(k)} \setminus \Lambda_8^{(k)})$. If $P^{(k)}(\Lambda_7^{(k)}, \theta_k A^{(k)}, \varphi)$ in [2, (2.57)] is understood as the polynomial already built with the gated field $g_k a$, there is no such term; otherwise the gating is a move, whose cost is the following proposition.

Proposition F.139 (the terms of $P^{(k)}$ removed by g_k; the $P^{(k)}$-half of the second sentence of [2, p. 579])**.** *For $1 \le k \le K-1$, under* [2, (2.55)] *and the caps of Step 3(D_2), for every $\beta > 0$,*

$$|T_{\mathrm{gate}}^{(k)}| \le C_{\mathrm{HP}} t_k^{1/2-\beta} |\mathrm{Bad}_k| \le C_{\mathrm{HP}} (2M+1)^3 t_k^{1/2-\beta} \big(|\mathrm{Col}_k| + |\Lambda_7^{(k)} \setminus \Lambda_8^{(k)}|\big),$$

with C_{HP} depending on the data of theorem F.128 and on $c_\theta, c_{\theta 2}, c_g$, and free of ε, k, ε_0, the torus and the history. The remainder $R_a^{(k)}$ (the R-vertices and the orders $> \bar{n}$ in a) is $O(t_k^\kappa)|\Lambda_2^{(k)}|$ for every κ (proposition F.105(iii)), the first half of the sentence.

Proof. Step 1: the weight. Count every a-leg in d_v: the a-dependence of each parallel transporter is $U(\tilde{B}+a) = U(\tilde{B})U(a)$ (the group is abelian), so each a-insertion carries its own factor e_k, and $W := 2d_v + 4d_s - n_{\mathrm{ext}}$ is defined for a family with a-legs as in lemma F.132, the a-legs included in d_v. Inserting one a-leg into a family raises W by at least 2: on an existing gauge vertex,

$d_v(v)$ rises by 1 and $X(v)$ by 2; as a new vertex of the type [3, (1.8)] with $n = 1$ (one vector leg, two scalar legs) splitting a scalar line, whether an internal line or the line $G_k(\Omega_A, \tilde{B} + a)$ of an external field Φ, $X(\text{new}) = 2 - 2 = 0$ and l_s rises by 1, while n_{ext} is unchanged (the part of the split line beyond the new vertex is the external field $\Phi(\tilde{B})$); as an insertion on the block average $Q_k^*(\tilde{B} + a)$ of an external field, d_v rises by 1 and nothing else changes: the external field becomes $a_k G_k(\Omega_A, \tilde{B}) f_a$ with the source $f_a := [Q_k^*(\tilde{B} + a) - Q_k^*(\tilde{B})]\Lambda_6^{(k-1)'}\varphi$, $\|f_a\|_\infty \le e_k(L-1)d \sup|a| \sup|\varphi|$ (the transporter of Q_k^* runs along a contour of η-length at most $(L-1)d$ in scale-k units, [2, p. 559], and q is antisymmetric), which the proof of lemma F.131 bounds by $e_k(L-1)d\, 2C_{2.91}p \cdot C_\Phi p \lambda_k^{-1/4}$, the factor $e_k C_a p$ of Step 3 up to $O(1)$. By lemma F.132 every family of $P^{(k)}$ has $W \ge 0$, so every term of $T^{(k)}_{\text{gate}}$ has $W \ge 2$ (W stays even). In particular the $W = 0$ objects of lemma F.134 (the lone quartic and the single-line terms) occur in $T^{(k)}_{\text{gate}}$ only with an a-leg attached, hence with $W \ge 2$, and the cancellation of lemma F.134(b) is not needed.

Step 2: the fields. The external vector field a obeys lemma F.137: $|a| \le 2C_{2.91}p$ on $\operatorname{supp}\theta_{k+1}$ [2, (2.90)–(2.91)], $|\partial^\eta a| \le (C'_{2.91} + 1)p$, Hölder quotient $\le C^H_{2.91}p$; so it is an admissible external vector field of [3, p. 412] with norm $\le C_a p$. The external scalar fields are those of lemma F.131. The background $\tilde{B}$ is regular by [2, (2.94)–(2.98)], as in proposition F.105(iii).

Step 3: the bound per family. Proposition 1 with the external vector field a (the legs of [3, (1.10)] with $n' \ge 1$; its $O(1)$ is independent of ε, k, the domains and the background, [3, p. 421]) bounds each localised family by

$$O(1)\, e^{d_v} \lambda^{d_s - n_{\text{ext}}/4} (C_\Phi p)^{n_{\text{ext}}} (C_a p)^{n_a}\, t^{W/4} e^{-\delta_0 d(\{\square\})},$$

by the computation of lemma F.133 in $d = 3$ ($e_k^{d_v}\lambda_k^{d_s}(p\lambda_k^{-1/4})^{n_{\text{ext}}} = e^{d_v}\lambda^{d_s - n_{\text{ext}}/4} p^{n_{\text{ext}}} t^{W/4}$, the e_k of the a-legs included in $e^{d_v} t^{d_v/2}$); the [3, (1.7)] vertices with the finite parts δm^2_{fin}, $\delta m^2_{K,k}$ enter through theorem F.99 and lemma F.101. By Step 1, $t^{W/4} \le t^{1/2}$; $n_{\text{ext}} + n_a \le C(\bar{n})$, so $p^{n_{\text{ext}} + n_a} \le C_\beta t^{-\beta}$ ($P(\beta, n) < \infty$); hence each family is $\le C_\beta t^{1/2-\beta} e^{-\delta_0 d(\{\square\})}$.

Step 4: the sum. Root the tree of each family at the cube of a leg carrying $(1 - g_k)a$: at most $|\text{Bad}_k|$ choices; sum the remaining cubes by lemma F.26; there are finitely many families of order $\le \bar{n}$ with $n_a \le \bar{n}$. This gives the first bound, and lemma F.138 the second.

Step 5: what is not gated. The R-vertices and the orders $> \bar{n}$ in a are not terms of $T^{(k)}_{\text{gate}}$; they are bounded in proposition F.105(iii) by $O(t_k^\kappa)|\Lambda_2^{(k)}|$, Proposition 1 with the field restrictions and $\bar{n} + 1$ legs a. □

Corollary F.140 (the two parts in the ledger)**.** *The part of $T^{(k)}_{\text{gate}}$ with a gated leg in $B^k(\Lambda_7^{(k)c})$ is $\le C_{\text{HP}}(2M+1)^3 t_k^{1/2-\beta}|\text{Col}_k|$: a collar term of the form of [2, (2.57)]'s remainder, which enters [2, (2.117)] with the collar terms (it is the $P^{(k)}$-part of the printed $O((L^k\varepsilon)^{\kappa_0})|\Lambda_2^{(k)} \cap \Lambda_7^{(k)c}|$). The part with a gated leg in $B^k(\Lambda_7^{(k)} \setminus \Lambda_8^{(k)})$ is $\le C_{\text{HP}}(2M+1)^3|\Lambda_8^{(k)c}|$ ($t_k^{1/2-\beta} \le 1$): a term of the form of proposition F.126's moves and of proposition F.127's bound, which enters the augmented exponent of theorem F.125 with the same sign and in the same place. Theorem F.125 holds for every k-free $c_8 \ge 0$; the three terms of this kind are carried together by taking*

$$c_8 \;\ge\; Q_1(C_{\text{D22}} + C_{\text{D23}}) + (2M+1)^3\Big(\sup_{0<t\le 1} F_H(t) + C_{\text{HP}}\Big),$$

a constant free of k, ε, K, the torus and the history, fixed before ε_0, so that (E.ent) of theorem F.125 is one cap on ε_0 as before.

Proof. Lemma F.138 splits Bad_k into the cubes within M of $\Lambda_2^{(k)} \cap \Lambda_7^{(k)c} \subset \text{Col}_k$ and those within M of $\Lambda_7^{(k)} \setminus \Lambda_8^{(k)} \subset \Lambda_8^{(k)c}$; the rooting of Step 4 is done in the respective set. The bookkeeping is that of the proof of proposition F.126, and lemma F.113 carries the terms through [2, Sect. 3A]; F_H is bounded on $(0, 1]$ (proposition F.127). □

The final step. Let K be the last step, $l_K = t_K = L^K\varepsilon \in (\varepsilon_0/L, \varepsilon_0]$ (theorem F.4), in scale-K units, with the Setting of theorem F.128 at $k = K$: $\Omega_A = B^K(\Lambda_2^{(K-1)'})$, $\Omega_1 = B^K(\Lambda_7^{(K-1)'})$,

$\tilde B = A^{(K)} = \theta_K A^{(K)}$ on Ω_A, $\Phi = \phi^{(K)}$ of [2, (2.56)] at $k = K$, under [2, (2.55)] at step K. No set $\Lambda_i^{(K)}$ is defined at the final step: [2, (2.116)] bounds $P^{(K)}$ on its whole domain Ω_1, and its right side is the volume $|\Lambda_7^{(K-1)'}|$ of that domain, as in the last term of [2, (2.117)]. The chain is run with $E_0 + E_K$ (lemma F.3), so no step polynomial contains $E_{\rm fin}$, which is the global factor bounded by theorem F.4.

Proposition F.141 (the final step, [2, (2.116)]). *For every $\beta > 0$,*

$$P^{(K)}(\Omega_1, \tilde B, \varphi) = -\lambda_K \sum_{x\in\Omega_1} \eta^3 |\Phi(x)|^4 + R_{\rm field} + c_K(\Omega_1), \qquad |R_{\rm field}| \le C_\beta t_K^{1/2-\beta} |\Lambda_7^{(K-1)'}|,$$

with $c_K(\Omega_1)$ the field-independent part, $|c_K(\Omega_1)| \le C t_K (1+\ln t_K^{-1})^5 |\Lambda_7^{(K-1)'}| \le C_\beta t_K^{1-\beta} |\Lambda_7^{(K-1)'}|$. Hence $P^{(K)}(\Omega_1, \tilde B, \varphi) \le C_\beta t_K^{1/2-\beta} |\Lambda_7^{(K-1)'}|$, which is [2, (2.116)] *with $\kappa_0 = \frac12 - \beta$. The constants are those of theorem F.128, free of ε, K, ε_0, the torus and the history.*

Proof. Run the proof of theorem F.128 at $k = K$ with the inner domain $\Omega_1' := \emptyset$.

Step 1: the empty domain. By lemma F.129 the vertex domain enters only through $\mathbf{1}_{\Omega_1'}$ on the vertices [3, (1.6)–(1.7)], through the gate on the A'-legs ($\operatorname{supp} g \subset \Omega_1'$) and through the localised counterterms. With $\Omega_1' = \emptyset$ every vertex vanishes and every A'-leg with it, so $P^{(K)}(\emptyset, \tilde B, \varphi)$ is field-independent and equals $c_K(\emptyset)$, which is 0: on the chain run with $E_0 + E_K$, $c_k(\Omega') = -E_{K,k} + a_k(\Omega')$ (lemma F.102), $E_{K,K} = E_K - E_K = 0$, and $a_K(\emptyset)$ has no pair vertex to place. In lemma F.130, $D_{\rm good} = \emptyset$ and $\widetilde{\rm Col} = \Omega_1$: every localised family is marked, and $P^{(K)}(\Omega_1) = P^{(K)}(\Omega_1) - P^{(K)}(\emptyset)$ is the sum over all localised families, rooted at a cube of Ω_1 ($|\Omega_1| = |\Lambda_7^{(K-1)'}|$ choices).

Step 2: the three kinds of family. The families with $W \ge 2$ are bounded by lemma F.133 at $k = K$: Proposition 1 on the lines $G_K(\Omega_A, \tilde B)$ with the external scalar fields of lemma F.131 at $k = K$ (which uses [2, (2.55)] at step K and [1, (2.24)–(2.25)] at points of Ω_1, at depth $\ge r(t_K) - 2 \ge R_0$ in Ω_A, $\tilde B$ regular by [2, (2.60)]), the [3, (1.7)] vertices through theorem F.99 and lemma F.101, the sum by lemma F.26 with the root in Ω_1: in all $\le C_\beta t_K^{1/2-\beta} |\Lambda_7^{(K-1)'}|$. The families with $W = 0$ are lemma F.134: the lone [3, (1.6)] vertex, whose sum over the vertex domain is sharp, gives exactly $-\lambda_K \sum_{x\in\Omega_1} \eta^3 |\Phi(x)|^4$; the single-line terms give $\le C_{\rm SL} e^2 \lambda^{-1/2} p^4 t_K^{1/2}$ per cube of Ω_1 by [2, Lemma 2.4, (2.65)–(2.66)] at every $y \in \Lambda_7^{(K-1)'}$ (theorem F.2(c)) and the slice kernels of lemma F.88, in all $\le C t_K^{1/2-\beta} |\Lambda_7^{(K-1)'}|$. The field-independent families are lemma F.135(ii) with $\Omega_1' = \emptyset$: the vacuum families with a cube meeting Ω_1 have $W \ge 4$, and $|c_K(\Omega_1)| = |c_K(\Omega_1) - c_K(\emptyset)| \le C t_K (1 + \ln t_K^{-1})^5 |\Lambda_7^{(K-1)'}|$; at zero background this is corollary F.29, and on the chain of lemma F.3 it is the bound of theorem F.103(a) on $a_K(\Omega_1)$, whose proof does not use $k < K$.

Step 3: the inputs at $k = K$. Every input of theorem F.128 is available at the final step: [2, (2.55)] holds at every step, K included; Proposition 1's $O(1)$ does not depend on the step; the layer distances used are those of the step-$(K-1)$ sets, $\operatorname{dist}(\Omega_1, \Omega_A^c) \ge r(t_K) - 2$; and no set $\Lambda_i^{(K)}$ enters, since the inner domain is empty. The one-sided bound follows by dropping the quartic term, which is ≤ 0. □

The same bound, with $|\Lambda_7^{(K-1)'}|$ replaced by $L^3 |T^{(K)}| \ge |\Lambda_7^{(K-1)'}|$, is the $k = K$ term of the exponents of [2, (3.42)] as proposition F.123 carries it. This completes the field-dependent part of the second sentence of [2, p. 579] (proposition F.139) and of the final step (proposition F.141); with theorem F.128 (the collar), the three transfers of the mechanism of [2, Prop. 2.1] that [2, p. 570] announces ("similar conclusions hold for other expressions which will be included in the action in the later stages of the procedure") are written.

Step 3(A): the propagator estimates of group (A). This part proves the estimates of group (A) that Steps 3(B)–3(D) consume: the coercivity [4, (1.8)] on a general region and for the mixed-pin operators [4, (3.5)], the Combes–Thomas form of [4, Cor. 2.3] for every pin structure, the two-sided bounds [4, (1.15)], the decay [4, (1.16)], the conditioning-region difference [4, (1.18)],

the domain differences [4, (5.5), (1.20)], [1, Prop. 2.2, (2.27)–(2.29)], the two-sided form of the depth factor of [4, Thm., (1.12)], and their validity on the torus with torus-free constants. The notation is that of Steps 3(B)–3(D): $\eta = L^{-k}$, $e_k = et_k^{1/2}$, $a_k \in [a_*, a]$ with $a_* = a(1-L^{-2})$, $Q_k(A)$, $P_k(A) = Q_k^* Q_k$, $G_k(\Omega, A)$, $\Delta^{(k)}(\Omega, A)$, $C_\Lambda^{(k)} = C_\Lambda$, and M the large-block side. Distances are the sup-distances of [1, (1.3)] (on a torus, the quotient distances); $|x-x'|$ is the sup-distance of unit-lattice points, and kernels of N-component operators are $N \times N$ matrices, $|K(x,x')|$ their operator norm.

Setting. Functions on the fine lattice $\eta\mathbb{Z}^d$ carry the η^d-weighted ℓ^2 product $\langle \phi, \psi\rangle = \sum_x \eta^d \phi(x) \cdot \psi(x)$, functions on the unit lattice the unweighted one; bond functions are weighted at the initial point b_- of the bond. $U(A_b) := \exp(qe_k \eta A_b)$ with q a fixed real antisymmetric $N \times N$ matrix [4, (1.2)]; all $U(A_b)$ commute and are orthogonal, and $U(A(\Gamma)) := \prod_{b\in\Gamma} U(A_b)$ for a contour Γ. For a union Ω of unit blocks, $(D_A\phi)(b) := \eta^{-1}(U(A_b)\phi(b_+) - \phi(b_-))$ on the bonds $b \subset \Omega$, so that $\langle \phi, -\Delta_{A,\Omega}^{\eta,N}\phi\rangle = \|D_A\phi\|^2$ is the Neumann covariant Dirichlet form [4, (1.3)]. $Q_k(A)$ averages over the unit blocks $B^k(y)$, $y \in \Omega^{(k)}$, along the contours $\Gamma_{y,x}^{(k)} \subset B^k(y)$ of [1, (2.1)–(2.2)]; $q_k(y) := Q_k(A)^* \delta_y$ is supported in $B^k(y)$, $|q_k(y)(x)| = 1$ there, $\|q_k(y)\| = 1$; $P_k(A) = Q_k^* Q_k$ is the orthogonal projection onto the transported constants $x \mapsto U(A(\Gamma_{y,x}^{(k)}))v$ of the unit blocks. $P(A) := Q(A)^* Q(A)$, with $Q(A)$ the covariant average over the L-blocks of the unit lattice, is the orthogonal projection onto the transported constants of the L-blocks, with $|P(A)(y,y')| \le 1$ and $P(A)(y,y') = 0$ unless y, y' lie in one L-block. The operators of this part are

$$G_k(\Omega, \Lambda, A) := T_\Lambda^{-1}, \quad T_\Lambda := -\Delta_{A,\Omega}^{\eta,N} + m_k^2 + a_k P_k(A)(\Omega \setminus B^k\Lambda) + a_{k+1} L^{-2} P_{k+1}(A)(B^k\Lambda)$$

[4, (3.5)], for $\Lambda \subset \Omega^{(k)}$ a union of L-blocks ($P_k(A)(Y)$ is the sum of the block projections of the unit blocks in Y, $P_{k+1}(A)(Y)$ that of the L-blocks), so that $T_\emptyset = G_k(\Omega, A)^{-1}$ and $T_{\Omega^{(k)}}$ is the operator of $G_{k+1}^\eta(\Omega, A)$; and

$$X_\Lambda := (\Delta^{(k)}(\Omega, A) + aL^{-2}P(A))|_\Lambda, \qquad C_\Lambda = X_\Lambda^{-1}$$

[4, (1.13)], with $X|_\Lambda := \Lambda X \Lambda$ on $\ell^2(\Lambda)$. Three facts about Bałaban's blocks and contours are used. (H1) $\Omega^{(k)}$ and Λ are unions of large blocks, and every large block is a union of L-blocks ([1, p. 607]: the large-block partition is the block partition with L replaced by ML). (H2) The contours are composed, $\Gamma_{z,x}^{(k+1)} = \Gamma_{z,y} \cup \Gamma_{y,x}^{(k)}$ for $x \in B^k(y)$, $y \in B(z)$ [1, (2.2)]; hence $Q_{k+1}(A) = Q(A)Q_k(A)$ and $P_{k+1}(A) = Q_k^* P(A) Q_k$ is the orthogonal projection onto the transported constants of the L-blocks $B^{k+1}(z)$. (H3) Each contour is a union of axis-ordered monotone lattice paths inside its block [1, (2.1)]; the η-length of $\Gamma_{y,x}^{(k)}$ is at most $d(L-1)L^{-1}(1 + L^{-1} + \dots) \le d$, that of $\Gamma_{z,x}^{(k+1)}$ at most $d(L-1) + d \le d(L+1)$. A vector field A is *regular* on Ω when $|\partial_\mu^\eta A_\nu(x)| \le c e_k^{\beta-1}$ for $x \in \Omega$, some $\beta > 0$ and a universal c [4, (1.7)]; its curvature $F_{\nu\mu} := \partial_\nu^\eta A_\mu - \partial_\mu^\eta A_\nu$ [1, (1.6)] is gauge invariant and then obeys $|F_{\nu\mu}| \le 2ce_k^{\beta-1}$. Every smallness condition on $e_k = et_k^{1/2} \le e\varepsilon_0^{1/2}$ below is a cap on ε_0 depending on $(d, N, \|q\|, c, \beta, a, L)$ only. For $\theta > 0$ put $S(\theta) := \sum_{n\in\mathbb{Z}^d} e^{-\theta|n|} < \infty$.

Lemma F.142 (cube coercivity)**.** *Let $\Delta \subset \eta\mathbb{Z}^d$ be a cube of side $\ell \in \{1, L\}$ that is a union of unit blocks, A a bond field on Δ with $|F_{\nu\mu}(A)| \le f$ on the plaquettes of Δ, Π^A the orthogonal projection onto $\{x \mapsto U(A(\Gamma_x))v : v \in \mathbb{R}^N\}$ for contours $\Gamma_x \subset \Delta$ from a root, each of η-length at most λ_Γ, and $c_p > 0$. Put $\alpha := (d-1)\ell f$ and assume $\|q\| e_k \max\{\lambda_\Gamma, 1\}\alpha \le \ln 2$. Then, with $\kappa := 2\|q\|e_k\alpha$ and $\varepsilon_g := 2\|q\|e_k\lambda_\Gamma\alpha$,*

$$-\Delta_{A,\Delta}^{\eta,N} + c_p\Pi^A \;\ge\; \min\{2\ell^{-2}, c_p\} - 2c_p\varepsilon_g - d\kappa^2.$$

If A is regular on Δ, then $f \le 2ce_k^{\beta-1}$ and $\kappa, \varepsilon_g = O(e_k^\beta)$, so there is $e_1(d, N, \|q\|, c, \beta, L, a) > 0$ such that for $e_k \le e_1$ the right side is at least $\frac{1}{2}\min\{2\ell^{-2}, c_p\}$ for every $c_p \le a$ and every $\lambda_\Gamma \le d(L+1)$.

Proof. Gauge covariance. For $\lambda\colon \Delta \to \mathbb{R}$ the gauge transform $A_b^\lambda := A_b + \eta^{-1}(\lambda(b_+) - \lambda(b_-))$ gives $U(A_b^\lambda) = U(A_b)e^{qe_k(\lambda(b_+)-\lambda(b_-))}$. With the orthogonal multiplication operator $G := e^{-qe_k\lambda}$

(acting on bond functions through $\lambda(b_-)$), $D_{A^\lambda} = GD_AG^{-1}$, since all the matrices commute; and $U(A^\lambda(\Gamma_x)) = e^{qe_k\lambda(x)}U(A(\Gamma_x))e^{-qe_k\lambda(x_0)}$ for the root x_0, so the range of Π^{A^λ} is G^{-1} of the range of Π^A and $\Pi^{A^\lambda} = G\Pi^AG^{-1}$. Hence $-\Delta^{\eta,N}_{A^\lambda,\Delta} + c_p\Pi^{A^\lambda} = G(-\Delta^{\eta,N}_{A,\Delta} + c_p\Pi^A)G^{-1}$, and the lower bound is gauge invariant; F is unchanged.

Axial gauge. Choose λ so that $A' := A^\lambda$ has $A'_1 \equiv 0$, $A'_2 = 0$ on the bonds of one face $\{x_1 = x_1^0\}$, $A'_3 = 0$ on the bonds of one line $\{x_1 = x_1^0, x_2 = x_2^0\}$, and so on in d dimensions (the bonds on which a component is prescribed form a tree, so λ exists). Then $A'_\mu(x)$ is reached from a bond where it vanishes by at most $d-1$ axis-parallel segments of length at most ℓ, along each of which it changes by η times the curvature: on a segment in direction $\nu < \mu$, $\partial^\eta_\nu A'_\mu = F_{\nu\mu} + \partial^\eta_\mu A'_\nu$, and A'_ν vanishes identically on the set over which the segment runs. Hence $|A'_\mu| \le (d-1)\ell f = \alpha$ on Δ.

Bond and pin defects. $|U(A'_b) - 1| \le \|q\|e_k\eta|A'_b|e^{\|q\|e_k\eta|A'_b|} \le 2\|q\|e_k\alpha\eta$, so $\kappa \ge \sup_b |U(A'_b) - 1|/\eta$. The transported constants are $g(x) := U(A'(\Gamma_x)) = \exp(qe_k\sum_{b\in\Gamma_x}\eta A'_b)$, and $\sum_{b\in\Gamma_x}\eta|A'_b| \le \lambda_\Gamma\alpha$ gives $|g(x) - 1| \le 2\|q\|e_k\lambda_\Gamma\alpha = \varepsilon_g$.

Error operator. Write $D_{A'} = D_1 + E$ with $(D_1\phi)(b) = \eta^{-1}(\phi(b_+) - \phi(b_-))$ and $(E\phi)(b) = \eta^{-1}(U(A'_b) - 1)\phi(b_+)$. Each site is the final point of at most d bonds of Δ, so $\|E\phi\|^2 \le \kappa^2\sum_b\eta^d|\phi(b_+)|^2 \le d\kappa^2\|\phi\|^2$, and $\|D_{A'}\phi\|^2 \ge \frac12\|D_1\phi\|^2 - d\kappa^2\|\phi\|^2$.

Neumann gap. Δ is a product of d paths with ℓ/η sites and spacing η; the Neumann Laplacian of a path has eigenvalues $4\eta^{-2}\sin^2(\pi j\eta/(2\ell))$, $0 \le j < \ell/\eta$, so $\|D_1\phi\|^2 \ge \lambda_\ell\|(1-\Pi_1)\phi\|^2$ with Π_1 the projection onto the constants and $\lambda_\ell = 4\eta^{-2}\sin^2(\pi\eta/(2\ell)) \ge 4\ell^{-2}$, by $\sin t \ge 2t/\pi$ on $[0,\pi/2]$.

Pin comparison. $\Pi^{A'} = V_gV_g^*$ with $(V_gv)(x) := g(x)v/\ell^{d/2}$, an isometry $\mathbb{R}^N \to \ell^2(\Delta)$, and $\Pi_1 = V_1V_1^*$; $\|V_g - V_1\| \le \sup_x|g(x) - 1| \le \varepsilon_g$, so $\|\Pi^{A'} - \Pi_1\| \le 2\varepsilon_g$.

Combination. $-\Delta^{\eta,N}_{A',\Delta} + c_p\Pi^{A'} \ge \frac12\lambda_\ell(1-\Pi_1) + c_p\Pi_1 - 2c_p\varepsilon_g - d\kappa^2 \ge \min\{2\ell^{-2}, c_p\} - 2c_p\varepsilon_g - d\kappa^2$. The last sentence of the statement follows since $\alpha \le 2(d-1)Lce_k^{\beta-1}$ and $\kappa, \varepsilon_g \le 4d(L+1)\|q\|(d-1)Lc\,e_k^\beta$. □

Lemma F.143 (coercivity with mixed pins; [4, (1.8)] on a general region)**.** *Assume (H1)–(H3), A regular on Ω and $e_k \le e_1$. Then for every $\Lambda \subset \Omega^{(k)}$ that is a union of L-blocks ($\Lambda = \emptyset$ and $\Lambda = \Omega^{(k)}$ included)*

$$T_\Lambda \;\ge\; \gamma_L := \tfrac14\min\{2L^{-2}, a_*L^{-2}\},$$

and $T_\emptyset = G_k(\Omega, A)^{-1} \ge \gamma_\emptyset := \frac14\min\{2, a_*\}$, which is [4, (1.8)] on every union Ω of unit blocks, with $\gamma_0 = \gamma_\emptyset$ independent of L, η, k, Ω, A and the torus.

Proof. Partition Ω into the unit blocks of $\Omega \setminus B^k\Lambda$ and the L-blocks of $B^k\Lambda$ (cells). Dropping the bonds between cells removes non-negative terms from the Dirichlet form, so $-\Delta^{\eta,N}_{A,\Omega} \ge \bigoplus_{\text{cells}}(-\Delta^{\eta,N}_{A,\text{cell}})$. The pins are block-diagonal on this partition: $P_k(A)(\Omega\setminus B^k\Lambda)$ is the sum of the projections Π^A of the unit blocks, with contours of η-length at most d (H3), and $P_{k+1}(A)(B^k\Lambda)$ is the sum of the projections of the L-blocks (H2), with contours of η-length at most $d(L+1)$ (H3). On each cell apply lemma F.142 with $(\ell, c_p) = (1, a_k)$, resp. $(L, a_{k+1}L^{-2})$, and $c_p \le a$: the cell operator is at least $\frac12\min\{2, a_*\} \ge \gamma_L$, resp. $\frac12\min\{2L^{-2}, a_*L^{-2}\} \ge \gamma_L$. Drop $m_k^2 \ge 0$. For $\Lambda = \emptyset$ all cells are unit blocks and the bound is $\frac12\min\{2, a_*\} \ge \gamma_\emptyset$. □

Lemma F.144 (Combes–Thomas bound; the L^2 form of [4, Cor. 2.3] for every pin structure)**.** *Let $T := D_A^*D_A + V + \sum_B c_B\Pi_B$ on $\ell^2(\Omega, \eta^d)$, where $V \ge 0$ is a multiplication operator, the Π_B are orthogonal projections whose ranges are supported in pairwise disjoint sets B of diameter at most ℓ_*, $0 \le c_B \le a$, and $T \ge \gamma > 0$. Put*

$$\theta_0(\gamma, \ell_*) := \min\Big\{\frac{1}{\ell_*},\ \frac{\gamma^{1/2}}{36\sqrt{d}},\ \frac{\gamma}{12\mathrm{e}^2a\ell_*}\Big\},$$

e *Euler's number. Then for $f, f' \in \ell^2(\Omega)$ and bond functions g, g', with δ the distance of the supports (bonds located at b_-) and $\theta := \theta_0(\gamma, \ell_*)$:*

(i) $|\langle f, T^{-1}f'\rangle| \le 2\gamma^{-1}e^{-\theta\delta}\|f\|\|f'\|$;

(ii) $|\langle g, D_A T^{-1} f'\rangle|$ *and* $|\langle f, T^{-1} D_A^* g'\rangle|$ *are at most* $(2\gamma^{-1/2} + 6\sqrt{d}\,\theta\gamma^{-1})e^{\theta}e^{-\theta\delta}$ *times the product of the norms;*
(iii) $|\langle g, D_A T^{-1} D_A^* g'\rangle| \le 4e^{\theta}e^{-\theta\delta}\|g\|\|g'\|$.
The constants do not depend on η, Ω, A *or the torus.* $\theta_0(\gamma, \ell_*)$ *is increasing in* γ *and decreasing in* ℓ_**; in particular* $\theta_0(\gamma_\emptyset, 1) \ge \theta_0(\gamma_L, L)$.

Proof. Let $\rho := \operatorname{dist}(\cdot, \operatorname{supp} f')$ (for bond functions evaluated at b_-), a 1-Lipschitz function along bonds, and $T_\theta := e^{\theta\rho} T e^{-\theta\rho}$ for $0 < \theta \le \theta_0(\gamma, \ell_*)$; then $T^{-1} = e^{-\theta\rho} T_\theta^{-1} e^{\theta\rho}$ and $e^{\theta\rho} f' = f'$.

Hopping. $(e^{\theta\rho} D_A e^{-\theta\rho}\phi)(b) = (D_A\phi)(b) + \eta^{-1}(e^{-\theta(\rho(b_+)-\rho(b_-))} - 1)U(A_b)\phi(b_+)$. Since $|\rho(b_+) - \rho(b_-)| \le \eta$ and $\theta\eta \le 1$, $|\eta^{-1}(e^{-\theta s} - 1)| \le 3\theta$ for $|s| \le \eta$, so $e^{\theta\rho} D_A e^{-\theta\rho} = D_A + \theta B_1$ with $\|B_1\| \le 3\sqrt{d}$ (each site is the final point of at most d bonds). Likewise $e^{\theta\rho} D_A^* e^{-\theta\rho} = (e^{-\theta\rho} D_A e^{\theta\rho})^* = (D_A - \theta B_1')^*$ with $\|B_1'\| \le 3\sqrt{d}$. Hence $e^{\theta\rho} D_A^* D_A e^{-\theta\rho} - D_A^* D_A = \theta(D_A^* B_1 - B_1'^* D_A) - \theta^2 B_1'^* B_1$, and

$$|\langle \phi, (e^{\theta\rho} D_A^* D_A e^{-\theta\rho} - D_A^* D_A)\psi\rangle| \le 3\sqrt{d}\,\theta(\|D_A\phi\|\|\psi\| + \|\phi\|\|D_A\psi\|) + 9d\theta^2\|\phi\|\|\psi\|.$$

V commutes with $e^{\theta\rho}$.

Pins. Write $\Pi_B = VV^*$ with V an isometry onto the range, and let ρ_B be the value of ρ at a point of B. Then $e^{\theta\rho}\Pi_B e^{-\theta\rho} = (e^{\theta(\rho-\rho_B)}V)(e^{-\theta(\rho-\rho_B)}V)^*$, and $|\rho - \rho_B| \le \ell_*$ on B, so $\|e^{\pm\theta(\rho-\rho_B)}V - V\| \le \theta\ell_* e^{\theta\ell_*}$ and $\|e^{\theta\rho}\Pi_B e^{-\theta\rho} - \Pi_B\| \le 2\theta\ell_* e^{\theta\ell_*} + \theta^2\ell_*^2 e^{2\theta\ell_*} \le 2\theta\ell_* e^{2\theta\ell_*}$, using $te^t \le 2(e^t - 1)$ for $t = \theta\ell_* \le 1$. The sets B are disjoint, so the pin changes are block-diagonal and their sum has norm at most $2a\theta\ell_* e^{2\theta\ell_*} \le 2\mathrm{e}^2 a\theta\ell_*$.

Perturbation. Put $K := T^{-1/2}(T_\theta - T)T^{-1/2}$. Since $D_A^* D_A \le T$, $\|D_A T^{-1/2}\| \le 1$, and $\|T^{-1/2}\| \le \gamma^{-1/2}$, so

$$\|K\| \le 6\sqrt{d}\,\theta\gamma^{-1/2} + 9d\theta^2\gamma^{-1} + 2\mathrm{e}^2 a\theta\ell_*\gamma^{-1} \le \tfrac{1}{6} + \tfrac{1}{144} + \tfrac{1}{6} < \tfrac{1}{2}$$

for $\theta \le \theta_0(\gamma, \ell_*)$. Hence $T_\theta = T^{1/2}(1 + K)T^{1/2}$ is invertible, $T_\theta^{-1} = T^{-1/2}(1 + K)^{-1}T^{-1/2}$, and $\|T_\theta^{-1}\| \le 2\gamma^{-1}$, $\|D_A T_\theta^{-1}\| \le 2\gamma^{-1/2}$, $\|D_A T_\theta^{-1} D_A^*\| \le 2$.

Bounds. (i) $\langle f, T^{-1} f'\rangle = \langle e^{-\theta\rho} f, T_\theta^{-1} f'\rangle$ and $\|e^{-\theta\rho} f\| \le e^{-\theta\delta}\|f\|$. (ii) $\langle g, D_A T^{-1} f'\rangle = \langle e^{-\theta\rho} g, (D_A + \theta B_1)T_\theta^{-1} f'\rangle$, bounded by $(2\gamma^{-1/2} + 3\sqrt{d}\,\theta \cdot 2\gamma^{-1})e^{-\theta\delta}\|g\|\|f'\|$; the adjoint form follows from the symmetry of T with ρ the distance to $\operatorname{supp} f$. (iii) With ρ the distance to $\operatorname{supp} g'$, $e^{\theta\rho} g' = g'$ and $\langle g, D_A T^{-1} D_A^* g'\rangle = \langle e^{-\theta\rho} g, (D_A + \theta B_1)T_\theta^{-1}(D_A - \theta B_1')^* g'\rangle$, bounded by $(2 + 12\sqrt{d}\,\theta\gamma^{-1/2} + 18d\theta^2\gamma^{-1})e^{-\theta\delta}\|g\|\|g'\| \le 4e^{-\theta\delta}\|g\|\|g'\|$. The factors e^{θ} in (ii)–(iii) are ≥ 1. The monotonicity of θ_0 is read off its definition, and $\gamma_\emptyset \ge \gamma_L$, $1 \le L$. □

With lemma F.143 ($\ell_* = L$, $\gamma = \gamma_L$: the pins of T_Λ are the unit-block and L-block projections, of diameters ≤ 1 and $\le L$, with $c_B \in \{a_k, a_{k+1}L^{-2}\} \subset [0, a]$), lemma F.144 gives [4, Cor. 2.3] for $G_k(\Omega, \Lambda, A)$ and $G_{k+1}^{\eta}(\Omega, A)$, with constants depending on L as [4, p. 588] says; with $\Lambda = \emptyset$ ($\ell_* = 1$, $\gamma = \gamma_\emptyset$) it is a proof of the first clause of [4, Cor. 2.3], the bounds for $G_k(\Omega, A)$ (its δG_k clause is not used in this appendix). Throughout Steps 3(B)–3(D), $\theta_0 := \theta_0(\gamma_L, L)$; by the last sentence of lemma F.144 every Combes–Thomas rate of this part is at least θ_0.

Lemma F.145 (block inversion; [4, (3.8), (1.15), (1.16)])**.** *Assume (H1)–(H3),* A *regular on* Ω, $e_k \le e_1$, *and let* $\Lambda \subset \Omega^{(k)}$ *be a union of* L*-blocks. Put* $P_\Lambda := \Lambda P(A)\Lambda$ *(an orthogonal projection on* $\ell^2(\Lambda)$*, since* $P(A)$ *is block-diagonal over the* L*-blocks) and* $W := Q_k^*\Lambda(1 - (a_{k+1}/a_k)L^{-2}P_\Lambda)$. *Then:*

(a) X_Λ *is invertible and* $X_\Lambda^{-1} = W^* G_k(\Omega, \Lambda, A)W + a_k^{-1} - (a_{k+1}L^{-2}/a_k^2)P_\Lambda$*; that is,* $W\delta_y = q_k(y) - (a_{k+1}/a_k)L^{-2}q_{k+1}(y)$ *with* $q_{k+1}(y) := Q_k^* P(A)\delta_y$*, which is* [4, (3.8)].
(b) $\gamma_X \le X_\Lambda \le a_k + aL^{-2} \le \gamma_1 := a + aL^{-2}$*, with* $\gamma_X := (a_*^{-1} + \gamma_L^{-1})^{-1}$*: this is* [4, (1.15)].
(c) $|C_\Lambda(y, y')| \le c_2 e^{-\theta_0|y-y'|}$ *for* $y, y' \in \Lambda$*, with* $c_2 := (2\gamma_L^{-1} + 2a_*^{-1})e^{2\theta_0 L}$ *and* $\theta_0 = \theta_0(\gamma_L, L)$*: this is* [4, (1.16)]*, also for* $C^{(k)}(\Omega, A) = C_{\Omega^{(k)}}$.

The constants depend on $(d, N, \|q\|, a, L)$ *only.*

Proof. On $\ell^2(\Lambda)\oplus\ell^2(\Omega)$ let $\mathcal{M}:=\begin{pmatrix}A_\psi & -B\\ -B^* & D_\phi\end{pmatrix}$ with $A_\psi := a_k + aL^{-2}P_\Lambda$, $B := a_k\Lambda Q_k$, $D_\phi := H + a_kP_k(A)$, $H := -\Delta^{\eta,N}_{A,\Omega} + m_k^2$. Its quadratic form is

$$a_k\|\psi-\Lambda Q_k\phi\|^2 + a_k\|(1-\Lambda)Q_k\phi\|^2 + aL^{-2}\|P_\Lambda\psi\|^2 + \langle\phi, H\phi\rangle.$$

Schur complement in ϕ. $D_\phi = G_k(\Omega,A)^{-1} \geq \gamma_\emptyset$ by lemma F.143, and $A_\psi - BD_\phi^{-1}B^* = \Lambda(a_k - a_k^2Q_kG_k(\Omega,A)Q_k^* + aL^{-2}P(A))\Lambda = X_\Lambda$ by [4, (1.14)].

Schur complement in ψ. $A_\psi \geq a_k > 0$, and with $b := aL^{-2}$, $(a_k+bP_\Lambda)^{-1} = a_k^{-1}(1-P_\Lambda)+(a_k+b)^{-1}P_\Lambda$, so $a_k - a_k^2(a_k+bP_\Lambda)^{-1} = \frac{a_kb}{a_k+b}P_\Lambda = a_{k+1}L^{-2}P_\Lambda$ by the recursion $a_{k+1} = aa_k/(aL^{-2}+a_k)$. Hence

$$D_\phi - B^*A_\psi^{-1}B = H + a_kQ_k^*(1-\Lambda)Q_k + a_{k+1}L^{-2}Q_k^*P_\Lambda Q_k = T_\Lambda,$$

since $Q_k^*(1-\Lambda)Q_k = P_k(A)(\Omega\setminus B^k\Lambda)$ and $Q_k^*P_\Lambda Q_k = P_{k+1}(A)(B^k\Lambda)$ by (H2). This is the computation [4, (3.4)–(3.5)] done as algebra.

Block inverse. $A_\psi > 0$ and $T_\Lambda \geq \gamma_L > 0$ (lemma F.143), so $\mathcal{M}$ is positive definite, and its inverse's $\psi\psi$ block is both $(A_\psi - BD_\phi^{-1}B^*)^{-1} = X_\Lambda^{-1}$ and $A_\psi^{-1} + A_\psi^{-1}BT_\Lambda^{-1}B^*A_\psi^{-1}$. Now $A_\psi^{-1}B = (1-\frac{b}{a_k+b}P_\Lambda)\Lambda Q_k = W^*$, because $\frac{b}{a_k+b} = \frac{a_{k+1}}{a_k}L^{-2}$, and $A_\psi^{-1} = a_k^{-1} - \frac{b}{a_k(a_k+b)}P_\Lambda = a_k^{-1} - \frac{a_{k+1}L^{-2}}{a_k^2}P_\Lambda$. This is (a); $P(A)\delta_y$ is supported in the L-block of y, which lies in Λ by (H1), so $\Lambda P_\Lambda\delta_y = P(A)\delta_y$ and $W\delta_y$ is as stated.

(b). $\|Q_k\| \leq 1$ (Cauchy–Schwarz on blocks of volume 1) and $\|1-tP_\Lambda\| \leq 1$ for $t = (a_{k+1}/a_k)L^{-2} \in [0,1]$ (as $a_{k+1}\leq a_kL^2$), so $\|W\|\leq 1$ and $\|X_\Lambda^{-1}\| \leq a_k^{-1}+\gamma_L^{-1} \leq a_*^{-1}+\gamma_L^{-1}$. The upper bound holds because $a_k^2Q_kG_kQ_k^* \geq 0$ and $\|P_\Lambda\|\leq 1$.

(c). $W\delta_y$ is supported in the L-block B^{k+1} containing y, a cube of side L containing y, and $\|W\delta_y\|\leq 1$. Lemma F.144(i) for T_Λ (pins of diameter $\leq L$, $\gamma=\gamma_L$) gives $|\langle W\delta_y, T_\Lambda^{-1}W\delta_{y'}\rangle| \leq 2\gamma_L^{-1}e^{-\theta_0(|y-y'|-2L)}$. The term $a_k^{-1}\delta_{yy'}$ is at most a_*^{-1} and vanishes unless $y=y'$; the term $(a_{k+1}L^{-2}/a_k^2)P_\Lambda(y,y')$ has $a_{k+1}L^{-2}/a_k^2 = \frac{b}{a_k(a_k+b)} \leq a_*^{-1}$, $|P_\Lambda(y,y')|\leq 1$, and vanishes unless $|y-y'|\leq L$. Summing, $|C_\Lambda(y,y')| \leq (2\gamma_L^{-1}e^{2\theta_0L} + a_*^{-1} + a_*^{-1}e^{\theta_0L})e^{-\theta_0|y-y'|} \leq c_2e^{-\theta_0|y-y'|}$. $\Lambda=\Omega^{(k)}$ is allowed. □

Lemma F.146 (the resolvent difference; [4, (3.9), (3.10)])**.** *Under the hypotheses of lemma F.145:*

(a) $G_k(\Omega,\Lambda,A) - G^\eta_{k+1}(\Omega,A) = G_k(\Omega,\Lambda,A)\,M'\,G^\eta_{k+1}(\Omega,A)$ *with* $M' := [a_{k+1}L^{-2}P_{k+1}(A) - a_kP_k(A)](\Omega\setminus B^k\Lambda)$.

(b) *For* $f,f'\in\ell^2(B^k\Lambda)$, *with* $d_f := \operatorname{dist}(\operatorname{supp} f, \Omega\setminus B^k\Lambda)$,

$$|\langle f, (G_k(\Omega,\Lambda,A) - G^\eta_{k+1}(\Omega,A))f'\rangle| \leq c_3\|f\|\|f'\|\exp\big[-\tfrac{\theta_0}{6}\big(\operatorname{dist}(\operatorname{supp} f,\operatorname{supp} f') + d_f + d_{f'}\big)\big],$$

with $c_3 := 8a\gamma_L^{-2}e^{\theta_0L/3}S_L(\theta_0/3)S_L(\theta_0/6)$, $S_L(\theta) := \sum_{n\in\mathbb{Z}^d} e^{-\theta L(|n|-1)_+}$. *This is stronger than* [4, (3.10)], *whose* $B^k(\Lambda^c)$ *contains* $\Omega\setminus B^k\Lambda$.

Proof. (a) is the second resolvent identity, since $T_{\Omega^{(k)}} - T_\Lambda = M'$. (The bracket of [4, (3.9)] is printed with the opposite sign; only norms of it are used there.)

(b) M' is block-diagonal over the L-blocks $\Delta\subset\Omega\setminus B^k\Lambda$ (by (H2) each P_{k+1} summand and the P_k summands it contains live on one L-block), with $\|M'_\Delta\| \leq a_{k+1}L^{-2} + a_k \leq 2a$. Let first f, f' be supported in single L-blocks $B,B'\subset B^k\Lambda$, and write $u_\Delta := \operatorname{dist}(\Delta,\operatorname{supp} f)$, $v_\Delta := \operatorname{dist}(\Delta,\operatorname{supp} f')$. Both T_Λ and $T_{\Omega^{(k)}}$ are at least γ_L (lemma F.143), so lemma F.144(i) gives $\|\mathbf{1}_\Delta G_k(\Omega,\Lambda,A)f\| \leq 2\gamma_L^{-1}e^{-\theta_0u_\Delta}\|f\|$ and the same for $G^\eta_{k+1}(\Omega,A)f'$ with v_Δ; hence

$$|\langle f, (G_k(\Omega,\Lambda,A) - G^\eta_{k+1}(\Omega,A))f'\rangle| \leq 2a(2\gamma_L^{-1})^2\sum_\Delta e^{-\theta_0(u_\Delta+v_\Delta)}\|f\|\|f'\|.$$

Since $\Delta\subset\Omega\setminus B^k\Lambda$, $u_\Delta\geq d_f$ and $v_\Delta\geq d_{f'}$; since Δ has diameter L, $\operatorname{dist}(\operatorname{supp} f,\operatorname{supp} f') \leq u_\Delta + L + v_\Delta$. Therefore $u_\Delta+v_\Delta \geq \frac13(\operatorname{dist}(\operatorname{supp} f,\operatorname{supp} f') - L) + \frac13(d_f+d_{f'}) + \frac13(u_\Delta+v_\Delta)$, and $\sum_\Delta e^{-(\theta_0/3)u_\Delta} \leq S_L(\theta_0/3)$, a sum over the L-block lattice, uniform in the torus (on a torus, each block has a nearest lift, and distinct blocks have distinct lifts). So for block-supported f, f'

the bound holds with the rate $\theta_0/3$ and the constant $8a\gamma_L^{-2}e^{\theta_0 L/3}S_L(\theta_0/3)$. For general f, f' decompose $f=\sum_B f_B$, $f'=\sum_{B'} f'_{B'}$ over the L-blocks of $B^k\Lambda$; restricting supports increases all three distances, so $\operatorname{dist}(\operatorname{supp} f_B,\operatorname{supp} f'_{B'})+d_{f_B}+d_{f'_{B'}}\geq \operatorname{dist}(\operatorname{supp} f,\operatorname{supp} f')+d_f+d_{f'}$, and splitting the rate $\theta_0/3$ into two halves, one half gives the exponential of the statement and the other leaves the kernel $e^{-(\theta_0/6)\operatorname{dist}(\operatorname{supp} f_B,\operatorname{supp} f'_{B'})}\leq e^{-(\theta_0/6)\operatorname{dist}(B,B')}$, whose Schur norm on ℓ^2 of the block indices is at most $S_L(\theta_0/6)$; since $\sum_B\|f_B\|^2=\|f\|^2$, the Schur test gives (b). □

Lemma F.147 (the conditioning-region difference; [4, (1.18)])**.** *Under the hypotheses of lemma F.145, for $y, y'\in\Lambda$,*

$$|C_\Lambda(y,y')-C^{(k)}(\Omega,A;y,y')|\leq c_4\exp\big[-\tfrac{\theta_0}{6}\big(|y-y'|+\operatorname{dist}(y,\Omega^{(k)}\setminus\Lambda)+\operatorname{dist}(y',\Omega^{(k)}\setminus\Lambda)\big)\big],$$

with $c_4:=c_3e^{(\theta_0/6)(4L+2)}$. The bound decays a fortiori in the distances to the whole complement Λ^c.

Proof. $C^{(k)}(\Omega,A)=C_{\Omega^{(k)}}$. For $y\in\Lambda$, $W_{\Omega^{(k)}}\delta_y=W_\Lambda\delta_y=Q_k^*(\delta_y-(a_{k+1}/a_k)L^{-2}P(A)\delta_y)$, because $P(A)\delta_y$ lives on the L-block of y, which lies in Λ (H1); and $P_\Lambda(y,y')=P(A)(y,y')$ for $y,y'\in\Lambda$. By lemma F.145(a) for both Λ and $\Omega^{(k)}$ the constant terms cancel and

$$C_\Lambda(y,y')-C_{\Omega^{(k)}}(y,y')=\langle W\delta_y,(G_k(\Omega,\Lambda,A)-G_{k+1}^\eta(\Omega,A))W\delta_{y'}\rangle.$$

Apply lemma F.146(b) with $f=W\delta_y$, $f'=W\delta_{y'}$, both of norm ≤ 1 and supported in B^k of the L-blocks of y, y', which lie in $B^k\Lambda$. Every point of $\operatorname{supp} f$ is within L of y, and $\Omega\setminus B^k\Lambda=B^k(\Omega^{(k)}\setminus\Lambda)$ lies within 1 of $\Omega^{(k)}\setminus\Lambda$; so $\operatorname{dist}(\operatorname{supp} f,\operatorname{supp} f')\geq|y-y'|-2L$ and $d_f\geq\operatorname{dist}(y,\Omega^{(k)}\setminus\Lambda)-L-1$, and the shifts total at most $4L+2$. □

The remaining estimates use the printed theorem of [4], which is [1, Prop. 2.1]: for $\alpha<1$ there are $\delta_0,c_0,R_0>0$, independent of A, k, Ω, depending on d, M (and c_0 on α), such that for e_k small and $f\colon\Omega\to\mathbb{R}^N$: the $C^{1,\alpha}$ bound [4, (1.9)] for $x,x'\in\Omega$ with $\operatorname{dist}(\{x,x'\},\Omega^c)\geq R_0$; $|(D^\eta_{A,\mu}G_k(\Omega,A)f)(x)|$, $|(G_k(\Omega,A)f)(x)|\leq c_0e^{-\delta_0\operatorname{dist}(x,\operatorname{supp} f)}\|f\|_\infty$ for $x\in\Omega$ with $\operatorname{dist}(x,\Omega^c)\geq R_0$ [4, (1.10)]; and, for $\Omega\subset\Omega_0$ and $\delta G_k(\Omega,\Omega_0,A):=G_k(\Omega,A)-G_k(\Omega_0,A)$ [4, (1.11)], the same bounds with the same restrictions and an additional factor [4, (1.12)], printed in [1, (2.26)] as $\exp(-\delta_0\operatorname{dist}(\operatorname{supp} f,\Omega^c)-\delta_0\operatorname{dist}(\{x,x'\},\Omega^c))$. When Ω is the whole torus the depth condition is void [1, p. 611]. Only this restricted form (points at depth at least R_0, or Ω the whole torus) is used anywhere in this appendix; the clause of [1, p. 611] and [4, p. 573] that drops the depth condition on rectangular parallelepipeds is not used.

Lemma F.148 (the depth factor, two-sided)**.** *Let $\Omega\subset\Omega_0$ be unions of large blocks, A regular on Ω_0, e_k small, $\operatorname{supp} f$ of diameter at most s, and $x\in\Omega$ with $\operatorname{dist}(x,\Omega^c)\geq R_0$. Then*

$$|(\delta G_k(\Omega,\Omega_0,A)f)(x)|\leq c_0e^{\delta_0 s/2}\exp\big[-\tfrac{\delta_0}{2}\big(\operatorname{dist}(x,\operatorname{supp} f)+\operatorname{dist}(x,\Omega^c)+\operatorname{dist}(\operatorname{supp} f,\Omega^c)\big)\big]\|f\|_\infty,$$

and the same for $D^\eta_{A,\mu}\delta G_k(\Omega,\Omega_0,A)f$, and for the Hölder quotient of [4, (1.9)] with $|x-x'|\leq 1$ and $\operatorname{dist}(\{x,x'\},\Omega^c)\geq R_0$, with $\operatorname{dist}(\{x,x'\},\cdot)$ in place of $\operatorname{dist}(x,\cdot)$. For general f (no bound on the diameter) the bounds hold at the rate $\delta_0/4$ with the constant $c_0e^{3\delta_0/4}S(\delta_0/4)$ ($c_0e^{\delta_0}S(\delta_0/4)$ for the Hölder quotient).

Proof. $\operatorname{dist}(x,\Omega^c)\leq\operatorname{dist}(x,\operatorname{supp} f)+s+\operatorname{dist}(\operatorname{supp} f,\Omega^c)$, so $\operatorname{dist}(x,\operatorname{supp} f)+\operatorname{dist}(x,\Omega^c)+\operatorname{dist}(\operatorname{supp} f,\Omega^c)\leq 2\operatorname{dist}(x,\operatorname{supp} f)+2\operatorname{dist}(\operatorname{supp} f,\Omega^c)+s$. The printed factor, read with both of its terms as $\operatorname{dist}(\operatorname{supp} f,\Omega^c)$, gives $c_0e^{-\delta_0\operatorname{dist}(x,\operatorname{supp} f)}e^{-2\delta_0\operatorname{dist}(\operatorname{supp} f,\Omega^c)}\|f\|_\infty$, which is at most the right side; read as printed in [1, (2.26)], it gives the right side without the factor $e^{\delta_0 s/2}$ and at the rate δ_0. So the bound holds under either reading. For the Hölder quotient the printed factor carries $\operatorname{dist}(\{x,x'\},\cdot)$ and the same arithmetic applies, because $\operatorname{dist}(\{x,x'\},\Omega^c)\leq\operatorname{dist}(\{x,x'\},\operatorname{supp} f)+s+\operatorname{dist}(\operatorname{supp} f,\Omega^c)$ (the inequality for x and for x' separately, then the minimum over the two points). For general f decompose $\operatorname{supp} f$ into the unit blocks $\square$ it meets, apply the bound to $f\mathbf{1}_\square$ ($s\leq 1$; $\operatorname{dist}(x,\square\cap\operatorname{supp} f)\geq\operatorname{dist}(x,\operatorname{supp} f)$ and $\operatorname{dist}(\square\cap\operatorname{supp} f,\Omega^c)\geq\operatorname{dist}(\operatorname{supp} f,\Omega^c)$), and sum: with $\square_n$ the unit block of $n\in\mathbb{Z}^d$ and $x\in\square_{n_x}$, $\operatorname{dist}(x,\square_n)\geq(|n-n_x|-1)_+$, so $\sum_\square e^{-(\delta_0/2)\operatorname{dist}(x,\square\cap\operatorname{supp} f)}\leq e^{-(\delta_0/4)\operatorname{dist}(x,\operatorname{supp} f)}\sum_{n\in\mathbb{Z}^d}e^{-(\delta_0/4)(|n-n_x|-1)_+}\leq$

$e^{-(\delta_0/4)\,\mathrm{dist}(x,\mathrm{supp}\,f)}e^{\delta_0/4}S(\delta_0/4)$; for the Hölder quotient $\mathrm{dist}(\{x,x'\},\square_n)\ge\mathrm{dist}(x,\square_n)-1$ costs one more factor $e^{\delta_0/4}$. □

Lemma F.149 (domain differences; [4, (5.5), (1.20)]). *Let $\Omega\subset\Omega_0$ be unions of large blocks, (H1)–(H3), A regular on Ω_0, $e_k\le e_1$ and small enough for the printed theorem. Put*

$$\delta_7:=\min\{\delta_0/2,\,\theta_0(\gamma_\emptyset,1)\},\qquad \delta_7^S:=\min\{\theta_0,\,\delta_7\}=\min\{\theta_0(\gamma_L,L),\,\delta_0/2\}.$$

(a) *For $y,y'\in\Omega^{(k)}$, with $d(y):=\mathrm{dist}(B^k(y),\Omega^c)$,*

$$|(\Delta^{(k)}(\Omega,A)-\Delta^{(k)}(\Omega_0,A))(y,y')|\le c_6\exp\big[-\delta_7\big(|y-y'|+d(y)+d(y')\big)\big],$$

$c_6:=\max\{a^2c_0e^{\delta_0/2},\,4a^2\gamma_\emptyset^{-1}e^{2\theta_0(\gamma_\emptyset,1)R_0}\}e^{\delta_7}$: this is [4, (5.5)].

(b) *For every $\Lambda\subset\Omega^{(k)}$ that is a union of L-blocks and $x,x'\in\Lambda$, with $\tilde d(x):=\mathrm{dist}(x,\Omega^{(k)c})$,*

$$|C_\Lambda(\Omega,A;x,x')-C_\Lambda(\Omega_0,A;x,x')|\le c_{20}\exp\big[-\tfrac{\delta_7^S}{4}\big(|x-x'|+\tilde d(x)+\tilde d(x')\big)\big],$$

$c_{20}:=c_2^2c_6e^{\delta_7^S/2}S(\delta_7^S/2)^2$: this is [4, (1.20)].

Proof. (a) $Q_k(A)$ on Ω and on Ω_0 agree on the blocks $B^k(y)$, $y\in\Omega^{(k)}$, so by [4, (1.14)] the kernel is $\delta\Delta(y,y')=-a_k^2\langle q_k(y),\delta G_k(\Omega,\Omega_0,A)q_k(y')\rangle$, with $q_k(y),q_k(y')$ supported in Ω. Three cases. If $d(y)\ge R_0$, every $x\in B^k(y)$ has $\mathrm{dist}(x,\Omega^c)\ge R_0$, and lemma F.148 with $f=q_k(y')$ ($\|f\|_\infty=1$, $s\le1$) gives $|(\delta G_kq_k(y'))(x)|\le c_0e^{\delta_0/2}e^{-(\delta_0/2)(\mathrm{dist}(B^k(y),B^k(y'))+d(y)+d(y'))}$ for $x\in B^k(y)$; pairing with $q_k(y)$, whose entries have modulus 1 on a block of volume 1, gives $|\delta\Delta(y,y')|\le a_k^2c_0e^{\delta_0/2}e^{-(\delta_0/2)(\mathrm{dist}(B^k(y),B^k(y'))+d(y)+d(y'))}$. If $d(y')\ge R_0$, use $\delta\Delta(y,y')=\delta\Delta(y',y)^{\mathsf T}$. If $d(y),d(y')<R_0$, apply lemma F.144(i) to $G_k(\Omega,A)$ and to $G_k(\Omega_0,A)$ separately (lemma F.143 with $\Lambda=\emptyset$ on Ω and on Ω_0; $\ell_*=1$, $\gamma=\gamma_\emptyset$): $|\delta\Delta(y,y')|\le 4a_k^2\gamma_\emptyset^{-1}e^{-\theta_0(\gamma_\emptyset,1)\,\mathrm{dist}(B^k(y),B^k(y'))}\le 4a_k^2\gamma_\emptyset^{-1}e^{2\theta_0(\gamma_\emptyset,1)R_0}e^{-\theta_0(\gamma_\emptyset,1)(\mathrm{dist}(B^k(y),B^k(y'))+d(y)+d(y'))}$. Finally $\mathrm{dist}(B^k(y),B^k(y'))\ge|y-y'|-1$.

(b) The inverses are bounded by lemma F.145(b) on both domains, so $C_\Lambda(\Omega,A)-C_\Lambda(\Omega_0,A)=X_\Lambda(\Omega,A)^{-1}[X_\Lambda(\Omega_0,A)-X_\Lambda(\Omega,A)]X_\Lambda(\Omega_0,A)^{-1}$, and $X_\Lambda(\Omega_0,A)-X_\Lambda(\Omega,A)=-\delta\Delta|_\Lambda$, because $\Lambda\subset\Omega^{(k)}$ and $P(A)$ is the same on Λ for both domains. With lemma F.145(c) for both domains and (a),

$$|\delta C(x,x')|\le c_2^2c_6\sum_{y,y'\in\Lambda}e^{-\delta_7^S\Sigma},\qquad \Sigma:=|x-y|+|y-y'|+d(y)+d(y')+|y'-x'|.$$

Since $\Omega^c=B^k(\Omega^{(k)c})$ and $\mathrm{dist}(B^k(y),B^k(z))\ge|y-z|-1$, $\mathrm{dist}(y,\Omega^{(k)c})\le d(y)+1$; so $\tilde d(x)\le|x-y|+d(y)+1$, $\tilde d(x')\le|x'-y'|+d(y')+1$, and $|x-x'|+\tilde d(x)+\tilde d(x')\le2\Sigma+2$. Hence $e^{-\delta_7^S\Sigma}\le e^{\delta_7^S/2}e^{-(\delta_7^S/4)(|x-x'|+\tilde d(x)+\tilde d(x'))}e^{-(\delta_7^S/2)(|x-y|+|y'-x'|)}$, and the sums over y and y' give $S(\delta_7^S/2)^2$. □

Proposition F.150 ([1, Prop. 2.2, (2.27)–(2.29)]). *Assume (H1)–(H3), A regular on $\Omega_0\supset\Omega$, e_k as in lemma F.149. Then for $y,y'\in\Omega^{(k)}$*

$$|\Delta^{(k)}(\Omega,A;y,y')|\le c_7e^{-\delta_7|y-y'|},\qquad |\Delta^{(k)}(\Omega,A;y,y')-\Delta^{(k)}(\Omega_0,A;y,y')|\le c_7\exp\big[-\delta_7\big(|y-y'|+\mathrm{dist}(y,\Omega^{(k)c})+c$$

with $c_7:=\max\{a+2a^2\gamma_\emptyset^{-1}e^{\theta_0(\gamma_\emptyset,1)},\,c_6e^{2\delta_7}\}$. The first bound holds on every union Ω of unit blocks and uses only lemmas F.143 and F.144; its rate is in fact $\theta_0(\gamma_\emptyset,1)\ge\delta_7$.

Proof. $\Delta^{(k)}(y,y')=a_k\delta_{yy'}-a_k^2\langle q_k(y),G_k(\Omega,A)q_k(y')\rangle$ [4, (1.14)]. Lemma F.143 with $\Lambda=\emptyset$ and lemma F.144(i) with $\ell_*=1$ give $|\langle q_k(y),G_kq_k(y')\rangle|\le2\gamma_\emptyset^{-1}e^{-\theta_0(\gamma_\emptyset,1)\,\mathrm{dist}(B^k(y),B^k(y'))}$, and $\mathrm{dist}(B^k(y),B^k(y'))\ge|y-y'|-1$; this is the first bound. The second is lemma F.149(a) with $d(y)\ge\mathrm{dist}(y,\Omega^{(k)c})-1$. □

Theorem F.151 (the propagator estimates of group (A)). *Let $d=3$, $L=5$, (H1)–(H3), $\Omega\subset\Omega_0$ unions of large blocks of $T_1^{(k)}$ (the whole torus included), $\Lambda\subset\Omega^{(k)}$ a union of L-blocks, A regular on Ω_0, and ε_0 below the caps of lemmas F.142 and F.149. Then [4, (1.8)] on Ω (lemma F.143); the G_k clause of [4, Cor. 2.3] for $G_k(\Omega,A)$, $G_k(\Omega,\Lambda,A)$ and $G_{k+1}^\eta(\Omega,A)$ (lemma F.144); [1, Prop. 2.3], i.e. [4, (1.15), (1.16), (1.18), (1.20)] (lemma F.145(b), (c),*

lemma F.147, lemma F.149(b)); [4, (5.5)] *(lemma F.149(a)); and* [1, Prop. 2.2] *(proposition F.150) hold with the constants*

$$\gamma_\emptyset,\ \gamma_L,\ \gamma_X,\ \gamma_1,\ \theta_0=\theta_0(\gamma_L,L),\ \theta_0(\gamma_\emptyset,1),\ c_2,\ c_3,\ c_4,\ c_6,\ c_7,\ c_{20},\ \delta_7,\ \delta_7^S$$

defined above. They depend on $(d,N,\|q\|,c,\beta,a,L)$*, on* M *through the constants* δ_0,c_0,R_0 *of* [4, Thm.] *(which enter lemmas F.148 and F.149, proposition F.150 and nothing else), and on the cap on* ε_0*; they do not depend on* η*,* k*,* Ω*,* Ω_0*,* Λ*,* A*,* ε *or the torus.*

Proof. Each estimate is the lemma named after it; the dependences are read off the definitions of the constants. The order of the arguments is not circular: lemmas F.142 to F.147 and the first bound of proposition F.150 use no estimate of [4] beyond its definitions, while lemmas F.148 and F.149 and the second bound of proposition F.150 use the printed theorem of [4] in its restricted form (supplied on a torus, with the constants of [4], by lemma F.154 below), whose proof [4, Sect. 2] rests on [4, Lemmas 2.1, 2.2, 2.4]; the j-uniform form of [4, Lemma 2.4] is theorem F.7, whose clause [4, (2.37)] is lemma F.145(c) at $A=0$, $\Lambda=\Omega^{(k)}$. □

Remark F.152 (dependence on L; the depth margin). (i) $\gamma_\emptyset=\frac14\min\{2,a_*\}$ and the constants of theorem F.7 do not depend on L, with $a_*\ge 3a/4$ for $L\ge 2$; γ_L, γ_X, θ_0 and c_2 do, through $\gamma_L\propto L^{-2}$, as [4, p. 588] says of the constants of [4, (2.37)] and [1, Prop. 2.3]. At the fixed value $L=5$ all are numbers. (ii) The depth condition of the printed theorem is met wherever it is invoked in Steps 3(B)–3(D): with $r(t)=R(1+\ln t^{-1})^\varrho$ and $R>R_0$ [2, (2.7)], every point within distance 1 of a set at distance more than $2r(t)$ from a boundary has depth more than $2R-1\ge R_0$ (as $R_0\ge 1$ may be assumed, enlarging R_0 weakening nothing in the theorem); pairs x,x' with $|x-x'|>1$ in [4, (1.9)] reduce to the sup bounds [4, p. 578].

Lemma F.153 (the polylogarithmic layer)**.** *Let* $r(t)=R(1+\ln t^{-1})^\varrho$ *with* $\varrho>1$*, and* $\delta,R,\kappa>0$*. Then* $e^{-\delta r(t)}\le t^\kappa$ *for all* $t\in(0,1]$ *if* $\delta R\ge\kappa$*, and in general for* $t\le t_0:=\exp(-(\kappa/(\delta R))^{1/(\varrho-1)})$*. Consequently* $e^{-\delta r(t)}\le C_\kappa t^\kappa$ *on* $(0,1]$ *with* $C_\kappa:=t_0^{-\kappa}=\exp(\kappa(\kappa/(\delta R))^{1/(\varrho-1)})$*.*

Proof. With $u:=\ln t^{-1}\ge 0$, $\delta R(1+u)^\varrho\ge\delta R\max\{1+u,u^\varrho\}$. This is at least κu if $\delta R\ge\kappa$, or if $u^{\varrho-1}\ge\kappa/(\delta R)$, i.e. $t\le t_0$. For $t\in[t_0,1]$, $e^{-\delta r(t)}\le 1\le t_0^{-\kappa}t^\kappa$. □

Lemma F.154 ([1, Prop. 2.1] on a torus, with torus-free constants)**.** *Let* T *be a torus* $\eta\mathbb{Z}^d/\Gamma$ *whose period lattice* Γ *is contained in* $M\mathbb{Z}^d$ *(so the large blocks of the unit lattice tile* T*) and whose sup-systole is at least* $3M$*; let* $\Omega\subset T$ *be a union of large blocks (*$\Omega=T$ *included),* A *regular on* Ω*, and* e_k *small. Then the theorem of* [4] *(*[1, Prop. 2.1, (2.24)–(2.26)]*) holds on* T *in its restricted form, with distances the quotient sup-distances and with the constants* δ_0*,* c_0*,* R_0 *of* [4]*, which do not depend on* Γ*; for* $\Omega=T$ *it holds for all* $x,x'\in T$*. The same is true for tori of sup-systole less than* $3M$ *(but at least* M*) whose period lattice lies in* $M\mathbb{Z}^d$*. Consequently every estimate of theorem F.151 holds on* T *with the constants of theorem F.151.*

Proof. The proof of [4, Sect. 2] (pp. 574–580) is carried onto T step by step; it uses the geometry of the lattice in the following places, and nowhere else.

(1) *The local cubes.* Let J index the vertices Mj of the large-block partition of T and $C_j:=Mj+[-M,M]^d$ (in unit-lattice units). Since $2M$ is less than the sup-systole, the projection $\eta\mathbb{Z}^d\to T$ is injective on each C_j; it preserves bonds, unit blocks, L-blocks and the contours of Q_k inside blocks. So $\Pi_j:=\Omega\cap C_j$ [4, p. 575] is a union of at most 2^d large blocks inside an embedded cube of side $2M$, a cube when Mj is not a boundary point of Ω, and the decomposition $A=A_0+A'$ on C_j with the constant $A_0:=A(Mj)$ and the local field $A_j:=A_0+\theta_jA'$ of [4, p. 575] is built from $A|_{C_j}$ alone, with $|A'|,|\partial^\eta A'|\le c'e_k^{\beta-1}$, $c'=dMc$. Hence [4, Lemma 2.1] applies to every $G_k(\Pi_j,A_j)$ and [4, Lemma 2.2] to the interior cubes (for Π_j meeting $\partial\Omega$, $A_j:=A$ [4, p. 575]), with their constants, which depend on d, M, α, c only; [4, Lemma 2.4], used in the proof of [4, Lemma 2.2], is theorem F.7 on parallelepipeds of the fine lattice.

(2) *The partition of unity.* With $h\in C_0^\infty((-\frac23,\frac23))$, $h=1$ on $[-\frac13,\frac13]$, $\sum_{n\in\mathbb{Z}}h(\cdot-n)^2=1$, and $h_j(x):=\prod_\mu h(x_\mu/M-j_\mu)$ [4, p. 575]: because $\Gamma\subset M\mathbb{Z}^d$ the family $\{h_j\}_{j\in J}$ is well defined on T, $\sum_{j\in J}h_j^2=1$ on T and $\operatorname{supp}h_j\subset C_j$. So the approximate inverse $G_0=\sum_j h_jG_k(\Pi_j,A_j)h_j$

[4, (2.2)] and the identities [4, (2.3)–(2.9)] hold on T as on $\eta\mathbb{Z}^d$; for $\Omega = T$ the boundary of Ω is empty and the boundary term of [4, (2.6)] vanishes.

(3) *The remainder.* Each j has at most $3^d - 1$ neighbours and each point lies in at most 2^d supports, so the operator R of [4, (2.11)] has $\|R\|_{2,2} \le c_d O(M^{-1}) < 1$ by [4, Lemma 2.1], as on [4, p. 577]; the Neumann series [4, (2.12)] converges and gives the walk representation [4, (2.13)], the walks being sequences of indices $j \in J$ with consecutive terms adjacent, and the count $2^d(3^d)^{n-1}2^d$ of walks of length n [4, p. 579] is unchanged.

(4) *The metric.* Consecutive cubes of a walk have centres at quotient distance at most M, so a walk of length n from a cube containing x, x' to a cube containing points of $\operatorname{supp} f$ has $\operatorname{dist}(\{x, x'\}, \operatorname{supp} f) \le (n+2)M$, i.e. $n \ge M^{-1}\operatorname{dist}(\{x, x'\}, \operatorname{supp} f) - 2$, which is the restriction used in [4, (2.22)]; with (3), the geometric sum [4, (2.22)] gives δ_0 and c_0 as on $\eta\mathbb{Z}^d$, once M is fixed with $3^d c_{2.1} O(1) M^{-1} \le \mathrm{e}^{-1}$, $c_{2.1}$ the constant of [4, Lemma 2.1, (2.15)] (not the c_2 of lemma F.145). The definition [4, (2.19)] of R_0 (the diameter of the union of the first n_0 cubes of a walk, plus $2M$) and the implication "$\operatorname{dist}(\{x, x'\}, \Omega^c) \ge R_0$, hence the first n_0 cubes Π_{ω_i} of every walk from x, x' are cubes contained in Ω" use only the quotient metric and (1); on these cubes A_j is constant near $\partial\Pi_j$ and [4, Lemma 2.2] applies, on the others only [4, Lemma 2.1] is used, as in [4, (2.20)–(2.21)]. The cancellation of the terms with interior cubes in the two representations of $\delta G_k(\Omega, \Omega_0, A)$ [4, p. 579] uses (1)–(2) only. If $\Omega = T$, every $\Pi_j = C_j$ is a cube with A_j constant near its boundary, so no restriction on x, x' is needed [1, p. 611].

(5) *Small tori.* If the sup-systole of $T = \eta\mathbb{Z}^d/\Gamma$ is less than $3M$, let $\tilde{T} := \eta\mathbb{Z}^d/3\Gamma$, a covering torus of systole at least $3M$ with deck group $(\mathbb{Z}/3)^d$. The operators $-\Delta^{\eta,N}_{A,\Omega} + m_k^2 + a_k P_k(A)$ lifted to $\tilde{T}$ commute with the deck translations and map deck-invariant functions to deck-invariant functions, so for f on Ω the lift of $G_k(\Omega, A)f$ is $G_k(\tilde{\Omega}, \tilde{A})\tilde{f}$, with $\tilde{\Omega}, \tilde{A}, \tilde{f}$ the lifts; $\|\tilde{f}\|_\infty = \|f\|_\infty$, and the quotient distances in $\tilde{T}$ from a lift of x to $\operatorname{supp}\tilde{f}$ and to $\tilde{\Omega}^c$ equal those in T. So the bounds on $\tilde{T}$, proved in (1)–(4), descend to T with the same constants. The same applies to δG_k.

No other constant of [4, Sect. 2] involves the global geometry. Finally, lemmas F.142 to F.147, the first bound of proposition F.150 and lemma F.153 are proved above on an arbitrary union of blocks of an arbitrary torus (their only global ingredients are the sums $S(\theta)$ and $S_L(\theta)$ over the block lattice, bounded by their values on $\mathbb{Z}^d$), and lemmas F.148 and F.149 and the second bound of proposition F.150 use the printed theorem only in the form just established. Hence theorem F.151 holds on T with the same constants. □

Step 4: the mass. The scalar mass is a parameter of the theorems: [1, (1.14)] and [2, (1.3)] are stated for every $M^2 > 0$, with constants depending on M, and nothing in Steps 1–3 distinguishes m from M_1. The one printed clause on the mass, [2, Prop. 3.1] "we assume $m^2 \le O(1)$ also", is the cap $(L^k\varepsilon)^2 M^2 \le c_m(d, a)$ on the dimensionless mass at the scales at which the proposition is applied (remark 22.2), one more one-sided cap on ε_0; taking ε_0 below the caps for both masses gives theorem F.1 at m and at M_1 with one stopping scale. This proves theorem F.1, and proposition 3.8 follows as stated there.

F.3. **The volume limit at fixed spacing.** All quantities of lemma 12.9 are built from finitely many torus Riemann sums $\mathcal{R}_T[F]$ of smooth periodic functions, and the volume limit at fixed ε is the convergence of Riemann sums to integrals. We record the rate, which is also used in appendix F.4.

Lemma F.155 (torus Riemann sums)**.** *Let $d_0 = 3L_0$, let $F \in C^\infty(\mathbb{R}^{d_0})$ be $(2\pi/\varepsilon)$-periodic in every coordinate, and let $\mathcal{R}_T[F]$ be its Riemann sum over $(T^*)^{L_0}$ for a torus of side $\ell \in \varepsilon\mathbb{N}$. For every $r \in \mathbb{N}$,*

$$\Big|\mathcal{R}_T[F] - \int_{\mathsf{B}^{L_0}} F\Big| \;\le\; \sum_{n \in \mathbb{Z}^{d_0}\setminus\{0\}} \Big(\frac{\pi}{2\ell|n|}\Big)^r \big\|\partial^r_{n/|n|} F\big\|_{L^1(\mathsf{B}^{L_0})},$$

∂_v the directional derivative; in particular, for $r \ge d_0+1$, $|\mathcal{R}_T[F] - \int F| \le C_{d_0,r}\,\ell^{-r} \max_{|\alpha|=r} \|\partial^\alpha F\|_{L^1(\mathsf{B}^{L_0})}$ with $C_{d_0,r} = (\pi/2)^r d_0^{r/2} \sum_{n\neq 0} |n|^{-r}$. If $\int_{\mathsf{B}^{L_0}} F = 0$, the same bounds hold for $|\mathcal{R}_T[F]|$.

Proof. Expand F in its Fourier series on the period cell, $F(p)=\sum_j c_j e^{i\varepsilon j\cdot p}$. With $h=2\pi/\ell$ and $N_\ell=\ell/\varepsilon$, $\sum_{p\in(T^*)^{L_0}} e^{i\varepsilon j\cdot p}=N_\ell^{d_0}\mathbf{1}[j\in N_\ell\mathbb{Z}^{d_0}]$, so $\mathcal{R}_T[F]=\sum_{n\in\mathbb{Z}^{d_0}}\hat F(n)$ with $\hat F(n):=\int_{\mathsf{B}^{L_0}}F(p)e^{-i\ell n\cdot p}dp$ (Poisson summation); the term $n=0$ is $\int F$. For $n\neq 0$ put $\delta:=\pi n/(\ell|n|^2)$; periodicity gives $\int F(p-\delta)e^{-i\ell n\cdot p}dp=e^{-i\ell n\cdot\delta}\hat F(n)=-\hat F(n)$, so $2\hat F(n)=\int(F-F(\cdot-\delta))e^{-i\ell n\cdot p}$ and, iterating r times, $|\hat F(n)|\le 2^{-r}\|\Delta_\delta^r F\|_1$. The difference quotient is $\Delta_\delta G(p)=\int_0^1(\delta\cdot\nabla G)(p-\theta\delta)\,d\theta$, so by Fubini and the shift invariance of the L^1 norm on the period cell $\|\Delta_\delta G\|_1\le|\delta|\,\|\partial_{\delta/|\delta|}G\|_1$; applied r times in the same direction, $|\hat F(n)|\le(\pi/(2\ell|n|))^r\|\partial^r_{n/|n|}F\|_1$. The second form follows from $\partial_v^r=\sum_{|\alpha|=r}\frac{r!}{\alpha!}v^\alpha\partial^\alpha$ and $\sum_{|\alpha|=r}\frac{r!}{\alpha!}|v^\alpha|\le d_0^{r/2}$ for $|v|=1$, and $\sum_{n\neq0}|n|^{-r}<\infty$ for $r>d_0$. □

Nothing in the lemma refers to ε except through the period and the norms: $C_{d_0,r}$ is absolute, and every ε-dependence sits in the L^1 norms of the derivatives. At fixed ε those norms are finite numbers, continuous in every mass parameter of F.

Proof of lemma 12.9, with the boundedness clause. Fix k. We show that each quantity converges as $J\to\infty$, uniformly for $M\in[M_1,m]$, to its value on $\varepsilon\mathbb{Z}^3$; the limits named in the lemma are these values, and boundedness in $J\ge J_*(k)$ follows from convergence.

The counterterm. By (i) of appendix F.1, $\delta m^2{}_{(ab),T}(M)$ is defined order by order as minus the sum of the one-particle-irreducible scalar two-point graphs at zero momentum of order (a,b), in which the lower-order counterterms appear as vertex coefficients. Each graph with L_0 loops is $\mathcal{R}_T[F_G]$ for a smooth $(2\pi/\varepsilon)$-periodic F_G (products of periodic propagators $(\hat p^2+M^2)^{-1}$, $(\hat q^2+\mu_0^2)^{-1}$ at the masses M, μ_0, all positive, times trigonometric polynomials; at fixed ε there is no singularity); by lemma F.155 with any $r\ge 3L_0+1$, $\mathcal{R}_T[F_G]\to\int_{\mathsf{B}^{L_0}}F_G$ as $\ell\to\infty$ at the rate ℓ^{-r}, uniformly on $[M_1,m]$ because the L^1 norms are continuous in M on that compact interval. By induction on the order, the lower-order coefficients converge, so $\delta m^2{}_{(ab),T}(M)\to\delta m^2{}_{(ab)}(M)$, the coefficient defined by the same recursion with $\int_{\mathsf{B}}$ in place of $\mathcal{R}_T[\cdot]$, which is King's free-boundary counterterm on $\varepsilon\mathbb{Z}^3$ [14, p. 330]: $\alpha^{\mathrm{tor}}_{k,T_J}(M)\to\alpha^\infty_k(M)$ uniformly on $[M_1,m]$. Hence $s_{k,T_J}(M_1)\to s^\infty_k(M_1)$, and the first offset $\sigma_{k,T_J}=\frac12(\alpha^{\mathrm{tor}}_{k,T_J}(m)-\alpha^F)$ tends to 0, since $\alpha^F=m^2+\delta m^2{}_\varepsilon=\alpha^\infty_k(m)$ (definition 12.1).

The Wick constant. $c_{k,T}(M)=\mathcal{R}_T[(\hat p^2+M^2)^{-1}]\to\int_{\mathsf{B}}(\hat p^2+M^2)^{-1}=\mathsf{W}_\varepsilon(M)$, and $\Delta c_{T_J}=\mathsf{W}_\varepsilon-c_{k,T_J}(m)\to 0$; this is also the image-sum statement, $c_{k,T}(M)=\sum_{n\in\mathbb{Z}^3}\mathsf{W}_\varepsilon(M;n\ell)$, each image decaying exponentially at fixed ε.

The constants. $E_0(M)/|T|$ is a Riemann sum of the smooth periodic function $\frac N2\ln(\hat p^2+M^2)$ plus an M-independent photon term, so $[E_0(M_1)-E_0(m)]/|T|$ converges to $\frac N2\int_{\mathsf{B}}\ln\frac{\hat p^2+M_1^2}{\hat p^2+m^2}$. $E_1(M)$ is a finite sum of connected vacuum graphs on T; by translation invariance each equals $|T|$ times a Riemann sum $\mathcal{R}_T[F_G]$ of a smooth periodic integrand (the graph with one vertex fixed), with the counterterm vertices $\delta m^2{}_{(ab),T}(M)$ as coefficients, so $E_1(M)/|T|$ converges by lemma F.155 and the first step. Hence

$$|T_J|^{-1}B^z_{k,T_J}(M_1)=\frac{E(M_1)-E(m)}{|T_J|}-\tfrac12\big(2c_{k,T_J}(m)+z_k\big)\big(\alpha^{\mathrm{tor}}_{k,T_J}(m)-\alpha^{\mathrm{tor}}_{k,T_J}(M_1)\big)$$

converges to a finite limit $b^\infty_k(M_1)$, every term converging and z_k being independent of J. □

F.4. **The c-number lemma.** Lemma 3.9 says that $\sigma_1(M_1)=\liminf_k s^\infty_k(M_1)$ and $\beta_1(M_1)=\limsup_k b^\infty_k(M_1)$ are finite. Both are statements about the $\varepsilon\mathbb{Z}^3$ values of appendix F.3, uniformly in $\varepsilon=\varepsilon_k$, and both come from one source: a mass derivative of King's renormalised two-point graphs lowers every lattice degree by two and leaves a convergent lattice integral.

Proposition F.156 (the derivative identity)**.** *Fix ε and a King torus T (or $\varepsilon\mathbb{Z}^3$, with $\int_{\mathsf{B}}$ in place of $\mathcal{R}_T[\cdot]$ and densities in place of extensive quantities). Coefficientwise in (e,λ),*

$$\frac{d}{dM^2}B^z_{k,T}(M)=\tfrac12|T|\,\Psi_{k,T}(M),\qquad \Psi_{k,T}(M):=\big(1+\delta m^{2\prime}{}_{k,T}(M)\big)\big[N\big(c_{k,T}(m)-c_{k,T}(M)\big)+z_k-z_{k,T}(M)\big]-\big[\big(1+$$

where $' = d/dM^2$, $Y_T(M) := |T|^{-1}\langle\sum_x \varepsilon^3|\phi_x|^2\rangle^{(M)}_{k,T} - Nc_{k,T}(M)$ *is the connected one-point function of* $|\phi|^2$ *in the torus model at mass* M, $z_{k,T}(M)$ *its order-*e^2 *part,* $\hat{Y}_T := Y_T - z_{k,T}(M)$, *and* $[\cdot]_{\le 13}$ *the truncation at total order* 13 *in* (e,λ). *Consequently* $|T|^{-1}|B^z_{k,T}(M_1)| \le \frac{1}{2}|m^2 - M_1^2| \sup_{M\in[M_1,m]} |\Psi_{k,T}(M)|$.

Proof. Write $\ln Z^{(M)} = \mathcal{L}(M) - E_1(M)$ as in appendix F.1. The vertices depend on M only through $\delta m^2{}_{k,T}(M)$, and for the normalised Gaussian measure $\frac{d}{dM^2}\int d\mu_M G = -\frac{1}{2}\langle G; \sum_x \varepsilon^3|\phi_x|^2\rangle_{\mu_M}$; so $\mathcal{L}' = -\frac{1}{2}|T|\big[Y_T + \delta m^{2\prime}{}_{k,T}(Nc_{k,T} + Y_T)\big]$. Further $E_0' = -\frac{N}{2}|T|c_{k,T}(M)$, and $E_1' = [\mathcal{L}']_{\le 13}$ since E_1 is the Taylor part of $\mathcal{L}$ and the derivatives commute at fixed ε. With $B^z = E(M) - E(m) - \frac{1}{2}(Nc_{k,T}(m) + z_k)(\alpha(m) - \alpha(M))|T|$ and $\alpha' = 1 + \delta m^{2\prime}$,

$$\frac{dB^z}{dM^2} = \tfrac{1}{2}|T|\Big\{(1+\delta m^{2\prime})\big(Nc_{k,T}(m) + z_k\big) - Nc_{k,T}(M) - \big[Y_T + \delta m^{2\prime}(Nc_{k,T}(M) + Y_T)\big]_{\le 13}\Big\}.$$

The terms $\delta m^{2\prime}\, Nc_{k,T}(M)$ and $(1+\delta m^{2\prime})z_{k,T}(M)$ have order at most 6 and pass through the truncation unchanged (δm^2 has orders $a+2b \le 4$); collecting gives Ψ. Odd powers of e vanish by charge conjugation, and the order-λ part of Y_T vanishes because $\delta m^2{}_{(01),T} = -2\lambda(N+2)c_{k,T}(M)$ uses the same torus tadpole. Since $B^z(m) = 0$, $B^z(M_1) = -\frac{1}{2}|T|\int_{M_1^2}^{m^2}\Psi\, d\mu^2$, which gives the last claim. □

By lemma 12.9, $b_k^\infty(M_1) = \lim_J |T_J|^{-1}B^z_{k,T_J}(M_1) = -\frac{1}{2}\int_{M_1^2}^{m^2}\Psi_k^\infty(\mu)\, d\mu^2$ with Ψ_k^∞ the $\varepsilon\mathbb{Z}^3$ form of Ψ, so $\beta_1(M_1) < \infty$ follows from a bound on Ψ_k^∞ uniform in k and in $M \in [M_1, m]$; and $s_k^\infty(M_1) = \frac{1}{2}(m^2 - M_1^2) + \frac{1}{2}\int_{M_1^2}^{m^2}\delta m^{2\prime}{}_k(\mu)\, d\mu^2$, so $\sigma_1(M_1)$ is controlled by the mass derivative of the counterterm. Both are finite sums of renormalised lattice Feynman integrals at zero external momentum, and the tool is Reisz's power counting.

Reisz's theorems in three dimensions. Reisz's power counting theorem [20, Thm. 1] states that a lattice Feynman integrand of his class $\mathcal{F}$ (numerators trigonometric polynomials in the momenta, denominators $e(l;\varepsilon) + \mu^2$ with $e(l;\varepsilon) = \varepsilon^{-2}\eta(l\varepsilon)$, $\eta \ge 0$, $\eta(x) \to x^2$) whose lattice degrees of divergence (ultraviolet degrees in the sense of his Definition 2.3, computed with the lattice behaviour of the propagators) are negative for the integral and for every Zimmermann subspace has a convergent continuum limit, and its proof gives the absolute bound $\sup_{\varepsilon\le\varepsilon_0}\int_{\mathsf{B}^{L_0}}|F| \le K$. His renormalisation theorem [21] states that the forest formula with his periodic subtraction operators, applied to a graph all of whose one-particle-irreducible subgraphs have degree $\omega \ge 0$ subtracted to order ω, produces an integrand in $\mathcal{F}$, periodic in all loop momenta, with all Zimmermann degrees negative, provided the degrees are additive over disjoint subgraphs and the subtraction degrees satisfy his conditions (4-3) and (4-4) [21, Lemma 5.2]. Both are printed for $d = 4$; the dimension enters only through the degrees, and the proofs apply verbatim in $d = 3$ with the degrees $\omega = 3L_0 - 2I_\phi - 2I_A +$ (numerator degree). King's graphs satisfy the hypotheses: his vertices are the quartic vertex, the hopping vertices with one or two photon legs (trigonometric numerators), and the counterterm vertices; the only divergent one-particle-irreducible subgraphs are the scalar two-point function ($\omega = 1$ at one loop, $\omega = 0$ at two loops) and the photon two-point function ($\omega = 1$), and King's recursive counterterm is exactly the forest-scheme counterterm: on $\varepsilon\mathbb{Z}^3$ the order-1 Taylor term of the scalar two-point subtraction vanishes by parity and the photon two-point function at zero momentum vanishes by the lattice Ward identity, so the subtracted integrands are King's graphs with his $\delta m^2{}_{(ab)}$ inserted. We write $\hat{R}F_G$ for the subtracted integrand of a graph G with these rules, and $\hat{R}'F_{G,l}$ for the mass-derivative integrand defined next.

Lemma F.157 (derivative integrands)**.** *Let* G *be one of King's two-point or vacuum graphs on* $\varepsilon\mathbb{Z}^3$ *with its scalar two-point subgraphs subtracted to order* 1 *(one loop) or* 0 *(two loops) and its photon two-point subgraphs to order* 1. (i) $\frac{d}{dM^2}\int_{\mathsf{B}^{L_0}}\hat{R}F_G = \sum_l \int_{\mathsf{B}^{L_0}}\hat{R}'F_{G,l}$, *the sum over the scalar lines* l *of* G, *where* $F_{G,l}$ *has the line* l *doubled (a factor* $-(\hat{p}_l^2 + M^2)^{-1}$*) and* $\hat{R}'$ *uses the same forests with the* effective *degrees: unchanged for subgraphs not containing* l; 0 *(the order-*1

Taylor term dropped) for a one-loop two-point subgraph containing l; no subtraction for a two-loop two-point subgraph containing l. (ii) *$\hat{R}'F_{G,l}$ lies in $\mathcal{F}$, is periodic in all loop momenta, and has all Zimmermann degrees negative; the same holds for every momentum derivative $\partial_p^\alpha \hat{R}'F_{G,l}$ and $\partial_p^\alpha \hat{R}F_G$, with no degree increased.*

Proof. (i) The subtraction operators act on the external momentum of a subgraph and do not depend on M, so d/dM^2 commutes with them and acts on the propagators, $\frac{d}{dM^2}(\hat{p}^2+M^2)^{-1} = -(\hat{p}^2+M^2)^{-2}$: this gives $\sum_l \hat{R}^{\text{formal}}F_{G,l}$ with the original degrees. Doubling a line lowers the degree of every subgraph containing it by 2, so a one-loop two-point subgraph containing l has degree -1 and a two-loop one degree -2; the Taylor terms of order $>$ (new degree) are over-subtractions, which Reisz's conditions do not allow. They integrate to zero: the order-1 term of a one-loop subgraph is $\sum_\rho P(p_\rho;\varepsilon)\,\partial_{p_\rho}F_\gamma(q,0)$ with q the loop momentum of the subgraph, a free integration variable at fixed outer momenta, and $\partial_p F_\gamma(q,0)$ is odd under $q\to -q$ while the rest of the integrand does not depend on q; the order-0 term of a two-loop subgraph containing the doubled line is its value at zero momentum, which is the mass derivative of the two-loop counterterm and is cancelled exactly by the derivative of the counterterm vertex that the forest formula carries. Dropping them gives $\hat{R}'$. (ii) With the effective degrees, every subtracted subgraph is subtracted to its (new) degree and every unsubtracted one has negative degree, which is Reisz's condition (4-3) with equality and (4-4); his Lemma 5.2 gives negative Zimmermann degrees. Momentum derivatives lower the degree of the differentiated propagator by one and keep the class $\mathcal{F}$ and the periodicity. □

Proposition F.158 (uniform bounds)**.** *For each of King's graphs G, each scalar line l and each multi-index α there are $\varepsilon_0>0$ and $K<\infty$ with*

$$\sup_{\varepsilon\le\varepsilon_0}\sup_{M\in[M_1,m]}\int_{\mathsf{B}^{L_0}}\left|\partial_p^\alpha \hat{R}F_G(p;M)\right|dp\le K,\qquad \sup_{\varepsilon\le\varepsilon_0}\sup_{M\in[M_1,m]}\int_{\mathsf{B}^{L_0}}\left|\partial_p^\alpha \hat{R}'F_{G,l}(p;M)\right|dp\le K,$$

and for $\varepsilon_0<\varepsilon\le 1/5$ the same integrals are bounded with an ε_0-dependent constant.

Proof. By lemma F.157 the integrands are in $\mathcal{F}$ with negative Zimmermann degrees, so Reisz's theorem gives the absolute bound at each fixed mass. Uniformity on the compact mass interval: every term of the integrand is a mass-free trigonometric numerator over a product of propagators $(e_L+\mu_L^2)^{-n_L}$; bringing the terms to the common denominator $\prod_L(e_L+\mu_L^2)^{n_L}$, the numerator is a polynomial in the squared masses with coefficients that are mass-free trigonometric polynomials, $\sum_i P_i(\mu^2)V_i(p;\varepsilon)$ with finitely many distinct monomials P_i; each $V_i/\prod_L(e_L+\mu_L^2)^{n_L}$ has the same or lower degrees, so is bounded in L^1 by Reisz's theorem at the fixed mass $\mu_{\min}=\min(M_1,\mu_0)$, and dominates the integrand at every mass in the interval up to the factor $\sup_i\sup_{[M_1,m]}|P_i|\prod_L(\mu_{\min}^2/(\dots))$; since all masses are bounded above and below on the interval, the constant is uniform. For $\varepsilon>\varepsilon_0$ the integrals are finite numbers of a finite family, continuous in M. □

Theorem F.159 (the c-number lemma)**.** *Let $0<M_1<m$.* (a) *There is $C=C(e,\lambda,\mu_0,m,M_1)$ with $\sup_k\sup_{M\in[M_1,m]}|\Psi_k^\infty(M)|\le C$; hence $\sup_k|b_k^\infty(M_1)|\le\frac12(m^2-M_1^2)C$ and $\beta_1(M_1)<\infty$.* (b) *For every $M>0$ the limit $\alpha_\infty(M)-\alpha_\infty(m)=\lim_k[\alpha_k^\infty(M)-\alpha_k^\infty(m)]$ exists and is a polynomial of degree ≤ 2 in (e^2,λ) with finite coefficients; hence $\sigma_1(M_1)=\lim_k s_k^\infty(M_1)=\frac12(\alpha_\infty(m)-\alpha_\infty(M_1))$ exists and is finite. In particular lemma 3.9 holds, and so does the statement* (CT_∞) *of the coupling-range section (before lemma 22.1).*

Proof. (a) By proposition F.156, Ψ_k^∞ is a polynomial of degree ≤ 13 in (e,λ) whose coefficients are finite combinations of four kinds of quantities, each to be bounded uniformly in $\varepsilon=\varepsilon_k\le\varepsilon_0$ and $M\in[M_1,m]$: (Q1) the coefficients $x_{ab}(M)$ of $\hat{Y}(M)$ at orders $a/2+b\ge 2$; (Q2) the coefficients $y_{a'b'}(M)$ of $\delta{m^2}'_k(M)$ at the five orders $(0,1),(2,0),(0,2),(2,1),(4,0)$; (Q3) $c(m)-c(M)=\mathsf{W}_\varepsilon(m)-\mathsf{W}_\varepsilon(M)$; (Q4) $z_k-z_k(M)$. (Q1): $x_{ab}(M)$ is a sum over the connected two-point graphs of $\langle|\phi|^2\rangle$ (rings of scalar lines with the vertices of the action, the subdivergences subtracted by King's counterterms) of $\int_{\mathsf{B}^{L_0}}\hat{R}F_G$ with overall degree

$\omega = 3-2(a/2+b)-\ldots \le -1$ and negative proper degrees; proposition F.158 with $\alpha = 0$ bounds each. (Q2): $y_{a'b'}(M) = -\frac{d}{dM^2}$ of the zero-momentum one-particle-irreducible two-point function at order (a', b') without its own counterterm $= -\sum_l \int \hat{R}' F_{G,l}$ by lemma F.157(i), bounded by proposition F.158. (Q3): $\mathsf{W}_\varepsilon(m) - \mathsf{W}_\varepsilon(M) = (M^2 - m^2)\int_{\mathsf{B}} (\hat{p}^2 + m^2)^{-1}(\hat{p}^2 + M^2)^{-1}$, a degree -1 integral, bounded. (Q4): $z_k = z_k(m)$ is the reference scheme of remark 2.4, the order-e^2 coefficient of Y at mass m (lemma 6.8); $z_k(m) - z_k(M) = -\int_{M^2}^{m^2} \frac{d}{d\mu^2} z_k(\mu)\, d\mu^2$, and $\frac{d}{d\mu^2} z_k(\mu) = e^2 \sum_l \int \hat{R}' F_{(20),l}$ is bounded by proposition F.158 (the degree-0 one-loop photon-exchange and seagull rings, each lowered to -2 by the doubled line); integrate over the interval. For the finitely many k with $\varepsilon_k > \varepsilon_0$ each quantity is a finite number, continuous in M. (b) For each order (a', b'), $y_{a'b'}(\mu;\varepsilon) = -\sum_l \int_{\mathsf{B}^{L_0}} \hat{R}' F_{G,l}$ converges as $\varepsilon \to 0$ by Reisz's renormalisation theorem applied to the effective integrand (lemma F.157(ii) provides its hypotheses) and is bounded uniformly in μ on compact intervals (proposition F.158); dominated convergence in $\delta m^2{}_k(m) - \delta m^2{}_k(M) = \int_{M^2}^{m^2} \delta m^{2\prime}{}_k(\mu)\, d\mu^2$ gives the limit, a polynomial of degree ≤ 2 in (e^2, λ) since δm^2 has orders $a+2b \le 4$ and odd powers of e vanish. The divergent parts of the counterterm, $1/\varepsilon$ at one loop and $\ln(1/\varepsilon)$ at two loops, are mass-independent and cancel in the difference; this is what the negative degrees of the derivative integrands express. □

The one-loop evaluation of the limit in (b) is the closed form of lemma 22.1(b)(ii); the two-loop coefficients are the finite numbers of (b), bounded in proposition 22.15 by explicit majorants. Nothing in this subsection uses the torus beyond lemma 12.9: the torus forms of (Q1)–(Q4), with $\mathcal{R}_T[\cdot]$ in place of $\int_{\mathsf{B}}$ and the additional holonomy vertices (the torus photon two-point function at zero momentum, which vanishes on $\varepsilon\mathbb{Z}^3$ by the Ward identity but is $O(\ell^{-r})$ on T by lemma F.155), are bounded uniformly in ε and in the torus by the same argument, but the Wick leg does not need them.

Appendix G. Two steps of the renormalisation-group analysis

This appendix proves the two statements of section 5.2. Both concern Bałaban's renormalised classes of graphs [3, Sects. 1–3] at the moment when some of their lines are *cut*: replaced by a pair of external legs carrying a two-variable kernel. This happens when two propagator domains are compared ([3, p. 414]: "we will have also the propagators $\delta G_k(\Omega, \Omega_2, \tilde{B})$", which "will be treated as an external field") and whenever a line object is telescoped. Appendix G.1 proves ($\mathrm{REN}_{\mathrm{orb}}$) (lemma 5.2): the cut terms of a renormalised family regroup into complete renormalised classes of the cut graphs. Appendix G.2 proves (BAL_5)[int] (lemma 5.1), the interaction-term half of [2, Prop. 2.6] at a nonzero background, by an exact regrouping of a second kind, into full classes of the cut components, to which Bałaban's renormalisation bound [3, Prop. 1, (1.33)] applies at every background.

G.1. **Orbit regrouping of cut renormalised families.**

Graphs and labels. A *renormalised family* is a finite sum $\mathcal{R} = \sum_{\Gamma\in\mathcal{G}} c_\Gamma E(\Gamma)$ over a finite set $\mathcal{G}$ of graphs built from the vertices [3, (1.6)–(1.15)] (the seagulls included), from mass-counterterm vertices [3, (1.7)] with the coefficients [3, (1.23), (1.29)], and from source vertices; c_Γ collects the couplings, the combinatorial weight and the counterterm values, and every counterterm is a vertex of the action with a field- and domain-independent coefficient ([3, (1.29)–(1.31), p. 419]: the counterterms are built from torus propagators at zero background). Every vertex carries *labels* [3, p. 420]: each vector-field leg the support box (a grid box of side ≤ 2) of its partition function, each vertex of type (1.6)/(1.7) its unit cube, each block vertex (1.13)–(1.15) its block $B^k(y)$; a vertex is identified by its type and its labels. The counterterm vertex that replaces a divergent two-scalar-leg subgraph γ carries γ's labels ([3, p. 417]: "the same graph G but with both external legs localized in x", and [3, p. 420]: the localisation is performed "for the terms of δm_k^2 also"), so that a graph and its counterterm partner have the same label multiset: this is Bałaban's "localizations in these graphs should be in agreement" [3, p. 432]. A *line* is a pair of legs with a kernel type (scalar or photon, with its domain and background); an unpaired leg is an *external leg* and carries an external field, which may be a function of several variables

[3, p. 419]. The lines are full lines: the scale decomposition is internal to the bounds that are applied to the classes.

Subgraphs and degrees. A *subgraph* H of Γ is a connected set of lines of Γ (a line set, not a vertex set); $V(H)$ is the set of its vertices, and its *slots* are the legs of $V(H)$ that are not on lines of H: legs on other lines, external legs and cut ends. Its *type* is the number and kind (scalar or photon) of its slots, with the differentiation of its scalar slots; its order is the sum of the orders of its vertices; its degree $D(H)$ is Bałaban's [3, (2.1)–(2.3)]: $D_H(v)$ counts the factors η of v, the legs of v "belonging to internal lines" of H (with weight $(2-d)/2$), minus the differentiations of v acting on internal lines, plus the vector legs on internal lines when v is a block vertex; "external fields have a dimension equal to 0"; $D(H) = \sum_v D_H(v) - d$; and a counterterm vertex counts as the graph it replaces. Hence $D(H)$ *depends only on* $(V(H), H,$ *which legs are slots*), and so does the type. The divergent subgraphs and their completions [3, pp. 420, 429–432] are: a two-scalar-leg, one-particle-irreducible H with undifferentiated slots and $D(H) \leq 0$, completed by its *counterterm partner*; a two-vector-leg H with $D(H) \leq 0$, completed by the *two-vector-leg class*, in Bałaban's maximal form "all possible connected graphs of the considered type ... of a fixed order" [3, p. 431]; a $\phi\phi A$ subgraph with $D(H) = 0$, one-particle-irreducible ([3, (2.18)–(2.20)]) or reducible ([3, (2.21)–(2.22)]), completed by the *permutation* of its two external scalar legs [3, p. 429]. Graphs with a block vertex have $D > 0$ [3, Cor. 2.3], so no divergent subgraph contains one; vacuum subgraphs do not occur below (every member and every cut component carries an external field).

Moves. Each completion is a *move* μ attached to a subgraph H of its source Γ: it replaces H by a graph H^* together with a bijection s_μ from the slots of H to the slots of H^*, and whatever sits at a slot of H (an outer line end, an external field, a cut end) is re-attached at the image slot; nothing else changes. (P) the counterterm partner: $V(H)$ and the lines of H are replaced by one counterterm vertex carrying H's labels, the two slots going to its two legs; (P′) its inverse, the expansion of a counterterm vertex; (W) the two-vector-leg completion: $H \to H'$ for every H' in the per-label two-vector-leg class of H (all connected graphs with H's label multiset and order whose two free photon legs carry H's two slot labels, with their counterterm members), the slots mapped by their vector-leg labels; it is applied at *every* divergent two-vector-leg subgraph, maximal or not (substituting at a non-maximal one gives a member of the class of the maximal one, so the classes are the same); (X) the exchange: the attachments at the two scalar slots of a divergent $\phi\phi A$ subgraph are exchanged (if the two slots are joined by one line, that line is mapped to itself with its ends exchanged; if both carry external fields of one kind, their arguments are exchanged). Let M be a set of moves. The moves (P), (W), (X) have the following five properties, which are all that the proofs below use: (A1) a move changes only the lines and vertices of H and the attachment of the slots, and does not depend on what is attached at the slots; (A2) whether a move is available at H, and its results (H^*, s_μ), depend only on H with its slots (lines, vertices, labels, slot types), because $D(H)$ and the slot types are intrinsic and the per-label two-vector-leg class depends only on H's labels; (A3) every move has an inverse in M at H^* with slot map s_μ^{-1}; (A4) a move keeps the label multiset, the kind of every slot and the kernel type of every surviving line; (A5) $c_{\mu\Gamma}/c_\Gamma$ depends only on (H, H^*, s_μ) (the exchange keeps the labelled vertex multiset, and (P), (W) change it only inside H).

Definition G.1 (classes)**.** The *class* $\mathrm{Cl}_M(\Gamma)$ of a graph is its connected component under the moves of M, the moves at Γ itself included when Γ is of a primitively divergent type (its one-vertex counterterm partner [3, p. 430], its permuted graph [3, p. 429], all two-vector-leg graphs of its order [3, p. 431]). With M the moves (P), (W), (X), $\mathrm{Cl}_M(\Gamma)$ is Bałaban's class G_{ren} of Γ: "the smallest sets of graphs, symmetric with respect to permutation of external scalar field legs and with each divergent subgraph renormalized" [3, p. 429] are the connected components, since the moves are invertible. A *per-label full class* is the set of all connected graphs of one type, one order and one label multiset, with their counterterm members [3, p. 431]; it is closed under every move. A family is *closed under* M if $\mathcal{G}$ is a finite union of classes; Bałaban's renormalised families are.

Definition G.2 (cut terms)**.** Fix a set of *decomposed kinds* of objects of the family (for instance all lines and legs on a domain, or all restricted lines), and write every object o of such a kind as $o = o' - \Delta o$, with Δo a symmetric kernel, $\Delta o(x,y) = \Delta o(y,x)^T$ (true for every difference of two symmetric resolvents and for the kernels used below). An *oriented cut term* is $\tau = (\Gamma, S, \theta)$ with S a set of decomposed objects of Γ and θ an orientation of every line of S; its weight is $2^{-|S_L|}c_\Gamma$, S_L the set of lines of S. By the symmetry of Δo the two orientations of a line add up to Δo, so multilinearity gives the exact expansion

$$\mathcal{R} \;=\; \sum_{\tau} 2^{-|S_L|} c_\Gamma \prod_{o\in S} (-\Delta o)_\theta \prod_{o\notin S} o'. \tag{84}$$

The *cut graph* $\mathrm{cut}(\tau) = \Gamma \setminus S$ is Γ with the lines of S removed, their ends carrying the two arguments of Δo (end 0 and end 1 by θ), each leg of S carrying Δo, and the objects of S labelled. This is Bałaban's "we replace these lines by pairs of external legs" [3, p. 432], with external fields "in the form of functions of many variables" [3, p. 419]. A subgraph H of Γ *avoids* S if it contains no line of S. For a move μ at a subgraph avoiding S put $\mu\cdot\tau := (\mu\Gamma, s_\mu(S), s_\mu(\theta))$: the ends of S-lines and the S-legs sitting at slots of H are moved to the image slots, everything else is fixed. The *orbits* are the classes of the equivalence relation generated by $\tau \sim \mu\cdot\tau$.

Lemma G.3 (orbit regrouping, ($\mathrm{REN}_{\mathrm{orb}}$))**.** *Let $\mathcal{R}$ be closed under M, where M satisfies* (A1)–(A5)*. Then:*

(a) *A connected line set H of Γ avoids S if and only if it is a connected line set of* $\mathrm{cut}(\tau)$*; its vertices, slots, slot types, order and degree are the same in Γ and in* $\mathrm{cut}(\tau)$*, and so are the moves available at it and their results.*

(b) *$\mu\cdot\tau$ is an oriented cut term of $\mathcal{R}$; μ^{-1} acts at H^*, which avoids $s_\mu(S)$, and $\mu^{-1}\cdot(\mu\cdot\tau) = \tau$. The orbits partition the cut terms; each is finite, and the number of S-objects of each kind is constant on it.*

(c) $\mathrm{cut}(\mu\cdot\tau) = \mu(\mathrm{cut}\,\tau)$*. Hence* cut *maps the orbit of τ onto* $\mathrm{Cl}_M(\mathrm{cut}\,\tau)$*, and re-joining the labelled S-objects maps* $\mathrm{Cl}_M(\mathrm{cut}\,\tau)$ *onto the orbit exactly r_τ-to-one, where $r_\tau \geq 1$ is the order of the group of relabellings of identical S-objects (same kind, same kernel, same orientation pattern) that map* $\mathrm{Cl}_M(\mathrm{cut}\,\tau)$ *to itself; and* $\mathrm{Cl}_M(\mathrm{cut}\,\tau) = \prod_K \mathrm{Cl}_M(K)$ *over the connected components K of* $\mathrm{cut}\,\tau$*. In particular the cut graphs of one orbit form exactly one class of the (possibly disconnected) cut graph.*

(d) *At fixed labels, after the sum over the separable indices of the cut kernels (write $\Delta o(x_0,x_1) = \sum_a c_a F_a(x_0) H_a(x_1)^*$),*

$$\mathcal{R}^{(O)} \;:=\; \sum_{\tau'\in O} 2^{-|S_L|} c_{\Gamma'} \prod_{S'} (-\Delta o) \prod_{O_{\Gamma'}\setminus S'} o' \;=\; (-1)^{|S|} 2^{-|S_L|}\, c_*\, r_O^{-1} \prod_K E\big(\,\mathrm{Cl}_M(K);\, \textit{its fields}\big),$$

where $E(\mathrm{Cl}_M(K); \textit{fields})$ is the class sum of K with the fields F_a, H_a, the S-legs' kernels and the original fields at its slots, and c_* is a constant. (At fixed indices, when $r_O > 1$ and two identical S-objects carry different indices, the left side is the r_O-symmetrisation of the right side.)

(e) $\sum_S \mathcal{R}^{(S)} = \sum_O \mathcal{R}^{(O)}$*:* (84) *summed in another order.*

(f) *All members of an orbit carry the same labels, and an end of an S-object moves only from a slot of H to a slot of H^*: under* (X) *to the other scalar slot, under* (P) *onto the counterterm vertex (which carries H's labels), under* (W) *to the leg with the same vector-leg label. No component is created, merged or split, and the label set of each component is kept.*

(g) *Each orbit lies in the cut terms of one class of $\mathcal{G}$; the cut terms of one class with a fixed number of S-objects of each kind are a finite disjoint union of orbits.*

Proof. (a) "H avoids S" says that H is a set of lines of Γ none of which is in S, that is, a set of lines of $\Gamma\setminus S$; connectedness and $V(H)$ are read from H alone. The slots of H in Γ are the legs of $V(H)$ not on lines of H; in $\mathrm{cut}(\tau)$ they are the same legs (a leg on an S-line in Γ is an external leg of $\mathrm{cut}(\tau)$ carrying an end of Δo; a leg on a non-S line outside H is on the same line in both; an external leg is the same). The kind and the differentiation of a slot are properties of the leg (on a cut end the derivative acts on the external field), and $D(H)$ depends only on

$(V(H), H, \text{slots})$; so the type of H is the same in both graphs, and by (A2) the available moves and their results coincide.

(b) By (A2) and the closure of $\mathcal{G}$, $\mu\Gamma \in \mathcal{G}$. By (A1) every object of Γ outside H survives in $\mu\Gamma$, with the same kernel by (A4); S consists of objects outside H, so $s_\mu(S)$ is a set of decomposed objects of $\mu\Gamma$ of the same kinds and $s_\mu(\theta)$ orients its lines: $\mu\cdot\tau$ is a cut term of $\mathcal{R}$. By (A3), μ^{-1} is available at H^*; H^* avoids $s_\mu(S)$ because the S-objects of $\mu\cdot\tau$ are images of objects outside H, none internal to H^*; applying μ^{-1} undoes the slot map. So $\sim$ is generated by a symmetric relation on a finite set, and s_μ keeps the numbers and kinds of the S-objects.

(c) In $\mathrm{cut}(\tau)$, H is a subgraph with the same slots, by (a). The move μ applied to $\mathrm{cut}(\tau)$ re-attaches at the image slots whatever sits at the slots of H, outer lines, external fields and cut ends alike (A1). In $\mu\cdot\tau$ the S-ends at the slots of H are moved by the same s_μ, and the other S-ends are untouched. Removing the S-lines of $\mu\cdot\tau$ therefore gives the same graph as moving the cut ends in $\mathrm{cut}(\tau)$: $\mathrm{cut}(\mu\cdot\tau) = \mu(\mathrm{cut}\,\tau)$. Conversely every move of $\mathrm{cut}(\tau)$ is at a subgraph H of $\mathrm{cut}(\tau)$, which avoids S by (a), so it is the cut of an orbit move. Hence cut maps the orbit onto the connected component of $\mathrm{cut}\,\tau$ under M, which is $\mathrm{Cl}_M(\mathrm{cut}\,\tau)$. A move acts at a connected H, which lies in one component K of $\mathrm{cut}\,\tau$, and H^* is connected and attached at the same slots; so a move changes only K, keeps the number of components and the label set of each (A4). Moves in different components act on disjoint data and commute; hence $\mathrm{Cl}_M(\mathrm{cut}\,\tau) = \prod_K \mathrm{Cl}_M(K)$, every combination being reached by moving one component at a time. Finally, re-joining the two ends of each labelled S-line and restoring each S-leg's field inverts cut on labelled objects; two labelled cut graphs of the class give the same unlabelled cut term if and only if they differ by a permutation of identical S-objects; the set of such permutations preserving the class is a group, since relabelling commutes with every move by (A1); so every cut term of the orbit has exactly r_τ preimages.

(d) By (A5), $c_{\Gamma'} = c_\Gamma \prod_{\text{moves}} w(H, H^*, s)$, and the moves in different components multiply; in the label picture $c_{\Gamma'} = c_* \prod_K c(K')$ with $c(K')$ the product of the vertex weights of K' (repeated labels are handled by the ordered-partition count at a constant cost). For fixed labels, $E(\mathrm{cut}\,\tau';\text{fields}) = \prod_K E(K';\text{fields})$: positions are summed within the fixed label boxes, the external fields are one-variable fields at fixed separable indices, and the lines of different components are different. The separable index of an oriented S-line is kept by every move (s_μ moves ends and keeps the line). Hence $\sum_{\mathrm{Cl}_M(\mathrm{cut}\,\tau)} cE = c_* \prod_K E(\mathrm{Cl}_M(K))$; the left side counts each cut term of the orbit r_τ times by (c), and the sign $(-1)^{|S|}$ and the weight $2^{-|S_L|}$ are common to the orbit. When $r_O > 1$ and identical S-objects carry different indices, the relabellings permute the indices, which gives the symmetrisation; after the sum over the indices the identity holds as stated.

(e) The orbits partition the index set of the finite sum (84), by (b).

(f) (A4) gives the labels; the slot maps of (P), (W), (X) are as listed; by (c) no component is created, merged or split.

(g) Every move is at a divergent subgraph of a member of a class of $\mathcal{G}$, so it is a move of that class; fixing the numbers of S-objects by kind is compatible with $\sim$ by (b). □

Remark G.4. (i) The orientation device is needed exactly when a move maps a cut term to itself with the two ends of an S-line exchanged (both ends of one S-line at the two scalar slots of a $\phi\phi A$ subgraph): with unoriented terms the orbit is a class only at the level of the symmetric two-variable kernel. (ii) If S contains no line, the cut graph is Γ itself with its external fields changed, and the moves at Γ itself (the permutation of a top-level $\phi\phi A$ graph, which moves a difference leg from one slot to the other; the one-vertex partner of a two-scalar-leg graph) are orbit moves. (iii) A move at a divergent subgraph that contains an S-line is not an orbit move; the cut graph needs no completion there, and the S-terms of Γ and of its partners at that subgraph fall into different orbits, each complete. (iv) The per-object grouping (collecting, for a fixed S, the cut terms of all members of a class) is *not* closed under the moves: with the pair [3, (2.18)] inside a larger member, an external line at the (1.8) slot cut, the exchange at the pair moves the cut end to the seagull, so the per-object group contains the cut graph but not its partner, which is still needed, since the pair is still a divergent $\phi\phi A$ subgraph of the cut graph.

Only the counterterm moves keep the per-object grouping, when the legs of a counterterm vertex are read as the legs of the graph it replaces. This is why the regrouping is by orbits.

Lemma G.5 (cut data of a per-label full class)**.** *Let $\mathcal{G}$ be a per-label full class of type T, order o and label multiset $\mathcal{L}$, and group the unoriented cut terms by their* cut datum *δ: the components K of $\Gamma\setminus S$ with their label sets $\mathcal{L}_K$ and their orders, at each K the numbers of S-ends of each kind and of original external fields of each kind, and the incidence of the S-objects (which components each S-line joins, in which component each S-leg sits). Then:*

(a) *Label the S-objects and fix one matching of their ends. The cut terms with datum δ correspond, s_δ-to-one, to the tuples $\prod_K \mathcal{K}_K$, where $\mathcal{K}_K$ is the per-label full class of the component (all connected graphs with label multiset $\mathcal{L}_K$ and the order of K whose unpaired legs carry the prescribed fields, S-end kernels and original fields, in all assignments, with their counterterm members,* members without internal line and of order 0 included*), and s_δ is the order of the group of relabellings of identical S-objects of δ, the exchange of the two ends of an S-line with both ends in one component included.*

(b) *At fixed labels and separable indices, $\mathcal{R}^{(\delta)} = (-1)^{|S|} c_* s_\delta^{-1} \prod_K E(\mathcal{K}_K; \text{fields})$.*

(c) *Every orbit of lemma G.3 lies in one group; every group is closed under the exchange of the two scalar legs of every two-scalar-leg or $\phi\phi A$ subgraph, the component itself included, and under the completion of a one-vector-leg subgraph to the two-vector-leg class of its first background-Taylor coefficient. Hence each group is a union of orbits also for the moves enlarged by these two.*

(d) $\sum_S \mathcal{R}^{(S)} = \sum_\delta \mathcal{R}^{(\delta)}$.

Proof. (a) Cutting lands in $\prod_K \mathcal{K}_K$: each component is connected, has the label set $\mathcal{L}_K$ and the prescribed unpaired legs, and every counterterm vertex of Γ lies in one component with labels inside $\mathcal{L}_K$. Conversely, given members $K_i \in \mathcal{K}_K$, an assignment of the labelled stub fields to their unpaired legs (included in $\mathcal{K}_K$) and the fixed matching, gluing the S-lines gives a graph with label multiset $\bigcup_K \mathcal{L}_K = \mathcal{L}$, the uncut type T and the order o, connected because the incidence multigraph of δ is connected (the original Γ was); it is therefore a member of the full class $\mathcal{G}$, with S among its lines and legs. A two-scalar-leg subgraph of K_i avoids S, so by lemma G.3(a) it is a two-scalar-leg subgraph of Γ, and its counterterm member corresponds to Γ's; conversely a counterterm vertex of Γ is a vertex of one component. Two labelled tuples give the same unlabelled cut term if and only if they differ by a relabelling of identical S-objects, and such a relabelling maps the product of full classes to itself (each $\mathcal{K}_K$ contains all assignments of its stub fields); so the map is s_δ-to-one. (b) As in lemma G.3(d). (c) The moves keep δ: labels by (A4), stubs by lemma G.3(f), incidence because S-lines survive. A full class contains every pairing of its legs, and the two further moves are re-pairings inside a component with the same labels and stubs (the exchange is a bijection of Wick pairings; the first background-Taylor coefficient of the full connected generating function of a one-vector-leg type is the full two-vector-leg class, since the generating function depends on the background and on the other vector legs only through their sum). (d) The groups partition the unoriented cut terms. □

Remark G.6 (how the two lemmas are used). When a renormalisation bound ([3, Prop. 1, (1.33)] or its cube form) is applied to a telescoped family, it is applied per orbit: by lemma G.3(c), (d) each orbit is a product of complete classes of the components, each carrying at least one external field (an S-end or an original field), with the cubes of the original graph, so the bound applies to each component with its tree decay, and the components are joined along a spanning tree of the cut graph's component multigraph by the decay of the cut kernels ([3, (3.2), p. 432]: "the exponential factor in (1.33) is obtained from the estimates of external fields"). The gain of a cut kernel is read at the maximum over the end cubes of the orbit, which is legitimate because the kernel bounds hold at every pair of cubes at the required depth (lemma G.10); where a consumer needs the gain at one member's cubes, any two cubes of the fixed vertex set W are within $d(W)$, so the conversion costs $e^{2|S|\delta'_\sigma d(W)}$ with the gain taken at the reduced rate $\delta'_\sigma = \delta'/(4|S|)$, which half of the tree factor $e^{-\delta' d(W)}$ absorbs. At zero background the orbits of (P), (W), (X) suffice. At a nonzero regular background the renormalisation bound is proved for per-label full classes (the touched two-scalar-leg subgraph becomes a $\phi\phi A$ node whose partner is

its mirror, a move not among (P), (W), (X)); there the regrouping is by the groups of lemma G.5, and the members of the component classes without internal line or of order 0 are treated as in the proof of lemma G.14.

G.2. **The domain-change estimate at a nonzero background.**
The setting. We are at the step $k \geq 1$ of [2, Sect. 2], $d = 3$, $L = 5$, $t := 5^k\varepsilon \leq \varepsilon_0$, $\eta := 5^{-k}$, in scale-k units (unit blocks on $\eta\mathbb{Z}^3$; dist is the sup-metric [1, (1.3)], and $|\cdot|$ counts unit blocks). The couplings are $e_k = et^{1/2}$, $\lambda_k = \lambda t$; $p = p(t) = b_0(1 + \ln t^{-1})^{\tilde{p}}$ is the small-field threshold and $r = r(t) = R(1 + \ln t^{-1})^{\varrho}$, $\varrho > 1$, $R > R_0$, the layer width [2, (2.7), p. 558]; R may be enlarged, and $r \geq R$. The large-field regions $\Lambda_0^{(k)} \supset \Lambda_1^{(k)} \supset \cdots \supset \Lambda_7^{(k)}$ are those of [2, (2.8), pp. 558, 566]: $(\Lambda_{i+1}^{(k)})^c$ is the union of the large blocks within distance r of $(\Lambda_i^{(k)})^c$; $\Lambda_0^{(k)} \subset \Lambda_7^{(k-1)\prime}$, primed sets being the scale-$(k-1)$ sets read at scale k; $B^k(S)$ is the set of fine-lattice points of the blocks of S. The two propagator domains of [2, Prop. 2.6] are

$$\Omega_A := B^k(\Lambda_2^{(k-1)\prime}) \;\supset\; \Omega_C := B^k(\Lambda_2^{(k)}),$$

the inclusion because $\Lambda_2^{(k)} \subset \Lambda_0^{(k)} \subset \Lambda_7^{(k-1)\prime} \subset \Lambda_2^{(k-1)\prime}$; both are unions of large blocks. The background on $B^k(\Lambda_0^{(k)})$ is $\tilde{B} = (1 - \theta_{k+1})A^{(k)} + \theta_{k+1}B^{(k+1),\eta}$ [2, p. 579], where θ_{k+1} equals 1 on $B^k(\Lambda_2^{(k)})$ and varies from 1 to 0 on a slice of thickness $< M$ around it ([2, p. 567] at the step k, [2, (2.96)] at the step $k+1$); $\tilde{B}$ is regular on Ω_A in the sense of [1, (2.23)], with $|\partial\tilde{B}| \leq O(p)$ ([2, (2.60)] for $A^{(k)}$ on Ω_A, [2, (2.94)–(2.98)] for $\tilde{B}$ on $B^k(\Lambda_0^{(k)})$). For a union Ω of large blocks and a regular field A on it, $G_k(\Omega, A) = (-\Delta_{A,\Omega}^N + m^2t^2 + a_kP_k(A))^{-1}$ is the scalar propagator of the step on Ω with Neumann conditions [1, (2.22)], [4, (1.3)], $P_k(A)$ the projection onto the transported block constants. For a function F of one or two lattice variables on a set S, $\|F\|_{1,\alpha;S}$ is the natural two-variable extension of Bałaban's norm [3, (1.32), p. 420]: the maximum, over the variables, of the sup of F, of $|D_AF|$ in each variable, of $|D_xD_{x'}F|$, and of the transported Hölder quotients of exponent α of these derivatives in each variable; for a localisation h ($h = 1$ on a cube, smooth, supported in a fixed neighbourhood), $\|hF\|_{1,\alpha} \leq C_h\|F\|_{1,\alpha;\operatorname{supp} h}$. We write $\sigma, \sigma' \in \{0, D, HD\}$ for "no derivative", "the covariant derivative" and "the Hölder quotient of the covariant derivative", with $|0| = 0$, $|D| = 1$, $|HD| = 1 + \alpha$.
Standing lower bounds. $R \geq R_0 + 3$ and

$$D_7 \;:=\; 5(r-1) - 4 \;\geq\; \max(R_0 + 2,\, r), \tag{85}$$

where $R_0 \geq 1$ is the depth constant of [1, Prop. 2.1]; since $r \geq R$ these are one more lower bound on R or, equivalently, one more cap on ε_0 of Bałaban's kind.
Printed inputs. (P1) [1, Prop. 2.1, (2.24)–(2.26)] (which is the theorem [4, (1.9)–(1.12)]): for a regular A on Ω and a bounded source f, bounds on $G_k(\Omega, A)f$, $DG_k(\Omega, A)f$ and the Hölder quotient of the latter, with decay $c_0e^{-\delta_0 \operatorname{dist}(x,\operatorname{supp} f)}$, at points at distance $\geq R_0$ from Ω^c (2.24)–(2.25), and for $\Omega_2 \subset \Omega$ the difference $(G_k(\Omega, A) - G_k(\Omega_2, A))f$ with the additional factor $e^{-\delta_0 \operatorname{dist}(\operatorname{supp} f, \Omega_2^c) - \delta_0 \operatorname{dist}(\{x,x'\}, \Omega_2^c)}$ (2.26), δ_0, c_0, R_0 independent of A, k and Ω; these hold at every scale $j \leq k$ after rescaling, by the same statement. (P2) [1, Props. 2.2–2.3] and [4, (1.13)–(1.20)]: the exponential decay of the kernel of the block-spin operator $\Delta^{(j)}(\Omega, A)$ and of its domain difference $\Delta^{(j)}(\Omega, A) - \Delta^{(j)}(\Omega_0, A)$, the latter with the two-sided factor $e^{-\delta_0(\operatorname{dist}(x,\Omega^{(j)c}) + \operatorname{dist}(x',\Omega^{(j)c}))}$ [1, (2.27)–(2.29)]; the positivity and exponential decay of the restricted block covariances $C_\Lambda^{(j)}(\Omega, A) = \big((\Delta^{(j)}(\Omega, A) + aL^{-2}P_L)|_\Lambda\big)^{-1}$ with the fixed bare a [1, (2.31)–(2.34)], [4, (1.13), (1.15)–(1.16)]; and the two-sided decay of their domain differences $C_\Lambda^{(j)}(\Omega, A) - C_\Lambda^{(j)}(\Omega_0, A)$ [4, (1.19)–(1.20)]; Λ a union of big blocks, constants c_2, θ_0 free of j, Ω, Ω_0, Λ and A. (P3) the Combes–Thomas bound of lemma G.7 below for the fixed-spacing resolvent, with no restriction near the boundary; the positivity it needs is [4, (1.8)]. Below, θ_0 denotes the smaller of the rates of (P2) and (P3).

Lemma G.7 (a Combes–Thomas bound at fixed spacing)**.** *Let Ω be a subset of a lattice of spacing $\eta \leq 1$ with the sup-metric, and on $\ell^2(\Omega)$ (weights η^d) let $T = D_A^*D_A + V + \sum_B c_B\Pi_B$, where D_A is the covariant difference on the bonds of Ω, $V \geq 0$ is a multiplication operator, the*

Π_B are orthogonal projections whose ranges are supported in pairwise disjoint sets B of diameter $\le \ell_*$, $\eta \le \ell_*$, $0 \le c_B \le \bar a$, and $T \ge \gamma > 0$. Put $\theta_0 := \min\{\ell_*^{-1}, \gamma^{1/2}/(36\sqrt{d}), \gamma/(12e^2\bar a \ell_*)\}$. Then for all $f, f' \in \ell^2(\Omega)$ and all bond functions g, g', with δ the distance of the supports (a bond located at its initial point),

$$|\langle f, T^{-1} f'\rangle| \le 2\gamma^{-1} e^{-\theta_0 \delta} \|f\| \|f'\|,$$
$$|\langle g, D_A T^{-1} f'\rangle|,\ |\langle f, T^{-1} D_A^* g'\rangle| \le (2\gamma^{-1/2} + 6\sqrt{d}\,\theta_0 \gamma^{-1})\, e^{\theta_0} e^{-\theta_0 \delta}\, \|\cdot\| \|\cdot\|,$$
$$|\langle g, D_A T^{-1} D_A^* g'\rangle| \le 4 e^{\theta_0} e^{-\theta_0 \delta} \|g\| \|g'\|,$$

with no restriction on the position of the supports relative to $\partial\Omega$ (the factor e^{θ_0} covers locating a bond at either end). In particular $\|\mathbf{1}_Y T^{-1} f\| \le 2\gamma^{-1} e^{-\theta_0 \operatorname{dist}(Y, \operatorname{supp} f)} \|f\|$ and $\|\mathbf{1}_Y D_A T^{-1} f\| \le (2\gamma^{-1/2} + 6\sqrt{d}\,\theta_0 \gamma^{-1}) e^{\theta_0} e^{-\theta_0 \operatorname{dist}(Y, \operatorname{supp} f)} \|f\|$ for every set Y. For the operators of lemma G.10 in scale-1 units ($\eta = 1/L$, unit blocks, $\ell_* = 1$, $c_B = a$, $V = m^2 t^2 l_1^2$) the lower bound $T \ge \gamma$ is [4, (1.8)]*, with γ free of η, Ω and A.*

Proof. Weight. Let $\rho := \operatorname{dist}(\cdot, \operatorname{supp} f')$ on the sites, $\rho(b) := \rho(b_-)$ on the bonds, and $T_\theta := e^{\theta\rho} T e^{-\theta\rho}$; then $T^{-1} = e^{-\theta\rho} T_\theta^{-1} e^{\theta\rho}$ and $e^{\theta\rho} f' = f'$. *Hopping.* The ends of a bond are at distance η and ρ is 1-Lipschitz, so for $\theta\eta \le 1$, $e^{\theta\rho} D_A e^{-\theta\rho} = D_A + \theta B_1$ with $(B_1 u)(b) = \theta^{-1}\eta^{-1}\big(e^{\theta(\rho(b_-) - \rho(b_+))} - 1\big) U_b u(b_+)$; since $|\theta^{-1}\eta^{-1}(e^{\theta s} - 1)| \le e \le 3$ for $|s| \le \eta$ and each site is the final point of d bonds, $\|B_1\| \le 3\sqrt{d}$. Likewise $e^{\theta\rho} D_A^* e^{-\theta\rho} = (D_A - \theta B_1')^*$ with $\|B_1'\| \le 3\sqrt{d}$. *The hopping change.* $e^{\theta\rho} D_A^* D_A e^{-\theta\rho} - D_A^* D_A = \theta(D_A^* B_1 - B_1'^* D_A) - \theta^2 B_1'^* B_1$, so $|\langle \phi, (\cdots)\psi\rangle| \le 3\sqrt{d}\,\theta(\|D_A\phi\|\|\psi\| + \|\phi\|\|D_A\psi\|) + 9d\theta^2\|\phi\|\|\psi\|$. *Pins.* Write $\Pi_B = WW^*$ and let ρ_B be the value of ρ at a point of B; then $e^{\theta\rho}\Pi_B e^{-\theta\rho} = (e^{\theta(\rho-\rho_B)}W)(e^{-\theta(\rho-\rho_B)}W)^*$, and since $|\rho - \rho_B| \le \ell_*$ on B this differs from Π_B by at most $(2e^x + xe^{2x})x \le 2e^2 x$ in norm, $x := \theta\ell_* \le 1$; the B being disjoint, the change of the pin term has norm $\le 2e^2 \bar a\, \theta\ell_*$. *Perturbation.* Put $K := T^{-1/2}(T_\theta - T)T^{-1/2}$. Since $D_A^* D_A \le T$, $\|D_A T^{-1/2}\| \le 1$, and $\|T^{-1/2}\| \le \gamma^{-1/2}$; hence $\|K\| \le 6\sqrt{d}\,\theta\gamma^{-1/2} + 9d\theta^2\gamma^{-1} + 2e^2\bar a\theta\ell_*\gamma^{-1} \le \frac{1}{6} + \frac{1}{144} + \frac{1}{6} < \frac{1}{2}$ for $\theta \le \theta_0$. *Bounded inverse.* $T_\theta = T^{1/2}(1+K)T^{1/2}$, so $T_\theta^{-1} = T^{-1/2}(1+K)^{-1}T^{-1/2}$ with $\|T_\theta^{-1}\| \le 2\gamma^{-1}$, $\|D_A T_\theta^{-1}\|, \|T_\theta^{-1} D_A^*\| \le 2\gamma^{-1/2}$ and $\|D_A T_\theta^{-1} D_A^*\| \le 2$. *Bounds.* $\langle f, T^{-1} f'\rangle = \langle e^{-\theta\rho} f, T_\theta^{-1} f'\rangle$ with $\|e^{-\theta\rho} f\| \le e^{-\theta\delta}\|f\|$; $\langle g, D_A T^{-1} f'\rangle = \langle e^{-\theta\rho} g, (D_A + \theta B_1) T_\theta^{-1} f'\rangle$ with $\|(D_A + \theta B_1) T_\theta^{-1}\| \le 2\gamma^{-1/2} + 6\sqrt{d}\,\theta\gamma^{-1}$, and the adjoint form follows from the symmetry of T; for the third form take $\rho := \operatorname{dist}(\cdot, \operatorname{supp} g')$, so that $\langle g, D_A T^{-1} D_A^* g'\rangle = \langle e^{-\theta\rho} g, (D_A + \theta B_1) T_\theta^{-1} (D_A - \theta B_1')^* g'\rangle$ and $\|(D_A + \theta B_1) T_\theta^{-1} (D_A - \theta B_1')^*\| \le 2 + 12\sqrt{d}\,\theta\gamma^{-1/2} + 18d\theta^2\gamma^{-1} \le 4$. Take $\theta = \theta_0$. □

Lemma G.8 (the background move is exact)**.** *On every bond of Ω_C, $\tilde B = B^{(k+1),\eta}$. Hence $G_k(\Omega_C, \tilde B) = G_k(\Omega_C, B^{(k+1),\eta})$, $\Delta^{(k)}(\Omega_C, \tilde B) = \Delta^{(k)}(\Omega_C, B^{(k+1),\eta})$ and $C_\Lambda^{(k)}(\Omega_C, \tilde B) = C_\Lambda^{(k)}(\Omega_C, B^{(k+1),\eta})$ for every $\Lambda \subset \Omega_C^{(k)}$.*

Proof. $\theta_{k+1} = 1$ on $\Omega_C = B^k(\Lambda_2^{(k)})$, and $\Omega_C \subset B^k(\Lambda_0^{(k)})$, so $\tilde B = B^{(k+1),\eta}$ there. The three operators depend on the field only through the transporters on the bonds of their domain (the Neumann Laplacian, [4, (1.3)]) and on the contours $\Gamma_{y,x}^{(k)}$, which lie in $B^k(y) \subset \Omega_C$ [4, (1.4)]. □

So the replacement of [2, Prop. 2.6] is a pure domain change: $G_k(\Omega_A, \tilde B) \to G_k(\Omega_C, \tilde B)$, with $\delta G := G_k(\Omega_A, \tilde B) - G_k(\Omega_C, \tilde B)$ on $\Omega_C \times \Omega_C$ the $\delta G_k(\Omega, \Omega_2, \tilde B)$ of [3, p. 414]. The terms containing $\tilde B^{(k)}$ were estimated before it ([2, p. 579]: "Proposition I.3.1 can also be applied … the terms containing the configuration $\tilde B^{(k)}$ … by $O((L^k\varepsilon)^\kappa)|\Lambda_2^{(k)}|$"), as were the terms of order $> \bar n$ and the terms localised in $B^k((\Lambda_7^{(k)})^c)$; they are not part of the proposition.

Lemma G.9 (three elementary facts)**.** (a) (layer distances) *For $i \ge 0$, $j \ge 1$: $\operatorname{dist}(\Lambda_{i+j}^{(k)}, (\Lambda_i^{(k)})^c) \ge j(r-1)$; $\operatorname{dist}(\Lambda_7^{(k-1)\prime}, (\Lambda_2^{(k-1)\prime})^c) \ge r - 1$; and $\operatorname{dist}(B^k(S), B^k(S')) \ge \operatorname{dist}(S, S') - 1$ on $\eta\mathbb{Z}^3$. In particular every point within distance $2 + \frac{1}{4} + \eta < 3$ of $B^k(\Lambda_7^{(k)})$ lies at depth $\ge D_7$ in Ω_C.*

(b) (smallness) *For $\delta > 0$, $n, m \geq 0$ and $\kappa > 0$ there is $t_0 = t_0(\delta, n, m, \kappa, R, \varrho, b_0, \tilde{p}, \lambda)$ with $p(t)^n \lambda(t)^{-m} e^{-\delta r(t)} \leq t^\kappa$ for $t \leq t_0$; and for every $\beta > 0$, $p(t)^n \leq C_{\beta,n} t^{-\beta}$ for $t \leq 1$.*

(c) (tree sums) *Let $d(\{\square\})$ be the length of a shortest tree connecting given cubes of side ≤ 2* [3, p. 420]. *For a fixed $\square_1$, $\sum_{\square_2,\dots,\square_n} e^{-\delta d(\{\square_1,\dots,\square_n\})} \leq C_T(n,\delta)$, and for a set S of unit cubes $\sum_{\square} e^{-\delta \operatorname{dist}(\square,S)} \leq C(\delta)|S|$.*

Proof. (a) If $\operatorname{dist}(w, \Lambda_i^c) \leq r$, the large block of w lies within r of Λ_i^c, so the closed r-neighbourhood of Λ_i^c lies in Λ_{i+1}^c. Induction on j: for $x \in \Lambda_{i+j+1}$ and $z \in \Lambda_i^c$ nearest to x, the point w on a sup-metric geodesic from z to x with $\operatorname{dist}(z,w) = \lfloor r \rfloor \geq r-1$ lies in Λ_{i+1}^c, so $\operatorname{dist}(x,z) \geq j(r-1)+(r-1)$. The second claim is the first at scale $k-1$ ($5(r(t/5)-1)$, divided by 5, and $r(t/5) \geq r(t)$). The third holds because a block has sup-diameter ≤ 1. For the last claim, the first claim with $i=2$, $j=5$ and the third give $\operatorname{dist}(B^k(\Lambda_7^{(k)}), \Omega_C^c) \geq 5(r-1)-1$, hence depth $> 5(r-1)-4 = D_7$ at distance < 3 from $B^k(\Lambda_7^{(k)})$. (b) Split $e^{-\delta r} = e^{-\delta r/2} e^{-\delta r/2}$. With $l = \ln t^{-1} \geq 0$: $e^{-\delta r/2} = e^{-(\delta R/2)(1+l)^\varrho} \leq e^{-(\kappa+m+1)l} = t^{\kappa+m+1}$ once $(1+l)^{\varrho-1} \geq 2(\kappa+m+1)/(\delta R)$, that is, for $t \leq t_1$; and $b_0^n (1+l)^{n\tilde{p}} e^{-(\delta R/2)(1+l)^\varrho}$ is bounded in l; absorb this bound and λ^{-m} by $t \leq t_0 \leq t_1$. The second clause: $(1+l)^{n\tilde{p}} e^{-\beta l}$ is bounded in l. (c) A minimal spanning tree of the cube centres has length $\leq 2d(\{\square\}) + 4n$, so $e^{-\delta d} \leq e^{2\delta n} \sum_\tau \prod_{uv\in\tau} e^{-(\delta/2)|c_u - c_v|}$ over the n^{n-2} labelled trees; sum the positions leaf by leaf with $\sum_z e^{-(\delta/2)|z|} \leq (1+4/\delta)^3$. The second clause is $\sum_{s\in S} \sum_{\square} e^{-\delta \operatorname{dist}(\square,s)} \leq |S|(1+2/\delta)^3$. □

Lemma G.10 (the difference kernel as a two-variable external field)**.** *Let $\Omega_2 \subset \Omega$ be unions of large blocks, A regular on Ω, $\delta G := G_k(\Omega, A) - G_k(\Omega_2, A)$ on $\Omega_2 \times \Omega_2$, and $d(x) := \operatorname{dist}(x, \Omega_2^c)$. Let $D \geq R_0 + 2$, and let every evaluation point (x, x' and the Hölder partners) have $d \geq D-1$. Then for $\sigma, \sigma' \in \{0, D, HD\}$*

$$\big|\partial_x^\sigma \partial_{x'}^{\sigma'} \delta G(x,x')\big| \;\leq\; C_{\delta G} \exp\big[-\delta_{\delta G}\big(d(x) + d(x') + |x-x'|\big)\big],$$

and consequently, for localisations h, h' supported at depth $\geq D$,

$$\|h\, \delta G\, h'\|_{1,\alpha} \;\leq\; C_h C_{\delta G} \exp\big[-\delta_{\delta G}\big(\operatorname{dist}(\operatorname{supp} h, \Omega_2^c) + \operatorname{dist}(\operatorname{supp} h', \Omega_2^c) + \operatorname{dist}(\operatorname{supp} h, \operatorname{supp} h')\big)\big].$$

$C_{\delta G}$ and $\delta_{\delta G}$ depend on the constants of (P1)–(P3)*, on L and on α, and are free of k, η, Ω, Ω_2 and A. This is the two-variable, two-sided form of the domain-difference clauses* [1, (2.26)], [4, (1.20)], [3, (2.5)].

Proof. Slices. For $1 \leq j \leq k+1$ put $G_j^{[k]}(\Omega, A) := (-\Delta_{A,\Omega}^N + m^2 t^2 + a_j l_j^{-2} P_j(A))^{-1}$, $l_j := 5^{j-k}$, $P_j(A)$ the projection onto the transported constants of the scale-j blocks; $G_k^{[k]} = G_k(\Omega, A)$. Rescaling $x = l_j \hat{x}$, $\hat{A} = l_j^{1/2} A$ keeps the transporters and gives $G_j^{[k]}(x,x') = l_j^{-1} \hat{G}_j(\hat{x}, \hat{x}')$ with $\hat{G}_j$ the scale-j propagator in scale-j units, $\hat{A}$ regular at scale j. The block identity at scale j (the $\phi\phi$ block of the Schur inverse of the block operator, with Bałaban's recursion $a_{j+1} = a a_j/(aL^{-2} + a_j)$ [1, p. 609]) gives, for $1 \leq j \leq k$,

$$S_j := G_{j+1}^{[k]} - G_j^{[k]}, \qquad S_j(x,x') = a_j^2 l_j^{-1} \sum_{\hat{y},\hat{y}' \in \Omega^{(j)}} \mathsf{L}_j(\hat{x},\hat{y})\, C^{(j)}(\hat{y},\hat{y}')\, \mathsf{L}_j(\hat{x}',\hat{y}')^T,$$

with the leg $\mathsf{L}_j(\hat{x},\hat{y}) := (\hat{G}_j\, Q_j^*(\hat{A})\delta_{\hat{y}})(\hat{x})$ and the block covariance $C^{(j)} = (\Delta^{(j)}(\Omega, \hat{A}) + aL^{-2} P_L)^{-1}$ of (P2). Telescoping, $G_k(\Omega, A) = G_1^{[k]} + \sum_{j=1}^{k-1} S_j$; this is Bałaban's [3, (2.6), p. 424] in kernel form.

Slice bounds. At points at distance $\geq R_0$ from Ω^c, (P1) at scale j applied to the bounded source $Q_j^* \delta_{\hat{y}}$ (sup 1, support $B_j(\hat{y})$) gives $|\partial^\sigma \mathsf{L}_j| \leq c_0 e^{-\delta_0 \operatorname{dist}_j(\hat{x}, B_j(\hat{y}))}$ for each σ (the depth hypothesis holds since $\operatorname{dist}_j \geq \operatorname{dist} \geq R_0$; the Hölder clause (2.24) has no restriction on the separation), and (P2) gives $|C^{(j)}(\hat{y},\hat{y}')| \leq c_2 e^{-\theta_0 |\hat{y} - \hat{y}'|}$. Since $|\hat{x} - \hat{x}'| \leq \operatorname{dist}(\hat{x}, B(\hat{y})) + |\hat{y} - \hat{y}'| + \operatorname{dist}(\hat{x}', B(\hat{y}')) + 2$, keeping half of each exponent for the sums over $\hat{y}, \hat{y}'$ (lemma G.9(c)) and rescaling (each D is $l_j^{-1}\hat{D}$, each Hölder quotient $l_j^{-1-\alpha}$ times the scale-j one) gives

$$|\partial_x^\sigma \partial_{x'}^{\sigma'} S_j(x,x')| \;\leq\; C_K l_j^{-1-|\sigma|-|\sigma'|} e^{-\delta_K |x-x'|/l_j}, \qquad \delta_K := \tfrac{1}{2}\min(\delta_0, \theta_0),$$

with C_K free of k, j, η, Ω, A; and $|\partial^\sigma\partial^{\sigma'}G_1^{[k]}| \le C_1(L)l_1^{-1-|\sigma|-|\sigma'|}e^{-\theta_0|x-x'|/(2l_1)}$ by (P3) in scale-1 units (spacing $1/L$; point and bond masses of norm $L^{3/2}$; Hölder quotients over separations $\ge 1/L$ bounded by $2L^\alpha \sup|D\cdot|$).

Difference of slices. Apply the slice identity on Ω and on Ω_2 with the same A: $\delta G = \delta G_1^{[k]} + \sum_{j=1}^{k-1}\delta S_j$. By multilinearity each δS_j is $a_j^2 l_j^{-1}$ times the sum of three terms, $\delta\mathsf{L}\,C\,\mathsf{L}^T$, $\mathsf{L}_2\,\delta C\,\mathsf{L}^T$, $\mathsf{L}_2 C_2\,\delta\mathsf{L}^T$, with $\mathsf{L} = \mathsf{L}_j(\Omega)$, $\mathsf{L}_2 = \mathsf{L}_j(\Omega_2)$ (extended by 0 for $\hat y \notin \Omega_2^{(j)}$), $\delta\mathsf{L} = \mathsf{L} - \mathsf{L}_2$, $C = C^{(j)}(\Omega)$, $C_2 = C^{(j)}(\Omega_2)$ (extended by 0), $\delta C = C - C_2$. Write $d_j := l_j^{-1}d$. *(i)* $\delta\mathsf{L}$. For $\hat y \in \Omega_2^{(j)}$, (P1)'s clause (2.26) at scale j gives $c_0 e^{-\delta_0(\mathrm{dist}(\hat x,B(\hat y))+\mathrm{dist}(B(\hat y),\Omega_2^c)+d_j(\hat x))}$; for $\hat y \notin \Omega_2^{(j)}$, $\delta\mathsf{L} = \mathsf{L}(\hat x,\hat y) \le c_0 e^{-\delta_0\,\mathrm{dist}(\hat x,B(\hat y))}$ by (2.25), and $\mathrm{dist}(\hat x,B(\hat y)) \ge d_j(\hat x)$ since $B(\hat y)\subset\Omega_2^c$; in all cases $|\partial^\sigma\delta\mathsf{L}| \le c_0 e^{-(\delta_0/2)(\mathrm{dist}(\hat x,B(\hat y))+d_j(\hat x))}$. *(ii)* δC. For $\hat y,\hat y' \in \Omega_2^{(j)}$ split $\delta C = [C_{\Omega^{(j)}}(\Omega) - C_{\Omega_2^{(j)}}(\Omega)] + [C_{\Omega_2^{(j)}}(\Omega) - C_{\Omega_2^{(j)}}(\Omega_2)]$, both restricted to $\Omega_2^{(j)}$. The first bracket is $-(PXP)^{-1}PXQX^{-1}P$ with $X = \Delta^{(j)}(\Omega) + aL^{-2}P_L$, $P = \mathbf{1}_{\Omega_2^{(j)}}$, $Q = 1-P$ (Schur identity); P_L has no entries across $\Omega_2^{(j)}$, a union of L-blocks, so $PXQ = P\Delta^{(j)}(\Omega)Q$, whose kernel decays exponentially by (P2) (2.27), and both inverses decay at rate θ_0; hence the bracket is $\le Ce^{-\delta_C(|\hat y-\hat y'|+\tilde d(\hat y)+\tilde d(\hat y'))}$ with $\tilde d(\hat y) := \mathrm{dist}(\hat y,(\Omega_2^{(j)})^c)$, because the path through Q has length $\ge \tilde d(\hat y)+\tilde d(\hat y')$. The second bracket is the domain difference of the restricted block covariance, which has the same form by (P2), [4, (1.20)]. If $\hat y \in \Omega_2^{(j)}$ and $\hat y' \notin \Omega_2^{(j)}$ (the only other pattern meeting a nonzero L_2), $\delta C = C(\hat y,\hat y') \le c_2 e^{-(\theta_0/2)(|\hat y-\hat y'|+\tilde d(\hat y))}$. *(iii) Assembly.* Proceed as for S_j, carrying the decay factors and using $d_j(\hat x') \le d_j(\hat x) + |\hat x - \hat x'|$ and its symmetric form: each term is $\le Ce^{-\delta'(d_j(\hat x)+d_j(\hat x')+|\hat x-\hat x'|)}$, and after rescaling $|\partial^\sigma\partial^{\sigma'}\delta S_j| \le Cl_j^{-1-|\sigma|-|\sigma'|}e^{-\delta'\mathsf{S}/l_j}$ with $\mathsf{S} := d(x)+d(x')+|x-x'|$. *(iv)* $\delta G_1^{[k]}$. In scale-1 units use the layer identity $H_\Omega = (H_{\Omega_2}\oplus H_{\Omega\setminus\Omega_2}) + D_\Sigma^* D_\Sigma$, Σ the bonds of Ω with exactly one end in Ω_2 and D_Σ the covariant difference on them (valid because P_j and m^2 are block-diagonal and both pieces are unions of blocks), which gives $(G(\Omega)f)|_{\Omega_2} - G(\Omega_2)f = -(G(\Omega)D_\Sigma^* D_\Sigma(G(\Omega_2)f\oplus 0))|_{\Omega_2}$; (P3) on Ω and on Ω_2 bounds the two factors pointwise at the fixed spacing, with no restriction near $\partial\Omega_2$, for each of the four derivative patterns, and the layer weight L^{-2} per bond of Σ summed gives $|\partial^\sigma\partial^{\sigma'}\delta\hat G_1| \le C(L)e^{-(\theta_0/2)(d_1(\hat x)+d_1(\hat x')+|\hat x-\hat x'|)}$; rescale with l_1. *(v) Sum over j.* With $l_j = 5^{-n}$, $n = k-j \ge 1$, and $\mathsf{S} \ge 2D-2 \ge 2$: $\sum_n 5^{n(3+2\alpha)}e^{-\delta'5^n\mathsf{S}} \le e^{-\delta'\mathsf{S}/2}\sum_n 5^{n(3+2\alpha)}e^{-\delta'5^n/2} \le Ce^{-\delta'\mathsf{S}/2}$, since $5^n\mathsf{S} \ge (5^n+\mathsf{S})/2 + 5^n/2$ for $\mathsf{S}\ge 2$. The localised form follows from the definition of the norm. □

The interaction terms. The interaction terms of the stage of [2, p. 579], after the removals made there (the R-vertices, the terms of order $> \bar n$, the terms containing $\tilde B^{(k)}$, the terms localised in $B^k((\Lambda_7^{(k)})^c)$), are a finite sum, over label multisets $\mathcal{L}$ whose boxes lie within $O(1)$ of $B^k(\Lambda_7^{(k)})$ (the terms localised in $B^k((\Lambda_7^{(k)})^c)$ having been estimated separately on [2, p. 579], and [2, Prop. 2.6] being stated for the terms localised in $B^k(\Lambda_7^{(k)})$), over types T and over orders $o \le \bar n$, of per-label full classes $\mathcal{C}$ (definition G.1): each $\mathcal{C}$ is the set of all connected graphs of that type, order and label multiset built from the vertices [3, (1.6)–(1.15)] of the stage, with its Wick and Taylor weights, with the counterterm vertices [3, (1.7)] carrying their fixed coefficients [3, (1.23), (1.29)] and labels, *including the whole-graph counterterm members* (Bałaban's G' with $G_0 = G$, [3, p. 420]: the lone (1.7) vertex with two external legs is a member of the two-scalar-leg classes). This is the connected expansion of the effective action at fixed order and labels; the one-particle-irreducibility of [3, p. 430] concerns the counterterm-defining classes, whose coefficients are fixed, not the members. The terms may also be finite products of such classes contracted by fixed many-variable external fields [3, p. 419]. The scalar lines are $G_A := G_k(\Omega_A,\tilde B)$; the external scalar fields are $E_A := a_k G_A f$ with $\|f\|_\infty \le Cp\lambda(t)^{-1/4}$, and the other external fields have norms $\le Cp^a\lambda(t)^{-b}$ with a,b bounded by $\bar n$. All vertices lie in $B^k(\Lambda_7^{(k)}) \subset \Omega_C$.

Lemma G.11 (the expansion)**.** *Put $G_C := G_k(\Omega_C,\tilde B)$ $(= G_k(\Omega_C, B^{(k+1),\eta})$ by lemma G.8), $\delta G := G_A - G_C$ on $\Omega_C\times\Omega_C$ with G_C extended by 0, $E_C := a_k G_C f$ and $\delta E := E_A - E_C =$*

$a_k\delta G f_{\mathrm{in}} + a_k G_A f_{\mathrm{out}}$ *with* $f_{\mathrm{in}} := \mathbf{1}_{\Omega_C} f$, $f_{\mathrm{out}} := f - f_{\mathrm{in}}$. *For a class* $\mathcal{C}$, *or a product of classes contracted by fixed fields,*

$$\mathcal{C}[G_A; E_A] - \mathcal{C}[G_C; E_C] \;=\; \sum_{(n,m)\neq(0,0)} \mathcal{C}^{(n,m)},$$

where $\mathcal{C}^{(n,m)}$, *the coefficient of* $s^n u^m$ *in* $\mathcal{C}[G_C + s\,\delta G; E_C + u\,\delta E]$, *is the sum over* $\Gamma \in \mathcal{C}$, *over sets* Λ *of* n *scalar lines of* Γ *and over sets* M *of* m E*-legs of* Γ, *of the value of* Γ *with* δG *on* Λ, G_C *on its other scalar lines,* δE *on* M *and* E_C *on its other* E*-legs.*

Proof. A class value is a polynomial, multilinear in the scalar line kernels and in the E-legs (Wick expansion); put $s = u = 1$. The counterterm coefficients are vertex data built from torus propagators at zero background [3, (1.29)–(1.31)] and are not expanded: the replacement of propagators acts on lines only. □

Definition G.12 (cut skeletons and σ-classes)**.** Fix $\mathcal{C}$ (type T, order o, labels $\mathcal{L}$) and (n, m). A *cut skeleton* σ consists of a partition $\mathcal{L} = \mathcal{L}_1 + \cdots + \mathcal{L}_\rho$ into non-empty sub-multisets with orders $o_1 + \cdots + o_\rho = o$, $o_c \geq 0$; a connected multigraph Q on $\{1, \ldots, \rho\}$ with n edges (loops and multiple edges allowed); and an assignment of the external legs of $\mathcal{C}$ to the components, each E-leg marked δE or E_C with m marked δE. For each component c, $\mathcal{C}_{\sigma,c}$ is the per-label full class of all connected configurations with labels $\mathcal{L}_c$ and order o_c, whose external legs are the legs assigned to c together with $\deg_Q(c)$ scalar *stub* legs, with scalar lines G_C and vector lines as in $\mathcal{C}$, *members without internal line and of order* 0 *included*: equivalently, the coefficient of the monomial of the labels $\mathcal{L}_c$ in the stub-Taylor expansion of the full connected generating function $\ln \int e^{-V(\cdot+\chi+\psi)} d\mu_{G_C}$, ψ the stub field. The stub and E-legs may sit at any scalar slot of any vertex, differenced or not; the class sums over all of them. The *σ-class* is the contraction $\Sigma_\sigma(\mathcal{C}) := \big\langle \bigotimes_c \mathcal{C}_{\sigma,c}, \bigotimes_{e\in Q} \delta G_e \big\rangle$: an edge $\{c, c'\}$ contracts one stub of c with one stub of c' by δG, a loop two stubs of c. It carries the Taylor weight w_σ, with $0 \leq w_\sigma \leq C(\bar{n})$.

Lemma G.13 (regrouping into σ-classes)**.** $\mathcal{C}^{(n,m)} = \sum_\sigma w_\sigma E(\Sigma_\sigma(\mathcal{C}))$, *the sum over the cut skeletons of* $(\mathcal{C}; n, m)$.

Proof. The bijection. To (Γ, Λ, M) assign the components of $\Gamma \setminus \Lambda$ (connected through the kept lines, vector lines included, since Λ consists of scalar lines), their label multisets and orders, the multigraph Q that Λ induces on them (connected, since Γ is), the assignment of the external legs, and for each c the restriction Γ_c of Γ to c with the ends of Λ-lines at c turned into stubs. Then $\Gamma_c \in \mathcal{C}_{\sigma,c}$: it is connected, with labels $\mathcal{L}_c$, order o_c and its external legs; it may be a single vertex all of whose legs are stubs or external fields (when every line at that vertex is in Λ), or a path of order-0 vertices (when Λ cuts out a stretch of an expanded line), which is why these members are in $\mathcal{C}_{\sigma,c}$ (with [3, p. 415]'s "there is at least one internal line" the map would not land in the product). Conversely, given σ, members $\Gamma_c \in \mathcal{C}_{\sigma,c}$ and a pairing of all stubs compatible with Q, joining the paired stubs by lines gives a connected graph of type T, order o and labels $\mathcal{L}$, hence a member of the full class $\mathcal{C}$, with Λ the set of joining lines; the two maps are inverse. *The weights.* In the labelled form with vertex labels $1, \ldots, N$ and weight $\frac{1}{N!}\prod_v(-c_v)$, the components take disjoint sets of vertex labels, in $N!/\prod_c N_c!$ ways, so the weight is $\prod_c \frac{1}{N_c!}\prod_{v\in c}(-c_v)$ times the symmetry factor of identical components, parallel edges and loops of Q, which is that of the Taylor coefficients. *Generating-function form* (the same identity, weights included): with $W(K;\chi)$ the connected generating function at fixed labels, scalar covariance K and external field χ, $W(G_C + s\,\delta G; \chi) = \ln \mathbb{E}_{s\delta G}\big[\exp W(G_C; \chi + \psi)\big]$, a formal Gaussian convolution in the stub field ψ (the moments of covariance $K_1 + K_2$ are the convolution of those of K_1 and K_2, an identity of polynomials in the kernels; no positivity of δG is used). By the linked-cluster theorem [9] the right side is a sum of connected graphs whose vertices are the ψ-Taylor coefficients of $W(G_C; \chi + \psi)$, the classes $\mathcal{C}_{\sigma,c}$ with their stubs, and whose lines are δG; taking the coefficient of the labels and of s^n, with $\chi = E_C + u\,\delta E$, and then of u^m, gives the claim. *Products.* For a product of classes contracted by fixed many-variable fields [3, p. 419], apply the above factor by factor; the fixed fields stay external fields of the components they touch and join the components in step 5 below exactly as the δG fields do. □

The classes $\mathcal{C}_{\sigma,c}$ are full per-label classes, not orbits: the finer groupings (lemma G.3) fix slot data, which is needed where a bound depends on slot types; here only the $\|\cdot\|_{1,\alpha_0}$ norms of δG and δE are consumed, which control both slot types (lemma G.10 has $\sigma, \sigma' \in \{0, D, HD\}$ in each variable), so the full grouping is admissible, and it is the form in which the renormalisation bound is available at a nonzero background. Lemma G.13 is Bałaban's own step at [3, p. 432] for the lines δG_k ("we replace these lines by pairs of external legs ... G' may be unconnected ... it can be represented as a sum of classes G'_{ren}"), which for full classes is exact.

Lemma G.14 (the σ-classes at a nonzero background)**.** *For every cut skeleton σ and component c:*

(a) *$\mathcal{C}_{\sigma,c}$ is a family of the form of* [3, Prop. 1] *on the domain Ω_C at the background $\tilde{B}$: vertices* [3, (1.6)–(1.15)] *localised in $B^k(\Lambda_7^{(k)})$, scalar lines $G_k(\Omega_C, \tilde{B})$, vector lines as printed, and external fields of several variables (the stub kernels δG, "treated as an external field"* [3, pp. 414, 419]*; the fields δE, E_C; the remaining fields of $\mathcal{C}$).*

(b) *Bałaban's classes are sets of graphs built from the vertices of* [3, (1.5)]*, and $\tilde{B}$ enters only the line operators and the transporters of* [3, (1.12)–(1.15)]*; so $G_{\mathrm{ren}}(G)$ is the same set of graphs at $\tilde{B} = 0$ and at $\tilde{B} \neq 0$. Every completion keeps type, order and label multiset; hence $G_{\mathrm{ren}}(G) \subset \mathcal{C}_{\sigma,c}$ for every member G, and $\mathcal{C}_{\sigma,c}$ is the union of the $G_{\mathrm{ren}}(G)$ over its members.*

(c) *$\mathcal{C}_{\sigma,c}$ is closed under the exchange of the two external scalar legs of every two-scalar-leg or $\phi\phi A$ subgraph, the whole graph included (lemma G.5(c)), exchanges that move a stub or change its slot type included; and its Taylor coefficients in the fluctuation of the background are again per-label full classes, every touched two-scalar-leg subgraph having its mirror.*

(d) *Split $\mathcal{C}_{\sigma,c} = \mathcal{C}^{\mathrm{ren}}_{\sigma,c} + \mathcal{C}^{0}_{\sigma,c}$, where $\mathcal{C}^{\mathrm{ren}}_{\sigma,c}$ consists of the members of order ≥ 1 which are graphs in the sense of* [3, p. 415] *together with the lone counterterm vertex* [3, (1.7)] *(the counterterm member of the whole-graph pairing), and $\mathcal{C}^{0}_{\sigma,c}$ of the bare single vertices with every leg a stub or an external field and of the order-0 paths of $n = 0$ vertices joined by G_C lines. Then $\mathcal{C}^{\mathrm{ren}}_{\sigma,c}$ is a per-label full class of the form of* [3, Prop. 1]*, to which the bound* [3, (1.33)] *applies at block factor 5 with constants "independent of ε, k, the domains $\Omega, \Omega_1, \Omega_2$, the vector field $\tilde{B}$"* [3, p. 421] *(theorem F.2; the unprinted steps of that proposition are those listed in Step 3 of the proof of theorem F.1), with the localisation h of each external field equal to 1 on the label boxes of every vertex at which the field sits in some member of the class, the class being split by the cluster at which each leg sits (at most $V(\bar{n})^{\#\mathrm{legs}}$ pieces). Each member of $\mathcal{C}^{0}_{\sigma,c}$ obeys a bound of the form of* [3, (1.33)] *directly: a bare vertex is at most its coupling times $\prod_i \|h_i \Phi_i\|_{1,\alpha_0}$ summed over its cube or block (*[3, p. 421]*: "the vertex itself with all external legs ... is treated separately"); an order-0 path is a tree whose only pieces of degree ≤ 0 are the pairs VG_CV of* [3, (2.4)]*, so that* [3, Prop. 2.1] *applies to it: each pair is treated by* [3, (2.8)–(2.9)] *at one vertex, the derivative moved onto a stub, an E-leg or the external vector leg (allowed since the norm* [3, (1.32)] *contains one derivative, and lemma G.10 covers the stub), after which all degrees are positive. No member of $\mathcal{C}^{0}_{\sigma,c}$ is a partner that the bound needs: partners have the same order ≥ 1, and a zero-line member of order ≥ 1 other than a counterterm vertex is a (1.6) vertex with four external legs, alone in its class. The factors of the background in these bounds are $O(e(t)p)$* [3, p. 412].

Proof. (a) By construction and lemma G.8. (b) At [3, p. 414] "the vertices of (1.5) are given by (1.6)–(1.8), (1.10), (1.13), and (1.14) with $n' = 0$ and $\tilde{B} = A^{(k)}$", and the lines are "$G_k(\Omega, \tilde{B})$, $G_k(\Omega_2, \tilde{B})$, $\delta G_k(\Omega, \Omega_2, \tilde{B})$ or the operator (1.16)"; the completions (the counterterm graph of a divergent two-scalar-leg subgraph [3, p. 420], the permutation [3, p. 429], the two-vector-leg class in its maximal form [3, p. 431]) keep type, order and labels ("localizations in these graphs should be in agreement"), and a per-label full class contains all graphs of its type, order and labels. (c) The first clause is lemma G.5(c); the second is the identity of generating functions in its proof: the expansion in the fluctuation $\tilde{B}'$ of the background is the Taylor expansion of the full connected generating function, whose coefficients are full classes with $\tilde{B}'$-legs carrying their partition labels, and the mirror of a touched two-scalar-leg subgraph is an exchange of the first clause. (d) The members of $\mathcal{C}^{\mathrm{ren}}_{\sigma,c}$ form the per-label full class of [3, Prop. 1]'s form by (a)–(c);

the lone (1.7) vertex is the counterterm member of the whole-graph pairing ([3, p. 417]: "the same graph G but with both external legs localized in x"). The localisation rule for a field that sits at several vertices of a class is the reading of [3, p. 420] ("$h = 1$ on box(v)") for classes; per piece the stub bound of step 4 below holds with the piece's h, whose supports are at least as far apart as the original ones. The bounds on $\mathcal{C}^0_{\sigma,c}$ are as stated: a bare vertex carries no line and no loop; an order-0 path consists of $n = 0$ vertices, each with at most two scalar legs, so it is a tree, and [3, (2.8)] is the one-derivative move of [3, Sect. 2] at a vertex of a (2.4)-type pair. The last two clauses are read off the orders. □

Theorem G.15 ((BAL$_5$)[int]: the interaction-term half of [2, Prop. 2.6] at every background)**.** *In the setting above, for every $\kappa > 0$ there are $C_\kappa < \infty$ and $t_0(\kappa) > 0$ such that for $t \le t_0(\kappa)$*

$$\sum_{\mathcal{L}} \sum_{\mathcal{C}} \sum_{(n,m)\neq(0,0)} \sum_{\sigma} w_\sigma \big| E\big(\Sigma_\sigma^{(n,m)}(\mathcal{C})\big)\big| \;\le\; C_\kappa\, t^\kappa\, |\Lambda_7^{(k)}|,$$

the sums over every localisation $\mathcal{L}$ of the stage (vertex cubes in $B^k(\Lambda_7^{(k)})$, label boxes within $O(1)$ of it), every class $\mathcal{C}$ with labels $\mathcal{L}$, every $(n,m) \neq (0,0)$ and every cut skeleton σ. By lemmas G.11 and G.13 the left side dominates $\sum_{\mathcal{L}} \sum_{\mathcal{C}} |\mathcal{C}[G_A; E_A] - \mathcal{C}[G_C; E_C]|$: this is [2, Prop. 2.6] *for the interaction terms, with error $O((L^k\varepsilon)^\kappa)|\Lambda_7^{(k)}|$ for every κ, at every background $\tilde B$, $\tilde B = 0$ included; C_κ and $t_0(\kappa)$ depend on κ, on $(R, \varrho, b_0, \tilde p, \bar n, \alpha_0)$, on the couplings and on the constants of* (P1)–(P3) *and of* [3, (1.33)] *only. This proves lemma 5.1.*

Proof. Step 1 (domain change only). By lemma G.8 the replacement is $G_A \to G_C$ at the one background $\tilde B$; $\Omega_C \subset B^k(\Lambda_0^{(k)}) \subset \Omega_A$, and $\tilde B$ is regular on Ω_A.

Step 2 (expansion and regrouping). By lemmas G.11 and G.13 the difference is $\sum_{(n,m)\neq(0,0)} \sum_\sigma w_\sigma E(\Sigma_\sigma^{(n,m)}(\mathcal{C})$

Step 3 (depth). Every vertex, every label box and every localisation support lies within $2 + \frac14 + \eta < 3$ of $B^k(\Lambda_7^{(k)})$, hence at depth $\ge D_7 = 5(r-1) - 4$ in Ω_C by lemma G.9(a), with $D_7 \ge R_0 + 2$ by (85). The domain hypothesis of [3, Prop. 1] for $\Omega_2 = \Omega_C$ ("the same conditions as Ω" [3, p. 414], including $\operatorname{dist}(\operatorname{supp}\tilde A, \partial\Omega_2) > 2r$ [3, p. 412]) holds for the $\tilde A$-legs actually present: with $\tilde A = \theta_{k+1} A'^{(k)}$ the raw support reaches $\partial\Omega_C$, but after [2, p. 579] every vertex and every localised leg lies within $O(1)$ of $B^k(\Lambda_7^{(k)})$, at depth $\ge D_7 > 2r$ (since $r \ge R \ge R_0 + 3 > 3$); multiplying $\tilde A$ by a cut-off equal to 1 on that neighbourhood and supported at depth $> 2r$ changes no term.

Step 4 (external fields). For the localisations h, h' of the two ends of an edge of Q ($h = 1$ on the label boxes of the vertices at which the stub can sit in its component, supported in their $\frac14$-neighbourhood), lemma G.10 with $(\Omega, \Omega_2, A) = (\Omega_A, \Omega_C, \tilde B)$ gives

$$\|h\, \delta G\, h'\|_{1,\alpha_0} \;\le\; C_h C_{\delta G}\, e^{-2\delta_{\delta G} D_7}\, e^{-\delta_{\delta G} \operatorname{dist}(\operatorname{supp} h, \operatorname{supp} h')},$$

with the mixed clauses $\sigma, \sigma' \in \{0, D, HD\}$ that the joining of components needs. For δE: $\|h\, \delta E\|_{1,\alpha_0} \le 2c_0 e^{-\delta_0 D_7} \|f\|_\infty$, since (P1) (2.26) bounds $a_k \delta G f_{\text{in}}$ and (2.25) bounds $a_k G_A f_{\text{out}}$ at points at distance $\ge D_7$ from $\operatorname{supp} f_{\text{out}} \subset \Omega_C^c$. The norms of E_C and of the other external fields are $\le C p^a \lambda(t)^{-b}$ ((P1) (2.24)–(2.25) on Ω_C at depth D_7).

Step 5 (the renormalisation bound per σ-class). By lemma G.14(d), [3, (1.33)] or the direct bound applies to each $\mathcal{C}_{\sigma,c}$ at fixed labels with the external fields of Step 4. The components are joined along a spanning tree of Q by the decay of the stub kernels, the non-tree edges and loops entering as many-variable external fields ([3, (3.2), p. 432]: "the exponential factor in (1.33) is obtained from the estimates of external fields in (3.2)"). With $e_k, \lambda_k \le 1$,

$$\big|E\big(\Sigma_\sigma^{(n,m)}(\mathcal{C}); \mathcal{L}\big)\big| \;\le\; O(1)\, e^{-\delta'' d(\mathcal{L})} \big(C' e^{-2\delta_{\delta G} D_7}\big)^n \big(2c_0 C p \lambda^{-1/4} e^{-\delta_0 D_7}\big)^m \big(C p^a \lambda^{-b}\big)^{n_{\text{ext}}},$$

where $d(\mathcal{L})$ is the tree length of all localisation cubes over all components and $\delta'' := \frac12 \min(\delta_0, \delta_{\delta G}, \delta_{\text{ren}})$, δ_{ren} the rate of [3, (1.33)] (called δ_0 there; it is not the δ_0 of (P1)): the trees inside the components come from (1.33), the edges between components from the stub decay, and the $\frac14$-neighbourhoods change the tree length by at most a constant times the number of cubes, a constant of $\bar n$ absorbed in $O(1)$.

Step 6 (size). Since $n+m \geq 1$ the product carries $e^{-\delta_* D_7}$ at least once, $\delta_* := \min(2\delta_{\delta G}, \delta_0)$, and $D_7 \geq r$; by lemma G.9(b), $p^a \lambda^{-b} e^{-\delta_* D_7} \leq t^\kappa$ for $t \leq t_0(\kappa)$, the background factors $O(e(t)p)$ of lemma G.14(d) being absorbed the same way.

Step 7 (sum). Fix one cube in $B^k(\Lambda_7^{(k)})$ ($|\Lambda_7^{(k)}|$ unit cubes, and every cube lies there); lemma G.9(c) sums the others with $e^{-\delta'' d}$. The numbers of types, of orders $\leq \bar{n}$, of pairs (n,m) and of skeletons σ are bounded by constants of $\bar{n}$ ($\rho \leq V(\bar{n})$, n at most the number of scalar lines, Q and the leg assignments finite). □

Remark G.16. (i) The proof uses no telescoping in a fixed line order and no forest families: the terms with one difference line, grouped per line, carry their counterterm partners but not the two-vector-leg and permutation partners of the cut graph (remark G.4(iv)), and a lone one-loop two-vector-leg subgraph is $O(\eta^{-1})$ per graph, which $e^{-2\delta_{\delta G} D_7} = O(t^\kappa)$ does not cure; the regrouping of lemma G.13 restores the complete classes, at every background. (ii) The corresponding replacement in the polynomial $Q^{(k)}$ of [2, p. 579], the other half of [2, Prop. 2.6], is not used through lemma 5.1. (iii) On the torus skew in the rotated frame (the instance consumed by the rotation part through [13, Thm. 3.1] there), lemmas G.11, G.13 and G.14 are combinatorics without geometry, [3, (1.33)] transfers as the other torus estimates do, and the proof of lemma G.10 uses no cubic symmetry, its inputs (P1)–(P3) being available on that torus: lemmas G.18 and G.19 below carry (P1)–(P3), lemma G.10 and theorem G.15 to every lattice torus fitted to the blocks, these tori included.

G.3. **The difference-kernel bound on a lattice torus fitted to the blocks.** The rotation part applies King's [13, Thm. 3.1], and through it (BAL_5)[int], on the tori $T_1 = D_J$ and $T_2 = R^{-1} D_J$ of definition 17.5, each carrying both lattices Λ_k and $R^{-1}\Lambda_k$ of definition 15.6(b). In the frame of the lattice $R^{-1}\Lambda_k$ the torus D_J is the quotient of that lattice's own axis lattice by $R(50MJ\,\mathbb{Z}^3)$, and in the frame of Λ_k the torus $R^{-1}D_J$ is the quotient by $R^{-1}(50MJ\,\mathbb{Z}^3)$: lattices of translations which are not rectangular in those frames. The statements of this appendix were written for subsets of $\eta\mathbb{Z}^3$. This subsection records that their proofs use the global geometry only through the two properties of definition G.17, which the tori of the rotation part have, so that lemmas G.7 and G.10 and theorem G.15 hold on them with constants that do not depend on the torus.

Definition G.17 (a lattice torus fitted to the blocks)**.** In scale-k units let $T = \eta\mathbb{Z}^3/\Gamma$, where $\Gamma \subset \mathbb{Z}^3$ is a lattice of rank 3 (so T carries the unit blocks and, since $\Gamma \subset \mathbb{Z}^3$, the blocks of every scale $j \leq k$, of side $l_j = 5^{j-k}$). Let $M_b \in \mathbb{N}$ be the side of a large block. T is *fitted to the blocks* if $\Gamma \subset M_b\mathbb{Z}^3$ and its sup-systole $\mathrm{sys}_\infty(T) := \min\{|\gamma|_\infty : \gamma \in \Gamma \setminus \{0\}\}$ is at least $3M_b$. Distances on T are the quotient sup-distances $\mathrm{dist}_T(x,x') := \min_{\gamma\in\Gamma} |x - x' - \gamma|_\infty$; $\Omega \subset T$ denotes a union of large blocks, T itself included, and "regular on Ω" is [1, (2.23)] read on the bonds of Ω.

If T is fitted to the blocks in scale-k units, so is its image in scale-j units for every $j \leq k$: the lattice becomes $l_j^{-1}\Gamma \subset l_j^{-1}M_b\mathbb{Z}^3 \subset M_b\mathbb{Z}^3$ (as $l_j^{-1} = 5^{k-j} \in \mathbb{N}$) and the systole grows by the factor l_j^{-1}. For the tori of the rotation part, in the frame of either lattice, the lattice of translations is $50MJ\,\mathbb{Z}^3$ or $R^{\pm1}(50MJ\,\mathbb{Z}^3) \subset 10MJ\,\mathbb{Z}^3$ in physical units (lemma 15.3(b)). At Bałaban's step k on the lattice of spacing $\varepsilon_K = 5^{-K}$ the scale-k unit is $t = 5^k\varepsilon_K = 5^{k-K} \leq 1$, so in scale-$k$ units these lattices lie in $10MJ\,5^{K-k}\mathbb{Z}^3$ and the systole is at least $10MJ\,5^{K-k}$. They are therefore fitted to the blocks at every step whenever the large-block side M_b divides $10M$ and $3M_b \leq 10MJ$; in King's flow, the one run on these tori through [13, Thm. 3.1], the large block is King's and $M_b = M$.

Lemma G.18 (the printed inputs on a torus fitted to the blocks)**.** *Let T be fitted to the blocks, $\Omega \subset T$ a union of large blocks and A regular on Ω. Then the bounds* (P1) *(*[1, Prop. 2.1, (2.24)–(2.26)], [4, (1.9)–(1.12)]*),* (P2) *(*[1, Props. 2.2–2.3], [4, (1.13)–(1.20)]*) and the positivity* [4, (1.8)] *hold for the operators $G_k(\Omega,A)$, $\Delta^{(k)}(\Omega,A)$, $C_\Lambda^{(k)}(\Omega,A)$ on T, with the distances of definition G.17, with constants c_0, δ_0, R_0, c_2, θ_0, γ that depend on d, a, M_b and L (c_0 also on α) and not on Γ, k, Ω, Λ or A. For $\Omega = T$ the restriction to points at distance $\geq R_0$ from Ω^c is void. The same holds in scale-j units for every $j \leq k$.*

Proof. Bałaban's proofs [4, Sect. 2, pp. 574–586] are written for a union Ω of large blocks in $\eta\mathbb{Z}^d$ ("the torus identified with a rectangular parallelepiped"). They use the geometry of the ambient lattice in exactly the following places, each of which we verify on T.

(1) The covering cubes. Let v_j run through the vertices of the large-block grid of T and $C_j := v_j + [-M_b, M_b)^3$, the union of the 2^3 large blocks with vertex v_j ([4, p. 578]: "a cube with center Mj and size equal to $2M$"). Since $2M_b < \mathrm{sys}_\infty(T)$, the projection $\eta\mathbb{Z}^3 \to T$ is injective on each lift of C_j; it preserves bonds, unit blocks and the contours $\Gamma^{(k)}_{y,x}$ of the averaging operators, which lie inside blocks [4, (1.4)]. Hence $\Pi_j := \Omega \cap C_j$ is a union of at most 2^3 large blocks in an embedded cube, the constant gauge $\lambda_0(x) = A_0 \cdot (x - c_j)$ is defined on C_j, and [4, Lemmas 2.1, 2.2, 2.4], which are statements about $G_k(\Pi_j, A_j)$ on a parallelepiped of $\eta\mathbb{Z}^d$ with a local field A_j built from $A|_{C_j}$, apply with their constants, which depend on d, M_b, α and the bare a only.

(2) The partition of unity. Let $h \in C^\infty_c((-\frac{2}{3}, \frac{2}{3}))$ with $h = 1$ on $[-\frac{1}{3}, \frac{1}{3}]$ and $\sum_{n\in\mathbb{Z}} h(\cdot - n)^2 = 1$, and $h_j(x) := \prod_\mu h((x_\mu - (v_j)_\mu)/M_b)$ [4, p. 575]. Because $\Gamma \subset M_b\mathbb{Z}^3$, the family $\{h_j\}$ is Γ-periodic, descends to T, and $\sum_j h_j^2 = 1$ on T with $\operatorname{supp} h_j \subset C_j$; so [4, (2.2)–(2.9)] hold on T as on $\eta\mathbb{Z}^d$ (if $\Omega = T$, then $\partial\Omega = \emptyset$ and (2.6) is trivial).

(3) The counts. Each j has at most 3^3 neighbours and each point lies in at most 2^3 of the supports, so the remainder of the parametrix has $\|R\|_{2,2} \le c_d O(M_b^{-1}) < 1$ as on [4, p. 577], and the number $2^d(3^d)^{n-1}2^d$ of walks of length n [4, p. 579] is unchanged.

(4) The metric. Consecutive cubes of a walk have centres at quotient distance $\le M_b$, so a walk of length n from a cube meeting $\operatorname{supp} f$ to a cube containing x has $n \ge \mathrm{dist}_T(x, \operatorname{supp} f)/M_b - 2$; the geometric series [4, (2.22)] then gives δ_0, c_0 as on $\eta\mathbb{Z}^d$. The definition [4, (2.19)] of R_0 and the implication "$\mathrm{dist}_T(\{x, x'\}, \Omega^c) \ge R_0$ implies that the first n_0 cubes of the walk lie in Ω" use only this. The cancellation of the interior-cube terms in $G_k(\Omega, A) - G_k(\Omega_0, A)$, which gives the two-sided factor of [1, (2.26)], uses (1) and (2). If $\Omega = T$, every $\Pi_j = C_j$ is a full cube and no restriction on x, x' is needed [1, p. 611].

(5) Positivity. For [4, (1.8)], $P_k(A)$ is block-diagonal, so dropping the bonds between unit blocks bounds the form from below by a sum over the unit blocks Δ of $\langle\phi, (-\Delta^N_{A,\Delta} + a_k P_k(A))\phi\rangle_\Delta$; on each block the gauge of (1) and the expansion in the field reduce the bound to $A = 0$ [4, p. 580], where it is the Poincaré inequality on a unit block, which embeds in T. So γ is Bałaban's.

The block operators of (P2) are built from $G_k(\Omega, A)$ and the averaging operators, and the proofs of [1, Props. 2.2–2.3], [4, Sect. 3] use the same cubes, partition and walk and no further property of the ambient lattice; the restricted covariances $C^{(k)}_\Lambda$ for a union Λ of large blocks are handled by the same walk on Λ. Nothing else in [4, Sects. 2–3] refers to the global geometry. The last sentence of the lemma follows because the rescaled torus is fitted to the blocks (the remark after definition G.17) and the statements are the same at every scale. □

Lemma G.19 (lemmas G.7 and G.10 and (BAL$_5$)[int] on a torus fitted to the blocks)**.** *Let T be fitted to the blocks. Then:*

(a) *lemma G.7 holds for subsets of T with the quotient metric, with the same constants.*

(b) *lemma G.10 holds for unions $\Omega_2 \subset \Omega$ of large blocks of T and A regular on Ω, with* dist_T *in place of* dist *and with $C_{\delta G}$, $\delta_{\delta G}$ depending on d, a, M_b, L and α only.*

(c) *theorem G.15 holds on T, with the propagator domains $\Omega_A \supset \Omega_C$ and the background $\tilde{B}$ of Bałaban's stage on T ([2, Sect. 2] run on T, as in theorem F.2) and with the renormalisation bound [3, Prop. 1, (1.33)] on T; its constants C_κ, $t_0(\kappa)$ are those of the statement on $\eta\mathbb{Z}^3$ and do not depend on Γ.*

In particular all three hold on the tori T_1, T_2 of definition 17.5, in the frame of either lattice.

Proof. (a) The proof of lemma G.7 uses the weight $\rho = \mathrm{dist}(\cdot, \operatorname{supp} f')$ only through two properties: ρ is 1-Lipschitz along bonds (the ends of a bond are at distance η), and each site is the final point of d bonds. Both hold on T with $\rho := \mathrm{dist}_T(\cdot, \operatorname{supp} f')$, since $\mathrm{dist}_T(\cdot, S)$ is 1-Lipschitz for the quotient metric and the bond structure is that of $\eta\mathbb{Z}^3$ locally. The lower bound $T \ge \gamma$ it needs is [4, (1.8)] on T (lemma G.18). The constants are unchanged.

(b) The proof of lemma G.10 consists of the slice decomposition $G_k(\Omega, A) = G_1^{[k]} + \sum_{j=1}^{k-1} S_j$, an operator identity which needs only that the blocks of every scale $j \le k$ partition Ω (true on T: a

union of large blocks is a union of unit blocks of every scale $j \le k$, and these descend to T because $\Gamma \subset M_b\mathbb{Z}^3 \subset l_j\mathbb{Z}^3$); the rescaling to scale-j units, under which T stays fitted to the blocks; the slice bounds, which are (P1) at scale j for the legs, (P2) at scale j for the block covariances and (P3) at scale 1, all available on T with Γ-free constants by lemma G.18 and (a); the sums over the block centres $\hat{y}, \hat{y}'$, which use lemma G.9(c) in the form $\sum_{\square} e^{-\delta\, \mathrm{dist}_T(\square, S)} \le C(\delta)|S|$, true on T because the projection of the cubes of $\eta\mathbb{Z}^3$ onto those of T is onto, so the sum on T is at most the sum on $\eta\mathbb{Z}^3$ over a lift of S; and the triangle inequality for the metric. Every step is therefore available on T with the quotient metric, and the proof goes through word for word. The constants $C_{\delta G}$, $\delta_{\delta G}$ are built from those of (P1)–(P3), L and α.

(c) In the proof of theorem G.15: lemmas G.11, G.13 and G.14 are statements about the expansion and regrouping of graph sums and about the sets of graphs forming a class; they do not refer to the geometry. lemma G.8 uses only that the three operators depend on the field through the bonds of their domain and on contours inside blocks. lemma G.9(a) uses the construction of the layers $\Lambda_i^{(k)}$ by sup-neighbourhoods of unions of large blocks and sup-metric geodesics, which exist in the quotient metric (a shortest path lifts to $\eta\mathbb{Z}^3$); (b) is a scalar inequality; (c) holds on T as shown in (b) above, the number of cubes of T at a given tree distance from a fixed cube being at most the number on $\eta\mathbb{Z}^3$. Steps 1–3 of the proof use (a) of lemma G.9 and the definitions of the stage; Step 4 uses (b) above and (P1) on T; Step 5 uses [3, Prop. 1, (1.33)] on T, which is Bałaban's renormalisation bound for a torus aligned with the blocks (theorem F.2); Steps 6 and 7 use lemma G.9(b), (c). The constants are those of the statement on $\eta\mathbb{Z}^3$.

For the tori of the rotation part, both lattices of translations are contained in $10MJ\,\mathbb{Z}^3$ in the frame in question (lemma 15.3(b)), so the tori are fitted to the blocks with $M_b = M$ by the remark after definition G.17. □

Remark G.20. (i) The two conditions of definition G.17 are what Bałaban uses of the rectangular geometry: that the large blocks tile the torus at every scale (the partition of unity and the block operators), and that a cube of two large blocks' side embeds (the local parametrix and the local gauge). A rectangular torus of sides that are multiples of M_b and at least $3M_b$ satisfies both. (ii) The systole condition can be removed by passing to the covering torus $\eta\mathbb{Z}^3/3\Gamma$ and pulling back, at the price of the factor 3^3 in the kernel bounds from the image sum over the deck group; it is not needed for the tori of the rotation part, whose systole is $10MJ$ at least.

Appendix H. Bałaban's Proposition 1 at block factor 5: complete proofs

This appendix writes out the proof of Bałaban's perturbative estimate for localised renormalised classes, [3, Prop. 1, (1.33)], at block factor 5, in the form used in appendix F. The printed proof on [3, pp. 420–445] consists of twenty sketched steps and seven further sketched places; each is stated here in the form in which the proposition uses it and proved, with constants independent of the lattice spacing, the step, the domains, the background and the torus. Appendix F cites these steps; nothing in this appendix is used elsewhere.

H.1. The sketched steps of Proposition 1, I: slices, external-field kernels, the summation by parts and the induction.

Appendix F reads the perturbative estimate [3, Prop. 1, (1.33)] at block factor $L = 5$ and uses it in the form

$$|E(G_{\mathrm{ren}}, \Lambda, \Phi)| \;\le\; O(1)\, e_k^{d_v} \lambda_k^{d_s}\, e^{-\delta_0 d(\Lambda)} \prod \|h\Phi\|_{1,\alpha_0},$$

for a localised renormalised class G_{ren} with localisation $\Lambda = \{\square(v)\}$, $d(\Lambda)$ the length of a shortest tree through the cubes of Λ. Its proof on [3, pp. 420–445] consists of twenty sketched steps and seven further sketched places. This appendix writes them out. We number the sketched steps S1–S21 in the order in which they are invoked in the printed proof (S12 is the printed cube split of [3, p. 433], which needs no further argument), and the seven further places F1–F7. This part contains the first five:

step	[3] page	printed object
S1	414	the kernel [3, (1.16)], $[G_k(\Omega,\tilde B)V_k]^n G_k(\Omega,\tilde A+\tilde B)[V_k G_k(\Omega,\tilde B)]^{n'}$, as an external field
S2	414, 424	$\delta G_k(\Omega,\Omega_2,\tilde B)=G_k(\Omega,\tilde B)-G_k(\Omega_2,\tilde B)$ as an external field: [3, (2.5)], first alternative
S3	424, 426	the slice decomposition [3, (2.6)] and the slice bounds [3, (2.10)–(2.12)]
S4	425	the summation by parts [3, (2.8)–(2.9)] at a vertex of the graph [3, (2.4)]
S5	428	the induction [3, (2.16)]

S1 and S2 rest on S3, so the order below is S3 (appendix H.1.2), two common tools (appendix H.1.3), S1 (appendix H.1.4), S2 (appendix H.1.5), S4 (appendix H.1.6) and S5 (appendix H.1.7). Everything is at $L=5$, $d=3$, on the scale-k lattice with $k\ge 1$, in Bałaban's own model (no shell mass).

H.1.1. *Setting and inputs.*
Units and scales. $\eta := L^{-k}$ with $L=5$ and $k\ge 1$, so $\eta\le\frac15$; $t := t_k = L^k\varepsilon$; $e_k := e\,t^{1/2}$ is the running charge [1, (2.23)]. The slice scales are $s_j := L^j\eta = L^{j-k}$, $0\le j\le k-1$, so $\eta=s_0\le s_j\le\frac1L$. Scalar fields take values in $\mathbb{R}^N$; $U(A_b)=\exp(qe_k\eta A_b)$ with q antisymmetric, $\|q\|\le 1$ [1, p. 605]; all U commute and are orthogonal. The covariant derivative is $(D_{A,\mu}\phi)(x)=\eta^{-1}(U(A_{x,\mu})\phi(x+\eta e_\mu)-\phi(x))$ [1, (1.7)]; inner products carry the weight η^d and kernels are taken with respect to η^d. $|\cdot|_\infty$ is the torus sup metric [1, (1.3)], used for every distance, depth and exponential; $|\cdot|$ is the Euclidean torus norm, used only in the Hölder quotients of [3, (1.32)]; $|\cdot|_\infty\le|\cdot|\le\sqrt{3}|\cdot|_\infty$.
Blocks and domains. Unit blocks $B^k(y)$ have side 1; large blocks have side M [1, p. 607]. $\Omega=B^k(\Omega^{(k)})$ with $\Omega^{(k)}$ a union of large blocks; Ω_1, Ω_2 are of the same kind [3, p. 414]. The depth of x in X is $\rho_X(x):=\operatorname{dist}_\infty(x,X^c)$.
Averages and propagators. $(Q_k(A)\phi)(y)=\sum_{x\in B^k(y)}\eta^d U(A(\Gamma^{(k)}_{y,x}))\phi(x)$ along the composite contours of [1, (2.1)–(2.2)]; $P_k=Q_k^*Q_k$; $H_X(A):=-\Delta^{\eta,N}_{A,X}+m^2t^2+a_kP_k(A)$ and $G_k(X,A):=H_X(A)^{-1}$ [1, (2.22)], where X is a union of unit blocks with Neumann conditions or the torus T_η. We assume $0\le m^2t^2\le c_m$. The vector field is the case $N=d$, $A=0$, $X=T_\eta$, with μ_0 in place of m [1, p. 608]. $a_j=a(1-L^{-2})/(1-L^{-2j})\in[\frac{24}{25}a,a]$ [1, (2.15)]; put $a_*:=\frac{24}{25}a$.
The hypotheses of [3, p. 412]. $|\tilde A|$, $|d^\eta\tilde A|$, $|d^\eta\tilde B|$ and their Hölder norms of exponent α_0 are $\le c_1p(L^k\varepsilon)$, and $\operatorname{dist}(\operatorname{supp}\tilde A,\partial\Omega)>2r(L^k\varepsilon)$. Put

$$\sigma := e_k(\|\tilde A\|_\infty+\|d\tilde A\|_\infty)\le 2c_1e_kp(t),\qquad \theta(\tilde B):=e_k\|d\tilde B\|_\infty\le c_1e_kp(t),\qquad \theta(\tilde A+\tilde B)\le 2c_1e_kp(t).$$

Smallness. We assume throughout

$$2c_1e_k^{1-\beta}p(t)\le\min\{c,1\},\tag{E}$$

with $\beta\in(0,1)$ and c the constants of [1, (2.23)]. It holds for $t\le t_0$, since $e_k^{1-\beta}p(t)=e^{1-\beta}t^{(1-\beta)/2}b_0(1+\ln t^{-1})^{\bar p}\to 0$; it is the "$e(L^k\varepsilon)$ sufficiently small" of [1, Prop. 2.1].
Depth hypotheses. The lemmas below are stated with explicit depth hypotheses on the supports of the vertex localisations and of $\tilde A$ (S1: $\operatorname{supp}\tilde A$ at depth $\ge R_0+2$ and vertex supports at depth $\ge R_0+3$ in Ω; S2: vertex supports at depth $\ge R_0+12$ in Ω_2; S3: scalar line ends at depth $\ge R_0$). At the uses these depths are supplied by the nesting of Bałaban's regions: the vertex localisations lie within 1 of Ω_1 or of $\operatorname{supp}\tilde A$ ([3, p. 412]: the A'-legs carry g_k with $\operatorname{supp}g_k\subset\Omega_1$; [3, p. 414]: the block vertices [3, (1.13)–(1.15)] are paired at a common y), $\operatorname{dist}(\operatorname{supp}\tilde A,\partial\Omega)>2r(t)\ge 2R\ge R_0+2$ under $R\ge 3R_0+15$, and at the domain changes of [2, pp. 579–580] the vertex supports have depth $\ge 5r(t)-9$ in the smaller domain. This supply is part of the reading of the step in appendix F; here the depths are hypotheses.
Inputs. Three propagator estimates are used, all of them statements about $G_k(X,A)$ and the fluctuation covariance at a single scale, proved in appendix F (group (A) there, theorem F.151) from [4]:

(i) **(P)** [1, Prop. 2.1] = the theorem [4, (1.9)–(1.12)]: for $\Omega=B^k(\Omega^{(k)})$ a union of large blocks and A regular in the sense of [1, (2.23)], $|(D^\eta_{A,\mu}G_k(\Omega,A)f)(x)|$ and $|(G_k(\Omega,A)f)(x)|$ are

$\le c_0 e^{-\delta_0 \operatorname{dist}(x,\operatorname{supp} f)}\|f\|_\infty$ for $\operatorname{dist}(x,\Omega^c) \ge R_0$ [1, (2.25)], and the Hölder quotient $|x-x'|^{-\alpha}|U(A(\Gamma_{x,x'}))(DGf)(x') - (DGf)(x)|$ obeys the same bound with $\operatorname{dist}(\{x,x'\},\operatorname{supp} f)$ in the exponent, for $\operatorname{dist}(\{x,x'\},\Omega^c) \ge R_0$ [1, (2.24)]. The constants δ_0, c_0, R_0 are independent of A, k, Ω and of the torus, and the depth restriction is omitted for $\Omega = T_\eta$ [1, pp. 610–611].

(ii) **(C)** [1, Prop. 2.3, (2.33)–(2.34)] at $L = 5$: $|C_\Lambda^{(k)}(X, A; y, y')| \le c_2 e^{-\theta_0|y-y'|_\infty}$ with no restriction on the points, and $(C_\Lambda^{(k)})^{-1} \ge \gamma_X > 0$; the constants depend on L (through γ_L) and not on k, X, ε or the torus.

(iii) **(H0)** [4, (1.8)] on every union X of unit blocks and on T_η: $H_X(A) - m^2t^2 \ge \gamma_0 := \frac{1}{4}\min\{2, a_*\}$, for A regular, which (E) supplies.

The norm [3, (1.32)] for kernels. For a kernel $Y(x,x')$ "extended in a natural way" to two variables [3, p. 420] we use the strongest natural reading: $\|Y\|_{1,\alpha}$ is the maximum over $a, b \in \{0,1\}$ of the suprema of $\|Y_{ab}\|$, $Y_{ab} := D^a Y D^{b*}$, and of the α-Hölder quotients in x (for $a = 1$), in x' (for $b = 1$) and of the mixed second difference (for $ab = 1$), with partners $0 < |x - \tilde{x}| \le 1$ and $\tilde{B}$-transport along shortest contours. Quotients over larger distances are at most twice a supremum, and the summed form of [3, (1.32)] is at most 16 times this maximum.

H.1.2. *S3: the slice decomposition* [3, (2.6)] *and the slice bounds* [3, (2.10)–(2.12)]. [3, p. 424] decomposes every scalar line of G' by

(86)
$$G_k(\Omega,\tilde{B}) = C^{(0),\eta}(\Omega,\tilde{B}) + \sum_{j=1}^{k-1} a_j^2 (L^j\eta)^{-4} G_j^\eta(\Omega,\tilde{B}) Q_j^*(\tilde{B}) C^{(j),L^j\eta}(\Omega,\tilde{B}) Q_j(\tilde{B}) G_j^\eta(\Omega,\tilde{B}) = \sum_{j=0}^{k-1} G_{(j)}^\eta(\Omega,\tilde{B}),$$

"and the similar equality for the vector field propagator", and [3, p. 426] asserts the slice bounds [3, (2.10)–(2.12)] "by rescaling from the η-lattice to the L^{-j}-lattice and application of Propositions I.2.1 and I.2.3". The consumed form is [3, (2.13)–(2.14)]: every line l of G' carries one slice index j_l and contributes a factor $(L^{j_l}\eta)^{-(d-2)}$, times $(L^{j_l}\eta)^{-1}$ per derivative and $L^{j_l}\eta$ per contour leg, and an exponential with rate $\delta_1(L^{j_l}\eta)^{-1}$.

Definitions in scale-k units. For $1 \le j \le k-1$ and $s := s_j$: $(Q_j^\eta(A)\phi)(z) := s^{-d}\sum_{x\in B_s(z)} \eta^d U(A(\Gamma_{z,x}^{(j)}))\phi(x)$, the average over the s-block $B_s(z)$; it acts on $\ell^2(\eta^d)$, its range carries the weight s^d, and $Q_j^\eta Q_j^{\eta *} = 1$. $P_j^\eta := Q_j^{\eta *} Q_j^\eta$. $H_0 := -\Delta_{A,\Omega}^{\eta,N} + m^2t^2$, $G_j^\eta := (H_0 + a_j s^{-2} P_j^\eta)^{-1}$ (this is [1, (2.20)] with $\varepsilon \to \eta$, $L^k\varepsilon \to s$), $\Delta^{(j),s} := a_j s^{-2} - a_j^2 s^{-4} Q_j G_j^\eta Q_j^*$ [1, (2.21)], $C^{(j),s} := (a(Ls)^{-2}P^{(j)} + \Delta^{(j),s})^{-1}$ [1, (2.30)], where $P^{(j)}$ is the L-block projection on the s-lattice. $C^{(0),\eta} := G_1^\eta$ (this is [1, (2.30)] at $k = 0$ with [1, (2.17)]: $C^{(0)} = (-\Delta + m^2 + a(L\varepsilon)^{-2}P)^{-1} = G_1$, the identity "$G_1 = C^{(0)}$" of [1, p. 612]). Finally $G_{(0)}^\eta := C^{(0),\eta}$ and $G_{(j)}^\eta := a_j^2 s^{-4} G_j^\eta Q_j^* C^{(j),s} Q_j G_j^\eta$.

Lemma H.1 (composite contours)**.** $Q_{j+1}^\eta(A) = Q^{(j)}(A) Q_j^\eta(A)$, *with* $Q^{(j)}$ *the* L*-block average on the* s_j*-lattice. Hence* $P_{j+1}^\eta = Q_j^{\eta *} P^{(j)} Q_j^\eta$.

Proof. By [1, (2.2)], $\Gamma_{z,x}^{(j+1)} = \Gamma_{z,x_j} \cup \Gamma_{x_j,x}^{(j)}$ with $x \in B^j(x_j)$, $x_j \in B(z)$; U is a homomorphism, so $U(A(\Gamma_{z,x}^{(j+1)})) = U(A(\Gamma_{z,x_j}))U(A(\Gamma_{x_j,x}^{(j)}))$; the sum over $x \in B^{j+1}(z)$ splits as a sum over $x_j \in B(z)$ and $x \in B^j(x_j)$, and $(Ls)^{-d} = L^{-d}s^{-d}$. The second claim is $Q^{(j)*}Q^{(j)} = P^{(j)}$. □

Lemma H.2 (one renormalisation step; [1, (2.42)] at scale s)**.** *For* $1 \le j \le k-1$ *every inverse below exists and* $G_{j+1}^\eta = G_j^\eta + a_j^2 s^{-4} G_j^\eta Q_j^* C^{(j),s} Q_j G_j^\eta$.

Proof. On $\ell^2(\Omega^{(j)}, s^d) \oplus \ell^2(\Omega, \eta^d)$ let $\mathsf{M} := \begin{pmatrix} a_j s^{-2} + a(Ls)^{-2}P^{(j)} & X \\ X^* & H_0 + a_j s^{-2} P_j \end{pmatrix}$, $X := -a_j s^{-2} Q_j$.

Its quadratic form is $a_j s^{-2}\|\psi - Q_j\phi\|^2 + a(Ls)^{-2}\langle\psi, P^{(j)}\psi\rangle + \langle\phi, H_0\phi\rangle$, since $\langle Q_j\phi, Q_j\phi\rangle = \langle\phi, P_j\phi\rangle$. The (ϕ,ϕ) block $H_0 + a_j s^{-2} P_j = (G_j^\eta)^{-1}$ is $\ge \gamma_0 s^{-2}$ by (H0) after the rescaling of lemma H.4; its Schur complement is $a(Ls)^{-2}P^{(j)} + \Delta^{(j),s} = (C^{(j),s})^{-1} \ge \gamma_X s^{-2} > 0$ by (C), rescaled. So $\mathsf{M} > 0$. The other Schur complement: on $\ell^2(\Omega^{(j)})$, $(a_j s^{-2} + a(Ls)^{-2}P^{(j)})^{-1} =$

$s^2[a_j^{-1}(1-P^{(j)})+(a_j+aL^{-2})^{-1}P^{(j)}]$, hence

$$a_j s^{-2}P_j - a_j^2 s^{-4}Q_j^*(a_j s^{-2}+a(Ls)^{-2}P^{(j)})^{-1}Q_j = s^{-2}Q_j^*P^{(j)}Q_j\Big(a_j - \frac{a_j^2}{a_j+aL^{-2}}\Big) = a_{j+1}(Ls)^{-2}P_{j+1},$$

by the recursion $a_j - a_j^2/(a_j+aL^{-2}) = aa_jL^{-2}/(a_j+aL^{-2}) = a_{j+1}L^{-2}$ [1, (2.13)], lemma H.1 and $s_{j+1} = Ls$; so this complement is $(G_{j+1}^\eta)^{-1}$. The (ϕ,ϕ) entry of M^{-1} is both the inverse of this complement and $D^{-1}+D^{-1}X^*(A-XD^{-1}X^*)^{-1}XD^{-1}$ with D the (ϕ,ϕ) block and $A - XD^{-1}X^* = (C^{(j),s})^{-1}$, which is the claim. □

Corollary H.3 ((86), scalar and vector)**.** *Iterating lemma H.2 from $j=1$ to $k-1$ gives $G_k^\eta = \sum_{j=0}^{k-1} G_{(j)}^\eta$, and $G_k^\eta = G_k(\Omega,\tilde B)$ since $s_k = 1$ [1, (2.22)]. The vector propagator is $G_k(T_\eta,0)\otimes \mathbf{1}_d$ with mass $\mu_0^2 t^2$ ([1, p. 608], the Feynman-gauge action [1, (1.11)] being componentwise), so it has the same decomposition, with slices $G_{(j)}^{\mathrm{vec},\eta}$. These are the covariances of the independent slice fields $A'^{(j),\eta} = a_j(L^j\eta)^{-2}G_j^\eta Q_j^* A'_j$ of [3, (1.2)], which is [3, p. 414]: "the field A' has the covariance G_k".*

Lemma H.4 (rescaling)**.** *Fix $0\le j\le k-1$, $s := s_j$, and let $(S_sf)(x) := f(x/s)$ map functions on the L^{-j}-lattice (scale-j units) to functions on the η-lattice. Give the rescaled background $\tilde B^{[j]}$ the same bond phases, $e_jL^{-j}\tilde B^{[j]}_{b/s} := e_k\eta\tilde B_b$, so $U^{[j]}_{b/s} = U_b$ and $\tilde B^{[j]}(y) = s^{1/2}\tilde B(sy)$ (as $e_k/e_j = s^{-1/2}$); rescale $\Omega^{[j]} := \Omega/s$ and $m^2t_j^2 = s^2m^2t^2$. Then $D^\eta S_s = s^{-1}S_sD^{(j)}$, $-\Delta^\eta S_s = s^{-2}S_s(-\Delta^{(j)})$, $Q_j^\eta S_s = TQ_j$, $Q_j^{\eta *}T = S_sQ_j^*$ (T the corresponding map on coarse functions, Q_j Bałaban's average over unit blocks at scale j), and*

$$G_j^\eta(x,x') = s^{2-d}G_j(x/s,x'/s),\qquad G_{(j)}^\eta(x,x') = s^{2-d}K_j(x/s,x'/s),\quad K_j := a_j^2G_jQ_j^*C^{(j)}Q_jG_j\ (K_0 := C^{(0)}),$$

with $G_j = G_j(\Omega^{[j]},\tilde B^{[j]})$ [1, (2.22)] and $C^{(j)} = C^{(j)}(\Omega^{[j]},\tilde B^{[j]})$ [1, (2.31)]; kernels with respect to η^d on the left and L^{-jd} on the right. $\Omega^{[j]}$ is again a union of large blocks of side M at scale j (the pavings nest), and depths scale as $\rho_{\Omega^{[j]}}(x/s) = s^{-1}\rho_\Omega(x)$.

Proof. Each identity is a one-line computation from $\eta = sL^{-j}$, equal bond phases and $\langle S_sf,S_sh\rangle_{\eta^d} = s^d\langle f,h\rangle_{L^{-jd}}$. Composition: $a_j^2s^{-4}(s^2SG_jS^{-1})(SQ_j^*T^{-1})(s^2TC^{(j)}T^{-1})(TQ_jS^{-1})(s^2SG_jS^{-1}) = s^2SK_jS^{-1}$, and the kernel of s^2SKS^{-1} with respect to η^d is $s^{2-d}K(x/s,x'/s)$. □

Lemma H.5 (regularity transfers to every slice)**.** *Under (E), $\tilde B^{[j]}$ and $(\tilde A+\tilde B)^{[j]}$ satisfy [1, (2.23)] at index j, $|d^{L^{-j}}A^{[j]}|\le ce_j^{\beta-1}$, and the coercivity hypothesis of (H0) at index j.*

Proof. $d^{L^{-j}}\tilde B^{[j]}(y) = s^{3/2}(d^\eta\tilde B)(sy)$, so $|d\tilde B^{[j]}|\le s^{3/2}c_1p(t)$. With $e_j = e_ks^{1/2}$, condition [1, (2.23)] reads $s^{3/2}c_1p(t)\le ce_k^{\beta-1}s^{(\beta-1)/2}$, i.e. $c_1e_k^{1-\beta}p(t)s^{2-\beta/2}\le c$, which (E) gives since $s\le 1$. For $\tilde A+\tilde B$ use $2c_1$. □

Lemma H.6 (unit-scale slice kernels)**.** *Let $1\le j\le k-1$, $n,n'\in\{0,1\}$, $0\le\alpha<1$, $\delta_* := \min\{\delta_0,\theta_0\}$, $\delta_1 := \delta_*/2$. On the scale-j lattice, for y,y' with $\rho_{\Omega^{[j]}}\ge R_0$ at y, y' and at the Hölder partners:*

$$|D_y^nD_{y'}^{n'}K_j(y,y')|\le c_3e^{-\delta_1|y-y'|_\infty},\qquad |y_1-y_2|^{-\alpha}\big|U(\tilde B^{[j]}(\Gamma_{y_1,y_2}))D_{y_2}D_{y'}^{n'}K_j(y_2,y') - D_{y_1}D_{y'}^{n'}K_j(y_1,y')\big|\le c_3e^{-\delta_1|}$$

the same in y' and for the mixed second difference, with $c_3 := 2a^2c_0^2c_2e^{2\delta_0}S(\delta_1)^2$, $S(\delta) := \sum_{n\in\mathbb{Z}^d}e^{-\delta(|n|_\infty-1)_+}$. For $j=0$ the same holds for $K_0 = C^{(0)}$ with no depth restriction and constants from the unit-lattice coercivity and decay of (C).

Proof. Let δ_w be the unweighted coarse delta at $w\in\mathbb{Z}^d$; $Q_j^*\delta_w = U(\tilde B(\Gamma_{w,\cdot}))^T\mathbf{1}_{B(w)}$ has sup norm 1 and support $B(w)$; G_j is real symmetric. So $K_j(y,y') = a_j^2\sum_{w,w'}(G_jQ_j^*\delta_w)(y)\,C^{(j)}(w,w')\,[(G_jQ_j^*\delta_{w'})(y')]^T$. (P) at index j with $f = Q_j^*\delta_w$ and lemma H.5 gives $|D^n(G_jQ_j^*\delta_w)(y)|\le c_0e^{-\delta_0\,\mathrm{dist}_\infty(y,B(w))}\le c_0e^{\delta_0}e^{-\delta_0|y-w|_\infty}$, and [1, (2.24)] the Hölder quotient with $\mathrm{dist}(\{y_1,y_2\},B(w))$ in the exponent; the same at y'. (C) gives $|C^{(j)}(w,w')|\le c_2e^{-\theta_0|w-w'|_\infty}$. Since $|y-w|+|w-w'|+|w'-y'|\ge|y-y'|$,

half of the exponents pays $e^{-\delta_1|y-y'|}$ and the other half the two lattice sums, each $\le S(\delta_1)$; mixed derivatives and quotients act on different outer factors. For $j=0$, $C^{(0)}=(-\Delta_A^{(0)}+m^2\varepsilon^2+aL^{-2}P(A))^{-1}$ on the unit lattice is the fluctuation covariance at $k=0$, so (C) gives $|C^{(0)}(y,y')|\le c_2 e^{-\theta_0|y-y'|_\infty}$ with no depth restriction; a derivative at unit spacing is a difference of two kernel values (U orthogonal), and a Hölder quotient at unit spacing has $|y_1-y_2|\ge 1$. □

Lemma H.7 ([3, (2.10)–(2.11)] in scale-k units)**.** *For $0\le j\le k-1$, $n,n'\in\{0,1\}$, $x,x'\in\Omega$ with $\rho_\Omega\ge R_0$ at x, x' and at the Hölder partners (no restriction for $j=0$, none on $\Omega=T_\eta$):*

$$|D_x^n D_{x'}^{n'} G^\eta_{(j)}(x,x')|\le c_3 s_j^{-d+2-n-n'} e^{-\delta_1 s_j^{-1}|x-x'|_\infty},\quad |x_1-x_2|^{-\alpha}\big|U(\tilde B(\Gamma_{x_1,x_2}))(DG^\eta_{(j)})(x_2,x')-(DG^\eta_{(j)})(x_1,x')\big|\le$$

and the same with the quotient in x' and for the mixed second difference (power $s_j^{-d-2\alpha}$). The constant c_3 is independent of ε, k, j, Ω, $\tilde B$ and the torus side.

Proof. Lemma H.4 then lemma H.6: $D^\eta=s^{-1}D^{(j)}$, $|x_1-x_2|^{-\alpha}=s^{-\alpha}|y_1-y_2|^{-\alpha}$, contours and phases correspond. The depth needed at scale j is R_0 there, i.e. $R_0 s_j\le R_0/5$ in scale-k units, so $\rho_\Omega\ge R_0$ suffices for every j. The constants of (P) and (C) do not depend on the index nor on the torus side. □

Lemma H.8 (vector slices and [3, (2.12)])**.** (a) *The vector slices $G^{\mathrm{vec},\eta}_{(j)}$ obey lemma H.7 at every point of T_η, with no depth condition, including one derivative at each end.* (b) *For a contour leg put $G^\eta_{(j)}(\Gamma,b):=\sum_{b'\in\Gamma}\eta\, G^{\mathrm{vec},\eta}_{(j)}(b',b)$ with $\Gamma=\Gamma^{(j+1)}_{x_{j+1},x}$ [1, (2.2)], [3, (1.3)]. Then for $n\in\{0,1\}$*

$$|D_b^n G^\eta_{(j)}(\Gamma^{(j+1)}_{x_{j+1},x},b)|\le c_3\, dL\, e^{\delta_1 L}\, s_j^{-d+3-n} e^{-\delta_1 s_j^{-1}|x-b|_\infty}.$$

(c) *With contour legs at both ends the power is s_j^{-d+4} and the constant $c_3(dLe^{\delta_1 L})^2$. These imply* [3, (2.12)]*, since* $\mathrm{dist}(B^j(x),b)\le|x-b|_\infty$*; at $L=5$ the constant is $15e^{5\delta_1}c_3$ (resp. its square).*

Proof. (a) is lemma H.7 at $A=0$, $N=d$, $X=T_\eta$, μ_0 for m. (b) $\Gamma^{(j+1)}_{x_{j+1},x}=\Gamma_{x_{j+1},x_j}\cup\cdots\cup\Gamma_{x_1,x}$; the piece Γ_{x_{i+1},x_i} lies on the $L^i\eta$-lattice and has d straight segments of length $\le(L-1)L^i\eta$ [1, (2.1)], so the total length is $\le d(L-1)\sum_{i\le j}L^i\eta\le dLs_j$; every point of the contour lies in $B^{j+1}(x_{j+1})$, of sup-diameter $\le Ls_j$, so $|z-b|_\infty\ge|x-b|_\infty-Ls_j$ on Γ. Bound each term by (a) and sum: the factor η per bond times the length gives dLs_j, the "additional factor $L^j\eta$". (c) is the same at both ends. Since the slice fields are independent [3, p. 414], a contour leg $A'^{(j),\eta}(\Gamma^{(j+1)})$ contracts only with slice j: the contour index and the line index coincide, which is how [3, (2.14)] books the factor at the line's scale. □

Remark H.9. (i) [3, (2.12)] prints $\mathrm{dist}(B^j(x),b)$ in the exponent; the bound above has the stronger $|x-b|_\infty$ at the cost of $e^{\delta_1 L}$, and since contour points lie up to Ls_j from x the printed form also needs a constant of this type inside its $O(1)$. (ii) The $j=0$ slice $C^{(0),\eta}=G^\eta_1$ is a unit-lattice inverse with an L-block pin; it needs neither (P) nor a depth condition. (iii) Scalar lines need depth $\ge R_0$ at both evaluation points and at the Hölder partners; vector lines and contour legs live on the torus and need none.

H.1.3. *Two tools: weighted ℓ^2 bounds and summed pointwise bounds.* Conventions. X is a union of unit blocks with Neumann conditions, or $X=T_\eta$ (for the pointwise bounds, a union of large blocks, as (P) requires). Bond functions are weighted at b_-. ρ is any function which is 1-Lipschitz for $|\cdot|_\infty$, so $|\rho(b_+)-\rho(b_-)|\le\eta$ on every bond and $|\rho(x)-\rho(x')|\le 1$ inside a unit block. For an operator Y put $Y^\beta:=e^{\beta\rho}Ye^{-\beta\rho}$.

Lemma H.10 (real-part Combes–Thomas bound)**.** *Let $Y=D^*D+v+aP$ on $\ell^2(X,\eta^d;\mathbb{R}^N)$, with arbitrary bond set and orthogonal transporters, $v\ge 0$ a multiplication operator, $a\ge 0$, $P\ge 0$ with $P(x,x')=0$ unless x,x' lie in one unit block and $\sup_x\sum_{x'}\eta^d\|P(x,x')\|\le 1$, and $Y\ge\gamma>0$. Let $0\le\beta\le 1$ with $c(\beta):=3\beta^2\cosh\beta+a(e^\beta-1)\le\gamma/2$. Then for every F the solution w of $Y^\beta w=F$ exists, $\|w\|\le 2\gamma^{-1}\|F\|$ and $\|Dw\|^2+\langle w,vw\rangle\le 4\gamma^{-1}\|F\|^2$.*

Proof. Negative β is the case $-\rho$; all operators are real, so F and w may be taken real. For a bond b put $\tau := \beta(\rho(b_+) - \rho(b_-))$, $|\tau| \le \beta\eta$, and $z := w(b_-) \cdot U_b w(b_+)$. The contribution of b to $\langle w, e^{\beta\rho} D^* D e^{-\beta\rho} w\rangle = \langle De^{\beta\rho}w, De^{-\beta\rho}w\rangle$ is $\eta^{d-2}[|w(b_+)|^2 + |w(b_-)|^2 - 2\cosh(\tau)z]$; at $\tau = 0$ this is $\eta^d |Dw(b)|^2$, and the difference is at most $(\cosh\tau - 1)(|w(b_+)|^2 + |w(b_-)|^2)\eta^{d-2} \le \frac{1}{2}\beta^2\cosh(\beta)\eta^d(|w(b_+)|^2 + |w(b_-)|^2)$. Each site lies in at most $2d = 6$ bonds, so the bond part is $\ge \|Dw\|^2 - 3\beta^2\cosh(\beta)\|w\|^2$. v commutes with $e^{\beta\rho}$. $P^\beta - P$ has kernel $(e^{\beta(\rho(x)-\rho(x'))} - 1)P(x,x')$ with $|\rho(x) - \rho(x')| \le 1$, so $\|P^\beta - P\| \le e^\beta - 1$ by Schur's test. Hence $\langle w, Y^\beta w\rangle \ge \langle w, Yw\rangle - c(\beta)\|w\|^2 \ge \frac{\gamma}{2}\|w\|^2$, and $\langle w, Y^\beta w\rangle = \langle w, F\rangle \le \|w\|\|F\|$ gives the first bound; then $\|Dw\|^2 + \langle w, vw\rangle \le \langle w, Yw\rangle \le \|w\|\|F\| + \frac{\gamma}{2}\|w\|^2 \le 4\gamma^{-1}\|F\|^2$. □

Lemma H.11 (the weighted bounds (W))**.** *Assume (H0) for A on X, let $m^2t^2 \ge 0$ and $G := G_k(X, A)$. Fix $0 < \beta_* \le 1$ with $c(\beta_*) \le \gamma_0/4$ (at $a = a_k$) and put $c' := \sqrt{d}\,\beta_* e^{\beta_*}$. Then for every 1-Lipschitz ρ and $|\beta| \le \beta_*$: $\|G^\beta\| \le 2/\gamma_0$, and $\|(DG)^\beta\|$, $\|(GD^*)^\beta\|$, $\|(DGD^*)^\beta\|$ are $\le K_W := \max\{2\gamma_0^{-1/2} + 2c'\gamma_0^{-1}, (1 + c'^2)/\min\{\frac{1}{2}, \gamma_0/4\}\}$. The bounds are uniform in η, k, ε, $m^2 \ge 0$, the shape of X (wrapping allowed) and the boundary condition. With $\mathfrak{D} := (1, D_1, D_2, D_3)$, $\|(\mathfrak{D}G\mathfrak{D}^*)^\beta\| \le c_W := 2\gamma_0^{-1} + 3K_W$.*

Proof. Lemma H.10 with $Y = H_X(A)$, $v = m^2t^2$, $P = P_k(A)$, $a = a_k$, $\gamma = \gamma_0$ gives $\|G^\beta\| \le 2/\gamma_0$ and $\|DG^\beta\| \le 2\gamma_0^{-1/2}$. Next $(DG)^\beta = D^\beta G^\beta$ with $D^\beta = D + \beta K_\beta$, $(\beta K_\beta\phi)(b) = \eta^{-1}(e^{-\beta(\rho(b_+)-\rho(b_-))} - 1)U_b\phi(b_+)$; the coefficient is at most βe^β in modulus and $\phi \mapsto (\phi(b_+))_b$ has norm $\sqrt{d}$, so $\|\beta K_\beta\| \le c'$. G is real symmetric, so $(GD^*)^\beta = ((DG)^{-\beta})^*$. For $(DGD^*)^\beta$ let $w := G^\beta(D^*)^\beta g$; the adjoint of $(D^*)^\beta$ is $D^{-\beta}$. By the argument of lemma H.10 with $H \ge D^*D$ and $H \ge \gamma_0$, $\mathrm{Re}\langle w, H^\beta w\rangle \ge \frac{1}{2}\|Dw\|^2 + \frac{\gamma_0}{4}\|w\|^2$, while $\mathrm{Re}\langle w, H^\beta w\rangle = \mathrm{Re}\langle D^{-\beta}w, g\rangle \le (\|Dw\| + c'\|w\|)\|g\|$; with $s^2 := \|Dw\|^2 + \|w\|^2$ this gives $\min\{\frac{1}{2}, \gamma_0/4\}s \le (1 + c'^2)^{1/2}\|g\|$ and $\|D^\beta w\| \le (1 + c'^2)^{1/2}s$. Only $|\rho(b_+) - \rho(b_-)| \le \eta$ and the block range of P_k were used, and (H0) holds on every union of unit blocks and on T_η. □

At $\beta = 0$ and $\rho = \mathrm{dist}(\cdot, \mathrm{supp}\, f')$ this is [4, Cor. 2.3, (2.30)] for DGD^*, with no restriction on the supports.

Lemma H.12 (slice sums at $L = 5$)**.** *Let $p > 0$, $\delta > 0$, $z \ge 0$, $r := \max(z, \eta)$. Then $\sum_{j=0}^{k-1} s_j^{-p} e^{-\delta z/s_j} \le c_\Sigma(p, \delta)\, r^{-p} e^{-\delta z/2}$ with $c_\Sigma(p, \delta) := (1 - 5^{-p})^{-1} + \sup_{1\le u\le 5}\sum_{m\ge 0}(u5^m)^p e^{-\delta u 5^m/2}$.*

Proof. Slices with $s_j \ge r$ contribute at most $r^{-p}\sum_m 5^{-mp}e^{-\delta z}$. Slices with $s_j < r$ occur only when $r = z$; then $u_j := z/s_j > 1$ has the form $u5^m$ with $u \in (1, 5]$, $m \ge 0$, each term is $z^{-p}u_j^p e^{-\delta u_j}$, and $e^{-\delta u_j} \le e^{-\delta u_j/2}e^{-\delta z/2}$ because $u_j \ge z$. □

Kernel classes. For $0 < \kappa < 3$ put $K_\kappa^\delta(z) := \max(|z|_\infty, \eta)^{\kappa-3}e^{-\delta|z|_\infty}$; $K_3^\delta(z) := (1 + \log_+(1/\max(|z|_\infty, \eta)))e^{-\delta|z|_\infty}$; $K_\kappa^\delta(z) := e^{-\delta|z|_\infty}$ for $\kappa > 3$.

Lemma H.13 (the summed pointwise bounds (K))**.** *Let X be a union of large blocks or T_η, $A \in \{\tilde{B}, \tilde{A} + \tilde{B}\}$ (regular by (E)), $G := G_k(X, A)$, $0 < \alpha < 1$, $\delta_K := \delta_1/2$. For x, w with $\rho_X(x), \rho_X(w) \ge R_0$ (on T_η: all x, w):*

(K0) $\|G(x, w)\| \le c_K K_2^{\delta_K}(x - w)$;

(K1) $\|D_x G(x, w)\| \le c_K K_1^{\delta_K}(x - w)$, *and* $\|(GD^*)(w, x)\|$ *obeys the same bound;*

(K2) *for $|x - \tilde{x}| \le 1$ with $\rho_X(\tilde{x}) \ge R_0$: $\|U(A(\Gamma_{\tilde{x},x}))D_x G(x, w) - D_{\tilde{x}}G(\tilde{x}, w)\| \le c_K(\alpha)|x - \tilde{x}|^\alpha\big(K_{1-\alpha}^{\delta_K}(x - w) + K_{1-\alpha}^{\delta_K}(\tilde{x} - w)\big)$;*

with $c_K := c_3 \max\{c_\Sigma(1, \delta_1), c_\Sigma(2, \delta_1)\}$ and $c_K(\alpha) := 2c_3c_\Sigma(2 + \alpha, \delta_1)$.

Proof. $G = \sum_j G_{(j)}^\eta$ by corollary H.3. (K0), (K1): sum lemma H.7 ($d = 3$: powers s^{-1}, s^{-2}) with lemma H.12 at $p = 1, 2$. (K2) slice by slice: for $s_j \ge |x - \tilde{x}|_\infty$ use the Hölder clause of lemma H.7 with $\mathrm{dist}(\{x, \tilde{x}\}, w) \le \min(|x - w|, |\tilde{x} - w|)$; for $s_j < |x - \tilde{x}|_\infty$ bound the two terms separately (U orthogonal) with $|x - \tilde{x}|^{-\alpha} \le s_j^{-\alpha}$; in both cases the slice quotient is $\le c_3 s_j^{-2-\alpha}(e^{-\delta_1|x-w|/s_j} + e^{-\delta_1|\tilde{x}-w|/s_j})$, and lemma H.12 at $p = 2 + \alpha$ sums it. The column bound: G is real symmetric, so $(GD^*)(w, x) = (D_x G(x, w))^T$. □

The mixed kernel $D_x G D_w^*$ has s^{-3} per slice, which is not summable uniformly in η at $|x-w| \lesssim \eta$. It never enters pointwise below: it enters only through lemma H.11.

H.1.4. *S1: the kernel* [3, (1.16)] *as an external field.* [3, p. 414] expands $G_k(\Omega, \tilde{A}+\tilde{B})$ by [1, (3.44)–(3.45)] and treats the last term,

$$K_{n,n'} := [G_k(\Omega,\tilde{B})V_k(\tilde{A},\tilde{B})]^n\, G_k(\Omega,\tilde{A}+\tilde{B})\, [V_k(\tilde{A},\tilde{B})G_k(\Omega,\tilde{B})]^{n'}, \tag{87}$$

as an external field: "for n, n' sufficiently large, a kernel of the operator (1.16) is a sufficiently regular function of both variables ... uniformly bounded by $O(1)(e(L^k\varepsilon)^{1-\alpha})^{n+n'}$". The consumed form is the second alternative of [3, (2.5)]: $\|hK_{n,n'}h'\|_{1,\alpha} \le O((e(L^k\varepsilon)p(L^k\varepsilon))^{n+n'})e^{-\delta_0 \operatorname{dist}(\operatorname{supp} h, \operatorname{supp} h')}$ for the localisation functions h, h' of the two end vertices. Here h, h' are the smooth localisers of [3, (1.33), p. 420] ($h = 1$ on $\square(v)$, $h = 0$ outside a neighbourhood, $|dh|, |ddh| \le c_h$), not the sharp cubes of the vertex localisations. At every use $m := n + n' \ge \bar{n} + 1 \ge 14$, since [1, (3.45)] stops at $(n, n') = (\bar{n}+1, 0)$ and $\bar{n} > 12$ [3, p. 421]; the proof needs only $m \ge 6$ (or $m \ge 4$ with $\alpha_0 < \frac{1}{2}$).

Write $G := G_k(\Omega, \tilde{B})$, $G' := G_k(\Omega, \tilde{A}+\tilde{B})$, $V := G^{-1} - G'^{-1} = V_k(\tilde{A}, \tilde{B})$ (the resolvent identity), so $K_{n,n'} = L_0 V L_1 V \cdots V L_m$ with $L_n = G'$ and $L_i = G$ for $i \ne n$.

Lemma H.14 (the vertex operator)**.** *Put $D := D_{\tilde{B}}$, $\phi_\mu(x) := \eta^{-1}(1 - U(\tilde{A}_{x,\mu})^{-1})$, $\psi_\mu := \phi_\mu^*\phi_\mu$, $\Pi := P_k(\tilde{A}+\tilde{B}) - P_k(\tilde{B})$, $\mathfrak{D} := (1, D_1, D_2, D_3)$, $\operatorname{supp}\tilde{A}$ the set of endpoints of bonds with $\tilde{A}_b \ne 0$, and $\mathcal{U} := \{y : \operatorname{dist}_\infty(y, \operatorname{supp}\tilde{A}) \le 1\}$.*

(i) *$V = -[\sum_\mu (D_\mu^*\phi_\mu + \phi_\mu^* D_\mu + \psi_\mu) + a_k\Pi] = -\mathfrak{D}^*\mathfrak{E}\mathfrak{D}$, with $\mathfrak{E}_{00} = \sum_\mu \psi_\mu + a_k\Pi$, $\mathfrak{E}_{0\mu} = \phi_\mu^*$, $\mathfrak{E}_{\mu 0} = \phi_\mu$, 0 otherwise.*

(ii) *(Right-pushed form.) $V = -(R_0 + \sum_\mu R_{1,\mu} D_\mu) =: -\mathfrak{r}\mathfrak{D}$, where $(R_{1,\mu}f)(x) := \phi_\mu(x)^* f(x) - \phi_\mu(x-\eta e_\mu) U(\tilde{B}_{x-\eta e_\mu,\mu})^{-1} f(x-\eta e_\mu)$ and $R_0 := \sum_\mu(\psi_\mu - \mathsf{m}_\mu) + a_k\Pi$, m_μ the multiplication by $(d_\mu^\eta \phi_\mu)(x - \eta e_\mu)$.*

(iii) *(Left-pushed form.) $V = V^* = -\mathfrak{D}^*\mathfrak{r}^*$.*

(iv) *Let $\sigma \le 1$. Then $|\phi_\mu|, |d_\mu\phi_\mu| \le \sigma$, $|\psi_\mu| \le \sigma^2$, $|\Pi(y,y')| \le 2d\sigma$ and $\sum_{y'} \eta^d |\Pi(y,y')| \le 2d\sigma$; hence $\|\mathfrak{r}\|, \|\mathfrak{E}\| \le c_V\sigma$ on $\ell^p(\eta^d)$, $p \in \{1, 2, \infty\}$, $c_V := 4d + 2da + 1$. The kernels have sup-range ≤ 1, and $V = \mathbf{1}_\mathcal{U} V \mathbf{1}_\mathcal{U}$, $\mathfrak{r} = \mathbf{1}_\mathcal{U}\mathfrak{r}\mathbf{1}_\mathcal{U}$, $\mathfrak{E} = \mathbf{1}_\mathcal{U}\mathfrak{E}\mathbf{1}_\mathcal{U}$.*

On $X = \Omega$ with Neumann conditions, (i)–(iv) hold if $\operatorname{dist}_\infty(\operatorname{supp}\tilde{A}, \Omega^c) \ge 2$.

Proof. $U(\tilde{A}+\tilde{B}) = U(\tilde{A})U(\tilde{B})$, so $D_{\tilde{A}+\tilde{B},\mu} f = U(\tilde{A})(D_\mu f + \phi_\mu f)$, and since U is pointwise orthogonal, $-\Delta_{\tilde{A}+\tilde{B}} = \sum_\mu (D_\mu + \phi_\mu)^*(D_\mu + \phi_\mu)$ on the same bond set; the masses cancel and the pins differ by Π: this is (i). (ii): $D_\mu^*(\phi_\mu f)(x) = -\phi_\mu(x-\eta e_\mu) U^{-1}_{x-\eta e_\mu,\mu}(D_\mu f)(x-\eta e_\mu) - (d_\mu\phi_\mu)(x-\eta e_\mu) f(x)$ (the computation of [4, (2.32)], redone in lemma H.22). (iii): both inverses are real symmetric. (iv): $|e^{q\theta} - e^{q\theta'}| \le |\theta - \theta'|$ gives $|\phi| \le e_k|\tilde{A}|$, $|d_\mu\phi_\mu| \le e_k|d\tilde{A}|$; for y, y' in one block $B(z)$, $\Pi(y,y') = U(\tilde{B}\cdots)^T[U(\tilde{A}(\Gamma_{z,y}))^T U(\tilde{A}(\Gamma_{z,y'})) - 1]U(\tilde{B}\cdots)$ with contours of length $\le d$, so $|\Pi| \le 2de_k\|\tilde{A}\|$; the Laplacian part lives on bonds with $\tilde{A}_b \ne 0$ plus one η-shift, Π on unit blocks whose contours meet $\operatorname{supp}\tilde{A}$, of sup-diameter $1 - \eta$. On Ω every such bond and block lies in Ω once the distance is ≥ 2. □

Lemma H.15 (convolution and smearing; $d = 3$, sup metric)**.** (a) *For $\kappa_1, \kappa_2 \in (0, 3)$ and $0 < \delta' < \delta \le 1$: $\sum_w \eta^d K^\delta_{\kappa_1}(z-w) K^\delta_{\kappa_2}(w) \le c(\kappa_1,\kappa_2)(1 + (\delta-\delta')^{-3}) K^{\delta'}_{\kappa_1+\kappa_2}(z)$.* (b) *$K^\delta_\kappa \le c(1 + (\delta - \delta'')^{-1}) K^{\delta''}_{\kappa'}$ for $\kappa \ge \kappa'$, $\delta'' < \delta$.* (c) *If R is a multiplication, a one-bond shift, or a kernel supported in single unit blocks, with $|R(z,z')| \le r_R$ and column sums $\le r_R$, then $(|R|K^\delta_\kappa(\cdot - y))(z) \le c(\kappa)e^{2\delta} r_R K^\delta_\kappa(z-y)$.*

Proof. (a) $e^{-\delta|z-w|-\delta|w|} \le e^{-\delta'|z|} e^{-(\delta-\delta')(|z-w|+|w|)}$. Put $r := \max(|z|_\infty, \eta)$ and split into $|w| \le r/2$, $|z-w| \le r/2$ and the rest. In the two inner regions one factor is $\le cr^{\kappa_i - 3}$ and the other is summed with $\sum_{|w|\le R} \eta^d \max(|w|,\eta)^{\kappa-3} \le c(\kappa)R^\kappa$ ($R \ge \eta$); the outer region is $\le c\int_{r/2}^\infty u^{\kappa_1+\kappa_2-4} e^{-(\delta-\delta')u} du$, which gives the stated classes (with the conventions at $\kappa_1 + \kappa_2 \ge 3$). (b) For $|z| \le 1$ the larger exponent gives the smaller kernel; for $|z| > 1$ the power is absorbed by

$e^{-(\delta-\delta'')|z|}$. (c) For $|z-y|_\infty \geq 2$ every z' with $R(z,z') \neq 0$ has $\max(|z'-y|,\eta) \geq \frac{1}{2}\max(|z-y|,\eta)$; for $|z-y|_\infty < 2$ the block sum of $K_\kappa(\cdot - y)$ is $\leq c(\kappa)$ and $K_\kappa(z-y) \geq ce^{-2\delta}$. □

Lemma H.16 (end factors)**.** *Let $L_0, L_1, L_2 \in \{G, G'\}$, $a \in \{0,1\}$, $E_a^{(2)} := D^a L_0 V L_1$, $E_a^{(3)} := D^a L_0 V L_1 V L_2$, every V in right-pushed form; $\delta' := \delta_K/2$, $0 \leq \beta \leq \delta_K/4$. Assume (K) at every pair of points used (automatic on T_η).*

(i) *For $y \in \mathcal{U}$: $\|E_a^{(2)}(x,y)\| \leq c_E \sigma K_2^{\delta'}(x-y)$ and $\|E_a^{(3)}(x,y)\| \leq c'_E \sigma^2 K_3^{\delta'}(x-y)$.*

(ii) *For $a=1$ and $|x-\tilde{x}| \leq 1$ the $\tilde{B}$-transported difference in x of $E_1^{(2)}$ is $\leq c_E(\alpha)\sigma|x-\tilde{x}|^\alpha (K_{2-\alpha}^{\delta'}(x-y) + K_{2-\alpha}^{\delta'}(\tilde{x}-y))$; the same for $E_1^{(3)}$ with σ^2 and $K_{3-\alpha}$.*

(iii) *Put $J_\kappa := (\sum_y \eta^d K_\kappa^{\delta'}(y)^2 e^{2\beta|y|_\infty})^{1/2}$. Then $\|e^{\beta|\cdot - x|}\mathbf{1}_{\mathcal{U}} E_a^{(2)}(x,\cdot)\|_2 \leq c_E J_2 \sigma$; the transported difference of (ii) has weighted ℓ^2 norm $\leq 2e^\beta c_E(\alpha) J_{2-\alpha}\sigma|x-\tilde{x}|^\alpha$; for $E^{(3)}$ the same with J_3, $J_{3-\alpha}$, σ^2. $J_\kappa < \infty$ uniformly in η and in the torus side iff $\kappa > \frac{3}{2}$: so $J_{2-\alpha} < \infty$ iff $\alpha < \frac{1}{2}$, while J_3, $J_{3-\alpha}$ are finite for every $\alpha < 1$.*

Proof. (i) $E_a^{(2)} = -(D^a L_0)R_0 L_1 - \sum_\mu (D^a L_0) R_{1,\mu}(D_\mu L_1)$: every line carries at most one derivative (L_0 the external one, L_1 the vertex one). For a line of type G' write $D_{\tilde{B},\mu} = U(\tilde{A})^{-1} D_{\tilde{A}+\tilde{B},\mu} - \phi_\mu$, so (K) for G' gives $\|D_{\tilde{B}} G'(x,w)\| \leq c_K(K_1 + \sigma K_2) \leq cK_1^{\delta'}$ by lemma H.15(b). Lemma H.15(c) moves R ($r_R \leq c_V\sigma$) onto the kernels and (a), (b) give $K_{2-a} * K_1 \to K_{3-a} \leq cK_2$, $K_{2-a} * K_2 \to K_2$; for $E_a^{(3)} = (D^a L_0)\mathfrak{r}(\mathfrak{D}L_1)\mathfrak{r}(\mathfrak{D}L_2)$ one gets $K_{2-a} * K_1 * K_1 \to K_3$. (ii) The difference acts on the first factor only. For $L_0 = G$ use (K2), then $K_{1-\alpha} * K_1 \to K_{2-\alpha}$ (resp. $K_{1-\alpha} * K_1 * K_1 \to K_{3-\alpha}$). For $L_0 = G'$ write $U(\tilde{B}(\Gamma)) = U((\tilde{A}+\tilde{B})(\Gamma))U(\tilde{A}(\Gamma))^{-1}$: the quotient is at most the (K2) quotient of $D_{\tilde{A}+\tilde{B}}G'$, plus $3\sigma c_K K_1$ from $|U(\tilde{A}(\Gamma))^{-1}U(\tilde{A}_x)^{-1} - U(\tilde{A}_{\tilde{x}})^{-1}| \leq e_k(\|\tilde{A}\||\Gamma| + \eta|\tilde{A}_x - \tilde{A}_{\tilde{x}}|) \leq 3\sigma|x-\tilde{x}|$, plus the ϕ-terms, telescoped along Γ as in lemma H.18, which give $c\sigma|x-\tilde{x}|(K_1(x-\cdot) + K_1(\tilde{x}-\cdot))$; since $|x-\tilde{x}| \leq |x-\tilde{x}|^\alpha$ and $K_1 \leq cK_{1-\alpha}$, (K2) holds for $D_{\tilde{B}}G'$ with constant $c_K(\alpha) + 10\sigma c_K$. (iii) Square and sum, with $|y-x| \leq |y-\tilde{x}| + 1$ for the $\tilde{x}$-term. Near 0, $\sum_y \eta^d \max(|y|,\eta)^{2\kappa-6}$ is bounded uniformly in η iff $2\kappa - 6 > -3$; the tail converges because $\delta' - \beta \geq \delta_K/4$. □

Proposition H.17 (the ℓ^2 sandwich)**.** *Assume (E); then (H0), (K) for G and G', and $\sigma \leq 1$ hold. Let $0 < \beta \leq \beta_U := \min\{\beta_*, \delta_K/4\}$, and either $m \geq 4$ and $0 < \alpha < \frac{1}{2}$, or $m \geq 6$ and $0 < \alpha < 1$. Put $\mathfrak{N}_{\alpha,\beta}(Y) := \max_{a,b} \sup_{x,x'} e^{\beta|x-x'|_\infty} \max\{\|Y_{ab}(x,x')\|, aH_x, bH_{x'}, abH_{xx'}\}$, H the α-Hölder quotients of Y_{ab} with $\tilde{B}$-transport.* (a) *On $X = T_\eta$: $\mathfrak{N}_{\alpha,\beta}(K_{n,n'}) \leq C_U(\alpha)(C_V\sigma)^m$, with $C_V := e^{2\beta}c_V \max\{c'_W, 1\}$, $c'_W := (1+\sigma)^2 c_W$.* (b) *On $X = \Omega$ with Neumann conditions and $\operatorname{dist}_\infty(\operatorname{supp}\tilde{A}, \Omega^c) \geq R_0 + 2$ the same bound holds with the supremum restricted to x, x' such that x, x', their bond partners, their Hölder partners and the points of the contours $\Gamma_{\tilde{x},x}$, $\Gamma_{\tilde{x}',x'}$ have depth $\geq R_0$. $C_U(\alpha)$ depends only on $d, a, M, L, \gamma_0, K_W, c_K, c_K(\alpha), \alpha, \beta$; the bound is uniform in η (hence k, ε), in the torus side, in $0 \leq m^2 t^2 \leq c_m$ and in the backgrounds within (E).*

Proof. $K_{n,n'}$ has m vertices and $m+1$ lines, and each vertex is first order on each side (lemma H.14(i)), so $D^a K D^{b*}$ carries $m+2$ derivatives on $m+1$ lines and some line carries two; that line is put in the middle factor, which enters only through lemma H.11. No pointwise bound on DGD^* is used.

Two-line ends ($m \geq 4$). $Y := D^a K D^{b*} = E_L \mathcal{N} E_R$ with $E_L := E_a^{(2)} = D^a L_0 V L_1$, $\mathcal{N} := V L_2 V \cdots V L_{m-2} V$ (vertices $2, \ldots, m-1$) and $E_R := L_{m-1} V L_m D^{b*} = (D^b L_m V L_{m-1})^*$ (by $V = V^*$ and the symmetry of the L_i). Put the first vertex of $\mathcal{N}$ in right-pushed, the last in left-pushed and the others in symmetric form: $\mathcal{N} = (-1)^{m-2}\mathfrak{r}(\mathfrak{D}L_2\mathfrak{D}^*)\mathfrak{E}\cdots\mathfrak{E}(\mathfrak{D}L_{m-2}\mathfrak{D}^*)\mathfrak{r}^*$. By lemma H.14(iv) $\mathcal{N} = \mathbf{1}_{\mathcal{U}}\mathcal{N}\mathbf{1}_{\mathcal{U}}$, so $E_L(x,\cdot)$, $E_R(\cdot,x')$ are needed only on $\mathcal{U}$. For $L = G'$ each $\mathfrak{D}_{\tilde{B}} = U(\tilde{A})^{-1}\mathfrak{D}_{\tilde{A}+\tilde{B}} - \Phi$ costs $(1+\sigma)$ per side; the kernels of $\mathfrak{r}, \mathfrak{E}$ have sup-range $\leq 1 + \eta \leq 2$, so their conjugates by $e^{\pm\beta\rho}$ are bounded by $e^{2\beta}c_V\sigma$; lemma H.11 bounds the blocks $\mathfrak{D}L\mathfrak{D}^*$. Hence $\|e^{\mp\beta\rho}\mathcal{N}e^{\pm\beta\rho}\| \leq (e^{2\beta}c_V\sigma)^{m-2}c'^{m-3}_W$. Fix x, x', $\rho := |\cdot - x|_\infty$, and write $Y(x,x') = \langle e^{\beta\rho}\mathbf{1}_{\mathcal{U}}E_L(x,\cdot)^*, (e^{-\beta\rho}\mathcal{N}e^{\beta\rho})e^{-\beta\rho}\mathbf{1}_{\mathcal{U}}E_R(\cdot,x')\rangle$; Cauchy–Schwarz, $e^{-\beta|y-x|} \leq e^{-\beta|x-x'|}e^{\beta|y-x'|}$ and lemma H.16(iii) (for E_R by transposition) give $e^{\beta|x-x'|}\|Y(x,x')\| \leq (c_E J_2\sigma)^2(e^{2\beta}c_V\sigma)^{m-2}c'^{m-3}_W$.

Y is linear in $E_L(x,\cdot)$ and in $E_R(\cdot,x')$ separately, so H_x is the same pairing with $E_L(x,\cdot)$ replaced by $U(\tilde B(\Gamma_{\tilde x,x}))E_L(x,\cdot)-E_L(\tilde x,\cdot)$, similarly $H_{x'}$, $H_{xx'}$; lemma H.16(iii) bounds these with $2e^{\beta}c_E(\alpha)J_{2-\alpha}$, finite iff $\alpha<\frac12$.

Three-line ends ($m\ge 6$, any $\alpha<1$). Repeat with $E_L:=E_a^{(3)}$, $\mathcal N:=VL_3V\cdots VL_{m-3}V$ (vertices $3,\dots,m-2$, at least two since $m\ge6$) and $E_R:=L_{m-2}VL_{m-1}VL_mD^{b*}$; now J_3, $J_{3-\alpha}$ are finite. Since $c_V\ge1$ both cases give $\le C_U(\alpha)(C_V\sigma)^m$: (a).

(b) (H0) and lemma H.11 hold on Ω with no depth condition, and lemma H.14 holds since dist ≥2. Every pointwise use of (K) above is at a point of $\mathcal U$ or a bond partner of one (depth $\ge R_0+2-1-\eta\ge R_0$), or at $x,x',\tilde x,\tilde x'$, their bond partners and the contour points of the telescoping in lemma H.16(ii), which are deep by the restriction. So (K) applies and the argument is verbatim. □

Lemma H.18 (localisation of two-variable bounds)**.** *Let Y be a kernel, $\beta\ge0$, $M_Y\ge0$, and $\mathcal P,\mathcal P'$ sets of lattice points such that $e^{\beta|y-y'|_\infty}\max\{\|Y_{ab}(y,y')\|,aH_y,bH_{y'},abH_{yy'}\}\le M_Y$ for all a,b, all directions, all $y,\tilde y\in\mathcal P$, $y',\tilde y'\in\mathcal P'$ with partners $0<|y-\tilde y|,|y'-\tilde y'|\le1$. Let $0\le h,h'\le1$ with $|d_\mu h|,|d_\nu d_\mu h|\le c_h$ (and the same for h'), and assume that every point at sup-distance $\le1+\eta$ from* supp h *lies in $\mathcal P$, and likewise for h', $\mathcal P'$. Then $\|hYh'\|_{1,\alpha}\le e^{3\beta}(1+3\sqrt3c_h)^2M_Ye^{-\beta\,\mathrm{dist}_\infty(\mathrm{supp}\,h,\mathrm{supp}\,h')}$.*

Proof. Leibniz: $D_\mu(hf)=F_1D_\mu f+F_2f$ with $F_1:=h(\cdot+\eta e_\mu)$, $F_2:=d_\mu h$; for the second variable, $(ZD_\mu^*)(x,x')=((D_\mu Z^\dagger)(x',x))^*$. Along a shortest monotone contour $\Gamma_{\tilde x,x}$, of ℓ^1-length $\le\sqrt3|x-\tilde x|$, every φ with $|d\varphi|\le c$ obeys $|\varphi(x)-\varphi(\tilde x)|\le\sqrt3c|x-\tilde x|$, and $U(\Gamma)W(x)-W(\tilde x)$ is a sum over the bonds of Γ of transported $\eta D_\nu W$ terms, so $\|U(\Gamma)W(x)-W(\tilde x)\|\le|x-\tilde x|_1\max_{z\in\Gamma,\nu}\|D_\nu W(z)\|$. In one variable, $U(hY)_1(x)-(hY)_1(\tilde x)$ splits into $F_1(x)[UY_1(x)-Y_1(\tilde x)]$, $(F_1(x)-F_1(\tilde x))Y_1(\tilde x)$, $F_2(x)[UY_0(x)-Y_0(\tilde x)]$ (telescoped, $D_\nu Y_0$ being a component of Y_1) and $(F_2(x)-F_2(\tilde x))Y_0(\tilde x)$; with $|x-\tilde x|\le|x-\tilde x|^\alpha$ the quotient of $(hY)_1$ is $\le(1+3\sqrt3c_h)$ times the largest component or quotient of Y at points of $\{x,\tilde x\}\cup\Gamma$, and a term vanishes unless x or $\tilde x$ is within η of supp h, so every evaluated point is within $1+\eta$ of supp h and lies in $\mathcal P$. The expansions in x and x' commute (different variables, transports from opposite sides); the factors are $(1+c_h)^2$ per component, $(1+3\sqrt3c_h)(1+c_h)$ per one-variable quotient and $(1+3\sqrt3c_h)^2$ for the mixed one. Finally $|y-y'|_\infty\ge\mathrm{dist}_\infty(\mathrm{supp}\,h,\mathrm{supp}\,h')-2(1+\eta)$. □

Corollary H.19 ([3, (2.5)] for [3, (1.16)] at $L=5$)**.** *Assume* (E) *and* dist$_\infty$(supp $\tilde A,\Omega^c)\ge R_0+2$*; let $m=n+n'\ge6$, $\alpha_0\in(0,1)$, $0<\beta\le\beta_U$, and let h,h' be as in lemma H.18 with supports at depth $\ge R_0+3$ in Ω (on T_η: no condition). Then*

$$\|hK_{n,n'}h'\|_{1,\alpha_0}\le e^{3\beta}(1+6c_h)^2C_U(\alpha_0)\,(2c_1C_Ve_kp(t))^{n+n'}\,e^{-\beta\,\mathrm{dist}_\infty(\mathrm{supp}\,h,\mathrm{supp}\,h')}.$$

Proof. Apply lemma H.18 with $\mathcal P=\mathcal P':=\{z:\rho_\Omega(z)\ge R_0+1+\eta\}$: for points of $\mathcal P$ the bond partners and the Hölder partners (and their bond partners, and the contour points between them) have depth $\ge R_0$, so proposition H.17(b) gives the hypothesis with $M_Y=C_U(C_V\sigma)^m$. A point within $1+\eta$ of a set of depth $\ge R_0+3$ has depth $\ge R_0+2-\eta\ge R_0+1+\eta$ since $\eta\le\frac15$. Finally $\sigma\le2c_1e_kp(t)$ and $3\sqrt3\le6$. □

Remark H.20. (i) This is the second alternative of [3, (2.5)] with $\delta_0:=\beta\le\beta_U=\min\{\beta_*,\delta_K/4\}$; shrinking the rate is free. Since [3, p. 426] extracts half of each line's rate for $\exp[-\frac12\delta_1\,\mathrm{dist}(\square(v),\square(v'))]$, the rate δ_0 of Proposition 1 is $\min\{\delta_1/2,\beta\}>0$. (ii) The corollary holds for every $\alpha_0<1$, in particular $\alpha_0=\frac12$; for $m\in\{4,5\}$ it holds for $\alpha_0<\frac12$. It implies the [3, p. 414] form read as $(e(L^k\varepsilon))^{1-\alpha'}$ raised to $n+n'$: $(2c_1C_Ve_kp(t))^m\le(e_k^{1-\alpha'})^m$ as soon as $2c_1C_Vp(t)\le e_k^{-\alpha'}$, i.e. for $t\le t_0(\alpha')$, since p is polylogarithmic and $e_k=et^{1/2}$. (The other parsing, $e\,(L^k\varepsilon)^{1-\alpha}$, is not consumed: [3, (1.33)] counts powers of $e(L^k\varepsilon)$.) (iii) "n,n' sufficiently large" is necessary: at $m=0$, with $g:=\eta^{-d}\mathbf1_x$ and $\Lambda:=4d\eta^{-2}+m^2t^2+a\ge\|H\|$, $(D_\mu G'D_\mu^*)(x,x)=\langle D_\mu^*g,G'D_\mu^*g\rangle\ge\Lambda^{-1}\|D_\mu^*g\|^2=2\eta^{-2-d}/\Lambda\sim\eta^{-3}/6$, not bounded uniformly in η. (iv) The depths used are supp $\tilde A$ at $\ge R_0+2$ and vertex supports at $\ge R_0+3$, both in the sup metric and uniform in $k\ge1$.

H.1.5. *S2: the two-domain propagator $\delta G_k(\Omega,\Omega_2,\tilde B)$ as an external field.* [3, p. 414] replaces Ω by $\Omega_2\subset\Omega$ "satisfying the same conditions as Ω", and treats $\delta G_k(\Omega,\Omega_2,\tilde B)=G_k(\Omega,\tilde B)-G_k(\Omega_2,\tilde B)$ as an external field; [3, (2.5)], first alternative, asserts $\|h\,\delta G_k(\Omega,\Omega_2,\tilde B)\,h'\|_{1,\alpha}\le O(e^{-\delta_0\operatorname{dist}(\Omega_2,\partial\Omega)})e^{-\delta_0\operatorname{dist}(\operatorname{supp}h,\operatorname{supp}h')}$, and [3, p. 426] uses it in [3, (2.13)] as an $O(1)$ bound with the two-point decay; the small depth factor is what the domain changes of [3, pp. 421–422] use. δG is defined on $\Omega_2\times\Omega_2$, $G_k(\Omega_2,\tilde B)$ being extended by 0.
The reading of the depth factor. The printed factor $e^{-\delta_0\operatorname{dist}(\Omega_2,\partial\Omega)}$, read as a distance between the sets Ω_2 and $\partial\Omega$, cannot be what is meant: δG is of order one at points near $\partial\Omega_2$ whatever the distance to $\partial\Omega$. (For $\Omega=T_\eta$ there is no $\partial\Omega$ at all; in one dimension at $\eta=\frac15$, $A=0$, $m=0$, $a_k=1$, with Ω_2 a union of 20 blocks, $\delta G(x,x)$ takes the values -0.43, -0.28, -0.05, -0.0008 at depth 0, 0.2, 1, 3 in Ω_2, against $G(x,x)=0.52$; the same values arise with Ω a Neumann segment at distance 90 from Ω_2.) The factor holds in the form of [1, Prop. 2.1, (2.26)], where the extra decay is in the depth of the arguments in the *smaller* set: here, the depth of the vertex supports in Ω_2. This is the form proved below, and it is the form the uses need.
Setting. $X_{\rm in}\subset X_{\rm out}$ are unions of large blocks of T_η, $X_{\rm out}$ may be T_η, $X_{\rm in}\neq T_\eta$; A is real and regular by (E), $0\le m^2t^2\le c_m$; $H_{\rm in/out}:=H_{X_{\rm in/out}}(A)$, $G_{\rm in/out}:=H^{-1}_{\rm in/out}$, $\delta:=G_{\rm out}-G_{\rm in}$, $\rho:=\operatorname{dist}_\infty(\cdot,X_{\rm in}^c)$; then $\rho_{X_{\rm out}}\ge\rho$.

Lemma H.21 (collars)**.** *Fix $f\in C^\infty(\mathbb R)$ with $0\le f\le1$, $f=0$ on $(-\infty,0]$, $f=1$ on $[1,\infty)$, and $\varphi\in C_c^\infty([-\frac14,\frac14]^3)$, $\varphi\ge0$, $\int\varphi=1$. For a unit block b with lowest corner y_b put $\hat b:=y_b+[0,1)^3$ and $\hat X_{\rm in}:=\bigcup_{b\subset X_{\rm in}}\hat b$; let ρ^c be the continuum sup-distance to the complement of $\hat X_{\rm in}$, $\tilde\rho:=\rho^c*\varphi$, and $\chi_s:=f(\tilde\rho-s)$ at lattice points; $c_1':=\|f'\|_\infty$, $c_2':=\|f''\|_\infty+\|f'\|_\infty\max_\nu\|\partial_\nu\varphi\|_{L^1}$. Then:* (i) *$\tilde\rho$ is 1-Lipschitz for $|\cdot|_\infty$ and $\rho-\frac14-\eta\le\tilde\rho\le\rho+\frac14$ at lattice points;* (ii) *$\chi_s=0$ on $\{\rho\le s-\frac14\}$ and $\chi_s=1$ on $\{\rho\ge s+\frac54+\eta\}$;* (iii) *$|d^\eta_\mu\chi_s|\le c_1'$ and $|d^\eta_\mu d^\eta_\nu\chi_s|\le c_2'$, uniformly in η, s, $X_{\rm in}$;* (iv) *if $\chi_s(u)\neq\chi_s(u')$ and $|u-u'|_\infty\le l$, then $\tilde\rho(u),\tilde\rho(u')\in(s-l,s+1+l)$.*

Proof. (i) A lattice point outside $X_{\rm in}$ lies in a cube $\hat b'$ with $b'\not\subset X_{\rm in}$, so $\rho^c\le\rho$; the infimum defining $\rho^c(z)$ is attained on the closure of such a cube, whose lattice points are η-dense for $|\cdot|_\infty$, so $\rho\le\rho^c+\eta$; the mollifier preserves the Lipschitz constant and moves values by at most $\frac14$. (ii) follows from (i). (iii) Lattice differences are averages of derivatives along segments; $|\partial_\mu\tilde\rho|\le1$ and $\partial_\mu\partial_\nu\tilde\rho=(\partial_\mu\rho^c)*\partial_\nu\varphi$. (iv) If $\tilde\rho(u)\le s-l$ then $\tilde\rho(u')\le s$ and both values of χ_s are 0; symmetrically both are 1 if $\tilde\rho(u)\ge s+1+l$. □

Lemma H.22 (commutators)**.** *Let $s\ge1$, $\chi:=\chi_s$, $\phi_\mu:=d^\eta_\mu\chi$.* (i) *$H_{\rm out}(\chi g)=H_{\rm in}(\chi g)$ for every g on $X_{\rm in}$ (extended by 0).* (ii) *$C_\chi:=H_{\rm out}\chi-\chi H_{\rm in}=[H_{\rm in},\chi]=-\mathfrak r_\chi\mathfrak D=\mathfrak D^*\mathfrak r_\chi^*$, where $\mathfrak r_0:=\sum_\mu(d_\mu\phi_\mu)(\cdot-\eta e_\mu)-a_k[P_k(A),\chi]$ and $(\mathfrak r_\mu g)(x):=\phi_\mu(x)g(x)+\phi_\mu(x-\eta e_\mu)U_{x-\eta e_\mu,\mu}(A)^{-1}g(x-\eta e_\mu)$.* (iii) *$\sum_{i=0}^3\sup_z\sum_w\eta^d\|\mathfrak r_i(w,z)\|\le c_{\mathfrak r}:=6c_1'+3c_2'+ac_1'$.* (iv) *If $\mathfrak r_i(w,z)\neq0$ then $|w-z|_\infty\le1$ and $w,z\in S_s:=\{s-\frac54-\eta<\rho<s+\frac94+2\eta\}$.*

Proof. (i) If $\chi(y)\neq0$ then $\rho(y)>s-\frac14\ge\frac34>\eta$, so every bond at y and the unit block of y lie in $X_{\rm in}$; a bond of $X_{\rm out}$ which is not a bond of $X_{\rm in}$ has both ends at depth $\le\eta$, outside $\operatorname{supp}\chi$; P_k acts inside unit blocks. (ii) $D_\mu(\chi f)=\chi(\cdot+\eta e_\mu)D_\mu f+\phi_\mu f$ and $D^*_\mu(\chi(\cdot+\eta e_\mu)g)=\chi D^*_\mu g-\phi_\mu g$, so $[D^*_\mu D_\mu,\chi]=D^*_\mu(\phi_\mu\,\cdot)-\phi_\mu D_\mu$, and $D^*_\mu(\phi_\mu f)(x)=\eta^{-1}(\phi_\mu(x-\eta e_\mu)U^{-1}f(x-\eta e_\mu)-\phi_\mu(x)f(x))=-\phi_\mu(x-\eta e_\mu)U^{-1}_{x-\eta e_\mu,\mu}(D_\mu f)(x-\eta e_\mu)-(d_\mu\phi_\mu)(x-\eta e_\mu)f(x)$ ([4, (2.32)]); collecting terms gives $-\mathfrak r_\chi\mathfrak D$, and $C^*_\chi=-C_\chi$ since $H_{\rm in}$ is self-adjoint and χ real. (iii) The two parts of each $\mathfrak r_\mu$ give $2c_1'$, so $6c_1'$ over three directions; the multiplication part of $\mathfrak r_0$ gives $3c_2'$; $[P_k,\chi](w,z)=P_k(w,z)(\chi(z)-\chi(w))$ with $|\chi(z)-\chi(w)|\le c_1'|\tilde\rho(z)-\tilde\rho(w)|\le c_1'$ and $\sum_w\eta^d\|P_k(w,z)\|\le1$. (iv) Lemma H.21(iv) with $l=2\eta$ (bond and multiplication parts) or $l=1-\eta$ (block part) gives $\tilde\rho\in(s-1+\eta,s+2-\eta)$; convert to ρ by lemma H.21(i). □

Choices. $\chi:=\chi_{R_0+3}$, $\psi:=\chi_{R_0+13/2}$, $\mathcal P_D:=\{z:\rho(z)\ge R_0+10+2\eta\}\supset\{\rho\ge R_0+10.4\}$. With $\eta\le\frac15$, lemmas H.21 and H.22 give: $S_\chi\subset\{R_0+1.55<\rho<R_0+5.65\}$, $S_\psi\subset\{R_0+5.05<\rho<R_0+9.15\}$; $\psi=0$ on $\{\rho\le R_0+6.25\}\supset S_\chi$; $\chi=1$ on $\{\rho\ge R_0+4.45\}$ and $\psi=1$ on $\{\rho\ge R_0+7.95\}$, both containing $\mathcal P_D$ and every bond partner of a point of $\mathcal P_D$; for $z\in\mathcal P_D$,

$\operatorname{dist}_\infty(z,S_\chi) > \max\{\rho(z)-R_0-7, 4.75\}$ and $\operatorname{dist}_\infty(z,S_\psi) > \max\{\rho(z)-R_0-10, 1.25\}$; every point of S_χ, S_ψ and its bond partners has depth $> R_0+1.5$ in X_{in}, hence in X_{out}.

Proposition H.23 (the two-domain theorem)**.** *Let $0<\alpha<1$, $0<\beta\le\beta_D:=\min\{\beta_*,\delta_K/4\}$, $a,b\in\{0,1\}$, $Y:=D^a\delta D^{b*}$. For $x,x'\in\mathcal{P}_D$ with Hölder partners in $\mathcal{P}_D$,*

$$e^{\beta|x-x'|_\infty}\max\{\|Y(x,x')\|, aH_x, bH_{x'}, abH_{xx'}\} \le C_D(\alpha)\exp\Big(-\frac{\delta_K}{2}\big[(\rho(x)-R_0-7)+(\rho(x')-R_0-10)\big]\Big),$$

with $C_D(\alpha):=c_W c_{\mathrm{end}}(\alpha)^2$, $c_{\mathrm{end}}(\alpha):=c_{\mathfrak{r}}e^{2\delta_K}\max\{c_K, 2c_K(\alpha)\}c_3'^{1/2}$, $c_3':=\sup_\eta\sum_z\eta^d e^{-(\delta_K/2)|z|_\infty}\le(\frac15+4/\delta_K+1)^3$. The constants depend only on $d,a,M,L,\gamma_0,K_W,c_K,c_K(\alpha),f,\varphi$, not on η,k,ε, the torus side, X_{in}, X_{out}, A, $m^2t^2\in[0,c_m]$ or R_0.

Proof. $G_{\mathrm{out}}C_\chi G_{\mathrm{in}}=\chi G_{\mathrm{in}}-G_{\mathrm{out}}\chi$, so at y,y' with $\chi(y)=\chi(y')=1$, $\delta(y,y')=-(G_{\mathrm{out}}C_\chi G_{\mathrm{in}})(y,y')$, the inner variable ranging over S_χ only (lemma H.22(iv)), where $\psi=0$. Likewise $G_{\mathrm{in}}C_\psi G_{\mathrm{in}}=\psi G_{\mathrm{in}}-G_{\mathrm{in}}\psi$ gives $G_{\mathrm{in}}(z,y')=-(G_{\mathrm{in}}C_\psi G_{\mathrm{in}})(z,y')$ when $\psi(z)=0$, $\psi(y')=1$. Inserting and using lemma H.22(ii), at all points of $\mathcal{P}_D$ and their bond partners (the only points at which D^a, D^{b*} evaluate δ),

$$\delta=-G_{\mathrm{out}}\mathfrak{r}_\chi(\mathfrak{D}G_{\mathrm{in}}\mathfrak{D}^*)\mathfrak{r}_\psi^*G_{\mathrm{in}},\qquad Y(x,x')=-\langle E_L(x,\cdot),\mathfrak{M}E_R(\cdot,x')\rangle,$$

$E_L:=D^aG_{\mathrm{out}}\mathfrak{r}_\chi$, $\mathfrak{M}:=\mathfrak{D}G_{\mathrm{in}}\mathfrak{D}^*$, $E_R:=\mathfrak{r}_\psi^*G_{\mathrm{in}}D^{b*}$. *Left end.* $E_L(x,z)=\sum_w\eta^dD^a_xG_{\mathrm{out}}(x,w)\mathfrak{r}_\chi(w,z)$ with $w,z\in S_\chi$, $|w-z|_\infty\le1$; (K) applies on X_{out} at (x,w) (both deep, $|x-w|_\infty\ge4.75\ge1$), so $K^{\delta_K}_{2-a}(x-w)\le e^{-\delta_K|x-w|_\infty}$ and $\|E_L(x,z)\|\le c_{\mathfrak{r}}c_Ke^{\delta_K}e^{-\delta_K|x-z|_\infty}\mathbf{1}_{S_\chi}(z)$; with $2(\delta_K-\beta)\ge\frac32\delta_K$, $\|e^{\beta|\cdot-x|}E_L(x,\cdot)\|_2\le c_{\mathfrak{r}}c_Ke^{\delta_K}e^{-(\delta_K/2)\operatorname{dist}(x,S_\chi)}c_3'^{1/2}\le c_{\mathrm{end}}e^{-(\delta_K/2)(\rho(x)-R_0-7)}$; for $a=1$ and a partner $\tilde x\in\mathcal{P}_D$, (K2) with $|\tilde x-w|\ge1$ gives the transported difference with the factor $2c_K(\alpha)e^{\delta_K}|x-\tilde x|^\alpha$. *Right end.* $(G_{\mathrm{in}}D^{b*})(v,x')=(D^b_{x'}G_{\mathrm{in}}(x',v))^T$ with $v\in S_\psi$, and (K) on X_{in} at (x',v) gives $\|e^{\beta|\cdot-x'|}E_R(\cdot,x')\|_2\le c_{\mathrm{end}}e^{-(\delta_K/2)(\rho(x')-R_0-10)}$, and the same times $|x'-\tilde x'|^\alpha$ for the transported difference. *Middle.* Lemma H.11 on X_{in} with weight $|\cdot-x|_\infty$ gives $\|e^{-\beta|\cdot-x|}\mathfrak{M}e^{\beta|\cdot-x|}\|\le c_W$; then $e^{-\beta|u-x|}\le e^{-\beta|x-x'|}e^{\beta|u-x'|}$ and Cauchy–Schwarz give the bound. The quotients follow by replacing one or both ends by their transported differences, Y being linear in each end. No kernel is evaluated at points less than 1 apart, and the doubly differentiated line enters only through its operator norm. □

Corollary H.24 (the consumed form at $L=5$)**.** *Let $0\le h,h'\le1$ with $|dh|,|ddh|\le c_h$ (the same for h'), the smooth localisers of* [3, (1.33)]*; put $d_h:=\operatorname{dist}_\infty(\operatorname{supp}h, X_{\mathrm{in}}^c)$, $d_{h'}$ likewise, and assume $d_h,d_{h'}\ge R_0+12$. Then, at any mutual distance of the supports (overlap included),*

$$\|h\,\delta\,h'\|_{1,\alpha}\le C_{\mathrm{loc}}\exp\Big(-\frac{\delta_K}{2}(d_h+d_{h'}-2R_0-19.4)\Big)e^{-\beta\operatorname{dist}_\infty(\operatorname{supp}h,\operatorname{supp}h')},\qquad C_{\mathrm{loc}}:=e^{3\beta}(1+6c_h)^2C_D(\alpha).$$

Since $d_h+d_{h'}-2R_0-19.4\ge4.6>0$, this is the first alternative of [3, (2.5)] *in the $O(1)$ form, and it carries the depth factor $\exp(-\frac{\delta_K}{2}(d_h+d_{h'}))$, the form of* [1, (2.26)] *with the complement of the smaller set.*

Proof. Lemma H.18 with $\mathcal{P}:=\{\rho\ge d_h-1-\eta\}$, $\mathcal{P}':=\{\rho\ge d_{h'}-1-\eta\}$, both contained in $\mathcal{P}_D$ since $R_0+12-1.2=R_0+10.8\ge R_0+10.4$; on $\mathcal{P}\times\mathcal{P}'$ proposition H.23 gives the hypothesis with $M_Y:=C_D\exp(-\frac{\delta_K}{2}[(d_h-1.2-R_0-7)+(d_{h'}-1.2-R_0-10)])$. □

Remark H.25. The three instances in [1, 2] are $X_{\mathrm{out}}=\Omega$, $X_{\mathrm{in}}=\Omega_2$; $X_{\mathrm{out}}=\Omega$, $X_{\mathrm{in}}=$ the cube; and $X_{\mathrm{out}}=T_\eta$, $X_{\mathrm{in}}=$ the cube, $A=0$. In the first the vertex supports have depth $\ge5r(t)-9$ in Ω_2 (the nesting of [2, pp. 579–580]: the summations in the vertices are restricted to $\Lambda_7^{(k)}$ while the propagator domain is $B^k(\Lambda_2^{(k)})$), which exceeds R_0+12 under $R\ge3R_0+15$; at $L=5$ the depth R_0+12 is uniform in $k\ge1$, there being no separate $k=1$ case ($\eta_1=\frac15$). Near $\partial\Omega_2$ nothing is proved, and nothing follows from (P) alone (Neumann boundary regularity at the edges of a union of large blocks).

H.1.6. *S4: the summation by parts* [3, (2.8)–(2.9)] *at a vertex of* [3, (2.4)]. The graph [3, (2.4)] is two vertices [3, (1.8)] with $n+n'=1$, joined by one scalar line carrying the derivative at both ends, with two external scalar and two external vector legs; its degree is 0. [3, p. 425] replaces such graphs by graphs of positive degree "by integration by parts" at one of the two vertices, [3, (2.8)], applied "for all graphs $G_1, G_2^{(2)}, G_3^{(3)}, \dots$" of a fixed ordering, so that G' becomes a sum of graphs $\{G'^*\}$ whose components $G_1, \dots, G_m = G'^*$ along the ordering have positive degrees. Two things are to be proved: the identity, as an identity of multilinear forms in the legs, and the degree statement, which is the hypothesis of S5.

Lemma H.26 (the identity)**.** *Let $X \subset T_\eta$, $\phi_1, \phi_2 \colon X \to \mathbb{R}^N$, and h_μ real functions with $\operatorname{supp} h_\mu \subset \{x : \operatorname{dist}_\infty(x, X^c) \geq 2\eta\}$; $\tilde{B}$ arbitrary and real. Put $D_\mu^{\leftarrow}\phi(x) := U(\tilde{B}_{x-\eta e_\mu,\mu})^{-1}(D_\mu\phi)(x-\eta e_\mu) = \eta^{-1}(\phi(x) - U^{-1}\phi(x-\eta e_\mu))$ and $d_\mu^* h(x) := \eta^{-1}(h(x-\eta e_\mu) - h(x))$. Then*

$$(-e)\sum_\mu\sum_x \eta^d[(D_\mu\phi_1)(x)\cdot q\phi_2(x)]h_\mu(x) = (-e)\sum_x \eta^d[\phi_1(x)\cdot q\phi_2(x)]\Big(\sum_\mu d_\mu^* h_\mu\Big)(x) + e\sum_x\sum_\mu \eta^d[\phi_1(x)\cdot q(D_\mu^{\leftarrow}\phi_2)$$

With $h_\mu = gA'_\mu$, $d_\mu^*(gA'_\mu)(x) = g(x)(d_\mu^* A'_\mu)(x) + (d_\mu^* g)(x)A'_\mu(x-\eta e_\mu)$ gives the first two terms of [3, (2.8)] *exactly; the third printed term carries the derivative at the bond $(x-\eta e_\mu, x)$ transported to x, and its sign is as above.*

Proof. For $F(x) := q\phi_2(x)h_\mu(x)$, supported where both bonds at x in direction μ lie in X, $\sum_x \eta^d(D_\mu\phi_1)(x)\cdot F(x) = \sum_x \eta^d \phi_1(x)\cdot(D_\mu^* F)(x)$ with $(D_\mu^* F)(x) = \eta^{-1}(U^{-1}_{x-\eta e_\mu,\mu}F(x-\eta e_\mu) - F(x))$, no boundary term arising because $\operatorname{supp} F$ is at depth $\geq 2\eta$. U commutes with q, so $D_\mu^*(q\phi_2 h_\mu) = qD_\mu^*(\phi_2 h_\mu)$, and $D_\mu^*(\phi_2 h_\mu)(x) = -(D_\mu^{\leftarrow}\phi_2)(x)h_\mu(x-\eta e_\mu) + \phi_2(x)(d_\mu^* h_\mu)(x)$. Multiply by $-e$ and sum over μ. □

Remark H.27. With $D^\eta_{\tilde{B},\mu}$ the forward covariant difference [1, (1.7)] evaluated at x, the printed third term of [3, (2.8)] has the opposite sign; it is exactly right with $D^{\eta *}_{\tilde{B},\mu}$, the symbol of the first line of [3, (2.8)], in place of $D^\eta_{\tilde{B},\mu}$, so the print most likely dropped an asterisk. This is harmless for Proposition 1, which takes absolute values of every factor [3, p. 426], the graphical rule [3, (2.9)] recording only where the derivative sits.

Which vertex, and why the identity holds graph by graph. The vertex of [3, (2.4)] is [3, (1.8)] with $n+n'=1$: $-e(L^k\varepsilon)\sum_b \eta^d[(D^\eta_{\tilde{B}}\phi')(b)\cdot q\phi'(b_-)](g_k A'_b)^n(\tilde{A}_b)^{n'}$; after the localisation of [3, p. 420], g carries the partition-of-unity factor of the vector leg, so $h_\mu = gA'_\mu$ ($n=1$) or $h_\mu = \tilde{A}_\mu$ ($n'=1$). The supports are at depth $\geq 2\eta$ in Ω: $\operatorname{supp} g \subset \Omega_1$ at distance $\geq r(t)-2$ from Ω^c, and $\operatorname{dist}(\operatorname{supp}\tilde{A}, \partial\Omega) > 2r(t)$. $E(G')$ is a sum over pairings; for a fixed pairing the value is obtained by substituting into each vertex form a function for each leg (a slice $G^\eta_{(j)}(\cdot, y)$ on Ω for a scalar leg, a vector slice for an A'-leg, or an external field). The vertex form is multilinear in its legs and lemma H.26 is an identity of multilinear forms, so it holds after substitution, graph by graph, for every fixed ordering and index set $J(\tilde{l})$, even though $\phi'(x)\cdot q\phi'(x) = 0$ as a polynomial (the two legs are distinct in each graph).

Lemma H.28 (degrees after [3, (2.8)])**.** *Use the degree count* [3, (2.1)–(2.2)]*: $D_G(v) =$ (number of factors η in v) $\frac{2-d}{2}$(legs of v on internal lines of G)$-$(differentiations in v acting on internal lines of G)$+$(contour credit), $D(G) = \sum_v D_G(v) - d$.* (a) *$D($*[3, (2.4)]*$) = 0$.* (b) *Let v be a vertex of a* [3, (2.4)]*-subgraph $G_0 = \{l_0\}$ and apply lemma H.26 at v. In each of the three terms the graph has the same vertices and lines; only the place of the derivative at v changes, from l_0 to the A'-leg (or $\tilde{A}$-leg), to g, or to the ϕ_2-leg. For every connected $H \ni l_0$: $D_{\text{new}}(H) = D_{\text{old}}(H)+1$ if the new place is not an internal line of H, and $= D_{\text{old}}(H)$ otherwise; in particular $D_{\text{new}}(G_0) = 1$.* (c) *For a connected H containing v but not l_0, and containing the line of the new place: $D_{\text{new}}(H) = D_{\text{old}}(H) - 1$.* (d) *Fix an ordering $\tilde{l}$ and call the component $G_i^{(i)}$ created by $l(i)$* fresh *if $l(i)$ joins two vertices not met by $l(1), \dots, l(i-1)$ and these with $l(i)$ form a* [3, (2.4)] *graph. Apply lemma H.26 at one vertex of each fresh component. If every component $G_i^{(\alpha)}$ of G' along $\tilde{l}$ has $D > 0$ except the fresh ones, then in each resulting G'^* every component $G_i^{(\alpha)}$ along $\tilde{l}$ has $D > 0$.*

Proof. (a) Each vertex of [3, (2.4)] has η-count d, one internal scalar leg and one derivative on it, so $D(v) = d + \frac{2-d}{2} - 1 = \frac{d}{2}$ and $D = 2 \cdot \frac{d}{2} - d = 0$. (b) The three terms keep the η-count and the legs; only "differentiations acting on internal lines" changes: for $H \ni l_0$ the old derivative was on an internal line, the new one is on an internal line exactly when H contains the A'-line or the ϕ_2-line at v (the g-term never is). (c) For $H \not\ni l_0$ the old derivative was not on an internal line of H and the new one may be. (d) A vertex [3, (1.8)] has exactly one derivative leg; a fresh component uses the derivative legs of both its vertices, so each vertex lies in at most one fresh component, the fresh components are vertex-disjoint and the transformations commute. If v lies in the fresh component created by $l_0 = l(i_0)$, then v is met by no earlier line, so every component $G_i^{(\alpha)}$ with $i \geq i_0$ containing v contains l_0, and none with $i < i_0$ contains v. By (b) every component containing a transformed vertex has $D_{\text{new}} \geq D_{\text{old}} > 0$ or is fresh with $D_{\text{new}} = 1$; the others are unchanged. □

Remark H.29. [3, p. 425] states "the degrees of these graphs are $\geq D(G_0)$" for a graph G_0 containing the [3, (2.4)] subgraph, where it holds by (b), and claims positivity only for the subgraphs $G_1, \dots, G_m = G'^*$ along the ordering, which is (d). By (c), a connected subgraph through the transformed vertex avoiding l_0 can lose one unit, so G'^* need not have every connected subgraph of positive degree; the induction of S5 uses only (d). The transformation depends on the ordering, so the sum over orderings of [3, (2.7)] is taken first, as printed. The new vertices are admissible for the bounds of S3 and S5: a derivative on A' is a derivative at one end of a vector slice (lemma H.8(a), one extra s^{-1}); $|d^*g| \leq c$ since g_k is bounded with its derivatives up to second order [3, p. 412]; a derivative on ϕ_2 is a derivative at one end of a scalar slice (lemma H.7) or of the external field (bounded by $\|h\Phi\|_{1,\alpha}$); the shifts by ηe_μ keep every point within η of the vertex localisation; the η-count is unchanged, and no line carries two derivatives at one end, the ϕ_2- and A'-legs of [3, (1.8)] carrying none.

H.1.7. *S5: the induction* [3, (2.16)]. After [3, (2.13)–(2.14)] the quantity to bound is, for a graph G with lines l, slice indices $j \in J(\tilde{l})$ and cubes $\Delta(v)$,

$$\tilde{E}(G(j), \{\Delta(v)\}) = \sum_{\Delta(v)} \prod_v s_{j(v)}^{d+e_v} \prod_l s_{j_l}^{d_l} \prod_l \exp\big[-\delta\, s_{j_l}^{-1} \operatorname{dist}_\infty(\Delta(v_l), \Delta(v_l'))\big], \tag{88}$$

with $\delta := \delta_1/2$, $s_j = L^j\eta$, $j(v) := \min_{l \ni v} j_l$, $e_v \geq 0$ the explicit η-power of a vertex [3, (1.8)–(1.11)] $(n+n'-1, n+n'-2, \dots$ from [3, p. 413]; 0 for the other vertices), the bound $\eta^{e_v} \leq s_{j(v)}^{e_v}$ being the "proper power" of [3, (2.14)], and $d_l := -(d-2) - (\text{derivatives on } l) + (\text{contour legs of } l)$, each line having two legs. Put $D(H) := \sum_{l \in H} d_l + \sum_{v \in V(H)} e_v + d(|V(H)| - 1)$ for a connected subgraph H.

Lemma H.30 (dictionary)**.** *$D(H)$ is Bałaban's degree* [3, (2.2)], *$\sum_{v \in H} D_H(v) - d$, with the contour credit going to legs of lines internal to H at the line's own scale (lemma H.8(b)). $D(H)$ is an integer, so $D(H) > 0$ implies $D(H) \geq 1$.*

Proof. Sum [3, (2.1)] over $v \in H$: each internal line contributes two legs, $-(d-2)$ in total, its derivatives and its contour legs; each vertex contributes d (the η^d of its sum) and e_v. □

Theorem H.31 ([3, (2.16)] for all i, and [3, (2.15)])**.** *Fix an ordering $\tilde{l} = (l(1), \dots, l(m))$ of the m lines of a connected graph G; for $j \in J(\tilde{l})$, $j_{l(1)} \leq \dots \leq j_{l(m)} \leq k-1$. Let $G_i := \{l(1), \dots, l(i)\}$ with endpoints, $G_i^{(\alpha)}$ its components, $W_i := \sum_\alpha D(G_i^{(\alpha)})$, and assume $D(G_i^{(\alpha)}) > 0$ for all i, α, so $W_i \geq 1$. At stage i (Bałaban's G/G_i) every component of G_i is shrunk to one vertex u carrying only its cube factor $s_{j(u)}^d$, every vertex not yet met keeps $s_{j(v)}^{d+e_v}$, $j(u)$ is the minimum index of the remaining lines at u, $\Delta(u)$ has side $s_{j(u)}$, and the remaining lines keep $s_{j_l}^{d_l}$ with rates $\delta_i > 0$; call the resulting expression $\tilde{E}_i(G/G_i)$. Then for every fixed $j_{l(i+1)} \leq \dots \leq j_{l(m)}$,*

$$\sum_{j_{l(1)} \leq \dots \leq j_{l(i)} \leq j_{l(i+1)}} \sum_{\Delta\text{'s of } G} \tilde{E} \;\leq\; C_i\, s_{j_{l(i+1)}}^{W_i} \sum_{\Delta\text{'s of } G/G_i} \tilde{E}_i(G/G_i), \tag{89}$$

which is [3, (2.16)]*: the single factor* $(L^{j_{l(i+1)}}\eta)^{\sum_\alpha D(G_i^{(\alpha)})}$ *and a constant depending on the number of lines (hence on* $\bar n$*), on* δ_1 *and on* L*. The constants are* $C_0 = 1$*,* $C_{i+1} := C_i e^{\delta_i' n_i} S_d(\delta_i')(1 - L^{-1})^{-1}$*,* $\delta_{i+1} := \delta_i' := \delta_i/(n_i + 1)$*,* n_i *the number of other remaining lines at the far endpoint of* $l(i + 1)$*,* $S_d(\delta) := \sum_{n\in\mathbb{Z}^d} e^{-\delta(|n|_\infty - 1)_+}$*; at* $L = 5$*,* $(1 - L^{-1})^{-1} = \frac{5}{4}$*. At* $i = m$*:* $\sum_{j\in J(\tilde l)} \sum_\Delta \tilde E \le C_m 5^d$*, which is* [3, (2.15)].

Proof. Induction on i; at $i = 0$ the statement is the definition. Assume $(89)_i$, let $l := l(i + 1)$, $j' := j_l$, and u, u' its endpoints in G/G_i. (1) l is the lowest remaining line, so $j(u) = j(u') = j'$ and $\Delta(u)$, $\Delta(u')$ have side $s_{j'}$. (2) *Case* $u \neq u'$. Split $\exp[-\delta_i s_{j'}^{-1} \operatorname{dist}(\Delta(u), \Delta(u'))]$ into $n_i + 1$ equal factors of rate $\delta' := \delta_i/(n_i + 1)$. For another line l' at u' with far end u'' and $j_{l'} \ge j'$: $s_{j'}^{-1} \ge s_{j_{l'}}^{-1}$ and $\operatorname{dist}(\Delta(u), \Delta(u'')) \le \operatorname{dist}(\Delta(u), \Delta(u')) + \operatorname{diam}_\infty \Delta(u') + \operatorname{dist}(\Delta(u'), \Delta(u''))$ with $\operatorname{diam}_\infty \Delta(u') \le s_{j'} \le s_{j_{l'}}$, so one share times l''s factor is $\le e^{\delta'} \exp[-\delta' s_{j_{l'}}^{-1} \operatorname{dist}(\Delta(u), \Delta(u''))]$: the line l' now ends at u with rate δ'. The last share sums $\Delta(u')$ over the paving of side $s_{j'}$: $\sum_{\Delta(u')} \exp[-\delta' s_{j'}^{-1} \operatorname{dist}(\Delta(u), \Delta(u'))] \le S_d(\delta')$, scale-free since the cubes are labelled by $s_{j'}\mathbb{Z}^d$ and $\operatorname{dist} \ge s_{j'}(|n|_\infty - 1)_+$. The factor of u' stays ($s_{j'}^d$ for a shrunk component, $s_{j'}^{d+e_{u'}}$ for a vertex not yet met). (3) *Shrinking.* Merge u, u' into $\hat u$ with $j(\hat u)$ the minimum index of the remaining lines at $\hat u$, $\ge j_{l(i+2)} \ge j'$ (if no line remains, $\hat u$ is the last vertex). Fix $\Delta(\hat u)$ of side $s_{j(\hat u)}$ in the nested paving and sum $\Delta(u) \subset \Delta(\hat u)$: every remaining line at $\hat u$ has $\exp[-\cdots \operatorname{dist}(\Delta(u), \cdot)] \le \exp[-\cdots \operatorname{dist}(\Delta(\hat u), \cdot)]$, independent of $\Delta(u)$; there are $(s_{j(\hat u)}/s_{j'})^d$ cubes $\Delta(u)$ in $\Delta(\hat u)$, which with u's factor $s_{j'}^d$ (times $s_{j'}^{e_u}$ if u was not yet met) gives $\hat u$'s cube factor $s_{j(\hat u)}^d$. What is left at scale j' is $s_{j'}^{d_l + d + E}$, E the sum of e_v over the endpoints of l not met before. (4) *Case* $u = u'$ (l closes a loop or is a tadpole). There is no $\Delta(u')$ sum; what is left at scale j' is $s_{j'}^{d_l}$ if u was met before and $s_{j'}^{d_l + e_u}$ for a tadpole at a vertex not met before; shrinking is as in (3). (5) *Degrees.* With the stage-i factor $s_{j'}^{W_i}$, the power of $s_{j'}$ is W_{i+1}: for $u \neq u'$, $D(\text{new}) = D(\text{comp } u) + D(\text{comp } u') + d_l + d$ with $D(\{v\}) := e_v$ for a vertex not yet met (the vertex count adds $|V| + |V'| - 1$); for $u = u'$, $D(\text{new}) = D(\text{old}) + d_l$; all other components are unchanged. (6) *The scale sum.* The only dependence on j' is now $s_{j'}^{W_{i+1}}$ with $W_{i+1} \ge 1$ (the $j(\cdot)$ of the vertices involve only $l(i + 2), \dots,$ independent variables $\ge j'$), so $\sum_{j'\le j_{l(i+2)}} s_{j'}^{W_{i+1}} \le (1 - L^{-1})^{-1} s_{j_{l(i+2)}}^{W_{i+1}}$; if no line remains, the sum over $j' \le k - 1$ is $\le (1 - L^{-1})^{-1}$. This is $(89)_{i+1}$. (7) *The end.* At $i = m$ one vertex remains with its cube factor; its cube sum, over cubes of side s meeting a region of sup-diameter ≤ 4 at weight s^d, is $\le 5^d$. The rates obey $\delta_m \ge \delta/\prod_i (n_i + 1) > 0$, the n_i being bounded by the number of legs, a function of $\bar n$. □

Remark H.32. (i) The induction uses only $W_{i+1} \ge 1$ at each stage, which is weaker than the positivity of each component that lemma H.28(d) supplies. (ii) Booking each component's degree at its own next scale, as a vertex weight D at $j(u) \ge j_{l(i+1)}$, gives a larger right side (since $s_{j_{l(i+1)}} \le s_{j(u)}$ and $D > 0$); the printed common factor $s_{j_{l(i+1)}}^{W_i}$ is the sharper bookkeeping, and [3, (2.16)] is correct as stated. (iii) The theorem is stated for general line dimensions d_l and vertex powers e_v, which covers the lines of [3, Prop. 2.2] as far as they are of the slice type; the generalised graphs themselves are the next step (part II of this appendix, appendix H.2). (iv) The hypothesis "every component along the ordering has positive degree" is exactly what S4 gives for the graphs G'^*; it is not the hypothesis that every connected subgraph has nonnegative degree, which the G'^* need not satisfy (remark H.29): e.g. two [3, (1.8)] vertices v, v' with $n = 1$ joined by a scalar line l' carrying both derivative legs, with v's A'-leg contracted by a vector line l_A to a third [3, (1.8)] vertex, all other legs external, have $D(\{l'\}) = 0$, $D(\{l_A\}) = 2$, $D(\{l', l_A\}) = 2$; in the ordering (l_A, l') the line l' is not fresh, is not transformed, and $G'^* = G$ keeps the zero-degree connected subgraph $\{l'\}$, while the components along this ordering have degrees 2, 2 and the theorem gives $O(1)$.

H.2. **The sketched steps of Proposition 1, II: generalised graphs, power counting, Ward–Takahashi identities and the estimate (3.1).** This part contains the next four

sketched steps of the proof of [3, Prop. 1, (1.33)], in the order in which the printed proof invokes them, and the degree facts about cut lines on which the regrouping of [3, p. 432], proved in theorem F.57, rests:

step	[3] page	printed object
S6	428	[3, Prop. 2.2]: the induction [3, (2.16)] for "generalized expressions and graphs"
S7	428–429	the estimate [3, (2.17)] and [3, Cor. 2.3]: which graphs have $D(G) \le 0$
S8	430–431	the Ward–Takahashi identities [3, (2.23)–(2.28)] "for free boundary conditions ... by taking a
S9	432	the estimate [3, (3.1)] for a line between separated localisations
—	432	degrees of a graph after a line is cut (appendix H.2.5)

The order is S6 (appendix H.2.1), S7 (appendix H.2.2), S8 (appendix H.2.3), S9 (appendix H.2.4) and the cut-line degrees (appendix H.2.5). S6 and S9 are implications from the slice bounds of appendix H.1.2 (and, for the subgraphs [3, (2.4)], the summation by parts of appendix H.1.6); S7, S8 and the cut-line degrees are self-contained.

Conventions for this part.
Scales. As in appendix H.1.1: $d = 3$ unless a statement says otherwise (the lemmas of appendix H.2.1 and appendix H.2.3 hold for general d with the same proofs), $L = 5$, $\eta = \eta_k = L^{-k}$, $k \ge 1$, and the slice scales are $s_j := L^j \eta \in (0, \frac{1}{L}]$, $0 \le j \le k-1$. The running charge is e_k, and $U(a) := \exp(e_k \eta q a)$ with q real antisymmetric, $\|q\| \le 1$; all U are orthogonal and commute.
Vertices and lines. The vertices are those of [3, (1.6)–(1.15)]: the quartic vertex [3, (1.6)]; the mass vertex [3, (1.7)], with a finite counterterm ($\delta m^2_{\mathrm{fin}}$ or $\delta m^2_{K,k}$) or with a counterterm of δm^2_k carrying its own graph G_0; the expansion vertices [3, (1.8)–(1.11)] with n legs A' and n' legs $\tilde{A}$ (the $\tilde{A}$ are external fields inside the vertex, not legs), the R-vertices [3, (1.9), (1.11)] having $n' = \bar{n} + 1$; and the block vertices [3, (1.13)–(1.15)], each counted as a vertex of its own with one scalar leg ϕ' and n vector legs ([3, p. 414]: "we will treat (1.13)–(1.15) as the vertices"; [3, p. 423 (v)]). A graph is connected and has at least one line [3, p. 415]. We write V, I for its numbers of vertices and lines, $\ell := I - V + 1 \ge 0$ for its number of independent loops, E for its number of external legs among the ϕ' and A' legs, E_s, E_v for the scalar and vector parts of E, Δ_{ext} for the number of differentiations acting on external legs, and l_v for the number of ϕ', A' legs of the vertex v (so $\sum_v (l_v - 2) = 2I + E - 2V$).
Degrees. The degree is that of [3, (2.1)–(2.3)]: a scalar or vector leg lying on a line of the graph has dimension $-\frac{d-2}{2}$; a vector leg of [3, (1.14)] or [3, (1.15)] lying on a line has $-\frac{d-2}{2} + 1$ ([3, p. 422] prints "(1.4) and (1.15)" for (1.14) and (1.15); [3, (2.1)] itself has (1.14) and (1.15)); a differentiation acting on a line of the graph has -1; each factor η has $+1$; external fields have 0. For a connected subgraph H (a set of lines with their end vertices) $D(H) := \sum_{v \in H} D_H(v) - d$, where $D_H(v)$ counts only the legs of v which lie on lines of H. The degree $D(v)$ of a vertex is $D_H(v)$ with all its legs on lines. A mass vertex with a counterterm of δm^2_k is counted by the convention [3, (2.3)], through its attached graph.

H.2.1. *S6: generalised graphs,* [3, Prop. 2.2]. After the proof of [3, Prop. 2.1] the text says ([3, p. 428]): "It is easily seen from the proof that the theorem can be generalized to a much wider class of graphs and expressions. We can have vertices with an arbitrary number of legs and arbitrary power of η. We can have lines with the same exponential factors but with arbitrary dimensions instead of $-d + 2$. The only thing which matters is that propagators have representations corresponding to (2.6) with the estimates corresponding to (2.10)–(2.12), so that we have the inequality (2.13) with the proper generalization of (2.14)." The induction step of [3, Prop. 2.1] itself is "We can prove (2.16) for $i+1$ repeating the same reasoning as in the proof of (2.15)." Both are written out here once, for the generalised class. What is used later is: the sum over indices and localisations of the bound function is $O(1)$, with $O(1)$ depending on the number of lines, the decay rate and the minimal positive degree only, and, with the top index held fixed, the bound carries a factor $(L^{j_*}\eta)^{D(G)}$ without a sign condition on $D(G)$ ([3, (3.4)–(3.5)]).

Definition H.33 (generalised bound function)**.** The data are: a finite connected multigraph G (self-loops allowed) with vertex set $\mathcal{V}$, line set $\mathcal{L}$, $m := |\mathcal{L}| \ge 1$, and $N_G :=$ the largest number

of line-ends at a vertex of any graph obtained from G by contracting lines ($N_G \le 2m$); reals ω_l ($l \in \mathcal{L}$, the line dimension) and p_v ($v \in \mathcal{V}$, the vertex power); localisation boxes $\square(v)$, each a union of unit cubes of the integer grid with $|\square(v)| \le 2^d$ ([3, p. 420]: unit cubes, or cubes of side 2 for the partition-of-unity localisations; a side-2 support which is not grid-aligned is replaced by the at most 3^d unit cubes meeting it, which changes constants only); an index assignment $j = (j_l)$, $j_l \in \{0, \dots, k-1\}$, with $s_l := L^{j_l}\eta$, $j(v) := \min\{j_l : l \text{ has an end at } v\}$ and $s(v) := L^{j(v)}\eta$; a localisation $\Delta = (\Delta(v))_v$, $\Delta(v)$ a cube of the grid $s(v)\mathbb{Z}^d$ contained in $\square(v)$ (possible since $s(v) = L^{j(v)-k}$ divides 1); and a rate $\delta_0 > 0$ ([3, (2.14)] has $\delta_0 = \delta_1/2$). The bound function is

$$\tilde{E}(G(j), \Delta) := \prod_{v \in \mathcal{V}} s(v)^{d+p_v} \prod_{l \in \mathcal{L}} s_l^{\omega_l} \exp\big[-\delta_0 s_l^{-1} \operatorname{dist}(\Delta(v_l), \Delta(v_l'))\big],$$

v_l, v_l' the ends of l and dist the sup-distance of sets. The degree of a subgraph H (a set $\mathcal{L}_H$ of lines with their end vertices $\mathcal{V}_H$) is $D(H) := \sum_{v \in \mathcal{V}_H}(d + p_v) + \sum_{l \in \mathcal{L}_H} \omega_l - d$.

Lemma H.34 (Bałaban's graphs are instances)**.** (a) *For a graph of* [3, Sect. 1] *with lines* $G^{\eta}_{(j)}$ *(scalar or vector) put, for each line* l, $\omega_l := -(d-2) - (\text{number of differentiations acting on } l) + (\text{number of ends of } l \text{ which are vector legs of (1.14)/(1.15) vertices})$, *and for each vertex* $p_v :=$ *the number of factors* η *in* v *other than the* η^d *of its summation. Then the bound* [3, (2.14)] *is* $\tilde{E}(G(j), \Delta)$ *with* $\delta_0 = \delta_1/2$, *and* $D(H)$ *of definition H.33 is the degree* [3, (2.1)–(2.2)]. (b) $p_v \ge 0$ *for every vertex* [3, (1.6)–(1.15)]*:* [3, (1.8)] *has* $\eta^{n+n'-1}$ *with* $n + n' \ge 1$, [3, (1.9)] $\eta^{n+\bar{n}}$, [3, (1.10)] $\eta^{n+n'-2}$ *with* $n + n' \ge 2$, [3, (1.11)] $\eta^{n+\bar{n}-1}$, *and* [3, (1.6), (1.7), (1.13)–(1.15)] *none.*

Proof. (a) In [3, (2.14)] each vertex carries $(L^{j(v)}\eta)^d$ and "a proper power of $L^{j(v)}\eta$"; the latter is the bound $\eta^{p_v} \le s(v)^{p_v}$, valid because $p_v \ge 0$ by (b) and $\eta \le s(v)$. Each leg of l carries $s_l^{-(d-2)/2}$, each differentiation on l carries s_l^{-1}, each vector leg of l at a (1.14)/(1.15) vertex carries s_l ([3, (2.12)]), and each line carries $\exp[-\frac{1}{2}\delta_1 s_l^{-1} \operatorname{dist}(\Delta(v), \Delta(v'))]$; collecting the factors of l gives $s_l^{\omega_l}$. For the degree, sum [3, (2.1)] over $v \in \mathcal{V}_H$: the η-factors give $\sum(d + p_v)$, each line of H contributes its two leg dimensions, minus its differentiations, plus the (1.14)/(1.15) bonus, that is ω_l, and legs on lines not in H do not enter. (b) is read off [3, (1.8)–(1.15)]. □

Lemma H.35 (one shrinking step)**.** *Fix an ordering* $l(1), \dots, l(m)$ *of* $\mathcal{L}$ *and let* $J(l)$ *be the set of index assignments with* $j_{l(1)} \le \cdots \le j_{l(m)}$. *For* $0 \le i \le m$ *let* G_i *be the subgraph formed by* $l(1), \dots, l(i)$, *with connected components* $G_i^{(\alpha)}$, *and* G/G_i *the graph* G *with each* $G_i^{(\alpha)}$ *contracted to one vertex. For a vertex* u *of* G/G_i *put* $p_u := p_v$ *if* u *is an original vertex* $v \notin G_i$ *and* $p_u := 0$ *if* u *is a contracted component;* $\square(u) :=$ *the box of a representative original vertex of* u *(when two vertices are merged below, the merged vertex keeps the representative of the end which is not summed);* $j(u)$, $s(u)$, $\Delta(u)$ *as in definition H.33 with the lines of* G/G_i *only. For* $0 \le i < m$ *and* $j \in J(l)$ *let*

$$F_i(j, \Delta) := \prod_{\alpha} s_{l(i+1)}^{D(G_i^{(\alpha)})} \prod_{u} s(u)^{d+p_u} \prod_{l \notin G_i} s_l^{\omega_l} \exp\big[-\delta_i s_l^{-1} \operatorname{dist}(\Delta(u_l), \Delta(u_l'))\big], \qquad \delta_i := \delta_0 (2N_G)^{-i},$$

Δ *ranging over the localisations of the vertices of* G/G_i *(at* $i = 0$ *there are no components and* $F_0 = \tilde{E}$*). Assume*

$$\text{(PD)} \qquad D(G_i^{(\alpha)}) \ge \theta > 0 \qquad \text{for every } 1 \le i \le m-1 \text{ and every } \alpha.$$

Then for $0 \le i \le m-2$ *and fixed* $j_{l(i+2)}, \dots, j_{l(m)}$,

$$\sum_{j_{l(i+1)} \le j_{l(i+2)}} \sum_{\Delta} F_i(j, \Delta) \;\le\; c_{i+1} \sum_{\Delta'} F_{i+1}(j, \Delta'), \qquad c_{i+1} := c_d(\delta_i/2)\, e^{d\delta_i}\, (1 - L^{-\theta})^{-1}, \quad c_d(r) := \sum_{n \in \mathbb{Z}^d} e^{-r(|n|_\infty - 1}$$

Proof. Write $l := l(i+1)$, $j := j_l$, $s := s_j$. Since j is the smallest index among the lines of G/G_i, every end u of l has $j(u) = j$ and $s(u) = s$.

Step 1: l joins two distinct vertices $u \ne u'$ of G/G_i. (i) *Moving the other lines of u'.* Let $l' \ne l$ be a line of G/G_i with an end at u' and other end u'', and put $a := \operatorname{dist}(\Delta(u), \Delta(u'))$,

$b := \operatorname{dist}(\Delta(u'), \Delta(u''))$, $c := \operatorname{dist}(\Delta(u), \Delta(u''))$. The sup-diameter of $\Delta(u')$ is s, so $c \le a + s + b$. First lower the rate, $e^{-\delta_i b/s_{l'}} \le e^{-\delta_{i+1} b/s_{l'}}$ $(b \ge 0)$, then insert $b \ge c - a - s$ and use $s_{l'} \ge s$:

$$e^{-\delta_i b/s_{l'}} \le e^{-\delta_{i+1} c/s_{l'}}\, e^{\delta_{i+1} a/s}\, e^{\delta_{i+1}}.$$

(The order matters: the triangle inequality at the rate δ_i would cost $e^{\delta_i a/s}$ per moved line, which half of the exponential of l does not pay.) There are at most $N_G - 1$ such l', and their total cost $e^{(N_G-1)\delta_{i+1} a/s} \le e^{(\delta_i/2)a/s}$ is paid by half of the exponential of l; this is Bałaban's "we represent the exponential factor for the line $l(1)$ as a product of as many equal factors as there are legs in v"' [3, p. 427]. If $u'' = u$, the line l' becomes a self-loop after the merge and its exponential is bounded by 1. (ii) *Summing* $\Delta(u')$. The remaining factor $e^{-(\delta_i/2)a/s}$ gives $\sum_{\Delta(u')} e^{-(\delta_i/2)\operatorname{dist}(\Delta(u),\Delta(u'))/s} \le c_d(\delta_i/2)$, uniformly in s, because the cubes of the grid $s\mathbb{Z}^d$ are at sup-distance $s(|n|_\infty - 1)_+$ from $\Delta(u)$. (iii) *Relocalising the merged vertex.* Let β be the component of G_{i+1} containing l. Every line of G/G_{i+1} at β has index $\ge j_{l(i+2)} \ge j$, so $s(\beta) \ge s$, and the cube $\Delta(\beta)$ of side $s(\beta)$ containing $\Delta(u)$ lies in $\square(u)$. For the lines at β, $\exp[-\delta s_{l''}^{-1} \operatorname{dist}(\Delta(u), \cdot)] \le \exp[-\delta s_{l''}^{-1} \operatorname{dist}(\Delta(\beta), \cdot)]$ because $\Delta(u) \subset \Delta(\beta)$. The summand then depends on $\Delta(u)$ only through $s(u)^d = s^d$, and $\sum_{\Delta(u)\subset\Delta(\beta)} s^d = s(\beta)^d$ is the volume factor of β in F_{i+1}. (iv) *Powers of s.* Not yet used are $s^{d+p_{u'}}$ ($p_{u'} = 0$ if u' is contracted), s^{p_u}, s^{ω_l} and the prefactors $s^{D(\alpha)}$ of the components merged into β. Their product is $s^{D(\beta)}$, because $D(\beta) = D(\alpha_u) + D(\alpha_{u'}) + d + \omega_l$, where $D(\alpha_w)$ is the degree of the component w if w is contracted and $D(\alpha_w) := p_w$ if w is an original vertex (a one-vertex "component" without lines has degree $(d + p_w) - d$): this is the additivity of the degree of definition H.33 under adding one line.

Step 2: l is a self-loop at a vertex u of G/G_i. There is no summation; only (iii) (with $\Delta(u)$ itself) and (iv) apply, now with $D(\beta) = D(\alpha_u) + \omega_l$, the factor s^{d+p_u} supplying the volume $s(\beta)^d$ after relocalisation and s^{p_u} going into the degree.

Step 3: untouched components. A component α of G_i not touched by l is a component of G_{i+1}; its prefactor satisfies $s_{l(i+1)}^{D(\alpha)} \le s_{l(i+2)}^{D(\alpha)}$, since $j_{l(i+1)} \le j_{l(i+2)}$ and $D(\alpha) > 0$ by (PD).

Step 4: the index sum. After Steps 1–3 the only dependence on $j_{l(i+1)}$ is $s^{D(\beta)}$ with $D(\beta) \ge \theta$ (β is a component of G_{i+1}, $i + 1 \le m - 1$), and $\sum_{j=0}^{j_{l(i+2)}} (L^j\eta)^{D(\beta)} \le (1 - L^{-\theta})^{-1} (L^{j_{l(i+2)}}\eta)^{D(\beta)}$, the prefactor of β in F_{i+1}. Lowering all other rates from δ_i to δ_{i+1} only increases the bound. The factors collected are $c_d(\delta_i/2)$, at most $e^{N_G\delta_{i+1}} \le e^{\delta_i}$ from Step 1(i), and the geometric sum; $e^{d\delta_i} \ge e^{\delta_i}$ covers the second. □

Theorem H.36 ([3, Prop. 2.2], with the top index free or fixed)**.** *Assume* (PD). (i) *For each fixed $j_* \in \{0, \dots, k-1\}$,*

$$\sum_{j \in J(l),\, j_{l(m)} = j_*} \sum_{\Delta} \tilde{E}(G(j), \Delta) \le C_A\, (L^{j_*}\eta)^{D(G)},$$

with no sign condition on $D(G)$. (ii) *If in addition $D(G) \ge \theta$, then $\sum_{j\in J(l)} \sum_\Delta \tilde{E}(G(j), \Delta) \le C_A(1 - L^{-\theta})^{-1}$.* (iii) *Summed over the $m!$ orderings (the sets $J(l)$ cover all index assignments) the same bounds hold with an extra factor $m!$. Here $C_A := 2^d \prod_{i=0}^{m-2} c_{i+1} \cdot c_d(\delta_{m-1}/2) e^{d\delta_{m-1}}$ depends only on d, L, δ_0, θ and m (through $N_G \le 2m$); it is independent of η, k, the boxes $\square(v)$, the torus and the domain.*

Proof. Apply lemma H.35 for $i = 0, \dots, m-2$. At the last line $l(m)$ (index j_*, scale s_*) do Steps 1–2 of lemma H.35 once more: afterwards G/G_m is a single vertex $\beta = G$ without lines, with the factor $s_*^{D(G)}$ (Step 1(iv) with $\beta = G$) and the volume s_*^d of the representative vertex, whose sum over Δ inside $\square$ is $|\square| \le 2^d$. That is (i). For (ii), sum (i) over $j_* \le k - 1$: $\sum_{j_*} (L^{j_*}\eta)^{D(G)} \le (1 - L^{-\theta})^{-1} (L^{k-1}\eta)^{D(G)} \le (1 - L^{-\theta})^{-1}$, since $L^{k-1}\eta = L^{-1} < 1$. (iii) is the finite union over orderings. □

Remark H.37 (scope of theorem H.36). (a) For Bałaban's own lines the hypothesis of [3, Prop. 2.2], the generalised [3, (2.13)–(2.14)], is the slice bounds of appendix H.1.2; for subgraphs [3, (2.4)] the summation by parts of appendix H.1.6 is applied first, as on [3, p. 426]. The induction

[3, (2.16)] itself (S5) is the instance $\omega_l = -(d-2)$, p_v as in lemma H.34. (b) The proof uses lines only through $s_l^{\omega_l}$ and the exponential, and vertices only through $s(v)^{d+p_v}$; so "arbitrary dimensions" (ω_l real) and "arbitrary number of legs" are free. A line of dimension 0 whose kernel is bounded by $e^{\delta L\sqrt{d}}e^{-\delta L\,\mathrm{dist}}$ (the contact joining the two factors of one product [3, (1.18)] at the top scale $s = 1/L$, convention (CV) of Step 3(D_3) in appendix F) is such a line. A Hölder factor $|x-x'|^\alpha$ between two vertex positions enters as $\omega_l \to \omega_l + \alpha$ with δ_0 halved, by $|x-x'|^\alpha e^{-\delta|x-x'|/s} \le (2\alpha/(e\delta))^\alpha s^\alpha e^{-\delta|x-x'|/(2s)}$. (c) "Arbitrary power of η" means $p \ge 0$: the bound $\eta^p \le s(v)^p$ which turns a vertex factor into [3, (2.14)] needs $p \ge 0$, and for $p < 0$ the conclusion of theorem H.36 does not control η^p (which can be as large as $L^{k|p|}$). Vertex factors which are powers of the vertex scale $s(v)$ (any real exponent) are also covered, since they enter $s(v)^{d+p_v}$ directly. This covers every use in [3]: lemma H.34(b) for [3, (1.6)–(1.15)], and in [3, Sect. 3] the new vertices carry powers of the lowest external index $(L^{j_0}\eta)^{\cdots}$ (for instance [3, (3.30), p. 442]), that is of $s(v)$. (d) A mass vertex [3, (1.7)] with a counterterm of δm_k^2 from a graph G_0 has degree $2 + D(G_0)$ by [3, (2.3)], and $D(G_0)$ can be -1; its bound is not of the form η^p with $p \ge 0$. Such a vertex is estimated through its attached graph, whose lines have their own indices [3, (1.27)–(1.28)]; that bookkeeping is part of the renormalisation step (Step 3(D_4) of appendix F), not of theorem H.36. Thus graphs containing δm_k^2 vertices are not instances of theorem H.36 as single graphs (their attached G_0 is divergent, so (PD) fails for it); only their classes are bounded.

H.2.2. *S7: power counting,* [3, (2.17)] *and* [3, Cor. 2.3]. [3, pp. 428–429] states: a case analysis for the vertices [3, (1.13)–(1.15)] ("analyzing more carefully the possibilities we arrive at the conclusion $D(G) > 0$ again"); "if a graph G has an R-vertex ... then it has positive degree"; for graphs without [3, (1.13)–(1.15)] vertices the estimate

$$D(G) \;\ge\; -2\tfrac{d-2}{2} - 1 + (V-1)\tfrac{4-d}{2} + (E-1)\tfrac{d-2}{2}, \tag{2.17}$$

"from which it follows that the graphs with more than four external legs have positive degree in $d = 3$"; and [3, Cor. 2.3]: "If a graph G has a vertex of the form (1.13)–(1.15), or an R-vertex, or a vertex (1.7) with finite counterterms, or it has more than four external legs, then $D(G) > 0$." What is used afterwards ([3, pp. 429–432]) is the classification of the graphs with $D(G) \le 0$, corollary H.43 below.

Lemma H.38 (vertex degrees)**.** (a) $D(v) = 4 - d$ *for* [3, (1.6)]*;* $D(v) = 2$ *for* [3, (1.7)] *with a finite counterterm (whose coefficient has degree* 0 *by the convention of* [3, p. 423]*) and* $D(v) = 2 + D(G_0)$ *for* [3, (1.7)] *with a counterterm of* δm_k^2 *attached to the graph* G_0 [3, (2.3)]*;* $D(v) = n\frac{4-d}{2} + n'$ *for* [3, (1.8)–(1.11)] *($n' = \bar{n} + 1$ for the R-vertices);* $D(v) = \frac{d+2}{2} + n\frac{4-d}{2}$ *for* [3, (1.13)–(1.15)]. (b) *For v not of type (1.14)/(1.15),* $D_G(v) = D(v) + e_v\frac{d-2}{2} + \delta_{\mathrm{ext}}(v) \ge D(v)$*, with e_v the number of external legs of v and $\delta_{\mathrm{ext}}(v)$ the number of its differentiations acting on them. For v of type (1.13)–(1.15),* $D_G(v) = d - [\phi' \text{ on a line}]\frac{d-2}{2} + n_{\mathrm{int}}(v)\frac{4-d}{2} \ge \frac{d+2}{2}$*, with $n_{\mathrm{int}}(v)$ the number of its A' legs on lines.* (c) *At $d = 3$: $D(v) \ge \frac{1}{2}$ for every vertex not of type (1.13)–(1.15), $D_G(v) \ge \frac{1}{2}$ for every vertex of every graph, and $D_G(v) \ge \frac{5}{2}$ if v is of type (1.13)–(1.15). Moreover $D(G_0) \ge -1$ for every counterterm graph G_0 of δm_k^2, so a δm_k^2 vertex has $D(v) \ge 1$.*

Proof. (a), (b) follow from [3, (2.1)]: count η^d and the explicit η-powers, the legs ($-\frac{d-2}{2}$ each, $+1$ more for A' legs of (1.14)/(1.15)) and the differentiation of [3, (1.8)]/[3, (1.9)]; a leg not on a line has dimension 0 and its differentiation is not counted. For (1.13)–(1.15) a leg made external removes $-\frac{d-2}{2}$ (ϕ') or $+\frac{4-d}{2}$ (A'), whence the separate formula. (c) The first part is (a), (b) at $d = 3$ with $n + n' \ge 1$ for [3, (1.8)] and $n + n' \ge 2$ for [3, (1.10)]. For $D(G_0)$: G_0 is one-particle-irreducible with two external legs [3, pp. 416–417], built of [3, (1.6), (1.7), (1.8), (1.10)] with $n' = 0$; so $\ell(G_0) \ge 1$ (a tree has bridges), and lemma H.39 below, applied to G_0 (by induction on the order for its own counterterm vertices), gives $D(G_0) = \ell(G_0) + 2 - 4 + \sum\kappa + \Delta_{\mathrm{ext}} \ge -1$. □

The vertex (1.13)–(1.15) is read as Bałaban's one-scalar-leg vertex ([3, p. 423 (v)], $n+1$ legs). Under the alternative reading, in which the two factors of a product [3, (1.14)]·[3, (1.15)] at a

common y form one bilinear vertex, every statement $D > 0$ below still holds, because each factor carries its own $\sum_{x\in B^k(y)} \eta^d$ [3, p. 414], that is $D_G \geq \frac{5}{2}$ per factor; only $D > 0$ is used.

Lemma H.39 (the exact power-counting identity)**.** *Let G have no vertex of type (1.13)–(1.15), and put $\kappa_v := D(v) - (l_v - 2)\frac{4-d}{2}$, so that $\kappa = 0$ for* [3, (1.6)]*, $\kappa = n'$ for* [3, (1.8)–(1.11)]*, $\kappa = 2$ for* [3, (1.7)] *with a finite counterterm, and $\kappa = 2 + D(G_0)$ (≥ 1 at $d = 3$) for* [3, (1.7)] *with δm_k^2. Then*

$$D(G) = (4-d)\,\ell + E - 4 + \sum_v \kappa_v + \Delta_{\text{ext}}.$$

At $d = 3$, $D(G) = \ell + E - 4 + \sum \kappa + \Delta_{\text{ext}}$ with every term on the right nonnegative except -4.

Proof. By lemma H.38(b), $D(G) = \sum_v D_G(v) - d = \sum_v D(v) - d + E\frac{d-2}{2} + \Delta_{\text{ext}}$ (the counterterm degrees are inside $D(v)$ by [3, (2.3)]). Counting leg-ends, $\sum_v (l_v - 2) = 2I + E - 2V$, so $\sum_v D(v) = (2I + E - 2V)\frac{4-d}{2} + \sum \kappa$. Hence $D(G) = (I - V)(4 - d) + E + \sum \kappa - d + \Delta_{\text{ext}} = (\ell - 1)(4 - d) + E - d + \sum \kappa + \Delta_{\text{ext}}$. The values of κ follow from lemma H.38(a) with $l_v = 4$ for (1.6), 2 for (1.7) and $n + 2$ for (1.8)–(1.11). □

Lemma H.40 ((2.17) and its use)**.** *Let G have no vertex of type (1.13)–(1.15), $d \in \{2, 3\}$.* (a) *$D(G) \geq V\frac{4-d}{2} + E\frac{d-2}{2} - d$, and the right side equals the right side of (2.17) identically.* (b) *At $d = 3$ and $(V, E) = (1, 5)$, (2.17) gives only $D(G) \geq 0$, so the printed inference "graphs with more than four external legs have positive degree" does not follow from (2.17) in this one case; it does follow from lemma H.39: $E \geq 5$ implies $D(G) \geq 1$ (and at $V = 1$ even $D(G) \geq 2$, since then $\ell = I \geq 1$).*

Proof. (a) $D(G) = \sum D_G(v) - d \geq \sum D(v) + E\frac{d-2}{2} - d$ by lemma H.38(b), and $D(v) \geq \frac{4-d}{2}$ for every vertex ((a) of lemma H.38 at $d \in \{2, 3\}$; for a δm_k^2 vertex $2 + D(G_0) \geq 1 \geq \frac{4-d}{2}$ by the identity of lemma H.39 applied to G_0). Expanding, $-(d-2) - 1 + (V-1)\frac{4-d}{2} + (E-1)\frac{d-2}{2} = V\frac{4-d}{2} + E\frac{d-2}{2} - d$. (b) Insert $V = 1$, $E = 5$, $d = 3$: $-1 - 1 + 0 + 2 = 0$. The rest is lemma H.39 with $\kappa \geq 0$. □

Lemma H.41 ([3, Cor. 2.3], with its exact scope)**.** *Let $d = 3$ and G connected with at least one line.*

(a) *If G has a vertex of type (1.13)–(1.15), then $D(G) \geq 1$.*
(b) *If G has an R-vertex* [3, (1.9)] *or* [3, (1.11)] *and $\bar{n} \geq 3$, then $D(G) \geq \bar{n} - 2 \geq 1$ ([3] takes $\bar{n} \geq 12$,* [3, p. 421]*).*
(c) *If G has a vertex* [3, (1.7)] *with a finite counterterm, then: $E \geq 2$ implies $D(G) \geq 1$; $E = 1$ implies $D(G) \geq 0$, with $D(G) = 0$ exactly for one graph G_{tad}, the vertex $(1.7)_{\text{fin}}$ and a vertex* [3, (1.8)] *with $n = 1$, $n' = 0$ joined by two scalar lines, the A' leg external; $E = 0$ allows $D(G) = -1$ (the single $(1.7)_{\text{fin}}$ vertex with its two legs contracted, $D = 2 - d$) and $D(G) = 0$. So the third clause of* [3, Cor. 2.3] *holds for $E \geq 2$ and fails as printed for vacuum graphs and for G_{tad}.*
(d) *If $E \geq 5$, then $D(G) \geq 1$.*

Proof. Degrees lie in $\frac{1}{2}\mathbb{Z}$ at $d = 3$. Let $\mathcal{V}_5$ be the set of (1.13)–(1.15) vertices and $\mathcal{V}'$ the rest.

(a) By lemma H.38(b) at $d = 3$, $D_G(v) = \frac{5}{2} + \frac{1}{2}e_v^\phi + \frac{1}{2}n_{\text{int}}(v)$ for $v \in \mathcal{V}_5$, where $e_v^\phi = 1$ if the ϕ' leg of v is external and 0 otherwise, and $D_G(v') \geq \frac{1}{2} + \frac{1}{2}e_{v'}$ for $v' \in \mathcal{V}'$. Put $E' := \sum_{\mathcal{V}'} e_{v'} + \sum_{\mathcal{V}_5} e_v^\phi$. Then $D(G) \geq -3 + \frac{5}{2}|\mathcal{V}_5| + \frac{1}{2}|\mathcal{V}'| + \frac{1}{2}E' + \frac{1}{2}\sum n_{\text{int}}$. If $|\mathcal{V}_5| \geq 2$, $D(G) \geq 2$. Let $\mathcal{V}_5 = \{v\}$. Every vertex of $\mathcal{V}'$ has 2 or 4 scalar legs and every scalar line uses two, while v has one ϕ' leg; so the number of external scalar legs, which are counted in E', is odd, and $E' \geq 1$. If $|\mathcal{V}'| \geq 2$: $D(G) \geq -3 + \frac{5}{2} + 1 + \frac{1}{2} = 1$. If $\mathcal{V}' = \emptyset$: all lines are self-loops at v, hence vector self-loops (v has one scalar leg), so $n_{\text{int}} \geq 2$ and the ϕ' leg is external: $D(G) \geq -3 + \frac{5}{2} + \frac{1}{2} + 1 = 1$. If $\mathcal{V}' = \{v'\}$: $D(G) = D_G(v) + D_G(v') - 3$. If $D(v') \geq 1$, then $D(G) \geq \frac{5}{2} + 1 + \frac{1}{2}E' - 3 \geq 1$. If $D(v') = \frac{1}{2}$, then v' is [3, (1.8)] with $n = 1$, $n' = 0$ (two scalar legs and one A' leg; lemma H.38(a)). If the ϕ' leg of v is external, v and v' are joined by a vector line only, so $n_{\text{int}} \geq 1$, both scalar legs of v' are external, and $D(G) \geq \frac{5}{2} + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} - 3 = 1$. If the ϕ' leg of v lies on the line to

v', the other scalar leg of v' is external and its A' leg is external ($e_{v'} = 2$) or joins v ($e_{v'} = 1$, $n_{\text{int}} = 1$); in both cases $D(G) \geq \frac{5}{2} + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} - 3 = 1$.

(b) If G has a vertex $w \in \mathcal{V}_5$, then with u the R-vertex $D(G) \geq D_G(w) + D_G(u) - 3 \geq \frac{5}{2} + (\bar{n}+1) - 3 = \bar{n} + \frac{1}{2}$, all other $D_G \geq \frac{1}{2}$ being positive. Otherwise lemma H.39 with $\kappa_u \geq \bar{n}+1$ and all other terms ≥ 0: a tree has $E \geq 2$ ($\sum(l_v - 2) = E - 2 \geq 0$), so $\ell + E \geq 1$ and $D(G) \geq 1 - 4 + \bar{n} + 1 = \bar{n} - 2$.

(c) If G has a vertex $w \in \mathcal{V}_5$, then with u the $(1.7)_{\text{fin}}$ vertex ($D_G(u) \geq D(u) = 2$), $D(G) \geq \frac{5}{2} + 2 - 3 = \frac{3}{2}$, which implies every clause. Otherwise lemma H.39 gives $D(G) \geq \ell + E - 2 + \sum_{v \neq u} \kappa_v + \Delta_{\text{ext}}$. $E \geq 3$ gives $D(G) \geq 1$. $E = 2$: $\ell \geq 1$ gives $D(G) \geq 1$; if $\ell = 0$, then $\sum(l_v - 2) = 0$, every vertex has two legs, every two-leg vertex has $\kappa \geq 1$ ((1.7): ≥ 1; (1.8) with $n = 0$: $\kappa = n' \geq 1$; (1.10) with $n = 0$: $\kappa \geq 2$; R-vertices: $\bar{n} + 1$), and $V \geq 2$, so $D(G) \geq 1$. $E = 1$: a tree is impossible ($\sum(l_v - 2) = -1$); $\ell \geq 2$ gives $D(G) \geq 1$; if $\ell = 1$, then $\sum(l_v - 2) = 1$, one vertex has three legs and all others two, and $D(G) = \sum_{v \neq u} \kappa_v + \Delta_{\text{ext}} = 0$ forces $\kappa_v = 0$ at every other vertex; no two-leg vertex has $\kappa = 0$, and the only three-leg vertex with $\kappa = 0$ is [3, (1.8)] with $n = 1$, $n' = 0$. So $V = 2$, $G = G_{\text{tad}}$, its external leg being the undifferentiated A' leg ($\Delta_{\text{ext}} = 0$; E_s is even since every vertex has 2 or 4 scalar legs and every scalar line uses two, so the external leg is a vector leg). $E = 0$: the one-vertex ring has $\ell = 1$ and $D = 1 - 4 + 2 = -1$; the pair $(1.7)_{\text{fin}}$, [3, (1.6)] joined by three lines has $\ell = 2$ and $D = 0$.

(d) If G has a vertex of type (1.13)–(1.15), (a) gives $D(G) \geq 1$. Otherwise lemma H.39 with $\kappa \geq 0$ (lemma H.38(c) for the δm_k^2 vertices) gives $D(G) \geq E - 4 \geq 1$. □

Lemma H.42 (the two exceptions are immaterial)**.** (a) *At background $\tilde{B} = 0$ (after the gauge transformation removing B_0, [3, p. 433]) every localised value of G_{tad}, and of every graph containing G_{tad} as a subgraph, is* 0. (b) *Vacuum graphs never rely on* [3, Cor. 2.3]*: by* [3, p. 432] *"all vacuum energy counterterms corresponding to these graphs are added" to every vacuum class, whatever its degree.*

Proof. (a) At $\tilde{B} = 0$ the scalar propagator is a scalar kernel times $\mathbf{1}_N$. The vertex [3, (1.7)] is $\phi' \cdot \delta m^2 \phi'$, where δm^2 is a matrix: [3, (1.22)–(1.23), pp. 416–417] contain q^2 and q^4, so δm^2 is an even polynomial $P(q^2)$ in q, real symmetric and commuting with q; this holds for δm^2_{fin} and $\delta m^2_{K,k}$ alike. The vertex [3, (1.8)] with $n + n' = 1$ is $[D\phi'(b) \cdot q\phi'(b_-)]A'_b$. Contracting the scalar loop gives a scalar kernel times $\operatorname{Tr}(qP(q^2)) = 0$, the trace of an antisymmetric matrix times a symmetric one commuting with it. The same trace occurs in every graph containing G_{tad}. (b) is the printed rule. □

At $\tilde{B} \neq 0$, before the cube step, G_{tad} need not vanish; it is not used there, because the one-vector-leg classes are formed "by graphs necessary to renormalize all divergent proper subgraphs" [3, p. 431] and the $\tilde{B}'$-expansion vertices carry $\kappa \geq 1$, which makes $D > 0$. By the same count, the statement of [3, p. 431] that graphs with "one external vector field leg ... do not introduce any new divergences" is also true only after the gauge transformation, by lemma H.42(a).

Corollary H.43 (the classification used in [3], $d = 3$)**.** *Let $D(G) \leq 0$, G not a vacuum graph and not G_{tad}. Then G has no vertex of type (1.13)–(1.15), no R-vertex and no finite-counterterm vertex, and $E \leq 4$; $E = 4$ only for the graph* [3, (2.4)]*; $E = 3$ only with $(E_s, E_v) \in \{(0,3), (2,1)\}$, and then $\ell \leq 1$; $E = 2$ only with $(E_s, E_v) \in \{(2,0), (0,2)\}$ and $V \leq 4$; $E = 1$ only with $E_v = 1$.*

Proof. The first statement is lemma H.41. $E = 4$: $D = \ell + \sum \kappa + \Delta_{\text{ext}} \leq 0$ forces $\ell = 0$, $\kappa = 0$ everywhere and $\Delta_{\text{ext}} = 0$. A tree with four legs has $\sum(l_v - 2) = 2$, and a $\kappa = 0$ vertex has $l_v \geq 3$ ((1.8) with $n \geq 1$, $n' = 0$: $n + 2$ legs; (1.10) with $n' = 0$: $n \geq 2$; (1.6): four legs); so $V \leq 2$, and $V = 2$ because G has a line. Both vertices are [3, (1.8)] with $n = 1$, $n' = 0$. A vector line between them would leave both differentiated scalar legs external ($\Delta_{\text{ext}} = 2$), so they are joined by a scalar line carrying both differentiations: this is [3, (2.4)]. $E = 3$: $D = \ell - 1 + \cdots \leq 0$ gives $\ell \leq 1$, and E_s is even. $E = 2$: lemma H.40(a) gives $D \geq \frac{V}{2} - 2$, and E_s is even. $E = 1$: E_s is even. □

H.2.3. *S8: the Ward–Takahashi identities, and the limit from periodic to free boundary conditions.* [3, pp. 430–431] prints "the basic identity" [3, (2.23)], its first-order differential [3, (2.24)]

and the examples [3, (2.25)–(2.28)], and continues: "Taking other functions F, or differentiating (2.24) to higher order in A, we can get all necessary Ward–Takahashi identities. They hold for free boundary conditions also by taking a limit of the identities with periodic boundary conditions." ("Free boundary conditions" is the infinite lattice, as in [4, p. 584].) The identities are consumed on [3, pp. 441–442], on $\xi\mathbb{Z}^d$ with $C^\xi = (-\Delta^\xi + 1)^{-1}$, after "we resum over proper orderings and j-indices, we replace the propagators ... by C^{ξ}": the square bracket of [3, (3.29)] is [3, (2.26)] with A concentrated at one point y and $\lambda(y') = \sum_{\mu'} g'(y) A'_{\mu'}(y)(y'_{\mu'} - y_{\mu'})$, a linear function whose gradient is the constant $g'(y)A'(y)$; the same zeroth-moment identity at the next order is "apply a proper Ward–Takahashi identity" in [3, (3.31)–(3.32)]. So the consumed form is: the identities on the infinite lattice, for full propagator lines, with a constant gauge gradient, for sums over all connected graphs of a fixed order (the vertex positions and the internal photon lines summed). This section proves exactly that.

Setting. X is the infinite lattice $\eta\mathbb{Z}^d$ or a torus $T_\ell := \eta\mathbb{Z}^d/\ell\mathbb{Z}^d$, $\ell \in \eta\mathbb{N}$. Fields $\phi\colon X \to \mathbb{R}^N$; q real antisymmetric; $U(a) := \exp(e_k \eta q a)$. B is a fixed real bond field (periodic on T_ℓ), A a bond field, $\langle \phi, -\Delta_{B+A}\phi\rangle := \sum_b \eta^{d-2} |U(B_b + A_b)\phi(b_+) - \phi(b_-)|^2$, and $H(A) := -\Delta_{B+A} + M^2$ with $M^2 > 0$. For $\lambda\colon X \to \mathbb{R}$ put $(d\lambda)(b) := (\lambda(b_+) - \lambda(b_-))/\eta$ and $(T_\lambda\phi)(x) := \exp(e_k q \lambda(x))\phi(x)$. A polynomial F in finitely many $\phi(x)$ is gauge invariant if $F(T_\lambda\phi) = F(\phi)$ for all λ; polynomials in $\phi(x)\cdot P(q)\phi(x)$, P a polynomial, are examples, and they include the Wick-ordered $:|\phi(x)|^4:$, $:|\phi(x)|^2:$ of [3, (2.24)–(2.28)] (the Wick matrix is a function of q, since every kernel below is built from the U) and the mass term $-\frac{1}{2}\delta m^2 :|\phi|^2:$ (δm^2 is an even polynomial in q, [3, (1.23)] and lemma H.42).

Lemma H.44 (exponential decay of the kernels)**.** *Let $H = -\Delta_B + M^2$ on X with any real bond field B, and let $C := H^{-1}$ have kernel $C(u,u')$ with respect to the weight η^d. There are $\theta, c > 0$ depending on d, η and M only (not on B, X or ℓ) such that $|C(u,u')| \le c\, e^{-\theta \operatorname{dist}_X(u,u')}$, dist_X the sup-distance on X.*

Proof. $H \ge M^2$ since $-\Delta_B \ge 0$. Fix u' and let $\rho(x) := \operatorname{dist}_X(x,u')$, which is 1-Lipschitz for the sup-distance. The conjugate $H_\theta := e^{\theta\rho} H e^{-\theta\rho}$ differs from H only in the hopping terms, each multiplied by $e^{\pm\theta(\rho(b_+)-\rho(b_-))} \in [e^{-\theta\eta}, e^{\theta\eta}]$; hence $\|H_\theta - H\| \le 2d\eta^{-2}(e^{\theta\eta} - 1) \le M^2/2$ for $\theta \le \theta_0(d,\eta,M)$, and H_θ is invertible with $\|H_\theta^{-1}\| \le 2/M^2$. Then $C(u,u') = e^{-\theta\rho(u)}(H_\theta^{-1})(u,u')e^{\theta\rho(u')}$ with $\rho(u') = 0$, and a kernel entry is bounded by η^{-d} times the operator norm. □

Lemma H.45 (gauge covariance)**.** *$H(A + d\lambda) = T_\lambda^{-1} H(A) T_\lambda$ on X, for every λ (finitely supported if $X = \eta\mathbb{Z}^d$; arbitrary on T_ℓ).*

Proof. With $\psi = T_\lambda\phi$: $U(B_b + A_b + d\lambda(b))(T_\lambda^{-1}\psi)(b_+) - (T_\lambda^{-1}\psi)(b_-) = e^{-e_k q \lambda(b_-)}[U(B_b + A_b)\psi(b_+) - \psi(b_-)]$, because $U(\eta\, d\lambda(b)) = \exp(e_k q(\lambda(b_+) - \lambda(b_-)))$ and all these matrices commute. The prefactor is orthogonal, so the quadratic forms agree, and M^2 commutes with T_λ. □

Lemma H.46 (the identities on a torus: exact, every order, connected form)**.** *On T_ℓ let $F_1, \dots, F_p$ be gauge-invariant polynomials and, for a multi-index β, $Z_\beta(A) := \int d\phi\, e^{-\frac{1}{2}\langle\phi, H(A)\phi\rangle} F^\beta(\phi)$, $F^\beta := \prod_r F_r^{\beta_r}$; let $Z(A,\tau) := \sum_\beta \tau^\beta Z_\beta(A)/\beta!$ as a formal power series in τ. Then $Z_\beta(A + d\lambda) = Z_\beta(A)$ for every β and λ. Consequently, for bond fields $A^{(1)}, \dots, A^{(n)}$ and every β, the coefficient*

$$W_{n,\beta}(A^{(1)}, \dots, A^{(n)}; \lambda) := \partial_s \partial_{t_1} \cdots \partial_{t_n} [\tau^\beta] \log Z\Big(\sum_i t_i A^{(i)} + s\, d\lambda, \tau \Big)\Big|_{t=s=0}$$

vanishes ($[\tau^\beta]$ denoting the coefficient of $\tau^\beta/\beta!$ in the formal logarithm). $W_{n,\beta}$ is a finite sum of connected graphs with lines $C := H(0)^{-1}$, vertices from the Taylor coefficients of $H(A)$ in A (the vertices [3, (1.8), (1.10)] with $g = 1$ at background B) at the bonds of $\operatorname{supp} A^{(i)}$, one vertex linear in $d\lambda$, and β_r vertices F_r. The identities [3, (2.25)–(2.28)] are the instances $(n,\beta) = (0,0)$, $(1,0)$, $(1,e_r)$ with $F = -\lambda_k \sum_{x\in\Delta} \eta^d :|\phi(x)|^4:$ and $B = 0$.

Proof. Finiteness. $H(A) \ge M^2$ for real A, and $\operatorname{Re} H(A) \ge M^2/2$ for A in a complex neighbourhood of the real bond fields; so each Z_β is a finite-dimensional Gaussian integral of a polynomial,

analytic in A there, and $Z_0 > 0$ on real A. (τ is a formal parameter only; $e^{\tau F}$ is never integrated.) *Invariance.* Substitute $\phi = T_\lambda^{-1}\psi$: the Jacobian is 1 (orthogonal), the quadratic form changes by lemma H.45, and F^β is unchanged; so $Z_\beta(A + d\lambda) = Z_\beta(A)$. *Logarithm.* Every coefficient of the formal series $\log Z(A,\tau)$ is a polynomial in the $Z_\beta(A)/Z_0(A)$, hence gauge invariant in A, and its ∂_s-derivative at $s = 0$ vanishes identically in t. *Graph form.* By Wick's theorem the A-Taylor coefficients at $A = 0$ of these coefficients are the connected Gaussian moments of the polynomials $\partial_A^a H(A)|_{A=0}$ and of the F_r: finite sums of connected graphs with lines C. *Instances.* [3, (2.25)] is $n = 0$; [3, (2.26)–(2.27)] is $n = 1$ at $B = 0$ (the two bubbles, the seagull loop and the η-loop); [3, (2.28)] adds one F-vertex. □

Lemma H.47 (free boundary conditions as the limit of the periodic ones)**.** *On $\eta\mathbb{Z}^d$ let $C_\infty := H(0)^{-1}$ (bounded on ℓ^2, $H \geq M^2$) with B bounded, let λ and the $A^{(i)}$ be finitely supported and the F_r polynomials in the $\phi(x)$, $x \in X_F$ finite. Define $W^\infty_{n,\beta}$ by the finite graph sum of lemma H.46 with C_∞ in place of C. Then $W^\infty_{n,\beta} = 0$.*

Proof. Finitely many positions. Every vertex of the graphs sits at a fixed position: the $A^{(i)}$-vertices at bonds of $\operatorname{supp} A^{(i)}$, the λ-vertex at bonds of $\operatorname{supp} d\lambda$, the F-vertices in X_F; no position is summed, because $H(A) - H(0)$ vanishes off $\operatorname{supp} A$. So W^∞ is a fixed polynomial in the finitely many numbers $C_\infty(u,u')$, $u,u' \in S$, S the union of the supports and their η-neighbours. *Embedding in a torus.* Choose $\ell > \operatorname{diam} S + 2\eta$ and B_ℓ periodic of period ℓ agreeing with B on a box of side ℓ around S (the periodisation of its restriction; it changes H only at distance $\geq \ell/2 - \operatorname{diam} S$ from S). The projection $\eta\mathbb{Z}^d \to T_\ell$ is injective on the convex hull of S, which contains $\operatorname{supp}\lambda$ (a finitely supported λ has its support in the hull of $\operatorname{supp} d\lambda$), so λ, the $A^{(i)}$ and the F_r lift to T_ℓ, and lemma H.46 gives $W^{T_\ell} = 0$. *Convergence of the kernels.* Let $\tilde C_\infty := (-\Delta_{B_\ell} + M^2)^{-1}$ on $\eta\mathbb{Z}^d$. The torus kernel is the image sum $C_\ell(u,u') = \sum_{n\in\mathbb{Z}^d} \tilde C_\infty(u, u'+n\ell)$: the sum converges absolutely by lemma H.44, it is ℓ-periodic, and $(-\Delta_{B_\ell} + M^2)$ applied to it gives $\sum_n \delta_{u'+n\ell}$, so it solves the periodic equation on T_ℓ, whose solution is unique since $H_\ell \geq M^2 > 0$. By the second resolvent identity and lemma H.44, $|\tilde C_\infty - C_\infty|(u,u') \leq c' \exp(-\frac{\theta}{2}(\ell/2 - \operatorname{diam} S))$ on $S \times S$ (one half of each exponential pays the sum over the support of $B_\ell - B$), and the images $n \neq 0$ are $\leq c\, e^{-\theta(\ell - \operatorname{diam} S)}$ each, summable. *Limit.* Hence $C_\ell(u,u') \to C_\infty(u,u')$ for $u,u' \in S$ as $\ell \to \infty$, and $W^\infty = \lim_\ell W^{T_\ell} = 0$, W being a fixed polynomial in these entries. □

Lemma H.48 (the constant-gradient form)**.** *In lemma H.47 let $n \geq 1$ or $\beta \neq 0$, and write $W^\infty = \sum_{b'} \eta^d K(b')(d\lambda)(b')$, which is possible since W^∞ is linear in $d\lambda$. Then:* (a) $|K(b')| \leq Ce^{-\theta \operatorname{dist}(b', S_0)}$, *with S_0 the union of the supports of the $A^{(i)}$ and X_F;* (b) *for every $c \in \mathbb{R}^d$, $\sum_{b'} \eta^d K(b')\, c_{\mu(b')} = 0$, $\mu(b')$ the direction of b'. This is the zeroth-moment identity of* [3, (3.29)] *($c = g'(y)A'(y)$) and of* [3, (3.31)–(3.32)].

Proof. (a) $K(b')$ is the sum of the connected graphs with the λ-vertex at b'; all other positions lie in S_0 and its η-neighbourhood, and in each connected graph some line joins a neighbour of b' to S_0, so each monomial carries a factor $|C_\infty(u,u')| \leq ce^{-\theta|u-u'|}$ with u near b' and u' near S_0 (lemma H.44); the number of graphs is finite. (b) Put $\lambda(x) := \sum_\mu c_\mu x_\mu$ and $\chi_R(x) := \min\{1, \max\{0, 2 - |x|_\infty/R\}\}$. Then $\lambda_R := \chi_R\lambda$ is finitely supported, $(d\lambda_R)(b') = c_{\mu(b')}$ if both ends of b' lie in $\{|x|_\infty \leq R\}$, and everywhere $|d\lambda_R(b')| \leq |c|_\infty + 2|c|_1 + O(\eta/R)$, since $|\lambda| \leq 2R|c|_1$ on $\operatorname{supp}\chi_R$ and $|d\chi_R| \leq 1/R$. Hence $|\sum_{b'} \eta^d K(b')(c_{\mu(b')} - d\lambda_R(b'))| \leq C(|c|_1) \sum_{|b'|_\infty \geq R-\eta} \eta^d |K(b')| \to 0$ as $R \to \infty$ by (a), S_0 being fixed; and $\sum_{b'} \eta^d K(b')\, d\lambda_R(b') = 0$ for every R by lemma H.47. □

Lemmas H.47 and H.48 are stated for finitely supported $A^{(i)}$, a finite X_F and no photon integration. The uses need more: [3, (2.28)] has $F = -\lambda_k \sum_{x\in\Delta} \eta^d : |\phi(x)|^4 :$ summed over all cubes Δ of the class, and the degree-0 two-vector-leg classes of [3, (3.31)–(3.32)] ("all possible connected graphs … of a fixed order", [3, p. 431]) have $\ell = 2$ (lemma H.39), so they contain internal vertices and internal photon lines at all positions. The next lemma supplies this.

Lemma H.49 (summed internal positions and internal photon lines)**.** *Work on $\xi\mathbb{Z}^d$ (so $\eta = \xi$) at $B = 0$ with the scalar covariance $C^\xi = (-\Delta^\xi + M^2)^{-1}$ and the photon covariance $C^\xi_{\mathrm{ph}} =$*

$(-\Delta^\xi + M^2)^{-1} \otimes \mathbf{1}_d$, $M^2 = 1$ ([3, p. 442]: *"replace the propagators $G_{j_0}(0)$, G_{j_0} by C^ξ"*). *Fix finitely many external vector legs at bonds of a finite set S_0, gauge-invariant vertex densities f_r (polynomials in $\phi(x) \cdot P(q)\phi(x)$, for instance $-\lambda_k : |\phi|^4 :$ and $-\frac{1}{2}\delta m^2 : |\phi|^2 :$), and an order in e_k and in the f_r. Let W^{sum} be the sum, over all connected graphs of that order with full lines C^ξ (scalar) and C^ξ_{ph} (photon), of the graphs with one λ-vertex (linear in $d\lambda$), the external legs at S_0, the f_r-vertices summed over all $x \in \xi\mathbb{Z}^d$ with weight ξ^d, and the A-vertices of $H(A)$ (the vertices [3, (1.8), (1.10)] at $B = 0$) summed over all bonds, their A-legs either external (at S_0) or contracted in pairs by C^ξ_{ph}. Then:* (a) *for finitely supported λ, $W^{\mathrm{sum}} = 0$;* (b) *if $S_0 \neq \emptyset$, then writing $W^{\mathrm{sum}} = \sum_{b'} \xi^d K^{\mathrm{sum}}(b')(d\lambda)(b')$ one has $|K^{\mathrm{sum}}(b')| \leq Ce^{-\theta' \operatorname{dist}(b', S_0)}$ for some $\theta' > 0$, and $\sum_{b'} \xi^d K^{\mathrm{sum}}(b') c_{\mu(b')} = 0$ for every $c \in \mathbb{R}^d$.*

With $S_0 = \emptyset$ the class is translation invariant, $K^{\mathrm{sum}}(b')$ depends only on the direction of b', and (b) is not meaningful as written; (a) needs no such hypothesis. The uses on [3, p. 442], [3, (3.29), (3.31)–(3.32)], have external vector legs.

Proof. Step 1: the polynomial identity, pointwise in the photon field. On a torus T_ℓ ($\ell \in \xi\mathbb{N}$) let A be any real bond field. For real e_k near 0 the functions $e_k \mapsto Z_\beta(A; e_k)$ of lemma H.46 are smooth ($H(A) \geq M^2$), and their Taylor coefficients $P_{r,\beta}(A) := \frac{1}{r!}\partial^r_{e_k} Z_\beta(A; e_k)|_{e_k=0}$ are polynomials in the finitely many A_b (the expansion of $U(A) = \exp(e_k \xi q A)$ gives vertices with A-legs, each e_k^r coming with A^r). The invariance $Z_\beta(A + d\lambda; e_k) = Z_\beta(A; e_k)$ for all real e_k near 0 and all A, λ (lemma H.46, T_λ depending on e_k) gives $P_{r,\beta}(A + d\lambda) = P_{r,\beta}(A)$ as polynomial identities, for every r, β, A and λ on T_ℓ. *Step 2: integrating the photon.* Put $A = A_{\mathrm{ext}} + A_{\mathrm{int}}$ with A_{int} Gaussian of covariance C^ℓ_{ph}, the torus photon covariance at $M^2 = 1$ (a nondegenerate finite-dimensional Gaussian). The A_{int}-expectation of the identity of Step 1 (a polynomial in A_{int}, so a finite Wick sum) gives $\mathbb{E}[P_{r,\beta}(A_{\mathrm{ext}} + d\lambda + A_{\mathrm{int}})] = \mathbb{E}[P_{r,\beta}(A_{\mathrm{ext}} + A_{\mathrm{int}})]$, and by Wick's theorem $\mathbb{E}[P_{r,\beta}(\cdot)]$ is the sum over graphs with internal photon lines C^ℓ_{ph}. Every coefficient of the formal logarithm in (e_k, τ) is a polynomial in the $\mathbb{E}[P_{r,\beta}]/\mathbb{E}[P_{0,0}]$ ($P_{0,0} > 0$ is A-independent), so it is gauge invariant in A_{ext} and its mixed A_{ext}-$d\lambda$ derivatives vanish. These derivatives are the connected graphs of the class with scalar lines C^ℓ, photon lines C^ℓ_{ph} and all internal positions summed over T_ℓ (the f_r-vertices entering through $F_r = \sum_x \xi^d f_r(\phi(x))$). This is (a) on T_ℓ, with λ finitely supported and lifted as in lemma H.47. *Step 3: absolute convergence, uniformly in ℓ.* By lemma H.44, $|C^\xi(u,u')|, |C^\xi_{\mathrm{ph}}(u,u')| \leq ce^{-\theta|u-u'|}$ and $|C^\ell(u,u')|, |C^\ell_{\mathrm{ph}}(u,u')| \leq ce^{-\theta d_\ell(u,u')}$ uniformly in ℓ, d_ℓ the torus distance. Every graph of the class is connected and has a fixed vertex (an external leg at S_0, or the λ-vertex on the finite $\operatorname{supp} d\lambda$). Choose a spanning tree rooted at a fixed vertex and bound the non-tree lines by c; changing variables to the tree differences z_i (minimal representatives on T_ℓ) bounds each graph by $\prod_i ce^{-\theta|z_i|}$ times constants, summable over $(\xi\mathbb{Z}^d)^{\#\text{tree lines}}$ uniformly in ℓ. *Step 4: the limit $\ell \to \infty$.* For fixed z's all positions stay within a fixed distance of the root, so for ℓ large the torus distances equal the lattice distances and $C^\ell \to C^\xi$, $C^\ell_{\mathrm{ph}} \to C^\xi_{\mathrm{ph}}$ there (lemma H.47). By dominated convergence (Step 3) each graph converges to its value on $\xi\mathbb{Z}^d$, with absolutely convergent position sums; the number of graphs of the fixed order is finite. So $W^{\mathrm{sum}} = \lim_\ell W^{\mathrm{sum},T_\ell} = 0$, which is (a). *Step 5: (b).* As in lemma H.48(a): in each connected graph the λ-vertex at b' is joined to S_0 by a path of lines; summing the intermediate positions along this path with $\sum_z e^{-\theta|b'-z|}e^{-\theta|z-s|} \leq C_{\theta'}e^{-\theta'|b'-s|}$ ($\theta' < \theta$) and the other positions by Step 3 gives the decay of K^{sum}. The cut-off $\lambda_R = \chi_R\lambda$ of lemma H.48(b) and dominated convergence then give the zeroth-moment identity exactly as there. □

Remark H.50 (scope of the identities). (a) *Why the limit is needed.* On T_ℓ a linear λ is not a function on the torus, and a constant $d\lambda = c$ is a flat connection with holonomy, not gauge-trivial; so lemma H.48(b) is false on a torus in general, the defect being a sum over images winding around the torus, exponentially small in ℓM. Bałaban's "by taking a limit" is essential, and it is exactly lemmas H.47 to H.49: the torus identity with finitely supported λ, the limit, then the extension to linear λ on $\xi\mathbb{Z}^d$. (b) *Full lines only.* Lemmas H.46 to H.49 are identities between sums over all graphs of a given order with every line the full propagator. They say

nothing about a single scale assignment or a fixed ordering of slices; after the decomposition [3, (2.6)] the per-ordering, per-slice analogue is false. [3] applies [3, (2.26)] only after "we resum over proper orderings and j-indices, we replace the propagators $G_{j_0}(0)$, G_{j_0} by C^{ξ}" [3, p. 442], which is the full-line form; that this resummation and replacement is legitimate at the places of [3, Sect. 3] is part of the renormalisation step (Step 3(D_4) of appendix F), not of S8. (c) *Background and averaging terms.* Lemmas H.45 to H.48 hold at any background B; [3] uses $B = 0$, after the gauge transformation removing B_0. If $H(A)$ contains a covariant averaging term $a_k P_k(B + A)$ (the propagators G_k), lemma H.45 still holds, since $Q_k(A + d\lambda)T_\lambda^{-1} = T_\lambda^{\mathrm{unit}} Q_k(A)$ by transport along the contours; so the identities hold with the vertices [3, (1.13)–(1.15)] added. The printed and the consumed identities use C_{M^2}, which needs no such remark. (d) The constants of lemma H.49 depend on ξ and the order; only the limits $\ell \to \infty$ and $R \to \infty$ are taken, at fixed ξ, which is how the identities are used.

H.2.4. *S9: the estimate* [3, (3.1)] *for lines between separated localisations.* [3, p. 432]: "Now if two vertices v, v' have localizations satisfying $\mathrm{dist}(\square(v), \square(v')) \geq 1$, then we consider every propagator corresponding to a line connecting these vertices as an external field also. Such a possibility is assured by the following estimates

$$\|hG_k(\Omega, \tilde{B})h'\|_{1,\alpha} \leq O(1)e^{-\delta_0 \,\mathrm{dist}(\square(v),\square(v'))}, \tag{3.1}$$

and similarly for the vector field propagator, h, h' are localization functions." No argument is printed. What is used is (3.1) with $O(1)$ and δ_0 independent of ε, k, Ω, $\tilde{B}$ and the torus: it turns every line between separated localisations into an external field of exponentially small $(1, \alpha)$-norm, which with [3, (2.5)] and [3, (3.2)] gives [3, (1.33)]. In appendix F the estimate is applied to the cluster localisation functions h_K, $h_{K'}$ (Step 3(D_3)); proposition H.54 below covers them.

Definition H.51 (the two-variable norm)**.** For a kernel $f(x, x')$ on η-lattice points, an operation in one variable is one of $O_0 :=$ the identity, $O_1 := D^{\eta}_{\tilde{B},\mu}$ (a covariant forward difference), $O_2 :=$ the Hölder quotient of [3, (1.32)], $H_\mu f(x_1, x_2) := |x_2 - x_1|^{-\alpha}[U(\tilde{B}(\Gamma_{x_1 x_2}))(D_\mu f)(x_2) - (D_\mu f)(x_1)]$, $x_1 \neq x_2$; put $|O_0| = 0$, $|O_1| = 1$, $|O_2| = 1 + \alpha$, and

$$\|f\|_{1,\alpha} := \max_{(a,a')\in\{0,1,2\}^2,\ \mu,\mu'} \sup \big|(O_a \text{ in } x)(O_{a'} \text{ in } x')f\big|.$$

This is the strongest natural extension of [3, (1.32)] to two variables ([3, p. 420]: "extends in a natural way"), the reading of appendix H.1.1: every operation a vertex can apply to one leg of a two-leg external field, independently at the two legs. Any weaker reading follows from it; the summed form of [3, (1.32)] in two variables is at most 9 times the maximum above.

Input: the slice bounds. [3, (2.6)] writes $G_k(\Omega, \tilde{B}) = \sum_{j=0}^{k-1} G^{\eta}_{(j)}(\Omega, \tilde{B})$, and appendix H.1.2 proves constants $c_s, \delta_1 > 0$, independent of ε, k, j, Ω, $\tilde{B}$ and the torus, such that for every j, with $s := L^j\eta$, and all evaluation points at the depth required there (the line ends and their Hölder partners at depth $\geq R_0$ in Ω),

$$\big|(O_a \text{ in } x)(O_{a'} \text{ in } x')\, G^{\eta}_{(j)}(x, x')\big| \leq c_s\, s^{-(d-2)-|O_a|-|O_{a'}|} \exp\big(-\delta_1 s^{-1}\,\mathrm{dist}(X, X')\big), \tag{SL}$$

where $X = \{x\}$ or $\{x_1, x_2\}$ is the set of points used by O_a, and similarly X'. This is [3, (2.10)–(2.11)] read as stated there ("for each differentiation, there is an additional factor $(L^j\eta)^{-1}$ … This applies also to Hölder norms", [3, p. 426]), applied at each end; the two-ended Hölder case $(a, a') = (2, 2)$ is the mixed second difference of appendix H.1.2.

Lemma H.52 (summing the slices at a fixed separation)**.** *Let $\rho_0 > 0$, $P \geq 0$. For $\eta = L^{-k}$ and all $k \geq 1$, all $\rho \geq \rho_0$,*

$$\sum_{j=0}^{k-1} (L^j\eta)^{-P} \exp\big(-\delta_1\rho/(L^j\eta)\big) \leq K(P, \delta_1, \rho_0)\, e^{-(\delta_1/2)\rho}, \qquad K(P, \delta_1, \rho_0) := \Big(\frac{2P}{e\delta_1\rho_0}\Big)^P \Big(1 - e^{-\delta_1\rho_0(L-1)/2}\Big)^{-1},$$

independent of k. (Where the lemma is used with several P at once, K is read as the maximum over those P.)

Proof. Write $s_j = L^{j-k} = L^{-m}$ with $m := k - j \geq 1$, and split the exponential into two equal halves. With $u := \rho/s$, $s^{-P}e^{-\delta_1\rho/(2s)} = \rho^{-P}u^P e^{-\delta_1 u/2} \leq \rho^{-P}(2P/(e\delta_1))^P \leq (2P/(e\delta_1\rho_0))^P$. The remaining factors sum to $\sum_{m\geq 1}\exp(-\frac{\delta_1}{2}\rho L^m)$; since $L^m \geq 1 + m(L-1)$, this is $\leq e^{-\delta_1\rho/2}\sum_{m\geq 1}\exp(-\frac{\delta_1}{2}\rho_0(L-1)m)$, bounded by the geometric factor. □

Lemma H.53 (localisation functions)**.** *Let h, h' be real functions with supports in the w-neighbourhoods of $\square(v)$, $\square(v')$, where the collar w is fixed independently of η and k with $\eta \leq w \leq \frac{1}{4}$, $h = 1$ on $\square(v)$, $h' = 1$ on $\square(v')$, and $|h|$, $|d^\eta h|$ and the Lipschitz constant of $d^\eta h$ bounded by c_h (likewise h'); for Bałaban's smooth h with derivatives up to second order bounded* [3, pp. 412, 420]*, $c_h \sim w^{-2}$. Let $\tilde{B}$ satisfy $e_k\|d^\eta\tilde{B}\|_\infty \leq 1$ (the hypothesis (E) of appendix H.1.1 gives this). Then*

$$\|hfh'\|_{1,\alpha} \leq C(c_h, d) \max_{(a,a'),\mu,\mu'} \sup\big|(O_a)(O_{a'})f\big|,$$

the supremum over the points of $\operatorname{supp} h \times \operatorname{supp} h'$ *and their η-neighbours.*

Proof. Product rule. $D_{\tilde{B},\mu}(hf)(x) = (d^\eta_\mu h)(x)\,U(\tilde{B}_b)f(x+\eta e_\mu) + h(x)(D_\mu f)(x)$. *Hölder quotient of the first term.* It is controlled by the Lipschitz bound of $d^\eta h$ together with the transported difference $|U(\tilde{B}(\Gamma_{x_1x_2}))f(x_2+\eta e_\mu) - f(x_1+\eta e_\mu)|$. That difference is at most $|x_2-x_1|_1 \sup|Df|$ plus the change of transport between the two shortest contours, which by Stokes' theorem is at most $c(d)\,e_k\|d\tilde{B}\|_\infty|x_2-x_1|^2\sup|f| \leq c(d)|x_2-x_1|\sup|f|$ for $|x_2-x_1| \leq 1$ (the area enclosed by the two shortest contours is at most $c(d)|x_2-x_1|^2$). For $|x_2-x_1| > 1$ the quotient is at most $2\sup|D(hf)|$. If one of the two points, say x_1, lies outside $\operatorname{supp} h$, the term $h(x_1)(Df(x_2) - Df(x_1))$ vanishes, and the remaining terms are bounded by $|h(x_2)-h(x_1)|\sup|Df| \leq c_h|x_1 - x_2|\sup|Df|$ and by the dh-terms; all of them use f and Df only at points within $w+\eta$ of $\square(v)$. *The second term* is h times a quotient of f. *Both variables.* The same in x'. □

Proposition H.54 ([3, (3.1)], scalar lines)**.** *Let $L \geq 5$, $k \geq 1$, and let h, h' be as in lemma H.53 with $r := \operatorname{dist}(\square(v), \square(v')) \geq 1$. Assume (SL) at the points used (they lie within $w+\eta$ of $\square(v)$, $\square(v')$; the depth there is supplied by the hypotheses of appendix H.1.1). Then, with $\rho_0 := 1 - 2(w+\eta) > 0$,*

$$\|hG_k(\Omega, \tilde{B})h'\|_{1,\alpha} \leq C_9\, e^{-(\delta_1/2)(r-2(w+\eta))}, \qquad C_9 := C(c_h, d)\, c_s\, K(d+2\alpha, \delta_1, \rho_0),$$

with C_9 independent of ε, k, Ω, $\tilde{B}$ and the torus. So [3, (3.1)] *holds with $\delta_0 := \delta_1/2$ and $O(1) = C_9 e^{\delta_1(w+\eta)}$; for $w = \frac{1}{4}$, $\eta \leq \frac{1}{5}$ one has $\rho_0 \geq \frac{1}{10}$, and the bound reads $C_9 e^{-(\delta_1/2)(r-\frac{1}{2}-2\eta)}$, the form used for the cluster functions h_K, $h_{K'}$ of appendix F.*

Proof. By [3, (2.6)] the kernel is a sum of slices. The points X, X' used lie within $w+\eta$ of $\square(v)$, $\square(v')$, so $\operatorname{dist}(X, X') \geq r - 2(w+\eta) \geq \rho_0$. (SL) for each slice, with $P := (d-2) + |O_a| + |O_{a'}| \leq d + 2\alpha$, and lemma H.52 with $\rho = \operatorname{dist}(X, X')$ give, for each pair (a, a'), the bound $c_s K(P, \delta_1, \rho_0)e^{-(\delta_1/2)(r-2(w+\eta))}$ (Hölder pairs x_1, x_2 with $|x_1 - x_2| > 1$ are bounded by twice the derivative entry). Lemma H.53 multiplies by $C(c_h, d)$. The sum over j converges because the points are separated: at separation $\geq \rho_0$ the factors $(L^j\eta)^{-P}e^{-\delta_1\rho/(L^j\eta)}$ are uniformly summable in j, and the diagonal singularity of the mixed derivative never occurs. At $k = 0$ there is no slice sum. □

Proposition H.55 ([3, (3.1)], vector lines)**.** *The A' covariance is G_k on the torus* [3, p. 414]*, with the slice decomposition "the similar equality for the vector field propagator"* [3, p. 424]*; its slices satisfy* [3, (2.10)–(2.11)] *("the same inequalities hold for vector field propagators",* [3, p. 426]*) with no depth condition, and a leg in a vertex (1.14)/(1.15) is a contour sum $A'(\Gamma^{(k)}_{y,x})$ inside the unit block $B^k(y) \subset \square(v)$, whose slice carries the extra factor $L^j\eta$ of* [3, (2.12)] *(appendix H.1.2). The statement and proof of proposition H.54 then hold verbatim, with P lowered by 1 at a contour leg and with ordinary differences ($\tilde{B} = 0$) in definition H.51.*

Remark H.56 (collars and depth). (a) The vertex positions are sharp (unit cubes, side-2 supports), but the localisation functions h, h' of a cut line must be smooth for lemma H.53: with a sharp indicator ($w = 0$) the Hölder quotient of hf jumps at the edge of the box. Any fixed

collar $\eta \le w \le \frac{1}{4}$ works, since $h = 1$ on $\square(v)$, where all vertex positions lie; [3, p. 420] ("$h = 0$ outside some neighborhood of box(v)") leaves the collar unspecified, and the constants depend on it. For wider collars [3, (3.1)] holds for $r \ge 1 + 2w + 2\eta$ instead of $r \ge 1$, and lemma H.52 holds for any $\rho_0 > 0$. (b) The only depth used is that of (SL) at points within $w + \eta \le \frac{9}{20}$ of the two boxes; the vertex supports lie within 1 of Ω_1 or of supp $\tilde{A}$ (appendix H.1.1), so the depth hypothesis holds once the layer width satisfies $r(t) \ge R_0 + 5$, which the choice of R in [2, (2.7)] secures. The block vertex [3, (1.13)] never stands alone: at least one factor of each product [3, (1.18)] is (1.14) or (1.15) [3, p. 414], and those carry $g_k A'$ or $\tilde{A}$, so they are deep. (c) In contrast with [3, (2.5)], which bounds δG_k and [3, (1.16)] without any separation and therefore needs the two-domain theorem (S2), [3, (3.1)] needs only separated points and the slice bounds.

H.2.5. *Degrees after a line is cut.* [3, p. 432]: "if we replace a line in an arbitrary graph G by a pair of external fields, then the degree of the new graph is greater or equal to the degree of G, thus a convergent graph is transformed into a convergent one. From our analysis in previous chapters it follows that most of the divergent graphs are transformed into convergent ones, with the possible exception of graphs with one external vector field leg and vacuum graphs." The regrouping into classes which this sentence prepares is proved in theorem F.57; the three degree facts it uses are proved here. Throughout, a cut set S is a set of lines of G, and $G|_S$ denotes G with every line of S replaced by its two legs, which become external fields (the kernel of the line, an external field by propositions H.54 and H.55). Degrees are those of [3, (2.1)–(2.3)], $d = 3$ where stated.

Lemma H.57 (degrees are intrinsic)**.** (a) *The connected subgraphs of $G|_S$ are exactly the connected subgraphs H of G with no line in S, and $D(H)$ is the same whether computed in $G|_S$ or in G.* (b) *Hence, if every connected subgraph of G has positive degree (the hypothesis of* [3, Prop. 2.1]*, with or without the exception* [3, (2.4)]*), so does every connected subgraph of $G|_S$; and the divergent connected subgraphs of $G|_S$ are exactly the divergent connected subgraphs of G which avoid S.*

Proof. A subgraph is given by its lines and their end vertices. [3, (2.1)] for $D_H(v)$ counts only legs on lines of H, and a leg on a line of S is on no line of H, both in G and in $G|_S$. □

Lemma H.58 (the degree of the whole graph)**.** *Apply* [3, (2.2)] *verbatim to $G|_S$, with one $-d$ even if $G|_S$ is disconnected. Then $D(G|_S) - D(G) = \sum_{l \in S} \big[\sum_{\text{ends of } l} c_{\text{end}} + \delta_l \big]$, with $c_{\text{end}} = \frac{d-2}{2}$ for an ordinary leg, $c_{\text{end}} = \frac{d-4}{2}$ for a vector leg of a vertex (1.14)/(1.15), and δ_l the number of differentiations acting on l. So the printed sentence "the degree of the new graph is greater or equal to the degree of G" holds whenever no line of S ends at a vector leg of a (1.14)/(1.15) vertex, and at $d = 3$ the degree drops by $\frac{1}{2}$ per such end (two (1.14) vertices with $n = 2$, ϕ' external, joined by their two vector lines have $D = 5$; cutting one line gives $D = 4$). What the regrouping uses is lemma H.57, together with lemma H.41(a), which keeps every component containing a (1.13)–(1.15) vertex at $D \ge 1$.*

Proof. By lemma H.38(b), an ordinary leg leaving a line changes $D_G(v)$ by $+\frac{d-2}{2}$, a (1.14)/(1.15) vector leg by $-\frac{4-d}{2}$, and a no longer counted differentiation by $+1$. In the example, $D_G = 3 + 2 \cdot \frac{1}{2} = 4$ per vertex before and $3 + \frac{1}{2}$ after; with -3 once, 5 and 4. □

Lemma H.59 (cutting a divergent graph; $d = 3$)**.** *Let γ be connected without (1.13)–(1.15) vertices, with $E(\gamma) \ge 2$, and let S' be a set of $t \ge 1$ lines of γ whose removal leaves $\gamma|_{S'}$ connected (for $t = 1$: a line which is not a bridge; every line, if γ is one-particle-irreducible) and such that $\gamma|_{S'}$ still has a line. Then*

$$D(\gamma|_{S'}) = D(\gamma) + t + (\textit{differentiations on } S') \;\ge\; \max\{0, D(\gamma) + t\},$$

and $D(\gamma|_{S'}) = 0$ only if $t = 1$ and $\gamma|_{S'}$ is the graph [3, (2.4)]*. If $E(\gamma) \ge 3$, then $D(\gamma|_{S'}) \ge 1$.*

Proof. Exact change. By lemma H.39 for the connected $\gamma|_{S'}$: ℓ drops by t, E rises by $2t$, Δ_{ext} rises by the differentiations on S', and κ is unchanged. *Lower bound.* Since the removal of S' leaves a connected graph, γ has a cycle, $\ell(\gamma) \ge 1$ and $D(\gamma) \ge \ell + E - 4 \ge -1$; hence $D(\gamma|_{S'}) \ge 0$,

with equality only if $t = 1$, $D(\gamma) = -1$ and S' carries no differentiation. *Equality.* Then $\gamma|_{S'}$ has $\ell = 0$, $E = 4$, all $\kappa = 0$ and $\Delta_{\text{ext}} = 0$, so it is [3, (2.4)] by corollary H.43, which requires a line; this is where the last hypothesis enters (without it, a self-loop cut of a one-vertex γ leaves a bare vertex with four or two external legs and $D = 0$: the [3, (1.6)] tadpole and the [3, (1.10)] tadpoles with $n + n' = 2$). $E \geq 3$. $D(\gamma) \geq 1 + 3 - 4 = 0$, so $D(\gamma|_{S'}) \geq 1$. □

The cut of [3, p. 432] removes only lines joining distinct vertices whose localisations are at distance ≥ 1, never a self-loop, and a connected $\gamma|_{S'}$ with at least two vertices has a line; so the last hypothesis of lemma H.59 holds at every use. The outcome [3, (2.4)] arises from the e^2 self-energy with its photon line cut and from the one-loop vacuum polarisation with one scalar line cut; it needs no partner, since the summation by parts of appendix H.1.6 treats it. The other two possible outcomes of cutting a divergent subgraph, a vacuum graph and a one-vector-leg graph, are Bałaban's exception list: vacuum classes always receive their counterterms [3, p. 432], and the one-vector-leg loop through a finite-counterterm vertex vanishes after the gauge transformation by lemma H.42(a) (its coefficient is an even polynomial in q). The regrouping itself, with these inputs, is Step 3(D_3) of appendix F (theorem F.57).

H.3. **The sketched steps of Proposition 1, III: the component cube, the constant field, the expansion in the new background and the old-vertex components.** This part contains the steps of [3, pp. 432–433] that prepare the passage from the domain Ω to a cube $\square$ of constant side and from the background $\tilde{B}$ to a constant background B_0:

step	[3] page	printed object
S11	432	"the localizations of each connected component are contained in some cube of a linear size de
		a number of vertices, hence on $\bar{n}$ only"
S12	433	the cube split $G_k(\Omega, \tilde{B}) = G_k(\square, \tilde{B}) + \delta G_k(\Omega, \square, \tilde{B})$ (printed; only its three properties (P-i)–(
S13	433	the constant field B_0 with $\lvert\tilde{B} - B_0\rvert \leq O(r(L^k\varepsilon)p(L^k\varepsilon))$ on $\square$
S14	433	the expansion in $\tilde{B}' = \tilde{B} - B_0$ by [1, (3.14), (3.44), (3.45)], and its decomposition into renorn
S15	433	"If a class contains an old vertex ..., then all vertices of this class are localized in $\square_1$"

The order is S11 (appendix H.3.1), the properties of the cube split (appendix H.3.2), S13 (appendix H.3.3), the expansion S14 (appendices H.3.4 and H.3.5) and S15 (appendix H.3.6). Everything is at $L = 5$, $d = 3$, on the scale-k lattice with $k \geq 1$; the geometric lemmas hold for every d with the same proofs. The step S10 (the regrouping after the separated lines are cut) is proved in Step 3(D_3) of appendix F (theorem F.57); its cluster form is recalled below, because S11 and S15 are statements about the components it leaves. The chains without an old vertex (S16), the gauge covariance and the gauge-out of B_0 (S17–S18), the cube propagator (S19), the norms (S20), the renormalisation (S21) and the closure lemmas of the class family are in the fourth part.

Conventions for this part.
Scales and fields. As in appendix H.1.1: $\eta = L^{-k}$, $t = L^k\varepsilon$, $e_k = e\,t^{1/2}$; $p(t) = b_0(1 + \ln t^{-1})^{\bar{p}}$ with $\bar{p} > 2$ and $r(t) = R(1 + \ln t^{-1})^{\varrho}$ with $\varrho > 1$, $R > R_0$ [2, (2.3), (2.7)]; q is the fixed real antisymmetric $N \times N$ matrix, $U(A_b) := \exp(qe_k\eta A_b)$, and for an oriented contour Γ, $A(\Gamma) := \sum_{b\subset\Gamma} \eta A_b$ and $U(A(\Gamma)) := \exp(qe_k A(\Gamma))$; all U are orthogonal and commute, $U(A + A') = U(A)U(A')$. The covariant derivative is $(D^\eta_A\phi)(b) := \eta^{-1}(U(A_b)\phi(b_+) - \phi(b_-))$ on the bond $b = (b_-, b_+)$, $Q_k(A)\phi(y) := \sum_{x\in B^k(y)} \eta^d U(A(\Gamma^{(k)}_{y,x}))\phi(x)$ with the contours $\Gamma^{(k)}_{y,x} \subset B^k(y)$ of η-length $\leq d$ [3, (1.13)], $P_k(A) = Q_k(A)^*Q_k(A)$, and on a union X of unit blocks $H_X(A) := -\Delta^{\eta,N}_{A,X} + m^2t^2 + a_kP_k(A)$, $G_k(X, A) := H_X(A)^{-1}$, with $-\Delta^{\eta,N}_{A,X} = (D^\eta_A)^*D^\eta_A$ over the bonds of X and $m^2t^2 > 0$. The hypotheses of [3, p. 412] on $\tilde{B}$ are those of appendix H.1.1: $\lvert\partial^\eta_\nu \tilde{B}_\mu\rvert \leq c_1p(t)$ on every pair of bonds of Ω, and the Hölder quotients of exponent α_0 of $\partial^\eta\tilde{B}$ are $\leq c_1p(t)$ on pairs of points at distance ≤ 1 at depth $\geq R_{\text{Ba}} + 2$ in Ω, where $R_{\text{Ba}} \geq R_0$ is the depth parameter of the background regularity of appendix F. The constant c_1 is independent of k (this is the k-uniformity of the background bounds proved in Step 3(C) of appendix F); nothing else about $\tilde{B}$ is used.

Grid boxes and label boxes. $|\cdot|_\infty$ is the sup norm; "side", "diameter" and dist are sup-norm notions unless said otherwise. A *grid box* is a closed set $\prod_\mu[\alpha_\mu,\beta_\mu]\subset\mathbb{R}^d$ with all $\alpha_\mu\le\beta_\mu$ in the block-corner grid (a translate of $\mathbb{Z}^d$). A *label box* is a grid box with all sides ≤ 2. The localisation of [3, p. 420] is read as in Step 3(D_3) of appendix F (its convention (Loc)): a label is attached to each leg of a vector field (the support box of its partition function $H_y\subset y+[-1,1]^d$, the partition $\{H_y\}_{y\in\mathrm{grid}}$ being smooth with $\operatorname{supp}H_y\subset y+[-\frac34,\frac34]^d$ and derivatives bounded independently of η; a bond b with $H_y(m_b)\neq 0$ at its midpoint m_b has both ends in $y+[-1,1]^d$), to each [3, (1.6)] or [3, (1.7)] vertex (the unit grid cube of its block), and to each block factor [3, (1.12)–(1.15)] (the unit grid cube containing $B^k(y)$, which contains the contours $\Gamma^{(k)}_{y,x}$). The *localisation box* Q_v of a vertex v is the intersection of the boxes of its labels; it is a label box (possibly degenerate) containing every summation point of the vertex. A counterterm vertex carries the labels of the anchor vertex of the subgraph it replaces (the reading (Agr) of appendix F), so its localisation box contains its point. The distance "$\mathrm{dist}(\square(v),\square(v'))$" of [3, pp. 432–433] is $\mathrm{dist}(Q_v,Q_{v'})$; the point-set distance of the summation points is at least this, so the rule "distance $\ge 1\Rightarrow$ external field" is applied only where the separation that [3, (3.1)] needs holds for the actual points.

Classes and the cut rule. The classes are the per-label classes of appendix F: all graphs of a type and order with a given multiset $\mathcal{B}$ of label boxes, with the counterterm graphs of the divergent subgraphs; every member of a class has the same $\mathcal{B}$. The *clusters* of $\mathcal{B}$ are the connected components of the graph on $\mathcal{B}$ in which two boxes are joined when they intersect, and U_K is the union of the boxes of the cluster K. The cut rule of [3, p. 432] in the form of appendix F: a line is *cut* (made an external field by [3, (3.1)], appendix H.2.4) when its two end vertices lie in different clusters, and *kept* otherwise. Every vertex lies in the intersection of its label boxes, so its labels lie in one cluster and "the cluster of a vertex" is well defined. A *component* is a connected component of a graph with respect to its kept lines.

Vertex counts. Three counts occur. The *merged* count V_{mer} counts a product [3, (1.18)] of two block factors at a common y as one vertex; Bałaban's count $V_{\mathrm{Bal}}\le 2V_{\mathrm{mer}}$ counts each of [3, (1.13)–(1.15)] separately ([3, p. 414]; this is the count in which degrees are computed, appendix H.2); the *label* count $V_{\mathrm{lab}}\le(2\bar n+1)V_{\mathrm{Bal}}$ is the number of label boxes, since a vertex has at most $2\bar n+1$ labels (at most $n+n'\le 2\bar n+1$ vector legs for [3, (1.8)–(1.11)], the R-vertices having $n'=\bar n+1$ and $n\le\bar n$; one unit cube for [3, (1.6)–(1.7)]; one unit cube and $n\le\bar n$ legs for a block factor). Lemma H.62 bounds V_{mer} by a function of $\bar n$; with it $V_{\mathrm{lab}}(\bar n):=(2\bar n+1)\cdot 2\bar n(4N_A+5)$ and

$$l_N:=2V_{\mathrm{lab}}(\bar n)$$

are the constants of appendix F. Here N_A is the fixed maximal number of insertions in the expansion of a scalar line or external scalar field in the fluctuation of the background, [1, (3.45)] to order $\bar n$ on a line ($N_A=\bar n$) and the fixed odd order $n_{\mathrm{loop}}\ge\bar n$ of [1, (3.47)] on a closed scalar loop; $N_A:=\max\{\bar n,n_{\mathrm{loop}}\}$ depends on $\bar n$ only.

The cube step. $r_\square$ denotes the side parameter of the cube step: $r_\square=r^\flat$, the constant of the block-factor-5 reading, theorem F.2 (its cube step replaces the cubes of sides $r(t)$, $3r(t)$ by concentric cubes of constant sides $r^\flat$, $3r^\flat$), and $r_\square=r(t)$ wherever the printed step is applied as printed. M is the side of the large blocks. Rounding a set outward to large blocks enlarges each side by at most $2M$.

H.3.1. *S11: a connected component lies in a cube of side l_N.* [3, p. 432]: "each line of each connected component of G'_{ren} is localized in cubes $\square(v)$, $\square(v')$ with $\mathrm{dist}(\square(v),\square(v'))=0$ (by the definition of new graphs it should be <1, but by the construction of localizations it is then $=0$). From this it follows that the localizations of each connected component are contained in some cube of a linear size depending on a number of vertices, hence on $\bar n$ only." The consumed form is corollary H.63: after the cut, every component of every graph of a class has all its label boxes in one grid cube Q_0 of side l_N, and Q_0 can be chosen common to the class component. Three facts enter: the dichotomy for grid boxes (the parenthesis of the quotation), the diameter bound for a chain of intersecting boxes, and the bound on the number of vertices in terms of $\bar n$

([3, p. 414]: "The maximal number of vertices depends in a simple way on $\bar{n}$, but it is unessential here").

Lemma H.60 (the dichotomy "$< 1 \Rightarrow = 0$")**.** *Let Q, Q' be grid boxes. Then for every ℓ^p distance, $1 \le p \le \infty$, either $Q \cap Q' \neq \emptyset$ (so $\operatorname{dist}(Q, Q') = 0$) or $\operatorname{dist}(Q, Q') \ge 1$. The same holds on a torus whose side is an integer number of unit blocks, with the torus distance.*

Proof. Write $Q = \prod_\mu I_\mu$, $Q' = \prod_\mu I'_\mu$ with $I_\mu = [\alpha_\mu, \beta_\mu]$, $I'_\mu = [\alpha'_\mu, \beta'_\mu]$, all endpoints in one translate of $\mathbb{Z}$. If $I_\mu \cap I'_\mu \neq \emptyset$ for every μ, pick $x_\mu \in I_\mu \cap I'_\mu$; then $(x_\mu)_\mu \in Q \cap Q'$. Otherwise there is a μ with, say, $\alpha'_\mu > \beta_\mu$; the gap $g := \alpha'_\mu - \beta_\mu$ is a difference of two points of a translate of $\mathbb{Z}$, so $g \in \mathbb{Z}$ and $g \ge 1$, and for $x \in Q$, $x' \in Q'$, $|x - x'|_p \ge |x_\mu - x'_\mu| \ge g \ge 1$. On the torus the distance is the minimum over the lifts of Q' by the period lattice; integer periods map grid boxes to grid boxes, so the argument applies to each pair of lifts. □

With the labels of (Loc) this is the printed parenthesis: every line kept by the threshold 1 of [3, p. 432] joins two vertices whose localisation boxes intersect, and under the cluster form of the rule every kept line joins two boxes of one cluster.

Lemma H.61 (chains of intersecting boxes)**.** *Let $Q_1, \dots, Q_V$ be label boxes (sides ≤ 2), and let the graph on $\{1, \dots, V\}$ in which i and j are joined when $Q_i \cap Q_j \neq \emptyset$ be connected.*

(a) *If i and j are joined by a path of m edges, then $|x - z|_\infty \le 2(m+1)$ for all $x \in Q_i$, $z \in Q_j$, and $\operatorname{dist}_\infty(z, Q_i) \le 2m$ for all $z \in Q_j$.*
(b) *$\bigcup_i Q_i \subset Q$ for a grid cube Q of side $2V$.*
(c) *On a torus of side $\ell_T > 2V$ (in unit blocks) the same holds, with Q a cube of the torus that does not wrap.*

Proof. (a) Let $i = i_0, i_1, \dots, i_m = j$ be the path and pick $p_s \in Q_{i_{s-1}} \cap Q_{i_s}$, $s = 1, \dots, m$. Consecutive points of the sequence $x, p_1, \dots, p_m, z$ lie in a common box ($x, p_1 \in Q_{i_0}$; $p_s, p_{s+1} \in Q_{i_s}$; $p_m, z \in Q_{i_m}$), each of sup-diameter ≤ 2, so $|x - z|_\infty \le 2(m+1)$ (≤ 2 when $m = 0$). Dropping the first segment, $\operatorname{dist}_\infty(z, Q_i) \le |z - p_1|_\infty \le 2m$.

(b) Any two indices are joined by a path of $\le V - 1$ edges, so by (a) the union has sup-diameter $\le 2V$. For each μ the projection of the union onto the μ-axis is a union of intervals with grid endpoints; its hull $[a_\mu, b_\mu]$ has grid endpoints and $b_\mu - a_\mu \le 2V$. Put $Q := \prod_\mu [a_\mu, a_\mu + 2V]$.

(c) Choose a spanning tree of the intersection graph and a lift of Q_{i_0} to $\mathbb{R}^d$. Lift the boxes along the tree, each to the lift that meets the lift of its parent (it exists because the boxes meet on the torus). Tree paths have $\le V - 1$ edges, so (a) bounds the sup-diameter of the lifted union by $2V$; every box is lifted once, so the projection of the lifted union is $\bigcup_i Q_i$. It lies in the projection of a cube of side $2V$ of $\mathbb{R}^d$, which is injective and does not wrap when $\ell_T > 2V$. □

Lemma H.62 (the number of vertices depends on $\bar{n}$ only)**.** *Every graph of the expansion* [3, (1.5)]*, and every graph obtained from one by cutting lines, by passing to the graphs of its renormalised class, or by localisation, has at most*

$$V_{\mathrm{mer}}(\bar{n}) := \bar{n}(4N_A + 5)$$

vertices in the merged count, hence at most $2V_{\mathrm{mer}}(\bar{n})$ in Bałaban's count and at most $V_{\mathrm{lab}}(\bar{n}) = (2\bar{n} + 1) \cdot 2V_{\mathrm{mer}}(\bar{n})$ label boxes.

Proof. The facts about the expansion that are used, all from [3, pp. 412–415] and [1, (3.44)–(3.47)]:

(E1) The expressions are the derivatives [3, (1.5)], $\sum_{1 \le \alpha+\beta \le \bar{n}} (\alpha!\beta!)^{-1} \partial_{e'}^\alpha \partial_{\lambda'}^\beta E_k$ at $e' = \lambda' = 0$, of the connected functional $E_k = -\log[\dots]$ [3, (1.4)]. So each graph is connected, has at least one internal line [3, p. 415], and has total order $\alpha + \beta \in [1, \bar{n}]$ in (e', λ').
(E2) e' multiplies only the fluctuation field $g_k A'$ in [3, (1.4)], λ' the quartic term and the counterterms ([3, p. 419]: the dependence of δm^2, E_1 on e', λ' "is obtained by replacing e, λ by $e'e, \lambda'\lambda$"). So a vertex with $n \ge 1$ legs A' carries e'^n, a [3, (1.6)] vertex carries λ', and a [3, (1.7)] vertex carries a counterterm all of whose Taylor terms have order ≥ 1 (the counterterms are sums of graphs). Call these vertices *order-carrying*. Every other vertex

has $n = 0$ and no A'-leg; it is two-legged (the types [3, (1.8)–(1.11)] at $n = 0$ and the products [3, (1.18)] of two block factors at $n = 0$, merged) or one-legged (the product of [3, (1.12)] with a block factor at $n = 0$).

(E3) The $n = 0$ vertices arise only from the expansion [1, (3.45)] of the propagator $G_k(\Omega, \tilde{A}+\tilde{B})$ and of the external scalar field $a_k G_k(\Omega, \tilde{A}+\tilde{B})Q_k^*(\tilde{A}+\tilde{B})\phi$ in $\tilde{A}$ ([3, p. 414]: the vertices of [3, (1.5)] "are given by (1.6)–(1.8), (1.10), (1.13), and (1.14) with $n' = 0$"), taken to the fixed number N_A of insertions with the remainder [3, (1.16)] treated as an external field. So the scalar path that replaces one propagator, or one external scalar field, carries at most N_A two-legged $n = 0$ vertices and at most one one-legged one. On a closed scalar loop the expansion [1, (3.46)–(3.47)] is taken to the fixed odd order $n_{\mathrm{loop}} \geq \bar{n}$, which N_A dominates.

(E4) Every vertex has at most four scalar legs: [3, (1.6)] has four, all others at most two.

(E5) The expressions of [3, p. 419] "obtained by an integration and a cumulant expansion in kth step, where some external legs are contracted by propagators $C^{(k)}_{\Lambda_5^{(k)}}$" are handled there with "external fields in the form of functions $\Phi_{\mathrm{ext}}(x, x', x'', \dots)$, $A_{\mathrm{ext}}(y, y', \dots)$ of many variables". With this device the contractions are external fields, not internal lines, and (E1) holds for each factor of the cumulant separately; the count below applies to each factor.

Order-carrying vertices. Each consumes at least one unit of the total order $\alpha + \beta \leq \bar{n}$ by (E1)–(E2), so there are $V_c \leq \bar{n}$ of them.

Chains. A *chain* is a maximal path or cycle, along scalar lines, of $n = 0$ vertices whose interior vertices are two-legged. Every $n = 0$ vertex lies in exactly one chain: a two-legged one has both its lines in the chain, a one-legged one ends its chain. A cycle of $n = 0$ vertices only would be a whole component of order 0, excluded by (E1); a cycle through an order-carrying vertex uses two of that vertex's legs and is counted below like a chain that uses those legs.

Every chain ends at an order-carrying vertex. $n = 0$ vertices have no A'-leg (E2), so a chain meets the rest of the graph only through the scalar lines at its ends. If neither end were a leg of an order-carrying vertex, the chain would be a connected component of order 0, contradicting (E1).

Number of chains. Each scalar leg of an order-carrying vertex ends at most one line, so distinct chains use distinct legs; by (E4) there are at most $4V_c \leq 4\bar{n}$ chains.

Vertices per chain. Each chain replaces one propagator or one external scalar field of the unexpanded expression, so by (E3) it has at most $N_A + 1$ vertices.

Total. $V \leq V_c + 4V_c(N_A + 1) \leq \bar{n}(4N_A + 5)$.

Stability. Cutting lines can only disconnect, and does not change the vertex set; the counterterm graphs of a class replace a divergent subgraph by one counterterm vertex, which lowers the count; localisation does not change the vertex set. Bałaban's count at most doubles the merged one, since only the products [3, (1.18)] are merged, and the label count is at most $2\bar{n} + 1$ per vertex (conventions of this part). □

Corollary H.63 (S11, consumed form)**.** *Let $\mathcal{C}$ be a class (every member carries the same multiset $\mathcal{B}$ of label boxes), cut by the cluster rule. Then:*

(a) *every cluster K of $\mathcal{B}$ has* $\operatorname{diam}_\infty U_K \leq 2|K| \leq l_N$*, and U_K lies in a grid cube Q_K of side l_N;*

(b) *every component of every $G \in \mathcal{C}$ has all its label boxes in one cluster K, hence in Q_K;*

(c) *a connected component of the class (the members of $\mathcal{C}$ restricted to the components lying in a fixed cluster K, including the counterterm graphs) has all its label boxes, for all its graphs, in the one cube $Q_0 := Q_K$ of side l_N.*

On a torus the same holds when its side exceeds l_N. The side l_N depends on $\bar{n}$ only, not on ε, k, Ω, Ω_1, Ω_2, $\tilde{B}$ or the torus.

Proof. (a) A cluster is a family of $|K| \leq V_{\mathrm{lab}}(\bar{n})$ label boxes (lemma H.62) whose intersection graph is connected, so lemma H.61(b), (c) give the cube of side $2|K| \leq l_N$ (enlarge the cube of side $2|K|$ to one of side l_N on the grid). (b) A kept line joins two vertices of one cluster, and

every vertex lies in the intersection of its own label boxes, so all boxes along a kept-connected component lie in one cluster. (c) $\mathcal{B}$, hence the clusters and the cubes Q_K, are common to all members of the class; a counterterm vertex carries the labels of its anchor, which lie in the cluster of the subgraph it replaces. □

Remark H.64. (a) The cube of (c) is what the cube step centres $\square_1$ on ([3, p. 433]: "The localization cubes of each component are contained in some cube of a size $O(\bar{n})$. We take a cube $\square_1$ ... containing the above cube in the center"). It must be common to the class component, since the cancellations inside a class (a graph against its counterterm graph, the leg-exchange and Ward partners) are between different graphs of the class, and the split $G_k(\Omega, \tilde{B}) = G_k(\square, \tilde{B}) + \delta G_k$ must be the same for all of them; with per-label classes this is automatic, as (c) shows. (b) For a single graph cut by the threshold 1 on single lines, lemmas H.60 and H.61 give the per-graph cube of side $2V_{\mathrm{Bal}}(\bar{n}) \le l_N$ directly: a kept line joins intersecting boxes, and a component has at most $V_{\mathrm{Bal}}(\bar{n})$ vertices. (c) The parenthesis "it is then = 0" of [3, p. 432] is true because the label boxes lie on a common grid (lemma H.60); for half-open blocks of the η-lattice adjacent blocks are at distance η. Bałaban's freedom in choosing the partitions of unity allows the grid choice, and it is made in (Loc).

H.3.2. *The cube split: the three properties used.* [3, p. 433]: "We take a cube $\square_1$ of size $r(L^k\varepsilon)$ containing the above cube in the center, and next we take a cube $\square$ of size $3r(L^k\varepsilon)$ and with $\square_1$ in the center. We assume that $\square_1$, $\square$ are sums of big blocks of the unit lattice. Now we replace each scalar field propagator $G_k(\Omega, \tilde{B})$ [or $G_k(\Omega_2, \tilde{B})$] in the expression by $G_k(\square, \tilde{B})$, more exactly we represent $G_k(\Omega, \tilde{B}) = G_k(\square, \tilde{B}) + \delta G_k(\Omega, \square, \tilde{B})$, and we treat $\delta G_k(\Omega, \square, \tilde{B})$ as an external field. Thus again we drop the lines corresponding to these operators and decompose the obtained set of graphs into renormalized classes." This step is printed. Appendix F performs it with the constant side $r^\flat$ in place of $r(t)$ (the containment $\square \subset \Omega$ fails at $L = 5$ for the printed side) and proves the bound on $\delta G_k(\Omega, \square, \tilde{B})$ there. The steps S13–S15 use only the following three properties, which hold for $r_\square = r^\flat$ under the cube step of theorem F.2 and for $r_\square = r(t)$ wherever the printed step applies.

(P-i) $\square_1 \subset \square$ are unions of large blocks, obtained by rounding outward the cubes of sides $r_\square$ and $3r_\square$ concentric with the class-component cube Q_0 of corollary H.63(c). So $\square_1$ contains the cube of side $r_\square$ concentric with Q_0, $\square$ contains the cube of side $3r_\square$ concentric with Q_0, and $\mathrm{side}(\square) \le 3r_\square + 2M$.

(P-ii) $\square \subset \Omega$ (resp. $\square \subset \Omega_2 \subset \Omega$), so $\tilde{B}$ is defined, and satisfies the hypotheses of [3, p. 412], on every bond of $\square$; and $\square$ does not wrap around the torus.

(P-iii) After the split every scalar line is $G_k(\square, \tilde{B})$; $\delta G_k(\Omega, \square, \tilde{B})$, $\delta G_k(\Omega, \Omega_2, \tilde{B})$ and the kernels [3, (1.16)] are external fields.

H.3.3. *S13: the constant field B_0.* [3, p. 433]: "The field $\tilde{B}$ is regular in the sense that $\|\partial_\mu^\eta \tilde{B}\|_{0,\alpha} \le O(p(L^k\varepsilon))$. Hence there exists a constant field B_0 such that $|\tilde{B} - B_0| \le O(r(L^k\varepsilon)p(L^k\varepsilon)) \le O(p(L^k\varepsilon)^2)$ on the cube $\square$." The consumed form: a constant vector field B_0 with $|\tilde{B} - B_0| \le Cr_\square p(t)$ on the bonds of $\square$, such that $\tilde{B}' := \tilde{B} - B_0$ has the derivative and Hölder bounds of $\tilde{B}$ and is an external vector field of norm [3, (1.32)] $O((r_\square + 1)p(t))$ on $\square$: $O(p(t))$ at the constant side $r^\flat$.

Lemma H.65 (the constant field)**.** *Assume (P-i), (P-ii), and let c be a site of $\square$ with $c + \eta e_\mu \in \square$ for every μ and $|x - c|_\infty \le \frac{1}{2}\,\mathrm{side}(\square) + \eta$ for all $x \in \square$ (a site next to the centre). Put $B_{0,\mu} := \tilde{B}_\mu(c)$ for every μ (a constant field) and $\tilde{B}' := \tilde{B} - B_0$ on the bonds of $\square$. Then for every bond $(x, x + \eta e_\mu) \subset \square$ and every contour $\Gamma \subset \square$:*

(a) $|\tilde{B}'_\mu(x)| \le c_1 p(t)\,|x - c|_1 \le C_{13}\,p(t)$, *with* $C_{13} := c_1 d\big(\frac{3}{2}r_\square + M + 1\big)$;

(b) $\partial^\eta \tilde{B}' = \partial^\eta \tilde{B}$ *on* $\square$, *so* $|\partial_\nu^\eta \tilde{B}'_\mu| \le c_1 p(t)$, *and the Hölder quotients of* $\partial^\eta \tilde{B}'$ *are those of* $\partial^\eta \tilde{B}$;

(c) $|\tilde{B}'(\Gamma)| \le |\Gamma|_\eta\, C_{13}\, p(t)$, *with* $|\Gamma|_\eta$ *the length of* Γ *($\le d$ for the contours $\Gamma_{y,x}^{(k)}$);*

(d) *the norm* [3, (1.32)] *of* $\tilde{B}'$ *as an external vector field ("with* $\tilde{B} = 0$", [3, p. 420]*) over* $\square$ *satisfies* $\|\tilde{B}'\|_{1,\alpha_0;\square} \le (C_{13} + 3c_1)p(t)$ *wherever the Hölder clause on* $\tilde{B}$ *holds, i.e. on pairs of points at depth* $\ge R_{\mathrm{Ba}} + 2$ *in* Ω*;*

(e) *at* $r_\square = r^\flat$*,* C_{13} *is a constant and* $\tilde{B}' = O(p(t))$*, the size class of* $\tilde{A}$*; at the printed* $r_\square = r(t)$*,* $|\tilde{B}'| \le c_1 d(\frac{3}{2}R(1 + \ln t^{-1})^{\varrho} + M + 1)p(t)$*, which is* $\le C(R/b_0 + 1)p(t)^2$ *uniformly in* $t \le 1$ *if and only if* $\varrho \le \bar{p}$*.*

Proof. A path inside $\square$. Fix a bond $(x, x+\eta e_\mu) \subset \square$. Go from c to x by changing one coordinate at a time, $\nu = 1, \dots, d$, in steps of $\pm\eta$; every intermediate site z has z_ν between c_ν and x_ν for the coordinates already moved and equal to c_ν or x_ν otherwise. $\square$ is a product of intervals of sites (a cube that does not wrap, (P-ii)); both $c + \eta e_\mu$ and $x + \eta e_\mu$ lie in $\square$, so $\max(c_\mu, x_\mu) + \eta$ is at most the last site coordinate of $\square$ in direction μ, and $z + \eta e_\mu \in \square$ for every intermediate z. Hence every pair of bonds $(z, z + \eta e_\mu)$, $(z + \eta e_\nu, z + \eta e_\nu + \eta e_\mu)$ used below lies in $\square \subset \Omega$, where the hypothesis on $\partial^\eta \tilde{B}$ holds.

(a) Telescoping. $\tilde{B}_\mu(x) - \tilde{B}_\mu(c) = \sum_{\text{steps}} \pm\eta\, \partial^\eta_\nu \tilde{B}_\mu(z)$ over $|x - c|_1/\eta$ steps, each bounded by $\eta c_1 p(t)$. So $|\tilde{B}'_\mu(x)| \le c_1 p(t)|x - c|_1 \le c_1 p(t)\, d(\frac{1}{2}\,\mathrm{side}(\square) + \eta) \le C_{13} p(t)$, using $\frac{1}{2}\,\mathrm{side}(\square) \le \frac{3}{2} r_\square + M$ (P-i) and $\eta \le 1$.

(b) B_0 is constant, so $\partial^\eta B_0 = 0$.

(c) $\tilde{B}'(\Gamma) = \sum_{b \subset \Gamma} \eta \tilde{B}'_b$ has $|\Gamma|_\eta/\eta$ terms, each bounded by $\eta C_{13} p(t)$ by (a).

(d) The norm is the sum of the sup (a), the derivative bound (b) and the Hölder bound: for $|x - x'| \le 1$ the quotient is $\le c_1 p(t)$ by (b) and the hypothesis; for $|x - x'| > 1$ it is $\le 2 \sup |\partial^\eta \tilde{B}'| \le 2 c_1 p(t)$.

(e) Insert $r_\square$. At the printed cube, $r(t)p(t) = (R/b_0)(1 + \ln t^{-1})^{\varrho - \bar{p}} p(t)^2$, which is $O(p(t)^2)$ uniformly in $t \le 1$ exactly when $\varrho \le \bar{p}$. $\square$

Remark H.66. (a) *The inequality "*$O(rp) \le O(p^2)$*".* [2, (2.3), (2.7)] fix only $\bar{p} > 2$ and $\varrho > 1$; the printed inequality needs $\varrho \le \bar{p}$, which holds for the values used throughout ($\varrho = 2$, $\bar{p} = 6$) and is not needed at the constant side, where $\tilde{B}' = O(p)$ by (e).

(b) *Where the Hölder clause is used.* The norm (d) is needed at the vertices of the $\tilde{B}'$-expansion (appendix H.3.5). At the vertices of the components containing an old vertex the depth is $\ge r_\square$ (lemma H.73), and the clause holds there when $r^\flat \ge R_{\mathrm{Ba}} + 2$. At the vertices of the chains without an old vertex (S16, the fourth part), which may lie anywhere in $\square$, a point $z \in \square$ has depth $\ge \rho - D_\square$, with ρ the depth of the old vertices and $D_\square := \frac{3}{2} r_\square + 2M + l_N$; the cube step of theorem F.2 places Q_0 at depth $> D^\flat = \frac{3}{2} r^\flat + 2M + l^\sharp$, so $\mathrm{depth}(z) > l^\sharp - l_N = 4V^\sharp_{\mathrm{lab}}$ (appendix H.3.6). The parameter row $R \ge R_{\mathrm{flat}} + R_{\mathrm{Ba}} + 2$ of appendix F (R_{flat} the depth parameter of its cube step) makes every point of $\square$ lie at depth $\ge R_{\mathrm{Ba}} + 2$ in Ω, so (d) holds on all of $\square$; it is a lower bound on R of Bałaban's kind, fixed after R_{Ba} and $l^\sharp$. The chain step of the fourth part in fact uses only $\sup|\tilde{B}'|$ and $\sup|\partial^\eta \tilde{B}'|$ at the chain vertices, so the row is a convenience there.

(c) *On a torus* the same proof works for a cube $\square$ that does not wrap (P-ii), so that the coordinate path is defined.

(d) Nothing in the lemma depends on k except through c_1, whose k-uniformity is the input named in the conventions.

H.3.4. *S14, I: the expansion of the vertices and of the propagator in* $\tilde{B}'$. [3, p. 433]: "We have $\tilde{B} = B_0 + \tilde{B}'$, and we expand in $\tilde{B}'$ the expressions connected with each renormalized class of graphs using the formulas (I.3.14), (I.3.44), and (I.3.45). After this expansion we get a sum of expressions corresponding to some new big set of graphs, which can be decomposed into a sum of renormalized classes." The consumed form has three parts: (i) the expression of a class after the cube split equals a finite sum of expressions of graphs with scalar lines $G_k(\square, B_0)$, old vertices of the same types and localisations with B_0 in place of $\tilde{B}$ and extra external $\tilde{B}'$-legs, new vertices of the types [3, (1.8)–(1.11)] and [3, (1.13)–(1.15)] "with external vector field $\tilde{B}'$ only", at most N_B of them on each old line, each with two scalar legs, no A'-leg and a factor e_k per $\tilde{B}'$-leg, and remainder operators treated as external fields; (ii) the number of terms is

bounded in terms of $\bar n$, N_B, d only, and the new factors obey lemma H.65; (iii) the terms can be regrouped into renormalised classes. This subsection proves the two identities behind (i); appendix H.3.5 proves (i)–(ii) and (iii).

The Taylor remainder. For z in the Lie algebra spanned by q (real multiples of q, antisymmetric) put

$$R_{\bar n+1}(z) := (\bar n+1)\int_0^1 (1-s)^{\bar n} e^{sz}\,ds, \qquad \text{so that} \qquad e^z = \sum_{j=0}^{\bar n}\frac{z^j}{j!} + \frac{z^{\bar n+1}}{(\bar n+1)!}R_{\bar n+1}(z)$$

(Taylor's formula with integral remainder); e^{sz} is orthogonal, so $\|R_{\bar n+1}(z)\| \le (\bar n+1)\int_0^1(1-s)^{\bar n}ds = 1$. Define, for a bond value A and a contour value $A(\Gamma)$,

$$F_{1,k}(A) := \eta^{-1}\big(U(A)-1\big) = \sum_{j=1}^{\bar n}\eta^{j-1}\frac{(qe_kA)^j}{j!} + \eta^{\bar n}\frac{(qe_kA)^{\bar n+1}}{(\bar n+1)!}R_{\bar n+1}(\eta q e_k A), \qquad U(A(\Gamma))-1 = \sum_{j=1}^{\bar n}\frac{(qe_kA(\Gamma))^j}{j!} + \cdots$$

These are [1, (3.14)–(3.15)] (with the sign convention of this appendix; every identity below holds for either sign of the exponent). $F_{1,k}(A)$ is a sum of $\bar n+1$ terms, the jth homogeneous of degree j in e_kA with the factor η^{j-1}, and the last carrying the matrix $R_{\bar n+1}$ of norm ≤ 1.

Lemma H.67 (the vertex factors)**.** *On $\square$, with $\tilde B = B_0 + \tilde B'$ as in lemma H.65, for every bond $b \subset \square$ and contour $\Gamma \subset \square$:*

(a) $U(\tilde B_b) = U(B_{0,b})U(\tilde B'_b)$ *and* $U(\tilde B(\Gamma)) = U(B_0(\Gamma))U(\tilde B'(\Gamma))$;

(b) $U(\tilde B'_b) = 1 + \eta F_{1,k}(\tilde B'_b)$ *and* $U(\tilde B'(\Gamma)) = 1 + [U(\tilde B'(\Gamma)) - 1]$, *with the finite expansions displayed above;*

(c) $D^\eta_{\tilde B}\phi(b) = D^\eta_{B_0}\phi(b) + F_{1,k}(\tilde B'_b)U(B_{0,b})\phi(b_+)$;

(c′) *equivalently, in the form of* [1, (3.44)] *with* $(A,B) = (\tilde B', B_0)$: $D^\eta_{\tilde B}\phi(b) = U(\tilde B'_b)\big[D^\eta_{B_0}\phi(b) - F_{1,k}(-\tilde B'_b)\phi(b_-)\big]$;

(d) $Q_k(\tilde B)\phi(y) = Q_k(B_0)\phi(y) + F_{2,k}(\tilde B', B_0)\phi(y)$, *with* $F_{2,k}(A,B)\phi(y) := \sum_{x\in B^k(y)}\eta^d\,[U(A(\Gamma^{(k)}_{y,x}))-1]\,U(B(\Gamma^{(k)}_{y,x}))\phi(x)$.

Proof. (a) The U are exponentials of real multiples of the fixed matrix q; they commute and multiply, and $A \mapsto A(\Gamma)$ is additive. (b) is the definition of $F_{1,k}$ and the Taylor formula. (c) Insert (a)–(b) into $D^\eta_{\tilde B}\phi(b) = \eta^{-1}(U(\tilde B_b)\phi(b_+) - \phi(b_-))$: $U(\tilde B_b)\phi(b_+) = U(B_{0,b})\phi(b_+) + \eta F_{1,k}(\tilde B'_b)U(B_{0,b})\phi(b_+)$. (c′) Write $U(\tilde B_b) = U(\tilde B'_b)U(B_{0,b})$ and $\phi(b_-) = U(\tilde B'_b)U(-\tilde B'_b)\phi(b_-)$ with $U(-\tilde B'_b) = 1 + \eta F_{1,k}(-\tilde B'_b)$; then $\eta^{-1}(U(\tilde B_b)\phi(b_+) - \phi(b_-)) = U(\tilde B'_b)\eta^{-1}\big(U(B_{0,b})\phi(b_+) - \phi(b_-) - \eta F_{1,k}(-\tilde B'_b)\phi(b_-)\big)$. (d) Insert (a) into the definition of $Q_k(\tilde B)$. □

Lemma H.68 (the propagator expansion and the vertex operators)**.** *Let $H_\square(A) = -\Delta^{\eta,N}_{A,\square} + m^2t^2 + a_kP_k(A)$ be the Neumann operator on the sites of $\square$ and $G_k(\square,A) = H_\square(A)^{-1}$; both $H_\square(\tilde B)$ and $H_\square(B_0)$ are invertible since $m^2t^2 > 0$ (the identities below need nothing more; the uniform lower bound* [4, (1.8)] *enters only the estimates of the fourth part). Put $G_0 := G_k(\square, B_0)$, $D_0 := D^\eta_{B_0}$, $Q_0 := Q_k(B_0)$, $(F\phi)(b) := F_{1,k}(-\tilde B'_b)\phi(b_-)$, $F_2 := F_{2,k}(\tilde B', B_0)$, and $V_k := H_\square(B_0) - H_\square(\tilde B)$. Then:*

(a) *(*[1, (3.44)] *with $(A,B) = (\tilde B', B_0)$.) $V_k = F^*D_0 + D_0^*F - F^*F - a_k\big[F_2^*Q_0 + Q_0^*F_2 + F_2^*F_2\big]$, adjoints with respect to the η^d-weighted inner products on sites and bonds of $\square$.*

(b) *(*[1, (3.45)]*.) For every $N \ge 0$, $G_k(\square,\tilde B) = \sum_{m=0}^N G_0(V_kG_0)^m + \mathcal{R}_N$ with $\mathcal{R}_N := G_0(V_kG_0)^N V_kG_k(\square,\tilde B)$.*

(b′) *The two-sided form, for $n_1, n_2 \ge 1$:*

$$G_k(\square,\tilde B) = \sum_{m<n_1} G_0(V_kG_0)^m + \sum_{m'<n_2}(G_0V_k)^{n_1}G_0(V_kG_0)^{m'} + (G_0V_k)^{n_1}G_k(\square,\tilde B)(V_kG_0)^{n_2}.$$

Its last term is the analogue on $\square$ of the kernel [3, (1.16)]*; its main terms have at most $N_B := n_1 + n_2 - 1$ insertions of V_k.*

(c) *Inserting the finite expansions of* $F_{1,k}$ *and* $U(\tilde B'(\Gamma))-1$ *into (a) writes* V_k *as a sum of at most* $6(\bar n+1)^2$ vertex operators*, each a bilinear form in* (ϕ,ψ) *with exactly two scalar legs and no* A'*-leg, every factor* $\tilde B'$ *accompanied by* e_k*,* B_0 *standing where* $\tilde B$ *stood in* [3, (1.8)–(1.15)] *and* $\tilde B'$ *where* $\tilde A$ *stood:*

- *the terms of* F^*D_0*,* D_0^*F *are* $\sum_b \eta^d\big[D_0\psi(b)\cdot\eta^{j-1}\frac{(-qe_k\tilde B'_b)^j}{j!}\phi(b_-)\big]$ *and their adjoints,* $1\le j\le\bar n$*, with the R-term for* $j=\bar n+1$*: the forms* [3, (1.8)] *at* $n=0$*,* $n'=j$*, and* [3, (1.9)]*;*
- *the terms of* F^*F *are* $\sum_b\eta^d\eta^{j+j'-2}(e_k\tilde B'_b)^{j+j'}\big[\psi(b_-)\cdot c_{jj'}q^{j+j'}\phi(b_-)\big]$ *with scalar coefficients* $c_{jj'}$*: the forms* [3, (1.10)]*, and* [3, (1.11)] *when an R-factor occurs.* The terms with $j+j'$ odd cancel as operators*: the* (j,j') *and* (j',j) *terms are* $(-1)^j\frac{(z)^{j+j'}}{j!j'!}$ *and* $(-1)^{j'}\frac{(z)^{j+j'}}{j!j'!}$ *with* $z=-\eta qe_k\tilde B'_b$*, of opposite signs when* $j+j'$ *is odd. Equivalently* $F_{1,k}(A)^TF_{1,k}(A)=\eta^{-2}(2-e^z-e^{-z})$ *with* $z=\eta qe_kA$*, which is even in* z*. This is the parity condition "*$n+n'$ *even" of* [3, (1.10)]*; the terms with an R-factor carry no parity, as* [3, (1.11)] *carries none. A single odd* (j,j') *term is a nonzero operator, so the cancellation is between the two orderings, not termwise;*
- *the terms with* F_2 *are block sums* $\sum_y$ *of products of a block factor* [3, (1.13)] *at* B_0 *with a factor* [3, (1.14)] *or* [3, (1.15)] *at* $n=0$*, and of two such factors: the products* [3, (1.18)].

The η*-powers per leg are those of* [3, (1.8)–(1.15)]*:* η^{j-1} *at a term of type* [3, (1.8)] *and* $\eta^{j+j'-2}$ *at a term of type* [3, (1.10)].

Proof. (a) By lemma H.67(c′) and (d), as operators on the fields of $\square$, $D^\eta_{\tilde B}=U(\tilde B')(D_0-F)$ with $U(\tilde B')$ the bondwise orthogonal multiplication by $U(\tilde B'_b)$, and $Q_k(\tilde B)=Q_0+F_2$. Hence $(D^\eta_{\tilde B})^*D^\eta_{\tilde B}=(D_0-F)^*(D_0-F)=D_0^*D_0-D_0^*F-F^*D_0+F^*F$ and $P_k(\tilde B)=(Q_0+F_2)^*(Q_0+F_2)$. Subtracting $H_\square(\tilde B)$ from $H_\square(B_0)$, the mass terms cancel and V_k is the stated bracket. (The factorisation of lemma H.67(c), $D^\eta_{\tilde B}=D_0+\mathfrak F$ with $(\mathfrak F\phi)(b):=F_{1,k}(\tilde B'_b)U(B_{0,b})\phi(b_+)$, gives the different exact form $V_k=-[D_0^*\mathfrak F+\mathfrak F^*D_0+\mathfrak F^*\mathfrak F]-a_k[F_2^*Q_0+Q_0^*F_2+F_2^*F_2]$; the two brackets are equal as operators but not term by term. The form (a) is the one whose terms are the printed vertices.)

(b) $G_0^{-1}-G_k(\square,\tilde B)^{-1}=V_k$ gives the second resolvent identity $G_k(\square,\tilde B)=G_0+G_0V_kG_k(\square,\tilde B)$; iterate it N times.

(b′) Apply the identity on the other side, $G_k(\square,\tilde B)=G_0+G_k(\square,\tilde B)V_kG_0$, n_2 times to the factor $G_k(\square,\tilde B)$ of the remainder of (b) at $N=n_1-1$, i.e. to $(G_0V_k)^{n_1}G_k(\square,\tilde B)$.

(c) Distribute (a) over the $\bar n+1$ terms of F and of F_2: at most $2(\bar n+1)+(\bar n+1)^2$ terms from the three D-terms and the same from the three Q-terms, at most $6(\bar n+1)^2$ in all. Each term has one ϕ and one ψ leg (two scalar legs) and no A'. The forms are read off: $(F\phi)(b)=F_{1,k}(-\tilde B'_b)\phi(b_-)$ has the undifferentiated leg at b_- and no $U(B_0)$ factor, so F^*D_0 pairs $D_0\psi(b)$ with $q^j\phi(b_-)$, the printed shape "$[(D^\eta_{\tilde B}\phi')(b)\cdot q^{n+n'}\phi'(b_-)]$" of [3, p. 413]; F^*F pairs $\psi(b_-)$ with $\phi(b_-)$ through $F_{1,k}(-\tilde B'_b)^TF_{1,k}(-\tilde B'_b)$, whose (j,j') term is $\eta^{j+j'-2}\frac{((-z)^T)^j(-z)^{j'}}{j!j'!}$ with $z=\eta qe_k\tilde B'_b$, and $(q^T)^j=(-1)^jq^j$ gives the parity statement. The F_2 terms are the definition of $F_{2,k}$ expanded. □

Remark H.69. The identities of this subsection are exact finite-dimensional identities, valid for any invertible $H_\square(\tilde B)$, $H_\square(B_0)$ and any bond field $\tilde B'$; the regularity of lemma H.65 is used only in lemma H.70(v) and in the estimates of the fourth part. [13, (3.47)–(3.53)] prints the same expansion on the torus.

H.3.5. *S14, II: the expanded expression and its classes.*

Lemma H.70 (the expanded expression)**.** *Let* $\mathcal C$ *be a connected component of a class after the cube split, with all scalar lines* $G_k(\square,\tilde B)$ *(P-iii) and all old vertices localised in the cube* $Q_0\subset\square_1$ *of corollary H.63(c). Expand every scalar line by lemma H.68(b′) with fixed* $n_1,n_2\ge2$ *(the orders for which the last term of (b′) is a regular kernel of both variables, as in the fourth part), so with at most* $N_B=n_1+n_2-1$ *insertions in a main term, and every old vertex by*

lemma H.67: its dependence on $\tilde{B}$ is through $D^{\eta}_{\tilde{B}}$ in [3, (1.8)–(1.9)] *and through $U(\tilde{B}(\Gamma))$ in* [3, (1.13)–(1.15)], *while* [3, (1.6), (1.7), (1.10), (1.11)] *and the counterterm vertices do not depend on $\tilde{B}$. Then*

$$E(\mathcal{C}) = \sum_{G\in\mathcal{C}} \sum_{\tau\in T(G)} E(G_\tau),$$

a finite sum, where a pattern τ *chooses, for each scalar line, one main term of (b') together with one vertex operator of lemma H.68(c) per insertion, or the remainder term, and for each old vertex one term of its expansion. Each G_τ has:*

(i) *scalar lines $G_k(\square, B_0)$ and vector lines unchanged (they do not depend on $\tilde{B}$); the external fields unchanged; each line with the remainder choice replaced by a pair of external scalar legs carrying the two-variable field $(G_0V_k)^{n_1}G_k(\square,\tilde{B})(V_kG_0)^{n_2}$;*

(ii) *old vertices of the same type and the same localisation box Q_v, with B_0 in place of $\tilde{B}$, and extra $\tilde{B}'$-legs at the vertex's own bond or contour, inside Q_v, so that no relocalisation is needed. "Same type" is meant collectively, as* [3, p. 433] *says it ("Vertices are now of the same type (1.6)–(1.11) and (1.13)–(1.15), but with B_0 instead of $\tilde{B}$ and with the new external vector field $\tilde{B}'$ beside the old one"): the term $F_{1,k}(\tilde{B}'_b)U(B_{0,b})\phi(b_+)\cdot q^m\phi(b_-)$ of an expanded* [3, (1.8)] *or* [3, (1.9)] *vertex has no derivative leg and $n'+j$ vector legs, i.e. the shape of* [3, (1.10)] *or* [3, (1.11)]*; its degree is the old one plus j;*

(iii) *new vertices, at most N_B on each old scalar line, each of the form of lemma H.68(c): exactly two internal lines, both on the chain of that old line, no A'-leg, lying on bonds and blocks of $\square$ (their label boxes may leave $\square$ by at most 1). An old scalar line may have one old end and one external scalar leg; its chain then has a single old end;*

(iv) *$|T(G)| \le \big[(N_B{+}2)(6(\bar{n}{+}1)^2)^{N_B}\big]^{L(\bar{n})}\,(2(\bar{n}{+}1))^{2dV_{\mathrm{mer}}(\bar{n})}$, with $L(\bar{n}) \le 2V_{\mathrm{mer}}(\bar{n})$ the maximal number of scalar lines: a bound in terms of $\bar{n}$, N_B, d only;*

(v) *every $\tilde{B}'$-factor is $e_k\tilde{B}'_b$ or $e_k\tilde{B}'(\Gamma)$, with $|e_k\tilde{B}'_b| \le e_kC_{13}p(t)$ and $|e_k\tilde{B}'(\Gamma)| \le e_kdC_{13}p(t)$ (lemma H.65); the R-factors have norm ≤ 1;*

(vi) *$V_{\mathrm{mer}}(G_\tau) \le V_{\mathrm{mer}}(\bar{n}) + N_BL(\bar{n}) \le V_{\mathrm{mer}}(\bar{n})(1+2N_B)$; the label count after the expansion is bounded by a function $V^{\sharp}_{\mathrm{lab}}(\bar{n})$ of $\bar{n}$ (and the fixed N_B) only, since each new vertex carries at most $\bar{n}+1$ labels.*

Proof. $E(G)$ is multilinear in the factors that depend on $\tilde{B}$: each scalar line, and each $D^{\eta}_{\tilde{B}}$ or $U(\tilde{B}(\Gamma))$ at an old vertex. Replace each by its finite expansion (lemmas H.67 and H.68) and distribute; this is the sum over τ. (i)–(iii) read off the terms: an inserted vertex operator of lemma H.68(c) lies on a bond or block of $\square$ (the operators act on fields of $\square$), its two scalar legs are joined to its neighbours on the chain, and its label boxes, of side ≤ 2 around a point of $\square$, leave $\square$ by at most 1. (iv) Per scalar line there are $n_1+n_2 = N_B+1$ main terms and one remainder, and each insertion has at most $6(\bar{n}+1)^2$ vertex operators; per old vertex at most $2d$ bond or contour factors of $\tilde{B}$-type, each with at most $\bar{n}+2 \le 2(\bar{n}+1)$ terms; a graph has at most $4V_{\mathrm{mer}}/2$ scalar lines. (v) is lemma H.65(a), (c) with $|\Gamma|_\eta \le d$. (vi) Each line gains at most N_B vertices. □

The regrouping into renormalised classes. The class rules of [3, Sect. 2] (appendix H.2.2): divergent graphs ($D \le 0$) are built of [3, (1.6)–(1.8), (1.10)] only, and a graph with an R-vertex, a block vertex [3, (1.13)–(1.15)], a [3, (1.7)] vertex with finite counterterm or more than four external legs has $D > 0$ [3, Cor. 2.3]; the two-scalar-leg and $\phi\phi A$ classes are "the smallest sets of graphs, symmetric with respect to permutation of external scalar field legs and with each divergent subgraph renormalized" [3, p. 429], two-scalar-leg classes with their mass counterterm [3, p. 430]; the $2A$ classes contain "all graphs connected with these graphs by Ward–Takahashi identities", or, "if we refrain from taking the smallest classes", "all possible connected graphs of the considered type ... of a fixed order" [3, p. 431]; $1A$ and $3A$ graphs and differentiated two-scalar-leg graphs "do not introduce any new divergences" [3, p. 431]; vacuum classes carry their vacuum counterterms [3, p. 432]; and "for an arbitrary graph G we form a class G_{ren} containing this graph adding all graphs necessary to renormalize all divergent proper subgraphs. Of course

localizations in these graphs should be in agreement" [3, p. 432]. For $G \in \mathcal{C}$, a pattern τ and a subgraph S of G, say τ *touches* S if τ has an insertion on an internal scalar line of S or a nonzero $\tilde{B}'$-order at a vertex of S; S_τ is the corresponding subgraph of G_τ.

Lemma H.71 (the expanded class is a sum of renormalised classes)**.** *Let $\mathcal{C}$ be a per-label class (every member has the label multiset $\mathcal{B}$) at background $\tilde{B}$ on $\square$.*

(a) Untouched subgraphs. *Let S be a divergent proper subgraph of $G \in \mathcal{C}$ and $G' \in \mathcal{C}$ a graph that differs from G only inside S (the counterterm graph G/S, or a leg-exchange or Ward partner inside S). The patterns of G not touching S correspond bijectively to the patterns of G' trivial inside S, and $(G_\tau)' = (G')_\tau$. So inside G_τ the subgraph $S_\tau = S$ is renormalised by terms of the expansion of $\mathcal{C}$.*

(b) Touched subgraphs change type. *If τ touches S, then S_τ has $j \geq 1$ more external vector legs than S and $D(S_\tau) \geq D(S) + j$; S_τ is convergent if τ puts an R-term or an F_2-term inside S. The only partner sets a touched divergent S_τ can require are: (b1) S of two-scalar-leg type with $j = 1$: the partner with the two external scalar legs of S_τ exchanged; (b2) S of type $1A$ with $j = 1$, or a vacuum component with $j = 2$: the Ward class of type $2A$; (b3) the four-leg graph [3, (2.4)], handled by the summation by parts of appendix H.1.6. Touched $1A$ and $3A$ subgraphs need nothing.*

(c) Closure. *The per-label class $\mathcal{C}$ is closed under the exchange of the two external scalar legs of any two-scalar-leg or $\phi\phi A$ subgraph, the whole graph included (so a touched two-scalar-leg graph, now of type $\phi\phi A$ and degree 0, finds its mirror partner in the expansion of $\mathcal{C}$), and the first (resp. second) $\tilde{B}'$-Taylor coefficient of the per-label $1A$ (resp. vacuum) class of an order is the per-label $2A$ class of that order with one (resp. two) external legs equal to $\tilde{B}'$, with the weights of the Taylor coefficients. Hence the partners of (b1)–(b2) are terms of the expansion of $\mathcal{C}$, and $\sum_{G,\tau} E(G_\tau)$ is a sum of per-label renormalised classes at background B_0, with $\tilde{B}'$ among the external fields.*

Proof. (a) Counterterm vertices [3, (1.26)–(1.31)] and the vertices [3, (1.6), (1.7), (1.10), (1.11)] do not depend on $\tilde{B}$. A pattern not touching S consists of choices on the lines and vertices outside S, which are the same data for G and G'; the expansion acts factor by factor (lemma H.70), so the two expansions agree outside S, and the labels of the new vertices are fixed by their own bonds and blocks, which also agree.

(b) The vertex degrees of [3, p. 423] with the definition [3, (2.2)]: a vertex of type [3, (1.8)] or [3, (1.10)] has degree $n(4-d)/2 + n'$, so each added $\tilde{B}'$-leg raises the degree of the subgraph containing it by 1, and an inserted vertex of lemma H.68(c) has $n = 0$, $n' = j \geq 1$ and adds j to the degree of the line it sits on (two legs of dimension $-\frac{d-2}{2}$ each and one new line, net j); an R-term or a block factor inside S makes $D(S_\tau) > 0$ by [3, Cor. 2.3] (lemma H.41). The list of divergent types with at most four external legs and their partner sets is [3, pp. 429–431] (corollary H.43): a two-scalar-leg subgraph of degree -1 touched once becomes $\phi\phi A$ of degree 0, whose class is its symmetrisation; a $1A$ subgraph of degree -1 (resp. a vacuum component of degree -2) touched once (resp. twice) becomes $2A$ of degree 0, whose class is the Ward class; a four-scalar-leg subgraph touched becomes [3, (2.4)] with a vector leg, of degree ≥ 1, and is handled by lemma H.26; everything else touched has $D > 0$.

(c) The two closure statements are the two lemmas on the class family in the fourth part of this appendix (appendix H.4) (the exchange closure of per-label classes, which is a bijection of Wick pairings preserving orders and labels, and the identity of generating functions $\partial/\partial\tilde{B}' = \partial/\partial X$ for the Taylor coefficients of log of the Gaussian integral at background B_0, the vertices and lines depending on $\tilde{B}'$ and the other vector legs X only through $B_0 + \tilde{B}' + X$). Given them: for (b1), the exchange acts on the external connections of S and the insertion on its internal lines, so $(S_\tau)^\pi = (S^\pi)_\tau$ with $S^\pi \subset G^\pi \in \mathcal{C}$; for (b2), the $2A$ Ward class at the order of S_τ is the set of all single (resp. double) $\tilde{B}'$-attachments, by line insertion or vertex expansion, to all graphs of S's type and labels, which are exactly the touched terms of the expansion of $\mathcal{C}$. Together with (a) every divergent subgraph of every G_τ is renormalised, or completed to its Ward class, inside

$\sum_{G,\tau} E(G_\tau)$, and the labels of the partners agree, so the sum is a sum of per-label renormalised classes. □

Remark H.72. (a) The sentence "which can be decomposed into a sum of renormalized classes" of [3, p. 433] is a closure property of the class family under differentiation in the background; it holds for per-label classes because these are complete at fixed labels ("if we refrain from taking the smallest classes", [3, p. 431]), and [3, Prop. 1] for these classes is all that its consumers use, since they sum over all graphs of each order and over the localisations ([3, p. 421]: "Proposition I.3.1 is completely obvious if we sum over localizations"). (b) The regrouping and the renormalisation of [3, pp. 434–445] are applied in the fourth part to per-label classes, so the classes produced here are the ones they take.

H.3.6. *S15: old-vertex components lie in $\square_1$.* [3, p. 433]: "We repeat some of the operations done before, i.e. we localize new vertices in the way described in Chap. 1 and we include the operators (1.16) and propagators with localizations $\square(v)$, $\square(v')$ satisfying the condition $\mathrm{dist}(\square(v), \square(v')) \geq 1$ into the external fields. These operations give us again a sum of renormalized classes. If a class contains an old vertex coming from the expression before the expansion, then all vertices of this class are localized in $\square_1$." The consumed form: after the expansion (lemma H.70), the localisation of the new vertices by (Loc), and the cut on $\square$ by the cluster rule on the label boxes of the expanded class (the estimate [3, (3.1)] on $\square$ at the background B_0 is the repetition of appendix H.2.4 proved in the fourth part), every component that contains an old vertex has all its vertices in $\square_1$, at distance $\geq r_\square$ from $\square^c$; the components without an old vertex are the chains of S16.

Lemma H.73 (old-vertex components)**.** *Assume (P-i), with the old vertices of all graphs of the class component in the cube Q_0 of side l_N concentric with $\square_1$ (corollary H.63(c)); let the lines of every G_τ of lemma H.70 be cut by the cluster rule on the label multiset $\mathcal{B}^\sharp$ of the expanded class (common to the expanded class by lemma H.71), with $|\mathcal{B}^\sharp| \leq V^\sharp_{\mathrm{lab}}(\bar n)$. Put*

$$l^\sharp := l_N + 4V^\sharp_{\mathrm{lab}}(\bar n), \qquad Q_0^+ := \{x : \mathrm{dist}_\infty(x, Q_0) \leq 2V^\sharp_{\mathrm{lab}}(\bar n)\},$$

the cube of side $l^\sharp$ concentric with Q_0. If $r_\square \geq l^\sharp$, then for every G and every pattern τ:

(a) *every component of G_τ containing an old vertex has all its label boxes in Q_0^+; hence it lies in $\square_1$, at sup-distance $\geq \frac{1}{2}(3r_\square - l^\sharp) \geq r_\square$ from $\square^c$;*

(b) *every component of G_τ without an old vertex consists of new vertices of a single old scalar line: a sub-chain of at most N_B consecutive new vertices of that line, whose label boxes lie in one cluster, so of sup-diameter $\leq 2V^\sharp_{\mathrm{lab}}(\bar n)$; with the threshold applied to single lines (lemma H.60) its diameter is also $\leq 2N_B$.*

Proof. Old vertices. They are the vertices of the pre-expansion component with their boxes unchanged (lemma H.70(ii)), so their boxes lie in Q_0.

New vertices. By lemma H.70(iii) a new vertex lies on the chain $u = n_0, n_1, \dots, n_m, n_{m+1} = w$ ($m \leq N_B$) of exactly one old scalar line $\ell = (u, w)$ with a main-term choice; its only internal lines are its two chain lines, its $\tilde B'$-legs are external and it has no A'-leg. If ℓ has one old end u and an external scalar leg, the chain is $u = n_0, \dots, n_m$ with u its only old end.

(a) Let K be a component containing an old vertex v. Every kept line joins two vertices of one cluster of $\mathcal{B}^\sharp$, and every vertex lies in the intersection of its label boxes, so all label boxes of K lie in the cluster K_v of v. The intersection graph of K_v is connected with $|K_v| \leq V^\sharp_{\mathrm{lab}}(\bar n)$ boxes, and one of them (a box of v) meets Q_0; by lemma H.61(a), every point z of every box of K_v has $\mathrm{dist}_\infty(z, Q_v) \leq 2(|K_v| - 1)$, hence $\mathrm{dist}_\infty(z, Q_0) \leq 2V^\sharp_{\mathrm{lab}}(\bar n)$, i.e. $z \in Q_0^+$. If $r_\square \geq l^\sharp$, the cube Q_0^+ of side $l^\sharp$ concentric with Q_0 lies in the cube of side $r_\square$ concentric with Q_0, which lies in $\square_1$ (P-i; rounding outward only enlarges). For the distance to $\square^c$: $\square$ contains the cube of side $3r_\square$ concentric with Q_0 (P-i), so every point of Q_0^+ is at sup-distance $\geq \frac{1}{2}(3r_\square - l^\sharp) \geq r_\square$ from $\square^c$.

(b) A component without an old vertex consists of new vertices only; by the structure of the new vertices its lines are chain lines of one old line, so it is a sub-chain $n_i, \dots, n_j$ with at

most N_B vertices. Its label boxes lie in one cluster (all its lines are kept), so lemma H.61(b) with $|K| \le V^\sharp_{\mathrm{lab}}(\bar n)$ boxes bounds its diameter by $2V^\sharp_{\mathrm{lab}}(\bar n)$. Under the single-line threshold, consecutive boxes of the sub-chain intersect (lemma H.60), and lemma H.61(b) on the at most N_B boxes gives $2N_B$. □

Remark H.74. (a) *Per component.* The sentence "If a class contains an old vertex ..., then all vertices of this class are localized in $\square_1$" is read per connected component: a graph of the expanded class can have an old-vertex component and, beside it, a chain component anywhere in $\square$ (part (b)). Bałaban treats the components separately ([3, p. 433], before (3.3): "Considering each connected component of G_{ren} separately we can assume additionally that G_{ren} is connected"), so the reading costs nothing. Part (a) is also what places the old-vertex components at distance $\ge r_\square$ from $\square^c$, where the bounds on $\delta G_k(\Omega, \square, \tilde B)$ and on the cube propagator of the fourth part are evaluated.

(b) *The parameter condition.* The side $l^\sharp = l_N + 4V^\sharp_{\mathrm{lab}}(\bar n)$ is "the side of the localisation cube of a connected renormalised class after the expansion" with which the cube step of theorem F.2 defines its constant, $r^\flat \ge 3l^\sharp + R_0 + 12$; so $r_\square = r^\flat \ge l^\sharp$ and the hypothesis of lemma H.73 holds. It depends on $\bar n$ only: a chain of new vertices advances by at most 2 per vertex in the sup metric (the side of a label box), not by less than 1, which is why the side is $l_N + 4V^\sharp_{\mathrm{lab}}$ and not l_N plus the number of new vertices. At the printed cube, $r(t) \ge R$ and the condition is $R \ge l^\sharp$.

(c) *Counterterm graphs.* The counterterm graphs of the regrouped classes are members of the expanded class (lemma H.71(a)), and lemma H.73 holds per (G, τ) for every member; with the class-common cube Q_0 their counterterm vertices, which carry the labels of their anchors, lie in $Q_0 \subset Q_0^+$.

(d) *"These operations give us again a sum of renormalized classes."* This is the regrouping after the cut (S10, theorem F.57, in its cluster form), applied on $\square$ to the per-label classes of lemma H.71: the clusters of $\mathcal{B}^\sharp$ are common to the expanded class, a touched two-scalar-leg subgraph and its mirror, and a touched $1A$ subgraph and its $2A$ completion, have the same labels, so partners are kept or cut together.

(e) *A second printed source.* [13, Sect. 3.4] reaches the same conclusion on the torus by a different device, a cut-off θ supported within $\frac{2}{3}p(L^k\varepsilon)$ of the inner cube ("The result is a sum of new graphs, all with vertices inside $\square'$"); his mechanism does not use the distance rule and is not needed here.

H.4. **The sketched steps of Proposition 1, IV: the cube at zero field, the norms, the identities of the renormalisation, the further places and the classes after the expansion.**

This part contains the sketched steps of the proof of [3, Prop. 1, (1.33)] that follow the cube split of [3, p. 433] and the $\tilde B'$-expansion (S13–S15 of part III), the identities and the one-loop classes of the renormalisation step of [3, pp. 434–445] on which the renormalisation theorem of appendix F (Step 3(D_4) there, theorem F.70) rests, the seven further sketched places F1–F7, and the closure of the classes under the $\tilde B'$-expansion (S14.4):

step	[3] page	printed object
S16	433	the classes without old vertex (chains): "the estimate (3.2) for them is a consequer
S17	433	gauge covariance of the expressions on the cube $\square$
S18	433	the gauge transformation removing the constant field B_0
S19	433	$G_k(\square,0) = G_k(0) + \delta G_k(\square, \eta\mathbb{Z}^d, 0)$, with δG_k an external field
S20	434	the norms [3, (1.32)] at $\tilde{B} = B_0$ and at $\tilde{B}$: "we get (3.2) exactly"
S21	434–445	the identities and the one-loop classes of the renormalisation (appendices H.4.6 an
F1	433	the Ω_2 branch of the cube split: "[or $G_k(\Omega_2, \tilde{B})$]"
F2	418–419, 433–445	torus propagators in the counterterms against $\eta\mathbb{Z}^d$ propagators in the classes
F3	433	"We repeat some of the operations done before": [3, (1.16)] and [3, (3.1)] on $\square$
F4	430–431	"we can get all necessary Ward–Takahashi identities"
F5	427	"the remaining powers of $L^j\eta$ ... give us $(L^j\eta)^{D(G_1)}$"
F6	428	"In the case $v = v'$ we make the same operations"
F7	433	components separately; "We can extract also all coupling constants"
S14.4	433	"which can be decomposed into a sum of renormalized classes": the closure of the

The order is: the zero-field kernel bounds at every point of a box, the common tool of S16, S19, F2, F3 and S21 (appendices H.4.1 and H.4.2); S16 (appendix H.4.3); S17–S19 (appendix H.4.4); S20 (appendix H.4.5); the identities and one-loop classes of S21 (appendices H.4.6 and H.4.7); F1–F7 (appendix H.4.8); S14.4 (appendix H.4.9). The renormalisation of the divergent classes beyond these identities (the two-loop remainder terms, the transport of legs across the subgraphs [3, (2.4)] and the induction of [3, pp. 444–445]) is proved in Step 3(D_4) of appendix F and is not repeated here.

Conventions for this part.

Scales, fields, propagators. As in appendix H.1.1: $d = 3$ (the kernel bounds of appendix H.4.1 hold for general d), $L = 5$, $\eta = L^{-k}$, $k \geq 1$, $t = L^k\varepsilon$, $e_k = e\,t^{1/2}$, and $s_i := L^i\eta$. Scalar fields take values in $\mathbb{R}^N$; $U(A_b) = \exp(e_k\eta q A_b)$ with q a fixed real antisymmetric $N \times N$ matrix, $\|q\| \leq 1$; all transporters commute and are orthogonal. $(D_A\phi)(b) = \eta^{-1}(U(A_b)\phi(b_+) - \phi(b_-))$; on a union X of unit blocks with Neumann conditions $\langle\phi, -\Delta^{\eta,N}_{A,X}\phi\rangle = \sum_{b\subset X}\eta^d|D_A\phi(b)|^2$, and $G_k(X,A) := (-\Delta^{\eta,N}_{A,X} + m_k^2 + a_kP_k(A))^{-1}$ with $P_k = Q_k^*Q_k$, $Q_k(A)$ the covariant unit-block average along the contours of [1, (2.1)–(2.2)], $0 \leq m_k^2 = m^2t^2 \leq c_m$ and $a_k \in [a(1-L^{-2}), a]$. Vertices are those of [3, (1.6)–(1.15)]; the vertices [3, (1.8)–(1.11)] carry $e_k^{n+n'}$ from the expansion of U, which fixes the factor e_k inside U. Degrees are those of [3, (2.1)–(2.3)] (appendix H.2).

The cube. $\square \subset \Omega$ is the cube of the cube split of [3, p. 433] in the constant-side form of theorem F.2: a non-wrapping parallelepiped of the lift of the torus, a union of large blocks (side M), with $3r^\flat \leq \mathrm{side}(\square) \leq 3r^\flat + 2M$, containing the concentric cube $\square_1$ of side $r^\flat$ in which every old vertex is localised; so $\operatorname{diam}\square \leq \sqrt{3}(3r^\flat + 2M)$ and every point of $\square_1$ has depth $\geq r^\flat$ in $\square$. On $\square$ the background is $\tilde{B} = B_0 + \tilde{B}'$ with B_0 constant, and

$$\sigma_0 := \sup_{b\subset\square}|\tilde{B}'_b| \leq c\,(r^\flat + M)\,p(t), \qquad \sigma_1 := \sup_{b\subset\square}|\partial^\eta\tilde{B}'_b| = \sup_{\square}|\partial^\eta\tilde{B}| \leq c_1 p(t),$$

by the choice of B_0 in the cube step. Throughout we assume the cap

$$\text{(cap)} \qquad e_k(\sigma_0+\sigma_1)\,\big(1+\operatorname{diam}\square\big) \leq 1,$$

which holds for $t \leq t_0$ since $e_k(\sigma_0+\sigma_1)(1+\sqrt{3}(3r^\flat+2M)) \leq C\,e\,t^{1/2}(1+\ln t^{-1})^{\bar{p}} \to 0$; it is a condition of the kind of [3, p. 412], fixed after L, $\bar{n}$, R_0 and $r^\flat$ and before ε_0. Only $e_k(\sigma_0+\sigma_1) \leq 1$ is used outside appendix H.4.5.

Kernel notation. For a kernel $Y(x_1,x_2)$ on a lattice of spacing $\xi \in (0,1]$ and $S,T \subset \{1,2\}$, D_S applies one forward difference (any direction, bonds of the domain) in each variable x_i, $i \in S$, and Δ_T applies in each variable x_i, $i \in T \subset S$, a Hölder difference $Y(\dots,\tilde{x}_i,\dots) - Y(\dots,x_i,\dots)$ at a partner with $0 < |x_i - \tilde{x}_i| \leq 1$ (at zero field no transport is needed). Put $p(S,T) := d - 2 + |S| + \alpha|T|$ and $\rho := \min\{|u-w| : u \in \{x_1\}\cup\{\tilde{x}_1 \text{ if } 1\in T\},\ w \in \{x_2\}\cup\{\tilde{x}_2 \text{ if } 2\in T\}\}$; $h_\gamma(r) := 1$ for $\gamma < 0$, $1+|\log r|$ for $\gamma = 0$, $r^{-\gamma}$ for $\gamma > 0$. In appendices H.4.1 and H.4.2 distances

are ℓ^1; elsewhere they are the sup-distances of appendix H.1.1 ($|\cdot|_\infty \le |\cdot| \le |\cdot|_1 \le d\,|\cdot|_\infty$, so the choice affects constants only). A quotient over two points at distance > 1 is at most the sum of the two values, so every bound below is needed only at $|x_i - \tilde{x}_i| \le 1$; this is how the Hölder supremum of [3, (1.32)], which runs over all pairs of points, is reached from bounds at partners within distance 1: for $|x - x'| > 1$ the quotient is $\le 2\sup|D\cdot|$, so the all-pairs supremum is $\le \max\{\text{local supremum}, 2\sup|D\cdot|\}$.

The norm [3, (1.32)]. $\|f\|_{1,\alpha;A} := \sup|f| + \sup|D_A f| + \sup_{x\neq x'} |x-x'|^{-\alpha}|U(A(\Gamma_{x,x'}))D_A f(x') - D_A f(x)|$ for a function of one scalar slot, $\Gamma_{x,x'}$ a shortest contour; for kernels of several variables the maximal reading of appendix H.1.1, slot by slot. Vector fields carry the norm at $A = 0$ [3, p. 420].

Inputs. Besides the inputs (P), (C), (H0) of appendix H.1.1 and the lemmas of parts I–III, this part uses:

(i) **(A5)** the scale-uniform form of [4, Lemma 2.4] proved in theorem F.7: for $\xi = L^{-j}$, $j \ge 0$, $m^2 \ge 0$, $a \in [a_-, a_+]$, and X a rectangular parallelepiped of $\xi\mathbb{Z}^d$ with Neumann conditions, a periodic box or $\xi\mathbb{Z}^d$, each a union of unit blocks, the columns $G_j\mathbf{1}_{B(y)}$ of $G_j := (-\Delta_X^\xi + m^2 + aQ_j^*Q_j)^{-1}$ obey $|(G_j\mathbf{1}_{B(y)})(x)|$, $|D(G_j\mathbf{1}_{B(y)})(x)| \le c_0 e^{-\delta_0|x-y|}$ and $|x-\tilde{x}|^{-\alpha}|D(G_j\mathbf{1}_{B(y)})(\tilde{x}) - D(G_j\mathbf{1}_{B(y)})(x)| \le c_1(\alpha)e^{-\delta_0\min(|x-y|,|\tilde{x}-y|)}$ for $|x-\tilde{x}| \le 1$, with $c_0, c_1(\alpha), \delta_0$ depending only on $d, a_\pm, \alpha$. Its proof (reflection and periodisation of the full-lattice kernel) also gives the image identities used in lemma H.89 and lemma H.109: for a Neumann box X that is a union of unit blocks, $G_j(X,0)(x,x') = \sum_\gamma G_j(\xi\mathbb{Z}^d,0)(x,\gamma x')$ over the reflection group of the box, and for a periodic box the sum over its period lattice, both absolutely convergent.

(ii) **(C′)** the bound (C) of appendix H.1.1 at every scale: for the unit-lattice block operators $C^{(j)}$ of lemma H.75, $|C^{(j)}(y,y')| \le c_2 e^{-\theta_0|y-y'|_\infty}$ with c_2, θ_0 depending on $d, L, a_\pm, a'$ only. The operators at scale j are the rescaled operators at scale k with the same unit-block structure, so (C) applies verbatim.

(iii) **(cube)** the cube split of appendix F (the replacement (a) of theorem F.2): the properties of $\square$, $\square_1$, B_0, $\tilde{B}'$ listed above, and the fact that after the $\tilde{B}'$-expansion and the cut of separated lines every class component containing an old vertex is localised in $\square_1$ (S15).

(iv) **(expansion)** the $\tilde{B}'$-expansion of part III (S14): $G_k(\square,\tilde{B}) = \sum_{m=0}^{\bar{n}}(-1)^m(G_0V)^mG_0 +$ remainders of the type [3, (1.16)], $G_0 := G_k(\square, B_0)$, $V := G_k(\square,\tilde{B})^{-1} - G_0^{-1}$ expanded in $\tilde{B}'$ into the vertices [3, (1.8)–(1.11)] ($n = 0$, $\tilde{A} \to \tilde{B}'$, $\tilde{B} \to B_0$) and the block pairs [3, (1.13)–(1.15)] ($A' \to 0$, $\tilde{A} \to \tilde{B}'$); the new vertices are localised by a partition of unity per $\tilde{B}'$-leg with supports in cubes of side 2 (block vertices by y), and the remainders and the lines between localisations at distance ≥ 1 become external fields.

H.4.1. *Zero-field kernel bounds at every point of a box, I: one scale and the cumulative propagators.* After the gauge-out of B_0 (S18) every scalar line of a class on the cube is a zero-field propagator on a Neumann box, on $\eta\mathbb{Z}^d$ or on the torus. The following bounds hold at *every* pair of points of the box, with no depth condition, which is what S16, S19, F2, F3 and the renormalisation step need. Distances in this and the next subsection are ℓ^1; $\alpha \in (0,1)$; S, T, $p(S,T)$, ρ are as in the conventions of appendix H.4.

Lemma H.75 (one scale, all points)**.** *Fix $L \ge 3$ odd, $0 < a_- \le a_+$, $a' > 0$ and $\alpha \in (0,1)$. For $j \ge 0$ let $\xi := L^{-j}$ and let X be* (N) *a Neumann parallelepiped of $\xi\mathbb{Z}^d$ that is a union of L-blocks of the unit lattice,* (P) *a periodic box $\xi\mathbb{Z}^d/K\mathbb{Z}^d$ with $K \in (L\mathbb{Z}_{\ge 1})^d$, or* (Z) *$\xi\mathbb{Z}^d$. Let $m^2 \ge 0$ and $a \in [a_-, a_+]$.*

- *For $j \ge 1$ put $\mathsf{T} := -\Delta_X^{\xi,N} + m^2 + aQ_j^*Q_j$ on $\ell^2(X,\xi^d)$, $G_j := \mathsf{T}^{-1}$, $\Delta^{(j)} := a - a^2Q_jG_jQ_j^*$, $C^{(j)} := (\Delta^{(j)} + a'L^{-2}P)^{-1}$ on the unit lattice $X^{(j)}$ (P the projection onto functions constant on L-blocks), and $S_j := a^2G_jQ_j^*C^{(j)}Q_jG_j$.*
- *For $j = 0$ ($\xi = 1$) put $S_0 := (-\Delta_X^{1,N} + m^2 + a'L^{-2}P)^{-1}$.*

Then there are $C_Z, \delta_Z > 0$, depending only on $(d, L, a_\pm, a', \alpha)$, such that for all j, X, m^2, a, x_1, x_2, S, T:

$$|\Delta_T D_S S_j(x_1, x_2)| \le C_Z e^{-\delta_Z \rho} \prod_{i \in T} |x_i - \tilde{x}_i|^\alpha. \tag{90}$$

Proof. Step 1 (the kernel, $j \ge 1$). $Q_j^* \delta_y = \mathbf{1}_{B(y)}$ and G_j is real symmetric, so $(Q_j G_j f)(y') = \langle G_j \mathbf{1}_{B(y')}, f \rangle$. Hence, with respect to the weight ξ^d,

$$S_j(x_1, x_2) = a^2 \sum_{y, y' \in X^{(j)}} (G_j \mathbf{1}_{B(y)})(x_1)\, C^{(j)}(y, y')\, (G_j \mathbf{1}_{B(y')})(x_2).$$

Each variable sits in its own factor, so D_S and Δ_T act factor by factor.

Step 2 (the factors). By (A5), on all three kinds of X, $|(G_j \mathbf{1}_{B(y)})(x)|$, $|D(G_j \mathbf{1}_{B(y)})(x)| \le c_0 e^{-\delta_0 |x-y|}$ and $|x - \tilde{x}|^{-\alpha} |D(G_j \mathbf{1}_{B(y)})(\tilde{x}) - D(G_j \mathbf{1}_{B(y)})(x)| \le c_1 e^{-\delta_0 \min(|x-y|, |\tilde{x}-y|)}$. By (C′), $|C^{(j)}(y, y')| \le c_2 e^{-\theta_0 |y-y'|_\infty} \le c_2 e^{-(\theta_0/d)|y-y'|}$.

Step 3 (the sum). Put $\delta := \min\{\delta_0, \theta_0/d\}$. For the points u, w that carry the factors (a partner where the Hölder clause is used), $|u - w| \le |u - y| + |y - y'| + |y' - w|$, so the product of the three exponentials is $\le e^{-(\delta/2)\rho} e^{-(\delta/2)(|u-y| + |y-y'| + |y'-w|)}$. The unit-lattice sums are bounded by $\sum_{y \in \mathbb{Z}^d} e^{-c|x-y|} \le (2/(1 - e^{-c}))^d$ for every real x (a minimum over two points costs a factor 2). This gives (90) with $\delta_Z := \delta/2$.

Step 4 ($j = 0$). On the unit lattice, with $\mathsf{T}_0 := -\Delta_X^{1,N} + m^2 + a' L^{-2} P$: for a single L-block Q with Neumann conditions the spectral gap of $-\Delta_Q^{1,N}$ above the constants is $4 \sin^2(\pi/2L) \ge 4L^{-2}$ (the one-dimensional free-end path on L sites has eigenvalues $2 - 2\cos(\pi m/L)$, $m = 0, \dots, L-1$), so $-\Delta_Q^{1,N} + a' L^{-2} P_Q \ge L^{-2} \min\{4, a'\}$; dropping the bonds between blocks only lowers the quadratic form, so $\mathsf{T}_0 \ge \gamma' := L^{-2} \min\{4, a'\}$ on X. The argument of lemma H.10 with $Y = \mathsf{T}_0$, $v = m^2$ and P of range L (so that $|\rho(x) - \rho(x')| \le L$ on a block and $c(\beta) = 3\beta^2 \cosh \beta + a' L^{-2}(e^{\beta L} - 1)$) gives, for $\theta' > 0$ with $c(\theta') \le \gamma'/2$, $\|e^{\theta' \rho} \mathsf{T}_0^{-1} e^{-\theta' \rho}\| \le 2/\gamma'$ for every 1-Lipschitz ρ in the sup-metric; with $\rho = |\cdot - x'|_\infty$ and $\|\delta_x\| = 1$ this is $|S_0(x, x')| \le 2\gamma'^{-1} e^{-\theta' |x - x'|_\infty}$. A forward difference on the unit lattice is a difference of two neighbouring values, and two distinct partners are at distance ≥ 1, so every quotient is at most twice a supremum. This gives (90) at $j = 0$ with $\delta_Z \le \theta'/d$. □

Lemma H.76 (geometric sums)**.** *For $p > 0$, $\delta > 0$, $\eta \in (0, 1]$, $s_i := L^i \eta \le 1$ and $\rho \ge 0$: $\sum_i s_i^{-p} e^{-\delta \rho / s_i} \le C_g(p, \delta, L)(\eta + \rho)^{-p}$.*

Proof. If $\rho \le \eta$ the sum is $\le \eta^{-p}/(1 - L^{-p}) \le 2^p (\eta + \rho)^{-p}/(1 - L^{-p})$. If $\rho > \eta$, the terms with $s_i \ge \rho$ sum to $\le \rho^{-p}/(1 - L^{-p})$. The terms with $s_i < \rho$ equal $\rho^{-p} u_i^p e^{-\delta u_i}$, where the $u_i = \rho/s_i > 1$ form a geometric sequence of ratio L; at most one u_i lies in each interval $[L^n, L^{n+1})$, $n \ge 0$, and $t^p e^{-\delta t} \le L^{(n+1)p} e^{-\delta L^n}$ there, so these terms sum to $\le \rho^{-p} \sum_{n \ge 0} L^{(n+1)p} e^{-\delta L^n}$. Finally $\rho^{-p} \le 2^p (\eta + \rho)^{-p}$. □

Corollary H.77 (cumulative propagators at zero field, all points)**.** *Let $k \ge 1$, $\eta = L^{-k}$, $s_i := L^i \eta$, and let $X \subset \eta \mathbb{Z}^d$ be of type* (N), (P) *or* (Z) *and a union of unit blocks. Let $m^2 \ge 0$, $a_1 := a'$, $a_{i+1} := a' a_i / (a' L^{-2} + a_i)$ (so $a_i \in [a'(1 - L^{-2}), a']$), and $\mathsf{T}_i := -\Delta_X^{\eta, N} + m^2 + a_i s_i^{-2} P^{(i)}$, $P^{(i)}$ the projection onto functions constant on the blocks of side s_i, for $1 \le i \le k$, and also for $i = k + 1$ when X is a union of L-blocks of the unit lattice. Then:*

(i) *for $1 \le J \le k$, and for $J = k + 1$ under that hypothesis, $\mathsf{T}_J^{-1} = \sum_{i=0}^{J-1} G_{(i)}$, where $G_{(0)} := \mathsf{T}_1^{-1}$, $G_{(i)} := \mathsf{T}_{i+1}^{-1} - \mathsf{T}_i^{-1}$, and $G_{(i)}(x, x') = s_i^{2-d} S_i(x/s_i, x'/s_i)$ with S_i the kernel of lemma H.75 at $j = i$ on X/s_i with mass $m^2 s_i^2$ and $a = a_i$;*

(ii) *for the same J and all S, T:*

$$|\Delta_T D_S \mathsf{T}_J^{-1}(x_1, x_2)| \le C_1 (\eta + \rho)^{-p(S,T)} e^{-\delta_Z \rho/2} \prod_{i \in T} |x_i - \tilde{x}_i|^\alpha, \tag{91}$$

with C_1 depending only on (d, L, a', α) (for $d = 3$, $p(S, T) > 0$).

In particular $G_k(X,0) = \mathsf{T}_k^{-1}$ ($s_k = 1$) at $a' = a$, and (i) is the zero-field case of the slice decomposition [3, (2.6)]. *The constants do not depend on k, J, X, $m^2 \geq 0$, nor on a' within a fixed range.*

Proof. Step 1 (rescaling). For $s > 0$ let $(U_s f)(y) := f(sy)$. If $A = U_s^{-1}\hat{A}U_s$, then $A(x,x') = s^{-d}\hat{A}(x/s, x'/s)$ for the kernels with respect to the respective lattice weights. Since $U_s(-\Delta^\eta)U_s^{-1} = s^{-2}(-\Delta^{\eta/s})$, $\mathsf{T}_i = U_{s_i}^{-1}s_i^{-2}\hat{\mathsf{T}}_i U_{s_i}$ with $\hat{\mathsf{T}}_i = -\Delta^{\xi_i} + m^2 s_i^2 + a_i\hat{P}$ on $\xi_i\mathbb{Z}^d$, $\xi_i = L^{-i}$, $\hat{P}$ the unit-block projection. Hence $\mathsf{T}_i^{-1}(x,x') = s_i^{2-d}\hat{\mathsf{T}}_i^{-1}(x/s_i, x'/s_i)$, and the same for T_{i+1} at the same rescaling, whose block term is $a_{i+1}L^{-2}\hat{P}_{(L)}$.

Step 2 (composition). In the units $s_i = 1$, the block inversion of the operator $\hat{\mathsf{T}}_i + a_{i+1}L^{-2}\hat{P}_{(L)}$, read in its $\phi\phi$-block (the inverse of the Schur complement in the block variable), gives $\hat{\mathsf{T}}_{i+1}^{-1} = \hat{\mathsf{T}}_i^{-1} + a_i^2\hat{\mathsf{T}}_i^{-1}Q^*C^{(i)}Q\hat{\mathsf{T}}_i^{-1}$ with $C^{(i)} = (a_i - a_i^2 Q\hat{\mathsf{T}}_i^{-1}Q^* + a'L^{-2}P)^{-1}$: this is the identity $(M^{-1})_{\phi\phi} = D^{-1} + D^{-1}B^*(A_\psi - BD^{-1}B^*)^{-1}BD^{-1}$ for $M = \begin{pmatrix} A_\psi & -B \\ -B^* & D \end{pmatrix}$ with $D = \hat{\mathsf{T}}_i$, $B = a_iQ$, $A_\psi = a_i + a'L^{-2}P$ (the quadratic form $\langle\phi, (-\Delta^{\xi_i} + m^2 s_i^2)\phi\rangle + a_i\|Q\phi - \psi\|^2 + a'L^{-2}\langle\psi, P\psi\rangle$ in (ϕ,ψ)). Its ϕ-block Schur complement is $D - B^*A_\psi^{-1}B = \hat{\mathsf{T}}_i - a_i^2 Q^*(a_i + a'L^{-2}P)^{-1}Q = -\Delta^{\xi_i} + m^2 s_i^2 + \frac{a_i a'L^{-2}}{a_i + a'L^{-2}}Q^*PQ$, since $Q^*(a_i + bP)^{-1}Q = a_i^{-1}Q^*(1-P)Q + (a_i+b)^{-1}Q^*PQ$ and $Q^*Q - Q^*(1-P)Q = Q^*PQ$; this is $\hat{\mathsf{T}}_{i+1}$ (block term $a_{i+1}L^{-2}\hat{P}_{(L)}$, $\hat{P}_{(L)} = Q^*PQ$) exactly when $a_{i+1} = a'a_i/(a'L^{-2} + a_i)$, the recursion which reproduces $a_k = a(1-L^{-2})/(1-L^{-2k})$ of [1, (2.15)] and is the zero-field form of [3, (1.27)]. So $G_{(i)} = s_i^{2-d}S_i(\cdot/s_i, \cdot/s_i)$, and the sum telescopes to (i). X/s_i is a union of unit blocks and of L-blocks of its unit lattice since $Ls_i = s_{i+1} \leq 1$, so lemma H.75 applies. For $J = k+1$ the last slice is $i = k$ ($s_k = 1$, $X/s_k = X$), and lemma H.75 at $j = k$ needs X to be a union of L-blocks of the unit lattice; this holds for $\xi\mathbb{Z}^d$, for periodic boxes with sides in $L\mathbb{Z}$, and for Neumann boxes that are unions of L-blocks. (Rescaling the statement at $J \leq k$ is not enough for this case: its exponential $e^{-\delta\rho_\eta/2}$ would become $e^{-\delta s\rho_\xi/2}$.)

Step 3 (bounds). Under the rescaling a difference costs s_i^{-1} and $|x-\tilde{x}|^\alpha = s_i^\alpha|y-\tilde{y}|^\alpha$. For $|x-\tilde{x}| \leq s_i$, (90) applies; for $s_i < |x-\tilde{x}| \leq 1$ bound the difference by two values, which is $\leq 2(|x-\tilde{x}|/s_i)^\alpha$ times a value. So $|\Delta_T D_S G_{(i)}| \leq 2^{|T|}C_Z s_i^{-p(S,T)}e^{-\delta_Z\rho/s_i}\prod|x_i - \tilde{x}_i|^\alpha$. Write $e^{-\delta_Z\rho/s_i} \leq e^{-\delta_Z\rho/2}e^{-\delta_Z\rho/(2s_i)}$ and apply lemma H.76. □

H.4.2. *Zero-field kernel bounds at every point of a box, II: the free propagator and the remainder $\breve{G} - C^\xi$.*

Lemma H.78 (free massive propagator smeared over a block)**.** *Let $\xi = L^{-J}$, $J \geq 0$, L odd, $\mu^2 \in [\frac{1}{2}, 2]$, and $C_\mu := (-\Delta^\xi + \mu^2)^{-1}$ on $\xi\mathbb{Z}^d$. Put $\delta_C := (88d)^{-1/2}$. Then for every unit block $B(y)$, every x, $\tilde{x}$ with $|x-\tilde{x}| \leq 1$ and every direction,*

$$|(C_\mu \mathbf{1}_{B(y)})(x)|,\ |D(C_\mu\mathbf{1}_{B(y)})(x)| \leq c_C e^{-\delta_C|x-y|}, \qquad |x-\tilde{x}|^{-\alpha}|D(C_\mu\mathbf{1}_{B(y)})(\tilde{x}) - D(C_\mu\mathbf{1}_{B(y)})(x)| \leq c_C(\alpha)e^{-\delta_C \min(}$$

with c_C, $c_C(\alpha)$ depending only on (d,α).

Proof. Let $N := 1/\xi = L^J$ (odd) and $B(y) = y + ([0,1)\cap\xi\mathbb{Z})^d$. With $\hat{f}(p) := \sum_x \xi^d f(x)e^{-ip\cdot x}$ on $B_\xi := [-\pi/\xi, \pi/\xi]^d$, the geometric sum $\sum_{n=0}^{N-1}\xi e^{-it\xi n} = e^{-i(1-\xi)t/2}g(t)$, $g(t) := \xi\sin(t/2)/\sin(\xi t/2)$, gives $\hat{\mathbf{1}}_{B(y)}(p) = e^{-ip\cdot y}u(p)$ with $u(p) := e^{-i(1-\xi)(p_1+\cdots+p_d)/2}\prod_\nu g(p_\nu)$. Hence, for $M \in \{1, \hat{\partial}_\nu, \hat{\partial}_\nu(e^{ip\cdot v'} - 1)/|v'|^\alpha\}$ ($\hat{\partial}_\nu(p) = (e^{i\xi p_\nu} - 1)/\xi$, $v' := \tilde{x} - x \in \xi\mathbb{Z}^d$, $0 < |v'| \leq 1$), the three quantities of the statement are

$$(2\pi)^{-d}\int_{B_\xi} e^{ip\cdot(x-y)}\,\frac{M(p)\,u(p)}{\Delta^\xi(p) + \mu^2}\,dp, \qquad \Delta^\xi(p) = \sum_\nu \frac{4}{\xi^2}\sin^2\frac{\xi p_\nu}{2}.$$

Step 1 (periodicity). Every factor is $(2\pi/\xi)$-periodic in each p_ν: $x - y$, $v' \in \xi\mathbb{Z}^d$; Δ^ξ and $\hat{\partial}$ are trigonometric in ξp; and u is periodic because N is odd: $g(t + 2\pi/\xi) = \xi(-1)^N\sin(t/2)/(-\sin(\xi t/2)) = g(t)$, and the phase gains $e^{-i\pi(N-1)} = 1$. The integrand is entire in each p_ν (g is entire).

Step 2 (the denominator on the strip). Let $p = q + i\sigma$, $q \in B_\xi$, $|\sigma|_\infty \leq \delta_C \leq \frac{1}{4}$. From $\sin^2(a+ib) - \sin^2 a = \sinh^2 b\,(\sin^2 a - \cos^2 a) + \frac{i}{2}\sin 2a \sinh 2b$ with $a = \xi q_\nu/2$, $b = \xi\sigma_\nu/2$, and

$\sinh x \le 1.13x$, $\sinh 2x \le 3.1x$ for $0 \le x \le \frac12$: $|\Delta_\nu^\xi(p) - \Delta_\nu^\xi(q)| \le 1.28\sigma_\nu^2 + 3.1|\sigma_\nu|\Delta_\nu^\xi(q)^{1/2}$. Summing over ν and using $\sum_\nu \Delta_\nu^{1/2} \le \sqrt{d}\,\Delta^{1/2}$ and $3.1\delta\sqrt{d}\,\Delta^{1/2} \le \frac14\Delta + 9.7d\delta^2$: $|\Delta^\xi(p) - \Delta^\xi(q)| \le \frac14\Delta^\xi(q) + 11d\delta_C^2 \le \frac14\Delta^\xi(q) + \frac18 \le \frac14(\Delta^\xi(q)+\mu^2)$. Hence $|\Delta^\xi(p)+\mu^2| \ge \frac34(\Delta^\xi(q)+\mu^2) \ge c\,(1+|q|)^2$, since $\Delta^\xi(q) \ge (4/\pi^2)|q|^2$ on B_ξ.

Step 3 (the numerator). For $z = t+i\sigma$, $|t| \le \pi/\xi$, $|\sigma| \le \frac14$: $|\sin(z/2)|^2 = \sin^2(t/2) + \sinh^2(\sigma/2)$ and $|\sin(\xi z/2)|^2 = \sin^2(\xi t/2) + \sinh^2(\xi\sigma/2) \ge (\xi t/\pi)^2 + (\xi\sigma/2)^2$. For $|t| \le \pi$ this gives $|g(z)|^2 \le \max\{(\pi/2)^2, 1.03\} \le 2.5$; for $|t| > \pi$, $|\sin(z/2)| \le \cosh\frac18 \le 1.01$ gives $|g(z)| \le 3.2/|t|$. In both cases $|g(z)| \le 10/(1+|t|)$. The phase of u has modulus $\le e^{d\delta_C/2} \le 1.2$, $|\hat\partial_\nu(p)| \le 1.3|p_\nu| \le 1.3(1+|q|)$, and $|e^{ip\cdot v'} - 1|/|v'|^\alpha \le C\min\{1, |q||v'|\}^\alpha |v'|^{-\alpha} + C|v'|^{1-\alpha} \le C(1+|q|)^\alpha$ on the strip. So $|M(p)| \le C(1+|q|)^{1+\alpha}$.

Step 4 (integrability). By Steps 2–3 the modulus of the integrand on the shifted contour is $\le e^{-\sigma\cdot(x-y)}\,C(d)\,(1+|q|)^{\alpha-1}\prod_\nu(1+|q_\nu|)^{-1} \le e^{-\sigma\cdot(x-y)}C(d)\prod_\nu(1+|q_\nu|)^{-1-(1-\alpha)/d}$, using $(1+|q|)^d \ge \prod_\nu(1+|q_\nu|)$; this is integrable over $\mathbb{R}^d$ uniformly in ξ.

Step 5 (the shift). Shift each p_ν to $p_\nu + i\delta_C\,\mathrm{sgn}(x-y)_\nu$; the two vertical sides of each rectangle cancel by Step 1, and the shifted integral carries $|e^{ip\cdot(x-y)}| = e^{-\delta_C|x-y|_1}$. For the Hölder quotient, $|x-y| \ge \min(|x-y|, |\tilde x - y|) - 1$ costs a factor e^{δ_C}. □

Lemma H.79 (the free propagator $C^\xi := (-\Delta^\xi + 1)^{-1}$ on $\xi\mathbb{Z}^d$, $\xi = L^{-J}$)**.** *For all S, T: $|\Delta_T D_S C^\xi(x_1,x_2)| \le C'_C(\xi+\rho)^{-p(S,T)}e^{-\delta'\rho}\prod_{i\in T}|x_i - \tilde x_i|^\alpha$, with C'_C, δ' depending only on (d, L, α).*

Proof. Let $G := \mathsf{T}_J^{-1}$ of corollary H.77 on $\xi\mathbb{Z}^d$ ($\eta \to \xi$, $k \to J$) with $m^2 = 1$; its block term is $a_J Q^*Q$ on unit blocks. Then $(C^\xi)^{-1} = G^{-1} - a_J Q^*Q$, so $C^\xi = G + a_J G Q^*Q C^\xi$, and the kernel of the second term is $a_J\sum_y (G\mathbf{1}_{B(y)})(x_1)(C^\xi\mathbf{1}_{B(y)})(x_2)$, since $(QC^\xi f)(y) = \langle C^\xi\mathbf{1}_{B(y)}, f\rangle$. G obeys (91). By (A5) for $G\mathbf{1}_{B(y)}$ and lemma H.78 for $C^\xi\mathbf{1}_{B(y)}$, and the unit-lattice sum as in lemma H.75, Step 3, the second term obeys $|\Delta_T D_S \cdot| \le Ce^{-\delta''\rho}\prod|\cdot|^\alpha \le C'(\xi+\rho)^{-p}e^{-\delta''\rho/2}\prod|\cdot|^\alpha$. □

Lemma H.80 (lattice convolution, $d = 3$)**.** *For $p_1, p_2 \in [0,3)$, $\delta > 0$, $\xi \in (0,1]$ and $x, w \in \xi\mathbb{Z}^3$:*

$$\sum_{z\in\xi\mathbb{Z}^3} \xi^3(\xi+|x-z|)^{-p_1}(\xi+|z-w|)^{-p_2}e^{-\delta(|x-z|+|z-w|)} \le C_K\, h_{p_1+p_2-3}(\xi+|x-w|)\,e^{-\delta|x-w|/2}.$$

Proof. Factor out $e^{-\delta|x-w|/2}$ by the triangle inequality and let $r := \xi + |x-w|$. On $\{|x-z| \le |x-w|/2\}$ we have $\xi + |z-w| \ge r/2$, and $\sum_{|u|\le R}\xi^3(\xi+|u|)^{-p_1} \le C(R+\xi)^{3-p_1}$ since $p_1 < 3$; this region gives $\le Cr^{3-p_1-p_2}$. The region $\{|z-w| \le |x-w|/2\}$ is symmetric. On the rest, $|x-z| \le 3|z-w|$ and the sum is $\le C\sum_{|u|\ge|x-w|/2}\xi^3(\xi+|u|)^{-p_1-p_2}e^{-\delta|u|/2} \le Ch_{p_1+p_2-3}(r)$. For $r \ge 1$ the powers are absorbed by the remaining exponential. □

Lemma H.81 (cumulative minus free: the gain of the resolvent comparison, $d = 3$)**.** *Let $\breve G := (-\Delta^\xi + \mu^2 + bP_Y)^{-1}$ on $\xi\mathbb{Z}^3$, with $\mu^2 \in [0,1]$, $b \in [b_-, b_+] \subset (0,\infty)$, P_Y the projection onto constants on the blocks Y of side $s_Y \in \{1, L\}$, and assume that $\breve G$ obeys (91) (with ξ in place of η). This holds for the rescaled cumulative propagators T_J^{-1} ($s_Y = 1$) and T_{J+1}^{-1} ($s_Y = L$; the case $J = k'+1$ of corollary H.77 on the ξ-lattice, $\xi\mathbb{Z}^3$ being a union of L-blocks) of corollary H.77, Step 1. Put $R := \breve G - C^\xi$. Then*

$$|\Delta_T D_S R(x_1,x_2)| \le C_R\, h_{|S|+\alpha|T|-1}(\xi+\rho)\,e^{-\delta_R\rho}\prod_{i\in T}|x_i - \tilde x_i|^\alpha. \tag{92}$$

So R and its Hölder quotients are bounded, one derivative costs at most a logarithm, and two derivatives (one per variable) cost $(\xi+\rho)^{-1}$: a gain of 2 (up to the logarithm) over C^ξ.

Proof. $\breve G - C^\xi = \breve G\big((C^\xi)^{-1} - \breve G^{-1}\big)C^\xi = (1-\mu^2)\breve G C^\xi - b\,\breve G P_Y C^\xi$; this is the identity of [3, p. 437].

Block part. $P_Y = \sum_Y |Y|^{-1}\mathbf{1}_Y\langle\mathbf{1}_Y, \cdot\rangle$, so the kernel of $\breve G P_Y C^\xi$ is $\sum_Y |Y|^{-1}(\breve G\mathbf{1}_Y)(x_1)(C^\xi\mathbf{1}_Y)(x_2)$. $\breve G\mathbf{1}_Y$ obeys the bounds of (A5): for $s_Y = 1$ directly, and for $s_Y = L$ after rescaling by L (the blocks become unit blocks), with constants depending on L. $C^\xi\mathbf{1}_Y$ is a sum of s_Y^3

unit-block terms $C^{\xi}\mathbf{1}_B$, $B \subset Y$, each bounded by lemma H.78. So this part is bounded by $Ce^{-\delta''\rho}\prod|\cdot|^{\alpha} \le Ch_{\gamma}(\xi+\rho)e^{-\delta''\rho}\prod|\cdot|^{\alpha}$ for every γ.

Convolution part. The kernel is $\sum_z \xi^3 \breve{G}(x_1,z)C^{\xi}(z,x_2)$; differences and quotients in x_1 fall on $\breve{G}$ and those in x_2 on C^{ξ}. By (91) and lemma H.79 the factors are bounded by $(\xi+|\cdot|)^{-p_i}e^{-\delta|\cdot|}$ with $p_1 = 1 + [1 \in S] + \alpha[1 \in T] < 3$ and likewise p_2 (a quotient gives a minimum over two points, bounded by the sum of the two terms). Lemma H.80 gives $h_{p_1+p_2-3} = h_{|S|+\alpha|T|-1}$. □

H.4.3. *S16: the classes without old vertex (chains),* [3, p. 433]. [3, p. 433]: after the $\tilde{B}'$-expansion "we get classes without old vertices. Such a class is a chain of propagators $G_k(\square, B_0)$ with new vertices in between … For such chains every subgraph has positive degree or contains (2.4), so the estimate (3.2) for them is a consequence of Proposition 2.1." (The subscript of the chain propagator is printed as a Q-like glyph; the expansion $G_k(\square, B_0+\tilde{B}') = \sum (G_0V)^m G_0 + \dots$ forces B_0.) Setting: (expansion) of appendix H.4, $G_0 := G_k(\square, B_0)$, $e_k(\sigma_0+\sigma_1) \le 1$.

Lemma H.82 (structure of a chain)**.** *A connected component that contains no old vertex is a path $v_1 - v_2 - \cdots - v_m$ of new vertices of one product $(G_0V)^m G_0$, joined by lines G_0. Its end vertices v_1, v_m each carry one external scalar field (a cut line), and otherwise the component has only external $\tilde{B}'$-legs; if $m = 1$ it has no internal line and is a product of external fields. Chains contain no field* [3, (1.12)]*: their vertices are of the forms* [3, (1.8)–(1.11), (1.13)–(1.15)] *"with external vector field $\tilde{B}'$ only"* [3, p. 433]*, which is what makes "the end field is a cut line" exact.*

Proof. Lines between new vertices arise only inside one product $(G_0V)^m G_0$, where consecutive V's are joined by one G_0; new vertices of different products are joined only through old vertices. So a component without old vertices lies in one product. Cutting lines of a path leaves sub-paths. Each end of a sub-path was joined in the product either to an old vertex or to another new vertex, and in both cases that line was cut (otherwise the component would be larger), so it is an external scalar field. □

Lemma H.83 (degrees of chains)**.** *Let $d = 3$. Every connected subgraph H of a chain component is a sub-path with $m_H \ge 2$ vertices, and $D(H) > 0$, except when $m_H = 2$ and H is the graph* [3, (2.4)]*: two* [3, (1.8)]*-vertices with $n' = 1$ whose common line carries the derivative at both ends. Every sub-path properly containing a* [3, (2.4)] *has $D \ge 1$. In particular "every subgraph has positive degree or contains (2.4)" holds in the sharper form "or is (2.4)".*

Proof. By [3, (2.1)], with external $\tilde{B}'$-legs of dimension 0 that still carry their factor η, $s \in \{1,2\}$ the number of internal scalar legs at v, and $\delta \in \{0,1\}$ with $\delta = 1$ only if the differentiated scalar leg is internal: [3, (1.8)] ($n = 0$, $n' \ge 1$; η-power $3 + n' - 1$) has $D_H(v) = 2 + n' - s/2 - \delta$; [3, (1.9)] ($n' \to \bar{n} + 1$) has $D_H(v) \ge \bar{n} + 1$; [3, (1.10)] ($n' \ge 2$; η-power $3 + n' - 2$) has $D_H(v) = 1 + n' - s/2 \ge 2$; [3, (1.11)] has $\ge \bar{n}$; a block pair w ([3, (1.18)], one vertex made of two factors, each with η^d and one scalar leg, the $\tilde{B}'$-contour legs external) has $D_H(w) = 6 - s/2 \ge 5$, which is [3, p. 423 (v)] summed over the two factors (the reading under which the proof of [3, Cor. 2.3] on [3, p. 428], "if both endpoints of l belong to v, then $D_G(v) - d > 0$", holds). Hence an interior vertex ($s = 2$) has $D_H(v) \ge 1$, and an end vertex ($s = 1$) has $D_H(v) \ge \frac{3}{2}$, with equality only for [3, (1.8)] with $n' = 1$ and $\delta = 1$; every other end vertex has $\ge \frac{5}{2}$. With $D(H) = \sum_v D_H(v) - 3$: for $m_H = 2$, $D(H) \ge 0$ with equality iff H is [3, (2.4)], otherwise $D(H) \ge \frac{3}{2} + \frac{5}{2} - 3 = 1$; for $m_H \ge 3$, $D(H) \ge 3 + (m_H - 2) - 3 \ge 1$. □

Lemma H.84 (the chain bound)**.** *Let H be a chain component with vertices $v_1, \dots, v_m$ ($2 \le m \le \bar{n}+1$), end fields Φ_1 at v_1 and Φ_m at v_m, and localisation cubes $\square(v)$. Assume $e_k(\sigma_0+\sigma_1) \le 1$, and let h be smooth with $h = 1$ on the 2η-neighbourhood of the end cube and $|\partial^{\eta} h| \le c_h$. Then for every $\delta_0 \in (0,1]$*

$$|E(H)| \le C_H \prod_v (e_k\|q\|\sigma_0)^{n'_v}\, \|h\Phi_1\|_{1,0}\|h\Phi_m\|_{1,0}\, e^{-\delta_0 d(\{\square(v)\})},$$

with C_H depending only on $(d, N, \|q\|, a, L, \bar{n}, c_h, c_m)$. This is [3, (3.2)] *for the chains, with sup-norms in place of $\|\cdot\|_{1,\alpha_0}$. No slice decomposition, no summation by parts and no depth inside $\square$ is used.*

Proof. Step 1 (coercivity). B_0 is constant, hence regular, so (H0) on $\square$ gives $G_0^{-1} \geq \gamma_0 > 0$. Put $\|\phi\|_E := \langle \phi, G_0^{-1}\phi\rangle^{1/2}$. Since $G_0^{-1} \geq D_{B_0}^* D_{B_0}$ and $G_0^{-1} \geq \gamma_0$, $\|D_{B_0}\phi\| \leq \|\phi\|_E$ and $\|\phi\| \leq \gamma_0^{-1/2}\|\phi\|_E$.

Step 2 (each vertex is a bounded form in the energy norm). Let $\beta_v(\phi,\chi)$ be the bilinear form of v in its two scalar legs. For [3, (1.8)]: $\beta = \kappa \sum_{b\subset\square} \eta^d [(D_{B_0}\phi)(b) \cdot M(b)\chi(b_-)]$ with $\kappa = e_k^{n'}\eta^{n'-1}/n'!$ and $|M(b)| \leq \|q\|^{n'}\sigma_0^{n'}$, so by Cauchy–Schwarz over the d bonds at a site, $|\beta| \leq \sqrt{d}(e_k\|q\|\sigma_0)^{n'}\gamma_0^{-1/2}\|\phi\|_E\|\chi\|_E$ ($\eta \leq 1$); the same with the derivative on χ. For [3, (1.9), (1.11)]: the same, using $|R_{\bar n+1}| \leq 1$ [3, p. 414]. For [3, (1.10)]: $\beta = \kappa' \sum_b \eta^d[\phi(b_-)\cdot M(b)\chi(b_-)]$ with $\kappa' = e_k^{n'}\eta^{n'-2}/n'!$, so $|\beta| \leq d(e_k\|q\|\sigma_0)^{n'}\gamma_0^{-1}\|\phi\|_E\|\chi\|_E$. For a block pair: a_k times a product of two factors [3, (1.13)–(1.14)] at one y; the contour factors obey $|\tilde B'(\Gamma)| \leq 2d\sigma_0$ (unit-block contours of η-length $\leq 2d$), $|U| = 1$, and $\sum_{x\in B(y)} \eta^d|\phi(x)| \leq \|\phi\|$, so $|\beta| \leq a(2de_k\|q\|\sigma_0)^{n'}\gamma_0^{-1}\|\phi\|_E\|\chi\|_E$. Altogether $|\beta_v| \leq c_v\|\cdot\|_E\|\cdot\|_E$ with $c_v \leq C(e_k\|q\|\sigma_0)^{n'_v}$.

Step 3 (sandwich). On the finite-dimensional space $\ell^2(\square,\eta^d)\otimes\mathbb{R}^N$ let K_v represent β_v ($\langle\phi, K_v\chi\rangle = \beta_v(\phi,\chi)$) and put $\hat K_v := G_0^{1/2}K_vG_0^{1/2}$, so $\|\hat K_v\| \leq c_v$ by Step 2. The component is $E(H) = \langle \bar\Phi_1, K_1G_0K_2G_0\cdots G_0K_m\bar\Phi_m\rangle$ with $\bar\Phi_i = h\Phi_i$: the end forms see their external field only within η of the end cube (transposes, for the other orientation of a leg, have the same norms). Since $K_1G_0K_2\cdots G_0K_m = G_0^{-1/2}\hat K_1\hat K_2\cdots\hat K_mG_0^{-1/2}$, we get $|E(H)| \leq \|h\Phi_1\|_E\prod_v c_v\|h\Phi_m\|_E$.

Step 4 (end fields). $h\Phi$ is supported in a cube of side ≤ 4, so $\|h\Phi\|_E^2 = \|D_{B_0}(h\Phi)\|^2 + m_k^2\|h\Phi\|^2 + a_k\|Q_k(B_0)h\Phi\|^2 \leq C(1+c_h)^2\|h\Phi\|_{1,0}^2$ by the Leibniz rule $D(h\Phi) = h(\cdot+\eta e)D\Phi + (\partial^\eta h)\Phi$; the constant uses $m_k^2 \leq c_m$.

Step 5 (exponential factor). Consecutive cubes $\square(v_i)$, $\square(v_{i+1})$ are at distance 0, because the line between them was not cut [3, pp. 432–433]. So $d(\{\square(v)\}) \leq \sum_i \operatorname{diam}\square(v_i) \leq 2\sqrt{3}m$, and $1 \leq e^{2\sqrt{3}\delta_0 m}e^{-\delta_0 d(\{\square(v)\})}$. □

Remark H.85 (why the chains are bounded directly). (a) The operator form of Step 3 is the ℓ^2 sandwich of proposition H.17 in its simplest case: every vertex is first order on each side and G_0 is never needed pointwise, so the subgraphs [3, (2.4)] need no summation by parts and the vertices may lie up to $\partial\square$. The printed route through [3, Prop. 2.1] would need the summation by parts [3, (2.8)] at [3, (2.4)]-subgraphs whose vertices may sit near $\partial\square$; [3, (2.8)] is printed with $\sum_{b\subset T_\eta}$, on the torus [3, p. 425], and on the Neumann box $\square$ the function $\psi = q\phi' g\tilde B'$ extended by zero jumps at the faces wherever the $\tilde B'$-localisation g does not vanish there, so [3, (2.8)] has a boundary term; the reason that removes it in S4 (lemma H.26: the supports of the localisation functions lie at depth $\geq 2\eta$, $\operatorname{supp} g \subset \Omega_1$) is absent here. Lemma H.84 makes the point moot. (b) [3, (3.2)] carries no exponential factor (it comes from the external-field estimates [3, (2.5), (3.1)]); the factor $e^{-\delta_0 d(\{\square(v)\})}$ in lemma H.84 is harmless, Step 5 showing that it costs only $e^{2\sqrt{3}\delta_0 m}$. (c) Under the cluster cut of appendix F, a chain component lies in one cluster of label boxes, so its exponential factor is bounded below by $e^{-2\delta_0 V^\sharp_{\mathrm{lab}}}$, a constant depending on $\bar n$ only.

S16, consumed form. [3, p. 433]: "the estimate (3.2) for them is a consequence of Proposition 2.1". Delivered: [3, (3.2)] for every chain class, with sup-norms (stronger), constants independent of η, k, $\square$, B_0 and the torus, by lemmas H.82 to H.84. The chains are bounded at $G_0 = G_k(\square, B_0)$ directly; they do not pass through S17–S19, whose $O(1)$ bound on $\delta G_k(\square,\eta\mathbb{Z}^d,0)$ holds only at depth (lemma H.90).

H.4.4. *S17–S19: gauge covariance on $\square$, the gauge-out of B_0, and the field $\delta G_k(\square,\eta\mathbb{Z}^d,0)$.*
S17: gauge covariance, [3, p. 433].

Lemma H.86 (gauge transformation of the operators)**.** *Let $\square$ be a union of unit blocks with Neumann conditions, A a vector field on the bonds of $\square$, and $\lambda\colon \square \to \mathbb{R}$. Put $(V_\lambda\phi)(x) := e^{-e_kq\lambda(x)}\phi(x)$ (orthogonal, commuting with q) and $A^\lambda_b := A_b + \eta^{-1}(\lambda(b_+) - \lambda(b_-))$. Then:* (a) *$D_{A^\lambda}(V_\lambda\phi)(b) = e^{-e_kq\lambda(b_-)}(D_A\phi)(b)$, hence $-\Delta^{\eta,N}_{A^\lambda,\square} = V_\lambda(-\Delta^{\eta,N}_{A,\square})V_\lambda^*$;* (b) *$(Q_k(A^\lambda)V_\lambda\phi)(y) = e^{-e_kq\lambda(y)}(Q_k(A)\phi)(y)$, hence $P_k(A^\lambda) = V_\lambda P_k(A)V_\lambda^*$ and $G_k(\square,A^\lambda) = V_\lambda G_k(\square,A)V_\lambda^*$.*

Proof. (a) $U(A_b^\lambda) = U(A_b)e^{e_k q(\lambda(b_+)-\lambda(b_-))}$, all exponentials of multiples of q commuting; insert into the definition of D. Both Laplacians sum over the same bonds $b \subset \square$, so the forms agree. (b) For the contour $\Gamma_{y,x} \subset B^k(y) \subset \square$ of [1, (2.1)], $A^\lambda(\Gamma_{y,x}) = A(\Gamma_{y,x}) + \lambda(x) - \lambda(y)$ by telescoping, hence $U(A^\lambda(\Gamma))e^{-e_k q\lambda(x)}\phi(x) = e^{-e_k q\lambda(y)}U(A(\Gamma))\phi(x)$. Then $P_k = Q_k^* Q_k$, the orthogonal factor at y cancels, and the mass term commutes. □

Lemma H.87 (gauge invariance of the expressions)**.** *For every expression E of a class after the $\tilde{B}'$-expansion, with scalar lines $G_k(\square, B_0)$, vertices* [3, (1.6)–(1.11)] *and* [3, (1.13)–(1.15)] *at B_0, and any vector fields (A'-lines, $\tilde{A}$, $\tilde{B}'$, photon lines),*

$$E(B_0^\lambda; V_\lambda \Phi_{\mathrm{ext}}) = E(B_0; \Phi_{\mathrm{ext}}),$$

where V_λ acts on every scalar slot of every external field (kernels in several variables included: δG-fields, [3, (1.16)]*-kernels, the fields $\phi(y)$ of* [3, (1.12)]*) and the vector fields are not transformed. This is* [3, p. 433]*: "the expression corresponding to any such renormalized class is gauge-invariant ... if the external scalar fields are simultaneously transformed".*

Proof. [3, (1.6), (1.7)] and $(\phi \cdot \phi)^2$ are invariant under pointwise orthogonal maps. [3, (1.8)]: $[D_{B_0^\lambda}(V_\lambda\phi')(b)\cdot q^m (V_\lambda \phi'')(b_-)] = [e^{-e_k q\lambda(b_-)} D_{B_0}\phi'(b)]\cdot q^m e^{-e_k q\lambda(b_-)}\phi''(b_-) = D_{B_0}\phi'(b)\cdot q^m \phi''(b_-)$ by lemma H.86(a) and the orthogonality of $e^{-e_k q\lambda(b_-)}$, which commutes with q^m. [3, (1.9)–(1.11)]: the same; $R_{\bar{n}+1}(-\eta q e_k \tilde{A}_b)$ is a function of q and commutes. [3, (1.13)–(1.15)]: by lemma H.86(b) each factor at y acquires $e^{-e_k q\lambda(y)}$, and the vector contour factors are scalars times powers of q and commute; a pair at the same y is invariant, and a pair with [3, (1.12)] is invariant when $\phi(y) \mapsto e^{-e_k q\lambda(y)}\phi(y)$. Lines: $G_k(\square, B_0^\lambda) = V_\lambda G_k(\square, B_0) V_\lambda^*$. The expression is a full contraction of vertex tensors with line kernels and external fields; every V_λ either cancels against the V_λ^* of the line at the same slot or is absorbed into the transformed external field. □

S18: the gauge-out of B_0, [3, p. 433].

Lemma H.88 (removing the constant field)**.** *Let $\square = \prod_\mu \{c_\mu, c_\mu + \eta, \dots, c_\mu + K_\mu - \eta\}$ be non-wrapping, i.e. a box of the lift of the torus (the no-wrap property of the cube step), let $B_0 \in \mathbb{R}^d$ be constant and $\lambda_0(x) := -B_0\cdot(x-c)$ in the box coordinates. Then $\eta^{-1}(\lambda_0(x+\eta e_\mu) - \lambda_0(x)) = -B_{0,\mu}$ on every bond of $\square$, so $B_0^{\lambda_0} = 0$, $G_k(\square, B_0) = V_{\lambda_0}^* G_k(\square, 0) V_{\lambda_0}$ and $E(B_0; \Phi_{\mathrm{ext}}) = E(0; V_{\lambda_0}\Phi_{\mathrm{ext}})$.*

Proof. λ_0 is single-valued and linear on $\square$ because the sites of a non-wrapping box have unique box coordinates. Apply lemmas H.86 and H.87 with $\lambda = \lambda_0$. □

This is the exact content of the gauge transformation "described more precisely in the proof of Lemma II.2.4" ([2, (2.69)–(2.74)]); nothing beyond lemmas H.86 and H.87 is needed, and no smallness is used.
S19: the field $\delta G_k(\square, \eta\mathbb{Z}^d, 0)$, [3, p. 433].

Lemma H.89 (images)**.** *Let $\square$ be as in lemma H.88 with $c \in \mathbb{Z}^d$ and $K \in \mathbb{Z}^d_{\geq 1}$ (a union of unit blocks), Neumann, and let $\mathcal{G}$ be the group generated by the reflections in the planes $x_\mu = c_\mu - \eta/2$ and $x_\mu = c_\mu + K_\mu - \eta/2$. Then for $x, x' \in \square$*

$$G_k(\square, 0)(x, x') = \sum_{\gamma \in \mathcal{G}} G_k(\eta\mathbb{Z}^d, 0)(x, \gamma x'),$$

absolutely convergent. Hence $\delta G_k(\square, \eta\mathbb{Z}^d, 0) := G_k(\square, 0) - G_k(\eta\mathbb{Z}^d, 0)|_{\square\times\square} = \sum_{\gamma \neq \mathrm{id}} G_k(\eta\mathbb{Z}^d, 0)(\cdot, \gamma\cdot)$.

Proof. This is the image identity of (A5) at $j = k$, $\xi = \eta$, $m^2 = m_k^2$, $a = a_k$: the reflections preserve the block structure (the box is a union of unit blocks), the reflected sum $\sum_\gamma G_k(\eta\mathbb{Z}^d, 0)(x, \gamma x')$ is $\mathcal{G}$-symmetric, hence satisfies the Neumann conditions, and it inverts $-\Delta_\square^{\eta,N} + m_k^2 + a_k P_k$ on functions on $\square$, so it is $G_k(\square, 0)$. The absolute convergence follows from (91): the images $\gamma x'$ lie one in each tile $\gamma\square$. □

Lemma H.90 (bound)**.** *For $u \in \square$ let $\varrho(u) := \operatorname{dist}_\infty(u, \square^c) \geq \eta$. For $\gamma \neq \mathrm{id}$ and $u, w \in \square$: $|u - \gamma w| \geq \varrho(u) + \varrho(w) - \eta \geq (\varrho(u) + \varrho(w))/2$. Consequently, for $x_1, x_2 \in \square$ with $\varrho(x_i) \geq 2$ and Hölder partners in $\square$ with $\varrho \geq \varrho(x_i) - 1$,*

$$|\Delta_T D_S\, \delta G_k(\square, \eta\mathbb{Z}^d, 0)(x_1, x_2)| \leq C_\delta e^{-(\delta_Z/8)(\varrho(x_1)+\varrho(x_2))} \prod_{i \in T} |x_i - \tilde{x}_i|^\alpha,$$

with C_δ depending only on (d, L, a, α). At points of the 1-neighbourhood of $\square_1$ and their partners, where $\varrho \geq r^\flat - 2 - \eta \geq 2$ (so $r^\flat \geq 5$ suffices), and with the localisation functions h, h' of [3, p. 432] *($|dh|, |d^2 h| \leq c_h$),*

$$\|h\, \delta G_k(\square, \eta\mathbb{Z}^d, 0)\, h'\|_{1,\alpha} \leq C(1 + 3\sqrt{3}c_h)^2 e^{-(\delta_Z/4)(r^\flat - 1)}.$$

Proof. Step 1 (distance to images). Let $\gamma \neq \mathrm{id}$ and let μ be a coordinate in which γ acts nontrivially. In that coordinate the group acts freely and each orbit meets the interval $I_\mu = \{c_\mu, \dots, c_\mu + K_\mu - \eta\}$ once, so $z := (\gamma w)_\mu \notin I_\mu$. The segment from u_μ to z crosses an end plane b of I_μ, so $|u_\mu - z| = |u_\mu - b| + |b - z|$ with $|u_\mu - b| \geq \varrho(u) - \eta/2$. If z lies in the copy adjacent across b, then $z = 2b - w_\mu$ and $|b - z| = |b - w_\mu| \geq \varrho(w) - \eta/2$; points of farther copies are at distance $\geq K_\mu - \eta/2 \geq |b - w_\mu|$ from b. Hence $|u - \gamma w| \geq |u_\mu - z| \geq \varrho(u) + \varrho(w) - \eta$.

Step 2 (termwise bounds). γ is a lattice isometry mapping bonds of $\square$ to bonds (possibly reversed) and partners to partners at the same distance. So each term $\Delta_T D_S[G_k(\eta\mathbb{Z}^d, 0)(x_1, \gamma x_2)]$ is bounded by (91) at the image points, where $\eta + \rho_\gamma \geq 1$: $\leq C_1 e^{-\delta_Z \rho_\gamma/2} \prod |\cdot|^\alpha$, ρ_γ the minimal distance between the points involved.

Step 3 (sum). $e^{-\delta_Z \rho_\gamma/2} \leq e^{-(\delta_Z/8)(\varrho(x_1)+\varrho(x_2))} e^{-(\delta_Z/4)\rho_\gamma} e^{\delta_Z}$, the shifts by η and by the partners being absorbed in the last factor. The images γx_2 lie one in each tile $\gamma\square$, and the tiles have sides ≥ 1, so $\sum_\gamma e^{-(\delta_Z/4)\operatorname{dist}(x_1, \gamma\square)} \leq C_d(\delta_Z)$.

Step 4 (localised norm). Lemma H.18 converts the pointwise bound at the points within $1 + \eta$ of $\operatorname{supp} h \cup \operatorname{supp} h'$ into the $(1, \alpha)$-norm; the Hölder supremum of [3, (1.32)] over all pairs is $\leq \max\{\text{local supremum}, 2 \sup |D \cdot|\}$ (conventions of appendix H.4), so the localised bound covers it. □

S19, consumed form. [3, p. 433]: "we substitute $G_k(\square, 0) = G_k(0) + \delta G_k(\square, \eta\mathbb{Z}^d, 0)$ and we treat δG_k as an external scalar field". Delivered: the substitution is the image identity (lemma H.89), and the external field has $(1, \alpha)$-norm $O(e^{-(\delta_Z/4)(r^\flat - 1)}) \leq O(1)$ at the localisations of the classes with an old vertex, all of which lie in $\square_1$ (S15). The field is *not* $O(1)$ at face-adjacent points: for $\varrho(x_1) = \varrho(x_2) = \eta$ the face image is at distance η, and (91) gives only $\sim (2\eta)^{-1}$; so the $O(1)$ norm needs depth of order ≥ 1, which the old-vertex classes have ($\varrho \geq r^\flat - 1$), while the chain classes, whose localisations are not confined to $\square_1$, are bounded by lemma H.84 directly at $G_k(\square, B_0)$, without S19. No auxiliary torus and no depth input beyond the containment in $\square_1$ is used.

H.4.5. *S20: the norms* [3, (1.32)] *at B_0 and at $\tilde{B}$,* [3, p. 434]. [3, p. 434]: after S17–S19 the expressions are estimated by [3, (3.3)] "with norms equal to (1.32) with $\tilde{B} = 0$" of the transformed external scalar fields, and "Thus we get the estimate (3.2) exactly". The transformed fields are $V_{\lambda_0}\Phi$; their norms at 0 are the norms of Φ at B_0 (lemma H.91), and the norms at B_0 are compared with those at $\tilde{B}$ (lemma H.92). Here $\|f\|_{1,\alpha;A}$ is the sum form of [3, (1.32)] with the Hölder supremum over all pairs.

Lemma H.91 (covariance of the norm)**.** *For λ on $\square$ and any function f of one scalar slot, $\|V_\lambda f\|_{1,\alpha;A^\lambda} = \|f\|_{1,\alpha;A}$. In particular $\|V_{\lambda_0} f\|_{1,\alpha;0} = \|f\|_{1,\alpha;B_0}$, slot by slot for functions of several variables.*

Proof. The sup term is unchanged, V_λ being orthogonal. Derivative term: by lemma H.86(a), $|D_{A^\lambda}(V_\lambda f)(b)| = |D_A f(b)|$. Hölder term: $U(A^\lambda(\Gamma_{x,x'})) = e^{-e_k q\lambda(x)} U(A(\Gamma_{x,x'})) e^{e_k q\lambda(x')}$ by telescoping along Γ; with the factor $e^{-e_k q\lambda(x')}$ from lemma H.86(a) at the bond starting at x', the transported difference equals $e^{-e_k q\lambda(x)}$ times the old one. □

Lemma H.92 (B_0 versus $\tilde B$)**.** *Let $B = B_0 + B'$ on $\square$ with B_0 constant, $\sigma_0 := \sup|B'|$, $\sigma_1 := \sup|\partial^\eta B'|$ over the bonds of $\square$, and $e_k(\sigma_0+\sigma_1) \le 1$. Then for every f supported in $\square$ (one slot),*

$$\|f\|_{1,\alpha;B_0} \le \big(1+C_{20}(f)\,e_k(\sigma_0+\sigma_1)\big)\|f\|_{1,\alpha;B}, \qquad C_{20}(f) := C(d,\|q\|)\,(1+D_f)^{1-\alpha}, \quad D_f := \operatorname{diam}\operatorname{supp} f,$$

and the same with B and B_0 exchanged. For several slots the factor is raised to the number of slots. The diameter factor cannot be dropped from this form of the bound: for B' constant and $D_B f$ a transported constant the B-Hölder term vanishes while the B_0-quotient at distance D is of order $e_k\sigma_0 D^{1-\alpha}|Df|$.

Proof. Write $D := D_B$, $D_0 := D_{B_0}$. For the bond b at x in direction μ,

$$D_0 f(x) = Df(x)+E(x), \qquad E(x) := U(B_b)\Phi(B'_b)f(x+\eta e_\mu), \qquad \Phi(\beta) := \eta^{-1}(U(-\beta)-1) = \eta^{-1}(e^{-e_k\eta q\beta}-1).$$

For antisymmetric X, $|e^X-1| \le |X|$, and $d\Phi/d\beta = -e_k q e^{-e_k\eta q\beta}$; so $|\Phi(\beta)| \le e_k\|q\||\beta|$ and $|\Phi(\beta)-\Phi(\beta')| \le e_k\|q\||\beta-\beta'|$.

Step 1. $\sup|D_0 f| \le \sup|Df| + e_k\|q\|\sigma_0 \sup|f|$.

Step 2 (the Hölder term, pairs with $|x-\tilde x| \le 1+D_f$). Let $\Gamma = \Gamma_{x,\tilde x}$ be shortest, $U_0 := U(B_0(\Gamma))$ and $U := U(B(\Gamma)) = U_0 U(B'(\Gamma))$. Then

$$U_0 D_0 f(\tilde x) - D_0 f(x) = [UDf(\tilde x) - Df(x)] + (U_0-U)Df(\tilde x) + [U_0 E(\tilde x) - E(x)].$$

The first bracket is $\le |x-\tilde x|^\alpha$ times the Hölder term of $\|f\|_{1,\alpha;B}$. Next, $|U_0-U| = |U(B'(\Gamma))-1| \le e_k\|q\|\sigma_0|x-\tilde x|_1$. For the last bracket, with $\tilde b \parallel b$ at $\tilde x$, $|B'_{\tilde b} - B'_b| \le \sigma_1|x-\tilde x|_1$; write $U_0 U(B_{\tilde b})f(\tilde x+\eta e_\mu) = U(-B'(\Gamma))[U(B(\Gamma))U(B_{\tilde b})f(\tilde x+\eta e_\mu)]$. The bracket differs from $U(B_b)f(x+\eta e_\mu)$ by at most $|x-\tilde x|_1 \sup|Df| + |\mathsf{H}-1|\sup|f|$, where H is the holonomy of the loop $b \cup \Gamma' \cup \tilde b^{-1} \cup \Gamma^{-1}$, Γ' the translate of Γ by ηe_μ, and $|\mathsf{H}-1| \le 2e_k\|q\|\sigma_1\eta|x-\tilde x|_1$, since the plaquette flux is $\eta^2(\partial B'_\nu - \partial B'_\mu)$, B_0 being constant. Altogether $|U_0E(\tilde x) - E(x)| \le Ce_k\|q\|(\sigma_0+\sigma_1)|x-\tilde x|_1(\sup|f| + \sup|Df|)$. For $|x-\tilde x| \le 1+D_f$ one has $|x-\tilde x|_1 \le \sqrt{d}\,|x-\tilde x|^{1-\alpha}|x-\tilde x|^\alpha \le \sqrt{d}(1+D_f)^{1-\alpha}|x-\tilde x|^\alpha$, so the second and third terms contribute at most $C(1+D_f)^{1-\alpha}e_k(\sigma_0+\sigma_1)(\sup|f|+\sup|Df|)$ to the B_0-quotient.

Step 3 (pairs with $|x-\tilde x| > 1+D_f$). Df, E and $D_0 f$ vanish outside $S_\mu := \{y : y \in \operatorname{supp} f$ or $y+\eta e_\mu \in \operatorname{supp} f\}$, and $\operatorname{diam} S_\mu \le D_f+\eta \le 1+D_f$, so one of the two points, say x, lies outside S_μ. Then $U_0 D_0 f(\tilde x) - D_0 f(x) = U_0 Df(\tilde x) + U_0 E(\tilde x)$, so the B_0-quotient at this pair is at most the B-quotient at the same pair ($= |Df(\tilde x)|/|x-\tilde x|^\alpha$) plus $|E(\tilde x)|/|x-\tilde x|^\alpha \le e_k\|q\|\sigma_0\sup|f|$ (Step 1 and $|x-\tilde x| \ge 1$); if $\tilde x$ lies outside S_μ the roles are exchanged.

Step 4. In both cases the Hölder term of $\|f\|_{1,\alpha;B_0}$ is at most that of $\|f\|_{1,\alpha;B}$ plus $C(1+D_f)^{1-\alpha}e_k(\sigma_0+\sigma_1)(\sup|f|+\sup|Df|)$. Adding Step 1 gives the claim; the argument is symmetric in B, B_0. □

Remark H.93 (a diameter-free form). Bounding instead the third bracket of Step 2 by $2\sup|E| \le 2e_k\|q\|\sigma_0\sup|f|$ for $|x-\tilde x| \ge 1$, and the second by $\sup|Df|\min\{2, e_k\|q\|\sigma_0|x-\tilde x|_1\}/|x-\tilde x|^\alpha \le 2^{1-\alpha}(\sqrt{d}e_k\|q\|\sigma_0)^\alpha\sup|Df|$, gives $\|f\|_{1,\alpha;B_0} \le (1+C(d,\|q\|)[(e_k\sigma_0)^\alpha + e_k(\sigma_0+\sigma_1)])\|f\|_{1,\alpha;B}$, symmetric in B_0, B, with no diameter; the order $(e_k\sigma_0)^\alpha$ is sharp by the example in lemma H.92. The weakest form sufficing for corollary H.94 is that the all-pairs Hölder supremum is $\le \max\{\text{local supremum}, 2\sup|D_0 f|\}$, so that the factor is $\le 3+Ce_k(\sigma_0+\sigma_1) \le 3+C$. Lemma H.92 is the form used.

Corollary H.94 ("Thus we get the estimate (3.2) exactly")**.** *If* [3, (3.3)] *holds for the transformed expressions, with the norms* [3, (1.32)] *at $\tilde B = 0$ of the external scalar fields $V_{\lambda_0}\Phi$, then* [3, (3.2)] *holds with the norms* [3, (1.32)] *at $\tilde B$, and the constant is multiplied by $(1+C_{20}(f)e_k(\sigma_0+\sigma_1))^s$, s the number of external scalar slots. For the h-localised external fields of* [3, (3.2)]*, $D_f = O(1)$:* [3, (1.33)] *multiplies the external legs by h, with $h = 1$ on $\square(v)$ and $h = 0$ outside a neighbourhood. In general $D_f \le \operatorname{diam}\square \le \sqrt{3}(3r^\flat + 2M)$, and the factor is $\le (1+C(d,\|q\|))^s = O(1)$ under* (cap)*, which reads $e_k(\sigma_0+\sigma_1)(1+\sqrt{3}(3r^\flat+2M)) \le 1$; with $\sigma_0 \le c(r^\flat+M)p(t)$ and $\sigma_1 \le c_1 p(t)$ this is $Ce\,t^{1/2}(1+\ln t^{-1})^{\bar p} \le 1$ for $t \le t_0$, a cap of the kind of* [3, p. 412].

Proof. Lemma H.88, then lemma H.91 (the 0-norm of $V_{\lambda_0}\Phi$ is the B_0-norm of Φ), then lemma H.92 (the B_0-norm is at most the factor times the $\tilde{B}$-norm). External vector fields carry the norm with $\tilde{B} = 0$ by definition [3, p. 420] and are untouched. □

The cost of this step is thus a size factor $(1 + O(e_k r^{\flat} p(t)))^{\#\text{slots}} \to 1$ of the printed kind; with the factor e_k inside U it is not $(1 + O(r^{\flat} p))^{\#\text{slots}}$.

H.4.6. *S21, I: the identities of the renormalisation step,* [3, pp. 434–445]. [3, p. 434] states [3, (3.5)]: for each class G_{ren} (after S13–S20, so with scalar lines $G_k(0)$ on $\eta\mathbb{Z}^d$, photon lines on the torus, vertices localised in $\square_1$ and all other objects external),

$$\sum_{G \in G_{\text{ren}}} \sum_{\tilde{l}} \sum_{j \in J(\tilde{l})} E(G(j)) = \sum_{G' \in G'_{\text{fin}}} \sum_{\tilde{l}'} \sum_{j \in J(\tilde{l}')} \gamma(G', \tilde{l}') E'(G'(j)),$$

with generalised graphs G' and sets of orderings (indicator γ) along which every subgraph of the sequence has positive degree; [3, Prop. 2.2] (theorem H.36) then gives [3, (3.3)]. Bałaban writes out the lowest-order members of each divergent type and says "the same procedure" [3, p. 438], "the same applies" [3, p. 437], "in the way described several times" [3, p. 444] for the rest. This subsection and the next prove the identities and the one-loop classes which the renormalisation theorem of appendix F (Step $3(\mathrm{D}_4)$, theorem F.70) uses; the two-loop remainder terms, the transport of legs across the subgraphs [3, (2.4)] and the induction of [3, pp. 444–445] are proved there. Throughout $d = 3$, and $\alpha \in (0, \frac{1}{2})$ is Bałaban's Taylor exponent [3, p. 435], so that a loss of α never makes a half-integer degree non-positive. Every Ward–Takahashi identity below is applied to full lines (cumulative propagators summed over all internal scales below the external scale, then replaced by the free C^{ξ}), never to a fixed ordering or a fixed scale assignment, and every remainder of the replacement is bounded explicitly.

Lemma H.95 (degree formula and the divergent types)**.** *Let G be connected and built from* [3, (1.6), (1.8), (1.10)] *only. Let m_v be the number of vector legs at v (internal and external), $\delta_v \in \{0, 1\}$ with $\delta_v = 1$ iff the differentiated scalar leg of a* [3, (1.8)]*-vertex is internal, V_6 the number of* [3, (1.6)]*-vertices, and $E = E_s + E_v$ the number of external scalar and vector legs. Then*

$$D(G) = V_6 + \sum_{(1.8)} \left(1 + \tfrac{m_v}{2} - \delta_v\right) + \sum_{(1.10)} \tfrac{m_v}{2} + \tfrac{E}{2} - 3.$$

The vertex weights are $\geq \frac{1}{2}$ for [3, (1.8)] *(equality iff $m_v = 1$, $\delta_v = 1$) and ≥ 1 for* [3, (1.6), (1.10)]*; E_s is even. Mass-counterterm vertices count as the graph they replace* [3, (2.3)]*, and graphs containing a block vertex* [3, (1.13)–(1.15)]*, an R-vertex or a finite counterterm have $D > 0$ (corollary H.43). Hence $D(G) \leq 0$ only in the following cases: $E \geq 5$ never; $E = 4$ exactly the graph* [3, (2.4)]*; $E = 3$ with $(E_s, E_v) = (0, 3)$, which vanishes by lemma H.96, or $(2, 1)$, the $\phi\phi A$ class* [3, (2.18)–(2.21)]*; $E = 2$ with $(2, 0)$, the scalar self-energies, $D = -1$ only for* [3, (3.6)] *(the two tadpoles and* [3, (3.8)]*), and $(0, 2)$, the vector self-energies, $D = -1$ only for the four graphs* [3, (3.25)]*; $E = 1$ with $(0, 1)$ vanishes by lemma H.96, and $(1, 0)$ is impossible; $E = 0$ the vacuum graphs. Divergent graphs with $E \geq 2$, in particular every proper divergent subgraph, have at most four vertices; vacuum graphs can have up to six (six* [3, (1.8)]*-vertices with $m = 1$ on one scalar loop with three photon lines have $D = 0$ at order e^6), which matches the vacuum counterterms up to $\alpha + 2\beta \leq 6$ of* [3, p. 418].

Proof. $D_G(v) = (\eta\text{-power}) - (\text{internal legs})/2 - \delta_v$ in $d = 3$, with η-powers $3, 3+m-1, 3+m-2$ for [3, (1.6), (1.8), (1.10)] (lemma H.38). Use $\sum_v (\text{internal legs}) = \sum_v (\text{legs}) - E$ with $4, 2+m, 2+m$ legs; then $D = \sum_v D_G(v) - 3$ gives the formula. $D \leq 0$ forces $\sum_v w_v \leq 3 - E/2$, and $w \geq \frac{1}{2}$ bounds the number of vertices by $6 - E$: by 4 for $E \geq 2$ and by 6 for $E = 0$. $E = 4$, $\sum w \leq 1$: a single vertex with a self-line has $E \leq 2$; two vertices need $w = \frac{1}{2}$ each, i.e. [3, (1.8)] with $m = 1$, $\delta = 1$, joined by the scalar line that carries both derivatives, which is [3, (2.4)]. $E = 5$, $\sum w \leq \frac{1}{2}$: one [3, (1.8)]-vertex with $\delta = 1$ needs an internal scalar line, hence a second vertex or a self-line, both impossible. $E = 2$, $D = -1$, $\sum w = 1$: a single [3, (1.6)] with a self-line; a single [3, (1.10)] with $m = 2$ (the photon tadpole); or two [3, (1.8)] with $m = 1$,

$\delta = 1$, i.e. [3, (3.8)] for $E_s = 2$ and the bubbles of [3, (3.25)] for $E_v = 2$; a single [3, (1.8)] with $m = 2$ has $\delta = 0$ when both scalar legs are external ($w = 2$) and $\delta = 1$ when they form a self-line ($E_v = 2$, the fourth graph of [3, (3.25)]); [3, (1.10)] with $m = 2$, $E_v = 2$ is the third graph of [3, (3.25)]. □

Lemma H.96 (C-parity; [3, p. 434]: "$\operatorname{tr} q^{2n+1} = 0$")**.** *After S18–S19 every scalar line is a scalar kernel times $\mathbf{1}_N$. If a graph without R-vertices has no external scalar leg and an odd number of external vector legs, its expression vanishes.*

Proof. At [3, (1.8)]/[3, (1.10)] the two scalar legs are joined by q^{m_v}; at [3, (1.6)] ($(\phi\cdot\phi)^2$) and [3, (1.7)] the legs are paired by δ_{ab}. With no external scalar legs the $\mathbb{R}^N$-index contractions close into loops, and each loop gives $\operatorname{Tr}\prod_v q^{m_v} = \operatorname{Tr} q^{M_\ell}$, M_ℓ the number of vector legs on the loop (all transporters are 1 at $B_0 = 0$). $\sum_\ell M_\ell = \sum_v m_v = 2I_v + E_v$ is odd, so some M_ℓ is odd, and $\operatorname{Tr} q^{\mathrm{odd}} = \operatorname{Tr}((q^{\mathrm{odd}})^T) = -\operatorname{Tr} q^{\mathrm{odd}} = 0$. □

Lemma H.97 (Taylor's formula [3, (3.10)])**.** *For f on $\eta\mathbb{Z}^d$ and a shortest (monotone) contour $\Gamma_{x,y}$ from x to y,*

$$f(y) = f(x) + \sum_{\mu} (y_\mu - x_\mu)(\partial^\eta_\mu f)(x) + \sum_{b\subset\Gamma} s_b \eta \big[(\partial^\eta_{\mu_b} f)(b^{\mathrm{lo}}) - (\partial^\eta_{\mu_b} f)(x)\big],$$

b^{lo} the lower endpoint of b, μ_b its direction, $s_b = \operatorname{sgn}(y-x)_{\mu_b}$. The remainder is bounded by $|y-x|_1^{1+\alpha}$ times the α-Hölder constant of ∂f on the box spanned by x, y.

Proof. Telescoping along Γ: all μ-bonds of a monotone Γ have the same orientation, and there are $|y_\mu - x_\mu|/\eta$ of them. (Bałaban's $(b)_x$ is the forward bond at x parallel to b.) □

Lemma H.98 (max-scale regrouping)**.** *Let $\mathcal{L}$ be the line set of a graph, $I \subset \mathcal{L}$ the internal lines of a subgraph G_0, and $\mathcal{E} \subset \mathcal{L}\setminus I$, $\mathcal{E} \neq \emptyset$, its external lines (the lines of the graph with an end at a vertex of G_0). Fix a strict total order $\prec$ of $\mathcal{L}$. For an index assignment $j\colon \mathcal{L} \to \{0,\dots,k-1\}$ let $\tilde l(j)$ order $\mathcal{L}$ by increasing (j_l, l) lexicographically, and put $J(\tilde l) := \{j : \tilde l(j) = \tilde l\}$; this is an admissible choice of the sets of [3, (2.7)] (they partition all assignments, and $J(\tilde l) \subset \{j_{l(1)} \le \cdots \le j_{l(m)}\}$). Let $\mathcal{M}$ be any set of orderings of $\mathcal{L}\setminus I$ and $\mathcal{L} := \{\tilde l : \tilde l|_{\mathcal{L}\setminus I} \in \mathcal{M}$, and every $l \in I$ precedes every $e \in \mathcal{E}\}$. Then for every function F of the assignment*

$$\sum_{\tilde l\in\mathcal{L}} \sum_{j\in J(\tilde l)} F(j) = \sum_{j_{\mathcal{L}\setminus I}:\ \tilde l(j_{\mathcal{L}\setminus I})\in\mathcal{M}} \ \sum_{j_I \in \prod_{l\in I}\{0,\dots,J_l-1\}} F(j),$$

where $e^* := \operatorname{argmin}_{e\in\mathcal{E}}(j_e, e)$ is the lexicographic minimum in $(j_e, \prec)$, which depends on $j_{\mathcal{E}}$, and $J_l := j_{e^*} + [l \prec e^*] \in \{j_{e^*}, j_{e^*}+1\}$. If F is multilinear in the slice propagators of the lines of I, the inner sum replaces every $l \in I$ by the cumulative propagator $\sum_{i<J_l} G_{(i)} = \mathsf{T}_{J_l}^{-1}$ (corollary H.77(i); for the photon lines the same with the vector covariance). This is the form of "the same expression appears for all orderings ... with the only condition that they are earlier than the external lines" [3, p. 437] and "we sum over admissible orderings and indices" [3, p. 444].

Proof. The $J(\tilde l)$ partition the assignments, so the left side is the sum over $\{j : \tilde l(j) \in \mathcal{L}\}$. The order induced on $\mathcal{L}\setminus I$ depends on $j_{\mathcal{L}\setminus I}$ only. "Every $l \in I$ precedes every $e \in \mathcal{E}$" is equivalent to $(j_l, l) <_{\mathrm{lex}} (j_{e^*}, e^*)$ for all $l \in I$, and given $j_{\mathcal{E}}$ this is the product set $j_l \le J_l - 1$. □

The lemma is a statement about sums over complete index ranges of the internal lines below the external scale; it asserts no identity for a fixed ordering or scale assignment.

Lemma H.99 (rescaling of the regrouped lines and coefficients)**.** *Let $j_0 := j_{e^*}$, $s := L^{j_0}\eta$ and $\xi := \eta/s = L^{-j_0}$. Rescaled by s (corollary H.77, Step 1), a regrouped scalar line $\mathsf{T}_{J_l}^{-1}$ becomes $s^{2-d}\breve G_l(\cdot/s, \cdot/s)$, where $\breve G_l$ is of the type of lemma H.81: mass $m^2 s^2 \in [0,1]$, block term $a_{j_0}P$ on unit blocks if $J_l = j_0$, and $a_{j_0+1}L^{-2}P$ on L-blocks if $J_l = j_0 + 1$ (the case $J = k'+1$ of corollary H.77). The same holds for the photon lines (componentwise; mass $\mu_0^2 t^2 s^2$). A local coefficient of degree D built from such lines (a sum over all but one vertex of a product of line kernels, derivatives and moments $(x'-x)^\beta$) equals s^D times the same expression on the ξ-lattice; in $d = 3$ with $D = 0$ it is "the same expression but on the scale ξ" [3, p. 444].*

Proof. Each line contributes s^{2-d} times its rescaled kernel, each derivative s^{-1}, each moment factor s, each vertex sum $\eta^d = s^d\xi^d$ and each vertex η-power s; the total exponent is D by [3, (2.1)–(2.3)]. □

Lemma H.100 (the Ward–Takahashi identities, generated at full lines)**.** *Let Y be $\xi\mathbb{Z}^d$ or a periodic box, $C := C^\xi \otimes \mathbf{1}_N$ (lemma H.79), $C_{\mathrm{ph}} := C^\xi \otimes \mathbf{1}_d$, A a finitely supported external bond field, and F any polynomial in the $|\phi(x)|^2$ (for instance $\lambda\sum :|\phi|^4:$ and $\frac{1}{2}\sum \delta m^2(x) :|\phi(x)|^2:$, Wick-ordered with respect to any constants; mass counterterm vertices included). Put $Z(A) := \int d\mu_{C_{\mathrm{ph}}}(A') \int d\phi \exp[-\frac{1}{2}\langle\phi, (-\Delta^\xi_{A+A'}+1)\phi\rangle - F(\phi)]$ and $W := \log Z$ as formal power series in $(e, \lambda, \delta m^2, A)$. The Taylor coefficients of W are the sums of all connected graphs of the given order and external structure with full lines C, C_{ph} (the vertices are those of [3, (1.8)–(1.11)] with unrestricted n, n', which agree with Bałaban's below order $\bar{n}+1$). Then:* (a) *on a periodic box, $Z(A+\partial\lambda) = Z(A)$ for every periodic λ;* (b) *every Taylor coefficient W_n (symmetric n-linear in A at fixed order in the couplings) obeys $W_n(\partial\lambda, A_2, \dots, A_n) = 0$;* (c) *(b) holds on $\xi\mathbb{Z}^d$ for finitely supported λ;* (d) *the two-vector-leg kernel $\Pi_{\mu\nu}(y, y')$ of every order obeys $\sum_{y'} \xi^d \Pi_{\mu\nu}(y, y') = 0$ for all μ, ν, y.*

Proof. (a)–(c) are lemmas H.45 to H.47 with $M^2 = 1$, the block term absent and $B = 0$ (remark H.50(c)): substituting $\phi = V_\lambda\psi$ has Jacobian 1, $-\Delta_{A+\partial\lambda+A'} = V_\lambda(-\Delta_{A+A'})V_\lambda^*$ (lemma H.86(a); a periodic box has no boundary), and F is invariant, which gives (a); comparing powers of s in $W((sA) + s\partial\lambda) = W(sA)$ and polarising gives (b); at fixed order a coefficient is a finite sum of connected graphs whose lines are periodisations of the $\xi\mathbb{Z}^d$ lines (exponential decay, lemma H.79), and with A, λ finitely supported each graph sum converges absolutely to its $\xi\mathbb{Z}^d$ value as the box grows, which gives (c). (d) is lemma H.48(b) for Π: by (c), $\sum_{y',\nu} \xi^d \Pi_{\mu\nu}(y, y')(\lambda(y' + \xi e_\nu) - \lambda(y'))/\xi = 0$ for all finitely supported λ, i.e. $\sum_\nu [\Pi_{\mu\nu}(y, z - \xi e_\nu) - \Pi_{\mu\nu}(y, z)] = 0$ for every z; multiply by z_κ and sum over z (absolutely convergent by the exponential decay); the shift gives $\xi \sum_w \xi^d \Pi_{\mu\kappa}(y, w) = 0$. □

Lemma H.101 (lattice reflections; the symmetry behind [3, (3.27)–(3.28), (3.37), (3.38)])**.** *For the theory of lemma H.100 on $\xi\mathbb{Z}^d$ with constant coefficients, the reflection R_κ ($x_\kappa \mapsto -x_\kappa$) acts by $\phi \mapsto \phi \circ R_\kappa$, $A_\mu(b) \mapsto A_\mu(R_\kappa b)$ ($\mu \neq \kappa$) and $A_\kappa(b) \mapsto -A_\kappa(b^r)$, where for $b = (x, x + \xi e_\kappa)$ we put $b^r := (R_\kappa x - \xi e_\kappa, R_\kappa x)$. This leaves Z invariant. Consequently, for the full fixed-order classes with lines C, C_{ph}:* (a) *the zero-momentum $\phi\phi A$ coefficient $\sum_{x_1,x_2} \xi^{2d} \Gamma_\kappa(x_1, x_2; b)$ vanishes;* (b) *if $\sum_{y'} \xi^d \Pi_{\mu\nu}(y, y') = 0$, then the first moment $\sum_{y'} \xi^d \Pi_{\mu\nu}(y, y')(y'_\lambda - y_\lambda)$ vanishes for all μ, ν, λ ($d \geq 2$).*

Proof. Invariance. $|U(A_b)\phi(b_+) - \phi(b_-)| = |U(-A_b)\phi(b_-) - \phi(b_+)|$ by orthogonality, so reversing a bond with $A \mapsto -A$ preserves the action; C is even, and C_{ph} is invariant under $A_\kappa \mapsto -A_\kappa$. *(a)* $\Gamma_\kappa(R_\kappa x_1, R_\kappa x_2; b^r) = -\Gamma_\kappa(x_1, x_2; b)$, and by translation invariance $\sum_{x_1,x_2} \Gamma_\kappa(x_1, x_2; b)$ does not depend on b; so the sum equals minus itself. *(b)* Write $m(b)$ for the bond midpoint. Translation invariance gives $\Pi_{\mu\nu}(b, b') = \pi_{\mu\nu}(m(b') - m(b))$ and the invariance gives $\pi_{\mu\nu}(R_\kappa u) = s^\kappa_\mu s^\kappa_\nu \pi_{\mu\nu}(u)$ with $s^\kappa_\mu = -1$ iff $\mu = \kappa$. Since $y'_\lambda - y_\lambda = (m(b') - m(b))_\lambda + \frac{\xi}{2}(\delta_{\mu\lambda} - \delta_{\nu\lambda})$ and $\sum_u \pi_{\mu\nu}(u) = 0$ by hypothesis, the first moment is $M := \sum_u \xi^d \pi_{\mu\nu}(u) u_\lambda$. Reflection in $\kappa = \lambda$ gives $M = -s^\lambda_\mu s^\lambda_\nu M$, so $M = 0$ unless exactly one of μ, ν equals λ; reflection in $\kappa \neq \lambda$ gives $M = s^\kappa_\mu s^\kappa_\nu M$, so $M = 0$ unless, for every $\kappa \neq \lambda$, $\mu = \kappa \Leftrightarrow \nu = \kappa$. Both conditions together are impossible: if $\mu = \lambda \neq \nu$, then $\kappa = \nu$ violates the second. □

[3, (3.27)–(3.28)] (a computation for the one-loop bubble) and [3, (3.37), (3.38)] ("the functions which we are summing are odd") are instances of (b) and (a).

H.4.7. *S21, II: the one-loop classes and the subtraction identities.* Notation: j_0 is the lowest index of the external legs of the class, $\xi = L^{-j_0}$, and after lemmas H.98 and H.99 every internal line is $\breve{G}_l = C^\xi + R_l$ with R_l as in lemma H.81; $\hat{G}_s$, $\hat{G}_{\mathrm{ph}}$ denote the rescaled regrouped scalar and photon lines. Sums over y' carry the weight ξ^3. The bounds $|C^\xi|, |\hat{G}| \leq C(\xi + \rho)^{-1} e^{-\delta\rho}$ (lemma H.79, corollary H.77), $|\partial R| \leq C(1 + |\log(\xi + \rho)|)e^{-\delta\rho}$ and $|\partial R \partial^*| \leq C(\xi + \rho)^{-1} e^{-\delta\rho}$ (lemma H.81) are used throughout.

Lemma H.102 (scalar self-energies with $D=-1$, [3, (3.6)–(3.17)]). (a) *The two tadpoles of* [3, (3.6)] *and the last two counterterms of* [3, (3.7)] *cancel exactly ("Wick ordering"): the class graph has the loop line $G_k(0)(x,x)$ or the torus photon line at (x,x) and the localisation of* [3, (1.6)] *to unit cubes of Ω_1, resp. the factors $g_k(x)^2$; the counterterms* [3, (1.29)] *carry the same localisation and the same lines once the torus scalar line is identified with $G_k(0)$, the difference being an image term, an external field of norm $O(e^{-\delta S_k})$ (lemma H.109).* (b) *For* [3, (3.8)] *minus its counterterm, i.e. the expression* [3, (3.9)]*: Taylor's formula (lemma H.97) applied to the leg with the smaller index* [3, p. 435] *gives* [3, (3.11)]*. Its second term is the generalised graph* [3, (3.12)]*, of degree $-1+1+\alpha=\alpha$; the factor $|x'-x|^{1+\alpha}$ multiplies every subgraph containing a line of G_0, since both lines join x and x', so its* [3, (2.4)]*-subgraph has degree $1+\alpha>0$. Its first term, regrouped (lemma H.98) and rescaled (lemma H.99, $D=0$), is the vertex* [3, (3.15)] *with coefficient*

$$c_\mu(y)=\sum_{y'}\xi^3\sum_{\nu} q(\partial_\nu\hat G_s\partial_\nu^*)(y,y')\,q\,g(y)\hat G_{\rm ph}(y,y')g'(y')\,(y'_\mu-y_\mu),$$

and $|c_\mu|\le C$ uniformly in ξ, j_0, k. This is [3, (3.17)].

Proof. (a) is the identity of the lines and localisations stated. (b): write $\hat G_s=C^\xi+R_s$. Since C^ξ is translation invariant, $\sum_\nu\partial_\nu C^\xi\partial_\nu^*=-\Delta^\xi C^\xi=\delta^\xi-C^\xi$ and $\delta^\xi(y'-y)(y'_\mu-y_\mu)=0$, so

$$c_\mu=-q^2\sum_{y'}\xi^3C^\xi(y-y')\,g\hat G_{\rm ph}g'\,(y'-y)_\mu+q^2\sum_{y'}\xi^3\Big(\sum_\nu\partial_\nu R_s\partial_\nu^*\Big)(y,y')\,g\hat G_{\rm ph}g'\,(y'-y)_\mu,$$

which is [3, (3.16)] (whose first line prints $g(y')$ for $g'(y')$). The first sum is $\le C\sum\xi^3(\xi+\rho)^{-2}\rho e^{-\delta\rho}\le C$, and by lemma H.81 the second is bounded in the same way. □

Lemma H.103 (scalar self-energies with $D=0$, [3, (3.18)–(3.20)]). *If $\Sigma(x,x')$ is the expression of a graph G_0 of the class with external legs at x, x', and its counterterm graph is G_0 with both legs at x ([3, p. 417]: "represented by the same graph G but with both external legs localized in x"; so it has the same lines, slices and orderings), then* [3, (3.19)] *holds exactly:*

$$\sum\eta^{2d}\phi(x)\cdot\Sigma(x,x')\phi'(x')-\sum\eta^d\phi(x)\cdot\Big(\sum_{x'}\eta^d\Sigma(x,x')\Big)\phi'(x)=\sum\eta^{2d}\phi(x)\cdot\big(\Sigma(x,x')|x'-x|^\alpha\big)\frac{\phi'(x')-\phi'(x)}{|x'-x|^\alpha}.$$

Along a chain of vertices $x=x_0,\dots,x_l=x'$ of G_0, $|x'-x|^\alpha\le\sum_i|x_{i+1}-x_i|^\alpha$ (concavity), which distributes the gain $+\alpha$ to the lines of a path from x to x'. A subgraph containing no line of the path is a proper subgraph of G_0, hence positive when G_0 is primitively divergent within the ordering; the Hölder quotient of the external leg costs α on that leg, and $D\ge\frac12>\alpha$ for every positive subgraph containing it.

Proof. The identity is the definition of the counterterm graph; the degree bookkeeping is as stated. □

Lemma H.104 (the differentiated-leg graphs [3, (3.21)–(3.24)]). (a) *The first graph of* [3, (3.21)] *(a* [3, (1.8)]*-vertex with $n=2$ whose two A'-legs are contracted) has a coefficient of the type $\eta e_k^2G_{\rm ph}(b,b)$, bounded uniformly: by* (91) *at $\rho=0$, $|G(x,x)|\le C_1\eta^{-1}$ in $d=3$; this is the printed "$\eta G_k(x,x)$ is convergent".* (b) *The second graph, after Taylor at order 0, is* [3, (3.22)]*: a remainder of degree α plus the vertex* [3, (3.23)]*, whose coefficient (of the vertex $\phi\cdot\partial_\mu\phi'$) is bounded uniformly in ξ, j_0, k.*

Proof. (b) Write $g'(x')=g'(x)+(g'(x')-g'(x))$; the difference carries $|\partial g'||x'-x|$ (degree +1) and is convergent. The main term, rescaled and with $\hat G=C^\xi+R$, is $q^2g(x)g'(x)[\sum_y\xi^3(\partial_\mu C^\xi)(y)C^\xi(y)+$ (terms with an R)]. Exactly, $\sum_y\xi^3(\partial_\mu C^\xi)(y)C^\xi(y)=-\frac\xi2\sum_y\xi^3|(\partial_\mu C^\xi)(y)|^2$, by symmetrising with $\sum_y C(y+\xi e)^2=\sum_y C(y)^2$; by lemma H.79 this is $\le\frac\xi2C\sum\xi^3(\xi+|y|)^{-4}\le C$. The R-terms are $\le C\sum\xi^3[(1+|\log(\xi+\rho)|)(\xi+\rho)^{-1}+(\xi+\rho)^{-2}]e^{-\delta\rho}\le C$ by lemmas H.79 and H.81. ([3, (3.24)] shows a finite limit; only boundedness is used.) □

Lemma H.105 (vector self-energies with $D=-1$, [3, (3.25)–(3.30)])**.** *Let the class be all order-e_k^2 graphs with two external vector legs: the two bubbles and the two tadpoles of* [3, (3.25)]*. Apply Taylor to first order on the leg $g'A'$ at x' around x (*[3, (3.26)]*). The Hölder remainder is a generalised graph of degree $-1+1+\alpha=\alpha$. After regrouping and rescaling the zero-order coefficient is $\Pi^{(\eta)}_{\mu\mu'}(x)=(L^{j_0}\eta)^{-1}\Pi^{(\xi)}_{\mu\mu'}(y)$ and the first-moment coefficient is $\Pi^{(\eta)}_{\mu\mu'\nu}(x)=\Pi^{(\xi)}_{\mu\mu'\nu}(y)$, with lines $\breve{G}_l=C^\xi+R_l$. Then:* (i) *the all-C part of $\Pi^{(\xi)}_{\mu\mu'}$ vanishes (lemma H.100(d); Bałaban's* [3, (3.29)] *via* [3, (2.26)] *at $M^2=1$), and the all-C part of $\Pi^{(\xi)}_{\mu\mu'\nu}$ vanishes (lemma H.101(b);* [3, (3.27)–(3.28)]*);* (ii) *the terms with at least one R-line are bounded uniformly in ξ, j_0, k, both for $\Pi^{(\xi)}_{\mu\mu'}$ and for $\Pi^{(\xi)}_{\mu\mu'\nu}$. So* [3, (3.30)] *holds: the class equals two generalised graphs of degree α, plus the vertex $A\cdot\Pi_{\mu\mu'\nu}\partial A$ with a bounded coefficient, plus the vertex $A\cdot\Pi_{\mu\mu'}A$ with coefficient $(L^{j_0}\eta)^{-1}O(1)$. The last is a zero-momentum remainder vertex of power -1 at the scale $L^{j_0}\eta$ of the external legs; it is kept and bounded, not claimed to vanish.*

Proof. (i) is as stated. (ii) The R-terms of the bubbles are of four kinds. $(C\partial^*)(R\partial^*)$ and $(R\partial^*)(R\partial^*)$: bounded by $C\sum\xi^3(\xi+\rho)^{-2}(1+|\log(\xi+\rho)|)e^{-\delta\rho}\le C$. $C\cdot(\partial R\partial^*)$: bounded by $C\sum\xi^3(\xi+\rho)^{-2}e^{-\delta\rho}\le C$. $R\cdot(\partial R\partial^*)$: bounded likewise. $R(y,y')(\partial_{\mu'}C\partial^*_\mu)(y'-y)$: here absolute values give only a logarithm. Write $R(y,y')=R(y,y)+[R(y,y')-R(y,y)]$. Then $\sum_{y'}\xi^3(\partial_{\mu'}C\partial^*_\mu)(y'-y)=0$ exactly (the sum over y' of a forward difference in y' of a decaying function), and $|R(y,y')-R(y,y)|\le C'\rho(1+|\log(\xi+\rho)|)$ for $y'\ne y$, by integrating the pointwise bound $|\partial R(y,z)|\le C(1+|\log(\xi+|z-y|)|)$ of lemma H.81 along a monotone path from y to y' (the integral is $\le C(\xi+\rho)(2+|\log(\xi+\rho)|)\le C'\rho(1+|\log(\xi+\rho)|)$ since $\rho\ge\xi$; the cruder bound $|y'-y|_1\sup|\partial R|$ would give a $\log(1/\xi)$). This leaves $\sum\xi^3\rho(1+|\log(\xi+\rho)|)(\xi+\rho)^{-3}e^{-\delta\rho}\le C$. The tadpoles with R are bounded by $|R(y,y)|\le C$ and $\xi|(R\partial^*)(y,y)|\le C\xi(1+|\log\xi|)$. For the first moment there is an extra factor ρ, and all four kinds are bounded by absolute values; the fourth gives $\sum\xi^3\rho(\xi+\rho)^{-3}e^{-\delta\rho}\le C$. □

Remark H.106. Bałaban's "If at least one propagator $G_{j_0}(0)$ is replaced by $G_{j_0}(0)(1-m^2-aP)C^\xi$, then we get a convergent expression" [3, p. 441] is true, but for the term $R\cdot\partial C\partial^*$ of the bubble it needs the cancellation $\sum_{y'}\partial_{\mu'}(\cdots)=0$ above, not power counting with absolute values: in position space R is bounded but does not vanish at coincident points. The cancellation uses only the bounds of lemma H.81 on the R-line and the forward-difference structure of the other line.

Lemma H.107 (the subtraction identities of the $D=0$ vector and $\phi\phi A$ classes, [3, (3.31)–(3.38)])**.** (a) *For the remaining divergent two-vector-leg graphs (orders e_k^4, $e_k^2\lambda$, with counterterm insertions; at most four vertices by lemma H.95),* [3, (3.31)] *is an exact identity: the Hölder term has degree $+\alpha$, and the local term has coefficient $\sum_{x'}\eta^d\Pi_{\mu\mu'}(x,x')$. After regrouping and rescaling, the all-C^ξ part of this coefficient vanishes by lemma H.100(d) for the full class of that order (the classes "containing all possible connected graphs ... of a fixed order"* [3, p. 431]*); the localisation functions of internal photon legs are first moved to x by Taylor in g, which costs a degree $+1$ (*[3, p. 444], [3, (3.36)]*). The terms with at least one R-line gain at least 1 in the superficial degree: $|R|\le C$ against $(\xi+\rho)^{-1}$, $|\partial R|\le C(1+|\log|)$ against $(\xi+\rho)^{-2}$, $|\partial R\partial^*|\le C(\xi+\rho)^{-1}$ against $(\xi+\rho)^{-3}$.* (b) *For the $\phi\phi A$ class:* [3, (3.34)] *is an exact rewriting (transport the legs with the lowest indices to the vertex with the highest-index leg, through Hölder differences); the localisation functions are moved to the vertex x' by Taylor in g, costing degree $+1$ (*[3, (3.36)]*); after regrouping, rescaling and C-replacement the all-C zero-momentum coefficient $\sum_{y,y''}\xi^{2d}\Gamma''_\mu$ vanishes for the full class of each order by lemma H.101(a), of which Bałaban's class-by-class vanishings are instances or consequences:* [3, (2.21a)] *by lemma H.96;* [3, (2.18)] *"the same expressions with opposite signs";* [3, (2.19)] *by* [3, (3.37)]*;* [3, (2.20)] *by* [3, (3.38)]*;* [3, (2.21c)] *by translation invariance;* [3, (2.21b), (2.21g)] *"with the proper counterterm". The R-terms gain ≥ 1 as in (a).*

Proof. The identities are Taylor's formula (lemma H.97) at order 0 and the definitions; the vanishings are lemma H.100(d) and lemma H.101; the gains of the R-terms are lemma H.81 against lemma H.79. □

What remains of S21, and where it is proved. The identities above reduce [3, (3.5)] to three statements about the regrouped totals: that the R-terms of the two-loop $D = 0$ classes (the superficial gain of lemma H.107 made into a bound with the inner subtractions) are bounded uniformly in ξ; that the legs can be transported across the subgraphs [3, (2.4)] so that each is either made convergent by a Hölder factor or admits the summation by parts [3, (2.8)] without a derivative on an external field ([3, p. 443]); and the induction of [3, pp. 444–445] over the orderings, with the vacuum classes cancelled exactly by the vacuum counterterms [3, (1.24)] up to the image terms of lemma H.109 (vacuum graphs of at most six vertices, lemma H.95). These are the renormalisation theorem of appendix F (Step 3(D_4), theorem F.70), which takes lemmas H.95 to H.105 and H.107 and corollary H.77 and lemmas H.79 to H.81 as its inputs. In it, every inner processing is an exact rewriting (lemmas H.97 and H.98) and the counterterm graphs carry the same lines ([3, p. 417]), so the total over the regrouped index range is the original total and lemma H.100(d), lemma H.101 and the exact counterterm cancellation apply to it.

H.4.8. *The seven further sketched places F1–F7.*

Lemma H.108 (F1: the Ω_2 branch of the cube split, [3, p. 433]: "[or $G_k(\Omega_2, \tilde{B})$]")**.** *If every old vertex has depth $> D^\flat$ in Ω_2 (the depth parameter of the cube step), then $\square \subset \Omega_2$ and $G_k(\Omega_2, \tilde{B}) = G_k(\square, \tilde{B}) + \delta G_k(\Omega_2, \square, \tilde{B})$. At points of $\square_1$, $\delta G_k(\Omega_2, \square, \tilde{B})$ obeys the two-domain estimate* [1, (2.26)] *in its restricted form for the pair $(\square, \Omega_2)$ (proposition H.23 and corollary H.24), exactly as the cube step does for $(\square, \Omega)$. Every later step (S13–S21, F2–F7) treats the split line $G_k(\square, \tilde{B})$ and never refers to which of Ω, Ω_2 was split.*

Proof. The cube step of theorem F.2 is stated and proved for both branches ("depth $> D^\flat$ in Ω (resp. Ω_2)"): the containment $\square \subset \Omega_2$ follows from the depth hypothesis and the side of $\square$, and the two-domain estimate for $(\square, \Omega_2)$ is proposition H.23 with Ω_2 in the role of the larger domain. No new step is involved. □

Lemma H.109 (F2: torus versus $\eta\mathbb{Z}^d$ propagators in the counterterms, [3, pp. 418–419] against [3, pp. 433–445])**.** *Let S_k be the torus side in scale-k units, a multiple of L, with $S_k \geq \max\{6\sqrt{3}\, r(t_k) + 2,\ 8/\delta_Z\}$, δ_Z the constant of lemma H.75. (The second entry can be dropped at the cost of a constant depending on δ_Z, since $\sum_{m \neq 0} e^{-\delta_Z S_k |m|/4} \leq C(d)(1 - e^{-\delta_Z/4})^{-d} e^{-\delta_Z S_k/4}$ for all $S_k \geq 1$; it is kept so that the constants below are those of corollary H.77.) Then:*

(a) Periodisation. *$G_k(T_\eta, 0)(x, x') = \sum_{n \in S_k \mathbb{Z}^d} G_k(\eta\mathbb{Z}^d, 0)(\tilde{x}, \tilde{x}' + n)$ for lifts $\tilde{x}, \tilde{x}'$, and the same for each slice and for the photon covariance; the sum converges absolutely.*

(b) The image term on a cube. *On a non-wrapping cube $\square$ of side $3r(t_k)$, $\delta K := G_k(T_\eta, 0) - G_k(\eta\mathbb{Z}^d, 0)$ obeys $\|\delta K\|_{1,\alpha} \leq C e^{-\delta_Z S_k/4}$ on $\square \times \square$.*

(c) The counterterms in class form. *Expand every scalar line of a counterterm graph* [3, (1.26)–(1.31)]*, transformed as on* [3, p. 433]*, as $G_k(\eta\mathbb{Z}^d, 0) + \delta K$. The term without δK is the counterterm in the class form used in* [3, (3.7)–(3.9), (3.19)] *and on* [3, pp. 444–445]*: same lines, vertices and localisation, the photon lines being torus lines on both sides. Every other term carries at least one external field δK of norm $\leq C e^{-\delta_Z S_k/4} \leq C e^{-\delta_Z r(t_k)}$, and with that line removed it is bounded by* [3, (3.3)] *for the classes into which it decomposes (the regrouping after a cut, theorem F.57), by induction on the number of lines, lower-order counterterms being treated first* [3, p. 417].

Proof. (a) is the periodisation identity of (A5) for the periodic box, with the absolute convergence from (91) (the images lie one in each period cell). (b) Every image point is at distance $\geq S_k/2 \geq 1$ from $\square$, so (91) applies termwise with $\rho \geq S_k/2$, and the sum over $n \neq 0$ is bounded as in lemma H.90, Step 3; the pointwise bounds give the $(1, \alpha)$-norm since the all-pairs Hölder supremum is $\leq \max\{\text{local supremum}, 2 \sup |D \cdot|\}$. (c) is the stated expansion and induction. □

Lemma H.109 makes the cancellations "exact" in lemma H.102(a), lemma H.103 and in the vacuum clause of the induction, up to these image terms. The condition $S_k \geq 8/\delta_Z$ is a lower bound on the torus side of the same kind as $S_k \geq 6\sqrt{3}\, r(t_k) + 2$, fixed after L and before ε_0; appendix F lists both.

Lemma H.110 (F3: "We repeat some of the operations done before", [3, p. 433])**.** (i) *([3, (3.1)] on $\square$ at the constant field.) For localisation functions h, h' with* $\operatorname{dist}(\operatorname{supp} h, \operatorname{supp} h') \geq 1$ *and* $|dh|, |d^2 h| \leq c_h$*:* $\|hG_k(\square, B_0)h'\|_{1,\alpha} \leq C(1 + 3\sqrt{3}c_h)^2 e^{-(\delta_Z/4)\operatorname{dist}(\operatorname{supp} h, \operatorname{supp} h')}$*.* (ii) *([3, (1.16)] on $\square$.) For the remainders $K = [G_0 V]^n \tilde{G} [V G_0]^{n'}$ of the $\tilde{B}'$-expansion, with $G_0 := G_k(\square, B_0)$, $\tilde{G} := G_k(\square, \tilde{B})$, $V := G_0^{-1} - \tilde{G}^{-1}$, $n, n' \geq 2$ ([3, p. 414]: "for n, n' sufficiently large"), and either $m := n + n' \geq 4$ and $\alpha < \frac{1}{2}$, or $m \geq 6$ and $\alpha < 1$: $\|hKh'\|_{1,\alpha} \leq C(C\sigma)^{n+n'} e^{-\beta \operatorname{dist}(\operatorname{supp} h, \operatorname{supp} h')}$ with $\sigma := e_k(\sigma_0 + \sigma_1) \leq 1$.*

Proof. (i) $G_k(\square, B_0) = V_{\lambda_0}^* G_k(\square, 0) V_{\lambda_0}$ (lemma H.88), and the $(1, \alpha)$-norms at B_0 equal those of the gauge-transformed kernel at 0 (lemma H.91 in each slot). (91) on $X = \square$ (type (N)) gives the mixed bounds at every pair of points with $\rho \geq 1$, and lemma H.18 localises them. No depth inside $\square$ is needed, because $\square$ is a parallelepiped at zero field. (ii) The ℓ^2 sandwich (proposition H.17) applies on $X = \square$ with $(\tilde{A}, \tilde{B}) \to (\tilde{B}', B_0)$, all its inputs being available at every point of $\square$: (H0) from the coercivity of appendix H.1.1; the weighted bounds (W) of lemma H.11 for any regular field and any boundary condition; the pointwise bounds (K), needed only for the end factors, which are G_0 (since $n, n' \geq 2$), from lemma H.88 and (91) on $\square$ at every point; and the symmetry. The vertex lemma lemma H.14 holds with $(\tilde{A}, \tilde{B}) \to (\tilde{B}', B_0)$, both operators living on the same bonds and blocks of $\square$, with $\sigma = e_k(\sigma_0 + \sigma_1) \leq 1$. If an end factor were $\tilde{G}$ (n or $n' \leq 1$), pointwise bounds for $\tilde{G}$ at a non-constant field near $\partial\square$ would be needed; this case does not occur, the remainders being taken in the two-sided form with $n, n' \geq 2$ (part III, S14). □

Lemma H.111 (F4: "all necessary Ward–Takahashi identities", [3, pp. 430–431])**.** *The identities are those of lemma H.100 (appendix H.2.3): they cover every gauge-invariant polynomial F, every order in A, the limit of free boundary conditions (part (c)) and the zero-momentum form used on* [3, pp. 441–442] *(part (d)). The classes to which they are applied are the full fixed-order classes: all connected graphs of a given order and external-leg type with agreeing localisations (per label, as in appendix H.4.9).* [3, p. 431] *allows this explicitly for the vector self-energies, and every smallest class of* [3, pp. 429–431] *is contained in one; the consumers of Proposition 1 ([1, 2] and [13]) use only sums over all graphs, so Proposition 1 for the full classes is what is consumed.*

Proof. Lemma H.100; the class choice is a definition. □

Lemma H.112 (F5: "the remaining powers of $L^j\eta$ connected with the graph G_1 give us $(L^j\eta)^{D(G_1)}$", [3, p. 427])**.** *In* [3, (2.14)] *let $G_1 = \{l(1); v, v'\}$, $v \neq v'$, $j := j_{l(1)}$ the lowest index, so $j(v) = j(v') = j$ and $|\Delta(v)| = |\Delta(v')| = (L^j\eta)^d$. Perform $\sum_{\Delta(v')}$ against the exponential of $l(1)$ (an $O(1)$ sum, the cubes having side $L^j\eta$), then $\sum_{\Delta(v) \subset \Delta(v_1)}$ using the factor $(L^j\eta)^d$ of v, which produces the factor $(L^{j(v_1)}\eta)^d$ of the new vertex v_1 (the cubes are L-adically nested). The factors left over from G_1 are $(L^j\eta)^{d+p_{v'}}$ from v' and $(L^j\eta)^{p_v}$ from v, where p_v is the η-power of v excluding the volume factor η^d ("a proper power of $L^{j(v)}\eta$"); $(L^j\eta)^{-(d-2)/2}$ for each of the two legs of $l(1)$; and $(L^j\eta)^{-1}$ for each derivative on $l(1)$. The legs of the other lines at v, v' keep their own factors and now belong to v_1. The product is $(L^j\eta)^e$ with $e = d + p_v + p_{v'} - (d-2) - \delta_v - \delta_{v'} = D_{G_1}(v) + D_{G_1}(v') - d = D(G_1)$ by* [3, (2.1)–(2.2)]*. The general step* [3, (2.16)] *is the same count with v, v' replaced by shrunk vertices whose factors are those of* [3, (2.14)] *for the shrunk graph (lemma H.35, Step 1(iv)).*

Lemma H.113 (F6: "In the case $v = v'$ we make the same operations beginning with the summation over $\Delta(v)$", [3, p. 428])**.** *If $l(1)$ is a self-line at v, its exponential factor is 1 and there is no $\Delta(v')$-sum. $\sum_{\Delta(v) \subset \Delta(v_1)}$ uses $(L^j\eta)^d$ and produces $(L^{j(v_1)}\eta)^d$. The leftover is $(L^j\eta)^{p_v - (d-2) - \delta_v} = (L^j\eta)^{D_{G_1}(v) - d} = (L^j\eta)^{D(G_1)}$. Then $\sum_{j \leq j_{l(2)}} (L^j\eta)^{D(G_1)} \leq (1 - L^{-D})^{-1} (L^{j_{l(2)}}\eta)^{D(G_1)}$, since $D(G_1) \geq \frac{1}{2}$. This is* [3, (2.15)] *in that case (lemma H.35, Step 2).*

Proof of lemmas H.112 and H.113. Bookkeeping on [3, (2.14)], as stated; the two lemmas are the first step of lemma H.35 in Bałaban's notation. □

Lemma H.114 (F7: components separately; "We can extract also all coupling constants", [3, p. 433])**.** (a) *If the graphs of a class are unions of components sharing no line, the expression is the product of the component expressions: the vertex sums, the lines and the external fields factorise, and the class of a union is the product of the component classes by construction* [3, pp. 431–432]. (b) *Every vertex carries the monomial* $e_k^{d_v(v)}\lambda(t)^{d_s(v)}$ *times a bounded combinatorial factor depending on* $\bar{n}$ *only; all graphs of a class have the same orders* [3, p. 420]*, so* $e_k^{d_v(G_{\mathrm{ren}})}\lambda^{d_s(G_{\mathrm{ren}})}$ *factors out, the counterterm coefficients being defined graph by graph with the order of their graph* [3, p. 417]. (c) *For a connected class localised in a cube of side* $O(\bar{n})$ *(the cluster side* l_N *of appendix F),* $d(\{\square(v)\}) \le O(\bar{n})$*, so* $e^{-\delta_0 d} \ge e^{-O(\bar{n})\delta_0}$*. Hence* [3, (3.3)] *for the components, times their coupling monomials, gives* [3, (3.2)]*, and then* [3, (1.33)] *with the exponential factor from* [3, (3.1)] *(proposition H.54;* [3, p. 432]*).*

Proof. Bookkeeping, as stated. □

H.4.9. *S14.4: the classes after the* $\tilde{B}'$*-expansion,* [3, p. 433]. [3, p. 433]: the $\tilde{B}'$-expansion of a renormalised class gives an expression "which can be decomposed into a sum of renormalized classes". Part III proves (S14) that the untouched divergent subgraphs of an expanded graph are renormalised by terms of the expansion of the class, and that a touched divergent subgraph changes type: a 2ϕ subgraph touched once becomes a $2\phi 1A$ subgraph of degree 0, which needs the partner with its two external scalar legs exchanged; a $1A$ subgraph touched once, or a vacuum component touched twice, becomes a $2A$ subgraph of degree ≤ 0, which needs its Ward–Takahashi class. The closure property this requires of the class family is proved here.
Classes. Localisation labels are attached to each leg of a vector field (the support box of its partition function, a grid box of side ≤ 2), to each [3, (1.6)]/[3, (1.7)] vertex (its unit cube) and to each block factor [3, (1.13)–(1.15)] (the block $B^k(y)$); the counterterm graph $T_\gamma G$ of a 2ϕ subgraph γ carries the labels of G ([3, p. 417]: the legs are attached at one point, the internal vertices keep their localisations). Two graphs are "in agreement" [3, p. 432] if they carry the same label multiset, and the *per-label full class* of a given external type and order is the set of all connected graphs of that type and order with a given label multiset; equivalently, with the partition functions written as multipliers $t_y H_y$ on the vector legs and $t_Q \mathbf{1}_Q$ on the [3, (1.6)]/[3, (1.7)] vertices, it is the coefficient of one monomial $\prod t$ in the expansion of the generating functional [3, (1.5)]. This is the class choice of lemma H.111, permitted by [3, p. 431], and it is the one appendix F uses for the regrouping after the cut and for the renormalisation. After the $\tilde{B}'$-expansion the new vertices are labelled in the same way (one label per $\tilde{B}'$-leg).

Lemma H.115 (exchange of the external scalar legs)**.** *A per-label full class is closed under the exchange of the two external scalar legs of any* 2ϕ *or* $\phi\phi A$ *subgraph, the whole graph included.*

Proof. The class is a t-monomial coefficient of [3, (1.5)], and the Wick expansion sums over all pairings of legs. Exchanging which of two legs is paired with which outside slot is a bijection of pairings; it preserves orders and labels (labels sit on legs and vertices, not on pairings), and it maps graphs of the class to graphs of the class. For the whole graph the two external slots are the arguments of the external field $\Phi_{\mathrm{ext}}(x, x')$, and the sum over pairings contains both assignments. □

Lemma H.116 (the $\tilde{B}'$-Taylor coefficients of a class)**.** *Let* $\mathcal{C}$ *be the per-label full class of type* $1A$ *(resp. vacuum) and order* o*, at background* $\tilde{B} = B_0 + \tilde{B}'$ *on* $\square$*. Its first (resp. second)* $\tilde{B}'$*-Taylor coefficient is the per-label full class of type* $2A$ *and the same order, with one (resp. two) external legs equal to* $\tilde{B}'$ *carrying their partition labels; the combinatorial weights are those of the Taylor coefficients.*

Proof. On $\square$ the action $S(\phi; B_0 + \tilde{B}' + X)$ depends on $\tilde{B}'$ and on the other gauge-field legs X (A' with its g and partitions, $\tilde{A}$) only through the total field $B_0 + \tilde{B}' + X$ ([3, (1.4)]: both the kinetic term and the averaging term depend on the total field only), and $\tilde{B}'$-legs carry partition labels like every vector leg. Let $W(\tilde{B}', X) :=$ log of the Gaussian integral at background B_0; W is a function of $\tilde{B}' + X$, so $\partial/\partial\tilde{B}'(y) = \partial/\partial X(y)$ on every Taylor coefficient, label by label.

By the linked-cluster theorem (lemma H.46) the Taylor coefficients of W are the sums of all connected graphs, whose vertices are the joint Taylor coefficients of S in the total field and whose lines are $G_k(\Box, B_0)$. The $\tilde{B}'$-expansion of part III produces exactly these joint Taylor coefficients: insertions on lines are the Taylor coefficients of $G_k(\Box, B_0 + \tilde{B}')$, and expanded old vertices are the $\tilde{B}'$-Taylor coefficients of the old vertices; the factorised form of part III and the resolvent expansion [1, (3.44)] are two groupings of the same polynomials. So the first $\tilde{B}'$-coefficient of the $1A$ class is the coefficient of one X-leg and one $\tilde{B}'$-leg of $\partial^2 W/\partial X^2$, which is the $2A$ class, and the second coefficient of the vacuum class is the same with two $\tilde{B}'$-legs. (The parity statement that odd total orders in $(A, \tilde{B}')$ vanish plays no role here: the bijection holds for the full polynomials.) □

Lemma H.117 (compatibility with the cut and with the renormalisation)**.** (a) *The expanded class is again a per-label full class. Its cut on* □ *(the lines between different clusters of label boxes become external fields) is the cluster cut of appendix F on its label boxes, and the regrouping theorem there applies verbatim. A touched* 2ϕ *subgraph and its mirror (lemma H.115) have the same labels, as do a touched* $1A$ *subgraph and its* $2A$ *completion (lemma H.116); so partners stay in one block of the cut or are split together.* (b) *The renormalisation step (appendices H.4.6 and H.4.7 and Step 3(*D_4*) of appendix F) is applied to per-label full classes, the class choice of lemma H.111. The* $\tilde{B}'$*-legs are external fields* A *in the Ward–Takahashi identities (lemma H.100), which hold for every external field and every additional field* X*; after the gauge-out S18,* $B_0 = 0$ *and the lines are* $G_k(0)$*.*

Proof. (a) The clusters are determined by the label multiset, which the exchange and the completion preserve (the moved ends stay at vertices whose labels lie in the same cluster), so the cut set is unchanged. (b) As stated. □

S14.4, consumed form. With lemmas H.115 to H.117, the partners required by the touched subgraphs of part III are terms of the expansion of the class: the exchange partner of a touched 2ϕ subgraph S is $(S^\pi)_\tau = (S_\tau)^\pi$ with $S^\pi \subset G^\pi$ in the class (lemma H.115), and the $2A$ Ward–Takahashi class at the order of a touched $1A$ subgraph or vacuum component is the set of all single (resp. double) $\tilde{B}'$-attachments to all graphs of its type (lemma H.116). The expansion of a per-label full class is therefore a sum of per-label full classes, which is the decomposition "into a sum of renormalized classes" of [3, p. 433]; S15 ("again a sum of renormalized classes") inherits it. Bałaban's one sentence hides a closure property of the class family under differentiation in the background, which the per-label full classes have.

Appendix I. Corrections to King's papers used in the rotation part

We list the points in [13, 14] that affect the use of King's theorems in the rotation part. None of them affects the existence theorem (E_C) (Theorem 3.2), the existence of King's cube limits at zero scalar source, which is proved in section 5 (Remark 15.5). The point concerning the Wick square and the constant $\ln 5$ was treated in Remark 16.1.

I.1. **The large-field term in Theorem 3.1 of [13].** Theorem 3.1 of [13] compares the lattice generating functional on a torus T with the small-field integral $I^{(k)} := \int (dA_k)(d\phi_k)\, \chi_k \exp[-S^{(k),1}]$ of the effective action after k block-spin steps, $\eta := 5^k \varepsilon_K$. As printed on p. 655 the upper bound reads

$$Z^{\varepsilon_K}(T_{\varepsilon_K}, g, h) \le \int (dA_k)(d\phi_k)\, \chi_k \exp\big[-S^{(k),1} + C\eta^\sigma |T|\big] + \exp\big[-p(\eta)^2 + C|T|\big], \qquad (3.4)$$

with $p(\eta) = b_0(1 + \log \eta^{-1})^p$, followed by "Once again, C depends on g, h and $|T|$"; the lower bound (3.3) has the small-field integral with the exponent $\mp C\eta^\sigma |T|$ (in a lower bound the sign must be $-$). The second term of (3.4) is an additive large-field remainder; it is used in [13] only at fixed volume ((3.10)–(3.13) there), and in the same way in King's torus comparison [14, (4.7)], where it is harmless.

Proposition I.1 (the large-field term)**.** *Suppose (3.3)–(3.4) hold with C independent of $|T|$, and $I^{(k)} \geq e^{-C''|T|}$. Then*

$$-C\eta^\sigma \leq |T|^{-1} \log\left(Z/I^{(k)}\right) \leq \max\{C\eta^\sigma,\ C + C'' - p(\eta)^2/|T|\} + |T|^{-1} \log 2.$$

At fixed η the right side tends to $C + C''$ as $|T| \to \infty$, not to $O(\eta^\sigma)$.

Proof. $Z/I^{(k)} \leq e^{C\eta^\sigma|T|} + e^{-p(\eta)^2 + (C+C'')|T|}$ by (3.4) and the lower bound on $I^{(k)}$, and $\log(x+y) \leq \max(\log x, \log y) + \log 2$ for $x, y > 0$. The lower bound is (3.3). □

Hence the printed bounds do not give a two-sided representation of the free energy per unit volume by a small-field effective integral at a fixed coarse scale, uniformly in the volume, and arguments that need such a representation cannot be based on them. In the rotation part the volume is fixed wherever bounds of this type enter (the one-sided bounds (F) of Definition 17.5 are posed on one torus at a time), and the large-field term is absorbed on the left by a polynomial lower bound for the walled partition function (Remark 17.9). The constant of [13, Thm. 2.1(ii)] likewise depends on $|T|$, so King's theorems give no bound on torus functionals uniform in the torus; where such a bound is used (Lemma 17.3(d)) it is taken on a fixed torus and comes from the existence of the torus limit.

I.2. **King's torus conjecture: Theorem 2.4 of [14] as printed.** In [13, 14] the curvature source is $F(g) = \sum_p \varepsilon^3 g(p) F(p)$ with one function g on all plaquettes [14, (2.12)], and in Theorem 2.4 of [14] the source is rotated as a scalar. For a lattice Λ with frame (f_α) descending to D_J (Definitions 15.4 and 15.6) and a real function g on D_J put $F^{sc}_\Lambda(g) := \sum_p \varepsilon^3 g(c(p)) F(p)$, the sum over the plaquettes $p = (x; \alpha, \beta)$ of one fundamental cell, $c(p)$ their centres.

Proposition I.2 (the scalar reading)**.** *Let $e = 0$, $R := R_{12}(\theta_0)$, $\ell := 50MJ$ and $g(x) := \cos(2\pi x_1/\ell)$. Then*

$$\lim_{k\to\infty} \left\langle e^{F^{sc}_{\Lambda_k}(g)} \right\rangle_{\Lambda_k, D_J} \neq \lim_{k\to\infty} \left\langle e^{F^{sc}_{R\Lambda_k}(g)} \right\rangle_{R\Lambda_k, D_J},$$

where $R\Lambda_k$ carries the frame (Re_α) and descends to D_J by Lemma 15.3(b).

Proof. At $e = 0$ the photon is the centred Gaussian with covariance $\delta_{\alpha\beta}(-\Delta_\varepsilon + \mu_0^2)^{-1}$ on the torus, so $\langle e^{F^{sc}(g)} \rangle = e^{\frac{1}{2} \operatorname{Var} F^{sc}(g)}$. On a lattice with frame (f_α), $\sum_{\alpha<\beta} F(f_\alpha, f_\beta) = \langle u, B \rangle$ for a continuum field strength $F = dA$, $B = \star F$, where $u := \sum_{\alpha<\beta} f_\alpha \times f_\beta$. For the plane wave g only the momenta $\pm k_0$, $k_0 = (2\pi/\ell)e_1$, contribute, and the lattice symbols converge, so

$$\lim_{k\to\infty} \operatorname{Var} F^{sc}(g) = \frac{\ell^3}{2} \frac{|k_0|^2 \left(|u|^2 - u_1^2\right)}{|k_0|^2 + \mu_0^2}.$$

For Λ_k, $u = (1, -1, 1)$ and $|u|^2 - u_1^2 = 2$; for $R\Lambda_k$, $u = R(1, -1, 1) = (\frac{7}{5}, -\frac{1}{5}, 1)$ and $|u|^2 - u_1^2 = \frac{26}{25}$. The limiting variances differ by the factor $\frac{25}{13}$. □

With 2-form sources sampled in each lattice's own frame (Definition 15.4) the limit on either lattice is the variance of $\int \langle G, F \rangle$, which does not depend on the frame, and the obstruction disappears. This reading is consistent with King's proof, which treats the source components separately (Remark 15.5); it is the form of the torus comparison (K_0), $(\mathrm{K})_a$ (Theorem 17.1 and definition 17.2). The second error in the printed statement concerns the Wick constant (Remark 16.1). Since the printed statement is false on both counts, we refer to it as King's torus conjecture; what the rotation part uses is the torus comparison (K_0) (Theorem 17.1), in the corrected form. We note also that Theorem 2.4 is not, by itself, a statement about King's infinite-volume state: the passage from tori to the state [14, (2.24)–(2.25)] is explicitly conjectural there ("we can expect that a cluster expansion would give the following result"), and the rotation part does not use it (Remark 17.12).

I.3. **The bound (4.33) of [13].** In the proof of [13, Lemma 4.5] (p. 674) the positivity (4.33) is obtained through (4.36)–(4.37), which treat the block average as a Fourier multiplier; that step is false as printed. The conclusion holds. We use King's setting at the unit scale of step k; in this subsection a and a_j are King's coefficients of [13, (4.5)], not a scheme, and η is King's unit scale, not a tolerance. Ω is a periodic box of the unit lattice $\mathbb{Z}^3$ that is a union of blocks of side 5, the background vanishes, $\Delta^{(k)}$ and $\Delta^{(k+n)}$ are the translation-invariant operators of the flow on Ω, Q is the block average, so that $Q^*Q = P$ is the orthogonal projection onto block-constant functions, and $-\Delta$ is the lattice Laplacian of Ω. King's coefficients satisfy $a_j \geq \frac{24}{25}a$, and $\Delta^{(j)}(p)^{-1} = a_j^{-1} + \sum_l |u(p+l)|^2/\Delta^\eta(p+l)$ [13, (4.5)], where u is the symbol of the average over a unit block of the fine lattice of spacing η, $\sum_l |u(p+l)|^2 = 1$, the sum runs over $l \in 2\pi\mathbb{Z}^3$ with $p+l$ in the Brillouin zone of the fine lattice, and $\Delta^\eta(q) = \sum_\mu 4\eta^{-2}\sin^2(\eta q_\mu/2) + m_k^2$ with $m_k^2 \geq 0$.

Proposition I.3 (the bound (4.33))**.** *For $s \in [0,1]$, all k, n and $m_k^2 \geq 0$, the operator $X(s) := s\Delta^{(k)} + (1-s)\Delta^{(k+n)} + \frac{a}{25}Q^*Q$ on $\ell^2(\Omega)$ satisfies $X(s) \geq \gamma_0$, where*

$$\gamma_0 := \min\Big(4c_\Delta \sin^2\frac{\pi}{10},\ \frac{a}{25}\Big), \qquad c_\Delta := \Big(\frac{25\cdot 3\pi^2}{24a} + \frac{\pi^2}{4}\Big)^{-1}.$$

Proof. (i) For $|t| \leq \pi$, $4\sin^2(t/2) \geq 4t^2/\pi^2$, so $\Delta^\eta(q) \geq 4|q|^2/\pi^2$ on the fine Brillouin zone; and $|p_\mu + l_\mu| \geq |p_\mu|$ for $|p_\mu| \leq \pi$, $l_\mu \in 2\pi\mathbb{Z}$. With $\sum_l |u(p+l)|^2 = 1$ this gives $\Delta^{(j)}(p)^{-1} \leq a_j^{-1} + \frac{\pi^2}{4}|p|^{-2}$, and since $|p|^2 \leq 3\pi^2$ and $a_j \geq \frac{24}{25}a$, $\Delta^{(j)}(p) \geq c_\Delta|p|^2 \geq c_\Delta \sum_\mu 4\sin^2(p_\mu/2)$, the symbol of $-\Delta$. So $s\Delta^{(k)} + (1-s)\Delta^{(k+n)} \geq c_\Delta(-\Delta)$. (ii) On one block $B = \{0,\dots,4\}^3$ the Neumann Laplacian is a sum of three path-graph Laplacians, whose smallest non-zero eigenvalue is $\lambda_1 = 4\sin^2\frac{\pi}{10}$; hence $\sum_{x\in B}|\phi(x)-\phi_B|^2 \leq \lambda_1^{-1}\sum_{\{x,x'\}\subset B}|\phi(x)-\phi(x')|^2$, the sum over nearest-neighbour pairs, ϕ_B the block mean. Summing over the blocks, whose inner bonds are bonds of Ω, gives $\|(1-P)\phi\|^2 \leq \lambda_1^{-1}\langle\phi, -\Delta\phi\rangle$. (iii) Therefore $\langle\phi, X(s)\phi\rangle \geq c_\Delta\lambda_1\|(1-P)\phi\|^2 + \frac{a}{25}\|P\phi\|^2 \geq \gamma_0\|\phi\|^2$. For the photon the same holds componentwise. □

I.4. **Minor points.**

Remark I.4 (minor points). (i) In [14, (3.41)] the real part of a scalar source is split as $h_+ - h_-$ with $h_\pm \geq 0$ and no further regularity stated; the argument of Theorem 16.6(iii) uses smooth non-negative parts. (ii) The inequalities [14, (3.24)–(3.29)] are stated for ratios of expectations of $e^{r^2(h)+\mathrm{i}A(g)}$; the rotation part uses them in the form of [6, Thms. 6.1, 6.2], Lemma 2.7, which also covers killed sites and the two-level walls. (iii) King's bound [14, (3.43)] is asserted for the Neumann–Dirichlet cubes by "a standard argument (see [GJ1])", with no argument printed; it is not a bound on torus functionals, which lie above the cube functionals on the cone (Lemma 15.8). The torus moment constant of Lemma 17.3(d) is therefore taken at a fixed torus and comes from the existence of the torus limit under $(\mathrm{K})_a$, not from (3.43).

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